SHRI BRAHMAGUPTA VIRACITA

BRĀHMA-SPHUṬA SIDDHĀNTA

WITH

Vāsanā, Vijñāna and Hindi Commentaries

Vol. I

FOREWORD

By

DR. SAMPURNANANAD

Governor-Rajasthan

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Shri S. K. Patil

Union Minister for Railways

श्रीब्रह्मगुप्ताचार्य-विरचित

ब्राह्मस्फुटसिद्धान्तः

(संस्कृत-हिन्दी-भाषायां वासनाविज्ञानभाष्याभ्यां समलंकृतः सोपपत्तिकः)

प्रथमो भागः

प्राक्कथन :

डा० सम्पूर्णानन्द-राज्यपाल-राजस्थान

प्रधानसम्पादक :

आचार्यवर पंडित रामस्वरूप शर्मा

(सञ्चालक-इण्डियन इंस्टीटयूट आफ् ऐस्ट्रानौमिकल एण्ड संस्कृत रिसर्च)

प्रकाशक :

इण्डियन इंस्टीट्यूट आफ़ ऐस्ट्रानौमिकल एण्ड संस्कृत रिसर्च

२२३९, गुरुद्वारा रोड, क़रौलबाग़, न्यू देहली-५

प्रकाशक—
इंडियन इंस्टीट्यूट आफ़ ऐस्ट्रानौमिकल एण्ड संस्कृत रिसर्च
२२३६, गुरुद्वारा रोड, क़रौल बाग़,
नई दिल्ली-५ (भारत)

⊙

भारत सरकार के शिक्षा मन्त्रालय द्वारा
प्रदत्त अनुदान से प्रकाशित ।

⊙

सम्पादक मण्डल—
श्री रामस्वरूप शर्मा सञ्चालक
प्रधान सम्पादक
डा० सत्यप्रकाश डी० एस-सी०
प्रस्तावना-लेखक
श्री मुकुन्दमिश्र ज्योतिषाचार्य
श्री विश्वनाथ झा ज्योतिषाचार्य
श्री दयाशंकर दीक्षित ज्योतिषाचार्य
श्री ओंदत्त शर्मा शास्त्री
एम. ए., एम. ओ. एल.

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प्रथम संस्करण
१९६६

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मूल्य रु० ८०.००

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मुद्रक :
पद्म श्री प्रकाशन एण्ड प्रिण्टर्स
१२ चमेलियन रोड,
दिल्ली ।

समर्पण :

श्रीयुत एस० के० पाटिल

यूनियन मिनिस्टर फ़ार रेल्वेज़

को

सादर समर्पित

Foreword

The Indian Institute of Astronomical & Sanskrit Research has done a very useful piece of work by publishing the Brāhma Sphuṭa Siddhānta. This is an original work on Astronomy written about twelve hundred years ago. Mainly devoting himself to Astronomy as was natural, the author, Brahma Gupta, has also given considerable space to such branches of Mathematics as in his opinion are particularly applicable to Astronomy. Brahma Gupta is not a mere theoriser. His work bears ample evidence of close observation of astronomical phenomena, though he did not have at his disposal any large and well equipped Vedhālaya or Observatory. He has had to find fault with some of his predecessors. The main reason for his criticism was as stated by him:

ब्रह्मोक्तं ग्रहगणितं महता कालेन यत् श्लथीभूतम् ।
अभिधीयते स्फुटं तज्जिष्णुसुतब्रह्मगुप्तेन ॥

The creator Brāhma himself had given certain calculations in his book namely Brahma Siddhānta. But in the course of many years these calculations have been found to have bocome inaccurate. Of course, in referring to calculations, the author is referring to the bases of these calculations, the main data of astronomy. The excellent and fairly exhaustive introduction by Dr. Satya Parkash is of very great help in understending the book because it brings out clearly the various points on which Brahma Gupta has laid stress, particularly those on which he differs from other great astronomers.

A great controversy still rages between two schools of Indian astronomers which for want of better names may be called the schools of Arya Bhata and Brahma Gupta. Some years ago the Indian Institute of Astronomical & Sanskrit Research published the Vateshwar Siddhānta which is supposed to be associated with the Arya Bhata School. It is, therefore, in the fitness of things that it should now publish the present standard volume of the other school.

To my mind the Publication of such books is of great value from two points of view. It helps to remove the ignorance of average educated Indian of today about the achievements of his ancestors. In the field of two important branches of science, Mathematics and Astronomy, it is really surprising how those ancient astronomers could reach such heigts of accuracy with the help of instruments which seem to be laughably crude compared to those that modern astronomers have at their disposal. At the same time a study of such literature is a very salutary corrective

to the vanity of those who feel that in astronomy as in certain other subjects the last word was said by those writers who have written in Sanskrit. We learn how far behind we are in certain very important directions in the field of knowledge.

It is interesting by the way to find that the author of this great work was a Vaishya. This shows how diffused knowledge even of the most abstract subjects was in those days and how Scholarship was honoured irrespective of caste. Not one the critics of Brahma Gupta has taunted him on the ground of his not being a Brahmana.

Sampurnanand
Governor-Rajasthan

Raj Bhavan,
Jaipur,
Dated May 4, 1966.

—:o:—

Observations

of

Dr. K. M. Munshi

President of the Reception Committee

The Author Shri Ram Swaroop Sharma has given to the world of scholars an excellent and valuable treatise on the famous Astronomer Brahma-Gupta, son of Jisnugupta, born in 589 A. D. in Bhillamala or Srimala now in Southern Marwad close to the northern frontier of modern Gujarat. The work is divided into 13 chapters and at the end of each, the learned author has given reference to other works and treatises for further study as also in support of his various conclusions. He has described Astronomy in Ancient India in the light of the knowledge of the subject available in different countries in his time.

After giving personal details of Brahma-gupta, the author starts on a critical study of the works of that Astronomer-particularly his *Brahma-sphuta-siddhānta* and *Khandana-Khādya*. The author has also discussed the Indian Luni-solar Astronomy, the Greek and Hindu methods in spherical Astronomy and the Epicyclic theory of ancient Indians. Discussing in detail the contribution of Brahma Gupta in Arithmetic, Algebra, in Astronomy and the Astronomical instruments that he used, the author has brought out the various highlights of his work and achievements in comparison with those of his predecessors, successors and contemporaries in other countries.

The author deserves to be congratulated for this scholarly contribution on the subject of Ancient Indian mathematics and Astronomy.

—*K.M. Munshi*

Bhartiya Vidya Bhawan
Bombay

Publisher's Note

It is with pride and pleasure that the Indian Institute of Astronomical & Sanskrit Research brings out its second monumental work 'Brāhma Sphuṭa Siddhānta'. The Institute, set up on the 23rd November, 1957, with the object of promoting research into ancient Indian manuscripts on Astronomy and allied Sciences, started its programme of publication with the preparation of a critical edition of Vateshwara Siddhānta running into more than 700 pages. It was far back in early forties that the idea of reviving the treasures of knowledge hidden in ancient manuscripts came to my mind. In 1945 I could succeed in editing and publishing 'Trailokya Prakāśa' of Acharya Hema Prabha Sūri.

Ālbruni's Travel Accounts of India contain a reference to Vateshwara Siddhānta and this fact was responsible for prompting me to arrange publication of Vateshwar Siddhanta. It was rather difficult to search out this manuscript and the clue given by Mahamahopadhyaya Sudhākar Dwivedi (a great Mathematician and Astronomer of 19th century) that the manuscript was available in Gwalior did not lead me to any results. I kept my search on and found a copy of this text in the collection of manuscripts inherited by a Brahman widow of Gujranwala District in west Pakistan. The same style of copy was later seen by me in the Panjab University Library at Lahore.

Vateshwarāchārya, the great writer of Vateshwara Siddhānta, has criticised Brāhma Sphuṭa Siddhānta of Brahma Gupta, in his work. This naturally aroused my curiosity to procure and publish Brahma Sphuṭa Siddhānta so as to provide right perspective to the students of ancient Indian Scientific literature. Moreover, in one of my discussions with the late Dr. K. S. Krishnan, Director, National Physical Laboratories, New Delhi, he referred to Dr. Colebrook (a German Writer) as having translated into English two chapters of 'Brāhma Sphuṭa Siddhānta', namely, Vyakta (Arithmetic)and Ayakta(Algebra). The late Dr. Krishnan showed keen interest in this manuscript and encouraged me in taking up preparation of a critical edition of this text. The Institute is grateful to the Govt. of India (Department of cultural Affairs) for meeting part of the expenditure on this publication and thereby enabling us to complete our task.

I shall be failing in my duty if I do not mention the great and invaluable contribution made for the success of the Institute by its founder President the late Shri Brijlal Nehru. It was under his able guidance that the Institute started functioning and its various programmes were finalised. His passing away has been a big and irreparable loss to us The Institute will keep his memories always alive by following his ideals and by executing with zeal the programmes inspired by him.

I must take this opportunity to express my heartfelt gratitudes to Shri B. B. Varma, Member Parliament, who has never shirked in shouldering the responsibilities of managing the affairs of the Institute first as a member of its Executive Council and now as its President. I am also thankful to Dewan Hari Krishan Das, Chairman of the Executive Council and Shri F.C. Bedi Treasurer of the Institute for their ungrudging support to the cause of the Institute.

My thanks are also due to Dr. Sampurnanand, Governor Rajasthan, who has blessed the Institute by inaugurating it and has now been kind to write a foreword for the present book. I am also indebted to Dr. Satya Prakash, Head of the Deptt. of Chemistry, Allahabad University for his having spared time to write an introduction in English for this publication. He has always been cheerfully allowing us to draw on his rich experience and wealth of knowledge. I am also grateful to Pt. Mukund Mishra, Pt. Vishwa Nath Jha, Shri Daya Shankar Dixit and Pt. Om Datt Sharma, the learned scholars who helped me in preparing this work.

The book has been published in four volumes. Volume I includes introduction in English by Dr. Satya Prakash followed by the text as given in the copy of the manuscript obtained from the Bhandarkar Research Institute, Poona, with footnotes to collate the various versions found in the copies of the manuscripts procured from the Royal Asiatic Society, Bombay, the Oriental Research Institute, Baroda and the Vishweshwaranand Vedic Research Institute, Hoshiarpur. The volume also contains a Sanskrit Bhumika from the Chief Editor and an Index of Shlokas. Volume II comprises of the first nine chapters of Brāhma Sphuṭa Siddhānta containing only those readings which we have considered as correct. It also includes Vāsanā commentary of Pṛthūdaka Swāmi and Vigyān Bhashya in Sanskrit with Hindi translation. Volume III includes chapters 10 to 16. This volume also contains Nutan Tilak commentary of Sudhakar Dwivedi and Vigyan Bhashya in Sanskrit with Hindi translation. Volume IV comprises of chapters 17 to 24 and also an Index of Shlokas contained in Volumes II. III and IV. It also includes an appendix on Dhayāna Grhaopadeshādadhyāya. Another appendix in this volume contains Vāsanā commentary on Golādhyāya.

The Institute has planned to add to its list of publications, shortly, Samrāṭ Siddhānta (3000 pages and already in press). Bṛhad Yavan Jātaka (2000 pages) and Panch Siddhāntikā (1600 pages).

The Indian Institute of Astronomical & Sanskrit Research considers it a proud priviledge to dedicate Brāhma Sphuṭa Sidhānta to Shri S. K. Patil, Union Minister for Railways. I am highly grateful to him for his having consented to this dedication. I am also grateful to Shri K. M. Munshi and Shri Wadilal Chaturbhuj Gandhi for their kindly agreeing to my request to be President and Chairman respectively of the Reception Committee. My heart felt thanks are also due to Shri Kanti Lal H. Shah. Hony. Secy. of the Organising Committee and to all the members of the organising and the Reception Committees for their help and cooperation in my work. I also owe a debt of grati-

tude to the Patrons of the Institute : His Majesty the King of Nepal, Abhinaya Samrat, Shri Prithvi Raj Kapoor, Shri Dhairyasinh, R. Morarji, Chairman Dharamsi Morarji & Co., Bombay, Shri Pranlal Bhogilal, Patel Bombay for their invaluable help. I am also grateful to the life members of the Institute : H.H. Maharaja Manabendra Shah of Tehri Garhwal, Yuvraj P.C Dev, Shri B.N. Bhaskar, Shri R.K. Batra, Shri Sada Jivatlal Chandulal, Shri Shiv Kumar Bhuvalka, Shri Jamna Das H. Moorjani, Shri Shantilal K. Somaiya, Shri Kisonlal M. Diwanji, Solicitor, Shri N. K. Jalan, Shri D. R. Nayar, Shri Vadilal C. Gandhi, Shri Devi Prasad Khandelval, Shri Ram Prasad Khandelval, Shri B. K. Jalan, Shri Amir Chand T. Gupta, Shri Laxman Vaman Apte, Smt. Padma Koregaonkar, Shri Yodh Raj Bhalla for their support to the Institute.

I must make a special mention here of shri A. N. Jha, Chief Commissioner, Delhi, who has greatly encouraged me in my task of publishing old manuscripts. His help in solving the various problems faced by me, has been invaluable. I am highly grateful to him for his support in my cause. I am also grateful to H. H. Maharaja Manabedra Shah of Tehri Garhwal, Seth Sadajivat Lal Smti. Padma Koregaonkar, Lala Yodh Raj Bhalla, Shri F.A Fazalbhai, Shri Murar J. Vaidya, Shri N. J. Aggarwal, Shri P. S. Naulakha, Shri R. L. Maheshwari, Shri D. N. Bhattacharjee, Shri P. A. Narialwala, Shri P. M. Aggrawal, Shri Shiv Kumar Bhuvalka, Shri Navin Bhai Khandwala and Smti. Nirmala Gomte, Lala Jagan Nath Ji, Shri L.S. Aggarwal, Shri B.N. Saxena, Shri R.B. Shah, Shri L.O. Joshi, Shri T.S. Krishnamurti, Shri R. S. Chitkara, Dr. Ram Karan Sharma, Shri V.P. Agnihotri, Shri K.L. Handa and Shri K.G. Somayya for their help and advice at various occasions.

Ram Swarup Sharma

Bombay
17-5-1966.

—:o:—

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भूमिका

ब्रह्मगुप्तकृतो ब्राह्मस्फुटसिद्धान्तः

श्रीचापवंशतिलके श्रीव्याघ्रमुखे नृपे शकनृपाणाम्।
पञ्चाशत्संयुक्तैर्वर्षशतैः पञ्चभिरतीतैः ॥
ब्राह्मः स्फुटसिद्धान्तः सज्जनगणितज्ञगोलवित्प्रीत्यै।
त्रिंशद्वर्षेण कृतो जिष्णुसुतब्रह्मगुप्तेन ॥

इति ब्राह्मस्फुटसिद्धान्तीयसंज्ञाध्याये आचार्योक्त्या ५२० शाकवर्षे-आचार्यस्य (ब्रह्मगुप्तस्य) जन्म समयस्ततः परं त्रिंशति वर्षेषु गतेषु तेन ब्राह्मस्फुटसिद्धान्तसंज्ञको ज्यौतिषसिद्धान्तग्रन्थो विरचितः[1]। गुर्जरदेशमध्ये भिनमाल नाम ग्राम एवास्य जन्मस्थानं बहूनां पाश्चात्यानां गवेषणया सिध्यति। गुर्जरदेशीयज्योतिर्विदां मुखकथातोऽपि भिल्लमालकोऽधुना भिनमालनाम्ना प्रसिद्धो ग्राम एवाऽऽचार्यस्य जन्मस्थानं सिध्यति। गुर्जरदेशोत्तरसीमायां मालव (माड़वार) देशतो दक्षिणस्यां दिशि आबूपर्वतलूणीनद्योर्मध्ये तत्पर्वताद्वायुकोणेऽयं भिनमालनामा ग्रामोऽधुना प्रसिद्धोऽस्ति।

विष्णधर्मोत्तरपुराणान्तर्गतं ब्रह्मसिद्धान्तमागमीकृत्य ब्राह्मस्फुट सिद्धान्तो ब्रह्मगुप्तेन विरचितः। नलिकादिवेधद्वारेण दृग्गणितैक्यकारि—ग्रहादि साधनकारणात्प्राचीनं ब्रह्मसिद्धान्तं संशोध्य नवीनं ब्राह्मस्फुटसिद्धान्तं रचितवान् ब्रह्मगुप्तः।

ब्रह्मोक्तं ग्रहगणितं महता कालेन यत् खिलीभूतम्।
अभिधीयते स्फुटं तज्जिष्णुसुतब्रह्मगुप्तेन ॥
संसाध्य स्पष्टतरं बीजं नलिकादियन्त्रेभ्यः।
तत्संस्कृतग्रहेभ्यः कर्त्तव्यौ निर्णयादेशौ ॥

इत्याद्युक्त्या स्फुटं ज्ञायते। अस्य चतुर्वेदाचार्यपृथूदककृता तिलकसंज्ञिका टीका प्रसिद्धाऽऽसीत् साऽधुना सम्पूर्णा सम्प्रति नोपलभ्यते। 'कोलब्रूक' साहेबमहोदयैः सम्पूर्णा सा टीको (पृथूदक कृता) पलब्धा तद्वलेनास्य ग्रन्थस्य द्वादशाष्टादशाध्याययोर्व्यक्ताव्यक्तगणितसंज्ञयोराङ्गलभाषायामनुवादः १७६९ शाकवर्षे

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१. गुप्तपदान्ताद्वैश्यकुलोत्पन्नो रीवांनरेशस्य व्याघ्रभटेश्वरस्य प्रधानज्यौतिषिक आसीदयमिति बहूनां मतमस्ति।

(१८१७ खीष्टाब्दे) कृत इत्युपलभ्यते। अत्र ब्राह्मस्फुटसिद्धान्तेऽष्टाधिक सहस्र-१००८मिता आर्याः सन्ति, तथा पूर्वार्धे १. मध्यगतिः। २. स्पष्टगतिः। ३. त्रिप्रश्नाध्यायः। ४. चन्द्रग्रहणाध्यायः। ५. सूर्यग्रहणाध्यायः। ६. उदयास्तमयाध्यायः। ७. चन्द्रशृङ्गोन्नत्यध्यायः। ८. चन्द्रच्छायाध्यायः। ९. ग्रहयुत्यध्यायः, १० भग्रह-युत्यध्यायः, इति दशाध्यायाः सन्ति। परार्धे च १. तन्त्र परीक्षाध्यायः। २. गणिताध्यायः। ३. मध्यगत्युत्तराध्यायः। ४. स्फुटगत्युत्तराध्यायः। ५. त्रिप्रश्नोत्तरध्यायः। ६. ग्रहणोत्तराध्यायः। ७. छेद्यकाध्यायः। ८. [illegible]-त्युत्तराध्यायः। ९. कुट्टाकाराध्यायः। १०. छन्दश्चित्युत्तराध्यायः। ११. गोला-ध्यायः। १२. यन्त्राध्यायः। १३. मानाध्यायः। १४. संज्ञाध्यायः। इति [illegible]-ध्यायाः सन्ति, पूर्वार्धपरार्धाध्याययोर्योगेन चतुर्विंशतिरध्याया अत्र ग्रन्थे सन्ति। तेषु तन्त्रपरीक्षाध्यायो विचारार्हः। अत्र बहूनामाचार्याणां नामानि मतानि चाचार्येणोल्लिखितानि। यथा—

लाटात् सूर्यशशाङ्की मध्याविन्दूच्च चन्द्रपातौ च।
कुजबुध शीघ्रबृहस्पति सितशीघ्र शनैश्चरान् मध्यान्॥
युगयातवर्षभगणान् वासिष्ठाद्विजयनन्दिकृतपादात्।
मन्दोच्चपरिधि-पात-स्पष्टीकरणाद्यमार्यभटात्॥
श्रीषेणेन गृहीत्वा रक्षोच्चयरोमकः कृतः कन्था।
एतानेव गृहीत्वा वासिष्ठो विष्णुचन्द्रेण॥
अनयोर्न कदाचिदपि ग्रहणादिषु भवति दृष्टिगणितैक्यम्।
यद्भवति तद्घुणाक्षरमतोऽस्फुटाभ्यां किमेताभ्याम्॥

एभिः श्लोकैः श्रीषेणाचार्यकृतो रोमकसिद्धान्तः, विष्णुचन्द्रकृतश्च गो वासिष्टसिद्धान्तस्तयोर्दूषणमाहेति टीकाकृतश्चतुर्वेदाचार्यस्य कथनमस्ति। पञ्च-सिद्धान्तिकायां श्रीषेण विष्णुचन्द्रयोरनुल्लेखात् वराहमिहिराचार्यानन्तरं ब्रह्म-गुप्ततः पूर्वमेतौ ज्यौतिषसिद्धान्तग्रन्थरचयितारावास्ताम् ४२७ शाकवर्षात् पश्चात् ५५० शाकवर्षात्पूर्वं तयोः (श्रीषेण विष्णुचन्द्रयोः) रोमकसिद्धान्तो वासिष्ट-सिद्धान्तश्चातिगणितस्थौल्यमापन्नाविति स्वयं वेधद्वारा स्थिरीकृत्यैवाऽऽचार्येण (ब्रह्मगुप्तेन) प्रौढोक्त्या 'यद्भवति तद्घुणाक्षर'मित्यादि कथ्यते। अत्रार्यभटस्य सिद्धान्तः सर्वत्रैव दोषावह एवेति वदन्नप्याचार्यो (ब्रह्मगुप्तः) बहुधैव तदुक्तीरेव खण्डयितुं ग्रन्थमरचयत्। यथा भूभ्रमणं खण्डयति—

प्राणेनैति कलां भूर्यदि तर्हि कुतो व्रजेत् कमध्वानम्।
आवर्त्तनमुर्व्याश्चेन्न पतन्ति समुच्छ्रयाः कस्मात्॥

पृथिव्याश्चलत्वं भगणानां स्थिरत्वं स्वीकृत्य—अहोरात्रासुभिः स्वाक्षोपरि पृथिव्या भ्रमणं मन्यते आर्यभटेनेति–आवर्त्तनमुर्व्याश्चेदित्याद्युक्तिः। अन्यत्रानेका अत्युक्तयो दुराग्रहवशात् कुत्रचिद्वाग्बलेन कथिता ब्रह्मगुप्तेन यथा—

स्वयमेव नाम यत्कृतमार्यभटेन स्फुटं स्वगणितस्य ।
सिद्धं तदस्फुटत्वं ग्रहणादीनां विसंवादात् ॥
जानात्येकमपि यतो नार्यभटो गणितकालगोलानाम् ।
न मया प्रोक्तानि ततः पृथक् पृथक् दूषणान्येषाम् ॥
आर्यभटदूषणानां संख्या वक्तुं न शक्यते यस्मात् ।
तस्मादयमुद्देशो बुद्धिमताऽन्यानि योज्यानि ॥

स्वयं ग्रहग्रहणादिवेधकर्त्ता ब्रह्मगुप्तः प्राचीनाचार्येभ्योऽनेकान् विशिष्टान् ग्रहादिसाधनविधीन् गणितसत्यासत्यपरीक्षार्थं वेधविधींश्च स्वग्रन्थे प्रौढोक्त्या प्रतिपादयति—

ज्ञातं कृत्वा मध्यं भूयोऽन्यदिने तदन्तरं भुक्तिः ।
त्रैराशिकेन भुक्त्या कल्पग्रहमण्डलानयनम् ॥
यदि भिन्नाः सिद्धान्ता भास्करसंक्रान्तयोऽपि भेदसमाः ।
स स्पष्टः पूर्वस्यां विषुवत्यर्कोदयो यस्य ॥

इत्यादिना वास्तवविचारासक्तः सर्वप्रथमं सर्वतो विशिष्टं विवेचनाबहुलं सिद्धान्तग्रन्थं ब्रह्मगुप्त एव रचितवानित्युपलब्धज्यौतिषसिद्धान्तग्रन्थेभ्यो ज्ञायते। 'कृती जयति जिष्णुजो गणकचक्रचूडामणि' रितिस्वसिद्धान्तशिरोमणेर्गणिताध्यायारम्भे भास्कराचार्येण ब्रह्मगुप्तमभिवन्द्य तत्परमपि बहुत्रस्थले ब्रह्मगुप्तमतोल्लेखं कुर्वताऽपि 'यथाऽत्रग्रन्थे ब्रह्मगुप्तस्वीकृतागमोऽङ्गीकृत' इत्युक्त्या तद्ग्रन्थानुसारी ग्रन्थो रचित इति प्रतीयते। ब्रह्मगुप्तेनायनचलनं नोपलब्धमिति ब्रह्मस्फुटसिद्धान्ततो ज्ञायते। अयनचलनोपलब्धिः प्रत्युत खण्डितेति दृश्यते यथा—

परमल्पामिथुनान्ते द्युरात्रिनाड्योऽर्कगतिवशाद्गतवः ।
नायनयुगमयनवशात् स्थिरमयनद्वितयमपि तस्मात् ॥

वराहमिहिरोऽयनचलनसम्बन्धे सन्दिहान आसीदिति 'नूनं कदाचिदासीद्येनोक्तं पूर्वशास्त्रेषु' इति तत्कथनेन ज्ञायते। तत्समयेऽश्विन्यादौ क्रान्तिपात आसीदित्यश्विन्यादि नक्षत्रगणना प्रवृत्ता सैव ब्रह्मगुप्तात्परमद्यावधि प्रचलति।

पञ्चषष्टचा वर्षैः क्रान्तिपातः प्राय एकमंशं पश्चिमदिश्यपसरति तज्ज्ञानं स्वल्पकालेनासम्भव इवेति वेधविज्ञेनापि ब्रह्मगुप्तेनायनचलनं नोपलब्धमिति । आर्यभटविरोधी भूत्वापि ब्रह्मगुप्तो ब्राह्मस्फुटसिद्धान्तरचनानन्तरं सप्तत्रिंशत्तमे वर्षे खण्डखाद्यकं नामकरणग्रन्थं रचितवान् यथा तत्प्रारम्भे—

प्रणिपत्य महादेवं जगदुत्पत्तिस्थितिप्रलयहेतुम् ।
वक्ष्यामि खण्डखाद्यकमाचार्यार्यभटतुल्यफलम् ॥
प्रायेणार्यभटेन व्यवहारः प्रतिदिनं यतोऽशक्यः ।
उद्ग्राहजातकादिषु तत्समफललघुतरोक्तिरतः ॥

इति तदुक्तग्रन्थपर्यालोचनया ज्ञायते यत्सर्वत्र व्यवहारकृतां मानवानां मध्ये प्रचरितार्यभटमत निराकरणमतीव कठिनमासीदिति—'आर्यभटमतानुसारिव्यवहरतां तात्कालिकजनानामुपकारायैव व्यावहारिकः करणग्रन्थः खण्डखाद्यकनामको रचितो ब्रह्मगुप्तेन । यथोपलब्धेषु प्राचीनज्यौतिषसिद्धान्तग्रन्थेषु ब्राह्मस्फुटसिद्धान्त आदर्शरूपो ग्रन्थस्तथैव सर्वेषां करणग्रन्थानामादर्शरूपमादिमं वा इतस्त्रयोदशशतवर्षैः प्राचीनं खण्डखाद्यकं करणम् । अथ 'ब्रह्माह्वयश्रीधरपद्मनाभबीजानि यस्मादति विस्तृतानि' इति बीजगणिते भास्करोक्त्या ज्ञायते यद्ब्रह्मगुप्तस्य विपुलो बीजगणित ग्रन्थ आसीत्परमयं ग्रन्थः कुत्रापि नोपलभ्यते न वा श्रूयते । ब्रह्मगुप्त एव सर्वापेक्षया श्रीपतेरादर्शभूतः । ब्राह्मस्फुटसिद्धान्तसिद्धान्तशेखरयोः पर्यालोचनया ज्ञायते यद् ब्रह्मगुप्तोक्ताः संक्षिप्ता बहुलार्थयुक्ता आर्या एव बृहदाकारैश्छन्दोभिरनूदिताः श्रीपतिना । वस्तुतो ब्रह्मगुप्तोक्तं ग्रहगणितं सूक्ष्मं ज्ञात्वा सत्यमित्र तदेव स्वीकुर्वन् श्रीपतिस्तदुक्तिवैषम्यं स्वसुन्दररचनाभिरपहरन् सुगमतरं ग्रन्थांतरं (सिद्धान्त शेखरं) चकार, नात्र केषामपि विप्रतिपत्तिः । ग्रन्थरचनासम्बन्धे श्रीपतेर्विशेषतो लल्लाचार्य एवादर्शः । ये केचन विषया ब्रह्मगुप्तेन न कथिता अथ लल्लेन कथितास्तान् सर्वानेव नियतमेव श्रीपतिः श्लोकान्तरेण तथैव कथितवान् । वस्तुतो द्वयोर्ग्रन्थयोः (ब्राह्मस्फुट सिद्धान्तशिष्यधीवृद्धिदयोः) परिशीलनं कृत्वा श्रीपतिना सिद्धान्तशेखरो रचितः । ब्राह्मस्फुट सिद्धान्ते येषामाचार्याणां नामानि समागतानि सन्ति तेषां सम्बन्धे किञ्चिद्विविच्यते । सर्वेषां सिद्धान्तानामादिमः सर्वेभ्यः प्राचीनो वा ब्रह्मसिद्धान्त एव, स एव कैश्चित् पितामहसिद्धान्तनाम्नाप्युच्यते । पञ्चसिद्धान्तिकायां वराहमिहिरेण द्वादशोऽध्याय आर्यापञ्चात्मकः पैतामहसिद्धान्तः कथितो यथा—

रविशशिनोः पञ्चयुगं वर्षाणि पितामहोपदिष्टानि ।
अधिमासस्त्रिंशद्भिर्मासैरवमस्त्रिषष्ट्याऽह्नाम् ॥

द्व्यूनं शकेन्द्रकालं पञ्चभिरुद्धृत्य शेषवर्षाणाम् ।
द्विगुणं माघसिताद्यं कुर्याद् युगणं तदन्ह्युदयात् ॥२॥
सैक षष्ट्यंशे गणे तिथिर्भमार्कं नवाहतेऽक्ष्यर्कैः ।
दिग्रसभागैः सप्तभिरूनं शशिभं धनिष्ठाद्यम् ॥३॥
प्रागर्धे पर्व यदा तदोत्तराऽतोऽन्यथा तिथिः पूर्वा ।
अर्कघ्ने व्यतिपाता युगणे पञ्चाम्बरहुताशैः ॥४॥
द्व्यग्निनगेषूत्तरतः स्वमितमेष्य दिनमपि याम्यायनस्य ।
द्विघ्नं शशिरसभक्तं द्वादशहीनं दिवसमानम् ॥५॥

एतदनुसारेणैकस्मिन् युगे सौरवर्षाणि=५, सौरमासाः=५×१२=६० । अधिमासौ=२ । चान्द्रमासाः=६२ एते त्रिंशद्गुणितास्तिथयः=१८६०, अवमानि=३०, एभी रहितास्तिथयोऽहर्गणः=१८३० इति ॥

पञ्च सिद्धान्तिकायां 'पौलिश सिद्धान्तः' इति नाम्ना एकोऽध्यायो वर्त्तते, ब्राह्मस्फुट सिद्धान्तटीकायां पृथूदकेन बहून्येव पौलिशसिद्धान्तवचनानि प्रमाणत्वेनोद्धृतानि सन्ति । अस्य सिद्धान्तरचयितुः सम्बन्धे बहूनि मतान्तराणि सन्ति । वराहोक्तपौलिशसिद्धान्ते यवनपुरात् उज्जयिन्या वाराणस्याश्च देशान्तरमुल्लिखितमस्ति । यथा —

भवनाच्चरजा नाडयः सप्तावन्त्यां त्रिभागसंमिश्राः ।
वाराणस्यां त्रिकृतिः साधनमन्यत्र वक्ष्यामि ॥

शाकल्यसंहितोक्त ब्रह्मसिद्धान्ते पुलिशसिद्धान्तोल्लेखो वर्त्तते । ब्राह्मस्फुटसिद्धान्तटीकायां पृथूदकेन 'देशान्तर रेखा च पौलिशे पठ्यते' इत्युक्तम् । तथा च पुलिशाचार्यः—

उज्जयिनी रोहीतककुरुयमुनाहिमनिवासमेरूणाम् ।
देशान्तरं न कार्यं तल्लेखामध्यसंस्थदेशेषु ॥'

इत्यादि विचारेण पौलिशसिद्धान्तः सर्वमान्य आसीदिति प्रतिभाति । परमयं सिद्धान्तः साम्प्रतं नोपलभ्यते इति ।

सूर्य सिद्धान्त एव प्राचीनतमः सर्वप्रथमः सिद्धान्तग्रन्थ इति बहूनां विदुषां मतम् ।

'केचित्प्रत्यक्षसूर्याच्च भिन्नोऽयमिति यद्वलात् ।
वदन्ति मूढवादस्याप्रामाण्यात्तदसद्ध्रुवम् ॥'

इति कमलाकरोक्तया स्वयं भगवान् सूर्य एवास्य रचयिता। आश्चर्यस्य विषयोऽयमस्ति यदत्रा (सूर्यसिद्धान्ते) यनांशानयनम्।

'त्रिंशत्कृत्यो युगे भानां चक्रं प्राक् परिलम्बते।
तद्गुणाद्भूदिनैर्भक्ताद् द्युगणाद्यदवाप्यते॥'
'तद्दोस्त्रिघ्ना दशाप्तांशा विज्ञेया अयनाभिधाः।
तत्संस्कृताद् ग्रहात् क्रान्तिच्छाया चरदलादिकम्॥'

इत्यनेन कृतमस्ति। परं ब्राह्मस्फुटसिद्धान्तनिर्मात्रा ब्रह्मगुप्तेन तस्य (अयनांशस्य) चर्चाऽपि न कृता। कथं ब्रह्मगुप्तेन तच्चर्चा न कृतेति न ज्ञायते। सूर्य सिद्धान्तस्योदयास्ताधिकारे 'अभिजिद् ब्रह्महृदयं स्वातीवैष्णव वासवाः' इत्यादिना सदोदित नक्षत्राणि भगवता सूर्येण कथितानि सन्ति, अस्य श्लोकस्य सुधावर्षिणी टीकायां 'देशज्ञानं विना सदोदित नक्षत्राणां ज्ञानं न भवति निरक्षे च सौम्यध्रुवोऽप्यदृश्योऽतः केनचिद्गोलानभिज्ञेनायं श्लोकः प्रक्षिप्त, इति यल्लिखितमस्ति तत्समीचीनं नास्ति। पाताधिकारे पातस्थितिकालस्य फलं—

आद्यन्तकालयोर्मध्यः कालो ज्ञेयोऽतिदारुणः।
प्रज्वलज्ज्वलनाकारः सर्वकर्मसु गर्हितः॥
एकायनगतं यावदर्केन्द्वोर्मण्डलान्तरम्।
सम्भवस्तावदेवास्य सर्वकर्मविनाशकृत्॥
स्नानदानजपश्राद्धव्रतहोमादिकर्मभिः।
प्राप्यते सुमहच्छ्रेयस्तत्कालज्ञानतस्तथा॥

इत्यनेन पातस्थित्रिकालः सर्वकर्मविनाशकृदुक्तः। पातकाले स्नानदानजपश्राद्धव्रतहोमादिकर्मभिर्महत्कल्याणं प्राप्यते लोकैरिति। तथा च—

रवीन्द्वोस्तुल्यता क्रान्त्योर्विषुवत्सन्निधौ यदा।
द्विर्भवेद्धि तदा पातः स्यादभावो विपर्ययात्॥

इत्यनेनापूर्वविषयः कथितोऽर्थाद्रविगोलसन्धिसमीपे यदा रविचन्द्रयोः क्रान्तिसाम्यं भवेत्तदाऽल्पेनैव कालेन द्विवारः पातः स्यात्। यदा रव्ययनसन्धिसमीपे क्रान्तिसाम्याभावस्तदा बहुकालपर्यन्तं क्रान्ति साम्याभावः स्यादिति।

ब्राह्मस्फुटसिद्धान्तेऽपि पातस्थितिकालफलं सूर्यसिद्धान्तोक्तवत्कथितम्। सिद्धान्तशेखरे कियत्कालपर्यन्तं पातफलमिति 'भानोर्बिम्बं तुहिनकिरणापक्रमैरेकमार्गे यावत्तावन्मुनिभिरुदितसंभवस्तत्फलस्य। तस्याभावे भवति नियतं

तत्फलस्याप्यभावो यात्रोद्वाहादिषु पुनरिह द्युत्रयं नैव दुष्टम् ॥' अनेन कथितम्, अर्थात्सूर्यविम्बं तुहिनकिरणापक्रमेण (चन्द्रस्य स्पष्टक्रान्त्या सह) यावत्काल-पर्यन्तमेकमार्गे (एकस्मिन्नहोरात्रवृत्तेऽर्धाद्यावत् क्रान्त्योर्विवरं मानैक्यार्धादल्पं भवति यथा बिम्बैकदेशजक्रान्त्योः साम्यं भवतीत्यर्थः ।) तावत् मुनिभिः फलादेश-कृद्भिः पातसंजातफलस्य संभवः कथितः । तस्य सूर्यस्य चन्द्रस्य स्पष्टक्रान्त्या सहैकमार्गावस्थानाभावे तत्फलस्याप्यभावो भवति । इह पुनः पातस्थितिकाले यात्रोद्वाहादिषु मङ्गलकार्येषु दिनत्रयं दुष्टं नैव । केषुचित्फलग्रन्थेषु व्यतीपात-वैधृतयोः सतोस्तद्दिनं तत्पूर्वदिनमपरदिनं चेति दिनत्रयं शुभकार्ये निषिद्धमिति कैश्चिदुक्तं तत्परिहारार्थमाह—द्युत्रयं नैव दुष्टम् । सूर्यसिद्धान्तादिषु पातकाल एव शुभकार्येषु दोषभाक् गते पातकाले च दोषो न भवतीति । प्रसङ्गादत्र सिद्धान्तशेखरवचनोल्लेखः कृतः । यतः श्रीपतिराचार्या(ब्रह्मगुप्तात्) दर्वाचीनोऽस्ति । ब्राह्मस्फुट सिद्धान्ते तन्नामोल्लेखो नास्ति ।

भारतीयानां ज्योतिर्विदां मध्ये आर्यभट एव सर्वप्रथमं दिनरात्र्योः कारण-स्वरूपं पृथिव्या आवर्त्तनं कथयति । यथा गीतिकापादस्य प्रथमश्लोके एकस्मिन् महायुगे ४३२०००० भूमेर्भगणाः १५८२२३७५०० एतावन्तो भवन्तीति प्रथमं कथयित्वा दृष्टान्तद्वारेण भूभ्रमणं—

अनुलोमगतिर्नौस्थः पश्यत्यचलं विलोमगं यद्वत् ।
अचलानि भानि तद्वत् समपश्चिमगानि लङ्कायाम् ॥

अनेन दृढीकरोति । परमत्र विचित्रमेतदवलोक्यते यदार्यभटीय टीका-कारेण परमेश्वरेणैतत्-श्लोकस्यावतरणं 'भूमेः प्राग्गमनं नक्षत्राणां गत्यभावञ्चेच्छन्ति केचित्तन्मिथ्याज्ञानवशादुत्पन्नां प्रत्यग्गमन प्रतीतिमङ्गीकृत्य भूमेः प्राग्ग-तिरभिधीयते । परमार्थतस्तु स्थिरैवभूमिः' इत्युक्तम् । स्वयमप्यार्यभटः—

उदयास्तमय निमित्तं नित्यं प्रवहेण वायुना क्षिप्तः ।
लङ्कासमपश्चिमगो भपञ्जरः सग्रहो भ्रमति ॥

इत्यनेन भूभ्रमणमस्वीकरोतीति दृश्यन्ते । आर्यभटस्य मनसि निश्चयो नासीद्यत् पृथिवी चलति-नवा चलति । ब्रह्मगुप्तेनैकमपूर्वं वस्तु 'नतकर्म' प्रति-पादितम् । मन्दफल-शीघ्रफल-भुजान्तरादि संस्कारेण यो हि स्पष्टग्रहः समा-गच्छति स स्वगोलीयः (ग्रहगोलीयः) स ग्रहोऽस्माकं यत्र प्रत्यक्षीभूतो भवति स एवास्माकं स्पष्टग्रहो भवितुमर्हति । स्वगोलीयस्पष्टग्रहे यावता संस्कारेणास्माकं स्पष्टग्रहो भवति तस्यैव संस्कारस्य नाम 'नतकर्म' ब्रह्मगुप्ततः केऽपि प्राचीना आचार्या एतस्य (नतकर्मणः) नामोल्लेखं न कृतवन्तः । भास्कराचार्येण सिद्धान्त-

शिरोमणेर्गणिताध्यायस्य स्पष्टाधिकारे-एतस्याऽनयन प्रकारोऽभिहितः । 'मुहुः स्फुटातो ग्रहणे रवीन्द्वोस्तिथिस्त्विदं जिष्णुसुतो जगाद' इति भास्करोक्तेर्ज्ञायते यदेतस्याविष्कर्त्ता ब्रह्मगुप्त एवास्ति । भास्कराचार्येण भोग्यखण्डस्पष्टीकरणं यदभिहितं तन्मूलमपि ब्राह्मस्फुटसिद्धान्तस्य ध्यानग्रहोपदेशाध्याये ब्रह्मगुप्तोक्तमेव, न ह्यन्यैराचार्यैस्तत्सम्बन्धे किमपि लिखितम् । कमलाकरेण तु सिद्धान्ततत्वविवेके भास्करोक्तभोग्यखण्डस्पष्टीकरणस्य खण्डनमेव कृतम् । वस्तुतः कमलाकरोक्तं खण्डनं न समीचीनम् । ब्राह्मस्फुटसिद्धान्ते त्रिप्रश्नाधिकारे दिक्साधने 'पूर्वापरयोर्बिन्दू तुल्यच्छायाग्रयोर्दिगपराद्यः । पूर्वान्यः क्रान्तिवशात् तन्मध्याच्छङ्कुतल मितरे' इत्यत्र क्रान्तिवशाद्दिक्साधने कथं भेद उत्पद्यते तदर्थं चतुर्वेदाचार्येण कर्णवृत्ताग्रान्तरं यत्साधितं तदेव 'छायानिर्गमनप्रवेशसमयार्क-क्रान्तिजीवान्तरमि' त्याद्युक्त्या श्रीपतिना, तदनु 'तत्कालापम जीवयोस्तुविवरादि' त्यादिना भास्करेण च गृहीतम् । मन्दफलानयने वस्तुतो मन्दकर्णानुपातेनैव मन्दफलं सिध्यति । यद्यप्यत्र भास्करेण स्वमतं न प्रतिपादितं तथापि चन्द्रग्रहणे स्फुटरविचन्द्रकर्णसाधने 'मन्दश्रुतिर्द्राक्श्रुतिवत् प्रसाध्या' इत्यादिना ब्रह्मगुप्तस्यैव मतं स्वीकृतमित्यपि ब्रह्मगुप्तोक्तेर्वैलक्षण्यमस्ति । लल्लाचार्येण वलनदृक्कर्मणोरानयनमुत्क्रमज्यया कृतमिति ब्रह्मगुप्तोक्तौ 'अत्र ज्याशब्देनोत्क्रमज्या ग्राह्येति' चतुर्वेदाचार्यव्याख्यानमेव लक्ष्यीकृत्य भास्करेण 'ब्रह्मगुप्तकृतिरत्न-सुन्दरी साऽन्यथा तदनुर्गैविचार्यते' इत्युक्तम् ।

'युगमन्वन्तरकल्पाः कालपरिच्छेदकाः स्मृतावुक्ताः ।
यस्मान्न रोमके ते स्मृतिवाह्यो रोमकस्तस्मात् ॥'
(ब्राह्मस्फुट सिद्धान्त अध्याय १)

इति ब्रह्मगुप्तोक्त्या 'रोमक' इति नाम्ना चायं कस्यचित्पाश्चात्यज्योतिर्विदो मूलमादाय रोमकसिद्धान्तो रचित इति स्फुटं भवति । प्राचीनाचार्याणां मध्ये केवलं ब्रह्मगुप्त एव रोमकमतं खण्डयति । वराहमिहिरस्तु रोमक सिद्धान्तमतेनाहर्गणादीनां बहूनामेव सिद्धान्तोक्तविषयाणां साधनं स्वीकरोति । ब्रह्मगुप्तः—

ब्रह्मोक्तं ग्रहगणितं महता कालेन यत् खिलीभूतम् ।
अभिधीयते स्फुटं तज्जिष्णुसुतब्रह्मगुप्तेन ॥

इत्युक्त्या ब्रह्मसिद्धान्तमेव संस्कृतवानिति स्फुटमेव । तदेतेषां मूलरूप एको ब्रह्मसिद्धान्तोऽतिप्राचीन आसीदिति सिद्धयति । ब्रह्मगुप्तस्तु—

'येऽज्ञानपटलरुद्धदृशोऽन्यं ब्राह्माद्वदन्ति सिद्धान्तम् ।
तेषां युगादिभेदे ये दोषास्तान् प्रवक्ष्यामि ॥'

इत्युक्त्याऽन्यान् सर्वान् सिद्धान्तान्निन्दन्ति । ब्राह्मस्फुटसिद्धान्ते बहुषु स्थलेषु स्थौल्यं त्वस्त्येव तथापि बहवो विषया अत्र निवेशिताः सन्ति, तस्मात्सर्वश्रेष्ठो ब्राह्मस्फुटसिद्धान्तोऽस्तीतिकथने न काचिद्विप्रतिपत्तिः प्रतिभाति । अत्रैकोऽध्यायो 'छन्दश्चित्युत्तराध्यायः' अस्ति, यत्र लिखितानां श्लोकानां व्याख्योपपत्तिश्चाद्यावधि केनापि न कृता तेषु तादृशं काठिन्यमस्ति यदर्थो न मनसि समागच्छति, उपपत्तेश्च कथैव का ?, प्रश्नाध्यायो यादृशोऽत्र ग्रन्थेऽस्ति न तादृशोऽन्येषु सिद्धान्तग्रन्थेषु । अत्र ग्रन्थे मध्यगत्यादीनां पञ्चाध्यायानां प्रश्नाः सोत्तराः पृथक् पृथक् लिखिताः सन्ति । येषामभ्यासेन पाठका अतीव सिद्धान्तग्रन्थे निपुणा भवितुमर्हन्ति । सिद्धान्तशिरोमणेर्भूमिकायां 'जीवासाधनं विनैव यद्भुजज्यानयनं कृतवान् श्रीपतिस्तत्त्वपूर्वमेव स्यात् । यथा तत्प्रकारो विदां विनोदाय प्रदर्श्यते —

दोः कोटिभागरहिताभिहताः खनागचन्द्रास्तदीयचरणेन शरार्कदिग्भिः ।
ते व्यासखण्डगुणिता विहृताः फलं तु ज्याभिर्विनापि भवतो भुजकोटिजीवे ॥

इति केनापि लिखितमस्ति तन्नैव युक्तियुक्तं यतो ज्याभिर्विना भुजकोटिज्ययोरानयनं ज्यातश्चापानयनं च सर्वप्रथमं ब्रह्मगुप्तेनैव कृतम् यथा तदुक्तप्रकारः —

भुजकोटिज्यं शोनगुणा भार्धांशास्तच्चतुर्थभागोनैः ।
पञ्चद्वीन्दुखचन्द्रैर्विभाजिता व्यासदलगुणिता ॥
तज्ज्ये परमफलज्या सङ्गुणिता तत्फले विना ज्याभिः ।
इष्टोच्चनीचवृत्तव्यासार्धं परमफलजीवा ॥[1]

इष्टज्यातश्चापानयनञ्च

इष्टज्यासङ्गुणिताः पञ्चकयमलैकशून्यचन्द्रमसः ।
इष्टज्यापादयुतव्यासार्धं विभाजिता लब्धम् ॥
नवतिकृतेः प्रोह्यपदं नवतेः संशोध्य शेषभागकलाः ।
एवं धनुरिष्टाया भवति ज्याया विना ज्याभिः ॥

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१. चिरादेव प्रकारोऽयं श्रीपत्युक्तोऽस्तीति ज्योतिर्विदां मध्ये प्रसिद्धोऽस्ति । तथैव ज्यातश्चापानयनमपि । एतदवलम्ब्यैव ग्रहलाघवे करणग्रन्थे गणेशदैवज्ञेन बहवः प्रकारा विलिखिताः ।

एतदनुरूप एव बटेश्वरसिद्धान्ते बटेश्वरेण स्वीयप्रकारोऽभिहितः । सिद्धान्तशेखरे सर्वत्र श्रीपतेः स्वकीयः प्रकारोऽल्पीयानेवास्ति, ब्रह्मगुप्तोक्त-प्रकारा एव छन्दोऽन्तरेण लिखिताः सन्ति[1] वराहोक्त पञ्चसिद्धान्तिकायां त्रयोदशभिरार्याभिरुक्तो वासिष्ठसिद्धान्तः पैतामहसिद्धान्तश्च वराहोक्त्यैव दूरभ्रष्टाविति तत्रापि ब्रह्मसिद्धान्तापेक्षया यत्किञ्चित् वासिष्ठसिद्धान्तः सूक्ष्मताभिधायि । तन्त्रपरीक्षाध्याये—

लाटात् सूर्यशशाङ्कौ मध्याविन्दूच्च चन्द्रपातौ च ।
कुजबुधशीघ्रबृहस्पति सितशीघ्रशनैश्चरान् मध्यान् ॥
युगजातवर्ष भगणान् वासिष्ठाद्विजयनन्दिकृत पादात् ।
मन्दोच्चपरिधिपातस्पष्टीकरणाद्यमार्यभटात् ॥
श्रीषेणेन गृहीत्वा रत्नोच्चयरोमकः कृतः पन्थाः ।
एतानेव गृहीत्वा वसिष्ठो विष्णुचन्द्रेण ॥

इति ब्रह्मगुप्तोक्त्या वसिष्ठसिद्धान्तरचयिता विष्णुचन्द्रनामकः कश्चिदासीत् । सम्भवतोऽयं विष्णुचन्द्रः प्राचीनं वसिष्ठसिद्धान्तं संस्कृतवानिति 'एतानेव गृहीत्वा वासिष्ठो विष्णुचन्द्रेण' इत्युक्त्या स्फुटं भवति । परमयं ग्रन्थोऽधुना कुत्रापि नोपलभ्यते । एतस्यैव विष्णुचन्द्रस्य मतं यत्र तत्र श्रीषेणार्यभटाभ्यां सह ब्रह्मगुप्तेन खण्डितम् ।

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१. यथोदाहरणार्थं किञ्चित्प्रोच्यते । सिद्धान्तशेखरस्य सूर्यग्रहणाधिकारे ।
तिथ्यन्तात् स्थितिखण्डहीन सहितात् प्राग्वत्ततो लम्बनं
कुर्यात् प्रग्रहमोक्षयोः स्थितिदलं युक्तं विधायासकृत् ।
तन्मध्यग्रहणोत्थलम्बनभुवा विश्लेषणानेहसा
मर्दार्धोनयुताथितेरपि तथा संमीलनोन्मीलने ॥
अधिकमृणयोराद्यं मध्यात्तथाऽन्त्यमिहाल्पकं भवति
धनयोश्चाद्यं हीनं यदाऽधिकमन्तिमम् ।
नमनविवरेणैवं कुर्याद्विहीनमतोऽन्यथा स्थितिदल-
मृणस्वस्थे भेदे तदैक्ययुतं पुनः ॥

इति ब्रह्मगुप्तोक्तस्या—
प्राग्वल्लम्बनमसकृत् तिथ्यन्तात् स्थितिदलेन हीनयुतात् ।
अधिकोनं तन्मध्याद्दृणयोरूनाधिकं धनयोः ॥
मध्यधिकं स्थित्यर्धं तदाऽन्तरेणान्यथोनमृणमेकम् ।
अन्यद्धनं तदैक्येनाधिकमेवं विमर्दार्धे ॥

(शेष पृष्ठ ११ पर)

न च श्रीपतिरेव निजपूर्ववर्त्तिनां ग्रन्थकाराणां ग्रन्थेभ्यस्तदुक्तविषयान् छन्दोऽन्तरेण तथैव निबध्य स्वग्रन्थे स्वोक्त्या लिखितवानपितु तत्पूर्ववर्तिनां ग्रन्थकाराणामपि सैव रीतिः। परवर्त्तिनो भास्कराचार्यादयोऽपि न तां रीतिममुञ्चन्निति प्रत्यक्षदर्शनादेव स्फुटीभवति।

यथा भास्कराचार्यः—गणिताध्यायस्य मध्यमाधिकारे सिद्धान्तग्रन्थलक्षणं ज्योतिः शास्त्रस्य वेदाङ्गत्व निरूपणं वेदाङ्गानां नामानि वेदाङ्गेषु ज्योतिः शास्त्रस्य प्राधान्यं तच्च द्विजैरेव पठनीयमिति सर्वं परतोऽपि भचक्रचलनं काल-

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ऽस्य सर्वथा सदृशमेव तथा च प्रकारान्तरेण स्फुट स्थिति दल साधनं श्रीपत्युक्तम्।

स्पित्यर्धोनयुतात् परिस्फुटतिथेः स्याल्लम्बनं पूर्ववत्
तन्मध्यग्रहवेच मध्यमतिथो ततस्तु तिथौ।
स्थित्यर्धेन परिस्फुटेषु जनितेनोनाधिकाद्वाऽसकृत्
तत्तिथ्यन्तर नाडिकाः स्थितिदले स्तः स्पर्शमुक्त्योः स्फुटे॥

अस्य श्लोकस्य द्वितीयं चरणं शुद्धं नास्ति। प्रकारोऽयं ब्रह्मगुप्तोक्तस्या—

स्फुटतिथ्यन्ताल्लम्बनमसकृत् स्थित्यर्धहीनयुक्ताद्वा।
तत्स्फुट विक्षेपकृत स्थित्यर्धोनयुततिथ्यन्तात्॥

तत्स्पष्टतिथिच्छेदान्तरे स्फुटे दिनदले विहीनयुतात्।
स्व विमर्दार्धेनासकृदेवं स्पष्टे विमर्दार्धे॥

ऽस्य पुनरुक्तिरेव। सिद्धान्तशिरोमणौ—

तिथ्यन्ताद्गणितागतात् स्थितिदलेनोनाधिकाल्लम्बनं
तत्कालोत्थनतींषु संस्कृति भव स्थित्यर्धहीनाधिके।
दर्शान्ते गणितागते धनमृणं वा तद्विधायासकृज्
ज्ञेयौ ग्रग्रहमोक्ष संज्ञसमयावेवं क्रमात् प्रस्फुटौ॥

भास्करोक्तमपि सर्वथैव तदनुरूपमेवास्ति। एवं सिद्धान्तशेखरस्य सूर्यग्रहणाध्यायोपसंहारे स्फुटं भवति पञ्चजीवया लम्बनं नहि यतस्ततः कृतम्। युक्तमुक्तमिति जिष्णुसूनुना तन्मयाऽपि कथितं परिस्फुटम् इति ब्रह्मगुप्तोक्तस्या—

दृग्गणितैक्यं न भवति यस्मात् पञ्चज्यया रविग्रहणे।
तस्माद्यथा तदैक्यं तथा प्रवक्ष्यामि तिथ्यन्ते॥

ऽस्य सदृशमेव। मध्यगत्यध्यायतो ग्रन्थ समाप्ति यावत्सादृश्यस्यैवमेव स्थितिरिति द्वयोर्ग्रन्थयो 'ब्रह्मिस्फुट सिद्धान्त सिद्धान्तशेखरयोः' श्लोकनेन स्फुटं भवतीति।

प्रवृत्तिः। कालमानानां परिभाषाः सर्वाः, ग्रहाणां भगणाः, युगानां मन्वादीनां नामानि मानानि च ब्रह्मणो गतवर्षादेः प्रयोजनाभाव इत्यादि सर्वमपि मध्यमाधिकारोक्तं श्रीपतेः साधनाध्यायोक्तश्लोकानां श्लोकान्तरमात्रमेवाकरोत्। सुधियो ग्रन्थाभ्यन्तरे ग्रन्थकारयोरानुरूप्यमवलोकयन्तु। एवं प्राचीनकृतेरनेकान् विशेषान् प्रवक्तुमेव श्रीपतिः प्रथमं (साधनाध्यायं) ग्रहभगणाध्यायं वा कृतवान्। तत्परं मध्यमाध्यायेसप्तभिः प्रकारैरहर्गणानयनं, कदा प्रभृति वारप्रवृत्तिरित्यत्र बहूनामाचार्याणां मतानि, तद्दूषणपुरःसरस्वाभिमतवारप्रवृत्तिकथनं, मध्यमग्रहसाधनार्थं बहून्येव नूतनानि प्रकारान्तराणि, रव्यादीनां सर्वेषामपि ग्रहाणां राश्यादिमन्दोच्चकथनमित्यादयो बहवोऽपि श्रीपतिकृताप्राचीनकृतेर्विशेषा वर्त्तन्ते।

ब्राह्मस्फुट सिद्धान्ते बहुभिरेव प्रकारैरहर्गणानयनं लघ्वहर्गणानयनं ब्रह्मगुप्तेन कृतं, आचार्योक्ताहर्गणानयनस्यानुकरणमेव श्रीपतिना कृतम्। परं सिद्धान्तशेखरे लघ्वहर्गणानयनस्य चर्चा ग्रन्थकृता न कृता। अहर्गणाद्वारज्ञानार्थमहर्गणः सैकः कार्य इति ब्रह्मगुप्तेन श्रीपतिना च कथितः। परमहर्गणो निरेकोऽपि कर्त्तव्यो वारज्ञानार्थमिति सिद्धान्तशिरोमणौ 'अभीष्टवारार्थमहर्गणश्चेत्सैको निरेकस्तिथयोऽपि तद्वदि'त्यनेन भास्कराचार्यः कथयति। वटेश्वरसिद्धान्तेऽपि बहुभिः प्रकारैरहर्गणानयनं लघ्वहर्गणानयनं च ब्रह्मग्रन्थकृता कृतमस्ति। भास्कराचार्यस्तु महदहर्गणानयनं 'कथितकल्पगतोऽर्कसमागण' इत्यादिना, लघ्वहर्गणानयनं च 'चैत्र सितादिगतस्थिति संघ' इत्यादिना कृतम्। यद्यपि लघ्वहर्गणानयने स्थौल्यं वर्त्तते तथाप्येकमपूर्वं वस्तु प्रतिपादितम्। वटेश्वरकृतं लघ्वहर्गणानयनं स्थौल्यरहितं नास्ति, एतदतिरिक्तैः प्राचीनाचार्यैर्नवीनैश्च लघ्वहर्गणानयनं न कृतं प्रत्युत कमलाकरेण भास्करकृतलघ्वहर्गणानयनस्य खण्डनमेव कृतम्। स्फुटगत्यध्याये सर्वैरेवार्यभटब्रह्मगुप्त लल्लाचार्यादिभिर्वृत्तचतुर्थांशे चतुर्विंशतिः क्रमज्या उत्क्रमज्याश्च तत्त्वास्त्रि २२५ कलावृद्ध्या साधितास्तत्र आर्यभटस्य लल्लस्य च त्रिज्या=३४३८, ब्रह्मगुप्तस्य खमुनिरद ३२७०मिता त्रिज्या, श्रीपतिना चैतद्भिन्ना ३४१५ त्रिज्या गृहीता ब्रह्मगुप्तोक्तभूपरिधिः=५०००, भास्कराचार्यमतेन पादोनगोक्षधृतिभूमितयोजनानीत्यनेन ग्रहाणां योजनगतिः=११८५८।४५। गतियोजनतिथ्यंशः कुदलस्य यतोमितिरित्यनेन भूव्यासः=१५८१, भूपरिधिः=४९६७ ग्रहाणां भुजान्तरकर्म प्रतिपादितमस्ति, सूर्यसिद्धान्तोक्त भुजान्तरकर्मवदेवास्ति, सिद्धान्तशेखरे, सिद्धान्तशिरोमणावपि भुजान्तरकर्मण उपपादनमाचार्योक्तवदेवास्ति, अन्येऽपि बहवो विषयादर्शनीयाः पठनयोग्याश्चेति।

त्रिप्रश्नाधिकारे रवेर्मध्यान्हकालिकनतांशान् ज्ञात्वा तद्वशतो रव्यानयनार्थं प्रथमतः क्रान्तिज्यासमागच्छति। ततोऽनुपातेन रवेर्भुजांशज्ञानं भवति।

भुजांशतो राश्यादिरवेर्ज्ञानं पदाधीनं तत्र पदज्ञानोपायः प्राचीनैः कैश्चिन्न कृतः, यथाऽत्राचार्येण—

'क्रान्तिर्व्यासार्धगुणा जिनभागज्याहृता धनुर्जादौ।
कर्क्यादौ चक्रार्धात्प्रोह्य तुलादौ सचक्रार्धम् ॥

चक्रार्धात्प्रोह्यमृगादौ स्फुटो सकृत् व्यस्तमृणं धनं मध्यम्। अर्कोऽस्मादिति' एतेन रवेरानयनं कृतम्। श्रीपतिना 'अजतुलादिगतस्य विवस्वतो दिनदलप्रभयो-र्युतिरर्धिता। भवति वैषुवती निजदेशजे' त्यनेन पलभामानं ज्ञात्वा—
'आद्येपदेऽपचयिनी पलभाऽल्पिका स्यात् छायाल्पिका भवति वृद्धिमती द्वितीये।
छायाधिका भवति वृद्धिमती तृतीये तुर्येपुनः क्षयवती तदनल्पिका च ॥
वृद्धिं प्रयान्ती यदि दक्षिणाग्रच्छाया तथापि प्रथमं पदं स्यात्।
ह्रासं व्रजन्तीमथ तां विलोक्य रवेर्विजानीहि पदं द्वितीयम् ॥'

इत्यनेन गोलयुक्तिसिद्धं पदज्ञानं कृतम्। अत्र भास्कराचार्यः—

क्रान्तिज्या त्रिज्याघ्नी जिनभागज्योद्धृता दोर्ज्या।
तद्धनुराद्ये चरणे वर्षस्यार्कः प्रजायतेऽन्येषु ॥
भार्धाच्च्युतः सभार्धो भगणात्पतितोऽब्द चरणानाम्।
ऋतुचिन्हैर्ज्ञानं स्यादृतुचिन्हान्यग्रतस्ततो वक्ष्ये ॥'

इत्यनेनाचार्योक्तवदेव कथितवान् केवलं 'ऋतुचिन्हैर्ज्ञानं स्यादिति' विशेषं कथितवान्। पदज्ञानार्थमृतुवर्णननामकमेकमधिकारं सिद्धान्तशिरोमणेर्गोला-ध्यायेऽभिहितम्। भास्करतो नवीनाः कमलाकरतः प्राचीनाः सर्वेऽपि सिद्धान्त-ग्रन्थकर्तारो ज्यौतिषसिद्धान्तस्यैकमङ्गं ज्ञात्वा स्वस्वसिद्धान्तग्रन्थे निश्चितरूपेण 'ऋतुवर्णनाध्यायं,' प्रोक्त'वन्तः। सिद्धान्ततत्त्वविवेके आद्ये पदेऽपचयिनी पलभाऽल्पिका स्या' दित्यादि श्रीपत्युक्त पदज्ञानबोधक श्लोकद्वयं लिखित्वा कमलाकरेण—

'ऋतुचिह्नैरिदं पूर्वैरुक्तं सर्वत्र तन्न हि।
केवलं कुकविप्रीत्यै पदज्ञप्तयै न तद्भवेः ॥'

इत्यनेन भास्करोक्त ऋतुवर्णनं निन्दितम्। वस्तुतः 'सर्वत्र तन्नही'ति कमलाकरोक्तं यथार्थमेव। परं पदज्ञान प्रकारोऽयं श्रीपत्युक्त इति कमलाकरेण नोक्तः। सिद्धान्तशेखरस्याप्राप्तौ प्रकारोऽयं कमलाकरस्यैवेत्याधुनिका ज्योतिर्विदो जानन्त आसन्। यदि रवेः पदज्ञानार्थं कोऽपि प्रकारः प्राचीनैर्यथार्थतो नोक्तस्तदा प्रकारस्यास्य रचयिता श्रीपतिरवश्यमेव सर्वेषां ज्योतिर्विदां प्रशंसापात्रमित्यत्र न कोऽपि सन्देहः। महदाश्चर्यं चैतद्यत् श्रीपतिकृतमिदं गोलयुक्तियुक्तं पदज्ञानं

त्यक्त्वा भास्कराचार्येण ऋतुवर्णनद्वारा पदज्ञानं समीचीनं ज्ञात्वा कृतमिति। चन्द्रग्रहणाध्याये रविचन्द्रभुवां योजनबिम्बानि, रविचन्द्रयोर्योजनात्मककर्ण-स्पष्टीकरणं, भूभा बिम्बानयनं, ग्रासमानाद्यानयनं, परिलेखप्रकारश्चाचार्येरुक्तः, श्रीपतिना भास्कराचार्येण च कथनक्रममादाय विशेषतया तथैवानूदितः। ब्रह्म-गुप्तकृत एव सूर्यग्रहणाधिकारः श्रीपतिना प्रायः श्लोकान्तरैरुक्तः। उदयास्तमया-ध्याये आयनदृक्कर्मसाधनं कृतं परं तन्न समीचीनं श्रीपतिनाऽपि प्राचीनोक्तं तदानयनं कृत्वा—

खनभोधृतिभिः समाहतं प्रथमं दृक्फलमायनाह्वयम्।
द्युचराश्रितभोदयासुभिर्विहृतं स्पष्टमिह प्रजायते॥

इत्यनेन तत्स्फुटीकरणं कृतं, एतदवलोक्य भास्कराचार्येण 'आयनं वल-नमस्फुटेषुणा सङ्गुणमि' त्यादिना तदेवोक्तम्। चन्द्राध्याये ब्रह्मगुप्तेन बहवो-विषयाः प्रतिपादिताः सन्ति, श्रीपतिना तु वराहब्रह्मगुप्तलल्लाचार्याणां बहवः श्लोका अनूदिताः। वस्तुतो नास्ति कश्चिद्विशेषः। केवलं चन्द्रस्य स्पष्टचरा-नयने परिलेखसूत्रप्रमाणानयने च बहून्येव प्रकारान्तराणि स्फुटोक्त्या लिखि-तानि सन्ति। वराहब्रह्मगुप्तलल्लाचार्यैरुदयान्तरकर्म नोक्तं ग्रहयुत्यध्याये (ग्रह-युद्धाध्याये-ग्रहयोगाध्याये वा)—

अन्त्यभ्रमेणगुणिता रविबाहुजीवाऽभीष्टभ्रमेण विहृता फलकार्मुकेण।
बाहोः कलासु रहितास्ववशेषकं ते यातासवो युगयुजोः पदयोर्धनर्णम्॥

इत्यनेन श्रीपत्युक्तं दृग्गणितैक्यकृत् कर्मैव भास्कराचार्येण उदयान्तर-कर्मेति नाम्नोक्तम्। सिद्धान्तशेखरस्याप्राप्तौ भास्कराचार्येणैवानुभूतमिदं कर्मे-त्याधुनिकानां गणकानां प्रतीतिरासीत्। इदमुदयान्तरकर्म प्रथमं श्रीपतिरेव स्वकीय विचारेण कथयामासेति ज्ञायते। तथा च—

त्रिभविरहितचन्द्रोच्चोनभास्वद्भुजज्या गगननृपविनिघ्नी भत्रयज्या विभक्ता।
भवति चरफलाख्यं तत्पृथक्स्थं शरघ्नं हृतमुडुपतिकर्णत्रिज्ययोरन्तरेण॥१॥
परमफलमवाप्तं तद्धनर्णं पृथक्स्थे तुहिन किरणकर्णे त्रिज्यकोनाधिकेऽथ। स्फुट-दिनकर हीनादिन्दुतो या भुजज्या-स्फुटपरमफलघ्नी भाजिता त्रिज्ययाऽऽप्तम्॥२॥

शशिनिचरफलाख्यं सूर्यहीनेन्दुगोलात् तदृणमुतधनं चेन्दूच्चहीनार्कगोलम्।
यदि भवति हि साम्यं व्यस्तमेतद्विधेयं स्फुटगणितदृगैक्यं कर्त्तुमिच्छद्भिरत्र॥३॥

श्लोकत्रयेणानेन दृग्गणितैक्यार्थं श्रीपतिना चन्द्रे संस्कारविशेषो दत्तः। सिद्धान्तेऽपि प्राचीनग्रन्थे नायं संस्कारो लिखितो वर्त्तते। यद्यपि—

इन्दूच्चोनार्ककोटिघ्ना गत्यंशा विभवा विधोः ।
गुणो व्यर्केन्दुदोः कोटघोरूपपञ्चाप्तयोः क्रमात् ।।
फले शशाङ्कतद्गत्योर्लिप्ताद्ये स्वर्णयोर्वधे ।
ऋणं चन्द्रे धनं भुक्तौ स्वर्णसाम्यवधेऽन्यथा ।।

इत्यनेनैतादृश एव चन्द्रसंस्कारो मुञ्जालाचार्येण 'लघुमानस'नामके करणग्रन्थे कथितः । परमेतयोः सदृशत्वाभावात्--श्रीपतिना वेधेन दृष्ट्वा ततो भिन्नोऽयं कथित इति प्रतिभाति । श्रीपत्युक्तोऽयं संस्कारो भास्कराचार्येणासकृद्-दृष्ट्वा विवेचितस्तत्र स्वोपलब्धेर्विस्तारपूर्वकं प्रतिपादनार्थमेको 'बीजोपनय' नामको ग्रन्थः ५९ श्लोकात्मकः सिद्धान्तशिरोमणि रचनैकवर्षानन्तरं—

मयाय बीजोपनये यदन्ते सूर्योक्तमाद्यं परमं रहस्यम् ।
प्रकाशये गोप्यमपीह देवं प्रणम्य बीजं जगतां हितार्थम् ।।१।।
यद्यपि पूर्वमपीदं संक्षेपादुक्तमागमोक्तदिशा ।
नैतावतैव कश्चित् दृक्करणैक्याय कल्पते गणकः ।।२।।
दृक्करणैक्यविहीनाः खेटाः स्थूला न कर्मणामर्हाः ।
अत इह तदर्हतायै तात्कालिकबीजविस्तरं वक्ष्ये ।।३।।
पाता रवेस्तामसकीलकाख्यास्तेषां समाकर्षणतः शशाङ्कः ।
तत्तुङ्गशक्तिश्च निजस्वभावं विहाय नित्यं विषमत्वमेति ।।४।।
चन्द्राच्च तद्योगवियोगतश्च साध्यं हि भाद्यं विषमं यतः स्यात् ।
तस्माद्विधोरत्र विशुद्धिशुद्धयै विस्तार्यते बीजफलक्रियेयम् ।।५।।
एकेन पुंसा निखिलग्रहाणामन्तं प्रबोधो नहि शक्यतेऽतः ।
व्यासात्समासाच्च यथोपलब्धं प्रोक्तं मयेत्यादरणीयमेतत् ।।६।।

इत्यादिना सिद्धान्तशिरोमणिवद्वासनाभाष्यसहितो विरचित इति । भग्रहयोगाध्याये—

कृत्वापि दृष्टिकर्म श्रीषेणार्यभटविष्णुचन्द्रोक्तम् ।
प्रतिदिनमुदयेऽस्ते वा न भवति दृग्गणितयोरैक्यम् ।।१।।
भमुनिमृगव्याधानां यतस्ततो दृष्टिकर्म वक्ष्यामि ।
दृग्गणितसमं देयं शिष्याय त्रिरोषितादेयम् ।।२।।

इति ब्रह्मगुप्तेन पाण्डित्यपूर्णमुक्तम् । एतदेव यथार्थं बुध्वा तदुक्तो (ब्रह्म-गुप्तोक्तः) भग्रहयुत्यध्यायः सम्पूर्णोऽपि स्फुटोक्त्याऽनूदितः श्रीपतिनेति ।

ब्राह्मस्फुटसिद्धान्तोत्तरार्धे परिकर्मविंशतिः (सङ्कलितम्, व्यवकलितम्, प्रत्युतभिन्नो गुणनम्। भागहारः, वर्गः, वर्गमूलम्, घनः, घनमूलम्। पञ्चजातयः, त्रैराशिकम्, व्यस्तत्रैराशिकम्, पञ्चराशिकम्, सप्तराशिकम्, नवराशिकम्, एकादशराशिकम्, भाण्डप्रति भाण्डं चेति) कथिताऽस्ति, सर्वत्रैव चतुर्वेदाचार्योक्ता उद्देशकाः (उदाहरणानि) सन्ति। सिद्धान्तशेखरेऽपि परिकर्मविंशतिः (अभिन्नाङ्कानां गुणन-भजन-वर्ग-वर्गमूल-घन-घनमूलानीति षट् ६, भिन्नाङ्कानां योगान्तर-गुणन-भजन-वर्ग-वर्गमूलानीति षट् ६, भाग प्रभाग-भागानुबन्ध-भागापवाहाख्य-जाति चतुष्टयम् ४, विलोमकर्म १, त्रैराशिकम् १, व्यस्त त्रैराशिकम् १, पञ्च-राशिकम् १, इति) ब्रह्मगुप्तश्रीपत्युक्त विंशत्यां परिकर्मसु विषयवर्णने महानेव भेदोऽस्ति तद्विंशतेः परिकर्मणां नामानि बहुधा भिन्नानि सन्ति। भास्करोक्त-प्रकीर्णविषयाः (सङ्कलिततो भाण्डप्रतिभाण्डं यावत्) यादृशाः स्फुटरूपेण वर्णिताः सन्ति न तादृशा ब्रह्मगुप्तश्रीपत्योः परिकर्मविंशत्युक्तविषयाः। विषयास्तु ब्रह्मगुप्तौ-श्रीपत्युक्तौ-भास्करोक्तौ-समाना एव किन्तु तत्प्रति पादनरीतयो भिन्ना भिन्नाः सन्ति। एवमष्टौ व्यवहाराः (मिश्रक व्यवहारः, श्रेढी व्यवहारः, क्षेत्रव्यवहारः, खातव्यवहारः, चिति व्यवहारः, क्राकचिक व्यवहारः, राशिव्यव-हारः, छायाव्यवहारः) ब्राह्मस्फुटसिद्धान्ते, सिद्धान्तशेखरे भास्करीय लीलावत्यां च सन्ति, एतेष्वष्टसु व्यवहारेष्वपि बहुधैवान्यादृशत्वमस्ति, सर्वेषु व्यवहारेषु ब्रह्मगुप्तोक्तश्रीपत्युक्तव्यवहाराभ्यां भास्करोक्तव्यवहारेषु-विषयाधिक्यं-विषय-प्रतिपादन स्फुटत्वं चास्ति, ग्रन्थत्रया (ब्राह्मस्फुटसिद्धान्तः, सिद्धान्तशेखरः, लीला-वती च) वलोकनेनेति स्फुटं भवति, एतत्परं प्रश्नाध्यायोऽस्ति यत्र मध्यगत्युत्तरा-ध्यायः, स्फुटगत्युत्तराध्यायः। त्रिप्रश्नोत्तराध्यायः। ग्रहणोत्तराध्यायः। शृङ्गो-न्नत्युत्तराध्यायः। एतेषु पञ्चसूत्तराध्यायेषु सोत्तराः प्रश्नाः सन्ति, प्रश्नाश्चातीव विलक्षणाः सन्ति, येषामभ्यासेन पाठका अतीव ज्यौतिषसिद्धान्तविषयेषु-निपुणा भवितुमर्हन्ति प्रत्यध्यायमीदृशः सोत्तरप्रश्नक्रमलेखः किञ्चित् किञ्चिद्वटेश्वर-सिद्धान्ते-सिद्धान्तशेखरे चावलोक्यते। सिद्धान्तशिरोमण्यादौ नायं क्रमोऽस्ति। एतत्परं प्रश्नसहितः कुट्टकाध्यायोऽस्ति, ब्रह्मगुप्तोक्तकुट्टकाध्याये[1] श्रीपत्युक्त-भास्करोक्ताभ्यां विषयाधिक्यमस्ति किन्तु विषयकथनस्फुटता भास्करोक्ता-वेवास्ति, धनर्णादीनां सङ्कलितव्यवकलितादि-भास्करोक्तवदेवास्ति-सिद्धान्त-शेखरेऽपि, ततः परमेकवर्णसमीकरणं बीजमस्ति, भास्करोक्तैकवर्ण समीकरण-बीजतोऽल्पमेवास्ति। ततः परमनेकवर्णसमीकरणबीजमस्ति, ब्रह्मगुप्तमिदमपूर्व-

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१. ब्रह्मगुप्तोक्त कुट्टकाध्याये बहवः प्रश्नास्तादृशाः सन्ति येषामुत्तरमतीवाऽऽनन्द-जनकमस्ति, येषामभ्यासेन पाठकास्तत्सम्बन्धिविषयाणामतिज्ञातारो भवितु-मर्हन्तीति।

मस्ति, अत्र विषयाश्चाप्यधिकाः सन्ति, भास्करोक्तमप्यनेकवर्णसमीकरणबीजं विषयाधिक्येन सहितं विद्यते। किन्तु सिद्धान्तशेखरेऽतीव स्वल्पं तद्विद्यते। भावित-बीजं ब्रह्मगुप्तोक्ताच्छ्रीपत्युक्तं स्वल्प भास्करोक्तं चाधिकं विद्यते, सर्वत्रैव भास्करोक्तौ वैशद्यं वर्त्तते। एतत्परं वर्गप्रकृतिरस्ति, कनिष्ठे ज्येष्ठक्षेपाणां योगभावनाऽन्तरभावना च ब्रह्मगुप्तोक्तादेवाऽऽदाय श्रीपतिना भास्करेण च विलिखिता, परं श्रीपतिना भावनास्वरूपं न प्रतिपादितम्। वर्गात्मकप्रकृतौ कनिष्ठज्येष्ठयोरानयनं ब्रह्मगुप्तोक्तमेव बीजगणिते भास्करेण लिखितम्। परं श्रीपतिना तत्सम्बन्धे किमपि न लिखितम्। शङ्कुच्छायादिज्ञानाध्यायो ब्रह्मगुप्तस्यापूर्वं वस्तु वर्त्तते सिद्धान्तशेखरे भास्करीय ग्रन्थे चैतदध्यायोक्ता विषया न सन्ति, वस्तुतो दर्शनीयोऽयमध्यायः। छन्दश्चित्युत्तराध्यायस्त्वेतादृशोऽस्ति, यत्रस्थश्लोकोपपत्तीनां का कथा तद्व्याख्याऽप्यतिकठिना तेनैव हेतुनाऽद्यावधि तद्व्याख्योपपत्तिश्च कैरपि न लिखिता। गोलाध्याये-भूगोलसंस्थानं-देवासुरसंस्थानं-चक्रभ्रमणव्यवस्थादेवादीनां रविभ्रमणस्थितिः, देवदैत्यानां राशिसंस्थानं-देवादीनां रविदर्शनकालः, भूगोले लङ्कावन्तीसंस्थानमित्यादयो विषयाः सन्ति—

भूपरिधितुर्यभागे लङ्काभूमस्तकात् क्षितितलाच्च।
लङ्कोत्तरतोऽवन्ती भूपरिधेः पञ्चदशभागे॥

इत्यनेन लङ्कोत्तरतो भूपरिधिपञ्चदशभागेऽवन्ती वर्त्तत इत्याचार्येण कथ्यते, आचार्यानुयायी भास्कराचार्यः सिद्धान्तशिरोमणेर्गोलाध्याये 'निरक्षदेशात् क्षितिषोडशांशे भवेदवन्ती' इति कथितवान्, चतुर्वेदाचार्यसम्मतः पाठश्च 'पञ्चदशभागे, अयमेव, अत्राध्याये-आचार्योक्ता बहवो विषयाः सूर्यसिद्धान्तीयभूगोलाध्याये तथैव सन्ति, मध्ये मध्ये उभयत्र (आचार्योक्ताध्याये-सूर्यसिद्धान्तीयभूगोलाध्याये) विषयान्तराण्यपि सन्ति, सिद्धान्तशेखरस्य गोलाध्याये श्रीपतिनाऽपि कियन्तो विषया आचार्योक्तसदृशा एव कथिताः। 'यन्मूले तद्व्यासो मण्डलिप्ताकृतेर्दशहृतायाः' इत्यनेन परिधितो व्यासानयनं सूर्यसिद्धान्तोक्त 'तद्वर्गतो दशगुणात्पदं परिधिरि'त्यादेर्विलोमेन परिधितो व्यासानयनवदस्ति, श्रीपतिनापि 'कालः स्यात्परिधेर्वर्गाद्दिग्भक्ताच्च पदं त्विहे'ति प्रकारानुकूलैव त्रिज्यातिथियुगाग्नि ३४१५ मिताऽस्तीति स्वीकृता भास्कराचार्येण 'व्यासे भनन्दाग्निहते विभक्ते खवाणसूर्यैः परिधिः सुसूक्ष्मः' इत्यनेन व्यासात्परिध्यानयनं कथ्यते। एतद्विलोमेन परिधितो व्यासानयनं भवेत्। परं केषामपि व्यासात्परिध्यानयनं परिधितो व्यासानयनं वा न समीचीनं तयोः (परिधव्यासयोः) सम्बन्धस्यास्थिरत्वादिति।

ज्याप्रकरणे—आचार्येण यथा चापार्धाशज्याद्यानयनं कृतं तथैव सिद्धान्तशेखरे श्रीपतिना सिद्धान्तशिरोमणेर्गोलाध्याये भास्करेण च कृतमिति तद्ग्रन्था-

वलोकनात्स्फुटमस्ति । भास्करेण 'अन्त्यज्योत्पत्त्यौ' बहवो विशेषाः प्रतिपादिताः सन्ति ।

स्फुटगतिवासनायां—मन्दफलसाधनेऽपि कर्णानुपातेन यत्फलं तदेव समीचीनमिति कर्णः कथं न कृत इत्यस्य कारणम्—

त्रिज्याभक्तः परिधिः कर्णगुणो बाहुकोटिगुणकारः ।
असकृन्मान्दे तत्फलमाद्यसमं नात्रकर्णोऽस्मात् ॥

कथितम् । सिद्धान्तशेखरे—

त्रिज्याहृतः श्रुतिगुणः परिधिर्यतोदोः कोटचोर्गुणो मृदुफलानयनेऽसकृत्स्यात् ।
स्यान्मान्दमाद्यसममेव फलं ततश्च कर्णः कृतो न मृदुकर्मणि तन्त्रकारैः ॥

इति श्रीपत्युक्तश्लोक आचार्यो (ब्रह्मगुप्तो)क्त श्लोकस्यानुवादरूप एव, भास्कराचार्येणापि—

स्वल्पान्तरत्वान्मृदुकर्मणीह कर्णः कृतो नेति वदन्ति केचित् ।
त्रिज्योद्धृतः कर्णगुणः कृतेऽपि कर्णे स्फुटः स्यात्परिधिर्यतोऽत्र ॥
तेनाद्यतुल्यं फलमेति तस्मात् कर्णः कृतो नेति च केचिदूचुः ।
नाशङ्कनीयं न चले किमित्थं यतो विचित्रा फलवासनाऽत्र ॥

इह कर्णेन यत्फलमानीयते तदेव समीचीनम् । यन्मन्दकर्मणि कर्णो न कृतस्तत्स्वल्पान्तरादिति कथ्यते । मन्दकर्मणि मन्दकर्णतुल्येन व्यासार्धेन यद्वृत्तं तत्कक्षावृत्तम् । तेन ग्रहो गच्छति । यो मन्दपरिधिः पाठपठितः स त्रिज्यापरिणतः । अतोऽसौ कर्णव्यासार्धे परिणाम्यते । यदि त्रिज्यावृत्तेऽयं परिधिस्तदा कर्णवृत्ते क इति स्फुटपरिधिः । ततः स्वेनाहते परिधिना भुजकोटिजीवे भांशैरित्यादिना यत्फलमागच्छति तत्त्रिज्यया गुणितं कर्णेनभक्तं यल्लब्धं तत्पूर्वफलतुल्यमेव फलमागच्छतीति ब्रह्मगुप्तमतम् । अथ यद्येवं परिधेः कर्णेन स्फुटत्वं तर्हि किं शोघ्रकर्मणि न कृतमित्याशङ्क्य चतुर्वेदः कथयति—ब्रह्मगुप्तेनान्येषां प्रतारणपरमिदमुक्तमिति । तदसत् । चले कर्मणीत्थं किं न कृतमिति नाशङ्कनीयम् । यतः फलवासना विचित्राऽस्तीति । अन्यत्सर्वं पूर्वकथितमेवेति ।

ग्रहणवासनायां—छादकनिर्णयं कृत्वा राहुकृतं ग्रहणं न भवतीति वराहमिहिरादीनां मतं प्रतिपाद्य संहितामतमवलम्ब्य तन्निराकृतम् ।

राहुकृतं ग्रहणद्वयमागोपालाङ्गनादि सिद्धमिदम् ।
बहुफलमिदमपि सिद्धं जपहोमस्नानफलमत्र ॥

लोक प्रथेयमिति कथयित्वा राहुकृतं ग्रहणं भवतीत्यत्र स्मृति वाक्यं वेदवाक्यं च प्रतिपादितम्। युक्त्या राहुकृतं ग्रहणं न भवति। परन्तु स्मृतिषु-पुराणेषु वेदेषु च राहुकृतं ग्रहणं प्रतिपादितमस्त्यतोऽत्र द्वयोर्मतयोः समन्वयः—

राहुस्तच्छादयति प्रविशति यच्छुक्लपञ्चदश्यन्ते।
भूछाया तमसीन्दोर्वरप्रदानात् कमलयोनेः॥
चन्द्रोऽम्बुमयोऽधःस्थो यदग्निमयभास्करस्य मासान्ते।
छादयति शमिततापो राहुश्छादयति तत् सवितुः॥

आचार्येण कृतः। सिद्धान्तशिरोमणेर्गोलाध्याये भास्कराचार्येण भूभाकृतं चन्द्रग्रहणम्। चन्द्रकृतं सूर्यग्रहणं प्रतिपाद्य स्मृतिपुराणादिमतेन साकं समन्वयार्थं—

राहुः कुभामण्डलगः शशाङ्कं शशाङ्कगश्छादयतीव बिम्बम्।
तमोमयः शम्भुवरप्रदानात्सर्वागमानामविरुद्धमेतत्॥

इति कथितम्। सिद्धान्तशेखरे राहुकृतग्रहणखण्डनार्थमेको 'राहुनिराकरणाध्यायः' नामकोऽध्यायोऽस्ति।[1] गोलबन्धाधिकारे महद्वृत्तानां (पूर्वापरवृत्त-याम्योत्तरवृत्त क्षितिजवृत्तादीनां) लघुवृत्तानां (मेषादिद्वादशराशीनामहोरात्र-वृत्तानि) रचनां विधाय परमलम्बनावनत्योः स्वरूपं प्रतिपाद्य दृक्कर्मानयनं कृत-माचार्येण, यथाऽत्र ब्राह्मस्फुटसिद्धान्ते ब्रह्मगुप्तेन गोलबन्धः प्रतिपादितस्तथैव सिद्धान्तशिरोमणौ चास्ति। ग्रहर्क्षगोलयोः पञ्च स्थिरवृत्तानि (पूर्वापरम्, याम्योत्तरम्, क्षितिजम्, उन्मण्डलम्, विषुवन्मण्डलम्) इति प्रतिपादितानि, एतानि कक्षामण्डलतुल्यानि महद्वृत्तानि ज्ञेयानि। ग्रहाणां चलवृत्तानि च—

मन्दनीचोच्चवृत्तानि = ७, भौमादीनां शीघ्रनीचोच्चवृत्तानि = ५। मन्द-प्रतिवृत्तानि—७, शीघ्रप्रतिवृत्तानि = ५ दृग्मण्डलं, दृक्क्षेपमण्डलं कक्षामण्डलं

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१. शिद्धान्तशेखरे सूर्यचन्द्रयोर्ग्रहणयोराहोरकारणत्वेऽपि लोकमतश्रुति स्मृति संहि-तानां यथैक्यं भवति तथा तत्प्रतिपादनाय राहोरेव ग्राहकत्वं कथ्यते यथा—

विष्णुलूनशिरसः किल पङ्क्तोर्दत्तवान् वरमिमं परमेष्ठी।
होमदानविधिना तव तृप्तिस्तिग्मशीतमहसोरुपरागे॥
भूमेश्छायां प्रविष्टः स्थगयति शशिनं शुक्लपक्षावसाने।
राहुर्ब्रह्मप्रसादात् समधिगतवरस्तत्तमो व्यासतुल्यः॥
ऊर्ध्वस्थं भानुबिम्बं सलिलमयतनोरप्यधोवर्त्ति बिम्बं
संसृत्यैवं च मासव्युपरतिसमये स्वस्य साहित्यहेतोः।

चेति सप्तानां ग्रहाणाम्=२१, चन्द्रादीनां विमण्डलानि=६ सर्वेषां योगश्चल-वृत्तान्येक पञ्चाशत् सन्तीति, सिद्धान्त शेखरेऽप्येवमेवास्ति यथा—

मन्दोच्चनीचवलयानि भवन्ति सप्त शैघ्र्याणि पञ्च च तथा प्रतिमण्डलानि।
दृक्क्षेप दृष्टचपमजानि च खेचराणामर्कं विनैव खलु षट् च विमण्डलानि ॥
पञ्चाशदेकसहितानि च मण्डलानि पूर्वापरं वलयमुत्तरदक्षिणं च।
क्ष्माजं तथा विषुयदुद्वलयाभिधाने पञ्च स्थिराणि कथितान्युडु खेचराणाम् ॥

इति यन्त्राध्याये—

सप्तदश कालयन्त्राण्यतो धनुस्तुर्यगोलकं चक्रम्।
यष्टिः शङ्कुर्घटिका कपालकं कर्त्तरी पीठम् ॥
सलिलं भ्रमोऽवलम्बः कर्णश्छाया दिनार्धमर्कोऽक्षः।
नतकालज्ञानार्थं तेषां संसाधनान्यष्टौ ॥

इत्यने धनुर्यन्त्रम्। तुरीयम्। चक्रयन्त्रम्। यष्टिः, शङ्कुः, घटीयन्त्रम्। कपालयन्त्रम्। कर्त्तरीयन्त्रम्। पीठसंज्ञं यन्त्रम्। सलिलं (जलम्), भ्रमः (शराः), अवलम्बसूत्रम्, छायाकर्णः, शङ्कुच्छाया, दिनार्धमानम्, सूर्यः, अक्षः (पलांशाः) नतकालज्ञानार्थं सप्तदश काल यन्त्राणि सन्ति, तेषां यन्त्राणां मध्ये सलिलादीन्यष्टौ यन्त्ररचना मूलकानि सन्ति, सिद्धान्तशेखरे 'गोलश्चक्रं कार्मुकंकर्त्तरी च कालज्ञाने यन्त्रमन्यत् कपालम्। पीठं शङ्कुः स्याद् घटी यष्टिसंज्ञं गन्त्री यन्त्राण्यत्र दिक् सम्मितानी' त्युक्त्य १ गोलयन्त्रम्, २ चक्रयन्त्रम्, ३ धनुर्यन्त्रम्। ४ कर्त्तरी नामक-यन्त्रम्। ५ कपाल यन्त्रम्। ६ पीठ (फलक) यन्त्रम्, ७ शङ्कुनामकयन्त्रम्। ८ घटीनामक यन्त्रम्, ९ यष्टि यन्त्रम्। १० गन्त्री (शकट) यन्त्रम्। इति दश-मितानि यन्त्राणि श्रीपत्युक्तानि सन्तीति। शिष्यधीवृद्धिद तन्त्रे द्वादश यन्त्रा-ण्युक्तानि यथा—

गोलो भगणश्चक्रं धनुर्घटी शङ्कुशकटकर्त्तर्यः।
पीठकपालशलाका द्वादश यन्त्राणि सह यष्टया ॥
कर्णच्छाया द्युदलं रविरक्षो लम्बको भ्रमः सलिलम्।
स्युर्यन्त्रसाधनानि प्रज्ञा च समुद्यमाश्चैवम् ॥

भास्कराचार्येण सिद्धान्त शिरोमणौ दशैव यन्त्राण्युक्तानि। यथा—

गोलो नाडीवलयं यष्टिः शङ्कुर्घटी चक्रम्।
चापं तुर्यं फलकं धीरेकं पारमार्थिकं यन्त्रम् ॥ इति।

सूर्य सिद्धान्तोक्त यन्त्राणि—

तुङ्गबीजसमायुक्तं गोलयन्त्रं प्रसाधयेत् ।
गोप्यमेतत्प्रकाशोक्तं सर्वगम्यं भवेदिह ।।

कालसंसाधनार्थाय तथा यन्त्राणि साधयेत् ।
एकाकी योजयेद्बीजं यन्त्रे विस्मयकारिणि ।।

शङ्कुयष्टिधनुश्चक्रैश्छायायन्त्रैरनेकधा ।
गुरूपदेशाद्विज्ञेयं कालज्ञानमतन्द्रितैः ।।

तोययन्त्रकपालाद्यैर्मयूरनरवानरैः ।
ससूत्ररेणुगर्भैश्च सम्यक् कालं प्रसाधयेत् ।।

पारदाराम्बुसूत्राणि शुल्बतैलजलानि च ।
बीजानि पांसवस्तेषु प्रयोगास्तेऽपि दुर्लभाः ।।

ताम्रपात्रमधश्छिद्रं न्यस्त कुण्डेऽमलाम्भसि ।
षष्टिर्मज्जत्यहोरात्रे स्फुटं यन्त्रं कपालकम् ।।

नरयन्त्रं तथा साधु दिवा च विमले रवौ ।
छाया संसाधनैः प्रोक्तं कालसाधनमुत्तमम् ।।

मानाध्याये—

मानानि सौरचान्द्रार्क्षसावनानि ग्रहानयनमेभिः ।
मानैः पृथक् चतुर्भिः संव्यवहारोऽत्र लोकस्य ।।

इत्यनेन सौरचान्द्रनाक्षत्रसावनमानानि कथितानि, एभिरेव चतुर्भिर्मानैर्लोकिानां व्यवहारा भवन्ति। केन केन मानेन के के पदार्था गृह्यन्ते इति सर्वे प्रतिपादिताः सन्ति, ब्राह्मं, दिव्यं, पित्र्यं, प्राजापत्यं, बार्हस्पत्यं, सौरं, सावनं, चान्द्रं, नाक्षत्रमिति नव मानानि सन्ति, एतेषु नवमानेष मनुष्यलोके मानचतुष्टयानां (सौर चान्द्रार्क्ष सावनानां) प्राधान्यम्, यतस्तैरेव तेषां व्यवहारा दृश्यन्ते । सूर्यसिद्धान्त-सिद्धान्तशेखर-सिद्धान्तशिरोमण्यादिषु सर्वेषु ग्रन्थेषुमानानां सम्बन्धे समानरूपेणैव सर्वं प्रतिपादितमस्ति। ब्राह्मस्फुटसिद्धान्तेऽत्राध्याये भूभादैर्घ्यं भूभामानञ्चापि प्रतिपादितमस्ति।

संज्ञाध्याये—संज्ञाकथनकारणम्। सिद्धान्त एक एवास्ति, कस्मिन्नंशे सूर्यसिद्धान्तादयो भिन्नाः सन्तीति प्रतिपाद्य स्वसिद्धान्तोत्तरार्धेऽनुक्रमणिकाः कथिता आचार्येण, नान्येषु-सूर्यसिद्धान्त-सिद्धान्तशेखरादिसिद्धान्तग्रन्थेषु संज्ञाध्यायः। अध्यायोपसंहारात्पूर्वमेकः प्रश्न विशेषः—

आग्नेये नैर्ऋत्ये वेष्टदिने संस्थितस्य योऽर्कस्य ।
शङ्कुच्छाये कथयति वर्षादपि वेत्ति सूर्यं सः ॥
कथितोऽस्ति यदुत्तरं कोणाशङ्कोरानयनेत स्फुटमस्ति ।
संज्ञाध्यायकथनस्य किमपि प्रयोजनं नासीत् ॥
आचार्येण कथमेतस्याध्यायस्योल्लेखः कृत इति न जानीमः ।

ध्यानग्रहोपदेशाध्याये—चैत्रादौ मासगणानयनम् । चैत्रादौ दिनादिकं तिथि ध्रुवसाधनम् । इष्टमासादौ रव्यानयनम् । प्रतिमासं शशिकेन्द्रतिथिध्रुवक्षेपयोगानयनम् । प्रतिदिनचालनम् । चन्द्रसाधनमौदयिकरविसाधनञ्च । ज्याखण्डकानि केन्द्रज्या साधनञ्च यथा —

त्रिंशत् सनवरसेन्दुर्जिनतिथिविषया गृहार्धचापानाम् ।
अर्धज्याखण्डानि ज्याभुक्तैक्यं स भोग्य फलम् ॥
गतभोग्यखण्डकान्तरदलविकलवधाच्छतैर्नवभिराप्तैः ।
तद्युतिदलं युतोनं भोग्याद्दूनाधिकं भोग्यम् ॥

इत्याचार्योक्तभोग्यखण्डस्पष्टीकरणमेव सिद्धान्तशिरोमणेर्ग्रहगणिताध्यायस्य स्पष्टाधिकारे 'यातैष्ययोः खण्डकयोर्विशेषः शेषांशनिघ्नः' इत्यादिना भास्कराचार्येण कथितम् । भास्कराचार्येण खार्क-१२० मिता त्रिज्या गृहीता, अत्राचार्येण खतिथि १५० मिता त्रिज्या गृहीता । इत्यतीववैचित्र्यं यत्सर्वत्रैव श्रीपतिनाऽऽचार्यश्लोकोक्तविषया एव छन्दोन्तरेण लिखिताः परं किं कारणं यदपूर्वं भोग्यखण्डस्पष्टीकरणं न लिखितम् । चन्द्रे भुजफलसंस्कारः, तिथौ फलसंस्कारश्च । इत्यादयः सर्वे विषया अपूर्वाः सन्ति । अत्राध्याये ये केचन विषया लिखिताः सन्ति ते सूर्यसिद्धान्ते सिद्धान्तशेखरे, सिद्धान्तशिरोमणौ न सन्ति ।

अनुगृहीतोऽस्म्यहं श्रीडाक्टर सत्यप्रकाश डी० एस० सी० महोदयानां यैराङ्गलभाषायां कृपया ग्रन्थस्यास्य प्रस्तावना समरचि ।

सम्पादकमण्डलस्यान्ये सहयोगिनः ज्योतिषाचार्याः श्रीमुकुन्दमिश्राः, श्रीविश्वनाथ झा, श्रीदयाशंकरदीक्षिताः, शास्त्रिणः श्रीओंदत्तशर्माणश्च सर्वे एव मम धन्यवादस्य पात्रतामर्हन्ति । एतेषाममूल्यसहयोगेनैवायं महान् ग्रन्थः सुचारुरूपेण सम्पूर्णतामगच्छत् ।

'पद्मश्रीप्रकाशनालयस्य' स्वामिने रमेशचन्द्रशर्मणेऽपि धन्याञ्जलिर्भवतु यस्य महता परिश्रमेण ग्रन्थस्य प्रकाशनं कालेनैवाभवत् ।

अन्येभ्यः सर्वेभ्योऽपि धन्यवादान् प्रददे, यैरल्पमपि साहाय्यं विधायाहं कृतार्थीकृतोऽस्मि, इति शम् ।

भृगु-आश्रमः
३०-३-६६

विदुषामनुचरः
रामस्वरूप शर्मा

श्रीब्रह्मगुप्ताचार्य विरचितः

विभिन्नपाठान्तरसहितो

ब्राह्मस्फुटसिद्धान्तः

CHAPTER I

Astronomy in Ancient Nations

Brahmagupta's great works like the *Khaṇḍakhādyaka* and *the Brāhma-sphuṭa-siddhānta* took astronomy to Arabs through whom it spread to many countries of Europe. Al Beruni records this testimony in his great book on India. It is doubtful that astronomy had its birth in Greece and China. From remote ages China, India, Greece, Arabia and Egypt developed the entire system in close cooperation. This knowledge must have spread from their common cradle home where man for the first time developed his culture and civilisation. In this chapter we propose to give a review of astronomy as developed in many of these ancient lands, especially Arabia, people of which land came in close contacts with India much before any recorded time.

Dawn of Astronomy

The earliest man must have been the primitive astronomer. The striking spectacles presented to him by the varied appearances of a sky covered with thousands of twinkling and non-twinkling objects of different degrees of brightness, apparently revolving round the Earth, and the daily changing phases of the Moon must have raised strange feelings of the most primitive man also. Then he must have in course of time observed the bright morning and evening stars, and at a considerably late stage the comets and shooting stars and then on occasions eclipses of the Sun and the Moon. These phenomena not only raised feelings of admiration, but in different sections of human society often feelings of superstitious alarm. By and by stars became guides for the traveller by land and sea. In the midst of these observations, one discovered various cycles : cycle of day and night, cycle of seasons and cycle of other details. Then there was a striking observation of the tides in a sea changing with the phases of the Moon.

Earliest Discoveries

We shall briefly sketch out the order of astronomical discoveries. The first phenomena to be noted must have been the regularly recurring dawn (for this one may refer to the Uṣā Sūkta of the *Ṛgveda*), the sunrise and sunset (which led to *prātaḥ* and *sāyaṃ*, *i.e.* morning and evening prayers of the Vedic times), daylight, twilight and night concerning which we have numerous Vedic hymns. Next it led to the measurement of a day (which was of a short duration in winters and of a long duration in summers). The Vedic Aryans also discovered the variations in the duration of the day along different latitudes, and the time of sunrise in places of different longitudes. In fact the idea of longitudes and latitudes came sufficiently afterwards. Man discovered month as related to the variation of light with the Moon's phases. In temperate regions, where probably the first astronomical observations were systematically made, the changing length of the day or the direction of the Sun at rising or setting or the lengths of shadows cast at midday, would show that the Sun's daily path in the sky altered throughout the year, a time interval which was already marked by the changing vegetation. According to Sir W.C. Dampier, "attempts were made to determine the number of months in the cycle of the seasons in Babylonia about 4000 B. C. and in the China soon after. About 2000 B. C. the Babylonian year settled down to one of 360 days or twelve months, the necessary adjustments being made from time to time by the interposition of extra-months." In India, this concept is of even much earlier origin. The old inspired sages like Dīrghatamas discovered for the observing man the Vedic Era and intercalation. I have described this discovery in a special chapter on the subject in my book the *Founders of Sciences in Ancient India* and a reference may be made to the *Asya Vāmasya Sūktam* of the *Ṛgveda*. It is impossible to assign an age to these old traditions. Round the *Yajña*, developed the science of astronomy, mathematics, anatomy and medicine in this ancient land of ours, which in fact was the common heritage of a large number of people of the modern world.

One might also say that a considerable period might have well elapsed before it was noticed that at a particular season of

the year, the same stars are seen at corresponding hours of the night. Of course this circumstance was less conspicuous than the regular variation of the Sun's altitude in the sky as the year progresses. It is the surmise that the striking naked-eye cluster, the Pleiades, must have been one of the earliest noted star-groups, and it became the first star-group for providing the first fairly close determination of the length of the year as approximately 365 days. The rising of this cluster in the evening was a mark of the coming winter to primitive man ; and the husbandman judged the time of reaping by its rising, and of ploughing by its setting in very ancient times ; Sirius, Arcturus, the Hyades and Orion were similarly equally useful to him. The passages in the *Taittirīya Saṁhitā* and in the *Śatapatha Brāhmaṇa* clearly indicate the confusion once created by following the concept of lunar months without further adjustments :

> "Now the seasons were desirous to have a share in the sacrifice among the gods and said, 'Let us share in the sacrifice. Do not exclude us from the sacrifice ! Let us have a share in the sacrifice !' The gods, however, did not approve of this. The gods, not approving, the seasons went to the Asuras, the malignant, spiteful enemies of the gods. Those (Asuras) then throve in such a manner that they (the gods) heard of it, for even while the foremost (of the Asuras) were still ploughing and sowing, those behind them were already engaged in reaping and threshing : indeed even without tilling, the plants ripened forthwith for them. (*ŚBr.* I.6.1.1-3)

The Zodiac

It is difficult to say how much time it must have taken, but in fact, it was eventually noted that the Sun and Moon travel over very similar paths among the stars during their circuit of the sky. This led to the formation of the Zodiac and its constellations, the centre of this zone, a belt about 16° broad, being the annual path of the Sun or Ecliptic. The division into twelve parts, each corresponding to a month of the Sun's movement, was made ; and their connection with the solar course during the year was found by observations of heliacal risings or settings. These were the times of the year when certain bright stars would first be seen to rise before the Sun, or when

they were last seen to set after sunset. In the case of Śirius, the brightest fixed star, these would happen when the Sun was about ten degrees below the horizon. For the less bright stars the angle would be a larger one.

It must have been almost simultaneously observed that the Moon in going like the Sun round the heavens always in the same direction from west to east (i. e., opposite to the diurnal motion which she shares with the other bodies), kept in general to the same track in the sky. After a time, however, it must have been noted by careful observers that this path was not constant, but deviated from the centre line of the Zodiac, getting away from that line up to a maximum deviation on either side but slowly returning to it. In the course of a number of years, it must have become evident that the Moon's path among the stars does not lie always in the same line on the celestial sphere, but in a zone or band about twenty moon breadths (10°) wide, occupying the middle of the Zodiacal zone itself.

Among the bright stars Mercury, Venus, Mars, Jupiter and Saturn (the first two of which are never seen very far from the Sun in the sky) soon must have been noted to be moving in the Zodiac with varying periods. The English name *planet* is derived from Greek *planetes*, meaning a wanderer, since the planets change their positions among the Zodiacal stars.

There is a word *Stṛ* which in the *Ṛgveda* always occurs in the instrumental plural, *Stṛbhiḥ*. The English word star is derived from this word. Parāśara and Gṛtsamada, I have shown elsewhere, were the first amongst the great observers, inspired by the Ṛgvedic hymns, and Vāmadeva identified Bṛhaspati or the Jupiter planet and Vena Bhārgava discovered tht planet Venus which still bears the name of its discoverer.

Constellations

Long before the Zodiacal belt was divided into "signs" (700 B. C.), a number of asterisms, or the configuration of stars in the sky had been arranged, the brighter stars of these configurations, thus identified, proved very useful in indicating the seasons of the year by the times of their rising or setting, and also in locating the positions on the celestial vault of such moving objects as planets, comets and shooting stars and in helping the traveller by land or sea to determine direction.

These named constellations date back to very early period. In India, Gārgya is the name of an astronomer who is associated with a hymn of the *Atharvaveda* which for the first time enumerates constellations. In many of these constellations, the stars form a well marked group, clearly separated from other groups, and the names given to these formations are supposed to have been suggested by a resemblance to the shapes of certain familiar objects. Of course, the resemblance is usually very slight, and depended merely on a fancy.

It is remarkable that different countries devloped almost similar notions regarding these constellations. The late Dr. A. C. D. Crommelin considered that there is a reason to believe that the stars may have been grouped to some extent by the Egyptians as early as 4000 B. C., and he remarked on their use of the then Pole Star for orienting the Great pyramid, Again, Chinese are said to have mapped out the sky into many divisions of stars by 2500 B. C., if one can rely on their records.

The idea of constellations takes us to a date much earlier than 2500 B. C. even. In total forty-eight have come down from extremely ancient times, but these do not cover the entire extent of the sky. The part not occupied by any of them evidently did not rise above the horizon where the early astronomers to whom we owe their naming lived; and the stars concerned were therefore not included in their constellation schemes. The centre of this part (near the bright star Achernar) must have been near the South Pole of the heavens of the time, and its angular radius from the Pole gives us roughly the latitude of their homes. The date appears to have been about 2800 B. C., when, owing to the precession of the equinoxes, the South celestial pole was in the position indicated. The latitude seems to have been about 38° North. These are the findings of E. W. Maunder (*Astronomy without a Telescope*, p.5, 1902); but from the same considerations Dr. Crommelin assigns a latitude of 36° and a date 2460 B.C. and Proctor 2200 B. C. Maunder also suggested that the presence of the Lion and Bear among the stellar configurations and the absence of Elephant, Tiger, Camel and Crocodile seem to exclude India towards the East and the countries towards the West, the latitude and the longitude indicated being those of Asia Minor or Armenia. The suggestion that the blank area in

the sky referred to gave an approximate date for the formation of the constellations appears to have beeen first put forward in 1807 by Carl Schwartz, for some time Swedish Consul at Baku.[1]

Indo-Greek Contacts

It is highly improbable that before Alexander, there had been absolutly no contacts between India and the distant nations. Even in pre-Alexandrian era, there had been such migration is clearly evinced by the philological and mythological studies. But we do not possess historic record of it.

The conquests of Alexander the Great made the Greeks acquainted with the Eastern world, which had up to that time been visited probably by very few Europeans, and it likewise spread Greek culture to all the countries which the victorious Macedonian had been able to reach. The Indian province of his Empire became independent soon after Alxander's death and though the spread of Buddhism in the third century B. C. checked the progress of Hellenism in Northern India, the rise of the Greek kingdom of Bactria and its gradual extension south and east continued for a long time to keep alive the connection between India and the West. Not only (as has beeen asserted) the Greek and Indian drama and architecture have been strongly influenced by Hellenistic and Indian contacts, it is beyond a doubt that the entire astronomy of the two great nations is the offspring of these mutual contacts.

In earliest times astronomy had only been cultivated in India and in no other country. Some idea had been acquired during those days of the periods of the Sun and Moon and the planet Venus and Bṛhaspati (Jupiter), which were used for chronological purposes, the lunar motions being specially connected with the proper times for sacrificial acts. The Vedic era was discovered during this period by Viśvāmitra, and Gārgya enumerated the Nakṣatras. Lagadha composed his *Vedaṅga Jyotiṣa*, which is the *first* book on astronomy written in human literature. India developed her geometry in connection with the construction of sacrificial altars, and its account is found in the *Śulba Sūtras* of Baudhāyana, Āśvalāyana and Kātyāyana. Āryabhaṭa laid the foundations of algebra. One might still say that there is no sign of

1. See, Peter Doig : *A concise History of Astronomy*, London, 1950.

accurate knowledge of the planetary motions earlier than about the century of the Christian era. From thenceforth astronomy, hitherto confined to rituals, appears as a science, treated in the course of the next thousand years in a series of text-books, the *Siddhāntas*,[1] the contents of which, though supposed to be derived from divine sources are strongly influenced by Greek authors. Prior to the Greek influence, we had ceremonies like dvādaśāha (lasting for twelve days), ṣaḍaha (lasting for six days), tryaha (lasting for three days) besides darśa-pūrṇamāsa ceremonies connected with the New Moon and Full Moon. But the week of seven days (*Saptāha*) was unknown in India. This concept of week and the dedication of each day to the deity of one of the seven planets, now appears for the first time. It is difficult to say whether names of the planets were borrowed by Greeks from India or vice versa, but they became common, e.g., Aśvajit or *Asphudit* (Aphrodite). *Dyugatiḥ* or *dyaus* or *Jiva* (Zeus), Heli (Helios), &c., while the zodiacal signs have superseded the earlier but totally different twelve star-groups connected with the Sun's motion, and proclaim their origin by their names:

Kriya, Tāvuri, Jituma, Karkin, Leya, Pāthena, Jūka, Kaurpya, Taukshika, Ākokera, Hridroga, Ittha,

corresponding to *Κριός, Ταῦρος, Δίδυμος, Καρκῖνος, Λέων, Παρθένος, Ζυγόν, Σκορπίος, Τοξότης, Αἰγόκερως, Ὑδροχόος, Ἰχθύς*[2].

A great many other Greek terms connected with geometry, astronomy and astrology have also been transferred from Sanskrit works to Greek and vice versa, This conclusively shows the mutual influence on astronomy. Indian authors never failed to acknowledge the ideas they borrowed from Greeks, e.g. Varāhamihira quotes the Yavanas or peoples of the west as authorities for some of the scientific statements he makes. The name of the Romaka Siddhānta (which is at least as old as A.D. 400) also points in an unmistakable manner to its origin in one of the provinces of the Roman Empire.

1. The *Romaka* or *Pauliśa Siddhānta* (before 400 A. D.), See Varāhamihira's the *Pañcasiddhāntikā* (about 570 A.D., Varāhamihira died in 587 A.D.). The original *Sūrya-siddhānta* was prior to Varāhamihira, the modern edition is perhaps of the 13th century. See J. Burgess "*Notes on Hindu Astronomy*" J. R. A. S., October 1893, p. 742.

Earth as a Sphere

The astronomers of the *Siddhāntas* taught that the Earth is a sphere, unsupported in space, and they reject the ancient mythological notion that it is supported by some animal like śeṣanāga (serpent), kacchapa (tortoise), or diggajas (elephants) which in turn rest on another, and so on, until the support of the last one after all has to be left unexplained. Bhāskara II, about A.D. 1150, who comments on the absurdity of this, also rejects the idea that the Earth is perpetually falling, since it would fall faster than an arrow shot upwards, on account of being heavier, so that an arrow could never again reach the Earth.[1] Round the Earth the planets are moving, all with the same linear velocity. The diameter of the Earth is 1600 yojanas, the distance of the Moon is 51,570 yojanas (or 64.5 times the radius of the Earth, nearly equal to Ptolemy's greatest distance, 64⅛), while the distances of the other planets result from the assumption of equal velocities.[2] The equation of centre of the planets is found by an epicycle and to this arrangment the Hindus add one of their own invention, by assuming that the epicycle had a variable circumference, greatest when the planet is at apogee or perigee and least at 90° from these, when the equation reaches its maximum. This contrivance of an oval epicycle was by some astronomers applied to all the planets, by others (Brahmagupta and Bhāskara) only to Mars and Venus, by others it was altogether rejected.[3] Why they complicated the calculation in this way is not clear. Āryabhaṭa I of Kusumapura or Pāṭaliputra, born A.D. 476, made another deviation from the Alexandrian doctrines, as appears in the *Brāhma-sphuṭa-siddhānta* of Brahmagupta, wherein he quotes the following from Āryabhaṭa: "The sphere of the stars is stationary, and the Earth, making a revolution, produces the daily rising and setting of stars and planets." Brahmagupta rejects this idea, saying: "If the Earth moves a minute in a prāṇa, then whence and what route does it proceed? If it revolves, why do not lofty objects fall?" But his commentator Caturveda Pṛthūdaka Svāmī replies: "Āryabhaṭa's

1. *As. Res.* XII. p. 229 (*Essays*, II, p. 394).
2. The distances are proportional to the orbital periods of revolution, but for Mercury and Venus to the periods in the epicycles.
3. For further details see *As. Res.* II. p. 251 (Davis) and XII. p. 236 (Colebrooke, also *Essays*, II. p. 401).

opinion appears nevertheless satisfactory, since planets cannot have two motions at once : and the objection, that lofty things would fall, is contradicted; for every way the under part of the Earth is also the upper; since wherever the spectator stands on the Earth's surface, even that spot is the uppermost spot.[1]

Earth rotation by a current of aerial fluid

It is very interesting to see the theory once advocated by Herakleides of Pontus transplanted on Indian soil, especially when we remember that Seleukus, the Babylonian, had adopted that theory. From Babylon the theory might easily find its way to India, though it is of course equally possible that Āryabhaṭa, quite independently of his Greek precursors, hit on the same idea. He appears to have accounted for the Earth's rotation by a wind or current of aerial fluid, the extent of which, according to the orbit assigned to it by him, corresponds to an elevation of little more than a hundred miles (114) from the surface of the Earth, or fifteen yojana's while he put the diameter of the Earth equal to 1050 yojanas (of 7.6.miles each[2]). This was in accordance with the general opinion of the Indians, that the planets are carried along their orbits by mighty winds with the same velocity and parallel to the ecliptic (while one great vortex carries all stars round the Earth in twenty-four hours, but that the planets are deflected from these courses by certain invisible powers having hands and reins, with which they draw the planets out of their uniform progress. The power at the apogee, for instance constantly attracts the planet towards itself, alternately with the right and left hand (like Lachesis in Plato's *Republic*), while the deity at the node diverts the planet from the ecliptic first to one side and then to the other. And lastly the deity at the conjunction causes the planet to move with variable velocity and to become occasionally stationary and even retrograde. This is gravely set forth in the *Sūrya-siddhānta* ,and even Bhāskara gives the theory in his notes, though he omits it from his text. Similarly Brahmagupta, although he gives the theory of eclipses, affirms the existence of an eighth planet, Rāhu, which is the immediate cause of eclipses; and he blames Varāhamihira,

1. *Asiat. Res*, XII. p. 227; Colebrooke's *Essays*, II. p. 392.
2. Colebrooke, *Notes and Illustrations to the Algebra of Brahmagupta*. p. xxxviii.' *Essays*. II. p. 467.

Āryabhaṭa and others for rejecting this orthodox explanation of the phenomenon.[1]

Indian astronomy some times appears to be a curious mixture of old fantastic ideas and sober geometrical methods of calculation. But it is wrong to presume that these geometric calculations were derived from froeign contacts. Indians have always been fond of geometry (from the earliest times of the Vedic rituals), and they from the very beginning realised the importance of applying geometry to astronomy. Side by side we find Greek contacts also. As remarked by Colebrooke, the absence of the most characteristic parts of Ptolemy's system, the equant and the details of the theories of the Moon and Mercury seems to indicate that Greek planetary theory must have been introduced in India between the times of Hipparchus and Ptolemy; and with the exception of the epicycle from the circular form, the Hindus did not modify the theory or perfect it in any way. The precession of the equinoxes they held to consist in a liberation within the limits of 27° (Āryabhaṭa says 24°) east and west of its mean position, but they came much nearer to the truth than Ptolemy did as regards the annual amount, as they supposed the space travelled over in a century to be $1\frac{1}{2}°$.

Contacts with Arabs

Notwithstanding some isolation of India from Europe during the Middle Ages, her astronomy was destined to exercise an indirect influence on the progress of astronomy. Through the conquest of Persia in the seventh century, the Arabs, like the Greeks a thousand years earlier, came in contact with India, from whence physicians and astrologers found their way to the court of the Caliph already before the reign of Harun al Rashid. We possess a detailed account of the manner in which the Indian astronomy was introduced at Baghdad, from the pen of the astronomer Ibn al Adami (who died before 920), confirmed by the celebrated memoir on India by Al Beruni, written in 1031[2]. In the year 156 of the Hijra (A. D. 773), there appeared before the Caliph Al Mansur a man who had come from India; he was skilled in

1. *Asiat. Res.* XII, pp. 233, 241; *Essays*, II, pp. 398, 407.
2. Hankel, *Zur Geschichte der Mathematik im Alterthum* and *Mittelalter*, Leipzig, 1874, p. 229; Cantor, *Gesch. d. Math.* I, p. 656.

the calculus of the stars known as the Sindhind (i. e. Siddhānta), and possessed methods for solving equations founded on the kardagas (i. e. kramajyā, sines) calculated for every half degree, also methods for computing eclipses and other things. Al Mansur ordered the book in which all this was contained to be translated into Arabic, and that a work should be prepared from it which might serve as a foundation for computing the motions of the planets. This was accordingly done by Muhammed ben Ibrahim Al Fazari, whose works the Arabs call the great *Sindhind*, and from it an abstract was afterwards made for Al Mamun by Abu Giafar Muhammed ibn Musa al Kwarizmi, who made use of it to prepare his tables, which obtained great renown in the lands of Islam. But when Al Mamun became Caliph, he promoted these noble studies and called in the most learned men in order to examine the *Almagest* and make instruments for new observations.

Arabs and Greeks

The account of which the above is an abstract shows us clearly the origin of the study of astronomy and mathematics under the Abbasid Caliphs. But though the first impulse came from India the further development of Arabian science was to a considerable extent founded on that of Greece and Alexandria. It was through the court physicians from the flourishing medical school kept up by Nestorian Christians of Khusistan that a knowledge of Greek Philosophy and science was first spread among the subjects of the Caliphs; and by degrees the works of Aristotle, Archimedes, Euclid, Apollonius, Ptolemy, and other mathematicians were translated into Arabic. Fresh translations of Ptolemy were made from time to time in the various kingdoms into which the vast empire of the Caliph was soon split up,[1] and a thorough knowledge of Ptolemaic astronomy was thus spread from the Indus to the Ebro. There were several special inducements for Muhamedans to pay attention to astronomy, such as the necessity of determining the direction in which the

1. The earliest is probably that of Al Haǧǧag ben Jūsuf ben Matar early in the ninth century. See Suter, *Die Mathematiker und Astronomen der Araber und ihre Werke*, Leipzig, 1900 (p. 9), which valuable bibliograpical summary has been follwed by J. L. E. Dreyer as regards names and dates (J. L. E. Dreyer's *A History of Astronomy*, 1953; we have reproduced this account from his chapter XI.)

faithful had to turn during prayers, also the importance of the lunar motions for the calendar, and the respect in which judicial astrology was held all over the East. The Caliph Al Mamun, son of Harun Al Rashid (813-833) is the first great patron of science, although the Omayyad Caliphs had much earlier an observatory near Damascus, and the Jew Mashallah (who died about 815) had already before the reign of Al Mamun won a name as an observer and astrologer. But the Damascus observatory became quite eclipsed by that erected at Baghdad in 829 where continuous observations were made and tables of the planetary motions constructed while an important attempt was made to determine the size of the Earth. Among the astronomers of Al Mamun and his successors one of the greatest was Ahmed ben Muhammed Al Fargani (afterwards known in the West as Alfraganus), whose *Elements of Astronomy* were translated into Latin in the twelfth century and contributed greatly to the revival of science in Europe.[1] Tabit ben Korra (826–901) was a most prolific writer and translator, but is chiefly known in the history of astronomy as a supporter of the erroneous idea of the oscillatory motion of the equinoxes. A younger contemporary of his, Muhammed Al Battani (died 929), was the most renowned of all the Arabian astronomers and became known in the West in the twelfth century (under the name of Albategnius) by the translation of the introduction to his tables.[2] Already in his time the power of the Caliphs had commenced to decline, and they soon lost all temporal power. The study of astronomy was, however, not influenced by this loss of patronage, as the Persian family of the Buyids, who in 946 obtained possession of the post of Amir-al-Omara (corresponding to the Frankish Major Domus) took over the *role* of patrons of science, so long and so honourably carried on by the Abbasid Caliphs. Sharaf al Daula built in 988 a new observatory in the garden of his palace, and among the astronomers who worked there was Muhammed Abu 'l Wefa al Buzjani (959-998), who wrote an *Almagest* in order to

1. First printed at Ferrara in 1493. See the edition of Golius, Amsterdam, 1669.

2. Translated by Plato of Tivoli. First Printed in 1537 after the book of Alfargani. Dreyer has used the edition of Bologna, 1645, and a new edition which is now being published by C. A. Nallino, of which the Arabic and a Latin translation of the text have already appeared (*Pubbl. d. R. Osservatorio di Brera in Milano*, No. 40, 1899-1903).

make the contents of Ptolemy's work accessible to the less learned. In the nineteenth century this book gave rise to a long controversy, which we shall presently consider somewhat in detail.

Western Countries under Islam

In the eleventh and twelfth centuries we do not find any names of conspicuous astronomers in Muhammedan Asia. But the western countries under Islam had in the meantime become ready to do their share of the work of keeping the mathematical sciences alive. In the Fatimite kingdom of Egypt Ali ben Abi Said Abderrahman ben Ahmed ben Jūnis, generally called Ibn Jūnis (died 1009), was distinguished both as an astronomer and a poet. At Cairo a liberally equipped observatory enabled him to verify the planetary theories which had once been developed in the neighbouring Alexandria, and in token of his gratitude to the reigning sovereign, Al Hakim, he named his work the Hakemite Tables.[1] We have to pass to the farthest west to find the next astronomer of mark in the person of Ibrahim Abu Ishak, known as Al Zarkali (in Europe afterwards called Arzachel). He was a native of Cordova, lived about 1029-1087, and edited planetary tables called the Toledo Tables.[2] In the following century we find two celebrated astronomers of Seville, Gabir ben Aflah, known as Geber (died 1145, often mistaken for the great alchemist, Gabir ben Haijan, in the eighth century),[3] and Nur ed-din al Betrūgi (Alpetragius), both of whom raised objections to the planetary theories of Ptolemy, though they failed to produce anything better of their own. Spanish astronomy continued to flourish for a while, although the power of the Arabs in the Peninsula was rapidly declining, and it produced in the thirteenth century a very remarkable man, who, although a Christian king, must be included in this account of Arabian astronomy,

1. Caussin has published an extract in vol. *vii* of the *Notices et Extraits des manuscrits* (*Le livre de la grande table Hakemite*). Other Chapters, translated by the elder Sedillot but never published, are reviewed by Delambre, *Hist, de l'astr. du Moyen Age*. p. 95 sqq.
2. Never published. Delambre, l. c. p 176, and Steinschneider, *Etudes sur Zarkali*. *Bullettino Boncompagni* T. *xx*. p. 1.
3. The word algebra has also sometimes erroneously been connected with his name.

as he owed all he knew about the science to the example and the teaching of Muhammedans and Jews. King Alfonso X, of Castille named el Sabio (1252-1284), followed the example of the Caliphs and called astronomers to his court to assist in the preparation of the renowned Alfonsine Tables.

With Alfonso the study of astronomy disappeared from Spain, but not before it had been revived in the East. In 1258 the still existing but shadowy Caliphate of Baghdad was swept away by the Mongol conqueror Hulagu Khan, grandson of Genghis Khan; but already in the following year this great warrior listened to the advice of his new vazier, Nasir ed-din al Tūsi (born at Tūs in Khorasan in 1201, died in 1274), and founded a great and magnificient observatory at Merāgha, in the north-west of Persia. In this observatory, which was furnished with a large number of instruments, partly of novel construction, Nasir ed-din and his assistants observed the planets diligently and produced after twelve years labour, the "Ilokhanic Tables." Among the astronomers of Merāgha seems to have been Juhanna Abu 'l Faraj, called Bar Hebrāyā, or the son of a Jew. He was a Christian, born in 1226 and from 1264, till his death in 1286 Maphrian or Primate of the Eastern Jacobites. He left a well-known chronicle and an astronomical work, both written in Syriac, as well as other writings.[1] The observatory at Merāgha had not a long life, and Asiatic astronomy had to wait a century and a half, until the grandson of another terrible conqueror erected another observatory. Ulug Begh, grandson of Tamerlan, drew learned men to Samarkand and built an observatory there about the year 1420, where new planetary tables and a new star catalogue, the first since Ptolemy's, were prepared. Ulug Begh died in 1449, he was the last great Asiatic protector of astronomy; but just as the Eastern countries saw the star of Urania setting, it was rising again for Europe.

In this review of Arabian astronomers we have only mentioned a few, omitting several names of distinction, whose

1 *Le livre de l' ascension de l' esprit sur la forme du ciel et de la terre.* Cours d' Astronomie redige en 1279 par Gregoire Aboulfarag, dit Bar Hebraeus. Publie par F. Nau, Paris, 1899-1900 (2 parts, Syriac and French). His chronicle is the chief authority for the fable about the burning of the Alexandrian Library by order of the Caliph Omar. For a very thorough refutation of this see Butler, *The Arab Conquest of Egypt*, Oxford, 1902 pp 401-426.

owners devoted themselves to other branches of astronomy. Though Europe owes a debt of gratitude to the Arabs for keeping alive the flame of science for many centuries and for taking observations, some of which are still of value, it cannot be denied that they left astronomy pretty much as they found it. They determined several important constants anew, but they did not make a single improvement in the planetary theories. It will therefore be sufficient to enumerate the improvements attempted and the opinions held by Arabian astronomers without keeping strictly to the chronological order, although we are here dealing with a period of about six hundred years and men belonging to very different nations, who had little in common except their religion and the language in which they wrote.

Figure of Earth

Turning first to the question of the figure of the Earth, we find a remarkable contrast between Europe and Asia. In the world under Islam there was an entire absence of that hostility to science which distinguished Europe during the first half of the Middle Ages. Though we learn from Kazwini's *Cosmography*[1] that some of the earlier Arabs believed the Earth to be shaped like a shield or a drum, still there is no record of any Arabian having been persecuted for asserting that the Earth is a sphere capable of being inhabited all over. Whether this was in consequence of the warriors of the Caliphs having carried their arms to the centre of France on one side and to the borders of China on the other while their merchants travelled southward to Mazambique and northward to the centre of Asia, is another question : anyhow, the fact of the Earth being a sphere of very small dimensions in comparison to the size of the universe was accepted without opposition by every Arabian scholar, and the very first scientific work undertaken after the rise of astronomy among them was a determination of the size of the Earth. It was carried out by order of the Caliph Al Mamun in the plain of Palmyra. According to the account given by Ibn Jūnis, the length of a degree was measured by two observers between Wamia and Tadmor and by two others in another locality, we are not told where. The first measure gave a degree

1. *Zakarija Ben Muhammed Ben Mahmūd El Kazwini's Kosmographie*, deutsch von H. Erbe, Leipzig, 1868, p. 295.

equal to 57, the second one equal to $56\frac{1}{4}$ Arabian miles of 4000 black cubits, and the approximate mean, $56\frac{2}{3}$ miles, was adopted as the final result, the circumference of the Earth being 20,400 miles and the diameter 6500 miles. Another report, by Ahmed ben Abdallah, called Habash, an astronomer under Al Mamun (quoted by Ibn Jūnis), states that a party of observers (no names given) proceeded along the plain of Sinjar until they found a difference in meridian altitudes, measured the same day, equal to one degree, while the distance travelled over was found to be $56\frac{1}{4}$ miles[1]. Probably two different determinations were made. If the "black cubit" is the Egyptian and Babylonian cubit of 525 mm.[2], the mile would be=2100 m. and $56\frac{2}{3}$ miles=119,000 meters, rather a large result.

The doctrine of the spherical earth remained undisputed in the Muhammedan learned world, though the curious error of assuming that the level of the sea was higher on some parts of the Earth than on others appears to have found some adherents among Arabian writers as well as in Europe.[3] We may, therefore, at once pass on to the motions of the heavenly bodies. Al Battani determined the longitude of the Sun's apogee and found it=82° or 16° 47[1] more than Ptolemy had given. As he believed

1. Caussin, *Not. et Extraits.* vii. pp. 94-96; Delambre, *Hist. de l'astr du Moyen Age.* pp. 78 and 97; Shems ed-din, *Manuel de la cosmographie*, traduit par Mehren, Copenhague. 1874 p. 6. Suter, p. 209, mentions a third report (from Ibn Challikān's *Biographical Dictionary*), according to which the sons of Musa first measured in the plain of Sinjar and afterwards as a test at Kūfa, by order of Al Mamun. The eldest of the sons of Musa died 41 years after Al Mamun, and the names of the observers in the first report are different, so that the third report is not to be relied on. Al Fargani merely gives $56\frac{2}{3}$ miles as the result of Al Mamun. According to Shah Cholgii *Astronomica . . . . studio et opera Ioh. Gravii*, London, 1652, p. 95, Ala ed-din Al Kūsgi (one of the Ulug Begh's astronomers) gives the circumference of the earth=8000 parasangs. As a persian parasang =30 stadia (Hultsch; *Griech u. Rom. Metrologie.* p. 476) this would seem to be the value of Posidonius, 240,000 stadia. Kazwini (p. 298) gives the circumference=6800 parasangs on the authority of Al Berūni.
2. Hultsch p. 390.
3. It deserves to be mentioned that Shems ed-din of Damascus (1256-1327) explains the great Preponderance of dry land in the northern hemisphere by the attraction of the Sun on the water, which is the greatest when the Sun is in perigee, at which time it is nearly at its greatest south declination. That this accumulation of water would not be a permanent one does not occur to him (*Cosmographie*, p. 4).

that Ptolemy's value had been found by himself,[1] and as he adopted 54″ (or 1° in 66 years) as the annual amount of precession, there remained (assuming that 760 years had passed since the time of Ptolemy) an outstanding error of 79″—54″=25″ per annum. In reality the annual motion of the solar apsides is 11½″; still we may say that the discovery of this motion is due to Al Battani, though he did not announce it as such; in fact he merely gives his own value as an improvement on that of Ptolemy. Even Ibn Jūnis (who found 86° 10′) did not suspect that the apogee was steadily moving but merely says that it must be corrected for precession (1° in 70 years), and remarks that the longitude of the apogee is very difficult to determine accurately.[2] On the other hand, Al Zarkali found a smaller value, 77° 50′ and as he also found a smaller value of the eccentricity he thought it necessary to let the centre of the Sun's eccentric orbit describe a smaller circle, after the example set by Ptolemy in the case of Mercury.[3] The inclination of the ecliptic which the Greeks had found—23° 51 20″ was by the astronomers of Al Mamun found—23° 33′ (in 830), by Al Battani (in 879), and by Ibn Jūnis 23° 35′[4]. When Al Zarkali found 23° 33′,he, and afterwards Abu 'l Hassan Ali of Morocco, concluded that the obliquity oscillated between 23° 53 and 23° 33′, an idea to which the prevailing belief in the "trepidation" of the equinoxes lent countenance.[5]

Moon and its orbit

If we now turn to the Moon, we do not find that the Arabs made any advance on Ptolemy. Several of them noticed that the inclination of the lunar orbit was not exactly 5°, as stated by Hipparchus. Thus, Abu 'l Hassan Ali ben Amagiur early in the tenth century says that he had often measured the greatest latitude of the Moon and found results greater than that

1. *Scient. Stell.* Cap. xxviii. Bologna, 1645, p. 72; Nallino. p. 44. At the end of Cap. xlv. he says the apogees of the Sun and Venus are both in 82° 14′ and Ibn Jūnis also gives 82° 14′ as the value found by Al Battani (Caussin, p. 154).
2. Caussin, pp. 232 and 238. Abu'l Faraj gives 89° 28′ for the year 1279 (p. 22).
3. Sedillot, *Prolegomenes aux tables astron. d' Olough Beg* (1847), pp. lxxx-lxxxii. Riccioli, *Almag. Novum*, I. p. 157.
4. Caussin. p. 56. For A.D. 900 Newcomb gives 23° 34′ 54″, with a diminution of 46″ per century, so that the Arabian astronomers erred less then 1′.
5. Aboul Hassan Ali, *Traite des Instruments astron. des Arabes*; T.I.p. 175; Sedillot, *Memoire sur les instr. astr. des Arabes*, p. 32.

of Hipparchus, but varying considerably and irregularly. Ibn Jūnis, who quotes this, adds that he has himself found 5° 3′ or 5° 8′, while other observers are said to have found from 4° 58′ to 4° 45′.[1] Want of perseverance and of accurate instruments caused them to miss a remarkable discovery, that of the variation of the lunar inclination.

Abu 'l Wefa and his Almagest

But an even more remarkable discovery has been claimed for an Arabian astronomer. In 1836 the younger Sedillot announced that he had found the third inequality, the variation, distinctly announced in Abu 'l Wefa's *Almagest*. A fierce controversy raged for a number of years as to the reality of this discovery, Sedillot alone defending his hero with desperate energy and refusing to listen to any arguments, while Biot, Libri and others as strenuously maintained that Abu 'l Wefa simply spoke of the second part of the evection, the prosneusis of Ptolemy. The fight had died out when, in 1892, Chasles suddenly took up the cudgels for Sedillot and pointed out what seemed to him to be some contradictions in Ptolemy's statement.[2] Nobody answered this until Bertrand did so in 1871; he called attention to several inaccuracies in the text of Abu 'l Wefa as we possess it now, and also showed that Abu 'l Wefa did not add his "mohazat" to the prosneusis, the latter not being included in his "second anomaly."[3] It is unnecessary to enter into a more detailed account of the controversy; but to show that any weapon was considered good enough with which to defend Abu 'l Wefa, it may be mentioned that Sedillot and Chasles tried to prove that Tycho Brahe must have copied his discovery from Abu 'l Wefa, because he calls it *hypothesis redintegrata*. Tycho used this same phrase in speaking of his own planetary system, which he most emphatically claimed as

1. Sedillot, *Prolegomenes*. p. *xxxviii*. *Materiaux pour servir a l' hist. des sciences chez les Grecs et les Orientaux*, T. I. p. 283. The sons of Mūsā ben Sakir (about 850) seem to have been the first to find a value differing from that of the ancients. Abraham ben Chija, a Jewish writer who lived about A. D. 1100 says that Ptolemy found 5°, but that according to the opinion of the Ishmaelites it is 4½° (*Sphaera mundi*. Basle. 1546 p. 102).
2. *Lettre a M. Sedillot sur la question de la variation lunaire*. Paris, 1862, 15 pp. 4° and *Comptes Rendus*, vol. 54, p. 1002.
3. *Comptes Rendus*. vol. 73, pp. 581, 756, 889 ; *Journal des Savants*. 11 Oct. 1871.

an original discovery, and which he vigorously defended against other claimants. In future it will be hopeless for anybody to claim the discovery for Abu 'l Wefa as the matter has now been thoroughly sifted, both by mathematicians and orientalists.

The *Almagest* of Abu 'l Wefa has never been published in full, but there are three translations of the chapters in question,[1] which only differ in some trivial points. In no part of the book does he make any advance on Ptolemy or claim to have made any new discovery, and in speaking of three inequalities he merely does what the other Arabian astronomers do.[2] He begins by describing the first (equation of the centre) and the second (evection) and states when they reach their maxima. He then says that we have found[3] a third inequality, which takes place when the centre of the epicycle is between the apogee and the perigee of the eccentric, and which reaches its maximum when the Moon is about a *tathlith* or a *tasdis* from the Sun, while it is insensible in syzygy and quadrature. The maximum is $\frac{3}{4}°$. He explains that this is caused by a deviation of the line or apsides of the epicycle, and he describes quite correctly the construction adopted by Ptolemy (whose name he does not mention), letting the line of apsides be directed, not to the Earth but to another point on the line of apsides of the eccentric. It is difficult for an unbiassed reader to understand how anyone could fail to see that Abu 'l Wefa is simply copying Ptolemy. Sedillot maintained that the words *tathlith* and *tasdis* mean the octants (where the variation reaches its maximum); but every other orientalist who has expressed an opinion, states that by their roots the words correspond to the numbers 6 and 3, in other words, to elongations 60° and 120° from the Sun. This is in accordance with

1. By Reinaud, Munk, and de Slane(for Biot)in the *Journal des Savants*, March, 1845 (14 pp. the whole section on the Moon) ; by Sedillot, *Materiaux*, i. pp. 45--49 ; and by Carra de Vaux, "L' almageste d' Abū 'l Wefa Albūzdjani," *Journal asiatique*, 8° Serie, T. *xix.* (1892), pp. 408--471 (translation on pp. 443-44). Most of the chapters on the planets are lost.

2. The unknown author of a short resume of astronomy (in the *Bibl. Nationale*) even calls the inequality of prosneusis the first equation (Carra de Vaux, l.c. p. 460). This is not unreasonable, since the equation of the centre must be taken from the lunar tables, using as argument not the mean anomaly but the latter corrected for the effect of the prosneusjs.

3. He uses exactly the same expression when speaking of the first and second inequalities.

facts, as Biot has shown from the Ptolemy's numerical data that the deviation of the line of apsides reaches its maximum value of $\pm$ 13° 8′ .9 in elongations 90° $\mp$ 32° 57′.5.[1] But it must be acknowledged that the words in question are also used very vaguely, e. g. by Abu 'l Wefa himself, who says that the velocity of the superior planets after emerging from the Sun's rays diminishes gradually till their distance from the Sun is about a *tathlith*, when they become stationary. It looks almost as if these words might be used to denote any elongation outside syzygy and quadrature.[2]

If Abu 'l Wefa had made a new discovery, we should have expected later Arabian astronomers to have alluded to it. But not one of them gives anything but interpretations of the lunar theory of Ptolemy, and in expressions very similar to those employed by Abu 'l Wefa. Attention was at once called to this fact, and Isaac Israeli of Toledo (about 1310) and Geber of Seville were quoted as examples,[3] though it would, of course, have been quite possible for these two writers to have remained ignorant of whatever progress astronomy might have made in the school of Baghdad. But this objection does not apply to Nasir ed-din al Tūsi, in whose review of the Almagest and Memorial of Astronomy the inequalities known to Ptolemy and no others, are described and credited to Ptolemy[4]; not to Mahamud al Jagmini (about 1300), who wrote a compendium (*mulachchas*)

1. *Journal des Savants*, 1843, p. 701 ("Sur un traite arabe relatif a l' astronomie," Reprint, p. 47). This deviation does not represent the amount of the correction to the Moon's place as seen from the Earth, so that there is not any contradiction in Ptolemy's account.

2. Carra de Vaux, l. c. p. 466. The Arabs had no word for "octants." Nasir-ed-din on one occasion wants to mention them, and has to call them "the points midway between syzygy and quadrature."

3. Isaac Israeli repeatedly speaks of these inequalities discovered by Ptolemy, two of which are not found at conjunction and opposition. *Liber Jesod, Olam seu Fundamentum Mundi auctore R. Isaac Israeli Hispano*, section III. ch. 8 and sect. v. ch. 16, Part I. p. xxiv Part II. p. xxxi (Berlin, 1848 and 1846; this publication is not mentioned by Carra de Vaux).

4. C. de Vaux, "Les spheres celestes selon Nasir Eddin Attūsi", Appendix to P. Tanhery's *Recherches sur l' astr. anc.* p. 342, and *Journ. asiat.* 1892, p. 459: "The third anomaly is that of the prosneusis; it is called the equation of the proper motion" (i.e. of the motion on the epicycle).

of astronomy.[1] Nor can any objection be raised to Abu'l Faraj (Bar Hebraeus), and it would be impossible to explain more clearly than he does the effect of the prosneusis. He says: " The third inequality is the angle formed at the centre of the epicycle by two lines which are drawn, one from the centre of the universe and the other from the point called the prosneusis, at the end of which is the apogee of the epicycle, at which commences the proper motion, and which is called the mean apogee. The apogee which is at the end of the line drawn from the centre of the universe is called the apparent one. The point proseneusis is on the side of the perigee of the eccentric, 10 parts 17 minutes from the centre of the world[2] which is itself at the same distance from the centre of eccentric. The maximum value of this angle is 13 parts 9 minutes when the Moon is a crescent or ¾ gibbous, that is, near the hexagon or trigon with the Sun. In fact, when the epicycle is four or eight signs distant from the apogee of the eccentric, the Sun is itself two or four signs distant from [the centre of the epicycle] because it is half way between this centre and the apogee. In the tables, this inequality of the two apogees is called the first angle and is included in the motion of the centre."[3] While this describes the construction of Ptolemy as clrealy as possible, at the same time the agreement of the account with that of Abu' l Wefa is perfect. Abu'l Faraj even (like Nasir ed-din) describes as a fourth inequality in longitude that caused by the motion along an orbit inclined to the ecliptic, so that he would not have neglected to describe the variation, if it had been found by an astronomer of Baghdad. We may add that the Jewish writer Abraham ben Chija (A. D. 1100), in his *Sphaera Mundi*, also describes the "aberration" of the apside of the epicycle, chiefly "in sexta et tertia parte mensis."[4]

1. Translated by Rudloff and Hochheim, *Zeitschrift der Deutschen Morgenland Ges.* XLVII pp. 213—275. He describes (p. 249) how the line of apsides is directed to a point called "the corresponding point," and gives its position correctly. The inequality he calls the deviation.
2. Nasir ed-din gives 10° 9'.
3. *Le livre de l' ascension*, & c. T. II. PP. 29-30. Two codices add after the word prosneusis : "This is the point mohazat."
4. *Sphaera Mundi* (1546, ed. Schreckenfuchs), p. 75. Munster's commentary to the Hebrew text (p. 116) has "cum centrum est in sextili aut trino aspectu [id est, quando abest a sole duobus signis aut quatuor]" ; the words in brackets are not in the Hebrew original. The words "sixth" and "third" are unmistakable (shithith and shelishith). Apparently no one has hitherto thought of consulting Abraham ben Chija.

Abu 'l Wefa and Ptolemy

Therefore, Abu 'l Wefa did not know a single thing about the motion of the Moon which he had not borrowed from Ptolemy. But the prosneusis of Ptolemy is not the variation discovered by Tycho Brahe. The latter depends solely on the elongation of the Moon from the Sun, as it is = $+39'.5 \sin 2\varepsilon$, while it is beyond the power of mortal man to express the effect of the prosneusis without the anomaly. Ptolemy's exprssion for all the inequalities in longitude assumed by him, when developed analytically, found to contain, in addition to terms represening the equation of the centre and the evection, the latter being

$$+1°19'.5 \sin (2\varepsilon - m),$$

a very considerable term

$$+17.8 \sin 2\varepsilon [\cos (2\varepsilon + m) + 2 \cos (2\varepsilon - m)],$$

where ε is the elongation and m the mean anomaly.[1]

Obviously this term has nothing in common with the variation, except that it disappears in the syzygies and quadratures. Tycho Brahe did not hang his new term on to the unaltered lunar theory of Ptolemy, and by doing that we should in fact only spoil the latter and make its maximum error rise to more than a degree.[2] Owing to the insufficiency of the observations at his disposal, Ptolemy could only perceive that there was some outstanding inequality after allowing for the evection, only appearing outside the syzygies and quadratures, but he was neither able to find the law which governed the phenomenon, nor was he aware what a large quantity it represented; he could only tinker up his constructions a little, and in this he was most faithfully followed by the Arabs, who added nothing to what he had done and left it to the reviver of practical astronomy to discover the third lunar inequality.

Al Fargani and others on Planets

Passing to the five planets, we find that, generally speaking, very few attempts were made to improve the work of Ptolemy. But the Arabs were not content to consider the Ptolemaic system

P. Tannery, *Recherches*, p. 213. Another expansion of Ptolemy's lunar inequalities in a series was given by Biot, *Journal des Savants*, 1843, p. 703 (Reprint, p. 49).

P. Kempf, *Untersuchungen uber die Ptolemaische Theorie der Mondbewegung*, Berlin, 1878 (Inaug. Diss.), p. 37.

merely as a geometrical aid to computation; they required a real and physically true system of the world, and had therefore to assume solid crystal spheres after the manner of Aristotle. Above the Moon is the Alacir, the fifth essence, which is devoid of lightness and heaviness, and is not perceptible to the human senses; of this substance the spheres and planets are formed.[1] Already in the book of Al Fargani we find the principle adopted which we have seen dates from the fifth century (Proklus) and which became universally accepted in the Middle Ages, that the greatest distance of a planet is equal to the smallest of the planet immediately above it, so that there are no empty spaces between the spheres.[2] The semidiameter of the Earth is by Al Fargani given as 3250 miles, which corresponds very nearly to Al Mamun's $56\frac{2}{3}$ miles to a degree, if we put $\pi = \frac{22}{7}$. Starting from Ptolemy's distances of

Greatest Distance of	Al Fargani	Al Battani	Abu 'l Faraj[3]
Moon	$64\frac{1}{6}$	$64\frac{1}{6}$	$64\frac{1}{6}$
Mercury	167	166	174
Venus	1,120	1,071	1,160
Sun	1,220	1,146[4]	1,260
Mars	8,876	8,022	8,820
Jupiter	14,405	12,924[5]	14,259
Saturn	20,110	18,094	19,963

the Moon and the Sun, it was easy to express the other distances in semidiameters of the Earth, the ratios between the greatest and

1. Al Battani, cap. 50 (p. 195).
2. Al Fargani, cap. 21 (ed. Golius, p. 80). Much later, Maurolycus in his *Cosmographia* (Venice, 1543, f, 20a) proves that Mercury and Venus must be below the Sun, by pointing out that there would otherwise be a large vacant space between the Sun and the Moon.
3. pp. 189-191.
4. So in Nallino's ed. (Milan, 1903 p. 121) the ed. of 1645 has 1176.
5. The ed. of 1645 has 12, 420; obviously an error, as the ratio of greatest to smallest distance is given as 37:23 for Saturn 7:5 (misprinted 7:2), or "quantitas unius et duarum quintarum ad unum" (p. 199). Nallino's ed (Milan 1903) has 12,924. Abraham ben Chija has 12,400.

smallest distances being in substantial agreement with the theory of Ptolemy. Al Battani also gives a similar set of figures, though with some slight differences. He does not mention peculiar treatment given by Ptolemy to the theory of Mercury. The above table gives the distance expressed in semidiameters of the Earth.

Al Kusgi and diameters of planets

Al Kūsgi, one of the astronomers of Ulug Begh, gives a list of the semidiameters of the "concavities" of the planetary spheres (i.e. the smallest distances of the spheres) expressed in parasangs, the diameter of the Earth being 2,545 parasangs.[1] Expressed in semidiameters of the Earth, the figures turn out somewhat differrent from those given above, e.g. the smallest distance of the Sun being 1,452 and the greatest of Saturn 26,332, but he does not supply any means of making out how these figures were found.

Before leaving this subject, we shall also give the diameters of the planets according to Al Fargani, as they became known in Europe at an early date and were quoted by Roger Bacon and others.[2] With trifling variations the same values are given by Al Battani, Abu 'l Faraj, and Abraham ben Chija.

	Apparent Diameter	True Diameter (Earth's=1)
Moon in apogee	$31\frac{2}{3}'$ ...	1 : $3\frac{2}{3}$
Mercury, mean dist.	$\frac{1}{18}$ of Sun's ...	$\frac{1}{18}$
Venus " "	$\frac{1}{170}$ " ...	1 : $3\frac{1}{3}$
Sun " "	$31\frac{2}{3}'$ ...	$5\frac{1}{2}$
Mars " "	$\frac{1}{20}$ of Sun's ...	$1\frac{1}{6}$
Jupiter " "	$\frac{1}{12}$ of Sun's ...	$4\frac{1}{2} + \frac{1}{18}$
Saturn " "	$\frac{1}{18}$ " ...	$4\frac{1}{2}$

Al Kazwini, Abu'l Faraj and Al-Jagmini on Excentric Spheres of the Sun

The system of the spheres is set forth in greatest detail in three treatises of later date, the cosmograpy of Zakarija ben Muhammed ben Mahmud al Kazwini (about 1275), the astronomy

1. *Astronomica Shah Cholgii*, pp. 95-97.
2. There are some slight differences between the figures given in the various editions (J.L.E. Dreyer has compared those of 1493, 1546, and 1669), but those given above agree with the cubic contents according to Al Fargani. The figures of Kazwini seem to have been greatly corrupted.

of Abu 'l Faraj, written in 1279, and that of Mahmud ibn Muhammed ibn Omar al Jagmini, whose date and nationality are equally uncertain, but who probably wrote in the thirteenth or fourteenth century. We find in these text-books an elaborate system of spheres designed to account for every particular of planetary motion, in perfect agreement with each other as to the general arrangement of the spheres, and offering nothing new as to lunar or planetary theory. The accompanying figures (taken from Jagmini) will illustrate the ideas better than a lengthy description.[1] The Sun is a solid sperical body, fitting between two excentric spherical surfaces, which touch two other surfaces, in the

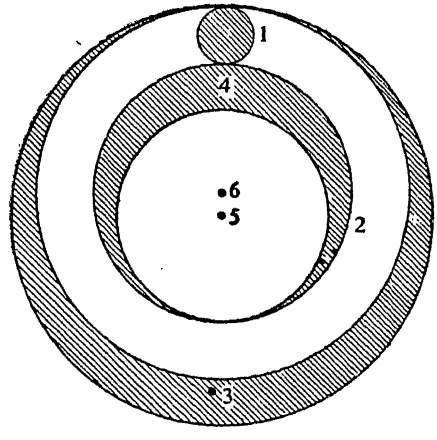

Fig. 1.—Planetary motions and system of spheres.

1. The Sun. 2. Excentric sphere. 3. The surrounding spheres. 4. The complement of the surrounding sphere. 5. Centre of the world. 6. Centre of the excentric sphere.

common centre of which the Earth is situated, and which between them enclose a space (or intersphere, as Abu 'l Faraj calls it), named by Jagmini al-mumaṭṭal, or the equably turning sphere, which has the same motion from west to east as the fixed stars, i.e. precession. The spheres of the three outer planets and Venus are arranged on the same plan, except that the place of the body of the Sun is taken hy the epicycle-sphere of each planet, to the inner surface of which the planet (a solid spherical body) is attached or (as Abu 'l Faraj says[2]) "fixed like a pearl on a ring, touching the

1. Al Kusgi gives very similar diagrams of the spheres of the Saturn, Mercury, and the Moon.

2. Precession is supposed to be included in this, "the first motion." The second

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surface in one point." The axis of the excentric sphere is inclined to that of the mumaṭṭal sphere, which causes the motion in latitude. The lunar system comprises an additional sphere outside the others, the centre of which coincides with the centre of the world, and which is called *al-gauzahar*, signifying the constella-

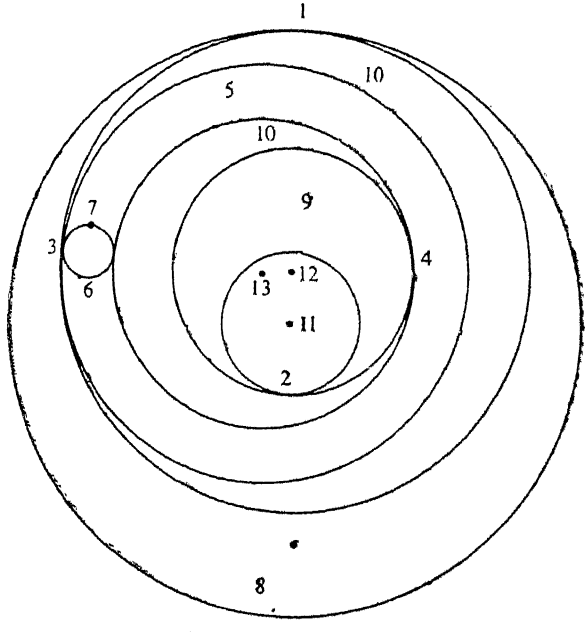

Fig. 2.—Spheres of Mercury

1. Upper Apsis. 2. Lower Apsis. Upper Apsis of deferent sphere. 5. Deferent sphere. 4 Lower Apsis of deferent sphere. 6. Epicycle. 7. Mercury. 8. Surrounding complement. 9. Surrounded part of Mumaṭṭal sphere. 10. Mūdir sphere. 11. Centre of the world. 12. Centre of Mūdir. 13. Centre of deferent sphere.

tion Draco, as this sphere provides for the revolution of the lunar nodes ("the head and tail of the dragon") round the zodiac. The inner one of the two concentric spherical surfaces, between which the excentric sphere lies, surrounds immediately the fire sphere of the Earth. The system of Mercury is more complicated, as a space had to be provided for the revolution of the centre of the

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one is that of the concentric oblique intersphere (called the mail sphere or the sphæra deflectens) round the centre of the world, 11°9′ per day, by which amount the lunar apogee moves towards the west. The third motion is that of the excentric, carrying the centre of the epicycle 24°22′ towards the east. The fourth is the motion on the epicycle. Abu 'l Faraj, p. 27.

excentric sphere. The figure shows the excentric sphere enclosed in a sphere, *al-mūdir* or the turning one, which allows the upper apsis or apogee of the excentric or deferent sphere (3 in the figure) to move right round the outer surface of the *mūdir*. The inner surface of the mumaṭṭal sphere immediately surrounds the *gauzahar* sphere of the Moon.

It was a necessary consequence of the large solar parallax of 3' accepted by Ptolemy, that Mercury and Venus must be very near the Earth, since they are assumed to be nearer than the Sun. Thus Abraham ben Chija says that the shadow or the Earth extends beyond the orbit of Mercury but does not reach that of Venus.[1] Ptolemy never mentions the parallaxes of Mercury and Venus, as to which nothing was known, though they ought, of course, to be greater than 3'. But on the assumption that the smallest distance of Mercury is equal to the distance of the Moon at apogee, the parallax of Mercury ought to rise to 54', which must have been felt to be too large a quantity, though it does not seem to have struck Al Battani as anything surprising, perhaps because Mercury cannot be seen when in inferior conjunction. It may have been this necessarily large parallax of Mercury, which induced Ibn Jūnis (without any explanation) to reduce the solar parallax from 3 to 2', or rather to 1' 57".[2] Geber[3] blames Ptolemy for having said that the parallaxes of the planets are insensible, and remarks that he ought, therefore, logically to have placed Venus and Mercury above the Sun. He takes great pains to show that Venus may be exactly on the line joining the Sun and the Earth. Indeed, Geber neglects no opportunity of criticising Ptolemy's methods of finding the elements of the orbits,[4] and he is generally very unjust to him but he does not venture to

1. *Sphaera mundi*, ed. Osw. Schreckenfuchs, Basle, 1546 pp. 84, 86.

2. Unpublished chapters of Ibn Jūnis, reviewed by Delambre, *Hist. de l'astr. du Moyen Age*. p. 101.

3. *Instrumentum primi mobilis a P. Apiano. Accedunt ijs Gebri filii Affla Hispalensis. . . libri* IX. *de astronomia*. Norimbergæ, 1534, fol. (Introd. p. 3 and lib. VII. p. 104).

4. See the long indictment on pp. 2-3 of his introduction. He blames Ptolemy among other things for assuming that the centre of the deferent is half-way between the centres of the zodiac and of the equant, while he himself deduces this from the movements.

substitute any other system and does not object to the general principles of the Ptolemaic system,[1]

Three great names :
Ibn Badja, Ibn Tofeil (Abubacer) and Abu Welid (Averroes)

Geber's attempts to pick holes in the work of Ptolemy were, perhaps, not unconnected with the rapid rise of Aristotelean philosophy in Spain in the twelfth century, which, though not destined to last long, nevertheless exercised a considerable influence on the spread of knowledge of Aristotle in the Christian world, while it cast a halo round the Caliphate of Cordova, which at that time, under the enlightened rule of the Almohades, seemed to have reestablished the glory of the best days of the Moslem world. Three names are specially associated with this movement :

(i) Abu Bekr Muhammed Ibn Jahya al Sayeg, called Ibn Badja (of Saragossa, died 1139), known as Avempace among the Scholastics ;

(ii) his pupil Muhammed ben Abdelmelik Ibn Tofeil (of Granada, died 1185-1186), called Abubacer by the Scholastics ;

(iii) and finally the greatest philosopher of Islam, Ibn Rosd Abu Welid, known as Averroes (1126-1198).

In studying Aristotle they laid special stress on his scientific works, and did not, like their Christian successors, think of little but dialectics. The acceptance of the system of homocentric spheres or some modification of it must, therefore, have seemed a necessity to the Arabian philosophers and this, of course, led them to reject the theory of epicycles.. The little we know of the opinions of Ibn Badja on this subject is found in the famous work *The Guide of the Perplexed* of the great Jewish scholar Moses ben Maimun of Cordova, better known as Maimonides, who tells us that he had his information from a pupil of Ibn Badja. Like Geber (with whose son he had been familiar), Maimonides doubted that Mercury and Venus were nearer than

1. Copernicus possessed a copy of Geber's book; which is now in the University library at Upsala. On the title page, after the author's name, he has written : "Egregii, Calumiatoris Ptolemaei," while a number of marginal notes show that he has read the book carefully. Curtze, *Mittheilungen des Coppernicus Vereins*, I, p. 37.

the Sun, though he would not venture to say how they actually moved.[1] But what is more important, he declared the motion of a planet on an epicycle to be contrary to physical principles, because there are only three motions possible in this world : around its centre, or towards it, or away from it; while he also maintained that according to Aristotle circular motion can only take place round a real, central body.[2] Though Aristotle in reality did not object to epicyclic motion with a mathematical point as centre, for the simple reason that it had not been proposed when he wrote, while, as we have seen, his moving principle had nothing to do with the centre of motion, it is easy to see that Ibn Badja's real difficulty was the same which afterwards produced so many obstacles to the advance of science in Europe : whatever could not be found in Aristotle's book must be unworthy of notice. According to Maimonides (who, however, makes the reservation that he had not heard it from disciples), Ibn Badja constructed a system of his own, in which he only admitted excentric circles but no epicycles. We are not given any particulars as to this system but there can hardly be any doubt that its author confined himself to generalities and did not attempt to represent phenomena like the lunar inequalities by it. Maimonides remarks that there is nothing gained by Ibn Badja's reform, since the excentric hypothesis is as objectionable as the epicyclic one, as it also supposes motion round an imaginary point outside the centre of the Earth. The centre of the excentric, on which the Sun is supposed to move, is outside the convexity of the lunar sphere and inside the concavity of that of Saturn's excentric is between the spheres of Mars and Jupiter. He adds that the revolution of a number of concentric spheres around a common axis is conceivable, but not the revolution round different axes inclined to each other, as the spheres would disturb each other unless there are other spherical bodies between them. This attempt to revive and modify the system of (movable ?) excentrics did, therefore, not mend matters.[3]

1. *Rabbi Mosis Majemonidis Liber . . . . Doctor Perplexorum.* Basileæ, 1629, Pars II. cap. IX.

2. Ibid., Pars II. cap. XXIV.

3. Maimonides also remarks (in the same chapter) that the supposed inclinations of Mercury and Venus in the Ptolemaic system are difficult or impossible to comprehend or imagine as really existing. Therefore, if what

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Ibn Tofeil

Ibn Tofeil, the second of the three Moslem philosophers of Spain, vizier and physician at the court of Jusuf ben Abd el Mumin of Morocco, seems to have walked in the footsteps of his master ; but the only extant work of his, a kind of religious mystic romance about the emancipation of a soul from the trammels of this material world, does not give any clue to his ideas as to the planetary system. But Averroes, who also objected to the excentrics and epicycles says in his commentary to Aristotle's *Metaphysics* that Ibn Tofeil possessed on this subject excellent theories [2], and Ibn Tofeil's pupil, the astronomer Al Betrugi, in the introduction to his theory of the planets, says of him : "You know that the illustrious judge Abu Bekar Ibn Tofeil told us that he had found an astronomical system and principles of the various movements different from those laid down by Ptolemy and without admitting either excentrics or epicycles, and with this system all the motions are represented without error." Ibn Tofeil was therefore probably the real author of the fairly elaborate system, which his pupil worked out and handed down to us in a work on the planets, which was translated into Hebrew in the following century and from that again into Latin, and published in 1531 [3].

The object of this system was to explain the constitution of the universe as it really is, and not merely to represent the motions of the planets geometrically, so as to be able to foretell their places in the heavens at any time ; and the author (be he Ibn Tofeil or Al Betrugi alias Alpetragius) specially disclaims any intention of testing the theory by comparing it with observations

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Aristotle says is true, there is neither epicycle nor excentric, and everything turns round the centre of the Earth.

2. Munk ; *Melanges de philosophie juive et arabe.* Paris, 1859, p. 412.

3. *Alpetragii Arabi Planetarum theorica phisicis rationibus probata, nuperrime latinis litteris mandata a Calo Calonymos, Hebreo Neapolitano,* Venice 1531, 28 ff. folio (published with Sacrobosco's *Sphaera*). A translation by the famous Micheal Scot has never been printed, but is still extant in Paris (Munk, *Melanges*, p. 519). The principle of the system is described by Isaac Israeli, who, however, does not mention the author's name (*Liber Jesod Olam*, II. 9. Part I. p. XI

or of accounting for minor details of the motions.[1] The leading idea is that of the homocentric spheres, each star being attached to a sphere, and the motive power is the ninth sphere, the sphere outside that of the fixed stars. The Spanish philosopher ought, therefore, to have been content with the system of Eudoxus or its modification by Aristotle (whom he never mentions by name, but only as "the sage"), but unfortunately he became possessed with the notion that the prime mover must everywhere produce only a motion from east to west, and he had, therefore, to reject the independent motion of the planets from west to east, and revert to the old Ionian idea that the seven planets merely perform the daily revolution with a speed slightly slower than that of the fixed stars. The true speed of the *primum mobile* is a little faster than this ; the eighth sphere performs a revolution in a slightly longer period (24 hours), and the effect of the prime mover is gradually weakened more and more, with increasing distance, until we find the sphere of the Moon, being furthest from the prime mover, taking nearly twenty-five hours to complete a revolution. This was the old primitive Ionian idea, but Al Betrugi (or his teacher) saw that this was not sufficient, as not only is the pole of the ecliptic different from that of the equator, which prevents the planets from moving in closed orbits, but the planets do not even keep at the same distance from the pole of the ecliptic but have each their motion in latitude, as well as variable velocity in longitude ; and all this had yet to be accounted for. The ninth sphere has but one motion, but the eighth has two, that in longitude (precession) and another which is caused by the pole of the ecliptic describing a small circle round a mean position, thereby producing the supposed oscillation or trepidation of the equinoxes.[2] Similarly, the pole of each planet describes a small circle round a mean position (i. e. the pole of the ecliptic), thereby producing inequalities in longitude and motion in laritude[3] Whenever the actual orbit-pole of a planet is on the parallel of the mean pole, it is obvious that the planet will perform its daily revolution with its mean velocity, while the velocity is increased or lessened when the actual pole is respectively at ts minimum or maximum distance from the pole of the heavens

1. Fol. 8 b. 2. Fol. 9 b.

3. Fol. 14 b;. sq.

(the motion of the pole of the orbit being added to or subtracted from the motion of the planet), so that the epicycle is hereby rendered superfluous. The lengths of the radii of these small circles are not given, except in the case of Saturn, where the radius is 3° 3′,[1] while the mean pole of the moon is 5° (the inclination of the lunar orbit) distant from the pole of the ecliptic,[2] and the small circle is so exceedingly small as to produce no retrograde motion, which is also the case with the Sun. The periods of the poles of the outer planets are given by the following figures Saturn makes 57 revolutions in 59 years and $1\frac{1}{2}+\frac{1}{4}$ days, in which period the mean pole lags behind 2 revolutions $1\frac{2}{3}°+\frac{2}{3}°$. Jupiter makes 65 revolutions in 71 years, the mean pole lagging behind 6 revolutions. Mars makes 37 revolutions in 79 years and $3\frac{1}{6}+\frac{1}{12}$ days, the pole lagging behind 42 revolutions and $3\frac{1}{7}°$.

In other words, the motion on these small circles are completed in the synodic periods of planets. Similarly, the pole of Venus makes 5 revolutions in the 8 years less $2\frac{1}{4}$d$+\frac{1}{2\,\cdot\,0}$, lagging $1\frac{5}{8}$ revolutions in one year; and Mercury 145 revolutions in 46 years and $1\frac{1}{8\,\cdot\,0}$d[4]. It is curious that Alpetragius alters the order of the planets, placing Venus between Mars and the Sun, because the defectus (lagging) of Venus smaller than that of the Sun.[5] He also says that nobody has given any valid reason for accepting the usually assumed order of the planets, and that Ptolemy is wrong in stating that Mercury and Venus are never exactly in a line with the Sun (a remark already made by Geber); and as they shine by their own light they would not appear as dark spots, if passing between us and the Sun. That they do not receive their light from the Sun is proved, he thinks, by the fact that they never appear crescent-shaped.[6]

There is no need to dwell any longer on this quaint theory

1. Fol. 16 a.
2. Fol. 25 a.
3. Fol. 16 a, 18 a, 19 b.
4. Fol. 21 b, 24 b.
5. "Nam reperimus defectum eius primum minorem defectu orbis solis et maiorem defectu orbis martis, et sequitur juxta radices nostras ut sit inter eos ambos."
6. Fol. 21 a

of spiral motion, as it has been rather improperly called.[1] It represented a retrograde step of exceedingly great magnitude, totally unjustified as the theory could not seriously pretend to be superior to the Ptolemaic system, which had only become so very simple if one was content with representing only the principal phenomena. We are told by the Jewish astronomer Isaac Israeli of Toledo, that the new system made a great sensation, but that it was not sufficiently worked out to be taken seriously, and that the system of Ptolemy, founded on the most rigorous calculations, could not be superseded by it.[2] Another Jewish author, Levi ben Gerson, in a work written in 1328, entered into a lengthy refutation of the hypotheses of Al Betrugi.[3] But the latter certainly represented a general desire on the part of the Spanish Aristoteleans to overcome the physical difficulties in accepting the Ptolemaic system; thus Averroes says that the astronomy of Ptolemy is merely a convenient means of computing, and that he himself in his youth had hoped to prepare a work on the subject.

Nasir ed-din Al Tūsi

While ineffectual attempts were being made in the far west. to devise a new astronomical theory, the astronomers of the east did not remain blind to the desirability of finding a system, in which the planets were not supposed to move unsupported in space in such a wonderfully complicated manner; and in the thirteenth century we find one of the greatest astronomers, Nasir ed-din Al Tūsi, advocating a system of spheres which he supposed to be more acceptable than excentrics and epicycles.[4] In addition to a review or digest of the *Syntaxis* of Ptolemy he wrote a shorter work entitled *Memorial of Astronomy*, in various

1. e.g. by Riccioli, *Almag. Nov.* T. I. p. 504, where Kepler's figure of the real motion of Mars in space from 1580 to 1596 (supposing the earth to be at rest) is copied, as if that had anything to do with the "*Spirals*" of Alpetragius.

2. He adds that he was not qualified himself to sit in judgment on the proposed system (*Liber Jesod Olam*, II. 9, p. XI.)

3. Munk, *Melanges* pp. 500 and 521.

4. "*Less spheres celestes selon Nasir-Eddin Al tūsi. Par M. Carra de Vaux.*" Appendix VI. *to Tannery's Recherches sur l'astr. anc.* pp. 337—360. Includes a translation of the chapter in which the new theory is set forth.

passages of which he shows his dissatisfaction with the Ptolemaic system. In the chapter on the Moon (to which we have already alluded) he counts up the various anomalies, among which he mentions the anomaly of illumination, that is, the spots on the Moon, which he believes to be caused by other bodies moving in the lunar epicycle and unequally exposed to the Moon's light. He then says that we should expect in a simple theory to find the centre of the epicycle in equal times describing equal arcs on the deferent, and the diameter of the epicycle joining the pericentre and the apocentre pointing to the centre of the deferent. But neither of these conditions is fulfilled. In the theories of the planets he makes the same objections, which it must be said are very just, since the introduction of the equant was a very unnatural arrangement. But this is nothing to the artificial machinery designed by Ptolemy to account for the motion in latitude of the five planets, especially of Mercury and Venus. Nasir ed-din describes the marvellously complicated movements of the deferents and epicycles of these planets, and remarks that "these motions require the introduction of a system of guiding spheres, about which the ancients have not said anything". He next proceeds in the following chapter to explain a system of his own which allows us to discard these combinations.

First he proves that if there are two circles in one plane, one touching the other internally and of a diameter equal to half that of the other, and if the greater one rotates, and a point moves along the circumference of the smaller one in the opposite direction with twice the velocity and starting from the point of contact, then that point will move along a diameter of the greater circle.[1] These two circles may now be assumed to be the equators of two spheres, and for the point we may substitute a sphere representing the Moon's epicycle (1 in the figure), Nasir ed-din assumes another sphere (2) surrounding the epicycle and destined to keep the diameter from apogee to perigee in its place always coinciding with the diameter of the sphere (4) "let us give it a suitable thickness, but not too great, so as not to take up too much space." He next assumes two more spheres, one (3) which corresponds to the smaller sphere in the distance of the centre of the deferent in the Ptolemaic system from the

1. Compare Copernicus, *De revolutionibus*, III. 4 (Secular ed. 1873, p. 166).

centre of the Earth; and another sphere (4) with a diameter twice as great. Finally (4) is placed in the interior of a carrying sphere (5) concentric with the world and occupying the concavity of the

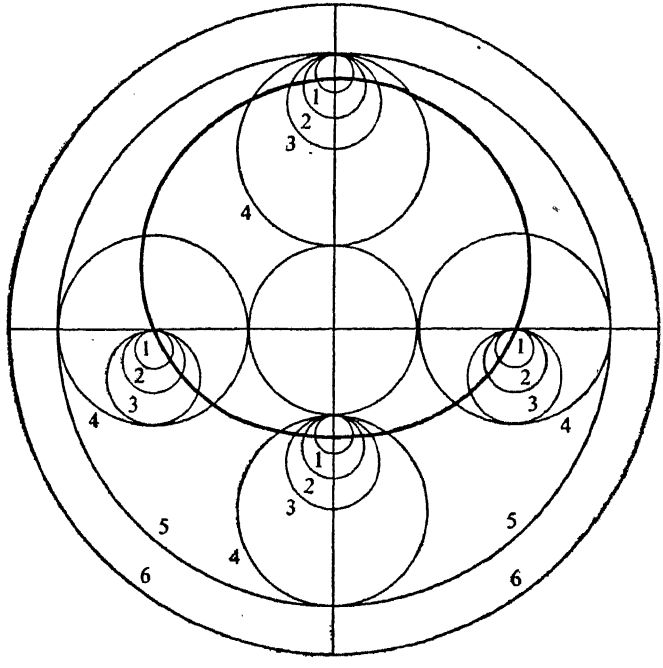

Fig. 3.—Movements of deferents and epicycles of planets.
The thickline is *not* a circle. All others are circles.

sphere (6), the equator of which is in the plane of the lunar orbit. (2) and (4) and (5) revolve in the same period, that in which the centre of the epicycle performs a revolution; (3) revolves in half that time, while (6) revolves in the opposite direction with the same speed as the apogee of the excentric. The figure now shows how the epicycle moves to and fro along the diameter of (4) and during the revolution of the circle (5) describes a closed curve, about which Nasir ed-din justly says that it is somewhat like a circle but is not really one, for which reason it is not a perfect substitute for the eccentric circle of Ptolemy. He estimates the greatest difference between the lunar places given by the two theories as one-sixth of a degree, half-way between syzygy and quadrature. Except for the action of the guiding sphere (2), it would not be the centre of the epicycle but the point of contact of circles (3) and (4), which describes the curve resembling a circle. The same

method may be adopted for Venus and the three outer planets, and Nasir ed-din promises to explain the new theory of Mercury in an appendix, but this appears to have been lost.

Nasir ed-din also endeavours to improve on the machinery proposed by Ptolemy to illustrate the manner in which the epicycle remains parallel to the plane of the ecliptic. He mentions that the celebrated Ibn al Haitham (afterwards known in the west as Alhazen, author of a well-known book on optics) had written a chapter on this subject, adding to each epicycle two spheres to account for the inclination of the diameter perigee-apogee, and two additional ones for the inferior planets for the diameter at right angles thereto.[1] Nasir ed-din makes use of the same principle which guided him in his demonstration about the motion in longitude, and he shows how in this way we may by means of two spheres make the extremities of the diameter of the epicycle move backwards and forwards along an arc of a sphere.[2] He claims that this arrangement is superior to that of Ptolemy by not introducing any error in longitude,[3] but he acknowledges that he has not been able to get rid of the strong objection to Ptolemy's auxiliary circle, viz. that the irregular motion in longitude with regard to the centre of the deferent necessitates the introduction of a corresponding irregularity in the motion on the auxiliary circle by letting the motion be uniform with regard to an equant. It baffled Nasir ed-din's ingenuity to find an arrangement of spheres which could obviate the necessity of having recourse to this expedient.

All the attempts at rebellion against the Ptolemaic system had thus turned out failures. And they deserved nothing else, since it was impossible to find anything better than what Ptolemy had produced, until it was perceived that where Ptolemy was wrong was not in his mathematical methods, which were perfect, but in the fundamantal idea of the Earth being at rest. The time

1. Ibn al Haitham said that by using discs instead of spheres one might complete the demonstration; but Nasir ed-din objects to the arrangement (about which he gives no details) that a non-spherical system is not in accordance with the principles of astronomy.
2. It is not quite clear whether this plan is his own or is the same as Ibn al Haitham's.
3. Due to disturbance of the position of the diameter from perigee to apogee, from which the anomaly is counted.

was apparently not ripe for a radical change with regard to this idea. Though the doctrine of the Earth's motion does not seem to have been mentioned by Arabian writers, we have evidence that the hypothesis of the daily rotation of the Earth was not unknown among them, a natural consequence of their familiarity with the writers of antiquity. One of Nasir ed-din's fellow-workers at the Meragha observatory, Ali Negm ed-din al Katibi, who died in 1277, wrote a book, the *Hikmat al-ain*, on philosophy, in which he combats this opinion, which he attributes to "some philosphers." "I do not," he says, "advance as an argument against it that, if this were the case, a bird flying in the direction of the motion of the Earth would not be able to keep up with it, because the motion of the Earth would be much faster than that of a bird, inasmuch as it returns to its place in a day and a night. Such an argument is not conclusive, because it may be urged that the atmosphere which is close to the Earth partakes of its motion as the ether partakes of the motion of the heavenly sphere. But I reject this theory, because all terrestrial motions take place in a straight line, and therefore we cannot admit that the Earth should move in a circle."[1]

What reformation of astronomy could be hoped for, as long as this kind of argument could be used? We cannot see from this remark of Katibi's whether there really were any Arabian philosophers who believed in the rotation of the Earth. It is, however, stated in the *Zohar*, the great Kabbalistic work attributed to Mosheh ben Shemtob of Leon (died 1305), that a certain Rabbi Hamnuna the Elder (otherwise unknown) taught that "the Earth turns like a sphere in a circle and that some people are above and others below."[2] Though this passage as well as others in the Zohar may have been interpolated much later, it would

1. A. Sprenger, "The Copernican System of Astronomy among the Arabs." *Journ. Asiat. Society of Bengal*, Vol. XXV. (1857), p. 189. Katibi's contemporary, Abu 'l Faraj (II. p. 10) deems it necessary to prove that the Earth cannot be in motion, neither rectilinear nor circular, but his arguments (about birds and stones flung upwards) seem merely taken from Ptolemy, lib, 1. cap. 6, Kazwini (*Kosmographie*, p. 296) says that among the ancients there were some adherents of Pythagoras who maintained that the Earth continually moves round in a circle; but whether these adherents were Greeks or Arabians cannot be seen from the context.

2. *Sohar*, Amsterdam, 1718, T. III. f. 10a; Gunther, *Studien Z. Gesch. d. math, Geogr.*, p. 113.

after all not be very surprising if some learned Jews had been influenced by the opinion of Herakleides, since it is an established fact that the doctrines of the Kabbalists were intimately connected with the later Greek philosophy. But any how nothing came of this isolated case, and the daily rotation of the heavens continued to be universally accepted as a self-evident fact.

Arabian astronomers and Ptolemaic system—

Arabian astronomers who really wished to follow in detail the celestial motion were therefore obliged to adopt the Ptolemaic system altogether. New planetary tables had long been found to be a necessity, and this important work was at last undertaken by King Alfonso X. of Castille and several Jewish and Christian astronomers working under him at Toledo, who prepared the celebrated Alfonsine Tables, Apparently the King must have had his doubts about the physical truth of the system, judging from his well-known saying that if God had consulted him when creating the world, he would have given Him good advice. The tables were prepared under the direction of the Jew Ishak ben Said, called Hasan, and a physician, Jehuda ben Mose Cohen, and were finished in 1252, the year in which Alfonso ascended the throne of Castille. They continued in great repute for three hundred years as the best planetary tables; they were first printed in 1483, but had been spread all over Europe long before that time in numerous MS copies, many of which are still in existence, Twenty-six codices are counted up in the *Libros del Saber de Astronomia del Rey D. Alfonso X.de Castella,* Madrid, 1863-67 (5 vols. fol.). This compilation, a series of chapters on spherical and theoretical astronomy followed by tables, must have been made up from several codices, as there are numerous repetitions even of very elementary matters. In the third volume the theories of the planets are dealt with, but one looks in vain for any improvement on Ptolemy; on the contrary, the low state of astronomy in the Middle Ages is nowhere better illustrated. In general the elements of the orbits are those of Ptolemy, though sometimes only approximations are given, while different values are given in different chapters, though Ptolemy places the centre of the deferent midway between the centre of the equant and the Earth, the *Libros del Saber* places the centre of the equant (*cerco del alaux.*[1]) midway between the Earth and the centre of the deferent

1. *al* is the Arbic article, *aux* (apside) is a corruption of the Arabic *Oudj* (Abu 'l Faraj, II. p. 25). The equant is called the *cerco del Y guador.*

(*cerco del levador*[1]), as in Ptolemy's theory of Mercury, which the authors would seem to have extended to the planets, omitting the motion of the centre of the deferent on a small circle; this, they have, however, correctly given in the case of Mercury.[2] There is a very curious figure[3] of the deferent of Mercury in the form of an ellipse (the axes being as 6 to 5 nearly), with what looks like the Sun in the centre. This curve has been constructed from a number of small circular arcs,[4] and it is obviously nothing but the curve described by the centre of the epicycle of Mercury in Ptolemy's theory. For according to the latter the centre of the deferent describes a small circle with radius=$\frac{1}{21}$ of that of the deferent, in direction from east to west, in the same time which the centre of the epicycle takes to pass round the circumference of the deferent from west to east. This makes the centre of the epicycle describe a closed curve resembling an ellipse, the axes of which are in the ratio 11: 10, almost exactly the same as in the Spanish diagram, and there is therefore in the latter no anticipation whatever of Kepler's great discovery, since in the case of the inferior planets it is the epicycle which is the real orbit.[5] The small sun-like object in the centre of the ellipse represents the centre of Ptolemy's small circle, and it has either been inserted in the manuscript centuries after the essay had been written, or, more likely, it has been caused by a small blot on the place in the parchment where the stationary leg of the draughtsman's compasses had made a small hole. An oval deferent of Mercury occurs in several books published in the sixteenth and seventeenth centuries.[6]

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1. Vol. III. pp. 246—253.
2. Vol. III, pp. 253 and 278. In the the latter place the radius of the small circle is $\frac{1}{21}$, as in the "*Hypotheses*" of Ptolemy.
3. Vol. III. p. 282.
4. See the lengthy description on pp, 278—280.
5. The editor, Don Manuel Rice y Sinobas, on p. xxxiii. of his preface, even goes so far as to suggest that Kepler may have known of this great discovery of Alfonso's or rather of Arzachel's, as the text attributes the construction to him. This and other similar diagrams were intended to be used instead of planetary tables in the manner afterwards adopted by Apianus.
6. First (about 1460) in Purbach's *Theoricae novae Planetarum* (ed. of Basle, 1573, p. 82): "Ex dictis apparet manifeste, centrum epicycle Mercurij,

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Though the somewhat confused collection of essays entitled the *Libros del Saber* would not, if published in the thirteenth century, have advanced astronomical science, it cannot be denied that the Alfonsine Tables were very useful in their day. The actual elements are not given, nor is any thing said about any observations by which somewhat more correct values of the mean motions must have been found.[1]

Arabs on motions of fixed stars.

Thus we finish our review of the planetary theories of the Arabs. Now we shall say a few words about their ideas as to the nature and motion of the fixed stars. The exaggerated notion which prevailed before the invention of the telescope with regard to the apparent angular diameters of the stars naturally, led to erroneous estimates of their actual size, founded on the assumption that the sphere of the fixed stars (the eighth sphere) was immediately outside that of Saturn.[2] The stars of the first magnitude were supposed to have an apparent diameter equal to $\frac{1}{20}$ of that of the Sun, from which it followed that their actual diameters were about $4\frac{3}{4}$ times that of the Earth, or about

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propter motus supradictos non (ut in alijs planetis fit) circumferentiam deferentis circularem, sed potius figuræ, habentis similitudinem cumplana ovali peripheriam describere." Next by Albert of Brudzew in 1482 in his *Commentariolum super theoricas novas*, printed at Milan in 1495 (ed. Cracow, 1900, p. 124), where it is remarked that the centre of the lunar epicycle describes a similar figure. This is also stated by E. Reinhold in his commentary to *Purbach*, 1542, fol. p. 7 verso (ed. of Paris, 1558, fol. 78.) by Vurstisius in his *Questiones novae in theoricas*, & c., Basle, 1573, p. 233; and in Riccioli's *Almagestum novum* T. I. p. 564. The last three writers (who give a figure) also take the equable angular motion round the centre of the equant into account, which centre lies on the point of the circumference of the small circle nearest the Earth. The curve described by the centre of the epicycle thus becomes egg-shaped, and not like an ellipse.

1. The tables in vol. v. of the *Libros del Saber* are quite different from the Alfonsine Tables, and are apparently only intended for astrological purposes.
2. Al Battani (cap. 50) gives the greatest distance of Saturn = 18,094. and the distance of the fixed stars = 19,000 semidiameters of the Earth. Al Fargani (p. 82) puts them exactly equal. Al Kūsgi gives the diameters in parasangs, of the concavity of the stellar sphere = 33,509,180 of the ninth sphere 33,524,309, of its convexity "no one but God knows" (Shah Cholgi, p. 97).

twice that of Mars.[1] As to the nature of the stars, they seem generally to have been assumed self-luminous, being condensed parts of the sphere, though Abraham ben Chija says that the eighth sphere does not shine with a uniform light, but has denser spots, which are illuminated by the Sun and appear to us as the fixed stars.[2]

To account for the apparent slow motion of the stars paralleled to the ecliptic, from west to east, whereby their longitudes increase while their latitudes remain unaltered, it became necessary to introduce a ninth sphere (*primum mobile*), turning in twenty-four hours and communicating this motion to the eighth sphere, while the latter moved extremely slowly round its own axis forming an angle of 23° 35′ with that of the ninth.[3] But the simple phenomenon of precession was by many Arabian astronomers complicated by being assumed variable. It may be mentioned that according to Theon and Proklus it had been assumed by some astronomers apparently before the time of Ptolemy, that the precessional motion of the stars was not progressive, but was confined to an oscillation along an arc of 8°, along which the equinoctial points moved backwards and forwards on the ecliptic, always at the same rate of 1° in 80 years. The absurdity of the sudden change of direction must have become obvious as soon as astronomy began to be cultivated among the Arabs, for we find that one of the earliest astronomers, Tābit ben Korra, substituted a physically less objectionable theory.[4]

1. Al Fargani (p. 85, Golius) gives the cubic contents of the six spheres as 107, 90, 72, 54, 36, 18 times that of the Earth. Abu 'l Faraj, p. 199, gives a similar series from 93 to 15¼ for the average star of each class. Shems ed-din of Damascus in his *Cosmography* (p. 3) merely says that the smallest fixed star is much larger than the Earth.

2. According to Suter, p. 77, a writer called Ibn Zura wrote a treatise "On the cause of the light of the stars, though they and the spheres consist of one single substance."

3. The outermost sphere is by the philosopher Ibn Sina (Avicenna) defined as a spherical, single (not composite) body, emanating directly from God and subject to dissolution, endowed innately with circular motion as an expression of its praise of the Creator (Mehren in *Oversigt, K. Danske Vid. Selskab*, 1883, p. 70).

4. The treatise "*On the motion of the 8th sphere*" has never been printed; an abstract is given in Delambre's *Hist. de l'astr. du Moyen Age*, p. 73. Compare a quotation by Ibn Junis, Caussin, *Notices et Extraits*, VII, p. 116.

He imagines a fixed ecliptic (in the ninth sphere) which intersects the equator in two points (the mean equinoxes) under an angle of 23° 33′ 30″, and a movable ecliptic (in the ninth sphere), attached at two diametrically opposite points to two small circles, the centres of which are in the mean equinoxes and the radii of which are=4° 18′ 43″. The movable tropical points of Cancer and Capricorn never leave the fixed ecliptic, but move to and fro to the extent of 8° 37′ 26″, while two points on the movable ecliptic 90° from the tropical points move on the circumferences of the small circles, so that the movable ecliptic rises and falls on the fixed one, while the points of intersection of the equator and the movable ecliptic advance and recede to the extent of 10° 45′ either way, This is a motoin of the eighth sphere, common to all stars, and the Sun will, therefore, sometimes reach its greatest declination in Cancer, sometimes in Gemini. Tabit does not say that the obliquity of the ecliptic is variable, and perhaps it did not occur to him that this would be a necessary consequence of his theory; he only notices the change in direction and amount of the motion of the equinoxes, which, he says, has increased since the days of Ptolemy, when it was only 1° in 100 years, while later observers have found 1° in 66 years. The erroneous value given by Ptolemy was, therefore, mainly responsible for the continuance of the imaginary theory. It is to be observed that Tabit expresses himself with a certain reservation, and seems to think that further observations are necessary to decide if the theory is true or not. His younger and greater contemporary Al Battani was even more cautious, for though he repeats the account of the trepidation given by Theon (which he says that Ptolemy *manifeste in suo libro declarat*[1]) he does not make use of it, but simply adopts 1° in 66 years (or 54″.5 a year), which he finds by a comparison between his own observations and some made by Menelaus. In rejecting the erroneous value of Ptolemy, which Al Fargani alone had accepted,[2] Al Battani was followed by Ibn Jūnis, who came still nearer to the truth by adopting 1° in 70 years or 51″.2 a year, and who does not allude to trepidation. It is greatly to the credit of several other Arabian writers that they were not led astray by this imaginary phe-

1. Cap. 52, (.205). Plato's translation gives the period as 84 years, but Nallino's ed. has 80 (p. 127).
2. c. 13, p. 49.
3. Schjellerup, *Descr. des etoiles fixes*, p. 43,

nomenon; among them are Al Sūfi, the author of the only uranometry of the Middle Ages[3], who followed Al Battani, also Abu 'l Faraj and Jagmini,[1] while Nasir ed-din mentions it but seems to doubt its reality.[2] By others it was willingly accepted, for instance by Al Zarkali, who made the period of oscillation of 10° either way equal to 2000 Muhammedan years (or 1940 Gregorian years, i.e. 1° in 97 years or 37″ a year). The motion is in a circle of 10° radius; at the Hijra the movable equinox was it 40′ in increasing precession, and in A: D. 1080 at 7° 25′.[3]. The diminution of the inclination of the ecliptic, which the astronomers of Al Mamun had found=23° 33′, no doubt lent countenance to the idea of trepidation, and the next step in the development of this curious theory was the combination of progressive and oscillatory motion. Al Betrugi, who gives a sort of history of the theory, beginning with a mythical H mes, makes out that Theon (or Taun Alexandrinus as he calls him) combined the motion of 1° in 100 years with the oscillation[3]. A century later this was actually done, and the theory received its last development by King Alfonso or his astronomers, who[4] perceived that the equinoxes had receded much further than Tabit's theory allowed. The equioxes were now supposed to pass right round the heavens in 49,000 years (annual motion=26″.45), while the period of the inequality of trepidation was 7000 years, so that in a sort of Great Jubilee year everything was again as it had been in the beginning.[5] The progressive motion belongs to

1. Abu 'l Faraj, p. 12, simply says that the motion is 1° in 100 years according to Ptolemy, or 1° in 66 years according to others. But on p. 18 he says that if the ancient Chaldeans gave the tropical points a motion backwards and and forwards, and if ancient astrologers adopted this, then the motion of the fixed stars must have been unknown to them. Jagmini (p. 229) says that most people adopt 1° in 66 solar years.
2. *Spheres celestes*, p. 347.
3. Sedillot, *Memoire sur les instr. astr. des Arabes*, pp. 31, 32. Abraham ben Chija (p. 196 of Munster's *Sphaera mundi*. Basle, 1546) gives the period as 1600 years without quoting any authority. He adds that the ancient Indians, Egyptians, Chaldeans, Greeks, and Latins first proposed the theory—Ptolemy neither approved nor disapproved of it, but Al Battani confuted it.
4. Alpetragius, f. 12a. He says that Al Zarkali did the same.
5. A later writer, Augustinus Ricius. *De motu octave sphaere*, Paris, 1521, who traces the theory back to Hermes, 1985 years before Ptolemy (!) credits

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the ninth sphere; the annual precession varies between 26″.45± 28″.96, or from + 55″.41 to −2″.51.[1] It was now necessary to assume the existence of a tenth sphere, which as primum mobile communicated the daily rotation to all the others, while the ninth produced the progressive and the eighth periodical motion on the small circles, which are situated "in the concavity of the ninth sphere." This was a nice and comfortable theory on account of the long periods involved and the slow changes it produced in the amount of annual precession; and oblivious of the fact that the theory had no foundation except the circumstance that the obiquity of the ecliptic was now about 20′ less than it had been stated to be by Ptolemy, and that he had given the amount of precession as 36″ a year instead of about 50″, and often shutting their eyes to several of the necessary consequences of it, such as the changes in the latitudes of stars which it ought to produce[2], astronomers continued to accept the theory until at last a real observer of the stars arose and wiped it out by showing that the obliquity of the ecliptic had steadily diminished, and that the amount of annual precession had never varied. We have in this place only alluded to it because it involved some rearrangement of the spheres and because it is eminently characteristic of the period during which no persistent observations were taken, and hardly an attempt was made to improve the theories of Ptolemy. The theory of *trepidatio* or *titubatio*, as it was sometimes called, was one attempt and it would have been better left alone. But it forms a not uninteresting chapter in the history of astronomy.

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this development to a Jew of Toledo, Isaac Hassan (see above. p. 38), adding that Alfonso four years after the completion of the tables became convinced of the futility of the theory by reading the book on the fixed stars by Al Sufi. Riccioli, *Almag. novum*, I. p. 166.

1. In the Alfonsine Tables the maximum took place at the birth of Christ. In Essler's *Speculum astrologicum*, p. 224 (appended to Purbach's *Theoricae novae*, Basle, 1573) the epoch is A.D. 15, diebus 137 completis. Reinhold in his commentary to Purbach (Paris, 1558, f. 163b) explains that 26″.45 is the space passed over by the Sun in 10 mins. 44 secs., by which amount the Alfonsine Tables made the tropical year smaller than 365¼ days.
2. Abraham ben Chija (p. 103, *Schrackenfuchs*) says that trepidation does not change the latitudes. Perhaps he refers to the earliest form of the notion, that described by Theon of Alexandria.

Here we finish our review of ancient astronomy. We have omitted as not coming within our province several valuable contributions to science which did not deal with cosmology or planetary theory. But even with this limitation enough has been said to show that when Europeans again began to occupy themselves with science they found astronomy practically in the same state in which Ptolemy had left it in the second century. But the Arabs had put a powerful tool into their hands by altering the calculus of chords of Ptolemy into the calculus of sines or trigonometry, and hereby they influenced the advancement of astronomy in a most important manner.

References

1. Peter Doig : *A Concise History of Astronomy* London 1950.
1. J.L.E. Dreyer : *A History of Astronomy from Thales to Kepler Dover publications*, 1953 (chapter XI reproduced).
3. Satya Prakash : *Founders of Sciences in Ancient India*, Delhi, 1965.

CHAPTER II

Personal References of Brahmagupta

Sudhākara Dvivedī in his *Gaṇaka Taraṅgiṇī*, a small book on biographical skethes of astronomers and astrologers of this country gives a brief account of Brahmagupta thus :

Brahmagupta was born in 520 Śaka (655 Vikramī or 589 A.D.) in the reign of King Vyāghramukha, belonging to the *Cāpa* family ; his father was Jiṣṇugupta ; and at the age of 30, he wrote in 550 Śaka (628 A. D.) his well known treatise on Astronomy known as the *Brāhmasphuṭasiddhānta*, which is corroborated by the statement in the *Viṣṇudharmottara Purāṇa*, (Chapter on the *Brahma-siddhānta*), His other treatise, entitled the *Khaṇḍa-Khādyaka*, which is a *karaṇa* book, was completed in 587 Śaka (665 A. D.). According to some authorities, Brahmagupta was the grandson of Viṣṇugupta, and the family suffix (*Gupta*) indicated that he belonged to the Vaiśya family, and he was in the service of the King of Rewah, known as Vyāghrabhaṭa.

Brahmagupta was a great critic ; he did not spare any of his predecessors like Āryabhata, Varāhamihira, Śrīṣeṇa, Viṣṇucandra and others. Later on, his influence on the writing of the succeeding generations has been immense. Bhāskarācārya II in his *Bījagaṇita* has acknowledged him as a great authority on algebra and has given him as the first place amongst the galaxy consisting of Brahmagupta, Śrīdhara, Padmanābha etc. The Eighteenth Chapter of the *Brāhmasphuṭasiddhānta*, known as Kuṭṭaka Chapter (on Pulveriser) has been translated by H. T. Colebrooke in English in 1817. The English translation of the Twelfth Chapter on Gaṇita or Calculations from the *Brāhmasphuṭasiddhānta* is also available in English. (See Colebrook's *Algebra with Arithmetic and Mensuration from the Sanskrit of Brahmagupta and Bhāskara*, London, 1817).

The *Vāsanā Commentary* on the Brāhmasphuṭasiddhānta by Pṛthūdakasvāmī (860 A. D.) is also available though with difficulty (as indicated by Sudhākara Dvivedi) ; its incorrect manus-

cript is available in the Library of the King of Banaras (Kāśirāja) which has the colophony at the end as :

श्री चापवंशतिलके श्री व्याघ्रमुखे नृपे शकनृपालात्,
पञ्चाशत्संयुक्तैर्वर्षशतैः पञ्चभिरतीतैः।
ब्राह्मस्फुटसिद्धान्तः सज्जनगणितज्ञगोलवित्प्रीत्यै,
त्रिंशद्वर्षेण कृतो जिष्णुसुतब्रह्मगुप्तेन ॥

Bhāskara II has written the famous treatise *Siddhāntaśiromaṇi* (1150), which is almost based on the *Brāhmasphuṭasiddhānta*. It has been edited by the author's own gloss (*Vāsanābhāṣya*) by Bāpu Deva Śāstrī (Vārāṇasī) ; by Murlīdhar Jha with the commentaries, *Vāsanāvārttika* of Nṛsiṁha (1621) and Marīci of Munīśvara (1635), vol. I (containing chapter 1 of the *Gaṇitadhyāya*) (Vārāṇasī, 1917); by Girija Prasad Dvivedi with original commentaries in Sanskrit and Hindi, vols. I and II (Lucknow, 1911, 1926) ; English translation of the text only by Bāpu Deva Śāstrī and Wilkinson (Calcutta, 1861).

In the very first Chapter (verse 2), Brahmagupta writes : The old calculations dealing with planets (i.e. the old astronomy), based on the system of Brahmā have become erroneous in course of past ages and therefore, I, the son of Jiṣṇugupta would like to clarify them.

Brahmagupta was not a mere theorist, he based his calculations on direct observations with the help of instruments or devices (*nālikādi yantra*) ; he was in favour of making corrections on the basis of these observations. He was himself an expert *observer*. In his *Khaṇḍakhādyaka* also he has emphasised the need of direct observation.

At many places, Brahmagupta has severely criticised the Romaka and Pauliśa systems of astronomy which were introduced in this country by Lāṭadeva and Śrīṣeṇa, There are many passages where this criticism would be available with vehemance.

Brhamagupta was opposed to the system of Āryabhaṭa I. He never spares the school of Āryabhaṭa which was regarded as the most authoritative then. Sudhākara Dvivedi says that as Brahmagupta was opposed to the system of Āryabhaṭa, so the Vaṭeśvara Siddhānta was opposed to that of Brahmagupta. The Institute has already published the *Vaṭeśvara Siddhānta* and now it has the privilege of publishing the *Brāhmasphuṭasiddhānta*.

A Note on Bhillamāla

It is said that Brahmagupta completed his *Brāhmasphuṭa-siddhānta* in Śaka 550, and he has come to be known as Bhillamālakācārya or a teacher residing in "Bhillamālaka." In this connection, therefore, it would be interesting to reproduce a note on *Bhillamāla* from G. Bhūler's article on *Gurjara Inscriptions*, No. III, published in the *Indian Antiquary*, July 1888, vol. 17, p. 192 :

With a single exception all the complete inscriptions call the princes enumerated above, scions of the Gurjara race; and *Khe* I. and II. highly extol the greatness and wide extent of this family. *Na.* alone names the Mahārāja Karṇa as their ancestor. With respect to this personage it is for the present impossible to say whether the famous hero of the *Mahābhārata* may be meant, or some real historical king. But the name *Gurjara* makes it evident that this dynasty belonged to the great tribe which is still found in Northern and Western India and after which two provinces, one in the Bombay Presidency and one in the Pañjāba, have been named. The Gurjaras or *Gūjars* are at present pretty numerous in the western Himālaya, in the Pañjāba and in Eastern Rājpūtānā. In Kachh and Gujarāt their number is much smaller. It would, therefore, seem that they came into Western India from the north. Their immigration must have taken place in early times, about the beginning of our era or shortly afterwards. In Western India they founded, besides the kingdom of Broach, another larger state which lay some hundred miles further north. Hiuen Tsiang mentions in his travels[1] the kingdom of *Kiu-che-lo* and its capital *Pi-lo-mi-lo*. It has been long known that the former word corresponds to Gurjara.

But the name of the town has been incorrectly connected by the French scholars with Bālmer in the Jēsalmīr territory, and this identification has been accepted in Mr. Beal's new translation of *Siyuki*. As I have stated already formerly[2] following Colonel. J. Watson, *Pilomilo* corresponds exactly to Bhillamāla

1. Beal, *Siyuki*, Vol. II, p. 269f. Hiuen Tsiang assigns to the northern Gurjara State an extent about double of that given for the kingdom of Broach.

9. *Ante*. Vol. VI. P. 63.

one of the old names of the modern Bhīnmāl or Śrīmāl[1] in southern Mārvāḍ close to the northern frontier of Gu'arāt. Another work, which was composed a few years before Hiuen Tsiang's visit to Gujarāt, contains likewise a notice of this northern kingdom of the Gurjaras. The astronomer, Brahmagupta, who completed his *Siddhānta* in Śaka-Samvat 550 or 628 A.D. calls himself Bhillamālakakācārya[2], "the teacher residing in Bhillamālaka and is called so by his commentator Pṛthūdakasvāmin. He further states that he wrote under king Vyāghramukha who was 'an ornament of the Cāpa race.' This family, whose name recurs in the Haḍḍāla grant of Dharaṇivarāha[3] prince of Vadhvān, thus seems to have been the reigning house of Bhillamāla. It is most probably identical with the Cāuḍās, Cāvóṭakas[4] or Chāpótkaṭas, who from 756 to 941 A.D. held Aṇhilvāḍ and still possess various small districts in northern Gujarāt. The Gurjara kingdom of Broach was without a doubt an offshoot of the larger State in the north, and it may be that its rulers, too, belonged to the Cāpa family.

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1. Bhillamāla means etymologically 'the field of the Bhil' and Śrīmāla 'the field of Śrī'. The latter name must also be ancient, as the Śrīmāli Brāhmaṇ as are called after it. The Jainas narrate various, of course incredible, legends, which explain how Śrīmāla came to be called Bhillamāla. Merutuṅga says that king Bhoja invented the latter name, because the people of Śrīmāla let the poet Māgha die of starvation. According to another authority, the town had a different name in each Yuga. It is in India very common for ancient towns to have two or even more names. Thus Kanauj was called, Kānyakubja, Gādhipura, and Mahodaya.

2. See Professor A. Weber, *Die Sanskrit und Prakrit Handschriften der Berliner Bibliothek* Vol. II. pp. 297-298. In the first passage the MSS. offers incorrectly Bhilamācārya; in the second which occurs in the commentary on the Khaṇḍakhādyaka, we have Bhillamālavakācārya, a slightly corrupt reading. This latter varia lectio occurs also in other MSS., see Weber, *Indische Streifen*. Vol. III, p. 90, and has given rise to erroneous suppositions regarding Brahmagupta's home. The Gujarāti Joshis still preserve the tradition that Brahmagupta was a native of Bhīnmāla.

3. Ante, Vol. XII. p. 190ff. The remark which I have made there that the Cāpas are not named elsewhere, of course requires correction.

4. The form Cāvoṭaka, which occurs in Dr. Bhagavanlal's grant of the Gujarāt Cālukya king Pulakeśin of Samvat 490, is the immediate predecessor of the word Cāuḍā. Its Sanskrit original is certainly not cāpotkaṭa which probably has been coined in comparatively speaking modern times, in order to explain the difficult Prakrit word, just as the bards of Rajputana have invented Rāṣṭrauḍha as etymon for Rāthoḍ.

Brahmagupta's own References

In the Twenty-fourth Chapter (*Sañjñādhyāya*), of the *Brāhmasphuṭasiddhānta*, Brahmagupta has made a reference to his own biography: In the reign of Vyāghramukha belonging to the family of Cāpa, in the year 550 Śaka the treatise *Brāhmasphuṭasiddhānta* was composed for the benefit of benevolent astronomers by Brahmagupta, son of Jiṣṇugupta at the age of 30. (*BrSpSi.* XXIV. 7. 8)

Then again he says: The *Brāhmasphuṭasiddhānta* has been written by Brahmagupta, son of Jiṣṇu, in 1008 verses of Āryāchanda. (ibid 10)

In the beginning of this Sañjñādhyāya, he refers to the differences in fundamental notions created by the various existing systems of astronomy as the *Sūrya-siddhānta, Puliśa-siddhānta. Romaka-siddhānta, Vasiṣṭha-siddhānta* and other *Yavana-siddhāntas,* which have caused anomalies in the calculations of eclipses. He also refers to the anomalies due to the calculations based on midnight day-reckoning and sunrise day-reckoning.

From the point of view of own references, the following would be of interest:

Brahmagupta, son of Jiṣṇugupta (Jiṣṇusuta-Brahmagupta): *BrSpSi.* I. 2; XVI. 35. 37; XXIV. 8. 10; XXV. 73.

It is strange that in the *Khaṇḍakhāḍyaka*, Brahmagupta has not given his name nor his father's name anywhere. At least the reading of the *Khaṇḍakhāḍyaka* as given by Pṛthūdakasvāmī does not contain this name. In the edition of Bhaṭṭotpala, there are three more chapters in the *Khaṇḍakhādyaka* (Chapters IX, X and XI). In the Chapter XI (known as *Pātādhikāra*), we have 21 verses and in the last 21st verse we find the name of Brahmagupta,[1] son of Jiṣṇu mentioned:

> Those who are eager to have the knowledge of the motion of stars and planets, for them and for the benefit of disciples in this field, Brahmagupta son of Jiṣṇu has composed this *Khaṇḍakhādyaka.*

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1. खण्डखाद्यकमिदं तृप्त्यर्थं ग्रहगतिज्ञधार्त्तानाम् ।
शिष्याणां हितार्थं प्रोक्तं जिष्णुसुतब्रह्मगुप्तेन ॥ —*KK. XI. 21*

Reference to Āryabhaṭa

We have said that it appears that Brahmagupta was a bitter opponent of Āryabhaṭa in his younger days (628 A. D.), but later on (in 665 A. D.), he climbed down to describe and teach one of the Āryabhaṭa's system of astronomy. Āryabhaṭa was universally revered. and it was difficult for Brahmagupta to have ignored him and thus he has to refer to this great authority some times to oppose some of his views and some times to expound his views. The following are the pasages in the *Brāhmasphuṭasiddhānta*, where the author has referred to the name of Āryabhaṭa. There are many other passages where the name "Āryabhaṭa" does not occur but where Brahmagupta indirectly means to quote the views of this great master.

BrSpSi. I. 9. 12. 28. 32. 60. 61 ; II. 33. 46; V. 21.25; VI. 13; IX.10; XI. 4. 9. 10. 12. 25. 29. 33. 41. 42. 43. 44, 46, 47. 49, 62; XIII, 27; XIV. 45; XVI. 37. 46; XXI. 39.

The largest references are in Chapter XI, where Brahmagupta has made an attempt to show the discrepancies of the Āryabhatiaya system of reckoning astronomical observations and constants.

In the *Khaṇḍakhādyaka* also there is a reference to the name of Āryabhaṭa but on very few occasions. In Chapter I, we have this reference at three places-

> Having made obeisance to God Mahādeva, who is the great cause of this world's rise (i.e. creation), existence and destruction, I shall declare the *Khaṇḍakhādyaka* (i.e. a short treatise on astronomy, which is as pleasant as food prepared with sugarcandy),which will yield the same results as the great astronomical treatise of Āryabhaṭa. As in most cases calculation by the great work of Āryabhata, for (the knowlege of time and longitude of planets etc. at) marriage, nativity and the like, is impracticable for common use every day, this smaller treatise is made so as to yield the same results as that. [1]

1. प्रणिपत्य महादेवं जगदुत्पत्तिस्थितिप्रलयहेतुम् ।
वक्ष्यामि खण्डखाद्यकमाचार्य्यार्य्यभट तुल्यफलम् ॥
प्रायेणार्य्यभटेन व्यवहारः प्रतिदिनं यतोऽशक्यः ।
उद्वाहजातकादिषु तत्समफललघुतरोक्तिरतः ॥ —*KK. I. 1-2*

These two verses show that Brahmagupta was not hostile to Āryabhaṭa when he wrote this *Khaṇḍakhādyaka* ; he merely intends presenting the subject matter on simple lines and furnishing the results obtained by Āryabhaṭa in a simpler way.

In the following verse of the same chapter, he refers to Āryabhaṭa's midnight day-reckoning system :

> The mean Saturn diminished by 3 seconds, the Śighrocca of Mercury diminished by 22 seconds, the mean Mars increased by 2 seconds and the mean Jupiter increased by 4 seconds are equal to the respective mean planets of Āryabhaṭa's "midnight-system".[1]

In the Appendix of the *Khaṇḍakhādyaka* known as *Khaṇḍakhādyakottara*, we again find a few verses where the name of Āryabhaṭa occurs; Three of these verses have been reproduced from the *Brāhmasphuṭasiddhānta (BrSpSi* I. 62, 63; II.47) :

> Āryabhaṭa made the apogee of the Moon as moving more quickly and the node as moving more slowly than their actual motions ; if his constants give correct results in relation to the end of *tithis* [i. e. conjunction etc.] or eclipses, they must be considered as accidental as are the letters cut into wood by weevils. (*KK*. IX.1; *BrSpSi*. I. 62)
>
> On seeing me, who possess the most accurate knowledge of mean motions, men who have learnt from the works of Śrīṣeṇa, Viṣṇucandra and Āryabhaṭa, cannot face me in any meeting just like deer on seeing a lion (*KK*. IX. 2; *BrSpSi*. I. 63)
>
> As the apparent planets beginning with Mars as derived from the works of Śrīṣeṇa, Āryabhaṭa and Viṣṇucandra, are far deviated from their true places, the works of these authors are therefore not valued among the learned.[2] (*KK*. IX. 3; *BrSpSi*. II. 47)

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1. तिसृभिः शनिर्ज्ञशीघ्रं द्वाविंशत्या कुजोऽधिको द्वाभ्याम् ।
चतसृभिरधिको जीवोऽर्द्धं रात्रिकार्य्यभटमध्यसमाः ॥ —*KK.I. 7*

2. अकृतार्य्यभटः शीघ्रगमिन्दूच्चं पातमल्पगं स्वगतेः ।
तिथ्यन्त ग्रहणानां घृणाक्षरं तस्य संवादः ॥१॥
मध्यगतिज्ञं वीक्ष्य श्रीषेणार्य्यभट विष्णुचन्द्रज्ञाः ।
सदसि न भवन्त्यभिमुखाः सिंहं दृष्ट्वा यथा हरिणाः ॥२॥
दूरभ्रष्टाः स्पष्टाः श्रीषेणार्य्यभटविष्णुचन्द्रे षु ।
यस्मात् कुजादयस्ते विदुषां नैवादरस्तस्मात् ॥३॥ —*KK. IX. 1-3*

In the Appendix of the *Khaṇḍakhādyaka*, there is another verse which also speaks in the same strain against Āryabhaṭa (this verse does not occur in the *Brāhmasphuṭasiddhānta*) :

> As the process of finding the apparent places of planets as given by Āryabhaṭa does not make them agree with observation, I shall, therefore, speak of this process. Of the Sun the apogee is at two signs and seventeen degrees (*KK.* IX. 4)[1]

Brahmagupta opposed to Śrīṣeṇa—

Viṣṇucandra, Lāṭa, Vijayanandī and others

Brahmagupta was a great critic; he did not spare Āryabhaṭa, and along with him he was vehemently opposed to the doctrine of Śrīṣeṇa, Viṣṇucandra, Lāṭadeva, Vijayanandī and Pradyumna also. He was opposed to the *Romaka* and *Pauliśa Siddhāntas*, which were the systems of foreign astronomy, derived from Greece, Babylonia and other centres of learning. He did his best to resist the foreign influences on astronomy.

The following are the verses in the *Brāhmasphuṭasiddhānta*, where Brahmagupta expressed his note of discord against the systems or notions of Āryabhaṭa, Śrīṣeṇa and Viṣṇucandra :

BrSpSi. I.60; II 46, 47; X 13, 62; XI.31, 46, 47, 48-50, 55, XVI. 36, 46; XXI.39; XXII. 2.

In two of the verses, he refers to Lāṭasiṃha :

BrSpSi. XI. 46, 48.

In the following verse, he refers to Aṅkaciti, Vijayanandī, Pradyumna and others: *BrSpSi.*XI. 58.

In the *Khaṇḍakhādyaka* we do not find the names of these adversaries of Brahmagupta ; we, however, have a reference of Śrīṣeṇa, Āryabhaṭa and Viṣṇucandra in the verses already quoted, occurring in the Appendix of the *Khaṇḍakhādyaka* (*Khaṇḍakhādyakottara*), Chapter IX. 2 and 3. As we have said before, these verses of the Appendix have been reproduced from the *Brāhmasphuṭasiddhānta.* (I.62 and II.47).

Reference to Romaka and Pauliśa systems

Brahmagupta speaks of his system as if expounded by Brahmā himself for the first time, later on deteriorated, and then revived by Brahmagupta himself. The very second verse of the *Brāhma-*

1. न स्फुटमार्क्षभटोक्तं स्पष्टीकरणं यतस्ततो वक्ष्ये ।
भानुमतो मन्दोच्चं राशिद्वयमंशकाश्च सप्तदश ॥ —*KK. IX. 4.*

sphuṭasiddhānta (I. 2) substantiates this view :

The science of astronomy (or the calculations of heavenly bodies) in course of long duration became ineffective or erroneous; this was revived by Brahmagupta, son of Jiṣṇu.[1] (*BrSpSi.* I.2).

The system of astronomy which goes by the name Brahma (*Brahmasiddhānta*) has been handed down to us in three forms : (i) one is as treated by the *Śākalya Saṃhitā*, (ii) one as described in prose in the *Viṣṇudharmottara Purāṇa*, and (iii) the one described by Varāhamihira in the *Pañcasiddhāntikā*, which recognises the *yuga* of the duration of five years. Which of these three was accepted by Brahmagupta is not clear. But from the measures of the number of revolutions performed by a planet in a given period (*graha-bhagaṇa*) etc., it is clear that Brahmagupta acknowledges the system as propounded in the *Viṣṇudharmottara Purāṇa*. In his chapter "*Tantra-parīkṣādhyāya* or *Dūṣaṇādhyāya*, he contradicts the notions of the *Vedāṅga Jyotiṣa* which accepts the *yuga* of five years.

We may further emphasize the fact that Brahmagupta has not clearly detailed out the errors to which the Brahma-siddhānta succombed in course of time, and how these errors were eradicated by him. During the days of Brahmagupta, Romaka and Pauliśa systems were getting currency in this country. Reference to these two are found in the *Brāhmasphuṭasiddhānta* at several places as follows :

ROMAK : I.13; XI.50; XXIV.3
PAULIŚA : XIV.45; XXIV.3

In fact, *BrSpSi.* XXIV.3, we find the line "*Sūryendu-Puliśa-Romaka-Vaśiṣṭha-Yavanādyaiḥ*", where we have a reference to all the then existing systems : *Sūrya-siddhānta*, *Indu-siddhānta*, *Puliśa-siddhānta*, *Vasiṣṭha-siddhanta*, and other *Yavana-siddhāntas*. Just as the Sun is one, so the astronomical system is also one ; this is a different thing that calculations in different systems may vary according to different sunrises in different places.

Brahmagupta refers at one place to Varāhamihira in *BrSpSi.* XXI.39, where he has been spoken of in connection with a list of

१. ब्रह्मणोक्तं ग्रहगणितं महता कालेन यत् खिलीभूतम् ।
अभिधीयते स्फुटं तज्जिष्णुसुतब्रह्मगुप्तेन ॥ —*BrSpSi. I. 2.*

anti-authoritative versions of astronomical systems :

Evam Varāhamihira-Śrīṣenācāryabhaṭa-Viṣṇucandrādyaiḥ

Lokaviruddhamabhihitam Veda-smṛtisaṃhitābāhyam.

At one place, he mentions the difference between the calculations based on the system of Āryabhaṭa and the Pañca-siddhāntas (Five systems) : Pauliśa, Romaka, Vasiṣṭha, Saura, and Paitāmaha. (*BrSpSi.* XIV.46).

Brahmagupta was also familiar with the Jaina systems of astronomy; for example, at one place he uses the term "Jinoktam" (i.e., one propounded by the Jainas) : He repudiates the concept of *two Suns and two Moons* "*Dvāvarkaindavau*" (do canda do sujja) (*BrSpSi.* XI.3) as enunciated by the Jainas.

At several places we find a reference to the *Vaśiṣṭha-siddhānta* (*BrSpSi.* XI.49, 50; XXIV.3)

Wherever, Brahmagupta has to press for his views in preference to the views of others, he uses the words *Brāhma* or *Brahmokta* : *BrSpSi.*

Brāhma : I.32; X.62; XI, 61; XVI.37

Brahmokta : II.31, 33; X.63, 69; XV.59; XVI.38

Reference

Sudhākara Dvivedī : *Gaṇaka Taraṅgiṇi*

G. Bühler : Gurjara Inscriptions, *Indian Antiquary*, 1888.

P. C. Sengupta : *Khaṇḍakhādyaka*, Calcutta, 1934.

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CHAPTER III

Manuscripts of the Brāhmasphuta Siddhānta

Sudhākara Dvivedī has given an account of some of the manuscripts of the *Brāhmasphuṭasiddhānta* in his *Bhūmikā* or Introduction appended to the edition published in the PANDITA, Vol.XXIV, 1902 (New Series) : (i) available in the Library of the Government College, Kashi (Vārāṇasī) i. e. Kāśika-Rājakīya-Pāṭhālaya,(ii) Dr.Thibaut's Manuscript,(iii) the Manuscript in possession with Yajñadatta Sharmā, the Chief Astronomer attached to the Prince of Ayodhyā. It is further mentioned that Dr. Thibaut's Manuscript was a copy of a Manuscript available in the Deccan College, Poona. The Manuscripts (ii) and (iii) were identical. The Manuscripts (iii) was very faulty and incorrect.

The Manuscript from which Colebrooke translated out in English the Twelfth and the Eighteenth Chapters on *Gaṇita* or Mathematics and *Kuṭṭaka* (the Chapter on Pulveriser) respectively, appeared to be different from the three Manuscripts described above. The readings differed considerably. The *Kuṭṭaka*-Chapter of that book is, writes Sudhākara Dvivedī, still available in the India Office Library. (See Catalogue of the Sanskrit[1] Manuscripts in the Library of India Office, Part V.p. 995).[1]

1. एतत्कृतस्यास्य सिद्धान्तग्रन्थस्यैका प्रतिः काशिकराजकीय पाठालयतो द्वितीया डा० थिबौ साहिब महाशयतस्तृतीया चायोध्यानरेशप्रधानज्योतिर्विच्छ्रीयज्ञदत्तशर्मणो मया लब्धा। डा० थिबौमहाशयस्य पुस्तकं कथञ्चिद्दक्षिणदेशीय (Deccan College, Poona) डेका पुस्तकस्य प्रत्यन्तरम्। इदं पुस्तकं तथा पं० श्री यज्ञदत्त पुस्तकं चैकमातृकमेव। इदं पुस्तकत्रयमतीवाशुद्धं बहु च स्खलितं चास्ति।

यत्पुस्तकानुसारेण व्यक्ताध्यायोर्द्वादशाष्टादशसंख्ययोरंगलभाषायामनुवादः कोलब्रूकसाहिबेन कृतस्तत्पुस्तकमेतत्त्रयतो भिन्नमित्यसंशयं विभाति पाठविभेदात्। तत्पुस्तकस्य कुट्टकाध्यायः संप्रति इण्डिया-आफ़िस-सरस्वती भवने वर्तते (See Catalogue of the Sanskrit Manuscripts in the Library of India Office, Part V, Page 995)

पृथूदककृताऽस्य सिद्धान्तस्य टीका या कोलब्रूकसाहिबेन भारतवर्षे ह युपलब्धा सा च संप्रति लण्डननगरे इण्डिया-आफिस-सरस्वती भवने वर्तते। तस्या एका प्रतिर्मद्द्वारेण श्री डा० थिबौ साहिबमहाशयेनोत्पादिता सापि संप्रति मन्निकटेऽस्ति। अहो इण्डिया-आफिस-सरस्वती

(Continued on next page)

Colebroke procured a Manuscript of the Pṛthūdaka Svāmī's Annotation on the *Brāhmasphuṭasiddhānta* from somewhere in India. This is also available in the India office Library, London. Undoubtedly it appears that this Manuscript is a copy from one written in the *Maithila* script. Dvivedī describes a few characteristics of this Manuscript. For example, on the Folio 11. 7 (७) is written instead of 9 (९). At a few other places also, the same has happened. Following a visarga, (*kva* क्व) is inscribed instead of '*ka*' (क), for example refer to Folio 12, line 6. At some places instead of *visarga* (:) we find ṣa (ष) inscribed, for example on Folio 12, line 1. Sometimes we find a *sandhi* at the *virāma* or end of a sententence ; e. g. in the Golādhyāya, Folio 21, line 2: *sarvamupapannamuktamakhaṇḍena* (सर्वमुपपन्नमुक्तमखण्डेन). At some places we find *nū* (न्) written in place of *nta* (न्त). In one of the Folios, *Śri Gaṇeśāya namaḥ.* श्रीगणेशायनमः is written in Maithila script.[1] Sudhākara Dvivedī prepared a copy of this Manuscript for Dr. Thibaūt, and this copy was available with Sudhākara Dvivedī when he published his commentary with Text in the *Paṇḍita.* During the course of binding, on account of carelessness, many of the Folios got misarranged, and many of them got fragmented. Sudhākara Dvivedī emphasises in his Introduction the need of careful research on the arrangement of these Folios and their readings. (See *Catalogue of the Sanskrit Manuscripts* in the Library of India Office, Part V,p. 993-995).

Sudhākara Dvivedī with considerable efforts could rearrange the Text :

Golādhyāya-Bhāṣyam 1 to 45, mutilated at the beginning
Madhyamādhikāra-Bhāṣyam 45 to 59

(Continued from previous page)

भवने अस्य पुस्तकस्य पुटकबन्धनकालेऽनवधानतया पत्राण्यसंगतानि जातानि, बहुत्र खण्डितानि च सन्ति । तानि कदाचिदनुपयुक्तपत्राणां मध्ये स्युरिति तेषां सम्यगन्वेषणं समुचितम् । (See Catalogue of the Sanskrit Manuscripts in the Library of India Office Part, V, Page 993-995)

1. इदं पुस्तक कस्यचिन्मिथिलाक्षरैर्लिखितस्य पुस्तकस्य प्रत्यन्तरमिति निःसंशयं प्रतिभाति । ११ पत्रे, ९ इत्यस्य स्थाने ७ इति लेखात् । एवमन्यत्रापि । विसर्गात् परतः 'क' स्थाने 'क्व' इति लेखात् । यथा १२ पत्रे, ६ पंक्तौ । तथा विसर्गस्थाने 'ष' इति लेखात् , यथा १२ पत्रे, १ पंक्तौ । वाक्यविरामेपि सन्धिकरणात् । यथा गोलाध्यायस्य २१ पत्रे, २ पंक्तौ सर्वमुपपन्नमुक्तमखण्डेन । कुत्रचित् 'न्त' स्थाने 'नू' इति लेखात् । एकस्मिन् पत्रे मैथिलाक्षरैः श्री गणेशायनमः इति लेखाच्च ।

In this, the commentary is up to verse 31. In the *Spaṣṭādhikāra*, the commentary begins from verse 29, Folio 60. At the top of the Folio, we have the old numbering 1 and 115.

After this, 68 folios are mis-arranged, and there the old numbering is marked 9.

After that, up to 87 folio, we have in the bound volume a commentary up to the verse 6 of *Tripraśnādhikāra*. Here the old numbering is marked 28. Then we have the commentary up to verse 27 of the *Tripraśnādhikāra*. Here the numbering of folios is marked 1 and 159 (old numbering) and 218 new. After this, then we have in the bound volume, commentary up to verse 33 of the Tripraśnādhikāra. On the last folio, the old numbering is marked 5 and 163, the new numbering 222. After this begins the commentary on the *Candragrahaṇādhikāra* from verse 4, old numbering 1 and 297 and new numbering 257. Then follows the commentary on *Sūryagrahaṇa* up to verse 23. The last old folio numbering is 36 and 232, the new numbering 292. This is the last folio-numbering of the volume in the India Office Library. After this, begins the commentary on *Grahayutyadhikāra*, verse 11, old numbering 1 and 164 and the new numbering 223.

Then follows the *Madhyagatyuttarādhyāya* up to verse 40, old folio numbering 119, and new numbering 178. Then we have the commentary of the *Madhyagatyuttarādhyāya* beginning from the verse 45; the old folio numbering 120 and the new numbering 179. Then we have the commentary on *Tripraśnottarādhyāya* up to verse 56. Here the last old folio number is 48 and 158, and the new numbering 217.

In this way, we have the commentary on : the multilated *Golādhyāya*, mutilated *Madhyamādhikāra*, mutilated *Spaṣṭādhikāra*, mutilated *Tripraśnādhikāra*, mutilated *Candra-grahaṇādhikāra*, mutilated *Sūrya-grahaṇādhikāra*, mutilated *Grahayūtyadhikāra*, *Bhagrahayutyadhikāra*, *Tantra-parīkṣādhyāya* (also known as *Dūsaṇādhikāra*), *Gaṇitādhyāya* (Arithmetic), mutilated *Madhyagatyuttarādhyāya*, *Sphuṭagatyuttarādhyāya*, mutilated *Tripraśnottarādhyāya*.[1]

1. मया महताऽऽयासेन तत्प्रति-पाठक्रम एवं नियोजितः ।
गोलाध्यायभाष्यम् १-४५ आदौ खण्डितम् ।
मध्यमाधिकारभाष्यम् ४५-५९ ।

(Continued on next page)

Sudhākara Dvivedī says that nowhere in the *Brāhma-Sphuṭasiddhānta, Madhyamādhikāra,* is found the verse :

Saṁsādhya spaṣṭataram bījam nalikādiyantreṇa.
Tat-saṁsṛtagrahebhyaḥ kartavyau nirṇayādeśau.

(This verse has been quoted by Dvivedī in the *Gaṇaka-Taraṅgiṇī,* page 19).[1]

This verse has been quoted by Dvivedī from the Translation of *Grahalāghava* by Mallāri.

Dvivedi further says that in the Manuscripts available, there is mentioned a Twenty-fifth Chapter under the title "*Dhyānagrahopadeśādhyāya*". Dvivedi thinks that this Chapter does not constitute the *Brāhmasphuṭasiddhānta* proper, which ends in fact with twenty-four chapters.[2] In his commentary and Edition in the *Paṇḍita,* he has published it as a separate treatise of Brahmagupta. Thus he has named his Edition as :

अत्र मध्यमाधिकारे ३१ श्लोकपर्यन्तमेव भाष्यम् ।

अतः स्पष्टाधिकारस्य २९ श्लोकतष्टीकाऽऽरब्धा ५० पत्रतोऽत्र प्राचीन संख्या पत्रोपरि १, तथा ११५ ।

अग्रेऽत्र ६८ पत्रमसंगतम् , यत्र प्राचीन संख्या ९ ।

ततः ८७ पत्रपर्यन्त संलग्नग्रन्थस्त्रिप्रश्नाधिकारस्य ६ श्लोकपर्यन्तं टीका । अत्र प्राचीनपत्रसंख्या २८ । ततस्त्रिप्रश्नाधिकारस्य २७ श्लोकतष्टीका, पत्रसंख्या प्राचीना १ तथा १५९, नवीना २१८ । अग्रे संलग्न ग्रंथस्य त्रिप्रश्नाधिकारस्य ३३ श्लोकपर्यन्तं टीकान्तिमपत्रप्राचीनसंख्या ५ तथा १६३, नवीना संख्या च २२२ । ततश्चन्द्रग्रहणाधिकारस्य ४ श्लोक टीकारब्धा प्राचीनपत्रसंख्या १ तथा २९७ । नवीना संख्या च २५७ । ततः सूर्यग्रहणस्य २३ श्लोकपर्यन्तं टीका । अन्तिम प्राचीन पत्रसंख्या ३६ तथा २३२, नवीना संख्या च २९२ । इण्डिया-आफिस पुस्तकपुटके चेयमन्तिमपत्रसंख्या । ततो ग्रहयुत्यधिकारस्य ११ श्लोकतष्टीकारब्धा यत्र प्राचीनपत्रसंख्या १ तथा १६४, नवीना संख्या २२३ । ततो मध्यमगत्युत्तराध्यायस्य ।

४० श्लोकपर्यन्तं संलग्नग्रन्थौ यत्रान्तिमप्राचीनपत्र संख्या ११९ नवीना संख्या च १७८ । ततो मध्यगत्युत्तराध्यायस्य ४५ श्लोकतष्टीकारब्धा । अत्र प्राचीनपत्रसंख्या १२० नवीनासंख्या च १७९ । ततस्त्रिप्रश्नोत्तराध्यायस्य ५६ श्लोकपर्यन्तं टीका । अत्रान्तिमप्राचीनपत्रसंख्या ४६ तथा १५८, नवीना संख्या च २१७ ।

खण्डितगोलाध्यायस्य । खण्डितमध्यमाधिकारस्य । खण्डितस्पष्टाधिकारस्य । खण्डितत्रिप्रश्नाधिकारस्य । खण्डितचन्द्रग्रहणाधिकारस्य । खण्डितसूर्यग्रहणाधिकारस्य । खण्डितग्रहयुत्यधिकारस्य । भग्रहयुत्यधिकारस्य । तन्त्रपरीक्षाध्यायस्य (दृग्गणाधिकारस्य) । गणिताध्यायस्य पाटीगणितस्य) । खण्डितमध्यगत्युत्तराध्यायस्य । स्फुटगत्युत्तराध्यायस्य । खण्डितत्रिप्रश्नोत्तराध्यायस्य च टीका वर्त्तते ।

1. सर्वेष्वपि पुस्तकेषु 'संसाध्य स्पष्टतरं बीजं नलिकादियंत्रेण' इत्यादिश्लोको मध्यमाधिकारे नास्ति । मया गणकतरंगिण्यां मल्लारिकृतग्रहलाघवटीकातो व्यलेखि (द्रष्टव्या गणकतरंगिणी पृ० -१८१९) ।

Brāhmasphuṭasiddhānto Dhyānagrahopadeśādhyāyaśca or Brāhmaphuṭasiddhānta and *Dhyānagrahopadeśādhyāya by* Brahmagupta (1902)[1]

The manuscript of this small treatise was also mutilated, and Dvivedī took special pains in editing it, and he revised the calculations also incorporated in this treatise.

The small treatise, *Dhyānagrahopadeśādhyāya* must have been composed prior to the *Brāhmasphuṭasiddhānta*, since we find a verse in the last Chapter (the 24th Chapter also known as the *Sañjñādhyāya* of the *Brāhmasphuṭasiddhānta)*, verse 9, a reference to this book :

How could this result be obtained in a simple way has been shown by me in the *Dhyānagrahopadeśādhyāya* of 72 Āryā verses, and therefore, it is not repeated here. *BrSpSi.* XXIV. 9)

In the *Dhyānagrahopadeśādhyāya*, we have a verse 61, which is also found in the *Khaṇḍakhādyaka* (KK. I. 21) :

Navatithayaḥ (159) divided by *aṣṭi* (16), pañcarasāḥ (65) divided by *vasu* (8), 10 divided by 3, each multiplied by the equinoctial shadow are the (tabular differences of) ascensional difference expressed in *vināḍis*. (*KK.* I. 21; *DhGr.* 61)

This verse then indicates that the *Dhyānagrahopadeśādhyāya* has been composed after the *Khaṇḍakhādyaka*. It may also be possible that the *Brāhmasphuṭasiddhānta* and the *Dhyānagrahopadeśādhyāya* were simultaneously written, and the above verse (*DhGr.* 61) was repeated again in the *Khaṇḍakhādyaka*.

Sudhākara Dvivedī has taken the help from the commentary of Pṛthūdaka Svāmī in the Chapters on *Pāṭigaṇita* (Arithmetic), and has quoted the examples from this commentary. At many places he has corrected the readings which were mutilated in manuscripts.

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1. उपलब्धमूलपुस्तकत्रये पंचविंशतितमेध्याये वस्तुतो ब्रह्मगुप्तकृतो ध्यानग्रहोपदेशाध्यायो वर्त्ततेऽतो मयायं पृथक्त्वेन तन्नाम्ना मुद्रितः । अत्र बहुत्र स्खलितानि पदानि तानि गणितेन संशोध्य मुद्रिताग्यपि सुधीभिर्भृशं विचिन्त्यानि ।

अस्य सिद्धान्तस्य चतुर्विंशतितमेऽन्तिमे संज्ञाध्याये (पृ० ४०८)
गणितेन फले सिद्धिर्ब्राह्मे ध्यानग्रहे यतोऽध्याये ।
ध्यानग्रहो द्विसप्ततिरार्याणां न लिखितोऽत्र मया ॥

इति नवमश्लोकेन ध्यानग्रहोपदेशाध्यायस्य रचनेतसिद्धान्तरचनातः पूर्वं विभाति परन्तु तत्राहर्गणसाधनप्रकारेण सिद्धान्तरचनाकाल एवास्य रचना सिध्यति तथा 'नवतिथयोष्टिविभक्ता' इत्यादिगद्येन खण्डखाद्यरचनातः पश्चात् सिध्यति ।

Reference

Sudhākara Dvivedī : His *Bhūmikā* on *BrSpSi.*, *Pandita*, Vol. XXIV, 1902 (New Series)

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Chapter IV

Subject Matter Classified in the Brāhmasphuṭasiddhānta & Khaṇḍakhādyaka

It needs no emphasis that Āryabhaṭa commanded a great influence as an astronomer not only prior to Brahmagupta but during his days also, and we have seen how Brahmagupta quoted this great authority in his writings, sometimes borrowing from him and sometimes contradicting him or improving upon his calculations. Āryabhaṭa's great work is known as the *Aryabhaṭiyam* written in 499 A. D. Āryabhaṭa was born in 476 A. D. (Kaliyuga Saṃvat 3577). The *Aryabhaṭiyam* is also known as the *Aryasiddhānta*. For details about Āryabhaṭa the reader is referred to the Chapter entitled" *Aryabhaṭa lays Foundations of Algebra*" (*Founders of Sciences in Ancient India*, 1965, Chapter X). This great work was written in Kusumapura, (modern Patna, Bihar).

Divisions of Āryabhaṭiyam

The *Aryabhaṭiyam* is divided into four chapters called *Pāda*: (i) the *Gītikā Pāda*, with ten verses; (ii) the *Gaṇita Pāda* with 33 verses; (iii) the *Kālakriyā Pāda* with 25 verses and (vi) the *Gola Pāda* with 50 verses.

In the *Gaṇita Pāda*, a chapter on mathematics, we have such subjects : squares (*varga*), cubes (*ghana*) (verse 3), square-root (*vargamūla*) (4), cube-root (*ghanamūla*) (5) ; area of a triangle, volume of a prism (6), area of a circle, volume of a sphere (7), area of a quadrilateral (*viṣamacaturasra*) (8); circumference of a circle (10), Rsine (radius x sine) (*Jīvā*) (11); determination of the Rsine of the zenith distance, the base (*bāhu*) of a right-angled triangle, and the upright (*koṭi*) of a right-angled triangle (16) ; hypotenuse (*karṇa*) of a right-angled triangle and *ardhajyā* (17), Reversed-sine (*śara*) (18), areas of series-figures (*śreḍhīphala*) (19), rule of three (*trairāśika*) (23), reduction of fractions (*savarṇīkaraṇa* of *bhinna*) (27) ; inverse-rule of three (*vyasta*) (28) ;

evaluation of unknown values (*mūlya-pradarśana* of *avyakta mūlya* (30) ; and the theory of pulveriser (*Kuṭṭaka*) (32, 33). In this chapter, Āryabhaṭa gives the solutions of quadratic equations, and thus he earns for himself the credit of founding the science of algebra.

(iii) The *Kālakriyā Pāda* with 25 verses, which enumerates the units of time [1 year (*varṣa*)=12 months (*māsa*) ; 1 month= 30 days (*divasa*) ; 1 day =60 *nāḍīs* ; 1 *nāḍī*=60 *vināḍī* ; *vināḍī* or *vināḍikā* is the same as *vighaṭikā* equivalent to our 24 seconds ; *nāḍī* or *nāḍikā* or *ghaṭī* is equal to 24 minutes][1] ; correlation of time division with the *kṣetra*—division or angular division,[2] Twelve signs of Zodiac or *rāśis* go to constitute a *bhagaṇa*[3] ; solar day (*ravimāsa*), lunar day (*śaśi māsa*) ; additional month or intercalary month (*adhimāsa*) ; various kinds of years : the solar year is human or *manuṣya* year ; 30 human years=1 *pitṛ* year ; 12 *pitṛ* years=1 *divya* year (divine year) ; 12,000 *divya* years constitute a *yuga* (6, 7, 8) ; the first half of the *yuga* is *utsarpiṇi kāla*, and the latter half is *avasarpiṇi kāla* and they are calculated from the apex of the Moon (*candrocca*) (this is not very clear) (9) ; a *yuga* is of 60 years, and such 60 *yugas*, that is 3,600 years had passed away since the *kaliyuga*, when the author was of 23 years of age, (10) ; the count of a *yuga*, year, month and

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१. वर्षं द्वादशमासास्त्रिंशद्दिवसो भवेत् स मासस्तु ।
षष्टिर्नाड्यो दिवसष् षष्टिस्तु विनाडिका नाडी ॥

—*Ārya. III. 1.*

२. गुर्वंचराणि षष्टिर्विनाडिकार्द्धं षडेव वा प्राणाः ।
एवं कालविभागः क्षेत्रविभागस्तथा भगणात् ॥

—*Ārya. III. 2.*

यावता कालेन षष्टिगुर्वंचराण्युच्चरति मध्यमया वृत्या पुरुषः ।
तावान्काल आर्द्धं विनाडिका । ...यावता कालेन पुरुषः
षडुच्छ्वासान् करोति, तावान्कालश्चार्द्धं विनाडिका स्यात् ।
(परमादीश्वर)

Just as we have the time division, similarly we have the kṣetra division or the circular angular division. A year has twelve months, so do we have 12 *rāśi* in a *bhagaṇa*. One-thirtieth of a *rāśi* is one *bhāga* ; one-sixtieth of a *bhāga* is one *liptā* ; one-sixtieth of a *liptā* in one *viliptā* ; and one-sixtieth of a *viliptā* is one *tatparā*.

३. चन्द्रोऽशैर्द्वादशभिरविक्षिप्तोऽर्कान्तरस्थितैर्दृश्यः ।
नवभिर्भुगुर्भगोस्तैर्द्व्यधिकैर्यथाश्लक्ष्णाः ॥

—*Ārya. IV. 4.*

day should begin from the month *Caitra*, *Śukla Pratipadā* (the first day of the brighter half of the month *Caitra*) (11) ; *mandocca* (apex of slowest motion) and *śīghrocca* (apex of fastest motion) (17—24).

(iv) The *Gola Pāda* with 50 verses. In Verse 1, there is a reference to a point in the Sun's path, the commencement of *meṣa* (*meṣādi*) ; this must have been *vasanta*-equinox. The ascending nodes (*pāta*) of planets and the shadow of the Earth move on the path of the Sun (*arka-apamaṇḍala*) (2-3) ; the angular difference in relation to the Sun at the appearance of Moon (12 degree or *aṁśa*), of the Venus (9 degrees or 9 *vināḍikā*) ; of the Jupiter (2 more than the Venus, i.e. 11 *vināḍikās*) ; the Mercury or *Budha* (13 *vināḍikās*), of the Saturn or *Śani* (15 *vināḍikās*) ; and of Mars or *kuja* (17 *vināḍikās*).

The half of the Earth, Moon, planets and stars is dark, since those parts happen to be under their own shadow ; the other half is bright as it faces the Sun (this is not true with respect to stars—author) (5). The Earth is surrounded with an atmosphere of air and water (6, 7). In the *Brahma-Divasa* (Brahma's day), the sphere of the Earth is increased by one yojana and decreased by this amount during the Brahmā's night (8). Just as a person sitting on a moving boat sees the stationary trees etc. on the bank of a river moving in the opposite direction, similarly the stationary stars are seen moving from Laṅkā (or equator) moving towards the West(9). On account of *pravahavāyu* (air) the *nakṣatra* system and planets rise and set receding towards the West (10). The dimensions of *Sumeru Parvata* (North pole) is given to be one *yojana*, and it shines like a jewel (11) and in the next verse is given the position of Sumeru and the *Bāḍavamukha* (South pole) (12). The four cities situated at a difference of 90° each on the equator are given (13). The distance of Ujjayinī from Laṅkā (thus giving the latitude of Ujjayinī) is given (14).

On account of the thickness of the Earth-sphere, the *Khagola* (celestial sphere) is seen less than the hemisphere (15). The next verse describes how moving appears the Khagola on the North and South poles (16). Then is given the measure of day and night of *devas* (gods), *pitar* (fathers), *asura* (demons) and *manuja* (men) (17). Then are given a few technical definitions

of celestial mathematics (18—21) like *dṛṣṭi-sthāna* (intersection of horizontal and vertical axis—*pūrvapardiggatā rekhā* and *adha-ūrdhva-diggatā rekhā*). *Dṛṅ-maṇḍala*, *dṛgkṣepamaṇḍala* (*dṛgkṣepa* is the zenith distance of that point of a planet's orbit which is at the shortest distance from the zenith). Then the *Bhabhagola* instrument is described (22, 23). Then follow the formulæ for calculating *lagna* (the horizon ecliptic point in the East), *kuja* etc. comprising the *Tripraśnādhikāra* (24—33). In the next verses, we have *lambaka* or Rsine of colatitude (34); *dṛk-karma* (35) and *ayana-dṛk-karma* (36). Then follow the calculations of lunar and solar eclipses (37—47). Verse 48 describes the coordinates of the Sun which are determined by the conjunction of horizon with the Sun, of the Moon by the conjunction of the Sun and the Moon, of planets by the conjunction of the Moon and planets or stars. Verse 49 describes how this jewel treatise has been procured out of an ocean of true and untrue knowledge with the help of a boat of intellect. This means that the author has taken special pains in discriminating true knowledge from falsehood with respect to the prevalent notions of astronomy. In the last verse he says, that he has not given anything new; he has given that very knowledge which was imparted by the *Svayambhu* in the earliest times (50).

The *Pañchāṅgas* prepared according to the rules and formulæ of the *Āryabhaṭiyam* are still regarded with reverence by the *Vaiṣṇavas* in the South. Brahmagupta was a great critic of Āryabhaṭa, but finally he wrote his treatise, the *Khaṇḍakhādyaka*, on the basis of this very *Āryabhaṭiya* (this treatise is a *karaṇagrantha* i.e. one containing a principal element of the Indian Calendar). The four commentaries of the *Āryabhaṭiya* available in Sanskrit are of Bhāskara I, Sūryadeva Yajvā, Parameśvara and Nīlakaṇṭha. Two English translations by P.C. Senagupta (1927) and W.E. Clark (1930) are also available.

Major Landmarks

Indian astronomy which reached its zenith in the times of Āryabhaṭa and Brahmagupta shows its evolution in the following stages:

1. Rudiments of astronomy in the *Vedas* and *Brāhmaṇa* Books like the *Taittiriya Saṁhitā* and the *Śatapatha Brāhmaṇa*. This period is associated with the dis-

covery of the Vedic era, some of the planets, the twenty-four *nakṣatras*, cycle of seasons, concept of leap year, the dimensions of a yuga, solar and lunar years and the like.

II. Astronomy of the *Vedāṅga*, also known as the *Vedāṅga Jyotiṣ*. Lagadha is the most prominent figure of this period (1400 B.C. to 850 B.C.) ; he is the first compiler of a text on astronomy. In his work, we for the first time in history find a reference to the *Jñeya Rāśi* (the knowable or the unknown quantity) and the *Jñāna-Rāśi* (known quantity). He lays the foundations of astronomical calculations. In his treatise we find a mention of such subjects as : Solstices (northern and southern journey of the Sun), increase in days and nights in the *ayanas* or solstices, solstitial *tithis* omission of *tithis*, *parva rāśi* , acceptable and non-acceptable *parvas*, addition of day, acceptable *parvas*, concept of *yoga* (a term applied to the joint space which would be travelled by the Sun and the Moon in a given period of time on the presumption that these two bodies have travelled in directions opposite to each other), method of finding out a *nakṣatra* on any *parva* day, distinction between *parva-nakṣatra* and *tithi-nakṣatra*, correlation of solar and lunar dates, measure of a *nāḍikā* (unit of time), *nakṣatra* of the Sun, *yoga* and its *nakṣatra*, *parva-bhaśeṣa* and equivalent *kalās*, solar year, lunar revolutions (risings of *nakṣatras*) deities of *nakṣtras*, lunar and *sāvana day* differences (*ahika māsa*), divisions of a *sāvana day* and length of day in two *ayanas*. This author probably belonged to Kashmir.

III. The period of *Siddhāntas* : Varāhamihira in his well known treatise, the *Pañcasiddhāntikā*, refers to five *Siddhāntas* or systems of Astronomy : Paitāmaha, Vasiṣṭha, Romaka, Pauliśa and Sūrya (Saura). As regards its importance, he gives the first place to the Sūryasiddhānta, places next the Romaka and Pauliśa, and declares the remaining two to be definitely inferior to the former. We do not possess the full treatise of these Siddhāntas, except the *Sūryasiddhānta*. Here too,

we have difficulty. The *Sūryasiddhānta*, as summarised by Varāhamihira in his *Pañcasiddhāntika*, in many essential features differs from the system prescribed by the Text of the *Sūryasiddhānta* now available. So we have two versions, the one with which Varāhmihira was familiar and the modern one.

The present *Sūryasiddhānta* comprises of fourteen chapters called *adhyāyas*. We have an authentic commetary on it by Parameśvara. The first two chapters of the *Sūryasiddhānta* have no special name. The classification of chapters is as follows :

Modern Sūrya Siddhānta

Chapter	Name of the Chapter	Number of verses	Subject
I	—	69	Mean longitude of planets.
II	—	68	True motion and true longitude of planet ; and elements of *Pañcāṅga*.
III	Tripraśnādhyāya	50	Directions, place and time.
IV	Candra-grahaṇādhyāya	29	Lunar eclipse.
V	Sūrya-grahaṇādhyāya	17	Solar eclipse.
VI	Chedyakādhyāya	24	Projection of eclipse on a plane surface.
VII	Graha-samāgama-yuddhādhyāya	24	Conjunction of one planet with another.
VIII	Tārāviṣayodhyāya	21	Conjunction of a Planet with the junction star of a nakṣatra.
IX	Udayāstamaya-viṣayodhyāya	18	Heliacal rising and setting of planets.
X	Candrāstamayādi viṣayāḥ	16	Moon rise and elevation of Moon-horn.
XI	Vyatipāta-viṣayāh	23	Pāta (*vyatipāta*)
XII	—	87	Cosmogony and geography

Chapter	Name of the Chapter	Number of verses	Subject
XIII	Vyatipāta-viṣayāh	26	Armillary sphere and astronomical instruments
XIV	—	28	Modes of reckoning time
	Total	500	

From the contents of the Modern *Sūryasiddhānta*, it appears that its great author (or rather *compiler*, since this *Siddhānta* has been existing much before he compiled this treatise) intends giving a complete but brief account of the entire Indian astronomy. He has been careful enough to avoid controversies, and he has omitted the methods occurring in other *Siddhāntas* and *Mahā-tantras* as alternative rules. He describes only those which are most important and of general nature. Part II of the book dealing with cosmogony and geography etc.. is very brief , a nd the entire subject has been summarised in three chapters (Chapters XII to XIV). The Chapter on Astronomical instruments (XIII) is sketchy and rather incomplete; the author has given the names only of the instruments, as if the purpose was to keep the details secret and guarded. The whole text of the *Sūryasiddhānta* is finished in 500 verses (compare this with the length of other Indian books on Astronomy: the *Brāhmasphuṭasiddhānta* 1008 verses, the *Siddhānta-śekhara* 890 verses and *Siddhānta-śiromaṇi* proper 962 verses).

The Modern *Sūryasiddhānta* is a composition or compilation of a period when the *Āryabhaṭiyam* and the *Brāhmasphuṭasiddhānta* had already become popular. This *Sūryasiddhānta* is indebted to both these systems, though it has nowhere acknowleged them in the Text. We shall refer to this matter later at an appropriate stage of our discussion.

It may casually be mentioned here that Alberuni has ascribed the authorship of the *Sūryasiddhānta* to Lāṭadeva while Munīśvara (603 A. D.) has ascribed it to Āryabhaṭa I. Though there is little or no support to these views, it is not improbable, that the works of Āryabhaṭa I and Lāṭadeva which followed midnight-day-reckoning were based on the *Sūryasiddhānta* prevalent at the time. P. C. Sengupta on the other hand is of the opinion that " the old *Sūryasiddhānta* was made up-to-date by Varāhamihira by replacing the old constants in it by new ones from Ārya-

bhaṭa I's midnight system. " In this connection, it must be remembered that Varāhamihira nowhere expresses his indebtedness to Āryabhaṭa I.

The *Sūryasiddhānta* undoubtedly is the most popular book on Astronomy in this country. It has been so for the last 1000 years as is seen from the list of commentators on its text (K. S. Sukla has given a list of 28 commentators includiug those who wrote commentaries in the South Indian languges like Telugu and Kannaḍa.) :

Allanarya Suri
Amaredya
Bhaṭṭotpala (966 A. D.)
Bhūdhara (1572 A. D.)
Bhūtiviṣṇu
Caṅḍeśvara (1178)
Cola
Dādābhai (1719 A. D.)
Devīdāsa
Kamala Kara Daneśvarah (1618 A. D.)
Kāmabhaṭṭa
Kṛṣṇa Daivajña
Madanapāla
Mādhavācārya
Maheśvara
Mallikārjuna Sūri (1178 A. D.)
Nārāyaṇa
Nrsinha Daivajña (b. 1586 A.)
Nṛsiṃhadeva
Parameśvara (1432 A. D.)
Rāghava Śarmā (1592 A. D.)
Ṛamakṛṣṇa Ārādhya (1472 A. D.)
Raṅganātha (1603 A. D.)
Sārvabhauma
Tamma Yajva (1599 A. D.)
Yallaya (1472 A. D.)
Viśvanātha (1628 A. D.)

A large number of astronomical books in this country were written on the basis of the *Sūryasiddhānta*, as the *Gaṇakānanda* by Sūrya (1387-1447 A. D.), *Gaṇitādarśa* by Dharmapathin, *Makaranda Sāriṇi* by Makaranda (1478 A D.), *Grahacakra* by Kucanācārya (1299 A.D.), *Viṣṇukaraṇa* by Viśnu (1556 A.D.) besides many others with indefinite dates such as the *Sūryasiddhānta nayana-prakāraḥ*, *Sūryasiddhānta-gaṇita Sūryasiddhānta-saṁgraha* by Viśvanātha Sūri, and Sūryasiddhānta Sāriṇi by Rāmadatta Daivajña. K. S. Sukla has given a list of this literature in his Introduction to the *Sūryasiddhānta*.

(iv) *Period of Bhāskara* I : The author or the compiler of the *Sūryasiddhānta* is not known nor its date of composition. Bhāskara and Brahmagupta are the brilliant names of a contemporary period. Bhāskara I lived in the seventh century of the Christian era and was a contemporary of Brahmagupta (628

A.D.). He wrote three works on Astronomy which were most likely composed in the following order : (i) the *Mahā-Bhāskarīya.* (ii) a commentary on the *Āryabhāṭiya*, and (iii) the *Laghu-Bhāskarīya*, His commentary on the *Āryabhātīya* was written in 629, i,e., only one year after the *Brāhmasphuṭasiddhānta.* His commentary on the *Āryabhatīya* was written in 629 A.D., i,e, one year after the completion of the *Brāhmasphuṭasiddhānta.* Undoubtedly Bhāskara was the follower of Āryabhaṭa I. Shukla says, that his works provide us with a detailed exposition of the astronomical methods taught by Āryabhaṭa I and throw light on the development of Astronomy in India during the sixth and early seventh centuries A. D. which was the most brilliant period in the history of Indian Astonomy-Shukla has brought a critical edition with English translation of the *Mahābhāskarīya* and the *Laghu-Bhāskarīya* and he proposes to bring out a volume on the life and works of this great astronomer,

Division of Mahābhāskarīya

The *Mahābhāskarīya* is divided into Chapters called the *adhyāyas.* The *adhyāyas* have not been named as some of the Chapters of the *Sūryasiddhānta* or the *Brāhmaspuhṭasiddhānta.* The number of verses in the *Mahābhāskarīya* is as follows :

	Subject	*Number of verses*
Adhyāya I	Mean Longitude of a Planet and Planetary Pulveriser	52
Adhyāya II	The Longitude Correction	10
Adhyāya III	Direction, Place and Time. Junction-stars of the Zodiacal Asterisms and conjunction of Planets with them.	75
Adhyāya IV	The Longitude of a Planet	64
Adhyāya V	Eclipses	78
Adhyāya VI	Rising, Setting and Conjunction of Planets	62
Adhyāya VII	Astronomical Constants	35
Adhyāya VIII	Examples	27

These titles to the subject-matter given in the above table have been assigned by Dr. Kripa Shanker Shukla in his critical

edition. The total mumber of verses in this work is 403. The Sūryasiddhānta has, as stated above, 500 verses in all.

Laghu-bhāskarīya

The *Laghu-Bhāskarīya* is also divided into eight chapters, each chapter is known as *Adhyāya.* The subject matter in the book has been dealt with as follows :

Subject			*Number of verses*
Adhyāya	I	Mean longitudes of the Planets	37
Adhyāya	II	True Longitudes of the Planets	41
Adhyāya	III	Direction, Place and Time from Shadow	35
Adhyāya	IV	The Lunar Eclipse	32
Ādhyāya	V	The Solar Eclipse	15
Adhyāya	VI	Visibility, Phases and Rising and Setting of the Moon	25
Adhyāya	VII	Visibility and Conjunction of the Planets	10
Ādhyāya	VIII	Conjunction of a Planet and a Star	19

The total number of verses in this text is 214. The *Laghu-Bhāskarīyam* is thus an abridged edition of the *Mahā-Bhāskarīyam.* From the closing stanza of this work, it is clear that the author wrote this work for the benefit of young students with immature mind by condensing and simplifying the contents of his larger work, the *Mahā-Bhāsarīyam* (also known as *Karma-nibandha).* Thus we have a little of the parallelism: Brahmagupta, after finishing his bigger treatise, the *Brāhma-sphuṭasiddhānta,* wrote a minor abridged work, the *Khaṇḍa-khādyaka,* as a *karaṇa grantha.* This latter work of Brahmagupta, however, incorporates some original ideas not included in the earlier work. Shukla has given an analytical table indicating the rules of the *Mahā-Bhāskarīyam* incorporated in the *Laghu-Bhāskarīyam* also in an abridged or modified form, and also a list of the rules which have been omitted in the *Laghu-Bhāsakrīya.* There are a few rules in the *Laghu-Bhāskarīya* also which have no counterpart in the

Mahā-Bhāskarīya. Shukla further says that the arrangement of the contents of the *Laghu-Bhāskarīya* is more systematic and logical than that of the *Mahā-Bhāskarīya*, and is, at the same time, in keeping with the general practice followed by the other Hindu astronomers. Numerous quotations of this work occur in the annotative works of Sūryadeva (b. 1191 A.D.), Yallaya (1480 A.D.), Nīlakaṇṭha (1500 A.D.), Raghunātha Rāja (1597 A.D.), Govinda Somayājī and Viṣṇu Śarmā and in the *Prayoga-racanā*, an anonymous commentary on the *Mahā-Bhāskarīyam*. We find the commentaries of this abridged work in Malayalam and Tamil also. All this speaks of the great popularity of this work.

There are circumstantial evidences to show that Bhāskara I had associations with the countries of Aśmaka and Surāṣṭra. His commentary on the *Āryabhaṭīya* was probably written in the city of Valabhī in Surāṣṭra. Perhaps Bhāskara I was born and educated in Aśmaka and later on he migrated to Valabhī, where he wrote his commentary on the *Āryabhaṭīya* or that he was a native of Valabhī and got his education in the Aśmaka country. Perhaps there was a strong school of Astronomy in the Aśmaka country, which was founded by the followers of Āryabhaṭa, so much so that at places, Bhāskara I has also called Āryabhaṭa as Āśmaka, his *Āryabhaṭīyam* by the name *Aśmaka-Tantra* or the *Āśmakīya* and the followers of Āryabhaṭa as Āśmakīyāḥ. This Aśmaka country or Aśmaka Janapada is mentioned in the Buddhist literature also. It was somewhere either in the north-west of India, or was situated between the rivers Narmadā and Godāvarī. Bhāskara I was evidently a resident of the latter Aśmaka (which was between the Narmadā and Godāvarī).

Brahamagupta and Bhāskara I were contemporaries. Both of them developed their systems in the earlier part of the seventh century A.D. (3700 years of *Kaliyuga*). *Brahāmasphuṭasiddhānta* was written in 628 A.D. and the commentary on the Āryabhaṭīyam by Bhāskara I was composed in 629 A. D. Bhāskara closely followed Āryabhaṭa, but Brahmagupta had the guts to oppose the views expressed by this great master and he not only contradicted him at places, but also propounded many new ideas, methods of calculation and constants of greater accuracy.

The classification of the contents of Astronomy in *adhikāras* appears to be the original concept of *Brahmagupta* ; this system was to some extent adopted in the modern *Sūrya-Siddhānta* in the case of a few chapters. The *Vaṭeśvara-Siddhānta* by Vaṭeśvarācārya (born 802 Śaka or 880 A.D. in Ānandapura city, Punjab, son of Mahādatta) also adopts the terminology: *Madhyamādhikāra, Spaṣṭādhikāra* and *Tripraśnādhikāra.* We owe this type of caption-nomenclature to Brahmagupta.

Contents of the Brahmasphuṭasiddhānta

Now we shall summarise the contents of the Chapters of this great treatise and also enumerate the number of verses in each chapter. The author has himself given the total number of verses in the Chapter in the ending verse of each chapter. Sometimes, the verse specifying this number itself is not taken into account while giving the total number of verses in that chapter, and therefore, there occurs a minor discrepancy in the actual number and the number specified by the author himself. The following table records both these numbers separately :

Chapter	Title	Number of verses indicated by the author	Actual number of verses
PŪRVA-DAŚĀDHYĀYI (First Ten Chapters)			
I	Madhyāmadhikāraḥ (Madhyamagati-radhyāyaḥ)	63	63
II	Spaṣṭādhikāraḥ (Sphuṭagati-radhyāyaḥ)	67	68
III	Tripraśnādhikāraḥ (Tripraśnādhyāyaḥ)	66	66
IV	Candragrahaṇādhikāraḥ (Candragrahaṇādhyāyaḥ)	20	20
V	Sūryagrahaṇādhikāraḥ (Ārka-grahaṇam or Ravigrahaṇādhyāyaḥ)	26	27
VI	Udayāstādhikāraḥ (Udayāstamayādhyāyaḥ)	12	13

VII	Candraśṛṅgonnatyadhikāraḥ (Candraṣṛṅgonnati-adhyāyaḥ)	18	18
VIII	Candracchāyādhikāraḥ (Candracchāyā-adhyāyaḥ)	9	9
IX	Grahayutyadhikāraḥ (Grahamelanādhyāyaḥ)	26	26
X	Bhagrahayutyadhikāraḥ (Bhagrahayutiḥ-adhyāyaḥ)	70	70
	Total (of Daśādhyāyī)	377*	380*
XI	Tantra-Parīkṣādhyāyaḥ (Dūṣaṇādhyāyaḥ)	63	63
XII	Gaṇitādhyāyaḥ		
	Miśraka-vyavahāraḥ 1—16		
	Śreḍhī-vyavahāraḥ 17—20		
	Kṣetra-vyavahāraḥ 21—39		
	Vṛttakṣetra-gaṇitam 40—43		
	Khāta-vyavahāraḥ 44—46		
	Citi-vyavahāraḥ 47—		
	Kākacika-vyavahāre karaṇasūtre 48—49		
	Rāśi-vyavahāraḥ 50—51		
	Chāyā-vyavahāraḥ 52—66	66	66
XIII	Praśnādhyāyaḥ (Madhyagatyuttarādhyāyaḥ)	49	48
XIV	Sphuṭagatyuttarādhyāyaḥ	54	55
XV	Tripraśnottarādhyāyaḥ	60	60
XVI	Grahaṇottarādhyāyaḥ	46	47
XVII	Śṛṅgonnatyuttarādhyāyaḥ	10	10
XVIII	Kuṭṭādhyāyaḥ (Kuṭṭakādhyāyaḥ)	103	102
	Kuṭṭākāraḥ 1—29		
	Dhanarṇa-śūnyānām saṃkalanam 30—42		
	Ekavarṇa.samīkaraṇa-bījam 42—50		
	Anekavarṇa-samīkaraṇa-		

	bījam	51—59		
	Bhāvita-bījam	60—63		
	Varga-prakṛtiḥ	64—74		
	Udāharaṇāni	75—102		
XIX	Śaṅkucchāyādi-ñanādhyāyaḥ		20	20
XX	Chandaścityuttarādhyāyaḥ		20	19
XXI	Golādhyāyaḥ		70	70
	Sāmānya-golaprakaraṇam	1—16		
	Jyā-prakaraṇam	17—23		
	Sphuṭagati-vāsanā	24—30		
	Bimba-sādhanam	31—35		
	Grahaṇa-vāsanā	36—48		
	Golabandhādhikāraḥ	49—70		
XXII	Yantrādhyāyaḥ		53	57
XXIII	Mānādhyāyah		12	12
XXIV	Sañjnādhyāyaḥ		13	13
		Total	1016	1022

In one of the verses, Brahmagupta states that he has composed the treatise containing 1008 verses. Sudhākara Dvivedī has given the total as 1021, whereas he says, this number according to Brahmagupta's own statement should be 1020. If one deducts the concluding 12 verses of the *Sañjñanādhyāya*, the number should be 1008.

Sudhākara Dvivedī, in his addition of the *Brāhma-sphuṭa-siddhānta* (published in the *Paṇḍita*, 1901 and 1902) gives as a supplement a small treatise of Brahmagupta known as *Dhyānagrahaṇopadeśādhyāya* or *Dhyānagrahaṇādhikāraḥ* which has 72 verses.

It would be worthwhile to give here the details of the *Khaṇḍakhādyaka* also, a book of Brahmagupta about which we have spoken so much. The titles to the chapters have not been indicated in the Text ; most likely they have been assigned by the commentator, Pṛthūdaka Svāmī known as the *Khaṇḍakhādyaka-vivaraṇam*.

Chapter	Title	Number of verses
I	Tithi-nakṣatrādhikārādhyāyaḥ (On tithīs, nakṣatras etc.)	32
II	Grahagatyadhyāyaḥ (On the mean and true places of 'star' planets)	19
III	Tripraśnādhyāyaḥ (On the three problems relating to diurnal motion)	16
IV	Candragrahaṇādhyāyaḥ (On lunar eclipses)	7
V	Sūryagrahaṇādhyāyaḥ (On solar eclipses)	6
VI	Udayāstādhikāraḥ (On the rising and setting of planets)	7
VII	Candrasṛṅgonnatyadhāyaḥ (On the position of the Moon's cusps)	4
VIII	Samāgamādhyāyaḥ (On conjunction of planets)	6
	UTTARA KHAṆḌAKHĀDYAKA—APPENDIX	
IX	Corrections and new methods	14
X	On conjunction of stars and planets	16
	Total	127

Bhaṭṭotpala, in his commentary on the *Khaṇḍakhādyaka* has given several additional verses in the main or proper treatise and also in its Uttara portion or the Appendix. P.C. Sengupta's edition (Sanskrit Text 1941) has given at the end of this publication the account of these additional verses. The English

edition (1934) classifies the *Uttara-Khaṇḍakhaḍyaka* into two chapters (which the author calls as Chapters IX and X), the Sanskrit edition gives 3 verses in Chapter IX, 21 verses in Chapter X and 24½ verses in Chapter XI. Of these three chapters, the Chapter X has been given the title "*Patādhikāra*" and Chapter XI the title "*Parilekhādhyāya*" by Bhaṭṭotpala.

TABLE

Arrangement of contents in different treatises

Topic	*BrSpSi*	*SūSi*	*MBh*	*MSi*	*SiŚe*	*SiŚi*
Mean longitudes of the planets	I,XIII	I	I	I.II	I,II	I
True longitudes of the planets	II,XIV	II	IV	III	III	II
Direction, place and time	III,XV	III	III	IV	IV	III
Computation of a lunar eclipse	IV,XVI	IV	V	V	V,VII	IV,V
Computation of a solar eclipse	V,XVI	V	V	VI	VI	VI
Projection of an eclipse	XVI	VI	V	VIII		V
Conjunction of a planet with another planet	IX	VII	VI	XI	XI	X
Conjunction of a planet with a star	X	VIII		XII	XII	XI
Heliacal rising of planets	VI	IX	VI	IX,X	IX	VII, VIII

Topic	*BrŚpSi*	*SūSi*	*MBh*	*MSi*	*SiŚe*	*SiŚi*
Moonrise and elevation of lunar horns	VII,XVII	X	VI	VII, VIII	X	IX
Pāta	XIV	XI	VII	XIII	VIII	XII
Cosmogony and geography	XXII	XII		XVI		II.iii
Astrono. instruments	XXIII	XIII	III		XIX	II.xi
Time reckoning	XXIV	XIV				I.i

Āryabhaṭa and Brahmagupta Controversy

The scientific Indian astronomy was more or less created by Āryabhaṭa I (476 A. D.). It is said that he was the teacher of two distinct systems of astronomy, one of which is called the *audayika* system, and the other the *ardharātrika* system. In the first, the astronomical day is taken to begin at *sunrise* at Laṅkā, and in the other, the same begins at the *midnight* of the same place. In the *Khaṇḍakhādyaka* Brahmagupta gives compendious rules for the calculation of longitudes, etc, of planets according to the *ardharātrika* system of Āryabhaṭa I. In this connection, he refers to Āryabhaṭa in the following words in his *Khaṇḍakhādyaka :*

> Having made obeisance to God Mahādeva, who is the great cause of this world's rise (i. e. creation), existence, and destruction, I shall declare the *Khaṇḍakhādyaka* which will yield the same results as the great astronomical treatise of Āryabhaṭa.[1]
>
> As in most cases calculation by the great work of Āryabhaṭa, for (the knowledge of time and longitude of planets etc. at) marriage, nativity and the like is impracticable for common use every day, this smaller

1. प्रणिपत्य महादेवं जगदुत्पत्तिस्थितिप्रलयहेतुम् ।
वक्ष्यामि खण्डखाद्यकमाचार्य्यार्य्यभटतुल्यफलम् ॥

treatise (i.e. *Khaṇḍakhādyaka*, literally meaning food prepared from sugar-candy) is made so as to yield the same results as that.[1]

The mean saturn diminished by 3 seconds, the *Śighrocca* of Mercury diminished by 22 seconds, the mean Mars increased by 2 seconds and the mean Jupiter increased by 4 seconds are equal to the respective mean planets of Āryabhaṭa's *midnight* system.[2]

In the *Brāhmasphuṭasiddhānta*, Brahmagupta accepts the astronomical day to begin with the sunrise at Laṅkā, and the calculations of days, months, years, Yugas, and Kalpas all begin from Caitra Śukla Pratipadā (the first *tithi* of the month Caitra in the bright-half of the Moon) and the first day is regarded as Sunday.[3]

Varāhamihira in his epicyclic cast to the *Sūryasiddhānta* in his *Pañcasiddhāntikā* adopts the *ardharātrika* system or the system of reckoning days from midnight. The question why Brahmagupta who was so bitter an opponent of Āryabhaṭa I in his younger days (628 A.D.) claimbed down to describe and teach one of the systems of Āryabhaṭa's astronomy in his sixty-seventh year (665 A.D.) is difficult to explain. In fact so great was Āryabhaṭa's reputation and fame that in spite of Brahmagupta's severe ricticisms of the former in Chapter XI of the *Brāhmasphuṭasiddhānta*, it perhaps was undiminished and it was Āryabhaṭa who continued to be universally followed.

Some authorities have thus expressed the view that to meet the popular demand Brahmagupta in the *Khaṇḍakhādyaka* took upon himself the task of simplifying Āryabhaṭa's *ardharātrika* system and in this task he became eminently successful. But it has been supposed that in this task he could not be a mere simplifier or expounder.

प्रायेणार्य्यभटेन व्यवहारः प्रतिदिनं यतोऽशक्यः ।
उद्ग्राहजातकादिषु तत्समफल लघुतरोक्ति रतः ॥
त्रिसृभिः शनिर्द्वशीघ्रं द्वाविंशत्या कुजोऽधिको द्वाभ्याम् ।
चतसृभिरधिको जीवोऽर्द्धरात्रिकार्य्यभट मध्य समाः ॥

—*KK*. I. 1,2,7

1. वैत्रसितादेरुदयाद्भानोर्दिनमासवर्षयुगकल्पाः ।
सृष्ट्यादौ लंकायां सम प्रवृत्तः दिनेऽर्कस्य ॥

—*BrSpSi I.4*

The minor work of Brahmagupta known as *Khaṇḍakhādyaka* has two distinct parts : *Khaṇḍakhādyaka* proper and the *Uttara Khaṇḍakhādyaka*. In the first part the astronomical constants are the same as those of Āryabhaṭa's *ardharātrika* system, but the methods of spherical astronomy, calculation of eclipses and other topics are almost the same as in tbe *Brāhmasphuṭasiddhānta*. The corrections for parallax in calculating a solar eclipse is here an important illustration.[1]

In the *Uttarakhaṇḍakhādyaka*, Brahmagupta gives corrections to the *Khaṇḍakhādyaka* proper. In it are to be found the neat and original methods of interpolation and correction to the longitudes of the aphelia, as also to the dimensions to the epicycles of apsis of the Sun and the Moon.[2] while a few additional chapters supply what else is necessary to the first seven chapters of the first part, to make the whole a complete treatise on Indian scientific astronomy. It was perhaps through the influence of this supplementary part of the *Khaṇḍakhādyaka* that Brahmagupta's great work, the *Brāhmasphuṭasiddhānta* came to be valued among a distinct school of Indian astronomers. For long in this country India, this *Siddhānta* of Brahmagupta has been forming the basis for the calculation of almanacs by astronomers of the orthodox school of Rājasthān, Bombay and others.

We might at this stage take up the question : Was Āryabhaṭa the author of two distinct systems of astronomy ? Undoubtedly he was. Several authors have written on this subject. I may specially mention the name of Prabodha Chandra Sengupta (*Journal of the Department of Letters*, Calcutta University, vol. XVIII; *Bulletin, Calcutta Mathematical Society*, vol. XXII. Nos. 2 and 3). The reasons advanced by him may be restated in slight details thus. In his *Brāhma-*

1. व्यासार्धेन विभक्ता दृग्गतिजीवा चतुर्गुणा लब्धम् ।
लम्बननाड्यः पञ्चदश गुणितया त्रिज्यया भक्ता ॥
दृक्क्षेपज्या भुक्त्यन्तरा हता लब्धमवनतिर्भवति ।
स्फुटयोजनकर्णाभ्यां भू-व्यासेन च विना स्पष्टे ॥
आर्यभटेनास्मिन् सति लघुनि किमर्थं महत् कृतं कर्म ।
गणिताज्ञानाज् जाड्यं विज्ञानता यदि ततः सुतराम् ॥

—*BrSpSi.* XI. 23-25, also *KK.* V.

2. See UKK. 9.

sphuṭasiddhānta, Brahmagupta thus speaks of the two works of Āryabhaṭa:

> As in both the works the number of the Sun's revolutions is spoken of as 432,000 years, their planetary cycle is clear, i. e., of 4,320,000 years. Why then is there difference of 300 civil days in the same cycle of the two books ?[1]

Again, he says :

> In 14,400 years elapsed of the *Mahāyuga*, there is produced a difference of one day in the *audayika* and *ardharātrika* systems.[2]

Varāhamihira in the *Pañcasiddhāntikā* writes :

> Āryabhaṭa maintains that the beginning of the day is to be reckoned from midnight at Laṅkā; and the *same* teacher again says that the day begins from sunrise at Laṅkā. [3]

Thus from the writings of Brahmagupta and Varāhamihira, it is clear that Āryabhaṭa I was the author of both the *audayika* and *ardharātrika* systems of astronomy. In Varāhamihira's verse the phrase *sa eva* (स एव), meaning "he undoubtedly" is of speeial significance. It removes the least doubt as to Āryabhaṭa's authorship of both these systems. The *audayika* and *ardharātrika* astronomical constants are respectively to be found from the Āryabhatīya and may be deduced from the *Khaṇḍakhādyaka* as well. The following is the comparative view of the constants of Varāhamihira and of the present day *Sūryasiddhānta*.

TABLE I

Planetary revolutions in a mahāyuga of 4,320,000 years, according to various authorities.

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1. युगरविभगणाः रव्युद्धृति क्त् प्रोक्तं तत् तवोयुगं स्पष्टम् ।
त्रिशती रव्युदयानां तदन्तरं हेतुना केन ॥
—*BrSpSi XI. 5*

2. अधिकैः शतैश्चतुर्भिर्वर्षं सङ्ख्यैश्चतुर्दशभिरेकः ।
युगपातैर्दिनवारान्तर मौदयिकार्धं रात्रिकयोः ॥
—*BrSpSi XI. 13*

3. लङ्कार्द्धरात्रसमये दिनप्रवृत्तिं जगाद चार्यभटः ।
भूयः स एव सूर्योदयात् प्रभृत्याह लङ्कायाम् ॥
—*PSi. XV. 20*

Planet	*BrSpSi*	*Āryabhaṭīya*	*Khaṇḍa-khādyaka*	Varāha-*Sūrya-siddhānta* (PSi.)	Later or modern *Sūrya-siddhānta*
Moon	57,753,300	57,753,336	57,753,336	57,753,336	57,753,336
Sun	4,320,000	4,320,000	4,320,000	4,320,000	3,320,000
Mars	2,296,828.522	2,296,824	2,296,824	2,296,824	2,296,832
Jupiter	364,226.455	364,224	364,220	364,220	364,220
Saturn	146,567.298	146,564	146,564	146,564	145,568
Moon's apogee	—	488,219	488,219	488,219	488,203
Venus	7,022,389.492	7,022,338	7,022,388	7,022,388	7,022,376
Mercury	17,936,998.984	17,937,020	17937,000	17,937,000	17,937.060
Moon's nodes	232,311.168	232,226	232,226	232,226	232,238

TABLE II

Longitudes of the apogees of the orbits of Planets.

Planets	*Āryabhaṭīya*	*Khaṇḍa-khādyaka*	Varāha *Sūrya-siddhānta.*	Modern *Surya-siddhānta*
Sun	78°	80°	80°	77° 07′
Mercury	210°	220°	220°	220° 26′
Venus	90°	80°	80°	79° 49′
Mars	118°	110°	110°	130° 00′
Jupiter	180°	160°	160°	171° 16′
Saturn	236°	240°	240°	236° 37′

TABLE III

Dimensions of the epicycles of Apsis

Planets	*Ārybhaṭīya*	*Khaṇḍa-khādyaka*	Varāha. *Sūrya-siddhānta*	Modern *Sūrya-siddhānta*
Sun	13°-30′	14°	14°	$13\frac{2}{3}$°-14°
Moon	31°-30′	31°	31°	$31\frac{2}{3}$°-32°
Mercury	$22\frac{1}{2}$°-$31\frac{1}{3}$°	28°	28°	28°-30°
Venus	9°-18′°	14°	14°	11°-12°
Mars	63°-81°	70°	70°	72°-75°
Jupiter	$31\frac{1}{3}$-$36\frac{1}{2}$°	32°	32°	32°-33°
Saturn	$40\frac{1}{2}$-$58\frac{1}{3}$	60°	60°	48°-49°

Table IV

Dimensions of the Śīghra epicycles (i.e. conjunctions)

Planet	*Āryabhaṭīya*	*Khaṇḍa-Khādyaka*	Varāha *Sūrya-siddhānta*	Modern *Sūrya-siddhānta*
Saturn	$36\frac{1}{2}°$- 40°	40°	40°	39°- 40°
Jupiter	$67\frac{1}{2}°$- 72°	72°	72°	70°- 72°
Mars	$229\frac{1}{2}°$-$239\frac{1}{2}°$	234°	234°	232°-235°
Venus	$256\frac{1}{2}°$-$265\frac{1}{2}°$	260°	260°	262°-262°
Mercury	$130\frac{1}{2}°$-$139\frac{1}{2}°$	132°	132°	132°-133°

Table V

Longitudes of the nodes of the orbits of planets

Planets	*Āryabhāṭīya*	*Khaṇḍa-khādyaka siddhānta*	Varāha *Sūrya-siddhānta*	Modern *Sūrya-siddhānta*
Saturn	40°	40°	Not stated	Have to be
Jupiter	20°	20°	in the	calculated
Mars	80°	80°	Text	from the
Venus	60°	60°		data of the
Mercury	100°	100°		text

Table VI

Orbital inclinations (geocentric) to the ecliptic

Planets	*Āryabhaṭīya*	*Khaṇḍa-khādyaka*	Varāha *Sūrya-siddhānta*	Modern *Sūrya-siddhānta*
Mars	90′	90′	10′	90′
Mercury	120′	120′	135′	120′
Jupiter	60′	60′	101′	60′
Venus	120′	120′	101′	120′
Saturn	120′	120′	135′	100′

The *Mahābhāskarīya* of Bhāskara I (522 A. D.) contains a passage which corroborates the fact that Āryabhaṭa I was the author of both the *audayika* and the *ardharātrika* systems of Indian Astronomy. According to Pṛthūdakasvāmin, whose

commentary on the *Brāhmasphuṭasiddhānta* we have the privilege of presenting to the public, it is clear that in certain respects Bhāskara and others may be wrong but the Āryabhaṭa's authenticity cannot be questioned. Pṛthūdakasvāmin while commenting on the *Brāhmasphuṭasiddhānta*, XI. 26 writes:

> Such a mistake may have been made by Bhāskara and others; they have not understood his (Āryabhaṭa's) intention.

The passage in the *Mahābhāskarīya* giving constants of the *ardharātrika* system runs as follows (we are giving the translation from Kripa Shankar Shukla's edition on the *Mahābhāskarīya*) :

> The astronomical processes which have been set forth above come under the sunrise day - reckoning (*audayika* system). In the midnight day-reckoning (*ardharātrika* system) too, all this is found to occur: the difference that exists is being stated (below). [1]

The next fourteen stanzas relate to the midnight day-reckoning of Āryabhaṭa I.

(i) Civil days and omitted lunar days in a *yuga* and revolution numbers of Mercury and Jupiter are thus given :

(To get the corresponding elements of the midnight day-reckoning) add 300 to the number of civil days (in a *yuga*) and subtract the same (number) from the number of omitted lunar days (in a *yuga*); and from the revolution numbers of (the *śighrocca* of) Mercury and Jupiter subtract 20 and 4 respectively, [2]

Thus according to the midnight day-reckoning, we get

civil days in a *yuga*	=	1,577,917,800
omitted lunar days in a *yuga*	=	25,082,280
revolution number of the *śighrocca* of Mercury	=	17,937,000
revolution number of Jupiter	=	364,220

---

1. निबन्धः कर्मणां प्रोक्तो योऽसावौदयिको विधिः ।
अर्धरात्रे त्वयं सर्वो यो विशेषः स कथ्यते ॥ —*MBh. VII. 21*

2. त्रिंशती भूदिने क्षेप्या ह्ययमेभ्यो विशोध्यते ।
ह गुर्वोर्भगणेभ्योऽपि विंशतिश्च ततोऽब्धयः ॥ —*MBh. VII. 22*

(ii) Diameters of the Earth, the Sun and the Moon are thus given :

(In the midnight day-reckoning) the diameter of the Earth is (stated to be) 1,600 *yojanas;* of the Sun 6,480 (*yojanas*) and of the Moon, 480 (*yojanas*)[1].

(iii) Mean distances of the Sun and the Moon are as follows :

The (mean) distance of the Sun is stated to be 689,358 (*yojanas*), and of the Moon 51,566 (*yojanas*).[2]

(iv) Longitudes of the apogees of the planets are as follows: 160, 80, 240, 110, and 220 are in degrees the longitudes of the apogees of Jupiter, Venus, Saturn, Mars and Mercury respectively.[3]

(v) *Manda* and *Śighra* epicycles of the planets are as follows :

The *Manda* epicycles (of the same planets) are 32, 14, 60, 70, and 28 (degrees) respectively; and the *Śighra* epicycles are 72, 260, 40, 234, and 132 (degrees) respectively. The Sun's apogee and epicycle are the same as those of Venus (i. e. 80° and 14° respectively).

The Moon's epicycle in the midnight day-reckoning is stated to be 31 (degrees). [4]

(vi) The positions of the so called *manda* and *śighra pātas* of the planets are given below:

(The following directions for) the degrees of the (*manda* and *śighra*) *pātas* of the planets as devised

---

1. अष्टिश्शतगुणा व्यासो योजनानां भुवो रवेः ।
खाष्टाब्ध्यङ्कानि शीतांशोः शून्यवस्वब्धयस्तथा ॥ —23

2. वस्विन्द्रिय गुणच्छिद्रवस्वङ्कानि विभास्वोः ।
अङ्काङ्गोष्वेक भूतानि चन्द्रकर्णः प्रकीर्तितः ॥ —24

3. अष्टिरष्टौ जिनारुद्रा विंशतिर्द्व्यधिकाः क्रमात् ।
दशघ्ना गुरुशुक्रार्किं भौमज्ञांशाः स्वमन्दजाः ॥ —25

4. मन्दवृत्तानि द्वात्रिंशन्मनवः षष्टिरेव च ।
खाद्रयो वसुदस्राः स्युः शीघ्रवृत्तान्यथ क्रमात् ॥ —26
द्व्यद्रयः खाङ्गनेत्राणि खाब्धयोऽब्ध्यग्निदस्रकाः ।
द्व्यग्नीन्दवो रवेर्मन्दं शुक्रवद् वृत्तमेव च ॥ —27
एकत्रिंशत्तपाभर्तुर्ध्रुवरात्रे विधीयते । —*MBh. VII 26-28 (i)*

(under the midnight day-reckoning) should be noted carefully by learned scholars :

Add 180° to the longitudes of the *mandoccas* (apogees) apd *śighroccas* of Mercury and Venus, and subtract 3 signs from the *mandoccas* (apogees) and *śighroccas* of the remaining planets. Then are obtained the longitudes of the *manda* and *śighra pātas* of the planets. (Also) add 2 degrees to the longitudes of the *manda pātas* and *śighroccas* of Venus, Saturn and Jupiter, and $1\frac{1}{2}$ degrees to those of Mars and Mercury. (It should be noted that) the *śighra pātas* have been stated for all the planets excepting Mercury. (Mercury does not have a *śighra pāta*).[1]

That is to say, the longitudes of the *manda pātas* of Mars, Mercury, Jupiter, Venus, and Saturn are 21.5, 41.5, 72, 262 and 152 degrees respectively; and the longitudes of the *śighra pātas* of Mars, Jupiter, Venus and Saturn are (*śighrocca*−88.5°), (*śighrocca* −88°), (*śighrocca*+182°) and (*sighrocca*−88°) respectively.

(vii) A rule for finding the celestial latitude of a planet is as follows :

(From the longitude of a planet severally) subtract the longitudes of its (*manda* and *śighra*) *pātas* and therefrom calculate (as usual) the corresponding celestial latitude of that planet. Add them or take their difference according as they are of like or unlike directions. Then is obtained the true celestial latitude of that particular planet. The true celestial latitude of any other planet is also obtained in the same way. The remaining (astronomical) determinations are the same as stated before. This all is in brief the difference of the other *tantra* (embodying the midnight day-reckon-

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1. पातभागाश्च विज्ञेयाः पण्डितैः परिकल्पिताः ॥ —28
मन्द शीघ्रोच्चयोः क्षेप्यं चक्रार्धं बुधशुक्रयोः ।
राशित्रयं तु शेषाणां पात्यते पातसिद्धये ॥ —29
शुक्रार्किदेव पूज्यानां भागौ द्वावेव संयुतौ ।
मन्दपाताच्च शीघ्रोच्चात् सार्धाशस्तु कुजज्ञयोः ॥ —30
विबुधानां च सर्वेषां शीघ्रपाताः प्रकीर्तिताः । *MBh. VII. 28 (ii) 31 (i)*

ing of Āryabhaṭa I).[1]

(viii) A rule for finding the longitude of the true mean planet according to midnight-day reckoning is as follows :

Apply half the *Śighraphala* and (then) half the *mandaphala* to the longitude of the planet's own *mandocca* (reversely).

From the resulting longitude of the planet's *mandocca* calculate the (*mandaphala* and apply it to mean longitude of the planet : the resulting longitude of planet is stated to be) the true-mean longitude of the planet. This is stated to be another difference (of the midnight day-reckoning) [2]

(ix) Length of the circle of the sky and derivation of the lengths of the orbits of the planets are given as follows :

Multiply the revolutions of the Moon (in a *yuga*) by 3, 240,000 and then discard the zero in the unit's place : (this is the length of the circle of the sky in terms of *yojanas*). (Severally) divide that by the revolutions of the planets (in a *yuga*): thus are obtained the lengths of the orbits of the respective planets in terms of *yojanas*.[3]

From these stanzas (from 20-35), it is evident that one *yojana* of the sunrise day-reckoning is one and a half times that of the midnight day-reckoning.

Now from stanza 22, it appears that 300 is to be added to the number of civil days in a *Mahāyuga*. According to the *Āryabhaṭīya*. the number of civil days in this cycle is 1,577,917, 500, which increased by 300, becomes 1,577,917,800, the number of

1. शोधयित्वा क्रमात् पातान् विक्षेपांशान् प्रसाधयेत् । —*31*
योगविश्लेषनिष्पत्तिरेकानेकस्वदिग्वशात् ।
विक्षेपः स स्फुटो ज्ञेयो ग्रहस्यैकस्य कीर्त्तितः ॥ —*32*
अन्यस्याप्येवमेव स्याच्छेषाः प्रागुक्त कल्पनाः ।
एतत्सर्वं समासेन तन्त्रान्तरमुदाहृतम् ॥ —*MBh. VII. 31 (ii)-33*

2. शीघ्रमन्दोच्चचापार्धसंस्कृतात् स्वीयमन्दतः ।
स्फुटमध्यग्रहाः स्युर्वै विशेषः परिकीर्तितः ॥ —*MBh. VII. 34*

3. वेदाश्विरामगुणितान्य युताहतानि ।
चन्द्रस्य शून्यरहितान्यथ मण्डलानि ॥
रवेः स्वैर्हृतानि भगणैः क्रमशो ग्रहाणां ।
कक्ष्या भवन्ति खलु योजनसंख्यया ॥ —*MBh. VII. 35*

civil days in a *Mahāyuga* according to Brahmagupta as referred to in the *Khaṇḍakhādyaka.*

Again, the same stanza tells us to subtract 20 and 4 respectively from the revolutions of Mercury and Jupiter, and we arrive at the figures 17,937,000 and 364, 220, which are the revolutions of Mercury and Jupiter in a *Mahāyuga* according to Brahmagupta as given in the *Khaṇḍakhādyaka.*

Again we can compare the figures for the diameters of the Earth, the Sun and the Moon given in the *Mahābhāskarīya,* in the present day *Sūrya-siddhānta* and the *Āryabhaṭīya.*

Diameter of	*Mahābhāskarīya*	Modern *Sūrya-siddhānta*	*Āryabhaṭīya*
Earth	1,600 yojanas	1,600 yojanas	1,050, yojanas
Sun	6,480	6,500	4,410
Moon	480	480	315

Then in stanza 24, we are given the distances of the Sun and the Moon as 689, 358 and 51, 566 yojanas respectively. The same figures are worked out by Lalla according to the *Āryabhaṭīya* and quoted in the *Śiṣyadhīvṛddhida* (IV.3.4) and they come to be 459,585 and 34,377.

The stanza 25 states the longitudes of the aphelia of planets these figures tally with the corresponding figures given by Brahmagupta in the *Khaṇḍakhādyaka* :

> Longitude of aphelion of Jupiter 160°, of Venus 80°, of Saturn 240°, of Mars 110° and of Mercury 220°.

Similarly the stanza 26 gives the peripheries of planets' epicycle of apsis, which also is in concordance with the values given by Brahmagupta :

> Periphery of epicycle of apsis of Jupiter 32°, of Venus 14°, of Saturn 60°, of Mars 70° and of Merury 28°.

In the stanza 27 of the *Mahābhāskarīya,* we have the dimensions of the epicycles of conjunction for planets; these figures are also the same as given by Brahmagupta :

> Epicycles of conjuction for Jupiter 72°, for Venus 260°, for Saturn 40°, for Mars 234° and for Mercury 132°.

In the stanza 28, we have the Sun's epicycle having a periphery of 14° and the Moon's epicycle 31° ; the longitudes of

the nodes of the planets to be the same as in the *Āryabhaṭīya*. All these are the same as given by Brahmagupta in the *Khaṇḍakhādyaka*.

In the stanza 33 we have rules for finding the geocentric longitudes of planets which may be taken to be the same as in the *Khaṇḍakhādyaka*[1]; compare these values with those in the *Sūryasiddhānta* of Varāhamihira in the Pañcasiddhāntikā, XVII. 6, but slight different from the *Āryabhaṭīya*.[2]

The last stanza of the *Mahābhāskarīya* (35) gives the dimensions in *yojanas* of the orbits of planets ; these are the same as in the modern *Sūryasiddhānta*.

Thus we find a great semblance in the constants as given by the *Mahābhāskarīya* of Bhāskara I, of the Sūryasiddhānta as given by the *Pañcasiddhāntikā* and understood by Varāhamihira, and also the constants as given by Brahmagupta in his treatises, specially the *Khaṇḍakhādyaka*. It must not be forgotten that the same Āryabhaṭa I who is the celeberated author of the *Āryabhaṭīya* is also the author of another treatise very often referred to as the *Tantra*.

I shall quote Prabodha Chandra Sengupta in connection with these similarities, and the great influence of Āryabhaṭa on Indian Astronomy. He writes in his Introduction to the *Khaṇḍakhādyaka* as follows :

> We have shown that there is much resemblance in the constants between the *Sūryasiddhānta* of Varāha and the *Khaṇḍakhādyaka* and for the matter of that with the *Tantrāntara* of Āryabhaṭa I. In my papers "Āryabhaṭa and Āryabhaṭa's Lost Work", I have established the fact that the *Sūryasiddhānta* as it existed before the time of Varāha, was made more accurate by him by borrowing the constants from Āryabhaṭa's *ardharātrika* system. That there was a *Sūryasiddhānta*

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१. शीघ्रफलार्द्धं मध्ये मन्दफलार्द्धं च मन्दशीघ्रफले ।
सकले मध्ये स्पष्टः शीघ्रं मध्योनकं केन्द्रम् ॥ —*KK. II. 18*

२. मन्दोच्चाच्छीघ्रोच्चादर्धमृणधनं ग्रहेषु मन्देषु ।
मन्दोच्चात्स्फुट मध्याश्शीघ्रोच्चाच्च स्फुटा ज्ञेयाः ॥
शीघ्रोच्चादर्धोनं कर्तव्यमृणं धनं स्वमन्दोच्चे ।
स्फुटमध्यौ तु भृगुबुधौ सिद्धान्मन्दात्स्फुटौ भवतः ॥ —*Arya. III. 23-24*

before the time of Varāha, is seen from Section 6 of the Table on page xii given before. This point is made clear from another consideration, viz., the star table in the modern *Sūryasiddhānta*, which unmistakably points to the conclusion that the longitudes of some stars, e.g., Spica *etc.*, correspond to a time much anterior to that of Āryabhaṭa I. The great fame of Āryabhaṭa I induced Varāha, the first maker of a *neo-Sūryasiddhānta* to use the elements of Āryabhaṭa's *ardharātrika* system to supplant the older materials in it. No wonder, therefore, that there is an opinion in favour of the hypothesis that Āryabhaṭa I was the author of the *Sūryasiddhānta*. If there were a shadow of truth in it, Varāha would have admitted it. Alberuni indeed says that the *Sūryasiddhānta* was composed by *Lāṭa* (Alberuni's India, translated by Sachau, Vol. I.p. 153). We now know that this Lāṭa or Lāṭadeva was one of the first pupils of Āryabhaṭa I. He was the expounder of the Romaka and Paūliśa *Siddhāntas* as we learn from Varāhamihira's *Pañcasiddhāntikā*, (I.3). As Alberuni's statement is not corroborated by Varāha, we are not inclined to take it as correct. None of the earlier writers suggest that the *Sūryasiddhānta* was in any way modified or changed by Āryabhaṭa I.

It has now been established beyond doubt that the same Āryabhaṭa was the author of the *Aryabhaṭiya* and another *Tantra* which is now lost. There is reason in support of hypothesis that this *Tantra* itself was the first work of Āryabhaṭa I and that the *Aryabhaṭiya* was the second work from the order in which Varāha mentions them in the Stanza quoted earlier. If this hypothesis be true, the stanza in the *Aryabhaṭiya*[1], which was translated by me as :

"Now when sixty times sixty years and three quarter *yugas* also have elapsed, twenty increased by three years have elapsed since my birth."

1. षष्ट्यब्दानां षष्टिर्यदा व्यतीतास्त्रयश्च युगपादाः ।
व्यधिका विंशतिरब्दास्तदेह मम जन्मनोऽतीताः ॥ —*Arya. III. 10.*

should now be translated thus :

"In this *Mahāyuga* when sixty times sixty years and three quarter yugas also had passed, twenty increased by three years had elapsed since my birth."

Now Bhāskara I the author of the *Mahābhāskarīya* and the *Lagubhāskarīya*, wrote a commentary on the *Āryabhaṭīya*. The author commenting on this stanza observes that :

> "Or this was addressed by Āryabhaṭa when expounding the science to Pāṇḍuraṅgasvāmin, Lāṭadeva Niḥśanku and other pupils."[1]

This direct pupil of Āryabhaṭa I also says that this stanza does not show that the *Āryabhāṭīya* was composed when Āryabhata I was only 23 years old, but refers to the time when he probably began his career as a teacher of Astronomy.

Senagupta out of his discussion concludes that we are not justified in accepting that the Āryabhaṭīya was composed when Āryabhaṭa was only 23 years of age. This treatise as it exists in the present form must have been the composition of a mature age; it is a treatise highly finished in form; the date mentioned in this great work refers to a date when its author became a reputed *guru* or teacher.

Alberuni and Brahmagupta

Dr. E.C. Sachau in his translation of Alberuni's India (vol. II, p. 304) speaks of Brahmagupta in the following words :

> Brahmagupta holds a remarkable place in the history of Eastern civilization. It was he who taught the Arabs astronomy before they became acquainted with Ptolemy; for the famous *Sindhind* of Arabian literature, frequently mentioned but not yet brought to light, is a translation of his *Brāhmasiddhānta*; and the only other book on Indian astronomy, called Atarkand, which they knew, was a translation of his *Khaṇḍakhādyaka*.

Brahmagupta, the celebrated author of the *Brāhmasphuṭasiddhānta*, has another great work as we have said before to his credit which goes by the name *Khaṇḍakhādyaka*. This has

1. एतदेवाचार्य्यार्य्यभटस्य शास्त्रव्याख्यान समये वा पाण्डुरङ्गस्वामिलाटदेवनिःशङ्कुप्रभृतिभ्यः प्रोवाच ।

already been said that perhaps to meet the popular demand, Brahmagupta in this treatise took upon himself the task of simplifying Āryabhaṭa's *ardharātrika* system or the system of midnight day reckoning. Alberuni, the author of the *Indika* has made several references or quotations from the *Khaṇḍakhādyaka* proper and also its supplement, known as *Uttara-Khaṇḍakhādyaka.*

(a) There is a reference to the accepted circumference of the Earth, as given in the *Khaṇḍakhādyaka* (Sachau's *Alberuni*, Vol. I,p. 312)

Multiply the difference in longitude (from Ujjayinī) by the (mean) daily motion of a planet (in minutes) and divide by 4,800; apply the quotient taken as minutes negatively in places east of the meridian line of Ujjayinī and positively in places lying west.[1]

(b) The rules for finding the *ahargaṇa* as given in the *Khaṇḍakhādyaka* in I. 3-5 (Sachau's Alberuni, Vol. II, 46-47), to which Dr. Schram adds a valuable annotation, the constants being taken from the later Pauliśa Tantra as known to Bhaṭṭotpala. This Pauliśa astronomy is derived from Āryabhaṭa I's *ardharātrika* system.[2]

(c) A quotation from the Uttara *Khaṇḍakhādyaka* (Sachau's Alberuni, Vol. II, pp. 84-86) which Sengupta has given in his translation, Chapter X, pp. 148-152.

(d) A quotation also probably from the *Uttara Khaṇḍakhādyaka* (Sachau's Alberuni, Vol. II. p. 87). These stanzas are found in the *Brāhmasphuṭasiddhānta*, XIV, 47-52, also quoted by Bhaṭṭotpala as occurring in the *Brahma Siddhānta* in his commentary on the Bṛhat-Samhitā, IV, 7. The manuscripts which Sengupta used did not show them as occurring in the *Uttara Khaṇḍakhādyaka.* These relate to the dimensions of the *nakṣatras* as seen, as distinguished from the same as calculated.

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1. उज्जयिनी-याम्योत्तर-रेखायाः प्रागृणं धनं पश्चात् ।
देशान्तर भुक्तिवधात् ख खाष्टवेदैः कलाद्यप्तम् ॥
– *KK. I. 15.*

2. – *KK, Pt. B.* Misra's edition, *p. 145.*

(e) Two quotations from the *Uttara Khaṇḍakhādyaka* relating to the celestial co-ordinates of Canopus and Sirius (Sachau's Alberuni, Vol. II, p. 91). Present manuscripts do not show these stanzas, which are probably the same as stanzas 35-36 and 40 of Chapter X of the *Brāhmasphuṭasiddhānta.*

(f) Two quotations from the *Khaṇḍakhādyaka* proper as alleged by Alberuni (Sachau's Alberuni, Vol.II, p. 116). According to Āmarāja, the first is a couple of stanzas of which the author is Bhaṭṭotpala and not Brahmagupa. The second quotation cannot be traced. These relate to finding the possibility of an eclipse whether of the Sun or of the Moon.

(g) Two quotations from the *Khaṇḍakhādyaka* proper as asserted by Alberuni (Sachau's Alberuni, Vol. II, p. 119). These relate to finding the Lords of the year and of the month. According to Āmarāja the rules in question were given by Bhaṭṭotpala and not by Brahmagupta. Pṛthūdaka in his commentary on the first chapter at its concluding portion says ;

> "In this work the *Khaṇḍakhādyaka*, the teacher (Brahmagupta) has not given the rules for finding the Lords of the year and the month[1]."

1. अथाऽत्र खण्डखाद्यके वर्षाधिपमासाधिपा नयनमाचार्य्येण नाभिहितम् ।

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Reference

P.C. Sengupta : *The Khaṇḍakhādyaka, 1934.*
K.S. Shukla : *Mahābhāskarīya, 1960.*
K.S. Shukla : *Sūrya-Siddhānta, 1957.*

Chapter IV

Brahmagupta's Originality in the Khaṇḍakhādyaka

Sengupta in his Indroduction to the Commentary of the *Khaṇḍakhādyaka* has discussed this point. We shall reproduce here some of the points mentioned by him.

Brahmagupta's Khaṇḍakhādyaka

(i) Brahmagupta does not accept the system of Āryabhaṭa but has simplified it in the *Khaṇḍakhādyaka* proper ; and here he has given the system which he thinks to be correct.

Uttar Khaṇḍa Khādyaka

(ii) In the *Uttara-Khaṇḍakhādyaka*, he has further corrected some of his results, given earlier in the *Khaṇḍakhādyaka* proper. In the proper *Khaṇḍakhādyaka*, Brahmagupta assignes to the longitude of the Sun's apogee the value 80°, whereas in the *Uttara* text he corrects it to 77° (UKK. 4) :

> As the process of finding the apparent places of planets as given by Āryabhaṭa does not make them agree with observation, I shall, therefore, speak of this process. Of the Sun the apogee is at two signs and seventeen degrees (2 signs 17°=60 plus 17 degrees=77°).[1]

Compare this with the value given in the *Khaṇḍakhādyaka* proper (I.13)[2] :

The longitude of the Sun's apogee is 80° [KK.I,13] (The Sun's apogee is 80° or two signs plus 20 degrees) *inocco* means

1. न स्फुटमार्य्यभटोक्तं स्पष्टीकरणं यतस्ततो वक्ष्ये ।
   भानुमती मन्दोच्चं राशिद्वयमंशकाश्च सप्तदश ॥ *UKK. IX 4*
2. भागाशीतिरिनोच्चं शशिनः पादोनकृत शरकृतोनाद् ।
   भगणादि द्वित्रिरदैर्वसुनव यम नव गुणैः सकलम् ॥ *KK. I, 13*

mandocca of the Sun). The value given in the *Pañcasiddhāntikā*, IX.7-8) is also the same.

Let us compare it with the present value. According to the astronomical constants as given in the *Conn. des Temps.*, the longitude of the Sun's apogee in 499 A.D. (i.e. 1,400 years before 1900 A.D.) was

—77° 19′19.44″ according to *Conn.des Temps'* equation.

—76° 40′37.22″ according to Newcomb's equation.

The mean of these two values is very nearly 77° as given by Brahmagupta in the *Uttara* text. Thus the value given by Brahmagupta is more correct than the value given by Āryabhaṭa. The *Āryabhaṭīya* gives the value 78° which is less correct.

Brahmagupta more correct than Āryabhaṭa

(iii) Brahmagupta detected that Āryabhaṭa had made the Moon's apogee quicker and nodes slower, than they really are. In both the cases, Brahmagupta made rather an over-correction. We shall give the extract from *Uttara-Khaṇḍakhādyaka* in this connection :

> Multiply the *ahargaṇa* by 110, increase the product by 511 and divide by 30, 31; subtract the result taken as revolutions, etc., from the mean Moon; the final result is the Moon's apogee.[1]

Evidently Brahmagupta assumes that the anomalistic month=3031/110 days. This convergent to the anomalistic month was known to the author of the *Vasiṣṭha Siddhānta* as summarised in the *Pañcasiddhāntikā*[2] (II-2-6).

According to Brahmagupta, the length of the anomalistic month

$$= \frac{1582236450000-4320000000}{57753300000-488105858} \text{ days. } \underbrace{(BrSpSi.\ \text{I.15,16,18. and 20})}$$

= 27.55454641 days which is for 1900 A. D.

= 27.5545502 days according to Radau.

= 27.554602 according to the *Āryabhaṭīya.*

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1. द्युगणात् ख रुद्र गुणिता द् भवशरयुक्ताच्छशिन्निखाग्नि हृतात् ।
भगणादि फलं शोध्यं धमचन्द्राच्छशाङ्कोच्चम् ॥ (*UKK.* IX. 5)

Here also Brahmagupta is more accurate.

Again, the length of the sidereal period of the Moon's apogee

$$= \frac{1577918450000}{488105858} \text{ days}$$

$$= 3232.732048 \text{ days.}$$

Āryabhaṭa's value of the same is 3231.987844 days, and the modern value is 3232.3754 days. Hence Brahmagupta's result is by 0.3566 of a day out, while Āryabhaṭa's is by 0.3876 of a day in.

> Further in the *Uttara-Khaṇḍakhādyaka*(IX.10) we have : Deduct 354½ from the *ahargaṇa*, divide the remainder by 6792 ; subtract the quotient that is obtained in revolutions etc. from the circle : the result is the longitude of the ascending node.[1] (IX.10)

Here Brahmagupta gives the approximate period of the sidereal revolution of the Moon's node to be=6792 days. This according to his *Brāhmasphuṭasiddhānta*$=\frac{1577916450000}{232311168}$ days

= 6792.25396 days, which according to Lockyer would be 6793.39108 days and according to the *Khaṇḍakhādyaka* proper is 6794.75083 days. Hence Brahmagupta's attempt to correction makes the node quicker than it actually is.

Brahmagupta corrects Mars's Aphelion Point

(iv) Again Brahmagupta states that the longitude of Mars's aphelion should be increased by 17° and that of Jupiter by 10°. Evidently here too, Brahmagupta is more correct than Āryabhaṭa. The passage in the *Uttara Khaṇḍakhādyaka* is as follows in this connection :

> Of Mars the apogee (the aphelion point) is to be increased by 17°, that of Jupiter by 10° ; from the *śighra* of Venus 74′ are to be subtracted ; Saturn's equation of apsis should be decreased by its one-fifth ; the *śighra* equation of Mercury should be increased by

1. सार्द्धकृतेषु गुणोनादहर्गणाद् द्विनवमुनि-रसैर्भक्तात् ।
यन्मण्डलादि लब्धं चक्रात् संशोध्य तत् पातः ॥ —*UKK. IX 10*

one-sixteenth.[1]

This stanza says that in 499 A.D., Mars's aphelion point had a longitude of 127°; of Jupiter the longitude of the aphelion was 170°. (*KK.II.* 6[2])

According to Newcomb's rule, the longitude of the aphelion point of Mars in 499 A. D. works out to have been = 128°28'12". According to the *Conn. des. Temps'* rule, the same was 128°27'51". Hence Brahmagupta's determination of Mars's aphelion is correct within 1°30' and is therefore, quite satisfactory. According to the *Khaṇḍakhādyaka* proper it was 110°, and according to the Āryabhaṭīya 118°.

Of planets, beginning with Mars, the degrees of longitude of the apogees are respectively 11, 22, 16, 8 and 24, each multiplied by 10. (*KK .II. 6*)

Thus the longitudes of apogees of Mars = 110 (3 signs 20°); of Mercury = 220° (7 signs 10°); of Jupiter = 160° (5 signs 10°); of Venus = 80° (2 signs 20°) and of Saturn = 240° (8 signs). Compare these values with those given in the *Pañcasiddhāntikā* XVII. 2 (the *Sūrya-siddhānta).*

Again according to this stanza Jupiter's aphelion had a longitude of 170° in 499 A.D.

According to *Conn. des. Temps'* rule the same was 170°25'. Thus here too, Brahmagupta is very accurate. According to the *Khaṇḍakhādyaka* proper, Jupiter's aphelion had a longitude of 160° (*KK.* II.6) and according to the *Āryabhaṭiya*, the value was 180°.

Brahmagupta First to Use Second Differences

All these illustrations reproduced here very well establish the point that the great Indian astronomers from Āryabhaṭa I to Brahmagupta were aware of the methods of separating the two distinct planetary inequalities, viz., that of the apsis and of conjunction in the cases of the five 'star' planets (*PSi.* Introduction Lii). In the *Khaṇḍakhādyaka*, Brahmagupta having given the "sines" and the equations of the Sun and the Moon

1. सप्तदशांशैरधिकं भौमस्योच्चं गुरोर्द्वाशभिरंशैः ।
सितशीघ्रात् कृतमुनयो लिप्ताः शोध्याः शनेः फलं मान्द्यम् ॥
पञ्चांशोनं शैघ्र्यं षोडशभागाधिकं बुधस्य फलम् ॥ —*UKK.* IX. 11

2. मन्दांशा दशगुणिता रुद्रा द्वियमाश्च षोडशाष्टजिनाः । —*KK. II.* 6

at the interval of 15° of arc of the mean anomaly, in the *Uttara Khaṇḍakhādyaka* teaches, for the first time in the history of mathematics, the improved rules for interpolation by using the *second difference*. This very important feature I am reproducing here from the translation by Senagupta of the verse [1] (*UKK. 8*) :

> Multiply the residual arc left after division by 900′ (i.e by 15°), by half the difference of the tabular difference passed over and that to be passed over and divide by 900′ (i. e. 15°) : by the result increase or decrease, as the case may be, half the sum of the same two tabular differences ; the result which, whether less or greater than the tabular difference to be passed, is the true tabular difference to be passed over. (*UKK. 8*)

The rule given here applies to the case of all functions hitherto considered in the *Khaṇḍakhādyaka*, which are tabulated at the difference of 15° of arc of the argument. They are :

(i) the tabular differences of the Sun's equation,

(ii) the tabular differences of the Moon's equation.

(iii) the tabular differences of the 'sines'.

Sengupta has illustrated the rule by an example belonging to the table of sines.

Illustration—To find the 'sine' of 57°.

Brahmagupta's table of sines in the *Khaṇḍakhādyaka* is as follows :

> Thirty increased severally by nine, six and one; twenty-four, fifteen and five, are the tabular differences of sines at intervals of half-a-sign. For any arc, the 'sine' is the sum of the parts passed over, increased by the proportional part of the tabular difference to be passed over.[2] (*KK.*I.30 ; also III.6)

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1. गतभोग्य खण्डकान्तरदलविकलवधात् शतैर्नवभिराप्त्या ।
तद्युतिदलं युतोनं भोग्यादूनाधिकं भोग्यम् ॥ —*UKK. IX. 8*

2. त्रिंशत् सनवरसेन्दुजिन (तिथि) विषयागृद्धार्द्ध चापानां ।
अर्द्धज्या खण्डानि ज्या भुक्तैक्यं सभोग्यफलम् ॥ —*KK. I. 30; also III. 6*

This can be shown in the tabular form thus :

Arc	'Sine'	Tabular difer-ence	Second difference
0°	0		
15°	39	39	
30°	75	36	—3
45°	106	31	—5
60°	130	24	—7
75°	145	15	—9
90°	150	5	—10

Now 57° = 3420 minutes=900′×3+720′. Thus three of the tabular differences are considered as passed over ; the last one being 31 and the one to be passed over is 24.

The true tabular difference by the rule, for arc 57°,

$$=\frac{31+24}{2}-\frac{720}{900}\times\frac{31-24}{2}$$

Hence the 'sine' of 57°

$$=39+36+31+\frac{720}{900}\left[\frac{31+24}{2}-\frac{720}{900}\times\frac{31-24}{2}\right]$$

$$=125.76$$

As worked out from the logarithm tables the same comes out to be 125.80.

Again 'sine' of 57° from Brahmagupta's formula

$$=106+\frac{720}{900}\times24+\frac{31-24}{2}\times\frac{720}{900}-\left[\frac{720}{900}\right]\times\frac{31-24}{2}$$

$$=106+\frac{720}{900}\times24+\frac{720}{900}\left[\frac{720}{900}-1\right]\times\frac{24-31}{2}$$

This in fact is the modern form the interpolation equation up to the term containing the second difference. Brahmagupta thus takes a decidedly improved step here and is undoubtedly the *first man in the history of mathematics* who has done this. One should also remember that in the case where the function is not tabulated at a constant interval, Brahmagupta's rule is remarkable.

Brahmagupta First to Introduce Sine Rule in Indian Plane Trigonometry

In this connection, we shall reproduce the following verse from the *Khaṇḍakhādyaka :*

> Multiply the 'sine' of the *(Śīghra)* anomaly by the 'sine' of the maximum *Śī*ghra equation and divide by the 'sine' of the corresponding *Śīghra* equation, the result is the *'Śīghra* hypotenuse' when the *(Śīghra)* anomaly is half a circle, this śīghra hypotenuse is equal to the radius diminished by the 'sine' of the maximum equation ; when the anomaly is equal to the whole circle, the same is equal to the radius increased by the same 'sine' of the maximum equation.[1]

Let S, E and P be the positions of the Sun, the Earth and the planet, say Mars, respectively. Complete the parallelogram SEMP ; with M as centre and MP as radius, describe a circle. This circle is the epicycle of conjunction of Mars. Produce EM to cut this circle at K. The ∠PMK = ∠S'SP (the point S' is on ES produced), the angle gained by the Earth over Mars since the preceding conjunction. The ∠PMK is called the śīghra anomaly or anomaly of conjunction. We take EM=360, and MP=234. The ∠PEM, which is equal to ∠PES, the annual parallax of Mars, is called the *śighra* equation. The ∠MPE is equal to the ∠SEP, the elongation. The ∠PMK is given, and PM and ME are also given. Hence in the triangle MPE, we have

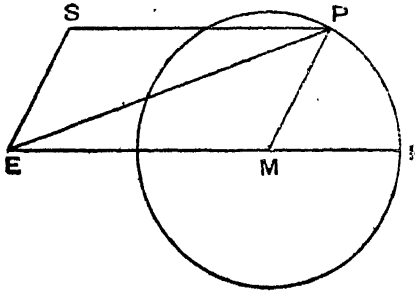

Fig. 4

$$\tan \tfrac{1}{2}(P-E) = \frac{EM-MP}{EM+MP} \tan \tfrac{1}{2} PMK$$

$$= \frac{126}{594} \tan \tfrac{1}{2} PMK$$

$$\therefore L \tan \tfrac{1}{2}(P-E) = \log \left[\frac{126}{594}\right] + L \tan \tfrac{1}{2} PMK$$

We have also $\frac{1}{2}(P+E) = \frac{1}{2} \angle PMK$

1. केन्द्रज्याऽन्त्यफलज्या गुणिता फलजीवया हृताकर्णः ।
त्रिज्याऽन्त्य फलज्योना चक्रार्द्धे संयुता चक्रे ॥ *KK. VI. 1.*

$$\text{Now } \log \left[\frac{126}{594}\right] = 1.3265841.$$

The values of the ∠PMK and the ∠PEM and Brahmagupta's values as given in the verse[1] are presented below in a tabular form :

∠PMK=	28°	60°	90°	121°	135°	148°	164°	173°
∠PEM=	10°58′	23°1′	33°1′	39°56′	40°23′	37°31′	25°32′	12°35
Brahmagupta's ∠PEM=	11°	23°	33°	40°	40°30′	37°30′	25°30′	12°30′

It will be seen that Brahmagupta gives the values of the equation within 1/8th of a degree. It seems inexplicable why such discrepancies should remain in Brahmagupta's calculations. It is probable that he wanted to state his equations to the nearest half a degree.

Now we shall take up the Sighra equations of Mars, and then revert to the Sine Rule. We have in the *Khaṇḍakhādyaka :*

Mars, by the degrees of *Śighra* anomaly (i.e. anomaly of conjunction) of 28 getting at the corresponding equation of 11° rises (heliacally) in the east ; by the next 32° gets 12° more of the equation ; by the next 30°, 10° more ; by the next 31°,7°, more ; by next 14°, half a degree ; these are positive ; by the next 13°, negative 3° ; by the next 16° ; negative 12° after this he is retrograde ; by the next 9°, negative 13° ; by the next 7°, negative 12½°. After this the parts of the equations occur in the reverse order[1].

On the basis of this we have the following table of the *Śighra* equations for Mars ;

Degrees of anomaly of conjunction	Equation of conjunction	Phenomena
0°	0°	Motion direct.
28°	+11°	Rises in the east.

1. भौमोऽष्टयमै रुद्रान् भुक्त्वा पूर्वोदितोरदैरर्कान् ।
ख_ णैर्दशरूपगुणैः सप्तांशा मनुभिरर्द्धशान् ॥
धनभृ मग्नि शशाङ्कै स्त्रीनष्टथा भास्करानतो वक्री ।
नवभिः योदशनगैर्द्वादशसार्द्धान् विलोमोऽतः ॥ *KK.II.8-9*

Degrees of anomaly of conjunction	Equation of conjunction	Phenomena
60°	11+12=+23°	
90°	23+10=+33°	
121°	33+ 7=+40°	
135°	40+ ½=+40°30'	
148°	40°30'—3°=+37° 30'	
164°	37°30'—12°=+25°30'	Retrograde motion begins.
173°	25°30'—13°=+12°30'	
180°	12°30—12°30'=0°0'	
187°	—12°30'	
196°	—25°30'	Direct motion begins.
212°	—37°30'	
225°	—40°30'	
239°	—40°	
270°	—33°	
300°	—23°	
332°	—11°	Sets in the west.
360°	0°	

Now we come back to our discussion on the verse VI.1.

The *Śighra* hypotenuse spoken of here is EP, when SP or EM is taken to be R; when ∠PEM is a maximum, PM is its 'sine'.

It would be seen from the figure that

$$EP = \frac{R \sin PMK \times PM}{R \sin PEM}$$

This may again be written as

$$\frac{EP}{\sin PMK} = \frac{PM}{\sin PEM}$$

This is equivalent to the sine rule for a triangle in plane trigonometry. Brahmagupta is here seen to be the first person to give it in Indian mathematics, This expression reminds us of the famous relationship in respect to triangle ABC :

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Brahmagupta corrects Dimensions of the Epicycle of Apsis

Brahmagupta corrects the dimensions of the epicycles of apsis of the Sun and Moon by—1/42 nd part and 1/48th parts respectively, The reference may be made to the following verse in the *Uttara Khaṇḍakhādyaka :*

> The Sun's equations are to be made less by *dvikṛtāṁśonam* (1/42nd) and the Moon's equations, increased by *vasuvedabhāgayutam* (1/48th). Multiply the Sun's equation a planet's daily motion in minutes and divide by the number of minutes of a whole circle and this is called *Bhujāntara* correction and applied in the same way to the planet as the equation is applied to the Sun.[1]

The Sun's epicycle of apsis has the dimension 14° in the *Khaṇḍakhādyaka* proper. With the correction introduced here, the value becomes $14^\circ \left[1-\frac{1}{42}\right] = 13^\circ\ 40'$.

The correction to the Moon's equations would make the epicycle's dimension changed from 31° to $31^\circ \left[1+\frac{1}{48}\right] = 31^\circ 38' 45''$.

Pṛthūdaka's commentary further corrects it to $31^\circ\left(1+\frac{1}{52}\right) = 31^\circ 35'$

Brahmagupta 's correction to Saturn's epicycle of apsis is—1/5th part and that to the *Śighra* epicycle of Mercury 1/16th part as seen in the verse :

> Of Mars the apogee (the aphelion point) is to be increased by 17°, that of Jupiter by 10° ; from the *Śighra* of Venus 74' are to be subtracted ; Saturn's equation of apsis should be decreased by its one-fifth and the *Śighra* equation of Mercury should be increased by one-sixteenth.[2]

In the *Khaṇḍakhādyaka* proper (II.6), we have been given the longitudes of the apogee of planets : Mars 11°, Mercury 22°, Jupiter 160°, Venus 80° and Saturn 240°. Now with these corrections introduced in the *Uttara Khaṇḍakhādyaka* in the above

1. द्विकृतांशोनं रविफलमिन्दोर्वसुवेदभागयुतम् ।
   भकैफलभुक्तिघाताद् भगणकलाप्तं भुजान्तरं रविवत् ॥ *UKK. IX. 9*
2. सप्तदशांशैरधिकं भौमस्योच्चं गुरोर्दशभिरंशैः ।
   सितशीघ्रात् कृतमुनयो लिप्ताः शोध्याः शनेः फलं मान्द्यम् ।
   पञ्चांशोनं शैघ्र्यं षोडशभागाधिकं बुधस्य फलम् ॥ *UKK.IX.11*

verse, the aphelion point of Mars in 499 A.D. had a longitude of 110 plus 17=127°; of Jupiter 160 plus 10=170°.

According to Newcomb's rule, the longitude of the aphelion point of Mars in 499 A.D. works out to have been 128°28′12″. The same according to *Conn. des Temps* rule would be 128°27′51″. Hence Brahmgupta's determination of Mars's aphelion is correct within 1°30′, and may be, therefore, regarded as very satisfactory. According to the *Khaṇḍakhādyaka* proper this value was, as already said, 110°, while according to the *Āryabhaṭīya*, it was 118°.

The same may be said regarding Jupiter's aphelion. According to the *Khaṇḍakhādyaka* proper the value of its longitude is 160°, according to the *Āryabhaṭīya* it was 180°; according to *Conn. des. Temps'* rule, it would be in 499A. D, 170°25′, and the value given by Brahmagupta in the *Uttara Khaṇḍakhādyaka*, it is 170°.

We have thus shown by many illustrations the important corrections introduced by Brahmagupta in his *Khaṇḍakhādyaka* specially the *Uttara* part. Brahmagupta was highly original in his methods of calculations, accuracies and interpolations. He introduced new ideas in mathematics. He went much ahead Āryabhaṭa in many details. He so many times did not follow Āryabhaṭa in calculations. In the *Khaṇḍakhādyaka* proper, his treatment of parallax in the calculation of solar eclipses is different from that of Āryabhaṭa. The methods followed here are the same as propounded by him in the *Brāhmasphuṭasiddhānta*.[1]

Senagupta is right when he says : As has already been remarked, these corrections and innovations in the *Uttara Khaṇḍakhādyaka* paved the way for the acceptance of his great work the *Brāhmasphuṭasiddhānta* as a standard work on astronomy by the western Indian school of astronomers. The directness of the treatment of topics, and the simplicity of calculations taught in the *Khaṇḍakhādyaka* made it very neat handbook for the beginner. These two works of Brahmagupta were perhaps the only astronomical works in circulations in western India when the Arabs conquered Sind early in the eighth century

1. *On Parallax—*

नाडी चतुष्कविधिना सर्वत्र समो यतस्ततः स्थूलः ।
मानार्थं कर्म महत् कृतमार्यभटेन लघुनि सति ॥ *BrSpSi.* XI. 33

(712 A.D.) and the new conquerers learnt Indian astronomy and mathematics from these works as has been observed by Sachau. Alberuni who came to India early in the 11th centuty of the Christian era, learnt Indian astronomy chiefly by studying the *Khaṇḍakhādyaka* and the *Bṛhat-Saṁhitā* of Varāhamihira, and both of them with the help of commentary of Bhattotpala.

—:c:—

Reference

P.C. Sengupta : *The Khaṇḍakhādyaka, 1934.*

CHAPTER VI

Indian Luni–Solar Astronomy

In this chapter, it is proposed to give an account of astronomical constants and the equations in Indian luni-solar astronomy and to present a comparative view of these quantities with the corresponding ones in Greek and modern Astronomy. This account has been reproduced from P. C. Sengupta's. Appendix I of the *Khaṇḍakhādyaka* It has been shown that in many cases the Indian values of these constants are more accurate than the Greek values. and in Indian lunar astronomy the equations or inequalities discovered are the most startling.

Solar Astronomy

In solar astronomy the length of the year was determined by Āryabhaṭa[1] from the heliacal risings of some bright star at the intervals of 365 and 366 days.

(1) The year according to the Āryabhaṭīya

$$= \frac{1577917500}{4320000} \text{ days} = 365.2586805 \text{ days},$$

=365 da. 6 hrs. 12 mins. 29.64 secs.

(2) The same

$$= \frac{1577917800}{4320000} \text{ days} = 365.25875 \text{ days}.$$

=365 da. 6 hrs. 12 mins. 36 secs., according to the *Khaṇḍakhādyaka*, the *Sūryasiddhānta* of Varāha and the modern *Sūryasiddhānta*.

(3) It is

$$= \frac{1577916450}{4320000} \text{ days} = 365.2584375 \text{ days},$$

=365 da. 6 hrs. 12 mins. 9 secs., according to the

1, P. C. Sengupta, "Āryabhata's Method of determining the Mean Motions of Planets," *Bulletin of the Calcutta Mathematical Society*. Vol. XII, No. 3.

Brāhmasphuṭa Siddhānta of Brahmagupta.

Now the mean sidereal year

=365 da. 6 hrs. 9 mins. 9.3 secs. (Lockyer).

The mean anomalistic year

=365 da. 6 hrs. 13 mins. 49.3 secs. (Lockyer).

The mean tropical year

=365 da. 5 hrs. 48 mins. 46.054 secs. (Lockyer).

Though we take that Indian year was designed to be the sidereal year, it approached most closely the anomalistic year ; and its excess over the sidereal year was about 3 minutes. From this consideration it appears that the Indian astronomers were justified in taking the Sun's apogee to be fixed.

Against the error of+3 min. in the Indian sidereal year, we may point out that—

(1) The Hipparchus-Ptolemy tropical year

=365 da. 14' 48" in sexagesimal units,[1]

=365 da. 5 hrs. 55 min. 12 secs., which has an error of about+6 min.

(2) Meton's sidereal year

$$= \left[365+\tfrac{1}{4}+\frac{1}{76}\right] \text{ days}^{2}$$

=365 da. 6 hrs. 18 min. 57 secs. which has an error of +9 min. 48 secs. nearly.

(3) The Babylonian sidereal-year was 4½ min. too long.[3] Thus the Indian value of it is closer to the true value.

Again in 150 A.D. the longitude of the Sun's apogee according to the *Conn. des Temps* was

$$=101^\circ\ 13'\ 15''.17-6189''.03\left[\frac{1900-150}{100}\right]$$

$$=1''.63\times\left[\frac{1900-150}{100}\right]^{2}$$

$=71^\circ\ 16'\ 26''.37$

while Ptolemy states it to be 65° 30′ which was wrong by—5° 36′ 27″ :

1. *Syntaxis*, Karl Manitius's edition, Vol. I. p. 146.
2. Ibid, p. 145.
3. *Encyclopaedia Britannica*, History of Astronomy.
4. *Syntaxis*, Vol. I. p. 148. The Romaka Siddhānta of the *Pañca-siddhāntikā*, VIII. 2, indicates the Sun's apogee to be at longitude of 75° ; this was perhaps a correction made by Lāṭadeva to the Greek constant.

In 500 A. D (Āryabhaṭa's time) the longitude of the Sun's apogee by the same rule works out to be=77°19′19.44″.

Āryabhaṭa states this to be 78° in the Āryabhaṭīya, Brahmagupta in the Uttarādhyāya of the *Khaṇḍakhādyaka* states the same to be 77°, while the Khaṇḍakhādyaka gives it as=80°. Hence the Indian findings of the longitude of the Sun's apogee were more accurate.

Again as to the Sun's equations of the centre we find that the *Āryabhaṭīya* states the periphery of the Sun's epicycle to be 13°30′, The *Khaṇḍakhādyaka* gives it as 14° ; while according to the Indian form, Ptolemy's value of the same is 15°. Hence according to these writers, the Sun's equations at 90° of the mean anomaly are :—

According to the *Āryabhaṭīya*	=2° 8′ 54″.
,, ,, *Khaṇḍakhādyaka*	=2° 1′ 40″.
,, ,, Brahmagupta	=2° 7′ 20″.
,, ,, Ptolemy	=2° 23′ 3″.
The modern value	=1° 55′ 97″.

Thus the Indian equations of the Sun are in general by more correct than the Greek ones. The Indian constants in solar astronomy are thus, generally, more accurate than the Greek ones. We now turn to the Indian Lunar astronomy.

Lunar Astronomy

Before discussing the constants in Indian lunar astronomy it is necessary to state something as to the time when the Moon was observed by our ancient astronomers and the astronomers from Āryabhaṭa I (499 A.D. to Pṛthūdaka Svāmī (864 A.D.). The months were reckoned from the first visibility of the crescent at the time of the *Mahābhārata* (1400 B. C.). We have a passage in the Bhīṣmaparva where Vyāsa speaks of the evil omens on the eve of the Kurukṣetra war thus—

चन्द्रसूर्य्यावुभौ ग्रस्तावेकामासीं त्रयोदशीम् ।

"That the Moon and the Sun have been both eclipsed on the 13th days of the light and dark halves of the same month."

The eclipses could not take place on the 13th days of the month unless the months were reckoned from the first visibility of the crescent. This was the custom in Babylonia and it has still survived in the Mahomedan world. Even in the *Pañca siddhāntikā* of Varāhamihira (540 A. D.), there is a special

chapter on शशि-दर्शनम् or the fiirst visibility of the crescent. It is thus clear that the practice was to observe the Moon when very near the Sun.

Again Āryabhaṭa says that 'रवीन्दुयोगात् प्रसाधितश्चेन्दुः', "the Moon was determined from her conjunctions with the Sun." The Moon was observed by him at the time of solar eclipses, or at the time of the first visibility of the crescent.

Even up to the time of Pṛthūdaka, the accuracy in lunar astronomy was chiefly aimed at the time of eclipses. Thus in his commentary on the *Khaṇḍakhādyaka* IV, he makes the following introductory remarks :—

"All knowledge relating to (luni-solar) astronomy is desired by the wise (or cultured) specially for knowing the right instants of opposition or conjunction ; these instants are, however, not visible to the eye. Of other things such as *tithis*, *nakṣatras* and *Karaṇas*, as the planets, the Sun and the Moon, are not clearly observed, their beginnings and ends are not visible. Men see the agreement between calculation and observation at the times of solar and lunar eclipses. Hence the word of the astronomers is esteemed amongst men even in respect to such things as *tithis*, etc."[1]

Thus the chief aim of the ancient Indian astronomers was to calculate the eclipses accurately and the Moon was observed chiefly at lunar or solar eclipses, though the time for observation related also to the fiinding of the first visibility of the crescent. This latter phenomena did not perhaps lead them to directly observing the Moon's position at such times by using instruments.

Moon's Mean Motion

The practice of observing the Moon at the time of the eclipses alone led to the determination of the synodic month with the following results :—

(i) Mean synodic month according to the *Āryabhaṭiya*

$$= \frac{1577917500}{57753336 - 4320000} \text{ days,}$$

$$= 29.530582 \text{ days.}$$

---

1. बाहुल्येन पर्व्वज्ञानार्थं सकलं ज्ञानमिष्यते शिष्टैः । तेषां च पर्व्वाणां दर्शनं नास्ति । अन्येषामपि तिथिनक्षत्रकरणानां तस्मात् तेषां शशिभास्करयोरव्यक्तित्वस्मात् । शशिभास्कर-ग्रहणयोर्ग् गणितैक्यं लोकः पश्यति । तस्मात् तिथ्यादिष्वप्यर्थेषु दैवज्ञं वाक्यं लोके आद्रियते ।

(ii) The same according to the *Khaṇḍakhādyakā* =29.5305874 days.

(iii) The same according to the *Brāhma-sphuṭa-siddhānta* =29.530582 days.

(iv) The same according to Ptolemy=29 da. 31′ 50″ 8″ 20‴ in sexagesimal units=29.5305927 days

The modern value according to Newcomb and Radau =29.5305881 days.

Hence the *Khaṇḍakhādyaka* mean length appears to be the closest approximation.

The mean sidereal month must have been deduced from the mean synodic month and the year adopted. Hence no comparison need be made of this element here.

We will now consider the sidereal periods, the nodes and the apogee of the Moon. These are shown below :-

According to	Sid. Per. of Moon's Apogee	Sid. per. of the Ascending Node
Āryabhaṭīya	3231.987079 da.	6794.749511 da.
Khaṇḍakhādyaka	3231.987844 da.	6794.750834 da.
Brāhma-sphuṭa - Siddhānta	3232.73411 da.	6792.25396 da.
Ptolemy	3232.617656 da.	6796.45571 da.
Modern values (Lockyer)	3232.37543 da.	6793.39108 da.

Here also the Indian values show a closer approximation to the true values, Brahmagupta's figures representing the nearest approach.

Other Constants

So far the Indian values of the constants have been more accurate than the Greek ones ; but as to the inclination of the Moon's orbit the Greek value is more accurate than the Indian value.

Inclination of the lunar orbit

Indian value=4°30′.

Greek value=5°0'.

Modern mean value=5°8'43".427 (Brown)

This discrepancy confirms the conclusion, that the observation of the Moon was restricted to the time when she was near a node, either at solar or lunar eclipses, where a small error of observation magnified itself into about half a degree.

We now turn to the parallaxes of the Sun and the Moon :-

	Sūn's Mean Hor. Parallax	*Moon's mean Hor. Parallax*
Āryabhaṭīya	3'55".62	52'30"
Kaṇḍakhādyaka	3'56"5.	52'42".3
Ptolemy	2'51"	53'34"
Modern values	0'8".806	57'2".79

As to the Sun's horizontal parallax, the ancients were of course totally wrong, but in respect to that of the Moon their values were fairly approximate.

We next consider the angular semi-diameters of the Sun and the Moon. These are :-

	Moon's Mean Semi-diameter	*Sun's Mean Semidiameter*
Āryabhaṭīya	15'45"	16'29".4
Khaṇḍakhādyaka (Brāhma-sphuṭa-siddhānta)	16'0".22	16'15"
Ptolemy	17'40"	15'40"
Modern values	15'33".60	16'1".8

Here also the Indian values are more accurate than the Greek values.

Moon's Equations. The First Equation.

It remains now to consider the Moon's equations in ancient Indian astronomy. As has been pointed out before, obser-

vation was up to the time of Brahmagupta, restricted to the time of eclipses perhaps also of syzygies.

The modern form of the Moon's equations is

$$=377' \sin (nt-\alpha)+13' \sin 2(nt-\alpha)+.....$$

$$+76' \sin [2(nt-\theta)-(nt-\alpha)]+40' \sin 2(nt-\theta).....^{1}$$

where nt=mean longitude of the Moon, α the longitude of the perigee, θ=longitude of the Sun.

Here the first two terms, viz., $377' \sin (nt-\alpha) + 13' \sin 2(nt-\alpha)$, are due to elliptic motion about the Earth in one focus; the term $76' \sin [2(nt-\theta)-(nt-\alpha)]$ is known as the evection. We combine a part of the first term with the evection term and the expression for the equation of centre becomes $=301' \sin (nt-\alpha)+13' \sin 2(nt-\alpha)+.....+152 \sin (nt-\theta) \cos (\theta-\alpha)+40' \sin 2(nt-\theta)$.

Now at syzygies and eclipses $\sin (nt-\theta)$ and $\sin 2(nt-\theta)$ will very nearly vanish. Hence according to modern astronomy at the syzygies and eclipses, the chief term of the Moon's equation $=301' \sin(nt-\alpha)$.

This according to the *Āryabhaṭīya*

$$=300' 15'' \sin (nt-\alpha)$$

" " *Khaṇḍakhādyaka*

$$=296' \sin (nt-\alpha)$$

" " *Uttara Khaṇḍakhādyaka*

$$=301'.7 \sin (nt-\alpha)$$

" " *Brāhmasphuṭasiddhānta*

$$=293' 31'' \sin (nt-\alpha)$$

" " Greek astronomy

$$=300' 15'' \sin (nt-\alpha)$$

very nearly.

Hence both the Greek and the ancient Indian astronomers were very near the true value of the Moon's equation at the syzygies and eclipses. Godfray in his *Lunar Theory*, page 107, observes, "the hypothesis of an excentric, whose apse has a progressive motion as conceived by Hipparchus served to calculate with considerable accuracy the circumstances of eclipses; and observations of eclipses, requiring no instruments, were then the only ones which could be made with sufficient exactness to test

1. The accurate values of the coefficients appear to be 377' 19".06, 12' 57".11, 76' 26" and 39' 30".

the truth or fallacy of the supposition." We next consider the second inequality of the Moon.

Moon's Second Inequality or Equation

In ancient times it was Prolemy who first really found a second inequality of the Moon. According to Godfray (*Lunar Theory*, p. 107) "by dint of careful comparison of observations he (Ptolemy) found that the value of this second inequality in quadrature was always proportional to that of the first in the same place, and was additive or subtractive according as the first was so; and thus, when the first inequality was at its maximum or 5°1′, the second increased it to 7° 40′ which was the case when the apse line happened to be in syzygy at the same time."

It is well known that though Ptolemy discovered the second inequality in the Moon's motion he was not able to ascertain its true nature. His corrections in this case are true when at the quadrature the Moon's apse line passes through the Sun or it is at right angles to the line joining the Earth and the Sun.[1] In the general case his construction does not lead to the elegant form of the evection term as we know it now, nor does it lead to the nice form in which it was given by later Indian astronomers from the time of Mañjula (or Muñjāla, 854 Śaka era=932 A.D.).

As has already been pointed out, the early Indian astronomers from Āryabhaṭa to Brahmagupta aimed at accuracy in lunar calculation only for the eclipses and syzygies, and did not interest themselves about the Moon's longitude at the quadratures. Hence this second inequality is absent in the works of these makers of Indian astronomy, as also in the Pre-Ptolemaic Greek astronomy. This points to the conclusion that in both the earlier Indian and Greek systems of astronomy, the modes of observation of the Moon were copied from an earlier system of astronomy whether Babylonian or Chaldean. Even in the *Romaka Siddhānta* of the *Pañcasiddhāntikā*, there is no mention of evection.[2] Thus inspite of the transmission of a vague system of Greek astronomy, Indian astronomy as developed by Āryabhaṭa and Brahmagupta must be regarded as independent and

1. Godfray's *Lunar Theory*. pp. 108-110.

2. Vide the Summary in P.C. Sengupta's paper, "Āryabhaṭa the Father of Indian Epicyclic Astronomy." *Journal of the Department of letters*, vol. XVIII, Calcutta University Press.

original not only from this but also from other considerations. It sought to correct the constants as were obtained from the Babylonian and the Greek systems as has in some cases been shown already.

Mañjula's Second Equation of the Moon (932 A.D.)

We now take up in detail Mañjula's second equation of Moon. It is, however, necessary to say something about his first inequality.

This is given in the form

$$\frac{-488 \sin (nt-\alpha')}{96+\frac{488}{120}\cos (nt-\alpha')} \text{ degrees,}$$

where nt stands for the Moon's mean longitude, α''—that of the apogee.

Hence when $nt-\alpha'=90°$, the equation $= \frac{488°}{96} = 5°4' = 304'$ showing an excess of 4′ over the modern value.

It is further necessary to modify the expression for the Moon's modern form of the equation by changing α to $180°+\alpha$, as in ancient Indian astronomy anomaly is measured not from the perigee but from the apogee.

The modified form is

$$=-301' \sin (nt-\alpha)+13 \sin 2(nt-\alpha).....$$
$$-152' \sin (nt-\theta) \cos (\theta-\alpha)+40' \sin 2(nt-\theta)+...$$

Mañjula's lines giving the second equation are—

> The (mean) daily motion of the Moon diminished by 11° and multiplied by the "cosine" of the longitude of the Sun diminished by that of the Moon's apogee is the multiplier of the "sine" and the "cosine" of the longitude of the Moon diminished by that of the Sun, divided severally by 1 and 5. The results taken as minutes are to be applied negatively and positively to the Moon and to her daily motion if the quantities multiplied together are of opposite signs and in the reverse order if they are of the same sign.[1]

1. इन्दूच्चोनार्ककोटिघ्ना गत्यंशा विभवा विधोः ।
गुणो व्यर्केन्दुदोः कोट्योरूप पंचाप्तयोः क्रमात् ।।
फले शशाङ्क--तद्गत्योर्लिप्ताद्ये स्वर्णयोर्वधे ।
ऋणं चन्द्रे धनं भुक्तौ स्वर्णं सान्य वधेन्यथा ।। —11, 12

As to the positive or negative character of the "sine" and the "cosine" he gives the rule:—

> The mean planet diminished by its *ucca*, the apogee, aphelion or the Śīghra, is called *Kendra* or mean anomaly; its "sine" from above six signs (180°) arises from half circles and are respectively positive or negative, and its "cosine" in different quadrants are respectively positive, negative, negative, and positive.[1]

The convention followed is that the "sine" is negative from 0° to 180° and positive from 180° to 360° of the arc and that the cosine is positive betwen 0° and 90°, negative between 90° and 270° and positive between 270° and 360°.

We may now symbolically express Mañjula's second inequality thus :-

$-(13^\circ\ 11'\ 35''-11^\circ)\times 8^P\ 8'\cos(\theta-\alpha)\times 8^P\ 8'\sin(D-\theta)$ where D stands for the Moon as corrected by the 1st equation; we leave out the correction to the Moon's daily motion as given in the stanzas quoted above.

The moon's new equation comes out to be

$$=-143'58''\cos(\theta-\alpha)\sin(D-\theta).$$

This, it will be seen, is exactly the modern form of the evection as combined with a part of the equation of apsis shown before. The difference in the main is that Mañjula's constant is 144', a quantity less by 8'. *In form the equation is most perfect, it is far superior to Ptolemy's, it is above all praise.* It is from this inequality, we trust, that Mañjula should have an abiding place in the history of astronomy. The next writer who gives the second equation is Śripati (1028 A.D.).

Śripati's Second Inequality of the Moon

The following stanzas from Śripati's *Siddhānta Śekhara*, it is said, were communicated to Sengupta by Pandit Babua Misra. Though they are probably not very correct still the general meaning is clear. They carry the following sense :

> "From the Moon's apogee subtract 90°, diminish the Sun by the remainder left; take the "sine" of the

1. ग्रहः स्वोच्चोनितः केन्द्रं षड्भोद्दलजो भुजः ।
धनर्णः पदशः कोटिर्धनर्णं धनात्मिका ॥

result; multiply it by 160′ and divide by the radius; the result is called *caraphala*. Put it down in another place, multiply it by *śara*(i.e., R vers $(D-\alpha)$ or versed sine of the Moon's distance from the apogee) and divide by the difference between the Moon's distance (hypotenuse) and the radius; the result is called *parama (cara) phala*, which is to be considered positive or negative according as the hypotenuse put down in another place is less or greater than the radius. Multiply the "sine" of the Moon which has been diminished by the apparent Sun, by the apparent *paramaphala* and divide by the radius; the final result is to be called *caraphala* to be applied to Moon negatively or positively as the Moon minus the Sun and the Sun minus the Moon's apogee (diminished by 90°) be of opposite signs; if these latter quantities be of the same signs, the new equation should be applied in the inverse order by those who want to make the calculation of the apparent Moon agree with observation.[1]

Symbolically :—

$$\frac{160'\ R\sin[\theta-(\alpha-90°)]}{R}=caraphala$$

$$\mp\frac{160'\ R\sin[\theta-(\alpha-90°)]}{R}\times\frac{R\ \text{vers}\ (D-\alpha)}{H-R}$$

$=$ *paramaphala*, according as $H>$ or $<R$

The new equation

$$=\mp\frac{R\sin(D-\theta)}{R}\times paramaphala$$

1. त्रिभविरहितचन्द्रोच्चो नभास्वद् भुजज्या ।
गगननृपविनिघ्नी भत्रयज्या विभक्ता ॥
भवति चरफलाख्यं तत् पृथक् स्थं शरघ्नं ।
हृतमुडपतिकर्णत्रिज्ययोरन्तरेण ॥
परमफलमवाप्तं तद्धनर्णं पृथक्स्थे ।
तुहिनकिरणकर्णौ त्रिज्यकोनाधिकेऽथ ।
स्फुटदिनकरहीनादिन्दुतो या भुजज्या ।
स्फुटपरमफलघ्नी भाजिता त्रिज्ययाप्तम् ॥
शशिनि चरफलाख्यं सूर्यहीनेन्दुगोलात् ।
तदृणमुतधनं चेन्दूच्चहीनार्कगोलम् ॥
यदि भवति हि साम्यं व्यस्तमेतद् विधेयम् ।
स्फुटगणितदृगैक्यं कर्तुमिच्छद्भिरत्र ॥

$$= \mp \frac{160' \, R \sin[\theta - (\alpha - 90^\circ)] \; R \text{ vers } (D-\alpha) \times R \sin (D-\theta)}{R\,(H-R) \times R}$$

$$= \mp \frac{160' \, R \cos(\theta-\alpha) \times R \sin(D-\theta)}{R \times R} \times \frac{R \text{ vers } (D-\alpha)}{H-R}$$

This equation is a slightly modified one but practically the same in form as that of Mañjula, except that the constant here is 160', greater than his by 16'. The constant is 160' also in Candraśekhara's form as we shall see later on. We next consider the Moon's inequalities as given by Bhāskara II in his *Bijopanaya*,* a special work on these inequalities composed in the Saka year 1074 (=1152A.D.) two years after he had composed the *Siddhānta Śiromaṇi*.

Bhāskara II on Moon's Inequalities

His preliminary statement runs thus .—

> 112' positive or negative representing the maximum difference. have been found by me in the daily observed Moon (as calculated and as observed) at that point of the ecliptic where the arc from the *kadamba* (i.e., its pole) passing through the zenith cuts it.[1]

Thus for observing the Moon he selected the nonagesimal as the suitable point where the uncertainty about her parallax is zero, and found ∓112' of arc to be the maximum difference between her calculated and observed places.

Mallabhaṭṭa, perhaps a contemporary of Bhāskara II, ascribed this difference to a supposed *Śighrocca* of the Moon. Bhāskara in stanzas 9-13, refutes the existence of the Śighra in the case of the Moon, the substance of his argument begins (i) that it is against the teaching of the *Sūrya-siddhānta* and other accepted authorities, (ii) that there is no variation of the apparent angular diameter of the Moon corresponding to this alleged *Śighra*, and (iii) that planets having a *Śighra* have retrograde motion which is never the case with the Moon.

*There is some uncertainty, about this new fraction introduced by Śripati.

1. लिप्ता विधोरर्क महीमिता मे दृग्गोचराः प्रत्यहमीक्षितस्य ।
कदम्ब गोलगत-सूत्रपाते क्रान्तौ धनर्णत्वजुषो भमध्यात् ॥

—*Bijopanaya*, stanza 8

The reasons for his new equations are stated as follows:—

> When the Moon is situated at a quadrant ahead of the apogee and with the Sun at half a quadrant ahead of her, the maximum discrepancy (of 112′) is seen in the negative character.
>
> When the Moon is situated at three quadrants ahead of the apogee and with the Sun at half a quadrant behind her, the maximum discrepancy (of 112′) is seen in the positive character.
>
> When the eclipses of the Sun and the Moon take place at the apogee or the perigee of the Moon, the Moon as corrected by the equation of apsis is seen to be without any new correction called *bīja*.
>
> When the eclipses of the Sun and the Moon take place at the ends of the odd quadrants of the Moon's anomaly (measured from the apogee), the discrepancy is seen to be less by 34′.
>
> When the Moon is at the apogee, whether the Sun be ahead or behind her by half a quadrant, the discrepancy amounts to be 34′.
>
> The same discrepancy of 34′ is observed when the Moon is at the perigee and the Sun is ahead or behind her by the same distance.
>
> Thus by analysis and synthesis, and by repeated observations, this variable correction has been devised by me; let it be seriously considered by the learned.[1]

Bhāskara here speaks of six cases and we consider them one after another :—

The Moon's equations as modified to suit *siddhāntas* are given by

$$-301' \sin (nt-\alpha)+13' \sin 2(nt-\alpha).....$$

$$-152' \sin (nt-\theta) \cos (\theta-\alpha)-40' \sin 2(nt-\theta)+.....$$

According to Bhāskara's *Siddhānta - śiromaṇi*. the Moon's equation of apsis

$$= -\frac{31^\circ\ 36'}{360^\circ} \times 3438' \sin (nt-\alpha)$$

$$= -301'\ 46''.8 \sin (nt-\alpha),$$

this agrees well with the corresponding term of the modern equation. As Bhaskara takes in all the six cases, $nt-\alpha=90°$, 270°, 0° or 180°, the second term of the equation of apsis vanishes.

Case I.

$$nt-\alpha=90°,\ nt-\theta=-45°,\ \theta-\alpha=135°;$$

Here the total equation of the Moon

$$=-301'-(76'+40')=-301'-116'.$$

This fairly agrees with Bhāskara's observation, the difference being only of 4'.

Case II.

$$nt-\alpha=270°,\ nt-\theta=45°,\ \theta-\alpha=225°;$$

the total equation of the Moon

$$=301'+76'+40'=301'+116'.$$

This also agrees with Bhāskara's observation.

Case III.

$nt-\alpha=0°$ or 180°, $nt-\theta=0°$ or 180°, $\theta-\alpha=0°$ or 180°, the total equation=0', this also agrees with Bhāskara's observation.

Case IV,

$nt-\alpha=90°$ or 270°, $nt-\theta=0°$ or 180°, $\theta-\alpha=90°$ or 270°,

---

1. तुङ्गादाद्यपदान्तस्थाद् विधोरर्कं पदार्द्धतः ।
परमं चन्द्रवैषम्यं ऋणत्वेन समीक्ष्यते ॥ 20 ॥
तत् तृतीय पदान्तस्थात् पृष्ठगेऽर्कं पदार्द्धतः ।
परमं चन्द्रवैषम्यं धनत्वेन समीक्ष्यते ॥ 21 ॥
चन्द्रतुङ्गे च नीचे च शशाङ्कार्कग्रहौ यदि ।
मन्दस्फुटगतश्चन्द्रो निर्जीवतुल्यमीक्ष्यते ॥ 22 ॥
ओजान्त्योर्विधोस्तुङ्गाच्छशाङ्कार्कौ ग्रहौ यदि ।
चतुस्त्रिंशत्-कलाहीनं वैषम्यं तु समीक्ष्यते ॥ 23 ॥
अग्रतः पृष्ठतो वाऽपि रवेश्चन्द्रे पदार्द्धगे ।
तुङ्गतुल्ये चतुस्त्रिंशत् कलावैषम्यमीक्ष्यते ॥ 24 ॥
एवं तन्नीचतुल्येऽपि वैषम्यं तावदेव हि ।
एवं व्यासात् समासाच्च पौनःपुन्येन वेदनात् ॥
चरबीजमिदं कल्पितं मया सद्भिः समीक्ष्यताम् ॥ 25 ॥

the total equation $= \mp 301'$. This does not agree with Bhāskara's statement that the total equation

$= \mp(301' \pm 78')$.

Case V.

$nt - \alpha = 0$, $nt - \theta = \pm 45°$, $\theta - \alpha = \pm 45°$,

the total equation

$= 0' - 76' + 40' = -36'$ or $0' + 76' - 40' = +36'$.

This fairly agrees with Bhāskara's observation.

Case VI.

$nt - \alpha = 180°$, $nt - \theta = \pm 45°$, $\theta - \alpha = 180° \mp 45°$

the total equation

$= 0' + 76' + 40' = 0' + 116'$, or $0 - 76 + 40' = 0' - 36'$.

This does not agree with Bhāskara's statement.

Bhāskara then states his first system of 24 equations corresponding to 24 sines in a quadrant to be 6', 13', 21', 27', 33', 39',45', 51', 56', 61', 65', 68', 70', 72', 74', 75', 75', 76', 76' 77', 77', 78', 78', 78'.[1]

These equations, he says—"are negatively added to the equation of apsis when that is negative and positively added to the same when that is positive"[2]. In other words his new equations are complements of the equation of apsis, the two together being represented by

$-301'\ 46''.\ 8 \sin (nt - \alpha) - 78' \sin (nt - \alpha)$

i.e., by $-379'\ 46''.\ 8 \sin (nt - \alpha)$.

Hence next states his second set of equations depending on $\theta - D$, to be 6', 9', 13', 17', 22', 24', 27', 30', 32', 33', 34', 34', 34', 33', 31', 29', 26', 24', 20', 16', 11', 8', 3', 0'[3] and says :

"These minutes are negative in the odd quadrants of the argument and are positive in other quadrants.[4]"

When the value of the argument is 15°, the equation is 17',

" " " 45° " 34',

" " " 90° " 0',

1. *Bījopanaya*, 26-28.

2. "...फलं ऋणे ऋणं ।
धने धनं मन्दफलेन संयुतम् ॥ 28 ॥

3. *Bījopanaya*, 29-32.

4. एताः कला ओजपदे ऋणं स्युर्धनं तदन्यत्र भवन्ति भूयः ।

Hence the new equation $= -34' \sin 2(\theta - D)$,
$= 34' \sin 2(D - \theta)$.

Here the symbol D stands for the Moon as corrected by the ancient Indian equation of apsis and its complement as given by Bhāskara. It is readily seen that Bhāskara is the first of all the Indian astronomers to detect the equation known as "Variation" His constant, 34', is less than the modern value by about 6', and cannot be considered as a serious error.

We now see that the sum-total of the Moon's equation as given dy Bhāskara

$$= -379' 46''.8 \sin (nt - \alpha) + 34' \sin 2(D - \theta),$$

the evection term being totally absent. This is a serious defect, and Bhāskara's new equations would make the Moon generally more incorrect at the syzygies and eclipses than what the ancient Indian equation of apsis would do.

Perhaps late in life when he was 69 years old in 1105 of Śaka era (=1183 A.D.) he discovered the inapplicability of his new equations at the times of eclipses and in his *Karaṇa-kutūhala* he altogether omitted these new equations which he had given in his *Bṛjopanaya*.

As to Bhāskara's second inequality which is really the complement of the equation of apsis without the evection term, it is far inferior to that of Mañjula and of Śrīpati; as we have seen their form of the second inequality combines the complement of the equation of apsis and evection in the mathematically correct form. For the discovery of such a form of the equation as of these authors, very patient, careful and frequent observation must have been coupled with very careful and nice comparison of observed facts.

As to "variation" it was first discovered by Abul-Wefa in 976 A.D.[1] which was quite forgotten when Tycho-Brahe re-discovered it in 1580 A.D. Hence Bhāskara, in 1152 A.D., re-discovered it in India four centuries before Tycho.

Candraśekhara of Orissa on the Moon's Inequalities

In connection with lunar inequalities it is nesefsary here to record what were the equations discovered or verified by M.M. Candraśekhera Siṁha of Orissā in the later half of the last century. He was educated in the orthodox Sanskrit fashion

1. Godfray's *Lunar Theory*, p. 114.

and had no acquaintance with English education. His work *Siddhānta-darpaṇa* was edited by Prof. Jogeschandra Ray, late of the Cuttack College, in 1899.[1] Candraśekhara in his work gives four equations of the Moon which are :-

(1) The equation of apsis.

(2) The Tungāntra equation or the complement of the equation of apsis in combination with evection.

(3) The fortnightly equation or variation.

(4) The Digaṁśa equation or the annual equation (i. e., $\frac{1}{10}$ of the Sun's equation).

(5) The first equation is of the form

$$= \frac{[31^\circ\ 30' - 30'\cos(nt-\alpha)]3438 \times \sin(nt-\alpha)}{360^\circ}$$

$$= -300'\ 49''.5 \sin(nt-\alpha) + 4'\ 46''.5 \sin(nt-\alpha)\cos(nt-\alpha)$$

$$= -300'\ 49''.5(\sin nt-\alpha) + 2'\ 23''.25 \sin 2(nt-\alpha)$$

It is seen that Candraśekhara wanted to correct the equation of apsis to the second order of small quantities as in all the Indian authors from Brahmagupta but Candraśekhara's form is correct though his contant is wrong.

(2) His second equation is of the form

$$\frac{160' \times 3438 \sin[\alpha - (\theta + 90^\circ)]}{3438} \times \frac{3438 \sin(D-\theta)}{3438}$$

$$\times \frac{\text{Moon's appt. daily motion}^2}{\text{Moon's mean motion}},$$

$$= -160' \cos(\theta-\alpha) \sin(D-\theta)$$

$$\times \frac{\text{Moon's appt. daily motion}}{\text{Moon's daily mean motion}}$$

Here the constant is the same as that of Śrīpati discussed before. The symbol means the Moon as corrected by the equation of apsis. It is readily seen that the constant of the first term of the equation of apsis is increased by 80'. and that the constant of evection is taken at 80'. In both the cases the error is about+4'.

(3) Candraśekhara's third equation or Variation

$$= \frac{3438' \sin 2(D'-\theta)}{90} = 38'\ 12'' \sin 2(D'-\theta).^3$$

1. *Siddhānta-darpaṇa*, V, 100-114.
2. Ibid, VI, 7-9
3. *Siddhānta-darpaṇa* VI. 11-12.

where D' means the Moon as corrected by the 1st and the 2nd equations. Here the constant is wrong by −1' 18″.

(4) His fourth equation or the annual equation $=\pm\frac{1}{10}$ of the Sun's equation of apsis,[1]

$$=\pm\frac{1}{10}\times\frac{12\times3438}{360}\sin\text{ (Sun's distance from the apogee).}$$

$=\pm11'27''.6$ sin (Sun's distance from the apogee).

The modern value of the constant is 11' 10″. Tycho found it to be 4' 30″. Horrocks' (1639) co-efficient was 11' 51″.

As Candraśekhara was aware of Bhāskara's *Bijopanaya*, as also of the work of Śrīpati, his merit here lies in the discovery of the annual equation, and correction to the constant of variation.

Thus we have seen that so far as the luni-solar astronomy is concerned Indian astronomy is independent of Greek astronomy in respect of astronomical constants, that Indian astronomy is generally more accurate than Greek astronomy and that Indian astronomers were not mere "calculators"[2]. There were observers who verified and corrected the old astronomical constants as they came down from Āryabhaṭa and Brahmagupta, who also found independently all the principal equations of the Moon.

1. Siddhānta darpaṇa VI. 13.
2. G.R. Kaye *Hindu Astronomy, p. 60*

—: o :—

Reference

P.C. Sengupta : *The Khaṇḍakhādyaka, 1934*
Karl Manitius's edition : *Syntaxis.*
Godfray : *Lunar Theory.*

CHAPTER VII

Greek and Hindu Methods in Spherical Astronomy

Here we shall reproduce from Sengupta's paper a comparative account of the Greek and ancient Indian methods in Spherical Astronomy and to bring out the independence of the Indian Astronomers on this subject. The views on this subject would necessarily differ from those of many European scholars such as Colebrooke and Bentley (early 19th century) to Kaye (early 20th century). Kaye wrote as follows in the *Journal of Asiatic Society of Bengal* 1919, No. 3.

> The methods by which (the rules) were obtained are buried in obscurity. Braunmühl[1] has stated "that the Indians were the first to utilise the method of projection in the Analemma of Ptolemy." It is intended to present the Hindu methods as clearly as possible and to show that Braunmühl has not done sufficient justice to the Indian astronomers.

As to Kaye, we shall show that his remark quoted above is due to the fact that he had to rely mostly on the English translation of the *Sūryasiddhānta* of Burgess, and perhaps he had no access to the works of Bhāskara II (1150 A.D.), who was the first to explain the ancient Indian methods clearly.

Greek and Hindu Methods in Spherical Astronomy

Of the Greek methods in Spherical Astronomy, the history begins with elementary principles only from Euclid (300 B.C.). Even in Theodosius' *Sphaerica*[2] (about 153 B. C.) "there is nothing that can be called trigonometrical." Heath again says,

1. Heath, *Greek Mathematics*, Vol. II. p. 291. Braunmuhl, *Geschichte der Trigonometrie*, pp. 38-42.
2. Heath, *Greek Mathematics*, Vol. II, p. 250.

"the early spheric did not deal with the geometry of the sphere as such, still less did it contain anything of the nature of the spherical trigonometry. (This deficiency was afterwards made good by Menelaus's *Sphaerica*).[1] Hence the Greek spherical trigonometry began with Menelaus (90 A.D.). His theorem in geometry is well-known—"If the sides of a plane triangle be cut by a transversal into six segments, the continued product of any three alternate segments, is equal to the continued product of the remaining three." From this proposition he deduced the so-called "*regula sex quantitatum*" or the theorem, if the sides of a spherical triangle be cut by an arc of a great circle into six segments, the continued product of the chords of the doubles of any three alternate segments is equal to the continued product of the chords of doubles of the remaining three segments." In plane geometry if the sides BC, CA, AB of a triangle be cut by any transversal at L, M, N, respectively, then we have

$$\frac{BL}{LC} \cdot \frac{CM}{MA} \cdot \frac{AN}{NB} = 1.$$

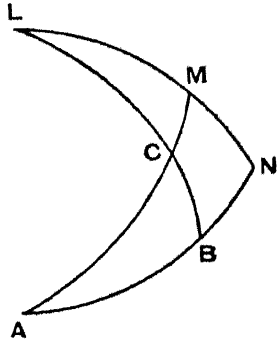

Fig. 5

In spherics the theorem is :

$$\frac{\text{Chord } 2\,BL}{\text{Chord } 2\,LC} \cdot \frac{\text{Chord } 2\,CM}{\text{Chord } 2\,MA} \cdot \frac{\text{Chord } 2\,AN}{\text{Chord } 2\,NB} = 1$$

Both these theorems are proved in Ptolemy's *Syntaxis* (Karl Manitius's edition, Vol. I, pp. 45-51).

If R be the radius of the sphere on which the spherical triangle ABC is constructed, then the chord of the arc 2 BL= 2 R sin BL. Hence Menelaus's theorem in spherics may be expressed as follows :

$$\frac{\sin BL}{\sin LC} \cdot \frac{\sin CM}{\sin MA} \cdot \frac{\sin AN}{\sin NB} = 1$$

This theorem is true for any spherical triangle.

If $\angle B = AN = AM = 90°$ and L the pole of AB, then LMN is a secondary to the arc AB. There are four arcs of great circles; taking any three as forming a spherical triangle and the fourth as the transversal we readily get for the right-angled

1. A.A. Bjornbo, "*Studien über Menelaos' Sphärik*" in Abhandlungen Zur Geschichte der Mathematischen Wissenchaften for 1902, pp. 89 *et seq.*; also Heath, *Greek Mathematics*, vol. II p. 261-73.

triangle ABC, the relations :—

(i) $\sin a = \sin b \sin A$

(ii) $\sin c = \tan a \cot A$

(iii) $\cos b = \cos a \cos c$

(iv) $\tan c = \tan b \cos A$

The above are some of the Napier's rules for a right-angled spherical triangle, deducible from Menelaus's theorem[1]. They are generally sufficient in the case of such triangles. In any spherical triangle, however, this theorem of Menelaus does not in any single step lead to any of the equivalents of the time-altitude or altazimuth equations in spherical astronomy. The ancient Indian methods, though none of them are so highly finished as Menelaus's theorem, yet are not less powerful in tackling the problems that arise in astronomy in connection with the apparent diurnal motion of the heavens. The Greek or Ptolemaic method presents no further points of interest except in its application. We now proceed to illustrate the ancient Indian methods and shall refer to the Ptolemaic method as occasion arises.

Ancient Indian Methods in Spherical Astronomy

In the Indian methods there is no general rule to follow. It is by properties of similar right-angled triangles that a fairly complete set of accurate formulae are obtained. These right-angled plane triangles are classified under the names,—'Krānti-kṣetras' (triangles of declination) and 'Akṣa-kṣetras' (triangles of latitude). We consider the following problems :—

Problem :- To find the time of rising on the equator of a length l, of arc of the ecliptic measured from the first point of Aries.

Let ω be the obliquity of the ecliptic and R. A. the right ascension corresponding to the longitude l, and δ the corresponding declination. The Indian form of the equation is :

1. Three more can be deduced similarly, namely,

(v) $\sin c = \sin b \sin C$

(vi) $\sin a = \tan c \cot C$

(vii) $\tan a = \cos C \tan b$.

*R sin R.A.$= \frac{R \sin l \times R \sin \omega}{R \cos \delta}$, where R is the radius of the sphere.

Note:— If R be the radius of the circle of reference, the Indian trigonometrical functions for the arc θ, are (1) the 'sine,' (2) the 'cosine' and (3) the versed sine. They are respectively equal to R sin θ, R cos θ and R vers θ.

In the adjoining figure, O is the centre of the armillary sphere, ϒQ.ϒC are quadrants of the equator and the ecliptic, respectively. P is the celestial pole, PCQ the summer solstitial colure. Join Oϒ, OQ, OP and OC.

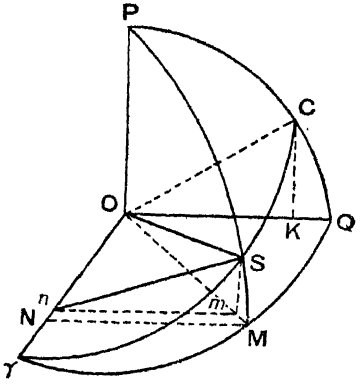

Fig. 6

Let ϒS be$=l$, ϒM=R.A., CQ$=\angle$SϒM$-\omega$, SM$=\delta$.

Join OS, OM. PSM is the secondary to the equator.

From C draw CK perpendicular to OQ. From S draw S*m* and S*n* perpendicular to OM and Oϒ, respectively. Join MN and from M draw MN perpendicular to Oϒ.

Then the triangles S*mn* and CKO are similar. They are called '*Krānti-kṣetras*'[1] or declination triangles,—similar right-angled triangles having one acute angle$=\omega$.

$\therefore$ S*m* : S*n*=CK : OC

or R sin δ : R sin l=R sin ω : R

$$R \sin \delta = \frac{R \sin l \times R \sin \omega}{R} \quad \ldots\ldots(I)$$

The *Aryabhaṭiya*, *Gola*, 25. Varāhamihira, in the *Pañcasiddhāntikā* (IV. 92) states it in the form $2R \frac{\sqrt{(R^2 \sin^2 l) - R^2 \sin^2 \delta}}{2R \cos \delta}$ =R Sin R.A., which is evident from figure. Brahmagupta's equation is identical with that of Āryabhaṭa, (*BrSpSi.* III. 15, *Sūrya-siddhānta* III. 40-41.

Also Bhāskara II, *Grahagaṇita* cap. VIII, stanzas 54-55, is in agreement with Varāhamihira's forms.

Greek Method

In the same figure[1] let PSC be the triangle and γMQ be the transversal. Then Menelaus's theorem gives

$$\frac{\sin PM}{\sin MS}\times\frac{\sin S\gamma}{\sin \gamma C}\times\frac{\sin CQ}{\sin QP}=1$$

or $$\frac{1}{\sin\delta}\times\frac{\sin l}{1}+\frac{\sin \omega}{1}=1$$

or $$\sin \delta=\sin l\times\sin \omega.$$

Indian Method

Again by the Indian method from the same two similar triangles we get

$$mn : nS=OK : OC$$

or, $$mn : R\sin l=R\cos \omega : R$$

$$\therefore \quad mn=\frac{R\sin l\times R\cos \omega}{R}$$

Again $MN : mn=OM : Om$

i.e., $R\sin R.A. : mn=R : R\cos \delta$

$$\therefore \quad R\sin R.A.=\frac{R\sin l\times R\cos \omega}{R\cos \delta}$$

Greek Method

Take PQM for the triangle and γSC for the transversal.

Then, $$\frac{\sin PC}{\sin CQ}\times\frac{\sin Q\gamma}{\sin \gamma M}\times\frac{\sin MS}{\sin SP}=1$$

or $$\frac{\cos \omega}{\sin \omega}\times\frac{1}{\sin R.A}\times\frac{\sin \delta}{\cos \delta}=1$$

or $$\sin R.A.=\tan \delta \cot \omega,$$

The Indian form of the equation is different from that of Ptolemy's. It is also better for the purpose of calculation.

Note :—From the same two similar triangles we have

$$On : ON=R\cos \delta ; R$$

$$\therefore \quad On : R\cos l=\frac{R\cos R.A.\times R\cos \delta}{R} \qquad .....(3)$$

Again, $$\tan R.A.=\frac{mn}{on}$$

$$=\frac{R\sin l\times R\cos \omega}{R\times R\cos l} \qquad .....(4)$$

Again, $mn : Sm=OK : KC$

1. Manitius' Edition of *Syntaxis*, I. 51-53.

$$\text{or} \quad mn = \frac{R \sin\delta \times R \cos\omega}{R \sin \omega}$$

$$\therefore \quad R \sin R.A. = \frac{MN}{mn} \times mn = \frac{R}{R \cos\delta} \times \frac{R \sin\delta \times R \cos\omega}{R \sin \omega} \quad (5)$$

Problem II :— **Sidereal Time-intervals**

Indian Method

The problem discussed above provides the method of finding the sidereal time-intervals in which the signs of the zodiac rise on the equator. To find the corresponding times at any latitude ϕ, it is necessary to calculate and apply what is the ascensional difference due to the elevation of the celestial pole. This ascensional difference is called '*carakāla*' or the variation in the length of half the day. The 'sine' of this '*carakāla*' is called '*carajyā*.' If *ch* denotes this '*carakāla*,'

then,[1] $R. \text{Sin } ch = \frac{R \sin\phi \times R \sin \delta \times R}{R \cos \phi \times R \cos \delta}$

Just as in the solution of the previous problem, the declinational triangles or '*Krānti Kṣetras*' were constructed and used, so in the solution of this and other problems another set of similar triangles were conceived and constructed and were given the name '*Akṣa kṣetras*.'[2]

Let NPZH be the meridian (Fig.7), NOH the north-south line passing through the observer O, P the celestial pole, OQ the trace of the equator on the meridian plane, Z the zenith. Join OZ. From Q draw QM perpendicular to OZ. Then the triangle QOM is an '*Akṣa-kṣetra*' or a latitudinal right-angled triangle, as $\angle QOM = \phi$, the latitude of the station. Another '*Akṣakṣetra*' is thus conceived, in the same figure, let P, P' be the north and south celestial poles, N, the north point, AB A'B' the diurnal

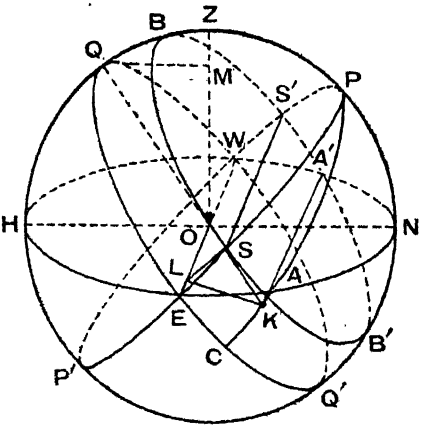

Fig 7

1. *Āryabhaṭīya*, Gola, 26; *Pañca-siddhāntikā*, IV, 34; *Brāhmasphuṭasiddhānta*, II, 57-58; *Sūrya-siddhānta*, II, 91; *Grahagaṇita*, VIII, 48-49.

2. Bhāskara, *Goladhyāya* (Wilkinson and Bapūdeva Śāstri's tr.) PP. 173-76; also, Bhāskara, Grahaganita, Ch. IX. 13-17.

circle of a heavenly body with declination δ, NEHW the horizon, PEP′ W the six O′ clock circle. Here AA′ the line of intersection of the diurnal circle with the horizon is called the "*udayāsta-sūtra*"[1] (or the thread joining the rising and setting points). SS′ the line of interesection of the diurnal circle and the six o′ clock circle, is the horizontal diameter of the diurnal circle. From S draw SK and SL perpendiculars respectively to AA′ and EW. Join KL.

Now since PN=ϕ, the latitude of the station, in the small right-angled triangle KLS, the $\angle$ KLS is also=ϕ.

$$\therefore \quad SK : SL = QM : MO$$

$$\text{or} \quad SK = \frac{SL \times QM}{MO} = \frac{R \sin \delta \times R \sin \phi}{R \cos \phi}$$

Now SK[2] is a "sine" in the small circle AB A′B′ of which the radius is R cos δ; this "sine" reduced to the equator (radius R) is the 'sine' of *cara*.

$$\therefore \quad R \sin ch = R \sin EPA$$

$$= \frac{R \sin \delta \times R \sin \phi \times R}{R \cos \phi \times \cos \delta}$$

Greek Method

Let[3] the arc PA be produced to meet the equator at C. Take PCQ′ for the triangle and EAN for the transversal. Then we get,

$$\frac{\sin PA}{\sin AC} \times \frac{\sin CE}{\sin EQ'} \times \frac{\sin Q'N}{\sin NP} = 1$$

$$\text{or} \quad \frac{\cos \delta}{\sin \delta} \times \frac{\sin CE}{1} \times \frac{\cos \phi}{\sin \phi} = 1$$

$$\therefore \quad \sin CE = \sin ch = \frac{\sin \phi \times \sin \delta}{\cos \phi \times \cos \delta}.$$

Note—The perpendicular distance between AA′ and Ew is called the 'sine' of the amplitude or the '*Agrā*' which is thus calculated :—

$$KL : LS = QO : OM$$

$$\therefore \quad [4]R \sin \text{amplitude} = \text{'}Agrā\text{'} = KL$$

$$= \frac{LS \times QO}{OM} \qquad = \frac{R \sin \delta \times R}{R \cos \phi}$$

It is now evident that the ancient Indian method is different

1. Bhāskara, *Gola*, VII, 39.
2. This is called by the name '*kujyā*' or '*kśitijyā*'. i, e.. earth-sine. Āryabhaṭa, *Gola*, 26, Brahmagupta, II, 57, *Sūrya-siddhānta, II, 61 etc.*
3. Manitius, ibid, p. 84.
4. Āryabhaṭa, *Gola*. 30, etc.

from the Greek method in this case also. As the triangle KLS is difficult to show in the diagram, it is shown in its projection on the meridian plane in Burgess's translation of the "*Sūrya-siddhānta*,"(page 232)and also in Wilkinson and Bāpūdeva Śāstrī's translation of the '*Siddhānta Śiromaṇi*,' p. 175. This has led Braunmühl to assume that the Indian method of arriving at the equation of ascensional difference and some other equations of spherical astronomy has its origin in the Analemma of Ptolemy. A careful study, however, does not justify the identification of Indian methods with the graphic method of the Analemma, which is deduced from the projections of the position of a heavenly body on the meridian prime vertical and the horizon. It is being presently shown that what was done out of difficulty in drawing the figures properly has been taken by Braunmühl as a Greek connection.

Problem III[1] :—

To find the "Time-altitude" Equation

If from any point S on the diurnal circle a perpendicular be drawn to the *Udyāsta-Sūtra* spoken of before, this perpendicular is called the *cheda* or *'iṣṭahṛti.'* The perpendicular from S on the horizon is called *'Śaṅku'*[2] the sine of the altitude. The line joining the foot of the *'Śaṅku'* and that of the perpendicular on the *'Udayāsta-Sūtra'* goes by the name of *'Śaṅkutala'* and this *Śaṅkutala* lies to the south of the *'Udayāsta-Sūtra'* during the day.

In this figure (Fig 8) if AA′ be the 'Udayāsta-Sūtra' or the intersection of the diurnal circle and the horizon, and S a point on the diurnal circle denoting a position of the Sun, SK, SL perpendiculars on AA′ and the horizon respectively; SL is called the *'Śaṅku,'* SK the *'cheda'* and LK, the *'Śaṅkutala'*. In this triangle KSL, the angle KSL was recognised to be the latitude of the station.

Thus the triangle SKL is not taken in its projection on the meridian plane. The side SK is taken as formed of two parts.

1. Āryabhaṭa could not arrive at the true equation. Cf. Gola 28. The correct rules occur in *Pañcasiddhāntikā*, IV, 42, 44; *Brāhmasphuṭasiddhānta*, III, 36-38, 26-40; *Sūryasiddhānta*, III, 34-35.

2. Bhāskara says : महस्थानाल्लम्बः शंकुः । तस्यतलमुदयास्तसूत्राद्दक्षिणतो भवति ॥ "*Gola*, VIII-39-41, Āryabhaṭa uses the term शङ्क्वग्रम्" *Gola*, 29.

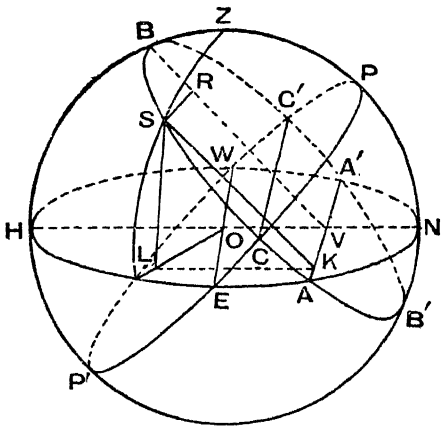

Fig. 8

Let CC′ be the line of intersection of the diurnal circle and the 'six o'clock 'circle EPW. Let SK cut CC′ in M. Then.

$$SK=SM+MK$$

Here SM, the 'sine' in the diurnal circle of the complement of the hour angle is given a distinct name '*Kalā*'[1] and MK as explained before is known by the name '*Kujyā*.' This '*Kalā*' is constructed from the point S in the diurnal circle. Thus the triangles like SKL were not taken in their projections on the meridian plane as Braunmühl would suggest.

From the triangle KSK, we get,

'*Cheda*' : '*Śaṅku*'=R : R cos ϕ where ϕ is the latitude of the observer;

'*Śaṅku*' is here=R cos Z, Z being fhe Sun's zenith distance.

$$\therefore \quad \text{'}cheda\text{'}=\frac{R\cos Z\times R}{R\cos\phi}$$

Now '*Cheda*'=radius of the diurnal circle+*Kujyā*—versed sine of the hour-angle in the diurnal circle O′ B+O′ V−BR,

$$=R\cos\delta+\frac{R\sin\delta\times R\sin\phi}{R\cos\phi}-\frac{R\,\text{vers}\,H\times R\cos\delta}{R}$$

As in the previous problem, $Kujyā=SK=\frac{R\sin\delta\times R\sin\phi}{R\cos\phi}$

or $$\frac{R\cos Z\times R}{R\cos\phi}=\frac{R\cos\delta}{R}\left\{R+\frac{R\sin\delta\times R\sin\phi}{R\cos\phi}\times\frac{R}{R\cos\delta}-R\,\text{vers}\,H\right\}$$

The above equation simplified becomes

$\cos Z=\sin\delta\sin\phi+\cos\delta\cos\phi\cos H$.

In this connection we consider the altazimuth equation by the Indian method.

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1. Bhāskara's *Grahagaṇita*, VIII, 55.
O′ is the middle point of CC′ or it is the centre of the diurnal circle ABB′.

[1]*Problem IV* :—

The Altazimuth Equation

Indiad Method

Let α denote the azimuth of the Sun from the south. In the same triangle SKL in the same figure, we have,

$$LK : SL = R \sin \phi : R \cos \phi$$

or, $$\text{'Śaṅkūtala'} : \text{'Śaṅku'} = R \sin \phi : R \cos \phi$$

$$\therefore \quad \text{'Śaṅkutala'} = \frac{R \cos Z \times R \sin \phi}{R \cos \phi}$$

Now '*Śaṅkutala*' is made up of two parts, namely, '*Bāhu*' and '*Agrā*,' of which the former is the distance of L from the observer's East-West line; the 'Agrā' has been already found.

$$\text{Here 'Bāhu'} = \frac{R \sin Z \times R \cos \alpha}{R} \text{ and 'Agrā'} = \frac{R \sin \delta \times R}{R \cos \phi}$$

$$\therefore \quad \text{'Saṅkutala'} = \text{'Bāhu'} + \text{'Agrā'}$$

or $$\frac{R \cos Z \times R \sin \phi}{R \cos \phi} = \frac{R \sin Z \times R \cos \alpha}{R} + \frac{R \sin \delta \times R}{R \cos \phi}$$

$$\text{or } R \sin \delta = \frac{R \cos \phi}{R} \left(\frac{R \cos Z \times R \sin \phi}{R \cos \phi} - \frac{R \sin Z \times R \cos \alpha}{\phi} \right)$$

which is easily seen to be equivalent to

$$\sin \delta = \cos Z \sin \phi - \sin Z \cos \phi, \cos \alpha$$

Greek Method

Ptolemy[2] has also a method of finding the Sun's altitude at any hour of the day. His method is as follows :—

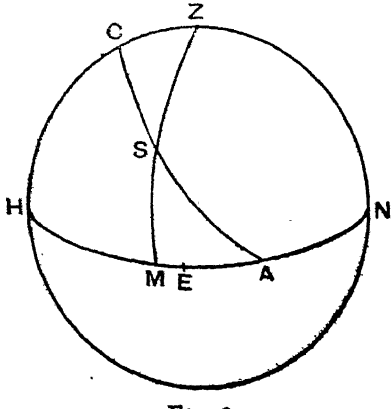

Fig. 9

(i) He would find by means of his tables for the times of risings of the signs of the zodiac, the orient ecliptic point. (ii) He would then find the culminating point of the ecliptic. (iii) He would finally apply Menelaus's theorem in spherics thus :—

Let ASC be any position of the ecliptic, (Fig. 9) NZC the

1. The equivalent of this, in a particular case, is first found in *Brāhma-sphuṭasiddhānta*, Ch. III, 54-56 Cf. *Sūryasiddhānta*, III, 28-31, also Bhāskara *Grahagaṇita*, IX. 50-52.

2. Manitius, ibid, pp. 118, 19.

meridian, NAMH the horizon, Z, the zenith and S the Sun. Here the celestial longitudes of C, S and A are taken to be known; hence ZC and CH are also known.

Now take ZCS for the triangle and HMA to be the transversal ; we then have by Menelaus's theorem.

$$\frac{\sin ZH}{\sin HC}\times\frac{\sin CA}{\sin AS}\times\frac{\sin SM}{\sin MZ}=1$$

$$\text{or } \sin SM=\frac{\cos CZ\times\sin AS}{\sin CA}$$

It is thus clear that Ptolemy had no direct method for connecting the Sun's altitude and the hour-angle. This method is workable for the problem "given time, find the altitude" but is not workable in the converse problem ; besides, the calculation of the longitudes of A and C is very cumbrous.

Again, when EA has been found out, taking ZHM for the triangle and CSA for the transversal, we get,

$\frac{\sin HA}{\sin Am}\times\frac{\sin MS}{\sin SZ}\times\frac{\sin ZC}{\sin CH}=1$, whence and thence HM, the azimuth can be found. The method is here also cumbrous, there being no direct connection between altitude and azimuth ; besides the time-element is not avoided.

The Analemma of Ptolemy and the Indian Method.

When the Sun's declination is zero and his hour-angle, is H, Zeuthen[1] following the method of the 'Analemma' of Ptolemy, as explained by Braunmühl[2] has deduced the following equations :

(1) $\cos Z=\cos H.\cos\phi$

(2) $\tan\alpha=\frac{\tan H}{\sin\phi}$

To these two, Heath following Braunmühl, adds

(3) [3]$\tan ZQ=\frac{\tan H}{\cos\phi}$

1. Heath, *Greek Mathematics*, Vol. II. pp, 290-91.
Zeauthen, *Bibliotheca Mathematica*, 1s, 1900, pp. 23-27.
2. Braunmühl , ibid, pp. 12-13.
3. The Indian form of this equatiom is $R\ Sin\ ZQ=\frac{R\ Sin\ H\times R}{\sqrt{R^2-R^2\cos^2 H\times R^2\ Sin^2\phi}}$
R

Bhāskara's, *Golādhyāya*, Com. on VIII, 67.

where Z is the zenith and Q is the point of intersection of the prime vertical and its secondary passing through the Sun and the north-south points.

Zeuthen[1] points out that later in the same treatise Ptolemy finds the arc 2β described above the horizon by a star of given declination δ′ by a procedure equivalent to the formula.

(4) $\cos \beta = \tan \delta' \tan \phi$.

With regard to the 'Analemma' of Ptolemy. it may be noted, as Heath[2] says, that "the procedure amounts to a method of graphically constructing the arcs required as parts of an auxiliary circle in one plane." Many things may be, in practice, done graphically far more easily than by the theoretical method. Besides, no theoretical calculations occur in the 'Analemma'. Zeuthen[2], following the method of this work, has deduced in the general case, the two equations.

(5) $\cos Z = (\cos \delta, \cos H + \sin \delta . \tan \phi) \cos \phi$.

(6) $$\tan \alpha = \frac{\cos \delta . \sin H}{\frac{\sin \delta}{\cos \delta} + (\cos \delta . \cos H + \sin \delta . \tan \phi) \sin \phi}$$

These equations are suggested to a modern reader from a study of the figures in the 'Analemma.' But neither in this work nor in the '*Syntaxis*' are they to be found. With regard to the first four formulae, it is possible that they were recognised by Ptolemy. With regard to the last two, Zeuthen[3] remarks "mais le texte nen contient rien,' and they were certainly not recognised by Ptolemy.

Besides the tangent function is wholly absent in Greek trigonometry. They are also different in form from those arrived at by the Indian method as explained before. Thus, it is clear that the Indian methods are in no way connected with the method of the 'Analemma.'

Even taking for granted that the Indians followed a method of projection much allied to the method of the Analemma' there is no adequate reason for assuming that their method is derived from any Greek source. Analogy and precedence do not necessarily constitute originality—there is still the chance of a remoter origin from which both the systems drew their inspiration. The method of the 'Analemma,' as has been already stated, presents a

1, 2, Bjornbo, *loc. cit.* p. 86.

3. Zeuthen, *loc, cit.* p. 27.

graphical method for constructing the Sun's altitude and azimuth from the hour angle when the Sun's declination is zero but such a graphical method is generally complex as compared with the elegant Indian method. An astronomer who constructs and uses an armillary sphere to arrive at his equations in spherical astronomy and who has not a well-developed spherical astronomy at his command must have to draw perpendiculars from the positions of the heavenly body, not only on the meridian plane, the horizon or on the prime vertical, as the occasion arises, but also on the line of intersection of the diurnal circle with the horizon. Hence Braunmühl's statement that the Indian methods of spherical astronomy have their origin in the 'Analemma's, in spite of his admitting that Indians were first to utilise its methods, is rather far-fetched and tends to take away the honour from the great Indian astronomers, who devised the beautiful methods. The 'Analemma' as it now exists is a Latin translation from an Arabic version of the original Greek[1]. We may reasonably doubt that the Arabic version was greatly influenced by the ancient Indian system.

We now pass on to the consideration of other allied or similar problems in the two systems of astronomy.

Problem V—

To find the Angle between the Ecliptic and the Meridian

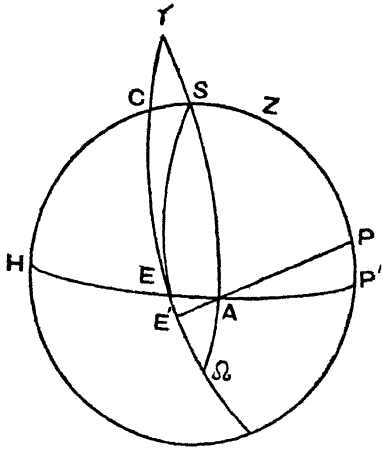

Fig. 10

Indian Method[2]

Let γSA be thee cliptic, γCE the equator, E the east-point of the horizon (Fig 10). Cut off SH=90° and draw the great circle HEAP′ cutting the meridian P′SCH at the points P′ and H. The aim is to find AP′ but it is enough to find EA since AP′ is the complement of EA′.

Both Āryabhaṭa and Brahmagupta were unable to find EA correctly. Let P be the celestial pole and let PAE′ be

1. On the influence of the ancient Indians on Arab mathematics and astronomy; see Alberuni's India, translated by Dr. E. Sachau, Vol. II, p. 304.

2. Āryabhaṭa, Gola, 45; *BrSpSi*, IV. 17; *Sūrya-siddhānta*, IV. 25; Bhāskara's *Golādhyāya*, VIII, 21-74, first example in his own commentary.

the secondary to the equator cutting it at E'. Both the above astronomers were content with the idea that AE=AE', or that AE=the declination of the point A of the ecliptic which is 90° ahead of S in the above figure. This idea continued till the time of Bhāskara II (1150 A. D.) who found out the correct equation.

He recognised that CS, the declination of S=PP'; P'EH is then the horizon of the station whose north geographical latitude is CS. Also, the 'sine' of EA is the '*Agrā*' or the sine of the amplitude of the point A for the latitude CS.

$$\therefore R \sin EA = \frac{R \sin AE' \times R}{R \cos CS} = \frac{R \sin (90° + \gamma S) \times R \sin \omega}{R} \times \frac{R}{R \cos CS}$$

$$\text{or } R \sin EA = \frac{R \sin (90° + l) \times R \sin \omega}{R \cos \delta}$$

where l stands for γS and δ for CS.

Greek Method :

We give below the Ptolemy's method in a slightly modified form[1]. Let SHA be the triangle and γCE be the transversal ; then we have,

$$\frac{\sin SC}{\sin CH} \times \frac{\sin HE}{\sin EA} \cdot \frac{\sin A\gamma}{\sin \gamma S} = 1$$

$$\text{or } \frac{\sin \delta}{\cos \delta} \times \frac{\sin 90°}{\sin EA} \times \frac{\sin (90° + l)}{\sin \iota} = 1$$

$$\therefore \sin EA = \frac{\sin \delta \times \sin (90° + l)}{\cos \delta \times \sin l},$$

which is readily transformed into Bhāskara's equation. The originality of Bhāskara would be readily admitted.

Problem VI—

To find the Angle between the Ecliptic and the Horrizon

Indian Method :

(A) Āryabhaṭa's method. It consists of the following[2] steps :—

(1) Determination of the orient point of ecliptic.

(2) Finding the sine of its amplitude.

1. Manitius, *ibid*, Book I, pp. 104-06.
2. Āryabhaṭa, *Gola*, 33 : *Sūryasiddhānta*, V. 5-6.

(3) Determination of the culminating point of the ecliptic from the hour-angle of the Sun.

(4) Finding the declination of the culminating point of the ecliptic.

Having obtained the above elements, his rule can be followed thus :

In this Fig. 11 NZH is the meridian, HMEAN the horizon, CN'A the ecliptic. If N' be the nonagesimal or the highest point of the ecliptic, the altitude of N' is the inclination of the ecliptic to the horizon.

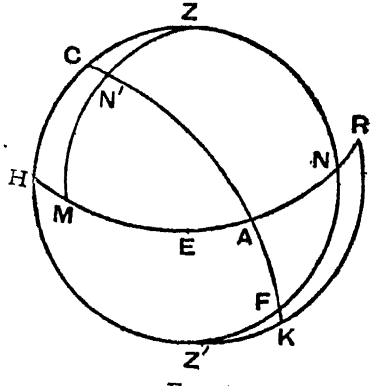

Fig. 11

Let ZN'M be the vertical through N', meeting the horizon at M.

When the time is given, the longitudes of A and C can be found out, from which CZ the zenith distance of C and EA the amplitude of the orient ecliptic point can be determined.

Here HM=EA.

According to Āryabhaṭa,

$$R \sin CN' = \frac{R \sin CZ \times R \sin HM}{R}$$

and $R \sin ZN' = \sqrt{(R \sin CZ)^2 - (R \sin CN')^2}$

This is only an approximate rule. As expressed here,

$$R \sin ZN' = \frac{R \sin CZ \times R \cos HM}{R} \text{ approximately.}$$

$$= \frac{^1R \sin CZ \times R \cos HM \times R}{R \times R \cos CN'} \text{ accurately.}$$

$$= \frac{R \sin CZ \times R \cos HM}{R \cos CN'}.$$

(B) The method of Brahmagupta[2] :

Brahmagupta would also first determine the orient ecliptic

1. This correction was perhaps first noticed by Raṅganātha (1603 A. D.) in his commentary in the *Sūryasiddhānta*.

2. *BrSpSi.* V 3.

point A. Then he subtracts 90° from the longitude of A. Thus having the longitude of N', he next finds the part of the day elapsed of N'; from which by the time-altitude equation discussed above, he finds ZN'. This is of course more accurate than that of Āryabhaṭa. Bhāskara[1] here follows Brahmagupta.

Greek Method :

Let the ecliptic CN'A cut the lower half of the meridian at F. Ptolemy takes AK along the ecliptic=90° and AR along the horizon=90°; then the great circle passing through R and K passes through the nadir Z'. Now take Z'FK for the triangle and ANR for the transversal, then by Menelaus's theorem.[2]

$$\frac{\sin FN}{\sin NZ'} \times \frac{\sin Z'R}{\sin RK} \times \frac{\sin KA}{\sin AF} = 1$$

$$\therefore \sin RK = \frac{\sin FN}{\sin AF} = \frac{\cos FZ'}{\sin AC} = \frac{\cos CZ}{\sin AC} = \frac{\sin CH}{\sin AC}$$

$$\text{or } \sin MN' = \frac{\sin CH}{\sin AC}.$$

Here Ptolemy's equation is simpler than that of Āryabhaṭa; hence they must be independent of each other.

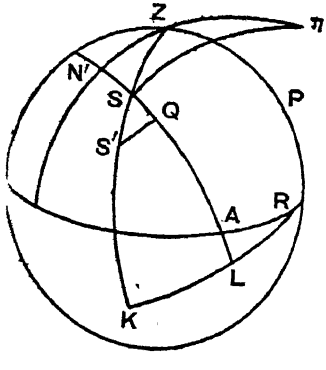

Fig. 12

Problem VII:—

To find the Angle made by the Vertical through any Point of the Ecliptic with the Latter

This problem is considered by Ptolemy but it is not considered separately in Indiam Astronomy, but from the rule for parallax in longitude, the rule for its calculation can be deduced.

Indian Method :

In Fig. 12 S represents the true position of the Sun and S' the Sun's position as depressed by parallax. N'SA is the ecliptic. If from S', S'Q be drawn perpendicular to the ecliptic, then, if P is the horizontal parallax,

1. *Grahaganita*; XII, 3-4.
2. Manitius, *ibid.* pp. 110-111.

$$SQ = SS' \times \frac{R \cos S'SQ}{R} = \frac{P \times R \sin ZS}{R} \times \frac{R \cos S'SQ}{R}$$

$$^{1} = \frac{P}{R} \sqrt{(R \sin ZS)^2 - (R \sin ZN')^2}$$

$$^{2} = \frac{P}{R^2} \times R \sin N'S \times R \cos ZN', \text{ where N' is the nonagesimal}$$

Thus R cos S'SQ is seen to be

$$= \frac{R \sin N'S \times R \cos ZN'}{R \sin ZS}$$

The Indian method is fully described by Bhāskara in his *'Goladhyāya.* VIII. 12-25. The truth of the Indian rule for R cos S'SQ is easily seen from the spherical triangle AZS, where A is the pole of the ecliptic.

Greek Method :

[3]Ptolemy takes SK and SL 90° each, along the vertical circle ZSEK and the ecliptic N'SA. The great circle through K and L cuts the horizon at R which is the pole of the vertical circle. He takes SKL for the triangle and EAR for the transversal, then

$$\frac{\sin SE}{\sin EK} \times \frac{\sin KR}{\sin LR} \times \frac{\sin LA}{\sin AS} = 1$$

or $$\sin LR = \frac{\cos ZS \times \cos AS}{\sin ZS \times \sin AS}$$

or $$\cos S'SQ = \cot ZS \times \cot AS = \tan SE \times \cot AS.$$

The Indian and the Greek rules are altogether different both in form and method. There can, therefore, be no question of any connection between them.

Problem VIII :—

To convert the Celestial Longitude of a Heavenly Body into its Polar Longitude

If σ be the position of a (Fig.13), γK and σK are the celestial longitude and the celestial latitude, respectively ; γM and σM are the polar longitude and polar latitude ; γN and σV are the right ascension and declination of the star.

Indian Method :

All Indian astronomers attempt at finding MK which sub-

1. Āryabhaṭa, *Gola*, 34; *Pañcasiddhāntika*, IX. 22 *BrSpSi*, XI. 23.
2. *BrSpSi*, V, 4-5 ; *Sūryasiddhānta*, V, 7-8 Bhāskara, *Grahaganita*, XII, 4.
3. Manitius, ibid, p. 119.

point A. Then he subtracts 90° from the longitude of A. Thus having the longitude of N′, he next finds the part of the day elapsed of N′ ; from which by the time-altitude equation discussed above, he finds ZN′. This is of course more accurate than that of Āryabhaṭa. Bhāskara[1] here follows Brahmagupta.

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Let the ecliptic CN′A cut the lower half of the meridian at F. Ptolemy takes AK along the ecliptic=90° and AR along the horizon=90°; then the great circle passing through R and K passes through the nadir Z′. Now take Z′FK for the triangle and ANR for the transversal, then by Menelaus's theorem.[2]

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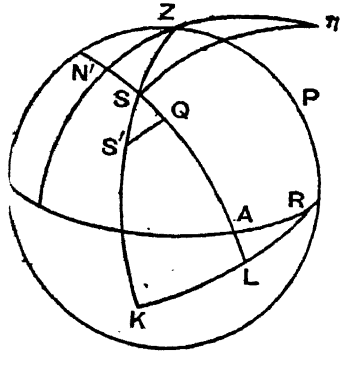

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1. *Grahaganita*; XII, 3-4.
2. Manitius, *ibid.* pp. 110-111.

$$SQ=SS'\times\frac{R\cos S'SQ}{R}=\frac{P\times R\sin ZS}{R}\times\frac{R\cos S'SQ}{R}$$

$$^{1}=\frac{P}{R}\sqrt{(R\sin ZS)^2-(R\sin ZN')^2}$$

$$^{2}=\frac{P}{R^2}\times R\sin N'S\times R\cos ZN',$$ where N' is the nonagesimal.

Thus R cos S'SQ is seen to be

$$=\frac{R\sin N'S\times R\cos ZN'}{R\sin ZS}$$

The Indian method is fully described by Bhāskara in his '*Golādhyāya*. VIII, 12-25. The truth of the Indian rule for R cos S'SQ is easily seen from the spherical triangle πZS, where π is the pole of the ecliptic.

Greek Method :

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1. Āryabhaṭa, *Gola*, 34; *Pañcasiddhāntikā*, IX, 22 *BrSpSi*. XI. 23.
2. *BrSpSi*. V, 4-5 ; *Sūryasiddhānta*, V, 7-8 Bhāskara, *Grahagaṇita*, XII, 4.
3. Manitius, ibid, p. 119.

tracted from, or added to, γK the celestial longitude, gives γM the polar longitude.

According to Āryabhaṭa[1],

$$MK = \frac{\sigma K \times R \text{ vers } \gamma K \times R \sin \omega}{R^2}.$$

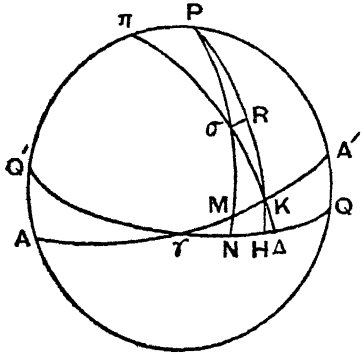

Fig- 13

Brahmagupta[2] makes a distinct improvement on Āryabhaṭa and gives his rule for finding the projection MK on the celestial equator.

If P be the celestial pole, PKH the secondary to the equator, Brahmagupta says that,

$$NH = \frac{\sigma K \times R \sin (\gamma K + 90°) \times R \sin \omega}{R}$$

If from σ, σR is drawn perpendicular to PKH, it is evident that,

$$R \sin \sigma R = \frac{R \sin \sigma K \times R \sin \sigma KR}{R}.$$

According to Āryabhaṭa and Brahmagupta, as explained before,

$$R \sin \sigma KR = \frac{R \sin (\gamma K + 90°) \times R \sin \omega}{R}$$

Hence Brahmagupta intends that,

$$NH = \sigma R = \frac{\sigma K \times R \sin \sigma KR}{R}$$

which is rather a big assumption. He then directs the finding of the part of the ecliptic of which σR or NH is the projection on the equator thus approximately to MK.

Āryabhaṭa, Brahmagupta[3] and the modern *Sūryasiddhānta* take the declination $\sigma N = \sigma K + KH$ where σK is small. They do not consider the case where σK is large.

Bhāskara alone gives us fairly correct rules for this transformation of co-ordinates.

1. Āryabhaṭa, *Gola*, 36.
2. *BrSpSi* X. 17.
3. *BrSpSi*, X. 15, *Sūryasiddhānta*, II, 58.

In order to find σN, he would multiply

σK by $\frac{R \cos \sigma KP}{R}$; according to him,

$$\sigma N = \frac{\sigma K \times R \cos \sigma KP}{R} + KH^1$$

This is a decided improvement on Brahmagupta's corresponding rule. The declination σN obtained would be very nearly accurate.

Having obtained σN, Bhāskara[2] then directs the finding of NH, thus,

$$NH = \frac{\sigma K \times R \sin \sigma KP}{R \cos \sigma N}$$

He then directs the finding of MK on the ecliptic of which NH is the projection by means of the times of rising of the signs of the zodiac on the equator.

Thus, the Indian methods show a beginning and development only. The Greek method as given by Ptolemy is mathematically accurate.

Greek Method[3]

To transform the celestial longitude and celestial latitude to right ascension and declination.

Let the great circle $\pi\sigma K$ meet the equator at Δ. Ptolemy would then from the given value of γK, find $\gamma\Delta$ and ΔK by using his tables for the rising of signs of the zodiac on the equator. He then takes $\pi P\sigma$ for the triangle and $\gamma N \Delta Q$ for the transversal. The Menelaus' Equation, then, is

$$\frac{\sin \pi Q}{\sin QP} \times \frac{\sin PN}{\sin N\sigma} \times \frac{\sin \sigma\Delta}{\sin \Delta\pi} = 1^2$$

Here $\pi Q = 90° + \omega$, $QP = 90°$, $PN = 90°$; $\sigma\Delta = \sigma K + K\Delta$.

$\pi\Delta = 90° + K\Delta$, whence $N\sigma$ is obtained.

He next takes PNQ for the triangle and $\pi\sigma\Delta$ for the transversal,

$$\therefore \frac{\sin P\pi}{\sin \pi Q} \times \frac{\sin Q\Delta}{\sin \Delta N} \times \frac{\sin N\sigma}{\sin \sigma P} = 1$$

Here $P\pi = \omega$, $\pi Q = 90° + \omega$, $Q\Delta = 90° - \gamma\Delta$.

Hence the above equation gives him ΔN. Now,

$\gamma N = \gamma\Delta - \Delta N$,

1. Bhāskara, *Grahagaṇita* XIII, 3.
2. Ibid. XIII, 4.
3. Manitius, Ibid, Vol, II. *Achtes Buch*, pp. 84-85.

It is almost needless to say that neither in the method nor in the rules is there any agreement between the Indian and Greek spherical astronomy in the solution of this problem.

Kaye's view[1]

As to Kaye, it appears that he has not been able to find a method in the translation of the *Sūyrasiddhānta* by Burgess. The figures of his paper referred to before do not show the "*Akṣakṣetras*" even in their projections on the meridian place. He refers to Braunmühl's *History of Trigonometry* but does not appear to have been able to follow him in his "*Methode der indischen Trigonometrie.*" Kaye, however, is not slow in belittling Indian trigonometry when he says :—The Indian astronomers employed the sine function principally and the versed sine occasionally ; they never employed the tangent function; and generally, but not always, preferred to employ the sine of the complementary angle rather than the cosine functions."

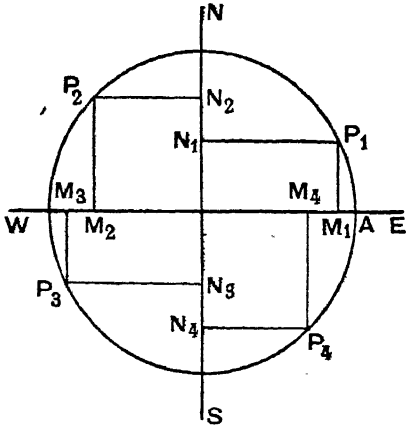

Fig. 14

It is evident that Kaye never understood the meaning of the Indian functions of 'sine' and 'cosine.' These functions are fully explained by Bhāskara[2] when he says :—

"Of that point the distance from the east-west line is the sine and the distance of the point from the north-south line is the "cosine".

In (Fig. 14), of the arc AP_1, P_1M_1 is the "sine" and P_1N_1 is the "cosine" of AP_2, P_2M_2 is the "sine" and P_2N_2 is the "cosine"; of AP_3, P_3M_3 is the "sine" and P_3N_3 is the "cosine"; etc. It is evident that a better definition of these functions was never given.

We have thus seen that some of the solutions of Āryabhaṭa

1. J. A. S. B., N. S., XV, p. 154.
2. तस्य बिन्दोः प्राच्य-परायाश्च यदन्तरं सादोर्ज्या ।
बिन्दोर्याम्योत्तरायाश्च यदन्तरं सा कोटिर्ज्या ॥
—(Bhāskara, *Grahaganita*, commentary. II. 88-21

are imperfect, of Brahmagupta the solutions are more accurate, while those of Bhāskara are generally mathematically correct. The date of the scientific ancient Indian Astronomy is indeed 499 A. D., while that of the Syntaxis is about 150 A. D. It is by these shortcomings and differences in the methods, new ideas (e.g., the idea of the differential calculus)* and the like, that we can safely say that Indian Astronomy in its scientific form, although of a later date than the "Syntaxis" of Ptolemy, is original and not borrowed from foreign source. There is evidence that some crude form of Greek astronomy was transmitted to India and went by the name of the "*Romaka*" or the "*Pauliśa*" Siddhānta, prior to the time of Āryabhaṭa but our great Indian astronomers, Āryabhaṭa with his pupils, Varāha-mihira and Brahmagupta, had to construct a new science altogether.

(This Chapter is almost a reproduction of the paper by P. C. Sengupta, as acknowledged earlier).

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Reference

P.C. Sengupta : The *Khaṇḍakhādyaka*, 1934

CHAPTER VII

Epicyclic Theory of Ancient Indians

We shall give here some details of the Indian concepts regarding the motion of planets or wandering bodies among the stars. The *Vedāṅga Jyotiṣa* (1400 B. C. or earlier) does not speak of this, A comparison of the astronomical constants of the Greek and the ancient Indian systems, points unmistakably to the conclusion that the Indian constants as determined by Āryabhaṭa I and his successors, are almost in all cases different from those of Greeks. Indian astronomers were highly original in their concepts and treatment. The originality of Āryabhaṭa I and other astronomers would be seen from what we are discribing below.

Apparent Motions of the Sun and Moon

We have the following passages from Āryabhaṭa :

All planets move in eccentrics to their orbits at the mean rates of angular motion, in the direction of the signs of the zodiac from their apogees (or aphelia) and in the opposite directions from their *Śighroccas.*

The eccentric circles of planets are equal to their concentrics and the centre of the eccentric is removed from the centre of the Earth.

The distance between the centre of the Earth and the centre of the eccentric is equal to the radius of the planet's epicycle; on the circumference (whether of the epicycle or of the eccentric) the planet undoubtedly moves with the mean motion.

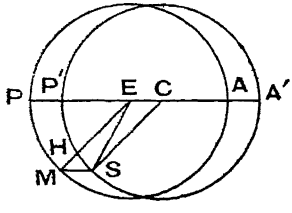

Fig. 15

Here the central idea was that undoubtedly planets moved unifomly in circles round the Earth; if the motion appeared to be variable, it was due to the fact that the centres of such circle (i. e. the concentric circles) did not coincide with the centre of the Earth.

Let E represent the centre of the Earth(Fig.15), APM the Sun's circular orbit or concentric; let A and P be the apogee and the perigee respectively. From EA, cut off EC equal to the radius of the Sun's epicycle. With centre C and radius equal to EA describe the eccentric A'P'S cutting AP and AP produced at P' and A'. Here A' and P' are the real apogee and perigee of the Sun's orbit. Let PM and P'S be any two equal arcs measured from P and P.'

The idea is that the mean planet M and the apparent Sun S move simultaneously from P and P' in the counterclockwise direction along the concentric and the eccentric circles. They move with the same angular motion and arrive simultaneously at M and S.

Here EM and CS are parallel and equal, hence MS is also equal and parallel to EC. Let SH be drawn perpendicular to EM.

The angle PEM is the mean anomaly and the angle P'ES the true anomaly; the angle SEM is the equation of the centre, is readily seen to be plus (+) from P' to A' and minus (−) from A' to P'. Thus as regards the character of the equation, the eccentric circle is quite right. We now turn to exmine how far it is true as to the amount.

Let the angle SEM denoted by E and the angle $\angle$PEM $=\angle$P'CS$=\theta$; EP$=$CP'$=a$; EC$=$MS$=p$, then

$$\tan E=\frac{SH}{HE}=\frac{p\sin\theta}{a-p\cos\theta}$$

$$\therefore \quad E=\frac{p}{a}\sin\theta-\frac{p^2}{2a^2}\sin 2\theta+\frac{p^3}{3a^3}\sin 3\theta\ldots\ldots$$

Now the true value of E in elliptic motion is given by

$$E=\left(2e-\frac{e^3}{4}\right)\sin\theta+\frac{5}{4}e^2\sin 2\theta+\frac{13e^3}{12}\sin 3\theta^*;\ldots$$

It we now put $\frac{p}{a}=2e-\frac{e^3}{4}$, as a first appoximation $\frac{p}{a}=2e$.

Hence $\frac{p^2}{2a^2}=2e^2$, which is greater than $\frac{5}{4}e^2$ by $\frac{3}{4}e^2$. In the case of the Sun if the value of p be correctly taken the error in the coefficient of the second term becomes+3'; similarly in the case of the Moon, the corresponding error becomes+8'.

*Godfray's *Astronomy*, p. 149.

Again if $\frac{p}{a}=2e$, what is the centre of the eccentric circle is the empty focus of the ellipse or that the ancient astronomers practically took the planets to be moving with uniform angular motion round the empty focus. This was not a bad approximation.

Also ES=r=EH approximately,

$$\therefore \quad r=a\left(1-\frac{p}{a}\cos\theta\right)$$

but in the elliptic motion

$$r=a(1-e\cos\theta).*$$

Hence the error is not very considerable here also.

This is the way in which the ancient astronomers, both Greek and Hindu, sought to explain the inequalities in the motion of the Sun and the Moon. In the case of the Moon, these astronomers took the coefficient $2e-\frac{e^3}{4}=300'$ nearly; the modern value is 377′ nearly. The reason for this has been pointed out to be that the Moon was observed correctly only at times of eclipses. At the eclipses of sygygies, the evection term of the Moon's equation diminishes (numerically) the principal ecliptic term by about 76′.

We have thus far explained the idea of planetary motion of the ancients under the eccentric circle costruction. The same, however, is explained under the epicyclic construction.

Let AMP be the circular orbit of the Sun, having E the centre of the earth for the centre. (Fig. 16) Let the diameter AEP be the apse line. A the apogee and P the perigee. Let M be the mean position of the Sun in the orbit. With M as the centre, describe the epicycle UNS. Let EM cut the epicycle at N and U. Now the construction for finding S the apparent Sun is thus given:—

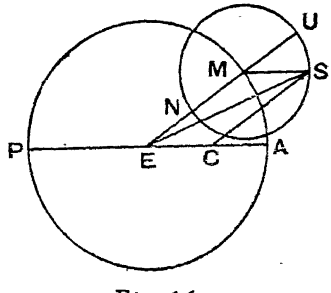

Fig. 16

Make ∠UMS=∠MEA, the arc US is measured clockwise whereas the arc A to M is measured counterclockwise.

From this construction MS is parallel to EA. If EC be measured equal to MS, the radius of the epicycle, along EA to-

* Godfray's *Astronomy*, p. 149.

wards the apogee, then CS is a constant length and C is a fixed point. Hence the locus of S is an equal circle with the centre at C. Thus both the eccentric, and the epicycle and the concentric combined, led to the same position.

It was thus usual to explain the planetary motion under both the assumed constructions; and both gave the position for a planet. The eccentric circle construction appears to be the earlier in the history of astronomy and the latter was later. If the former construction can be traced to Apollonius of Perga who did so much to develop the "conic sections" as science, the reason why he preferred the eccentric circle to the ellipse, appears to be that either that this planetary construction was always deep-rooted in the minds of men or that he was carried by the idea that "the circle was the most perfect curve." We are inclined to the view that the eccentric circle idea was transmitted from Babylonia to Greece. We now pass on to consider the Indian construction for the possition of superior and inferior planets.

Superior Planets

With regard to the five planets, Mercury, Venus, Mars, Jupiter and Saturn, the Indian astronomers give only one construction for finding the apparent geocentric position. Each of these "star planets" is conceived as having twofold planetary inequalities : (i) the inequality of apsis, (ii) the inequality of the *śīghra*. With regard to the superior planets, the *śīghra* apogee or the *śīghrocca* coincided with the mean position of the Sun. As Varāhamihira observed, of the other planets beginning with Mars, the Sun is the so-called *śīghra*. (*PSi*. XVII. 1)

Let AMSP be the concentric of which the centre E is the same as that of the earth(Fig.17); $A'M_1P'$ the eccentric circle of apsis of a superior planet, of which the centre is C; A, M, S, P be respectively the apogee, the Mars planet, the direction of the *śīghra* and the perigee of the concentric; A', M_1, P' be the apogee, the planet as corrected by the equation of apsis, P' the perigee in the eccentric. The arc

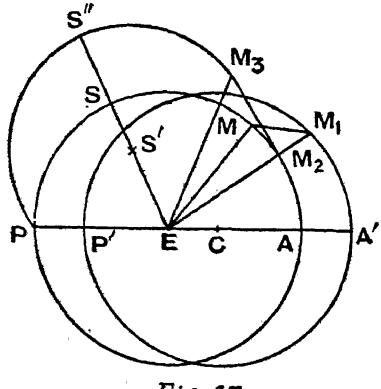

Fig. 17

AM=arc A'M_1; MM_1 is parallel and equal to EC. As used before, both the concentric and the eccentric are of the same radius.

Here the mean planet M in the concentric is taken to be deflected to M_1 due to the true motion in the eccentric circle. Join EM_1 cutting the concentric at M_2. Now let ES be joined and let S' be taken along ES, such that

$$\frac{ES'}{ES} = \frac{\text{Śighra periphery of the planet in degrees}}{360}$$

$$= \frac{\text{Sun's mean distance from the Earth}}{\text{Planets mean distance from the Sun or the Earth;}}$$

ES' thus determined is called the radius of the *śighra* epicycle of the superior planet.

With S' as the centre and the radius equal to ES or EA describe another circle which is called the *śighra* eccentric cutting ES produced at S''. Now measure the arc S''M_3 in the concentric=SM_2 in the concentric. The apparent superior planet is seen in the direction EM_3 from the Earth. This is the construction used in Hindu astronomy calculating the geocentric longitude of any star planet.

It is evident in the case of a superior planet that the eccentric having S' for the centre and whose radius=EA=R the standard radius for any circular orbit, is the mean orbit of the planet and S' the mean position of the Sun. In other words, in the case of a superior planet, the *śighra* eccentric represents the mean orbit round the Sun. If the parallelogram CES'C' be constructed, then an equal circle described with C' as the centre is the apparent eccentric orbit of the superior planet.

In the actual method of calculating the geocentric longitude of a "star planet" there are four operations given, the first two of which have the effect of changing the arc MA or rather the point A. The last two operations relate to the two displacements MM_1 and M_2M_3. We have here followed solely the construction of the eccentric circles; the same geocentric position of a superior planet could be equally well obtained by the epicyclic construction. In describing the constructions for finding the position, of an inferior planet we shall follow the epicyclic construction only

Inferior Planets

Let E be the centre of the Earth (Fig. 18), AMS the orbit of

a mean inferior planet or the mean Sun. EA the direction of the apogee of apsis and ES that of the *śighra*. The inequality of the apsis takes the mean geocentric planet from M to M_1, such that MM_1 is parallel to EA. Let EM_1 be joined cutting the concentric at M_2 ; M_2 is taken as the centre of the *śighra* epicyle or the real circular orbit in which the apparent planet moves.

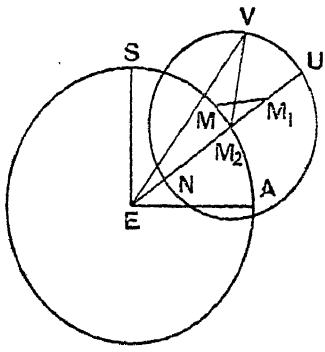

Fig. 18

With M_2 as the centre and the radius of the inferior planents' *śighra* epicycle as radius, describe the circle NVU which is here the *śighra* epicycle or the real circular orbit. In it draw the radius M_2V parallel to ES; then VP is the geocentric position of the inferior planet.

Here the first displacement MM_1 is due to the inequality of apsis and is for finding the position of M_2 the centre of the real circular orbit. The idea was that the apparent planet moved in a circular orbit of which the centre was very near the mean position of the Sun, the first operation in this construction was calculated to determine the centre of this so-called circular orbit of an inferior planet.

The *śighra* of an inferior planet moves round the Earth at the same mean rate in which the inferior planet moves round the Sun; hence the line ES in this figure is always parallel to the line joining the Sun to the mean heliocentric inferior planet, and in our construction, it is parallel to M_2V.

This in brief is an outline of the Indian idea of planetary motion as taught by Āryabhaṭa I, Brahmagupta and Bhāskara II.

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Reference

P.C. Sengupta : The *Khaṇḍakhādyaka*, 1934.

CHAPTER VIII

Brahmagupta and Arithmetic

Scope of Gaṇita

The word *Gaṇita* means the science of calculation. The term occurs in the *Vedāṅga Jyautiṣa* (c. 1200 B. C.) :

> Just as the crest is to the peacocks, and just as the head-gem is to the snakes, so the Gaṇita among the Vedāṅga Śāstras stands at the head.[1] (VJ. 4)

In the ancient Buddhistic literature, we find mention of three classes of *gaṇita* : (i) *mudrā* (finger arithmetic). (ii) *gaṇanā* (mental arithmetic), and (iii) *saṁkhyāna* (higher arithmetic in general). In the *Brāhmasphuṭasiddhānta*, Brahmagupta uses the word *gaṇita* in the sense of entire calculations. His *gaṇitādhyāya* (Chapter XII) includes ;

> (i) *Miśraka* (mixtures). (ii) *Śreḍhī* (series). (iii) *Kṣetra* plane figures), (iv) *Vṛtta-kṣetra* (circles), (v) *Khāta* (excavations), (vi) *Citi* (piles of bricks), (vii) *Krākacika* (sawn pieces of timber), (viii) *Rāśi* (heaps or mounds of grain, and (ix) *Chāyā* (shadow).

Brahmagupta also uses the term *Dhūlikarma* (literally meaning "dustwork") for higher mathematics :

> The one learned man who knows the *dhūlikarma* or the science of mathematics as propounded by Brahmagupta would far excell them in learning who are taught the calculations according to Āryabhaṭa. Viṣṇucandra and others.[2]

In these ten chapters of the *Brahmasiddhānta* has been given the *dhūlikarma* or the science of entire calculations

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1. यथा शिखा मयूराणां नागानां मणयो यथा ।
   तद्वद् वेदांग-शास्त्राणां गणितं मूर्धनि स्थितम् ॥ *VJ. 4.*
2. नावायों ज्ञातेरपि तन्त्रैरार्यभटविष्णुचन्द्राद्यैः ।
   यो ब्रह्म धूलिकर्मविदाचार्यत्वं भवति तस्य ॥ —*BrSpSi. X. 62.*

which is faultless.[1]

This science of *dhūlikarma* has not been imparted by great teachers for blasphemy. One who would be using it for this purpose would lose all good name.[2]

Brahmagupta uses the term *gaṇita* only for those calculations which are of arithmetical in nature. The science of algebra, the foundations of which was laid by Āryabhaṭa I, was named as *kuṭṭaka* or *kuṭṭākāra* by Āryabhaṭa, and in the *Brāhmasphuṭa-siddhānta* also it is separately dealt with under *Kuṭṭādhyāya* or *kuṭṭakādhāya* (Chapter XVIII). Later on the term *bījagaṇita* was specifically given to the science of algebra.

The *Kuṭṭādhyāya* of the *Brāhmasphuṭasiddhānta* deals with the (i) concept of *kuṭṭaka* (pulveriser), addition of positive and negative as well as zero quantities, equations in one unknown (*eka-varṇa samīkaraṇa*), equations in several unknowns (*aneka-varṇa samīkaraṇa*), equations involving products of unknowns (*bhāvita*) and quadratic equations (*varga-prakṛtiḥ*) (Chapter XVIII of the *Brāhmasphuṭasiddhānta*).

Āryabhaṭa, Bhāskara and Brahmagupta use Place Value Notations.

In Europe the first definite traces of the place-value numerals are found in the tenth and eleventh centuries, but the numerals came into general use in mathematical text books only in the seventeenth century. In India, however, Āryabhaṭa I (499), Bhāskara I (522), Lalla (c. 598) and Brahmagupta (628) all use the place value numerals. There is no trace of any other system in their works. Perhaps in this country we had the place value system as early as 200 B.C. if not earlier. The use of a symbol for zero is found in Piṅgala's *Chandaḥ Sūtra* (perhaps of 200 B.C.). In literature, we have an indication of the place value from about 100 B.C. and later in the Purāṇas from the second to the fourth century A.D. The *Bakhaśālī Manuscript* (perhaps of 200 A.D.) uses the place-value notations. The earliest use of the place value principle with the letter numerals

1. ग्रहयोगोत्र ग्रहयुतिरार्यात्रिशतियुताष्टसप्तत्या ।
अध्यायैर्दशभिर्धूलिकर्म वोच्चैर्विना ब्रह्म ॥ —*BrSpSi. X. 66.*

2. गुरुणा न धूलिकर्म प्रतिकंचुककारिणे प्रदातव्यम् ।
दत्त मुक्तप्रणाशं कुरुते प्रतिकंचुकं यस्य ॥ —*BrSpSi. X. 67*

is found in the works of Bhāskara I about the beginning of the sixth century A.D. Thus for 3179, the expressive words are *Navādrirūpāgni*[1] (*nava* 9, *adri* 7. *rupa* 1 and *agni* 3). Similarly in the *Brāhmasphuṭasiddhānta*, for a large number like 2296828522, the expressive terms are DVIYAMAŚARĀṢṬAPAKṢAVASŪRASANAVADVIYAMĀḤ(*Dviyama* two twos 22, *Śara* 5,aṣṭa 8, *pakṣa* 2, *vasū* 8, *rasa* 6. *nava* 9, *dviyamāḥ* 22).[2] Such usages are to be found in all works, which clearly state the place value concept was popular as a routine. From India, this system reached Arabia. During the reign of the Khalif Al-Mansur (753-774 A.D.) there came embassies from Sindh to Baghdad, and among them were scholars, who brought along with them several works on mathematics including the *Brāhmasphuṭasiddhānta* and the *Khaṇḍakhādyaka* of Brahmagupta: With the help of these scholars, Alfazari, perhaps also Yakub ibn Tarik, translated them into Arabic. Both works were largely used and exercised great influence on Arab mathematics. It was on that occasion that the Arabs first became acquainted with a scientific system of astronomy. It is acceptable to all writers on the subject that it was at that time that the Hindu numerals were first definitely introduced amongst the Arabs. Arabs at first adopted the *ghobar* form of numerals which they had already obtained (but without zero) from the Alexandrians or from the Syrians. This they continued for about two centuries, but since they were not suited to their right-to-left script, they gave them up and adopted the more convenient ones. For a detailed discussion on how numerals went to the west from India and spread in Europe one is referred to this discussion in the *History of Hindu Mathematics*. Part I by Datta and Singh (1935, Single volume Edition, 1962, pp. 83-104). It is remarkable that Brahmagupta's works like the *Brāhmasphuṭasidhānta* and the *Khaṇḍakhādyaka* became instrumental in the spread of the place-value notation in the neighouring countries of the Middle East, and from their this system spread into Europe.

Operations and Determinations in Pāṭīgaṇita

The word *Pāṭīgaṇita* is a compound formed from the words *pāṭī*, meaning 'board', and *gaṇita*, meaning 'science of calculation',

1. *MBh. 1. 4;*
2. *BrSpSi 1-16.*

hence it means the science of calculation which requires the uses of writing material (the board). The word *pāṭi* is not Sanskrit (it originated in the non-Sanskrit literature in India); the oldest term in Sanskrit for the board is *Phalaka* or *paṭṭa*. However this term got currency in the Sanskrit literature also about the beginning of the seventh century. Brahmagupta does not use the term *pāṭigaṇita*: he favours the use of the term *dhūlikarma* or writing figures on dust spread on a board or on the ground. The word *pāṭīgaṇita* was translated into Arabic as *ilm-hisab-al-takht* (calculation on board) and the word *dhūlikarm a* as *hisab-al-ghobār* (calculation on dust).

Brahmagupta, in the very first verse in the Chapter XII (*Gaṇitādhyāya*) refers to twenty operations (*parikarma*) and eight determinations :

> He who distinctly and severally knows the twenty logistics, addition etc.. and the eight determinations (*vyavahāra*) inculding (measurement by) shadow is a *gaṇaka* (mathematician).[1]

The commentators have given the list of these logistics (*parikarma*) and determinations (*vyavahāra*) as follows;

(A) *Parikarma* or logistics

1. Saṁkalitam (addition)
2. Vyavakalitam (subtraction)
3. Gaṇanam (multiplication)
4. Bhāgahāraḥ (division)
5. Vargaḥ (square)
6. Vargamūlam (square-root)
7. Ghanaḥ (cube)
8. Ghanamūlam (cube root)
9. 13. Five standard forms of fractions (Pañca-jāti)
14. Trairāśikam (the rule of three)
15. Vysta-trairāśikam (the inverse rule of three)
16. Pañca-rāśikam (the rule of five)
17. Sapta-rāśikam (the rule of seven)
18. Nava-rāśikam (the rule of nine)

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1. परिकर्म विंशतिं यः संकलितप्रथां पृथक् विजानाति ।
अष्टौ च व्यवहारान् छायान्तान् भवति गणकः सः ॥ —*BrSpSi. XII. 1*

19. Ekādaśa-rāśikam (the rule of eleven)
20. Bhāṇḍa-pratibhāṇḍam (barter and exchange)

(B) *Vyavahāra* or determinations

1. Miśrakaḥ (mixture)
2. Śreḍhī (progession or series)
3. Kṣetram (plane figures)
4. Khātam (excavation)
5. Citiḥ (stock)
6. Krākacikaḥ (saw)
7. Rāśiḥ (mound)
8. Chāyā (shadow)

Of the operations enlisted here, the first eight have been considered fundamental by later writers as Mahāvīra. The operations of duplation and mediation (doubling and halving) were considered fundamental by Arabs, Greeks and Egyptians; since they were not familiar with the place-value system.

Mathematics in this country developed as an aid to astronomy, and therefore, for the first time we find Āryabhṭa(499AD.) in his *Āryabhaṭiya* describing as a special section (*Gaṇitapāda*). Brahmagupta (628 A.D.) also followed Āryabhaṭa in this respect and gave the science of calculation (*gaṇita*) a special place in his treatise on astronomy. The *Siddhānta* treatises, earlier than those of Āryabhaṭa and Brahmagupta do not contain a chapter exclu- ʌɪsʌɪᵊ devoted to *gaṇita* (the *Sūrya-Siddhānta* and the Siddhāntas of *Vasiṣṭha,Pitāmaha* and *Romaka* are thus without *gaṇita* chapetrs). Later on Bhāskara I and Lalla also did not include *gaṇita* as a section or chapter in their treatises.It is said, however, that Lalla wrote a separate treatise on *Pāṭīgaṇita.*

It may further be remarked here that Āryabhaṭa I gives the rules for finding the square and cube-roots only whilst Brahmagupta gives the cube-root rule only (*BrSpSi.* XII. 7).

Multiplication

Undoubtedly the common Indian name 'multiplication' is *guṇana*', this term occurs in the Vedic literature also. The other terms for this logistics are *hanana, vadha, kṣaya* etc., which all mean 'killing' or 'destroying.' The synonyms of '*hanana*' (killing) for multiplication have been used by Āryabhaṭa I (499). Brahmagupta (628), Śrīdhara (c.750) and later writers, and these terms

also occur in the Bakhaśālī Manuscript.

Āryabhaṭa I does not mention the everyday methods of multiplication in his *Āryabhaṭiya* probably because they were too elementary to be included in a Siddhānta work. Brahmagupta, however, in a supplement to the section on mathematics in his *Siddhānta*, gives the names of some methods with very brief descriptions of the processes:—

> The multiplicand repeated, as in *gomūtrikā* as often as there are digits in the multiplier, is severally multiplied by them and (the results) added according to places; this gives the product. Or the multiplicand is repeated as many times as there are component parts in the multiplier.[1]

(the word *bheda* occurring in the verse has been translated as "integrant portions" by Colebrooke p. 319. Again by the term *bheda* are meant portions which added together make the whole, or aliquot parts which multiplied together make the entire quantity.

> The multiplicand is multiplied by the sum or the difference of the multiplier and an assumed quantity and, from the result the product of the assumed quantity and the multiplicand is subtracted or added.[2]

(Colebrooke thinks that this is a method to obtain the true product when the multiplier has been taken to be too great or too small by mistake.[3] Datta and Singh think, however, that this is not correct.[4]

Thus Brahmagupta mentions four methods of multiplication: (i) *gomūtrikā*, (ii) *khaṇḍa*, (iii) *bheda*, and (iv) *iṣṭa*. The common and the well known method of *kapāṭa-sandhi* has been omitted by him.

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1. गुणकारखण्डतुल्यो गुण्यौ गोमूत्रिकाकृतो गुणितः ।
सहितः प्रत्युत्पन्नो गुणकारकभेदतुल्यो वा ॥ —*BrSpSi XII. 55*

2. गुण्यो राशिर्गुणकारराशिनेष्टाधिकोनकेन गुणः ।
गुरुष्टघ्नो न युतो गुणकेऽभ्यधिकोनके कार्यः ॥ —*BrSpSi. XII. 56*

3. Colebrooke, T. H., *Hindu Algebra*, p. 320.

4. Datta, B. and Singh, A. N., *History of Hindu Mathematics* Pt. I (Arithmetic), p. 135 (1962).

(i) *Gomūtrika*-method or zig-zag method. The word *gomūtrikā* means "similar to the course of cow's urine", hence "zigzag". This method in all essentials is the same as the *sthāna-khaṇḍa* method. The following illustration is based on the commentary of Pṛthūdaka Svāmī :

Example : To multiply 1223 by 235.

The numbers are written thus :

2 1223
3 1223
5 1223

The first line of figures is t hen multiplied by 2, the process beginning at units place, thus : $2\times3=6$; 3 is rubbed out and 6 substituted in its place, and so on. After all the horizontal lines have been multiplied by the corresponding numbers on the left in the vertical line, the numbers on the *pāṭī* stand thus :

2446
3669
6115
287405

after being added together as in the present method.

The *sthāna-khaṇḍa* and the *gomūtrikā* methods resemble modern plan of multiplication most closely.

(ii) *Khaṇḍa* Method or Parts Multiplicatian Method : Since the days of Brahmagupta, this method of multiplication also became very popular. We have two methods under this head :

(i) The multiplier is broken up into two or more parts whose sum is equal to it. The multiplicand is then multiplied severally by these and the results added.

To take an example :

$$13\times158=(6+7)\times158=(6\times158)+(7\times158)$$
$$=948+1106$$
$$=2054$$

(ii) The multiplier is broken up into tw o more aliquot parts. The multiplicand is then multiplied by one of these, the resulting product by the second and so on till all the parts are exhausted. The ultimate product is the result.

Thus for example :

$$96\times237=(4\times4\times6)\times237$$
$$=(4\times237)\times4\times6=948\times4\times6$$
$$=(4\times948)\times6=3792\times6$$
$$=22752$$

These methods of multiplication are found among the Arabs and the Italians, having obtained from people of India. They were known as the "Scapezzo" and "Repiego" methods respectively amongst Italians.

(iii) *Iṣṭa-guṇana* Method or the Algebraic Method.

We have already quoted the relevant verse from the *Brāhmasphuṭa-siddhānta* in this connection; (XII. 56) :

> The multiplicand is multiplied by the sum or the difference of the multiplied and an assumed quantity and from the result the product of the assvmed quantity and the multiplicand is subtracted or added.[1]

This method is of two kinds according as we (i) add or (b) subtract an assumed number. The assumed number is so chosen as to give two numbers with which multiplication will be easier than with the original multiplier. The two ways are illustrated below :

(i) $93\times13=(93+7)\times13-7\times13=1300-91=1209.$

(ii) $93\times13=(90+3)13=90\times13+3\times13=1170+39$
$=1209$

This method was in use among the Arabs and in Europe, obviously having gone out from this country.

This process has been regarded as an inverse of multiplication. The terms used for this operation are *bhāgahāra, bhājana, haraṇa, chedana*, etc., all these terms more or less carrying the sense "to break into parts", "to divide" etc., excepting "*haraṇa*" which denotes "to take away". This term shows the relation of division to the operation of subtraction. The dividend is termed as *bhājya, hārya* etc., the divisor is known as

1. गुरवो राशिगुणकारराशिनेष्टाधिकोनकेन गुणः ।
गुण्येष्टघ्नो न युतो गुणकेऽभ्यधिकोनके कार्यः ॥ —*BrSpSi. XII. 56*

bhājaka, bhāgahāra or simply *hara*; quotient is known as *labdhi* or *labdha* (or "what is obtained").

India never regarded this operation as a difficult one; in Europe, this operation was regarded as a tedious one till the 15th century or so. Division was such a common operation that Āryabhaṭa did not regard it as worth being included in his treatise. But since he has given the methods of extracting square-roots and cube-roots, which obviously depend on division, we conclude that the method of division was known to him. Most *Siddhānta* writers have followed Āryabhaṭa I in omitting this operation from their texts, this being regarded too elemantary to be included. Brahmagupta does not give details of this operation. The later treatises on Arithmetic as Śrīdhara's *Triśatikā* and the *Pāṭīgaṇita* (I.20) and Āryabhaṭa II (c.950 A.D.) have given the details of this operation.

Square

The Sanakrit term for square is *varga* or *kṛti* (*varga* literally means "rows" or "troops" of similar things). In mathematics, it usually means the square power and also the square figure or its area. Thus we find in the *Āryabhaṭiya* :

> A square figure of four equal sides (and the number representing its area) are called *varga*. The product of the two equal quantities is also *varga*[1].

The term *kṛti* means "doing", "making" or "action". It carries with it the idea of specific performance probably the graphical representation.

For the first time we have a definite rule for squaring in the writings of Brahmagupta. But it does not mean that prior to him it was not known. It must have been known to Āryabhaṭa I since he has given the square-root method.

Brahmagupta gives his method of squaring briefly as follows :

> Combining the product, twice the digit in the less (lowest) place into the several others (digits) with its (i.e. of the digit in the lowest place) square (repeatedly) gives the square.[2]

1. वर्गस्समचतुरश्रः फलञ्च सदृशद्वयस्य संवर्गः । —*Ārya. II.* 3.
2. राशेरूनं द्विगुणं बहुतरगुणमूनकृतियुतं वर्गः । —*BrSpSi. XII.* 63.

The method has been more clearly enunciated by Mahāvīra (850 A.D.) in the *Gaṇitasārasaṁgraha :*

> Having squared the last (digit), multiply the rest by the digits by twice the last, (which) is moved forward (by one place). Then moving the remaining digits continue the same operation (process), This gives the square.[1]

Brahmagupta's method of squaring is shown by the following example :

To square 125.

The number is written down

125

The square of the digit in the last place, i. e., $5^2=25$ is set over it thus :

25
125

Then, $2 \times 5=10$ is placed below the other digits, and 5 is rubbed out, thus :

25
12
10

Multiplying by 10 the rest of the digits, i.e., 12 and setting the product over them (the digits), we have.

1225
12
10

Then rubbing out 10 which is not required and moving the rest of the digits, i. e. 12 we, have

1225
12

Thus one round of operations is completed.

Again as before, setting the square of 2 above it and $2 \times 2=4$ below 1. we have

1625
1
4

1. GSS. P. 12.

Multiplying the remaining digit 1 by 4, and setting the product above it, we have

5625
1

Then moving the remaining digit 1, we obtain

5625
1

Thus the second round of operations is completed.

Next setting the square of 1 above it the process is completed for there are no remaining figures, and the result stands thus :

15625

Algebraic Method of Squaring

Brahmagupta in his *Brāhmasphuṭasiddhānta* gives a minor method of squaring thus :

> The product of the sum and the difference of the number (to be squared) and an assumed number plus the square of the assumed number give square[1].

This may be represented by the following identity :

$$n^2=(n-a)\ (n+a)+a^2$$

This identity has been used for squaring by most of the Indian mathematicians. Thus

$$15^2=(15-5)\ (15+5)+5^2=225$$

We are not giving here other identities which have been used by latter mathematicians of India in getting the squares of numbers; for example, when Mahāvīra says :

> The sum of the squares of the two or more portions of the number together with their products each with the others multiplied by two gives the square[2] :

he obviously refers to the identity

$$(a+b+c.....)^2=a^2+b^2+c^2+.....+2ab+.....$$

Cube

The Sanskrit term for cube is *ghana*. It when used in the geometrical sense also means the solid cube. In the arithmetical sense, it means the continued product of the same number taken three times. Thus we have the definition in the *Ārya-*

1. राशेरिष्ट्युतोनाद्वधः कृतिर्वेष्टकृतियुक्तः । — *BrSpSi. XII. 63*
2. *GSS.* p. 13.

bhaṭya :

> The continued product of three equals and also the solid having twelve (equal) edges are called *ghana*.[1]

A method of cubing applicable to numbers written in the decimal place-value notation, has been in use in this country from before the 5th century A.D. Āryabhaṭa I (499 A.D.) had the familiarity with this method; he, however, does not give the method of cubing in his treatise, though he describes the inverse process of extracting the cube-root.

Brahmagupta gives the method of cubing in the following verse :

> Set down the cube of the last (*antya*); then place at the next place from it, thrice the square of the last multiplied by the succeeding; then place at the next place thrice the square of the succeeding multiplied by the last, and (at the next place) the cube of the succeeding. This gives the cube.[2]

The rule may be illustrated by an example.

Example : To cube 1357.

The given number has four places, i.e., four portions. First we take the last digit 1 and the succeeding digit 3, i.e. 13 and apply the method of cubing thus :

(i) Cube of the last (1^3)	=	1
(ii) Thrice the square of the last (3.1^2) multiplied by the succeeding (3) gives ($3.3.1^2$)	=	9 (placing at the next place)
(iii) Thrice the square of the succeeding, multiplied by the last gives ($3.3^2.1$)	=	27 (placing at the next place)
(iv) Cube of the succeeding (3^3)	=	27 (placing at the next place)
Thus 13^3 is the sum		2197

1. सदृशत्रयसंवर्गो घनस्तथा द्वादशाग्रस्यात् ॥ —*Āryа, II. 3.*
2. स्थान्योऽन्त्यघनोऽन्त्यकृतिस्त्रिगुणोत्तरसंगुणा च तत्प्रथमात् ।
उत्तरकृतिरन्त्यगुणा त्रिगुणा चोत्तर घनश्च घनः । —*BrSpSi. XII, 6.*

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उत्तरकृतिरन्त्यगुणा त्रिगुणा चोत्तर घनश्च घनः । —*BrSpSi. XII, 6.*

After this we take the next figure, 5, i.e., the number 135, and in this consider 13 as the last and 5 as the succeeding. Then the method proceeds thus :

(i) The cube of the last (13^3) as already obtained	= 2197	
(ii) Thrice the square of the last multiplied by the succeeding, i.e. $3.13^2.5$	= 2595	(placing at the next place)
(iii) Thrice the square of the succeeding multiplied by the last, i.e, $3.5^2.13$	= 975	(placing at the next place)
(iv) Cube of the succeeding, i.e. 5^3	= 125	(placing at the next place)
Thus 135^3 is the sum	2460375	

Now the remaining figure 7 is taken, so that the number is 1357, of which 135 is the last and 7 the succeeding. The method proceeds thus :

(i) Cube of the last, i.e. (135^3) as already obtained	= 2460375	
(ii) Thrice the square of the last into the succeeding, i.e, $3.(135)^2.7$	= 382725	(placing at the next place)
(iii) Thrice the square of the succeeding into the last, i.e. $3.7^2.135$	= 19845	(placing at the next place)
(iv) Cube of the succeeding i.e. 7^3	= 343	(placing at the next place)
Thus $(1357)^3$ is the sum	2498846293	

Evidently these methods of cubing are based on the identity:

$$(a+b)^3=a^3+3a^2b+3ab^2+b^3$$

and keeping in mind the place values of numerals in a given

number (this accounts for keeping the results of each of the four operations at the next place).

Square-Root

Indian synonyms for square-root are *vargamūla* or *pada* of a *kṛti*. The word *mūla* means the "root" of a tree, which may also mean the "foot" or the lowest part or bottom of a thing and hence "pada" or foot also became a synonym of root. Brahma-Gupta defines square-root as follows :

> The *pada* (root) of a *kṛti* (square) is that of which it is the square.[1]

While the word *mūla* for *root* is the oldest in Indian literature (it occurs in *Anuyogadvāra-sūtra*, c. 100 B.C.), the word *pada* for root probably for the first time occurs in the writings of Brahmagupta. The term *mūla* was borrowed by the Arabs who translated it by *jadhr*, meaning "basis of square". The Latin term *radix* also is a translation of the term *mūla*. In the Śulba literature and in the Prākṛta texts, we find a term *karaṇī* for square-root. In geometry, this term *karaṇī* means a "side", In later days, the term *karaṇī* was reserved for surds, i.e. a square-root which cannot be exactly evaluated, but which may be represented by a line.

We would like to quote here a rule for determining square-root of numbers from the *Āryabhaṭīya* :

> Always divide the even place by twice the square-root (upon the preceding odd place); after having subtracted from the odd place the square (of the quotient), the quotient put down at the next place (in the line of the root) gives the root[2].

As an illustration, we shall proceed to find the square-root of 18225.

The odd and even places are marked out by vertical (I) and horizontal (—) lines : The other steps are as follows :

---

1. पदं कृतिर्यत् तत् — *BrSpSi.* XVIII. 35

2. भागं हरेदवर्गान्नित्यं द्विगुणेन वर्गमूलेन ।
वर्गाद्वर्गे शुद्धे लब्धं स्थानान्तरे मूलम् ॥ — *Arya*, II. 4.

	1 - 1 - 1 1 8 2 2 5	
Subtract square	1	root = 1
Divide by twice the root	2) 8 (3 6	placing quotient at the next place, the root=13
	22	
Subtract square of quotient	9	
Divide by twice the root	26)132(5 130	placing quotient at the next place, the root=135
Subtract square of the quotient	25 25	

The process ends. The square-root of 18225 is thus 135.

It has been stated by Kaye, that Āryabhaṭa's method of finding out the square-root is algebraic in character, and that it resembles the method given by Theon of Alexandria. Āryabhaṭa's method is purely atithmetic and not algebraic is the view of Datta and Singh who do not agree with Kaye on this point.

Cube Root

The Sanskrit term for cube-root is *ghanamūla* or *ghanapada.* The first mention of the operation of cube-root is found in the *Āryabhaṭiya* of Āryabhaṭa I (499 A.D.), though the operation is given in only a concise form :

> Divide the second *aghana* place by thrice the square of the cube-root; subtract from the first *aghana* place the square of the quotient multiplied by thrice the preceding (cube-root); and (subtract) the cube (of the quotient) from the *ghana* place; (the quotient put down) at the next place (in the line of the root) gives (the root).[1]

As has been explained by all the commentators on the *Āryabhaṭiya*, the units place is *ghana*; the tens place is first *aghana*, the hundreds place is the second aghana, the thousands place is ghana, the ten thousands place is first *aghana*, the hun-

1. अघनाद् भजेद् द्वितीयात् त्रिगुणेन घनस्य मूलवर्गेण ।
वर्गस्त्रिपूर्व गुणितश्शोध्यः प्रथमाद् घनश्च घनात् ॥ —*Arya.* II. 5

dred-thousands place is second *aghana*, and so on. Thus to find out the cube-root, one has to mark out the *ghana*, first *aghana* and second *aghana* places, then the process of finding out the cube-root begins with the subtraction of the greatest cube number from the figures up to the last *ghana* place. Though this has not been explicitly mentioned in the rule, the commentators say that it is implied in the expression *ghanasya mūla-vargeṇa* etc. ("by the square of the cube-root etc.")

We are reproducing here an illustration given by Datta and Singh.

Example. Find the cube-root of 1953125.

The places are divided into groups of three by marking them as below [*ghana* (|) first *aghana* (—) and second *aghana* (—)]:

	| — — | — — |		
	1 9 5 3 1 2 5		
Subtract cube	1		(c) Root=1
Divide by thrice square of root, i.e. 3.1^2	3)9(2	...	(a) Placing quotient after the root 1 gives the root 12
Subtract square of quotient multiplied by thrice the previous root, i.e. $2^2.3.1$	6 35 12	...	(b)
Subtract cube of quotient, i.e. 2^3	233 8	...	(c)
Divide by thrice square of the root, i.e. 3.12^2	432)2251(5	...	(a) Placing quotient after the root 12 gives the root 125
Subtract square of quotient multiplied by thrice the previous root, i. e. $5^2.3.12$	2160 912 900		
Subtract cube of quotient, i.e. 5^3	125 125	...	(b) (c)

Thus the cube-root=125.

From the details given, it would be clear that the present

method of extracting the cube-root is almost a contraction of the method first given by Āryabhaṭa I (499 A.D.)

The method of Āryabhaṭa has been invariably followed by Indian mathematicians. Brahmagupta in his *Brāhmasphuṭa-siddhānta* repeats the method in the following words :

> The divisor for the second *aghana* place is thrice the square of the cube-root; the square of the quotient multiplied by three and the preceeding (root) must be subtracted from the next (*aghana* place to the right). and the cube (of the quotient) from the *ghana* place (the procedure repeated gives) the root.[1]

Śridhara and Āryabhaṭa II have further improved on the method of extracting cube-root proposed by Āryabhaṭa I and followed by Brahmagupta. Rule for finding the cube-root as given by Śridhara in his *Pāṭigaṇita* is as follows :

> (Divide the digits beginning with the units' place into periods of) one *ghana*-pada (one "cube" place) and two *aghana-padaṣ* (two "non-cube" places). Then subtracting the (greatest possible) cube from the (last) *ghana-pada* and placing the (cube) root underneath the third place (to the right of the last *ghana-pada*), divide out the remainder up to one place less (than that occupied by the cube root) by thrice the square of the cube-root, which, is not destroyed. Setting down the quotient (obtained from division) in the line (of the cube-root), (and designating the quotient as the 'first' (*ādima*) and the cube-root as the 'last' (*antya*), subtract the square of that quotient, as multiplied by thrice the 'last' (*antya*) from one place less than that occupied by the quotient (*uparima-rāśi*) as before, and the cube of the 'first' (*ādima*) from its own place.
>
> (The number now standing in the line of cube-root is the cube-root of the given number up to its last-but one *ghana-pada* (cube place) from the left).
>
> Again apply the rule, "(placing cube-root) under the third place" etc. (provided there be more than two *ghana-padas* (cube places) in the given number; and

1. छेदो घनाद् द्वितीयाद् घनमूलकृतिस्त्रिसंगुणाप्तकृतिः ।
शोध्या त्रिपूर्वगुणिता प्रथमाद् घनतो घनो मूलम् ॥ —*BrSpSi.* XII. 7

continue the process till all *ghana-padas* (cube-places) are exhausted). This will give the (cube) root (of the given number).[1]

K.S. Shukla in his translation and commentary of this book has given the illustration of extracting cube-root as follows :

Example :- To find the cube root of 277167808.

Let us indicate *ghana-padas* or 'cube' places by "*c*" and *aghana-padas* or non-cube places as "*n*" :

n n c n n c n n c
2 7 7 1 6 7 8 0 8

Subract the greatest possible cube (i.e. 6^3 or 216) from the last 'cube' place (i.e. from 277) and place the cube-root (i.e. 6) underneath the third place to the right of the last 'cube' place; thus we have

n n c n n c n n c
6 1 1 6 7 8 0 8 (remainder)
6 (line of cube-root)

Dividing out by thrice the square of the cube-root (i.e. by 3.6^2 or 108) the remainder up to one place less than that occupied by the cube-root (i.e. 611) and setting down the quotient in the line of the cube-root (to the right of the cube-root), we have

n n c n n c n n c
7 1 6 7 8 0 8 (remainder)
6 5 (line of cube-root)

Let now quotient 5 be called the 'first' (*ādima*) and the cube-root 6 the 'last' (*antya*). Then subtracting the square of the 'first' (*ādima*) as multiplied by thrice the 'last' (*antya*) (i.e. $3\times6\times5^2$ or 450) from one place less than that occupied by the quotient (i.e. from 716), we get

1. घनपदमघनप्रदे द्वे घन (पद) तोऽपास्य घनमदो मूलम् ।
संयोज्य तृतीयपदस्याधस्तदनष्टवर्गेण ॥ २९ ॥
एकस्थानोनतया शेषं त्रिगुणेन (सं) भजेत्तस्मात् ।
लब्धं निवेश्य पङ्क्त्यां तद्वर्गं त्रिगुणमन्यहतम् ॥ ३० ॥
जह यादुपरिमराशेः प्राग्वद् घनमादिमस्य (च) स्वपदात् ।
भूयस्तृतीय पदस्याध इत्यादिक विधिर्मूलम् ॥ ३१॥

—Śridhara, *Pāṭīgaṇita*, 29-31

n n c n n c n n c
2 6 6 7 8 0 8 (remainder)
6 5 (line of cube-root)

And subtracting the cube of the 'first' (*ādima*) (i.e. 5^3 or 125) from its own place (i.e. from 2667), we get

n n c n n c n n c
2 5 4 2 8 0 8 (remainder)
6 5 (line of cube-root)

One round of the operation is now over; and the number 65 standing in the line of the cube-root is the cube-root of the given number (277167808) up to its last-but-one 'cube' place (*ghana pada*) from the left (i.e. of 277167),

As there is one more 'cube' place (*ghana-pada*) on the right, the process is repeated. Thus placing the cube-root (i.e. 65) under the third place beginning with the last-but-one 'cube' place (*ghana-pada*), we have

n n c n n c n n c
2 5 4 2 8 0 8 (remainder)
6 5 (line of cube-root)

Dividing out 25428 by 3.65^2 (=12675) as before, and placing the quotient in the line of the cube-root, we have

n n c n n c n n c
7 8 0 8 (remainder)
6 5 2 (line of cube-root)

Subtracting $3\times65\times2^2$ (=780) we get.

n n c n n c n n c
8 (remainder)
6 5 2 (line of cube-root)

Finally subtracting 2^3=8 from 8, we get

n n c n n c n n c
0 (remainder)
6 5 2 (line of cube-root)

The second round of operation is now over. There being no more of *ghana-pada* ('cube' place) on the right, the process ends, The quantity in the line of cube root, viz., 652, is the cube-root of the given

number. The remainder being zero, the cube-root is exact.

Fractions

The concept of fractions in India can be traced to very early times. In the *Ṛgveda*,[1] we find such terms as one-half (*ardha*) and three-fourths (*tri-pāda*). In a passage of the *Maitrāyaṇī Saṁhitā*[2] are mentioned the fractions one-sixteenth (*kalā*), one-twelfth (*kuṣṭha*), one-eighth (*śapha*) and one-fourth (*pāda*). In the *Śūlba Sūtras*[3] we have not only a mention of fractions, but they have been used in the statement and solution of problems of geometric nature. Here in the *Śulba*, unit fractions are denoted by the use of cardinal number with the term *bhāga* or *aṁśa*; thus *pañca-daśa-bhāga* (literally "fifteen parts") is equivalent to one-fifteenth, *sapta-bhāga* (literally, "seven parts") is equivalent to one-seventh, and so on... The use of ordinal numbers with the term *bhāga* or *aṁśa* is also quite common: thus *pañcama bhāga* stands for one-fifth. The composite fractions like *tri-āṣṭama* stands for three-eighths and *dvi-saptama* for two-sevenths. In the *Bakhaśālī Manuscript*, the term *tryaṣṭa* occurs for 3/8 and $3\frac{3}{8}$ is called *trayastrayasta* (three-three-eighths).

The Sanskrit term for fraction is *bhinna* (literally meaning 'broken'). Obviously the European terms as *fractio*, *fraction*, *roupt*, *rotto* or *rocto* are translations of the same term; they are derived from the Latin *fractus* (*frangere*) or *ruptus* meaning 'broken'. The Indian term *bhinna* has a few more connotations; it stands for such numbers of the form:

$$\left(\frac{a}{b}\pm\frac{c}{d}\right),\ \left(\frac{a}{b}\text{ of }\frac{c}{d}\right),\ \left(\frac{a}{b}\pm\frac{c}{d}\text{ of }\frac{a}{b}\right)\text{or}\left(a\pm\frac{b}{c}\right).$$

These forms were termed *jāti*' i.e., 'classes', and the Indian treatises contain special rules for their reduction to proper fractions. Śrīdhara and Mahāvīra each enumerate six *jātis*. while our author, Brahmagupta, gives only five (Bhāskara II gives only four). The need for division of fractions in 'classes' arose out of the lack of proper symbolism to indicate mathematical operations. (Datta and Singh *Arithmetic*, p. 188). The only operational symbol in use was a dot, standing for the negative sign.

1. *Rv.* X. 90.4
2. *Mait* S. III. 7.7.
3. B. Datta, *Śulba*, pp. 212ff.

Reduction to lowest terms.—A non-mathematical work, *Tattvārthādhigama-Sūtra-Bhāṣya* by Umāsvāti (c.150A.D.) casually mentions as follows in the context of a philosophic discourse:

> Or, as when the expert mathematician, for the purpose of simplifying operations, removes common factors from the numerator and denominator of a fraction, there is no change in the value of the fraction, so....[1]

Reduction to common denominator. Whenever we have to add or subtract fractions, we follow this reduction operation to a common denominator. Brahmagupta gives the reduction along with the similar processes :

> By the multiplication of the numerator and denominator of each of the (fractional) quantities by other denominators, the quantities are reduced to a common denominator. In addition, the numerators are united, In subtraction their difference is taken.[2]

Fractions in combination :—Since there was no proper symbolism available to these early Indian mathematicians, they divided combination of fractions into four classes :

Bhāga, prabhāga, bhāgāpavāha and *bhāga-bhāga.*

(i) *Bhāga* has been mentioned by Brahmagupta (*BrSpSi.* XII. 8) thus : $\left(\frac{a}{b} \pm \frac{c}{d} \pm \frac{e}{f} \pm \ldots\ldots\right)$ usually written as

$$\left|\begin{array}{c|c|c} a & c & e \\ b & d & f \end{array}\right| \text{ or } \left|\begin{array}{c|c|c} a & .c & .e \\ b & d & f \end{array}\right|$$

where the dots denote subtraction.

(ii) *Prabhāga* : The form $\left(\frac{a}{b} \text{ of } \frac{c}{d} \text{ of } \frac{e}{f} \ldots\ldots\right)$

This is written as

$$\left|\begin{array}{c|c|c} a & c & e \\ b & d & f \end{array}\right|$$

(iii) *Bhāgānubandha* :The form $\left(a + \frac{b}{c}\right)$ is written as

$$\left|\begin{array}{c} a \\ b \\ c \end{array}\right|$$

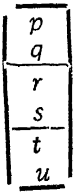
1. II, 52.

2. विपरीतच्छेदगुणा : राश्योश्छेदांशकाः समच्छेदाः ।
संकलितेऽशा योज्या व्यवकलितेऽशान्तरं कार्यम् ॥ —*BrSpSi.* XII. 2.

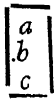
and the form

$$\frac{p}{q}+\frac{r}{s} \text{ of } \frac{p}{q}+\frac{t}{u}\text{of}\left(\frac{p}{q}+\frac{r}{s}\text{ of}\frac{p}{q}\right)+\ldots\ldots\ldots$$

is written as

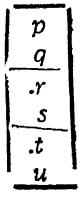


(iv) *Bhāgāpavāha*, i.e., the form $\left(a-\frac{b}{c}\right)$ is written as



and the form $\frac{p}{q}-\frac{r}{s}$ of $\frac{p}{q}-\frac{t}{u}$ of $\left(\frac{p}{q}-\frac{r}{s}\text{ of }\frac{p}{q}\right)-\ldots\ldots\ldots$

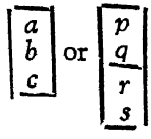
is written as



(v) Bhāga-bhāga : The form

$$\left(a\div\frac{b}{c}\right) \text{ or} \left(\frac{p}{q}\div\frac{r}{s}\right)$$

There does not appear to have been any notation for division, such compounds being written as



just as for *bhāgānubandha*. That division is to be performed was known from the problem, e.g., $1\div\frac{1}{6}$ was written as ṣaḍ-bhāga-bhāga, i.e., "one-sixth *bhāga-bhāga*" or "one divided by one-sixth". It is only in the Bakhśālī Manuscript that the term *bhā* is sometimes placed before or after the quantity affected.

(vi) *Bhāga-mātṛ*, i.e., combinations of forms enumerated above. Mahāvīra, the author of the *Gaṇitasārasaṁgraha* (850

A.D.) gives twenty-six variations of this class. We shall illustrate it by the following example from Śrīdhara :

> What is the result when half, one-fourth of one-fourth, one divided by one-third, half-plus half of itself, and one-third diminished by half of itself, are added together ? (*Triśatikā*, p. 12).

A modern writer would have written it as :

$\frac{1}{2}+(\frac{1}{4}$ of $\frac{1}{4})+(1\div\frac{1}{3})+(\frac{1}{2}+\frac{1}{2}$ of $\frac{1}{2})(\frac{1}{3}-\frac{1}{2}$ of $\frac{1}{3})$

In the old Indian notation, it is written as

1	1	1	1	1	1
2	4	4	1	2	3
			3	1	.1
				2	2

The defect of the notation is obvious: $\left|\begin{matrix}1\\4\end{matrix}\right|\left|\begin{matrix}1\\4\end{matrix}\right|$ can be read also as $\frac{1}{4}+\frac{1}{4}$ and $\left|\begin{matrix}1\\1\\3\end{matrix}\right|$ can also be read as $1\frac{1}{3}$.

And therefore the original meaning is inferred from the context or from the enunciation of the problem.

The rules for reduction of the first two classes (*bhāga* and *prabhāga)* are those of addition or subtraction and multiplication. The rule for the reduction of the third (*bhāgānubandha)* and fourth (*bhāgāpavāha)* classes are given by Brahmagupta in the *Brāhmasphuṭa-siddhānta* thus :

> The (upper) denominator is multiplied by the denominator and the upper numerator by the same (denominator) increased or diminished by its own numerator.[1]

"Numerator" is known as "*aṁśa*" and the "denominator" as "*cheda.*"

We give here from Śrīdhara's *Pāṭigaṇita* (about 900 A.D., according to K.S. Shukla, 750 A.D. according to Datta and Singh) a rule for reducing a fraction of the *bhāgānubandha* class (i.e., a whole number increased by a fraction or a fraction increased by a fraction itself) :

1. ऊर्ध्वं रागश्छेदगुणास्तृतीयजातौ द्वयोः पृथक्परयोः ।
छेदैश्छेदा गुणिताः स्वांशयुतोनैरुपरिमांशाः ॥ —*BrSpSi. XII.* 9.

> In the *bhāgānubandha* class, the whole number (*rūpa-gaṇa*) is multiplied by the denominator (of the fraction) should be increased by the numerator (of the fraction) or the upper denominator having been multiplied by the lower denominator, the initial numerator (i.e. the upper numerator) should be multiplied by the sum of the lower numerator and denominator.[1]

(*Pāṭīgaṇita*, 39 *cf. BrSpSi. XII. 9 (ii)*; *GSS.* (iii) 113

This means that

(i) $a+\frac{b}{c}=\frac{ac+b}{c}$

(ii) $\frac{a}{b}+\frac{c}{d}$ of $\frac{a}{b}$ (which was written by Indians in the style

$$\left|\begin{matrix}\frac{a}{b}\\ \frac{c}{d}\end{matrix}\right|$$

is equal to $\frac{a(d+c)}{bd}$

Addition and Subtraction of Fractions

In the *Brāhmasphuṭa-siddhānta*, Brahmagupta gives the rule for the addition and subtraction of fractions :

> If the denominators (*cheda*) of fractions are different then reduce these fractions to a common denominator. Now for the additions, unite the numerators and take their difference in case of subtraction.[2]

Brahmagupta and Mahāvīra give the method under *Bhāgajāti*.

Multiplication

Brahmagupta says :

The product of the numerators divided by the pro-

1. भागानुबन्धजातौ रूपगणश्छेद सङ्गुणः सांशः ।
अवरहरघ्नोर्ध्व हरेऽधोंशाऽयुतहरघ्न आद्यंशः ॥ —*Pāṭīgaṇita 39.*
2. विपरीतच्छेदगुणाः राश्योश्छेदांशकाः समच्छेदाः ।
संकलितेंऽशा योज्या व्यवकलितेंऽशान्तरं कार्यम् ॥ —*BrSpSi. XII. 2*

duct of the denominators is the (result of) multiplication of two or more fractions.[1]

While all other writers give the rule in the same way as Brahmagupta, Mahāvira in the *Gaṇitasārasaṁgraha* refers to cross reduction in order to shorten the work :

> In the multiplication of fractions, the numerators are to be multiplied by the numerators and the denominators by denominators, after carrying out the process of cross reduction, if that be possible.[2]

Division of Fractions

The *Āryabhaṭiya* does not explicitly give the rule of division, but under the Rule of Three, we have an indication of this operation. The Rule of Three states the result as $\frac{f \times i}{p}$, where f stands for *phala* i.e. "fruit", i for *icchā*, i.e., demand or requisition, and p for *pramāṇa* i.e. argument. When these quantities are fractional, we get an expression of the form

$$\frac{\frac{a}{b} \times \frac{c}{d}}{\frac{m}{n}}$$

for the evaluation of which Āryabhaṭa I states :

> The multipliers and the divisor are multiplied by the denominators of each other.

These quantities are written in the following way

a	m
b	n
c	
d	

Transferring the denominators we have

a	m
n	b
c	d

Performing multiplication, the result is $\frac{anc}{mbd}$. The above interpretation of the obscure line in the *Āryabhaṭiya* is based

1. रूपाणिच्छेद गुणान्यंशयुतानि द्वयोर्बहूनां वा ।
प्रत्युत्पन्नो भवति च्छेदवधेनोद्धृतोऽशवधः ॥ —*BrSpSi.* XII. 3
2. GSS. p. 25. (2)

on the commentaries of Sūryadeva and Bhāskara I (the commentary of Parameśvara on this line is vague and misleading). Sūryadeva in this connection says :

> Here by the word *guṇakāra* is meant the multiplier and multiplicand, i.e., the *phala* and *icchā* quantities that are multiplied together. By *Bhāgahāra* is meant the *pramāṇa* quantity. The denominators of the *phala* and *icchā* are taken to the *pramāṇa*. The denominator of the *pramāṇa* is taken with the *phala* and *icchā*. Then multiplying these, i.e., (the numerators of) the *phala* and *icchā* and this denominator, and dividing by (the product of) the numbers standing with the *pramāṇa* the result is the quotient of the fractions.

Brahmagupta gives the method of division as follows :

> The denominator and numerator of the divisor having been interchanged, the denominator of the dividend is multiplied by the (new) numerator. Thus division of *proper fractions* is performed.[1]

Square and Square-Root of Fractions

Brahmagupta says as follows in this connection:—

> The square of the numerator of a proper fraction divided by the square of the denominator gives the square[2].

This rule of Brahmagupta has been followed by other authors also. The rule regarding the square-root as given by Brahmagupta is as follows :

> The square-root of the numerator of a proper fraction divided by the square-root of the denominator gives the square-root.[3]

The Rule of Three :

The Indian term in Sanskrit for the Rule of Three is *Trairāśika* (literally, "three terms"). The term occurs in the *Bakhśālī Manuscript* also, and also in the *Āryabhaṭīya*, indicating the

1. परिवर्त्य भागहारच्छेदांशौ छेद संगुणच्छेदः ।
   अंशोऽशगुणो भाज्यस्य भागहारः सवर्णितयोः ॥ —*BrSpSi.* XII. 4
2. संवर्गितांशवर्गश्छेदकृतिविभाजितो भवति वर्गः । —*BrSpSi.* XII. 5 (1)
3. संवर्गितांशमूलं छेदपदेनोद्धृतं मूलम् । —*BrSpSi.* XII. 5 (2)

antiquity of the term. Bhāskara in his commentary of the *Āryabhāṭiya* gives a justification of the use of this term for the Rule of Three thus :

> Here three quantities are needed (in the statement and calculation) so the method is called *trairāśika* (meaning thereby the "rule of three terms").

The problem of the Rule of Three has the form :

> If *p* (*pramāṇa*) yields *f* (*phala*), what will *i* (*icchā*) yield ?

Āryabhaṭa II (the author of the *Mahāsiddhānta*, 950 A.D.) uses the terms *māna*, *vinimaya* and *icchā*, instead of *pramāṇa*, *phala* and *icchā* respectively. It has also been pointed out by several authors that the first and third terms are similar, i.e., of the same denomination.

We shall give here the Rule of Three as given by Āryabhaṭa I and Brahmagupta :

> In the Rule of Three, the *phala* ("fruit"), being multiplied by the *icchā* ("requisition") is divided by the *pramāṇa* ("argument"). The quotient is the fruit corresponding to the *icchā* The denominators of one being multiplied with the other give the multiplier (i.e. numerator) and the divisor (i.e. denominator).[1]
>
> In the Rule of Three *pramāṇa* ("argument"), *phala* ("fruit") and *icchā* ("requisition") are the (given) terms; the first and the last terms must be similar. The *icchā* multiplied by the *phala* and divided by the *pramāṇa* gives the fruit (of the demand).[2]

Śrīdhara also gives the Rule of Three almost in the same words. Bhāskara II, Nārāyaṇa and others follow Brahmagupta and Śrīdhara in the *Trairāśika* operation. Śrīdhara in his *Pāṭīganita* says :

---

1. त्रैराशिकफलराशिं तमथेच्छाराशिनाहतं कृत्वा ।
लब्धं प्रमाणभजितं तस्मादिच्छाफलमिदं स्यात् ॥
छेदाः परस्परं हता भवन्ति गुणकार भागहाराणां ।
छेदगुणं सच्छेदं परस्परं तत्सवर्णत्वम् ॥ —*Arya*. II 26-27.

2. त्रैराशिके प्रमाणं फलमिच्छाचन्त्योः सदृशराशी ।
इच्छाफलेन गुणिता प्रमाणभक्ता फलं भवति ॥ —*BrSpSi* XII. 10

> In (solving problems on) the Rule of Three, the argument (*pramāṇa*) and the requisition (*icchā*), which are of the same denomination, should be set down in the first and last places; the fruit (*phala*), which is of a different denomination, should be set down in the middle, (this having been done) that (middle quantity multiplied by the last quantity should be divided by the first quantity.[1]

We shall illustrate the Rule of Three by an example from the *Pāṭīgaṇita* (Example 25) :

> Example, If 1 *pala* and 1 *karṣa* of sandalwood are obtained for ten and a half *paṇas*, then for how much will nine *palas* and one *karṣa* (of sandalwood) be obtained ?[2]

Here in this Example.

argument=1 *pala* and 1 *karṣa*=$1\frac{1}{4}$ or 5/4 *palas*; fruit=$10\frac{1}{2}$ or 21/2 *paṇas*;

and requisition=9 palas and 1 karṣa=$9\frac{1}{4}$ or 37/4 palas. According to the Rule we shall write them as :

$$\begin{vmatrix} 1 & 10 & 9 \\ 1 & 1 & 1 \\ 4 & 2 & 4 \end{vmatrix}$$

Converting these into proper fractions we have

$$\begin{vmatrix} 5 & 21 & 37 \\ 4 & 2 & 4 \end{vmatrix}$$

Then applying the rule, (i.e. multiplying the second and the last and dividing by the first), we have

$$\begin{vmatrix} 21 & 5 \\ 2 & 4 \\ \hline 37 & \\ 4 & \end{vmatrix} \qquad \frac{21}{2} \times \frac{37}{4} = \frac{\quad}{\frac{5}{4}}$$

Or transferring denominators $\begin{vmatrix} 21 & 5 \\ 4 & 2 \\ 37 & 4 \end{vmatrix} = \frac{21.4.37}{5.2.4}$ *pala*

---

1. आद्यन्त्योस्त्रिराशावभिन्नजाती प्रमाणमिच्छा च।
पाठ्यमध्येऽन्त्यमृल्यं तदन्त्यगुणमादिना विभजेत् ॥ —*Pāṭīgaṇita* 43.

2. चन्दनपलं सकर्षं सार्धैर्यदि लभ्यते पणैर्दशभिः।
तत्किं पलानि लभ्यन्ते पलानि नव कर्षयुतानि ॥ —*Pāṭīgaṇita*. Ex. 25.

=4 purāṇa, 13 paṇas, 2 kākiṇīs and 16 varāṭakas. (One purāṇa is equivalent to 16 paṇas; one paṇa is equivalent to 4 kākiṇīs, and one kākiṇī is equivalent to 20 varāṭakas or cowries.

Inverse Rule of Three

This is known as *vyasta-trairāśika* (literally meaning "inverse rule of three terms)". After having described the rule of three, Brahmagupta proceeds to give an account of this inverse rule of three :

> Divide the *phala* with *icchā* and multiply by *pramāṇa*; this gives the *vyasta-trairāśika* inverse rule of three[1].

Here *pramāṇa* is the argument also known as the first term and, and *phala* is the fruit also known as the middle term and *iccha* is known as requisition or the last term. As Bhāskara II clearly states, this rule is applied where with the increase of the *icchā*, the *phala* decreases or with its decrease the *phala* increases (*Līlāvatī*).

Rule of Compound Proportion

Brahmagupta and other writers call the rule of compound proportions as *pañca-rāśika, sapta-rāśika* etc., meaning the rule of five terms, rule of seven terms etc. depending on the number of terms involved the problems. These are sometimes grouped under the general application of the "Rule of Odd Terms". Āryabhaṭa I (499 A.D.) though actually gives the rule of three appears to have been quite familiar with the rule of compound proportion also. In fact the difference between the rule of three and compound proportion is more or less artificial. This view was expressed by Bhāskara I (525 A.D.) in his commentary on the *Āryabhaṭīya* :

> Here Ācārya Āryabhaṭa has described the Rule of Three only. How the well-known Rules of Five etc. are to be obtained ? I say thus : The Ācārya has described only the fundamentals of *anupāta* (proportion). All others such as the Rule of Five etc. follow from that fundamental rule of proportion. How ? The Rule of Five etc. consist of combinations of the Rule of Three. ……In the Rule of Five, there are two Rules of

1. व्यस्तं त्रैराशिकं फलमिच्छा भक्तः प्रमाण फलघातः ।
त्रैराशिकादिषु फलं विद्यमेष्वेकादशान्तेषु ॥ —*BrSpSi* XII. 11

Three, in the Rule of Seven three Rules of Three, and so on. This I shall point out in the examples.

Brahmagupta gives the following rule relating to the solution of problems in compound proportion :

> In the case of odd terms beginning with three terms up to eleven, the result is obtained by transposing the fruits of both sides, from one side to the other, and then dividing the product of the larger set of terms by the product of the smaller set. In all the fractions, the transposition of denominators, in like manner, takes place on both sides.[1]

This may be illustrated by taking an example from the commentary of Pṛthūdaka Svāmī on the *Brāhmasphuṭasiddhānta* :

> *Example* —If there is an increase of 10 in 3 months on 100 (*niṣkas*), what would be the increase on 60 (*niṣkas*) in 5 months.

Here the *Pramāṇa pakṣa* (the first set of terms) is
100 niṣkas, 3 months, 10 niṣkas (phala)

The second set or the *iccha pakṣa* is
60 *niṣkas*, 5 months, *x niṣkas*

The terms are written in compartments as below :

100	60
3	5
10	0

In the above 10 (written lowest) is the *fruit* of the first side (*pramāṇa pakṣa*), and there is no *fruit* on the second side or the *iccā pakṣa*. Interchanging the *fruits* we get

100	60
3	5
0	10

The larger set of terms is on the second side (*iccha pakṣa*). The product of the numbers is 3,000. The product of the

1. व्यस्त त्रैराशिक फलमिच्छा भक्तः प्रमाणफलघातः ।
त्रैराशिकादिषु फलं विषमेष्वेकादशान्तेषु ॥
फलसंक्रमणमुभयतो बहुराशि वधोऽल्पवधहृतो ज्ञेयम् ।
सकलेष्वेवं भिन्नेष्वपि तच्छेदसंक्रमणम् ॥ —*BrSpSi.* XII. 11-12.

number on the side of the smaller set of terms is 300. Therefore the required result is $\frac{3000}{300}=10$.

Rule of Three as a Particular Case

According to Brahmagupta, the above method of "compound proportion" may be applied to the Rule of Three. Taking the example solved under the Rule of Three:

> If one *pala* and one *karṣa* of sandal wood are obtained for ten and a half *paṇas*, for how much will be obtained nine *palas* and one *karṣa*?
>
> (4 karṣas=1 pala).

We shall represent them according to the Rule of Compound Proportion as

Pramāṇa pakṣa : 1 *pala*, 1 *karṣa*, $10\frac{1}{2}$ *paṇa*
or $\frac{5}{4}$ *pala* , $\frac{21}{2}$ *paṇa*

Icchā pakṣa : 9 *pala*, 1 *karṣa*, *x paṇa*
or 37/4 *pala* , *x paṇa*

This we shall represent as

5	37
4	4
21	0
2	

Transposing the fruits, we have

5	37
4	4
0	21
	2

Transposing denominators

5	37
4	4
0	21
2	

The product of numbers on the side of the larger set is divided by the product of the numbers on the side of the smaller set. 0 in this case is not a number. It is the symbol for the unknown or absence, Hence the result is:

$$\frac{37.4.21}{5.4.2} \text{ paṇas}$$

The above method of working Rule of Three is found among Arabs, although it does not seem to have been used in India after Brahmagupta.

Problem Containing Quadratic Equation

Perhaps Āryabhaṭa I is the first man in the history of mathematics to give a solution of a quadratic equation (499 A.D.). In his *Āryabhaṭiva*, he gives a rule for the solution of the following problem (I am reproducing it as described by Datta and Singh):

> The principal sum p (=100) is lent for one month (interest unknown=x). This unknown interest is then lent out for t(=six) months. After this period, the original interest (x) plus the interest on this interest amounts to A(=16). The rate-interest (x) on the principal (p) is required.

This problem requires the solution of the quadratic equation:—

$$tx^2+px-Ap=0$$

which gives $x=\dfrac{-p/2\pm\sqrt{(p/2)^2+Ap}}{t}$

The negative value of the radical does not give a solution of the problem; so that the result is

$$x=\frac{\sqrt{Apt+(p/2)^2}-p/2}{t}$$

This solution is stated by Āryabhaṭa I in the following words:

> Multiply the sum of the interest on the principal and the interest (A) by the time (t) and by the principal (p). Add to this result the square of half the principal $(p/2)^2$. Take square-root this. Subtract half the principal ($p/2$) and divide the remainder by the time (t). The result will be the (unknown) interest (x) on the principal.[1]

Here the Sanskrit terms are *mūla* for principal and *phala* for interest.

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1. मूलफलं सफलं कालमूलगुणमर्धमूलकृतियुक्तं ।
मूलं मूलार्धोनं कालहृतं स्यात्समूलफलम् ॥ *Ārya. II. 25,*

Brahmagupta (628 A.D.) gives a more general rule : He enunciates his problem thus :

> The principal (p) is lent out for t_1 months and the unknown interest on this ($=x$) is lent out for t_2 months at the same rate and becomes A. To find x.

This evidently gives the quadratic :

$$x^2+\frac{pt_1}{t_2}\ x-\frac{Apt_1}{t_2}=0$$

whose solution is

$$x=\pm\sqrt{\frac{Apt_1}{t_2}+\left(\frac{pt_1}{2t_2}\right)^2}-\frac{pt_1}{2t_2}$$

The negative value of the radical does not give a solution of the problem, so it is discarded.

Brahmagupta states the formula thus :

> Multiply the principal (p) by its time (t_1) and divide by the other time (t_2) (placing the result) at two places. Multiply the first of these by the mixture (A). Add to this the square of half the other. Take the square-root of this (sum). From the result subtract half the other. This will be the interest (x) on the principal.[1]

A Problem on Interest

Brahmagupta gives a solution of a problem on interest :

> In what time will a given sum s, the interest on which for t months is r, become k times itself ?

The rule for the solution of this problem as given by Brahmagupta is :

> The given sum multiplied by its time and divided by the interest (*phala*), being multiplied by the factor (*guṇa*) less one, is the time (*required*).[2]

Miscellaneous Problems

Brahmagupta in his *Gaṇitādhyāya* of the *Brāhmasphuṭa-siddhānta* gives numerous solutions in relation to miscellaneous problems. Here I shall be quoting a few of the problems which

1. कालप्रमाणघातः परकालहृतो द्विधाऽऽद्यमिश्रवधात् ।
अन्यार्धकृतियुतात् पदमन्यार्धोनं प्रमाणफलम् ॥ —*BrSpSi.* XII. 15.
2. कालगुणितं प्रमाणं फलभक्तं व्येकगुणहृतं कालः ।
स्वफलयुतरूपभक्तं मूलफलैक्यं भवति मूलम् ॥ —*BrSpSi.* XII. 14.

have been quoted by his commentator Pṛthūdaka Svāmī in connection with one of his karaṇa-sūtra.[1]

1. A horse was purchased by (nine) dealers in partnership, whose contributions were one, etc. up to nine; and was sold by them for five less than five hundred. Tell me what was each man's share of the sale proceed[2]

2. Four colleges (mathas), containing an equal number of pupils, were invited to partake of a sacrificial feast. A fifth, a half, a third and a quarter (of the total number of pupils in the college) came from the respective colleges to the feast; and added to one, two, three and four, they were found to amount to eighty-seven; or, with those deducted, they were sixty seven. Find the actual number of the pupils that came from each college.[3]

3. Three (unequal) jars of liquid butter, of water and of honey, contained thirty-two, sixty and twenty-four *Pala* respectively ; the whole was mixed together and the jars filled again. Tell me the quantity of butter, of water and of honey in each jar[4].

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1. प्रक्षेपयोगहृतया लब्ध्वा प्रक्षेपका गुणा लाभाः ।
ऊनाधिकोत्तरास्तद्युतोनया स्वफलमूनयुत ॥ *BrSpSi.* XII. *16.*

2. एकादिनैव पर्यन्तैर्वर्गिजैर्मूलराशिभिः ।
क्रीतो हयोऽसौ विक्रीतः पञ्चोनैः पञ्चोनैः पंचभिः शतैः ।
किमेकैकस्य तत्रासीद् ब्रूहि त्वं मिश्रकान् मम ॥

3. मठस्थानानि चत्वारि छात्राणां समसंख्यया ।
भोक्तुं संमन्त्रितान्यासन् दीक्षायां किल यज्वना ॥
पंचार्धत्रिचतुर्थांशास्तेभ्यो भोक्तुं समागताः ।
एकद्वित्रिचतुर्युक्ता दृष्टाशीतिः ससप्तका ॥
एकोत्तरैरथवा हीना सप्तषष्टिश्चतेंऽशकाः ।
मठेभ्यश्छात्रसंख्यां मे ब्रूहि ये चागता यतः

4. घृतोदक मधूनां ये त्रयः कलसकाः पलैः ।
द्वात्रिंशष्टिजिनैः पूर्णा एकीभूतास्ततः पुनः ॥
मिश्रेण पूरिता यावत् तावत् संख्यां न वेद्म्यहम् ।
घृतोदकमधूनां तामेकैकत्र गतां वद ॥

—: o :—

Reference

B. Datta and A.N. Singh : *History of Hindu Mathematics*, Part I (1962).

K.S. Shukla : The *Pāṭīgaṇita* of Śrīdharācārya (1959).

Kern : The *Āryabhaṭīya. Bhaṭadīpikā* of Parameśvara (1875).

CHAPTER IX

Brahmagupta as an Algebraist

Ancient Indian name for algebra is *Bījagaṇita* where *bīja* means *element* or *analysis* and *gaṇita* stands for the *science of calculation*. As early as 860 A.D., Pṛthūdaka Svāmi used this epithet for algebra in his commentary. Brahmagupta calls algebra as *Kuṭṭakagaṇita* or merely *kuṭṭaka*, a term which was later on used for "*pulveriser*" which deals with that special section of algebra which is connected with indeterminate equations of the first degree. Algebra is often also known as *avyakta-gaṇita* or the calculations with *unknowns*, in contrast to arithmetic which was known as *vyakta-gaṇita* or the calculations with *knowns*.

Algebra goes to Europe from India

In the history of mathematical sciences, as Colebrooke rightly remarks, it has long been a question to whom the invention of algebraic analysis is due. There is no doubt that Europe got algebra from Arabs mediately or immediately. But the Arabs themselves scarcely pretend to the discovery of algebra. Colebrooke says that they were not in general inventors but scholars during the short period of their successful culture of the sciences; and the germ at least of the algebraic analysis is to be found among the Greeks in an age not precisely determined, but more than probably anterior to the earliest dawn of civilisation among the Arabs; and this science in a more advanced state subsisted among the Hindus prior to the earliest disclosure of it by the Arabians to modern Europe. (Colebrooke: *Dissertation on the Algebra of the Hindus*)[1].

Colebrooke based his observations on the texts he could procure for his studies. These were: Bhāskara II's *Bījagaṇita* or *Vījagaṇita* (1150A.D.) and *Līlāvatī* (1150A.D.), the *Gaṇitādhyāya* and *Kuṭṭakādhyāya* of Brahmagupta in his famous treatise the *Brahma Siddhānta* or rather the *Brāhmasphuṭasiddhānta* (628

1. Colebrooke, H. T., *Miscellaneous Essays*, Vol. II, 1872, p. 418.

A.D.). There can be no doubt regarding the age of these two authors. Bhāskara II completed his great work on the *Siddhānta-śiromaṇi* in 1072 Śaka, and *Karaṇa-kutūhala* a practical astronomical treatise in 1105 Śaka; these dates are based on the passages given by Bhāskara himself in his works. The *Bīja-gaṇita* and the *Līlāvatī* form parts of the great treatise, the *Siddhānta-śiromaṇi*. The genuineness of the text is established, as Colebrooke says, with no less certainty by numerous commentators in Sanskrit, besides a Persian version of it. Those commentaries comprise a perpetual gloss, in which every passage of the original is noticed and interpreted : and every word of it is repeated and explained. From comparison and collation of various texts, it appears then that the work of Bhāskara, exhibiting the same uniform text which the modern transcripts of it do, was in the hands of both Muhammedans and Hindus, between two or three centuries ago : and numerous copies of it having been diffused throughout India, at an earlier period, as of a performance held in high estimation, It was the subject of study and habitual reference in countries and places so remote from each other as the north and west of India and the Southern Peninsula.

This though not marking any extraordinary antiquity, nor approaching to that of the author himself, was a material point to be determined : as there will be in the sequel, so says Colebrooke, occasion to show, that modes of analysis, and in particular, general methods for the solution of indeterminate problems both of the first and second degrees, are taught in the *Bīja-gaṇita*, and those for the first degrees repeated in the *Līlāvatī*, which were unknown to the mathematicians of the West, until invented anew in the last two centuries by algebraists of France and England.[1] Bhāskara who himself flourished more than six hundred and fifty years ago, was in this respect a compiler and took those methods from Indian authors as much more ancient than himself.

Regarding the age of the precursors of Bhāskara II, Colebrooke says : The age of his precursors cannot be determined with equal precision. He then proceeds to examine the evidence as follows :

1. Colebrooke, H. T., *Miscellaneous Essays*, p. 421.

Towards the close of his treatise on Algebra, Bhāskara II informs us, that it is compiled and abridged from the more diffuse works on the same subject, bearing the names *Brāhme* (meaning no doubt Brahmagupta), Śrīdhara and Padmanābha; and in the body of his treatise, he has cited a passage of Śrīdhara's algebra and another of Padmanābha. He repeatedly adverts to preceding writers and refers to them in general terms, where his commentators understand him to allude to Āryabhaṭa, to Brahmagupta to the latter's scholiast Caturveda *Pṛthūḍaka Svāmī* and to the other writers above mentioned.

Most, if not all, of the treatises, to which he thus alludes, must have been extant, and in the hands of his commentators, when they wrote; as appears from their quotations of them; more especially those of Brahmagupta and Āryabhaṭa, who are cited, and particularly the first mentioned, in several instances. A long and diligent research in various parts of India, has, however, failed of recovering any part of the *Padmanābha Bīja* (or the algebra of Padmanābha) and of the algebraic and other works of Āryabhaṭa.

But the translator has been more fortunate in regard to the works of Śrīdhara and Brahmagupta, having in his collection Śrīdhara's compendium of arithmetic, and a copy incomplete however, of the text and scholia of Brahmagupta's *Brahmasiddhānta* comprising among other no less interesting matter, a chapter treating of arithmetic and mensuration; and another, the subject of which is algebra : both of them fortunately complete. The commentary is a perpetual one; successively quoting in length each verse of the text; procceeding to the interpretation of it, word by word; and subjoining elucidations and remarks; and its colophon, at the close of each chapter, gives the title of the work and the name of the author. Now the name which is there given, Caturveda Pṛthūdaka Svāmī, is that of a celebrated scholiast of Brahmagupta, frequently cited as such by the commentaries of

Bhāskara and by other astronomical writers; and the title of the work, *Brāhmasiddhānta* or sometimes *Brāhmasphuṭasiddhānta*, corresponds, in the shorter form, to the known title of Brahmagupta's treatise in the usual references to it by Bhāskara's commentators, and answers, in the longer form, to the designation of it, as indicated in an introductory couplet which is quoted from Brahmagupta by Lakṣmīdāsa, a scholiast of Bhāskara II. Remarking this coincidence, the translator proceeded to collate, with the text and commentary, numerous quotations from both, which he found in Bhāskara's writings or in those of his expositors. The result confirmed the indication and established the identity of both text and scholia as Brahmagupta's treatise, and the gloss of Pṛthūdaka. The authenticity of the *Brāhmasiddhānta* is further confirmed by numerous quotations in the commentary of Bhaṭṭotpala on the *Saṁhitā* of Varāhamihira : as the quotations from the *Brāhmasiddhānta*, in that commentary, (which is the work of an author who flourished eight hundred and fifty years ago) are verified in the copy under consideration. A few instances of both will suffice, and cannot fail to produce conviction.

It is confidently concluded, that the chapters on arithmetic and algebra, fortunately entire in a copy in many parts imperfect, of Brahmagupta's celebrated work as here described, are genuine and authentic. It remains to investigate the age of the author.

Mr. Davis, who first opined to the public a correct view of the astronomical computations of the Hindus, is of opinion, that Brahmagupta lived in the seventh century of the Christian era. Dr. William Hunter, who resided for some time with a British Embassy at Ujjayinī, and made diligent researches into the remains of Indian science at that ancient seat of Hindu astronomical knowledge, was there furnished, by the learned astronomers whom he consulted, with the ages of the principal ancient authorities. They assigned to Brahmagupta the date of 550 Saka; which answers to A.D.

628. The grounds on which they proceeded are unfortunately not specified: but as they gave Bhāskara's age correctly, as well as several other dates right, which admit of being verified; it is presumed that they had grounds, though unexplained, for the information which they communicated.

Mr. Bentley, who is little disposed to favour the antiquity of an Indian astronomer, has given his reasons for considering the astronomical system which Brahmagupta teaches, to be between twelve and thirteen hundred years old (1263 years in A.D. 1799). Now as the system taught by this author is professedly one corrected and adapted by him to conform with the observed positions of the celestial objects when he wrote, the age, when their positions would be conformable with the results of computations made as by him directed, is precisely the age of the author himself: and so far as Mr. Bentley's calculations may be considered to approximate the truth, the date of Brahmagupta's performance is determined with like approach to exactness, within a certain latitude however of uncertainty for allowance to be made on account of the inaccuracy of Hindu observations.

The translator has assigned on former occasions the grounds upon which he sees reason to place the author's age, soon after the period when the vernal equinox coincided with the beginning of the lunar mansion and zodiacal asterism *Aśvinī*, where the Hindu ecliptic now commences. He is supported in it by the sentiments of Bhāskara and other Indian astronomers, who infer from Brahmagupta's doctrine concerning the solistitial points, of which he does not admit a periodical motion, that he lived when the equinoxes did not, sensibly to him, deviate from the beginning of *Aśvinī* and middle of *citrā* on the Hindu sphere. On these grounds it is maintained, that Brahmagupta is rightly placed in the sixth or beginning of the seventh century of the Christian era, as the subjoined calculations will more particularly show. The age when Brahmagupta

flourished, seems then, from the concurrence of all these arguments, to be satisfactorily settled as antecedent to the earliest dawn of the culture of sciences among the Arabs; and *consequently establishes* the fact, that *the Hindus were in possession of algebra before* it was known to the Arabians.

Brahmagupta's treatise, however, is not the earliest work known to have been written on the same subject by an Indian author. The most eminent scholiast of Bhāskara II (Gaṇeśa) quotes a passage of Āryabhaṭa specifying algebra under the designation of *Bīja*, and making separate mention of *Kuṭṭaka*, which more particularly intends a problem subservient to the general method of resolution of indeterminate problems of the first degree : he is understood by another of Bhāskara's commentators to be at the head of the elder writers, to whom the text then under consideration adverts, as having designated by the name of *Madhyamāharaṇa* the resolution of affected quadratic equations by means of the completion of the square. It is to be presumed, therefore, that the treatise of Āryabhaṭa then extant, did extend to quadratic equations in the determinate analysis, and to indeterminate problems of the first degree; if not to those of the second likewise, as most probably it did.

This ancient astronomer and algebraist, so says Colebrooke, was anterior to both Varāhamihira and Brahmagupta, being repeatedly named by the latter; and the determination of the age when he flourished is particularly interesting as his astronomical system, though on some points agreeing, essentially disagreed on others, with that which the Hindu astronomers still maintain.

He, as Colebrooke says, is considered by the commentators of the *Śūryasiddhānta* and *Śiromaṇi*, as the earliest of uninspired and mere human writers on the science of astronomy, as having introduced requisite corrections into the system of Parāśara, from whom he took the numbers for the planetary mean motions; as

having been followed in the tract of emendation, after a sufficient interval to make further correction requisite, by Durgāsinha and Mihira; who were again succeeded after a further interval by Brahmagupta, son of Jiṣṇu.

In short, says Colebrooke, Āryabhaṭa was founder of one of the sets of Indian astronomers, as Puliśa, an author likewise anterior to both Varāhamihira and Brahmagupta, was of another : which were distinguished by names derived from the discriminative tenets respecting the commencement of planetary motions at sunrise according to the first, but at midnight according to the latter, on the meridian of Laṅkā, at the beginning of the great astronomical cycle. A third sect began the astronomical day, as well as the great period, at noon.

Āryabhaṭa's name accompanied the intimation which the Arab astronomers (under the Abbasside Khalifs, as it would appear,) received, that three distinct astronomical systems were current among the Hindus of those days : and it is but slightly corrupted, certainly not at all disguised, in the Arabic representation of it *Arjabahar*, or rather *Arjabhar*, (corrupted form of Āryabhaṭa). The two other systems were, first, Brahmagupta's *Siddhānta* which was the one they became best acquainted with, and to which they apply the denomination of the *sind-hind*; and second, that of *Arca*, the Sun, which they write *Arcand* a corruption still prevalent in the vulgar Hindi.

Āryabhaṭa appears to have had more correct notions of the true explanation of celestial phenomena than Brahmagupta himself, so says Colebrooke; who in a few instances, correcting errors of his predecessor, but oftener deviating from that predecessor's juster views, has been followed by the herd of modern Hindu astronomers, in a system not improved, but deteriorated, since the time of the more ancient author.

Considering the proficiency of Āryabhaṭa in astronomical science, and adverting to the fact of his having

written algebra, as well as to the circumstance of his being named by numerous writers as the founder of a sect, or author of a system in astronomy, and being quoted at the head of algebraists, when the commentators of extant treatises have occasion to mention early and original writers on this branch of science, it is not necessary to seek further for a mathematician qualified to have been the great improver of the analytic art, and likely to have been the person by whom it was carried to the pitch to which it is found to have attained among the Hindus, and at which it is observed to be nearly stationary through the long lapse of ages which have since passed : the later additions being few and unessential in the writings of Brahmagupta, of Bhāskara and of Jñānarāja, though they lived at intervals of centuries from each other.

Āryabhaṭa, Colebrooke rightly says, then being the earliest author known to have treated of Algebra among the Hindus, and being likely to be, if not the inventor, the improver of that analysis, by whom too it was pushed nearly to the whole degree of excellence which it is found to have attained among them; it becomes in an especial manner interesting to investigate any discoverable trace in the absence of better and more direct evidence, which may tend to fix the date of his labours; or to indicate the time which elapsed between him and Brahmagupta, whose age is more accurately determined.

Taking Āryabhaṭa, for reasons given, to have preceded Brahmagupta and Varāhamihira by several centuries; and Brahmagupta to have flourished more than twelve hundred years ago, and Varāhamihira, concerning whose works and age, Colebrooke has given a few notes, and has placed him at the beginning of the sixth century after Christ, it appears probable that this earliest of known Hindu algebraists wrote as far back as the fifth century of the Christian era; and perhaps in an earlier age. Hence it is concluded that he is nearly as ancient as the Greecian algebraist Diophantus, sup-

posed on the authority of Abulfaraj, to have flourished in the time of the Emperor Julian or about A. D. 360.

Colebrooke further says : Admitting the Hindu and Alexandrian authors to be nearly equally ancient, it must be conceded in favour of the Indian algebraist, that he was more advanced in the science; since he appears to have been in possession of the resolution of equations involving several unknowns, which it is not clear, nor fairly presumable, that Diophantus, knew; and a general method of indeterminate problems of at least the first degree, to a knowledge of which the Greecian algebraist had certainly not attained ; though he displays infinite sagacity and ingenuity in particular solutions ; and though a certain routine is indiscernible in them.

Colebrooke appears to be of the view that Greeks were the first to discover the solution of equations involving one unknown; and this knowledge was passed to ancient Indians by their Greek instructors in improved astronomy. But "by the ingenuity of the Hindu-scholars, the hint was rendered fruitful and the algebraic method was soon ripened from that slender beginning to the advanced state of a well arranged science, as it was taught by Āryabhaṭa, and as it is found in treatises compiled by Brahmagupta and Bhāskara."

We do not agree with this analysis in entirety. Indian algebra is entirely of Indian roots. It had its beginning in the times of Saṃhitās and Brāhmaṇas. Some of the equations and problems were solved by geometric methods. It must have had its origin in the Śulba period if not before. Āryabhaṭa undoubtedly was the discoverer of many algebraic solutions of equations of the first and higher order with one and more unknowns. It is rather too much to trace the influence of Diophantus on Indian algebra which developed in this country independently. Brahmagupta is one of the most brilliant algebraists we ever had in the entire history of mathematics.

Technical Terms

Coefficient—

In the ancient Indian algebra, there is no systematic term for the coefficient. Usually, the power of the unknown is mentioned when the reference is to the coefficient of that power. At one place, for example, we find Pṛthūdaka Svāmī (the commentator of Brahmagupta's *Brāhmasphūṭasiddhānta*) writing "the number (*aṅka*) which is the coefficient of the square of the unknown is called the '*square*' and the number which forms the coefficient of the (simple) unknown is called the 'unknown quantity' (*avyaktamāna*)."[1] However, at many places, we find the use of a technical term also. Brahmagupta once calls the coefficient *saṁkhyā*[2] (number) and on several other occasions *guṇaka*[3] or *guṇakāra*[4] (multiplier). Pṛthūdaka Svāmī (860 A.D.) calls it *aṅka* (number) or *prakṛti* (multiplier). These terms may also be seen in the works of *Śrīpati*[6] (1039) and Bhāskara II[7] (1150 A.D.). The former also used the word *rūpa* for the same purpose.[8]

Unknown Quantity

The unknown quantity has been termed as *yāvat-tāvat* (meaning so-much-as or as-many-as) in literature as early as 300 B. C. (vide the *Sthānāṅga-sūtra*[9]). In the *Bakhaśālī Manuscript*, it has been termed as *yadṛcchā*. *vāñchā* or *kāmikā* (or any desired quantity)[10]. Āryabhaṭa I in one of his verses calls the unknown as *gulikā*[11] (literally meaning a shot) From the early seventh century A.D., the word *avyakta* was used for unknown quantities. Brahmagupta uses this term in his *Brāhmasphuṭasiddhānta*[12]

1. *BrSpSi. XVIII. 44* (Com.)
2. वर्णप्रमाण भावितघातो भवतीष्ट वर्ण संख्यैवम् । —*BrSpSi. XVIII 63*
3. मूलं द्विघ्नेष्ट वर्गाद् गुणक गुणादिष्ट युत विहीनाच्च । —*BrSpSi. XVIII 64*
   वर्गच्छिन्ने गुणके प्रथमं तन्मूल भाजितं भवति । —*BrSpSi. XVIII 70*
4. प्रथमोन्यमूलमन्यो गुणकार पदोद्धृतः प्रथमः । —*BrSpSi. XVIII 69*
5. *BrSpSi XVIII. 44* (Com.)
6. *Śiśe XIV. 33-5.*
7. *Bijaganita*
8. *Siśe XIV. 33-5.*
9. *Sūtra 747.*
10. *BMs.* Folio 22, verso; 23, recto and verso.
11. गुलिकान्तरेण विभजेद् द्वयोः ।
12. अव्यक्तवर्ग घनवर्ग वर्गपंक्गत षड्गतादीनाम् ।

Power

Since long, the word *varga* has been used for the second power; the word also stands for *square* (*Uttarādhyayana Sūtra*[1], B. C. c. 300). The third power is similarly known as *ghana*, the fourth power as *varga-varga* (square-square), the sixth power as *ghana-varga* (cube-square) and the twelfth power as *ghana-varga-varga* (cube-square-square). In later days, the fifth power was called *vargaghana ghāta* (here the word *ghāta* means product; the term means product of cube and square). The former system was multiplicative, rather than additive; wheras the latter was on the additive system. The seventh power on the additive system was known as *varga-varga-ghana-ghāta* (product of square-square and cube). Brahmagupta, however, uses a more scientific system for expressing the powers more than four. He calls the fifth power as *Pañca gata* (literally meaning, raised to the fifth), the sixth power as *ṣaḍ-gat* (raised to the sixth) and so on, thus adding the suffix *gata* to the name of the number indicating that power.[2] Bhāskara II has followed the system of Brahmagupta almost consistently for powers one and upwards.

Equation

Perhaps Brahmagupta has for the first time used the term *samakaraṇa* or *samīkaraṇa* (literally meaning making equal) or simply *sama* (equal or equation)[3]. Pṛthūdaka Svāmī (860) employs the term *sāmya* (equality or equation) for equation[4]. The equation is said to possess two *Pakṣas*[5] (sides) *Itara-Pakṣa* and *apara-pakṣa.*

Absolute Term

Brahmagupta uses the term *rūpa* (literally meaning appearance) for an absolute term. It represents the visible or known

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1. Chapter XXX, 10. 11.

2. अव्यक्तवर्ग घनवर्ग-वर्ग-पंचगत-षड्गतादीनाम् ।
सदृश द्विवधो वर्गस्त्र्यादि वधस्तद्गतोऽन्य जातिवधः ॥

3. वर्ण प्रमाण भावित-घातो भवतीष्टवर्ण संख्यैवम् ।
सिध्यति विनाऽपि भावित-समकरणात् किं कृतं तदतः ॥
अव्यक्तान्तर भक्तं व्यस्तं रूपान्तरं समेऽव्यक्तः ।

4. *Śiśe.* XIV. 19.

5. *Bījaganita.*

portion of the equation whilst its other part is practically invisible or unknown[1].

Unknowns and Symbolism

Āryabhaṭa I (499 A. D.) probably used coloured *gulikās* or shots for repreststing different unknowns. Brahmagupta mentions *varṇa* as the symobls for unknown. He has, however, not indicated how these *varṇas* or colours were used as symbols for unknowns. Perhaps we might conculude from this that the method of using colours as symbols for unknown quantities was very common and familiar to the algehraists. Datta and Singh say that the Sanskrit word *varṇa* denotes colour as well as a letter of alphabet, and therefore, letters of alphabet came into use for unknown quantities : *kālaka* (black), *ṇilaka* (blue), *pītaka* (yellow), *lohita* (red), *haṛitaka* (green), *śvetaka* (white), *citraka* (variegated), *kapilaka* (tawny), *piṅgalaka* (reddish-brown), *dhūmraka* (smoke-coloured), *pāṭalaka* (pink), *śaṛalaka* (spotted), *śyāmalaka* (blackish) *ṃecaka* (dark blue) etc[2].

It should be further noted that the first unknown quantity *yāvat-tāvat* is not a *varṇa* or colour. It thus clearly indicates that the use of colours as symbols came at a later stage, whilst the word *yāvat-tāvat* was in currency from much earlier times. Some authorities think that the term *yāvat-tāvat* is a corrupted form of *yāvakastāvat* (where *yāvaka* means red). Pṛthūdaka Svāmī has sometimes used the term *yavaka* for an unknown quantity[3].

Laws of Signs

Brahmagupta has in his Chaper XVIII devoted a special section entitled "*Dhanarṇa Śunyānām Samkalanam*" or calculations dealing with quantities bearing positive and negative signs and zero,

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1. अव्यक्तान्तरं भक्तं व्यस्तं रूपान्तरं समेऽव्यक्तः । —*BrSpSi* XVIII. 43
वर्गं चतुर्गुणितानां रूपाणां मध्यवर्गं सहितानाम् ॥ —*BrSpSi* XVIII. 44

2. यावत्-तावत् कालको नीलकोऽन्यो वर्णः पीतो लोहितश्चैतदाद्याः ।
अव्यक्तानां कल्पिता मानसंज्ञास्तत्संख्यानं कर्तुमाचार्यवर्यैः ॥
यावत्-तावत् कालक नीलक पीतास्च लोहितो हरितः ।
श्वेतक चित्रक कपिलक पाटलकाः पण्डु धूम्र शबलाश्च ॥
श्यामलक-मेचक-धवलक-पिशङ्ग-शारङ्ग-वभ्रु गौराद्याः । —Nārāyaṇa, *Bijaganita*

3. *BrSpSi XII. 15.* (Com.) ; XII 18.

Regarding *addition*, Brahmagupta says :

> The sum of two positive numbers is positive, of two negative numbers is negative; of a positive and negative number is the difference[1].

Regarding *subtraction*, Brahmagupta further says :

> From the greater should be subtracted the smaller; (the final result is) positive, if positive from positive, and negative, if negative, from negative. If, however, the greater is subtracted from the less, that difference is reversed (in sign), negative becomes positive and positive becomes negative. When positive is to be subtracted from negative or negative from positive, then they must be added together[2].

Mahāvīra (850 A. D.), Bhāskara II (1150 A.D.) and Nārāyaṇa (1350 A. D.) have also given similar rules regarding addition (*Saṁkalanam*) and the subtraction (*vyavakalanam*).

Again, the rule given by Brahmagupta regarding *Multiplication* is as follows :

> The product of a positive and a negative (number) is negative; of two negatives is positive; positive multiplied by positive is positive[3].

His rule regarding *division* is as follows ;

> Positive divided by positive or negative divided by negative becomes positive. But positive divided by negative and negative divided by positive remains negative[4].

Similar rules for multiplication and division were provided by later authorities as Mahāvīra and Bhāskara II.

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1. *BrSpSi XII. 15. (Com.); XII 18. (Com.)*
2. धनयोर्धनमृणमृणयोर्धनर्णं योरन्तरं समैक्यं खम् ।
   ऋणमैक्यं च धनमृणधन शून्ययोः शून्ययोः शून्यम् ॥ —*BrSpSi.* XVIII-30
3. ऊनमधिकाद्विशोध्यं धनं धनादृणमृणादधिकमूनात् ।
   व्यस्तं तदन्तरं स्याद्दृणं धनं धनमृणं भवति ॥
   शून्यविहीनमृणमृणं धनं धनं भवति शून्यमाकाशम् ।
   शोध्यं यदा धनमृणाद्दृणं धनाद्वा तदा क्षेप्यम् ॥ —*BrSpSi* XVIII. 31-32
4. ऋणमृणधनयोर्घातो धनमृणयोर्धनवधो धनं भवति ।
   शून्यर्णयोः ख धनयोः ख शून्ययोर्वा वधः शून्यम् ॥ —*BrSpSi* XVIII. 33
5. धनभक्तं धनमृणहृतमृणं धनं भवति खं खभक्तं खम् ।
   भक्तमृणेन धनमृणं धनेन हृतमृणमृणं भवति ॥ —*BrSpSi.* XVIII. 34

Brahmagupta lays down the rules regarding *evolution* and *involution* as follows :

> The square of a positive or a negative number is positive .....The (sign of the) root is the same, as was that from which the square was derived[1].

As regards the latter portion of this rule, Pṛthūdaka Svāmī has the following comment to make : "The square-root should be taken either negative or positive, as will be most suitable for subsequent operations to be carried on."

It will be interesting to observe the following observation of Mahāvīra (850 A. D.) regarding square-root of a negative quantity "Since a negative number by its own nature is not a square, it has no square-root."[2] So says Śrīpati : "A negative number by itself is non-square, so its square-root is unreal; so the rule (for the square-root) should be applied in the case of a positive number."[3]

Algebraic Operations

Brahmagupta and other algebraists recognise six operations as fundamental in algebra : addition, subtraction, multiplication, division, squaring and the extraction of the square-root.

Regarding *addition* and *subtraction* Brahmagupta says :

> Of the unknowns, their squares, cubes, fourth powers, fifth powers, sixth powers, etc., addition and subtraction are (performed) of the like; of the unlike (they mean simply their) statement severally.[4]

In place of "of the like", Bhāskara II uses the term "of those of the same species (*jāti*) amongst unknowns" :

> Addition and subtraction are performed of those of the same species (*jāti*) amongst unknowns; of different species they mean their separate statement.[5]

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1. खोद्धृत मूलं धनं वा तच्छेदं खमृणधनविभक्तं वा ।
   ऋणधनयोर्वर्गः स्वं खं स्वस्य पदं कृतिर्यत् तत् ॥ —*BrSpSi*, XVIII. 35
2. *GSS*. I. 52.
3. *Śiśe*. XIV. 5.
4. अव्यक्तवर्ग घनकौ वर्ग पंचगत षड्गतादीनाम् ।
   तुल्यानां संकलित व्यवकलिते पृथगतुल्यानाम् ॥ —*BrSpSi.XVIII. 41.*
5. योगोऽन्तरं तेषु समान जात्योर्विभिन्न जात्योश्च पृथक् स्थितिश्च ।
   — *Bhāskara* II, *Bījagaṇita*.

This means that the numerical coefficients of x cannot be added to or subtracted from the numerical coefficients of y or x^2 or x^3 or xy and so on because these terms belong to different *jāti* or they do not belong to the category of the "*like*".

Again, regarding *multiplication*, Brahmagupta says :

> The product of the two like unknowns is a square; the product of three or more like unknowns is a power of that designation. The multiplication of unknowns of unlike species is the same as the mutual product of symbols; it is called *bhāvita* (product or factum).[1]

Having given the rules of the operations for addition, subtraction and multiplication, Brahmagupta does not think it necessary to deal with other operations. His section on the calculations with zero, negative and positive quantities ends here.

How is an Equation Formed?

Pṛthūdaka Svāmī while commenting on a verse in *Brāhma-sphuṭasiddhānta* speaks as follows :

> In this case, in the problem proposed by the questioner, *yāvat-tāvat* is put for the value of the unknown quantity. Then performing multiplication, division etc. as required in the problem the two sides shall be carefully made equal. The equation being formed in this way, then the rule (for its solution) follows.[2]

Plan for Writing Equations

When in regards to a given problem, an equation has been formed, it has to be written down for further operations. This writing down of an equation is technically known as *nyāsa* Perhaps the oldest record of *nyāsa* is to be found in the *Bakhaśālī Manuscript*. According to the procedure prescribed in this work, the two sides of an equation are put down one after the other in the same line without any sign of equality being interposed. Thus the equations :

$$\sqrt{x+5}=s,\ \sqrt{x-7}=t$$

appear as

1. सदृशद्विवधो वर्गस्त्र्यादिवधस्तद् गतोऽन्यजातिवधः ।
   अन्योऽन्यवर्णघातो भावितकः पूर्ववच्छेषम् ॥ —*BrSpSi.*XVIII. 42.
2. *BrSpSi.* XVIII. 43 (com)

0	5	*yu*	*mū*	0	*sa*	0	7	+	*mū*	0
1	1			1		1	1			1

Here *yu* (यु) stands for *yuta* (युत), meaning added, subtraction is+sign, derived from Kṣaya or (क्षय) meaning diminished, *gu* (गु) for guṇa or *guṇita*, meaning multiplied; *bhā* (भा) for division from *bhājita* and *mū* (मू) for square-root, from *mūla* meaning root; *zero* (०) was used to mark a vacant place.

Again, the following equation

$$x+2x+3\times 3x+12\times 4x=300$$

is represented as ;

0	2	1	3	3	12	4	*dṛśya* 300
1	1	1	1	1	1	1	

There is no sign for unknown in the *Bakhaśālī Manuscript.*

Later on this plan of writing equations as adopted in *Bakhaśālī Manuscript* was abandoned in India; a new one was adopted in which the two sides are written one below the other without any sign of equality. It must be stated that in this new plan the term of similar denominations are usually written one below the other and even the terms of absent denominations on either side are clearly indicated by putting zeros as their coefficients. We find a reference to this new plan in the algebra of Brahmagupta.

> From which the square of the unknown and the unknown are cleared, the known quantities are cleared (from the side) *below* that[1].

Here in this verse, the words "*adhastāt*" clearly indicate that one side of the equation is written below the other., As an illustration, Pṛthūdaka Svāmī represented the equation[2] :—

$$10x-8=x^2+1$$

as :

yā	*va*	0	*yā*	10	*rū*	$\dot{8}$	$(x^2.0+x.10-8)$
yā	*va*	1	*yā*	0	*rū*	1	$(x^2.1+x.0+1)$

which means, x^2 was written as *yāvat-varga* (*yā va*) and x was written as *yāvat* or *yā*. The minus sign was represented by a dot at the top of the number.(—8 was written as $\dot{8}$). We shall take another illustration from Pṛthūdaka Svāmī.

He would write the equation

$$197x-1644\ y-z=6302$$

as

1. *BrSpSi.* XVII. 43; compare also Bhāskara II, *Bījagaṇita.*
2. *Br SpSi.* XVIII. 49 (*com.*)

yā	197	*kā*	1644	*nī*	1	*rū*	0
yā	0	*kā*	0	*nī*	0	*rū*	6302

Here the first unknown x is represented by *yā(vat)*, the second unknown y by *kā(laka)* and the third unknown z by *nī-(laka)* and the term without unknown, a mere number is wriiten by *rū(paka)*. The two sides, one written below the other if written in the present form, would appear as :

$$197x—1644y—z+0=0x+0y+0z+6302.$$

The *Bījagaṇita* of Bhāskara II also follows the same procedure. One instance from it would be quoted here to illustrate the method of expressing equations.

$$8x^3+4x^2+10y^2x=4x^3+12y^2x$$

or $8x^3+4x^2+10y^2x=4x^3+0x^2+12y^2x$

is written as follows on Bhāskara's or Brahmagupta's plan :

x^3 is *ghana* of *yāvat* (abbreviated as *yā gha*)
x^2 is *varga* of *yāvat* (abbreviated as *yā va*)
y^2 is *varga* of *kālaka* (abbreviated as *kā va*)
the coefficients 10 and 12 are *bhāvita* (abbreviated as *bhā*).

The equation is :

yā gha 8 *yā va* 4 *kā va yā. bhā* 10
yā gha 4 *yā va* 0 *kā va yā. bhā* 12

Datta and Singh state that the use of the old plan of writing equations is sometimes met with in later works also. For instance, in the MS. of Pṛthūdaka Svāmī's commentary[1] on the *Brāhmasphuṭasiddhānta*, an incomplete copy of which is preserved in the library of the Asiatic Society of Bengal (No. I B6), we find a statement of equations thus : "first side *yāvargaḥ* 1 *yāvakaḥ* 200 *rū* 0; second side *yāvargaḥ* 0 *yāvakaḥ* 0 *rū* 1500.

Śodhana or Clearance of an Equation

After *nyāsa* or statement of an equation, the operation to be performed is known as *śodhana* (clearance) or *saṁśodhana* (equi-clearance or complete clearance). The nature of this clearance varies according to the kind of equation In the case of an equation in one unknown only, whether linear,

1. *BrSpSi.* XII. 15 (*com.*)

quadratic or of higher powers, one side of it is cleared of the unknowns of all denominations and the other side of it of the absolute terms, so that the equation is ultimately reduced to one of the form

$$ax^2+bx=c,$$

where a, b, c may be positive or negative; some of them may even be zero. Thus Brahmagupta observes :

> From which the square of the unknown and the unknown are cleared, the known quantities (*rūpāṇi*) are cleared (from the side) *below* that.[1]

On this Pṛthūdaka Svāmi comments as follows :

> This rule has been introduced for that case in which the two sides of the equation having been formed in accordance with the statement of the problem, there are present the square and other powers of the unknown together with the (simple) unknown. The absolute terms should be cleared off from the side opposite to that from which are cleared the square (and other powers) of the unknown and the (simple) unknown. When perfect clearance (*saṁśodhana*) has been thus made...[2]

Śrīdhara and Bhāskara II have also given the rules of clearance almost on the same lines. Thus the equation

$$yā\ va\ 0\ yā\ 10\ rū\ \dot{8}$$
$$yā\ va\ 1\ yā\ \ 0\ rū\ 1$$

after perfect clearance having been made will be (according to Pṛthūdaka Svāmī)

$$yā\ va\ 1\ yā\ 1\ 0\ rū\ \dot{9}$$

i.e. the equation $10x-8=x^2+1$
after clearance would become

$$x^2-10x=-9.$$

Classification of Equations

Usually equations are classified as :
simple equation : *yāvat-tāvat*
quadratic : *varga*

1. वर्गाव्यक्ताः शोध्या यस्माद् रूपाणि तदधस्तात् ॥ —*BrSpSi.* XVIII. 43.
2. *BrSpSi.* XVIII. 43 (*com.*)

cubic : *ghana*
biquadratic : *varga-varga*

Brahmagupta classified them as

(i) equations in one unknown quantity : *eka-varṇa samīkaraṇa.*

(ii) equations in several unknowns : *aneka-varṇa samīkaraṇa.*

(iii) equations involving products of unknowns : *bhāvita.*

Eka-varṇa samīkaraṇas (equations with one unknown) are further divided into (i) linear equations, and (ii) quadratic equations (*avyakta-varga samīkaraṇa*).

Pṛthūdaka Svāmī has classified equations in a different manner as follows :

(i) linear equations with one unknown : *eka-varṇa samīkaraṇa.*

(ii) linear equations with more unknowns : *aneka-varṇa samīkaraṇa.*

(iii) equations with one, two or more unknowns in their second or higher powers : *madhyamāharaṇa.*

(iv) equations involving products of unknowns : *bhāvita.*

As the method of solution of an equation of the third class (i.e. equations with one or several unknowns in their second or higher powers) is based upon the principle of the elimination of the middle term, that class is called by the term *madhyama* (middle) *āharaṇa* (elimination). The classification of Brahmagupta and Pṛthūdaka Svāmī more or less received recognition by later writers on algebra as Bhāskara II and others.

Linear Equations with One Unknown and Their Solutions

The first solution of a linear equation with one unknown is obtainable in the *Śulba Sūtras* but not through an algebraic process,—the Śulba process is geometrical. It is said that there is a reference in the *Sthānāṅga Sūtra* (c. 300 B.C.) to a linear equation by its name *yāvat-tāvat*. There has been a good deal of

controversy regarding the date of the *Bakhaśalī Manuscript* where we have definitely a method of solving linear equations by the *Rule of False Position.* It would be interesting to give an account of this rule here by taking an illustration from the *Bakhaśālī Manuscript.*

Problem :

> The amount given to the first is not known. The second is given twice as much as the first; the third thrice as much as the second; and the fourth four times as much as the third. The total amount distributed is 132. What is the amount of the first ?
>
> (*BMS.* Folio 23, recto)

In modern algebraic language, the solution of the problem would be given by the equation

$$x+2x+6x+24x=132$$

where x is the amount given to the first.

The solution of this equation is given as follows in the *Bakhaśālī Manuscript* :

> 'Putting any desired quantity in the vacant place'; any desired quantity is ‖ 1 ‖ , 'then construct the series'
>
> | 1 | 2 | 2 | 3 | 6 | 4 |
> |---|---|---|---|---|---|
> | 1 | 1 | 1 | 1 | 1 | 1 |
>
> 'multiplied' ‖ 1 | 2 | 6 | 24 | ; 'added' 33. 'Divide the visible quantity' | 132 / 33 | ; which) on reduction becomes | 4 / 1 |. (This is) the amount given (to the first)
>
> (*BMS.* Folio 23, recto)

The Rule of False Position may be regarded as an early stage of the development of the science of algebra, since no symbol could have been evolved for an unknown quantity. As soon as the system of notations was introduced, the application of this Rule was no longer considered as necessary. Thus we find that Āryabhaṭa I does not mention of this Rule.

Āryabhaṭa I states as follows regarding the solution of linear equations :

> The difference of the known 'amounts' (*rūpaka*) relating to two persons should be divided by the difference

> of the coefficients on the unknown (*gulikā*). The quotient will be the value of the unknown (*gulikā*), if their possessions be equal.[1]

The original verse contains the term "*gulikāntara*" which has been here translated as the difference of the coefficients of the unknowns. We have already stated earlier that Āryabhaṭa uses the term *gulikā* or shot for an unknown quantity.(*gulikāntara* literally means only the difference of unknowns). This practice is also followed by other Indian algebraists. Pṛthūdaka Svāmī rightly observed that according to the usual practice in this country, "the coefficient of the square of the unknown is called the square (of the unknown) and the coefficient of the (simple) unknown is called the unknown.[2]

The rule given by Āryabhaṭa, then, contemplates a problem of this kind :

> Two persons, who are equally rich, possess respectively a, b times a certain unknown amount together with c, d units of money in cash. What is that amount ?

Now if x be the amount unknown, then according to the problem

$$ax+c=bx+d$$

Thence

$$x=\frac{d-c}{a-b}$$

Āryabhaṭa has merely expressed this solution in his language.

Regarding the solution of linear equations, Brahmagupta says :

> In a (linear) equation in one unknown, the difference of the known terms taken in the reverse order, divided by the difference of the coefficients of the unknown (is the value of the unknown).[3]

---

1. गुलिकान्तरेण विभजेद् द्वयोः पुरुषयोस्तु रूपकविशेषम् ।
लब्धं गुणिकामूल्यं यद्यर्थं कृतं भवति तुल्यम् ॥ —*Ārya.* II. 30

2. *BrSpSi*, XVIII. 44 (*com.*)

3. अव्यक्तान्तरभक्तं व्यस्तं रूपान्तरं समेऽव्यक्तः ।
वर्गाव्यक्ताः शोध्या यस्माद् रूपाणि तदधस्तात् ॥ —*BrSpSi.* XVIII. 43

Similar solutions have been offered by the other Indian algebraists who followed Brahmagupta like Śrīpati, Bhāskara II and Nārāyaṇa. Here again, we take a problem proposed by Brahmagupta in this connection :

Problem :

Tell the number of elapsed days for the time when four times the twelfth part of the residual degrees increased by one, plus eight will be equal to the residual degrees plus one.[1]

Pṛthūdaka Svāmī has solved this problem as follows :

Here the residual degrees are (put as) *yāvat-tāvat, yā*; increased by one, *yā* 1 *rū* 1; twelfth part of it, $\frac{yā\ 1\ rū\ 1}{12}$; four times this, $\frac{yā\ 1\ \ rū\ 1}{3}$; plus the absolute quantity eight, $\frac{yā\ 1\ \ rū\ 25}{3}$. This is equal to the residual degrees plus unity. The statement of both sides tripled is

yā 1	*rū* 25
yā 3	*rū* 3

This difference between the coefficients of the unknown is 2. By this the difference of the absolute terms, namely 22, being divided, is produced the residual of the degrees of the Sun, 11. These residual degrees should be known to be irreducible. The elapsed days can be deduced then, (proceeding) as before.

If put in the modern notations, it means the solution of the equation :

$$\frac{4}{12}(x+1)+8=x+1,$$

from which we have

$$x+25=3x+3$$

or $$2x=22$$

or $$x=11.$$

Rule of Concurrence or Saṁkramaṇa

Brahmagupta has included this rule in algebra, whereas other Indian mathematicians included it in arithmetic. *Saṁ-*

1. सैकादंशकशेषाद् द्वादशभागश्चतुर्गुणोऽष्टयुतः ।
सैकांशशेषतुल्यो यदा तदाऽहर्गणं कथय ॥ —*BrSpSi. XVIII.* 46

kramaṇa is the solution of the simultaneous equations of the type :

$$x+y=a$$
$$x-y=b$$

Brahmagupta's rule for solution is:

> The sum is increased and diminished by the difference and divided by two; (the result will be the two unknown quantities) : (this is) concurrence (*Saṁkramaṇa*).[1]

Brahmagupta restates this rule in the form of a problem and its solution :

> The sum and difference of the residues of two (heavenly bodies) are known in degrees and minutes. What are the residues? The difference is both added to and subtracted from the sum, and halved; (the results are) the residues.[2]

Linear Equations with Several Unknowns

The first mention of a solution of the problem with more than one unknown is found in the *Bakhaśālī Manuscript*, and a system of linear equations of this type is solved in the Bakhaśālī treatise substantially by the False Position Rule.

A generalised system of linear equations will be

$$b_1\Sigma x-c_1x_1=a_1,\ b_2\Sigma x-c_2x_2=a_2,\ldots\ldots$$
$$b_n\Sigma x-c_nx=a_n$$

Therefore

$$\Sigma x=\frac{\Sigma(a/c)}{\Sigma(b/c)-1}$$

Hence

$$x_r=\frac{b_r}{c_r}\times\frac{\Sigma(a/c)}{\Sigma(b/c)-1}-\frac{a_r}{c_r}$$
$$r=1, 2, 3,\ldots\ldots, n$$

One particular case, where $b_1=b_2=b_3=\ldots\ldots=b_n=1$ and $c_1=c_2=c_3=\ldots\ldots=c_n=c$ has been treated by Brahmagupta at one place. He gives the rule as follows :

1. योगोऽन्तरयुतहीनो द्विहृतः संक्रमणमन्तरविभक्तं वा ।
   वर्गान्तरमन्तरयुतहीनं द्विहृतं विषमकर्म ।। —*BrSpSi. XVIII. 36*
2. भागकला विकलैक्यं दृष्ट्वा विकलान्तरं च के शेषे ।
   ऐक्यं द्विधाऽन्तराधिक हीनं च द्विभाजितं शेषे ।। —*BrSpSi. XVIII. 96*

> The total value (of the unknown quantities) plus or minus the individual values (of the unknowns) multiplied by an optional number deing severally (given), the sum (of the given quantities) divided by the number of unknowns increased or decreased by the multiplier will be the total value; thence the rest (can be determined).[1]

$$\Sigma x \pm cx_1 = a_1,\ \Sigma x \pm cx_2 = a_2,\ \Sigma x \pm cx_3 = a_3, \ldots\ldots,$$
$$\Sigma x \pm cx = a_n$$

Therefore

$$\Sigma x = \frac{a_1 + a_2 + a_3 + \ldots\ldots + a_n}{n \pm c}$$

Hence

$$x_1 = \frac{1}{c}\left(\pm a_1 \mp \frac{a_1 + a_2 + a_3 + \ldots\ldots + a_n}{n \pm c} \right);$$

and so on for x_2, x_3 etc.

Now we shall give the rule enunciated by Brahmagupta for solving linear equations involving several unknowns :

> Removing the other unknowns from (the side of) the first unknown and dividing by the coefficient of the first unknown, the value of the first unknown (is obtained). In the case of more (values of the first unknown), two and two (of them) should be considered after reducing them to comon denominators. And (so on) repeatedly. If more unknowns remain (in the final equation), the method of the pulveriser (should be employed). (Then proceeding) reversely (the values of other unknowns can be found).[2]

Pṛthūdaka Svāmī has commented on this rule as follows :

> In an example in which there are two or more unknown quantities, colours such as *yāvat-tāvat*, etc., should be assumed for their values. Upon them should

---

1. गच्छधनमिष्ट गुणितैर्धनैर्युतोनं पृथक् पृथक् सहितम् ।
गुणकयुतोन पदहृतं सर्वधनमतोऽवशेषाणि ॥ —*BrSpSi.* XIII. 47

2. आद्यादवर्णादन्यान् वर्णान् प्रो ह्याद्यमानमाद्यहृतम् ।
सदृशच्छेदाक्षसकृद् द्वौ द्वौ व्यस्तौ कुट्टको बहुषु ॥ —*BrSpSi.* XVIII. 51

be performed all operations conformably to the statement of the example and thus should be carefully framed two or more sides and also equations. Equi-clearance should be made first between two and two of them and so on to the last: from one side one unknown should be cleared, other unknowns reduced to a common denominator and also the absolute numbers should be cleared from the side opposite. The residue of other unknowns being divided by the residual coefficient of the first unknown will give the value of the first unknown. If there be obtained several such values, then with two and two of them, equations should be formed after reduction to common denominators. Proceeding in this way to the end find out the value of one unknown. If that value be (in terms of) another unknown then the coefficients of those two will be reciprocally the values of the two unknowns. If, however, there be present more unknowns in that value, the method of the pulveriser should be employed. Arbitrary values may then be assumed for some of the unknowns.

Datta and Singh have said that the above rule of Brahmagupta, and also the one indicated in the commentary of Pṛthūdaka Svāmī, embraces the solution of indeterminate as well as the determinate equations. In fact, all the examples given by Brahmagupta in illustration of the rule are of indeterminate character. So far as the determinate simultaneous equations are concerned, Brahmagupta's method for solving them will be easily recognised to be the same as our present one.

Quadratic Equations

The geometrical solution of a quadratic equation in this country would take us to the Vedic Śulba period. The *Bakhaśālī Manuscript* also contains certain problems which need the solving of quadratic equations. I shall quote one out of the numerous available :

> A certain person travels *s yojana* on the first day and *b yojana* more on each successive day. Another who travels at the uniform rate of *S yojana* per day, has a start of *t* days. When will the first man overtake the second ?

This problem would today be expressed in terms of the following equation :

$$S(t+x)=x\{s+\left(\frac{x-1}{2}\right)b\},$$

where x is the number of days after which the first overtakes the second. We may write this equation as

$$bx^2-\{2(S-s)+b\}x=2tS$$

whence the value x would be after solving the quadratic :

$$x=\frac{\sqrt{\{2(S-s)+b\}^2+8bts}+\{2(S-s)+b\}}{2b}$$

The *Bakhaśālī Manuscript* gives this solution as follows :

> The daily travel (S) diminished by the march of the first day (s) is doubled; this is increased by the common increment (b). That (*sum*) multiplied by itself is designated (as the *kṣepa* quantity). The product of the daily travel and the start (t) being multiplied by eight times the common increment, the *kṣepa* quantity is added. The square-root of this (is increased by the *kṣepā* quantity; the sum divided by twice the common increment will give the required number of days), (*BMS*. Folio 5, recto)

Āryabhaṭa I (499 A.D.) is regarded as the founder of algebra, since he gives the solutions of a few quandratic problems. For example, to find the number of terms of an arithmetical progression (A.P.), he gives the following rule :

> The sum of the series multiplied by eight times the common difference is added by the square of the difference between twice the first term and the common difference: the square-root (of the result) is diminished by twice the first term and (then) divided by the common difference : half of this quotient plus unity is the number of terms.[1]

In the modern notations of algebra, the solution would be expressed as follows :

1. गच्छोऽष्टोत्तर गुणिताद् द्विगुणाद्युत्तर विशेषवर्गयुतात् ।
मूलं द्विगुणाद्यूनं स्वोत्तर भजितं सरूपार्धं ॥ —*Āryа.* II, 20

$$n=\tfrac{1}{2}\left\{\frac{\sqrt{8bs+(2a-b)^2}-2a}{b}+1\right\}$$

There is another certain interest problem[1], the solution of which has been provided in the *Āryabhaṭīya* as

$$x=\frac{\sqrt{Apt+(p/2)^2}-p/2}{t}$$

which is the solution of the quadratic equation :

$$tx^2+px-Ap=0$$

Āyabhaṭa I has thus given the solutions of a few quadratic equations, but he nowhere gives the procedure of solving these equations.

We give here the Rules of Brahmagupta for the solution of quadratic equations. He undoubtedly is not the discoverer of these rules; but perhaps for the first time in the history of algebra we find the process of solving a quadratic equation so clearly indicated.

First Rule :

> The quadratic : the absolute quantities multiplied by four times the coefficient of the square of the unknown are increased by the square of the coefficient of the middle (i.e. unknown); the square-root of the result being diminished by the coefficient of the middle and divided by twice the coefficient of the square of the unknown, is (the value of) the middle."[2]

This expressed in the modern notations would mean

$$x=\frac{\sqrt{4ac+b^2}-b}{2a}$$

It would be noted that in this rule, Brahmagupta has employed the term *madhya* (middle) to imply the simple unknown as well as its coefficient. The origin of the term is doubtless connected with the mode of writing the quadratic equation in the form

$$ax^2+bx+0=0x^2+0x+c$$

so that there are three terms on each side of the equation.

1. मूलफलं सफलं कालमूल गुणमर्धमूल कृति युक्तम् ।
   मूलं मूलार्धोनं कालहृतं स्यात्स्वमूलफलम् ॥ —*Ārya. II. 25.*
2. वर्ग चतुर्गुणितानां रूपाणां मध्यवर्गसहितानाम् ।
   मूलं मध्येनोनं वर्ग द्विगुणोद्धृतं मध्यः ॥ —*BrSpSi.* XVIII. 44.

Second Rule :

> The absolute term multiplied by the coefficient of the square of the unknown is increased by the square of half the coefficient of the unknown; the square-root of the result diminished by half the coefficient of the unknown and divided by the coefficient of the square of the unknown is the unknown.[2]

This when expressed in the modern algebraic notations would be

$$x=\frac{\sqrt{ac+(b/2)^2}-(b/2)}{a}$$

Here if the quadratic equation is

$$ax^2+bx+c=0$$

the 'absolute term' is c (the one without the unknown x), 'the coefficient of the square of unknown' means the coefficient of x^2, i.e. a, and the 'coefficient of the unknown' means the coefficient of x, i.e. b.

The above two methods of Brahmagupta are exactly the same as were suggested by Āryabhaṭa I.

The root of the quadratic equation for the number of terms of an arithmetic progression (A.P.) is given by Brahmagupta according to the first rule [2] :

$$n=\frac{\sqrt{8bs+(2a-b)^2}-(2a-b)}{2b}$$

Third Rule :

> Brahmagupta also suggests a Third Rule which is very much the same as is used commonly now. Though it has not been expressedly suggested as a new rule, we find its application in a few instances. For example this rule has been suggested in connection with the following problem on interest :
>
> > A certain sum (p) is lent out for a period (t_1); the interest accrued (x) is lent out again at this

---

1. वर्गाहत रूपाणामव्यक्तार्धकृति संयुतानां यत् ।
   पदमव्यक्तार्धोनं तद्वर्ग विभक्तमव्यक्तः ॥ —*BrSpSi.* XVIII. 45
2. उत्तरहीनद्विगुणादि शेषवर्गं धनोत्तराष्टवधे ।
   प्रक्षिप्य पदं शेषोनं द्विगुणोत्तरहृतं गच्छः ॥ —*BrSpSi.* XII. 18

rate of interest for another period (t_2) and the total amount is A. Find x.

The equation for determining x is

$$\frac{t_2}{pt_1}x^2+x=A.$$

The solution of this equation would be :

$$x=\sqrt{\left(\frac{pt_1}{2t_2}\right)^2+\frac{A}{t_2}\cdot 1}-\frac{pt_1}{2t_2}$$

Brahmagupta has stated the result in exactly the same form. Pṛthūdaka Svāmī has illustrated it in solving the following problem of interest :

Problem :

A sum of five hundred *paṇas* (p) is lent out for a period of 4 months (t_1); the interest accrued (x) is lent out again at this rate of interest for another period of 10 months (t_2) and the total amount is 78 (A). Give the *pramāṇa-phala*, i.e., the interest accrued x.

Here *pramāṇa-kāla* (t_1)=4 months

pramāṇa-dhana (p)=500 paṇas

para-kāla (t_2), the subsequent period=10 months

miśra dhana or the total interest accrued (A) =78 *paṇas*.

Brahmagupta states his solution of such quadratics like this :

Take the product of the *pramāṇa-dhana* (p) or the sum originally lent out and *pramāṇa-kāla*, i.e. the period for which originally lent out (t_1); and divide by the *para-kāla* or the subsequent time (t_2); place this result at two places. Multiply the one placed at the first place with the *miśra-dhana* (A), that is with the total interest accrued; in this product add the square of half the one placed in the second place; now take the square-root of it, and from it subtract half of the one placed at the second place.[1]

---

1. कालप्रमाणघातः परकालहृतो द्विधाऽऽत्वमिश्रवधात् ।
अन्त्यार्धकृति युतात् पदमन्त्यार्धोनं प्रमाण फलम् ॥ —*BrSpSi. XII. 15*.

Thus in the above example the product of *pramāṇa-dhana* and *pramāṇa kāla* divided by *parākāla* is (pt_1/t_2) is $\frac{500\times4}{10} = 200$. This is first multiplied by the total interest accrued (A); it becomes $200\times78=15600$. To this is now added square of half of 200 (which is 10000); it becomes 15600 plus 10000=25600. Its square-root is taken which is 160. From this is subtacted half of the quantity (i.e. half of 200 which is 100). Thus 160–100=60, which is the answer. It was the interest which first accrued (x).

Another Quadratic Problem:

Brahmagupta refers to an astronomical problem which involves the quadratic equation

$$(72+a^2)x^2 \mp 24\, apx = 144\left(\frac{R^2}{2}-p^2\right),$$

where $a=agra$ (the sine of the amplitude of the Sun), $b=palabhā$ (the equinoctial shadow of a gnomon 12 anguli long), R=radius, and $x=koṇaśaṅku$ (sine of the altitude of the Sun when his altitude is 45°). Dividing out by $(72+a^2)$ we have

$$x^2 \mp 2mx = n,$$

where

$$m=\frac{12\,ap}{72+a^2},\quad n=\frac{144(R^2/2-p^2)}{72+a^2}$$

Therefore we have

$$x=\sqrt{m^2+n}\pm m,$$

as stated by Brahmagupta. We find the same result in the *Sūrya-siddhānta* and in the text of Śrīpati. Āryabhaṭa II (1150) also followed the method of Āryabhaṭa I and Brahmagupta in solving a quadratic equation in connection with finding out the number of terms in an arithmetical progression (A.P.) whose first term is (a), common difference is b and the sum is s. The number of terms n is given by[1]

$$n=\frac{\sqrt{2bs+(a-b/2)^2}-a+b/2}{b}$$

Two Roots of a Quadratic Equation and Brahmagupta

A quadratic equation has two roots. This must have been known to Indian algebraists even at a very early stage. Bhāskara II in his *Bijaganita* has quoted a rule ascribed to an ancient writer Padmanābha whose works are not available now :

1. *Mahāsiddhānta*, Bhāskara II, XV. 50

> If (after extracting roots) the square-root of the absolute side (of the quadratic) be less than the negative absolute term on the other side, then taking it negative as well as positive two values (of the unknown) are found[1].

The term used here is *dvividhotpadyate mitih* which means that two values are obtained.

The existence of two roots of a quadratic equation appears to have been known also to Brahmagupta (628 A.D.). In illustration of his rules for the solution of a quadratic he has stated two problems involving practically the same equation :

> *Problem I* : The square-root of the residue of the revolution of the Sun less 2 is diminished by 1, multiplied by 10 and added by 2; when will this be equal to the residue of the revolution of the Sun less 1, on Wednesday ?[2]
>
> *Problem II* : When will the square of one-fourth the residue of the exceeding months less three be equal to the residue of the exceeding months ?

We shall follow Pṛthūdaka Svāmi in solving the Problem I. In this problem the residue of the revolutions of the Sun may be supposed to be x^2+2; then by the question, we hgve

$$10\,(x-1)+2=x^2+1,$$
$$\text{or } x^2-10x=-9$$

Again in Problem II, if we put $4x$ for the residue of the exceeding month, then we have

$$(x-3)^2=4x$$
$$\text{or } x^2-10x=-9.$$

Now by the second rule of Brahmagupta, retaining both the signs of the radical, we get :

$$x=5\pm\sqrt{25-9}=9 \text{ or } 1,$$

---

1. व्यक्त पक्षस्य चेन्मूलमन्यपक्षर्णरूपतः ।
   अल्पं धनर्णगं कृत्वा द्विविधोत्पद्यते मितिः ॥ —Bhāskaka, *Bijaganita*
2. मण्डलशेषाद् द्व्यूनान्मूलं च्येकं दशाहतं द्वियुतम् ।
   मण्डलशेषं च्येकं भानोर्ऽदिने कदा भवति ॥ —*BrSpSi.* XVIII. 49
3. अधिमासशेषपादात् त्र्यूनाद्वर्गोऽधिमासशेषसमः ।
   अवमावशेषतो वावमशेषसमः कदा भवति ॥ —*BrSpSi.* XVIII. 50.

As shown by Pṛthūdaka Svāmī, the first value is taken by Brahmagupta for the Problem I and second value for the problem II. Thus it is quite clear that Brahmagupta uses sometimes the positive and at other times the negative sign with the radical. Hence we shall say that Brahmagupta knew that a quadratic equation would have two roots, and according to the requisiteness of the problem, one value out of the two would be utilised.

Simultaneous Quadratic Equations

Indian authors usually treated problems involving various forms of simultaneous quadratic equations.

$$\text{(i)}\ \left.\begin{matrix} x-y=d \\ xy\ \ =b \end{matrix}\right\} \qquad \text{(ii)}\ \left.\begin{matrix} x+y=a \\ xy\ \ =b \end{matrix}\right\}$$

$$\text{(iii)}\ \left.\begin{matrix} x^2+y^2=c \\ xy=b \end{matrix}\right\} \qquad \text{(iv)}\ \left.\begin{matrix} x^2+y^2=c \\ x+y=a \end{matrix}\right\}$$

For the solution of the combination (i), Āryabhaṭa I gives the following rule in his *Āryabhaṭiya*.

> The square-root of four times the product (of two quantities) added with the square of their difference, being added and diminished by their difference and halved gives the two multiplicands.[1]

This means that

$$x=\tfrac{1}{2}(\sqrt{d^2+4b}+d),\quad y=\tfrac{1}{2}(\sqrt{d^2+4b}-d)$$

For the solution of the same combination, Brahmagupta states as follows :

> The square-root of the sum of the square of the difference of the residues and two squared times the product of the residues, being added and subtracted by the difference of the residues, and halved (gives) the desired residues severally.[2]

(Here by difference of the residues is mesnt $x-y$; and by product of the residues is meant xy.)

Brahmagupta does not seem to give the solution for simultaneous equations of the combination (ii). Mahāvīra (850 A.D.)

1. द्विकृति गुणात्संवर्गाद् द्वयन्तरवर्गेण संयुतान्मूलम् ।
   अन्तरयुक्तं हीनं तद्गुणकारद्वयं दलितम् ॥ —*Ārya. II. 24*
2. शेषवधाद् द्वि कृति गुणात् शेषान्तर वर्ग संयुतान्मूलम् ।
   शेषान्तरोन युक्तं दलितं शेषे पृथगभीष्टे ॥ —*Br SpSi. XVIII. 99*

has given the solution :

> Subtract four times the area (of a rectangle) from the square of the semi-perimeter then by *saṅkramaṇa* between the square-root of that (remainder) and the semi-perimeter, the base and the upright are obtained.[1] (*GSS.* VII. 129½)

This expressed in the modern notations would be :

$x=\frac{1}{2}(a+\sqrt{a^2-4b})$, $y=\frac{1}{2}(a-\sqrt{a^2-4b})$

For the combination (iii), Mahāvīra in his *Gaṇita-Sāra-Saṃgraha* gives the following rule :

> Add to and subtract twice the area (of a rectangle) from the square of the diagonal and extract the square-roots. By *saṅkramaṇa* between the greater and lesser of these (roots), the side and upright (are found).[2]

This put in modern notations would be :

$$x=\tfrac{1}{2}\sqrt{c+2b}+\sqrt{c-2b}).$$
$$y=\tfrac{1}{2}(\sqrt{c+2b}-\sqrt{c-2b}).$$

For the combination (iv), Āryabhaṭa I gives the following rule :

> From the square of the sum (of two quantities) subtract the sum of their squares. Half of the remainder is their product.[3]

The remaining operations will be similar to those for the equations (ii); so that

$x=\frac{1}{2}(a+\sqrt{2c-a^2})$, $y=\frac{1}{2}(a-\sqrt{2c-a^2})$.

Brahmagupta in this connection says :

> Subtract the square of the sum from twice the sum of squares; the square-root of the remainder being added to and subtracted from the sum and halved, (gives) the desired residues.[4]

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1. *GSS.* VII. 129½
2. *GSS.* VII. 127½
3. संपर्कस्य हि वर्गाद्विशोधयेदेव वर्गसंपर्कम् ।
   यत्तस्य भवत्यर्धं विद्याद् गुणकारसंवर्गम् ॥ —*Āaya.* II. 23
4. कृति संयोगाद् द्विगुणाद्युति वर्गं प्रोह्य शेष मूलं यत् ।
   तेन युतोनो योगो दलितः शेषे पृथगभीष्टे ॥ —*BrSpSi. XVIII.* 98

These equations have also been treated by Mahāvīra, Bhāskara II and Nārāyaṇa. Nārāyaṇa has attempted two other forms of quadratic equations :

(v) $\left.\begin{matrix} x^2+y^2=c \\ x-y=d \end{matrix}\right\}$ (vi) $\left.\begin{matrix} x^2-y^2=m \\ xy=b \end{matrix}\right\}$

For their solutions, see Datta and Singh, *Algebra*, P. 84.

Rule of Dissimilar Operations :

Datta and Singh say that the process of solving the following two particular cases of simultaneous quadratic equations was distinguished by most Indian mathematicians by the special designation *viṣama-karma* or dissimilar operation :

(i) $\left.\begin{matrix} x^2-y^2=m \\ x-y=n \end{matrix}\right\}$ (ii) $\left.\begin{matrix} x^2-y^2=m \\ x+y=p \end{matrix}\right\}$

These equations have been regarded by these mathematicians as if of fundamental importance. They have given the following solutions (expressed in modern algebraic symbols) :

For the combination (i) :

$$x=\tfrac{1}{2}\left(\frac{m}{n}+n\right),\ y=\tfrac{1}{2}\left(\frac{m}{n}-n\right),$$

For the combinaticn (ii) :

$$x=\tfrac{1}{2}\left(p+\frac{m}{p}\right),\ y=\tfrac{1}{2}\left(p-\frac{m}{p}\right).$$

We shall express these solutions as follows in the words of Brahmagupta :

> Thc difference of the squares (of the unknowns) is divided by the difference of the unknowns and the quotient is increased and diminished by the difference and divided by two ; (the results will be the two unknown quantities) ; (this is) dissimilar operation.

The same rule is restated by him on a different occasion in the course of solving a problem.

> If then the difference of their squares, also the difference of them (are given) ; the difference of the squares

1. योगोऽन्तरयुन्हीनो द्विहतः संक्रमणमन्तर विभक्तः वा ।
वर्गान्तरमन्तर-युतहीनं द्विहतं विषमकर्म ।

is divided by the difference of them, and this (latter) is added to and subtracted from the quotient and then divided by two ; (the results are) the residues whence the number of elapsed days (can be found).[1]

This *viṣama-karma* or dissimilar operation has been described by other Indian algebraists also, as Āryabhaṭa II (*Mahāsiddhānta*, XVII, 22); Śrīpati (*Siddhānta-śekhara*, XIV. 13) ; Bhāskara II (*Līlāvatī*) and Nārāyaṇa (*Gaṇita-kaumudī*, I, 32).

Indeterminate Equations of the First Degree

Āryabhaṭa I should be given the credit of giving for the first time a treatment of the indeterminate equation of the first degree. In his *Āryabhaṭīya*, we find a method for obtaining the general solution in positive integers of the simple indeterminate equation :

$$by-ax=c$$

for integral values of a,b,c, and further indicated how to extend it to get positive integral solutions of simultaneous indeterminate equations of the first degree. His disciple Bhāskara I (522) showed that the same method might be applied to solve :

$$by-ax=-c$$

and further that the solution of this equation would follow that of $by-ax=-1$. These methods of Āryabhaṭa I and Bhāskara I have also been adopted by Brahmagupta, and in certain cases, the improvement were suggested by Āryabhaṭa II in the middle of the tenth century A.D.

The problems which were treated by ancient Indian algebraists and which led them to the investigation of the simple indeterminate equation of the first degree may be classified under three heads :

Class I : To find a number N which being divided by two given numbers (a,b) will leave two given remainders (R_1, R_2).

Thus we have :

$$N=ax+R_1=by+R_2$$

---

1. तद्वर्गान्तरभाजे तदन्तरं चान्तरोद्धृतयुतोनम् ।
वर्गान्तरं विभक्तं द्वाभ्यां शेषे ततोद्गुणः । *BrSpSi*, XVIII, 97

Hence $by-ax=R_1-R_2$

Putting $c=R_1 \sim R_2$

we get $by-ax=\pm c$

the upper or lower sign being taken according as R_1 is greater than or less than R_2.

Class II : To find a number (x) such that its product with a given number (α) being increased or decreased by another given number (δ) and then divided by a third given number (β) will leave no remainder.

This means that in other words, we shall have to get the solution of :

$$\frac{\alpha x \pm r}{\beta}=y$$

in positive integers.

Class III : Here we have to deal with an equation of the form :

$$by \div ax = \pm c$$

Kuṭṭaka, Kuṭṭākāra and *Kuṭṭa :* These are the three terms which Brahmagupta has used in regards to the subject of indeterminate analysis of the first degree. Āryabhaṭa I has also described this method in brief, but he does not use the word *kuṭṭaka.* In the *Mahābhāskarīya* of Bhāskara I we have the terms *kuṭṭākāra* and *kuṭṭa* (522 A.D.) *MBh.* I. 41,49). These words have been translated into English as *pulveriser* or *grinder.* According to Datta and Singh, the Hindu method of solving the equation $by-ax= \pm c$ is essentially based on a process of deriving from it successively other similar equations in which the values of the coefficients (a,b) become smaller and smaller. Thus the process is indeed the same as that of breaking a whole thing into smaller pieces, and this accounts for its name *kuṭṭaka* or '*pulveriser*'.

In the problems of the Class I, the quantities (a and b) are called' divisors' *bhāgahāra, bhājaka, cheda* etc.) and R_1 and R_2 as 'remainders' (*agra* or *śeṣa* etc.). while in a problem of the Class II ; β is ordinarily called the 'divisor' (*bhāgahāra* or *bhājaka*) and γ the 'interpolator' *kṣepa, kṣepaka* etc.) ; here α is called the 'dividend' (*bhājya*), the unkown quantity to be found (x) is called the 'multiplier' or (*guṇaka* or *guṇakāra* etc) and y the

quotient or *phala*. In later years, Mahāvīra has called the unknwn number (x) as *rāśi*.

Preliminary Operations in Kuṭṭaka-Karma

Usually it has been suggested that in order that an equation of the form

$$by - ax = \pm c \text{ or } by + ax = \pm c$$

may be amenable to solution, the two numbers a and b must not have a common divisor; for otherwise, the equation would be absurd, unless the number c had the same common divisor. So before the rules which we shall give hereafter, could be applied, the numbers a, b, c must be made prime (*dṛḍha* or firm; *niccheda* or having no divisor, or *nirapavarta*, meaning irreducible to each other.

In this connection Bhāskara I writes :

> The dividend and divisor will become prime to each other on being divided by the residue of their mutual division. The operation of the pulveriser should be considered in relation to them.[1]

Similarly we find in the writings of Brahmagupta :

> Divide the multiplier and the divisor mutually and find the last residue; those quantities being divided by the residue will be prime to each other.[2]

Āryabhaṭa's Rule : Āryabhaṭa I is probably the first Indian writer on this subject, but the operation given by him is rather obscure. His disciple Bhāskara I has given the solution of indeterminate equations of the first degree in more satisfactory language. We shall give here the translation of Āryabhaṭa's verse from the *Āryabhaṭīya*, as rendered by Bibhutibhusan Datta, because other translations of this verse do very often confuse the sense :

> Divide the divisor corresponding to the greater remainder by the divisor corresponding to the smaller remain-

1. भूदिनेष्टगणान्योन्य भक्तशेषेण भाजितौ ।
हारभाज्यौ दृढौ स्यातां कुट्टाकारं तयोर्विदुः —*MBh*, I. 41

2. हृतयोः परस्परं यच्छेषं गुणकारभागहारकयोः ।
तेन हृतौ निश्छेदौ तावेव परस्परं हृतयोः । —*BrSpSi*. XVIII. 9.

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2. हृतयोः परस्परं यच्छेषं गुणकारभागहारकयोः ।
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der. The residue (and the divisor corresponding to the smaller remainder) being mutually divided, the last residue should be multiplied by such an optional integer that the product being added(in case the number of quotients of the mutual division is even) or subtracted (in case the number of quotients is odd) by the difference of the remainders (will be exactly divisible by the last but one remainder. Place the quotients of the mutual division successively one below the other in a column; below them the optional multiplier and underneath it the quotient just obtained). Any number below .: . the penultimate) is multiplied by the one just above it and then added by that just below it. Divide the last number (obtained by doing so repeatedly) by the divisor corresponding to the smaller remainder; then multiply the residue by the divisor corresponding to the greater remainder and add the greater remainder. (The result will be) the number corresponding to the two divisors.[1]

There is an alternative rendering of this passage also as follows :

Divide the divisor corresponding to the greater remainder by the divisor corresponding to the smaller remainder. The residue (and the divisor corresponding to the smaller remainder) being mutually divided (until the remainder becomes zero), the last quotient should be multiplied by an optional integer and then added (in case the number of quotients of the mutual division is even) or subtracted (in case the number of quotients is odd) by the difference of the remainders. (Place the other quotients of mutual division successively one below the other in a column; below them the result just obtained and underneath it the optional integer). Any

1. अधिकाग्रभागहारं छिन्द्यादूनाग्रभागहारेण।
शेषपरस्परभक्तं मतिगुणमग्रान्तरे क्षिप्तम् ॥
अथ उपरि गुणितमन्त्यगूनाग्रच्छेद भाजिते शेषम्।
अधिकाग्रच्छेदगुणं द्विच्छेदाग्रमधिकाग्रयुतम्। —*Arya. II.* 32-33

number below (i.e. the penultimate) is multiplied by the one just above it and then added by that just below it. Divide the last number (obtained by doing so repeatedly) by the divisor corresponding to the smaller remainder; then multiply the residue by the divisor corresponding to the greater remainder and add the greater remainder. (The result will be) the number corresponding to the two divisors.

Āryabhaṭa's problem may be enunciated thus :

To find a number (N) which being divided by two given numbers (a, b) will leave two given remainders (R_1, R_2).

This gives :

$$N=ax+R_1=by+R_2$$

(where R_1 is a greater remainder and R_2 lesser remainder, and a is the divisor corresponding to greater remainder and b the divisor corresponding to the lesser remainder.)

Denoting as before by c the difference between R_1, and R_2, we get

(i) $by=ax+c$, if $R_1>R_2$

(ii) $ax=by+c$, if $R_2>R_1$

the equation being so written as to keep c always positive.

Hence the problem now reduces to making either

$$\frac{ax+c}{b} \text{ or } \frac{by+c}{a}$$

according as $R_1>R_2$ or $R_2>R_1$, a positive integer. So Āryabhaṭa says : Divide the divisor corresponding to the greater remainder etc."

Now we shall proceed with the details of the operation as proposed by Datta and Singh in his *History of Hindu Mathematics, Part II. Algebra* :

Suppose $R_1>R_2$; then the equation to be solved will be

$$ax+c=by \qquad \text{...(i)}$$

a, b being prime to each other.

Let

$$
\begin{array}{l}
b)\ a\ (q \\
\quad bq \\
\hline
r_1)\ b\ (q_1 \\
\quad r_1q_1 \\
\hline
r_2)\ r_1\ (q_2 \\
\quad r_2q_2 \\
\hline
\quad r_3 \\
\quad \cdots \\
\hline
r_{m-1})\ r_{m-2}\ (q_{m-1} \\
\quad r_{m-1}\ q_{m-1} \\
\hline
r_m)\ r_{m-1}\ (q_m \\
\quad r_mq_m \\
\hline
\quad r_{m+1}
\end{array}
$$

Then we get (when $a<b$, we shall have $q=0$, $r_1=a$)

$$
\begin{aligned}
a &= bq+r_1 \\
b &= r_1q_1+r_2 \\
r_1 &= r_2q_2+r_3 \\
r_2 &= r_3q_3+r_4 \\
&\cdots \quad \cdots \quad \cdots \\
r_{m-2} &= r_{m-1}\ q_{m-1}+r_m \\
r_{m-1} &= r_mq_m+r_{m+1}
\end{aligned}
$$

Now, substituting the value of a in the given equation (i), we get

$$by=(bq+r_1)x+c$$

Therefore

$$y=qx+y_1$$

where

$$by_1=r_1x+c$$

In other words, since $a=bq+r_1$, on putting

$$y=qx+y_1 \qquad \text{(ii)}$$

the given equation (i) reduces to

$$by_1=r_1x+c \qquad \text{(iii)}$$

Again, since $b=r_1q_1+r_2$

putting similarly $x=q_1y_1+x_1$
the equation (iii) can be further reduced to

$$r_1x_1=r_2y_1-c \qquad \text{(iv)}$$

and so on.

Writing down the successive values and reduced equations in columns, we have

(1) $y=qx+y_1$	(I.1) $by_1=r_1x+c$
(2) $x=qy_1+x_1$	(I.2) $r_1x_1=r_2y_1-c$
(3) $y_1=q_2x_1+y_2$	(I.3) $r_2y_2=r_3x_1+c$
(4) $x_1=q_3y_2+x_2$	(I.4) $r_3x_2=r_4y_2-c$
(5) $y_2=q_4x_2+y_3$	(I.5) $r_4y_3=r_5x_2+c$
(6) $x_2=q_5y_3+x_3$	(I.6) $r_5x_3=r_6y_3-c$
.....	.....
$(2n-1)$ $y_{n-1}=q_{2n-2}\,x_{n-1}+y_n$	(I. $2n-1$) $r_{2n-2}\,y_n=r_{2n-1}\,x_{n-1}+c$
$(2n)$ $x_{n-1}=q_{2n-1}\,y_n+x_n$	(I. $2n$) $r_{2n-1}\,x_n=r_{2n}\,y_n-c$
$(2n+1)$ $y_n=q_{2n}\,x_n+y_{n+1}$	(I. $2n+1$) $r_{2n}\,y_{n+1}=r_{2n+1}\,x_n+c$

Now the mutual division can be continued either (i) to the finish or (ii) so as to get a certain number of quotients and then stopped. In either csse the number of quotients found, neglecting the first one (q). as is usual with Āryabhaṭa, may be even or odd.

Case (i) First suppose that the mutual division is continued until the zero remainder is obtained. Since a, b are prime to each other, the last one remainder is unity.

Subcase (*i*.1.). Let the number of quotients be even. We then have

$$r_{2n}=1,\ r_{2n-1}=0,\ q_{2n}=r_{2n-1}$$

The equations (1,2n) and (I.2n+1), therefore become

$$y_n=q_{2n}\,x_n+c$$

and

$$y_{n+1}=c$$

respectively. Giving an arbitrary integral value (t) to x_n we get an integral value of y_n. From that we can find the value of x_{n-1} by the equation ($2n$). Procceding backwards step by step we ultimately find the values of x and y in positive integers. So that the equation (I) is solved.

Subcase (*i*. 2) : If the number of quotients be odd, we shall have

$$r_{2n-1}=1,\ r_{2n}=0,\ q_{2n-1}=r_{2n-2}.$$

The equations $(2n+1)$ and (I. $2n+1$) will then be absent and the equations (I. 2n—1) and (I. 2n) will be reduced respectively to

$$x_{n-1}=q_{2n-1}\,y-\ c$$

and $x_n=-c$

Giving an arbitrary integral value (t') to y_n we get an integral value of x_{n-1}. Then proceeding backwards as before we calculate the values of x and y.

Case (ii) : Next suppose that the mutual division is stopped after having obtained an even or odd number of quotients.

Subcase (*ii*.1) : If the number of quotients obtained be *even* the reduced form of the original equation is

$$r_2\ \ y\ +_1=r_{2n}+_1\,x_n+c$$

or $$y_n+_1=\frac{r_{2u}+_1\,x_n+c}{r_2}$$

Giving a suitable integral value (t) to x_n as will make

$$y_{n+1}=\frac{r_{2n+1}\,t+c}{r_{2n}}=\text{an integral number,}$$

we get an integral value for y_n by $(2n+1)$. The values of x and y can then be calculated by proceeding as before.

Subcase (*ii*.2) : If the number of quotients be *odd* the reduced form of the quotient is

$$r_{2n-1}\,x_n=r_{2n}\,y_n-c$$

or $$x_n=\frac{r_{2n}\,y_n-c}{r_{2n-1}}$$

Putting $y_n=t'$, where t' is an integer, such that

$$x_n=\frac{r_{2n}\,t'-c}{r_{2n-1}}=\text{a whole number,}$$

we get an integral value of x_{n-1} by $(2n)$. Whence can be calculated the values of x and y in integers.

If $x=\alpha$ and $y=\beta$ be the least integral solution of $ax+c=by$, we shall have

$$a\alpha+c=b\beta$$

Therefore $a(bm+\alpha)+c=b\,(am+\beta)$,

m being any integer. Therefore, in general,

$$x=bm+\alpha$$

But we have calculated before that

$$x=q_1y_1+x_1\ ;$$

$$\therefore\ q_1y_1+x_1=bm+\alpha$$

Thus it is found that the minimum value α of x is equal to the remainder left on dividing its calculated value by b. whence we can calculate the minimum value of $N\ (=a\alpha+R_1)$. This will explain the *rationale* of the operations described in the latter portion of the rule of Āryabhaṭa I.

Bhāskara I and Kuṭṭaka Operation

In Chapter I of the *Mahābhāskarīya*, Bhāskara I has described the preliminary operation to be performed on the divisor and dividend of a pulveriser. We shall quote it from the edition of K.S, Shukla :

> The divisor (which is "the number of civil days in a *yuga*) and the dividend (which is "the revolution number of the desired planet") become prime to each other on being divided by the (last non-zero) residue of the mutual division of the number of civil days in a *yuga* and the revolution number of the desired planet. The operations of the pulveriser should be performed on them (i. e. on the abraded divisor and abraded dividend). So has been said.[1]

An indeterminate equation of the first degree of the type

$$\frac{ax-c}{a}=y$$

(with x and y unknown) is known in Hindu mathematics by the name of "pulveriser"—*kuṭṭākāra*). In this equation, a is called the "dividend" (*bhājya*), b the "divisor" (*bhāgahāra*), c the interpolator (*kṣepa*), x the "multiplier" (*guṇakāra*), and y the "quotient" (*labdha*).

In the pulveriser contemplated in the above stanza :

a=revolution number of a planet.
b=civil days in a *yuga*,
c=residue of the revolutions of the planet (*Śeṣa*)

1. भूदिनेष्टगणान्योन्य भक्तशेषेण भाजि ।
हारभाज्यौ दृढौ स्यातां कुट्टाकारं तयोर्विदुः ।
—*MBh.* I. 41

x=*ahargaṇa* ,

and y=complete revolutions performed by the planet.

The text says that as a preliminary operation to the solution of this pulveriser, *a* and *b*. i.e., civil days in *yuga* and revolution-number of the planet, should be made prime to each other by dividing them out by their greatest common factor, That is to say, in solving a pulveriser, one should always make use of abraded divisor and abraded dividend.

The interpolator, i.e., the residue, should also be divided out by the same factor. (This instruction is not given in the text, but it is implied that the residue should be computed for the abraded dividend and abraded divisor).

> Set down the dividend above and the divisor (*hāra*) below that. Divide them mutually and write down the quotients (*labdha*) of division one below the other (in the form of a chain). (When an even number of quotients is obtained) think out by what number the (last) remainder be multiplied so that the product being diminished by the (given) residue be exactly divisible (by the divisor corresponding to that remainder). Put down the chosen number called *mati* below the chain and then the new quotient underneath it. Then by the chosen number multiply the number which stands just above it, and to the product add the quotient (written below the chosen number). (Replace the upper number by the resulting sum and cancel the number below). Proceed afterwards also in the same way (until only two numbers remain). Divide the upper number (called the "multiplier") by the divisor by the usual process and the lower one (called the "quotient") by the dividend : the remainders (thus obtained) will respectively be the *ahargaṇa* and the revolutions etc. or what one wants to know.[1]

We shall illustrate the operation by taking a problem from the *Laghu-Bhāskarīya* (VIII. 17) :

> The sum, the difference, and the product increased by one, of the residues of the revolution of Saturn and Mars—each is a perfect square. Taking the equations

furnished by the above and applying the method of such quadratics, obtain the (simplest) solution by the substitution of 2, 3 etc. successively in the general solution). Then calculate the *ahragaṇa* and the revolutions performed by Saturn and Mars in that time together with the number of solar years elapsed.[1]

Let x and y denote the residues of the revolution of Mars and Saturn respectively. Then we have to find out two numbers x and y such that each of the expressions $x+y$, $x-y$ and $xy+1$ may be a perfect square.

Let $x+y=4P^2$ and $x-y=4Q^2$, so that

$$x=2P^2+2Q^2$$

$$y=2P^2-2Q^2$$

and therefore $xy+1=(2P^2-1)^2+4(P^2-Q^4)$

Hence the condition that $xy+1$ be a perfect square is that $P^2=Q^4$. Substituting these values, we have

$$x=2(Q^4+Q^2)$$

$$y=2(Q^4-Q^2)$$

where Q may possess any of the values 2, 3, 4.....but not 1. (We neglect the case when x or y is zero).

---

1. भाज्यं न्यसेदुपरि हारमधश्च तस्य ।
खण्डयात्परस्परमधो विनिधाय लब्धम् ।
केनाऽऽहतोऽयमपनीय यथाऽस्य शेषं ।
भागं ददाति परिशुद्धमिति प्रचिन्त्यम् ॥

आप्तां मतिं तां विनिधाय वल्ल्यां
नित्यं ह्यधोऽधः क्रमशश्च लब्धम् ।
मत्या हृतं स्यादुपरिस्थितं य
ल्लब्धेन युक्तं पातश्च तद्वत् ॥

हारेणभाज्यो विधिनोपरिस्थो
भाज्येन नित्यं तदधः स्थितश्च ।
अहांगणोऽस्मिन् भगणादयश्च
तद्वा भवेद्वरय समीहितं यत् ॥ —*MBh*, I, 42-44

2. शेषौ मण्डलजौ यमक्षितिजयोः संयुक्तविश्लेषिता
वन्योन्याहृतविग्रहौ च पददौ रूपेण संयोजितौ ।
एवं साधु विचिन्त्य वर्गविधिना द्वित्रिक्रमाद्वत्स रैः ।
संगण्या द्युगणार्कजक्षिति सुताः कालेन कालोद्भवाः ॥ —*LBh*. VIII. 17

Putting $Q=2$, we get $x=40$ and $y=24$, which is the least solution.

Assuming now that the residues of the revolution (*maṇḍalaja-śeṣa*) of Saturn and Mars are 24 and 40 respectively, we have to obtain the *ahargaṇa* (which means the number of mean civil days elapsed since the beginning of Kaliyuga, or, in fact, any epoch).

The revolution-number of Saturn is 146564, and the number of civil days in a *yuga* is 1,577,917,500. In the present problem, these are respectively the dividend and the divisor. Their H.C.F. is 4, so that dividing them out by 4 we get 36641 and 394,479,375 as the abraded dividend and abraded divisor respectively. We have, therefore, to solve the pulveriser

$$\frac{36641x-24}{394479375}=y$$

where x and y denote the *ahargaṇa* and the revolutions respectively made by Saturn.

Mutually dividing 36641 and 394479375, we get

36641) 394479375 (10766
394477006

2369) 36641 (15
35535

1106) 2369 (2

157) 1106 (7
1099

7) 157 (22
154

3) 7 (2
6

$1\times27-24=3$)3(1
3

0

We have chosen here the number 27 as the optional number (*mati*). In fact, *mati* may be chosen at any stage after an even number of quotients are obtained.

Writing down the quotients one below the other as prescribed in the rule, we get the chain

10766
15
2
7
22
2
(*mati*) 27
1

Reducing the chain, we successively get

10766	10766	10766	10766	10766	10766	3108044439 (multiplier)
15	15	15	15	15	288689	288689 (quotient)
2	2	2	2	18665	18665	
7	7	7	8714	8714		
22	22	1237	1237			
2	55	55				
(*mati*)						
27	27					
1						

(it would be seen in this reduction of chain that *mati* or 27×2 plus 1 is 55; 55×22 plus 27 is 1237; 1237×7 plus 55 is 8714; 8714×2 plus 1237 is 18665; 18665×15 plus 8714 is 288689; and finally 288689×10766 plus 18665 is 3108044439 which is the multiplier).

Dividing 3108044439 by 394479375, and 288689 by 36641, we obtain 346688814 and 32202 respectively as remainders, (This division is performed only when the multiplier and quotient are greater than the divisor and dividend respectively). These are the minimum values of x and y satisfying the above equation.

Therefore, the required *ahargaṇa*=346688814. and the revolutions performed by Saturn=32202.

To obtain the *ahargaṇa* and the revolutions of Mars, one has to solve the equation :

$$\frac{191402\,z-40}{131493125}=w$$

where z and w denote the *ahargaṇa* and the revolutions performed by Mars respectively.

The general solution of this equation is

$$z = 131493125\ s + 118076020$$
$$w = 191402\ s + 171872$$

where $s = 0, 1, 2, 3, \ldots\ldots$ When $s = 0$, we have the least solution.

Brahmagupta's Rules Concerning Indeterminate Analysis of the First Degree

For the solution of Āryabhaṭa's problem, Brahmagupta gives the following rule :

> What remains when the divisor corresponding to the greater remainder is divided by the divisor corresponding to the smaller remainder—that (and the latter divisor) are mutually divided and the quotients are severally set down one below the other. The last residue (of the reciprocal division after an even number of quotients has been obtained) is multiplied by such an optional integer that the product being added with the difference of the (given) remainders will be exactly divisible (by the divisor corresponding to that residue). That optional multiplier and then the (new) quotient just obtained should be set down (underneath the listed quotients). Now, proceeding from the lower-most number (in the column), the penultimate is multiplied by the number just above it and then added by the number just below it. The final value thus obtained (by repeating the above process) is divided by the divisor corresponding to the smaller remainder. The residue being multiplied by the divisor corresponding to the greater remainder and added to the greater remainder will be the number in view.[1]

1. अधिकाग्रभागहारादूनाग्रच्छेद भाजिताच्छेषम् ।
यत् तत् परस्परहृतं लब्धमधोऽधः पृथक् स्थाप्यम् ॥
शेषं तदेष्टगुणितं यथाऽग्रयोरन्तरेण संयुक्तम् ।
शुध्यति गुणकः स्थाप्यो लब्धं चान्त्यादुपान्त्यगुणः ॥
स्वोर्ध्वोऽन्त्य युतोऽग्रान्तो हीनाग्रच्छेदभाजितः शेषम् ।
अधिकाग्रच्छेदहृत मधिकाग्रयुतं भवत्यग्रम् ॥ —*BrSpSi. XVIII*, 3-5

Brahmagupta further observes :

> Such is the process when the quotients (of mutual division) are even in number. But if they be odd, what has been stated before as negative should be made as positive, or as positive should be made negative.[1]

Regarding the direction for dividing the divisor corresponding to the greater number by the divisor corresponding to the smaller remainder, Pṛthūdaka Svāmī (860A.D.) observes that it is not absolute, rather optional; so that the process may be conducted in the same way by starting with the division of the divisor corresponding to the smaller remainder by the divisor corresponding to the greater remainder. But in this case of inversion of the process, he continues, the difference of the remainders, must be negative.

That is to say, the equation

$$by=ax+c$$

can be solved by transforming it first to the form

$$ax=by-c$$

so that we shall have to start with the division of b by a,

For the details of the "Theory of the pulveriser" as applied to the problems in Astronomy, the reader is referred to the writings of Bhaṭṭa Govind, translated by K.S. Shukla, and given as an Appendix to the edition of the *Laghu-Bhāskarīya*. For the *rationale* of the rules in relation to *kuṭṭaka* or the pulveriser operation, one may also refer to the chapters by Datta and Singh in the *History of Hindu Mathematics: Algebra*.

Solution of $by=ax \pm 1$.

This simple indeterminate equation has a special use in astronomical calculations and therefore, Indian algebraists have paid special attention to it. In fact, this equation is solved exactly in the same way as the equation $by=ax \pm c$; it is a parti-

1. एवं समेषु विषमेष्वृणं धनं धनमृणं यदुक्तं तत् ।
ऋणधनयोर्व्यस्तत्वं गुण्य प्रक्षे पयोः कार्यम् ॥ —*BrSpSi.* XVIII. 13.

cular case only of the more general latter equation. Of course, there is a little justification also for treating it separately, since both the types of equations represent two different physical conditions of the astronomical problems. In the case of $by=ax\pm c$, the conditions are such that the value of either y or x, more particularly of the latter, has to be found and the rules for solution formulated with that objective. But in the case of the equation $by=ax\pm 1$, the physical conditions require the values of both y and x.

The equation $by=ax\pm 1$ is usually known by the name *sthira-kuṭṭaka*, literally meaning the '*constant pulveri*ser' Pṛthūdaka Svāmī also names it as *dṛḍha-kuṭṭaka* meaning firm-pulveriser. Later on this term *dṛḍha*-was confined to another sense, equivlent to *nicched* (having no divisor) or *nirapavarta* (irreducible). The origin of the name *sthira-kuṭṭaka* or constant pulveriser has been explained by Pṛthūdaka Svāmī as being due to the fact that the interpolator (± 1) is here invariable.

For the solution of this equation, we shall quote Bhāskara I's rule and the rule by Brahmagupta. Bhāskara I writes in this connection as follows :

> The method of the pulveriser is applied also after subtracting unity. The multiplier and quotient are respectively the numbers above and underneath. Multiplying those quantities by the desired number divide by the reduced divisor and dividend; the residues are in this case known to be the (elapsed) days and (residues of) revolutions respectively[1].

The pulveriser

$$\frac{ax-c}{b}=y \qquad \ldots (1)$$

may be written as

$$\frac{aX-1}{b}=Y \qquad \ldots (2)$$

where $x=cX$ and $y=cY$. If $X=\alpha$, $Y=\beta$ is a solution of (2), then $x=c\alpha$, $y=c\beta$ will be a solution of (i). Hence the above rule.

---

1. रूपमेकमपास्यापि कुहाकारः प्रसाध्यते ।
गुणकास्तेऽथ लब्धं च राशी स्यातामुपर्यधः ॥ —*MBh. I.* 45

Brahmagupta's Rule in this connection is as follows :

Solution of $by=ax-1$:

> Divide them (i.e., the abraded coefficient of the multiplier and the divisor) mutually and set down the quotients one below the other. The last residue (or the reciprocal division after an even number of quotients has been obtained) is multiplied by an optional integer such that the product being diminished by unity will be exactly divisible (by the divisor correponding to that residue). The (optional) multiplier and then this quotient should be set down (underneath the listed quotients). Now proceeding from the lower most term to the uppermost, by the penultimate multiply the term just above it and then add the lowermost number. (The uppermost number thus calculated being divided by the reduced divisor, the residue (is the quantity required. This is the method of the constant pulveriser[1].

Solution of

$$by+ax=\pm c$$

Indian algebraists usually transformed this equation as $by=-ax+c$, so that it appeared as a particular case of $by=ax+c$, in which a was negative. Brahmagupta has been the first person to solve this equation, but the rule given by him is obscure :

> The reversal of the negative and positive should be made of the multiplier and interpolator.[2]

Pṛthūdaka Svāmī has tried to explain it, but he too is not very clear. He says :

---

1. हृतयोः परस्परं यच्छेषं गुणकार भागहारकयोः ।
   तेन हृतौ निश्छेदौ तावेव परस्परं हृतयोः ॥
   लब्धमधोऽधः स्थाप्यं तथेष्ट गुणकारसङ्गुणं शेषम् ।
   शुद्ध्यति यथैकहीनं गुणकः स्थाप्यः फलं चान्त्यात् ॥
   अग्रान्तमुपान्त्येन स्वोर्ध्वो गुणितोऽन्त्य संयुतो भक्तम् ।
   निःशेषभागहारेणैवं स्थिरकुट्टकः शेषम् ॥ —*BrSpSi.* XVIII. 9-11

2. एवं समेषु विषमेष्वृणं धनं धनमृणं यदुक्तं तत् ।
   ऋणधनयोर्व्यस्तत्वं गुण्यप्रक्षेपयोः कार्यम् ॥ *BrSpSi* XVIII. 13

If the multiplier be negative, it must be made positive; and the additive must be made negative : and then the method of the pulveriser should be employed.

Pṛthūdaka Svāmī, however, does not indicate how to derive the solution of the equation .

$$by=-ax+c \qquad ...(1)$$

from that of the equation

$$by=ax-c \qquad ...(2)$$

The method, however, seems to have been this :

Let $x=\alpha$, $y=\beta$ be the minimum solution of (2). Then we get

$$b\beta=a\alpha-c$$

or $$b(a-\beta)=-a(\alpha-b)+c$$

Hence $x=\alpha-b$, $y=a-\beta$ is the minimum solution of (1). This rule is very clearly indicated by Bhāskara II and others.

We shall give two examples from Bhāskara II (*Bījagaṇita*) to illustrate the rule :

Example I.

$13y=-60x+3$

By the method described before, we find that the minimum solution of

$$13y=60x+3$$

is $x=11$, $y=51$. Subtracting these values from their respective abraders, namely 13 and 60, we get 2 and 90. Then by the maxim : "In the case of the dividend and divisor being of different signs, the results from the operation of division should be known to be so", making the quotient negative we get the solution of

$$13y=-60x+3$$

as $x=2$, $y=-9$. Subtracting these values again from their respective abraders (13, 60), we get the solution of

$$13y=-60x-3$$

as $x=11$, $y=-51$.

Example II.

$11y=18x+10$

Proceeding as before, we find the minimum solution of

$$11y=18x+10$$

to be $x=8$, $y=14$. These will also be the values of x and y in the case of the negative divisor but the quotient for the reasons stated before should be made negative. So the solution of

$$-11y=18x+10$$

is $x=8$, $y=-14$. Subtracting these (i.e., their numerical values) from their respective abraders, we get the solution of

$$-11y=18x-10$$

as $x=3$, $y=-4$,

> "When the divisor is positive or negative the numerical values of the quotient and multiplier remain the same : when either the divisor or the dividend is negative, the quotient must always be known to be negative"[1].

One Linear Equation in More Than Two Unknowns

Whenever a linear equation involves more than two unknown's the Indian algebraists used to assume arbitrary values for all the unknowns except two and then to apply the method of *kuṭaka* or "pulveriser". In this connection, Brahmagupta says :

> The method of the pulveriser (should be employed if there be present many unknowns (in any equation)[2],

---

1. Bhāskara II gives the following rule :

> "Those(the multiplier and quotient)obtained for a positive dividend being treated in the same manner give the results corresponding to a negative dividend."

The treatment alluded to in this rule is that of subtraction from the respective abraders. He has further elaborated it thus :

The multiplier and quotient should be determined by taking the dividend, divisor and interpolator as positive. They will be the quantities for the additive interpolator. Subtracting them from their respective abraders, the quantities for a negative interpolator are found. If the dividend or divisor, be negative, the quotient should be stated as negative, the quotient should be stated as negative.

—*Bījaganita*

1. आद्यादवर्णादन्यान् वर्णान् प्रोह्यावमानमाद्यहृतम् ।
सदृशच्छेदावसकृद् द्वौ व्यस्तौ कुट्टको बहुषु ॥ — *BrSpSi. XVIII.* 51

We shall take up one of the problems posed by Brahmagupta concerning astronomy and leading to the equation :[1]

$$197x-1644\,y-z=6302.$$

Hence

$$x=\frac{1644\,y+z+6302}{197}$$

The commentator assumes z=131. Then

$$x=\frac{1644\,y+6433}{197};$$

hence by the usual method of the pulveriser

$$x=41;\ y=1.$$

General Problem of Remainders

A certain type of simultaneous indeterminate equations of the first degree arise out of the general problem of remainders which may thus be stated : To find a number N which being severally divided by $a_1, a_2, a_3 \ldots\ldots a_n$, leaves as remainders $r_1, r_2, r_3 \ldots\ldots r_n$ respectively.

While dealing with such a case, we shall have the following series of equations :

$$N=a_1x_1+r_1=a_2x_2+r_2=a_3x_3+r_3=\ldots\ldots=a_n\,x_n+r\ .$$

We have reasons to believe that the method of solution of these equations was known to Āryabhaṭa I. In the translation of the verse in the *Āryabhaṭiya*, II. 32-33 (the translation of which we have already given), the term *dvicchedāgram* should be translated as "the result will be the remainder corresponding to the product of the two divisors", instead of "the result will be the number corresponding to the two divisors." (the last line of the translation). This explanation is in fact given by Bhāskara I, the direct disciple and earliest commentator of Āryabhaṭa I. Such a rule is clearly stated by Brahmagupta[2].

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1. अंशकशेषेण युतात् लिप्ताशेषात्तदन्तरादथवा ।
भानोर्ब्दिने युगार्धं यः कथयति कुट्टकज्ञः सः ॥ —*BrSpSi.* XVIII. 55

2. स्वोर्ध्वोऽन्त्ययुतोऽन्तो हीनाग्रच्छेदभाजितः शेषम् ।
अधिकाग्रच्छेदहृतमधिकाग्रयुतं भवत्यग्रम् ॥ —*BrSpSi. XVIII. 5*

The rationale of this method is not difficult. I shall quote it from the book of Datta and Singh: Starting with the consideration of the first two divisors, we have

$$N=a_1x_1+r_1=a_2x_2+r_2.$$

By the method described before, we can find the minimum value α of x_1 satisfying this equation. Then the minimum value of N will be $a_1\alpha+r_1$. Hence the general value of N will be given by

$$N=a_1\ (a_2t+\alpha)+r_1$$
$$=a_1a_2\ t+a_1\alpha+r_1$$

where t is an integer. Thus $a_1\alpha+r_1$ is the remainder left on dividing N by a_1 a_2 as stated by Ārybhaṭa I and Brahmagupta, Now taking into consideration the third condition, we have

$$N=a_1a_2t+a_1\alpha r_1=a_3x_3+r_3$$

which can be solved in the same way as before. Proceeding in this way successively, we shall ultimately arrive at a value of N satisfying all the conditions ;

Pṛthūdaka Svāmī remarks :

> Wherever the reduction of two divisors by a common measure is possible, there 'the product of the divisors' should be understood as equivalent to the product of the divisor corresponding to the greater remainder and quotient of the divisor corresponding to the smaller remainder as reduced (i.e. divided) by the common measure.[1] When one divisor is exactly divisible by the other, then the greater remainder is the (required) remainder and the divisor corresponding to the greater remainder is taken as 'the product of the divisors'. (The truth of) this may be investigated by an intelligent mathematician by taking several symbols.

As an illustration we shall take up a problem quoted by Bhāskara II in his *Bījagaṇita*, and which in its solution follows the method of Āryabhaṭa 1. Pṛthūdaka Svāmī while commenting on serveral verses from Brahmagupta (*BrSpSi.* XVIII. 3-6)

1. i.e., if p be the L.C.M. of a_1 and a_2, the general value of N satisfying the above two conditions will be
$$N=pt+a_1\alpha+r_1$$
instead of $N=a_1a_2t+a_1\alpha+r_1$.

observes that such problems were very popular amongst the ancient Indian mathematicians.

Problem : To find a number N which leaves remainders 5, 4,3,2 when divided by 6,5,4,3 respectively.

That is to solve the equations :

$$N=6x+5=5y+4=4z+3=3w+2.$$

We have since $N=6x+5=5y+4$,

$$x=\frac{5y-1}{6}$$

But x must be integral, so $y=6t+5$, $x=5t+4$

Hence $N=30t+29$

Again $N=30t+29=4z+3$

$$\text{Therefore, } t=\frac{2z-13}{15}$$

Since t must be integral, we must have $z=15s+14$; hence $t=2s+1$. Therefore

$$N=60s+59.$$

The last condition is identically satisfied. The method given here is the one followed by Pṛthūdaka Svāmī.

Thus when $N=60s+59=6x+5$

$$x=\frac{60s+54}{6}=10s+9 \qquad \ldots(1)$$

Again, when $N=60s+59=5y+4$,

$$y=\frac{60s+55}{5}=12s+11$$

Again when $N=60s+59=4z+3$

$$z=\frac{60s+56}{4}=15s+14$$

Lastly, when $N=60s+59=3w+2$,

$$w=\frac{60s+57}{3}=20s+19,$$

Varga Prakṛti or Kṛti Prakṛti or Square-Nature

The word *varga-prakṛti* (literally meaning 'square-nature') has been given by Indian algebraists to the indeterminate quadratic equation

$$Nx^2\pm c=y^2$$

Here in this equation the absolute number c should be *rūpa* (or unity), which means the equation

$$Nx^2 \pm 1 = y^2$$

or it may be any absolute number. The most fundamental equation of this class has been regarded as

$$Nx^2 + 1 = y^2$$

where N is a non-square integer.

This branch of mathematics has originated from the number which is the *prakṛti* of the square of *yāvat*, etc. (the unknown x etc.), and therefore, it is called *varga- prakṛti. The quantity N of the above equation is known as Prakṛti.* Brahmagupta uses the term GUṆAKA (*multiplier*) for the same purpose[1].

This term *guṇaka* together with its variation *guṇa* appears occasionally also in the writings of later authors. For example, Śrīpati (*Siddhānta-śekhara*. XIV. 32) employs the term *guṇaka* where as Bhāskara II and Nārāyaṇa use the term *guṇa* in their *Bījagaṇitas.*

In this connection, we would now like to quote from Pṛthūdaka Svāmī (863 A.D.) from his commentary on the *Brāhmasphuṭasiddhānta :*

> Here are stated for ordinary use the terms which are well known to people. The number whose square, multiplied by an optional multiplier and then increased or decreased by another optional number, becomes capable of yielding a square-root, is designated by the term the "lesser root" *kaniṣṭha pada* or the "first root" *ādya-mūla*). The root which results, after those operations have been performed is called by the name the "greater root" (*jyeṣṭha pada*) or the "second root" (*anya-mūla*). If there be a number multiplying both these roots, it is called the "augmenter" (*udvartaka*); and on the contrary, if there be a number dividing the roots, it is called the "abridger" (*apavartaka*).[2]

Thus in the equation

$$Nx^2 \pm c = y^2,$$

---

1. मूलं द्विघेघ्नं वर्गाद् गुणक गुणादिघ्नयुत विहीनाच्च ।
   श्रावयवो गुणकगुणः सहानयघातेन कृतमन्त्यम् ॥ —*BrSpSi*. XVIII. 64
2. *BrSpSi. XVIII.* 64 (*Com.*)

x is known as the lesser root, y is the greater root, N is the multiplier (*guṇaka*) and c is interpolator or *kṣepaka*. Bhāskara II has used the word "*hrasvamūla*" for *kaniṣṭha pada* or *ādya-mūla* literally meaning "lsser root". The earlier terms, the "first root" (*ādyamūla*) for the value of x and the "second root" or the "last root" *antya-mūla* for the value of y are quite free from ambiguity Their use is found in the algebra of Brahmagupta. The later terms appears in the works of his commentator Pṛthūdaka Svāmī.

Brahmagupta uses the term *kṣepa*, *prakṣepa* or *prakṣepaka* in the sense of "interpolator." Again, when negative, the interpolator is sometimes distinguished as the "subtractive" or *śodhaka* and the positive interpolator is then called "the additive."

Lemmas of Brahmagupta

Prior to our giving the general solution of the Square-nature or *Varga-Prakṛti*, it would be better to give two Lemmas established by Brahmagupta. We have the following in the *Brāhma-sphuṭasiddhānta :*

> Of the square of the optional number multiplied by the *guṇaka* and increased or decreased by an other optional number, *iṣṭa*, (extract) the square root. (Proceed) twice. The product of the first roots multiplied by the *guṇaka* together with the product of the second roots will give a (fresh) second root; the sum of their cross-products will be a (fresh) first root. The (corresponding) interpolator will be equal to the product of the (previous) interpolators.[1]

There is a little difficulty in ascertaing the real sense of the rule given in these lines since the word *dvidhā* (twice) has two implications. Firstly, it may mean that the earlier operations of finding roots are made on two optional numbers with two optional interpolators, and with the results thus obtained the subse-

1. मूलं द्विधेष्टवर्गाद् गुणक गुणादिष्टयुत विहीनाच्च ।
आद्यवधो गुणकगुणः सहान्त्यघातेन कृतमन्त्यम् ।।
कज्ञवधैक्यं प्रथमं प्रक्षेपः क्षेपवध तुल्यः ।
प्रक्षेपशोधकहृते मूले प्रक्षेपके रूपे ॥ —*BrSpSi. XVIII* 64-65

quent operations of their composition are performed. Secondly, it may also mean that the earlier operations are made with one optionally chosen number and one interpolator, and the subsequent ones are carried out after the repeated statement of those roots for the second time. It is also implied that in the composition of the quadratic roots, their products may be added together or subtracted from each other.

In other words, if $x=\alpha$, $y=\beta$ be a solution of the equation :

$$Nx^2+k=y^2,$$

and $x=\alpha'$, $y=\beta'$ be a solution of

$$Nx^2+k'=y^2.$$

then according to the above

$$x=\alpha\beta'\pm\alpha'\beta,\ y=\beta\beta'\pm N\alpha\alpha'$$

is a solution of the equation

$$Nx^2+kk'=y^2.$$

In other words, if

$$N\alpha^2+k=\beta^2$$
$$N\alpha'^2+k'=\beta'^2$$

then

$$N(\alpha\beta'\pm\alpha'\beta)^2+kk'=(\beta\beta'\pm N\alpha')^2 \qquad \text{(I)}$$

In particular, taking $\alpha=\alpha'$, $\beta=\beta'$ and $k=k'$, Brahmagupta finds from a solution $x=\alpha$, $y=\beta$ of the equation

$$Nx^2+k^2=y^2$$

a solution $x=2\alpha\beta$, $y=\beta^2+N\alpha^2$ of the equation

$$Nx^2+k=y^2$$

That is, if

$$N\alpha^2+k=\beta^2$$

then

$$N(2\alpha\beta)^2+k^2=(\beta^2+N\alpha^2)^2 \qquad \text{(II)}$$

This result will be hereafter called *Brahmagupta's Corollary.*

Thus *Brahmagupta's First Lemma* says that if two solutions of the equation (of the Square-nature) $Nx^2+1=y^2$ are known, then any number of other solutions can be found. For example if two solutions of the Square—nature are (a, b) and also (a', b'), then two other solutions will be :

$$x=ab'+a'b,\ y=bb'\pm Naa'.$$

We can compose this solution with the previous ones, and get another solution, and thus proceed on to innumerable solutions. From *Brahmagupta's Corollary to First Lemma* we get another set of solutions. If (a, b) be solution of the Square-nature, then another solution of it is

$$x=2ab, \text{ and } y=b^2+Na^2$$

Thus even if we have only one solution, we can get the other solution also (since N is known), and thus we can get any number of solutions one after the other by this Principle of Composition.

Brahmagupta's Lemmas have been described by Bhāskara II (1150 A.D.) in the following words :

> Set down successively the lesser root (*hrasva*), greater root (*jyeṣṭha*) and interpolator (*kṣepaka*); and below them should be set down in order the same or an another (set of similar quantities). From them by the Principle of Composition (Bhāvanā) can be obtained numerous roots. Therefore the Principle of Composition will be explained here. (Find) the two cross-products (*vajrābhyāsa*) of the two lesser and the two greater roots; their sum is a lesser root. Add the product of the two lesser roots multiplied by the *prakṛti* to the product of the two greater roots, the sum will be a greater root. In that (equation) the interpolator will be the product of the two previous interpolators. Again the difference of the two cross-products is a lesser root. Subtract the product of the two lesser roots multiplied by the *prakṛti* from the product of the two greater roots; (the difference) will be greater root. Here also the interpolator is the product of the two (previous) interpolators.[1]

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1. हस्वज्येष्ठ क्षेपकान् न्यस्य तेषां तानन्यान् वाऽधो निवेश्य क्रमेण ।
साध्यान्येभ्यो भावनाभिर्बहूनि मूलान्येषां भावना प्रोच्यतेऽतः ॥

वज्राभ्यासौ ज्येष्ठलघ्वोस्तदैक्यं हस्वं लघ्वोराहतिश्च प्रकृत्या ।
क्षुण्णा ज्येष्ठाभ्यास युग् ज्येष्ठमूलं तत्राभ्यासः क्षेपयोः क्षेपकः स्यात् ॥

हस्वं वज्राभ्यासयोरन्तरं वा लघ्वोर्घातो यः प्रकृत्या विनिघ्नः ।
घातो यश्च ज्येष्ठयोस्तद्वियोगो ज्येष्ठं क्षेपोत्रापि च क्षेपघातः ॥

Bhāskara *II*, *Bijaganita*, *VargaPrakṛti*.2-4

Principle of Composition

The above results have been technically known amongst Indian algebraists as *Bhāvanā* (demonstrated or proved, hence theorem or lemma). The word *bhāvanā* also means "composition or combination" in algebra. Bhāvanā may be of two types: *Samāsa Bhāvanā* (or addition Lemma, or additive composition) and *Antara Bhāvanā* (or subtraction Lemma or subtractive composition). Whenever, again, the *bhāvanā* is made with two equal sets of roots and interpolators, it is technically named as *Tulya Bhāvanā* (or composition of equals), and when with two unequal sets of values then it is known as *Atulya Bhāvanā* (or composition of unequals).

Proof of Brahmagupta's Lemmas

It is significant to be indicated that Brahmagupta's Lemmas were rediscovered by Euler in 1764 and by Lagrange in 1768, and a considerable importance was attached to them. Kṛṣṇa, (1580 A.D.) the commentator on the *Bījaganita* of Bhāskara II gives the following proof of Brahmagupta's Lemmas:

Let (α,β) and (α',β') be the two solutions of the equation

$$nx^2+k=y^2.$$

we have

$$N\alpha^2+k=\beta^2$$

$$N\alpha'^2+k'=\beta'^2$$

Multiplying the first equation by β'^2, we get

$$N\alpha^2\beta'^2+k\beta'^2=\beta^2\beta'^2$$

Now, substituting the value of factor $\beta^{2\prime}$ of the interpolator from the second equation, we get

$$N\alpha^2\ \beta'^2+k\ (N\alpha'^2+k')=\beta^2\beta'^2$$

or $N(\alpha^2\beta'^2+Nk\alpha'^2+kk'=\beta^2\beta'^2$

Again, substituting the value of k from the first equation in the second term of the left-hand side expression, we have

$$N\alpha^2\beta'^2+N\alpha'^2(\beta^2-N\alpha^2)+kk'=\beta^2\beta'^2$$

or $N(\alpha^2\beta'^2+\alpha'^2\beta^2)+kk'=\beta^2\beta'^2+N^2\alpha^2\alpha'^2$

Adding $\pm 2N\alpha\beta\alpha'\beta'$ to both sides, we get

$$N(\alpha\beta'\pm\alpha'\beta)^2+kk'=(\beta\beta'\pm N\alpha\alpha')^2$$

Brahmagupta's Corollary also follows at once from the above by putting $\alpha'=\alpha$, $\beta'=\beta$ and $k'=k$.

$$N(2\alpha\beta)^2+k^2=(\beta^2\pm N\alpha^2)^2$$

Thus the roots are $x=2\alpha\beta$ and $y=\beta^2\pm N\alpha^2$ which is the Corollary.

It would be seen that modern historians of mathematics are incorrect when they say that Fermat (1657) was the first to state that the equation $Nx^2+1=y^2$, where N is a non-square integer has an unlimited number of solutions in integers. For this assertion, history takes us to the early Seventh Century A.D. when Brahmagupta wrote his classical treatise, the *Brāhmasphuṭasiddhānta*, and gave the well known two Lemmas and the Corollary to the first Lemma.

Second Lemma of Brahmagupta

In the *Brāhmasphuṭa siddhānta*, we find another important Lemma by Brahmagupta stated as follows :

> On dividing the two roots (of a square-Nature) by the square-root of its additive or subtracrive, the roots for interpolator unity (will be found).[1]

This Lemma when expressed in the modern language of algebra would mean that if $x=\alpha, y=\beta$ be a solution of the equation.

$$Nx^2+k^2=y^2$$

then $x=\alpha/k$, $y=\beta/k$ is a solution of the equation

$$Nx^2+1=y^2.$$

This rule, at another place, has been re-enunciated as follows :

> If the interpolator is that divided by a square then the roots will be those multiplied by its square-root.[2]

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1. प्रक्षे पशोधक हृते मूले प्रक्षे पकें रूपे ।

—*BrSpSi.* XVIII. 65

2. वर्गच्छिन्ने क्षे पे तत्पदगुणिते तदा मूले ।

—*BrSpSi.* XVIII. 70

This rule may be expressed in terms of symbols as follows. Suppose the *Varga-prakṛti* (Square-nature) to be

$$Nx^2 \pm p^2 d = y^2,$$

so that its interpolator (*kṣepa*) $p^2 d$ is exactly divisible by the square p^2. Then, putting therein $u = x/p$, $v = y/p$, we derive the equation

$$Nu^2 \pm d = v^2$$

whose interpolator is equal to that of the original Square-nature divided by p^2. It is clear that the roots of the original equation are p times those of the derived equation.

Rational Solution

Indian algebraists have usually suggested the following method to obtain a first solution of $Nx^2+1=y^2$:

> Take an arbitrary small rational number, α, such that its square multiplied by the *guṇaka* N and increased or diminished by a suitably chosen rational number k will be an exact square.

In other words, we shall have to obtain empirically a relation of the form

$$N\alpha^2 \pm k = \beta^2$$

where α, k, and β are rational numbers. Let us call this relation as the *Auxiliary Equation.* Then by Brahmagupta's Corollary, we get from it the relation

$$N(2\alpha\beta)^2 + k^2 = (\beta^2 + N\alpha^2)^2,$$

or

$$N\left(\frac{2\alpha\beta}{k}\right)^2 + 1 = \left(\frac{\beta^2 + N\alpha^2}{k}\right)^2$$

Hence, one rational solution of the equation $Nx^2+1=y^2$ is given by

$$x = \frac{2\alpha\beta}{k}, \quad y = \frac{\beta^2 + N\alpha^2}{k}$$

Work on the rational solution of the Square-nature has been also done by Śrīpati. In fact, his solution, given in 1039 A.D. is of historical significance. He derives the rational solution without the aid of the "auxiliary equation." He gives the following rule :

> Unity is the lesser root. Its square multiplied by the *prakṛti* is increased or decreased by the *prakṛti* combined with an (optional) number whose square-root will be the greater root. From them will be obtained two roots by the Principle of Composition[1]

Thus if m^2 be the rational number optionally chosen, one shall have the identity :

$$N.1^2+(m^2-N)=m^2,$$

or $$N.1^2-(N-m^2)=m^2$$

Then by applying Brahmagupta's Corollary we get

$$N(2m)^2+(m^2 \backsim N)^2=(m^2+N)^2$$

$$\therefore N\left(\frac{2m}{m^2 \backsim N}\right)+1=\left(\frac{m^2+N}{m^2 \backsim N}\right)^2$$

Hence

$$x=\frac{2m}{m \backsim N} \quad y=\frac{m^2+N}{m^2 \backsim N}$$

where m is any rational number, is a solution of the equation $Nx^2+1=y^2$.

This rationol solution of the *varga-prakṛti* which was used by Śrīpati in 1039 A.D. was rediscovered in Europe by Brouncker in 1657.

We shall close this discussion by taking an illustration from Bhāskara II :

> *Problem :* Tell me, O mathematician, what is that square which multiplied by 8 becomes, together with unity, a square; and what square multiplied by 11 and increased by unity, becomes a square.

This means that we have to solve the equations :

$$8x^2+1=y^2 \quad \text{.....(i)}$$

$$11x^2+1=y^2 \quad \text{.....(ii)}$$

In the second example, let us assume 1 as the lesser root. Following the method of Śrīpati, let us multiply its square by the *prakṛti* (here in eq. ii, prakṛti is 11), then let us subtract 2 (an optional number) and then extracting the square-roots we

1. Śrīpati, *Śiddhānta-śekhara* XIV. 33

get the greater root as 3. Hence the statement for the composition is

$m=11 \quad l=1 \quad g=3 \quad i=-2$

$l=1 \quad g=3 \quad i=-2$

Here m=multiplier (*guṇaka* or *prakṛti*), l=lesser root (*kaniṣṭha-mūla*), g=greater root (*jyeṣṭha-mūla*) and i=interpolator (*kṣepa*).

Here we have set down successively the lesser root, greater root and interpolator, and below them again set down the same (See Brahmagupta's Lemmas described by Bhāskara II). Now proceeding as before we obtain the roots for the additive 4 :

$$l=6,\ g=20,\ \text{(for)}\ i=4.$$

Then by the rule :

> "If the interpolator (of a *varga-prakṛti* or Square-nature) divided by the square of an optional number be the interpolator (of another Square-nature), then the two roots (of the former) divided by that optional number will be the roots (of the other). Or, if the interpolator be multiplied, their roots should be multiplied."[1]

are found the roots for the additive unity

$$l=3,\ g=10\ \text{(for)}\ i=1.$$

Whence by the Principle of Composition of Equals, we get the lesser and greater roots : $l=60$, $g=199$ (for) $i=1$. In this way an infinite number of roots can be deduced.

Alternative method:–Bhāskara II has given another method for finding the two roots for the additive unity :

> Or divide twice an optional number by the difference between the square of that optional number and the *prakṛti*. This (quotient) will be the lesser root (of a Square-nature) when unity is the additive. From that (follows) the greater root.[2]

---

1. इष्टवर्गहृतः क्षेपः क्षेपः स्यादिष्टभाजिते ।
मूले ते स्तोऽथवा क्षेपः च्युएणः च्युएणे तदा पदे ॥

Bijaganita II. 5.

2. *Siddhānta-śekhara*, XIV. 32.

Let us solve the first example $8x^2+1=y^2$. We assume the optional number to be 3. Its square is 9; the *prakṛti* of multiplier is 8, their difference is 9-8=1. Dividing by this twice the optional number (2×3, i.e. 6), namely 6, we get the lesser root for the additive unity as 6. Whence proceeding as before, we get the greater to be 17, Thus here $x=6$ and $y=17$.

Let us use this method for the equation $11x^2+1=y^2$. Let the optional number be 3. Its square is 9: multiplier or *prakṛti* is 11; the difference is $11-9=2$; dividing by this twice the optional number (2×3), namely 6, we get $6/2=3$, which is the lesser root. Consequently the greater root would be 10, Thus for this equation $x=3$ and $y=10$.

Solution in Positive Integers

The Indian algebraists usually aimed at obtaining solutions of the *varga-prakṛti* or Square-nature in positive integers or *abhinna*. The tentative methods of Brahmagupta and Śrīpati always did not furnish solutions in positive integers. These authors, however, discovered that if the interpolator of auxiliary equation in the tentative method be ±1, ±2 or ±4, an integral solution of the equation $Nx^2+1=y^2$ can always be found. Thus Śrīpati says :

> If 1, 2 or 4 be the additive or subtractive (of the auxiliary equation), the lesser and greater roots will be integral (*abhinna*)[1].

(i) If $k=\pm1$, then the auxiliary equation will be

$$N\alpha^2\pm1=\beta$$

where α and β are intergers. Then by Brahmagupta' Corollary we get

$$x=2\alpha\beta \text{ and } y=\beta^2+N\alpha^2$$

as the required first solution in positive integers of the equation $Nx^2+1=y^2$

---

1. इष्टवर्ग प्रकृत्योर्यद्विवरं तेन वा भजेत् ।
द्विघ्नमिष्टं कनिष्ठं तत् पदं स्यादेक संयुतौ ।
ततो ज्येष्ठमिहानन्त्यं भावनाभिस्तथेष्टतः ॥

Bījaganita, Varga-Prakṛti, 5-6

(ii) Let $k=\pm 2$; then the auxiliary equation is

$$N\alpha^2 \pm 2=\beta^2$$

By Brahmagupta's Corollary, we have

$$N(2\alpha\beta)^2+4=(\beta^2+N\alpha^2)^2$$

or $$N(\alpha\beta)^2+1=\left(\frac{\beta^2+N\alpha^2}{2}\right)^2$$

Hence the required first solution is

$$x=\alpha\beta,\ y=\tfrac{1}{2}(\beta^2+N\alpha^2)$$

Since $$N\alpha^2=\beta^2 \mp 2,$$

we have $\frac{1}{2}(\beta^2+N\alpha^2)=\beta^2 \mp 1=\alpha$ whole number.

(iii) Now suppose $k=+4$: so that

$$N\alpha^2+4=\beta^2$$

With an auxiliary equation like this, the first integral solution of the equation $Nx^2+1=y^2$ is

$$x=\tfrac{1}{2}\alpha\beta$$
$$y=\tfrac{1}{2}(\beta^2-2);$$

if α is even; or

$$x=\tfrac{1}{2}\alpha(\beta^2-1)$$
$$y=\tfrac{1}{2}\ \beta\ (\beta^2-2);$$

if β is odd.

Thus we find Brahmagupta saying:

> In the case of 4 as additive the square of the second root diminished by 3, then halved and multiplied by the second root will be the (required) second root: the square of the second root diminished by unity and then divided by 2 and multiplied by the first root will be the (requircd) first root (for the additive unity).[1]

Datta and Singh has given the following *rationale* of this solution.

Since $N\alpha^2+4=\beta^2$ (i)

we have $N(\alpha/2)^2+1=(\beta/2)^2$, (ii)

Then by Brahmagupta's Corollary. we get

$$N(\alpha\beta/2)^2+1=\left(\frac{\beta^2}{4}+N\frac{\alpha^2}{4}\right)^2$$

---

1. चतुरधिकेऽन्त्यपदकृतिस्त्र्यूनादलिताऽन्त्यपद गुणाऽन्त्यपदम् ॥
अन्त्यपद कृतिर्त्र्येकाद्विहताऽऽपद्यदाहताऽऽद्यपदम् ॥ —*BrSpSi*. XVIII. 67

Substituting the value of N in the right-hand side expression from (i), we have

$$N.\left(\frac{\alpha\beta}{2}\right)^2+1=\left(\frac{\beta^2-2}{2}\right)^2 \qquad \text{(iii)}$$

Composing (ii) and (iii),

$$N\left\{\frac{\alpha}{2}(\beta^2-1)\right\}^2+1=\left\{\frac{\beta}{2}(\beta^2-3)\right\}^2$$

Hence $x=\frac{1}{2}\alpha\beta$, $y=\frac{1}{2}(\beta^2-2)$;

and $x=\frac{1}{2}\alpha(\beta^2-1)$, $y=\frac{1}{2}\beta(\beta^2-3)$;

are solutions of $Nx^2+1=y^2$.

If β be even, the first values of (x,y) are integral. If β be odd, the second values are integral.

(iv) Finally, suppose $k=-4$; the auxiliary equation is

$$N\alpha^2-4=\beta^2$$

Then the required first solution in positive integers of $Nx^2+1=y^2$ is

$x=\frac{1}{2}\alpha\beta(\beta^2+3)(\beta^2+1)$

$y=(\beta^2+2)\{\frac{1}{2}(\beta^2+3)(\beta^2+1)-1\}$.

Brahmagupta says:

> In the case of 4 as subtractive, the square of the second is increased by three and by unity; half the product of these sums and that as diminished by unity (are obtained). The latter multiplied by the first sum less unity is the (required) second root; the former multiplied by the product of the (old) roots will be the first root corresponding to the (new) second root.[1]

The rationale of this solution, as given by Datta and Singh is as follows:

$$N\alpha^2-4=\beta^2 \qquad \text{(i)}$$

$$N(\alpha/2)^2-1=(\beta/2)^2$$

Hence by Brahmagupta's Corollary, we get

$$N\left(\frac{\alpha\beta}{2}\right)^2+1=\left(\frac{\beta^2}{4}+N\frac{\alpha^2}{4}\right)^2$$

1. चतुरूनेऽन्त्यपदं कृती त्र्येकयुते वधदलं पृथग्त्र्येकम्।
त्र्येकाघाहतमन्त्यं पदवध गुणमाद्यमान्त्यपदम्॥ *BrSpSi. XVIII. 68*

$$=\{\tfrac{1}{2}(\beta^2+2)\}^2 \qquad \text{(ii)}$$

Again applying the Corollary, we get

$$N\{\tfrac{1}{2}\alpha\beta(\beta^2+2)\}^2+1=\{\tfrac{1}{2}(\beta^4+4\beta^2+2)\}^2 \qquad \text{(iii)}$$

Now by the Lemma we obtain from (ii) and (iii)

$$N\{\tfrac{1}{2}\alpha\beta(\beta^2+3)\ (\beta^2+1)\}^2+1$$
$$=[(\beta^2+2)\{\tfrac{1}{2}(\beta^2+3)\ (\beta^2+1)-1\}]^2$$

Hence $x=\frac{1}{2}\alpha\beta(\beta^2+3)\ (\beta^2+1)$,

$y=(\beta^2+2)\ \{\frac{1}{2}(\beta^2+3)\ (\beta^2+1)-1\}$

is a solution of $Nx^2+1=y^2$

This can be proved without difficulty that these values of x and y are integral. Since If β is even, β^2+2 is also even. And hence the above values of x and y are integral. On the other hand, if β is odd, β^2 is also odd; under these conditions β^2+1 and β^2+3 are even. In this also, therefore, the above values must be integral.

Putting $p=\alpha\beta$, $q=\beta^2+2$, we can write the above solution in the form

$$x=\tfrac{1}{2}p\,(q^2-1).$$
$$y=\tfrac{1}{2}q(q^2-3).$$

This was the form in which the solution was found by Euler.

Cakravāla or Cyclic Method

We have shown in the preceding articles that the most fundamental step in Brahmagupta's method for the general solution in positive integers of the equation

$$Nx^2+1=y^2$$

where N is a non-square integer, is to form an auxiliary equation of the kind

$$Na^2+k=b^2$$

where a and b are positive integers and $k=\pm1$, ±2 or ±4. From this auxiliary equation, by the Principle of Composition, applied repeatedly whenever necessary, one can derive, as we have already shown above, one positive integral solution of the original *Varga-prakṛti* or Square-Nature. And thence again, by means of the same principle, an infinite number of other solutions in integers can be obtained. How to form an auxiliary equation of

this type was a problem, write Datta and Singh, which could not be solved completely nor satisfactorily by Brahmagupta. In fact, Brahmagupta had to depend on trial. Success in this direction was, however, remarkably attained by Bhāskara II. He evolved a simple and elegant method which assisted in deriving an auxiliary equation having the required interpolator $\pm 1, \pm 2$, or ± 4, simultaneously with its two integral roots, from another auxiliary equation empirically formed with any simple integral value of the interpolator, positive or negative. This method has been technically known as *Cakravāla* or the *cyclic method*. This is so called because it proceeds as in a circle, the same set of operations being applied again and again in a continuous round. For the details of this method, our reader is requested to consult the Algebra of Bhāskara II and the narrative on this method as given by Datta and Singh under the title "Cyclic Method" in their *History of Hindu Mathematics: Algebra*, 1962 Edition, pp. 161-72.

Solution of Indeterminate Quadratic Equation

It is remarkable to see that Brahmagupta was the first algebraist in the history of mathematics to find a general solution of the indeterminate quadratic equation

$$Nx^2 \pm c = y^2$$

in positive integers. We have the following verse in the *Brāhmasphuṭasiddhānta* in this connection:

> From two roots (of a Square-nature or *varga-prakṛti*) with any given additive or subtractive, by making (combination) with the roots for the additive unity other first and second roots (of the equation having) the given additive or subtractive (can be found).[1]

Let us take the following two equations:

$a_1k = an + b$; and $b_1k = bn + Na$

From them we get: by eliminating n

$$a_1b - ab_1 = 1$$

---

1. रूप प्रक्षे पपदे पृथगिष्टक्षे प्यशोच्यमूलाभ्याम् ।
कृत्वाऽन्त्याद्यपदे ये प्रक्षे पे तेऽने ॥ —*BrSpSi.* XVIII. 66

Hence $b_1=\frac{a_1b-1}{a}$ = a whole number.

Now $n^2-N=\frac{(a_1k-b)^2-Na^2}{a^2}$

$$=\frac{a_1^2k^2-2bka_1+k}{a^2}$$

$$=\frac{k(a_1^2k-2ba_1+1)}{a^2}$$

Therefore $\frac{k}{a^2}(a_1^2k-2ba_1+1)$ is a whole number.

Since a, k have no common factor, it follows that

$$\frac{a_1^2k-2ba_1+1}{a^2}=\frac{n^2-N}{k}=k_1=\text{an integer.}$$

Also $k_1=\frac{n^2-N}{k}$ $= \frac{a_1^2k-2ba_1+1}{a^2}$

$$= \frac{a_1^2(b^2-Na^2)-2ba_1+1}{a^2}$$

$$= \left(\frac{a_1b-1}{a}\right)^2-Na_1.$$

Thus having known a single solution in positive integers of the equation $Nx^2\pm c=y^2$, says, Brahmagupta, an infinite number of other integral solutions can be obtained by making use of the integral solutions of $Nx^2+1=y^2$. If (p,q) be a solution of the former equation found empirically and if (α, β) be an integral solution of the latter, then by the principle of Composition

$$x=p\beta\pm q\alpha;\ y=q\beta\pm Np\alpha$$

will be a solution of the former. Repeating the operations, we can easily deduce as many solutions as we like.

FORM $Mn^2x^2\pm c=y^2$:

In this connection, Brahmagupta says :

If the remainder is that divided by a square, the first root is that divided by its root[1].

This seems to mean that if we have the equation

$$Mn^2x^2\pm c=y^2 \qquad \text{(i)}$$

such that the multiplier (i.e. the coefficient of x^2) is divisible

1. वर्गोच्छिन्ने गुणके प्रथमं तन्मूल भाजितं भवति । —*BrSpSi.* XVIII.70

by n^2, then we are justified in saying that if we put $nx=u$, the equation (i) becomes $Mu^2 \pm c = y^2$ (ii),
and clearly the first root of (i) is equal to the first root of (ii) divided by n. The corresponding second root will be the same for both the equations.

FORM $a^2x^2 \pm c = y^2$:

We find Brahmagupta giving the following rule in this connection : This is a solution of a particular form of a *varga-prakṛti* or Square-nature.

> If the multiplier be a square, the interpolator divided by an optional number and then increased and decreased by it, is halved. The former (of these results) is the second root; and the other divided by the square-root of the multiplier is the first root.[1]

Thus the solutions of the equation

$$a^2x^2 \pm c = y^2$$

are :

$$x = \frac{1}{2a}\left(\frac{\pm c}{m} - m\right)$$

$$y = \frac{1}{2}\left(\frac{\pm c}{m} + m\right)$$

where m is an arbitrary number.

Bhāskara II and Nārāyaṇa have also given the same solutions as proposed by Brahmagupta.

Rational Geometrical Figures

In the days of the *Taittirīya Saṁhitā* and the *Śatapatha Brāhmaṇa*, Indian mathematicians got familiarity with the solution of such equations

$$x^2 + y^2 = z^2$$

and the results were arrived geometrically on the basis of the law of rectangle as propounded by Baudhāyana in the Śulba Sūtras and which goes by his name. The reader is referred to the Chapter on Baudhāyana, the first Geometer in the author's "*Founders of Sciences in Ancient India*". Baudhāyana (c. 800 B.C.) gave a

1. वर्गे गुणके क्षेपः केनचिदुद्धृत्युतोनितो दलितः ।
प्रथमोऽन्त्यमूलमन्त्ये गुणकारपदोद्धृतः प्रथमः ॥ —*BrSpSi.* XVIII. 69

method of transforming a rectangle into a square, which is equivalent to the algebraic identity :

$$mn=\left(m-\frac{m-n}{2}\right)-\left(\frac{m-n}{2}\right)$$

where m, n, are any two arbitrary numbers.

Brahmagupta in connection with the solution of rational triangles says :

> The square of the optional (*iṣṭa*) side is divided and then diminished by an optional number; half the result is the upright, and that increased by the optional number gives the hypotenuse of a rectangle.

We shall put these statements of Brahmagupta in the algebraic language thus : If m, n be any two rational numbers, then the sides of a right-angled triangle will be

$$m,\ \tfrac{1}{2}\left(\frac{m^2}{n}-n\right),\ \tfrac{1}{2}\left(\frac{m^2}{n}+n\right)$$

This Sanskrit term *iṣṭa* may either mean "given" or "optional". With the former meaning the rule would imply the method of finding rational right angles having a given leg.

Brahmagupta was the first to give a solution of the equation $x^2+y^2=z^2$ in integers. His solution is

$$m^2-n^2,\ 2mn,\ m^2+n^2.$$

m and n being two unequal integers.[2]

Thus if $m=7$ and $n=4$, then $m^2-n^2=33$, $2mn=56$ and $m^2+n^2=65$; then the three numbers 33, 56 and 65 bear the relation $33^2+56^2=65^2$.

Mahāvīra (850 A.D.) also states

> The difference of the squares (of two elements) is the upright, twice their product is the base and the sum of their squares is the diagonal of a *generated* rectangle.[2]

Isosceles Triangles with Integral Sides : The following statement of Brahmagupta in this connection is very significant :

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1. इष्टस्य भुजस्य कृतिर्भक्तो नेष्टेन तद्दलं कोटिः ।
   आयतचतुरस्रस्य च त्रस्येष्टाधिका कर्णः ॥ —*BrSpSi.* XII. 35

2. *GSS.* VII. 90½

> The sum of the squares of two unequal numbers is the side; their product multiplied by two is the altitude, and twice the difference of the squares of those two unequal numbers is the base of an isosceles triangle.[1]

Thus if m,n be two integers such that m is not equal to n, the sides of all rational isosceles triangles with integral sides are given by

$$m^2+n^2,\ m^2+n^2,\ 2(m^2-n^2)$$

and the altitude of the triangle is $2mn$.

This method was also followed by Mahāvīra and other Indian mathematicians. In fact, their solutions are based on the juxtaposition of two rational right triangles, equal so that they have a common leg. It is remarkably a powerful device, for every rational triangle or quadrilateral may be formed by the juxtaposition of two or four rational right triangles.

Isosceles Triangles with a Given Altitude

Here we have a rule given by Brahmagupta for finding out all rational isosceles triangles possessing the same altitude :

> The (given) altitude is the producer (*karaṇī*). Its square divided by an optional number is increased and diminished by that optional number. The smaller is the base and half the greater is the side.[2]

Thus if m be any rational number then for a given definite altitude a, the sides of the rational isosceles triangles are $\frac{1}{2}\left(\frac{a^2}{m}+m\right)$ each and the base is $\frac{a^2-m^2}{m}$. We shall illustrate it by an example taken from the commentary of Pṛthūdaka Svāmī The given altitude is 8; let us take any rational number $m=4$ then the two equal sides of the isosceles are given by $\frac{1}{2}\left(\frac{8^2+4^2}{4}\right)=$ 10 each and the base is $\frac{8^2-4^2}{4}=12$. Thus the three sides of the

---

1. कृति युतिर् सदृशराश्योर्बाहुर्घातो द्विसंगुणो लम्बः ।
कृत्यन्तरमसदृशयोर्द्विगुणं द्विसमत्रिभुज भूमिः ॥ —*BrSpSi.* XII. 33

2. करणी लम्बस्तत्कृतिरिष्टहृतेष्टोन संयुताऽल्पा भूः ।
अधिको द्विहृ तो बाहुः संज्ञे ष्वो यद्वधो वर्गः ॥ —*BrSpSi. XII.* 37.

rational isosceles triangle with altitude 8 are (10,10, 12).

Rational Scalene Triangles: Brahmagupta lays down the following rule in the case of rational scalene triangle ;

> The square of an optional number is divided twice by two arbitrary numbers; the moieties of the sums of the quotients and (respective) optional numbers are the sides of a scalene triangle; the sum of the moities of the differences is the base.[1]

In other words, if m, p, q are any rational numbers, then the sides of a rational scalene triangle are :

$$\tfrac{1}{2}\left(\frac{m^2}{p}+p\right),\ \tfrac{1}{2}\left(\frac{m^2}{q}+q\right),$$

$$\tfrac{1}{2}\left(\frac{m^2}{p}-p\right)+\tfrac{1}{2}\left(\frac{m^2}{q}-q\right)$$

Here the altitude (m), area and segments of the base of this triangle are all rational.

Thus putting $m=12$, $p=6$, and $q=8$ in Brahmagupta's general equation, Pṛthūdaka Svāmī derives a scalene triangle with sides (13,15) and (14) altitude (12), area (84 and the segments of the base (5) which are all integral numbers.

$$\tfrac{1}{2}\left(\frac{m^2}{p}+p\right)=\tfrac{1}{2}\left(\frac{12^2}{6}+6\right)=15;$$

$$\tfrac{1}{2}\left(\frac{m^2}{q}+q\right)=\tfrac{1}{2}\left(\frac{12^2}{8}+8\right)=13$$

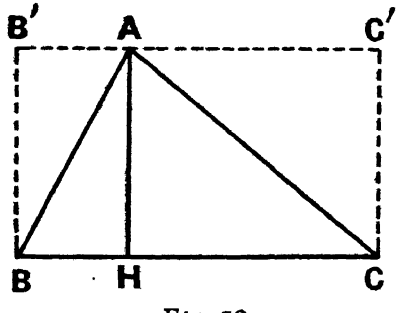

Fig. 19

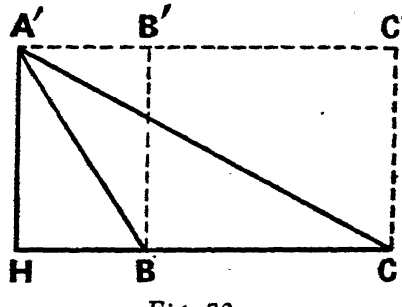

Fig. 20

1. इष्टद्वयेन भक्तो द्विघ्नेष्ट वर्ग फलेष्टयोगार्धे ।
विषमत्रिभुजस्य भुजाविष्टोनफलार्धयोगो भूः ॥

—*BrSpSi.* XII, 34.

Thus the two sides of the rational scalene triangle are 15 and 13. The base is.

$$\tfrac{1}{2}\left(\frac{12^2}{6}-6\right)+\tfrac{1}{2}\left(\frac{1}{8}^{2}-8\right)=9+5=14$$

The altitude is $m=12$; area is equal to $\frac{\text{base}\times\text{altitude}}{2}$

$=\frac{14\times 12}{2}=$ and the segments are $\sqrt{13^2-12^2}=5$ and $\sqrt{(15^2-12^2)}=9$. Thus they are all integers.

Rational Isosceles Trapeziums

Brahmagupta has given us a method of obtaining such isosceles trapeziums whose sides, diagonals, altitude, segments and area are all rational numbers. His rule is as follows:

The diagonals of the rectangle (generated) are the flank sides of an isosceles trapezium; the square of its side is divided by an optional number and then lessened by that optional number and divided by two; (the result) increased by the upright is the base and lessened by it is the face.[1]

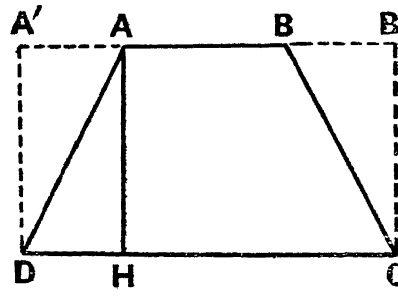

Fig. 21

Here in the figure, we have the isosceles trapezium ABCD of which C D is the base and A B is known as the fase. According to Brahmagupta's rule, we have (p being the optional number).

$$CD=\tfrac{1}{2}\left(\frac{4m^2n^2-p}{p}\right)+\left(m^2-n^2\right) \qquad \text{(base)}$$

$$AB=\tfrac{1}{2}\left[\frac{4m^2n^2}{p}-p\right]-\left[m^2-n^2\right] \qquad \text{(face)}$$

$$DH=(m^2-n^2) \qquad \text{(upright)}$$

1. आहतकर्णौ बाहू भुजकृतिरिष्टेन भाजितेष्टोना।
द्विहृता कोट्यधिका भूर्मुखमूना दिसमचतुरस्रे ॥ —*BrSpSi.* XII. 36

$AD = BC = m^2+n^2$ (the sides of the trapezium)

HC = base−upright = $\frac{1}{2}\left[\frac{4m^2n^2}{p} - p\right]$ (segment)

$AC = BD = \left[\frac{4m^2n^2}{p} + p\right]$ (diagonal)

$AH = 2mn$ (altitude)

$ABCD = mn\left[\frac{4m^2n^2}{p} - p\right]$ (area)

By chosing the values of m n and p suitably, the values of all the dimensions of the isosceles trapezium can be made integral. Pṛthūdaka Svāmī starts with the rectangle (5, 12, 13) and suitably takes p as 6; then he calculates out the dimensions of the trapezium: flank sides (AD and BC) = 13, base =14, and base = 4, altitude (AH) = 12, segments of base (DH and HC) = 5, and 9, diagonals (AC and BD) = 15, area ABCD = 108. All these values are integers.

In this example, the rectangle chosen is (5, 12, 13) which is AA' DH, where $AD = m^2 + n^2 = 13$

and $DH = m^2 - n^2 = 5$

whence by adding the two we have

$2m^2 = 18$

This gives the value of $m = 3$, and hence $n = 2$. Pṛthūdaka Svāmī has taken the value of $p = 6$ by choice. Putting these values of m, n and p, the values for the dimensions of the isosceles trapezium follow from the expressions given by Brahmagupta.

$$CD = \frac{1}{2}\left(\frac{4.3^2.2^2}{6} - 6\right) + \left(3^2 - 2^2\right) = 9+5 = 14 \text{ (base)}$$

Face = 9−5=4

Sides $AD = BC = 3^2+2^2 = 13$

and so on for the other dimensions.

Rational Trapeziums With Three Equal Sides

This problem is very much the same as one of the rational isosceles tpapezium with the only difference that in this case one of the parallel sides is also equal to the slant sides. We

have the following solution of this problem from Brahmagupta :

> The square of the diagonal (of a *generated* rectangle) gives three equal sides; the fourth (is obtained) by subtracting the square of the upright from thrice the square of the side (of that rectangle). If greater, it is the base; if less, it is the face.[1]

As before, the rectangle generated from m, n is given by $(m^2-n^2, 2mn, m^2+n^2)$, that is these are the three sides of the right triangle, which correspond to the two sides and the diagonal of the rectangle generated by them. Let us suppose, we have a trapezium ABCD whose sides AB, BC and AD are equal, then

$$AB = BC = AD = (m^2+n^2)^2$$

$$CD = 3(2mn)^2 - (m^2-n^2)^2 = 14\,m^2n^2 - m^4 - n^4$$

$$\text{or } CD = 3(m^2-n^2)^2 - (2mn)^2 = 3m^4+3n^4 - 10\,m^2n^2.$$

Pṛthūdaka Svāmī has taken an illustration, where $m=2$, $n=1$ and he deduces two rational trapeziums with three equal sides (25, 25, 25, 39) and (25, 25, 25, 11).

The segment (CH), altitude (AH), diagonals (AC, BD) and area of this trapezium are also rational, and given by :

CH (segment) $= 6\,m^2n^2 - m^4 - n^4$

AH (altitude) $= 4\,mn\,(m^2-n^2)$

$AC = BD$ (diagonals) $= 4\,mn\,(m^2+n^2)$

$ABCD$ (area) $= 32m^3n^3\,(m^2-n^2)$.

Rational Inscribed Quadrilaterals

We find in the *Brāhmasphuṭasiddhānta* a remarkable proposition formulated by Brahmagupta :

> To find all quadrilaterals which will be inscribable within circles and whose sides, diagonals, perpendiculars, segments (of sides and diagonals by perpendiculars from vertices as also of diagonals by their intersection), areas, and also the diameters of the

1. कर्णकृतिस्त्रिसम भुजास्त्रयश्चतुर्थो विशोध्य कोटि कृतिम् ।
बाहुकृतेस्त्रिगुणाया बधिको भूमुंखं हीनः ॥ —*BrSpSi.* XII. 37

circumscribed circles will be expressible in integers. Such quadrilaterals we shall call as *Brahmagupta Quadrilaterals.*

The solution of this formidable problem has been given by Brahmagupta as follows :

> The upright and bases of two right-angled triangles being reciprocally multiplied by the diagonals of the other will give the sides of a quadrilateral of unequal sides : (of these) the greatest is the base, the least is the face, and the other two sides are the two flanks.[1]

Taking Brahmagupta's integral solution, the sides of the two right triangles of reference are given by :

(1) $m^2-n^2,\ 2mn,\ m^2+n^2$;

(ii) $p^2-q^2,\ 2pq,\ p^2+q^2$;

where m, n, p, q are integers. Then the sides of the *Brahmagupta Quadrilateral* are.

$$(m^2-n^2)(p^2+q^2),\ (p^2-q^2)(m^2+n^2),$$
$$2mn(p^2+q^2),\ 2pq(m^2+n^2) \qquad \text{(Arrangement A)}$$

Pṛthūdaka Svāmī has illustrated the rational inscribed quadrilateral by taking an example of the right angle triangles.

(i) (3,4,5) ($m^2-n^2=3$, $m^2+n^2=5$, whence $m=2, n=1$)

(ii) (5,12,13)($p^2-q^2=5$, $p^2+q^2=13$, whence $p=3, q=2$)

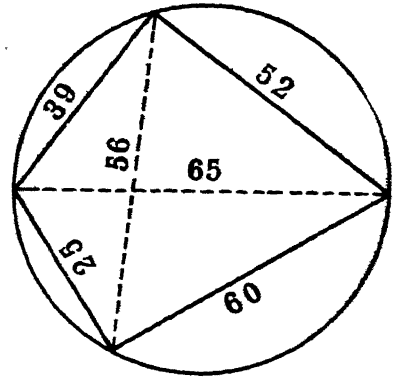

Fig. 22

Substituting these values in the above equations, we get the sides of the quardilateral as (39, 25, 52 and 60).[2]

1. जात्यद्वय कोटिभुजाः परकर्णगुणा भुजाश्चतुर्विषमे ।
अधिको भूर्मुखहीनो बाहुद्वितयं भुजावन्यौ ॥ —*BrSpSi.* XII. 38

2. The diagonals of this quadrilateral are given by Bhāskara II as 56 (=3.12+4.5) and 63 (=4.12+3.5).

(Cont. on page 268)

Put in other words, this means that one has to solve the following equations :

(i) $5x-25 = y^2$

(ii) $10x-100 = y^2$

(iii) $83x-7635 = y^2$

Pṛthūdaka Svāmī, the commentator on the *Brāhmasphuṭa-siddhānta* proceeds to solve these equations as follows :

(1.1) Suppose $y = 10$; then $x = 125$. Or put $y = 5$; then $x = 10$.

(2.1) Suppose $y = 10$; then $x = 20$.

(3.1) Assume $y = 1$; then $x = 92$.

He then remarks that by virtue of the multiplicity of suppositions there will be an infinitude of solutions in every case, But no method has been given either by Brahmagupta or his commentator to obtain the general solution.

Double Equations of the First Degree

Perhaps we have the earliest reference of the simultaneous indeterminate quadratic equations of the type

$$x \pm a = u^2$$
$$x \pm b = v^2$$

in the *Bhakaśālī Manuscript* (Folio 59, recto).

Brahmagupta gives the solution of such simultaneous indeterminate quadratic equations of a general case as follows :

> The difference of the two numbers by the addition or subtration of which another number becomes a square, is divided by an optional number and then increased or decreased by it. The square of half the result diminished or increased by the greater or smaller (of the given number) is the number (required).[1]

Expressed in the language of algebra, shall have :

$$= \left\{ \frac{1}{2}\left(\frac{a-b}{m} \pm m\right) \right\}^2 \mp a$$

1. यास्यां कृतिरधिको नस्तदन्तरं हृत युतो न मिष्टेन ।
तद्दल कृतिरधिकोनाऽधिकयो रधिको न यो राशिः ॥ —*BrSpSi.* XVIII. 74

$$\text{or } x=\left\{\tfrac{1}{2}\left(\frac{a-b}{m}\mp m\right)\right\}^2\mp b$$

where m is an arbitrary number.

Datta and Singh has given the *rationale* of this method as follows :

$$u^2=x\pm a;\ v^2=x\pm b,$$

From them, we have $u^2-v^2=\pm a\mp b$

Therefore $u-v=m$

$$\text{and } u+v=\frac{\pm a\mp b}{m},$$

where m is arbitrary. Hence

$$u=\tfrac{1}{2}\left(\frac{\pm a\mp b}{m}+m\right)=\pm\tfrac{1}{2}\left(\frac{a-b}{m}\pm m\right)$$

Since it is obviously immaterial whether u is taken as positive or negative, we have

$$u=\tfrac{1}{2}\left(\frac{a-b}{m}\pm m\right)$$

Similarly $v=\tfrac{1}{2}\left(\frac{a-b}{m}\mp m\right)$

$$\text{Therefore } x=\left\{\tfrac{1}{2}\left(\frac{a-b}{m}\pm m\right)\right\}^2\mp a,$$

$$\text{or } x=\left\{\tfrac{1}{2}\left(\frac{a-b}{m}\mp m\right)\right\}^2\mp b,$$

where m is an arbitrary number.

Now we shall take up another particular case, for which Brahmagupta has given a rule :

> The sum of the two numbers the addition and subtraction of which make another number (severally) a square, is divided by an optional number and then diminished by that optional number. The square of half the remainder increased by the subtractive number is the number (required)[1].

In the algebraic notations, we shall express it as follows :

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1. यैरूनो यैश्च युतो रूपैर्वर्गस्तदैक्यमिष्ट हृतम् ।
इष्टोनं तद्दल कृतिरूनाऽभ्यधिका भवति राशिः ॥ —*BrSpSi. XVIII. 71*

$$y = \frac{1}{a}\left(\frac{ad+bc}{m} + b.\right)$$

if $b > c$ and $m > \frac{ad+bc}{m}$. If these conditions be reversed then x and y will have their values interchanged.

Datta and Singh have given the following *rationale* of these solutions :

$$axy = bx + cy + d,$$

or $$a^2xy - abx - acy = ad,$$

or $$(ax - c)(ay - b) = ad + bc.$$

Suppose $ax - c = m$, a rational number;

then $ay - b = \frac{ad+bc}{m}$.

Therefore

$$x = \frac{1}{a}(m + c)$$

$$y = \frac{1}{a}\left(\frac{ad+bc}{m} + b\right)$$

Or, we may put $ay - b = m$;

in that case, we shall have $ax - c = \frac{ad+bc}{m}$;

whence $x = \frac{1}{a}\left(\frac{ad+bc}{m} + c\right)$,

$$y = \frac{1}{a}(m + b).$$

Brahmagupta's own rule.

Whilst the rule given above is ascribed to an unknown author, Brahmagupta's own rule for the solution of a quadratic indeterminate equation involving a factum is as follows :

> With the exception of an optional unknown, assume arbitrary values for the rest of the unknowns, the product of which forms the factum. The sum of the product of these (assumed values) and the (respective) coefficients of the unknowns will be absolute quantities. The continued products of the assumed values and of the coefficient of the factum will be the coefficient of the optionally (left out) unknown. Thus the solution

is effected without forming an equation of the factum Why then was it done so ?[1]

Datta and Singh think that the reference in the latter portion of this rule is to the method of the unknown author :

"*Kiṁ kṛtaṁ tadataḥ*" ? The principle underlying Brahmagupta's method is to reduce, like the Greek Diophantus (c.275 A.D.), the given indeterminate equation to a simple determinate one by assuming arbitrary values for all the unknowns except one. So undoubtedly it is inferior to the earlier method.

We now take an illustrative example from Brahmagupta :

> On subtracting from the product of signs and degrees of the Sun, three and four times (respectively) those quantities, ninety is obtained. Determining the Sun within a year (one can pass as a proficient) mathematician.

If we presume x to denote the signs and y the degrees of the Sun, then the equation would be :

$$xy-3x-4y=90$$

Pṛthūdaka Svāmī solves it in two ways :

(i) Let us assume the arbitrary number to be 17. then

$$x=\tfrac{1}{1}\left(\frac{90.1+3.4}{17}+4\right)=10$$

$$y=\tfrac{1}{1}\ (17+3)=20$$

(ii) Let us assume arbitrarily $y=20$. On substituting this value of y in the above equation, we get

$$20x-3x=170$$

whence $x=10$.

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1. भावितके यद्घातो विनष्टवर्णेन तत्प्रमाणानि ।
कृत्वेष्टानि तदाहत वर्णैक्यं भवति रूपाणि ॥
वर्ण प्रमाणभावित घातो भवतीष्ट वर्ण संख्यैवम् ।
सिध्यति विनाऽपि भावित-त्रमकरणात् किं कृतं तदतः ॥ —*BrSpSi. XVIII. 62-63*

2. भानो राश्यंशवधात् त्रिचतुर्गुणितान् विशोध्य राश्यंशान् ।
नवतिं दृष्ट्वा सूर्यं कुर्वन्नावत्सराद् गणकः ॥ —*BrSpSi. XVIII. 61.*

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Reference

H.T. Colebrooke : *Algebra with Arithmetic and Mensuration from the Sanscrit of Brahmagupta and Bhascara*, London, 1817.

B. Datta and A.N. Singh : *History of Hindu Mathematics* Pt. I and II, 1962.

CHAPTER X

Arabic and Indian Divisions of the Zodiac

It has long been a debated question whether the Indian and Arabian divisions of the zodiac had a common origin. Sir William Jones thought that they had not, but Colebrooke holds a contrary view. The coincidence, in the two systems of division is so exact that he thinks, it could not be due to chance. Colebrooke has discussed this point in details in one of his Papers entitled "On the Indian and Arabian divisions of the zodiac", *Asiatic Researches* Vol. ix.. p. 323-376, reproduced in the *Miscellaneous Essays*, Vol. II., p. 321-373, 1872.

1. *Aśvini*, now the first *nakṣatra*, but anciently the last but one, probably obtained its present situation at the head of the asterisms, when the beginning of the zodiacwas referred to the first degree of Meṣa (the Ram). As measuring a portion of the zodiac, it occupies the first 13°20′ of Meṣa; and its beginning follows immediately after the principal star in the last *nakṣatra* Revati, reckoned by some exactly, by others nearly, opposite to the very conspicuous one, which forms the fourteenth asterism. As a constellation, Aśvini comprises three stars (Aries α, β, γ) figured as a horse-head; and the principal, which is also the northern one, is stated by all ancient authorities, in 10°N and 8°E. from the beginning of the Meṣa.

According to Arabs, the first manzil or lunar mansion is entitled *Sheratan* (by Persians *Sheratain*), and comprises two stars of the third magnitude on the head of Aries. in lat. 6°36′ and 7°51′ N and long. 26°13′ and 27°7′. With the addition of a

third, also in the head of the Ram, the asterism is denominated *Āshrait*. The bright star of the second or third magnitude which is out of the figure of the Ram, according to Ulugh Beg, but on the nose according to Hipparchus, cited by this author from Ptolemy, is determined *Nātih* : It is placed in lat. 9°30'N and long. 1 0° 43', and is apparently the same with the principal star of the Indian asterism; for Muhammad of Tizin, in his table of declination and right ascension, expressly terms it the first star of the *Sheratain*.

2. *Bharaṇī*, the second asterism, comprises three stars (35, 39, 41 Aries) figured by the *yoni* or *pudendum muliebre* and the principal and southern star of this nakṣatra is placed in 12°N. On the Arabian system, the second manzil, entitled *Butain* is placed by Ulugh Beg in lat. 1°12' and 3°12', and this cannot possibly be reconciled with the Indian constellation. But Muhammad of Tizin assigns to the bright star of *Butain* a declination of 23°N exceeding by nearly 2° the declination allotted by him to *Nātih* or his first star in *Sheratain*. This agrees with the difference between the principal stars of *Aśvinī* and *Bharaṇī*; and it may be inferred, that some among the Mohammadan astronomers have concurred with the Hindus, in referring the second constellation to stars that form Musca.

3. *Kṛttikā*, now the third, formerly the first, *nakṣatra* consists of six stars figured as knife or razor, and the principal and southern star is placed in $4\frac{1}{2}$ or 5°N and in 65 sixths of degrees (or 10°50') from its own commencement (cf. the *Sūryasiddhānta*). or 37° 28' to 38° from the beginning of the Meṣa (the *Siddhānta-Śiromaṇi* or the *Graha Lāghava*) respectively. This longitude of the circle of declination corresponds nearly with that of the bright star in the Pleiades, which is 40° of longitude distant from the principal star of *Revatī*.

The stars indicated by Ulugh Beg for *Thurayyā*, also correspond exactly with the Pleiades.

4. *Rohiṇī*, is the fourth *nakṣatra*, the Arabic name for the fourth mansion is *Debarān* (or with the article *Aldebarān*). It corresponds to the bright star called the Bull's eye, and which is unquestionably the same with the principal and eastern star of *Rohiṇī*, placed in $4\frac{1}{2}$° or 5°S and $49\frac{1}{2}$°E by the Hindu writers on Astronomy. This *nakṣatra* is

figured as a wheel cart, and comprises five stars, out of the seven which the Greeks named the Hyades. The Arabs, however, like the Hindus, reckon five stars only in the asterism. Sir William Jones supposes them to be in the head and neck of the Bull; they probably are α, ρ, γ, δ, ε Tauri, agreeably to Mons. Bally's conjecture.

5. *Mṛgaśirā*, the fifth *nakṣatra*, represented by an antelope head, contains three stars; the same which constitute the fifth lunar mansion *Hakāh*; for the distance of 10°S assigned to the northern star of this *nakṣatra*, will agree with no other but one of the three in the head of Orion. The difference of longitude (24° to $25\frac{1}{2}$°) from Kṛttikā corresponds with sufficient exactness; and so does the longitude of its circle of declination (62° to 63°) from the end of Rewatī; since the true longitude of λ Orionis, from the principal star in Revatī (ζ piscium) is $63\frac{1}{2}$°.

6. *Ārdrā*, the sixth *nakṣatra*, consists of a single bright star, described as a gem, and placed in 9°S (by some in 11°) and at the distance of $4\frac{1}{3}$ to 4° in longitude from the last asterism. This indicates the star in the shoulder of Orion (α orionis). The sixth lunar mansion is named by the Arabs as *Hanāh*; and comprises two stars in the feet of the second twin, according to Ulugh Beg, though others make it to be a shoulder. Mohammad of Tizin allots five stars to this constellation; and the Kāmūs, among various meanings of Hanāh, says, that it is a name for five stars in the left arm of Orion; remarking also, that the lunar mansion is named *Tahāyi*, comprising three stars called Tahyāt. Obviously here the Indian and Arabian asterisms are irreconcileable.

7. *Punarvasū* (used in a dual number) is the seventh *nakṣtra*, and is represented by a house, or even a bow, and it includes four stars, among which the principal and eastern one is 30° or 32° from the fifth asterism; but has been placed by all authorities in 6°N. This agrees with (β Geminorum) one of the two stars in the heads of the Twins, which together constitute the seventh lunar mansion *ziraā*, according to Mohammad of Tus and Mohammad of Tizin and other Arabian authorities. The seventh lunar mansion of Arabs is named *ziraā ul ased* according to Jauhari and other cited by Hyde in his Commen-

tary on Ulugh Beg. and that the Kāmūs makes this term to be the name of eight stars in the form of a bow.

8. *Puṣya*, the eighth asterism, is described as an arrow, and consists of three stars the chief of which being also the middle most, has no latitude, and is 12° to 13° distant from the seventh asterism, being placed by Hindu astronomers in 106° of longitude. This is evidently δ Cancri; and does not differ widely from the eighth lunar mansion *Nethrah*, which according to Ulugh Beg and others consists of two stars, including the nebula of Cancer. The Indian constellation comprises two other stars besides δ Cancri, which are perhaps γ and β of the same constellation.

9. *Aśleṣa*, the ninth asterism, contains five stars figured as a potter's wheel, and of which the principal or eastern one is placed in 7°S. and according to different tables, 107°, 108° or 109° E. This appears to be intended for the bright star in the southern claw of Cancer (α Cancri), and cannot be reconciled with the lunar mansion *Tarf* or *Tarfah*, which comprises two stars near the lion's (simha) eye, the northernmost being placed by Mohammad of Tizin in 24° of N. declination.

10. *Maghā*, the tenth asterism, contains like the last, five stars, but which are figured as a house. The principal of the Southern one has no latitude; and according to all authorities, has 129° longitude. This is evidently Regulus (α Leonis) : which is exactly 129⅔° distant from the last star in *Revati*. The tenth lunar mansion of Arabians is *Jebhah*, which comprises three (some say, four) stars, nearly in the longitude of the lion's heart. In this instance, therefore, the Indian and Arabian divisions of the zodiac coincide. This *nakṣatra* consists of α γ ζ η and ν Leonis.

11. *Pūrva-Phālguni* is the eleventh *nakṣatra* and is represented by a couch or bedstead; it consists of two stars determined by the place of the chief star (the northernmost, according to the *Sūrya-Siddhānta*) in 12°N and 144°E, or according to Brahmagupta, the *Śiromaṇi* and the *Grahalāghava* 147° or 148° E. They are probably δ and θ Leonis. The Arabian name for this lunar mansion is *zubrah* or *Khertān*.

It may be mentioned here that Brahmagupta and Bhāskara selected the southern for the principal star; while the *Sūrya-Siddhānta* took the northern. Hence the latitude stated by several Hindu authorities is the mean between both stars; and the difference of longitude, compared to the preceding and subsequent asterisms, may be exactly reconciled upon this supposition.

12. *Uttara-Phālgunī*, which is the twelfth *nakṣatra*, consists of two stars, and is figured as a bed or cot. These stars are ascertained by the place of one of them (the northernmost) 13°N. and 155° E. This indicates β Leonis; the same which singly constitutes the Arabian Lunar mansion *Serfah*, though Mohammad of Tīzīn seems to hint that it consists of more than one star.

13. *Hasta*, the thirteenth *nakṣatra*, has the name and figure of a hand; and is suitably made to contain five stars. The principal one towards the west, next to the north-western star, is placed according to all authorities in 11° and 170° E. This can only belong to the constellation Corvus ; and accordingly five stars in that constellation (α, β, γ, δ and ε Corvi). The thirteenth lunar mansion of Arabs is *Awwā*, which is also described to contain five stars, situated under Virgo and so disposed as to resemble the letter *Alif*. They are placed by Ulugh Beg in the wing. Here obviously, there is nothing common between the Hindu and Arabian specification of the asterism. The agreement is only in the number of stars and in the longitude.

14. *Citrā*, the fourteenth *nakṣatra*, is figured as pearl. It is placed by the *Sūrya-Siddhānta* in 2° S and 180° E. and by Brahmagupta, the *Śiromaṇi* and *Grahalāghava* in $1\frac{3}{4}$ or 2°S and 183°E. This agrees with the Virgin's spike (α Virginis). The same star constitutes the fourteenth lunar mansion of the Arabs named from it *Simāc ul aāzi*.

15. *Svāti*, the fifteenth *nakṣatra*, is represented by a coral bead. The *Sūrya-Siddhānta*, Brahmagupta, the *Śiromaṇi* and *Grahalāghava*, all concur in placing it at 37°N. They differ one degree in longitude of its circle of declination, three of them

making it 199° and the other 198°. The Indian asterism totally disagrees with the lunar mansion *Ghafr* which is the fifteenth Arabian mansion. and which consists of three stars in the Virgin's (*Kanyā*) foot, according to Ulugh Beg. but in or near the balance (*Tulā*), according to others.

16. *Viśākhā*, the sixteenth *nakṣatra*, consists of four stars described as a festoon. All the authorities place the principal and northernmost star in 1°, 1°20′ or 1°30′ S and in 212°, 212°5′ or 213° E. The latitude seems to indicate the bright star in the Soutehrn Scale (α Librae), though the longitude disagrees (suggesting possibly a remote star κ Librae). Colebrooke suggests the four stars to be α ν ι Librae and γ Scorpii. The sixteenth lunar mansion according to Arabs is *Zubanah* or *Zubāniyah* according to Mohammad of Tīzīn, the bright star in the northern scale (β Librae).

17. *Anurādhā*, the seventeenth *nakṣatra*, consists of four stars and is described as a row of oblations in a right line. Its chief or middlemost star is placed in 3°, or 2° or 1°45′ S and in 224° or 224°5′E, thus placing it near the head of the Scorpion (*Vṛścika*) (δ Scorpionis) and the asterism comprises β, δ, π, and ρ Scorpionis. The seventeenth lunar mansion of Arabs is called *Iklīl* or *Iklīlu'l-jebhah*, which is said to contain 4, 3, or 6 stars lying in a straight line. Those assigned by Ulugh Beg for this mansion are β, δ, ν and π Scorpionis. Thus here the Indian and Arabian astronomers both concur exactly.

18. *Jyeṣṭha*, the eighteenth *nakṣatra*, comprises three stars figured as a ring. The principal and middlemost star is placed in 4°,3½° or 3° S and in 229°,229°5′ or 230°E; this position indicates Antares or the Scorpion's heart (α Scorpionis), which is also the eighteenth lunar mansion, named *Kalb* or *Kalbul'akrab*. The three stars of Indian asterism may be α, σ and τ Scorpionis.

19. *Mūla*, the nineteenth *nakṣatra*, is represented by a lion's tail, and it contains eleven stars, of which the characteristic one, the easternmost, is placed in 9°, 8½° or 8° S and in 241° or 242° E. This probably (not exactly) indicates ν Scorpionis. This agrees with the eighteenth lunar mansion of Arabs known as *Shaulah*, consisting of two stars near the Scorpion's

sting. The Hindu asterism probably includes all the stars in the Scorpion's tail (ε , μ, ζ, η, θ, ι, κ, λ, υ and ν Scorpionis).

20. *Pūrva-Āṣāḍha*, the twentieth *nakṣatra*, is figured as an elephant's tooth or as a couch, and it consists of two stars, of which the most southern one is placed in $5\frac{1}{2}°$, $5\frac{1}{3}$ or 5° S and 254° or 255° E. This corresponds well with δ Sagittarii, and which also corresponds with the twentieth lunar mansion of Arabs called *Nāaim*. The Arabian mansion consists of four, or according to some eight, stars. The Indian *nakṣatra* corresponds to δ and ε Sagittarii.

22. *Uttara-Āṣāḍha*, the twenty-first *nakṣatra*, is represented by a couch or by an elephant's tooth. The principal or the most northerly star is placed in 5° S and 260° or 261° E, agreeing with a star in the body of Sagittarius (τ Sagittarii), and the other star is perhaps the one marked ζ. The Arabian lunar mansion corresponding to it is *Baldah*, consisting of six stars, two, of which are placed by Mohammad of Tīzīn in declination 21° and 16°. One of these must be a star in the head of Sagittarius. Some authors, on the contrary, describe the lunar mansion as destitute of stars. Here the Arabs and Hindus do not show reconciliation.

22. *Abhijit*, the twenty-second asterism, consists of three stars figuring as a triangle or as a nut of floating Trapa (in modern Indian astronomy, it does not occupy an equal portion of the ecliptic with other *nakṣatras*). Its brightest star is very remote from the zodiac, being in 60° or 62° N. The longitude of its circle of declination is 265°, 266° 40′ or 268° according to different authorities. The corresponding lunar mansion of Arabs is *Zābih*, consisting of two stars (according to some, four) in the horns of Capricorn. This totally disagrees with Indian asterism.

20. *Śravaṇa*, the twenty-third *nakṣatra*, is represented by three footsteps, and contains three stars of which the middlemost is placed in 30° N (all authorities agree), and longitude 280° (*Sūrya-Siddhānta*) or 278° (Brahmagupta and *Śiromaṇi*), or 275° (*Grahalāghava*). The assigned latitude indicates the bright star in the Eagle, whence the three may be inferred to be α, β

and γ Aquilae. According to Arabs, the twenty-third lunar mansion is *Balā*, which consists of two stars in the left hand of Aquarius. Here again Arabian and Hindu divisions are at variance.

24. *Dhaniṣṭhā*, the twenty-fourth *nakṣatra*, is represented by a drum or tabor. It comprises four stars, the westernmost of which is placed in 36° N and according to Brahmagupta, *Śiromaṇi* and the *Śūrya-Siddhānta* in 290° E (*Grahalāghava* gives 286°). This longitude of the circle of declination and the distance of the star on it from the ecliptic indicate the Dolphin : and the four stars are α, β, γ and δ Dolphini. The corresponding lunar mansion of Arabs is *Sāud*, which comprises two stars in Aquarius (β and ζ Aquarii). Here again the two divisions disagree completely.

24. *Śatahbiṣak*, the twenty fifth *nakṣatra*, is a cluster of 100 stars figured by a circle. The principal or the brightest has no latitude; or only a third, or at utmost half, a degree of south latitude; and longitude 320°. This best corresponds with λ Aquarii. According to Arabs, the twenty-fifth lunar mansion is known as *Akhbiyah* which consists of three stars only, placed in the wrist of the right hand of Aquarius. However, it appears from Ulugh Beg's tables, as well as from Mohammad of Tīzīn's, that four stars are assigned to this mansion. The Indian and Arabian systems of division differ considerably but less widely according to some.

26. *Pūrva-Bhādrapada*, the twenty-sixth *nakṣatra*, consists of two stars represented by a couch or bed, or else by a double headed figure, one of which is placed in 24° N and 325° or 326° E. The only conspicuous star nearly in that position is the bright star in Pegasus (α Pegasi) and the other may be the nearest considerable star in the same constellation (ζ Pegasi). The twenty-sixth Arabian lunar mansion is *Mukaddim*, consisting of two brightest stars in Pegasus (α and β). Here the Indian and Arabian divisions show concurrence.

27. *Uttar-Bhādrapada*, the twenty-seventh *nakṣatra*, consists of two stars, figured as a twin or a person with double face, or else as a couch. The position of the most northerly of the two

is in 26° or 27°N and 337° E. which probably indicates the bright star in the head of Andromeda, and the other star to be the one in the extremity of the wing of Pegasus (γ Pegasi). This exactly agrees with the twenty-seventh lunar mansion of Arabs named as *Muakkher*. Ulugh Beg assigns those stars to it.

28. *Revatī*, the twenty-eighth *nakṣatra*, comprises thirty-two stars figured as a tabor. The principal star is the southernmost one, it has no latitude, and two of them assert no longitude. but some make it ten minutes short of the origin of the ecliptic, viz. 359° 50'. This clearly marks the star on the ecliptic in the string of the Fishes (ζ Piscium). The ascertainment of this star is important in regard to the adjustment of the Hindu sphere. The Arabic name for this mansion is *Rishā*, signifying a cord. But the constellation as described by Jauhari and cited by Golius, consists of a multitude of stars in the shape of a fish and termed *Betnu'lhūt*; in the navel of which is the lunar mansion. Mohammad of Tīzīn also makes this lunar mansion to be the same with *Betnu'lhūt*, which appears, however, to be the bright star in the girdle of Andromeda (β Andromedae) though others describe it as the northern fish, extending, however, to the horns of Ram. The lunar mansion and the Indian asterism, therefore, are not reconcileable in this last instance.

I leave it to the readers to draw an inference as to the concurrence of the divisions of zodiac in Indian and Arabian systems. I would personally agree with Sir William Jones that the agreements are by chance. Arabs derived the idea of dividing zodiac in 27 or 28 mansions from Indians, or may have got it from Greeks, and then they proceeded in their own way for details. I do not agree with those scholars who sometimes state that the Hindus took the hint of dividing the ecliptic from Greeks. The *Atharvaveda* devotes a number of Sūktas or hymns on Nakṣatras, and I have shown elsewhere that inspired by these hymns, Gārgya was the first Ṛṣi who detailed out the *nakṣatras*. This happened much before Greeks developed even their first notions of astronomy. While the concept of 27 *nakṣatras* is Vedic and most ancient and of purely Indian origin the concept of 12 Rāśis (signs) or twelve constellations is probably inspired from Greeks. [The names Kanyā, (virgo), Tulā

(Libra), Vṛścika (Scorpio), Dhanu, (Sagittarius), Makara, (Capricorn), Kumbha (Aquarius), Mīna (pisces), Meṣa (Aries), Vṛṣa (Taurus), Mithuna (Gemini), Karka (Cancer), and Siṁha (Leo) were not used for Rāśis or signs in the Vedic times]. I shall conclude this description with a passage from Colebrooke :

> The result of comparison shows, I hope satisfactorily, that the Indian asterisms, which mark the divisions of the ecliptic, generally consist of nearly the same stars, which constitute the lunar mansions of the Arabians : but in a few instances, they essentially differ. The Hindus have likewise adopted the division of the ecliptic and zodiac into twelve signs or constellations, agreeing in figure and designation with those of the Greeks; and differing merely in the place of the constellations, which are carried on the Indian sphere a few degrees further west than on the Grecian. That the Hindus took the hint of this mode of dividing the ecliptic from the Greeks, is not perhaps altogether improbable; but if such be the origin of it they have not implicitly received the arrangement suggested to them, but have reconciled and adapted it to their own ancient distribution of the ecliptic into twenty-seven parts.
>
> In like manner, they may have either received or given the hint of an armillary sphere as an instrument for astronomical observation ; but certainly they have not copied the instrument which was described by Ptolemy, for the construction differs considerably.

Names, Shapes, and Number of the Stars of the Nakṣatras

The *Muhūrta-cintāmaṇi* provides a list of shapes associated to the nakṣatras (*MuC.* II. 59-60), In this list we are giving the number of stars as indicated by Varāhamihira, Brahmagupta and Lalla. The identification given here is as indicated by E. Burgess, in his Translation of the *Sūrya-Siddhānta* 1935, p. 378. (Calcutta). This table has been reproduced here from the *Mahābhāskarīya* of Bhāskara I, edited by K.S. Shukla.

Nakṣatra	Shape	Number of stars			Identification
		Varāha	Brahma	Lalla	
Aśvinī	Head of a horse	3	2	3	α,β,γ Aries
Bharaṇī	Yoni	3	3	3	35,39,41 Aries
Kṛttikā	Razor	6	6	6	η Tauri etc. (Pleiades)
Rohiṇī	Cart	5	5	5	$\alpha,\theta,\gamma,\delta,\varepsilon$ Tauri (Hyades)
Mṛgaśirā	Head of a deer	3	3	3	$\lambda,\phi_1\phi_2$ Orionis
Ārdrā	Jewel	1	1	1	α Orionis
Punarvasū	House	5	2	4	β,α,i,ν,μ Geminorum
Puṣya	Arrow-head	3	1	3	θ,δ,γ Cancri
Āśleṣā	Wheel	6	6	5	$\varepsilon,\delta,\sigma,\eta,\rho$ Hydrae
Maghā	House	5	6	5	$\alpha,\eta,\xi,\zeta,\eta,\varepsilon$ Leonis
P-Phālgunī	Mañca	8	2	2	δ,θ Leonis
U-Phālgunī	Cot	2	2	2	β, 93 Leonis
Hasta	Hand	5	5	5	$\delta,\gamma,\varepsilon,\alpha,\beta$ corvi
Citrā	Pearl	1	1	1	α Virginis (Spica)
Svātī	Coral bead	1	1	1	α Bootis (Arcturus)
Viśākhā	Arched doorway	5	2	4	$i,\gamma,\beta,\alpha,$ Librae
Anurādhā	Heaps of offerings to gods	4	4	4	δ,β,π Scopionis
Jyeṣṭhā	Pendent	3	3	3	α,σ,τ Scorpionis
Mūla	Tail of a lion	11	2	11	$\lambda,\nu,\kappa,i,\theta,\eta,\zeta,\mu,\varepsilon$ Scorpionis
P-Āṣāḍha	Tusk of elephant	2	4	2	δ,ε Sagittarii
U-Āṣāḍha	Mañca	3	4	2	σ,ξ Sagittarii
Śravaṇa	Three feet	3	3	3	$\alpha\,\beta,\gamma$ Aquilae
Dhaniāṣṭh	Drum	5	5	4	$\beta,\alpha,\gamma,\delta$ Delphini
Satabhiṣak	Circle	100	1	100	λ Aquarii etc.
P-Bhādra.	Mañca	2	2	2	α,β Pegasi
U-Bhādra	Pair	8	2	2	γPegasi; α Andromedae
Revati	Drum	32	1	32	ζPiscium etc.

Reference

H.T. Colebrooke : *Miscellaneous Essays*, Vol. II. 1872.
K.S. Shukla . The *Mahābhāskarīya*.

CHAPTER XI

Brahmagupta's Astronomy : Its Highlights

Beginning or Starting Point

Very often in Indian astronomy, we come across a term *ahargaṇa* (literally meaning collection of days), which means the number of mean civil days elapsed at mean Sunrise at Laṅkā on a given lunar day (*tithi*), since the beginning of *Kaliyuga*. It is the beginning of *Kaliyuga*, which is taken as the starting point for the reckoning of *ahargaṇa*. This happened on Friday, February 18, B.C. 3102, at mean sunrise at Laṅkā, when the Sun, Moon, and the planets are supposed to have been in conjunction at the first point of the *nakṣatra Aśvini* (which is a fixed point situated near the star ζ-Piscium). According to Āryabhaṭa and Bhāskara I, the duration of Kaliyuga is 1,080,000 solar years. Four times this (4,320,000) is the duration in solar years of a bigger unit called *Mahāyuga* or even *yuga*.

Laṅkā in Indian astronomy is a hypothetical place where the meridian of Ujjain (latitude 23° 11′ N, longitude 75° 52′ E from Greenwich) intersects the equator. It is one of the four hypothetical cities on the equator called Laṅkā. Romaka, Siddhapur and Yamakoṭi (or Yavakoṭi). The *Sūrya-siddhānta* describes Laṅkā as a great city (mahāpurī) situated on an island to the south of Bhāratavarṣa.[1] The present Ceylon is not the

1. समन्तान्मेरुमध्यात्तु तुल्यभागेषु तोयधेः ।
द्वीपेषु दिक्षु पूर्वादिनगयों देवनिर्मिताः ॥
भूवृत्त पादे पूर्वस्यां यवकोटीति विश्रु ता ।
भद्राश्व वर्षे नगरी स्वर्णप्राकारतोरणा ।
याम्यायां भारतेवर्षे लङ्का तद्वन्महापुरी ॥

(Cont. on page 290)

astronomical Lankā, as it is about six degrees to the north of equator. The astronomical Laṅkā is mentioned by Brahmagupta in the beginning of his very first Chapter[1].

According to Brahmagupta all the four yugas of a *Caturyuga* or *mahāyuga* are not of the equal duration :

> *Kaliyuga* is of 432,000 years. *Dvāpara* of 864,000 years, *Tretā* of 1,296,000 years and *Kṛtayuga* of 1,728,000 years; total of the four is 4,320,000 years. Āryabhaṭa regards all *yugas* of equal duration, 1,080,000 years[2].

The Saka era, which is usually used in Indian astronomy for the reckoning of years commenced 3179 years after the beginning of *Kaliyuga*.

The number of lunar months in a *yuga* does not coincide with the number of solar months. Thus we have the conception of the *Intercalary months* : the number of intercalary months in a *yuga* denotes the excess of the number of lunar months in a *yuga* over the number of solar months in a *yuga*. Thus in a *yuga* we have

Lunar months	53,433,336
Solar months	51,840,000
Intercalary months	1,593,336
Lunar days	1,603,000,080
Civil days	1,577,917,500
Omitted lunar days	25,082,580

The number of omitted lunar days in a *yuga* is equal to the number of lunar days in a *yuga* minus the number of civil days in a *yuga*.

(*Cont. from page 289*)

पश्चिमेकेतुमालाख्ये रोमकाख्या प्रकीर्तिता ।
उदक्सिद्धपुरी नाम कुरुवर्षे प्रतिष्ठिता ।। —*Sūrya*. XII. 36-39

1. चैत्रसितादेरुदयाद्भानोर्दिनमासवर्षयुगकल्पाः ।
सृष्ट्यादौ लंकायां समं प्रवृत्ता दिनेऽर्कस्य ॥ —*BrSpSi*. I. 4

2. युगदशभागो गुणितः कृतं चतुर्भिस्त्रिभिर्गुणस्त्रेता ।
द्विगुणो द्वापरमेकेन संगुणः कलियुगं भवति ॥
युगपादानार्यभटश्चत्वरि समानि कृतयुगादीनि ।
यदभिदितवान् न तेषां स्मृत्युक्तसमाननेकमपि ॥ —*BrSpSi*. I. 8-9

Units of time

For the measurements of durations, it is necessary to have units of time. Brahmagupta gives the following units :[1]

6 prāṇas or Asus	=1 Ṛkṣa-vināḍikā or nakṣatra-vighaṭikā or one *pala* (24 seconds)
60 palas	=1 ghatikā (24 minutes)
60 ghaṭikās	=1 divasa or dina (day) (24 hours)
30 dinas	=1 māsa (month)
12 māsas	=1 varṣa or year

Similar to the divisions of time, we have the divisions of an arc :[2]

Vikalā (or viliptā or viliptikā) =second of arc.

60 vikalās	=1 kalā (minute of arc)
60 kalās	=1 aṁśa (degree of arc)
30 aṁśas	=1 rāśi
12 rāśis	=1 bhagaṇa (complete circle, 360°)

Unlike Āryabhaṭa and others, who take kali, dvāpara, tretā, and kṛta of equal number of years, Brahmagupta regards kali consisting of 432,000 years, dvāpara twice of it, consisting of 864,000 years, tretā thrice of Kali consisting of 1,296,000 years and kṛta four times of kali conisting of thus 1,728,000 years, all the four to be making a *yuga* of 4,320,000 years.[3] Further in the beginning of kṛta there is a *saṇdhyā* of 1728000/12 years (=144,000 years), and at the end of Kṛta, there is a saṇdhyāṁśa of 144,000 years; similarly in the beginning of tretā, we have a

1. प्राणैर्विनाडिकार्द्धं षड्भिर्घटिका षष्ट्या ।
   घटिका षष्ट्या दिवसो दिवसानां त्रिंशता मासाः ॥ —*BrSpSi*. I. 6
2. मासा द्वादशवर्षं विकलालिप्तांशराशिभगणांतः ।
   क्षेत्र विभागस्तुल्यः कालेन विनाडिकाद्येन ॥ —*BrSpSi*. I. 6
   वर्षं द्वादश मासास्त्रिंशद्दिवसो भवेत्समासस्तु ।
   षष्टि र्नाड्यो दिवसष्षष्टिस्तु विनाडिका नाडी ॥ —*Arya*. III. 1
   गुर्वक्षराणि षष्टिर्विनाडिकार्द्धं षडेव वा प्राणाः ।
   एवं काल विभागः क्षेत्र विभागस्तथा भगणात् ॥ —*Arya*, III. 2
3. स्वचतुष्टयरदवेदा रवि वर्षाणां चतुर्युगं भवति ।
   सन्ध्या सन्ध्यतांशैः सह चत्वारि पृथक् कृतादीनि ॥
   युगदशभागो गुणितः कृतं चतुर्भिस्त्रिभिर्गुणस्त्रेता ।
   द्विगुणो द्वापरमेकेन संगुणः कलियुगं भवति ॥ —*BrSpSi*. I. 7. 8

sandhyā of 1,296,000/12; i. e. 108,000 years and at the close of tretā a sandhyāṁśa of 108,000 years. Again, in the beginning of dvāpara we have a sandhyā of 864,000/12, i. e. 72,000 years and at the close of dvāpara a sandhyāṁśa of 72000 years; and similarly at the heginning a sandhyā and at the close a sandhyāṁśa of 432,000/ 1, i. e. of 36,000 years in the case of kali. In this respect Brahmagupta appears to follow Manu. the first author or giver of law. He regards further the following divisions of time :[1]

71 yugas =1 manu
14 manus =1 kalpa

Again, in the beginning, at the middle and at the close of each manu, there are sandhis, each equal to the measure of kṛta. Thus, taken as a whole

1 kalpa=71×14 yugas+15 sandhyā-sandhyāṁśa
=994 yugas+15×duration of kṛta
=994 yugas+15×(4×432,000) years
=994 yugas + 6 yugas =1000 yugas
= 1 Brahma-dina (Brahmā's day)

Thus Brahmā's day is regarded as 1 kalpa or one thousand caturyugis or 1000 *yugas* or the same as 1000 *mahāyugas*).

Āryabhaṭa regards a manu to consist of 72 yugas and therefore a kalpa according to him would be of 14×72 yugas, or 1008 yugas.[2] Since in the foreign Siddhāntas like Romaka, there is no reference to yuga, manu and kalpa. Brahmagupta regards these systems to be unauthoritative.[3]

We have said that our starting point was the beginning of Kaliyuga, Friday February 18, B.C. 3102, at mean rise at Laṅkā, when the Sun, Moon and the planets are supposed to have been in conjunction at the first point of the Nakṣatra Aśviṇī. This type of conjunction would again happen after a period of *kalpa*.

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1. मनुरेकसप्ततियुगः कल्पो मनवश्चतुर्दश मनूनाम् ।
आद्यन्तरान्त सन्धिषु कृतकालोऽस्माद्युग सहस्रम् ॥ —*BrSpSi.* I. 10

2. दिव्यं वर्ष सहस्रम् ग्रहसामान्यं युगं द्विषट्क गुणम् ।
अष्टोत्तरसहस्रं ब्राह्मो दिवसो ग्रहयुगानाम् ॥ —*Arya.* III. 8

3. युगमन्वन्तरकल्पाः कालपरिच्छेदकाः स्मृतावुक्ताः ।
यस्मान्नरोमके ते स्मृतिबाह्य। रोमकस्तस्मात् ॥ —*BrSpSi.* I- 13

We shall have the same type of conjunction of *grahocca, mandocca śighrocca* and *pāta* after a complete cycle of kalpa as we had in the beginning of creation. This is a natural observable cycle which is recognised in Indian astronomy and in no other foreign system; and hence only the Indian system recognises the time measure of *kalpa.*

Ucca or apex is of two kinds : *mandocca* (apex of the slowest motion) and *śighrocca* (apex of the fastest motion). The *mandocca* is that point of a planet's orbit which is at the remotest distance and where the motion of the planet is slowest. In the case of the Sun and the Moon, it is the apogee; and in the case of other planets, it is the apogee or aphelion, the geocentric longitude of the apogee being equal to the helio-centric longitude of the aphelion. The *śighrocca* of a superior planet (Mars, Jupiter and Saturn) is defined as the mean Sun; that of an inferior planet Mercury or Venus) *śighrocca* is an imaginary body which is supposed to move in such a way that its direction from the earth is always approximately the same as that of the actual planet from the Sun.

The *bhagaṇas* or the revolution numbers of a planet have been given by Āryabhaṭa and Brahmagupta[1] both: they mean the number of revolutions that a planet performs in a certain period, say a kalpa of 4,320,000,000 years.

The *bhagaṇas* of planets are given as follows :

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1. कल्पेऽर्कबुधसिताना भगणाः शून्यानि सप्त रदवेदाः ।
प्राग्व्रजता कुजगुरुशनि शीघ्रोच्चानां स्वकक्षासु ॥
पञ्चान्वराणि गुणा-गुणा पञ्चमुनि स्वशरैर्मिताः शशिनः ।
भौमस्य द्वियमशराष्टपक्षवसुरसनवद्वियमाः ॥
कृतवसुनवाष्टनद-नवषट्त्रि नवागेन्दवोब्जशीघ्रस्य ।
जीवस्य शरेषूदधि षट्पक्षद्वि कृतरसरामाः ॥
सितशीघ्रस्य यमलगोवेदनवाष्टाग्निपक्षयमखनगाः ।
अष्टनवपक्ष मुनि रसशररस मनवोऽर्कपुत्रस्य ॥
खाष्टाब्धयो वसुशरवसुपञ्चखचन्द्रवसुवसुसमुद्राः ।
द्विनवयमा द्वित्रिगुणाः शरेषु वसवस्त्रि पञ्चरसाः ॥
शशिवेदा मन्दानामर्कादीनां विलोमपातानाम् ।
वसुरसरुद्रेन्दुगुणा द्वित्रियमाः सप्तरसपक्षाः ॥
शशियमशरा गुणारसास्त्रिनन्दवसवः समुद्रवसु विषयाः ।
चन्द्रादीनां पश्चात् व्रजतोऽश्विन्यादि भगणस्य ॥ *BrSpSi.* I. 15-21

Planet or a body	Bhagaṇas
Ravi or Sun	4,320,000,000
Budha or Mercury	4,320,000,000
Śukra or Venus	4,320,000,000
Candra or Moon	57,753,300,000
Kuja or Bhauma or Mars	2,296,828,522
Budha-śīghrocca	17,936,998,984
Bṛhaspati or Jupiter	364,226,455
Śukra-śighrocca	7,022,389,492
Śani or Saturn	146,567,298
Arka or Ravi-mandocca	480
Candra mandocca	488,105,858
Kuja or Bhauma mandocca	292
Budha-mandocca	332
Bṛhaspati or Jīva-mandocca	855
Śukra-mandocca	653
Śani-mandocca	41
Candra-pāta	232,311,168
Kuja or Bhauma-pāta	267
Budha-pāta	521
Bṛhaspati or Guru-pāta	63
Śukra-pāta	893
Śani-pāta	584

By pāta is meant the ascending node of a planet's orbit (on the ecliptic).

In a kalpa, the number of *bha-bhramas* (sidereal days) or also known as *bha-parivartas* is 51,040,000,000. If we subtract out from this number the *bhagaṇa* of the Sun, we get what is known as *ku-dinas* or *Sāvana* days or the solar or sacrificial days. (51,040,000,000—4,320,000,000=46,720,000,000 Sāvana days or kudinas).

In a kalpa, the number of Ravi-bhagaṇas also correspond to the number of solar years (Saura-varṣas). i.e., 4,320,000,000; this number multiplied by 12 gives the number (i.e. 51,840,000,000) of solar months.

The difference between the candra-bhagaṇas and the Ravi-bhagaṇas in a kalpa gives the number of lunar months (Cāndra-

māsa) in a kalpa (57,753,300,000-4,320,000,000=53,433,300,000 lunar months).

By subtracting the number of solar months from the number of lunar months in a kalpa, one gets the number of *adhi-māsas* (additional-months) : 53,433,300,000-51,840,000,000= 1,593,300,000 adhimāsas. This multiplied by 30 gives the number of lunar days (śaśi-divasa) in a kalpa ; 53,433,300,000×30= 1,602,999,000,000 lunar days. The difference between the lunar days and *kudinaṣ* in a kalpa gives the number of *avama-dinas* in a kalpa : 1,602,999,000,000-46,720,000,000=1,556,279,000,000.[1]

Brahmagupta calculates out the sṛṣṭi-saṁvatsara or the Creation Era during his year of composition of the Treatise. He says : Six manus have gone in the kalpa ; the seventh manu is now running of which have lapsed 27 caturyugis ; of the twenty eighth caturyugī, the three yugas, kṛta, dvāpara and tretā have gone by and also of the present kaliyuga 3179 years have lapsed. The total period thus lapsed on calculation comes to be 1,972,947,179 years :[2]

Total Period=6 manus+7 manu−sandhis+27 yugas +kṛta+dvāpara+tretā+3179 years of kali.
=(6×71×4,320,000 years) +(7×4×432,000 years)+ (27×4,320,000 years)+(1,728,000+1,296,000+864,000)+ 3,179=1,972,947,179 years.
=1,840,320,000+12,096,000+116,640,000+3,888,000+ 3,179=1,972,947,179 years.

Calculation of Ahargaṇa : The method of calculating ahargaṇa (number of days elapsed since the beginning of kaliyuga)

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1. परिवर्त्ताः स्वचतुष्टयाराब्धि रसगुणयमद्विवसुतिथयः ।
रवि भगणोना भानोः सावनदिवसाः कुदिक्स्तास्ते ॥
रवि भगणारब्ध्वा द्वादशगुणिता भवन्ति रविमासाः ।
भगणान्तरं रवीन्द्वोः शशिमासाः सूर्यमासोनाः ॥
अधिमासाः शशिमासास्त्रिंशद्गुणिता भवन्ति शशिदिवसाः ।
शशिसावनदिवसान्तरनवमानि तिथिः शशांकदिनम् ॥ —*BrSpSi.* I. 22-24

2. कल्पपरार्द्धमनवः षट्कस्य गताश्चतुर्युगत्रिघनाः ।
त्रीणिकृतादीनिकलेर्गोऽगैक गुणाः शकान्तेऽब्दाः ॥
भवनगशशि मुनिकृत नव यमनगनन्देन्दवः शकनृपान्ते ।
सार्धमतीतमनूनां सन्धिभिराद्यन्तरान्तगैः ॥ *BrSpSi. I.* 26-27

has been given by Brahmagupta and Bhāskara I is almost identical. The rule given in the *Brāhmasphuṭasiddhānta*[1] may be compared with the following given by Bhāskara I in the *Laghu-Bhāskarīya*:

> Add 3179 to the (number elapsed) years of the Śaka era, (then) multiply (the resulting sum) by 12, and (then) add the (number of lunar) months (expired) since the commencement of Caitra. Set down (the result thus obtained) at (two) separate places; multiply (one) by (the number of) intercalary months in a *yuga*, which are 1,593,336 in a yuga : and divide (the product) by 5,184×10,000 (i.e.) by 51,840,000). Add the (resulting complete) intercalary months to the result placed at the other place. Then multiply (that sum) by 30 and (to the product) add the (lunar) days (i.e. *tithis*) expired of the current month. Set down (the result thus obtained) in two places; multiply (one) by the (number of) omitted lunar days in a *yuga* i.e. by 25,082,580 and divide by 1,603,000,080. The resulting (complete) omitted lunar days when subtracted from the result put at the other place give the (required) *ahargaṇa*. The remainder obtained on dividing (the *ahargaṇa*) by 7 gives the day beginning with Friday at sunrise (at Laṅkā)[2]

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1. कल्पगतान्द द्वादशघातश्चैत्रादिमास युक्तोऽधः ।
गुणितो युगाधिमासै रविमासाप्ताधिमास युतः ॥
त्रिंशद्गुणास्तिथियुत्य पृथग् युगावमगुणो युगेन्दु दिनैः ।
भक्तः फलावमोनोऽर्कं सावनाहर्गणोऽर्कादिः ॥ — *BrSpSi.* I. 29-30

2. नवाद्रयेकाग्नि संयुक्ताः शकाब्दा द्वादशाहताः ।
चैत्रादिमास संयुक्ताः पृथग् गुण्या युगाधिकैः ॥
ते च षट् त्रिकरामाहित नव भूतेन्दवो युगे ।
भागहारोऽब्धि वस्त्येक शरास्युर् युताहताः ॥
अधिमासाःपृथक्स्थेषु प्रक्षिप्य त्रिंशताहते ।
युक्तवादिनाग्नि यातानि प्रतिराश्य युगावमैः ॥
संगुणव्याःवराष्टेषुद्वयष्टशून्यशरःखिवेभिः ।
छेदः खष्टवियद् व्योमख खाग्नि खरसेन्दवः ॥
लब्धान्यवमराठाग्नि तेषु शुद्धे ष्वहर्गणः ।
वारः सप्तहृते शेषे शुक्रादिर्भौमकारोदयात् । —*LBh. I. 4-8*

Addendum : The mean lunar day (*madhyama tithi*) may, however, differ from a true lunar day (*spaṣṭa tithi*) by one, so that the *ahargaṇa* obtained by the above process may sometimes be in excess or defect by *one*. To test whether the *ahargaṇa* (obtained by the above process) is correct, it is divided by seven and the remainder counted with Friday. If this leads to the day of calculation, the *ahargaṇa* is correct; if it leads to the preceding day, the *ahargaṇa* is in defect; and if that leads to the succeeding day, the *ahargaṇa* is in excess. When the *ahargaṇa* is found to be in defect. it is increased by *one*; when it is found to be in excess, it is diminished by *one*. (K.S. Shukla : *MBh.* p. 4-5)

Example—Calculate the *ahargaṇa* on October 1,1955.

From Indian Calendar we find that October 1,1965 falls on Friday. 7th lunar day (tithi) in the light half of the 7th month Āśvina in the Saka year 1887 (elapsed). Let us proceed as follows :

Adding 3,179 to 1,887. we get 5,066. (1)

Multiplying this by 12 and adding 6 (i. e. the number of lunar months elapsed since the beginning of Caitra) we get 60,798. ... ... (2).

Multiplying this by 1,593,336 and dividing the product by 51,840,000, we get 1,868 as quotient. (The remainder is discarded as unnecessary) (3)

Adding this number (i.e. 1,868) to the previous one (i.e.) 60,798) we get 62,666. (4)

Multiplying this by 30 and adding 6 (i.e. the number of lunar days elapsed since the beginning of the current month) to the product, we get 1,879,986. (5)

Multiplying this by 25,082,580 and dividing the product by 1,603,000,080, we get 29,416 as the quotient. (The remainder is discarded as not necessary). (6)

Subtracting this number (i.e. 29,416) from the previous one (i.e. 1,879,986) we get 1,850,570. (7)

This is the required *ahargaṇa*. Since division by 7 leaves

1 as the remainder. we subtract one from it, and get 1,850,569 as the correct *ahargaṇa* for the day.

An Alternative Rule for Ahargaṇa

Both Bhāskara I and Brahmagupta give an alternative rule for calculating out *ahargaṇa*[1] :

> Multiply the number of (solar months) elapsed since the beginning of kaliyuga by the number of lunar months (in a *yuga*) and divide by the number of solar months (in a *yuga*). Reduce the quotient to days (and add the number of lunar days elapsed since the beginning of the current lunar month); then multiply by the number of civil days (in a *yuga*) and divide by the number of lunar days (in a *yuga*); the quotient denotes the *ahargaṇa*.

Mean Longitude of a Planet

(i) The mean longitude of a planet in *revolutions* is given by the expression : (Brahmagupta[2] and also Bhāskara[3]).

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1. शशांकमासैरभिताडितान् हरेदतीतमासानथ वार्कसम्भवैः ।
दिनीकृतान् भूमिदिनैर्हतान् दिनैर्विभज्य लब्धशशिजैरहर्गणः ॥ *MBh. I. 7*
युगगतशशिमासवधाद्रविमासाप्तं दिनीकृतं सदिनम् ।
भूदिनगुणितं शशिदिनहृतमाप्तमहर्गणः सैकः ॥ —*BrSpSi.* XIII. 18

2. इष्टग्रह भगण गुणादहर्गणात् कल्पसावन द्युहृतात् ।
भगणादि फलं मध्यो लंकायां भास्करौदयिकः ॥ *BrSpSi. I. 31*

3. उदाहरितान् यान् भगणान् क्षमादिनैर्लंभामहे कान् कलियातवासरैः ।
इति प्रलब्धा भगणास्ततः क्रमाद् गृह्णांशलिप्ता विकलाः सतत्परा ॥ —*MBh. I.* 8
पर्यायाहर्गणाभ्यासो ह्रियते भूदिनैस्ततः ।
लभ्यन्ते पर्यायाः शेषाद्राशि भागकलादयः ॥
भास्करै स्त्रिंशता षष्ट्या सङ् गुणय्य पृथक् पृथक् ।
तेनैव भागहारेण लभ्यन्तेऽर्कोदयावधेः ॥ —*LBh. I. 15-17*

(Divide the product of the revolution-number of a planet and the *ahargaṇa* by the (number of) civil days (in a *yuga*); thus are obtained the (number of) revolutions (performed by that planet). From the (successive remainders multiplied respectively by 12,30 and 60 and divided by the same divisor (i.e. the number of civil days in *yuga*) are obtained the signs, degrees and minutes etc. (of the mean longitude of that planet) for (mean) sunrise (at Lanka).

$$\text{Mean longitude} = \frac{\text{revolution number of planet} \times \textit{ahargaṇa}}{\text{civil days in a } \textit{yuga}}$$

Similar expression is given by more recent Indian astronomers also.

(ii) Mean longitude of desired planets in minutes

$$= \frac{\text{(mean longitude of the known planet in revolutions etc. reduced to minutes)} \times \text{(revolution number of the desired planet)}}{\text{revolution number of the known planet.}}$$

This rule is common to Brahmagupta[1] and Bhāskara I[2].

(iii) An alternative rule for deriving the mean longitude of the Moon from that of the Sun and vice versa has been given by Bhāsakara I and Brahmagupta both.

> Multiply the *ahargaṇa* by the number of intercalary months in a yuga and divide (the product) by the number of civil days (in a yuga) : the result is in the terms of revolutions etc. Add that to thirteen times the mean longitude of the Sun. (This is the process) to obtain the mean longitude of the Moon[3].

Mean longitude of the Moon

$$= \frac{\text{(intercalary months in a yuga)} \times \textit{ahargaṇa}}{\text{civil days in a } \textit{yuga}} \text{ revolutions}$$

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1. ज्ञातभगणादिभुक्तं सविकलमिष्टयुग भगणसंगुणितम् ।
ज्ञात युगभगणभक्तं मध्यो भगणादि फलमिष्टः ॥ —*BrSpSi.* XIII. 27

2. निशाकरं वाग्रहमुच्चमेव वा कलीकृतं तत्त्हयातमण्डलैः ।
यथेष्ट नक्षत्रगणैर्हतं हरेत् तदीयनक्षत्र गणैस्ततः कलाः ॥ *MBh.* I. 10

> The (mean) longitude of the Moon, the planet, or the *Ucca* (whichever is known) together with the revolutions performed should be reduced to minutes. The resulting minutes should then be multiplied by the revolution-number of the desired planet and (the product obtained should be) divided by the revolution-number of that (known) planet. The result is (the mean longitude of the desired planet) in minutes.

3. युगाणां युगाधिमासैर्गुणितं युग भूदिनैर्भजेल्लब्धम् ।
भगणादि मध्यमार्कं त्रयोदश गुणाधिकं चन्द्रः ॥ —*BrSpSi.* XIII. 33

युगाधिमासैर्युगणं हतं हरेत् क्षमादिनैर्वा भगणादि लभ्यते ।
त्रयोदशघ्ने सवितर्यथा क्षिपेन्निशीथिनीनां पतिचारसिद्धये ॥ —*MBh.* I. 11

$$+13 \text{ (Sun's mean longitude)}$$

This expression may be rearranged to get the mean longitude of the Sun from the mean longitude of the Moon[1].

Mean longitude of the Sun

$$= \frac{1}{13}\left[\text{mean longitude of the Moon} - \frac{(\text{intercalary months in a yuga}) \times \textit{ahargaṇa}}{\text{civil days in a yuga}} \text{ revolutions}\right]$$

Calculating the Mean Longitudes of the Sun and the Moon without using Ahargaṇa

Bhāskara I follows the method of Āryabhaṭa I and the same method more or less has been adopted by Brahmagupta in calculating the mean longitudes of the Moon and the Sun without the use of *ahargaṇa*. The method may be described thus :

> Reduce the years elapsed since the beginning of kali-yuga to months and add to them elapsed months of the current year. Then multiply the sum by 30 and add the product to the number of lunar days elapsed since the beginning of the current month. Multiply that sum by the number of intercalary months in a *yuga* and divide by the number of solar months in a *yuga* reduced to days; the quotient denotes the number of intercalary months elapsed. The remainder is the *adhimāsaśeṣa*. Multiply the complete intercalary months thus obtained by 30 and to the product add the number of solar days elapsed since the beginning of kaliyuga[2] : then multiply that sum by the number of omitted lunar days in a *yuga* and divide by the number of lunar days in a *yuga* ; the remainder obtained is the *avamaśeṣa* called *āhnika*. Then multiply the *avamaśeṣa*

1. कुमुद्वतीनां सुहृदोऽथवाऽऽगतं विशोध्य शेषस्य लवस्त्रयोदशः ।
स मध्यमार्कौ गणकैर्निरूप्यते गुरुप्रसादात्प्रति बुद्ध बुद्धिभिः ॥ *MBh.* I. 11-12

2. By the number of solar days here is meant the number obtained above by reducing the years elapsed since the beginning of kaliyuga to months, then adding to them the number of months elapsed since the beginning of the current year, then multiplying the sum by 30, and then adding to the product thus obtained the number of lunar days elapsed of the current month.

(also called *āhnika*) by the number of intercalary months in a *yuga* and divide by the number of civil days (in a *yuga*). Add the resulting quotient to the *adhimāsaśeṣa* and divide the sum by the number of lunar months in a *yuga* : this gives degrees etc. (This is the total *adhimāsaśeṣa*). Next multiply again the *avamaśeṣa* called *āhanika* by 60 and divide by the number of civil days in a *yuga* : the result is in minutes, seconds, thirds etc. The number of months elapsed (since the beginning of Caitra) are to be taken as signs and the number of lunar days elapsed of the current month as degrees. The sum of these signs and degrees and the minutes, seconds etc. corresponding to the avamaśeṣa is the *grahatanu*. From thirteen times and from one time that *grahatanu* severally subtract the degrees, minutes etc. corresponding to the total *adhimāsaśeṣa* : the remainders thus obtained are the mean longitudes of the Moon and the Sun respectively[1].

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1. गुणिताद्युगाधिमासैर्युगभूदिवसैर्हृतादवमशेषात् ।
फलयुक्तमधिकमासकशेषं मध्यावतोऽर्केन्दू ॥
अधिमासावमशेषे युगशशि भूदिनहृते पृथग्लब्धेः ।
मासदिनाद्ये स्थाप्ये गतमासदिनानि चैत्रादेः ॥
अवमासशेषलब्ध्या सहितानि पृथक् त्रयोदश गुणानि ।
अधिमास शेषलब्ध्या हीनानि पृथक् रविशशांकौ ॥ —*BrSpSi.* XIII. 20-22
विनाद्युराशेरपि चन्द्रभास्करौ प्रकुर्वतो वा किधिरेव कश्चन ।
समासु मासीकृतविग्रहासु ये ह्यतीतमासा विनियोज्य तान् पुनः ॥
खरामनिघ्नान् दिवसेषु योजयेद् गतेषु मासस्य ततोऽधिमासकैः ।
निहत्य सर्वं विभजेत् सर्वदा युगार्कमासैर्दिवसत्वमागतैः ॥
भवन्ति लब्धास्त्वधिमासकाः पुनस्ततोऽपनीयाशु च भागहारकम् ।
भजेत शेषं शशिमास संख्यया ततोऽशलिप्ता विकलाः स तत्पराः ॥
ततोऽधिमासान् प्रविहृत्य खाग्निभिर्नियोज्य सम्यग्गतवासरैः क्रमात् ।
युगावमघ्नाब्धिशिवासरैर्हरेत् तमत्र शेषं प्रवदन्ति चाह्निकम् ॥
हृत्वाऽधिमासैरवमस्य शेषं छित्वा धराया दिवसैः प्रलब्धम् ।
संयोज्य नित्यं त्वधिमासशेषे कार्यं पुनस्तत् करणैर्यथोक्तम् ॥
युगप्रसिद्धैर्धरणी दिनैर्हरेन्निहत्य षष्टं यावमशेषमान्हिकम् ।
कलाविलिप्ताः क्रमशस्तत्परास्त्वतीतमासा दिवसा गृहांशकाः ॥
त्रयोदशघ्नादपि रूपताडितादिशोधयेच्चत्वधिमासशेषजम् ।
निशाकरार्कौ गणकैः प्रकीर्तितौ भट प्रणीताविति बुद्धिमत्तमैः । —*MBh.* I. 13-19
(Here verse 17 should follow verse 15—*K.S. Shukla*)

K. S. Shukla has provided the following rationale to the rule cited above :

The fraction of the intercalary month (obtained in the rule) $= \frac{\text{adhimāsaśeṣa}}{\text{solar days in a } yuga}$, in mean lunar months.

$= \frac{\text{adhimāsaśeṣa}}{\text{lunar days in a } yuga}$, in mean solar months. (i)

The fraction of the omitted lunar day (obtained in the rule)

$= \frac{\text{avamaśeṣa or } \bar{a}hnika}{\text{lunar days in a } yuga}$, in mean civil days.

$= \frac{\text{avamaśeṣa}}{\text{civil days in a } yuga}$, in mean lunar days.

$= \frac{\text{avmaśeṣa} \times 60}{\text{civil days in a } yuga}$, in mean lunar *ghaṭis*. (ii)

The fraction of the intercalary month corresponding to the above fraction of the omitted lunar day

$= \frac{(\text{intercalary months in a yuga}) \times (\text{avamaśeṣa})}{(\text{lunar days in a } yuga) \times (\text{civil days in a } yuga)}$

in mean solar months. (iii)

Adding (i) and (iii) and multiplying by 30, the total fraction of the intercalary month

$$= \left\{ \frac{\text{adhimāsaśeṣa}}{\text{lunar months in a } yuga} + \frac{(\text{intercalary months in a } yuga) \times (\text{avamaśeṣa})}{(\text{lunar months in a } yuga) \times (\text{civil days in a } yuga} \right\}$$

in mean solar days, (iv)

Suppose that m lunar months and d lunar days have elapsed since the beginning of Caitra. Then, treating them as mean lunar months and mean lunar days, m months and d days denote the time elapsed since the beginning of mean Caitra up to the beginning of the current lunar day (treated as mean lunar day). As (ii) is the interval, in mean lunar *ghaṭis*, between the beginning of the current lunar day and the mean sunrise on that day, therefore

m months $+ d$ days $+$ (ii)

denotes the time in mean lunar months, days, ghaṭīs[1] elapsed

1. 1 hour $= 2\frac{1}{2}$ ghaṭīs; 1 ghaṭī $= 60$ vighaṭīs; 1 vighaṭī $= 60$ pravighaṭīs.

since the beginning of mean Caitra up to the mean sunrise on the current lunar day.

Like wise

m months+d days+(ii)−(iv)

denotes the time in mean solar months, days, ghaṭīs etc. elapsed since the beginning of the current mean solar year up to the mean sunrise on the current lunar day[1].

Let M, D, G, V, and P denote respectively the mean solar months, mean solar days, mean solar ghaṭīs mean solar vighaṭīs and mean solar pravighaṭīs elapsed since the beginning of the current mean solar year up to the mean sunrise on the current lunar day. Then evidently mean longitude of the Sun

=M signs, D degrees, G minutes, V seconds and P thirds.

=(m signs and d degrees)+[minutes, seconds etc. corresponding to (ii)]−[degrees, minutes etc. corresponding to (iv)].

and mean longitude of the Moon

=13 [m signs and d degrees+(minutes, seconds, etc. corresponding to (ii)]−[degrees, minutes etc. corresponding to (iv)]

because

[(1/12) mean longitude of the Moon−mean longitude of the Sun.

=m signs+d degrees+(minutes, seconds etc. corresponding to (ii)]

(This equality is based on the fact that the left hand side denotes the mean lunar date also known as *madhyama tithi*).

A similar rule of these calculations of the mean longitude

1. Because (iv) is equal to fraction of a lunar month between the beginning of Caitra and the beginning of the current mean solar year fraction of an intercalary month corresponding to the tithis elapsed up to the beginning of the current mean lunar day since the beginning of Caitra fraction of an intercalary month corresponding to the avamaśeṣa, i.e., the lunar portion between the beginning of the current lunar date and the following sunrise.

of the Sun and the Moon without basing on *ahargaṇa* has also been given by Brahmagupta in the *Khaṇḍakhādyaka*[1].

Concordance of Working Rules

There has been a good deal of agreement on various rules of astronomical constants from the time of Āryabhaṭa I (499 A. D.) to the Bhāskara II (1150 A. D.) or even later to the days of Munīśvara (1620 A, D). Earliest concepts were formulated during the days of the Vedāṅga-Jyautiṣa and the Siddhāntas of Indian and the western origin, for example of Brahma, Vasiṣṭha, Pitāmaha, Romaka and Puliśa. We in this section are giving some important concordances which we find common in the writings of Brahmagupta and his predecessors, contemporaries and successors as listed below. The list is not exhaustive. Only a a few illustrations have been cited.

1. Ārya.—*Āryabhaṭīya.* Āryabhaṭa I, 499 A. D.
2. BrSpSi. —*Brāhmasphuṭasiddhānta,* Brahmagupta, 628 A. D.
3. K. K.—*Khaṇḍakhādyaka,* Brahmagupta, 628 A. D.
4. KKu.—*Karṇa-kutūhala,* Bhāskara II, 1150 A. D.
5. LBh.—*Laghu-Bhāskarīya,* Bhāskara I, 522 A. D.
   MBh.—*Mahā-Bhāskarīya,* Bhāskara I, 522 A. D.
   MSi.—*Mahā-siddhānta,* Āryabhaṭa II, 950 A. D.
   PSi.—*Pañcasiddhāntikā,* Varāhamihira, 505 A. D.
   ŚiDVṛ.—*Śiṣyadhīvṛddhida,* Lalla, 598 A. D.
   SiSā.—*Siddhāntasārvabhauma,* Munīśvara, 1620 A. D.
   SiŚe—*Siddhāntaśekhara,* Śripati, 1039 A. D.
   SiŚi.—*Siddhānta-Śiromaṇi,* Bhāskara II, 1150 A. D.
   SūSi.—*Sūryasiddhānta,* Modern, 6th or 7th Century.

1. Rule for finding the mean longitudes of the Sun, Mercury and Venus : *BrSpSi.* I. 44.

Also *MBh.* 1. 31. *MSi* I. 26 ; SiŚe II. 42, 43 ; *SiŚi.* I i, (d). 15 ; *SiSā* I. 105 ; *KKu* I. 7.

2. Rule for finding the mean longitude of the Moon's ascending node : *BrSpSi,* XXV. 35.

1. द्विनवरसभक्तमवमावशेषमाप्त दिनादि तत्सहितात् ।
   अधिमासशेषकाच्च त्रिंशद् गुणितादतुखदिग्भिः ॥
   मासदिन प्रथमैक्यं पृथक् त्रयोदशगुणं द्वितीयोनौ ।
   द्वाक्ष्येवं मध्यौ राश्याद्यावर्कचन्द्रौ वा ॥ =*KK.* I. 11–12

Also *MBh.* I. 33 ; *ŚiDVṛ.* I. i. 52 (ii)

3. Rule for finding the mean longitude of the *Śighrocca* of Venus and also giving the additives for the *Śighrocca* of Mercury and Moon : *BrSpSi.* XXV. 36.

Also *MBh.* I. 35 ; *ŚiDVṛ.* I. i, 57 (ii)

4. Rule for finding the mean longitude of the *Śighrocca* of Mercury : *BrSpSi.* XXV. 34.

Also *MBh.* I. 36 ; *ŚiDVṛ.* I. i. 50 (ii)

5. Rule for finding the mean longitude of Saturn : *BrSpSi.* XXV. 35.

Also *ŚiDVṛ.* I. i. 52 (i) ; *MBh.* I. 37.

6. Rule for finding the mean longitude of Mars : *BrSpSi.* XXV. 33.

Also *ŚiDVṛ* I. i. 50 (i) *MBh.* I. 38.

7. Rule for finding the mean longitude of Jupiter : *BrSpSi.* XXX. 35.

Also *MBh.* I. 39 ; *ŚiDVṛ.* I. i. 51 (i).

8. Rule for finding the distance of a place from the prime meridian : *BrSpSi.* I. 36.

Also *MBh.* II. 3-4 ; *LBh.* I. 25-26 ; *ŚiDVṛ.* I. 57-58 (i) ; *SiSa.* I. 143-144.

9. Rule for finding the directions : *BrSpSi.* III. 1.

Also *MBh.* III. 2 ; *SūSi.* III. 1-4 ; *LBh.* III. 1; *SiDVṛ.* I. iii. 1 ; *MSi.* IV. 1-2 ; *SiŚe.* IV 1-3 ; *SiŚi.* I. iii. 8-9.

Alternative rule : *BrSpSi.* III. 2.

Also *MBh.* III. 3 ; *PSi.* XIV. 14-16 ; *ŚiDVṛ.* I. iii. 2 ; *SiŚe.* IV. 4.

10. Rule for finding the latitude and colatitude and the zenith distance and altitude of the Sun : *BrSpSi.* III. 10.

Also *MBh.* III 5; *SūSi* III. 13-14; *LBh.* III 2-3 *ŚiDVṛ* I. iii. 4-5 ; *SiŚe.* IV. 7; *SiŚi* I. iii. 18.

11. Rule for determining the declination, day—radius, earth sine and ascensional difference (for the Sun or a point on the ecliptic) : *BrSpSi.* II. 55.

Also *SūSi* II. 28 ; *LBh.* II. 16; . I. ii *ŚiDVṛ.* 17; *SiŚe.* III.

63-64 ; *SiŚi*. I. ii. 47 (ii) (For RSine of the Declination).

BrSpSi. II. 56 ; also *Ārya*. IV 24 ; *MBh*. III, 6; *LBh*. II. 17 ; *ŚiDVṛ*. I. ii. 18 ; *SiŚe*. III. 66 ; . *SiŚi* I. ii. 48 (For day-radius).

BrSpSi. II. 57-58 ; also *MBh*. III. 7 ; *LBh*. II. 18 ;

SūSi .II. 61; *ŚiDVṛ*. I. ii. 18 ; *SiŚe* III. 67 ; *SiŚi*. I. ii. 49 (i) (For the ascensional difference).

12. For finding the times of rising of the sāyana signs at the equator : *BrSpSi*. III. 15.

Also *MBh*. III. 9 ; *SūSi*. III. 42–43 : *ŚiDVṛ*. I. iii. 8 ; *SiŚe*. IV. 15 ; *SiŚi* I. 11. 51.

13. Rule for finding the ascensional differences of the Sāyana signs Aries. Taurus and Gemini : *KK*. 1. 21.

Also *MBh*. III. 8 ; *PSi*. III. 10 ; *ŚiDVṛ*. 1, XIII. 9 ; *SiŚi*. I. ii. 50-51.

14. Rule for the determination of the meridian zenith distance and meridian altitude of the Sun with the help of the Sun's declination and the latitude of the place ; *BrSpSi*. III, 47.

Also *MBh*. III. 11 ; *LBh*. III. 27 ; *ŚiDVṛ*. I. iii. 16 ;*SiŚe*. IV. 42.

15. Rule for determination of the latitude with the help of the Sun's meridian zenith distance and declination :

BrSpSi. : III. 13.

Also *MBh*. III. 17: *LBh*. III. 34; *SūSi*. III. 15-16; *SiŚe*. IV. 51,

16. Rule for finding out the Rsine of the Sun's altitude for the given time in *ghaṭis* : *BrSpSi*. III. 25-26 Also *Ārya* IV. 28; *MBh*. III. 18-20; *LBh*. III. 7-10: *ŚiDVṛ*. I. iii. 24-25; *SiŚe*. IV. 32,34; *SiŚi*. I. iii. 53-54.

17. Rule for finding out the Sun's altitude : RSin $\alpha =$

$$\frac{M \times \text{day radius}}{R} \times \frac{\text{gnomon}}{\text{hypotenuse of equinoctial midday shadow}}$$

where M=R Sin (given *ghaṭis* ∓ asc. diff.) Rsin (asc. diff.), the upper or lower sign being taken according as the Sun is the northern or southern hemisphere, α is the Sun's altitude. *BrSpSi*. III. 27 (i).

Also *MBh.* III 24; *ŚiDVṛ.* I. iii. 27; *SiŚe* IV. 37

18. Rule for finding the Sun's altitude when the Sun's ascensional difference is greater than the given time : *BrSpSi.* III. 33 :

Also *MBh.* III 25, *LBh.* III. 11; *ŚiDVṛ.* I. iii. 29 *SiŚe.* IV. 41.

19. Rule for finding the Sun's altitude in the night : *BrSpSi.* III. 63.

Also *MBh.* III. 26; *LBh.* III. 11; *SiŚe.* IV. 89.

The Sun's altitude for the night has been called *Pātāla Śaṅku* by Brahmagupta (*BrSpSi.* XV.9)

20. Rule for finding the longitude of the rising point of the ecliptic with the help of (i) the instantaneous sāyana longitude of the Sun and (ii) the civil time measured since sunrise, or with the help of (i) the Sun's sāyana longitude at sunrise and (ii) the sidereal time elapsed since sunrise : *BrSpSi* III. 18-20.

Also *MBh.* III. 30-32: *LBh.* III. 17-19; *SūSi.* III. 46-48;

ŚiDVṛ. I. iii. 11-12: *SiŚe* IV. 18-19 (i) : *SiŚi* I. iii. 2-4.

21. Rule for obtaining the civil time measured since sunrise with the help of (i) the Sun's instantaneous sāyana longitude and (ii) the sāyana longitude of the rising point of the ecliptic, or the sidereal time elapsed since sunrise with the help of (i) the Sun's sāyana longitude at sunrise and (ii) the sāyana longitude of the rising point of the ecliptic: *BrSpSi* III. 21-23.

Also *SūSi.* III.50-51; *MBh.* III. 34-36; *LBh.* III. 20;

ŚiDVṛ. I. iii. 13; *SiŚe* IV. 19 (ii)—22 (i); *SiŚi* I. iii. 5-7. (i).

22. Rule for determining the R Sines of the Sun's prime vertical altitude : *BrSpSi* III. 52.

Also *Ārya.* IV; *MBh.* III.37-38; *LBh.* III. 52.

(An error created by Āryabhaṭa has been criticised by Brahmagupta.)

23. Construction of the locus of the end of the shadow of a gnomon : *BrSpSi.* III. 2-3.

Also *MBh.* III 52; *ŚiDVṛ.* I. iii. 3; *SiŚe,* IV. 5;

24. Rule for finding the Sun's mean anomaly: *BrSpSi.* II. 12 (i).

Also *MBh.* IV. 1; *SūSi.* II. 29 ; *ŚiDVṛ* I. ii. 10; *SiŚe.* iii. 12; *SiŚi.* I. ii. 18-19 (i).

25. Rule for finding the RSine (Reversed sine) of an arc (<90°) : *BrSpSi* II. 10.

Also *SūSi.* II. 31-32 ; *MBh.* IV. 3-14; *LBh.* II, 2 (11)-3 (i); *ŚiDVṛ.* I. ii. 12 ; *SiŚe.* III. 15; *SiŚi.* I, ii. 10 (ii)-11.

(We shall discuss it separately in the light of Brahmagupta formula.)

26. Rule for finding the Sun's equation of the centre: *BrSpSi.*. II. 15 (ii);

Also *MBh.* IV. 4 (ii); . III *SiŚe.* 27

27. Rule for determining the Sun's true longitude: *BrSpSi.* XIV. 17-18.

Also *MBh.* IV. 21-23; *SiŚe.* III. 52.

28. Rule for finding the Sun's *bhujāntara* correction under the eccentric theory : *BrSpSi.* XIV. 19.

Also *MBh.*-IV. 24.

29. Rule for determining the *cara-saṁskāra* or *cara* correction : *KK.* I. 22.

30. Rule for finding the semi-durations of the day and night : *BrSpSi.* II. 60 ; *KK.* I. 23.

Also *SuSi.* II. 62-63 ; *ŚiDVṛ.* I. ii. 20-21 ; *SiŚe.* III. 70 ; *SiŚi.* I. ii. 52.

31. Rule for calculating the *tithi* : *BrSpSi.* II. 62 ; *KK.* I. 25.

Also *SūSi.* II. 66 ; *ŚiDVṛ* I. ii. 22 ; *SiŚe.* III. 71 ; *SiŚi.* I. ii. 66.

32. Rule for calculating the *Karaṇa : KK.* I. 27.

Also. *ŚiDVṛ.* I. ii. 24 ; *SiŚe.* III. 77 ; *SiŚi.* I. ii. 66.

33. Rule for calculating *nakṣatra : BrSpSi.* II. 62 ; *KK.* I. 24.

Also *SūSi.* II. 64 ; *ŚiDVṛ.* I. ii. 23 (i) ; *SiŚe.* III. 75; *SiŚi.* I. ii. 67.

34. Rule pertaining to direct and retrograde motions of a planet : *BrSpSi.* II. 50-51.

Also *MBh.* IV. 56-57 ; *SiŚe.* III. 59 ; *ŚiDVṛ.* I. ii. 42 ;

35. A rule for converting true distances known in minutes into true distances into yojanas : for example : Sun's true distance in yojanas

$$\frac{\text{Sun's mean distance in } yojanas \times \text{Sun's true dist. in minutes}}{\text{Radius}}$$

BrSpSi. XXI. 31 (ii).

Also *MBh.* V. 3 ; *ŚiDVṛ.* I. iv. 5 (i) ; *LBh.* IV. 3 ; *SiŚe.* V. 4 (ii) ; *SiŚi* I.v. 5 (i) ;

36. Rule for finding angular diameters of the Sun and the Mcon : *BrSpSi.* XXI. 34 (ii) ;

Also *MBh.* V. 5 : *ŚiDVṛ.* I. iv. 8 ; *SiŚe.* V. 6 ; *SiŚi.* I. v. 7,

37. Formulae for the true (i. e. angular) diameters of the Sun, the Moon and the shadow in terms of the true daily motions of the Sun and the Moon (Here by shadow is meant the section of the cone of the Earth's shadow at the Moon's distance.) : *BrSpSi.* IV. 6 (i) : *KK.* IV. 2 (i).

Also *MBh.* V. 6-7 ; *ŚiDVṛ.* I. iv. 9 ; *MSi.* V. 5 (ii) ;

SiŚe V. 9 ; *SiŚi.* I. v. 8-9 ;

38. Rule for finding the *spaṣṭa-valana* (resultant *valana*) for the circle drawn with half the sum of the diameters of the eclipsed and eclipsing bodies as radius : *BrSpS.* IV. 18 (i).

Also *MBh.* V. 46 ; *ŚiDVṛ.* I. iv. 26.

39. Method for calculating the phase of the eclipse for the given time : *BrSpSi.* IV. 11-12.

Also *MBh.* V. 62-63 ; *ŚiDVṛ.* I. iv. 19-20 ; *SiŚe.* V. 14.

40. Rule for the determination of the diameter of the shadow i.e., the diameter of the Section of the Earth's shadow where the Moon crosses it : *BrSpSi.* XXIII. 8-9.

Also *MBh.* V. 71-73 ; *Āryc.* IV. 39-40 ; *ŚiDVṛ.* I. iv. 6 (ii)-7.

41. Process of successive approximations in connection with calculations of a lunar eclipse : *BrSpSi.* IV. 8-9.

Also *MBh.* V. 75-76 ; *LBh.* IV. 10-12 ; *ŚiDVṛ.* I. iv. 14-16 *SiŚe.* V. 12-13; *SiŚi.* I, v. 12-13.

42. Rule relating to the visibility-correction known as *akṣa-dṛkkarma* : *BrSpSi*. VI. 4.

Also *MBh*. VI. 1-2 ; *ŚiDVṛ*. I. vii. 3 (ii) ; *MSi*. VII. 4 ; *SiŚe*. IX. 7.

43. Rule relating to the visibility correction known as *ayanadṛkkarma* : *BrSpSi*. VI. 3 ; X. 17.

Slightly modified in *MBh*. VI. 2 (ii)-3 ; *ŚiDVṛ*. I. vii. 2-3 (i) · *SiŚe*. IX. 4-5 ; similar in *MSi*. VII. 2-3 ; more accurate in *SiŚi*. I. viii. 4-5.

44. Rule relating to the visibilily of moon : *BrSpSi*. VI. 6 ; X. 32.

Also *MBh*; VI. 4-5 (i) *PSi*. V. 3 : *ŚiDVṛ*. I. vii. 5 ; *SiŚe*. IX. 8 (i). 13.

45. Rule for calculating the phase of the Moon : *BrSpSi*. VII. 11 (ii)-12.

Also *MBh*. VI. 5 (ii)-7 ; *ŚiDVṛ*. I. ix. 12.

46. Rule for the determination of the Moon's true declination (i.e. the declination of the centre of the Moon's disc) : *BrSpSi*. VII. 5.

Also *MBh*. V1. 8 ; *ŚiDVṛ*. 1. viii. 2 ; *SiŚe*. X. 7. (these are approximate rules ; a more accurate rule occurs in *SiŚi*. I. vii. 3 and 13).

47. Graphical representation of the elevation of the lunar horns in the first quarter of the month at sunset : *BrSpSi*. V1I. 7-10.

Also *MBh*. V1. 13-17 ; *ŚiDV*. I. ix ; *SiŚi*. I. ix.

48. Minimum distances of the planets from the Sun when they are visible : *BrSpSi*. vi 6 ; X. 32.

Also *MBh*. VI. 44 ; *ŚiDVṛ*. I. vii. 5 (i) ; *SiŚe*. IX. 8 (i). 12.

49. Rule relating to the determination of the time and the common longitude of two planets when they are in conjunction in longitude : *BrSpSi*. IX. 5-6.

Also *MBh*. VI. 49-51 ; *ŚiDVṛ*. I. x. 7-9 (i) ; *SiŚe*. XI. 12-12.

50. Rule relating to the distance between two planets

which are in conjunction in longitude : *BrSpSi*. IX. 11.

Also *MBh*. VI. 54 ; *ŚiDVṛ*. I. x. 11 ; *SiŚe*. XI. 10.

51. Rule For finding the *Bhujaphala* and *Koṭiphala* etc. without the use of the RSine-difference table : *BrSpSi*. XIV. 23-24.

Also *MBh*. VII, 17-19 ; *SiŚe*. III. 17.

52. To obtain the Sun's mean true longitude derived from the midday shadow of the gnomon : *BrSpSi*. XIV. 28; III. 61-62.

Also *MBh*. VIII. 5; *SiŚi* I. ii. 45.

53. Rule to find the arc corresponding to a given RSine : *BrSpSi* II. 11.

Also *MBh*. VIII. 6; *SūSi*. II. 33, *ŚiDVṛ*. I. ii. 13, *SiŚe*. III. 16, *SiŚi*. I. ii. 11 (ii)—12 (i).

For the concordance given here we express our indebtedness to the work of K.S. Shukla on the *Mahābhāskarīya*.

Tables of Constants

Some of the Tables of Constants have been given in an earlier chapter. We here give a few more tables which would indicate how far Brahmagupta introduced new concepts in evaluating these constants of greater accuracy and refinement.

TABLE I

Position of Planets for the Beginning of *kaliyuga*

In this Table are given the positions of the planets, including the Moon's apogee and ascending node, for the beginning of *Kaliyuga*. The calculations of Brahmagupta are different from those of the *Sūryasiddhānta* and of Āryabhaṭa I,

Planet	Positions *BrSpSi*.	according *Ārya*.	to *SūSi*
1	2	3	4
	s ° ′ ″	*s* ° ′ ″	*s* ° ′ °
Sun	0 0 0 0	0 0 0 0	0 0 0 0
Moon	0 0 0 0	0 0 0 0	0 0 0 0

1	2	3	4
	s ° ′ ″	s ° ′ ″	s ° ′ ″
Moon's apogee	4 5 29 46	3 0 0 0	3 0 0 0
Moon's asc. node	5 3 12 58	6 0 0 0	6 0 0 0
Mars	11 29 3 50	0 0 0 0	6 0 0 0
Mercury	11 27 24 29	0 0 0 0	0 0 0 0
Jupiter	11 29 27 36	0 0 0 0	0 0 0 0
Venus	11 28 42 14	0 0 0 0	0 0 0 0
Saturn	11 28 46 34	0 0 0 0	0 0 0 0

TABLE II

Diameters of the Sun, the Moon and the Earth in *yojanas* and the distances of the Sun and the Moon from the Earth

	BrSpSi	Bhāskara I	Śrīpati	Bhāskara II	Modern (in miles)
1	2	3	4	5	6
Sun's diameter in *yojanas*	6,522	4,410	6,522	6,522	86,400
Sun's distance in *yojanas* (midnight reck.)	689,358	459,585	684,870	689,377	92,900,000
Ratio		0·009596	0,009,523	009,461	00,093
Moon's diameter in *yojanas*	480	315	480	400	2,160
Moon's distance in *yojanas* (midnight reck.)	51,566	34,377	51,566	51,566	2,389,000
Ratio Earth's diameter in *yojanas*	1,581	0,009,163	0,009,308	009,308	0,009

TABLE III

Sidereal Revolutions of the Apogees of the Planets in a Kalpa

Apogee of	*BrSpSi.*	According to *Sūrya-Siddhānta*	*Āryabhaṭīya*
1	2	3	4
Sun	480	387	not given
Mars	292	204	
Mercury	332	368	
Jupiter	855	900	
Venus	653	535	
Saturn	41	39	

TABLE IV

Sidereal Revolutions of the Nodes of the Planets in a Kalpa

(Not given in the *Āryabhaṭīya*)		*Sūrya-Siddhānta*
Node of	*BrSpSi.*	
1	2	3
Mars	267	214
Mercury	521	488
Jupiter	63	174
Venus	893	903
Saturn	584	662

TABLE V

Peripheries of the Epicycles of the Planets

Planet	*BrSpSi.*		*Sūrya-Siddhānta*		*Āryabhaṭīya*	
	odd quad.	even quad.	odd quad.	even quad.	odd quad.	even quad.
1	2		3		4	
(a) *Manda* epicycles						
Sun	13°40′		13°40′	14°	13°30′	
Moon	31°36′		31°40′	32°	31°30′	
Mars	70°		72°	75°	63°	81°
Mercury	38°		28°	30°	31°30′	22°30′
Jupiter	33°		32°	33°	31°30′	36°
Venus	9°	11°	11°	12°	18°	9°
Saturn	30°		48°	49°	40°30′	58°30′
(b) *Śighra* epicycles						
Mars	243°40′[1]		232°	235°	238°30′	229°30′
Mercury	132°		132°	133°	139°30′	130°30′
Jupiter	68°		72°	70°	72°	67°30′
Venus	263°	258°	260°	262°	265°30′	256°30′
Saturn	35°		40°	39°	40°30′	36°

1. In the middle of quadrants. it is 237°

TABLE VI

Mean Diameters of the Planets

Planet	*BrSpSi.*	*Sūrya-Siddhānta*	*Āryabhaṭiya*	Modern
1	2	3	4	5
Sun	32′31″	32′24″	33′ approx.	32′2′36″
Moon	32′ 1″ approx.	32′	31′30″	31′8″
Mars	4′46″	2′	1′17″	9′36″
Mercury	6′14″	3′	2′8″	6′68″
Jupiter	7′22″	3′30″	3′12″	3′14.72″
Venus	9′	4′	6′24″	16.8″
Saturn	5′24″	2′30″	1′36″	2′49.5″

TABLE VII

Inclination of the Orbits of the Planets to the Ecliptic

Planet	*BrSpSi.*	*Sūrya-Siddhānta*	*Āryabhaṭiya*	Modern (Jan.00,195°)
1	2	3	4	5
Moon	4°30′	4°30′	4°30′	5°8′40″
Mars	1°50′	1°30′	1°30′	1°51′0″
Mercury	2°32′	2°	2°	7°0′14″
Jupiter	1°16′	1°	1°	1°18′21″
Venus	2°16′	2°	2°	2°23′39″
Saturn	2°10′	2°	2°	2°29′25″

TABLE VIII

Longitudes of the Junction Stars according to Different Authorities

Junction-star of	Longitude (polar) according to		Longitude according to		
	BrSpSi.KK. SiŚi.	*SiŚe.SūSi*	*MBh.*	*LBh.*	*SiDVṛ.*
1	2	3	4	5	6
Aśvinī	8°	8°	8°	8°	8°
Bharaṇī	20°	20°	27°	26°30′	20°
Kṛttikā	37°28′	37°30′	36°	36°	36°
Rohiṇī	49°28′	49°30′	49°	50°	49°
Mṛgaśirā	63°	63°	62°	62°	62°
Ārdrā	67°	67°20′	70°	70°	70°
Punarvasū	93°	93°	92°	92°	92°
Puṣya	106°	106°	105°	105°	105°
Āśleṣā	108°	109°	114°	114°	114°
Maghā	129°	129°	128°30′	128°30′	128°
P-Phālgunī	147°	144°	141°	141°	139°20′
U-Phālgunī	155°	155°	154°	154°	154°
Hasta	170°	170°	173°	173°	173°
Citrā	183°	180°	185°	185°	184°20′
Svāti	199°	199°	197°	197°	197°
Viśākhā	212°5′	213°	212°	212°	212°
Anurādhā	224°5′	224°	222°	222°	222°
Jyeṣṭhā	229°5′	229°	228°	228°	228°
Mūla	241°	241°	241°	241°30′	241°
P-Āṣāḍhā	254°	254°	254°	254°30′	254°

1	2	3	4	5	6
U-Āṣāḍhā	260°	260°	267°	266°30′	267°20′
Śravaṇa	278°	280°	285°	284°30′	284°10′
Dhaniṣṭhā	290°	290°	296°	295°30′	296°20′
Śatabhiṣak	320°	320°	307°	307°	313°20′
P-Bhādra.	326°	326°	328°	328°	327°
U-Bhādra	337°	337°	345°	345°	335°20′
Revatī	0°	359°50′	360°	360°	359°

TABLE IX

Celestial Latitudes of the Junction-Stars

Junction star of	Polar latitude given in			Latitude given in		
	BrSpSi.KK. SiSe.	*SūSi.*	*SiŚi*	*MBh.*	*LBh.*	*ŚiDVr.*
1	2	3	4	5	6	7
Aśvinī	10°*N*	10°*N*	10°*N*	10°*N*	10°*N*	10°*N*
Bharaṇī	12°*N*	12°*N*	12°*N*	12°*N*	12°*N*	12°*N*
Kṛttikā	4°31′*N*	5°*N*	4°30′*N*	5°*N*	5°*N*	5°*N*
Rohiṇī	4°33′*S*	5°*S*	4°30′*S*	5°*S*	5°*S*	5°*S*
Mṛgaśirī	10°*S*	10°*S*	10°*S*	10°*S*	10°*S*	10°*S*
Ārdrā	9°*S*	9°*S*	9°*S*	9°*S*	9°*S*	9°*S*
Punarvasū	6°*N*	6°*N*	6°*N*	6°*N*	6°*N*	6°*N*
Puṣya	0	0	0	0	0	0
Āśleṣā	7°*S*	7°*S*	7°*S*	7°*S*	7°*S*	7°*S*
Maghā	0	0	0	0	0	0
P-Phālgunī	12°*N*	12°*N*	12°*N*	12°*N*	12°*N*	12°*N*
U-Phālguni	13°*N*	13°*N*	13°*N*	13°*N*	13°*N*	13°*N*

1	2	3
Hasta	11°*S*	11°*S*
Citrā	1°45′*S*	2°*S*
Svāti	37°*N*	37°*N*
Viśākhā	1°23′*S*	1°30′*S*
Anurādhā	1°44′*S*	3°*S*
Jyeṣṭhā	3°30′*S*	4°*S*
Mūla	8°30′*S*	9°*S*
P-Āṣāḍhā	5°20′*S*	5°30′*S*
U-Āṣāḍhā	5°*S*	5°*S*
Śravaṇa	30°*N*	30°*N*
Dhaniṣṭhā	36°*N*	36°*N*
Śatabhiṣak	18′*S*	30′*S*
P-Bhādra	24°*N*	24°*N*
U-Bhādra	26°*N*	26°*N*
Revatī	0	0

backing to the west) in one day. It is sufficient to postulate the existence of only one Sun, one Moon and twenty-seven nakṣatras to explain the astronomical phenomena.[1]

3. Brahmagupta differs from Āryabhaṭa I in the length of the four *yugas*. Ārvabhaṭa regards all the four *yugas* of equal lengths, i. e. 1,080,000 years; the *caturyuga* being of 4,320,000 years. Brahmagupta regards *Kaliyuga* to be of 432,000 years, *Dvāpara* to be twice of it, *Tretā* to be thrice of it and *Kṛtayuga* of four times of the length of the *Kaliyuga*. Both have the *caturyuga* of the same length.[2]

4. Āryabhaṭa was not clear with respect to the number of civil days (*sāvana dina*) in a *yuga* ; in one of his treatises he gives this number to be 1,577,917,800 and in the other 1,577,917, 500 with a difference of 300 days. though in both the treatises, Āryabhaṭa I regards the number of solar years to be 4,320,000 in a *Caturyuga* or *Mahāyuga*. Why is this difference ? asks Brahmagupta.[3]

5. Āryabhaṭa regards *mandocca* (the apogee) and *pāta* (the ascending node of the orbit on the ecliptic) as constant or stationary; then how could he propound a *sphuṭa-yuga* or the concept of true *yuga* with the concurrence of year, month and day on the Caitra Śukla Pratipadā (the first day of bright half of the month Caitra) at the same time as indicated by Āryabhaṭa in his *Laghvāryabhaṭiya Tantra*.[4]

Āryabhaṭa was not clear in respect to the variance in the pāta. In the Āryāṣṭa-śata (in the Aryabhaṭiva which has 108 Āryā verses), Aryabhaṭa states that the *pāta* of all the planets

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1. भानिचतुस्पञ्चाशद्द्वौ द्वावर्कैन्दवौ जिनोक्तं यत् ।
ध्रुवमस्यस्यावर्त्तौ भवति यतोऽन्हा ततस्तदसत् ॥ *BrSpSi.* XI. 3

2. आर्यभटोयुगपादांस्त्रीन् यातानाह कलिय गादौ यत् ।
तस्यकृतान्तर्यस्मात् स्वयुगाद्यन्तौ न तत् तस्मात् ॥ —*BrSpSi.* XI. 4

3. युगरविभगणाः स्युर्धृतिं यत्प्रोक्तं तत् त्वोर्युगं स्पष्टम् ।
त्रिंशती स्युदयानां तदन्तरं हेतुना केन ॥ —*BrSpSi.* XI. 5

4. युगवर्षादीन् वदताचैत्रसितादेः समं प्रवृत्तान् यत् ।
तदसत् यतः स्फुट्युगं तत् स्वैर्वान्मन्दपातानाम् ॥ —*BrSpSi.* XI. 6

show variance of movement, but in the Daśagītikā (a chapter of ten Āryā verses), he states that with the exception of the *pāta* of the *Moon*, the *pāta's* of all other planets are stationary or constant. Brahmagupta points out this anomaly in the concept of Āryabhaṭa.[1]

6. Brahmagupta points out a self-contradiction in Āryabhaṭa. At one place he says that the Moon covers the Sun during the solar eclipse and similarly the shadow of the Earth covers the Moon during the lunar eclipse (and he does not mention Rāhu) in this connection (*Āryabhaṭiya*, Gola. 37). At the same time it is said that Aryabhaṭa was familar with the movement of "*eight*" planets, and thus postulating the presence of Rāhu; in fact, the *pātas* of planets are responsible for their eclipse, and the eighth planet Rāhu is not present.[2]

7. Āryabhaṭa gives measures to Manus. Yugas and Kalpas different from what have been given in the recognised *Smṛtis*.[3]

8. Āryabhaṭa regards *Guruvāra* or Thursday to be the first day of the *Kalpa*, and not Sunday, which is wrong according to Brahmagupta.[4]

9. Brahmagupta unnecessarily criticises Āryabhaṭa in respect to the order of days. The motion of the planets decreases in the following order : Moon, *Budha* (Mercury), *Śukra* (Venus), Sun, *Kuja* (Mars), *Guru* (Jupiter) and *Śani* (Saturn). Āryabhaṭa in one of his verses states that starting from the Sun and proceeding in the increasing order every fourth is the *dinapati* or the "lord of the day" (*Āryabhaṭiya*, Kāla. 16) ; this gives the order : Sun, Moon. Mars, Mercury, Venus and Saturn and hence the order of days as *Ravivāra* (Sunday); *Candravāra* (Monday,)

1. आर्य्याष्टशते पाता भ्रमन्ति दशगीतिके स्थिराः पाताः ।
मुक्त्वेन्दु पातमपमण्डले भ्रमन्ति स्थिरा नातः ॥ —*BrSpSi.* XI. 8

2. आर्यभटो जानाति ग्रहाष्टगतिं यदुक्तवांस्तदसत् ।
राहुकृतं न ग्रहणं तत्पातो नाष्टमो राहुः ॥ —*BrSpSi.* XI. 9

3. समामनुयुगकल्पाः कल्पादिगतं कृतादियातं च ।
स्मृत्युक्तैरार्यभटो नातो जानाति मध्यगतिम् ॥ —*BrSpSi.* XI. 10

4. ओङ्कारो दिनवारो गुरुरौदयिकोऽस्य भवति कल्पादौ ।
न भवत्यर्को यस्मादोङ्कारो विस्तरस्तस्मात् ॥ —*BrSpSi.* XI. 11

Maṅgalavāra (Tuesday), (*Budhavāra*) (Wednesday), *Guruvāra* (Thursday) Sukravāra (Friday) and Śanivāra (Saturday). Thus Āryabhaṭa gives the same order of days as other authorities and there is no reason why he should be criticised.

Certainly what is Sunday for Laṅkā, may not be Sunday at the same time for Siddhapura; and Āryabhaṭa should have emphasised that the day (Monday, Tuesday etc.) is not constant for all the places.[1]

10. Brahmagupta expresses surprise why Āryabhaṭa, in two of his treatises propounds two different systems of reckoning one from the Sunrise in Laṅkā and the other from the Midnight in Laṅkā.[2] This causes a difference of one-fourth of the daily-motion in the two reckonings of the motion of planets.[3]

11. Brahmagupta criticises Āryabhaṭa on the point of diameter of the Earth. In the *Gitikāpāda*, 5 and 6, Āryabhaṭa states that one yojana=8,000 × puruṣa, and 1 puruṣa=4 hasta, thus 1 yojana=32,000 puruṣas, and the diameter of the Earth is 1050 yojanas. Brahmagupta further says that an error in the diameter of the Earth would cause an error in deśāntara or longitude and thus also in the true *tithi* and consequently in the calculation of eclipses also.[4]

12. Āryabhaṭa has rightly stated that the Earth is in motion and the *Bhagaṇas* are stationary. Brahmagupta's objection is that if the Earth is in motion, birds would not be able to return to their nests, and if the Earth's motion is upside-down,

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1. सूर्य्योदयश्चतुर्थो दिनवारा यदुक्तं तदसदार्यभटः ।
लङ्कोदयो यतोऽर्कं स्यात्समयं प्राह सिद्धपुरे ॥ —*BrSpSi.* XI. 12

2. अधिकैः शतैश्चतुर्भिर्वर्षसहस्रैश्चतुर्दशभिरेकः ।
युगपादैर्दिनवारान्तरं मौदयिकार्धं रात्रिकयोः ॥ —*BrSpSi.* XI. 13.

3. औदयिकादिनभुक्तेस्तुर्यांशो नार्धं रात्रिको भवत्यूनः ।
कतरं स्फुटं न निश्चितमनयोः स्फुटमेकमपि नातः ॥ —*BrSpSi.* XI. 14

4. षोडशगतिर्योजनरिधिं प्रतिभूव्यासं पुलावदता ।
आत्मज्ञानं रव्यापितमनिश्चयस्तनि कृतकन्यात् ॥
मूल्यासस्थानाद् व्यर्थं देशान्तरं तदज्ञानात् ।
स्फुटतिथ्यज्ञानं तिथिनाशाद् ग्रहणयोर्नाशः ॥ —*BrSpSi.* XI. 15-16

then the roof and hills would come down, which is contrary to our observation.[1] Obviously Brahmagupta is not justified in his criticism.

13. Brahmagupta points out to the differences in his calculations and the calculations of Āryabhaṭa in the peripheries of the *manda* and *śighra* epicycles of planets in the odd and even quadrants. (This difference we have shown in Table V, p. 313)[2]

14. Āryabhaṭa I and Bhāskara I have both given a rule for the determination of the *dṛkkṣepajyās* of the Sun and the Moon :

> Take the product of the Sun's or Moon's own *madhyajyā* and *udayajyā*, then divide the product by the radius and then take the square of the quotient. Subtract that from the square of the own *madhyajyā* ; the square-root of that difference is known as the Sun's or Moon's *dṛkkṣepajyā*.[3]

The Sun's *dṛkkṣepajyā* is the Rsine of the zenith distance of that point of the ecliptic which is at the shortest distance from the zenith (this point is called nonagesimal or the central ecliptic point). The Moon's dṛkkṣepajyā is the Rsine of the zenith distance of that point of the Moon's orbit which is at the shortest distance from the zenith. The rule given above is only approximate and has been criticised by Brahmagupta.[4]

15. In the *Āryabhaṭīya*, there is a rule in the Golapāda for finding the Rsine of the *agra* of the true Sun; and also for the

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1. प्राग्गैति कलां भूर्यदि तर्हि कुतो व्रजेत् कमध्वानम् ।
आवर्त्तनमुर्व्याश्चेन्न पतन्ति समुच्छ्रयाः कस्मात् ॥ —*BrSpSi.* XI. 17

2. औदयिको यः परिधिर्विषमेऽन्योमेऽन्यः समे भुजस्य गुणः ।
तदसद्विषमान्तफलं यतो न युग्मादि फलतुल्यम् ॥
विषमेऽन्योऽन्यो युग्मे परिधिर्गुणकः क्रमोत्क्रमज्यानाम् ।
चक्रार्धे फलनाशो न भवति यस्मादसत् तदपि । —*BrSpSi. XI.* 18-19

3. स्वमध्यज्योदयाभ्यासं विष्कम्भार्धाप्तवर्गितम् ॥
मध्यज्यावर्गतोऽपास्य स्वदृक् क्षेपं पदं विदुः ॥ —*Mbh. V.* 19

4. विन्निभलग्ने दृक्क्षेपमण्डलं तदपमण्डलयुतौ ज्या ।
मध्याद्दृक्क्षेपज्या नार्यभटोक्ताऽनया तुल्या ॥
दृक्क्षेपज्याऽतोऽसत् तन्नाशादवनतेर्नाशः ।
अवनतिनाशात् ग्रासस्योनाधिकता रविग्रहणे ॥ —*BrSpSi.* XI. 29-30

Rsine of the Sun's prime vertical altitude.[1] Bhāskara also gives the rule in his *Mahābhāskarīya* :

> Multiply the Rsine of the (Sun's) greatest declination by the Rsine of the Sun's true (*sāyana*) longitude; then divide (the product) by the Rsine of the colatitude, the result is (the Rsine of) the *agra* of the true Sun. When that (*agra*) is less than the latitude and when the Sun is also in the northern hemisphere, multiply (the Rsine of the Sun's *agra*) by (the Rsine of) the colatitude; the result is the Rsine of the Sun's prime vertical altitude.[2]

The condition laid down in the rule that the "Sun's *agra* should be less than the latitude" is incorrect. The error was originally comitted by Āryabhaṭa and Bhāskara followed it. This error was noticed by Brahmagupta[3]. Bhāskara I, however, corrected the error in the *Laghu-Bhāskarīya*. There he gives the correct conditions. It is not the *agra* that should be less than the latitude, it is the Sun's declination (or rather the Rsine of the Sun's northern declination as we have in the *Laghu-Bhāskarīya*), which should be less than the latitude (or rather the Rsine of the latitude). This condition is necessary for the existence of the prime vertical shadow of the gnomon.

It may be pointed out that the commentators of Āryabhaṭa I have interpreted the rule given by Āryabhaṭa I as conveying the correct meaning; they say that Āryabhaṭa also meant declination when he used the term *agra*.

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1. परमापक्रमजीवाभिष्टघ्न्यार्धाहतां ततो विभजेत् ।
   ज्यालम्बकेन लब्धार्काग्रा पूर्वापरे क्षितिजे ॥
   सा विषुवज्ज्योना चेद् विषुवदुदग्लम्बकेन संगुणिता ।
   विषुवज्ज्यया विभक्ता लब्धः पूर्वापरे शंकुः ॥ —*Ārya.* V. 30-31
2. स्फुटरविभुजनिघ्नां यां परां क्रांतिजीवां ।
   हरतु समवलम्बज्या फलापेन भूयः ॥
   स्फुट दिक्स्थ कराग्रा सा यदाऽक्षांशहीना ।
   रविरपि यदिगोले चोत्तरे लम्बकघ्नाम् ॥
   अवलम्ब्या हरेद् भूयः शंकुः स्यात् सममण्डले ।
   तद् वर्गं व्यास कृत्योर्यद् विश्लेषं तत्पदं प्रभा ॥ —*MBh.* III. 37-38
3. उत्तरगोलेऽग्राघ्नं विषुवज्ज्यातो यदुत्क्रमज्यायाम् ।
   सममण्डलभस्तदसत् क्रान्तिज्यायां यतो भवति ॥ —*BrSpSi.* XI. 22

Usually by *agra*, we mean the arc of the celestial horizon lying between the east point and the point where a heavenly body rises; or between the west point and the point where a heavenly body sets. Declination is *krānti*.

16. Brahmagupta has criticised Āryabhaṭa and his group for their expressions for determining *lambana* (i. e. the difference of the parallaxes, in longitude, of the Sun and the Moon), and the rule for determining the *avanati* or *nati* (i.e. the difference in parallaxes, in latitude of the Sun and Moon).[1]

Lambana is obtained with the help of the five Rsines : (i) *madhya-jyā*, (ii) *udaya-jyā*, (iii) *dṛk-kṣepa-jyā*, (iv) *dṛg-jyā*, and (v) *dṛg-gati-jyā*.

(i) The *madhyajyā* is the Rsine of the zenith distance of meridian ecliptic point :

madhyajyā=Rsin ($\phi \pm$ declination of the meridian ecliptic point).[2]

In this expression ϕ is the latitude of the place, and by R sin is meant R $\times$sine, R being the radius of the celestial sphere.

(ii) The *udayajyā* is the Rsine of the arc of the horizon intervening between the equator and the ecliptic, and is given by :

$$udayajyā = \frac{R \sin L \times R \sin \varepsilon}{R \cos \phi}$$

where L is the longitude of the horizon ecliptic point in the east and ε the obliquity of the ecliptic.

(iii) The *dṛkksepajyā* is the R sine of the zenith distance of the central ecliptic point[3], and is given by :

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1. व्यासार्धेन विभक्ता दृग्ज्ञति जीवा चतुर्गुणा लब्धम् ।
लम्बननाड्यः पञ्चदशगुणितया त्रिज्यया भक्ता ॥
दृक्क्षेपज्या भुक्तयन्तराहता लब्धमवनतिर्भवति ।
स्फुटयोजनकर्णाभ्यां भूव्याहतेन च विना स्पष्टे ॥
आर्यभटेनास्मिन् सति लघुनि किमर्थं महत् कृतं कर्म ।
गणिता ज्ञानाज्जाड्यं विजानता यदि ततः सुतराम् ॥ —*BrSpSi*. XI. 23-25

2. The meridian ecliptic point is the point of the ecliptic on the meridian.

3. The central ecliptic point is the central point of the portion of the ecliptic lying above the horizon.

dṛkkṣepajyā

$$=\left[(madhyajyā)^2-\left\{\frac{udayajyā\times madhyajyā}{R}\right\}^2\right]^{\frac{1}{2}}$$

where R is the radius of the celestial sphere.

(iv) The *dṛgjyā* is the Rsine of the zenith distance (of the Sun) and is given by :

$$dṛgjyā=\left[R^2-\left\{\frac{dṛggatijyā\times R\ sin\ (L-\theta)}{R}\right\}\right]^{\frac{1}{2}}$$

where L is the longitude of the horizon ecliptic in the east and θ the longitude of the Sun.

(v) The *dṛggatijyā* is the Rsine of the altitude of the central ecliptic point, and is given by :

$$dṛggatijyā=[R^2-(dṛkkṣepajyā)^2]^{1/2}$$

where R is the radius of the celestial sphere.

In the *Mahābhāskarīya*[1], the expression for the Sun's *dṛggatijyā* is :

$$(\text{Sun's } dṛggatijyā)^2$$
$$=(\text{Sun's } dṛgjyā)^2-(\text{Sun's } dṛkkṣepajyā)^2$$

and similar is the expression for the the Moon's *dṛggatjyā*.

Now *lambana*, which is the difference of the parallaxes, in longitude, of the Sun and the Moon, is given by the expression :

Lambana=Moon's *lambana*−Sun's *lambana*.

Sun's *lambana*

$$=\frac{\text{Sun's } dṛggatijyā\times\text{Earth's semidiameter}}{\text{Sun's true distance in yojanas}}$$

Moon's *lambana*

$$=\frac{\text{Moon's } dṛggatijyā\times\text{Earth's semi-diameter}}{\text{Moon's true distance in yojanas}}$$

These *lambanas* are in terms of minutes of arc etc[2].

1. स्वदग्दच्छे प गुणयोर्वर्गं विश्लेषजे पदे ।
   द्भ्यातिच्ये भवेतां ते भास्करामृत तेजसः ॥ —*MBh. V*, 23
2. स्वदग्भातिच्छमा व्यास भेद संवर्गं संभवम् ।
   पृथग्योजनं कर्णाप्तं लिप्ताद्यं लम्बनं विदुः ॥

(Cont. on Page 225)

Thus *lambana* is given by subtracting the Sun's *lambana* from Moon's *lambana*.

This *lambana* is also expressed in the following way.

$$\text{Lambana} = \left[\frac{\{(dṛgjyā)^2-(dṛkkṣepajyā)^2\}^{1/2}\times 18}{\text{Moon's true distance}} - \frac{\{(dṛgjyā)^2-(dṛkkṣepajyā)^2\}^{1/2}\times 18}{\text{Sun's true distance}} \right]$$

in minutes.

$= \frac{60}{d} \times$ (*lambana* calculated in minutes) is the *lambana* in *ghaṭis*, where *d* denotes the difference between the daily motions of the Sun and the Moon.

Aryabhaṭa I has given his description of the determination of *lambana* and *avanati* in the Golapāda 33, 34 of the *Āryabhaṭīya*[1], and Bhāskara has followed his rules in the *Mahābhāskarīya*[2]. Brahmagupta criticises them in his *Brāhma-sphuṭasiddhānta*[3].

(vii) We shall now take up *nati* or *avanati* (both the terms mean the same). *Nati* is the difference of the parallaxes in *latitude*, of the Sun and the Moon and is given by :

$$nati = \left[\frac{dṛkkṣepajyā\times 18}{\text{Moon's true dist.}} - \frac{dṛkkṣepajyā\times 18}{\text{Sun's true dist.}} \right]$$

minutes.

(*Cont. from Page 324*)

तद्विशेषो हतः षष्ट्या स्फुटमुक्त्यन्तरोद्धृतः ।
घटिकादिस्तिथेः प्राह्णे शुद्धिः च्छेपोऽपरे मतः ॥
दिनार्धकालनिष्पन्नं लम्बनं शोध्यते तिथेः ।
उदगिन्दूदयज्यायां दीयते तत्र दक्षिणे ॥
एवं पुनः पुनः कर्म यावत्तदविशिष्यते ।
तिथिवच्चन्द्रतीक्ष्णांशू सञ्चार्यैव परिग्रहैः ॥ —*MBh.* V. 24-27

1. मध्यज्योदयजीवासंवर्गे व्यासदलहृते यत् स्यात् ।
तन्मध्यज्याकृत्योर्विशेषमूलं स्वदृक् च्छेपः ॥
दृग्दृक्च्छेप कृति विशेषितस्यमूलं स्वदृग्गतिः कुवशम् ।
क्षितिजे स्व दृक् छाया भूव्यासार्धं नभोमध्यात् ॥ —*Arya.* IV. 33-34

2. loc. cit. *MBh.* V. 24-27

3. loc. cit. *BrSpSi.* XI. 23-25

(viii) Moon's true latitude=Moon's latitude×*nati*.

The present Sūryasiddhānta and Brahmagupta both utilise the following expressions for *lambana* and *nati* which give more accurate values :

$$lambana=\frac{\text{Rsin}\,(M-\odot)\times drggati\text{-}yā}{\{\text{Rsin}\,(30^\circ)\}^2}\ \text{ghaṭis}$$

where M and $\odot$ denote the longitudes of the meridian ecliptic and the Sun respectively.

$$nati=\frac{dṛkkṣepajyā\times d}{15\times R}$$

where R is the radius of the celestial sphere and d denotes the difference between the daily motions of the Sun and the Moon[1].

Brahmagupta has raised objections to the Āryabhaṭa system regarding *lambana* (XI. 26-28), *dṛkkṣepa* (XI. 30-31) *ayanadṛk-karma* (XI. 35), elevation of Moon's horns *(śṛṅgonnati)* (XI. 39) and similar several other points. He is so vehemently opposed to Āryabhaṭa that finally he declares :

> "It is beyond my capacity to enumerate all the defects of Āryabhaṭa. Only a few have been given here as illustration. Intelligent people can easily find out others[2].

He also says :

> Āryabhaṭa is neither conversant with the Gaṇita (mathematics), nor Kāla (time calculations) nor Gola (celestial or spherical calculations). It is difficult to enumerate separately the fallacies committed by him in the respective chapters of the *Gaṇitapāda, Kālakri-yāpāda* and *Golapāda*[3].

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1. loc. cit. *BrSpSi.* XI. 23.

2. आर्यभटदूषणानां संख्या वक्तुं न शक्यते यस्मात् ।
तस्मादस्मद्दर्शो बुद्धिमताऽन्यानि योज्यानि ॥ —*BrSpSi.* XI. 44

3. ज्ञानात्येकमपि यतो नार्यभटो गणितकालगोलानाम् ।
न मया प्रोक्तानि ततः पृथक् पृथग् दूषणान्येषाम् ॥ —*BrSpSi.* XI. 43.

Brahmagupta and Śrīṣeṇa

In Varāhamihira's *Pañcasiddhāntikā* we have a critical review of the five *Siddhāntas* or five systems of astronomical study : Puliśa Siddhānta, Romaka Siddhānta, Vasiṣṭha Siddhānta, Sūrya Siddhānta and Brāhma Siddhānta. Colebrooke in his Paper "*On the notion of the Hindu Astronomers concerning the precession of the equinoxes and motions of the planets*", published in the *Asiatic Researches*, vol. xii. p. 209-250, Calcutta. 1816,4 to, reproduced in the *Miscellaneous Essays*. Vol. II. 1872, says the following in regards to the authorship of these schools of astronomy :

> All these books are frequently cited in the astronomical compilations and are occasionally referred to their real or supposed authors. The first is everywhere assigned to Puliśa, whose name it bears. The *Romaka Siddhānta* is ascribed by the scholiast of Brahmagupta and by a commentator of the *Sūrya Siddhānta* to Śrīṣeṇa. The Vāsiṣṭha Siddhānta is by the same authority given to Viṣṇucandra. Both these authors are repeatedly mentioned with censure by Brahmagupta ; and it is acknowledged that they are entitled to no particular deference.
>
> The *Brāhmṇa Siddhānta*, which is the basis of Brahmagupta's work, is not anywhere attributed to a known author ; but referred to in all quotations of it which have fallen under observation, either to the *Viṣṇudharmottara Purāṇa*, of which it is considered as forming a part, or to Brahmā (also called Pitāmaha) who is introduced into it as the speaker in a dialogue with Bhṛgu, or it is acknowledged to be the work of some unknown person. The true author it may be now impracticable to discover, and would be vain to conjecture.
>
> The *Sūrya Siddhānta* (if the same which we now possess) is in the like manner ascribed to no certain author unless in the passage cited by my colleague Mr. Bently (*Asiatic Researches*, vol. vi. p. 572) who says that "in the commentary of the *Bhāsvatī*, it is declared, that

> Varāha was the author of the *Sūrya Siddhānata*", and who adds, that "Satānanda, the author of the *Bhāsvatī* was a pupil of Varāha under whose directions, he himself acknowledges, he wrote that work".
>
> This concluding remark alludes to the following verse of the *Bhāsvatī Karaṇa* : "Next I will propound succinctly, from Mihira's instruction, (the system) equal to the *Sūrya Siddhānta*[1], (*Miscellaneous Essays*, p. 388-90) (The word 'Mihira' has double meaning : it might be an abbreviation of *Varāhamihira*, or it may mean sun or *Sūrya*).

Thus on the authority of Colebrooke, Srīṣeṇa may by regarded as the initiator of the Romaka system. Brahmagupta himself mentions in one of his passages the name of Srīṣeṇa in connection with the Romaka system, and further the conceptions of the Romaka system came down as Vāsiṣṭha system through Viṣṇucandra,[2] Lāṭadeva also derived from Srīṣeṇa the concepts of the mean motions of the Sun, the Moon, the Moon's apogee and her node and the mean motions of Mars. Mercury's. *Śighra*, Jupiter, Venus' *śigrha* and Saturn. I have indicated elsewhere, which is also the view of Sankara Bālakṛṣṇa Dikṣita, that the original Romaka and Pauliśa Siddhāntas were introduced to Indians by Lāṭadeva, and the latter Romaka Siddhānta by Srīṣeṇa (Original Romaka Siddhānta was prevalent before Śaka 427 and this is the one which is mentioned by Varāhamihira who makes no reference to Srīṣena and Viṣṇucandra in the *Paṇcasidhāntikā*, and the latter Romaka Siddhānta was introduced by Śrīṣeṇa as is indicated by Brahmagupta. Thus we have two Vāsīṣṭha Siddhāntas and two Romaka Siddhāntas). My personal view is that Lāṭadeva, Śrīṣeṇa and possibly Viṣṇucandra also, were naturalised Greeks, settled in India and they had adopted themselve to Indian life. They were conversant in Greek and Indian Astronomy both and had contributed substantially to Indian astronomy. Brahmagupta was opposed to any of these f oreign influences dominating Indian

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1. अथ प्रवक्ष्ये मिहिरोपदेशात् तत्सूर्यसिद्धान्त समं समासात् ॥

—*Bhāsvatī Karaṇa*

2. श्रीषेणेन गृहीत्वा रचोच्चयरोमकः कृतः कन्था ।
पद्मनेव गृहीत्वा वासिष्ठो विष्णुचन्द्रेण ॥

—*BrSpSi.* XI. 50

systems, and he very much resented such interferences in pure academic life of this country. He was opposed to Āryabhaṭa for a different reason. Āryabhaṭa was universally regarded as an authority in this country, and the conservativism was so deep that even where it could be shown by direct observation or on valid theoretical grounds, that a particular concept was erroneous or less accurate, people still chose to adhere to it, since they had the backing of Āryabhaṭa's authority. Brahmagupta was against this nonscientific attitude. Needless to say, Brahmagupta was not always fair to Āryabhaṭa in his criticism ; he overdid in enumerating the shortcomings of Āryabhaṭa's system, as if he was personally jealous of his wide popularity.

Brahmagupta's feelings against Lāṭadeva, Śrīṣeṇa, Viṣṇucandra and others would be seen from the following passage in the *Brāhmasphuṭasiddhānta* :

> From the fact that Śrīṣeṇa, Viṣṇucandra, Pradyumna Āryabhaṭa, Lāṭa, and Siṁha contradict one another regarding eclipses and similar topics, their ignorance, is proved daily. The criticisms which I have passed on Āryabhaṭa are, with the requisite modifications, to be applied to the doctrines of each of these teachers as well. I will, however, make some further critical remarks on Śrīṣeṇa and others.
>
> Srīṣeṇa took from Lāṭa the rules concerning the mean motions of the Sun, and the Moon, the Moon's apogee and her node, and the mean motions of Mars, Mercury's Śīghra, Jupiter, Venus's Śīghra, and Saturn ; he took elapsed years and the revolutions of *yuga* (*yuga-yāta-varṣa-bhagaṇa*) from Vasiṣṭha and the *Pādakaraṇa* of Vijayānandi; further took from Āryabhaṭa the rules concerning the apogee, epicycles and nodes, and those referring to the true motions of the planets and thus the Romaka Siddhānta which was (or is) a heap of jewels (as it were) has, by Śrīṣeṇa, been made into a patched rag (as it were)[1].

1. श्रीषेणविष्णुचन्द्र प्रद्युम्नार्यभटलाटसिंहानाम् ।
ग्रहणादिक्रियासंवादात् प्रतिदिवसं द्विगुणमभवत्वम् ॥ [*Cont. on Page 330*]

Brahmagupta very emphatically says about his system that so long as people would be finding concordance between the observed and theoretical results *(dṛgganitaikyam)* in respect of solar and lunar eclipses, his Brāhma Siddhānta would be held in esteem[1].

In other systems, whatever concordance appears to be between the observation and calculation, of eclipses etc., it is, Brahmagupta says, merely accidental or by chance, as the maxim of letters bored by an insect in wood or paper[2].

युक्तयाऽऽर्यभटोक्तानि प्रत्येकं दूषणानि योज्यानि । [*Cont. from Page 329*]
श्रीषेणप्रभृतीनां कानि चिदन्यानि वक्ष्यामि ॥

लाघात् सूर्यशशांकौ मध्याविन्दूच्च चन्द्रपातौ च ।
कुजबुधशीघ्रबृहस्पति सितशीघ्र शनैश्चरान् मध्यान् ॥

युगयातवर्षभगणान् वासिष्ठाद्विजयनन्दि कृतपादात् ।
मन्दोच्च परिधिपातस्पष्टीकरणाद्यमार्यभटात् ॥

श्रीषेणेन गृहीत्वा रव्वोच्चयरोमकः कृतः कन्या ।
एतानेव गृहीत्वा वासिष्ठो विष्णुचन्द्रेण ॥ —*BrSpSi.* XII. 46-50

2. चन्द्ररवि ग्रहणेन्दुच्छायादिषु सर्वदा यतो ब्राह्मे ।
दृग्गणितैक्यं भवति स्फुटसिद्धान्तस्ततो ब्राह्मः ॥ —*BrSpSi.* XI. 61

3. अनयोर्नं कदाचिदपि ग्रहणादिषु भवति दृष्टिगणितैक्यम् ।
यद्भवति तद् घुणाक्षरमतोऽस्फुटाभ्यां किमेताभ्याम् ॥ —*BrSpSi.* XI, 51.

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Reference

Brahmagupta : *Tantraparikṣādhyāya in BrSpSi.*

K.S. Shukla : The *Mahābhāskarīya* and the *Laghubhāskarīya.*

H.T. Colebrooke : *Miscellaneous Essays,* Vol. II., 1872.

G. Thibaut and Sudhākara Dvivedī : The *Pañcasiddhāntikā,* Preface, 1889.

CHAPTER XIII

Brahmagupta and Astronomical Instruments

The Twenty-second Chapter of the *Brāhmasphuṭasiddhānta* is known as the *Yantrādhyāya* or a chapter on instruments. There is a description of seventeen types of time-reckoning instruments (*Kāla-yantra*)[1] :

1. Dhanuryantra—Bow instrument.
2. Turyagolaka yantra—Quadrant (one-fourth sphere)
3. Cakra yantra=wheel or circle.
4. Yaṣṭi yantra—a pole or staff instrument.
5. Śaṅku yantra—Gnomon.
6. Ghaṭikā yantra—a clock or pot instrument.
7. Kapāla yantra—Bowl or potsherd instrument.
8. Karttarī yantra—Scissor or knife ; cutter.
9. Pīṭha yantra=Pedastal or seat instrument.
10. Salila yantra—Water-leveller.
11. Brahma or Śāṇa yantra—For describing circles.
12. Avalamba Sūtra—Threads with plumbs (Plumb lines).
13. Karṇa or chāyā-karṇa—A set of squares for diagonals.
14. Chāyā or śaṅku-chāyā—Sundial.

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1. सप्तदश कालयन्त्राण्यतो धनुस्तुर्यगोलकं चक्रम् ।
यष्टिः शंकुर्घटिका कपालकं कर्त्तरी पीठम् ॥
सलिलं भ्रमीऽवलम्बः कर्णश्छाया दिनार्धमर्कोऽब्जः ।
नतकालज्ञानार्थं तेषां संसाधनान्यष्टौ ॥ —*BrSpSi. XXIII*, 5-6

15. Dinārdha yantra—Midday measure instrument.
16. Arka yantra—Sun-instrument.
17. Akṣa or Palāṅśa yantra—Small degree measure arc instrument.

Salila yantra is used for levelling; since a liquid such as water seeks its own level, it can be utilised to know whether a surface has been levelled or not.[1] Bhrama or *Śāṇa* is used for drawing circles. Avalambaka or plambline is used for adjusting vertical line. *Karṇa* is used in connection with angles and diagonals. From *Salila* (no. 10) to the last (no. 17); these eight are used for adjustments and are basically important.

The *dhanuryantra* is used for *nata* and *unnata* kāla *ghaṭikās*.

On the *paridhi* or the circumference of the *cakra-yantra* are indicated the twelve *rāśis*. ending up to Mīna (XXII. 18). Brahmagupta has described the *yaṣṭi yantra* and shown how it could be used to give time at different parts of the day, and from its shadow *dṛgjyā* and other characteristics can be calculated. This instrument can also be used for ascertaining the solar-lunar differences, and for fixing up the directions. It can be used for determining various heights and altitudes.

The *karttarī yantra* is of the shape of a pair of scissors with two semi-circular blades, fastended to a string at the centre ; at the centre is fixed a pin or a pole which casts shadows.

Setting up of the Gnomon

Here it would be interesting to describe the setting of a gnomon, which K.S. Shukla has given in details while commenting on the *Mahābhāskarīya* (IV. 1):

> After having tested the level of the ground by means of water, draw a neat circle with a pair of compasses (*karkaṭa*) (At the centre of that circle, set up a vertical gnomon). The gnomon should be large, cylindrical, massive, and tested for its perpendicularity by means of four threads with plumbs (*avalmbaka*) tied to them.

1. सलिलेन समं साध्यं भ्रमेण वृत्तमवलम्बकेनोर्ध्वम्।
तिर्यक् कर्णोन्नान्यैः कथितैश्च नव प्रवक्ष्यामि ॥ —*BrSpSi* XXII. 7.

Bhāskara I in his commentary on the *Āryabhaṭiya* tells us that there was a difference of opinion amongst astronomers in his time regarding the shape and size of gnomon (also called style). Some astronomers prescribed a gnomon with its one-third in the bottom of the shape of a prism on a square base (*caturasra*), one-third in the middle of the shape of a cow's tail (*go-pucchākāra*). and one-third at the top of the shape of a spear-head (*Śūlākāra*) and some others prescribed a square prismoidal (*samacaturasra*), gnomon. The followers of Āryabhaṭa I, he informs us, prescribed the use of a broad (*pṛthu*), massive (*guru*), and large (*dirgha* cylindrical gnomon, made of excellent timber and free from any hole, a scar or knot on its body. In the above stanza, Bhāskara I prescribes this last kind of gnomon : the other two kinds he proves in the commentary to be defective and so he rejects them.

For getting the shadow end easily and correctly the cylindrical gnomon was surmounted by a fine cylindrical iron or wooden nail fixed vertically at the centre of the upper end. The nail was taken to be longer than the radius of the gnomon, so that its shadow was always seen on the ground.

Certain writers, Bhāskara I tells us in the commentary, prescribed a gnomon of half a cubit (=12 *aṅgulas*) in length and having twelve divisions. But according to Bhāskara I (although it was the usual custom) there was no such hard and fast rule. The gnomon could be of any length and any number of divisions. The gnomon should, however, be large enough, so that the rings of graduation on the gnomon may be clearly seen on the shadow. A broad and massive gnomon was preferred because it was unaffected by the wind.

Brahmagupta describes gnomon which at the bottom is two *aṅgulas* wide, pointed as a needle., 12 *aṅgulas* in length, and full of holes from the basic circular part to the pointed extremity. (*BrSpSi*. XXII. 39).

As regards testing the level of the ground, Bhāskara I observes :

> When there is no wind, place a jar (full) of water upon a tripod on the ground which has been made plane by means of eye or thread, and bore a (fine) hole (at the bottom of the jar) so that the water may

> have continuous flow. Where the water falling on the ground spreads in a circle, there the ground is in perfect level ; where the water accumulates after departing from the circle of water, it is low ; and where the water does not reach, there it is high. (Bhāskara's Commentary on the *Āryabhaṭīya*, II. 13).

After the ground was levelled, a prominently distinct circle was drawn on the ground as stated in the text (*MBh.* III. 1). In the time of Sankaranārāyaṇa (869 A.D.), there it seems that all lines were drawn on the ground with sandal paste (candanākṣodārdra). The above circle having been thus drawn and coated with sandal paste, another small concentric circle was drawn with the radius of the gnomon. The gnomon was then placed vertically with the periphery of its base in coincidence with that circle. The gnomon was thus set up exactly in the middle of the bigger circle. The verticality of the gnomon was tested by means of four plumb lines hung on the four sides of the gnomon

Gnomon Used for Finding the Directions

The rule in this connection has been described by Brahmagupta in *BrSpSi.* III. 1. The same rule in other words has been described by Bhāskara I in *MBh.* III. 2. In the *Vāsanā Bhāṣya*, Pṛthūdaka Svāmī describes the details of determining the directions. The level of the ground is ascertained by means of water and a gnomon of 12 aṅgulas is set up. Find out two points where the shadow of the gnomon enters into and passes out of the circle. Bhāskara prescribes drawing out a fish figure with these points. The thread line which goes through the mouth and tail of the fish figure indicates the north and south directions with respect to the gnomon. Brahmagupta says that if the Sun is on the eastern side, then where the shadow-point enters circle (in the forenoon) that point would be the west, and the point where it emerges out (in the afternoon) is the east.

As the Sun moves along the ecliptic, its declination changes. By the the time the shadow moves between the forenoon and afternoon points as given above, the Sun traverses some distance of the ecliptic and, so, theoretically speaking, its

declination gets changed. It follows, therefore, that the East-West line in the above determination is not the true position of the actual East-West line. *Brahmagupta* (628 A. D.) *was the first Hindu astronomer who prescribed the determination of the East-West line with proper allowance for the change in the Sun's declination.* (Shukla) The details of the method intended by him have been supplied by his commentator Pṛthūdaka Svāmī (860 A. D.).

Bhāskara and Brahmagupta both give another method of determining directions : (*BrSpSi.* III. 2 ; *MBh.* III. 3) : With the three points (at the ends of the three shadows of the gnomon) corresponding to (any three) different times (in the day), draw two fish-figures (each with two of the three points) in accordance with the usual method. From the point of intersection of the lines passing through the mouth and tail (of the two fish-figures), determine the north and south directions. (*MBh.* III. 3).

Brahmagupta in his rule is more precise :

> The point where the lines passing through the two fish-figures, which are drawn by means of three shadow ends (of the gnomon), intersect each other is for places in the northern hemisphere, the south direction, (if the midday shadow falls to the north of the foot of the gnomon). If the midday shadow falls towards the south of the foot of the gnomon, it is the north direction. (*BrSpSi.* III. 2).

This rule is obviously based on the assumption that the the locus of the end of the shadow of the gnomon is a circle. In fact the locus for places whose latitude is less than (90°—the obliquity of the ecliptic), this locus is a hyperbola.

Brahmagupta has made numerous uses of gnomon. He and Bhāskara, for example, both give the rules for finding the latitude and colatiude and the zenith distance and altitude of the Sun by finding out the length of the shadow and the length of the gnomon (*BrSpSi.* III. 10 ; *MBh.* III. 5) ; also rule for the determination of the latitude with the help of the Sun's meridian zenith distance and declination (*BrSpSi.* III. 13 ; *MBh.*

III. 17) ; also rule for finding the Sun's altitude (*BrSpSi*, III. 27 ; *MBh*. III. 24) (The Sun's altitude for the night has been called by Brahmagupta as *pātāla-śaṅku*, *BrSpSi*. XV. 9).

Golayantra or Armillary Sphere

The first mention of the Golayantra or the armillary sphere is in the *Aryabhaṭiya* (Golapāda. 22)[1] which was a uniformly round circle made of wood or of bamboo and which was of uniform weight or density alround. It was levelled with mercury, oil or water. A śalākā or pin (or rod) was fixed in it in the south-north direction. Its description from the commentary *Bhaṭadīpikā* of Paramādīśvara is given here :

> A sphere of wood. uniformly round on all sides and with uniform density, and also light is made to revolve round an iron axis fixed north-south without friction (oil may be introduced to avoid friction). To the backside of the sphere, is fixed a *nālaka* full of water which has the length equal to the circumference of the sphere: and which has a hole at the bottom.
>
> Now a thread, connected to the hook of the wooden ball (on the top side) passing over another small ball (in the same axis of the wooden ball) is attached to the mercury lobe by its other end. The mercury lobe is placed on the level of water and water is allowed to flow through the bottom hole and with water mercury lobe also goes down. The time in which the above hook of ball comes to bottom (180°) is noted. The experiment is repeated with oil. The use of this mechanism is to revolve the ball by water or oil[1]

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1. काष्ठमयं समवृत्तं समन्ततस्सम गुरु लघुं गोलम् ।
पारतं तैलजलैस्तं भ्रमयेत्स्वधिया च कालसमम् ॥ *Arya. IV. 22*

काष्ठमयं वंशादि काष्ठेन निर्मितं समवृत्तं सर्वतोवृत्तं समन्ततस्सम गुरुं सर्वावयवेषु समं गुरुत्वं यथा भवति तथा कृतं । लघुभगुरं एवं भूतं गोलं कृत्वा पारतादिभिस्तं स्वधिया च कालसमं भ्रमयेत् । अयमर्थः । भूमिष्ठ दक्षिणोत्तरस्तम्मयोरुपरि गोलप्रोतायशलाकाया अग्रे स्थापयेत् । गोलदक्षिणोत्तराच्छिद्रे च तैलेन सिञ्चेत् यथा निस्तंङ्को गोलो भ्रमति । गोलस्यापरतो गोलपरिधिसंमित दैर्घ्यं साधश्छिद्रं जलपूर्णं नलकं निदध्यात् ततो गोलस्यापरस्वस्तिक कीलकं निधाय तस्मिन्सूत्रस्यैक मग्रं बद्ध्वादो विलुम्बन्मस्थलपृष्ठेन प्राङ्मुखं नीत्वा तदग्रबद्धं पारतपूर्णमलाबु जलपूर्णे नलके निदध्यात् ततो नलकस्याध

[*Cont. on Page 337*]

In the Arabic epitome of the Almagest entitled *Tahriru'l mejesti*. the armillary sphere, *Za ul halk*, is thus described :

> Two equal circles are placed at right angles, the one representing the ecliptic, the other the solstitial colure. Two pins pass throught the poles of the ecliptic and two other pins are placed on the poles of the equator. On the first two pins are suspended a couple of circles moving, the one within, the other without, the first mentioned circles, and representing two secondaries of the ecliptic. On the two other pins a circle is placed, which encompasses the whole instrument, and within which the different circles turn; it represents the meridian. Within the inner secondary of the ecliptic, a circle is fitted to it, in the same plane and turning in it. This is adapted to measure latitudes. To this internal circle, two apertures or sights, opposite to each other, and without its plane are adapted like the sights of an instrument for altitudes. The armillary sphere is complete when consisting of these six circles. The ecliptic and secondaries are to be graduated as minutely as may be practicable. It is best to place both secondaries, as by some directed, within the ecliptic (instead of placing one of them without it), that the complete revolution of the outer secondary may not be obstructed by the pins at the poles of the equator. The meridian likewise should be doubled, or made to consist of two circles; the external one graduated and the internal one moving within it. Thus the pole may be adjusted at its proper elevation above the horizon of any place. The instrument so constructed consists of seven circles.
>
> It is remarked that when the circle representing the meridian, is placed in the plane of the true meridian, so

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स्छिद्रं विकृतं कुर्यात् तेन जलं निस्स्रवति। नलकस्य जलमधो गच्छति। तद्वशाच्च तत्रस्थमलावु पारतपूर्यां गुरुत्वज्जलेन सहाधो गच्छद् गोलं प्रत्यतङ्मुखमाकर्षति। एवं त्रिंशद् घटिकाभिरर्वसंमितं यथा जलं भवति गोलस्य चार्धं भ्रमति तथा स्वबुद्धया जलनिस्रावो योज्यः। इति।

अमृतस्रावयोगेन कालभ्रमण साधनम्।
गुणबीजसमाकृष्टं गोलयंत्रं प्रकल्पयेत्॥ *SūSi. XIII, 16-17*

that it cuts the plane of the horizon at right angles, and one of the poles of the equator is elevated above the horizon conformably with the latitude of the place; then the motions of all the circles round the poles represent the motions of the universe.

After rectifying the meridian, if it be wished to observe the Sun and Moon together, the outer secondary of the ecliptic must be made to intersect the ecliptic at the Sun's place for that time: and the solstitial colure must be moved until the place of intersection be opposite to the Sun. Both circles are thus adjusted to their true places; or if any object but the Sun, be observed, the colure is turned until the object be seen in its proper place, on that secondary referred to the ecliptic: the circle representing the ecliptic being at the same time in the plane of the true ecliptic and in its proper situation. Afterwards, the inner secondary is turned towards the Moon (or to any star intended to be observed), and the smaller circle within it, bearing the two sights is turned, until the Moon, (or to any star intended to be observed), and the smaller circle within it, bearing the two sights, is turned, until the Moon be seen in the line of the apertures. The intersection of the secondary circle and ecliptic is the place of the Moon in longitude: and the arc of the secondary, between the aperture and the ecliptic, is the latitude of the Moon on either side (North or South). (From Colebrooke's *Miscellaneous Essays*).

The same instrument, as described by Montucla from the text of Ptolemy (1. 3. c. 2) consists of six circles: first a large circle representing the meridian; next four circles united together, representing the equator, ecliptic and two colures, and turning within the first circle on the poles of the equator, lastly a circle turning on the poles of the ecliptic, furnished with sights and nearly touching, on its concave side, the circumference of the ecliptic.

The armillary sphere described by the Arabian epitomiser, differs, therefore, from Ptolemy's in omitting the equator and

equinoctial colure, and adding an inner secondary of the ecliptic, which as well as the meridian, is doubled.

According to Lalande, the astrolobe of Ptolemy, from which Tycho Brahe derived his equatorial armillary, consisted only of four circles: two placed at right angles to represent the ecliptic and solstitial colure; a third turning on the poles of the ecliptic and serving to mark longitudes; and a fourth, within the other three, furnished with sights to observe celestial objects and measure their latitudes and longitudes.

Whether the ancient Greeks had any more complicated instrument formed on similar principles, and applicable to astronomical observations, is perhaps uncertain. We have no detailed description of the instrument which Archimedes is said to have devised to represent the phenomena and motions of the heavenly bodies; nor any sufficient hint of its construction; nor does Cicero's account of the sphere exhibited by Posidonius suggest a distinct notion of its structure.

Among the Arabs, no addition is at present known to have been made to the armillary sphere; between the period when the Almagest was translated and the time of Alhazen, who wrote a treatise of optics, in which a more complicated instrument than that of Ptolemy, is described; Alhazen's armillary sphere is stated to have been the prototype of Tycho Brahe's; but neither the original treatise, nor the Latin translation of it, are procurable and one is therefore unable to ascertain whether the sphere, mentioned by the Arabian author, resembled that described by Indian astronomers. At all events, says Colebrooke, he is more modern than the oldest of the Hindu writers.

Here we give the literal translation of the passage on armillary sphere or *Golayantra* occurring in the *Sūrya-Siddhānta* :

> Let the astronomer frame the surprising structure of the terrestitial and celestial spheres.
>
> Having caused a wooden globe to be made (of such size) as he pleases; to represent the Earth : with a staff for the axis passing through the centre, and exceeding the globe at both ends; let him place the supporting hooks, as also the equinoctial circle.
>
> Three circles must be prepared, (divided for signs and degrees), the radius of which must agree with the

respective diurnal circles, in proportion to the equinoctial: the three circles should be placed for the Ram (*Meṣa*) and following signs. respectively, at the proper declination in degrees N. or S. ; the same answer contrariwise for the Crab (*Karkaṭa*) and other signs. In like manner three circles are placed in the southern hemisphere, for the Balance (*Tulā*) and the rest, and contrariwise for Capricorn (*Mṛgā*) and remaining signs. Circles are similarly placed on both hoops for the asterisms in both hemispheres, as also for *Abhijit* and for the *Seven Ṛṣis*, *Agastya*, *Brahmahṛdaya* and other stars.

In the middle of all these circles is placed the equinoctial. At the intersection of that and supporting hoops, the distant from each other half the signs, the two equinoxes should be determined; and the two solstices, at the degrees of obliquity from the equinoctial; and the the places of the Ram (*Meṣa*) and the rest, in the order of the signs, should be adjusted by the strings of the curve. Another circle thus passing from equinox to equinox, is named the ecliptic; and by this path, the Sun illuminating worlds, for ever travels. The Moon and other planets are seen deviating from their nodes in the ecliptic to the extent of their respective greatest latitudes (within the zodiac).[1]

1. भूलोकस्य रचनां कुर्यादाश्चर्यकारिणीम् ।
अभीष्टं पृथिवीगोलं कारयित्वा तु दारवम् ॥
दण्डं तन्मध्यगं मेरोरुभयत्र विनिर्गतम् ।
आधारकक्ष्याद्वितयं कक्ष्यां वैषुवतीं तथा ॥
भगणांशुलैः कार्या दलितास्तिस्र एव ताः ।
स्वाहोरात्रार्धकर्णैश्च तत्प्रमाणानुपाततः ॥
क्रान्तिविक्षेपभागैश्च दलिता दक्षिणोत्तरा ।
स्वैस्स्वरपक्रमैः कार्या मेषादं नामपक्रमात् ॥
कक्ष्याः प्रकल्पयेत्ताश्च कर्क्यादीनां विपर्ययात् ।
तद्वत्तिस्रस्तुलादीनां मृगादीनां विलोमतः ॥
याम्यगोलाश्रिताः कुर्यात् कक्ष्याधारद्वयोपरि ।
याम्योदग्भाग संस्थानां भानामभिजितस्तथा ॥
सप्तर्षीणामगस्त्यस्य ब्रह्महृदादीनां प्रकल्पयेत् ।
मध्ये वैषुवती कक्ष्या सर्वासामेव संस्थिता ।

[*Cont. on page 341*]

The author of the *Sūrya-Siddhānta* then proceeds to notice the relation of the great circles before mentioned to the horizon, and observes that whatever place be assumed for the apex of the sphere, the middle of the heaven for that place is its horizon. He concludes by showing, that the instrument may be made to revolve with regularity, by means of a current of water; and hints, that the appearance of spontaneous motion may be given, by a concealed mechanism, for which quicksilver is to be employed. There is a hint of secrecy also in one of the lines, and it has, therefore, been stated that the construction and the mechanism of working should be learnt under the guidance of a teacher.

How to Observe Places of Stars

Details are not available in this connection. The *Sūrya-Siddhānta* only hints "that the astronomer should frame a sphere and examine the apparent longitude and latitude (*sphuṭavikṣepa and sphuṭadhruvaka*)". The commentators, however, describe the manner of making the observation. They direct a spherical instrument (*Golayantra*) to be constructed as described above. This instrument is very much similar to the armillary sphere. An additional circle graduated for degrees and minutes, is directed to be suspended on the pins of the axis as pivots. It is named as *Vedhavalaya* or intersecting circle, and appears to be a circle of declination. After noticing this addition to the instrument, the instructions proceed to the rectifying of the *Golayantra* or armillary sphere which is said to be placed, so that the axis shall point to the pole: and the horizon be true by a water level.

The instrument being thus placed, the observer is instructed to look at the star *Revatī* through a sight fitted to an orifice at the centre of the sphere; and having found the star, to adjust by it the end of the sign Pisces on the ecliptic. The observer is then to look through the sight, at the *yoga* star of *Aśvinī*, or at

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तदाधारयुतेः भार्धमयने विषुवद्वये ।
अयनादयने चैव कक्ष्या तिर्यक्नथाऽपरा ॥
क्रान्तिसंज्ञा तया सूर्यः सदा सदापर्येति भासयन् ।
चन्द्राद्याश्च स्वरकैः पादैपमण्डलमाश्रितैः ॥
ततोऽपकृष्टादृश्यन्ते विक्षेपाग्रेष्वपक्रमात् । *SuSi. XIII. 3-12*

some other proposed object; and to bring the moveable circle of declination over it. The distance in degrees, from the intersection of this circle and ecliptic, to the end of *Mīna* or Pisces, is its longitude (*dhruvaka*) in degrees; and the number of degrees on the moveable circle of declination, from the same intersection to the place of the star, is its latitude (*vikṣepa*) North or South.

The commentators have rightly remarked that the "latitude so found is *sphuṭa* or apparent, being the place intercepted between the star and the ecliptic, on a circle passing through the poles; but the true latitude (*asphuṭa*) is found on a circle hung upon the poles of the celestial sphere as directed in another place". (From Colebrooke's Paper on the Indian and Arabian Divisions of the Zodiac. *Miscellaneous Essays*, Vol. II, 324-326).

For the details of the *Golayantra*, readers are requested to refer to the description in the *Siddhānta-Śiromaṇi* of Bhāskara II.

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Reference

Brahmagupta : yantrādhyāya in —*BrSpSi.*
K.S. Shukla : The *Mahābhāskarīya.*
Bhāskara I : Commentary on the *Āryabhāṭīya.*
Paramādīśvara : *Bhaṭadīpikā*, a commentary on the *Āryabhāṭīya*,
H.T. Colebrooke : *Miscellaneous Essays*, Vol. II., 1872.

॥ श्रीगणेशायनमः ॥

अथ ब्राह्मस्फुट सिद्धान्तस्य

पूर्वादशाध्याय्यां मध्यमाधिकारः

जयति प्रणतसुरासुरकिरीटरत्नप्रभाच्छुरितपादः ।
कर्ता जगदुत्पत्तिस्थितिविलयानां महादेवः ॥१॥
ब्रह्मणोक्तं ग्रहगणितं महता कालेन यत् खिलीभूतम् ।
अभिधीयते स्फुटं तज्जिष्णुसुतब्रह्मगुप्तेन ॥ २ ॥
ध्रुवताराप्रतिबद्धं ज्योतिश्चक्रं प्रतिक्षणगमादौ ।
पौष्णाश्विन्यन्तस्थैः[१] सह ग्रहैर्ब्रह्मणा सृष्टम् ॥ ३ ॥
चैत्रसितादेरुदया[४]द्भानोर्दिनमासवर्ष युगकल्पाः[१] ।
सृष्ट्यादौ लंकायां समं[३] प्रवृत्ता दिनेऽर्कस्य[२] ॥ ४ ॥
प्राणैर्विनाडिकार्क्षी षड्भिर्घटिका विनाडिका षष्ट्या ।
घटिका षष्ट्या दिवसो दिवसानां त्रिंशता मासाः ॥ ५ ॥

(ग) १. श्री नमः ॥ परमात्मने । श्री रामाय सीतालक्ष्मणसहिताय: ॥ श्री गुरुभ्यो नमः ॥ for (श्री.....मध्यमाधिकारः)

(च) १. "ॐ नमः परमात्मने श्रीरामाय सीतालक्ष्मणसहिताय श्री गुरुभ्यो नमः । for (श्री.....मध्यमाधिकारः)"

३. (ग) १. पौष्णाश्विनांतरस्थैः for (पौष्णाश्विन्यन्तस्थैः)

(च) १. पौष्णाश्विन्यंतरास्थैः for (पौष्णाश्विन्यन्तस्थैः)

४. (ग) १. द्दिन (च) for (दिन)

२. ऽर्क्कस्य for (र्कस्य) (ख) दिनेर्कस्य for (दिनेऽर्कस्य)

(क) श्लोक उपलब्ध नहीं

(ख) ३. समप्रवृत्ता for (समं प्रवृत्ता)

(च) ४. चित्र for (चैत्र) २. दिनेऽर्क्कस्य for (दिनेऽर्कस्य)

५. (क) श्लोक उपलब्ध नहीं ।

मासा[१] द्वादशवर्षं विकलालिप्तांशराशिभगणांतः[२]
क्षेत्रविभागस्तुल्यः कालेन विनाडिकाद्येन[३] ॥ ६ ॥
स्वचतुष्टयरदवेदा[१] रविवर्षाणां चतुर्युगं भवति । ४३२०००[२]
सन्ध्या सन्ध्यांशै[३] सह चत्वारि पृथक्कृतादीनि ॥७॥
युगदशभागो[१] गुणितः कृतं चतुर्भिस्त्रिभिर्[२]गुणस्त्रेता[३] ।
द्विगुणो[४] द्वापरमेकेन संगुणः कलियुगं भवति ॥ ८ ॥
१७२८०००।१२९६०००।८६४०००।४३२०००[५]
युगपादानार्यभटश्चत्वारि[१] समानि[२] कृतयुगादीनि[५] । १०००००
यदभिहितवान्[४] न तेषां स्मृत्युक्तसमानमेकमपि[५] ॥ ९ ॥

६. (ख) १. मास for (मासा)
२. भगणांत for (भगणांतः)
३. द्येवम् for (द्येन)

(च) २. भगणांताः for (भगणांतः)

७. (ग) १. खचतुष्टय for (स्वचतुष्टय) (क) (ख) (च)

(क) २, संख्या लुप्त है। टीका में उपलब्ध है।

(ख) ३. संध्यांशैः for (सन्ध्यांशै)

८. (ग) १+१०+(च)
२+४+(च)
३+३+(क) गुणोत्रेता for (गुणस्त्रेता) (च) +३+
४+२+(च)
५ ८६४००० for (८६४०००) (क) टीका में उपलब्ध है। (च) ८६४०००

(क) संख्याएं लुप्त हैं, परन्तु टीका में दी हुई हैं। (ख)

९. (घ) १. नार्यः for (नार्य) (क) नार्यभट्ट for (नार्यभट)
(ग) २. समानिष्ट (ख) सभाति (च) समानिष्ट for (समानि)
(क) ३. संख्या लुप्त है।
४. हतवान्न for (हितवान् न)
५. समानामेकमपि (च) for (समानमेकमपि)

मनुरेक सप्ततियुगः कल्पो मनवश्चतुर्दश मनूनाम् ।
आद्यंतरांतसंधिषु कृतकालोऽस्माद्युगसहस्रम् ॥ १० ॥

४३२०००००००

आद्यंतरातसंधिषु कल्पमनूनां कृताब्दसमकालम् ।
नेच्छंति ये षडूनं तेषां कल्पो युगसहस्रम् ॥ ११ ॥

४२९४०८०००० ७

मनुसंधियुगमिच्छत्यार्यभटस्तन्मनुर्यतस्खयुगः ।
कल्पश्चतुर्युगानां सहस्रमष्टाधिकं तस्य ॥१२॥

४३५४५६०००० ४

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१०. (घ) २. ४३२०००००० for (४३२००००००००) (क) संख्या लुप्त है। टीका में अंकित है।
(ग) २+७१+(च)
३ कल्पो नवचतुर्दश १४ (च) for (कल्पोमनवश्चतुर्दश)
४ +१५+(ख) कृतकल्पो for (कृतकालो) (च)+१५+
(ख) ५. अद्यंतगतसंधियु for (आद्यंतरांतसंधिषु)
६. स्मुद्युगसहस्रम् for (स्माद्युगसहस्रम्)

११. (घ) १. डूनं for (षडूनं) (च) षडूनां for (षडूनं)
२. ४२६४०८०००० for (४२९४०८०००००) (क) संख्या लुप्त है
(क) ३. युगसहश्र for (युगसहस्रम्)
(ख) ४. अद्यंतरांत for (आद्यंरांत)
५. कल्पमनुना for (कल्पमनूनां)
६. कृताब्दाः for (कृताब्द)

१२. (घ) १. (अस्पष्ट)
२. चतुर्यगाणां for (चतुर्युगानां)
(ग) ३. सहश्रमष्टाधिकं for (सहस्रमष्टाधिकं) (ख)
४. संख्या लुप्त है।
(ख) ५. कृतमिच्छात्यार्य for (युगमिच्छत्यार्य)
६. कल्पाश्चतुर्गुणां for (कल्पश्चतुर्युगानां)
(च) ७. +७२+

युगमन्वन्तरकल्पाः कालपरिच्छेदकाः[२] स्मृतावुक्ताः ।
यस्मान्न[१] रोमके ते स्मृति[३]बाह्यो रोमकस्तस्मात् ॥ १३ ॥
कालर्क्षदेशयोगाद्भू[१]यो[२] ग्रहमन्दशीघ्र[५]पातानाम् ।
कल्पेन यतो योगस्ततः[३] स्फुटं ग्रहयुगं[४] कल्पः ॥ १४ ॥
कल्पेऽर्कबुध[१] सितानां भगणाः[४] शून्यानि सप्तरदवेदाः ।
प्राग्व्रजता[२] कुजगुरुशनिशीघ्रोच्चानां स्वकक्षासु[५] ॥ १५ ॥
४३२०००००००[३]

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१३. (घ) १. न रोमके ते (च) न रौनुके ते for (न रोमके ते)

(ख) २. कलापरिछेदकाः for (कालपरिछेदकाः)

(च) ३. वि० इस प्रति में "·····परिछे" के पश्चात् लेखक ने भूल से फिर "मनुरेक सप्तति युग" से लिखना आरंभ कर दिया। इस प्रकार ५½ पंक्तियां पुनः लिखी गईं! उसके पश्चात् फिर क्रमशः लिखता गया।

४. स्मृतबाह्यो for (स्मृतिबाह्यो)

१४. (घ) १. कालर्क्ष्य (ख) कालक्ष (च) कालर्क्ष्य for (कालर्क्ष)

२. दूद्भूयो (च) for (द्भूयो)

३. तत (च) for (ततः)

४. ग्रहयुगकल्पः (ग) ग्रहयुतं कल्पः for (ग्रहयुगं कल्पः)

(ग) ५. ग्रहशीघ्रमंद पातानां (क) (ख) for (ग्रहमन्द शीघ्रपातानां)

(च) ४. ग्रहयुग कल्पः for (ग्रहयुगं कल्प.)

१५. (घ) १. बुध (च) for (बुध)

२. व्रजनां (ग) व्रजती (ख) व्रजतां for (व्रजता)

(क) २. प्रागव्रजतां for (प्राग्व्रजता)

३. संख्या लुप्त है। टीका में अंकित है।

(ख) ४. भगणः for (भगणाः)

५. सा for (सु) (च) स्व कक्षासु for (स्व कक्षासु)

पंचांबराणि[3] गुणरामपंच[४] सप्तस्वरेषवः शशिनः ।
[१]५७७५३३००००
भौमस्य द्वियमशराष्टपक्षवसुरसनवद्वियमाः[५] ॥ १६ ॥
[२]२२९६८२८५२२
[१] [४]
९३६९९८९८४
कृतवसुनवाष्टनवनव[१] षड्नवागेन्द्वो[२] ज्ञशीघ्रस्य ।
[५]१७९३६९९८९८४
जीवस्य शरेषूदधि षडयक्षि[3] द्विकृतरसरामाः ॥ १७ ॥
[६]३६४२२६४५५
सितशीघ्रस्य यमलगो[१] वेदनवाष्टाग्नि[६] पक्षयमखनगाः ।
[४]७०२२३८९४९२
अष्टनवपक्षमुनिरसशररसमनवोऽर्कपुत्रस्य[२] [3] ॥ १८ ॥
[५]१४६५६७२९८

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१६. (क) १. संख्या लुप्त है । टीका में अंकित है ।
२. संख्या लुप्त है । टीका में अंकित है ।
(ख) ३. पंच वाराणि for (पंचांबराणि)
४. गुणरामं for (गुणराम)
(च) ५. वृद्धि for (वद्धि)

१७. (घ) १. १७९३६९९८९८४ for (१७९३६९९८९८४) (च)
२. वांगे (ग) षड्त्रिनव (क) षट्त्रिन (ख) षट्त्रिनवा for (षड्नवागेन्दवो)
३. पद्यक्ष (ग) षट्द्वयक्षि (क) षट्पक्ष (ख) षष्ट्यक्ष for (षडपक्षि)
(ग) ४. +२२९६८२८५२२ (क) 'ग' में अंकित संख्या १६वें श्लोक की टीका के अन्त में है ।
(क) ५. संख्या मूल में नहीं, टीका में अंकित है ।
६. संख्या मूल में नहीं, टीका में अंकित है ।
(ख) ७. वसू for (वसु)
(च) २. षड्नवांगेदवो for (षड्नवांगेवो) ३ पद्यक्ष for (षडयक्षि)

१८. (घ) १. लागो for (लगो) (च) यमलागो for (यमलगो)
२. यक्ष for (पक्ष)
३. ऽक्र्क (ग) ऽक्र्क for (ऽर्क)
(क) ४. मूल में संख्या लुप्त है । टीका में अंकित है ।
५, मूल में संख्या लुप्त है । टीका में अंकित है ।
(च) ६. चाष्टाग्नि for (वाष्टाग्नि)

[1]खाष्टाब्धयो ४८० [7]वसुशर वसुपंचखचन्द्रवसुवसुसमुद्राः ।
[8]४८८१०५८५८
[4]द्विनवयमा २९२ [5]द्वित्रिगुणा ३३२ शरेषुवसव [9]८५५
[2][6]स्त्रिपञ्चरसाः । १९ ॥ ६५३ ॥ १९ ॥
शशिवेदा ४१ मन्दानामर्कादीना[1] विलोमपातान्तम्
[2][3]वसुरसरुद्रेन्द्रगुण द्वित्रियमा २३२३१११६८ [4]सप्तरसपक्षाः
२६७ ॥ २० ॥

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१९. (घ) १. षष्ठौदधया (ग) खाष्टोदधयो for (खाष्टाब्धयो)
२. स्त्रिपंचसाः (च) स्त्रिपंचरसाः (स्त्रिपञ्चरसा)
(वि० चिह्नित संख्याएं यहां उपलब्ध नहीं हैं)

(ग) ४. द्विनवयमाः for (द्विनवयमा)
५. द्वित्रिगुणाः for (द्वित्रिगुणा)

(क) १. खाष्टोदधयो for (खाष्टाब्धयो)
६. +शशिवेदाः+
वि० श्लोकांकित कोई भी संख्या मूल में नहीं है। हाँ, टीका में विद्यमान है।

(ख) खाष्टोदधयो for (खाष्टाब्धयो)

(च) १. षाष्टोदधयो for (खाष्टाब्धयो)
७, संख्या लुप्त ८. संख्या लुप्त ९. शरेषुवस for (शरेषुवसव)
वि० श्लोकांकित संख्याएं लुप्त हैं।

२०. (घ) यहाँ श्लोक में निर्दिष्ट संख्याएं लुप्त हैं।

(ग) १. मर्क्कादीनां (क) चद्रादीनां (च) for (मर्कादीना)
२. द्वित्रियमाः (क) (ख) द्वित्रियशः for (द्वित्रियमा)

(क) ३. संख्या लुप्त है, परन्तु ठीका में अंकित है। (ख)
४. संख्या लुप्त है, परंतु टीका में अंकित है। (ख)

(च) वि० श्लोकांकित संख्याएँ लुप्त हैं।
२ द्वित्रियमाः for (द्वित्रियमा)

शशियमशरा ५२१[४] गुणरसा ६३[३] स्त्रिनंदवसवः
८९३[४] समुद्रवसुविषयाः ५८४[५]।
चन्द्रादीनां पश्चात् व्रजतोऽश्विन्यादिभगणस्य[१] ॥ २१ ॥[२]
परिवर्त्ता[३] रवचतुष्टय शराब्धिरसगुणयम[५] द्विवसु
तिथयः । १५८२२३ । ६४५०००० [१]
रविभगणोना भानोः सावन दिवसाः कुदिवसास्ते[२] ॥ २२ ॥
४३२०००००००[३]
१५७७९१६४५०००

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२१. (घ) १. व्रजनोश्विन्यादि for (व्रजतोऽश्विन्यादि)

२. (१९, २०. तथा २१ श्लोकों के अन्तर्गत आई हुई संख्याएं यहाँ एक ही स्थान पर दे दी गई हैं)

(ग) ३. स्त्रिनंदवसव for (स्त्रिनंदवसवः)

४. संख्या मूल में लुप्त हैं, परन्तु टीका में अंकित हैं।

५. संख्या मूल में लुप्त हैं, परन्तु टीका में ५८५ अंकित हैं।

(च) वि० श्लोकांकित संख्याएं लुप्त हैं।

१. (१९, २०, तथा २१ वें श्लोकों के अन्तर्गत आई हुई संख्याएं यहाँ एक ही स्थान पर अंकित हैं यथा ४८०, ४८८१०५८५८ २९२ ३३२ ८५५ ६५३ ॥ ४१ : २३२३१११६८ २६७ ॥ ५२१, ६३ ८९३, ५८४।

२२. (घ) १. १५८२२३ ॥ ६४५०००० (च) १५८२२३ ॥ ६४५०००

२. सावा (ग) सा वा (ख) (च) दिवसां वा for (दिवसास्ते)

(ग) ३. परिवर्त्ताः (च) for (परिवर्त्ता)

४. यह संख्या यहाँ नहीं दी गई है।

(क) ५. शरब्धि for (शराब्धि)
वि० मूल की तीनों संख्याएं लुप्त हैं, परन्तु टीका में क्रमशः अंकित हैं।

रविभगणा रव्यब्दा द्वादशगुणिता भवन्ति रविमासाः ।
                                ४
                                ५१०४०००००००
भगणांतरं[२] रवींद्वोः[३] शशिमासाः [१]५३४३३३००००० सूर्यमासोनाः ॥ २३ ॥

अधिमासाः १५९३३००००० शशिमासास्त्रिंशद्गुणिता
                                [१]
                                १६०२९९९०००००० भवन्ति शशिदिवसाः ।

शशिसावनदिवसान्तरमवमानि तिथिः शशांकदिनम् ॥ २४ ॥
                                २
                                २५०८२५५०००

सावनमुदयादुदयो[१] भानां चार्क्षं[२] नृवत्सरोर्कांश्च[३] ।
पितृदिवसाः[५] शशिमासा दिव्यानि दिनानि रविभगणाः ॥ २५ ॥

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२३. (घ) १. यह संख्या यहाँ श्लोक की समाप्ति के उपरांत रक्खी गई हैं। (ग) (च)
(क) २. भगणांतर (च) for (भगणांतरं)
३. रवींदोः for (रवींद्वोः)
४. संख्या मूल में लुप्त है, परन्तु टीका में अंकित है।
१. मूल में संख्या लुप्त है, परन्तु टीका में अंकित है।
(च) ३. रविंदोः for (रवींद्वोः)
(ख) "रविभगणरव्यब्दा द्वादशगुणिता भवंति ।
रविमासाः भगणांतरे रवींद्वोः शशिमासाः सूर्यमासो ना ॥ २३ ॥"
यह श्लोक अधिक है।
(ङ) यह श्लोक अंकित है।

२४. (घ) १. यह संख्या यहाँ पहली पंक्ति के तमाप्त होने पर लिखी गई हैं (ग) (च)
२. २५०८२५५०००० for (२५०८२५५०००)
(क) मूल की तीनों संख्याएं लुप्त हैं, टीका में क्रमशः निम्नांकित हैं —
५३४३३००००, १६०२९९९०००००, २५०८२५५०००

२५. (घ) १. सवन (च) for (सावन)
२. चार्क्ष्यं (ग) (क) वार्क्षं (ख) वक्षा for (चार्क्षं)
३. कब्दिः (ग) (क) (ख) कब्दिः for (कर्काश्च)
(ग) ४. पितृदिवसा for (पितृदिवसाः)
(क) ५. दुदयं for (दुदयो)
(च) २. चार्क्ष्यं for (चार्क्षं) ३ क्रांश्चः for (कर्काश्च)

कल्प परार्धमनवः षट्कस्य गताश्चतुर्युगत्रिघनाः ।
त्रीणि कृतादीनि कालो गोगैकगुणाः शकान्तेऽब्दाः ॥ २६ ॥
१८५२४१६०००,११६६४००००,३१७६

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२६. (घ) १. परार्द्ध (ग) कल्पापरार्द्ध ४३२०००००००००००००००००००००००० for (परार्धमनवः)
२. द्यता (ग) घनः ११६६४०००० (क) घनः (ख) त्रिघनम् for (घनाः)
३. काले (ग) कले (क) कलौ (ख) कलेर्गोगैकगुणः for (कालो गोगैकगुणाः)
४. गोगैः (ग) गोगैक गुणः ३१७६ for (गोगैकगुणाः)
५. अतिरिक्त संख्या + ३८८८००० +
(ग) ६. १८५२४१६०० for (१८५२४१६०००)
७. गताश्वचतुर्युग for (गताश्चतुर्युग)
८. +३८८८०००+
(वि० — उपर्युक्त संख्याएं 'ग' में श्लोक के अन्तर्गत ही हैं)
(क) १. परार्द्ध for (परार्ध)
६. ऽन्धः for (ऽब्दाः) (ख) सकनृपांते ऽब्दाः for (शकान्ते ऽब्दाः)
वि० मूल में संख्याएं लुप्त हैं, टीका में क्रमशः हैं—
१०००००००००००००००००, २७, १७२८०००, ३१२६६०००,
८६४०००, ३१७६.
(ख) १. कल्पपरार्द्ध for (कल्पपरार्ध)
(च) १. परार्द्ध ७, गताश्चतुर्युगत्रिद्यत for (गताश्चतुर्युगत्रिघनाः)
३. काले for (कालो) ४ गोगैः कगुणाः for (गोगैकगुणः) ६ शकांतेऽब्दाः for शकांतेऽब्दाः)
(घ) ग्रहनक्षत्रोत्पत्ति ब्रह्मदिनादौ दिनक्षये प्रलयः ।
यस्मात्कल्पस्तस्माद्ग्रहगणिते कल्पयाताब्दाः ॥ २७ ॥
(वि०) (यह श्लोक १६०२ ई० में प्रकाशित 'ब्राह्मस्फुट सिद्धान्त' में नहीं है। यह संस्करण क्वींस कालेज के प्रोफेसर महामहोपाध्याय सुधाकर द्विवेदिकृत नूतन-तिलक समेत 'मैडिकल हाल प्रैस में मुद्रित हुआ — ब्राह्मस्फुट सिद्धान्त के साथ इसमें ध्यानग्रहोपदेशाध्याय' भी प्रकाशित है)
(वि०) यह श्लोक (ग) तथा (क) पांडुलिपियों में निम्नांकित पाठान्तरों के साथ उपलब्ध है—
(ग) १. ग्रहः for (ग्रह); (क) २. +स्युः+

नवनगशशिनुनिकृतनवय[1] नगनन्देन्दवः[4]
[2]१९७२९४७१७९ शकनृपांते ॥
सार्धमतीत[3] मनूनां सन्धिभिराद्यन्तरान्तगतैः ॥ २७[6] ॥[5]
अधिकः स्मृत्युक्तमनोरार्यभटोक्तश्चतुर्युगेन मनुः ।
अधिकं विंशांश युतैस्त्रिभिर्युगैस्तस्य[1] कल्पगतम् ॥ २८ ॥[2]
[3]१८६१२७१९
कल्पगताब्दद्वादशघात[2]श्चैत्रादिमासयुक्तोऽधः[4] ।[1]
गुणितो युगाधिमासै रविमासाप्ताधिमासयुतः[3] ॥ २९ ॥

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२७. (घ) १. यमांगतांदेन्दव (ग) यमाग (ख) यमागनंदेदवः for (यनगनन्देन्दवः)
२. यह संख्या यहाँ श्लोक के अन्त में दी है, (ग)
(ग) ३. सार्ध प्रतीत (ख) सार्धमतीत (च) सार्द्धमतिव for (सार्धमतीव)
(क) १. यमागणंदेदंवः for (यनगनन्देन्दवः)
२. यह संख्या मूल में लुप्त है। टीका में उपलब्ध है।
(च) १. यमांगतं for (यनगनन्)—"देन्दवः" उत्तरार्ध का आरंभ है
४. उत्तरार्ध का आरंभ—देंदवः for (देंदवः)
५. +शकनृपांते+
६. +१९७२९४७१७९+

२८. (घ) १. युगैस्तस्य कल्पगतम् for (युगैस्तस्तस्य कल्पगतम्) (ग) (क) (ख)
(ग) २. विंशांक (क) (विंशांश)
३. १९८६१२३१७९ for (१८६१२७१९)
(क) ३. यह संख्या मूल में लुप्त है, टीका में निम्नांकित है—१९८६१२३७९
(ख) ४. अधिकस्मृत्युक्त for (अधिकः स्मृत्युक्त)
(च) ३. स्मृत्युत्त for (स्मृत्युक्त)

२९. (घ) १. ब्दा for (ब्द) (च) कल्पगताब्दा for (कल्पगताब्द)
२. घाता for (घातश्) (क)
(ग) ३. युताः (for युतः)
(क) ४. युक्तोऽध । (ख) युक्तोघ (च) युक्तोधः for (युक्तोऽधः)
(च) २. द्वादशघातश्चित्रादि for (द्वादशघातश्चैत्रादि)

त्रिंशद्गुणास्तिथियुतः पृथग्युगावमगुणो युगेंदुदिनैः ।
भक्तफलावमोनोऽर्कसावनाहर्गणोऽर्कादिः ॥ ३० ॥
इष्टग्रहभगणगुणादहर्गणात् कल्पसावनद्युहृतात् ।
भगणादि फलं मध्यो लंकायां भास्करौदयिकः ॥ ३१ ॥
आनयति दिवसवारं स्मृत्यभिरोधेन मध्यमानथवा ।
ब्राह्मादन्यैस्तन्त्रैरार्यभटाद्यैर्न्न कश्चिदपि ॥ ३२ ॥
जगति तमोभूतेऽस्मिन् सृष्ट्यादौ भास्करादिभिः स्पृष्टैः ।
यस्माद्दिनप्रवृत्ति दिन वारोऽर्कोदयात्तस्मात् ॥ ३३ ॥

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३०. (घ) १. विमानो (च) for (वमोनो)
२. सावनोः (ग) सावनो (क) सावनो (ख) (च) for (सावना)
३. हर्ग्रणो (ग) (क) ऽहर्गणो (च) for (हर्गणो)
(ग) ४. +३०+
५. भक्तः (क) (ख) for (भक्त)
६. ऽर्कादि for (ऽर्कादिः)
(क) ४. त्रिंशत् गुणास्तिथियुतः for (त्रिंशद्गुणस्तिथियुतः)
(ख) ५. मगुणे for (भक्तफला)
(च) ६. ऽर्क्क for (ऽर्क) ६ ऽर्क्कादिः for (ऽर्कादिः)
३१. (घ) १. भास्करोदायिकः । (ग) (च) for (भास्करौदयिकः)
(ग) २. बधादहर्गणात् for (गुणादहर्गणात्)
(क) १. भास्करोदयिकः for (भास्करौदयिक.)
(ख) ३. इष्टग्रहणभगणा for (इष्ट ग्रहभगण)
(च) ४. दर्गणात् for (दहर्गणात्)
५. द्युहृता for (द्युहृतात्)
३२. (ग) १. नं कश्चदपि for (न्नं कश्चिदपि) (क) (ख) नं for (न्नं)
(क) २. ब्रह्मा (ख) for (ब्राह्मा)
३३. (घ) १. तसोभूते for (तमोभूते) (ख) तमोभूतो for (तमोभूते)
२. सृष्टैः (ग) (क) (च) for (स्पृष्टैः)
(ग) ३. दिनवारो (ख) for (दिनवारो)
(ख) ४. ऽर्कोदयस्तस्मात् for (ऽर्कादयात्तस्मात्)
(च) १. तसोभूते for (तमोभूते) ३. दिनवारोर्क्का for (दिनवारोऽर्को)

लङ्कासमयाम्योत्तररेखायां[१] भास्करोदये मध्याः ।
देशान्तरोनयुक्ता रेखा[२] प्रागपरदेशेषु ।। ३४ ।।
दिनवारादिः पश्चादुज्जयनी दक्षिणोत्तरायाः[१] प्राक् ।
देशान्तरघटिकाभिः पश्चात्प्राक्[२] भवति रव्युदयात् ।। ३५ ।।
भूपरिधिः[४] खखखशरारेखा स्वाक्षान्तरांश[६] संगुणिताः[५] ।
५०००० [१]
भगणांश ३६० हृता[२] फलकृतिहीनो[३] देशान्तरस्य कृतिः ।। ३६ ।।[७]
शेषपदगुणभुक्तिर्भूपरिधि[१] हृता[२] कलादि[३] लब्धमृणम् ।
उज्जयनी याम्योत्तर रेखायाः[४] प्राग्धनं[५] पश्चात् ।। ३७ ।।

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३४. (य) १. रेषायाँ for (रेखायां)
(ग) २. रेखा (क) (ख) for (रेषा)

३५. (घ) १. दक्षणोत्तरायाः for (दक्षिणोत्तरायाः) (च)
२. प्राग्भवति (ग) (क) (च) for (प्राक्भवति)

३६. (घ) १. ५००० (ग) for (५००००)
२. 'हृता' से पूर्व दी गई ३६० की संख्या यहाँ 'हृता' के पश्चात् दी गई है। (ग) (ख) हृल (च)
(ग) ३. हीना (क) (ख) for (हीनो)
(क) ४. परिधि for (परिधि)
५. संगुणिता for (संगुणिताः)
वि० श्लोकान्तर्गत संख्याएं लुप्त हैं। (ख)
(ख) ६. स्वक्षांतरांस for (स्वाक्षांतरांश)
(च) ३. हिनो for (हीनो)
७. क्रमसंख्या लुप्त

३७. (घ) १. गुणा (ग) (क) (च) for (गुण)
२. परिधिताकलादि for (परिधि हृताकलादिं)
(ग) ३. कलादिलब्धमृणम् for (कलादिं लब्धमृणम्)
(क) ४. रेखाया for (रेखायाः)
५. प्राग्रणं धनं (ख) for (प्राग्धनं)
(च) ३. कलादिलब्धमृणं for (कालादिंलब्धमृणम्)

मध्यग्रहे[१] स्फुटे[२] वा भूपरिधिहृतात्पदाद्[५] गुणात् षष्ट्या[७]।
लब्धं घटिकाद्यथवा[३] कर्म तिथि[६] ऋण[४] धनं ग्रहवत् ॥ ३८ ॥
कल्पगताब्दा गुणिता रूपाष्टजिनैर्नवाग्निसतगैः[१][३]
२४८१७७३९।
खखरसनवभि ९६००[२] भक्ता[४] दिनावमान्यंशकाः[५] शेषाः ॥ ३९ ॥
तद्दिकगुणा[१] १० योगादधिस्त्रिशता[३] धृताल्लब्धम्[२]।
शेषास्तिथयः[४] शुद्धिदिनानि विकलं दिनांशेभ्यः ॥ ४० ॥

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३८. (घ) १. मध्ये (ग) (च) for (मध्य)
२. स्फटे (च) for (स्फुटे)

(क) ३. 'द्य' लुप्त है—'छूट गया' का चिह्न है, आगे 'थावा' है।
४. मृण for (ऋण)

(ख) ५. भूपरिहृतात् for (भूपरिधिहृतात्)
६. तिथिषु for (तिथि)

(च) ७. +॥ ६ ॥

३९. (घ) १. जनैर्नवाग्नि for (जिनैर्नवाग्नि) (ग) जिनै २४२४८१
२. यह संख्या यहाँ 'भक्ता' के पश्चात् अंकित है।

(ग) ३. सप्तनगैः ७७३९ (क) सप्तनगैः (ख) for (सतगैः)
४. यहाँ 'भक्ता' से आगे ९६०० की संख्या है।

(क) ५. मानांशकाः for (मान्यंशकाः)

(च) १. रूपाथजनैर्नवाग्नि for (रूपाष्टजिनैर्नवाग्नि)

४०. (घ) १. गुणाः for (गुणा १०) (ग) तद्दिग्गुणा १० ब्द for (तद्दिकगुणा १०)
२. हृता (ग) (क) हृताल्लब्धां (ख) हतालब्धम् for (धृताल्लब्धम्)

(ग) ३. दधिमासास्त्रिशता (क) (ख) दधिमासा for (दधिस्त्रिशता)

(क) १. तद्दशगुणाब्दयोगा for (तद्दिकगुणा १० योगा)

(ख) १. तद्दिग्गुणाब्द for (तद्दिकगुणा १०)
४. दिनानि for (दिनानि)

(च) १. तद्दिकगुणाः for (तद्दिकगुणा)
२. द्धताल्लब्धं for (धृताल्लब्धं)

कल्पगताक्षदिनयुतेः सूर्याद्योऽब्दाधिपो गताब्दभगणवधः ।
कल्पाब्दभूतो गणादिमध्यमः सूर्यभगणान्ते ॥ ४१ ॥
चैत्रसिताद्यास्तिथयः शुद्धिविहीनाः पृथग्गुणा रुद्रैः ।
अवमांशेभ्यो यमनवरस ६९२ गुणितेभ्यो विभक्तेभ्यः ॥ ४२ ॥
स्वच्छेनफलयुता हृतास्त्रिखागैः ७९३ फलावमविहीनाः ।
रविमेषादि द्युगणो मुनिहृत् शेषोब्दपत्यादिः ॥ ४३ ॥

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४१. (घ) १. सूर्यो (च) for (सूर्या)
२. योगताब्द for (पोगताब्द) (ग) ऽब्दभगण (क) धियोऽब्दभगण for (धिपोगताब्दभगण)
(ग) ३. गताब्ददिनयुतैः for (गताक्षदिनयुतेः) (क) गताब्द for (गताक्ष)
४. हृतो (क) (ख) हृतो for (भूतो)
५. भगणादि (क) (ग) भगणादि for (गणादि)
(ख) ३. कल्पगताब्ददिनयुतेः for (कल्पगताक्षदिनयुतेः)
२. धिपोब्धभगणवधः for (धिपोगताब्द भगणवधः
(च) ६. अवग्रहचिह्नलुप्त

४२. (घ) १. प्रथगुणा (च) for (पृथग्गुणा)
(ग) २. गुणिते विभक्तेभ्यः for (गुणितेभ्यो विभक्तेभ्यः)
(क—यहां के मूलपत्र उपलब्ध नहीं हैं)
(च) ३. रुद्रैः for (रुद्रैः)

४३. (घ) १ त्रिखगैः for (त्रिखागैः) (ख) हृतास्त्रिखागैः for (हृतास्त्रिखागैः)
२. ७०३ (ग) for (७९३)
३. फलाऽवम for (फलावम)
४. मेखादि (ग) (च) for (मेषादि)
(ग) ५. स्वच्छेदेन ९६०० (ख) स्वछेदन for (स्वछेन)
(क) (क—यहां का मूलपत्र उपलब्ध नहीं है)
(ख) ६. मुनिहत for (मुनिहृत्)
(च) १. हृतास्त्रखगैः for (हृतास्त्रिखागैः) ३ फलाऽवमविहीनाः for (फलावमविहीनाः)
६. मुहृत for (मुनिहृत्)

द्युगणात्सप्त्यंशं स्वनवार्काशाधिकं १२९ विशोध्यांशाः ।
मध्याः सूर्यबुधसिताः शीघ्रोच्चं कुजगुरुशनीनाम् ॥ ४४ ॥
त्रिगुणमवमाशेषं विभजेद्गुणसप्तशशिः १७३ ।
पृथगधिकोऽर्को रविगुणतिथ्यंशैः संयुतश्चन्द्रः ॥ ४५ ॥
एकादशलिप्तांशा भौमः शरसप्तवसुभिरिन्दुयमैः ।
कृतगुणितोध द्युगणोशाः पंचरसै षट्क बुधशीघ्रम् ॥ ४६ ॥

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४४. (ग) १. यहाँ '१२९' संख्या नहीं दी गई ।
(क) क—यहाँ का मूलपत्र उपलब्ध नहीं है ।
(ख) २. सिता for (सिताः)
(च) ३. सप्रत्यंशं for (सपत्यंशं) ४ वशोध्यांशाः for (त्रिशोध्यांशा)

४५. (घ) १. सप्तशशिरामांशैः (ग) सप्तशशिभिरामांशैः for (सप्तशशिः १७३)
(ग) २. भावशेषं (च) for (भाशेषं)
(क) (क—यहाँ का मूल पत्र उपलब्ध नहीं है ।
(ख) १. सप्तशशिभिरामांशैः for (सप्तशशिः १७३)
३. रविगुणातिथ्यंशैः for (रविगुणतिथ्यंशैः)
(च) १. सप्तशशिभिः for (सप्तशशिः १७३)
४. +सप्तांशैः+
५. 'ण' लुप्त
६. र्क्का for (र्क्को)
३. रविगुणतिथ्यशै for (रविगुणतिथ्यंशैः)

४६. (घ) १. गणांशाः for (गणोशाः) (ख) द्युगणांशाः for (द्युगणोशाः)
(ग) २. लिप्तांशाः (ख) लिप्ताशन् for (लिप्तांशा)
३. ८७५ रिपुयमैः ॥ २१ ॥ for (रिन्दुयमैः)
४. गुणितो द्युगणोंशाः for (गुणितोवद्युगणोशाः) (ख) कृतगुणिता for (कृतगुणितोध)
५. रसैः ६५ (ख) रसैः (च) for (रसै)
(क) (कि—यहां का मूलपत्र उपलब्ध नहीं है)
(च) ३. रिन्दुयमै ११ for (रिंदुयनैः)
१. द्युगणोंशाः for (द्युगणोशाः)

द्युगणेषु[६] बधो लिप्ताजीवः कृतशरगुणैः[१] शरकालोनः[२] ॥
भागकलाः सितशीघ्रं[४] विषयैर्वसवो[५] द्विषष्टचाष्टौ[३] ॥ ४७ ॥
द्विगुणाः[१] कलाः[२] दिनगणस्तिथिरामैर्द्वे[३] काल[६] च सूर्यसुतः[७] ।
नवभिर्भागः साग[४] ११२१ खशून्यवेदैश्चन्द्रोच्चम्[५] ॥ ४८ ॥

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४७. (ग) १. गुणैः (ख) (च) for (गुणैः)

२. कलोनः । ३५४ । ५ ॥ for (कालोनः) (ख) शरकलोनः for (शरकालोनः)

३. ष्टौ ।८।८। for (ष्टौ) (ख) द्विषष्टमासौ for (द्विषष्टचाष्टौ) ।५।६२

(क) (कि—यहां का मूलपत्र उपलब्ध नहीं है ।

(ख) ४. शित for (सित)

५. विषवै for (विषयै)

(च) ६. द्युगणेशु for (द्युगणेषु) ३ द्विषष्टचष्टौ for (द्विषष्टचाष्टौ)

४८. (घ) १. द्विगुणः for (द्विगुणाः)

२. कला for (कलाः) (ख) कलादिनगणास् for (कलादिनगणस्) (च) कला for (कलाः)

३. द्वे कले । ३१५ । २ । for (द्वे काल)

४. सागर—for (साग ११२१) (यहां कोई संख्या नहीं दी गई है) (ख) सागर

५. + । ६ । १ । ४००४ । १० । +

(क) (क यहां का मूलपत्र उपलब्ध नहीं है)

(ख) ६. कले for (काल)

७. ंचचन्द्रोचः for (श्चन्द्रोच्चम्)

(च) ८. +१११+

भागो नन्दशशांकैः ११९८ शशिशून्य ८७५२१ स्वरयामै
२७।९४००४।०११
रविमंडलांतिकयुता मध्य भगणान्ति ३५६ काशेषाः ॥ ४९ ॥
पादार्द्ध विपाददिनै रात्र्यर्द्धास्तमयदिनदलौदयिकाः । ३१५ ।
ऊनीकृत्वा तिथयो देशान्तरनाडिकानयुताः ॥ ५० ॥

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४९. (घ) १. यमैश्च शशिपातः (ग) यमैश्च ७७०१ शशिपातः for (यामै २७ । ९४००४।०११)

२. मध्या (ग) (ख) for (मध्य)

३. ३५८ for (३५६) (ग) यहां यह संख्या नहीं दी गई ।

(ग) ४. भोगानन्द for (भागोनन्द)

५. १९ for (११९८)

६. भगणांतिका शेषाः for (भगणान्ति ३५६ का शेषाः)

(क) (क—यहां का मूलपत्र उपलब्ध नहीं है ।)

(ख) ७. सूर्य for (शून्य)

१. यमैश्च शशिपातः for (यामै २७ । ९४००४।०११)

(च) १. स्वरयामैश्च शशिपातः for (स्वरयामै)

३. ३५६ । ५ । for (३५६)

८. काः for (का)

५०. (ग) १. दलोदयिकाः for (दलौदयिकाः)

२. नाडिकोनयुताः for (नाडिकानयुताः) (ख) नाडिकेन for (नाडिकान)

(क) (कि -यहां का मूलपत्र उपलब्ध नहीं है)

(ख) १. दिनदलौदयिक्ता for (दिनदलौदयिकाः)

कलिगतशुद्धिः प्राग्वच्छुक्राद्योब्धाधिपोक्षभगण वध्यतक्षितिजस्य ।
खत्रिचाष्ट रससप्त वसुरवाग्निवेदयुतात् ॥ ५१ ॥

४३०८७६८०००

बुधशीघ्रस्य खरवांबर रसनन्द दृष्टोष्ट वसुयमो दधिभिः ।

४२८८८९६००

खचतुष्टययमशरगुण शशिवेदैः ४३१३५२०००० सुरेन्द्रगुरोः ॥ ५२ ॥

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५१. (घ) १. ब्दा for (क्ष) (ग) व्द भगणबधात् for (क्षभगणवध्यत) (ख) ब्द for (क्ष)
२. वध्यत् for (वध्यत) (पहली पंक्ति का अन्तिम शब्द) (ख) वधात् for (वध्यत)
३. 'क्षितिजस्य' दूसरी पंक्ति का आरम्भिक पद । (ग) (ख)
४. खत्र (ग) खत्रयाष्ट (ख) खत्रयाष्ट for (खत्रिचाष्ट)

(ग) ५. रसप्त for (रससप्त)

(क) (क—यहां का मूलपत्र उपलब्ध नहीं है।)

(ख) ६. शुधि for (शुद्धिः)
७. दृधियो for (ब्दाधिपो)

(च) ७. छुक्राद्योब्दाधियो for (छुक्राद्योब्धाधिपो)
२. वध्यत् for (वध्यत)
४. खत्रचाष्ट for (खत्रिचाष्ट)
८. दैदयुतात् for (वेदयुतात्)

५२. (घ) १. रसनं दृष्टो (च) for (रसनन्द दृष्टो)
२. यह संख्या यहाँ श्लोक के अन्त में दी है । (ग) (च)

(ग) ३. नन्दाष्टाष्ट (क) (ख) नन्दाष्टा for (नन्दद्दष्टोष्ट)
४. ४२८८८९६००० for (४२८८८९६००)

(क) ४. मूल में संख्या लुप्त है । टीका में ४२८८८९६००० अंकित है ।
२. मूल में संख्या लुप्त है, टीका में ४३१३५२०००० अंकित है ।

भार्गवशीघ्रस्यांबर खखाष्टं वेदाब्धिवेदखाग्नि कृतैः ४३०४४४८०००
भास्कर सुतस्य खत्रय रविगुणशरखगुण समुद्रैः ॥ ५३ ॥

४३०५३१२०००

शून्यचतुष्टय पक्षेदु रामगुण नवभिरर्कमंदस्य । ६३३१२०००००
इन्दोः खमय यमशरनव पंचव्योम शरचन्द्रैः ॥ ५४ ॥ १५०५६२०००
खत्रय यमनवपंचाष्टरामधृतिभिः शशांकपातस्य । १८३८५६२०००
कल्पगतभगणघातात् कुजादिमंदोच्चपातानाम् ॥ ५५ ॥

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५३. (घ) १. भाग्रव (च) for (भार्गव)
२. शरखद for (शरखगुण) (ख) शरखदहन सुमुद्रैः for (शरखगुणसमुद्रैः)
(ग) ३. खखाष्ट for (खखाष्टं) (क) (ख) खाखाष्ट for (खखाष्टं)
४. दहनसमुद्रैः (क) for (खगुणसमुद्रैः)
(ख) ५. वेदाश्च for (वेदाब्धि)
६. वेदखाब्दि for (वेदखाग्नि)
७. भागस्कर for (भास्कर)
(च) ३. खखाष्ट for (खखाष्टं)
४. खद for (खगुण)

५४. (घ) १. त्रयं for (खमय) (ख) त्रयमशर for (खमययमशर) (च) खत्रये for (खमय)
(ग) २. पक्षेन्दु (क) (च) for (पक्षेदु)
३. चन्द्रसमः शून्यत्रय for (इन्दोः खमय)
४. यमशनवशरखशरचन्द्रैः for (यमशरनवपंचव्यो मशरचन्द्रैः)
५. १५०५६५२००० (च) for (१५०५६२०००)
(क) ३. चंद्रमसः शून्यत्रय for (इन्दोः खमय)
६. शरख for (पंचव्योम)
वि० मूलांकित दोनों संख्याएँ मूल में नहीं है। टीका में अंकित हैं।
(ख) ७. सून्य for (शून्य)

५५. (क) मूल में संख्या उपलब्ध नहीं। टीका में अंकित है।
(ख) १. गणोध्ययातात्कुंजादि for (गणघातात् कुजादि)

भगणादिकल्पवर्षैं[५] लब्धं[१] रविमंडलान्तिकामध्या[६][४] ।
मेषादिद्युगुणफलाधिका[२][३] भवन्तीष्टदिनमध्याः ॥ ५६ ॥
शुद्धीश ११ बधे शुद्धे[१] देवमशेषात् सावनद्युगण[४] शुद्धिः ।
व्येकावमं[२] गृहीत्वा गुणस्वमुने[३] युतान्न[५] शुध्यति[६] चेत् ॥ ५७ ॥

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५६. (घ) १. लंब्धम् for (लब्धं) (ग) (क) (ख) लब्ध for (लब्धं)

२. मेखादि for (मेषादि)

३. द्युगण (ग) (क) for (द्युगुण)

(क) ४. मध्याः (ख) मध्यः (च) मध्य for (मध्या)

(ख) ५. वर्षैः for (वर्षैं)

६. लान्तिको for (लान्तिका)

(च) १. ब्धं for (लब्धं) ३ क्क (द्यु) गण for (द्युगुण)

५७. (घ) १. शुद्धेऽवशेषा (ग) शुद्धेऽवमशेषात् (ख) शुद्धेवमशेषात् for (शुद्धेदेवमशेषात्)

२. व्येकामवं (ख) एकावमं for (व्येकावमं)

३. खमुने (क) खमुनि for (स्वमुने)

(ग) शुद्धीश के पश्चात् ११ की संख्या नहीं दी गई है । (क)

(क) १. ऽवमशेषोत्सावन for (देवमशेषात्सावन)

४. सिद्धिः (ख) सुद्धिः for (शुद्धिः)

(ख) ५. युतानि for (युतान्न)

६. सुध्यति for (शुध्यति)

(च) १. शुद्धेइवमशेषात् for (शुद्धेदेवमशेषात्)

२. व्येकामवं for (व्येकावमं)

चैत्रसिताद्योब्द पतिद्धू नाया दिनाब्दरूपयुतेः ।
तद्युगणादिनवारान् मुहुर्मध्यमा प्राग्वत् ॥ ५८ ॥
अकृतार्यभटः शीघ्रगमिन्दुच्चं पातमल्पगं स्वगतेः ।
तिथ्यन्तग्रहाणानां घुणाक्षरं तस्य संवादः ॥ ५९ ॥

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५८. (घ) १. शुद्धचूनाया for (पतिद्धूनाया) (ग) (क) पतिः शुद्धचूनाया for (पतिद्धू-नाया)
२. वारन् (ग) वारः (क) वारः (ख) वारः for (वारान्)
३. मध्यमा (ग) (क) मध्यमा for (र्मध्यमा)
४. मुह्यू (शुद्धचू) (ग) शुद्धचूना (क) शुद्धचवमा for (मुहुर्मध्यमा)
(ग) ५. चैत्रसिताद्योद्योब्दपतिः for (चैत्रसिताद्योब्दपति)
(क) ६. तद्युगणात् for (तद्युगणा)
(ख) १. पतिः शुध्यूनाया for (पतिद्धूनाया)
४. शुध्यूना for (मुहुर्मध्यमा)
(च) १. पतिः शुद्धचूनाया for (पतिद्धूनाया)
६. तत् द्युगणा for (तद्युगणा)
३. मध्यमाः for (मध्यमा)

५९. (घ) १. स्वगतैः (ख) स्वगते for (स्वगतेः)
२. घुणाक्षरं (च) for (घुणाक्षरं)
(ग) ३. दूच्चं (क) (ख) दूचं for (दुच्चं)
(क) ४. ग्रहणानां (ख) (च) for (ग्रहाणानां)
५. मल्यगं for (मल्पगं)
६. 'संवादः' पद लुप्त है ! श्लोक तो 'तस्य' पर समाप्त है।
(च) ७. शिघ्र for (शीघ्र)

(घ) खखखार्क १२००० हृताब्देभ्यो, गतगम्याल्पाः खशून्ययमल २०० हृताः ।
लब्धं त्रि ३ सायक ५ हतं कलाभिरूनौ सदार्केन्दू ॥ ५९ ॥
शशिवत् जीवे द्वि २ हतं, चन्द्रोच्चे तिथि १५ हतं तु सितशीघ्रे ।
द्वीषु ५२ हतं च बुधोच्चे, द्वि २ कु १ वेद ४ हतं च पात कुजशनिषु ॥ ६०
वि० उपर्युक्त दोनों श्लोक १९०२ ई० में प्रकाशित ब्राह्मस्फुटसिद्धान्त में "अतिरिक्त" हैं ।

मध्यगतिज्ञं वीक्ष्य श्री षेणार्यभटविष्णुचन्द्रज्ञाः ।
सदसि न भवन्त्यभिमुखाः सिंहं दृष्ट्वा यथा हरिणाः ॥ ६० ॥
युगभगणमान यातार्हर्गण दिनवारमध्यमाद्येषु ।
मध्यमगति द्विषष्टचार्याणां प्रथमकृतो है ॥ ६१ ॥

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६०. (ग) १. वीक्ष for (वीक्ष्य)
२. दिष्ट्वा for (दृष्ट्वा)
(क) ३. श्रीहर्षेणार्य for (श्रीषेणार्य)
(ख) ५. विभुचन्द्राद्या for (विष्णुचन्द्रज्ञाः)
६. हरिणः for (हरिणाः)

६१. (घ) १. हर्ग्रण for (हर्गण)
२. द्विषष्टचार्यणां (ग) द्विषष्टचार्यां (क) द्विषष्टर्याणां for (द्विषष्टचार्याणां)
३. प्रथमः कृतोध्यायः (ग) (क) प्रथमाः for (प्रथमकृतो है)
(ग) वि०+इति श्री ब्रह्मसिद्धाते मध्यमाधिकारः प्रथमः+
(ख) ५. यतो for (याता)
२. द्विषष्टार्याणां for (द्विषष्टचार्याणां)
(ख) की दृष्टि से ६३ श्लोक हैं, २३ वां अधिक है।
(च) ३. प्रथमः for (प्रथम)
४. कृतोध्यायः for (कृतोहै)

॥ श्रीगणेशायनमः ॥

यस्मान्न मध्यतुल्यः प्रतिदिवसं दृश्यते ग्रहो भगणे ।
तस्माद् कल्पकरं वक्ष्ये मध्यस्फुटकरणम् ॥ १ ॥
अर्धज्याभूयमला मुनियमवेदा वसुऽनलषट्का ।
रसकृतवसवः शशिपंचखेन्दवश्चन्द्रशरसूर्याः ॥ २ ॥
षटुदधिमनवो भूवाग्निरसशशांका मुनींदवसुचन्द्राः ।
इन्दुनवनन्दचन्द्रा रसतिथय यमला रवित्रियमाः ॥ ३ ॥

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१. (घ) १. सस्माद्द for (तस्माद्) (ग) तस्माद्दक् तुल्यकरं (क) तस्माद्दक् तुल्यकरं for (तस्माद्कल्पकरम्)
२. स्फुटं for (स्फुट) (ग) स्फुटीकरणम् (क) स्फुटीकरणम् for (स्फुटकरणम्)
(ग) ३. वक्षे for (वक्ष्ये)
(क) ४. मध्यम for (मध्य)
(ख) १. तस्माद्दक् तुल्यकरं for (तस्माद् कल्पकरं)
२. स्फुटीकरणम् for (स्फुटकरणम्)
(च) ५. "मंगलसूचक शब्द" लुप्त है !
१. तस्मात् द्द for (तस्माद्) २ स्फुटं for (स्फुट)

२. (घ) १. मनुयमला for (भूयमला (ग) (क) अर्धज्यामनुयमला for (अर्धज्या-भूयमला)
२. वसुज्वलखट्काः (ग) (क) वसुज्वलनषट्काः for (वसुऽनलखट्का)
(ख) १. अर्धज्यामनुयमला for (अर्धज्याभूयमला)
२. वसुज्वलनषट्काः for (वसुऽनलखट्का)
३. चन्द्रममूर्या for (चन्द्रशरसूर्याः)
(च) १. अर्द्धज्यामनुयमला for (अर्धज्याभूयमाला)
२. वसुज्वलनषट्काः for (वसुऽनलषट्का)

३. (घ) १. तिधिय for (तिथय) (ग) (क) रसतिथि (ख) रसतिथि for (रसतिथय)
२. यम्नाः for (यमाः) (ख) for (यमाः)
(ग) ३. भूताग्नि (क) (ख) (च) for (भूवाग्नि)
४. मुनींदु (क) (ख) (च) for (मुनींद)
(क) ५. षडवब्धि for (षटुदधि)
६. चन्द्रा for (चन्द्राः)
(ख) ७. चन्द्राः for (चन्द्रा) (च) १. तिथि for (तिथय)

ऋतुनवखगुणः[३] नवशरचन्द्रगुणाः[१] सप्तशून्ययमदहना[४] ।
द्विजिनगुणा स्त्रिरसरदा[५] खस प्रयम छिद्रेषु[६] जिनाः[७] ॥ ४ ॥

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४ (घ) १. चन्द्रागुणा (च) for (चन्द्रगुणाः)
२. सप्त for (सप्त) (ग) सप्तयम वह्नयो व्यस्ताः ॥ ५ ॥ for (सप्रयम-छिद्रेषु जिनाः ॥ ४ ॥)

(ग) ३. गुणा for (गुणः)
४. दहनाः for (दहना)
५. रदाः for (रदा)

(क) 'क' में अंकित नहीं। इसके स्थान में दूसरा श्लोक अंकित है जो अगले पृष्ठ पर दिया है।

(ख) यहां यह श्लोक अंकित नहीं है।

(च) २. सप्त for (सप्र)
६. छिद्रेशु for (छिद्रेषु)
७. क्रमसंख्या लुप्त ।

(क. ख. ग.) में अतिरिक्त श्लोक —

छिद्रेषु[१] जिना[२] कृतनव पंचयमा दं नंद चन्द्रमुनि पक्षाः ।
दन्ताष्टयमागुण[५] रामनवयमाः[६] शशिख[३] रामाः ॥ ४ ॥
ऋतुनव खगुणा नवशरचंद्रगुणाः सप्तशून्ययमदहनाः ।
द्विजिन[४] गुणास्त्रिरसरदाः खसप्तयम वह्नयो व्यस्ताः[७] ॥ ५ ॥

(क) १. यमाः for (यमा)
२. 'दं' लुप्त पद है। (ख)
३. शशियमखरामाः ॥ ४ ॥ (ख) शिशिगुणखराम for (शशिखरामाः)
४. जित for (जिन)

(ख) ५. दताष्ट for (दन्ताष्ट)
६. यमगुण for (यमागुण)
७. व्यास्ताः for (व्यस्ताः)

(च) वि० इस स्थान पर यह श्लोक लुप्त हैं।
पृष्ठ ८b पर नीचे की ओर अन्य लेख में "ऋतुनव.....से-खसप्रयम" तक विद्यमान है। इस पर क्रम संख्या कोई नहीं।

कृतनव पंचयमा[२] नन्द चन्द्र[३] दन्ताष्ट यमगुण[१] रामनवयमाः[४] ।
शशियम खरामाः[५] मुनिपक्षाः वह्नयो व्यस्ताः ॥ ५ ॥
मुनयोष्टयमास्त्रिरसा रुद्र[४] शशांकाः समुद्रमुनिचन्द्राः[१] ।
नववेदयमामुनिगुणा[३] हुताशना[२] वसुगुणसमुद्राः ॥ ६ ॥
रूपेन्द्रियेषवोरसनगर्तवश्चन्द्रशीतकरवसवः[१] ।
वसुशरनन्दाः[३] सागररुद्रशशांकाः[२] नवागार्काः[४] ॥ ७ ॥
त्रिविषयवेदशशांकाःपंचत्रिरसेंदवोब्धियमधृतयः[४] ।
अतिधृतिस्वयमानवशशियमपक्षा[३] सागर द्विजिना[२] ॥ ८[१] ॥

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५. (घ) १. यमा (च) for (यम)
(ग) वि० यहां चौथे और पांचवें श्लोक की चारों पंक्तियों को कुछ भिन्न प्रकार से लिखा है। (क)
(क) १. क+षट् छिद्रेषु जिना २४५९+
२. यमाः+२५९४+for (यमा)
३. चन्द्रमुनिपक्षाः २७१९ for (चन्द्रदन्ताष्ट)
१. यमाः+२८३२ for (यम)। इसके पश्चात् दूसरी पंक्ति प्रारम्भ
४. +२९३३+
५. +३०२१+
(ख) यहां पर यह श्लोक 'लुप्त' है।

६. (ग) ३. गुण (च) for (गुणा)
(क) ४. रुद्राः for (रुद्र)
(च) २. कुताशना for (हुताशना)

७. (घ) १. र्त्ववश्चन्द्र for (र्तवश्चन्द्र)
(क) २. शशांका (च) for (शशांकाः)
(ख) ३. नन्दा for (नन्दाः)
४. नवागार्का for (नवागार्काः)

८. (घ) १. द्विजिनाः (ग) (च) for (द्विजिनाः)
२. ख (क) (च) for (स्व)
(ग) ३. पक्षाः (क) for (पक्षा)
(ख) ४. भि for (त्रि)
(वि० देखिए पृष्ठ २६—सारणी के लिए)

रदरसयमला गुणवेदवसुयमाः षट्क[3] विषयशून्यगुणाः ।
खमुनिरसा[1] व्यासार्द्ध[4] नवरदचन्द्रा जिनांशज्याः[2] ॥ ९ ॥ २४।१३।२९

२१४	१८८७[1]	२९३३	७	५५१	१८२४
४२७	१९९१	३०२१	२८	६७६	२०१९
६३८	२१५६	३०९६	६३	८११	२२१९
८४६	२३१२	३१५९	१११	९५८	२४२४
१०५१	२४५९	३२०७	१७४	१११४	२६७२[3]
१२५१	२५९४	३२४२	२४९	१२७९	२८४३
१४४६	२७१४[2]	३२६३	३३७	१४५३	३०५६
१६३५	२८३२	३२७०	४३८	१६३५	३२७०

९. (घ) १. रदा (ग) रदा (क) (च) for (रसा)

२. २४ ॥ १३२९ for (२४।१३।२९) ॥ (ग) १३२९+॥ २१४[1] ॥ ४२७[2] ॥ ६३८[3] ॥ ८४६[4] ॥ १०५१[5] ॥ १२५१[6] ॥ १४४६[7] ॥ १६३५[8] ॥ १८१७[9] ॥ १९९१[10] ॥ २१५६[11] ॥ २३१२[12] ॥ २४५९[13] ॥ २५९४[14] ॥ २७१९[15] ॥ २८३२[16] ॥ २९३३[17] ॥ ३०२१[18] ॥ ३०९६[19] ॥ ३१५९[20] ॥३२०७॥३२४२॥ ॥ ३२६३[21] ॥ ३२७०[22] ॥ एतेक्रमस्थाः ॥ अत उत्क्रमस्थाः[1] ॥ ७ ॥ २८[2] ॥ । ६३[3] । १११[4] । १७४[5] । २४९[6] । ३३७[7] । ४३८[8] । ५५१[9] । ६७६[10] । ८११[11] । ९५७[12] । १११४[13] । १२७९[14] । १४५३[15] । १६३५[16] । १८२४[17] । २०१९[18] । २२१९[19] । २४२४[20] । २६३२[21] । २८४३[22] । ३०५६[23] । ३२७०[24] । उत्क्रमज्या इति ॥+

(ख) ४. जिनांशज्या for (जिनांशयाः)

(सारणी) (घ) १. १८१७ (ग) (च) for (१८८७)
२. २७१९ (ग) (च) for (२७१४)
३. २६३२ (ग) (च) for (२६७२)
(क) यहां यह सब लुप्त हैं।

लिप्तास्तत्त्वयमहृता २२५ लब्धं द्याषष्ट्यंतराहता छेषात् ।
तिथिकृति २२५ हृत्फलयुता लब्धषष्ट्या द्वा ग्रहणमेवम् ॥ १० ॥
ज्यां प्रोह्य शेषगुणितास्तत्त्वयमा ज्यांतरा धृतालब्धम् ।
क्षेप्यं विशुद्धजीवासंख्यातिथि कृतिवधे चापम् ॥ ११ ॥

१०. (घ) १. ज्याष्टचांतरा (ग) ज्याज्यांतरा (क) ज्याज्यांतरा (च) for (द्याषष्टचांतरा)
२. हृतात् (क) for (हृत्)
३. ष्टचा (ग) ज्या (क) (ख) षष्ट for (षष्टचा)
४. ज्या (ग) ज्या (क) (ख) ज्या for (द्वा)

(ग) ५. मलहृता (ख) यमाहुता for (यमहृता)
६. २२५ हृतात्फल for (हृत्फल)
७. लब्धं for (लब्ध)

(क) २. हृतात्फलयुतात् for (हृत्फलयुता)

(ख) १. लब्धद्याधृचां for (लब्धद्याषष्टचां)
८. ग्रहणमेव for (ग्रहणमेवम्)

(च) ६. २., २२५ हृतात्फलयुता for (२२५ हृत्फलयुता)
३. ष्टचा for (षष्टचा)
४. छा for (द्वा)

११. (घ) १. ज्यांतरो (क) ज्यांतरोद्धृता (च) ज्यांतरोधृता for (ज्यांतराधृता)
२. संक्ष्या for (संख्या)

(ग) ३. गुणितात्तत्वयमा २२५ for (गुणितास्तत्त्वयमा)
४. +न+
५. +२२५+

(क) ६. सेष (ख) सेष for (शेष)

(ख) ७. प्रयोज्यय for (प्रोह्य)
८. क्षेपं for (क्षेप्यं)

(च) ७. प्रोज्य for (प्रोह्य) ८. क्ष्येप्यं for (क्षेप्यं)

मध्याद्विशोध्य मंदं शीघ्रात्संशोध्य मध्यमं केन्द्रम् ।
अयुतिगता यया यो युति पदेऽन्यथा बाहुकोटिज्ये ।। १२ ।।
त्रिज्याहता भुजज्या २२७० युगयुक्परिधि द्व्यंतर गुणाप्त्या ।
युग्मांत परिधिरधिको हीनोऽधिकः हीनोधिकः स्पष्टः ।। १३ ।।
तद्गुणितेज्ये भांशैः ३६२ हृते फले कोटिफलयुता त्रिज्या ३२७० ।
आद्यन्तयोर्निहीनो पदयोर्द्वितृतीययोः कोटिः ।। १४ ।।

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१२. (घ) १. व्ययुयगतयेयोर्युजि (ग) अयुजिगतयेयोर्युजि for (अयुतिगतायययायो युति)

(क) १. अयुजिगतैर्योज्ययोर्युजि for (अयुतिगतायययायोयुति)

(च) १. अयुतिगर्त for (अयुतिगता) २ या for (यो) ३ र्युति for (युति)

४. अवग्रहचिह्नलुप्त

१३. (घ) १. द्वयांतर (ग) द्वयांर (क) द्वयांतर (ख) द्वध्वयांतर for (द्व्यांतर)

२. गुणात्प्त्या (ग) गुणाप्त्या for (गुणात्प्या)

३. परिधिको हीनो हीनाधिकः for (परिधिरधिको हीनोऽधिकः हीनोऽधिकः)

(ग) ४. ३२७० हृता (क) त्रिज्याहृता (च) for (त्रिज्याहता)

५. यह संख्या अंकित नहीं ।

६. 'ऽधिकः' शब्द अंकित नहीं (क)

(क) ७. 'युग्मांतर for (युग्मांत)

(ख) ६. 'हीनोऽधिकः' पद लुप्त हैं

८. हीनाधिकः for (हीनोधिकः)

(च) ५. २७७० for (२२७०) २+८+

३. परिधिकोहीनो for (परिधिरधिको हीनोऽधिकः)

१४. (घ) १. विहीना (ग) विहीना (क) विहीना for (निहीनो)

(ग) २. हृंते (क) for (हृते)

३. द्वित्रितीययोः for (द्वितृतीययोः)

(ख) १. विहीनो for (निहीनो)

(च) १. विहीना for (निहीनो)

तद्भुजफल कृतियोगान्मूलं कर्णः[१] पदेष्वयुग्युक्षु[५] ।
स्वपरिधिगुणा[२] क्रमोत्क्रमजीवा[५] भांशै[३] ३६० र्हृतान्मन्दे[४] ॥ १५ ॥
क्षयधनधनक्षयास्तत्फलानि शीघ्रेऽन्यथा धनं धनयोः ।
ऋणमृणयोर्योगान्तर[२]मृणधनयोस्तुल्ययोः[१] शून्यं[३] ॥ १६ ॥
तच्चापं मन्दफलं फलयोगान्तरवशाद्धनमृणं[३] वा ।
शीघ्रफलं तद्गुणिताद्व्यासार्द्धात्[२] कर्णलब्धधनुः[१] ॥ १७ ॥ ३२७०
देशान्तरे खमध्ये[१] भुजफलचापे भुजांतरे च कृते ।
उन्मण्डलेऽर्क[२] चन्द्रौ स्पष्टौ रविचरदले क्षितिजे ॥ १८ ॥
अर्कोदयास्तमययोर्विना[१] चरार्द्धेन रात्रिदिनदलयोः[२] ।
न स्फुटमार्यभटोक्तं स्पष्टीकरणं स्फुटोक्तिरतः ॥ १९ ॥

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१५. (घ) १. कर्णा पटेष्वग्युक्षुः for (कर्णः पदेष्वयुग्युक्षु)
२. गुणाः (च) for (गुणा)
३. म्यांशे for (भांशै ३६०) (ग) र्हृताफलं मंदे for (र्हृतामन्दे)
४. मांदैः (क) र्हृतामांदो (ख) र्हृतामन्दे for (र्हृतान्मन्दे)
(क) ५. पदेष्वयुक् युक् स्यात् for (पदेष्वयुग्युक्षु)
(ख) ६. क्रमोक्रमजीवा for (क्रमोत्क्रमजीवा)
(च) १. कर्णपदेष्वयुक्षः for (कर्णः पदेष्वयुग्युक्षु)
३. ज्यां (भां) शै for (भांशै) ४ हृतामंदै for (हृतान्मन्दे)
१६. (घ) १. धनयोः for (धनयोस्)
(ग) २. योगोन्तर for (योगान्तर)
३. तुल्ययो for (तुल्ययोः)
१७. (घ) १. गुणुता for (गुणिताद्)
(ग) २. व्यासार्द्धत् (क) व्यासोर्द्धात् for (व्यासार्द्धात्)
(क) ३. मृणंवति for (मृणंवा)
(च) १. तद्गुणितोद् for (तद्गुणिताद्)
१८. (घ) १. स्वमध्यो (ग) स्वमध्यै (क) स्वमध्यौ (ख) स्वमध्यो for (खमध्ये)
(च) १. स्वमध्यो for (खमध्ये)
२. क्कौ for (ऽर्के)
१९. (ख) १. अर्कोदयास्तममयोर्विना for (अर्कोदयास्तमययोर्विना)
२. राभि for (रात्रि)
(च) १. अक्कोदयास्तमययोर्विना for (अर्कोदयास्तमययोर्विना)

सूर्यस्य मनुद्वितयं त्रिंशोनं दिनदलेन तस्य प्राक् । $\frac{१}{४}\frac{३}{०}$
तिथिघटिकाभिस्त्रयंशाधिकोनसूनाधिकं पश्चात् । $\frac{०}{३०}$ ॥ २० ॥
द्युदले जिनलिप्तोनं दशनद्वितयं $(\frac{३}{३}\frac{१}{६})$ द्विशर ५२ कलोनु प्राक् ।
पश्चाद्युतोनमिंदोः सूर्यवहृण धन परिध्यंशाः ॥ २१ ॥

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२०. (घ) १. $\frac{१}{४}\frac{३}{०}$।$\frac{१}{४}\frac{३}{०}$ (ग) for $(\frac{१}{४}\frac{३}{०})$

२. तिथि +१५+(ग) for (तिथि)

३. त्र्यंशा (ग) स्त्रियशाधिकोन ।१४।०॥१३।२०॥ for (स्त्रयंशाधिकोन)

(ग) ४. त्र्यंशोनं (क) for (त्रिंशोनं)

५. पश्चात् व for (पश्चात्)

(च) २. +१५+ १ संख्या लुप्त

३. त्र्यं $\frac{३}{४}\frac{३}{०}$॥$\frac{१}{४}\frac{३}{०}$॥ शाधिको for (त्रयंशाधिको)

२१. (घ) १. कलोनं (ग) (क) (ख) कालोनं for (कलोनु)

२. यहां '५२' संख्या जो 'द्विशर' के बाद अंकित है, लिखी है।

३. मंदोः for (मिंदोः)

(ग) ५. संख्या अंकित नहीं है।

६. संख्यां अंकित नहीं है।

(क) ७. सूर्याहृण for (सूर्यवहृण)

(ख) ८. द्युदलेतिन for (द्युदलेजिन)

(च) ६. संख्यालुप्त १ कलोनं for (कलोनु)

२. +५२+ ७ सूर्यवहृण for (सूर्यवहृण)

तद्युदलपरिध्यंतर गृणाकृता त्रिज्यया स्वनतजीवा ।
ऊने धनमृणमधिके दिनार्द्धं परिधौ स्फुटः परिधिः ॥ २२ ॥
भुजफलचापं केन्द्रे षड्राश्यूने खावृणं मध्ये ।
स्वभुजबलचापमिन्दौ षड्राश्यधिके धनं भवति ॥ २३ ॥
देशान्तराद्यमेवं स्पष्टीकरणं दिनार्द्धपरिधिभ्याम् ।
कृत्वा तत्तिथ्यन्त स्फुटपरिधिभ्यां स्फुटावसकृत् ॥ २४ ॥

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२२. (घ) १. हृता (ग) (क) (च) for (कृता)

२. तृज्यया for (त्रिज्यया) (क) त्रिज्ययाश्च नतजीवा for (त्रिज्यया स्वनत-जीवा)

(ग) यहां निम्नांकित संख्या अधिक है—

३१	३१	५२	३०	३२	३०
३६	३६		४४	२८	४४
				८	

(क) ३. तत्

४. मधिका for (मधिके)

(ख) ३. तद्युदलं for (तद्युदल)

(च) ६. परिघो for (परिधौ)

२३. (घ) १. षड्राशून्ये (च) for (षड्राश्यूने)

२. फल (च) for (बल)

(ग) ३. षड्राशधिके (क) षड्रास्यधिके (ख) षड्राश्याधिके for (षड्राश्यधिके)

२४. (क) १. तत्तिथ्यन्तः (ख) तत्तिथ्यन्तं for (तत्तिथ्यन्त)

प्राक् पश्चाद्वा याभिर्घटिकाभिर्दिनदलान्नतः[3] सूर्यः[4] ।
तिथ्यंते तद्रहिते त्रिंशत्[2] घटिकावशेषाभिः[5] ॥ २५ ॥
विपरीतमर्धरात्राच्चन्द्रग्रहणे[2] शशी रविग्रहणे ।
सूर्योयतस्ततस्ताभिरेव[3] घटिकाभिरिंदुरपि[1] ॥ २६ ॥
दिनदल[1] परिधि स्फुट तिथिनतकेन्द्रज्यावधो[2] गुणोर्केन्दोः[3] ।
इंदू[4] धृति धृतिभि १९१ नवनववेदै[5] ४९९ व्यासार्द्धकृतिर्भक्तः[7] ॥ २७ ॥
१०६९२९००[6]

२५. (घ) १. तद्रहिव for (ग) (तद्रहिते) (ग) तद्रहितस्त्रिंशद् for (तद्रहिते त्रिंशत्)(क) तद्रहितं for (तद्रहिते)
२. त्र्यंशद् (क) त्रिदशत for (त्रिंशत्)
(ग) ३. नतसूर्यः for (नतः सूर्यः)
(ख) ४. घटिकाभि for (र्घटिकाभि)
५. त्रिंशदधिकावशेषाभिः for (त्रिंशत्घटिकावशेषाभिः)
(च) ३. द्दिनदलान्नत् सूर्यः for (दिनदलान्नतः सूर्यः)
२. त्रिंशद् for (त्रिंशत्)
२६. (घ) १. यतस्वषः for (यतस्ततस्)
२. चन्द्रग्रहणे (क) चन्द्रग्रहादिनिग्रहाणः for (चन्द्रग्रहणे)
३. रेवमथ for (रेव)
(च) २. चन्द्रग्रहणो for (चन्द्रग्रहणे)
२७. (घ) १. दज for (दल)
२. व्या for (ज्या) (क) ज्यावाधो for (ज्यावधो)
३. कैंन्दोः (ग) (ख) कैंन्द्रो for (कैंन्दोः)
४. इंद्वृतिधृतिभि (ग) (क) इंदुविधृतिभि for (इंदूधृतिधृतिभि)
५. भक्तः for (र्भक्तः) (ग) (ख) कृतिभक्ति for (कृतिर्भक्तः)
६. १०६९ ॥ २९०० for (१०६९२९००)
(ग) ७. व्यासार्द्धकृति (क) व्यासार्द्धकृते for (व्यासार्द्धकृति)
वि० रेखाङ्कित दोनों संख्याएं "वेदै" के पश्चात् एक स्थान पर हैं। (१९१, ४९९)
(क) ३. गुणौर्केस्तः for (गुणोर्केन्दोः)
८. नंवनववेदै (च) for (नवनववेदै)
वि० संख्याएं मूल में लुप्त हैं।
(च) ३. कैंद्वोः for (कैंन्दोः) ५. भक्तः for (र्भक्तः)

फलविकला[४] वा सूर्ये[५] प्रागृणमसकृन्नते[६] धनं पश्चात् ।
केन्द्रफलमृणं[७] चन्द्रेऽन्यथा[३] धनं चेहृणं[२] स्पष्टौ ॥ २८ ॥
अर्कफलभुक्तिघाताद्भगणकलाप्तं[१] भुजांतरं[२] रविवत् ।
स्फुटभुक्तिरतीतैष्य[४] ग्रहान्तरं वर्तमानेऽह्नि[३] ॥ २९ ॥
क्षयधनहानिधनानि[१] प्राक् पश्चादन्यथा खेरिंदोः ।
प्राग्वत्पश्चात्स्वगतौ[३] धनक्षयक्षयधनानि प्राक्[२] ॥३०॥
ब्रह्मोक्तमध्यरविशशिततदुच्चतत्परिधिभिः[४] स्फुटीकरणम् ।
कृत्वैन[१] स्पष्टतिथिर्दृ[२]रभ्रष्टान्यतंत्रोक्तैः[३] ॥ ३१ ॥

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२८. (ग) १. धननं for (धनं)
२. चेहृणं (क) मे ऋणं (ख) चेहृणं for (चेहृणं)
(क) ३. चा प्रात्यपा for (चन्द्रेऽन्यथा)
(ख) ४. विकल for (विकला)
५. सूर्यो for (सूर्ये)
६. मासत्कृते for (मसकृन्नते)
७. धृतिभिर्नवनववेदौ for (केन्द्रफलमृणं)
(च) २. चेहृणं for (चेहृणं)

२९. (घ) १. भरणे for (भगण)
२. +२१६०+(च)+२१६००+
(क) ३. ह्नि: for (ह्नि)
(ख) ४. रतीतैष्यद् for (रतीतैष्य)
३. ऽहनि for (ऽह्नि)
(च) ५. अर्क्क for (अर्क)
४. रतीतैष्टचद्ग्रहांतरं for (रतीतैष्य ग्रहान्तरं)
३. ह्नि: for (ऽह्नि)

३०. (क) १. 'धनानि' पद लुप्त है ।
२. प्राग्वत् for (प्राक्)
(ख) ३. +च+

३१. (घ) १. कृत्वैवं (ग) (क) (च) for (कृत्वैनं)
२. स्पष्टतिथे for (स्पष्टतिथि)
३. च for (भ्र)
(क) ४. स्फुटाकरणम् for (स्फुटीकरणम्)
५. तंत्रौक्तो for (तंत्रोक्तैः)

स्वदिनार्द्धं परिधिभुजफलचापं मध्यार्कचन्द्रयोः[२] कृत्वा ।
पूर्ववदन्यत्स्पष्टं[१] संव्यवहारार्थमेवं वा ॥ ३२ ॥
आर्यभटस्याज्ञानान्मध्यममन्दोच्चशीघ्रपरिधिना[१] ।
न[२] स्पष्टा भौमाद्याः[३] स्पष्टा[४] ब्रह्मोक्तमध्याद्यैः[५] ॥ ३३ ॥
मंदोच्चनीचवृत्तस्य परिधिभागाः सितस्य विषमांते ।
नव ९ युग्मांते[४] रुद्राः ११ शीघ्रौ[३] जान्तेऽग्निरसयमलाः[१][२] ॥ २६३ ॥३४॥
युग्मांतेष्टशरयमाः[४][५] २५८ मंदफलात्मध्यमः[१][२] स्फुटो मध्यः ।
शीघ्रफलात्स्पष्टो सकृदेवं स्वफलैर्ज्ञगुरुसौराः[३] ॥ ३५ ॥

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३२. (घ) १. स्पष्टो (ग) स्पष्टौ (क) स्पष्टौ for (स्पष्टं)
(च) २. मध्यार्क्कं for (मध्यार्क)

३३. (घ) १. परिधीनाम् (ग) (क) (च) for (परिधिना)
२. त for (न)
३. भौमाछा for (भौमाद्याः)
(क) ४. स्पष्ट for (स्पष्टा)
(च) २. ते for (न) ३ भौमाद्या for (भौमाद्याः)
५. क्रमसंख्या लुप्त

३४. (घ) १. यमलाः (च) for (यमलाः)
२. ३४।२६३ for (२६३।३४) (क) संख्या लुप्त है।
(क) ३. शीघ्रो (च) for (शीघ्रौ)
(ख) यह श्लोक यहां ३० वें श्लोक के पश्चात् है। यद्यपि इस प्रति में श्लोक संख्या नहीं दी गई है।
(च) ४. विसर्गलुप्त ५. अवग्रहचिह्न लुप्त

३५. (घ) १. फलान् (ग) (क) (ख) मंदफला for (मंदफलात्)
(ग) २. मध्यम for (मध्यमः)
(क) ३. सौरः for (सौराः)
(ख) ४. युग्मांतिष्ठ for (युग्मांतेष्ट)
५. शरयमा for (शरयमाः)
(च) १. फलान् for (फलात्)

बुधमंदपरिधिभागावसुरामाः ३८ सुरगुरोखयंखिशत् ।
रविजस्यशून्यरामा ३० सशीघ्रपरिधिद्विगुणचन्द्राः ॥ ३६ ॥
देवगुरोरष्टरसा ६८ भास्कर पुत्रस्य शरगुणाः ३५ स्पष्टाः ।
कुजशीघ्रकेन्द्रपदगत ये अल्पज्याभिभागोनैः ॥ ३७ ॥

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३६. (घ) १. बुमंदपरिधि for (बुधमंदपरिधि)

२. +३३+(ग) (च)

३. ज्ञ (ग) (क) (ख) (च) for (स)

४. द्विगुण (ग) for (द्विगुण)

५. +१३२+(ग) (च)

(ग) ६. रामा for (रामाः)

(क) वि० मूल में अंकित संख्याएं यहां लुप्त हैं ।

(ख) ७. रंशा for (भागा)

८. त्रयस्त्रिंशत् for (त्रयस्त्रिंशत्)

९. चन्द्रा for (चन्द्राः)

(च) १०. रामाः for (रामा)

३७. (घ) १. रस्य (क) रस for (रसा)

२. शरगुणा स्पष्टाः । ३५ for (शरगुणाः ३५ स्पष्टाः)

३. याल्पज्या (ग) (क) for (अल्पज्या)

४. त्रि (ग) (क) (च) for (त्रि)

५. नौः for (नैः)

(च) २. शरगुणस्पष्टाः ३५ for (शरगुणाः ३५ स्पष्टाः)

३. याल्पद्या for (अल्पज्या)

सप्तभिरंशैर्गुणिता[१] ६।४०[६] दिवर्द्ध[७] राशिज्यया हृताप्तांशैः[८]।
[१]
२३१
अधिकोन[३]कुजमंदो मृगकर्कादौ[९] स्फुटौ[४] भवतीति[५] ॥ ३८ ॥
तत्स्फुट[८]परिधिः खनगाः ७०[१] शीघ्रस्फुटपरिधिराप्त[६]भागोनाः[४]।
वेदजिनार्कांशोनाः[५] २४३।३०[२] स्फुटीकरणं[३] कुजस्यैवम् ॥ ३९ ॥

३८. (घ) १. गुंणिताः (च) for (गुंणिता)
२. २३१२ (ग) (च) for (२३१)
३. नः (ग) (क) (च) for (न)
४. स्फुटो (ग) (क) (ख) (च) for (स्फुटौ)
५. भवति (ग) (क) भवंति (ख) भवति (च) for (भवतीति)
(ग) ६. यहाँ यह संख्या नहीं है (च) '४०' संख्या लुप्त
(क) ७. दिवार्द्ध for (दिवर्द्ध)
वि० मूल में दी गई संख्याएं लुप्त हैं।
(ख) ८. हतासाशः for (हृताप्तांशैः)
९. ष्टगकवचादौ for (मृगकर्कादौ)
(च) ९. मृगकङ्कादौ for (मृगकर्कादौ)

३९. (घ) १. ७०४० (च) for (७०)
२. २४३।४० (ग) (च) for (२४३।३०)
३. स्पटीकरणं (ग) स्पष्टीकरणं (ख) for (स्फुटीकरणं)
(ग) ४. रासभागोना for (रासभागोनाः)
५. स्त्रिशोनाः (ख) स्त्रांशोना for (स्त्रिशोनाः)
(क) ६. भागाः स्पष्टाः। (कुजशीघ्रस्फुट परिधिभागाभवंति)
७. दूसरी पंक्ति—के ते वेदस्त्रंशोना एते $\frac{२४३}{३०}$ परं आप्तं भागोनाः ॥
वि० ऐसा प्रतीत होता है कि लिपिकार मूल श्लोक लिखना भूल गया।
केवल टीका उपलब्ध है।
(ख) ६. रासो ना for (रासभागोनाः)
(च) ८. तेत्सुट् for (तत्स्फुट) ९ परिधिः for (परिधि)
५. स्त्र्यंशोनाः for (स्त्रिशोनाः)

मन्दफलं मध्येऽर्द्ध तच्छीघ्रफल च मध्यमे सकले ।
मध्ये सकृत् क्षितिसुतः स्पष्टो भुक्ति स्फुटाग्रहवत् ॥ ४० ॥
ग्रहकेन्द्रभुक्तिज्याकरगुणिताद्यजीवया भक्ता २१४ ।
लब्धं स्फुट परिधिगुणं भगणांशहृतं कलाभिस्तु ॥ ४१ ॥

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४०. (घ) १. फलं (ग) (क) (ख) फलस्य for (फल)

२. सकलो (ख) (च) for (सकले)

३. मध्येऽसकृत् (क) मध्यशकृत for (मध्येसकृत्)

४. भुक्तिः (ग) (च) for (भुक्ति)

६. क्षितिसुते (ख) for (क्षितिसुतः)

७ स्फुटौ for (स्फुटा)

(ख) ८. धर्ा (च) र्द्ध for (र्द्ध)

(च) ९. तत्शीघ्र for (तच्छीघ्र)

४१. (घ) १. ग्रहमंदकेन्द्र (ग) (क) ग्रहमंद्रकेंद्रा (ख) ग्रहमंदकेन्द्र for (ग्रहकेन्द्र)

२. ज्यतिर (ग) ज्यांतर (क) ज्यतिर (ख) ज्यांतर for (ज्याकर)

३. फलकबाभिः । (ग) (क) फलकलाभिः for (कलाभिस्तु)

(ग) ४. द्युजीवया (क) (ख) गुणिताद्या for (द्यजीवया)

(ख) ५. भक्ताः for (भक्ता)

६. स्फुटि for (स्फुट)

३. फलकलाभि for (कलाभिस्तु)

(च) १. ग्रहमंदकेंद्र for (ग्रहकेन्द्र)

२. ज्यांतर for (ज्याकर)

३. फलकबभिः for (कलाभिस्तु)

मृगकर्काद्यादूनाधिका स्वमध्यमगतिः स्फुटार्केंदोः ।
शीघ्रगतिमंदफलस्फुटभुक्तच्युनां कुजादिना ।। ४२ ।।
शीघ्रफलं भोग्यजीवागुणिता माघजीवया विभजेत् ।
फलगुणितं व्यासार्द्धं विभाजयेत् शीघ्रकरणेन ।। ४३ ।।

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४२. (घ) १. मृगकर्क्यादावूना (ग) (क) मृगकर्काद्यवूनाधिका for (मृगकर्काद्यादूनाधिका)
२. स्फुटार्केंद्वोः (ग) (क) स्फुटार्केंदोः for (स्फुटार्केन्दोः)
३. शीघ्रगतिं (क) शीघ्रगतिः for (शीघ्रगति)
४. भक्तूनां (ग) भुक्तच्यूनां (क) भुक्तच्यूनां (ख) for (भुक्तच्युनां)
५. कुजादीनाम् । (ग) (क) (ख) कुजादीनाम् for (कुजादिना)

(ग) ६. स्वमध्या for (स्वमध्यम)
७. मंदफल (ख) मदे for (मंदफल)

(ख) इस श्लोक का पूर्वार्ध लुप्त है—
८. शीघ्रगतिः for (शीघ्रगति)
९. 'फल' पद लुप्त है ।

(च) २. स्फुटाङ्केंद्वोः for (स्फुटार्केंदाः)
४. भुक्तच्यूनां for (भुक्तच्युनां)
५. कुजादीनां for (कुजादिना)

४३. (घ) १. शीघ्रफल (ग) (क) (ख) शीघ्रफल (च) for (शीघ्रफलं)
२. शृंगुणिता (ग) संगुणिता (क) संगुणिता (ख) for (गुणिता)
३. भजेत् for (विभजेत्)

(ग) ४. च्छीघ्र (च) for (त् शीघ्र)
५. विभजयेत् (ख) विभाजये (च) for (विभाजयेत्)

(ख) ६. व्यासाघ्यं for (व्यासार्द्धं)
७. शीघ्रकरणेना for (शीघ्रकरणेन)

(च) २. शृंगुणिता for (गुणिता) ८. माद्य for (माघ)

लब्धोनाशीघ्रगतिस्फुटभुक्तिर्लब्धमधिकं चेत् ।
शीघ्रगतेः शीघ्रगतिः लब्धा संशोध्य वक्रगतिः ॥ ४४ ॥
देयमसुताय नेदं शपथैरपि दत्तसुकृतनाशाद्यैः ।
यात्राविवाहजातक फलस्फुटत्वं यतः स्पष्टैः ॥ ४५ ॥
मेषादितः प्रवृत्तानार्यभटस्य स्फुटाः युगस्यादौ ।
श्रीषेणस्य कुजाद्याः सूर्याद्यविष्णुचन्द्रस्य ॥ ४६ ॥
दूरभ्रष्टाः स्पष्टा श्रीषेणार्यभटविष्णुचन्द्रेषु ।
यस्मात् कुजादयस्तेषु न विदुषामादरस्तस्मात् ॥ ४७ ॥

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४४. (घ) १. शीघ्रगतिः (ग) (क) गति (ख) शीघ्रगतिः (च) for (शीघ्रगनि)
२. भवति लब्ध (ग) भवति लब्ध (ख) भवनि लब्ध for (लब्ध)
३. शीघ्रगति (ग) (क) (ख) शीघ्रगतं for (शीघ्रगतिः)
(क) ४. लब्धात् for (लब्धा)
(ख) ५. संसोध्य for (संशोध्य)
६. वक्तगतिः for (वक्रगतिः)
(च) २. +भवति+
३. शीघ्रगति for (शीघ्रगतिः)

४५. (घ) १. जातकं (ख) जातव for (जातक)
२. स्पष्टै for (स्पष्टैः)
(क) ३. नाशोचैः (ख) नशाद्यैः for (नाशाद्यैः)
४. यत्र for (यात्रा)
(च) ५. नेदशपंथै for (नेदं शपथै)

४६. (घ) १. स्फुटा (ग) (क) (ख) स्फुटा (च) for (स्फुटाः)
२. ज्या (ग) द्या (ख) द्याः for (द्य)
(ख) ३. विष्नचंद्रस्य for (विष्णुचन्द्रस्य)
(च) २. सूर्याद्या for (सूर्याद्य)

४७. (घ) १. (चट) for (भट)
(ग) २. स्पष्टाः (ख) for (स्पष्टा)
(ख) ३. दूरम्यष्टः for (दूरभ्रष्टाः)
४. 'आर्य' लुप्त है
५. चन्देषु for (चन्द्रेषु)

अग्न्यष्टभिरिषुमनुभिः शरसूर्यैरिषु रसेंदुभिस्त्रिभवैः ।
शीघ्रान्त्यकेन्द्रभागैर्भौमादीनां भवति वक्रम् ॥ ४८ ॥
वक्रांशकैस्तदूनैः ३६० रनुवक्रं तदधिकोन भागकलाः ।
मंदफलस्फुट भुक्तच्यून शीघ्रभुक्तच्याहृता दिवसाः ॥ ४९ ॥
शीघ्रस्फुटाग्रहोना छेषे मध्यस्फुटांतरार्द्धं वा ।
अधिकेधनमृणमूने स्फुटग्रहा मध्यमे कृत्वा ॥ ५० ॥

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४८. (ग) १. अग्निष्टिभि (क) अग्न्यष्टिभि (च) for (अग्न्यष्टभि)
२. रिषुमुनिभिः for (रिषुमनुभिः)
३. +भौ० व० कें० ॥ ५ । १३ । ० ॥ बु० व० कें० ॥ ४ । २५ । ० । ० ॥ बृ० व० कें० ॥ ४ । ५ । ० । ० ॥ शु० व० कें० ॥ ५ । १५ । ० । ० ॥ श० व० कें० ॥ ३ । २३ । ० । ० ॥+

(ख) १. अग्न्याष्टिभि for (अग्न्यष्टभि)
४. रसेद्रुभि for (रसेंदुभि)
५. क्रर्म for (वक्रम्)

(च) २. रिषुमनुभि for (रिषुमनुभिः)

४९. (क) १. चक्रांशकै for (वक्रांशकै)

(ख) २. वक्तं for (वक्रम्)
३. भुक्तच्यान for (भुक्तच्यून)

५०. (घ) १. स्फुट (ग) (क) for (स्फुटा)
२. ग्रहान् (ग) (क) ग्रहान् for (ग्रहा)

(क) ३. शीघ्रात् (ख) for (शीघ्र)
४. ग्रहोनात् for (ग्रहोना)

(ख) ५. च्छेषे for (छेषे)
६. स्फुटार्धं वा for (स्फुटांतरार्द्धं वा)
७. नष्टृणमूने for (नमृणमूने)

(च) २. स्फुटग्रहान् for (स्फुटग्रहा)
८. यहां क्रम संख्या भूल से "१५०" अंकित है।
वि० यहां पर जो सारणी दी गई है। वह मूल पाठ में ५२ वें श्लोक में दी गई है।

राशिषु चतुर्षु वक्रं षट्स्वति वक्रमनुवक्रमष्टासु । ४ । ५ । ८
अप्राप्ता प्रीतीतकाला भुक्त्या सैवोद्धृता दिवसाः ॥ ५१ ॥
अष्टयमैः कृतचन्द्रैर्मुनीन्दुभिर्भौमजीव रविजानाम् ।
उदयप्रागस्तमयस्तदूनवक्रांशक पंचात् ॥ ५२ ॥

५	४	४	४	३
१३	२५	५	१५	२३
०	०	०	०	०
०	०	०	०	०

५१. (घ) १. वंक्रं for (वक्रं)
२. ४ । ६ । ८ (ग) (च) for (४ । ५ । ८)
३. अप्राप्तातीतकला (ग) अप्राप्तातातकला (ख) अप्राप्तातीतकाला for (अप्राप्ताप्रीतीतकाला)
४. स्यैवौद्धृता (ग) (क) (ख) for (सैवोद्धृता)
(ग) ५. दिवसा (ख) (च) for (दिवसाः)
(क) ३. अप्राप्ताऽतीतकला for (अप्राप्ताप्रीतीतकाला)
(ख) ६. षटत्वति for (षट्स्वति)
७. 'नु' पद लुप्त है ।
८. भक्त्या for (भुक्त्या)
(च) ६. 'षट्स्व' लुप्त ० अप्राप्तातीतकला for (अप्राप्ताप्रीतीतकाला)

५२. (घ) १. वक्रांशकैः (ग) (क) चक्रांशकैः (ख) वक्तांशकैः for (वक्रांशकैः)
२. पश्चात् (ग) (क) (ख) पश्चात् for (पंचात्)
३. यहां यह तालिका ५०वें श्लोक के साथ है (ख) लुप्त है
(ग) ४. अष्टमैः (ख) अष्टयमैं for (अष्टयमैः)
५. भौमजा (क) भौम for (भौम)
६. +२८ । १४ । १७ +
७. उदयः (ख) for (उदय)
३. यहां यह तालिका नहीं दी हुई है (क)
(क) ८. रविजानाम् for (रविजानाम्)
(ख) ९. कृतचन्द्रैः for (कृतचन्द्रैं)
१०. मुनिदुभौम for (मुनीन्दुभिर्भौम)
११. तादृन for (तदून)
(च) १. वक्रांशकैः for (वक्रांशकैः) २ पश्चात् for (पंचात्)

८ ५
खशरैर्जिनैर्ज्ञसितयोः २८ । १४ । १७ । ५० । २४ रिषुतिथिभिर्मुनिनगै-
१
दुभिः पश्चात् । १५५, ११७७
६ २ ३ ७ ४
उदयास्तमयो व्यस्तौ मंडलभागैस्तदूनैः प्राक् ॥ ५३ ॥
१ २ ६
स्पष्टाद्युराभिदलयोरव्यूदयान्तमये यौ रविचरार्द्धात् ।
७ ८ ३ ५ ४
एष्यत्यधिके तीतादवक्रिते हीनौ ॥ ५४ ॥

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५३. (घ) १ १५५११७७ (ग) १५५ । १७७ for (१५५, ११७७)

२. न्यस्तौ for (व्यस्तौ)

३. मंडभागैः for (मंडलभागै)

४. प्राकः for (प्राक्)

(ग) ५. ज्ञसितयोः । ५० । २४ ॥ (ख) ज्ञासितयो (च) for (ज्ञसितयोः)

६. उदयास्तमयौ (क) (ख)

७. +३६०+

(क) वि० संख्याएँ मूल में नहीं दी गई हैं।

(ख) ८. स्वशरै जिनै for (खशरैर्जिनै)

५४. (घ) १. त्रि for (भि) (ग) (क) (ख) रात्रि for (राभि)

२. रव्युदयोस्तमययो (ग) रव्युदयास्तमययो (ख) for (रव्यूदयान्तमये यौ)

३. क्रितो (ग) (क) (च) for (क्रिते)

४. हीने (ग) (क) for (हीनौ)

(क) २. रव्युदयास्तमययो for (रव्यूदयान्तमये यौ)

५. +वक्रितो+

(ख) ६. रविचराघात् for (रविचरार्द्धात्)

७. एह्यात्यधिके for (एष्यत्यधिके)

८. तीतादेवक्रितो for (तीतादवक्रिते)

(च) २. रव्युदयास्तमये for (रव्यूदयान्तमये)

जिनभागज्या[६] गुणिता सूर्याद्या[७] व्यासदलहृतालब्धम्[२] । ३२७०[३]
इष्टापक्रमजीवा[४] विषुवदुग्[५] दक्षिणा[८] सवितुः[६] ॥ ५५ ॥
इष्टापक्रमवर्गमिज्यावर्गाद्विशोध्य[१] १०६९१९०० शेषपदम्[५] ।
विषुवदुदग्[७] दक्षिणतः[३] स्वाहोरात्र्यार्द्ध[४] विष्कंभः[८] ॥ ५६ ॥

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५५. (घ) १. +१३२९+(च)
२. +३२७०+(ग) (च)
३. यह संख्या यहाँ पर 'हृता' के पश्चात् है ।
४. दृष्टा for (इष्टा)
५. विषुवदुदिग्दक्षिणा (ग) विषुवदुदिग्दक्षिणे for (विषुवदुग्दक्षिणा)
६. सदितु (ख) सदितु (च) for (सवितुः)

(ग) ७. +१३२९+सूर्यज्या (क) सूर्याज्या (ख) सूर्यज्य for (सूर्याद्या)

(क) ८. दक्षिणे (ख) दक्षिण for (दक्षिणा)

(ख) ९. जिनभागाज्या for (जिनभागज्या)

(च) ५. विषुवदुदग्दक्षिणा for (विषुवदुग्दक्षिणा)

५६. (घ) १. वर्ग त्रिज्या (ग) (क) (ख) वर्गत्रिज्या for (वर्गमिज्या)
२. वर्ग्राद्विशोध्य (ख) वर्गाद्विशेषपदम् for (वर्गाद्विशोध्य)
३. दक्षणतः (च) for (दक्षिणतः)
४. स्वाहोरार्द्ध (ग) स्वाहोरात्रार्द्ध (क) स्वाहोरात्रार्द्ध for (स्वाहोरात्र्यार्द्ध)

(ग) ५. +१०६९२९००+

(ख) ६. इष्टमपक्रम for (इष्टापक्रम)
७. 'ग्' लुप्त
४. स्वाहोरात्रार्घ for (स्वाहोरात्र्यार्द्ध)
८. विष्कम्रः for (विष्कंभः)

(च) १. वर्ग त्रिज्या for (वर्गमिज्या)
४. स्वाहोरार्द्ध for (स्वाहोरात्र्यार्घ)

क्रान्तिज्या विषुवच्छायया[१] गुणा[६] द्वादशो[२] धृता क्षितिजा ।
स्वाहोरात्रेऽनष्टा[३] व्यासार्धेनाहता[५] भक्ता[४] ॥ ५७ ॥
स्वाहोरात्रार्द्धेन[४] क्षयवृद्धिज्याधनुश्चरप्राणाः ।
षट्कोधृता[१] विनाड्यो[३] विनाडिका[५] नाडिका[६] षष्ट्या[२] ॥ ५८ ॥
चरदलघटिका[३] गुणिता[६] भुक्तिः[७] षष्ट्याहता[१] कलाद्याप्तम् ।
ऋणमुदयेऽस्तमये[४] धनमुत्तरगोलेऽन्यथा[५] याम्ये[२] ॥ ५९ ॥

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५७. (घ) १. त्सायाया (ग) छायया (ख) विषुचछाया for (विषुवच्छायया)
२. द्धृता (ग) (ख) हृता for (धृता)
३. स्वाहोरात्रेनष्टा (ख) (क) स्वाहारात्रेनष्टा for (स्वाहोरात्रेऽनष्टा)
४. भक्ताः (ग) for (भक्ता)
(ग) ५. व्यार्द्धेनाहता (ख) सार्धेन हता for (व्यासार्धेनाहता)
(ख) ६. गुण for (गुणा)
(च) २. द्धृता for (धृता) ३. स्वाहोरात्र for (स्वाहोरात्रे)
५. व्यासार्द्धे for (व्यासार्धे) ४. भक्ताः for (भक्ता)

५८. (घ) १. द्धृता (ग्र) (क) (ख) (च) for (धृता)
२. षष्टयाः (ख) षष्टा for (षष्टया)
(ग) ३. विड्यो for (विनाड्यो)
(ख) ४. शर for (चर)
५. +भवंति नाड्यो+
६. 'डका' लुप्त है
(च) २. षष्टयाः for (षष्टया)

५९. (घ) १. घाप्तम् (ग) घातम् (ख) द्यप्तम् for (द्याप्तम्)
२. गोलेन्यथा (ग) (च) for (गोलेऽन्यथा)
(ग) ३. पल for (दल)
४. ऋण for (ऋण)
५. ऽस्तसमये for (ऽस्तमये)
(क) ६. भुक्ति for (भुक्तिः)
७. हृता (च) for (हृता)
८. मुत्तरं for (मुत्तर) (ख) धनुत्तरगोलेन्यथा for (धनमुत्तरगोलेऽन्यथा)
(च) ५. अवग्रहचिह्नलुप्त

दिनरात्रिमानघटिकाश्चरार्द्धनाडीभिरुत्तरगोले ।
पंचदश १५ युक्तहीना याम्ये हीनाधिका द्विगुणाः ॥ ६० ॥
भान्यश्विन्यादीनि ग्रहलिप्ता खखवसु ८०० द्धृताल्लब्धम् ।
भुक्तिहृते गतगम्ये दिवसाः दिवसा षष्ट्याहृते घटिका ॥ ६१ ।
अर्कोन चन्द्र लिप्ताः खयम स्वर ७२० भाजिता फलं तिथयः ।
गतगम्ये षष्टिगुणे भुक्तयन्तर भाजिते घटिका ॥ ६२ ॥

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६०. (ग) १. रुत्तरे (क) (ख) कत्तरे for (रुत्तर)
२. यह संख्या लुप्त है
३. यहां (:) लुप्त हैं (क)
(क) ४. युक्ति for (युक्त)
५. ६ 'याम्ये हीना' पद लुप्त हैं। (ख)
(च) १. रुत्तरे for (रुत्तर)

६१. (घ) १. खखवसूद्धृता (क) (ख) खखावस्तूद्धृता for (खखवसु ८०० द्धृता)
२. यह '८००' संख्या यहां न होकर 'लब्धम्' के पश्चात् है (ग)
३. 'दिवसा' इस प्रति में विद्यमान नहीं (क) (ख) विवसाः for (दिवसा)
(ग) ४. यह पद यहां लुप्त है (ख)
(क) ५. भान्यस्विन्यादीनि (ख) भन्याश्विन्यादीनि for (भान्यश्वन्यादीनि)
६. लिप्ताः for (लिप्ता)
(ख) ७. भुक्तिहते for (भुक्तिहृते)
८. गुणे च्च घटिका for (षष्ट्याहते घटिका)
(च) १. वसूद्धृता for (वसुद्धृता)

६२. (घ) १. लिप्ता (क) for (लिप्ताः)
२. गुण (ग) for (गुणे)
३. घटिकाः (च) for (घटिका)
(ग) ४. यह संख्या 'भाजिता' और 'फलं' के बीच में है
५. तिथियः (ख) तिथ्ययः for (तिथयः)
(क) ६. खयमा for (खयम)
(ख) ७. खर for (स्वर)
(च) ८. अर्क्कों for (अर्कों) ४. संख्या लुप्त

रविचन्द्रयोगलिप्ताः खयमस्वरभाजिताः फलं योगः[१] ॥
गतगम्ये षष्टिगुणे भुक्तिसमासाद्धृते[२] नाड्यः ॥ ६३ ॥
राश्यंशकला[४] विकलाः[५] स्फुटमासांते[३] लिप्तिका[६] विकला[१] ।
पक्षांते[८] तिथ्यंते समा रवीन्दोः[२] कला विकलाः[७] ॥ ६४ ॥

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६३. (घ) वि० यह श्लोक इस प्रति में उपलब्ध नहीं। न इस संख्या का कोई दूसरा श्लोक ही। ऐसा प्रतीत होता है कि लिपिकार द्वारा यह श्लोक लिखने से रह गया।

(ग) यह श्लोक इस प्रति में भी उपलब्ध नहीं है।

(क) इस प्रति में उपलब्ध नहीं।

(च) १. योगा: for (योगः) २. समासाधृते for (समासाद्धृते)
वि० यह श्लोक इस प्रति में १५A पृष्ठ पर ऊपर की ओर किसी अन्य हाथ से लिखित है।

६४. (घ) १. विकलाः (ग) (क) (च) (विकला)
२. रवीन्द्वोः (ग) (क) रवीस्ताः (ख) रवीन्द्वोः (च) for (रवीन्दोः)

(ग) ३. मासांतेश (घ) (क) (ख) मासांतेशा for (मासांते)
वि० इसकी श्लोक संख्या ६३ है।

(क) ४. राश्यंश (ख) राशांश for (राश्यंश)

५. विकला for (विकलाः)

६. लिप्तिको for (लिप्तिका)

७. विवरलाश्चा for (विकलाः)
इसकी श्लोक संख्या ६३ है।

(ख) ८. पक्षातो for (पक्षांते)

(च) ३. मासांतेश for (मासांते)

कृष्णचतुर्दश्यन्ते[१] शकुनि[४] पर्वणि[२] चतुष्पदप्रथमे ।
तिथ्यर्द्धं[३] ते नागं[५] किंस्तुघ्नं[७] प्रतिपदाद्यर्द्धं[६] ॥ ६५ ॥
व्यकदु[७] कला भक्ता[३] खरसगुणैः[४] ३६० लब्धमूनमेकेन[५] ।
चलकरणानि[८] ववादीन्यग[९] हृतशेषे तिथिवदन्यत्[६] ॥ ६६ ॥[१०]

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६५. (घ) यह श्लोक इस प्रति में उपलब्ध नहीं है ।

(ग) १. दृश्यंते शशकुनिः for (दृश्यन्ते शकुनि) (ख) चतुर्दशांते for (चतुर्दश्यन्ते)
२. चतुष्पदं (क) (ख) (च) for (चतुष्पद)
३. तिथ्यर्द्धेत्येनागं (क) तिथ्यर्द्धेऽत्ये (ख) तिथ्यर्धेत्यनागः for (तिथ्यर्द्धते नागं)

(क) ४. शकुनिः (ख) for (शकुनि)
५. मार्गं for (नागं)

वि० इसकी श्लोक संख्या ६४ है ।

(ख) ७. किस्तुघ्नप्रति for (किंतुघ्नं प्रति)

(च) ३. तिथ्यर्द्धेते for (तिथ्यर्द्धते)

६६. (घ) १. वदन्यतः for (वदन्यत्)
इस श्लोक की क्रमसंख्या ६५ है। (ग) (क)

(ग) २. +७+(च)

(क) ३. भक्ता for (भक्ताः)
४. त्खर (स्व) स्वरस for (खरस)
५. लंब्ध (च) for (लब्ध)
६. तिथिन्यत् ॥ ६६ ॥ (ख) तिथि वदन्यात् for (तिथिवदन्यत्)

(ख) ७. व्यर्केंद्रकला for (व्यर्केंदुकला)
८. वल for (चल)
९. च for (व)

(च) १०. क्रमसंख्या ॥ ६५ for (६६)

इह नोक्तानि बहुत्वात् स्पष्टगतेरुत्तरेभिधास्यामि ।
संक्रांतिभतिथिकरण व्यतिपाताद्यं त गणितानि ॥ ६७ ॥
ज्या परिधि स्पष्टीकरण दिनगतिचरार्द्धं करणेषु ।
स्फुटगतिरध्याय सप्तषष्टिरार्या द्वितीयोऽयम् ॥ ६८ ॥
इति ब्रह्मगुप्ते २योऽध्यायः ॥

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६७. (घ) १. थिमकरण for (भतिथिकरण)
वि० इसकी श्लोक संख्या ६६ है । (ग) (क)

(क) २. बहुत्वत् for (बहुत्वात्)
३. स्पष्टगंते for (स्पष्टगते)
४. ऽभिधास्यामि for (भिधास्यामि)
६. 'करण' पद लुप्त है ।
७. व्यतीपाताद्य for (व्यतिपाताद्यंत)

(च) ४. वधास्यामि for (भिधास्यामि)
८. ॥ ६६ ॥ for (॥६७॥)

६८. (घ) १. +भतिथि+ (ग) (ख)
२. गति रध्यायः (ग) (क) (ख) गत्यध्यायः for (गतिरध्याय)
३. द्वितीयोयम् (ग) (च) for (द्वितीयोऽयम्)
इसकी श्लोक संख्या ६७ है । (च)
४. 'इति'—से 'अध्यायः' तक यहां अंकित नहीं इस अधिकार में केवल ६७ श्लोक हैं । (क)

(ग) ५. या for (ज्या)
६. करणं (ख) करणा for (करण)
७. दिनमानचरार्द्ध (ख) दिनरात्रिचरार्द्ध for (दिनगतिचरार्द्ध)
४. 'इति श्रीब्रह्मसिद्धान्ते स्फुटिकर्णाधिकारो द्वितियः' यह नोट और ही लेखनी का है, काग़ज़ के ऊपर दिया गया है ।

(क) ८. परिधिः for (परिधि)

(च) १. +भतिथि २. रध्यायः for (रध्याय)
४. 'इति' से 'अध्यायः' तक अंकित नहीं है ।

पूर्वापरयोर्विद्धो[1] तुल्य छायाग्रयो दिगपराद्याः[2] ।
पूर्वान्यः[5] क्रान्तिवशात्तन्मध्यात्[4] शंकुतलमितरे[3] ॥ १ ॥
नृछायाग्रगजमत्स्य[1] द्वयमध्यगसूत्रयोर्युतिर्यत्र[2] ।
सोत्तरगोले याम्या[5] शंकुतला[3] दक्षिणे सौम्या ॥२ ॥
छायाग्रभ्रमरेखा[6] सूत्रद्युतेर्वृ[1]तपरिधिरग्रस्पृक्[2] ।
मध्यच्छायान्तरमुदगीतरं[3] वा शंकुमण्डलयोः[4] ॥ ३ ॥

१. (घ) १. विद्ध (ग) विद्व (क) बिन्दु (ख) विन्दु for (विद्धो)
२. द्दिगपराधः (ग) दिगपरार्धः (क) दिगपराधः for (दिगपराद्याः)
(ग) ३. छंकुतलमितरे (क) (ख) for शंकुतलमितरे)
(क) ४: तन्मध्या (ख) तन्मध्य for (तन्मध्यात्)
(ख) २. दिग्पराद्यः for (दिगपराद्याः)
५. मूर्धन्याः for (पूर्वान्यः)
(च) २. द्दिग for (दिग)

२. (घ) १. नृछायाग्रजमत्स्य (क) त्रिछायाग्रजमस्य (ख) त्रिछायाग्रजमत्स्य for (नृछाया-ग्रगजमत्स्य)
(ग) १. त्रिछायाग्रजमछय for (नृछायाग्रगजमत्स्य)
२. युति for (र्युति)
३. छंकुतला (क) कुंतला for (शकुंतला)
(क) ४. यत्र for (र्यत्र)
५. याम्यात् for (याम्या)
(क्षेप) क्षेपक तत्काले द्विवरं क्रान्त्यो लंबेन भाजितं गुणयेत् ।
तत्करणेन छायायाः प्राचीलब्धां गुणैरयनैः ॥ २ ॥ इतिक्षेपः
(ग) यह श्लोक इसी 'प्रति' में वह भी 'क्षेप' के नाम से उपलब्ध है ।
(क) यह श्लोक इस प्रति में नहीं है ।
(च) यह श्लोक इस प्रति में नहीं है ।

३. (घ) १. युते (ग) (क) युते (ख) (च) for (द्युते)
२. वृ्त्तस्पक् (ग) वृत्त (ख) वृत्त (च) for (वृ्त)
३. सुदगितरं (ग) मुदगिरद्वा (क) मुदगितरथा for (मुदगीतरं)
४. मंडलयो (च) for (मंडलयोः)
(क) ५. 'वा' पद लुप्त है । (ख) द्वा
(ख) ६. 'ग्र' लुप्त है
७. मध्या for (मध्य) ३. मुद्गितर for (मुदगीतरं)

छायावृत्तेऽर्काग्रा कर्णगुणा व्यासदल हृतार्काग्रा ।
विषुवच्छाया याम्या तदंतरैक्यं भुजस्याग्रे ।। ४ ।।
शंकुप्राच्यपरा या छाया भुजकृतिविशेषमूलं नयत् ।
तत्प्राच्यपरछाया भुजाग्रयोरन्तरं कोटिः ।। ५ ।।
दिग्मध्ये छायाग्रं कृत्वा शंकोर्यथादिशं भ्रमणम् ।
दिग्मध्यस्थितशकोः छायाग्रं भ्रमति विपरीतम् ।। ६ ।।

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४. (घ) १. वृत्तेर्काग्रा (ग) (ख) वृत्तेर्काग्र for (वृत्तेऽर्काग्रा)
२. विषुवच्छायाम्या (ग) विषुवच्छायायाम्या for (विषुवच्छाया याम्या)
३. भुजोस्याग्रे (ग) भुजोस्याग्रे (क) भुजोस्याग्रो for (भुजस्याग्रे)
(ग) ५. कार्णनुणा for (कर्णगुणा)
(क) २. विषुवछाया for (विषुवच्छाया)
(ख) ४. हतार्काग्रा for (हृतार्काग्रा)
३. भुजोस्याग्रे for (भुजस्याग्रे)
(च) १. वृत्तेर्क्काग्रा for (वृत्तेऽर्काग्रा) ४. हृतार्क्काग्रा for (हृतार्काग्रा)
२. विषुवछा for (विषुवच्छाया) ३. भुजोस्याग्रे for (भुजस्याग्रे)

५. (घ) १. परयो (ग) परायाः (क) परायाः for (पराया)
२. 'न' इस प्रति में नहीं है। (ग) (क) (च)
४. भुजग्रेयो for (भुजाग्रयो)
(ग) ५. शंकुः (ख) शकुः for (शंकु)
(क) ६. कौशि for (कोटिः)
(ख) ७. प्राच्या for (प्राच्य)
८. रन्तरे for (रन्तरं)
(च) ६. तत्प्राच्यां for (तत्प्राच्य) ४. भुजग्रयो for (भुजाग्रयो)

६. (घ) १. भ्रवति (क) for (भ्रमति)
(क) २. दिग्मध्य (ख) दिङ्मध्ये for (दिग्मध्ये)
३. शंकेश् (ख) शंको for (शंकोः)
(ख) ४. शांकोर्यद्यथादिशं for (शंकोर्यथादिशं)
५. दिङ्मध्ये for (दिग्मध्य)
(च) ३. शकोः for (शंकोः)
६. विपरितं for (विपरीतं)

शंकुर्लंबछायाक्षज्या तद्वर्गसंयुते मूलम् ।
विषुवत्कर्णच्छाया कर्णोत्यदा शंकुः ॥ ७ ॥
उन्नतजीवाकोटिः छाया दृग्ज्या भुजो नतज्या वा ।
कर्णच्छायावृत्तं व्यासार्द्धं द्वयमतोन्यत्र ॥ ८ ॥

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७. (घ) १. क्षय्या (ग) क्षज्य (ख) वज्या for (क्षज्या)
२. मूलम् (च) for (मूलम्)
३. विषुव तिथिषुव (ग) विषुवति विषुवति for (विषुवत्)
४. कर्णोन्यदा (ग) (क) कर्णोऽन्यदा for (कर्णोत्यदा)

(ग) ५. शंकुर्लम्बः (क) शंकुर्लंबश (ख) शंकुलंब for (शंकुर्लंब)
६. कर्णः छाया (क) कर्णश्छाया for (कर्णच्छाया)

(क) ७. +विषुवति+(च)

(ख) ७. +विषुवति+
४. कर्णोन्यदा for (कर्णोत्यदा)

(च) १. क्षय्या for (क्षज्या) ४. कर्णोन्यदा for (कर्णोत्यदा)

८. (घ) १. कोटिछाया for (कोटिः छाया) (ख) उत्ततजोवकोटि for (उन्नतजीवा-कोटिः)
२. यत्रः for (यत्र)

(ग) ३. द्रिक्ज्या (क) शध्या for (दृग्ज्या)
४. भुजानतज्यावा for (भुजोनतज्यावा) (ख) भुज्या for (भुजो)
५. कर्णः for (कर्ण)

(क) ६. वा (छ) यः for (छाया)
७. तस्या (ज्या) या for (तज्या वा)
८. वृत्तव्यवसार्द्ध for (वृत्तं व्यासार्द्धं)
९. द्वयासनोऽस्यत्रम् for (द्वयमतोन्यत्र)
इसकी श्लोक संख्या ६ है, संभवतः यह मूल है ।

(ख) १०. सार्द्ध for (सार्धं)

(च) १. कोटि for (कोटिः) ३. दृग्द्या for (दृग्ज्या)

शंकु ७छाया कृत्योस्त्रिज्या ५कृतितत्समास गुणहृतया १।
२मूलं ३लम्बाक्षज्येस्तदंशकास्तद्धनुर्भागाः ॥ ९ ॥
विषुवत्कर्णहृते ६वा शंकु छायागते ३पृथक् ।
५त्रिज्ये त्रिज्येतर जीवा १लंबाक्षांसो २क्रमज्योना ॥ १० ॥

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९. (घ) १. हृतयो (ग) हृतयोः (ख) हृतयोः for (हृतया)

२. मूले (क) (ख) for (मूलं)

३. लंबाक्ष्यज्येतदंश (क) लबाक्षस्थे for (लंबाक्षज्येस्तदशं)

४. तर्द्धनुर्भागाः (क) सद्वक्रभागाः (ख) तद्धनुभागः for (तद्धनुर्भागाः)

(क) ५. नृत्यमासनस्य सूत्रयोः for (तत्समासगुणहृतया)

६. तदसंकाश (ख) तदशकास्व for (तदंशका)

इसकी श्लोक संख्या ८ है

(ख) ७. क्वाया for (छाया)

(च) ३. लम्बाक्ष्यजे for (लम्बाक्षजे) ६. तदंशका for (रतदंशका)

१०. (घ) १. जीवा वालंबाक्षांशो (क) (ख) जीवावलंबाक्षाशो for (जीवालंबाक्षांसो)

२. त्क्रमज्योना (क) कामावना (ख) कमज्योना for (क्रमज्योना)

३. छायाहृते (ग) (क) छायाहते (ख) हते for (छायागते)

(ग) ४. लंबाक्षांशो for (लंबाक्षांसो)

(क) ५. त्रिद्येतर (ख) तिज्ये for (तिज्येतर) इसकी श्लोक संख्या ११ दी है। परन्तु यह भूल प्रतीत होती है क्योंकि इसकी टीका के अन्त में १० लिखा है।

(ख) ६. विषुत्वार्याहते for (विषुवत्कर्णहृते)

(च) ३. छायाद्गते for (छायागते)

२. लंबांक्षांशोत्क्रमज्योना।

नवते ९० लंम्बांशात्प्रोज्य ज्यावेतराक्षलंम्बज्ये ।
शंकुच्छायागुणिते छाया द्वादशहृते वान्ये ॥ ११ ॥
लंबाक्षज्यावर्गं प्रोह्यज्या कृतेः पदं वान्याः ।
अन्यत्र सर्वदोन्नत जीवांशा नयनमेव ॥ १२ ॥
इष्टदिनार्द्धनतांश क्रान्त्यंशैक्यांतरं क्रियतुलादौ ।
अक्षांशा याम्यायां छायायामंतरमजादौ ॥ १३ ॥

११. (घ) १. लंबाक्षांशात्प्रात्सज्या (ग) लंबाक्षांशान् (क) लंबाक्षांशान् (त्) for (लंबांशात् प्रोज्यज्या)
(ग) २. यह संख्या लुप्त है (क)
३. प्रोह्य ज्यवितराक्ष (क) प्रोह्यज्या (ख) प्राज्याज्यार्ध for (प्रोज्यज्यावेतराक्ष)
(क) ४. नवतै for (नवते)
(ख) १. लंबक्षाशान् for (लम्बांशात्)
५. लंबज्यो for (लंबज्ये)
(च) १. लंबाक्षांशात् for (लंम्बांशात्)

१२. (घ) १. प्रोक्तज्याकृतेः (ग) प्राज्यत्रिज्योकृतेः (क) प्रोह्य त्रिज्या for (प्रोह्यज्याकृतेः)
२. वान्या (ग) (क) वान्यः for (वान्याः)
३. नतजीवांशा (ग) (क) (ख) नजीवांशा for (न्नतजीवांशा)
४. मेवम् (ग) (च) for (मेव)
(ग) ५. वग्रं (ख) वर्ग for (वर्गं)
(ख) १. प्रोज्या त्रिज्या for (प्रोह्यज्या)
४. नयनमवम् for (नयनमेव)
(च) १. प्रोज्य for (प्रोह्य)
२. वान्या for (वान्याः)

१३. (घ) १. दिनार्द्धन्नतांश (ख) दिनार्द्धनातांश for (दिनार्द्धनतांश)
(ग) २. क्रान्ति (क) क्रान्त्यां (ख) क्रान्त्यै for (क्रान्त्यं)
(ख) ३. शैक्यंतर for (शैक्यांतरं)
४. तुल्यादौ for (तुलादौ)
५. अक्षांश for (अक्षांशा)
६. षाम्यायां for (याम्यायां)
७. 'या' लुप्त है
(च) १. दिनार्द्धन्नतांश for (दिनार्द्धनतांश)

मेषवृषमिथुनजा[१] स्वाहोरात्र्यार्द्ध[२] चरदलप्राणान्[३] ।
प्राग्वत्कृत्वा स्वाधो[४] विशोध्य चरखंडकप्राणाः[५] ।। १४ ।।
मिथुनाहोरात्रार्द्ध[१] क्रियाहोरात्रदलहत[३] गुणितम्[५] ।
तज्ज्याभिराप्तं[६] चापान्तराणि[७] लंकोदयप्राणाः[८] ।। १५ ।।
ज्यावर्गात्तत्क्रान्तिज्या[५] वर्गोनात्तत्पदाहृता[१] त्रिज्या[२] ।
स्वाहोरात्रार्द्ध[३] हताश्चापांतराण्यथवा[४] ।। १६ ।।

१४. (घ) १. जीवा (ग) (क) (ख) जीवा (च) for (जा)
२. रात्रार्द्ध (ग) (क) रात्रर्ध (ख) रात्रार्ध for (रात्र्यार्द्ध)
(ख) ३. चरदप्राण (वा) न् । for (चरदलप्राणान्)
४. स्वाषो for (स्वाधो)
५. च खंडंक प्राणाः for (चरखंडकप्राणाः)
(च) २. रात्रार्द्ध for (रात्र्यार्द्ध)
५. चरषंडक प्राणाः for (चरखंडकप्राणाः)
१५. (घ) १. रात्रार्द्ध क्रिय।द्यहो for (रात्रार्द्धक्रियाहो) (ख) रात्रार्ध for (रात्रार्द्ध)
२. रात्र (क) (ख) रात्र for (राप्तं)
(ग) ३. क्रियाद्य (क) क्रियाद्यहो (ख) क्रियाद्यहो for (क्रियाहो)
(क) ४. हृतं (च) for (हतं)
(ख) ५. रात्रदलं for (रात्रदल)
६. तज्याभि for (तज्ज्याभि)
७. चापंतराणि for (चापान्तराणि)
८. प्राणः for (प्राणाः)
(च) ३. क्रियाद्यहोरात्र for (क्रियाहोरात्र)
६. तद्याभिरात्र for (तज्ज्याभिराप्तं)
१६. (घ) १. वर्गोनात्पदाहता (ग) वर्गो ना तत्पदाहता for (वर्गोनात्तत्पदाहता)
२. त्र्यज्या for (त्रिज्या)
३. हताः (ख) हतास्वार्धाश् for (हता)
४. ण्यथावाः for (ण्यथवा) (ख) चापांतराण्यथवा for (चापांतराण्यथवा)
(क) मूल श्लोक यहां उपलब्ध नहीं है । केवल टीका अंकित है ।
(ख) १. नात्यादा for (नात्तत्पदा)
(च) ५. ज्यावर्गान्त क्रान्तिज्या for (ज्यावर्गात्तत्क्रांतिज्या)
१. वर्गोनत्पदाहता
३. हताश्चाधश् for (हताश्)

स्वचरासुभिसूनयुता १६१७ क्रमोत् क्रमस्थैः क्रमोक्रमन्यस्ताः ।
उदयप्राणा व्यस्ताश्चार्कं तात्कालिकं कृत्वा ॥ १७ ॥
रविणा भुक्त्याराशेः कला गुणाः स्वोदयासुभिर्भक्ताः ।
राशिकलाभिर्लब्धा प्रश्नासुभ्योऽशवः शोध्याः ॥ १८ ॥

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१७. (घ) १. रूनयुताः (ग) रूनयुताः (क) (ख) नूनयुता for (सूनयुता)
२. +१७९५, १९३५, १९३५, १३९५, १६००+
३. क्रमोत्क्रमन्यस्ताः (ग) (क) क्रमोत्क्रमास्ते (ख) क्रमोक्रमभ :स्ताः for (क्रमोक्रमन्यस्ताः)
(ग) २. यहाँ कोई संख्या नहीं दी गई ।
(ख) ४. स्ववरासुभिः for (स्वचरासुभि)
५. क्रमाक्रमो for (क्रमोत्क्रमस्थै)
६. : विसर्ग लुप्त हैं
७. श्चार्क for (श्चार्कं)
८. 'कृत्वा' पद लुप्त है ।
(च) १. रूनयुताः for (सूनयुता)
३. क्रमोत्क्रमन्यस्ताः for (क्रमोक्रमन्यस्ताः)
७. चाक्रं for (चार्कं)

१८. (घ) १. शवशोध्याः for (ऽशवःशोध्याः) (ग) सवः (क) ऽसवः (ख) सवं for (ऽशवः)
२. +१८००(ख) + : +
(ग) ३. भुक्ताराशेः (ख) for (भुत्तचाराशेः)
४. +१६७०, १७९५, १९३५, १९३५, १७९५, १६७०+
५. लंब्धम् १८०० for (लंब्धा) (क) लब्धाः for (लंब्धा)
(क) ६. रविभुक्तहीनराशेः for (रविणा भुत्तचा राशेः)
(ख) ७. गुणा for (गुणाः)
८. सुभिः भक्ताः for (सुभिर्भक्ताः)
(च) ९. ९+१७९५ २+१८००+
१९३५
१९३५
१३९५
१५००+
१०. ग्रवग्रहचिह्नलुप्त

प्रक्षिप्य[६] राशय[२] भुक्तं शेषा सुभ्यः क्रमेण[३] यावंतः[४] ।
शुद्धंत्फदया[१] सूर्ये तावंतो राशयः क्षेप्याः[५] ॥ १९ ।
शेषांत्रिंशत्[१] गुणितानस्तन[२] शुद्धोदया[४] सुभिर्विभजेत्[५] ।
लब्धंभागादिरवौ[६] प्रक्षिप्य[३] तथा कृते लग्नम्[७] ॥ २० ॥

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१९. (घ) १. शुध्यंत् (ग) शुध्यत्युदयाः (क) शुध्यंत्युदयाः (ख) for (शुद्धंत्फदया)

(ख) २. राश for (राश्य)
३. क्रमण for (क्रमेण)
४. यावतः for (यावंत)
५. 'क्षेप्याः' पदलुप्त है ।

(च) ६. +१७९५
१९३५
१९३५
१३९५
१५०९
} पिछले श्लोक के फुटनोट में संख्याएं दर्शा दीं गई हैं ।

१. शुद्धंत्युदयाः for (शुद्धंत्फदया)

२०. (घ) १. शेषांस्तृंशद् (ग) शेषास्त्रिंशद् for (शेषांत्रिंशत्)
२. गुणिताः श्येनस्तन for (गुणितानस्तन) (ग) नसून for (नस्तन)
(क) दविशुद्धस्योदयाः for (नस्तनशुद्धोदया)

(क) ३. प्रक्षेप्यं for (प्रक्षिप्य)

(ख) १. शेषास्त्रिंशद्गुणा for (शेषांत्रिंशत् गुणिता)
२. तसून for (नरतन)
४. शुद्धासुभि for (शुद्धोदयासुभिर्)
५. वित्तजेत् for (विभजेत्)
६. लब्धा for (लब्धं)

(च) १. शेषांत्रिंशद्गुणिताः ३० नस्तन for (शेषां त्रिंशत् गुणितानस्तन)
७. क्रम संख्या लुप्त

रविराशयभुक्तलिप्तास्तदुदय गुणिता हता ग्रहकलाभिः १८०० ।
लब्धं प्राणास्थाप्याः प्रक्षिप्यार्के ग्रहाभुक्तम् ॥ २१ ॥
तावत्सूर्ये राशि न क्षिपेत्समो लग्नराशिभिर्यावत् ।
क्षिप्रग्रहाणां प्राणान् प्रक्षिप्य स्थापितेष्वसुषु ॥ २२ ॥
तदधिककालोदयवधराशि कलाभिर्भजेत् फलं प्राणान् ।
प्रक्षिप्य प्राणेषु प्राणाः सूर्योदयादसकृत् ॥ २३ ॥

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२१. (घ) १. हृतो (क) हृता (ख) कृता for (हतो)
२. गृहकलाभिः (ग) (क) (ख) गृहकालाभिः for (ग्रहकलाभिः)
३. गृहाभुक्तम् (ग) (क) (ख) गृहामुक्तम् for (ग्रहाभुक्तम्)
४. रविराश्यभुक्त (ग) (क) (ख) रविराशिभुक्त for (रविराशयभुक्त)
(ग) ५. लब्धप्राणाः (ख) लब्धां प्राणाः for (लब्धंप्राणा)
(क) ६. प्राणाः (ख) प्राणः for (प्राणा)
(च) ४. रविराश्यभुक्त for (रविराशयभुक्त) १ हृता for (हता) २ गृह for (ग्रह)
६. प्राणाः for (प्राणा) ३. गृहा for (ग्रहा)

२२. (ख) १. राशीन् (क) (ख) राशी for (राशिन)
२. क्षिप्तगृहाणां (ग) क्षिप्तगृणां (क) क्षिप्तगृहाणाँ for (क्षिप्रग्रहाणां)
(क) ३. समं for (समो)
(ख) ४. 'न' लुप्त
५. यावत् for (यर्वित्)
२. क्षिप्तगृहाणां for (क्षिप्रग्रहाणां)
(च) १. राशीन for (राशिन) २. गृहाणां for (ग्रहाणां)
६. स्थापितेष्वसुषुः for (स्थापितेष्वसुषु)

२३. (घ) १. कलोदय (ग) (क) (ख) कलोदय (च) for (कालोदय)
२. प्राणात् for (प्राणान्)
(ग) ३. वधं (क) (ख) for (वध)
४. भजेजेत् for (भंजेत्)
५. प्राणः (ख) for (प्राणाः)
(क) ६. फलप्राणान् (ख) फलप्राणान् for (फलंप्राणान्)
७. (……) ॥ २३ ॥
(ख) ८. राशिभिर्यावत् क्षिप्य गृहाणां प्राणान्प्रभिभजत् for (राशिकलाभिर्भजेत् फलं प्राणान्)
९. प्राणषु for (प्राणेषु)

प्रागुदयैः प्रश्नासुभिरूनोर्को भुक्तराशिभिर्लग्नम् ।
कृत्वैवमूनमर्कं लग्नसमं प्राग्भवेत्कालः ॥ २४ ॥
गतशेषाल्पास्याह्नः सौम्योत्तरगोलयोश्चरार्द्धेन ।
ऊनाधिकस्य जीवा स्वाहोरात्र्यार्द्ध संगुणिता ॥ २५ ॥
त्रिज्याहता युतोना क्षितिज्यया सौम्ययाम्ययोश्छेदः ।
छेदोवलम्बकगुणो व्यासार्द्ध विभाजितः शंकुः ॥ २६ ॥

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२४. (घ) १. भुक्तः (ग) भुक्ति for (भुक्त)

(ग) २. प्रश्नाशुभि for (प्रश्नासुभि)
३. मूनमर्कं (ख) मूनमर्कलग्नसमं for (मूनमर्कं लग्नसमं)

(ख) ४. रुभोको for (रूनोर्को)
५. लग्नम् for (लंग्नम्)
६. कलः for (कालः)

(च) ४. रूनोर्क्कों for (रूनोर्को)[३] मर्क्कं for (मर्कं)

२५. (घ) १. गत शेषाल्पस्याह्नः (ग) गतशेखाल्पस्याह्नः for (गतशेषाल्पास्याह्नः)
२. सौम्येतर (ग) (क) (ख) सौम्यैतर for (सौम्योत्तर)

(ग) ३. रात्रार्द्ध (क) (ख) (च) for रात्र्यार्द्ध)

(क) १. गतशेषाल्पस्याह्नः (ख) गतशेषाल्पस्याह्नाः for (गतशेषाल्पास्याह्नः)

(ख) ४. चरार्द्धेनः for (चरार्द्धेनः)
५. ऊनाधिकास्य for (ऊनाधिकस्य)

(च) १. गतशेषाल्पस्याह्नः for (गतशेषाल्पास्याह्नः) २. सौम्येतर for (सौम्योत्तर)

२६. (घ) १. विभाजितशंकुः (च) for (विभाजितःशंकुः)
२. त्रिज्याहृता (च) for (त्रिज्याहता)
३. छेदावलम्बकगुणो (च) for (छेदोवलम्बकगुणो)

(ग) ४. छेद (क) याम्ययोश्छेदः (ख) for (याम्ययोछेदः)

(ख) ५. व्यासार्धं for (व्यासार्द्ध)

विषुवत्कर्णविभक्तः[1] छेदो वा द्वादशाहृतः[3] शंकुः ।
शकुकृतिर्विहीनाया[2] व्यासार्द्धकृतेः[4] पदं दृग्ज्या ।। २७ ।।
दृग्ज्या द्वादश गुणिता[3] विभाजिता शंकुना[1] फल छाया ।
व्यासार्द्धं[4] छेदहतं[5] विषुवत्कर्णाहितं[6] कर्णः[2] ।। २८ ।।
गुणितं वा द्वादशभिर्व्यासार्द्धं[2] शंकुनाहतं[1] कर्णः ।
जीवा क्षयवृद्धिज्यायुतहीना ज्या[3] क्रियतुलादौ ।। २९ ।।

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२७. (घ) १. विभक्तं (च) for (विभक्तः)

२. शंकुकृतिविहीनाया (ग) (क) (ख) for (शकुकृतिर्विहीनाया)

(क) ३. हृतः (ख) हतशंकुः for (हृतःशंकुः)

(ख) ४. फलं दृग्जा for (पदंदृग्ज्या)

(च) द्वादशहृतः for (द्वादशाहृतः)

२. शंकुकृतिविहीनाया for (शंकुकृतिर्विहीनाया)

२८. (घ) १. फलं (ग) (क) (ख) फलं for (फल)

२. कर्णं for (कर्णः)

३. +१२+(च)+१२+

(ग) ४. व्यासार्द्धछेदहृतं for (व्यासार्द्धं छेदहृतं)

(क) ५. हृतं (च) for (हृतं)

(ख) ७. शांकुला for (शंकुना)

२९. (घ) १. हृतं for (हृतं) (ग) (क) (ख) हृतकर्णं for (हृतंकर्णः)

(क) २. व्यासार्द्धं (ख) व्यासार्ध for (व्यासार्द्धं)

(ख) ३. जा for (ज्या)

स्वाहोरात्र्यर्द्ध[१] गुणा व्यासार्द्ध[२] विभाजिताऽथवा छेदः ।
शंक्वादि[६] प्राग्वज्ज्या स्वाहोरात्र्यर्द्ध[३] घात[५] कृता[४] ॥ ३० ॥
व्यासार्द्धकृति[१] गुणिता[२] विषुवत्कर्णेन वा भवेत्कर्णः[३] ।
लम्बगुणो घातः[४] शंकु[६] व्यासार्द्धकृतिभक्तः[५] ॥ ३१ ॥

३०. (घ) १. रात्रार्द्ध (ग) (क) (ख) रात्रार्ध for (रात्र्यर्द्ध)

२. व्यासार्ध (ग) (ख) for (व्यासार्द्ध)

३. रार्द्ध (ग) (क) रात्रार्ध (ख) for (रात्र्यर्द्ध)

४. हृता (ग) (क) हृता (ख) हता for (कृता)

(ग) ५. घात (क) (ख) for (घात)

६. शंकवादि for (शंक्वादि)

(ख) ७. गुण for (गुणा)

(च) १. रात्रार्द्ध for (रात्र्यर्द्ध) ४. हृता for (कृता)

३१. (घ) १. व्यासार्वकृति १०६९२९०० (ग) व्यासार्द्धकृति १०६९२९०० for (व्यासार्द्धकृति)

२. गुंणिता (ग) (क) (च) for (गुणिता)

३. भवति (ग) (क) (ख) (च) for (भवेत्)

४. व्याघातः (ग) (क) वा घातः (ख) वाघतः for (घातः)

५. कृतिवक्तः (ग) कृतिभुक्तः for (भक्तः)

(ग) ६. शंकुर्व्यासार्द्ध (क) (ख) शंकुव्यासार्ध for (शंकुव्यासार्द्ध)

(ख) १. व्यासार्धकृति १०६९२९०० for (व्यासार्द्धकृति) २. गुंणिता for (गुणिता) ४. व्याघातः for (घातः)

घातोवार्कगुणा[४] १२ स्त्रिज्या विषुवत्कर्णवंधः[२] हृतः[३] शंकुः ।
कर्णकृते[५] संशोध्य[७] द्वादशवर्ग[८] १४४ पदं छाया[६] ॥ ३२ ॥
अल्पप्रश्नासूनां[२] यदि बहवश्चरदलासवः क्षितिजा ।
हृतयोना[१] जीवोना क्षयवृद्धिज्योक्तवच्छेषं[३] ॥ ३३ ॥
स्वाहोरात्रार्द्धमुदकदक्षिणयोः[१] क्षितिज्यया[६] युतविहीनं ।
द्युदलान्त्यज्याभिज्य[३] क्षयवृद्धिज्या[४] युतोनांत्या[५] ॥ ३४ ॥

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३२. (घ) १. कर्णः (च) for (कर्ण)
२. वधः (ग) वध (क) for (बंधः) (ख) 'बंध' पद लुप्त है।
३. हृतः (ग) (क) (च) for (हतः)
(ग) ४. गुणास्त्रिज्या (क) (ख) for (कृते)
५. कृतेः (क) (ख) गुणास्त्रिज्या for (गुणा १२ स्त्रिज्या)
६. छायाः for (छाया)
(क) संख्याएँ मूल में अंकित नहीं हैं ।
(ख) ७. संशोध्या for (संशोध्य)
८. वर्गगयछाया for (वर्ग १४४ पदं छाया)
(च) ४. घातोवार्क्कगुणा १२ for (घातोवार्कगुणा १२) २. वधः for (बंधः)
१. कर्णः for (कर्ण)

३३. (घ) १. हृतयोना (ग) (च) for (हतयोना)
(ग) २. अल्पप्रश्नासूना for (अल्पप्रस्नासूनां)
(क) मूल श्लोक अंकित नहीं है—केवल टीका लिखी हुई है ।
(ख) ३. छाषम् for (छेषम्)

३४. (घ) १. उदग्दक्षिणयोः (ग) स्वाहोरात्रामुदग्दक्षिणयोः (ख) for (स्वाहोरात्रार्द्धमुदकदक्षिणयोः)
(ग) २. द्युदलांतिज्या for (द्युदलान्त्यज्या)
३. त्रिज्या (क) (ख) त्रिज्या (च) for (भिज्या)
४. वृध्य for (वृद्धिज्या)
५. नांत्याः for (नांत्या)
(क) १. स्वाहोरात्रार्धमुदग्दक्षिणयोः for (स्वाहोरात्रार्द्धमुदकदक्षिणयोः)
६. क्षितिजया (ख) षितिजयया for (क्षितिज्यया)
(च) १. मुदग्दक्षिणयोः for (मुदकदक्षिणयोः)

छेदहता[1] द्युदलांत्या दिनार्द्ध[4]कर्णेन वा[2] कर्णः ।
भक्ता ज्यया ऽथवाऽन्त्या दिनार्द्ध कर्णा हृता[3] कर्णः ॥ ३५ ॥
द्युदलान्नतो[1] क्रमज्या स्वाहोरात्र्यार्द्ध[2] संगुण[4] विभजेत् ।
व्यासार्द्धेन[5] फलोना द्युदलान्त्ययाथवा[3] छेदः ॥ ३६ ॥
अन्त्यानतो[1] क्रमज्या हीनां ज्या षट्[2] पृथक्[3] छेदः ।
ज्याभ्यां[4] च सह[5] फलानि छाया नयनानि षट्[6] त्रिंशत् ॥ ३७ ॥

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३५. (घ) २. वा गुणा for (वा) (ग) संगुणा for (वा) (क) संगुणा for (वा) (ख) कर्णाः for (वाकर्णः)
१. छेदहृता (ग) (क) (च) for (छेदहता)
(ग) ३. कर्णहता (क) कर्णाहिता (ख) कर्णाः हृता for (कर्णाहृता)
(ख) ४. नवागुणाः for (र्द्धकर्णेन)
(च) २. वागुणाकर्णः for (वाकर्णः)

३६. (घ) १. द्युदलान्ततोत्क्रमज्यां (ग) (क) (च) for (द्युदलान्नतो क्रमज्या)
२. स्वाहारात्रार्ध संगुणं विभजेत् । (ग) स्वाहोरात्रार्द्धसंगुणां विभजेत् for (स्वाहोरात्र्यार्द्ध संगुणा विभजेत्)
३. दलान्त्य ज्याथवा for (द्युदलान्त्ययाथवा)
(ग) ३. दलान्त्यज्या यथा (क) द्युदलान्त्यज्याथवा (ख) for (द्युदलान्त्ययाथवा)
(क) ४. संगुणाम् (ख) संगुणं for संगुणा
(ख) १. द्युदलन्नतोत्कमजां for (द्युदलान्नतो क्रमज्या)
२. स्वाहोरात्रार्द्ध for (स्वाहोरात्र्यार्द्ध)
५. फलेना for (फलोना)
(च) २. स्वाहोरात्रार्द्ध for (स्वाहोरात्र्यार्द्ध)
४. संगुणां for (संगुणा)

३७. (घ) १. अन्त्यानतोत्क्रमज्याहीना (ग) (क) (ख) for (अन्त्यानतो क्रमज्याहीनां)
३. छेदाः (ग) (ख) छेदा for (छेदः)
(च) ४. 'च' अतिरिक्त पाठ
५. फलनि for (फलानि)
६. त्प्रट्त्रिशत् for (षट्त्रिंशत्)
(च) १. अन्त्यानतोत्क्रमज्याहीन for (अन्त्यानतो क्रमज्याहीनां) ३ छेदाः for (छेदः)

छाया कर्णविभक्ता विषुवत्कर्णेन संगुणा नृज्या ।
लब्धं सौम्येतरयोः क्षितिज्यया हीनसंयुक्तं ॥ ३८ ॥
गुणितं व्यासार्द्धेन स्वाहोरात्रार्द्ध भक्तमाप्तधनुः ।
उत्तर गोले युक्तं याम्ये हीनं चरप्राणैः ॥ ३९ ॥
दिनगतशेषछेदप्राणाः प्रागपरदिनार्द्धयोर्विशोध्याप्तम् ।
व्यासार्द्ध छेषो क्रमजीवा चापं नताः प्राणाः ॥ ४० ॥
स्वाहोरात्रार्द्धेन छायाकर्णाहतेन भक्ताया ।
विषुवत्कर्ण गुणाया व्यासार्द्धकृतेः फलं सौम्ये ॥ ४१ ॥

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३८. (ग) १. त्रिज्या (क) (ख) त्रिज्या for (नृज्या)
(क) २. संगुण for (संगुणा)
३. संयुतम् (ख) for (संयुक्तं)
(ख) ४. विभक्त for (विभक्ता)
५. सौम्येतरयो for (सौम्येतरयोः)
६. ज्याया for (ज्यया)

३९. (ख) १. रात्रार्द्ध for (रात्रार्द्ध)
२. भक्तमाप्तधनुः for (भक्तमाप्तधनुः)
३. प्राणै for (प्राणैः)

४०. (घ) १. व्यासार्द्धछिषोत्क्रम (ग) व्यासार्द्धातिशेषोत्क्रम for (व्यासार्द्ध छेषोक्रम)
२. नता (क) नत for (नताः)
(ग) ३. शेषप्राणः (क) शेषप्राणाः for (शेषछेद प्राणाः)
(क) ४. 'छेद' पद लुप्त है । (ख)
५. शेषोत्क्रम (ख) छेसोत्क्रम for (छेषोक्रम)
(ख) ६. विशोध्यप्तम् for (विशोध्याप्तम्)
(च) १. व्यासार्द्धछिषोत्क्रम for (व्यासार्द्ध छेषो क्रम)

४१. (ग) १. भक्तायाः (क) (ख) for (भक्ताया)
(ख) २. कर्णाहतेन for (कर्णाहतेन)
(ख) ३. विषुवत्कर्णा for (विषुवत्कर्ण)
४. गुणाय for (गुणाया)
५. कृते for (कृतेः)

क्षयवृद्धिज्याहीनं युक्तं याम्ये[२] धनुश्चर[३]प्राणैः[४]।
सौम्येयुतं विहीनं याम्ये प्रागपरयोः[५] प्राणाः[१] ॥ ४२ ॥
अह्नोगताऽवशेषाः[३] फलमन्त्याया विशोध्य शेषस्य।
धनुरुक्रमजीवाभिः[१] पूर्वापरयोर्नताः[२] प्राणाः[४] ॥ ४३ ॥
दिनदलकर्ण[३]गुणांत्या छायाकर्णाद्धृता[४] फलोनान्या[२]।
शेषस्योत्क्रमजीवा धनुर्दिनार्द्ध[५]नताः[१] प्राणाः ॥ ४४ ॥
चरदल जीवोनाधिक फल क्रमज्यो[१] धनुश्चरार्द्धेन।
युतहीन पूर्वाह्ण[३] दिवसगतं शेषं[४] अपरार्द्धे[२] ॥ ४५ ॥

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४२. (क) १. प्राणैः (ख) प्राणः for (प्राणाः)
(ख) २. याम्यो for (याम्ये)
३. धनुश्चर for (धनुश्चर)
४. प्रणै for (प्राणैः)
५. प्रागपरयो for (प्रागपरयोः)

४३. (घ) १. धनुरुत्क्रमजीवाभिः (ग) (क) (ख) (च) for (धनुरुक्रमजीवाभिः)
(ग) १. नंता (क) नत for (नंताः)
(क) ३. अह्नोगतावशेषात् (ख) अह्नोगतावशेषा for (अह्नोगताऽवशेषाः)
(ख) ४. प्राणः for (प्राणाः)
(च) ३. अवग्रहचिह्न लुप्त

४४. (घ) १. दिनाद्धान्नताः for (दिनार्द्धनताः) (ग) (क) दिनार्द्धनत for (दिनार्द्धनताः)
(ग) २. नान्त्या (ख) नान्त्या for (नान्या)
(ख) ३, गुणांत्या
४. कर्णाद्धृता for (कर्णोद्धृता)
५. जीव for (जीवा)
१. दिनार्धोन्नतप्राणः for (दिनार्द्धनताः प्राणाः)
(च) १. धनुर्दिनार्द्धन्नताः for (धनुदिनार्द्धनताः)

४५. (घ) १. क्रमज्या (ग) (क) (ख) क्रमज्या for (क्रमज्यो)
२. शेषमपराह्णे (ग) (ख) शेषमपराह्णे (च) for (शेषं अपराह्णे)
(ग) ३. पूर्वाह्णे (ख) पूर्वाह्ण for (पूर्वाह्णे)
(ख) ४. विवसगतं for (दिवसगतं)

उत्क्रमजीवाभ्यधिक क्रमज्यया संयुक्तं (तं) धनुर्धनुषां ।
व्यस्तविशुद्धौ हीनाश्चरांसवः पूर्ववच्छेषं ॥ ४६ ॥
दिनमध्यार्का क्रांत्याक्षभागयोगांतरं सामान्यदिशोः ।
नतभागा नवते प्राक् ज्योन्नताः शेषाः ॥ ४७ ॥
नतभागज्या द्वादशगुणोन्नतांशज्यया हृतात्लब्धम् ।
इष्टदिनार्द्धछाया यथोक्तकरणैर्दिनार्द्धद्विा ॥ ४८ ॥

४६. (घ) १. संयुतम् (ग) (क) (ख) संयुते for (संयुक्तं)
२. चरासवः (ग) (क) (ख) (च) for (चरांसवः)
३. पूर्ववछेषं (ग) पूर्ववत्सेषम् for (पूर्ववच्छेषं)
५. व्यक्त for (व्यस्त)
(ज) १. संयुतं for (संयुक्तं) ६. धनुर्द्धनुषा for (धनुर्धनुषां)
४७. (घ) १. दिनमध्यार्क (ग) (क) (ख) for (दिनमध्यार्का)
२. क्रान्त्यक्ष (ग) (क) (ख) for (क्रांत्याक्ष)
६. नतभागानतभान्नवते ९० (ग) नतभागानतभागान्नवतेः for (नतभागानवते)
३. प्रोन्नताः (ख) प्रोज्योन्नतभागास्तु for (ज्योन्नताः)
(ग) ४. समान्यदिशोः (क) समानदिशोः (ख) समान्यादिशोः for (सामान्यदिशोः)
५. प्रोह्योन्नताः शेषा (क) (ख) देखो ३. for (प्राक्ज्योन्नताः शेषाः)
(क) ६. नतभागा नतभागोन्नवतेः (ख) नतभागान्नवतेः for (नतभागानवते)
(ख) ७. शाय्याः स्युः for (शेषाः)
(च) १. दिनमध्यार्क्का for (दिनमध्यार्का)
२. क्रांत्यक्ष for (क्रांत्याक्ष)
४. समान्यदिशोः for (सामान्यदिशोः)
६. नतभागानतभागान्नवते ९० for (नतभागानवते)
४८. (घ) १. हृता (ग) (क) (ख) हतालब्धम् for (हृतात्लब्धम्)
२. करणै (ख) करणे for (करणैं)
३. दिनार्धा (ग) दिनार्द्धद्विा (क) दिनार्धाद्विा for (दिनार्द्धद्विा)
(ग) ४. द्वादगुणोन्नतांश (ख) द्वादशगुणोन्नतांश for (द्वादशगुणोन्नतांश)
(ख) ५. ज्याया for (ज्यया)
३. दिनार्धद्विा for (दिनार्द्धद्विा)
(च) १. हृतालब्धं for (हृतात्लब्धम्)
२. कर्णै for (करणैं) ३. दिनार्द्धा for (दिनार्द्धद्विा)

उन्नतजीवाभक्तं[४] व्यासार्द्धं[५] द्वादशाहतं[३] कर्णैः ।
मध्यच्छायाकर्णं[१] द्वादशक्रान्त्यन्तरं[२] पदं वा ॥ ४९ ॥
कुदलांत्याज्या[१] छेदोमध्यच्छाया यथोक्तकरणैर्वा[३] ।
अन्त्याज्या[४] छेदाद्यैर्मध्यच्छायाऽथवा[५] बहुधा[६] । ५० ॥
विषुवत्कर्णेन[६] गुणा विषुवच्छाया[१] हतोत्तरा क्रान्तिः[२] ।
यद्युनाक्षक्षज्याया[३] शंकुसममंडलस्थेऽर्के[७] ॥ ५१ ॥

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४९. (घ) १. छायाम् for (छाया)
२. कृत्यंतरपदम् (ग) (क) कृत्यन्तरपदं वा for (क्रान्त्यतरं पदं वा)
३. द्वादशाहतं ११ for (द्वादशाहतं)
(ग) ४. भुक्तं for (भक्तं)
(ख) ५. व्यासार्ध for (व्यासार्द्धं)
१. 'मध्यछायाकर्णं' पद लुप्त हैं।
२. कृतंतर for (क्रान्त्यतरं) यहां दूसरी पंक्ति 'द्वादश' से आरम्भ है।
(च) १. छायां for (छाया) २. कृत्यंतरपदं वा for (क्रान्त्यतरं पदं वा)
५०. (घ) १. द्युदलां (ग) (क) (ख) (च) for (कुदलां)
(ग) २. त्यज्या (क) (ख) for (त्याज्या)
३. वरणैर्वा (ख) करणै वाः for (करणैर्वा)
(क) ४. अन्त्यज्या (ख) अत्याज्याः for (अन्त्याज्या)
(ख) ५. मध्यछाया for (मध्यछाया)
(च) ६. अवग्रह चिह्न लुप्त
५१. (घ) १. विषुवछाया (ग) (ख) विषुवच्छाया for (विषुवत्छाया)
२. हृतोत्तरा for (हतोत्तरा) (क) छायोद्धृतोत्तरा for (छायाहतोत्तरा)
३. यद्यूनाक्षय्याया (ग) द्यूनाक्षज्यायाः (क) यद्यूनाक्षज्यायाः for (यद्युनाक्ष-क्षज्यायां)
४. समं (च) for (सम)
(ग) ५. शंकुः (क) (ख) 'शंकु' लुप्त पद for (शंकु)
(क) ६. बिषुवत्कर्णेन for (विषुवत्कर्णेन)
३. यद्यूनाक्षज्यायाः for (यद्युनाक्षक्षज्यायां)
७. मंडलास्थेर्के for (मंडलस्थे ऽर्के)
(च) १. विषुवछाया for (विषुवत्छायां)
२. हृतोत्तरा for (हतोत्तरा) ३ यद्यूनाक्षज्याया for (यद्युनाक्षक्षज्यायां)

सूर्यज्या जिनभागज्यया गुणाक्षज्ययाऽथवा भक्ता ।
अग्रा द्वादशगुणिता विषुवच्छाया विभक्ता वा ॥ ५२ ॥
द्वादश विषुवच्छाया गुणिते पृथगक्षलंबजीवे वा ।
क्रान्तिहृते समभण्डलकर्णौ प्राग्वत् पृथक् छाया ॥ ५३ ॥
अर्क्राग्रा वर्गोनं नृज्या वर्गार्धं ५३४६४५० मर्क १४४ कृतिगुणितं ।
आद्यान्योग्रा द्वादशविषुवच्छाया वधो हृतयोः ॥ ५४ ॥

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५२. (घ) १. ज्ययाथवा (च) for (ज्ययाऽथवा)
२. विषुवछाया (ग) for (विषुवच्छाया)
(ग) ३. दिनभागज्यागुणा for (जिनभागज्यया गुणा) (ख)
(क) ४. अग्रात् (ख) अग्र for (अग्रा)
(ख) ५. गुणाक्षयाथवा for (गुणाक्षज्ययाऽथवा)
६. 'ज्यया' लुप्त पद है।
७. विभक्त for (विभक्ता)
(च) २. विषुववछाया for (विषुवच्छाया)

५३. (घ) १. विषुवछाया (ग) (च) for (विषुवच्छाया)
२. हृते (ग) (क) (च) for (हृते)
३. छाये for (छाया) (ग) (ख) पृथकार्ये for (पृथक् छाया)
(ख) ४. पृस्थगक्ष for (पृथगक्ष)
५. कर्णौ for (कर्णौ)
(च) ३. छायो for (छाया)

५४. (घ) १. वर्ग्रार्धा ५३४६४५० for (वर्गार्धं ५३४६४५०) (ग) वर्ग्रार्धं (ख) वर्गा for (वर्गार्धं)
२. मर्ककृतिगुणितं १४४ (ख) धर्मकृति for (मर्क १४४ कृतिगुणितं)
३. आद्योन्योग्रा (ग) (क) (ख) आद्योन्योग्रा (च) for (आद्यान्योग्रा)
४. विषुवछाया (ग) (च) for (विषुवच्छाया)
५. हृतयोः (ग) (क) (ख) हृतयो for (हृतयोः)
(ग) ६. वर्ग्रोनं for (वर्गोनं)
७. त्रिज्या (क) (ख) for (नृज्या)
वि० श्लोक में दी गई संख्याएं इस प्रति में नहीं है।
(क) १. वर्गार्धमर्क कृति for (वर्गार्धं ५३४६४५० मर्क १४४ कृति)
(च) ८. अर्क्राग्रा for (अर्क्राग्रा) ६ वर्गोनं for (वर्गोनं)
१. वर्ग्रार्द्धं ५३४६४५० for (वर्गार्धं ५३४६४५०) २ मर्क्क कृतिगुणितं १४४ for (मर्क १४४ कृति गुणितं)
५. हृतयोः for (हृतयोः)

विषुवच्छाया कृत्याद्वा संगयुतया ७२ न्यकृतियुतादाद्यात् ।
पदमन्ययुतविहीनं सौम्येत्तरगोलयोः शंकुः ।। ५५ ।।
विदिशोः सौम्येत्तरयोरुत्तरगोले पदोनयुक्तो न्यः ।
सममण्डल दक्षिणतोतच्छाया नाडिकाः प्राग्वत् ।। ५६ ।।
प्राच्यपरा शंकुतलान्तरद्वयव्यस्तकर्णवधविवरं ।
समदिग्विषुवत्छायान्यदिगैक्यं कर्णविवरहृतम् ।। ५७ ।।

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५५. (घ) १. विषवछाया, (ग) for (विषुवच्छाया)
२. गसंयुतया ७२ for (संगयुतया ७२)
(ख) ३. कृत्याः व्यागसंयुतया for (कृत्याद्वा संगयुतया ७२)
(ग) ३. कृत्याद्वचग ७२ संयुतया (क) for (कृत्या द्वासंगयुतया)
४. परमन्यतयुत (ख) पदमन्ययुतं for (पदमन्ययुत)
(ख) ६. युतयाद्यात् for (युतादाद्यात्)
(च) १. विषुवछाया for (विषुवच्छाया) २. गसंयुतया ७२ for (संगयुतया ७२)

५६. (घ) १. विदिशो सौम्येतरयो (ग) विदशो सौम्येतरयो (क) for (विदिशोः सौम्येतरयो)
२. रुत्तरागाले (क) रुत्तरागोले for (रुत्तरगोले) (ख) गोपदोन for (गोलेपदोन)
३. तच्छाया (क) नच्छाया (ख) तच्छाया for (तच्छाया)
(ग) ४. दक्षिणागतछाया (क) दक्षिणगे (ख) दक्षिणगो for (दक्षिणतो)
(क) ५. ऽन्यः । (ख) युक्ता यः for (युक्तोन्यः)
(ख) १. वेदि for (विदिशोः)
६. शौम्येतरयो for (शौम्येतरयो)
७. नाहिका for (नाडिकाः)
(च) १. विदिशो for (विदिशोः) ३. तछाया for (तच्छाया)

५७. (घ) १. विषुवछाया (ग) (क) विषुवच्छाया for (विषुवत्छाया)
२. कर्णविवरहृतम् (ग) कर्णविवरहृतम् for (कर्णविवरहृतम्)
३. यात्यपरा (ग) प्राच्यापरा for (प्राच्यपरा)
(च) १. विषुवच्छाया for (विषुवत्छाया) २. कर्णविवरहृतम् for (कर्णविवरहृतं)

छायावृत्तागोमा[3] सौम्येन युगान्तरेण[5] याम्येन ।
विषुवच्छायाज्जादिषु[2] तुलादिषु तथान्तरं[4] हीनं ॥ ५८ ॥
इष्टाछाया[4] वृत्ते[5] तदग्रयो[6] यदुदयास्तमयसूत्रम् ।
अनुपातात[1] छंको विषुवत्छायांतरमिहाग्रा[2][3] ॥ ५९ ॥
शंकुप्राच्यपरांतरविषुवत्छायांऽगुलातरं[2] याम्ये ।
उदगैक्यं लम्बगुणं छाया-कर्णोद्धृतं क्रान्तिः[5] ॥ ६० ॥

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५८. (घ) १. विषुवछाया (ग) (च) for (विषुवच्छाया)
२. जादिषु (क) (च) for (ज्जादिषु)

(ग) ३. ग्रोना (क) for (ग्रोमा)
४. तयान्तरं for (तथान्तरं)

(क) ५. युतांऽतरेण for (युतांतरेण)

५९. (घ) १. अनुपातात्तच्छंको (ग) अनुपातात्तच्छंको for (अनुपातातछंको)
२. विषुवछाया for (विषुवत्छाया) (ग) विषुछायांतरमिहाग्रा (क) for (विषुवत्छायांतरमिहऽग्रा)
३. इहाग्रा for (इहऽग्रा)
४. इष्टछाया (ग) (क) for (इष्टछाया)

(ग) ५. तदग्रया for (तदग्रयो)

(क) ६. यंदुदयास्तमयं for (यदुदयास्तमय)

(च) १. अनुपातात्तछंको for (अनुपातातछंको)
२. विषुवछायांतर for (विषुवत्छायांतर)
३. मिहाग्रा for (मिहऽग्रा)

६०. (घ) १. विषुवछायां for (विषुवत्छायां) (ग) (क) विषुवच्छायांतरं for (विषुवत्छायांऽगुलातरं)
२. छायांगुलांतरम् (ग) for (छायांऽगुलांतरं)
४. क्राँति for (क्रान्तिः)

(क) १. विषुवछायां for (विषुवत्छायां) २. गुलांतरं for (ऽगुलांतरं)

क्रान्तिव्यासार्द्धगुणा ३२७० जिनभागज्या १३२९४ हतातुरजादौ ।
कर्कादौ चक्रार्द्धात्प्रोक्ततुलादौ स चक्रार्द्धं ॥६१॥
चक्रात्प्रोह्य मृगादौ स्फुटो सकृत् व्यस्तमृणंधनंमध्यम् ।
अर्कास्माद्देशान्तरयुगयाते मध्यमा प्राग्वत् ॥ ६२ ॥

६१. (घ) १. क्रान्तिर्व्यासार्घ गुणाः (ग) क्रान्तिर्व्यासार्द्ध गुणा (क) for (क्रान्तिव्यासार्द्ध गुणा)

२. कर्क्यादौ (ग) (क) (च) for (कर्कादौ)

३. प्रोत्तुलादौ (ग) प्रोह्य (क) for (प्रोक्ततुलादौ)

४. वक्रार्घम् (क) चक्रार्द्धं (सभार्धेन) for (चक्रार्द्धं)

५. १३२९ for (१३२९४)

६. धनुरजादौ (ग) हृता धनुरजादौ (क) for (हतातुरजादौ)

(च) १. क्रांतिर्व्यासार्द्धं for (क्रान्तिव्यासार्द्धं)

७. गुणाः for (गुणा) ६. हृता १३२९४ नुरजादौ for (१३२९४ हतातुर-जादौ)

६२. (घ) १. प्रामृगादौ for (प्रोह्यमृगादौ) (क) sssत्प्रास्य for (प्रोह्य)

२. अर्को (ग) (क) मध्यर्कोऽस्य for (अर्कास्माद्)

३. मध्यमाः for (मध्यमा)

४. मध्यः (ग) (च) for (मध्यम्)

(ग) ५. sसकृत् (क) sशकृद् for (सकृत्)

(क) ६. व्यस्तधनयुषं for (व्यस्तमृणंधनं) (च) मृण for (मृणं)

७. हेशातरयगया तेन for (देशान्तरयुगयाते)

(च) १. चक्रात्प्रोज्य for (चक्रात्प्रोह्य)

२. अर्क्कोस्मादेशांतर for (अर्कास्माद्देशांतर)

८. प्रागवत for (प्राग्वत्)

शशिमृगोन्यत्यर्धे[१] रात्रेर्गतशेषनाडिका[२] शंकु[३]।
विपरीतगोलविधिना[५] रात्र्यर्द्धा[४] क्रान्तिराभिर्वा ॥ ६३ ॥
क्षितिजेऽग्रा प्राच्यपराक्रान्तिस्त्रिज्यागुणावलम्बहता[१][२]।
द्विगुणमुदयास्तसूत्रं[४] तत्त्रिज्या कृतिविशेषपदम्[३] ॥ ६४ ॥
अक्षज्या शंकुवधालम्बकलब्धोदयास्तमयसूत्रात्[३][१][२]।
दक्षिणतः शंकुतलं दिवसो रात्रौ तदुत्तरतः ॥ ६५ ॥
दिग्लम्बाक्षस्वोदयलग्नच्छायादिषु[१][२] यदिष्टेषु।
षट् षष्टचार्याणां त्रिप्रश्नाध्यायस्तृतीयोयम्[३][४] ॥ ६६ ॥

इति ब्रह्मगुप्ते तृतीयोऽध्यायः

६३. (घ) १. शृंगोन्नत्यर्थं (ग) शृंगोन्यत्यर्थं (क) शृगोन्नत्यर्थं for (शशिमृगोन्यत्यर्धे)
२. नाडिकाम् for (नाडिका)
३. शंकुः (ग) (क) for (शंकु)
४. रात्र्यार्धाक्रंतिराभिर्वा (ग) क्रंतिराभिर्वा (क) कोन्तराभिर्वा for (रात्र्यर्द्धा क्रान्तिराभिर्वा)
(ग) ५. विपरीतगेलविधि for (विपरीतगोलविधिना) (च) विपरित for (विपरीत)
(च) १. शृंगोन्यत्यर्थं for (शशिमृगोन्यत्यर्धे)
४. रात्र्यार्धाक्रांतिराभिर्वा for (रात्र्यर्द्धाक्रान्तिराभिर्वा)
६४. (क) १. हृता (क) (च) for (हता)
(ग) २. गुणवलंबहृता for (गुणावलम्बहता)
(क) ३. वियुक्तपदम् for (विशेषपदम्)
(च) ४. त्सूत्रं for (सूत्रं)
६५. (घ) १. शंकुवधाल्लंबक for (शंकुबधालम्बक) (ग (क)
(ग) २. लब्धोदयस्तसूत्रात् for (लब्धोदयास्तमयसूत्रात्)
(क) ३. शंकुः for (शंकु)
६६. (घ) १. दिल्लंवाक्ष for (दिग्लम्बाक्ष)
(ग) २. छायादिषूपदिष्टेषु (क) (च) for (छायादिषु यदिष्टेषु)
३. त्रिप्रस्नाध्यायः for (त्रिप्रश्नाध्यायस्)
(क) ४. तृतीयोऽयम् for (तृतीयोयम्)
(च) ४. त्रतीयोयं ॥३॥ for (तृतीयोयं ॥६६॥)
वि० श्लोक क्रमांक ६६ के स्थानपर 'तीन' का अंक अध्याय की क्रम संख्या को दर्शाता है।

कालज्ञानं प्रायः पर्वज्ञानार्थमिष्यते सद्भिः ।
शशिभास्करग्रहणयोस्तद्भि[१] व्यक्तिः स्फुटा[२]भेदैः ॥ १ ॥
दिग्वर्णवलनवेलायां[१] निमिलनोन्मीलनस्थितिविमर्दाः[३] ।
स्पर्श-छाया-मोक्ष-ग्रासेष्ट-ग्रास-परिलेखा[२] ॥ २ ॥
भेदाश्चतुर्दश[४] तयोरिन्द्वर्कग्रहणयोः[५] परिज्ञानात्[१] ।
यस्माद्भेदज्ञानं तस्माद् ग्रहणं[२] प्रवक्षामि[३] ॥ ३ ॥
तिथिगतगम्ये भुक्ति[१] भुक्त्यंतरहते[४] फलोनयुतौ[२] ।
रविशशिनौ समलिप्तौ यातस्तात्कालिको[३] भवति ॥ ४ ॥

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१. (घ) १. तदभिव्यक्तिः (ग) for (तद्भिव्यक्तिः)

(क) १. स्वदभिव्यक्तिः for (तद्भिव्यक्तिः)

(च) १. तदभिव्यक्तिः for (तद्भिव्यक्तिः)

२. (घ) १. खला निमीलानान्मीलन (ग) वेला निमीलनोन्मीलन for (वेलायां निमिलनोन्मीलन)

(क) मूलश्लोक यहाँ उपलब्ध नहीं, हाँ उसकी टीका दी हुई है ।

(च) ३. निमीलनोन्मीलन for (निमिलनोन्मीलन)

३. (घ) १: परिज्ञानातः for (परिज्ञानात्)
२. ग्रहणे (ग) (क) for (ग्रहणं)
३. प्रविक्ष्यामि (ग) (क) प्रवक्ष्यामि for (प्रवक्षामि)

(ग) ४. भेदश्चतुर्दश for (भेदाश्चतुर्दश)
५. ग्रहणयो for (ग्रहणयोः)

(च) २. द्ग्रहणो for (द्ग्रहणं) ३ प्रविक्ष्यामि for (प्रवक्षामि)

४. (घ) १. गुणेभुत्त्यंतर for (भुत्त्यंतर)
२. हृते (क) (च) for (हते)

(क) १. यहाँ 'भुक्ति' के आगे 'गुणे' अधिक पद है
३. पातस् (च) for (यातस्)

(च) ४. +गुणे+

खत्रिघन (घ्न) गुणा व्यासार्द्ध[४] भाजिता[२] ३२७० चन्द्रपातयोगज्या[३]।
विक्षेपकालाः[५] सौम्याः[६] षट्[१] राश्यूनधिके याम्याः ॥ ५ ॥
रविशशिभुक्ती भवदशगुणे[५] नख[१] स्वरजिनैः[६] हते[२] नाते[३]।
तत्वाष्टगुणितभुक्तचोर्विवरं[८] षष्ट्याहतं[९] तमसः[४] ॥ ६ ॥
छाद्यं[१] छादक मानैक्यार्धं विक्षेप हनिकं[२] हन्नं[३] $\frac{११}{२०}$ $\frac{१०}{२४}$।
सर्वग्रहणं ग्राह्यादधिके खण्डग्रहणमूने[४] ॥ ७ ॥

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५. (च) १. षड्राश्यूने धिके (ग) षड्राशूनेधिके (क) षड्राश्यूनेऽधिके for (षट्राश्यूनधिके)
(ग) २. विभाजिता for (भाजिता)
३. ज्याः for (ज्या)
४. +२७०+
५. कलाः (क) for (कालाः)
(च) ६. सौम्या for (सौम्याः)

१. षड्राशूनधिके for (षट् राश्यूनधिके)

६. (घ) १. नखैः (क) (च) for (नखै)
२. हृते (ग) (क) (च) for (हते)
३. माने (ग) (क) (च) for (नाते)
४. षष्ट्याहृतमंतरं तमसः (ग) षष्ट्याहतं तसमः for (षष्ट्याहृतं तमसः)
(ग) ५. गुणैः ११।१०। for (गुणे)
६. जिनै २०।२४७ (क) जिनै for (जिनैः)
७. तत्वाष्ट २५।८ (क) तत्त्वाष्ट for (तत्वाष्ट)
(क) ८. भुक्तयो विवरं for (भुक्तचोर्विवरं)
९. षष्ट्याहृतं for (षष्ट्याहतं)

७. (घ) १. छाद्य-(ग) (क) for (छाद्यं)
२. हीनकं (ग) (क) (च) for (हनिकं)
३. छन्नं। (ग) छिन्नम् (क) छन्नम् (च) for (हन्नं)
४. षण्ड (क) for (खण्ड)

(च) वि० इस श्लोक के बाईं ओर "$\begin{vmatrix} ११ & १० \\ २० & २४ \end{vmatrix}$" संख्यासारणी अंकित है।

छाद्यं[४] न युतोनस्य छादकमानस्य[१] तद्दलकृतिभ्याम् ।
विक्षेपकृत्ति[२] प्रोह्य पदेति व[३] स्थिति विमर्दार्द्धे[५] ॥ ८ ॥
षष्ट्या विभाजिता स्थितिविमर्दार्द्ध[१] नाडिका गुणाः[५] स्वगतिः ।
आदौ[६] रविशशिपातेषूणमसकृत्तेषु[३] धनमंते[४] ॥ ९ ॥
स्पर्शान्निमीलनं[१] स्थितिदले विमर्दार्द्ध हीनके पश्चात् ।
मोक्षादर्वागुन्मीलनं[३] विमर्द्दस्तयोरैक्यम्[२][४] ॥ १० ॥
भुक्त्यन्तरमिष्टोनस्थितिदलघटिका गुणहतं[४][१] षष्ट्या ।
बाहुः[२] प्राग्वत् तत्फलहीनयुतैः[५] सूर्यशशिः[३] पातैः ॥ ११ ॥

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८. (घ) १. छदक for (छादक)
२. कृति (ग) for (कृत्ति)
३. तिथिवत् (ग) (क) for (तिवस्थिति)
(क) ४. युतोनास्य for (युतोनस्य)
५. विमर्दाधे for (विमर्दार्द्धे)
(च) २. कृति for (कृत्ति) ३ थिवस्थिति for वस्थिति)

९: (घ) १. विमर्ददलनाडिका (ग) (क) for (विमर्दार्द्धनाड़िका)
२. ३. पातेष्वृणमसकृत्तेषु (ग) (क) (च) for (पातेषूणमसकृतेषु)
४. धनमन्तेः for (धनमंते)
(ग) ५. गुणास्वतिः । (क) गुणास्वगतिः for (गुणाः स्वगतिः)
(क) ६. रवींदु for (रविशशि)

१०. (घ) १. निमीलिनं (च) for (न्निमीलनं)
(ग) २. तदर्द्धैक्यम् (क) तदैक्यार्धः for (तयोरैक्यम्)
(च) ३. माक्षा for (मोक्षा)
४. रेक्यम् for (रैक्यम्)

११. (घ) १. हृतं (ग) (क) (च) for (हृतं)
२. बहुः for (बाहुः)
३. सूर्यशशिपातैः (ग) (क)
(ग) ४. गुणं for (गुण)
(क) ५. तत्फलं for (तत्फल)
३. सूर्यशशिपातैः for (सूर्यशशिः पातैः)

तात्कालिकविक्षेपः कोटिस्तद्वर्गयुतिपदं कर्णः ।
मानैक्यार्द्धात्कर्णं विशोध्य तात्कालिको ग्रासः ॥ १२ ॥
असकृत् ग्रासकालोनप्रमाणयुतिदलकृते विशोध्य कृतिं ।
तत्कालिकविक्षेपस्य शेषमूलकृतं तिथिवत् ॥ १३ ॥
प्रग्रहण स्थित्यार्द्धात् प्राज्यप्रग्रहणतो भवति कालः ।
मौक्षे विशोध्य मोक्षस्थित्यर्द्धात्प्राग् भवेत् मोक्षम् ॥ १४ ॥
स्फुटतिथ्यंते मध्यं प्रग्रहणं स्थितिदलोनकेभ्यऽधिके ।
मोक्षो निमीलनोन्मीलने विमर्दार्धहीनयुते ॥ १५ ॥

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१२. (घ) १. तद्वर्ग्र युति (ग) तद्वर्ग्र संयुति for (तद्वर्गयुति)

१३. (घ) १. असकृद्ग्रासकलोन (ग) असकृग्रास कलोन for (असकृत्ग्रासकालोन)
२. तात्कालिक (ग) (च) for (तत्कालिक)
३. शेषमूलं (च) for (शेषमूल)

(ग) ४. कृति for (कृतिं)
५. तितिवत् for (तिथिवत्)

(क) १. असकृद्ग्रासकालोन for (असकृत् ग्रासकालोन)

(च) १. असकृद्ग्रासकलोन for (असकृत्ग्रासकालोन)
६. विशोध्य for (विशोध्य)

१४. (घ) १. प्रग्रणतो (ग) प्राज्यप्रग्रहणतो (क) प्रोज्य प्रग्रहणवो for (प्राज्यप्रगणतो)
२. मोक्षात् (ग) (च) for (मोक्षम्)

(ग) ३. भवेन् (क) for (भवेत्)

(क) ४. स्थित्यर्धात् for (स्थित्यर्द्धात्)
५. मोक्षं for (मोक्षे)

(च) १. प्राज्य प्रग्रहणतो for (प्राज्यप्रगणतो)

१.५ (घ) १. दलोनक्ये (क) दलोनकेऽभ्यधिके for (दलोनकेभ्यऽधिके)

(ग) २. प्रग्रहणं for (प्रग्रहणं)
३. स्थितिददलोनकेभ्यधिके for (स्थितिदलोनकेभ्यऽधिके)

(च) १. अवग्रहचिन्हलुप्त ४. विमर्द्धार्द्ध for (विमर्दार्ध)

प्राग्[१] पश्चान्नतविषुवज्यो[७] र्वधा त्रिज्ययाप्तचापांशैः[२] ।
उत्तरयाम्यैः[८] पूर्वा विषुवद्[४] वृतत्रिभे[५] ग्राह्यैः[६] ॥ १६ ॥
सममंडलविषुवतो[१] ग्राह्या त्त्रिगृह्[२]धाधिकादुदग्याम्यैः ।
क्रान्त्यंशैरपममंडलपूर्वास्याञ्[३]चन्द्रविक्षेपः ॥ १७ ॥
एकान्यदिशायुतिवियुतेर्ज्या प्रगृहणमध्यं मोक्षेषु ।
वलनं[४] निमीलनोन्मीलनेष्टकालेष्टवो[२] ऽन्येदिशां ॥ १८ ॥

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१६. (घ) १· प्राक् (ग) (क) (च) for (प्राग्)
२. विषुवज्ययो (ग) (क) (च) for (विषुवज्यो)
३. वंधात्त्रिज्यया (ग) (क) (च) for (वंधात्रिज्यया)
४. पूर्वादिषु for (पूर्वाविषुवद्)
५. वृत्त (ग) वृत् for (वृत)
६. ग्राह्या (क) for (ग्राह्याः)

(क) ७. पश्चात् विषुवज्ययो for (पश्चान्नतविषुवज्यो)
८. याम्यै for (याम्यैः)
१. त्रिमे for (त्रिभे)

१७. (घ) १. सममंलाद्विषुवतो (ग) for (सममंडलविषुवतो)
२. त्रिगृहाधिका (ग) for (त्रिग्रहाधिका)
३. पूर्वास्याश्चन्द्र (ग) (क) (च) for (पूर्वास्याञ्चन्द्र)

(क) १. सममंडलाद्विषुवतो for (सममंडलविषुवतो)

१८. (घ) १. न मीलानान् for (निमीलनोन्)
२. कालेष्वतोऽन्यदिशाम् (ग) (च) for (कालेष्टवोऽन्येदिशां)

(ग) ३. एकान्यदिशां (च) for (एकान्यदिशा)
४. विमलं for (वलनं)

(क) अवराम्यदिशाँ ज्यातिहि कुतज्यप्रग्रहणमध्यमोक्षेषु ।
वलनं निमीलनोन्मीनानष्टकालेष्टतोऽभ्यदिशांम् ॥ १८ ॥

(च) २. कालेष्वतोन्यदिशां for (कालेष्टवोऽन्येदिशां)

आद्यन्त्ययोः[१] सधूम्रः[८] कृष्णः[२] खण्डग्रहेऽर्द्धतोभ्यधिके[३] ग्रासः[४] ।
सकृष्ण (ष्ण) ताम्रः[५] सर्वग्रहणे शशिकपीलवर्णः[६][७] ॥ १९ ॥
मानविमर्दस्थितिदलवलनेष्ट[१] ग्रास समकलाद्येषु ।
चन्द्रग्रहणाध्याय[२] विंशतिरार्या इचतुर्थोयां[३] ॥ २० ॥

इति ब्रह्मगुप्ते चतुर्थोऽध्यायः

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१९. (घ) १. आद्यन्तयोः (ग) (क) for (आद्यन्त्ययोः)

२. षण्डग्रहे (ग) for (खण्डग्रहे)

३. ऽर्केतोभ्यधिके for (ऽर्द्धतोभ्यधिके)

४. ग्रासः (दूसरी पंक्ति का आरम्भ) (ग) (क)

५. शशी (ग) (क) (च) for (शशि)

६. कपिलाँ (ग) कपिलवर्णः (क) कपिलः for (कपिलवर्णः)

७. वर्णः (यह शब्द नहीं है) (क) 'वर्ण' पद लुप्त

(ग) ८. सुधूम्र for (सधूम्रः)

(च) ३. द्धेतोभ्यधिके for (द्धतोभ्यधिके) ६. कपिल for (कपील)

२०. (घ) १. दवलनेष्ट for (दलवलनेष्ट)

२. चन्द्रग्रहणाध्यायो (ग) (क) for (चन्द्रग्रहणाध्याय)

३. चतुर्थोयम् । (ग) (क) for (चतुर्थोयां)

५. 'इति' से 'अध्यायः' तक अंकित नहीं । केवल अध्याय समाप्ति सूचक "छः" अंकित है ;

(च) २. चंद्रग्रहणाध्याया for (चन्द्रग्रहणाध्याय) ३. चतुर्थोयं for (चतुर्थोयां)

५. 'इति' से 'अध्यायः' तक लुप्त हैं। अध्याय समाप्ति सूचक कोई और चिह्न भी विद्यमान नहीं।

दृग्गणितैक्यं न भवति यस्मात्पंचज्यया रविग्रहणे ।
यस्माद्यथा तदैक्यं तथा प्रवक्ष्यामि तिथ्यन्ते ॥ १ ॥
वित्रिभलग्नसमेऽर्केन लम्बनं तदधिकोनके भवति ।
तत्संक्रान्ति ज्योदक् यदाक्षजीवा समा न तदा ॥ २ ॥
अवनतिरतोन्यथा भवति संभवेत्तदुदयैर्विलग्नसमं कृत्वा ।
तदुदितघटिकातः छंकुस्तच्चरप्राणैः ॥ ३ ॥
त्रिज्याकृते चतुर्गुणशंकुहताया फलेन भक्तायाः ।
तात्कालिकार्कराशित्रयोनलग्नान्तरज्यायाः ॥ ४ ॥

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१. (घ) १. दृग्दणितैक्यं (च) for (दृग्गणितैक्यं)
(ग) ३. तस्माद्यथा for (यस्माद्यथा)
(क) २. 'न' पद लुप्त है ।
३. तस्माद्यथा for (यस्माद्यथा)

२. (घ) १. वित्रिग्नसमे for (वित्रिभलग्नसमे)
२. द्योदक् (ग) ज्योदग (क) ज्यादक् for (ज्योदक्)
(ग) ३. तत्स्वक्रांति (क) तन् खक्रान्ति for (तत्संक्रान्ति)
(च) ४. क्रेंन for (र्केन) ३ तत्सक्रांतिस्योदक् for (तत्संक्रान्तिज्योदक्)

३. (घ) १. स भवे तदुदयै (ग) संभवेतदुयै (क) संभवेतदुदयै for (संभवेत्तदुदगै)
२. विलग्न समं for (विलग्नसमं)
३. घटिकात्तछंकुस् (ग) घटिकास्त छंकुं (क) घटिकास्तच्छंकुस् for (घटिकातः छंकुस्)
(क) ५. 'कृत्वा' यह पद पूर्वार्ध का अंतिम न होकर उत्तरार्ध का आरम्भिक है (च)

४. (घ) १. कृतेश्चतुर्गुण (ग) कृतेश्चतुर्गुणं १०६९२९०० for (कृते चतुर्गुणं)
२. हृताया (ग) हृतायाः (च) हृताया for (हताया)
३. राशित्रं for (राशित्र)
४. लग्नांतरद्यायाः (च) for (लग्नांतरज्यायाः)
(क) १. कृतेः श्चतुर्गुण for (कृते चतुर्गुण)
(च) १. कृतेश्चतुर्गुण for (कृतेचतुर्गुण)

लम्बनघटिकालग्नात् तात्कालिकात्त्रिराश्यूनात् ।
ऋणं (ऋणां) अधिके ऽर्के हीने धनमसकृत्पंच दृश्यन्ते ॥ ५ ॥
कर्णं गुणाद् व्यासार्द्धाद्दसुवेद ४८ विभाजिता फलविभक्ता ।
लम्बननाडचो भास्करविन्त्रिभलग्नान्तरं वा ॥ ६ ॥
रविशशिपातगतिकला लम्बनघटिका गुणषष्ट्या ।
यदि लम्बनमृणमूना धनमधिकाः स्वफललिप्ताभिः ॥ ७ ॥
अक्षज्याया वित्रिभलग्नात्स्वक्रान्तिरूत्तरार्कस्य ।
इन्दोर्वा यद्यधिका न नतिः सौम्या ऽन्यथा याम्या ॥ ८ ॥

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५. (घ) १. लब्धं लंबन (ग) लब्धं लंबन् (च) for (लम्बन)
२. ऋणमधिके (ग) रुणमधिकेर्क्कं हीने (क) ऋणमधिकेऽर्क्कं हीने for (ऋणं अधिकेऽर्के हीने)
२. ससत्कृत्पंच (क) मसकृत्यं च for (मसकृत्पंच)
४. दश्यंते (क) दश्यन्ते (च) दश्यंते for (दृश्यंते)
(क) १. लब्धं लंबन for (लम्बन)
(च) ५. राश्रनात् for (राश्यूनात्) १ ऋणमधिकेर्के for ऋणं अधिकेऽर्के)

६. (घ) १. विभाजितात्पलविभक्ता (ग) विभाजितात्फलविभक्ता (क) (च) for (विभाजिता फलविभक्ता)
२. लंबतन्याडचो for (लम्बननाडचो)
३. वित्रिवजग्नांतर (ग) वित्रिलग्नांतर (क) बित्रिभलग्नांतर for वित्रिभलग्नांतरं)
४. घावा (ग) ज्यावा (क) ज्यावा for (वा)
(च) ४. लग्नांतरद्यावा for (लग्नांतरं वां)

७. (घ) १. गुणाद्धृता-गुणा हृता (ग) (क) गुणाहृता षष्टचा for (गुणषष्टचा)
(ग) २. शशिपातागति for (शशिपातगति)
३. स्वफललिप्तानिः for (स्वफललिप्ताभिः)
(च) १. गुणा हृताषष्टचा for (गुणषष्टचा)

८. (घ) १. अक्षय्याया (ग) अक्षच्छाया for (अक्षज्याया)
२. अवनतिः (ग) (क) ऽवनतिः for (ननतिः)
(क) ४. स्वाक्रान्ति for (स्वक्रान्ति)
(च) ३. वनतिः for (ननतिः)
५. अवग्रहचिह्न लुप्त

वित्रिभलग्नादुत्तरदक्षिणविक्षेपहीनसंयुक्तम् ।
शंकुधनुरूत्तरायामधिकोनं दाक्षणावनतौ ॥ ९ ॥
तज्ज्येंदु (तज्जेंदु) शंकुराद्यः सवितुः दृक्षेप मंडले युक्तं ।
अपमण्डलेन भानोश्चन्द्रस्य विमण्डलेन युते ॥ १० ॥
त्रिज्या वर्गाविुनौ स्वशंकुवर्गेण तत्पदे दृज्ये ।
रविशशिमध्यगतिगुणे तिथिगुणितव्यासदलभक्ते ॥ ११ ॥
४९०५० स्वच्छायागुणिते वा मध्यगतितिथिगुणस्वकर्णहृते ।
फलयोर्दिक् याम्येतरमवनतिरैक्यं दिगन्यत्वे ॥ १२ ॥
संयोगान्तरमवनतिशशांकविक्षेपयोः सामान्यदिशोः ।
स्फुटविक्षेपः शशिवत् स्थित्यर्द्धं विमर्ददलनाडयः ॥ १३ ॥
प्राग्वल्लम्बनमसकृत्तिथ्यांतात् स्थितिदलेन हीनयुतात् ।
अधिकोनं तन्मध्या ऋणयोरूनाधिकं धनयोः ॥ १४ ॥

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९. (घ) १. दक्षण for (दक्षिण)
२. दक्षिणावनतो (ग) (क) दक्षिणावनतौ for (दाक्षणावनतौ)
(च) १. दक्षण for (दक्षिण) २ दक्षिणावनतो for (दाक्षणावनतो)
१०. (घ) १. दृक् (क) दृक्क्षेप for (दृक्षेप)
(च) १. दृंक्षेप for (दृक्षेप)
११. (घ) १. वर्गाबूनौ (ग) वर्गानूनौ (क) वर्गाबूनौ for (च) वर्गाविुनौ (वर्गाविुनौ)
१२. (घ) १. गती (ग) (क) for (गति)
२. साम्येंतर (ग) साम्येतर (क) साम्येऽन्तर for (याम्येतर)
(ग) ३. यह संख्या इस प्रति में अंकित नहीं है।
४. स्वछया (क) तच्छाया for (स्वछाया)
१३. (घ) १. समान्यदिशोः (ग) (क) समान्यदिशोः (च) for (सामान्यदिशोः)
२. स्फुटिविक्षेपः for (स्फुट विक्षेपः)
३. शशिन for (शशिवत्)
(ग) ४. स्थित्यर्द्धं (क) for (स्थित्यर्द्धं)
(च) ३. शशिव for (शशिवत्)
५. विमर्द्द for (विमर्द)
१४. (घ) १. असकृत्तिथ्यंतात् (ग) (च) for (असकृत्तिथ्यांतात्)
२. मध्याद्दणयो (ग) (क) मध्याद्दणयो (च) for (मध्याऋणयो)

यद्यधिकं स्थित्यर्द्धं तदन्तरेणान्यथोनमूणमेकम् ।
अन्यद्धनं तदैक्येनाधिकमेवं विमर्दार्द्धः ॥ १५ ॥
स्फुटतिथ्यंताल्लम्बनमसकृतस्थित्यर्धहीनयुक्ताद्वा ।
तत्स्फुटविक्षेपकृतिस्थित्यर्द्धोनयुततिथ्यन्ते ॥ १६ ॥
तत्स्पष्टतिथिछेदांतरे स्फुटे तिथिदले विहीनयुतात् ।
स्वविमर्दार्द्धेनाऽसकृदेवं स्पष्टो विमर्दार्द्धं ॥ १७ ॥
शशिवद्वाहुः स्फुटविक्षेपकृतं स्थितिदलेन संगुणिता ।
स्पष्टः स्थित्यर्धहतो भवति भुजः पूर्ववच्छेषम् ॥ १८ ॥

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१५. (घ) तदैक्येनाधिकमेवं (ग) (च) for (तदेक्येनाधिकमेवं)
(क) २. विमर्दार्द्धे (च) for (विमर्दार्द्धः)

१६. (घ) १. कृत (ग) कृतः (च) कृत for (कृति)
(ग) २. तालंबन for (ताल्लंबन)
३. मसकृत्स्थित्यर्द्ध for (मसकृतस्थित्यर्ध)
(क) यहाँ पूरा श्लोक अंकित नहीं। लिपिकार की भूल से। केवल अंकित है—
"स्फुटे तस्य ताल्ल तत युत तिथ्यंते" ॥ १६ ॥
(च) ४. तिथांता for (तिथ्यंता)

१७. (घ) १. स्थिति (ग) (क) (च) for (तिथि)
२. स्पष्टे (ग) (क) (च) for (स्पष्टो)
(ग) ३. तत्स्फुट for (तत्स्पष्ट)
४. विमर्दार्द्धं for (विमर्दार्द्धे)
(च) ५. विमर्द्धार्द्धेनासक्तदेवं for (विमर्दार्द्धेनाऽसकृदेवं)

१८. (घ) १. कृतस्थितिर्दलेन for (कृतंस्थितिदलेन) (ग) कृतः (क) कृतः for (कृतं)
२. हृतो (ग) (क) (च) for (हतो)
(ग) ३. संगुणितः (च) for (संगुणिता)
४. पूर्ववछेषम् (क) पूर्ववत् शेषम् ॥ १८ ॥ for (पूर्ववच्छेषम्)
(क) ५. विसर्गलुप्त
६. विसर्गलुप्त
(च) १. कृतस्थिति for (कृतंस्थिति)
४. पूर्ववछेषम् for (पूर्ववच्छेषम्)

ग्रासात्कालः प्राग्वत्स्पष्टस्थितिदलगुणो सकृद्धृक्तः ।
स्फुटविक्षेपऋतस्थितिदलेन शोध्यः स्थितिदलात्स्वात् ॥ १९ ॥
वलनादिशशिवदन्यत् ग्रहणं तैक्ष्णाद्रवेरनादेश्यम् ।
द्वादशभागादूनं स्वच्छत्वात्षोडशादिदे ॥ २० ॥
न स्फुटमार्यभटादिष्टर्कग्रहणं यतस्ततः स्पष्टम् ।
शंकुज्यया कृतं लघु लघुतरमेवं रविग्रहणम् ॥ २१ ॥
लग्नात्त्रिराशिहीनादपक्रमक्षांशयुतिविशेषोनात् ।
भतृतया तज्जया भक्ता त्रिज्याद्धं कृत्तिः ३६७३२२५ फलेनहृता ॥२२॥

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१९. (घ) १. विक्षेकृत (ग) विक्षेपकृतः for (विक्षेपऋत)
(ग) २. आसात्कालः (क) आसीत्कालः for (ग्रासात्कालः)
३. शशिवत् (क) स्पर्शवन् for (प्राग्वत्)
(क) ४. ऽसकृद्धृक्तः for (सकृद्धृक्तः)
५. कृत (च) for (ऋत)
६. सोध्यः for (शोध्यः)
७. स्यात् for (स्वात्)
२०. (घ) १. तैक्ष्णाद्रवेरनादेश्यम् (ग) (क) (च) for (तैक्ष्णाद्रवरनादेश्यम्)
(ग) २. वदन्यग्रहणम् (क) वदन्यद्ग्रहणम् for (वदन्यत् ग्रहणम्)
३. दिदोः (क) for (दिदे)
२१. (घ) १. भटादिष्वर्क (ग) for (भटादिष्टर्क)
(ग) २. तत्स्पष्टम् (क) ततः स्पृष्टम् for (ततः स्पष्टम्)
(क) ३. रवेर्ग्रहणम् for (रविग्रहणम्)
(च) १. भटादिष्वर्क्कग्रहणं for (भटादिष्वर्कग्रहणं)
२२. (घ) १. हींदप for (हीनादप)
२. क्रमाक्षांश (ग) (क) (च) for (क्रमक्षांश)
३. तज्या for (तज्जया) (ग) ज्याछेदस्त्रिज्या (क) ज्ज्याछेदस्त्रिज्या for (भतृतया तज्जया)
४. त्रिज्यार्धकृतिः (ग) र्द्धकृते २६७३२२५ (क) कृतेः for (त्रिज्यार्द्ध कृतिः)
५. हृता (क) for (हृता)
(ग) ६. भत्रितया (क) for (भतृतया)
(च) ६. भत्रीतया for (भतृतया) ३ तज्या for (तज्जया)
४. त्रिज्यार्द्धकृतिः for (त्रिज्यार्द्धकृत्तिः)

त्रित्रिभलग्नार्कांतरजीवा[3] धटिकादिलम्बनं लब्धम् ।
वित्रिभलग्नापक्रमविक्षेपाक्षांशयुतिवियुति[2] (ते)[4] ॥ २३ ॥
जीवा शशांकभास्करमध्यमभुक्तांतरेण[1] संगुणिता[3] ।
पंचदश-भिर्गुणितं[2] या विभाजिता त्रिज्यया वनति[4] ॥ २४ ॥
४९०५० पूर्ववदन्यस्पष्टं[1] ब्रह्मोक्तस्पष्टसूर्यशशिपातैः[3] ।
नार्यभटादिभिरुक्तै[5] र्या तो स्फुटास्ते[4] ततो स्पष्टम्[6] ॥ २५ ॥

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२३. (घ) १. भितृ लग्नार्कांतर (ग) वित्रिभ लग्नार्कांतरे for (त्रित्रिभलग्नार्कांतर)
२. वियुते (ग) (च) for (वियुति (ते))

(ग) ३. जीवाद्य for (जीवाघ)
४. विक्षपाक्षांश for (विक्षेपाक्षांश)

(क) यहाँ श्लोक की प्रथम पंक्ति लिपिकार लिखना भूल गया। दूसरी है—
"वित्तिभलग्नापक्रमविक्षे पक्षांशयुतिवियुतेः ॥ २३ ॥"

(च) १. वित्रिभलग्नार्क्कांतर for (त्रित्रिभलग्नार्कांतर)

२४. (घ) १. मध्यमभुक्त्यंतरेण (ग) (क) (च) for (मध्यभुक्तांतरेण)
२. गुणितं (ग) गुणितया (क) for (गुणितं)

(ग) ३. संगणिता for (संगुणिता)
४. वनतिः (क) for (वनति)

२५. (घ) १. पूर्ववदन्यत्स्पष्टं (क) (च) for (पूर्ववदन्यस्पष्टं)
२. यतो (ग) यतोऽस्फुटा (क) यतोऽस्फुटस्ते for (यातो स्फुटास्ते)

(ग) ३. ब्रह्मोक्तं for (ब्रह्मोक्त)
४. स्तैततोऽस्पष्टाम् (क) स्तेततोऽस्पष्टम् for (स्तेततो स्पष्टम्)

(क) ५. रुचेक्तै for (रुक्तै)

(च) २. र्यंतो for (र्यातो)
६. वि०—यहां चिह्न लगाकर ऊपर के रिक्तोपान्त पर निम्नलिखित श्लोक—
किसी अन्य हाथ से लिखा हुआ विद्यमान है—
"अंगुलमात्र विरते रक्तः शशिमंडले भवेत्कर्णः ।
भानोस्तु पुन कुदमो वर्ण सर्वत्र निर्दिष्टः ॥"

इष्टग्रासविमर्दस्थित्यर्द्धा वनति लम्बा[1] नाद्येषु ।
आर्या षड्विंशत्यर्क[2]ग्रहण[3]पंचमोऽध्यायः ॥ २६ ॥
इत्ति ब्रह्मगुप्ते[4] पंचमोऽध्यायः ॥

२६. (घ) २. लंबनाद्येषु (ग) for (लम्बानाद्येषु)

२. षड्विंशत्याऽर्क (ग) (च) for (षड्विंशत्यर्क)

३. ग्रहणं (ग) for (ग्रहण)

(ग) ४. 'इति' से 'अध्यायः' तक इस प्रति में लुप्त। केवल समाप्ति सूचक छः, छः, छः, अंकित है

यह श्लोक इस प्रति में नहीं है।

(च) १. लंबनाद्युं for (लंबनाद्येषु)

३. 'इति' से 'अध्यायः' तक लुप्त है।

ग्रहाभास्करांतरैः प्राक् पश्चादर्कग्रहान्तरैर्यस्मात् ।
स्वांशैर्दृश्या दृश्यास्तस्माद्वक्ष्ये तदानयनम् ॥ १ ॥
प्रागूनभुक्तिरूनो दृश्ये दृश्यो रविरधिकभुक्तिः ।
पश्चादृश्योऽधिकगतिरधिको दृश्यो ग्रहो ऽल्पगतिः ॥ २ ॥
विक्षेपसत्रिराशिक्रान्तिवधो व्यासदलहृतो लिप्ताः ।
शोध्यास्तयोः समदृशोर्यदन्यदिशो ग्रहे क्षेपाः ५ ॥ ३ ॥

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१. (घ) १. ग्रहभास्करान्तरैः (ग) ग्रहभास्करांतरैः for (ग्रहाभास्करांतरैः)

(ग) २. वक्षे for (वक्ष्ये)

(क) यह श्लोक उपलब्ध नहीं है ।

(च) १. ग्रहभास्करांतरैः for (ग्रहाभास्करांतरैः) ३ प्राक्पश्चादर्क्कग्रहांतरै for (प्राक् पश्चादर्कग्रहाँतरै)

४. स्वाशै for (स्वांशै) ५. दृश्या for (दृंश्या)

२. (घ) १. रवेरधिकभुक्तिः (च) for (रविरधिकभुक्तिः)

२. गति (च) for (गतिः)

(ग) ३. दृश्योश्यो for (दृश्येदृश्यो)

४. पश्चाद्रिश्योरधिक for (पश्चादृश्योऽधिक)

(क) यह श्लोक उपलब्ध नहीं है ।

(च) ५. दृंश्यो for (दृश्यो)

३. (घ) १. र्यद्यन्यदिशो (ग) (च) for (र्यदन्यदिशो)

२. ग्रंहे (ग) for (ग्रहे)

(ग) ३. हृतो for (हृतो)

४. समर्दृशो (च) for (समदृशो)

(क) यह श्लोक उपलब्ध नहीं है ।

(च) ५. क्षेप्याः for (क्षेपाः)

विषुवच्छाया[१] गुणिताद्विक्षेपा द्वादशोद्धृता[४][५] सोम्यात्[२] ।
फलमृणधनं धनर्णं[६] याम्यादुदयास्तमन्यलग्ने[३] ॥ ४ ॥
प्रागूनमाद्यमधिकं पश्चाल्लग्नाद्ग्रहोदयोस्तमयषड्भयुतम्[५] ।
अन्यदुदयैः[१] घटिकाः कृत्वा[२] नमधिकं समम् ॥ ५ ॥

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४. (घ) १. विषुवछाया (ग) (च) for (विषुवच्छाया)
२. सौम्यात् (ग) त्सौम्यात् (क) for (सोम्यात्)
३. उदयास्तमयलग्ने (ग) उदयास्तमलग्ने for (उदयास्तमन्यलग्ने)

(क) ०. विक्षेपात् । तद्वादशो for (विक्षेपा द्वादशो)
५. हृतात् for (द्धृता)
६. धनंर्ण for (धनर्णं)
३. उदयास्तमयलग्नौ for (उदयास्तमन्यलग्ने)

(च) २. सौम्यात् for (सोम्यात्)
३. याम्यादुदयास्तमयलग्ने for (याम्यादुदयास्तमन्यलग्ने)

५. (घ) १. उदयैर्घटिकाः (क) उदयैर्घटिकाः for (उदयैः घटिकाः)
२. कृत्वोनमधिक (ग) (क) कृत्वोनमधिक for (कृत्वानमधिकं समं)

(ग) ३. प्रागूना for (प्रागून)
४. लग्नात्पश्चाद्ग्रहोदयोस्तमयम् for (पश्चाल्लग्नाद्ग्रहोदयोस्तमय)
५. षट्कयुतमन्यदुदयैर्घटिकाः (वि० दूसरी पंक्ति का आरम्भ) for (षड्भयुतम्—द्वि० पं०—अन्यदुदयैर्घटिकाः)

(क) ४. पश्चाल् लग्नान् ग्रहोदयोस्तमयः for (पश्चाल्लग्नाद्ग्रहोदयास्तमय)
५. 'षड्भसंयुतं' for (षड्भयुतम्) दूसरी पंक्ति का आरम्भ

(च) ५ षड्भूयुतं for (षड्भयुतम्)
१. अन्यदुदयैर्घटिकाः for (अन्यदुदयैः घटिकाः)
२. कृत्वोनमधिकसमं for (कृत्वा नमधिकं समं)

द्वादशभिः शितांशुसितजीवज्ञशनिभूमिजा नवभिः ।
द्युत्तर वृद्धयैरंतरधटिका षड्गुणिता कलांशैः ॥ ६ ॥
१२।९।११।१५।१७ दृश्यादृश्यायुतिवद्ग्रहार्क भुक्त्यंतरैक्यलब्धदिनैः ।
ऊनाधिकलिप्ताभ्यः प्राग्वत् तात्कालिकैरसकृत ॥ ७ ॥
प्रतिदिनमुदयास्तमयावसकृत् तात्कालिक गृह विलग्नैः ।
सूर्यास्तमयोदयिकैः शीतांशौ पौर्णिमास्यन्ते ॥ ८ ॥

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६. (घ) १. शीतांशुः (क) (ग) शीतांशुः for (शितांशु)
४. भूमिजातवभिः for (भूमिजानवभिः)
३. द्वयुत्तरवृद्धे (ग) द्वयूत्तरवृद्धि (क) द्वयुत्तरवृद्धै for (द्युत्तरवृद्धयै)
४. कालांशैः (ग) (क) for (कलांशैः)
६. निभूमिजा नवभिः ।९।११।१३।१५।१७।। for (शनिभूमिजा नवभिः)
(क) ७. षड्गुणित for (षड्गुणिता)
(च) १. शीतांशु for (शितांशु) ३ द्वयुत्तरद्वृद्धैरंतर for (द्युत्तरवृद्धयैरंतर)
४. कालाशैः for (कलांशैः)
७. (घ) १. रसकृतः (ग) रसकृत् (क) रसकृत् for (रसकृत)
(ग) २. यह संख्याएं पूर्वश्लोक के पूर्वार्द्ध के पश्चात् अंकित हैं ।
३. दृश्याग्रहयुति (क) दृश्यादृश्यैर्युति for (दृश्यादृश्यायुति)
(क) ४. ग्राहार्क for (ग्रहार्क)
५. ६. भवति for (लब्धदिनैः)
७. नैकेनाधिक for (ऊनाधिक)
८. लिप्ताभ्यां for (लिप्ताभ्यः)
(च) ४. ग्रहार्क्क for (ग्रहार्क) ७ उनाधिक for (ऊनाधिक)
१. रसकृत् for (रसकृत)
८. (च) १. तालिक for (तात्कालिक)
२. शीतांशो (ग) शीतांशोः (क) for (शीतांशौ)
३. पौर्णिमास्यां च (ग) (क) पूर्णिमास्या च (पौर्णिमास्यन्ते)
(ग) ४. वसकृ (क) च सकृत् for (वसकृत्)
५. दयकैः (क) दयकैः for (दयिकैः)
(क) ६. ग्रहाविलग्नैः for (ग्रहविलग्नैः)
४. वसकृत्तात्कालिक for (वसकृत् तात्कालिक)
३. पोर्णिमास्यांत for (पौर्णिमास्यन्ते)

उदयास्तमयावदिदोः[१] कलांशैरर्कसंमिते कार्यम् ।
हीनत्वं त्वधिकत्वंतरे योग काल स्यात् ॥ ९ ॥
उदयास्तविधौ विवद्बुध्य शीघ्रादाप्तफल युतोनज्ञः[१] ।
अस्मद्गुरू विहीन[२] समद्ग्र[३]····भागेन ॥ १० ॥
मानाऽल्पत्वात्पश्चादुदयोस्तमयः[७] सितस्य[८] दशभि[५] प्राक् ।
पश्चान्मानं[२] हत्वा दष्टमयोस्ताभि[४]रुदयः[६] प्राक्[९] ॥ ११ ॥

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९. (क) यह श्लोक मूल पाठ में नहीं है । (ग)

(क) यह श्लोक उपलब्ध नहीं है ।

(च) वि० यहाँ यह श्लोक निम्न उपान्त पर अन्यहस्तलेख में दिया गया है—

१. उदयस्तमयवदिदोः for (उदयास्तमयावदिदोः)

१०. (घ) यह श्लोक मूल पाठ में नहीं है । (ग) (क)

(च) वि० यह श्लोक २५b पृष्ट पर निम्नोपात पर अन्यपस्त से दिया हुआ है !

१. नोज्ञः for (नज्ञः) २ विहिनः for (विहीन)

३. ससम for (समद्ग्र)

११. (घ) इसका संख्याक्रम ९ है (ग)

१. माता for (माना)

२. मानमहत्वा (क) (क) for (मानं हत्वा)

३. दस्तमयो (ग) (क) for (दष्टमयो)

४. ष्टाभि (ग) (क) for (स्ताभि)

(ग) ५. दशभिः (क) (च) for (दशभि)

६. रुदय (क) for (रुदयः प्राक्)

(क) ७. दुदयो for (पश्चादुदयो)

८. ऽस्तमयो for (स्तमयः)

(च) २. मानमहत्त्वादस्तमयो for (मानं हत्वादष्टमयो)

४. ष्टाभिरुदयः for (स्ताभिरुदयः)

९. क्रमसंख्या '९' है !

ज्ञस्यैवं मनुसूर्ये पठितैः कुजजीवपुत्राणाम् ।
उदय प्राग् ऽस्तमयो मानसमत्वाद्भवति पश्चात् ॥ १२ ॥
आर्यभटः क्षेत्रांशैर्दृश्यान्यदुक्तवास्तदसत् ।
दृग्गणितैक्यं विसंवाहा दृग्गणितैक्यं स्वकालाशैः ॥ १३ ॥

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१२. (घ) इसका संख्याक्रम १० है (ग)

१. ज्ञास्येवं (ग) ज्ञस्यैव (क) ज्ञास्यवं for (ज्ञस्यैवं)

२. पठितै (क) पवितै for (पठितैः)

३. जीवसूर्यपुत्राणाम् (वा) (क) जीवसूर्यपुत्राणां for (जीवपुत्राणाम्)

४. उदयः (ग) (च) for (उदय)

५. प्रागस्तमयो (ग) प्रागस्तमयौ (क) प्रागस्तमयो for (प्राग्ऽस्तमयो)

(ग) ६. माहनमहत्त्वाद्भवति पश्चात् for (मानसमत्त्वाद्भवति पश्चात्)

(क) ७. ॥ १० ॥ (च) for (॥ १२ ॥)

(च) २. पठितै for (पठितैः) ३ + सूर्य +

५. प्रागस्तमयो for (प्राग् ऽस्तमयो)

१३. (घ) इसकी क्रमसंख्या ११ है (ग)

१. दुक्तवांस्तदसत् (ग) (च) for (दुक्तवास्तदमत्)

२. स्वकलांशैः । ११ । (ग) स्वकाल्पांशैः for (स्वकालाशैः)

३. रेखांकित पाठ मूल में नहीं है ।

(ग) ४. दृश्याद्ययुक्त for (दृश्यान्ययुक्त)

५. दृग्गणित विसंवादा (क) दृग्जगितं विसंविवादात् for (दृग्गणितैक्यं विसंवाहा)

(क) ६. आर्यभट्ट for (आर्यभटः)

७. अदुक्तवान् तदसत् for (अदुक्तवास्तदसत्)

८. सूपवित्तकालाशै र्दृग्जगितैक्यं भवति ॥ ११ ॥ for (विसंवाहा दृग्गणितैक्यं स्वकालाशैः ॥ १३ ॥)

(च) २. स्वकालांशैः for (स्वकालाशैः)

८. द्रग्गणितैक्यं for (दृग्गणितैक्यं)

दृग्लग्न[1] दृष्टिभाग ग्रहोदयास्तमयनाडिकाद्यैषु[4] ।
मयाध्यायः[2] षष्ठो[6] चतुर्दशभिरार्याभिः[3] ॥ १४ ॥
इति[5] ब्रह्मगुप्ते षष्ठोऽध्यायः ।

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१४. (घ) इसकी क्रमसंख्या १२ है (ग)

१. दृल्लग्न दृष्टिभाग (ग) दृग्लग्नद्रिष्टिभाग for (दृग्गलग्न दृष्टिभाग)
२. उदयास्तमयाध्यायः षष्टो (ग) (क) उदयास्तमयो for (मयाध्यायः)
३. द्वादशभिरार्याभि । द्वादशभिरार्याभिः । for (चतुर्दशभिरार्याभिः)
५. 'इति' से 'अध्यायः' तक मूल प्रति में नहीं है । केवल समाप्तिसूचक "छः" चिन्ह है ।

(क) ६. ऽध्यायाषष्टो द्वादशाभिरा र्याभिः for (अध्यायः चतुर्दशभि रार्याभिः ॥१४॥)

(च) २. उदयास्तमयाध्यायः षष्टो for (मयाध्यायः)
३. द्वादशभि for (चतुर्दशभि)
४. 'इति' से 'अध्यायः' तक लुप्त है ।

सितवृद्धि[६] हानिर्वा यदि शापाज्जायते[१] कथम[५] गणितात्[२] ।
उपरि खेरिन्दुश्चेदंर्धागर्धं[३] सदा शुक्लां[४] ॥ १ ॥
रविद्रष्टं[१] सितमर्द्धं[२] कृष्णमदृष्टं[३] यथा तपस्थस्य[५] ।
कुम्भस्य तथासन्नं खेरर्धस्थस्य[४] चन्द्रस्य ॥ २ ॥

(घ) १. (ग) शापात् ज्ञायते for (शापाज्जायते)

५. कथं (ग) 'अनंकित' है for (कथम)

३. दर्वागर्धं (ग) for (दंर्धागर्धं)

४. शुक्लम् (ग) for (शुक्लां)

(ग) २. गणितान् for (गणितात्)

(च) ५. कथं for (कथम)

३. चेदर्वागर्द्धं for (चेदंर्धागर्धं)

(क) १. २. गणितान् for (गणितात्)

३. दर्वागर्द्धं for (दंर्धागर्धं)

४. शुक्लं for (शुक्लां)

६. वृद्धिर्हानि for (वृद्धिहानि)

२. (घ) १. दृष्टं (ग) द्रिष्टं for (द्रष्टं)

४. रघस्थस्य (ग) for (रर्धस्थस्य)

(ग) ३. मकृष्णं for (मदृष्टं)

५. तस्य for (थस्य)

(च) १. दृष्टं for (द्रष्टं)

(क) १. दृष्टं for (द्रष्टं) २. शितमर्द्धं (सितमर्द्धं) ३. दृश्यं for (दृष्टं) ४. रघः for (रर्धं)

सिममुण्णतं यतोर्कं सितासितं शुक्लकृष्णपक्षांते ।
अर्धागर्द्धं पश्चाद्गणिता छ्रृंगोन्नतिस्तस्मात् ॥ ३ ॥
रविचन्द्रपात लग्नै स्वक्रान्त्युदयास्तलग्नगतशेषाः ।
घटिकाः स्वचरार्द्धस्तौ स्वेष्टकालिकैः शीतगो कृत्वा ॥ ४ ॥

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३. (घ) १. मुन्नतं (ग) (च) for (मुण्णतं)
२. यतोर्कः (ग) (च) for (यतोर्क)
६. सुक्ल for (शुक्ल)
३. अर्वागर्द्धं (ग) (च) for (अर्धागर्द्धं)
४. यस्मादगणिता (ग) for (पश्चादगणिता)
५. छ्रृंयोन्नतिस्तस्मात् (ग) for (छ्रृंगोन्नतिस्तस्मात्)

(क) १. मुन्नतं for (मुण्णतं) २. यतोऽर्कः for (यतोर्क) ३. अर्वागर्द्धं for (अर्धागर्द्धं)

४. यस्मान् गणितान् for (पश्चाद्गणिता) ५. श्रृंगोन्नति तस्मान् for (छ्रृंगोन्नतिस्तस्मात्)

४. (घ) २. लग्नैः (च) for (लग्नै)
३. स्वचरास्त for (स्वचरार्द्धस्तौ)

(ग) ४. पाता for (पात)
५. गतः for (गत)
६. घटिका स्वचरास्तं for (घटिकाः स्वचरार्द्धस्तौ)
७. श्वेष्ट for (स्वेष्ट)
८. शीतगोः (च) for (शीतगो)

(क) १. रवि चन्द्रपातलग्नैः श्रृंङ्गोन्नतिकालिकै स्वक्रान्ति उदयस्तलग्नगतशेषघटिकाः।
स्वचरांशून कृत्वा प्रति घटिका श्रृंङ्गोन्नति गणनीयः ॥

for

(रविचन्द्रपातलग्नै स्वक्रान्त्युदयास्तलग्नगतशेषाः ।
घटिकाः स्वचरार्द्धस्तौ स्वेस्टकालिकैः शीतगो कृत्वा ॥४॥)

शशिविक्षेपाक्रमधनुषो योगान्तरं समान्यदिशोः ।
सज्ज्येद्वयक्रमज्या स्वाहोरात्राद्यतो रविवत् ॥ ५ ॥
स्वक्रान्तिज्ये त्रिज्यागुणे हते लम्बकेन रविशशिनोः ।
अग्रे पृथक् स्वशंकुतल तुल्ययुक्ते विदिक् वियुते ॥ ६ ॥

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५. (घ) ६. धनुखोर्योगाँत for (धनुषोयोगांत)

३. तज्येंद्वपक्रमय्या (ग) तज्येंद्वपक्रमज्या for (सज्ज्येद्वयक्रमज्या)

(ग) १. विक्षेपापक्रम for (विक्षेपाक्रम)

७. योंगांन्तरं for (योगान्तरं)

(च) ६. धनुरवोर्योगांतरं for (धनुषो योगांतरं)

३. तज्येंद्वपक्रमज्या for (सज्ज्येद्वयक्रमज्या)

(क) १. विक्षेपराव्यपक्रम for (शशिविक्षेपाक्रम) २. समाना for (समान्य)

३. तज्ज्ये for (सज्ज्ये) ४. क्रमज्यात् for (क्रमज्या) ५. यातौ for (यतो)

६. (घ) २. हृते (च) for (हते)

६. शनिनोः for (शशिनोः) (च)

७. अग्रें for (अग्रे)

(ग) ४. युते for (युक्ते)

५. न्यदिग्वियुते for (विदिक् विद्युते)

(क) १. ज्योतिज्ये for (क्रान्तिज्ये) २. वृते for (हते) ३. लावकेन for (लम्बकेन)

४. युते for (युक्ते) ५. विदिम् for (विदिक्)

पृथगन्तरं संयोगौ भुजो युतोर्काच्छशी समान्यदिशो ।
दिग्ज्या वर्गात्स्वात् स्वपृथक्स्थं वर्गं विशोध्य पदे ॥ ७ ॥
वियुतसहिते रविन्द्वोरेकान्यकपालसंस्कृतयोराद्य ।
रविशशिदृक् शंक्वन्तरमन्योऽदृक् दृश्यशंककैक्यम् ॥ ८ ॥

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७. (घ) १. पृथगंतर (ग) (च) for (पृथगन्तरं)
३. च्छसी (च) for (छशी)
४. समान्यदिशोः (ग) (च) for (समान्यदिशो)
५. दृग्ज्या (ग) दृक्ज्या वर्ग्रात् for (दिग्ज्यावर्गात्)
७. स्वात्स्वं (ग) स्वात्स्वपृथक् स्थवग्रं for (स्वपृथक्स्थंवर्गं)

(ग) ८. भुजौ for (भुजौ)
९. विशोध्यशेषपदे for (विशोध्यपदे)

(च) २. युतोऽर्का for (युतोर्का) ५. दृग्ज्या for (दिग्ज्या)
७. स्वात्स्वं for (स्वात्स्व) ६. पृथक् स्वर्गं for (पृथकस्थं वर्गं)

(क) १. पृथगंतरं for (पृथगन्तरं) २. युतोऽर्कात् for (युतोर्का)
३. शशी for (छशी)
४. सामान्य दिशोः for (समान्यदिशो) ५. दृग्ज्यावर्ग्रात् स्वंस्वं for (दिग्ज्या-वर्गात्स्वात् स्व) ६. पृथक् स्ववर्ग्र for (पृथकस्थंवर्गं)

८. (घ) ६. रवींद्वो (ग) (च) for (रविन्द्वो)

२. संस्थयोराद्यः(ग) (च) for (संस्कृतयोराद्यः)

७. शक्वंवर (ग) शंक्वंतर for (शंक्वन्तर)

४. दृग (ग) दृश्यशंकैक्यम् for (दृश्यशंङ्कैक्यम्)

(च) ४. दृग् for (ऽदृक्)

(क) १. सहितै for (सहिते) २. संस्थयोराद्यः for (संस्कृतयोराद्यः)
३. वृक् for (दृक्)
४. ऽदृक for (ऽदृक्) ५. शङ्कैक्यम् for (शंङ्कैक्यम्)

आद्यन्तवर्गयोगान्मूलं पूर्वापरात् भुजात् कोटिः ।
भुजकोटिकृतियुतिपदं तिर्यक्कर्णस्य चन्द्राग्र ॥ ९ ॥
एवं तावाद्यावन्यदयोराद्यन्तयो शशिन्यर्कः ।
रविरर्द्धचक्रयुक्त कल्पो द्वितृतयोरर्कः ॥ १० ॥
व्यकन्दुदलभुजाशाः शशिमानगुणाः शितं नवतिभक्ताः ।
द्विगुणांशो क्रमजीवाद्यावन्नवतिरंशाः ॥ ११ ॥

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९. (घ) १. आद्यान्यवर्ग (ग) आद्यान्यवर्ग्र for (आद्यन्तवर्ग)
२. पूर्वापरा (ग) for (पूर्वापरात्)
३. चन्द्रोग्रे (च) for (चन्द्राग्रे)
(ग) ४. भुजाः for (भुजात्)
५. तिर्यक्कर्णोस्य (च) for (तिर्यक्कर्णस्य)
(च) १. आद्यांत्य for (आद्यन्त) २. पूर्वापरा for (पूर्वापरात्)
(क) १. आज्यात्प for (आद्यन्त) २. पूर्वोपराभुजाकोटि for (पूर्वापरात् भुजात् कोटिः)

१०. (घ) १. तावाद्यावत्पदयो (ग) तावद्यावत्पदयो for (तावाद्यावन्यदयो)
३. शशीव्यर्क्कः (ग) (च) for (शशिन्यर्कः)
५. द्वितृतीययोरर्कः (ग) for (द्वितृतयोरर्कः)
(च) १. तावाद्यावत्पदयो for (तावाद्यावन्पदयो)
५. द्वित्रीतीययोरर्क्कः for (द्वितृतयोरर्कः)
(क) १. तावद्यावत्यक्षरयो for (तावाद्यावन्यदयो) २. राज्यतयो for (राद्यन्तयो)
३. शशीज्यर्क्क for (शशिन्यर्कः) ४. युक्तः for (युक्त) (ग) युक्तः ६ कलो for (युक्तकल्पो)
५. द्वितृतीयोरर्कः for (द्वितृतयोरर्कः)

११. (घ) १. भुजांशाः (ग) (च) for (भुजाशाः)
३. नवतिभक्ता (च) for (नवतिभक्ताः)
५. द्विगुणांशोत्क्रमजीवा (च) for (द्विगुणांशो क्रमजीवा)
४. तावाद्यावन्नवतिरंशाः (ग) तावद्यावन्नवतिरंशाः for (द्यावन्नवतिरंशाः)
(घ) ६. सितं for (शितं)
(च) ७. व्यक्लेंदु for (व्यर्कंदु) ३ नवतिभक्ता for (नवतिभक्ताः)
(क) १. भुजांशाः for (भुजाशा) २. शशिमानागुणासितं for (शशिमानगुणाःशितं)
३. भवति for (नवति) ४. जीवातावद् यावन्नवतिरशाः for (जीवाद्यावन्नवतिरंशाः)

नवतेरधिकांशाना क्रमज्यया संयुतो क्रमत्रिज्या ।
चन्द्रप्रमाणगुणिता द्विगुणा व्यासार्द्धभक्तान्यत् ॥ १२ ॥ ६५ । ४०
प्रथमं शुक्लं रात्रौ दिवसेन्यसंध्ययोस्तदैक्यार्धम् ।
वर्णोज्यासवर्दिदावसितस्य सितं श्रवणं गत्या ॥ १३ ॥
शशिमानपादवर्गो नामार्द्धं सितांतरार्द्धभक्तयुतः ।
परिलेखसूत्रमध्ये शुक्लेर्द्धेन्दैव परिलेखः ॥ १४ ॥

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१२. (घ) ३. रधिकांशा: नां (ग) रधिकांशानां for (रधिकांशाना)
२. संयुतोत्क्रम (ग) संयुतोत्क्रमे (च) for (संयुतोक्रम)
४. द्विगुण for (द्विगुणा)
(ग) ५. ६५४० for (६५।४०)
(च) ३. रधिकांशा: नां for (रधिकांशानां)
(क) नचते रविकांशानां for (नवतेरधिकांशाना) २. क्त्रुम for (क्रम)

१३. (घ) २. दिन्य for (दिवसेन्य)
३. वर्णोग्रास for (वर्णोज्यास)
४. श्रवण (च) for (श्रवणं)
(ग) ५. कर्णेन चन्द्रबिंबे शेषस्य ग्रासवद्वर्ण: for (वर्णो ज्यासवर्दिदावसितस्य)
(च) ६. तदैक्यार्द्ध for (तदैक्यार्धम्)
(क) १. शुक्त (शुक्लं) for (शुक्लं) २ दिवसेन्यत् for (दिवसेन्य)
३. वर्णोग्रासवर्दिदावसितेस्य for (वर्णोज्यासवर्दिदासितस्य) ४. श्रवण for (श्रवणं)

१४. (घ) ४. परिलेख द्वे शुल्केर्द्धाद्येव (ग) परिलेखसूत्रमर्द्धे शुल्केर्द्धज्यैव for (परिलेख-सूत्रमध्ये शुक्लेर्द्धेन्दैव)
३. शुल्केर्द्धज्येव for (शुल्केर्द्धेन्दैव)
५. परिलेष: (च) for (परिलेख:)
(ग) २. पादवर्ज्योगानार्द्ध for (पादवर्गोनामार्द्ध)
६. वरार्द्ध for (तरार्द्ध)
७. भुक्तयुत: for (भक्तयुत:)
(च) ४. सूत्रमद्धेच for (सूत्रमध्ये)
३. शुक्लेर्द्धेष्टैव for (शुक्लेर्द्धेन्दैव) ५ परिलेष: for (परिलेख:)
(क) १. शैशमान for (शशिमान) २. पाक्ष्वर्गोमानार्द्ध for (पादवर्गोनामार्द्ध)
३. शुक्लेर्द्धज्येव for (शुक्लेर्द्धेन्दैव)

त्रिप्रश्नोत्तच्या[५] शंकोः[१] पूर्वापरतो निधाय[४] दिग्मध्ये ।
छायाग्रं[३] छायाग्रछंकुं[२]ग्र दीर्घसूत्रवशात् ॥ १५ ॥
कृत्वा विंशाद्वितयं[१] दृगुद्धृतं[२] पृथग्मुच्चरतमन्यत्[३] ।
प्रथमाग्रस्थितदृष्ट्या[५] द्वितीयवंशाग्रगं[४] चन्द्रम्[६] ॥ १६ ॥

१५. (घ) ३. छायाग्रा (ग) छायाग्रात् for (छायाग्र)
२. छंक्वग्र (क) शक्वग्रग for (छंकुंग्र)

(ग) १. शंकुं for (शंकोः)
४. नधाय for (निधाय)

(च) ५. त्रीप्रश्न्योत्तच्या for (त्रिप्रश्नोत्तच्या)
२. छंकुग्र for (छंकुंग्र)

(क) १. शंकुं for (शंकोः) २. त्यूंक्कग्रगत for (छंकुंग्र)

(ग) अवर्केंदुदलभुजांशार्द्ध ज्याकर्णो द्विगुणिता भुजस्य कृति ।
प्रोह्य पदं·····भुजाग्रचंद्रांतरं कोटिः ॥ १६ ॥
यह श्लोक 'ग' में अतिरिक्त पाया गया है ।

१६. (घ) ३. प्रथम्नुच्चतर for (पृथग्मुच्चरत)
४. वंशागं for (वंशाग्रगं)

(ग) १. वंश for (विंशा)
२. दृगुछ्तम् for (दृगुद्धृतं)
५. द्रिष्टचा for (दृष्टचा)
६. चन्द्रे for (चन्द्रम्)
(वि० इसकी क्रमसंख्या १८ है)

(च) ३. प्रथममुच्चरतमन्यत् for (पृथग्मुच्चरतमन्यत्)

(क) १. वंश for (विंशा) (ग) दृगुछितं (ग) for (दृगुद्धृतं) ३. प्रथम-मुच्चतरमन्यत् for (पृथग्मुच्चरतमन्यत्)

ग्रहणं ग्रहयोगं[२] वा विस्मयकरणाय[४] दर्शयेद् गणकः ।
लोकस्य नरपतेर्वा दुष्करमन्यैर्गणितविद्भिः[३] ॥ १७ ॥
इति[३] बाहुकर्णकोटिस्फुटसितपरिलेख[४]सूत्रकर्णेषु[१] ।
आर्याष्टादश चन्द्रशृंगोन्नतिरिह[५] सप्तमोऽध्यायः ॥ १८ ॥
इति[६] श्रीब्रह्मगुप्ते सप्तमोऽध्यायः

(ग) लब्धेवं वाततयोः पूर्वापरयो खेस···चक्रात् ।
अर्ग्राक्केंद्वोः शुक्लं पश्चाच्छुक्ले सिते पूर्वात् ॥ १७ ॥
यह श्लोक 'ग' में अतिरिक्त पाया गया है।

१७. (घ) २. ग्रहयो योगं for (ग्रहयोगं)
३. गंणिविद्भिः (ग) गंणितषड्भिः for (गंणितविद्भिः)
(ग) इसकी क्रमसंख्या १६ है ।
४. विश्मय for (विस्मय)
(क) १. नरपतेर्दुष्कर for (नरपतेर्वादुष्कर)

१८. (ग) वि०—इसकी क्रमसंख्या २० है।
३. इतिपादकोटिकर्णः for (इतिबाहुकर्णकोटि)
४. परिलेष for (परिलेख)
२. विंशतिरार्या शृंगोन्नतिरिंदोः सप्तमोध्यायः for (आर्याष्टादशचन्द्रशृंगोन्नतिरिह सप्तमोऽध्यायः)
(च) ५. अवग्रहचिह्न लुप्त
६. "इति" से "अध्यायः" तक लुप्त है।
(क) १. कोटिकर्ण for (कोटि) २. आर्योष्टादशभिः for (आर्याष्टादश)

प्राक्चन्द्रलग्नयोरस्तलग्नशशिनो यतोन्तरात्पश्चात् ।
प्रतिदिनमिन्दुच्छाया यतस्तदानयनमभिधास्ये ॥ १ ॥
प्रगृहणकालिकैरिष्टकालिकैर्वार्कलग्नशशिपातैः ।
कृत्वोदयास्तलग्ने स्वचरप्राणान् शंशाकस्य ॥ २ ॥
यद्यधिकमुदयलग्नादूनं षड्गृहयुतास्तमयलग्नात् ।
लग्नं तदा शशांको दृश्यः सवितुर्दर्शने छाया ॥ ३ ॥
लग्नसममुदयलग्नं षड्गृहयुतास्त लग्नं समम् ।
पूर्वापरयोः कृत्वा गतशेषाः स्वोदयैर्घटिका ॥ ४ ॥

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१. (घ) १. यतोंतरात् (ग) र्यतोंतरात् (च) for (यतोन्तरात्)
२. प्रतिदिनछाया for (प्रतिदिनमिन्दुच्छाया)
(ग) ३. रभ्र for (रस्त)
४. ततस्तानयनमभिधास्ये for (यतस्तदानयनमभिधास्ये)
(च) १. र्यतोंतरात् for (यतोंतरात्)
(क) यतोऽंतरात् for (यतोन्तरात्)

२. (घ) १. शशांकस्य for (शंशाकस्य)
(ग) २. काकालिकै for (कालिकै)
(च) ३. वर्क्ष for (वर्कं)
(क) १. शशांकस्यात् for (शंशाकस्य)

३. (घ) २. षड्गृह् (ग) for (षड्ग्रह्)
१. सविदर्शने (ग) सतिदर्शने for (सवितुर्दर्शने)
(ग) ३. छायाः for (छाया)
(च) ४. दुनं for (दूनं) २. षड्गृह् for (षड्ग्रह्)
१. सवितुदर्शने for (सवितुर्दर्शने)
(क) १. सति for (सवितु)

४. (घ) ३. षड्गृह्युता (ग) (च) for (षड्ग्रहयुता)
४. लग्नसमं (ग) (च) for (लग्नंसमं)
(ग) ५. गशेषाः for (गतशेषाः)
६. घटिकाः for (घटिका)
(क) १. लग्न लग्न for (लग्नं) २. युताश्च for (युतास्त)

गतघटिकाः शेषा वा स्वदिनार्द्धं समावंदी[१]दुरद्धार्द्धे ।
गतशेषनाडिकाभिर्नताभि[२]रथवार्कवच्छंकु[३]ः ॥ ५ ॥
शंकुधनुषोऽधिकस्य स्फुटप्रमाणार्द्धं लिप्तिकाभिर्वा[१] ।
रविशशिमध्यगतिकला[२] तिथ्यंशज्यो[३]नितालेदः ॥ ६ ॥
द्वादशभिर्गुणिताया[७] दृज्या[१] लब्धमंगुलाद्य[२] यत् ।
तत्प्रग्[३]हे छाया[५] स्फुटा न्यादावान्यथाऽऽसन्ना[४] ॥ ७ ॥

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५. (घ) १. यदींदुमह्नोर्द्धे (ग) यदींदुरह्नोर्द्धे for (वंदीदुरद्धार्द्धे)
२. नवाभि for (र्नताभि)
३. रथवार्कवच्छंकुः (ग) for (रथवार्कवछंकुः)

(च) १. यदींदुरह्नोर्द्धे for (वंदीदुरद्धार्द्धे)
२. रथवार्क्क for (रथवार्क)

(क) १. समायदादुराह्नोऽर्द्धे for (समावंदीदुरद्धार्द्धे) २. रस्त्र for (रथ)
३. वत् शंकुः for (वछंकुः)

६. (घ) १. ज्या (ग) र्ज्या for (र्वा)
३. शद्योनिपाछेदः (ग) श ज्योनिवच्छिदः for (शज्योनिताछेदः)

(क) १. र्या for (र्वा) २. तला for (कला) ३. सज्योतिताछेदः for (शज्योनिता-छेदः)

७. (घ) १. १२. दृग्ज्याया (ग) दृक्ज्यांया for (दृज्या)
३. तत्प्रग्रहणे (ग) सा प्रग्रहणे for (तत्प्रग्रहे)
५. छा for (छाया)
४. स्फुटान्य (च) स्फुटान्य for (स्फुटान्या)
६. यथासन्ना (ग) कन्ययासन्ना for (यथाऽऽसन्ना)

(च) ७+१२+ १. दृग्ज्याया for (दृज्या)
३. तत्प्रग्रहणे for (तत्प्रग्रहे) ८. अवग्रहचिह्न लुप्त

(क) १. दृग्ज्याया for (दृज्या) २. मंगुलाप्तं for (मंगुलाद्यं)
प्रग्रहणे for (प्रग्रहे) ४. न्यदा (ग) for (न्यादा)

ज्येष्टापक्रमभागैर्मध्यच्छायार्कवच्छशांकस्य ।
शशिवमादातामृक्षाणां तु स्वकांत ध्रुवकात् ॥ ८ ॥
इह नोदृष्टं यत्तद्रविवद्गतशेषनाडिकाद्येषु ।
आर्याभिर्नवाभिरयं चन्द्रछायाऽष्टमोऽध्यायः ॥ ९ ॥
इति श्री ब्रह्मगुप्ते अष्टमोऽध्यायः ॥

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८. (घ) १. स्वेष्टा (ग) for (ज्येष्टा)
२. वछशांकस्य for (वच्छशांकस्य)
३. शशिवद्भौमादीनामृक्षाणां (ग) शशिवद्भौमादीनां नक्षत्राणां च for (शशिवमादातामृक्षाणां)

(ग) ४. स्वकाध्रुवकात् for (स्वकांतध्रुवकात्)

(च) ५. छायार्क्क for (छायार्क) ४. स्वकात् for (स्वकांत)

(क) १. स्वेष्टा for (ज्येष्टा) ३. शशिवत्भौमादीनां नक्षत्राणां स्वकान् ध्रुवकान् for (शशिवमादाता मृक्षाणां तु स्वकांतध्रुवकात्)

९. (घ) २. नवभि for (नंवाभि)
३. चन्द्रछायां ष्टमोध्यायः for (चन्द्रछायाऽष्टमोऽध्यायः)

(ग) ४. हनोद्दिष्टं for (इहनोद्दिष्टं)
१. यत्तद्रविवत्तच्छेष for (यत्तद्रविवद्गतशेष)
५. आर्यानवभिरयं for (आर्याभिर्नवाभिरयं)
समाप्तिसूचक छः ॥ छः ॥ छः रेखांकित के स्थान में

(च) २. नंवभिरयं for (नंवाभिरयं)
३. अवग्रह चिन्हलुप्त
६. छाः श्री for ("इति·····अध्यायः")

(क) १. यद्वद्रविवत् for (यत्तद्रविवद्गत) २. नवभिरियं for (नंवाभिरयं)
३. चन्द्रच्छाया for (चन्द्रछाया)

शून्येशा[७] यमतिथय[६]: षडगा: षट्भिन्दव[१]: स्वगुण[८] चन्द्रा: ।
क्रान्ति[२]र्विक्षेपकाला[३]: कुजबुधगुरुशुक्र[४]रविजानां[५] ॥ १ ॥
११० । १५२ । ७६ । १३६ । १३०
व्यासार्द्धं संयुक्तं त्रिगुणान्त्य फलज्ययान्त्यकर्णोन[१] ।
त्रिधन[५]गुणं २७ स्वदशांशै[७] गुणायांत्य[२] फलज्ययाभक्तं[३] [४] ॥ २ ॥
स्फुटमानकलाभूमिजबुधसुरगुरुशुक्रसूर्यपुत्राणां ।
नाध:स्वयोर्द्रसितयोरासन्नत्वाद्रवेरसितम्[३] ॥ ३ ॥

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१. (घ) ६. मतिथिय: (ग) यमातथिय: १५२ for (यमतिथय:)
१. षट्त्रींदव: (ग) for (षट्भिन्दव:)
(ग) ७. शून्येंशा ११० for (शून्येशा)
८. १३६ खगुणचन्द्रा: for (स्वगुणचन्द्रा:)
२. ३. क्रान्तिविक्षेपकला for (क्रान्तिर्विक्षेपकाला:)
(च) ६. तिथिय: for (तिथय:) १. षट्त्रींदव: for (षड्भिन्दव:)
(क) १. षट्त्रींदव: for (षट्भिन्दव:) २. क्रान्ते for (क्रान्ति) ३. कला for (काला:)
४. शुक्र: for (शुक्र) ५. योनाम् for (जानां)

२. (घ) ५. त्रिद्यन (ग) त्रिधनगुणं for (त्रिधनगुणं)
२. स्वद्दशांशै for (स्वदशांशै)
३. गुंणायांत्य for (गुणायांत्य)
(ग) ६. स्वयांत्यकर्णोनम् for (ज्ययान्त्यकर्णोनं)
७. त्यद्दगं दौर्गुणायांत्यफलभक्तम् ॥ २ ॥ for (स्वदशांशैगुणयान्त्यफलज्या-भक्तम् ॥२॥)
(च) ५. त्रिद्यन for (त्रिधन) ७. स्वद्दशंशै for (स्वदशांशै)
३. गुंणायांत्य for (गुणायांत्य)
(क) १. कर्णात् for (कर्णोन) २. द्दगंशै: for (दशांशै) ३. गुणितांत्य for (गुणायांत्य)
४. भक्ता for (भक्तं)

३. (घ) २. स्थयो (ग) नाधस्थयोर्ज्ञ for (नाध: स्वयोर्द्र)
(च) २. स्थयोर्द्र for (स्वयोर्द्र)
(क) १. गुरुदैत्येज्य for (सुरगुरुशुक्र) २. स्थयोर्ज्ञ for (स्वयोर्द्र) ३. रशितम् for (रसितम्)

भुक्त्यंतरेण[१] भुक्तं ग्रहान्तरं[२] फलदिनैरधिकं[६] भुक्तौ जीवरविजानां ।
प्रागूनगतौ[३] पश्चाद्धृतिरधिके[४] वक्रिणोऽर्व्यस्ता[५] ॥ ४ ॥
एको वक्रिभुक्त्योर्युत्योनः[१] प्रागथादधिकः[२] ।
पश्चात्[४] ग्रहयोरन्तर लिप्तास्तथैवभक्ताः[५] स्वभुक्तिः[६] गुणाः ॥ ५ ॥
स्वफलमृणं प्राक् पश्चाद्युतौ धनं वक्रिणी[३] ग्रहे[१] व्यस्तम्[४] ।
समलिप्तौ बुधसितशीघ्रचन्द्रपाते[२] च स्वफलम् ॥ ६ ॥

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४. (घ) ६. दिवैरधिकभुक्तौ (ग) दिनैरधिकभुक्तौ for (दिनैरधिकं भुक्तौ)
४. पश्चाद्युति (ग) पश्चाधृति for (पश्चाद्धृति)
५. व्यस्ता भुक्तिगुणाः (ग) व्यस्ताः for (व्यस्ता)
(ग) १, भक्तं for (भुक्तं)
इस श्लोक की प्रथम पंक्ति का अन्तिम शब्द "जीवरविजानां" इस प्रति में उपलब्ध नहीं ।
(च) ६. रधिक for (रधिकं) ४ पश्चाद्युति for (पश्चाद्धृति)
५. व्यस्ता for (व्यस्ता)
(क) १. भक्तं for (भुक्तं) २. ग्रहंतरं for (ग्रहान्तरं) ३. गतै for (गतौ) ४. पश्चात्द्युति for (पश्चाद्धृति)
५. व्यस्तम् for (व्यस्ता)

५. (घ) ३. वक्री (ग) (च) for (वक्रि)
४. पश्चाञ्च्यात् for (पश्चात्)
(ग) १. युत्योनः for (र्युत्योनः)
२. प्रागवाधिकः पश्चात् for (प्रागथादधिकः । पश्चात्)
५. भक्ता for (भक्ताः)
६. भुक्ति (च) for (भुक्तिः) इस श्लोक की प्रथमपंक्ति का अंतिम पद 'पश्चात्' है । अतः दूसरी पंक्ति 'ग्रहयो' से आरम्भ होती है ।
(च) २. प्रागथाऽधिकः for (प्रागथादधिकः)
(क) १. युत्तयेन for (र्युत्योनः) २. प्रागथाधिकः for (प्रागथादधिकः)

६. (घ) ३. वक्रिणि (ग) for (वक्रिणी)
२. पातेषु च (ग) for (पातेच)
४. व्यस्तः for (व्यस्तम्)
(क) १. ग्रहो for (ग्रहे) २. पातेषु च फलम् for (पाते च स्वफलम्)

बुधसितपाते[१] व्यस्तं मन्दफलमुपान्त्यशीघ्रफलम्[२] ।
शेषाणां स्फुटपातात् विक्षेपो मध्यमात्[३] योगात्[५] ॥ ७ ॥
मदंफलस्फुट शशिनो विक्षेपो[५] भौमगुरुशशिनां[१] च ।
मदंफलव्यस्तस्फुटशीघ्रात्[४] बुधशीघ्रयोरथवा[३] ॥ ८ ॥
समलिप्ता[३] स्फुटमध्यात् स्वपातायुक्तात्[५] ज्ञः शुक्रयोः[४] शीघ्रात् ।
जीवविक्षेपगुणा[१] हृतात्कर्णेन[२] विक्षेपः[६] ॥ ९ ॥

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७. (घ) ४. विक्षेपोन्मध्यमात् (च) for (विक्षेपोमध्यमात्)
५. पातात् (ग) for (योगात्)
(च) ४. विक्षेपोन् for (विक्षेपो)
(क) १. सितप्ततै व्यस्तम् for (सितपातेव्यस्तं) २. सद for (मन्द) ३. मध्यमात्यात् for (मध्यमात्)

८. (घ) १. शनीनां (च) for (गुरुशशिनां च) (ग) जीवर विजानां for (गुरु-शशिनां च)
४. शीघ्राद्बुध (ग) for (शीघ्रात्बुंध)
३. शुक्रयोरथवा (ग) for (शीघ्रयोरथवा)
(ग) ५. शुक्रयोरथवा (ग) for (शीघ्रयोरथवा)
(च) १. शनीनां for (शशिनाँ)
२. शीघ्राद्बुधशुक्रयो for (शीघ्रात्बुंधशीघ्रयो)
'क' १. जीवरविजानां for (गुरुशशिनां च) २. फला for (फल) ३. शुक्रयो for (शीघ्रयो)

९. (घ) ३. समलिप्त (ग) (च) for (समलिप्ता)
४. ज्ञशुक्रयोः (ग) (च) for (ज्ञः शुक्रयोः)
१. जीवा विक्षेप (ग) (च) for (जीवविक्षेप)
२. गुणाहृतात्फ (ग) गुणाहतीत्य for (गुणाहतात्)
(ग) ५. युक्ता for (युक्तात्)
६. विक्षेपाः for (विक्षेपः)
(चं) २. हृतांत्य for (हृतात्)
(क) २. हृतान्य for (हृतात्)

अन्तरयोगो[१] तुल्यान्यदिशोर्विक्षेपयोर्ग्रहांतरकं ।
आर्यभटादिष्ट्रे[२]वं समलिप्तिकयोर्युतिर्ग्रहयोः ॥ १० ॥
चित्रास्वातिवदुदये[१]ऽन्यथान्यथास्ते तथायुतौ[२] ग्रहयोः ।
न भवति दृग्गणितैक्यं यथा तदैक्यं तदुक्तिरतः[३] ॥ ११ ॥
ग्रहयो[१]ः स्वोदयलग्ने समलिप्तिकयो[२]स्तदस्तलग्ने[३] च ।
उदये[४] स्वोदयलग्ने स्वषड्गृह[५] स्वास्तलग्नसमे[६] ॥ १२ ॥
कृत्वे[४]वं दिनघटिकाः ग्रहयोः स्वोदयविलग्नयो[५]रूनं ।
अनस्यास्तविलग्नादेष्यताधिकं युतिरविता[३] ॥ १३ ॥

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१०. (घ) २. आर्याभटादिष्वेवं (च) for (आर्यभटादिष्ट्रेवं)
(ग) १. अंतरयोगौ for (अन्तरयोगो)
(च) १. योगौ for (योगो)
(क) १. योगौ for (योगो) २. ष्वेवं for (ष्ट्रेवं)

११. (ग) ३. तदुक्तितरः for (तदुक्तिरतः)
(क) १. दयो for (दये) २. युतिः for (युतौ)

१२. (घ) ४. उदयैः (ग) for (उदये)
(ग) ५. सषड्गृहे for (स्वषड्गृह)
६. स्वास्तमयलग्ने for (स्वास्तलग्नसमे)
(च) ७. ग्रहयो for (ग्रहयोः) ४ उदयैः for (उदयै)
(क) १. +स्फुट ग्रह युति संसाधानाय सम+ २. समा for (सम)
३. लग्नं for (लग्ने) ४. उदयैः for (उदये) ५. स for (स्व)
६. समो for (समे)

१३. (घ) १. ऊनस्वास्त (ग) ऊनं स्वास्त for (अनस्यास्त)
२. देष्यत्यधिकं (ग) दिष्यत्यधिकं for (देष्यतधिकं)
३. युतिरवीता (ग) युतिरतीता for (युतिरविता)
(ग) ४. कृत्वैवं for (कृत्वेवं)
५. लिलग्नयोरूनम् for (विलग्नयोरूनं)
(च) ऊनस्वास्त for (अनस्यास्त)
३. रवीता for (रविता)
(क) १. ऊनस्वास्त for (अनस्यास्त) २. त्यधिक for (तधिकं)
३. रतीता for (रविता)

ऋणमूनं धनमधिकं स्वोदयलग्नात् [१]समस्तलग्नं चेत्भक्ताः [४]।
भक्ता स्तन्दरकलाः [२]पृथक् पृथक् [३]स्वदिननाडीभिः ॥ १४ ॥
ऋणयोर्वा [३]धनयोर्वान्तरेण [१]मुत्याधनर्णयोर्भक्ताः [२]।
अन्तरलिप्ताः [४]स्वोदय[५]विलग्नयोर्लब्धघटिकाभिः ॥ १५ ॥
उदयास्तविलग्नान्तरकलागुणाः स्वदिननाडिका [२]भक्ताः ।
लब्धकलादिक[१]मूनं स्वास्त विलग्नादुदयलग्नम् ॥ १६ ॥
यद्यधिकमूनमेवं समलिप्तौ स्वोदयाद्यु[१]तौ गृह्यो: ।
रात्रिविलग्नादूनावधिकौ षड्[१]ग्रहादृश्यौ ॥ १७ ॥

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१४. (घ) १. स्वमस्त for (समस्त)
२. तदंतरकलाः for (तन्दरकलाः)
(ग) ३. 'स्व' यह पद यहाँ अंकित नहीं।
(च) ४. 'भक्ताः' द्वितीय पंक्ति का "भक्तास्" रूप में आरंभिक शब्द है।
२. मक्तास्तदन्तर कलाः for (भक्तास्तन्दरकलाः)
'क' २. स्वमस्त for (समस्त)
(ग) २. तदंतर कालाः for (तन्दरकलाः)

१५. (घ) ३. वोधनयो for (वधिनयो)
२. युत्या (ग) युत्तचा for (मुत्या)
(ग) ४. अन्तरलिप्ता for (अन्तरलिप्ताः)
५. स्वाद for (स्वोदय)
(च) ३. वोंधनयो for (वधिनयो)
१. वतिरेण for (वन्तिरेण)
२. युत्या for (मुत्या)
(क) १. वऽन्तिरेण for (वन्तिरेण) २. युत्तचा for (मुत्या)

१६. (घ) २. नाडिकभक्ताः for (नाडिका भक्ताः)
१. लब्धकलाधिकमूनं (च) for (लब्धकलादिकमूनं)
(क) २. कलाधिकमूनं (कलादिकमूनं)

१७. (घ) १. षड्गृहयुताद् (ग) षड्गृहायुता for (षड्ग्रहा)
(च) १. षड्ग्रहयुता for (षड्ग्रहा)
(क) १. षड्ग्रहयुता for (षड्ग्रहा)

एवं मानैक्यार्द्धादधि[१]के मध्यांतरे युति[४]र्ग्रहयोः ।
स्थित्यर्द्ध[२]विमर्द[५]दले हीनेतरा [३]ग्रहेंदुयुतौ ॥ १८ ॥
लम्बनमर्क[५]ग्रहणवद[१]सकृत्[२]स्वावनतिलिप्ति[४]कास्पष्टौ ।
तात्कालिकविक्षेपौ तदन्तरै[३]क्यं समान्यदिशोः ॥ १९ ॥
विक्षेपो मध्यान्तरमु[१]र्ध्वस्वच्छादको ग्रहोऽध[२]स्थः ।
मानैक्यार्द्धादधिके नातिस्पष्टास्फुटोक्तिरतः ॥ २० ॥

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१८. (घ) ३. ताराग्रहेंदुयुतौ (ग) for (तराग्रहेन्दुयुतौ)

(ग) ४. युतिग्रहयोः for (युतिर्ग्रहयोः)

(च) ५. विमर्द्द for (विमर्द)

(क) १. दधि for (दधिके)
२. स्थित्यर्ध for (स्थित्यर्द्ध)
३. तारा for (तरा)

१९. (घ) ३. तदन्तरैक्यम् for (तदन्तरैक्यम्)

(ग) ४. लिप्ता for (लिप्तिका)

(च) ५. मक्कं for (मर्क)

(क) १. वत् for (वद)
२. दश कृत for (दसकृत्)
३. तदन्तारैक्यम् for (तदन्तरैक्यम्)

२०. (घ) १. मूर्ध स्वछादको ग्रहोधस्थः (ग) मूर्धस्य छादको ग्रहोधस्थः for (मुर्ध्वस्वछादको ग्रहोऽधस्थः)

(च) १. मूर्द्ध for (मुर्ध्व)
२. धस्थ for (ऽधस्थः)

(क) १. मूर्धस्य for (मुर्ध्वस्य)
२. ऽधः स्थः for (ऽधस्थः)

ऊनदिनोदितगुणितादधिका दिनादूनदिनाहृतालब्धम् ।
अधिकं प्राग्युतिरूनं यद्यधिक दिनोदिनात्पश्चात् ॥२१॥
अन्तरमाद्यो भूयो यदिष्टघटिका फलोनयुतयोश्च ।
प्राक्पश्चाद्वान्तरयोस्तदन्तरेणोद्धृता तदाद्यात् ॥ २२ ॥
युतान्यथेष्ट घटिका गुणितात्फलनाडिकाभिराद्यवशात् ।
प्राक् समलिप्तिकलात्पश्चाद्वा ग्रहयुतिर्भवति ॥ २३ ॥
स्वदिनघटिका विभक्तस्तदुदितपरदिवसनाडिकाद्यात् ।
तुल्यः परोदिताभिर्घटिकाभिर्यदियुतिर्ग्रहयोः ॥ २४ ॥

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२१. (घ) ३. अधिक दिनादूनदिनहृताल्लब्धम् (ग) for (अधिकादिनदूनदिनाहृतालब्धम्)
(ग) ४. गणिताद् for (गुणिताद्)
२. दिनोतात्पश्चात् for (दिनोदिनात्पश्चात्)
(च) ३. दधिकदिना for (दधिकादिना)
१. दूनदिनहृताल्लब्धं for (दूनदिनाहृतालब्धं)
(क) १. हृताल्लब्धम् for (हृतालब्धम्) २. सदिनोदिनात्पूर्वे for (दिनोदिनात्पश्चात्)

२२. (घ) १. अन्तमाद्यं for (अन्तरमाद्यो)
(ग) २. न्यदिष्टघटिका for (यदिष्टघटिका)
४. तदन्तरेणोद्धृता for (तदन्तरेणोद्धृता)
५. दाद्यात् for (तदाद्यात्)
(क) १. नाद्यं for (माद्यो) २. ज्यदि घटिका for (यदिष्टघटिका)
३. पश्चात् ।। द्वयन्तरयोः for (पश्चाद्वान्तरयोस्) ४. तदन्तारे for (तदन्तरे)
५. दाधान for (तदाद्यात्)

२३. (घ) १. युत्पान्यथेष्ट (ग) युक्तान्यथेष्ट for (युतान्यथेष्ट)
३. लिप्तिक कालात् (ग) for लिप्तिकलात्)
(च) १. युत्यान्यथेष्ट for (युतान्यथेष्ट)
१. द्युत्कान्यथेष्ट for (युतान्यथेष्ट) २. ऽराद्यवशात् for (राद्यवशात्)
(क) ३. लिप्तिक कालात् for (लिप्तिकलात्)

२४. (ग) ३. विभक्तात्तदुदित for (विभक्तस्तदुदित)
१. घातः for (घात्)
(क) १. घातः for (घात्) २. परोदितताभि for (परोदिताभि)

विक्षेपमानसमकलदिनघटिकाः[3] स्वोदयास्तलग्नाद्यैः[1] ।
(पंचविंशति) षड्विंशति[2] आर्याणां नवमो ग्रहमेलकोऽध्यायः[1] ॥ २५ ॥
इति ब्रह्मगुप्ते[4] नवमोऽध्यायः ॥

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२५. (घ) २. षड्विंशत्यार्याणां (च) for (पंचविंशति आर्याणां)

(ग) ३. घटिका for (घटिकाः)

दूसरी पंक्ति निम्नांकित है—

'नवमोध्यायोग्रहयुतिरार्याणां पंचविंशत्याः' for (पंचविंशति आर्याणां नवमो-ग्रहमेलकोऽध्यायः)

(च) ४. श्रीः for ("इति...नवमोऽध्यायः")

(क) १. सकल for (समकल)

२. पंचविंशत्यार्यश्च for (पंचविंशति आर्याणां)

अष्टनखैर्मेषगविरदलिप्तोनै गुणैः स्वरै मिथुने ।
ककटके गुणषोडशं धृतिभिः सिंहेन व त्रिधनैः ॥ १ ॥
कन्यायां पंचनखैस्तुलिनि अधिधृतिभिरलिनी सेषुकाले ।
द्विचतुर्दशाति धृतिभि धनुषि शंशाक मनुतत्वैः ॥ २ ॥

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१. (घ) ५. मोखग (ग) मोषग for (मेंषग)
१. लिप्तोनैर्गण (ग) लिप्तोनैर्गुण for (लिप्तोनैगुणैः)
२. स्वस्वरैंमिथुने (ग) for (स्वरै मिथुने)
६. धृतिभि (च) for (धृतिभिः)
७. च (च) for (व)

(ग) ३. कवर्कटके गुणषोडश for (कर्कटके गुणषोडशं)
४. त्रिघनैः for (त्रिधनैः)

(च) ५. मेंखे for (मेंष)
१. गुणस्वरैंमिथुने for (गुणैः स्वरैंमिथुने)
८. षोडश for (षोडशं)
४. त्रिद्यनैः for (त्रिधनैः)

(क) १. गुण for (गुणैः)
३. कर्कटे for (कर्कटके)
४. त्रिघनैः for (त्रिधनैः)

२. (घ) १. व्यतिधृतिभिरलिनि सषुकालेः (ग) व्यतिधृतिभिरलिनिसेषुकलैः for (अधिधृतिभिरलिनी । सेषुकाले)
४. धृतिभिर्धनुषि (ग) (च) for (धृतिभिधनुषि)
३. मनुखतत्त्वै (ग) मनुनखतत्त्वैः for (मनुतत्त्वैः)

(ग) २. द्विचतुर्दशानि for (द्विचतुर्दशाति)

(च) १. अतिधृ तिभि रलिनि (अधिधृतिभिरलिनी)
५. क्रमसंख्या लुप्त

(य) १. व्यतिधृतिभि रलितेषुकलैः for (अधिधृतिभिरलिनी सेषुकालै)
२. दशा विधृतिः for (दशातिधृतिभिः)
३. मनुनखै तत्वैः for (मनुतत्त्वैः)

मकरेष्टनखैः कुम्भे नखषड्विंशै ऋंणे मुनित्रिंशैः ।
पृथगश्विन्यादीनां ध्रुवकाशैर्योगताराश्वः ॥ ३ ॥
ध्रुवकादूनः पश्चादधिकः प्राग्वत् कृतेऽन्यथायोगः ।
अन्यत् गृहमेलकवत् ध्रुवक्रान्तेर्भविक्षेपाः ॥ ४ ॥
सौम्यादशार्कविषया याम्या शरदशभवारसासौम्याः ।
खसप्त दक्षिणाः खं सौम्याः सूर्य त्रयोदशकाः ॥ ५ ॥

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३. (घ) ४. नखड्विंशै for (नखषड्विंशै)
१. ऋंषे (ग) ऋंषे for (ऋंणे)
५. मुनिर्त्रिंशैः for (मुनित्रिंशैः)
२. ध्रुवकांशै (ग) (च) for (ध्रुवकाशै)
(ग) ३. स्वैः for (श्वैः)
(च) १. ऋंषे (ऋंषे) for (ऋंणे)
५. मुनित्रिंशैः ॥३०७॥ for (मुनित्रिंशैः)
३. योगताराश्वैः for (योंगताराश्वैः)
(क) १. ऋषे for (ऋंणे) २. कांशै for (काशै) ३. चैष्ट for (श्वैः)

४. (घ) २. प्राग्वत्कृतेन्यथायोगः (ग) प्राग्वत्कृतेऽथायोगः for (प्राग्वत्कृतेऽन्यथायोगः)
३. ध्रुववक्रांन्तेर्भ for (ध्रुवक्रान्तेर्भ)
(ग) ४. दूने for (दूनः)
५. दधिके for (दधिकः)
१. अन्यद्ग्रहमेलकवद्ध्रुववकक्रान्तेर्भविक्षेपः for (अन्यत् ग्रहमेलकवत् ध्रुवक्रान्तेर्भविक्षेपाः)
(च) २. अवग्रहचिह्नलुप्त १. अन्यद्ग्रह for (अन्यत्ग्रह)
३. ध्रुववक्रांन्ते for (ध्रुवक्रान्ते) ६. विसर्गलुप्त

५. (घ) ४. षं (ग) खं for (ख)
(ग) १. याम्याः for (याम्या)
२. ३. शरदिग्भवारसाः for (शरदशभवारसा)
५. सौम्यां for (सौम्याः)
(च) ६. दशाङ्कं for (दशार्कं) ४. षंसप्त for (खसप्त)
(क) १. याभ्याः for (याम्या) २. सर for (शर) ३. दिग् for (दश)
४. खं for (ख)

दक्षिणतोभयमलाः सप्त त्रिंशदुदगंशका याम्याः ।
अध्यर्धत्रिचतुष्कार्द्ध नवमसप्तत्रिंशविषयशराः ।। ६ ।।
सौम्यादब्धिका यष्टि त्रिंशत् षट्त्रिंशदितरतो लिप्ताः ।
अष्टादशोत्तरजिनाः षड्विंशत्यंतराण्यंशाः ।। ७ ।।
प्राजेशयोगन्तरा विक्षेपांशै कलात्रिघनहीनैः ।
आग्नेयस्य कलामेकोन त्रिंशताहीनैः ।। ८ ।।

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६. (घ) ३. अमला for (यमलाः)
४. शरा (च) for (शराः)
(ग) १. भव for (भ)
२. सत्र्यंश for (सप्तत्रिंश)
५. विखय for (विषय)
(च) ३: विसर्गलुप्त ६. अध्यर्द्ध for (अध्यर्ध)
(क) १. भव for (भ) २. शशंश (शत्र्यंश) for (सप्तत्रिंश)
७. (घ) १. सौम्याद्वचधिका (ग) (च) for (सौम्यादब्धिका)
३. षष्टि (ग) for (यष्टि)
५. षड्विंश for (षट्त्रिंश)
६. षड्वच'श त्यंतराण्यंशाः for (षड्विंशत्यंतराण्यशाः)
(ग) ७. स्त्रिंशत् for (त्रिंशत्)
(च) २. षष्टि for (यष्टि) ७. त्रिंश for (त्रिंशत्)
५. षड्त्रिंश for (षट्त्रिंश)
६. षड्वच'शत्य'तराण्य'शाः for (षड्विंशत्य'तराण्य'शाः)
(क) १. सौम्यद्वचधिका for (सौम्यदब्धिका) २. षष्टि for (यष्टि) ३. दिव for (दित)
४. बराण्य'शाः (ग) for (तराण्य'शाः)
८. (घ) १. योगतरा (ग) योगतारा for (योगन्तरा)
३. विक्षेपांशैः (ग) for (विक्षेपांशै)
(ग) ४. आग्रे यस्य for (आग्नेयस्य)
(च) १. योगतारा for (योगन्तरा) ३. विक्षेपांशैः for (विक्षेपांशै)
५. +२७+ ६. +३०+
(क) १. योगतारा for (योगन्तरा)
२. कलानामेकोन (ग) for (कलामेकोन)

पंचदशकला हीनैश्चित्रायाः[१] सप्ततिर्विशाखायाः[२][३] ।
षट् सप्तत्या मैत्रस्यैंद्रस्य[४] त्रिंशता हीनैः[५] ॥ ९ ॥
छादय[१] योगतारां मानार्द्धोनाधिकाद् विक्षेपात्[२] ।
स्फुट विक्षेपो यस्याधिकोनको[३] भवति समदिक्सु ॥ १० ॥
विक्षेपोंश[३] द्वितीयादधिको[४] वृषभस्य[१] सप्तदशभागे ।
यस्य गृहस्य याम्यो भिनत्ति शंकट[२] स रोहिण्याः[५] ॥ ११ ॥

९. (घ) ३. विशाषायाः (च) for (विशाखायाः)
४. मैत्रसौन्द्रस्य (ग) मैत्रस्येन्द्रस्य for (मैत्रस्यैंद्रस्य)
(ग) २. सप्तभि for (सप्तति)
(च) ५. +३०+
(क) १. श्चिताया: for (श्चित्रायाः)
२. सप्तभिः for (सप्तति)

१०. (घ) २. भविक्षेपात् (ग) भवति विक्षेपात् for (विक्षेपात्)
३. समदिक्सुः for (समदिक्सु)
(ग) १. छादयति for(छादय)
(च) २. भविक्षेपात् for (विक्षेपात्)
३. समदिक्सुः for (समदिक्सु)
(क) १. छादयसि for (छादय)
२. भविक्षेपात् for (विक्षेपात्)
३. दिक्स्थः (ग) for (दिक्सु)

११. (घ) ३. विक्षेपौंश for (विक्षेपोंश)
४. द्वितयादधिको (ग) (च) for (द्वितीयादधिको)
(ग) १. वृषस्य for (वृषभस्य)
५. रोहिण्या for (रोहिण्याः)
(च) २. शकटं for (शंकट)
(क) १. मृषभस्य for (वृषभस्य)
२. शकटं (ग) for (शंकट)

विक्षेपेंत्ये सौम्ये तृतीयतारां भिनत्ति पित्र्यस्य ।
इन्दुर्भिन्नति पुष्पं पौष्णं वारुणमविक्षिप्तः ॥ १२ ॥
कृत्वापि दृष्टिकर्म श्रीषेणाचार्यभट विष्नुचन्द्रोक्तम् ।
प्रतिदिनमुदयेऽस्ते वा न भवन्ति दृग्गणितयोरैक्यम् ॥ १३ ॥

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१२. (घ) १. विक्षोपंत्ये for (विक्षेपेंत्ये)
४. सौम्ये । २७० । (च) for (सौम्ये)
५. भिनति for (भिनत्ति)
६. इंदुर्भिनत्ति (ग) (च) for (इन्दुर्भिन्नति)

(ग) ७. व्यस्य for (पित्र्यस्य)

(क) १. विक्षेपोंऽत्ये for (विक्षेपेंत्ये)
२. पुषां for (पुष्पं)
३. करूण for (वारुण)

१३. (घ) ३. विष्णु (ग) (च) for (विष्नु)
५. मुदयेस्तेवा (ग) for (मुदयेऽस्तेवा)
४. भवति (ग) (च) for (भवन्ति)

(ग) ६. कृत्वा (यहां 'अपि' नहीं है) for (कृत्वापि)
२. श्रीषेणार्यभट for (श्रीषेणाचार्यभट)
७. दृग्गणितरैक्यम् for (दृग्गणितयोरैक्यम्)

(च) ८. प्रतितिदिन for (प्रतिदिन) ५. मुदयेस्ते for (मुदयेऽस्ते)

(क) १. कर्म्म for (कर्म)
२. श्रीषेणार्थ for (श्रीषेणाचार्य)
३. विष्णु for (विष्नु)
४. भवति for (भवन्ति)

भमुनिमृगव्याधानां यतस्तददृष्टिकर्म वक्ष्यामि।
दृग्गणित समदेयं शिष्याय चिरोषितायेदम् ॥ १४ ॥

अ	भ	कृ	रो	मृ.	आ.	पु.	पु.	अ.	म.	पू	उ	ह.	चि	स्वा	वि	अ.	जे	र
००	००	१	१	२	२	३	३	३	४	४	५	५	६	६	७	७	७	८
८	२०	७	१९	३	७	३	१६	१८	९	२७	५	२०	३	१९	२	१४	१९	१
००	००	२८	२८	००	००	००	००	००	००	००	००	००	००	००	००	००	००	००
००	००	००	००	००	००	००	००	००	००	००	०२	००	००	००	००	००	००	००

श	उ.	अ.	अ	घ	श	पू	उ.	रे
८	८	८	९	९	१०	१०	११	१२
१४	२०	२५	८	२०	१०	२६	७	००
००	००	००	००	००	००	००	००	००
००	००	००	००	००	००	००	००	००

१.४ (घ) १. ततो (च) त for (तद)
३. वक्षामि (ग) कक्षामि for (वक्ष्यामि)
४. समं (च) for (सम)
(ग) ५. दृष्टकर्म for (दृष्टिकर्म)
२. दस्पात्र for (दृग्गणित)
६. विरोषितायेदम् for (चिरोषितायेदम्)
(च) ७. ब्याध्यानां for व्याधानां ३. वक्षामि for वक्ष्यामि
८. सिष्याय for (शिष्याय)
(क) १. ततो for (तद) २. न माणित for (दृग्गणित)

१. ज्ये ३. मू ६. पू
७ ४. पू
१९
५
०

७. वि ८. अ० ५. श.
७ ७ १०
२ १४ २६
५ ५ ०
० ०

(च) ७. वि. ८. अ. १. ज्ये० ३. मू for (र) ४. पू for (श) ५. श०

७	७	१	८	८	१०
२	१४	१९	१	१४	२६
५	५	५	०	०	०
०	०	०	०	०	०

(घ) केवलमात्र परिवर्तित रूप—शेष सब ठीक है।

वि. ज्ये० २. मू. ४. श. ५. श. ६. पू

७	७	८	८	१०	,
२	१६	१	१४	२६	,
५	५	०	०	०	,
०	०	०	०	०	,

अतिरिक्त—

विक्षेपाः

अ	भ	कृ	रो	मृ	आ	पु	पु	अ	म	पू	उ	ह	त्रि
१०	१२	५	५	१०	११	६	०	७	०	१२	१३	१३	२
१													
स्वा	वि	अ	ज्ये	मू	पू	उ	उ	श्र	ध	श	पू	उ	रे
२७	१	०	४	५	५	५	१२	३०	३६	०	२४	२६	०
	३०			२०	२०					१८			

(ग) अश्विन्यादिध्रुवकाः

अ	भ	कृ	रो	मृ	आ	पु	पु	अ	म	पू	उ	ह	चि
०	०	१	१	३	२	३	३	३	४	४	५	५	६
८	२०	७	२६	०	७	३	१६	१०	६	२७	५	२०	३
		२८	२८		०	०	०	०	०	०	०	०	०

स्वा	वि	अ	ज्ये[७]	मू[८]	पू[९]	उ[८]	उ[८]	श्र[९]	ध[९]	श	मू	उ[११]	रे[१२]
६	७	७	६	१	२४	२०	२५	८	२०	२०	२०	७	३०
१६	२	१४	५	०	०	०	०	०	०	०	०	०	०
०	५	५	६०	०	०	०	०	०	०	०	०	०	०

क्रांतिज्या[१] तत्क्रान्तिविक्षेप[५]क्रान्तिचापा[२]नाम् ।
संयोगा[३]न्तरजीवा स्वक्रान्ति[४]ज्यैकभिन्न[६]दिशां ।। १५ ।।
एवं भमुनिध्रु[१]वयो ग्रहतत्क्रान्त्या चन्द्राष्टकं[२] कर्माद्यं ।
कृत्वा ग्रहे भमुनिवत् तस्मात्स्वक्रान्ति जीवा वा[३] ।। १६ ।।

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१५. (घ) २. चापभागानाम् (ग) for (चापानाम्)

(ग) ५. विक्षेप for (विक्षेप)

६. दिशोः for (दिशां)

(च) २. चापनां for (चापानां)

६. द्यैकभिन्नदिशां for (ज्यैकभिन्नदिशां)

(क) १. क्रांति ज्यतिक्राति for (क्रान्तिज्यातत्क्रान्ति)

२. चापभोगाना for (चापानाम्)

३. सोयोगांचर for (संयोगान्तर)

४. क्रातज्यैः कविभिर्मादशाम् for (क्रातिज्यैकभिन्नदिशाम्)

१६. (घ) १. ध्रुवयोर्ग्रह for (ध्रुवयोग्रह)

२. च दृष्टि कर्माद्यं (ग) for (चन्द्राष्टकं कर्माद्यं)

(ग) १. एवं भमुनेर्ध्रुवयोर्ग्रहवत्क्रान्त्या for (एवं भमुनिध्रुवयोग्रहतत्क्रान्त्या)

३. च for (वा)

(च) २. च दृष्टि for (चन्द्राष्टकं)

(क) १. भमुनिध्रुवका प्राक् प्रदर्शिता त्विषा ।
सेषामेवकार्यं ग्रहस्य पुनातत् क्रान्ते प्रथमं वृत्कर्णा
कृत्वा पश्चात् तज्ज्ञाति स्वक मुनिध्रुवकवत् ।

for

एवं भमुनिध्रु[१]वयो ग्रहतत्क्रान्त्या चन्द्राष्टकं[२] कर्माद्यं ।
कृत्वा ग्रहे भमुनिवत् तस्मात्स्वक्रान्ति जीवा वा[३] ।। १६ ।।

त्रिज्याप्तांशुभिरुदयैर्घाता द्विक्षेप सत्रिभक्रान्त्ये ।
ऋणधनमेकान्यदिशोस्तयोर्ग्रहेन भमुनि ध्रुवके ॥ १७ ॥
तत्स्वक्रान्तिज्याभ्यां चन्द्रादीनां पृथक् चरप्राणान् ।
कृत्वार्कवत्तदंतरसंयोगो तुल्यभिन्नदिशाम् ॥ १८ ॥
तत्प्राणैर्विक्षेपे सौम्ये हीनो ग्रहोऽधिके याम्ये ।
उदयैर्भध्रुवको वागस्त्यध्रुवकोऽथवा लग्नम् ॥ १९ ॥

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१७. (घ) ५. द्विक्षेप for (द्विक्षेप)
२. क्रान्त्यो (ग) क्रान्त्योः for (क्रान्त्ये)
(ग) १. त्रिज्याप्तासुभिरुदयैर्घाता ३२७० for (त्रिज्याप्ताँशुभिरुदयैर्घाता)
३. रुणधन for (ऋणधन)
(च) इस प्रति में यहां 'विक्षेप' शीर्षक से अतिरिक्त निम्न सारणी दी गई है।

विक्षेपाः

अ	भ	कृ	रो	मृ	आ	पु	पु	अ	म	पु	उ	ह
१८	१२	५	५	१०	११	५	०	७	०	१२	१३	१३
१												

चि०	स्वा.	वि	अ	ज्ये	मू	पू	उ	उ	श्र	ध	श	पू	उ	रे
२	२७	१	०	४	५	५	५	१२	३०	३५	०	२४	२६	०
		३०			२०	२०					१८			

(क) १. सुभिरुदयै for (शुभिरुदयै) २. भक्तम् for (क्रान्त्ये)
३. कर्णमनसे काम्य for (ऋणधनमेकान्य)
४. ग्रहेण for (ग्रहेन)

१८. (घ) १. संयोगौ (ग) संयोगस् for (संयोगो)
(च) ३. कृत्वार्क्कं for (कृत्वार्कं)
(क) १. संयोगौ for (संयोगो)
२. दिशम् for (दिशाम्)

१९. (घ) ३. ग्रहोधिको याम्ये (ग) (च) for (ग्रहोऽधिके)
४. ध्रुवकोथवा लग्नम् (ग) (च) for (ध्रुवकोऽथवा लग्नम्)
(क) १. तत्क्रांति ऋणो for (तत्प्राणै)
२. हीने for (हीनो)

उदये गृहभमुनीनामस्तमये षड्गृहाधिकः सौम्ये ।
अधिको याम्ये हीनः षड्राशियुतोस्तमयलग्नम् ॥ २० ॥
उदयविलग्नादधिके षड्राशियुतास्ते लग्नतो हीने ।
रात्रिविलग्ने दृश्यादिनेऽपि चन्द्रोन्यथा दृश्यः ॥ २१ ॥
प्रागुदय लग्न मूनं लग्नादधिकं गृहोदयः पश्चात् ।
ऊनमधिकेन तुल्यं कृत्वा घटिकाः स्वराश्युदयैः ॥ २२ ॥
प्रागस्तमयो लग्नादूनं षड्गृहयुतोस्तमयलग्नम् ।
अधिकं पश्चात् घटिकाः कृत्वा सममूनमधिकं चेत् ॥ २३ ॥

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२०. (घ) २. षड्गृहाधिकः (ग) (च) for (षड्ग्रहाधिकः)

(ग) ३. मुनीनां for (मुनीना)
४. ऽस्तमयलग्नम् for (स्तमयलग्नम्)
(क) १. दृक्कर्म कार्यम् for (सौम्ये)

२१. (घ) २. त (ग) (च) for (ते)
१. दृश्यो दिनेपि (ग) दृश्यो दिनेऽपि for (दृश्यादिनेऽपि)
(घ) ३. षड्राश for (षड्राशि)
(च) ४. लग्नांतो for (लग्नतो) ।
(क) दृश्योदिमेपि for (दृश्यादिनेऽपि)

२२. (च) १. राश्युदयो for (राश्युदयैः)

२३. (घ) १. षड्गृह (ग) (च) for (षड्ग्रह)
२. युतास्तमय (ग) for (युतोस्तमय)
४. पश्चाद्घटिकाः (ग) for (पश्चात्घटिकाः)
(ग) ३. अधिकेन for (अधिकं चेत्)
(च) ५. दुनं for (दूनं)
४. पश्चाध्यघटिकाः for (पश्चात् घटिकाः)
६. क्रत्वा for (कृत्वा)
(क) १. षड्गृह for (षड्ग्रह)
२. युतास्तमय for (युतोस्तमय)
३. अधिकेन for (अधिकं चेत्)

तात्कालिकोपकरणैः सकृदागतनाडिकाभिरह्नींदौ ।
रात्रौ वा प्रति घटिकं प्राग्वच्छेषोन्नतिः धार्या ॥ २४ ॥
मध्यछाया रविवत् स्वक्रान्त्या दर्शनेमनीष्टाच ।
एवं भ मुनीनामन्तर घटिका गुणं भुक्तिः ॥ २५ ॥
षष्ट्या विभजेत् लब्धं प्रागुदयास्तमययोः गृहे शोध्यम् ।
पश्चात् क्षेप्यं प्रतिदिनमुदयस्तमयासकृदेवं ॥ २६ ॥

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२४. (घ) १. रह्नींदो (ग) रह्नींदोः for (रह्नींदौ)
४. कार्याः (च) for कार्या
(ग) १. सकृतागत for (सकृदागत)
५. तात्कालिकोपकरणे for (तात्कालिकोपकरणैः)
(क) १. शकृद् for (सकृदागत) २. रह्नीदोः for (रह्नींदौ) ३. रात्रो for (रात्रौ)
३. घटिका for (घटिकं)

२५. (घ) ३. कांत्या for (क्रान्त्या)
१. सतीष्टा (ग) सती च for (मनीष्टा)
५. एवं ग्रह (ग) for (एवंभ)
६. मनीना (ग) मुनीना for (मुनीना)
७. गुणां for (गुणं) (ग) गुणाभक्तम् for (गुणं भुक्तिः)
(च) १. सतीष्टा for (मनीष्टा)
५. एवंग्रहभ for (एवंभ)
(क) १. सतीष्टा for (मनीष्टा) २. ग्रह for (भ) ३. भुक्तिम् for (भुक्तिः)

२६. (घ) ३. विभजेल्लब्धं (ग) for (विभजेत्लब्धं)
४. ग्रंह (ग) ग्रंहे (च) for (ग्रहे)
५. उदयास्त (ग) for (उदयस्त)
६. मयावसकृदेवं (ग) for (मयासकृदेवं)
(च) ३. विभजेलब्धं for (विभजेत्लब्धं)
७. विसर्गलुप्त
१. पश्चाक्षेप्यं for (पश्चात्क्षेप्य)
६. मयावसक्रदेवं for (मयासकृदेवं)
(क) ९. क्षेप्यत्र for (क्षेप्यं)
२. सकृदवम् for (सकृदेवं)

इष्टात्कालात् भानोरुदयास्ताद्वा ग्रहोदयास्तमयैः ।
तात्कालिकै र्विलग्नग्रहोदयास्तमयलग्नाद्यैः ।। २७ ।।
प्रागुदयलग्नमुदयैर्लग्नसमं लग्नमुदयलग्नेन ।
पश्चात् तत् घटिकाभिः कृत्वा तात्कालिकैरसकृत् ।। २८ ।।
उदयं प्रागस्तमयो लग्नेन पश्चानषड्ग्रहास्तमयः ।
लग्नं पश्चाल्लग्नचक्रार्द्धं संयुतास्तमये लग्नेन ।। २९ ।।
उनोल्पभुक्तिरुदितः प्रागथवो देष्यति ग्रहः सूर्यात् ।
पश्चादधिकोधिकगतिरल्पगती वक्रितौ ज्ञसितौ ।। ३० ।।

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२७. (घ) २. कालाद्भानो (ग) (च) for (कालात् भानो)
१. मयै (ग) मयौ for (मयैः)
३. र्विलग्न (ग) (च) for (र्विलग्नं)

(ग) ४. इष्टा for (इष्टात्)
५. होदयास्तमय.....for (ग्रहोदयास्तमय)

(च) १. मयै for (मयैः)

(क) १. मयौ for (मयैः)

२८. (घ) ३. पश्चात्तद्घटिकाभिः for (पश्चात् तत् घटिकाभिः:)
५. पश्चात्त for (पश्चात् तत्)
(च) ५. पश्चाद्घटिकाभिः for (पश्चात्तत् घटिकाभिः) ६ क्रत्वा for (कृत्वा)
(क) १. मदयै for (मुदयै) २. मदय for (मुदय)

२९. (घ) १. सषड्ग्रहास्तमयलग्नम् (ग) (च) for (पश्चानषड्ग्रहास्तमयः । लग्नम्)
३. पश्चाल्लग्नं (ग) for (पश्चाल्लग्न)
(ग) ४. उदयः for (उदयं)
२, मय for (मये)
(क) १. सषट् ग्रहास्तमय लग्नम् for (पश्चानषड्ग्रहास्तमयः लग्नम्)
२. मय for (मये)

३०. (घ) १. ऊनोल्पभुक्तिरुदितः (ग) (च) for (उनोऽल्पभुक्तिरुदितः)
(क) १. दर्ष्यति for (देष्यति)

प्रागूनाधिकभुक्तिः पश्चादधिकोल्पभुक्तिरस्तमितः ।
यास्यत्यथवास्तमयं यतस्ततो दृश्यघटिकोक्तिः ॥ ३१ ॥
घटिकाद्वयेन चन्द्रो दृश्योर्कात् सितगुरुज्ञ शनिभौमाः ।
अध्यर्द्धया घटिकाया त्रिभागघटिको ज्ञ (त्त) राधिकाया ॥ ३२ ॥
गूहसूर्यान्तरघटिका स्वभुक्ति लिप्तानि स्वनयुतयोश्च ।
प्राग्वदनंतर हृताहीनाधिकनाडिका दिवसाः ॥ ३३ ॥

चं	शु	बृ	शु	श	मं
२	३०	१	२	१	२
००	००	५०	००	३०	५०

३१. (घ) १. प्रागूनोधिकभुक्तिः for (प्रागूनाधिकभुक्तिः)
(क) २. संमितः for (स्तमितः)

३२. (घ) १. घटिकया (ग) (च) for (घटिकाया)
३. घटिकोत्तराधिकया (ग) (च) for (घटिकोत्तराधिकाया)
(ग) ४. अध्यर्द्धय for (अध्यर्द्धया)
(च) ५. दृश्योर्क्कात् for (दृश्योर्कात्)
(क) १. शित for (सित) २. घटिकया for (घटिकाया) ३. शशिकोत्तरा for (घटिकोत्तरा)

३३. (घ) ४. घटिका: (ग) (च) for (घटिका)
५. रूनयुतयोश्च (च) for (स्वनयुतयोश्च)

चं.	शु०	बृ०	बु०	श०	मं
३	१	१	२	२	२
	३	५०	१०	३०	५०

२. प्राग्वत्तदंतर (च) for (प्राग्वदनंतर)
३. हृता (ग) कृता for (हृता)
६. दिवसा (च) for (दिवसाः)
(ग) ८. युततयोश्च for (युतयोश्च)
(च) ३. हृता for (हृता)
(क) १. लिप्ताभिरून for (लिप्तानिस्वन.)
(ग) २. तदंतर for (दनंतर)
३. हवा for (हृता)

चं	शु०	बृ०	बु०	श०	मं०
३	१	१	२	२	२
	३	५०	१०	३०	५०

विपरितधनमृणं सौम्यशुक्रयोर्वक्रिणोः स्वभुक्तिकलाः ।
एवमुदयास्तमययो विपरितं वक्रिणः स्वफलम् ॥ ३४ ॥
विक्षेपो दक्षिणतस्तक्रांतेर्भाग सप्तसप्तत्या ।
मिथुनस्य सप्तविंशे भागेनस्त्योन्नतैर्भागैः ॥ ३५ ॥

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३४. (घ) ६. विपरीत (ग) for (विपरित)
७. विक्रणोः for (वक्रिणो)
३. एवनुदयास्तमययो (ग) एवमुदयास्तमययो for (एवमुदयास्तमययो)
८. विपरीतं (ग) for (विपरितं)

(ग) १. रुणधनं for (धनमृणं)

(च) ७. विक्रिणों for (वक्रिणोः)
८. विपरितं for (विपरितं)

(क) १. ऋणधनं for (धनमृणं)
२. कालः for (कलाः)
२. उदक for (उदया)
४. वक्रिणि (ग) for (वक्रिणः)
५. फलाम् for (फलम्)

३५. (घ) १. विक्षिप्तो (ग) for (विक्षेपो)
२. वक्रांते (ग) तत्क्रान्ते for (तक्रान्ते)
३. नस्त्योन्नतै भागैः (ग) गस्त्योनतैर्भागैः for (नस्त्योन्नतैर्भागैः)

(ग) ४. 'सप्त' यह पद यहां अंकित नहीं ।

(च) १. विक्षेप्रो for (विक्षेपो)
२. तत्क्रांते for (तक्रान्ते)
३. नस्त्योन्नतैभागैः for (नस्त्योन्नतैर्भागैः)

(क) १. विक्षिप्तो for (विक्षेपो)
२. स्व for (स्त)
३. भागेरगेस्त्योर्नतैः for (भागेनस्त्योन्नतै)

नवतेस्तनैर्दृश्यो[1] घटिकाद्वितयेन[3] तदुदय[2]विलग्नम्
उदयैरधिकं[6] कृत्वा[7] तदुदय[4]सूर्योस्तमयलग्नम् ॥ ३६ ॥
षड्भयुत[5]मूनमुदयैः षड्राशियुतं[1] तदस्तमय[2]सूर्यः ।
घटिकाद्वितयेनैवं[3] षड्भाग[4]युतेन[6] मृगेहंतु[5] ॥ ३७ ॥
एवं[4] नक्षत्राणां[1] घटिकाद्वितयेन[5] सत्रिभागेन ।
उदयार्कोस्तमयार्का[2] व्यस्तोनस्तत्सदादृश्यम्[3] ॥ ३८ ॥

३६. (घ) १. रूनैदृश्यो (ग) रूनैर्दृश्यो for (स्तनैर्दृश्यौ)
३. ९० घटिका for (घटिका)
(ग) ४. तदूदय for (तदुदय)
(घ) ५. रूनैदृश्यो ९० for (स्तनैर्दृश्यो)
६. रुदयै for (उदयै) ७. क्रत्वा for (कृत्वा)
(क) १. रूनै for (स्तनै) २. विलग्नम) for (तदुदयविलग्नम्)

३७. (घ) २. तदस्तमसूर्यः for (तदस्तमयसूर्यः)
३. घटिद्वितयेनैवं for (घटिद्वितयेनैवं)
४. षड्भागायुतेन (च) for (षड्भागयुतेन)
(ग) ५. भार्द्धयुत for (षड्भयुत)
६. हंतुः for (हंतु)
(च) ७. मृगहंतु for (मृगेहन्तु)
(क) १. रासि for (राशि)

३८. (घ) १. सतृभागेन (च) for (सत्रिभागेन)
३. व्यस्योनस्तत्सदा (च) for (व्यस्तोनस्तत्सदा)
(ग) ४. नक्षत्राणामेवं for (एवं नक्षत्राणां)
५. द्वितेन for (द्वितयेन)
२. उदयार्को मस्तयर्काद्यस्योनस्तदादृश्यम् for (उदयार्कोस्तमयार्का व्यस्तोनस्तत्सदादृश्यम्)
(च) २. उदयार्क्कोस्तमयार्क्का for (उदयार्कोस्तमयार्का)
(क) १. त्रिभागेन for (सत्रिभागेन २. उदय (लक्षय) लग्नमधिकं कृत्वा पूर्वं वदुदयास्तार्या भवति for (उदयार्कोस्तमयार्काव्यस्तौनस्तदादृश्यम्)

उदयास्तसूर्योरन्तरे रवौ दृश्यतेऽन्यथास्तमितः ।
उनाधिकारविकला रविभुक्त्या भाजिता दिवसाः ॥ ३९ ॥
षड्विंशो मिथुनांशेः शकचत्वारिंशता ४० मृग व्याधः ।
तत्क्रांतेर्दक्षिणतो विक्षिप्तागस्त्यवच्छेषम् ॥ ४० ॥
गुणिता व्यासार्द्धेन स्वक्रान्तिज्यावलंबकहृताग्रा ।
प्रतिदिनमुदयास्तमया वर्गाग्रे भग्नहमुनीनो ॥ ४१ ॥

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३९. (घ) १. उदयास्तसूर्यायारंतरे (ग) उदयास्तासूर्ययो for (उदयास्तसूर्योरंतरे)
२. स्तमित for (स्तमित) (ग) न्यथास्तमितिः for (ऽन्यथास्तमितः)
३. ऊनाधिकार (ग) for (उनाधिकार)
(ग) ४. दृश्यतेः for (दृश्यते)
(च) १. उदयास्तसूर्यायोरन्तरे for (उदयास्तसूर्योरन्तरे)
२. न्यथास्तमित for (ऽन्यथास्तमित)
(क) ९. सूययोरन्तर for (सूर्योरन्तरे)

४०. (घ) १. षड्विंशे (ग) (ग) for (षड्विंशो)
५. तत्क्रांतेर्दक्षिणतो (ग) तत्क्रांतोर्दक्षिणतो for (तक्रान्तेर्दक्षणतो)
४. विक्षिप्तोगस्त्यवच्छष (ग) विक्षिप्तोगस्त्यवछेषम् for (विक्षिप्तागस्त्यवच्छेषम्)
(ग) ६. मिथुनांशेऽशक for (मिथुनांशेः शक)
(च) ५. तत्क्रांते for (तक्रांते)
३. र्दक्षिणतो for (र्दक्षणतो)
(क) १. विंशे for (विंशो) २. व्याधेः for (व्याधः) ३. दक्षिण for (र्दक्षणतो)
४. विक्षेपोऽगस्त्य for (विक्षिप्तागस्त्य)

४१. (घ) १. ज्यावलंबलकहृताग्रा (ग) ज्यावलंबकोद्धृता स्वाग्रा for (ज्यावलंबकहृताग्रा)
२. वग्राग्रे (ग) for (वर्गाग्रे)
३. मुनीनाम् (ग) भग्नहमुनीनाम् (च) for (भग्नहमुनीनो)
(च) १. हृताग्रा for (हृताग्रा)
(क) १. प्रतिदिनमुदयास्तमयावयात्रे वग्रहमुनीरमां
गुणिता व्यासार्द्धेन स्वक्रान्तिज्या साद्धेनानेन
for (गुणिता व्यासार्द्धेन स्वक्रान्तिज्या वलंबकहृताग्रा । प्रतिदिनमुदयास्तमया वर्गाग्रे भग्नहमुनीनो ॥)

अग्राशंकुतलैक्यं[१] तुल्यदिशोरन्तरं तथान्यदिशो[४]।
प्राच्यपरायाः[५] शंकोः[६] तलं[३] दग्रे[२] गृहो नं[२] च ॥ ४२ ॥
शंकुतल प्राच्यपरांतरं भुजो दक्षिणोत्तरं[२] कर्णः।
दग्ज्या[१] तद्वर्गांतरमूलं दिग्मध्यतः[३] कोटिः ॥ ४३ ॥
उनाधिकशंकुगुणाः[३] स्वशंकुभक्ताः पृथक् स्वदग्याग्रे[१]।
कृत्वोनाधिकशंकुर्दृष्टि[४] कृत्वोनशंक्वग्रे[५] ॥ ४४ ॥

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४२. (घ) ३. शंकोस्वलं तदग्रे for (शंकोः तलं दग्रे)
२. भं वा (ग) for (नं च)

(ग) ४. तथान्य दिशोः for (तथान्यदिशो)
५. परयोः for (परायाः)

(ब) ६. शंको for (शंकोः)
३. तलं तदग्रे for (तलंदग्रे)

(क) १. तलेकां for (तलैक्यं) २. भ वा for (नं च)

४३. (घ) २. दक्षिणोत्तरः (ग) (च) for (दक्षिणोत्तरं)
१. दृग्ज्यातद्वर्गांतर मूलं (ग) (च) for (दग्ज्यावर्गान्तरमूलं)
३. दिग्मध्यत (च) for (दिग्मध्यतः)

(क) ९. दुप उप स्वर्गांतर मूलं for (दग्ज्यातद्वर्गांतरमूलं)

४४. (घ) ३. ऊनाधिक (ग) (च) for (उनाधिक)
१. दृग्याग्रे (ग) सुदृग्ज्याग्रे for (स्वदग्याग्रे)
४. शंक for (शंकु)
२. शंक्वाग्रं (शंक्वग्रे) (ग) शक्वग्रे for (शंक्वग्रे)

(ग) ५. दृश्यं for (दृष्टिं)

(च) १. स्वदृग्याग्रे for (स्वदग्याग्रे)
६. कृत्वो for (कृत्वो)
२. शंक्वग्रे for (शंक्वग्रे)

(क) १. दृश्या for (स्वदग्याग्रे) २. शंकुः for (शंक्वग्रे)

प्रतिघटिकमधिकशंकोर्ग्रहमध्ये दर्शयेत् भुवा ।
त्रिप्रश्नोत्तया रविवच्छंकु भ्रमणादिक्रमशेषम् ॥ ४५ ॥
शंकु प्राच्यपरांत्तरं विषुवच्छायैक्य मुत्तरे नृतले ।
याम्योतरं गुणहृतं स्वक्रान्तिर्लम्ब कर्णाभ्याम् ॥ ४६ ॥
तच्चापांशा सदशैः मंधुवकापक्रमांशकैरूना ।
विक्षेपांशे व्यस्ता व्यस्तविशुद्धा व्यसदृश्यांशैः ॥ ४७ ॥

४५. (घ) ५. मग्रे (ग) ग्रहमग्रे
६. दर्शयेनि (ग) दर्शयेन्मुनि भंवा for (दर्शयेत् भुवा)
३. रविवत्शंकु (च) for (रविवच्छंकु)
४. कमशेषम् (ग) (च) for (क्रमशेषम्)
(ग) २. त्रिप्रस्नोक्तं for (त्रिप्रश्नोत्तया)
(च) २. मग्रे for (मध्ये) ६. दर्शयेनि for (दर्शयेत्)
(क) १ अधिघटिकादिसमधिशंका ग्रहणं मग्रे दर्शयन् मुनिनंद for (प्रतिघटिकमधिकशंकोर्ग्रहमध्ये दर्शयेत् भुवा)
२. तृं: for (त्रि) ३. विवन् for (रविव) ४. कमा for (क्रम)

४६. (घ) २. परांतर (ग) (च) for (परांत्तरं)
३. नृतरे for (नृतले)
१. याम्येंतरं (ग) याम्यैरंतर for (याम्योतरं)
(च) ४. मुत्तर for (मुत्तरे)
५. गुणाहृतं for (गुणाहतं) १. याम्योत्तरं for (याम्योतरं)
(क) १. याम्येंतरे for (याम्योत्तरं)

४७. (ट) ४. चापांशाः (ग) (च) for (चापांशा)
१. सदृशैः (ग) (च) for (सदशैः)
५. रूनाः (ग) for (रूना)
३. विक्षेपांशा (ग) विक्षेपांशाद्व्यस्ता for (विक्षेपांशे व्यस्ता)
६. व्यस्तविशुद्धाव्यसदृश्यांशैः (ग) विस्तविशुद्धा for (व्यस्तविशुद्धाव्यसदृश्यांशैः)
(ग) २. भंध्रुवका (च) for (मंधुवका)
७. वसद्रिशांशैः for (व्यसदृश्यांशैः)
(क) १. सदृशै for (सदशैः) २. ध्रुवकः for (मंधुवकः) ३. पांशा for (पांशे)

सहिता विक्षेपांशस्तच्छापांशकवशादुदग्याम्या ।
एवं विक्षेपांशाः तक्रांत्यंशैर्ध्रुवो रविवत् ॥ ४८ ॥
उनैमानैक्यार्द्धात् गृहयोमध्यांतरे युतिर्गृहयोः ।
समलिप्तिकयो गृहणदधिके स्फुट मानयोगार्द्धात् ॥ ४९ ॥
समलिप्तिकालिकार्कात् कृत्वा लग्नं स्वदेशराश्युदयैः ।
गृहयोः समलिप्तिकयोः स्वदिनोदिननाडिका प्राग्वत् ॥ ५० ॥
अधिकदिनोदितघटिकाभिरूनदिननाडिका गुणा भक्ताः ।
अधिकदिननाडिकाभिः फलनाड्यो यदि भवन्त्यूनाः ॥ ५१ ॥

४८. (घ) ४. विक्षेपांशा (ग) for (विक्षेपांश)
१. याम्याः (ग) (च) for (याम्या)
५. तत्क्रांत्यंशै (च) for (तक्रांत्यंशै)
(ग) २. विक्षेपांशैस्तत्क्रान्त्यशा for (विक्षेपांशाः त क्रांत्यंशै)
(क) १. यामाः for (याम्या) २. विक्षेपाश for (विक्षेपांशाः)
३. जातैविक्षेपानयनम् for (ध्रुवोरविवत्)

४९. (घ) २. ऊनेमानैक्यार्द्धा (ग) (च) for (उनैमानैक्यार्द्धात्)
३. ग्रहयोर्म्मध्यांतरे (ग) (च) for (ग्रहयोमध्यांतरे)
४. ग्रहयैः for (ग्रहयोः)
(ग) १. ग्रहणवदधिके for (ग्रहणदधिके)
५. योगार्द्धा for (योगार्द्धात्)
(च) ४. ग्रहयौः for (ग्रहयोः)
(क) १. वदधिके for (दधिके)

५०. (घ) २. राश्युदयै (ग) for (राश्युदयैः)
१. नाडिकाः (ग) स्वदिनोदितनाडिका for (स्वदिनोदिननाडिका
(ग) ३. स्वदेदेश for (स्वदेश)
(च) ४. कालिकाक्कर्कात् for (कालिकार्कात्)
१. स्वदिनोदिननाडिकाः for (स्वदिनोदिननाडिका)
५. प्रग्वत् for (प्राग्वत्)
(क) ९. स्वदितनाडिका for (स्वदिनोदिननाडिका) ४. कति for (कर्कात्)

५१. (घ) २. नाड्यं for (नाड्यो)
(ग) ३. भक्ता for (भक्ताः) ४. भवत्यूनम् for (भवन्त्यूनाः)
(क) १. गुण for (दिन)

उनदिवसो दिताभ्यो घटिकाभ्यः प्रागथाधिकाः पश्चात् ।
योगस्तत् घटिकांतरमाद्यं कृत्वेषु घटिकाभिः ॥ ५२ ॥
गुणिता स्वभुक्तलिप्ता षष्टिहृताः प्रागृणं धनं पश्चात् ।
आद्यवदन्तरमन्यत्प्राक् पश्चाद्धान्तिर द्वितयम् ॥ ५३ ॥
आद्यघटिकान्तरं वशात्प्राक् पश्चाद् वा युतिः प्रथमकलात् ।
कृत्वग पृथकदिनोदित घटिकास्तत्कालिक ग्रहयोः ॥ ५४ ॥

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५२. (घ) २. ऊन (ग) ऊनादिवसो (च) for (उनदिवसो)
(ग) ३. प्रागधिकाः for (प्रागथाधिकाः)
४. कृत्वेष्ट for (कृत्वेषु)
(च) ५. योगस्तद्घटिकांतरमाद्यं for (योगस्तत् घटिकांतरमद्यं)
(क) १. घटिकांदिफलं ग्राह्यं for (घटिकांतमाद्यं)
२. यदिभरमाद्यं कृत्वेष्ट for (घटिकांतरमाद्यं कृत्वेषु)

५३. (घ) २. गुणिताः for (गुणिता)
३. लिप्ता (ग) (च) for (लिप्ता)
१. प्रागृणं (ग) (च) for (प्रागणं)
(ग) ४. षष्टिहृताः (च) for षष्टिहृताः)
५. पश्चाद्वान्तर for (पश्चाद्धांतिर-द्वांतर-)
(क) १. प्रागृण for (प्रागणं) २. गुणिताः for (गुणिता) ३. लिप्ताः for (लिप्ता) ५. पश्चाद्वांतर for (पश्चाद्धान्तिर-द्वांतर)
६. मन्यत्प्रा for (मन्यत्प्राक्)

५४. (घ) ३. अद्य (च) for (आद्य)
४. घटिकान्तरवशात् (ग) आद्यघटिकांतरवशा for (घटिकांतरं वशात्)
५. पृथगुदितोदित (ग) पृथग्दिनोदित for (पृथकदिनोदित)
(ग) २. तात्कालिकग्रहयोः for (तत्कालिकग्रहयोः), इसकी श्लोक संख्या ५५ है।
(च) ४. घटिकांतरवशात् for (घटिकांतरवशात्)
५. पृथगुदिनोदित for (पृथक्दिनोदित)
(क) १. कालात् for (कलात्) (ग) २. कोलिक for (कालिक) इसकी श्लोक संख्या ५५ है।
(ग) यद्याद्यान्यांतरयोरन्तर हृत मन्यथा तदैक्येन ।
गाद्यन्तरष्टाभिर्घटिका निर्गुणित माप्ताभिः ॥ ५४ ॥
इसमें यह अतिरिक्त श्लोक पाया गया है।
(क) १. आद्यंतर for (ग्राद्यंतर)

उदितघटिका यदिहृता गुणिताश्चाद्यान्यदिवसघटिकाभिः ।
यद्याद्यानान्तरयोरंतर हतमन्यथा तदैक्येन ॥ ५५ ॥

५५. (क) १. हृता for (हता) २. स्वान्य for (द्यान्य)

३. अन्योदितटिकाभिस्तुल्यो योगोभवेदसकृत् for (यद्याद्यानांतरयोरंतरहतमन्यथा तदैक्येन) इसकी श्लोक संख्या ५६ है।

(क) यद्याद्यान्यान्तरयोरन्तर हृतमन्यथा तदक्येन ।
आद्यंतरमिष्टाभिर्घटिकाभि गुणितमाप्ताभिः ॥ ५५ ॥

for

उदितघटिका यदि हृता गुणिताश्चाद्यान्यदिवसघटिकाभिः ।
यद्याद्यानाँतरयोरंतर हतमन्यथा तदैक्येन ॥ ५५ ॥

(घ) १. हृता (ग) (च) for (हता)

२. गुणिताश्चश्चा (ग) गुणिताश्चस्वान्य for (गुणिताश्चाद्यान्य)

३. यद्याद्यान्यांतर for (यद्याद्यानांतरयोरंतर)

वि०—इस श्लोक में एक और पंक्ति है और उसके पश्चात् क्रमसंख्या ५४ लिखी है। पंक्ति है —

"आद्यंतरमिश्चादांतरै द्वितयं । ष्टाभिर्घटिकाभिर्गुणितमास्ताभिः ॥ ५४ ॥
for (यद्याद्यानाँतरयोरन्तरहतमन्यथा तदैक्येन)

(ग) ५. उदकैत for (उदित)
वि० इसका उत्तरार्ध निम्नांकित है—श्लोक संख्या ५६ है।
'अन्योदितघटिकाभिस्तुल्यायोगो न चेदसकृत' for (यद्याद्यानांतरयोरंतर हतमन्यथा तदैक्येन)

(च) २. गुणिताश्चश्चान्यदिवसघटिकाभि for (गुणिताश्चाद्यान्यदिवसघटिकाभिः)

३. यद्याद्यान्यांतरं for (यद्याद्यानांतरयोरंतर)

६. हृतमन्यथा for (हतमन्यथा)

आद्यंतरमिष्टाभिर्घटिकाभिर्गुणितमाप्ताभिः ।
अन्योदित घटिकाभिस्तुल्यो योगो भवेदसकृत् ॥ ५६ ॥
अन्येष्टनाडिकाभिः कृत्वा तुल्या यदा तदा योगः ।
कार्यो अंगोन्नतिवत् ग्रहमध्यान्तरे योगे ॥ ५७ ॥

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५६. (क) उदितघटिका यदि हृता गुणिताश्च स्वान्यदिवस घटिकाभिः ।
अन्योदित घटिकाभिस्तुल्यो योगो भवेदसकृत् ॥ ५६ ॥

for

आद्यंतरमिष्टाभिर्घटिकाभिर्गुणितमाप्ताभिः ।
अन्योदित घटिकाभिस्तुल्यो योगो भवेदसकृत् ॥ ५६ ॥

(वि) इस श्लोक का पूर्वार्ध –अतिरिक्त ५४ का उत्तरार्ध भी है।

(घ) वि० इस श्लोक की पहली पंक्ति पिछले श्लोक की अंतिम पंक्ति दर्शाई गई है। क्रम संख्या ५४ लिखी है।

१. 'भिश्चार्द्धांतरे द्वितयं, ष्टाभिर्घटिकाभिर्गुणितमास्ताभिः ॥ ५४ ॥' इस पंक्ति के पश्चात् ५५ संख्या कहीं नहीं है । हां "अन्योदित ·····असकृत" लिखकर ॥ ५६ ॥ संख्या लिखदी है।

(ग) यह श्लोक इस प्रति में नहीं है।
इसकी पहली पंक्ति—ग के ५४ संख्यावाले श्लोक की दूसरी पंक्ति है। तथा ग के ५६ संख्या वाले श्लोक की दूसरी पंक्ति, इसकी दूसरी पंक्ति है। देखो पिछला श्लोक।

(च) २. यहां पर भी क्रमसंख्या ५६ दी हुई है।

५७. (घ) २. शृंगोन्नतिवद्ग्रहमध्यांतरे (ग) शृंगोन्नतिवद्ग्रहयो for (अंगोन्नतिवत् ग्रहमध्यान्तरे)
१. कार्यो for (कार्यो)
३. र्मध्यान्तरे for (मध्यांतरे)
इसकी श्लोक संख्या मिलती जुलती है।

(च) ४. क्रत्वा for (कृत्वा) शृंगोन्नतिवद्ग्रहमध्यांतरे योगे for (अंगोन्नतिवत् ग्रहमध्यांतरे योगे)

(क) १. कायो for (कार्यो) २. शृङ्गान्नति for (अंगोन्नति)
३. ग्रहयोम् for (ग्रहमध्यांतरे) ४. योगः for (योगे)

बाहु[५] संयोगांतरमन्नाशंक्वग्रयोः[१] समान्यदिशोः[२] ।
कर्णोदग्ज्येकोटि[३] स्वकर्णभुजकृतिविशेषपदे[४] ॥ ५८ ॥
कोटिभुजकर्णशंकु[६] षष्टि[२] गुणाध्वाहृता[३] मध्यात् ।
कोटि[४] प्रथक्चसार्ये[८] प्राच्यां प्रागपरयोः[७] पश्चात्[५] ॥ ५९ ॥

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५८. (घ) १. माग्रो शंकुग्राह्या: (ग) मग्राशंक्वग्रयो for (मन्नाशंक्वग्रयोः)

२. सामान्यदिशो: for (सम्मान्यदिशोः)

३. दृज्ज्येकोटि (ग) कर्णो दृग्येज्येकोटी for (कर्णोदग्ज्येकोटि)

४. वेशेषयदे for विशेषपदे)

(ग) ५. वाहुः for (बाहु)

(च) १. माग्राशंकुग्रह्यो: for (मन्नाशंक्वग्रयोः)

२. सामान्यदिशो: for समान्यदिशोः)

३. दृज्येकोटी for (कोटि) ४. वशेषपदे for (विशेषपदे)

(क) १. मग्रासंक्व for (मन्नाशंक्वग्रयोः) २. दृश्यौ for (दिशोः) ३. कौटी for (कोटि)

५९. (घ) ३. गुणाव्याहृता हृतान्मध्यात् (ग) गुणान् व्यासदलहृतान् for (गुणाध्वाहृता मध्यात्)

३. कोटी पृथक् पृसार्ये (ग) कोटीपृथक् प्रसार्ये for (कोटि प्रथक्चसार्ये)

(ग) ६. शंकून् for (शंकु)

७. प्रागपरया; for (प्रागपरयोः)

(च) ३. गुणा व्याहृतान्मध्यात् for (गुणाध्वाहृतामध्यात्)

४. कोटी for (कोटि)

८. पृथक्पृसार्ये for (प्रथक्चसार्ये)

(क) १. काटिभुजर्ण for (कोटि भुजकर्ण) २. यष्टि (ग) for (षष्टि)

३. गुणात् व्यासिक्तात्म for (गुणाध्वाहृता मध्यात्)

४. क्वाऽग्रपृथक् प्रसयो for (कोटिप्रथक्चसार्ये) ५. मध्यात् for (पश्चात्)

कोटचग्राम्यां बाहुकर्णे दिग्मध्यतो भुजाग्रांतौ ।
बाहुग्राह्यो स्वशंकुषष्टि मध्यान्तरागांते ॥ ६० ॥
दिग्मध्ये स्थित द्रष्टृचा पृथग् गृहौ दर्शयेत् स्वशंक्वग्रे ।
योगे शंक्वग्रान्तरमंतरमथवान्यदागृह्ययोः ॥ ६१ ॥
नावार्यो ज्ञातेरपि तन्त्रैरार्यभटविष्णुचन्द्राद्यैः ।
यो ब्रह्मधूलिकर्म विदाचार्यत्वं भवति तस्य ॥ ६२ ॥

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६०. (घ) ४. ग्राह्या for (ग्राह्यो) (ग) बाहुग्रयोः for (बाहुग्राह्यो)
५. शंकूयष्टी (ग) for (शंकुषष्टि)
(ग) १. कर्णो for (कर्णे)
६. मध्यात्तदाग्रांते for (मध्यांतराग्रांते)
(च) ७. कोद्यग्राम्यां for (कोटचग्राम्यां) बाहुकर्णो for (बाहुकर्णे)
३. भुजाग्रांतो for भुजाग्रांतौ
४. बाह्वग्राह्यो for (बाहुग्राह्यो) ५. स्वशंकूयष्टी for (शंकुषष्टि)
६: मध्यांतराश्रांते for (मध्यांतराग्रांते)
(क) १. कर्णात् for (कर्णे) २. दृग् for (दिग्) ३. भुजाग्रातौ for (भुजाग्रांतौ)
४. बाह्वग्रयोः for (बाहुग्राह्यो) ५. यष्टि for (षष्टि)
६. मध्यत्तदग्रते for (मध्यांतराग्रांते)

६१. (घ) १. दृष्टचा (ग) (च) for (द्रष्टृया)
३. शक्वयांतर (ग) शंक्वग्रांतर for (शंक्वग्रंतर)
(ग) ४. दिग्मध्यस्थित for (दिग्मध्येस्थित)
(च) ३. शंक्वग्रांतर for (शंक्वग्रन्तर)
(क) १. दृष्टचा for (द्रष्टृया) २. ग्रहोदशयन्न for (ग्रहौ दर्शयेत्)
३. स्वस्वाप्रतिर्मततो सलभ्य for (शंक्वग्रन्तरमन्तर)

६२. (घ) १. न वार्यो (ग) नाचार्यो for (नावार्यो)
२. ज्ञातैरपि (ग) (च) for (ज्ञातेरपि)
३. ब्राह्मधूलि (च) ब्राह्मधूलिकर्म्म for ब्रह्मधूलिकर्म)
(ग) ४. राचार्यभट for (रार्यभट)
(क) १. नावार्यो for (नावार्यो) २. जातै for (ज्ञातेरपि) ३. ब्राह्म for (ब्रह्म)

रविशशितमस्त्रिवरितं ब्रह्मोक्तं पुण्यमङ्गुतं ज्ञात्वा ।
रविचन्द्रराहु लोकात् प्राप्नोति पुमानीह यशश्च ॥ ६३ ॥
अन्यैरप्युक्तमिदं योयं सम्यक् ग्रहं विजानाति ।
याति सद्रलोकं ग्रहाष्टकज्ञं परं ब्रह्म ॥ ६४ ॥
मध्यमगतिः स्फुटगति स्त्रिप्रश्नश्चन्द्रभास्कर ग्रहणे ।
उदयास्तमयप्रतिघटिकामिदोः शृङ्गोन्नतिछाये ॥ ६५ ॥

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६३. (घ) १. तूवरितं (ग) तमस्त्रिचरितं for (तमस्त्रिवरितं)
३. पुमानिह (च) for (पुमानीह)
(ग) ४. श्रुत्वा for (ज्ञात्वा)
२. लोकान् for (लोकात्)
(च) १. तमसूचरितं for (तमस्त्रिचरितं)
(क) १. चरितं for (वरितं) २. लोकान् (ग) for (लोकात्)
३. पुमानिह (ग) for (पुमानीह)

६४. (घ) ४. अन्यैरप्युक्तमिदं (ग) (च) for (अन्यैरप्यूक्तमिदं)
५. सम्य (ग) सम्यग्ग्रहं for (सम्यक्ग्रहं)
२. सतद्ग्रहलोकं (ग) (च) for (सद्रलोकं)
३. ग्रहाष्टज्ञः (ग) ग्रहाष्टकज्ञः for (ग्रहाष्टकज्ञं)
(ग) ४. ब्रह्मः for (ब्रह्म)
(च) ५. सम्यग्रहं for (सम्यक्ग्रहं)
३. ग्रहाष्टकज्ञः for (ग्रहाष्टकज्ञं)
(क) २. सतग्रह for (सद्रलोकं)
३. ग्रहाष्टज्ञः for (ग्रहाष्टकज्ञं)

६५. (घ) १. भास्करं (च) for (भास्कर)
२. उदयास्तमयः (ग) उदयास्तमयौ for (उदयास्तमय)
३. प्रतिघटिकांनिदो (ग) प्रतिघटिकांमिदु for (प्रतिघटिकामिदोः)
(ग) ४. मध्यगतिः for (मध्यमगतिः)
५. स्पष्टगति for (स्फुटगति) ६. त्रिप्रस्नचन्द्र for (त्रिप्रश्नश्चन्द्र)
(च) २. उदयास्तमयः for (उदयास्तमय) ३. प्रतिघटिकांमिदो for (प्रतिघटिकामिदोः)
(क) १. मध्यमगतिस्त्रिप्रश्नश्चंद्र भास्कर ग्रहणे । उदयास्तमयः । चंद्रग्रहणं चतुर्थः सूर्यग्रहणं पचमः । उदयास्तमय षष्ठाप्रति घटिका मिंदोः शृंगोन्नति छाये । for (मध्यमगतिः स्फुटगतिस्त्रिप्रश्नश्चन्द्रभास्करग्रहणे । उदयास्तमय प्रतिघटिकांमिदोः शृङ्गोन्नतिछाये ।)

ग्रहयोगोत्र ग्रहयुतिरार्यात्रिंशति युताष्ट सप्तत्या ३७८
अध्यायैर्दशभिर्धूलिकर्म वोच्चैर्विना ब्रह्म ॥ ६६ ॥
गुरुणा नधूलिकर्म प्रतिकंचुककारिणे प्रदातव्यम् ।
दत्त सुकृत्प्रणाशं कुरुते प्रतिकंचुकं यस्य ॥ ६७ ॥
ग्रहमेलके यदुक्तं तत्स्थूलं स्पष्टमिह यदुक्तं तत् ।
ग्रहभमुनींदु छाया शृङ्गोन्नतिभोदयाद्येषु ॥ ६८ ॥
सूर्यास्तमयादिष्टाद्रातिगताछेष्ट भोदयास्तमयौ ।
जानाति न कश्चिदपि ब्राह्मनं मुच्चान्यतन्भज्ञः ॥६९॥

६६. (घ) १. भ (ग) (च) for (त्र)
(ग) ३. त्रिशती for (त्रिशति)
४. चाद्यै for (वोच्चै)
(च) ५. ईशभि for (दशभि) ६. धूंलिकर्म्म for (धूंलिकर्म)
(क) १. भ for (त्र) २. ब्राह्मी (ग) ब्राह्म्ये for (ब्रह्म)
६७. (घ) ३. योस्य (ग) (च) for (यस्य)
(ग) १. प्रतिकंचककारिणे for (प्रतिकंचुककारिणे)
४. सुकृतः प्रणाशः for (सुकृत्प्रणाशः)
(च) ५. कर्म्म for (कर्म)
(क) १. प्रातस्थर्क्क कारिणो for (प्रतिकंचुककारिणे)
२. सुर्क योस्या for (चुकंयस्य)
६८. (घ) १. ग्रेहभमुनींदु (ग) for (ग्रहभमुनींदु)
(ग) ३. स्पष्टमिदं for (स्पष्टमिह)
(च) ५. तस्थूलं for (तत्स्थूलं)
(क) १. ग्रहण मनींदुछायो for (ग्रहभमुनींदुछाया) २. लोदयाद्येषु for (भोदयाद्येषु)
४. ग्रहामलके for (ग्रहमेलके) ३. स्पष्ट इह for (स्पष्टमिह)
यह यहीं समाप्त है ।
६९. (घ) १. दिष्टचाद्ग्रारिगता for (दिष्टाद्रातिगता) (ग) सूर्यास्तमयाद्वारात्रिगता for (सूर्यास्तमयादिष्टाद्रातिगता)
२. द्वेष्ट (ग) स्वेष्टभोदयास्तमयोः for (छेष्टभोदयास्तमयौ)
३. ब्राह्मं मुत्वान्यतंत्रज्ञः (ग)ब्राह्मं मुक्तान्यतंत्रज्ञः for (ब्राह्मनं मुच्चान्यतन्भज्ञः)
(क) यह श्लोक उपलब्ध नहीं,
(च) १. दिष्टचाद्ग्रारिगता for (दिष्टाद्रातिगता) २. द्वे ष्ट for (छेष्ट)
३. ब्राह्मं मुच्चान्यतंत्रज्ञः for (ब्राह्मनंमुच्चान्यतंत्रज्ञः)

भमुनीध्र ु व विक्षेप ग्रहोदयास्त मयनाडिकाद्येषु ।
अध्यायोभग्रह युतिरार्याणां सप्तति दर्श ॥ ७० ॥
इत्ति ब्रह्मगुप्ते दशमोऽध्यायः ॥
इति श्री ब्राह्मस्फुटसिद्धान्ते ग्रहतारादिक्षेपाधिकारोदशमः ॥ १० ॥
सम्पूर्णा पूर्वादशाध्यायी

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७०. (घ) १. भमुनि (ग) (च) for (भमु)

२. दंशमा for (दंश)

(ग) ३. कालिकाद्येषु for (नाडिकाद्येषु)
आर्याणां सप्तत्या दशमोभग्रहयुतिरध्यायः ॥ ७० ॥
इति श्री ब्रह्मगुप्ताचार्य विरचिता दशाध्यायी समाप्ता ॥

for (अध्यायोभग्रहयुतिरार्याणां सप्ततिर्दश ॥७०॥ इति ब्रह्मगुप्ते दशमोऽध्यायः)

(क) यह श्लोक उपलब्ध नहीं है।

(च) २. ईंश for (दंश)
४. 'इति' से 'अध्यायः' तक लुप्त

(ग) वि०-अतिरिक्त पाठ—
इति श्री ब्रह्मगुप्ताचार्य विरचिता दशाध्यायी समाप्ता । अथ स्वस्ति श्री संवत् १६७७ वर्षे शाके १५४३ प्रवर्तमाने ॥ ज्येष्ठबदि १० सोमे । रेवती घटी १८ पल १७ नक्षत्रे । अत्र ग्रन्थ लिषित समये श्री राजनगर मध्यवास्तव्यं ।श्री·····ज्ञातीय । ज्योति श्री ···कस्य इयं पुस्तिकालिषितमस्ति । उक्तं च । यादृिशं पुस्के दृिष्ट्वा तादृिशं लिखितं मया । यदिशुद्धम-शुद्धं वा मम दोषो न दीयते ॥ पुनः ॥ भग्नदृष्टि कटिग्रीवाबद्धमुष्टिरधोमुखम् । कष्टेन लिखितं शास्त्रं यत्नेन परिपालयेत् । लेषकपाटकयोः शुभं भवतु । स्वयं पठनार्थं तथापुत्रपौत्रादिपठनार्थं । परोपकारार्थं । श्री कृष्णार्पणमस्तुः ॥ श्रीरस्तु मंगल मस्तु । शुभंभवतु । जयोस्तु । छः छः छः छः छः छः । उक्तं च । ब्रह्मा प्राह त नारदाय हिमगुर्य शौनकायालं मांडव्याय वसिष्ठसंज्ञकमुनिः सूर्यामियायं हि यत् । सत्पक्षागमशास्त्र युक्तममिदं शास्त्रं विहायान्यथा । ये कुर्वंति नरान् निर्वहंति तो विज्ञानशून्याश्चिरम् ॥

अथ तन्त्र परीक्षा नामैकादशोध्यायः

ये[१] ज्ञान पटल[२]रूद्धदशोन्यब्रह्म[३]द्वदन्ति सिद्धान्तम्[४]।
तेषां युगादिभेदेद्य[५] दोषास्तान्प्रवक्षामि[६] ॥ १ ॥
युगमाहुः पञ्चाब्दं[१] रविशशिनोः संहितांग[३] कारये।
अधिमासा वमरात्र स्फुट[४]तिथ्यज्ञान[२] तदसत् ॥ २ ॥

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१. (घ) १. येऽज्ञान for (ये ज्ञान)
२. रुद्धदशो (ग) रुद्धदशो for (रूद्धदशो)
३. न्यद्बाह्याद्वदन्ति (ग) ऽन्यद्ब्राह्माद्वदन्ति for (न्यब्रह्माद्वदंति)
४. सिद्धांतात् (ग) (च) for (सिद्धान्तम्)
५. भेदाद्ये (ग) (च) for (भेदेद्ये)
६. प्रवक्ष्यामि (ग) प्रविक्ष्यामि for (प्रवक्षामि)

(च) २. रुद्धदशो for (रूद्धदशो)
३. द्ब्राह्माद्वदन्ति for (ब्रह्माद्वदन्ति) ४. सिद्धान्तात् for (सिद्धान्तम्)
६. प्रवक्ष्यामि for (प्रवक्षामि)

(ग) वि०-अतिरिक्त पाठ—
अथ ब्रह्मगुप्तसिद्धांते एकादशमोऽध्यायः प्रारभ्यते ॥ खखखार्क १२००० दृवृताब्देभ्यो गतगम्याल्पा-त्खशून्य यमल । २०० हृतात् । लब्धं त्रि ३ सायक हतं कलाभिरूनौ सदार्कैंदू ॥ १ ॥ विधु वज्जीवेद्वि २ हतं चन्द्रोच्चं तिथि १५ गुणं च सितशीघ्रे ॥ द्वीषु ५२ हतं च स्वांर द्वि २ कु १ वेद ४ हतं च पात कुजशनिषु ॥ २ ॥ ग्रहबीजानि ब्रह्मसिद्धांते ॥

२. (घ) १. पंचाक्षरं for (पञ्चाब्दं)
२. ज्ञानतस्तदसत् (ग) तस्तदसत् for (ज्ञानतदसत्)

(ग) ३. —-रयैं for (कारये)
४. स्फुतिथ्य for (स्फुटतिथ्य)
५. ज्ञान् for (ज्ञान)

(च) ३. काराय for (कारये)

भानि ध्रुवपंचाशत् द्वावर्कादयो जिनोक्तं यत् ।
ध्रुवमसस्यावर्णो भवति यतोह्ना ततो सतत् ॥ ३ ॥
आर्यभटो युगपादांस्त्रीन् पातानाह कलियुगादौ यत् ।
३२४०००० तस्य कृतां तर्यस्मात् स्वयुगाद्यंतौ न तस्मात् ॥ ४ ॥
रवियुगभगणां रव्युधृति ४३२०००० यत्प्रोक्तं तत्रयोर्युगं स्पष्टम् ।
त्रिशतीरव्युदयानां तदन्तरं हेतुना केन ॥ ५ ॥

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३. (घ) १. पंचांशत् for (पंचाशत्) (ग) चतुः पंचाशत् द्वौ for (ध्रुवपंचाशत् द्वा)
२. ध्रुवमत्सस्या (ग) ध्रुवमत्सस्यावर्त्तो for (ध्रुवमसस्यावर्णो)
३. यतोह्ना ततोसत्तत् (ग) for (यतोह्नाततो सतत्)

(च) ४. द्वावर्क्कादयौ for (द्वावर्कादयो)
२. ध्रुवमत्सस्यावर्णो for (ध्रुवमसस्यावर्णो)
३. यतोह्ना ततो सत्तत् for (यतोह्ना ततो स तत्)

४. (घ) १. अर्यभटो for (आर्यभटो) (ग) आर्यभटोसुत for (आर्यभटोयुग)
२. यस्मात्तं for (र्यस्मात् स्व)
३. तत्तस्मात् (ग) for (न तस्मात्)

(ग) ४. यतोनाह for (पातानाह)

(च) १. अर्यभटो for (आर्यभटो)
३. न तत्तस्मात् for (न तस्मात्)

५. (घ) १. ४३२००० (च) for (४३२०००००)
२. तंत्रयो for (तत्रयो)
३. स्पष्टान्यत् for (स्पष्टम्)
वि०—इस श्लोक की दूसरी पंक्ति यहां लिपिकार लिखना भूल गया प्रतीत होता है। यही नहीं छठे श्लोक की पहली पंक्ति भी यहां लुप्त है।

(ग) ४. युगरविभगणाः for (रवियुगभगणां)
५. रव्युद्रिति for (रव्युधृति)
६. युगं for (र्युगं)
७. +३०० ॥०॥+

(च) १. ४३२०० यत्पोक्तं for (४३२०००००

युगवर्षादिनवदच्चैत्र सितादेः समं प्रवृत्तान्यत् ।
तदसत् यतः स्फुटयुगं तस्थैर्यान्मंदपातानाम् ॥ ६ ॥
ग्रहभुक्तेरूनाया मंदोच्चं भवति शीघ्रमधिकायाम् ।
उच्चगतौ मंदोच्चं न विना भुक्तेर्दु वजमेतः ॥ ७ ॥
आर्याष्टशते पाता भ्रमंति दशगीतके स्थिराः पठिताः ।
मुक्तेर्दुपातमंडपमण्डले भ्रमति स्थिरानातः ॥ ८ ॥
आर्यभटो जानाति ग्रहाष्टाति यदुक्तवांस्तदसत् ।
राहुकृतं न ग्रहणं तत्पातो नाष्टमो राहुः ॥ ९ ॥

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६. (घ) यहां श्लोक का पूर्वार्ध लुप्त है।
१. तदराज् (ग) तदसद्युतः for (तदसत्—तदसत् यतः)
२. तत्स्थैर्यान् (ग) स्थैर्यान् for (तस्थैर्यान्)
(ग) ३. युगवर्षादी for (युगवर्षादि)
४. सप्रवृत्तानयत् for (समं प्रवृत्तान्यत्)
५. युतं गं न for (युगं)
७. (घ) १. रूनायां for (रूनाया)
२. भुक्तचेन्दु for (भुक्तेंदु)
३. वर्जमतः for (वर्जमेतः)
(ग) ४. परिधि फले व्यावहारिके त्रिगुणे for (मंदोच्चं भवति शीघ्रमधिकायाम्)
५. दूसरी पंक्ति बिल्कुल भिन्न है—
'तद्वर्गाभ्यां दशभिः संगुणिताभ्यां सूक्ष्मे'

वि०—तीसरी पंक्ति यह है—
उच्चं भवति शीघ्रमधिकायां । उच्चगतौ मंदोच्चं न विना भुक्त्येंदुवर्जमनः ॥७॥
अर्थात् रेखांकित दो पंक्तियां अधिक हैं।

८. (घ) १. आर्याऽष्टशते (च) for (आर्याष्टशते)
२. दशगीतेके (ग) दशगीतिके for (दशगीतके)
३. मुक्तचेन्दु (ग) for (मुक्तेंदु)
४. भ्रमंति (ग) for (भ्रमति)
(ग) ५. मपमंडले for (मंडपमण्डले) (च)
(च) २. दश२गीतके for (दशगीतके)
९. (घ) १. ग्रहाष्टकगति (ग) for (ग्रहाष्टाति)
२. तस्पातो (ग) तस्यातो for (तत्पातो)

न समायुगमनुकल्पाः कल्पादिगत कृतादियातं च ।
स्मृत्युक्तैरार्यभटो ना तो जानाति मध्यगतिं ।। १० ।।
उँकारो दिनवारो गुरुरदयिकोस्य भवति कल्पादौ ।
न भवत्यर्को यस्मादोंकारे विस्वरस्तस्मात् ।। ११ ।।
सूर्योदयाच्चतुर्थात् दिनपात् यदुवाच तदसदार्यभटः ।
लंकोदयो यतोर्क स्यास्तमयं प्राह सिद्ध पुरे ।। १२ ।।
आधिकैः शतैश्चतुर्भिवर्ष सहस्रैश्चतुर्दशाभिरेकः १४४०० ।
युगपातैर्दिनवारोत्तरयोदयिकर्द्ध रात्रिकयोः ।। १३ ।।

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१०. (घ) १. कल्पादिगतं (ग) (च) for (कल्पादिगत)
२. स्मृत्युक्तैर्यभटो for (स्मृत्युक्तैरार्यभटो)
(ग) ३. समायुतं गमनुकल्पाः for (समायुगमनुकल्पाः)

११. (घ) १. ओंकारो for (उँकारो)
२. रौदयिको (ग) रोदयिक्येस्य for (रदयिकोस्य)
(ग) ३. दिनचरै for (दिनवारो)
(च) २. गुरुरौदयिको for (गुरुरदयिको)
४. न भवत्यक्र्को for (न भवत्यर्को)

१२. (ग) १. चतुर्थान् for (चतुर्थात्)
२. दिनपान for (दिनपात्)
३. तव for (तद)
४. लंकोदये for (लंकोदयो)
५. यतोर्क्कस्या for (यतोर्कस्या)

१३. (घ) १. अधिकैः (ग) (च) for (आधिकैः)
२. चतुर्भिर्वर्ष (ग) (च) for (चतुर्भिवर्ष)
३. चतुर्दशभिरेकः (ग) (च) for (चतुर्दशाभिरेकः)
४. मौदयिकार्द्ध (ग) for (योदयिकर्द्ध)
(ग) ५. युगजातै for (युगपातै)
६. दिनवारोंत्तर for (दिनवारोत्तर)
(च) ४. मोदयिकर्द्ध for (योदयिकर्द्ध)

उदयिकाद्दिन[१] भुत्तचार्द्ध[२] रात्रिको मध्यमोभवेत्[३] पूतः ।
कतरं[४] स्फुटं न निश्चितमनयोः[५] स्फुटमेकमपि नातः[६] ॥ १४ ॥
षोडशगियिवि[१] योजन परिधि[२] षडम्दव्यासं पुनर्लावदता[३] ।
आत्मज्ञानं[४] ख्यापितमनिश्चयः स्वमतिकृतकत्वात् ॥ १५ ॥
भुव्यासस्य[१] ज्ञाना[२]ध्यर्थं देशांतरं दज्ञानात्[३] ।
स्फुटतिथ्यंताज्ञान[४] तिथिनाशाद् ग्रहणयोनाशः[५] ॥ १६ ॥

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१४. (घ) १. औदयिकाद्दिन (ग) औदयिकादिन for (उदयिकाद्दिन)
२. भत्तचार्द्ध for (भुत्तचार्द्ध)
३. भवत्यूनः (ग) (च) for (भवेत्पूतः)
४. कतरत् (ग) कतर for (कतरं)
५. निश्चत for (निश्चित) (च)

(ग) ६. मतिनातः for (मपिनातः)

(च) १. औदयिकाद्दिन for (उदयिकाद्दिन) ४. कतरत् for (कतरं)

१५. (घ) १. षोडशगियियोजनपरिधि (ग) षोडासगियियोजनपरिधि for (षोडशगियिवि-योजन परिधि)
२. षडि भूव्यासं (ग) निभूव्यासं for (षडम्दव्यासं)
३. पुर्नजिलावदता (ग) पुना लावदता for (पुनर्लावदता)
४. आत्माज्ञानं (ग) (च) for (आत्मज्ञानं)

(च) २. षड्भिभूव्यासं for (षडम्दव्यासं)
३. पुर्नकिलावदता for (पुनर्लावदता)
४. यहाँ क्रमसंख्या लुप्त है ।

१६. (घ) १. भूव्यासस्या (ग) (च) for (भुव्यासस्य)
२. ज्ञानाद्वचर्थं (ग) (च) for (ज्ञानाध्व्यर्थं)
३. तदज्ञानात् (ग) for (दज्ञानात्)
४. ज्ञानम् (ग) for (ज्ञान)
५. नाशाद्ग्रणयोर्नाशः (ग) for (नाशाद्ग्रहणयो नाशः)

(च) ५. नाशाद्ग्रणयोर्नाशः for (नाशाद्ग्रहणयो नाशः)

प्राणेनैतिकलाभूर्यदि तत्कुतो व्रजेत्किमध्यानम् ।
आवर्तनमुर्व्याश्चेत्त पतंति समुच्छ्रयः कस्मात् ।। १७ ।।
उदयिके यः परिधिर्विषमोन्योन्य समेन्यभुजस्य गुणः ।
तदसद्विषमांतफलं यतोन्ययुग्मादि फल तुल्यम् ।। १८ ।।
विषमोन्योन्ये युग्मे परिधिर्गुणकः क्रमोक्रमाद्यानाम् ।
चक्रार्द्धे फलनाशो न भवति यस्मादसत्तदपि ।। १९ ।।
व्यासार्द्धहतो बाहुः परिधिविशेषाहृतः फलोनयुतः ।
प्रथमोधिकोनकोय तदसत्पदयोः परिधिपाठात् ।। २० ।।

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१७. (घ) १. कलां (ग) (च) for (कला)
२. तत्कत्कुतो (ग) तत्क्व कुतो (तत्ककुतो)
३. चेन्न for (चेत्त)
४. समुछ्रयः (ग) समुछ्रयाः for (समुछ्रयः)
(ग) ५. भूयदि for (भूर्यदि)
६. आवर्तनमुर्व्याश्चेन्न (च) for (आर्वतनमुर्व्याश्चेत्त)
(च) ४. समुछ्रयः for (समुछ्रयः) २. तत्क्वकुतो for (तत्ककुतो)

१८. (घ) १. औदयिके (ग) (च) for (उदयिके)
२. द्विषमेन्योन्य (ग) विषमेन्योन्यः for (विषमोन्योन्य)
३. समे भुजस्य गुणः (ग) समे भुजस्य गुणाः for (स मेन्यभुजस्य गुणः)
४. यग्मादि for (युग्मादि)
(ग) ५. यतोन for (यतोन्य)
(च) २. र्विषमे for (विषमो) ३. समे भुजस्य for (समेन्यभुजस्य)

१९. (घ) १. विषमे (ग) (च) for (विषमो) २. न्यो (ग) for (न्योन्ये)
३. क्रमोत्क्रमज्यानाम् (ग) for (क्रमोक्रमाद्यानाम्)
(च) १. विषमेन्योन्योयुग्मे for (विषमोन्यान्ये युग्मे)
३. क्रमोत्क्रमाद्यानां for (क्रमोक्रमाद्यानां)

२०. (घ) १. कोयत्तदसत् for (कोयतदसत्) (ग) २. हृतौ for (हृतो)
(च) २. हृतो for (हृतो) १. यत्तदसत्पदयोः for (यतदसत्पदयोः)

विषमसमयोर्यदि द्वौ परिधि[१] किं सूर्य चन्द्रयोर्नोक्तौ[२] ।
घटते न कथंचिदियं स्फुटक्रियोदयिकतन्त्रोक्ता ॥ २१ ॥
उत्तरगोले[२] गायां विषुवद्यातो[३] यदुक्तमुनायाम्[१] ।
सममंडलगस्तदसत् क्रान्तिज्यायां यतो भवति ॥ २२ ॥
व्यासार्द्धेन[३] भविभक्ता दग्गति[२]जीवा चतुर्गुणा[४] लब्धम् ।
लंबननाडयः[१] पंचदशगुणितया त्रिज्यया भक्ताः[५] ॥ २३ ॥
दृक्षेपज्या भुक्तितराहता[१] लब्धमवनतिर्भवति[२] ।
स्फुटयोजनकर्णाभ्यां भूव्यासार्द्धेन[३] च विना स्पष्टे ॥ २४ ॥
आर्यभटेनास्मि मतिलघुनि[४] किमर्थं महत्कृतं कर्म[२] ।
गणिताज्ञानाजाडयं[३] विजानयता[६] यदि ततः सुतराम् ॥ २५ ॥

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२१. (घ) १. परिधी (ग) (च) for (परिधि)
२. योर्नोक्तौ (ग) योनोक्तौ for (योर्नोक्तौ)
२२. (घ) १. मूनायाम् (ग) (च) for (मुनायाम्)
(ग) २. गोणे for (गोले)
३. ज्यातो for (द्यातो)
२३. (घ) १. ताडयः for (नाडयः) २. दृग्गति (ग) दृगाति for (दग्गति)
(ग) ३. 'भ' लुप्त है।
४. चतुगुणा for (चतुर्गुणा)
५. भक्ता for (भक्ताः)
(च) २. दृग्गति for (दग्गति)
२४. (घ) १. तराद्धिता for (तराहता)
(ग) २. भुत्तयंतरा for (भुक्तितरा)
३. भूव्यासे for (भूव्यासार्द्धेन)
(च) १. हृता for (हता)
२५. (घ) १. आर्यभटेनास्मिं सति (ग) आर्यभटेनास्मि न सति for (आर्यभटेनास्मिमति)
२. कर्मः for (कर्म)
३. ज्जाडयं for (जाडयं)
(ग) ४. लघुति for (लघुनि)
५. ज्ञातानाज्जाडयं for (ज्ञानाजाडयं)
६. विजानता for (विजानयता)
(च) २. कर्म्म for (कर्म)

लंबनमृणधनमुक्तं पूर्वापरयोस्तिथौ दिनार्द्धस्य ।
युक्तो भावो यद्भवति तदृणधनयोस्थिोकधरम् ॥ २६ ॥
दृक्षेपज्या बाहु दृग्ज्याकर्णोनयोः क्तितिशेषात् ।
मूलंदृग्गतिजीवा संस्थानमयुक्तमेतदपि ॥ २७ ॥
लंबनघटिका लिप्ता दृशज्ययार्केंदुदृग्गतिकलानाम् ।
यस्मान्न समस्तस्मात् दृशज्यया लंबनं स्थूलम् ॥ २८ ॥
वित्रिभलग्ने दृक्षेपमंडलयुतौ ज्या मध्या ।
मध्यादृक्षे पज्यानार्यभटोक्तनया तुल्या ॥ २९ ॥

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२६. (घ) १. स्थितौ (ग) for (स्तिथौ)
२. कतरत् (ग) for (कधरम्)
(ग) ३. पूर्वापरास्थितौ for (पूर्वापरयोस्तिथौ)
(च) १. स्तिथोकतरत् for (स्थिोकधरम्)

२७. (घ) १. दृक्षेपज्या (ग) क्षेपज्या for (दृग्क्षेपज्या)
२. कृतिविशेषात् for (क्तितिशेषात्)
(ग) ३. दृंगत्या for (दृग्ज्या)
२. कृतिविशेषात् (च) for (क्तितिशेषात्)
(च) १. दृक्षेपज्या for (दृग्क्षेपज्या) ५. दृंग्ज्या for (दृग्ज्या)

२८. (घ) १. दश (ग) द्रश for (दृश)
२. समा (च) for (सम)
(ग) ३. यस्मा समास्तस्मा for (यस्मान्न)
(च) १. दशज्ययार्केंदु for (दृशज्ययार्केंदु)
४. द्रग्गति for (दृग्गति) ५. दशज्यया for (दृशज्यया)

२९. (घ) १. मंडलं तदप (ग) मंडलं तप तदप for (मंडल युतो)
२. युतौ (ग) for (युतो)
३. (लुप्त)
४. भटोक्तातया (ग) for (भटोक्तनया)
(च) ३. "मध्या" लुप्त ५. दृक्येपज्या for (दृक्षेपज्या)
४. भटोक्ता तया for (भटोक्तनया)

दृक्षेपज्यातो सत्तन्नाशादवनते भवति ।
नाशः अवनतिनाशाभ्रामस्यो नाधिकवार विगृह्णे ॥ ३० ॥
पंचज्यया यतोर्कगृह्णं श्रीषेणविष्णुचन्द्रकृतम् ।
आर्यभटोक्तान्यनयोरर्क गृह दुषणानि ततः ॥ ३१ ॥
एवं विचार्यमाणं पंचज्यालम्बनं महास्थुलम् ।
स्थुलावनतिश्च तथा दृशज्यया लंबनावनति ॥ ३२ ॥
नाडीचुतुष्टुविधिना सर्वत्रसमो यतस्तत स्थुलः ।
मानार्थकर्ममहकृत् मार्यभटेन लघुनि सति ॥ ३३ ॥

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३०. (घ) १. पहली पंक्ति का अंतिम शब्द 'नाशः' ।
२. नाशाद्ग्रासस्यो for (नाशाभ्रामस्यो)
३. नाधिकता रवि for (नाधिकवारवि)
(ग) ४. भवति नाशः for (भवति)
५. दूसरी पंक्ति का आरम्भ 'अवनतिनाशाद्ग्रासस्यो ना'
(च) २. नाशाद्ग्रासस्यो for (नाशाभ्रामस्यो)

३१. (घ) १. विष्णु चं कृतं for (विष्णुचन्द्रकृतम्)
२. दूषणानि (ग) for (दुषणानि)
(च) ३. यतोर्क्क for (यतोर्क) १. चन्द्रकृतं for (चन्द्रकृतम्)
४. रर्क्क for (रर्क)

३२. (घ) १. स्थूलं (ग) माहास्थूलं for (महास्थुलम्)
२. स्थूलाऽवनतिश्च (ग) for (स्युलावनतिश्च)
३. वनती (ग) चनती for (वनति)
(च) १. महास्थूलं for (महास्थुलम्)
२. स्थूलावनतिश्च for (स्थुलावनतिश्च)
४. दशज्यया for (दृशज्यया)
३. लंबनावनती for (लंबनावनति)

३३. (घ) १. चतुष्क (ग) (च) for (चुतुष्टु)
२. स्थूलः (ग) (च) for (स्थुलः)
३. मानार्थं (ग) (च) for (मानार्थ)
४. महत्कृत (ग) (च) for (महकृत्)

विक्षेपगुणाक्षज्या लंबकभक्ता गृहे मृणधनं यत् ।
उक्तमुदयास्तसमयोन्नं प्रतिघटिकं ततस्तदसत् ॥ ३४ ॥
त्रिज्या कृत्या भक्ता विक्षेपापक्रमगुणो क्रमज्यैंदोः ।
अयनांतेमृणधनं न तदयनादौ ततौ सा तत् ॥ ३५ ॥
दक्कर्ण विज्ञानात् कालाविज्ञानमकथितत्वाच्च ।
कालज्ञानाच्छंकोरज्ञानं कोटिनाशोतः ॥ ३६ ॥
शशिशंकोः शच्यपराकोटिभुजवर्गयुतिमूलम् ।
तिर्यक्कर्णो न भवति यतो चन्द्रांतरं कर्णः ॥ ३७ ॥

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३४. (घ) १. गुण for (गुणा)
२. गृहे (ग) ग्रहेष्वृण for (ग्रहे)
३. मययो (ग) for (समयो)
(ग) ४. य्या for (ज्या)
५. नंत्रतिघटिकं for (न्नंप्रतिघटिकं)
(च) १. गुणाक्ष्यज्या for (गुणाक्षज्या) २. गृहे for (ग्रहे)

३५. (घ) १. यद्दण (ग) (च) for (मृण)
२. ततो (ग) for (ततौ)
३. सत्तत् (च) for (सातत्)
(ग) ४. विक्षेपायनगुणोत् for (विक्षेपापक्रमगुणो)
५. तत्तस्यादौ for (नतदयनादौ)
(च) ६. क्रत्या for (कृत्या) ७. गुणोत्क्रम for (गुणोक्रम)

३६. (घ) १. दक्कर्मा (ग) (च) for (दक्कर्ण)
२. विज्ञातात् for (विज्ञानात्)

३७. (घ) १. प्राच्य (ग) (च) for (शच्य)
२. कोटिकोठि (ग) कोटिः कोटि for (कोटि)
३. र्यंतोर्कचंद्रांतरं (ग) for (यतोचंद्रांतरं)
(ग) ४. वर्ग्र for (वर्ग)
(च) ३. र्यंतो for (यतो)

कोटिश्रवणज्ञानात् शशिना शृङ्गोन्नतिर्विसंवदति ।
उदयास्तमयो दिनकृतः प्रतिघटिकं माती च वाज्ञानात् ॥ ३८ ॥
अर्केंद्वंतरघटिका व्यस्तज्या चन्द्रमानगुणिता यत् ।
व्यास विभक्ता शुद्धं यतो न द्रक् सममसत्तस्मात् ॥ ३९ ॥
प्राक् गुदितोभ्यधिकः पश्चादुदिनकोयरेव्यस्तः ।
कालो यः छायार्थं तदसत्स्फुटभुक्तिमान् प्राक् ॥ ४० ॥
उदितामुदितास्तमितावशेषकालान्न वेत्ति यः स कथम् ।
आर्यभटज्ञः शशिनः छाया शृङ्गोन्नति वेंति ॥ ४१ ॥

३८. (घ) १. ज्ञाना छशिन for (ज्ञानात् शशिना) (ग) छशिनः for (शशिना)
२. प्रतिघटिकमतीव वा ज्ञानात् (ग) प्रतिघटिकमतीव चाज्ञानात् for (प्रतिघटिकं माती च वा ज्ञानात्)
(ग) ३. श्रवणा for (श्रवण)
४. विसंवदतिः for (विसंवदति)
५. दिनकृतः for (दिनकृतः)
(च) १. ज्ञाना छशिना for (ज्ञानात् शशिना)
२. प्रतिघटिकमती च for (प्रतिघटिकं माती च)

३९. (घ) १. अर्क्केंद्वंतर (ग) (च) for (अर्केंद्वतर)
२. द्दक्सम (ग) (च) for (द्रक्सम)
(ग) ३. गुणिना for (गुणिता)
४. शुक्लं (च) for (शुद्धं)

४०. (घ) १. प्राक्च गुदितोऽभ्यधिकः (ग) प्राक् प्रागुदितोभ्यधिकः for (प्राक् गुदितोभ्यधिकः)
२. दुदितो न कोपरे व्यस्तः (ग) (च) for (दुदिनकोयरे व्यस्तः)
३. कालो छायार्थं (ग) कालौ य स्थार्यर्थं for (कालो यः छायार्थं)
४. भुक्तिमानात् प्राक् (ग) भुक्तिगमना प्राक् for (भुक्तिमान् प्राक्)
(च) १. प्राक्यगुदितोऽभ्यधिकः for (प्राक् गुदितोभ्यधिकः)

४१. (घ) १. नुदिता (ग) (च) for (मुदिता)
२. कालान for (कालान्न)
३. शशिन (ग) शशिनस्छाया for (शशिनः छाया)
४. शृंगोन्नती (ग) (च) for (शृङ्गोन्नति)
५. वेत्ति (ग) (च) for (वेंति)
(ग) ३. शशिन for (शशिनः)

स्वयमेव नामय कृतमार्यभटेन स्फुटं स्वगणितस्य ।
सिद्धांतदस्फुटत्वं गृहणार्दानां विसंवदति ॥ ४२ ॥
जानात्येकमपि यतो नार्यभटोगणितकालगोलानाम् ।
न मया प्रोक्तोनि ततः पृथक् पृथक् दुषणात्येषाम् ॥ ४३ ॥
आर्यभटदुषणानां संख्या वक्तुं न शक्यते यस्मात् ।
तस्मादयमुद्देश्यो बुद्धिमतान्यानि योज्यानि ॥ ४४ ॥
कालांतरेण दोषा येन्यैः प्रोक्ता न ते मयाभिहिताः ।
किं ते दूष्येष्वथ दूषकेषु कोशोऽत्र पेयः स्थात् ॥ ४५ ॥
श्रीषेणविष्णुचन्द्रप्रद्युमनार्यभटलाटसिंहानाम् ।
ग्रहणादिविसंवादात् प्रतिदिवसं सिद्धमकृत्वम् ॥ ४६ ॥

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४२. (घ) १. यत्कृत (ग) (च) for (यकृत)
२. सिद्धं (ग) (च) for (सिद्धांत)
३. ग्रहणादीनां (च) for (ग्रहणार्दानां)
४. विसंवादात् (च) for (विसंवदति)

४३. (घ) १. जातात्येकमपि (च) for (जानात्येकमपि)
२. प्रोक्तानि (ग) for (प्रोक्तोनि)
३. पृथग्दूषणान्येषाम् (ग) पृथग्दूषणान्येषाम् for (पृथक्दुषणात्येषाम्)
(ग) ४. तार्य for (नार्य)
५. काकाल for (काल)
(च) ३. पृथग्दूषणान्येषां for (पृथक्दुषणात्येषाम्)

४४. (घ) १. दूषणानां (ग) पूषणानां for (दुषणानां)
२. मुद्देशो (ग) (च) for (मुद्देश्यो)
(च) १. दूषणानां for (दुषणानां)

४५. (घ) १. दूष्येद्यथ (ग) दूष्येप्यथ for (दूष्येष्वथ)
२. स्यात् (च) for (स्थात्)
(ग) ३. कालानांतरेण for (कालांतरेण)
४. दूबकेषु for (दूषकेषु)
(च) १. दूष्येष्वथ for (दृष्येष्वथ)

४६. (ग) १. सिद्धमत्कृत्वम् for (सिद्धमकृत्वम्)
(च) १. मकृत्वं for (मकृत्वम्)

युक्तधार्यभटोक्तानि प्रत्येकं दुषणानि[१] योज्यानि ।
श्रीषेण[२] प्रभृतीनां कानिचिदन्यानि वक्ष्यामि ॥ ४७ ॥
लाटात्[३]सूर्यशशांकौ मध्यार्विदुच्च[१] चंद्रपातौ च ।
कुजबुधशीघ्रबृहस्पतिसितशीघ्रशनैश्चरान्मध्या[२] ॥ ४८ ॥
युगयातवर्षभगणान् वाशिष्ठान्[४] विजयनंदिक्रतपादान्[२][५] ।
मंदोच्चपरिधिपात[३]स्पष्टीकरणाद्यमार्यभटात् ॥ ४९ ॥
श्रीषेण[१] ग्रहित्वा[४] चन्द्ररत्नोच्चयरोमकः कृतः[५] ।
कथा[२] एतान्येव गृहीत्वा[५] वसिष्ठो[३] विष्णुचन्द्रेण ॥ ५० ॥
अनयोर्न कदाचिदपि ग्रहणादिषु भवति दृष्टिगणितैक्यम् ।
यद् भवति तद्युगुणाक्षरमतो[१] स्फुटाभ्यां किमेताभ्याम् ॥ ५१ ॥

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४७. (घ) १. दूषणानि (ग) (च) for (दुषणानि)
२. कानिचदन्यानि for (कानिचिदन्यानि)

४८. (घ) १. बिंदूच्च (ग) (च) for (विन्दुच्च)
२. मध्यात् (ग) for (मध्या)
(च) ३. लाटा for (लाटात्)

४९. (घ) १. यान for (यात) २. कृत (ग) कृतम् for (क्रत)
३. मंदोच्चपरिपात (ग) मंदोच्चपरिधिपातः for (मंदोच्चपरिधिपात)
(ग) ४. वाशिष्टाद्विजयनंदि for (वाशिष्टान् विजयनंदि)
५. पादम् for (पादान्)
(च) १. युगपात for (युगयात)

५०. (घ) १. श्रीषोएन गृहीत्वा (ग) श्रीषेणगृहीत्वा for (श्रीषेण ग्रहित्वा)
२. कंथा (पहली पंक्ति के अन्त में) for (कथा)
३. वासिष्ठो (ग) (च) for (वसिष्टो)
(ग) ४. 'चन्द्र' यहां अंकित नहीं ५. ग्रहीत्वा for (गृहीत्वा)
(च) १. श्रीषेणेन गृहीत्वा for (श्रीषेण ग्रहित्वा)
५. क्रतः for (कृतः)

५१. (घ) १. तद्घुणाक्षर (ग) तत् द्युगणाक्षर for (तद्युगुणाक्षर)
(च) १. तद्युगणाक्षर for (तद्युगुणाक्षर)

नाचोच्चवृत्तमध्यस्य गोलबाहुचेन नामकृतमुच्चम् ।
तस्थो न भवत्युच्चो यतस्ततो वेत्ति नोच्चमपि ॥ ५२ ॥
अन्या विक्षेपकला मंदान्यत्वात्फलाधिकाः स्पष्टाः ।
यस्मान्माहायुगादौ न राहुमंदाः स्फुटात्तस्मात् ॥ ५३ ॥
परमालाब्दमिथुनांते द्युरात्रि नाडचोर्कगतिवशाद्दृतवः ।
नायनयुगमयनवशात् स्थिरमयी तद्वितयमपि तस्मात् ॥ ५४ ॥
तद्युगवधो महायुगमुक्तं श्रीषेणविष्णुचन्द्राद्यैः ।
तत्स्थुलं दृग्लिप्ता माहायुगादौ ग्रहेषु यतः ॥ ५५ ॥
कदिनादौ स्मृतियुक्तं ग्रहभोत्पतिर्दिनक्षये प्रलयः ।
तान्यपि बहुनि यस्मान्महायुगेऽतो प्रसिद्धमिदम् ॥ ५६ ॥

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५२. (घ) १. नीचोच्च (ग) नीचोच्चावृत्त for (नाचोच्चवृत्त)
२. तास्था for (तस्थो)
(च) १. नीचोच्च for (नाचोच्च)

५३. (घ) १. फलाधिका (च) for (फलाधिकाः)
२. महा (ग) (च) for (माहा)
३. स्फुटाः त्तस्मात् (ग) स्फुटास्तस्मात् for (स्फुटात्तस्मात्)
(च) ३. स्फुटाः त्तस्मात् for (स्फुटात्तस्मात्)

५४. (घ) १. परमाल्पा (ग) परिमाल्पा for (परमालाब्द)
२. स्थिरमय (ग) for (स्थिरमयी)
(ग) ३. न द्वितयमपि for (तद्वितयमपि)
(च) १. परमाल्पाब्द for (परमालाब्द) २. स्थिरमपि for (स्थिरमयी)

५५. (घ) १. स्थूलं (ग) for (स्थुलं) २. महा for (माहा)
(ग) ३. मयाहायुग for (महायुग)
(च) ३. तत्स्थूलं for (तत्स्थुलं) २. महायुगादौ for (माहायुगादौ)

५६. (घ) १. स्मृतिषुक्तं (ग) स्मृतिषूक्तं for (स्मृतियुक्तं)
२. बहूनि (ग) for (बहुनि)
(ग) ३. पर्त्तिदिनेक्षये for (पतिर्दिनक्षये) ४. युगेतो for (युगेऽतो)
(च) १. स्मृतिषुक्तं for (स्मृतियुक्तं)
३. ग्रहभोत्पत्तिर्दिनक्षये for (ग्रहभोत्पतिर्दिनक्षये)

प्रतिदिवस विसंवादात्[१] गृहतिथिकरणर्क्ष दिवसमानानाम् ।
ग्रहणग्रहयोगादिषु पादं पादेन कः स्पृशति ॥ ५७ ॥
अंगचिति[१]विजयनंदिप्रद्युम्नादीनि पादकरणानि ।
यस्मात्तस्मात्तेषां दूषणान्यत्र[२] लिखितानि ॥ ५८ ॥
इत्ति बहुधा विवदंति गृहर्थिनः[१] सद्गृहा[२] इव प्रसभम् ।
ब्राह्मे स्फुटसिद्धान्ते रवींदु भूयोगमज्ञात्वा[३] ॥ ५९ ॥
तंत्रभ्रंशं प्रतिदिनमेवं विज्ञाय[१] धिमता[२] यत्नः[३] ।
कार्यस्तस्मै[४] दृग्गणितैक्यं[५] सदा भवति ॥ ६० ॥
चन्द्ररविग्रहणेंदु छायादिषु सर्वदा यतो ब्राह्मे ।
दृग्गणितैक्यंभवति स्फुटसिद्धान्तस्ततो ब्राह्मः ॥ ६१ ॥
योयज्जनात्तेषां[१] स वेत्ति तदुषणानि[२] कथितानि ।
आर्यभटाद्युक्तानां तन्त्राणां दुषणाध्याये[३] ॥ ६२ ॥

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५७. (घ) १. विसंवादादग्रह for (विसंवादात्ग्रह) (च) विसंवादादग्रह for (विसंवादात्ग्रह)

५८. (घ) १. अंकचिति (ग) अंकयिति for (अंगचिति)
(ग) २. 'न' यहां अतिरिक्त है ।
(च) १. अंकचिति for (अंगचिति) २. +न+

५९. (घ) १. ग्रहार्चिनः (ग) ग्रहार्थिनः for (ग्रहर्थिनः)
(ग) २. सग्रहा for (सद्ग्रहा) ३. मत्कृत्वा for (मज्ञात्वा)

६०. (घ) १. विज्ञाया (च) for (विज्ञाय) २. धीमता (ग) (च) for (धिमता)
३. यत्ना (ग) यत्तः for (यत्नः) ४. तस्मिं (ग) तस्मिन्यस्मिन् for (तस्मै)
५. दृग्गणितै किं for (दृग्गणितैक्यं)
(च) ४. कार्यस्तस्मिं for (कार्यस्तस्मै)

६२. (घ) १. यं जानात्तेषां (ग) यं जानात्येषां for (यज्जनात्तेषां)
२. दूषणानि (ग) for (दुषणानि)
३. दूषणाध्याये ग) (च) for (दुषणाध्याये)
(च) १. यो यजनात्तेषां for (यो यज्जनातेषां)
२. तदूषणानि for (तदुषणानि)

इत्ति[४] कथिततन्त्राणा[१]काम्प[२]ठतिरपि दूषणैः करोत्यज्ञानात्[५] ।
तन्त्र परीक्षार्यणां[३] त्रिषष्टिरेकादशोध्यायः ॥ ६३ ॥
इति[६] श्री ब्रह्मगुप्तसिद्धान्ते एकादशोऽध्यायः ।

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६३. (घ) १, २. तन्त्रगणकान्यदितै (ग) तन्त्रगणकान् पठितैरपि for (तन्त्राणाकान्पठतिरपि)

३. परीक्षार्याणाम् (ग) for (परीक्षार्यणां)

(ग) ४. इति for (इत्ति) ५. करोत्यज्ञान् for (करोत्यज्ञानात्)
समाप्तिसूचक चिह्न छः छः छः

(च) १. तन्त्रगणकान्यदितैरपि for (तन्त्राणाकान्यठतिरपि)

५. करोत्यज्ञान् for (करोत्यज्ञानात्)

२. परिक्षार्याणां for (परीक्षार्यणां) ६. 'इति' से 'अध्यायः' तक सब लुप्त ।

अथ गणिताध्यायः

द्वादशः

परिकर्म विंशति यः संक(व)लिताद्यं पृथक् विजानाति ।
अष्टौ च व्यवहारान् छायातान् भवति गणकः सः ॥ १ ॥
विपरीत छेदगुणाः राश्योः छेदांशकाः समछेदाः ।
संकलितेशा योज्या व्यवकलितेंशांतरं कार्यम् ॥ २ ॥
रूपाणि छेदगुणान्यंशयुतानि द्वयोर्बहुनां वा ।
प्रत्युत्पन्नो भवति छेदवधेनोधृतांशवधः ॥ ३ ॥
प्रत्युत्पन्नः परिवर्त्य भागहारछेदांशौ छेदसंगुणछेदः ।
अंशांशगुणो भाज्यस्य भागहारः सवर्णितयोः मागहारः ॥ ४ ॥

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१. (घ) १. विंशति (ग) for (विंशति) २. संकलिताद्यां (ग) (च) for (संकलिताद्यं)
३. पृथग्विजानाति (ग) (च) for (पृथक् विजानाति)
४. व्यवहां (ग) व्यवहारांस् for (व्यवहारान्)
५. छायांतान् (ग) (च) for (छायातान्)
(ग) ६. परिकर्म्म for (परिकर्म)

२. (घ) १. संकलितेंऽशा (ग) (च) संकलितेंशा for (संकलितेशा)
(ग) २. गुणा for (गुणाः) ३. राश्योश्छेदांशकाः for (राश्योः छेदांशकाः)

३. (घ) १. बहूनां (ग) for (बंहुनां)
२. वधेनोद्धृतांशवधः (ग) बधेनोद्धृतोंशवधः for (वधेनोधृतांशवधः)
(च) १. बंहूनां for (बंहुनां) ३. प्रत्युत्पत्पन्नो for (प्रत्युत्पन्नो)
२. वधेनोद्धृतांशवधः for (वधेनोधृतांशवधः)

४. (घ) १. शंगुण for (संगुण) २. भाद्यस्य (च) for (भाज्यस्य)
३. सवणिवयोः (ग) सवर्णितयोः for (सवर्णितयोः)
वि० यहाँ श्लोक समाप्त ।
४. भागहारः (च) for (मागहारः)
(ग) ५. परिवृत्य (इसी पद से श्लोकारंभ होता है)
६. भाहारः for (भागहारः)
वि०-इस श्लोक में "प्रत्युत्पन्नः"..... और "भागहारः" पद अंकित नहीं है ।
(च) ७. छेदंशौ for (छेदांशौ)

सर्वाणतांशवर्गछेदकृति विभाजितो भवति [२]वर्गः ।
सर्वाणतांशमूलं छेदपदेनो[१]धृते मूलम् ॥ ५ ॥
[१]वर्गमूलो स्थाप्योंत्पद्यनो[३]न्यकृतिस्त्रिगुणोत्तरसंगुणा [८]च यत्प्रथमा ।
नोत्तर[५]कृतिं [६]रंत्यगुणा त्रि[६]गुणा चोत्तर घनश्च घनः ॥ ६ ॥ घनः [७]समाप्तः
छेदो[४]घना द्वितीयात् घनमूल[१]कृतिस्त्रिसंगुणा [२]सकृतिः ।
शोध्या त्रिपूर्वगुणिता प्रथमात् घनतो [५]घनो मूलं ॥ ७ ॥ घनमूलं [३]समाप्तः
[६]घनमूलं समाप्तः

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५. (घ) १. पदेनोद्धृतं (ग) (च) for (पदेनोधृते)
(ग) २. वर्ग्रः for (वर्गः)

६. (घ) १. वर्ग्रं ('वर्ग्रं मूलो' यह शब्द पांचवें श्लोक का अंतिम पद है)
२. स्थाप्योंत्यघनो (ग) स्थाप्योंत्यघनोत्यकृति for (स्थाप्योंत्पद्यनो)
३. न्यत्कृति for (न्यकृति) ४. तत् (ग) तत्प्रथमान् for (यत्प्रथमा)
५. उत्तरकृति (ग) उत्तरकृतिरंत्यगुणा for (उत्तरकृतिं)
६. त्रिणा (ग) त्रिगुणाच्चोत्तर for (त्रिगुणा)
७. समाप्तः for (घनः समाप्तः)
(ग) १. यह पद यहाँ लुप्त है। ८. संगुणो for (संगुणा)
(च) २. स्थाप्योंत्यघनोन्यकृति for (स्थाप्योंत्पद्यनोन्यकृति)
५. उत्तरकृतिं for (नोत्तरकृतिं) ६. रत्यगुणा for (रंत्यगुणा)
७. समाप्तः for (घनः समाप्तः)

७. (घ) १. द्वितीयाद्धनमूल (ग) दृद्वितीयाद्धनमूल for (द्वितीयात् घनमूल)
२. संगुणाप्तकृतिः (ग) for (संगुणासकृतिः)
३. समाप्तम् for (समाप्तः)
(ग) ४. छेदाघना for (छेदोघना)
५. प्रथमाद्धनोमूलं for (प्रथमात् घनतो घनमूलं)
वि०-'घनमूलं·····समाप्तः" शब्द अंकित नहीं है।
(च) १. द्वितियाद्यघनमूलकृति for (द्वितीयात् घनमूलकृति)
२. संगुणाप्तकृतिः for (संगुणा सकृतिः)
५. प्रथमाद्धनतो for (प्रथमात् घनतो)
३. घनमूलं समाप्तं for (घनमूलं समाप्तः)
६. 'घन' से 'समाप्तः' तक लुप्त

सदृशछेदांशयुतिछेदविभक्ताफलं प्रथमजातौ ।
अंशैरेशागुणिता छेदै छेदा द्वितीयाभ्याम् ॥८॥ प्रथमद्वितीयजाति समाप्तः
ऊर्द्धांशा छेदगुणास्तृतीयजातौ द्वयोः पृथक् परयोः ।
छेदै छेदा गुणिताः स्वांशयुतोनैरूपरिमांशाः ॥ ९ ॥
तृतीय चतुर्थीजाति स० ।
त्रैराशिके प्रमाणफलमिच्छाद्यं तयोः सदृशराशिः ।
इच्छाफलेन गुणिता प्रमाणभक्ताफलंभवति ॥ १० ॥ त्रैराशिकं स०
व्यस्तत्रैराशिकफलामिच्छाभक्तः प्रमाणफलघातः । व्यस्तत्रैराशिकं स०
त्रैराशिकादिषु फलं विषमेष्वेकादशांतेष ॥ ११ ॥

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८. (घ) १. रंशा (ग) for (रेशा)
(ग) २. सद्रिश for (सदृश) ३. युतिश्छेद for (युतिछेद)
४. छेदैश्छेदा for (छेदैछेदा)
५. "प्रथम·····समाप्तः" शब्द अंकित नहीं है ।
(च) १. अंशैरंशा for (अंशैरेशा)

९. (घ) १. रुपरिमांशाः (च) for (रूपरिमांशाः) २. समाप्तेः for (स०)
(ग) ३. ऊर्ध्वांशाश्छेद for (उर्द्धांशाछेद) ४. छेदैश्छेदा for (छेदैछेदा)
५. युरेता for (युतो)
वि० "तृतीय···से जाति स०" तक अंकित नहीं है ।
(च) २. जाती समाप्ते for (जाति स०)

१०. (घ) १. राशी (ग) (च) for (राशिः) २. प्रमाणक्ता for (प्रमाणभक्ता)
३. समाप्तम् for (स०)
(ग) "त्रैराशिकं समाप्तं" शब्द अंकित नहीं है ।
(च) समाप्तं for (स०)

११. (घ) १. फलमिच्छा (ग) for (फलामिच्छाभक्तः)
२. दश्यंतेषु for (दशांतेषु) ३. समाप्तम् for (स०)
(ग) ३. 'व्यस्त' से 'स०' तक यहां अंकित नहीं है ।
(च) २. दृश्यंतेषु for (दशांतेषु)

फलं[१] संक्रमणमुभयतो[२] बहुराशिवधे तुलवधह्रतो[३] ज्ञेयः ।
सकलेष्वेवं भिन्नेषु[४] भयतच्छेदसंक्रमणं ॥ १२ ॥

पंचसप्तनवैकादशराशि[५] समातानि ।
प्राग् मूल्यव्यत्यासो भांड[१]प्रतिभांडिकेत्यदुक्तः[२] ।
समपरिकर्मा[३]ण्यष्टानां[४] व्यवहाराणामभिहितानि ॥ १३ ॥
कालगुणितं प्रमाणं फलभक्तं व्येकगुणं[१]हतं कालः ।
स्वफलयुतरूपभक्तं मूलफलैक्यं भवति मूलम् ॥ १४ ॥
कालप्रमाणघातः परकालह्रतो द्विधाद्यमिश्रवधात् ।
अन्यार्द्धकृ[२]तियुतात्पद[३]मन्यार्द्धोन प्रमाणफलम् ॥ १५ ॥

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१२. (घ) १. फल for (फलं) २. मुयतो for (मुभयतो)
३. वधेऽल्पवध (ग) वधाऽल्पवधह्रतो (कृतो) for (वधेतुलवधह्रतो)
४. भन्नेषू (ग) भिन्नेषूभयतश्छेद for (भिन्नेषुभयतच्छेद)
५. राशिकानि for (राशि) (च)

(ग) ५. 'पंच' से 'समाप्तानि' तक पद यहां अंकित नहीं है ।

(च) १. फल संक्रमण for (फलं संक्रमण)
३. तुल्यवधह्रतो for (तुलवधह्रतो)

१३. (घ) १. भ्यंडप्रतिभ्यंडिके for (भांडप्रतिभांडिके)
२. इत्युदुक्त (ग) न्यदुक्तसमम् for (त्यदुक्तः)
३. सम (ज्ञ) for (सम)
४. परिकर्माण्यिष्टानां (ग) परिकर्माण्यिष्टनां for (परिकर्मान्यिष्टानां)

(ग) ४. परिकर्माण्यिष्टान्यां for (परिकर्मान्यिष्टानां)

१४. (ज्ञ) १. हृतं (च) for (हतं)

१५. (ग) १. मन्यार्द्धोनम् for (मन्यार्द्धोन)

(च) २. कति for (कृति) ३. पदमन्याः द्धोन for (पदमन्यार्द्धोन)

प्रश्नेपयोगहृतया लब्धा प्रश्नेपका गुणा लाभाः ।
ऊनाधिकोत्तरार्द्धं स्तद्युतोनया स्वफलमूनं युतम् ॥ १६ ॥

मिश्रक व्यवहार समाप्तः

पदमेकहीनमुत्तरगुणितं संयुक्तमादिनांत्यधनम् ।
आदि युतान्त्यधनार्द्धमध्यधनं पदगुणं गुणितम् ॥ १७ ॥
उत्तरहीनाद्विगुणादिशेशवर्गं धनोत्तराष्टघ्ने ।
प्रक्षिप्य पदं शेषोनं द्विगुणोत्तर हूतं गच्छः ॥ १८ ॥
एकोत्तरमेकाद्यं यदीष्टगच्छस्य भवति संकलितम् ।
तद् वियुत् गच्छ गुणितं तृ हूतं संकलितम् ॥ १९ ॥

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१६. (घ) १. प्रक्षेपयोगहृतया (ग) (च) for (प्रश्नेपयोगहृतया)
२. लब्ध्वा for (लब्धा) ३. प्रक्षेपका (ग) प्रक्षेपक for (प्रश्नेपका)
४. त्तरास्तद्युतो (ग) (च) for (त्तरार्द्धस्तद्युतो)
५. मूनयुतम् (ग) (च) for (मूनंयुतम्)
(ग) 'मिश्रक' से 'समाप्तः' तक अंकित नहीं है ।
(च) २. प्रक्षेपका for (प्रश्नेपका)
६. व्यवहारः for (व्यवहार)

१७. (ग) १. पदगुणं for (पदगणं) २. गणितम् for (गुणितम्)

१८. (घ) १. उत्तरहीन (ग) (च) for (उत्तरहीना)
२. शेषवग्रं (ग) (च) शेषवग्रं for (शेशवर्गं)
३. प्रक्षिप्य (ग) (च) for (प्रक्षिप्य)
४. हृतं (ग) यह पद अंकित नहीं है । for (हूतं)
(ग) ५. द्विगुणोत्तरं for (द्विगुणोत्तर)
(च) ४. हृतं for (हूतं)

१९. (घ) १. यदिष्ट (ग) (च) for (यदीष्ट) २. गच्छस्य (ग) (च) for (गच्छस्य)
३. हृतं (ग) त्रिहृतं for (तृहूतं)
(ग) ४. एकोत्तमेकाद्यं for (एकोत्तरमेकाद्यं)
५. गच्छ for (गच्छ) ६. संकलितं सकलितम् for (संकलितम्)
(च) ५. वियुतगच्छ for (वियुत्गच्छ) ३. त्रिहृतं for (तृहूतं)

द्विगुणपदसैकगुणितं तत्[1]हृतं भवति वर्गसंकलितम्[5] ।
[2]घूनसंकलितं [3]तक्रतिरेषां समगोलकैश्चितयः ॥ २० ॥
[4]श्रेष्टीव्यवहार समाप्तः ॥

स्थूलफलं त्रिचतुर्भुजबाहुप्रतिबाहुयोगदलघातः ।
भुजयोगार्द्धचतुष्टवभुजो न घातात्पदं [1]सुक्ष्मम् ॥ २१ ॥
भुजकृत्यंतरभूं [1]हूं तहीनयुताभूर्द्विभाजिता [3]बाधे ।
स्वाबाधावर्गोना[2]द्भुजवर्गान्फलमवलंब ॥ २२ ॥
[4]अविषमचतुस्त्रभुजगतिभुजवधयोर्[1]युते पदं कर्णः ।
[3]कर्णकृतिर्भूमुखयुतिदल व(प)र्गो नापदं लंब ॥ २३ ॥

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२०. (घ) १. तत्रिहृतं (ग) तत् त्रिहृतं for (तत्रृहतं)
२. घन (ग) घनसंकलित for (घूनसंकलितं)
३. तत्कृति (ग) for (तक्रति) ४. श्रेढी व्यवहारः for (श्रेष्टी व्यवहार)
(ग) ५. संलितम् for (संकलितम्)
४. वि० 'श्रेष्टी' से 'समाप्तः' तक यहाँ अंकित नहीं है ।
(च) १. तत्रृहृतं for (तत्रृहतं) २. घनसंकलितं for (घूनसंकलितं)
३. तत्कृति for (तक्रति) ४. श्रेढी for (श्रेष्टी)

२१. (घ) १. सूक्ष्मम् (ग) for (सुक्ष्मम्)
(च) १. शूक्ष्मं for (सुक्ष्मम्)

२२. (घ) १. भूर्हृत (ग) भूहृत्हीन for (भूर्हृत)
२. मूलमवलंबः (ग) मूलमवलंब for (फलमवलंब)
(ग) ३. भूद्विभाजिता बाधे for (भूर्द्विभाजितां बाधे)
(च) १. भूर्हृत for (भूर्हृत) २. मूलमवलंबः for (फलमवलंब)

२३. (घ) १. भुजगतिभुजगति (ग) भुजप्रतिभुजवधयो for (भुजगतिभुजवधयो)
२. भुजवधयोर्युतेः (ग) र्युतेः for (भुजवधयोर्युते)
३. लंबः (ग) (च) for (लंब)
(ग) ४. चतुरस्त्र for (चतुस्त्र)
(च) २. र्युतेः for (र्युते)

कर्णकृतेः कोटिकृतं विशोध्य मूलं भुजो भुजस्य कृतिं ।
प्रोह्य यदं कोटिबाहु कृतियुतिपदं कर्णः ॥ २४ ॥
कर्णयुतादुर्द्धाधरखण्डे कर्णावलंबयोगे ।
वास्वाबाधे स्वयुतिहृते द्विधा प्रथक्कर्णलंबगुणे ॥ २५ ॥
अविषमपार्श्वभुजगुणकर्णो द्विगुणोवलंबकविभक्तः ।
हृदयं विषमस्यभुजः प्रतिकृतिभुजकृतियोगमूलार्द्धं ॥ २६ ॥
त्रिभुजस्य बधोभुजयोर्द्विगुणितलंबोद्धृतो हृदयरजः ।
नृचतुर्भुजहृदय द्विगुणं वहिर्वृत विष्कुंभः ॥ २७ ॥

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२४. (घ) १. क्रिंति (ग) कृंति for (कृतं)
२. प्रोह्य पदं कोटिः (ग) for (प्रोह्ययंदं कोटि)
(ग) ३. विशोध्यं for (विशोध्य)
(च) १. क्रांति for (कृतं)

२५. (घ) १. वा (इस पंक्ति का अंतिम शब्द 'वा' है) (ग)
२. (दूसरी पंक्ति 'स्वा—' से आरम्भ होती है) (ग)
३. हिते (ग) हते द्विधा पृथक् for (हृतेद्विधा प्रथक्)
(ग) ४. मूर्ध्वा for (दुर्द्धा)
(च) ४. दूर्द्धाधर for (दुर्द्धाधर) ३. हते for (हृते)

२६. (घ) १. गुणः (ग) (च) for (गुण)
२. (यहां 'कृति' शब्द मूल पाठ में नहीं है) (ग)
(ग) ३. द्विगुणावलंबक for (द्विगुणोवलंबक)
४. विषमभुजः for (विषमस्यभुजः)
(च) २. 'कृति' लुप्त

२७. (घ) १. हृदयं (ग) (च) for (हृदय)
२. वृत्तं (ग) वृत्त for (वृत)
३. विष्कंभः for (विष्कुंभः)
(ग) ४. त्रुजयो for (भुजयो) ५. लंबोहृतो for (लंबोद्धृतो)
६. त्रि for (नृ) ७. द्विगुणितं for (द्विगुणं)
(च) २. वृत्तं for (वृत्त)

कर्णास्त्रितभुजघातैकामुभयथान्योन्यभाजितं गुणयेत् ।
योगेन भुजप्रतिभुजवधयोः कर्णोपदे विषमे ॥ २८ ॥
विषमचतुरस्त्रमध्ये विषमत्रिभुजद्वयं प्रकल्प पृथक् ।
कर्णद्वयेन पूर्ववदाबाधे लंबकौ च पृथक् ॥ २९ ॥
विषमभुजांतस्त्रिभुजे प्रकल्प्य कर्णौ भुजो तदाबधे ।
पृथगूर्द्धा धरखण्डे कर्णयुतौ कर्णयोरधरे ॥ ३० ॥
त्रिभुजभुजौभूर्भुमिस्तलल्लंबो उर्द्धमवलंब खण्डम् ।
लंबकयोगार्द्धं मधरोनम् ॥ ३१ ॥

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२८. (घ) १. कर्णासृत (ग) कर्णाश्रित for (कर्णास्त्रित)
२. घातैक्य (ग) घातैः क्य for (घातैका)
(ग) ३. 'ज' अंकित नहीं है ४. कर्णौ for (कर्णो)
(च) १. कर्णासृत for (कर्णास्त्रित) २. घातैक्यमुभयथा for (घातैकामुभयथा)

२९. (घ) १. चतुरस्त्रमध्ये (च) for (चतुरस्त्रमध्ये)
(ग) २. विखम for (विषम)

३०. (घ) १. वाधे (ग) तदा माबाधे for (तदावधे)
(ग) २. ये for (जे)
३. कर्णो भुजौ । (यहां पहली पंक्ति समाप्त—दूसरी पंक्ति 'तदा' से आरम्भ)
४. चरषंडे for (धरखण्डे)
(च) २. स्त्रिभुजे for (स्त्रिभुजे)
३. कर्णौभुजौ for (कर्णौभुजो) ४. पृथगूर्धाधरखण्डे for (पृथगूर्द्धाधरखण्डे)

३१. (घ) वि०—यहां इस श्लोक का उत्तरार्ध लिखना लिपि कार भूल गया प्रतीत होता है, '३१' का संख्याक्रम भी नहीं मिलता ।
१. भूमि for (भुमि) २. लंबयोरधरे (ग) लंबकाधरं for (उर्द्धमवलंब)
(ग) १. भूमि for (भुमि) ३. दूसरी पंक्ति – ऊर्ध्वमवलंबखंडं लंबकयोगार्द्धंमधरोनम्
(च) १. भूंमिस् for (भुंमिस्)
वि०—यह श्लोक यहां ३१ व ३२ मिला जुला इस प्रकार है—
त्रिभुलभुजौ भूर्भूमिस्तल्लंबोऊर्द्ध मवलम्बखंडं लंबकयोगार्द्धंमधरोनं ॥ ३१ ॥
वि० रेखांकित पद से आगे का समस्तभाग ऊपर अन्यहस्त से लिखा हुआ है ।
(ङ) ४. त्रिभुजे for (त्रिभुज) १. तुभूमि for (भूर्भूमि)
२. लम्बकाधरं for (उर्द्धमवलंब) ३. + ऊर्ध्वमवलम्बखण्डम् +

कर्णावलकयुतौ खंडे कर्णाविलंबयोरधरे ।
अनुपातेन तुद्दने ऊर्द्ध शून्यांस पाटायाम् ।। ३२ ।।
कृतियुरसदृशराश्योर्बाहुधातो द्विसंगुणो लंबः ।
क्रत्यन्तरमसद्रशयोर्द्विगुणं द्विसमत्रिभुजम्दमिः ।। ३३ ।।
इष्टद्वयेन भक्तो द्विधेष्टवर्गफलेष्टयोगार्द्धे ।
विषमत्रिभुजस्य भुजा विष्टोनफलार्द्ध योगा भूः ।। ३४ ।।
इष्टस्य भुजस्य क्रतिर्भक्तो नेष्टेन तद्वलं कोटिः ।
आयनचतुरस्त्रस्य क्षेत्रस्येष्टाधिका कर्णः ।। ३५ ।।

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३२. (घ) (वि०—पहली पंक्ति लिपिकार लिखना भूल गया प्रतीत होता है)
१. सून्यांस (ग) तव्यांस for (शून्यांस)
(ग) २. कर्णाविलंबकयुतौ (च) for (कर्णावलकयुतौ)
३. अमुपातेन for (अनुपातेन) ४. तदूने (ङ) for (तुद्दने)
५. ऊर्द्ध वेस् for (उर्द्ध) ६. पारायाम् for (पाटायाम्)
(च) ७. यहां क्रमसंख्या ३१ ही अंकित है ।
(ङ) २. कर्णाविलंबकयुतौ for (कर्णावलकयुतौ)
५. ऊर्ध्वे for (उर्द्ध) १. सून्यांस for (शून्यांस)

३३. (घ) १. कृतियुतिर (ग) (च) for (कृतियुर)
२. कृत्यंतर (ग) (ङ) for (क्रत्यंतर)
३. मसदृशयो (ग) (च) (ङ) for (मसद्रशयो)
४. त्रिभुज भूमिः for (त्रिभुजम्दमि)
(ग) ५. घातो (च) (ङ) for (घातो)
६. द्विद्विसमत्रिभुजभूमिः for (द्विसमत्रिभुजम्दमिः)
(च) ४. त्रिभुजभूमिः (ङ) for (त्रिभुजम्दमिः)
(ङ) १. कृतियुतिर for (कृतियुर)

३४. (घ) १. ष्टद्वयेन for (इष्टद्वयेन)
(ग) २. वर्ग्रः for (वर्ग) ३. विषत्रिभुजस्य for (विषमत्रिभुजस्य)
(ङ) २. वर्गः for (वर्ग) ४. योगोभूः for (योगाभूः)

३५. (घ) १. कृति (ग) (च) (ङ) for (क्रति)
२. तद्लं (ग) (ङ) for (तद्वलं) ३. आयत (च) (ङ) for (आयन)
४. चतुरस्रस्य (ग) (च) (ङ) for (चतुरस्त्रस्य)

आयतकर्णौ बाहु भुजकृतिरिष्टेन भाजितेष्टोना ।
द्विहृता कोद्यधिकाभूर्मूखमूलानाद्विसमचतुरस्रे ॥ ३६ ॥
कर्णकृतिस्त्रिसमभूजास्त्रयश्चतुर्थो विशोध्य कोटिकृति ।
बाहुकृतेस्त्रिगुणार्या यद्यधिकोभूर्मुखं हीनः ॥ ३७ ॥
जात्यद्वयकोटिभुजाः परकर्णगुणाः भुजाश्चतुर्विषमे ।
अधिकोभूर्मुखमूनो बाहु द्वितयं भुजा अन्यौ ॥ ३८ ॥
इष्टगुणकारगुणितो गिर्युच्छ्रायः पुरांतरमनष्टम् ।
द्वियतं गुणाकरभाजितमुत्पातोन्यस्य समगत्योः ॥ ३९ ॥
व्यासव्यासार्द्धकृतिपरिधिफले व्यासहारिके त्रिगुणे ।
तद्वर्गाभ्यां दशभिः संगुणिताभ्यां पदे सुक्ष्मे ॥ ४० ॥

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३६. (घ) १. बाहू (ङ) for (बाहु) २. हृता (ङ) for (हृता)
३. मुंखभूना (ग) for (मूंखमूलाना)
४. चतुर (यहां 'चतुर' के पश्चात् 'स्रे' और संख्या क्रम लिखना लिपिकार भूल गया है) ।
(ग) ५. आयनकर्णौ for (आयतकर्णौ)
(च) ६. क्रति for (कृति) ३. भूमुंखमूलाना for (भूमूंखमूलाना)
(ङ) ७. कोट्यधिका for (कोद्यधिका) ३. भूमुंखमूना for (भूमूंखमूलाना)
३७. (घ) १. कृतिम् (ग) कृति for (क्रति) २. भुजा (ग) (ङ) for (भूजा)
३. कृतिम् (ग) (ङ) (क्रति) ४. कृते (ङ) for (क्रते)
५. गुणायायद्यधिको (ग) (ङ) for (गुणार्यायद्यधिको)
(च) ३. कृतिं for (क्रति)
(ङ) १. कृतिस् for (क्रतिस्)
३८. (घ) १. कर्णगुणा (ग) हता for (कर्णगुणाः) २. वन्यो for (वन्यौ)
(च) १. गुणा for (गुणाः)
(ङ) १. कर्णगुणा for (कर्णगुणाः)
३९. (घ) १. द्वियुत (ग) (ङ) for (द्वियुतं)
(ग) २. गुणकार (च) (ङ) for (गुणाकर) ३. समगत्यौ for (समगत्योः)
४०. (घ) १. कृती (ग) कृती (ङ) for (कृति)
२. व्यावहारिके (ग) (च) (ङ) for (व्यासहारिके)
३. तद्वर्गाभ्यां (ग) तद्वर्ग्राभ्यां (च) तद्वर्गाभ्यां for (तद्वर्गाभ्यां)
४. सूक्ष्मे (ग) (च) (ङ) for (सुक्ष्मे)

वृत्ते शरोनगुणितात् व्यासाच्चतुराहतात्पदं जीवा ।
ज्यावर्गश्चतुराहतशरभक्तः शरयुतो व्यासः ।। ४१ ।।
ज्याव्यासकृतिविशेषान्मूलव्यासांतरार्द्धमिषुरल्पः ।
व्यासो ग्रासोनगुणो ग्रासो नैक्योद्धतौ बाणौ ।। ४२ ।।
इष्टशरस्य भक्ते ज्यार्द्धे कृन्तिशरफले युतौ व्यासौ ।
शरयोः फलयोरैक्यं ग्रासौ ग्रासद्यदुनं तत् ।। ४३ ।।

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४१. (घ) १. शरोत for (शरोन)
२. व्यासाच्चाँतु (ग) द्वचासा for (व्यासाच्चतु)
३. ज्यावर्गस्य for (ज्यावर्गश्च) ४. तुराहह for (तुराहत)

(ग) ५. गुणिता for (गुणितात्)

(च) ५. गुणिताद्वचासाच्च (ङ) for (गुणितात् व्यासाच्च)

४२. (घ) १. व्यासौ (ग) (ङ) for (व्यासो)
२. ग्रासोनैक्याधृतौ (ग) ग्रासोतैक्योद्ध ृतौ for (ग्रासोनैक्योद्धतौ)

(ग) ३. विशेषा मूलं for (विशेषान्मूल)

(च) २. ग्रासोनैक्योधृतौ for (ग्रासोनैक्योद्धतौ)

४३. (घ) १. शरद्वयभक्ते (ग) (च) (ङ) for (शरस्यभक्ते)
२. कृतीशर (ग) (ङ) for (कृंतिशर) ३. युते (ग) (च) for (युतौ)
४. ग्रासाद्यदूनं (ग) ह्यदनंतत् for (ग्रासद्यदुनं)
५. +क्षेत्र व्यवहारः समाप्तः+

(ग) ६. पले for (फले) ७. व्यासो for (व्यासौ)
८. पलयो for (फलयो) ९. ग्रासो (ङ) for (ग्रासौ)

(च) २. कृती for (कृंति) ४. ग्रामाद्यदूनं for (ग्रासद्यदुनं)

(ङ) ६. युते for (फले) ३. फले for (युतौ)
४. ग्रासोनमैक्यं for (ग्रासद्यदुनं)

क्षेत्रव्यवक्षेत्रफलं वेधगुणं समखातफलहृतं त्रिभिः शून्या ।
मुखतुल्यतुल्यभुजैक्यानेकाग्रहृतानि समरस्त्रः ॥ ४४ ॥
मुखतुलयुतिः दलगुणितं वेधगुणं व्यासहारिकं गणितम् ।
मुखतलगुणं व्यार्द्धं वेधेन गणितमौच्चम् ॥ ४५ ॥
उद्रगणिताद्विशोध्य व्यवहारफलंत्रिभिर्भजेच्छेशम् ।
लब्धं व्यवहारफले प्रक्षिप्य फलं भवति सूक्ष्मं ॥ ४६ ॥

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४४. (घ) १. +अथ खातः+ २. ('क्षेत्र व्यव'—यहाँ लुप्त है)२. (ग)+क्षेत्रव्यवहारः+
३. हृतं (ग) for (हतं) ४. शून्याः (ग) सूध्याः for (शून्या)
५. मुषतुल (ग) मुखतुल (च) for (मुखतुल्य)
६. न्येका (ग) न्यैकाग्र for (नेकाग्र)
७. समरज्जुः (ग) (ङ) for (समरस्त्रः)
(ग) ८. फलं (च) (ङ) for (फल)
(च) २. क्षेत्रे व्यवहारः समाप्तः, अथ खातः for (क्षेत्रव्यव)
४. शून्याः for (शून्या) ६. न्येकाग्रहृतानि for (नेकाग्रहृतानि)
७. समरस्रः for (समरस्त्रः)
(ङ) २. 'क्षेत्रव्यव' लुप्त ४. सूच्याः for (शून्या)
५. मुखतलतुल्य for (मुखतुल्यतुल्य)
४५. (घ) १. तल (ग) मुखतलयुति (च) for (मुखतुलयुतिः)
२. गणितं (ङ) for (गुणितं)
३. व्यावहारिकं (ग) (च) (ङं) for (व्यासहारिकं)
४. गणितैक्यार्द्धं (ग) मुखवगतैक्यार्द्धं for (गुणैक्यार्द्धं)
५. मौद्रम् (ड्रम्) (ग) मौत्र for (मौच्चम्)
(ग) ६. गुणम् for (गणित)
(च) १. मुखतलयुति for (मुखतुलयुतिः)
४. गणितैक्यार्द्धं (ङ) for (गुणैक्यार्द्धं) ५. मौध्रम् for (मौच्चम्)
(ङ) १. मुखतलयुति for (मुखतुलयुतिः)
६. +स्याद्+५ मौत्रम् for (मौच्चम्)
४६. (घ) १. छेषम् (ग) च्छेषम् for (छेशम्)
(ग) २. औद्र for (उद्र)
(च) २. तुद्र for (उद्र) १. छेषं for (छेशम्)
(ङ) २. औत्र for (उद्र) ३. भजेत् त्रिभिः शेषम् for (त्रिभिर्भजेच्छेशम्)
४. भवति फलं सूक्ष्मम् for (फलंभवतिसूक्ष्मम्)

खात[१] व्यवहारश्चतुर्थः

आकृतिफलमौव्याहतमग्रतलैक्यार्द्धमौच्यदैर्ध्यगुणम्[२][३][४] ।
घनगुणितमिष्टकाघनफलेन हूतमिष्टकागणितम्[५][६] ॥ ४७ ॥

चितिव्यवहारः समाप्तः ॥

विस्तारायांमांगुलघातो मार्गाहतो द्विवेदहृतः[७][२][३] ।
किष्कं गुलग्निलब्धं तत्पणवतिर्भवति कर्मः[४][५][६] ॥ ४८ ॥

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४७. (घ) १. ख्याव्यवहार (ग) श्री खातव्यवहारश्चतुर्थः for (खातव्यवहारश्चतुर्थः)
२. मौन्य (ग) मौव्य for (मौच्य)
३. दैघ्य for (दैर्ध्य) ४. गुणाः (च) for (गुणम्)
५. गणित (ग) (च) (ङ) for (गुणित) ६. हृत (ग) (च) (ङ) for (हूत)

(ग) ७. चितव्यवहारः for (चितिव्यवहारः)

(च) १. ख्याव्यवहारश्चतुर्थः for (खातव्यवहारश्चतुर्थः)

(ङ) १. "खात—चतुर्थः" लुप्त ७. मौच्च्याहृत for (मौव्याहत)
२. मौच्च्य for (मौच्य) ७. इति चितिव्यवहारः for (चितिव्यवहारः समाप्तः)

४८. (घ) १. विस्तराया for (विस्तारायां)
२. मागहितो (ग) माग्रहितो for (मार्गाहतो)
३. द्विवेहृतः (ग) द्विवेदहृतः (च) (ङ) for (द्विवेदहृतः)
४. किष्कं (ग) (च) for (किष्कं)
५. गुलानि (ग) (च) (ङ) for (गुलग्नि) ६. कर्म्मः (ग) for (कर्मः)

(ग) ७. विस्तारायामंगुल for (विस्तारायांमांगुल)

(च) ७. विस्तारायां for (विस्तारायां) ६. कर्म्म (ङ) for (कर्मः)

(ङ) ७. विस्तारायामाङ्गुल for (विस्तारायांमांगुल)
८. षण्णवति for (पणवति)

शाकादिषु शाल्मल्यां[1] शतद्वयं जीवके[4] शतं विंशं शालः[5] ।
सरलादिषु[2] सर्वविहग्रूसु[3] चतुः षष्टिः ।। ४९ ।।
क्रवच[6] व्यवहार समाप्तः ।

नवम[1] सूकिषु[2] दशमस्थूलेष्वेकादशो भवत्यणुषु ।
परिधेर्वेधपरिधिः[4] षडंशवर्गो[5] हतो[3] गणितम् ।। ५० ।।
द्विचतुः सत्र्यंशगुणोमित्यंतर[1] बाह्यकोणगः ।
परिधिः[2] प्राग्वत्कृत्वा गणितं तद् गणितं स्वगुणकारहूतं[3] ।। ५१ ।।
राशिव्यवहारः[4] समाप्तः ।

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४९. (घ) १. शालाल्पांश for (शाल्मल्यांश)
२. सरलादिषुशतं सर्व (ग) for (सरलादिषुसर्व)
३. विदारूषु (ग) विवहारुषु for (विहग्रूसु)
(ग) ४. बीजकेन for (जीवके)
५. यहां 'विंशं' पर श्लोकार्ध समाप्त करके 'शाल' से दूसरी पंक्ति आरम्भ है।
(ङ) ६. क्रकचव्यवहारः (च) for (क्रवचव्यवहार)
(च) ५. वीशंशालः for (विंशंशालः) २. +शतं+
३. विदारुषु for (विहग्रुसु)
(ङ) ५. विंशं for (विंशंशालः) २. ३. शतमथाविदारुषु for (सर्वविहग्रुषु)
६. लुप्त ।

५०. (घ) १. नवमं शूकिषु (ग) नवमः सूकिषु for (नवमसूकिषु)
२. दशमः (ग) (ङ) for (दशम) ३. वर्गाहतो (ग) वर्ग्राहतो for (वर्गोहतो)
(ग) ४. परिधेर्वेधः (च) (ङ) for (परिधेर्वेधः)
(च) १. नवमंशूकिषु for (नवमसूकिषु) ३. वर्गाहतो (ङ) for (वर्गोहतो)
(ङ) १. नवमः शूकिषु for (नवमसूकिषु) ५. परिधेः for (परिधिः)

५१. (घ) १. भित्त्यंतर (ग) (च) for (मित्यंतर)
२. परिधिः (यह शब्द पहली पंक्ति का अन्तिम शब्द है) (ङ)
३. हृतम् (ग) (च) (ङ) for (हूतं)
(ङ) १. भित्त्यन्तर् for (मित्यंतर)
४. इति राशिव्यवहारः for (राशिव्यवहारः समाप्तः)

छायांतरसैकहूतं द्युदलं प्रागपरयो द्युगतशेषम् ।
दिनगतं शेषांशहूतं द्युदलं छायांतरव्येकम् ॥ ५२ ॥
दीपतलशंकुतलयोरंतरमिष्टप्रमाणशंकुगुणम् ।
दीपशिखोच्या शंकुं विशोध्य शेषोद्धृतं छाया ॥ ५३ ॥
छायाग्रांतरगुणिता छाया छायांतरेण भक्ता भूः ।
भूः शंकुगुणा छाया विभाजिता दीप शिखौच्यम् ॥ ५४ ॥
छाया व्यवहार समाप्तः
गुणकार खण्डतुल्यो गुण्यौ गोमूत्रिकाकृतो गुणितः ।
सहितः प्रत्युत्पन्नो गुणकारकभेदतुल्यो वा ॥ ५५ ॥

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५२. (घ) १. +अथछाया+ (आरंभिक प्रकरण का नाम)
२. छायानरसैकहूतं (ग) (ङ) for (छायांतरसैकहूतं)
३. द्युगतशेषम् (ङ) (ग) द्युंगशेषम् for (द्युगतशेषम्)
४. दिनगत (ग) (च) (ङ) for (दिनगतं)
५. हूतं (ग) (च) (ङ) for (हूतं)
६. छायानरव्येकम् (ग) छायामराव्येकम् for (छायांतरव्येकम्)
(च) १. +अथछाया+ ७. हूतं for (हूतं)
३. द्युगतशेषं for (द्युगतशेषम्)
(ङ) १. छायानरव्येकम् for (छायांतरव्येकम्)
५३. (घ) १. छंकु (ग) च्यं छं कुं for शंकुं २. दोषोद्धृतं for (शेषोद्धृतं)
(ग) ३. दीपशिखौ (च) for (दीपशिखो)
(च) १. व्याछंकुं for (च्याशंकुं)
(ङ) ३. शिखौच्याच्छंकुं for (शिखोच्याशंकुं)
५४. (घ) १. दीपक (च) for (दीप)
(ग) २. भक्ता (अतिरिक्त अंकित) ३. छायाव्यवहारः for (छायाव्यवहार)
(ङ) १. दीपशिखयौच्यम् for (दीपशिखौच्यम्)
५५ (घ) १. स्वंज for (खण्ड) २. ण्गुण्यो (ग)गणो for (गुण्यौ)
३. कृतो (ग) (ङ) for (कृतो)
४. प्रत्युत्पन्नौ (ग) प्रत्युत्पन्नो for (प्रत्युत्पन्नो)
(ग) ५. गणितः for (गुणितः)
(ङ) २. गुण्यो for (गुण्यौ)

गुण्यो राशिर्गुणकारराशिनेष्टधिकोनकेन गुणः ।
गुण्येष्टवधो न यतो गुणकेभ्यधिकोनके कार्यः ॥ ५६ ॥
छेदेनेष्ट युतोनेना संभाज्यादनष्टमिष्टगुणम् ।
प्रकृतिस्थछेदहूतं लब्धायुतहीनकमनष्टम् ॥ ५७ ॥
गुणछेदफलवधो गुणकहतो गुण्यभाजितो गुणकः ।
छेदो धृतः फलं गुण्यगुणवधः फलहूतः छेदः ॥ ५८ ॥
गुण्यगुणकारयोः छेदलब्धयो द्वयो द्वयोर्नाशः ।
तेषां दृश्यो व्यस्तौ कृत्वा तस्थानयो रिष्टौ ॥ ५९ ॥

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५६. (घ) १. राशिनेष्टा (ग) (च) (ङ) for (राशिनेष्ट)
२. युतो (ग) (च) (ङ) for (यतो)
(ग) ३. गुण्यष्ट for (गुण्येष्ट) ४. गुणिकेभ्य for (गुणकेभ्य)
(च) ३. गुणेष्ट for (गुण्येष्ट)
(ङ) ४. गुणकेऽभ्यधिको for (गुणकेभ्यधिको)

५७. (घ) १. छेदो नष्टयुतो for (छेदेनेष्टयुतो)
२. नेनाप्तं (ग) (च) (ङ) for (नेनासं)
३. हृतं (ग) (च) (ङ) for (हूतं)
(ग) ४. दिनष्ट for (दनष्ट) ५. लब्ध्वायुतकम् for (लब्ध्वायुतं)
(ङ) ५. लब्ध्या for (लब्ध्वा)

५८. (घ) १. गुण्य (ग) (ङ) for (गुण)
२. फलहृतः (ग) फलहृत (च) for (फलहूतः)
(ग) ३. हृतो for (हतो)
(ङ) ३. गुणकहृतो for (गुणकहतो) ४. द्धतः for (धृतः)
२. फलहृतश्छेदः for (फलहूतः छेदः)

५९ (घ) १. गुणागुण for (गुण्यगुण)
१. द्वियोर्द्वयोर्नाशः (ग) वा द्वयोर्द्वयोर्नाशः for (द्वयोद्वयोर्नाशः)
(ग) ३. दृश्यो (च) for (दश्यो) ४. व्यस्तो for (व्यस्तौ)
(च) १. विसर्ग लुप्त २. द्वयोर्द्वयोर्नाशः for (द्वयोद्वयोर्नाशः)
५. तस्थानयौ for (तस्थानयौ)
(ङ) ६. कारयोश्छेद for (कारयोः छेद) ७. लब्धयोर्यदि द्वयो for (लब्धयोद्वयो)
२. द्वयोर्द्वयो for (द्वयो द्वयो) ३. दृश्यौ for (दश्यो)
८. श्चेष्टौ for (रिष्टौ)

गुण्यं गुणकारं वा गुणये छेदेन भागहरस्य ।
गुण्यगुणकारराश्यो छेदगुणो भागहरश्च ।। ६० ।।
छेदस्य छेदरूपं कृत्वा न्यदूक्तवत्सर्वम् ।
अपवर्त्यौ तुल्यो न छदगुणौ छेदगुण्यौ वा ।। ६१ ।।
स्वविकलषष्टचंशगुणं सकलस्त्र्यंशोद्धतो विकलवर्गः ।
प्रक्षेप्यः सकलकृतौ वर्गघातौ द्वित्रितुल्यवधौ ।। ६२ ।।
राशेखनं द्विगुणं बहुतरगुणमूनकृतियुतं वर्गः ।
राशेरिष्टयुतोनाद्वधः कृतिवेष्टकृतियुक्तः ।। ६३ ।।

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६०. (घ) १. भागहारस्य (ग) (च) (ङ) for (भागहरस्य)
२. भागहारश्च (ङ) for (भागहरश्च)
(ग) ३. गुणयो for (गुणये)
(च) २. भागहारश्च for (भागहरश्च)
(ङ) ३. गुणयेच्छेदेन for (गुणयेछेदेन) ४. राश्योश्छेद for (राश्योछेद)

६१. (घ) १. अच्छेदस्य छेदं (ग) for (छेदस्य छेदरूपं)
२. अपवत्यौ for (अपवर्त्यौ) ३. तुल्येन (ग) for (तुल्योन)
४. छेदगुणौ (ग) for (छदगुणौ)
(ग) ५. दुक्तवत्सर्वम् (ङ) for (दूक्तवत्सर्वम्) ६. छेदगुण्यो वा for (छेदगुण्यौ वा)
(च) ५. न्यदुक्तवत् for (न्यदूक्तवत्) २. अपवर्त्यौ for (अपवर्त्यौ)
४. छेदगुणौ for (छदगुणौ)
(ङ) १. अच्छेदस्य for (छेदस्य)
३, ४, ६. छेदगुणौ तुल्येनेष्टेन for (तुल्यो न छदगुणौ छेद)

६२. (घ) १. सकलस्त्रिंशोद्धतो (ग) सकलस्त्रिंशोहृतो for (सकलस्त्र्यंशोद्धतो)
२. प्रक्षेप्यः (ग) (च) (ङ) for (प्रश्नेप्यः)
३. वर्गघनौ (ग) वर्ग्घनौ for (वर्गघातौ)
(ग) ४. वश्यंशगुणः for (षष्टचंशगुणं) ५. वर्ग्गः for (वर्गः)
(च) १. सकलस्त्रिशो हृतो for (सकलस्त्र्यंशोद्धतो)
(ङ) ६. गुणः for (गुणं) १. सकलास्त्रिंशोद्धतो for (सकलस्त्र्यंशोद्धतो)
३. वर्गघनौ for (वर्गघातौ)

६३. (घ) १. रूनं (ग) (च) (ङ) for (खनं) २. वेंष्ट (ग) (च) for (वेष्ट)
(ग) ३. कृवि for (कृति) ४. बुतं वर्ग्गः for (युतं वर्गः)
५. र्युक्तः for (युक्तः)
(च) ३. क्रति for (कृति) (ङ) २. कृतिर्वेष्ट for (कृतिवेष्ट)

इष्टाल्पराशिवर्गौ युक्तोना वितरविकलवर्गाभ्याम् ।
द्विगुणोत्तरराशिभ्याम् भक्तौ तेनाधिकाभ्याम् ॥ ६४ ॥
स्थानांतरेषु लब्धं येन समं फलयुतोनकच्छेदः ।
दलितकृतियोगान्तरपदमितरो वा फलयुतो नः ॥ ६५ ॥
दिग्मात्रमेतदन्या ज्योत्पत्तौ कुट्टके च कथयिष्ये ।
संकलितादिष्टार्याः षट्षष्टिर्द्वादशोऽध्यायः ॥ ६६ ॥

इति ब्रह्मगुप्ते द्वादशोऽध्यायः ।

इति श्री ब्राह्मस्फुट सिद्धान्ते द्वादशोध्यायः ।

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६४. (घ) १. द्विगुणोतर (ङ) for (द्विगुणोत्तर)
२. तेनाधिकोनाभ्याम् (ग) (च) (ङ) for (तेनाधिकाभ्याम्)
(ग) ३. भक्तो for (भक्तौ)
(च) १. द्विगुणोतर for (द्विगुणोत्तर)

६५. (घ) १. दलितः (ग) (च) (ङ) for (दलित)
(ग) २. 'समं' यहां अंकित नहीं है ।
(ङ) ३. युतोनकच्छेदः for (युतोनकछेदः)

६६. (घ) १. संकलितादिष्वार्याः (ग) for (संकलितादिष्टार्याः)
२. +६६+
३. यह पंक्ति इस प्रति में लुप्त है । (ग)
(ग) ४. दिग्मात्रामेतदत्य for (दिग्मात्रमेतदन्या)
(च) ३. 'इति' से 'अध्याय' तक समाप्तिसूचक वाक्य लुप्त ।
(ङ) ४. दिङ्मात्रमेतदन्यज् for (दिग्मात्रमेतदन्या)
१. दिष्वार्या for (दिष्टार्याः)

अथ प्रश्नाध्यायः

त्रयोदशः

तत्र तावन्मध्यगत्युत्तराध्यायः

प्रश्नध्यायान् प्रवक्षामि सोत्तरान् गणकबुद्धिवृद्धिकरान् ।
यैर्ज्ञातैस्तन्त्रविदामाचार्यो भवति बुद्धिमताम् ॥ १ ॥
अधिमासकैः सविकलैर्दृष्टैर्युगयातमवमरात्रैर्वा ।
द्युगणेन वायुगगतयो वेत्ति स कालतंत्रज्ञः ॥ २ ॥
अवमाति यः सविकलैरधिमासैरधिकमासकानवमैः ।
ग्रहमिष्टं ताभ्यां वायो वेति स कलतंत्रज्ञ ॥ ३ ॥
द्युगुणं विनाधिमासावमैर्विनादिन गणेन चन्द्रार्कौ ।
ताभ्यां विनास्फुटतिथि यो वेत्ति स कालतंत्रज्ञः ॥ ४ ॥
इष्टान्मध्यादन्यांस्तिथिमिष्टान्मध्यमाच्छशिग्रहणम् ।
इष्टाद्रविंदुपातै र्यो वेत्ति बिना स तंत्रज्ञः ॥ ५ ॥

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१. (घ) १. प्रश्नाध्यायान् (ग) प्रस्नाध्यायान for (प्रश्नध्यायान्)
२. बुद्धिमत्यम् (ग)बुधिमताम् for (बुद्धिमताम्)
(ग) ३. वक्ष्यामि for (प्रवक्षामि) ४. 'वृद्धि' पद अंकित नहीं है ।
(च) १. प्रश्नाध्यायान् for (प्रश्नध्यायान्)

२. (घ) १. वायुगगं (ग) वायुगगतं for (वायुगगत)
(ग) २. अधिमासिकैः for (अधिमासकैः) ३. मवमरात्रैर्वा for (मवमररात्रैर्वा)
४. तत्रज्ञः for (तंत्रज्ञः)
(च) ३. मवमरात्रैर्वा for (मवमररात्रैर्वा)

३. (घ) १. वेत्ति (च) for (वेति)
२. कालतंत्रज्ञः (ग) सकलतंत्रज्ञः for (सकलतंत्रज्ञ)
(ग) ३. अवमानि for (अवमाति) ४. 'यः' पद यहाँ अंकित नहीं है ।
(च) २. सकालतंत्रज्ञः for (सकलतंत्रज्ञ)

४. (घ) १. द्युगणं (ग) (च) for (द्युगुणं)
(ग) २. 'विना' अंकित नहीं ३. 'सकाल' अंकित नहीं

५. (घ) १. रवींदु (ग) for (रविंदु) २. यो for (यों)
(ग) ३. दन्यास्तिथि for (दन्यांस्तिथि)
४. 'बिनास' पद अंकित नहीं । इसके स्थान में यहाँ 'सकल' अंकित है ।
सकल for (बिनास)

त्रिगुणः शनिरिन्दुनोन्यभगणलब्धैर्ग्रहादिभिः सहितः ।
भौमोऽर्कोगुरुरिंदुच्चं वान्यभगणाः के ॥ ६ ॥
द्वित्रिगुणयो रविर्द्वोर्युति कुजोनान्यभगणलब्धेन ।
राश्यादिनाधिकगुरूना शशिरन्यभगणाः के ॥ ७ ॥
द्रष्टोदयिकानश्विन्यौ दयिकात्वाकरो तिथौ मध्यात् ।
मध्यार्कं गुणकं गुणमिष्टं मध्यं स तंत्रज्ञः ॥ ८ ॥
पातोनतुल्योमगतीन् प्रतिलोमगतीन् ।
ग्रहान् दिवसं वारं विपरीतैः शन्याद्यैर्यो वेत्ति स कालतंत्रज्ञः ॥ ९ ॥

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६. (घ) १. रिन्दुनोऽन्य (ग) रिन्दूनोन्य for (रिन्दुनोन्य)
२. गृंहादिभिः (ग) for (ग्रंहादिभिः)
३. भौमो हीनोर्का (ग) भौमो हीनोर्को for (भौमोऽर्को)
४. वाऽन्यभगणाः for (वान्यभगणाः)
(ग) ५. रिंदूच्चं for (रिंदुच्चं)
(च) ३. भौमोर्क्को for (भौमोऽर्को)

७. (घ) १. रवींद्वोर्युतिः (ग) रवीद्वोर्युतिः (च) for (रविर्द्वोर्युति)
२. नाधिका (ग) for (नाधिक) ३. शनि (ग) for (शशि)
(च) ३. शनि for (शशि)

८. (घ) १. इष्टौ (ग) for (द्रष्टो)
२. यो (ग) (च) for (ो)
(ग) ३. दयिकोनश्विन्यौ for (दयिकानश्विन्यौ)
४. दयिको न वा for (दयिकात्वा)
(च) १. इट्टोदयिका for (द्रट्टोदयिका) ५. मध्यार्क्कं for (मध्यार्कं)
६. गुंणमिष्टं for (गुणमिष्टं)

९. (घ) १. पातानतु लोमगतीन् (ग) पातानन्नु लोमगतीन् for (पातोनतुल्योमगतीन्)
२. दिवस (ग) for (दिवसं)
(च) १. पातोनतुलो for (पातोनतुल्यो)
३. वत्ति for (वेत्ति)

व्यतिपातवैधृतबृहस्पतिवर्षस्वोच्चनीचपरिवर्तनात् ।
द्विग्रहयोगाश्च युगे यो वेत्ति स कालतंत्रज्ञः ॥ १० ॥
सावनमासाब्दाधिप होरेशानिष्टमध्यसंयोगात् ।
इष्टगुणितैर्युतोना दिष्टान्यो वेत्ति गणकः सः ॥ ११ ॥
युगरविदिवसैर्गुणिता गताधिमासाः स्वशेषसंयुक्ता ।
भक्ता युगाधिमासैः फलं युगादेर्गता दिवसाः ॥ १२ ॥
गुणितानि चन्द्रदिवसैर्गतावमानि स्वशेषसहितानि ।
विभजेद्युगावमैः फलमनष्टमधिमासकैर्गुणितम् ॥ १३ ॥
हृतमिदुदिनैर्लब्धाधिमासदिवसैर्विहिनकमनष्टम् ।
युगयातदिनान्यधिमासदिनगणेष्टग्रहाद्यमत ॥ १४ ॥
द्युगुणेंदुदिवसघातात् कुदिनाप्तमधोयुगादिमासगणम् ।
युगशशिदिनभक्तं फलमासदिनोनं युगगतदिनानि ॥ १५ ॥

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१०. (घ) १. परिवत्तात् (ग) परिवर्तान् for (परिवर्तनात्)
२. योगांश्च (च) for (योगाश्च)
(ग) ३. व्यतीपात for (व्यतिपात) ४. सकलतंत्रज्ञः for (सकालतंत्रज्ञः)
(च) १. परिवर्त्तान् for (परिवर्तनात्)
११. (ग) १. गुणिति for (गुणितै)
१२. (घ) १. गुणिताधिमासाः for (गुणितागताधिमासाः)
२. संयुक्ताः (ग) (च) for (संयुक्ता)
(च) १. गुणिता for (गुणिता)
१३. (घ) १. मानिः for (मानि)
२. गुणिता (ग) गुणितम् for (गुणितम्)
(च) २. गुणितं for (गुणितम्)
१४. (घ) १. हृत (ग) (च) for (हृत) २. लब्धा for (लब्धा)
३. विहीन (च) for (विहिन) ४. दिनाऽन्यधि for (दिनान्यधि)
(ग) ५. मिदू for (मिदु) ६. ग्रहाद्यमतः for (ग्रहाद्यमत)
१५. (घ) १. गणेंदु (ग) (च) for (गुणेन्दु)
(ग) २. कुदिनाप्तं for (कुदिनाप्त) ३. मधोर्यु for (मधोयु)
४. गुणम् for (गणम्)
(च) ४. युगाधिमासगणं for (युगादिमासगणम्)

गुणमधिमासकशेषं युगकुदिनै खमशेषमधिमासैः ।
तद्द्युतिरिंदुदिनहूताधिमासशेषं स्फुटं भवति ॥ १६ ॥
कुदिनगताधिकमासकघातः स्पष्टाधिमासशेषयुतः
भक्तो युगाधिमासैरहर्गणः पूर्ववन्मध्याः ॥ १७ ॥
भूदिनगता इवमवधः शेषयुक्तो युगावमविभक्तः ।
लब्धं भवति द्युगुणो युगयातो मध्यमाः प्राग्वत् ॥ १८ ॥
युगगतराशिमासवधाद्रविमासाप्तं दिनाकृतिं सदिनम् ।
भूदिनगुणशशिदिनहूतपाप्तमहर्गणः मैकम् ॥ १९ ॥

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१६. (घ) १. दित for (दिन) २. हृताधि (च) for (हूताधिमास)
(ग) ३. शेषस्फुटं for (शेषं स्फुटं)

१७. (ग) १. 'क' अंकित नहीं
२. मासरहर्ग्रणः for (मासैरहाणिः)
३. पूर्ववत् मध्या for (पूर्ववन्मध्या)
(च) २. युगाधिमासैरहर्गणः for (युगाधिमासै रहाणिः)

१८. (ग) १. वधः स्वशेष for (वधः शेष)
२. द्युगणो (ग) (च) for (द्युगुणो)
(ग) ३. मध्यमा for (मध्यमाः)
(च) १. +स्व+

१९. (घ) १. युगगराराशि (ग) युगतराशि for (युगगतराशि)
२. दिनीकृतं (ग) (च) for (दिनाकृतं)
३. गुणं for (गुण)
४. शशिदिनैर्हृता (ग) शशिदिनहृत for (शशिदिनहूत)
५. माप्तमहर्गणः (ग) माप्तमहर्ग्रणः for (पाप्तमहर्गणः)
६. सैकम् (ग) (च) for (मैकम्)
(ग) ७. मासास्त for (मासाप्तं)
(च) ५. माप्तमहर्गणाः for (पाप्तमहर्गणः)
४. दिनहृत for (दिनहूत)

गतदिवसा[५] पृथक्[१] विमासक गुणितारविदिनाप्तमासदिनैः[७] सहिता[२]।
पृथग्वमगुणा[३] शशिदिवसाप्तोनकाकपणाः[४] ॥ २० ॥

गुणिताद्युगाधिमासैर्युगभू[२] दिवसैर्हृ[३]तादवमशेषात्।
फलयुक्तमधिकमासकशेषं मध्यावतोऽर्केंदू[१] ॥ २१ ॥
अधिमासोवमशेष[१]युगशशिभू[२]हृते पृथग्लब्धे
मासदिनाद्ये स्याप्ये गतमासदिनानि चैत्राद्यैः[४] ॥ २२ ॥
अवमावशेषलब्धा[२] सहितानि[३] पृथक् त्रयोदशगुणानि[५]।
अधिमास शेषलब्धा हीनानि[१] पृथग् रविशशांकौ[६] ॥ २३ ॥

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२०. (घ) १. पृथगविमासकगुणिता (ग) पृथगधिमासकगुणिता for (पृथक् विमासक-गुणिता)
२. सहिताः (ग) for (सहिता)
वि० यह दूसरी पंक्ति का आरंभिक शब्द है
३. पृथगवमगुणाः (ग) for (पृथग्वमगुणा)
४. द्युपणाः for (कपणाः)
(ग) ५. गतदिवसाः for (गतदिवसा)
६. दिनसाप्तोन·····॥ २० ॥ for (दिवसाप्तोन)
(च) १. पृथगविमासक for (पृथक् विमासक)
७. मासदिनैसहिताः for (मासदिनैः सहिता)
३. पृथग् वमगुणाः for (पृथग्वमगुणा) ४. काद्युषणाः for (काकपणाः)

२१. (घ) १. र्कोन्दुः for (र्कंदू)
(ग) २. (भू) अंकित नहीं २. हृ॑ता for (हृ॑ता)
४. फलयुक्तं मधिमासशेषं for (फलयुक्तमधिकमासकशेषं)
(च) ३. हृता for (हृ॑ता) १. मध्यावतोऽङ्कंदू for (मध्यावतोऽर्कंदू)

२२. (घ) १. शेषे (ग) for (शेष)
२. भूदिनहृते (ग) भूदिनहृता for (भूहृते)
(ग) ३. मधिमासावम for (अधिमासोवम)
४. चैत्रदै for (चैत्राद्यैः)
(च) २. +दि+

२३. (घ) १. हीनातिपृथग् for (हीनानि पृथग्)
(ग) २. लब्धाः for (लब्धा) ३. सहिता for (सहितानि)
४. तयोर्दश for (त्रयोदश) ५. गुणाश्चेनि for (गुणानि)
६. शशिकौ for (शशांकौ)

गतमासदिनावशेषलब्धियोगाः त्र्योदशगुणाच्च ॥
अधिमासशेषलब्धारविचन्द्रोच्चैं युते शोध्ये ॥ २४ ॥
केन्द्रे पृथक् फले द्वादशोद्धृते न्यस्तमृणधनं सौरम् ।
अनुलोममैदवं मासदिनावमशेषलब्धियुतौ ॥ २५ ॥
तिथिविकलषष्टिघाताद्रविहृतभुक्त्यंतराप्तघटिकासु ।
देशांतरमनुलोमं भुजांतरं चार्कं फलमसवः ॥ २६ ॥
अवमविकलं न सावनमेभिः परिकल्पितं यतश्चांद्रम् ।
नार्यभटाद्यैः प्रश्नो मध्यान्यत्वा कृतो ज्ञातः ॥ २७ ॥
ज्ञातभुगणादिभुक्तं सविकलमिष्ट युगभगणसंगुणितम् ।
ज्ञातयुगभगणभक्तं मध्योभगणादिफलमिष्टः ॥ २८ ॥

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२४. (घ) १. चन्द्रोच्चै for (चन्द्रोच्चै)
(ग) २. दिनावमशेष for (दिनावशेष)
३. योगा for (योगाः) ४. त्रयोदश for (त्र्योदश)
(च) ४. त्रयोदश for (त्र्योदश)

२५. (घ) १. मैंदवं (ग) मैवं for (मैंदवं)
२. लब्धयुतो for (लब्धियुतौ)
(ग) ३. केन्द्रं for (केन्द्रे) ४. प्रथक् for (पृथक्)
५. बस्तमृण for (न्यस्तमृण)

२६. (ग) १. 'ष्टि' for (षष्टि)
२. द्रविहित for (द्रविहृत) ३. भुक्तांतराप्त for (भुक्त्यंतराप्त)
४. घटीकासु for (घटिकासु) ५. मनुलोमौ for (मनुलोमं)
६. भुजांतर for (भुजांतरं) ७. वाऽर्क for (चार्कं)
(च) २. द्रविहृत for (द्रविहृत)
७. चार्क्कं for (चार्कं)

२७. (घ) १. त्कृतोऽज्ञातः for (कृतोज्ञातः)
(ग) २. यतश्चान्द्र for (यतश्चान्द्रम्) ३. प्रस्नो for (प्रश्नो)

२८. (घ) १. भगणादि (ग) (च) for (भुगणादि) २. मिष्टम् for (मिष्ट)
(वि०)—इस श्लोक की दोनों पंक्तियों के मध्य में निम्नांकित तीसरी पंक्ति भी दी गई है—
"ज्ञात युगभगणादिभुक्तं सविकलमिष्ट युग भगण संगुणितम्"
(ग) ३. संयुणितम् for (संगुणितम्)

इष्टाहतभक्तानां ध्व्यादिनां[१] संयुतेर्द्वयोरथवा ।
इष्टगुणकारगुणयोर्विभक्तयोर्वांतरादथवा[२] ॥ २९ ॥
ज्ञानैकभगण[१][४] भुक्तिस्तद्भगमाप्त[२][५] यदिष्टभगणेभ्यः ।
भगण[६][३] दिवसविकलं सविकलेष्टभगणादि गुणमिष्ट[७] ॥ ३० ॥
शशिदिनगुणं सविकलं वद्भुक्तं[२] मंडलादितद्भुगणैः ।
विभेज[१]फलं सविकलास्थितयः[३] प्राग्वत् स्फुटीकरणम् ॥ ३१ ॥
पातेंदुयोगलब्धौ कृत्वा[२] देशांतराद्यमनुलोमम् ।
विक्षेपो स्यात्[१] सूर्यादिभिर्विनेंदुग्रहणमेवम् ॥ ३२ ॥
कुदिनदुतमवशेषं[१][३] द्वादशभिर्गुणितं[४] मासमशद्यम्[२][५] ।
द्वादशगुणतिथ्यंशैर्युतं धनं भास्करे चंद्रः ॥ ३३ ॥

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२९. (घ) १. नांद्यादीनां (ग) नां द्वचादीनां for (ध्व्यादिनां)
(ग) २. वांतरदथवा for (वांतरादथवा)
(च) १. द्वचादीनां for (ध्व्यादिनां)

३०. (घ) १. ज्ञातैक (च) for (ज्ञानैक)
२. भाप्तं (च) for (माप्त)
३. भगणादिसविकलं for (भगणदिवसविकलं)
(ग) ४. भगणस्तद्भगणाप्त for (भगणभुक्ति) ५. यं for (य)
६. भगणादि दिसविकलं for (भगणदिवसविकलं)
७. गुणमिष्टम् for (गुणमिष्ट)
(च) ६. भगणा for (भगण)

३१. (घ) १. विभजेत्फलं (च) for (विभेजफलं)
(ग) २. यद्भुक्तं for (वद्भुक्तं)
३. स्तिथयः (च) for (स्थितयः)

३२. (घ) १. विक्षेपोस्मात् (ग)(च) for (विक्षेपोस्यात्)
(च) २. क्रत्वा for (कृत्वा)

३३. (घ) १. हूत for (दुत) २. माप्त (ग) (च) for (मास)
(ग) ३. मवमशेषं for (मवशेषं) ४. गुंणित for (गुंणितं)
५. मंशाद्यम् for (मशद्यम्) (च)
(च) १. हूत for (दुत)

द्युगुण[१] युगाधिमासैर्गुणितं युगभूर्दिनैर्भजेल्लब्धम्[२] ।
भगणादिमध्यमार्कस्त्र्योदश गुणाधिकश्चन्द्रः[३][४] ॥ ३४ ॥
इष्टगुणकारगुणितं ग्रहभगणैक्यांतर[१][३] यथा निहितम्[४] ।
कृत्वा कुदिनैर्विभजे[५] छेषो न युतानि[६] कुदिनानि ॥ ३५ ॥
अघनभगणलब्धं[१] घनर्णमिष्टग्रहस्य[२] युगभगणैः ।
अन्यफलमृणधनं चेद्ऊनयुतान्यन्ययुगभगणाः ॥ ३६ ॥
भदिनानि ग्रहभगणैरूनानि भवंति सावनदिनानि[२] ।
इष्टाश्विन्योदयिकाः[१] स्वसावनैः पूर्ववन्मध्याः ॥ ३७ ॥
रविभगणाप्तं[१] दिसविकलं ज्ञेयमंडलेभ्यो यत् ।
मध्यार्कसविकलकलासंगुणितं[३] ज्ञेयमध्यमकलाः[२] ॥ ३८ ॥

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३४. (घ) १. द्युगणं (ग) द्युगुणं for (द्युगुण) २. भीजल्लब्धम् for (भर्जेल्लब्धम्)
३. मध्यमार्कत्रयोदश (ग) for (मध्यमार्कस्त्र्योदश)
४. गुणाधि चेंदुः (ग) गुणाधिकं चन्द्रः for (गुणाधिकश्चन्द्रः)
(ग) ५. भूदि नै (च) for (भूर्दिनै)
(च) १. द्युगणं for (द्युगुण) ३. मध्यमार्क्कत्रयोदश for (मध्यमार्कस्त्र्योदश)
४. गुणाधिकश्चेंदुः for (गुणाधिकश्चन्द्रः)

३५. (घ) १. गुणित for (गुणितं)
२. छेषोऽनयुतानि for (छेषो न युतानि)
(ग) ३. क्यांतरं for (क्यांतर)
४. भिहितम् for (निहितम्)
५. विभजेत् शेषो for (विभजे छेषो)
(च) ६. क्रत्वा for (कृत्वा)

३६. (घ) १. यद्यन्य (ग) for (अघन) २. धनर्ण (ग) for (घनर्ण)
(च) १. अद्य for (अघ) २. धनर्ण्य for (धनर्ण)

३७. (घ) १. इष्टाश्विन्यौदयिकाः (ग) for (इष्टाश्विन्योदयिकाः)
(ग) २. सवनदिनानि for (सावनदिनानि)

३८. (घ) १. रविभगणाप्तलिप्तादि for (रविभगणाप्तम्)
(ग) २. मध्यकलाः for (मध्यमकलाः)
(च) ३. मध्यार्क्क for (मध्यार्क)

इष्टभगोन[१] भूदिनशेषैर्भगणैः क्रतो[२] मध्यः ।
अनुलोगमगो विलोमो भवति विलोमो[३]नुलोमगतिः ।। ३६ ।।
द्युगुणेन[२] कुदिनशेषैर[३]नुलोमगो विलोमगतिः ।
भवति विलोमो मध्योर[२]नुलोमगो वाकृतः प्राग्वत्[४] ।। ४० ।।
कल्पदिनसप्तकवधात् कल्पगताहर्गणो[२]नकच्छेषात् ।
सप्तहृ[१]ता दिनवारः शेषं[३] शन्यादिको भवति ।। ४१ ।।
व्यतिपा[३]त वैधृतान्यर्क[५] चंद्रभगण[४]युताद्विसंगुणिता[१] ।
गुरूवर्षाण्याश्रा युजा द्वादशगुणिताः[२] गुरोर्भगणाः ।। ४२ ।।
स्वोच्चग्रहयुगभगणा विशेषिता[२]स्वोच्चनीचपरिवर्ताः ।
भगणांतरं वियो[१]गा कार्यस्त्रैराशिकेन गताः ।। ४३ ।।

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३६. (घ) १. इष्टभगणोन (ग) (च) for (इष्टभगोन)
२. कृतोग्रहो मध्यः for (क्रतोमध्यः)
३. लोमोऽनुलोमगतिः (ग) तिलोमोनुलोमगतिः for (विलोमोनुलोमगतिः)
(च) २. क्रतो ग्रहोमध्या for (क्रतोमध्यः)

४०. (घ) १. द्युगणोन (ग) (च) for (द्युगुणेन)
२. मध्योऽनुलोमगो (ग) मध्योनुलोमगो for (मध्योरनुलोमगो)
(ग) ३. शेषैर्भगणैं for (शेषैरनुलोमगो) ४. प्रग्वत् for (प्राग्वत्)
(च) २. मध्योऽनुलोमगो for (मध्योरनुलोमगो)

४१. (घ) १. सप्तहृतादिनवारः (ग) सप्तहृतादिनवारः for (सप्तहृतादिनवारः)
(ग) २. हर्ग्रणो for (हर्गणो) ३. शेषः for (शेषं)
(च) १. सप्तहृतादिनवारः for (सप्तहृतादिनवारः)

४१. (घ) १. संगुणिताः (ग) (च) for (संगुणिता)
२. गुणिता गुरोर्भगणाः (ग) for (गुणिताः)
(ग) ३. व्यतीपात for (व्यतिपात) ४. भगणा for (भगण)
(च) ५. वैधृतान्यर्क्क for (वैधृतान्यर्क) २. गुणिता for (गुणिताः)

४३. (घ) १. द्वियोगाः (ग) (च) द्वियोगा for (वियोगा)
(ग) २. विशेषिताः for (विशेषिता)

द्युगुणार्स्त्रिशद् भक्ताद्यलब्धं द्विगुणितं सरूपं तत् ।
सप्तविभक्तं शेषः सावनमासाधिपोर्कादि ॥ ४४ ॥
षष्टिशतत्रयभक्तात् कल्पगताहर्गणात्फलं त्रिगुणम् ।
सैकं सप्तविभक्तं सायनवर्षाधिपोर्कादि ॥ ४५ ॥
अर्कोनलग्नहोरा पंच गुणाः सविकलाः यदि सरूपाः ।
सप्तविभक्ता शेषो दिनाधिपात् कालहोरेशः ॥ ४६ ॥
त्रिचतुरनंतर षष्टयः सावनमासाब्ददिनसहोरेशाः ।
दिनगतघटिका द्विगुणा पंच हृताश्चान्यमतमेतत् ॥ ४७ ॥
गच्छधनमिष्ट गुणितैधनैर्युतोनं पृथक् पृथक् सहितम् ।
गुणक युतोन पदहृत सर्वधनमतोवशेषाणि ॥ ४८ ॥

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४४. (घ) १. द्युगणार्स्त्रिशद् (ग) (च) for (द्युगुणार्स्त्रिशद्)
२. यल्लब्धं (ग) भक्तात् यल्लब्धं for (भक्ताद्यलब्धं)
३. र्कादिः for (कादि)
(ग) ४. सरूप for (सरूपं)
(च) २. भक्ताद्यल्लब्धं for (भक्ताद्यलब्धं) ३. र्क्कादिः for (कादि)

४५. (घ) १. कादिः for (कादि)
(ग) २. हर्ग्रणा for (हर्गणा) ३. सैक for (सैकं)
(च) १. र्क्कादिः for (कादि)

४६. (घ) १. सप्तविभक्ताः (ग) (च) for (सप्तविभक्ता)
(ग) २. होराः (च) for (होरा) ३. सविकला for (सविकलाः)
४. दिनपाद्यः काल for (दिनाधिपात् काल)
(च) ५. अर्क्कोन for (अर्कोन)

४७ (घ) १. षष्टाः (ग) (च) for (षष्टयः) २. दिवस(ग) for (दिनस)
३. द्विगुणाः (ग) (च) for (द्विगुणा)
(च) २. मासाब्ददिवसहोरे रेशाः for (मासाब्ददिनसहोरेशाः)

४८ (ग) १. गुणतै for (गुणितै) २. हृतं for (हृत)
(च) ३. शाषेणि for (शेषाणि)

मध्योत्तरमेकोनार्याः[2] पंचाशस्त्रयोदशोध्यायः[1]।
ज्ञात्वैनं तंत्रविदामाचार्यो भवति मध्यगतौ ॥ ४९ ॥

इति[1] त्र्योदशोऽध्यायः

इति ब्राह्मस्फुटसिद्धान्ते मध्यगतिप्रश्नोत्तराध्यायः
त्रयोदशः।

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४९ (व) १. पंचाशत्त्रयोदशोध्यायः (ग) पंचशत्त्रयोदशोध्यायः for (पंचाशस्त्रयोदशोध्यः)

(ग) २. नार्या for (नार्याः)

३. 'इति' से 'अध्यायः' तक अंकित नहीं, केवल समाप्तिसूचक ॥ छः ॥ ॥छः ॥ ॥ छः ॥ अंकित है।

(च) १. पंचाशत्त्रयोदशोध्यायः for (पंचाशस्त्रयोदशोध्यायः)

३. 'इति' से 'अध्यायः' तक लुप्त.

अथ स्फुटगत्युत्तराध्यायः

चतुर्दशः

भुजभागैः कोटिज्यां कोद्यं[१]शैर्यः करोति बाहुज्याम् ।
कोटिं[३] भुजेन बाहुं[४] क्वोस्त्रावा[२] स्फुटं[५]गतिज्ञः सः ॥ १ ॥
परमफलकेंद्रविकः[१] करोति कोटिछ्या[२] स्फुटं कर्णम्[४] ।
कर्णांतकोटिकोद्या[३] बाहुं वा स्फुटं[५]गतिज्ञः सः ॥ २. ॥
केंद्रभुजकोटिजीवा परमफलज्ञः करोति यः कर्णम्[१] ।
स्वोच्चं स्वफल[२]स्पष्टं करोति यः स्फुटगतिज्ञः सः ॥ ३ ॥
द्युगुण[१]स्फुट[३]ग्रह[२]यो भुजकोटिज्ये फले विना ज्याभिः ।
ज्याभिर्विना फलधनुः करोति वा स्फुटं[४]गतिज्ञः ॥ ४ ॥
इष्ट[१]श्विन्योदयिकान् करोति यो मध्यमान्[४] ग्रहान्[२] स्पष्टान् ।
स्वोच्चस्फुटं[३]ग्रहं यः करोति वा स्फुटगतिज्ञः सः ॥ ५ ॥

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१. (घ) १. कोटच शै (ग) (च) for (कोद्यंशै) २. कोद्यावा (ग) (च) for (क्वोस्त्रावा)
(ग) ३. कोटिभुजेन for (कोटिंभुजेन) ४. बाहु for (बाहुं)
(ङ) २. कोटचा वा for (क्वोस्त्रा वा) ५. स्फुटगतिज्ञः for (स्फुगतिज्ञः)
२. (घ) १. विद्युः (ग) वित् यः for (केन्द्रविकः) २. ज्यया (ग) (ङ) for (छ्या)
(ग) ३. कोटिम् for (कोटि)
(च) १. विद्युः for (विकः) ४. कर्ण्यं for (कर्णम्)
३. कर्ण्यांतकोटिकोद्या for (करणांतकोटिकोद्या) ५. ज्ञतिज्ञः for (गतिज्ञः)
(ङ) १. विद्यः for (विकः) ३. कोटिकोटचा for (कोटिकोद्या)
३. (च) १. कर्ण्यं for (कर्णं)
(ङ) २. सफलं for (स्वफलं)
४. (घ) १. द्युगणा (ग) सुगुणात् for (द्युगुण) २. ग्रहं (ग) (च) (ङ) for (ग्रह)
(ग) ३. स्फुटं (ङ) for (स्फुट) ४. +सः+ (च) (ङ)
(च) १. द्युगणा for (द्युगुण)
(ङ) १. द्युगणात् for (द्युगुण)
५. (घ) १. इष्टाश्विन्यौदयिकान् (ग) इष्टाश्विन्योदयिकान् for (इष्टश्विन्योदयिकान्)
२. मध्यमान् स्पष्टान् for (मध्यमान् ग्रहान् स्पष्टान्)
(घ) ३. स्पुगतिज्ञः for (स्फुटगतिज्ञः)
(च) १. इष्टाश्विन्यो for (इष्टश्विन्यो)
(ङ) ४. मध्यान् for (मध्यमान्)

सङ्क्रान्तेरा[२]द्यंतौ ग्रहस्य यो राशिभतिथिक[३]रणांता[१]न् ।
व्यतिपाताद्यंतौ वा यो वेत्ति स्फुटगतिज्ञः सः ।। ६ ।।
व्यासदलमितरजीवाभुजकोद्यं[३]शो क्रमज्यया[१] हीनम् ।
कोटिभुजज्या व्यासार्द्धकृतिविशेषात्पदं चान्या[२] ।। ७ ।।
कोटिज्यया द्विगुणया गुणा[१]त्यफलज्यता[२] युतोनायाः ।
मृगक[६]र्क्यादौ त्रिज्यां [३]तत्फलकृतियुते [४]पदं [५]कर्णः ।। ८ ।।
त्रिज्यां[१]त्यफल[३]त्वतियुतेः कर्णकृतेश्चांतरेवशेषं यत् ।
द्विगुणात्यफल[४]हृतं तत्कोटी[२]ज्या बाहु[६]जीवातः ।। ९ ।।

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६. (घ) १. करणान्तात् (ग) (च) for (करणांतान्)
(ग) १. राद्यंतो for (राद्यंतौ) ३. तिथिक for (तिथि)

७. (घ) १. ज्येयाहीनाम् (च) for (ज्यया हीनम्) २. वान्या for (चान्या)
(ग) ३. कोटद्यंशोत्क्रम (च) for (कोद्यंशोक्रम)
(ङ) ३. कोटद्यंशोत्क्रम for (कोद्यंशोक्रम)

८. (घ) १. गुणयांत्यफल (ग) (च) for (गुणांत्यफल)
२. ज्यया (ग) (च) for (ज्यता)
३. त्रिज्यांत्यफल (ग) for (त्रिज्यां तत्फल) ४. युतेः (ग) (च) for (युतेपदं)
(च) ६. मृगकङ्क्यादौ for (मृगकर्क्यादौ) ३. त्रिज्यांत्यफल for (त्रिज्यांतत्फल)
५. कर्ण्यः for (कर्णः)
(ङ) १. ऽन्त्यफलज्या गुणितया for (गुणांत्यफलज्यता)
३. त्रिज्यान्त्यफलज्या for (त्रिज्यांतत्फल) ४. युतेः for (युते)

९. (घ) १. त्रिज्यांत्यफत्वति for (त्रिज्यांत्यफलत्वति)
२. तत्कोटिज्या (च) (ङ) for (तत्कोटीज्या)
(ग) ३. कृति for (त्वति) ४. हृतं for (हृतं)
(च) १. त्रिद्यांत्य for (त्रिज्यांत्य) ३. फत्वतियुतेः for (फलत्वतियुतेः)
(ङ) ३. कृतियुतेः for (त्वतियुतेः) ५. तरेऽवशेषं for (तरेवशेषं)
६. जीवाऽतः for (जीवातः)

कक्ष्यामंडलतुल्यं प्रतिमंडलमध्यमवनिमध्यात्स्वेत् ।
तत्स्वोच्चनीचवृत्तव्यासार्द्धेभिमुखमुच्चास्य ॥ १० ॥
प्रतिमंडलस्य परिधौ मध्यमभुक्त्या स्फुटग्रहो भ्रमति ।
मंदोच्चादनुलोमं शीघ्रात्प्रतिलोममवनिस्थाः ॥ ११ ॥
स्पष्टं पश्यति यस्मात् मध्याद्धनाधिकं स्वकक्षायाम् ।
तस्मात्तदन्तरफलमृणं धनं धने वा ग्रहे मध्ये ॥ १२ ॥
अंत्यफलज्ययात्स्वान्यदयोराद्यंतयोरुपरिकोटिः ।
द्वितीययोर्यतोधस्तदंतरैक्यं ततः कोटिः ॥ १३ ॥

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१०. (घ) १. कक्षा (ग) (च) (ङ) for (कक्ष्या) २. मंलमध्य for (मण्डलमध्य)
३. खे (ग) (ङ) for (स्वेत्) ४. तत्स्वोच्चनीच (ग) (ङ) for (त्स्वोच्चनीच)
५. मुच्चस्य (ग) (च) for (मुच्चास्य)

(ग) ६. तुल्य (ङ) for (तुल्यं)

(च) ३. त्स्वेत for (त्स्वेत्)

(ङ) ५. ऽभिमुखमुच्चस्य for (भिमुखमुच्चास्य)

११. (घ) १. अवनिस्थः (ग) for (अवनिस्थाः)

(ग) २. स्पुटग्रहो for (स्फुटग्रहो)

(च) १. मवनिस्थः (ङ) for (मवनिस्थाः)

१२. (घ) १. मध्यादूनाधिकं (ग) (च) (ङ) for (मध्याद्धनादिकं)

(ग) २. 'धने' लुप्त है, (ङ) ।

(च) ३. यस्मान् for (यस्मात्) ।

१३. (घ) १. ज्याग्रात्स्यात् (ग) for (ज्ययात्स्वान्)
२. द्वितृतीययोः (ग) for (द्वितीययो)

(ग) ३. पदयो for (यदयो) ४. कोटि for (कोटिः)

(च) १. अंत्फ (त्यु) फलज्याया for (अंत्यफलज्यया)
३. त्स्वात्यदयो for (त्स्वान्यदयो)

(ङ) १. ज्याग्रात् for (ज्ययात्) ३. स्यात्पदयो for (स्वान्यदयो)
२. द्वितृतीययो for (द्वितीययो) ३. रधस्तात् for (र्यतोधस्)

कोद्यांत्यफलाद्यैक्यं[2] मकरादावंतरं कुलिरादौ[3]।
तद्वाहुज्याकृत्योः संयोगपदं भवति कर्णः[4] ॥१४॥
प्रतिमंडलपदमाद्यं ग्रहत्रयं[1] सांत्यधनुरतोन्यच्च[2]।
चक्रार्द्धांतमतोन्यद्याद्यभचक्रांतं[4] मंत्यमतः[5] ॥ १५ ॥
त्रिमनेवांत्यफलधनुर्युतमाद्यं[1] नवमं[2] तृतीयपदमूनं।
द्विचतुर्थेषु द्वादशभाग्निप्रतिमंडलपदानि[4] ॥१६॥

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१४. (घ) १. कोटघंत्य (ग) for (कोद्यांत्य)
२. फलद्यैक्यं (ग) फलज्यैक्यं for (फलाद्यैक्यं)
३. कुलीरादौ (ग) (च) (ङ) for (कुलिरादौ)

(च) १. कोटघांत्य फलाद्यैकं for (कोद्यांत्यफलाद्यैकं) ४. कर्ण्यः for (कर्णः)

(ङ) १. कोटचन्त्य for (कोद्यांत्य) २. फलज्यैक्यं for (फलाद्यैक्यं)

१५. (घ) १. ग्रहत्रयं (ग) गृहत्रयं for (ग्रहत्रयं)
२. रतोऽन्यच्च (ग) रतोन्यश्च for (रतोन्यच्च)
३. मतोन्यद्यद्याद्यभ (ग) for (मतोन्यद्याद्यभ)
४. चक्रांतमंत्यमतः (ग) for (चक्रांतंमंत्यमतः)

(ग) ५. चक्रार्धंत for (चक्रार्द्धांत)

(च) १. गृहत्रयं (ङ) for (ग्रहत्रयं)

(ङ) ५. चक्र र्धमनेनोनं द्वितृतीयं चतुर्थमाद्यसमम् for (चक्रार्द्धांतमतोन्यद्याद्यभचक्रांतं मंत्यमतः)

१६. (घ) १. त्रिभमेवांत्य for (त्रिमनेवांत्य) २. नवमभम् for (नवमं)
३. मूनम् (ङ) for (मून) ४. भानि for (भाग्नि)

(ग) यह श्लोक इस प्रति में नहीं है।

(च) १. त्रिभमेवांत्य for (त्रिमनेवांत्य) ४. द्वदशभानि for (द्वदशभाग्नि)

(ङ) १. त्रिभमन्त्य for (त्रिमनेवांत्य) २. नवमं for (नवमं)
४. षड्द्वादशभानि for (षुद्वादशभाग्नि)

कर्णाहृते[1] व्यासार्द्धं भुजज्यया गुणितमाप्तधनुराद्यैः[2]।
प्रोह्यदलाद्वितये[4] षड्भाशियुतं[5] तृतीयपदे ॥ १७ ॥
चक्रत्प्रोह्य[1] चतुर्थे स्फुटोच्चयोरंतरं स्वमंदोच्चे[2] क्षेप्यम्[5]।
शीघ्रे शोध्यं तत्स्पष्टं[6] पूर्ववच्छेषम्[3] ॥ १८ ॥
मध्यस्फुटांतरकला बाहुफलं भवति तीक्ष्णकिरणस्य।
स्फुटभुक्तघार्कादीनां[1] भुजान्तरं क्षयधनं रविवत् ॥ १९ ॥

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१७. (घ) १. कर्णाहृतं (ग) (ङ) for (कर्णाहृते) २. राद्यैः (ग) राद्ये for (राद्यैः)
३. भदला (ग) (ङ) for (दला)

(ग) (वि० यहां यह श्लोक १६ की संख्या पर अंकित है।)

४. द्वितीये (ङ) for (द्वितये) ५. षड्भाशयुतं for (षड्भाशियुतं)

(च) १. हृतं for (हृते) ३. प्राकरभदला for (प्रोह्यदला)

(ङ) २. राद्ये for (राद्यैः)

१८. (घ) १. चक्रात्प्रोह्य (ग) (ङ) for (चक्रत्प्रोह्य)
२. मंदोच्चोत्क्षेप्यम् (ग) for (मंदोच्चे)
(वि० 'मंदोच्चे' पर श्लोकार्ध समाप्त।)
३. छेषम् for (वच्छेषम्)

(ग) ४. स्पुटोच्च for (स्फुटोच्चयो)
५. 'क्षय्यं' for (क्षेप्यं)
(वि० इससे द्वितीय पंक्ति का आरम्भ)
६. स्पष्टः (ङ) for (स्पष्टं)
(वि० इसकी श्लोक संख्या १७ है।)

(च) १. चक्रात्प्रोक्ष्यं for (चक्रत्प्रोह्य) ६. तत्स्पष्ट for (तत्स्पष्टं)

(ङ) ३. पूर्ववत्शेषम् for (पूर्ववच्छेषम्)

(वि० श्लोक की दूसरी पंक्ति "क्षेपम्" शब्द से आरम्भ होती है।)

१९. (ग) (वि० इसकी श्लोक संख्या १८ है)

(च) १. भुक्तघार्क्कादीनां for (भुक्तघार्कादीनां)

(ङ) १. भुक्तघार्ऽकादीनां for (भुक्तघार्कादीनां)

मंडलशेषात्[३] स्वोच्चं विशोध्य शेषं[४] चतुर्गुणालब्धम् ।
केंद्रं[५] यदा निजिनगुणा[१] जीवा गतयेयगतयो[२] यैः ॥ २० ॥
छेदचतुर्थे बाहुर्ये[१]यागतगतैस्तथा कोटिः[२] ।
प्राग्वदिनेन्दुभुजफलं[४] तथाफलं[५] भूसुतादिनाम्[३] ॥ २१ ॥
फलचापकला गुणिते छेदे मंडलकलाहृत[१] लब्धम् ।
मंडलशेषे पूर्ववद्रणधनमस्माद्[२] ग्रहः स्पष्टः[३] ॥ २२ ॥

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२०. (घ) १. गुणात् (ङ) for (गुणा) २. गतयेयैः (ग) गयेयैः for (गतयोयैं)
(ग) ३. शेखात् for (शेषात्) ४. शेखाश्च for (शेषं च)
५ पदानि for (यदानि) (च) (ङ)
(वि० इसकी श्लोक संख्या १६ है ।)
(च) १. यैं for (यैः)
(ङ) ४. सदृशं for (शेषं) ६. चतुर्गुणाल्लब्धम् for (चतुर्गुणालब्धम्)
२. गतयेयैः for (गतयोयैः)

२१. (घ) १. चतुर्थेबर्हुर्ये (ग) चतुर्थेबाहु for (चतुर्थे बाहुर्ये)
२. यगतागतगवे (गे) यगतागतगतैं for (यागतगतैं)
३. सुतादीनाम् (ग) (ङ) for (सुतादिनाम्)
(ग) ४. 'भु' लुप्त है । (वि० इसकी श्लोकसंख्या २० है ।)
(च) १. चतुर्थेबर्हुर्ये for (चतुर्थे बाहुर्ये) २. यगतागतगतैंस्तथा (यागतगतैंस्तथा)
३. सुतादीनां for (सुतादिनाम्)
(ङ) १. चतुर्थेबर्हो for (चतुर्थेबाहु) २. यैंयगता for (यैंया)
५. मन्दफलं for (तथाफलं)

२२. (ग) १. कलाहृते ल्लब्धम् (ग) for (कलाहृत लब्धम्)
२. पूर्ववद्दणधन (ग) (ङ) for (पूर्ववद्रण)
(वि०—यहाँ संख्याक्रम २२ के स्थान में २३ लिखा हुआ है)
(ग) यहाँ श्लोक संख्या २१ है । २२ वाँ श्लोक इसके स्थान पर निम्नांकित है—
भुजकोटयंशो न गुणाभार्द्धांशातश्चतुर्थ भागोनैः ।
पंच द्वींदुखचंद्रैर्विभाजिता व्यासदलगुणिताः ॥ २२ ॥
(च) १. हृतेल्लब्धं for (हृतलब्धम्) २. पूर्ववद्दणधन for (पूर्ववद्रणधन)
३. (वि० क्रमसंख्या २३ अंकित है ।)
(ङ) १. कलाहृते for (कलाहृत)

तज्ये[3] परमफलज्या संगुणिताद्[1] फ फले विनाद्याभिः[2] ।
इष्टोच्चनीचवृत्तव्यासार्द्धं परमफलजीवा[4] ।। २३ ।।

२३. (घ) वि० अतिरिक्त श्लोक —

तज्ये परमफलज्या संगुणितात्फले विभुजकोटच्यंशो न गुणा भार्द्धांशास्तच्चतुर्थे भागो नेः । पंचद्वींदुखचंद्रै विभाजिता व्यासदलगुणिताः ।। २३ ।।

(वि० यह श्लोक यहाँ २३ संख्याक्रम पर पाया गया)

२३. (घ) (वि० इसकी क्रम संख्या २४ है ।)

१. तत्फले (ग) संगुणितास्तत्फले for (संगुणितात्फफले)

२. ज्याभिः (ग) (ङ) for (द्याभिः)

(ग) वि० अतिरिक्त श्लोक –

तत्ज्ये परमफलज्या संगुणितास्तत्फले विना ज्याभिः ।
इष्टोच्चनीचवृत्तः व्यासार्द्धं परमफलजीवा ।। २३ ।।

३. तत् ज्ये for (तज्ये)

(वि० इसकी क्रम संख्या २३ है ।)

(च) यहाँ पर यह पंक्तियाँ लुप्त हैं—इनके स्थान में निम्नांकित शब्द किसी अन्यलेख में-निम्नोपांत में अंकित हैं ।

"भुजकोटच्यंशो गुणा भार्द्धांशास्तच्चतुर्थं भांतैः ।
पंचार्द्धां दुखचन्द्रै विभाजिता व्यासदलगुणिता ।।"

३. तद्ये for (तज्ये) १. संगुणितात्फले for (संगुणितात् फ फले)

(वि० क्रमसंख्या २४ अंकित है ।)

(ङ) २३ वाँ श्लोक निम्नांकित है—

:भुजकोटच्यंशोनगुणा भार्द्धांशास्तच्चतुर्थभागोनैः ।
पञ्चद्वींदुखचन्द्रैविभाजिता व्यासदलगुणिता ।। २३ ।।"

'तज्ये·····जीवा' की क्रमसंख्या २४ है ।

३. तज्ज्ये for (तज्ये) १. तत्फले for (त् फले)

इष्टज्या संगुणिता[१] पंचकयमलैक शून्यचन्द्रमसा[२]।
इष्टज्यापादयुतव्यासार्द्धविभाजितालब्धम् [३][४] ॥ २४ ॥
नवतिकृतेः प्रोह्यपदं नवतेः संशोध्य शेषकला[१]।
एवं धनुरिष्टाया[२] भवति ज्यया[३] विना ज्याभिः[४] ॥ २५[५] ॥
इष्टोदयिकभुजांतरमित[१] व[२] स्फुट मध्यमांतरकलाभिः।
वाश्विन्यौदयिकेषु[३] स्वचरप्राणैः[५] स्वफलमेवम् ॥ २६[६] ॥

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२४. (घ) (वि० इसकी क्रमसंख्या २५ दी गई है।)
१. संगुणिताः (ग) (च) (ङ) for (संगुणिता)
२. चन्द्रमसः (ग) शून्यचन्द्रसमः for (चन्द्रमसा)

(च) २. चन्द्रमसः for (चन्द्रमसा) ३. इष्टघ्ना for (इष्टज्या)
४. (वि०क्रमसंख्या २५ अंकित है।)

(ङ) (वि० इस श्लोक की क्रमसंख्या २५ है)
२. चन्द्रमसः for (चन्द्रमसा)

२५. (घ) १. कलाः for (कला) (ग) शेषभागकलाः or (शेषकला)
२. धनुरिष्टयो for (धनुरिष्टाया) ३. ज्याया (ग) (च) (ङ) for (ज्यया)
(वि०—इसकी संख्या २६ अंकित है)

(ग) ४. विज्याभिः for (विनाज्याभिः)

(च) १. शेषभागकलाः for (शेषकला) २. रिष्टया for (रिष्टाया)
५. (वि० क्रमसंख्या २६ अंकित है)

(ङ) १. शेषभागकलाः for (शेषकला) ४. विनाज्याभि for (विनाज्याभिः)
(वि० इसकी क्रमसंख्या २६ है)

२६. (घ) १. इष्टौदयिक (ङ) for (इष्टोदयिक)
२. मिनवत (ग) मिनवत् for (मितव)
३. नाश्विन्यौदयिकेषु (ग) नाश्विन्योदयिकेषु for (वाश्विन्यौदयिकेषु)

(ग) ४. भुजातर for (भुजांतर) ५. स्वरप्राणैः for (स्वचरप्राणैः)

(च) ६. (वि० यहां क्रमसंख्या २७ अंकित है) (ङ)

(ङ) २. वि० मिनवत् for (मितव) ७. भिनवत् for (मेवम्)
३. नाश्विन्यौदयिकेषु for (वाश्विन्यौदयिकेषु)

स्वोच्चं विशोध्य कृत्वा प्राग्वत्फलमृणधनं विपर्यस्वम् ।
कार्यामनष्टेस्पष्टे पुनः पुनर्निश्चलो मध्यः ॥२७॥
मानार्द्धात् षष्टिगुणाद्भुक्ति ह्रुतान्नाडिकादिलब्धेन ।
राश्यंतान् प्राणादि पश्चादेतोर्कसंक्रांतेः ॥ २८ ॥
संक्रांति पुष्यकालो यल्लब्धं नाडिकादि तद्विगुणम् ।
स्नानजपहोमादिकोत्र धर्मो विशिष्टफल ॥ २९ ॥

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२७. (घ) १. विविशोध्य (ग) स्वोच्चविशोध्य for (स्वोच्चं विशोध्य)
(वि० इसकी क्रमसंख्या २८ अंकित है)

(ग) २. विपर्यस्तम् (ङ) for (विपर्यस्वम्) ३. कार्यमनष्टे for (कार्यामनष्टे)

(च) ४. क्रत्वा for (कृत्वा) ५. षुनः for (पुनः)
६. (वि० यहां क्रमसंख्या २८ अंकित है) (ङ)

(ङ) १. स्वोच्चाद् for (स्वोच्चं) ३. कार्यमनर्ष्टस्पष्टे for (कार्यामनष्टेस्पष्टे)

२८. (घ) १. प्रागादिः (ग) for (प्राणादि) २. पश्चांदंतोर्क (ग) for (पश्चादेतोर्क)
(वि० इसकी क्रमसख्या २९ अंकित है)

(ग) ३. हृता (ङ) for (ह्रुता) ४. संक्रान्ते for (संक्रान्तेः)

(च) ५. षष्टी for (षष्टिः) १. प्रागादिः (ङ) for (प्राणादि)
२. पश्चादंतोर्क्कसंक्रांतेः for (पश्चादेतोर्कसंक्रांतेः)
६. (वि० यहां क्रमसंख्या २९ अंकित है) (ङ)

(ङ) २. पश्चादंतोऽर्क for (पश्चादेतोर्क) ७. मानार्धात् for (मानार्द्धात्)

२९. (च) १. पुण्यकालो (ग) (च) for (पुण्यकालो)
(वि० इसकी क्रमसंख्या ३० अंकित है)

(ग) २. होमदानादि (ङ) for (होमादि) ३. फलम् for (फल)

(च) ४. (वि० यहां क्रमसंख्या ३० अंकित है) (ङ) ।

(ङ) १. पुण्यकालो for (पुण्यकालो) ४. तद्द्विगुणम् for (तद्विगुणम्)
२. होमदानादिकोऽत्र for (होमादिकोत्र)
३. विशिष्टफलः for (विशिष्टफल)

एवं नक्षत्रांतातिथिकरणतो छशिप्रमाणाद्धात् ।
षष्टिगुणात् रविशशिनोभुत्तचंतर लब्धघटिकाभिः ॥ ३० ॥
संक्रांतिस्थो यावत्करोति मिश्रं फलं ग्रहस्तावत् ।
यस्मात्तस्मादिष्टो राद्यंतौ परिदरति लोकः ॥ ३१ ॥
चक्रार्द्धेऽर्कशशियुतौ भिन्नायनयोरपक्रमच्चे ।
रविशशिनोः सममधुधृतयोगादिषु वच्यतिपातः ॥ ३२ ॥

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३०. (घ) १. त्तिथि (ग) (च) for (तिथि) २. छसि (ग) शिखि for (छशि)
(वि० इसकी क्रमसंख्या ३१ है)
(ग) ३. करणी for (करणतो) ४. (च) (ङ) द्रवि for (त्ररवि)
५. भुत्तचतर for (भुत्तचंतर)
(च) ३. करणांताछसि for (करणतोछशि) ६. क्रमसंख्यालुप्त
(ङ) ७. नक्षत्रान्तात् for (नक्षत्रान्ता) ३. करणान्ता for (करणतो)
२. च्छशि for (छशि) ५. भुत्तचन्तर for (भुत्तचंतर)
(वि० इसकी क्रमसंख्या ३१ है) ।

३१. (घ) (वि० इसकी क्रमसंख्या ३२ है) १. विमिश्रं for (मिश्रं)
२. तस्मादिष्टे (ग) तस्मादिष्टे for (तस्मादिष्टो)
३. परिहरति (च) (ङ) for (परिदरति)
(ग) ४. ष्वाद्यंतौ for (राद्यंतौ) ५. (वि० क्रमसंख्या ३२ अंकित है) । (ङ)
(ङ) २. तस्मादिष्टे for (तस्मादिष्टो)

३२. (घ) (वि० इसकी संख्या ३३ अंकित है) ।
१. कक्रार्द्धे (ग) चक्रार्धेर्क for (चक्रार्द्धेऽर्कं)
२. रपक्रमसमच्चे (ग) रूपक्रमसमत्वे for (रपक्रमच्चे)
३. योगद्विषवद्वातीपातः (ग) तद्वषवद्वचतीपातः for (योगादिषुवच्व्यतिपातः)
(च) २. रपक्रममच्चे for (रपक्रमच्चे)
३. योगादिषुवद्वचतीपातः for (योगादिषुवच्व्यतिपातः)
४. (वि० क्रमसंख्या ३३ अंकित है) (ङ)
(घ) २. रपक्रमसमत्वे for (रपक्रमच्चे) ३. योगाद्विषदो for (योगादिषु)
४. व्यतीपातः for (वच्व्यतिपातः)

चक्रं[१] वैधृतमेकायतस्थयोः क्रांति[२]जीवयोः[५] साम्ये[६] ।
धनरविगणि[३]योगादग्नि[४] वदूनाधिक[७] कलभ्यः[८] ॥ ३३ ॥
भुक्तैक्य[१]लब्धदिवसैः[२] रवींदुपातायुतोनकाः स्वफलैः[३] ।
अर्कक्रान्तिज्याधनुरिदो[४] विक्षेपयुक्तोनम् ॥ ३४ ॥
त्रिनवग्रहेंदु[१] क्रान्तिर्मेव[२] तुलादौ[३] दिवाकरक्रांतः[४] ।
उनायावदभावस्तावद्भावो[५] न्यथार्केन्द्वोः[६] ॥ ३५ ॥[७]

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३३. (घ) १. मेकायस्थयोः (ग) मेकायनस्थयोः for (मेकायतस्थयोः)
२. इंधन (ङ) (ग) (च) for (धन)
३. मणि (ग) (च) (ङ) for (गणि)
४. कलाभ्यः (ग) (च) (ङ) for (कलभ्यः)
(वि० इसकी क्रमसंख्या ३४ अंकित है) (ङ)
(ग) ५. जीवाया: for (जीवयोः) (प्रथम पंक्ति समाप्त)
६. दूसरी पंक्ति का अंतिम पद ।
७. वरूनाधिक for (वदूनाधिक)
(च) ८. क्रमसंख्या ३४ अंकित है (ङ) ।
(ङ) १. मेकायनस्थयोः for (मेकायतस्थयोः)

३४. (घ) १. भुक्तयैक्य (ग) (च) (ङ) for (भुक्तैक्य)
२. विक्षेप (ग) (च) (ङ) for (विक्षेप)
(वि०-इसकी क्रम संख्या ३५ अंकित है)
(च) ३. अर्क्क for (अर्क)
४. (वि०—यहां क्रमसंख्या ३५ अंकित है) (ङ)
(ङ) ५. दिवसै for (दिवसैः)

३५. (घ) १. त्रिनवगृहेंदु (ग) (च) (ङ) for (त्रिनवग्रहेंदु)
२. मेंष (ग) (च) (ङ) for (मेंव)
४. ऊना (ग) (च) (ङ) for (उना) ५. ऽन्यथा (ङ) for (न्यथा)
(वि०—इसकी क्रमसंख्या ३६ अंकित है।)
(च) ६. क्केंद्वोः for (केंन्द्वोः)
७. (वि०—यहां क्रमसंख्या ३६ अंकित है) (ङ)
(ङ) ३. दिवाकरक्रान्तेः for (दिवाकरक्रान्तः)

व्यतिपातयोपक्रमयोर्दिवसाम्ये वैधृतं दिगन्यत्वे ।
अधिको द्यूनः कल्पो दिग्भेदेपक्रमः शशिनः ॥ ३६ ॥
मेषतुलादाविन्दोरपक्रमेख्यपक्रमादुने ।
एष्यत्यधिके तीतो विपरीतः कार्किमकरादौ ॥ ३७ ॥
क्रान्त्योर्युतिरन्यदिशोरेकदिशोरन्तरं व्यतिपाते ।
एकादशो युतिरंतरमन्यदिशो वैधृते प्रथमः ॥ ३८ ॥

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३६. (घ) (वि०—इसकी क्रमसंख्या ३७ अंकित है)
१. व्यतिपातोपक्रमयो (ग) व्यतीपातोपक्रमयो for (व्यतिपातयोपक्रमयो)
२. दिक्साम्ये (ग) दिक्साम्ये (च) for (दिवसाम्ये)
३. दिग्भेदे (ग) (च) for (दिग्भेदे)
(ग) ४. अधिकाप्यूनः for (अधिकोद्यूनः)
(च) १. व्यतिपातोपक्रमयो for (व्यतिपातयोपक्रमयो)
५. वैद्धृतं for (वैधृतं) ६. (क्रमसंख्या ३७ अंकित है) (ङ)
(ङ) १. व्यतिपातोऽपक्रमयो for (व्यतिपातयोपक्रमयो)
२. दिक्साम्ये for (दिवसाम्ये) ५. वैधृतो for (वैधृतं)
४. न्यूनः for (द्यूनः) ७. कल्प्यो for (कल्पो)
३. ऽपक्रमः for (पक्रमः)

३७. (घ) (वि०—इसकी क्रमसंख्या ३८ है)
१. दूने (ग) द्यूने for (दुने) २. ऽतीतो for (तीतो)
३. कर्क्वि (ग) कर्क्कि for (कार्कि)
४. मेतुला for (मेषतुला)
(च) १. दूने for (दुने) ३. कर्क्किमकरादौ for (कार्किमकरादौ)
५. (वि०—यहां क्रमसंख्या ३८ अंकित है) (ङ)
(ङ) १. दूने for (दुने) २. त्यधिकोऽतीतो for (त्यधिकेतीतो)
३. कर्क for (कार्कि)

३८. (घ) (वि०—इसकी क्रमसंख्या ३९ अंकित है)
१. एकदिशो (ग) एदिशोर्युति for (एकादशोयुति)
(ग) २. व्यतीपाते (ङ) for (व्यतिपाते)
(घ) १. एकदिशोर्युति (ङ) for (एकादशोयुति)
३. (वि०—यहां क्रमसंख्या ३९ अंकित है) (ङ)
(ङ) ४. वैंधृते for (वैधृते)

एवं द्वितीयराशियुतिहीनै रिष्टनाडिकास्वफलैः ।
एष्यदतीत वा यदिराशिद्वयमपि तदंतरकम् ॥ ३९ ॥
छेदोन्यथा तदैक्यं घातस्येष्टघटिका प्रथमराश्योः ।
फलघटिकाभिर्मध्यं द्वयोरपि प्रथमराशिवशात् ॥ ४० ॥
तात्कालिकैर्ग्रहैरसकृदिष्टघटिकाफलोनयुक्तैस्तैः ।
प्राग्वत्प्रथमे छेदः प्रमाणयोगार्द्धलिप्तानाम् ॥ ४१ ॥
इष्टघटिकागुणानामसकृत फलनाडिकाभिराद्यंतैः ।
व्यतिपातवैधृतानयनमन्यतंत्रेषु न ब्राह्म्यात् ॥ ४२ ॥

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३९. (घ) (वि०—इसकी क्रमसंख्या ४० अंकित है)
१. एष्यदतीनं for (एष्यदतीतं)
(च) २. नाडिका: for (नाडिका) १. ऐष्यदतीतं for (एष्यदतीतं)
३. (वि०—यहां क्रमसख्या ४० अंकित है) (ङ)
(ङ) ४. र्युतहीनै for (युतिहीनै)

४०. (घ) (वि०—इसका संख्याक्रम ४१ है)
१. प्रमराशिवशात् for (प्रथमराशिवशात्)
(ग) २. घतस्येष्ठ for (घातस्येष्ट) ३. द्वरपि for (द्वयोरपि)
(च) ४. (वि०—यहां क्रमसंख्या ४१ अंकित है) (ङ)

४१. (घ) (वि०—इसकी क्रमसंख्या ४२ है)
१. फलोनक्तैस्तैः for (फलोनयुक्तैस्तैः)
२. प्रथमछेदः (ग) प्रमछेदः for (प्रथमेछेदः)
(च) ३. रसक्रदिष्ट for (रसकृदिष्ट) ४. (वि०—यहां क्रमसंख्या ४२ अंकित है) (ङ)
(ङ) २. प्रथमश्छेदः for (प्रथमेछेदः)

४२. (घ) (वि०—इसकी क्रमसंख्या ४३ है)
१. असकृत्फल (ग) for (असकृतफल) २. राद्यंतो (ग) for (राद्यंतैः)
३. व्यतिपात (ग) व्यतीपात (ग) व्यतीपात for (व्यतिपाप)
४. ब्राह्मात् (ग) ब्रह्मात् for (ब्राह्म्यात्)
(ग) ५. गुणनाम् for (गुणानाम्)
(च) १. मसक्रत्फल for (मसकृतफल) ४. ब्राह्मात् for (ब्राह्म्यात्)
६. (वि०—क्रमसंख्या ४३ अंकित है) । (ङ)
(ङ) १. असकृत् for (असकृत) २. राद्यन्तौ for (राद्यंतैः)
३. व्यतिपात for (व्यतिपाप) ४. ब्राह्मात् for (ब्राह्म्यात्)

रविबिंबमेकमार्गाच्छशिबिंबापक्रमे भवति यावत् ।
तावत्फलं तदुक्तं तद्भावो तत्फलाभावः ॥ ४३ ॥
ग्रहकक्ष्यायनतुल्याः कक्षास्तन्मंदशीघ्रपातानाम् ।
यस्मात्तस्मान्न पृथक्कक्षाः कल्प्याश्चलोच्चाद्याः ॥ ४४ ॥
पौलिशरोमकवाशिष्ट सौरपैतामहेषु यत्प्रोक्तम् ।
तन्नक्षानयनं नार्यभटोक्तं तदुक्तिरतः ॥ ४५ ॥

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४३. (घ) (वि०—इसकी क्रमसंख्या ४४ है)
१. मार्ग्रा (ग) मार्ग्रात शशि for (मार्गाच्छशि)
२. तदभावे (ग) तद्भावे for (तद्भावो)

(ग) ३. बिंबापक्रमो for (बिंबापक्रमे) ४ तावत्पलं for (तावत्फलं)
५. तत्फलोभाव: for (तत्फलाभावः)

(च) २. तदभावे for (तद्भावो) ६. (वि०—क्रमसंख्या ४४ अंकित है) (ङ)

(ङ) १. मार्गाच्छशि for (मार्गाच्छशि) २. तदभावे for (तद्भावो)

४४. (घ) (वि०—इसकी क्रमसंख्या ४५ है)
१. पृथक्कक्षाः (ग) पृथक् कक्ष्या for (पृथकक्षाः)
२. कल्प्याश्चलोच्चाद्याः (ग) कल्पाचलोच्चाद्यात् for (कल्प्याश्चलोच्चाद्याः)

(ग) ३. ग्रहकक्षायन तुल्या for (ग्रहकक्ष्यायनतुल्याः) ४. तस्मात्तते for (तस्मान्न)

(च) १. पृथक्कक्षाः for (पृथकक्षाः) ५. (वि० क्रमसंख्या ४५ अंकित है) (ङ)

(ङ) ३. कक्षयैव for (कक्ष्यायन) १. पृथक् कक्षा for (पृथकक्षाः)
७. तुल्या कक्षा for (तुल्याःकक्षा) २. कल्प्या for (कल्प्याश्)
७. स्तनमंद शीघ्र for (तन्मन्द) ८. चलोचाद्या for (चलोच्चाद्याः)

४५. (घ) (वि०—इसकी क्रमसंख्या ४६ है)
१. तन्नक्षत्रानयनम् (ग) for (तन्नक्षानयनं)

(ग) २. पौलश for (पौलिश) ३. सौर्य for (सौर)

(च) १. तन्नक्षत्रानयनं for (तन्नक्षानयनं) ४. (यहां क्रमसंख्या '४६' अंकित है।)

(ङ) ५. वासिष्ठ for (वाशिष्ठ)

अध्वर्द्धार्द्धं क्षेत्रान्यूडुनि षट् षट् भवन्ति पञ्चदशाः ।
ऋक्षाणि समक्षेत्राण्यभिजिद्भोगो भवत्येकः ॥ ४६ ॥
केशादित्यविशाखा प्रोष्ट पदार्यम्ल वैश्वदेवाग्नि ।
षट् षट् ज्येष्टाभरणी स्वात्यर्द्धं त्रारूणाश्लेषा ॥ ४७ ॥
पंचदशात्रानुक्ता न्येको भिजदुक्तरक्षभोगोन्यः ।
तन्मानं नाक्षत्रं दुरधिगमं मंदबुद्धिनाम् ॥ ४८ ॥

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४६. (घ) (वि० इसकी क्रमसंख्या ४७ है)
१. अध्यर्द्धार्द्धं (ग) (च) for (अध्वर्द्धार्द्धं)
२. क्षेत्राण्युडूनि (ग) for (क्षेत्रान्यूडुनि) ३. पट् (ग) षड् for (षट्)
(ग) ४. 'षट्' पद लुप्त है ५. दशः for (दशाः)
(च) २. क्षेत्राण्युडूनि for (क्षेत्रान्यूडुनि) ३. पट् for (षट्)
६. (यहां क्रमसंख्या ४७ अंकित है।) (ङ)।
(ङ) अध्यर्धानि भवन्ति षड् नक्षत्राण्युडूनि षडर्धानि ।
पञ्चदश समक्षेत्राण्यभिजिद्भोगो भवत्येकः ॥४७॥
for
"अध्वर्द्धार्द्धं क्षेत्रान्यूडुनि षट् षट् भवन्ति पञ्चदशाः ।
ऋक्षाणि समक्षेत्राण्यभिजिद्भोगो भवत्येकः ॥ ४६॥

४७. (घ) (वि० इसकी क्रमसंख्या ४८ है)
१. वैश्वदेवानि (ग) (च) (ङ) for (वैश्वदेवाग्नि)
२. स्वात्यार्द्रा (ग) for (स्वात्यर्द्धा) ३. श्लेषाः (ङ) for (श्लेषा)
(ग) ४. मदार्यम्ल for (पदार्यम्ल)
(च) ५. विशाषा for (विशाखा) ६. प्रौष्ट for (प्रोष्ट)
२. स्वात्यार्द्धा for (स्वात्यर्द्धा) ३. श्लेषाः (ङ) for (श्लेषा)
७. (वि० – यहां क्रम संख्या अंकित नहीं है)
५. पदार्यम्ण for (पदार्यम्ल) २. स्वात्यार्द्रा for (स्वात्यर्द्धा)

४८. (घ) (वि०—इसकी क्रमसंख्या ४९ है)
१. ऽभिजिदुक्त (ङ) for (भिजिदुक्त)
२. भोगोऽन्यः (ग) भोगान्यः for (भोगोन्यः) ३. तात्मानं for (तन्मानं)
४. मंदबुधीनाम् (ग) (च) मंदबुद्धीनां for (मंदबुद्धिनाम्)
(ग) ५. ऋक्ष (ङ) for (रक्ष) (च) ३. तत्मानं for (तन्मानं)
६. (वि०—यहाँ क्रमसंख्या ४९ अंकित है) (ङ)
(ङ) ३. यस्मात् तन्नक्षत्रं for (तन्मानं नाक्षत्रं) ४. मन्दबुद्धीनाम् for (मंदबुद्धिनाम्)

अध्यर्द्धा[२] समक्षेत्राणां मध्यगते[३] लिप्तिकाः शशिनः ।
अध्यर्द्धार्द्धैकगुणा[४] भभोगलिप्तास्तदैक्ययोना[१] ॥ ४६ ॥[५]
मंडललिप्ताः शेषो भोगोभिजित्[१] शशांक[२] भगणा[३] वा ।
त्रिघनगुणाः संशोध्याः कल्पदिनेभ्यो यदवशेषम् ॥ ५० ॥[४]
तद्भगणैर्दिनभोगो[१] भिजितो भभोगलिप्तोनाः[३] ।
भानि ग्रहभुक्तकला[४] गतगम्या गतिहूता[२] दिवसाः[५] ॥ ५१ ॥[६]
भफलं प्रोक्तमभिजितो मंगलपात्रासु[३] संहिताकारैः ।
यैस्तद्भोगोनोक्तास्ते[१] गणकाः संहिताबाह्याः[२] ॥ ५२ ॥[४]

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४६. (घ) (वि०—इसकी श्लोक संख्या ५० अंकित है)
१. तदैक्योनाः (च) (ङ) for (तदैक्योना)
(ग) २. अध्यद्धार्द्ध for (अध्यर्द्धा) ३. मध्यगति (ङ) for (मध्यगते)
(च) ४. अध्यर्ध्यार्द्धैक for (अध्यर्द्धार्द्धैक)
५. (वि०—यहां क्रमसंख्या ५० अंकित है) (ङ)
(ङ) २. अध्यर्धार्ध for (अध्यर्द्धा) ४. अध्यर्धार्धैक for (अध्यर्द्धार्द्धैक)

५०. (घ) (वि०—इसकी क्रमसंख्या ५१ है)
१. भोगोऽभिजित् (ग) भोगोभिजितः for (भोगोभिजित्)
(ग) २. शशी for (शशांक) ३. भगणैर्वा for (भगणावा)
(च) ४. (वि०—यहां क्रमसंख्या ५१ है) (ङ)
(ङ) १. ऽभिजितोभोगः for (भोगोभिजित्)

५१. (घ) (वि०—इसकी क्रमसंख्या ५२ है)
१. भुक्तिर्भोगाभिजितो (ग) दिनभुक्तिर्भोगो for (दिनभोगो)
२. हृता (ग) (च) for (हूता)
(ग) ३. लिप्तोनाम् for (लिप्तोनाः) ४. भुक्ति for (भुक्त)
५. दिवसा for (दिवसाः)
(च) ६. (वि०— यहां क्रमसंख्या ५२ है) (ङ)
(ङ) १. ऽभिजितो for (भिजतो) ७ + भोगो +

५२. (घ) (वि० इसकी क्रमसंख्या ५३ है) १. नोक्तस्ते (ग) (ङ) for (नोक्तास्ते)
(ग) २. सहिता for (संहिता)
(च) ३. यात्रासु for (पात्रासु) ४. (वि०—यहां क्रमसंख्या ५३ अंकित है) (ङ)
(ङ) ३. यात्रासु for (पात्रासु)

अर्ध्यार्द्धक्षेत्राणि संहिता स्वभिहितानि गर्गाद्यैः ।
यस्मादुडुनितस्मान्नार्यभटोक्तं तदानयनम् ॥ ५३ ॥
आर्याणां पञ्चांशच्चतुभिरधिका चतुर्दशोऽध्यायः ।
चतुर्दशोध्यायः समाप्तः

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५३. (घ) (वि०—इसकी क्रमसंख्या ५४ है)
१. अध्यर्द्धा (ग) अध्यर्द्धादि for (अर्ध्यार्द्ध) २. क्षत्रणि for (क्षेत्राणि)

(ग) ३. यस्मादुडूति (च) for (यस्मादुडुनि)

(च) १. अध्यर्द्धार्द्धक्षेत्राणि for (अर्ध्यार्द्धक्षेत्राणि)
४. (वि०—यहां क्रमसंख्या ५४ अंकित है) ।

(ङ) १. अध्यर्धादि for (अर्ध्यार्द्ध) ३. यस्मादुडूनि for (यस्मादुडूनि)

५४. (घ) १. चतुर्भि (ग) च्चतुर्भि for (चतुभि) २. चतुर्दृशो (ग) for (चतुर्दशो)

(ग) 'चतुर्दशो' से 'समाप्तः' तक अंकित नहीं है । इसकी दूसरी पंक्ति निम्नांकित है
'स्फुटगतिरुत्तरमन्या दिशा व्यूहयत्प्रस्नात् ॥५४॥

(च) १. पञ्चाशच्चतुर्भिरधिका for (पञ्चांशच्चतुभिरधिका)
२. चतुर्दृशोध्यायः for (चतुर्दशोऽध्यायः)
३. 'छ' for (चतुर्दशोध्यायः समाप्तः)

(ङ) १. च्चतुर्भि for (च्चतुभि)
४. +स्फुटगत्युत्तरमन्यान् दिशाऽनयया ऽभ्युपगमेत् प्रश्नान् ॥५५॥
३. इति स्फुटगत्युत्तराध्यायश्चतुर्दशः for (चतुर्दशोऽध्यायः समाप्तः)

अथ त्रिप्रश्नोत्तराध्यायः

पञ्चदशः

स्फुटगत्युत्तरमन्या दिशानयाप्यूहयेत्प्रश्नम्।

योऽह्नः पूर्वापरयोस्तुल्यच्छायांगुलाग्रयो बिन्दुं।
वीक्ष्याक्रान्त्याक्षांशै विना दिशो वेत्ति गणकः सः ॥ १ ॥
त्रिच्छायाग्रज्ञो यः क्रान्त्यक्षार्के विना दिशो भ्रमणम्।
छायाग्रस्य दिनार्द्धे छायां वा वेत्ति गणकः सः ॥ २ ॥
यः छायाग्रं दृष्ट्वा क्रांत्यक्षज्ञो दिशो विजानाति।
शंकुछायाभ्रमणे दिक्ज्ञो वा वेत्ति गणकः सः ॥ ३ ॥

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१. (घ) १. प्रश्नात् (च) for (प्रश्नम्) २. (वि०—यहाँ पर समाप्तिसूचक चिह्न है)
३. योऽह्न (ग) येह्नः for (योह्न) ४. बिंदू (ग) बिंदुः for (बिंदु)
५. वीक्ष्य (ग) for (विक्ष्य) ६. विना (ग) for (विना)
(ग) ७. स्फुटगतिरुत्तर for (स्फुटगत्युत्तर)
८. व्यूयत्प्रस्नात् ।। ५४ ।। for (प्यूहयेत् प्रश्नम्)
(च) ३. योऽह्न for (योह्न) ५. विक्ष्य for (विक्ष्या)
४. बिंदू for (बिंदु) ६. विना for (विना)
(ङ) ३. योऽह्न for (योह्न) ४. विन्दु for (बिंदु)
५. बीक्ष्य for (विक्ष्या) ७. क्रान्त्यक्षांशै for (क्रान्त्याक्षांशै)
६. विना for (विना)

२. (घ) १. त्रछाया for (त्रिछाया) २. दि अर्द्ध छायां (ग) दिनार्द्ध for (दिनार्द्धे)
(ग) ३. क्रांति क्षार्के विना for (क्रान्त्यक्षार्के विना)
(च) ३. क्रान्त्यक्षाक्र्के for (क्रान्त्यक्षार्के)
(ङ) १. त्रिच्छाया for (त्रिछाया) ३. क्रान्त्यक्षार्के विना for (क्रान्त्यक्षार्के विना)
२. दिनार्धच्छायां for (दिनार्द्धे छायां)

३. (ग) १. दिज्ञो for (दिशो) २. "सः" यहाँ यह पद लुप्त है
३. यश्छायाग्रं for (यः छायाग्रं)
(वि० इसकी श्लोक संख्या ४ है। तीन की संख्या का श्लोक लुप्त है।)
(च) ४. क्रत्यक्षज्ञो for (क्रांत्यक्षज्ञो)
(ङ) ३. यश्छायाग्रं for (यः छायाग्रं) १. दिग्ज्ञो for (दिक्ज्ञो)
५. शंकुच्छाया for (शङ्कु छाया)

दृष्ट्वा विषुवच्छायां लम्बश्नेज्ये करोति यो बहुधा ।
मध्यच्छायार्कज्ञो श्नांशान्यो वेत्ति गणकः सः ।। ४ ।।
यश्चरखंडकलंकोदयाद्विजानाति लग्नमुदयैः स्वैः ।
लग्ना घटिका छायांगत शेषनताच्च गणकः सः ।। ५ ।।
गतशेषनता घटिका छायातो भिष्टदिनदलच्छायां ।
बहुधार्ककान्त्यश्नान दृष्ट्वा यो वेत्ति गणकः सः ।। ६ ।।

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४. (घ) १. लंबक्षोज्य (ग) for (लंबश्नेज्ये)
२. ऽक्षांशान् (ग) क्षांशान् for (श्रांशान्)

(ग) (वि० इसकी श्लोक संख्या ५ है)

(च) १. लंवक्षेज्ये for (लंबश्नेज्ये) ३. मध्यछायाऽर्कज्ञो for (मध्यछायार्कज्ञो)
२. क्षांशान् for (श्रांशान्)

(ङ) ४. विषुवच्छायां for (विषुवछायां) १. लंबाक्षज्ये for (लंबश्नेज्ये)
३. मध्यच्छाया for (मध्यछाया) २. ऽक्षांशान् for (श्रांशान्)

५. (घ) १. खंड (ग) for (खंडकलं) २. लग्नाद्घटिका (ग) for (लग्नाघटिका)

(ग) (वि० इसकी श्लोक संख्या ६ है)
३. श्छाया for (छायांगत) ४. नताश्च (च) for (नताच्च)

(च) २. लग्नाद्घटिका for (लग्नाघटिका) ४. नत्ताच्च for (नताच्च)

(ङ) १. खण्डक for (खंडक) ५. लङ्कोदयान् for (लङ्कोदयाद्)
२. लग्नाद् for (लग्ना) ३. घटिकाश्छायां for (घटिकाछायां)

६. (घ) १. ऽभीष्ट (ग) भीष्ट for (भिष्ट) २. क्रान्त्यक्षान् (ग) for (क्रांत्यश्नान)

(ग) ३. श्छायातो (ङ) for (छायातो) ४. छायाः for (छायां)
(वि० इसकी श्लोक संख्या ७ है)

(च) १. ऽभीष्ट (ङ) for (भिष्ट)
२. बहुधार्क्कक्रान्त्यक्षान् for (बहुधार्ककान्त्यश्नान)

(ङ) ४. दलच्छायाम् for (दलछालां)
२. बहुधा क्रान्त्यर्काक्षान् for (बहुधार्क क्रान्त्यश्नान)

क्रांतिज्ञः सममंडल शंकुकर्णं वयोश्नलंबज्ञः ।
जानाति कोणशंकुछाया घटिकाः स तंत्रज्ञः ॥ ७ ॥
शंकुतलप्राच्यपरांतरद्वयं विक्षयो विजानाति ।
विषुवच्छायामेकं द्रष्ट्वादित्यं च गणकः सः ॥ ८ ॥
पातलशंकुमुदयेंदुस्ते वा दृग्ज्यां खे विजानाति ।
द्रक् पाताल द्रक्शंकोः पृथक् तले वा स तंत्रज्ञः ॥ ९ ॥

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७. (घ) १. कांतिज्ञः for (क्रांतिज्ञः) २. शंकुं (च) (ङ) for (शंकु)
३. कर्णः (च) for (कर्णं)
४. वयोक्षलंबज्ञः (ग) च योक्षलंबज्ञः for (वयोश्नलंबज्ञः)
५. जानाति जानाति for (जानाति) ६. शंकुं for (शंकु)
(ग)(वि०— इसकी श्लोक संख्या ८ है)
(च) ४ वयोक्षलंबज्ञः for (वयोश्नलंबज्ञः)
(ङ) ३. कर्णं च for (कर्णं व) ४. योश्नलंबज्ञः for (योश्नलम्बज्ञः)
६. शंकुच्छाया for (शंकुछाया)

८. (घ) वीक्ष (ग) वीक्ष्य for (विक्षयो)
२. दृष्ट्वादित्यं (ग) (च) for (द्रष्ट्वादित्यं)
(ग) (वि०—इसकी श्लोकसंख्या ९ है)
(च) १. वीक्ष for (विक्ष) ३. व for (च)
(ङ) १. वीक्ष्य for (विक्ष) ४. विषुवच्छाया for (विषुवछाया)
२. दृष्ट्वाऽऽदित्यं for (द्रष्ट्वादित्यं)

९. (घ) १. मुदयेऽस्ते (ग) मुदयेस्ते for (मुदयेंदुस्ते)
२. दृग्पावालगशंक्वोः (ग) दृग्पातालगशंकुकोः for (द्रक्पाताल द्रक्शंकोः)
(ग) ३. विजानाति for (विजानाति)
४. तत्रज्ञः for (तंत्रज्ञः) ५. पाताल for (पातल)
(वि०—इसकी भी श्लोकसंख्या ९ है)
(च) ५. पाताल for (पातल) २. दृग्पाताल for (द्रक्पाताल)
६. द्रग्शंक्कोः for (द्रक्शंकोः)
(ङ) ५. पाताल for (पातल) २. दृक्पाताल for (द्रक्पाताल)
१. मुदयेऽस्तो for (मुदयेन्दुस्ते) ३. खेविजानाति for (खे विजानाति)
६. गशङ्क्कोः for (द्रक्शंकोः)

दिनगतशेषप्राणैरिष्टार्को दिनदलान्नतैरथवा ।
भवति सममंडले यैर्यस्तान् कथयति स तंत्रज्ञः ॥ १० ॥
सममंडलगः प्राणैर्दिनगत शेषं नतै र्द्विनार्द्धाद्वा ।
यैं भवति ज्ञात्वा तान्योऽर्कं कथयति स तंत्रज्ञः ॥ ११ ॥
यः सममंडलशंकुकर्णं व वीक्ष सूर्यमानयति ।
रविसममंडलशंकुज्ञो क्षं कथयति स तंत्रज्ञः ॥ १२ ॥
रविलग्नांतर घटिका विनोदयैर्लग्नमिष्टघटिकाभिः ।
वेत्ति चरार्द्धदिक्षं योर्कज्ञो वा स तंत्रज्ञः ॥ १३ ॥
अक्षचरार्द्धज्ञार्कं छायातो यश्चरार्द्धमिष्टायाः ।
इष्टचरार्द्धदथवा छायां कथयति स तंत्रज्ञः ॥ १४ ॥

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१०. (घ) १. रिष्टाऽर्को (ग) रिष्टाऽर्को for (रिष्टार्को)
(ग) २. दिनगते for (दिनगत) ४. दिनदलान्नतेरथवा for (दिनदलान्नतैरथवा)
५. समंडले for (सममंडले) ६. तत्रज्ञः for (तंत्रज्ञः)
(च) १. ऽर्को for (र्को)
११. (घ) १. नंतै (ग) (च) (ङ) for (नतै) २. दिनार्द्धाद्वा (च) for (दिनार्द्धाद्वा)
(ग) ३. सममंडलगं for (सममंडलगः) ४. यैंभवति (च) for (यैंभवति)
५. न्योर्कं for (न्योऽर्कं)
(च) ५. तान्योऽर्क्कं for (तान्योऽर्कं)
(ङ) १. दिनार्धाद्वा for (दिनार्द्धाद्वा) ६. योऽर्कं for (योऽर्कं)
१२. (घ) १. शंकु (ग) (ङ) for (शंकु) २. वा (च) (ङ) for (व)
३. वीक्ष्य (च) (ङ) for (वीक्ष) ४. कथति (च) for कथयति)
(ग) ५. समंडल for (सममंडल)
(ङ) ६. ऽक्षं for (क्षं)
१३. (ग) १. घटिको for (घटिका) २. लग्न for (र्लग्न)
(च) ३. योऽर्क्कज्ञो for (योर्कज्ञो)
(ङ) ३. योऽर्कज्ञो for (योर्कज्ञो)
१४. (घ) १. यस्यरार्द्धं (ग) यश्चचरार्धं for (यश्चरार्द्धं)
२. कथति for (कथयति)
(ग) ३. चरार्द्धज्ञो for (चरार्द्धज्ञार्कं) ४. चरार्द्धदथवा for (चरार्द्धादथवा)
(च) ३. ज्ञार्क्कं for (ज्ञार्कं)
(ङ) ३. चरार्धज्ञोऽर्कं for (चरार्द्धज्ञार्कं) ४. चरार्धादथवा for (चरार्द्धादथवा)

मध्यच्छायातो[५] क्षविदानयति रविवाक्[१] रज्ञोश्नं[६] ।
योग्रार्कज्ञो[२] लंबाक्षज्ये[३] कथयति स तंत्रज्ञः ॥ १५ ॥
उदयेस्तमये[६] चाग्रां[१] वेत्ति दिनार्द्धे[२] नतोन्नतज्ययः ।
ताभिर्विषुवच्छाया[४] श्नज्यालंबान्[३] स[५] तंत्रज्ञः ॥ १६ ॥
दिनरात्र्यर्धे[४] यश्चरदलाद्[५] विना स्वेकरोति[६] वाताभ्यां ।
अक्षावलंबकौ[१] वा नस्तमयार्कात्स[३] तंत्रज्ञः ॥ १७ ॥

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१५. (घ) १. छायातोक्षविदानयति (ग) तोऽक्षविदानब्दि for (छायोतोक्षविदानयति)
२. रविवाकरज्ञोक्षम् (ग) रविदिवाकरज्ञोऽक्षम् for (रविदिवाकरज्ञोऽक्षम्)
३. लोबाक्षज्ये for (लम्बाक्षज्ये)

(च) २. रविवाक्रज्ञोक्षं for (रविवाक्रज्ञोश्नं) ३. योग्रार्क्कज्ञो for (योग्रार्कज्ञो)
४. लंबाक्ष्यज्ये for (लंबाक्षज्ये)

(ङ) ५. मध्यच्छायातो for (मध्यछायातो)
१. ऽक्ष for (क्ष) ३ योऽग्रार्कज्ञो for (योग्रार्कज्ञो)
६. रवि for (रवि) २. दिवाकरज्ञोऽक्षम् for (वाक्रज्ञोश्नं)
४. लम्बाक्षांशान् for (लम्बाक्षज्ये)

१६. (घ) १. वाग्रां (ग) (ङ) for (चाग्रां) २. ज्ये (ग) ज्येयः for (ज्ययः)
३. क्षज्या (ग) क्षप्या for (श्नज्या)

(ग) ४. विषुवा for (विषुवच्छाया) ५. लंबात् for (लंबान्)

(च) ३. क्षज्या for (श्नज्या)

(ङ) ६. उदयेऽस्तमये for (उदयेस्तमये) २. ज्ये यः for (ज्ययः)
४. विषुवच्छाया for (विषुवच्छाया) ३. क्षज्या for (श्नज्या)

१७. (ग) १. नस्तमयार्के for (नस्तमयार्कात्) २. नस्तमयार्कौ for (नस्तमयार्कात्)
३. न स for (त्स)

(च) ४. दिनरात्र्यार्धे for (दिनरात्र्येर्धे) १. नस्तमयार्क्कात्स for (नस्तमयार्कात्स)

(ङ) ४. 'लुप्त' for (दिनरात्र्येर्धे) ५. दलंविना for (दलादविना)
६. +दिनरात्र्यर्धे+१. ऽनस्तमयार्कान् for (नस्तमयार्कात्)

आय[१] निवक्त्योनां प्रश्नानामुत्तरं चतुःषष्टि[५][३] ।
आयास्त्रिप्रश्नोक्ताशेषप्रश्नोत्तरोक्तिरतः ।। १८ ।।
विषुवच्छाया[५] गुणिता[१] स्वाहोरात्र्यार्द्ध[२] भाजिता त्रिज्या[४] ।
क्रान्तिद्वादशगुणिता फलचापकला[३] श्रुभिः सहितैः ।। १९ ।।
स्वचरप्राणैर्दिनगतशेषैः सममंडले रविर्भवति[३] ।
फल चापाशून्यभिस्तिथि[२] घटिकाभि[२] नताभिर्वा ।। २० ।।
उदयसममंडलान्तरं[५] घटिकाज्यालंबकाहतं[१][२] गुणयेत् ।
अक्षज्यया[३] हृतां तत[४] क्रांन्त्या व्यासार्द्धसंगुणया ।। २१ ।।

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१८. (घ) १. आर्यानवकोक्तानां (ग) (ङ) for (आयनिवक्त्योनां)
२. आर्यास्त्रि (ग) (च) for (आयास्त्रि)
(ग) ३. प्रस्नाना for (प्रश्नाना) ४. प्रस्नोक्ता: for (प्रश्नोक्ता)
(च) १. आर्यानवक्त्यानां for (आयनिवक्त्योनां)
(ङ) ५. षष्टया for (षष्टिः) २. आर्याणां for (आयास्त्रि)
४. प्रश्नोक्त्या for (प्रश्नोक्ता)

१९. (घ) १. गुणित (ग) गुणित (ङ) for (गुणिता)
२. रात्रार्ध (ग) (ङ)for(रात्र्यार्द्ध)
(ग) ४. त्रिज्या: for (त्रिज्या)
(च) १. गुणित for (गुणिता) २. रात्रार्द्ध for (रात्र्यार्द्ध)
३. कलासुभिः for (कलाश्रुभिः)
(ङ) ५. विषुवच्छाया for (विषुवच्छाया)

२०. (घ) १. चापाशूनाभि (ग) चापास्तनुभि for (चापाशून्यभि)
२. न्नतानिर्वा: (ग) नंताभिर्वा for (नताभि)
(च) ३. द्दिन for (दिन) १. शून्याभि for (शून्यभिः)
२. न्नताभिर्वा: for (नताभिर्वा)
(ङ) १. चापन्यूनाभि for (चापाशून्यभि) २. नंताभि for (नताभि)

२१. (घ) १. ज्यां (ङ) for (ज्या) २. लंबकाहता (ङ) (ग) लंबकाहलांfor (लंबकाहतं)
३. अज्यया (ग) अक्षया for (अक्षज्यया)
४. हृत्यांतक्रान्त्या (ग) हृतांत्यक्रान्त्या for (हृतां ततक्रान्त्या)
(ग) ५. मडलांतरघटिकाज्यां for (मंडलान्तरं घटिकाज्या)
(च) २. हतां for (हतं) ४. हृत्यांतक्रांत्या for (हृतांततत्क्रान्त्या)
(ङ) ४. हृताज्त्यक्रान्त्या for (हृतां ततत्क्रान्त्या)

लब्धघनुरविरजाक्ष कर्कटकादौ विशोद्ध चक्रार्द्धात् ।
तज्या तदुदय सममंडलांतरा क्षुज्ययाभक्ता ।। २२ ।।
प्रश्न सममण्डला सुक्रमज्यया संगुणा सकृत्सूर्यः ।
प्रश्न घटिकाभिरेवं गतघटिकाभिर्भवत्यहः ।। २३ ।।
त्रिज्यादिनार्द्धसममंडलांतरा सुज्यायाः कृतिविशेषः ।
स्वविषयविषुवच्छाया वर्गेण गुणो द्विधा प्रथमाः ।। २४ ।।
व्यासार्द्धवर्गभक्तौ लब्धं ध्वादशकवर्ग संयुक्तम् ।
छेदो द्वितीयराशे लब्धं पदं क्रान्तिरर्कातः ।। २५ ।।

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२२. (घ) १. रजादौ (ग) (च) for (रजाक्षै) २. विशोध्य (ग) (च) for (विशोद्ध)
(च) ३. कर्क्कट for (कर्कट) ४. सुज्यया for (क्षुज्यया)
(ङ) ५. रिनो for (रवि) १. ऽजादौ for (रजाक्षै)
३. कर्क्याकौ यदि for (कर्कटकादौ) २. विशोध्य for (विशोद्ध)
६. तज्ज्या for (तज्या)

२३. (घ) १. गत शेषोभिर्भवत्यह्नः (ग) गतशेषाभिर्भवत्यह्नः for (गतघटिकाभिर्भवत्यहः)
(ग) २. प्रस्न for (प्रश्न) ३. प्रस्न for (प्रश्न)
(च) ४. सकृसूर्यः for (सकृत्सूर्य) ४. गतशेषाभि (ङ) for (गतघटिकाभि)
५. भवत्यह्नः (ङ) for (भवत्यहः)
(ङ) ४. ऽसकृत् सूर्यः for (सकृत्सूर्यः)

२४. (घ) १. सुज्ययोः (ग) (ग) (ङ) for (सुज्यायाः)
२. विषुछाया (ग) (च) for (विषुवछाया)
३. प्रथमः (ग) (च) (ङ) for (प्रथमाः)
(ग) ४. दिनासम for (दिनार्द्धसम) ५. स्वविष for (स्वविषय)
(च) २: विषुवच्छायावर्गेण for (विषुवछायावर्गेण)

२५. (घ) १. वर्ग्र for (वर्ग) (ग) वर्ग्रभक्तो for (वर्गभक्तौ)
२. द्वादश (ग) for (ध्वादश) ३. संयुक्ताम् for (संयुक्तम्)
४. लेंब्धपदं (ग) (च) (ङ) for (लब्धंपदं)
(च) ५. रक्कातिः for (रकातिः)
(ङ) १. व्यासार्ध वर्गभक्तो for (व्यासार्द्धवर्ग भक्तौ)
२. द्वादशज for (ध्वादशक) ४. रर्कोऽतः for (रकातिः)

सममण्डल शंकुगुणाक्षज्या[3] जीनभागजीवया[1] भक्ता[2] ।
फलधनुरर्कों[4] जादौ कर्क्यादौ प्रोह्य चक्रार्द्धात् ॥ २६ ॥
द्वादशगुणिताक्षज्या[2] विषुवच्छायो[3] गुणावलंबज्या ।
सममंडल कर्कादौ[1] क्रान्तिज्ये भास्करः प्राग्वत् ॥ २७ ॥
परमापक्रमजीवा तत्कालिकसूर्यबाहुसंगुणिता[1] ।
सममंडलशंकुह्रताक्षज्या[2] तच्चापमक्षांशाः[3] ॥ २८ ॥
लंकोदयचरदलवद्रविलग्नाभ्यां पृथक् पृथक् प्राणाः[1] ।
कृत्वा तदंतरैक्यं मृगकर्क्यादिषु[2] पृथक्[3] लिप्ताः ॥ २९ ॥

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२६. (घ) १. दिन (ग) जिन for (जीन) २. भक्ताः (ग) for (भक्ता)
(ग) ३. क्षय्या for (क्षज्या)
(च) ३. जिनभाग for (जीनभाग) ४. धनुरर्क्कों for (धनुरर्कों)
(ङ) ३. ऽक्षज्या for (क्षज्या) १. जिन for (जीन)
४. रर्कोऽजादौ for (रर्कोजादौ)

२७. (घ) १. कर्णाकृते (ग) कर्णाहृते (ङ) for (कर्कादौ)
(ग) २. क्षय्या for (क्षज्या)
३. विषुवच्छाया for (विषुवच्छायो (च) for (विषुवच्छायो)
(च) १. कर्णाहृते for (कर्कादौ)
(ङ) २. ऽक्षज्या for (क्षज्या) ३. विषुवच्छाया for (विषुवच्छायो)

२८. (घ) १. तात्कालिक for (तत्कालिक)
(ग) २. हृता for (ह्रता) ३. क्षय्या for (क्षज्या)
(ङ) ३. ऽक्षज्या for (क्षज्या)

२९. (घ) १. प्राणान् (ग) (च) for (प्राणाः)
२. मृगकक्ष्यादिषु (ग) मृगकर्कादिषु for (मृगकर्क्यादिषु)
३. पृथग्लिप्ताः (ग) (च) (ङ) for (पृथक्लिप्ताः)
(ग) ४. लंकोदचरदलद्रवि for (लंकोदयचरदलवद्रवि)
(च) २. मृगकक्ष्यादिषु for (मृगकर्क्यादिषु)
(ङ) १. प्राणान् for (प्राणाः)

मेषादिषु[१] कर्कादिषु[२] विशोध्य[३] भार्द्धा[४] तुलादिषु[५] सभार्द्धा[६]।
मकरादिषु[७] संशोध्याश्चक्रविं[८] भुक्ता[९] लिप्ता वा ॥ ३० ॥
लग्नकालायद्यूना[१] सचकालिप्ता[२] विना[३] स्वराश्यूदयैः[४]।
एवं स्फुटाभवंत्यर्क[५] लग्नयोरंतरप्राणाः[६] ॥ ३१ ॥
अष्टयमाश्रुत्य[१] गुणा[२] दिगिषु कालोना[३] रदाः सतिथिलिप्ताः।
स्वचरार्द्धांशैरूना[४] विपरीता[५] संयुता व्यस्तै[६] ॥ ३२ ॥

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३०. (घ) १. कक्यादिषु (ग) कर्क्यादिषु (ङ) for (कर्कादिषु)
२. भार्द्धात् (ग) for (भार्द्धा) ३. सभार्द्धाः (ग) for (सभार्द्धा)
४. चक्राद्रवि (ग) (च) (ङ) for (चक्ररवि)
५. भुक्तलिप्तोनाः (ङ) (ग) भुक्तलिप्तोना for (भुक्तालिप्तावा)

(च) १. कक्वर्चादिषु for (कर्कादिषु) ६. तुलादीषु for (तुलादिषु)
३. "सभार्द्धा" लुप्त ५. भुक्तलिप्तोनाः for (भुक्तालिप्तावा)

(ङ) ६. शोध्या for (विशोध्य)

३१. (घ) १. लग्नकला (ग) (च) (ङ)for (लग्नकाला)
२. यद्यूनाः (ग) (च) (ङ)for (यद्यूना)
३. सचक्रलिप्ता (ग) (ङ) for (सचकालिप्ता)
४. राश्युदयैः (ग) रास्युदयैः for (स्वराश्यूदयैः)

(ग) ५. दिना for (विना)

(च) ३. सचक्रलिप्ताः for (सचकालिप्ता) ४. स्वराश्युदयैः for (स्वराश्यूदयैः)
६. भवंत्यर्क्क for (भवंत्यर्क)

(ङ) ४. स्वराश्युदयैः for (त्वराश्यूदयैः) ७. रंतते for (रंतर)

३२. (घ) १. यमाः (ग) (च) (ङ) for (यमा)
२. शून्य (ग) (च) (ङ) for (श्रुत्यगुणा)
३. कलोना (ग) (च) (ङ) for (कालीना) ४. रूनाः (च) for (रूना)
५. व्यस्तैः (ग) (च) (ङ) for (व्यस्तै)

(ग) ६. विपरीताः (ङ) for (विपरीता)
(च) ६. विपरिताः for (विपरीता)

व्यस्ता[१] चा जादीनां[२] कलांशलग्नमिष्टघटिकांशैः ।
लग्ना[३] घटिकाः कालांशै[४] विनैवं[५] स्वराश्युदयैः[६] ॥ ३३ ॥
इष्टार्क[४] चरार्द्धं[२] ज्या[१] क्षयवृद्धिज्याद्युरात्रिदलगुणिता[३] ।
व्यासार्द्धेन विभक्ता क्षितिजा द्वादशगुणा भक्ता ॥ ३४ ॥
क्रांत्याविषुवच्छाया क्षितिजेष्ट क्रान्तिवर्ग[१]योगं पदम् ।
अग्रे[२] क्षितिजापक्रमजीवे[३] त्रिज्या गुणो[४]भक्ता ॥ ३५ ॥
अर्काग्रयाक्षकलंबकजीवे[४] दिनकृतचरासु[१]विज्ञाने ।
अर्कज्ञाने[२] ज्ञाने[५] विषुवच्छाया चराभू[३]नाम् ॥ ३६ ॥

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३३. (घ) १. व्यस्तश्चाजादीनां (ग) (च) (ङ) for (व्यस्ताचाजादीनां)
२. कालाँशा (ग) कालांशौ for (कलाँश)
३. लग्नाद्घटिकाः (ग) (च) (ङ) for (लग्नाघटिकाः)
४. कालांशकै (ग) (च) (ङ) for (कालांशै)
५. विनैवं (ग) (च) (ङ) for (विनैवं)
६. स्वराश्युदयैः (ग) (च) (ङ) for (स्वराश्यूदयैः)
(च) २. कालांश for (कलांश) (ङ) २. काजांशैर्लग्न for (कलांशलग्न)
३४. (घ) १. क्षयं for (क्षय)
(ग) २. चरार्द्धज्या (ङ) for चरार्द्धज्या) ३. गुणिताः for (गुणिता)
(च) ४. इष्टार्क्क for (इष्टार्क) २. द्याक्षय for (ज्याक्षय)
(ङ) ५. द्यु रात्रदल for (द्यु रात्रिदल)
३५. (घ) १. वर्गयोगपदम् (ग) (च) for (वर्ग्ययोगंपदम्)
२. अग्ना (ग) अग्रा for (अग्रे) ३. जीव (ग) जीवा for (जीवे)
४. गुणेभक्ते (ग) (च) (ङ) for (गुणोभक्ता)
(च) २. अग्रक्षिति for (अग्रे क्षिति) ३. जीवत्रीज्या for (जीवेत्रिज्या)
(ङ) १. वर्गयोगपदम् for (वर्ग्ययोगंपदम्) २. अग्रा for (अग्रे)
३६. (घ) १. कृचरा for (कृतचरा) (ग) दिनक्रच्चरासुविज्ञाते for (दिनकृतचरासुविज्ञाते)
२. अर्का (ग) for (अर्क) ३. चरासूनाम् (ग) (ङ) for (चराभूनाम्)
४. 'क' यहां 'क' लुप्त है (च) (ङ) ५. ज्ञाते ज्ञाते for (ज्ञाने ज्ञाने)
(च) ६. अक्कर्ग्र for (अर्काग्र) ४. लंबक for (कलंबक)
१. दीनकृच्चरा for (दिनकृतचरा) २. अक्कर्ज्ञाने for (अर्कज्ञाने)
३. चरासूनां for (चराभूनाम्)
(ङ) ६. अर्काग्रयाऽक्ष for (अर्काग्रयाक्ष) १. कृच्चरासु for (कृतचरासु)
२. अर्कज्ञाने for (अर्कज्ञाने) ७. विषुवच्छाया for (विषुवच्छाया)

अष्टचरार्द्धयस्य[१] ज्या क्षयवृद्धि ज्यातदर्कवधकृत्या[२]।
त्रिज्याविषवच्छाया[५] वधवर्गो[३] युतवधछेदः[४] ॥ ३७ ॥
व्यासार्द्धक्रते[१] मूलं क्रांतिज्यावासदलगुणा[२] भक्ता ।
जीनभागजीवया[३] लब्धचापमर्कपादैः[४] प्राग्वत् ॥ ३८ ॥
विषुवच्छाया[२] भक्ता स्वचरार्द्धज्येष्टयात्यया[१] भक्ता गुणिता ।
लब्धस्य चापमिष्ट[३] छायायाश्चरदलप्राणाः ॥ ३९ ॥
स्वचरार्द्धज्या[१]भक्ता विषुवच्छायेष्ट[४] चरदल[२] श्रुनाम् ।
गुणिताज्येष्ट[३] चरदल विषुवच्छाया[४] फलं भवति ॥ ४० ॥

३७. (घ) १. चरार्धस्य (ग) इष्टचरार्द्धस्य for (अष्टचरार्द्धयस्य)
२. कृत्याः (ग) क्रत्या for (कृत्या) ३. ववर्गो for (वधवर्गो)
४. हृताछदः (घ) हृतछेदः for (युतवधछेदः)
(ग) १. अष्टचरार्द्धस्य ज्या for (अष्टचरार्द्धयस्य ज्या) २. क्रत्या for (कृत्या)
४. हृतछेदः for (वधछेदः)
(ङ) १. इष्टचरार्धस्य for (अष्टचरार्द्धयस्य) ५. विषुवच्छाया for (विषूवछाया)
४. युतहृतश्छेदः for (युतवधछेदः)

३८. (घ) १. कृतेर्मूलं (ग्र) भुक्तेर्मूलं for (क्रतेमूलं) २. व्यास (ग) (ङ) for (वास)
३. जिन (ग) (च) (ङ) for (जीन) ४. मर्कः for (मर्क)
५. पदैः (ग) (ङ) for (पादैः)
(ग) ६. लब्धं for (लब्ध)
(च) १. कृते for (क्रते) ४. मर्क्कः for (मर्क)
(ङ) १. कृतेर्मूलं for (क्रतेमूलं)

३९. (घ) १. व्यया गुणिता (ग) न्यया गुणिता for (त्यया भक्ता गुणिता)
(च) १. ज्येष्टयान्यया गुणिता for (ज्येष्टयात्ययाभक्ता गुणिता)
(ङ) २. विषुवच्छाया for (विषुवछाया) १. ऽन्यया for (त्यया)
३. मिष्टच्छायाया for (मिष्टछायाया)

४०. (घ) १. चरार्द्धज्या (ग) व्यचरार्द्धज्या for (चरार्द्धेज्या)
२. चरदलासूनाँम् (ग) (ङ) for (चरदलश्नुनाम्)
(ग) ३. ज्ययैष्ट for ज्ययेष्ट)
(च) २. चरदलामूनां for (चरदलश्नुनाम्) ३. द्येष्ट for (ज्येष्ट)
(ङ) १. चरार्धज्या for (चरार्द्धेज्या) ३. ज्ययेष्ट for (ज्येष्ट)
४. विषुवच्छायेष्ट for (विषुवछायेष्ट)
५. विषुवच्छाया for (विषुवछाया)

मध्यमछायाग्रमुदक् शंकुतलादक्षनतभागः ।
दक्षिणतो यदि सौम्याः स्वाक्षांशा सर्वदा याम्याः ॥ ४१ ॥
द्युदलनतोक्षांशानामेक दिशानामंतरं युतिर्भेदे ।
क्रांत्यंशाः प्रागदिनः क्रांत्यंशोरेव मक्षांशाः ॥ ४२ ॥
उदये ज्येष्ठापक्रम जीवा कृत्यंतरात्पदं क्षितिजा ।
व्यासार्द्ध गुणा क्षितिजा भक्तोदयजिवया कक्षः ॥ ४३ ॥

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४१. (घ) १. मध्यछाया (ग) for (मध्यमछाया)
२. तलादृक्षिणे (ग) तलादृक्षिणोनतभागाः for (तलादृक्षिणा नता भागाः)
३. नता (ङ) for (नता) ४. भागाः (ङ) (भागः)
५. स्वाक्षांशाः (च) (ङ) for (स्वाक्षांशा)
(ग) ६. दक्षणतो for (दक्षिणतो) ७. याम्या for (याम्याः)
(च) १. मध्यछायाग्रमुदक् for (मध्यमछायाग्रमुदक्)
६. दृक्षिणतो for (दक्षिणतो)
(ङ) १. मध्यच्छाया for (मध्यमछाया) २. दृक्षिणा for (दक्ष)

४२. (घ) १. नतांक्षा शानामेक (ग) नतः क्षांशानामेक for (नतोक्षांशानामेक)
२. भेदो (ग) भेदै for (भेदे)
३. प्राग्वदिनः (ग) प्राग्वदितः for (प्रागदिनः)
४. क्रांत्यंशैरेव (ग) (च) (ङ) for (क्रान्त्यंशोरेव)
(ग) ५. 'युति' लुप्त है
"क्रांत्यंशाः प्राग्वदितः" दो बार लिखा गया है।
(च) १. द्युदलनताक्षांशानामेक for (द्युदलनतोक्षांशानामेक)
३. प्राग्वदिनः for (प्रागदिनः)
(ङ) १. नताक्षांशानामेक for (नतोक्षांशानामेक)
६. दिशामन्तरं for (दिशानामंतरं) ३. प्राग्वदनः for (प्रागदिनः)

४३. (घ) १. उदय (ङ) for (उदये) २. क्षितिजाल्पा for (क्षितिजा)
३. जीवयाक्षज्या (च) for (जिवयाकक्षः)
(ग) 'म' लुप्त
४. पक्षज्यात् शेषाः ॥४३॥ for (कक्षः)
(च) ३. जीवयाक्षज्या for (जिवयाकक्षः)
(ङ) ५. व्यासार्ध for (व्यासार्द्ध) ३. जीवयाऽक्षज्या for (जिवयाकक्षः)

उदय ज्ययाविभक्ता क्रांतिज्या व्यासगुणालंबः ।
द्वादशगुणिता क्षितिजा विषुवच्छाया हृता क्रान्तिः ॥ ४४ ॥
यष्टिव्यासार्द्धेग्राप्राच्यपरा भास्करांतरांशज्या ।
द्विगुणमुदया सूत्रं तत्त्रिज्या कृतिविशेषपदम् ॥ ४५ ॥
द्यु दले शंकु न तज्ये प्राच्यपराया यदि स्थितः शंकुः ।
उदगूना दक्षिणतः स्तदंतरेणाधिकार्काग्रा ॥ ४६ ॥
उत्तरगोले नं तदं याम्य गोलगे सूर्ये ।
शंकुतल शंकु हूतं विषुवच्छाया द्विषट्क गुणम् ॥ ४७ ॥

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४४. (घ) १. हृता (ग) (च) for (हूता)
(ग) २. ज्याया for (ज्यया) ३. +दल+ (ङ) (च)
४. लब्धाम् for (लम्बः)
(ङ) ५. विषुवच्छाया for (विषुवच्छाया)

४५. (घ) १. ऽग्रा (च) for (ग्रा)
(ग) २. 'तरां' पद लुप्त है
३. मुदयास्तसूत्रं (ङ) for (मुदयासूत्रम्)
४. क्रांति for (कृति)
(च) २. ज्याः for (ज्या)
(ङ) १. व्यासार्धेऽग्रा for (व्यासार्द्धेग्रा)

४६. (घ) १. दक्षिणत (ग) for (दक्षिणतः)
२. ऽधिका for (धिका)
(ग) ३. द्यू दले for (द्यु दले) ४. शंकुतज्ये for (शंकुनतज्ये)
५. उदगुना for (उदगूना) ६. काकाग्रा for (कार्काग्रा)
(च) ७. तच्छे for (तज्ये) १. दक्षिणांत for (दक्षिणतः)
६. ऽधिकाक्कार्काग्रा for (धिकार्काग्रा)
(ङ) १. दक्षिणत for (दक्षिणतः)

४७. (घ) १. गोलेग्रों (ग) उत्तगोलेग्रो for (उत्तर गोले)
२. तलं (ग) (ङ) for (तल)
३. हृतं (ग) (ङ) (च) for (हूतं)
(ग) ४. मंतदंतरं for (नं तदंतरं) ५. गोले for (गोलगे)
(च) १. गोलग्रोंनं for (गोलेनं) ३. तदंतरं for (तदं)
७. विषुवच्छाया for (विषुवछाया)

शंकुतलशंकुगणिते त्रिज्ये तद्वर्गयुतिपदविभक्ते ।
अक्षावलंबज्ये द्युदलस्थेऽर्कन्यदा द्युदलात् ॥ ४८ ॥
छायावृतेऽर्कग्रा कर्णगुणा व्यासदल हृतार्काग्रा ।
प्राच्यपरा शंक्वंतरमुत्तरयाम्यं तदून युता ॥ ४९ ॥
उत्तर गोले याम्ये विषुवच्छायाग्रयांतरं हीनम् ।
एवं विषुवच्छाया युक्तिविहीनांतरेणाग्रा ॥ ५० ॥
बाहुक्रांतिः कोटिः क्षितिजातद्वर्गयुतिपदं कर्णः ।
अग्रोदयास्त सूत्राद्दक्षिणतो द्रश्य शंकु तलम् ॥ ५१ ॥

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४८. (घ) १. तर्द्वर्ग for (तद्वर्ग)
२. अक्षावलंबकज्ये (ग) अक्षावलंबकज्ये (ङ) for (अक्षावलंबज्ये)
३. ऽर्केऽन्यदा (ग) स्छेऽकेन्यद for (ऽर्कन्यदा)
(ग) ४. द्यूदलात् for (द्युदलात्)
(च) १. तर्द्वर्ग for (तद्वर्ग) २. अक्षावलंबकद्ये for (अक्षावलंबज्ये)
३. ऽर्क्केऽन्यदा for (ऽर्कादान्यदा)
(ङ) ३. ऽर्क्केऽन्यदा for (ऽर्कन्यदा)

४९. (घ) १. वृत्तऽर्काग्रा (ग) (ङ) for (वृतेऽर्कग्रा)
२. हृता for (हृता)
(ग) ३. शंकुतर for (शंक्वन्तर) ४. तदूनं for (तदून)
५. युतम् for (युता)
(च) १. छयावृत्तेऽक्कार्काग्राकर्णा for (छायावृत्तेऽर्कग्राकर्णा)
२. हृताक्कार्ग्रा for (हृतार्काग्रा) ४. तदूना for (तदून)
(ङ) २. हृताऽर्काग्रा for (हृतार्काग्रा)

५०. (घ) १. युक्त (ग) (च) (ङ) for (युक्ति)
(ग) २. विहिनां वा रणाग्रा (विहीनांतरेणाग्रा)
(ङ) ३. विषुवच्छाया for (विषुवछाया) ४. ऽग्रयान्तरं for (ग्रयांतरं)
५. विषुवच्छाया for (विषुवछाया)
२. विहीनाऽन्तरेणा for (विहीनांतरेणाग्रा)

५१. (घ) १. द्रश्य (ग) for (द्रश्य)
(ग) २. क्षितिजां for (क्षितिजा) ३. अग्रोदयास्त्र for (अग्रोदयास्त)
४. दक्षणतो for (दक्षिणतो)
१. द्रश्य (ङ) for (द्रश्य)
(ङ) ५. बाहुः for (बाहु)

त्रिज्याक्षयवृद्धिघ्नैः कांतर मुदगितरयोर्दिनार्द्धात्या ।
व्यासार्द्धचरार्द्ध ज्यांतर संयोगोर्द्धरात्र्यंत्या ॥ ५२ ॥
उक्रमजीवा चापं क्रमजीवाचापसहितमधिकं यत् ।
दिनरात्र्यर्द्ध प्राणाः पृथग्दिना चरदलप्राणैः ॥ ५३ ॥
दिवसार्द्धोत्क्रमजीवाधिकक्रमज्याधिकादिनार्द्धत्या ।
व्यासार्द्ध दिनार्द्धात्यांतर चरज्याक्षयजीवातः ॥ ५४ ॥

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५२. (घ) १. द्यैक्यांतर (ग) ज्यैक्यांतर for (द्यैकांतर)

(ग) २. दिनार्द्धाद्धा for (दिनार्द्धात्या) ३. दिनार्द्ध for (चरार्द्ध)
४. ज्यांतरं for (ज्यांतर) ५. संयोगेर्द्ध for (संयोगोर्द्ध)

(च) २. द्विनार्द्धात्या for (दिनार्द्धात्या)

(ङ) १. ज्यैक्यान्तर for (द्यैकान्तर) २. दिनार्धान्त्या for (दिनार्द्धात्या)
३. व्यासार्ध चरार्ध for (व्यासार्द्ध चरार्द्ध)
५. संयोगेऽर्धरात्र्यन्त्या for (संयोगोर्द्ध रात्र्यंत्या)

५३ (घ) १. उत्क्रम (ग) (च) for (उक्रम)

(ग) २. तीवा for (जीवा) ३. मधीकं चेत् for (मधिकं यत्)
४. विमा for (दिना)

(ङ) १. उत्क्रम for (उक्रम) ५. चेत् for (यत्)
४. पृथग् विना for (पृथग्दिना)

५४. (घ) १. नार्द्धात्या (ग) (च) for (नार्द्धात्या)
२. त्यांतरं (ग) (च) for (त्यातर)

(ग) ३. दिवासार्कोत्क्रम for (दिवसार्द्धोत्क्रम) ४. ज्यया for (ज्या)
५. 'य' लुप्त है (ङ)

(च) ५. चरद्याक्षजीवातः for (चरज्याक्षयजीवातः)

(ङ) ३. दिवसार्धोत्क्रम for (दिवसार्द्धोत्क्रम) ४. ज्याऽधिका for (ज्याधिका)
१. दिनार्धान्त्या for (दिनार्द्धात्या)
२. व्यासार्ध दिनार्धार्द्धन्त्यान्तरं for (व्यासार्द्ध दिनार्द्धात्यांतर)
५. ऽक्षजीवाऽतः for (क्षयजीवातः)

स्वाहोरात्रार्द्धसम[१] यत्राक्षज्याबलंबकः[२]।
क्रान्त्या[३] ह्येषा[४] दिगम्यता[५] वद्यावत्[६] कर्क्यादिगस्य खेः ॥ ५५ ॥
नास्तमयस्तत्र[३] तुला मकरादिस्थस्य[४] नोदयोर्कस्य[२][१]।
तन्मध्यांतरलिप्ता मध्यमभुक्त्या हृता[२] दिवसाः ॥ ५६ ॥
कोणछाया[८] क्रतिदल[७] यदविषुवछायाया[१] रूदानृतबलं[५] प्राच्य[२] परयोः[३]।
यद्यैक्यंतरं[४] याम्यदिक्स्थं[६] चेत् ॥ ५७ ॥

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५५. (घ) १. समा (ग) for (सम)
२. ज्यावलंबकः क्रान्त्या for (ज्याबलंबकः क्रान्त्या)
३. यह पद 'पहली पंक्ति के अन्त में है।
४. मेषादिगस्य (ग) for (ह्येषा दिगम्यता)
५. तावद्यावत् for (वद्यावत्) ६. कर्कादिगस्य for (कर्क्यादिगस्य)
(च) १. समा यात्रा for (सम यत्रा) ४. मेषादिगस्य for (ह्येषादिगम्य)
६. कर्क्यादिगस्य for (कर्क्यादिगस्य)
(ङ) १. रात्रार्धसमा for (रात्रार्द्धसम)
३. 'क्रान्या' यह शब्द पहली पंक्ति का अन्तिम है।
४. ५. मेषादिगस्य तावद्यावत् for (ह्येषा दिगम्यता वद्यावत्)

५६. (च) १. नोदयोऽर्कस्य (ग) नोदयेऽर्कस्य for (नोदयोर्कस्य)
२. हृता (ग) (ङ) for (हता)
(ग) ३. सूत्रतुला for (स्तत्र तुला) ४. छस्य for (स्थस्य)
(च) १. नोदयोऽङ्कस्य for (नोदयोर्कस्य) २. हृता for (हता)
(ङ) १. नोदयोऽर्कस्य for (नोदयोर्कस्य)

५७. (घ) १. पद (ग) (च) (ङ)for (यद) २. रुदग्नृतलं (ग) (ङ) for (रुदानृतल)
वि० श्लोक की पहली पंक्ति समाप्त।
३. 'प्राच्य परयोः' दूसरी पक्ति के आरंभिक पद हैं (ग)
४. यद्यैक्यमंतरं (ग) (च) (ङ) for (यद्यैक्यंतरं)
(ग) ५. छाययो for (छायाया) ३. प्राच्य पराया (ङ) for (प्राच्यपरयोः)
वि० दूसरी पंक्ति का आरम्भ।
६. स्छ for (स्थं)
(च) ७. कृतिदल (ङ) for (क्रतिदल) २. रुदग्नृतबलं for (रुदानृतबलं)
(ङ) ८. कोणच्छाया for (कोणछाया) ५. विषुवच्छाययो for (विषुवछायाया)

कोणच्छायाकर्णेन भक्तमवलंबकेन संगुणितम् ।
इष्टापक्रमजीवा त्रिप्रश्नोक्त्या स्फुटोर्कोतः ॥ ५८ ॥
मध्यमगति स्फुटगति त्रिप्रश्नात् सोत्तरान् विजानाति यः ।
स भवत्याचार्यो ब्रह्मोक्ता ज्ञानतंत्रज्ञाः ॥ ५९ ॥
अध्यायः पंचदशस्त्रिप्रश्नस्योत्तरं यदीह नोक्तम् ।
आर्याषष्टचोक्तं तद्गोलादुत्प्रेक्ष्य बुद्धिमता ॥ ६० ॥
इत्तिब्रह्म गुप्ते त्रिप्रश्नाध्यायः
पंचदश अध्यायः समाप्तः

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५८. (ङ) २. कोणच्छाया for (कोणछाया) १. sकोंऽतः for (कोंतः)

५९ (घ) १. 'यः' दूसरी पंक्ति का पहला पद है ।
२. तंत्रज्ञः (च) for (तंत्रज्ञाः)

(ग) ३. मध्यगतिः for (मध्यमगति)
४. स्पष्टगतिः for (स्फुटगति) १. 'यः' लुप्त है
५. वै ब्रह्मोक्तानान्य तंत्रज्ञा for (ब्रह्मोक्ताज्ञानतंत्रज्ञाः)

(ङ) ३. मध्यगति for (मध्यमगति) ४. स्पष्टगति for स्फुटगति)
६. त्रिप्रश्नान् for (त्रिप्रश्नात्) १. 'यः' लुप्त ।
५. ब्रह्मोक्तान् for (ब्रह्मोक्ता) २. योऽन्यतंत्रज्ञः for (ज्ञानतंत्रज्ञाः)

६०. (घ) १. यदिह (ग) (ङ) for (यदीह) २. दुत्प्रेक्ष्य for (दुत्प्रेक्ष्य)
३. बुद्धिमताः for (बुद्धिमता)
वि०—४. 'इति' से आरंभ कर 'समाप्त' तक मूल में लुप्त है ।

(ग) ५. त्रि for (स्त्रि) ६. तच्चार्याषष्टचो यं for (आर्याषष्टचोक्तं)
४. इति ब्रह्मसिद्धांते त्रिप्रश्नोत्तर पंचदशोध्यायः for (इत्ति ब्रह्मगुप्ते त्रिप्रश्नाध्यायः)

(च) १. यदिह for (यदीह)
४. 'इति' से 'समाप्तः' तक समस्त लुप्त

(ङ) १. तच्चार्याषष्टचाऽयं for (आर्याषष्टचोक्तं)
२. गोलादुत्प्रेक्ष्य for (तद्गोलादुत्प्रेक्ष्य)

अथ ग्रहणोत्तराध्यायः

षोडशः

ग्रहणग्रहसंयोगग्रहः क्षीयोगेषु सर्वतन्त्रविदाम् ।
आचार्यछेद्यक विद्यतस्ततः छेदकं वक्षे ।। १ ।।
दुर्जनकृतघ्नशत्रु प्रतिकंचुक करिपतितमुर्खेभ्यः ।
छेदकमदेयम्योददतः सुकृतायुषोनाशः ।। २ ।।
उषिताय दीर्घकालशिष्याय गुणान्विताय भक्ताय ।
भ्रात्रे वास्तुह देवा सुताय वा छेदकं देयम् ।। ३ ।।

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१. (घ) १. ग्रह (ङ) (ग) for (ग्रहः) २. छेद्यकं (च) (ङ) for (छेदकं)

(ग) ३. क्षतंत्रेषु for (क्षीयोगेषु) ४. 'तंत्र' लुप्त है, (ङ)
५. आचार्यः for (आचार्य) ६. छेदक for (छेद्यक)
७. तत् for (ततः) ८. वक्ष्ये (ङ) for (वक्षे)

(ङ) ३. क्षतंत्रेषु for (क्षीयोगेषु) ५. आचार्यश् for (आचार्य)
७. ततश् for (ततः)

२. (घ) १. कारि (ग) (ङ) for (करि) २. मूर्षेभ्यः (ग) for (मुर्खेभ्यः)
३. छेद्यक (ग) (ग) (ङ) for (छेदक) ४. यमेभ्यो (ग) (ङ) for (यम्यो)
५. नशि (ग) नाशः for (नशिः)

(ग) ६. दहतः for (ददतः)

(च) १. कारिपतित for (करिपतित) २. मूर्षेभ्यः for (मुर्खेभ्यः)
४. मदेयमेभ्यो for (मदेयम्यो)

(ङ) २. मूर्खेभ्यः for (मुर्खेभ्यः)

३. (घ) १. दीर्घकालं (ग) (च) (ङ) for (दीर्घकाल)
२. सुहृदेवा (ग) for (स्तुहृदे) ३. छेद्यकम् (च) (ङ) for (छेदकं)

(ग) ४. गुणाधिकाय (ङ) for (गुणान्विताय)

(च) २. वासुहृदेवा for (वास्तुहृदेवा)

(ङ) ५. पात्रे for (भ्रात्रे) २. सुहृदे वा for (स्तुहृदे वा)

विषुवदयमंडलदिशो वलनज्याभि ग्रहास्त्रिग्रहवृत्ते ।
संपर्के ग्राह्यवायो वेत्ति छेदकज्ञः सः ॥ ४ ॥
संपर्कमण्डले यः प्रग्रहमोक्षोपृथक् स्वविक्षेपात् ।
मध्यान्मध्यग्रासं परिलिखित छेदकज्ञः सः ॥ ५ ॥
परिलितीष्टग्रासं तत्कालिक बाहुकोटिकर्णो यः ।
अथवा निमिलनोन्मीलनद्वयं छेदकज्ञः सः ॥ ६ ॥

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४. (घ) १. वदपमंडल (च) (ङ) for (वदयमंडल)
२. ग्रहात्रि (ग) गृत्रिगृहवृत्ते for (ग्रहास्त्रिग्रहवृत्ते)

(ग) ३. ग्रास्यै for (ग्राह्य)

(च) २. ग्रहात्रिगृहवृत्ते for (ग्रहास्त्रिग्रहवृत्ते) ४. छेद्यकज्ञः (ङ) for (छेदकज्ञः)

(ङ) २. भिस्त्रिगृहवृत्ते for (भिग्रहास्त्रिग्रहवृत्ते) ५. संपर्के for (संपक्के)
३. ग्रासं for (ग्राह्य)

५. (घ) १. परिलिषिति (ग) पिरिलिषति for (परिलिखित)

(ग) २. मोक्षे for (मोक्षो) ३. प्रथक् for (पृथक्)

(च) ४. संपक्के for (सम्पर्क) १. परिलिषिति for (परिलिखित)

(ङ) २. मोक्षौ for (मोक्षो) १. परिलिखति for (परिलिखित)
५. च्छेदकज्ञः for (छेदकज्ञः)

६. (घ) १. परिलिषतीष्ट (ग) परिलिषतीष्ट (च) for (परिलितीष्ट)
२. कर्णैर्यः (च) (ङ) for (कर्णोयः)
३. निमीलनोन्मीलन (ग) (च) for (निमिलनोन्मीलन)
४. छेद्यकज्ञः (च) (ङ) for (छेदकज्ञः)

(ग) ५. तात्कालिबाहुकोटि (ङ) for (तात्कालिकबहुकोटि)

(च) ५. तत्कालि for (तत्कालिक)

(ङ) १. परिलिखतीष्ट for (परिलितीष्ट)

ग्राह्यं परिलिखितैक्यं[१] परिलिखति[२] ग्रहादिकं[३] तत्र ।
भूमौ यफलके[६] वा पर्वर्त[४] छद्म(द्य)कज्ञः[५] सः ॥ ७ ॥
देशांतरं यथा[१] द्रक्[२] प्रग्रहणांतराद्विजानाति ।
यो रेखातोध्वानं[६] पर्वेष्टुदिनात्[४] स तत्रज्ञः[५] ॥ ८ ॥
योवेति[१] राहुमार्गं तेनेष्टकालमिष्टकालाद्वा[३] ।
ग्रांसं ग्रासात्कालं[२] जानाति छेद्यकज्ञः[४] सः ॥ ९ ॥
व्यासप्रमाणयोगग्राह्यग्रहाकदलानि[५] वलनाद्या[१] ।
विक्षेपाश्चापरतः[२] पभृतिखे[४] पूर्वतः[३] शशिनः ॥ १० ॥

७. (घ) १. परिलिख्यैकं (ग) परिलेख्यैक्यं (ङ) for (परिलिखितैक्यं)
२. लिषति (ग) for (लिखति) ३. प्रग्रहादिकं for (ग्रहादिकं)
४. पर्वर्त्तंव (ग) परिवर्त्य (ङ) for (पर्वर्त)
५. छेद्यकज्ञः (ग) छेदकज्ञः for (छद्यकज्ञः)
(ग) ६. यः (ङ) for (य)
(च) १. परिलिख्यैकं for (परिलिखितैक्यं) ४. पर्वर्त्त for (पर्वर्त)
५. वछेद्यकज्ञः for (छद्यकज्ञः)
(ङ) ३. ग्रहगृहादिकं for (ग्रहादिकं) ५. च्छेद्यकज्ञः for (छद्म(द्य)कज्ञ)
८. (घ) १. यथागत (ग) ययागत for (यथा)
२. दृक्यग्रहणां (ग) for (द्रक्प्रग्रहणांतरा)
३. ऽध्वानं (ग) (च) (ङ) for (ध्वानं) ४. पर्केष्ट (च) for (पर्वेष्टु)
५. तंत्रज्ञः (ग) (ङ) for (तत्रज्ञः)
(ग) ६. योरखातो for (योरेखातो)
(च) १. यथागत for (यथा) २. दृक्थग्रहणांतरा for (द्रक्पग्रहणांतरा)
(ङ) १. यथागत for (यथा) २. दृक् for (द्रक्)
९. (घ) १. वेत्ति (च) (ङ) for (वेति) २. वाः for (वा)
(ग) ३. ग्रास (ङ) for (काल) ४. छेदकज्ञः for (छद्य(द्य)कज्ञः)
(च) ३. ग्रास for (काल) २. मिष्टकलाद्वाः for मिष्टकालाद्वा)
(ङ) ४. च्छेद्यकज्ञः for (छद्य(द्य)कज्ञः)
१०. (घ) १.ग्राहक (ग) (ग) (ङ) for (ग्रहाक)
२. वलनद्या (ग) चलनज्या for (वलनाद्या) ३. रवेः (ग) (च) (ङ)(रवे)
(ग) ४. प्रभृति for (पभृति) (च) २. चलनद्यां for (वलनाद्या)
(ङ) ५. ग्रास for (व्यास) २. वलनज्या for (वलनाद्या)
६. विक्षेपश्चापरतो for (विक्षेपाश्चापरतः) ४. भवति for (प्रभृति)

दिनदलविभक्तजित्तगुणदिनगतशेषाल्पजीवयेष्टगुणम् ।
त्रिज्यार्द्धमधिकमंगुललिप्तास्त्रिग्रहज्यया भक्तम् ।। ११ ।।
योनचेत्या द्वितया द्वितया दंगुललिप्ता स्त्रिसंगुणाष्टहृतात् ।
ज्याद्वितीययुक्तिभक्तात्तद्वितुष वयो दरैः षड्भिः ।। १२ ।।
व्यास वलनापर्वन मेकेनेष्टेना चार्यमितरेषाम् ।
अंगुलकलाभिरेवं शशिसितपरिलेख सूत्राणाम् ।। १३ ।।

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११. (घ) १. जिन (ग) (च) (ङ) for (जित्त) २. स्त्रिगृह (च) for (स्त्रिग्रह)
(ग) ३. जीवयेषु (ङ) for (जीवयेष्ट) ४. तृज्यार्द्ध for (त्रिज्यार्द्ध)
५. मंडल for (मंगुल)
(ङ) ४. त्रिज्यार्ध for (त्रिज्यार्द्ध)

१२ (घ) १. चेत्याद्वितया for (चेत्याद्वितयाद्वितया)
२. हृतात् (ग) स्त्रिहृतात् for (हृतात्)
३. द्वितय (ग) for (द्वितीय)
४. यवोदारैः (ग) यवोदरैः (ङ) for (वयोदरैः)

(ग) ५. ज्योना for (योन) १. चेज्याद्वितीयदंगुल for (चेत्या द्वितया द्वितया)
६. भक्तात् (ङ) for (भक्तात्त) ७. वितुषक for (द्वितुष)

(च) १. चेनत्याज्या for (चेत्या) ५. ज्योना for (योन)

(ङ) ५. ज्याना for (योन) ७. द्वितीय for (द्वितयाद्वितया)
८. संगणात् for (संगुणाष्ट) २. त्रिहतात् for (हृतात्)
३. ज्याद्वितययुक्ति for (ज्याद्वितीययुक्ति)

१३. (घ) १. वलनापवर्त्तन (ग) वलनाप्रवर्त्तन for (वलनापर्वन)
२. नवार्य for (नाचार्य) (ग) न कार्यमीतरेषाम् for (नाचार्यमितरेषाम्)
३. परिलेष (ग) (च) for (परिलेष)

(च) १. पर्वत for (पर्वन) २. मेकेनेष्टेनवर्य for (मेकेनेष्टेनाचार्य)
४. नित for (सित)

(ङ) १. वलनापवर्त्तन for (वलनापर्वन) २.मेकेनेष्टेन for (मेकेनेष्टेना)
५. कार्य for (चार्य)

प्रथमे वलनज्याभिर्[3]दिशो द्वितीयं[1] यथादिशं भानोः ।
आद्यंतौ विक्षेपौ मध्यान्मध्योन्यथा शशिनः[2] ॥ १४ ॥
विक्षेपाग्रात्[1] ग्राह्यं परिलिष्य[3] ग्राहकप्रमाणेन[4] ।
प्रग्रहमोक्षग्रासामुपरिलेख[2] भवंत्येवम् ॥ १५ ॥
पश्चापग्रहणे[1] प्राग्मोक्षे[4] रविबिम्बं[2] मध्यतो बाहुः ।
स्ववलन सिद्धायां[5] दिशि विपरीतशीतकरणमध्यात्[6][3] ॥ १६ ॥
भानुमते[1] बाह्वग्राद्यधा[2] दिशं कोटिरन्यथा[3] शशिनः[4] ।
रविशशिमध्यात् कर्णस्तिर्यक्कर्णाग्रकोटियुतेः[5] ॥ १७ ॥

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१४. (ग) १. द्वितीये (ङ) for (द्वितीयं) २. राशिनः for (शशिनः)
(ङ) ३. द्दिशो for (दिशो)

१५. (घ) १. विक्षेपाग्रादग्राह्यं (ग)·····विक्षेपाग्राह्यं for (विक्षेपाग्रात्ग्राह्यं)
२. भूपरिलेखे (ग) भूपरिलेषे for (भुपरिलेख)
(ग) ३. पिरिलिख्य for (परिलिष्य)
४. यहां 'प्रमाणेन' के साथ विसर्ग भी लगे हैं ।
(च) १. विक्षेपाग्रादग्राह्यं for (विक्षेपाग्रात्ग्राह्यं)
२. भूपरिलेषे for (भुपरिलेखे)
(ङ) १. शशिविक्षेपाग्रेभ्यः ग्राह्यं for (विक्षेपाग्रात्ग्राह्यं)
३. परिलिख्य for (परिलिष्य) २. भूपरिलेखे for (भुपरिलेखे)

१६. (घ) १. पश्चात्प्रग्रहणे (ग) पश्चात्प्रग्रहणात् for (पश्चापग्रहणे)
२. बिंब (ग) (च) (ङ) for (बिम्बं)
३. शीतकत (ग) शीतकरमध्यात् (ङ) for (शीतकरणमध्यात्)
(ग) ४. प्राक् मोक्षे for (प्राग्मोक्षे) ५. सिद्धज्या (ङ) for (सिद्धायां)
६. विपरीतः (ङ) for (विपरीत)
(च) १. पश्चाप्रग्रहणे for (पश्चापग्रहणे)
(ङ) १. पश्चात्प्रग्रहणे for (पश्चापग्रहणे)

१७. (घ) १. भानुमतो (ग) (च) (ङ) for (भानुमते)
२. बाह्वग्रा (ग) (च) for (बाह्वग्राद्यथा)
(ग) ३. कोटिरं for (कोटि) ५. तियकर्णा for (तिर्यक्कर्णा)
(ङ) भानुमतो for (भानुमते)

परिलेख[१] ग्राह्यस्य ग्राहकमानेन पूर्ववत्कृत्वा ।
तात्कालिकसंस्थानं निमीलनोन्मीलने चैवम् ॥ १८ ॥
विक्षेपगुण[१]त्रिज्या मानैक्यार्द्धो द्धृताप्तचापांशाः ।
आद्यंतयोर्यथादिशमर्कस्येंदौ[२] विपर्यस्था[३]ः ॥ १९ ॥
तत्स्ववलनांशक[१]योगांतर जीवाग्राह्यमानदलघ्नात्[२] ।
त्रिज्यालब्धज्याग्रे[३] ग्रग्रहमोक्षौ[४] प्राग्वर्केंद्वोः ॥ २० ॥
हुतया[४] व्यासार्द्धेनार्कचन्द्रमानार्द्धलिप्तिका[१] गुणया ।
मध्यवलनछाया[५] दक्षिणोत्तरा दिगतवा[२]मध्यात्[३] ॥ २१ ॥

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१८. (घ) १. परिलेषं (ग) (च) for (परिलेखं)

१९. (घ) १. गुणा (ग) गुण्या for (गुण)
२. मर्कस्येंदो (ग) (ङ) for (मर्कस्येंदौ)
३. विपर्ययस्ता (ग) विपर्यस्ताः (ङ) for (विपर्यस्थाः)

(च) १. गुणा for (गुण) २. मर्क्कस्येंदौ for (मर्कस्येंदौ)
३. विपर्ययस्ताः for (विपर्यस्थाः)
१. विक्षेपगुणा for (विक्षेपगुण)

२०. (घ) १. वलनांशयोगान्तर (ग) (ङ) for (वलनांशकयोगान्तर)
२. घातात् (च) (ङ) for (घ्नात्)
३. ग्रे (च) (ङ) for (ग्रे) ४. प्राग्वदर्केंद्वोः (ग) for (प्राग्वर्केन्द्वोः)

(च) १. वलनांशयोगांतर for (वलनांशकयोगांतर)
४. प्राग्वदर्क्केंद्वोः for (प्राग्वर्केंद्वोः)

(ङ) ५. ग्रहमोक्षौ (ग्रग्रहमोक्षौ) ४. प्राग्वदर्केन्द्वोः for (प्राग्वर्केंद्वोः)

२१. (घ) १. लिप्तका (च) for (हुतया) २. ज्यया (ग) for (छाया)
२. दिगनया (ग) दिगमनया मध्या for (दिगतवा)

(ग) ४. हतव्यासार्द्धेनार्कं for (हुतयाव्यासार्द्धेनार्कं)

(च) ४. दूतया व्यासार्द्धेनार्क्कं for (हुतया व्यासार्द्धेनार्कं)
३. दिगनया for (दिगतवा)

(ङ) १. हूतया for (हुतया) ५. मध्यमवलनज्या for (मध्यवलनछाया)
३. दिग्गमनया for (दिगमया) ६. मध्या for (मध्यात्)

प्राग्वत् प्रसार्य विक्षेपलिप्तिका ग्राहकप्रमाणेन ।
विक्षेपायाद्ग्राह्यं परिलिख्य ग्राह्यसंस्थानम् ॥ २२ ॥
त्रिज्या विक्षेपगुणा भक्तेष्टग्रासकर्णलिप्ताभिः ।
प्राग्वत् फलचाप स्ववलनांश योगांतरं तज्या ॥ २३ ॥
मानार्द्ध गुणाव्यासार्द्धभाजिता पूर्ववत् प्रसार्या स्यात् ।
कर्णं प्रसार्य मध्यादग्रं मध्येन कर्णाग्रात् ॥ २४ ॥
तात्कालिकसंस्थानं परिलिख्य ग्राहकप्रमाणेन ।
ग्राह्येन मीलोन्मीलने परिलेख एवं वा ॥ २५ ॥

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२२. (घ) १. विक्षेपाग्रादग्राह्यं (ग) for (विक्षेपायादग्राह्यं)
२. परिलिष्य (ग) परिलेक्ष for (परिलिख्य)
३. ग्रास (ग) (च) (ङ) for (ग्राह्य)
(ग) ४. प्राग्प्रसार्य for (प्राग्वत् प्रसार्य)
(च) १. विक्षेपाग्रादग्राह्यं for (विक्षेपायादग्राह्यं)
२. परिलिष्य for (परिलिख्य)
(ङ) १. विक्षेपाग्रात् for (विक्षेपायाद्)

२३. (घ) १. गुणाः (च) for (गुणा)
(ग) २. व्यास for (ग्रास) ३. चार्या for (चाप)
४. यथाजिवा for (तज्या)
(च) ५. त्रिद्या for (त्रिज्या) ४. तद्या for (तज्या)
(ङ) ४. तथा जीवा for (तज्या)

२४. (ग) १. दग्रकर्णेन for (दग्रं मध्येन) २. मध्याग्रात् for (कर्णाग्रात्)
३. ३+गुणा व्यासार्द्ध+
(ङ) १. दग्रं कर्णेन for (दग्रं मध्येन) २. मध्याग्रात् for (कर्णाग्रात्)

२५. (घ) १. परिख्य (च) for (परिलिख्य)
२. निमीलनोन्मीलने च (ग)..... नमीलनोन्मीलने च for (ग्राह्येन मीलोन्मीलने च)
(ग) ३. परिलेष for (परिलेख)
(च) ४. +व+
(ङ) २. एवं निमीलनोन्मीलने for (ग्राह्येन मीलोन्मीलने)
४. +च+

प्राच्यपरिविपरिते विपरीतं मध्यवलनमर्कंद्वोः ।
पूर्वदन्यत् सर्वं फल कोष्टौ ग्रहणपरिलेखः ॥ २६ ॥
दृग्गणितग्रग्रहयोरंतरघटीका फलं ग्रहे मध्यौ ।
देशांतरं धनं प्राक् प्रग्रहणे त क्षयः पश्चात् ॥ २७ ॥
प्रग्रहणांतर घटिकाभूपरिधिहता विभाजयेत् षष्टया ।
फलयोजनेश्चवंत्या प्राग्वत् प्रागपरयोर्देशः ॥ २८ ॥
पातोनखेभार्द्धात् च्चक्रोच्चोनाधिकाः कलाभक्ताः ।
तद्गतियुत्याप्तदिनै रासन्ने कर्स्य मासांते ॥ २९ ॥

२६. (घ) १. विपरीते (ङ) for (विपरिते) २. फलकेष्टौ (ग) for (फलके स्वे।
३. परिलेखाः (ग) (च) (ङ) for (परिलेखः)
(ग) ४. पार for (परि) ५. विपरितं for (विपरीतं)
(च) ४. पर for (परि) १. विपरीते for (विपरिते)
५. मर्क्कंद्वोः for (मर्कंद्वोः) ७. स्थर्वं for (सर्वं)
२. फलकौष्टौ for (फलकोष्टौ)
(ङ) ४. प्राच्यपरे for (प्राच्यपरि) ८. पूर्वंवदन्यत् for (पूर्वंदन्यत्)
२. फलके स्वे for (फलकोष्टौ)

२७. (घ) १. प्रग्रयो for (प्रग्रहयो) २. घटिका (च) for (घटीका)
३. मध्ये (ङ) for (मध्यौ) ४. तत्क्षयः for (तक्षयः)
(ग) ५. फलग्रहमध्ये for (फलंग्रहेमध्यौ) ६. 'त' लुप्त है।
(च) १. प्रग्रयो for (प्रग्रहयो) ३. मध्यो for (मध्यौ)
(ङ) २. घटिका for (घटीका) ४. क्षयं for (तक्षयः)
७. +तत्+

२८. (घ) १. योजनेष्ववंत्याः (ङ) for (योजनेश्चवंत्या)
(च) १. योजनेश्ववंत्याः for (योजनेश्चवंत्या)

२९. (घ) १. प्रतौनखेभार्द्धा (ग) पातार्कयुतेभार्द्धाच्च for (पातोनखेभार्द्धात्)
२. रासन्नेऽर्कस्य for (रासन्नेर्कस्य)
(ग) ३. क्रार्द्धो (घ) चक्राचौ for (चक्रोच्चो)
४. कलाभक्ता (ङ) for (कलाभक्ताः) ५. मासांत्ये for (मासांते)
(च) ३. चक्रार्च्चेनाधिकाः for (चक्राच्चोनाधिकाः) २. ऽर्क्कस्य for (कर्स्य)
(ङ) १. पातार्क युतिर्भार्धात् for (पातोनखेभार्द्धात्)
३. चक्राद्धोनाधिका for (चक्रोच्चोनाधिकाः)
२. रासन्नोऽर्कस्य for (रासन्नेर्कस्य)

पर्वेंदोः[१] पक्षांते प्रागाधिकोना[२] युतिर्भवति[४] पश्चात् ।
तन्मध्ये न ग्रहणं यदि भानोः पंच जिनभरसाः[३] ॥ ३० ॥
इंदोर्विषया[३] द्वितमा[१] दिवाकरा त्रिविषयास्तदुच्चस्य[४][५] ।
व्योमातिधृतिद्वियुगानि[६][२] रसशरांश्चन्द्रपातस्य[७] ॥ ३१ ॥
ख[५] नंदा द्वियमाः[४] खाब्धयो[५] ग्रहाद्या[१] यथेष्ट[२] पर्वगुणाश्रेप्याः[२] ।
पर्वाण्यिष्यति[३] शोध्याः[६] पातेऽन्यथातीते[७] ॥ ३२ ॥

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३० (घ) १. पर्वेंदा: for (पर्वेंदोः) २. प्रागधिको (ग) (च) for (प्रागाधिको)
(ग) ३. रसा for (रसाः)
(ङ) २. प्रागधिकोना for (प्रागाधिकोना) ४. "युतिर्" लुप्त

३१ (ङ) १. द्वियमा (ग) (च) (ङ) for (द्वितमा)
२. द्वियुगानि (ङ) द्वियुगानि for (द्वियुगानि)
(ग) ३. विषया for (विषया) ४. स्त्रिविषया (ङ) for (त्रिविषया)
५. स्तदुस्तदुच्चस्य । ५ । २२ । १२ । ४३ । for (स्तदुच्चस्य)
६. धृते for धृति
७. सरा-। १६ । ४२ । ५६ श्चन्द्र for (शरांश्चन्द्र)
(ङ) ५. स्तदुच्चस्य । ५ । १२ । १२ । ५३ for (स्तदुच्चस्य)
७. रसशराश्चन्द्रपातस्य ० । १६ । ४२ । ५६ । ३१ for (रसशरांश्चन्द्रपातस्य)

३२. (घ) १. गृहा (ग) (च) for (ग्रहा)
२. क्षेप्याः (ग) for (श्रेप्याः) द्वितीय पंक्ति का आरम्भिक पद
१. पर्वण्ये (ग) पर्वणोष्यति for (पर्वाण्यिष्यति)
(ग) ४. ':' विसर्ग लुप्त है
५. खध्वयो० । ६ । २२ । ४० for (खाब्धयो)
१०. स्तथेष्ट for (यथेष्ट)
६. शोध्या मध्ये क्षतिक्रांते for (शोध्याः पातेऽन्यथातीते)
७. 'शोध्याः' से आगे के पद लुप्त हैं ।
(च) २. पर्वगुणाः क्षेप्याः ıor (पर्वगुणा श्रेप्याः)
३. पर्वाण्येष्यति for (पर्वाण्यिष्यति)
(ङ) ८. खं for (ख) ५. खाब्धयो० । ६ । २२ । ४० for (खाब्धयो)
१. गृहाद्या for (ग्रहाद्या) १०. स्तथेष्ट for (यथेष्ट)
६. गुणाः for (गुणा) वि० यहां श्लोकार्ध समाप्त
२. क्षेप्याः for (श्रेप्याः) (वि० श्लोक की दूसरी पंक्ति का आरम्भ)
३. पर्वण्येष्ये for (पर्वाण्यिष्यति) ६. शोध्या for (शोध्याः)
७. मध्ये for (पाते) ११. त्वतिक्रांते for (ऽन्यथातीते)

ग्रहणे यथा[१] रवींद्वोः स्पष्टीकरणाद्य मुक्तवत्कृत्वा[२]
एवं पर्वज्ञानं ग्रहणज्ञानं[३] स्फुटं गणितात् ॥ ३३ ॥
चत्वारि त्रयुवर्त्त[१] ग्रहणात्यर्कस्य[२] सप्तचन्द्रस्य[४] ।
द्रष्टोदयास्तमययोर्कराात्रिदलयोश्च केंद्रस्य[३] ॥ ३४ ॥[५]
सर्वपादानामंतो[१] तिथ्यंतज्ञानमिंदुभास्करयोः[३] ।
ग्रहणे च[२] क्तते स्पष्टे जिष्णुसुतब्रह्मगुप्तेन ॥ ३५ ॥

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३३. (ग) १. यथरविद्वोः for (यथा रवींद्वोः)
२. ज्ञात्वा for (कृत्वा) ३. "ग्रहणज्ञानं" लुप्त पद हैं

(च) ३. ग्रहणज्ञानस्फुटं for (ग्रहणज्ञानं स्फुटं)

(ङ) २. मुक्तवज् ज्ञात्वा for (मुक्तवत् कृत्वा)

३४. (घ) १. त्रपुवर्त्त (ग) त्रपुवत्त for (त्रयुवर्त्त)
२. ग्रहणान्यर्कस्य (ग) (च) (ङ) for (ग्रहणात्यर्कस्य)
३. द्रष्टो (ग) इंद्वोदयास्त समयायार्क for (द्रष्टोदयास्त)
४. शीतरश्मेश्च (ङ) for (सप्तचन्द्रस्य)

(च) १. त्रपुवर्त for (त्रयुवर्त्त) ३. द्रष्टो for (द्रष्टो)

(ङ) १. चत्वारोऽत्रपर्त्त for (चत्वारित्रयुवर्त्त)
३. द्रष्ट्वोदयास्त for (द्रष्टोदयास्त)
५. समया तथाऽर्क for (मययोर्क)

३५. (घ) १. मंते (ग) पदानामन्ते (च) for (पादानामंतो)
२. कृते (ग) चक्रते for (चक्तते)

(घ) ३. तिथ्ये for (तिथ्यन्त)

(च) २. कृते (ङ) for (क्तते)

(ङ) १. सर्वपदानामन्ते for (सर्वपादानामन्तो)
३. तिथ्यन्ते for (तिथ्यंत)

दुरभ्रष्टे[1] ग्रहणे श्रेषेणार्यभट[2]विष्णुचन्द्रस्य[3] ।
दृग्गणित[5]विसंवादः काकताली[6]यम् ।। ३६ ।।
स्फुटतिथ्यंतज्ञानं यन्नार्य[1]भटादिभिः कृतमतीतैः ।
ब्राह्मे स्फुटं कृतं तज्जिष्णुसुतब्रह्मगुप्तेन ।। ३७ ।।
ब्रह्मोक्तार्के[2]न्दु तदुच्चपातदेशांतरस्फुटी[3]करणैः ।
स्फुटसिद्धर्कं[1] ग्रहण[4]द्वयं स्फुटमतीतोक्त[5]म् ।। ३८ ।।
विक्षेपाग्रेषु त्रीन्बिन्दून् प्रग्रहणमध्यमोक्षेषु ।
कृत्वा[1] तन्मन्य[2]द्वयमध्य[3]गयोः सूत्रयोर्योगात् ।। ३९ ।।

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३६. (घ) १. दूरभ्रष्टे (च) (ङ) for (दुरभ्रष्टे)
२. श्रीषेणार्य (ग) खेणार्य (ङ) for (श्रेषेणार्यभट)
३. चंन्द्रेषु (ग) (च) (ङ) for (चन्द्रस्य)
४. विसंवादात्संवादः (च) for (विसंवादः)
(ग) ५. ग्रहगणितविसंवादात्संवादः for (दृग्गणितविसंवादः)
(च) २. श्रीषेणार्य for (श्रेषेणार्य)
(ङ) ५. ग्रहगणित for (दृग्गणित) ४. विसंवादात् संवादः for (विसंवादः)
६. काकतालीयः for (काकतालीयम्)

३७. (घ) १. यत्रार्य for (यन्नार्य)

३८. (घ) १. मिद्वर्कं (घ) मिद्वर्कं for (सिद्धर्कं)
(ग) २. ब्रह्मोक्तार्केंदुस्तदुच्च for (ब्रह्मोक्तोर्केंदु)
३. स्फुटिकरणैः for (स्फुटीकरणैः) ४. द्वितयं न for (द्वयं)
५. स्फुटमतीवोक्तम् for (स्फुटमतीतोक्तम्)
(च) २. ब्रह्मोक्ताङ्कें दु for (ब्रह्मोक्तोर्केन्दु) १. मिद्वङ्कं for (सिद्धर्कं)
(ङ) २. ब्राह्मोक्तार्केन्दु for (ब्रह्मोक्तोर्केंदु) १. मिद्वर्कं for (सिद्धर्कं)
४. ग्रहणद्वितयं for (ग्रहणद्वयं)

३९. (घ) १. कृत्पा for (कृत्वा)
२. तन्मत्स्यद्वय (ग) तन्मत्सद्वय for (तन्मन्यद्वय)
३. मध्यमयोः for (मध्यगयोः)
(च) १. क्रत्या for (कृत्वा) २. तन्मत्स्य for (तन्मन्य)
(ङ) २. तन्मत्स्य for (तन्मन्य)

बिंदु परिलेखरेखा[१] ग्राहकमार्गः प्रसार्य सूत्रे द्वे ।
आद्यंताभ्यां मध्यममाछेद्य[४] स्थुल[२] एवं वा[३] [५]।। ४० ।।
बिन्दु द्वयांतरं[२] स्थितिदलेन हृतमिष्टनाडिका[३]गुणितम् [४] ।
ग्राह्यफलांगुलस्थं[६] ग्राहकमानेन परिलेखः[५] [१]।। ४१ ।।
इष्ट ग्रासोर्केंद्वोर्[१]निमीलनोन्मीलनं[२] च भानुमतः[५] ।
त्रषुवर्तः[३] प्राग्मध्यात् पश्चाद्विष्टांगुले[४] स्थेन ।। ४२ ।।

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४०. (घ) १. परिलेषरेषा (ग) for (परिलेखरेखा)
२. माछोद्य (ग) माछाद्य for (माछेद्य)
३. स्थूल (ग) (च) (ङ) for (स्थुल)

(च) १. परिलेषरेषा for (परिलेखरेखा)

(ङ) ४. मध्यग for (मध्यम) २. मुच्छाद्य for (माछेद्य)
५. मेवं वा for (एवं वा)

४१. (घ) १. परिलिख्य (ग) परिख्यः for (परिलेखः)

(ग) २. बिंदुद्वयान्तर for (बिन्दुद्वयांतरं) ३. हृत (च) (ङ) for (हृत)
४. नालिका for (नाडिका) ५. छा for (स्थं)

(च) १. परिलेख्य for (परिलेखः)

(ङ) १. परिलिख्य for (परिलेखः) ६. ग्राह्यं for (ग्राह्य)

४२. (घ) १. ऽर्केंद्वो (ग) इष्टग्रार्कंदो for (इष्टग्रासोर्केंद्वो)
२. मीलने (ग) मिलने for (मीलनं)
३. त्रपुवर्त्तः (ग) (च) for (त्रषुवर्तः)
४. पश्चाद्द्वेष्टांगुल (ग) (च) for (पञ्चाद्विष्टांगुले)

(च) १. ऽर्क्केंद्वो for (ऽर्केंद्वो) २. मीलने (ङ) for (मीलनं)

(ङ) १. ऽर्केन्द्वो for (र्केंद्वो) ५. भानुमतोः for (भानुमतः)
३. उर्वरितः for (त्रषुवर्तः)
४. पश्चाद्वेष्टाङ्गुल for (पश्चाद्विष्टांगुले)

मध्यस्य[१] दिनांत्य नवांतरं[२] मिष्टघटिकाभिः[४]।
स्थित्यर्द्धनाडिकाद्रतमृणधनमूनाधिके[५] मध्ये[३] ॥ ४३ ॥
आद्यन्ते[२] वान्कृत्वा विक्षेपः[१] कोटिरूक्तवद्ग्रासः।
विक्षेपान्तरमेवं गुणमिष्टग्रासलिप्ताभिः[३] ॥ ४४ ॥
मध्यग्रासकला[१] हृतमृणधन[४] चोक्तवत्स्वविक्षेपः[५]।
तेन ग्रासात्कालः[६] कालादसकच्च[३] विक्षेपः ॥ ४५ ॥
गृहणोत्तरं न देयं शपथैरपि दत्त सुकृत्[४] नाशाद्यैः।
गृहणं स्फुटमिह[१]स्फुटमार्यभटाद्यैर्यतस्तंत्रैः[२][३] ॥ ४६ ॥

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४३. (घ) १. मध्यस्याद्येनांत्ये (ग) मध्यस्याद्येनांते for (मध्यस्यदिनांत्य)
२. गुणितमिष्ट (ग) गुणीतमिष्ट for (मिष्ट)
३. द्धृत (ग) हृत for (द्रत)
(ग) ४. घटिभि for (घटिकाभिः) ३. धनमृण for (मृणधन)
(च) १. मध्यस्याद्येनांत्यनवांतरं for (मध्यस्य दिनांत्य नवांतरं)
२. गुणितमिष्टघटिकाभिः for (मिष्टघटिकाभिः) ३. हृत for (द्रत)
(ङ) १. मध्यस्याद्येनान्तेन for (मध्यस्यदिनांत्य न)
२. वाऽन्तरं गुणित for (वांतरं) ३. हृत for (द्रत)

४४. (घ) १. विक्षे (ज्ञे) यः for (विक्षेपः)
(ग) २. वा कृत्वा for (वान्कृत्वा) ३. विलिकाभिः for (लिप्ताभिः)
(ङ) २. च पृषत्के for (वान्कृत्वा)

४५. (घ) १. हृतमृणंधनं (ग) हृतमृणधन (च) for (हृतमृणधन)
२. विक्षेण (ग) विक्षेपं for (विक्षेपः)
३. दसकृच्च (ङ) (च) for (दसकच्च)
(ग) ४. वोक्त for (चोक्त) ५. ख for (स्व)
६. कालदसत्कृच्चविक्षेप for (कालादसकच्च विक्षेपः)
(च) २. विक्षे for (विक्षेपः)
(ङ) १. हृतमृणंधनं for (हृतमृणधन)

४६. (ग) १. 'स्फुट' लुप्त है। २. 'मिहा' लुप्त है।
३. तंतैः for (तंत्रैः)
(च) ४. सुक्रत for (सुकृत) १. स्पष्ट for (स्फुट)
(ङ) ४. सुकृतनाशाय for (सुकृत्नाशाद्यैः) २. मार्यभट for (मिहास्फुट)
५. श्रीषेणाद्यै for (मार्यभटाद्यै) ३. र्यतस्तन्त्र for (र्यतस्तंत्रैः)

परिलेखो[१] बलिनज्या[५] विक्षेपाद्येषु षोड़षोऽध्यायः[२]।
ग्रहणोत्तरमर्केंद्वोः[३] षट्चत्वारिंशदार्याणाम् ॥ ४७ ॥
इत्ति[४] श्री ब्रह्मगुप्ते ग्रहणाधिकारः
षोडशोऽध्यायः समाप्तः

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४७. (घ) १. परिलेषो (ग) परिलेष for (परिलेखो)
२. षोडशोऽध्यायः (ङ) for (षोडशोऽध्यायः)
३. मर्केंद्वो for (मर्केंद्वोः)
४. 'इति' से 'समाप्तः' तक पाठ अंकित नहीं।

(ग) ५. वलन for (बलिन)
४. इति श्री ब्रह्मसिद्धांते ग्रहणोत्तराध्यायः षोडशः for (इति श्री ब्रह्मगुप्ते ग्रहणाधिकारः षोडशोऽध्यायः समाप्तः)

(च) १. परिलेषो for (परिलेखो) ५. वलनद्या for (वलिनज्या)
२. षोडशो for (षोडषो) ३. मर्क्केंद्वोः for (मर्केंद्वोः)
४. 'इति' से 'समाप्तः' तक लुप्त।

(ङ) १. परिलेख for (परिलेखो) ५. वलनजीवा for (बलिनज्या)
४. इति श्री ब्राह्मस्फुट सिद्धान्ते ग्रहणोत्तराध्यायः षोडशः for (इति श्री ब्रह्मगुप्ते ग्रहणाधिकारः षोडषोऽध्यायः समाप्तः)

अथ शृङ्गोन्नत्युत्तराध्यायः

सप्तदशः

भुजकोटिशशिमान शुक्लपरिलेख सूत्रपरिलेखान् ।
प्रतिदिवसं प्रतिघटिकं यो वेत्तींदुद यज्ञः सः ॥ १ ॥
पराच्य परादिगभिमुखं शुक्लेतरपक्षयोर्लिखेत भूमौ ।
अपवर्त्तेष्टेनैकेन राशिना कोटिभुजकर्णात् ॥ २ ॥
परिकल्पार्कं बिन्दु तस्माद्बाहुं यथादिशं कृत्वा ।
बाहुग्रात्प्राच्यपरां कोटिं तिर्यक्स्थं कर्णः ॥ ३ ॥

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१. (घ) १. छेदभुजकोटिकर्णशशि (ग) भुजकोटिकर्णशशि for (भुजकोटिशशि)
२. परिलेष for (परिलेख) (ग) 'परिलेख' लुप्त है ।
३. परिलेषान् (ग) परिलेखात् for (परिलेखान्)
(ग) ४. वेत्तं for (वेत्ती) ५. हृदयज्ञः for (दुंदयज्ञः)
(च) १. भुजकोटि कर्णशशिमान for (भुजकोटिशशिमान)
२. परिलेष for (परिलेख) ३. परिलेषान् for (परिलेखान्)
४. वेत्तीदूंदं for (वेत्तींदुद)
(ङ) २. सित for (परिलेख) ३. लेखात् for (लेखान्)
४. वेत्ति for (वेत्ती) ५. स तन्त्रहृदयज्ञः for (दुंदयज्ञः सः)

२. (घ) १. प्राच्य (ग) (च) (ङ) for (पराच्य)
२. मुषं for (मुखं) ३. लिषेत (ग) लिखेद्भूमौ for (लिखेत भूमौ)
४. कर्णान् (ङ) (च) for (कर्णात्)
(ग) ५. पदिगभि मुखं for (परादिगभिमुखं)
६. तरतर for (तर) ७. अपवर्त्यैकेनेष्टेन (ङ) for (अपवर्त्तेष्टेनैकेन)
८. वाशिना for (राशिना)
(च) २: मुषं for (मुखं) ३. लिषेत् for (लिखेत)
(ङ) ४. लिखेत् (लिखेत)

३. (घ) १. बाह्वग्रात् (ग) (ङ) for (बाहुग्रात्)
२. तिर्यक् स्थतं (ग) तिर्यक्स्थितं कर्णम् ॥ ३ ॥
(ग) ३. बिंदुं (ङ) for (बिन्दु) ४. यथादिशि (यथादिशं) ५. कोटि for (कोटिं)
(च) ६. परिकल्पार्कं for (परिकल्पार्कं) २. तिर्यक्स्थं for (तिर्यक्स्थं)
(ङ) ६. परिकल्प्यार्कं for (परिकल्पार्कं)
७. दत्त्वा for (कृत्वा) २. तिर्यक्स्थितंकर्णं for (तिर्यक्स्थं कर्णः)

कर्णाग्रे[२] चन्द्रमसं परिलेख्य[१] सितं प्रवेश्यकर्णेन ।
शशिबिंबे[३] शुक्लाग्रात्परिलेखसमेन[४] सूत्रेण ॥ ४ ॥
कर्णगतस्थेनेंन्दो[१] शुक्ल[३] परिलेख्य[४] पश्चिमाभिमुखः[५] ।
राशिषु[२] मेषतुलादिषु संशोध्य दिवाकरं[६] चन्द्रात् ॥ ५ ॥
पूर्वाभिमुखः[३] कर्कटमकरादिषु[२] भवति शुक्लसंस्थानम् ।
एवं वा संस्थानं परिलेख्येन्दुं[१] प्रसाध्य दिशः ॥ ६ ॥
बाहुज्यें[२]दुदलगुणाकर्णविभक्ता भुजोन्यदिक् चन्द्रे ।
कर्णोभुजागतश्चन्द्रमध्यगः[३] पूर्वछेषम्[१] ॥ ७ ॥
प्राच्यपरे[१] विपरीते[२] फलकेन्यत्सर्वमुक्तवत्कार्यम्[३] ।
शृङ्गोन्नतिपरिलेखाश्चत्वारः शीतकिरणस्य[४] ॥ ८ ॥

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४. (घ) १. कर्णाग्र for (कर्णाग्रे) २. परिख्य (ग) परिलिख्य for (परिलेख्य)
(ग) ३. बिंबे सूत्रां for (बिंबे शुक्ला)
(च) २. परिलेष्य for (परिलेख्य) ४. परिलेष for (परिलेख)
(ङ) २. परिलिख्य for (परिलेख्य)

५. (घ) १. कर्ण गतिस्थे (ग) (ङ) for (कर्णगतस्थे) २. मेख for (मेष)
(ग) ३. नेंशे for (नेंन्दो) ४. परिलिख्य (ङ) for (परिलेख्य)
५. मुखम् (ङ) for (मुखः) ६. दिकारं for (दिवाकरं)
(च) १. कर्णगतिस्थेनेदो for (कर्णगतस्थेनेंदो)
४. शुक्लं परिलेख्य for (शुक्लं परिलेख्य)
(ङ) ३. नैशे for (नेंन्दो) ७. शुक्ले for (शुक्ल)

६. (ग) १. परिलेखेन्दुं for (परिलेख्येन्दुं)
(च) २. कर्क्कट for (कर्कट) १. परिलिख्येंदुं for (परिलेख्येन्दु)
(ङ) ३. मुखं for (मुखः) १. परिलिख्येन्दुं for (परिलेख्येन्दुं)

७. (घ) १. पूर्ववछेषम् (ग) पूर्ववछेशम् for (पूर्वछेषम्)
(ग) २. भुजात्य दिक्केद्रे for (भुजोन्यदिक्चन्द्रे) ३. मध्यतः (ङ) for (मध्यगः)
(ङ) २. भुजान्यदिक् for (भुजोन्यदिक्) १. पूर्ववच्छेपम् for (पूर्वछेषम्)
(च) १. पूर्ववछेषं for (पूर्वछेषम्)

८. (ग) १. प्राच्यररे for (प्राच्यपरे) २. विपरिते for (विपरीते)
३. मुक्तवछेषम् for (मुक्तवत्कार्यम्) ४. सीत for (शीत)
(च) ५. फलकेन्य सर्वमुक्त for (फलकेन्यत्सर्वमुक्त)
(ङ) ६. वच्छेषम् for (वत्कार्यम्)

ग्रहयोगेंदुछाया ग्रहदयोस्तमयभगृहयुतीनाम् ।
तत्स्वक्रान्तिज्याद्युत्तराणि भगृहयुतौ न पृथक् ॥ ९ ॥
इति परिलेखाध्यायः शशांक भृंगोन्नतेर्भुजाद्येषु ।
शशिभृङ्गोन्नत्युत्तरमार्यादशकेन सप्तदशः ॥ १० ॥
इत्ति सप्तदशाध्यायः समाप्तः

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९. (ग) १. ग्रहोदयास्त (च) (ङ) for (ग्रहदयोस्त)
२. मुनिनां for (युतीनाम्) ३. तत्क्रान्तिज्याप्पा (घ) for (तत्क्रान्तिज्याद्यु)
४. प्रथक् for (पृथक्)

(च) ५. द्या for (ज्या)

(ङ) २. ग्रह मुनीनां for (ग्रह युतीनां)
३. तत्क्रान्तिज्या for (तत्स्वक्रान्तिज्या)
५. प्रश्नोत्तराणि for (द्युत्तराणि)

१०. (घ) १. भुजोज्येषु (ग) भुजाद्येषु for (भुजाद्येषु)
२. 'इति' से 'समाप्त' तक अंकित नहीं है ।

(ग) ३. : लुप्त हैं ५. भृंगोन्युत्तर for (भृङ्गोन्नत्युतर)
६. मार्यादसकेन for (मार्यादशकेन)
२. इति ब्रह्मसिद्धांते सप्तदशोध्यायः for (इत्ति सप्तदशाध्यायः समाप्तः)
४. राशि for (शशि)

(च) २. 'इति' से 'समाप्तः' तक लुप्त

(ङ) २. इति श्रीब्राह्मस्फुटसिद्धान्ते भृङ्गोन्नत्युत्तराध्यायः सप्तदशः
for
(इति सप्तदशाध्यायः समाप्तः)

अथ कुट्टकाध्यायः

अष्टादशः

प्रायेण यतः प्रश्नाः[४] कुदाकाराद्रतेन[१,२] शक्यंते ।
ज्ञातुं वक्षामि[३] ततः कुदाकारं[५] सह प्रश्नैः ॥ १ ॥
कुदकर्णं[१] धनाव्यक्त मध्यमाहरणं[६] वर्ण्यता[२] वितकैः ।
आचार्यस्तंत्रविदां[३] ज्ञातैर्वर्गं[४] प्रकृत्याच्च[५] ॥ २ ॥
अधिकाग्रभागहारा[४] दूनाग्रच्छेदभाजिता[६] शेषम्[१] ।
यत्तत्परस्य[२] हृतं लब्धमध्योधः[३] पृथक्[५] स्थाप्यम् ॥ ३ ॥
शेषं तथेष्टगुणितं यथाग्रयोरंतरेण[२] संयुक्तम् ।
शुध्यति गुणकः स्थाप्यो लब्धं चांत्यादुपांत्यगुणाः[१] ॥ ४ ॥

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१. (घ) १. २ कुट्टाकाराट्टते (ग) कुट्टाकाराट्टते for (कुदाकाराद्रते)
३. वक्ष्यामि (ग) (च) (ङ) for (वक्षामि)
(ग) ४. प्रश्ना for (प्रश्नाः) ५. कुट्टाकारं (ङ) for (कुदाकारं)
(च) २. ट्टतेन for (द्रतेन)
(ङ) १. कुट्टाकाराट्टते for (कुदाकाराद्रते)

२. (घ) १. कुट्टकर्ण for (कुदकर्ण)
(ग) १. कुट्टकखर्णधना for (कुदकर्णधना)
२. कर्णभावितकैः (ङ) for (वर्ण्यतावितकैः)
३. विदा for (विदां) ४. वर्ग्र for (वर्गं)
(च) ५. च for (च्च)
(ङ) १. कुट्टकखर्ण for (कुदकर्ण) ६. हरणैक for (माहरण)

३. (घ) १. छेषम् (च) (ङ) for (शेषम्)
२. यत्तत्परस्परहृतं (ग) यत्परस्परहृतं (ङ) for (यत्तत्परस्यहृतं)
३. मध्येधः (ग) मधोऽधः (ङ) for (मध्योधः)
(ग) ४. भावहारा for (भागहारा) ५. प्रथक् for (पृथक्)
(च) ३. मध्येधः for (मध्योधः)
(ङ) ६. दूनाग्रच्छेद for (दूनाग्रे छद)

४. (ग) १. गुणः (ङ) for (गुणाः)
(ङ) २. यथाऽग्रयो for (यथाग्रयो)

स्तोर्द्धात्ययुतोग्रातो हीनाग्रच्छेदभाजितः शेषम् ।
अधिकाग्रच्छेदाहतमधिकाग्रयुतं भवत्यग्रम् ॥ ५ ॥
छेदवधस्य द्वियुगच्छेद वधोयुगतं द्वयोरग्रम् ।
बुधकारेणैवं ज्यादिग्रहयुगगतानयनम् ॥ ६ ॥
यो जानाति युगादिं ग्रहयुगपातैः पृथक् पृथक् तिथितैः ।
द्वित्रिचतुः प्रभृत्तिनां कुट्टाकारं स जानाति ॥ ७ ॥

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५. (घ) १. स्वोर्द्धातियुतो (ग) स्वोर्धात्ययुतो for (स्तोर्द्धात्ययुतो)

(ग) २. ग्रांतो for (ग्रातो) ३. हृत for (हत)

(च) १. स्तोर्द्धोत for (स्तोर्द्धात्य)
(ङ) १. स्वोर्ध्वोऽन्त्ययुतो for (स्तोर्द्धात्ययुतो)
२. ऽग्रान्तो for (ग्रातो) ३. च्छेदहृत for (छेदाहत)

६. (घ) १. द्वियुगं (ग) (च) (ङ) for (द्वियुग)
३. रग्रंम् for (रग्रम्) २. युगगतं (ग) युगमतं for (युगतं)
४. कुधकारेणैवं (ग) कुट्टाकारेणैव (ङ) for (बुधकारेणैवं)

(ग) ५. छोदवहो for (छेदवधो)

(च) २. युगगतं (ङ) for (युगतं) ४. कुद्यकारेणैवं for (बुधकारेणैवं)

७. (घ) १. प्रभृतीनां (ङ) (ग) प्रभृतिनां for (प्रभृत्तिनां)

(ग) (वि०—यह श्लोक इस प्रति में श्लोक संख्या १५ पर है)
२. युगादीग्रह for (युगार्दिग्रह) ३. युगजातैः for (युगपातैः)
४. प्रथक् २ for (पृथक् पृथक्) ५. कथितैः for (तिथितैः)
६. कुट्टकारं for (कुट्टाकारं) ७. जाति for (जानाति)

(च) ३. युगः यातैः for (युगपातैः)
४. पृथक्प्रथक्कृतिथितैः for (पृथक् पृथक् तिथितैः)
१. प्रभृतीनां for (प्रभृत्तिनां)
२. युगादि for (युगार्दि) ३. ग्रहयुगयातैः for (ग्रहयुगपातैः)
(वि०—श्लोक संख्या १५ है)

भगणादिशेषमग्रं छेदहतं ख चर दिनजशेषहतम् ।
अनयोरग्रं भगणादि दिनजशेषोद्धृतं द्यगुणः ॥ ८ ॥
अथवा जिनजभगणादिशेषं येन गुणं मंडलादि शेषोनं ।
शुध्यति विभाजितं स्वेन भागहारेण सद्यगुणः ॥ ९ ॥
मंडल राश्यंशकला विकलाशेषादभिष्टतः कथितान ॥
आनयति दिनगणं यः कुठाकारं स जानाति ॥ १० ॥

८. (घ) १. छेदहृतं (ग) (च) (ङ) for (छेदहतं)
२. खंच (यहां मूल में 'र' नहीं है) for (खचर)
३. हृतम् (च) (ङ) for (हृतम्)
४. क्वगणः (ग) द्युगणः for (द्यगुणः)
(ग) ५. च for (चर)
६. भगणादिधृतं for (भगणादिदिनजशेषोद्धृतं)
(वि०—इसकी श्लोक संख्या ७ है) (ङ)
(च) २. खच for (खचर) ४. द्युगणः (ङ) for (द्यगुणः)
(वि०—इसकी क्रमसंख्या ७ है)

९. (घ) १. गणः for (गुणः)
(ग) २. यह श्लोक 'दिनज' से आरम्भ होता है, (ङ)
३. शेषकयोः (ङ) for (शेषोनं)
दूसरी पंक्ति बिल्कुल भिन्न है—
"सदृशछेदो धृतयोस्तद्धातमहर्गण द्यमतः" ॥८॥
(च) १. सद्युगणः for (सद्यगुणः)
(ङ) २. 'अथवा' लुप्त, दिनज for (जिनज)
दूसरी पंक्ति निम्नांकित—
सदृशच्छेदोद्धृतयोस्तद्घातमहर्गणाद्यमतः for (शुध्यति विभाजितं स्वेन भागहारेण सद्यगुणः)
(वि—इसकी क्रमसंख्या ८ है)

१०. (घ) १. दभीष्टतः कथितान् for (दभिष्टतः कथितान)
२. कुट्टाकारं for (कुठाकारं)
(च) १. दभीष्टतः for (दभिष्टतः) २. कुदाकारं for (कुठाकारं)
(ङ) श्लोक उपलब्ध नहीं।

यो जानाति[२] युगगतं कथितादयिमास[४] शेषकादिष्टात् ।
अवभाव[१] शेषतो वा तद्योगाद्वासकुट्टज्ञः[३] ॥ ११ ॥
एकेष्टदिवसघटिका विनाडिका मंडलादिशेषकयोः ।
सद्रशछेदो[१] धृतयोस्तघात्[२] महर्गणाद्यमतः[३] ॥ १२ ॥
हृतयोपरस्परं यछेषं[४] गुणकारभागहारकयोः[३] ।
तेन हृतौ[२] निछेदौ[५] तावेव परस्परं हृतयोः[१] ॥ १३ ॥
लब्धमध्येधः[३] स्थाप्यं तथेष्ट गुणकारसंगुणं शेषम्[४] ।
शुध्यति तथैकहीनं गुणकः[१] स्थाप्यः[२] पलं चांत्यात्[५] ॥ १४ ॥

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११. (घ) १. अवमाव (ग) (च) (ङ) for (अवभाव)
(ग) (वि०—इसकी श्लोक संख्या २५ है)
२. यो युगगतं for (यो जानाति युगगतं)
(च) ३. कुदज्ञः for (कुट्टज्ञः)
(ङ) (वि---श्लोक संख्या २५ है)
२. जानाति यो for (यो जानाति) ४. दधिमास for (दयिमास)

१२. (घ) १. सद्दश (च) for (सद्रश) २. हृतयो (च) for (धृतयो)
३. स्तद्यात for (स्तघात्)
(ग) यह श्लोक इस प्रति में उपलब्ध नहीं है ।
(च) ३. तद्यात् for (तघात्)
(ङ) श्लोक अनुपलब्ध है ।

१३. (घ) १. हृतयोः (ग) (ङ) for (हृतयोः)
२. हृतौ (ग) (च) for (हृतौ)
(ग) ३. हृतयोः (च) for (हृतयो)
(वि०—इसकी श्लोक संख्या ६ है) (ङ)
(ङ) ३. हृतयोः for (हृतयो) ४. यच्छेषं for (यछेषं)
५. निश्छेदौ for (निछेदौ)

१४. (घ) १. यथैक (च) (ङ) for (तथैक)
२. स्थाप्यपलं (ग) स्थाप्यफलं for (स्थाप्यः पलं)
(ग) ३. लब्धमध्योध for (लब्धमध्येधः) ४. शेषे for (शेषं)
५. वांत्यात् for (चांत्यात्)
(वि०—इसकी श्लोक संख्या १० है) (ङ)
(च) ५. चाप्तात् or चात्यात् for (चांत्यात्)
(ङ) ३. लब्धमधोऽधः for (लब्धमध्येधः)

अग्रांत्यमुपांत्येना स्वार्द्धो गुणितोंत्य संयुतो भगणः ।
छेदभागहारेणैवं स्थिरकटुकः शेषम् ॥ १५ ॥
इष्टभगणादिशेषात् स्वकुटकगुणात् स्वाभागहार हृताद् ।
शेषं द्युगणोहृत निरपवर्त गुणभागहारं युतः ॥ १६ ॥

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१५. (घ) १. अग्रांत (ग) (च) (ङ) for (अग्रांत्य)
३. भक्तः (ग) (च) for (भगणः)
४. निछेदभागहारे for (छेदभागहारे)
५. कुद्दकः (ङ) for (कटुकः)
७. गुणितोप्त (त्प) (च) for (गुणितोंत्य)

(ग) २. स्वेर्धो for (स्वार्द्धो)
६. मुपांत्येन (ङ) for (मुपांत्येना)
(वि०—इसकी श्लोक संख्या ११ है)
४. निछेद for (छेद) ५. कुदकः for (कटुकः)

(ङ) २. स्वोर्ध्वो for (स्वार्द्धो) ३. भक्तः for (भगणः)
४. निः शेष for (छेद)

१६. (घ) १. कुद्दक (ङ) for (कुटक)
२. गुणत्स्वभाग (ग) गुणान् स्वभाग for (गुणात्स्वाभाग)
३. हृताद् (ग) हृतान् for (हृताद्)
४. गत (ग) भगणोगत for (द्युगणोहृत)
५. निरपवर्त्त (ग) (च) (ङ) for (निरपवर्त)
६. भागहारयुतः (ग) गुणभागहारयुता for (गुणभागहारंयुतः)

(ग) (वि०—इसकी श्लोक संख्या १२ है)

(च) ७. दष्ट for (इष्ट) १. कुदक for (कुटक)
२. गुणत्स्वभाग for (गुणात्स्वाभाग)
३. हृताद् for (हृताद्) ४. गत for (हृत)
६. गुणभागहारयुतः (ङ) for (गुणभागहारंयुतः)

(ङ) २. गुणात्स्वभाग for (गुणात्स्वाभाग) ४. गत for (हृत)

एवं समेषु विषमेष्वृणं धनं धनमृणं यदुक्तं तत् ।
ऋणधनयोर्व्यस्तत्वं गुण्यप्रक्षेपयोः कार्यम् ॥ १७ ॥
गुणकछेदः छेदो गुणको धनमृणधनं कार्यम् ।
वर्गे पदं पदकृति रंताद्वैपरित माद्यं तत् ॥ १८ ॥
अंशक शेषा ऽन्यूनास्तप्त हृतान्मूल मूनमष्टाभिः ।
नवभिर्गुणं सरूपं कदा शतं बुधदिने सवितुः ॥ १९ ॥

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१७. (घ) १. विषमेष्वृणं (ग) पुणधन for (विषमेष्वणं)
(ग) ४. व्यस्तत्वं for (व्यस्तत्वं) ५. गुण्यम् for (गुण्य)
(वि०—इसकी श्लोक संख्या १३ है) (ङ)
(च) १. विषमेष्वृणं for (विषमेष्वणं) २. प्रक्षेपयो for (प्रक्षेपयोः)
(ङ) १. विषमेष्वृणं for (विषमेष्वणं)

१८. (घ) १. छेदः (ग) छेद (च) for (छेदः छेदो)
२. वैपरीत for (वैपरित)
(ग) ३ मृणमृणं for (मृण) ४. वागपदं for (वर्गेपदं)
५. रंत्याद्विपरित for (रंताद्वैपरित)
(वि०—इसकी श्लोक संख्या १४ है) (ङ)
(ग) +तिथिमानदिनेष्विष्टा ये ऽर्काद्यास्ते पुनः कदा तेषु ।
इष्टग्रहवारे यः कथयति कुट्टकज्ञः सः ।। ८ ।।+
इसकी श्लोक संख्या १८ के स्थान में ८ लिखी गई प्रतीत होती है ।
(च) 'श्लोक' लुप्त
(ङ) (वि०—इसकी क्रमसंख्या १८ है)
१. ग्रहवारेषु for (ग्रहवारे)
(च) ४. वर्गे पदं for (वर्गेपदं) ६. पदेकृति for (पदकृति)
५. रंताद्वैपरीत for (रंताद्वैपरितं)
(ङ) १. गुणकश्छेदश्छेदो for (गुणकछेदः छेदो)
४. वर्गः पदं पदं for (वर्गेपदं पद)
५. रन्त्याद्विपरीत for (रंताद्वैपरित)

१९. (घ) १. हृतान् for (हृतान्)
(ग) (वि०—इसकी श्लोक संख्या २७ है)
(ङ) (वि०—इसकी श्लोक संख्या २७ है)
२. शेषात्न्यूनात् for (शेषान्यूनाम्)
१. सप्तहृतात् for (तप्तहृतान्)

ऊनाधिमासः शेषात् मूलध्व्यधिकं विभाजितं षड्भिः ।
ऊनं वर्जित मधिकं नवभि र्नवति कदा भवति ॥ २० ॥
अवमावशेष वर्गोन्येकोंवशति विभाजितोध्व्यधिक ।
अष्टगुणे दशभक्ते द्वियुते ष्टादशकदा भवति ॥ २१ ॥
इष्टभगणादिशेषा ह्युगणस्तत्कुट्टकेन संयुक्तः
तच्छेदेनैस्तावद्दिनवारो यावदिष्टस्य ॥ २२ ॥

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२०. (घ) १. शेषान्मूलं (ग) for (शेषात् मूल) २. द्वयधिकं (ग) (च) for (ध्व्यधिकं)
३. वर्गित (ग) (च) for (वर्जित) ४. न्नवतिः (ग) नवभिः for (नवतिः)
(ग) (वि०—इसकी श्लोक संख्या २८ है)
५. न्यूनाधिमास for (ऊनाधिमास) ६. द्वचूनं २ for (ऊनं)
(च) १. शषान्मूलं for (शेषात्मूल) २. द्वयधिकं for (ध्व्यधिकं)
४. न्नवतिः for (नवति)
(ङ) (वि०—इसकी श्लोक संख्या २८ है)
५. च्यूनाधिमास for (ऊनाधिमासः)
१. शेषान्मूलं for (शेषात्मूल) २. द्वयधिकं for (ध्व्यधिकं)
६. द्वचूनं for (ऊनं) ३. वर्गित for (वर्जित)
४. नवतिः for (नवति)

२१. (घ) १. द्वयधिकः (ग) (च) (ङ) for (ध्व्यधिकः)
१. देशभक्तो (ग) (च) for (दशभक्तो)
३. द्वियुतोऽष्टादश (ग) (च) for (द्वियुते)
(ग) (वि०—इसकी श्लोक संख्या २६ है)
४. शेषे for (शेष) ५. व्येको for (न्येको)
६. गुणो for (गुणे)
(ङ) ५. व्येको for (न्येको)
(वि०—इसकी श्लोक संख्या २६ है)

२२. (ग) १. कुटकेन for (कुट्टकेन) २. दिष्टः for (दिष्ट)
३. स्यात् for (स्य)
(वि०—इसकी श्लोक संख्या १६ के स्थान में ६ है)
(च) १. कुदकेन for (कुट्टकेन)
(ङ) (वि०—श्लोक संख्या १६ है)
४. शेषाद् for (शेषा) ५. तच्छेद for (तच्छेद)
२. ३. यावदिष्टः स्यात् for (यावदिष्टस्य)

भगणाद्यमिष्टशेषं कदेंदुदिवसे खेर्गुरू दिने वा ।
ज्ञदिने वायः कथयति कुदाकारं स जानाति ॥ २३ ॥
ज्ञदिने यदंशशेषं विकलाशेषं कदा न दिंदु दिने ।
भानोरथवा शशिनो यः कथयति कुटकज्ञः सः ॥ २४ ॥
इष्टेषुमान दिवसे द्विमासान्मून रात्रिशेषे वा ।
भूयस्ते यः कथयति पृथग् पृथग वा कुदज्ञः ॥ २५ ॥

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२३. (घ) १. कथति (च) for (कथयति)
२. कुट्टाकारं (ग) कुट्टकारं for (कुदाकारं)
(ग) ३. ज्ञदिनैराशयः for (ज्ञदिनेवायः)
(वि०—इसकी श्लोकसंख्या १६ के स्थान में केवल ६ लिखी है)
(च) १. कुट्टाकारं ror (कुदाकारं)
(ङ) (वि०—श्लोकसंख्या १६ है)
३. राशीन् for (वायः) २. कुट्टाकारं for (कुदाकारं)

२४. (घ) १. तर्दिदु (ग) (च) (ङ) for (नर्दिदु)
२. कुट्टकज्ञः (ग) for (कुटकज्ञः)
(ग) (वि०—इसकी श्लोक संख्या १७ के स्थान में ७ है)
(च) २. कुदकज्ञः for (कुटकज्ञः)
(ङ) (वि०—क्रमसंख्या १७ है)
१. कुट्टकज्ञः for (कुटकज्ञः)

२५. (घ) १. मासन्मून for (मासान्मून)
२. पृथगपृथग्वा (ग) प्रथक् पृथग् वा स for (पृथग् पृथग वा)
३. कुट्टज्ञः (ग) for (कुदज्ञः)
(ग) (वि०—इसकी श्लोक संख्या २६ है)
१. न्यून for (न्मून)
४. दिवसेष्वधिमास for (दिवसे द्विमास)
(च) १. मासन्मून for (मासान्मून) २. पृथगपृथग्वा for (पृथग् पृथगवा)
(ङ) (वि०—इसकी क्रमसंख्या २६ है)
४. दिवसेष्वधि for (दिवसे द्वि) १. मासन्यून for (मासान्मून)
५. रात्र शेषे for (रात्रिशेषे) २. पृथक् for (प्रथग्)
२. पृथग् वा for (पृथग वा) ३. स कुट्टज्ञः for (कुदज्ञः)

निच्छेदभागहाराद्राश्यादि कलानाहताद्भुक्तात् ।
भगणकलाभिर्लब्धं मंडलशेषं दिनगणोस्मात् ॥ २६ ॥
एवं राश्यंश कलाशेषाण्यहर्गणः प्राग्वत् ।
नष्टस्थानेष्विष्टानं कान् कृत्वोक्त वच्छेशषम् ॥ २७ ॥
यो राश्यादीन् द्रष्ट्वा मध्यस्येष्टस्य कथयति द्युगुणम् ।
द्व्यधिग्रहयोगाद् ग्रहांतराद्वा स कुटज्ञः ॥ २८ ॥

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२६. (घ) १. कलादिनाहताद्भुक्तात् (ग) कलादिना हृताद्भुक्तात् for (कलानाहताद्भुक्तात्)

(ग) (वि०—इसकी श्लोक संख्या २१ है ।) २. (लंब्धं भमंडल for (लंब्धं मंडल)

(च) १. कलादिनाहताद्भुक्तात् for (कलानाहताद्भुक्तान्)

(ङ) (वि०—इसकी क्रम संख्या २१ है

४. निश्छेद for (निछेद) १. कलादिना for (कलाना)

३. हताद्भुक्तात् for (हताद्भुक्तात्)

५. दिनगणोऽस्मात् for (दिनगणोस्मात्)

२७. (घ) १. हर्गऽण: for (हर्गण:) २. छेषम् for (छेशषम्)

(ग) (वि—इसकी श्लोक संख्या २२ है ।) ३. कलाविकला (ङ) for (कला)

४. शेषादहर्गण: (ङ) for (शेषाण्यहर्गण:)

(च) २. वछेषं for (वछेशषम्) १. हऽर्गण: for (हर्गण:)

(ङ) (वि० इसकी क्रम संख्या २२ है)

५. नष्टस्थेष्विष्टान् for (नष्टस्थानेष्विष्टानं)

६. तान् for (कान्) ७. कृत्वा भक्त्वोक्त for (कृत्वोक्त)

२. वच्छेषम् for (वछेशषम्)

२८. (घ) १. द्युगणम् (ग) (च) (ङ) for (द्युगुणम्)

२. द्वचादि for (द्व्यधि) (ग) द्वचादिग्रहसंयोगात् for (द्व्यधिग्रहयोगाद्)

३. गुहांततरात् (ग) ग्रहांतरद्वा for (ग्रहांतराद्वा)

४. कुदकज्ञ: (ग) for (कुटज्ञ:)

(ग) (वि—इसकी श्लोक संख्या २० है परंतु लिखी हुई १० है ।)

५. दृष्ट्वा (च) (ङ) for (द्रष्ट्वा)

(च) २. द्वचादिग्रह for (द्व्यधिग्रह) ४. कुदज्ञ: for (कुटज्ञ:)

(ङ) (वि०—इसकी क्रम संख्या २० है)

२. द्वचादिग्रह संयोगात् for (द्व्यधिग्रहयोगाद्)

४. कुदज्ञ: for (कुटज्ञ:)

येन गुणे[१] शेषयुते[२] छेदः शुध्यति हृत[४] स्वगुणकेन[५] ।
तद्भक्तं[६] शेषं फलमेव[७] शेषाद् ग्रहद्युगुणैः[३] ॥ २९ ॥
राश्यंशकला[३] विकला शेषात्कथिताद्भीष्टतोनष्टात्[७] ।
यः साधयद्युपरिमान[१] समध्यमा[४] कुदकज्ञः[२] स ॥ ३० ॥
इति श्री कुट्टक प्रकरणम्[५][६]

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२९. (घ) १. गुण (ग) गुणः for (गुणे) २. युतछेद (ग) हृताछेदः for (युते छेदः)
३. गणैः (ग) गणौ for (गुणैः)

(ग) (वि०—इसकी श्लोक संख्या २४ है) ४. हृदः for (हृत)
५. न तदुक्तं for (तद्भक्तं) ६. फलमेवं for (फलमेव)

(च) १. गुणों शेषयुतां for (गुणे शेषयुते)
४. हृतः for (हृत) ३. द्युगणैः for (द्युगुणैः)

(ङ) १. (वि०—इसकी क्रम संख्या २४ है)
१. गुणः for (गुणे) २. युतश्छेदः for (युते छेदः)
४. हृतः for (हृत) ७. शेषाद्ग्रह for (शेषाद्ग्रह)
३. द्युगणौ for द्युगुणैः)

३०. (घ) १. त्युपरिमान् (ग) त्युपरिघमान् for (द्युपरिमान्)
५. वि०—मूल में 'इति श्री' पद नहीं है ।
२. कुट्टकज्ञः (ग) for (कुदकज्ञः स)

(ग) (वि—इसकी श्लोक संख्या २३ है)
३. विकलांशशेषात् for (विकलाशेषात्)
४. समध्यमान् for (समध्यमा)

(च) १. साधयत्युपरिमान् for (साधयद्युपरिमान्)
४. समध्यमात् for (समध्यमा) ५. 'इति श्री' लुप्त
६. कुदक for (कुट्टक)

(ङ) (वि०—इसकी क्रमसंख्या २३ है)
७. नष्टान् for (नष्टात्) १. साधयत्युपरितनान् for (साधयद्युपरिमान्)
४. समध्यमान् for (समध्यमा)
२. कुट्टकज्ञः for (कुदकज्ञः) ५. 'इति - से—राम्' तक सब लुप्त ।

५ अथ धनर्णं १ श्रुत्यानां ६ संकलना

धनयोर्धनमृणयोर्धनर्णयोरंतरं ७ समैक्यखं ८ २ ।
वर्णैक्यमृणं ३ धनशून्ययोः ४ दूनं शून्ययोः शून्यम् ।। ३१ ।।
ऋणमधिकाद् १ विशोध्यं धनं धनाद्दणमृणादधिकमूनात् ।
व्यस्तं तदंतरं वा ऋणं २ धनं ४ न मृण ३ भवति ।। ३२ ।।

३१· (घ) १. शून्यानां for (श्रुत्यानां)
२. समैक्यं (ग) for (समैक्य)
३. खर्णैक्य (ग) स्वर्णैक्यमृण for (वर्णैक्यमृणं)
४. द्धूनं (ग) for (दूनं)

(ग) (वि०—इसकी श्लोक संख्या ३० है)
५· 'अथ' से 'संकलना' तक लुप्त है।
६· स्वं for (खं)

(च) ५. अथधनर्ण्य for (अथधनर्णं)
१. शून्यानां संकलनाः for (श्रुत्यानां सकलना)
७. धनर्ण्ययो for (धनर्णयो) २. समैक्यं for (समैक्य)
३. खवर्ण्यैक्य for (वर्णैक्य) ४. द्धूनं for (दूनं)

(ङ) १. शून्यानां for (श्रुत्यानां) ४. "दूनं" लुप्त
६. सङ्कलनम् for (संकलना)
८.+मृण+ २. समैक्यं for (समैक्य)
३. ऋणमैक्यं च for (वर्णैक्य) १. धनमृण for (मृणं)

३२. (घ) १. ऊन (ग) (च) (ङ) for (ऋण)
२. खाद्दणं (ग) (च) for (वा ऋणं)
३. 'ण' को लिखना भूल गया

(ग) (वि०—इसकी श्लोक संख्या ३१ है)
४. धनं धनमृणं भवति for (धनं न मृणभवति)

(च) ३. नमृभति for (न मृणभवति)

(ङ) २. स्याद्दणं for (वा ऋणं) ३. धनमृणं for (नमृण)

शून्यविहीनमृणमृणं धनं धनं भवति शून्यमाकाशम् ।
शोध्ये यदा धनमृणादृणं धद्धातदश्रेय्यम् ॥ ३३ ॥
प्रत्युत्पन्नः ऋणमृणधनयोर्घाते धनमृणयोर्धन वधोधनं भवति ।
शून्यण्ययोः स्वधनयोः खशून्ययो र्वा वध शून्यम् ॥ ३४ ॥

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३३. (घ) १. दृणं (ग) (च) (ङ) for (दणं)
२. धनाद्धा (ग) (ङ) for (धद्धा)
३. क्षेप्यम् (ग) तदा क्षेप्यम् for (तदश्रेय्यम्)
(वि०—इसकी क्रमसंख्या ३४ लिखी है)

(ग) ४. विहिन for विहीन ५. शोध्यं (च) (ङ) for (शोध्ये)
(वि०—इसकी संख्या ३२ है)

(च) २. धट्वा for (धद्धा) ३. तदक्षेप्यं for (तदश्रेय्यम्)
(वि०—यहां क्रमसंख्या ३४ अंकित है)

(ङ) ७. तदा for (तद) ३. क्षेप्यम् for (श्रेय्यम्)

३४. (घ) १. शून्यर्णयोः (ग))ङ) for (शून्यण्योः)
२. वधः (ग) (च) (ङ) for (वध)
(वि०—इसका संख्याक्रम ३५ लिखा है ।)

(ग) (वि०—इसकी श्लोक संख्या ३३ है, परन्तु 'प्रत्युत्पन्न' यहां लुप्त है) (ङ)
ऋणमृणधनयोर्धातौ धनमृणयोर्धनवधो धनं भवति ।
शून्यर्णयोः खधनयोः खशून्ययोर्वा वधः शून्यम् ॥ ३३ ॥
३. घातो (ङ) for (घति) ४. खधनयोः for (स्वधनयोः)

(च) ३. घतो for (घते) ५. द्धेन for (धन)
१. शून्यर्ण्ययोः for (शून्यण्योः)
५. (वि०—यहां क्रमसंख्या ३५ अंकित है)

(ङ) ७. 'प्रत्युत्पन्नः' लुप्त

धनभक्तं[1] धनमृणहृतं[2] मृणं धन भवति खंखभक्तम्[5] ।
खंभक्तमृणे[6] धनधनमृणधनेन[3] हतमृणमृणं[4] भवति[5] ॥ ३५ ॥[8]
खोद्धृतमृणधनं[4] वा तच्छेदं[5] खमृणधन विभक्तं[1] वा ।
धनमृणधनयोर्वर्गः[6] खं खस्यपदं[7] कृतियत्तत[2] ॥ ३६ ॥[3]
योगोन्तर युतहीनो[5] द्विहतः[1] संक्रमणमंतरविभक्तम्[7] ।
वार्गान्तरमंतर[2] युतहीनं[3] द्विहतं[4] विषमकर्म[2] ॥ ३७ ॥[6]

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३५. (घ) (वि०—इस श्लोक की दूसरी पंक्ति का आरंभिक 'ख' पहली पंक्ति का अंतिम शब्द है)

१. हृतमृणं (ग) (ङ) (च) for (हतंमृणं)

२. धनं (ग) (च) (ङ) for (धन) ३. मृणं (ग) (च) (ङ) for (मृण)

४. हृत (ग) (च) (ङ) for (हत)

(वि०—इसका संख्याक्रम ३६ लिखा है)

(ग) ५. भक्तं खम् । यहां 'खं' पहली पंक्ति का अंतिम पद है ।

दूसरी पंक्ति 'भक्त' से आरम्भ होती है । इसकी क्रमसंख्या ३४ है । (ङ)

६. मृणेन (ङ) for (भक्तमृणे) ७. 'धन' यह पद लुप्त है (ङ)

८. मृणधनं भवति for (मृणमृणं)

(च) १. यहां क्रमसंख्या '३६' अंकित है ।

३६. (घ) १. वाः for (वा)

(वि०—इसका संख्याक्रम ३७ लिखा है)

(ग) २. कृतिर्यकृत् for (कृतियत्तत्)

(वि०—इसकी श्लोक संख्या ३५ है)

(च) ३. (वि०—यहाँ क्रमसंख्या ३७ अंकित है)

(ङ) ४. मृणं for (मृणं) ५. तच्छेदं for (तछेदं)

६. ऋणधनयो for (धनमृणधनयो)

७. स्वं खं for (खं) २. र्यत्तत् for (यत्तत)

३७. (घ) १. हृतः (ग) (च) (ङ) for (हतः)

२. वर्गान्तर (ग) (च) (ङ) for (वार्गान्तर)

३. हृतं (ग) (च) (ङ) for (हतं) ४. कर्म्मः (ग) (च) for (कर्म)

(वि०—इसका संख्याक्रम ३८ लिखा है)

(ग) ५. युतहिनो for (युतहीनो) (वि०—इसकी श्लोक संख्या ३६ है)

(च) ६. (वि०—यहां क्रमसंख्या ३८ अंकित है)

(ङ) ७. +वा+

करार्णलंबस्तत्कृति रिष्टहूतेष्टोन संयुतोऽल्पोभूः ।
अधिको द्विहतो बाहु संक्षेप्यावद्वधो वर्गः ॥ ३८ ॥
इष्टोद्धृत करार्णी पदक्रतियुतिरिष्टगुणितांतरकृतिर्वा गुण्यः ।
स्तिर्यगधोधो गुणकः समसद्गुणः सहितः ॥ ३९ ॥
सेष्टार्णछेदगुणो भाज्यछेदौ पृथक् युतावसकृत् ।
छेदैकगत हूतौद्धौ भाज्यो वर्गः समद्विवधः ॥ ४० ॥

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३८. (घ) १. करणी (ग) (ङ) for (करार्ण)
२. हूतो (ग) (च) for (हूतो) ३. बाहू (ग) बाहुः (ङ) for बाहु
(वि०—इसका संख्याक्रम ३९ है ।)
(ग) ४. हूते (च) for (हूते) ५. संयुताल्पोभूः for (संयुतोऽल्पोभूः)
६. संक्षेप्या यद्वधो for (संक्षेप्यावद्वो)
(च) १. करणी for (करार्ण) ७. (यहां क्रमसंख्या ४९ अंकित है)
(ङ) ५. संयुताऽल्पा भूः for (संयुतोऽल्पोभूः) ६. संक्षेप्पो यद्वधो for (संक्षेप्यावद्वधो)

३९. (घ) १. करणी (ग) (च) (ङ) for (करार्णी)
२. 'गुण्य' यह पद दूसरी पंक्ति के आरम्भ में है (ग)
(वि०—इसका संख्याक्रम ४० है)
(ग) ३. पदयूतिक्रति for (पदक्रतियुति) २. गुणास्ति for (गुण्यः)
४. गुणक (ङ) for (गुणकः) ५. समस्तद्गुण सहितः ॥ ३८ ॥ for
(समसद्गुणः सहितः ॥ ३९ ॥)
(च) २. वर्गुण्य for (वर्गुण्यः) ६. (यहां क्रमसंख्या ४० अंकित है)
(ङ) ३. पदयुतिकृति for (पदक्रतियुति) २. 'गुण्यः' दूसरी पंक्ति का आरम्भिक पद ।
५. समस्तद्गुणः for (समसद्गुणः)

४०. (घ) हूतो (ग) (च) (ङ) for (हूतौ)
२. समद्विवधः (ग) समिद्विविधः for (समद्विवधः)
(वि०—इसका संख्याक्रम ४१ है)
(ग) ३. गुणौ for (गुणो) ४. प्रथक् for (पृथक्) ५. वा भाज्यो (ङ) for (द्धौभाज्यो)
(वि०—इसकी क्रमसंख्या ३९ है) (ङ)
(च) ३. सेष्टूर्ण्यछेदगुणो for (सेष्टूर्णछेदगुणो) ६. भाद्यछेदौ for (भाज्यछेदौ)
५. द्धौभाद्यो for (द्धौभाज्यो) २. समद्विवधः for (समद्विवधः)
७. (वि०—यहां क्रमसंख्या ४१ अंकित है)
(ङ) ३. च्छेदगुणौ for (छेदगुणो) ७. युजावसकृत् for (युतावसकृत्)
२. समद्विवधः for (समद्विवधः)

इष्टकरन्यूनाया[१] रूपकृतेः[४] पदयुतोनरूपार्द्धे[५] ।
प्रथमे[२] रूपाण्यन्यहिना[३] द्वितीयं[६] करण्यसकृत् ।। ४१ ।।[७]
अव्यक्तवर्गधनवर्गवर्ग[१][२] पंचगतषड्धतादीनाम्[३] ।
तुल्यानां संकलित[४] व्यवकलिते[५] पृथगतुल्यानाम्[६] ।। ४२ ।।[७]
सदृश द्विवधो[२] वर्गव्यादिवधस्तद्गतोऽन्यरूपवधः[३] ।
अन्योन्यवर्गघातो[४] भावितकः[१] पूर्ववच्छेषम्[६] ।। ४३ ।।[५]

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४१. (घ) १. करण्यूनाया (ग) (च) (ङ) for (करन्यूनाया)
२. प्रथमं (ग) (ङ) for (प्रथमे) ३. द्विना (ग) द्वितिय for (हिना)
(वि०—इसका संख्याक्रम ४२ है)
(ग) ४. रूपकृतेः (ङ) for (रूपकृतैः) ५. यूतो for (युतो)
६. 'द्वितीयं' लुप्त है
(वि०—इसकी श्लोक संख्या ४० है)
(च) ३. रूपाण्यन्यद्विना for (रूपाण्यन्यहिना)
(वि०—यहां क्रमसंख्या ४२ अंकित है)
(ङ) १. ण्यूनाया for (न्यूनाया) ३. त्ततो for (हिना)

४२. (घ) (वि०—इसका संख्याक्रम ४३ है।)
(ग) १. घनवर्ग (ङ) for (धनवर्ग) २. यहां एक 'वर्ग' लुप्त है।
३. गतः for (गत) ४. संकलिते for (सङ्कलित)
५. 'व्यवकलिते' लुप्त है
६. प्रथग for (पृथग)
(वि०—इसकी श्लोक संख्या ४१ है)
(च) ७. (वि०—यहां क्रमसंख्या ४३ अंकित है)
(ङ) (वि०—क्रमसंख्या ४१ है)

४३. (घ) (वि०—इसका संख्याक्रम ४४ है)
१. वर्ण (ग) (ङ) for (वर्ग)
(ग) २. द्विधो for (द्विवधो) ३. न्यवधः for (ऽन्यरूपवधः)
४. सन्योन्य for (अन्योन्य)
(वि०—इसकी श्लोकसंख्या ४२ है)
(च) १. वर्ण्य for (वर्ग)
५. (वि०—यहां क्रमसंख्या ४४ अंकित है)
(ङ) ३. ऽन्यजातिवधः for (ऽन्यरूपधः) ६. पूर्ववच्छेषम् for (पूर्ववंछेषम्)

'इति धनर्णादीनां सङ्कलित व्यवकलितादि'

मंडलशेषाद्युनान्मूलं व्येकं दशाधृतं द्वियुतम् ।
मंडलशेषं व्येक भानोर्ज्ञदिने कदा भवति ॥ ५० ॥
अधिमास शेषपादा न्यूनाद्वर्गोधिमासशेषसमः ।
अवमावशेषतो वा वमवशेषसमः कदा भवति ॥ ५१ ॥
आद्याद्वर्णादन्यान् व्यस्तान्प्रोह्याद्यमानमाद्यहृतात् ।
सद्रशा छेदान् सकृत् द्वौ व्यस्तौ उदकोबप्पू ॥ ५२ ॥

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५०. (घ) (वि०—इसकी क्रमसंख्या ५१ है)
१. द्वचू (ग) द्युनामूलं for (द्युनान्मूलं)
२. हृतं (ग) हतं for (धृतं) ३. व्येकं (ग) (ङ) for (व्येक)
(ग) . (वि०—इसकी श्लोक संख्या ४६ है)
(च) १. द्वचूनान् for (द्युनान्) २. हतं for (धृतं)
४. (वि०—इसकी क्रमसंख्या ५१ अंकित है)
(ङ) १. द्वचूनान्मूलं for (द्युनान्मूलं) २. दशाहतं for (दशाधृतं)

५१. (ग) (वि०—इसकी क्रमसंख्या ५२ है)
(ग) १. न्यूनाद्वर्गा for (न्यूनाद्वर्गो) २. यमः for (समः)
६. अमावशेषतो for (अवमावशेषतो) ४. 'व' लुप्त है (ङ)
(वि०—इसकी श्लोक संख्या ५० है)
(च) १. द्वर्गोऽधिमासशेषः for (द्वर्गोधिमासशेष)
५. (वि०—क्रमसंख्या ५२ अंकित है)
(ङ) ६. पादात् for (पादा) १. न्यूनाद्वर्गो for (न्यूनाद्वर्गो)
१. हृतात् (च) for हूतात्) २. सदृस (ग) सदृशा for (सद्रशा)
३. नसकृत (ग) for (नसकृत्)
४. कुदको (ग) (ङ) for (उदको) ५. बहुषु (ग) (ङ) for (बप्पू)
(वि०—इसकी श्लोक संख्या ५३ है)
(ग) (वि०—इसकी श्लोक संख्या ५१ है)
६. मद्य for (माद्य) ६. तत् for (तात्)
(च) २. सदृस for (सद्रशा) ३. नसकृत् for (नसकृत्)
५. बप्पु for (वप्पू) (वि०—यहां श्लोक संख्या ५३ अंकित है)
(ङ) शीर्षक+इदानीमनेकवर्णसमीकरणमाह+
८. वर्णान् for (व्यस्तान्) २. सदृश for (सद्रशा)
१. हृतम् for (हूतात्) ३. च्छेदावसकृदद्वौ for (छेदान् सकृत्द्वौ)

५ १ २ ३ ६
गतभगणयुता द्यु गुणा तद्वे षयुता तदैक्ययुताद्वा ।
७ ४ ८ ६
तद्योगाद्वा द्यु गुणो यः कथयति कुट्टक्ज्ञः सः ।। ५३ ।।
३ १ २ ५
गतभगणोना द्यु ना गणातच्छेषो ना तदैक्यहीनाद्वा ।
४ ६ ७
तद्विवराद्वा द्विगुणं यः कथयति कुट्टकज्ञः सः ।। ५४ ।।

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५३. (घ) १. गणात्त (ग) द्युगुणात् for (द्युगुणा)
२. छेष (ग) for (तद्वेष) २. त्तदैक्य (ग) for (तदैक्य)
४. गणं for (गुणो)
(वि०—इसकी क्रमसंख्या ५४ लिखी है)
५. गतभंगयुतात् for (गतभगणयुता)
६. संयुक्तात् (ङ) for (युताद्वा) ७. तद्योगाद्यु गुणं for (तद्योगाद्वाद्यु गुणो)
८. कुट्टकज्ञः (ङ) for (कुट्टक्ज्ञः)
(वि०—इसकी श्लोक संख्या ५२ है)

(च) १. द्यु गणा for (द्यु गुणा) २. त्तच्छेषयुता for (तद्वे षयुता)
६. युक्ताद्वा for (युताद्वा) ७. ताद्योगाद्वा for (तद्योगाद्वा)
४. द्यु गणं for (द्यु गुणो) ८. कुदकज्ञ for (कुट्टक्ज्ञः)
१. (वि०—यहां श्लोक संख्या ६४ अंकित है)

(ङ) ५. भगणयुताद् for (भगणयुता) १. द्युगणात् for (द्यु गुणा)
२. तच्छेषयुतात् for (तद्वे षयुता)
४. द्यु गणं वा for (वा द्यु गुणो)

५४. (घ) (वि०—इसकी क्रमसंख्या ५५ है)
१. द्यु गणात्त (ग) द्युगणात्त्त for (द्युनागणा)
२. छोषानात्तदैक्य (ग) छेशोना for (छेषोना)

(ग) ३. नात् (ङ) नाद् for (ना) ४. तद्विवराद्युगणं for (तद्विवराद्वा)
(वि०—इसकी श्लोकसंख्या ५३ है)

१
राश्याद्यैस्त छेशैश्चैवं भुक्ताधिमास हीनदिनै ।
२
तच्छेषैश्चयुगगतं य कथयति कुट्टकज्ञः सः ।।५४।।

(ङ) १. दिनहीनैः for (हीनदिनैः) २. तच्छेषैश्च for (तच्छेषैश्च)

(च) १. द्यु गणा for (द्यु नागणा) ५. त्तदैक्य for (तदैक्य)
६. कुदकज्ञ for (कुट्टकज्ञ) ७. (वि०—यहां क्रमसंख्या ५५ अंकित है)

(ङ) ३. द्युगणात् for (द्यु नागणा) २. तच्छेषोनात् for (तच्छेषोन)
४. द्युगणं वा for (वाद्विगुणं)

अंशक शेषेण युता स्त्रिप्ता शेषात्र तदंतरादथवा ।
भानोर्ज्ञदिने द्युगुणं यः कथयति कुटकज्ञः सः ।। ५५ ।।
अंशक शेष त्रियुतं लिप्ता शेषं कदा रवेर्ज्ञ दिने षट् ।
सप्ताष्टोन वा कुर्वन्नावत्सराद्गणकः ।। ५६ ।।
अंशसम मंशके कलासमं वा कला कदा शेषम् ।
दिवसकरस्येष्ट दिने कुर्वन्नावत्सराद्गणकः ।। ५७ ।।

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५५. (घ) (वि०—इसकी क्रमसंख्या ५६ है)
१. लिप्ता (ग) (च) for (स्त्रिप्ता) ३. द्युगणं (च)(ङ) for (द्युगुणं)
२. शेषात्तदंतरा (ग) (च) for (शेषात्रतदंतरा)
४. कुट्टकज्ञ (ग) कुट्टकज्ञः for (कुटकज्ञः)
(ग) ५. हता for (युता) (वि०—इसकी श्लोक संख्या ५५ है)
(च) ४. कुदकज्ञः for (कुटकज्ञः) (वि०—इसकी क्रमसंख्या ५६ अंकित है)
(ङ) ५. युतात् for (युता) लिप्ता for (स्त्रिप्ता)
२. शेषात् for (शेषात्र)

५६. (घ) (वि०—इसकी क्रमसंख्या ५७ है) १. नं वा (ग) नावां for (नवा)
(ग) २. शेषं (ङ) for (शेष)
३. 'षट्' दूसरी पंक्ति का प्रथम पद है (ङ)
४. सप्तष्टौ for (सप्ताष्टो)
(वि०—इसकी श्लोकसंख्या ५६ है)
(च) १. सप्ताष्टोनं for (सप्ताष्टोन)
५. (वि०—इसकी क्रमसंख्या ५७ अंकित है)
(ङ) १. नव वा for (न वा) ४. सप्ताष्टौ for (सप्ताष्टो)

५७. (घ) (वि०—इसकी क्रमसंख्या ५८ है)
१. मंशाशकज्ञ for (मंशके) २. कदा (ग) for (कला)
६. कला (ग) for (कदा) ४. दिवसरस्येष्ट for (दिवसकरस्येष्ट)
५. त्स्थराद्गणकः for (त्सराद्गणकः)
(ग) ६. अंशयुगमंशशेषं for (अंशसममंशके) ७. कलासमांश for (कलासमं)
(वि०—इसकी क्रमसंख्या ५८ है, संख्या ५७ अंकित करना भूल गया)
(च) १. मंशशे for (मंशके) २. ३. कदा कलाशेषं for (कलाकदाशेषम्)
४. दिवसरस्येष्ट दिने for (दिवसकरस्येष्ट दिने)
५. स्थराद्गणकः for (सराद्गणकः)
(ङ) १. मंशशेषं for (मंशके) ३. 'कदा' लुप्त

अवमावशेषमवमैरधिमासकशेषमधिकमासैर्वा[3] ।
इष्टयुतोनं तुल्यं कुर्वन्नावत्सरगणकः[1][2] ॥ ५८ ॥
निच्छेद[3] भागहारो[1] भानो[4] सप्ततिगुणोंशशेषोनः[2] ।
शुध्यत्ययुत विभक्तः कुर्वन्नावत्सराद्गणकः[5] ॥ ५९ ॥
भावितकरूप गुणना साव्यक्तवधेष्टभाजिताप्वासौ[1] ।
अल्पाधिकाधिकोल्पेक्षेप्यौ[2][3] भावितहूतौ[4] व्यस्तौ[5] ॥ ६० ॥[6]

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५८ (घ) (वि०—इसकी क्रमसंख्या ५९ है)
१. सराद्ग णकः for (सरगणकः)

(च) १. सद्गणकः for (सरगणकः)
२. (वि०—यहां क्रमसंख्या ५९ अंकित है) ३. र्वाः for (र्वा)

(ङ) ३. मधिमासैः for (मधिकमासैर्वा) १. सराद्गणकः for (सरगणकः)

५९. (घ) (वि०—इसकी क्रमसंख्या ६० है)
१. भागहारो (ङ) (मागहारो) २. गुणोऽंश for (गुणोंश)

(ग) ३. निश्छेद (ङ) for (निच्छेद) ४. भानोः (च) (ङ) for (भानोः)
(वि०—क्रमसंख्या ५९ है)

(च) २. गुणोऽंशशोषोनः for (गुणोंशशेषो नः)
(वि०—क्रमसंख्या ६० अंकित है)

६० (घ) (वि० – इस श्लोक की क्रमसंख्या ६१ है)
१. भाजितेऽष्टचाप्तौ (ग) भाजितेष्टाप्त्योः (ङ) for (भाजिताप्वासौ)
२. अल्पऽधिकोऽधिको (ग) अल्पेऽधिकल्पः for (अल्पाधिकोधिकोल्पे)
३. थौ (ग) क्षेप्यो for (क्षेप्यौ) ४. हूतौ (च) (ङ) for (हूतौ)

(ग) ५. व्यस्तम् (ङ) for (व्यस्तौ)

(च) १ भाजिष्टचाप्तौ for (भाजिताप्वासौ)
६. (वि०—यहां क्रमसंख्या ६१ अंकित है)

(ङ) शीर्षक—+अथ भावितबीजम्+
२. अल्पेऽधिकोऽधिकेऽल्पः for (अल्पाधिकोधिकोल्पे)
३. क्षेप्या for (क्षेप्यौ)

भानोराश्य शवधा[२] त्रिचतुर्गुणितान्विशोध्यराश्यंशात्[३] ।
नवति[१] द्रष्ट्वा[४] सूर्यं कुर्वन्नावत्सराद्गणकः[५] ॥ ६१ ॥
भावितके[३] तद्घातो[६] विनष्ट[१] वर्णेन तत्प्रमाणानि ।
ऋत्वेष्टानि[२] तदाहत[४] वर्णैक्यं[१] भवति रूपाणि[५] ॥ ६२ ॥
वर्णप्रमाणभावितघातो[४] भवतीष्टवर्णसंख्यैवम्[२] ।
सिध्यति विनापि भावितसमकरणात् किंकृतं तदतः[३] ॥ ६३ ॥[५]
मूलद्विधेष्ट[१]वर्गागुण[६]गुणा[२]दियुत[७]विहीनाच्च[५] ।
आध्यवधो[३] गुणक[४] गुणः[६] संहात्पवातेन[४] पमत्यम्[५] ॥ ६४ ॥[१०]

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६१. (घ) (वि० इस श्लोक की क्रमसंख्या ६२ है) १. नवतिं (ग) नतिं for (नवति)
(ग) २. वधात्रि (ङ) for (वधात्रि) ३. राश्यंशान् for (राश्यंशात्)
(च) १. नवतिं for (नवति) ४. दृष्ट्वा (ङ) for (द्रष्ट्वा)
५. (वि०—यहां क्रमसंख्या ६२ अंकित है) (ङ) १. नवतिं for (नवति)

६२. (घ) (वि०—इसकी श्लोकसंख्या ६३ है) १. विनष्टे (च) for (विनष्ट)
२. कृत्वेष्टानि (ग) (च) (ङ) for (ऋत्वेष्टानि)
(ग) ३. भाविक यद्घातो for (भावितके तद्घातो)
(च) ५. (वि०—इसकी क्रमसंख्या ६३ अंकित है)
(ङ) ६. यद्घातो for (तद्घातो)

६३. (घ) (वि०—इसकी क्रमसंख्या ६४ है)
(ग) १. भवतिष्ट for (भवतीष्ट) २. संख्यैक्यम् for (संख्यैवम्) ३. काम for (सम)
(च) ४. द्यातो for (घातो) ५. (वि० इसकी क्रमसंख्या ६४ अंकित है)

६४. (घ) (वि०—इस श्लोक की क्रम संख्या ६५ है)
१. मूलं (ग) for (मूल) २. वर्गाद्गुण (ग) वर्गात् गुणक for (वर्गागुण)
३. आद्य (ग) (च) (ङ) for (आध्यवधो)
४. सहांत्यघातेन (ग) संहत्य घातेन for (संहात्पवातेन)
५. पमंत्यम् (ग) मन्त्यम् for (पमत्यम्)
(ग) ३. द्विधेष्ठ for (द्विधेष्ट) ७. गुणादिष्टयुत for (गुणादियुत)
८. विहिनाच्च for (विहीनाच्च) ९. गुण for (गुणः)
(च) २. वर्गाद्गुण for (वर्गागुण) ४. सहांत्यघातेन for (संहांत्यवातेन)
५. पमंत्यं for (पमत्यम्) १०. (वि०—इसकी क्रम संख्या ६५ अंकित है)
(ङ) श्री—+अथ वर्ग प्रकृतिः+
२. वर्गाद् गुणक for (वर्गागुण) ७. गुणादिष्ट for (गुणादि)
४. सहान्त्यघातेन for (संहांत्पवातेन) ५. कृतमन्त्यम् for (पमत्यम्)

वज्ञावाधेक्यं[१] प्रथमं प्रक्षेपः क्षेप्य[३] शोध्यतुल्यवधः ।
प्रक्षेपशोधकहूते[२] मूले प्रक्षेपके रूपे[४] ॥ ६५ ॥
रुपप्रक्षेपपदे[३] पृथगीष्टक्षेप्यशोध्यमूलाभ्याम्[१] ।
कृत्वाद्योंत्पाद्यपदे[२] प्रक्षेपे शोधने वेष्टे[५] चतुरधिके[६] ॥ ६६ ॥
इत्यपदकृतिस्त्र्यूना[४] दलितांत्य पदगुणात्पदम्[१] ।
अंत्यपदकृतिव्येका[५] द्विहताद्यपदा[३] हताद्यपदम्[६] ॥ ६७ ॥[८]

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६५. (घ) (वि०—इस श्लोक की क्रम संख्या ६६ है)
१. वज्ञवधैक्यं (ग) वज्रमधैक्यं for (वज्ञावाधेक्यं)
२. हूते (ग) तुहूते for (हूते)
(ग) ३. 'शोध्य' पद लुप्त है।
(च) १. वज्ञवाधेक्यं for (वज्ञावाधेक्यं)
२. हूते for (हूते) ४. (वि०—यहाँ क्रमसंख्या ६६ अंकित है।)
(ङ) १. वजवधैक्यं for (वज्ञावाधेक्यं)
३. क्षेपवधतुल्यः for (क्षेप्यशोध्यतुल्यवधः) २. हूते for (हूते)

६६. (घ) (वि०—इस श्लोक की क्रम संख्या ६७ है)
१. पृथगिष्ट (ग) (च) (ङ) for (पृथगीष्ट)
२. ऽद्योत्पाद्य (ग) कृत्वाद्यांशाद्य पदे for (कृत्वाद्योत्पाद्यपदे)
(ग) ३. रूपं for (रूप) ४. 'क्षेप्य' पद लुप्त है।
५. वेष्टे (यहाँ श्लोक समाप्त है), 'चतुरधिके' ६७ वें श्लोक का आरंभ है।
(च) ४. रूपप्रक्षेपदे for (रूपप्रक्षेपपदे) २. ऽद्योत्पाद्य for (द्योत्पाद्य)
५. वेष्टे for (वेष्टे) ६. रधिकें for (रधिके)
(ङ) २. कृत्वाऽऽन्त्याद्यपदे for (कृत्वाद्योंत्पाद्यपदे)
७. +ये+ ५. वेष्टे for (वेष्टे) ६. "चतुरधिके" लुप्त

६७. (घ) (वि०—इसकी श्लोक संख्या ६७ है)
१. गुणांत्यपदं (ग) for (गुणात्पदम्)
२. व्यैका (च) (ङ) for (व्येका) ३. हूता (ग) (च) (ङ) for (हता)
(ग) ४. चतुरधिकेत्य for (इत्य) ५. पदं for (पद)
६. पादा for (पदा)
(च) ९. स्त्र्यना for (स्त्र्यूना) १. गुणांत्यपदं for (गुणात्पदं)
७. व्यैका for (व्येका) ८. (वि०—क्रमसंख्या ६८ अंकित है)
(ङ) ४. चतुरधिकेऽन्त्य for (इत्य) १. गुणाऽन्त्यपदम् for (गुणात्पदम्)

चतुरूनेंत्यपद कृताग्रेकयुतेवधदलं प्रथग्वेकम् ।
व्येकाध्व्यहृतमंत्यं पदवधगुण मन्यदाद्यहदम् ।। ६८ ।।
वग गुणकः क्षेप; केनचिदुद्धृत युतोन को दलितः ।
प्रथमोंत्ये मूलमन्यो गुणरपदोद्धृत प्रथमम् ।। ६९ ।।
गुणके वर्गछिन्ने छेदपदे यो धृतपदं प्रथमवर्गः ।
छेत्रे क्षेपे तन्मूले छेदपद गुणिते ।। ७० ।।

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६८. (घ) (वि०—इसकी श्लोक संख्या ६९ है) १. श्यैक for (अेक)
२. पृथग् (ग) for (प्रथग्) (च) ३. द्याह् (ग) (ङ) for (ध्व्यह्)
(ग) ४. चतुरूनोंत्यपदकृतित्येकयुते for (चतुरूनेंत्यपदकृता अेकयुते)
५. मन्यथाद्यपदम् for (मन्यदाद्यपदम्)
(च) १. श्येकयुते for (अेकयुते) ३. द्याहत for (ध्व्यहत)
(ङ) ७. कृती for (कृता) १. श्येकयुते for (अेकयुते)
२. पृथग्व्येकम् for (प्रथग्वेकस्) ५. माद्यमान्त्य for (मन्यदाद्य)

६९. (घ) (वि०—क्रम संख्या ७० है) १. गुणकार (ग) (ङ) for (गुणर)
२. पदोद्धृतः प्रथमम् (ग) धृतप्रथमः for (पदोद्धृत प्रथमम्)
(ग) ३. वर्गेण for (वर्गे) ४ गुणके (च) (ङ) for (गुणकः)
५. दुद्धृतो नितोदलतः for (दुद्धृ तयुतोनको दलितः)
६. प्रथमोंत्य (ङ) for (प्रथमोंत्ये)
(च) २. पदोद्धूतः for (पदोद्धृत) ७. (वि०—क्रम संख्या ७० अंकित है)
(ङ) ७. युतोनितोदलितः for (युतोनको दलितः)
२. पदोद्धृतः प्रथमः for (पदोद्धृत प्रथमम्)

७०. (घ) (वि०—इसकी क्रमसंख्या ७१ है)
१. धृतं for (धृतपदं) २. प्रथमं for (प्रथम)
३. 'वर्ग' (यह शब्द दूसरी पंक्ति का आरंभिक पद है) ४. छिन्ने for (क्षेत्रे)
(च) ६. वर्गछोन्ने for (वर्गछिन्ने) १. द्धृतपदं for (धृतपदं)
३. वर्ग for (वर्गः) ४. छेन्ने for (छेत्रे)
७. (वि०—यहाँ क्रम संख्या ७१ अंकित है)
(ङ) 'वर्गच्छिन्ने गुणके प्रथमं तन्मूलभाजितं भवति ।
वर्गच्छिन्ने क्षेपे तत्पदगुणिते तदा मूले ।। ७० ।।
for
गुणके वर्गच्छिन्ने छेदपदे यो धृतपदं प्रथमवर्गः ।
क्षेत्रे क्षेपे तन्मूले छेद पद गुणिते ।। ७० ।।

शशिलिप्ता शेषकृतिद्विनवति गुणिता अशिति गुणितं वा ।
सैकाज्ञ दिने वर्गं कुर्वन्नावज्ञपराद् गणकः ॥ ७१ ॥
इष्टभगणादि शेषं द्विनवत्यूनं व्यशिति संगुणितम् ।
रूपेण युतेन वर्गं कुर्वन्नावत्सराद् गणकः ॥ ७२ ॥
भगणादिशेषवर्गं चतुर्गुणं पंचषष्टिसंयुक्तम् ।
षष्टूनं वा वर्गं कुर्वन्नावत्सराद् गणकः ॥ ७३ ॥

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७१. (घ) १. कृतिः (ग) कृति for (कृति) (वि०—इसकी क्रम संख्या ७२ है)
२. अशीति (ग) व्यशीति for (अशिति) ३. गुणितां for (गुणितं)
४. कुर्वन्नावत्सरादृगणकः (ग) for (कुर्वन्नावज्ञपरादृगणकः)
(ग) ५. द्विनवगुणितं for (द्विनवतिगुणिता) ६. सैकं for (सैक)
७. वर्गे for (वर्ग)
वि०—इसके आगे "गुणितं वा । सैकंज्ञ दिने वर्ग कुर्वन्ना० ॥ ७५ ॥" लिखा है ।
(च) १. कृतिः for (कृति) ५. द्विनवति गुणितां for (द्विनवतिगुणिता)
२. अशीति for (अशिति) ३. गुणितां वा for (गुणितं वा)
४. कुर्वन्नावत्सरादृगणकः for (कुर्वन्नावज्ञपरादृगणकः)
८. (वि०—यहाँ क्रमसंख्या ७२ अंकित है)
(ङ) वि यह श्लोक इस प्रति में उपलब्ध नहीं है ।

७२. (घ) (वि०—इसकी क्रमसंख्या ७३ है)
१. व्यशीति (ग) व्यशीति for (व्यशिति)
२. युतं (ग) (ङ) for (युतेन)
(ग) ३. वर्ग for (वर्ग) (वि०—इसकी श्लोकसंख्या ७७ हैं)
(च) १. व्यशीति for (व्यशिति) २. युतं for युतेन
४. (वि०—क्रम संख्या ७३ अंकित है)
(ङ) १. व्यशीति for (व्यशिति)

७३. (घ) (वि०—इसकी क्रम संख्या ७४ है)
यह श्लोक १९०२ में बनारस से मुद्रित प्रति में ७८ संख्या पर है
(ग) (वि०—यह श्लोक 'ग' में उपलब्ध नहीं है)
(च) (वि०—क्रमसंख्या ७४ अंकित है)
(ङ) (वि०—इसकी क्रमसंख्या ७८ अंकित है)

भगणादिशेष वर्गं त्रिभिर्गुणं संयुतं शतैर्नवभिः ।
वर्गोष्टशतोनं वा कुर्वन्नावत्सराद् गणकः ।। ७४ ।।
अधिमासशेषवर्गं त्र्योदशगुणं त्रिभिर्युक्तम् ।
त्रिघनोनं वा वर्गं कुर्वन्नात्सराद् गणकः ।। ७५ ।।
अवमावशेषवर्गं द्वादशगुणितं शतेन संयुक्तम् ।
त्रिभिरूनं वा वर्गं कुर्वन्नावत्सराद् गणकः ।। ७६ ।।

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७४. (घ) (वि०—इसकी क्रमसंख्या ७५ है)

(ग) १. शनैर्न्नवभिः for (शतैर्नवभिः)
(वि०—इसकी क्रमसंख्या ७८ है)

(च) २. (वि०—क्रमसंख्या ७५ अंकित है)

(ङ) ३. कृतिमष्ट for (वर्गोष्ट)

७५. (घ) (वि०—इसकी क्रमसंख्या ७६ है)
१. त्रयोदश (ग) for (त्र्योदश) २. कुर्वन्नावत् (ग) (ङ) for (कुर्वन्नात्)

(ग) ३. वर्ग्रं for (वर्गं) ४. त्रिभिः शतैर्युक्तम् (ङ) for (त्रिभिर्युक्तम्)
५. वर्ग्रं for (वर्गं)
(वि०—इसकी क्रमसंख्या ७६ है)

(च) १. त्रयोदश (ङ) for (त्र्योदश)
६. (वि०—यहां क्रमसंख्या ७६ अंकित है)

७६. (घ) (वि०—इसकी क्रमसंख्या ७७ है)

(च) (वि०—इसकी क्रमसंख्या ७७ है)

(ङ) (वि०—इसकी क्रमसंख्या ८२ है)

(ग) (वि०—इसकी क्रमसंख्या ८० है)

७६. संख्या पर अंकित निम्नांकित अतिरिक्त श्लोक है—

सूर्यादि विलिप्ताशेषं पंचभिरूनादृगतं तथा दशभिः ।
वर्ग्रो बृहस्पति दिने कुर्वन्नावत्सराद् गणकः ।। ७६ ।।

(वि०—यह श्लोक इस प्रति में क्रमसंख्या ७६ पर है)
१. सूर्यं for (सूर्यादि) २. रूनाहतं for (रूनादृगतं)
३. वर्गं for (वर्ग्रो)

गुणकयुतिरष्टगुणिता गुणिकांतरवर्गभाजिता राशिः ।
गुणकौ त्रिगुणौ व्यस्ताधिकौ हृतावंतरेण पदे ॥ ७७ ॥
इन्दुविलिप्ताशेषं सप्तदशगुणं च यो दशगुणं च ।
पृथगेयुतं वर्गं कुर्वन्नावत्सराद् गणकः ॥ ७८ ॥
वर्गोनकृतियुतो नानगास्तत्संयोंतरार्द्धकृतिभक्तः ।
तद्गुणकौ युतिवियुतौ वर्गो घाते च रूपयुते ॥ ७९ ॥

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७७ (घ) (वि०—इसकी क्रमसंख्या ७८ है)
१. गुणकांतर (ग) (ङ) (च) for (गुणिकांतर)
३. हृतावतरेण (ग) हृतावंतरेण (च) (हृतावंतरेण)
(ग्र) (वि०—यह श्लोकसंख्या इस प्रति में ७० पर, तथा बनारस में मुद्रित प्रति में ७१ पर अंकित है)
(च) (वि०—यहां क्रमसंख्या ७८ अंकित है)
(ङ) (वि०—इसकी क्रमसंख्या ७१ है)

७८. (ध) (वि०—इसकी क्रमसंख्या ७९ है)
१. त्रयोदशं for (चयोदश) (ग) 'च'से 'च' तक लुप्त
(ग) ३. गुणितं वा for (गुणं च) ३. पृथगेक (ङ) for (पृथगे)
(वि०—इसकी क्रमसंख्या ८१ है)
(च) १. त्रयोदश (ङ) for (चयोदश)
४. (वि०—क्रमसंख्या ७९ अंकित है)
(ङ) ५. चापि for (च)

७९.(घ) (वि०—इसकी क्रमसंख्या ८० है)
१. नगांस्तत्सांयातरार्द्ध (ग) नस्तत् संयोगांतरार्द्ध for (नानगास्तत्संयोतरार्द्ध)
(ग) २. न्यकृति for (नकृति) ३. कृतिर्भक्तः for (कृतिभक्तः)
४. तद्गुणितौ (ङ) for (तद्गुणकौ)
(च) २. वर्गोन्यकृति for (वर्गोनकृति)
१. नगांस्तत्संयोतरार्द्ध for (नानगास्तत्संयोतरार्द्ध)
४. तद्गुणको for (तद्गुणकौ)
५. (वि०—यहां क्रमसंख्या ८० है)
(ङ) (वि०—इसकी क्रमसंख्या ७२ है)
२. वर्गोऽन्यकृति for (वर्गोनकृति) १. नस्तत् for (नानगास्तत्)
५. वर्गौ for (वर्गो) ६. संयोगान्तरार्ध for (संयोंतरार्द्ध)

यैरूनो यैश्चयुतो रूपैवर्गस्तदैक्यमिष्टहूतम् ।
इष्टोनं तद्दलकृतिरूनाभ्यधिका भवन्ति राशिः ॥ ८० ॥
याभ्यां कृतिरधिकोनस्तदंतरं हूतयुतो न मिष्टेन ।
तद्दलकृतिरधिके न्यूनाभ्यधि भवति राशि ॥ ८१ ॥
ज्ञदिनेऽर्क कलाशेषं गुरू दिनविकलावशेष युक्तोनम् ।
वर्गोविधं च सैकं कुर्वन्नावत्सराद् गणकः ॥ ८२ ॥

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८०. (घ) (वि०—इसकी क्रमसंख्या ८१ है)
१. वर्ग (ग) (ङ) for (वर्ग) २. हृतं (ग) (ङ) for (हूतम्)
३. भवति (ग) (च) (ङ) for (भवन्ति)
४, राशिम् (च) for (राशिः)
(ग) (वि०—इसकी क्रमसंख्या ७२ है)
(च) १. वर्गस्तदैक्य for (वर्गस्तदैक्य) ५. (वि०—इसकी क्रमसंख्या ८१ अंकित है)
(ङ) (वि०—इसकी क्रमसंख्या ७३ है) १. वर्गस् for (वर्गस्)

८१. (घ) १. रधिकेन for (रधिके) (वि०— इसकी क्रमसंख्या ८२ है)
२. राशिः (ग) (च) (ग) for (राशि)
(ग) १. रधिकोना धिकयोरधिकोनयो राशिः for (रधिके न्यूनाभ्यधि भवति राशि)
३. हृत (ङ) (च) for (हूत) ४. मीष्टेन for (मिष्टेन)
५. ६. 'न्यूनाभ्यधि' 'भवति' पद लुप्त हैं।
(च) १. रधिकेन for (रधिके) ५. न्यूनाभ्यधिका for (न्यूनाभ्यधि)
७. (वि०—क्रमसंख्या ८२ अंकित है)
(ङ) १. रधिकोना for (रधिके)
५. ऽधिकयोरधिको for (न्यूनाभ्यधि) ६. नयो for (भवति)

८२. (घ) (वि० इसकी क्रमसंख्या ८३ है)
(ग) (वि०—इसकी क्रमसंख्या ८२ है)
१. ज्ञदिनेर्क for (ज्ञदिनेऽर्क) २. दिने for (दिन)
३. वर्ग धन for (वर्गोविधं)
(च) १. ज्ञदिनेर्क्क for (ज्ञदिनेऽर्क) २. दिनविवला for (दिनविकला)
३. वर्गोवधं for (वर्गोविधं) ४. (वि०—क्रमसंख्या ८३ अंकित है)
(ङ) (वि०—इसकी क्रमसंख्या ८३ है)
३. वर्ग वधं च for (वर्गो विधंच)

विकलाशेषं[3] सहितं त्रिवत्याऽ[1]३ सप्तषष्टिहीनं[2] च
भानोर्ज्ञदिने वर्गं[4] कुर्वन्नावत्सराद् गणकः ॥ ८३ ॥
ज्ञदिनेऽर्क[1] कलाशेषं[2] वर्गो[3] द्वादशभिः संयुतं त्रिषष्टया च ।
षष्टयाष्टाभिश्चोनं[5] कुर्वन्नावत्सराद् गणकः ॥ ८४ ॥
इन्दु[2] विलिप्ता[5] शेषं[1] मंत्राशेषं वा ।
अथवा मध्यमिष्टं[3] कुर्वन्नावत्सराद् गणकः ॥ ८५[6] ॥
जीवविलिप्ता[2] शेषा कुजमिदुं[3] भौमलिप्तिकाशेषात् ।
रविमिदभागशेषा[1] कुर्वन्नावत्सराद्[4] गणकः[5] ॥ ८६[6] ॥

८३. (घ) (वि०—इसकी क्रमसंख्या ८४ है)
१. ९३ (च) for (ऽ३) २. +६७+(च)
(ग) ३. शेषसहितं for (शेषं सहितं)
(च) ४. (वि०—क्रमसंख्या ८४ अंकित है)
(ङ) १. 'ऽ३' लुप्त ।

८४. (घ) (वि०—इसकी क्रमसंख्या ८५ है)
(ग) १. २. 'ऽर्ककला' पद लुप्त है । ३. वर्ग for (वर्गो)
(च) १. ऽर्क्क for (ऽर्क) ४. (वि०—क्रमसंख्या ८५ अंकित है)
(ङ) ३. 'वर्गो' (लुप्त)

८५. (घ) (वि०—इसकी क्रमसंख्या ८६ है)
१. शेषामंशशेषं वा (ग) शेषाद्रविलिप्ताशेषमंशशेषं वा for (शेषं मंत्राशेषं वा)
(ग) २. इदं for (इंदु)
(च) १. शेषामंशशेषं वा for (शेषं मंत्राशेषं वा)
३. मध्यममिष्टं for (मध्यमिष्टं) ४. (वि०—यहां क्रमसंख्या ८६ अंकित है)
(ङ) ५. +शेषाद् रविलिप्ता+ १. शेषमंशशेषं for (शेषं मंत्राशेषं)
३. मध्यममिष्टं for (मध्यमिष्टं)

८६. (घ) (वि०—इसकी क्रमसंख्या ८७ है)
१. मिंदु (ग) (ङ) for (मिंद)
(ग) २. जीवविप्ताशेषं for (जीवविलिप्ताशेषा)
३. विलिप्तिका for (लिप्तिका) ४. शेषात् (ङ) for (शेषा)
(च) १. द्रविमिंदुभाग for (रविमिंदभाग)
५. कुर्वन्नोवत्सरादगणकः for (कुर्वन्नावत्सरादगणकः)
७. (वि०—क्रमसंख्या ८७ अंकित है)
(ङ) २. विलिप्ताशेषात् for (विलिप्ताशेषा)

इष्टग्रहेष्टशेषा द्युगणो गतनिरपर्वर्तयं संगुणितैः ।
छेदहतैरधिकोस्मादन्यग्रहशेषमिष्टो वा ॥ ८७ ॥
निच्छेदभागहारौ ग्रहयोर्भगणादि शेषयोर्द्युगुणात् ।
यस्मात्तं निच्छेदेनोद्धृतयोर्लब्धयोर्गुणितौ ॥ ८८ ॥
निच्छेदभागहारौ विपरीतौ तद्युतात्फलस्तस्मात् ।
शेषेद्युगुणादेवं व्यादिना प्राग्वदिष्टदिने ॥ ८९ ॥

८७. (घ) (वि०—इसकी क्रमसंख्या ८८ है)
१. 'यं' यह मूलपाठ में उपलब्ध नहीं (ग)
(ग) २. निरवर्त for (निरपवर्तयं) ३. संगुणतैः for (संगुणितैः)
(च) २.निरपवर्त्तसंगुणितैः for (निपर्वर्तयं संगुणितैः)
४. (वि०—क्रमसंख्या ८८ है)
(ङ) ५. शेषाद् for (शेषा)

८८. (घ) (वि०—इसकी क्रमसंख्या ८९ है)
१. र्कगणात् for (र्द्युगुणात्) २. द्धृतयो (ग) धृताया for (द्धृतयो)
(ग) ३. शेषवो द्युगणात् for (शेषयोर्द्युगणात्)
४. यस्मात्तन्न for (यस्मात्तं)
५. लब्धसंगुणितौ for (लब्धयोर्गुणितौ)
(च) १. र्द्युगणात् (ङ) for (र्द्युगुणात्) २. नोद्धृतयो for (नोद्धृतयो)
६. (वि०— क्रमसंख्या ८९ अंकित है)
(ङ) ७. निश्छेद for (निच्छेद) १. निश्छेदे for (निच्छेदे)
८. विपरीतौ for (भंगणादि)
५. लंब्धसंगुणितौ for (लंब्धयोर्गुणितौ)
३. ग्रहयो for (शेषयो)

८९. (घ) (वि०—इसकी क्रमसंख्या ९० है)
१. पुन (ग) पुनः शेषे for (फलस्तस्मात्)
२. द्युगणादेवं (च) (ङ) for (द्युगुणादेवं)
३. व्यादीनां (ग) (ङ) for (व्यादिना)
४. ३ द्युगणादैवत्यादीनां for (शेषेद्युगुणादेवं व्यादिना)
(ग) (वि०—दूसरी पंक्ति "द्युगणा" से आरम्भ होती है)
(च) १. पुनस्तस्मात् (ङ) for (फलस्तस्मात्)
५. (वि०— क्रमसंख्या ९० अंकित है)
(ङ) ६. निश्छेद for (निच्छेद)

द्यु[१]गुणमवशेषाद्रविचंद्रौ मध्य[२]मौ स्फुटा[३]दथवा ।
एवं तिथिं ग्रहं वा कुर्वन्नावत्सराद् गणकः ॥ ९० ॥[४]
एक[३]दिनावमशेषं षट्[१] गणमेकरविचन्द्र[२]भगणोनम् ।
शुध्यति भूदिन[४]भक्तं व्येकं चांद्रै[५]स्तदुक्तिरियम् ॥ ९१ ॥[५]
इषु[४]शरकृताष्टदिग्भिः १०८४५५ संगुणितादवमशेषा[१]द् भक्तात् ।
रूपाष्टरसवेदर[२]सशून्यशरगुणैः[६] ३५०६४८१ दिनगण[३]ः शेषम् ॥ ९२ ॥[५]

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९०. (घ) (वि०—इसकी क्रमसंख्या ९१ है)
१. गण for (गुण) (ग) द्यु गणमवमासशेषा for (द्यु गुणमवशेषा)
२. मध्यमौ (च) for (मध्यमौ)
(ग) ३. स्फुटावथवा for (स्फुटादथवा) (वि०—इसकी क्रमसंख्या ९० है)
(च) १. द्यु गण for (द्यु गुण)
४. (वि०—क्रमसंख्या ९१ अंकित है)
(ङ) १. गुणमवमावशेषा for (गुणमवशेषा)

९१. (घ) (वि०—इसकी क्रमसंख्या ९२ है)
(ग) १. चद्गुण for (षट्गुण) २. सगणोना for (भगणोनम्)
३. एकादिना for (एकदिना) ४. भूदिभक्तं for (भूदिनभक्तं)
(च) ५. (वि०—यहां क्रमसंख्या ९२ अंकित है)
(ङ) ३. एकदिनमवमशेषं for (एकदिनावमशेषं)
१. यद्गुणमेक for (षट्गणमेक)

९२. (घ) (वि०—इसकी क्रमसंख्या ९३ है)
१. दवमवशेषकाद् for (दवमशेषाद्भक्तात्)
२. रूपाष्ट वेदरसशून्य (ग) for (रूपाष्टरसवेदरसशून्य)
३. दिनगणः (ङ) (च) for (दिनगणः)
(ग) ४. इष्टशरकृताष्टदिग्भिः for (इषुशरकृताष्टदिग्भिः)
५. देवमशेषकाभक्तात् for (दवमशेषाद् भक्तात्)
६. रगुणो for (शरगुणैः)
(च) १. दवमवशेषकाद्भक्तात् for (दवमशेषाद्भक्तात्)
२. वेदरस (ङ) for (रसवेदरस)
८. (वि०—क्रमसंख्या ९३ अंकित है)
(ङ) १. शषेकाद्भक्तात् for (शेषाद्भक्तात्)

जिनरस गोब्धिरद्गुणा ३२४९६२४ छशि वसुकृत रसखभुतरामहूतान् ।
इष्टावमशेषाद्यछेषं रविभगण शेषं तत् ॥ ९३ ॥
गोगेंदुखेश ११०१७९ गुणितद्धुक्तान्नखपक्षयम रसेषु गुणैः ३५६२२२० ।
शेषमवमावशेषात्तिथयो वमशेषकाद्विकलम् ॥ ९४ ॥
भागकलाविकलैक्यं दृष्ट्वा विकलांतरं न के शेषैः ।
ऐक्यं द्विधांतराधिकहीनं द्विविभाजितं शेषम् ॥ ९५ ॥

९३. (घ) (वि०—इसकी क्रमसंख्या ९४ है)
१. रदगुणा (ग) रसगुणा for (रद्गुणा)
२. भूत (ग) (ङ) for (भुत)
३. हृतान् । ३५६४८१ (ग) राहृतात् for (रामहूतान्)
(ग) ४. (वि०—संख्या के अंक लुप्त हैं) ५. शशि (ङ) for (छशि)
५. शेषाद्या for (शेषाद्य)
(च) १. रदगुणा for (रद्गुणा) २. भूत for (भुत)
३. रामहृतान् for (रामहूतान्) ७. वि०—यहां क्रमसंख्या ९४ अंकित है)
(ङ) १. गोऽब्धिरद ३२४९६२४ गुणात् for (गोब्धिरद्गुणा ३२४९६२४)
३. हृतात् for (हूतान्) ६. शेषाद्य शेषं for (शेषाद्यछेषं)

९४. (घ) (वि०—इसकी क्रमसंख्या ९५ है)
१. गुणिताद्धुक्ता (ग) (च) (ङ) for (गुणितद्धुक्ता)
२. '३५६२२२०' दूसरी पंक्ति का आरंभ (य) संख्या लुप्त
(ग) ३. गोगेंदुखेश for (गोगेंदुखेश) ४. संख्या के अंक लुप्त हैं
५. नाख for (न्नख) ६. शेषावमाव for (शेषमवमाव)
(च) ७. गणैः for (गुणैः)
(वि०—यहां क्रमसंख्या ९५ अंकित है)
(ङ) ३. गोऽगेंदु for (गोगेन्दु)

९५. (घ) (वि०—इसकी क्रमसंख्या ९६ है)
१. वके शेषे (ग) च के शेषे for (नके शेषैः)
२. शेषे (ग) (ङ) for (शेषम्)
(ग) ३. द्विघातरा for (द्विधांतरा) ४. द्विभाजितं for (द्विविभाजितं)
(च) १. वकेशेषे for (नके शेषैः) २. शेषे for (शेषम्)
(वि०—क्रमसंख्या ९६ अंकित है)
(ङ) १. च for (न) ६. शेषे for (शेषैः)
३. द्विघाऽन्तरा for (द्विधांतरा) ४. च द्विभाजितं for (द्विभाजितं)

तद्वर्गांतरमाद्यं[५] तदंतरं[१] चांतरोद्धृतयुतोनम् ।
वर्गांतरं विभक्तं[२] ताभ्यां[३] शेषोन्यतो[६] द्युगणः[४] ।। ९६ ।।
कृतिसंयोगाद्विगुणं[१] छेषयुतिकृति[२] विशोध्य मूलेन ।
शेषं[३] युतिर्युत[५] हीनाद्विहता[६] शेषे पृथक् युक्त्या[७] ।। ९७ ।।
शेषवधाद्विऋृति[१] गुणा छेषांतरवर्गसंयुतान्मूले[२] ।
शेषांतरीनयुक्तं[३] दलितं[४] शेषे पृथगभिष्टे[५] ।। ९८ ।।

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९६. (घ) (वि०—इसकी क्रमसंख्या ९७ है)
१. चांतरोद्धृत (ग) (ङ) for (चांतरीद्धृत्र)
२. द्वा द्वाभ्यां for (ताभ्यां) ३. शेषे ततो (ग) (ङ) for (शेषोन्यतो)
(च) १ चांतरोद्धृत for (चांतरीद्धृत) ४. (वि०—क्रमसंख्या ९७ है)
३. शेषो ततो द्युगणः for (शेषोन्यतो द्युगणः)
(ङ) ५. माद्यं for (माद्यं) २. द्वाभ्यां for (ताभ्यां)
९७. (घ) (वि०—इसकी क्रमसंख्या ९८ हैं)
१. द्विगुणा (ग) (च) for (द्विगुण)
२. छष for (छेष) (ग) छेष कृतियुति for (छेषयुतिकृति)
३. विहृता (ग) विज्ञता for (द्विहृता)
(ग) ४. कृत संयोगा for (कृतिसंयोगा) ५. पृथग्युक्त्या for (पृथक् युक्त्या)
(च) ६. द्विहृता for (द्विहता)
६. (वि०— क्रमसंख्या ९८ अंकित)
(ङ) कृतिसंयोगाद् द्विगुणाद्युतिवर्गं प्रोह्य शेषमूलं यत् ।
तेन युतोनो योगो दलितः शेषे पृथगभींष्टे ।। ९८ ।।
for
(कृतिसंयोगाद्विगुण छेषयुतिकृति विशोध्य मूलेन
शेषयुतिर्युतहीना द्विहता शेषे पृथक् युक्त्या ।। ९७ ।।)
९८. (घ) (वि—इसकी क्रमसंख्या ९९ है)
१. विकृति (ग) for (द्विऋृति) मूलम् (ग) (ङ) for (मूले)
३. शेषांतरोन (ग) (ङ) for (शेषांतरीन युक्तं)
४. पृथगगभीष्टे (ग) (ङ) for (पृथगभिष्टे)
(ङ) १. कृति for(ऋृति) २. मूलं for (मूले)
४. पृथगभीष्टे for (पृथगभिष्टे)
५. (वि०— क्रमसंख्या ९९ अंकित है)
(ङ) १. वधाद् द्विकृतिगुणात् for (वधाद्विऋृतिगुणा) ६. शेषांतर for (छेषान्तर)

सुखमात्रममी प्रश्ना प्रश्नान्यात्सहस्रशः कुर्यात् ।
अन्यैर्दत्तात्प्रश्नानुक्तैं वा साधयेत्करणैः ॥ ९९ ॥
जन संसदि दैवविदां तेजो नाशयति भानुरिव भानाम् ।
कुट्टाकारप्रश्नैः पठितैरपि किं पुनर्ज्ञातैः ॥ १०० ॥
प्रतिसूत्रममी प्रश्नाः पठिताः सोद्देशकेषु सूत्रेषु ।
आयाणां व्यधिकशते कुट्टकोष्टादशोऽध्यायः ॥१०१॥
इति श्री ब्रह्मगुप्ते अष्टादशोऽध्यायः ।

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९९ (घ) (वि०—इसकी क्रमसंख्या १०० है)
१. प्रश्नानन्या (ग) प्रश्नानन्यान् (ङ) for (प्रश्नान्यात्)
(ग) देव्यात्र for (सुखमात्र) ३. दंत्तान् for (दंत्तात्) ४. त्करणैः for (करणैः)
(च) १. प्रश्नानन्यान्सहस्रशः (प्रश्नान्यात्सहस्रशः)
(वि०—इसकी क्रमसंख्या १०० अंकित है)
(ङ) २. हृदि घात्रममी for (सुखमात्रममी) ६. प्रश्नाः for (प्रश्ना)
३. अन्यैर्दत्तान् for (अन्यैर्दत्तात्) ७. प्रश्नानुक्त्यैवं for (प्रश्नानुक्तैं वा)
४. करणैः for (करणैः) (वि०—३. 'हृतान्मात्रममी' इति सुधाकरः)

१००. (घ) (वि०- इसकी क्रम संख्या १०१ है)
१. पुनर्ज्ञातैः (ग) (च) for (पुनर्ज्ञातैः) २. भारिव for (भानुरिव)
३. कुदाकारं for (कुट्टाकार) ४. प्रस्नैः for (प्रश्नैः)
(च) ३. कुदाकार प्रश्नैः for (कुट्टाकार प्रश्नैः)
५. (वि०—यहाँ क्रम संख्या १०१ अंकित है)
(ङ) १. पुनः शतशः for (पुनर्ज्ञातैः)

१०१. (घ) (वि०—इसकी क्रम संख्या १०२ है)
१. आर्याणाम् (ग) आर्या for (आयाणां) २. शतेन कुष्ट for (शते कुट्ट)
३. 'इति' से 'अध्यायः' तक अंकित नहीं है ।
(ग) २. समी प्रश्नाः for (ममी प्रश्नाः) २. शतेन for (शते)
३. इति श्रीब्रह्मसिद्धांते कुट्टकाध्यायाष्टदशमः ॥ १८ ॥ for (इति श्रीब्रह्मगुप्ते-अष्टादशोऽध्यायः)
(च) १. आर्याणां for (आयाणां)
३. "इति" से "अध्यायः" लुप्त (केवल "श्री" अंकित है)
५. (वि०—क्रमसंख्या १०२ अंकित है)
(ङ) १. आर्या for (आयाणां) २. व्यधिकशतेन for (व्यधिकशते)
६. कुट्टश्चाष्टादशो for (कुट्टकोष्टादशो)

अथ शंकुच्छायादिज्ञानाध्यायः

एकोनविंशः

दृष्ट्वा दिनार्द्धघटिका योर्कजो क्षांशकान्बीजानाति ।
उदयांतरघटिकाभिर्ज्ञाता ज्ञेयं स तन्त्रज्ञः ॥ १ ॥
अस्तांतर्घटिकाभिर्यो ज्ञाताज्ञेयमानयति तस्मात् ।
मध्यगति ग्रहभगणाननयति तया स तंत्रज्ञः ॥ २ ॥
आनयति यस्तमो रविशशांक मानानि दिप कौच्यतलात् ।
शंकुतलांतरभूमिं ज्ञाने छायां स तन्त्रज्ञः ॥ ३ ॥
छायाद्वितीयाग्रांतर विज्ञाने वेत्ति दीपौच्ये ।
शंकु छायाज्ञो वा यच्छायौच्ये स तंत्रज्ञः ॥ ४ ॥

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१. (घ) १. ऽर्कजो (ग) (ङ) for (योर्कजो) २. विजानाति (ग) for (बीजानाति)
(च) १. योऽर्क्कजो for (योर्कजो) २. विजानाति for (बीजानाति)
(ङ) ३. ऽक्षांशकान् for (क्षांशकान्) २. विजानाति for (बीजानाति)
४. ज्ञाताज्ञेयं for (ज्ञाता ज्ञेयं)

२. (घ) १. अस्तांतर (ग) for (अस्तांतर्) २. मध्यगतिं (च) (ङ) for (मध्यगति)
(ग) ३. युगभ (ङ) for (ग्रहभ) ४. नानयति (च) (ङ) for (ननयति)
५. ततः (ङ) for(तया) (वि०—इसकी क्रम संख्या १ लिखी है)
(च) १. अस्तांतर घटिकाभिर्यो (ङ) for (अस्तांतर्घटिकाभिर्यो)
(ङ) ६. ज्ञाताज्ञेय for (ज्ञाताज्ञेय)

३. (घ) १. दीपाक्चे क्चच्यतलात् (ग) दीपक शिखोन्याः for (दिपकौच्यतलात्)
(ग) २. भूमि (ङ) for (भूमिं) (वि०— इसकी क्रमसंख्या २ लिखी है)
(ङ) १. दीपकशिखोच्च्यात् for (दिपकौच्यतलात्)

४. (घ) १. छायाद्वितया (ग) for (छायाद्वितीया)
२. भूमिदीपौच्ये (ग) दीप्यौच्ये for (दीपौच्ये)
(ग) ३. परभूमेत्छायां for (यच्छायौच्ये) (वि०—इसकी श्लोक संख्या ५ है)
(च) १. द्वितया for (द्वितीया)
(ङ) १. छायाद्वितीयभाग्रान्तर for (छायाद्वितीयाग्रांतर)
४. विज्ञानेन for (विज्ञाने) २. दीपौच्यम् for (दीपौच्ये)
५. शङ्कुच्छायाज्ञो for (शंकुछायाज्ञो) ३. भूमेश्छायां for (यच्छायौच्ये)

दृष्टि गृहौच्यज्ञो यः स्तदंतरज्ञो नराक्रते नुजले ।
गृहभित्यग्रं दर्शयति दर्प्पणे वा स तंत्रज्ञः ॥ ५ ॥
हग्गृहतलांतरजले यो दृष्ट्वाग्रं गृहस्य भूमिज्ञः ।
वेत्ति गृहौच्यं दृष्ट्वा तैलस्थं वा स तंत्रज्ञः ॥ ६ ॥
वीक्ष गृहाग्रं सलिले प्रसार्य सलिलं पुनश्चभूज्ञाने ।
आनयति जलाङ्गुमि गृहस्य चौच्यं स तंत्रज्ञः ॥ ७ ॥
ज्ञातै छाया पूरुषै विज्ञाते तोय कुघयो विवरे ।
कुट्टेर्क तेजसो यो वेत्थारुढ़ेः स तंत्रज्ञः ॥८॥

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५. (घ) १. नरात्कृते (ग) for (नराक्रते) २. तु (ग) (ङ) for (नु)
३. गृहभित्यर्गं (ग) गृहभित्यग्र (ङ) for (ग्रहभित्यग्रं)
(ग) ४. इष्टग्रहौच्यज्ञो for (दृष्टिगृहौच्यज्ञो) ५. दर्पणे (ङ) for (दर्प्पणे)
(वि०—इसकी श्लोक संख्या ४ है)
(च) १. नराकृते for (नराक्रते)
(ङ) ४. इष्टगृहौच्यज्ञो for (दृष्टिगृहौच्यज्ञो) ६. यस्तदन्तरज्ञो for (यः स्तदंतरज्ञो)
१. निरीक्ष्यते for (नराक्रते)

६. (घ) १. नृगृह हग्ग्रहनलांतर (ङ) for (ह्ग्रह तलांतर)
२. गृहस्य (ग) (च) (ङ) for (ग्रहस्य)
३. गृहौच्यं (ग) गृह्यौच्यं (च) (ङ) for (ग्रहौच्यं)
(ग) ४. जलयो for (जले यो) ५. दृष्ट्वाग्रं (ङ) for (दृष्ट्वाग्रं)
६. व for (वा)
(च) १. दृगृहतलांतर for (हग्रहतलांतर)
(ङ) १. दृष्ट्वा गृहतलान्तर for (हग्रहतलांतर) ४. जालभो for (जले यो)

७. (घ) १. गृहाग्रं (ग) (च) (ङ) for (ग्रहाग्रं) २. भूमि (ग) (च) (ङ) for (भूमि)
३. गृहस्य (ग) (च) (ङ) for (ग्रहस्य)
(ग) ४. पुनः स्वभूज्ञाने (ङ) for (पुनश्च भू ज्ञाने) ५. वौच्यं for (चौच्यं)
(ङ) ६. वीक्ष्य for (वीक्ष)

८. (घ) १. कुडचयोर्विवरे (ङ) for (कुघयोर्विवरे)
२. कुडर्घेर्क (ग) कुघेर्का for (कुट्टेर्क)
३. वेत्यारूढ़ि (ग) वेत्यारूढ़ेः for (वेत्थारूढ़ेः) (ग) ४. ज्ञाते for (ज्ञातै)
५. पुरुषै (ङ) for (पूरुषै) ६. विज्ञातै for (विज्ञाते)
(च) १. कुडचयो for (कुघयो) २. कुट्टेऽर्क for (कुट्टेर्क)
(ङ) १. ज्ञातैश्छाया for (ज्ञातै छाया)
२. कुडर्घेर्क for (कुट्टेर्क) ३. वेत्यारूढ़ि for (वेत्थारूढ़ेः)

इष्ट दिवसार्द्ध घटिका घटिका च[१] दशकांतरं[२] प्राणाः ।
तद्दिवसचरप्राणास्तैरक्षं साधयेत्प्राग्वत्[३] ।। ९ ।।
ज्ञातः सभार्द्ध उदयैरस्तांतरनाडिकाभिरधिको नः ।
ज्ञातात्पूर्वापरयोर्ज्ञेयो[१] भार्द्धोनके[२] ज्ञयः[३] ।। १० ।।
ज्ञातज्ञेयग्रहयो[२]रुदयांतरनाडिकाभिरधिकोन[४]: ।
उदयैर्ज्ञातो[५] ज्ञाताज्ञेये[६] प्रागधरयोर्ज्ञेयः[७] ।। ११ ।।
ज्ञातं कृत्वा मध्यं भूयोन्यदिने तदंतरं भुक्तिः ।
त्रैराशिकेन मुत्तचा[१] कल्पग्रहमंडलानयनम् ।। १२ ।।
स्थित्यर्द्धाद्विपरीतं तमः प्रमाणं स्फुटं[१] ग्रहणे ।
मानोदयाद्रवौद्वोर्घटिकावयवेन भोदयतः ।। १३ ।।

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९. (घ) १. पंचदशकांतर for (चदशकांतरं)
(ग) २. दशांतर for (दशकांतरं)
३. प्रास्नात् for (प्राग्वत्)
(ङ) शी०+अथोत्तराणि+
१. २. पंचदशांतर for (चदशकांतरं)

१०. (च) (वि०—यह श्लोक यहां ११ वीं क्रमसंख्या पर अंकित है)
१. ज्ञेये for (ज्ञेयो) २. भार्द्धोनको for (भार्द्धोनके)
३. ज्ञेयः (ङ) for (ज्ञयः)
(ङ) (वि०—यहां इसकी क्रमसंख्या ११ है)

११. (घ) १. परयो for (धरयो) (ग) प्रागपरो for (प्रागधरयो)
(ग) २. स्थान रिक्त है, ३. 'ज्ञेय' अक्षर लुप्त है ४. नम् for (नः)
५. ज्ञातौ for (ज्ञातो) ६. ज्ञाताज्ञेय for (ज्ञाताज्ञेये)
७. ज्ञेयः for (ज्ञेयः)
(वि०—इसकी क्रमसंख्या १० है)
(च) १. प्रागपरयो for (प्रागधरयो)
८. (वि०—इसकी क्रमसंख्या १० अंकित है)
(ङ) ६. ज्ञाताज्ञेयः for (ज्ञाताज्ञेये)

१२. (घ) १. भुत्तचा (ग) (च) (ङ) for (मुत्तचा)

१३. (घ) १. शशिग्रहणे for (स्फुटं ग्रहणे)

दीपतलशंकुतलयोरंतरमिष्टप्रमाणशंकुगुणम् ।
दीपशिखौच्याच्छंकु विशोध्य शेषोद्धृतं छाया ॥ १४ ॥
शंक्वंतरेण गुणिता छाया छायांतरेण भक्ता भूः ।
स छाया शंकुगुणा दीपोच्यं छायया भक्ता ॥ १५ ॥
ज्ञात्वा शंकुछाया मनुपातात्साधयेत्समुछ्रायान् ।
गृहचैत्यतरू नगानामौच्यं विज्ञायवछायाम् ॥ १६ ॥
दृष्टि गृहौच्यैक्य हूतातदन्तरधरादगौच्य गुणिता ।
फलभून्यस्ते तोये प्रतिरूपाग्रं गृहस्य नरात् ॥ १७ ॥

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१४. (घ) १. छंकु for (च्छंकु)
(ग) १. शिखौच्यात् शंकुं for (शिखौच्याच्छंकु)
२. हृतं (च) for (द्धृतं) ३. छायाः for (छाया)
(च) ४. दीपशिषौच्या for (दीपशिखौच्या)
(ङ) ४. शिखौच्च्याच्छङ्कुं for (शिखौच्याच्छंकु)

१५. (घ) १. दीपौच्यं (ग) (ङ) for (दीपौच्च्यं)
(ग) २. भक्ताः (च) for (भक्ता)
(ङ) ३. सच्छाया for (सछाया)

१६. (घ) १. वा (ग) (च) (ङ) for (व)
(ग) २. मनुपातान् for (मनुपातात्)
(ङ) ३. शङ्कु॰च्छाया for (शंकुछाया)
४. समुच्छ्रायान् for (समुछ्रायान्)
५. मौच्च्यं for (मौच्यं)

१७. (घ) १. गृहौच्चैक्य (ग) ग्रहौच्येक for (गृहौच्यैक्य)
(ग) २. हृता (च) (ङ) for (हूता)
३. दगौच्यमं for (धरादगौच्य)
(च) १. गृहौच्चैक्य for (गृहौच्यैक्य)
३. दगौच्य for (दगौच्य) (ङ) ४. युतदृष्टि for (दृष्टि)
१. गृहौच्च्य for (गृहौच्यैक्य) ५. ह्यन्तर for (तदन्तर)
३. भूमिर्दगौच्च्य for (धरादगौच्य) ६. संगुणिता for (गुणिता)
७. न्यंस्ते for (न्यस्ते)

गृह पुरूषांतर सलिले विक्ष्य गृहाग्रं दृगौच्य संगुणितम् ।
गृहतोयांतरमौच्यं गृहस्य नृजलांतरेण हृतम् ॥ १८ ॥
प्रथम द्वितीय नृजलांतरांतरेणोद्धृता जलाय सूतिः ।
दृष्टौच्य गुणोछ्रायस्तोयानृजलांतरगुणा भूः ॥ १९ ॥
छाया पुरूषछिन्नं जल कुधांतर मवाप्तमारूढ़िः ।
अध्याये विंशत्यार्याणामेकोनविंशोयम् ॥ २० ॥

इत्ति ब्रह्म गुप्ते छायाध्यायः
एकोनविंशोऽध्यायः समाप्तः

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१८. (घ) १. वीक्ष्य (ग) (च) (ङ) for (विक्ष्य)
२. गृहाग्रं (ग) (च) (ङ) for (ग्रहाग्रं)
३. हृतम् (ग) (च) (ङ) for (हृतम्)
(ग) ४. सलिल for (सलिले)
५. व्यसंगुणितम् for (दृगौच्यसंगुणितम्)
६. नृतलांतरेण for (नृजलोन्तरेष)
(च) ५. दृगौच्च्य for (दृगौच्यं)
७. मौच्च्यं for (मौच्यं)

१९. (घ) १. जलापसूतिः (ग) (ङ) for (जलाय सूतिः)
(ग) २. नृजलांतरांतरा for (नृजलांतर) ३. छाया for (छ्राय)
४. स्तोयन्नृजलांतर for (स्तोयानृजलांतर)
(च) ५. प्रथ for (प्रथम) ६. द्वितिय for (द्वितीय)
(ङ) ७. दृष्टचौच्च्य for (दृष्टौच्य) ३. गुणोच्छ्राय for (गुणोछ्रांय)

२०. (घ) १. बल for (जल) ३. कुटचांतर (च) for (कुधांतर)
३. अध्यायो (ग) अध्याय for (अध्याये)
४. 'इति' से 'समाप्त' तक पाठ उपलब्ध नहीं (ग)
(ग) ५. मवामारूदिः for (मवाप्तमारूदिः) ६. विशत्यार्याणा for (विंशत्यार्याणां)
७. मेकान विंशोयम् for (मेकोनविंशोयम्)
(च) ३. अध्यायो (ङ) for (अध्याये) ४. 'इति' से 'समाप्त' तक पाठ लुप्त ।
(ङ) ८. च्छिन्नं for (छिन्नं) २. कुडचान्तर for (कुधांतर)
४. इति श्री ब्राह्मस्फुटसिद्धान्ते शंकुच्छायादिज्ञानं नामैकोनविंशोऽध्यायः for (इति ब्रह्मगुप्ते छायाध्यायः)

अथ छन्दश्चित्युत्तराध्यायो

विंशतितमः

ऋग्वर्गः पर्यायः समूह योगो वयुक्षु युग्मेषु ।
सोपाः प्राग्वत्पादाश्चतुष्कका शेष युक्तोंत्यः ॥ १ ॥
एकैक युत विहीना वाद्यंतौ तद्विपर्ययो यावत् ।
वर्गादिष्व समयुजः क्रमोक्रमाद्वर्द्धयेत्पादान् ॥ २ ॥
एकैकेनद्वाद्या न सोयेष्टधिकेषु तत्प्रविष्टेषु ।
वर्गादिरीष्टांत प्रस्तारो भवति यवमध्यात् ॥ ३ ॥

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१. (घ) १. रिग्वर्गः (ग) ऋग्वर्क्कः for (ऋग्वर्गः)
२. पर्यायः (ग) (च) (ङ) for (पर्यायः) ३. योगा (ङ) for (योगो)
४. वयुत्कु (ग) वपुक्ष for (वयुक्षु)
५. चतुष्काः (ग) (च) (ङ) for (श्चतुष्कका)
(ग) ६. प्रागृत for (प्राग्वत्) ७. युक्तोत्यः for (युक्तोंत्यः)
(ङ) ८. सोयाः for (सोपाः) ६. प्राप्तादा for (पादा)

२. (घ) १. विहीनाद्यंतौ (ग) वाद्यंतै for (वाद्यंतौ)
२. क्रमोत्क्रमाद् (ग) क्रमात् (ङ) for (क्रमोक्रमाद्)
(ग) ३. एकादि (ङ) for (एकैक) ४. तद्विपर्यायौ for (तद्विपर्ययो)
५. वर्गादिषु (ङ) for (वर्गादिष्व) ६. विषमयुजाक्रमो for (समभुजः)
(च) २. क्रमोत्क्रमाद्वर्द्धयेत् for (क्रमोक्रमाद्वर्द्धयेत्)
(ङ) ४. तद्विपर्ययौ for (तद्विपर्ययो) ६. विषमयुजां for (समभुजः)

३. (घ) १. सोपेश्वधिकेषु for (न सोयेष्टधिकेषु)
२. भीष्टां (ग) वर्गादिरभीष्टांतः for (वर्गादिरीष्टांत)
(ग) ३. द्वचाव्याः for (द्वाद्या) ४. सोपेप्पधिकेषु for (न सोयेष्टधिकेषु)
५. तत्प्रतिष्ठेषु (ङ) for (तत्प्रविष्टेषु)
६. मध्यः (ङ) for (मध्यात्)
(च) १. नसोपेष्वधिकेषु for (नसोयेष्टधिकेषु)
(ङ) ३. द्वचाद्वचाः for (द्वाद्या) १. सोप्पप्पधिकेषु for (न सोपेष्टधिकेषु)
२. वर्गादिरभीष्टान्तः for (वर्गादिरीष्टांत)

न्यूनोचोद्विपदाग्रं विपदाद्या नामतः पृथक् संख्या ।
तछोधोथेव्येकः पृथगंतापमूर्द्ध युतम् ॥ ४ ॥
यावत्पादाव्येका गछद्वर्णेष्वथैक वृद्धेषु ।
रूपाद्युद्धृतघाते वर्गाद्यानां परा संख्या ॥ ५ ॥
रूपाधिकयोर्द्ध विषमे पूर्वः समेषु पादार्द्धे ।
आधा द्विगुणाव्येकां गुलान्यधस्तस्य सर्वेषाम् ॥ ६ ॥
मध्येस्तथार्द्धहीनैः क्रमपादैर्व्यस्त तुल्य पांदाद्यैः ।
विषमेष्वेकं मध्ये प्रोह्याद्यान्यतः कुर्यात् ॥ ७ ॥

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४. (घ) १. त्र्यूनोन्यो (ग) त्र्यूनोद्विपादाग्रं for (त्र्यूनो चो)
२. ताश्चांधोध्योव्येक for (तछोधोथेव्येकः) ३. 'धे' लुप्त
(ग) ४. त्रिपदाद्यानाममधः for (विपदाद्यानामतः) २. व्यव्येकः for (व्येकः)
५. पृथगंताद्रूपमूर्द्धयुतम् for (पृथगंतापमूर्द्ध युतम्)
(च) १. त्र्यूनोंचो for (त्र्यूनोचो) २. ताँछोधोथे for (तछोधोथे)
५. पृथगंताद्रूमपूर्द्धयुतं for (पृथगंतापमूर्द्धयुतम्)
(ङ) १. सूनोन्त्यो for (त्र्यूनो चो)
४. त्रिपदाद्यानामधः for (विपदाद्यानामतः) २. तच्छोध्यो for (तछोधोथे)
५. पृथगन्ताद्रूपमूर्ध्वयुतम् for (पृथगंतापमूर्द्धयुतम्)

५. (घ) १. चैक for (थैक)
(ग) २. व्येको for (व्येका)
३. गच्छद्वर्णेप्यथैकवृद्धेषु for (गच्छद्वर्णेष्वथैकवृद्धेषु)
(ङ) ४. गच्छाद् for (गच्छेद्) ५. रूपाद्युतघाते for (रूपाद्युद्धृतघाते)

६. (घ) १. रूपाधिकपादार्धे (ङ) (ग) रूपाधिकापादार्धे for (रूपाधिकयोर्द्ध)
३. अज्या for (आधा)
(ग) २. विषमेषूर्द्धस्मसेषु पादा for (विषमेपूर्वः)
३. अर्द्धाद्विगुणाव्येकां (ङ) for (आधाद्विगुणाव्येकां)
(च) १. रूपाधिकपादार्द्धे for (रूपाधिकयार्द्धे) ३. अद्या for (आद्या)
(ङ) २. विषमेषूर्ध्वः for (विषमेपूर्वः) ४. युलान्यधस्तस्य for (गुलान्यधस्तस्य)

७. (घ) १. (मध्यै) (ग) मध्यै for (मध्ये) २. मधे for (मध्ये)
२. प्रोच्याद्यान्यन्यतः for (प्रोह्याद्यान्यतः) ४. र्यात् for (कुर्यात्)
(ग) ५. व्यस्तुल्य for (व्यंस्ततुल्य)
(ङ) १. माध्यै for (मध्ये) ६. पादाद्यः for (पादाद्यैः)
७. विषमेरव्येकं for (विषमेष्वेकं)

सैवक्रम[3] तुलाद्यैर्न्यासोप्यधिको[1][2] विशोधितश्चाधः ।
सांख्यैक्यं[4] तादृक्[5] तादृक्[7] प्रथमस्त्रिरहितो[6] नष्टे ॥ ८ ॥
माध्यैः[1] कृतैश्चदलितै[4] सम संख्यायां क्रमोक्रमाक्षेप्यम्[2][5] ।
विषमायां व्येकायां दलं क्रमादुक्रमात्[3] सैकम् ॥ ९ ॥
समसंख्यायां[1] सोपात्[7] क्रमोक्रमाभ्यां[2] तथैव विषमायाम्[8] ।
कल्पोपविते[3] दृष्टे[5] प्रथमः शेषाक्षराण्यंते[6][4] ॥ १० ॥
समदलसम विषमाणां संख्यापादार्द्धं[1] सर्वकल्पवधः ।
स्वाद्य[2] वधोन्यैः पादैः स्वपरस्य प्राग्वधः सैकैः ॥ ११ ॥

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८. (घ) १. तुल्याद्यै (ग) (ङ) for (तुलाद्यै)
२. व्यधिको (ग) नसोभ्यधिको (ङ) for (न्यासोप्यधिको)
(ग) ३. सैकक्रमं for (सैवक्रम) ४. संख्यैक्यं (ङ) for (सांख्यैक्यं)
५. तादृक् यादृक्च for (तादृक् तादृक्) ६. यस्त्रि for (प्रथमस्त्रि)
(च) ३. सैकक्रम (ङ) for (सैवक्रम) ४. सोख्यैक्यं for (सांख्यैक्यं)
(ङ) ७. यादृक् for (तादृक्)

९. (घ) १. माध्येः (ग) मध्यैः for (माध्यैः)
२. क्रमोत्क्रमा (ग) (च) for (क्रमोक्रमा)
३. दुत्क्रमात् (च) for (दुत्क्रमात्)
(ग) ४. दलितैः (च) (ङ) for (दलितै) ५. दक्षेप्यम् for (क्षेप्यम्)
(ङ) २. क्रमोत्क्रमात् for (क्रमोक्रमा) ५. क्षेप्पम् for (क्षेप्यम्)

१०. (घ) १. संक्षाया for (संख्यायां)
२. क्रमोत्क्रमाभ्यां (ङ) (ग) क्रमत्क्रमाभ्यां for (क्रमोक्रमाभ्यां)
३. कल्पोपवितो (ग) कल्पोपचिते for (कल्पोपविते)
४. शेषात्कराण्यंते for (शेषाक्षराण्यन्ते)
(ग) ५. दृष्टे for (दृष्टे) ६. शेषाः for (शेषा)
(च) २. क्रमोत्क्रमाभ्यां for (क्रमोक्रमाभ्याम्)
(ङ) ७. सोपान for (सोपात्) ८. विषमाभ्याम् for (विषमायाम्)
३. कल्प्यापचिते for (कल्पोपविते)

११. (घ) १. संक्षापार्द्ध (ग) सख्यापा for (संख्या)
(ग) २. स्वाद्य for (स्वाद्य)
(ङ) १. संख्यापादार्ध for (संख्यापादार्द्ध)
२. स्वाद्यवधोऽन्यैः for (स्वाद्यवधोन्यैः)

आद्यादनंतरोधः कल्पोत्य तुल्य माद्य कल्प प्राक् ।
न्यासो वर्गोन्योन्य प्रस्तारेर्द्ध समविषमाणाम् ॥ १२ ॥
नष्टांत्पास्वाधस्थो न कल्प घातो अर्द्ध तुल्य विषमाणाम् ।
व्येकः पृथक् स्ववर्गोद्धृतः फलं तुल्य कल्पानाम् ॥ १३ ॥
उद्दिष्टे कल्प कृतो तीतै प्रथम फलस्वरूपेन्यः ।
असकृद्वर्गांश युते सैके वार्द्ध सम विषमाणाम् ॥ १४ ॥

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१२. (घ) १. कल्पोऽनु (ग) कल्पोन्य for (कल्पोत्य)
२. न्यास्पोवर्ग्रो (ग) न्यासो च गोन्यो न for (न्यासोवर्गो)
३. न्योत for (न्योन्य)
४. प्रस्तारेऽर्ध (ग) (च) for (प्रस्तारेर्द्ध)
(ग) ५. कल्पः for (कल्प)
(च) १. काल्पोऽन्यतुल्य for (कल्पोत्यतुल्य) ३. न्योन for (न्योन्य)
(ङ) ६. दनन्तरोऽधः for (दनंतरोधः)
१. कल्प्पोऽन्यतुल्यमाद्यः for (कल्पोत्यतुल्यमाद्य)
५. 'कल्प' लुप्त ।
४. प्रस्तारोऽर्ध for (प्रस्तारेर्द्ध)
३. ऽन्योनः for (न्योन्य)

१३. (घ) १. नष्टांत्याश्चोस्थोन for (नष्टांत्पास्वाधस्थोन)
२. र्द्ध (ग) (च) (ङ) ऽर्ध for (अर्द्ध) ३. वग्रों for (वर्गो)
(ग) १. नाष्टंत्यात् for (नष्टांत्) ५. ककल्प for (नकल्पा)
६. तुविषमाणाम् for (तुल्यविषमाणाम्) ७. द्धृतफलः for (द्धृतः फलं)
(ङ) १. नष्टेऽन्त्यात् for (नष्टांत्पा)

१४. (घ) १. कृतेऽतीतैः (ग) हृतेतीतैः (ङ) for (कृतोतीतै)
२. फले (ङ) for (फल)
३. वग्रांश (ग) व्यसकृद्वर्गांश for (असकृद्वर्गांश)
(ग) ४. उद्दिष्ट for (उद्दिष्टे) ५. चार्ध for (वार्द्ध)
(च) १. कृतेऽतीतैः for (कृतोतीतै)
(ङ) ६. प्रथमः for (प्रथम) ५. वार्ध for (वार्द्ध)

कल्पेषु[१] पृथुगुरू[४] लघुतुल्यवधैक्यं[५] गुरु[२] लघुनि यवमध्ये ।
दलसम[६] विषम लघूनि[३] स्वद्य लघूनानि विषमेषु ॥ १५ ॥
ध्व्यापाः[१] कोष्टक वृद्धचा[२] लघु[३] संख्यैकादि भाजिताः प्राग्वत् ।
विषमे ज्याघल[४] द्यूनो लघूनो[५] लघुभिर्मेरूः समादीनाम् ॥१६॥
एकद्विकयोः[२] परतो द्वि[४] संगुणो नंतरो द्विरूपोधः ।
वर्गापराद्योनोदल[१] समविषमाणां[३] ध्वजो लघुभिः ॥ १७ ॥
लघु संख्यादौ[२] परतो दलिता धोधश्च[१] शुध्यति हताद्यैः[३] ।
द्विगुणांतैः[४] संख्यै शुद्धैर्वर्ग[५] परैर्मंदरो[१] लघुभिः ॥ १८ ॥

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१५. (घ) १. कल्पपृथुगुरू for (कल्पेषु) २. लघूनि for (लघुनि)
३. स्वाद्य (च) for (स्वद्य)
(ग) १. कल्प्पेषु for (कल्पेषु) ४. पृथग्गुरुलघु (ङ) for (पृथुगुरुलघु)
५. संख्यैकादिभाजिताः प्राग्वत् (ङ) for (तुल्यवधैक्यं गुरुलघुनि यवमध्ये)
६. विषमेष्वाद्यलघूनो लघुभिर्मेरुः समादीनाम् ॥ १५ ॥ (ङ)
for
(दलसम विषमलघूनि स्वद्य लघूनानि विषमेषु)
(च) २. लघूनि for (लघुनि)

१६. (घ) १. द्वचाद्याः (च) for (ध्व्यापाः) २. वृद्धा (च) for (वृद्धचा)
३. लघू (च) for (लघु) ४. ज्याद्य for (ज्याघ)
(च) ४. द्याद्यल for (ज्याद्यल) ५. 'लघूनो' लुप्त ।
(ङ) यह श्लोक यहां उपलब्ध नहीं ।

१७. (घ) १. वर्ग्र (ग) for (वर्गा)
(ग) २. द्वितयोः (ङ) for (द्विकयोः) ३. चराद्यो for (पराद्यो)
(च) १. वर्ग (ङ) for (वर्गा) ४. परार्द्धां for (पराद्यो)
(ङ) (वि०—इसकी क्रमसंख्या १६ है) ४. ऽनन्तरा for (नंतरो)

१८. (घ) १. यरै for (परै)
(ग) २. संख्यापद दलिता परतो (ङ) for (संख्यादौ परतो दलिता)
३. हृताद्यैः for (हताद्यैः) ४. द्विगुणानांतैः for (द्विगुणान्तैः)
५. शुद्धैः वर्ग्र for (शुद्धैर्वर्ग)
(च) ६. ध्येधश्च for (धोधश्च) ७. लघुभि for (लघुभिः)
(वि०—इसकी क्रमसंख्या १७ है) ३. हृतायैः for (हताद्यैः)
६. ऽधोऽधश्च for (धोधश्च) ८. 'संख्यै' लुप्त ।

कृत्वाधोधः कल्पान्ये[१] काद्येकोत्तरानधः[२] शेषात् ।
खात्परतां[३] तैकमधः प्रस्तारा द्युक्त[४] वदीहाद्यैः ।। १९ ।।
गुरूषष्टूचैका[७] विघटि[१] द्विगुणा व्येकांगुलानि[८] संख्याद्वा[२] ।
विंशति रार्या विंश[३] छंदश्चितुत्तराध्यायः[४] ।। २० ।।
इत्ति[५] ब्रह्म गुप्ते छंद श्चित्युत्तराध्यायः ।।२०।। अध्यायः समाप्तः
विशंतितमः समाप्तः ।

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९. (घ) १. कल्पानेका for (कल्पान्येका)
२. धनः (ग) नघस्तेषाम् (ङ) for (नधः शेषात्)
३. खात्पतोंत्यैक (ग) खात्परतोत्यैक्यमधः for (स्वात्परतो ऽन्यैक्यमधः)
४. वदिहाक्षैः (ग) वदेहाद्यैः for (वदीहाद्यैः)

(च) ५. ध्येधः for (धोधः)
१. कल्पानेकाद्येकोत्तरानधः for (कल्पान्येकाद्येकोत्तरानधः)
३. परतांत्यैकमधः for (परतांतैकमधः)
४. वदिहाद्यैः for (वदीहाद्यैः)

(ङ) (वि०—इसकी क्रमसंख्या १८ है)
१. कल्प्यान्येका for (कल्पान्येका) ४. दुक्तवदिहाद्यैः for (द्युक्तवदीहाद्यैः)

२०. (घ) १. विघटी द्विगुणाद्वचे (ग) विघटी for (विघटि) २. स्या (ग) for (द्वा)
३. विशश्छंद (ग) विशच्छंद for (विंशच्छैद)
४. श्चित्युत्तरोऽध्यायः (ग) (ङ) (श्चित्युत्तरोऽध्यायः)
५. 'इति' से 'समाप्त' तक पाठ उपलब्ध नहीं है । (ग)

(ग) ६. विशति for (विंशति)

(च) १. विघटी for (विघटि) २. संख्याध्या for (संख्याद्वा)
३. श्चित्युत्तराध्यायः for (श्चित्तुत्तराध्यायः)
५. 'इति' से 'समाप्त होने तक लुप्त ।

(ङ) (वि०—इसकी क्रससंख्या १९ है)
७. गुरुषष्टचेका for (गुरुषष्टूचैका)
१. निघटीद्विगुणा for (विघटिद्विगुणा)
८. न्येकांगुलानि for (व्येकांगुलानि) २. संख्या स्यात् for (संख्याद्वा)
६. द्वाविंशतिरार्याणां for (विंशतिरार्याविंशं)

अथ गोलाध्यायो नाम

एकविंशतितमः

ग्रह नक्षत्रभ्रमणं न समं सर्वत्र भवति[3] भूस्थानम्[1] ।
तद्विज्ञानं गोलाद्यतस्ततो गोलमभिधास्ये[2] ।। १ ।।
शशि बुध सितार्क कुज गुरु शनि कक्षविष्टितोभ[1,2,3] कक्षांतः[4] ।
भूगोलः सत्वानां शुभाशुभैः कर्मभिरूपात्तः ।। २ ।।
खेभूगोलस्तदुपरि मेरौ देवाः स्थितास्तले[3] दैत्याः
खे[1] भगणाक्षाग्रस्तावु[2] पर्यूधश्चा ध्रुवो[4] तेषाम् ।। ३ ।।
ध्रुवयो विद्वंस[1] व्यगम मराणां क्षितिजसंस्थमुडुचक्रम्[2] ।
अपसव्यगमसुराणां भ्रमति प्रवहानि लाक्षिप्तम् ।। ४ ।।

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१. (घ) १. भूस्थानाम् (ङ) for (भूस्थानम्)
२. धास्येः for (धास्ये)
(ग) ३. 'भवति' पद यहां अंकित नहीं है ।

२. (घ) १. कक्ष्या (ग) कक्षा for (कक्ष)
२. चेष्टितो (ग) वेष्टितो for (विष्टितो)
(ग) ३. भक्षांत for (भकक्षांतः)
(च) ४. सिताङ्क for (सितार्क)
१. कक्ष्यावेष्टितो for (कक्षविष्टितो)
(ङ) १. कक्षावेष्टितो for (कक्षविष्टितो)

३. (घ) १. षे for (खे)
२. स्थावुपर्यधश्च (ङ) (ग) तावुपर्यधश्च for (स्तावुपर्यूधश्चा)
(ग) ३. स्थिताः स्तल for (स्थितास्तले) ४. ध्रुवौ for (ध्रुवो)
(च) २. क्षाग्रस्थावुपर्युधश्चा for (क्षाग्रस्तावुपर्यूधश्चा)

४. (घ) १. वंद्वंस (ग) बंद्धः for (विद्वंस)
(ग) २. क्षिति for (क्षितिज)
(च) १. वंद्वंस for (विद्वंस)
(ङ) १. बंद्ध for (विद्वं)

अन्यत्र सर्वतो दिशि[१]मुन्नमति[२]भयं जरोध्रुवो भ्रमति[३] लंकाया[४]: ।
मुहु[५] चक्रं पूर्वापरगं ध्रुवौ क्षितिजे ।। ५ ।।
देवा[१] सव्यगम सुरा:[५] पश्यंत्यपसव्यगं[२] रविं क्षितिजे ।
विषुवति समपश्चिमगं निरक्ष[३]देशास्थिता:[४] पुरुषा:[६] ।। ६ ।।
सौम्यमपमण्डलार्द्धं[२] मेषाद्यं[३] सव्यंग[४] सदादेवा: ।
पश्यंति तुलाद्यर्द्धं दक्षिणमप सव्यगं दैत्या:[१] ।। ७ ।।
पश्यंति देवदैत्या रविवर्षार्द्ध[३]मुदितं सकृत्सूर्यम् ।
शशिगा: स[१]शिमासार्द्धं पितरोभूस्था नरा: स्वदिवम्[२] ।। ८ ।।
भूपरिधि तूर्य[१]भागै लंका[४]भूमस्तका[६]क्षिति[५]तलाच्च ।
लंकोत्तरतोवंती[२] भूपरिधे: पंच दशभागै:[३] ।। ९ ।।

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५. (घ) १. दिश (ग) दिशिषुन्नमति for (दिशिमुन्नमति)
२. तपंजरो for (भयंजरो) २. नमति (ग) (ङ) for (भ्रमति)
४. 'लंकाया' दूसरी पंक्ति का पहलापद (ग) (ङ)
५. मुडुचक्रं (च) (ङ) for (मुहुचक्रं)
(च) ३. नमति for (भ्रमति)
४. लंकाया (ङ) for (लंकाया:) ५. मुडुचक्रं for (मुहुचक्रं)
(ङ) १. दिशमुन्नमति for (दिशिमुन्नमति) २. भपंजरो for (भयंजरो)
६. (घ) १. देवा: (ग) (च)(ङ) for (देवा) २. सव्य गरविं for (सव्यगं रविं)
३. तिरक्ष for (निरक्ष) ४. देशस्थिता: (ग) (च) (ङ) for (देशास्थिता:)
(ग) ५. सुरा for (सुरा:) (च) ६. पुरुषा: for (पुरूषा:)
७. (घ) १. दैत्या (च) for (दैत्या:) (ग) २. नंगुलार्द्ध for (मंडलार्द्ध)
३. मेखाद्यं for (मेषाद्यं) ४. मंसदा for (गंसदा)
(ङ) २. मपमंडलार्धं for (मपमंडलार्द्धं)
८. (घ) १. शशि (ग) शशि (च) (ङ) for (सशि)
२. स्वदिनम् (ङ) for (स्वदिवम्) (ङ) ३. वर्षार्ध for (वर्षार्द्धं)
९. (घ) १. चतुर्भागे (ग) तुर्य for (तूर्यभागै)
२. ऽवंती (ङ) (ग) लंकोत्तरोवन्ती for (लंकोत्तरतोवन्ती)
३. भगै: (ग) for (भागै:) (ग) ४. लंका for (लङ्का)
५. क्षिति for (क्षति) (च) १. चतुर्भागे for (तूर्यभागै)
२. ऽवन्ती for (वन्ती) (ङ) १. तुर्यभागे for (तूर्यभागै)
६. भूमस्तकात् for (भूमस्तका) ३. दशभागे for (दशभागै:)

अक्षांशभूपरिधि[1] वधान्मंडलभागा[2] सयोजनै[3] र्विषुवत् ।
तत्भाग[4] योजनैरेवमुपरिस्तयोन्य[5] दनुपातात्[6] ॥ १० ॥
अंबरयोजनपरिधिः शशिभगणाः शून्यखखजिनाग्नि गुणाः[2] ।
यस्य भगणै[1] र्विभक्तास्तत्कक्षार्कोभषष्टयंशे[3] ॥ ११ ॥
भूपरिधि[3] समान[4] षष्टया[5] ख[1] परिधि तुल्यानि कल्पविवषैः ।
गच्छंति योजनानि ग्रहाः स्वकक्षास्तु[2] तुल्यानि ॥ १२ ॥
भगणस्याधः[2] शनिगुरु भूमिजरविशुक्र सौम्यचंद्राणाम्[1] ।
कक्षाक्रमेण[3] शीघ्राः शनैश्चराद्याः कलाभुक्त्या ॥ १३ ॥

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१०. (घ) १. कुपरिधि (ग) (च) (ङ) for (भूपरिधि) २. प्त (ग) (ङ) for (स)
३. विषुवत् (ग) (ङ) for (विषुवत्) ४. नत (ग) (ङ) for (तत्)
५. सूर्योन्य (च) for (स्तयोन्य)
(ग) ६. दनुपात् for (दनुपातात्)
(च) २. भागाप्तयोजनै for (भागासयोजनै)
३. विषुवत् for (विषुवत्) ४. नतभाग for (तत्भाग)
(ङ) (ङ) सूर्योऽन्य for (स्तयोन्य)

११. (घ) १. विभक्तस्तत् कक्षार्कोभ for (विभक्तास्तत् कक्षार्कोभ)
(ग) २. +३३२४०००+
(च) ३. कक्षार्क्कोभषष्टयंशे for (कक्षार्कोभषष्टयंशे)
(ङ) ३. कक्षाऽर्को for (कक्षार्को) ४. भषष्ठयंशः for (भषष्टयंशे)

१२. (घ) १. भ for (ख) २. स्वकक्षासू for (स्वकक्षास्तु)
(ग) ३. भपरिधि for (खपरिधि)
४. समानि (ङ) for (समान)
५. षष्टय for (षष्टया)
(च) ५. खष्टयाष for (षष्टया ख)
(ङ) २. स्वकक्षासु for (स्वकक्षास्तु)

१३. (घ) १. चंप्राणां for (चन्द्राणाम्)
(ग) २. भगणस्याधिः for (भगणस्याधः)
(ङ) ३. कक्षाः for (कक्षा)

लघवोल्पो[१] राश्यंशा[२] महति महतो[३] ऽल्पदृतमल्पेन[६][७]।
पूरयंस्तींदुर्महता[४] कालेन महाशनैश्चारी[५] ॥१४॥
यन्मूलं तध्वासो[१] मंडललिप्ता कृतेर्दशहतायाः[२]।
तस्यार्द्धं व्यासार्द्धं[३] योजनकर्णप्रमाथार्थम्[४][५][६] ॥ १५ ॥
भगणकलाव्यासार्द्धं भवति कलाभिर्यतो न[२] सकलाभिः[३]।
ज्यार्द्धानि न स्फुटानि[१] ततः कृतं व्यासदलमन्यत् ॥ १६ ॥
राश्यष्टांशेष्टांकांत्यदसंधिभ्यः[४] क्रमोक्रमाक्रत्वा[२]।
बध्नीयात् सुत्राणि[३] द्वयोर्द्वयो ज्यास्तदर्द्धानि ॥ १७ ॥

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१४. (घ) १. लघवोऽल्पे for (लघवोल्पो) २. राशंशा (च) for (राश्यंशा)
३. महांतो (ग) (ङ) for (महतो)
४. पूरयतींदुर्म्महता (ग) पूरयतींदुर्महता (ङ) for (पूरयंस्तींदुर्महता)
५. महशनैश्चारी (ग) महछनैश्चारी (ङ) for (महाशनैश्चारी)
(ग) ६. ल्पवृत्त for (ऽल्पवृत्त) ७. मन्येन for (मल्पेन)
(च) ४. पूरयंती दुर्म्महता for (पूरयंस्तींदुर्महता)
५. महशनैश्चारी for (महाशनैश्चारी)
(ङ) १. लघवोऽल्पे for (लघवोल्पो)

१५. (घ) १. तद्वघासो (ग) (ङ) for (तध्वासो) २. कृतेर्द्दश (ग) for (कृतेर्दशा)
३. नस्यार्द्धं for (तस्यार्द्धं) ४. व्यार्द्धं for (व्यासार्द्धं)
५. कर्णमाणार्थम् (ग) (च) कर्णं प्रमाणार्थम् for (कर्णं प्रमाथार्थम्)
(च) १. तद्वघासो for (तध्वासो) २. र्दशहृतायाः for (र्दशहतायाः)
४. दघासार्द्धं for (व्यासार्द्धं)
(ङ) ६. प्रमाणार्थम् for (प्रमाथार्थम्)

१६. (घ) १.+(च)+(ङ)
(ग) २. र्यतोर्यतो for (र्यं तो न)
(ङ) ३. सविकलं हि for (सकलाभिः)

१७. (घ) १. ष्वंकान्पद (ग) for (ष्टांकांत्यद)
२. क्रमोत्क्रमात्कृत्वा (ग) (ङ) for (क्रमोक्रमाकृत्वा)
३. सूत्राणि (ग) (च) (ङ) for (सुत्राणि)
(च) १. ष्वंकान्पद for (ष्टांकांत्यद) ग. क्रमोत्क्रमा कृत्वा for (क्रमोक्रमा क्रत्वा)
(ङ) १. राश्यष्टांशेष्वङ्कान् for (राश्यष्टांशेष्टांकांत्)
४. पदसन्धिभ्यः for (यदसंधिभ्यः)

ज्याद्धानि ज्याद्धानां ज्याखंडान्यंतराणि तन्ये च ।
व्यस्तान्यं तदेथ विषुरू क्रमज्या धनुस्ताभ्याम् ॥ १८ ॥
एक द्वित्रि गुणाया व्यासार्द्ध क्रतेः पृथक् चतुर्थेभ्यः ।
मूलान्यष्ट द्वादश षोडष षडान्यतो न्यानि ॥ १९ ॥
तुल्यक्रमोक्रम समज्या खंडक वर्ग युतिर्भागम् ।
प्रोह्यानष्टं व्यासार्द्ध वर्गतस्तत्पदे प्रथमं ॥ २० ॥
तद्दलखण्डानि तदुन जिन समानि द्वितीयमुत्पत्तौ ।
कृतप्रमलैक दिगीशेषु सप्तरसगुणा नवादीनाम् ॥ २१ ॥
एवं जीवा खंडान्यल्पानि बहुनि वाद्यखण्डानि ।
ज्याद्धानि वृत्तपरिधेः षष्टचतुर्थत्रिभागानाम् ॥ २२ ॥

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१८. (घ) १. तान्येव (ग) (ङ) (च) for (तन्ये च)
२. वेषु (ग) पादथतेषु for (विषुरूक्रमज्या)
(ग) ६. व्यस्तान्यत् for (व्यस्तान्यं) ३. व्यस्तान्यतादथ तेषु for (व्यस्तान्यं तदेथ विषु)
२. तेषुरुत्क्रमज्या for (विषुरुक्रमज्या)
(ङ) ३. व्यस्तान्यन्त्यादथ for (व्यस्तान्यं तदेथ)
२. वेषुरुत्क्रमज्या for (विषुरूक्रमज्या)

१९ (घ) १. कृतेः (ग) (च) (ङ) for (क्रतेः) २. षोडश (ग) (च) (ङ) for (षोडष)
(ग) ३. द्विगुणया for (द्वित्रिगुणाया) ४. मूलात्यष्ट for (मूलान्यष्ट)
५. षंडान्यतो for (षडान्यतो)
(ङ) ५. खण्डान्यतोऽन्यानि for (षडान्यतोन्यानि)

२०. (घ) १. तुल्यक्रमोत्क्रम (ग) (च) (ङ) for (तुल्यक्रमोक्रम)
२. चतुर्भागम् (ङ) for (भागम्)
३. प्राज्यानष्टं (ग) प्राह्यानष्टं for (प्रोह्यानष्टं)
(ग) ४. खंड for (खंडक) ५. वर्ग्रस्तत्पदे for (वर्गतस्तत्पदे)
(च) ३. प्रोज्यानष्टं for (प्रोह्यानष्टं) (ङ) ६. ज्यासम for (समज्या)

२१. (घ) १. तदून (ग) (च) (ङ) for (तदुन)
(ग) २. (वि०—'द्वितीय' पद लुप्त है)
(च) ३. गुणनवादीनाम् (ङ) for (गुणानवादीनाम्)

२२. (घ) १. बहूनि (ग) (च) (ङ) for (बहुनि) २. षंडानि (च) for (खण्डानि)
(ग) ३. ज्यार्धनिवृत्तिपरिधेः for (ज्याद्धानिवृत्तपरिधेः)
(ङ) २. वाऽऽद्यखण्डानि for (वाद्यखण्डानि) ३. ज्यार्धानि for (ज्याद्धानि)

उक्रम[४] समखण्ड[५] गुणाध्व्यासादथवा चतुर्थभागाद्यम्[१] ।
कृत्वोक्त खण्डकानि[२] ज्यार्द्धानयनं[६] नलध्वमस्मात्[३] ॥ २३ ॥
कक्षामंडलभूमध्ये[१] मध्यमः[२] स्वकक्षायाम्[३] ।
अनुलोमं मंदोच्चात्प्रतिलोमं भ्रमति शीघ्रोच्चात् ॥ २४ ॥
त्यक्तं[१] तत्परिधौ प्रतिलोमं मन्दोच्चा भ्रमति शीघ्रोच्चात् ।
अनुलोमं मध्यस्थसमं[२] भूस्थः पश्यति यतोनकक्षायाम् ॥ २५ ॥

२३. (घ) १. गुणाढ्यासादथवा (ग) (च) (ङ) for (गुणाध्व्यासादथवा)
२. षंडकानि (च) for (खंडकानि)
३. लद्यस्मात् (ग) लध्वमस्तात् for (लध्वमस्मात्)
(ग) ४. उत्क्रम (च) (ङ) for (उक्रम) ५. खण्डा for (खण्ड)
६. यामार्द्धनयनं for (ज्यार्द्धानयनं)
(च) २. नलध्वमस्मात् for (नलध्वमस्मात्)
(ङ) ७. भागाद्यत् for (भागाद्यम्) ३. लध्वस्मात् for (लध्वमस्मात्)
२४. (घ) १. कक्ष्या for (कक्षा) २. मंडलमध्यं (ग) for (मडलभूमध्ये)
(च) १. कक्ष्या for (कक्षा) २. मंडलमध्यं for (मंडल)
(ङ) शी०+अथ स्फुट गतिवासना+ ३. मध्यं भूमध्ये for (भूमध्ये)
२५. (घ) १. (वि०—यहाँ श्लोक की तीन पंक्तियाँ हैं।
"नीचोच्चवृत्तमध्यं मध्ये तद्भ्रमति मध्यमः स्वोच्चात्" (च) (ङ) for
(त्यक्तं तत्परिधौ प्रतिलोमं मंदोच्चा भ्रमति शीघ्रोच्चात्)
२. मध्यसमं (ग) (च) (ङ) for (मध्यस्थसमं)
(ग) १. "नीचोथवृत्तमध्यं मध्ये तद्भ्रवति मध्यमः स्वोच्चात् ।"
तत्परिधौ प्रतिलोमं मंदोच्चात् भवति शीघ्रोच्चात् ॥ २५ ॥
for
(त्यक्तं तत्परिधौ प्रतिलोमं मंदोच्चा भ्रमति शीघ्रोच्चात् । अनुलोमं मध्यस्थसमं भूस्थः पश्यति यतो न कक्षायाम् ॥ २५ ॥)
(च) इस प्रति में 'घ' की भाँति श्लोक में तीन पंक्तियाँ हैं—पहली पंक्ति है :—'नीचोच्चवृत्तमध्यं मध्ये तद्भ्रमति मध्यमः स्वोच्चात्'
(ङ) (वि०—इस प्रति की पहली पंक्ति 'नीचोच्चवृत्तमध्यं तद् भ्रमति मध्यगः स्वोच्चात्" उपर्युक्त प्रतियों में छूट गई। और दो श्लोकों की चार पंक्तियों में से केवल तीन को सम्मिलित करके एक श्लोक माना गया। आगे के श्लोकों में फलतः सर्वत्र इस प्रति की दृष्टि से पूर्व श्लोक का उत्तरार्ध+ अग्रिम श्लोक का पूर्वार्ध मिलाकर श्लोकों की व्यवस्था की गई है।

स्पष्टं[1] तन्मध्यांतरमृणं धनं वा तप्तो[2] मध्ये ।
कोटिफलं व्यासार्द्धात्पदयोराद्यं तयो भवत्युपरि[3] ।। २६ ।।
द्वितृतीययोर्यतोऽधस्ताष्टृक्तोनं[1] तरतः[2] कोटिः[3] ।
कर्णस्तद्भुजफलकृतिसंयोग[4] पदं तदुद्धृता त्रिज्या ।। २७ ।।
भुजफल[6] गुणिताप्तधनु गुणिते[1] चैवं[2] फलं शीघ्रे ।
त्रिज्याभक्तः कर्णः परिधिगुणो[4] बाहुकोटिः[7] गुणाकारः[5] ।। २८ ।।

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२६. (घ) १. अष्टं for (स्पष्टं)
२. ततो (ग) गृहे मध्ये ।। २६ ।। for (तप्तो मध्ये)
३. भवत्युपरि (ग) (च) (ङ) for (भवत्युपरि)
(ग) इस श्लोक की पहली पंक्ति २६ वें श्लोक की दूसरी पंक्ति है ।
इस श्लोक की दूसरी पंक्ति २७ वें श्लोक की पहली पंक्ति हैं—
(च) २. ततो for (तप्तो)
(ङ) दे० वि० नोट २५ वें श्लोक के अन्तर्गत टिप्पणी में ।
२. गृहे मध्ये for (तप्तो मध्ये)

२७. (घ) १. तद्युक्तोनं (ङ) for (ताष्टृक्तोनं) २. ततः (ङ) for (तरतः)
(ग) ३. कोटिः ।। २७ ।। (ङ) for (कोटिः)
(वि०—इस श्लोक की दूसरी पंक्ति—२८ वें श्लोक की प्रथम पंक्ति है—
४. फलसंयोगं for (फलकृति संयोग)
(च) १. ऽधस्तद्युक्तोनं for (ऽधस्ताष्टृक्तोनं)

२८. (व) १. धनुर्गुणिते for (धनुगुणिते) (ग) र्गणितेनैवं फलं शीघ्रे ।। २८ ।।
२. नैवं for (चैवं)
३. विज्या for (त्रिज्या)
४. गुणो (ग) (ङ) for (गुणो)
५. गुणकारः (ग) (ङ) for (गुणाकारः)
(ग) इस श्लोक की पहली पंक्ति २८ वें श्लोक की द्वितीय पंक्ति है ।
६. भुजफ for (भुजफल) ७. कोटि (ङ) for (कोटिः)
(च) १. गुणिते for (गुणिते) ४. परिधोगुणो for (परिधिगुणो)
७. कोटि for (कोटिः) ५. गुणकारः for (गुणाकारः)
(ङ) १. धनुर्गणिते for (धनुर्गुणिते) २. नैवं for (चैवं)

असकृत्मांदे[1] तत्फलमाद्यसमं[3] नात्र कर्णेस्मात्[4] ।
प्रतिपादनार्थमुच्चं[2] प्रकल्पितं ग्रहगतेस्तथा पातः ।। २९ ।।
भुक्तेरूनाधिकनामानस्यच[1] भवति कर्णवशात्[2] ।
कक्ष्या[3] व्यासार्द्धगुणा मंडललिप्ता विभाजिताकर्णः ।। ३० ।।
स्वकलाकर्णेन गुणः कर्णस्त्रिज्याहतः[1] स्पष्टः ।
मद्दहन[2] जलमयानं[4] विष्कंभा[5] योजनैः कनेंदुना[3] ।। ३१ ।।

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२९. (घ) १. असकृन्मां (ग) असकृन्मांद्ये for (असकृत् मांदे)
२. प्रतिपदेनार्थमुच्चं for (प्रतिपादनार्थमुच्चं)
(ग) ३. सम for (ममं)
४. कर्णेस्मात् ।। २९ ।। (ङ) for (कर्णेस्मात्)
इस श्लोक का उत्तरार्ध, ३० वें श्लोक का पूर्वार्ध है।
(च) १. असकृन्मांदे for (असकृत्मांदे)
४. कर्णेस्मात् for (कर्णेस्मात्)
(ङ) २. असकृन्मान्दे for (असकृत्मांदे)
३०. (घ) १. तामानस्य (ग) for (नामानस्य)
(ग) २. व स्यात् ।। ३० ।। for (वशात्)
३. कक्षा (ङ) for (कक्ष्या)
इस श्लोक का उत्तरार्ध, ३१ वें श्लोक का पूर्वार्ध है।
(च) १. तामानस्य for (नामानस्य)
(ङ) १. रूनाधिकता for (रूनाधिकना)
४. +इति स्फुटगतिवासना+
३१. (घ) १. हृतः (ग) हृतस्पष्टः ।। ३१ ।। for (हृतः)
२. मृद्दहन (ग) (ङ) for (मद्दहन)
३. क्विनेन्दूनाम् (ग) केनेदूनाम् for (कनेंदुना)
(ग) ४. मायानां for (मयानं)
इस श्लोक की दूसरी पंक्ति अगले श्लोक की पहली पंक्ति है।
(च) हृतः for (हृतः)
२. मृद्दहन (ङ) for (मद्दहन)
४. जलमयानां (ङ) for (जलमयानं)
३. क्विनेंदुनां for (कनेंदुना)
(ङ) ५. विष्कंभो for (विष्कंभा)
३. द्विनेंदूनां for (कनेंदुना)

शशिवसुतिथिभि[६] यमपक्षशररसैः[१] शून्य[२] वस्तु[२] वेदैः[२] ।
व्यर्क[३] व्याप्तांतर[४] गुणमिदु स्फुट कर्णमर्ककर्णहतम्[५] ॥ ३२ ॥
प्रोह्य[१] भुवो भुछाया[१] विष्कंभश्चंद्रकक्षायाम्[२] ।
तद्गुणितं व्यासार्द्धं शशिकर्णहतं[३] ततः प्रमाणकलाः ॥ ३३ ॥
एवं त्रिज्या[१] रविशशि विष्कंभगुणाः[६] स्वकर्णहताः[२] ।
भूछायेंन्दुश्चंद्रसूर्य[३] छादयति[४] मानयोगार्द्धात्[५] ॥ ३४ ॥

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३२. (घ) १. यंम (ग) यमपक्षरासैः ६५२२ for (यमपक्षशररसैः)
२. वसु (ग) वसुवेदैः (ङ) for (वस्तुवेदैः)
३. क्कर्क for (व्यर्क) ४. व्यासांतर (ग) for (व्याप्तांतर)
५. हृतम् (ग) for (हृतम्)
(ग) ६. राशिवसुतिथिभिः १५८१ for (शशिवसुतिथिभिः)
(वि०—इस श्लोक के पूर्वार्ध की समाप्ति पर ३२ संख्या होनी चाहिए परन्तु लिखी हुई ३३ है)
(च) १. यंम for (यम) २. वसु for (वस्तु)
३. व्यर्क्क for (व्यर्क) ४. व्यासांतर for (व्याप्तांतर)
५. मर्क्ककर्णहृतं for (मर्ककर्णहतम्)
(ङ) १. यंम for (यम)
३३. (घ) १. भूछाया (ग) for (भुछाया)
२. कक्ष्यायाम् (ग) कक्षायाम् ॥ ३४ ॥ for (कक्षायाम्)
३. तमः (ग) (च) (ङ) for (ततः)
(ग) इस श्लोक का उत्तरार्ध अगले श्लोक का पूर्वार्ध है ।
(च) १. प्रोज्यभुवोभूछाया for (प्रोह्यभुवोभुछाया)
(ङ) १. भूच्छाया for (भुछाया)
३४. (घ) १. तृज्याशशि for (त्रिज्यारविशशि) २. हृताः (ग) हृता ॥३५॥ for (हृताः)
३. सूर्यं (ग) (च) (ङ) for (मूर्यं)
४. भूछायेंदु चंद्रः (ग) भूछायेंदूंचंद्रः for (भूछायेन्दुश्चंद्र)
५. (वि०—यहां पर कोई अंक संख्या नहीं दी)
(ग) ६. गुणः for (गुणाः)
(वि०—इस श्लोक का उत्तरार्ध ३६ वें श्लोक का पूर्वार्ध है)
(च) २. हृताः for (हताः) ४. भूछायेंदुचंद्रः for (भूछायेंदुश्चन्द्र)
५. (वि०—यहां पर क्रमसंख्या लुप्त है)
(ङ) ४. भुच्छायेंदु चन्द्रः for (भूछायेंदुश्चन्द्र) ६. गुणा for (गुणाः)

विक्षेपो[६] यद्यूनः शुक्लेतर पंच[३] दृश्यंते[१] ।
महदिदो[७] रावणं[४] कुठं[५] विषाणो यतो[२] संछन्नः[८] ॥ ३५ ॥
अर्द्धछिन्नो[५] भानुस्तिक्षण[१] विषाणस्ततो[२] स्याल्पे ।
यदि राहुः प्राग्भादिंदु[३] छादयति किं तथा नार्कम् ॥ ३६ ॥[४]

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३५. (घ) १. दश्यंते (ङ) for (दृश्यन्ते)
(यहां ३५ की संख्या अङ्कित है)
२. र्द्धसंछन्नः (ग) र्धसंछन्नः (ङ) र्ध सञ्छन्नः for (संछन्नः)

(ग) ३. पंचं for (पंच)
१. दिश्यंते ॥३६॥ for (दृश्यंते)
४. रादरणं for (रावणं)
५. कुठ (ङ) for (कुठ)
वि०—इस श्लोक का उत्तरार्ध ३७ वें श्लोक का पूर्वार्ध है)

(च) ६. विक्ष्येपो for (विक्षेपो)
३. पंद्रश्यंते ३५ for (पंचदृश्यंते)
७. महदिदो (ङ) for (महदिदो)
२. यतोर्द्ध for (यतो)
८. (वि०—यहां कोई क्रमसंख्या अंकित नहीं है)

(ङ) ४. रावरणं for (रावणं)

३६. (घ) १. स्तीक्ष्ण (ग) (ङ) for (स्तिक्षण)
२. ततोऽस्याल्पम् (च) (ङ) for (ततोस्याल्पे)
(वि०—यहां क्रमसंख्या ३६ दी है)
३. प्राग्भागार्दिदु (ग) प्राग्भार्दिदुं for (प्राग्भार्दिदु)

(ग) २. ततोऽस्याल्पी ॥३७॥ for (ततोस्याल्पे)
(वि०—इस श्लोक का उत्तरार्ध ३८ वें श्लोक का पूर्वार्ध है)

(च) १. स्तीक्ष्ण for (स्तिक्षण)
४. (वि०—यहां कोई क्रमसंख्या नहीं दी, क्योंकि इस प्रति के अनुसार यहां का उत्तरार्ध ३७ वें श्लोक का पूर्वार्ध है।

(ङ) ५. अर्धच्छन्तो for (अर्द्धछिन्नो)
२. प्राग्भागादिन्दुं for (प्राग्भार्दिदु)

स्थित्यर्द्ध[1] महदिन्दो य[5]था तथा किन्न सूर्यस्य[2] ।
कि प्रतिविषयं[3] सूर्यो[4]राहुश्चान्यो यतो रविग्रहणे ॥ ३७ ॥[6]
ग्रासान्यत्वं न ततो राहु[1]कृतं ग्रहण मर्कद्धो[2]ः ।
एवं वराहमिहिर[3]श्रीषेणार्यभटविष्णुचंद्राद्यैः ॥ ३८ ॥[4]
लोक विरुद्धमभिहितं वेदस्मृति[1]संहिता[5]बाह्यम् ।
यद्य[5]वं ग्रहणफलं गर्गाद्यैः संहि[2]तामु[3]दयभिहितम् ॥ ३९ ॥

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३७. (घ) १. स्थित्यर्द्ध for (स्थित्यर्द्ध)
२. सूर्यः स्य (यहां क्रमसंख्या ३७ दी है) (ग) सूर्यस्य ॥३८॥ for (सूर्यस्य)
३. विषग्रं (ग) विखयं for (विषयं)
(ग) (वि०—इस श्लोक का उत्तरार्ध ३९ वें श्लोक का पूर्वार्ध है।
४. सूयो for (सूर्यो)
(च) ५. यंथा (ङ) for (यथा)
२. (वि०— यहां क्रमसंख्या ३७ अंकित है)
६. वि०—यहां कोई क्रमसंख्या नही दी है, क्यों कि इस श्लोक का उत्तरार्ध, ३९ वें श्लोक का पूर्वार्ध है। (ङ) १. स्थित्यर्धं for (स्थित्यर्द्ध)

३८. (घ) १. कृतं (ग) (च) (ङ) for (कृतं)
२. मर्कैद्वोः (ग) मर्कैद्वोः ॥३९॥ for (मर्कद्धो) (यहां ३८ संख्या दी है)
(ग) ३. श्रीषेणाचार्यं for (श्रीषेणार्यं)
इस श्लोक का उत्तरार्ध, ४० वें श्लोक का पूर्वार्ध है।
(च) २. मर्क्कैद्वोः for (मर्कद्धो.) (वि०—यहां श्लोक संख्या ३८ अंकित है)
४. यहां क्रमसंख्या लुप्त है, क्योंकि यह ३९ वें श्लोक का पूर्वार्ध है।
(ङ) २. मर्केन्द्वोः for (मर्कद्धोः)

३९. (घ) १. स्त्रिम्मृवि for (स्मृति) (यहाँ संख्या ३९ है)
२. संहितासु (ग) (ङ) for (संहितामु)
३. यदि भिहितम् (ग) for (दयभिहितम्)
(ग) ४. संहिताद्वाह्यम् ॥ ४० ॥ for (संहिताबाह्यम्)
५. यद्येव ग्रहफलं for (यद्येवं ग्रहणफलं)
(वि०—इस श्लोक का उत्तरार्ध ४१ वें श्लोक का पूर्वार्ध है।
(च) ४. यहाँ '३९' की क्रमसंख्या अंकित है। परन्तु उत्तरार्ध के अन्त में कोई क्रमसंख्या अंकित नहीं, क्योकि उत्तरार्ध, ४० वें श्लोक का पूर्वार्ध माना जाता है।
(ङ) ३. यदभिहितम् for (दयभिहितम्)

तदभावे[२] होमजप:[३][४] स्नानादि फलस्य[५] चाभाव:।[७]
राहुक्रतं[१] ग्रहणद्वय मागोपालांगनादि[२] सिद्धमिदम् ॥ ४० ॥[८]
बहुफलमिदमपि[१] सिद्धं जपहोमस्नानफलमत्र[२]।
स्मृतीषुक्तं[३] न स्नानं राहोरन्यत्र दर्शनाद्रात्रौ ॥ ४१ ॥
राहुग्रस्ते सूर्य[१] सर्वं गंगासमं तोयम्[२]।
स्वर्भानुरास्तुरिरिनं[३] तमसा वित्याधवेदवाक्यमिदम्[४] ॥ ४२ ॥

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४०. (घ) (वि०—यहाँ प्रथम पंक्ति के अन्त में ४० संख्या दी है)
१. कृतं (ग) for (क्रतं)
(ग) २. तदभावो for (तदभावे) ३. लुप्त पद है। ४. लुप्त पद है।
५. फलस्याभाव: ॥ ४१ ॥ for (फलस्य चाभाव:)
(वि०—इस श्लोक का उत्तरार्ध, अगले श्लोक का पूर्वार्ध है)
६. माग्गो पालाऽगना for (मागो पालांगनादि)
(च) ७. यहाँ '४०' क्रमसंख्या अंकित है। ८. यहाँ कोई संख्या अंकित नहीं, क्योंकि यह उत्तरार्ध भाग ४१ वें श्लोक का पूर्वार्ध माना जाता हैं।
(ङ) ४. जप for (जप:) ५. स्नानादीनां फलाभाव: for (स्नानादि फलस्य चाभाव)

४१. (घ) १. बाहु for (बहु)
२. मंत्र (यहाँ क्रमवंख्या ४१ दी है) (ग) मत्र ॥ ४२ ॥ for (मत्र ॥ ४१ ॥)
३. स्मृतिषूक्तं (ग) (ङ) for (स्मृतीषुक्तं)
(ग) इस श्लोक का उत्तरार्ध ३३ श्लोक का पूर्वार्ध है
(च) २. यहाँ '४१' क्रमसंख्या दी गई है। ३. स्मृतीषूक्तं for (स्मृतीषुक्तं)
(वि०—इस प्रति के अनुसार इस श्लोक का उत्तरार्ध ४२ वें श्लोक का पूर्वार्ध है अत: यहाँ अन्त में क्रमसंख्या अंकित नहीं है)

४२. (घ) १. सूर्ये (ग) (च) (ङ) for (सूर्य)
२. (वि०—यहाँ क्रमसंख्या ४२ (ग) ४३ दी है)
३. रासुरि (ग) रासुरि न तमसा for (रास्तु)
४. विच्याध (ग) विव्याध (ङ) for (वित्यधा)
(ग) इस श्लोक का उत्तरार्ध अगले श्लोक का पूर्वार्ध है।
(च) २. (वि०—यहाँ क्रमसंख्या ४२ अंकित है) ३. रासुरिरिन for (रास्तुरिरिनं)
४. विव्याध for (वित्याध)
यहाँ श्लोक के अन्त में कोई क्रमसंख्या अंकित नहीं क्योंकि यह उत्तरार्ध माना जाता है।
(ङ) ३. रासु for (रास्तु)

श्रुतिसंहितास्मृतीनां भवति तथैक्यं[1] तदुक्तिरतः[3][2] ।
राहुस्तच्छादयति प्रविशति यच्छुक्ल[4] पंच[5] दृश्यन्ते[1] ।। ४३ ।।
भूच्छाया[4] तमसींदो[5] वरप्रदानात्कमलजस्य[1] ।
चन्द्रोवुभयोधस्थो[2][3] यदग्निमय[6] भास्करस्य[4] मासांते[1] ।। ४४ ।।
छादयति शमिततयो[3] राहु[6] छादयति[4] तत्सवितुः[1] ।
भूच्छाया व्यासमः[2] शशिकक्षायां स्थितः शशिग्रहणे[5] ।। ४५ ।।

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४३. (घ) १. यथैक्यं (ग) (च) (ङ) for (तथैक्यं)
२. (वि०—यहाँ क्रमसंख्या ४३ दी है) (च) (ङ)
(ग) ३. रंतः ।। ४४ ।। for (रतः)
४. यत् शुक्ल पंच दृश्यंते for (यच्छुक्ल पंचदृश्यन्ते)
(च) (वि०—यहाँ इस श्लोक के अन्त में कोई क्रमसंख्या अंकित नहीं क्योंकि यह उत्तरार्ध यहाँ ४४ वें श्लोक का पूर्वार्ध माना जाता है)
(ङ) ४. यच्छुक्ल for (यच्छुक्ल)

४४. (घ) १. (वि०—यहां क्रमसंख्या ४४ दी है) (च) (ङ)
२. वुमयो (ग) चंद्रोबुमयोधस्थः for (चन्द्रोवुभयोधस्थो)
३. धस्थो for (ऽधः स्थो)
(ग) ४. नभसींदो for (तमसींदो)
५. त्कमलाया ने ।।४४।। for (कमलजस्य) ६. यदग्नियम for (यदग्निमय)
(वि०—इस श्लोक का उत्तरार्ध ४५ वें श्लोक का पूर्वार्ध है)
(च) (वि०—इस श्लोक का उत्तरार्ध ४५ वें श्लोक का पूर्वार्ध है)
अतएव यहां अन्त में कई क्रमसंख्या अंकित नहीं ।
(ङ) ५. कमलयोनेः for (कमलजस्य)
२. छन्द्रोऽम्बुमयोऽधः स्थो for (चन्द्रोवुभयोधस्थो)

४५. (घ) १. (वि०—यहां क्रमसंख्या ४५ अंकित है) (च) (ङ)
२. सम for (समः)
(ग) ३. शमिततापो (ङ) for (शमिततयो)
४. तच्छवितुः ।।४५।। for (तत्सवितुः)
(वि०—इस श्लोक का उत्तरार्ध, ४६ वें श्लोक का पूर्वार्ध है) (च)
(च) ३. शमिततायो for (शमिततयो) ५. क्रम संख्या लुप्त
(ङ) ६. राहुश्छादयति for (राहुछादयति)

राहु छादयतींदु[3] सूर्यग्रहणेर्कमिंदु[4] समः[1] ।
यत्तदधिकं तमोमयराहुव्यासस्य सूर्य[5] द्रष्टं[2] तत्[6] ॥ ४६ ॥
नश्यति भूछायेंदु[1] व्यास समोस्माद् भवति राहुः[3] ।
भूछायानेंदुमतो[2] ग्रहणे छादयति नार्कबिंदुर्वा[4] ॥ ४७ ॥[5]
तत्स्थस्तध्व्याससमो[1] राहु[2] छादयति शशिसूर्यौ[3] ।
प्राच्यपरं सममंडलमन्यद्याम्योत्तरं क्षितिजमन्यत्[4] ॥ ४८ ॥

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४६. (घ) १. मिंदुं समः (ग) (च) (for मिंदुसमः)
(वि०—यहां क्रमसंख्या ४६ अंकित है)
२. दृष्टं तत् (ग) for (द्रष्टं तत्)
(ग) ३. छादयतींदुं for (छादयतींदु) ४. सूर्यं for (सूर्य)
५. 'सूर्य' लुप्त पद है ।
(वि०—श्लोक का उत्तरार्ध ४७ वें श्लोक का पूर्वार्ध है) (च)
(च) २. दृष्टं for (द्रष्टं) ६. क्रमसंख्या लुप्त
(ङ) ३. राहुश्छादयतीन्दुं for (राहुछादयतीन्दु) २. दृष्टत्वात् for (दृष्टंतत्)
४७. (घ) १. समोऽस्माद् (च) for (समोस्माद्)
(वि०—यहां क्रमसंख्या ४७ दी है)
२. भूछाययेंदु for (भूछायानेंदुमतो)
(ग) ३. राहुः ॥४७॥ for (राहुः)
४. मिंदुर्वा for (बिंदुर्वा)
(वि०—इस श्लोक का उत्तरार्ध, ४८ वें श्लोक का पूर्वार्ध है) (च)
(च) २. भूछायातेंदुमतो for (भूछायानेंदुमतो) ४. नार्क्कमिंदुवति for (नार्कबिंदुर्वा)
५. क्रम संख्या लुप्त
(ङ) ६. भूछायेन्द्वोर्व्यास for (भूछायेंदुव्यास)
२. भूछायेन्दुमतो for (भूछायानेंदुमतो)
४८. (घ) १. तद्व्यास (ग) (ङ) for (तध्व्यास) २. राहुं for (राहु)
३. (वि०—यहां क्रमसंख्या ४८ अंकित है) (च)
(ग) ३. सूर्यैः ॥४८॥ for (सूर्यौ)
(वि०—इस श्लोक का उत्तरार्ध ४९ वें श्लोक का पूर्वार्ध है) (च)
(च) १. श्छस्तद्व्यास for (तत्स्थस्तध्व्यास) ४. क्रमसंख्या लुप्त
(ङ) (वि०—प्रथम पंक्ति समाप्ति पर 'ग्रहण वासना' समाप्त)
+अथगोलबन्धाधिकारः+आरम्भ ।
२. राहुश्छादयति for (राहुछादयति)

परिकरवत्तन्मध्ये भूगोलस्थितद्रष्टुः ।
पूर्वापरयोर्लग्नं याम्योत्तरयोर्न्नतोन्नतं क्षितिजात् ॥ ४९ ॥
स्वाक्षांशैरुन्मंडलमहर्निशो वृद्धिहानिकरम् ।
विषुवन्मंडलमूर्द्धं सममंडलस्थितं स्वकक्षांशैः ॥ ५० ॥
याम्येनोत्तरतोधः क्षितिजे प्राच्यपरयो लग्नम् ।
विषुवन्मंडललग्नं मेषतुलादावुदक्कुलीरादौ ॥ ५१ ॥

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४९. (घ) १. भूगोलस्तस्थित द्रष्टुः (ङ) for (भूगोलस्थितद्रष्टुः)
(वि०—यहां कोई क्रमसंख्या नहीं दी गई) (च)
(ग) १. स्थस्थितद्रष्टुः ॥४९॥ for (स्थितद्रष्टुः)
२. याम्योत्तरयोर्नतं क्षितिजात् for (याम्योत्तरयोर्न्नतोन्नतं क्षितिजात)
(वि०—इस श्लोक का उत्तरार्ध ५० वें श्लोक का पूर्वार्ध है) (च)
(च) १. स्तस्थित for (स्थित) ३. क्रमसंख्या लुप्त ।

५०. (घ) १. रुन्मंडल (ग) for (रून्मंडल)
२. (वि०—यहां क्रमसंख्या ५० दी हुई है) (ग) (च)
३. मंडलः (ग) मंडलवत्स्थितं for (मंडलस्थितं)
४. स्वकक्षोंशैः for (स्वकक्षांशैः)
(ग) ५. स्वक्षांशैः for (स्वाक्षांशैः)
(वि०—इस श्लोक का उत्तरार्ध, ५१ वें श्लोक का पूर्वार्ध है) (च)
(च) ३. सममंडलः for (सममंडल) ७. क्रमसंख्या लुप्त
(ङ) ६. महर्निशोर्हानिवृद्धिकरम् for (महर्निशोर्वृद्धिहानिकरम्)
३. सममंडलतः for (सममंडल)
४. स्वकाक्षांशै for (स्वकक्षांशैः)

५१. (घ) १. (वि०—यहां क्रमसंख्या ५१ दी है) (ग) (च)
(ग) २. याम्यो for (याम्ये) ३. क्षितिज for (क्षितिजे)
४. कुलीरादौ for (कुलीरादौ)
(वि०—इस श्लोक का उत्तरार्ध ५२ वें श्लोक का पूर्वार्ध है) (च)
(च) ४. वुदत्कुलीरादौ for (वुदकुलीरादौ) ५. क्रमसंख्या लुप्त।
(ङ) २. याम्येनोत्तरतोऽधः for (याम्येनोत्तरतोधः)
४. वुदक् कुलीरादौ for (वुदकुलीरादौ)

जिनभागैर्याम्येन[१] मृगादावपमंडलमिहार्कः।[२]
पातश्चंद्रादिनां[३] भ्रमंति[६] भार्द्धे[४] खेश्च[७] भूछाया ॥ ५२ ॥
पातदयमंडल[१] वह्नि[३] मंडलानि स्वविक्षेपैः।[२]
सौम्यं विमंडलार्द्धं प्रथमं याम्यं द्वितीयमेतेषु[४] ॥ ५३ ॥[५]
चन्द्रकुज जीवमंदा भ्रमंति शीघ्रेण[१] बुधशुक्रौ।
दृग्मंडलार्द्ध[३] मूर्द्ध[२] यत्तत्परिधि स्थितं ग्रहं द्रष्टाः ॥ ५४ ॥[४]
पश्यति यतः क्षितिस्थस्तद्भ्रमति[२] ततो ग्रहाभिमुखं[१]।
क्षितिजापमंडलयुतिर्लग्नं[३] लग्नाग्रतो[५] दिशंवलनं ॥ ५५ ॥[४]

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५२. (घ) १. यैमोन for (यांम्येन)
२. (वि० – यहां क्रमसंख्या ५२ दी हुई है) (ग) (च)
३. पाताश्चंद्रादीनां (ग) (ङ) for (पातश्चन्द्रादिनां)
४. भाद्धे for (भार्द्धे) (ग) ५. निज for (जिन)
६. भ्रमंते for (भ्रमंति)
(वि०—इस श्लोक का उत्तरार्ध ५३ वें श्लोक का पूर्वार्ध है) (च)
(च) ३. पाताश्चंद्रादीनां for (पातश्चंद्रादिनां) ७. क्रमसंख्या लुप्त
५३. (घ) १. पातादषमंडल (ग) पातादपमंडलवत् for (पातदयमंडलवद्)
२. (वि०—यहां क्रमसंख्या ५३ दी है) (ग) (च)
(ग) ३. विमंडलानि for (द्विमंडलानि)
(वि०—इस श्लोक का उत्तरार्ध ५४ वें श्लोक का पूर्वार्ध है) (च)
४. द्वितीयामातष्ट for (द्वितीयमेतेषु)
(च) १. पातदपमंडल for (पातदयमंडल) ५. क्रमसंख्या लुप्त
(ङ) १. पातादपमंडलवद् for (पातदयमंडलवद्)
५४. (घ) १. (वि०—यहां क्रमसंख्या ५४ दी है) (ग) (च)
२. द्रष्टा (ग) दृष्टा for (द्रष्टाः) (ग) ३. भूर्द्ध for (मूर्द्ध)
(वि०—इस श्लोक का उत्तरार्ध ५५ वें श्लोक का पूर्वार्ध है) (च)
(च) २. द्रष्टा for (दृष्टाः) ४. क्रमसंख्या लुप्त
(ङ) ३. मूर्ध्व for (मूर्द्ध) २. द्रष्टा for (द्रष्टाः)
५५. (घ) १. (वि०—यहां क्रमसंख्या '५५' दी गई है) (ग) (च)
(ग) २. तद्भमति for (तद्भ्रमति) ३. यादिशं for (तोदिशं)
(वि०—इस श्लोक का उत्तरार्ध ५६ वें श्लोक का पूर्वार्ध है) (च)
(च) ४. क्रमसंख्या लुप्त (ङ) ३. लग्नाग्रया for (लग्नाग्रतो)
५. दिशा लग्नम् for (दिशंवलनं)

२ ३ १ ४
हक्षेपं मंडलं दक्षिणोत्तरं वित्रिभं विलग्नम् ॥ ५६ ॥
३ १
विषुवदुदग्बध्नीया क्रान्त्यांशसमांतरेष्वजादीनाम् ।
४ २
वृत्ततृतीयं व्यस्तं कर्कादीनां तुलादीनाम् ॥ ५७ ॥
३ ४ २ १
विषुवद्दक्षिणतो न्यन्मकरादीनां तदेव विपरीतम् ।
स्वाहोरात्रान्येषां व्यासा पृगवेव भिष्टमपि ॥ ५८ ॥
१ ३
लंका समपश्चिमगं प्राणेन कलाभूमंडलं भ्रमति ।
४ २
अपमंडलस्थराशिर्द्वादशभागाः क्षितिजलग्नाः ॥ ५९ ॥

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५६. (घ) १. तृभ विलग्ने (ग) for (वित्रिभविलग्नम्)
(ग) २. हक्षेपमंडलं for (हक्षेपं मंडलं)
३. दक्षणोत्तरं for (दक्षिणोत्तरं)
(च) २. हक्षेपमंडलं for (हक्षेपं मंडलं)
४. विलग्ने for (विलग्नं)
(ङ) २. हक्क्षेपमंडलं for (हक्षेपं मंडलं)
१. वित्रिभविलग्ने for (वित्रिभ विलग्नम्)

५७. (घ) १. क्रांत्यंश (ग) (च) (ङ) for (क्रान्त्यांश)
२. कर्क्यादीनां (ङ) for (कर्कादीनां)
(ग) ३. विषविदुद्दग्बध्नीयात् for (विषुवदुदग्बध्नीया)
४. वृत्तत्रितयं (ङ) for (वृत्ततृतीयं)
(च) २. कर्क्कादीनां for (कर्कादीनां)
(ङ) ३. विषुबदुदग्बध्नीयात् for (विषुवदुदग्बध्नीया)

५८. (घ) १. रात्राण्येषां (ग) (च) (ङ) for (रात्रान्येषां)
२. पृथगेव (ग) (ङ) for (पृगवेव)
४. व्यासाः (ङ) for (व्यासा)
(च) २. पृथगेव for (पृगवेव)
(ङ) ३. दक्षिणतोऽन्यन् for (दक्षिणतोन्यन्)

५९. (घ) १. कलां (ग) (ङ) for (कला)
२. जलग्नः for (जलग्नाः)
(ग) ३. भमंडनं for (भूमंडलं) ४. भाग for (भागाः)
(ङ) ३. भमण्डलं for (भूमण्डलं) ४. भागः for (भागाः)

यात्यूदयं मेषाद्या यतस्ततस्तदुदयानकालसमाः ।
क्रांतिवशालंकायां तदूनताधिक्यमक्षवशात् ॥ ६० ॥
क्षितिजोन्मंडलयोर्यत्स्वाहोरात्रांतरं चरदलं तत् ।
क्षितिजेग्राप्राच्यपर स्वाहोरात्रांतरांशज्या ॥ ६१ ॥
स्वाहोरात्रे क्षितिजाद् दिनगतशेषोऽस्तारवेः शंकुः ।
तस्माद् दिनगतशेषं शंकु कुमध्यांतरं दृग्ज्या ॥ ६२ ॥
दृग्मंडले नतांशज्या दृग्ज्या शंकुरून्नतांशज्या ।
अर्कोदयास्तसूत्रादिनशंकोःदक्षिणे नृतलम् ॥ ६३ ॥
क्षितिजे भूदललिप्ताः कक्षायां दृग्गतिनभो मध्यात् ।
अवनति लिप्ता याम्योत्तरारर्क ग्रहवदन्यतः ॥ ६४ ॥

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६०. (घ) यांत्युदयं (ग) यात्युदयं (च) for (यात्यूदयं) २. ग्रत for (यत)
३. 'न' यहाँ लुप्त है ४. ल्लंकायां (ग) (ङ) for (लंकायां)
(ग) ५. 'ततस्' पद लुप्त है ६. मज्ञ for (मक्ष)
(च) ४. ल्लंकायां for (लंकायां)
(ङ) १. यान्त्युदयं for (यात्यूदयं) ५. तदुदया for (ततस्तदुदया)

६१. (घ) १. रात्रेतरं for (रात्रांतरं) २. रपस्वाहो for (परस्वाहो)
(ग) ३. यत for (यत्) (च) १. रात्रेतरं for (रात्रांतरं)
(ङ) ४. क्षितिजेऽग्रा for (क्षितिजेग्रा)

६२. (ग) १. जातुदिन for (जादिन) २. श्च for (च्च)
३. ताखे for (तारवे) ४ 'दृग्ज्या' पद लुप्त । यहाँ···कुछ अंश छूटा हुआ है
(च) ५. तस्मा दिनगत for (तस्मादिनगत)
(ङ) १. जितिजादिन for (क्षितिजादिन)

६३. (घ) १. दृग्मंडले (ग) दृग्मंडल for (दृग्मंडले)
२. शकोर्दक्षिणेन (ङ) (च) for (शंकोःदक्षिणेनृ)
(ग) ३. सूत्रान् for (सूत्रादिन) ४. दक्षिणानतलम् for (दक्षिणेनृतलम्)
(च) ५. अक्कर्कादयास्त for (अर्कोदयास्त)
(ङ) ३. सूत्रादिन for (सूत्रादिन)

६४. (घ) १. स्युरर्क (ग) रविग्रहवदन्यत् ॥ ६५ ॥ for (रर्कग्रहवदन्यतः)
(ग) (वि०—इसकी क्रमसंख्या ६५ है) २. 'भू' लुप्त है) ३. दृक्नति for (दृग्गति)
(च) ३. दृग्गतिन्नभोमध्यात् for (दृग्गतिनभोमध्यात्)
(ङ) ३. दृङ्नतिर् for (दृग्गति) १. रवि for (रर्क) ४. वदन्यत्र for (वदन्यतः)

दृश्यादृश्यं दृग्गोलार्द्धं भूव्यासदलविहीनयुतम् ।
दृष्टाभूगोलोपरि यतस्ततो लंबनावनति ॥ ६५ ॥
सत्रिग्रह क्रांतिरूदग्दक्षिणयोस्त्रिज्यया हतं वलनम् ।
विक्षेपगुणमृणधनं ग्रहेन्यदृक्कर्म्म चरदलवत् ॥ ६६ ॥
कक्षामंडलतुल्यं प्राच्यपरं दक्षिणोत्तरं क्षितिजम् ।
उन्मंडलविषुवन्मंडले स्थिराणि ग्रहर्क्षाणाम् ॥ ६७ ॥
मंदोच्याना सप्तोच्चनीच वृत्तानि पंचशीघ्राणाम् ।
प्रतिमंडलानि चैवं प्रत्येकां भास्करादीनाम् ॥ ६८ ॥
दग्मंडलदग्क्षेपापमन्डलानि क्षपाकरादीनाम् ।
षट्कं विमण्डलान्यं चलवृत्यानेक पंचाशत् ॥ ६९ ॥

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६५. (घ) १. दृश्यो for (दृश्यादृश्यं)
(ग) (वि०—इसकी क्रमसंख्या ६४ है)
२. गोलार्द्धं for (दृग्गोलार्द्धं) ३. विहीनम् for (विहीन)
४. दृष्टो for (दृष्टा) (वि०—दूसरी पंक्ति 'भूगोल' से आरम्भ है)
५. 'दृष्टा' लुप्त है ६. नती ॥६४॥ for (नति ॥६५॥)
(च) २. दृग्गोलार्द्धं for (दृग्गोलार्द्धं)
(ङ) (वि०—यह श्लोक उपलब्ध नहीं)
६६. (घ) १. गृह (च) (ङ) for (ग्रह) २. हृतं (च) (ङ) for (हतं)
३. गृहे (च) for (ग्रहे) ४. न्यदृक्कर्म (ग) (ङ) for (न्यदृक्कर्म)
(ग) ५. रुदक् for (रूदग्) ६. गुणधनं for (गुणमृणधनं)
६७. (ग) १. कक्ष्या for (कक्षा) २. प्राक्चरं for (प्राच्यपरं)
३. क्षिजम् for (क्षितिजं)
६८. (घ) १. मंदोच्चानां (ग) (च) (ङ) for (मंदोच्याना)
(ग) २. प्रत्येकं (च) for (प्रत्येकां)
६९. (घ) १. दृक्षेपाप (ग) विक्षेपाप) (ङ) for (दग्क्षेपाप)
२. विमंडलानां (ग) (ङ) for (विमंडलान्यं)
३. वृत्तान्येक (ग) वलवृत्तान्येक for (चलवृत्यानेक)
४. पंचाशतः for (पंचाशत्)
(ग) ५. दृग्मंडल (ङ) for (दग्मंडल)
६. क्षिपा for (क्षपा)
(च) १. विक्षेपाप for (दग्क्षेपाप) २. विमंडलानां for (विमंडलान्यं)

यत्स्पष्टीकरणाद्यं गोलादुत्क्षेपतत्कृतं[1] सर्वं[3] गोलाध्यायः[2]।
सप्तत्यार्याणामेक विंशोयम्[4] ॥ ७० ॥
मध्याद्यमिह यदुक्तं तत्प्रत्यक्षमिव दर्शयति यस्मात्।
तस्मादाचार्यत्वं गोलविदो[1] भवति नान्यस्य ॥ ७१ ॥
आचार्येण[1] ज्ञातः श्रीषेणार्यभटविष्णुचंद्राद्यैः[2]।
गोले यस्मात्तस्माद् ब्राह्मे[4] गोलः कृतः स्पष्टः[4] ॥ ७२ ॥
गणितज्ञो गोलज्ञो गोलज्ञो ग्रहगतिं विजानाति।
यो गणितगोलबाह्यो जानाति ग्रहगतिं[2] सः[3] कथम् ॥ ७३ ॥

इति[1] ब्रह्मगुप्ते गोलाध्यायः (२१ अध्यायः समाप्तः)

एक विंशतितमः समाप्तः

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७०. (घ) १. दुत्प्रेक्ष (ग) दुत्प्रेक्ष्य (ङ) for (दुत्क्षेप)
२. (वि०—यह दूसरी पंक्ति का आरम्भिक शब्द है) (ग) (ङ)
(ग) ३. सर्वां for (सर्वं) ४. समाप्ति सूचक ॥ छः॥ ॥छः॥ ॥छः॥
(च) १. दुत्प्रेक्षतत्कृतं for (दुत्क्षेपतत्कृतं)
(ङ) +इति श्री ब्राह्मस्फुट सिद्धान्ते गोलाध्यायो नामैकविंशतितमो ऽध्यायः+ (वि०—वस्तुतः गोलाऽध्याय यहां ७० वें श्लोक पर समाप्त हो गया है। परन्तु उपर्युक्त प्रति में यन्त्राध्याय के पहले तीन श्लोक मिलाकर ७३ पर समाप्त किया है)

७१. (घ) १. गोलो भवति for (गोलविदो भवति) (वि०—इसकी क्रमसंख्या १ है)
(ङ) वि०—यह 'यन्त्राध्याय' का प्रथम श्लोक है)

७२. (घ) १. आचार्यैर्न (ङ) for (आचार्येण) (ग) २. ज्ञाता for (ज्ञातः)
३. 'विष्णु' पद लुप्त है। ४. गोलो (ङ) (च) for (गोले)
५. ब्राह्मो (ङ) for (ब्राह्मे) (वि०—इसकी क्रमसंख्या २ है)
१. आचार्यैर्न for (आचार्येण) (ङ) (वि०—यह यन्त्राध्याय का द्वितीय श्लोक है)

७३. (घ) १. इति भट्ट जिष्णुसुतब्रह्मगुप्त विरचिते ब्रह्मस्फुट सिद्धांतो गोलाध्यायः एक विंशतितमः। for (इति ब्रह्मगुप्ते गोलाध्यायः)
(ग) २. ग्रहति for (ग्रहगति) ३. स (ङ) (च) for (सः)
१. ये पद लुप्त है (वि० – इसकी क्रमसंख्या ३ है)
(च) १. इति भट्ट जिष्णुसुत ब्रह्मगुप्त विरचिते ब्राह्मे स्फुटसिद्धांतो गोलाध्यायः एक विंशतितमः for (इति ब्रह्मगुप्ते गोलाध्यायः २१ अध्यायः समाप्तः)
(ङ) (वि०—यह यन्त्राध्यायः का तीसरा श्लोक है)

अथ यन्त्राध्यायो नाम

द्वाविंशः

कालस्य[२] परिछेदः[४] कर्तुं यंत्रं[१] विना[३] यतो शक्यः ।
संक्षिप्तं स्पष्टार्थं यंत्राध्यायं ततो वक्ष्ये ॥ १ ॥
सप्तदशकालयंत्राण्यतो धनुस्तुर्यगोलकश्चक्रम्[४] ।
यष्टिशंकुर्घटिका[५] कपालकं[१] कर्त्तरिकीटम्[२][३] ॥ २ ॥
सलिल[१]भ्रमोवलंब[२]ः कर्णछाया[३] दिनार्द्धमर्कोक्षः[६] ।
नवकाल[४] ज्ञानार्थं[५] तेषां संसाधानान्यष्टौ ॥ ३ ॥

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१. (घ) १. विना (ग) (च) (ङ) for (विना)
(ग) २. गोलस्य (ङ) for (कालस्य)
३. यतोऽशक्यः (ङ) for (यतोशक्यः)
(वि०—इसकी श्लोक संख्या ४ है)
(ङ) (वि०—श्लोक संख्या ४ है इसके पूर्ववर्ती तीन श्लोकों को गोलाध्याय के अंत में रख दिया गया है)
४. परिछेदः for (परिछेद)

२. (घ) १. वंटिका for (घंटिका) २. कर्त्तरी (ग) (च) (ङ) for (कर्त्तरि)
३. फीटं (ग) पीठम् for (कीटम्)
(ग) ४. गोलकं चक्रम् for (गोलतश्चक्रम्)
५. यष्टिः (च) (ङ) for (यष्टि)
(वि०—इसकी श्लोक संख्या ५ है)
(च) ३. फीटा for (कीटम्)
(ङ) ३. पीठम् for (कीटम्)

३. (ग) १. सलिलं (ङ) for (सलिल) २. वमो for (भ्रमो)
३. कर्णश्छाया (ङ) for (कर्णछाया) ४. नय for (नव)
५. यहां अनुस्वारलुप्त है ।
(वि०—इसकी श्लोक संख्या ६ है)
(च) २. ऽवलंबः for (वलंबः) ६. मर्क्षोक्षः for (मर्कोक्षः)
(ङ) २. भ्रमोऽवलम्बः for (भ्रमोवलंबः)
६. मर्कोऽक्षः for (मर्कोक्षः) ४. नत for (नव)

सलिले न समं साध्यं भ्रामणवृत्तमवलंब केनोर्द्धम् ।
तिर्यक कर्णोनान्यैः कथितैश्च नव प्रसाध्यति ॥ ४ ॥
धार्यं धनुस्तथाग्रछायासाम्यं यथोन्नता भागाः ।
दिनगतशेषा घटिकाश्चलंबमुक्ता धनुर्मध्यात् ॥ ५ ॥
धार्यं समं तथा वा ज्या छाया पृथगा यथा भवति अग्रादम् ।
शक घटिका ज्यामध्यछायया मुक्ता ॥ ६ ॥
घटिका स्वशंकुभागैः पृष्टगतैं लंब भूसमं समज्यार्द्धात् ।
सशिति शतांशकं चक्रास्यार्द्धं धनुर्यंत्रम् ॥ ७ ॥

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४. (घ) १. भ्रमेण (ग) (ङ) for (भ्रामण)
२. तिर्यङ्कर्णोनान्यैः (ग) (ङ) तिर्यक्कर्णोनान्यैः for (तिर्यककर्णोनान्यैः)
३. प्रसाध्यानि (ग) प्रवक्षामि for (प्रसाध्यति)
(ग) ४. केनोर्ध्वम् (ङ) for (केनोर्द्धम्) ५. तिर्यक् (ङ) for (तिर्यक)
(वि०—इसकी श्लोक संख्या ७ है)
(च) ५. तिर्यकर्णोनान्यैः for (तिर्यककर्णोनान्यैः)
३. प्रसाध्यानि for (प्रसाध्यति) (ङ) ३. प्रवक्ष्यामि for (प्रसाध्यति)

५. (ग) १. तथान्य for (तथाग्र) (वि०—इसकी श्लोक संख्या ८ है)
२. भागम् for (भागाः) ३. धनुर्मुक्तान् for (धनुर्मध्यात्)
(ङ) १. तथाऽन्यत् for (तथाग्र) ४. स्वलम्बभुक्ता for (श्चलंबमुक्ता)

६. (घ) १. पृष्टगा (ग) for (पृथगा) २. (वि०—दूसरी पंक्ति का आरम्भिक पद) (ग)
३. छायाया (ग) छाया यथा for (छायया) (ग) ४. धार्यं (ङ) for (धार्यं)
२. अग्रादश for (अग्रादम्) ५. भुक्ताः ॥६॥ (ङ) for (मुक्ता)
(वि०—इसकी श्लोक संख्या ९ है) (च) ३. छायाया for (छायया)
(ङ) १. मध्यगा for (पृथगा) २. अग्रादिष्टा for (अग्रादम्) ६. 'शक' लुप्त

७. (घ) १. सशीति (ग) साशीति (ङ) for (सशिति)
२. शतांशाकं (ग) शाकं (ङ) for (शतांशकं)
३. चक्रस्यार्द्ध (ग) for (चक्रास्यार्द्धं) (ग) ४. घटिकाः (च) for (घटिका)
५. लंब (ङ) for (लंब) ६. 'समं' पद लुप्त है ।
(वि० इसकी श्लोकसंख्या १० है)
(च) १. सशीति for (सशिति) २. शतांशाकं for (शतांशकं)
३. चक्रस्यार्द्ध for (चक्रास्यार्द्धं) (ङ) ७. पृथग्गतैं for (पृष्ठगतैं)
६. भू for (भूसमं)

मध्यादिनोन्नतांशै[3] र्दिनार्द्धं नाडयो[4] वदंति तुल्या[5] ये।
ते सूर्यास्त[6] च्छादष्ट[1] च्छाया[2] समान यतः ।। ८ ।।
जीवा[1] स्वाहोरात्रं परिकल्पाग्रा[4] नतोन्नतत्रिज्या[1] ।
अनुपातात्[5] तत्कार्या[2] सूर्यगोलके[3] चक्रे[4] चैवम् ।। ९ ।।
दिन घटिकांकितपृष्टे[3] व्यस्त[4] न तज्याग्रोन्नतज्या[5] ।
चदिग्मध्ये[2] च शलाकातच्छायाग्रं[6] न तानाडयः[7] ।। १० ।।

८. (घ) १. दृष्ट (ग) छायादृष्ट for (छादष्ट) २. समानयत for (समानयतः)
(ग) ३. मध्यदिनोन्नतांशै for (मध्यादिनोन्नतांशै)
४. नाडी for (नाडयो) ५. तुल्यायै for (तुल्याये)
६. सूर्यास्त for (सूर्यास्त) (वि०—इसकी श्लोक संख्या ११ है)
(च) १. छादृष्ट for (छादष्ट) २. युतः for (यतः)
(ङ) ३. मध्यदिवसोन्नतांशै for (मध्यादिनोन्नतांशै)
४. नाडीर्वदन्ति for (नाडयोवदन्ति) ६. ते सूर्यास् for (ते सूर्यास्)
१. तच्छाया इष्ट for (तच्छादष्ट)

९. (घ) १. श्नतो (ग) नतेन्नत for (नतोन्नत) २. स्तुर्या (ग) स्तुर्य for (सूर्य)
३. चक्रके (ङ) for (चक्रे)
(ग) ४. प्रकल्प्यग्रो for (परिकल्पाग्रा)
५. अनुपातातात्कार्या for (अनुपातात्तत्कार्या)
(वि०— इसकी श्लोक संख्या १२ है)
(च) १. न्नतोन्नतत्रिद्या for (नतोन्नतत्रिज्या) २. स्तुर्यगोलके for (सूर्यगोलके)
(ङ) ६. जीवां स्वाहोरात्रे for (जीवास्वाहोरात्रं)
४. (परिकल्याग्रा) for (परिकल्पाग्रा)
१. श्नतोन्नतविज्याः for (नतोन्नतत्रिज्या)
२. कार्यास्तुर्यगोलके for (कार्यासूर्यगोलके)

१०. (घ) १. (च) (वि०—पहली पंक्ति का अंतिम पद) (ङ)
२. 'दिग्मध्ये' से आरम्भ (ग) दिनमध्ये for (दिग्मध्ये)
(ग) ३. पृष्टि for (पृष्टे) ४. व्यस्तनज्याग्र for (व्यस्तनतज्याग्रो)
५. उन्नतज्या च for (ग्रोन्नतज्या) ६. शिलाका for (शलाका)
७. यान्नता नाडयः for (ग्रंनतानाडयः)
(वि०—इसकी श्लोक संख्या १३ है)
(ङ) ३. यष्टे for (पृष्टे)
५. ग्रमुन्नतज्यां for (ग्रोन्नतज्या) ७. तच्छायाग्राग्नता for (तच्छायाग्रंनता)

पृष्टो नेष्टो वेध्यो ज्यामध्यसंस्थाया ।
द्रष्टृचा द्रष्टांतरं न तज्या धनुषि छायोन्नत ज्याया ॥ ११ ॥
ज्यार्द्धं द्रष्टि द्रग्ज्यान्नत जीवा शंकुमुन्नतज्या च ।
धनुषि प्रकल्पयोज्यां यष्टयुक्तं नाडीकाद्यंता ॥ १२ ॥
अवलंबनं शलाकां ज्यार्द्धं यष्टिं प्रकल्पवा धनुषि ।
भुम्यूछ्रायं बहुशो यष्टयुक्तै रानयेत्कर्णैः ॥ १३ ॥

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११. (घ) १. धनुषपृष्टेनेष्टौ वेध्यो (ग) धनुष प्रिष्टे नेष्टौवेध्यो for (पृष्टोनेष्टौ वेध्यो)
२. संस्थया (ग) (च) (ङ) for (संस्थाया)
३. दृष्टृचा (ग) दृष्टृचा (ङ) for (द्रष्टृचा) (वि—पहली पंक्ति का अंतिम पद)
४. इष्टांतरं (ग) (ङ) for (द्रष्टांतरं) ५. ज्यायाः (ङ) for (ज्याया)

(ग) (वि०—इसकी क्रमसंख्या १४ है)

(च) १. धनुषपृष्टे for (पृष्टो) ३. दृष्टृचा for (द्रष्टृचा)
६. तद्या for (तज्या) ५. ज्यायाः for (ज्याया)

(ङ) १. धनुषः पृष्ठे द्रष्ट्रा for (पृष्टो नेष्टो) ७. वेध्या for (वेध्यो)
३. ४. इष्टांतरं for (द्रष्टृचा द्रष्टांतरं) ३. दृष्टृचा for (द्रष्टृचा)

१२. (घ) १. दृष्टिं (ग) यष्टिं for (द्रष्टिं) २. दृग्ज्या (ग) दृग्ज्यां (ङ) for (द्रग्ज्या)
३. नाडिकाद्यन्ता (ग) नाडिकाद्यं च for (नाडिकद्यंता)

(ग) ४. ज्यार्द्धं (ङ) for (ज्यार्द्धं) ५. नतजीवां for (नतजीवा)
६. ज्यां च for (ज्याच) ७. प्रकल्प्य for (प्रकल्प)
८. योज्यं for (योज्यां) (वि०—इसकी श्लोकसंख्या १५ है)

(च) १. दृष्टिं for (द्रष्टिं) २. दृग्ज्या for (द्रग्ज्या)
५. न्नतजीवाँ for (न्नतजीवा) ६. ज्याव for (ज्याच)

(ङ) १. दृष्टेर् for (द्रष्टिं) ६. यद्युक्तं for (यष्टयुक्तं)

१३. (घ) १. ज्यार्द्धं for (ज्यार्द्धं)
२. भूम्युछ्रायं (ग) भूम्युछ्रायां for (भूम्युच्छ्रायं)

(ग) २. अवलंबन for (अवलंबनं) ४. प्रकल्प्य (ङ) for (प्रकल्प)
५. द्वाहुःशो for (बहुशो) ६. येष्टयुक्तै for (यष्टयुक्तै)
(वि० —इसकी श्लोक संख्या १६ है)

(च) २. भूम्युछ्रायं for (भूम्यूछ्रायं) ७. करणैः (ङ) for (कर्णैः)

(ङ) २. भूम्युच्छ्रायां for (भूम्यूच्छ्रायं) ५. ल्लम्बो for (बहुशो)

यष्टि व्यासार्द्धं[४] तु विवृत्तं भगणांशांकितं[१] कृत्वा[७] ।
यष्टिकीलप्रोते[२] मूले[५] पृथग्भग्रयो[६] विद्धे[३] ॥ २१ ॥
ताभ्यां[१] सूर्यशशांकौ वेध्यावर्गस्थितेन[४] सूत्रेण ।
सूत्रज्यायांतरां[२] शायेतेऽर्कं[५] विभाजिता स्थितयः[३] ॥ २२ ॥
सूत्रार्द्धगुणा त्रिज्या यष्टिहृता[१] फलधनुर्द्विगुणितं[२] वा ।
रविचंद्रांतरमिष्टव्यासार्द्धोल्लितवृत्तस्य[३] ॥ २३ ॥
मध्यधृतायायष्टे लंबकशंकुप्रवेशनिर्गमने ।
क्रांतिवशात्प्राच्यपरे मत्स्याद्याम्योत्तरे[१] साध्ये ॥ २४ ॥

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२१. (घ) १. भगणांशकांकितं (ग) (च) for (भगगणांशांकितं)
२. यष्टी (ग) (ङ) (च) for (यष्टि)
३. विद्धे (ग) विधिः (विधेः) for (विद्धे)
(ग) ४. व्यासार्द्धाद्भ्रुवि (ङ) for (व्यासार्द्धांतुवि)
५. मूल for (मूले) ६. प्रथग्रयो (च) for (पृथग्भग्रयो)
(वि०—इसकी श्लोकसंख्या २४ है)
(च) ४. भुविवृत्तं for (तुविवृत्तं)
७. क्रत्वा for (कृत्वा) ३. विद्धे for (विद्धे)
(ङ) १. भगणांशकं for (भगणांशांकितं) ६. पृथगग्रयो for (पृथग्भग्रयो)
३. बंद्धे for (विद्धे)

२२. (घ) १. वाभ्यां for (ताभ्यां)
२. ज्ययांतरांशा (ग) (च) (ङ) for (सूत्रज्यायांतरांशा)
३. स्तिथयः (ग) (च) (ङ) for (स्थितयः)
(ग) ४. वग्र (च) (ङ) for (वर्ग) ५. येनेऽर्कं for (येतेऽर्कं)
(वि०—इसकी श्लोसंख्या २५ है)
(च) ५. येतेऽङ्कं for (येतेऽर्कं)

२३. (घ) १. हृता (ग) (ङ) (च) for (हता)
२. द्विगुणितं (ग) (ङ) (च) for (द्विगुणितं)
३. व्यासार्द्धोल्लिखित (ग) (ङ) (च) for (व्यासार्द्धोल्लित)
(ग) ४. रविचंद्रांतर (च) for (रविचंद्रातर) (वि०—इसकी श्लोकसंख्या २६ है)

२४. (घ) १. मत्स्याद्यामोत्तरे (ङ) (ग) मतस्याद्याग्योत्तरे for (मत्साद्याम्योत्तरे)
(ग) २. (वि०—इसकी श्लोकसंख्या २७ है)
(च) १. मत्स्याद्यामोत्तरे for (मत्साद्यामोत्तरे)

शंकुतलानांतरयुतिरन्यैकदिशोभुजो भुजस्य कृतिम् ।
द्दग्ज्या कर्णकृतेः प्रोह्यपदपूर्वापरा कोटिः ॥ २५ ॥
उदयास्तसूत्रशंक्वंतरं हृतं शंकुनार्कसंगुणितम् ।
विषुवच्छायैयं वा दिनोदयास्तमयसूत्रेण ॥ २६ ॥
प्राच्यपरा शंकुतलांतरयुति समान्यदिशो ।
द्वादशगुणिता विषुवच्छाया शंक्वंतरविभक्ता ॥ २७ ॥

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२५. (घ) १. शकुंतलाग्रांतर (ङ) (ग) शंकुतलग्नांतर for (शंकुतलानांतर)
२. दिशा (ग) दिशोर्भुजो (ङ) for (दिशोभुजो)
३. कृतिः (ग) कृतेः (ङ) for (क्रतेः) ४. पदं (ग) प्रोश्चपदं for (पद)

(ग) ५. कोटि for (कोटिः)
(वि०—इसकी श्लोकसंख्या २८ है)

(च) १. शंकुतलग्नांतर for (शंकुतलनांतर) २. दिशौ for (दिशो)
३. कृतेः for (क्रतेः) ४. प्रोज्यपद for (प्रोह्यपद)

(ङ) ४. पदं for (पद)

२६. (घ) १. शंक्वंतरहृतं (ग) हृतं for (हूतं)
२. नार्कं १२ संगुणितम् (ग) शंकुनार्कं संगुणितम् for (शंकुनार्कसंगुणितम्)
३. छायायैवं (ग) विषुवच्छायेव वा for (विषुवच्छायैयं)

(ग) (वि०—इसकी श्लोक संख्या २९ है)
४. विनोदयास्तमयसूत्रेण (ङ) for (दिनोदयास्तमयसूत्रेण)

(च) १. हृतं for (हूतं)
२. शंकुनार्क्कं १२ for (शंकुनार्कं)
३. विषुवच्छायैवं for (विषुवच्छायैयंवा)

(ङ) ३. विषुवच्छायैवं for (विषुवच्छायैयं)

२७. (घ) १. युतिः (च) (ङ) for युति)
२. दिशोः (च) (ङ) for (दिशो)
३. विभक्ताः for (विभक्ता)

(ग) (वि०—इसकी श्लोक संख्या ३० है)
४. द्वयांतरयुति (ङ) for (युति)
५. द्वादभक्ता for (द्वादश) ६. 'गुणिता' पद लुप्त है।

शंकुप्राच्यपरांतर शंक्वग्रं क्ष्मुदयांतरं याम्ये ।
लब्धगुणं यष्टिगुणं क्रांति ज्यातो रविः प्राग्वत् ॥ २८ ॥
अपसृतिरन्यशलाका गुणाशलाक्ष्मांतरेण भक्ता भूः ।
भू स्वशलाका गुणिता गुणिताद्रष्टि विभक्ता ॥ २९ ॥

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२८. (घ) १. हृतं (ग) for (गुणं)
(ग) २. प्राच्या for (प्राच्य) (वि०—इसकी श्लोक संख्या ३१ है)
३. मुदगंतरं यामे for (मुदयांतरं याम्ये)
(च) १. हृतं (ङ) for (गुणं)
(ङ) ३. मुदगंतरं for (मुदयांतरं)
४. लम्बगुणं for (लब्धगुणं)

२९. (घ) १. शलाक्ष्मंतरेण (ग) शलाकांतरेण (ङ) for (शलाक्ष्मांतरेण)
२. 'गुणिता' लुप्त
३. द्दष्टि (ग) द्रिष्टि for (द्रष्टि)
(वि०—इसकी क्रमसख्या ३० है)
(ङ) ६. भूः for (भू)
३. ४. यष्टि विभक्ता गृहाद्यौच्यम् for (गुणिता द्रष्टि विभक्ता)
(ग) (वि०—इसकी श्लोकसंख्या ३२ है)
४. 'गृहाद्यौच्यम्' पंक्ति का अंतिम शब्द । ३. द्दष्टि for (द्रष्टि)
५. (वि०—इसकी क्रमसंख्या ३० अंकित है)
४. + 'ग्रहाद्यौ च्यं' यह पद पृष्ठ के नीचे अन्य हाथ से लिखा हुआ है ।

द्दष्ट्या गुणितापसृति द्दष्टि विशेषभाजिता भूमिः ।
भूमिस्व द्रिष्टिभक्ता शलाकाया संगुणोत्छ्रायम् ॥ ३३ ॥

(च) यह अतिरिक्त श्लोक यहां पृष्ठ के निम्नोपांत पर अन्यहस्तसे लिखा हुआ है ।
१. द्दंष्टि for (द्दष्टि) २. भूः for (भूमिः)
३. द्रष्टि for (द्रिष्टि)
४. शलाकया (ङ) for (शलाकाया)
५. संगुणोछ्राय (ङ) for (संगुणोत्छ्रायम्)
(वि०—इसकी भी क्रमसंख्या ३० अंकित है)
(ङ) १. द्दंष्टि विशेषेण for (द्दष्टिविशेष) ७. भूमिः for (भूमि)
३. द्दष्टि for (द्रिष्टि)

लंबनिपातांतरकं लंबौच्यांतरविभक्तमधिकगुणम् ।
भूर्लंबांतरगुणिता[१] लबनिपातांतरविभक्ता[२][३] [४] ॥ ३० ॥
लब्धोनाद्रक् लम्बोदग्लम्बा[१][२] दग्रलंबके हीने ।
अधिकेऽधिको ग्रहोच्यं[३] तलाग्रवेध[४] द्वया[५] षष्टया[६] ॥ ३१ ॥
दृष्टिर्द्रक्[१] लंबगुणा विभाजिताद्यः[२] शलाकया भूमिः ।
सकलशलाका गुणिता[३] भूमिदृष्टया[४] हृतोछ्रायः[५] ॥ ३२ ॥[६]

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३०. (घ) १. भूर्ल्लंबांतर for (भूर्लंबांतर)
२. लंबनिपातांतर for (लंबनिपातांतरकं)
३. विभक्ताः for (विभक्ता)
(वि०—इसकी क्रमसंख्या ३१ है)
(ग) (वि०—इसकी श्लोकसंख्या ३४ है)
(च) १. भूर्ल्लंबातर for (भूर्ल्लंबातर)
४. (वि०—यहां क्रमसंख्या ३१ अंकित है)

३१. (घ) १. लब्धोनाद्दक् (ग) सशेनो द्रिग् for (लब्धोनाद्रक्)
२. लंबोद्दग्लंबा (ग) (च) (ङ) for (लंबोदग्लंबा)
३. गृहौच्यं (ग) (च) (ङ) for (ग्रहोच्यं)
४. तलग्रंवेध (ग) तलाग्रवे बध्यया for (तलाग्रवेध)
५. द्वया यष्टया (ग) द्रिष्टया ॥ ३५ ॥ for (द्वयाषष्टया)
(वि०—इसकी क्रमसंख्या ३२ है)
(ग) (वि०—इसकी श्लोक संख्या ३५ है)
(च) १. नाद्दक् for (नाद्रक्) ५. द्वया यष्टया for (द्वया षष्टया)
६. (वि०— क्रमसंख्या ३२ अंकित है)
(ङ) १. लब्धोनोद्दग् for (लब्धोनाद्रक्) ४. तलाग्रके for (तलाग्र)
५. विद्धया दृष्टया for (वेधद्वयाषष्टया)

३२. (घ) (वि०—इसकी क्रमसंख्या ३३ है)
१. यष्टियैलग्नंबगुणा (ग) दृष्टिर्द्दग्लंबगुणा for (दृष्टिर्द्रक्लंबगुणो)
२. विभाजिताधः (ग) (च) (ङ) for (विभाजिताद्यः)
३. गुणिताः (च) for (गुणिता) ४, भूमिर्द्दष्टया (ङ) for (भूमिदृष्टया)
५. हृतोछ्रायः (ग) (च) (ङ) for (हतोछ्रायः)
(ग) (वि०—इसकी श्लोकसंख्या ३६ है)
(च) १. द्रष्टिद्रक् for (दृष्टिर्द्रक्) ६. (वि०—यहाँ क्रमसंख्या ३३ अंकित है)
(ङ) १. दृष्टिर्द्दग्लम्ब for (दृष्टिर्द्रक्लम्ब)

मित्वाग्रहैकदेशे विद्वेष्ट शलाकया न्ययासर्वम् ।
प्रथमशलकाभक्तं मीत द्वितीया गुणितमौच्यम् ।। ३३ ।।
षष्टघाहूता शलाका त्रिज्याघात धनु गृहांतरकं ।
यैरूक्तं मूर्खास्ते यतो न द्रष्टघंतरं द्रग्ज्या ।। ३४ ।।
मूलेद्वंगुलविपुलः शुच्यग्रो द्वादशांगुलोच्छ्रायः ।
शंकुतलाग्रविद्धोग्रवेध लम्बाद्वार्यः ।। ३५ ।।

३३. (घ) १. गृहैक (ग) (च) (ङ) for (ग्रहैक) (वि०—इसकी क्रमसंख्या ३४ है)
२. शलाका (ग) शिलाका for (शलका)
३. मिति (ग) मितं (ङ) for (मीत) ४. द्वितीयया for (द्वितीया)
५. गुणिताम्यौच्यम् for (गुणितमौच्यम्)
(ग) ६. देशं (च) for (देशे) ७. विधेष्ट for (विद्वेष्ट)
८. गृहम् for (न्यया) (वि०—इसकी श्लोकसंख्या ३७ है)
(च) ३. मित for (मीत) ४. द्वितिया for (द्वितीया)
५. गुणितम्यौच्यं for (गुणितमौच्यं) (ङ) ७. विद्धेष्ट for (विद्वेष्ट्)
३४. (घ) १. यष्टघाहूता (ग) त्रिज्याहूत for (षष्टघाहूता)
२. घाता (ग) for (घात) ३. गृंहांतरकम् (च) for (गृहांतरकम्)
४. यैरुक्तं (च) for (यैरूक्तं) ५. दृष्टघंतरं (ग) (च) for (द्रष्टघन्तरं)
६. द्रग्ज्या (ग) द्रग्ज्याजुर्द्धा यः ।। ३८ ।। for (द्रग्ज्या)
(च) १. यष्टघाहूता for (षष्टघाहूता) ६. द्रग्ज्या for (द्रग्ज्या)
७. (वि०—क्रमसंख्या ३५ अंकित है) (घ)
(ङ) १. यष्टघाहूताच्छलाका for (षष्टघाहूताशलाका)
२. घाताद्धनुर्गृहान्तरकम् for (घातधनुगृहांतरकम्)
५. दृष्टांतरम् for (द्रष्टघंतरं) ७. द्रग्ज्या for (द्रग्ज्या)
३५. (घ) १. द्विगुल (ग) द्वघंगुल (ङ) for (द्वंगुल) (वि०—इसकी क्रमसंख्या ३६ है)
२. शूच्यग्रोः (ग) शूच्यग्रो for (शुच्यग्रो)
३. शंकुस्तलाग्र (ग) for (शंकुतलाग्र) ४. द्दजुर्धार्यः for (द्वार्यः)
(ग) ५. विपुल for (विपुलः) ६. विधोग्रवेध for (विद्धोग्रवेध)
७. लम्बाद्दजु र्द्धार्यः ।। ३९ ।। for (लम्बाद्वार्यः)
(वि०—इसकी श्लोक संख्या ३९ है)
(च) २. शूच्यग्रोः for (शुच्यग्रो) ८. छायः for (छायः)
७. लम्बादजुर्द्धार्यः (लम्बाद्वार्यः) (वि०—क्रमसंख्या ३६ अंकित है)
(ङ) २. सूच्यग्रो for (शुच्यग्रो) ६. विद्धा for (विद्धो) ४. भूद्दुद्वयोः for (द्वार्यः)

छायांदृग्ज्या यष्टि छयाकर्णे मवलम्बकम् शंकुं ।
परिकल्प्यां शंकु यंत्रे योज्यं घटिकादिषुष्टचुक्तम् ।। ३६ ।।
घटिकाकलशार्द्धकृतिताम्रं पात्रं तले गुरुछिद्रम् ।
मध्येतज्जलमज्जन षष्टया द्युनिशं यथा भवति ।। ३७ ।।
मध्याध्व्यस्तनतांशैः कपालकं दिवकत्थ सूत्रात् मग्रात् ।
व्यस्तोन्नतांशा विवरे सूत्रं क्या पाततो नाडयः ।। ३८ ।।

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३६. (घ) (वि०—इसकी क्रम संख्या ३७ है)
१. छायाकर्णे (ग) (च) (ङ) for (छयाकर्णे)
२. परिकल्प्या (ग) परिकल्प्य (ङ) for (परिकल्प्यां)
३. घटिकादि पुष्टचुक्तम् (ग) घटिकादि यद्युक्तम् for (घटिकादि पुष्टचुक्तम्)
(ग) ४. दृग्ज्यां (ङ) for (दृग्ज्या) (वि०—इसकी श्लोकसंख्या ४० है)
(च) २. परिकल्प्य शंकु for (परिकल्प्यांशंकु)
३. घटिकादिसंयुक्तम् for (घटिकादिषु ष्टचुक्तम्)
(वि०—क्रमसंख्या ३७ अंकित है) (ङ) ५. दृष्टि for (यष्टि)
३७. (घ) (वि०—इसकी क्रमसंख्या ३८ है)
१. कलशार्द्धाकृति (ग) कलाशतार्द्धा for (कलशार्द्धं)
(ग) २. कृतिताम्रं for (कृतिताम्रं) ३. पृथु for (गुरू)
४. मज्ञान for (मज्जन) (वि०—इसकी श्लोकसंख्या ४१ है)
(च) २. कृतिताम्रं (ङ) for (कृतिताम्रं)
(वि०—इसकी क्रमसंख्या ३८ अंकित है) (ङ) कलसार्धा for (कलशार्द्धं)
३. ऽपृथुच्छिद्रम् for (गुरुछिद्रम्)
३८. (घ) (वि०—इसकी क्रमसंख्या ३९ है)
१. द्वयस्त (ग) मध्यद्वयस्तनतांशै for (मध्व्यस्तनतांशैः)
२. दिवक्स्थ (ग) दिवस्थ (ङ) for (दिवकत्थ)
३. सूत्रमग्राग्रात् (ग) मूत्रमग्रात् for (सूत्रात्मग्रात्)
(ग) (वि०—इसकी श्लोकसंख्या ४२ है)
४. व्यस्तोन्नत्यंश for (व्यस्तोन्नतांशा) (च) १. द्वयस्त for (व्व्यस्त)
२. दिक्स्थ for (दिवकत्थ) ३. सूत्रमग्राग्रात् for (सूत्रात्मग्रात्)
४. व्यस्तोन्नतांश for (व्यस्तोन्नतांशा) ५. विववरे for (विवरे)
६. (वि०—क्रमसंख्या ३९ for (३८)
(ङ) १. मध्याद्यस्वनतांशैः for (मध्याध्व्यस्तनतांशैः)
३. सूत्रमध्याग्रात् for (सूत्रात्मग्रात्) ४. व्यस्तोन्नतांश for (व्यस्तोन्नतांशा)

अथवा कपालके नाडिकादिसर्वं[२] धनुष्फक्तं[३] कर्त्तरि यंत्रम्[४] ।
स्थुलं[१] कृतं यतोन्यैर्वदामि ततः ॥ ३९ ॥[५]

दिक्स्थितफलकद्वियुतिस्तले तदग्रस्य सूत्रयोर्मध्ये ।
कीलस्तछायाग्रात्[१] कर्त्तयानाडिका[२] स्थूलाः ॥ ४० ॥

दष्टौच्यं[१] समपठिं[२] यष्टि व्यासार्द्धं मंकितं[५] परिधौ ।
दिग्भगणांशै[५] मूर्द्धन्यग्रा[४] घटिकादि यष्टयुक्तम्[३] ॥ ४१ ॥

नलकोमूले विद्धस्तछ्रूति[१२] घटिकोद्धृत[४] समुछ्रायः[३] ।
लब्धांगुलैस्तुतैर्नाडिका क्रियायंत्रसिद्धिरतः ॥ ४२ ॥

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३९. (घ) १. स्थूलं (ग) (ङ) for (स्थुलं) (वि०—इसकी क्रमसंख्या ४० है)
(ग) २. नाडिकादिकं सर्वम् for (नाडिकादिसर्वं)
३. था धनुष्युक्तं for (धनुष्फक्तं) ४. पत्रम् for (यंत्रम्)
(वि०—इसकी श्लोकसंख्या ४३ है) (च) १. स्थूलं for (स्थुलं)
५. (वि० – क्रमसंख्या अंकित है)
(ङ) ३. यथा धनुष्युक्तं for (धनुष्फक्त)
४. 'कर्त्तरियंत्रम्' दूसरी पंक्ति का आरम्भिकपद ।

४०. (घ) १. स्वछायाग्रा for (स्तछायाग्रात्) (वि०—इसकी क्रमसंख्या ४१ है)
२. कर्त्तर्या (ग) for (कर्त्तया) ३. नाडिका: for (नाडिका)
(ग) (वि०—इसकी श्लोक संख्या ४१ है)
(च) २. कर्त्तर्यानाडिका: for (कर्त्तयानाडिका) (वि०—इसकी क्रमसंख्या ४१ है)
(वि०—इसकी क्रमसंख्या ४१ है)
(ङ) २. कर्त्तयां for (कर्त्तया)

४१. (घ) (वि०—इसकी संख्याक्रम ४२ है)
१. दृष्टौच्यसमं (ग) दृष्टौच्यं (ङ) for (दष्टौच्यं)
२. पीठं (ग) (ङ) for (पठिं) (ग) ३. ष्युक्तम् for (यष्टयुक्तम्)
(च) १. दृष्टौच्यं समं for (दष्टौच्यं सम) २. पीठं for (पठिं)
४. मूर्द्धन्यग्रा for (मूर्द्धन्यग्रा) (वि०—इसकी क्रम संख्या ४२ है)
(ङ) ५ मन्तिकं for (मंकितं) ३. रुक्तम् for (यष्टयुक्तम्)

४२. (घ) (वि०—इसकी क्रमसंख्या ४३ है)
१. विद्वस्त (ग) विद्धस्तछ्रूति for (विद्धस्तछ्रूति) २. छ्रति for (छ्रूति)
३. द्धृतस्तदुछ्रायः for (समुछ्राय) (ग) ४. घटिकोद्धृतः (ङ) for(घटिकोद्धृत)
(वि०—इसकी श्लोक संख्या ४६ है) (च) (वि०—श्लोक संख्या ४३ है)
(ङ) १. तत्स्रुति for (तछ्रूति)

घटिकांगुलांतरस्थै वर्यांगुटकै घंटीधृतैरन्यः।
उपरिनरोतः शुषिरस्तिर्यक् कीलोस्य मुखमध्ये ॥ ४३ ॥
कीलो परिगामिन्यां वीर्याबद्धं सपारतमलांबु।
स्त्रवति जलोक्षिपति नरो गुटिकां कुर्मोदयश्चैवम् ॥ ४४ ॥
जलपूर्णकृतघटीभिस्तनस्य कर्णादिभिर्जलं श्रिपति।
पुरुषोन्यर्द्धो सत्वं चक्रचतुष्कस्य कृतमुपरि ॥ ४५ ॥

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४३. (घ) (वि०—इसकी क्रमसंख्या ४४ है)
१. वीर्यांगुटकै (ग) वीर्यांगुडकै for (वर्यांगुटकै)
२. घंटीधृतीरन्यः for (घंटीधृतैरन्यः) ३. उपरिनरांत for (उपरिनरोतः)
(ग) ४. सुषिरस्तिर्यक् (ङ) for (शुषिरस्तिर्यक्)
(वि०—इसकी श्लोक संख्या ४७ है)
(च) १. वीर्यां for (वर्यां)
(वि०—इसकी श्लोकसंख्या ४४ है)
(ङ) १. वीर्यांङ्गुलकै for (वर्यांगुटकै)
३. नरोतः for (नरोनः)
४४. (घ) ((वि०—इसकी क्रमसंख्या ४५ है) १. चीर्यांबद्धं for (वीर्याबद्धं)
२. लावु for (लांबु) ३. स्रवति (ग) (च) (ङ) for (स्त्रवति)
४. जले (ग) (च) (ङ) for (जलो)
५. कूर्मादयश्चैवम् (ग) कूर्मोदयाश्चैवम् ॥४८॥ for (कुर्मोदयश्चैवम्)
(वि०—इसकी श्लोक संख्या ४८ है)
(च) २. लावु for (लांबु)
५. कूर्मोदयश्चैवं for (कुर्मोदयश्चैवम्)
(ङ) १. चीर्यांद्यं for (वीर्याबद्धं)
२. लाबुतु for (लांबु) ६. द्वपारदम् for (सपारतम)
४५. (घ) (वि०—इसकी क्रमसंख्या ४६ है)
१. स्तनास्य (ग) (ङ) for (स्तनस्य)
२. पुरुषोऽन्यद्वासत्वं (ग) पुरुषोन्यर्द्धासित्व for (पुरुषोन्यर्द्धोसत्वं)
(ग) ३. क्षिपितम् for (श्रिपति) ४. जनमुपरि ॥४९॥ for (कृतमुपरि ॥४५॥)
(वि—इसकी श्लोक संख्या ४९ है)
(च) ३. क्षिपति for (श्रिपति)
२. ऽन्यर्द्धा for (न्यर्द्धो) क्रमसंख्या ४६ है।
(ङ) ३. क्षिपति for (श्रिपति) २. पुरुषोऽन्यर्धासक्तं for (पुरुषोन्यर्द्धोसत्वं)

एवं वफवरं नाडिकांगुलैः संस्थितेऽतरे योज्यम् ।
युद्धानि मल्लगज महिषमेष विविधयुधनृतां च ॥ ४६ ॥
निगिरति च घटिकां गुलांकितै गंडिकै मंयुरोऽहीं ।
वीर्याभ्रुमेवं गुडकैरूपरिस्थै ब्रह्मचर्याद्यैः ॥ ४७ ॥
कालोक्षपाप्तिहितः पटहः शब्दकरोति घंटा वा ।
एवं यन्त्रं सहस्राण्यनेन बीजेन कार्याणि ॥ ४८ ॥

४६. (घ) १. वधूवरं (ग) (ङ) for (वफवरं) (वि०—इसकी क्रमसंख्या ४७ है ।)
(ग) २. संयुतं for (संस्थिते) ३. चरे (वरे) for (ऽतरे)
४. विवधायुधभृतां च ॥५०॥ (ङ) for (विविधयुधनृतां च)
(वि०— इसकी श्लोक संख्या ५० है)
(च) १. एवंच वरं for (एवं वफवरं)
४. विविधायुधभृतां च for (विविधयुधनृतां च) (वि० —श्लोक संख्या ४७ है)
(ङ) २. संयुता for (संस्थिते) ३. वरे योज्या for (ऽतरे योज्यम्)
४७. (घ) (वि०— इसकी क्रमसंख्या ४८ है)
१. निगिरति गिरति च (ग) निगिरति गिरति for (निगिरति)
२. गंडकैर्मयूरोऽहिम् (ग) खंडकै for (गंडिकै)
३. वीर्यामिवं (ग) for (वीर्याभ्रुमेवं)
४. ब्रह्मचर्याद्यैः (ग) ब्रह्मचार्याद्यैः ॥५१॥ (ङ) for (ब्रह्मचर्याद्यैः)
(ग) ५. गुलांकितैः for (गुलांकितै) ६. मयूरोहिम् for (मंयूरोऽहीम्)
(वि०—इस की श्लोक संख्या ५१ है)
(च) २. गंडकै for (गंडिकै) ६. मंयूरोऽहिं for (मंयुरोऽहीं)
३. वीर्यामिवं for (वीर्याभुमेवं) ४. ब्रह्मचर्याद्यैः for (ब्रह्मचर्याद्यैः)
(वि०—श्लोक संख्या ४८ है) (ङ) २. खण्डकैर for (गंडिकै)
६. मंयूरोऽहिम् for (मंयूरोऽहीं) ३. चीर्यामिवं for (वीर्याभ्रुमेवं)
४८. (घ) (वि०—इसकी क्रमसंख्या ४९ है)
१. कालोत्क्षपाप्तिहितः (ग) कीलीत्क्षेपाभिहतः for (कालोक्षपाप्तिहतः)
२. यंत्रसहस्राण्यनेन (च) (ग) for (यंत्रं सहस्राण्यनेन)
३. शब्दं (च) (ङ) for (शब्द)
४. 'वा' पद लुप्त है (वि०—इसकी श्लोकसंख्या ५२ है)
(च) १. कालोक्षेपाभिहतः for (कालोक्षपाप्तिहितः)
(वि०—इसकी श्लोकसंख्या ४९ है)
(ङ) १. कीलोत्क्षेपाभिहतः for (कालोक्षपाप्तिहितः)

लघुदारुमयं चक्रं समशुषिरांतरं पृथगराणाम् ।
अर्धं पारदपूर्णं मध्ये सुल्लिष्ट कृतसंध्यौ ।। ४९ ।।
तिर्यक् कीलोमध्येध्व्याधारस्थो स्यपारदे व्रजति ।
छिद्रान्यर्द्धं चक्रकर्मजभ्रमे वांबरे भ्रमति ।। ५० ।।

४९. (घ) (वि०—इसकी क्रमसंख्या ५० है)
१. लघुदारमयं for (लघुदारमयं)
२. शुषिरांतरां for (शुषिरांतरं)
३. पारतंपूर्णं (ग) परेतपूर्णं for (पारदपूर्णं)
४. सुश्लिष्ट (ग) सश्लिष्ट for (सुल्लिष्ट)
(ग) ५. चत्र for (चक्र)
६. दाम for (सम) ७. अर्द्धं for (अर्धं)
८. परिधौ for (मध्ये) १. कृतं संधिः ।।५३।। for (कृतसंध्यौ)
(वि०—इसकी श्लोकसंख्या ५३ है)
(च) १. लघुदारमयं for (लघुदारूमयं) ३. पारतपूर्णं for (पारदपूर्णं)
४. सुश्लिष्ट for (सुल्लिष्ट) (वि०—श्लोक संख्या ५० है)
(ङ) २. सुषिरारान्तरं for (सुषिरांतरं)
द्वितीयपंक्ति—अर्धेरसेन पूर्णे परिधौ संश्लिष्ट कृतसन्धिः ।
for
अर्धां पारदपूर्णं मध्ये सुल्लिष्ट कृतसंध्यौ ।।

५०. (घ) १. द्वघा (ग) (ङ) for (ध्व्या) (वि०—इसकी क्रमसंख्या ५१ है)
२. ऽस्य पारते for (स्यपारदे) ३. छिद्राण्यूर्ध्व (ग) (ङ) for (छिद्रान्यर्द्धं)
४. चक्रकमजस्रमेवांबरे (ग) चक्रकमजभ्रमेवांबरे भ्रमति ।।५४।। for (चक्रकर्मजभ्रमेवांबरे भ्रमति ।।५०।।)
(ग) ५. पारंते for (पारदे)
६. भ्रमति (ङ) for (व्रजति)
(वि०—इसकी श्लोकसंख्या ५४ है)
(च) १. द्वयाधारस्थो for (ध्व्याधारस्थो) २. ऽस्य for (स्य)
५. पारते for (पारदे) ३. छिद्राण्यर्द्धं for (छिद्रान्यर्द्धं)
(वि०—श्लोकसंख्या ५१ है)
(ङ) २. ऽस्य for (स्य)
५. पारदो for (पारदे) ७.+भ्रमति+
४. चक्रमजस्रमे for (चक्रकर्मजभ्रमे) ८. वाम्बरं for (वांबरे)

छिद्रे स्वधिया[५] क्षेप्यं[१] समं[६] तथा पारदं[२] यदा[३] भ्रमति ।
कालसममिष्टमानैश्चक्रकमुतानमूर्द्ध[४] वा[७] ॥ ५१ ॥
कीलस्योपरिगमिभिन[१] तिर्यक्[३] सूत्रके धृतमलाषु[२] ।
प्राग्वन्नलके प्रक्षिप्य नाडिका श्रवति[५] पानीये ॥ ५२ ॥

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५१. (घ) (वि०—इसकी क्रमसंख्या ५२ है)
१. ष्येप्यं for (क्षेप्यं) (ग) 'क्षेप्यं' लुप्त है ।
२. पारतं (ग) (च) for (पारदं)
३. यथा (ग) (च) for (यदा)
मुत्तान (ग) (च) for (कमुतान)

(ग) ५. स्वधियां समं for (स्वधिया क्षेप्यं)
(वि०—इसकी श्लोक संख्या ५५ है)

(च) १. ष्येप्यम् for (क्षेप्यम्)
(वि०—श्लोक संख्या ५२ अंकित है)

(ङ) १. क्षिप्त्वा for (क्षेप्यं)
३. 'यदा' लुप्त ।
६. यथा (for) तथा
४. समुत्तान for (कमुतान)
७. मूर्ध्वं वा for (मूर्द्धं वा)

५२. (घ) (वि०—इसकी क्रम संख्या ५३ है)
१. गामिनि न (ग) गामिनितत्पर्यय for (गमिभिन)
२. मलाबु (ङ) मलाबु for (मलाषु)

(ग) (वि०—इसकी श्लोक संख्या ५६ है)

(च) १. परिगामिनि (ङ) for (परिगमिभिन)
(वि०—श्लोक संख्या ५३ अंकित है)

(ङ) ३. तत्पर्य for (तिर्यक्)
५. स्रवति for (श्रवति)

६ ४ १ ७ २
कर्णेज्या क्षिप्रचलनमेव शरमोक्षशंस्वशबाश्च ।
५
अध्यायो द्वाविंशो यन्त्रेष्वार्यास्त्रिपंचाशत् ॥ ५३ ॥
३
इत्ति ब्रह्मगुप्ते यंत्राध्यायः (२२ वां अध्याय समाप्तः)
द्वाविंशः समाप्तः

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५३. (घ) (वि०—इसकी क्रमसंख्या ५४ अंकित है)

१. मेवं (ग) (च) (ङ) for (मेव) २. शंखशब्दाश्च (ग) for (शंस्वशबाश्च)

३. इति ब्राह्मस्फुटसिद्धांते यन्त्राध्यायो द्वाविंशतितमः (ग) for (इति श्री ब्रह्मसिद्धांते यन्त्राध्यायो द्वाविंशः)

(ग) ४. चालन for (चलन) ५. सप्त for (स्त्रि)
(वि०—इसकी श्लोक संख्या ५६ है)

(च) २. शंखशब्दाश्ब for (शंस्वशबाश्च)
(वि०—इसकी श्लोकसंख्या ५४ अंकित है)

३. इति ब्राह्मस्फुटसिद्धांते यंत्राध्यायो द्वाविंशतितमः for (इत्ति ब्रह्मगुप्ते यंत्राध्यायः (२२ अध्यायसमाप्तः)

(ङ) ६. करणैर्ज्या for (कर्णेज्या) २. खशब्दाश्च for (स्वशबाश्च)

७. शरमोक्षणं for (शरमोक्षशं)

३. इति ब्राह्मस्फुट सिद्धान्ते यन्त्राध्यायो नाम द्वाविंशोऽध्यायः for (इति ब्रह्मगुप्ते यन्त्राध्यायः)

अथ मानाध्यायो नाम

त्रयोविंशः

सौरेणाब्द मासतिथयश्चांद्रेण सावनैर्दिवसाः ।
दिनमासाब्द पमध्यानत द्विनार्कैंदु मानाम्याम् ॥ १ ॥
मानाति सौरचन्द्रार्क्षसावनानि ग्रहानयनमेभिः ।
मानेन पृथक्वत्रुभिर्व्यवहारोत्र लोकस्य ॥ २ ॥
युगववर्ष विषुवदयनत्वहर्निशोर्वृद्धिहानयः ।
सौरात्तिथि करणाधिकमासोन रात्रपर्वक्रियाश्चांद्रात् ॥ ३ ॥
यसस्य वन प्रमाण ग्रहगत्फपवास सूतक चिकित्साः ।
सावन मानात् ज्ञेया प्रायश्चित्त क्रियाश्चान्या ॥ ४ ॥

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१. (घ) १. सौरेणाब्दा (ग) सौरेणाब्दा (ङ) for (सौरेणाब्द)
२. मासास्तिथय (ग) (ङ) for (मासतिथय)
(ग) ३. मप for (पम) ४. तद्दिनार्कैंदु for (तद्दिनार्कैंदु)
(च) १. सौरेणाब्द्दा for (सौरेणाब्द)
२. मासास्तिथयश्चांद्रेण for (मासतिथयश्चांद्रेण) ५. दिवसा for (दिवसाः)
२. (घ) १. चान्द्रार्क्ष (ग) चान्द्राक्ष for (चन्द्रार्क्ष)
२. मानैः (ग) च) (ङ) for (मानेन)
३. व्यवहारोऽत्र (ग) संव्यवहाराश्च for (व्यवहारोत्र)
(ग) ४. पृथक् चतुर्भिः (ङ) for (पृथक्वत्रुभि)
(च) १. चान्द्रार्क्ष for (चन्द्रार्क्ष) ४. पृथक्चतुर्भि for (प्रथक्वत्रुभि)
(ङ) ५. मानानि for (मानाति) ३. संव्यवहारोऽत्र for (व्यवहारोऽत्र)
३. (घ) १. युगवर्षे (ग) युगं वर्ष for (युगववर्ष) २. नंशो for (निशो)
३. हानियः for (हानयः)
(ग) ४. सौरानतिथि for (सौरात्तिथि)
(च) १. युगवर्ष (घ) for (युगववर्ष) ५. विषुचदय for (विषुवदय)
२. नर्त्वर्हनिशो (ङ) for (नत्वहर्निशो)
४. (घ) १. यज्ञसवन (ग) for (यसस्य वन)
ग्रहत्युपवास (ग) ग्रहगत्युरुवास for (ग्रहगत्फपवास) ३. ज्ञेयाः (ग) for (ज्ञेया)
(ग) ४. प्रायश्चि for (प्रायश्चित्त) ५. चान्याः ॥ ४ ॥ for (चान्या)
(च) १. यज्ञस्य for (यसस्य) २. ग्रहगत्युपवास for (ग्रहगत्फपवास)
(ङ) १. यज्ञसवन for (यसस्य वन) २. गत्युपवास for (गत्फपवास)
३. सावनमानाज्ज्ञेयाः for (सावनमानात्ज्ञेया) ५. चात्र for (चान्या)

नक्षत्रसावनदिनात् सूर्यादीनां स्वसावन दिनानि ।
यस्मात्तस्मादार्क्षं दुरधिगमं मन्दबुद्धिनाम् ॥ ५ ॥
मानुष्य पित्रदेव ब्राह्म्यान्यष्टा वमूर्त्तकालस्य ।
उक्तानि ज्ञानार्थं बार्हस्पत्यं नवममन्यत् ॥ ६ ॥
द्वौ द्वौ राशि मकराहृतवः षट् सूर्यगति वशाद्योज्याः ।
शशिरवसन्त ग्रीष्मा वर्षा शरदौ सहेमन्ताः ॥ ७ ॥
भूव्यास गुणोभक्तः कर्क व्यासांतरेण रविकर्णं भूमध्या
द्भ छाया दिर्घत्वं चन्द्रकर्णोनं शेषम् ॥ ८ ॥

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५. (घ) १. मंदबुद्धीनाम् (ग) (ङ) for (मंदबुद्धिनाम्)
(च) १. बुद्धीनां for (बुद्धिनाम्)

६. (घ) १. पित्र्यदेव (ग) पित्र्यदिव्य for (पित्रदेव)
२. ब्राह्मचान्यष्टा (ग) for (ब्राह्मचान्यष्टा)
बर्हस्पत्यं (ग) वार्हस्पत्यं for (बार्हस्पत्यं)
(च) १. पित्र्यदेव for (पित्रदेव)
४. न्नवममन्यत् for (नवममन्यत्)
(ङ) २. दिव्यपित्र्य for (पित्रदेव) ब्राह्माण्यष्टा for (ब्राह्मचान्यष्टा)

७. (घ) १. राशी (ग) (ङ) for (राशि)
२. शशिबसंत (ग) शिशिरवसन्त (ङ) for (शशिरवसन्त)
(ग) ३. मकराहृतवः (ङ) (मकराहृतवः)
(च) १. राशी for (राशि)
३. मकराहृतवः for (मकराहृतवः)
४. सहेमन्ता for (सहेमन्ताः)
(ङ) ५. वशाद्भाज्याः for (वशाद्योज्याः) ६. शरदः for (शरदौ)

८. (घ) १. दीर्घवत्वं (ग) दीर्घत्वम् (ङ) for (दिर्घत्वं)
२. 'कर्णो' तक पहली पंक्ति समाप्त (ग) कर्णः (ङ) for (कर्णौ)
३. 'कर्णोनम्' तक दूसरी पंक्ति समाप्त (ग) (ङ)
४. 'शेषम्' अगले श्लोक का प्रथम पद है (ग) (ङ)
(च) १ दीर्घत्वं (ङ) for (दिर्घत्वम्)
४. ॥ ८ ॥ शेषं for (शेषं ॥ ८ ॥)

व्यासगुणं दीर्घत्वहृतं शशांककक्षायाम् ।
तमसो व्यास शशि कर्णद्धृतस्त्रिज्यया गुणालिप्ता ॥ ९ ॥
रविकर्ण द्धृता स्त्रिज्या कर्क व्यासान्तर हूता शोध्या ।
त्रिज्याभूव्यास वधा छशि कर्ण हूतातमो व्यासः ॥ १० ॥
भू व्यासेंदुगति वधात् कर्क व्यासांतरार्कभुक्ति वधम् ।
प्रोह्येंदुमध्यभुक्तघा तिथिगुणयाप्तं तमो व्यासः ॥ ११ ॥
योधिकमासावमरात्रसम्भवज्ञः सवेत्रिमानानि ।
आर्याभिर्द्वादशाभिर्मानाध्यायस्त्रियोविंशः ॥ १२ ॥
इत्ति ब्रह्म गुप्ते मानाध्यायः (२३ अध्याय समाप्तः)

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९. (घ) १. यह पंक्ति 'शेषं से आरम्भ है शेषं 'पूव्यास' (ग) 'शेषं भू' लुप्त ।
२. हृतं (ग) (च) (ङ) for (हतं) ३. कक्ष्यायाम् (च) for (कक्षायाम्)
४. गुणो (ग) (ङ) for (गुणो) ५. हृत (ग) (च) (ङ) for (द्धृत)
(ग) ६. व्यासः (ङ) for (व्यास) ७. स्त्रिज्यो for (स्त्रिज्यया)
८. लिप्ताः for (लिप्ता) ९. दीर्घत्वं for (दीर्घत्व)
(ङ) १. 'शेषं भू' लुप्त ७. स्त्रिज्या for (स्त्रिज्यया)
१०. (घ) ४. हृता (ग) (च) (ङ) for (द्धृता)
२. तृज्या (ग) त्रिज्या (ङ) for (स्त्रिज्या)
३. व्यासांतराहृता (ग) (च) for (व्यासांतरहूता)
४. हृतं (ग) हृतात्तमो for (हूतातमो)
(ग) ५. वधात् शशिकर्ण (ङ) for (वधाच्छशिकर्ण)
(च) २. त्रिज्या for (स्त्रिज्या) ४. हृतातमो for (हूतातमो)
(ङ) ६. व्यासांतराहता for (व्यासांतरहूता) ४. हृतात् for (हूता)
११. (घ) १. प्रोह्येन्दु for (प्रोह्येन्दु) (च) १. प्रोज्येन्दु for (प्रोह्येन्दु)
१२. (घ) १. योऽधिक (च) for (योधिक) २. वेत्ति (ग) (च) (ङ) for (वेत्रि)
३. द्वादशभि (ग) (च) for (द्वादशाभिर्)
४. स्त्रयोविंशः (ग) (च) (ङ) for (स्त्रयोविंशः)
५. 'इति' से 'समाप्त' तक लुप्त है ।
(ग) ६. सं भावज्ञः for (संभवज्ञः)
(च) ५. 'इति' से 'समाप्त' तक लुप्त है ।
(ङ) ३. आर्याद्वादशभिरयं for (आर्याभिर्द्वादशाभिर्)

अथ संज्ञाध्यायो नाम

चतुर्विंशः

यस्मात्संप्रति पत्तिर्न संज्ञया संज्ञितो[१] विना तस्मात् ।
लोकप्रसिद्धसंज्ञारूपादीनां शशांकाद्याः ॥ १ ॥
युगपद्युगारूदया[१] याम्यायां[३] भास्करस्य वारूण्याम् ।
रात्र्यर्धात्सौम्याया मस्तमया[२] दिनदलादैंद्याम् ॥ २ ॥
अपमेवकृतः[३] सूर्येंदु पुलिश[१] रोमक वशिष्ट[४] यवनाद्यैः ।
यस्मात्तस्मादेकः सिद्धान्तो ग्रन्थ[२]रचनान्या[४]याः ॥ ३ ॥
यदि भिन्नाः सिद्धान्तभास्कर संक्रान्तयोपि भेदसमाः[३] ।
सप्तष्टिः[२] पूर्वस्यां विषुवत्पार्कोदयो[१] यस्य ॥ ४ ॥
तन्त्र[१] पक्षागणितं मध्यम गत्युत्तरादयः पञ्च ।
कुट्टाकारो[३] वेधः[४] छन्दश्चित्यूतरं[२] गोलः ॥ ५ ॥

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१. (घ) १. संज्ञिनो (ग) संज्ञिने for (संज्ञितो) (च) १. संज्ञिनो for (संज्ञितो)

२. (घ) १. द्युगादि (ग) युगपत् युगादिरुदयात् for (युगपद्युगारूदया)
२. द्दिनदलादैन्द्र्याम् (ग) (ङ) for (दिनदलादैंद्याम्)
(च) १. युगपद्युगादि for (युगपद्युगा) २. द्दिनदलादै for (दिनदलाद्यै)
(ङ) १. द्युगादिरुदया for (द्युगारुदया) ३. द्याम्यायां for (याम्यायां)

३. (घ) १. पुलिस for (पुलिश) २. ग्रन्थः (च) for (ग्रन्थ)
(ग) ३. अयमेव for (अपमेव) ४. रचनान्या for (रचनान्याः)
(ङ) ५. वसिष्ठ for (वशिष्ट) २. विरचितो for (ग्रन्थरच)
३. नान्यः for (नान्याः)

४. (घ) १. विषुवत्यार्को (ग) (च) for (विषुवत्पार्को)
(ग) २. सिद्धान्ताः (ङ) for (सिद्धान्त) ३. समाम् for (समाः)
(ङ) ३. विभेद समाः for (पिभेद समाः) १. विषुवत्यर्कोदयो for (विषुवत्पार्कोदयो)

५. (घ) १. तन्त्रपरीक्षागणिते (ग) तन्त्रपरीक्षागणितम् for (तन्त्रपक्षागणितम्)
२. चित्युंत्तरं गोलः (ग) (च) for (चित्यूत्तरं गोलः)
(च) १. तन्त्रपक्षागणिते for (तन्त्रपक्षागणितं) ३. कुदाकरो for (कुट्टाकारो)
(ङ) १. तन्त्र परीक्षा for (तन्त्रपक्षा) ४. वेद्यश्छन्द for (वेधः छन्द)
२. श्चित्युत्तरं गोलः for (श्चित्यूत्तरं गोलः)

यंत्राणिमान संज्ञाश्चैवाध्यायाश्चतुर्दश ब्रह्मोध्यायः ।
चतुर्विंशतिरार्द्या र्दृशभिः सहाध्यायैः ॥ ६ ॥
श्री चापवंशतिलके श्री व्याघ्रमुखे नृपे शकनृपाणाम् ।
पंचाशत्संयुक्तैः वर्षशतैः पंचभिरतीतैः ।५५०। ॥ ७ ॥
ब्रह्मस्फुटसिद्धांतैः सज्जनगणितगोलविप्रित्यै ।
त्रिंशद्वर्षेण ततो जिष्णुसुतब्रह्मगुप्तेन ॥ ८ ॥
गणितेन फलं सिद्धं ब्राह्मे ध्यान ग्रहे यतोध्याये ।
ध्यानग्रहो द्विसप्तत्यार्याणां न लिखितोत्र मया ॥ ९ ॥

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६. (घ) १. ब्राह्मो अध्यायः for (ब्रह्मोध्यायः) (ग) ब्राह्मो for (ब्रह्मो)
(वि०—यहां प्रथम पंक्ति समाप्त)
२. र्दशभिः (ग) (च) (ङ) for (र्दृशभिः)
(ग) ३. संज्ञारब्यौ for (संज्ञाश्चैं) ४. चाध्याय for (वाध्याया)
५. 'अध्यायः' से दूसरी पंक्ति आरम्भ (ङ)
(च) ५. अध्याया: for (ध्यायः)
(ङ) ३. ख्याता for (श्चैवा) १. ब्राह्मे for (ब्रह्मो)
६. र्युताध्यायैः for (सहाध्यायैः)

७. (ग) १. नृपाना: for (नृपाणाम्) २. पंचशत्भंयुक्तैं for (पंचाशत्संयुक्तैः)
३. र्वषशतैः for (वर्षशतैः) ४. '५५०' संख्या लुप्त है।
(च) ५. श्रीचापविंशतिलके for (श्रीचापवंशतिलके)
६. पंचभिरतीतैं for (पंचभिरतीतैः)
(ङ) ३. संयुक्तैंर्वषशतैः for (संयुक्तैः वर्षशतैः)

८. (घ) १. विप्रीत्यैं (ग) वित्प्रीत्यैः for (विप्रित्यैं)
२. त्तो (ग) कृतो (ङ) for (ततो) (ग) ३. ब्राह्मः (ङ) for (ब्रह्म)
४. स्फुटसिद्धांतः (ङ) for (स्फुटसिद्धान्तैः) ५. गणितश for (गणित)
६. गुप्तेन (ङ) for (गुप्ते) (च) ५. सज्जनणित for (सज्जनगणित)
१. विप्रीत्यैं for (विप्रित्यैं) २. त्तो for (त्तो)
(ङ) १. वित्प्रीत्यैं for (विप्रित्यैं)

९. (घ) १. लिखितोऽत्र (ङ) for (लिखितोत्र) (ग) २. सिद्धं for (सिद्धं)
(च) ८. ब्रह्मे for (ब्राह्मे) (ङ) ४. फले for (फलं)
२. सिद्धिब्राह्मे for (सिद्धं ब्राह्मे) ६. यतोऽध्याये for (यतोध्याये)
५. द्विसप्ततिरार्याणां for (द्विसप्तत्यार्याणां)

तिथिभोगनाडिकासु द्विगुणास्त्रिगुणा खे शोध्याः ।
पंचाशीत्यंशोनास्तिथि नाडीशोधयेच्छशिनः ॥ १२ ॥
वेदहतौ नवभक्तौ राश्यादौ दिनकरोडुपकेंद्रम् ।
त्रिगुणं सप्त विभक्तं गजाद्रयोंशा रवेरुच्चम् ॥ १३ ॥
विकला संयुक्त नव वाणा लिप्तिकारवेर्भुक्तिः ।
खनवनगा शीतांशोः पंचत्रिंशद्विलिप्ताश्च ॥ १४ ॥
स्वोच्चोनकेंद्रमिनो नवभिर्लिप्ता शतैर्जीवा ।
विषमे भुक्तस्य समे भोग्य सदैव केंद्रपदे ॥ १५ ॥
त्रिंशत्सनवरसेंदुः ॥ १६ ॥
गतभोग्य खंडकांतरे दल ॥ १७ ॥

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१२. (ङ) १. रसगुणोद्धृताः शोध्याः for (स्त्रिगुणा खे शोध्याः)
२. पंचाशीत्यधिको for (पंचाशीत्यंशो) ३. नाड्यः for (नाडी)

१३. (ङ) १. (वि०—इस श्लोक का उत्तरार्ध—१४ वें श्लोक का पूर्वार्ध भी है)
त्रिगुणं सप्त विभक्तं नगाद्रयोऽशा रवेरुच्चम् ।
विकलाष्टक् संयुक्ता नवबाणा लिप्तिका ५९।८ रवेर्भुक्तिः ॥ १३ ॥
for
(वेदहतौ नवभक्तौ राश्यादौ दिनकरोडुपकेंद्रम् ।
त्रिगुणं सप्त विभक्तं गजाद्रयोंशारवेरुच्चम् ॥ १३ ॥)

१४. (ङ) (वि०—इस श्लोक का पूर्वार्ध, १३ वें श्लोक का उत्तरार्ध भी है)
१. विकलाष्टक् for (विकला)
२. संयुक्ता for (संयुक्त)

१५. (ङ) १. स्वोच्चोनं for (स्वोच्चोन) २. मितो for (मिनो)
३. शतैस्ततो for (शतैर्जीवा) ४. भोग्यस्य for (भोग्य)

१६. (ङ) १. त्रिंशत्सनवरसेन्दुर्जिनतिथि विषया गृहार्धचापानाम् ।
अर्धज्याखंडानि ज्याभुक्तैक्यं सभोग्य फलम् ॥ १६ ॥
for
(त्रिंशत्सनवरसेंदुः ॥ १६ ॥)
२. गतभोग्यखण्डकान्तरदलविकलवधाच्छतैर्नवभिराप्तैः ।
तद्युतिदलं युतोनं भोग्यादूनाधिकं भोग्यम् ॥ १७ ॥
for
(गतभोग्य खंडकांतरेदल ॥ १७ ॥)

स्पाप्तांशोना सवितु द्विगुणा ज्याशीतगौः फलं लिप्ताः ।
स्वफलमृणं चक्रार्द्धोने केंद्रे धिके धनं मध्ये ॥ १८ ॥
नगमूहृद्रविभोग्यं खंडं चांद्रं मत्त भागोनं ।
द्विगुणं भुक्तिफलं स्वमृणं स्यात्कुलीरमकरादिकेंद्रे ॥ १९ ॥
भांशोर्कफलस्येंदौ रविवद्दृशोद्धृतः स्वोर्के रवि ।
फलमिनवतिथौ चांद्रव्यस्तः स्फुटार्काप्तम् ॥ २० ॥
पंचेषु पंचयुगगुणयम चंद्राशुचंद्रकेन्द्र भफलानि ।
द्विद्विद्विद्विकुकुभुवः षटक्षर हिते त सूर्यम् ॥ २१ ॥
भान्यश्विन्यादीनि ॥ २२ ॥
अर्कोनचंद्रलिप्ता ॥ २३ ॥

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१८. (ङ) १. स्वाष्टांशोना for (स्पांप्तांशोना)
२. द्विगुणा for (द्विगुणा) ३. गौः for (गौः)
४. 'धनम्' पद लुप्त है ।

१९. (ङ) १. विव सुलवं for (मत्तभागोनं)
२. 'द्विगुणं' यहां यह पद प्रथम पंक्ति का समापक है ।
३. मकरादिके for (मकरादिकेंद्रे)

२०. (ङ) १. रविवद्दृग्वा द्विशोधिते तथा स्वोच्चे for (रविवद्दृशोद्धृतः स्वोर्के)
२. यहां 'रवि' पद दूसरी पंक्ति का आरंभक है ।
३. नवच्च तिथौ for (नवतिथौ) ४. चांद्रे for (चान्द्र)
५. व्यस्तं for (व्यस्तः)

२१. (ङ) १. चंद्राश्चन्द्र for (चन्द्राशुचन्द्र) २. ज for (भ)
३. द्विकुभुव खरहिते for (द्विद्विद्विद्वि कुकुभवः षटक्षरहिते)
४. तथा सूर्ये·····for (तसूर्यम्)

२२. (ङ) १. अर्कोनचन्द्रलिप्ताः खयमस्वरभाजिताः फलं तिथयः ।
गतगम्ये षष्टिगुणे भुक्तयन्तरभाजिते घटिकाः ॥ २२ ॥
for
(भान्यश्विन्यादीनि ॥ २२ ॥)
२. भान्यश्विन्यादीनि ग्रह लिप्ताः खखवसूद्धृता लब्धम् ।
भुक्तिहृते गतगम्ये दिवसाः षष्टघाहते घटिका ॥ २३ ॥
for
(अर्कोन चन्द्रलिप्ताः ॥ २३ ॥)

[1]व्यर्केंदुफलाक्ताः ॥ २४ ॥
[2]रविचन्द्र योग लिप्ताः ॥ २५ ॥
इति तिथ्यधिकारः ॥

अंगै रुद्रैः सिद्धैर्गजोजनैरर्के[1] वत्सरात् ॥ ६ ॥ ११ ॥ २४ ॥ ८ ॥ २४ ॥
शैलैर्विलैर्गु[2]णैरसवह्निभिर्योजयो[3]द्भौमः[4] ॥ ७ ॥ ९ ॥ ३२ ॥ २६ ॥
[1]शशिनोक्षितैः शराविरंकैः षड्वह्निभिर्हतानंदान् ।
शशिनाद्वियमैश्चतुरविभिर्दिनैर्यु[2]तं भवति बुधशीघ्रम् ॥ २७ ॥
रूपेण[1] खेनकुयमै[2]रंगै[3]र्नभसावकरणरव्यब्दात् ।
गुणिता युक्ता वैदैः[4] कुयमैस्त्रियमैर्गु[5]णैश्च गुरुः ॥ २८ ॥

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२४. (ङ) १. रविचन्द्र योगलिप्ताः खखवसुभिर्भाजिताः फलं योगः ।
गतगम्ये षष्टिगुणे गतियोनिभाजिते घटिकाः ॥ २४ ॥
for
(व्यर्केंदुफलाक्ताः ॥ २४ ॥)

२५. (ङ) १. व्यर्केन्दुकला भक्ताः खरसगुणैर्लब्धमूनमेकेन ।
चरकरणानि ववादीन्यगताच्छेषात् तिथिवदन्यत् ॥ २५ ॥
for
(रविचन्द्रयोगलिप्ताः ॥ २५ ॥)

२६. (ङ) १. र्यमैरर्कवत्सरान् गणयेत् for (जिनैरर्केवत्सरात्)
२. विश्वैर्गुणितै for (विलैर्गुणै) ३. रष्टवह्निभि for (रसवह्निभि)
४. र्योजयेत् for (र्योजयोद्भौमः)

२७. (ङ) १. शशिना जिनैः रंकैः षड्वह्निभिर्हतादब्दात्
for
(शशिनोक्षितैः शराविरंकैः षड्वह्निभिर्हतानंदान् ।)
२३. द्विपैर्यमैश्चतुरब्धिभिरन्वितं for (द्वियमैश्चतुरविभिर्दिनैर्यु[2]तं)

२८. (ङ) १. +१+ २. +२१+ ३. +६+
४. नवभिश्चकरणाब्दाः for (र्नभसावकरणरब्यब्दात्)
५. वेदैः for (वैदैः)
६. यमैश्च भवति गुरुः ॥ २८ ॥
for
(यमैर्गुणैश्च गुरुः ॥ २८ ॥)

शैलैस्तिथिभी रुद्रैर्यम विषयैरस्त[1] सागरैर्गुणिता ।
वसुभिरनलैर्जिनैः खड्गुणैश्च[3] शुक्रा[4] भृगोः शीघ्रम् ॥ २९ ॥
शून्येन द्वादशभिर्द्वादशभिः ददिशिभिः[1] खेषुभिस्त्रयोदशभिः ।
गुणिता युक्ता[2] रसरवितिथिविषयैर्वंशभिरर्कैः[3] ॥ ३० ॥
गगने ननंद चन्द्रैः[1] कुयमैरसाधिभिरं[2] बरेण[3] हृताः[4] ।
रुद्रैर्नृपैः[5] खवेदैर्युक्ता राश्यादिको राहुः[1] ॥ ३१ ॥
सर्वाणि स्थानानि क्रमात्[1] स्वहरैर्नंयेदुपर्युपरि[2] ।
एवं रव्यब्दांते ग्रहध्रुवास्[3]तत्परांशाः[4] स्युः ॥ ३२ ॥
पृथगर्को दिग्गुणितो[1] वसुशरचंद्रैर्हृतः[2] फलेन युतः ।
दलितो भौमध्रुवके क्षिप्तः[3] स्यान्मध्यमो भौमः ॥ ३३ ॥
चतुराहतो विवस्वान्[1] पृथक्सस्ताहतो[2] न[3] विधृति विभक्तः ।
फलसंयुक्तादेयो[4] ज्ञचलध्रुवके[5] ज्ञशीघ्रं स्यात् ॥ ३४ ॥

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२९. (ङ) १. 'रस्त' पद लुप्त है। २. रनिलै for (रनलै)
३. षड्गुणैश्च for (खड्गुणैश्च) ४. युक्तम् for (शुक्रा)

३०. (ङ) १. 'ददिशिभिः' पद लुप्त है। २. युता for (युक्ता)
३. रसैरब्धिभिस्त्रिविषयै for (रसरवितिथिविषयै)
४. रार्किः ॥ ३० ॥ for रर्कैः ॥ ३० ॥

३१. (ङ) १. गगनेन नवचंद्रैः for (गगने ननंद चन्द्रैः)
२. रसाब्धिभिः for (रसाब्धिभि) ३. संवरेण for (रंबरेण)
४. हृताः for (हृताः) ५. 'नृपैः' पद लुप्त है।
६. कः पातः ॥ ३१ ॥ for (कः पातः ॥ ३१ ॥)

३२. (ङ) १. क्रमतः for (क्रमात) २. नंयेदुपरि for (नंयेदुपर्युपरि)
३. ध्रुवा for (ध्रुवास्)
५. मध्यमाः स्युस्ते for (तत्परांशाः स्यु)

३३. (ङ) १. दशगुणितो for (दिग्गुणितो) २. चंद्रैर्हृतः for (चंद्रहृतः)
३. क्षेप्यः for (क्षिप्तः)

३४. (ङ) १. ऽब्धिगुणितः for (विवस्वान) २. च सप्ताहतो for (सस्ताहतो)
३. ऽब्धिधृतिभक्तः for (न विधृतिविभक्तः)
४. संयुतो विधेयो for (संयुक्तादेयो)
५. ध्रुवको for (ध्रुवके)

भट ब्रह्माचार्येण[४] जिष्णुतनयेन[६] गणितगोलविदा ।
आर्याष्टसहस्रेण[१] स्फुटसिद्धान्ते[२] ऋतो[३] ब्रह्मा[५] ।। १० ।।
भग्रह[१] युति[२] वतशंकु[२] वित्रिभलग्नाद्रविग्रहोक्तसमः[७] ।
शशिनः[३] कर्मबहुत्वान्न[४] कृतातो[५] भास्करग्रहणे ।। ११ ।।
आज्ञेयी[१] नैरूत्योरुद्दिष्टदिने[२] स्थितस्य[३] योऽर्कस्य[५] ।
शंकुच्छाये[४] कथयत्यब्दापि वेत्ति सूर्यशः[५] ।। १२ ।।
अत्र[४] मया यन्नोक्तं गोलादप्रेक्ष्यधिमतोह्यं[४]तत्[५] ।
आर्यात्र्योदशोऽयं[२] संज्ञाध्यायश्चतुर्विंशः[३] ।। १३ ।।

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१०. (घ) १. सहस्रेणः (ग) सहस्त्रेण for (सहस्रेण)
२. सिद्धंते (ग) सिद्धांतः(ङ) for (सिद्धान्ते) ३. कृतो (ग)(च)(ङ) for (ऋतो)
(ग) ४. ब्रह्माचायेण for (ब्रह्माचार्येण) ५. ब्राह्मः for (ब्रह्मा)
(च) १. सहस्रेणः for (सहस्रेण), (ङ) ६. जिष्णोस्तनयेन for (जिष्णुतनयेन)
११. (घ) १. नग्रह for (भग्रह) २. नार्च्छकु for (वतशंकु)
३. शशनः for (शशिनः) ४. बहुत्वन्ने for (बहुत्वान्न)
५. कृतोऽतो (ङ) for (कृतातो), (ग) ६. वित्रिभलग्नात् for (वित्रिभलग्नाद्)
(च) ५. बहुत्वान्नक्कतोऽतो for (बहुत्वान्नकृतातो)
(ङ) २. वच्छंकु for (वतशङ्कु) ७. ग्रहोक्तसमः for (ग्रहोक्तसमयः)
१२. (घ) १. आज्ञेयीनैऋत्यो (ग) आज्ञेयैनैऋत्यो for (आज्ञेयीनैरुत्यो)
२. रुद्दिष्टदिने for (रूद्दिष्टदिने) ३. स्वितस्य योऽर्कस्व for (स्थितस्य योऽर्कस्य)
४. कथयत्या ···(ग) कथयत्यब्दादपि for (कथयत्यब्दापि)
५. सः (ग) (ङ) for (शः), (च) १. आग्नेये for (आज्ञेयी)
६. नैर्ऋत्ये for (नैरुत्यो) २. वेष्टदिने for (रूद्दिष्टदिने)
३. संस्थितस्य for (स्थितस्य) ४. कथयति वर्षादपि for (कथयत्यब्दापि)
१३. (घ) १. दत्प्रेक्ष्य (ग) दुत्प्रेक्ष्य for (गोलादप्रेक्ष्य)
२. आर्यात्रयोदशो (ग) आर्यत्रयोदशायं for (आर्यात्र्योदशोऽयं)
३. चतुर्विंशः (ग) for (चतुर्विंशः)
(ग) ४. (वि०—यहाँ समाप्तिसूचक ।।छः।। ।।छः।। ।।छः।। अंकित है)
(च) १. गोलादत्प्रेक्ष्य for (गोलादप्रेक्ष्य)
४. धीमती for (धिमतो) २. आर्यात्रयोदशोयं for (आर्याश्योदशोऽयं)
३. चतुर्विंशः (ङ) for (चतुर्विंशः) (ङ) १. गोलादुत्प्रेक्ष्य for (गोलादप्रेक्ष्य)
४. धीमता for (धिमतो) ५. वोह्यम् for (ह्यं तत्)
(वि०+इति श्रीमदाचार्य जिष्णुसुतब्रह्मगुप्तविरचिते ब्राह्मस्फुटसिद्धान्ते संज्ञाध्यायश्चतुर्विंशतितमः सम्पूर्णतामगमत्+)

गणितं बहुप्रकारं गोलो यंत्राणि यत्र कथितानि ।
स ब्रह्मगुप्तविहितः[1] स्फुट सिद्धांतो[2] स्मृतो ब्रह्म[3] ॥ १४ ॥[4]
सिद्धांतेपि[1] स ब्राह्मघं तपसा भक्तधा[2] वयो महादेवम् ।
आराध्यकर[3] नमस्तस्मै श्री ब्रह्म गुप्ताय ॥ १५ ॥
इति श्री भिलमाचार्यभट[4] जिष्णु[5] सुत ब्रह्मगुप्तविरचिते ब्राह्म-
स्फुटसिद्धान्ते संज्ञाध्यायः
ब्रह्म गुप्त सिद्धांतः समाप्तः ॥

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१४. (घ) १. विहतः (च) for (विहितः) २. सिद्धांतः (च) for (सिद्धान्तो)
३. ब्रह्मः for (ब्रह्म) (ग) 'ग' में यह श्लोक अंकित नहीं है।

(च) ३. ब्रमः for (ब्रह्म) ४. क्रमसंख्या—१ for (१४)

(ङ) श्लोक अनुपलब्ध है।

१५. (घ) १. संब्राह्मघं for (सब्राह्मघं) २. (च) +च+
३. आराध्य च कर for (आराध्यकर) ४. भिल्लमा for (भिलमाचार्य)
५. भट्ट for (भट)

६. अतिरिक्तपाठ—चतुर्विंशतितमः समाप्तोयं श्रीब्रह्मगुप्तकृतो ब्रह्मस्फुट सिद्धांतः। ब्रह्मसिद्धांतबीजानि। खखखांङ्क हृताप्येभ्योः। १२००० गतगम्यात्पात एव शून्य यमलहृता। २००, तल्लब्धं त्रि ३ सायक ५ हतं कलाभिरूनौ। सदार्कंदु विधुवद् जीवद्धि।२। हतंचंद्रोस्वतिथि १५ गुणं च शतशीघ्रे द्वीषुं ५२ हतं स्वक् चलेद्विकु १ वेदधहतं च पातकुजशशनिषु २ कल्याणं भूयात् ॥१॥ संवत् १५५४ वर्षे फाल्गुणवदि ४ चतुर्थी रवौ लिखितं भयपुरवास्तव्य रेवातटे वास्तव्य ब्रह्मसिद्धांतः। ग्रंथ संज्ञा।

(ग) 'ग' में यह श्लोक अंकित नहीं है। (ङ) श्लोक अनुपलब्ध है।
चतुर्विंशतितमः समाप्तोऽयं श्रीब्रह्मगुप्तकृतो ब्रह्मस्फुटसिद्धांतः। ब्रह्मसिद्धांत-बिजानि खखखांङ्क हृताद्येभ्यो १२००० गतगम्यात्पात एव शून्य यमलहृता २०० तलब्धं त्रि ३ सायक ५ हतं कलाभिरूनौ सदार्कंदु विधुवत् जीवेद्धि २ हतं चन्द्रोस्वतिथि १५ गुणं च शतशीघ्रे द्वीषुं ५२ हतं स्वंक्चलेद्विकु १ वेद ४. हतं च पात कुजशनिषु। २ कल्याणं भूयात् १ श्री ॥ श्री ॥ for (ब्रह्मगुप्त सिद्धांतः समाप्तः)

अथ ब्रह्मगुप्तकृतो ध्यानग्रहोपदेशाध्यायः

॥ ॐ नमः श्री सरस्वत्यै ॥

पंचाशत्संयुक्तैर्वर्षशत[1] पंचभिर्विना शाकः ।
त्रिस्थोर्कैर्वसुवेदैर्नवचन्द्रै[2]स्ताडितः क्रमशः ॥ १ ॥
[1]समायकसंयुक्तः खजिन २४० विभागो नितः सयमवेदैः ।
मध्यमराशिः शशिविश्वभाजिततोह्यधिक मासा स्युः ॥ २ ॥
तैरुपरितनो युक्तो मासगणोभ्यधिक [1]शेषके[2] शुद्धे ।
घटिकादिक[3] भचक्राद्रविरव[4]शेषो भव[5]द्भादि ॥ ३ ॥
रूपेण रूपरामैः खसायकैस्ताडितो गुणो[1] युक्तः ।
षड्भिः [2]करैश्च दिग्भि[3]राघटिकां विघटिका स्युः ॥ ४ ॥
खखरसलब्धं घटिकासु नियोजये[1] तिथिघ्र[2] ।
रव्यादिकस्तदुदये [3]चैत्रदवेर्क [4]चन्द्रो [5]वा ॥ ५ ॥

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१. (ङ) १. शतैः for (शतै) २. त्रिष्ठोऽर्कै for (त्रिस्थोर्कै)

२. (ङ) १. पंचाब्धियुतोऽधः षष्टिभाजितो लब्धियुक् सरसवेदः ।
for
समायक संयुक्तः खजिन २४० विभागो नितः सयमवेदैः ।
२. मध्यमराशि विश्वैर्विभाजितोऽभ्यधिकमासाः स्युः ॥ २ ॥
for
मध्यमराशिः शशिविश्वभाजिततोह्यधिक मासा स्युः ॥ २ ॥

३. (ङ) १. शेषकः for (शेषके) २. शुद्धः for (शुद्धे)
३. घटादिको for (घटिकादिक) ४. रविशेषो for (खशेषो)
५. भवेद्भादिः ॥ ३ ॥ for (भवद्भादि) ॥ ३ ॥

४. (ङ) १. गणो for (गुणो) २. वेदैर्घृत्या for (करैश्च)
३. वासरघटिका for (दिग्भिराघटिकां)

५. (ङ) १. नियोजयेत् for (नियोजये) २. तिथिध्रुवकाः for (तिथिघ्र)
३. चैत्रादावर्क for (चैत्रदवेर्क) ४. चन्द्रौ for (चन्द्रो)
५. च for (वा)

मासगुणो[१] यमगुणितः पृथक् द्वितो[२]द्धृतः फलसमेतः ।
सार्धा[३]त्तयुतो वसुयम विभक्तः[४] शेषाधोः[५] केन्द्रम् ॥ ६ ॥
चैत्रादिमासगुणिते द्वे नक्षत्रे क्षिपेत्सहस्रांशौ ।
घटिकैकावशयुक्त[१]ः सार्द्धेन व[२] पलेन रहिते[३] वा[४] ॥ ७ ॥
नाडघर्द्धेन समेतं द्वितयं प्रक्षिपेच्च शशिकेन्द्रे ।
रूपं रूप[१] हुताशांखशराश्च तिथिध्रुवे क्रमशः ॥ ८ ॥
वारं दद्यात्प्रतिदिनमब्धिपलोनां परित्यजे[१]न्नाडीम् ।
केन्द्रे क्षिपे[२]द्भूमेकं सद्वियमलं घटीचतुष्कमितौ[३] ॥ ९ ॥
उज्जय[१]नी याम्योत्तर रेखायाः प्राग्धनं क्षयः पश्चात् ।
योजन[२]खाष्टि ८० नाडी चरदलमपि सौम्य दक्षिणयोः ॥ १० ॥
तिथि[१]यो दशभागो[२] रवि[३]धिष्ण्यसमन्विताः शशी[४]मध्यः ।
तिथिभोग नाडिकोनं कर्त्तव्यं शीतगोः केन्द्रः ॥ ११ ॥

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६. (ङ) १. मासगणो for (मासगुणो) २. कुतत्त्वोद्धृतः for (द्वितोद्धृतः)
३. सार्धाष्ट for (सार्धात्त)
४. ५. विभक्तशेषो विधोः for (विभक्तः शेषाधोः)

७. (ङ) १. युक्ते for (युक्तैः) २. फलेन for (वपलेन)
३. सहिते for (रहिते) ४. च for (वा)

८. (ङ) १. हुताशाः for (हुताशां)

९. (ङ) १. भद्वियमलं for (सद्वियमलं) २. मिते for (मितौ)

१०. (ङ) १. उज्जयिनी for (उज्जयनी)
२. षष्टया नाडी for (खाष्टि ८० नाडी)

११. (ङ) १. तिथयो for (तिथियो) २. भागोना for (भागो)
३. रविणा समन्विता for (रविधिष्ण्यसमन्विता)
४. शशी भवति मध्यः for (शशीमध्य)
दूसरी पंक्ति—तिथ्यंशाढया शोध्यास्तिथि भोगजनाडिकाः केन्द्रा[२] ।
for
तिथिभोगनाडिकोनं कर्त्तव्यं शीतगोः केन्द्रः ॥

सप्तहस्त्रिवसुहृतो गुरोः शनिर्द्विगुणितो नवेषु हृतः ।
दिग्गणितो रसधृतिहृद्राहो लिप्ता सुकृतलिप्ताः ॥ ३५ ॥
त्रिगुणो दलितः शुद्धादशयुकसित चलध्रुवे देयः ।
तात्कालिकं चलं स्याद्रविरन्येषां ज्ञशुक्रौ सहः ॥ ३६ ॥
मंदांशो नगरवयो भयमाः खनगेंदवः खनंदाश्च ।
यमतत्वानि तदूनान्यथाज्ञात्सूर्यवग्द्राह्याः ॥ ३७ ॥
रदगुणिता सप्तकृता कुजस्य सौम्यस्य युगगुणा गताः ।
द्विगुणैव फलं सूरेर्द्विगुणा त्रिभाजिताः स्फुजिताः ॥ ३८ ॥
त्र्यंशद्भागोनास्य त्रिगुणारविजाय मंदफलं लिप्ताः ।
मंदफलदलोनयुतं शीघ्रोच्चात् शोधयेन्मध्यः ॥ ३९ ॥
तस्मात् शीघ्रफलदलं स्वमृणं वामंदसंस्कृतं कृत्वा ।
प्राग्वनमंदफलमतः सकलं मध्यग्रहे कुर्यात् ॥ ४० ॥

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३५. (ङ) १. हृत for (हृस्त्रि) २. दिग्गुणितो for (दिग्गणितो)
३. लिप्तः for (लिप्ता)

३६. (ङ) १. स्वद्वादशांशयुक्तः for (शुद्धादशयुक)
२. सितचलं ध्रुवं स्यात् for (सितचलध्रुवे देयः)
३. स्तः for (सहः)

३७. (ङ) १. मंदाशा for (मंदांशो) २. तदूनान्मध्याज्ज्या for (तदूनान्यथाज्ञात्)
३. सूर्यवत् ग्राह्या for (सूर्यवद्ग्राह्याः)

३८. (ङ) १. हृता for (कृता) २. नगगुणा for (युगगुणा)
३. त्रिहृता for (गताः) ४. द्विगुणा हि for (द्विगुणैव)
५. गुणाग्निविभाजिता for (गुणात्रिभाजिताः)
६. स्फुजितः for (स्फुजिताः)

३९. (ङ) १. त्रिगुणा त्रिंशद्भक्ता रविजस्य फलस्य मंदफललिप्ताः ।
मन्दफलयुतोनं स्वशीघ्रोच्चाच्छोधयेन्मध्यम् ॥ ३९ ॥
for
त्र्यंशद्भागोनास्य त्रिगुणारविजाय मंदफलं लिप्ताः ।
मंदफल दलोनयुतं शीघ्रोच्चात् शोधयेन्मध्यः ॥ ३९ ॥

४०. (ङ) १. संस्कृते for (संस्कृतं) २. दत्त्वा for (कृत्वा)
३. प्राग्वन्मंद for (प्राग्वनमंद) ४. मान्दग्रहात् for (मध्यग्रहे)

तस्मत्पृथक् स्थितादपि शीघ्रोच्चाद्दषितशोधिताच्चफलेन ।
सकलेन संस्कृतो मृदुफलस्फुटो जायते स्पष्टः ॥ ४१ ॥
भागीकृतचलकेन्द्रत्रिगुणैश्चद्युद्धृते फलं पिंडा ।
षड्राशधिकेन चक्राद्विशुद्धशेषे तथैव स्यात् ॥ ४२ ॥
पिंडांतरेण गुणितं शाखं खाद्युद्धृतक्रमं वेयम् ।
उत्क्रमविधौ विशोध्यं गतपिंडे शीघ्रफलमेत् ॥ ४३ ॥
पिंडतवि विकणं गुणौ यदाद्येन पिंडके ततः ।
यल्लभ्यते खवेदैस्तदेव फलमत्र बोद्धयम् ॥ ४४ ॥
पिंडे चतुर्दशे विश्वपिंडगुणितान्मखोद्धृताद्धि कलात् ।
लब्धेन विश्वपिंडो रहितः शेषं फलं भवति ॥ ४५ ॥

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४१. (ङ) १. तस्मात् for (तस्मत्) २. सितादि for (स्थितादपि)
४. विवर्जितात् (स्फुटंकेन्द्रम्) for (आद्दषितशोधिताच्च फलेन)
५. तस्मात् शीघ्र फलेन संस्कृतः for (सकलेन संस्कृतो)
६. 'मृदुफल' पद लुप्त है।

४२. (ङ) १. केन्द्रे for (केन्द्र) २. त्रिगुणै for (त्रिगुणैश्)
३. खाग्न्युद्धृते for (च द्युद्धृते) ४. पिंडः for (पिंडा)
५. षड्राश्यधिके for (षड्राशधिकेन)
६. विशोध्य for (विशोध्य) ७. शेषेण for (शेषे)
८. पिंडः for (तथैव)

४३. (ङ) १. गुणिते for (गुणितं) २. शेषे for (शाखं)
३. खाब्ध्युद्धृते for (खाद्युद्धृत) ४. क्रमाद्देयम् for (क्रमंदेयम्)
५. विधौ for (विधो) ६. पिंडे for (पिंडे)
७. मेतत् for (मेत्)

४४. (ङ) १. पिंडाभावे for (पिंडतवि) विकलं for (विकणं)
२. गुणयेदाद्येन for (गुणौ यदाद्येन)
३. पिंडकेन for (पिंडके) ४. गण्यंतेतु for (यल्लभ्यते)
५. बोद्धव्यम् for (बोद्धयम्)

४५. (ङ) १. चतुर्दश for (चतुर्दशे) २. विश्वैगुंणिते for (विश्वपिंडगुणितान्)
३. नखोद्धृते विकलाः for (मखोद्धृताद्धि कालात्)

वसुवेदयुगेति—भौमस्य—४८ । ९४ । १४० । १८४ । २२८ । २७० ।३०६।
३३९ । ३५९ । ३६६ । ३४२ । २६८ । १०८ । ० ।

(नोट—रेखाङ्कित अङ्क कटे हुए हैं।)

बुधस्य—३३ । ६६ । ९८ । १२८ । १५४ । १७७ । १९४ । २०४ । २०४ । १८८ । १५७ । १०७ । ३६ । ० ॥

गुरोः—१८ । ३६ । ५० । ६६ । ७८ । ८६ । ९० । ८८ । ८४ । ७४ । ५८ । ३६ । १६ । ० ॥

शुक्रस्य—५० । १०० । १५० । १९९ । २४६ । २९६[१] । ३३३ । ३३१[२] ।४०० । ४१८ । ४०८ । ३४० । १५० । ० ॥

शनेः—११ । २२ । ३१ । ३८ । ४४ । ४६[३] । ४८ । ४६ । ४२ । ३४[४] । २०[५] । । १६ । ६ । ० ॥

रूपगुणा ३१ वाणजिना २४५ । शर ५ । षट्नव ९६ । यम २। गुणा ३ क्रमशः ॥ मध्यमभुक्तिकलाः ॥ स्फुटषट् द्वि २६ रदा ३२ वसु ८ रवेशा १७ ।

पिंडकफलन[१] विभागो भागादि फलगृहे[२] स्वमृणं[३] तत् ।
चलकेंद्रमेषतुलादि[४] संस्थिते[५] कारयेत्क्रमशः ॥ ४६ ॥
वसुवेदायुगनंदाः खवेदचन्द्राः समुद्रवसुचंद्राः ।
वसुयमभलागगनाद्भृत बाहवो रसनभो रामाः ॥ ४७ ॥

(ङ) १. २९० for (२९६) २. ३७१ for (३३१) ३. ४८ (४६)
४. ३५ for (३४) ५. २७ for (२०)

४६. (ङ) १. पिंडफल नवमभागो for (पिंडकफलनविभागो)
२. फलं for (फल) ३. ग्रहेषु वा for (गृहे)
४. स्वमृणम् for (स्वमृणंतत्)
५. चलकेंद्रे for (चलकेन्द्र) ६. मेषादौ for (मेष)
७. तुलादिके for (तुलादि)
८. 'संस्थिते' पदं लुप्त है।

४७. (ङ) १. वसुयमयमा रसनभोरामा नन्दाग्निरामाश्च ॥ ४७ ॥
for
(वसुयमभलागगनाद्भृत बाहवो रसनभो रामाः ॥ ४७ ॥)

[1]वेदसुरोगोक्षगुणा रसरसरामविलोचना द्विगुणाः ।
[2]वसुरसयमा वसुदिशो नभस्य कुजशीघ्रपिंडा स्युः ॥ ४८ ॥
गुणरामाः षट्करसा वसुनंदारामलोचन शशांकाः ।
सागर विषयशशांका नगनगचन्द्राः कृतांकभुवः ॥ ४९ ॥
वे वनखा जलधिनखा वसुवसुचंद्रास्तुरंग विषयभुवः ।
तुरगदिशो रसरामा नभश्चपिंडास्तशशिसूनोः ॥ ५० ॥
[1]द्वूरस्त्रिगुणाः खशराः [2]षड्रसा गजनगा रसाप्तौ [3]च ।
[4]खांका [5]भुजंगरसवः सागर वसवः समुद्रनगाः ॥ ५१ ॥
[1]भुजगशरारस रामारसेंदवश्चंद्रपिंडकाः सूरेः ।
[2]वक्राद्विशुद्धशेषः स्फुटो भवेत्सिंहिकासूनुः ॥ ५२ ॥
खशराः [1]शनंखति तिथयो [2]वसुवसुवंशः शरा विनाशत्याः ।
[3]शशिन वयमागुणसुराः [4]कुनवगुणाः शून्यखांबुधयः ॥ ५३ ॥

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४८. (ङ) १. मोक्षगुणा रसरसरामा विलोचनाब्धिगुणाः ।
for
(वेदसुरोगोक्षगुणा रसरसराम विलोचना द्विगुणाः ।)
२. वसुवसुयमा वसुदिशो नभश्च कुजशीघ्रपिंडाः स्युः ॥ ४८ ॥
for
वसुरसयमावसुदिशो नभस्य कुजशीघ्रपिंडा स्युः ॥ ४८ ॥

४९. (ङ) १. गजविलोचन for (रामलोचन)

५१. (ङ) १. धृतिरसगुणाश्च for (द्वूरस्त्रिगुणाः)
२. षट्क for (षड्)
३. रसाष्टौ for (रसाप्तौ)
४. खाङ्काश्च for (खांका)
५. भुजगवसवः for (भुजंगरसवः)

५२. (ङ) १. 'चन्द्र' पद लुप्त है ।
२. रसेन्दवः पिण्डकाः for (रसेंदवश्चन्द्रपिंडकाः)
३. चक्रा for (वक्रा)

५३. (ङ) १. शतंखतिथ्यः for (शनंखतितिथयो)
२. सागरनन्देन्दवोऽङ्कजिनाः for (वसुवसुवंशः शरा विनाशत्याः)
३. गुणगुणरामाः for (शशिन वयमागुणसुराः)
४. कुनगगुणाः for (कुनवगुणाः)

कुंजरचन्द्र समुद्रागजाभ्रवेदानभो[१] बुधिज्वलनाः ।
गगनशिलीमुखचन्द्रा वियच्चर्पिंडाः सुरारिगुरौ[२] ।। ५४ ।।
रुद्रायमाः[१] कु[२]गणा वसुरामाः सागराम्बुनिधयश्च[३]ः ।
रसवेदागजवेगजवेदाष[६]डवयो[५] लोचनांबुधयः[४] ।। ५५ ।।
पंचगुणा[१] सप्तयमा रसचंद्राः षण्ण[२]भश्च रविसूनोः ।
अद्यतनस्वसूनयोर्ग्रहयोर्विवरं भवे[३]द्भुक्तिः ।। ५६ ।।
रूपगुणा[१] बाणजिनाः शराः षडंकायमौ गुणाः क्रमशः ।
मध्यम[२]भुक्तिकलाः स्युरत्कृतिरदखास्तखेखरा विकलाः ।। ५७ ।।
मदस्फुट[१] खंडगुणा भुक्ति खखनवहृतागतिज्या स्यात् ।
ग्रहवत्[२]तत्फलविधिमृदुकेंद्रवशाद्गतौ स्वमृणम् ।। ५८ ।।

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५४. (ङ) १. नभोऽम्बुधि for (नभोबुधि) २. गुरोः for (गुरौ)

५५. (ङ) १. द्वियमाः for (यमाः) २. कुगुणा for (कुगणा)
३. च for (चः) ४. वसुवेदा for (रसवेदा) ५. 'वेगज' लुप्त है ।
६. गजवेदाः for (गजवेगजवेदा) ७. षडब्धयो for (षडवयो)

५६. (ङ) १. पंचगुणाः for (पंचगुणा) २. षड्नभश्च for (षण्णभश्च)
३. रूपगुणा ३१ बाणजिनाः २४५ शर ५ षण्णव ९६ यम २ गुणाः ३ क्रमशः ।। ५६ ।।
for
(अद्यतन स्वसूनयोर्ग्रहयोर्विवरं भवेद्भुक्तिः ।। ५६ ।।)

५७. (ङ) १. मध्यमभुक्तिकलाः स्युः षड् द्वि २६ रदाः ३२ खंवसु ८ शका ११ विकलाः
for
(रूपगुणा बाणजिनाः शराः षडंकायमा गुणाः क्रमशः ।)
२. मन्दगुणिताभुक्तिः खखनवविहृताभुक्तिः स्यात् ।। ५७ ।।
for
(मध्यमभुक्तिकला स्युरत्कृतिरदखास्तखेखरा विकलाः ।। ५७ ।।)

५८. (ङ) १. ग्रहवत् तन्मन्दफलं मृदु केन्द्रवशात् स्वमृणं तदूनां च ।
for
(मंदस्फुट खंड गुणा भुक्ति खखनवहृतागतिज्यास्यात् ।)
२. शीघ्रगतिं सङ्गुणयेदेवं शीघ्रस्य खण्डेन ।। ५८ ।।
for
(ग्रहवत्तत्फलविधिमृदुकेन्द्रवशाद्गतौ स्वमृणम् ।। ५८ ।।)

न[१] द्रिहता शीघ्रगतिः संगुणाय शीघ्रभोग्यखंडेन ।
खार्कैर्विभजेल्लिप्ता भौमादीनामतिशीघ्रम् ।। ५९ ।।
स्वमृणं क्रमोत्क्रमविधौ[१] चतुर्दशो विश्वपिंडको गुणकः ।
हारः[२] ख रसागगतिरे वक्रिणोद्बोधिता चक्रम् ।। ६० ।।
भुक्तेरपि प्रदलिते द्वे सकलफले द्वे च कारयेद् ग्रहवत् ।
मांदंफलं त्रितयं वक्रोच्चस्य द्वितो व्यस्तम् ।। ६१ ।।
नवतिथयो ६२ चरषंडा विज्यावत् ।। ६३ ।।

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५९. (ङ) १. पिंडान्तरेण खार्कैः १२० लिप्ताद्यं स्यात् फलं गतेः शीघ्रम् ।
for
(नद्रिहता शीघ्रगतिः संगुणाय शीघ्रभोग्य खंडेन ।)
२. स्वमृणं क्रमोत्क्रमविधौ चतुर्दश विधिश्च पिंडको गुणकः ।। ५९ ।।
for
(खार्कैर्विभजेल्लिप्ता भौमादीनामतिशीघ्रम् ।। ५९ ।।)

६०. (ङ) १. हरस्वगतिरेवं वह्वृणात्याज्ये भुक्ते ।
for
(स्वमृणं क्रमोत्क्रमविधौ चतुर्दशो विश्वपिंडको गुणकः ।)
२. पददलिते द्वे द्वे सुकाले कारयेत् स्फुटा भुक्तिः ।। ६० ।।
for
(हारः खरसगगतिरेवक्रिणोद्बोधिता चक्रम् ।। ६० ।।)
नोट—(नवतिपयोधिविभक्ता इत्यादि आर्यापट्कं खण्डखाद्याश्चित्यम्)

६१. (ङ) १. नवतिथयो १५९ ऽष्टि १६ विभक्ताः,
पंचरसा ६५ वसु ८ हृता दश १० त्रिहृताः ।
विषुवच्छगुणिताः स्वदेशजाश्चरदलविनाडयः ।। ६१ ।।
for
भुक्तेरपि प्रदलिते द्वे सकल फले द्वे च कारयेद् ग्रहवत् ।
मांदं फलं त्रितयं वक्रोच्चस्य द्वितो व्यस्तम् ।। ६१ ।।
(ङ) नोट—खण्डखाद्यस्य श्लोकाश्चैते ।

६२. (ङ) १. ज्या केन्द्रं स्फुटभानुं कृत्वा ये राशयश्चरार्धानि ।
भुक्तानि भोग्यगुणिताच्छेषात् खखधृतिहृतात्तुफलम् ।। ६२ ।।
for
(नवतिथयो ।। ६२ ।।)

६३. (ङ) गतिपादं पादोनां गतिं विशोध्यास्तत्काल उदये च ।
संसाधितस्य तस्य ग्रहस्य चरकर्म चान्यस्य ।। ६३ ।।
for
(चरषंडा विज्यावत् ।। ६३ ।।)

पंचदशहीनयुक्ता ६४ क्रांतिकालाद्विरसगुणाः ॥ ६५ ॥

त्रिज्यां विषुवच्छाया घातस्तच्छ्रुति विभाजिताश्चाप्तम् ।

अक्षो नित्यं याम्यो गोलवशाद्दिग्भवेत्क्रांतिः ॥ ६६ ॥

क्रांत्यक्षायुति वियोगादक्षणपदैः शोधितैर्दिनदलाभाः ।

भाश्रुतिकृत्योः कृतमनुयुतो नयोस्तत्पदे व्यस्ते ॥ ६७ ॥

षड्गुणितागत शेषा नाड्यो दिवसार्द्ध विभाजितात् स्यात् ।

दिनदलकर्णगुणाप्ता तया त्रिज्यया फलं कर्णः ॥ ६८ ॥

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६४. चरदल विनाडिकागति कलावधात् खखरसाग्नि ३६०० लब्धकलाः ।
ऋणमुदयेऽस्तमये धनमुत्तरगोलेऽन्यथा याम्ये ॥ ६४ ॥

for

(पंचदशहीनयुक्ता ॥ ६४ ॥)

६५. पंचदशहीनयुक्ताश्चरार्धं नाडीभिरुत्तरे गोले ।
याम्ये युक्त विहीना द्विसङ्गुणा रात्रिदिननाड्यः ॥ ६५ ॥

for

(क्रांतिकालाद्विरसगुणाः ॥ ६५ ॥)

६६. (ङ) १. मिश्रेशृन्तरगुणिता भुक्तिर्दिवसे निशाद्दले प्रथमे ।
षष्टया विभज्य लब्धं विशोध्य तात्कालिको भवति ॥ ६६ ॥

for

(त्रिज्यां विषुवच्छाया घातस्तच्छ्रुति विभाजिताश्चाप्तम् ।
अक्षो नित्यं याम्यो गोलवशाद्दिग्भवेत्क्रांतिः ॥ ६६ ॥)

६७. (ङ) १. 'क्षा' लुप्त है ।
२. दक्षपदैः for (दक्षणपदैः) ३. शोधिते for (शोधितै)
४. दिनदलेभा for (दिनदलाभाः) ५. नयाकृत्वकर्षः स्यात् ॥ ६७ ॥

for

(नयोस्तत्पदे व्यस्ते ॥ ६७ ॥)

(क्रान्त्यक्ष युति वियौगाच्चक्रपदात्शोधिते दिनदले भा ।
भाश्रुतिकृत्योः कृतमनुयुतोनयोः कृतिरकर्षस्य ॥)

अयं पाठः साधुः

६८. (ङ) १. दिवसविभाजिता ज्यातत् for (दिवसार्द्ध विभाजितात् स्यात्)
२. कर्मगुणाः for (कर्मगुणा) ३. स्वानया for (प्तातया)
४. त्रिभज्याभक्तं for (त्रिज्यया)

(षड्गुणिता गतशेषा नाड्यो दिवसार्धभाजिता तज्ज्या ।
दिनदलकर्णगुणाऽप्तानया त्रिभज्या फलं कर्णः ॥ ६८ ॥) अयं पाठः साधुः

दिनदले[1] कर्णागुणे[2] त्रिज्या निहते[3] श्रवणोधृते[4] फलस्य धनुः ।
द्यु[5]दलगुणं तिथिभक्तं दिनगतशेषावसः क्रमशः ॥ ६९ ॥
ज्या[1]खंडोने शेषे ७० चापानयने नवशत ७१ ।
तिथिनक्षत्र ग्रहसंक्षिप्तोतिस्फुटश्चैवम् ॥ ७२ ॥[2]
दुर्जनकृतघ्न शत्रु[1]प्रभृतीनामेषनिवदातव्यः[2] ।
ध्यानग्रहाधिकारो जिष्णुसुतब्रह्मगुप्तकृतः ॥ ७३ ॥
आर्याः १०९३ ॥ ॥ छः ॥ इति श्री ब्रह्मगुप्त सिद्धान्तः समाप्तः । श्रीरस्तु ॥ छः ॥ स्वस्ति श्री संवत् १६७८ वर्षे शाके १५४४ प्रवर्त्तमाने । उत्तरायने । वसंत ऋतौ । माहामङ्गलप्रदा । वैशाख शुदि १० बुधे, समये श्री राजनगरमध्य वातव्यं·····लिखितमस्ति । स्वयं पठनार्थं । पठनार्थं तथा पुत्रपौत्र पठनार्थं तथा परोपकाराय । श्री कृष्णार्पणमस्तु । याद्रिशं पुस्त० । इत्यादि । भग्न द्रिष्टि । त्यादि । श्रीरस्तु ॥

इति समाप्तोऽयं ग्रन्थः ।

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६९. (ङ) १. दिनदलकर्णे for (दिनदलकर्णगुणे) २. त्रिभज्यागुणे for (त्रिज्या)
३. 'निहते' पद लुप्त है ।
४. श्रवणोद्धृते for (श्रवणोधृते) ५. शेषासवः for (शेषावसः)

७०. (ङ) १. ज्याखण्डोने शेषे गुणिते नवभिः शतैरशुद्धहृते ।
क्षेप्याणि शुद्धखण्डैर्गुणितानि शतानि नवचापम् ॥ ७० ॥ for
(ज्याखंडोने शेषे ॥ ७० ॥)
इति तिथिनक्षत्रदिनमाद्यादिकसिद्धौ ब्रह्मगुप्तेन ।
द्वासप्तत्यार्याणां संक्षिप्तोऽतिस्फुटश्चैषः ॥ ७० ॥ for
(चापानयने नवशत ॥ ७१ ॥)
२. ७२ वीं संख्या का श्लोक ७३ अंकित है । अगला पृष्ठ देखिये ।

७३. (ङ) इसकी श्लोक संख्या ७२ है ।
वामपार्श्वे लिखितम्] प्रथम दशध्याई पत्र प्रथम लितम् । पश्चात् कियत् दिवसे पंचदशाध्या इति लिखितमस्ति एवं पत्र ॥
१. प्रतिकञ्चुककारिणे for (प्रभृतीनामेष नि) २. न for (व)
३. +इति श्री ब्रह्मगुप्तकृतो ध्यानग्रहोपदेशाध्यायः समाप्तः ॥ +

अकारादिक्रमेण श्लोकानुक्रमणिका

अंशकशेषत्रियुतम्	२५२	अध्यायः पञ्चदशः	२१५
अंशकशेषा न्यूनाः	२३८	अध्वर्द्धार्द्धं क्षेत्रान्यूडुनि	१९६
अंशकशेषेण युताः	२५२	अनयोर्न कदाचिदपि	१४९
अंशसममंशके कलासमं	२५२	अन्तरमाद्यो भूयो	१०८
अकृतार्यभटः शीघ्रगम्	२१	अन्तरयोगो तुल्यान्यदिशोः	१०५
अक्षचरार्द्धाज्ञार्कम्	२०२	अन्त्यफलज्ययात्स्वान्य	१८४
अक्षज्याया वित्रिभलग्नात्	७९	अन्त्यानतो क्रमज्या	६२
अक्षज्या शंकुवधालम्ब	७१	अन्यत्र सर्वतोदिशम्	२७९
अक्षांशभूपरिधि	२८०	अन्या विक्षेपकला	१५०
अग्न्यष्टभिरिषुमनुभिः	४०	अन्येष्टनाडिकाभिः कृत्वा	१३१
अग्रांत्यमुपांत्येना	२३७	अन्यैरप्युक्तमिदं योयं	१३४
अग्राशंकुतलैक्यं	१२६	अपमेंवकृतः सूर्येंदु	३१९
अघनभंगणलब्धम्	१७८	अपसृतिरन्यशलाका	३०६
अंगचितिविजयनंदि	१५१	अम्बरयोजनपरिधिः	२८०
अंगैः रुद्रैः सिद्धैर्गजैः	३२७	अर्कफलभुक्ति घाताद्	३३
अत्र मया यन्नोक्तम्	३२१	अर्काग्रयाक्षकलंबकजीवे	२०८
अथवा कपालके	३१०	अर्काग्रावर्गोनं नृज्या	६७
अथवा जिनजभगणादि	२३५	अर्केंद्वतरघटिका	१४७
अधिकदिनोदितघटिकाभिः	१२८	अर्कोदयास्तमययोर्विना	२९
अधिकः स्मृत्युक्तमनोरार्य	१०	अर्कोनचन्द्रलिप्ता	३२६
अधिकाग्रभागहारा	२३३	अर्कोनचन्द्रलिप्ता	४५
अधिकैः शतैश्चतुर्भिः	१४०	अर्कोनलग्नहोरा पंच	१८०
अधिमासकैः सविकलैः	१७१	अर्द्धच्छिन्नो भानुः	२८७
अधिमासशेषपादा	२५०	अर्धज्याभूयमला	२३
अधिमासशेषवर्गम्	२५८	अर्ध्यर्द्धक्षेत्राणि संहिता	१९८
अधिमासाः शशिमासाः	८	अल्पप्रश्नासूनां यदि	६१
अधिमासोवमशेष	१७५	अवनतिरतोन्यथा भवति	७८
अध्यर्द्धा समक्षेत्राणाम्	१९७	अवमविकलं न सावनमेभिः	१७६

अवमाति यः सविकलैः	१७१	आर्यभटस्याज्ञानात्-	३४
अवमावशेषमवमैः	२५३	आर्यभटेनाभिम मतिलघुनि	१४३
अवमावशेषलब्धा	१७५	आर्यभटो जानाति	१३६
अवमावशेषवर्गम्	२५८	आर्यभटो युगपादांस्त्रीन्	१३८
अवमावशेषवर्गो	२३९	आर्याष्टशते पाता	१३९
अवलंबनं शलाकां	३०१	इति कथिततन्त्राणा नाम्	१५२
अविषमचतुरस्रभुजगति	१५८	इति परिलेखाध्यायः	२३२
अविषम पार्श्वभुजगुणकर्णो	१५९	इति बहुधा विवदंति	१५१
अव्यक्तवर्गधनवर्गवर्ग	२४७	इति बाहुकर्णकोटि	९८
अव्यक्ताँतरभक्तम्	२४८	इत्यपदकृतिस्त्र्यूना	२५५
अष्टचरार्धयस्य ज्या	२०९	इन्दुविनिघ्नाशेषम्	२५६
अष्टनखैर्मेपगविरदलिप्तोनै	११०	इन्दुविनिघ्नाशेषम्	२६१
अष्टयमाश्व्युत्यगुणा	२०७	इंदोर्विषया द्वितमा	२२८
अष्टयमैः कृतचन्द्रैः	४१	इषुशरकृताष्टदिग्भिः	२६३
असकृत् ग्रासकालोन	७५	इष्टकरन्यूनाया	२८७
असकृत् मांदे तत्फल	२८५	इष्टगुणकारगुणितम्	१७८
अस्ताँतर्घटिकाभिर्यो	२६७	इष्टगुणकारगुणितो	१६२
अह्नोगताऽवशेषाः	६४	इष्टग्रहभगणगुणादहर्गणात्	११
आकृतिफलमोव्याहत	१६५	इष्टग्रहेष्टशेषाद्युगणो	२६२
आचार्येण ज्ञातः श्री	२९७	इष्टग्रासविमर्दस्थित्यर्द्धा	८४
आज्ञेयी नैरूत्यो	३२१	इष्टग्रासोकेंद्वोः	२२७
आद्यघटिकांतरं वशात्	१२९	इष्टघटिकागुणानामसकृत्	१९४
आद्यंतरमिष्टाभिर्घटिकाभिः	१३१	इष्टज्या संगुणिता	१८९
आद्यंतरातसंधिषु कल्प-	३	इष्टदिनार्द्धनतांश	५३
आद्यन्तवर्ग योगान्मूलं	९५	इष्टदिवसार्द्धघटिका	२६९
आद्यन्त्ययोः सधूम्रः कृष्णः	७७	इष्टद्वयेन भक्तो द्विघेष्ट	१६१
आद्यन्ते वान्कृत्वा विक्षेपः	२२८	इष्टभगणादिशेषं	२५७
आद्यादनन्तरोधः	२७५	इष्टभगणादिशेषाः	२३९
आद्याद्वगणादिन्यान् व्यस्तान्	२५०	इष्टभगणादिशेषात्	२३७
आनयति दिवसवारम्-	११	इष्टभगोन भूदिनशेषैः	१७९
आनयति यस्तमो	२६५	इष्टशरस्य भक्ते ज्यार्द्धं	१६३
आयतकर्णौ बाहू	१६२	इष्टश्विन्योदयिकान्	१८२
आयनिवक्र्योनाम्	२०४	इष्टस्य भुजस्य कृतिः	१६१
आर्यभटः क्षेत्रांशैः	८९	इष्टाछायावृत्ते तदग्रयो	६९
आर्यभट दूषणानां	१४८	इष्टात्कालात् भानोरुदया	१२१

इष्टान्मध्यादन्यांस्तिथिम्	१७१	उद्दिष्टे कल्पकृतो	२७५
इष्टापक्रमवर्गमिज्यावर्गाद्	४३	उद्गणिताद्विशोध्य	१६४
इष्टार्कचरार्द्ध ज्या	२०८	उनदिवसोदिताभ्यो	१२६
इष्टाल्पराशिवर्गो युक्तोना	१७०	उन्नतजीवाकोटिः छाया	५१
इष्टाहतभक्तानाम्	१७७	उन्नतजीवाभक्तव्यासार्द्ध	६६
इष्टेषु मानदिवसे	२४०	उषिताय दीर्घकाल	२१६
इष्टोदयिकभुजांतर	१८६	ऊनदिनोदितगुणितात्	१०८
इष्टोद्धृतकराणी	२४६	ऊनाधिकशंकुगुणा	१२६
इह नोक्तानि बहुत्वात्	४८	ऊने मानैक्यार्द्धात् ग्रहयो	१२८
इह नोद्दिष्टं यत्तद्रविवद्	१०१	ऊनोत्पभुक्तिरुदितः	१२१
उंकारो दिनवारो गुरुः	१४०	ऊर्द्धांशा छेदगुणाः	१५५
उक्रमजीवा चापं	२१३	ऋग्वर्गः पर्यायः	२७२
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श्रीब्रह्मगुप्ताचार्य-विरचित

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(संस्कृत-हिन्दी-भाषायां वासनाविज्ञानभाष्याभ्यां समलंकृतः सोपपत्तिकः)

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Dedicated to

Shri S. K. Patil

Union Minister for Railways

विषयानुक्रमणिका

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३. **त्रिप्रश्नाधिकारः** २५९-३६०

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४. चन्द्रग्रहणाधिकारः ३६३-३९७

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५. सूर्यग्रहणाधिकारः ४००-४५६

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६. उदयास्ताधिकारः ४५७-४७६

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७. चन्द्रशृंगोन्नत्यधिकारः ४७९-५००

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८. चन्द्रच्छायाधिकारः ५०१-५१४

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९. ग्रहयुत्यधिकारः ५१५-५६७

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ब्राह्मस्फुटसिद्धान्तः

७

मध्यमाधिकारः

❀ श्रीगणेशाय नमः ❀

सकलगणकसार्वभौम-सर्वतन्त्रस्वतन्त्र-ग्रहादिवेधविधिज्ञ-श्रीब्रह्मगुप्ताचार्यप्रणीतः

ब्राह्मस्फुटसिद्धान्तः

विज्ञानभाष्योपपत्त्या तद्विन्द्यनुवचनिकया च सहितः

मध्यमाधिकारः

तत्रादौ सकलगणकसार्वभौमः सर्वतन्त्रापरतन्त्रोऽद्वितीयो ग्रहादिवेधविधिज्ञो ब्रह्मगुप्ताचार्यो निर्विघ्नग्रन्थपरिसमाप्तये स्वेष्टदेमहादेवानुस्मरणरूपं मङ्गलाचरणमातनोति।

जयति प्रणतसुरासुरमौलिगरत्नप्रभाच्छुरितपादः।
कर्त्ता जगदुत्पत्तिस्थितिविलयानां महादेवः॥ १ ॥

अथ भाष्यकृद् मङ्गलं परामृशति

यद्दन्तोदन्तलक्ष्मीं दशसु विकरितुं दिक्षु सोमार्कतारा-
सारा स्फारप्रसारा अनुकलमुदयं यान्ति बिभ्राजमानाः।
विघ्नक्ष्मोणोध्रपक्षक्षपणविषमुं मङ्गलोदञ्चिताङ्घ्रि-
न्यासं सिन्दूरभासं विनयविरचनैर्मन्महे सिन्धुरास्यम्॥ १ ॥

सूर्याङ्गारबुधेन्द्रवन्द्यभृगुजच्छायासुतान् खेचरान्
वाग्देवीं च निधाय चेतसि मुदा रामस्वरूपो बुधः।
ब्राह्मग्रन्थपयोधिलङ्घनचिकीर्षूणां कृते सश्रमं
सेवायामुपदीकरोति सुधियां विज्ञानभाष्यप्लवम्॥ २ ॥

आर्यब्रह्मवराहभास्करमुखास्त्रिस्कन्धविद्याचणा
भट्टः श्रीकमलाकरोऽथ विबुधौ श्रीसेनलल्लाभिधौ।
भूपेन्द्रो जयसिंहनामभृदसौ यो यन्त्रशिल्पी नव-
स्ते चान्येऽपि मुनीश्वरप्रभतयः श्रद्धास्पदं नः परम्॥ ३ ॥

गूढार्थैर्नु विसंज्ञतामुपगता श्रीब्रह्मणो भारती
तामुज्जीवयितुं श्रमेण मथितो ज्योतिस्त्रिपर्वाम्बुधिः ।
एतद्ग्राह्यरुचि पीयतामम्भुवि बुधै रामस्वरूपोदितम्
भूयो विज्ञमतल्लिकाभिरनघं विज्ञानभाष्यामृतम् ॥ ४॥

क्षीरमेव हि गृह्णन्ति हंसा सारविवेकिनः ।
एष्टुकामाः परे पङ्कं कोला लोला जलेष्वपि ॥ ५॥

वि. भा.—महादेवः (शङ्करः) जयति (सर्वोत्कर्षेण वर्त्तते), कीदृशः प्रणतसुरासुरमौलिगरत्नप्रभाच्छुरितपादः (प्रणता नतमस्तका ये सुरासुरा देवराक्षसास्तेषां मौलिगानि शिरोगतानि यानि रत्नानि हीरकादीनि तेषां प्रभाभिर्ज्योतिर्भिः, छुरितौ संमिश्रितौ पादौ चरणौ यस्य सः) पुनः कीदृश. जगदुत्पत्तिस्थितिविलयानां (जगतः संसारस्योत्पत्तिः प्रादुर्भावः स्थितिः संरक्षणमवस्थानं वा विलयो नाशस्तेषां कर्त्ता कारकोऽर्थात्संसारोत्पत्तिस्थितिविनाशानां कारणभूत इत्येतावता ब्रह्मगुप्तो महादेवमीश्वरं स्वीकरोतीति सिद्धयति, ईश्वरमन्तराऽन्येषां जगदुत्पत्तिस्थितिविनाशकरणसामर्थ्याभावात् । विषयस्यास्य दर्शनशास्त्रेण सम्बन्धोऽस्त्यतोऽत्र विशिष्य तद्विचारस्याऽवश्यकता नास्ति ॥ १ ॥

हि. भा.—प्रणाम करते हुए देवों और राक्षसों के मस्तक (शिर) पर स्थित रत्नों (हीरा आदि) की ज्योति से मिश्रित (मिले हुए) हैं दोनों चरण (पाँव) जिनके ऐसे महादेवजी सब तरह के उत्कर्ष से विद्यमान हैं, पुनः संसार की उत्पत्ति, स्थिति (अवस्थान) और विलय (नाश) के करने वाले हैं। ब्रह्मगुप्ताचार्य की इस उक्ति से महादेव में ईश्वरत्व सिद्ध होता है, क्योंकि जगत् की उत्पत्ति, स्थिति और विनाश करने की शक्ति ईश्वर से भिन्न किसी में नहीं हो सकती। इस विषय का दर्शन शास्त्रों से सम्बन्ध है इसलिए यहाँ उस पर विशेष विचार करने की आवश्यकता नहीं है ॥ १ ॥

इदानीं ग्रन्थारम्भप्रयोजनमाह ।

ब्रह्मोक्तं ग्रहगणितं महता कालेन यत् श्लथीभूतम् ।
अभिधीयते स्फुटं तज्जिष्णुसुतब्रह्मगुप्तेन ॥२॥

वि. भा.—ब्रह्मोक्तं (ब्रह्मणा कथितं) ग्रहगणितं (ब्रह्मसिद्धान्तः) यदस्ति महता कालेन श्लथीभूतं जातमर्थाद्बहुषु समयेषु व्यतीतेषु तत्र खिलीभूतत्वं समागतम् । तत् (तस्मात्कारणात्) जिष्णुसुतब्रह्मगुप्तेन (जिष्णुपुत्रब्रह्मगुप्तनामकेन मया) स्फुटं (स्पष्टं श्लथीभूतत्वरहितं वा) अभिधीयते । अधुनोपलब्धेषु ब्रह्मसिद्धान्तेषु शाकल्यसंहितान्तर्गत एकः, विष्णुधर्मोत्तरपुराणान्तर्गतो गद्यमयो द्वितीयः, पञ्चवर्षमययुगवर्णनात्मको वराहमिहिररचितपञ्चसिद्धान्तिकान्तर्गतस्तृतीयः । एतेषां मध्ये ब्रह्मगुप्तेन कतमः

स्फुटोऽभिधीयते इति स्पष्टं न कथ्यते, तथापि ग्रहभगणादिपठितमानानां समत्वाद्विष्णुधर्मोत्तरपुराणान्तर्गत एव ब्रह्मसिद्धान्तः प्रमाणीभूतत्वेन (आगमत्वेन) स्वीकृतो ब्रह्मगुप्तेन, 'युगमाहुः पञ्चाब्द' मित्यादिना पञ्चवर्षमययुगस्य तन्त्रपरीक्षाध्याये संहिताकारमतं वर्णयितुर्ब्रह्मगुप्तस्य मते ज्यौतिषवेदाङ्गं ब्रह्ममतं नास्तीति स्पष्टम्। इदं वराहमिहिरमताद्भिन्नं सुगणकैर्बोध्यम्। सिद्धान्ततत्त्वविवेके 'अहो विष्णुधर्मोत्तरं चापि सम्यङ् न बुद्धमिति' कमलाकरोक्तादपि तदेव सिद्धचति। बहुषु समयेषु व्यतीतेषु तत्र ब्रह्मसिद्धान्ते कीदृशी श्लथीभूतता समागता तन्निराकृतिर्ब्रह्मगुप्तेन कीदृशी कृतेति ब्राह्मस्फुटसिद्धान्ते कुत्रापि नोपलभ्यते। तत्कथनं तथ्यमतथ्यं वेति विवेचका गाणितिका विवेचयन्त्विति ॥ २ ॥

अब ग्रन्थारम्भ करने के कारण को कहते हैं।

हि. भा.—ब्रह्मा से कथित ग्रहगणित (ब्रह्मसिद्धान्त) जो है, बहुत समय में (कालान्तर में) उसमें खिलत्व आ गया है इसलिए मैं जिष्णुपुत्र ब्रह्मगुप्त उसको स्पष्ट कहता हूँ। इस समय ब्रह्मसिद्धान्त तीन प्रकार के उपलब्ध होते हैं। एक शाकल्यसंहितान्तर्गत, द्वितीय विष्णुधर्मोत्तरपुराणान्तर्गत गद्यमय और तृतीय पञ्चवर्षमययुगवर्णनात्मक वराहमिहिरकृत पञ्चसिद्धान्तिकान्तर्गत, इन तीनों में ब्रह्मगुप्ताचार्य किसको स्फुट करते हैं इस बात को स्पष्टरूप से नहीं कहते हैं तथापि पठित ग्रहभगणादिमानों के समत्व के कारण विष्णुधर्मोत्तरपुराणान्तर्गत ही ब्रह्मसिद्धान्त को ब्रह्मगुप्त आगमत्व करके स्वीकार करते हैं, पञ्चवर्षमय युग के तन्त्रपरीक्षाध्याय में 'युगमाहुः पञ्चाब्दं' इत्यादि से संहिताकार के मत को वर्णन करते हुए ब्रह्मगुप्त के मत में ज्योतिष वेदाङ्ग ब्रह्ममत नहीं है, यह स्पष्ट है। वराहमिहिराचार्य के मत से यह विरुद्ध है इसका गणक लोग विचार करें। सिद्धान्ततत्त्वविवेक में 'अहो विष्णुधर्मोत्तरं चापि सम्यङ् न बुद्धम्' इत्यादि कमलाकरोक्ति से भी स्पष्ट है। बहुत समय में (कालान्तर में) उस ब्रह्मसिद्धान्त में कैसी श्लथीभूतता आ गई और ब्रह्मगुप्त ने उनके निराकरण किस तरह किये, ये बातें ब्राह्मस्फुटसिद्धान्त में कहीं भी नहीं पाई जाती हैं, उनके कथन ठीक हैं या नहीं, विवेचक ज्योतिषी लोग विचार करें ॥ २ ॥

इदानीं ज्योतिःशास्त्रमूलभूतस्य सग्रहस्य भचक्रस्य चलनमाह।

ध्रुवतारात्रप्रतिबद्धज्योतिश्चक्रं प्रदक्षिणगमादौ।
पौष्णाश्विन्यन्तस्थैः सह ग्रहैर्ब्रह्मणा सृष्टम् ॥३॥

वि. भा.—पौष्णाश्विन्यन्तस्थैः (पौष्णं रेवती, अश्विनी, तयोर्मध्य -(सन्धि-)-स्थितैः) ग्रहैः (सूर्यादिभिः) सह (साकं) ध्रुवतारात्रप्रतिबद्धज्योतिश्चक्रं (ध्रुवद्वयगतरेखा ध्रुवयष्टी ध्रुवाक्षो वा तन्निबद्धं ज्योतिश्चक्रं ज्योतींषि तेजोमयानि नक्षत्राण्यन्यानि विशिष्टानि च तेषां चक्रं समूहं (भचक्रमित्यर्थः)

प्रदक्षिणगम् (दक्षिणावर्त्तक्रमेण चलायमानम्) आदौ (सर्वप्रथमं) ब्रह्मणा (जगदुत्पादकेन) सृष्टम् (रचितम्) अर्थादश्विन्यादौ स्थितैश्चन्द्रादिभिस्तदुच्चपातादिभिश्च साकं ध्रुवयष्ट्याधारेण भ्रमणशीलं ज्योतिर्मयपदार्थानां नक्षत्रादीनां चक्रं (गोलं) ब्रह्मणा रचितं यद्यप्यत्राचार्येणोच्चादीनां चर्चा न क्रियते तेषां स्वरूपाभावात् किन्तु सृष्ट्यादिकाले तेषामपि स्थानाङ्कनरूपसर्जनं कृतमिति ॥३॥

अत्र ज्योतिश्चक्रशब्देन भचक्रम् । भानां चक्रं भचक्रमेतावता यत्र भानि सन्ति स च गोलाकारः पदार्थः तदेव भचक्रशब्देन व्यवह्रियते, भचक्रे कथं गोलत्वं, भचक्रचलनज्ञानं, ग्रहाणां कथं पूर्वाभिमुखी गतिः चन्द्रादिग्रहाणां (चन्द्रबुधशुक्ररविकुजगुरुशनीनां) कथमेवमूर्ध्वाधः क्रमेण स्थितिरित्यादि-विविधविषयाणां विचारार्थं वटेश्वरसिद्धान्तोऽप्यवलोकनीयः भास्कराचार्येणापि 'सृष्ट्वा भचक्रं कमलोद्भवेन ग्रहैः सहैतद्भगणादिसंस्थै' रित्यादिना ब्रह्मगुप्तोक्तानुरूपमेव सर्वं कथितमिति ॥ ३ ॥

अब ज्योतिःशास्त्र के मूलभूत ग्रहसहित भचक्रचलन को कहते हैं।

हि. भा.—रेवती नक्षत्र और अश्विनी के मध्य में (सन्धि में) ग्रहों के साथ ध्रुवाक्ष (दोनों ध्रुवों में गई हुई रेखा) में बंधे हुए दक्षिणावर्त क्रम से भ्रमणशील ज्योतिश्चक्र (भचक्र) को सब से पहले ब्रह्मा ने बनाया अर्थात् अश्विन्यादि में स्थित चन्द्रादिग्रह और उनके उच्चादियों के साथ ध्रुवयष्टि के आधार पर भ्रमणशील नक्षत्र आदि ज्योतिर्मय पिण्डों के चक्र (गोल) को ब्रह्मा ने बनाया। यद्यपि यहां आचार्य उच्च आदि की चर्चा नहीं करते हैं क्योंकि उनके स्वरूप ग्रहों की तरह नहीं हैं किन्तु सृष्टि के आदिकाल में उनका भी स्थानाङ्कन किया ॥ ३ ॥

यहाँ ज्योतिश्चक्र शब्द से भचक्र समझना चाहिए। भचक्र (नक्षत्रों के गोल) शब्द से ज्ञात होता है कि नक्षत्र और ग्रह, सब जहां देखा जाता है गोलाकार पदार्थ है, उसी को भचक्र कहा गया है। भचक्र में गोलत्व क्यों है, भचक्रचलन क्या है और उसका ज्ञान कैसे होता है, ग्रहों की पूर्वाभिमुख गति क्यों है, ग्रहों (चन्द्र, बुध, शुक्र, रवि, भौम, गुरु, शनि) की इस तरह ऊर्ध्वाधः क्रम से स्थिति क्यों है, इत्यादि अनेक विषयों के विचार के लिए वटेश्वर सिद्धान्त का मध्यमाधिकार देखना चाहिए। भास्कराचार्य भी 'सृष्ट्वा भचक्रं कमलोद्भवेन ग्रहैः सहैतद्भगणादिसंस्थैः' इत्यादि से ब्रह्मगुप्त के अनुरूप ही कहते हैं ॥३॥

इदानीमनाद्यनन्तस्य कालस्य प्रवृत्तिमाह ।

चैत्रसितादेरुदयाद् भानोर्दिनमासवर्षयुगकल्पाः ।
सृष्ट्यादौ लङ्कायां समं प्रवृत्ता दिनेऽर्कस्य ॥ ४ ॥

वा. भा.—चैत्रशुक्लप्रतिपत्प्रभृति लंकोपलक्षितभूप्रदेशेऽर्कोदयाद्दिनोदयाः प्रवृत्ताः रविदिनवारः सृष्ट्यादौ कल्पादौ । ननु चैत्रादेरिति सिद्धं सितग्रहणमतिरिच्यते इति चेत्सत्यं । यतोऽत्र विप्रतिपन्ना बहवः कृष्णप्रतिपदादिकं मासमिच्छन्ति तद्

व्युदासाय सितग्रहणम् ॥ ज्ञापकान्यत्र वेदस्मृतिवाक्यान्यपि योज्यानि "यासौ-वैशाखस्यामावास्या तस्यामादधीत सा रोहिण्या संपद्यते सोपरपक्षत्रैधीः प्रविशति, अपरपक्षे श्राद्धं कुर्वित्यादि" स्मृतिवाक्यानि योज्यानि । अथैवमुच्यते "एवमाह, संवत्सरस्य प्रथमरात्रिर्यत् फाल्गुनो पौर्णमासी, योत्तरा एषोत्तमा या पूर्वा" एतदपि वेदवाक्यमिति चेत्, अत्र पूर्वायाः फाल्गुन्याः पौर्णमास्या मासं प्रत्युत्तमत्वं न संभवति । यतो द्वितीयदिने या प्रतिपत्तस्या अपि फाल्गुनीत्वं न व्याहन्यते । तेन वाक्येन चाह । संवत्सरस्य प्रथमरात्रिर्यत्फाल्गुनी पौर्णमासी योत्तरेति । ततश्चादित्वमेवमस्यां वक्तुं शक्यते, द्वयोरपि तिथ्योः फाल्गुनीत्वात् । एतावदवगम्यते, गता फाल्गुनी पौर्णमासी नाद्यापि फाल्गुनो मासो गत इत्येवं स्थिते विचार्यते शेषवाक्यैः सह यत्राविरोधस्तत्रादित्वम् तदत्र स्वेच्छया न कल्पयितुं शक्यते । लङ्कायामित्यनेन ज्ञापयति क चेदृशे मध्याह्ने तत्रास्तमयेऽन्यत्रार्धरात्रेऽन्यत्राष्टघटिकाधिके शते काले कालादेः प्रवृत्त्या दिनप्रवृत्त्या मासादीनां प्रवृत्तिसिद्धौ पृथगुपादानं श्रावणादौ वर्षादिति वृत्त्यर्थम् ।

वि. भा.—सृष्ट्यादौ (सृष्ट्यादिकाले) चैत्रसितादेः (चैत्रशुक्लप्रतिपदादितः) लङ्कायां भानोरुदयात् (लङ्कासूर्योदयकालात्) अर्कस्य दिने (रविवारे) दिनमासवर्षयुगकल्पाः समं (एककालावच्छेदेन) प्रवृत्ता बभूवुरिति शेषः । व्यापकस्यानाद्यनन्तस्य कालस्य विभागो दिनमासवर्षयुगकल्पद्वारा लोकव्यवहारार्थं कृतोऽस्ति, एतेषामेव विभक्तकालावयवानां (दिनमासवर्षादीनां) सृष्ट्यादिकाले एककालावच्छेदेन प्रवृत्तिर्नामादिर्भवति, तथा च सृष्ट्यन्ते(प्रलयकाले) तेषामेव विभक्तकालावयवान्नामतो भवतीत्येतदनुसारेणैवावतरणे 'अनाद्यनन्तस्य कालस्य' इत्यादिलेखनं युक्तियुक्तं भवितुमर्हति, कथमन्यथा व्यापककालस्य प्रवृत्तिकथनं युक्तियुक्तं सङ्गच्छते । भास्कराचार्येणापि सिद्धान्तशिरोमणौ 'लङ्कानगर्यामुदयाच्च भानोस्तस्यैव वारे प्रथमं बभूव' इत्यादिना ब्रह्मगुप्तोक्तानुरूपमेव कथयत इति ॥ ४ ॥

अब अनादि (जिसका आदि नहीं) और अनन्त (जिसका अन्त नहीं है) इस काल की प्रवृत्ति कहते हैं।

हि. भा.—चैत्र शुक्ल प्रतिपदादि से लङ्का में सूर्योदयकाल से रवि के दिन में सृष्ट्यादि (चैत्रशुक्ल प्रतिपदा रविवार लङ्कासूर्योदयकाल) में दिन मास वर्ष युग और कल्प इन सब की प्रवृत्ति एक ही समय में हुई। यहाँ व्यापक काल (जिसका आदि नहीं है और अन्त नहीं है) का विभाग दिन, मास, वर्ष, युग, कल्प के द्वारा लोकव्यवहार के लिए किया गया है। इन्हीं विभक्त कालावयवों (दिनमास-वर्ष आदि) की प्रवृत्ति सृष्ट्यादि काल में एक कालावच्छेदेन होती है और सृष्टि के अन्त (प्रलयकाल)में उन्हीं कालों का अन्त इसी के अनुसार अवतरण में, 'अनाद्यनन्तस्य कालस्येत्या'दि लिखना युक्तिसङ्गत होता है। ऐसा यदि नहीं होगा तो फिर व्यापक काल की

प्रवृत्ति कैसे युक्तियुक्त हो सकेगी । भास्कराचार्य भी सिद्धान्तशिरोमणि में 'लङ्का-नगर्यामुदयाच्च भानोस्तस्यैव वारे प्रथमं बभूव' इत्यादि से ब्रह्मगुप्तोक्तानुरूप ही कहते हैं इति ॥ ४ ॥

वि. भा.—अथाऽत्र शास्त्रे पृथिव्याः सम्बन्धेनैव ग्रहादिकक्षादीनां सर्वोपयोगि-विषयाणां ज्ञानं भवत्यत एतस्या आकृतिः कीदृशी तत्परिमाणं च कियदिति निर्णयार्थं विचारः । कुत्रचिद् वृक्षादिविरहितसमभुवि कियद्दूरस्थेष्टिकास्तम्भाग्रस्थोद्दीपितशीशकघटप्रदीपं निशायां दृष्ट्वा किमिदमिति साशङ्को गतस्तत्संमुखम् । गत्वा चाऽसन्नं स्तम्भमूलेऽप्येकमन्यदीपमवलोक्य दृष्ट्यवरोधकाभावेऽपि कथं न दृष्टमिति विस्मितेन दृष्ट्यवरोधिका भूरेवेत्यनुमितमतो भूपृष्ठे वक्रत्वमस्तीति सिद्धम् ।

अथ सत्यपि वृक्षाग्राच्चतुर्दिक्षु ममाऽकाशे भुव्येव पक्वं फलमेकं बहुत्र पतदवलोक्य भूपृष्ठनिष्ठाखिलबिन्दुष्वाकर्षणशक्तिरस्तीत्यनुमितम् । तथा मापनेन वृक्षाग्रात्पतनबिन्दुं यावत्सूत्रं < पतनेतरबिन्दुषु बद्धसूत्रेभ्य इति निर्णयात् भुवि बहिःस्थबिन्दोः पृष्ठस्थबिन्दुगरेखाणां बहिः खण्डेभ्योऽल्पं केन्द्रगरेखाबहिःखण्डमिति गोलीयनैसर्गिकधर्मदर्शनाद् गोलत्वमस्ति कस्मिंदिति मतिः प्रासूत । अतस्तावद् गोलत्वं प्रकल्प्याऽत्र सन्ति गोलीयधर्मा न वेति परीक्षा क्रियते । कल्पनावशादुक्तालपतररेखा भूमध्यगैवातो भुवो मध्ये पृष्ठबिन्दुषु चाऽकर्षणशक्तिकल्पनया समकार्योत्पत्तेर्भूमध्य एव सर्वाधिका तच्छक्तिरस्तीति कल्प्यते । करोत्यवश्यं मतिमान् बहुभृत्यक्षमकार्यं यद्येकेन भवेत्तदा तत् । अतोऽवश्यं मतिमता ब्रह्मणा वस्त्वाकर्षणकार्ये भूपृष्ठबिन्दून् भृत्यान् विहायैक एव भूमध्यबिन्दुर्भृत्यो नियुक्तः ।

भुवि स्थानद्वये समस्तम्भद्वयमारोप्यैकस्तम्भस्य शीर्षशीर्षेतरबिन्दुभ्यां विद्धेऽन्यस्तम्भाग्रे जातव्यस्रे दृग्लग्नकोणे तदन्तःस्तम्भखण्डं च विज्ञाय कोणानुपातेन स्तम्भाग्रान्तरं विज्ञाय १८०—२×एकस्तम्भाग्रलग्नकोण=वर्धितस्तम्भद्वयोत्पन्नभूमध्यलग्नकोण, उक्तत्रिभुजस्य समद्विबाहुकत्वात् । ततः कोणानुपातेन

$$\frac{\text{स्तम्भाग्रबद्धरेखा}\times\text{स्तम्भाग्रलग्नकोणज्या}}{\text{भूकेन्द्रलग्नकोणज्या}}=\text{भूव्यासार्ध}+\text{स्तम्भः}=\text{ज्ञातबाहु}$$

एवमन्येऽपि विषयाः ।

अथ विषुवांशयोरन्तरं क्रान्तिद्वयञ्च ज्ञात्वेष्टक्रान्त्यानयनार्थं परमक्रान्त्यानयनम् ।

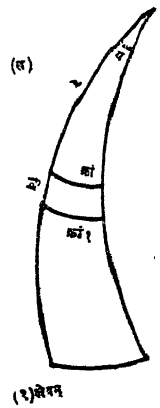
एतदानयनं वेधेन नव्यैः समुद्रयात्रादिना विलक्षणधियैव विहितं तत्तु सिद्धान्तसेतौ निवेशितम् । ततः ज्ञातबाह्वोः—स्तम्भौ=भूव्या $\frac{1}{2}$, एवं कृते

सर्वत्रैवमुपलब्धं फलसाम्यमतो भूर्गोलाकाराऽस्तीति सिद्धम् । न च यत्कल्पितं तत्सिद्धिरिति किं चित्रमतो नाऽस्या गोलत्वसिद्धिरिति वाच्यम् । गोलत्वकल्पनया केवलं स्तम्भविशिष्टत्रिभुजे समद्विबाहुकत्वम् किन्त्व सर्वत्र फलसाम्यं तु परीक्षामूलकमेव अतः शङ्का निरस्ता । वस्तुतस्तु भूर्दीर्घपिण्डानुकाराऽस्ति । परन्तु तत्र लघुव्यासबृहद्व्यासयोरत्यल्पान्तरत्वात्तयोः समत्वं कल्पितं सर्वैराचार्यवर्यैरिति ।

तत्र तावन्नाड़ीवृत्तं कालवृत्तं कथमित्युच्यते ।

प्रवहवायुना भ्राम्यमाणेऽपि भगोले बहुभिरपि वर्षैर्न खलु कासाञ्चित्तारकाणां स्थिरतयोपलब्धध्रुवताराङ्कितध्रुवस्थानाद् ध्रुव्याचापान्तरमुपलभ्यते । एतावतैवावगतं यद् वास्तवभगोलपृष्ठनिष्ठस्थिरकेन्द्रोत्पन्ननाड़ीद्युनिशवृत्तयोर्धरातलस्थैर्यम्, तत्रैकरूपोपलब्धप्रवहवायुभ्राम्यमाणोक्तमण्डलद्वयस्यैवावलम्बेन कालगणनोचिता, अनाद्यनन्तस्याच्युतोपमकालस्यागमनिर्णीतसर्वदैकरूपत्वात् । इयमेव युक्तिः प्राचीनार्वाचीनघटीयन्त्रादिभिः कालावबोधेऽपीति ।

कदम्बाख्यतारायां ध्रुव्याचापं स्थिरं कदम्बे ताराणां च चलं दृश्यते। तेन भचक्रस्य काचित्प्रवहेतरनिदानाऽपि गतिरस्तीत्यनुमितम् । सा च कदम्बोत्पन्नमहद्वृत्तरूपमार्गे स्यादिति गोलयुक्त्यैव स्फुटम् । अस्या आन्दोलिकाकारगतेः कारणं प्रवहाधिकरणकभचक्रत्यागकालिकस्रष्टृकराघातमेवेत्यनुमितम् । उक्तमहद्वृत्ते प्रवहप्रधानमार्गान्नाड़ीमण्डलात्प्रस्तुतगतिम् आन्दोलकं यावन्मितं भचक्रचलनसङ्कलनं तावदेवाचार्यैः प्रागपराख्या अयनांशाः परिभाषिताः । तत्साधनमुक्तमहावृत्ताधिकरणकसार्वदिकावस्थानविशिष्टस्य पूर्णप्रकाशवतो नक्षत्रबिम्बस्य ग्रहबिम्बस्य वाऽवलम्बेन कर्त्तुं शक्यम् अतस्तावत् सूर्यबिम्बस्यैव भचक्रचलनज्ञानं वेधेन निर्णीयते । तत्रोक्तमहावृत्तमार्गनिर्णयार्थं वेधगोलीयस्थिरगोलीय-(भगोलीय) नाड़ीवृत्तधरातलान्तरज्ञानेन वेधगोलीयक्रान्तिज्ञानेन च स्थिरगोलीयक्रान्तिज्ञानं कथं भवेदित्यस्यैव ग्रन्थस्य चन्द्रभगणोपपत्तौ द्रष्टव्यम् । द्वितीयदिने षष्टिदण्डात्मककालेऽर्काधिष्ठानबिन्दुर्याम्योत्तरे (ध्रुवप्रोतवृत्ते तत्रैवागतोऽनन्तरं यावता कालेनार्को याम्योत्तरवृत्ते समागतस्तत्कालमानं षड्गुणितं रवेर्निरक्षोदययोर्विषुवांशयोरन्तरं स्याद्याम्योत्तरवृत्तस्य निरक्षदेशीयक्षितिजत्वात् । क्रान्तिश्चोक्तयुक्त्या ज्ञाता । कृत्वैवं बहुषु दिनेषु गोलमेकं स्वाग्रे संस्थाप्य तत्र नाड़ीवृत्ताख्यं महद्वृत्तं विधाय तत्स्थेष्टबिन्दोः पूर्वपूर्वदिशि क्रमेण विषुवांशान्तरान् दत्वेष्टबिन्दौ प्रत्यग्र (दानाग्र) बिन्दौ च कृतध्रुवप्रोतवृत्तेषु तत्तत्क्रान्तो (प्रत्याह्निकक्रान्ती) दत्वा क्रान्तिद्वयाग्रलग्न महद्वृत्तकृतं तत्क्रान्त्यग्रेषु गतमित्युपलब्धम् । तेन रविभ्रमणमार्गो महावृत्तमिति सिद्धम् क्रान्त्यग्रे गतत्वात्तत्क्रान्तिवृत्तमिति संज्ञा शोभनेति ॥



नाड़ीक्रान्तिवृत्तयोरुत्पन्नकोणः=य=परमक्रान्तिः विषुवांशान्तरम्=वि. । मध्यावयवः=र तदा मध्यजा दोर्ज्यात्रिज्या गुणा प्रान्त्यस्पर्शरेखाहृतिर्भवेदिति नियमेन

ज्यार×त्रि=कोस्पय×स्पक्रां, तथा त्रि×ज्या (र+वि)=कोस्पय×स्पक्रां'

अतः $\frac{\text{ज्यार}\times\text{त्रि}}{\text{स्पक्रा}}=\frac{\text{ज्या}(\text{र}+\text{वि})\times\text{त्रि}}{\text{स्पक्रां}'}$

ततः $\frac{\text{ज्यार}\times\text{स्पक्रां}'}{\text{स्पक्रां}}=\text{ज्या}(\text{र}+\text{वि})$

चापगोरिष्टयोर्दोर्ज्ये मिथः कोटिज्यकाहृते, त्रिज्याभक्ततेतयोरित्यादिना

$$\text{ज्या}(\text{र}+\text{वि})=\frac{\text{ज्यार}\times\text{कोज्यावि}+\text{कोज्यार}\times\text{ज्यावि}}{\text{त्रि}}=\frac{\text{ज्यार}+\text{स्पक्रां}'}{\text{स्पक्रां}}$$

ज्यार×गु। अत्र $\frac{\text{स्पक्रां}'}{\text{स्पक्रां}}=\text{गु}$

पक्षौ त्रिगुणितौ तदा

ज्यार×कोज्यावि+कोज्यार×ज्यावि=ज्यार×गु×त्रि समशोधनेन

ज्यार×गु×त्रि—ज्यार×कोज्यावि=कोज्यार×ज्यावि=ज्यार (गु+त्रि—कोज्यावि)

ततः $\frac{\text{ज्यार}(\text{गु}\times\text{त्रि}-\text{कोज्यावि})}{\text{कोज्यार}}=\text{ज्यावि}$ अतः $\frac{\text{ज्यार}}{\text{कोज्यार}}$

$=\frac{\text{ज्यावि}}{\text{गु}\times\text{त्रि}-\text{कोज्यावि}}=\text{व्यक्त}$

पक्षौ द्वादशभिर्गुणितौ तदा $\frac{\text{ज्यार}\times १२}{\text{कोज्यार}}=$रतुल्याक्षदेशीयपलभा $=१२\times$ व्यक्त, अस्या येऽक्षांशास्तदेव रमानम्। वा तावेव पक्षौ यदि त्रिगुणितौ तदा

$\frac{\text{ज्यार}\times\text{त्रि}}{\text{कोज्यार}}=$रतुल्याक्षांशस्पर्शरेखा, स्पर्शरेखातः स्वापकरणेन रतुल्याक्षांशोऽर्थात्तद्देशीयाक्षांशमानमेव रमानम्। ततो य मानज्ञानं सुगममेवेति ॥

अथ यत्क्रान्तिवृत्ताधारं भचक्रचलनं तदेव निरूपितरविमार्गरूपक्रान्तिवृत्तमिति निर्णयः। ध्रुवस्थाने कदम्बं, याम्योत्तरवृत्तस्थाने कदम्बप्रोतवृत्तं, नाड़ीवृत्तस्थाने क्रान्तिवृत्तमक्षज्यास्थाने दृक्क्षेपञ्च नीत्वा चन्द्रभगणोपपत्तौ नाड़ीवृत्तधरातलान्तरज्ञानं ततो ग्रहगोलीयक्रान्त्यानयनार्थं या युक्तिः प्रदर्शिता सैवाऽत्राप्यनुसन्धेया

किन्त्वत्र 'लम्बरेखा ∽ अन्तर = ०' इत्युपलब्धमतः सिद्धम् ।

गोलद्वय(वेधगोलस्थिरगोल)केन्द्राभ्यां कदम्बे रेवत्याञ्च रेखे नीते तदा भूकेन्द्रलग्नकोणः = स्थिरगोलीया भगोलीया वा शरकोटिः = दृष्टिस्थानलग्नकोणः = वेधगोलीयशरकोटिः (कदम्बगतरेखयो रेवतीगतयोश्च रेखयोः समानान्तरत्वात्)। तदूनो नवत्यंशः = शरचापः = ०, इत्युपलब्धम् ।

अथ प्रकृतमनुसरामः ।

अथ गोलद्वयकेन्द्राभ्यां ध्रुवे रेवत्याञ्च रेखे नीते । तदा गोलद्वयकेन्द्रलग्नकोणमाने गोलद्वयीयद्युज्याचापमिते तुल्ये ध्रुवगतयो रेखयो रेवतीगतरेखयोश्च समानान्तरत्वात्, तेन ९०—रेवतीद्युज्याचापः = रेवतोक्रान्तिः

ततः $\frac{\text{त्रि} \times \text{ज्याक्रां}}{\text{ज्याजि}} = \text{ज्याभु}$.

अस्याश्चापं रेवतीभुजांशाः = अयनांशाः । एते परमा = २७' भवन्ति ।

अत्र प्रसङ्गागतानां गोलद्वयीलग्नत्रिभदृक्क्षेपचापाक्षांशचापादीनां समत्वोपपत्तिरूह्येति ।

अथ ग्रहाणां पूर्वाभिमुखगतिः कथमिति निर्णयः ।

प्रथमपदे ग्रहे तत्कालीनक्रान्तीनां वेधेन क्रमादधिकत्वं द्वितीयपदे ह्रासत्वं तृतीयपदे प्रथमपदवच्चतुर्थपदे च द्वितीयपदवद् दृश्यतेऽतो ग्रहाणां प्रागगतित्वं सिद्धम् । ग्रहाणां बहुभिर्दिनैः प्रवहस्य त्वेकेनैवाह्ना भगणपूर्त्तिरतो ग्रहाणां तदल्पगतित्वं सिद्धम् ।

ग्रहपिण्डे गोलत्वं नवेत्येतदर्थं ग्रहाणाञ्चोर्ध्वाधररूपेणावस्थानमेतदर्थञ्च विचारः ।

गोलमेकं क्वापि संस्थाप्य दृष्टिस्थाने समायष्टित्रयस्तथा स्थापिता यथा गोलस्पर्शकरणि दृष्टिसूत्राणि भवेयुस्तानि च दृश्यवृत्ताधारसमसूचीकर्णगतानि, आधारवृत्तधरातलसमानान्तरं यष्टचग्रेषु मियोबद्धरेखात्रयजनितत्रिभुजोपरिष्ठवृत्तमूक्तसूच्याः कर्णाग्रेषु लगतीति सुस्पष्टम् । कृतवृत्तकेन्द्रगदृष्टिसूत्रं वर्धितं सदाधारवृत्तकेन्द्रगतञ्चैते गोलधर्माः । अथ तावद् ग्रहपिण्डे गोलत्वं प्रकल्प्योक्तगोलधर्मा दृश्यन्तेऽतो ग्रहपिण्डे गोलत्वं सिद्धम् । वेधेन ग्रहबिम्बीयकर्णानयनं कथं भवतीति मङ्गलगुरुशनीनां शीघ्रोच्चोपपत्त्यवसरेऽत्रैव ग्रन्थे प्रदर्शितम् । सर्वेषां ग्रहाणां तन्मानमतुल्यमायाति तेनैव हेतुना ग्रहाणां कक्षानिवेश ऊर्ध्वाधरक्रमेण (यस्य ग्रहस्य कर्णमानं यस्माद् ग्रहकर्णमानादधिकं तदीया कक्षा तद्ग्रहकक्षात उपरिगता भवतीत्यनुसारेण चन्द्रबुधशुक्ररविकुजगुरुशनिभानां कक्षाश्चन्द्रत उपरिक्रमेण) सर्वैराचार्यवर्यैः कृत इति ।

अथ सहग्रहैरिति कथं तदुच्यते ।

भूगर्भादिष्टव्यासार्धको हि गोलो भगोलः । भचक्रभगोलयोर्ध्रुवसूत्रयष्टिप्रोतत्वेन सहैवागमनादिभवनाद् भगोलसंसक्तयोर्मन्दगोलशीघ्रगोलयोर्ग्रहाधिकरणकयोरपि तेन सहैव गमनमिति सिद्धम् ।

ध्रुवसूत्राधिकरणकं पश्चिमाभिमुखं भचक्रभ्रमणम् । तत्सूत्रमध्ये स्रस्ट्रा कदम्बसूत्रं तथा निबद्धं यथा कदम्बसूत्रं भचक्रस्य पश्चिमाभिमुखभ्रमे विघ्नं न कुर्वत् स्रष्टृकराघातजनितभ्रमे भचक्रपृष्ठे कदम्बस्थाने खचितं भूत्वा स्थिरं भवति । तेन ध्रुवसूत्रध्रुवस्थानादुक्तवेगावरामान्तं प्रागपरदिशि २७° पर्यन्तं भचक्रपृष्ठं घर्षति । तेन ध्रुवतारा न स्थिरा केवलं ध्रुवस्थानमेव स्थिरमिति सिद्धमतस्तदन्ततारे च तथा ध्रुवत्वे इति भास्करोक्तं, ध्रुवतारां स्थिरां ग्रन्थे मन्यन्ते ते कुबुद्धय इति कमलाकरोक्तं च संगच्छत इति ॥ एताभिरुपपत्तिभिः 'ध्रुवताराप्रतिबद्धज्योतिश्चक्रमि' त्याद्याचार्योक्तं सर्वं युक्तियुक्तमुपपद्यत इति ॥

हि. भा.—इस ज्योतिष सिद्धान्त में पृथिवी के सम्बन्ध ही से सर्वोपयोगी ग्रहादि कक्षा आदि का ज्ञान होता है, इसलिए इसकी आकृति कैसी है, उसका परिमाण कितना है, इन सब के निर्णय के लिए विचार करते हैं। किसी वृक्ष आदि से रहित समान पृथिवी में दूर में ईंटों के बने हुए खम्भ के अग्र में जलते हुए लालटेन को रात में दखकर 'क्या बात है' इस आशङ्का से उसकी तरफ चले, उसके समीप जाकर खम्भा के जड़ में भी जलती हुई एक लालटेन को देखकर दृष्टि के रोकने वाली चीजों के नहीं रहने पर भी क्यों देखने में नहीं आया यह शंका हुई और विचार करने पर मालूम हुआ कि दृष्टि को रोकने वाली पृथिवी ही है इसलिए पृथिवी के पृष्ठ में वक्रत्व है । यह सिद्ध हुआ ।

चारों तरफ आकाश के बराबर रहने पर पृथिवी ही के ऊपर बहुत जगह पके हुए फल को गिरता हुआ देखकर भूपृष्ठ-स्थित प्रत्येक बिन्दु में आकर्षण शक्ति है यह अनुमान किया गया । और मापन करने से वृक्ष के अग्र से गिरे हुए बिन्दु तक बद्धसूत्र < फलों के गिरने के स्थानों से भिन्न बिन्दुओं से वृक्षाग्र तक सूत्र, इस निर्णय से पृथिवी में बहिःस्थित बिन्दु से पृष्ठस्थ बिन्दुगत रेखाओं के बहिःखण्डों से केन्द्रगत रेखा बहिःखण्ड अल्प होता है यह गोलसम्बन्धी स्वाभाविक धर्म देखने से इसमें किसी तरह का गोलत्व है यह मन में आया । इसलिए पहले इसमें गोलत्व कल्पना कर के देखना चाहिए कि इसमें गोलीय धर्म है या नहीं । पृथिवी के ऊपर दो स्थानों में समान दो खम्भों को गाड़कर एक खम्भा के शीर्ष स्थान से और शीर्ष स्थान से कुछ हट कर उसी खम्भा में दृष्टि स्थान रखकर दूसरे खम्भा के अग्र को वेध किए, दोनों खम्भा के अग्र में सूत्र बांध दिये तब जो एक त्रिभुज बनता है उसमें स्तम्भाग्र प्रथम दृष्टिस्थान और उस से भिन्न स्थल में जो दृष्टि स्थान रखे हैं इन दोनों दृष्टिस्थान लग्न कोणों को मापन द्वारा जानकर तथा दृष्टिस्थान द्वयान्तर्गत रेखा को भी मापन से जानकर कोणानुपात से खम्भों के अग्रान्तर समझकर १८०—२× एक स्तम्भाग्रलग्नकोण = वर्धितस्तम्भद्वयोत्पन्नभूकेन्द्रलग्नकोण क्योंकि दोनों खम्भों से और खम्भों के अग्रान्तर से जो त्रिभुज बनता है वह समद्विबाहुक है। तब कोणानुपात करते हैं—

$$\frac{\text{खम्भों के अग्रगत रेखा} \times \text{खम्भा के, अग्रलग्न कोणज्या}}{\text{भूकेन्द्रलग्न कोणज्या}} = \text{भुव्यासार्ध} + \text{खम्भा} = \text{ज्ञातबाहु}.$$

इसी तरह दूसरे स्तम्भ प्रमाण के लिए भी करना, उसका प्रमाण भी इतना ही आता है अतः दोनों ज्ञातबाहु—दोनों खम्भा=भुव्यासार्ध, इस तरह प्रत्येक जगह में फल (भुव्यासार्ध) बराबर आता है इसलिए पृथिवी गोलाकार है यह सिद्ध है। आनीत भुव्यासार्ध को दूना कर देने से भुव्यास होता है तब 'व्यासे भनन्दाग्निहते विभक्ते खबाणसूर्यैः' इससे भूपरिधि का ज्ञान होगा। वस्तुतः पृथिवी का पिण्ड दीर्घपिण्ड के आकार का है लेकिन इसके लघु व्यास, और वृहद्व्यास में बहुत ही कम अन्तर होने के कारण दोनों व्यासों को आचार्य ने बराबर मान लिया।

नाड़ीवृत्त कालवृत्त क्यों है यह कहते हैं।

प्रवह वायुद्वारा भगोल को घुमाने पर भी बहुत वर्षों में भी स्थिरता से प्राप्त ध्रुवतारा से चिह्नित ध्रुवस्थान से किसी तारा के अन्तर ज्युयाचाप उपलब्ध नहीं होता है, इसी से समझा गया कि वास्तव भगोल पृष्ठनिष्ठ स्थिर केन्द्र से उत्पन्न नाड़ीवृत्त और अहोरात्र वृत्त के धरातल में स्थिरता है वहां एक रूप से प्राप्त प्रवह वायुद्वारा घुमाये हुए जन्हीं दोनों वृत्तों की सहायता से कालगणना उचित है, क्योंकि जिस काल का न आदि है न अन्त है ऐसे काल का रूप आगम प्रमाण से सदा एक रूप है। यही युक्ति प्राचीन और नवीन घटीयन्त्रादि के द्वारा कालज्ञान के लिए है।

कदम्ब तारा का ज्युया चाप स्थिर है और कदम्बस्थान में ताराओं को चल देखते हैं इससे सिद्ध होता है कि प्रवहगति के अलावा भी भचक्र की कोई गति है वह कदम्बोत्पन्न महद्वृत्त रूप मार्ग में होता है। इस आन्दोलिकाकार गति के कारण प्रवहाधार भचक्रत्यागकालिक ब्रह्मा के हाथ का आघात ही हो सकता है। उस महद्वृत्त में प्रवह के नाड़ीवृत्तरूप प्रधान मार्ग से प्रस्तुत गति के कारणभूत भचक्रचलन का सङ्क्रलन जितना होता है उतना ही आचार्य पूर्वायनांश और पश्चिमायनांश कहते हैं। उसके साधन पूर्वकथित महद्वृत्त के आधार पर बराबर रहने वाले अतिप्रकाशमान नक्षत्रबिम्ब के या ग्रहबिम्ब के अवलम्बन [आधार] से कर सकते हैं। अतः वेध से भचक्रचलन ज्ञान करते हैं। पूर्व-कथित महद्वृत्त मार्गनिर्णय के लिए वेधगोलीय और स्थिरगोलीय (भगोलीय) नाड़ीवृत्तधरातलान्तर ज्ञान से तथा वेधगोलीय क्रान्तिज्ञान से भगोलीय क्रान्तिज्ञानप्रकार इसी ग्रन्थ में चन्द्र भगण की उपपत्ति में देखना चाहिए।

द्वितीय दिन में षष्टि ६० दण्डात्मक काल में रवि जिस बिन्दु में प्रथम दिन में थे वह बिन्दु याम्योत्तरवृत्त (ध्रुवप्रोतवृत्त) में वहीं पर आया। उसके बाद जितने काल में रवि याम्योत्तरवृत्त में आये उस कालमान को छह से गुणने से रवि के निरक्षदेशीय उदयमानद्वय (विषुवांशद्वय) का अन्तर होता है, क्योंकि याम्योत्तरवृत्त निरक्षदेशीय क्षितिज वृत्त है। पूर्व युक्ति से क्रान्तिज्ञान भी कर लिया; इस तरह अनेक दिनों में करके अपने आगे

एक गोल को रखकर उसमें नाड़ीवृत्त संज्ञक महद्वृत्त की रचना कर उसके किसी इष्ट-बिन्दु से पूर्व पूर्वदिशाक्रम से विषुवांशान्तर दान देकर इष्टबिन्दु और प्रत्येक दानाग्र बिन्दुगत ध्रुवप्रोत वृत्तों में प्रत्येक दिन की क्रान्ति दान देकर दो क्रान्तियों के अग्रगत जो महद्वृत्त किया जायगा वह प्रत्येक क्रान्ति के अग्रगत होता है ऐसा उपलब्ध होता है इसलिए रवि का भ्रमण-मार्ग महद्वृत्त सिद्ध हुआ और वह वृत्त प्रत्येक क्रान्ति के अग्र में गया हुआ है, इस लिए उसका नाम क्रान्तिवृत्त है। इति।

अब विषुवांशद्वय के अन्तर और क्रान्तिद्वय जानकर इष्टक्रान्तिज्ञान के लिए परम क्रान्ति का आनयन करते हैं।

यहां संस्कृत में इसी की उपपत्ति में (१) क्षेत्र को देखिये। नाड़ीवृत्त और क्रन्तिवृत्त से उत्पन्न कोण=परमक्रान्ति=य विषुवांशान्तर = वि, मध्याक्षयव = र, तब 'मध्यज्या दोर्ज्या त्रिज्या गुणा प्रान्त्यस्पर्शहृतिर्भवेत्' इससे ज्यार×त्रि=कोस्पय×स्पक्रां, तथा त्रि×ज्या (र+वि)=कोस्पय×स्पक्रां,। यहां प्रथम क्रान्ति=क्रां। द्वितीय-क्रान्ति=क्रां,

$$\therefore \frac{\text{ज्यार}\times\text{त्रि}}{\text{स्पक्रां}}=\text{कोस्पय}=\frac{\text{त्रि}\times\text{ज्या}\,(\text{र}+\text{वि})}{\text{स्पक्रां},}\quad \text{तब}\ \frac{\text{ज्यार}\times\text{स्पक्रां},}{\text{स्पक्रां}}=\text{ज्या}\,(\text{र}+\text{वि}),$$

चापयोरिष्टयोर्ज्ये मिथः कोटिज्यकाहते इत्यादि से ज्या (र+वि)=

$$\frac{\text{ज्यार}\times\text{कोज्यावि}+\text{कोज्यार}\times\text{ज्यावि}}{\text{त्रि}}=\frac{\text{ज्यार}\times\text{स्पक्रां},}{\text{स्पक्रां}}=\text{ज्यार}\times\text{गु}\ \Big|\ \text{यहां}\frac{\text{स्पक्रां},}{\text{स्पक्रां}}=\text{गु}$$

दोनों पक्षों को त्रि से गुणने से

ज्यार×कोज्यावि+कोज्यार×ज्यावि=ज्यार× गु × त्रि, समशोधन करने से

ज्यार×गु×त्रि—ज्यार×कोज्यावि=ज्यार (गु×त्रि—कोज्यावि) = कोज्यार ×ज्यावि, परस्पर गुणकाङ्क से भाग देने से—

$$\frac{\text{ज्यार}}{\text{कोज्यार}}=\frac{\text{ज्यावि}}{\text{गु}\times\text{त्रि}-\text{कोज्यावि}}=\text{व्यक्त},$$ दोनों पक्षों को बारह से गुणने से $\frac{\text{ज्यार}\times १२}{\text{कोज्यार}}$

=१२×व्यक्त=रतुल्याक्ष देशीय पलभा इससे जो अक्षांश होगा वही 'र' का मान होगा। अथवा उन्हीं दिनों पक्षों को यदि त्रि से गुण देते हैं तो $\frac{\text{ज्यार}\times\text{त्रि}}{\text{कोज्यार}}=\text{त्रि}\times\text{व्य}=$ स्पर=रतुल्याक्षांशस्पर्शरेखा, स्पर्शरेखाखण्ड से चाप करने से 'र' का मान ही उस देश का अक्षांशमान होता है, इस पर से 'य' का मानज्ञान सुलभ ही है।।

अब जिस क्रान्तिवृत्त के आधार पर भचक्र का चलन होता है, वही पूर्वकथित रविभ्रमण मार्ग रूप क्रान्तिवृत्त है इसका निर्णय किया जाता है—ध्रुव के स्थान में कदम्ब, याम्योत्तर वृत्त के स्थान में कदम्बप्रोतवृत्त, नाड़ीवृत्त के स्थान में क्रान्तिवृत्त और अक्षज्या के

स्थान में दृक्क्षेप लेकर आगे चन्द्रभगण की उपपत्ति में प्रदर्शित नाड़ीवृत्तधरातलान्तर ज्ञान उससे ग्रहगोलीय क्रान्ति के आनयन के लिए जो युक्ति है वही यहां भी समझनी चाहिए किन्तु यहां 'लम्बरेखा ७ अन्तर=० उपलब्ध होता है; अतः सिद्ध हुआ।

अब रेवती में शराभाव क्यों होता है इसका निर्णय करते हैं।

गोलद्वय(वेधगोल और भगोल)के केन्द्रों से कदम्ब में और रेवती में जो रेखा लाये वे दोनों समानान्तर हैं अर्थात् कदम्बगत रेखाद्वय समानान्तर है तथा रेवतीगत रेखाद्वय भी समानान्तर है। इसलिए भूकेन्द्रलग्नकोण=स्थिरगोलीय या भगोलीय शरकोटि=दृष्टि-स्थान लग्न कोण=वेधगोलीय शरकोटि, अतः इसको नवत्यंश में घटाने से शरचाप=० यह उपलब्ध हुआ इस लिए रेवती का शराभाव सिद्ध हुआ।

अब प्रकृत विषय (अयनांश) का आनयन करते हैं।

गोलद्वय केन्द्रों (भूकेन्द्र और दृष्टिस्थान) से ध्रुव में और रेवती में रेखायें लाये तब गोलद्वय केन्द्रलग्न कोणमान ज्युज्याचाप बराबर होते हैं (क्योंकि गोलद्वय केन्द्रों से ध्रुवगत रेखाद्वय समानान्तर है तथा रेवतीगत रेखाद्वय भी समानान्तर है) इसलिए ९०—रेवती ज्युज्याचाप=रेवती क्रान्ति, तब $\frac{\text{त्रि}\times\text{ज्याक्रां}}{\text{ज्याज}}$=ज्याभु, इसके चाप करने से रेवती का भुजांश=अयनांश, यह परम (परमायनांश)=२७° होता है।

यहां प्रसङ्गवश आये हुए दोनों गोलों के लग्न, विक्षिप्तदृक्क्षेपचाप, अक्षांशचाप आदि की समत्व उपपत्ति स्वयमेव समझनी चाहिए।

अब ग्रहों की पूर्वाभिमुखगति क्यों है इसका निर्णय करते हैं।

प्रथमपद में ग्रह के रहने से उनकी तात्कालिक क्रान्ति की वृद्धि, द्वितीयपद में ह्रास (क्षीणत्व) तृतीयपद में प्रथमपद की तरह और चतुर्थपद में द्वितीयपद की तरह स्थिति देख से देखते हैं। इसलिए ग्रहों की पूर्वाभिमुख गति है यह सिद्ध हुआ। और ग्रहों की भगण-पूर्ति बहुत दिनों में होती है तथा प्रवह की भगणपूर्ति एक ही दिन में होती है इसलिए प्रवहगति की अपेक्षा ग्रहगति की अल्पता भी सिद्ध हुई।

अब ग्रहपिण्डों में गोलत्व है या नहीं और ग्रहों की स्थिति ऊर्ध्वाधर क्रम से क्यों है इनके लिए विचार करते हैं।

एक गोल को कहीं पर रखकर दृष्टिस्थान में मूल (जड़) मिलित समान यष्टित्रय को इस तरह रखना चाहिए जिस से दृष्टिसूत्र सब गोल के स्पर्शकारक (याने गोल स्पर्शरेखायें) हो और वे दृश्यवृत्ताधार समसूची कर्णगत हो। यष्टियों के अग्रों में परस्पर रेखा करने से जो त्रिभुज बनता है उसके ऊपर जो वृत्त होता है उसका धरातल आधारवृत्त धरातल के समानान्तर है और पूर्वकथित समसूची के कर्णाग्रों में जाता है।

परस्पर समयष्टित्रयाग्रगतरेखाजनितत्रिभुजोपरिगत वृत्तकेन्द्र में दृष्टिस्थान से जो दृष्टिसूत्र आवेगा उसको बढ़ाने से आधार वृत्त केन्द्रगत भी होता है ये सब गोल ही धर्म है क्योंकि ये सब बातें गोलात्मक पदार्थ ही में हो सकती हैं। आचार्य ग्रहपिण्डों में गोलत्व स्वीकार कर पूर्व-कथित गोलीय धर्म देखते हैं इसलिए ग्रहपिण्डों में गोलत्व सिद्ध हुआ। वेध से ग्रहबिम्बीय कर्णज्ञान कैसे होता है इसे इसी ग्रन्थ में मङ्गलगुरु और शनि की शीघ्रोच्चोपपत्ति स्थल में देखना चाहिए। सब ग्रहों के बिम्बीय कर्णमान अतुल्य उपलब्ध हुए इसीलिए ग्रहों के कक्षानिवेश ऊर्ध्वाधर क्रम से (जिस ग्रह का कर्णमान जिस दूसरे ग्रह के कर्णमान से अधिक उपलब्ध हुआ उसकी कक्षा उस दूसरे ग्रह की कक्षा से उपरिगत हुई इसके अनुसार) चन्द्र, बुध, शुक्र, रवि, कुज, गुरु, शनि और नक्षत्रों की कक्षायें चन्द्र से उपरिक्रम से सब आचार्यों ने अपने-अपने सिद्धान्त ग्रन्थों में लिखी हैं।

'सहग्रहैः' इसकी युक्ति।

भूगर्भ से इष्टत्रिज्या व्यासार्ध से जो गोल होता है, वह भगोल है। ध्रुवसूत्र रूप यष्टी में बन्धे हुए भचक्र और भगोल के साथ-साथ आने-जाने के कारण उनसे मिले हुए मन्दगोल और शीघ्रगोल (जो ग्रहों के आधार गोल हैं) के भी भ्रमणादि उनके साथ ही होते हैं यह सिद्ध हुआ ॥+

इन उपपत्तियों से 'ध्रुवतारा प्रतिबद्धज्योतिश्चक्रं' इत्यादि आचार्योक्त सब सिद्ध हुआ।

ध्रुवसूत्रयष्टी के आधार पर पश्चिमाभिमुख भचक्र भ्रमण होता है। उस (ध्रुवयष्टी) के मध्य में ब्रह्मा ने कदम्बसूत्र इस तरह बांध दिया जिससे वह कदम्बसूत्र भचक्र के पश्चिमाभिमुख भ्रमण में विघ्न नहीं करते हुए ब्रह्मा के हाथ के आघात से उत्पन्न भ्रमण में भचक्र के पृष्ठ में कदम्ब स्थान में खचित (जड़ा हुआ) होकर स्थिर हो इसलिए ध्रुवसूत्र ध्रुवस्थान से कथितवेग (गति) की समाप्ति तक पूर्व और पश्चिम तरफ २७° अंश पर्यन्त भचक्रपृष्ठ को घिसता है अतः ध्रुवतारा स्थिर नहीं है केवल ध्रुवस्थान ही स्थिर है यह सिद्ध हुआ, इसलिए 'तदन्तारे च तथा ध्रुवत्वे' यह भास्करोक्त 'ध्रुवतारां स्थिरां ग्रन्थे मन्यन्ते ते कुबुद्धयः' यह कमलाकरोक्त भी युक्तिसङ्गत है।

इदानीं कालेऽब्दस्य क्षेत्रस्य भगणस्य च तुल्यां विभागकल्पनां प्रदर्शयन्नाह।

प्राणैर्विनाड़िकार्क्षी षड्भिर्घटिका विनाड़िका षष्ट्या।
घटिका षष्ट्या दिवसो दिवसानां त्रिंशता भवेन्मासः ॥ ५ ॥

मासा द्वादश वर्षं विकला लिप्तांश-राशि-भगणान्तः।
क्षेत्रविभागस्तुल्यः कालेन विनाड़िकाख्येन ॥ ६ ॥

वा. भा.—प्राणैः षड्भिरार्क्षी विनाड़िका भवति। ऋक्षाणामियमृक्षाणां विषुवन्मण्डलसंबन्धिनीति यावत्। यतो विषुवन्मण्डलमेव प्राणेन कलामुदेति। नाक्षमण्डलमथवा सर्वस्यैव च नक्षत्रस्य स्वोदयाद् यावत् षष्टिघटिका भवन्ति, अगतिमत्त्वात्। ग्रहाणां पुनर्गतिवश्याद्भिद्यते। अतः शोभनमुक्तं विनाड़िकार्क्षी षड्भिः प्राणैर्भवति इति। प्राणश्च स्वस्थेन्द्रियस्य स्वपतो जाग्रतो वा गृह्यते। न रोगाद्यपहतस्य। तासां विनाड़िकानां षष्ट्या घटिका घटिकानां षष्ट्या दिवसो भवति। दिवसानां त्रिंशता मासो भवति।

मासैर्द्वादशभिः वर्षो भवति, एवं कालविभागो विनाडिका, नाडिका, दिवसः, मासो वर्षान्तः क्षेत्रविभागोऽप्येवम् यथा विनाडिकानां षष्ट्या घटिका। एवं विकलानां षष्ट्या लिप्ता भवति। यथा घटिकानां षष्ट्या दिवस एवं लिप्तानां षष्ट्यांशः, यथा दिवसानां त्रिंशता मास एवम्मासानां त्रिंशता राशिः, यथा मासैर्द्वादशभिः वर्षमेवं द्वादशभीराशिभिर्भगण इत्यर्थः। अत उक्तं विकलालिप्तांशराशिभगणान्तः क्षेत्रविभागतुल्यः कालेन विनाडिकाद्येन। अत्र विभागकल्पनया तुल्यत्वमुच्यते। अथवा कलया विनाड़िकास्ताः क्षेत्रे षड्लिप्ता भवन्ति विषुवन्मण्डले। एतद् गोले प्रदर्शयेदिति।

वि. भा.—षड्भिः प्राणैः (षड्भिरसुभिः) आर्क्षी विनाड़िका (नाक्षत्री विघटिका) भवति, विनाडिका षष्ट्यैका घटिका (एको दण्डः) घटिकाषष्ट्या (दण्डषष्ट्या) दिवसः (एकं दिनं) दिवसानां (दिनानां) त्रिंशता (त्रिंशत्तुल्येन) मासो भवति, द्वादशभिः मासैः वर्षं (सौरवर्षं) भवति, विनाड़िकाद्येन (पलदण्डदिनमासवर्षेण) कालेन तुल्यो विकला लिप्तांशराशिभगणान्तः (विकला-कलांशराशिभगणः) क्षेत्रविभागो ज्ञेयोऽर्थाद्यथैकवर्षस्य मासदिनादयो विभागास्तथैव भगणस्य राश्यंशादय इत्यधो विलिख्य प्रदर्श्यते—

६ असवः =१ पलम्	एतत्सदृशा एव क्षेत्रीय (कक्षा) विभागा यथा—
६० पलानि =१ घटिका	६० विकलाः=१ कला
६० घटिकाः=१ दिनम्	६० कलाः =१ अंशः
३० दिनानि =१ मासः	३० अंशाः =१ राशिः
१२ मासाः =१ वर्षम्	१२ राशयः =१ भगणः

सिद्धान्तशिरोमणौ भास्कराचार्येण 'क्षेत्रे समाद्येन समा विभागाः स्युश्चक्रराश्यंशकलाविलिप्ताः' इत्यनेन सिद्धान्तशेखरे श्रीपतिना च 'एवं चक्रर्क्षांशलिप्ताविलिप्तास्तुल्याः क्षेत्रेऽनेहसाऽब्दादिकेनेत्या'दिना ब्रह्मगुप्तोक्तानुरूपमेव कथ्यते, किन्तु 'अक्षणोनिमेषः कथितो निमेषस्त्रिंशद्विभागोऽस्य च तत्परा स्यादि'त्यादिना श्रीपतिना, योऽक्षर्णोनिमेषस्य खरामभागः स तत्परस्तच्छतभाग उक्त' इत्यादिना भास्करेण च ब्रह्मगुप्तोक्तादधिकं कथ्यते, वटेश्वरसिद्धान्ते निमिषादेर्या परिमितयः परिभाषितास्ततो भिन्ना एवोपर्युक्ताचार्यकथिताः। सर्वान् विलोक्य विवेचका गणितिका विवेचयन्त्विति ॥ ५-६ ॥

अब कालमानों की विभागकल्पना को कहते हैं।

हि. भा.—छह असु की एक नाक्षत्री विघटिका (पल) होती है, साठ पल की एक घटी (दण्ड) होती है। साठ घटी का एक दिन होता है, तीस दिनों का एक मास होता है। बारह मासों का एक वर्ष होता है। पल, दण्ड, दिन, मास, और वर्ष इन्हीं के बराबर क्षेत्रीय (कक्षा) विभाग विकला, कला, अंश, राशि और भगण हैं अर्थात् जैसे एक वर्ष का विभाग मास, दिन आदि हैं वैसे ही एक भगण का विभाग राशि, अंश आदि हैं। इनके स्पष्टीकरण के लिए नीचे लिखकर दिखलाता हूँ।

इन के बराबर ही क्षेत्रीय (कक्षा) विभाग होता है। जैसे—

६ असु = १ पल	विकला —
६० पल = १ घटी	६० विकला = १ कला
६० घटी = १ दिन	६० कला = १ अंश
३० दिन = १ मास	३० अंश = १२ राशि
१२ मास = १ वर्ष	१२ राशि = १ भगण

सिद्धान्तशिरोमणि में भास्कराचार्य 'क्षेत्रे समाद्येन समा विभागाः स्युश्चक्रराश्यंशकला-विलिप्ताः' इससे तथा सिद्धान्तशेखर में श्रीपति 'एवं चक्रर्क्षांशलिप्ता विलिप्तास्तुल्याः क्षेत्रेऽनेह-साऽब्दादिकेन, इससे ब्रह्मगुप्तोक्त के अनुरूप ही कहते हैं लेकिन, 'अक्ष्णोर्निमेषः कथितो निमेषस्त्रिंशद्विभागोऽस्य च तत्परा स्यात्' इत्यादि से श्रीपति तथा 'योऽक्ष्णोर्निमेषस्य खरामभागः स तत्परस्तच्छतभाग उक्ता' इत्यादि से भास्कराचार्य भी ब्रह्मगुप्तोक्त से अधिक कहते हैं। वटेश्वरसिद्धान्त में वटेश्वराचार्य निमिष आदि कालप्रमाण की जो परिभाषायें करते हैं वे श्रीपत्यादिकथित उनके मानों की परिभाषाओं से भिन्न ही हैं। इन सब बातों को देखकर विवेचक ज्योतिषी लोग विचार करें इति ॥५-६॥

चतुर्युगसंख्या युगचरणमानानि च प्रदर्शयन्नाभ्यामाह।

खचतुष्टयरदवेदा ४३२०००० रविवर्षाणां चतुर्युगं भवति।
सन्ध्यासन्ध्यांशैः सह चत्वारि पृथक् कृतादीनि ॥ ७ ॥

युगवशभागो गुणितः कृतं १७२८००० चतुर्भिस्त्रिभिर्गुण १२९६००० स्त्रेता।
द्विगुणो ८६४००० द्वापरमेकेन सङ्गुणः ४३२००० कलियुगं भवति ॥८॥

वा. भा.—रविवर्षाणां खचतुष्टयरदवेदसंख्यया चतुर्युगं भवति। खचतुष्टयरदवेदाश्चाथ प्रदक्षिणेन स्थाप्यमानाः त्रिचत्वारिंशल्लक्षाणि विंशतिश्च सहस्राणि भवन्ति ४३२०००० संध्यासंध्यांशैः सहेति येयं चतुर्युगसंख्या मयाभिहितैषा संध्या-संध्यांशैश्च सह। संध्या च कृतादीनां स्वद्वादशभागतुल्या संध्यांशाश्च तावानेव मानवे

धर्मशास्त्रे पठ्यते । चत्वारः पृथक् पृथक् । कृतत्रेताद्वापर-कलियुगानि च तत्रैव पठ्यन्ते इति चेह यथा भवन्ति तथार्यमाह ।

युगशब्देन चतुर्युगमुच्यते, तेनायमर्थः युगदशभागः खत्रिरदवेदाश्चतुस्थानस्थाः क्रमेण चतुर्भिस्त्रिद्व्येकगुणाः सन्तश्चत्वारि पृथक् कृतादीनि मानानि भवन्ति । तद्यथा वसुयमनगेन्दवः, रसनवनेत्रचन्द्राः, कृतषड्वसवः द्वित्रिवेदाः सर्वे सहस्राघ्नाः १७२८००० कृतमानम्, १२९६००० त्रेतामानम् ८६४००० द्वापरमानम्, ४३२००० कलियुगमानम् । एतावती वर्षसंख्या सौरेण मानेन कृतयुगादीनां भवति ।

वि. भा.—रविवर्षाणां (सौरवर्षाणां) खचतुष्टयरदवेदाः ४३२०००० चतुर्युगं (महायुगं) भवति, सन्ध्यासन्ध्यांशैः सह कृतादीनि (सत्ययुगादीनि) पृथक् चत्वारि युगचरणमानानि भवन्ति, युगदशभागः ४३२००० चतुर्भिर्गुणितस्तदा कृतं (सत्ययुगचरणमानं) १७२८००० भवति, त्रिभिर्गुणितः १२९६००० त्रेतायुगचरणमानम् । द्विगुणितः ८६४००० द्वापरयुगचरणमानं भवति । एकगुणितः ४३२००० कलियुगचरणमानं भवति । अत्रैतदुक्तं भवति युगचरणद्वादशांशसमस्तत्सन्ध्या, सा चरणादौ भवति तावानेव सन्ध्यांशः स च युगचरणान्ते भवति, सन्ध्यासन्ध्यांशैः सह एते युगचरणाः कथिताः ।

कृतादौ सन्ध्यावर्षाणि $=\frac{\text{कृतयुग}}{१२}=\frac{१७२८०००}{१२}=$ १४४०००, कृतान्ते-सन्ध्यांशः $=$ १४४०००, एवं त्रेतादौ सन्ध्यावर्षाणि $=\frac{१२९६०००}{१२}=$ १०८०००, त्रेतान्ते सन्ध्यांशः १०८०००, द्वापरादौ सन्ध्यावर्षाणि $=\frac{८६४०००}{१२}=$ ७२०००, द्वापरान्ते सन्ध्यांशः $=$ ७२०००, कलियुगचरणान्ते सन्ध्या $=\frac{४३२०००}{१२}=$ ३६०००, कलियुगचरणान्ते सन्ध्यांशः $=$ ३६०००, चतुर्णां युगचरणानां योगः पूर्वोक्तयुगमानं भवतीति ॥६-८॥

अब युगमान कहते हैं ।

हि. भा.—४३२०००० इतने सौर वर्ष का एक युग होता है । सन्ध्या और सन्ध्यांश सहित पृथक् सत्ययुगादि चार युग चरण होते हैं । युग के दशमांश को चार से गुणने से कृतयुगचरणमान १७२८००० होता है, तीन से गुणने से त्रेतायुगचरण १२९६००० होता है । दो से गुणने से द्वापरयुगचरण ८६४००० होता है, और एक से गुणने से कलियुगचरण ४३२००० होता है ।

युगचरणों का द्वादशांश अपनी-अपनी सन्ध्या और सन्ध्यांश होता है अर्थात् युगचरण के आदि में सन्ध्या और उतने ही युगचरण के अन्त में सन्ध्यांश होता है । सन्ध्या और सन्ध्यांश से सहित पूर्वकथित युगचरण मान होता है ।

कृतादि में सन्ध्यावर्ष $=\frac{१७२८०००}{१२}=$ १४४००० कृतान्त में सन्ध्यांश $=$ १४४०००, त्रेतादि में सन्ध्यावर्ष $=\frac{१२९६०००}{१२}=$ १०८०००, में सन्ध्यांश $=$ १०८०००, द्वापरादि में सन्ध्यावर्ष $=\frac{८६४०००}{१२}=$ ७२०००, द्वापरान्त में सन्ध्यांश $=$ ७२०००, कलियुगचरणादि

सन्ध्या।वर्ष=$\frac{४३२०००}{४}$=३६०००, कलियुगचरणान्त में सन्ध्यांश=३६००, चारों युगचरणों का योग पूर्वकथित युगमान होता है। इति ॥७-८॥

इदानीमार्यभटोक्तानां कृतादीनां यानि वर्षसंख्याप्रमाणानि तेषां स्मृतिविरोधाद् दूषणमाह ।

युगपादानार्यभटश्चत्वारि समानि कृतयुगादीनि ।
यदभिहितवान् न तेषां स्मृत्युक्तसमानमेकमपि ॥९॥

वा. भा. युगपदेनात्र चतुर्युगमुच्यते । तेन युगस्य पादा युगपादाः युगचतुर्भागा इत्यर्थः । तावत एव युगपादान् आर्यभटो यदभिहितवान् चत्वारि युगानि कृतयुगादीनि तदसदुक्तं भवति । यदार्यभटेनोक्तं दश गीतिकासु 'गतास्ते च मनुयुग(ख) कल्पादेर्युगपाद' इति तदयुक्तम् । यस्मात् तेषां कृतादियुगानामार्यभटोक्तानां स्मृत्युक्तत्वादियुगेन तुल्यमेकमपि न भवत्यत्र च भगवान्मनुः "चत्वार्याहुः सहस्राणि वर्षाणि च कृतं युगम् । तस्य तावच्छती संध्या संध्यांशश्च तथाविधः।" एतावद्दिव्येन मानेन, तद्यथा दिव्यवर्षैः कृतयुगपरिमाणम् ४००० अस्य संध्या ४०० संध्यांशैश्च ४०० एकत्र ८०० एतत्संध्यासंध्यांशैः सह कृतयुगपरिमाणम् । इतरेषां त्रेतादीनां त्रेता ३००० सन्ध्या ३०० संध्यांशः ३०० द्वापरः २००० संध्या २०० संध्यांशश्च २०० कलिः १००० संध्या १०० संध्यांशः १००, एकत्र त्रेता ३६००। द्वापरः २४०० कलिः १२०० एतानि षष्टिशतत्रयेण गुणितानि सौरमानेन कृतादीनां वर्षाणि भवन्ति, ब्रह्मगुप्तोक्तयुगानां तुल्यानि १७२८०००। १२९६०००। ८६४०००। ४३२००० नार्यभट्टोक्तसममपि । पौलिशे दिव्येन मानेन कृतादीनां प्रमाणाब्दाः "अष्टाचत्वारिंशत्पादविहीनाः क्रमात्कृतादीनाम् । अब्दास्ते शतगुणिता ग्रहतुल्ययुगं तदेकत्वम्" इति पौलिशसिद्धान्ते द्रष्टव्यम् ॥

वि. भा.—आर्यभटः चत्वारि समानि (तुल्यानि) कृतयुगादीनि युगपादान् महायुगचतुर्थांशमितान् युगचरणान् यदभिहितवान् (यत्कथितवान्) तेषां युगपादानां (युगचरणानां) मानमध्ये एकमपि स्मृत्युक्तसमानं (स्मृतिकथितसदृशं) न, स्मृतिकथितयुगचरणमानानि समानानि न सन्ति तस्मादार्यभटकथितानि तुल्ययुगचरणमानानि स्मृतिविरुद्धानि तेनोपेक्ष्याणीत्यर्थः । युगचरणसम्बन्धे आर्यभटवाक्यम् 'युगपादाः ग ३ च'; इति । तथा—

अष्टाचत्वारिंशत् पादविहीना क्रमात्कृतादीनाम् । अब्दास्ते शतगुणिता ग्रहतुल्ययुगं तदेकत्वम् । इति पौलिशसिद्धान्तोक्तं दिव्यमानेन कृतादियुगचरण-वर्षमानं स्मृत्युक्तसमानमवलोक्यते । नहि केनापि स्मृत्युक्तवचनेन पुराणोक्तवचनेन चार्यभटमतस्य पुष्टिर्भवत्यतस्तन्मतं न शोभनम् ज्यौतिषसिद्धान्तकारेषु केवलं वटेश्वरसिद्धान्तकारः (वटेश्वरः) आर्यभटस्येदं मतं स्वीकरोत्येतदर्थं किमपि प्रबलं प्रमाणं नोपस्थापयत्यतस्तन्मतमपि न शोभनमिति ॥९॥

अब आर्यभटोक्त युगचरणमान को कहते हैं, खण्डन भी करते हैं।

हि. भा.—आर्यभट ने चार बराबर कृतादि युगचरणों (महायुग के चतुर्थांशतुल्य) को जो कहा है, उन युगचरणों में एक भी स्मृतिकथित युगचरण के बराबर नहीं है, स्मृतिकथित युगचरणमान सब बराबर नहीं हैं। इसलिए आर्यभटोक्त तुल्य युगचरणमान स्मृति के विरुद्ध होने से उपेक्षणीय (त्याज्य) है, युगचरण के विषय में आर्यभटोक्त वाक्य है 'युगपादाः ग३च' तथा 'अष्टाचत्वारिंशत् पादविहीना क्रमात् कृतादीनाम्। अब्दास्ते शतगुणिता ग्रहतुल्ययुगं तदेकत्वम्' इस पौलिश-सिद्धान्तोक्त दिव्यमान से कृतादि युगचरणवर्षमान स्मृतिकथित वर्ष के बराबर ही देखने में आते हैं, आर्यभटमत की पुष्टि किसी स्मृतिवचन से या पुराणोक्त वचन से नहीं होती है, इसलिए उनका मत ठीक नहीं है। वटेश्वरसिद्धान्त में वटेश्वराचार्य ने आर्यभट के इस मत को स्वीकार किया है परन्तु विरोध में स्मृतिकारादियों के मत रहते हुए भी कोई प्रबल प्रमाण नहीं उपस्थापित किया है इसलिए उनका मत भी ठीक नहीं है ॥९॥

इदानीं मनुप्रमाणानि कल्पप्रमाणं चाह।

मनुरेकसप्ततियुगः कल्पो मनवश्चतुर्दश मनूनाम्।
आद्यन्तरान्तसन्धिषु कृतकालोऽस्माद्युगसहस्रम् ॥ १०॥

वा. भा.—मनुस्तावदेकसप्ततियुगैः। युगग्रहणेन चतुर्युगमुच्यते। एकसप्ततिचतुर्युगैः मन्वन्तरं भवतीत्यर्थः। कल्पस्तु मनवश्चतुर्दश, यद्येवं न तर्हि चतुर्युगसहस्रं कल्प इत्याशंक्याह, मनूनामाद्यंतरांतसंधिषु कृतकाल इति। मनूनामादिसंधिश्चांतरसन्धिश्चान्तसन्धिश्च ते भवन्त्याद्यन्तरांतसंधयः चतुर्दशानां मनूनां पञ्चदश सन्धयो भवन्तीत्यर्थः। तेषु कृतयुगतुल्यः काल एकैकस्मिन् संधौ कृतयुगतुल्यानि वर्षाणि भवन्तीति यावत्। अस्माद्धेतोश्चतुर्युगसहस्रेण स्मृतिषु कल्पोऽभिहितः। तद्यथा मन्वतरं चतुर्युगानि एतानि चतुर्दशगुणानि वेदनवनन्दा ९९४ कृताब्दाः १७२८००० पंचदश गुणा २५९२०००० चतुर्युगप्रमाणैरेतैः ४३२०००० विभज्यावाप्तं ६। इदं पूर्वन्यस्तेषु ९९४ एषु संयोज्य जातं सहस्रम् १००० यैस्तु पुनरष्टोत्तरेण चतुर्युगसहस्रेण कल्प उक्तस्तैः स्मृतिविरोधः कृत इत्यर्थः। यतो भगवान्मनुः "दैविकानां युगानान्तु सहस्रपरिसंख्यया, ब्राह्ममेकमहो ज्ञेयं तावती रात्रिरेव च" इति। तथा च व्यासमुनिः "सहस्रयुगपर्यन्तमहो ये ब्रह्मणो विदुः। रात्रिं युगसहस्रांतां तेऽहोरात्रविदो जनाः" इति ॥१०॥

वि. भा.—एकसप्ततियुगैः मनुरुक्तः। चतुर्दश मनवः कल्पः (अर्थाच्चतुर्दशमनूनामेकः कल्पः) भवति, मनूनामाद्यमध्यावसानसन्धिषु कृतकालः (कृतयुगसमानकालः) अस्मात् कारणाद्युगसहस्रं कल्प इति ॥

अत्रोपपत्तिः—युगसहस्रं कल्पः कथं भवतीति प्रदर्श्यते—

एकसप्ततियुगानामेको मनुः=७१ युगाः, परं कल्पे चतुर्दश मनवोऽतः १४ मनु=७१यु×१४=९९४ युः परं मनूनामाद्यमध्यावसानसन्धिषु कृतकाल इत्युक्तेः

सन्ध्यासन्ध्यांशाः=कृतयुगचरण×१५
$= \frac{४युग \times १५}{१०} = ६यु$

चतुर्दश मनुषु सन्धयः=१५
युगे धर्मचरणाः=१०
ततः= $\frac{४ \times युग}{१०}$ =कृतयुग

ततः १४ मनु+मनुसन्ध्यासन्ध्यांश=९९४यु+६यु=१०००यु=१ कल्पः=ब्रह्मदिनम्।

एतावता 'चतुर्युगसहस्रेण ब्रह्मणो दिनमुच्यते' इति पुराणोक्तं युगसहस्रं कल्प इति ब्रह्मगुप्तोक्त चाप्युपपद्यते। सूर्यसिद्धान्ते सूर्यसिद्धान्तकारेण 'इत्थं युगसहस्रेण भूतसंहारकारकः। कल्पो ब्राह्ममहः प्रोक्त' मित्यनेन, भास्करेणापि स्याद्युगानां सहस्रं दिनं वेधसः सोऽपि कल्पः' इत्यनेन तदेव कथ्यते इति ॥ १० ॥

अब मनुमान और कल्पमान को कहते हैं ।

हि. भा.—इकहत्तर युगों का एक मनु होता है। चौदह मनु कल्प हैं, मनुओं के आदि, मध्य और अन्त में सन्धियां कृतकाल के बराबर हैं इस कारण एक हजार युगों का कल्प होता है ॥१०॥

एक हजार युगों का कल्प क्यों होता है इसकी उपपत्ति ।

इकहत्तर युगों का एक मनु होता है परन्तु कल्प में चौदह मनु हैं अतः १४ मनु= ७१ यु×१४=९९४ यु.

लेकिन मनुओं के आदि, मध्य और अन्त में कृतयुग के बराबर सन्धि है इसलिए चौदह मनुसम्बन्धी सन्ध्यासन्ध्यांश=कृतयुग×१५ अतः चौदह मनुसम्बन्धिनी

सन्धि=१५
$= \frac{४युग \times १५}{१०} = ६यु$

युग में धर्मचरण=१०, कृतयुग में धर्मचरण=४
इसलिए $\frac{४ \times युग}{१०}$ =कृतयुग

अतः १४मनु+मनुसन्ध्यासन्ध्यांश=९९४यु+६यु=१०००यु=१कल्प=ब्रह्मदिन, इससे 'चतुर्युगसहस्रेण ब्रह्मणो दिनमुच्यते' यह पुराणोक्त और 'युगसहस्रम्' यह ब्रह्मगुप्तोक्त भी सिद्ध हो गया। सूर्यसिद्धान्त में सूर्यसिद्धान्तकार 'इत्थं युगसहस्रेण भूतसंहारकारकः। कल्पो ब्राह्ममहः प्रोक्तम्,' इससे सिद्धान्तशिरोमणि में भास्कराचार्य भी 'स्याद्युगानां सहस्रं दिनं वेधसः सोऽपि कल्प' इससे उसी विषय को कहते हैं ॥ १० ॥

इदानीं कल्पे विशेषं प्रतिपादयति ।

आद्यन्तरान्तसन्धिषु कल्पमनूनां कृताब्दसमकालम् ।
नेच्छन्ति ये षडूनं तेषां कल्पो युगसहस्रम् ४२९४०८०००० ॥११॥

वा. भा.—कल्पे मनवः कल्पमनवः तेषामाद्यन्तरान्तसन्धिषु कृतयुगतुल्यः कालो यैर्नेप्सितस्तेषां कल्पश्चतुर्युगशतैः नवभिश्चतुर्नवत्यधिकैः भवतीति किमत्रोच्यते । प्रागार्या व्याख्यानेनैव येषामार्या गतार्थेति ।

वि. भा.—ये कल्पमनूनां (चतुर्दशमितानां) कृताब्दसमकालं (कृतयुगवर्षं) आद्यन्तरान्तेषु (आदिमध्यावसानेषु) सन्धि नेच्छन्ति तेषां मते कल्पः षडूनं युगसहस्रं भवति ।

एकसप्ततियुगैरेको मनुर्भवति, परं कल्पे चतुर्दश मनवोऽतः १४ मनु ७१यु×१४ =९९४यु. अत्र चतुर्दशमनुसम्बन्धिसन्ध्यासन्ध्यांशमानं योज्यते तदा वस्तुतः कल्पप्रमाणं भवति, परं ये मनुसम्बन्धिसन्ध्यासन्ध्यांशमानं न गृह्णन्ति तन्मते तु ९९४यु.=कल्पः ॥११॥

अब कल्प के सम्बन्ध में विशेष कहते हैं ।

हि. भा.—आचार्य कल्पमनु (चतुर्दश-संख्यक) सम्बन्धी आदि, मध्य और अन्त में सन्धि को कृताब्द (सत्ययुगवर्ष) के बराबर नहीं मानते हैं, उनके मत में कल्पप्रमाण छः घटा हुआ एक हजार युग(९९४ युग) होता है ॥११॥

इकहत्तर युगों का एक मनु होता है, लेकिन कल्प में चौदह मनु हैं इसलिए. १४ मनु =१४×७१यु=९९४यु, इसमें मनुसम्बन्धी सन्ध्यासन्ध्यांश जोड़ने ही से वास्तवकल्प प्रमाण हो सकता है, जो उनके मान नहीं लेते हैं उनके मत में ९९४ यु=कल्प पर यह ठीक नहीं है ॥११॥

इदानीं कल्पसम्बन्धे आर्यभटमतं प्रदर्शयन्नाह

मनुसन्धि युगमिच्छत्यार्यभटस्तन्मनुर्यतः खयुगः ।
कल्पश्चतुर्युगानां सहस्रमष्टाधिकं तस्य ४३५४५६०००० ॥ १२ ॥

वा. भा.—मनोः सन्धिः मनुसंधिः । सधिमिच्छत्यार्यभटः युगतुल्यं प्रायेण । यतः तन्मनुः खयुगः तस्य मनुस्तन्मनुः स च संख्या निर्युगानि यत्र.....सम..... युग.....त्तदं.....सुक ...मन .....नुयु.....द्वास.....तस्य मन्वन्तरं भवतीत्यर्थः । एवं चतुर्युगानां सहस्रमष्टाधिकं तस्य कल्पः । तथा चाष्टशते अष्टोत्तरसहस्रं ब्राह्मो दिवसो ग्रहयुगानामिति ॥ यथा कुडव-प्रस्थ-द्रोणाद्यैः मेषो राशिः परिच्छिद्यते एवं युगमन्वन्तरकल्पैः काल इति स्मृतिषु पठ्यते । तथाचार्यश्रीषेणनिबद्धे रोमकसिद्धान्ते न पठितः ।

वि. भा.—आर्यभटो युगं (युगतुल्यं) मनुसन्धिमिच्छति, यतस्तत् (तस्याऽऽर्यभटस्य) मनुः खयुगोऽर्थाद् द्विसप्ततियुगैर्भवति अतस्तस्याऽऽर्यभटस्य मते चतुर्युगानां (महायुगानां) अष्टाधिकं सहस्रं (१००८) कल्पः (ब्रह्मदिनं) भवतीति ॥ १२ ॥

अत्रोपपत्तिः ।

वर्गाऽक्षराणि वर्गे इत्याद्यार्यभटोक्तेः श=७०, ख=२ एतयोर्योगः=७२, ब्रह्मगुप्तादिभिरेकसप्ततियुगैरेको मनुः कथ्यते, आर्यभटेन द्विसप्ततियुगैः कथ्यते । तेन तन्मते युगसम एव मनुसन्धिरिति स्पष्टं प्रतीयते । तन्मतेनापि कल्पमानम्=१४ मनु=१४×७२ युग=१००८ युग=कल्पः, एतेन 'कल्पश्चतुर्युगानां सहस्रमष्टाधिकं तस्ये'ति ब्रह्मगुप्तोक्तमुपपद्यते ।

दशगीतिकायाम् । "काहो मनवो ढ १४ मनुयुग ख ७२ गतास्ते च ६ मनुयुग छ्ना २७ च । कल्पादेर्युगपादा ग३च गुरुदिवसाच्च भारतात्पूर्वम्" इति ।

कालक्रियापादे च "दिव्यं वर्षसहस्रं ग्रहसामान्यं युगं द्विषट्कगुणम् । अष्टोत्तरं सहस्रं ब्राह्मो दिवसो ग्रहयुगानाम् ।।" इति च आर्यभटोक्तमस्ति । केवलमार्यभटेन ब्रह्मगुप्तमतविरोधिनाऽऽर्यभटमताश्रयिणा वटेश्वराचार्येण च द्विसप्ततियुगैरेकः कल्पो भवतीति कथ्यते परं तत्समर्थनमन्यैर्ज्यौतिषाचार्यैः स्मृतिकारैः पुराणैश्च न कृतं, तेन तन्मतं कथं शोभनमिति सुधियो विभावयन्त्विति ।। १२ ।।

अब आर्यभटोक्त कल्पमान को कहते हैं

हि. भा.—आर्यभट भी युगतुल्य मनुसंधि स्वीकार करते हैं क्योंकि उनके मत में एक मनु ख युग (७२ युग) बहत्तर युगों के होते हैं और एक हजार आठ युगों (कृतयुग त्रेता, द्वापर और कलियुगों के योग=युग) के कल्प (ब्रह्मदिन) होता है ॥ १२ ॥

उपपत्ति

ब्रह्मगुप्तादि आचार्य इकहत्तर युगों के एक मनु कहते हैं, आर्यभट बहत्तर युगों को एक मनु कहते हैं इसलिए उनके मत में युगसमान ही मनुसन्धि है यह स्पष्ट प्रतीत होता है । उनके मत में भी कल्प=१४ मनु=१४×७२युग=१००८ युग= ब्रह्मदिन, इससे 'कल्पश्चतुर्युगानां सहस्रमष्टाधिकंतस्य' यह ब्रह्मगुप्तोक्त उपपन्न होता है ।

आर्यभट दशगीतिका में कहते हैं

"काहो मानवो ढ १४ मनुयुग ख ७२ गतास्ते च ६ मनुयुग छ्ना ७२ च । कल्पादेर्युगपादा ग३ च गुरुदिवसाच्च भारतात्पूर्वम् ।"

कालक्रियापाद में—

"दिव्यं वर्षसहस्रं ग्रहसामान्यं युगं द्विषट्कगुणम् । अष्टोत्तरं सहस्रं ब्राह्मो दिवसो ग्रहयुगानाम् ।।" केवल आर्यभट और ब्रह्मगुप्त-मत विरोधी तथा आर्यभटमताश्रयी

वटेश्वराचार्य बहत्तर युगों का कल्पमान कहते हैं, लेकिन उनके मत का समर्थन अन्य किसी ज्यौतिषाचार्य, स्मृतिकार तथा पुराणों ने नहीं किया है इसलिए उनका मत कैसे ठीक है इस बात का विवेचक लोग विचार करें ॥ १२ ॥

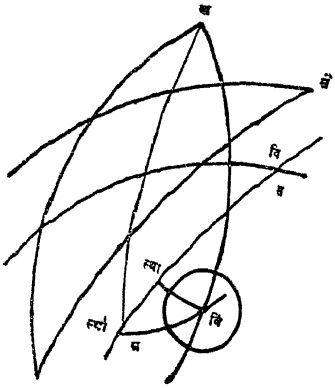
इदानीं रोमकसिद्धान्तमतं खण्डयति ।

युगमन्वन्तर कल्पाः कालपरिच्छेदकाः स्मृतावुक्ताः ।
यस्मान्न रोमके ते स्मृतिबाह्यो रोमकस्तस्मात् ॥ १३ ॥

वि. भा.—यस्मात् कारणात्स्मृतौ (वेदार्थप्रतिपादके ग्रन्थे) युगमन्वन्तर-कल्पाः (युगमन्वन्तरादयः) कालपरिच्छेदकाः (समयविभाजकाः) उक्ताः (कथिताः) अन्यथाऽनाद्यनन्तव्यापककालेन मानवानामेकमपि व्यवहारकार्यं न चलेत् । रोमके (रोमकसिद्धान्ते) ते (युगमन्वन्तरादयो) न सन्त्यर्थात्तेषां तेषां नामोल्लेखा न सन्ति तस्मात्कारणाद्रोमकः स्मृतिबाह्योऽतोऽत्र रोमकसिद्धान्तस्त्याज्य इति ॥ वस्तुतो ज्यौतिषसिद्धान्तग्रन्थेषु युगमन्वन्तरकल्पानां ग्रहादिसाधनार्थं कीदृशं प्राधान्यमिति तत्साधका एव ज्ञातुं शक्नुवन्ति । तान् विना सकलं ज्यौतिषशास्त्रं निरर्थकमेव भवेत् । अतो रोमकसिद्धान्ते तच्चर्चाऽकरणेन कीदृश्यस्त्रुटयः कृतास्त-त्सिद्धान्तकर्त्रेति मन्दमतयोऽपि ज्ञातुं शक्ष्यन्तीति ॥ १३ ॥

अब रोमकसिद्धान्त मत का खण्डन करते हैं ।

हि. भा.—जिस कारण से वेदार्थ के प्रतिपादन करने वाले स्मृतिग्रन्थों में युग-मन्वन्तर और कल्प को काल (समय) का परिच्छेदक (विभाजक) कहा गया है अर्थात् इन्हीं युग-मन्वन्तरादियों के द्वारा विभक्त अनाद्यनन्त व्यापक काल से मानवों के सब व्यवहार चलते हैं, यदि ऐसा नहीं होता तो अविभक्त व्यापककाल से एक भी कार्य होना असम्भव है इसलिए पूर्वोक्त विषयों के उल्लेख स्मृतिकारों ने अत्यावश्यक समझ कर किये हैं। रोमकसिद्धान्त में इन सब की चर्चा भी नहीं की गई है इसलिए वह सिद्धान्त स्मृतिबाह्य है अर्थात् स्मृतिशास्त्रों से बहिर्भूत है इसलिए वह त्याज्य है। वस्तुतः ज्यौतिषसिद्धान्त में युग-मन्वन्तरादियों की ग्रहादि साधनार्थ कैसी प्रधानता है यह विषय ग्रहादि साधन करने वाले ही जान सकते हैं । बिना उनके सम्पूर्ण ज्यौतिष-शास्त्र निरर्थक है इसलिए रोमक सिद्धान्त में उनकी चर्चा न करके बहुत बड़ी त्रुटि की गई है, इस बात को अल्पज्ञ भी समझ सकते हैं ॥ १३ ॥



ख = स्वस्वस्तिकम् ।
सं = गर्भीयगोलसन्धिः ।
सं′ = पृष्ठीयगोलसन्धिः ।
स्था = गर्भीयपरिणतबिम्बोपरिगतकदम्बवृत्त-तत्रत्यक्रान्तिवृत्तसंपातविन्दुः ।
स्थां = प्रैष्ठिकबिम्बीयकदम्बवृत्ततत्रत्यक्रान्ति-मण्डलसंपातः ।
खस्थां = स्थां ग्रहस्य पृष्ठीयनतभागाः ।
खग्र = स्थां ग्रहस्यैव नतांशसमाना गर्भगोले नतभागाः ।
ग्र = दार्ष्टिकगोलीयग्रहसमानो गर्भगोले दृङ्म-ण्डलीयो ग्रहः ।
विख = गर्भीयबिम्बग्रहनतांशाः ।
वि = विम्बभम् (पृष्ठगोले)

अत्र खविस्थां चापजात्यत्रिकोणे विख, स्थांविभुजकोटध्योर्ज्ञानेन त्रिकोणमि-त्या खस्थां कर्णचापमानं विधातव्यम् । तदेव खग्रमानं भवेत् । दृग्मण्डलस्यै-कत्वात्, तथा भूकेन्द्रात् 'ग्र' गतसूत्रदृष्टिस्थानात् स्थागतसूत्रयोः समानान्तरत्व-सिद्धेः । ततोऽनन्तरं खविग्रचापीयत्रिभुजे 'खग्र' 'खवि' भुजयोस्तदन्तर्गतकोणस्य दिगंशान्तरस्य च ज्ञानात् 'विग्र' चापमानं प्रजायते । ततो ग्रहान्तरं स्थाग्र-मानं ज्ञातव्यम् । इदमेव दृग्गर्भग्रहयोरन्तरमानं भवति यदर्थं विद्वांसो निमज्ज-न्ति गणिताम्बुधौ । किं बहुनेति ।

अथ रविबुधशुक्रभगणोपपत्तिः ।

अथ 'रविभगणा रव्यब्दाः' इत्याचार्योक्तेः कल्पे यावन्तो रविभगणास्ता-वन्त्येव सौरवर्षमानानि भवन्ति, परं कल्पवर्षप्रमाणं तु विदितमेवास्ति, तत्तुल्यरवि-भगणमानमपि विदितं जातं, बुधशुक्रयोरुदयसत्तायां रवेरुदयास्तकाले रविबुधयो रविशुक्रयोश्चान्तरांशाः प्रतिदिनं यन्त्रद्वारा ज्ञातव्यास्ते च राशित्रयान्यूना एवा-गच्छन्ति । वराहमिहिराचार्येणापि बृहज्जातके पूर्वाचार्योक्तवज्रादियोगलक्षणानां खण्डनावसरे 'पूर्वशास्त्रानुसारेण मया वज्रादयः कृताः । चतुर्थे भवने सूर्याज्ज्ञासतौ भवतः कथमि'त्यादिना रविणा सह तयोरन्तरमल्पमेव भवतीति कथ्यते, तेनैव हेतुना रवेरतिनिकटस्थितत्वात्कदाचिदग्रतः कदाचित्पृष्ठस्तस्यानुचराविव सदा तौ (बुध-शुक्रौ) व्रजन्तौ दृश्येते, तेन तयोरपि कल्पभगणा रविकल्पभगणतुल्या एवाचार्यैः स्वीकृता इति ।

अथ चन्द्रभगणोपपत्तिः।

अथ ग्रहवेधाथ समुपयुक्तस्थाने निर्मिते वेधालये नाड़ीवृत्तक्रान्तिवृत्तकदम्बप्रोतवृत्तादिरचितं विपुलं गोलयन्त्रं कार्यं, तत्र क्रान्तिवृत्ते चक्रांशाः ३६० राश्यंशकलादयो नाड़ीवृत्ते च दण्डपलादयोऽङ्कनीयाः, तद्यन्त्र केनचिदाधारद्वितयेन केन्द्रगतनलिकया च स्थिरीकृत्य गोलकेन्द्रे ध्रुवाभिमुखीं यष्टीं कृत्वा रात्रौ तद्गोलकेन्द्रगतदृष्ट्या रेवतीतारामवलोकयेत्। सा गोलयन्त्रीयक्रान्तिवृत्ते यत्र परिणता तत्रैव मेषादिरङ्कनीयः। तथा च गोलमध्यगतदृष्ट्यैव चन्द्रवेधकरणेन यत्र गोलयन्त्रे परिणतो भवेत्तदुपरिगतं कदम्बप्रोतवृत्तं (वेधवृत्तं) तद्गोलीयक्रान्तिवृत्ते यत्र लगति स एव वेधागतः स्पष्टचन्द्रस्तद्राश्यादिमानं मेषादेर्यद्भवति तद्विगणय्य ग्राह्यमेवं राश्यादिकः स्पष्टश्चन्द्रो विदितो जातः, एवं द्वितीयदिनेऽपि राश्यादिस्पष्टचन्द्रो वेदितव्यः। एताभ्यां विदितस्पष्टचन्द्राभ्यां विदितचन्द्रमन्दोच्चाच्च 'स्फुटं ग्रहं मध्यखगमि' त्यादिना दिनद्वयजौ मध्यमचन्द्रौ विदितौ भवेतां, तयोरन्तरमेकदिनजा चन्द्रमध्यमगतिर्भवेत्ततोऽनुपातेन 'यद्येकेन दिनेनेयं चन्द्रमध्यमगतिर्लभ्यते तदा कल्पकुदिनैः किम्' इत्यनेन समागच्छन्ति कल्पे चन्द्रभगणाः। परमत्रोपपत्तौ वेधद्वारा यः स्पष्टचन्द्रो गृहीतः स च वेधगोलीय (पृष्ठीयत्रिज्यागोलीय) स्पष्टचन्द्रः, परमपेक्षितस्तु भूकेन्द्रिकत्रिज्यागोलीयः, अतस्तयोर्वेधगोलीय (भूपृष्ठीयगोलीय) स्पष्टचन्द्रभूकेन्द्रिकत्रिज्यागोलीयस्पष्टचन्द्रयोरन्तरानयनं कृत्वा तेन सस्कृतो वेधगोलीयस्पष्टचन्द्रो भूकेन्द्रिकत्रिज्यागोलीय(भूगर्भगोलीय)स्पष्टचन्द्रो भवेत्। एवं वेधगोलीयद्वितीयदिनजस्पष्टचन्द्राद् भूगर्भगोलीयस्पष्टचन्द्रो वेदितव्यस्ततो विदितभूगर्भगोलीयदिनद्वयजस्पष्टचन्द्राभ्यां पृथक्-पृथक् 'स्फुटं ग्रहं मध्यखगमि' त्यादिना दिनद्वयजौ मध्यमचन्द्रौ भूगर्भगोलीयौ भवेतां, ततस्तदन्तर-(मध्यमचन्द्रान्तर)-वशात् पूर्ववत्कल्पे चन्द्रभगणा भवितुमर्हन्ति।

अथाधुना वेधगोलीयस्पष्टचन्द्राद् भूगर्भगोलीयस्पष्टचन्द्रज्ञानार्थमुपपत्तिः। पूर्वोपपत्तौ स्पष्टचन्द्रस्य चर्चाऽस्ति। तेन गोलद्वयीययोः(वेधगोलीय भूगर्भगोलीययोः स्पष्टचन्द्रयोरन्तरानयनं क्रियते। परमेतदर्थं योपपत्तिः सैव सर्वेषां ग्रहाणां (वेधगोलीयग्रहेभ्यो भूगर्भगोलीयग्रहाणां) ज्ञानार्थं भवतीति बोध्या।

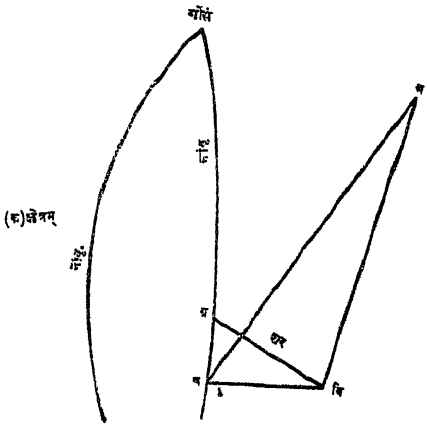
वेधगोले दृग्वशेन (दृष्ट्या) परिणतचन्द्रबिम्बस्य स्पष्टभोगचिह्नं (चन्द्रबिम्बोपरिगतकदम्बप्रोतवृत्तं क्रान्तिवृत्ते यत्र लगति स बिन्दुः) तद्गोलीयस्पष्टचन्द्रः। एवं भूगर्भगोलेऽपि स्पष्टचन्द्रस्थानं ज्ञेयम्।

अथ परिभाषाः

वेधगोलीयचन्द्रस्थानम् = स्थान, स्थानीयदृग्वृत्तधरातलेन च्छिन्नस्य भूगर्भगोलस्य च्छेदनं तद्गोलीयदृग्वृत्तम्। तस्य (तद्गोलीयदृग्वत्तस्य) भूगर्भ-

गोलीय क्रान्तिवृत्तस्य च योगबिन्दुः=ष, भूगर्भात् ष-बिन्दुगता रेखा = षसंज्ञका, दृष्टितः स्थानगता रेखा च फसंज्ञका,

अथ प फ, रेखे समान्तरे (रेखागणितैकादशाध्याययुक्त्या), भूगर्भदृष्टिस्थानाभ्यां रेखतीगते रेखे च समानान्तरे। अतो भूगर्भदृष्टिस्थानलग्नकोणयोः साम्यात् सिद्धं यद् भूगर्भगोले (भगोले) रेखतीतो ष बिन्दुपर्यन्तं वेधगोलीयस्पष्टचन्द्रतुल्यमर्थात् (भगोलीयरेखतीतो र बिन्दुपर्यन्तम् = वेधगोलीयरेखतीतः स्थानपर्यन्तम्) केन्द्रलग्नकोणस्य चापमानत्वात्। स्थानीयनतांशाः = ष बिन्दूत्पन्ननतांशाः (प,फ रेखयोः समानान्तरत्वात्) स च नतांशो वेधगोले मापनेन ज्ञातः, तथा चन्द्रबिम्बोयनतांश ष-बिन्दूत्पन्ननतांशचापाभ्यां जायमानः खस्वस्तिकलग्नकोणो यावान् वेधगोले तावानेव भूगर्भगोलेऽपि (गोलद्वये धरातलैकत्वात्) स च नतांशोत्पन्नकोणो वेधगोले मापनेन विज्ञेयः। अतो भूगर्भगोलपृष्ठे जायमानत्रिभुजे त्रिज्यागुणाद्धरणिकोटिगुणादित्यादि विलोमेन परिणतचन्द्रबिम्बकेन्द्र-ष-बिन्दुगतवृत्तीयाऽधारचापज्ञानं जातम्। तथा च वेधगोलीयशरक्रान्तिवृत्तधरातलान्तरयोर्ज्ञानाद् भूगर्भगोले शरज्ञानम् (वेधगोलीयभूगर्भगोलीययोर्नाड़ीवृत्तधरातलयोरन्तरस्य वेधगोलीयक्रान्तेश्च ज्ञानाद् भूगर्भगोलीयक्रान्तिज्ञानार्थं या युक्तिस्तादृश्येवात्र शरज्ञानार्थमस्ति) तेन चापजात्ययुक्त्या भूगर्भगोलीयस्पष्टचन्द्र-ष-बिन्दोरन्तरचापस्य संस्कारसंज्ञकस्य ज्ञानम्।



अन्तरम्=संस्कारचापः

अतः वेधगोलीय स्पचन्द्र ± संस्कारचा = भूगर्भगोलीयस्पचन्द्रः। नहि दार्ष्टिकगोलेऽपि स्थानोद्भवा नतांशा वेधेन ज्ञातुं शक्याः। स्थानस्याप्रत्यक्षत्वात्। वेधस्तु प्रत्यक्षे वस्तुनि भवति। अतस्तत्रत्यक्षितिभग्रहान्तर दृक्षेपवशात् त्रिकोणमित्या विज्ञेयम्।

अथ संस्कारचापस्य धनर्णव्यवस्था। तत्र परिभाषाः।

वेधगोलीय-क्रान्तिवृत्तम्=इष्टक्रान्तिवृत्तम्। भूगर्भगोलीयक्रान्तिवृत्तम्=वास्तवक्रान्तिवृत्तम्। बिम्बोयकर्णगोलीयक्रान्तिवृत्तम्=वास्तवक्रान्तिवृत्तम्। परेखा वर्धिता वास्तवक्रान्तिवृत्ते यत्र लगति तत्र ष-बिन्दुः। चन्द्रबिम्ब-केन्द्रादिष्टक्रान्तिवृत्तधरातले या शरज्या लम्बस्तन्मूलं=क्ष। अयं बिन्दुर्वर्धितायां फ-रेखायामेव स्यात्। फरेखा तु स्थानीयदृग्वृत्तधरातले उक्तशरज्या वर्धिताऽवर्धिता वा वास्तवक्रान्तिवृत्तधरातले लम्बः स्यात्। अत्रेदमुक्तं भवति स्थानीयदृग्वृत्तधरातल-

निष्ठः क्ष बिन्दोर्वास्तवक्रान्तिवृत्तधरातले लम्बः क्रियते तन्मूलं यस्यां दिशि स्थानीय-दृग्वृत्त-वास्तवक्रान्तिवृत्त-धरातलाभ्यामुत्पन्नकोणोऽल्पः स्यात्तद्दिश्येव पतिष्यति ।

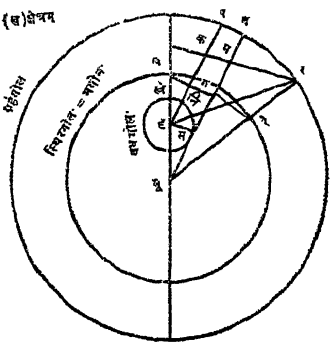
भूगर्भाद्विम्बीयकर्ण-व्यासार्धेन यो गोलस्तत्रोच्यते ।

ष-विन्दूत्थदृग्वृत्तवास्तव-क्रान्तिवृत्ताभ्यामुत्पन्नकोणो दक्षेपचापाभिमुखोऽल्पः स्यात्, क्ष-विन्दुस्तु वास्तवक्रान्तिवृत्तधरातलोर्ध्वाधरसूत्रयोर्मध्ये स्यात् फरेखाया मध्ये स्थितत्वात्, एतेन सिद्धं यद्दक्षेपवृत्तात्पूर्वकपाले चन्द्रे सति रेखातः पश्चिमायां दिश्येव लम्बः पतिष्यति, ष-रेखायाः स्थानीयदृग्वृत्तवास्तवक्रान्तिवृत्तधरातलयोर्योगरेखारूपत्वात् । भूगर्भाल्लम्बमूलगता रेखा ष' विन्दुतः पश्चिमायामेव दिशि क्रान्तिवृत्ते लगिष्यति, स एव विन्दुर्भूगर्भाभिप्रायिक चन्द्रस्थानम् । त्रिज्यागोलेऽपीयमेव स्थितिः। पश्चिमकपालेऽप्येवमेव विचारणीयम् । एतावता सिद्धं यद्वित्रिभादूने चन्द्रे संस्कारचापं धनमन्यथर्णमिति ॥

अथाऽधुना पूर्वोपपत्तौ वेधगोलीयभूगर्भगोलीययोर्नाड़ीवृत्तधरातलयोरन्तरस्य वेधगोलीयक्रान्तेश्च ज्ञानाद्भूगर्भगोलीयक्रान्तिज्ञानार्थं या युक्तिस्ताद्दृश्येव वेधगोलीयशरक्रान्तिवृत्तधरातलान्तरयोर्ज्ञानाद् भूगर्भगोलीयशरज्ञानार्थं भवतीति यल्लिखितं तदर्थं विचार्यते ।

दृष्टिस्थानान्निर्मितो गोलो वेधगोलो दृश्यगोलो वा, भूगर्भान्निर्मितो गोलः स्थिरगोलो भगोलो वा, भूगर्भाद् दृष्टिस्थानाच्च भचक्रस्थध्रुवतारागते रेखे यत्र यत्र स्वस्वगोले (स्थिरगोले-वेधगोले च) लग्ने तत्र तत्र तद्गोलद्वये परिणतध्रुवे, ताभ्यां (परिणतध्रुवाभ्यां नवत्यंशेन कृते वृत्ते गोलद्वये नाड़ीवृत्ते, भूगर्भदृष्टिस्थानाभ्यां भचक्रस्थध्रुवगतरेखाभ्यां भूकेन्द्रदृष्टिस्थानान्तररेखया च यत्त्रिभुजं तद्धरातलच्छिन्नगोलद्वयीमार्गे च गोलद्वये याम्योत्तरवृत्ते, स्वनाड़ीवृत्तयाम्योत्तरवृत्तधरातलयोर्योगरेखा स्वनिरक्षोर्ध्वाधरसूत्रम् । वर्धितभूकेन्द्रदृष्टिस्थानगतरेखा चोर्ध्वाधरसूत्रम् । ध्रुवसूत्रस्य नाड़ीवृत्तधरातलोपरिलम्बत्वाद् ध्रुवसूत्रयोः समानान्तरत्वाच्च स्थिरगोलीय (भगोलीय) वेधगोलीयनाड़ीवृत्तधरातले समानान्तरे (रेखागणितैकादशाध्याययुक्त्या) । अथ दृष्टिस्थानाद् भगोलीयनाड़ीवृत्तधरातलोपरिकृतो लम्बो वेधगोलीयभगोलीयनाड़ीवृत्तधरातलयोरन्तरम् । गोलद्वयेऽक्षांशयोः समत्वात्तद्धरातलान्तरज्ञानं क्रियते । भूगर्भाद् दृष्टिस्थानं यावदेको भुजः, दृष्टिस्थानाद् भगोलीयनाड़ीवृत्तधरातलोपरिकृतो लम्बो नाड़ीवृत्तधरातलान्तरं द्वितीयो भुजः । गर्भीयनिरक्षोर्ध्वाधररेखाखण्डं तृतीयो भुज इति भुजत्रयैरुत्पन्नजात्यत्रिभुजे भूगर्भलग्नकोणः=अक्षांशः, लम्बमूलबिन्दुलग्नकोणः=९०, भूगर्भदृष्टिस्थानान्तरम्=केन्द्रान्तरसंज्ञकम् । तदाऽनुपातो यदि त्रिज्यया कोणज्यया केन्द्रान्तरं लभ्यते तदाऽक्षज्यया किमित्यनेन समागतं नाड़ीवृत्तधरातलान्तरम् $=\frac{\text{अक्षज्या}\times\text{केन्द्रान्तर}}{\text{त्रि}}$ । अथ दृष्टिस्थानाद्विगतदृष्टि-

सूत्रस्य स्वनाड़ीवृत्त·(वेधगोलीयनाडीवृत्त·) धरातलस्य चान्तरं वेधगोलेऽन्तरम्=वेधगोलीयक्रांज्या वेधगोलीयक्रान्तिज्यामापनेन विदितैव, अतो $\frac{\text{'वेधगोलीयक्रांज्या}\times\text{दृष्टिकर्ण}}{\text{वेधगोलव्या}}=$ ग्रहाद् वेधगोलीयनिरक्षोर्ध्वाधरसूत्रोपरिलम्बरेखा=लम्ब,' ततो लम्ब ∽ नाड़ीवृत्तधरातलान्तर=लम्ब ∽ अन्तर= ग्रहगोलीयक्रान्तिज्या। एतज्ज्ञानेन $\frac{\text{ग्रहगोलीय क्रांज्या}\times\text{त्रि}}{\text{ग्रबिम्बीयकर्ण}}=$ भगोलीयक्रांज्या = भूगर्भगोलीयक्रांज्या = स्थिरगोक्रांज्या। अनयैव रीत्या वेधगोलीयशरज्ञानेन क्रान्तिवृत्तधरातलान्तरज्ञानेन च भूगर्भगोलीयशरज्ञानं भवेदेवेति।



र=रविः। ख=भगोलीय खस्वस्तिकम्। ख'=वेधगोले खस्वस्तिकम् दृर=रविदृष्टिकर्णः। भूर=रविकर्णः। दृस=नाड़ीवृत्तधरातलान्तरम् ± कम। रम=लम्बः। भूदृ=केन्द्रान्तरम्। भू=भूकेन्द्रम्। दृ=दृष्टिस्थानम्।
खनि=वेधगोलीयाक्षांशाः।
खनि=भगोलीयाक्षांशाः।
वेधगोलीयाक्षांश=, भगोलीयाक्षांश=ख'नि= खनिरंग=भगोलीयक्रान्तिज्या।

अथ रविमन्दोच्चोपपत्तिः।

भूकेन्द्राद्रविमन्दगोलकेन्द्रगता रेखा रविमन्दप्रतिवृत्ते ऊर्ध्वभागे यत्र लगति तदेव रविमन्दोच्चम् (भूकेन्द्रात्तद्विन्दोर्मन्दप्रतिवृत्तान्यबिन्दुभ्योऽतिदूरे स्थितत्वादुच्चमिति नाम सार्थकम्)। रविकर्णेन त्रिज्या लभ्यते तदा रविबिम्बव्यासार्धेन किमित्यनुपातेन यत्फलं तच्चापं द्विगुणितं तदा रविबिम्बकलाप्रमाणमागच्छति। प्राचीनाचार्यैरनुपातागतफलमेवं द्विगुणीकृत्य तच्चापं बिम्बकलामानं कथ्यते तन्न युक्तम्। एतस्या रविबिम्बकलायाः परमाल्पत्वे वेधागतरविरेव तत्र मन्दोच्चो भवितुमर्हति, बिम्बकलानयनप्रकारदर्शनादुच्चस्थान एव तत्कर्णस्य परमाधिकत्वात्तद्बिम्बकलायाः परमाल्पत्वात्। द्वितीयपर्ययेऽप्येवं रविमन्दोच्चज्ञानं कार्यम्। तयो रविमन्दोच्चयोरन्तरेण तद्दिनजा (प्रथमविरदितरविमन्दोच्चाद् द्वितीयपर्ययीयविमन्दोच्चज्ञानार्थं तदन्तरे यावन्ति

दिनानि) रविमन्दोच्चगतिर्भवेत् तदाऽनुपातो यद्येभिर्दिनैरियं रविमन्दोच्चगतिस्तदैकदिनेन किमित्यनुपातेनैकदिनजा तद्गतिः ततोऽनुपातो यद्येकेन दिनेनेयं रविमन्दोच्चगतिस्तदा कल्पकुदिनैः किमित्यनुपातेन कल्पे रविमन्दोच्चभगणाः समा गच्छन्तीति ।

अथवा ज्ञातरविमन्दोच्चा ७८' दष्टाद्रिमितादंशात्कुट्टकयुक्त्या तज्ज्ञानं प्रदर्श्यते ।

कल्परविमन्दोच्चभगणमानम् = य

तदा १८५३ शकान्ते कल्पादितः सौराब्दाः = १९७२९४१०३२ (नवनगशशिमुनिकृतनवेत्यादिब्राह्मस्फुटसिद्धान्तोक्तया, गोऽद्रीन्द्वद्रिकृताङ्कदस्रनगगोचन्द्रा इत्यादिभास्करोक्तया वा) । कल्पसौरवर्षैर्यदि कल्परविमन्दोच्चभगणा लभ्यन्ते तदा कल्पादितः शकान्तं यावत्पूर्वानीतसौरवर्षैः किमित्यनुपातेन सशेषा गतरविमन्दोच्चभगणा इष्टवर्षान्ते समागच्छन्ति, तत्स्वरूपम् =

$\frac{१९७२९४१०३२ \times य}{४३२०००००००} = ल + \frac{शे}{४३२०००००००}$ ततः १९७२९४१०३२ × य = ४३२०००००००० × ल + शे समशोधनेन,

१९७२९४१०३२ × य — ४३२०००००००० × ल = शे, एते चक्रांशैः ३६० गुणयित्वा कल्पवर्षैर्भज्यास्तदा फलं रविमन्दोच्चप्रमाणम् = ७८' ततः

$\frac{(१९७२९४१०३२ \times य — ४३२०००००००० \times ल) \times ३६०}{४३२०००००००} = ७८'$ छेदगमेन

(१९७२९४१०३२ × य — ४३२०००००००० ल) × ३६० = ४३२०००००००० × ७८ पक्षौ ३६० भक्तौ तदा १९७२९४१०३२ × य — ४३२०००००००० × ल = $\frac{४३२०००००००० \times ७८}{३६०}$

= १२००००००० × ७८ = ९३६०००००० समयोजनेन १९७२९४१०३२ × य = ९३६०००००० + ४३२०००००००० × ल समशोधनेन.

१९७२९४१०३२ × य — ९३६०००००० = ४३२०००००००० × ल

$\therefore \frac{१९७२९४१०३२ \times य — ९३६००००००}{४३२०००००००} = ल$ । अत्राष्टभिरपवर्त्तनेन

$\frac{२४६६१८६२९ \, य — ११७००००००}{५४०००००००} = ल$. अत्र ५५६७०१ एभिरपवर्त्तनेन

$\frac{४४३ \, य — २१०}{९७०} = ल$ (स्वल्पान्तरात्) तदारूपक्षेपे ऋणात्मके लब्धिगुणौ समानीयाभीप्सितक्षेपविशुद्धिनिष्पावित्यादिना लब्धिः = २१९, गुणकः = ४८० ते भाज्यतद्भाजकवर्णमाने इति भास्करोक्तया गुणक एव भाज्यवर्ण (य) मानं भवेत्तेन य = ४८० = कल्परविमन्दोच्चभगणाः ।।

अथ चन्द्रमन्दोच्चभगणोपपत्तिः ।

उच्चं द्विविधं मन्दशीघ्रभेदेनोच्चं भवत्यर्थान्मदोच्चं शीघ्रोच्चं च । शीघ्राख्यतुङ्गस्य तयोरभावादित्यादिभास्करोक्ते रविचन्द्रयोः केवलं मन्दोच्चमेव भवति । चन्द्रस्य बिम्बकलायाः परमाल्पत्वं तन्मन्दोच्चस्थाने भवेत्तत्र तदा यावान् वेधागतः स्पष्टचन्द्रस्तावदेव तन्मन्दोच्चम् । एवं द्वितीयपर्ययेऽपि चन्द्रमन्दोच्चं तयोश्चन्द्रमन्दोच्चयोरन्तरं तद्दिनजा (प्रथमपर्ययीयवेधविदितचन्द्रमन्दोच्चाद् द्वितीयपर्ययीयवेधविदितचन्द्रमन्दोच्चं यावद्यावन्ति दिनानि) चन्द्रमन्दोच्चगतिर्जाता ततोऽनुपातो यद्येभिर्दिनैरियं चन्द्रमन्दोच्चगतिस्तदैकेन दिनेन किमिति समागतैकदिनजा चन्द्रमन्दोच्चगतिस्ततो यद्येकेन दिनेनेयं चन्द्रमन्दोच्चगतिस्तदा कल्पकुदिनैः किमित्यनुपातेन कल्पे चन्द्रमन्दोच्चभगणा जायन्ते । बिम्बीयकर्णसम्बन्धेनापि चन्द्रमन्दोच्चभगणोपपत्तिर्भवितुमर्हति । यथा वेधेन प्रत्यहं चन्द्रस्य बिम्बीयकर्णज्ञानं कार्यम् । यदा चन्द्रस्य शराभावो भवेत्तदा चन्द्रबिम्बीयकर्णस्य परमत्वे वेधेन स्पष्टचन्द्रो बोद्धव्यस्तदेव तदा तन्मन्दोच्चं भवेत् । एवं द्वितीयपर्ययेऽपि चन्द्रमन्दोच्चं ज्ञात्वा तयोरन्तरेण तदन्तर्गतदिनप्रमाणेन च पूर्ववच्चन्द्रमन्दोच्चभगणज्ञानं भवेदिति ॥ बिम्बकलासम्बन्धेन भगणज्ञानार्थं लम्बनावनतिदर्शनार्थमियं भूरन्यथा केवलं बिन्दुरेव भूरिति भास्कराचार्योक्तिरेवाश्रयणीया अन्यथा भगणज्ञानमतीव दुर्घटमिति ॥

अथ चन्द्रपातभगणोपपत्तिः ।

अत्र दृष्टिस्थानाभिप्रायिकगोलस्य निर्माणं कार्यं स एव वेधगोलः, दृष्टिस्थानात् (वेधगोलकेन्द्रात्) चन्द्रो वेध्यो यत्र वेधगोले समुपलब्धस्तदुपरि वेधगोलीयं कदम्बप्रोतवृत्तं कार्यं, तत्तद्गोलीयक्रान्तिवृत्ते यत्र लगति ततश्चन्द्रबिम्बकेन्द्रं यावत्कदम्बप्रोतवृत्ते चन्द्रशरांशाः । ततः

$\frac{\text{वेधगोलीय शरज्या} \times \text{त्रि}}{\text{वेधगोल व्या}\frac{1}{2}} = \text{चन्द्रगोले शरज्या}$, वेधगोलीयभगोलीययोः क्रान्तिवृत्तधरातलयोरन्तरं यत्तस्य चन्द्रगोलीयशरज्यायां संस्करणेन यद् भवति तद्वशाद् भगोलीयशरज्ञानं भवेदेव । एवं प्रतिदिनं वेधेन भगोलीयशरज्याज्ञानं कार्यम् । यस्मिन् दिने दक्षिणशराभाव उपलब्धस्तत्र वेधेन यः साधितश्चन्द्रः स भगणा-(द्वादशराशितः) च्छोधितस्तदा चन्द्रपातो भवेत्, एवमेव द्वितीयपर्ययेऽपि तज्ज्ञानं कार्यं, स च पातः पूर्वपातात्पृष्ठ एव भवति, अतः पातस्य विलोमा गतिरस्तीति सिद्धम् । एतत्पातद्वयान्तरजनितपातगत्या वेधदिनान्तरवशेन चाऽनुपातेनैकदिनजा पातगतिर्भवेत्ततः पूर्ववत्कल्पे चन्द्रपातभगणा जायन्ते, परमेतदानयनं न समीचीनं, चन्द्रकर्णस्य त्रिज्यातोऽल्पत्वादुपर्युक्तोपपत्तौ तत्त्रिज्यासमग्रहणादिति ॥

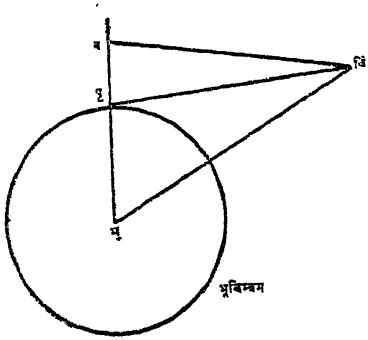
अथ बुधशुक्रयोः शीघ्रोच्चभगणोपपत्तिः

वेधगोलीयस्पष्टबुधशुक्राभ्यां भूगर्भगोलीयं तयोर्ज्ञानं (चन्द्रभगणोपपत्तौ

वेधगोलीयस्पष्टग्रहाद् भूगर्भगोलीयस्पष्टग्रहज्ञानं कथं भवेदिति यथास्थानं प्रदर्शितं तत्तत्रैव द्रष्टव्यम्) कार्यं, वेधेनान्त्यफलज्याज्ञानविधिना तयोः शीघ्रान्त्यफलज्याज्ञानं कृत्वा ग्रहगोलीयशरबिम्बीयकर्णवशेन तयोः स्पष्टकेन्द्रप्रमाणे विदिते भवतस्ततः स्वस्वस्पष्टकेन्द्राभ्यां संस्कृतौ बुधशुक्रौ तयोः शीघ्रोच्चे भवतः, एवं द्वितीयपर्ययेऽपि तयोः शीघ्रोच्चे ज्ञातव्ये । बुधशीघ्रोच्चयोरन्तरवशात्तदन्तरदिनैश्चैकदिनजां गतिमानीयाऽनुपातेन कल्पे तद्भगणाः जायन्ते । एवमेव शुक्रशीघ्रोच्चयोरन्तरवशात्तदन्तरदिनैश्च कल्पे तच्छीघ्रोच्चभगणा विदिता भवन्तीति ॥

अथ मङ्गलगुरुशनीनां शीघ्रोच्चोपपत्तिः

प्रथममेतेषां वेधेन बिम्बीयकर्णज्ञानं कार्यं, तथाशीघ्राऽन्त्यफलज्याज्ञानञ्च कार्यं, तदा भूकेन्द्राद्बिम्बगता रेखा बिम्बीयकर्णः एको भुजः । ग्रहगोलकेन्द्राद्बिम्बकेन्द्रगता त्रिज्या रेखा द्वितीयो भुजः । भूकेन्द्रग्रहगोलकेन्द्रयोरन्तरं शीघ्रान्त्यफलज्या तृतीयो भुज इति भुजत्रयैरुत्पन्नत्रिभुजे भुजत्रयज्ञानात्तत्कोणत्रयमपि विदितं भवेत् । वेधद्वारा विदितग्रहगोलीयशरात्स्थानीयकर्णस्य (भूकेन्द्राद् ग्रहस्थानगतरेखायाः) ज्ञानं सुलभं भवेत् । प्रत्यहं स्थानीयकर्णज्ञानमनयैव रीत्या कार्यं, यस्मिन् दिने तत्कर्णस्य परमत्वं भवेत्तद्दिने शीघ्रोच्चस्थाने एव ग्रहो भवेद्यत उच्चस्थाने ग्रहे तत्कर्णस्य परमत्वं भवति, तत्र यावान् स्फुटग्रहः स च पूर्ववेधविधिना विदितोऽस्ति, तेन तत्सममेव तदा शीघ्रोच्चं भवेत् । एवं द्वितीयपर्ययेऽपि शीघ्रोच्चज्ञानं कार्यं द्वयोर्विदितशीघ्रोच्चयोरन्तरं तद्दिनज–(प्रथमविदितशीघ्रोच्चदिनाद् द्वितीयशीघ्रोच्चज्ञानं यावद्भिर्दिनैर्जातं) शीघ्रोच्चगतिस्ततोऽनुपातो यद्येतावद्भिर्दिनैरियं शीघ्रोच्चगतिस्तदैकेन दिनेन किमित्यनुपातेनैकदिनजा शीघ्रोच्चगतिः, ततः पुनरप्यनुपातो यद्येकेन दिनेनेयं शीघ्रोच्चगतिस्तदा कल्पकुदिनैः किमित्यनुपातेन कल्पे शीघ्रोच्चभगणा जायन्ते। मङ्गलगुरुशनीनां शीघ्रोच्चोपपत्तिरनयैव रीत्या विधेयेति। अत्रोपपत्तौ ग्रहबिम्बीयकर्णज्ञानस्यातीवाऽवश्यकतास्ति तज्ज्ञानमन्तरेयमुपपत्तिर्निरर्थका भवेदतो वेधेन ग्रहबिम्बीय-कर्णज्ञानं क्रियते ।



विं=ग्रहबिम्बकेन्द्रम् । भूं=भूकेन्द्रम् । पृं= भूपृष्ठस्थानम् । चं=दृष्टिस्थानम् । पृच= दृष्ट्युच्छ्रायः । पृविं=पृष्ठकर्णः । भूविं=ग्रहबिम्बीयकर्णः । चविं=दृष्टिकर्णः । भूपृं=भूव्यासार्धम् । विंपृच, विंचंपृकोणौ तुरीययन्त्रद्वारा मापनेन विदितौ भवतः । तदा विंपृचत्रिभुजे १८०—(<विंपृच+<विंचपृं) =<पृंविंच पृच=दृष्ट्युच्छ्रायो विदित एवास्ति तदोक्तत्रिभुजेऽनुपातः ।

$\frac{\text{पृच}\times\text{ज्या}<\text{विचपृ}}{\text{ज्या}<\text{पृविच}}$ = पृवि, १८०—<विपृच=< भूपृवि अयमपि कोणो विदितो जातः, भूपृ=भूव्यासार्धं विदितमेवास्ति, तदा भूपृवि त्रिभुजे भूपृ, पृवि-भुजयोस्तदन्तर्गतकोणस्य च ज्ञानात्सरलत्रिकोणमित्या 'भूवि' इत्याधारज्ञानं भवेदयमेव ग्रहबिम्बीयकर्णः। अथ च ग्रहशीघ्रान्त्यफलज्याज्ञानं कथं भवतीति-प्रदर्श्यते । उपरिप्रदर्शितनियमेन ग्रहबिम्बीयकर्णज्ञानं कार्यं यदा ग्रहस्य शराभाव-स्तदा तस्य कर्णस्य यदा परमत्वं भवेत्तदा परमोच्चकर्णः=त्रि+शीघ्रान्त्यफलज्याः। एवं परमाल्पे कर्णः=त्रि — शीघ्रान्त्यफलज्या, अतः परमोच्चकर्णः—त्रि= शीघ्रान्त्यफलज्या। त्रि—परमनीचकर्णः=शीघ्रान्त्यफलज्यानयनं कार्यमिति।

अथ मन्दोच्चोपपत्तिः

वेधेन स्पष्टग्रहं ज्ञात्वाऽस्मात् स्फुटं ग्रहं मध्यखगं प्रकल्प्येत्यादिनाऽसकृन्मन्द-स्पष्टग्रहो वेदितव्यस्तस्मान्मन्दस्पष्टग्रहान्मध्यमग्रहश्च ज्ञातव्य एतयोर्मन्द-स्पष्टमध्यमग्रहयोरन्तरं मन्दफलम् भवेत् । यस्मिन् दिने तन्मन्दफला-भावो भवेत्तदा तत्र मन्दस्पष्टमध्यमग्रहयोः साम्यं भवेत्तदेव मन्दोच्चम्। एवं द्वितीयपर्ययेऽपि मन्दोच्चज्ञानं कार्यं तयोरन्तरं प्रथमविदितमन्दोच्चदिनाद् द्वितीयपर्यये विदितद्वितीयमन्दोच्चदिनं यावद्यावन्ति दिनानि तद्दिनजा मन्दोच्चगतिर्भवेत्ततोऽनुपातो यद्येभिर्दिनैरियं मन्दोच्चगतिस्तदैकदिने किं जातैक-दिनजा तद्गतिस्ततो यद्येकेन दिनेनेयं मन्दोच्चगतिस्तदा कल्पकुदिनैः किमित्य-नुपातेन कल्पे मन्दोच्चभगणा जायन्ते इति ।

अथवा

भूकेन्द्राद् ग्रहबिम्बस्य स्पर्शरेखा कार्या तथा ग्रहबिम्बकेन्द्रात् स्पर्शबिन्दुगता रेखा कार्या तद्ग्रहबिम्बव्यासार्धम् । भूकेन्द्राद् ग्रहबिम्ब-केन्द्रगता रेखा ग्रहकर्णस्तदा ग्रहकर्णबिम्बव्यासार्धस्पर्शरेखाभिर्जाय-मानत्रिभुजेऽनुपातो यदि ग्रहकर्णेन त्रिज्या लभ्यते तदा ग्रहबिम्बव्यासार्धेन किमित्यनुपातेन दृष्टिस्थानलग्नकोणार्धज्या बिम्बकलार्धज्या समागच्छति तत्स्वरूपम् $=\frac{\text{त्रि}\times\text{ग्रबिव्या}}{\text{ग्रकर्ण}}$ अत्र भाज्यस्य स्थिरत्वाद्यदा ग्रहकर्णमानं पर-माधिकं भवेत्तदा फलं परमाल्पं भवेदर्थादुच्चस्थाने ग्रहे तत्कर्णस्य परमाधि-कत्वाद् बिम्बाकलार्धज्यामानं परमाल्पं भवेत्तच्चापं द्विगुणितं तदा परमाल्पं ग्रह-बिम्बकलामानं भवेदतो बिम्बकलायाः परमाल्पत्वे उच्चस्थो ग्रहो भवति तत्र यावान् वेधागतस्फुटग्रहस्तावदेव तन्मन्दोच्चमपि एवं द्वितीयपर्ययेऽपि ज्ञेयं तयोरन्तरं तद्दिनजा मन्दोच्चगतिस्ततो विदितमन्दोच्चान्तरदिनैस्त-

द्दिनजमन्दोच्चगत्या चानुपातेन कल्पे मन्दोच्चभगणाः समागच्छन्नीति। अत्र बिम्बकलामानं भूकेन्द्रलग्नकोणमानमस्ति तज्ज्ञानं मापनेन कार्यं लम्बनावनतिदर्शनार्थमियं भूरन्यथा केवलं बिन्दुरेव भूरित्याचार्योक्तादत्र बिन्दुरूपभुवः कल्पने बिम्बकलाकोणमापने न काचिद्धानिरिति।

अथैषां पातभगणोपपत्तिः

अथैषां वेधगोलीयशरज्ञानतो भगोलीयशरान् ज्ञात्वा तदभावो यत्र भवेत्तत्र गणितागतान् तान् मन्दस्पष्टग्रहान् द्वादशराशिभ्यः शुद्धान् कृत्वा पातो ज्ञेयः। द्वितीयपर्ययेऽप्येवं तत्पातो ज्ञेयस्तदन्तरैस्तद्दिनान्तरैश्च पूर्ववत्तत्पातभगणा भवन्तीति॥१३॥

रवि, बुध और शुक्र की भगणोपपत्ति

'रविभगणा रव्यब्दाः' इस आचार्योक्ति से कल्प में जितने रविभगण होते हैं उतने ही सौर वर्षमान होते हैं, लेकिन कल्पवर्षप्रमाण विदित है इसलिए तत्तुल्य कल्परविभगण मान भी विदित हो गया, बुध और शुक्र के उदयलक्षण में रवि के उदयकाल और अस्तकाल में रवि और बुध के अन्तरांश तथा रवि और शुक्र के अन्तरांश भी प्रत्येक दिन यन्त्रद्वारा जानने चाहिएँ। वे अन्तरांश तीन राशि से अल्प ही आता है। वराहमिहिराचार्य भी 'बृहज्जातक में पूर्वाचार्योक्त वज्रादि योगों के खण्डन में 'पूर्वशास्त्रानुसारेण मया वज्रादयः कृताः। चतुर्थे भवने सूर्याज्ज्ञसितौ भवतः कथम्।" इत्यादि से रवि के साथ बुध और शुक्र का अन्तर अल्प ही होता है, ऐसा कहते हैं। इसलिए रवि के अतिनिकट (समीप) रहने के कारण कभी आगे कभी पीछे उनके नौकर की तरह बुध और शुक्र जाते हुए देखे जाते हैं। इसी कारण से बुध और शुक्र के कल्पभगण कल्परविभगण के बराबर ही आचार्यों ने स्वीकार किये हैं॥

चन्द्रभगण की उपपत्ति

ग्रहवेध के लिए हर तरह से उपयुक्त स्थान में वेधालय बनाना चाहिए। उसमें नाड़ीवृत्त, क्रान्तिवृत्त, कदम्बप्रोतवृत्त आदि वृत्तों से युक्त एक गोल यन्त्र बनाना चाहिए। क्रान्तिवृत्त में भगणांश ३६० और राशि-अंश-कला आदि अङ्कित करना तथा नाड़ीवृत्त में दण्ड, पल आदि चिह्नित करना, किन्हीं दो आधारों पर तथा केन्द्रगत नलिका से उस गोलयन्त्र को खूब दृढ़ कर गोलकेन्द्र में ध्रुवाभिमुख (ध्रुव की तरफ) यष्टि को करके रात्रि में उस गोलकेन्द्रगत दृष्टि के द्वारा रेवती तारा को देखने से गोलयन्त्रीय क्रान्ति वृत्त में जहाँ पर परिणत हुई वहीं पर मेषादि चिह्नित करना। तथा गोल केन्द्रगत दृष्टि ही से चन्द्र के वेध करने से गोलयन्त्र में जहाँ परिणत हुए उनके ऊपर गोलयन्त्रीय कदम्ब प्रोतवृत्त (वेधवृत्त) करने से वह वृत्त (वेधवृत्त) गोलयन्त्रीय क्रान्तिवृत्त में जहाँ पर लगता है वही वेधागत स्पष्ट चन्द्र है। मेषादि से उनके जितने राश्यादिमान हैं वही राश्यादि स्पष्टचन्द्र है। इस तरह स्पष्टचन्द्र का ज्ञान हो गया, इसी तरह द्वितीय दिन में भी स्पष्टचन्द्र का ज्ञान करना,

इन विदित स्पष्टचन्द्रद्वय से तथा विदित चन्द्रमन्दोच्च से 'स्फुटं ग्रहं मध्यखगं प्रकल्प्य कृत्वा फले मन्दचले यथोक्ते' इत्यादि से दोनों दिनों के मध्यम चन्द्र जान कर दोनों मध्यम चन्द्रों के अन्तर (चन्द्रमध्यमगति) जान लेना. इससे अनुपात करते हैं यदि एक दिन में यह चन्द्रमध्यम गति पाते हैं तो कल्प कुदिन में क्या इससे कल्प चन्द्रभगण मान आते हैं । लेकिन इस उपपत्ति में वेध द्वारा जो स्पष्टचन्द्र लिये गये हैं वे वेधगोलीय (दृग्गोलीय या भूपृष्ठस्थान में दृष्टिस्थान रखने से पृष्ठीय त्रिज्यागोलीय) हैं, लेकिन भूकेन्द्रिक त्रिज्यागोलीय स्पष्टचन्द्र अपेक्षित है, इसलिए वेधगोलीय स्पष्टचन्द्र और भूकेन्द्रिक त्रिज्यागोलीय (भूगर्भगोलीय) स्पष्टचन्द्र का अन्तरानयन करके वेधगोलीय स्पष्टचन्द्र में उस अन्तर को संस्कार करना तब भूगर्भगोलीय स्पष्टचन्द्र होते हैं । इसी तरह वेधगोलीय द्वितीय दिन के स्पष्ट चन्द्र से गर्भगोलीय स्पष्ट चन्द्र ज्ञात करना, तब इन विदित भूगर्भगोलीय स्पष्टचन्द्रद्वय से 'स्फुटं ग्रहं मध्यखगं प्रकल्प्य' इत्यादि से दोनों दिनों के मध्यम चन्द्र ज्ञात कर अन्तर करने से एक-दिन-सम्बन्धिनी चन्द्र-मध्यमगति होती है इससे पूर्ववत् कल्पचन्द्रभगण ज्ञात करना ।

उपर्युक्त उपपत्ति में वेधगोलीय स्पष्टचन्द्र से भूगर्भगोलीय स्पष्टचन्द्र ज्ञान का उल्लेख किया गया है परन्तु वह अवतरण रूप में कहा गया । अब यहां उस का साधनप्रकार लिखते हैं । यहां स्पष्टचन्द्र का प्रसङ्ग है इसलिए दोनों गोलीय (वेधगोलीय और भूगर्भगोलीय) स्पष्टचन्द्रों का अन्तरानयन करते हैं परन्तु जिस किसी वेधगोलीय स्पष्टग्रह से भूगर्भगोलीय स्पष्टग्रहज्ञान करना हो तो यही अधोलिखित उपपत्ति समझनी चाहिए । वेधगोल में दृष्टिवश परिणत चन्द्रबिम्ब के स्पष्टभोग चिह्न (परिणतचन्द्रबिम्बोपरिगतकदम्बप्रोतवृत्त क्रान्तिवृत्त में जहां लगता है वह बिन्दु) तद्गोलीय स्पष्टचन्द्र है । इस तरह भूगर्भगोल में भी स्पष्टचन्द्र स्थान समझना ।

उपपत्ति के लिए परिभाषाएँ

वेधगोलीय चन्द्रस्थान=स्थान स्थानीय दृग्वृत्त धरातल से कटित भूगर्भगोल का प्रदेश उस गोल का दृग्वृत्त होता है । भूगर्भगोलीय दृग्वृत्त और भूगर्भगोलीय क्रान्तिवृत्त के योग बिन्दु =ष, भूगर्भ से ष बिन्दुगत रेखा=प संज्ञक । दृष्टिस्थान से स्थानगत रेखा=फ संज्ञक । प और फ रेखा समानान्तर हैं (रेखागणित एकादशाध्याय युक्ति से), भूगर्भ से तथा दृष्टिस्थान से रेवतीगत रेखाद्वय समानान्तर है अतः भूगर्भलग्न कोणमान और दृष्टिस्थानलग्न कोणमान समान हुआ अर्थात् भूगर्भगोल (भगोल) में रेवती से ष बिन्दुपर्यन्त चाप वेधगोलीय स्पष्टचन्द्र के बराबर हुआ, अर्थात्

भगोलीय रेवती से र बिन्दुपर्यन्त= वेधगोलीय रेवती से स्थान पर्यन्त; क्योंकि केन्द्रलग्न कोण का मान तत्संमुखचाप होता है । स्थानीयनतांश= ष बिन्दूत्पन्न नतांश क्योंकि प, और फ रेखा समानान्तर हैं । वह नतांश वेधगोल में मापन

करने से विदित हुआ । तथा चन्द्रबिम्बीयनतांश और प-बिन्दूत्पन्न नतांश चापों से उत्पन्न खस्वस्तिक लग्नकोण जितने वेधगोल में होते हैं उतने ही भूगर्भगोल में भी, क्योंकि दोनों गोलों में धरातल एक ही है। अतः ग्रहों के दिगंशवश वह कोण का मान निकालना होगा तब भूगर्भगोल पृष्ठ पर जो त्रिभुज बना है उसमें 'त्रिज्या-गुणाद्धरणिकोटिगुणात्' इत्यादि के विलोम से पलिणत चन्द्रबिम्ब केन्द्र ष विन्दु-गतवृत्तीयाधार चापज्ञान हुआ । तथा वेधगोलीयशर, क्रान्तिवृत्तधरातलान्तर के ज्ञान से भूगर्भगोल में शरज्ञान (वेधगोलीय और भूगर्भगोलीय नाड़ीवृत्तधरातलान्तर और वेधगोलीय क्रान्तिज्ञान से भूगर्भगोलीय क्रान्तिज्ञान के लिए जो युक्ति है उसी तरह की युक्ति शरज्ञान के लिए है), इस लिए चापजात्य युक्ति से भूगर्भगोलीय स्पष्टचन्द्र और ष बिन्दु के अन्तर संस्कारसंज्ञक चाप का ज्ञात हो जायगा । अन्तर=संस्कारचाप ।

अतः वेधगोलीयस्प चन्द्र ± संस्कारचाप=भूगर्भगोलीय स्प चन्द्र ।

अब संस्कार चाप की धन और ऋण की व्यवस्था दिखलाते हैं ।

परिभाषाएँ

वेधगोलीयक्रान्तिवृत्त = इष्टक्रान्तिवृत्त । भूगर्भगोलीयक्रान्तिवृत्त=वास्तव-क्रान्तिवृत्त । बिम्बीय कर्णगोलीय क्रान्तिवृत्त=वास्तवक्रान्तिवृत्त । 'प' रेखा बढ़कर वास्तव क्रान्तिवृत्त में जहां लगती है वहां ष बिन्दु है, चन्द्रबिम्ब केन्द्र से इष्ट क्रान्तिवृत्त धरातल का जो शरज्यालम्ब है उसका मूल बिन्दु=क्ष, यह बिन्दु वर्धित फ रेखा ही में होता है। फ-रेखा स्थानीय दृग्वृत्त धरातल में है । पूर्वकथित शरज्या बढ़कर या नहीं बढ़कर वास्तव क्रान्तिवृत्त धरातल के ऊपर लम्ब है। स्थानीय दृग्वृत्तधरा-तलनिष्ठ 'क्ष' बिन्दु से वास्तव क्रान्तिवृत्त धरातल के ऊपर लम्ब करते हैं, उसका मूल बिन्दु स्थानीय दृग्वृत्त धरातल और वास्तव क्रान्तिवृत्त धरातल से उत्पन्न कोण जिस दिशा में अल्प है उसी दिशा में पतित होगा ।

अब भूगर्भ से बिम्बीय कर्णव्यासार्ध से जो गोल होता है उस पर विचार करते हैं।

ष विन्दुजनित दृग्वृत्त और वास्तवक्रान्तिवृत्त से उत्पन्न दृक्षेप चापाभिमुख कोण अल्प है, क्ष बिन्दु तो वास्तव कान्तिवृत्त धरातल और ऊर्ध्वाधर सूत्र के मध्य (बीच) में है क्योंकि फरेखा मध्य में है । इससे सिद्ध होता है कि दृक्षेप वृत्त से पूर्वकपाल में चन्द्र के रहने से रेखा से पश्चिम दिशा ही में लम्बमूल गिरेगा, क्योंकि स्थानीय दृग्वृत्त धरातल और क्रान्तिवृत्त धरातल की योगरेखा प-रेखा है । भूगर्भ से लम्बमूलगतरेखा ष बिन्दु से पश्चिम दिशा ही में क्रान्तिवृत्त में लगेगी वही बिन्दु भूगर्भाभिप्रायिक चन्द्र-स्थान है। त्रिज्यागोल में भी यही स्थिति है । पश्चिम कपाल में भी इसी तरह विचार करना। इससे सिद्ध होता है कि वित्रिभ से चन्द्र के अल्प रहने से संस्कारचाप धन होता है अन्यथा ऋण होता है इति ।।

अब पूर्वोपपत्ति में वेधगोलीय नाड़ीवृत्त धरातल और भूगर्भगोलीय नाडीवृत्त धरातलान्तर ज्ञान से तथा वेधगोलीय क्रान्तिज्ञान से भूगर्भगोलीय क्रान्तिज्ञान के लिए जो युक्ति है उसी तरह की युक्ति वेधगोलीय शर और क्रान्तिवृत्तधरातलान्तर ज्ञान से भूगर्भगोलीय शरज्ञान के लिए होती है यह हमने जो लिखा है उसके लिए विचार करते हैं । दृष्टिस्थान से जो गोल बनाया जाता है उसको वेधगोल या दृश्यगोल कहते हैं । और भूगर्भ से जो गोल बनता है उसे स्थिरगोल या भगोल कहते हैं । भूगर्भ से और दृष्टिस्थान से भचक्रस्थ ध्रुवतारागत रेखाद्वय अपने अपने गोल (भूगर्भगोल और वेधगोल) में जहां जहां लगता है वहां वहां दोनों गोलों में परिणत ध्रुव होता है । इन दोनों (परिणत ध्रुवबिन्दुओं से नवत्यंश जो वृत्त बनते हैं वे दोनों गोल में नाड़ीवृत्त होते हैं, भूगर्भ से और दृष्टिस्थान से ध्रुवगत रेखाद्वय और भूगर्भदृष्टिस्थान से जो त्रिभुज बनता है उस त्रिभुजरूपी धरातल से कटित गोलद्वय का प्रदेश गोलद्वय में याम्योत्तरवृत्त होता है, स्वनाड़ीवृत्त धरातल और याम्योत्तर वृत्त धरातल की योगरेखा स्वनिरक्षोर्ध्वाधरसूत्र है, वर्धित भूकेन्द्रदृष्टिस्थानान्तरेखा ऊर्ध्वाधर सूत्र है । नाड़ीवृत्त धरातल के ऊपर ध्रुवसूत्र लम्ब है परन्तु दोनों गोलों के ध्रुवसूत्र समानान्तर हैं इसलिए दोनों गोलों के नाड़ीवृत्त धरातल समानान्तर हुए (रेखागणित की एकादशाध्यायोक्ति से), दृष्टिस्थान से भगोलीय नाड़ीवृत्त धरातल के ऊपर जो लम्ब होता है वही वेधगोलीय और भगोलीय नाड़ीवृत्त धरातलान्तर है, दोनों गोलों में अक्षांश समान होने से नाड़ीवृत्त धरातलान्तर ज्ञान करते हैं ।

भूगर्भ से दृष्टिस्थान पर्यन्त (केन्द्रान्तर) एकभुज । दृष्टिस्थान से भगोलीय नाड़ीवृत्तधरातल के ऊपर लम्ब (नाड़ीवृत्तधरातलान्तर) द्वितीय भुज, गर्भीय निरक्षोर्ध्वाधररेखाखण्ड तृतीय भुज, इन तीनों भुजों से उत्पन्न जात्यत्रिभुज में भूगर्भलग्न कोण = अक्षांश, लम्बमूल बिन्दुलग्नकोण = ९०, अतः अनुपात करते हैं यदि त्रिज्या कोणज्या में केन्द्रान्तर पाते हैं तो अक्षज्या कोणज्या में क्या इस अनुपात से नाड़ीवृत्त धरातलान्तर प्रमाण आता है; $\frac{\text{अज्या} \times \text{केन्द्रान्तर}}{\text{त्रि}}$ = नाडीवृत्तधरातलान्तर । दृष्टिस्थान से स्वगोलस्थ रविगतरेखा और स्वनाड़ीवृत्त (वेधगोलीयनाड़ीवृत्त) धरातल के अन्तर वेधगोल में वेधगोलीय क्रान्तिज्या है । वेधगोलीय क्रान्तिज्या मापनद्वारा विदित है इसलिए $\frac{\text{वेधगोलीय क्रांज्या} \times \text{दृष्टिकर्ण}}{\text{वेधगोल व्या}\frac{1}{2}}$ = ग्रह से निरक्षोर्ध्वाधर सूत्र के ऊपर लम्बरेखा = लम्ब । अतः लम्ब ± नाड़ीवृत्त धरातलान्तर = लम्ब ± अन्तर = ग्रहगोलीय क्रान्तिज्या । इसके ज्ञानसे $\frac{\text{ग्रहगो कांज्या} \times \text{त्रि}}{\text{ग्रविकर्ण}}$ = भगोलीय क्रांज्या = भूगर्भगोलीय क्रांज्या = स्थिरगोलीय क्रांज्या, इस रीति से वेधगोलीय शरज्ञान से क्रान्तिवृत्त धरातलान्तरज्ञान से भूगर्भगोलीय शरज्ञान होता है ।

यहां (B) क्षेत्र देखिये ।

र = रवि । ख = भगोलीय खस्वस्तिक । खं = वेधगोलीय खस्वस्तिक, हर = रविदृष्टि कर्ण, भूर = रविकर्ण, हस = नाड़ीवृत्त धरातलान्तरम् = कम, रम = लम्ब भूह = केन्द्रान्तर, भू = भूकेन्द्र, ह = दृष्टिस्थान । खनि = वेधगोलीयाक्षांश खनि- = भगोलीयाक्षांश वेधगोलीयाक्षांश = भगोलीयाक्षांश = खनि = खनि । रग = भगोलीय क्रान्तिज्या ।

रविमन्दोच्चोपपत्ति

भूकेन्द्र से रविमन्दगोल केन्द्रगतरेखा रविमन्दप्रतिवृत्त में जहां लगती है वही रवि के मन्दोच्च है । (भूकेन्द्र से मन्दप्रतिवृत्तीय अन्य बिन्दुओं से वह बिन्दु अधिक उच्च में है इसलिए उसका उच्चनाम अनुगतार्थ है, रविकर्ण में यदि त्रिज्या पाते हैं तो रवि बिम्ब व्यासार्ध में क्या—इस अनुपात से फल आता है उसके चाप को द्विगुणित करने से रवि के बिम्बकला प्रमाण होता है । प्राचीनाचार्य लोग अनुपात से जो फल आता है उसीको द्विगुणित कर चाप करते हैं उसीको बिम्बकला-मान कहते हैं यह ठीक नहीं है, बिम्बकलानयन प्रकार के देखने से उच्चस्थान में रविकर्ण के परमत्व के कारण रविबिम्बकलाप्रमाण परमाल्प सिद्ध होता है । 'लम्बनावनतिदर्शनार्थमियं भूरन्यथा केवलं बिन्दुरेव भूः' इस भास्करोक्ति से यहां भूगर्भ और भूपृष्ठ में अभेद मानकर भूकेन्द्रलग्न कोण (रविबिम्बकला) को मापन कर जान लिया जाय तब रवि बिम्बकला के परमाल्पत्व वेधागत स्पष्ट रवि ही रवि-मन्दोच्च होंगे । द्वितीय पर्यय में भी इस तरह रविमन्दोच्च ज्ञान कर दोनों रवि-मन्दोच्चों के अन्तर करने से वेधदिनान्तरजनित रविमन्दोच्च गति होगी । तब अनु-पात करते हैं यदि इन वेधदिनान्तर में यह रविमन्दोच्च गति पाते हैं तो एक दिन में क्या इस अनुपात से एकदिनसम्बन्धिनी रविमन्दोच्च गति आई । फिर अनु-पात करते हैं यदि एक दिन में यह रविमन्दोच्च गति पाते हैं तो कल्पकुदिन में क्या इससे कल्परविमन्दोच्च भगण आया ।

अथवा विदित रविमन्दोच्च ७८° से कुट्टकयुक्ति से उसका (रविमन्दोच्च) आन-यन करते हैं ।

कल्पना करते हैं कल्परविमन्दोच्चभगणमान = य

तब १८५३ शकान्त में कल्पादि से सौर वर्ष = १९७२९४१०३२ (नव-नगशशिमुनिकृतनव इत्यादि ब्राह्मस्फुटसिद्धान्तोक्ति से या गोऽद्रीन्द्रद्विकृताङ्कदस्र-नगगोचन्द्रा इत्यादि भास्करोक्ति से) तब अनुपात करते हैं यदि कल्प सौर वर्ष में कल्परविमन्दोच्च भगण पाते हैं तो कल्पादि से शकान्त तक पूर्वानीत सौर वर्ष में क्या—इससे शेषसहित रविमन्दोच्चभगण इष्टवर्षान्त में आया ।

$$\frac{१९७२९४१०३२ \times य}{४३२०००००००} = ल + \frac{शे}{४३२०००००००} \therefore १९७२९४१०३२ \times य =$$

४३२०००००००×ल+शे समशोधन करने से १९७२९४१०३२×य—४३२०००००००

×ल=शे इसको चक्रांश (३६०) से गुणकर कल्प वर्ष से भाग देना तब फल रविमन्दोच्च प्रमाण = ७८° ∴ $\frac{(१९७२९४१०३२\times य-४३२०००००००\times ल)\times ३६०}{४३२०००००००}$

=७८° छेदगम से

(१९७२९४१०३२×य—४३२०००००००×ल)×३६०=४३२०००००००×७८-

दोनों पक्षों को ३६० इस से भाग देने से

१९७२९४१०३२×य—४३२०००००००×ल=$\frac{४३२०००००००\times ७८}{३६०}$=१२०००-

००×७८=९३६०००००० दोनों पक्षों में समान जोड़ने से १९७२९४१०३२×य=

४३२०००००००×ल+९३६०००००० ∴ $\frac{१९७२९४१०३२\times य-९३६०००००}{४३२०००००००}$=ल।

आठ से भाग देने से $\frac{२४६६१८६२९\times य-११७०००८००}{५४०००००००}$=ल, यहां ५५६७०१ इन

से अपवर्त्तन देने से $\frac{४४३य-२१०}{९७०}$=ल (स्वल्पान्तर से) ऋणात्मक रूपक्षेप में लब्धि और गुण लाकर 'अभीप्सितक्षेपविशुद्धिनिघ्नौ' इत्यादि से लब्धि=२१९, गुणक =४८० 'ते भाज्यतद्भाजकवर्णमाने' इस भास्करोक्ति से गुणक ही भाज्यवर्ण (य) का मान होता है इसलिए य=४८०=कल्परविमन्दोच्च भगण।

चन्द्रमन्दोच्चोपपत्ति

दो तरह के मन्द और शीघ्र भेद से मन्दोच्च और शीघ्रोच्च होता है। 'शीघ्राख्यतुङ्गस्य तयोरभावात्' इत्यादि भास्करोक्ति से रवि और चन्द्र का केवल मन्दोच्च ही होता है। चन्द्रबिम्ब कला की परमाल्पता चन्द्रमन्दोच्च स्थान में होती है वहां उस समय में जितने वेधागत स्पष्ट चन्द्र होते हैं उतने ही चन्द्रमन्दोच्च होते हैं। परमाल्पचन्द्रबिम्बकलाप्रमाण को मापन कर समझ लेना चाहिए इसके लिए भूगर्भ और भूपृष्ठ को अभेद मानना पड़ेगा, द्वितीय पर्यय में भी पूर्वोक्त युक्ति से चन्द्रमन्दोच्च जानकर दोनों चन्द्रमन्दोच्च के अन्तर करने से वेधकालान्तर दिन-सम्बन्धी चन्दगति होती है तब अनुपात करते हैं यदि वेधकालान्तर में चन्द्रगति पाते हैं तो एक दिन में इससे एकदिनसम्बन्धिनी चन्द्रमन्दोच्च गति आई, इससे पूर्ववत् कल्प चन्द्रभगण ले आना। बिम्बीय कर्णसम्बन्ध से भी चन्द्रमन्दोच्च भगणोपपत्ति हो सकती है। यथा

प्रत्येक दिन वेध से चन्द्रबिम्बीय कर्ण साधन करना, जब चन्द्र शराभाव होगा उस दिन में चन्द्रबिम्बीय कर्ण के परमत्व में वेधागत स्पष्ट चन्द्र ही मन्दोच्च होंगे। द्वितीय पर्यय में भी इसी तरह चन्द्रमन्दोच्चों के अन्तर पर से पूर्ववत् कल्पचन्द्र मन्दोच्च भगण आ जाएंगे।

चन्द्रपातभगणोपपत्ति

दृष्टिस्थानाभिप्रायिक गोल का निर्माण करना ही वेधगोल है। दृष्टिस्थान (वेध-

गोल केन्द्र) से चन्द्र को वेध करने से वेधगोल में जहाँ उपलब्ध होते हैं उनके ऊपर वेधगोलीय कदम्ब प्रोतवृत्त करने से वह कदम्ब प्रोतवृत्त वेधगोलीय क्रान्तिवृत्त में जहां लगता है वहां से चन्द्रविम्बकेन्द्र तक कदम्ब प्रोतवृत्त में चन्द्र का शरांश है, तब

$\frac{\text{वेधगोलीय शरज्या} \times \text{त्रि}}{\text{वेधगोलव्या } \frac{1}{2}}$ —चन्द्रगोलीयशरज्या। इसमें क्रान्तिवृत्त धरातलान्तर संस्कार करने से जो होता है उसके वश से भगोलीय शरज्ञान होगा ही, इस तरह प्रत्येक दिन वेध से भगोलीय शरज्या का ज्ञान करना ॥१३॥

इदानीं ग्रहयुगस्य परिभाषां करोति

कालर्क्षदेशयोगाद् भूयो ग्रहमन्दशीघ्रपातानाम्।
कल्पेन यतो योगस्ततः स्फुटं ग्रहयुगं कल्पः ॥१४॥

वा. भा.—कालयोगः चैत्रसितादेरुदयाद् भानोः। ऋक्षयोगः पौष्णाश्विनांतरस्थैः सह ग्रहैः। देशयोगो लङ्कायामिति। तेनायमर्थः, कालर्क्षदेशयोगाद् भूय एतावत्या सामग्र्या पुनरपि ग्रहशीघ्रमन्दपातानां यतो यस्मात्कल्पे योगो भवति तस्मात् कल्प एव ग्रहयुगं स्फुटम् ॥

वि. भा.—ग्रहमन्दशीघ्रपातानां (ग्रहाणां मन्दोच्चानां शीघ्रोच्चानां पातानां च) कालर्क्षदेशयोगात् (कालयोगश्चैत्रसितादेरुदयाद्भानोः) ऋक्ष(नक्षत्र)योगः पूर्वकथितैः पौष्णाश्विन्यन्तस्थैः सह ग्रहैः देशयोगः(लङ्कायाम्) एतस्मात्कालर्क्षदेशयोगाद्, भूयः (वारं वारं) यतः (यस्मात्कारणात्) कल्पेन (कल्पवर्षप्रमाणेन)योगो भवेन्नहि तत्पूर्वं पश्चाद्वा ततः(तस्मात्कारणात्)कल्पो ग्रहयुगमिति स्फुटमर्थाद्यदैकदा सृष्ट्यारम्भे कल्पादौ ग्रहमन्दशीघ्रपातानां कालदेशयोगो भवति ततोऽनन्तरं पुनः कल्पान्ते (द्वितीयसृष्ट्यारम्भे) तादृशयोगः कल्पवर्षैर्भवत्यतः कल्प एव ग्रहयुगमिति ॥१४॥

अब ग्रहयुग की परिभाषा करते हैं।

हि. भा.—ग्रहों, मन्दोच्चों, शीघ्रोच्चों और पातों के कालसम्बन्ध से योग हो (पहले सृष्ट्यारम्भ में जो कहा गया है चैत्रसितादेरुदयाद्भानोः) ऋक्ष (नक्षत्र) योग (पूर्वकथित भचक्रचलनसूत्रक श्लोक में पौष्णाश्विन्यन्तस्थैः सह ग्रहैः) देशयोग (पूर्वकथित काल प्रवृत्तिसूचक श्लोक में) 'लङ्कायाम्' इस तरह की स्थिति जब (कल्पादि में) हो उसके बाद फिर उन सब ग्रहमन्दशीघ्रपातों) के वह योग कल्पवर्ष में होता है इसलिए कल्प ही स्फुट ग्रह युग होता है। अर्थात् ग्रहादि के कालयोग, ऋक्षयोग और देशयोग, एकबार जब (कल्पारम्भ में) होता है फिर उन सब के उस तरह के योग कल्पवर्ष में (कल्पान्त में) होता है उससे पहले या पीछे नहीं होता है इसलिए कल्पादि से कल्पान्त पर्यन्त एक कल्पवर्ष ही ग्रहयुग होता है। इति ॥१४॥

इदानीं रविबुधसितानां कुजगुरुशनिशीघ्रोच्चानां च प्राग्व्रजतां कल्पभगणानाह।

कल्पेऽर्कबुधसितानां भगणाः शून्यानि सप्तरदवेदाः ४३२०००००००।
प्राग्व्रजतां कुजगुरुशनिशीघ्रोच्चानां स्वकक्षासु ॥१५॥

वा. भा.—कल्पाख्ये कालप्रमाणे, भगणाश्चक्रपरिवर्त्ताः, केषामर्क-बुधसितानां, कियन्तस्ते इत्याह शून्यानि सप्त रदवेदाः ४३२०००००००० प्राग्व्रजतां पूर्वाभिमुख गच्छतां स्वगत्येत्यर्थः, स्वकक्षासु स्वेषु भ्रमणप्रदेशेषु। न केवलमर्कादीनामेते भगणाः स्युः कुजगुरुशनिशीघ्रोच्चानाञ्च, यतो रविकक्षायामेव तेषां शीघ्रप्रतिमण्डलमध्यभ्रमणम्। (सर्वमेतद्गोले छेदके वा प्रतिपादयेत्) भयञ्जर-तुल्यया परगत्या, यतोऽहोरात्रेणैव स्वकक्षायां पूरयन्तो दृश्यन्ते। अतएवोक्तं प्राग्व्रजतामिति, एवं सर्वेषां वक्ष्यमाणानामपि योज्यम्। भगणाश्च खेरविभूयोगभगणभोगोपलब्धासकृद्रविचन्द्रयोगोपलब्धा चन्द्रभगणाःशेष ग्रहाणाम्। चन्द्रयोगोपलब्धा शीघ्रमंदानां परमफलोत्पत्त्यनुत्पत्तिभ्याम्। पातानां परमविक्षेपा विक्षिप्त्युपलब्ध्यभियोगाशयेन च ज्ञेयाः। सूक्ष्मावयवोपलब्धा च। अथवा भगणादिष्वस्माकमागम एव प्रमाणमिति।

पञ्चाम्बराणि गुणगुण-पञ्चमुनिस्वर-शरैर्मिताः शशिनः ५७७५३३०००००।
भौमस्य द्वियमशराष्टपक्षवसुरसनवद्वियमाः २२९६८२८५२२ ॥१६॥

चन्द्रभगणाः५७७५३३०००.०० ।

कुजभगणाः२२९६८२८५२२ एतेन चन्द्रभौमयोः कल्पे भगणसंख्या प्रतिपादिता।

कृतवसुनवाष्टनवनवषट्त्रिनवनगेन्दवोज्ञ शीघ्रस्य १७९३६९९८९८४।
जीवस्य शरेषूदधिषट्पक्षद्विकृतरसरामाः ३६४२२६४५५ ॥१७॥

बुधशीघ्रभगणाः १७९३६९९८९८४ ।
बृहस्पतिभगणाः ३६४२२६४५५ ।

इदानीं शुक्र शीघ्र शनैश्चरयोः कल्पभगणानाह

सितशीघ्रस्य यमलगोवेदनवाष्टाग्निपक्षयमखनगाः ७०२२३८९४९२।
अष्टनवपक्षमुनिरसशररसमनवोऽर्कपुत्रस्य १४६५६७२९८ ॥१८॥

शुक्र शीघ्रभगणाः ७०२२३=९४९२ ।
शनैश्चरभगणाः १४६५६७२९८।

इदानीं रव्यादिमन्दानां चन्द्रादिपातानां च कल्पभगणानाह ॥

खाष्टाब्धयो ४८० वसुशरवसुपञ्चखचन्द्रवसुवसुसमुद्राः ४८८१०५८५८।
द्विनवयमा २९२ द्वित्रिगुणाः ३३२ शरेषु वसव ८५५ स्त्रिपञ्चरसाः ६५३ ॥१९॥

खाष्टोदधयोर्कमन्दस्य ४८० । वसुशरपंचखचन्द्रवसुवसु-समुद्राः ४८८१०-५८५८ चन्द्रमन्दस्य ।

भौममन्दस्य द्विनवयमाः २९२ । बुधमन्दस्य द्वित्रिगुणाः ३३२ । जीवमन्दस्य शरेषुवसवः ८५८। शुक्रमन्दस्य ६५३ त्रिपञ्चरसाः कल्पे भगणाः भवन्ति ।

शशिवेदा ४१ मन्दानामार्कादीनां विलोमपातानाम् . ।
वसुरसरुद्रेन्दु-गुण द्वित्रियमाः २३२३१११६८ सप्तरसपक्षाः २६७ ॥२०॥

शशिवेदा ४१ इति शनिमन्दस्य कल्प भगणाः भवन्ति । विलोमपातानामिदानीं लिख्यते ।

वसुरसरुद्रेन्दु गुणाद्वित्रियमा इति चन्द्रपातस्य २३२३१११६८ । सप्तरसपक्षा २६७ इति भौमपातस्य भगणाः भवन्ति ।

शशियमशरा ५२१ गुणरसा ६३ स्त्रिनववसवः ८९३ समुद्रवसुविषयाः ५८४ ।
चन्द्रादीनां पश्चाद् व्रजतोऽश्विन्यादिभगणस्य ॥२१॥

शशियमशरा ५२१ इति बुधपातस्य । गुणरसा ६३ इति गुरुपातस्य, त्रिनन्दवसवः ८९३ इति शुक्रपातस्य । समुद्रवसुविषयाः ५८४ इति शनिपातस्य ।

एते यथाक्रमेण कल्पभगणा विलोमपातानामित्यर्थः । यतः सर्व एव पाताः मेषान्मीनं मीनात्कुम्भमित्याद्युत्क्रमेण भगणपरिवर्त्तं कुर्वते इति । पश्चाद् व्रजतोश्विन्यादिभगणस्य इत्युत्तरसंबन्धो भविष्यतीति ॥

वि. भा.—कल्पे (ब्राह्मदिने) अर्कबुधसितानां (रविबुधशुक्राणां) शून्यानि सप्तरदवेदाः (सप्तशून्यानि-रदा द्वात्रिंशत् वेदाश्चत्वारोऽर्थात् ४३२०००००० एतावन्तः) भगणा भवन्ति । किं विशिष्टानां कुजगुरुशनिशीघ्रोच्चानाम् (मङ्गलगुरुशनीनां शीघ्रोच्चरूपा ये तेषां) स्वकक्षासु (स्वभ्रमणवृत्तेषु) प्राग्व्रजतां (पूर्वाभिमुखं गच्छतां) शशिनश्चन्द्रस्य पञ्चाम्बराणि गुणगुणपञ्चमुनिस्वरशरैर्मिताः (पञ्चशून्यानि त्रित्रिपञ्चसप्तसप्तपञ्च ५७७५३३००००० तुल्याः) कल्पभगणा भवन्ति । द्वियमशराष्टपक्षवसुरसनवद्वियमाः (द्विद्विपञ्चाष्टद्वयष्टषड्नवद्वियमाः २२९६८२८५२२) भौमस्य (मङ्गलस्य) कल्पभगणाः । कृतवसुनवाष्टनवनवषट्त्रिनवागेन्दवः । (चतुरष्टनवाष्टनवनवषट् त्रिनव सप्तचन्द्राः १७९३६९९८९८४ (ज्ञशीघ्रस्य) बुधशीघ्रोच्चस्य) कल्पे भगणा भवन्ति शरेषूदधिषट्पक्षद्विकृतरसरामाः (पञ्चपञ्चचतुःषट्द्विद्विवेदषड्गनयः ३६४२२६४५५) जीवस्य (गुरोः) कल्पभगणा भवन्ति । यमलगोवेदनवाष्टाग्निपक्षयमखनगाः (द्विनवचतुर्नवाष्टत्रिद्विद्वशून्यसप्त (७०२२३८९४९२) सितशीघ्रस्य(शुक्रशीघ्रोच्चस्य,)अष्टनवपक्षमुनिररसशरसमनवः (अष्टनवद्विसप्तषट्पञ्चषट्चतुर्दश (१४६५६७२९८)अर्कपुत्रस्य(शनैश्चरस्य),खाष्टाब्धयः(शून्याष्टचत्वारः

४८०)वसुशरवसुपञ्चखचन्द्रवसुवसुसमुद्राः(अष्टपञ्चाष्टपञ्चशून्यैकाष्टाष्टचत्वारः४८८१०-५८५८), द्विनवयमाः (२९२), द्वित्रिगुणाः (३३२), शरेषुवसवः (पञ्चपञ्चाष्टौ ८५५) त्रिपञ्चरसाः(त्रिपञ्चषट्काः६५३), शशिवेदाः (एकचत्वारिंशत् (४१) एते क्रमशोऽर्कादीनां (रव्यादिग्रहाणां) मन्दानां (मन्दोच्चानां) कल्पभगणा भवन्ति, वसुरसरुद्रेन्दुगुणद्वित्रियमाः । अष्टषडेकादशैकत्रिद्वित्रियमाः२३२३१११६८), सप्तरसपक्षाः (सप्तषट्दस्राः २६७), शशियमशराः (एकद्विपञ्च ५२१), गुणरसाः (त्रिषष्टिः), त्रिनववसवः (त्रिनवाष्टौ (८९३), समुद्रवसुविषयाः चतुरष्टपञ्च (५८४) इति चन्द्रादीनां ग्रहाणां विलोमपातानां (विपरीतगतिकपातसंज्ञकानां), अश्विन्यादिभगणस्य पश्चात् व्रजतः कल्पे भगणा भवन्तीति ॥१५-२१॥

इदानीं भभ्रमान् कुदिनानि चाह

भपरिवर्त्ताः खचतुष्टय-शराब्धिरसगुणयमद्विवसुतिथयः १५८२२३६४५०००० ।
रविभगणोना भानोः सावनदिवसाः कुदिवसास्ते ॥२२॥

वा. भा.—पश्चाद् व्रजतः पश्चिमाभिमुखं भ्रमतः अश्विन्यादिभगणस्य कल्पे परिवर्त्ताः कियन्त इत्याह ।

खचतुष्टयशराब्धिरसगुण-यम-द्विवसुतिथयः १५८२२३६४५०००० एत एव रविभगणैरूनाः सन्तो भानोस्संबन्धिनः सावनदिवसाः भवन्ति । कल्पे एतावन्तोऽर्कोदया भवन्ति । कुदिवसाः भूमेः संबन्धिनो दिवसा वा एते । अयमभिप्रायो भूर्वा भ्रमति प्राङ्मुखा सा चावर्त्तवत्यैतावतो वारान् क्षितिजे रविणा सह युज्यते । तथाप्यर्कोदयः समो भवति, रविग्रहणमत्र ग्रहोपलक्षणार्थम् । तेन यस्यैव ग्रहस्य भगणैरूना नक्षत्रपरिवर्त्ताः क्रियन्ते तस्यैव सावनदिवसा भवन्ति । कल्पे तावन्त उदयास्तस्य, अर्थादेव नक्षत्रस्य परिवर्ततुल्या उदयाः, यतो नक्षत्रमगतिमतो ग्रहस्य स्वमुक्तितुल्यमन्तर-प्रतिदिनं नक्षत्रेण सह भवत्यतः कल्पेन स्वभगण-तुल्यमन्तरं भवतीति कृत्वा नक्षत्रपरिवर्त्तेभ्यो भगणान्संशोध्य तदुदयाः भवन्ति । सावनमुदयादुदय इति लक्षणेन सावनदिवसा उच्यन्ते, तद्यथा—नक्षत्रपरिवर्त्ता १५८२२३६४५०००० रविभगणैरमीभिः ४३२०००००० ऊना जाताः १५७७९ १६४५०००० खचतुष्टयशराब्धिरसचन्द्रनवागमुनितिथयः । तथा च बलभद्रः 'खचतुष्टयशराब्ध्यष्टिनवागागशरेन्दवः । कल्पे सूर्योदया ज्ञेयास्त एव च कुवासराः' ॥ इति ॥

वि. भा.—कल्पे १५८२२३६४५०००० एतावन्तो भभ्रमा भवन्ति । ते भभ्रमा रविभगणोनाः (रविभगणरहिताः) तदा भानोः (सूर्यस्य) सावनदिवसाः स्युः । ते कुदिवसाः (कुदिनानि) स्युः, कल्पे यावन्ति सूर्यसावनदिनानि तान्येव कुदिनसंज्ञकानि ॥२२॥

अत्रोपपत्तिः ।

एकस्मिन् दिने उदयकाले केनापि नक्षत्रेण सह रविरुदितो दृष्टो द्वितीयदिने नक्षत्रस्यगत्यभावात्प्रथमं तदुदयस्तदनन्तरं रविगतिकलोत्पन्नासुभिस्तदुदयोऽत एकस्मिन् नाक्षत्रदिने रविगति-कलोत्पन्नासुयुक्ते एक सावनदिनान्तःपातिनाक्षत्र-कालः । द्वितीयदिने नाक्षत्रदिनद्वये रविदिनद्वयगतियोगोत्पन्नासुयुक्ते सावनदिन-द्वयान्तःपातिनाक्षत्रकाल एवमग्रेपि अर्थाद्यस्मिन्निष्टदिने नाक्षत्रकालोऽपेक्षितस्तद्दिन-संख्यक-नाक्षत्रदिने इष्टदिनरविगतियोगासुयुक्ते सतीष्टदिनान्तःपातिनाक्षत्र-कालो भवेदेतेनैव नियमेन एकवर्षान्तःपातिसावनसंख्यातुल्ये नाक्षत्रदिने एकवर्ष-सम्बन्धिरविगतियोगा (क्रान्तिवृत्त) सु (नाक्षत्रदिनमेकं) युक्ते वर्षान्तःपाति नाक्षत्रदिनमर्थाद् वर्षान्तःपातिभभ्रमो भवेदतो वर्षान्तःपातिभभ्रम=वर्षान्तः पातिरविसावन सं+१ नाक्षत्रदिनम् । ततोऽनुपातेन, यद्येकेन वर्षेण वर्षान्तःपाति-भभ्रमा लभ्यन्ते तदा कल्पवर्षैः किमिति । कल्पे भभ्रमाः=(वर्षान्तःपातिरवि-सावन सं+१) क वर्ष=वर्षान्तःपातिरविसावनसं×कवर्ष+कवर्ष=कल्परवि-सावनदिन+कवर्ष=कल्पकुदिन+करविभगण=पठिताङ्काः, तथा कल्पभभ्रम—करविभगण=कल्पकुदिन कल्परविसावनदिन+कवर्ष=कल्पकुदिनम् भास्करा चार्येणापि खखेषुवेदषड्गुणाकृतीभभूतभूमयः शताहता भपश्चिमभ्रमा भवन्ति काहनी'त्यनेनेदमेव कथ्यत इति ॥२२॥

अब भभ्रम को और कुदिनों को कहते हैं ।

हि. भा.—कल्प में १५८२२३६४५०००० इतने भभ्रम होते हैं, भभ्रम में रविभगण को घटाने से रवि के सावन दिन होते हैं वह कुदिनसंज्ञक हैं अर्थात् कल्प या युग में जो सूर्यसावन दिन होते हैं उन्ही को कल्प या युग में कुदिन कहते हैं ॥२२॥

उपपत्ति

एक दिन में उदय काल में किसी नक्षत्र के साथ रवि उदित हुए, दूसरे दिन में नक्षत्र की गति नहीं रहने के कारण पहले नक्षत्र का उदय होता है उसके बाद रविगति-कलोत्पन्नासु करके रवि का उदय होता है इसलिए एक नाक्षत्र दिन में रविगतिकलोत्पन्नासु जोड़ने में एक सावन दिनान्तःपाती नाक्षत्र काल होगा । एवं दूसरे दिन में दो नाक्षत्र दिन में दो दिनों के रविगतियोगकालोत्पन्नासु से दो सावन दिनान्तःपाति नाक्षत्र काल होगा । इसी तरह तीसरे, चौथे आदि दिनों में भी विचार करना । इससे यह देखने में आता है कि जिस इष्टदिन में नाक्षत्र काल अपेक्षित हो उस दिन के संख्यातुल्य नाक्षत्र दिन में एक वर्ष सम्बन्धित रविगतियोगा (क्रान्तिवृत्त) सु (एक नाक्षत्र दिन) जोड़ने से एक वर्षान्तःपाति नाक्षत्र दिन अर्थात् एक वर्षान्तःपाति भभ्रम होता है । इसलिए वर्षान्तः-

पतिभभ्रम=वर्षान्तःपाति रविसावनसं+१ नाक्षत्र दिन तब अनुपात यदि एक वर्ष में यह वर्षान्तःपाति भभ्रम पाते हैं तो कल्पवर्ष में क्या, इससे कल्प में भभ्रम=(वर्षान्तः-पाति रसावन सं+१)×कवर्ष=वर्षान्तःपाति र सावन सं×कवर्ष=कल्परविसावनदि+कवर्ष=कल्पकुदिन+करविभगण=पठिताङ्क तथा कल्पभभ्रम—कल्परविभगण=ककुदिन (भास्कराचार्य भी खखेषुवेदषड्गुणा कृतीभभूतभूमयः। शताहता भपश्चिमभ्रमा भवन्ति काहनि' इससे वही कहते हैं इति ॥ २२ ॥

इदानीं कल्पे रविवर्षमासशशिमास-दिनाधिमासोनरात्राणां प्रतिपादनायार्यद्वियमाह—

रविभगणा रव्यब्दा द्वादशगुणिता भवन्ति रविमासाः=५१८४०००००००।
भगणान्तरं रवीन्द्वोः शशिमासाः ५३४३३३००००० सूर्यमासोनाः ॥२३॥

अधिमासाः१५९३३००००० शशिमासास्त्रिंशद्गुणिता १६०२९९९०००००० भवन्ति शशिदिवसाः।

शशिसावनदिवसान्तरमवमानि तिथिः शशाङ्कदिनम् ॥२४॥

वा. भा.—रवेर्भगणाः एव रव्यब्दा भवन्ति, यतः स्वभगणभोगेनैव तस्य वर्षं भवति। त एव भगणा द्वादशगुणिताः सन्तो रविमासा भवन्ति। वर्षद्वादशगुणं मासत्वमापद्यते। सर्वस्यैवमतः। तद्यथा रविभगणाः ४३२०००००० एते द्वादशहताः कृतवसुचन्द्रशराः शून्यसप्तकेनाहताः ५१८४०००००० तथा सप्त शून्यानि वेदाष्टनिशाकर शिलीमुखा भवन्ति, 'मासाः सावित्राब्राह्म्णाह्ना सदैव तु। भगणान्तरं रवीन्द्वोः शशिमासा' इति शशिमासाः। पुनर्भगणांतरं रवीन्द्वोः कृत्वा शशिमासा भवन्ति, यस्माद्रविचन्द्रयोः यावन्तः कल्पे योगाःतावन्त एव शशिमासास्तावन्त्य एव कल्पेऽमावास्या इत्यर्थः। वक्ष्यति यतः तिथिशशांकदिनमिति। तद्यथा रविभगण ४३२०००००० शशिभगणाश्च ५७७५३३०००००० एतेषामन्तरं ५३४३३३००००० पञ्चशून्यानि गुणरामाग्निवेदलोकशराः एते कल्पे शशिमासा एत एव सूर्यमासैरमीभिः १५९३००००० ऊना जाता कल्पाधिमासकाः त्रिगुणनवतिथयः पंचशून्येनाहताः १५९३३००००० तथा च पंचशून्यानि रामाग्नि-नवपञ्चनिशाकरकल्पाधिमासका ज्ञेया नित्यमेव विचक्षणैः॥

शशिमासाश्च त्रिंशद्गुणिताः शशिदिवसाः भवन्ति। न केवलं चन्द्रस्यान्यस्यापि मासाः त्रिंशद् गुणिताः दिवसत्वमापद्यन्ते। तद्यथा शशिमासास्त्रिंशद् गुणिता जाता नवनन्दनवखरसचन्द्राः शून्यषट्काहता १६०२९९९०००००० एते कल्पे चन्द्रदिवसाः। तथाच 'शून्यषट्कं च गोनन्दनवाश्विखरसेन्दवः। कल्पे चन्द्रदिनान्याहुर्नित्यं गणितपारगाः।' शशिसावनदिवसान्तरमवमानीति। शशिदिनानां सावनानां च यावत्यन्तरे दिनानि तावन्त्यवमानि, तावन्तः कल्पे तिथिलोपाः। तद्यथा कल्पे शशिदिनानि १६०२९९९०००००० अर्कसावनदिनानि च १५७७९१६४५

०००एतयोरन्तरं शरेषुयमवसु-ख-शराश्विनः खचतुष्टयैकाहताः—

२५०८२५५०००० तथा च बलभद्रः "खचतुष्कं शरार्थाश्विवसु-शून्यशराश्विनः। कल्पोनरात्रा विज्ञेया नित्यमेव मनीषिभिः"। तिथिशशाङ्कदिनमित्युत्तरत्र संबद्धं भवतीति।

वि. भा.—रविभगणाः (कल्पपठितरविभगणाः ४३२०००००००) रव्यब्दा (कल्पसौरवर्षाणि) भवन्त्यर्थात्कल्पे यावन्तो रविभगणास्तावन्त्येव कल्पसौरवर्षाणि भवन्ति। ते (रव्यब्दाः) द्वादशगुणितास्तदा रविमासा (सौरमासाः) भवन्ति। रवीन्द्वोर्भगणान्तरं (कल्पचन्द्रभगणरविभगणयोरन्तरं) शशिमासाः कल्पचान्द्रमासाः) भवन्ति, चान्द्रमासाः सूर्यमासोनाः(कल्पचान्द्रमासाः कल्पसौरमासरहिताः) तदाऽधिमासाः (कल्पाधिमासा.) भवन्ति। शशिमासाः(कल्पचान्द्रमासाः) त्रिंशद्गुणितास्तदा शशिदिवसाः(कल्पचान्द्रदिनानि)भवन्ति,शशिसावनदिवसान्तरं कल्पचांद्रदिनकल्पकुदिनयोरन्तरं कल्पावमानि भवन्ति) तिथिः शशाङ्कदिनं (तिथिश्चान्द्रदिनं) भवतीति ॥२३-२४॥

अत्रोपपत्तयः

सृष्ट्यादिकाले नाड़ीक्रान्तिवृत्तयोः सम्पाते (स्थिरमेषादौ) एव रविस्ततोऽनन्तरं रवेर्भ्रमणेन पुनर्यदा तद्विन्दौ (स्थिरमेषादौ) रविरागच्छति तदा तद्भगणपूर्तिर्भवति परमेतावति (द्वादशराशिभोग) काले तत्सम्पातस्यापि तु किमपि चलनं भवेत्तेन पूर्वोक्तरविभगणे (सौरवर्षे) सम्पातस्य यच्चलनं भवेत्तद्योज्यं तदा सम्पातात् सम्पातं यावत्सायनसौरवर्षमेकरविभगणभोगकालो वा भवति परमत्राचार्येण निरयणसौरवर्षमेव कथ्यते, कल्पेऽपि रविभगणतुल्यानि निरयणसौरवर्षाण्येव भवितुमर्हन्ति, आचार्येण तथैव कथ्यन्ते। कल्पसौरवर्ष × १२ = कल्पसौरमासाः एतावता 'रविभगणा रव्यब्दाः' इत्याचार्यकथनं निरयणसौरवर्षपरं बोध्यं, भास्कराचार्येणापि 'रवेश्चक्रभोगोऽर्कवर्षप्रदिष्टमि'त्यनेन निरयणसौरवर्षमेव कथ्यते, सर्वैरेवाचार्यैरयनगतिरत्र विषये शून्या कल्पितेति ॥

अथ चान्द्रमासोपपत्तिः

अमान्तकाले रविचन्द्रावेकत्रैव भवतः एव (दर्शः सूर्येन्दुसङ्गम इत्यमरोक्तेः) ततोऽनन्तरं स्वस्वगत्या तौ चलितौ तयोश्चन्द्रस्याधिकगतित्वाच्चन्द्रः पूर्वस्थानं (अमान्तबिन्दुं) गत्वा रविणा सह पुनरपि मिलितस्तदैकचान्द्रमासपूर्त्तिर्जाता। तत्र चन्द्रगतिः = १चन्द्रभगण + रविगतिः। अतः चन्द्रगति—रविगति = १चन्द्रभगणः, ततोऽनुपातो यदा रविचन्द्रयोर्गत्यन्तरमेकभगणतुल्यं तदैकश्चान्द्रमासस्तदा कल्पीयगत्यन्तरेण (कल्पीय-रविचन्द्रभगणान्तरेण) कियन्तो लब्धा कल्परविचन्द्रभगणान्तरतुल्याश्चान्द्रमासा एतावता 'भगणान्तरं रवीन्द्वोः शशिमासाः' इति आचार्योक्तमुपपद्यते। वटेश्वरश्रीपतिभास्करप्रभृतिभिराचार्यैरेतदनुरूपमेव कथ्यते। सूर्यसिद्धान्तेऽपि 'भवन्ति शशिनो मासाः सूर्येन्द्रभगणान्तरमि'त्यनेन तदेव कथ्यते सूर्यांशदेवेनेति ॥

अथाऽधिमासोपपत्तिः

अर्थैकसावनदिने $\left\{\begin{array}{l}\text{चान्द्रमध्यगतिः}=७९°।३५''\\ \text{रविमध्यग}=५९'।८''\end{array}\right\}$ अनयोरन्तरम्=७३१'।२७''
=१२°।११'।२७''

अथ यतः चंग−रग=१२°=१ तिथिस्तस्मात्सावनदिनपूर्तिकालात् पूर्वमेव चान्द्रदिनपूर्तिः सिद्धाऽतः चान्द्रदि<सावनदि<सौरदिन यतः सौरदि =६०', यदा रविगतिः षष्टिकला भवेत्तदा सौरदिनपूर्तिः, सावन-दिन-पूर्तिरस्ति ५९'।८'' एतत्तुल्यरविगतावेवातो दिनसंख्यया सौरदिसं<चान्द्र दिवसम्

ततः कचान्द्रमास−कसौरमास=कल्पाधिमास, कचांदिन−कसावनदि= कक्षयदिन=कल्पावमदिन एतेन 'शशिमासाः सूर्यमासोना' इत्यारभ्य 'शशिसावन-दिवसान्तरमवमानीत्यन्तमुपपद्यते । भास्कराचार्येणापि 'सौरान्मासादैन्दवः स्याल्लघीयान् यस्मात्तस्मात्संख्यया तेऽधिकाः स्युरि'त्यादिना तदेव कथ्यत इति ॥२४॥

अब सौरमास चान्द्रमास अधिमास और अवमदिन को कहते हैं

हि.भा.—कल्प पठित रविभगण ४३२०००००० कल्पसौरवर्ष होते हैं अर्थात् कल्प में जितने रविभगण हैं उतने ही कल्पसौरवर्ष होते हैं, कल्पसौरवर्ष को बारह से गुणने से कल्पसौरमास होते हैं, कल्पचन्द्रभगण और कल्परविभगण का अन्तर कल्प चान्द्रमास होते हैं। कल्पचान्द्रमास में कल्प सौरमास को घटाने से कल्पाधिमास होता है। कल्पचान्द्रमास को तीस से गुणने से कल्पचान्द्रदिन होते हैं, कल्पचान्द्रदिन और कल्पकुदिन का अन्तर कल्पावमदिन होते हैं। एक तिथि एकचान्द्रदिन है ॥२३-२४॥

इन सब की उपपत्ति

सृष्ट्यादि काल में नाड़ीवृत्त और क्रान्तिवृत्त के सम्पात (स्थिरमेषादि) में रवि थे। उसके बाद रवि भ्रमण करते हुए फिर जब उसी बिन्दु में आते हैं तब उनकी एक भगण (द्वादशराशिभोग) पूर्त्ति होती है लेकिन इतने समय में वह सम्पात भी कुछ पूर्व स्थान से चलेगा, इसलिए पूर्वकथित रवि के एकभगणभोग कालतुल्य सौरवर्ष में सम्पात चलन (अयनगति) जोड़ने से वास्तव सायन सौरवर्ष होगा. परन्तु यहां आचार्य निरयण सौरवर्ष ही लेते हैं, कल्पसौरवर्ष भी निरयण ही कहते हैं।

कल्पसौरवर्ष×१२=कल्पसौरमास 'इससे (रविभगणरव्यब्दाः) यह आचार्यकथन निरयणसौरवर्षपरक समझना चाहिए, भास्कराचार्य भी 'रवेश्चक्रभोगोऽर्कवर्ष-प्रदिष्टम्' इससे जो एक रविभगण भोगकाल को एक सौरवर्ष कहते हैं वह भी निरयण सौरवर्ष ही सिद्ध होता है, सब आचार्य यहां अयनगति को शून्य मानते हैं जो ठीक नहीं है ॥

चान्द्रमास की उपपत्ति

अमान्तकाल में रवि और चन्द्र एक ही स्थान में रहते हैं उसी का नाम दर्शान्त

(अमान्त) है उसके बाद रवि और चन्द्र अपनी-अपनी गति से चलने लगे, चन्द्र गति की अधिकता के कारण जिस स्थान (अमान्त बिन्दु) से चले थे वहां जाकर फिर रवि के साथ योग करते हैं तब एक चान्द्रमास (प्रथमामान्त से द्वितीयामान्त तक) भी पूरा हो जाता है यहाँ पर चन्द्रगति = १ चन्द्रभगण + रविगति हुई क्योंकि जिस स्थान से (प्रथमामान्तबिन्दु) चले थे वहाँ फिर जाने से चन्द्र की एक भगण पूर्ति होती है इसलिए चन्द्रगति—रविगति = चन्द्रभगण, तब अनुपात करते हैं यदि एक चन्द्रभगणतुल्य रविचन्द्र गत्यन्तर में एक चान्द्रमास पाते हैं तो कल्पीय रविचन्द्रगत्यन्तर (कल्पीय रविचन्द्रभगणान्तर) में क्या इस अनुपात से कल्पचन्द्र भगण और कल्परविभगण के अन्तर तुल्य ही कल्पचान्द्रमास सिद्ध हुआ इससे 'भगणान्तरं रवीन्द्वोः शशिमासाः' यह उपपन्न हुआ ॥

अधिमास की उपपत्ति

कल्पचान्द्रमास में कल्प सौरमास को घटाने से कल्पाधिमास क्यों होता है इसके लिए विचार करते हैं।

एक सावन दिन में चन्द्र मध्यमगति = ७९०′।३५″
रविमध्यमगति = ५९′।८″ दोनों के अन्तर = ७३१′।२७″ =
१२°११′।२७″ = चंमग — रमग तथा चंमग — रमग = १२° = १ तिथि इसलिए सावन दिन पूर्ति (पूरा) काल से पहले ही चान्द्रदिन पूर्ति सिद्ध हुई।

अतः चान्द्रदिन < सावनदि < सौरदिन, क्योंकि जब रवि की मध्यमगति साठ कला के बराबर होती है तब सौर दिन पूर्ति होती है और ५९′।८″ इतनी रवि मध्यमगति में सावन दिन पूर्ति होती है इसलिए दिन संख्या से सौ दिसं < चांदिसं अतः कल्पचान्द्रमास—कल्पसौर मास = कल्पाधिमास, तथा कल्पचांदि—कसावनदि = कक्षयदि = कल्पावमदि इससे शशिमासा रविमासोना—यहाँ से 'शशिसावनदिवशान्तरमवमानि' यहाँ तक उपपन्न हुआ ॥ २४ ॥

इदानीं सावनदिननाक्षत्रदिनमानववर्षापितृदिनदिव्यदिनान्याह

सावनमुदयादुदयं भानां चाक्षं नृवत्सरोऽर्कब्दिः।
पितृदिवसाः शशिमासा दिव्यानि दिनानि रविभगणाः ॥२५॥

वा. भा.—शशिमासा दिव्यानि दिनानि रविभगणाः। तिथिः शशांकदिनं तिथिरेव चन्द्रदिनम्। दिनग्रहणेनाहोरात्रोगृह्यते सर्वेष्वेव मानेषु, तेन यावदेव तिथिभोगप्रमाणं तावदेव चन्द्रमासमानेन दिनप्रमाणं भवतीति, तैस्त्रिंशता शशिमासा इत्यादि योज्यम्। एतच्च भगणान्तरं रवीन्द्वोः शशिमासा इत्येनेनैव सिद्धेः स्पष्टीकरणायोच्यते। सावनमुदयादुदयमित्यनेनैव सिद्धेः चन्द्रनक्षत्रभागावधिजस्य नाक्षत्रमानस्य व्युदासार्थमाचार्येणोक्तम्। भानां चाक्षमिति। तथा नक्षत्रसावन-

मपि नाक्षत्रमुच्यते यतोर्कसावनमेवमेवोपयोगि-ग्रहगत्यानयनेऽन्यत्सावनेन नृवत्सरोर्काब्दः मानुषवर्षमित्यर्थः, तस्य द्वादशभागः मध्यरविसंक्रान्त्यवधिजो रविमासः तस्यैव त्रिंशद्भागो रविदिवसः स च शशिभोगावधिज इत्यादि सौरमानमुक्तम्, पितृदिवसाः शशिमासाः इत्थं त एव शशिनो मासास्त-पितृमानेन एव दिवसाश्चन्द्रमासेनैकेन पितृणामहोरात्रो भवतीत्यर्थः। अत्र वासना पूर्वमेव गोलाध्याये प्रदर्शितेति। दिव्यानि तु पुनर्दिनानि रविभगणाः; मेरुवासिनां रविभगणभोगकालेनाहोरात्रं भवति। वडवामुखवासिनामप्यसुराणां रविभगणभोगकालेनैव त्वहोरात्रं भवति, रविमासैः षड्भिः तेषां दिनम् षड्भो रात्रिर्भवति एतत् गोलाध्याये सर्वव्याख्यातम्। सवासनिको ब्राह्मो दिवसः कल्पः। एवं मानेश्च यत्प्रयोजनम् तन्मानाध्याये वक्ष्यत्याचार्यः। वयमपि तत्रैव व्याख्यास्याम इति॥

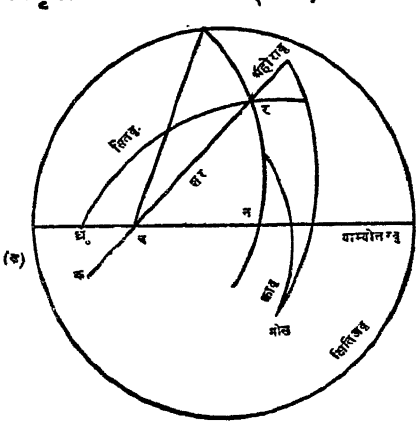
वि. भा.—रवेरुदयादुदयं सावनं (रविसावनं) भानां च (उदयादुदयं) आर्क्षं सावनं (नाक्षत्रसावनं दिनं) भवति। अर्काब्दः (सौरवर्षम्) नृवत्सरो (मानववर्षं) अर्थादेकसौरतुल्यं मानववर्षं भवति, पितृदिवसाः (पितृदिनानि) शशिमासाः (चान्द्रमासाः) भवन्त्यर्थाच्चान्द्रमासतुल्यानि पितृणां दिनानि भवन्ति, रविभगणाः दिव्यानि दिनानि (दैवदिनानि) भवन्त्यर्थादेकरविभगणतुल्यानि दैवदिनानि भवन्तीति॥२५॥

अत्रोपपत्तयः

रवेरुदयादुदयं यावद्रविसावनं कुदिनसंज्ञकम्। नक्षत्रोदयात्पुनस्तदुदयं यावन्नाक्षत्रदिनं, सौरवर्षतुल्यं मानववर्षमिति परिभाषारूपाः कथ्यन्ते। अथाधुना चान्द्रमासतुल्यं पितृदिनं कथं भवतीत्येतदर्थं विचार्यते।

विधूर्ध्वभागे पितरो वसन्तीति पुराणादिकथितमवलम्ब्य विचारः। तत्र विधूर्ध्वभाग (चन्द्रोर्ध्वभाग) शब्देन चन्द्रस्य कियान् भागो ग्रहीतव्य इति। दृष्टिस्थानात् (भूकेन्द्रात्) चन्द्रबिम्बस्यानेकाः स्पर्शरेखा कार्यास्तदा प्रतिस्पर्शबिन्दुजनितचन्द्रबिम्बप्रदेशो वृत्ताकारो भवति, चन्द्रबिम्बे एतस्य शोधनेन यच्छेषं स एव चन्द्रोर्ध्वभागस्तत्र पितरो निवसन्तीति पुरातनानां कथनेन ज्ञायते। अथ भूकेन्द्राच्चन्द्रकेन्द्रगता रेखा यत्र पितृत्रिज्यागोलीययाम्योत्तरवृत्ते लगति तत्रैव परिणतश्चन्द्रस्तदेव पितृ-ख-स्वस्तिकमप्यस्ति, तत्र यदा रविर्भवेत्तदा दर्शः सूर्येन्दुसङ्गम इत्युक्तेरमान्तो भवेत्तथा परिणतचन्द्ररूपपितरूर्ध्वखस्वस्तिकबिन्दौ रवेर्गमनात्तन्मध्याह्नकालोऽतः सिद्धं यदमान्तकाले पितृणां मध्याह्नकालो (दिनार्धं) भवति, पुनर्यदा तद् द्वितीयामान्तो भवेत्तदा पितृणा द्वितीयदिनार्धं भविष्यति तेन प्रथमामान्ताद् द्वितीयामान्तं यावच्चान्द्रमासतुल्यं पितृप्रथमद्वितीयमध्याह्नकालयोरन्तरं जातं परन्तु प्रथमद्वितीयमध्याह्नकालयोरन्तरं प्रथमद्वितीय-

सूर्योदययोः कालयोरन्तरं समं, प्रथमद्वितीयसूर्योदयकालयोरन्तरं सावनदिनम्, अतः पितृणां दिन (अहोरात्रं) चान्द्रमासतुल्यं सिद्धम्। परमत्रोपपत्तावमान्तकाले पितृमध्याह्नकालं स्वकृतौ विचारः कृतः, सिद्धान्तशिरोमणौ भास्करेणापि, 'दर्शे यतोऽस्माद् द्युदलं तदैवामि'त्यादिनाऽमान्तकाल एव पितृदिनार्धकालः स्वीकृतः स च न समीचीनः। पितृयाम्योत्तरेऽर्कात्चन्द्रोपरिगतध्रुवप्रोतवृत्ते यदा रविरागच्छेत्तदैव तन्मध्याह्नकालः, चन्द्रोपरिगतकदम्बप्रोतवृत्तमेव यदि पितृयाम्योत्तरवृत्तं भवेत्तदैवामान्तकालपितृमध्याह्नकालयोरभेदत्वं भवेत्तरमेवं चन्द्रशराभावस्थले, शरसत्तायामपि मिथुनान्ते चन्द्रे सति भवद्यथा (चन्द्रक्रेन्द्रोपरि ध्रुवप्रोतकदम्बप्रोतयोरभेदे सतीव स्थितिः स्याच्छरसत्वेऽपि। तत्तु मिथुनान्ते धनुरन्ते वा भवतीति।) भूकेन्द्राच्चन्द्रकेन्द्रगता रेखा पितृगोले यत्र लगेत्तत्रैव यदि रविकेन्द्रं स्यात्तदा तस्मिन्नेवामान्तबिन्दौ चन्द्रशराभावत्वान्मध्याह्नकालामान्तयोरभेदत्वं भवेत्। तथा च मिथुनान्ते विमण्डले चन्द्रे सति तस्मिन्नेवायनप्रोतवृत्त-क्रान्तिवृत्तयोर्योगबिन्दौ यदा रविः स्यात्तदैवामान्तकालमध्याह्नकालयोरभेदत्वं स्यादेतत्थितस्थानद्वयातिरिक्तस्थले सर्वदैवामान्तकालमध्याह्नकालयोर्भेदो भवेद्यथा, यदारविः पितृयाम्योत्तरे समागच्छेत्तदैव मध्याह्नकालः। चन्द्रोपरिगतकदम्बप्रोतवृत्तक्रान्तिवृत्तयोर्योगबिन्दौ (चन्द्रस्थाने) यदा रविरागच्छेत्तदाऽमान्तकालो भवेदमान्तबिन्दुनोऽर्वाञ्चन्द्रोपरिगतकदम्बप्रोतवृत्त-क्रान्तिवृत्तयोर्योगबिन्दुस्थरविबिन्दुतः पितृयाम्योत्तरवृत्तं प्राक् पश्चिमे वा भवेत्तत्रामान्तकालाद्यावता कालेन पितृयाम्योत्तरे रविरागच्छेत्तकालमानमायनदृक्कर्मासु तुल्यं तेन कालेन (आयनदृक्कर्मासुना) अमान्तकालो यदि संस्कृतो भवेत्तदा पितृयाम्योत्तरे रविर्भवेत् स एव वास्तवपितृमध्याह्नकालः।



रन=आयनदृक्कर्मकलाभवः
∴अमान्तकाल±आयन दृक्कर्मकलासु =वास्तवपितृदिनार्धं एतद्दृशेनैव रात्र्यर्धमपि बोध्यम्। पितृणामूर्ध्वखस्वस्तिके (परिणतचन्द्रबिन्दौ) रवावमान्तकाले तद्दिनार्धम्, परिणतचन्द्रात्षड्भान्तरेऽधः खस्वस्तिके रवौ पूर्णान्ते तद्रात्र्यर्धं भवतीति।

अमान्तकाले पितृदिनार्धं पूर्णान्तकाले च रात्र्यर्धं सिद्धं परं तदुदयास्तौ कदा कुत्र भवेतां तदर्थं विचार्यते।

भूकेन्द्राच्चन्द्रकेन्द्रगता रेखा पितृत्रिज्यागोले याम्योत्तरवृत्ते यत्र लगति तत्र परिणतचन्द्रः पितृखस्वस्तिकञ्च. । पितृखस्वस्तिकान्नवत्यंशवृत्तं तत्क्षितिज-

वृत्तम्। तत्स्थरवौ परिणतचन्द्रतद्विगतमिष्टवृत्तं (सितवृत्तं कार्यम्) तथा परिणतचन्द्रोपरिगतं कदम्बप्रोतवृत्तं च कार्यं, सितवृत्त-कदम्बप्रोतवृत्त-क्रान्तिवृत्तजचापैः कर्णभुजकोटिभिरुत्पन्नचापीयजात्ये सितवृत्तीयचापं सितवृत्तीय-रविचन्द्रान्तरं वा=९०, अतो गोलीयरेखागणितयुक्त्या क्रान्ति-वृत्तीयचापं क्रान्तिवृत्तीयरविचन्द्रान्तरं वा =९० ततस्तदुदयास्तयोः सर्वदैव रविचन्द्रान्तरं नवत्यंशसमं भवितुमर्हति। परिणतचन्द्रोपरिगतकदम्बप्रोतवृत्त-क्रान्तिवृत्तयोः सम्पातबिन्दोश्चन्द्रत्वात्. । तेन कृष्णपक्षसार्धसप्तम्यां तदुदयः शुक्लपक्षसार्धसप्तम्यां चास्तो ज्ञेयः। सिद्धान्तशिरोमणौ भास्करेण 'कृष्णो रविः पक्षदलेऽभ्युदेति शुक्लेऽस्तमेत्यर्थत एव सिद्धमि'त्यनेनैवमेव कथ्यते। परमेतत्कथितयोरुदयास्तकालयोः खण्डनं म० म० पण्डितसुधाकरद्विवेदिभिः क्रियते। यथा—

भूकेन्द्राच्चन्द्रकेन्द्रगता रेखा चन्द्रपृष्ठे यत्र लगति तद्बिन्दुतश्चन्द्रगर्भक्षितिज-धरातलस्य समानान्तरधरातलं कार्यं तदैकं त्रिभुजमुत्पद्यते। भूकेन्द्राच्चन्द्रपृष्ठं यावच्चन्द्रव्यासार्धयुतश्चन्द्रकर्णः कोटिरेको भुजः। रविकेन्द्रमुदयास्तकाले सर्वदा पृष्ठक्षितिज एव भवेत्तत्रत्यो रविकर्णःकर्णो द्वितीयो भुजः। पृष्ठक्षितिजधरातले भुजस्तृतीयो भुजोऽत्र कोटिकर्णभुजैरुत्पन्नत्रिभुजेऽनुपातः क्रियते यदि रविकर्णेन त्रिज्या लभ्यते तदा चन्द्रव्यासार्धयुतचन्द्रकर्णेन किमित्यनुपातेन रविलग्नकोणज्या समागच्छति तत्स्वरूपम्=$\frac{\text{त्रि}\times(\text{चंक}+\text{चंव्या}\frac{3}{5})}{\text{रा०क०}}$ अस्याश्चापम् =चा, नवत्यंशे विशोधितं तदा भूकेन्द्रलग्नकोणमान रविचन्द्रयोः सितवृत्तीयमन्तरं भवेत्, ९०—चा=सितवृत्तीयान्तरम्, ततो भक्ताब्यर्कविधोर्लवा यमकुभिरित्यादिना पितॄणामुदयकालिकगततिथिः=$\frac{९०-\text{चा}}{१२}=७\frac{१}{२}-\frac{\text{चा}}{१२}$ एतद्दर्शनेन स्पष्टमव-गीयते यत्कृष्णपक्षसार्धसप्तम्यां यत्तदुदयकालो भास्करेण कथितः स च न समीचीनः, सार्धसप्तम्यां चापस्य द्वादशांशप्रमाणं यदि शोध्यते तदोदय कालिकतिथिः समागच्छति पूर्वं पितॄणामुदयकालं मत्वा तत्कालीनतिथि-प्रमाणमानीतं तद् भास्करोक्तं नागच्छति, एवं पितॄणामस्तकालोऽपि शुक्ल-पक्ष-सार्धसप्तम्यां न भवत्यतो भास्करोक्तं 'कृष्णेरविपक्षदलेऽभ्युदेतीत्या'दि न समीचीनमिति। परं म० म० सुधाकरद्विवेदिकृतखण्डनमपि समीचीनं नास्ति। भक्ताब्यर्कविधोर्लवा इत्यादिना क्रान्तिवृत्तीयरविचन्द्रान्तरांशवशेन तिथ्यानयनं भवति, सितवृत्तीयरविचन्द्रान्तरवशेन नहि, परं पूर्वोपपत्तौ सितवृत्तीयान्तरवशे-नैव तिथ्यानयनं कृतमतस्तन्न तथ्यम्। अत्र वास्तवानयनं क्रियते। पूर्वोपपत्तिबलेन सितवृत्तीयान्तरं विदितमस्ति, चन्द्रशरोऽपिविदितोऽस्ति तदोपरिप्रदर्शितचापीय जात्ये (सितवृत्तीयान्तरं कर्णः, क्रान्तिवृत्तीयरविचन्द्रान्तरं कोटिः, चन्द्रशरो भुज इति कर्णकोटिभुजैरुत्पन्ने) भुजकोटिज्याकोटिकोटिज्ययोर्घातस्य त्रिज्याकर्णको-टिज्ययोर्घातसमत्वात्।

सितवृत्तीयान्तरकोटिज्या × त्रि=क्रान्तिवृत्तीयान्तरकोज्या × शरकोज्या, पक्षौ (शरकोज्या) भक्तौ तदा

$$\frac{\text{सितवृत्तीयान्तरकोज्या} \times \text{त्रि}}{\text{शरकोज्या}} = \text{क्रान्तिवृत्तीयान्तरकोज्या}$$

अस्याश्चापं नवतेर्विशोध्यं तदा क्रान्तिवृत्तीयान्तरांशा भवेयुस्तदैतद्वशेन पितृणामुदयास्तकालिकतिथी साध्येते वास्तविके भवेतामिति ॥

वस्तुतस्तु पितृणां दिनार्धरात्र्यर्धकालौ दृक्कर्मसुभिर्विभिद्येते इति गोलयुक्त्या स्फुटमेव । तेन तदीयोदयास्तकालावपि विभिन्नावेव । किं तत्र वैचित्र्यम् । अत्र बहवो विशेषाः प्रतिपादयितुं शक्यन्ते । किमत्र ग्रन्थविस्तरेण ।

दिव्यानि दिनानि रविभगणा इत्यादेरुपपत्तिः

उत्तरध्रुवो देवानामूर्ध्वखस्वस्तिकं, दक्षिणध्रुवश्च राक्षसानाम् । खस्वस्तिकात् (ध्रुवात्) नवत्यंशेन यद्वृत्तं तन्नाडीवृत्तं तत्क्षितिजवृत्तम् । नाड़ीवृत्तादुत्तरे (मेषादितः कन्यान्तं यावत्) स्थिते रवौ षण्मासं देवदिनं राक्षसरात्रिश्च, तथा नाडीवृत्तादक्षिणे (तुलादेर्मीनान्तं यावत्) स्थिते रवौ षण्मासं देवरात्रिः, राक्षसदिनञ्च (क्षितिजादूर्ध्वस्थे रवौ दिनं तदधःस्थे रवौ रात्रिरिति नियमात्) तेन मेषादितो द्वादशराशिभोगकालः (रवेरेकभगणभोगः) सौरवर्षमेकं देवराक्षसयोरहोरात्रं (दिनं) सिद्धम् । वस्तुतस्तु मेषादितो मीनान्तं यावद्रविर्गमिष्यति तावति काले सम्पातस्याऽपि किमपि चलनं भवेत्तदैकरविभगणभोगकाले देवराक्षसयोरहोरात्रान्तकालिकायनगत्युत्पन्नकालस्य संस्करणेन वास्तवं तदहोरात्रमानं भवेत् । सम्पातचलनमेवायनगतिः । आचार्येणायनगतिर्न स्वीकृता तेनतज्जन्यात्र त्रुटिरस्ति, भास्करेणापि 'रवेश्चक्रभोगोऽर्कवर्षं प्रदिष्टमित्यादिनैतदेव कथ्यते भास्करोक्तावपि सैव त्रुटिरस्ति । ब्रह्मगुप्तेन केवलं दिव्यानि दिनानि रविभगणतुल्यानि कथ्यन्ते, राक्षसदिनानां चर्चा न क्रियते, यदा देवानां दिनं भवति तदा राक्षसानां रात्रिः, यदा च देवानां रात्रिस्तदा राक्षसानां दिनं भवति, द्वयोः सहैव विलोमेन रात्रिदिने भवताऽतो मया दैवदिनेन साकं राक्षसदिनमपि प्रदर्शितमिति ॥ २५ ॥

अब सावन दिन नाक्षत्र दिन मानव वर्ष पितृदिन और दिव्य दिनों को कहते हैं

हि. भा.—रवि के उदय से द्वितीय उदय पर्यन्त रविसावन दिन है, नक्षत्रोदय से नक्षत्रोदय तक नाक्षत्रसावन दिन होता है; एवं किसी ग्रह के उदय से उदय तक उस ग्रह का दिन सावन होता है । मानव वर्ष सौरवर्ष के बराबर होता है । पितरों का दिन चान्द्रमास के बराबर होता है, रविभगण के बराबर दिव्य (देवताओं के) दिन होते हैं ॥ २५ ॥

इन सब की उपपत्तियाँ

सावनदिन, मानववर्ष ये परिभाषा रूप में कहे जाते हैं, अब पितरों का दिन (अहोरात्र) एक चान्द्रमास के बराबर क्यों होता है इसके लिए विचार करते हैं। चन्द्र के ऊर्ध्वपृष्ठ पर

पितर लोग बसते हैं ऐसा पुराणादियों में कहा गया है, इसी के अवलम्बन से विचार करते हैं। चन्द्र के ऊर्ध्वपृष्ठ (ऊर्ध्वभाग) से कितना भाग ग्रहण करना चाहिए। दृष्टिस्थान (भूकेन्द्र) से चन्द्रबिम्ब की अनेक स्पर्शरेखा करने से प्रत्येक स्पर्शबिन्दुजनित चन्द्रबिम्बप्रदेश वृत्त कार होता है; चन्द्रबिम्ब में इसको घटाने से जो शेष रहता है वही चन्द्र का ऊर्ध्व भाग है। वहां पितर लोग वास करते हैं, यह पुराणादि वचनों से विदित होता है। भूकेन्द्र से चन्द्रकेन्द्रगत रेखा पितरों के त्रिज्यागोलीय याम्योत्तरवृत्त में जहां लगती है वही पितरों के त्रिज्यागोल में परिणत चन्द्र है और पितरों का ऊर्ध्व खस्वस्तिक भी है। वहां जब रवि आयेंगे तो रवि और चन्द्र के एक स्थान में रहने के कारण अमान्तकाल होगा और वहीं परिणत चन्द्र पितरों का ऊर्ध्व खस्वस्तिक है और वहीं रवि भी है इसलिए ऊर्ध्व खस्वस्तिक में रवि के जाने से दोपहर (दिनार्ध वा मध्याह्न) होता है अतः सिद्ध हुआ कि अमान्तकाल में पितरों का दिनार्धकाल होता है। फिर दूसरा अमान्तकाल जब होगा तो पितरों के वहां दूसरा दिनार्धकाल होगा, अतः प्रथम अमान्त से द्वितीय अमान्त तक (एक चान्द्रमास) प्रथम दिनार्ध से द्वितीय दिनार्ध तक काल के बराबर हुआ। परन्तु प्रथम दिनार्ध से द्वितीय दिनार्ध तक काल प्रथम सूर्योदय स द्वितीय सूर्योदय तक काल (अहोरात्र) के बराबर होता है इसलिए पितरों का अहोरात्र एक चान्द्र मास के बराबर सिद्ध हुआ। परन्तु यहां अमान्तकाल में पितरों का मध्याह्नकाल स्वीकार कर विचार किया गया है, यह ठीक नहीं है। सिद्धान्त शिरोमणि में भास्कराचार्य ने भी 'दर्शे यतोऽस्माद् ध्रुवलं तदेषाम्' इससे अमान्तकाल ही में पितरों का दिनार्धकाल स्वीकार किया है, पितरों के याम्योत्तरवृत्त में अर्थात्चन्द्रोपरिगत ध्रुवप्रोतवृत्त में जब रवि आते हैं तभी उनका दिनार्धकाल होता है, चन्द्रोपरिगतकदम्ब-प्रोतवृत्त ही यदि पितरों का याम्योत्तरवृत्त हो तभी अमान्तकाल और पितृमध्याह्नकाल में अभेदत्व होगा। लेकिन ऐसी स्थिति चन्द्रशराभाव स्थान में और चन्द्र की सत्ता में भी चन्द्र के मिथुनान्त में रहने से होती है। यथा भूकेन्द्र से चन्द्रकेन्द्रगत रेखा पितृगोल में जहां लगती है वहीं यदि रविकेन्द्र होगा तभी उसी अमान्त बिन्दु में चन्द्र के शराभाव के कारण मध्याह्नकाल और अमान्तकाल में अभेदत्व होगा। और मिथुनान्त में विमण्डल में चन्द्र के रहने से उसी अयन प्रोतवृत्त और क्रान्तिवृत्त के योगबिन्दु में जब रवि होंगे तभी अमान्तकाल और पितृमध्याह्नकाल में अभेदत्व होता है। इन दोनों स्थानों से भिन्न स्थल में अमान्तबिन्दु से अर्थात् चन्द्रोपरिगतकदम्बप्रोतवृत्त और क्रांतिवृत्त के योगबिन्दुस्थ रविबिन्दु से पितरों के याम्योत्तरवृत्त पूर्व या पश्चिम में होता है। वहां अमान्तकाल से जितने काल में पितृयाम्योत्तरवृत्त में रवि होता है वह काल आयनदृक्कर्मासुतुल्य है, उस काल (आयनदृक्कर्मासु) करके यदि अमान्तकाल का संस्कार करते हैं तब पितृयाम्योत्तरवृत्त में रवि होता है वही वास्तव पितृमध्यान्हकाल है।

यहां (क) क्षेत्र देखिए। रन=आयनदृक्कर्मकलासु। अतः अमान्तकाल ± आयन-दृक्कर्मासु=वास्तव पितृदिनार्ध, इसी के वश से रात्र्यर्ध भी समझना चाहिए।

पितरों के ऊर्ध्व खस्वस्तिक में (परिणत चन्द्रबिन्दु में) रवि के रहने से अमान्तकाल में उनका दिनार्ध होता है। परिणत चन्द्र से छह राशि अन्तर पर अधः खस्वस्तिक में रवि

के रहने से पूर्णान्त काल में पितरों का रात्र्यर्ध सिद्ध होता है। अब पितरों का उदयकाल और अस्तकाल कहां कहां होता है इसके लिए विचार करते हैं। भूकेन्द्र से चन्द्रकेन्द्रगत रेखा चन्द्रपृष्ठ में जहां लगती है उस बिन्दु में परिणत चन्द्र है और वही पितरों का ऊर्ध्व खस्वस्तिक भी है। पितृखस्वस्तिक से नवत्यंश व्यासार्ध से जो वृत्त होता है वह पितरों का क्षितिजवृत्त है, उसमें रवि के रहने पर परिणत चन्द्र और उस रवि में गये हुए वृत्त करते हैं उसका नाम सितवृत्त है। परिणत चन्द्रोपरिगत कदम्बप्रोतवृत्त करने से सितवृत्त कदम्बप्रोतवृत्त और क्रान्तिवृत्त के चापों से एक चापीय जात्य त्रिभुज बनता है, जिस में सितवृत्तीयचाप (सितवृत्तीय रविचन्द्रान्तरांश) कर्ण है, क्रान्तिवृत्तीयचाप (स्थानीय रवि चन्द्रान्तरांश) कोटि है, कदम्बप्रोतवृत्तीय चाप शर भुज है. इस चापीयजात्य त्रिभुज में सितवृत्तीय रविचन्द्रान्तरांश कर्णचाप = ९० इसलिए गोलीयरेखागणित की युक्ति से क्रान्तिवृत्तीय रविचन्द्रान्तरांश कोटिचाप भी नवत्यंश के बराबर होगा, अतः पितरों के उदय और अस्त समय में सर्वदा रवि और चन्द्र का अन्तर नवत्यंश के बराबर होगा. क्योंकि परिणत चन्द्रोपरिगत कदम्बप्रोतवृत्त और क्रान्तिवृत्त का सम्पात बिन्दु चन्द्रस्थान है। इसलिए कृष्ण पक्ष के साढ़े सप्तमी में पितरों के उदय और शुक्ल पक्ष के साढ़े सप्तमी में अस्त समझना चाहिए। सिद्धान्तशिरोमणि में भास्कराचार्य 'कृष्णे रविः पक्षदलेऽभ्युदेति शुक्लेऽस्तमेत्यर्थत एव सिद्धम्' इससे यही बात कहते हैं, परन्तु भास्करकथित पितरों के उदयकाल और अस्तकाल का खण्डन म० म० पण्डित सुधाकर द्विवेदी ने किया है। जैसे—

भूकेन्द्र से चन्द्रकेन्द्रगत रेखा चन्द्रपृष्ठ में जहां लगती है उस बिन्दु से चन्द्रगर्भक्षितिज धरातल के समानान्तर करने से एक त्रिभुज बनता है. भूकेन्द्र से चन्द्रपृष्ठपर्यन्त चन्द्रव्यासार्धयुत चन्द्रकर्ण कोटि प्रथमभुज। उदय और अस्तसमय में रविकेन्द्र सर्वदा पृष्ठ क्षितिज ही में रहते हैं इसलिए वहां के रविकर्ण कर्ण द्वितीयभुज, पृष्ठक्षितिज धरातल में तृतीयभुज, इन कोटि-कर्ण भुजों से उत्पन्न जात्यत्रिभुज में अनुपात करते हैं यदि रविकर्ण में त्रिज्या पाते हैं तो चन्द्रव्यासार्धयुत चन्द्रकर्ण में क्या आजायगी रविकेन्द्रलग्न कोणज्या $= \frac{\text{त्रि} \times (\text{चन्द्रकर्ण} + \text{चन्द्रव्या} \frac{१}{२})}{\text{रविक}}$ इसके चाप = चा, नवत्यंश में घटाने से भूकेन्द्रलग्न कोणप्रमाण अर्थात् सितवृत्तीय रवि चन्द्रान्तरांश = ९०—चा, हुआ तब 'भक्तांव्यर्कविधोर्लवा यमकुभिः' इत्यादि से पितरों के उदयकालिक गत तिथिप्रमाण $= \frac{९०-\text{चा}}{१२}$ ७ $\frac{१}{२} - \frac{\text{चा}}{१२}$ इसको देखने से स्पष्ट है कि कृष्ण पक्ष की साढ़े सप्तमी में पितरों का उदयकाल नहीं होता है किन्तु साढ़े सप्तमी में $\frac{\text{चा}}{१२}$ इतना घटाने से जो होता है उसमें उदयकाल सिद्ध हुआ, इसी तरह शुक्ल पक्ष की साढ़े सप्तमी में उनका अस्तकाल भी नहीं होता है अतः भास्करोक्त 'कृष्णे रविः पक्षदलेऽभ्युदेति' इत्यादि ठीक नहीं है; परन्तु म० म० पण्डित सुधाकर द्विवेदीकृत खण्डन में भी त्रुटि है; उपर्लिखित खण्डनोपपत्ति में सितवृत्तीय रविचन्द्रान्तर पर से 'भक्तांव्यर्कविधोर्लवाः' इत्यादि से जो तिथिप्रमाण लाया गया है सो ठीक नहीं है। क्रान्तिवृत्तीय रविचन्द्रान्तर से तिथ्यानयन करना उचित है, इसलिए उक्त खण्डन भी दोषयुक्त है। अतः अब इसका वास्तवानयन प्रकार

दिखलाया जाता है । सितवृत्तीय रविचन्द्रान्तर कर्ण, क्रान्तिवृत्तीय रविचन्द्रान्तर कोटि, चन्द्र-शरभुज इस त्रिभुज में पूर्वोक्त नियम से सितवृत्तीय रविचन्द्रान्तरांश विदित है, और चन्द्रशर भी विदित है तब भुजकोटिज्या और कोटिकोटिज्या के घात त्रिज्या और कर्णकोटिज्या के घात के बराबर होता है इस नियम से सितवृत्तीयान्तर कोटिज्या × त्रि = क्रान्तिवृत्तीयान्तर कोज्या × शरकोज्या. दोनों पक्षों को (शरकोज्या) इस से भाग देने से—

$\frac{\text{सितवृत्तीयान्तर कोज्या} \times \text{त्रि}}{\text{शरकोज्या}}$ = क्रान्तिवृत्तीयान्तर कोज्या, इसके चाप को नवत्यंश में घटाने से क्रान्तिवृत्तीय रविचन्द्रान्तरांश होंगे, इसके वश से पितरों की उदयकालिक तिथि और अस्तकालिक तिथि साधन करना वह वास्तविक तिथि होगी । इति ।

अब दिव्य दिन रविभगण के बराबर होता है इसकी उपपत्ति दिखलाते हैं

देवों का ऊर्ध्व खस्वस्तिक उत्तर ध्रुव है, राक्षसों का ऊर्ध्व खस्वस्तिक दक्षिण ध्रुव है। खस्वस्तिक (ध्रुव) से नवत्यंश चाप व्यासार्ध से जो वृत्त (नाड़ीवृत्त) होता है वही उनका क्षितिज वृत्त है । नाड़ीवृत्त से उत्तर (मेषादि से कन्यान्त तक) जब रवि रहते हैं तब देवों का छः महीने का दिन होता है और राक्षसों की छः महीने की रात्रि होती है तथा नाड़ीवृत्त से दक्षिण (तुलादि से मीनान्त तक) रवि के रहने से देवों के छः महीने की रात्रि होती है और राक्षसों के छः महीने के दिन होते हैं । (क्षितिज से ऊपर रवि के रहने से दिन और उससे नीचे रहने से रात्रि होती है इस नियम से) इसलिए मेषादि से रवि के बारह राशि भोगकाल (एक रविभगण भोग) याने एक सौरवर्ष देव और राक्षस का अहोरात्र (दिन) सिद्ध हुआ । लेकिन रवि के एक भगण भोगकाल में सम्पात का भी कुछ चलन होगा उसी को अयन गति कहते हैं; इसलिए रवि के एकभगण भोगकाल में देव और राक्षसों के अहोरात्रान्त कालिक अयनगत्युत्पन्न काल का संस्कार करने से उन दोनों का वास्तव अहोरात्रमान होता है, यहां आचार्य ने अयनगति का ग्रहण नहीं किया है इसलिए उतनी त्रुटि है । भास्कराचार्य भी 'रवेश्चक्रभोगोऽर्कवर्षं प्रदिष्टम्' इत्यादि से वही बात कहते हैं इनमें भी वही त्रुटि है । यहां आचार्य 'दिव्यानि दिनानि 'रविभगणाः' इससे देव-सम्बन्धी दिन के विषय में कहते हैं, राक्षसों की चर्चा नहीं की है, देव और राक्षस का विलोम (उल्टा) करके रात्रि और दिन होते हैं लेकिन दिन और रात्रि दोनों की बराबर होती है. इसलिए देव अहोरात्र (दिन) के साथ ही राक्षस अहोरात्र भी दिखला दिये हैं । यदि दिव्य दिन से (राक्षस सम्बन्धी दिन भी) कहा जाय तब तो कोई बात कहने की जरूरत ही नहीं होगी । इति ॥ २५ ॥

इदानीं तत्सर्वस्यैव कोत्पत्तेरारभ्य गतकालस्य शककालस्य ग्रहगणितेऽहर्गणादयः प्राप्ता इत्येतदाशङ्क्य सयुक्तिकं परिहारमाह

कल्पपरार्धे मनवः षट्कस्य गताश्चतुर्युगत्रिघनाः ।
त्रीणि कृतादीनि कलेर्गोऽगैकगुणाः ३१७९ शकान्तेऽब्दाः ॥ २६ ॥

नवनगशशिमुनिकृतनवयमनगनन्देन्दवः १९७२९४७१७९ शकनृपान्ते ।
सार्धमतीतशनूनां सन्धिभिराद्यन्तरान्तगतैः ॥ २७ ॥

वा. भा.—कशब्देन ब्रह्मोच्यते, तस्य कस्य कल्पानां परार्द्धं गतम्। वर्त्तमाने च कल्पे मनवः षट् गताः। चतुर्युगानां त्रिघनाः सप्तविंशतिश्चतुर्युगानां गता। त्रीणि कृतादीनि कृतत्रेताद्वापराणि, कलेः कलियुगस्य गोगैकगुणाः ३१७९ एतावन्तोऽब्दाः, शकान्ते शककालकालात् प्राग्गताः इत्यर्थः। तद्यथा कल्पाब्दाः ४३२०००००० एतैः परार्धसंख्ययाष्टादशस्थानात्मिकया हृता जाताः द्वित्रिवेदाः खचतुर्विंशत्येकहृता जाता :—४३२०००००००००००००००००००००० वर्त्तमान-कल्पात्षण्मनवो गताः, चतुर्युगत्रिघनाः त्रीणि कृतादीनि सर्वेषां वर्षाणि कृत्वा कलिगतवर्षैः पूर्वस्थापितैश्च सह शकान्ते के जन्मनोतीतानि वर्षाणि नवनग-शशिमुनिकृतनवयमा नगनवेन्दुखचतुर्दशकद्विगुणाब्धयोमी ४३२०००००००००-०००० १९७२९४७१७९ एतावन्तोऽब्दाः सौरमानेन कस्य गताः, सोपि कलिनियता-युरिति अन्यैः पुनरेवं व्याख्यातं कल्पपरार्ध कस्य दिनम्, कशब्देन परः पुरुष उच्यते। तस्य कल्पानां परार्धेनाहोरात्रो भवति यस्मान्मानाधिकारे एतत् पठितमाचार्येण। वर्त्तमाने च कल्पे षण्मनवो गता इत्यादौ तुल्यव्याख्यानं तथा च पुराणकारःपरार्धसंख्यैः कल्पैस्त्रिदिवसाः पौकषः स्मृता इति यद्येवम् ॥२६-२७॥

वि. भा.—कस्य(ब्रह्मणः) कल्पपरार्धे षट् मनवो गताः (व्यतीताः), चतुर्युगत्रि-घनाः(चतुर्युगानां महायुगानां सप्तविंशतिः)गता अर्थात्सप्तविंशतिमितानि महायुगानि व्यतीतानि; कृतादीनि त्रीणि (सत्ययुगादीनि त्रीणि युगचरणमानानि) गतानि शकान्ते (शकनृपस्यान्ते) कलेः (कलियुगादितः) गोऽगैकगुणाः ३१७९ अब्दाः (वर्षाणि) गताः, अतीतमनूनां (गतमनूनां षड्मितानां) आद्यन्तरान्तगतैः (आदिमध्यावसानस्थितैः) सन्धिभिः सार्धं, शकनृपान्ते नवनगशशिमुनि-कृतनवयमनगनन्देन्दवः १९७२९४७१७९ एतावन्तोऽब्दा गता इति। ब्रह्मणो द्वितीयार्धे षड् मनवो व्यतीताः, वर्त्तमानसप्तममनोः सप्तविंशतिसंख्यकानि युगानि व्यतीतानि, अष्टाविंशतितमयुगस्य सत्ययुगादयस्त्रयो युगचरणा व्यतीताः, कलेः शकान्तं यावत् ३१७९ अब्दाः (सौरवर्षाणि) व्यतीताः। एतेषां योगः कियान् भवेदित्यानीयते।

श्लोकोक्त्या ६ मनु+७ मनुसन्धि+२७ युग+युगचरणत्रय+३१७९
=६ मनु+ ७ मनुसं+२७ युग+(युग-कलिच)+३१७९
=६×७१ यु+७×४×४३२००० +२७ युग+(युग-कलिच)+३१७९
=६×७१×४३२०००० +७×४×४३२००० +२७×४३२०००० (४३२०००० -४३२०००)+३१७९
=४२६×४३२०००० +२८×४३२००० +२७×४३२०००० +(४३२०००० -४३२०००)+३१७९

$=१८४०३२०००० + १२०९६००० + ११६६४०००० + ३८८८००० + ३१ ९$
$=१९७२९४७१७९ =$ आचार्यपठिताङ्काः

एतत्कथनस्येदं तात्पर्यं ग्रहादिचारसाधनं शकादेवार्यभटवटेश्वराचार्यौ विहाय सर्वे प्राचीना नवीनाश्चात्रत्याः (भारतीयाः) आचार्याः कृतवन्तः । कलियुगादित ३१७९ एतन्मितवर्षान्ते शकाब्दारम्भ इति गणकसमाजे जनश्रुतिः प्रसिद्धिरस्ति, तेन कलादितः शकान्तं यावत्कियन्ति सौरवर्षाणि गतान्येतत्प्रयोजनमत्यावश्यकमतः पूर्वोक्तानां 'कल्पपरार्धमनव इत्यादीनां योगकरणेन पूर्वोक्ता अङ्का जायन्त एतद्वशेनैवाहर्गणादीनां साधनं भवत्यत एतत्पाठकरणमतीवावश्यकत्वादाचार्येण तेऽङ्काः पठिता इति । भास्कराचार्येणापि 'याताः षड् मनवो युगानि' भमितानीत्यादिनैतदनुरूपमेव कथ्यते ॥ २६-२७ ॥

अब कल्पगत कहते हैं ।

हि. भा.—ब्रह्मा के कल्प के द्वितीयार्ध (परार्ध) में छः मनु गत हो गये । वर्त्तमान मनु के सत्ताइस महायुग बीत गये, अठाइसवें युग के सत्य युगादि तीन युग चरण बीत गये, शकान्त में कलियुगादि से ३१७९ इतने वर्ष बीत गये, गत छः मनुओं के आदि में मध्य में और अन्त में जो मनु सन्धि है उनके साथ, शकान्त में १९७२९४७१७९ इतने सौरवर्ष बीत गये, पूर्व कथित गतमनु-गतमहायुगादियों के योग करने से आचार्य पठिताङ्क आता है या नहीं इसके लिए गणित दिखलाते हैं ।

श्लोकोक्ति के अनुसार ६ मनु + ७मनुसन्धि + २७ युग + कृतादि युगचरणत्रय + ३१७९ = ६मनु + ७मनुसं + २७युग + (युग--कलिचरण) + ३१७९

$=७ \times ७१युग + ७ \times ४ \times ४३२००० + २७ \times ४३२०००० + (४३२०००० - ४३२०००) + ३१७९$

$=४२६युग \times २८ \times ४३२००० + २७ \times ४३२०००० + (४३२०००० - ४३२०००) + ३१७९$

$=४२६ \times ४३२०००० + १२०९६००० + ११६६४०००० + ३८८८००० + ३१७९$

$=१८४०३२०००० + १२०९६००० + ११६६४०००० + ३८८८००० + ३१७९$
$=१९७२९४७१७९ =$ आचार्य पठिताङ्क ।

यहाँ कहने का अभिप्राय यह है कि आर्यभट और वटेश्वराचार्य को छोड़कर जितने भारतीय ज्यौतिषाचार्य हुए हैं उन्होंने ग्रहादि चार साधन शक ही से किये हैं, कलियुगादि से ३१७९ एतन्मितवर्षान्त में शक वर्षारम्भ हुआ यह बात भारतीय गणक समाज में प्रसिद्ध है, इसलिए कलादि से शकान्त तक जितने सौरवर्ष बीते हैं इसकी बहुत आवश्यकता प्रतीत हुई अतः उपर्युक्त गतमनु, मनुसन्धि आदि का योग कर आचार्य ने उपरिलिखित अङ्क पठित

किये हैं, इसके बिना ग्रहगणितादि का साधन हो ही नहीं सकता। भास्कराचार्य भी 'याताः षड्मनवो युगादि भनितानि' इत्यादि से ब्रह्मगुप्तोक्त के अनुरूप ही कहते हैं। इति ॥२६-२७॥

ग्रहनक्षत्रोत्पत्तिर्ब्रह्मदिनादौ दिनक्षये प्रलयः ।
यस्मात्कल्पस्तस्माद् ग्रहगणिते कल्पयाताब्दाः ॥ २८ ॥

वा. भा.—ग्रहाणां नक्षत्राणां च सृष्टिः ब्रह्मदिनादौ कल्पादौ, दिनक्षये प्रलयः कल्पान्ते ग्रहनक्षत्राणां पुनरपि विनाशः, यस्मादेवं तस्मात्कल्प एव ग्रहगणिते उपयुज्यते। न ततोर्वाक् नवाग्रतो ग्रहगणयितुरभावादित्यर्थः। द्वितीयार्धे पाठोऽन्येषां यस्मात्कल्पस्तस्माद् ग्रहगणितं यत एव कालात्कल्पः प्रवृत्तस्तत एव कालात् ग्रहगणितमपि प्रवृत्तमित्यर्थः। कल्पयाताब्दाः इत्युत्तरत्र संबद्धं भविष्यतीति यदुक्तं प्राक्कल्पयाताब्दास्तानाह ॥

वि. भा.—ग्रहाश्च नक्षत्राणि च ग्रहनक्षत्रं तस्योत्पत्तिः सृष्टिः ब्रह्मदिनादौ कल्पादौ भवति। तथा दिनक्षये ब्रह्मदिनावसाने कल्पान्त इत्यर्थः। तेषां ग्रहनक्षत्राणां प्रलयः नाशो भवति। अर्थादेतदुक्तं भवति। कल्पप्रमाणं ब्रह्मणो दिनं भवति। "कल्पो ब्राह्ममहः प्रोक्त" मिति सूर्यसिद्धान्तोक्तेः। तत्प्रमाणा तस्य रात्रिः स्यात्। "सर्वान् पदार्थान् संहृत्य ब्रह्मा शेते" इत्यागमवचसा ब्रह्मा स्वदिनादौ कल्पादौ सृष्टिं रचयति। कल्पावसानेऽर्थाद्धि त्रिकल्पे सर्वान् संगृह्य शेते। अतः सृष्टिकल्प एव ग्रहनक्षत्रादयो वर्तन्ते। कल्पावसाने च सर्वाणि तानि ग्रहनक्षत्राणि तस्मिन् विलीयन्ते। "अव्यक्ताद्व्यक्तयः सर्वाः प्रभवन्त्यहरागमे। राज्यागमे विलीयन्ते तत्रैवाव्यक्तसंज्ञके" इति भगवद्वाक्यम्।

अथ यस्मात्कारणात् कल्पोऽस्त्यत्र कल्पशब्देन दिनकल्प एव विवक्षितः। अस्मिन्नेव ग्रहादीनां सद्भावात्। विद्यमानेषु तेषु ग्रहादिषु तेषां गत्यावगमो भवितुमर्हति। अतएव भास्कराचार्या अपि "यतः सृष्टिरेषां दिनादौ दिनान्ते लयस्तेषु सत्स्वेव तच्चारचिन्ता" इति निजे सिद्धान्तशिरोमणौ प्रोचुः। तस्मात् कारणात् ग्रहगणिते ग्रहादीनां गत्यावगमे कल्पयाताब्दाः साध्यन्त इत्यर्थः। नेयमार्या म० म० सुधाकरद्विवेदिसंकलितपुस्तके समुपलभ्यत इति म० म० मुरलीधरझा लेखादवगम्यते। किन्तु मुनीश्वरेण मरीचिभिधायां शिरोमणेर्निजटीकायां बहादरेण सन्निवेशिता समाहृता च। अतएव अस्यैव दिनकल्पस्य प्रयोजनं भवति नान्यस्येति। ग्रन्थकाराशयः ॥२८॥

हि. भा.—एक कल्प के बराबर ब्रह्मा का एक दिन होता है। ऐसा आगमशास्त्र का मत है। सूर्यसिद्धान्त का वचन है कि "कल्पो ब्राह्ममहः प्रोक्तम्" एक कल्प ब्रह्मा का दिन कहा

जाता है उतनी ही उनकी रात्रि होती है। अपने दिनादि में ब्रह्मा सब ग्रहनक्षत्रों की रचना करते हैं और दिनावसान में अपने सब को संहार करके सोते हैं। ऐसा ही भगवान् का वाक्य है। "अव्यक्ताद्व्यक्तयः सर्वाः प्रभवन्त्यहरागमे। रात्र्यागमे प्रलीयन्ते तत्रैवाव्यक्तसंज्ञके॥" दिन के आरम्भ में अव्यक्तरूप ब्रह्म से यह सब व्यक्त (जगतरूप में) निकलते हैं और रात्र्यारंभ में सब उसी अव्यक्त ब्रह्म में लीन हो जाते हैं। इसलिए यहां आचार्य का कथन है कि दिनकल्प मे ही ग्रहों के रहने का कारण उनकी गति होती है और उसके साधन के लिये यही दिनकल्प से वर्षमान की आवश्यकता है, अन्यकल्पवर्ष की आवश्यकता नहीं है। यह ग्रन्थकार का आशय है॥ २८॥

इदानीं कल्पगतसम्बन्धे आर्यभटमतं कथयति

अधिकः स्मृत्युक्तमनोरार्यभटोक्तश्चतुर्युगेन मनुः।
अधिकं विंशांशयुतैस्त्रिभिर्युगैस्तस्य कल्पगतम्॥ २९॥

वा.भा.—स्मृत्युक्तश्चासौ मनुश्च स्मृत्युक्तमनुः तस्मात् स्मृत्युक्तमनोः सकाशादार्यभटोक्तो मनुरधिकः, क्रियते इत्याह—चतुर्युगेन। यत एवमतः त्रिगुणयुगैः विंशांशसंयुतैः तस्याधिकं कल्पे गतं कल्पगतम्। तद्यथा मनवः षट् द्वासप्तत्या गुणिता ४३२ षण्णां मनूनामेतावन्ति चतुर्युगानि, चतुर्युगसप्तविंशत्या युतानि ४५, एतानि चतुर्युगगुणितानि १९८२८८०००० एतेषु युगपादेषु २०८०००० त्रिगुणः क्षिप्तोऽम्बर-चतुष्क-वेद-यम-रस-संख्या ६२४०००० कलियुगाब्दांश्च गोगैकगुणसंख्यानि क्षिप्त्वा जातोऽब्दराशिःशककालावधिजः कल्पादेरार्यभटमतेन नवनगशशिगुणद्विचन्द्ररसवसुनन्दशीतकराः १९८६१२३१७९ यत उक्तं दशगीतिकासु गतास्ते च मनुयुगछनाच" कल्पादेर्युगपादा च गुरुदिवसाश्च भारतात्पूर्वमिति। तस्य राशेः ब्रह्मोक्तकल्पगतकालस्य नवनगशशिमुनिकृतनवयमागनन्देन्दुसंख्यस्य चान्तरे च कृते, जाता अधिका अब्दाः खत्रयरसमुनिरूपगुणचन्द्राः १३१७६००० एतावन्तोऽब्दाः आर्यभटमतेनाधिका गताश्चतुर्युगत्रयं चतुर्युगविंशांशयुतमेतावानेव कालो भवति, तद्यथा चतुर्युगाब्दा ४३२०००० त्रिगुणाः १२९६०००० चतुर्युगविंशांशाः २१६००० चानेन युता जाता १३१७६००० एतेऽब्दा अधिककालसमाः अत उक्तं अधिकं विंशांशयुतैस्त्रिभिस्तस्य युगैः कल्पगतम्।

वि. भा.—आर्यभटोक्तो मनुः (आर्यभटकथितो मनुः) स्मृत्युक्तमनोः (स्मृतिकथितमनुमानात्) चतुर्युगेन (एकेन महायुगेन) अधिकोऽस्ति, मनुस्मृत्यादिकथितो मनुरेकसप्ततियुगसमः। आर्यभटोक्तमनुर्द्विसप्ततियुगैः समोऽतोऽनयोरन्तरमेकयुगसमोऽधिकः। तस्यार्यभटस्य मते कल्पादौ विंशांशयुतैः त्रिभिर्युगैरधिकं कल्पगतं भवति, द्वयोराचार्ययोः (ब्रह्मगुप्तार्यभटयोः) मतेन यत्कल्पगतं तदन्तरमेकयुगस्य विंशाधिकं युगत्रयं भवतीति॥ २९॥

अत्रोपपत्तिः

आचार्य मतेन सत्ययुगे युगमानम्=$\frac{१२}{४}$ गतषड्मनूनामादिमध्यावसानेषु सप्तसन्धिषु युगमानम्=$\frac{७\times४}{१०\times४}=\frac{२८}{४०}$।

हि. भा.—आर्यभट्ट तथा मनुस्मृति के महायुगों में एक महायुग का अन्तर है। मनुस्मृत्यनुसार ७१ महायुग और आर्यभट्ट के मतानुसार ७२ महायुग होते हैं। दोनों में कल्पादि मानने के दृष्टिकोण का ही अन्तर है ॥२९॥

इदानीं कल्पगतार्कसावनाहर्गणार्धमार्याद्वयमाह

कल्पगताब्दद्वादशघातश्चैत्रादिमासयुक्तोऽधः।
गुणितो युगाधिमासै रविमासाप्ताधिमासयुतः ॥३०॥
त्रिंशद्गुणस्तिथियुतः पृथग्युगावमगुणो युगेन्दुदिनैः।
भक्तः फलावमोनोऽर्कसावनाहर्गणोऽर्कादिः ॥३१॥

वा. भा.—अत्र युगग्रहणेन कल्पो गृह्यते। तेन कल्पादेरारभ्य गता ये रव्याब्दास्तेषां द्वादशानां च घातः परस्परगुणनं ततः स घातश्चैत्रादिमासैर्युतः। वर्तमानकल्पे येऽब्दाः गता नवनगशशिमुनिकृतनवयमागनन्देन्दवः तेऽतीत शककालाब्दाः द्वादशहताः। चैत्रसितादिगतचान्द्रैः मासैर्युताः गतरविमासगणः कल्प्यते कल्पादेरारभ्येत्यर्थः। यद्यपि संक्रान्त्यवधिजो रविमासः, तथापि न कश्चिद्दोषो यस्मादधिमासकावयवा गृह्यन्ते। यद्यपि संक्रान्त्यवधिजो रविः। अहर्गणानयनोऽन्यश्च यद्यधिमासपातकालो दूरे, तन्मासैरप्यन्तरं न भवति। निकटश्चेत्तदामावास्योद्देश एवार्कसंक्रान्तिः तथापि न दोषो यतो मासद्वयात्मक एव समासः तथा चान्द्रमासाः सर्व एवाचार्येण कर्त्तुमारब्धाः तन्मासांतावधित्वेनाधिमासोपलब्धये रवेर्मासगणः परिकल्पितोप्यतो रविमासाश्चान्द्रैर्मासैर्नीयमाना अधिका भवन्ति चन्द्रमासस्याल्पत्वात्। ततो रविमासगणोऽधः पृथक् कार्यः। ततो गुणितो युगाधिमासै रविमासाप्ताधिमासयुत इति त्रैराशिकमत्र यदि कल्परविमासैः कल्परविमासानां संबंधिनोऽधिकमासाः लभ्यन्ते तदैभीरविमासैः तांश्च गतरविमासेषु संयोज्य कल्पादेरारभ्यगतश्चांद्रमासगणो भवति। तन्मासान्तनिरोधेनाधिमासशेषाश्चातीता। नागतयोरधिमासपातकालयोः परिज्ञानं तद्यथा यदिकल्परविमासैः कल्पाधिमासाः लभ्यन्ते तदैकेन रविमासेन किमिति कल्पाधिमासतुल्यं, अधिमासशेषं भवति। ततो द्वितीयं यद्येतदेकस्य शशिमासस्याधिमासशेषं तत्कियन्तस्ते शशिमासा येषामिदमभीष्टमधिमासशेषमित्यत्र भागहारगुणकारयोः कल्परविमासाश्छेदोतस्तुल्यत्वान्नष्टेषु कल्परविमासेषु कल्पाधिमासका भागहारः। एकको गुणकारोऽभीष्टाधिमासशेषस्यफलम्। शशिमासाश्चान्द्रमासान्तावधित्वेन गतस्याधिकपातकालस्य यतोऽनंतरानीताधिमासपातका-

लाद्यारभ्य सर्वे एव मासाश्चान्द्राः प्रदत्ता। सौरकलाना च तेषामधिमासानयनत्रैराशिकसिध्यर्थं प्रागस्माभिः कृता तस्मादुपपन्नं फलं चन्द्रमासा इति, तथाविधमावशेषं कलारविमासेभ्यो विशोध्य रविशेषमधिमासशेषं परिकल्प्यमेव। एषः कालोऽधिमासपातस्य वाचः अथवा रविमासादिनीकृत्यैतत्कर्मक्रियते, तदभीष्टचन्द्रदिनान्तनिरोधेन, गतगम्यकालयोः परिज्ञानं भवति,

वा. भा.—ततो गतचन्द्रमासगणः त्रिंशद्गुणितः तिथियुतः कार्यः तिथिरेव चन्द्रदिनं यस्मादतः कल्पादारभ्य गतातिथगणोऽभीष्टतिथ्यन्तो मध्यमे भवति। चान्द्रोदिनगणः स एव भवति। स च पृथक् कार्यः ततो युगावमगुणैर्गुणितो युगेन्दुदिनैः भक्तः फलावमोनोऽकसावनोऽहर्गणोऽर्कादिरिति, अत्र त्रैराशिकं यदि कलाचन्द्रदिनानां सावनीक्रियमाणानां कलावमतुल्यानि दिनानि पूर्यन्ते। तद्गत चन्द्रदिनानां कियन्तीति फलं गतचन्द्रदिनानां सबन्धीन्यवमदिनानि चागतचन्द्रदिनगणात् संशोध्य कलागतरविसावनदिवसगणो भवति। लंकामध्यार्कोदयावधौ यस्मात्सावनचान्द्रमानयोरन्तरमवमानि स चाहर्गणोऽर्कादिरिति। यतोऽर्कवारे कलारम्भः। अवमशेषाच्च गतागतोन रात्रपातकालयो परिज्ञानं "तद्यथा" यदि कल्पचन्द्रदिनैः कलावमानि लभ्यन्ते, तदैकेन चन्द्रदिनेन किमिति.....(कलोन) योन रात्रि तुल्यं.. (अव) मशेषं लभ्यते, द्वितीयत्रैराशिकं यद्येतत् सावनमवमशेष...दिनस्य कि...ति तानि चन्द्रदिनानि येषामिदमभीष्टदैवसिकमवमशेषं सावनमित्यत्र भागहारगुणकारयोः तुल्यच्छेदः। कलादिनचन्द्रप्रमाणकः तस्मिन्नष्टोऽभीष्टावमशेषस्यैककागुणकारः कलोनरात्राणि भागहारः फलं चन्द्रदिनान्यतीतस्योनरात्रपातस्य, तदवमशेषं कलाचन्द्रदिनेभ्यो विशोध्यावशेषमवमावशेषं परिकल्प्योक्तवदेष्य कालः साध्य इति, अधिमासपातकाले मासद्वयेऽपि दत्ते एक एव मासो भवति। तिथिद्वयेऽपि दत्ते एक एव वारो लभ्यते इति। अथैवं ज्ञातुमिच्छति। कियद्भिर्दिनैरधिमासः पततीति। तद्यथा कलाधिमासैः कलारविदिनानि भवन्ति तदैकेनाधिमासेन कियन्तीति फलं रविदिनानि भवन्ति। तदैकेनाधिमासेन कियन्तोति फलं रविमासाः षड्गनन्दरव्यास्तदंशाश्च पंचशून्याश्विनवगुणचन्द्राः कलाधिमासच्छेदः $\frac{१३३२०००००}{१५३३०००००}$ अथवा यदि कलाधिमासैः कला चन्द्रदिनानि भवन्ति, तदैकेनाधिमासकेन किमिति फलं चन्द्रदिवसाः रसशून्यखचन्द्राः तदंशास्तुल्या एव १००६ $\frac{१३३२०००००}{१५३३०००००}$ अथवा यदि कलाधिमासैः कलासावनदिवसा भवन्ति तदैकेनाधिमासेन कियन्तः फलं सावनदिवसाः खनन्दानवसख्यास्तदंशाश्च खचतुष्टयशरचेदनवकृतेषवः छेदः स एव $\frac{५४६४५०००००}{१५३३०००००}$ एतेषु प्रकारेष्वधिमासोन्तर्गतस्तिष्ठतीति। अथवाऽवमशेषेतद्यथा—यदि कलोनरात्रः कलाचन्द्रदिनानि भवन्ति, तदैकेनोनरात्रेण कियन्तीति फलं चन्द्रदिनानि त्रिरससंख्यानि तदंशाश्च खचतुष्टयशरगुणवसुनन्दगद्वियमाः $\frac{३२७६८३५००००}{२८०८२५५०००००}$ अत्र छेदस्थं शानां चापवर्त्तनं वसुचन्द्राष्टेन्दुगुणयमखवसुद्वियमैः कृत्वा जातौ राशी उपरि दशाधः एकादशत्रिषष्टेरधः कृत्वा $\frac{६३}{११}$ दर्शनम्। अत्र

सावर्णित जातमुपरि त्रिखमुनयोऽधः एकादश । अत्रेतज्जातं त्रिखमुनिसंख्यं चान्द्रदिनैरेकादशावमानि । भवन्तीति यच्चापवर्तने क्रियमाणेतरमत्र सूक्ष्मं तदर्थमाचार्येण खण्डखाद्यवमुनिबद्धम् । ऋतिथिरुद्रैरवाप्तेन फलेनैकादशगुणश्चान्द्रदिनगणः ऊनः क्रियते इति. अथवा यदि कल्पोनरात्रैः कल्पसावनदिनानि भवन्ति तदेकेनोनरात्रेण किमिति फलं सावनदिवसा द्विरूपसंख्या । तदंशाश्च खतुष्टयशरगुणवसुनन्दागद्वियमाः ३३७६८५५४०००० अत्रापि तेनैवापवर्तनापवर्त्तच्छेदांशान् द्विपञ्च्या सहस्त्रवर्णने कृत जातम्, द्विनवरससख्यैः सावनैरेकादशावमानि भवन्तीति ।

वि. भा.—कल्पाद्यो गताब्दः शकनृपान्ते नवनगशशिमुनिकृत नवयमनगनन्देन्दु १९७२९४७०७९ संख्यकस्तस्य द्वादशस्य च घातो गतचत्रादिमासैर्युतः पृथक् स्थापितः, अधः स्थो राशिर्युगपठिताधिमासैर्गुणितोयुगपठितरविमासै (सौरमासैः) र्भक्तो लब्धाधिमासैः पृथक् स्थापितो युतः स त्रिंशद् गुणितः (शुक्लप्रतिपदादिगततिथिभि)र्युतः स पृथक् स्थापितः । अधः स्थो राशिर्युगपठितावमदिनैर्गुणितो युगपठितचान्द्रदिनभक्तो लब्धावमैरुपरिस्थोराशिरूनः (वर्जितः) तदा रविसावनाहर्गणो भवेत्. । तस्याहर्गणस्यार्कनारभ्य प्रवृत्तिर्भवत्यत एवार्कादिः कथ्यतेऽऽचार्येणेति ।।३०-३१।।

अत्रोपपत्तिः

अथ मध्यम ग्रहानयनं विवक्षुस्तत्रादौ तावत्तदुपयुक्तं कल्पादितः सावनाहर्गणं साधयत्याचार्यः ।

प्रागानीतेन प्रकारेणेष्टशकान्तं यावत्सौरवर्षमानान्यानीय द्वादशगुणानि-अभीष्टशकान्ते सौरा मासा भवन्ति । तत्र चैत्रामान्ततोऽभीष्टामान्तावधि ये चान्द्रा मासास्तत्समाः सौरा मासा एव क्षेप्यास्तदेष्टमासीयसंक्रान्ति यावत्कल्पादितः सौरमासाः स्युः । अत्र मध्यममेषसंक्रान्तिज्ञानाभावत् चैत्रामान्ततो मासग्रहणं विहितम् । अन्यविधानस्य प्रापकाभावात् । अयमेव कल्पादितोऽभीष्टमासीय संक्रान्तपर्यन्तमभीष्टः सौरमासगणः स्यात् ततस्त्रैराशिकेनैतत्सम्बन्धीयाधिमासमानं सशेषं साध्यते ।

$$\text{तथाहि} \quad \frac{\text{कल्प अधिमास} \times \text{इष्टसौरमास}}{\text{कल्पसौरमास}} = \text{इष्टाधिमास} + \frac{\text{अशे}}{\text{कसौमा.}}$$

$$\therefore \text{अतोऽभीष्टसंक्रान्तिकाले चान्द्रमासाः सावयवाः} = \text{इसौ} + \text{इअमा} + \frac{\text{अशे}}{\text{कसौ}}$$

अत्रा $\frac{\text{अशे}}{\text{कसौ}}$ मं चान्द्रात्मकोऽधिशेषः । स चामान्तसंक्रान्त्यन्तरं स्यात् ।

"दर्शाग्रतः संक्रमकालतः प्राक् सदैव तिष्ठत्यधिमासशेष" मिति भास्करोक्तेः। अत्राधिशेषखण्डस्य विशोधनेन तिथ्यन्तकालिकः स्यात् अतोऽभीष्टामान्तीयचांद्रमासगणः इसौ+इअमा अयं त्रिंशद्गुणोऽभीष्टतिथियुतः कार्यस्तदा तिथ्यन्ते चान्द्राहर्गणः स्यात्। अतोऽनुपातेनैतच्चान्द्रसम्बन्धीन्यवमान्यानीयात्रविशोधनेन तिथ्यन्ते सावयवः सावनाहर्गणः स्यात्।

$$\therefore \frac{\text{कअवम}\times\text{इचा}}{\text{कचा}} = \text{अवम} + \frac{\text{अशे}}{\text{कचा}}$$

$$\therefore \text{तिथ्यन्ते सावनाहर्गणः सावयवः} = \text{इचा} - \text{अवम} - \frac{\text{अवशे}}{\text{कचा}}$$ अत्रावमशेषस्य तिथ्यन्तोदयान्तरे वर्तमानत्वात् अवमशेषं योज्यते चेत्तदौदयिकः सावनाहर्गणो निरवयवः। अहर्गणः =इचा—अवम।

अत्र परिदर्शित समीकरण दर्शनेन विज्ञायते यत्केवलाधिमासैः सहितोऽभीष्टसंक्रान्तिकालिकः सौरमासगणस्तिथ्यन्ते चान्द्रमासगणः स्यात्। एवं च भवेलावमदिनरहितस्तिथ्यन्तकालिकश्चान्द्राहर्गणस्तिथ्यन्ताव्यवहितोत्तरौदयिकः सावनाहर्गणो जातः। अत एवाधिशेषावमशेषे त्यक्ते।

अथैतत्प्रतीत्यर्थं विचार्यते

क	द	च	व	अ	ति	उ	सं	उ

क=कल्पादिः।
च=चैत्रामान्तः।
व=मध्यमः सौरवर्षान्तः।
ति=तिथ्यन्तोऽभीष्टमासीयः।
उ=उदयकालः।
सं=अभीष्टसंक्रान्तिः।
अ=अभीष्टमासीयामान्तः।

क स्थानात् व स्थानपर्यन्तं सौरवर्षगणं समाननीय द्वादशगुणनेन तत्रत्यः सौरमासगणः स्यात्। अत्र चैत्रामान्ततो अ अमान्तावधि यावन्तश्चान्द्रमासास्तान् सौरान् प्रकल्प्य प्रागानीतसौरमासगणे क्षेप्याः। तथा कृते सति अभीष्टसंक्रान्तिबिन्दौ सौरमासगणः सिद्ध्यति। अत्र अ अमान्तकालिक चान्द्रमाससाधनार्थं तत्र तावत् क स्थानमारभ्य प्रतिसौरमाससंख्याकसमचान्द्रमासदानेन पूर्वमेव कुत्राप्यमान्ते तत्पर्यवसान भवेत्। संख्यया सौरमासगणस्याल्पत्वात्। कल्प्यते द बिन्दौ तत्पूर्त्तिर्जाता। अत्र द, अ बिन्द्वोरन्तर्गताश्चान्द्रात्मका अधिमासा निरवयवाः। अ, स, बिन्द्वोरन्तरे तदधिशेषमानम्। अतः क, द, बिन्द्वोरन्तर्गते सौरमासग-

णसंख्याकसमे चान्द्रमासगणे अ, द, बिन्द्वोरन्तर्गतचान्द्राधिमासगणस्य योगेन अ बिन्दौ चान्द्रमासगणो भवतीति स्फुटमेव। ततो दिनीकरणेन अ बिन्दौ चान्द्राहर्गणः स्यात्। अत्र तिथिसंख्याया योगेन ति बिन्दौ चान्द्राहर्गणः। अत्रापि ति स्थानीयचान्द्रदिनसंख्याक समं सावनमानं प्रकल्प्य क स्थानात् तादृश-प्रतिसावनानां समायोगेन उ स्थानात् क्वचिदग्रगतो भवेत्। सावनसंख्यापेक्षया चान्द्रदिनसंख्यायाः अधिकत्वात्। स च दानाग्रबिन्दुः उ कल्पितः। अत्र उ उ बिन्द्वन्तर्गतानि दिनानि सावनात्मकानीति स्वरूपेणैव स्फुटम्। तैः केवलदिनैः सावनसंख्यासमैश्चान्द्रैर्विशोध्यते तदा उ बिन्दौ सावनाहर्गणः स्यात्। अत्राधिशेषावमशेषे न गृहीते। अतो भास्करेण "ध्रुवटिकादिकमत्र न गृह्यते" इत्युक्तम्।

एवं च सौराच्चान्द्रावगमेऽधिमासाश्चान्द्रात्मकास्तथाधिशेषं च चान्द्रात्मकमिति। तथैव चान्द्रात्सावनागमेऽवमानि सावनात्मकानि। तथा तच्छेषं च सावनात्मकमित्यपि सिद्ध्यति।

एवमहर्गणात्कल्पगताब्दावगमे सावनदिनगणादनुपातेन यान्यवमदिनानि तानि चान्द्रजातीयानि भवन्ति। शेषं च तज्जातीयमेव। तत्तु प्रागानीतसावनात्मकावमशेषस्य सममेव स्यात्। एवमेव चान्द्रात्सौरावगमे येऽधिमासास्ते सौरजातीयास्तच्छेषमपि तथैव। तदपि प्रागानीतचान्द्राधिशेषेण सममवैत्यनन्तरप्रदर्शितोपपत्त्या स्पष्टमेव गणितपटूनाम्। अतएवाधिमासस्य चान्द्रत्वे सौरत्वे चाधिशेषं तुल्यमेव स्यात्। किन्त्वत्र सौरदिनानि हारः अन्यत्र चान्द्रदिनानीति स्वं 'सौरेभ्यः साधितास्ते चे"दित्यादिगोलीय ग्रन्थेन प्रपञ्चितं भास्कराचार्यैः।

अब अहर्गणानयन कहते हैं

हि. भा.—कल्पादि से जो गत वर्ष संख्या १९७२९४७१७९ है उसको बारह से गुणा देना, गत चैत्रादि मास संख्या जोड़कर जो हो उसको दो स्थान में स्थापित करना। एक स्थान में युग पठित अधिमास संख्या से गुणाकर युग पठित सौर मास संख्या से भाग देने से जो लब्धाधिमास हो उसको द्वितीय स्थान में स्थापित फल में जोड़कर जो हो उसको तीस से गुणा कर शुक्ल प्रतिपदादि से गत तिथि संख्या जोड़कर जो फल हो उसको दो स्थान में स्थापित करना, एक स्थान में उसको युगपठित अवमदिनों से गुणा कर युगपठित चान्द्र दिनों से भाग देने से जो लब्धि हो उसको (गतावमदिन) द्वितीय स्थान स्थित पूर्व फल में घटाने से रविसावनाहर्गण होता है; रवि से आरम्भ कर अहर्गण की प्रवृत्ति होती है। इसीलिये आचार्य पद्य में 'अर्कादिः' कहते हैं ॥ ३०-३१ ॥

उपपत्ति

शाकवर्ष-स्वष्टमास तिथि इन सबों के ज्ञान से अहर्गण का आनयन करते हैं। 'कल्प-पूर्वे मनवः षट्कस्य गताः' इत्यादि आचार्योक्त श्लोक से सृष्ट्यादि से गतवर्षान्त तक गत वर्ष संख्या विदित है, सृष्ट्यादि से चान्द्रवर्षतुल्य दान देने से जो दानान्त बिन्दु होता है वह किसी चैत्रामान्त बिन्दु ही पर होता है; फिर गत चैत्रामान्त से इष्टतिथ्यन्त पर्यन्त जो चान्द्रदिन संख्या है तत्संख्यक सौर दिन संख्या (इष्ट तिथि संख्यक सौर दिन संख्या) गत मेषादि से दान दिया वह सौरदानान्तबिन्दु इष्ट तिथ्यन्त से आगे होता है क्योंकि गतमेषादि बिन्दु गतचैत्रामान्त से आगे है फिर दानान्त बिन्दु रूप चान्द्रवर्षान्त से (दानान्त चैत्रामान्त से) इष्ट तिथि तुल्य चान्द्र दिन दिया तब वह दानान्तबिन्दु गतवर्षान्त से पहले ही कहीं इष्ट तिथ्यन्तसमबिन्दु ही में होता है. क्योंकि. चान्द्रदिन < सौरदि. तब सृष्ट्यादि से सौरवर्षादि संख्या और चान्द्रवर्षादि संख्या-समान ही दान दिया क्योंकि सृष्ट्यादि से गतवर्षान्त पर्यन्त सौर वर्ष संख्या जो है तत्संख्यक ही सृष्ट्यादि से चान्द्रवर्ष दिये उससे आगे फिर गत मेषादि से सौरान्तपर्यन्त इष्टतिथितुल्य सौर संख्या जो है उतने ही दानान्त चान्द्रवर्षान्त से इष्ट तिथि तुल्य चान्द्र दिन दिया, अतः इन दोनों (सृष्ट्यादि से सौरदानान्त बिन्दुपर्यन्त चान्द्रवर्षादि संख्या जो है तत्संख्यक ही सृष्ट्यादि से इष्टतिथ्यन्त सम दानान्त बिन्दु पर्यन्त चान्द्रवर्षादि संख्या) के अन्तर (इष्टतिथ्यन्तसमदानान्तबिन्दु से सौरान्तबिन्दुपर्यन्त) चान्द्रजातीय सावयवाधिमास है, उनमें इष्टतिथ्यन्तसमदानान्त बिन्दु से इष्टतिथ्यन्त पर्यन्त पूरा अधिमास है उसके बाद. (इष्टतिथ्यन्त से सौरान्त तक) अधिशेष है। सौरान्त में जो चान्द्र है उनका और सौरसंख्यक चान्द्र का अन्तर सावयवाधिमास जो है वही सौरान्त में जो सौर है उनका और उस सौरान्तःपाति चान्द्रसंख्यक सौर का अन्तर है, दोनों में संख्या की तुल्यता ही है किन्तु चान्द्रात्मक अधिमास सौरान्त से पहले और सौरात्मक अधिमास आगे होता है। इन दोनों में यहाँ पहला ही अधिमास लेना चाहिये। इसलिए उसका आनयन करते हैं। $\frac{\text{कल्पाधिमास} \times \text{इष्टसौर}}{\text{कल्पसौर}} = \text{गताधिमास} + \frac{\text{अधिशेष}}{\text{कल्पसौ}}$ इस तरह करके यहां अधिशेष को त्याग देना क्योंकि इष्ट तिथ्यन्त से सौरान्त तक अधिशेष ही है, इष्टाधिमाससंख्या (गताधिमाससंख्या) को तीस से गुणकर जो हो गताधिदिन उसमें इष्टसौर दिन संख्या जोड़ने से जो इष्ट तिथ्यन्त में चान्द्राहर्गण होता है। तब कल्पचान्द्रदिन में कल्पावमदिन पाते हैं तो इष्टचान्द्रदिन (आनीतचान्द्राहर्गण) में क्या इस अनुपात से अवमशेष सहित गतावमदिन आता है $\frac{\text{कल्पावम} \times \text{इष्टचान्द्रदिन}}{\text{कल्पचान्द्रदिन}} = \text{गतावमदिन} + \frac{\text{अवमशेष}}{\text{कल्पचान्द्रदिन}}$ इन अवमशेष सहित गतावमदिन को चान्द्राहर्गण में घटाने से तिथ्यन्त में सावनाहर्गण होता है चान्द्राहर्गण-गतावमदिन $-\frac{\text{अवमशेष}}{\text{कल्पचान्द्रदिन}}$ परन्तु तिथ्यन्त और सूर्योदय के मध्य में अवमशेष है इसलिए तिथ्यन्त कालिक सावनाहर्गण में अवमशेष को जोड़ने से सूर्योदयकालिक सावनाहर्गण होता है, चान्द्राहर्गण $-$ गतावमदिन $-\frac{\text{अवमशेष}}{\text{कल्पचान्द्रदिन}} + \frac{\text{अवमशेष}}{\text{कल्पचान्द्रदिन}} =$ चान्द्राहर्गण—

गतावमदि=तिथ्यन्तकालिक सावनाहर्गण+$\frac{\text{अवमशे}}{\text{कल्पचांदि}}$ =सूर्योदयकालिक सावनाहर्गण।

अहर्गण के दिन निश्चित रहने के कारण अहर्गण में सात से भाग देने से जो शेष रहता है तत्तुल्य रव्यादि दिन होता है क्योंकि कल्पादि में रविवार दिन था इसलिए रवि ही से गणना करनी चाहिये, इससे आचार्योक्त उपपन्न हुआ। आनीत अहर्गण मध्यम सावनाहर्गण है, क्योंकि इसके आनयन में अनुपात से काम लिया गया है, सिद्धान्तशिरोमणि में भास्कराचार्य ने भी "कथितकल्पगतोऽर्कसमागणः" इत्यादि से इसी तरह आनयन किये हैं, इति ॥३०-३१॥

इदानीं ग्रहमन्दशीघ्रपातानां मध्यमानां मध्यमानयनमाह

इष्टग्रहभगणगुणादहर्गणात्कल्पसावनघ्नहृतात्।
भगणादिफलं मध्यो लङ्कायां भास्करौदयिकः ॥३२॥

वा. भा.—ग्रहग्रहणेन ग्रहंमन्दशीघ्रपाताः गृह्यन्ते। तेनायमर्थः इष्टस्य ग्रहादेः भगणैरिष्टाहर्गणं सुगणय्य कल्परविसावनदिवसैर्विभजेत्। लब्धं भगणादिरिष्टो ग्रहादिरत्र त्रैराशिकं यदि कल्पाहर्गणेनेष्टकल्पानां भगणाः लभ्यन्ते तदिष्टाहर्गणेन किमिति भगणादि स एव लभ्यते। तत्र भगणांस्त्यक्त्वा राश्यादिकः स्थाप्यते स्फुटीकरणादिषु प्रश्नेषु पुनः सभगणश्च स्थाप्यते। शेषमतिप्रसिद्धत्वात् नोदाहृतं स च भगणादिको मध्यो भवति। कक्षामण्डलग इत्यर्थः। लंकायां भास्करौदयिकश्च भवति। अन्यत्र देशे यतो देशान्तरकर्मणा स्वौदयिको भविष्यतीति। स्वमध्यभुक्तिश्च ग्रहवदेकाहर्गणेन स्वभगणैश्च साध्या, मया च रव्यादिनां सिद्धा एव भुक्तयो लिख्यन्ते। तत्पराशेषैः सह बलभद्रकृतैः श्लोकैः तद्यथा—

एकोनषष्टिरष्टौ च दश चेति रवेर्गतिः।
खाष्टगोनन्दनन्दर्त्तुसंख्यं शेषश्च तत्परः॥
चन्द्ररामारसयमावसुपक्षाः कुजस्य तु।
द्विपञ्चरसशून्याश्विपक्षाः शेषश्च तत्परः॥
अर्थाश्विपक्षादशनाक्षशीघ्रस्याष्टा रात्रियाः।
वेदतत्वेन्दुखनवसंख्यशेषश्च तत्परः॥
वेदा नवार्थं नव च गतिजीवस्य कीर्तिता।
पञ्चागगोनवागाश्विसंखशेषश्च तत्परः॥
सितशीघ्रस्य षट् नन्दाः पर्वता कृतसागरा।
पक्षार्थाग्निरसाब्धीश संख्याशेषश्च तत्परः।
रसाखवेदा रामार्थाश्चन्द्रस्योच्चगतिः स्मृता।
गुणाष्टगोर्थाग्निवसुचन्द्राः शेषश्च तत्परः॥
चन्द्रपातस्य दहना दिशश्चाष्टार्णवास्तथा।
अष्टर्त्तुनवपक्षार्थरसाः शेषश्च तत्परः॥
शेषाणां तत्पराः शेषाः क्रमशश्चार्कमन्दतः।

खाष्टखर्त्तुकृताः शेषाः सूर्योच्चस्य तु तत्पराः ॥
भौमोच्चस्याश्विदहनाः शून्यवस्वश्विनः स्मृताः ।
बुधोच्चस्याश्विशैलाष्टपंचरामाः प्रकीर्त्तिताः ॥
खाष्टशून्याष्टवसवो जीवोच्चस्य प्रकीर्त्तिताः ।
शुक्रोच्चस्य तु वस्वष्टरसपक्षरसाः स्मृताः ।
ऋत्वग्निनन्ददहनाः सौरोच्चस्य प्रकीर्त्तिताः ॥
भौमपातस्य च तथा दशनर्तुशराश्विनः ।
बुधपातस्य षट् चन्द्रशून्यशून्यषवः स्मृताः ॥
जीवपातस्य वस्वश्विशून्यषट्काः प्रकीर्त्तिताः ।
शुक्रपातस्य वस्वश्विसप्तार्थवसवः स्मृताः ॥
सौरपातस्य वेदर्त्तुशून्यषट्पंचकाः स्मृताः ।
छेदस्तुतत्पराशेषे सर्वेषाञ्च निगद्यते ।
पंचदेवखवस्वब्धिनवप्रालेयरश्मयः ।
सर्वेषां तत्पराशेषाणां भूदिनानां चापवर्त्तकः ॥ ८१००० ।

वि. भा.—इष्टग्रहकल्पपठितभगणगुणितादहर्गणात्कल्पपठितसावनद्यु (कल्पकुदिनं) भक्तात्फलं भगणादि मध्यमो ग्रहो लङ्कासूर्योदयकालिको भवतीति ॥ ३२ ॥

अत्रोपपत्तिः

$\frac{\text{कल्पग्रहभगण} \times \text{अहर्गण}}{\text{कल्पकुदिन}} = \text{ग्रहभ} + \frac{\text{भगणादेः}}{\text{ककु.}}$ प्रतिदिनजगतिकलोत्पन्नासु वैषम्यमूलकप्रतिकुदिनवैषम्येनैतादृशानुपाताभावादेकवर्षान्तःपातिस्पष्टकुदिनानामेकत्रितानां कृतस्वसंख्यकसमखण्डानां मध्यसावनमेवं स्पष्टगतिकलाभ्यो मध्यगतिकलेति च कृत्वैकस्तादृशो ग्रहश्चेत्कल्पितो भवेद्यस्य कुदिनं मध्यमसावनं तद्गतिकला च मध्यमगतिकला भवेत्तदा तत्कुदिनेनैवमनुपातः स्यात् । परं नायं क्रान्तिवृत्ते चालितो भवेत्तत्र समचापजासूनामप्यसमत्वात् $\frac{\text{अथ वर्षान्तःपातिस्पष्टसावनयोग}}{\text{वर्षान्तःपातिस्पष्टसावनसं}} =$ मध्यमसावन, वर्षान्तःपातिस्पष्टसावनयोगसम्बन्धि नाक्षत्रम्=वर्षान्तः पातिस्पष्टसावनसंख्या+१ नाक्षत्र

अतः मध्यमसावन$= \frac{\text{वर्षान्तःपातिस्पष्ट सावनसं ना}+\text{१ ना}}{\text{वर्षान्तःपातिस्पष्टसावनसं}}$

$= \text{१ ना} + \frac{\text{१ ना}}{\text{वर्षान्तः पातिस्पसा}} = \text{१ ना} + \frac{\text{२१६०० असु}}{\text{वर्षान्तःपातिस्पष्टसावनसं}}$

परञ्च $\frac{\text{२१६०० कला}}{\text{वर्षान्तः पातिस्पष्टसावनसं}} =$ मध्यगतिक; अतः मध्यमगतिकलासमासु

$=\frac{\text{२१६०० असु}}{\text{वर्षान्तः पातिस्पष्टसावनसं}}$ अतः मध्यसा = १ ना + मध्यमगतिकलासमासु, परञ्च कलातुल्या असवो नाडीमण्डल एवातो नाडीमण्डल एवोक्तग्रहश्चालनीय इति सिद्धः। अतः स्वस्वभगणात्पूर्वोक्तानुपातेन नाडीमण्डलीय मध्यमार्कस्य काल्पनिकत्वात्कल्पिते क्रान्तिवृत्तीयमध्यमार्के आगतोऽयं मध्यमग्रह इत्यत्र 'दशशिरः पुरि मध्यमभास्करे क्षितिजसन्निधिगे इति वदति भास्करः' गोलसन्धेः प्रागाशाभिमुखं चालितयोः समगतिवेगवतोरुभयवृत्तीयमध्यमार्कयोर्यदा नाडीवृत्तीयमध्यमार्को लङ्कायां क्षितिजस्थस्तदा क्रान्तिवृत्तीयमध्यमार्कः पदवशेन क्षितिजोऽर्ध्वेऽधश्च कोटिकर्णान्तरस्थः, (नाडीक्रान्तिध्रुवप्रोतवृत्तैरुत्पन्नजात्यत्रिभुजीयकोटिकर्णयोरन्तरस्थः स्यादिति) अत्रोक्त ग्रहोदययोरन्तरमुदयान्तरं परममप्यल्पमेवातः क्षितिजसन्निधिग इति कथ्यते भास्करेण, परं ब्रह्मगुप्तेन तदन्तरं (उदयान्तरं) शून्यं मत्वा "लङ्कायां भास्करौदयिकः" कथ्यते, अत्र भास्करेण यदुदयान्तरं स्वीकृत्य 'क्षितिजसन्निधिगे' यत्कथ्यते तद्ब्रह्मगुप्तोक्ता- 'लङ्कायां भास्करौदयिकः' पेक्षया समीचीन इति मध्यस्थबुद्धया विवेचनीयं सुधीभिरिति।

अधुना प्रसङ्गादुदयान्तरसम्बन्धे किञ्चिद्विचार्यते

क्रान्तिवृत्त यत्र मध्यमरविस्तदुपरिगतं ध्रुवप्रोतवृत्तं (निरक्षक्षितिजं) यत्र नाडीवृत्ते लगति तस्माद् गोलसन्धि यावन्मध्यमरविगतिकलोत्पन्नासवो विषुवांशा वा, एतन्मध्यमरविगतिकलोत्पन्नासुप्रमाणं नाक्षत्रषष्टिघटिकया युक्तं तदा स्पष्टसावनदिनं भवेत् तथा गोलसन्धि केन्द्रं मत्वा क्रान्तिवृत्तीयमध्यमरविभुजांशव्यासार्धवृत्तं यत्र नाडीवृत्ते लगति ततो गोलसन्धि यावन्मध्यमरविगतिकलातुल्यासवो मध्यमरविभुजांशा वा, एतन्मध्यमरविगतितुल्यासुयुतं नाक्षत्रषष्टिप्रमाणं मध्यमसावनदिनं भवति, अनयोः स्पष्टसावनमध्यमसावनदिनयोरन्तरम् = ६० + मध्यमरविगतिकलातुल्यासु ~ (६० + मध्यमरविगतिकलोत्पन्नासु) = मध्यमरविगतिकलातुल्यासु ~ मध्यमरविगतिकलोत्पन्नासु = उदयान्तरासु, एतत्सम्बन्धिग्रहगतिप्रमाणमानीयते, यथा यद्यहोरात्रासुभिर्ग्रहगतिकला लभ्यन्ते तदोदयान्तरासुभिः किमित्यनुपातेनोदयान्तरसम्बन्धिनी ग्रहगतिरागच्छति तत्स्वरूपम् $=\frac{\text{ग्रहगतिकला} \times \text{उदयान्तरासु}}{\text{अहोरात्रासु}}$ अनया गत्या रहिताः सहिताश्चाहर्गणोत्पन्ना ग्रहाः (नाडीवृत्तीयमध्यमार्कोदयकालिकग्रहाः) क्रान्तिवृत्तीयमध्यमार्कोदयकालिका (निरक्षक्षितिजस्थाः) ग्रहा भवन्त्येतावता "मध्यार्कभुक्ता असवो निरक्षे ये ये च मध्यार्ककलासमाना इत्यादि" भास्करोक्तमुपपद्यते। परमुदयान्तरासुमध्येऽपि ग्रहस्य कापि गतिर्भविष्यति तद्ग्रहणं भास्करेण न कृतमतः पूर्वोक्तयुक्त्योदयान्तरासुसम्बन्धिग्रहगत्यानयनं विधाय तत्संस्कृतोऽहर्गणोत्पन्नमध्यमार्कोदयकालिकग्रहो नहि वास्तविको निरक्षक्षितिजे क्रान्तिवृत्तीयमध्यमार्कोदयकालिकग्रहो भवितुमर्हत्यतो भास्करोक्तमानयनं न समीचीनमित्यतो वास्तवं तदानयनं प्रोच्यते।

अथ कल्प्यते वास्तवमुदयान्तरप्रमाणम्=य, एतेनैवौदयिको ग्रहो भवितुमर्हति, अस्मिन्नुदयान्तरकाले ग्रहगतिर्या भवेत्तदुत्पन्नासुभिः संस्कृतं भास्करोक्तमुदयान्तरं वास्तवमुदयान्तरं भवेद्यथा यद्यहोरात्रासुभिर्ग्रहगतिकला लभ्यन्ते तदा वास्तवोदयान्तरा (य) सुभिः किं समागच्छति तत्सम्बन्धिनी ग्रहगतिकला तत्स्वरूपम् $= \frac{\text{ग्रहगतिक} \times \text{य}}{\text{अहोरात्रासु}} = १$ असुजग्रहगति×य, ततोऽनुपातो यदि राशिकलाभिरष्टादशशतकलाभिस्तद्राश्युदयासवो लभ्यन्ते तदा वास्तवोदयान्तरासुसम्बन्धिग्रहगतिकलाभिः किं समागच्छन्ति तत्सम्बन्धिनोऽसवः=

$$\frac{\text{राश्युदय} \times \text{वास्तवोदयान्तरासु सं ग्रहगतिकला}}{१८००}$$

$$= \frac{\text{राश्युदय} \times \text{ग्रहगतिकला} \times \text{य}}{१८०० \times \text{अहोरात्रासु}} = १ \text{ असुजग्रहगति} \times \text{य} \times १ \text{ कलोत्पन्नासु} \ldots (१)$$

$$\times \frac{\text{राश्युदय} \times १}{१८००} = १ \text{ कलोत्पन्नासु}$$

(१) एतेन संस्कृतं भास्करोक्तमुदयान्तरं वास्तवमुदयान्तरं भवेत्
अतः भास्करोक्तोदयान्तर∓१ असुजग्रहगति×य×१ कलोत्पन्नासु=वास्तवोदयान्तर=य

=पूर्वोदयान्तर∓१ असुजग्रहगति×य×१ कलोत्पन्नासु=य

समशोधनेन

य∓१ असुजग्रहग×य×१ कलोत्पन्नासु=पूर्वोदयान्तर

=य (१∓१ असुजग्रहग×१ कलोत्पन्नासु)=पूर्वोदयान्तर

$$\text{अतः} \frac{\text{पूर्वोदयान्तर}}{१ \pm १ \text{ असुजग्रहग} \times १ \text{कलोत्पन्नासु}} = \text{य}$$

एतावता "एकासुजेनगतिसंगुणितैकलिप्तोत्पन्नासुराश्युदययुक्तविहीनितेन रूपेण पूर्वमुदयान्तरमत्र भक्तं स्वर्णं ग्रहे युगयुजोः पदयोः क्रमेण"

म० म० पण्डितसुधाकरद्विवेदिसूत्रमुपपद्यते

अत्रेनगतिः=सूर्यगतिः, सूर्यसम्बन्धेनैव वास्तवोदयान्तरसाधनं प्रदर्शितमस्ति द्विवेदिमहोदयेन अन्येषां ग्रहाणामपि स्वस्वगतिसम्बन्धेन तथैव तत्साधनं भवेद्ययोपरि प्रदर्शितमस्तीति।

अधुनोदयान्तरस्य परमत्वं कुत्र भवेदिति विचार्यते

भुजांशविषुवांशयोरन्तरमुदयान्तरमित्युदयान्तरस्वरूपदर्शनात्स्फुटमस्त्यत उदयान्तरज्या=ज्या (भुजांश—विषुवांश) चापयोरिष्टयोरित्यादिना

$$\frac{\text{ज्याभु}\times\text{कोज्यावि}-\text{ज्यावि}\times\text{कोज्याभु}}{\text{त्रि.}}=\text{उदयान्तरज्या}=\text{ज्याउ},$$

$$\text{यतः}\ \frac{\text{कोज्याभु}\times\text{त्रि}}{\text{द्यु}}=\text{कोज्यावि.}$$

$$\text{तथा}\ \frac{\text{पद्यु}\times\text{ज्याभु}}{\text{द्यु}}=\text{ज्यावि, अत उदयान्तरज्यास्वरूपे समुत्थापनेन}$$

$$\text{ज्याउ}=\frac{\text{ज्याभु}\times\text{कोज्याभु}\times\text{त्रि}-\text{पद्यु}\times\text{ज्याभु}\times\text{कोज्याभु}}{\text{त्रि. द्यु}}$$

$$=\frac{\text{ज्याभु}\times\text{कोज्याभु}\ (\ \text{त्रि}-\text{पद्यु}\ )}{\text{त्रि. द्यु}}$$

$$=\frac{\text{ज्याभु}\times\text{कोज्याभु}\times\text{ज्याजिउ}}{\text{त्रि. द्यु}}$$

अत्र ज्याजिउ=जिनांशोत्क्रमज्या

हरभाज्यौ 'त्रि+पद्यु' गुणितौ तदा $\frac{(\text{त्रि}+\text{पद्यु})\,(\text{ज्याभु}\times\text{कोज्याभु}\times\text{ज्याजिउ})}{\text{त्रि. द्यु}\ (\ \text{त्रि}+\text{पद्यु}\ )}=$

$$\frac{(\ \text{त्रि}\times\text{ज्याभु}\times\text{कोज्याभु}+\text{ज्याभु}\times\text{कोज्याभु}\times\text{पद्यु}\ )}{\text{त्रि. द्यु}\ (\text{त्रि}+\text{पद्यु})}\frac{\text{ज्याजिउ}}{}=\text{ज्याउ}$$

$$=\frac{(\ \text{ज्याभु}\times\text{कोज्यावि}+\text{ज्यावि}\times\text{कोज्याभु}\ )\ \text{ज्याजिउ}}{\text{त्रि}\ (\text{त्रि}+\text{पद्यु})}$$

$$=\frac{\text{ज्या}\ (\text{भु}+\text{वि})\ \text{ज्याजिउ}}{\text{त्रि}+\text{पद्यु}}=\text{ज्याउ}=(१)$$

(१) एतेन 'विषुवांशभुजांशयोगजीवा जिनभागोत्क्रमजज्यया विनिघ्नी।
परमाल्पद्युज्यया विभक्ता त्रिभजीवायुतयोदयान्तरज्या॥'
म० म० सुधाकरोक्तसूत्रमुपपद्यते।

(१) अत्रै 'ज्याजिउ, त्रि+पद्यु' तयोर्गुणकहरयोः स्थिरत्वात्सिद्धं यद्यत्र ज्या (भु+वि) परमार्थात्त्रिज्यासमा भवेत्तत्रैवोदयान्तरज्या परमा भवेदर्थात्तत्रैवोदयान्तरस्य परमत्वं भवेदिति। गोलसन्धौ (नाडीक्रान्तिवृत्तयोः सम्पाते) तथाऽयनसन्धौ मध्यमार्के, उदयान्तराभाव इत्युदयान्तरस्य भुजांशविषुवांशयोरन्तररूपस्य स्वरूपदर्शनेनैव स्फुटमिति॥

अधुना परमोदयान्तरकालीनभुजांशविषुवांशयोरानयनं क्रियते

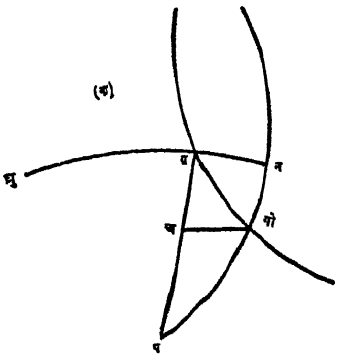
पूर्वं सिद्धं यद्यदा भुजांशविषुवांशयोर्योगज्या त्रिज्यासमार्थाद्भुजांशविषुवांशयोर्योगो नवत्यंशसमस्तदोदयान्तरस्य परमत्वं भवितुमर्हति तेन परमोदयान्तरे भुजांश+विषुवांश=९० तथा तदा भुजांशविषुवांशयोरन्तरम्=परमोदयान्तरम्,

ततः संक्रमणगणितेन $\frac{९०+\text{परमोदयान्तर}}{२}=४५+\frac{\text{परमोदयान्तर}}{२}=$

परमोदयान्तरकालीनभुजांशाः

तथा $\frac{९०-\text{परमोदयान्तर}}{२}=४५-\frac{\text{परमोदयान्तर}}{२}=$परमोदयान्तरकालीनविषुवांशाः

अथवा



ध्रु=ध्रुवः। ग्र=क्रान्तिवृत्ते मध्यमार्कः। गो=गोलसन्धिः । गोग्र=भुजांशाः । गोन=विषुवांशाः, गोग्रभुजांशसमं नाडीवृत्ते गोप छित्वा पग्रवृत्तं कार्यं तथा गोपवृत्तोपरि गोबिन्दुतो गोचलम्बवृत्तं कार्यं तदा गोप+गोन=परमोदयान्तरकालीनभुजांशविषुवांशयोर्योगः = ९०, <ग्रगोन=जिनांश; <ग्रगोप=१८०—जिनांश <ग्रगोच=$\frac{<\text{ग्रगोप}}{२}=\frac{१८०-\text{जिनांश}}{२}$

$=९०-\frac{\text{जिनांश}}{२}=$जिनांशार्धको ग्रनप-चापीय जात्यत्रिभुजे पनकोटिः=९० तदा पग्रकर्णोऽपि=९० तदा ग्रगोपचापीय त्रिभुजस्य समद्विबाहुकत्वात् ग्रच=चप=४५, <ग्रचगो=९० ततो गोचपचापीयजात्यैऽनुपातेन $\frac{\text{त्रि}\times\text{ज्या}\,४५}{\text{कोज्याजि}_{२}}=$ज्यागोप=परमोदयान्तरकालीनभुजज्या, अस्याश्चापम्=परमोदयान्तरकालीनभुजांशाः, कोज्याजि$_{२}$=जिनांशार्धकोटिज्या, एतावताऽधोलिखितसूत्रमुपपद्यते।

त्रिज्येषु वेदांशगुणेन ताडिता जिनार्धकोटच्युत्थगुणेन भाजिता।
तदीय चापेन समा भुजांशका यदा तदा तत्र परोदयान्तरम् ॥

एतेनैतन्मिते भुजभागे परमोदयान्तरं भवतीति सिद्धयति। वस्तुतस्तन्मानं कियदिति जिज्ञासायां उज्या=$\frac{\text{ज्या}\,(\text{वि}+\text{भु})\,\text{उज्याजि}}{\text{त्रि}+\text{पद्यु}}$

∴ परमोदयान्तर=$\frac{\text{त्रि}\times\text{उज्याजि}}{\text{त्रि}+\text{पद्यु}}$

∴ वा $\frac{\text{त्रि.}\,\text{उज्याजि}\times\text{त्रि}^{२}}{२\,(\text{त्रि}+\text{पद्यु})\times\text{त्रि}^{२}}=\frac{\text{ज्या}^{२}\,\frac{१}{२}\,\text{जि.}\,\text{त्रि}^{२}}{\text{कोज्या}^{२}\,\frac{१}{२}\,\text{जि.}\,\text{त्रि}^{२}}$(त्रिकोणगणितेन)

$$=\frac{\text{स्य}^{2}\frac{1}{2}\text{ जि.}}{\text{त्रि.}}$$

यतः $\frac{\text{त्रि. उज्याजि}}{२}=\text{ज्या}^{2}\frac{1}{2}\text{ जि.}$

$\frac{\text{त्रि }(\text{त्रि}+\text{पद्यु})}{२}=\text{कोज्या}^{2}\frac{1}{2}\text{ जि.}$

एतेन—

जिनांशार्धस्य यः स्पर्शरेखावर्गो विभाजितः।
परमोदयान्तरज्या स्याल्लब्धिस्त्रिज्यकया स्फुटा ॥

अत्र लघुरिकथने—

$\text{स्य}\frac{1}{2}\text{ जि}=९\cdot३१७८७८९$

$\therefore \text{स्य}^{2}\frac{1}{2}\text{ जि}=१८\cdot६६४७४७८$

$\therefore \frac{\text{स्य}^{2}\frac{1}{2}\text{ जि}}{\text{त्रि}}=८\cdot६६४७४७८$

∴ परमोदयान्तरासवः=२'। =२√'=१४√

षड्भिर्भक्ताः पलानि=२५,

अत्र भास्कराचार्येण २६ पलानि गृहीतानि। भास्करोक्तमुदयान्तरं व्यर्थं दुराग्रहेण प्रखण्डितं कमलाकरेण। तदर्थं तत्त्वविवेको द्रष्टव्यः।

एतद्बलेन "परमोदयान्तरज्ञानेनाहर्गणानयनं कथं भवेदेतस्य विलक्षणप्रश्नस्योत्तरसिद्धिर्भवति" यथा परमोदयान्तरज्ञानेन पूर्वोक्तसूत्र 'त्रिज्येषु वेदांशगुणेने-त्यादि' द्वारा तत्कालीनभुजांशज्ञानं भवेत्ततो "निरग्रचक्रादपि कुट्टकेनेत्यादि विलोमेन" ऽहर्गणज्ञानं सुखेनैव भवेदिति, उदयान्तरखण्डनं कमलाकरेण कृतं तत्समीचीनं नास्ति तथाऽन्येऽपि बहवो विशेषाः सन्त्युदयान्तरसम्बन्धे तेऽत्रविस्तृतिभयान्न लिख्यन्त इति ॥ ३१ ॥

अब ग्रहानयन कहते हैं

हि. भा.—अहर्गण को इष्टग्रह के कल्प में पठित भगण से गुण कर कल्प सावन (कल्प कुदिन) से भाग देने से फल भगणादि मध्यम ग्रह लङ्काक्षितिजोदय (लङ्का सूर्योदय) कालिक होते हैं ॥ ३१ ॥

उपपत्ति

$\frac{\text{कल्प ग्रहभगण} \times \text{अहर्गण}}{\text{कल्पकुदिन}} = \frac{\text{ग्रहभ}}{\text{ककु}} + \frac{\text{भगणशे}}{\text{ककु}}$ यहाँ अहर्गण के मध्यम सावनदिन समूह रूप होने के कारण उस पर से पूर्वोक्तानुपातद्वारा जो भगणादि मध्यम ग्रह आते हैं, वे भी मध्यम सावनान्त बिन्दुक ही होंगे, $\frac{\text{वर्षान्तःपातिस्पष्टसावनयोग}}{\text{वर्षान्तःपातिस्पष्टसावनसं}}$ = मध्यमसावन, वर्षान्तःपाति, स्पष्टसावनयोगसम्बन्धिनाक्षत्र = वर्षान्तःपातिस्पष्टसावनसंख्या + १ नाक्षत्र इसलिए

$\frac{\text{वर्षान्तः पातिस्पष्टसावन सं १ नाक्षत्र}}{\text{वर्षान्तःपातिस्पष्टसावनसं}} = १ \text{ ना} + \frac{१ \text{ ना}}{\text{वर्षान्तःपातिस्पसा}} = १ \text{ ना} +$

$\frac{२१६०० \text{ असु}}{\text{वर्षान्तःपातिस्पष्टसावनसं}}$ परं $\frac{२१६०० \text{ कला}}{\text{वर्षान्तःपातिस्पष्टसावनसं}}$ = मध्यगतिक; अतः मध्यमगति-

कलासमासु = $\frac{२१६०० \text{ असु}}{\text{वर्षान्तःपातिस्पष्टसावनसं}}$

इसलिए मध्यमसा = १ ना + मध्यगतिकलासमासु, लेकिन कला समान असु नाडीवृत्त ही में होती है इसलिए उक्त ग्रह नाडीवृत्त ही में चालनीय है, यह सिद्ध हुआ। इसलिए अपने कल्प पठित भगण से पूर्वोक्तानुपात से जो मध्यमग्रह आते हैं, वे क्रान्ति वृत्तीय भुजांशतुल्य नाडीवृत्तीय चाप के अग्रबिन्दुक होते हैं अतः 'दशशिरः पुरि मध्यम भास्करे क्षितिज सन्निधिगे सति मध्यमः' भास्कराचार्य कहते हैं, गोल सन्धि बिन्दु से पूर्वाभिमुख चालित समान गति वेगक नाडीवृत्तीय और क्रान्तिवृत्तीय मध्यमार्कों में जब नाडीवृत्तीय मध्यमार्क लङ्कास्थ क्षितिज में होते हैं तब क्रान्तिवृत्तीय मध्यमार्क पदवश से क्षितिज से ऊपर और नीचे (भुजांश और विषुवांश के अन्तर पर) होते हैं; दोनों ग्रहोदयों के अन्तर को उदयान्तर भास्कराचार्य कहते हैं परमोदयान्तर भी अल्प ही होता है इसलिए भास्कराचार्य 'क्षितिज सन्निधिगे' कहते हैं; इस अन्तर (उदयान्तर) को ब्रह्मगुप्त शून्य मानते हैं इसलिए 'लङ्कायां भास्करौदयिकः' कहते हैं यहां भास्कराचार्य का कथन ही ठीक है उदयान्तर नहीं मानना अनुचित है, भास्कराचार्य ने एक उदयान्तर रूप विलक्षण वस्तु दिखलाकर अपने अद्भुत पाण्डित्य का परिचय दिया है इस विषय पर ज्यौतिषिक लोग निष्पक्ष बुद्धि से विचार करें ॥ ३२ ॥

अब प्रसङ्ग से उदयान्तर के सम्बन्ध में कुछ विचार करते हैं

क्रान्तिवृत्त में जहां पर मध्यम रवि है उसके ऊपर ध्रुवप्रोतवृत्त (निरक्षक्षितिज) करने से नाडीवृत्त में जहां लगता है, वहां से गोलसन्धि पर्यन्त नाडीवृत्तीयचाप विषुवांश या मध्यमरविगतिकलोत्पन्नासु है, नाक्षत्र षष्टि (६०) घटी में मध्यमरविगतिकलोत्पन्नासु को

जोड़ने से स्पष्टसावन होता है, गोलसन्धि बिन्दु को केन्द्र मानकर क्रान्तिवृत्तीयमध्यमरवि भुजांश व्यासार्धवृत्त करने से नाडीवृत्त में जहां लगता है वहां से गोल सन्धि बिन्दु तक नाडीवृत्त में मध्यमरविगतिकलातुल्यासु है, नाक्षत्र षष्टि (६०) घटी में मध्यमरविगतिकला तुल्यासु को जोड़ने से मध्यम सावन होता है। इन दोनों (स्पष्टसावन और मध्यम-सावन) के अन्तर करने से मध्यमरविगतिकलातुल्यासु-मध्यमरविगतिकलोत्पन्नासु = उदयान्तरासु, एतत्सम्बन्धिग्रहगति प्रमाण लाते हैं जैसे यदि अहोरात्रासु में ग्रहगति कला पाते हैं तो उदयान्तरासु में क्या इस अनुपात से उदयान्तरासु सम्बन्धिनी ग्रहगति आती है $\frac{\text{ग्रहगतिकला} \times \text{उदयान्तरासु}}{\text{अहोरात्रासु}}$ = उदयान्तरासु सम्बन्धिनी ग्रहगति, इसको अहर्गणोत्पन्न ग्रह में घटाने से और जोड़ने से निरक्षक्षितिजस्थ (क्रान्तिवृत्तस्थ मध्यमरविगत ध्रुवप्रोतवृत्त नाड़ीवृत्त सम्पात बिन्दुक) ग्रह होते हैं। इसी विषय को सिद्धान्त शिरोमणि में भास्कराचार्य 'मध्यार्क भुक्ता असवो निरक्षे ये ये च मध्यार्ककला 'समाना' इत्यादि से कहते हैं। लेकिन उदयान्तरासु के मध्य में भी ग्रह की कुछ गति होगी। उस गति का ग्रहण भास्कराचार्य नहीं किये हैं, इसलिए पूर्व प्रदर्शित उदयान्तरासुसम्बन्धिनी ग्रहगतिसम्बन्ध से जो निरक्षक्षितिजोदय कालिक ग्रह लाये हैं सो ठीक नहीं है इसलिए भास्करोक्त उदयान्तरानयन ठीक नहीं है, यह सिद्ध हुआ।

अब वास्तव उदयान्तर साधन करते हैं

वास्तव उदयान्तर प्रमाण = य मानते हैं, इसी से औदयिक ग्रह होते हैं, इस उदयान्तर काल में ग्रह की जो गति होती है, तदुत्पन्नासु करके संस्कृत भास्करोक्त उदयान्तर वास्तव उदयान्तर होता है। यथा यदि अहोरात्रासु में रविगतिकला पाते हैं तो वास्तवोदयान्तरा (य) सु में क्या इस अनुपात से वास्तवोदयान्तरासु सम्बन्धिनी रविगति कला आती है, $\frac{\text{रविगकला} \times \text{य}}{\text{अहोरात्रासु}}$ = एकासुजरविग × य, फिर अनुपात करते हैं यदि राशि कला (१८००) में उस राशि का उदयासु पाते हैं तो वास्तवोदयान्तरासुसम्बन्धिनी रविगतिकला में क्या इस अनुपात से तत्सम्बन्धि असु प्रमाण आता है $\frac{\text{राश्युदयासु} \times \text{वास्तवोदयान्तर संरविगतिक}}{१८००} =$

$\frac{\text{राश्युदय} \times \text{रविगतिकला} \times \text{य}}{१८०० \times \text{अहोरात्रासु}}$ = एकासुजरविगति × १ कलोत्पन्नासु × य ···(१)

$\because \frac{\text{राश्युदय} \times १}{१८००} = १$ कलोत्पन्नासु, भास्करोक्तोदयान्तर में (१) इसको संस्कार करने से वास्तव उदयान्तर होता है, अतः भास्करोक्तोदयान्तर ∓ एकासुजरविग × १ कलोत्पन्नासु × य = य = पूर्वोदयान्तर ∓ एकासुजरविग × १ कलोत्पन्नासु × य समशोधन करने से य ± एकासुजरविग × १कलोत्पन्नासु × य = पूर्वोदयान्तर = य (१ ± एकासुजरविग × १कलोत्पन्नासु)

$$\therefore \text{य} = \frac{\text{पूर्वोदयान्तर}}{१ \pm \text{एकासुजरविग} \times १\text{कलोत्पन्नासु}} \cdots\cdots\cdots (\text{क})$$

यहां इनगति=सूर्यगति, सूर्यगतिकलासम्बन्ध से वास्तव उदयान्तर साधन किय गया है, अपनी-अपनी गति के सम्बन्ध से अन्य ग्रहों का साधन उसी तरह से होता है।

(क) इससे 'एकासुजेनगतिसङ्गुणितैकलिप्तोत्पन्नासु राश्युदययुक्तविहीनितेन' इत्यादि संस्कृतोपपत्तिस्थ म० म० पण्डित सुधाकर द्विवेदी जी का सूत्र उपपन्न होता है।

अब उदयान्तर का परमत्व कहां होता है, विचार करते हैं

भुजांश और विषुवांश का अन्तर उदयान्तर है; इसलिए उदयान्तरज्या=(भु—वि)

चापयोरिष्टयोः इत्यादि से $\frac{\text{ज्याभु}\times\text{कोज्यावि}-\text{ज्यावि}\times\text{कोज्याभु}}{\text{त्रि}}$=उदयान्तरज्या,

यतः $\frac{\text{कोज्याभु}\times\text{त्रि}}{\text{द्यु}}$= कोज्यावि

तथा $\frac{\text{पद्य}\times\text{ज्यासु}}{\text{द्यु}}$=ज्यावि, उदयान्तरज्या स्वरूप में उत्थापन से

$$\frac{\text{ज्याभु}\times\text{कोज्याभु}\times\text{त्रि}-\text{पद्यु}\times\text{ज्याभु}\times\text{कोज्याभु}}{\text{त्रि. द्यु}}=\frac{\text{ज्याभु}\times\text{कोज्याभु}\ (\text{त्रि}-\text{पद्यु})}{\text{त्रि. द्यु}}$$

$$=\frac{\text{ज्याभु}=\text{कोज्याभु}\times\text{ज्याजिउ}}{\text{त्रि. द्यु}}=\text{उदयान्तरज्या, ज्याजिउ}=\text{जिनांशोत्क्रमज्या।}$$

यहां हर और भाज्य को 'त्रि×पद्यु' इससे गुण देने से

$$\frac{(\text{त्रि}+\text{पद्यु})\ (\text{ज्याभु}\times\text{कोज्याभु}\times\text{ज्याजिउ})}{\text{त्रि. द्यु. (त्रि}+\text{पद्यु})}$$

$$=\frac{(\text{त्रि. ज्याभु. कोज्याभु}+\text{ज्याभु. कोज्याभु. पद्यु})\ \text{ज्याजिउ}}{\text{त्रि. द्यु (त्रि}+\text{पद्यु})}$$

$$=\frac{(\text{ज्याभु. कोज्यावि}+\text{ज्यावि. कोज्याभु})\ \text{ज्याजिउ}}{\text{त्रि. (त्रि}+\text{पद्यु})}$$

$$=\frac{\text{ज्या (भु}+\text{वि)}\ \text{ज्याजिउ}}{\text{त्रि}+\text{पद्यु}}=\text{उदयान्तरज्या...(१)}$$

(१) इससे "विषुवांशभुजांशयोगजीवा" इत्यादि संस्कृतोपपत्तिस्थ म० म० पण्डित सुधाकर द्विवेदी जी का सूत्र उपपन्न होता है॥

(१) इसमें ज्याजिउ, त्रि+पद्यु इन दोनों गुणक और हर के स्थिरत्व के कारण जहाँ ज्या (भु+वि) इसका परमत्व होगा वहीं पर उदयान्तरज्या का परमत्व होगा अर्थात् उदयान्तर का परमत्व होगा। परन्तु ज्या त्रिज्या से अधिक नहीं होती है इसलिए

जहां ज्या (भु+वि)=त्रि होती है, वहीं पर उदयान्तर का परमत्व होता है, यह सिद्ध हुआ। गोलसन्धि (नाड़ीवृत्त और क्रान्तिवृत्त के सम्पात) में भुजांश और विषुवांश के अभाव से उन दोनों के अन्तररूप उदयान्तर का अभाव होता है तथा अयन सन्धि में मध्यमार्क के रहने से भुजांश और विषुवांश के नवत्यंश के बराबर होने से दोनों के अन्तररूप उदयान्तर का अभाव होता है, यह सिद्ध हुआ।

अब परमोदयान्तरकालीन भुजांश और विषुवांश के साधन करते हैं

पहले सिद्ध हुआ है कि जब भुजांश और विषुवांश की योगज्या त्रिज्या के बराबर होती है अर्थात् भुजांश और विषुवांश का योग नवत्यंश के बराबर होता है तब उदयान्तर का परमत्व होता है, इसलिए परमोदयान्तर में भुजांश+विषुवांश=९०, और उस अवस्था में भुजांश और विषुवांश का अन्तर=परमोदयान्तर, तब संक्रमण गणित से

$\frac{९०+\text{परमोदयान्तर}}{२} = ४५ + \frac{\text{परमोदयान्तर}}{२}$ = परमोदयान्तर कालीन भुजांश

तथा $\frac{९०-\text{परमोदयान्तर}}{२} = ४५ - \frac{\text{परमोदयान्तर}}{२}$ = परमोदयान्तर कालीन विषुवांश।

अथवा

ध्रु=ध्रुव। ग्र=क्रान्तिवृत्त में मध्यमार्क, गो=गोलसन्धि, गोग्र=भुजांश, गोन=विषुवांश नाड़ीवृत्त में गोग्र भुजांश तुल्य गोप काटकर पग्र वृत्त बना दीजिये, गो बिन्दु से पग्र के ऊपर गोच लम्ब वृत्त कर दीजिये, गोन+गोप=विषुवांश+भुजांश यह जब नवत्यंश के बराबर होता है तब ही उदयान्तर का परमत्व होता है, इसलिए उदयान्तर के परमत्व में गोन+गोप=विषुवांश+भुजांश=९० अनप चापीय जात्यत्रिभुज में पनकोटि=९० अतः पग्रकर्णोऽपि=९०, तदा ग्रच=चप=४५ (ग्रगोपचापीय त्रिभुज के समद्विबाहुकत्व के कारण), <ग्रचगो=९०, <ग्रगोन=जिनांश <ग्रगोप=१८०-जिनांश, <ग्रगोच=$\frac{<\text{ग्रगोप}}{२}=\frac{१८०-\text{जिनांश}}{२}$

$= ९०-\frac{\text{जिनांश}}{२}$=जिनांशार्ध को तब गोचपचापीय जात्यत्रिभुज में अनुपात से

$\frac{\text{त्रि}\times\text{ज्या }४५}{\text{कोज्या}\frac{\text{जि}}{२}}$=ज्या गोप=परमोदयान्तर कालीन भुजज्या, इसके चाप करने से परमोदयान्तर कालीन भुजांश हुआ, कोज्या$\frac{\text{जि}}{२}$=जिनांशार्धकोटिज्या, इससे संस्कृतोपपत्तिस्थ 'त्रिज्येषु वेदांशगुणेन ताडिता' इत्यादि सूत्र उपपन्न हुआ। यहाँ संस्कृतोपपत्तिस्थ (क) क्षेत्र देखिये॥

उपर्युक्त उपपत्ति के बल से "परमोदयान्तर ज्ञान से अहर्गण ज्ञान कैसे होगा, इस विलक्षण प्रश्न का उत्तर सुलभ ही होता है" जैसे परमोदयान्तर ज्ञान से उपर्युक्तोपपत्ति द्वारा तत्कालीन भुजांश ज्ञान हो जायगा तब 'निरग्रचक्रादपि कुट्टकेन' इत्यादि के विलोम से अहर्गण ज्ञान सुलभेन हो जायगा, उदयान्तर का खण्डन सिद्धान्ततत्त्वविवेक में कमलाकर जी ने किया है सो ठीक नहीं है, उदयान्तर के सम्बन्ध में बहुत अन्य विशेष विषय है जो विस्तृति भय से नहीं लिखते है इति ॥३२॥

इदानीं स्वसिद्धान्तप्रशंसार्थमार्यामाह—

आनयति दिवसवारं स्मृत्यविरोधेन मध्यमथवा वा ।
ब्राह्मादन्यैस्तन्त्रैरार्यभटाद्यैर्न कश्चिदपि ॥ ३३॥

वा. भा.—ब्राह्मादन्यैस्तंत्रैः (मदुक्तब्रह्मसिद्धान्तभिन्नै :) आर्यभटाद्यैः आर्यभटैस्तदनुयायिभिस्तथाऽऽर्यभटमतं स्वीकृत्य ग्रन्थकृद्भिराचार्यैश्च भिन्नैः कोऽप्याचार्यः स्मृत्यविरोधेन स्मृतिसंगतेन दिवसवारं अहर्गणं, न केवलमहर्गणमपितु मध्यमग्रहान् अपि न आनयति । इत्यनेन स्वतन्त्रस्य प्रौढ़ित्वं प्रकटयति अन्यानधिक्षिपतिश्च ग्रन्थकारः ।

वि. भा.—स्मृत्यविरोधेन (स्मृतिशास्त्रानुकूलेन) कारणेन, ब्राह्मात् (आचार्यकथितात्तन्त्रादेव) दिवसवारं (अहर्गणं) अथवा मध्यं (मध्यग्रहान्) गणक आनयति, अन्यैः (भिन्नैः) आर्यभटाद्यैः (आर्यभटादिरचितैः) तन्त्रैः कश्चिदपि दिवसवारं मध्यग्रहांश्च नाऽऽनयति स्मृतिशास्त्रप्रतिकूलादिति,

आचार्येण कथ्यते यत्स्मृतिशास्त्रानुकूलान्मत्प्रणीततन्त्रादेव गणका अहर्गणमध्यग्रहादिसाधनं कुर्वन्ति, आर्यभटादिप्रणीततन्त्रात्स्मृतिशास्त्रप्रतिकूलात्कोऽपि दिवसवारमध्यग्रहादिसाधनं न करोतीति ॥३३॥

आचार्य अपने आनयन की प्रशंसा कहते हैं

हि. भा.—स्मृतिशास्त्रानुकूल हमारे तन्त्र ही से ज्यौतिषिक लोग अहर्गण और मध्यग्रहों को लाते हैं, स्मृतिशास्त्र प्रतिकूल आर्यभटादि आचार्य प्रणीत तन्त्रों से कोई भी अहर्गण-मध्यग्रहों को नहीं लाते हैं ॥३३॥

इदानीं येऽर्धरात्रे वारप्रवृत्तिमिच्छन्ति तान्प्रत्याह

जगति तमोभूतेऽस्मिन्सृष्ट्यादौ भास्करादिभिः सृष्टैः ।
यस्माद्दिनप्रवृत्तिर्दिनवारोऽर्कोदयात्तस्मात् ॥ ३४ ॥

वा. भा.—अयमर्थः सृष्ट्यादौ यत्रैव काले भास्करोदये ग्रहाः सृष्टास्तत एव कालादुद्वापरप्रवृत्तिः। यतः प्रथमं दिनं पश्चाद्रात्रिः। षट्प्रहराश्च वाररहिता भवन्तीत्येतदपि न शक्यते वक्तुम्। सृष्ट्यादेः पूर्वार्धरात्रे चेत्तदपि न यस्मात् "सति धर्मिणि धर्माश्चिन्त्यन्ते" इति यावान्नार्कोदयः सृष्टस्तावदियं रात्रिरयं दिवसः इति कथमुच्यते। सृष्टेश्चप्राक् सर्वमेव तमोभूतमासीत्। अत्र भगवान्मनुः—आसीदिदं तमोभूतमप्रज्ञातमलक्षणम्। अप्रतर्क्यमनाघृष्टं प्रसुप्तमिव सर्वतः इति। अत्र लंकासमयाम्योत्तर रेखापेक्षयोच्यते। अर्धरात्रे वारप्रवृत्तिर्न भवति अन्यथा रोमके सर्वदैव वारप्रवृत्तिरर्धरात्रेऽन्यत्रास्तमयेऽन्यत्रमध्याह्नेऽन्यत्रेष्टकाले इत्यादि योज्यम्।

वि. भा.—यस्मात् कारणात्–अस्मिन् तमोभूते प्राकृतिकप्रलये सूर्यादीनामभावादन्धकारमये जगति (संसारे) ब्रह्मणा सृष्टैः (रचितैः) भास्करादिभिः (सूर्याद्यैः) सृष्ट्यादौ (सृष्ट्यादिकाले) दिनप्रवृत्तिर्जाताऽर्थात्सृष्टैर्भास्करादिभिरित्यनेन सर्वप्रथमं सूर्यस्यैव रचना कृता तत्समय एव वारप्रवृत्तिरभवत्—तस्मात् कारणादर्कोदयादेव (सूर्योदयादेव) दिनवारो ज्ञातव्य इति, आचार्योक्तमिदं तदैव समीचीनं भवितुमर्हति यदा प्राकृतिकप्रलये सूर्यस्य लयो भवेत् परं 'सूर्याचन्द्रमसौ धाता यथा पूर्वमकल्पयदिति' वेदोक्त्या सूर्यस्य नित्यत्वं सिद्धयति, सृष्ट्यादौ ब्रह्मणा किञ्चित्प्रकाशवति सूर्येऽतिप्रकाशवर्धनार्थमेकः सीसकरूपपदार्थो निवेशितो यद्द्वारा सूर्येऽतीवप्रकाशः परिलक्ष्यते, 'सूर्य आत्माजगतस्तस्थुषश्चेति' वेदोक्त्या ब्रह्मा सूर्यस्य पुत्रोऽस्तीति सिद्धयति तर्हि ब्रह्मणा सूर्यस्यरचना कथं भवेत् पुत्रद्वारा पितुः सृष्टेरभावात् सूर्यरचनासम्बन्धे सूर्येण सह ब्रह्मणः केवलमेतावानेव सम्बन्धो यश्च सृष्ट्यादौ किञ्चित्प्रकाशवति सूर्ये प्रकाशवर्धनार्थमुपरि मया प्रदर्शित इति। अन्येषां मते—सृष्टैरित्यत्र सूर्यादीनां नवीना सृष्टिरासीदिति न भ्रमितव्यम्। न चात्र पितापुत्रयोः सम्बन्धोऽपेक्ष्यते येन सृष्टौ वैषम्यमापद्येत। अत्र सृजनं पार्थिवजीवानां भवति। तथा सति सृष्ट्यादौ नित्याम्बरवासिनां तेषां सूर्यादीनां प्रथमं संदर्शनं भवेदित्येव सृष्टिपदस्य तात्पर्यम्। "भूगतानां विनाशः स्यान्नो नित्याम्बरवासिनामिति कमलाकरोक्तेः"। सूर्य आत्मा जगतस्तस्थुषश्चेति वेदे उक्तः ॥३४॥

अब दिन प्रवृत्ति को कहते हैं

हि. भा.—जिस कारण से प्राकृतिक प्रलय में सूर्यादिग्रहों के अभाव से अन्धकारमय इस संसार में ब्रह्मा ने सूर्यादि की रचना की, इससे सृष्ट्यादि काल में दिन प्रवृत्ति हुई अर्थात् ब्रह्मा ने सबसे पहले सूर्य की रचना की, उसी समय वार प्रवृत्ति हुई, इस कारण से सूर्योदय ही से दिनवार समझना चाहिये,

आचार्य का यह कथन तब ही ठीक हो सकता है जब कि प्राकृतिक प्रलय में सूर्य का लय हो, लेकिन 'सूर्याचन्द्रमसौ धाता यथापूर्वमकल्पयत्' इस वेदोक्ति से सूर्य नित्य है यह

सिद्ध होता है । सृष्ट्यादि काल में किञ्चित्प्रकाशवान् सूर्य में अतिशय प्रकाश बढ़ाने के लिये ब्रह्मा ने एक सीसा रूप पदार्थ सूर्य मण्डल में दे दिया जिसके द्वारा सूर्य-मण्डल में अतीव प्रकाश फैलने की शक्ति हुई, 'सूर्य आत्मा जगतस्तस्थुषश्च' इस वेदोक्ति से ब्रह्मा सूर्य के पुत्र सिद्ध होते हैं तब पुत्र द्वारा पिता की सृष्टि कैसे हो सकती है इससे सिद्ध होता है कि ब्रह्मा द्वारा सूर्य की सृष्टि नहीं होती है, सूर्य रचना के विषय में सूर्य के साथ ब्रह्मा का सम्बन्ध वही है जो पहले सूर्य के प्रकाश बढ़ने के लिये कहा गया है ॥३४॥

इदानीं य एते मध्यमा आनीता तेषां देशनियमार्थमार्ययाह—

लङ्कासमयाम्योत्तररेखायां भास्करोदये मध्याः ।
देशान्तरोनयुक्ता रेखायाः प्रागपरदेशेषु ॥३५॥

वा. भा.—लंकाउत्तरेण मेरुं यावत् उज्जयिन्यादिदेशानां मध्येन या याम्योत्तररेखा सा लंकासमयाम्योत्तररेखा तत्र यो भास्करोदयकालः तत्रैव ते पूर्वानीता ग्रहा मध्याः। न तु पारे शेषदेशेष्वित्यर्थः किन्तु विषुवतीति वक्तव्यम् । यस्माच्चरदलवशात् अन्यत्रकालेऽर्कोदयरेखास्थदेशेष्वपि भिद्यते विषुवद्दिवसे, पुनः तुल्योऽर्कोदयो रेखावासिनामेतत्सर्वं गोले प्रदर्शयेत् । परिशेषदेशेषु विषुवद्दिवसे यथा मध्यमाः पूर्वानीताः स्वौदयिका भवन्ति । देशान्तरफलेन यथासंख्यमूनयुक्ताः संतः प्राग्परदेशेषु मध्या स्वोदये विषुवद्दिने भवन्ति । यस्माद्रेखातः पूर्वेण यो द्रष्टा स रेखास्थद्रष्टुः सकाशात्पूर्वमेव रविमुद्यन्तं पश्यति । अतो ग्रहफलं तत्र विशोध्यते । नागतदर्शनात्पश्चात्तु दीयते वैपरीत्वादुपपन्नमेतत् । प्रवृत्तस्वाद्भूमेः एतत्सर्वं गोले प्रदर्शयेत् । वारप्रवृत्तिरपि याऽर्कोदये पूर्वमभिहिता सापि लंकासमयाम्योत्तरे रेखायां, नान्यत्र तावता च कालेनान्यदेशेषु विषुवद्दिने भवति ।

वि. भा.—लङ्कासमयाम्योत्तररेखायां (लङ्कायाम्योत्तररेखायां) स्थितानां मानवानां भास्करोदये (मध्यमार्कोदयकाले) मध्याः (ग्रहर्गणसाधिता) मध्यमाः ग्रहा भवन्ति । रेखायाः प्रागपरदेशेषु (रेखातः पूर्वपश्चिमदेशेषु) गणितागतग्रहा देशान्तरोनयुक्ताः (देशान्तरफलेनोनयुताः) तदा स्वनिरक्षोदयकालिका भवन्तीति ॥३५॥

अत्रोपपत्तिः

एतेनाऽचार्येणो- (ब्रह्मगुप्तेन) दयान्तरं न स्वीक्रियते तदात्वहर्गणेन (मध्यमसावनदिनसमूहेन) साधिता ग्रहा वस्तुतोऽहर्गणान्तेऽर्थान्मध्यमसावनान्ते समागच्छन्तोऽपि लङ्कायाम्योत्तररेखायां समागच्छन्ति, रेखातः पूर्वदेशे पूर्वमेव ग्रहदर्शनं भवति तेन रेखादेशीयग्रहादिष्टदेशीयग्रहस्याल्पत्वाद्देशान्तरफलेन स्पष्टभूपरिधिना ग्रहगतिस्तदा देशान्तरयोजनैः किमित्यनुपातेन समागतदेशान्तरयोजनैः-

सम्बन्धिकलात्मकफलेन हीनो रेखादेशीयग्रहोऽभीष्टदेशीयग्रहो भवेत् रेखातः पश्चिम-देशे तु पश्चाद्ग्रहदर्शनं भवत्यतस्तत्र रेखादेशीयग्रहादभीष्टदेशीयग्रहस्याधिकत्वेन देशान्तरफलेन सहितो रेखादेशीयग्रहोऽभीष्टदेशीय ग्रहो भवेदिति ॥३५॥

अब अहर्गण से साधित ग्रह कहां आते हैं सो कहते हैं

हि. भा.—लङ्कायाम्योत्तररेखा में स्थित मनुष्यों के मध्यमरव्युदय काल में अहर्गण द्वारा साधित मध्यम ग्रह होते हैं, रेखा से पूर्व और पश्चिम देशों में क्रम से गणितागतग्रह में देशान्तर फल को घटाने और जोड़ने से अपने निरक्षोदयकालिक ग्रह होते हैं ॥३५॥

उपपत्ति

अहर्गण मध्यम सावन दिनों का समूह है इसलिए अहर्गण से साधित मध्यमग्रह अहर्गणान्त बिन्दुक होते हैं, वह बिन्दु (मध्यम सावनान्त बिन्दु) क्रान्तिवृत्तस्थ मध्यमार्कोपरिगत ध्रुव प्रोतवृत्त नाडीवृत्त के सम्पात बिन्दु (स्पष्ट सावनान्त बिन्दु) से पदवश से ऊपर और नीचा होता है दोनों बिन्दुओं (मध्यमसावनान्त बिन्दु और स्पष्ट सावनान्तबिन्दु) के अन्तर को भास्कराचार्य उदयान्तर कहते हैं परन्तु ब्रह्मगुप्त उदयान्तर नहीं स्वीकार किये हैं इसलिए इनके मत से अहर्गण द्वारा सिद्ध ग्रह लङ्कायाम्योत्तर रेखा में जो रहते हैं उनके मध्यमरव्युदय काल में होते हैं, रेखा से पूर्व देश में पहले ही ग्रह दर्शन होता है इसलिए रेखादेशीय ग्रह से अभीष्ट देशीय ग्रह के अल्पत्व के कारण देशान्तर फल को रेखादेशीय ग्रह में हीन करने से अभीष्ट देशीय ग्रह होते हैं। रेखा से पश्चिम देश में पीछे ग्रह दर्शन होता है इसलिए वहां रेखा देशीय ग्रह से अभीष्ट देशीय ग्रह के अधिकत्व के कारण रेखा देशीय ग्रह में देशान्तर फल को जोड़ने से अभीष्ट ग्रह होते हैं, इति ॥३५॥

इदानीं तत्प्रतिपादनार्थमार्यामाह—

दिनवारादिः पश्चादुज्जयिनी दक्षिणोत्तरायाः प्राक् ।
देशान्तरघटिकाभिः पश्चात्प्राग् भवति रव्युदयात् ॥३६॥

वा. भा.—दिनवारादिर्वारप्रवृत्तिरित्यर्थः, सा पश्चाद् भवति, स्वरव्युदयात् । क्व उज्जयिनी दक्षिणोत्तरायाः प्राक् लंकासमयाम्योत्तररेखातः पूर्वदेशेष्वित्यर्थः । कियता कालेन देशान्तरघटिकाभिरिति, यतः पूर्वमेव रेखार्कोदयादर्कोदयस्तेषु देशेषु, न च तत्र काले वारप्रवृत्तिरद्यापि रेखार्कोदयकाले यस्माद् भविष्यतीति दिनवारादिः। स च देशान्तरघटिकाभिः व्यवहितं इत्यत उपपन्नं भुवो वृत्तत्वात् । पश्चात् प्राग्भवति इति, उज्जयिनी दक्षिणोत्तररेखातः पश्चात्तु देशेषु प्राग्भवति, स्वरव्युदयात्, पूर्वदिनवारप्रवृत्तिः पश्चाद्रव्युदयस्तेषु यस्माद्रेखार्कोदयादतः प्राग्वारादिः तेषु देशान्तर-

घटिकाभिरेव विषुवद्दिवस एतत्सर्वं गोले प्रदर्शयेत् । चरदलं चान्यत्रकाले स्वधिया योज्यम् , यथास्वभावम् । तत्रायं प्रयोगः स्वदेशान्तरघटिकाः पञ्चदशभ्यो विशोध्यावशेषा या घटिकास्ताभिः मध्याह्नात् प्राग्वारादिः स्वदेशे नित्यं वक्तव्याः । रेखातः प्राक्पश्चात्स्वर्धरात्रादूर्ध्वं तावतीभिरेव घटिकाभिर्दिनवारादिः प्रथमंपश्चाद्वा भवति, तन्न ज्ञायते ।

वि. भा.—उज्जयिनीदक्षिणोत्तरायाः (रेखातः) प्राग्देशे (पूर्वदेशे) रव्युदयात् (सूर्योदयात्) देशान्तरघटिकाभिः पश्चात् दिनवारादिः (वारप्रवृत्तिः) भवति, रेखातः पश्चात् (पश्चिमदेशे) रव्युदयाद्देशान्तरघटिकाभिः प्राक् (पूर्वं) दिनवारादिर्भवतीति ॥३६॥

अत्रोपपत्तिः

यतो लङ्कोदये वारादिरितिनियमेन रेखातः पूर्वदेशे सूर्योदयाद्देशान्तरघटीभिः पश्चात् , पश्चिमदेशे तु ताभिरेव घटीभिः पूर्वं वारप्रवृत्तिर्भवितुमर्हति, भास्कराचार्येणापि सिद्धान्तशिरोमणौ "अर्कोदयादूर्ध्वमधश्च ताभिः प्राच्यां प्रतीच्यां दिनप्रवृत्तिरित्यनेन" ब्रह्मगुप्तोक्तमेव कथ्यते, कदाप्रभृति वारप्रवृत्तिर्भवत्येतद्विषये बहूनामाचार्याणां बहूनि भिन्नानि मतानि सन्ति यथा आर्यभटसिंहाचार्यादयो रवेरुदयात् (अर्धोदितरविबिम्बात्) दिनारम्भकालं कथयन्ति, अन्ये दिनार्धात्कथयन्ति, लाटदेवादयो रवेरर्धास्तमयकालमारभ्य तं कथयन्ति, यवननृपतिर्निशि दशभिर्मुहूर्त्तैस्तं कथयति, लाटाचार्यः पुनः स्वसिद्धान्तेऽर्धरात्रौ तं कथयतीति । पञ्चसिद्धान्तिकायां वराहमिहिरेण तथैवोक्तम्—

दिनवारप्रवृत्तिर्न समा सर्वत्र कारणे कथिता ।
नेहापि भवति यस्माद्विप्रवदन्तेऽत्र दैवज्ञाः ॥
द्युगणाद्दिनवाराप्तिर्द्युगणोऽपि च देशकालसम्बन्धः ।
लाटाचार्येणोक्तो यवनपुरे चास्तगे सूर्ये ॥
रव्युदये लङ्कायां सिंहाचार्येण दिनगणोऽभिहितः ॥
यवना निशीह दशभिर्गतैर्मुहूर्तैश्च तद् गुरुणा ।
लङ्काऽर्धरात्रसमये दिनप्रवृत्तिं जगाद चार्यभटः ।
भूयः स एव चार्कोदयात्प्रभृत्याह लङ्कायाम् ॥

सिद्धान्तशेखरे श्रीपतिनाऽऽर्यभटादिमतखण्डनपुरःसरं ब्रह्मगुप्तमतानुरूपमेव स्वमतं कथ्यते यथा—

सृष्टेर्मुखे ध्वान्तमये हि विश्वे ग्रहेषु सृष्टेष्विनपूर्वकेषु ।
दिनप्रवृत्तिस्तदधीश्वरस्य वारस्य तस्मादुदयात्प्रवृत्तिः ॥

सिद्धान्तशेखरे एवैतस्य व्याख्या तट्टीकाकर्त्रा मल्लिभट्टेनैवं क्रियते—सृष्टेर्मुखे पूर्वं विश्वे ध्वान्तमये तमोमये भूते सति पश्चादिनपूर्वकेष्वर्कपूर्वकेषु ग्रहेषु सत्सु दिनप्रवृत्तिर्यस्मात्तस्मात् तदधीश्वरो यो ग्रहः अर्कादिस्तत्सम्बन्धिनो वारस्योदयादर्कोदयमारभ्य प्रवृत्तिरित्यर्थः। विश्वशब्दस्य नामत्वसंज्ञात्वात्तेन विश्वस्मिन्निति प्रयोगो न भवति, एतदुक्तं भवति सृष्टेः पूर्वं विश्वमन्धकमभूत्।

यथाह भगवान् मनुः—

आसीदिदं तमोभूतमप्रज्ञातमलक्षणम्।
अप्रतर्क्यमनाधृष्यं प्रसुप्तमिव सर्वतः॥ इति

नह्येवंविधे काले वारप्रवृत्तिविचारः सम्भवति वाराधीश्वराणां ग्रहाणामभावात् तस्माद्रात्रौ वारप्रवृत्तिपक्षो न सम्भवति नापि मध्याह्नास्तमयकालावारभ्य वारप्रवृत्तिः। तथा सति तावन्तं कालं वारेण दिनेन भवितव्यम्। तथा च लोकव्यवहारलोपप्रसङ्गः। तस्मादर्कोदयपक्ष एव श्रेयानिति। अस्मिन्नपि पक्षे देशभेदाद्वारप्रवृत्तिभेदो भवति यथा—

वारप्रवृत्तिं मुनयो वदन्ति सूर्योदयाद्रावणराजधान्याम्।
ऊर्ध्वं तथाऽधोऽप्यपरत्र तस्याश्चरार्धदेशान्तरनाडिकाभिः॥

एतदेव स्पष्टयति—

लङ्कोदग्याम्यसूत्रात्प्रथममपरतः पूर्वदेशे च पश्चा—
दध्वोत्थाभिर्घटीभिः सवितुरुदयतो वासरेशप्रवृत्तिः॥
ज्ञेया सूर्योदयात् प्राक् चरशकलभवैश्चासुभिर्याम्यगोले।
पश्चात्तैः सौम्यगोले युतिवियुतिवशाच्चोभयोः स्पष्टकालः इति॥३६॥

अब वारादि को कहते हैं

हि. भा.—उज्जयिनी की दक्षिणोत्तर रेखा (रेखा) से पूर्वदेश में सूर्योदय के बाद देशान्तर घटी करके दिनवारादि वारप्रवृत्ति होती है, रेखा से पश्चिम देश में देशान्तर घटी करके सूर्योदय से पहले दिनवारादि होती है॥३६॥

उपपत्तिः

लङ्कोदय काल में वारादि होती है. इस परिभाषा से रेखा से पूर्वदेश में सूर्योदय के बाद देशान्तर घटी करके वारप्रवृत्ति होती है, रेखा से पश्चिम देश में सूर्योदय से पहले उतने (देशान्तर घटी) ही काल पश्चात् प्रवृत्ति होती है। सिद्धान्तशिरोमणि में भास्कराचार्य भी "अर्कोदयाद्ऊर्ध्वमधश्च तासां प्राच्यां प्रतीच्यां दिनप-

प्रवृत्तिः" इससे इसी ब्रह्मगुप्तोक्त बात को कहते हैं। वारप्रवृत्ति कब से होती है, इस विषय में बहुत आचार्यों के भिन्न-भिन्न बहुत मत हैं। जैसे आर्यभट, सिंहाचार्य आदि आचार्य अर्धोदित रवि बिम्ब से दिनारम्भकाल को कहते हैं। अन्य आचार्य दिनार्ध से दिनारम्भ कहते हैं। लाटदेव आदि आचार्य रवि के अधस्तिकाल से उसको कहते हैं। यवन राजा रात्रि में दश मुहूर्त करके उसको कहते हैं, लाटाचार्य अपने सिद्धान्त ग्रन्थ में अर्धरात्रि से उसको कहते हैं। पञ्चसिद्धान्तिका में वराहमिहिराचार्य संस्कृतोपपत्ति में लिखित 'दिनवारप्रवृत्तिर्न समा सर्वत्र कारणे कथिता' इत्यादि पद्यों से इन्हीं विषयों को कहते हैं। सिद्धान्तशेखर में श्रीपति ने उपर्युक्त आर्यभटादि आचार्यों के मत का खण्डन किया है और ब्रह्मगुप्त कथित बात को स्वीकार किया है। जैसे—

सृष्टेर्मुखे ध्वान्तमये हि विश्वे ग्रहेषु सृष्टेष्विनपूर्वकेषु।
दिनप्रवृत्तिस्तदधीश्वरस्य वारस्य तस्मादुदयात्प्रवृत्तिः॥

सृष्ट्यादि से पहले विश्व अन्धकारमय था जो संस्कृतोपपत्ति में लिखित भगवान् मनु 'आसीदिदं तमोभूतमप्रज्ञातमलक्षणम्' इत्यादि के इस वचन से विदित होता है, ऐसे समय में वारप्रवृत्ति के विचार सम्भव नहीं हैं क्योंकि उस समय में वारेश्वरग्रहों का अभाव रहता है। इसी कारण से रात्रि में वारप्रवृत्ति कहने वालों के मत ठीक नहीं कहे जा सकते हैं। मध्याह्न काल और अस्तमय काल से आरम्भ कर वारप्रवृत्ति कहना भी ठीक नहीं है, क्योंकि इस बात को स्वीकार करने से उतने काल तक दिन मानना पड़ेगा जो कि असङ्गत है। इसलिए रव्युदय काल से वारप्रवृत्ति मानने वाले आचार्य का मत ही ठीक है। इस मत में भी देश भेद से वारप्रवृत्ति में भेद होता है, जैसे—

संस्कृतोपपत्ति में लिखित 'वारप्रवृत्तिं मुनयो वदन्ति सूर्योदयाद्रावणराजधान्याम्' इत्यादि पद्य से विदित होता है, इसी को संस्कृतोपपत्ति में लिखित 'लङ्कोदग्याम्यसूत्रात्प्रथममपरतः पूर्वदेशे च पश्चात्'— इत्यादि पद्य द्वारा स्पष्ट किया है इति॥३६॥

इदानीं तदर्थं देशान्तरकर्मणा त्रयेण प्रदर्शयति

भूपरिधिः खखखशरारेखा स्वाक्षान्तरांशसङ्गुणिताः।
भगणांशहृताः फलकृतिहीना देशान्तरस्य कृतिः॥३७॥
शेषपदगुणाभुक्तिर्भूपरिधिहृता कलादिलब्धमृणम्।
उज्जयिनीयाम्योत्तररेखायाः प्राग् धनं पश्चात्॥३८॥
मध्यग्रहे स्फुटे वा भूपरिधिहृतात् पदात् गुणात् षष्ट्या।
लब्धं घटिका अथवा कर्मतिथावृणधनं ग्रहवत्॥३९॥

वा. भा.—भूपरिधिः भूगोलस्य परिणाहः कियानित्यत आह खखखशरा इति पञ्चसहस्राणि योजनानामित्यर्थः। अत्र च देशान्तरकक्ष्योन्नत्यादयः प्रमाणं

भुवो महत्वनिराकरणगोलाध्याये च मयाऽयमर्थोऽतिविस्तरेण व्यावर्णितः पूर्वमेव । देशान्तरकर्मैव साम्प्रतमुच्यते । भूपरिधेर्ये खखखशराः ते रेखाक्षस्वाक्षान्तरांशसंगुणिता भगणांशहृताश्च सन्तः फलत्वं व्रजन्ति । रेखाख्योभीष्ट एव रेखास्थस्य देशसंबंधी अक्षो गृह्यते यस्य देशस्य स्वदेशेन सहान्तरयोजनानि ज्ञायन्ते । स्वाक्षस्तु पुनः ज्ञायत एव तयोरन्तरे ये अंशास्तैः खखखशरान् संगुण्य षष्टिशतत्रयेण विभजेदत्रायमर्थः स्वदेशेन सह तुल्याख्यो यो रेखायां स्थितो देशः तस्याभीष्टाक्षकस्य रेखास्थदेशस्य चान्तरं कियति योजनानि । इति तदर्थत्रैराशिकं यदि षष्टिशतत्रयभागकल्पितस्य भूगोलस्य पंचयोजनसहस्राणि तदस्येष्टभूगोलभागस्य रेखास्थस्वाक्षांतरांशमितस्य कियंति योजनानि भवन्तीति । तदर्थमक्षान्तरांशैः खखखशरान्संगुण्य भांशैश्च विभज्य फलं दक्षिणोत्तरयोजनात्मिकाभुजा रेखास्थयोरन्तरम् । स्वदेशस्य ज्ञाताध्वरेखास्थदेशस्य चान्तरं कर्णः । कर्णकृतेर्भुजाकृतिं विशोध्य शेषपदं कोटिः योजनात्मिका पूर्वापरा । स्वदेशतुल्याक्षरेखास्थदेशयोरन्तरम् ॥३७॥

तेन शेषपदेनेष्टग्रहस्य गुणा भुक्तिः, भूपरिधिहृता फलं भवति, यस्मादुक्तं फलकृतिहीना देशान्तरस्य कृतिः । शेषपदगुणा भुक्तिः भूपरिधिहृता कलादिलब्धमिति । अत्र तावत् स्वदेशे भूपरिधिरेव देशान्तरकर्मायोग्यः साध्यते ग्रहदेशान्तरफलानयनम् त्रैराशिकार्थं तद्यथा यत्र व्यासार्धतुल्यो विषुवद्वलंबकः तत्र पञ्चसहस्रः परिधिः यत्रेष्टौ विषुवद्वलम्बकः, तत्रः कः परिधिरिति स्वदेशावलंबज्यया खखखशरान्संगुण्य व्यासार्धेन विभजेत् । लब्धं स्वदेशप्रदेशे देशान्तरपरिधिर्यस्मात्स्वदेशाक्षांशान्नवतेः प्रोह्य शेषभागतुल्येन सूत्रेण मेरुं मध्यं कृत्वा यद्वृत्तमुत्पद्यते स देशान्तरकर्मपरिधिः । तया गत्या निरक्षपूर्वापरगतेत्यर्थः । नायमर्थः आचार्येणोक्त इति चेत् । उक्त एव स्वाक्षतुल्यरेखास्थदेशस्वदेशयोरन्तरे देशान्तराभ्युपगमात् । तत्र स्फुटभूपरिधिना यदभीष्टग्रहभुक्तिर्भवति, तद्देशान्तरयोजनैः किमिति सर्वेषां फलानयनमेव तच्च कलादिकं भवति, तद्दृणमुज्जयिनीयाम्योत्तररेखाया एव, अत्र वासना पूर्वमेव व्याख्याता ॥३८॥

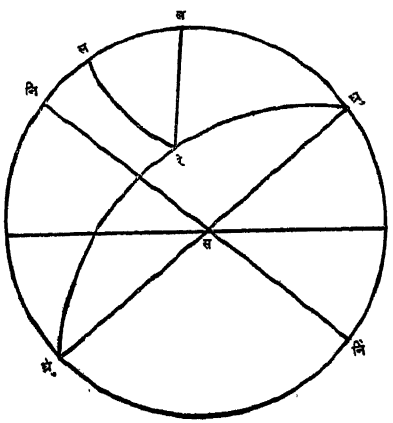
मध्यग्रहं स्फुटे चेति, यदि मध्यभुक्त्या तत् स्फुटग्रहे देशान्तरफलमित्यर्थः । एवं कृते स्वदेशे मध्यमो ग्रह उन्मण्डले भवति । अथवान्यदेशान्तरकर्मप्रकारः, भूपरिधिहृतात् यद्गुणात्षष्ट्या लब्धं घटिकाद्यथवा कर्मतिथिष्वृणं धनं ग्रहवत् । अत्र कालेन सह त्रैराशिकं यदि स्फुटपरिधियोजनैः षष्टिघटिका भवन्ति तदेष्टदेशांतरयोजनैः कियत्य इति लब्धाः घटिकाः कर्मयोग्यासु तिथिषु ग्रहवत्कार्याः यस्मात्ता एवतिथयः सावनी भविष्यन्त्यन्यथा सावनदिवसेषु तत्फलं भवति घटिकादिकम् । तथा च क्रियमाणेऽवमशेषान्तरं भवत्यत उक्तम् , कर्मतिथिष्विति । ततस्ताभिर्देशान्तरसंस्कृताभिः तिथिभिरहर्गणादिकं कृत्वा यद् ग्रहानयनं तद्देशान्तरकृतमेवागच्छति एतत् सर्वं देशान्तरकर्मयथास्थिते गोले प्रदर्शयेत् । यदत्र परिधिगतैर्भुजकोटिकर्णैः

कल्पना कृता सातिशोभनास्माकं न प्रतिभाति भुवो निम्नोन्नतत्वात्। इष्टरेखास्थदेशान्तरयोजनानां सम्यग् परिज्ञानाच्च महापर्वताद्यन्तरित्वाच्च, पथो वक्रत्वं यतः संभवति, लोकप्रसिद्धेरनैकांतिकत्वाच्च तस्माद्देशान्तरयोजनैर्ज्ञातैरेव कर्म कर्त्तव्यम्। देशान्तररेखा च योजने लिखे प्रथते। उज्जयिनीरोहीतककुरुयमुनाहिमनिवासिमरूणां देशान्तरं न कार्यम्। तल्लेखामध्यस्थत्वात्। अत्राचार्येणापि दृक्प्रग्रहणयोरन्तरादेशान्तरयोजनानयनमभिहितम्। ग्रहणोत्तरे यत्तदपि अस्माकं न प्रतिभाति। यतस्तिथ्यन्तचलनं बहुप्रकारमिति। एवं तावत्कल्पगताहर्गणेन ग्रहानयनमभिहितम् ॥३६॥

वि. भा.—भूपरिधिः खखखशराः ५००० रेखादेशस्वदेशयोरक्षांशान्तरेण गुणिता भगणांशै ३६० र्भक्ता सन्तो यत्फलं (लब्धं) तद्वर्गहीना देशान्तरस्य (योजनात्मकस्य रेखास्वदेशयोरन्तरस्य) कृतिः (वर्गः) शेषस्य पदेन (मूलेन) गुणा भुक्तिः (ग्रहगतिः) भूपरिधिहृता (स्पष्टभूपरिधिभक्ता) कलादिलब्धं यत्तत् उज्जयिनीयाम्योत्तररेखायाः पूर्वदेशे मध्यग्रहे स्फुटग्रहे वा ऋणं कार्यं, उज्जयिनीयाम्योत्तररेखायाः पश्चात् (पश्चिमदेशे) मध्यमग्रहे स्फुटग्रहे वा धनं कार्यम्। पदात् षष्ट्या गुणात् भूपरिधिहृतात् (स्पष्टभूपरिधिभक्तात्) लब्धं घटिकात्मकं फलं ग्रहवत्तिथावृणं धनं कार्यमिति ॥३७-३८-३९॥

अत्रोपपत्तिः

यदि व्यासमानम्=१५८१ तदा व्यासवर्गाद्दशगुणात्पदं भूपरिधिर्भवेदिति सूर्यसिद्धान्तोक्त्या भूपरिधिमान ५००० मागच्छति ब्रह्मगुप्तमते।



ध्रु-ख-ल-नि $\overset{1}{\text{ध्रु}}$=स्वयाम्योत्तरवृत्तम्। ध्रु, $\overset{1}{\text{ध्रु}}$ ध्रुवौ, ख=स्वखस्वस्तिकम्, नि=स्वनिरक्षखस्वस्तिकम्। ध्रु-रे $\overset{1}{\text{ध्रु}}$=लङ्कायाम्योत्तरवृत्ते रे कोऽपि रेखादेशः। नि-सनि'=नाडीवृत्तम्। रे-ल=नाडीवृत्तसमानान्तरं लघुवृत्तम्। ल-नि=तद्रेखादेशाक्षांश सम्बन्धीनि योजनानि। यदि भगणांशैर्भूपरिधियोजनानि लभ्यन्ते तदा स्वाक्षांशैः किमित्यनुपातेन स्वदेशाक्षांशयोजनानि=$\frac{\text{भूप}\times\text{अक्षांश}}{३६०}$, एवमेव यदि-

भगणांशैर्भूपरिधियोजनानि लभ्यन्ते तदा रेखादेशाक्षांशैः किमित्यनुपातेन

रेखादेशाक्षांशयोजनानि $= \frac{\text{भूप} \times \text{रेखाक्षांश}}{३६०}$, रे-ख=रेखास्वदेशयोरन्तरयोजनानि।

अत्राचार्येण रे-ल-ख त्रिभुजं सरलजात्यं स्वीकृतम्। तत्र रे-ल पूर्वापरान्तरं देशान्तरयोजनतुल्यम्। एतदानयनं क्रियते

$$\sqrt{\text{रेख}^२ - \text{लख}^२} = \sqrt{\text{रेख}^२ - \left\{\frac{\text{भूप}}{३६०}\left(\text{स्वाक्षांश} - \text{रेखाक्षांश}\right)\right\}^२} = \text{रे-ल},$$

ततोऽनुपातो यदि स्फुटपरिधिना ग्रहगतिर्लभ्यते तदा देशान्तरयोजनैः किमित्यनुपातेन यत्कलात्मकं फलमागच्छति तद्रेखातः पूर्वदेशे पश्चिमदेशे च मध्यग्रहे स्फुटग्रहे वा क्रमशो हीनं धनं कार्यं तदा स्वदेशीयग्रहो भवति। एवं यदि स्पष्टभूपरियोजनैः षष्टिघटिका लभ्यन्ते तदा देशान्तरयोजनैः किमित्यनुपातेन यद् घट्यात्मकं फलमागच्छति तत्पूर्वपश्चिमदेशवशेन ग्रहवत्तिथावृणधनं कार्यमिति,

ब्रह्मगुप्तेनात्र स्फुटभूपरिधेश्चर्चा न क्रियते, मध्यमभूपरिधिसम्बन्धेनैव देशान्तरघट्यानयनं कृतमतो न समीचीनमिति विवेचनीयं विज्ञैरिति ॥३७-३८-३९॥

अब देशान्तर कहते हैं

हि. भा.—भूपरिधि ५००० पांच हजार है, इसको स्वदेश और रेखादेश के अक्षांशान्तर से गुणा कर भगणांश (३६०) से भाग देने से जो फल हो उसके वर्ग को देशान्तर (रेखादेश और स्वदेश के योजनात्मक अन्तर) वर्ग में घटाकर जो शेष रहे उसके मूल से ग्रहगति को गुणकर भूपरिधि (स्पष्टभूपरिधि) से भाग देने से जो कलात्मक फल हो उसको उज्जयिनीयाम्योत्तररेखा से पूर्वदेश में मध्यमग्रह में या स्फुटग्रह में ऋण करना, पश्चिमदेश में धन करना चाहिये, पद याने मूल (देशान्तर योजन) को साठ से गुणा कर भूपरिधि (स्पष्टभूपरिधि) से भाग देने से जो घट्यात्मक फल हो उसको पूर्व और पश्चिम देश में ग्रहवत् तिथि में (तिथिभुक्त घटी) संस्कार करना चाहिये ॥३७-३८-३९॥

उपपत्ति

यदि व्यासमान=१५८१ मानते हैं तब 'व्यासवर्गाद्दशगुणात्पदं भूपरिधिर्भवेत्' इस सूर्यसिद्धान्तोक्त प्रकार से भूपरिधिमान=५००० आता है, यही ब्रह्मगुप्तमत में भूपरिधिमान है। संस्कृतोपपत्ति में जो क्षेत्र लिखा हुआ है, वही यहां देखना चाहिये।

ध्रु-ख-ल-ध्रु=स्वयाम्योत्तरवृत्त है, ध्रु, ध्रु दोनों ध्रुव हैं। ख=स्वखस्वस्तिक है, नि=स्वनिरक्ष खस्वस्तिक है, धु-रे ध्रु लंकायाम्योत्तरवृत्त में रे कोई रेखादेश है, नि-स-नि=नाडीवृत्त है, रे-ल=नाडीवृत्त के समानान्तर लघुवृत्त है, ल-नि=रेखादेशाक्षांश-

सम्बन्धियोजन है, अब अनुपात करते हैं यदि भगणांश में भूपरिधियोजन पाते हैं तो स्वाक्षांश में क्या इस अनुपात से स्वदेशाक्षांशयोजनमान $=\frac{\text{भूप}\times\text{अक्षांश}}{३६०}$, इसी तरह भगणांश में यदि भूपरियोजन पाते हैं तो रेखादेशाक्षांश में क्या इस अनुपात से रेखादेशाक्षांशयोजन $=\frac{\text{भूप}\times\text{रेखाक्षांश}}{३६०}$, रे-ख=रेखादेश और स्वदेश के अन्तर योजन हां आचार्य ने रे-ल-ख त्रिभुज को सरलजात्य मान लिया है, रेल पूर्वापरान्तर=देशान्तर-योजन इसका आनयन करते हैं

$$\sqrt{\text{रेख}^२-\text{लख}^२}=\sqrt{\text{रेख}^२-\left\{\frac{\text{भूप}}{३६०}(\text{स्वाक्षांश}-\text{रेखाक्षांश})\right\}^२}=\text{रेल},$$ अब अनुपात करते हैं, यदि स्फुटपरिधि में ग्रहगति कला पाते हैं तो देशान्तरयोजन में क्या इस अनुपात से जो कलात्मक फल आता है उसको रेखा से पूर्वदेश में और पश्चिमदेश में मध्यमग्रह में या स्फुटग्रह में क्रम से ऋण और धन करने से स्वदेशीय ग्रह होते हैं। इसी तरह यदि स्फुट भूपरियोजन में साठ घटी पाते हैं तो देशान्तरयोजन में क्या इस अनुपात से जो घट्यात्मक फल आता है उसको पूर्व और पश्चिम देशवश से ग्रह की तरह तिथि (तिथि भुक्तघटी) में ऋण और धन करना चाहिये, यहां आचार्य (ब्रह्मगुप्त) स्फुट-भूपरिधि की चर्चा नहीं करते हैं, मध्यमभूपरिधि के सम्बन्ध ही से देशान्तर घट्यादि साधन करते हैं इसलिये यह ठीक नहीं है, इस विषय को विज्ञ लोग सोचें इति ॥ ३७-३८-३९ ॥

इदानीं वर्षादौ दिनाद्यवमदिनादिसाधनमाह

कल्पगताब्दा गुणिता रूपाष्टजिनैर्नवाग्निसप्तनगैः ।
खखरसनवभिर्भक्ता दिनावमान्यंशकाः शेषाः ॥ ४० ॥

वा. भा.—अभीष्टे रविमण्डलान्ते ये कल्पगताब्दाः तेऽत्र गृह्यन्ते कल्पगताब्दाः गुणिताः कैरित्याह। रूपाष्टजिनैः २४८१ अन्यत्र नवाग्निसप्तनगैः ७७३९ तत उभयतोऽपि खखरसनवभिर्भक्ताः ९६०० फलानि यथासंख्यं दिनान्यवमानि चांशकाः शेषा द्वयोरपि स्थानयोः एतदुक्तं भवति। कल्पगताब्दा रूपाष्टजिनैः संगुण्य खखरसनवभिर्विभजेत्फलं दिनानि भवन्ति। तत्र यद्विकलं तदु दिनांशशब्देनोच्यते, तत्र दिनानि दिनांशश्च स्वच्छेदेन सहैकान्ते स्थापयेदत्रेयं वासना यदि कल्परविवर्षैः कल्पसौरसावनान्तरदिनानि लभ्यन्ते खचतुष्टयशरकृतरसेन्दुनगद्वियमसंख्यानि, २२७१६४५०००० तदेकेन रविवर्षेण किमितिलब्धं ५ विकलम् खचतुष्टयशरवेदरसरुद्रचन्द्रसंख्यं छेदश्च सप्तशून्यानि रदवेदा $\frac{११}{४३}\frac{१६४५०००००}{०००००००}$ अनयोश्छेद्यछेदकराश्योरपवर्त्तनं कृत्वा खचतुष्टयभूतवेदै जातौ राशी रूपाष्टजिना उपरि, खखरसनवाधः $\frac{३४८१}{६४००}$ एतावद्विकलं पंचानां दिना-

नामुपर्यकस्मिन् रविवर्षेऽतस्त्रैराशिकमाचार्येण प्रकल्पितम् । तद्यथा यद्येकस्य रविवर्षस्य दिनांशविकलमेतत्कल्पादेरारभ्य ये गताब्दाः तेषां किमिति

कल्पगताब्दा रूपाष्टजिनैः संगुण्यते रूपेण खखरसनवसंख्येन गुणकारच्छेदेन च विभज्यते । फलं दिनानि सविकलानि भवंति । ततः पुनरपि कल्पगताब्दान्नवाग्निसप्तनगैः संगुण्य खखरसनवभिर्विभजेत्, फलमवमानि भवन्ति यद्विकलमवमांशशब्देनोच्यते ऽन्यत्राथवा अवमाशांश्च स्वच्छेदेन सहैकांते स्थापयेत् । तत्रोपपत्तिर्यदि कल्परविवर्षैः कल्पोनरात्रा लभ्यन्ते तदेकेन वर्षेण किमिति फलं पंचावमानि लभ्यन्तेऽवमानि ५ शेषं विकलं खचतुष्टयशरबाणयमवसुकृताग्निसंख्यकल्परविवर्षच्छेद्यं तच्चेदं $\frac{३४५८३४५५}{४३२०००००}$ अनयो राश्योरपवर्त्तनं कृत्वा खचतुष्टयभूतवेदैर्जातौ राशी नवाग्निसप्तनगा उपरि खखरसनवाधः $\frac{७७३९}{४३२०}$ एतावता विकलेन पञ्चोनरात्राण्यधिका भवन्ति । रव्यब्देऽतस्त्रैराशिककल्पना कृताचार्येण । यद्येकस्य रविवर्षस्य पञ्चभ्योऽवमेभ्यः एतावदधिकमवशेषं तदा कल्पगतानां किमित्यतः कल्पगताब्दा नवाग्निसप्तनग संगुण्यन्ते रूपेण खखरसनवभिः विभज्यन्ते फलमवमानि शेषमवमानि दिनानि शेषं कल्पवर्षैः सावनदिनैः रविवर्षोत्थसावनदिनानयने प्रथमं त्रैराशिकं यदि कल्पसावनवर्षैर्दिनैः कल्परविदिनानि वर्षाणि लभ्यन्ते तदेकेन रविवर्षेण किमिति न्यासः ४३२०००००००। २५७७९१६४५०००० । १११६६४५०००००००० फलं ३६५ । १५७७९१६४५०००० । २५०८२५५०००० । ४३२०००००००० एभिः कल्पसावनदिनैः सहावमानयनार्थं द्वितीयत्रैराशिकम् । कल्पभूदिवसैः कल्पावमानि लभ्यन्ते तदेभिः रविवर्षोत्थसावनदिवसैः किमिति न्यासः सवर्णिता रविवर्षोत्थसावनाः अत्रोनरात्राणां भूदिवसतुल्यौ गुणकारभागहारौ तयोर्नाशे कृते कल्पोनरात्राणामेकको गुणकारः कल्परविवर्षाणि भागहारो फलं तावन्त्येवावमानि लब्धानि $\frac{७७३९}{४३२०}$ ॥४०॥

वि. भा.—कल्पगताब्दा (कल्पादितो गतवर्षाणि) एकत्र रूपाष्टजिन २४८१ रन्यत्र नवाग्निसप्तनगैः ७७३९ गुणिताः खखरसनवभिः ९६०० भक्तास्तदा फलं प्रथमस्थाने दिनानि (सावनदिनानि) द्वितीयस्थानेऽवमानि (अवमदिनानि) स्युः, शेषाः क्रमेणांशकाः (दिनांशका अवमांशकाश्च) बोध्या इति ॥४०॥

अत्रोपपत्तिः

गताब्दा विभक्ताः समुद्रैः खसूर्यैः खखाङ्काङ्ककैर्वा फलैक्यं दिनाद्यमिति भास्करोक्तसूत्रेण $\frac{\text{गव}}{४}+\frac{\text{गव}}{१२०}+\frac{\text{गव}}{९६००}=\frac{२४०० \text{ गव}+८० \text{ गव}+\text{गव}}{९६०}$

$=\frac{२४८१ \text{ गव}}{९६००}=$दिनाद्यम् ।

तथा स्वषष्ट्यंश हीनाब्दखाङ्गेन्दुभागः स्वपञ्चांशहीनाब्दयुग्बाक्षयाहा इति सिद्धान्तशिरोमणिस्थभास्करोक्तपद्येन गव— $\frac{\text{गव}}{५}+\frac{\text{गव}-\frac{\text{गव}}{६०}}{१६०}$=क्षयाहाद्यम्।

अत्र सवर्णेन $\frac{४\text{ गव}}{५}+\frac{६०\text{ गव}-\text{गव}}{९६००}=\frac{४\text{ गव}}{५}+\frac{५९\text{ गव}}{९६००}=\frac{७६८०\text{ गव}+५९\text{ गव}}{९६००}$

$=\frac{७७३९\text{ गव}}{९६००}$=क्षयाहादिः एतेनोपपन्नमाचार्योक्तमिति ॥४०॥

अब वर्षादि में दिनादि और अवमदिनादि साधन को कहते हैं

हि. भा.—कल्पादि से जो गतवर्ष हैं उनको एक स्थान में २४८१ इनसे और द्वितीय स्थान में ७७३९ इनसे गुणाकर दोनों स्थानों में ९६०० इनसे भाग देने से फल (लब्धि) प्रथम स्थान में दिन (सावनदिन) और द्वितीय स्थान में अवमदिन होते हैं, शेष क्रम से दिनांश और अवमांश समझने चाहियें इति ॥४०॥

उपपत्ति

'गताब्दा विभक्ताः समुद्रैः खसूर्यैः' इत्यादि सिद्धान्तशिरोमणिस्थभास्करोक्त सूत्र से

$\frac{\text{गव}}{४}+\frac{\text{गव}}{१२०}+\frac{\text{गव}}{९६००}=\frac{२४००\text{ गव}+८०\text{ गव}+\text{गव}}{९६००}=\frac{२४८१\text{ गव}}{९६००}$=दिनादि।

तथा 'स्वषष्ट्यंशहीनाब्दखाङ्गेन्दुभागः स्वपञ्चांशहीनाब्दयुग्बाक्षयाहा' इस भास्करोक्त सूत्र से गव—$\frac{\text{गव}}{५}+\frac{\text{गव}-\frac{\text{गव}}{६०}}{१६०}=\frac{४\text{ गव}}{५}+\frac{५९\text{ गव}}{६०\times १६०}=\frac{४\text{ गव}}{५}$

$+\frac{५९\text{ गव}}{९६००}=\frac{७६८०\text{ गव}+५९\text{ गव}}{९६००}=\frac{७७३९\text{ गव}}{९६००}$=क्षयाहादिः, इससे आचार्योक्त उपपन्न हुआ ॥४०॥

इदानीं वर्षादावधिमाससाधनमाह

तद्दिग्गुणाब्दयोगा अधिमासार्स्त्रिंशता हृता लब्धम्।
शेषास्तिथयः शुद्धिर्दिनानि विकलं दिनांशेभ्यः ॥४१॥

वा. भा.—तद्दिग्गुणाब्दयोगादधिमासार्स्त्रिंशता गुणा लब्धमिति तदित्यनेनांतरागतानां दिवसानामवमानां च परामर्शः। दिग्गुणाब्दाश्च कल्पगताब्दा दशगुणिता उच्यन्ते। तेनायमर्थः तेषां दिनावमानां दशगुणितकल्पगताब्दान यो योगस्तस्मादधिमासा भवन्ति। त्रिंशता हृता लब्धं यत्। इयमत्र युक्तिः

यानि पञ्चदिनानि सौरसावनयोरन्तरोत्पन्नानि प्रागार्यायां प्रदर्शितानि यानि च पञ्चावमानि सावनचान्द्रयोरंतरोत्पन्नानि रव्यब्दे तानि मिलितानि दश भवन्ति अतस्त्रैराशिकम्। यद्येकेन रविवर्षेण दशदिनानि, तत् कल्पगताब्दानां कियंतीत्यतो दशभिः कल्पगताब्दा गुण्यन्ते। एकेन च विभज्यन्ते, ततः पञ्चकराश्याप्तं विकल्पद्वयं। ततोऽपि प्रागार्योक्तविधिना यानि दिनावमानि चाप्तानि तेषु गुणिताब्देषु संयोज्य कल्पादेरारभ्य गताधिमासकदिनानि भवन्ति, यतः सौरमानमूलत्वेन सर्वमानानां परिच्छित्तिः। तत्प्राधान्याद् ग्रहगत्युपलब्धावतः सौरवर्षं यदा रविदिवसै परिच्छिद्यन्ते तदा षष्टिशतत्रयं भवन्ति। यदा सावनदिवसैः परिच्छिद्यन्ते तदा पंचषष्ट्यधिकानि त्रीणि शतानि रूपाष्टजिनतुल्यं विकल्पसहितानि भवन्ति यदा पुनश्चान्द्रैर्दिवसैः परिच्छिद्यन्ते, तदैकसप्तत्यधिकानि त्रीणि शतानि भवन्ति। सविकलसहितानि तद्विकलं च खयमरससंख्यं खखनवरसनवच्छेद्यमधिकदिनान्यपि सौरचान्द्रदिनानामन्तरस्थान्युच्यन्ते। अतस्तथैवाचार्येणैव निबद्धं; प्रथमं सौरसावनान्तरं साधितं। तच्च सौरे निक्षिप्य सावनं भवति। अतस्तेषां सावयवानां सावनानामूनरात्राण्यपि साधितानि। तानि सावने प्रक्षिप्य चान्द्रो भवति। सौरेण सहान्तरे कृतेऽधिमासदिनान्येव केवलान्यतिरिच्यन्ते तथैवास्माभिरथाधिकमेवोदाहृतमुपपन्नम्। ततोधिकदिनानां त्रिंशता भागे हृते कल्पगता अधिकमासा भवन्ति तैर्न प्रयोजनम् वर्षान्तिकाहर्गणे शेषास्तिथयः शुद्धिदिनानि विकलं दिनांशेभ्य इति चन्द्रदिनानि तिथयो भवन्त्येव शुद्धिरपि भवति। शोधनत्वात् विकलं दिनांशेभ्यः कृत्वा सावनदिनानि तान्येव भवन्ति, चैत्रादिप्रथमार्कोदयरविमण्डलान्तयोरन्तरे सावनोऽहर्गणो भवतीत्यर्थः। एतदुक्तं भवति, तद्दिग्गुणाष्टयोगे त्रिंशदुद्धृते शेषाण्यधिमासकदिनानि चान्द्राण्यवशिष्यन्ते। तान्येव सावनदिनानि चैत्राद्यार्कोदयात्प्रभृति कल्प्यन्ते। यतस्तेषां शुद्धिदिनानां सम्बन्धि यदवमशेषं तच्चैत्रादि तदर्कोदयान्तरमवशेषे मिलितं तिष्ठति। येऽवमांशाः भवद्भिरभिधीयन्ते। अन्यथामावस्यान्तामवशेषं तदर्कोदयामावास्यां तयोरन्तरं भवति तच्च पृथक्भूतामाचार्येणानीतम्। शुद्धिदिनान्यवमावनदिनानि कल्पितानि, एवं कृतेऽमावास्यानां तदार्कोदययोरन्तरं शुद्धं चान्द्रसावनदिनानामन्तरं चावमांशैः संगृहीतं भवति। तत ऊर्ध्वं शुद्धिसावनदिनैः संगृहीतं तावद्यावदर्कोदयदिने रविमण्डलान्ते भविष्यति। चैत्र प्रथमार्कोदयात्पूर्वं तत ऊर्ध्वं तत्रैव दिने या घटिकाः तदर्को रविमण्डलान्तरे स्थिताः ताश्च दिनांशेभ्यः षष्टिगुणेभ्यो विभक्तेभ्यः स्वच्छेदेन भवन्ति यस्मात् तद्दिग्गुणाब्दयोगो यदा कृतः तदा सावनदिनानामधो दिनांशा आसंस्तेभ्यश्च या घटिकास्ता अपि तावन्त्यो भवन्तीत्युपपन्नम्। एवं शुद्धि दिनघटिकांतिकां वर्षोपयोगिनीं स्थापयेदवमांशांश्च ॥४१॥

वि. भा.—तत् (तयोः) पूर्वसाधितदिनादिक्षयाहाद्योर्दशगुणितगतवर्षस्य च योगास्त्रिंशता हृताः (भक्ता) लब्धं गताधिमासा भवन्ति, दिनांशेभ्योऽवमदिनां-

शेभ्यो विकलं सद्दिनानि शुद्धिर्भवत्यर्धादधिशेषतिथयः क्षयशेषरहिताः सत्यः शुद्धिसंज्ञकं भवतीति ॥ ४१ ॥

अत्रोपपत्तिः

एकस्मिन् वर्षे सावनदिनानि=३६५।१५।३०।२२।३०=३६५+
एकवर्षसम्बन्धिदिनाद्यम् ।

एकस्मिन् वर्षे क्षयाहाद्यम् =५।४८।२२।७।३०=५+
एकवर्षसम्बन्धिक्षयघट्यादि ।

द्वयोर्योगेन

एकस्मिन् वर्षे चान्द्राहाः =३७१।३।५२।३०=३७०+एकवर्षसंदिनादि
+एकवर्षसम्बन्धिघट्यादि

एकस्मिन् वर्षे सौराहाः =३६० =३६०

अनयोरन्तरेण

एकस्मिन् वर्षेऽधिदिनानि =११।३।५२।३० = १०+एकवर्षसंदिनादि
+एकवर्षसम्बन्धिक्षयाहादि

ततोऽनुपातेन

$$\text{गताधिमासाः}=\frac{\text{१ वर्ष सं अधिदिन}\times\text{गतवर्ष}}{\text{१ वर्ष}\times३०}$$

$$=\frac{(\text{१०+एकवर्षसंदिनादि+एकवर्ष सम्बन्धिक्षयाहादि})\text{ गतवर्ष}}{३०}$$

$$=\frac{\text{१०गव+एकवर्षसंदिनादि}\times\text{गव+एकवर्षसंक्षयाहादि}\times\text{गव}}{३०}$$

$$=\frac{\text{१० गव+गतवर्ष सं दिनादि+गतवर्षसंक्षयाहादि}}{३०}$$

अत्र अधिशेषे—वर्षान्तक्षयघटी=शुद्धिरितिपरिभाषितमत उपपन्नमाचार्योक्तमिति ॥ ४१ ॥

अब वर्षादि में अधिमासानयन करते हैं

हि. भा.—पूर्वसाधित दिनादि—क्षयाहादि और दशगुणित गतवर्षों के योग को तीस से भाग देने से अधिमास होता है, दिनांश (अवमदिनांश) से जो शेष रहता है

वह दिनादि शुद्धि होती है अर्थात् अधिशेष तिथि में क्षयशेष को घटाने से शेष शुद्धि संज्ञक होता है ॥ ४१ ॥

उपपत्ति

एकवर्ष में सावन दिन = ३६५।१५।३०।२२।३० = ३६५ + १ वर्षसंदिनादि
एकवर्ष में क्षयाहादि = ५।४८।२२। ७।३० = ५ + १ वर्षसंक्षयघट्यादि

दोनों के योग करने से

एकवर्ष में चान्द्रदि = ३७१।३।५२।३० = ३७० + १ वर्षसंदिनादि + १ वर्षसंक्षयाहादि
एकवर्ष में सौरदिन = ३६० = ३६०

दोनों के अन्तर करने से

एकवर्ष में अधिदिन = ११।३।५२।३० = १० + १ वर्षसंदिनादि + १ वर्षसंक्षयाहादि

∴ अनुपात करने से

$$\text{गताधिमास} = \frac{\text{१ वर्षसंअधिदिन} \times \text{गतवर्ष}}{\text{१ वर्ष ३०}}$$

$$= \frac{(\text{१०} + \text{१ वर्ष सदिनादि} + \text{१ वर्षसंक्षयाहादि}) \text{ गव}}{\text{३०}}$$

$$= \frac{\text{१० गव} + \text{१ वर्ष संदिनादि} \times \text{गव} + \text{१ वर्षसंक्षयाहादि} \times \text{गव}}{\text{३०}}$$

$$= \frac{\text{१० गव} + \text{गतवर्ष संदिनादि} + \text{गतवर्षसंक्षयाहादि}}{\text{३०}}$$

यहां अधिशेष—वर्षान्तक्षयशे = शुद्धिसंज्ञक, इससे आचार्योक्त उपपन्न हुआ ॥४१॥

इदानीं वर्षेशानयनं लघ्वहर्गणानयनं चाह

कल्पगताब्ददिनयुतः सूर्याद्योऽब्दाधिपोऽब्दभगणवधः ।
कल्पाब्दहृतो भगणादिमध्यमाः सूर्यभगणान्ते ॥ ४२ ॥
चैत्रसिताद्यास्तिथयः शुद्धिविहीनाः पृथक्गुणा रुद्रैः ।
अवमांशेभ्यो यमनवरसगुणितेभ्यो विभक्तेभ्यः ॥ ४३ ॥
स्वच्छेदेन फलयुता हृतास्त्रिखाग्नैः ७०३ फलावमविहीनाः ।
रविमेषादिद्युगणो मुनिहृत् शेषोऽब्दपत्यादिः ॥ ४४ ॥

वा. भा.—कल्पगताब्दानामन्तरानीतदिनानां च या युति, ततो वाराधिपो वर्षान्तिकस्याहर्गणस्य रव्यादिको भवत्यत्रेयं वासना रविवर्षणसावनदिनानां पञ्चषष्ट्याधिकं शतत्रयं सविकलं भवति। तत्र पञ्चषष्ट्याधिकेन शतत्रयेण सप्तहृते सत्येकदिवसो ऽवशिष्यते। वाराधिपोऽपि सावनदिनमेकं भुंक्ते, अतोब्दसमवारा गृह्यन्ते। विकलोत्था अपि सावनदिवसा रूपाष्टजिनैरित्यादिना ये कृताः तावन्त एव वारास्तत्र गतवर्षतुल्यसंख्यावारेषु प्रक्षिप्यन्ते। ततो वारगणात् सप्तहृतावशेषः सूर्याद्यो दिनपः तद्दिनार्कोदये भुक्तो लभ्यते। तदनन्तरग्रहो वाराधिपतिः, तत्र दिने भवति स एवाब्दाधिपतिर्भवति, यतो रविमेषादिकस्थाहर्गणस्य स एव प्रथमो दिवसोऽतः कल्पगताब्दयुतो यो वाराधिपः ततोऽपि यो ग्रहो द्वितीयो भवति तदादिका वारगणना मेषाद्यहर्गणस्था कार्या यतो भुक्तो वाराधिपो लभ्यते। अत्र सिद्धान्ते मेषादिद्युगणस्य तदादिका वारगणना कार्येति। अब्दभगणावधः कल्पाब्दहृत इति। सामान्योक्ता अप्यब्दाः कल्पगताब्दा गृह्यन्ते। तेषां भगणानाञ्च यो वधोऽसौ कल्पाब्दहृतो रविमण्डलस्थान्तिको मध्यग्रहो भवति, त्रैराशिकमेतत्। एवं सर्वग्रहादयो रविमण्डलान्तिका वर्षोपयोगिनः स्थाप्यन्ते।

चैत्रसितप्रतिपदाद्यभीष्टे दिने या व्यतीततिथिसंहितास्ता अत्र गृह्यन्ते चत्रसिताद्यास्तिथयः। शुद्धिविहीनाः पृथग्गुणा रुद्रैरिति चैत्रसिताद्यास्तिथयः गताः ताभ्यः शुद्धि विशोध्य शेषास्तिथयः पृथग्तष्टाः रुद्रगुणकार्या इत्यर्थः। ततोऽवमांशेभ्यो यमनवरसगुणितेभ्यो विभक्तेभ्यः स्वच्छेदेन फलयुता रविमण्डलान्ते येऽनन्तरमेवानीता अवमांशाः तान्यमनवरसैः संगुण्य खखरसनवसंख्येन स्वच्छेदेन विभज्य यदवाप्तं फलं तेन युतास्ताः कार्या इति। तत एवंविधास्ता हृतास्त्रिखागैः ७०३ यत्फलं तान्यवमानि तैर्विहीना, पृथक्स्थो रविमेषादिद्युगणो भवति, यस्मिन् काले मध्यमो रवि मेषं याति तत्कालात्प्रभृतिसावनान्तोहर्गणो भवति सविकलो मुनिहृतशेषः सप्तविभक्तशेषेऽब्दपत्यादिरेव समादिनाधिपस्तत्र दिने भवतीति वाक्यशेषः यदा पुनश्च चैत्रसितादिभ्यस्तिथिभ्यः शुद्धिर्न शुध्यति तदा विपरीतशोधने कृते शेषो रविमण्डलान्ताद्विपरीताहर्गणो भवति, तमेकादशगुणं रविमण्डलान्तावमाद्विशोधयेत्। शेषं तत्र दिनेऽवमशेषं भवति, न शुध्यति चेत्तदा गुणखमुनियुताद्रविमण्डलान्तावमशेषादिको न विपरीताहर्गणं रुद्रगुण विशोधयेदेकोन एव यतस्तत्र विपरीताहर्गणो भवति। वारोऽपि रविमण्डलान्तवारविपरीतवारगणनया गणनीया इति, मध्यमाश्च भौमाद्या रविमण्डलान्तिका विपरीताहर्गणफलेनोना तद्दैवसिका भवन्तीत्यत्रेयं वासना चैत्रसिताद्यास्तिथयो यदा शुद्धिसावनदिनैः नूनं क्रियते। तदा चैत्राद्यवमशेषे रव्युदयामावस्यां तयोरन्तरं शुद्धिदिनावमशेषं चान्द्रसावनदिनानामन्तरं द्वे अप्येकत्र मिलितेऽवमांशत्वं गते भवतः अवमांशा अधिकाः शुद्धोनाः सतिथिभ्यो रविमण्डलान्तचैत्रसिताद्योरन्तरं चान्द्रशुद्धं भवति सर्वं केवलमवमांशांशं अद्यापि शुध्यतीत्यर्थः। पृथक्गुणा-

रित्यत्र त्रैराशिकं यदिति त्रिखमुनिसंख्यैश्चन्द्रदिनैरेकादशावमानि भवन्ति, तत्प्रतिमण्डलान्तादारभ्य यास्तिथयो गताः सविकलाः तासां किमित्यत एकादशगुणानां तासां त्रिखसंख्यो भागहारः। एवं स्थिते रविमण्डलान्ते यदवमशेषं तत्तत्र योज्यते यतः शुद्धिविशोधनकाले तन्न शोधितं यदेवावमशेषं तद्योज्यते, तदेव शुद्धं भवति, चन्द्रदिनान्युपरि शुद्धानि स्थितानि भवन्ति। अतोऽवमांशास्त्रिखागैर्गुणिताः सवर्णाः भवन्तीत्येवं सवर्णो योजयितुं प्राप्तः, एकादशगुणितासु तिथिषु यावदवमांशास्तेष्वेव तिथिष्वधिकास्तिष्ठन्ति ते च तिथिभिः सह एकादशगुणाः सम्पन्नाः अतस्त्रिखागेभ्य एकादशसंशाध्य द्विनवरसा एवावमांशानां गुणकारः स्वच्छेदो भागहारश्च फलं रुद्रगुणितासु तिथिषु प्रयोज्यावमशेषं राशिर्भवति। ततस्त्रिखागैर्विभज्योनरात्रा लभ्यन्ते शेषामिष्टदिने सावनमवशेषं लब्धोनरात्रांश्च रविमण्डलान्तात् गततिथिगणात् विशोध्य रविमेषाद्यहर्गणो भवतीत्युपपन्नं सचाहर्गणो मुनिहृतशेषोऽब्दपत्यादि यो ग्रहो भवति, ततोऽपि यो द्वितीयः सावनाधिपो भवति यतः सविकलोऽहर्गण इति ॥४२-४३-४४॥

वि. भा.—कल्पादेर्ये गताब्दास्तेषां गतवर्षसम्बन्धिदिनाद्यानाञ्च योगः सूर्याद्योऽब्दाधिपः (रव्यादिवर्षपतिः) भवति, अब्दभगणवधः (कल्पगतवर्षकल्पग्रहभगणघातः) कल्पाब्दहृतः (कल्पवर्षभक्तः) तदा सूर्यभगणान्ते (सौरवर्षान्ते) भगणादिमध्यमा ग्रहा भवन्ति चैत्रशुक्लप्रतिपदादितो वर्त्तमानमासेष्टतिथिपर्यन्तं यास्तिथयस्ताः शुद्धिरहिताः पृथक् (स्थानद्वये स्थापिताः) रुद्रैः (एकादशभिः) गुणिताः यमनवरस ६९२ गुणितेभ्योऽवमांशेभ्यः (वर्षान्तक्षयशेषेभ्यः) स्वच्छेदेन (९६००) विभक्तेभ्यः फलयुताः, त्रिखागैर्हृताः (७०३ एभिर्भक्ताः) फलावमविहीनाः पूर्वस्थापितास्तदा रविमेषादिद्युगणो (यस्मिन् काले रविर्मेषं याति तस्मात्कालादहर्गणः) भवति, मुनिहृत् (सप्तभक्तः) शेषोऽब्दपत्यादिर्भवतीति ॥४२-४३-४४॥

अत्रोपपत्तिः

एकस्मिन् वष सावनदिनादयः=३६५।१५।३०।२२।३० ततोऽभीष्टवर्षान्ते सावनदिनादयः=(३६५।१५।३०।२२।३०) गव=३६५ गव+गव (१५।३०।२२।३०)=३६५ गव+गतवर्षसंदिनादि=कल्पादित इष्टवर्षान्ते सावयवोऽहर्गणस्तत्र दिनादिबोधकं द्वितीयखण्डमेवास्ति, अत्र सप्तभक्तावशिष्टोऽहर्गणो वारनियामकस्तेनात्र ३६५ यावत्सप्तभिर्विभज्यते तावद्रूपसममेव शेषम्। तेन कल्पगताब्ददिनयुत इत्युक्तं युक्तम् रव्यादिको वर्षपतिर्भवेत्कल्पादौ रविवारस्य सद्भावादिति। अथ रव्यब्दान्ते ग्रहानयनं क्रियते यदि कल्पवर्षैः कल्पग्रहभगणा लभ्यन्ते तदा गतवर्षैः किमित्यनुपातेन रविवर्षान्ते (सूर्यभगणान्ते) मध्यमा ग्रहाः समागच्छन्तीत्यत आचार्योक्तमुपपन्नम् ॥४२॥

लघ्वहर्गणसाधनार्थं वर्षादितो गततिथयः = इष्टतिथि—अधिशेतिथि, तथा स्वल्पान्तरात् ७०३ चान्द्रदिनैरेकादश ११ मितान्यवमानि भवन्तीति स्वीकृत्यानुपातेन यदि ७०३ चान्द्रदिनैरेकादशमितान्यवमानि लभ्यन्ते तदा वर्षादितो गततिथिभिः किमित्यनुपातेन यत्फलं तत्र वर्षादिक्षयशेषयुतं तदाऽवमानि भवन्ति

$$\text{तत्स्वरूपम्} = \frac{११\ (\text{इष्टतिथि}—\text{अधिशेति})}{७०३} + \frac{\text{वर्षादिक्षशे}}{९६००}$$

$$= \frac{११\ (\text{इष्टति}—\text{अधिशेति}) + \frac{७०३\ \text{वर्षादिक्षशे}}{९६००}}{७०३}$$

$$= \frac{११(\text{इष्टति}—\text{अधिशेति}) + \frac{११\ \text{वक्षशे}}{९६००} + \frac{६९२\ \text{वक्षशे}}{९६००}}{७०३}$$

$$= \frac{११\left\{\text{इष्टति} - \left(\text{अधिशेति} - \frac{\text{वक्षशे}}{९६००}\right)\right\} + \frac{६९२\ \text{वक्षशे}}{९६००}}{७०३}$$

$$= \frac{११\ (\text{इष्टति} - \text{शुद्धि}) + \frac{६९२\ \text{वक्षशे}}{९६००}}{७०३} = \text{अवमानि}$$

ततो लघ्वहर्गणः = (इष्टति—शुद्धि)—अवम एतदनुरूप एव चैत्रसितादिगतस्थितिसंघ इति भास्कराचार्य प्रकारः। अत्रापि सैव स्थूलता या च भास्करानयनेऽपि वर्तते। एतेनाचार्योक्तमुपपद्यत इति ॥४३-४४॥

अथ लघ्वहर्गणः कदा सावयवो निरवयवश्च भवतीति विचार्यते यदाऽवमशेषाभावस्तदा सूर्योदयामान्तवर्षान्तिमेकत्र स्थितत्वात्सौराहर्गणचान्द्राहर्गणसावनाहर्गणानां निरवयवत्वमन्यथा सावयवत्वमिति। कल्पे निरग्रलक्षणं कियन्मितमित्येतदर्थं यदा निरग्रलक्षणमस्ति तदाऽहर्गणानां (सौरचान्द्रसावनाहर्गणानां) महत्तमापवर्त्तनाङ्कमन्विष्य तेनापवर्त्तितास्तेऽहर्गणाः कार्यास्तदा लब्धितुल्यवर्षैः पुनः पुनस्तेषां निरवयवत्वं भविष्यतीति। अथापवर्त्तितसौराहर्गणमानानि कियद्भिर्वर्षैर्वर्षान्ते भविष्यतीति विचारः महत्तमापवर्त्तनाङ्केनापवर्त्तनेन यावन्ति दिनान्यागच्छन्ति तान्ये ३६० भिर्भजनेन यानि शेषमानानि तानि येनाङ्केन गुणनेनै ३६० तद् भवेत्तैरेव गुणकतुल्यवर्षैस्तान्यपवर्त्तितसौराहर्गणमानानि वर्षान्ते भविष्यन्तीति सिद्धान्तितम्। एवं चापवर्त्तितचान्द्राहर्गणसावनाहर्गणमानेकियद्भिर्वर्षैर्वर्षान्ते भविष्यत इति विचार्यते। सौराहर्गणेन साकं चान्द्राहर्गणसावनाहर्गणयोर्महत्तमापवर्त्तनाङ्कमन्विष्य तेनापवर्त्तनाङ्केन चान्द्रसावनाहर्गण-

मानेऽपवर्त्तिते कार्ये तदा लब्धितुल्यवर्षैः पुनर्वर्षान्ते तावद्गणौ भविष्यत इति दिक् । विषयोऽयं वटेश्वरसिद्धान्तेऽपि मया प्रदर्शितोस्तीति ।

अब वर्षेशानयन और लघ्वहर्गणानयन कहते हैं

हि. भा.—कल्पादि से जो गतवर्ष है उसका और गतवर्षसम्बन्धि दिनाद्यों का योग सूर्यादि वर्षपति होते हैं । कल्पगतवर्ष और कल्पग्रहभगण के घात (गुणनफल) को कल्पवर्ष से भाग देने से जो फल होता है वह सूर्यभगणान्त में (रविवर्षान्त में) मध्यमग्रह होते हैं। चैत्रशुक्लप्रतिपदादि से वर्त्तमान मास की इष्टतिथि पर्यन्त जो तिथियां हैं उनमें शुद्धि को घटा देने से जो हो उसको दो स्थान में स्थापित करना, एक स्थान में उसे ग्यारह ११ से गुण देना, वर्षादिक्षयशेष को ६९२ इनसे गुणा कर अपने हर ९६०० से भाग देकर जो फल हो उसे ग्यारह गुणित शुद्धि रहित इष्टतिथि में जोड़ देना ७०३ इनसे भाग देने से जो लब्धि अवम हो उसको पूर्वस्थापित (द्वितीय स्थान स्थित शुद्धि रहित इष्टतिथि) में घटाने से रविमेषादि (जिस काल में रविमेष में जाते हैं उस काल) से अहर्गण (लघ्वहर्गण) होता है, इसको सात से भाग देने से जो शेष रहता है वह अब्दपत्यादि (वर्षपत्यादि) होते हैं ।।४२-४३।।

उपपत्ति

एक वर्ष में सावनदिनादि=३६५।१५।३०।२२।३०, इष्टवर्षान्त में सावनदिनादि=गव (३६५।१५।३०।२२।३०)=गव×३६५+गव (१५।३०।२२।३०)=गव ३६५ +गतवर्षसंदिनादि=कल्पादि से इष्टवर्षान्त में सावनाहर्गण इसमें दिनादिबोधक द्वितीय खण्ड ही है, आनीत सावनाहर्गण में सात से भाग देने जो शेष रहता है वह रव्यादि वर्षपति होते हैं कल्पादि में रविवार था इसलिये रवि ही से गणना करते हैं, अब रविभगणान्त (रविवर्षान्त) में ग्रहानयन के लिये अनुपात करते हैं यदि कल्पवर्ष में कल्पग्रह भगण पाते हैं तो गतवर्ष में क्या इस अनुपात से रविवर्षान्त में मध्यमग्रह आते हैं, इससे आचार्योक्त उपपन्न हुआ ।।४१।।

लघ्वहर्गण साधन के लिये वर्षादि से गततिथि=इष्टतिथि—अधिशेति, तथा स्वल्पान्तर से ७०३ चान्द्र दिनों में ग्यारह ११ अवम होते हैं यह मानकर अनुपात 'यदि ७०३ चान्द्रदिनों में ग्यारह ११ अवम पाते हैं तो वर्षादि से गततिथि में क्या' इससे जो फल आता है उसमें वर्षादिक्षय शेष जोड़ने से पूरे अवम होते हैं, उसका स्वरूप यह है जैसे—

$$\frac{११\ (\text{इष्टति}-\text{अधिशेति})}{७०३}+\frac{\text{वर्षादिक्षशे}}{९६००}$$

$$= \frac{११\ (\text{इष्टति}-\text{अधिशेति}) + \frac{७०३\ \text{वक्षशे}}{९६००}}{७०३}$$

$$= \frac{११\ (\text{इष्टति}-\text{अधिशेति}) + \frac{११\ \text{वक्षशे}}{९६००} + \frac{६९२\ \text{वक्षशे}}{९६००}}{७०३}$$

$$= \frac{११ \left\{ \text{इष्टति} - \left(\text{अधिशेति} - \frac{\text{वक्षशे}}{९६००} \right) \right\} + \frac{६९२\ \text{वक्षशे}}{९६००}}{७०३}$$

$$= \frac{११\ (\text{इष्टति}-\text{शुद्धि}) + \frac{६९२\ \text{वक्षशे}}{९६००}}{७०३} =$$

अवम

∴ लघ्वहर्गण = (इष्टति—शुद्धि)—अवम इससे आचार्योक्त उपपन्न हुआ ॥४३-४४॥

लघ्वहगण कब सावयव और निरवयव होता है इसके लिये विचार करते हैं। जब अवमशेष का अभाव होगा तब सूर्योदय—अमान्त और वर्षान्त के एक ही स्थान में रहने के कारण चान्द्राहर्गण-सौराहर्गण और सावनाहर्गण निरवयव होगें यह सिद्ध हुआ। कल्प में निरग्रलक्षण कितने होते हैं इसके लिये विचार—जब निरग्रलक्षण हैं तब अहर्गणों-(सौराहर्गण-चान्द्राहर्गण और सावनाहर्गण) के महत्तमापवर्त्तनाङ्क निकालना उससे पूर्वोक्त अहर्गणों को अपवर्त्तित करने से जो लब्धियां हों तत्तुल्य वर्षों में फिर फिर वे अहर्गण निरवयव होंगे। अपवर्त्तित सौराहर्गण मान कितने वर्षों में वर्षान्त में होता है उसके लिये विचार—सौराहर्गण को महत्तमापवर्त्तनाङ्क से अपवर्त्तित करने से जितने दिने आवें उनको तीन सौ साठ ३६० से भाग देने से जो शेष रहे उसको जिस अङ्क से गुणने से तीन सौ साठ ३६० हो उसी गुणक तुल्य वर्षों में अपवर्त्तित सौराहर्गण वर्षान्त में होगा यह सिद्ध हुआ। इसी तरह अपवर्त्तित चान्द्राहर्गण और सावनाहर्गण कितने वर्षों में वर्षान्त में होगें इसका भी विचार करना। जैसे सौराहर्गण के साथ चान्द्राहर्गण और सावनाहर्गण का महत्तमापवर्त्तनाङ्क निकालना तब उस अपवर्त्तनाङ्क से चान्द्राहर्गण और सावनाहर्गण को अपवर्त्तित करना तब लब्धि तुल्य वर्षों में फिर वे दोनों वर्षान्त में होगें। यह विषय वटेश्वर सिद्धान्त में भी मैंने लिखा है इति ॥४३-४४॥

इदानीमहर्गणग्रहानयनविवक्षुरादौ तावद्रविसितबुधानां कुजगुरुशनिशीघ्रोच्चानां चानयनायार्यामाह।

द्यगणात्सप्तत्यंशं स्वनवाकांशाधिकं विशोध्यांशाः।
मध्यबुधसूर्यसिताः शीघ्रोच्चं कुजगुरुशनीनाम् ॥४५॥

वा. भा.—द्युगणात् सप्ततितमोंशः स्वेन नवार्कांशेनाधिकः स्वनवार्कांशाधिकोऽतस्तं सप्तत्यंशं स्वनवार्कांशाधिकं विशोध्य द्युगणा देवांशाः भवन्ति, यतः सौरास्तावतैव सावनमधिकम्, अतएव सावनाद्विशोध्यतेऽधिकं चतुर्थी जातिरियं तद्यथा $\frac{७०}{१२९}$ सवर्णने समशून्यनिपाते च कृते जातमुपरि त्रयोदश, अधस्त्रिखनन्दाः अत्रैतज्जातमिष्टाहर्गणं रविमेषादिकं त्रयोदशभिः $\frac{१०३}{१३}$ संगुण्य त्रिखनन्दैर्विभज्य भागादिफलं यल्लब्धं तस्मादेवाहर्गणाद्विशोध्य भागा भवन्ति ९०३ मध्यमरविसितबुधानां शीघ्रोच्चस्य कुजगुरुशनीनां भागैश्च राश्यादिको ग्रहः कार्य इत्यत्रेयं युक्तिर्यदि कल्पार्कसावनदिनेभ्यः सौरसावनदिवसान्तरतुल्यानि दिनानि विशोध्य सावनदिवसा एव भागा रवेर्भवन्ति तदेकस्मात् सावनदिवसात् किं विशोध्यते येनासौ रविभागो भवतीत्यत्र सौरसावनदिवसांतरतुल्येन राशिनापवर्त्तने कृते त्रैगुणकारभागहारयोस्त्रैराशिकस्थापनभागहारः एकोनसप्ततिःविंशाधिकशतभागानां पंचपंचाशदधिकगुण्यगुणकारस्थापनयोरेकैकौ ६९ ॥ अत्र भागहारे सप्ततिः कियन्नूना अतो भागहारं सप्ततेर्विशोध्य शेषं $\frac{५}{१२}$ अनेन भागहारेण $\frac{४५}{१२०}$ विभज्य लब्धं नवार्काः १२९ । अतएव सप्तत्यंशः स्वनवार्कांशेनसंयुतः क्रियते । लघुत्वात्भागहारस्य सप्ततेरितिलब्धमेकसावनदिनस्य शोध्यं भवतीति ॥४५॥

वि. भा.—स्वनवार्कांशाधिकं (स्वकीयै १२९ तन्मितांशेन युतं) अहर्गणस्य सप्तत्यंशं (अहर्गणस्यै ७० तन्मितमंशं) द्युगणात् (अहर्गणात्) विशोध्यावशिष्टांशा मध्यमबुधसूर्यसिता भवन्ति, कुजगुरुशनीनां तदेव शीघ्रोच्चमपि भवत्यर्थान्मध्यमरविसमावेव मध्यमबुधशुक्रौ भवतस्तथा मध्यमरविरेव कुजगुरुशनीनां शीघ्रोच्चमस्ति तेन साधितमध्यमरविरेव मध्यमबुधशुक्रौ, कुजगुरुशनीनां च शीघ्रोच्चं भवेदिति ।

अब रवि का आनयन करते हैं

हि. भा.—अहर्गण के सत्तरवें अंश में अपने १२९ अंश को जोड़कर जो होता है उसको अहर्गण में घटाने से मध्यम बुध, मध्यमसूर्य और मध्यमशुक्र होते हैं, तथा वही कुज, गुरु और शनि इनके शीघ्रोच्च होते हैं, अर्थात् मध्यमरवि ही के बराबर मध्यमबुध और मध्यमशुक्र होते हैं और मध्यमरवि ही कुज, गुरु और शनि इन ग्रहों का शीघ्रोच्च है इसलिये साधित मध्यमरवि ही मध्यमबुध और शुक्र होते हैं और वही कुज, गुरु और शनि इनके शीघ्रोच्च होते हैं, गणितन्यास से अंशादि मध्यम (रवि, बुध और शुक्र)

$$=\text{अहर्गण}-\left(\frac{\text{अहर्गण}}{७०}+\frac{\text{अहर्गण}}{७०\times १२९}\right)=\text{कुज, गुरु और शनि इनके शीघ्रोच्च}$$

इति ॥४५॥

उपपत्ति

रूपतुल्य अहर्गण से आचार्य के मत में कलादिक रविगति=५९′८″।१०‴।-२२⁗ तब अनुपात करते हैं यदि एक दिन में यह रविगति पाते हैं तो अहर्गण में क्या

इससे मध्यमरवि आते हैं, यहां गति के अंकों के सवर्णन आदि करने से ऐसा स्वरूप होता है अहर्गण—$\left(\frac{\text{अहर्गण}}{७०}+\frac{\text{अहर्गण}}{७०\times १२९}\right)$ = मध्यमरवि = मध्यमबुध = मध्यमशुक्र भास्कराचार्य के मत से एकदिन में रविगति = ५९′।८″।१०‴।२१⁗ आती है इति ॥४५॥

प्रकारान्तरम्

कल्पे रविभगणाः = ४३२०००००० ३६० एभिर्गुणिता अंशा भवन्त्यतः

अंशाः = १५५५२०००००००

अतोऽनुपातेनाहर्गणान्ते भागात्मको रविः

$$= \frac{\text{अ.}\ १५५५२०००००००}{१५७७९१६४५०००}$$

$$= \text{अ.}\ \frac{१५५५२०००}{१५७७९१६४५}$$

अत्रासन्नमानग्रहणेन—

$$\frac{१५५५२०००}{१५७७९१६४५} = ०+\frac{१}{१+\frac{२२७१६४५}{१५५५२०००}}$$

$$= ०+\frac{१}{१+\frac{१}{६८+\frac{१०४८१४०}{२२७१६४५}}}$$

$$= ०+\frac{१}{१+\frac{१}{६८+१+\frac{१०४८१४०}{२२७१६४५}-१}}$$

$$= ०+\frac{१}{१+\frac{१}{६९-\frac{२२७१६४५-१०४८१४०}{२२७१६४०}}}$$

$$= ०+\frac{१}{१+\frac{१}{६९-\frac{१२२३५०५}{२२७१६४५}}}$$

अत्र र्ण खण्ड$\frac{१२२३५०५}{२२७१६४५}$ मिदं यदि त्यज्यते तदाऽऽसन्नमानग्रहणेन—

लब्धयः ०, १, ६६

आसन्नमानानि $\frac{०}{१}$, $\frac{१}{१}$, $\frac{६६}{७०}$

अत्राचार्येण $\frac{६६}{७०}$ आसन्नमानं गृहीतम् ।

अथ वास्तवभिन्नेऽ $\frac{१५५५२०००००}{१५७७६१६४५}$ स्मिन्

समयोगवियोगेन—

$$\frac{६६}{७०}+\frac{१५५५२०००००}{१५७७६१६४५}-\frac{६६}{७०}$$

$$=\frac{६६}{७०}-\frac{६६\times १५७७६१६४५-७०\times १५५५२०००००}{७०\times १५७७६१६४५}$$

$$=\frac{६६}{७०}-\frac{१०८८७६२३५०५-१०८८६४००००००}{७०\times १५७७६१६४५}$$

$$=\frac{६६}{७०}-\frac{१२२३५०५}{७०\times १५७७६१६४५}$$

$$=\frac{६६}{७०}-\frac{१}{७०\times १२६}$$

अत्राचार्येण $\frac{१५७७६१६४५}{१२२३५०५}$ स्वल्पान्तरेण अर्धाधिकग्रहणेन १२६ लब्धिः स्वीकृतेति ।

अतोंऽशादिको रविः $=अ\left(\frac{६६}{७०}-\frac{१}{७०\times १२६}\right)$

$$=अ\left(१-\frac{१}{७०}-\frac{१}{७०\times १२६}\right)^{०}$$

$$=अ\left\{१-\left(\frac{१}{७०}+\frac{१}{७०}\times\frac{१}{१२६}\right)\right\}^{०}$$

$$= \text{अ}^\circ - \text{अ}\left(\frac{१}{७०} + \frac{१}{७० \times १२९}\right)^\circ$$

उपपन्नमाचार्योक्तम्। रविमध्यमतुल्यावेव मध्यमबुधशुक्रौ भवतः। "रवेरग्रतः पृष्ठतो वा ह्यनुचराविव बुधशुक्रौ सदैव दृश्येते। अतो रविभगणसमा एव तयोर्भगणा भवन्ती"ति भगणाध्याये वासनाभाष्ये भास्कराचार्याः प्रोचुस्तदर्थं शिरोमणिर्द्रष्टव्यः। तथा च रविमध्यमसमा एव कुजगुरुशनीनां शीघ्रोच्चभगणाः। तेषां शीघ्रोच्चं मध्यमरविरेव भवतीत्यपि भगणाध्याये प्रतिपादितं भास्करेण ॥

इदानीं मध्यमचन्द्रानयनमाह

त्रिगुणमवमावशेषं विभजेद् गुणसप्तशशिभिराप्तांशैः।
पृथगधिकोऽर्को रविगुणतिथ्यंशैः संयुतश्चन्द्रः ॥४६॥

वा. भा.—त्रिगुणमवमावशेषं विभजेत्। कैरित्याह गुणसप्तशशिभिस्तत्र ये आप्ता अंशास्तैः पृथक् यथाधिकः कार्यः पूर्वमेव यदनष्टमवमावशेषं स्थापितं तत्त्रिगुणं कृत्वा गुणसप्तशशिभिर्विभजेत्। लब्धमंशाधिकं भवति द्वितीयस्थानस्थे रवौ योज्यमित्यर्थः। तत स तादृगर्को रविगुणिततिथ्यंशैश्च सहितश्चन्द्रो भवति। अहर्गणानयनवर्त्तमाने मासे सितप्रतिपदादिका यास्तिथयो दत्ताः ता रविगुणद्वादशगुणिताः सत्यस्तिथय एव चन्द्रांशा भवन्ति तैश्चांशै संयुतोऽर्कः पृथक्स्थोऽवमफलसंस्कृतः कार्य इत्यर्थः कृतोऽर्कश्चन्द्रो भवति। अत्रेयं वासना, रविचन्द्रौ द्वावपि अमावास्यान्ते तुल्यौ भवतः तत आरभ्य यावत्यस्तिथयोऽतीताः तावदन्तरं तयोः तिथौ द्वादशचन्द्रभागा भवन्ति, यस्माच्चन्द्रलिप्ताभिः खयमस्वरसंख्याभिस्तिथिः भचक्रत्रिंशांशेन चैतावत्य एव लिप्ता भवन्ति तिथयश्च त्रिंशत्। तस्मादुपपन्नं रविगुणास्तिथयश्चन्द्रभागा भवन्ति। तैर्युक्तश्चन्द्रस्तत्र दिने मध्यमः तिथ्यन्तकालिकः भवत्यतोऽवमशेषेणार्कोदयकालिकः क्रियते। तद्यथा यदि सावनदिवसशेषेण यमनवरससंख्येनैकं चन्द्रदिनं चन्द्रदिनञ्च द्वादशभागा भवन्ति। तदेष्टेनावमशेषेण किमित्यत्र गुणकारिभागहारयोः तुल्यछेदस्त्रिस्वमुनिसंख्यः तस्मिन् नष्टे प्रथमराशेश्चतुर्भागेन गुणसप्तशशिनो द्वितीयराशेश्चतुर्भागेन द्वादशानां त्रयो तत इष्टावमशेषस्य सर्वदा त्रिको गुणकारः। अग्रशशिनो भागहारः फलं भागादि चन्द्रमसः तत्तिथ्यन्तकालिके चन्द्रमसि संयोज्यार्कोदयकालिको संकायां भवतीत्युपपन्नम्। अनयैव विपरीतवासनया चन्द्रे ज्ञाते अवमावशेषमानयेत् यो मध्यमः इति ॥ ४६ ॥

वि. भा.—अवमावशेषं (क्षयशेषं) त्रिगुणं गुणसप्तशशिभिः (१७३ एभिः) विभजेत्, लब्धांशैः पृथगर्को युतो रविगुणतिथ्यंशैः (द्वादशगुणिततिथ्यंशैः) संयुतस्तदा चन्द्रो भवतीति ॥४६॥

उपपत्तिः

अत्राचार्यमतेनावमशेषम् $=\frac{\text{अंश}}{७०३}$,

इदं सावनं, चान्द्रकरणेन—

चान्द्रावमशेषम् $=\frac{\text{अंश}\times ७०३}{७०३\times ६९२}=\frac{\text{अंश}}{६९२}$

द्वादशगुणमंशात्मकम् $=\frac{\text{अंश}\times १२}{६९२}=\frac{\text{अंश}\times ३}{१७३}$

अथ चन्द्रार्कयोरन्तरभागा द्वादशभक्ता तिथिः स्यात्

∴ तिथ्यन्ते रविचन्द्रान्तरभागाः $=१२$ ति $=$ चा-र.

∴ च $=१२$ ति $+$ र अत्रावमभागयोगेनौदयिकश्चन्द्रः $=१२$ ति $+$ र $+\frac{३\text{ अंश}}{१७३}$ उपपन्नमाचार्योक्तम्।

अब चन्द्रानयन कहते हैं

हि. भा.—अवमावशेष (क्षयशेष) को तीन से गुणाकर एक सौ तेहत्तर १७३ से भाग देने से जो लब्धांश होता है रवि में जोड़ना और बारह गुणित तिथ्यंश को जोड़ने से अंशात्मक चन्द्र होते हैं इति ॥४६॥

हिन्दी में अन्य ढंग से उपपत्ति

अहर्गणान्त कालिक तिथि $=$ गति $+$ क्षयघटीसंचान्द्र, $\frac{\text{अंशात्मक चं}-\text{अंशात्मक र}}{१२}=$ तिथि

इसलिए अंशात्मक चं $=$ अंशात्मक र $+१२$ तिथि, अतः अहर्गणान्तकालिक चं $=$ अंशात्मक र $+१२$ (गति $+$ क्षघसंचां) $=$ अंशात्मक र $+१२$ गति $+१२$ क्षघसंचां $\frac{\text{१ क्षदि}\times\text{इचां}}{\text{६४ चान्द्र}}$ $=$ गक्षदि $+\frac{\text{क्षशे}}{६४}$ ∴ $\frac{\text{क्षयशे}}{६४}=\frac{\text{क्षघ}}{६०}$ इसलिए अनुपात से क्षयघटीसंचांदि $=\frac{६४\text{ चां}\times\text{क्षशे}}{६३\text{ सावन}\times ६४}=\frac{\text{क्षशे}}{६३}$ ∴ १२ क्षघसंचां $=\frac{१२\text{ क्षशे}}{६३}$ एतत्सम्बन्धी कलात्मक फल $=\frac{१२\times ६०\times\text{क्षशे}}{६३}=\frac{२०\times १२\times\text{क्षशे}}{२१}=\frac{२०\times ४\text{ क्षशे}}{७}=\frac{८०\text{ क्षशे}}{७}$ इसको अंशात्मक करने

से $\frac{८०\ क्षशे}{६०\times७}=\frac{८०\ क्षशे\times३}{६०\times७\times३}$ =परन्तु आचार्य कथित क्षयशेष को ग्यारह से भाग देने से भास्कर-

कथित क्षयशे $=\frac{क्षयशे}{११}$ $\therefore\frac{८०\ क्षशे\times३}{६०\times११\times७\times३}=\frac{क्षशे\times३}{\frac{७\times११\times६०\times३}{८०}}=\frac{क्षशे\times३}{\frac{१३८६}{८}}$

$=\frac{क्षशे\times३}{\frac{६९३}{४}}=\frac{क्षशे\times३}{१७३}$ =१२ क्षयघसंज्ञां अतः अहर्गणान्त कालिक अंशात्मक चन्द्र=

अंशात्मकरवि + १२ गति + $\frac{क्षशे\times३}{१७३}$ इससे आचार्योक्त उपपन्न हुआ इति ॥४६॥

इदानीं वर्षान्तिकादहर्गणात् भौमादिग्रहमन्दफलानयनार्थमार्याचतुष्टयमाह

एकादशलिप्ताशा भौमः शरसप्तवसुभि ८७५ रिन्दुयमैः २१ ।
कृतगुणितद्युगणांशाः पञ्चरसैः ६५ षट्बुधः शीघ्रम् ॥४७॥

वा. भा.—भागो नन्दशशांकैः शशिसूर्यस्वरयमैश्च शशिपातः रविमण्डलान्तिकयुता मध्या भगणान्तिका शेषाः। स्पष्टार्थमिदमार्याचतुष्टयम्। यतस्त्रैराशिकेन सर्वग्रहाणां फलानयनम्। भौमस्य तावद्यद्येकविंशकेनाहर्गणेनैकादशांशान्भौमो भुङ्क्ते तदिष्टाहर्गणेन किमिति तथा शरसप्त च त्रिभिश्च दिनैः सावनैर्यद्येकादशलिप्ताश्च भौमो भुङ्क्ते तदिष्टाहर्गणेन कियतीरिति। लब्धं फल द्वययोगो रविमण्डलाद् भवति भौमः। एवं बुधोऽपि यद्येकसंख्येनाहर्गणेन चत्वारोंऽशा बुधस्य भवन्ति तदिष्टाहर्गणेन किमिति पुनः पञ्चरसैर्दिने यदि षट्भागा भवन्ति, तदिष्टाहर्गणेन किमिति, पुनर्यदि कृतरसगुणतुल्यैर्दिनैः पञ्चलिप्ताः साध्यन्ते। लब्धफलद्वययोगो बुधः ॥४७॥

वि. भा.—कुजः $=\frac{११\ अहर्गण}{२१}$ अंशाः $+\frac{११\ अहर्गण}{८७५}$ कलाः। तथा बुध-

शीघ्रोच्चमंशाद्यम् $=४\ अहर्गण+\frac{६\ अहर्गण}{६५}$ ॥४७॥

अत्रोपपत्तिः

अत्र रूपमितमहर्गणं प्रकल्प्य 'महीमितादहर्गणात्फलानि यानीत्यादि' भास्करोक्तप्रकारेण कुजस्याऽऽचार्यप्रकारेण मध्यमा गतिर्यद्यानीयते तदा ३१'।२६''। २८'''।७'''' भवति, भास्कराचार्योक्तापोयमेव ततो विलोमविधिना खण्डगुणनयोगादिनाऽऽचार्योक्तमध्यमकुजसिद्धिर्भवतीति ॥४७॥

अथ बुधशीघ्रोच्चानयनार्थं

पूर्वोक्तप्रकारेणैकस्मिन् दिनेऽऽचार्योक्तप्रकारेण बुधशीघ्रोच्चगतिः= ४°।५'।३२''।१८'''।२८'''' ततोऽनुपातेन बुधशीघ्राच्चांशाः=अहर्गण (४°।५'।३२''। १८'''।२८'''')=४×अहर्गण+अहर्गण (५'।३२''।१८'''।२८'''') अथा ५'।३२''। १८'''।२८'''' स्य स्वरूपान्तरं क्रियते ५।३२।१८+$\frac{२८}{६०}$=५।३२।१८+$\frac{७}{१५}$=५।३२। $\frac{२७०+७}{१५}$=५।३२।$\frac{२७७}{१५}$=५।३२+$\frac{२७७}{६०\times१५}$=५। $\frac{२८८००+२७७}{९००}$

=५। $\frac{२९०७७}{९००}$= ५ + $\frac{२९०७७}{६०\times९००}$ = ५ + $\frac{२९०७७}{५४०००}$

=$\frac{२७००००+२९०७७}{५४०००}$=$\frac{२९९०७७'}{५\ ०००}$

ततः $\left(४°'\frac{२९९०७७'}{५४०००}\right)$अहर्गण=४° अहर्गण+$\frac{२९९०७७}{६०\times५४०००}$ × अहर्गण

=$\left(४+\frac{२९९०७७}{६०\times५४०००}\right)$° अहर्गण

=$\left(४+\frac{१\times१२}{\frac{६०\times५४०००\times१२}{२९९०७७}}\right)$° अहर्गण=$\left(४+\frac{१२}{१३०}\right)$° अहर्गण

=४ अहर्गण°+$\frac{६\ \text{अहर्गण}°}{६५}$=बुधशीघ्रोच्चमंशाद्यम्, एतावताऽऽचार्योक्तमुपपद्यते।

$\left(४+\frac{१२}{१३०}\right)$ अहर्गण=४ अहर्गण+$\frac{४\times३\ \text{अहर्गण}}{१३०}$=बुधशीघ्रोच्चमंशाद्यम्, एतेन 'दिनगणः कृतसङ्गुणित' इत्यादि भास्करोक्तमप्युपपद्यत इति ॥ ४७ ॥

अथवोपपत्तिः

४°।५'।३२''।१८'''२८''''=वास्तवबुधशीघ्रोच्चगतिः। ४°= अवास्तवबुधशीघ्रोच्चगतिः ३६०×कल्पबुधशीघ्रोच्चभगण=कल्पांशा बुधस्य, $\frac{४\times\text{ककु}}{१}$ =कल्पेऽवास्तवांशा बुधस्य।

अनयोरन्तरम्

$$\text{३६०} \times \text{कल्पबुश।उभ}-४\,\text{ककु} = \text{कल्पेऽन्तरांशाः, ततोऽहर्गणस-अन्तरांशाः} = \frac{(\text{३६०} \times \text{कल्पबुशीउभ}-४\text{ककु})\,\text{अहर्गण}}{\text{ककु}} = \frac{\text{१२ अहर्गण}}{\frac{\text{१२ ककु}}{\text{३६० कबुशीउभ}-४\,\text{ककु}}}$$

$$= \frac{\text{१२ अहर्गण}}{\text{१३०}}$$

$$\text{अतोऽहर्गणसम्बन्धिवास्तवबुधशीघ्रोच्चांशाः} = ४\,\text{अहर्गण} + \frac{\text{१२ अहर्गण}}{\text{१३०}}$$

$$= ४\,\text{अहर्गण} + \frac{६\,\text{अहर्गण}}{६५}$$ एतेनाप्याचार्योक्तमुपपन्नम् ॥४७॥

अब मध्यम कुज और बुधशीघ्रोच्च के आनयन को कहते हैं

हि. भा.—$\text{कुज} = \frac{\text{११ अहर्गण}}{२१}$ अंश $+ \frac{\text{११ अहर्गण}}{८७५}$ कला तथा अंशादिबुधशीघ्रोच्च

$$= ४\,\text{अहर्गण} + \frac{६\,\text{अहर्गण}}{६५}$$ ॥४७॥

उपपत्ति

रूपतुल्य अहर्गण मानकर 'महीमितादहर्गणात्फलानि यानि इत्यादि' भास्करोक्तविधि से कुज के आचार्योक्त प्रकार से यदि मध्यमगति लाते हैं तो ३१′।२६″।२८‴।७⁗ होती है, इतनी ही भास्करोक्त भी है, इस पर से विलोम विधि और खण्डगुणन सवर्णन आदि से आचार्योक्त मध्यम कुजानयन उपपन्न होता है ॥४७॥

बुधशीघ्रोच्चानयन के लिये विचार करते हैं

पूर्वोक्त प्रकार से आचार्योक्त विधि से एक दिन में बुधशीघ्रोच्च गति=४°।५′।३२″।१८‴।२८⁗ इससे अनुपात द्वारा बुधशीघ्रोच्चांश=अहर्गण (४°।५′।३२″।१८‴।२८⁗)=४° अहर्गण+अहर्गण (५′।३२″।१८‴।२८⁗) अब ५′।३२″।१८‴।२८⁗ इसका स्वरूपान्तर करते हैं $५।३२।१८ + \frac{२८}{६०} = ५।३२।१८ + \frac{७}{१५}$

$$= ५।३२।\frac{२७०+७}{१५} = ५।३२।\frac{२७७}{१५} = ५।३२ + \frac{२७७}{१५\times६०} = ५ + ३२ + \frac{२७७}{९००}$$

$$= ५।\frac{२८८००+२७७}{९००} = ५।\frac{२९०७७}{९००} = ५ + \frac{२९०७७}{६०\times९००} = ५ + \frac{२९०७७}{५४०००}$$

$=\frac{२७०००० + २६०७७}{५०००} = \frac{२६६०७७}{५४०००}$ ∴ $(४° + \frac{२६६०७७'}{५४०००} \times$ अहर्गण $= ४°$ अहर्गण

$+ \frac{२६६०७७ \text{ अहर्गण}}{६० \times ५४०००} = \left(४ + \frac{२६६०७७}{६० \times ५४०००}\right)° \times$ अहर्गण

$= \left(४ + \frac{१ \times १२}{\frac{६० \times ५४००० \times १२}{२६६०७७}}\right)° +$ अहर्गण $= \left(४ + \frac{१२}{१३०}\right)° \times$ अहर्गण

$= ४$ अहर्गण$° + \frac{६ \text{ अहर्गण}°}{६५} =$ अंशादिबुधशीघ्रोच्च, इससे आचार्योक्त उपपन्न हुआ।

$\left(४ + \frac{१२}{१३०}\right) \times$ अहर्गण $= ४$ अहर्गण $+ \frac{४ \times ३ \times \text{अहर्गण}}{१३०} =$ अंशादिबुशीउ, इससे 'दिनगणः कृतसङ्गुणितः' इत्यादि भास्करोक्त उपपन्न हुआ ।। ४७ ।।

अथवा

$४°।५'।३२''।१८'''।२८''''=$ वास्तवबुधशीघ्रोच्चगति, $४° =$ अवास्तव, बधशीघ्रोच्चग ३६० × कल्पबुधशीघ्रोच्चभगण = बुधशीघ्रोच्च के कल्पांश। $\frac{४ \times \text{ककु}}{१}$ = कल्प में अवास्तवबुधशीघ्रोच्चांश।

दोनों के अन्तर करने से

३६० × कल्प बुधशीउभ — ४ ककु = कल्प में वास्तव और अवास्तव के अन्तरांश

अतः अहर्गणसं अन्तरांश $= \frac{(३६० \times \text{कल्पबुशीउभ} - ४ \text{ ककु}) \text{ अहर्गण}}{\text{ककु}}$

$= \frac{१२ \text{ अहर्गण}}{\frac{१२ \text{ ककु}}{३६० \text{ कबुशीउभ} - ४ \text{ ककु}}}$

$= \frac{१२ \text{ अहर्गण}}{१२०} =$ अन्तरांश

इसलिए अहर्गणसंअवास्तवबुशीउच्चांश + अन्तरांश = वास्तवबुधशीघ्रोच्चांश-

$= ४$ अहर्गण $+ \frac{१२ \text{ अहर्गण}}{१३०} = ४$ अहर्गण $+ \frac{६ \text{ अहर्गण}}{६५}$ इससे भी आचार्योक्त उपपन्न हुआ ।।४७।।

प्रकारान्तरेणोपपत्तिः

अथ कल्पकुजभगणाः = २२९६८२८५२२,

कल्पकुदिनानि = १५७७९१६४५०००

ततोऽनुपातेन भागात्मिका दैनन्दिनी

कुजगतिः $= \frac{२२९६८२८५२२ \times १२ \times २०}{१५७७९१६४५०००}$

चतुर्भिरपवर्तनेन—

$$= \frac{२२९६८२८५२२ \times ३ \times ५०}{३९४४७९११२५००}$$

$$= \frac{२०६७१४५६६९८०'}{३९४४७९११२५००}$$

$$= ० + \cfrac{१}{१ + \cfrac{१}{१ + \cfrac{१}{९ + १ - १ + \frac{१७२१४३५२३८०}{१८९५००२१४६०}}}}$$

$$= ० + \cfrac{१}{१ + \cfrac{१}{१ + \cfrac{१}{१० - \left(१ - \frac{१७२१४३५२३८०}{१८९५००२१४६०}\right)}}}$$

$$= ० + \cfrac{१}{१ + \cfrac{१}{१ + \cfrac{१}{१० - \frac{१७३५६६९०८०}{१८९५००२१४६०}}}}$$

अत्रासन्नमानग्रहणेन—

लब्धयः क्रमेण = ०, १, १, १०

आसन्नमानानि $= \frac{०}{१}, \frac{१}{१}, \frac{१}{२}, \frac{११}{२१}$,

अत्राचार्येण $\frac{११'}{२१}$ इदं मानं संगृह्य कर्म कृतम्।

अत्र वास्तवभिन्नेऽ $\frac{२०६७१४५६६९८०}{३९४४७९११२५००}$ स्मिन्

समयोगवियोगेन—

भागात्मककुजः $=\frac{११^{\circ}}{२१}+\frac{२०६७१४५६६९८०}{३९४४७९११२५००}-\frac{११^{\circ}}{२१}$

$=\frac{११^{\circ}}{२१}+\frac{२१\times२०६७१४५६६९८०-११\times३९४४७९११२५००}{२१\times३९४४७९११२५००}$

$=\frac{११^{\circ}}{२१}+\frac{१७३५६६९०८०}{२१\times३९४४७९११२५००}$

$=\frac{११^{\circ}}{२१}+\frac{१७३५६६९०८०\times६०\times११}{२१\times३९४४७९११२५००\times११}$

$=\frac{११^{\circ}}{२१}+\frac{१७३५६६९०८०\times२०\times११}{७\times३९४४७९११२५००\times११}$

$=\frac{११^{\circ}}{२१}+\frac{१७३५६६९०८०\times१०\times१०}{८\times१९७२३९५५६२५०\times११}$

$=\frac{११^{\circ}}{२१}+\frac{१७३५६६९०८००\times११'}{१५१८७४४५८३१२५०}$

$=\frac{११^{\circ}}{२१}+\frac{११'}{८७५+\frac{३४१३८१२५०}{१७३५६६९०८००}}$

अत्र भिन्न $\frac{३४१३८१२५०}{१७३५६६९०८००}$ मिदं रूपाल्पत्वात् त्यक्तमाचार्येण।

अतो मध्यमः कुजोऽहर्गणान्ते भागादिकः

$=\left(\frac{११^{\circ}}{२१}+\frac{११'}{८७५}\right)\text{अ}$

$=\frac{११\text{अ}^{\circ}}{२१}+\frac{११\text{अ}'}{८७५}$ उपपन्नं कुजानयनम्।

अत्र भाष्यकर्त्रा चतुर्वेदाचार्येण कुजानयनार्थं विलक्षणैव रीतिः प्रतिपादिता स्ववासनायाम्। तेन कल्पकुजभगणभागानां खण्डद्वयं विहितम्। तत्रैकखण्डम्=

८२६५२७६६४२८१, द्वितीयखण्डम्=३३०६०३६३९ अनयोर्योगः कल्पभूदिनभक्तो भागात्मकः कुजो भवति।

$$\text{यथा } \frac{११\times८२६५२७६६४२८१}{११\times१५७७९१६४५०००० }+\frac{३३०६०३६३९\times६०\times११}{१५७७९१६४५००००\times११}$$

$$=\frac{११^{\circ}}{२१}+\frac{११'}{८७५}$$ उपपन्नं यथोक्तमिति।

अथवा प्रकारान्तरेणोपपत्तिः

अत्राप्यनुपातेन भागात्मिका दैनन्दिनी बुधशीघ्रोच्चगतिः

$$=\frac{१७९३६९९८९८४}{४३८३१०१२५०}$$

$$=४^{\circ}+\frac{४०४५९३९८४^{\circ}\times१२}{४३८३१०१२५०\times१२}$$

$$=४^{\circ}+\frac{१२^{\circ}}{१३०}$$ स्वल्पान्तरात्।

$$=४^{\circ}+\frac{६^{\circ}}{६५}\cdots\cdots\cdots\cdots(१)$$

अतोऽहर्गणान्ते भागादिकं बुधशीघ्रोच्चम्=४ अ°+$\frac{६\ अ^{\circ}}{६५}$ अत्र (१) समीकरणेन भास्कराचार्योक्तम्।

हि. भा.—आचार्य के मत से कल्प में पठित कुज का भगण—यहां भास्कराचार्य ने $\frac{1}{2}$ आसन्नमान स्वीकार कर यथोक्त रूप से क्रिया करने पर "दिनगणार्धमधो गुणसंगुणं" इत्यादि भास्कर का प्रकार और $\frac{11}{21}$ इस आसन्न मान पर से "अहर्गणे युगाहते" इत्यादि श्रीपति का राश्यादिक कुजानयन उपपन्न होता है।।४७।।

इदानीं गुरुशुक्रशीघ्रोच्चयोरानयनमाह

द्युगणेषु वधो लिप्ता जीवः कृतशरगुणैः शरकलोनः।
भागकलाः सितशीघ्रं विषयैर्वसवो द्विषष्ट्याष्टौ।। ४८ ।।

वा. भा.—तथा जीवस्य, यद्येकेन दिनेन पञ्चलिप्तास्तदिष्टाहर्गणेन किमिति। तदिष्टाहर्गणदिनैः कियत्य इति लब्धा फलद्वयान्तररविमण्डलान्तात् गुरुः। शुक्रस्यापि यदि पञ्चभिर्दिनैरष्टौ भागा भवन्ति तदिष्टाहर्गणेन किमिति। पुनर्यदीष्टदिनैरष्टौ लिप्ता भवन्ति तदिष्टाहर्गणेन किमिति फलद्वययोगः शुक्रः।।४८।।

वि. भा.—कलात्मको गुरुः=५ अहर्गण—$\frac{५\ \text{अहर्गण}}{३५४}$, महीमितादहर्गणादित्यादिना रूपतुल्याहर्गणेऽर्थदिकस्मिन् दिने कलादिका गुरुगतिः=०।४'।५९''।९'''।९'''' भास्करोक्तापीयमेव।

एवं शुक्रशीघ्रोच्चम्=$\frac{८\ \text{अहर्गण}}{५}$ अंशाः—$\frac{८\text{अहर्गण}}{६२}$ कलाः, अत्राप्येकस्मिन् दिनेऽशादिका शुक्रशीघ्रोच्चगतिः=१°।३६'।७''।४४'''।३१'''' अत्र भास्करमतेन ३१ स्थाने ३५ समागच्छन्ति सावयवे ॥४८॥

अत्रोपपत्तिः

एकस्मिन् दिने गुरोः कलादिकागतिः=४'।५९''।९'''।९'''', अत्र ५'=अवास्तवगतिर्गुरोः स्वीक्रियते तदा वास्तवावास्तवगत्योरन्तरम्=अवास्तवगुगति—वास्तवगुगति=५'—(४'।५९''।९'''।९'''')=०'।०''।५०'''।५१''''।०''।५०'''।५१'''' एतस्य स्वरूपान्तरं क्रियते ५०'''।५१''''=५०+$\frac{५१}{६०}$=५०+$\frac{१७}{२०}$=$\frac{१०००+१७}{२०}$

=$\frac{१०१७'''}{२०}$ ततः ०''+$\frac{१०१७}{२०\times६०}$=$\frac{१०१७''}{१२००}$ ∴ ०'+$\frac{१०१७}{१२००\times६०}$=$\frac{१०१७}{७२०००}$

=$\frac{१}{\frac{७२०००}{१०१७}}$=$\left(\frac{१}{७१}\right)'$ स्वल्पान्तरात्=$\frac{५}{७१\times५}$=$\frac{५}{३५५}$=$\frac{५'}{३५८}$ अतो गुरोर्वास्तवगतिः=५'—$\frac{५'}{३५४}$ ततोऽहर्गणसम्बन्धिकलात्मकगुरुः=५ अहर्गण—$\frac{५\ \text{अहर्गण}}{३५४}$

एतावताऽऽचार्योक्तं गुरोरानयनमुपपन्नम्। ५'—$\frac{१'}{७१}$=वास्तवगुरुगति। अत्र प्रथमखण्डस्यांशात्मककरणेन $\frac{५}{६०}-\frac{१'}{७१}=\left(\frac{१}{१२}\right)^{०}-\left(\frac{१}{७१}\right)'$ ततोऽहर्गणसम्बन्धिगुरुः=$\left(\frac{\text{अ}}{१२}\right)^{०}-\left(\frac{\text{अहर्गण}}{७१}\right)'$ एतावता 'द्युमणिभिः कुनगैरित्यादि' भास्करोक्तमप्युपपद्यत इति।

अब गुरु और शुक्रशीघ्रोच्च का आनयन करते हैं

हि. भा.—कलात्मक गुरु=५ अहर्गण—$\frac{५\ \text{अहर्गण}}{३५४}$, रूपतुल्य अहर्गण से कलादिक गुरुगति =४'।५९''।९'''।९'''' यही भास्करोक्त भी है। एवं शुक्रशीघ्रोच्च=$\frac{८\ \text{अहर्गण}}{५}$ अंश—

$\frac{८ \text{ अहर्गण}}{६२}$ कला, यहां भी एक दिन में शुक्रशीघ्रोच्चगति अंशादिक=१°।२६'।७''।४४'''। ३१'''' भास्कर मत से ३१ के स्थान में ३५ आता है।

गुरु के आनयन के लिये विचार करते हैं

एक दिन में गुरु की कलादिक गति=४'।५६''।६'''।६'''', यहां ५' गुरु की अवास्तव गति मानते हैं तब वास्तव और अवास्तव गतियों के अन्तर=अवास्तवगुरुगति—वास्तवगुरुगति=५'—(४'।५६''।६'''।६'''')=०'।०''।५०'''।५१'''' यहां ०''।५०'''। ५१'''' इसका स्वरूपान्तर करते हैं ५०'''।५१''''=५० $+\frac{५१}{६०}$=५० $+\frac{१७}{२०}$

$=\frac{१०००+१७}{२०}=\frac{१०१७'''}{२०}$ ∴ ०'' $+\frac{१०१७}{२०\times६०}=\frac{१०१७''}{१२००}$ कलात्मक करने से ०'+

$\frac{१०१७}{६०\times१२००}=\frac{१०१७}{७२०००}=\frac{१}{\frac{७२०००}{१०१७}}=\left(\frac{१}{७१}\right)'$ स्वल्पान्तर से, $\frac{१}{७१}=\frac{५}{७१\times५}=\frac{५}{३५५}$

$=\left(\frac{५}{३५४}\right)'$ ∴ गुरुवास्तवगति=५'—$\frac{५'}{३५४}$, अहर्गणसम्बन्धि कलात्मक गुरु= (५ अहर्गण)'—$\left(\frac{५ \text{ अहर्गण}}{३५४}\right)'$ इससे आचार्योक्त उपपन्न हुआ।

५' —$\frac{१'}{७१}$=वास्तव गुरुगति, यहां प्रथम खण्ड को अंशात्मक करने से $\frac{५}{६०}-\frac{१'}{७१}$

$=\left(\frac{१}{१२}\right)°-\left(\frac{१}{७१}\right)'$ ∴ अहर्गणसम्बन्धि गुरु=$\left(\frac{\text{अहर्गण}}{१२}\right)°-\left(\frac{१}{७१}\right)'$ इससे 'द्युमणिभिः कुनगैः' इत्यादि भास्करोक्त उपपन्न होता है।

शुक्रशीघ्रोच्चानयनार्थमुपपत्तिः

अत्रैकदिने शुक्रशीघ्रोच्चगतिः=१°।३६'।७''।४४'''।३५'''', अत्र स्वल्पान्तरात् १°।४०' इति गृह्यते तदा १°।४०'=अवास्तवशुशीउग। अथ वास्तवावास्तवशुक्रशीघ्रोच्चगत्योरन्तरम्=१°।४०—(१°।३६'।७''।४४'''।३५'''')=३'।५३''।१५'''।२५'''' अथ १°।४०'

$=१+\frac{४०}{६०}=१+\frac{२}{३}=\frac{३+२}{३}=\frac{५}{३}$=प्रथमखण्डम्। ३'।५२''।१५'''।२५''''=द्वितीयखण्ड, प्रथमखण्ड—द्वितीयखण्ड=१°।३६'।७''।४४'''।३५''''=वास्तवशुशीउगति।

$$३'।५२''।१५'''।२५''''=३।५२।१५+\frac{२५}{६०}=३।५२।१५+\frac{५}{१२}=३।५२।$$

$$\frac{१८०+५}{१२}=३।५२।\frac{१८५}{१२}=३।५२+\frac{१८५}{१२\times६०}=३।५२+\frac{३७}{१२\times१२}=३।५२$$

$$+\frac{३७}{१४४}=३।\frac{५२\times१४४+३७}{१४४}=३।\frac{७५२५}{१४४}=३+\frac{७५२५}{६०\times१४४}=३+\frac{१५०५}{१२\times१४४}$$

$$=\frac{३\times१२\times१४४+१५०५}{१७२८}=\frac{६६८६}{१७२८}\text{ ततः }\left(\frac{६६८६}{६०\times१७२८}\right)^{\circ}=\left(\frac{६६८६}{१०३६८०}\right)^{\circ}$$

$$=\frac{१०^{\circ}}{१०३६८००}=\frac{१०^{\circ}}{१५५}\text{ स्वल्पान्तरात् }\therefore\text{ प्रथमखं}-\text{द्वितीयखं}=\text{वास्तवशुशीउग}=\frac{५}{३}$$

$$-\frac{१०}{१५५}=\frac{१०}{६}-\frac{१०}{१५५}=\frac{१०\times८}{६\times८}-\frac{२}{३१}=\frac{८}{\frac{६\times८}{१०}}-\frac{२}{३१}=\frac{८}{\frac{४८}{१०}}-\frac{२}{३१}=\frac{८}{५}-\frac{२}{३१}$$

$$=\frac{८}{५}-\frac{४}{६२}\text{ अत्र द्वितीयखण्डस्य कलात्मककरणेन }\frac{८}{५}=\frac{४}{६२}\times६०=\frac{८}{५}-\frac{४\times२}{६२}$$

$$\times३०=\frac{८}{५}-\frac{८}{६२}\times३०\text{ पूर्वमाचार्योक्तशुक्रशीघ्रोच्चगतौ }१^{\circ}।३६'।७''।४४'''।३१''''\text{ ३१}$$

स्थाने ३५ गृहीत्वोपपत्तिः कृतेत्यतः स्वल्पान्तरात् $\frac{८}{५}-\frac{८}{६२}$ गृहीतम् ततोऽहर्गणसम्बन्धिशुक्रशीघ्रोच्चम् $-\frac{८\text{ अहर्गण}}{५}-\frac{८\text{ अहर्गण}}{६२}$ एतेनोपपन्नमाचार्योक्तम् । तथा च पूर्वोक्तोपपत्तौ प्रथमखं — द्वितीयखं $=\frac{५^{\circ}}{३}-\frac{१०^{\circ}}{१५५}=\frac{१०}{६}-\frac{१०^{\circ}}{१५५}=$ वास्तवशुशीउग, ततः शुक्रशीघ्रोच्चम् $=\frac{१०^{\circ}\text{अहर्गण}}{६}-\frac{१०^{\circ}\text{ अहर्गण}}{१५५}$ एतेन 'ऋतुभिरक्षदिनैरित्यादि' भास्करोक्तमपि शुक्रशीघ्रोच्चानयनमुपपद्यत इति ॥४८॥

शुक्रशीघ्रोच्चानयन के लिये उपपत्ति

एक दिन में शुक्रशीघ्रोच्चगति $=१^{\circ}।३६'।७''।४४'''।३१''''$ यहां अन्तिमावयव ३१ के स्थान में ३५ लेकर क्रिया करते हैं तब शुक्रशीउग $=१^{\circ}।३६'।७''।४४'''।३५''''$ यहां स्वल्पान्तर से $१^{\circ}।४०'$ ग्रहण करते हैं तब $१^{\circ}।४०'=$अवास्तवशुशीउग, अब वास्तव और अवास्तव शुक्रशीघ्रोच्चगति के अन्तर करने से $१^{\circ}।४०'-(१^{\circ}।३६'।७''।४४'''।३५'''')$ $=३'।५२''।१५'''।२५''''$, यहां $१^{\circ}।४०'=$प्रथमखण्ड, $३'।५२''।१५''।२५''''=$द्वितीयखण्ड तब प्रथम खण्ड – द्वितीय खण्ड $=१^{\circ}।४०'-(३'।५२''।१५''।२५'''')=१^{\circ}।३६'।७''।४४''$।

३५″″=वास्तवशुशीउग १°।४′०=१+$\frac{४०}{६०}$=१+$\frac{२}{३}$=$\frac{३+२}{३}$=$\frac{५°}{३}$= प्रथम खण्ड,

३′।५२″।१५‴।२५⁗=३।५२।१५+$\frac{२५}{६०}$=३।५२।१५+$\frac{५}{१२}$=३।५२।$\frac{१८०+५}{१२}$=३।५२।

$\frac{१८५}{१२}$=३।५२+$\frac{१८५}{६०\times१२}$=३।५२+$\frac{३७}{१२\times१२}$=३।५२+$\frac{३७}{१४४}$=३।

$\frac{५२\times१४४+३७}{१४४}$=३।$\frac{७५२५}{१४४}$=३+$\frac{७५२५}{६०\times१४४}$=३+$\frac{१५०५}{१२\times१४४}$

=$\frac{३\times१२\times१४४+१५०५}{१७२८}$=$\frac{६६८६}{१७२८}$ अंशात्मक करने से $\frac{६६८६}{६०\times१७२८}$=$\frac{६६८६}{१०३६८०}$

=$\frac{६६८६\times१०}{१०३६८०\times१०}$=$\frac{१०}{\frac{१०३६८०\times१०}{६६८६}}$=$\frac{१०}{१५५}$ स्वल्पान्तर से,

∴प्रथम खण्ड—द्वितीय खण्ड=वास्तव शुक्रशीउग=$\frac{५}{३}-\frac{१०}{१५५}$=$\frac{१०}{६}-\frac{१०}{१५५}$

=$\frac{१०\times८}{६\times८}-\frac{२}{३१}$=$\frac{८}{\frac{६\times८}{१०}}-\frac{२}{३१}$=$\frac{८}{\frac{४८}{१०}}-\frac{२}{३१}$=$\frac{८}{५}-\frac{२}{३१}$=$\frac{८}{५}-\frac{४}{६२}$ यहां द्वितीय खण्ड

को कलात्मक करने से $\frac{८}{५}-\frac{४}{६२}\times६०$=$\frac{८}{५}-\frac{८}{६२}\times३०$ पहले शुक्रशीघ्रोच्चगति १°।३६′।

७″।४४‴।३१⁗⁗ में ३१ के स्थान में ३५ ग्रहण कर उपपत्ति की गई है, इसलिये स्वल्पान्तर

से $\frac{८}{५}-\frac{८}{६२}$ यहीं ग्रहण करते हैं तब अहर्गण-सम्बन्धि-शुक्रशीक्रोच्च=$\frac{८\text{ अहर्गण}}{५}$

$-\frac{८\text{ अहर्गण}}{६२}$ इससे आचार्योक्त उपपन्न हुआ। पहले की उपपत्ति में प्रथमखं—द्वितीयखं

=$\frac{५}{३}-\frac{१०}{१५५}$=$\frac{१०}{६}-\frac{१०}{१५५}$= शुक्रशीघ्रोच्चगति ∴ शुक्रशीघ्रोच्च=$\frac{१०°\text{ अहर्गण}}{६}$

$-\frac{१०°\text{ अहर्गण}}{१५५}$ इससे 'ऋतुभिरक्षदिनैः' इत्यादि भास्करोक्त शुक्रशीघ्रोच्चानयन उपपन्न

हुआ ॥ ४८ ॥

प्रकारान्तरेणोपपत्तिः

कल्पशुक्रभगणैस्तत्रत्यैः कुदिनैश्चानुपातेन भागात्मिका दैनन्दिनी शुक्रगतिः=$\frac{७०२२३८९४९२°}{४३८३१०१२५०}$

$$= १ + \cfrac{१}{१ + \cfrac{१}{१ + \cfrac{१}{१ + \cfrac{१}{१ + \cfrac{४७१३७४६०}{८४८३३७७७४}}}}}$$

अत्रासन्नमानानयने क्रमे लब्धयः

लब्धयः = १, १, १, १, १,

आसन्नमानानि = $\frac{१}{१}, \frac{२}{१}, \frac{३}{२}, \frac{५}{३}, \frac{८}{५}$ अत्राचार्येणे $\frac{८}{५}$ दमासन्नमानं परिगृहीतम् । इदमासन्नमानं वास्तवभिन्नेऽ $\frac{७०२२३८९४९२}{४३८३१०१२५०}$ स्मिन् योजने वियोजने च तन्मौल्यं न हीयते ।

अतः शीघ्रोच्चगतिः शुक्रस्य

$$= \frac{७०२२३८९४९२}{४३८३१०१२५०} + \frac{८}{५} - \frac{८}{५}$$

$$= \frac{८^{०}}{५} + \frac{७०२२३८९४९२}{४३८३१०१२५०} - \frac{८}{५}$$

$$= \frac{८^{०}}{५} + \frac{५ \times ७०२२३८९४९२ - ८ \times ४३८३१०१२५०}{५ \times ४३८३१०१२५०}$$

$$= \frac{८^{०}}{५} + \frac{४७१३७४६० \times ६० \times ८}{५ \times ४३८३१०१२५० \times ८}$$

$$= \frac{८^{०}}{५} + \frac{८'}{६२}$$

अत्र $\frac{५ \times ४३८३१०१२५० \times ८}{४७१३७४६० \times ६०}$

$$= \frac{४३८३१०१२५० \times २}{४७१३७४६० \times ३}$$

= ६२ स्वल्पान्तरादर्धाधिके रूपग्रहणाच्च ।

अतोऽहर्गणान्ते

शुक्रः=$\frac{\text{अ ८}'}{५}+\frac{८\ \text{अ}'}{६२}$ उपपन्नमाचार्योक्तम् ।

इदानीं शनिचन्द्रोच्चयोरानयनमाह

द्विगुणः कला दिनगणस्तिथिरामैर्द्वे कले च सूर्यसुतः ।
नवभिर्भागः सागरखशून्यवेदैश्च चन्द्रोच्चम् ॥४९॥

वा. भा.—शनेरपि यद्येकेन दिनेन लिप्ताद्वयं तदिष्टाहर्गणेन किमिति पुनश्च तिथिरामसंख्यैर्दिनैर्यदि द्वे कले भवतः तदिष्टाहर्गणेन किमिति फलद्वययोगः शनिः । अथ चन्द्रोच्चस्य यदि नवभिर्दिनैः भागो भवति तदिष्टाहर्गणेन किमिति पुनश्च सागरखशून्यवेदैश्च यदि सावनैरेको भागः तदिष्टाहर्गणेन किमिति फलद्वययोगः चन्द्रोच्चम् ॥४९॥

वि. भा.—श्लोकोक्त्या कलात्मकशनिः=२ अहर्गण+$\frac{२\ \text{अहर्गण}}{३१५}$,
चन्द्रोच्चमंशाद्यम्=$\frac{\text{अहर्गण}}{९}+\frac{\text{अहर्गण}}{४००४}$ ॥४९॥

अत्रोपपत्तिः

अत्रैकदिने शनेः कलादिकागतिः=२′।०″।२२‴।५१⁗, एतावत्येव भास्करोक्ताप्यस्ति ।

अथ ०″।२२‴।५१⁗=०।२२+$\frac{५१}{६०}$=०।२२+$\frac{१७}{२०}$=०।$\frac{४४०+१७}{२०}$=०।$\frac{४५७}{२०}$=०+$\frac{४५७}{६०\times२०}=\frac{४५७}{१२००}=\frac{४५७\times२}{१२००\times२}=\frac{२}{\frac{१२००\times२}{४५७}}=\frac{२''}{५}$ कलात्मक करणेन$\frac{२}{५\times६०}=\frac{२}{३००}=\frac{२}{३१५}$ स्वल्पा ततः शनिगतिः २′+$\frac{२'}{३१५}$ ततः कलात्मकशनिः=२ अहर्गण′ +$\frac{२\ \text{अहर्गण}'}{३१५}$ एतेनाचार्योक्तमुपपद्यते । यदि चै$\frac{'२}{५}$ तत्कलात्मकं न क्रियते तदा २′+$\frac{२''}{५}$=शनिगतिः, ततः शनिः=२ अहर्गण′

$+\frac{२\ \text{अहर्गण}''}{५}$ एतेन 'द्विघ्नो दिनौघः पृथगक्षभक्तो लिप्ता विलिप्ता ध्रुवकेस्वमार्किरिति' भास्करोक्तमुपपद्यत इति ।

एवमेवैकदिनसम्बन्धिनीमाचार्योक्तां कलादिकां ६'।४०''।५३'''।५६'''' चन्द्रोच्चगतिं संगृह्य पूर्वोक्तवत्क्रियाकरणेन चन्द्रोच्चमंशाद्यम् $+\frac{\text{अहर्गण}}{९}=\frac{\text{अहर्गण}}{४००४}$ एतावताऽऽचार्योक्तं सर्वमुपपद्यत इति ।।४९।।

अब शनि और चन्द्रोच्च के आनयन को कहते हैं

हि. भा.—श्लोकोक्ति से कलात्मकशनि $=२$ अहर्गण $+\frac{२\ \text{अहर्गण}}{३१५}$, अंशादि-चन्द्रोच्च $=\frac{\text{अहर्गण}}{९}+\frac{\text{अहर्गण}}{४००४}$ ।।४९।।

उपपत्ति

एकदिन में शनि की कलादिक गति $=२'।०''।२२'''।५१''''$ भास्कराचार्योक्त भी इतनी ही है, $०''।२२'''।५१''''=०।२२+\frac{५१}{६०}=०।२२+\frac{१७}{२०}=०।\frac{४४०+१७}{२०}$

$=०।\frac{४५७}{२०}=०+\frac{४५७}{६०\times२०}=\frac{४५७}{१२००}=\frac{४५७\times२}{१२००\times२}=\frac{२}{\frac{१२००\times२}{४५७}}$

$=\frac{२'}{५}$ कलात्मक करने से $\frac{२}{५\times६०}=\frac{२}{३००}=\frac{२}{३१५}$ स्वल्पान्तर से $\therefore$ शनिगति $=२'+\frac{२'}{३१५}$ अतः कलात्मकशनि $=२$ अहर्गण$'+\frac{२\ \text{अहर्गण}'}{३१५}$ इससे आचार्योक्त उपपन्न हुआ । यदि $\frac{२''}{५}$ इसको कलात्मक नहीं करते हैं तो शनिगति $=२'+\frac{२''}{५}$ अतः शनि $=\text{अहर्गण}\left(२'+\frac{२''}{५}\right)=२\ \text{अहर्गण}'+\frac{२\ \text{अहर्गण}'}{५}$ इससे 'द्विघ्नो दिनौघः पृथगक्षभक्तो लिप्ता विलिप्ता इत्यादि' भास्करोक्त उपपन्न होता है ।। एवं एकदिनसम्बन्धिनी आचार्योक्त कलादिक चन्द्रोच्चगति ६'।४०''।५३'''।५६'''' से पूर्वोक्तवत् क्रिया करने से

अंशादि चन्द्रोच्च $=\frac{\text{अहर्गण}}{९}=\frac{\text{अहर्गण}}{४००४}$ होता है, इससे आचार्योक्त सब उपपन्न हुआ ॥४९॥

प्रकारान्तरेण वोपपत्तिः

अत्रापि कल्पशनिभगणैः कुदिनैश्च कलात्मिका दैनन्दिनी शनिगतिः

$$=\frac{१४६५६७२९८\times १२\times ३०\times ६०}{१५७७९१६४५०००}$$

$$=\frac{१४६५६७२९८०\times ६०}{४३८३१०१२५००}=\frac{८७९४०३७८८००}{४३८३१०१२५००}$$

$$=२'+\frac{२७८३५३८०\times २}{४३८३१०१२५०\times २}=२'+\frac{२'}{३१५}$$ स्वल्पान्तरात् अतोऽहर्गणान्ते

कलात्मिका शनिगतिः $=२\ अ'+\frac{२\ अ'}{३१५}$

अथवा प्रकारान्तरेण चन्द्रोच्चमध्यमानयनोपपत्तिः। अत्रानुपातेन भागात्मिका दैनन्दिनी चन्द्रोच्चगतिः

$$=\frac{४८८१०५८५८}{४३८३१०१२५०}$$

$$=०^{\circ}+\cfrac{१}{८+\cfrac{४५८२५४३८६}{४८८१०५८५८}}$$

$$=०^{\circ}+\cfrac{१}{८+\cfrac{१}{१+\cfrac{२९८५१४७२}{४५८२५४३८६}}}$$

अत्रासन्नमानानयने—

लब्धयः क्रमेण $=०, ८, १$

आसन्नमानानि $=\frac{०}{१}, \frac{१}{८}, \frac{१}{९}$ अत्राचार्येणे $\frac{१}{९}$ दमासन्नमानं स्वीकृतम्।

ततो वास्तवभिन्नेना $\frac{४८८१०५८५८}{४३८३१०१२५०}$ नेन सह योगान्तरेण लब्धौ विकाराभावात्

भागात्मक चन्द्रोच्चम्$=\frac{१^{\circ}}{९}+\frac{४८८१०५८५८}{४३८३१०१२५०}-\frac{१}{९}$

$=\frac{१^{\circ}}{९}+\frac{४३९२९५२७२२-४३८३१०१२५०}{९\times४३८३१०१२५०}$

$=\frac{१^{\circ}}{९}+\frac{९८५१२७२}{३९४४७९११२५०}$

$=\frac{१^{\circ}}{९}+\frac{१^{\circ}}{४००४}$ स्वल्पान्तरात् ।

अतोऽहर्गणान्ते चन्द्रोच्चम्

$=\frac{अ^{\circ}}{९}+\frac{अ^{\circ}}{४००४}$ उपपन्नम् ।

इदानीं चन्द्रपातानयनमिष्टदिने ग्रहानयनं चाह

द्युगणो नन्दशशाङ्कैः शशिशून्यस्वरयमैश्च शशिपातः ।
रविमण्डलान्तिकयुता मध्या भगणान्तगाः शेषाः ॥ ५० ॥

वा. भा.—अथ पातस्य यदि नन्दशशांकसंख्यैः सावनैरेको भागो भवति तदिष्टाहर्गणेन किमिति पुनश्च शशिशून्यस्वरयमसंख्यैर्यदि वासरैरेको भागो भवति तदिष्टाहर्गणेन किमिति फलद्वययोगश्चन्द्रपातः । एवं रविमण्डलान्ताहर्गणादिष्टादन्यस्मात् वा ग्रहात् फलानयनम् । वासना चात्र भूदिनैः कल्पभगणभागैश्च प्रदर्श्य सर्वेषां भौमस्य मया प्रदर्श्यते । तद्यथा कल्पार्कसावनदिनानां भौमकल्पभगणभागानां चापवर्त्तनं कथमपि न शक्यते कर्त्तुमतो भौमकल्पभगणभागस्य इष्टैः भागैस्तथोनः क्रियते यावद्भूदिनैः सहापवर्त्तनं प्रयच्छति ते चेष्टभागा नवगुणरसाग्निखरसशून्यत्रिगुणाः ३३०६०३६३९ एतैरूना कल्पे भौमभगणभागाः खयमनवागरसद्विवसुपंचाष्टरसद्विवसवः ८२६८५८२६७९२० जाताः शेषभागाः चन्द्राष्टद्विकृतरसषट्सप्तयमेषुरसद्विवसवः ८२६५२७६६४२८१ एतैः भूदिनानामपवर्त्तनं प्रयच्छन्ति अतस्तावदपवर्त्यते एत एव रूपागशरवसुत्रिचन्द्रशरागाः ७५१३८५७१ अपवर्त्तिता जाता एकादशभूदिवसाः । अथानेनैवापवर्त्तिता जाता एकविंशतिः एवमिन्दुयमैः सावनैरेकादशभागा भौमस्य भवन्तीदानीं पूर्वत्यक्तभागैः षष्टिगुणैः सह भूदिनानामपवर्त्तनं क्रियते । तद्यथा रसवसुखगुणाग्नित्रिवह्निशून्याष्टचन्द्रैः १८०३३३३०८६ भूदिवसाः अपवर्त्तिता जाताः शरसप्तवसवः पूर्वत्यक्तभागैश्च षष्टिगुणा जाताः खवेदाग्निवसुचन्द्रद्विरसगुणाष्टनवचन्द्राः १९८३६२१८३४० अपवर्त्तिताः तेनैवापवर्त्तकराशिना जाता एकादशलिप्ताः यतः षष्ठगुण-

भागा अपवर्त्तिता अतः शरसप्तवसुभिः सावनैरेकादशलिप्ताश्च भवन्ति। एवं सर्वग्रहाणां मन्दशीघ्रपातानामपि स्वधियाचार्येण त्रैराशिकवासना प्रकल्पिता लघ्वर्थं गणकैरपि यथासम्भवं सर्वेषां योज्यं ततोऽनेन प्रकारेण ग्रहाः राश्यादिकाः समागतास्ते रविमेषादेरारभ्यते। रविमण्डलान्तिकयुतमध्या इति रविमण्डलान्तावधिः यवस्वनेन मध्यमेन युताः सन्तोऽभीष्टदिनोन्मण्डलकालिका मध्यरेखायां भवन्तीत्यर्थः। भगणान्तिका शेषा इति येषां वर्षान्तिकादहर्गणादानयनं नोक्तं ते शेषाश्चन्द्रवर्जानां मध्यपाता रविमण्डलान्तावधिजा एव तेऽभीष्टदिने मध्यमाः भवन्ति यतस्तेषां वर्षमध्ये विशेषो नास्तीति स्वल्पत्वात्तद्गतेरतः किमायासेनेत्यर्थः। एवं तावदुन्मण्डलका मध्या आनीताः ॥५०॥

वि. भा.—अंशात्मकश्चन्द्रपातः $=\frac{\text{अहर्गण}}{१९}+\frac{\text{अहर्गण}}{२००१}$, रविमण्डलान्तिकयुताः रविभगणान्तेऽर्थाद्रविवर्षान्ते पूर्वं ये ग्रहा आनीतास्तैरहर्गणोत्पन्नग्रहा युक्ताः तदेष्टदिनोदयकाले मध्या ग्रहा भवन्ति, शेषाः (मन्दोच्चादयः) भगणान्तगा अर्थाद्रविवर्षान्तोत्पन्ना एवेष्टदिने बोध्यास्तेषां गत्यल्पत्वादिति ॥५०॥

अत्रोपपत्तिः।

आचार्यमतेनैकस्मिन् दिने कलादिका चन्द्रपातगतिः=३′।१०″।४८‴।२२⁗

एतद्वशेन पूर्वोक्तान्यग्रहादिसाधनोक्तक्रियाकरणेनांशात्मकश्चन्द्रपातः $=\frac{\text{अहर्गण}}{१९}$

$+\frac{\text{अहर्गण}}{२७०१}$ सिद्ध्यतीति। शेषं भाष्ये स्पष्टं प्रतिपादितमेवेति ॥ ५० ॥

अब चन्द्रपातानयन और इष्ट दिन में ग्रहानयन को कहते हैं

हि. भा.—श्लोकोक्ति से अंशात्मक चन्द्रपात $=\frac{\text{अहर्गण}}{१९}+\frac{\text{अहर्गण}}{२००१}$, अहर्गणोत्पन्नग्रहों में रविवर्षान्तकालिकग्रहों (जो कि पहले साधित हैं) को जोड़ने से इष्ट काल में मध्यम ग्रह होते हैं, शेष (मन्दोच्चादि) रविवर्षान्तकालिक जो हैं वही इष्टकालिक भी समझने चाहियें क्योंकि उनकी गति बहुत अल्प है इति ॥ ५० ॥

उपपत्ति

आचार्यमत से एक दिन में चन्द्रपातगतिकलादिक=३′।१०″।४८‴।२२⁗ इससे जैसे पहले तत्तद्ग्रहों की दैनन्दिनी गतिवश से साधन किया गया है, उसी तरह साधन

करने से अंशात्मक चन्द्रपात $=\frac{\text{अहर्गण}}{१९}+\frac{\text{अहर्गण}}{२७०१}$, शेष बातें भाष्य में प्रतिपादित हैं, इससे आचार्योक्त उपपन्न हुआ, इति ॥ ५० ॥

प्रकारान्तरेणोपपत्तिः

अत्रापि प्राग्वदनुपातेन भागात्मिका दैनन्दिनी विलोमा चन्द्रपातगतिः

$$=\frac{२३२३१११६८}{४३८३१०१२५०}$$

$$=०+\cfrac{१}{१८+\cfrac{१}{१+\cfrac{२०८१०९४२}{२११५००२२६}}}$$

अत्राप्यासन्नमानानयने

लब्धयः ०, ८, १

आसन्नमानानि $\frac{०}{१}$, $\frac{१}{१८}$, $\frac{१}{१९}$ अग्रिमेऽवयवत्यागात् ।

$$\therefore \text{ पातगतिः}=\frac{१^{\circ}}{१९}+\frac{२३२३१११६८}{४३८३१०१२५०}-\frac{१}{१९}$$

$$=\frac{१^{\circ}}{१९}+\frac{१९\times२३२३१११६८-४३८३१०१२५०}{१९\times४३८३१०१२५०}$$

$$=\frac{१^{\circ}}{१९}+\frac{४४१३९१२१९२-४३८३१०१२५०}{८३२७८९२३७५०}$$

$$=\frac{१^{\circ}}{१९}+\frac{३०८१०९४२}{८३२७८९२३७५०}$$

$$=\frac{१^{\circ}}{१९}+\frac{१}{२७०३}\text{ स्वल्पान्तरात्}$$

अत्राचार्येण ३ स्थाने १ गृहीतमतोऽस्यां पातगतौ किञ्चिदन्तरं भवितुमर्हति । विद्भिर्विवेचनीयम् । वस्तुतो विचार्यमाणे तथा सति तत्प्रतिविकलायां नान्तरं पतति, किन्तु तदग्रेऽन्तरमापद्यते । ग्रन्थकारेण तदुपेक्षितम् । विकलावध्येव ग्रहसाधनत्वविधानात् । मन्मते तु "शशिशून्यस्वरयमै" रित्यत्र गुणशून्यस्वरयमैरिति पाठः साधीयान् ।

इदानीं यथाऽभीष्टकालिका भवन्ति तदर्थमार्यामाह

पादार्धविपाददिनैं रात्र्यर्धास्तमय दिनदलौदयिकाः।
ऊनाः कृत्वा तिथयो देशान्तरनाडिकोनयुताः॥ ५१॥

वा. भा.—पादार्धविपाददिनैरूनीकृत्वा तिथयो यथासंख्यं रात्र्यर्धास्तमय-दिनदलौदयिका ग्रहा आगच्छन्ति। एतदुक्तं भवति, यास्तिथयोऽहर्गणानयने दीयन्ते पञ्चदशभिर्घटिकाभिरूना इत्यर्थः। तदात्र योऽहर्गणो भवति तेन ये ग्रहा आनीयन्ते तेऽतीतार्धरात्रकालिका भवन्ति। अथवार्धदिनेन तिथयः ऊनीकृत्वा दीयन्ते घटिकानां त्रिंशतेत्यर्थः तदा तदहर्गणादस्तमयका ग्रहा आगच्छन्ति वाऽतीते-दिनेऽस्तोन्मण्डले इत्यर्थः। अथ विपाददिनोना दीयन्ते तिथयो विगता पाददिनञ्च पञ्चचत्वारिंशद्घटिका इत्यर्थः। ताभिरूनास्तिथयो यदा दीयन्ते तदा तद-हर्गणाद्ये ग्रहा आगच्छन्ति, ततो दिनदलकालिका भवन्ति। अथ सकलेनैव दिनेनो-नास्तिथयो दीयन्ते तदा तदहर्गणाद्ये ग्रहा आगच्छन्ति, ततो दिवसोन्मण्डलकालिका आगच्छन्ति, किमेतावतैवोक्तकालिका भवन्ति नेत्याह, देशान्तरनाडिका फलोन-युता इति, स्वदेशे यावत्यो देशान्तरनाडिकास्ताभिर्यास्तिथयो यथासम्भवमून-युताश्च देशवसात् कृतास्ताः पादार्धविपाददिनैरूनीकृता यदाहर्गणे दीयन्ते तदा तदहर्गणादुक्तकालिका आगच्छन्ति ग्रहा अर्थादेव यदा देशान्तरनाडिकायुताः पादार्धविपाददिनैरधिकाः तिथयोऽहर्गणानयने दीयन्ते तदा गामिनीदिनदलास्त-मयार्धरात्रोदयादिषु कालेष्वागच्छन्ति ग्रहाः। एवमिष्टघटिकाभिरिष्टकालिका ग्रहाः आगच्छन्ति। एष्यदतीतयोरपि कालयोरित्यर्थः। अथवोपलक्षणार्थमेते तद्यथागतो एवान्यथादेशान्तरकृतमध्यास्तात्कालिकाः क्रियन्ते। अभीष्टे काले गते गम्ये वा स्वमध्यभुक्तिमिष्टनाडिकाहतां विभजेत् षष्ट्यावाप्तं ततो यत्फलं तेन ग्रहोऽतीतकाले ऊनः कार्यः, एष्यत्कालेऽधिकः एवंकृतस्तात्कालिको भवति।

वि. भा.—पादार्धविपाददिनैरूनीकृत्य तिथयः क्रमशो रात्र्यर्धास्तमयदिन-दलौदयिका ग्रहाः समागच्छन्त्यर्थादहर्गणानयने यास्तिथयो दीयन्ते यदि ता दिन-पादमितार्घटीभिरूना दीयन्ते तदा तत्र योऽहर्गणो भवति ततः समानीता ग्रहा गतरात्र्यर्धकालिका भवन्ति, यदि दिनार्धोनादहर्गणाद् ग्रहा आनीयन्ते तदा ते गतास्तमयकालिका भवन्ति, विपादोनादहर्गणाद्ये ग्रहाः समागच्छन्ति ते गत-दिनार्धकालिका भवन्ति, एवं दिनोनादहर्गणाद्ये ग्रहास्ते गतदिनौदयिका भवन्ति। एवं यद्यहर्गणानयने प्रागपरत्रदेशान्तरनाडिकोनयुतास्तिथयो दीयन्ते तदा स्वदेशे तत्तत्कालिका ग्रहा भवन्तीति॥ अत्राचार्यकथनमेव प्रमाणं नान्यत्कारणं वक्तुं शक्यत इति॥ ५१॥

अब ग्रहानयन में विशेष कहते हैं

हि. भा.—ग्रहगणानयन में जो तिथि जोड़ते हैं, उन तिथियों में दिन के पाद (चतुर्थांश) तुल्य घटी को घटा कर जोड़ा जाय तब उस पर से जो ग्रहगण होता है उस से साधित ग्रह गत रात्र्यर्धकालिक होते हैं। दिनार्धोन (दिनार्धरहित) तिथि वश से जो ग्रहगण होगा उस से साधित ग्रह गतास्तकालिक होते हैं। एवं विपाददिनोनतिथि वश से जो ग्रहगण होगा उस से साधित ग्रह गत दिनार्धकालिक होते हैं। दिनोनतिथिवश कर के जो ग्रहगण होता है उस से साधित ग्रह गत दिनोदयिक होते हैं। इसी तरह पूर्वापर देश वश से देशान्तर घटी कर के हीन और युत तिथि को यदि ग्रहगणानयन में जोड़ा जाय तब जो ग्रहगण होगा उस से साधित ग्रह अपने देश में तत्तत्कालिक होते है, इन सब विषयों में आचार्य कथन ही प्रमाण कहा जा सकता है, दूसरा कारण नहीं कह सकते। इति ॥५१॥

एवं कल्पगत कालात् तात्कालिकानयनमुक्त्वा इदानीं कलिगतकालात् दर्शयति आर्यापट्केन

कलिगतशुद्धिः प्राग्वत्-शुक्राद्योऽब्दाधिपोऽब्दभगणवधात् ।
क्षितिजस्य खत्रयाष्टरससप्तवसुसुखाग्निवेद ४३०८७९८०००युतात् ॥५२॥
बुधशीघ्रस्य खखाम्बररसनन्दाष्टाष्टवसुयमोदधिभिः ४२८८८९६००० ।
खचतुष्टययमशरगुणशशित्रिवेदैः ४३१३५२०००० सुरेन्द्रगुरोः ॥५३॥
भार्गवशीघ्रस्याम्बरखखाष्टवेदाब्धिखाग्निकृतैः ४३०४४८०८० ।
भास्करसुतस्य खत्रयरविगुणशरखदहनसमुद्रैः ४३०५३१२००० ॥५४॥
खचतुष्टयपक्षेन्दुगुणगुणनवभि ९३३१२०००० रर्कमन्दस्य ।
इन्दोः खत्रययमशरनवपञ्चव्योमशरचन्द्रैः १५०५९५२००० ॥५५॥
खत्रययमनवपञ्चाष्टरामधृतिभिः १८३८५९२००० शशाङ्कपातस्य ।
कल्पगतभगणघातात्कुजादिमन्दोच्चपातानाम् ॥५६॥
भगणादिकल्पवर्षैर्लब्धं रविमण्डलान्तिका मध्याः ।
मेषादिद्युगणफलाधिका भवन्तीष्टदिनमध्याः ॥५७॥

वा. भा.—कलियुगादेरारभ्य यो गतः कालोऽभीष्टरविमण्डलान्ते सकले गतास्तस्मात् शुद्धिः प्राग्वद्यथा-कल्पगतात्तद्वदित्यर्थः। एतदुक्तं भवति। यथा कल्पगताब्दा गुणिता रूपाष्टजिनैरित्यादिना दिनांशा अवमानि अवमांशा कल्पगताब्ददिनयुतेः। सूर्याद्योऽब्दाधिपतिर्द्विगुणाब्दयोगादधिमांशाः शेषास्तिथयः शुद्धिदिनानीति प्रागानीतं तद्वत्कलिगताब्दैरथानेयम्। किन्तु अत्रायं विशेषः कलिगतदिनयुते शुक्राद्योऽब्दपतिर्भवति यतो गुरुदिवसान्ते द्वापरयुगं परिसमाप्तं शुक्रदिनादौ कलियुगं प्रवृत्तं रविचन्द्रौ तत्र

द्वापरान्ते निःशेषावभूताधिमासावमा अपि निःशेषा अभूवन्। तत्रैव काले चैत्रादिः स एव ततः प्रभृति शकान्ते ये गता अब्दा गोगैकगुणाः तेष्वभीष्टशककालगतवर्षाणि दत्वा राशियों भवति। तेन कलिगताब्दराशिनैव कर्म्म कर्त्तव्यम्। तत्र दिनांशा अवमांशाश्च कल्पाब्दकृतानां तुल्या एवं भवन्ति कलिगताब्ददिनयुतैः सप्तहृताया: शेषशुक्राद्योऽब्दपतिरियान्विशेषः। भौमादिग्रहमन्दशीघ्रपातास्तत्र निःशेषा नाभूवन्। अतस्तेषां द्वापरान्ते कल्पगताब्दभगणा वधे कलाहृते भगणशुद्धशेषा ये तैरधिकाः कलिगताब्दस्वभगणवधादानयने आचार्येणैव निबद्धम्। अब्दभगणावधिरूपे क्षितिजस्य खत्रयाष्टरससप्तवसुखाग्निवेदयुतादित्याभिरार्यष्टाभिः तद्यथा द्वापरान्ते भगणशुद्धशेषं कुजस्य ४३०८७६८००० अथ बुधशीघ्रस्यास्य ४२८८८९६००जीवस्य ४३१३५२०००० शुक्रशीघ्रस्य ४३०४४४८००० अथ शनेः ४३०५३१२००० अथ रविमन्दस्य ९३३१२०००० चन्द्रमन्दस्य १५०५९५२३००० । चन्द्रपातस्य १८३८१५०००० शेषाणां पुनः कुजादिमन्दपातानामानयनं कल्पगताब्दैरेव कार्यम्। यतस्तेषां बहुभिरपि वर्षैरन्तरं न भवति। फलानयनेऽतो ग्रन्थगौरवभयान्न पठितास्तेषां द्वापरान्तभगणशेषशुद्धशेषानां कलिगताब्दस्वभगणवधात् स्वक्षेपकयुतात्कल्परविवर्षैर्यल्लब्धं भगणादि फलम्। पृथक्-पृथक् तेन रविमण्डलान्तिका ग्रहा मध्या भवन्ति शुद्धावमशेषादिकं सर्वमुक्तवत् कृत्वा ततश्चैत्राद्याः तिथयः शुद्धिविहीना इत्यादिना रविमेषादिकोऽहर्गणः कार्यः। तत इष्टदिने मेषाद्यहर्गणाद्यत्फलं द्युगणात् सप्तत्यंशमेकादशलिप्तांश भौम इत्यादिना ग्रन्थेनागच्छति तेन स्वेन फलेनाधिकाः सन्तः इष्टदिनमध्ये भवन्तीतिप्रागेवोक्तत्वात् गतार्थमिति वासनाप्यत्र प्रागेवोक्तेति।

वि. भा.—प्राग्वत् (पूर्ववत्) कलिगतशुद्धिः साध्याऽर्थात्पूर्वं यथा कल्पगतवर्षेभ्यः शुद्धिरानीता तथैव कलिगतवर्षेभ्यः साध्या तत्र शुक्राद्यो वारोऽब्दाधिपोऽर्थात्कलियुगादौ शुक्रवारसद्भावाद् वर्षपतिः शुक्रवाराद् भवति। उपरिलिखितश्लोकेषु पठिताङ्का मङ्गलादिग्रहाणां (मङ्गलगुरुशनीनां) रविचन्द्रयोर्मन्दोच्चयोः, बुधशुक्रयोः शीघ्रोच्चयोश्चन्द्रपातस्य च क्षेपसंज्ञकाः स्युः। अब्दभगणवधात् (कलिगतवर्षाणां ग्रहभगणानां च घातात्) पाठपठितस्वस्वक्षेपयुतात्कल्पवर्षैर्भक्ताद्यद् भगणादिलब्धं भवेत्ते रविमण्डलान्तिका (रविवर्षान्तकालिकाः) मध्यमग्रहा भवेयुस्ते मेषादिद्युगणफलाधिकाः (लब्धहर्गणोत्पन्नग्रहैर्युताः) सन्तोऽभीष्टदिने मध्यग्रहा भवेयुरिति॥ ५२-५३-५४-५५-५६-५७॥

अत्रोपपत्तिः

रविवर्षान्ते ग्रहानयनार्थमनुपातः क्रियते, यदि कल्पवर्षैः कल्पग्रहभगणा लभ्यन्ते तदा कल्पगतवर्षैः किं समागच्छतीष्टवर्षान्ते भगणादिग्रहस्तत्स्वरूपम् $=\frac{\text{कल्पग्रहभगण}\times\text{कल्पगतवर्ष}}{\text{कल्पवर्ष}}$ अत्राचार्येण कल्पगतवर्षाणां खण्डद्वयं (कल्पादितः

कल्यादिपर्यन्तमेकं, कल्यादित इष्टवर्षपर्यन्तं द्वितीयम्) कृत तदा तदुत्थापनेन

$$\frac{\text{कल्पग्रहभगण (कल्पादितः कल्यादिपर्यन्तवर्ष+कल्यादितो गतवर्षाणि)}}{\text{कल्पवर्ष}}$$

$$=\frac{\text{कल्पग्रभ}\times\text{कल्पादितः कल्यादिपर्यन्तवर्ष}}{\text{कवर्ष}}+\frac{\text{कल्पग्रभ}\times\text{कल्यादितो गतवर्षाणि}}{\text{कवर्ष}}$$

=इष्टवर्षान्ते भगणादिग्र। अत्र प्रथमखण्डे यद्भगणशेषमानं तस्य क्षेपसंज्ञा कृताऽऽचार्येणेत्येतावताऽऽचार्योक्तिमुपपन्नम् सिद्धान्तशेखरे श्रीपतिना प्रथमखण्डजनितफलस्यैव ग्रहध्रुवसंज्ञा कृता, यथा तदुक्तं, यातवर्षखगपर्ययाहते कल्पवर्षविहृते ग्रहध्रुवाः। ते भवन्ति रविमण्डलान्तिका इति ॥५२-५३-५४-५५-५६-५७॥

अब कलिगतवर्ष ही से शुद्धि आदि के आनयन के लिए विशेष बात और रविवर्षान्तकालिक ग्रहानयन के लिये प्रकारान्तर को कहते हैं

हि. भा.—पूर्ववत् कलिगत शुद्धि साधन करना अर्थात् पहले कल्पगत वर्षों से जिस तरह शुद्धि लायी गई है उसी तरह कलिगत वर्षों से साधन करना, किन्तु कलि के आदि में शुक्रवार था इसलिये वर्षपति की गणना शुक्रवार से होती है। श्लोकों में जो पठिताङ्क हैं वे मङ्गल, बृहस्पति, शनैश्चर इन ग्रहों के, रवि और चन्द्र के मन्दोच्चों के, बुध और शुक्र के शीघ्रोच्चों के तथा चन्द्रपात के क्षेपसंज्ञक हैं, कलिगत वर्षों के और ग्रह भगणों के घात में पाठपठित अपने-अपने क्षेप जोड़कर कल्पवर्ष से भाग देने से जो भगणादि लब्धिप्रमाण आता है वे रविमण्डलान्तिक (रविवर्षान्तकालिक) मध्यम ग्रह होते हैं उनमें मेषादि द्युगण फल (लघ्वहर्गणोत्पन्नग्रह) को जोड़ने से अभीष्ट वर्षान्त में मध्यम ग्रह होते हैं इति ॥ ५२,५३,५४,५५,५६,५७ ॥

उपपत्ति

रविवर्षान्त में ग्रहानयन के लिये अनुपात करते हैं, यदि कल्पवर्ष में कल्प ग्रह भगण पाते हैं तो कल्पगतवर्ष में क्या इस अनुपात से अभीष्ट वर्षान्त में भगणादि मध्यमग्रह आते हैं।

$$\frac{\text{कल्पग्रहभ}\times\text{कल्पगतवर्ष}}{\text{कल्पवर्ष}}=\text{अभीष्टवर्षान्त में भगणादिग्र,}$$

यहाँ आचार्य ने कल्पगतवर्ष के दो खण्ड 'कल्पादि से कल्यादि तक प्रथम खण्ड, और कल्यादि से इष्टवर्षान्त तक द्वितीय खण्ड' किये हैं, तब इनसे कल्पगतवर्ष के उत्थापन करने से

$$\frac{\text{कग्रभ}\times\text{कल्पादि से कल्यादितकवर्ष}}{\text{कल्पवर्ष}}+\frac{\text{कग्रभ}\times\text{कलिगतवर्ष}}{\text{कल्पवर्ष}}=\text{अभीष्टवर्षान्त में भगणादिग्र,}$$

यहाँ प्रथम खण्ड में जो भगण शेष रहता है उसी का नाम आचार्य ने क्षेप रक्खा है। इससे आचार्योक्त उपपन्न हुआ। पूर्वोक्त प्रथमखण्डागतग्रहों को सिद्धान्तशेखर में श्रीपति 'यातवर्ष खगपर्ययाहते कल्पवर्षविहृते ग्रहध्रुवाः। ते भवन्ति रविमण्डलान्तिकाः' इससे ग्रह ध्रुवा कहते हैं इति ॥ ५२-५३-५४-५५-५६-५७ ॥

इदानीं चैत्रसितादिना सावनेनाखण्डेनाहर्गणेन ग्रहानयनचिकीर्षुरादौ तावच्चैत्राद्यर्कोदययोरन्तरपरिज्ञानार्थमार्यामाह

शुद्धीशवधे शुद्धेऽवमशेषात्सावनद्युगणसिद्धिः ।
व्येकावमं गृहीत्वा गुणखमुनियुतान्न शुद्धिश्चेत् ॥५८॥

वा. भा.—शुद्धेरीशानां च वधः शुद्धीशवधस्तस्मिन् शुद्धे सति कुत इत्याह, अवमे यद्यपि सामान्येनावमशेषग्रहणं तथापि रविमण्डलान्तावमशेषात् अवमांशेभ्यो यमनवरसगुणितेभ्यो विकृतेभ्यः स्वच्छेदेन यत्फलं भवति, तस्मादित्यर्थः । एवं कृते शेषं यत्सावनद्युगणशुद्धिश्चैत्रादावमशेषं भवतीत्यर्थः अथ । रविमण्डलान्तावमशेषादेकादशगुणशुद्धिर्न शुध्यति, तदा व्येकावमं गृहीत्वा गुणखमुनियुतात्पूर्वमेवावमानि यान्यतीतानि । नवाग्निसप्तनगैः कल्पगताब्दान् संगुण्य खखरसनवभिर्विभज्य तेभ्यो रूपमेकं विशोधयेद्रविमण्डलान्तावमशेषं च गुणखमुनियुक्तं कृत्वा ततः शुद्धीशवधं विशोध्य चैत्राद्यवमशेषं भवति, तत्रेयं वासना रविमण्डलान्ताद्विपरीत्येन चैत्रादर्कोदयेऽवमशेषं क्रियते तत्र शुद्धिरेव चैत्रादर्कोदयरविमण्डलान्तरं भूः । तत एकैकस्मिन्सावनदिने चन्द्रदिनेन सहान्तरमेकादशसंख्यमवमशेषमतः शुद्धतुल्यैःसावनदिनै शुद्ध एकादशगुणायां तुल्यमवशेषं भवति । अतो रविमण्डलान्तावमशेषाद्विशोध्य शेषं चैत्राद्यर्कोदयेऽवमशेषं भवति सैव सावनद्युगणसिद्धिर्भवति । यस्मात् तत्र दिने तावदेवान्तरं चैत्राद्यर्कोदययोरथैकादशगुणशुद्धिर्न शुद्ध्यति । रविमण्डलान्तावमशेषात्तदा चैत्रादिरविमण्डलान्तयोरंतरं एवोनरात्रयातो ज्ञेयोऽतः पुनरथावमगणादवम संगृहीत्वा रविमण्डलान्तिके त्रिखमुनिसंख्यं योज्यते । यस्मात् त्रिखमुनिसंख्येऽवमशेषेणोनरात्रो भवति । अत उक्तमेकावमं गृहीत्वा गुणखमुनियुतान्न शुध्यति वोदितः किन्त्वत्र तद्द्विगुणाब्दयोगादिति क्रियमाणे शुद्धिरथैकोना भवति, सात्रैकादशगुण विशोध्यते, इत्युपपन्नम् ॥ ५८ ॥

वि. भा.—शुद्धेः (पूर्वपरिभाषितायाः) ईशानां (एकादशानां) च गुणनफलेऽवमशेषात् (रविवर्षान्ते साधितावमशेषात्) ६९२ गुणितादेभिः (९६००) भक्ताच्छुद्धे 'हृतास्त्रिखागैः फलावमविहीना' इत्यादिना चैत्रादौ सावनाहर्गणसिद्धिर्भवति, यदि शुद्धेरेकादशस्य च गुणनफलस्यावमशेषाच्छोधनं न भवेत्तदा रविवर्षान्ते कल्पतो यान्यवमानि तान्येकरहितानि गृहीत्वा ७०३ युतादवमशेषात्तद्गुणनफलं शोध्यमिति ॥ ५८ ॥

अत्रोपपत्तिः

चैत्रादितिथिः=० तदा विपरीतशोधनेन चैत्रादिर्गति—शुद्धि=शुद्धि—तदा 'अवमांशेभ्यो यमनवरसगुणितेभ्य इत्याद्याचार्योक्त्या' ऽवमशेषस्वरूपम् $\frac{६९२ \text{ वक्षयशे}}{९६००}$
७०३

$-११ शु = इष्टक्षयशेष$ । यदि $११ शु > \frac{६९२ वक्षयशे}{९६००}$ तदाऽवमशेषप्रमाणात्मकं भवेत्तदा लब्धिः $= -$ इष्टक्षयशे, धनात्मकार्थमेकयोजनेन

$$१ + \frac{\frac{६९२ वक्षशे}{९६००} - ११ शु}{७०३} = \frac{७०३ + \frac{६९२ वक्षयशे}{९६००} - ११ शु}{७०३} = चैत्रादावमशेषमानम्$$

। यदिचैत्रशुक्लादौ रविवर्षान्तात्पूर्वमेव भवेत्तदा विलोमाहर्गणो भवति लल्लाचार्यादिभिरयमेव चैत्रादावृणाहर्गणः कथ्यत इति । सिद्धान्तशेखरे 'शुद्धिमेव पृथगीश्वराहतां शोधयेदवमशेषकान्निजादित्यादिना' ज्यमेव ब्रह्मगुप्तोक्तप्रकारः श्रीपतिना लिखितो विवेचकैर्ज्ञेय इति ॥ ५८ ॥

अब रविवर्षान्त और चैत्रादि के मध्य में कितने सावनदिन हैं उनका साधन करते हैं

हि. भा.—अवमशेष रविवर्षान्त में जो क्षयशेष है उसको ६९२ इससे गुणाकर अपने हर (९६००) से भाग देने से जो फल होता है, में पूर्वकथित शुद्धि और ग्यारह के घात (गुणनफल) को घटा देने से सावनाहर्गण की शुद्धि होती है अर्थात् "हृतास्त्रिखागैः फलावमविहीना इत्यादि" आचार्योक्त प्रकार से चैत्रादि में सावनाहर्गण की सिद्धि होती है । यदि अवमशेष में शुद्धि और ग्यारह की घात न घटे तब रविवर्षान्त में कल्प से जितने अवम हों उनमें एक घटाकर अवमशेष में ७०३ जोड़कर जो हो उसमें उस घात (गुणनफल) को घटा देना चाहिये इति ॥५८॥

उपपत्ति

चैत्रादितिथि $= ०$ तब विपरीत शोधन से चैत्रादिगति—शुद्धि $= ०$—शुद्धि $= -$ शुद्धि, तब 'अवमांशेभ्यो यमनवरसगुणितेभ्य इत्यादि आचार्योक्त प्रकार से' अवमशेषस्वरूप $= \frac{\frac{६९२ वक्षयशे}{९६००} - ११ शुद्धि}{७०३}$, यदि $११ शुद्धि > \frac{६९२ वक्षयशे}{९६००}$ तब अवमशेष प्रमाण ऋणात्मक होता है, तब लब्धि $= -$ इष्टक्षयशे $= -$ अवमशे, धनात्मक के लिये एक जोड़ने से

$$१ + \frac{\frac{६९२ वक्षयशे}{९६००} - ११ शुद्धि}{७०३} = \frac{\frac{६९२ वक्षयशे}{९} - ११ शुद्धि + ७०३}{७०३} = चैत्रादि में$$

अवमशे, यदि चैत्र शुक्लादि रविवर्षान्त से पहले हो तब विलोमाहर्गण होता है, इसी को लल्लाचार्य आदि आचार्य चैत्रादि में ऋणाहर्गण कहते हैं । सिद्धान्तशेखर में श्रीपति 'शुद्धिमेव पृथगीश्वराहतां शोधयेदवमशेषकान्निजात् इत्यादि' से ब्रह्मगुप्तोक्त प्रकार ही कहते हैं इसको विवेचक लोग समझें इति ॥५८॥

अत्र विशेषविचारः

चैत्रामान्ततो वर्षान्ताव्यवहितपूर्वतिथ्यन्तावधि निरवयवा अधितिथयः इत्येकं खण्डम्। तिथ्यन्तात्सूर्योदयावधि वर्षान्तीयावमघटिका द्वितीयखण्डम्। एवं सूर्योदयाद्वर्षान्तावधि वर्षान्तीयदिनादिघटिका इति तृतीयं खण्डमिति। खण्डत्रय-योगे चैत्रसितादितो वर्षान्तावधि अधिशेषदिनानि सावयवानि। तत्रावमशेषं विशोध्य शेषस्या अति+दिनादिशे स्य शुद्धिसंज्ञा कृता ग्रन्थकृता। इयमेव शुद्धिर्भास्कराचार्यस्यापि।

अथ लघ्वहर्गणावगमे मध्यमार्कसञ्चारवशाद्वर्षान्तस्य ज्ञानाभावात् चैत्रामान्ततोऽभीष्टदैवसिकतिथ्यन्तावधितिथयो गृह्यन्ते। तत्र चैत्रामान्तवर्षान्तरे यदोष्टतिथ्यन्तं मन्यते तत्रेष्टतिथिभ्यः शुद्धेरधिकत्वात् "चैत्रसिताद्यास्तिथयः शुद्धिविहीनाः पृथक्गुणा रुद्रैः" इत्यादिविधानेनावमानयने विप्रतिपत्तिः संपद्येत। शुद्धेः ऋणगतत्वसिद्धेः। अतस्तदानयनार्थमन्यथा यतते।

तथाहि। लघ्वहर्गणानयनेह्याचार्यकृतावमशेषस्वरूपम्

$$= ११\,(\text{इति}-\text{शु}) + \frac{\text{६९२ वक्षशे}}{\text{९६००}}$$
$$\frac{\qquad\qquad\qquad\qquad\qquad\qquad}{७०३} \quad \ldots\ldots\ldots\ldots\ldots (१)$$

अत्र शु<इति कल्प्यते

∴ इति−शु= −शेषम्

$$\therefore (१) \text{ समीकरणस्वरूपम्} = \frac{\frac{\text{६९२ वक्षशे}}{\text{९६००}} - ११ \text{ शेष}}{७०३}$$

अत्रापि यदि $\frac{\text{६९२ वक्षशे}}{\text{९६००}} < ११$ शे।

तदाऽ $\frac{\text{६९२ वक्षशे}}{\text{९६००}} - ११$ शे स्याप्यर्णगतत्वात् तथा लब्धिरूपाल्पत्वाच्च तत्र रूपं प्रक्षिप्य धनात्मकमवममानं विहितमाचार्येण।

$$\therefore \text{धनावममानम्} = १ + \frac{\frac{\text{६९२ वक्षशे}}{\text{९६००}} - ११ \text{ शे}}{७०३}$$

$$= \frac{७०३ + \frac{\text{६९२ वक्षशे}}{\text{९६००}} - ११ \text{ शे}}{७०३}$$

अत्राचार्यमतेन चैत्रामान्तेऽवमशेषमानं समागतम् शुद्धीशवध इत्युक्तत्वात्। इष्टतिथिमानस्य शून्यत्वेनावगमाच्च।

अत्र प्रागानीतेन समीकरणेनार्कात् वर्षान्तकालिकावमशेषमानं ६९२ एभिः संगुण्य ९६०० एभिर्विभज्य फलं त्रिशून्यमप्तसंख्यायां संयोज्य हारेणानेन ७०३ नेन हृते सत्यवमानमहर्गणोपयुक्तं भवेदिति ग्रन्थकाराः प्रोचुः। तथामति दिनमेकमधिकं गृहीतं भवेत्।

तथाहि। कल्प्यते ११ शु > $\frac{६९२ \text{ वक्षशे}}{९६००}$

तदाऽवमदिनमानम् = $\frac{११ \text{ शु} - \frac{६९२ \text{ वक्षशे}}{९६००}}{७०३}$ इत्येव भवितुं युज्यते।

परमिहाचार्येण

$\frac{७०३ + \frac{६९२ \text{ वक्षशे}}{९६००} - ११ \text{ शु}}{७०३}$ —— इदमवमानं स्वीकृतम्।

द्वयोरन्तरेण—

$\frac{७०३ + \frac{६९२ \text{ वक्षशे}}{९६००} - ११ \text{ शु}}{७०३} - \frac{११ \text{ शु} - \frac{६९२ \text{ वक्षशे}}{९६००}}{७०३}$

= अन्तर = $\frac{७०३}{७०३}$ = १ अत आचार्येणैकमधिकमवमदिनमानं गृहीतम्। तदर्थं "व्येकावमं गृहीत्वे" ति ग्रन्थकाराः प्रोचुः।

एतेनायमर्थः पर्यवसितो भवति यल्लब्ध्वहर्गणानयने साधिताद $\frac{६९२ \text{ वक्षशे}}{९६००}$ स्मादवमशेषाद्यत्र शुद्धीशवधोऽधिकः स्यात्तत्रावमशेषं त्रिखागमिते संयोज्य तत्रैव शुद्धीशवधः शोधनीयः। ततो भागहारेण विभज्यावमशेषं साधनीयम्। तथा कृते सति रूपसमं दिनमन्तरमापद्यते। तदर्थमाचार्येण व्येकमवमं गृहीतमिति सुष्ठुतरम्।

अत्रैव सिद्धान्तशेखरकारा अपि

"शुद्धिमेव पृथगीश्वराहतां
शोधयेदवमशेषकान्निजात् ।
चेन्न शुध्यति च सत्त्रिखाचलात्
शोध्यमेकमपि शुद्धितो दिनम्"

इति प्राहुः। तत्रैव व्याख्यायां मल्लिकभट्टः। चेन्न शुद्धयतीति—एकादशगुणा शुद्धिरवशेषान्न शुद्धयति चेत् तर्हि सत्त्रिखाचलादवमशेषाच्छोधयेत्। अवमशेषे त्रिखाचलं संयोज्य पश्चाच्छुद्धिं शोधयेदिति यावत्। यदा त्रिखशैलयोजनेन शोधनं क्रियते तदा विशेषमाहशोध्यमिति। शुद्धितोऽप्येकं दिनं शोधयेत्। अत्र शुद्धिशब्देनावमदिनान्युच्यन्ते। एतदुक्तं भवति। कल्पगतानब्दान् कलिगतान् वा नवगुणादिनगैः संगुण्य खखर्तुनवभि विभज्य लब्धेभ्यो रूपमेकं विशोधयेत्। तदुक्तं ब्रह्मगुप्तेन "व्येकावमं गृहीत्वेति" इति सर्वं चर्वितचर्वणमेव।

भास्कराचार्यास्तु चैत्रादितिथिभ्योऽधिकायां शुद्धौ तिथिषु शुद्धिर्न शुद्धयत्यतः प्राक् चैत्रामान्ततोऽभीष्टदैवसिकदिनावधि तिथयो ग्राह्याः। तत्रैव प्राक् वर्षजातां शुद्धिं विशोध्याहर्गणः साधनीयः। तद्वशेन ये ग्रहाः सिद्धयन्ति तेऽपि प्राग्वर्षान्तकालिकध्रुवेषु क्षेप्या भवन्तीति प्राहुः।

अत्रैव लल्लाचार्येण विपरीतशोधनेनर्णाहर्गणमानीतम्। तद्वाक्यं शिष्यधीवृद्धिदे।

यावन्न मेषं व्रजति प्रभाकरस्तावन्न पूर्वध्रुवकान् परित्यजेत्
क्षेत्रे प्रविष्टेऽपि विलोमकर्म वा शुद्धचा विजह्यादगते क्रियं रवौ।

भास्वानृणाहर्गणतश्च सिद्धः पात्यो भचक्रात्स्वफलानि चैवम्।
स्वस्वध्रुवादप्यथखेचराणां शोध्यानि यत्नात् प्रवदन्ति सन्तः॥ इति।

इदानीं चैत्रादावब्दाधिपपरिज्ञानार्थं तत्रैव सर्वग्रहणमर्कोदयकाले मध्यमानयनं चार्यायाह

चैत्रसिताद्योऽब्दपतिः शुद्धयूनाया दिनाब्दरूपयुतेः।
तद्द्युगणाद्दिनवारः शुद्धयूना मध्यमाः प्राग्वत्॥५९॥

वा. भा.—चैत्रसिताब्दपतिः स कथं भवतीत्याह। शुद्ध्यूनायाः कस्याः दिनाब्दरूपयुतेः एतदुक्तं भवति कल्पगताब्ददिनयुतेः सरूपायाः शुद्धिदिनानि

सकलानि संशोध्य शेषस्य सप्तभिर्भागेहृ̄नोऽवशेषांकसमोऽर्कोदिश्चैत्रादौ वाराधिपतिर्भवति। यदि सविकलाशुद्धिः अथ शुद्धौ भाविकलं नास्ति तदाब्ददिनयुतौ रूपं न देयम्। अथवा कल्पगताब्दै शुक्राद्योऽब्दपतिः कार्यः तथापि स एव भवति तद्युगणादिनवारः शुद्ध्यूना मध्यमा प्राग्वत्। ततश्चैत्रसितादेर्यो द्युगणः क्रियते, तस्य तदादिका वा गणना कार्या। शुद्ध्यूनाश्च मध्यमा रविमण्डलान्तिकाः कृताः चैत्रादावर्कोदये मध्यमाः भवन्ति। अथ प्राग्वत् कार्याः। अयमर्थः शुद्धितोऽहर्गणं परिकल्प्य ततो द्विगुणान् सप्तत्यंशं स्वनवार्कांशाधिकमित्यादिना रवेरेकादशन्यिप्तांशा भौम इत्यादिना भौमादीनाञ्च यत्फलं भवति, तेन स्वफलेनोना रविमंडलान्तिकाः कार्याः कृतश्चैत्राद्यर्कोदये लंकायां मध्यमो भवति। एवं चैत्रादौ मध्यमाः सर्व एव ग्रहमन्दपाताः कार्याः। अवमशेषाद्याधिमित्यादिकञ्च संलिख्य वर्षोपयोगी स्थापयेत्। तत इष्टदिने चैत्रसिताद्यस्तिथयः पृथग्गुणा रुद्रैरित्यादिना योऽहर्गणो भवति स खण्डो भवति स्वावमशेषसहितः ततो द्युगणात्सप्तत्यंशमित्यादिना ग्रन्थेन प्राग्वत्, सर्वेषां ग्रहशीघ्रमन्दपातानां फलान्यानीय चैत्राद्यौदयिकेषु संयोज्येष्टदिने मध्या भवन्ति, लंकार्कोदये ग्रहर्गणस्य सप्तहृतस्य शेषांकसमग्रहो द्वितीयो ग्रहः। स तत्र दिने वाराधिपतिर्भवति यतो भुक्ता वारा अहर्गणे भवन्ति, तत्रैवं वासना कल्पगताब्दा दिनयुतौ वारगणस्तिष्ठति, वारश्चैककुसावनदिवसे न भवति शुद्धैरपि सावनदिवसात्मिका अतः शुद्धिरब्ददिनयुतेर्विशोधयेत् तावता वाराश्चैत्रादेरतीतस्य रूपञ्च शुद्धेः सकलत्वात्दीयते। ततः सप्तहृतशेषश्चैत्रादौ वाराधिपतिर्भवति। चैत्रादिकाहर्गणस्य तदादिका वारग्रहणमपि युज्यत एव शुद्धीनाञ्च रविमण्डलांतिकाश्चैत्रादौ भवन्ति, रव्युदये यस्माद्यथाहर्गणेन ग्रहा अग्रतो नीयन्ते। एवं पश्चादपि तुल्यत्वात् त्रैराशिकस्य शुद्धिश्चाहर्गणः एवं यतस्तस्मादुपपन्नम् ॥५९॥

वि. भा.—पूर्वसाधितदिनाद्यस्य कल्पगतवर्षाणां रूपस्य च युतेः शुद्धिरहितायाश्चैत्रसिताद्यो वर्षपतिः साध्योऽर्थाच्छुद्धिरहिताया दिनाद्यकल्पगतवर्षरूपसंयुतेर्यच्छेषं तत्सप्तभिर्भक्तं तदा चैत्रादौ रव्यादिवारो भवेत्ततश्चैत्रादितो योऽहर्गणो भवति तत्र चैत्रादिवाराद्दिनवारो ज्ञातव्यस्ततः साधिता ग्रहा शुद्धिदिनोत्पन्नैर्ग्रहै रहितास्तदा सौरवर्षान्तात्पूर्ववन्मध्यमग्रहा भवन्ति, यदि कोऽपीष्टवर्षे चैत्रादितोऽहर्गणज्ञानं ततो ग्रहान् ततश्च सौरवर्षान्मध्यमग्रहान् ज्ञातुमिष्यति तदा तेनोपरिलिखितप्रकारेण तत्साधनं कार्यमिति ॥ ५९ ॥

अत्रोपपत्तिः

रविवर्षान्तामान्तयोर्मध्यवर्त्तिन्यस्तिथयोऽविशेषतिथयस्ततो वर्षान्तक्षयशेषघटिकाः शोध्याः शेषस्य शुद्धिसंज्ञा सैव वर्षान्तामान्तयोर्मध्ये सावनदिनसंख्या। रविवर्षान्तामान्तयोरंतरे सावनदिनानि=शुद्धिः कल्पादित इष्टसौरवर्षान्तं यावत्सावनदिनानि=३६५ गव+दिनादि, ततः कल्पादित इष्टसौरवर्षान्तं

यावत्सावनदि—शुद्धि=३६५ गव+दिनादि—शुद्धि=चैत्रादौ सावनदिनानि, एतानि सप्तभिर्भक्तानि वर्तमानवारार्थं रूपयोजितानि तदा चैत्रसिताद्वारः= गव+दिनादि—शुद्धि+१ एतावताऽऽचार्योक्तमुपपद्यते । सिद्धान्ततत्त्वविवेके कमलाकरेण लघ्वहर्गणानयने वारगणनार्थं विशेषः प्रतिपादितोस्तीति ॥५९॥

अब चैत्रादि से अहर्गणानयन करके मध्यग्रहानयन को कहते हैं

हि. भा.—पूर्वसाधित दिनादि—कल्पगतवर्ष और रूप (एक) इन सबों के योग में शुद्धि को घटाकर जो हो उस पर से चैत्रसितादिवर्षपति साधन करना, अर्थात् दिनाद्य-कल्पगतवर्ष और रूप इन सबों के योग में शुद्धि को घटाकर जो शेष बचे उसको सात से भाग देने से चैत्रादि में रव्यादिवार होते हैं। चैत्रादि से जो अहर्गण होता है उसमें चैत्रादिवार से दिनवार समझना चाहिए, उससे जो ग्रह होते हैं उसमें शुद्धिदिनोत्पन्नग्रह को घटाने से सौरवर्षान्त से पूर्ववत् मध्यम ग्रह होते हैं ॥५९॥

उपपत्ति

सौरवर्षान्त और अमान्त के मध्य में जो तिथि है वह अधिशेष तिथि है, उसमें वर्षान्तक्षयशेष घटी को घटाने से जो शेष रहता है उसका नाम शुद्धि है, वही वर्षान्त और अमान्त के मध्य में सावनदिन है, रविवर्षान्त और अमान्त के मध्य में सावन दिन=शुद्धि, कल्पादि से इष्टसौरवर्षान्तपर्यन्त सावनदिन=३६५ गव+दिनादि,

अतः कल्पादि से इष्टसौरवर्षान्तपर्यन्तसावनदि-शुद्धि=३६५ गव+दिनादि—शुद्धि=चैत्रादि में सावनदिन इसको सात से भाग देना और वर्तमान वार के लिये रूप जोड़ देना तब चैत्रसितादि से वार होते हैं, चैत्रसितादि से वार=गव+दिनादि—शुद्धि+१ इससे आचार्योक्त उपपन्न हुआ । कमलाकर ने सिद्धान्ततत्त्वविवेक में लघ्वहर्गणानयन में वार गणना के लिये बहुत विशेष विचार किया है, कभी-कभी बिना रूप जोड़ने से भी चैत्रादि में वार होते हैं, वर्तमान वारज्ञानार्थ अहर्गण में सैक और निरेक किया जाता है जिसको भास्कराचार्य ने भी सिद्धान्तशिरोमणि में 'अभीष्ट वारार्थमहर्गणश्चेत्' इत्यादि से कहा है इति ॥५९॥

इदानीं बीजकर्माह

खखखार्कहृताब्देभ्यो गतगम्याल्पाः खशून्ययमलहृताः ।
लब्धं त्रिसायकहतं कलाभिरूनौ सदाऽर्केन्दू ॥ ६० ॥

शशिवत् जीवे द्विहतं चन्द्रोच्चे तिथिहतं तु सितशीघ्रे ।
द्वीषु ५२ हतं च बुधोच्चे द्वि २ कु १ वेद ४ हतं च पातकुजशनिषु ॥६१॥

वा. भा.—अनयोः श्लोकयोर्वासनाभाष्यं नास्ति ।

वि. भा.—द्वादशसहस्र १२००० भक्तेभ्यो गतवर्षे- (कल्पगतवर्षे) भ्यो-ये लब्धास्ते गताः, गता हारा - (१२०००) त्पतितास्तदा गम्याः (एष्याः) स्युरे-तयोर्मध्ये येऽल्पास्ते द्विशत्या २०० भक्ता यल्लब्धं तत्त्रिभिः, पञ्चभिर्गुणितं कलात्मकफलैः सदा क्रमेण रविचन्द्रौ हीनौ कार्यौ, जीवे (बृहस्पतौ) चन्द्रवत्फलं देयमर्थाच्चन्द्रे यत्कलात्मकं फलमृणं तदेव बृहस्पतावप्यृणं कार्यम्। तदेव 'खशू-न्ययमल २०० हृता इत्यनेनानीतं' फलं द्विहतं (द्वाभ्यां गुणितं) चन्द्रमन्दोच्चे हीनं कार्यं तदेव फलं पञ्चदशभिर्गुणितं सद्यद्भवेत्तच्छुक्रशीघ्रोच्चे हीनं कार्यम्। तदेव फलं द्विपञ्चाशता ५२ गुणितं यद्भवेत्तद्बुधशीघ्रोच्चे ऋणं कार्यं तथा तदेव पूर्वफलं द्वि २ कु १ वेद ४ गुणितं गुणनफलं क्रमेण पातमङ्गलशनिषु ऋणं कार्यम्। 'द्वीषुहतं च बुधोच्चे द्विकुवेदहतमित्यनेन' यादशो हि (ऋणात्मकरूपः) संस्का-रोऽभिहितो ब्रह्मगुप्तेन तद्विपरीत (धनात्मक) संस्कारः "इन्दुना दस्रबाणैः कराभ्यां कृतैरित्यादिना" सिद्धान्तशिरोमणौ भास्कराचार्येणाभिहित इति ॥६०-६१।

अत्रोपपत्तिः

इष्टग्रहभगणगुणादित्यादिब्रह्मगुप्तोक्तेन, द्युचरचक्रहतो दिनसंचय इत्यादि-भास्करोक्तेन वाऽऽनीतक्रान्तिवृत्तीयमध्यमग्रहतः स्फुटक्रियाकरणेन वास्तव-स्फुटग्रहो नायाति, परन्त्वस्मिन् मध्यमग्रहे बीजकर्मजनितफलसंस्कारे कृते यो मध्यमग्रहस्तस्मात्स्फुटक्रियाकरणेन वास्तवस्फुटग्रह आयातीत्यागमवादिनो वदन्ति नात्र वस्तुतः प्रामाण्यम्। तत्रोपपत्त्यऽऽनयनार्हः कतिचिद्दिनैरुपलब्धभूतः पदार्थो बीजशब्दवाच्यस्तत्कर्म बीजकर्म इति, अथ तत्तत्पदार्थस्य सृष्ट्यादित आरभ्य षट्-सहस्र ६००० वर्षपर्यन्तं वृद्धिस्ततोऽग्रे षट्सहस्रवर्षपर्यन्तं ह्रास इत्यत्रागम एव प्रमाणम्। तेन सृष्ट्यादितः षट्सहस्रवर्षान्ते परमवृद्धिः। द्वादशसहस्रवर्षान्ते परमह्रास इति फलितम्। एवं प्रतिद्वादशसहस्रवर्षे भवति, तेनेष्टकाले गता ये कतिचित्सौराब्दास्ते द्वादशसहस्रभक्ताः (प्रतिद्वादशसहस्रवर्षान्ते तत्प्रलयत्वात्) शेषादनुपातेन फलमानीय मध्यमग्रहे संस्कृते सति स्फुटक्रियाकरणार्हो मध्यग्रहो भविष्यतीति। अथ यदि शेष < ६००० वर्ष तदा वृद्ध्यभिमुखफलम्, यदि च शेष > ६००० वर्ष तदा ह्रासोन्मुखफलम्। यदा शेष < ६००० वर्ष तदा शेष < १२००० - शेष = शेष', परं द्वाभ्यामपि शेषाभ्यामनुपातेन फलमेककालीन-मेवातोऽङ्कलाघवार्थमत्रा 'शे' स्मादेवानुपातकरणं युक्तम्। यदा च शे > ६००० वर्ष तदा शे > १२००० - शेष = शेष' परन्त्वत्रापि द्वाभ्यामपि शेषाभ्यां फलमेककालीनमेवाऽत्राङ्कलाघवार्थमस्मादेवानुपातकरणं युक्तमतोऽनुपातः—

$$\frac{\text{परमोपचयफल} \times \text{शेष}}{६०००} = \frac{\frac{\text{परमोपचयफल}}{३०} \times \text{शेष}}{२००} = \text{फल, अत्र रव्यादीनां ग्रहाणां}$$

पातादीनां च परमबीजफलं त्रिंशता भक्तं सत्क्रमेण ३, ५, ५, १५, २, १, ५२, २, ४ भवन्ति, तत्रानुपातागतफलमेतद्यदि रविचन्द्रबृहस्पतिचन्द्रोच्चादिषु ऋणं क्रियेत तदा स्फुटक्रियाकरणयोग्यो मध्यमग्रहः स्यादित्यत्रागम एव प्रमाणम्, एव' शे', मस्मादपि तथैवानुपातो यतो द्वादशसहस्रवर्षान्ताद् विलोमक्रमेणायं कालः, तत्र फलं वृद्धच्युन्मुखमेव यतो द्वादशसहस्रवर्षान्ते फलस्य परमह्रासस्तत्कालाद् विलोमक्रमेण फलसत्ताक्रमेणाधिका एवातोऽनुपातेनोभयत्र फलसाम्यमेवातः

$$\frac{\text{परमबृफल}+\text{शेष}'}{६०००}=\text{फल}=\frac{\frac{\text{पबृफ}}{३०}\times\text{शेष}'}{२०००}$$ इतोऽग्रे पूर्ववदत्र वर्षशब्देन सर्वत्र सौरवर्षमेव ग्राह्यमिति, भास्कराचार्येण सिद्धान्तशिरोमणौ भानुचन्द्रेज्यशुक्रेन्दुतुङ्गेष्वनुपातागतफलमृणं तथा भौमसौम्येन्दुपातार्किषु धनमभिहितं, परं ब्रह्मगुप्तेन सर्वत्रानुपातागतफलमृणमेव कार्यमिति कथितं कमलाकरेण सिद्धान्ततत्त्वविवेके बीजकर्मसंस्कारस्य खण्डनमेव कृतं किमत्र युक्तमितिनिर्णायकयुक्त्यभावात्किमपि वक्तुं न शक्नोम्यहमिति ॥ ६०-६१ ॥

अब बीज कर्म कहते हैं

हि. भा.—कल्पगत वर्ष में बारह हजार (१२०००) से भाग देने से जो लब्धि होती है वह गत है, उसको हर १२००० में घटाने से जो शेष रहता है वह गम्य है, इन दोनों में जो अल्प हो उसे दो सौ से भाग देने से जो लब्धफल हो उसे तीन और पाँच से गुणने से जो कलात्मक फल हो उनको क्रम से रवि और चन्द्र में ऋण करना, चन्द्र में जो कलात्मक फल ऋण किया गया है वही बृहस्पति में भी ऋण करना चाहिये। उसी फल को दो से गुणाकर चन्द्रमन्दोच्च में ऋण कर देना चाहिये, उसी फल को पन्द्रह से गुणा कर जो हो उसे शुक्र के शीघ्रोच्च में ऋण करना। उसी फल को बावन ५२ से गुणाकर जो हो उसे बुधशीघ्रोच्च में ऋण करना चाहिये। उसी फल को दो, एक और चार से पृथक्-पृथक् गुणा करके जो हो उन्हें क्रम से पात, मङ्गल और शनि में ऋण करना चाहिये ॥६०-६१॥

उपपत्ति

'इष्टग्रहभगणगुणात्' इत्यादि ब्रह्मगुप्तोक्तप्रकार से या 'ध्रुवरचक्रहतो दिनसंचयः इत्यादि' भास्करोक्तप्रकार से साधित क्रान्तिवृत्तीय मध्यम ग्रह से स्फुट क्रिया करने से वास्तव स्फुट ग्रह नहीं आते हैं, लेकिन इस मध्यम ग्रह में बीजकर्म जनित फल को संस्कार करने से जो मध्यम ग्रह होते हैं उससे स्फुट क्रिया करने से वास्तव स्फुटग्रह आते हैं, यह बात आगमवादी लोग कहते हैं। वस्तुतः इसमें कुछ प्रामाणिकता नहीं है।

उपपत्ति से आनयन योग्य कितने दिनों में उपलब्ध भूत पदार्थ बीज शब्द से कहा जाता है उसका कर्म बीजकर्म कहलाता है, उन २ पदार्थों की सृष्ट्यादि से लेकर छः हजार वर्ष पर्यन्त वृद्धि होती है उससे आगे छः हजार वर्ष पर्यन्त ह्रास होता है इसमें आगम ही प्रमाण है, इसलिये सृष्ट्यादि से छः हजार वर्षान्त में परमवृद्धि होती है, बारह हजार वर्षान्त में परम ह्रास होता है यह निर्गलितार्थ हुआ; इस तरह प्रत्येक बारह हजार वर्ष में होता है, इसलिये इष्ट काल में जो गत सौर वर्ष हो उसे बारह हजार से भाग देने से जो शेष रहे उस पर से अनुपात द्वारा फल लेकर मध्यम ग्रह में संस्कार करने से स्फुट क्रिया करण योग्य मध्यम ग्रह होंगे, यदि शेष <६००० वर्ष तब फल वृद्धयभिमुख होता है, यदि शेष >६००० वर्ष तब फल ह्रासोन्मुख होता है। यदि शेष <६००० वर्ष तब

शेष $<१२०००-$शेष$=$शे लेकिन दोनों शेषों के अनुपात से फल एक ही होता है इसलिये अङ्कलाघव के लिये यहां 'शे' इसी से अनुपात करना ठीक है। यदि शे>६००० वर्ष तब शे$>१२०००-$शेष$=$शे यहां भी दोनों शेषों से फल एककालीन ही होता है, यहाँ अङ्क लाघव के लिये शे इसी से अनुपात करना ठीक है, इसलिये अनुपात करते हैं

$$\frac{\text{परमोपचयफल}\times\text{शेष}}{६०००}=\frac{\frac{\text{परमोपचयफल}}{३०}\times\text{शेष}}{२००}=\text{फल}$$ यहां रवि आदि ग्रहों के और पातादियों के परम बीज फलों को तीस से भाग देने से क्रम से ३,५,५,१५,२,१,५२,२,४ होते हैं, यहाँ अनुपातागत फलों को यदि रवि, चन्द्र, बृहस्पति, चन्द्रमन्दोच्चादि में ऋण करते हैं तब स्फुट क्रियाकरण योग्य मध्यम ग्रह होते हैं, इसमें आगम ही प्रमाण है।

इस तरह शे इससे भी उसी तरह अनुपात होता है क्योंकि बारह हजार वर्षान्त से विलोम क्रम से यह काल होता है वहाँ फल वृद्धि के तरफ होता है, क्योंकि बारह हजार वर्षान्त में फल का परमह्रास होता है, उस काल से विलोम क्रम से फल की सत्ता क्रम से अधिक ही होती है, इसलिये अनुपात से दोनों जगह फल की तुल्यता ही होती है, इसलिये

$$\frac{\text{परमवृफल}\times\text{शेष}}{६०००}=\text{फल}=\frac{\frac{\text{पवृफ}}{३०}\times\text{शेष}}{२००}$$ इससे आगे पूर्ववत् होता है, यहाँ वर्षशब्द से सौरवर्ष ही ग्रहण करना चाहिये। भास्कराचार्य सिद्धान्तशिरोमणि में रवि, चन्द्र, बृहस्पति, शुक्रशीघ्रोच्च, चन्द्रमन्दोच्चों में अनुपातागत फलों को ऋण कहते हैं तथा मङ्गल बुध, चन्द्र, पात, शनि इनमें अनुपातागत फल को धन कहते हैं, ब्रह्मगुप्त सबों में अनुपातागत फलों को ऋण ही कहते हैं, सिद्धान्ततत्त्वविवेक में कमलाकर ने बीज कर्म संस्कार का खण्डन किया है। इनमें क्या ठीक है, इस विषय में प्रबल युक्ति नहीं मिलती है, इसलिये इस विषय में हम कुछ नहीं कह सकते ॥६०-६१॥

इदानीं चन्द्रमन्दोच्चपातयोरार्यभटोक्त्या दूषणमाह

अकृतार्यभटः शीघ्रगमिन्दूच्चं पातमल्पगं स्वगतेः ।
तिथ्यन्तग्रहणानां घुणाक्षरं तस्य संवादः ॥ ६२ ॥

वा. भा.—स्पष्टार्थेयमार्या अकृतार्यभटः । शीघ्रगामित्वादुच्चं पातमल्पगं स्वगतेः । तिथ्यन्तग्रहणानां घुणाक्षरं तस्य, इयं वासना चन्द्रमन्दस्य यत्र ब्राह्मसिद्धान्ते भगणाः वसुशरवसुपंचखचन्द्रवसुवसुसमुद्रो ४८८१०५८५८ एतेभ्यरार्यभटोक्ता भगणाः सहस्रघ्नाः बहवोऽतो गम्यतेऽधिका मन्दगतिस्तस्य ब्राह्मोक्तगतेः यतो दशगीतिकासु उक्तं चन्द्रोच्चजस्त्रिघेति—४८८२१६ एते भगणाः सहस्रघ्ना जाता अधिका ब्रह्मभगणेभ्यः ४८८२१६००० तस्मादुपपन्नं शीघ्रगामीन्दूच्चपातब्रह्मभगणेभ्यश्चोना आर्यभटस्तस्य पात ब्रह्मभगणोऽतः कल्पगतस्तस्य तद्यथा ब्रह्मपातभगणाः वसुरसरुद्रेन्दुगुणद्वित्रियमा २३२३१११६८ दशगीतिकासु उक्तञ्च पातविलोम इति चतुर्युगस्य सहस्रघ्नाश्च जाता २३२२११६००० एते ब्रह्मभगणेभ्यरूनास्तस्मादुपपन्नपातमल्पगं स्वगतेः । यत एवातस्तस्यास्फुटत्वात्तिथ्यन्तग्रहाणां घुणाक्षरवत्संवादो युज्यते, इति स्वकृता या मध्यगतेः प्रशंसार्थमार्यामाह ॥६२॥

वि. भा.—आर्यभटश्चन्द्रमन्दोच्चं स्वगतेः (मत्कथितचन्द्रमन्दोच्चगतेः) शीघ्रमर्थान्मत्कथितचन्द्रमन्दोच्चगतेः स्वो (आर्यभट) क्त तद्गतिमधिकां कृतवान् तथा पातं (चन्द्रपातं), अल्पगं (अल्पगतिं) कृतवान्, तस्य (आर्यभटस्य) तिथ्यन्तग्रहणानां मध्ये यदि मया सह कदाचित् संवादो (ऐक्यं) भवेत्तद्घुणाक्षरं बोध्यम् । मदुक्तेन तिथ्यन्तादिना यदि कदाचिदार्यभटोक्तानां तिथ्यन्तादीनामैक्यं भवेत्तदा तद्घुणाक्षरन्यायभवं बोध्यम् । चन्द्रमन्दोच्चभगणा आर्यभटोक्ताः कल्पे= ४८८२१६००० मन्मते कल्पे चन्द्रमन्दोच्चभगणाः=४८८१०५८५८<४८८२१६००० तथा कल्पे आर्यभटोक्ताश्चन्द्रपातभगणाः=२३२२२६००० मन्मते कल्पे चन्द्रपातभगणाः=२३२३१११६८/२३२२२६००० अतो मदुक्तचन्द्रमन्दोच्चपातगतिभ्यामार्यभटोक्ततद्गत्योरधिकत्वाल्पत्वाच्च तन्मतं न समीचीनमित्याचार्यः (ब्रह्मगुप्तः) कथयति, ब्रह्मगुप्तमनसीत्थं धारणाऽस्ति यन्मदुक्तमेव कल्पचन्द्रमन्दोच्चभगणमानं कल्पचन्द्रपातभगणमानं च युक्तियुक्तं मदुक्तेन सहार्यभटोक्तस्य पार्थक्यमस्ति तेन तन्मतं न समीचीनमिति, ब्रह्मगुप्तकथनमिदं तथ्यमतथ्यं वेति परीक्षणार्थं गणितमेव शरणमिति ॥६२॥

अब आर्यभटोक्त चन्द्रोच्च और चन्द्रपात के दोष को कहते हैं

हि. भा.—आर्यभट ने मेरी चन्द्रमन्दोच्चगति से चन्द्रमन्दोच्च को शीघ्रगति किया है अर्थात् मेरी चन्द्रमन्दोच्च गति से अपनी (आर्यभटीय) चन्द्रमन्दोच्चगति को अधिक

किया है और मेरी चन्द्रपातगति से अपनी चन्द्रपातगति को अल्प किया है, इसलिये तिथ्यन्त और ग्रहणों में यदि मेरे मत के साथ आर्यभट मत का ऐक्य (एकता) हो तो उसको (मतैक्य को) घुणाक्षर समझना चाहिए अर्थात् काठ को घुण (कीड़ा विशेष) के खाने से कभी-कभी अक्षर का आकार बन जाता है तो उससे यह नहीं समझा जाता है कि घुण ने अपनी बुद्धि से अक्षर बनाया है, उसी तरह मतैक्य के सम्बन्ध में भी समझना चाहिये।

कल्प में आर्यभटोक्त चन्द्रमन्दोच्चभगण=४८८२१९०००, मेरे मत से कल्प में चन्द्रमन्दोच्चभगण=४८८१०५८५८<४८८२१९००० तथा कल्प में आर्यभटोक्त चन्द्रपातभगण=२३२२२६०००, मेरे मत से कल्पचन्द्रपातभगण=२३२३१११६८> २३२२२६००० इसलिये मेरी चन्द्रमन्दोच्चगति और पातगति से आर्यभटोक्त उनकी गति अधिक और अल्प होने के कारण आर्यभट का मत ठीक नहीं है यह ब्रह्मगुप्त कहते हैं, ब्रह्मगुप्त अपने मत को बिलकुल ठीक समझ कर अपने मत से आर्यभट मत के पृथक् होने के कारण उसका खण्डन करते हैं, ब्रह्मगुप्त का कथन ठीक है या नहीं इसके लिये गणित ही शरण है इति ॥६२॥

इदानीं स्वकृताया मध्यगतेः प्रशंसार्थमाह

मध्यगतिज्ञं वीक्ष्य श्रीषेणार्यभटविष्णुचन्द्रज्ञाः ।
सदसि न भवन्त्यभिमुखाः सिंहं दृष्ट्वा यथा हरिणाः ॥६३॥

वि. भा.—मध्यगतिज्ञं (मध्यग्रहानयनवेत्तारं) वीक्ष्य (दृष्ट्वा) श्रीषेणार्यभटविष्णुचन्द्रोक्तानां मध्यग्रहानयनानां ज्ञातारः सदसि (सभायां) अभिमुखा (संमुखाः) न भवन्ति, यथा सिंहं दृष्ट्वा हरिणास्तदभिमुखा न भवन्तीत्यनेन ब्रह्मगुप्तेन स्वकीयमध्यग्रहानयनस्य वास्तवत्वं श्रीषेणार्यभटादीनामाचार्याणां मध्यमग्रहानयनस्यावास्तवत्वं कथ्यतेऽर्थादेतद्व्याजेन स्वप्रशंसां क्रियत इति ॥ ६२ ॥

अब अपनी प्रशंसा को कहते हैं

हि. भा.—हमारे मध्यग्रहानयन के समझने वाले को देखकर श्रीषेण-आर्यभट-विष्णुचन्द्र इन आचार्योक्त मध्यग्रहानयन को समझने वाले सभा में उनके (हमारे मध्यग्रहानयन को समझने वालों के) सम्मुख नहीं होते हैं जैसे सिंह को देखकर हरिण उसके सम्मुख नहीं होता है। इससे आचार्य (ब्रह्मगुप्त) अपने मध्यग्रहानयन के वास्तवत्व और श्रीषेण-आर्यभट आदि आचार्योक्त मध्यग्रहानयन के अवास्तवत्व को कहते हैं अर्थात् इस व्याज से अपनी प्रशंसा करते हैं इति ॥ ६३ ॥

इदानीं मध्यमाधिकारोपसंहारमाह

युगभगणमानयाताहर्गणदिनवारमध्यमाद्येषु ।
मध्यमगतिस्त्रिषष्टचार्याणां प्रथमः कृतोऽध्यायः ॥ ६४ ॥

वा. भा—युगादिष्वर्थेषु द्विषष्टचार्याणां मध्यगत्याख्यो मध्ययोनि वद इति ।

इति ब्राह्मसिद्धान्ते वासनाभाष्ये मध्यगत्याख्यो ग्रन्थशतैः सप्तभिः सार्धैर्व्याख्यातम् ।

इति ब्राह्मसिद्धान्तवासनाभाष्ये मध्यगत्याख्यो मध्ययोनि बद्ध इति ।

वि. भा.—ग्रहादीनां युगे कियन्ति भगणमानानि गताहर्गणमानानि यानि, दिनवारादेर्ये विचाराः, मध्यमग्रहादिसाधनानि यानि, एतदाद्येषु विषयेषु आर्याणां (आर्याछन्दसां) त्रिषष्ट्या (त्रिषष्टिप्रमिताऽऽर्याछन्दसा) प्रथमो मध्यगतिरध्यायः (मध्यगतिनामकः प्रथमोऽध्यायः) मया कृतोऽर्थान्मध्यगतिनामकेऽध्याये कियन्तो विषयाः सन्ति तेषामुल्लेखः कृत इति ॥ ६४ ॥

इति ब्राह्मस्फुटसिद्धान्ते मध्यमाधिकारः प्रथमः ॥

अब मध्यमाधिकार के उपसंहार को कहते हैं

हि. भा.—ग्रहादियों के युग में जो भगणमान है, गताहर्गणमान जो है, दिनवारादि के जो विचार हैं, और मध्यम ग्रहादि साधन जो है एतदादिक विषयों में तिरसठ आर्याछन्दों के (तिरसठ आर्याछन्द श्लोक द्वारा) मध्यगति नाम का प्रथम अध्याय किया गया अर्थात् मध्यगति नामक अध्याय में कितने विषय हैं उनका उल्लेख किया गया इति ॥६४॥

इति ब्राह्मस्फुटसिद्धान्त में मध्यमाधिकार (प्रथम अधिकार) समाप्त हुआ ॥

ब्राह्मस्फुटसिद्धान्तः

स्पष्टाधिकारः

ब्राह्मस्फुट-सिद्धान्तः

स्पष्टाधिकारः

तत्रादौ स्फुटीकरणस्य प्रयोजनमाह

यस्मान्न मध्यतुल्यः प्रतिदिवसं दृश्यते ग्रहो भगणे ।
तस्माद् दृक्तुल्यकरं वक्ष्ये मध्यस्फुटीकरणम् ॥१॥

वा. भा.—अथ स्फुटगत्यध्यायो व्याख्यायते । तत्रारम्भप्रयोजनमाह । यस्मान्मध्यग्रहेण तुल्यः दृग्विषये ग्रहो न दृश्यते भगणे नक्षत्रचक्र प्रतिदिवसं दिवसे-दिवसे तस्मात् स्फुटीकरणं वक्ष्ये । मध्यस्य कीदृगित्याह दृक्तुल्यकरमभिप्रायो-मध्यमो ग्रहः कक्षामण्डले परिकल्पते । न च कक्षामण्डले पारमार्थिको ग्रहः प्रति-मण्डले मध्यभुक्त्या भ्रमति । यस्माद् दृश्यते कक्षामण्डलेऽतोऽहं दृक्स्फुटीकरणं वक्ष्ये येन प्रतिमण्डलस्थो ग्रहः कक्षामण्डले दृक्तुल्यो दृश्यते इत्यर्थः, एतत्सर्वं कक्षामण्डलं प्रतिमण्डलनीचोच्चवृत्तादीनि विन्यस्य गोले छेदके च प्रदर्शयेत् । मया च पूर्वमेव गोलाध्याये प्रदर्शितमिति ।

वि. भा.—यस्मात्कारणात् प्रतिदिवसं (प्रतिदिनं) भगणे (क्रान्तिवृत्ते) ग्रहः (पारमार्थिकः स्फुटो वा ग्रहः) मध्यतुल्यो (मध्यमेन ग्रहेण समः) न दृश्यते (नावलोक्यते), तस्मात्कारणात् दृक्समकारकं स्फुटीकरणं वक्ष्ये, अत्रैतदुक्तं भवति मध्यमो ग्रहः कक्षावृत्ते परिकल्पितः, नहि कक्षावृत्ते पारमार्थिको ग्रहः प्रतिवृत्ते मध्यगत्या भ्रमति किन्तु स्पष्टगत्या प्रतिवृत्ते परिभ्रमन् कक्षावृत्ते दृश्यते, अतो येन गणितेव कक्षावृत्ते प्रतिवृत्तस्थो ग्रहो दृक्समो भवेत्तादृशं स्फुटीकरणमहं वक्ष्ये इति ॥१॥

हि. भा.—जिस कारण से प्रत्येक दिन क्रान्तिवृत्त में स्पष्ट ग्रह मध्यम ग्रह के बरा-बर नहीं देखे जाते हैं उस कारण से दृक्तुल्य (दृग्गणितैक्य) कारक स्पष्टीकरण को मैं (ब्रह्मगुप्त) कहता हूं, यहाँ यह कहा जाता है कि कक्षावृत्त में मध्यम ग्रह परिकल्पित प्रतिवृत्त में स्पष्टगति से भ्रमण करते हुए ग्रह कक्षावृत्त में देखे जाते हैं इसलिए जिस गणित से कक्षावृत्त में प्रतिवृत्तस्थित ग्रह दृक्सम होते हैं उस स्पष्टीकरण को मैं कहता हूं ॥१॥

इदानीं स्पष्टीकरणादिसर्वग्रहगणितानां ज्यागणिताऽधीनत्वात्प्रथममर्ध-ज्यामुक्त्वोत्क्रमज्यामाह

अर्धज्या मनुयमला मुनियमवेदा वसुज्वलनषट्काः ।
रसकृतवसवः शशिपञ्चखेन्दवश्चन्द्रशरसूर्याः ॥२॥

षड्दधिमनवो भूताग्निरसशशाङ्का मुनीन्दुवसुचन्द्राः ।
इन्दुनवनन्दचन्द्रा रसतिथियमला रवित्रियमाः ॥३॥
छिद्रेषु जिनाः कृतनवपञ्चयमा नन्दचन्द्रमुनिपक्षाः ।
दन्ताष्टयमा गुणरामनवयमाः शशियमखरामाः ॥४॥
ऋतुनवखगुणा नवशरचन्द्रगुणाः सप्तशून्ययमदहनाः ।
द्विजिनगुणास्त्रिरसरदाः खसप्तयमवन्हयो व्यस्ताः ॥५॥
मुनयोऽष्टयमास्त्रिरसा रुद्रशशाङ्काः समुद्रमुनिचन्द्राः ।
नववेदयमा मुनिगुणहुताशना वसुगुणसमुद्राः ॥६॥
भूमीन्द्रियेषवो रसनगर्तवश्चन्द्रशीतकरवसवः ।
शरवसुनन्दाः सागररुद्रशशाङ्का नवाङ्कार्काः ॥७॥
त्रिविषयवेदशशाङ्काः पञ्चाग्निरसेन्दवोऽब्धियमधृतयः ।
अतिधृतिखयमा नवशशियमपक्षाः सागरद्विजिनाः ॥८॥
रदरसयमला गुणवेदवसुयमाः षट्कविषयशून्यगुणाः ।
खमुनिरदा व्यासार्धं नवरदचन्द्रा जिनांशज्या ॥९॥

(२, ३, ४, ५, ६, ७, ८, ९) श्लोकानां कृते

वा. भा.—अत्र अङ्कन्यास एव व्याख्यानम् । अर्धज्यकास्ता इति तद्यथा । मनुयमला २१४। एतत् प्रथमज्यार्धम् । मुनियमवेदा ४२७ द्वितीयम् । वसुज्वलनषट्काः ६३८ तृतीयम् । रसकृतवसवः ८४६ चतुर्थम् । शशिपञ्चखेन्दवः १०५१ पञ्चमम् । चन्द्रशरसूर्या १२५१ षट्कम् । षट्उदधिमनवः १४४६ सप्तमम् । भूताग्निरसशशांकाः १६३५ अष्टमम् । मुनीन्दुवसुचन्द्राः १८१७ नवमम् । इन्दुनवनन्दचन्द्राः १९९१ दशमम् । रसतिथियमला २१५६ एकादशमम् । रवित्रियमा २३१२ द्वादशम्: छिद्रेषुजिना २४५९ त्रयोदशम् । कृतनवपञ्चयमाः २५९४ चतुर्दशम् । नन्दचन्द्रमुनिपक्षाः २७१९ पञ्चदशमम् । दन्ताष्टयमाः २८३२ षोड़शम् । गुणरामनवयमाः २९३३ सप्तदशमम् शशियमखरामाः ३०२१ अष्टादशम् । ऋतुनवखगुणाः ३०९६ एकोनविंशम् । नवशरचन्द्रगुणाः ३१५९ विंशतिः । सप्तशून्यदन्ताः ३२०७ एकविंशम् । द्विजनगुणाः ३२४२ द्वाविंशतिः । त्रिरसरदाः ३२६३ त्रयोविंशतिः । खसप्तयमदहनाः ३२९० चतुर्विंशतिः अर्धज्या प्रथमं ज्यार्धमुत्क्रमेण मुनयः अष्टयमाः २८ द्वितीयः, त्रिरसाः ६३ तृतीयम् । रुद्रशशांकाः १११ चतुर्थम्, समुद्रमुनिचन्द्राः १७४ पञ्चमम् । नववेदयमाः २४९ षष्ठः, मुनिगुणहुताशाः ३३७ सप्तमम्, वसुगुणसमुद्राः ४३८ अष्टमम् । रूपेन्द्रियेषवः ५५१ नवमम् । रसनगर्त्तवः ६७६ दशमम् । चन्द्रशशिवसवः ८११ एकादशम् । शरवसुनन्दाः ९८५ द्वादशम् । सागररुद्रशशांकाः १११४ त्रयोदशम् । नवाङ्कार्काः १२९९ चतुर्दशम् । त्रिविषयवेदशशांका १४५३ पञ्चदशम् । पञ्चत्रि-

रसेन्दवः १६३५ षोडशम्, अब्धियमधृतयः १८२४ सप्तदशमम्। अतिधृतिखयमाः २०१९ अष्टादशम्। नवशशियमपक्षाः २२१९ एकोनविंशतिः। सागरद्विजिना २४२४ विंशतिः। रदरसयमलाः २६३२ एकविंशतिः। गुणवेदवसुयमाः २८४३ द्वाविंशतिः। षट्विषयशून्यगुणाः ३०५६ त्रयोविंशतिः। खमुनिरदा ३२७९ चतुर्विंशति, व्यासार्धं चैतदेव चकारोऽत्र द्रष्टव्यः। नवरदचन्द्राः १३२९ एतावती जिनांशज्या क्रमेणेति वाक्यशेषः। अयमर्थः भागचतुर्विंशतिज्याक्रमेण क्रियते सा नवरदचन्द्रसंख्या भवति। परमक्रान्तिज्या प्रदर्शनार्थमेव चतुर्विंशतिग्रहणम्। यतश्चतुर्विंशतिज्या पराक्रान्तिरतोऽनया नवरदचन्द्रसंख्यया ज्यया त्रैराशिकेनेष्टदिने क्रान्तिज्यासाधन वक्ष्यति चाचार्यः। आसामार्याष्टानां वासना गोलाध्याये पूर्वमेव अस्माभिः प्रदर्शितेति।

वि. भा.—वृत्तचतुर्थांशे मनुयमला मुनियमवेदा इत्याद्यर्धज्याश्चतुर्विंशतिसंख्यकाः सन्ति याश्चाधोलिखिताः स्युः। व्यस्ता (उत्क्रमज्या) इत्यस्याग्रे सम्बन्धः।

चतुर्विंशतिरर्धज्याः (क्रमज्याः)

२१४, ४२७, ६३८, ८४६, १०५१, १२५१, १४४६, १६३५, १८१७, १९९१, २१५६, २३१२, २४५९, २५९४, २७१९, २८३२, २९३३, ३०२१, ३०९६, ३१५९, ३२०७, ३२४२, ३२६३, ३२७०।

वि. भा.—वृत्तचतुर्थांशे मुनयोऽष्टमा इत्यादि चतुर्विंशतिसंख्यका उत्क्रमज्याः सन्ति याश्चाधोलिखिताः स्युः—

७, २८, ६३, १११, १७४, २४९, ३३७, ४३८, ५५१, ६७६, ८११, ९८५, १११४, १२९९, १४५३, १६३५, १८२४, २०१९, २२१९, २४२४, २६३२, २८४३, ३०५६, ३२७०।

नवरदचन्द्राः=१३२९=जिनांशज्या (जिनांशानां चतुर्विंशतिसंख्यकांशानां परमक्रान्त्यंशानां ज्या) अस्या बहुषु स्थलेषूपयोगित्वात्संख्याः पठिता इति॥३-९॥

अत्रोपपत्तिः

वृत्तपरिधि ३६० चतुर्थांशे ९० चतुर्विंशतिसंख्यकाः क्रमज्या उत्क्रमज्याश्च $\frac{९०\times६०}{२४}$=२२५′=प्रथमचापम्, २२५′, २२५′×२, २२५×३……२४×२२५′ चापानां, ज्योत्पत्तिविधिना ३२७० मितत्रिज्यायामाचार्येणाऽऽनीय पठिताः। यथा ३४३८ त्रिज्यायां भास्करमतेन स्वल्पान्तरात् २२५=प्रथमज्या, ततोऽनुपातो यदि

३४३८ त्रिज्यायां २२५ तुल्या प्रथमज्या लभ्यते तदा ३२७० त्रिज्यायां किमित्यनुपातेनाऽऽगच्छत्याचार्योक्तप्रथमज्या $=\frac{२२५\times३२७०}{३४३८}$ हरभाज्यौ नवभिरपवर्तितौ तदा

$\frac{२५\times३२७०}{३८२}=\frac{२५\times१६३५}{१९१}=\frac{४०८७५}{१९१}=२१४\frac{१}{१९१}=२१४$=आचार्योक्तप्रथमज्या स्वल्पान्तरात्, एवमेवान्याः क्रमज्या उत्क्रमज्याश्चाऽऽयान्तीति । एवं पठितज्याभिर्ज्यासाधनविधिना चतुर्विंशत्यंशानां ज्या साध्या सा चै १३२९ तन्मिता भवतीयमेव परमक्रान्तिज्येति । सिद्धान्तशिरोमणेष्टिप्पण्यां संशोधकेन पठितज्यास्विष्टज्याज्ञानात्तत्पूर्वाग्रिमज्ययो (पृष्ठज्याऽग्रज्ययोः) र्घातानयनं कृत्वा तत्र पृष्ठज्या भक्तेऽग्रज्या भवेदग्रज्याभक्ते पृष्ठज्या भवेदिति प्रदर्शितम् । यथा यथेष्टचापम् $=इ$; प्रथमचापम् $=प्र$, तदा ज्या $(इ-प्र)$ =पृष्ठज्या, ज्या $(इ+प्र)$ =अग्रज्या ।

अनयोर्घातः

ज्या $(इ-प्र)\times$ज्या $(इ+प्र)$ =पृष्ठज्या $\times$ अग्रज्या, चापयोरिष्टयोरित्यादिना,

$$\frac{(ज्याइ\times कोज्याप्र-ज्याप्र\times कोज्याइ)}{त्रि}\times$$

$$\frac{(ज्याइ\times कोज्याप्र+ज्याप्र\times कोज्याइ)}{त्रि}$$ योगान्तरघातस्य वर्गान्तरसमत्वात् ।

$$\frac{ज्या^२इ\times कोज्या^२प्र-ज्या^२प्र\times कोज्या^२इ}{त्रि^२}=$$

$$\frac{ज्या^२इ\,(त्रि^२-ज्या^२प्र)-ज्या^२प्र\,(त्रि^२-ज्या^२इ)}{त्रि^२}$$

$$=\frac{ज्या^२इ\times त्रि^२-ज्या^२इ\times ज्या^२प्र-ज्या^२प्र\times त्रि^२+ज्या^२प्र\times ज्या^२इ}{त्रि^२}=$$

$$\frac{ज्या^२इ\times त्रि^२-ज्या^२प्र\times त्रि^२}{त्रि^२}$$

$$=\frac{त्रि^२\,(ज्या^२इ-ज्या^२प्र)}{त्रि^२}=ज्या^२इ-ज्या^२प्र,$$ ३४३८ त्रिज्यायां स्वल्पान्तरात्

ज्या^{२}प्र $=५०५६०$ ततः ज्या^{२}इ $-५०५६०=$पृष्ठज्या $\times$ अग्रज्या $\therefore \frac{ज्या^२इ-५०५६०}{पृष्ठज्या}$ =अग्रज्या, वा $\frac{ज्या^२इ-५०५६०}{अग्रज्या}$ =पृष्ठज्या, एतेन 'ज्यावर्गस्वरसाक्षाभ्रवाणोनादित्यादि' संशोधकोक्तमुपपन्नम् । ज्या^{२}इ—ज्या^{२}प्र=पृष्ठज्या $\times$ अग्रज्या, अत्राऽऽचा-

र्योक्तज्या[२]प्रवशेन तत्पृष्ठज्याऽग्रज्ययोर्घातज्ञानं भवेत्तत्र पृष्ठज्याभक्तेऽग्रज्या भवेदग्रज्याभक्ते पृष्ठज्या भवेदिति ।

एतस्य प्रकारस्य खण्डनं म. म. सुधाकरद्विवेदिनैवं क्रियते यथा ज्या (इ—प्र) × ज्या (इ+प्र) = पृष्ठज्या + अग्रज्या, अत्र यदि इष्टचापम् = प्रथमचापम् ।

तदा ज्या (इ—प्र) × ज्या (इ+प्र) = ज्या[२]इ—ज्या[२]प्र = पृष्ठज्या × अग्रज्या = ० × अग्रज्या

$\therefore \frac{\text{ज्या}^2\text{इ}-\text{ज्या}^2\text{प्र}}{०} = \text{अनन्त} = \text{अग्रज्या}$, तदेष्टचापसमे प्रथमचापेऽग्रज्यामानमनन्तसमं । संशोधकप्रकारेण समागच्छत्यतस्तन्मतं न युक्तमिति ।

परं सुधाकरद्विवेदिखण्डनं न युक्तं, संशोधकप्रकारः समीचीन एवेति प्रदर्श्यते ।

ज्या[२]इ—ज्या[२]प्र = पृष्ठज्या × अग्रज्या, यदि इष्टचा = प्रथमचा तदा ज्या (इ—प्र) = पृष्ठज्या = ०,

∴ ज्या[२]इ—ज्या[२]प्र = ० × अग्रज्या वर्गान्तरस्य योगान्तरघातसमत्वात् ।

(ज्याइ—ज्याप्र) (ज्याइ+ज्याप्र) = ० × (ज्याइ+ज्याप्र) = ० × अग्रज्या,

$\therefore \frac{०\times(\text{ज्याइ}+\text{ज्याप्र})}{०} = \text{ज्याइ}+\text{ज्याप्र} = \text{अग्रज्या}$, इतिलुप्तभिन्नसमीकरणेनाग्रज्यामानं समीचीनमेवागतमेतेनसंशोधकप्रकारस्य समाधानं जातमिति ।

म. म. सुधाकरद्विवेदिमहानुभावैरग्रज्यापृष्ठज्ययोर्योगवशात्तज्ज्ञानं कृतं यथा इष्टचापम् = इ । प्रथमचापम् = प्र । ज्या (इ—प्र) = पृष्ठज्या, ज्या (इ+प्र) = अग्रज्या ततः ज्या (इ—प्र) + ज्या (इ+प्र) = पृष्ठज्या + अग्रज्या, चापयोरिष्टयोरित्यादिना

$$\frac{\text{ज्याइ}\times\text{कोज्याप्र}-\text{ज्याप्र}\times\text{कोज्याइ}}{\text{त्रि}}+\frac{\text{ज्याइ}\times\text{कोज्याप्र}+\text{ज्याप्र}\times\text{कोज्याइ}}{\text{त्रि}}=$$

$$\text{पृज्या}+\text{अग्रज्या}=\frac{२\ \text{ज्याइ}\times\text{कोज्याप्र}}{\text{त्रि}}=\frac{२\ \text{ज्याइ}(\text{त्रि}-\text{ज्याउप्र})}{\text{त्रि}}=$$

$$२\ \text{ज्याइ}-\frac{२\ \text{ज्याइ}\times\text{ज्याउप्र}}{\text{त्रि}}=२\left(\text{ज्याइ}-\frac{\text{ज्याइ}\times\text{ज्याउप्र}}{\text{त्रि}}\right)=२$$

$$\left(\text{ज्याइ}-\frac{\text{ज्याइ}}{\frac{\text{त्रि}}{\text{ज्याउप्र}}}\right)२\left(\text{ज्याइ}-\frac{\text{ज्याइ}}{४६७}\right)=\text{पृज्या}+\text{अग्रज्या, अत्र त्रि}=३४३८,$$

अत्र पृष्ठज्याशोधनेनाग्रज्या भवेदग्रज्याशोधनेन च पृष्ठज्या भवेदेतेन तदीय-सूत्रमवतरति।

जीवा स्वसप्तारियुगांशहीना द्विघ्नी च पूर्वज्यकया विहीना।
स्यादग्रजीवा वृहतीति सर्वा आसन्नजीवा द्वयतो भवन्ति॥

अस्याऽऽचार्यस्य मते त्रिज्या=३२७०, एतत्त्रिज्यावशेनापि प्रथमोत्क्रमज्या=३४३८ त्रिज्योत्पन्नप्रथमोत्क्रमज्या=७, अतः पूर्वोक्तसूत्रेणाऽऽचार्योक्तज्यासु कयापीष्टज्यया तत्पूर्वाग्रिमज्ययोर्योगज्ञानं भवेदेवेति॥३-९॥

अब स्पष्टीकरणादि सब ग्रहगणितों के ज्यागणित के अधीन होने के कारण पहले अर्धज्या को कह कर उत्क्रमज्या को कहते हैं

हि. भा.—वृत्त परिधि के चतुर्थांश में २२५′, २×२२५′, ३×२२५′..... चापों की चौबीस क्रमज्यायें और उत्क्रमज्यायें हैं जो संस्कृत विज्ञान भाष्य में लिखी गई हैं, उन्हीं को यहाँ भी देखिए। चौबीस अंश की ज्या परमक्रान्तिज्या है, इसकी उपयोगिता बहुत स्थानों में होने के कारण उसकी संख्या १३२९ पठित की गई है इति॥३-९॥

उपपत्ति

वृत्त परिध्यंश ३६० के चतुर्थांश ९०° में चौबीस संख्यक क्रमज्यायें और उत्क्रमज्यायें

$$\left(\frac{९०\times६०}{२४}=२२५′=\text{प्रथमचाप }२२५′,\ २\times२२५′,\ ३\times२२५′\cdots\cdots२४\times२२५′\right)$$

चापों की ज्योत्पत्तिविधि से ३२७० तुल्य त्रिज्या में आचार्य ने लाकर पाठ किया है, जैसे भास्कराचार्य के मत में ३४३८ त्रिज्या में स्वल्पान्तर से प्रथमज्या=२२५, तब अनुपात करते हैं यदि ३४३८ तुल्य त्रिज्या में २२५ तुल्य प्रथमज्या पाते हैं तो ३२७० त्रिज्या में क्या, इस अनुपात से आचार्योक्त प्रथमज्या आती है, $\frac{२२५\times३२७०}{३४३८}$=आचार्योक्त प्रथमज्या, यहां हर और भज्य में ९ इससे अपवर्त्तन करने से

$$\frac{२५\times३२७०}{३८२}=\frac{२५\times१६३५}{१९१}=\frac{४०८७५}{१९१}=२१४\frac{१}{१९१}=२१४ \text{ स्वल्पान्तर से आचार्योक्त}$$

प्रथमज्या, इसी तरह अवशिष्ट (द्वितीयादि) क्रमज्यायें और उत्क्रमज्यायें आती हैं, पठित-ज्यायों से ज्यासाधन विधि से चौबीस अंश परमक्रान्त्यंश की ज्या साधन करने से १३२९ एतत्तुल्य होती है। सिद्धान्तशिरोमणि की टिप्पणी में संशोधक 'पठितज्यासु में एकका अंक

से उससे पूर्व और अग्रिम (पृष्ठज्या और अग्रज्या) ज्याओं के घातानयन करके उस घात में पृष्ठज्या से भाग देने से अग्रिमज्या होती है और अग्रज्या से भाग देने से पृष्ठज्या होती है, दिखलाते हैं जैसे इष्टचाप=इ, प्रथमचाप=प्र, ज्या (इ—प्र)=पृष्ठज्या ज्या (इ+प्र)= अग्रज्या, दोनों के घात करने से ज्या (इ—प्र)×ज्या (इ+प्र)=पृष्ठज्या ×अग्रज्या

चापयोरिष्टयोर्दोर्ज्ये मिथः कोटिज्यकाहते इत्यादि से

$$\frac{(\text{ज्याइ}\times\text{कोज्याप्र}-\text{ज्याप्र}\times\text{कोज्याइ})}{\text{त्रि}}\times\frac{(\text{ज्याइ}\times\text{कोज्याप्र}+\text{ज्याप्र}\times\text{कोज्याइ})}{\text{त्रि}}$$ योगान्तर

घात वर्गान्तर के बराबर होता है, इस नियम से $\frac{\text{ज्या}^2\text{इ}\times\text{कोज्या}^2\text{प्र}-\text{ज्या}^2\text{प्र}\times\text{कोज्या}^2\text{इ}}{\text{त्रि}^2}=$

$$\frac{\text{ज्या}^2\text{इ}\ (\text{त्रि}^2-\text{ज्या}^2\text{प्र})-\text{ज्या}^2\text{प्र}\ \ (\text{त्रि}^2-\text{ज्या}^2\text{इ})}{\text{त्रि}^2}=$$

$$\frac{\text{ज्या}^2\text{इ}\times\text{त्रि}^2-\text{ज्या}^2\text{इ}\times\text{ज्या}^2\text{प्र}-\text{ज्या}^2\text{प्र}\times\text{त्रि}^2+\text{ज्या}^2\text{प्र}\times\text{ज्या}^2\text{इ}}{\text{त्रि}^2}=$$

$$\frac{\text{ज्या}^2\text{इ}\times\text{त्रि}^2-\text{ज्या}^2\text{प्र}\times\text{त्रि}^2}{\text{त्रि}^2}=\frac{\text{त्रि}^2(\text{ज्या}^2\text{इ}-\text{ज्या}^2\text{प्र})}{\text{त्रि}^2}=\text{ज्या}^2\text{इ}-\text{ज्या}^2\text{प्र}=\text{पृष्ठज्या}\times$$

अग्रज्या, ३४३८ त्रिज्या में स्वल्पान्तर से ज्या^{2}प्र=५०५६० ∴ ज्या^{2}इ—५०५६०=

पृष्ठज्या×अग्रज्या इसलिए $\frac{\text{ज्या}^2\text{इ}-५०५६०}{\text{पृष्ठज्या}}=$अग्रज्या, वा $\frac{\text{ज्या}^2\text{इ}-५०५६०}{\text{अग्रज्या}}=$पृष्ठज्या,

इस से 'ज्यावर्गात्त्वरमाक्षाभ्रबाणोनात् इत्यादि' संशोधकोक्त उपपन्न हुआ।

यहाँ आचार्योक्त प्रथमज्या वश से पृष्ठज्या और अग्रज्या के घात से पूर्ववत् पृष्ठज्या और अग्रज्या का ज्ञान हो जायगा। संशोधकोक्त प्रकार का खण्डन किसी ने अधोलिखित युक्ति से किया है, ज्या(इ—प्र)×ज्या (इ+प्र)=पृष्ठज्या×अग्रज्या=ज्या^{2}इ—ज्या^{2}प्र; यहाँ यदि इष्टचा=प्रचा तब ज्या^{2}इ—ज्या^{2}प्र=०×अग्रज्या ∴ $\frac{\text{ज्या}^2\text{इ}-\text{ज्या}^2\text{प्र}}{०}=$अग्रज्या= अनन्त; परन्तु इष्टचाप और प्रथमचाप के बराबर रहने से अग्रज्यामान अनन्त के बराबर नहीं होना चाहिए इसलिए यह प्रकार ठीक नहीं है। लेकिन यहाँ किसी ने जो खण्डन किया है वह ठीक नहीं है, संशोधकोक्त प्रकार ठीक ही है जैसे:—

ज्या^{2}इ—ज्या^{2}प्र=पृष्ठज्या×अग्रज्या, यदि इष्टचाप=प्रथमचाप तब ज्या (इ—प्र)=०=पृष्ठज्या, इसलिए ज्या^{2}इ—ज्या^{2}प्र=०×अग्रज्या, परन्तु वर्गान्तर योगान्तर घात के बराबर होता है।

अतः (ज्याइ—ज्याप्र) (ज्याइ+ज्याप्र)=०×(ज्याइ+ज्याप्र)=०×अग्रज्या

$\therefore \frac{०\times(\text{ज्याइ}+\text{ज्याप्र})}{०}$ =अग्रज्या=ज्याइ+ज्याप्र=अग्रज्या, लुप्तभिन्नसमीकरण

से अग्रज्या का मान समीचीन ही आता है, अतः संशोधकोक्त प्रकार समीचीन ही है, यह सिद्ध हुआ।

यहां म.म. सुधाकर द्विवेदी जी ने अग्रज्या और पृष्ठज्या के योग के सम्बन्ध से अग्रज्या के ज्ञानार्थ विधि दिखलायी है, जैसे इष्टचाप =इ। प्रथमचाप=प्र, ज्या (इ—प्र)=पृष्ठज्या ज्या (इ+प्र)=अग्रज्या तब ज्या (इ—प्र)+ज्या (इ+प्र)=पृष्ठज्या+अग्रज्या, चापतोरिष्टयोर्ज्ये मिथः कोटिज्यकाहते इत्यादि से $\frac{\text{ज्याइ}\times\text{कोज्याप्र}-\text{ज्याप्र}\times\text{कोज्याइ}}{\text{त्रि}}+$

$\frac{\text{ज्याइ}\times\text{कोज्याप्र}+\text{ज्याप्र}\times\text{कोज्याइ}}{\text{त्रि}}=\frac{२\ \text{ज्याइ}\times\text{कोज्याप्र}}{\text{त्रि}}$

$=\frac{२\ \text{ज्याइ}(\text{त्रि}-\text{ज्याउप्र})}{\text{त्रि}}=२\ \text{ज्याइ}-\frac{२\text{ज्याइ}\times\text{ज्याउप्र}}{\text{त्रि}}=२\left(\text{ज्याइ}-\frac{\text{ज्याइ}\times\text{ज्याउप्र}}{\text{त्रि}}\right.$

$=२\left(\text{ज्याइ}-\frac{\text{ज्याइ}}{\frac{\text{त्रि}}{\text{ज्याउप्र}}}\right)=२\left(\text{ज्याइ}-\frac{\text{ज्याइ}}{\frac{३४३८}{७}}\right)$

$=२\left(\text{ज्याइ}-\frac{\text{ज्याइ}}{४६७}\right)$=पृष्ठज्या+अग्रज्या, अतः $२\left(\text{ज्याइ}-\frac{\text{ज्याइ}}{४६७}\right)$—पृष्ठज्या=

अग्रज्या।

इससे द्विवेदी जी का सूत्र 'जीवा स्वसप्तारियुगांशहीना इत्यादि' जो संस्कृतोपपत्ति में लिखा गया है, उपपन्न हुआ। आचार्य (ब्रह्मगुप्त) के मत में त्रिज्या=३२७०, इस त्रिज्या से भी प्रथमोत्क्रमज्या=७ अतः पूर्वोक्त सूत्र से इनकी पठित ज्याओं में किसी इष्टज्या से अग्रज्या का ज्ञान पूर्ववत् होता है, इति ॥३-६॥

केषाञ्चिन्मतं तत्प्रकारश्च

"पृष्ठज्या यत्र शून्या प्रथमगुणसमाऽभीष्टचापज्यका स्यादग्रज्या नैव सिद्ध्यत्युदितगणिततस्तत्र संशोधकस्य। शून्यत्वादिष्टज्याप्रथमगुणवियोगैक्यघातस्य तस्मात् दुष्टोऽयं तत्प्रकारो गणितमतिमता वेदितव्यो बुधेने" ति केनाप्युक्तवचसा दुष्टोऽयं प्रकार इत्यधिक्षिपति कश्चित्।

वस्तुतो विचार्यमाणे यत्र भिन्ने भाज्यभाजकगताव्यक्तराशेर्यस्मिन्कस्मिन्नपि

व्यक्तमाने तन्मूल्यं शून्यसमं भवेत्तल्लुप्तसंज्ञकभिन्नमिति नवीनाः प्रवदन्ति। तत्र लुप्तभिन्ना÷दस्मात्प्रकृतराशिज्ञानं कथं स्यादिति समालोच्यते।

अथ लुप्तभिन्नस्य भाज्यभाजकावव्यक्तस्य व्यक्तपदे शून्यसमौ भवेतां तत्र व्यक्ताव्यक्तयोरन्तरेण वा तत्सजातीयेन केनाप्यङ्केन तौ भाज्यभाजकाववश्यमेव निःशेषं भजेताम्। अन्यथा तत्र लुप्तभिन्नत्वं न स्थास्यतीत्युच्चगणितेन स्फुटमेव विदाम्। अतोऽत्रानया दिशा संशोधकप्रकारः परीक्ष्यते, तथाहि संशोधकप्रकारेण—

$$\text{अग्रज्या} \times \text{पृष्ठज्या} = \text{ज्या}^2\text{इ} - \text{ज्या}^2\text{प्र}$$

$$\therefore \quad \text{अग्रज्या} = \frac{\text{ज्या}^2\text{इ} - \text{ज्या}^2\text{प्र}}{\text{पृष्ठज्या}}$$

$$= \frac{(\text{ज्याइ} - \text{ज्याप्र})\,(\text{ज्याइ} + \text{ज्याप्र})}{\text{ज्या}\,(\text{इ} - \text{प्र})} \quad \ldots\ldots(१)$$

इदमेव तावद्वास्तवभिन्नस्वरूपम्।

तत्र यदि ज्याइ=ज्याप्र।

वा इ=प्र कल्प्येत चेत् तदा

(१) समीकरणस्वरूपम्= ÷ अतो लुप्तभिन्नत्वं जातम्।

अतो लुप्तभिन्नसिद्धान्तेन (१) समीकरणे भाज्यभाजकौ ज्या (इ—प्र) अनेन निःशेषं भवत एव।

$$\text{तदर्थं} \quad \text{ज्याइ} - \text{ज्याप्र} = \frac{२\,\text{कोज्या}\frac{१}{२}\,(\text{इ}+\text{प्र}),\ \text{ज्या}\frac{१}{२}\,(\text{इ}-\text{प्र})}{\text{त्रि}}$$

$$\text{एवं} \quad \text{ज्याइ} + \text{ज्याप्र} = \frac{२\,\text{ज्या}\frac{१}{२}\,(\text{इ}+\text{प्र}),\ \text{कोज्या}\frac{१}{२}\,(\text{इ}-\text{प्र})}{\text{त्रि},}$$

$$\text{द्वयोर्घातः} \quad = \text{ज्या}\,(\text{इ}+\text{प्र})\ \text{ज्या}\,(\text{इ}-\text{प्र})$$

$$\text{अतो वास्तवभिन्नः} = \frac{\text{ज्या}\,(\text{इ}+\text{प्र})\ \text{ज्या}\,(\text{इ}-\text{प्र})}{\text{ज्या}\,(\text{इ}-\text{प्र})} = \text{ज्या}\,(\text{इ}+\text{प्र})$$

एतेनाग्रजीवा वास्तवैव सिद्ध्यतीति। अतोऽधिक्षेपो न युक्त इति। संशोधक. प्रकारस्य वास्तवत्वं न हीयते, किन्तु स्थास्यत्येव।

इदानीं चापज्यानयनमाह

लिप्तास्तत्त्वयमहृता लब्धज्या ज्यान्तराहताच्छेषात् ।
तिथिकृतिहृतात्फलयुता लब्धज्या ज्याग्रहणमेवम् ॥१०॥

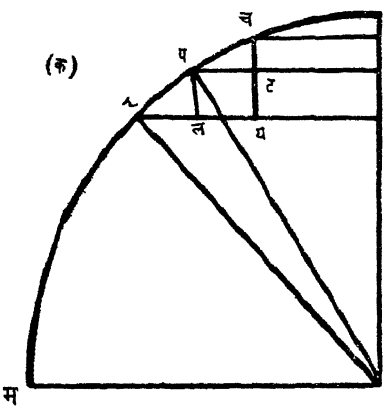
वा. भा.—इदानीं इष्टस्य धनुषो ज्याकरणायार्यामाह । इष्टचापस्य यस्य ज्याकर्तुमिष्यते तत्सम्बन्धिन्यो लिप्ता गृह्यन्ते, तेनायमर्थः । लिप्तास्तत्त्वयमहृताः कार्यास्ततो लब्धांकसमसंख्या ज्या स्थापयितव्या ज्यान्तरहृताच्छेषात् । इति लब्धज्यायाः तदग्रतो वर्त्तमानज्यायाश्च यदन्तरं तज्ज्यान्तरं तेन हृताच्छेषात् । प्रथममेव तत्त्वयमैर्यद्भागे हृते यदवशेषं तच्छेषमुच्यते । तस्मात् तिथिकृतिहृतादिति, तत् यमैरेव हृतादित्यर्थः । ततो यदवाप्तं फलं तेन युता लब्धज्या कार्या । एवं ज्याग्रहणमिष्टचापोत्क्रमोत्क्रमज्यामयीत्यर्थः । एतदुक्तं भवति वक्ष्यमाणविधिना ज्याकेन्द्रं कार्यम् । तल्लिप्तादिकेन्द्रं कृत्वा तत्त्वयमैर्विभजेत् । लब्धांकसमां ज्यां स्थापयेत् । मनुयमला मुनियमवेदा इत्यादिगणनया यदि क्रमेण ज्याग्रहणमथोत्क्रमेण तथा मुनयोऽष्टयमा इत्यादिकया गणनया ज्यां स्थापयेत् । शेषं विकलसंज्ञं भवति । ततो लब्धज्याया पुरतः स्थितया सहान्तरे कृते यदवशिष्यते । समगुणकारो भवति, तेन विकलं संगुण्य तत्त्वयमैर्विभजेत् । लब्धं लिप्तादिः पूर्वस्थापितज्यायां संयोज्य ज्या भवति क्रमोत्क्रमेण च ज्याग्रहणमेवं कार्यं । अत्रेयं वासना गोलाध्याये प्रदर्शितविधिना वृत्तक्षेत्रं दिगङ्कितं राश्यष्टांशेष्टकानित्यादिनोक्तवत् परिलिख्य प्रदर्शयेत् । तत्र खखषडिन्दुघनसंख्यापरिणाहे षण्णवतिहृते तत्त्वयमा भवन्ति । अतएवेष्टचापलिप्तानां तत्त्वयमा भागहारः । तस्मात्तत्त्वयमसंख्ये चापखण्डलके ज्यारेखा स्थिता, यावद्गुणश्च तत्त्वयमा लिप्ताभ्यो विशोध्यन्ते । तावत् संस्थज्या तासां शुद्धलिप्तानां ततो भवति ततः शुद्धलिप्ताभिः त्रैराशिकं यदि तत्त्वयमसंख्याभिर्लिप्ताभिर्लब्धाः भुक्तज्यायोज्यज्ययोरन्तरतुल्यं ज्याखण्डलकं भवति, तदाभिरिष्टलिप्ताभिः किमिति लब्धफलेनोपचीयते । पूर्वलब्धा ज्या यतः तद्रेखाक्रान्तं चापखण्डमतिक्रम्य पुरतः स्थितो ग्रहादिकस्तदवधिज्यया वास्माकं प्रयोजनमुपपन्नं ज्याग्रहणमेवेति ।

वि. भा.—यासां लिप्तानां (कलानां) ज्याः साध्यास्तास्तत्त्वयमहृता (२२५ एभिर्भाज्याः) लब्धज्या (लब्धसंख्यकगतज्या बोध्याः), शेषात् ज्यान्तराहतात् (गतैष्यज्ययोरन्तरगुणितात्) तिथि १५ कृतिहृतात्(२२५ एभिर्भक्तात्) यत्फलं (लब्धं) तेन युता लब्धज्या (गतज्या) कार्याः, एवं कृते ज्याग्रहणं (इष्टज्यामानं) भवेदिति ॥१०॥

अत्रोपपत्तिः

वृत्तपादे २२५′, २×२२५′, ३×२२५′.....२४×२२५′ चतुर्विंशतिसंख्यका ज्याः पूर्वं पठिताः सन्ति, पदादिबिन्दुतो यदि द्वयोरुक्तयोश्चापयो-

र्मध्ये इष्टचापाग्रं भवेत्तदा पदादिबिन्दुत इष्टचापाग्रं यावदिष्टचापकला यास्तास्तत्त्वाश्वि २२५ भिर्भाज्यास्तदा या लब्धिस्तत्संख्यका गतज्या भवेयुः, शेषचापादनुपातो 'यदितत्त्वाश्वि २२५ भिर्गतैष्यज्ययोरन्तरं लभ्यते तदा शेषचापेन किमिति, लब्धिः शेषचापसम्बन्धिनी ज्यावृद्धिरेतया युता गतज्येष्टा भवतीति परं शेषचापसम्बन्धिज्यावृद्ध्यर्थं यो ह्यनुपातः कृतः स च न समीचीनः कथमित्युच्यते ।



के=वृत्तकेन्द्रम् । नम=वृत्तपादः =९०, चज=गतज्या, रश=एष्यज्या, नप=इष्टचापम्, पस=इष्टज्या, चर =२२५, रय=एज्या—गतज्या, चप= शेषचापम् । पट=शेषचापसम्बन्धिनी ज्यावृद्धिः । <केरन=९०, <केपन=९० रकेशकोणः> पकेसकोणात् ∴ केरश कोण < केपस कोण तेन चरय कोण >चपट कोण अतः चरय, चपट त्रिभुजयोर्विजातीयत्वा—

$$\frac{(\text{एज्या}—\text{गज्या})\times\text{शेषचा}}{२२५}$$

देतादृशोऽनुपातो न भवितुमर्हत्यत आचार्योक्तमिष्टचापज्यानयनं न युक्तियुक्तमिति सिद्धम् । परमिष्टज्यानयनार्थं सूर्यसिद्धान्तकारसिद्धान्तशेखरकारप्रभृतिभिः प्राचीनाचार्यैर्भास्करप्रभृतिभिस्तदर्वाचीनैश्चायमेव विधिर्गृहीत ॥१०॥

अब इष्टचाप के ज्यानयन को कहते हैं

हि. भा.—जिन कलाओं का ज्या साधन करना हो उन चाप कलाओं को दो सौ पच्चीस २२५ से भाग देने से जो लब्धि होती है तत्संख्यक गतज्या होती है, शेषचाप का गतज्या और एष्यज्या के अन्तर से गुणा कर दो सौ पच्चीस से भाग देने से जो लब्धि होती है उसको गतज्या में जोड़ने से इष्टज्या होती है इति ॥१०॥

उपपत्ति

वृत्त के चतुर्थांश ९० में पदादि बिन्दु से २२५′ कला वृद्धि करके चापों की चौबीस संख्यक ज्या पहले पठित की गई है । यदि उन दो चापों के मध्य में इष्टचापाग्र हो अर्थात् उन दो ज्याओं के बीच में इष्टज्या हो तब उसका ज्ञान कैसे होगा तदर्थ विचार करते हैं, इष्टचाप कला को दो सौ पच्चीस २२५ से भाग देने से जो लब्धि होती है तत्संख्यक गतज्या समझनी चाहिए । शेष चाप से अनुपात करते हैं यदि दो सौ पच्चीस २२५ में गतज्या और एष्यज्या का अन्तर पाते हैं तो शेष चाप में क्या इससे लब्धि शेषचाप सम्बन्धिनी ज्यावृद्धि

होती है, इसको गतज्या में जोड़ने से इष्टज्या होती है। लेकिन शेष-चाप-सम्बन्धिनी ज्यावृद्धि के लिए जो अनुपात किया गया है सो ठीक नहीं है, इसके लिए विचार करते हैं।

यहाँ संस्कृतोपपत्तिस्थ (क) क्षेत्र को देखिये। के=वृत्तकेन्द्र, नम चाप=वृत्तपाद =९०, चज=गतज्या, रश=एष्यज्या, नप=इष्टचाप, पस=इष्टज्या, चर=२२५, रय =एज्या—गतज्या, चप=शेषचाप, पट=शेषचापसम्बन्धिनी ज्यावृद्धि<केरन=९०, < केपन=९०, रकेश कोण > पकेस ∴ केरश कोण<केपस कोण ∴ चरय कोण > चपट कोण इसलिए चरय, चपट दोनों त्रिभुजों के विजातीयत्व के कारण $\frac{(\text{एज्या}-\text{गज्या})\ \text{शेचा}}{२२५}$ यह अनुपात ठीक नहीं कहा जा सकता है इसलिए आचार्योक्त इष्ट-चापज्यानयन युक्तियुक्त नहीं है, यह सिद्ध हुआ। लेकिन इष्टज्यानयन के लिए सूर्य सिद्धान्त-कार प्रभृति प्राचीनाचार्य और भास्करप्रभृति उनसे अर्वाचीन आचार्य इसी (आचार्योक्त) विधि को अपनाये हुये हैं इति ॥१०॥

इदानीं ज्यातश्चापानयनमाह

ज्यां प्रोह्य शेषगुणितास्तत्त्वयमा ज्यान्तरोद्धृता लब्धम्।
क्षेप्यं विशुद्धजीवासंख्या तिथिकृतिवधे चापम् ॥११॥

वा. भा.—इदानीमिष्टज्यायाश्चापकरणार्थमार्यामाह। इष्टज्यायाः यस्याः चापं कर्त्तुमिष्यते। तस्या ज्यां प्रयोज्य इष्टज्यातो या ज्या ज्ञातसंख्या विशुध्यति तां विशोध्य यच्छेषं तेन गुणिताः तत्त्वयमा ज्यान्तरोद्धृताः कार्याः यल्लब्धं तत्तत्र क्षेप्यं क्वेत्याह, विशुद्धजीवा संख्यातिथिकृति-वधे:, एवं कृते चापं कृत्वा ज्यातो भवति। क्रमोत्क्रमाभ्यां अपि अयमर्थः यस्याः ज्यायाः चापं क्रियते। ततो मनुयमला इत्यादिकानां ज्यानां मध्याद्याजीवा विशु-ध्यति तां विशोध्य शेषं विकलं भवति, तत्त्वयमैः संगुण्य शुद्धजीवयोरन्तरेण विभ-जेत्। यल्लब्धं तच्चापखण्डके क्षेप्यं स्थापयेत्, ततो यावत्संख्या ज्या विशुध्यति तावत्संख्यागुणैस्तत्त्वयमैर्युक्तं तत्क्षेप्यं कार्यम्। एवं कृते इष्टज्यायाश्चापं कृतं भवति क्रमेण। अथोत्क्रमेण क्रियते तत्रेष्टज्यातो मुनयो यमादिकानामुक्तज्यानां मध्या या जीवा विशुध्यति तां विशोध्य शेषगुणितास्तत्त्वयमा : कार्यास्ततः शुद्धा शुद्धज्यान्तरेण विभज्य चापखण्डं कृत्वा तच्छुद्धजीवा संख्या तत्त्वयमवधे क्षिपेदेव-मुत्क्रमज्याभिश्चापं कृतं भवति। वासना चात्र प्रागाचार्योक्तवैपरीत्ये योज्या यस्माद्यया वासनया चापात् ज्याकरणं तयैव विपरीतं, ज्यातश्चापकरणमाचार्ये-णोपनिबद्धं तथा युञ्ज्यादिति।

वि. भा.—इष्टज्यातो या ज्या विशुद्धयन्ति ताः शोध्याः, ज्याशेषगुणिता-

स्तत्त्वयमा २२५ गतैष्या ज्यान्तरभक्ता लब्धं यत्संख्यका जीवा विशुद्धास्तासां पञ्चदशवर्गाणां २२५ च घाते योज्यं तदेष्टचापं भवेदिति ॥११॥

अत्रोपपत्तिः

इष्टज्यातो यत्संख्यका ज्या विशुद्ध्यन्ति ताः शोध्याः शेषा(गतज्येष्टज्ययोरन्तरात्)दनुपातो 'यदि गतैष्यज्यान्तरेण २२५ तत्तुल्यं चापं लभ्यते तदा गतज्येष्टज्ययोरन्तरेण किमिति' यल्लब्धं तद्विशुद्धज्यासंख्यागुणिततत्त्वाश्विं २२५ मिते योज्यं तदेष्टचापं भवेदित्येतदर्थमिष्टज्यानयने लिखितं क्षेत्रं विलोकयमित्यत्राप्यनुपातेऽप्यनौचित्यं पूर्ववदेव बोध्यमिति ॥११॥

अब ज्या से चापानयन करते हैं

हि. भा.—इष्टज्या में जितनी ज्यायें घटें उनको घटा देना, ज्याशेष और दो सौ पच्चीस २२५ के घात में गतज्या और एष्यज्या के अन्तर से भाग देने से जो लब्धि हो उसका विशुद्धज्या संख्यागुणित दो सौ पच्चीस २२५ में जोड़ने से इष्टचाप होता है, इति ॥११॥

उपपत्ति

इष्टज्या में जितनी ज्यायें घटें घटा देना, शेष (गतज्या और इष्टज्या के अन्तर) से अनुपात 'यदि गतज्या और एष्यज्या के अन्तर में दो सौ पच्चीस २२५ चाप पाते हैं तो गतज्या और इष्टज्या के अन्तर में क्या' से जो लब्धि आती है उसको विशुद्धज्या गुणित दो सौ पच्चीस २२५ में जोड़ने से इष्टज्या होती है। इसके लिये इष्टज्यानयन में लिखे हुए क्षेत्र को देखना चाहिये। यहाँ भी अनुपात में अशुद्धता पूर्ववत् समझनी चाहिये, इति ॥११॥

अत्र विशेषो विचारः

भास्कराचार्यास्तु प्राचीनोक्तदिशा ज्यानयनं ३४३८ त्रिज्यायां विहितवन्तः। किन्तु खार्कव्यासार्धवृत्ते दशभिरंशैर्ज्यां विधाय तदन्तरेण वृत्तपादे नवलघुज्याखण्डानि प्रोचुः। तत्राभीष्टज्यानयने यथोक्तदिशा त्रैराशिकक्रियया जीवायां महत्स्थौल्यमापद्यते तदर्थं स्फुटभोग्यखण्डानयनं विहितं भास्कराचार्यैः।

तद्यथा। कल्प्यतेऽभीष्टचापम्=चा, यस्य जीवा साधनीया। दशभक्तचापं गतसंज्ञकम्=ग शेषम्= शे. प्रथमचापम् प्र।

अत्र ग—प्र=पूर्वचापम्, ग+प्र=अग्रचापम्

$$\therefore \text{पूर्वज्या} = \frac{\text{ज्याग.कोज्याप्र}-\text{कोज्याग.ज्याप्र}}{\text{त्रि}} = \text{ज्यापू.}$$

$$\text{अग्रज्या} = \frac{\text{ज्याग.}\ \text{कोज्याग्र} + \text{कोज्याग.}\ \text{ज्याग्र}}{\text{त्रि}} = \text{ज्याग्र}$$

अत्र पूर्वज्यकायाः गतज्यकायां विशोधनेन गतखण्डं तथा अग्रज्यकायां गतज्यां विशोध्य भोग्यखण्डं भवतीत्यतः ।

$$\text{गतखण्डम्} = \text{गखं} = \frac{\text{ज्याग}\ (\text{त्रि}-\text{कोज्याग्र}) + \text{कोज्याग.}\ \text{ज्याग्र}}{\text{त्रि}}$$

$$\text{एवं भोग्यखण्डम्} = \text{भोखं} = \frac{\text{कोज्या.}\ \text{ज्याग्र} - \text{ज्याग}(\text{त्रि}-\text{कोज्याग्र})}{\text{त्रि.}}$$

गतैष्यखण्डयोर्योगान्तराभ्याम्—

$$\text{गखं} + \text{भोखं} = \frac{२\ \text{कोज्याग.}\ \text{ज्याग्र}}{\text{त्रि}}$$

$$\therefore \frac{\text{गखं} + \text{भोखं}}{२} = \frac{\text{कोज्याग.}\ \text{ज्याग्र}}{\text{त्रि}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (१)$$

$$\text{एवं} \quad \frac{\text{गखं} - \text{भोखं}}{२} = \frac{\text{ज्याग}\ (\text{त्रि}-\text{कोज्याग्र})}{\text{त्रि}}$$

$$= \frac{\text{ज्याग.}\ \text{उज्याग्र}}{\text{त्रि}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (२)$$

अथ $\text{ग} + \text{शे} = \text{इष्टचापम्} = \text{चा}$ ।

$$\therefore \quad \text{ज्याचा} = \frac{\text{ज्याग.}\ \text{कोज्याशे} + \text{कोज्याग.}\ \text{ज्याशे}}{\text{त्रि}}$$

अत्र शेषचापस्य दशभिरंशैरल्पत्वतः

$$\text{ज्याशे} = \frac{\text{ज्याग्र.}\ \text{शे}}{१०}, \quad \therefore \text{ज्या}^२\text{शे} = \frac{\text{ज्या}^२\text{ग्र.}\ \text{शे}^२}{१०२}$$

$$\therefore \text{कोज्या}^२\text{शे} = \text{त्रि}^२ - \frac{\text{ज्या}^२\text{ग्र.}\ \text{शे}^२}{१०२}$$

मूलेनासन्नमानेन—

$$\text{कोज्याशे} = \text{त्रि} - \frac{\text{ज्या}^२\text{ग्र.}\ \text{शे}^२}{\text{त्रि.}\ १०२}$$ स्वल्पान्तरादग्रिमावयवत्यागात् ।

यदि शे=प्र=१०

$$\text{कोज्याप्र}=\text{त्रि}-\frac{\text{ज्या}^2\text{प्र}}{\text{त्रि}}$$

$$\therefore \quad \frac{\text{ज्या}^2\text{प्र}}{\text{त्रि}}=\text{त्रि}-\text{कोज्याप्र}=\text{प्रथमोत्क्रमज्या}=\text{उज्याप्र}$$

$$\therefore \quad \text{कोज्याशे}=\text{त्रि}-\frac{\text{उज्याप्र.}\ \text{शे}^2}{१०२}$$

$$\therefore \quad \frac{\text{उज्याप्र.}\ \text{शे}^2}{१०२}=\text{त्रि}-\text{कोज्याशे}=\text{उज्याशे}$$

अथ पूर्वानीतचापज्यायां गतज्याया विशोधनेन

$$\text{ज्यान्तरम्}=\frac{\text{कोज्याग.}\ \text{ज्याशे}}{\text{त्रि}} \therefore \frac{\text{ज्याग.}\ (\text{त्रि}-\text{कोज्याशे})}{\text{त्रि}}$$

$$=\frac{\text{कोज्याग.}\ \text{ज्याप्र.शे}}{\text{त्रि}\times १०} \therefore \frac{\text{ज्याग.}\ \text{उज्याप्र.}\ \text{शे}^2}{\text{त्रि.}\ १०२}$$

$$=\frac{\text{कोज्याग.}\ \text{ज्याप्र}}{\text{त्रि}}\cdot\frac{\text{शे}}{१०} \therefore \frac{\text{ज्याग.}\ \text{उज्याप्र}}{\text{त्रि.}}\cdot\frac{\text{शे}^2}{१०२}$$

अत्र (१) (२) समीकरणाभ्यामुत्थापनेन—

$$\text{ज्यान्तरम्}=\frac{\text{गखं}+\text{भोखं}}{२}\cdot\frac{\text{शे}}{१०}-\frac{\text{गखं}-\text{भोखं}}{२}\cdot\frac{\text{शे}^2}{१०२}$$

$$=\frac{\text{शे}}{१०}\left(\frac{\text{गखं}+\text{भोखं}}{२}-\frac{(\text{गखं}-\text{भोखं})\text{शे}}{२०}\right)$$

अत्र कोष्ठकान्तर्गतखण्डं यदि भोग्यखण्डं मन्यते तर्हि प्रागुक्तानुपातेन वास्तवमेव ज्यान्तरं समागच्छतीति "यातैष्ययोः खण्डकयोर्विशेषःशेषांशनिघ्नो नखहृदित्यादि" भास्करोक्तं समीचीनमेव स्यात् व्यर्थं दुराग्रहेण प्रखण्डितं कमलाकरेण।

इदानीं मन्दशीघ्रकेन्द्रयोः केन्द्रभुजकोटिज्ययोश्च परिभाषामाह

मध्याद्विशोध्य मन्दं शीघ्रात्संशोध्य मध्यमं केन्द्रम् ।
अयुजि गतयेययोर्युजि पदेऽन्यथा बाहुकोटिज्ये ॥१२॥

वा. भा.—इदानीं मन्दशीघ्रकर्मणोः केन्द्रकरणायार्यार्धेनाह। अत्र करणागतो देशान्तरफलेन संस्कृतो मध्य उच्यते। मन्दस्फुटञ्च मध्यं भौमादीनां तेनायमर्थः मध्याद्विशोध्य मन्दं मध्याद् गृहात्स्वमन्दोच्चं विशोध्य केन्द्रं भवति। शीघ्रोच्चात् संशोध्य मध्यमं मन्दफलं स्फुटं शीघ्रकेन्द्रं भवति। केन्द्रशब्देन मध्यमुच्यते। तेन कक्षामण्डले प्रथमं केन्द्रं मन्दनीचोच्चवृत्तमध्यं द्वितीयं केन्द्रं शीघ्रनीचोच्चवृत्तमध्यं यथा न्यस्तेषु कक्षामण्डलप्रतिमण्डलादिषु कक्षामण्डले मध्यग्रहा चिन्हेत्। तत्प्रदेशे नीचोच्चवृत्तं मध्ये कृत्वा प्रदर्शयेत् तद्यथा मेषादेः प्रभृति। कक्षामण्डले यावतो राशिभागादयो भुक्ताः ग्रहास्तेभ्यः स्वमन्दोच्चभुक्ताः राशिभागादीन्विशोध्य शोधयेत् राशिभागादयस्तावद्भिर्मन्दोच्चभागावधेस्तन्मन्दोच्चनीचवृत्तमध्यं वर्त्तते। अत एव तदन्तरकराश्यादिकं केन्द्रं मध्यमुच्यते। स च रेखातस्तावत्प्रदेशे केन्द्रस्याग्रतो गतत्वात्। स्वमन्दोच्चात् शीघ्रगतिर्यतोऽतो ग्रहात् मन्दोच्चं विशोध्यतेऽधिकभुक्तिपरिज्ञानाय स्वशीघ्रोच्चान्मन्दो ग्रहः अतएव शीघ्रात् ग्रहो विशोध्यते, तदन्तरपरिज्ञानाय क्रियति दूरे राशिभागाधिके ग्रहोऽवलंबित इति। यावति च प्रदेशे राशिभागादिके स्वशीघ्रोच्चरेखातो ग्रहपश्चादवलम्बितः कक्षामण्डले तावति दूरे शीघ्रनीचोच्चवृत्तमध्ये वर्त्तते। अतएव तदन्तरकं शीघ्रकेन्द्रमुच्यते।

इदानीं द्वितीयार्यार्धेन तत एव केन्द्रात् भुजकोटिज्ययोः करणमाह। अयुजिगतये अयुजिपदेऽन्यथा बाहुकोटिज्ये।

अयुजि विषमपदस्थे केन्द्रे यथासंख्यगतये ययोर्बाहुकोटिज्ये कार्ये, युजिपदे समपदस्थे केन्द्रेऽन्यथा येयगतयोर्बाहुकोटिज्ये कार्ये इत्यर्थः। एतदुक्तं भवति। मध्यकर्मणि मध्यग्रहात् स्वमन्दोच्चं विशोध्य केन्द्रं कार्यम्। तद्यदि प्रथमे पदे भवति तदा केन्द्रेण यद्राश्यादिकं भुक्तं तल्लिप्तापि केन्द्रं कृत्वा तत्त्रयमसंख्याभागहारेण मनुयमला इत्यादौ ज्यार्धे या ज्या भवति सा भुजज्येत्युच्यते। अथ द्वितीये पदे केन्द्रे तदा द्वितीयपदस्य यत्केन्द्रं भुक्तं तेन या ज्या क्रियते सा कोटिज्या तस्य द्वितीयपदस्य यच्छेषं तेन या ज्या क्रियते सा भुजज्या एवं तृतीय पदस्थे केन्द्रे प्रथमपदवत्। चतुर्थपदस्थे केन्द्रे द्वितीयपदवद्भुजकोटिज्याकरणम्। अथ शीघ्रकर्मणि स्वशीघ्रान्मन्दफलस्फुटं ग्रहं विशोध्य केन्द्रं कार्यम्। तस्मान्मन्दकर्म्मवद्भुजकोटिज्याकरणं योज्यमिति। अत्र गोलाध्यायोक्तविधिना कक्षामण्डलादीनि विन्यस्य गोले वासनां प्रदर्शयेत्। तद्यथा प्रतिमण्डलपरिधेर्यत्र प्रदेशे नीचोच्चवृत्तेन सह संपातस्तत्र स्फुटग्रहः। तत्रसूत्रैकस्याग्रं बद्ध्वा स्वोच्चशलाकातः

द्वितीयस्यां दिशि तावत्येव प्रतिमण्डलपरिधेः भागे बध्नीयात्। तदर्धभुजज्या तस्य सूत्रस्य स्वोच्चशलाकया सह यत्र संपातस्तस्य प्रदेशस्य प्रतिमण्डलमध्यस्योच्चान्तरं कोटिज्या तद् दक्षिणोत्तरज्या भुजज्योच्यते । अतएव प्रतिमण्डले प्रथमप्रदेशोच्च-शलाकासंपाताद्यद्ग्रहेण भुक्तं तस्य भुजज्या यच्च पदशेषं तस्य कोटिज्या द्वितीये पदे वाधोगा कोटिज्या, पार्श्वस्था भुजज्या तस्मात्तत्र प्रतिमण्डलार्धवक्रादधिकं यद्भुक्तं तस्य भुजज्या पार्श्वस्था भवति। शेषस्याधोगा कोटिज्या भवति, द्वितीयपद-वासनायां एवं चतुर्थपदात् यच्छेषं तस्य पार्श्वस्था भुक्तस्य कोटिज्यार्धचक्रे स्थिता, प्रथमपदवासनयैव प्रदर्शयेत् ॥१२॥

वि. भा.—मध्यमग्रहान्मन्दोच्चं विशोध्य शेषं मन्दकेन्द्रम्। शीघ्रोच्चान्मध्यम-(मन्दस्पष्ट) ग्रहं विशोध्य शेषं शीघ्रकेन्द्रम्। अयुजि (विषम) पदे गतयेययोः (गतगम्ययोः) चापयोर्बाहुकोटिज्ये (भुजकोटिज्ये) साध्ये, अर्थात् गताद् भुजज्या, गम्यात्कोटिज्या साध्या, युजि (समे) पदेऽतोऽन्यथाऽर्थाद् गम्याद् भुजज्या, गतात्को-टिज्या साध्येति ॥१२॥

अत्रोपपत्तिः

अथ भूकेन्द्रान्मन्दप्रतिवृत्तस्य शीघ्रप्रतिवृत्तस्य च यो हि दूरतरप्रदेशोऽर्थादु-च्चरेखा (वर्धितभूकेन्द्रग्रहगोलकेन्द्रगतरेखा) मन्दप्रतिवृत्तसम्पातस्तथोच्चरेखाशीघ्र-प्रतिवृत्तसम्पातश्च मन्दोच्चं शीघ्रोच्चं चेति, मन्दप्रतिवृत्ते यत्र गणितागतो मध्यमग्रहोऽस्ति भूकेन्द्रात्तद्गता रेखा शीघ्रप्रतिवृत्ते यत्र लगति तत्र मन्दस्पष्टग्रहः। मन्दोच्चान्मध्यमग्रहावधिमन्दप्रतिवृत्ते मन्दकेन्द्रसंज्ञकम्। शीघ्रोच्चान्मन्दस्पष्ट-ग्रहावधिशीघ्रप्रतिवृत्ते शीघ्रकेन्द्रसंज्ञकम्। मध्यमग्रहगतितो मन्दोच्चगतेरल्पत्वात् मध्यमग्र—मन्दोच्च=मन्दकेन्द्रम्, तथा शीघ्रोच्चगतेर्मन्दस्पष्टग्रहगतेरल्पत्वात् शीघ्रोच्च—मन्दस्पग्र=शीघ्रकेन्द्रम्। ग्रहगोलकेन्द्रादुच्चरेखोपरि लम्बरेखा कार्या तदैताभ्यां रेखाभ्यां मन्दप्रतिवृत्तस्य समानि चत्वारि खण्डानि नवत्यंशमितानि भव-न्ति, उच्चरेखातः (उच्चबिन्दुतो वा) सव्यक्रमेण चत्वारि पदानि कल्पानि, प्रथमपदे मन्दप्रतिवृत्ते यत्र ग्रहस्तस्योच्चरेखायाश्चान्तरं भुजज्या। ग्रहात्तिर्यग्रेखो (मन्दगोल-केन्द्रादुच्चरेखोपरिलम्बरेखा) परिलम्बरेखा भुजकोटिज्या, यथा-यथा ग्रहो पदान्ता-भिमुखं गच्छेत्तथा तथा भुजज्योपचीयते, भुजकोटिज्या चापचीयते, पदान्ते भुजज्या परमा त्रिज्या समा भुजकोटिज्यायाश्चाभावः ततोऽग्रे द्वितीयपदे ग्रहो यथा-यथाऽग्रतो गच्छति तथा-तथा भुजज्याऽपचीयते, भुजकोटिज्योपचीयते द्वितीयपदान्ते भुजज्याऽ-भावः, भुजकोटिज्या च परमा त्रिज्या समा भवत्यतोऽयुजि गतयेययोर्युजि पदेऽन्यथा बाहुकोटिज्ये इत्याऽचार्योक्तं युक्तियुक्तं संगच्छते तृतीये चतुर्थे पदेऽप्येवमेव विचारः कार्यः। सूर्यसिद्धान्ते 'ग्रहं संशोध्य मन्दोच्चात्तथा शीघ्राद्विशोध्य चेत्यादिना, मध्यम-

मन्दोच्चाद्विशोध्य शेषं मन्दकेन्द्रं कथ्यते भगवता सूर्यसिद्धान्तेन, अन्यत्सर्वमाचार्योक्तवदेवास्ति सिद्धान्तशेखरे श्रीपतिना सिद्धान्तशिरोमणौ भास्कराचार्येण चाऽऽचार्योक्तमेव कथ्यत इति ॥१२॥

अब मन्दकेन्द्र और शीघ्रकेन्द्र की तथा केन्द्रज्या-और
केन्द्र कोटिज्या की परिभाषा कहते हैं

हि. भा.—मध्यमग्रह में मन्दोच्च को घटाने से शेष मन्दकेन्द्र होता है, शीघ्रोच्च में मन्द स्पष्ट ग्रह को घटाने से शेष शीघ्र केन्द्र होता है, विषम पद में गत चाप की ज्या भुजज्या कहलाती है, गम्य (एष्य) चाप की ज्या कोटिज्या कहलाती है, सम पद में इसके विपरीत अर्थात् गम्यचापज्या भुजज्या और गतचापज्या कोटिज्या होती है इति ॥१२॥

उपपत्ति

भूकेन्द्र से मन्द प्रतिवृत्त और शीघ्र प्रतिवृत्त का जो दूरतर प्रदेश (अर्थात् उच्चरेखा और वर्धित भूकेन्द्र ग्रहगोलकेन्द्रगतरेखा के योग बिन्दु) हैं वे क्रमशः मन्दोच्च और शीघ्रोच्च कहलाते हैं, मन्द प्रतिवृत्त में गणितागत मध्यम ग्रह जहाँ है भूकेन्द्र से तद्गत रेखा शीघ्र-प्रतिवृत्त में जहाँ लगती है वही मन्दस्पष्ट ग्रह है, मन्दोच्च से मध्यमग्रहपर्यन्त मन्द प्रतिवृत्तीय चाप मन्दकेन्द्र है, शीघ्रोच्च से मन्दस्पष्टग्रहपर्यन्त शीघ्र प्रतिवृत्तीय चाप शीघ्र-केन्द्र है, मन्दोच्च की गति मध्यमग्रह गति से अल्प होने के कारण मध्यमग्र—मन्दोच्च= मन्दकेन्द्र, तथा मन्दस्पष्टग्रह की गति से शीघ्रोच्च गति के अधिक होने के कारण शीघ्रोच्च —मन्दस्पष्टग्र=शीघ्रकेन्द्र, ग्रहगोल केन्द्र से उच्चरेखा के ऊपर लम्बरेखा करने से उच्च-रेखा और लम्ब रेखा से मन्द प्रतिवृत्त के समान चार खण्ड (नवत्यंश के बराबर) होते हैं, मन्दोच्च से सव्य क्रम से चार पद कल्पना करना।

प्रथम पद में मन्द प्रतिवृत्त में जहाँ ग्रह है उसका और उच्चरेखा का लम्बरूप अन्तर भुजज्या है, ग्रह से तिर्यक् रेखा (मन्दगोल केन्द्र से उच्चरेखा के ऊपर लम्ब रेखा) के ऊपर लम्बरेखा कोटिज्या है, ग्रह ज्यों-ज्यों पदान्ताभिमुख जाते हैं त्यों-त्यों भुजज्या उपचीयमान होती है और कोटिज्या अपचीयमान होती है, पदान्त में भुजज्या परम (त्रिज्यातुल्य) होती है और कोटिज्या का अभाव होता है, उससे आगे द्वितीय पद में ग्रह ज्यों-ज्यों आगे जाते हैं त्यों-त्यों भुजज्या अपचीयमान होती है और कोटिज्या उपचीयमान होती है, पदान्त में भुजज्या का अभाव होता है और कोटिज्या परम (त्रिज्या तुल्य) होती है इसलिए "अयुजि गतयेययोर्युजिपदेऽन्यथा बाहुकोटिज्ये" यह आचार्योक्त युक्तियुक्त है, इसी तरह तृतीय पद और चतुर्थ पद में भी विचार करना चाहिए, सूर्य सिद्धान्त में 'ग्रहं संशोध्य मन्दोच्चात्तथा शीघ्राद् विशोध्य च' इत्यादि से मन्दोच्च ही में मध्यम ग्रह को घटाकर शेष को मन्द केन्द्र कहते हैं, और सब कुछ आचार्योक्तवत् ही है, सिद्धान्तशेखर में श्रीपति और सिद्धान्तशिरोमणि में भास्कराचार्य भी आचार्योक्त ही को कहते हैं इति ॥१२॥

इदानीं स्पष्टपरिध्यानयनमाह

त्रिज्याहृता भुजज्या युगयुक्परिधिद्वयान्तरगुणाप्त्या ।
युग्मान्तपरिधिरधिको हीने हीनोऽधिके स्पष्टः ॥१३॥

वा. भा.—इदानीं शुक्रमन्दशीघ्रकर्मणोः परिधिसंस्कारार्थमार्यामाह । मन्दोच्चनीचवृत्तस्य परिधिभागाः सितस्य विषमान्ते नवचेत्यादिना ये परिधयः पठिताः मन्दपरिधिप्रथमपदान्ते तृतीयपदान्ते च नवरसयमाः, द्वितीयचतुर्थपदान्तयोरष्टभागाः द्वितीयपदान्ते चतुर्थपदान्ते चैकादशभागाः शीघ्रपरिधिप्रथमतृतीयपदान्तयोरग्निरसयमा भागा । एवं स्थिते चाभ्यन्तरे परिधेः साधनायेदमार्यासूत्रं त्रैराशिकात्मकं तद्यथा त्रिज्याहृता कासौ भुजज्या किं भूता, युगयुक्परिधिद्वयान्तरगुणा च शीघ्रकर्मणोरपि ततोऽवाप्तिः तया वाप्त्याधिको युग्मान्तरपरिधिः कर्त्तव्यः । यदि विषमान्तरपरिधेरूनः । अथ युग्मान्तरपरिधिरधिकस्तदा भागावाप्त्या हीनः कार्यः स मन्दनीचोच्चवृत्तस्य शीघ्रनीचोच्चवृत्तस्य च परिधिः स्फुटो भवति तेन वक्ष्यमाणविधिना फलानयनम् । अत्र मन्दप्रतिमण्डलोच्चप्रदेशे नीचोच्चवृत्तस्यैकादशकः परिधिः तत्क्रमेणोपचीयते । यावत्प्रथमं पदान्तं तत्र नवपरिधिः । तथा तत्र फलोपलब्धैस्ततः परं पुनरप्यपचीयते । यावत्प्रतिमण्डलार्धचक्रं तत्रैकादशको मन्दनीचोच्चपरिधिरुच्चप्रदेशवत्तत्रापि तथैव फलोपलब्धेः प्रथमद्वितीयपदयोस्त्रिचतुर्थपदे व्याख्याते । अथवान्तरे त्रैराशिकं यदि भुजज्या त्रिज्या यत्र तत्र परिध्यन्तरं द्वौ यत्र चेष्टौ भुजज्या तत्र किमिति लब्धेन युग्मान्तपरिधिरपचीयते । क्रमेण विषमान्तेन भविष्यन्ति । एवं शेषपदेषु मन्दकर्मणि योज्यं । शीघ्रकर्मणि परिधिसंस्कारो विपरीतः वासनावैपरीत्यात् तत्र यदि भुजज्यया त्रिज्यया तुल्यया पंचभागाः परिधिरन्तरं तदेष्टभुजज्यया किमिति फलेनोपचीयते । युग्मान्तरं परिधिर्यत्रोपचयक्रमेण विषमान्तेऽग्निरसयमलसंख्यो भविष्यति, तथा स्थिते एतत् गोले छेद्यके वा प्रदर्शयेत् । एवं शुक्रस्य मन्दनीचोच्चवृत्तशीघ्रनीचोच्चवृत्तयो परिधी संस्कार्यौ बुधगुरुशनीनां वक्ष्यमाणं यथा पठिता एवं परिधयः स्फुटाः मन्दशीघ्रयोरपि कुजस्यापि मन्दपरिधिर्यथा पठिता तच्छीघ्रपरिधेरपि संस्कारं वक्ष्यति शीघ्रस्फुटपरिधिनाप्तभागोना वेदजिनाः अंशोना इति स तथा एव स्फुटो भवति । इदानीं इष्टस्फुटः परिधिनाप्ता भागोना वेदजिना अंशौ, पूर्वानीतभुजकोटिभ्यां च भूमध्यप्रतिमण्डलस्थपारमार्थिक ग्रहान्तरकरणभुजकोटिज्ये क्रियेते तत्करणमार्यायाह ।

वि. भा.—भुजज्या (केन्द्रज्या) युगयुक्परिधिद्वयान्तरगुणा (समविषमपदान्तपरिध्योरन्तरेण गुणा) त्रिज्याहृता (त्रिज्याभक्ता) आप्त्या (लब्ध्या) युग्मान्तपरिधिः (समपदान्तपरिधिः) अधिकः कार्यो विषमपदान्तपरिधेः समपदान्तपरिधिमानेऽल्पे सति अधिके (विषमपदान्तपरिधेः समपदान्तपरिधिमानेऽधिके) लब्ध्या समपदान्तपरिधिर्हीनः कार्यस्तदा स्पष्टः परिधिर्भवेदिति ॥१३॥

अत्रोपपत्तिः

यदि त्रिज्यातुल्यया केन्द्रभुजज्यया समविषमपदान्तपरिध्योरन्तरं लभ्यते तदेष्टकेन्द्रभुजज्यया किमित्यनुपातेनागतं फलं (इष्टपरिध्यन्तरं) विषमपदान्तपरिधेः समपदान्तपरिधिमानस्याल्पत्वे समपदान्तीयपरिधिमाने योज्यं, विषमपदान्तपरिधेः समपदान्तपरिधिमानस्याधिक्ये समपदान्तीयपरिधिमाने वियोज्यं तदेष्टपरिधिः स्पष्टो भवेत् । सूर्यसिद्धान्तकारेण 'ओजयुग्मान्तरगुणा भुजज्या त्रिज्ययोद्धृतेत्यादिना' श्रीपतिना 'ओजयुक्परिधिजान्तरनिघ्नी दोर्ज्यकेत्यादिना' प्याचार्योक्तानुरूपमेव कथ्यत इति ॥१३॥

अब स्पष्टपरिध्यानयन को कहते हैं

हि. भा.—केन्द्रज्या को समपदान्तीय परिधि और विषमपदान्तीय परिधि के अन्तर से गुणाकर त्रिज्या से भाग देने से जो फल होता है उसको समपदान्तीय परिधि में जोड़ देना (यदि विषम पदान्त परिधि से समपदान्त परिधि अल्प हो तब), यदि विषम पदान्त परिधि से समपदान्त परिधि अधिक हो तब घटा देने से स्पष्ट परिधि होती है इति ॥१३॥

उपपत्तिः

यदि त्रिज्या तुल्य केन्द्र भुजज्या में सम-विषम पदान्तीय परिध्यन्तर पाते हैं तो इष्ट-केन्द्र भुजज्या में क्या इस अनुपात से जो फल आता है उसको विषम-पदान्त परिधि से सम-पदान्त परिधि के अल्प रहने से समपदान्तपरिधि में जोड़ने से इष्ट स्थानीय परिधि (स्पष्टपरिधि) होती है, विषम पदान्त परिधि से समपदान्त परिधि के अधिक रहने से समपदान्त परिधि में घटाने से स्पष्ट परिधि होती है । सूर्य सिद्धान्तकार 'ओजयुग्मान्तरगुणा भुजज्या त्रिज्ययोद्धृता इत्यादि से' तथा श्रीपति 'ओजयुक् परिधिजान्तरनिघ्नी' इत्यादि से आचार्योक्तानुसार ही कहते हैं । इति ॥१३॥

इदानीं भुजफलकोटिफलसाधनं स्पष्टां कोटिञ्चाह

तद्गुणिते ज्ये भांशैर्हृते फले कोटिफलयुता त्रिज्या ।
आद्यन्तयोर्विहीना पदयोर्द्वितृतीययोः कोटिः ॥१४॥

वा. भा.—तदितिनीचोच्चवृत्तस्फुटपरिधेः परामर्शः, स्वमन्दस्फुटपरिधिना स्वशीघ्रस्फुटपरिधिना वा गुणिते ये भुजकोटिज्ये, इत्यर्थः । भांशैर्हृते षष्टिशतत्रय-भक्ते फले यथा भुजकोटिज्ययोर्भवतः एतदुक्तं भवति, यानन्तरानीता भुजज्या मन्द-कर्मणि तां मन्दपरिधिना गुणयेत् । शीघ्रकर्मणि शीघ्रपरिधिना, तत् षष्टिशतत्रयेण विभजेत् । ततो लब्धं यत्फलं तत्कोटिफलमुच्यते । ततः कोटिफलयुतत्रिज्या कर्त्तव्या एवं कृता त्रिज्या प्रतिमण्डलप्राप्तिकर्णस्य कोटिर्भवति । भुजफलमेव भुजज्या । अत्र वासना पूर्वन्यस्तेषु कक्षाप्रतिमण्डलादिषु प्रदर्श्या तद्यथा यत्र प्रदेशे कक्षामण्डले मध्यमो

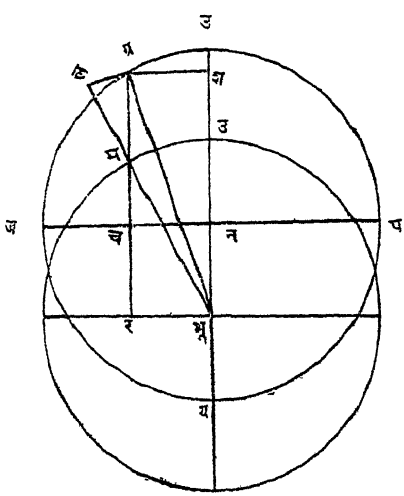
ग्रहस्तत्र नीचोच्चवृत्तमध्यं विन्यस्य शीघ्रनीचोच्चवृत्तस्य मध्यचक्रस्फुटगतप्रदेशे कक्षामण्डले विन्यस्य गोलविधिना ततस्त्रैराशिकं यदि भागषष्टिशतत्रयेण परिधेरेतावती भुजज्या तदस्य नीचोच्चवृत्तभागपरिधेः कियतीति फलं नीचोच्चवृत्तस्य भुजो भवति। यद्भुजफलमुच्यते। अतः पुनरपि यदि षष्टिशतत्रयपरिधेरेतावती कोटिज्या तत् नीचोच्चवृत्तपरिधेः कियतीति फलं नीचोच्चवृत्तस्य कोटिः यत्कोटिफलमभिधीयते भवद्भिः नीचोच्चवृत्तमध्यञ्च सर्वदा भूमध्यात् व्यासार्धतुल्यान्तरे स्थितः। यतस्तत्कक्षामण्डलं न त्यजति। नीचोच्चवृत्तकोटिश्च प्रथमचतुष्टयो केन्द्रपदयोरुपरि भवति। द्वितृतीययोरधस्थस्तन्मध्यादत आद्यन्तयोः केन्द्रपदयोर्नीचोच्चवृत्तकोट्या व्यासार्धमधिकं प्रतिमण्डलप्राप्तिकरणास्य कोटिर्भवति यतः कक्षामण्डलादुपरिग्रहस्तदास्थितो द्वितृतीययोः नूनं नीचोच्चवृत्तकोट्यैव व्यासार्धप्रतिमण्डलप्रापिकरणस्य कोटिर्भवति यस्मात्कक्षामण्डलादधो वर्त्तते गृहीतभुजज्यातो पुनः नीचोच्चवृत्तभुजज्ये वा यतो नीचवृत्तशलाकाग्रहयोस्तदेवान्तरमतएव तदप्युपपन्नमिति ॥१४॥

वि. भा.—ज्ये (केन्द्रभुजकोटिज्ये) तद्गुणिते (तेन स्पष्टपरिधिना गुणिते) भांशै ३६० र्भक्ते तदा फले (भुजफलकोटिफले) भवतः। आद्यन्तयोः पदयोः (प्रथमचतुर्थयोः पदयोरर्थान्मकरादिकेन्द्रे) त्रिज्या कोटिफलेन युता कार्या, द्वितृतीययोः पदयोः (कर्क्यादिकेन्द्रे) त्रिज्याकोटिफलेन विहीना कार्या तदा कोटिः (स्पष्टा कोटिः) भवेदिति ॥१४॥

अत्रोपपत्तिः

भू=भूकेन्द्रम्। भूबिन्दुतस्त्रिज्याव्यासार्धेन वृत्तं कार्यं तत्कक्षावृत्तम्। न बिन्दुतोऽपि तत्त्रिज्या व्यासार्धेनैव वृत्तं कार्यं तन्मन्दप्रतिवृत्तम्। न=मन्दगोलकेन्द्रम्। भून=मन्दान्त्यफलज्या, ऊर्ध्वाधरतिर्यग्रेखे कार्ये, उ=प्रतिवृत्ते उच्चस्थानम्। उ=कक्षावृत्ते उच्चस्थानम्। ग्र=मन्दप्रतिवृत्ते मध्यमग्रहः। म=कक्षावृत्ते मध्यम ग्रहः। वर्धितभूमरेखोपरि ग्रबिन्दुतो लम्बः=ग्रल=भुजफलम् मल=कोटिफलम्। ग्रम=मन्दान्त्यफलज्या=रच=भून, भूम=त्रिज्या, ग्रश=मन्दकेन्द्रज्या, ग्रच=मन्दकेन्द्रकोटिज्या,=मर तदा भूमर, मग्रल

(क)



त्रिभुजयोः सजात्यादनुपातः $=\frac{\text{भूर}\times\text{ग्रम}}{\text{भूम}}=\frac{\text{मन्दकेन्द्रज्या}\times\text{मन्दान्त्यफलज्या}}{\text{त्रि}}=$

ग्रल = मन्दभुजफलम् $=\frac{\text{मन्दकेज्या}\times\text{मन्दपरिधि}}{३६०}$

यतः $\frac{\text{मन्दान्त्यफलज्या}}{\text{त्रि}}=\frac{\text{मन्दपरिधि}}{३६०}$ म बिन्दुं केन्द्रं मत्वा मग्र. मन्दान्त्यफलज्या-

व्यासार्धेन यद् वृत्तं तन्मन्दनीचोच्चवृत्तं स एव मन्दपरिधिः तथा $\frac{\text{मर}\times\text{ग्रम}}{\text{भूम}}=\text{लम}=$

$\frac{\text{मन्दकेकोज्या}\times\text{मन्दान्त्यफलज्या}}{\text{त्रि}}=\frac{\text{मन्दकेकोज्या}\times\text{मन्दपरिधि}}{३६०}=$

मन्दकोटिफलम् । एवं शीघ्रकेन्द्रज्यादिवशेन शीघ्रभुजफलकोटिफले भवतः। उज=प्रथमपदम्, जय=द्वितीयपदम् । यप=तृतीयपदम् । उप=चतुर्थपदम् । जउप=मकरादिकेन्द्रम् । जयप=कर्क्यादिकेन्द्रम् । भूम+लम=त्रि÷मन्दकोटिफल =भूल=मन्दनीचोच्चवृत्तीयस्पष्टा कोटिः। एवं चतुर्थपदेऽपि भवति, द्वितीयतृतीय-पदयोः त्रि—मन्दकोटिफ=मन्दनीचोच्चवृत्तीय स्पष्टा कोटिरिति भुजफलकोटिफल-स्वरूपविन्यासेनैव तत्र स्फुटं भवत्येतावताऽऽचार्योक्तं सम्यगुपपद्यत इति ॥१४॥

अब भुजफल और कोटिफल के साधन तथा स्पष्टा कोटि को कहते हैं

हि. भा.—केन्द्रज्या और केन्द्र कोटिज्या को स्पष्ट परिधि से गुणा कर भांश ३६० से भाग देने से भुजफल और कोटिफल होता है। प्रथम पद और चतुर्थ पद (मकरादि केन्द्र) में त्रिज्या में कोटिफल को जोड़ने से स्पष्टा कोटि होती है, द्वितीय पद और तृतीय पद (कर्क्यादि केन्द्र) में त्रिज्या में कोटिफल को घटाने से स्पष्टा कोटि होती है। यहां मन्द-केन्द्रज्या, मन्दकेन्द्र कोटिज्या, और मन्द परिधि के सम्बन्ध से मन्दभुजफल और मन्द कोटि-फल होता है, तथा शीघ्रकेन्द्रज्या, शीघ्रकेन्द्र कोटिज्या और शीघ्रपरिधि के सम्बन्ध से शीघ्र भुजफल और शीघ्रकोटिफल होता है इति ॥१४॥

उपपत्ति

यहाँ संस्कृतोपपत्ति में लिखित (क) क्षेत्र को देखिये। भू=भूकेन्द्र, भू बिन्दु से त्रिज्या व्यासार्ध से वृत्त कक्षावृत्त होता है, न=ग्रहगोलकेन्द्र, न बिन्दु से उसी त्रिज्या व्यासार्ध से जो वृत्त होता है वह प्रतिवृत्त है, भून,=अन्त्यफलज्या भू, न बिन्दुओं से ऊर्ध्वाधर रेखा और तिर्यग्रेखायें कीजिये। उ=प्रतिवृत्त में उच्चस्थान। उ=कक्षावृत्त में उच्चस्थान । ग्र= प्रतिवृत्त में मध्यमग्रह । म=कक्षावृत्त में मध्यमग्रह । वर्धित भूम रेखा के ऊपर ग्र बिन्दु से लम्ब=ग्रल=भुजफल, मल=कोटिफल, ग्रम=अन्त्यफलज्या,=रच=भून, भूम=त्रिज्या, ग्रभ=केन्द्रज्या=भूर, ग्रच=केन्द्रकोटिज्या=मर भूमर, मग्रल दोनों त्रिभुज सजातीय हैं

इसलिए अनुपात करते हैं $\frac{\text{भूर}\times\text{ग्रम}}{\text{भूम}}=\frac{\text{केन्द्रज्या}\times\text{अन्त्यफज्या}}{\text{त्रि}}=\text{ग्रल}=\text{भुजफल}$

$=\frac{\text{केन्द्रज्या}\times\text{परिधि}}{३६०}$ क्योंकि $\frac{\text{अन्त्यफलज्या}}{\text{त्रि}}=\frac{\text{परिधि}}{३६०}$, म बिन्दु को केन्द्र मानकर 'मग्र' अन्त्य-फलज्या व्यासार्ध से जो वृत्त बनता है वही नीचोच्चवृत्त है उसी को स्पष्ट परिधि कहते हैं।

तथा $\frac{\text{मर}\times\text{ग्रम}}{\text{भूम}}=\frac{\text{केन्द्रकोज्या}\times\text{अन्त्यफज्या}}{\text{त्रि}}=\text{लम}=\frac{\text{केन्द्रकोज्या}\times\text{परिधि}}{३६०}=\text{कोटिफल}$, उज =प्रथमपद, जय=द्वितीयपद, यप=तृतीयपद, उप=चतुर्थपद, जउप=मकरादिकेन्द्र, जयप =कर्क्यादिकेन्द्र, भूम+लम=भूल=त्रि+कोटिफल=नीचोच्चवृत्तीय स्पष्टा कोटि, इसी तरह चतुर्थ पद में भी होता है, द्वितीय और तृतीय पद में त्रि—कोटिफल=नीचोच्चवृत्तीय स्पष्टा कोटि यह भुजफल और कोटिफल फलस्वरूप के विन्यास ही से स्पष्ट होता है, इससे आचार्योक्त उपपन्न होता है, सिद्धान्तशिरोमणि में भास्कराचार्य 'स्वेनाहते परिधिना' इत्यादि से, तथा त्रिज्या तथा कोटिफलेन युक्ता हीना च इससे इसी विषय को कहते हैं। सिद्धान्तशेखर में भी 'स्फुटनिजपरिणाहघ्नुण्ण दोःकोटिजीवे' इत्यादि से आचार्योक्त के अनुरूप ही कहा गया है इति ॥१४॥

इदानीं त्रिभिः श्लोकैः कर्णानयनं प्रकारान्तरेण भुजफलानयनं पूर्वानीत-फलस्यर्णत्वं धनत्वं, मन्दफलं शीघ्रफलंञ्च कथयति

तद्भुजफलकृतियोगान्मूलं कर्णः पदेष्वयुग्युक्षु ।
स्वपरिधिगुणा क्रमोत्क्रमजीवा भांशैर्हृता मन्दे ॥१५॥

क्षयधनधनक्षयास्तत्फलानि शीघ्रेऽन्यथा धनं धनयोः ।
ऋणमृणयोर्योगोऽन्तरमृणधनयोस्तुल्ययोः शून्यम् ॥१६॥

तच्चापं मन्दफलं फलयोगान्तरवशात् धनमृणं वा ।
शीघ्रफलं तद्गुणितात् व्यासार्धात्कर्णलब्धधनुः ॥ १७ ॥

वा. भा.—(१५, १६, १७ श्लोकानां कृते) इदानीं प्रतिमण्डलप्रापिकर्णस्यानयनमार्यायाह तदित्यनेन स्फुटकोटिभुजज्यातः पुनर्नीचोच्चवृत्तभुजज्येव। यतो नीच-उच्चशलाकाग्रहयोस्तदेवांतरमत एतदप्युपपन्नमिति। भुजफलञ्च नीचोच्चवृत्त-भुजज्योच्यते। तेनायमर्थः स्फुटकोटेः कृतिः कार्या, भुजफलस्य च तयोर्योगस्तस्मात्फलं यत्तद् भूमध्यप्रतिमण्डलस्थग्रहान्तरं कर्णः, स एवोच्यते, भुजकोटिकृतियोगपदं भवतीति किमत्रोच्यते यतः कोटिज्यातुल्यं फलं समचतुरस्रक्षेत्रस्य यत्फलं यच्च बाहुज्यातुल्यस्य समचतुरस्रस्य तयोर्फलयोरैक्यं। यत्तत्तुल्यफलं कर्णतुल्य-समचतुरस्रक्षेत्रस्य भवत्यत उपपन्नं पदेष्वयुग्युक्ष्विति। उत्तरोत्तरं सम्बद्धं भविष्यति। **वासना चात्र** यथा गोले स्थिते छेद्यके च प्रदर्शयेत्। तद्यथा नीचोच्चवृत्तशलाकानु-

सारेण कोटिज्या सर्वत्र भवति। कोट्यग्रग्रहान्तरभुजफलतुल्यभुजज्यया ग्रहाश्च भूमध्यं यावत्कर्णे इति।

अत्र केन्द्रे प्रथमतृतीये अष्टकपदे द्वितुर्ये युक्पदे तेषु तेषु अयुग्युक्पदेषु स्वपरिधिगुणा के त्यादिनाह क्रमोत्क्रमजीवा यथासंख्यं विषमसमपदेष्टभांशैर्हृताः षष्टिशतत्रयभक्ताः सन्तो मंदे मन्दकर्मणि क्षयधनधनक्षयाः भवन्ति। तासां फलानि तत्फलानि तानि शीघ्रेऽन्यथेति। शीघ्रकर्मणि तु पुनः अन्यथा वैपरीत्येन भवति। धनक्षयक्षयधनानीति यावती ततो धनं भवति। धनयोर्यथासंभवमृणमृणयोश्च तयोरन्तरमृणधनयोः कार्यम्। यदधिकं तत् ग्राह्यमित्यर्थः यत्फलं भवति तुल्यपक्षे तयोः धनयोः तुल्यफलाभावो भवतीत्यर्थः तस्यैवंविधस्य भुजफलस्य चापं तच्चापं मन्दफलं भवति। शीघ्रफलं तु पुनः तद्गुणितव्यासार्धलब्धधनुः तेन भुजफलेन व्यासार्धं निहत्य स्फुटकर्णेन विभजेत्। यतो यल्लब्धं तच्चापं कृत्वा शीघ्रफलं भवतीत्यर्थः।

तस्यैवंविधस्य भुजफलस्य एतदुक्तं भवति। मन्दकर्मणि शीघ्रकर्मणि वा यदि केन्द्रं प्रथमे पदे भवति, केन्द्रेण यद्भुक्तं तस्य क्रमज्या ग्राह्या सा भुजज्या भवति, तां स्वपरिधिहतां भांशैर्विभज्य ग्रहस्य भुजफलं भवति। द्वितीयपदे पुनर्यदि द्वितीयपदस्य भुक्तं तस्योत्क्रमज्या ग्राह्याः ततः प्रथमपदे परमभुजज्यां त्रिज्यातुल्यां ग्रहस्वमन्दपरिधिना स्वशीघ्रपरिधिना वा सगुणषष्टिशतत्रयेण विभजेत्। लब्धं ग्रहस्य परमं भुजफलम्। ततो द्वितीयपदात् क्रमज्यां स्वपरिधिना निहत्य भांशैर्विभज्य यल्लब्धं तद्ग्रहस्य परमं फलं ज्यातो विशोध्य शेषं ग्रहस्य भुजफलं भवति। यस्मादुक्तमन्तरमृणधनयोरिति, अथ तृतीये पदे केन्द्रं तदा पदभुक्तस्य क्रमज्यां कृत्वा स्वपरिधिना संगुण्य भांशैर्विभज्य लब्धं द्वितीयपदोत्पन्नपरमज्याफले योजयेत्। यत उक्तं धनं धनयोग इति। ततस्तस्माद्योगात्प्रथमपदोत्पन्नं परमभुजफलं विशोध्य शोधग्रहस्य भुजफलं भवति। अथ चतुर्थे पदे केन्द्रं भवति। तदा परमभुक्तस्योत्क्रमज्या स्वपरिधिहता भांशैर्विभजेत्। यल्लब्धं तत् प्रथमपदोत्पन्नग्रहपरमभुजज्याफले संयोजयेत्। यत उक्तमृणमृणयोर्योग इति, द्वितीयपदोत्पन्नयोः परमभुजफलयोर्धनात्मकयोर्योगस्तस्माद्गुणयोगं विशोध्य ग्रहस्य भुजफलं भवति। मन्दकर्मणि प्रथमो ज्योत्पन्नं भुजफलं क्षयो भवति। द्वितीयपदे उत्क्रमज्योत्पन्नं धनं भवति। तृतीयपदे क्रमज्योत्पन्नं धनमेव, चतुर्थपदे उत्क्रमज्योत्पन्नमृणम्भवति तत्फलानि शीघ्रमन्यथेति प्रथमपदे धनं द्वितीये क्षयं, तृतीये क्षयं, चतुर्थे धनमित्यर्थः। अत्र धनर्णविवक्षायां प्रतिमण्डलपदानि गृह्यन्ते। प्रथमं पदं राशित्रयस्य स्वपरमफलाधिकं तदेवार्धचक्राद्विशोध्य शेषं द्वितीयपदप्रमाणं प्रथमप्रमाणं चतुर्थपदं। द्वितीयप्रमाणं तृतीयपदं यस्मात्स्फुटगत्युत्तरे वक्ष्यत्याचार्यः एवं प्रतिमण्डलपदमाद्यं गृहत्रयं सान्यधनुरतोन्यादित्यादिना ग्रन्थेन तत उक्तवद्धनं धनयोः ऋणं ऋणयोर्योगोत्तरमृणधनयोरिति कृत्वा ग्रहभुजफलं कार्यम्। तुल्ययोस्तु पुनस्तयोः शून्यं भवतीति किमत्रोच्यते। एवं यथासम्भवं भुजफलमुत्पाद्य तस्य चापं

कृत्वा तन्मदकर्मणि फलं भवति । तस्य फलयोगान्तरवशाद्यदधिकं धनमृणं वा तद् भवतीत्यर्थः । शीघ्रकर्मणि तु पुनस्तद्गुणितात् व्यासार्धात्तेन भुजफलेन व्यासार्धं ताडयित्वा स्वकर्णेन यल्लभ्यते, तस्य चापं फलं भवति । तदपि फलयोगान्तरवशादेव धनमृणं वा कार्यम् । इतिवाक्यशेषः अत्र वासना तद्यथोक्तवत् । कक्षामण्डलादीनि च विन्यस्य केन्द्रं यदा शून्यं भवति तदोच्चरेखायामेव ग्रह स्थितो भवति । तत्र भुजाभावः ततः प्रतिमण्डलोच्चरेखातः क्रमेण ग्रहे विप्रकृष्टते प्रतिमंडलपरिधिं कृत्वा यत्र ग्रह स्थितो भवति तस्य प्रदेशस्योच्चरेखायाश्च यदन्तरं स भुजः असौ प्रथमपदे भुक्तस्य क्रमज्या भवति । तावदुपचीयते यावत्पदांतं तत्र त्रिज्या तुल्या भवति । ततो द्वितीयपदे भुक्तस्य क्रमज्या तावत् ग्रहेण भुक्तं तस्योत्क्रमज्या । तयापचिता त्रिज्या भुजज्या भवति । तस्मात् पुनारेखायाः सन्निकर्षो ग्रहो भवति । उत्क्रमज्याक्रमेण तावद्यावदर्धचक्रं तत्र नीचरेखायामेव ग्रहस्तस्मात्तयापि भुजज्या भवति। तृतीयपदे प्रथमपदवत् विप्रकर्षो ग्रहस्य योज्यश्चतुर्थपदे द्वितीयपदवत् सन्निकर्षो योज्यः, एवं व्यासार्धतुल्ये प्रतिमण्डले स्वनीचोच्चवृत्तेष्येकमतस्त्रैराशिकेन तद् भुजज्यानयनं चतुर्षु पदेषु, प्रदर्शितमाचार्येण, पदेष्वयुग्युक्स्वपरिधिगुणाः क्रमोत्क्रमजीवा भांशैः हृता इति तद्यथा भांशः षष्टिशतत्रयपरिणाहस्येयं भुजज्या,तत्स्वोच्चनीचपरिणाहस्य कियतीति फलं स्वोच्चनीचोच्चवृत्तभुजज्यया,प्रथमतृतीयपदयोः द्विचतुर्थपदयोस्तु पुनरनेनैव त्रैराशिकेन नीचोच्चवृत्तभुजज्याज्ञापनाय सिद्धिः ततः क्षयधनकल्पना मन्दोच्चरेखातः कक्षामंडले यावति प्रदेशे राशिभागादिकमध्ये ग्रहस्तत्र मन्दनीचोच्चवृत्तमध्यं विन्यसेत्ततो मन्दोच्चरेखातो यावति प्रदेशकक्षामंडले ततएव प्रतिमण्डलनीचोच्चवृत्तपरिध्योः संपातस्तत्र पारमार्थिको ग्रहस्तस्मात् भूमध्यं यावत्सूत्रं नीयते तत्कक्षामण्डलस्थमध्यग्रहात् पश्चिमे नयति यत्र सूत्रेण सह कक्षामण्डलस्य संपातस्तत्रस्थितं ग्रहं भूस्थो द्रष्टा पश्यति । अतस्तदनन्तरफलेन नापचीयमानः प्रतिमण्डलस्योपरिस्थितत्वान्मन्दफलं प्रथमे प्रतिमण्डलप्रदेशे क्षयो भवतीत्युपपन्नम् । द्वितीयपदे पुनरधः प्रतिमण्डलस्योपरिस्थितत्वादुपरि कक्षामण्डलं तत्रानेनोपचयविधिना यत्सूत्रं नीयते तत्प्रथमपदोत्पन्नस्यापचयस्यापचयमुत्क्रमेण धनमुदेति अतो यावदेवापचयस्यापचयस्तावदेव धनमुच्यते। द्वितीये प्रतिमण्डलपदेपि तदप्युपपन्नं तृतीयपदेऽपि कक्षामण्डलस्योपरिस्थितत्वाद्यावद् भूमध्याद्यत्सूत्रं पूर्ववत् प्रतिमण्डलस्थग्रहमध्येन कक्षामण्डलेन नीयते । तावत् कक्षामण्डले मध्यग्रहच्छिन्नतप्रदेशात्तत्पूर्वेण भवति । तत्रस्थं ग्रहं पश्यति भूस्थो द्रष्टा ततस्तदनंतरफलेनोपचीयते ग्रहः सोऽत्रापि उपपन्नः यथा द्वितीयपदे क्रमेणापचितमपचयफलम् । एवं चतुर्थपदोत्क्रमेणापचीयते धनफलमतस्तदपचयो यावानुत्क्रमेण तावद्दृणं चतुर्थपद इत्युच्यते तदप्युपपन्नं प्रतिमण्डलस्योपरिस्थितत्वादेवं स्थितानां मन्दफलानां यथासम्भवं योगान्तरं वा ग्रहफलं भवति एवं मन्दनीचोच्चवृत्तभुजज्येव ग्रहफलं । येन कारणेन स्फुटकोटिकर्णेन कृते स्फुटकर्णोत्पन्नभुजकोटिफलस्य च कक्षामण्डलपरिणाहाय त्रैराशिकं न कृतं तत्र गोलाध्याये कारणमाचार्येण प्रदर्शितम् । त्रिज्याभक्तकर्ण इत्यादिकया तया अस्माभिरुपरि तत्रैव परिहृतमिति शीघ्रकर्मणि धनक्षयोत्पत्तिः स्वशीघ्रप्रतिमण्डले स्वरेखातः

कक्षामण्डले यावति प्रदेशे पश्चादवलम्बितो मन्दस्फुटो ग्रहस्तत्तःशीघ्रनीचोच्चवृत्तं विधाय प्रदर्शयेत्। तत उच्चरेखातो मध्यमग्रहो यावत्कक्षामण्डले यावतो राशिभागादयस्तावन्तएव प्रतिमण्डलशीघ्रनीचोच्चवृत्तयोः संपातं यावत्प्रतिमण्डलोच्चरेखा तस्य एव तत्र शीघ्रफलस्फुटो ग्रहस्तस्मात् भूमध्यं प्रति यत्सूत्रं नीयते तत्र कक्षामण्डलस्थमन्दस्फुटग्रहात् पूर्वेण नीयात्। अतस्तदन्तरेणोपचीयते मन्दस्फुटो ग्रहः स कक्षामण्डले स्फुटो भवति तस्मादुपपन्नम्। प्रथमपदे धनमथद्वितीयपदे प्रथमपदोत्पन्नं धनं मुक्तं ज्याक्रमेणापचीयते स एवापचयो द्वितीये पदे भवतीत्युपपन्नम्। कक्षामण्डलस्योपरिस्थितत्वात् मन्दोच्चग्रहस्य स्वशीघ्रात् एवं तृतीयपदे क्षयोपचयौ योज्यौ चतुर्थे क्षयोपचयोत्क्रमज्याक्रमेण धनमुपपद्यते। इत्यादि स्वधिया योज्यम्। एतदार्यद्वयसार्धं फलोत्पन्नावुपचयापचया वासना प्रदर्शनार्थं न फलानयनार्थं, यदा फलानयनार्थं स्यात्तच्छुक्रस्य प्रथमपदान्तं मन्दपरिधिना नवकेन यत्फलमानीयते। यच्च द्वितीयपदांतपरिधिना एकादश तयोः ऋणधनयोः कथं तुल्यत्वं स्यात्। अतुल्यत्वे तु केन्द्रे चक्रावस्थोपि फलसम्भवः स्यात्। नवैव चक्रं योज्यम्। शीघ्रपरिध्योरप्येवमेव योज्यम्। आर्यभटस्य विषमेऽन्योन्यो युग्मे परिधिर्गुणकः क्रमोत्क्रमजानाम्, चक्रार्धे फलनाशो भवति न यस्मादसत्तदपि शेषाणामन्तरा ग्रहाणामेकत्वात्परिधेः फलानयनेऽपि न दोष इति। अथवा शुक्रस्यापि विषमपदादावेकादशकः परिधिः तदन्तरमेव नवक एवं समपदादौ नव तदन्ते चैकादशपरिधिरेवान्तरे त्रिज्याहता भुजज्येत्यादिकर्मणा वासनामानात् फलनाशोऽपि चक्रार्धे उपपद्यते इति।

वि. भा.—तद्भुजफलकृतियोगात् (तस्याःपूर्वानीतस्फुटकोटेर्भुजफलस्य च वर्गयोगात्) मूलं तदा कर्णो भवेत्। अयुग्युक्षु विषमसमपदेषु क्रमशः क्रमोत्क्रमजीवा (केन्द्रज्या, केन्द्रोत्क्रमज्या च) स्वपरिधिगुणा, भांशै ३६० र्भाज्या तदोत्क्रमज्यातो यत्फलं तत्परमे भुजफले हीनं कार्यं तदा वास्तवं भुजफलं भवेदित्यध्याहार्यम्, मन्दे इत्यस्याग्रिमश्लोकेन सम्बन्ध इति॥ १५॥

अत्रोपपत्तिः

१४तम श्लोकस्य संस्कृतोपपत्तिस्थक्षेत्रे भूग्र=कर्णः=$\sqrt{\text{ग्रल}^2+\text{भूल}^2}$=$\sqrt{\text{भुजफल}^2+\text{नाचाच्चवृत्तस्पष्टको}^2}$, प्रथमपदे गतकेन्द्रांशानां ज्या स्वपरिधिगुणा भांशै ३६० र्भाज्या तदा पूर्वोक्तप्रकारागतं भुजफलमेव भवेत्। द्वितीयपदे एष्यांशानां ज्या गतोत्क्रमज्योनत्रिज्यातुल्या तदा पूर्ववदेव भुजफलम्

$=\frac{\text{परिधि}\ (\text{त्रि}-\text{गतकेन्द्रांशोत्क्रमज्या})}{३६०}$, ∴ परमभुजफल

$-\frac{\text{परिधि}\ (\text{त्रि}-\text{गकेउज्या})}{३६०}$ अर्थात्समपदे केन्द्रांशोत्क्रमज्यातो यत्फलं समागच्छेत्तेन हीनं परमं भुजफलं वास्तवं भुजफलं भवेत्। विषमपदे (प्रथमपदे तृतीयपदे च)

$\frac{\text{केन्द्रज्या}\times\text{परिधि}}{३६०}$=भुजफलम्, पदान्ते केन्द्रज्यायाः परमत्वात्तत्र भुजफलस्यापि परमत्वं भवेत्। समपदे (द्वितीयपदे, चतुर्थपदे च), परम भुजफल $-\frac{\text{केउत्क्रमज्या}\times\text{परिधि}}{३६०}$=वास्तवभुजफल, पदान्ते भुजफलं शून्यं, अत्रोत्क्रमज्या-गतं फलं परमे भुजफले ऋणं कार्यमिति चतुर्वेदाचार्यमतानुकूलोऽध्याहार इत्याचार्यसूत्रतो नाऽऽयातीति बोध्यं विज्ञैरिति ॥ १५ ॥

अब कर्णानयन को तथा प्रकारान्तर से भुजफलानयन को भी कहते हैं

हि. भा.—पूर्वानीतस्फुटकोटि और भुजफल के वर्गयोग का मूल लेने से कर्ण होता है। विषम पदों में केन्द्रज्या को स्पष्ट परिधि से गुणा कर भांश ३६ से भुजफल होता है, समपदों में केन्द्रांशोत्क्रमज्या को स्पष्ट परिधि से गुणा कर भांश ३६० से भाग देने से जो फल हो उसको परमभुजफल में घटाने से वास्तव भुजफल होता है, यह अध्याहार करना चाहिये श्लोक में 'मन्दे' शब्द का सम्बन्ध अगले श्लोक के साथ है इति ॥१५॥

उपपत्ति

१४ वें श्लोक का संस्कृतोपपत्तिस्थ क्षेत्र देखिये, भूग्र=कर्ण=$\sqrt{\text{ग्रल}^२+\text{भूल}^२}$ $=\sqrt{\text{भुजफ}^२+\text{नीचोच्चवृत्तीय स्फो}^२}$ प्रथम पद में गत केन्द्रांशज्या को स्पष्ट परिधि से गुणाकर भांश ३६० से भाग देने से पूर्व प्रकारागत ही भुजफल होता है। द्वितीय पद में एष्यांशज्या गतोत्क्रमज्योन त्रिज्या के बराबर होती है तब उससे पूर्ववत् भुजफल=$\frac{\text{परिधि}\ (\text{त्रि}-\text{गतकेन्द्रांशोत्क्रमज्या})}{\text{त्रि}}$, अतः परम भुजफल—$\frac{\text{परिधि}\ (\text{त्रि}-\text{गतकेन्द्रांशोत्क्रमज्या})}{\text{त्रि}}$=वास्तव भुजफल अर्थात् सम पदों में केन्द्रांशोत्क्रमज्या से जो फल आवे उसको परमभुजफल में घटाने से वास्तव भुजफल होता है, विषमपद (प्रथम पद, तृतीय पद) में $\frac{\text{केज्या}\times\text{परिधि}}{३६०}$=भुजफल; पदान्त में केन्द्रज्या परमत्व के कारण परमभुजफल होता है, समपदों (द्वितीय पद चतुर्थ पद) में परम भुजफल—$\frac{\text{के उत्क्रमज्या}\times\text{परिधि}}{३६०}$=वास्तव भुफल; पदान्त में भुजफल शून्य होता है, यहाँ उत्क्रमज्या से आये हुये फल को ऋण करना चाहिये, यह चतुर्वेदाचार्य के मतानुकूल अध्याहार करना होता है, ये बातें आचार्य (ब्रह्मगुप्त) के सूत्र से नहीं आती हैं, यह समझना चाहिये इति ॥ १५ ॥

वि. भा.—मन्दे (मन्दकर्मणि) पदक्रमेण पूर्वानीतफलानि क्षयधनधनक्षयाः स्युः। शीघ्रे (शीघ्रकर्मणि) अन्यथाऽर्थात् धनक्षयक्षयधनानि स्युः। ततो धनयोर्योगो धनम्, ऋणयोर्योगश्च ऋणम्, ऋणधनयोरन्तरं संस्कारवशेन धनर्णं भवति, तुल्ययोः ऋणधनयोर्वशतो भुजफलं शून्यं भवतीति ॥ १६ ॥

अत्रोपपत्तिः

प्रथमद्वितीयपदयोर्मेषादिकेन्द्रम्, मन्दोच्चान्नीचपर्यन्तं स्थिते मध्यग्रहे मन्दोच्चमध्यग्रहान्तररूपस्य मन्दकेन्द्रस्य मेषादिषड्भ्यन्तर्गतत्वान्मन्दफलेन भुजफलचापसमेन हीनो मध्यग्रहो मन्दस्पष्टग्रहो भवेत्। तृतीयचतुर्थपदयोश्च तुलादिकेन्द्रम्, नीचान्मन्दोच्चावधिवर्त्तमाने मध्यग्रहे मन्दकेन्द्रस्य (मन्दोच्चमध्यग्रहान्तररूपस्य) तुलादिराशिषट्कान्तरे स्थितत्वान्मन्दफलेन युक्तो मध्यग्रहो मन्दस्पष्टग्रहो भवेदतो मन्दकर्मणि प्रथमपदे द्वितीयपदे च भुजफलचापमृणं तृतीयपदे चतुर्थपदे च भुजफलचापं धनं यतस्तत एव धनमन्दफलोत्पत्तिर्भवति, । द्वितीयपदे पूर्वागतं भुजफलम्=परमभुफ−$\frac{\text{परिधि}\times\text{केउज्या}}{३६०}$, धनर्णयोर्विलोमेन—परमभुजफल+ $\frac{\text{परिधि}\times\text{केउज्या}}{३६०}$, चतुर्थपदे पूर्वागतं भुजफलं धनात्मकम्=परमभुफ− $\frac{\text{परिधि}\times\text{केउज्या}}{३६०}$, मेषादिकेन्द्रत्वात्प्रथमे पदे ऋणं, तुलादिकेन्द्रत्वात्तृतीये पदे धनम्। एवं पदक्रमेण मन्दकर्मणि फलानि क्षयधनधनक्षयाख्यानि भवन्ति, शीघ्रकर्मणि मध्यग्रहः शीघ्रोच्चाल्पगतित्वेन पृष्ठतोऽवलम्बते प्रतिवृत्ते, तस्माच्छीघ्रोच्चान्नीचं यावद् ग्रहोच्चान्तररूपस्य शीघ्रकेन्द्रस्य मेषादिषड्भ्यन्तर्गतत्वाच्छीघ्रफलं धनात्मकं जायते, नीचादुच्चं यावच्छीघ्रकेन्द्रस्य तुलादिराशिषट्कान्तर्गतत्वाच्छीघ्रफलमृणात्मकं जायते तेनैव कारणेन शीघ्रकर्मणि भुजफलमन्यथा साध्यत इति, सिद्धान्तशेखरे श्रीपतिना।

'ऋणं क्रमादुत्क्रमतो धनज्या पुनः क्रमात् स्वं क्षय उत्क्रमाच्च।
क्रमोत्क्रमाभ्यां हि पदक्रमेण प्रसाध्य जीवां फलमानयेद्वा ॥

युतिः स्वयोः स्वं क्षययोः क्षयश्च धनर्णयोरन्तरतोऽधिकं यत्।
समानयोः स्वक्षययोश्च शून्यमृणं धनं शीघ्रफलेऽन्यथा स्यात् ॥

श्लोकाभ्यां ब्रह्मगुप्तोक्त 'तद्भुजफलकृतियोगान्मूलं कर्णः पदेष्वित्यादेः क्षयधनधनक्षयास्तत्फलानीत्यादि' श्लोकद्वयस्य पुनरुक्तीकरणमेव कृतमिति विवेचकैर्विवेचनीयम् ॥ १६ ॥

अब पूर्वानीत फल के ऋणत्व और धनत्व को उससे संस्कार वश भुजफल को कहते हैं

हि. भा.—मन्दकर्म में पदक्रम से पूर्वानीत फल ऋण, धन, धन, ऋण होते हैं, शीघ्र कर्म में इससे भिन्न होता है अर्थात् पद क्रम से धन, ऋण, ऋण, धन होते हैं, इस कारण से धन, धन का योग धन होता है, ऋण, ऋण का योग ऋण होता है, और धन, ऋण का अन्तर संस्कार वश से भुजफल धनात्मक और ऋणात्मक होता है, धन और ऋण के तुल्यत्व से भुजफलाभाव (शून्य) होता है इति ॥ १६ ॥

उपपत्ति

प्रथम पद और द्वितीय पद मेषादि केन्द्र है, तथा तृतीय पद और चतुर्थ पद तुलादि केन्द्र है, मन्द कर्म में भुजफल के चाप के बराबर मन्द फल मेषादि केन्द्र में ऋण होता है और तुलादि केन्द्र में धन होता है, इसलिये प्रथम पद और द्वितीय पद में भुजफल ऋण होता है, तृतीय पद में और चतुर्थ पद में भुजफल धन होता है क्योंकि उससे मन्द फल धनात्मक उत्पन्न होते हैं, द्वितीय पद में पूर्वागत भुजफल=परम भुजफल—$\frac{\text{परिधि}\times\text{केउज्या},}{३६०}$

धन और ऋण के विलोम से—परम भुजफल+$\frac{\text{परिधि}\times\text{केउज्या}}{३६०}$, इसी तरह चतुर्थ पद में

पूर्वागत भुजफल=परमभुफ—$\frac{\text{परिधि}\times\text{केउज्या}}{३६०}$ धनात्मक, मेषादि केन्द्र में प्रथम पद के होने के कारण प्रथम पद में भुजफल ऋण तथा तुलादि केन्द्र में तृतीय पद के होने के कारण वहां धन होता है, इस तरह पद क्रम से मन्द कर्म में फल ऋण, धन, धन, ऋण होते हैं, शीघ्र कर्म से इससे विलोम होता है क्योंकि शीघ्रकर्म में मेषादिकेन्द्र में शीघ्रफलधन होता है और तुलादिकेन्द्र में ऋण होता है इसी हेतु से वहाँ भुजफल भिन्न होता है, सिद्धान्त-शेखर में श्रीपति 'ऋणं क्रमादित्यादि, युतिः स्वयोः स्वं 'क्षययोः' इत्यादि संस्कृतोपपत्ति में लिखित श्लोकद्वय से आचार्योक्त 'तद्भुजफलकृतियोगात्' इत्यादि, 'क्षयधनधनक्षयास्तत्फलानि' इत्यादि श्लोकद्वय का पुनरुक्तीकरण ही किया है इति ॥ १६ ॥

वि. भा.—(तच्चापं मन्दभुजफलस्य चापं) मन्दफलं पूर्वसाधितफलयोर्योगान्तरवशात् धनं वा ऋणं भवति । तद्गुणितात् (पूर्वागतशीघ्रभुजफलेन गुणितात्) व्यासार्धात् (त्रिज्यातः) कर्णे (शीघ्रकर्णे) भक्ताद्यल्लब्धं तस्य धनुः (चापं) शीघ्रफलं भवतीति ॥१७॥

अत्रोपपत्तिः

तात्कालिककर्णानीयमन्दभुजफलचापं मन्दफलं न भवितुमर्हति, पठितमन्दकर्णानीयभुजफलस्य चापं मन्दफलं भवति कथमित्युच्यते ।

उ=मन्दोच्चम्। ग्र=मन्दप्रतिवृत्ते ग्रहः। भू=भूकेन्द्रम्। ज=ग्रहगोलकेन्द्रम्। भूज=मन्दान्त्यफलज्या। ग्रन=तात्कालिकान्त्यफलज्या=तच=भूज। भूग्र=तात्कालिकमन्दकर्णः। ग्रर=मन्दभुजफलम्। प बिन्दुतो भूर रेखायाः समान्तरा पग्र रेखा कार्या, प बिन्दुत एव ग्रन रेखायाः समान्तरा पश रेखा कार्या. भूग्र रेखा कार्या। पय=मन्दफलज्या, अथ भूग्रन, भूपश त्रिभुजयोः सजातीयात्

$$\frac{\text{ग्रन}\times\text{भूप}}{\text{भूग्र}}=\frac{\text{तात्कालिकान्त्यफज्या}\times\text{त्रि}}{\text{तात्कालिकमन्दकर्ण}}=\text{पश}=\text{पठितान्त्यफलज्या}=\text{ग्रन},$$

अतोभूग्र=पठितमन्दकर्णः। ग्रग=पठितमन्दकर्णाग्रीयभुजफलम्=पय=मंफलज्या ग्रर > पय वा तात्कालिकमन्दकर्णाग्रीयभुजफल > मन्दफलज्या अत एव मन्दभुजफलचापं मन्दफलं न भवितुमर्हति, किन्तु पठित-मन्दकर्णाग्रीयमन्दभुजफलचापं मन्दफलं भवितुमर्हत्यत आचार्योक्त 'तच्चापं मन्दफलं' मिदं न युक्तम्। सिद्धान्तशेखरे श्रीपत्यु 'दोः फलस्य च धनुः कलादिकं जायते मृदुफल नभः सदाम्' क्तमिदं 'मृदुदोः फलस्य चापं बुधा मन्दफलं वदन्ति' भास्करोक्तमिदं चाऽऽचार्योक्तानुरूपमेव। एवं $\frac{\text{शीघ्रभुजफल}\times\text{त्रि}}{\text{शीघ्रकर्ण}}=$ शीघ्रफलज्या, एतच्चापं शीघ्रफलं भवेत्। श्रीपतिना 'दोः फलत्रिगुणयोरभिघातात्कर्णलब्धधनुराशुफलं स्यात्' ऽप्यनेनाऽऽचार्योक्तानुरूपमेव कथ्यत इति ॥१७॥

अब मन्दफल और शीघ्रफल कहते हैं

हि. भा.—मन्द भुजफल का मन्दफल होता है, यह पूर्वसाधित फलद्वय के योग और अन्तर वश से धन वा ऋण होता है, पूर्वागत शीघ्रभुजफल को त्रिज्या से गुणा कर शीघ्र कर्ण से भाग देने से जो लब्धि होती है उसका शीघ्रफल होता है ॥ १७ ॥

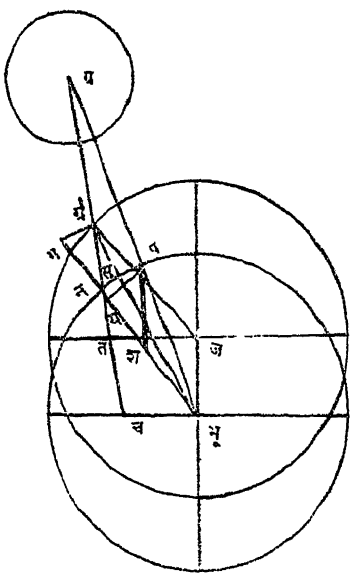
उपपत्ति

यहां संस्कृतोपपत्ति में लिखित (क) क्षेत्र को देखिये। तात्कालिक मन्दकर्णाग्रीय भुजफल का चाप मन्दफल नहीं होता है, किन्तु पठित मन्दकर्णाग्रीय भुजफल का चाप मन्दफल होता है इसके लिये अधोलिखित युक्ति है। उ=मन्दोच्च, ग्र=मन्दप्रतिवृत्त में ग्रह, भू=भूकेन्द्र, ज=ग्रहगोलकेन्द्र, भूज=मन्दान्त्यफलज्या, ग्रन=तात्कालिकमन्दा-न्त्यफज्या=तच=भूज, भूग्र=तात्कालिकमन्दकर्ण, ग्रर=मन्दभुजफल, प बिन्दु से भूर रेखा की समान्तरा रेखा पग्र, प बिन्दु से ग्रन रेखा की समान्तरा पश रेखा कीजिये, भूग्र रेखा कीजिये पय=मन्दफलज्या, भूग्रन, भूपश दोनों त्रिभुज सजातीय हैं इसलिये अनुपात करते हैं। $\frac{\text{ग्रन}\times\text{भूप}}{\text{भूग्र}}=\text{पश}=\frac{\text{तात्कालिकमन्दान्त्यफलज्या}\times\text{त्रि}}{\text{तात्कालिकमन्दकर्ण}}=\text{पठितमन्दान्त्यफलज्या}=$

ग्रन, अतः भूग्र=पठितमन्दकर्णः, ग्रग=पठितमन्दकर्णाग्रीयभुजफल=पय=मन्दफलज्या, ग्रर > पय वा तात्कालिकमन्दकर्णाग्रीय भुजफल > मन्दफलज्या इसलिये तात्कालिक मन्दकर्णाग्रीय भुजफल का चाप मन्दफल नहीं हो सकता है किन्तु पठितमन्दकर्णाग्रीय भुजफल का चाप मन्दफल के बराबर सिद्ध हुआ इसलिये 'तच्चापं मन्दफलं' यह आचार्योक्त ठीक नहीं है, सिद्धान्तशेखर में 'दोः फलस्य च धनुः कलादिकं इत्यादि' श्रीपत्युक्ति तथा 'मृदुदोः फलस्य चापं बुधा मन्दफलं वदन्ति' यह भास्करोक्ति आचार्योक्तानुरूप ही है, $\frac{\text{शीघ्रभुजफल} \times \text{त्रि}}{\text{शीघ्रकर्ण}}$=शीघ्रफलज्या; इसका चाप शीघ्रफल होता है, 'दोः फलत्रिगुणयोरभिघातात्' इत्यादि श्रीपत्युक्ति आचार्योक्त 'तद् मुणिताद् व्यासार्धात्कर्णलब्धधनु.' के अनुरूप ही है इति ॥ १७ ॥

अत्र विशेषविचारः

यथा-यथा कर्णो वर्धते तथा-तथाऽन्त्यफलज्यापि वर्धिता भवतीति मन्यते ब्रह्मगुप्तः। अतोऽभीष्टकाले पारमार्थिको ग्रहः प्रतिवृत्ते ग्र स्थाने न वर्तते, किन्तु ग्र स्थाने तिष्ठति। अतो वास्तवान्त्यफलज्या=ग्रन। अतो भू स्थानात् भूग्र व्यासार्धेन यो गोलस्तत्रैव प्रतिमण्डले ग्रहभ्रमणं भवतीति ब्रह्मगुप्तमतं साध्विति स्थापयन्ति भास्कराचार्याः। तदर्थमेव मन्दश्रुतिर्द्रार्क्श्रुतिवत्प्रसाध्येत्यादिवक्ष्यमाणविधिना भूग्र कलाकर्णमानं साधितं भास्करेण इत्यनया कल्पनया भुजफलचापसमं मन्दफलं भवति। अपूर्वेयं कल्पना ब्रह्मगुप्तस्य।



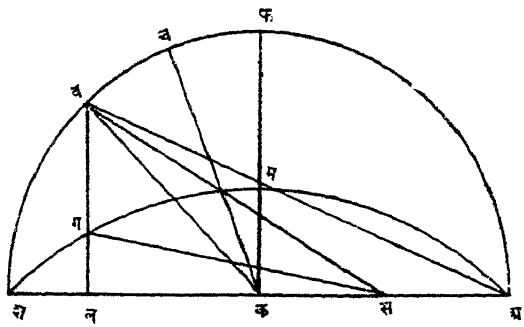
अथ नवीनानां मतेन सहेदं ब्रह्मगुप्तमतं समन्वेति। तदर्थं विदां विनोदाय नवीनमतं प्रतिपाद्यते।

तथाहि उपरि तनक्षेत्रदर्शनेन ग्रन पठितान्त्यफलज्या तद्वशेन गन कोटिफलम्। ततः स्पष्टाकोटिः=भूग। ग्रग=भुजफलम्। ततः कर्णः=भूग्र, तद्वशेन कक्षामण्डले स स्थानं एव स्फुटो ग्रहो भवितुमर्हति। किन्तु ब्रह्मगुप्तेन मन्दकर्मणि क्षेत्रभङ्गया साधित रविचन्द्राभ्यां वैषम्यं समुपलभ्य प स्थाने स्फुटो दृष्टः। तथा सति भूग्र तात्कालिककर्णानुपातेन साधितं फलं ग्रग भुजफलचापरूपं जायते।

अतो भास्कराचार्येणेह ब्रह्मगुप्तमतमेव समर्थितम्। मृदुदोः फलस्य चापं बुधाः मन्दफलं वदन्तीति प्राहुः भास्कराचार्याः। श्रीपत्यादयोऽप्येवमेव प्रोचुः।

नव्यास्तु वर्तुलाभासपिण्डे ग्रहान् परिभ्राम्य दीर्घवृत्तस्यैकस्यां नाभौ सूर्यं मत्वा तत्परितो दीर्घवर्तुले ग्रहो भ्रमतीति स्वीकृत्य सूर्यकेन्द्राद्ग्रहावधि मन्दकर्णस्तुल्यकाले समं क्षेत्रफलं समुत्पादयतीति केप्लरसिद्धान्तानुसारेण ग्रहाणां मन्दफलं साधयन्तीत्याहुः।

दीर्घवृत्तश्रेढीप्रकारेण फलानयनं नवीनमते



अत्र ब्रह्मगुप्तमतं नवीनगणितेन परीक्ष्यते। तथाहि कल्प्यते अश व्यासोपरि अगश भ्रमणेन दीर्घवृत्तखण्डम्। अश व्यासोपरि अपवश सहायकवृत्तार्धम्।

स=दीर्घवृत्तनाभिर्यत्र सूर्यः।
अ=नीचस्थानम्।
अक=महद्व्यासार्धम्=अ
कम=लघुव्यासार्धम्=क
क=दीर्घवृत्तकेन्द्रम्।

सहायकवृत्ते च स्थाने मध्यमो ग्रहः।
दीर्घवृत्ते ग स्थाने ग्रहः।

सहायकवृत्ते व स्थाने ग सम्बन्धी यो ग्रहः।

< अकच=मध्यमकेन्द्रम्=के।
< अकग=स्फुटकेन्द्रम्=ष.
< अकव=सहायककेन्द्रम्=के,

अथ अ स्थानात् ग स्थानं यावत् ग्रहगमनकालः=का,। एवं दीर्घमण्डलभ्रमणकालः=का.

अतः केप्लर द्वितीयसिद्धान्तेन

$$\frac{\text{का}_१}{\text{का}} = \frac{\text{क्षेफ असग}}{\text{दीवृफ}}.$$

अत्र मध्यममानेनैकस्मिन् दिने मन्दकेन्द्रगतिमानं न प्रकल्प्य ततोऽनुपातः। यदि भगणभोगकालेन भगणभागा लभ्यन्ते तदैकदिनेन किम्।

जातमेकस्मिन् दिने मध्यकेन्द्रमानम् $= \frac{२\pi}{\text{का}} = \text{न}$।

(अत्र π = रूपव्यासार्धेऽर्धपरिधिमानम्),

$$\therefore \text{का} = \frac{२\pi}{\text{न}}.$$

$$\therefore \frac{\text{का}_१ \cdot \text{न}}{२\pi} = \frac{\text{क्षेफ असग}}{\text{दीवृफ}}.$$

अत्र दीवृफ $= \pi \cdot$ अ $\cdot$ क $\cdot$

$$\therefore \frac{\text{का}_१ \cdot \text{न}}{२\pi} = \frac{\text{क्षेफ असग}}{\pi \cdot \text{अ} \cdot \text{क}}.$$

$$\therefore \text{का}_१ \cdot \text{न} = \frac{२ \cdot \text{क्षेफ} \cdot \text{असग}}{\text{अ} \cdot \text{क}}.$$

अत्र का$_१$ समे काले का$_१ \cdot$ न इदं मन्दकेन्द्रमानं भवति। अतः मन्दकेन्द्रम् $=$

$$\frac{२ \text{ क्षेफ} \cdot \text{असग}}{\text{अ} \cdot \text{क}} = \text{म}।$$

अथ दीर्घवृत्तसिद्धान्तेन—

$$\frac{\text{गल}}{\text{वल}} = \frac{\text{क}}{\text{अ}}.$$

परं च $\frac{\text{गल}}{\text{वल}} = \frac{\text{फ अलग}}{\text{फ अलव}}$

$$\therefore \frac{\text{फ अलग}}{\text{फ अलव}} = \frac{\text{क}}{\text{अ}}.$$

$$\therefore \text{फ अलग} = \text{फ अलव} \cdot \frac{\text{फ}}{\text{अ}}.$$

परं च फ अलव $= \triangle$अकव $+ \triangle$लकव $\cdot$

$$\therefore \text{फ अलग} = \frac{\text{क}}{\text{अ}} (\triangle \text{अकव} + \triangle \text{लकव})$$

$$= \frac{क}{अ}\left(\frac{के_१ \cdot अ^२}{२} + \frac{ज्याके_१ \cdot अ \cdot लक}{२}\right)$$

$$= \frac{क \cdot अ}{२}\left(के_१ + \frac{ज्याके_१ \cdot लक}{अ}\right)$$

$$= \frac{क \cdot अ}{२}(के_१ + ज्याके_१ \cdot कोज्याके_१)$$

एवं $\triangle$ सलग $= \frac{सल \cdot गल}{२}$

$$= \frac{सल \cdot वल \cdot क}{२ \cdot अ}$$

$$= \frac{अ \cdot क}{२} \cdot \frac{सल \cdot वल}{अ \cdot अ}$$

$$= \frac{अ \cdot क}{२} \cdot \frac{(लक + सक)\, वल}{अ \cdot अ}$$

$$= \frac{अ \cdot क}{२} \cdot \frac{लक \cdot वल}{अ \cdot अ} + \frac{सक \cdot वल}{अ^२}$$

$$= \frac{अ \cdot क}{२}(कोज्याके_१ \cdot ज्याके_१ + इ - ज्याके_१)$$

अनयोरन्तरेण—

$$\triangle असग = \frac{अ \cdot क}{२}(के_१ - इ \cdot ज्याके_१)$$

$$\therefore \quad म = के_१ - इ \cdot ज्याके_१ \quad \cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots (१)$$

एतेन च्युतिकेन्द्रमुखेन मध्यमकेन्द्रमानं सिद्ध्यति अतः परं स्फुटकेन्द्रद्वारा तन्मानमन्विष्यते।

अथ दीर्घवृत्तसिद्धान्तेन—

मक $\cdot$ कोज्याष $= अ \cdot$ कोज्याके$_१ - इ \cdot अ$

एवं मक $\cdot$ ज्याष $= क \cdot$ ज्याके$_१$

वर्गयोर्योगेन—

मक$^२ \cdot$ कोज्या^{२} ष $= अ^२ \cdot$ कोज्या^{२} के$_१ - २ \cdot अ \cdot$ कोज्याके$_१ \cdot इ \cdot अ + इ^२ \cdot अ^२$

एवं मक^{२}· ज्या^{२}ष=क^{२}· ज्या^{२}के$_{१}$

योगेन—

मक^{२}=अ^{२}· कोज्या^{२}के$_{१}$—२ अ· कोज्याके$_{१}$· इ· अ+इ^{२}· अ^{२}+ ज्या^{२}के$_{१}$· क^{२}

=अ^{२}· कोज्या^{२}के$_{१}$—२ अ· कोज्याके$_{१}$· इ· अ+इ^{२}· अ^{२}+ अ^{२} (१—इ^{२})· ज्या^{२}के$_{१}$

=अ^{२}· कोज्या^{२}के$_{१}$—२ अ· कोज्याके$_{१}$· इ· अ+अ^{२} इ^{२}+ (१—इ^{२}) ज्या^{२}के$_{१}$

=अ^{२} (कोज्या^{२}के$_{१}$—२ कोज्याके$_{१}$· इ+इ^{२}—इ^{२}· ज्या^{२}के$_{१}$ +ज्या^{२}के$_{१}$)

=अ^{२} (कोज्या^{२} के$_{१}$+ ज्या^{२}के$_{१}$—२ कोज्याके · इ +इ^{२}· कोज्या^{२}के$_{१}$)

=अ^{२} (१—इ· कोज्याके$_{१}$)^{२}

मूलेन—

मक=अ (१—इ· कोज्याके$_{१}$)·····(२)

अथ त्रिकोणगणितेन—

२ ज्या$^{२}\frac{१}{२}$ ष=१—कोज्याष

∴ २ मक· ज्या$^{२}\frac{१}{२}$ ष=मक (१—कोज्याष)

=मक—मक· कोज्याष

=अ (१—इ· कोज्याके$_{१}$)—(अ· कोज्याके$_{१}$ —इ ·अ)

=अ (१—इ· कोज्याके$_{१}$)—(अ· कोज्याके$_{१}$—इ· अ)

=अ १—इ· कोज्याके$_{१}$—(कोज्याके$_{१}$—इ)

=अ {१—इ· कोज्याके$_{१}$—कोज्याके$_{१}$ +इ}

एवं २ कोज्या$^{२}\frac{१}{२}$ष· मक=मक (१+कोज्याष)

=अ (१—इ· कोज्याके$_{१}$+कोज्याके$_{१}$—इ)

$$\therefore \frac{\text{ज्या}^{२}\frac{१}{२}\text{ ष}}{\text{कोज्या}^{२}\frac{१}{२}\text{ ष}}=\frac{(१-\text{इ}\cdot \text{कोज्याके}_{१}-\text{कोज्याके}_{१}+\text{इ})}{१-\text{इ}\cdot \text{कोज्याके}_{१}+\text{कोज्याके}_{१}-\text{इ}\cdot}$$

$$=\frac{(१+\text{इ})\ (१-\text{कोज्या के}_{१})}{(१-\text{इ})\ (१+\text{कोज्या के}_{१})}$$

$$\therefore \text{स्प}^2\tfrac{1}{2}\,\text{ष} = \frac{१+\text{इ}}{१-\text{इ}} \cdot \text{स्प}^2\tfrac{1}{2}\,\text{के}_1$$

$$\therefore \text{स्प}\tfrac{1}{2}\,\text{प} = \sqrt{\frac{१+\text{इ}}{१-\text{इ}}} \cdot \text{स्प}\tfrac{1}{2}\,\text{के}_1 \cdots\cdots\cdots\cdots\cdots\cdots (३)$$

एतेन च्युतिकेन्द्रमुखेन स्फुटकेन्द्रमानं सिद्धचति । पूर्वं तु (१) समीकरणेनापि च्युतिकेन्द्रमुखेन मध्यमकेन्द्रमानं सिद्धम् । अथेदानीं मध्यमस्फुटयोः सम्बन्धो गदेपणीयः ।

अत्र यदि इ=ज्यास, ष=य, के$_1$=र,

$$\text{तर्हि } \frac{१+\text{इ}}{१-\text{इ}} = \frac{१+\text{ज्यास}}{१-\text{ज्यास}} = \frac{१+२\,\text{कोज्या}\tfrac{1}{2}\text{स}\cdot\text{ज्या}\tfrac{1}{2}\text{स}\cdot}{१-२\,\text{कोज्या}\tfrac{1}{2}\text{स}\cdot\text{ज्या}\tfrac{1}{2}\text{स}\cdot}$$

$$= \frac{\text{कोज्या}^2\tfrac{1}{2}\text{स}+\text{ज्या}^2\tfrac{1}{2}\text{स}+२\,\text{कोज्या}\tfrac{1}{2}\text{स}\cdot\,\text{ज्या}\tfrac{1}{2}\text{स}\cdot}{\text{कोज्या}^2\tfrac{1}{2}\text{स}+\text{ज्या}^2\tfrac{1}{2}\text{स}-२\,\text{कोज्या}\tfrac{1}{2}\text{स}\cdot\,\text{ज्या}\tfrac{1}{2}\text{स}\cdot}$$

$$= \frac{(\text{कोज्या}\tfrac{1}{2}\text{स}+\text{ज्या}\tfrac{1}{2}\text{स})^2}{(\text{कोज्या}\tfrac{1}{2}\text{स}-\text{ज्या}\tfrac{1}{2}\text{स})^2}$$

$$= \frac{(१+\text{स्प}\tfrac{1}{2}\text{स})^2}{(१-\text{स्प}\tfrac{1}{2}\text{स})^2} \therefore \sqrt{\frac{१+\text{इ}}{१-\text{इ}}} = \frac{१+\text{स्प}\tfrac{1}{2}\text{स}}{१-\text{स्प}\tfrac{1}{2}\text{स}}$$

$$\therefore \text{स्प}\tfrac{1}{2}\text{ष} = \text{स्प}\tfrac{1}{2}\text{के}_1 \sqrt{\frac{१+\text{इ}}{१-\text{इ}}}$$

$$\therefore \text{स्प}\tfrac{1}{2}\text{य} = \frac{१+\text{स्प}\tfrac{1}{2}\text{स}}{१-\text{स्प}\tfrac{1}{2}\text{स}} \cdot \text{स्प}\tfrac{1}{2}\text{र}\cdot$$

अथ त्रिकोणमित्या—

$$\text{स्प}\tfrac{1}{2}\,\text{ष} = \frac{\text{इ}^{\text{iय}/२}+\text{इ}^{-\text{iय}/२}}{\text{इ}^{\text{iय}/२}-\text{इ}^{-\text{iय}/२}}$$

$$\text{एवं} \quad \text{स्प}\tfrac{1}{2}\,\text{र} = \frac{\text{इ}^{\text{ir}/२}+\text{इ}^{-\text{ir}/२}}{\text{इ}^{\text{ir}/२}-\text{इ}^{-\text{ir}/२}}$$

योगान्तरनिष्पत्या—

$$\text{इ}^{\text{iय}} = \text{इ}^{\text{ir}} \cdot \frac{१+३^{-\text{ir}^{\text{i}}} \cdot \text{स्प}\tfrac{1}{2}\,\text{स}}{१-३^{\text{ir}} \cdot \text{स्प}\tfrac{1}{2}\,\text{स}}$$

पक्षयोर्लघुरिक्थेन—

$$य=र+२\,(स्प\,\tfrac{१}{२}\,स\cdot ज्यार+\tfrac{१}{२}\,स्प^{२}\,\tfrac{१}{२}\,स\,ज्या\,२\,र+\cdots\cdots\cdots\cdots)$$

एवमेव $इ=ज्यास$।

$$\therefore \quad स्प\,\tfrac{१}{२}\,स=\frac{\sqrt{१-इ^{२}}}{इ}$$

$$=\frac{इ}{२}+\frac{इ^{२}}{८}+\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots$$

अत उत्थापनेन—

$$य=र+\left(इ+\frac{इ^{२}}{४}\right)ज्यार+\frac{इ^{२}}{४}ज्या\,२\,र+\cdots\cdots$$

वा $$ष=के_{२}+\left(इ+\frac{इ^{२}}{४}\right)ज्याके_{१}+\frac{इ^{२}}{४}ज्या\,२\,के_{१}+\cdots\cdots$$

अत्र (१) समीकरणेन—

$$म=के_{१}-इ\,ज्याके_{१}$$

यद्यत्र प्रथमवारं दक्षिणपक्षस्यैकखण्डं त्यज्यते तदा $म=के_{१}$ तत उत्थापनेन—

$$के_{१}=म+इ\cdot ज्याम\cdot$$

अनेन (४) समीकरणमुत्थाप्य जातम्

$$ष=म+इ\cdot ज्याम+\left(इ+\frac{इ^{२}}{४}\right)^{ज्या}(म+इ\cdot ज्याम)$$

$$=म+इ\cdot ज्याम+इ\cdot ज्या\,(म+इ\cdot ज्याम)+\cdots\cdots\cdots\cdots$$

$$=म+२\,इ\cdot ज्याम+\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots$$

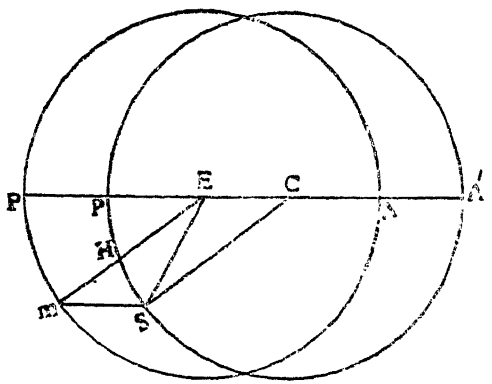
$\therefore$ $ष-म=२\,इ\cdot ज्याम$ अन्यपदत्यागात्।

अत्र म मध्यमकेन्द्रं, ष स्फुटकेन्द्रं, मध्यस्फुटकेन्द्रान्तरं मन्दफलं भवत्यतः

मन्दफलम्$=२\,इ\cdot ज्याम\cdot$ यद्यत्र $२\,इ=$अन्त्यफलज्या, तथा त्रि$=१$, तदा ब्रह्मगुप्तोक्तं मृदुदोः फलचापरूपं मन्दफलं नवीनमतेनापि सिद्धम्। अत्रान्त्यफलज्या $=२\,इ\cdot$ एतदर्थमन्यः प्रकारो द्रष्टव्यः तत्त्वग्रे वक्ष्ये।

अत्रैव केचन नवीनास्तु अन्यथैव क्षेत्रभङ्गीं प्रदर्श्य चन्द्रार्कयोर्मन्दफलं प्रसाध्य प्राचीनमतं समर्थयन्ति ।

तद्यथा



अथ E भुवो मध्यम् ।

C=ग्रहगोलकेन्द्रम् ।

EC=मन्दान्त्यफलज्या=p.

m=प्रतिवृत्ते मध्यमः

S=कक्षामण्डले सूर्यः स्पष्टः

Em=त्रिज्या=अ ।

∠mES=मन्दफलम्=E कल्प्यते ।

SH लम्ब : =Em रेखोपरि कृतः ।

$$\therefore \quad \text{स्प} \angle SEH = \frac{SH}{EH} = \angle mES.$$

∠EmS=मध्यकेन्द्रम्=म.

$$\text{स्प} \angle mES = \frac{p.\ \text{Sin}\ \text{म}}{\text{अ}-\text{कोज्याम}\cdot p.}$$

$$\therefore \quad E = \frac{P}{\text{अ}}.\ \text{ज्याम} + \frac{p^2}{2\text{अ}^3}.\ \text{ज्या}\ २\ \text{म} + \cdots\cdots\cdots$$

परमिह दीर्घवृत्तीयसिद्धान्तेन—

$$E = \left(२\ \text{इ} - \frac{\text{इ}^2}{४}\right)\ \text{ज्याम} + \frac{५}{४}\ \text{इ}^2\ \text{ज्या}\ २\ \text{म} +$$

$\frac{१३\ इ'}{१२}$. ज्या ३ म+.....

अतोऽत्र गोउक्रउमहाशयेन $\frac{p}{अ}$ = २ इ—$\frac{इ'}{४}$ प्रकल्पितम्, तत्र इ अस्य अत्यल्पत्वात् $\frac{इ'}{४}$ त्यज्यते तदा $\frac{p}{अ}$=२ इ.

$\therefore$ E=२ इ. ज्याम=$\frac{p}{अ}$. ज्याम.

अतोऽत्रापि ब्रह्मगुप्तमतेन—

$$\text{मन्दफज्या} = \frac{\text{ज्याअ. ज्याम}}{\text{अ}}$$

$$\text{अथवा} \quad = २\ इ.\ \text{ज्याम.},\ त्रि=१$$

$$२\ इ = \text{ज्यापरमफलम्}।$$

इदानीं रविचन्द्रयोः स्फुटत्वार्थमाह

देशान्तरे खमध्ये भुजफलचापे भुजान्तरे च कृते।
उन्मण्डलेऽर्कचन्द्रौ स्पष्टौ रविचरदले क्षितिजे ॥ १८ ॥

वा. भा.—इदानीं वा देशान्तरादिभिः संस्कारैः स्वदेशे यादृशौ रविचन्द्रौ भवतः यत्र प्रदेशे तत्प्रतिपादनायार्यामाह। लङ्कार्कोदयकालिकौ यौ रविचन्द्रौ मध्यौ तौ देशान्तरे कृते कर्मणि स्वदेशे मध्यौ भवतः यस्माल्लंकासमयाम्योत्तररेखातः प्राक् पश्चाद्वा स्वदेशेनैव भवितव्यम्। तत्र च प्रथमं पश्चाद्वार्कोदयो भवति। रेखार्कोदयादित्यत उपपन्नं भुजफलचापे भुजान्तरे च कृते, उन्मण्डलकेऽर्कचन्द्राविति भुजफलचापे स्वफलचापे स्वफले कृतेरपि स्फुटौ भवतः यस्मात् प्रतिमण्डलस्थौ स्वफलेनोपचितावपचितौ च कक्षामण्डले दृग्गतौ भवतः स्पष्टावपि तावेवाभिधीयेते। किन्तून्मण्डलसन्निधौ भवतः यस्मान्मध्यमरवेरुदयकालिकः सावनोऽहर्गणस्तस्माद्भुजान्तरे कृते उन्मण्डले स्पष्टौ भवतः यतो मध्यस्फुटरव्युदययोरन्तराद् भुजान्तरोत्पत्तिरुन्मण्डलञ्च निरक्षदेशक्षितिजमुच्यते। तस्मादुपपन्नम्।

रविचरदले क्षितिज इतिस्वसम्बन्धी यच्चरार्धं तस्मिन् कृते सति क्षितिजस्थे रवौ रविचन्द्रौ स्पष्टौ भवतः। यस्मात्स्वदेशे क्षितिजोन्मण्डलयोरन्तरं यच्चरदलं स्वाहोरात्रार्धवृत्ते तस्मादेतदप्युपपन्नम्। सर्वं गोले प्रदर्शयेत्। अत्रार्धार्या रविचन्द्रयोर्ग्रहणग्रहोपलक्षणार्था। तेनान्येऽपि ग्रहाः देशान्तरे कृते स्वदेशे मध्या भवन्ति, मन्दशीघ्रभुजफलचापद्वये भुजान्तरे च कृते उन्मण्डले

स्पष्टा भवन्ति । रविचरदले च यथासंभवं कृतेऽर्कोदयकालिका भवन्ति। यद्यर्कसावनेनानीताः अथान्यग्रहमधुनाहर्गणेनानीतास्तदा तत्सर्वं हि चरदले कृते तस्यैव ग्रहस्योदयकालिका भवन्तीति रव्यादयोऽथनक्षत्रसावनेनानीतास्ते चरदलं विना चोनाधिका भवन्तीति यदुक्तं रविचरदले कृते क्षितिजे रविचन्द्रौ भवतस्तल्लंकार्कोदयास्तमयिकावेव देशान्तरकृतौ नान्यावित्यत आर्यार्द्धेनाह ॥१८॥

वि. भा.—खमध्ये (खस्वस्तिके) ऽर्थात्स्फुटदिनार्धकाले यद्यर्कचन्द्रौ स्पष्टावपेक्षितौ तदा मध्यमरवौ चन्द्रे च भुजफलचापे (मन्दफले), भुजान्तरे च कृतेऽर्थात्संस्कृते सत्युन्मण्डले तौ स्फुटौ भवतः, तत्र रविचरदले (रविचरार्धे) संस्कृते सति क्षितिजे (स्वक्षितिजे) स्फुटौ रविचन्द्रौ भवेतामिति ॥१८॥

अत्रोपपत्तिः

यतो लङ्का याम्योत्तररेखातः पूर्वं पश्चाद्वा स्वदेशोऽतो देशान्तरसंस्कृतौ लङ्कार्कोदयकालिकौ रविचन्द्रौ स्वदेशमध्यमौ भवेताम् तत्र रेखार्कोदयात्प्रथमं पश्चाद्वा रव्युदयो भवति, तथा मन्दफलभुजान्तरयोः संस्करणेन स्वनिरक्षक्षितिजे रविचन्द्रौ स्फुटौ भवतस्तत्र रविचरार्धफलसंस्करणेन स्वक्षितिजे तौ स्फुटौ भवेतामिति । सिद्धान्तशेखरे श्रीपतिना 'अध्वकर्मणि कृते स्वमध्यमौ दोः फले रविफले इत्यादिना' ऽऽचार्योक्तानुरूपमेव रविचन्द्रयोः स्पष्टीकरणं प्रदर्शितमस्ति, भास्करतः प्राचीना आचार्याः स्वदेशोदयकालिकग्रहज्ञानार्थं पूर्वप्रतिपादितानि देशान्तरादित्रीण्येव कर्माणि कृतवन्तः, भास्करेणैकमपूर्वमुदयान्तरसंज्ञकं कर्म प्रतिपाद्य 'लङ्कायां भास्करोदये मध्याः' इति ब्रह्मगुप्तोक्तं तिरस्कृत्य 'यतोऽन्तरं तच्चलमल्पकं चे' त्यादिना पुनः समाधानं कृतमिति ॥ १८ ॥

अब रवि और चन्द्र के स्फुटत्व के लिये कहते हैं

हि. भा.—स्फुटदिनार्धकाल में यदि स्पष्टरवि और स्पष्ट चन्द्र अपेक्षित हों तो मध्यमरवि में और मध्यमचन्द्र में मन्दफल, भुजान्तर और देशान्तर इन तीनों कर्मों के संस्कार करने से उन्मण्डल में अर्थात् स्वनिरक्षक्षितिज में वे दोनों (रवि और चन्द्र) स्पष्ट होते हैं उनमें रविचरार्धफल संस्कार करने से स्वक्षितिज में स्पष्टरवि और स्पष्ट चन्द्र होते हैं इति ॥ १८ ॥

उपपत्ति

लङ्का याम्योत्तररेखा से स्वदेश पूर्व वा पश्चिम में है इसलिये लङ्कार्कोदयकालिक रवि और चन्द्र में देशान्तर संस्कार करने से स्वदेश में वे दोनों होते हैं, वहाँ रेखार्कोदय से पहले या पीछे रवि का [illegible] होता है, मन्दफल और भुजान्तर का संस्कार करने से स्वनिरक्षक्षितिज में स्पष्टरवि और स्पष्टचन्द्र होते हैं, उनमें रविचरार्ध का संस्कार करने से

स्वक्षितिज में स्पष्टरवि और स्पष्टचन्द्र होते हैं, सिद्धान्तशेखर में श्रीपति ने 'अध्वकर्मणि कृते स्वमध्यमौ' इत्यादि से आचार्योक्तानुरूप ही रवि और चन्द्र का स्पष्टीकरण किया है। भास्कराचार्य से प्राचीन आचार्यों ने स्वदेशोदयकालिक ग्रहज्ञान के लिये पूर्व प्रतिपादित देशान्तर आदि तीन ही कर्म किये हैं, भास्कराचार्य ने एक अपूर्व उदयान्तर संज्ञक कर्म कह कर 'लङ्कायां भास्करोदये मध्याः' इस ब्रह्मगुप्त कथन का अनादर कर 'यतोऽन्तरं तच्चलमल्पकं च' इससे पुनः उनके मत का समाधान किया है इति ॥१८॥

इदानीमार्यभटोक्तं स्फुटीकरणं न युक्तमित्याह

अर्कोदयास्तमययोर्विना चरार्धेन रात्रिदिनदलयोः।
न स्फुटमार्यभटोक्तं स्पष्टीकरणं स्फुटोक्तिरतः॥ १९॥

वा. भा.—लङ्कार्कोदयकालिकौ स्पष्टौ देशान्तरफलसंस्कृतौ चरदलेन यदा संस्क्रियेते तदा स्वदेशे प्राक् क्षितिजस्थे रवौ तात्कालिकौ भवतः। अथास्तमयिकौ यथोक्तौ तदपि रविचरदलकर्मणि कृते परक्षितिजस्थे रवौ तात्कालिकौ भवतः, सर्वं ग्रहाणां योज्यम्। विना चरार्धेन रात्रिदिनदलयोः यस्माद्याम्योत्तरमण्डलं चरदलवशान्न भिद्यते महत्याप्यक्षोन्नत्या सर्वमेतत् गोले प्रदर्शयेत्। इष्टकालिकास्तु ग्रहाश्चरदलकर्म विनापि स्फुटा भवन्तीति यस्मात् स्वदेशार्कोदयादस्तमयाद्वा स कालः कलित इति, आर्यभटोक्तं स्पष्टीकरणं स्फुटोक्तिरतः स्पष्टार्थमिदमार्यार्धम्।

वि. भा.—चरसंस्कारेण विनाऽर्धरात्रे दिनार्धे च रविचन्द्रौ स्फुटौ भवतः, आर्यभटकथितं स्पष्टीकरणं न स्फुटमतोऽस्मात्कारणात्स्फुटीकरणकथनं युक्तमस्तीत्याचार्येण स्वकीयस्पष्टीकरणकथने कारणं प्रतिपाद्यते॥१९॥

अत्रोपपत्तिः

स्वदेशनिरक्षदेशयोर्दिनार्धं रात्र्यर्धं चैककालावच्छेदेनैव भवत्यतस्तत्र चरार्धसंस्कारस्याऽवश्यकता न भवतीति॥१९॥

अब आर्यभटोक्त स्फुटीकरण ठीक नहीं है इसको कहते हैं

हि. भा.—चर संस्कार के विना अर्धरात्रि में और दिनार्ध में रवि और चन्द्र स्फुट होते हैं, आर्यभट का कहा हुआ स्पष्टीकरण स्फुट नहीं है। स्पष्टीकरण का कहना ठीक है इससे आचार्य अपने स्पष्टीकरण कथन में कारण कहते हैं, इति॥१९॥

उपपत्ति

स्वदेश में और निरक्ष देश में दिनार्ध और रात्र्यर्ध एक ही समय में होता है

इसलिये वहां चरार्ध संस्कार का अभाव होता है अर्थात् चरार्ध संस्कार की आवश्यकता नहीं होती है इति. ॥१९॥

इदानीं रविचन्द्रयोः स्पष्टीकरणार्थं मन्दपरिध्यंशानाह

सूर्यस्य मनुद्वितयं त्र्यंशोनं दिनदले नतस्य प्राक् ।
तिथिघटिकाभिस्त्र्यंशाधिकोनमूनाधिकं पश्चात् ॥२०॥
द्युदले जिनलिप्तोनं दशनद्वितयं द्विशरकलोनं प्राक् ।
पश्चात् युतोनमिन्दोः सूर्य इव ऋणे धने परिधिः ॥२१॥

वा. भा.—इदानीं स्फुटीकरणं विवक्षुरादौ तावद्रविचन्द्रयोः दिनदलेनार्धरात्रोदयास्तमयेषु मन्दनीचोच्चवृत्तस्य परिधिप्रमाणान्यार्याद्वयेनाह—

ऋणधनपरिध्यंशा इति सर्वत्र सम्बद्धो भवति । नायमर्थः सूर्यस्य मनुद्वितयं त्रिंशोनमृणधनपरिध्यंशाः द्युदले स्वमध्याह्ने एतदुक्तं भवति । मध्याह्ने रविमंदोच्चनीचवृत्तस्य परिधेस्त्रयोदशभागाः चत्वारिंशच्च लिप्ताःऋणकेन्द्रे धनकेन्द्रे च नतस्य प्राक्तिथिघटिकाभिरंशाधिकोनामिति स्वमध्याह्नात् पञ्चदशघटिकाभिर्यदि प्राग्नतोनो रविः स्वोन्मण्डलस्थित इत्यर्थः, तदा मुनिद्वितयमंशोनं यदुक्तं ऋणधनपरिध्यंशा द्युदलं तदेव त्रिंशाधिकोनं सत् यथासंख्यमृणधनपरिध्योरंशा भवन्ति। उन्मण्डलस्थस्य रवेर्ऋणपरिधिञ्च दशधनपरिधिस्त्रयोदश सन्ति भागा इत्यर्थः। ऊनाधिकस्वमध्यान्हात् पश्चात्कपालेन तस्य तिथिघटिकाभिरेव स्वास्तोन्मण्डलस्थितस्येत्यर्थः ।

तदेव त्रिंशोनं मनुद्वितयं त्र्यंशोनयुतं कृत्वा तत्रर्णधनपरिध्योरंशो यथासंख्यं भवति । ऋणपरिधिस्त्रयोदश सन्ति भागाः धनपरिधिश्चतुर्दशभागाः इत्यर्थः स्वार्धरात्रेऽपि द्युदलपरिधेस्तुल्या परिधिरिति चन्द्रस्यापि द्युदलं स्वमध्याह्ने जिनलिप्तोनं दशनद्वितयमिति चतुर्विंशतिलिप्तानामूनः। द्वात्रिंशद्भागाः स्वमन्दोच्चनीचवृत्तस्य ऋणधनकेन्द्रयोर्द्वयोरपि स्थितस्य परिधिर्भवति, द्विशरकलोनं प्रागिति नतस्य तिथिघटिकाभिरिति सूर्यवद्योज्यन्ते, न द्वात्रिंशद्भागा जिनलिप्तोनाद्वा पंचदशहीना लिप्तानां स्वोन्मण्डलस्थस्य चन्द्रमस ऋणधनकेन्द्रयोस्तन्मदोच्चपरिधयो भवन्ति । प्राक्पश्चाद्युतोनमिति पश्चात्पुनर्नते चन्द्रमसि घटिकाभिः स्वस्वोन्मण्डलस्येत्यर्थस्तदेव दशनद्वितयंशा भवन्ति, अतोऽपि स्वमध्यजिनलिप्तोनं द्वापञ्चाशद्धीना लिप्तानां स्वोन्मण्डलस्थस्य चन्द्रमस ऋणधनकेन्द्रयोस्तन्मन्दोच्चपरिधिभागा भवन्ति । प्राक् पश्चात् पुनर्नते चन्द्रमसि तिथिघटिकाभिः स्वास्तोन्मण्डलस्येत्यर्थः तदेव दशनद्वितयं जिनलिप्तोनं द्वापञ्चाशत्कलानां युतमूनं कृत्वा यथासंख्यं ऋणधनपरिध्योरंशाः भवन्ति ।

अत्रापि स्वमध्याह्नात् परिधिरेव स्वार्धरात्रपरिधिर्याम्योत्तरमण्डलस्यैक-

त्वात्तद्यथा सूर्यस्य ऋणकेन्द्रयोः द्युकलेर्ऽधरात्रे च परिधिः १३।४० स्वोन्मण्डले परिधिः १४ तत्रैव धनपरिधिः १३०२०। स्वास्तोन्मण्डले ऋणपरिधिः १३।२० तत्रैव धनपरिधिः १४ चन्द्रस्य स्वमध्याह्नार्धरात्रयोर्ऋणधनयोरपि परिधिः ३१३६ स्वोदयोन्मण्डले ऋणधनयोः परिधिः ३०४४ अस्तोन्मण्डलऋणपरिधिः ३२०२८ तत्रैव धनपरिधिः ३०४४ तत्र परिधेरूनाधिकत्वे उपलब्धिरेव वासना परमफलवशाद्यतो नीचोच्चवृत्तस्य महत्त्वाल्पत्वे भवति यादृगुदये फले न तादृग् मध्याह्ने नवास्तमये इष्टभूगोलोपरिस्थितत्वादित्येतद्गोले प्रदर्शयेत्। एवं द्युदलादिषु पञ्चदशघटिकान्तरितेषु कालेषु रविचन्द्रयोर्मन्दोच्चनीचवृत्तमध्यस्य स्फुटपरिधिरुक्तो वान्तरे यथास्फुटो भवति तथार्ययाह।

वि. भा.—मनुद्वितयं त्र्यंशोनं कार्यमर्थाच्चतुर्दशांशाः स्थानद्वये भागत्र्यंशेन रहितास्तदा ऋणे वा धने मन्दफले सूर्यस्य, दिनदले (मध्याह्ने) मन्दपरिध्यंशा भवन्ति, ऋणे धने वा मन्दफले दिनार्धात् प्राक्कपाले पञ्चदशघटीभिर्नतस्य सूर्यस्य दिनार्धपरिधिमानं क्रमेण भागत्र्यंशेनाधिकमूनं कार्यम्, दिनार्धात्पश्चिमकपाले पञ्चदशघटीभिर्नतस्य सूर्यस्य ऋणे धने वा मन्दफले दिनार्धपरिधिमानं क्रमेण भागत्र्यंशेनोनाधिकं कार्यं तदा प्राक्पश्चादुन्मण्डलस्थे सूर्ये तन्मन्दपरिध्यंशा भवन्ति। चन्द्रस्य ऋणे धने वा मन्दफले दशनद्वितयं जिनलिप्तोनं कार्यमर्थात्स्थानद्वये द्वात्रिंशदंशाश्चतुर्विंशतिकलाभिर्हीनास्तदा मध्याह्ने तन्मन्दपरिध्यंशा भवन्ति। सूर्य इव प्राक्कपाले पश्चिमकपाले च पञ्चदशघटीभिर्नतस्य चन्द्रस्य ऋणे मन्दफले मध्याह्ने परिधिमानं द्विशर ५२ कलोनं धने मन्दफले ताभिरेव घटीभिः प्राक्पश्चान्नतस्य चन्द्रस्य मध्याह्नपरिधिमानं द्विशर ५२ कलाभिः क्रमेण युतोनं कार्यं तदा प्राक्पश्चादुन्मण्डलस्थे चन्द्रे तन्मन्दपरिध्यंशा भवन्ति यथा—

रवेर्ऋणे मन्दफले	धने मन्दफले
मध्याह्ने मन्दपरिधिः=१३°।४०′	मध्याह्ने परिधिः=१३°।४०′
प्रागुन्मण्डलस्थे सूर्ये=१४′।०	प्रागुन्मण्डलस्थे सूर्ये=१३°।२०′
पश्चिमोन्मण्डलस्थे सूर्ये=१३°।२०′	पश्चिमोन्मण्डलस्थे सूर्ये=५४′।०′
चन्द्रस्य ऋणमन्दफले	धने मन्दफले
मध्याह्ने मन्दपरिधिः=३१°।३६′	मध्याह्ने मन्दपरिधिः=३१°।३६′
प्रागुन्मण्डलस्थे चन्द्रे=३०°।४४′	प्रागुन्मण्डलस्थे चन्द्रे=३०°।४४′
पश्चिमोन्मण्डलस्थे चन्द्रे=३२°।२८′	पश्चिमोन्मण्डलस्थे चन्द्रे=३०°।४४′

आर्यभटमतेन रवेर्मन्दपरिधिः=१३°।३०′, चन्द्रस्य मन्दपरिधिः=३१°।३०′,

एतयोर्ब्रह्मगुप्तार्यभटपठितपरिध्योर्दर्शनेन किञ्चिदन्तरं पतति, सूर्यसिद्धान्ते त्वे 'रवेर्मन्दपरिध्यंशा मनवः शीतगो रदाः, युग्मान्ते विषमान्ते च नखलिप्तोनितास्तयोः' वमस्ति, भिन्नभिन्नमन्दपरिध्यंशदर्शनेन ज्ञायते यन्मन्दान्त्यफलज्या सर्वदा न

स्थिरा यतो मध्यग्रहान्मन्दान्त्यफलज्याव्यासार्धवृत्तमेव मन्दपरिधिः, यस्याचार्यस्य समये या मन्दान्त्यफलज्योपलब्धा तदनुसारमेव मन्दपरिधिमानं स्वस्वसिद्धान्ते लिखितं तैरिति ॥२०-२१॥

अत्रोपपत्तिः

उपपत्तिसम्बन्धे चतुर्वेदाचार्येण 'अत्रोपलब्धिरेव वासनेत्यभिहितम्' भास्कराचार्येणाप्येतदनुकरणमेव कृतमिति ॥२०-२१॥

अब रवि और चन्द्र के स्पष्टीकरण के लिए मन्द परिध्यंशों को कहते हैं

हि. भा.—चौदह अंश में दो स्थानों में एक अंश के तृतीयांश (बीसकला) को घटाने से सूर्य के ऋणमन्द फल में वा धनमन्द फल में मध्यान्ह काल में मन्द परिध्यंश होता है, ऋणमन्दफल में वा धनमन्दफल में दिनार्ध से पूर्वकपाल में पञ्चदश १५ घटी कर के नत सूर्य के मध्यान्ह मन्दपरिधिमान में क्रम से एक अंश के तृतीयांश (२० कला) को युत और हीन करना चाहिए, दिनार्ध से पश्चिम कपाल में ऋणमन्दफल में और धनमन्दफल में मध्यान्ह परिधिमान में बीसकला को क्रम से हीन और युत करना तब पूर्व और पश्चिम उन्मण्डलस्थ सूर्य का मन्द परिध्यंश होता है। चन्द्र के ऋण या धन मन्दफल रहने पर बत्तीस ३२ में दो स्थानों में चौबीस २४ कला को घटा देना तब मध्यान्ह में उनके मन्द परिध्यंश होते हैं। सूर्य ही की तरह पूर्वकपाल में और पश्चिमकपाल में पञ्चदश १५ घटी करके नत चन्द्र के ऋणमन्दफल में मध्यान्ह परिधिमान में बावन ५२ कला घटा देना, धनमन्दफल में पञ्चदश घटी करके नतचन्द्र के मध्यान्ह परिधिमान में ५२ कला को क्रम से युत और हीन करना तब पूर्व और पश्चात् उन्मण्डलस्थ चन्द्र के मन्द परिध्यंश होता है। यथा.

रवि के ऋण मन्दफल में	धन मन्दफल में
मध्यान्ह में मन्द परिधि=१३°।४०′	मध्याह्न में मन्दपरिधि=१३°।४०′
प्राक् उन्मण्डलस्थ सूर्य में=१४°।०′	प्राक् उन्मण्डलस्थ सूर्य में=१३°।२०′
पश्चात् उन्मण्डलस्थ सूर्य में=१३°।२०′	पश्चात् उन्मण्डलस्थ सूर्य में=१४°।०′
चन्द्र के ऋण मन्दफल में	धन मन्दफल में
मध्याह्नमन्दपरिधि=३१°।३६′	मध्याह्नमन्दपरिधि=३१°।३६′
प्राक् उन्मण्डलस्थ चन्द्र में=३०°।४४′	प्राक् उन्मण्डलस्थ चन्द्र में=३०°।४४
पश्चात् उन्मण्डलस्थ चन्द्र में=३२°।२८′	पश्चात् उन्मण्डलस्थ चन्द्र में=३०°।४४

आर्यभट के मत से रवि की मन्द परिधि=१३°।३०′, चन्द्र की मन्द परिधि=३१°।३०′ ब्रह्मगुप्त पठित परिधिमान को और आर्यभट पठित परिधिमान को देखने से दोनों में कुछ अन्तर पड़ता है, सूर्यसिद्धान्त में—

'रवेर्मन्दपरिध्यंशा मनवः शीतगो रदाः युग्मान्ते। विषमान्ते च नखलिप्तोनितास्तयोः'

इस तरह है, भिन्न-भिन्न परिध्यंश देखने से मालूम होता है कि ग्रहों की मन्दान्त्य फलज्यायें सदा स्थिर नहीं हैं क्योंकि मध्यम ग्रह को केन्द्र मान कर मन्दान्त्य फलज्या व्यासार्ध से जो वृत्त बनता है वह मन्दनीचोच्चवृत्त परिधि है, जिन जिन आचार्य को जितनी-जितनी मन्दान्त्य फलज्या उपलब्ध हुई उसी के अनुसार मन्द परिधिमान अपने-अपने सिद्धान्त में भिन्न-भिन्न लिखे हैं इति ॥२०-२१॥

उपपत्ति

इसके विषय में चतुर्वेदाचार्य ने 'इस में उपलब्धि ही उपपत्ति है, कहा है' भास्कराचार्य ने भी इन्हीं का अनुकरण किया है इति॥२०-२१॥

इदानीमिष्टे नते स्फुटपरिध्यानयनमाह

तद्द्युदलपरिध्यन्तरगुणा हृता त्रिज्यया च नतजीवा ।
ऊने धनमृणमधिके दिनार्धपरिधौ स्फुटः परिधिः ॥ २२ ॥

वा. भा.—तदित्यनेनौदयिकस्यास्तमयिकस्य चोन्मण्डलपरिधेः परामर्शस्तस्य द्युदलपरिधेश्च यदन्तरं तत् द्युदलपरिध्यन्तरं तेन गुणा खनवजीवाहृता त्रिज्यया कार्या, एतदुक्तं भवति खे सदा परिध्यन्तरं विंशतिलिप्तास्ताभिस्त्रैराशिकं यदि त्रिज्यातुल्यया नतोत्क्रमज्यया स्वमध्याह्नेऽत्र रवौ प्राक् पश्चाद्वा परिध्यन्तमेतत्तदिष्टकालिकया नतोत्क्रमज्यया कः नते स्वमध्याह्नात् एव किमिति लब्धं यत्र फलं धनमृणं वा दिनार्धपरिधौ कार्यमनेन परिधिना सह दिनार्धपरिधेरन्तरे कृते तस्मादूने दिनार्धपरिधौ धनमधिके ऋणं कार्यम् । यस्मात् प्रतिक्षणमुपचीयतेऽपचीयते वा दिनपरिधिः, एवं कृते स्वमन्दोच्चनीचवृत्तस्य तत्र प्रदेशे स्फुटः परिधिर्भवति चन्द्रस्यापि स्वपरिध्यन्तरेण लिप्तानां द्वापञ्चाशता स्वमध्याह्नादिष्टकालनतोत्क्रमज्या च स्वमध्यपरिधिसंस्कृतः स्फुटो भवति । प्रागपरकपालयोर्धरात्राच्च नतज्यां गृहीत्वा स्वार्धरात्रपरिधिः संस्कृत इष्टकाले स्वमन्दोच्चनीचवृत्तस्य स्फुटो भवत्यतएव पञ्चदशेभ्यो घटिकाभ्योऽधिको नतः कालः त्रिंशतो विशोध्य शेषास्ते गृह्यन्ते, यस्मादुन्मण्डलमत्रावधिः परमोपचयापचययोः अतएव यत्र षट्षष्टिरक्षांशास्तत्रविमिथुनान्तार्को द्युदलपरिधेस्तुल्यस्तात्कालिकपरिधिरित्यादिकं गोले स्वाहोरात्रवृत्ते प्रदर्शयेत् सर्वत्र स्फुटपरिधिना च फलानयनं प्राग्वदिति ॥२२॥

वि. भा.—नतजीवा (इष्टनतकालज्या) तद्द्युदलपरिध्यन्तरगुणा (तच्छब्देनौदयिकस्यास्तमयिकस्य वोन्मण्डलपरिधेर्वा ग्रहणं तस्य दिनार्धपरिधेश्च यदन्तरं तेन गुणा) त्रिज्यया भक्ता लब्धं फलं दिनार्धपरिधौ धनमृणं वा कार्यं येन परिधिना सह दिनार्धपरिधेरन्तरं तस्मादूने दिनार्धपरिधौ धनमधिके ऋणं विधेयं तदाऽभीष्टस्थानीयः स्फुटः परिधिर्भवेदिति ॥ २२ ॥

अत्रोपपत्तिः

ये किल पूर्वं मन्दनीचोच्चवृत्तपरिधयः पठितास्ते दिनार्धकाल एव, ऋणे धने च फले पाक्पश्चिमोन्मण्डलस्थे रवौ चन्द्रे च ये परिधयस्तयोः (रविचन्द्रयोः), स्वस्वदिनार्धपठितपरिधिना सहैतत्परिधेर्यदन्तरं तद्वशादनुपाते-'यदि त्रिज्यातुल्यया नतकालज्यया इदं परिध्यन्तरं लभ्यते तदेष्टनतकालज्यया किमिति' नानेनेष्टपरिध्यन्तरं समागच्छति, पठितपरिधा-(दिनार्धपरिधौ) वेतद्ऋणं धनं कार्यं तदेष्टस्थानीयः स्फुटः परिधिर्भवेदेवेति, सिद्धान्तशेखरे श्रीपतिना 'तद्दिनार्धपरिधिद्वयान्तरेणाहृता स्वनतशिञ्जिनीहृता । त्रिज्ययाऽथपरिधौ दिनार्धजे हीनके स्वमधिके त्वृणं स्फुटम्' अनेनाऽऽचार्योक्तानुरूपमेव कथितमिति ॥ २२ ॥

अब इष्टनतकाल में स्फुट परिध्यानयन को कहते हैं

हि. भा.—इष्टनतज्या को प्राक् पश्चिम उन्मण्डल में रवि और चन्द्र के रहने से जो परिधि पहले कही गयी है पठित परिधि (दिनार्ध परिधि) के साथ उसका जो अन्तर है उससे गुणा कर त्रिज्या से भाग देकर जो फल हो उसको दिनार्ध परिधि में ऋण वा धन करना (जिस परिधि के साथ दिनार्ध परिधि का अन्तर करते हैं उस परिधि से दिनार्ध परिधि ऊन हो तब दिनार्ध परिधि में जोड़ना, दिनार्ध परिधि के अधिक रहने से दिनार्ध परिधि में पूर्वागत फल को ऋण करना) तब इष्टस्थानीय स्फुट परिधि होती है इति ॥२२॥

उपपत्ति

पहले जो मन्दनीचोच्चवृत्त परिधि पठित है वह दिनार्ध काल ही में, ऋण फल में और धन फल में प्राक् उन्मण्डल में और पश्चिमोन्मण्डल में रवि और चन्द्र के रहने से जो परिधि होती है उसको दिनार्ध पठित परिधि के साथ जो अन्तर होता है उसके वश से 'यदि त्रिज्यातुल्य नतकालज्या में यह परिध्यन्तर पाते हैं तो इष्टनतकालज्या में क्या' इस अनुपात से इष्टपरिध्यन्तर आता है, दिनार्ध परिधि (पठित परिधि) में इसको ऋण और धन करने से इष्टस्थानीय स्फुट परिधि होती है, सिद्धान्तशेखर में 'तद्दिनार्धपरिधिद्वयान्तरेणाहृता' इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से श्रीपति ने आचार्योक्तानुरूप ही कहा है इति ॥ २२॥

इदानीं मन्दफलस्य धनत्वमृणत्वञ्चाह

भुजफलचापं केन्द्रे षड्राश्यूने रवावृणं मध्ये ।
स्वभुजफलचापमेवं षड्राश्यधिके धनं भवति ॥२३॥

वा. भा.—इदानीं स्वमन्दोच्चनीचस्फुटपरिधिना आगतस्य फलस्य धनर्णप्रदर्शयन्नाह । मध्ये रवौ स्वकर्मोद्भवं भुजफलचापं प्रागेव प्रदर्शितम् । यत्तदृणं भवति, षड्राश्यूने स्वमन्दकेन्द्रे यस्मात् प्रथमकेन्द्रपदे फलमृणं

भवति। ततो द्वितीयपदोत्क्रमधनफलेनापचीयमानक्षयं यावत्। यावत्तावदर्धचक्रमत उक्तं षट्राश्यूने केन्द्रे स्वमृणमिति। अधिके तु पुनः केन्द्रेऽर्धचक्रवत् फलं धनं भवतीत्यनुक्तमपि ज्ञायते। ततस्तृतीयपदफलं धनम्। तच्चतुर्थपदोत्क्रमक्षयफलेनापचीयमानमपि चक्रं यावदतिरिच्यते। चन्द्रस्य तु पुनः षट्राश्यधिके केन्द्रे धनं भवति, वा तदूनं क्षयः स्वभुजफलचापं रविवासनयैव विस्तरेण पूर्वमेव क्षयधनोपपत्तौ प्रदर्शितेति, रविचन्द्रवन्नार्योयं ग्रहोपलक्षणार्था तेनान्येषामपि ग्रहाणां मन्दकर्मैव योज्यमिति।

वि. भा.—मन्दकेन्द्रे राशिषट्काल्पे भुजफलचापं (मन्दफलं) मध्ये रवावृणं भवति, एवं राशिषट्काधिके मन्दकेन्द्रे स्वभुजफलचापं (रविमन्दफलं) मध्ये रवौ धनं भवतीति ॥२३॥

अत्रोपपत्तिः

रविमन्दोच्चस्थानान्नीचपर्यन्तं स्थिते मध्यमरवौ तन्मन्दकेन्द्रस्य (रविमन्दोच्चमध्यमरवेरन्तररूपस्य) मेषादिराशिषट्कान्तर्गतत्वान्मन्दफलेन (मध्यमस्पष्टरव्योरन्तरेण) हीनो मध्यमरविः स्पष्टरविर्भवेत्। नीचादुच्चपर्यन्तं स्थिते मध्यमरवौ तत्केन्द्रस्य तुलादिराशिषट्कान्तरे विद्यमानत्वान्मन्दफलेन युक्तो मध्यमरविः स्फुटरविर्भवत्यतः सिद्धं यन्मेषादिकेन्द्रे मन्दफलमृणं तुलादिकेन्द्रे च मन्दफलं धनं भवत्यतः श्लोके षड्राश्यूने केन्द्रे (मेषादिकेन्द्रे) षड्राश्यधिके केन्द्रे (तुलादिकेन्द्रे) व्याख्येयः, सिद्धान्तशिरोमणौ भास्करेण 'तुलाजादिकेन्द्रेफलं स्वर्णमेवं मृदुज्ञेयमिति' आचार्योक्तानुरूपमेव कथ्यत इति ॥२३॥

अब मन्दफल के धनत्व और ऋणत्व को कहते हैं

हि. भा.—छः राशि से अल्प मन्द केन्द्र (अर्थात् मेषादि केन्द्र) में मन्दफल को मध्यमरवि में घटा देने से स्पष्ट रवि होते हैं। इसी तरह छः राशि से अधिक मन्द केन्द्र (अर्थात् तुलादि केन्द्र) में मध्यमरवि में मन्दफल को जोड़ने से स्पष्टरवि होते हैं इति ॥ २३ ॥

उपपत्ति

मन्दोच्च स्थान से नीच पर्यन्त मध्यमरवि के रहने से रवि मन्दकेन्द्र (मन्दोच्च और मध्यमरवि के अन्तर) के मेषादि छः राशि के अन्तर्गत होने के कारण रविमन्दफल को मध्यमरवि में घटाने से स्पष्टरवि होते हैं। नीच से उच्च पर्यन्त मध्यमरवि के रहने से रविमन्द केन्द्र के तुलादि छः राशि के अन्तर्गत होने के कारण रविमन्द फल को मध्यमरवि में जोड़ने से स्पष्टरवि होते हैं इससे सिद्ध होता है कि मेषादि केन्द्र में मन्दफल ऋण होता है और तुलादि केन्द्र में मन्दफल धन होता है, सिद्धान्तशिरोमणि में

भास्कराचार्य भी 'तुलाजादिकेन्द्रे फलं स्वर्णमेवं मृदुज्ञेयं' इससे आचार्योक्तानुरूप ही कहते हैं इति ॥ २३ ॥

इदानीं रविचन्द्रयोः स्पष्टीकरणे विशेषमाह

देशान्तराद्यमेवं स्पष्टीकरणं दिनार्धपरिधिभ्याम् ।
कृत्वा तत्तिथ्यन्तस्फुटपरिधिभ्यां स्फुटावसकृत् ॥२४॥

वा. भा.—इदानीमसकृत्कर्मप्रदर्शनार्थमार्यामाह । देशान्तराद्यं यदुक्तं तदेवमेव कृत्वा स्पष्टीकरणं तु पुनः प्रथमदिनार्धपरिधिभ्यां कृत्वा रविचन्द्रयोस्ततस्ताभ्यां तिथ्यन्तः साध्यते । तत्र तिथ्यन्ते स्वपरिधेः स्फुटीकृत्वा ताभ्यां पुनरसकृत् स्पष्टीकार्यौ । एवं तावद्यावदवशेषौ भवतः । तौ च कक्षामण्डलगौ भवत इत्यर्थः । अत्रोपपत्तिस्तात्कालिकेन परिधिना फलानयनं युज्यते । न तावत्तिथ्यन्तो ज्ञायते ततो दिनार्धपरिधिनैव स्फुटौ कृत्वा तिथ्यन्तः साध्यः । ततो ज्ञातो तात्कालिकेन परिधिना स्फुटीक्रियते, यतो नीचोच्चवृत्तपरिधिः प्रदक्षिणयाऽन्यद्देशे भवति । असकृत्कर्मवासना यथान्यस्तेषु कक्षामण्डलादिषु प्रदर्श्या तद्यथा कक्षामंडले मध्यग्रहचिन्हिते प्रदेशे यदा नीचोच्चवृत्तमध्यं क्रियते तदान्या नीचोच्चवृत्तभुजज्या भवति । यदा स्फुटग्रहप्रदेशे क्रियते तदान्यतो नीचोच्चवृत्तं पूरयति । तस्मात् प्रतिमण्डलस्थग्रहपरिज्ञानाय, प्रथममध्यमेनैवं ग्रहाणां फलानयनम् । ततो ज्ञात्वा प्रतिमण्डलस्थग्रहे तत्समसूत्रकक्षामण्डलस्थग्रहेणासकृत् यत्फलमागच्छति तन्मध्यस्फुटग्रहयोरन्तरं स्पष्टफलमपि तदेवाभिधीयते उपपन्नमिति ।

वि. भा.—रविचन्द्रयोर्दिनार्धपरिधिभ्यां देशान्तरादिस्फुटीकरणं कृत्वा ताभ्यां स्पष्टरविचन्द्राभ्यां ग्रहणे तिथ्यन्तः साध्यः, तत्तिथ्यन्तकालिकाभ्यां चन्द्रार्कयोः स्फुटपरिधिभ्यां स्फुटौ रविचन्द्रौ साध्यौ, ताभ्यां स्फुटरविचन्द्राभ्यां पुनस्तिथ्यन्तः साध्यः, पुनः स्पष्टपरिधिभ्यां स्फुटौ रविचन्द्रौ साध्यावेवमसकृत्करणेन ग्रहणोपयुक्तौ स्पष्टरविचन्द्रौ भवेतामित्याचार्याभिप्रायो भास्कराचार्येण प्रस्फुटीकृतोऽतएव सिद्धान्तशिरोमणौ 'मुहुः स्फुटाऽतो ग्रहणे रवीन्द्वोस्तिथिस्त्विदं जिष्णुसुतो जगाद' भास्करेण कथ्यत इति ॥ २४ ॥

अत्रोपपत्तिः

आचार्योक्तं नतकर्म सकृत्प्रकारेणापि भवितुमर्हति । यथा गणितागततिथ्यन्तकालासकृत्प्रकारसाधितनतकर्मसंस्कृतरविचन्द्रोत्पन्नतिथ्यन्तकालयोरन्तर्गतघटीमानम्=य, एतत्सम्बन्धिनोंऽशाः=६य, गणितागततिथ्यन्तकाले रविनतकालांशा=न, अनयोः संस्कारेण वास्तवनतकालांशाः=न± ६य, ततोऽनुपातो यदि घटीषष्ट्या रविचन्द्रयोर्गत्यन्तरकला लभ्यन्ते तदा 'य' घटीभिः काः समागता 'य' घटीषु रविचन्द्रान्तरकलाः=$\frac{\text{य (चंगक—रगक)}}{६०}$=य×ग

$\frac{\text{चंगक}-\text{रगक}}{६०}$ = ग· 'तिथ्यन्तनाडीनतबाहुमौर्व्या' इत्यादि

भास्करोक्तप्रकारेण सूर्यस्य नतकर्म = $\frac{\text{रफ}\times\text{ज्या}\ (\text{न}\pm\text{६य})}{४९२०}$ = रफ$_१$ × ज्या (न ± ६य) अत्र $\frac{\text{रफ}}{४९२०}$ = रफ$_१$ तथा चन्द्रस्य नतकर्म = $\frac{\text{चंफ}\times\text{ज्या}\ (\text{न}\pm\text{६य})}{४३७५}$ = चंफ$_१$ × ज्या (न ± ६य) अत्र $\frac{\text{चंफ}}{४३७५}$ = चंफ$_१$

अनयोः संस्कारः पूर्वसाधितान्तरेण तुल्यो भवेदन्यथा गणितागतयोर्नत-कर्मसंस्कृतयोश्च रविचन्द्रयोः 'य' घटयन्तरे कथं समानमन्तरमुत्पद्येत। ततः ग × य = चंफ$_१$ × ज्या (न ± ६य) − रफ$_१$ × ज्या (न ± ६य)

= (चंफ$_१$ − रफ$_१$) × ज्या (न ± ६य), अतः $\frac{\text{ग}}{\text{चंफ}_१-\text{रफ}}$ = $\frac{\text{ज्या}\ (\text{न}\pm\text{६य})}{\text{य}}$ = परः = प, ततोऽनुपातो यदि दशानामंशानां ज्या = २१ तदै '६य' तत्तुल्यांशानां का जाता ज्या ६य = $\frac{२१\times\text{६य}}{\text{य}}$, एवमेव $\frac{२१\times\text{३य}}{१०}$ = ज्या ३य = $\sqrt{\frac{\text{त्रि}\times\text{उज्या}\text{६य}}{२}}$ = $\sqrt{६०\times\text{उज्या}\text{६य}}$ अतो विलोमेन $\frac{२१^२\times३^२\times\text{य}^२}{१२^२\times६०}$ = उज्या ६य, तथा चापयोरिष्टयोरित्यादिना.

ज्या (न ± ६य) = $\frac{\text{ज्यान}\times\text{कोज्या}\text{६य}\pm\text{कोज्यान}\times\text{ज्या}\text{६य}}{\text{त्रि}}$ = ज्यान − $\frac{\text{ज्यान}\times\text{उज्या}\text{६य}}{\text{त्रि}}$ ± $\frac{\text{कोज्यान}\times\text{ज्या}\text{६य}}{\text{त्रि}}$ = ज्यान − $\frac{\text{ज्यान}\times२१^२\times३^२\times\text{य}^२}{१०^२\times६०\times१२०}$ ± $\frac{\text{कोज्यान}\times२१\times\text{६य}}{१०\times१२०}$

= ज्यान − $\frac{\text{ज्यान}\times\text{य}^२}{(\frac{२००}{३६})^२\times२}$ ± $\frac{\text{कोज्यान}\times\text{य}}{\frac{२००}{३६}}$ = ज्यान − $\frac{\text{ज्यान}\times\text{य}^२}{२\text{हा}^२}$ ± $\frac{\text{कोज्यान}\times\text{य}}{\text{हा}}$ अथ $\frac{\text{ज्या}\ (\text{न}\pm\text{६य})}{\text{य}}$ = प = $\frac{\text{ज्यान}}{\text{य}}$ − $\frac{\text{ज्यान}\times\text{य}}{२\text{हा}^२}$ ± $\frac{\text{कोज्यान}}{\text{हा}}$

पक्षौ 'हा' गुणितौ तदा प × हा = ध्रुवः = ध$_०$ = $\frac{\text{ज्यान}\times\text{हा}}{\text{य}}$ − $\frac{\text{ज्यान}\times\text{य}}{२\text{हा}}$ ±

कोज्यान,पक्षयोः समशोधनेन तथा ज्यान भक्तेन च $\frac{\text{ध्रु}\mp\text{कोज्यान}}{\text{ज्यान}}=\frac{\text{हा}}{\text{य}}-\frac{\text{य}}{२\,\text{हा}}$

छेदगमेन २ हा^{२}—य^{२}= ±अ×हा×य×२ अत्र $\frac{\text{ध्रु}\mp\text{कोज्यान}}{\text{ज्यान}}$ = ±अ= अन्यः समशोधनेन.

य^{२}±२ अ×हा×य=२ हा^{२} ततो वर्गपूर्त्तिकरणेन य^{२}±२ अ×हा×य +अ^{२}×हा^{२}=२हा^{२}+अ^{२}×हा^{२}=हा^{२} (अ^{२}+२)

मूलेन य±अ×हा=हा$\sqrt{\text{अ}^{२}+२}$

$$\therefore\ \text{य}=\text{हा}\,(\sqrt{\text{अ}^{२}+२}\mp\text{अ})$$

एतेन म.म.पण्डितसुधाकरद्विवेदिनः सूत्राण्यवतरन्ति--

गत्यन्तरकलाः षष्टिभक्ता गत्यन्तरं भवेत्, ।
फललिप्ताः स्वहाराप्ता रवीन्द्वोश्च फलं क्रमात् ॥
गत्यन्तरं फलवियोगहृतं विधोः प्राक्,
स्वे तत्फलेऽन्यसमये युतिहृत् परास्यम् ।
खाभ्राश्विनो विधुकरैर्विहृता हरस्तन्निघ्नं,
परेण भवति ध्रुवसंज्ञकं तत् ॥
स्वीय ध्रुवो नतजकोटिगुणेन हीनो,
मौर्व्या नतासु भवया विहृतोऽपरोऽस्य ।
वर्गात्पदं करयुतादपरो नितं तद्धारघ्न-
मेवमिह दण्डमुखं विधोः प्राक् ॥
अस्वे फले रविफलात् स्वफलस्य बाल्ये,
हीनान्यथा च सहितेष्टफलेन नूनम् ।
तिथ्यन्तदण्डमितिरत्र भवेत्स्फुटा सा,
प्राक् चेद्विधोर्घनफलाल्पमथात्र सौरम् ॥
तर्हि ध्रुवात् सनतकोटिगुणादिहान्यः,
साध्यो विदा गणितगोलविदा मुदैव ।
श्री ब्रह्मगुप्तनतकर्म भवेत् सुसूक्ष्ममेवं,
सकृत् सकलसज्जनरञ्जनार्थम् ॥

उदयान्तरभुजान्तरादिसंस्करणेन यो हि स्पष्टग्रहः समायच्छति वस्तुतो नहि स्पष्टग्रह इति ब्रह्मगुप्तोक्तनतकर्म पादानेन तत्साधनकर्तुर्भास्कराचार्यस्य लेखः

दपि ज्ञायते, सर्वे आकाशस्था ग्रहादयो भूवायुगोले परिणता अस्माकं प्रत्यक्षीभूताः स्पष्टा भवन्ति, तेन स्पष्टीकरणेन ये स्पष्टग्रहाः समागच्छन्ति तेषु यावता संस्कारेण भूवायुगोलपरिणताः स्पष्टग्रहा भवेयुस्तस्यैव नाम नतकर्म; ब्रह्मगुप्ततः प्राचीनाः सूर्यसिद्धान्तकाराऽऽर्यभटप्रभृतिभिरेतत्सम्बन्धे स्वस्वसिद्धान्तेन किमपि लिखितवन्तः। मन्मते नतकर्मसंस्कारस्यातीवाऽऽवश्यकता प्रतीयते, विषयेऽस्मिन् सारासारविचारदक्षा ज्यौतिषिका भृशं विचारयन्त्विति ॥२४॥

अब रवि और चन्द्र के स्पष्टीकरण के सम्बन्ध में विशेष कहते हैं।

हि. भा.—रवि और चन्द्र का दिनार्ध परिधियों से देशान्तरादि द्वारा स्फुटीकरण कर के उन स्पष्टरवि और स्पष्टचन्द्र से ग्रहण में तिथ्यन्त साधन करना,उस तिथ्यन्त कालिक रवि और चन्द्र के स्पष्ट परिधियों से स्पष्ट रवि और स्पष्ट चन्द्र साधन करना, उन स्पष्ट रवि और स्पष्ट चन्द्र से पुनः तिथ्यन्त साधन करना, पुनः स्पष्ट परिधियों से स्पष्टरवि और स्पष्ट चन्द्र साधन करना, इस तरह वार-वार करने से ग्रहण के लिये उपयुक्त स्पष्ट रवि और स्पष्ट चन्द्र होते हैं, आचार्य जी के इस अभिप्राय का भास्कराचार्य ने विशदरूप से प्रतिपादन किया है इसीलिये सिद्धान्त शिरोमणि में 'मुहुः स्फुटाऽतो ग्रहणे रवीन्द्वोस्तिथिस्त्विदं जिष्णुसुतो जगाद' भास्कराचार्य कहते हैं ॥२४॥

उपपत्ति

आचार्योक्त नतकर्म सकृत्प्रकार से भी हो सकता है जैसे गणितागत तिथ्यन्त काल और असकृत्प्रकार से साधित नत कर्म संस्कृत रवि और चन्द्र से उत्पन्न तिथ्यन्त काल के अन्तर घट्यात्मक मान मानते हैं (य) एतत्सम्बन्धि अंश $=६ य$, गणितागत तिथ्यन्त काल में रवि के नत कालांश $=न$, इन दोनों के संस्कार करने से वास्तवनतकालांश $=न\pm६य$, तब अनुपात करते हैं, यदि साठ घटी में रवि और चन्द्र की गत्यन्तर कला पाते हैं तो (य) घटी में क्या इस से य घटी में रवि और चन्द्र की अन्तर कला आती है,

$\frac{य\ (चंगक—रगक)}{६०}=य\times ग$, यहां $\frac{चंगक—रगक}{६०}=ग$. । 'तिथ्यन्तनाड़ीनतबाहुमौर्व्या इत्यादि' भास्करोक्तप्रकार से सूर्य के नतकर्म $=\frac{रफ\times ज्या\ (न\pm६य)}{४१२०}=रफ_१\times ज्या$ $(न\pm६य)$, यहाँ $\frac{रफ}{४१२०}=रफ_१$ तथा चन्द्र के नतकर्म $=\frac{चंफ\times ज्या\ (न\pm६य)}{४३७५}$ $=चंफ_१\times ज्या\ (न\pm६य)$, यहां $\frac{चंफ}{४३७५}=चंफ_१$ इन दोनों का संस्कार पूर्व साधित अन्तर के बराबर होता है अन्यथा गणितागत रवि और चन्द्र के अन्तर और नतकर्म संस्कृत रवि और चन्द्र के अन्तर य घटयन्तर में कैसे समान अन्तर को बनायेगा, अतः $ग\times य\ चंफ_१\times ज्या$

$(न \pm ६य) - {}_{१}रफ \times ज्या (न \pm ६य) = (चंफ_{१} - रफ_{१}) \times ज्या (न \pm ६य)$ इसलिये $\frac{ग}{चंफ_{१} - रफ_{१}}$

$= \frac{ज्या (न \pm ६य)}{य} = पर = प$, तब अनुपात करते हैं यदि दश अंश की ज्या = २१ पाते हैं तो '६ य' एतत्तुल्य अंश की क्या आ गयी '६य' इसकी ज्या, ज्या $६य = \frac{२१ \times ६य}{१०}$,

इसी तरह $\frac{२१ \times ३य}{१०} = ज्या ३य = \sqrt{\frac{त्रि \times उज्या ६य}{२}} = \sqrt{६० \times उज्या ६य}$ इसके

विलोम से $\frac{२१^{२} \times ३^{२} \times य^{२}}{१०^{२} \times ६०} = उज्या ६य$; चापयोरिष्टयोरित्यादि से ज्या $(न \pm ६य)$

$= \frac{ज्यान \times कोज्या ६य}{त्रि} \pm \frac{कोज्यान \times ज्या ६य}{त्रि} = ज्यान - \frac{ज्यान \times उज्या ६य}{त्रि}$

$\pm \frac{कोज्यान \times ज्या ६य}{त्रि} = ज्यान - \frac{ज्यान \times २१^{२} \times ३^{२} \times य^{२}}{१०^{२} \times ६० \times १२०} \pm \frac{कोज्यान \times २१ \times ६य}{१० \times १२०}$

$= ज्यान - \frac{ज्यान \times य^{२}}{\left(\frac{२००}{२१}\right)^{२} \times २} \pm \frac{कोज्यान \times य}{\frac{२००}{२१}} = ज्यान - \frac{ज्यान \times य^{२}}{२ हा} \pm \frac{कोज्यान \times य}{हा.}$,

$\frac{ज्या (न \pm ६य)}{य} = पर = प = \frac{ज्यान}{य} - \frac{ज्यान \times य}{२ हा^{१}} \pm \frac{कोज्यान}{हा}$ दोनों पक्षों को हा गुणने से

$प \times हा = ध्रुव = ध्रु = \frac{ज्यान \times हा}{य} - \frac{ज्यान \times य}{२ हा} \pm कोज्यान$ दोनों पक्षों में समशोधन

करने से तथा ज्यान से भाग देने से $\frac{ध्रु \mp कोज्यान}{ज्यान} = \frac{हा}{य} - \frac{य}{२ हा}$ छेदगम से $२ हा^{१} - य^{१}$

$= \pm अ \times हा \times य \times २$। यहां $\frac{ध्रु \mp कोज्यान}{ज्यान} = अन्य = \pm अ$ समशोधन करने से $य^{२} \pm २अ \times$

$हा \times य = २ हा^{१}$ वर्ग पूर्तिकरने से $य^{२} \pm २अ \times हा \times य + अ^{२} \times हा^{२} = २ हा^{२} + अ^{२} \times हा^{१}$

$= हा^{२} (अ^{२} + २)$ मूल लेने से $य \pm अ \times हा = हा \sqrt{अ^{२} + २}$ $\therefore य = हा (\sqrt{अ^{२} + २} \mp अ)$

इससे म. म. सुधाकर द्विवेदी जी के सूत्र उपपन्न होते हैं जो कि संस्कृतोपपत्ति में लिखे हुये हैं। उदयान्तर भुजान्तरादि संस्कार से जो स्पष्टग्रह होते हैं वे वस्तुतः स्पष्टग्रह नहीं होते हैं यह विषय ब्रह्मगुप्त के नतकर्म कहने से तथा नतकर्म साधन-

कारक भास्कराचार्य के लेख से मालूम होता है, आकाशस्थ ग्रहादि भूवायु गोल में परिणत होकर हम लोगों को प्रत्यक्षीभूत (स्पष्ट) होते हैं, इसलिये स्पष्टीकरण से जो स्पष्टग्रह आते हैं उनमें जितना संस्कार करने से भूवायु गोल परिणत स्पष्टग्रह हो उसी को नतकर्म कहना उचित है, ब्रह्मगुप्त से प्राचीन सूर्यसिद्धान्तकार, आर्यभट प्रभृति ने इसके विषय में अपने-अपने सिद्धान्त में कुछ नहीं लिखा है। मेरे मत में नतकर्म संस्कार की अतीव आवश्यकता है, इस विषय के उपर विज्ञ ज्यौतिषिक लोग विचार करें इति ॥ २४ ॥

इदानीं ग्रहणे सूर्याचन्द्रमसोर्नतकालमाह

प्राक् पश्चाद्वा याभिर्घटिकाभिर्दिनदलान्नतः सूर्यः।
तिथ्यन्ते तद्विहितं त्रिंशद्घटिकावशेषाभिः ॥ २५ ॥
विपरीतमर्धरात्रान्नेन्द्रग्रहणे शशी रविग्रहणे।
सूर्यो यतो नतस्ताभिरेव घटिकाभिरिन्दुरपि ॥ २६ ॥

वा. भा.—पूर्वेण परेण वा स्वदिनार्धाद्यावतीभिर्घटिकाभिर्नतः सूर्यः तिथ्यन्ते पौर्णमास्यन्ते ताभिः रहितास्तद्रहिताश्च तास्त्रिंशद्घटिकाश्च तद्रहित-त्रिंशद्घटिका ताभ्यो या अवशेषाः ताभिर्घटिकाभिश्चन्द्रग्रहणे शशी नतो भवति, विपरीतमर्धरात्राद्यदि रविः स्वदिनार्धपूर्वनतस्तदा चन्द्रो रविसम्बन्धिनोऽर्धरात्राद् परेण रविः पश्चाच्चन्द्रः पूर्वेणातः तुल्यो नतकालः, एकैव नतज्या स्वपरिधे. संस्कारार्थं चन्द्रग्रहणे कार्येत्यर्थः। रविग्रहणे पुनः सूर्यो यतः पूर्वेण परेण वा नतो यावतीभिर्घटिकाभिरमावस्यान्ते ततएवैतावतीभिरेव घटिकाभिश्चन्द्रो नतो भवत्यतोऽत्रापि नतकालस्तुल्य एव। तस्मादेकैव नतज्या परिध्योः संस्कारार्थं कार्येत्यत्र वासनागोले प्रदर्श्य स्वाहोरात्रवृत्तयोः स्वदिनार्धाद् याम्योत्तरमण्डल-स्वाहोरात्रवृत्तसंपातात्प्रभृति रविरर्धरात्रं यावत् पुनरपि स्वाहोरात्रयाम्योत्तर-संपातं यावद्यदा नतो भवति। तदा त्रिंशद्घटिकानतः सूर्यो भवति। चन्द्रश्च तदा स्वमध्याह्नात् याम्योत्तरमण्डलस्वाहोरात्रमण्डलयोः संपातो भवति। तत्र च त्रिंशद्घटिकाभ्योऽवशेषाः घटिकाः सूर्यनतो नास्त्यतो नताभावश्चन्द्रस्य रवेश्चार्धरात्रे नताभावोऽनुक्त एव ज्ञायते। स्वदिनार्धापेक्षया त्रिंशदुघटिकाभिर्नतः, इत्युच्यते। यदा तु पुनरर्धरात्रे रविः पश्चिमतो भवति। तदा चन्द्रोऽपि स्वमध्या-ह्नात्पूर्वेण तावत्येव प्रदेशे स्वाहोरात्रवृत्ते भवति। तुल्याभिरेव घटिकाभिः, तिथ्यन्ते चायं नियमो यस्मादर्धचक्रान्तरितौ तदा रविचन्द्रौ भवतोऽन्यत्र काले ऽन्यदर्धरात्रं रवेरन्यदिनार्धशशिनः तस्मादुक्तं तिथ्यन्तमिति। यदा रविः पूर्वेण स्वार्धरात्राद् भवति तदा स्वदिनदलाच्चन्द्रः पश्चिमतो भवति। तावतीभिरेव घटिकाभिरर्धचक्रान्तरितो यतः पौर्णमास्यन्ते अतः सर्वमुपपन्नम्। शशि-ग्रहणे रविग्रहणे च तयोर्लब्धे पृथक्कले एव स्वफलविकलानामागतानां धन-कलानां फलविकला चासुरेकत्र स्थितेः। मूर्खोऽपि जानाति एकएव नतकालो भवति। तस्मादत्राप्युपपन्नम्। यथा स्वे स्वाहोरात्रवृत्ते प्रदर्शयेदिति।

वि. भा.—चन्द्रग्रहणे तिथ्यन्ते (पूर्णान्तिकाले) दिनदलात् (मध्याह्न-कालात्) प्राक्पश्चाद्वा याभिर्घटिकाभिः (यस्मिन्नाभिर्घटीभिः) सूर्यो नतो भवति तत्सूर्यस्य नतकालमानं विहितं भवति । त्रिंशद्घटिकावशेषाभिस्ताभिर्घटिकाभि-र्विपरीतं (विलोमं) अर्धरात्रात् शशी (चन्द्रः) नतो भवति, अर्थाद्याभि-र्घटिकाभिस्तिथ्यन्ते रविर्नतस्ताभ्यस्त्रिंशद्घटिका विशोध्यावशिष्टं चन्द्रस्यार्धरात्रा-द्विपरीतं नतं भवति यदि रवेः प्राक् तदा चन्द्रस्य पश्चात् यदि रवेः पश्चात्तदा चन्द्रस्य प्रागर्धरात्रान्नतं भवति । रविग्रहणे याभिर्घटिकाभिः सूर्यो नतो भवति ताभिरेव घटिकाभिश्चन्द्रोऽपि तस्मिन्नेव कपाले नतो भवतीति ॥२५-२६॥

अत्रोपपत्तिः

पूर्णान्तिकाले चन्द्रग्रहणं भवति, पूर्णान्ते च रविचन्द्रौ षड्भान्तरितौ भवतोऽत ऊर्ध्वयाम्योत्तरवृत्ताद्याभिर्घटीभिर्नतो रविस्त्रिंशता रहिताभिस्ता-भिर्घटीभिर्विपरीतकपाले चन्द्रो नतो भवति, सूर्यग्रहणे सूर्याचन्द्रमसावेकराश्याद्य-वयवे स्थितौ भवतस्तेनैककपाले तुल्या एव नतघटिका भवन्तीति सर्वं ज्योतिर्विदा-मतिरोहितमेवेति सिद्धान्तशेखरे 'प्राक् पश्चाद्वा दिवसशकलाद्याभिरर्को घटीभि-स्तिथ्यन्ते स्यान्नत उडुपतिस्ताभिरेवार्धरात्रात् । व्यस्तं चन्द्रग्रहणसमये वाऽभ्रराम ३० च्युताभिः सूर्यग्रासे रविरिव विधुः स्यान्नतः प्राक्प्रतीच्योः श्रीपत्युक्तमिदमा-चार्योक्तानुरूपमेवेति गणकैर्विभाव्यम् ॥२५-२६॥

अब ग्रहण में सूर्य और चन्द्र के नतकाल को कहते हैं

हि. भा.—चन्द्रग्रहण में पूर्णान्तकाल में मध्याह्नकाल से पहले या पीछे जितनी घटी में रवि नत होता है वह सूर्य का नतकाल मान कथित है, तीस ३० में उस घटिका को घटाने से जो शेष रहता है उतनी ही संख्या करके चन्द्रनत होता है, किन्तु अर्धरात्र से विलोम अर्थात् तिथ्यन्त में जितनी घटी में रवि नत होता है उनमें तीस ३० घटी को घटाकर जो शेष रहता है वह चन्द्र का नत होता है, यदि रवि का प्राक्नत है तो चन्द्र का पश्चिमनत, यदि रवि का पश्चिमनत है तो चन्द्र का प्राक्नत अर्धरात्र से होता है, सूर्यग्रहण में रवि जितनी घटी करके नत रहता है उतनी ही घटी करके उसी काल में चन्द्र भी नत होता है इति ॥२५-२६॥

उपपत्ति

चन्द्रग्रहण पूर्णान्तिकाल में होता है। पूर्णान्तिकाल में रवि से चन्द्र छः राशि के अन्तर पर रहता है इसलिये ऊर्ध्वयाम्योत्तरवृत्त से जितनी घटी करके रविनत रहता है, उन घटी में तीस को घटा करके जो शेष रहता है उतनी घटी करके विपरीत कपाल में चन्द्रनत होता है। सूर्यग्रहण में सूर्य और चन्द्र एक ही राश्याद्यवयव में रहते हैं इसलिये एक कपाल में दोनों की तुल्य ही नत घटी होती है, सिद्धान्तशेखर में श्रीपति ने आचार्योक्तानु-रूप ही कहा है। उनके पद्य संस्कृतोपपत्ति में देखिये इति ॥२५-२६॥

इदानीं प्रकारान्तरेण नतकर्माह

दिनदलपरिधिस्फुटतिथिनतकेन्द्रज्यावधो गुणोऽर्केन्द्वोः ।
इन्द्वतिधृति १९१ भिर्नवनववेदै ४९९ र्व्यासार्धकृति १०६९२९०० भक्तः ॥२७॥
फलविकला वा सूर्ये प्रागृणमसकृन्नते धनं पश्चात् ।
केन्द्रफलमृणं चन्द्रेऽन्यथा धनं प्रागृणे स्पष्टौ ॥२८॥

वा. भा.—दिनदले यौ परिधी रविचंद्रयोरुक्तौ ताभ्यां यौ स्फुटौ ताभ्यां या तिथिस्तिथ्यन्त इत्यर्थः । तस्यास्तिथेर्यो नतः कालस्तस्य योत्क्रमज्या दिनदलपरिधिस्फुटतिथिः, नतज्या तस्याः तात्कालिककेन्द्रज्यायाश्च वधः केन्द्रसम्बन्धे यथासंख्यमिच्छति, इन्द्वतिधृतिगुणको यो नवनववेदैश्च पृथक् तदुभयतोपि व्यासार्धकृतिभक्तः कार्यः फलविकलात् पृथक् एतदुक्तं भवति । स्वमध्याह्नपरिधिना प्रथमचन्द्रार्कौ स्फुटौ कृत्वा ततस्ताभ्यां तिथ्यन्तं साधयेत् । तत्र तिथ्यन्ते स्वदिनदलादर्वाक् पुनरनयोर्वा यावत्यो नतघटिकाः तासां प्राणीकृतानामुत्क्रमज्या ग्राह्याः, यत्र पञ्चदशघटिका नतघटिका भवन्ति तदा त्रिंशतो विशोध्य शेषा नता स्वार्धरात्रात् परिकल्प्यास्तासामुक्तवज्ज्या कार्या सा नतज्या भवति । ततस्तिथ्यन्ते मध्यौ रविचन्द्रौ स्वोच्चसहितौ कृत्वा तद् दिवार्धपरिधिभ्यामेव स्फुटीकार्यौ तयोः स्फुटीक्रियमाणे ये केन्द्रे भवतस्ताभ्यां ज्ये ग्राह्ये । ततो रविकेन्द्रज्या नतजीवया गुणात् पुनरिद्वतिधृतिभिर्गुणयेत् ततो व्यासार्धकृत्या विभजेत् । लब्धं फलं विकला एवं चन्द्रकेन्द्रज्या नतज्याहृतान्नवनववेदैर्गुणयेत् । ततस्त्रिज्या कृत्या विभजेत् । फलं विकलाः एवं ततः स्वकेन्द्रज्याकरणे यज्ज्यांतरप्रभवाद् गुणकारः तेन स्वकेन्द्रभुक्ति संगुण्य तत्त्रयमैर्विभजेत् । फलं स्वकेन्द्रभुक्तिर्भवति । ततः केन्द्रभुक्तिज्ये पृथक् द्वेऽपि नतज्याहते स्वगुणकारगुणिते कृत्वा त्रिज्या कृत्या विभजेत् । लब्धे पृथक्कले विकलात्मके एव स्वफलविकलानामागतानां धनकलानां फलविकला सूर्ये प्रागृणमसकृन्नते धनं पश्चात् । वा शब्दोऽत्रचन्द्रार्कयोः स्फुटीकरणप्रकाराय । तेनायमर्थः प्राग्नते सूर्ये फलविकला ऋणं भवति । पश्चान्नते धनं प्राक्कपाले नते रवावृणम् । अपरकपालस्थे धनमित्यर्थश्चन्द्रे तु पुनर्यदिकेन्द्रफलमृणं कृतं तदन्यथा प्राग्धनमपरकपालस्थे क्षय इत्यर्थः । स्वफलविकलाः । अथ चन्द्रकेन्द्रफलं धनं कृतं । तर्कफलं विकला वा ऋणं भवति । सर्वथा कपालनिरपेक्षायां एवं कृते कर्मणि स्पष्टौ भवतः । एतच्च कर्म्मासकृत्कर्तव्यम् । तत्करणे स्फुटाभ्यां तिथ्यन्तः पुनर्साध्यः तिथ्यन्ते पुनः एतदेव कर्म तावद्यावद् विशेषो भवतः । अत्रोपपत्तिः—तद्यथा सर्वदा स्वे परिध्यंतरं लिप्ता विंशति(२०) एताभिः परमभुजज्या त्रिज्या तुल्या संगुणिता (६५४००)अतो भगणं भगणांशैर्भगे हृते लब्धं विकला(१८१)अंशा वापि विकला १९१० एतावद्भवेः परमं फलान्तरं चंद्रस्यापि परिध्यन्तरं द्वापंचाशल्लिप्ता ५२, एताभिर्व्यासार्धतुल्या परमभुजज्या गुणिता जाता १७०००४० अतो भगणां-

शैर्हृते लब्धं विकलाः ४७२ आसां धनुर्नवनवेदाः ४९९ एतावच्चन्द्रस्य परमं फलान्तरमेतच्च तदा संभवति, यदा रविचन्द्रयोरुन्मण्डलस्थयोः स्वकेन्द्रज्या त्रिज्या तुल्या भवति। नतज्या च त्रिज्यातुल्यैव। अतस्त्रैराशिका कल्पना वाचार्येण कृता। यदि त्रिज्या तुल्यैव, यदा स्वकेन्द्रनतज्ये भवतः तदा रवेर्विद्रतिधृतितुल्यविकलाः फलान्तरं भवति। यदा पुनरिष्टकालनतज्या स्वकेन्द्रज्ये इष्टप्रामाणिके तदा कियत् फलांतरमिति फलं विकला चन्द्रस्यापि नतकेन्द्रज्याबधे त्रिज्याकृति तुल्यछेदे नवनवेदसंख्या विकला भवन्ति। फले तावत्, इष्टनतकेन्द्रज्ययोर्वधेन किमिति फलं विकला भवन्ति। केन्द्रभुजज्ययोरप्येवमेव तस्मादुपपन्नम्। तद्विकलारूपफलं प्राक्कपालस्थे सूर्ये विशोधयेत्। यस्मात्तात्कालिकपरिधिदिनार्धपरिधेः-कोऽत्र फलं विशोध्यते। यतः प्रागृणमल्पं कृते तिष्ठति तदायचीयते येन तात्कालिकपरिधिना स्फुटो रविर्भवति। क्षयकेन्द्रधनकेन्द्रोज्यूना युतोदिनार्धपरिधेस्तत्परिधिरूनोऽधिकं धनकृते तिष्ठति। विशोध्यते पश्चात् कपालवैपरीत्येन रवेर्यतो धनं क्रियते चन्द्रस्यापि ऋणं केन्द्रमन्यथा प्राक्कपाले दिनार्धपरिधेरूनस्तात्कालिकः परिधिरतोऽधिकमृणं कृतं तिष्ठति पुनः दीयते। अपरकपाले वैपरीत्येन धने हीन इति। यतः प्राक्कपाले दिनार्धपरिधेरूनस्तात्कालिकपरिधिरतोऽधिकं धनं कृतं तिष्ठति तद्विशोध्यते। तस्मात्सर्वमुपपन्नम्। प्रागुक्तं भुजफलचापे भुजान्तरे कृते ग्रहे उन्मण्डले स्पष्टो भवति ॥२७-२८॥

वि. भा.—दिनदलपरिधिभ्यां यौ स्फुटरविचन्द्रौ ताभ्यां या तिथिः (तिथ्यन्तः) तत्र यो नतः (नतकालः) तस्य केन्द्रस्य च ज्ययोर्वधः (घातः) कार्यः स च यदि रवेस्तदै ५९१ भिर्गुणितः, यदि चन्द्रस्य तदै ४९९ भिर्गुणितस्त्रिज्यावर्ग १०६९२९०० भक्तः फलं विकला भवन्ति, एतत्फलं प्राक्नते सूर्ये ऋणं पश्चान्नते धनं कार्यम्। चन्द्रे तत्पूर्वागतं विकलात्मकं प्राक्कपाले मन्दफले ऋणात्मके धनं कार्यमन्यथा प्राक्पश्चिमे ऋणमेव कार्यमेवमसकृत्कार्यं तदा वा स्फुटौ रविचन्द्रौ भवतः, पूर्वं 'तद्द्युदलपरिध्यन्तरगुणा' इत्यादिना नतकर्मानीतमधुना पुनस्तत्साधितमतो 'वा' शब्दः प्रयुक्त इति ॥ २७-२८ ॥

अत्रोपपत्तिः

दिनार्धपरिधिना पूर्वप्रकारेण मन्दफलकला=$\frac{२२५ \text{ दिप·ज्याके}}{२१४\times३६०}$, अत्र दिप=दिनार्धपरिधिः ज्याके=मन्दकेन्द्रज्या.विकलात्मककरणेन$\frac{२२५\cdot\text{दिप}\cdot\text{ज्याके}\times६०}{२१४\times३६०}$,

रविपरिध्यन्तरम् =$\frac{\text{ज्यान}}{३\ \text{त्रि}\cdot}$ ततोऽनुपातो यदि दिनार्धपरिधिना मन्दफलविकला लभ्यन्ते तदाऽऽनीतपरिध्यन्तरेण किं समागच्छति रविपरिध्यन्तर-

सम्बन्धिनी फलविकला $= \frac{६० \text{ दिप} \cdot २२५ \cdot \text{ज्याके} \cdot \text{ज्यान}}{३ \times २१४ \times ३६० \times \text{त्रि} \times \text{दिप}} =$

$= \frac{२० \times २२५ \text{ ज्याके} \cdot \text{ज्यान} \cdot \text{त्रि}}{२१४ \times ३६० \times \text{त्रि}^२} = \frac{२२ \times ३२७० \text{ ज्याके} \cdot \text{ज्यान}}{१८ \times २१४ \times \text{त्रि}^२}$

$= \frac{२५ \times १६३५ \text{ ज्याके} \cdot \text{ज्यान}}{२१४ \times \text{त्रि}^२} = \frac{४०८७५}{२१४} \times \frac{\text{ज्याके} \cdot \text{ज्यान}}{\text{त्रि}^२} = \frac{१९१ \text{ ज्याके} \cdot \text{ज्यान}}{\text{त्रि}^२}$

एवमेव चन्द्रपरिध्यन्तरवशेन तद्गुणाकाङ्क्षा उत्पद्यन्ते, सिद्धान्तशेखरे 'तत्तिथ्यन्त-नतोद्भवोत्क्रमगुणः क्षुण्णः स्वकेन्द्रज्यया, गोऽक्षाभ्रेन्दुरसै ६१०५९ रवेः शररसा-ग्न्यब्ध्यश्विभिः २४३६५ शीतगोः। भक्तोऽथक्रमशो भवन्ति विकलास्ताः प्रागृणं स्वं रवौ पश्चाच्छीतकरेऽन्यथा क्षयफले हानिर्धनं चासकृत्' अनेन श्लोकेन श्रीपतिनाऽऽचार्योक्तानुरूप कथितः। केवलमाचार्येण ३२७० मिता त्रिज्या गृहीता, श्रीपतिना च ३४१५ मिता त्रिज्या गृहीता, तज्जन्यो भाज्यभाजकयोः स्वल्पभेदः समुचित एव, आचार्योक्तनतज्यास्थाने श्रीपतिना नतोत्क्रमज्या गृहीताऽत्रकारणं लल्लाचार्यानुयायिचतुर्वेदाचार्यव्याख्यानमेवेति ॥२७-२८॥

अब प्रकारान्तर से नतकर्म को कहते हैं

हि. भा.—पहले दिनदल (दिनार्ध) परिधियों से जो स्पष्टरवि और स्पष्टचन्द्र साधित हैं उनसे जो तिथ्यन्त हैं तात्कालिक नतज्या और केन्द्रज्या के घात करना, यदि वह रवि का है (अर्थात् रवि की नतज्या और रवि की केन्द्रज्या का घात है) तब उसको १९१ इससे गुणकर त्रिज्यावर्ग १०६९२९०० से भाग देना, यदि वह घात चन्द्र का है (अर्थात् चन्द्र-नतकालज्या और चन्द्रकेन्द्रज्या का घात है) तब उसको ४९९ इससे गुणाकर त्रिज्यावर्ग १०,६९,२९०० से भाग देना, दोनों स्थानों में जो विकलात्मक फल होता है पूर्वनत में सूर्य में ऋण करना, पश्चिम नत में सूर्य में धन करना, चन्द्र में प्राक् कपाल में ऋणात्मक मन्दफल रहने पर उस पूर्वागत केन्द्रफल को धन करना, अन्यथा प्राक् कपाल और पश्चिम कपाल में ऋण ही करना चाहिये, इस तरह असकृत्कर्म करने से स्फुटरवि और स्फुटचन्द्र होते हैं। पहले 'तद्द्युदलपरिध्यन्तरगुणा' इत्यादि से नतकर्म साधित है, यहां पुनः उसका साधन किया गया है इसलिये 'वा' शब्द का प्रयोग किया गया है इति ॥ २७-२८ ॥

उपपत्ति

दिनार्धपरिधि से पूर्व प्रकार से मन्दफलकला $= \frac{२२५ \times \text{दिप} \times \text{ज्याके}}{२१४ \times ३६०}$ विकलात्मक

करने से $\frac{२२५ \text{ दिप} \times \text{ज्याके} \times ६०}{२१४ \times ३६०}$, रवि के परिध्यन्तर $= \frac{\text{ज्यान}}{३ \text{ त्रि}}$ तब अनुपात करते हैं यदि दिनार्ध परिधि में पूर्वागत मन्दफल विकला पाते हैं तो आनीत (लाये हुये)

परिध्यन्तर में क्या इस अनुपात से रवि की परिध्यन्तर सम्बन्धिनी फल विकला

$$= \frac{६० \text{ दिप} \times २२५ \text{ ज्याके.ज्यान}}{३ \times २१४ \times ३६० \times \text{त्रि} \times \text{दिप}} = \frac{२० \times २२५ \text{ ज्याके त्रि. ज्यान}}{२१४ \times ३६० \times \text{त्रि}^२}$$

$$= \frac{२२५ \times ३२७० \text{ ज्याके. ज्यान}}{१८ \times २१४ \times \text{त्रि}^२} = \frac{२५ \times १६३५ \text{ ज्याके.ज्यान}}{२१४ \times \text{त्रि}^२} = \frac{४०८७५}{२१४}$$

$$\times \frac{\text{ज्याके.ज्यान}}{\text{त्रि}^२} = \frac{१९१ \text{ ज्याके.ज्यान}}{\text{त्रि}^२}$$, इसी तरह चन्द्र परिध्यन्तर से चन्द्र के गुणकाङ्क आते हैं इति। सिद्धान्तशेखर में 'तत्तिथ्यन्तनतोद्भवोत्क्रमगुणः इत्यादि' संस्कृतोपपत्ति में लिखित श्लोक से श्रीपति आचार्योक्तानुरूप ही कहते हैं, केवल आचार्य ने ३२७० एतत्तुल्य त्रिज्या ग्रहण किया है और श्रीपति ने ३४१५ त्रिज्याग्रहण किया है, तज्जन्य भाज्य और भाजक में थोड़ा अन्तर होना उचित ही है, आचार्योक्तनतज्या के स्थान में श्रीपति ने नतोत्क्रमज्या को ग्रहण किया है। इसका कारण केवल लल्लाचार्यानुयायिचतुर्वेदाचार्य का व्याख्यान ही है ॥२७-२८॥

इदानीं भुजान्तरकर्म स्पष्टगतिं चाह

अर्कफलभुक्तिघाताद् भगणकलाप्तं भुजान्तरं रविवत्।
स्फुटभुक्तिरतीतैष्यग्रहान्तरं वर्त्तमानेऽह्नि ॥ २९ ॥

वा. भा.—इष्टग्रहभुक्तिं कर्मफलेन संगुण्य भगणकलाभिर्विभजेत्। लब्धा भुजान्तराख्यं फलं तद्ग्रहे रविवद्धनमृणं वा कार्यं। यदि स्फुटभुक्तानीतं तत् स्फुटग्रहे मध्यभुक्तानीतं तन्मध्ये रवेर्भुजान्तरं कार्यं ग्रहवत्। यतो मध्यमार्कोदयकालिकाः स्पष्टग्रहा आगता भवन्ति। स्वफलेन ते च स्फुटार्कोदयकालिकाः क्रियन्ते। अर्कफलेन च त्रैराशिकाद्यस्मान्मध्यस्फुटार्कयोरन्तरं तत्कालमतः त्रैराशिककल्पना। तद्यथा यावत् स खषड्नसंख्याश्चक्रलिप्ता उदयं यान्ति। तावद् ग्रहः स्वभुक्तिं यदि भुंक्ते, तदर्कफललिप्तोदयेन किमिति स्वभुजान्तरं द्वितीयमार्यार्धमाह। स्फुटभुक्तिरतीतैव यद् ग्रहान्तरं वर्तमानेऽह्नि इष्टग्रहस्योक्तवत् स्फुटीकरणं कृत्वेष्टदिने तद्वदतीतदिनैव। तयोः स्फुटग्रहयोर्यदन्तरं सा ग्रहस्य स्फुटभुक्तिरेकदैवसिकी वर्तमाने दिने भवति।

अथवा आगामिदिने स्फुटं ग्रहं कृत्वा तेन सहान्तरं यद्वर्तमानदिने स्फुटग्रहस्य तुल्यकालस्य सा स्फुटभुक्तिर्भवत्यतश्च व्यवहारार्थमुच्यते। अन्यथा स्फुटभुक्तिः परमार्थतया न शक्यते वक्तुं प्रतिक्षणमन्यथात्वात् स्फुटभुक्तिर्यतः प्रतिमण्डलस्य कक्षामण्डलेन सहान्तरं स्वल्पक्षेत्रभागेप्यन्यथा भवति। यथास्थितेषु कक्षामण्डलादिषु प्रदर्शयेदिति ॥२९॥

वि. भा.- अर्कफलभुक्तिघातात् (रविमन्दफलग्रहगत्योर्वधात्) भगणकलाभिर्भक्तात्प्राप्तं फलं भुजान्तरं (भुजान्तरकलाः) रविवत् ग्रहेषु संस्कार्यमर्थादागतं फलमतो ऋणं धनं क्रियते यतो मध्यमार्कोदयात् प्राक् स्फुटार्कोदयः स्याद्ऋणे तत्फले स्वे यतोऽनन्तरम्, अतीतैष्यग्रहान्तरं (दिनान्तरस्पष्टग्रहान्तरं) वर्त्तमानेऽह्नि (वर्त्तमानदिने) स्फुटभुक्तिर्भवत्यर्थादद्यतनश्वस्तनस्फुटग्रहयोरन्तरमेव तत्समयान्तरे स्फुटगतिर्भवतीति ॥२९॥

अत्रोपपत्तिः

मध्यमार्कोदयकालीना ग्रहा येन कर्मणा स्फुटार्कोदयकालीना भवन्ति तदेव भुजान्तरकर्म, अहर्गणेन समानीता ग्रहा आचार्यमतेन लङ्कायां सूर्योदये मध्या भवन्त्यर्थान्मध्यमार्कोदयकालीनाः समागच्छन्ति, अपेक्षितास्तु स्फुटार्कोदयकालीनाः, मध्यमार्कस्फुटार्कयोरन्तरं रविमन्दफलकला, एतत्समा एव रविमन्दफलोत्पन्नासवः स्वीकृता आचार्येण ततोऽनुपातो यद्यहोरात्रासुभिर्ग्रहगतिकला लभ्यन्ते तदा रविमन्दफलकलासुभिः किं जाता रविमन्दफलकलासुसम्बन्धिनी ग्रहगतिः $=\frac{\text{ग्रग}\times\text{रविमन्दफलकलासु}}{\text{अहोरात्रासु}}$, मन्दफले ऋणे मध्यमार्कोदयात्प्रागेव स्फुटार्कोदयोऽतो समागतया रविमन्दफलकलासुसम्बन्धिन्या गत्या हीनो मध्यमार्कः स्फुटार्कोदये स्फुटार्को भवेत् । एवं तया गत्या हीना ग्रहाः स्फुटार्कोदयकालीना भवन्ति, धनात्मके रविमन्दफले मध्यमार्कोदयात्पश्चात्स्फुटार्कोदयो भवति तेन चालनफलेन युतो मध्यमार्कः स्फुटार्कोदये स्फुटार्को भवेत्, ग्रहा अपि चालनफलेन युता सन्तः स्फुटार्कोदये भवन्ति, अत्र स्थौल्यं स्पष्टमेवास्ति, रविमन्दफलकला रविमन्दफलासवोः समत्वस्वीकारात् । तथा च रविमन्दफलासुसम्बन्धिन्या गत्या संस्कृतो मध्यमार्कोदयकालिकग्रहो नहि वस्तुतः स्फुटार्कोदयकालिको भवितुमर्हति (मन्दफलासुमध्येऽपि ग्रहस्य कापि गतिर्भवेत्तद्ग्रहणमत्र न क्रियतेऽतस्तज्जन्यविकारसद्भावात्) वास्तवभुजान्तरज्ञानार्थं स एव विधिराश्रयणीयो यश्च पूर्वं वास्तवोदयान्तरज्ञानार्थं प्रदर्शितोऽस्ति तत्र प्राचीनोक्तोदयान्तरस्थाने प्राचीनभुजान्तरग्रहणं कर्त्तव्यमन्यत्सर्वं समानमेवेति । आचार्योक्तपद्ये भगणकलाशब्देनाहोरात्रासवो बोध्या इति, कस्याप्याचार्यस्य भुजान्तरकर्मसाधनं समीचीनं नास्ति पूर्वप्रदर्शितयुक्त्यैव स्फुटमिति ॥ २९ ॥

अब भुजान्तरकर्म को और स्पष्ट गति को कहते हैं

हि. भा.—रवि के मन्दफल और ग्रहणगति के घात में भगणकला से भाग देने से जो फल होता है, वह भुजान्तर कला है, ग्रह में उसको रवि की तरह संस्कार करना अर्थात् ऋणात्मक रविमन्दफल में मध्यमार्कोदय से स्फुटार्कोदय पहले होता है इसलिये

मध्यमार्कोदयकालिक ग्रह मे भुजान्तर फल को ऋण करने से स्फुटार्कोदयकालिक ग्रह होते हैं, धनात्मक रविमन्दफल से मध्यमार्कोदय से स्फुटार्कोदय पीछे होता है इसलिये मध्यमार्कोदय कालिक ग्रह में भुजान्तर फल को धन करने से स्पष्टार्कोदयकालिक ग्रह होते है, और गतदिन के ग्रह और एष्य दिन के ग्रह के अन्तर (दिनान्तर स्पष्ट ग्रहान्तर) वर्तमान दिन में स्पष्टगति होती है अर्थात् अद्यतन (आज के) स्पष्टग्रह और श्वस्तन (कल के) स्पष्टग्रह का अन्तर स्पष्टगति होती है, यहां भगण कलाशब्द से अहोरात्रासु समझना चाहिये इति ॥ २९ ॥

उपपत्ति

मध्यमार्कोदयकालिक ग्रह में जितना संस्कार करने से स्फुटार्कोदयकालिक ग्रह होते हैं उसी को भुजान्तरकर्म कहते हैं । ग्रहगणित से साधित ग्रह आचार्य के मत से लङ्का सूर्योदय काल में मध्यम होते है, अर्थात् मध्यमार्कोदयकालीन होते हैं, स्फुटार्कोदय कालीन ग्रह अपेक्षित हैं, मध्यमार्क और स्फुटार्क का अन्तर रविमन्दफल कला है, इसके बराबर ही रविमन्दफलोत्पन्नासु आचार्य ने स्वीकार किये हैं, अब अनुपात करते हैं यदि अहोरात्रासु में ग्रहगति कला पाते हैं तो रविमन्दफलासु में क्या इस अनुपात से रविमन्द-फलासुसम्बन्धिनी ग्रहगति आती है, मन्दफल ऋण रहने से मध्यमार्कोदय से पहले ही स्फुटार्कोदय होता है इसलिये अनुपातागत रविमन्दफलासुसम्बन्धिनी गति को मध्यमार्क में घटाने से स्फुटार्कोदय काल में स्फुटार्क होते हैं । इसी तरह उस गति को ग्रह में घटाने से स्फुटार्कोदयकालिक ग्रह होते हैं । धनात्मक रविमन्दफल में मध्यमार्कोदय से स्फुटार्कोदय पीछे होता है इसलिये पूर्वानीत चालनफल को मध्यमार्क में जोड़ने से स्फुटार्कोदयकालिक स्फुटार्क होते हैं, ग्रह में चालनफल को जोड़ने से स्पष्टार्कोदयकालिक ग्रह होते हैं, यहां स्थूलता स्पष्ट ही है क्योंकि रविमन्दफलकला के बराबर ही रविमन्दफलासु को आचार्य ने स्वीकार किया है । दूसरी त्रुटि इसमें यह है कि मध्यमार्कोदयकालिक ग्रह में पूर्वानुपातागत रविमन्दफलासुसम्बन्धिनी गति का संस्कार करने से स्फुटार्कोदयकालिक ग्रह नहीं हो सकते हैं क्योंकि रविमन्दफलासु के अन्तर्गत भी ग्रह की कुछ गति होगी उसका ग्रहण आचार्य ने नहीं किया है वास्तव भुजान्तर ज्ञान के लिये उसी विधि का आश्रयण करना चाहिये जो कि पहले वास्तवउदयान्तर ज्ञानार्थ दिखलायी गई है, वास्तव उदयान्तर साधन में प्राचीनोदयान्तर के स्थान पर प्राचीन भुजान्तर लेना चाहिये और सब विषय बराबर ही है, सूर्यसिद्धान्तकारादि किसी भी आचार्य के भुजान्तरकर्मसाधन ठीक नहीं है, पहले लिखी हुई युक्ति ही से स्पष्ट है इति ॥ २९ ॥

इदानीं नतकर्मवशेन रविचन्द्रयोर्गतिफलमाह

क्षयधनहानिधनानि प्राक् पश्चादन्यथा रवेरिन्दोः ।
प्राग्वत् पश्चात्स्वगतौ धनर्णयक्षयधनानि प्राक् ॥ ३० ॥

वा. भा.—इदानीं परिध्यन्तरोत्पन्नस्य केन्द्रफलभुक्तिफलस्य धनक्षयप्रतिपा-

दनार्थमार्यामाह। केन्द्रफलभुक्तिज्यानतज्ययोर्वधाद्रवेरिंदुतिथृतिगुणाच्छशिनो नवनववेदगुणा व्यासार्धकृत्या यत्फलं लब्धं तत्स्वभुक्तौ कदा धनं कदार्णं वा क्रियते तन्न ज्ञायते तदर्थमियमार्या। तद्यथा पूर्वकपाले क्षयः प्रथमे केन्द्रपदे द्वितीये धनं तृतीये हानिश्चतुर्थे धनं रविस्फुटभुक्तौ यस्माद्दिनार्धपरिधिना भुक्तफलमानीतं प्राग्दिनार्धपरिधेश्चाधिकस्तात्कालिकोऽत्र परिधिरतोऽल्पमृणं कृतमासीदधुना फलान्तरं विशोध्यते। प्रथमे पदे द्वितीये तु पुनर्भुक्तौ धनं कृतमासीत्तदपि दिनार्धपरिधिना ततश्च तात्कालिकपरिधिरधिकोऽतोऽल्पं धनं कृतमासीत्तद्दिनार्धपरिधिना धनं परिधिश्च तृतीयपदे सदोनो दिनार्धपरिधेरतोऽधिकं धनं तिष्ठति तद्विशोध्यते। चतुर्थपदे भुक्तावृणं कृतमासीत्। दिनार्धपरिधिना ततश्च तात्कालिकपरिधिरल्पः षड्राश्यधिकत्वात् केन्द्रस्यातोऽधिकमृणं प्राक्कृतं तत् पुनर्दीयते फलान्तरमेव पूर्वकपाले रवेः पश्चाद्यथा रवेः प्राक्कपाले धनर्णकल्पनैव चन्द्रस्यापरकपाले यदि स्यादपरकपाले चन्द्रस्य ऋणपरिधिः को भवति दिनार्धपरिधेर्यतोऽधिकमपि ऋणं कर्तुं युज्यते येन तात्कालिकपरिधिना संस्कृता भवति स्फुटभुक्तिः, द्वितीयपदे धनं यतो द्वितीयपदे भुक्तौ धनं कृतमासीत्तदधिकं कर्तुं युज्यतेऽधिकत्वात्तात्कालिकपरिधेस्तृतीयपदे भुक्तौ धनं कृतमासीद्दिनार्धपरिधिना। ततश्च धनपरिधिरूनस्तात्कालिकोऽतो बहुधनं कृतं तिष्ठति तत्पुनर्विशोध्यते फलान्तरं चतुर्थपदं चंद्रभुक्तावृणं कृतमासीत्, दिनार्धपरिधिना, ततस्तात्कालिकपरिधिनातः ऋणं बहुकृतमासीत्तत्पुनर्दीयते, फलान्तरमेवं। स्वकेन्द्रपदवशेन प्राक्कपाले तु पुनश्चन्द्रस्फुटभुक्तौ धनक्षयक्षयधनानि स्वकेन्द्रपदेषु यतः शशिनः ऋणं परिधिरूनो दिनार्धपरिधेः फलं च दिनार्धपरिधिना यदानीतं तद्भुक्तेर्विशोधितं तत्पुनर्दीयते उपपन्नं प्रथमे पदे धनं द्वितीये पदेऽपि दिनार्धपरिधिना धनं कृतमासीत्तदधिकं यस्मात् तात्कालिकः परिधिरूनोऽतो यदधिकः ॥३०॥

वि. भा.—दिनदलपरिधिस्फुटतिथिनतकेन्द्रज्येत्यादिना यद्विकलात्मकफलमानीतं तद्गतौ कदा धनं कदा क्षय इत्येतदर्थं कथ्यते, प्राक्कपाले रवेः केन्द्रवशेन तानि फलानि क्षयधनहानिधनानि स्वगतौ (पूर्वसाधितरविमन्दस्फुटगतौ) भवन्त्यर्थात्प्रथमे केन्द्रपदे क्षयः, द्वितीये केन्द्रपदे धनं, तृतीये हानिः (क्षयः) चतुर्थे धनं स्फुटगताविति्यर्थः। पश्चात्कपाले चान्यथार्थात् धनक्षयधनक्षयात्मकानि स्युः, इन्दोः (चन्द्रस्य) पश्चात्कपाले प्राग्वत् (पूर्ववत्) अर्थात् क्षयधनहानिधनानि भवन्ति, प्राक्कपाले धनक्षयक्षयधनानि भवन्ति चन्द्रमन्दस्पष्टगताविति ॥ ३० ॥

अत्रोपपत्तिः

प्रथमपदे तृतीयपदे च केन्द्रज्योपचीयते, द्वितीयपदे चतुर्थपदे चापचीयते, रवेः प्राक्कपाले नतकर्म ऋणं पश्चिमकपाले च धनं भवति, अतः प्रथमपदे तृतीयपदे च ऋणफलस्योपचयान्नतकर्मान्तरमृणं द्वितीयपदे चतुर्थपदे च ऋणफलस्यापचयान्नत-

कर्मान्तरं धनं भवति, पश्चिमकपाले नतकर्मणो धनत्वाद्विलोमं भवति। चन्द्रस्य पश्चिमकपाले मन्दफले ऋणे धने च नतकर्म ऋणमेव भवत्यतः केन्द्रपदवशात् क्षयधनहानिधनानि फलानि जायन्ते, पूर्वकपाले प्रथमपदे ऋणात्मके मन्दफले केन्द्रज्यायाः उपचयत्वान्नतकर्मणो धनत्वाच्च नतकर्मान्तरं धनं भवति, द्वितीयपदे केन्द्रज्यायाः अपचयत्वान्नतकर्मान्तरमृणं, तृतीयपदे मन्दफलं धनं नतकर्म ऋणं केन्द्रज्यायाश्चोपचयोऽतो नतकर्मान्तरमृणम्, चतुर्थे पदे केन्द्रज्यायाः अपचयत्वान्नतकर्मणश्च क्षयत्वान्नतकर्मान्तरं धनं भवतीति। दिनान्तरस्पष्टग्रहान्तरं स्पष्टा गतिर्भवतीति नियमाद् गतदिननतकाल एव द्वितीयदिनेऽपि नतकालस्तेन केन्द्रज्या एवोपचयापचयवशान्नतकर्मान्तररूपद्वितीयगतिफलस्य धनर्णत्वं समुचितमेवेति. ॥ ३० ॥

अब नत कर्मवश से रवि और चन्द्र के गतिफल को कहते हैं

हि. भा.—दिनदलपरिधिस्फुटतिथिनतकेन्द्रज्या इत्यादि से अद्यतन और श्वस्तन फल विकलाओं के अन्तर (विकलात्मक अन्तर) पूर्वकपाल में पूर्वसाधितरविमन्दस्पष्ट गति में रवि के केन्द्रपदवश से ऋण, धन, ऋण, धन होता है। पश्चिम कपाल में विलोम होता है अर्थात् धन, ऋण, धन, ऋण, चन्द्र के पश्चिम कपाल में पूर्ववत् होता है अर्थात् ऋण, धन, ऋण, धन, और पूर्व कपाल में चन्द्रमन्दस्पष्टगति में वे ही धन, ऋण, ऋण, धन होते हैं ॥३०॥

उपपत्ति

प्रथम पद में और तृतीय पद में केन्द्रज्या उपचीयमान रहती है और द्वितीय पद में तथा चतुर्थ पद में अपचीयमान रहती है; रवि के पूर्व कपाल में नतकर्म ऋण और पश्चिम कपाल में धन होता है इसलिये प्रथम पद में और तृतीय पद में ऋणफल के उपचय के कारण नतकर्मान्तर ऋण होता है। द्वितीय और चतुर्थ पद में ऋण फल के अपचयत्व के कारण नतकर्मान्तर धन होता है, पश्चिम कपाल में नतकर्म के धनत्व के कारण विलोम (उल्टा) होता है, चन्द्र के पश्चिम कपाल में मन्द फल के ऋण या धन रहने पर नतकर्म ऋण ही होता है। इसलिए केन्द्रपदवश से फल ऋण, धन, ऋण, धन होता है। द्वितीय पद में केन्द्रज्या के अपचयत्व के कारण नतकर्मान्तर ऋण होता है। तृतीय पद में मन्दफल धन, नतकर्म ऋण, केन्द्रज्या का अपचय रहता है। इसलिये नतकर्मान्तर ऋण होता है, चतुर्थ पद में केन्द्रज्या के अपचयत्व से और नतकर्म के क्षयत्व के कारण नतकर्मान्तर धन होता है। अद्यतन (आज के) श्वस्तन (कल के) स्पष्टग्रहों का अन्तर स्पष्ट गति है। इस नियम से गतदिन का नतकाल ही द्वितीय दिन में भी नतकाल होता है इसलिये केन्द्रज्या ही के उपचय और अपचय वश से नतकर्मान्तर रूप द्वितीय गति फल का धनत्व और ऋणत्व उचित ही है इति ॥३०॥

इदानीमाचार्यः स्वप्रशंसामाह

ब्रह्मोक्तमध्यरविशशितदुच्चतत्परिधिभिः स्फुटीकरणम् ।
कृत्वैवं स्पष्टतिथिर्दूरभ्रष्टान्यतन्त्रोक्तैः ॥३१॥

वा. भा.—श्लोकस्यास्य वासनाभाष्यं नोपलभ्यते ।

वि. भा.—ब्रह्मोक्त (ब्रह्मगुप्तकथित) मध्यरविचन्द्रतन्मन्दोच्चतत्परिधिभिः स्फुटीकरणं कृत्वा स्पष्टतिथिः साध्या यतोऽन्यतन्त्रोक्तैः (अन्येषामार्यभटादीनां तन्त्रेषु ये कथिता मध्यरविशशिपरिध्यादयस्तैः) स्पष्टतिथिर्दूरभ्रष्टा भवत्यर्थादत्यन्तनिषिद्धा भवत्येतावताऽऽचार्येण कथ्यते यन्मत्कथितस्पष्टीकरणसामग्रीभिरेव रविचन्द्रयोः स्पष्टीकरणं युक्तियुक्तं, ताभ्यां स्पष्टीकृताभ्यामेव रविचन्द्राभ्यां साधिता स्पष्टा तिथिः समीचीना भवति, आर्यभटादिकथितस्पष्टरविचन्द्रसामग्रीभिर्वस्तुतः स्पष्टौ रविचन्द्रौ न भवतोऽतस्ताभ्यां साधिता तिथिः स्पष्टा न भवतीति ॥३१॥

अब आचार्य अपनी प्रशंसा कहते हैं

हि. भा.—ब्रह्म (ब्रह्मगुप्त) कथित मध्यमरवि, मध्यमचन्द्र, उनके मन्दोच्च और उनकी मन्द परिधि इन सबों से रवि और चन्द्र के स्पष्टीकरण करके स्पष्ट तिथि का साधन करना चाहिये, क्योंकि आर्यभटादि अन्य आचार्यों के तन्त्रों में कथित मध्यमरवि, मध्यमचन्द्र और उनकी मन्द परिधियों से साधित स्पष्टतिथि वस्तुतः स्पष्टतिथि नहीं होती है इससे आचार्य कहते हैं कि मेरी स्पष्टीकरण सामग्रियों से ही रवि और चन्द्र का स्पष्टीकरण युक्तियुक्त होता है, उन स्पष्टीकृत रवि और चन्द्र से साधित स्पष्टतिथि यथार्थ स्पष्टतिथि होती है, आर्यभटादि आचार्यों से कथित स्पष्टरवि और स्पष्टचन्द्र की सामग्रियों से वस्तुतः स्पष्टरवि और स्पष्टचन्द्र ठीक नहीं होते हैं इसलिये उन रवि और चन्द्र से साधित स्पष्ट तिथि ठीक स्पष्ट तिथि नहीं होती है इति ॥३१ ॥

इदानीं व्यवहारोपयुक्तरविचन्द्रयोः स्पष्टीकरणमाह

स्वदिनार्धपरिधिभुजफलचापं मध्यार्कचन्द्रयोः कृत्वा ।
पूर्ववदन्यत् स्पष्टं संव्यवहारार्थमेवं वा ॥३२॥

वा. भा.—अस्य श्लोकस्य वासनाभाष्यं नास्ति ।

वि. भा.—स्वदिनार्धपरिधिना पूर्वकथितविधिना यद्भुजफलचापं (मन्दफलं) तन्मध्यमरविचन्द्रयोः संस्कृत्य अन्यद्देशान्तरादिसंस्कारं पूर्ववत् कृत्वैवं वा संव्यवहारार्थं स्पष्टीकरणं विधेयम् । नतकर्मसंस्कारं विनैवाऽऽचार्येणेदं स्पष्टीकरणं स्थूलं व्यवहारोपयोगिकथितम् । भास्कराचार्येण रविचन्द्रयोः स्थूलमेवेदं स्पष्टी-

करणं सर्वकर्मोपयुक्तं सूक्ष्मत्वेन स्वीकृतम्। 'मुहुः स्फुटाऽतो ग्रहणे रवीन्द्वोरित्यादिना' भास्करेण यदाचार्यमतं वर्णितं वस्तुतस्तत्तन्मतं नास्तीति विज्ञैर्बोध्यमिति ॥३२॥

अब व्यवहारोपयोगी रवि और चन्द्र के स्पष्टीकरण को कहते हैं

हि. भा.—स्वदिनार्धपरिधि से पूर्वकथित विधि के अनुसार जो मन्दभुजफल होता है उसके चाप (मन्दफल) को मध्यमरवि और मध्यमचन्द्र में संस्कार करके अन्य देशान्तरादि संस्कारों को पूर्ववत् करके व्यवहार के लिये वा इस तरह स्पष्टीकरण करना चाहिए। बिना नतकर्म संस्कार के आचार्य इस व्यवहारोपयोगी स्थूल स्पष्टीकरण को कहते हैं। 'भास्कराचार्य ने रवि और चन्द्र के इस स्थूल स्पष्टीकरण को ही सूक्ष्म सब कर्मों के लिये उपयुक्त स्वीकार किया है, 'मुहुः स्फुटाऽतो ग्रहणे रवीन्द्वोः इत्यादि से' भास्कराचार्य ने आचार्यमत का भिन्न तरह प्रतिपादन किया है इति ॥३२॥

इदानीं मङ्गलादिग्रहस्पष्टीकरणे कारणमाह

आर्यभटस्याज्ञानान्मध्यममन्दोच्चशीघ्रपरिधीनाम्।
अस्पष्टा भौमाद्याः स्पष्टा ब्रह्मोक्तमध्योच्चैः ॥३३॥

वा. भा.—नास्ति वासनाभाष्यमस्य श्लोकस्य।

वि. भा.— आर्यभटस्य मध्यममन्दोच्चशीघ्रपरिधीनामज्ञानात्कारणात्, भौमाद्या (मङ्गलादिकाः) ग्रहाः अस्पष्टा भवन्त्यतो ब्रह्मो (ब्रह्मगुप्तो) क्तमध्योच्चैर्भौमाद्याः स्पष्टाः कार्या अर्थादार्यभटस्य वास्तवमध्यममन्दोच्चादीनामज्ञानात्तन्मतेन भौमादिग्रहस्पष्टीकरणं न युक्तमतो मदुक्तवास्तवमन्दोच्चाद्यैस्तत्स्पष्टीकरणं विधेयमित्याचार्याभिप्राय इति ॥३३॥

अब मङ्गलादि ग्रहों के स्पष्टीकरण में कारण कहते हैं

हि. भा.—आर्यभट को वास्तव मध्यममन्दोच्च-शीघ्र परिधियों का ज्ञान नहीं था इसलिये भौमादि (मङ्गल आदि) ग्रह उनके मत से स्पष्ट नहीं होते हैं। ब्रह्मो (ब्रह्मगुप्तो) क्त मन्दोच्चादि से वे स्पष्ट होते हैं, आचार्य के कहने का तात्पर्य यह है कि आर्यभट को मध्यम मन्दोच्च शीघ्र परिधियों का ज्ञान नहीं था इसलिये हमारे मन्दोच्चादि से कुजादि ग्रहों के स्पष्टीकरण करने से वे ठीक स्पष्ट होते हैं इति. ॥३३॥

इदानीं मङ्गलादिग्रहाणां मन्दशीघ्रपरिध्यंशान् स्फुटीकरणञ्चाह

मन्दोच्चनीचवृत्तस्य परिधिभागाः सितस्य विषमान्ते।
नवयुग्मान्ते रुद्राः ११ शीघ्रौजान्तेऽग्निरसयमलाः २६३ ॥३४॥

युग्मान्तेऽष्टशरयमा २५८ मन्दफलान्मध्यमः स्फुटो मध्यः ।
शीघ्रफलात् स्पष्टोऽसकृदेवं स्वफलैर्ज्ञगुरुसौराः ॥३५॥

बुधमन्दपरिधिभागा वसुरामा ३८ सुरगुरोस्त्रयस्त्रिंशत् ।
रविजस्य शून्यरामाज्ञशीघ्रपरिधिद्विगुणचन्द्राः १३२ ॥३६॥

देवगुरोरष्टरसा ६८ भास्करपुत्रस्य शरगुणाः ३५ स्पष्टाः ।
कुजशीघ्रकेन्द्रपदगतये याल्पज्या त्रिभागोनैः ॥३७॥

सप्तभिरंशै ६।४० र्गुणिता दलाढ्यराशिज्ययाप्तांशैः ।
अधिकोनः कुजमन्दो मृगकर्क्यादौ स्फुटो भवति ॥३८॥

तत्स्फुटपरिधिः खनगाः ७० शीघ्रस्फुटपरिधिराप्तभागोनाः ।
वेदजिनास्त्र्यंशोनाः २४३।४० स्पष्टीकरणं कुजस्यैवम् ॥३९॥

मन्दफलं मध्येऽर्धं तच्छीघ्रफलस्य मध्यमे सकले ।
मध्येऽसकृत् क्षितिसुतः स्पष्टो भुक्तिः स्फुटा ग्रहवत् ॥४०॥

वा. भा.—मन्दोच्चनीचवृत्तस्य गोलाध्याये प्रदर्शितस्य शीघ्रोच्चनीचवृत्तस्य ये परिधिभागाः स्वकक्षा षष्टिशतत्रयकल्पना या स्वपरमफलवशादुपलब्धाः ते तु नीचोच्चवृत्तस्य परिधिभागा उच्यन्ते। सर्वेषां ग्रहाणां सितस्य वद्यथा विषमपदान्ते प्रथमतृतीययोः पदयोरन्तरेत्यर्थस्तत्र नवभागा मन्दनीचोच्चवृत्तस्य परिधियुग्मान्ते रुद्राः द्विचतुर्थयोरन्ते चैकादशभागाः, अन्तरे च त्रैराशिकमाचार्येण प्रागुक्तमेव, शीघ्रनीचोच्चवृत्तस्यौजान्ते प्रथमतृतीयपदान्ते रसयमलाः, युग्मान्ते द्विचतुर्थपदान्तेऽष्टशरयमाः अवान्तरे वापि त्रैराशिकम्। प्राग्वन्मन्दफलान्मध्यमस्फुटो मध्य इति करणागतो देशान्तरभुजान्तचरदलैः कृतैः मध्ये यः स्वदेशौदयिकः स्वमन्दफलेन स्फुटो मध्ये भवति। यतो मध्ये मन्दफलोच्चनीचवृत्तमध्यमः स्वशीघ्रनीचोच्चवृत्तमध्यमतएव शीघ्रफलेन मध्यमः स्फुटो भवति । यतस्तु परिधौ स्फुटो ग्रहः कक्षामण्डल एव दृश्यते अतस्तदुपलब्धये कर्म क्रियते शुक्रस्य तद्यथेष्टकेन्द्रे खस्वस्तिकेऽभीष्टकालिकं शुक्रं संस्थाप्य तस्मात्स्वमन्दोच्चं विशोध्य केन्द्रं भवति। ततः केन्द्राद् भुजज्योक्तवत्कार्या ततः त्रिज्याहता भुजज्येत्यादिना स्फुटमन्दपरिधिः कार्यः तेन भुजज्यां संगुण्य षष्टिशतत्रयेण विभजेत्। लब्धस्य चापं मन्दफलं भवति । षड्राश्यूने केन्द्रे तात्कालिकमध्ये क्षयः षड्राश्यधिके केन्द्रे धनमेव मध्यमो भवति। योऽस्माभिर्मन्दस्फुट इत्युच्यते। ततस्तं मध्यमं स्वशीघ्रात्तात्कालिकाद्विशोध्य शीघ्रकेन्द्रं भवति । तस्मादुक्तवद्भुजकोटिज्ये कार्ये तास्त्रिज्याहता भुजज्येत्यादिना शीघ्रपरिधिस्फुटः कार्यः तेन भुजकोटिज्ये संगुण्य भांशैर्विभजेत् लब्धे भुजकोटिफले पृथग्भवतस्ततः कोटिफलयुतात् त्रिज्यापदयोराद्यंतयोर्विहीना द्वितीयतृतीययोरिति स्फुटकोटिः कार्या ततः तद्भुजफलकृतियोगात्पदं स्फुटकर्णः कार्यः। ततस्तद्गुणितात् व्यासार्धात् लब्धकर्णधनुरिति शीघ्रफलं कृत्वा तन्मन्दस्फुटे शुक्रे षड्राश्यूने

शीघ्रकेन्द्रे धनं कार्यमधिके ऋणमेवं मन्दशीघ्रकर्मद्वयस्फुटः शुक्रो भवति। ततस्तं मध्यं परिकल्प्य तस्मात् स्वमन्दोच्चं विशोध्य मन्दफलमानयेत्तत्कारणागते मध्ये धनं ऋणं वा कृत्वा मन्दस्फुटो मध्यमः कार्यः। तदा शीघ्रं विशोध्योक्तवच्छीघ्रफलमानयेत्तदनन्तरं मन्दस्फुटे धनं ऋणं वा कृत्वा स्फुटः शुक्रो भवति। एवं तावद्यावद्विशेषः स्फुटो भवति। किन्त्वत्र प्रतिकर्मपरिघेः स्फुटीकरणं कार्यम्। एवं स्वफलैर्ज्ञगुरुसौरा इति। एवमनेनैव प्रकारेण बुधबृहस्पतिशनैश्चरा स्फुटाः कार्या असकृत् किन्तु स्वमंदशीघ्रपरिधिभिस्तेषां स्फुटीकरणं प्रागेव प्रदर्शितम्।

बुधमन्दनीचोच्चवृत्तस्य परिधिभागाः वसुरामा २८। सुरगुरोर्मन्दपरिधिभागास्त्रयस्त्रिंशत् ३३। रविज्यस्य मन्दपरिधिभागाः शून्यरामाः ३०। एतैर्मन्दफलानयनं शुक्रवत्। बुधशीघ्रनीचोच्चवृत्तस्य परिधिभागाः द्विगुणचन्द्राः १३२।

देवगुरोः शीघ्रपरिधिरष्टरसाः ६८। भास्करपुत्रस्य शरगुणाः ३५ एतैः शीघ्रफलानयनं शुक्रवत् त एव स्पष्टाः यत एषां भास्करा नास्ति। वासना चैषामुपलब्धिरेव परमफलवशादिति।

इदानीं कुजस्फुटीकरणार्थमार्यात्रयं सार्धमाह। मन्दफलमध्येऽर्धं तच्छीघ्रफलस्य मध्यमे सकलो मध्ये सकृदिति सुते स्पष्टो भुक्तिः स्फुटा ग्रहवत्। आदौ तावदिष्टदैवसिकमिष्टकालिकं भौमम्। तात्कालिकशीघ्राद्विशोध्य शीघ्रकेन्द्रं कार्यम्। तत् यस्मिन् पदे वर्त्तते तस्य पदस्य भुक्ता भुक्तयोर्यदल्पं तस्य ज्या ग्राह्या सा कुजकेन्द्रपदयोर्याल्पज्या सा च त्रिभागोनैः सप्तभिरंशैर्गुणिता कार्या ६।४० दलाढ्यराशिज्याहृता दलाढ्यराशि, तस्य ज्या रवित्रियमाः २३१२ अतो येंऽशा आप्तास्तैराप्तांशैरधिकोनो यथासंख्यं कुजमन्दो मृगकर्कादौ स्थिते केन्द्रः कार्यः एवं कृते स्फुटो भवति।

तत्स्फुटपरिधिः खनगा इति तदिति स्फुटपरिधिः। मन्दस्य परामर्शस्तस्य न संख्या सर्वदा स्फुटपरिधिः शीघ्रस्फुटपरिधिश्चाप्तभागैरूना। वेदजिनास्त्र्यंशोनाः २४३।४० कुजशीघ्रकेन्द्रपदगतये याल्पा ज्या त्रिभागोनैः सप्तभिरंशैर्गुणिता दलाढ्यराशिज्यया हृतेत्यतो ये भागा आप्तांशास्तैरूनास्त्रिशोना वेदजिना कुजस्य शीघ्रस्फुटपरिधिर्भवतीत्यर्थः स्पष्टीकरणं कुजस्यैवमिति। एवं कृतेऽस्य स्फुटीकरणमनन्तरोक्तोपकरणैः कार्यम्। तत्कथमिति तदर्थमाह।

मन्दफलमध्येऽर्धं प्रथमं तावन्मन्दफलं स्वोपकरणैरानीयते। अस्यार्धं मध्ये ग्रहे धनमृणं वा कार्यम्। तच्छीघ्रफलस्य मध्यम इति। तस्मान्मन्दफलार्धस्फुटान्मध्याच्छीघ्रफलं स्वोपकरणैरानीय तस्यार्धं मध्यमे मन्दार्धस्फुटे धनमृणं वा सकलं मध्ये। ततो द्वितीयसंस्कृताद्भौमान्मन्दफलमुक्तवत्कृत्वा तत्सकलं करणागते मध्ये धनमृणं वा कार्यम्। ततश्चोक्तवच्छीघ्रफलं तत्तत्रैव धनं कार्यम्।

एतच्चासकृत्कृत्वा क्षितिसुते कर्म स्फुटं भवति । कुजादीनां भुक्तिरपि ग्रहवत्स्फुटा कार्या । स्पष्टतरं कर्म प्रदर्श्यते तद्यथा स्वदेशे इष्टकालिकाद्भौमे शीघ्राद्भौमं विशोध्य कुजशीघ्रकेन्द्रपदेत्यादिना शीघ्रपरिधिं च स्फुटीकृत्वा ततः प्रथमं मन्दफलमानयेत् । स्वमन्दपरिधिना तदर्धपृथक्स्थे भौमे षड्राश्यूने मन्दकेन्द्रे ऋणमधिके धनं कृत्वा ततस्तं स्वशीघ्राद्विशोध्य शीघ्रफलमानयेत् । तस्याप्यर्धं तत्रैव षड्राश्यूने शीघ्रकेन्द्रे धनमधिके ऋणं कृत्वा तस्माद्द्विकर्मस्फुटाद्भौमात्स्वमंदोच्चं विशोध्य मन्दफलमानयेत् । तद्विकृते मध्यभौमसकलप्राग्वद्धनमृणं च कृत्वा मन्दस्फुटो मध्यमो भवति । ततस्तं स्वशीघ्राद्विशोध्य शीघ्रफलं स्वोपकरणैरानीय तदपि सकलमेव मन्दस्फुटप्राग्वद्धनमृणं वा कार्यम् । एवं कृते स्फुटो भौमो भवति । ततस्तं मध्यं परिकल्पयेत्तमेव स्वशीघ्राद्विशोध्य कुजशीघ्रकेन्द्रपदयोर्याल्पज्येत्यादिना मन्दः संस्कार्यः शीघ्रपरिधिश्च ततः स्फुटभौममन्दोच्चं विशोध्य मन्दफलमानयेत् । तस्यार्धकरणागतं मध्ये धनमृणं कृत्वा ततस्तं शीघ्राद्विशोध्य शीघ्रफलं प्राग्वत् तस्याप्यर्धं तत्रैव ततश्च मन्दशीघ्रफले सकले मध्ये भौमे कृत्वा स्फुटः कार्यः । एवं तावद्यावद्विशोध्यः, प्रतिस्फुटीकरणे शीघ्रपरिधिमन्दयोः संस्कारः कार्यः । एवं यस्य ग्रहस्य मंदकर्मणा केवलेन स्फुटीकरणं तस्य भुक्तिरपि एकेनैव कर्मणा ।

यस्य कर्मद्वयेन तस्य भुक्तिरपि कर्मद्वयेन, यस्य कर्मचतुष्टयेन तस्य भुक्तिरपि चतुष्टयेन स्फुटा कार्या । कर्मचतुष्टयेनैवात्रेयं वासना भौमः । शीघ्रमंडलादुच्चे यदा भवति तदा तन्मन्दोच्चं यथागतमेव भवति । ततः क्रमेणोपचीयते केन्द्रपदार्धं यावत्ततश्चापचीयते पदान्तं यावदेवं चतुर्थे पदेऽपि द्वितृतीययोश्च वैपरीत्येन पदसंधिषु । चतुष्टयेऽपि करणागतएवांतरे त्रैराशिककल्पना कृता । मन्दसंस्कारार्थं त्रिभागोनैः सप्तभागैः दलाढ्यराशिज्यया शीघ्रपरिधिचतुष्टय पदसंधिषु यथा पठितः । वा पदार्धे इत्यल्पः सर्वेषु तावद्भिरेव यदि भागैरतः सर्वंदोपचीयते । लब्धफलेन यच्च मन्दशीघ्रफलद्वयार्धेन प्रथमं कर्म तदपि मन्दफलसंस्कारार्थम् । न चात्र वासना गम्येत्येभिर्योगातिशयेनोपलब्धे रवि यदि कारणं स्यादथवागम एव प्रमाणम् । मन्दशीघ्रपरिध्यादिषु यतः शेषग्रहणातिरेकेण भौमस्यायं विधिरिति ।

वि. भा.—सितस्य (शुक्रस्य) विषमान्ते (विषमपदान्ते) मन्दोच्चनीचवृत्तस्य परिधिभागा (मन्दपरिध्यंशाः) नव ९ । युग्मान्ते (समपदान्ते) मन्दपरिध्यंशा रुद्राः (एकादश), ओजान्ते (विषमपदान्ते) शीघ्रपरिध्यंशाः २६३, युग्मान्ते (समपदान्ते) २५८, मन्दफलात्संस्कृतो मध्यमो मध्यः स्फुटः (मन्दस्पष्टः) ग्रहो भवति, शीघ्रफलात् संस्कृतो मन्दस्पष्टोऽसकृत् स्पष्टो भवेदर्थान्मध्यग्रहात्पुनर्मन्दफलमानीय तत्संस्कृतो मध्यमग्रहो मन्दस्पष्टो भवेत्तस्मात् (मन्दस्पष्टात्) पुनः शीघ्रफलमानीय तत्संस्कृतो मन्दस्पष्टग्रहः स्पष्टग्रहो

भवेत्, पुनरेतस्मात्स्पष्टग्रहात्पूर्ववन्मन्दस्पष्टस्पष्टग्रहौ साध्यौ यावदविशेष इति। एवं ज्ञगुरुसौराः (बुधबृहस्पतिशनैश्चराः) स्वफलैः (मन्दफलैः शीघ्रफलैश्च) पूर्ववदसकृत् स्फुटा भवन्ति, बुधमन्दपरिध्यंशाः =३८, सुरगुरोः (बृहस्पतेः) =३३, रविजस्य(शनैश्चरस्य)=३०, विषमपदान्ते, समपदान्ते चार्यादं बुधबृहस्पतिशनैश्चरमन्दपरिध्यशेषु संस्काराभावः बुधशीघ्रपरिध्यंशाः=१३२, बृहस्पतेः=६८, भास्करपुत्रस्य (शनेः) =३५, एषामपि बुधबृहस्पतिशनीनां शीघ्रपरिध्यंशेषु संस्काराभावः सर्वदैकरूपत्वात् कुजशीघ्रकेन्द्रपदगतये याल्पज्या (कुजशीघ्रकेन्द्रं यस्मिन् पदे भवेत्तत्र गतगम्ययोर्येऽल्पा अंशास्तेषां ज्या कार्या) सा त्रिभागोनैर्विंशत्यंशरहितैः सप्तभिरंशैः ६।४० गुणिता, दलाढ्यराशिज्यया (पञ्चचत्वारिंशदंशज्यया) भक्ता प्राप्तांशैर्मृगकर्क्यादिशीघ्रकेन्द्रे कुजमन्दः (कुजमन्दोच्चं) क्रमेणाधिको हीनश्च कार्यः, एवं स्पष्टीकरणोपयोगिकुजमन्दोच्चं स्फुटं भवति, तत्स्फुटपरिधिः (तस्य कुजस्य मन्दपरिध्यंशः) खनगाः =७० =त्र्यंशोना वेदजिना २४३।४० अंशा आप्तभागोना अर्थान्मन्दोच्चसंस्कारार्थं पूर्वयेंऽशा प्राप्तास्तैरूना - (हीनाः) स्तदा कुजस्य शीघ्रस्फुटपरिधिः स्यात्। एवं वक्ष्यमाणं कुजस्य स्पष्टीकरणं भवति। मध्ये (गणितागते मध्यमकुजे) ऽर्धं मन्दफलं धनमृणं वा यथागतं देयम्। तच्छीघ्रफलस्यार्धं (तस्मादर्धमन्दफलसंस्कृतान्मध्यकुजाद्यच्छीघ्रफलं तदर्धं) मध्यमे(अर्धमन्दफलसंस्कृतमध्यकुजे) देयम्। पुनरर्धफलद्वयसंस्कृतान्मध्यमाद्यन्मदफलं तत्संस्कृतान्मध्याद्यच्छीघ्रफलं च ते सकले (सम्पूर्णे) मध्ये (गणितागते) कुजे देये, ततो बुधबृहस्पतिशनिवदसकृत्कर्म कार्यं यावदविशेषः। तदा क्षितिसुतः (कुजः) स्फुटो भवति, स्फुटा भुक्तिः (स्पष्टगतिः) ग्रहवत्साध्या, दिनान्तरस्पष्टग्रहान्तरमेव तत्समयान्तराले ग्रहस्पष्टगतिर्भवतीति ॥३४-३५-३६-३७-३८, ३९, ४०॥

अत्रोपपत्तिः

ब्रह्मगुप्तोक्तशनेः शीघ्रपरिधिः=३५°, सूर्यसिद्धान्तकारोक्तशनिशीघ्रपरिधिः =४०°, भास्करोक्ततच्छीघ्रपरिध्यंशाः=४०° तथाऽऽचार्योक्तशनिमन्दपरिध्यंशाः =३०° भास्करोक्तशनिमन्दपरिध्यंशाः =५०°, शनिपरिध्यंशेषु भास्करेण कथं वैषम्यं कृतमिति त एव ज्ञातुं शक्नुवन्ति, सूर्यसिद्धान्तकारेण शीघ्रान्त्यफलज्याऽपि न स्थिरेति मनसि धृत्वा समपदान्तविषमपदान्तभेदेन भिन्ना भिन्नाः परिध्यंशाः पठिता यथा तदुक्तम्।

कुजादीनामतः शैघ्र्या युग्मान्तेऽर्थाग्निदस्रकाः।
गुणाग्निचन्द्राः खनगा द्विरसाक्षीणि गोऽब्धयः।
ओजान्ते द्वित्रियमला द्विविश्वे यमपर्वताः।
खर्त्तुदस्रा वियद्वेदाः शीघ्रकर्मणि कीर्त्तिताः॥

ब्रह्मगुप्तेन यथा परिधिस्फुटीकरणमभिहितं तदनुसारमेव भास्करेणापि कथितं मङ्गलस्य चतुर्ष्वपि शीघ्रकेन्द्रपदान्तेषु गणितागतमन्दोच्चमेकरूपकमेव भवति, पदमध्ये त्र्यंशोनसप्तभिरंशैः सर्वदा न्यूनं भवत्यतोऽवान्तरे ऽनुपातो यदि पञ्चचत्वारिंशच्छीघ्रकेन्द्रज्यया त्र्यंशोनसप्तांशा मन्दोच्चान्तरं लभ्यते तदेष्टशीघ्रकेन्द्र-पदगतगम्याल्पया शीघ्रकेन्द्रज्यया किमित्यनुपातेन यत्फलं तन्मकरादिकर्क्यादिकेन्द्रे गणितागतकुजमन्दोच्चे युतं हीनं कार्यं तदा स्फुटं तन्मन्दोच्चं भवितुमर्हति, तथा च मङ्गलस्य पदान्तेषु शीघ्रपरिधिरेकरूपक एव भवति, पदमध्ये त्र्यंशोनैः सप्तभिरंशैः सदा न्यून एव भवति, अवान्तरे पूर्वानुपातागतफलं तच्छीघ्रपरिधौ धनर्णं कार्यं तदा स्फुटः शीघ्रपरिधिर्भवेदिति, भास्करोक्तमप्याचार्योक्तानुरूपमेव, ग्रहस्पष्टीकरणे मङ्गलशुक्रयोरेवान्तरं तथा मङ्गलमन्दोच्चे वैषम्यं च ब्रह्मगुप्तेन दृष्टम्। काऽप्यत्र युक्तिर्न प्रतिपादिता केवलं वेध एव प्रमाणम् नान्यत्कारणं वक्तुं शक्यत इति भास्करोक्तेन ज्ञायत इति, सिद्धान्तशेखरे श्रीपतिना 'भौमो निते दिनकरे पदयातयेयन्यूनज्यका युगहताऽङ्ककृतेन्द्र १४४६ भक्ता। लब्धांशकैर्युत विहीनमसृङ्मृदूच्चं स्पष्टं भवेन्मकरकर्कटकादिकेन्द्रे' ऽनेनाचार्योक्तानुरूपमेव मङ्गलमन्दोच्चस्फुटीकरणं कथितम्। पदयातयेयन्यूनज्यकायास्त्र्यंशोनसप्तांशानां गुणकाङ्कानां स्थाने $\frac{३}{७}$=गुणकं, दलाढ्यराशिज्यात्मकभाजकस्थाने द्वादशी ज्या=२१४५ तुल्यं भाजकं $\frac{५}{७}$ अनेनापवर्त्यगुणकं चतुर्मितं भाजकं च १४४६ तुल्यं कृतम्। तथे- 'नीचोच्चवृत्तं क्षितिजस्य मान्दं स्फुटं वदन्तीह खशैल ७० भागान्। त्र्यंशोनिताम्भोधिजिना २४३।४० श्चशैघ्र्यमवाप्तभागै रहितं सदैव, ति श्रीपत्युक्त- 'तत्स्फुटपरिधिः खनगा इत्यादि' माचार्योक्तानुरूपमेव, तथा चे- 'मृदुफलदलमादौ मध्यमे मेदनीजे तदनुचलफलस्याप्यर्धमस्मिन् विधेयम्। पुनरपि परि-पूर्णं मान्दशैघ्र्ये च मध्ये ह्यसकृदवनिसूनोरेवमाहुः स्फुटत्वम्' ति श्रीपत्युक्ते- 'मन्दफलं मध्येऽर्धं तच्छीघ्रफलस्य मध्यमे सकल इत्यादि' राचार्योक्तमिदमेव मूलं 'दली-कृताभ्यां प्रथमं फलाभ्यामित्यादि' भास्करोक्तमपि बुधैश्चिन्त्यमिति ॥ ३४-३५-३६-३७-३८-३९-४० ॥

अब मङ्गलादि ग्रहोंके मन्द परिध्यंश, शीघ्र परिध्यंश
और स्पष्टीकरण को कहते हैं।

शुक्र के विषम पदान्त में मन्दोच्चनीचवृत्त परिध्यंश नव ९ होता है और सम पदान्त में ग्यारह होता है, विषम पदान्त में शीघ्र परिध्यंश २६३, और सम पदान्त में २५८ होते हैं। मध्यम ग्रह में मन्द फल के संस्कार करने से मन्द स्पष्ट ग्रह होता है। मन्द स्पष्ट ग्रह में शीघ्र फल संस्कार करने से असकृत्प्रकार से स्पष्ट ग्रह होते हैं अर्थात् मध्यम ग्रह से फिर मन्दफल लाकर मध्यमग्रह में संस्कार करने से मन्दस्पष्टग्रह होते हैं। उस (मन्दस्पष्टग्रह) से पुनः शीघ्र फल लाकर मन्दस्पष्टग्रह में संस्कार करने से स्पष्टग्रह होते हैं। पुनः इस स्पष्ट ग्रह से पूर्ववत् मन्दस्पष्टग्रह और स्पष्टग्रह साधन करना जब तक बिलकुल ठीक हो

जाय तब तक करना चाहिये, बुध, बृहस्पति और शनैश्चर मन्दफलों और शीघ्रफलों से पूर्ववत् असकृत् कर्म से स्फुट होते हैं। बुध के मन्दपरिध्यंश=३८°, बृहस्पति के=३३°, शनैश्चर के मन्दपरिध्यंश=३०°, विषमपदान्त में और समपदान्त में अर्थात् इन ग्रहों के मन्द-परिध्यंशों में संस्काराभाव होता है, बुध के शीघ्र परिध्यंश=१३२°, बृहस्पति के=६८°, शनि के=३५°, इन शीघ्र परिध्यंशों में भी संस्काराभाव होता है, अर्थात् बराबर स्थिर (एकरूपक) परिध्यंश होता है, मङ्गल का शीघ्रकेन्द्र जिस पद में हो उसमें गत केन्द्रांश और गम्य केन्द्रांश में जो अल्प हो उसकी ज्या को तृतीयांश रहित सात अंश ६°।४०′ से गुणा कर पैंतालीस अंश की ज्या से भाग देने से लब्ध जो अंश हो उस को मकरादिशीघ्र केन्द्र में मङ्गल के मन्दोच्च में जोड़ने से और कर्क्यादि केन्द्र में घटाने से स्पष्टीकरणोपयुक्त मङ्गल का मन्दोच्च होता है, उन (मङ्गल) के मन्दपरिध्यंश=७०°, और २४३°।४०′ इन में मन्दोच्च संस्कारार्थ पहले जो अंश प्राप्त हुये थे उनको हीन करने से मङ्गल की स्फुटशीघ्रपरिधि होती है। [मङ्गल का स्पष्टीकरण आगे लिखे हुए नियम के अनुसार होता है। गणितागत मङ्गल में यथागत मन्दफल के आधे को धन या ऋण करना, उस मन्द फलार्ध संस्कृत मध्यम मङ्गल से शीघ्रफल साधन करना, उस के आधे को मन्दफलार्ध संस्कृत मध्यम मङ्गल में देना, मन्दफलार्ध और शीघ्र फलार्ध संस्कृत मध्यम मङ्गल से जो मन्दफल हो मध्यम मङ्गल में उस को संस्कार कर के जो हो उस से शीघ्रफल साधन करना, वे दोनों (मन्दफल और शीघ्रफल) गणितागत मङ्गल में देना, बाद में बुध, बृहस्पति और शनि की तरह असकृत्कर्म तब तक करना चाहिये जबतक बिलकुल ठीक हो जाय, तब मङ्गल स्फुट होते हैं। ग्रह की तरह स्पष्टगति साधन करना, अद्यतन श्वस्तन ग्रहों का अन्तर स्पष्टग्रहगति होती है इति ॥३४-३५-३६-३७-३८-३९-४०॥

उपपत्ति

ब्रह्मगुप्तोक्त शनि की शीघ्रपरिधि=३५°, सूर्यसिद्धान्तोक्त शनि की शीघ्रपरिधि =४०°, भास्करोक्तशनिशीघ्र=४०, आचार्योक्त शनिमन्दपरिध्यंश=३०°, भास्करोक्तशनि-मन्दपरिध्यंश=५०°, शनि के परिध्यंशों में भास्कराचार्य आचार्योक्त से भिन्न क्यों कहे हैं इस को वे ही जान सकते हैं, सूर्यसिद्धान्तकार शीघ्रान्त्यफलज्या भी सदा स्थिर नहीं है इस बात को मन में रख कर समपदान्त और विषम पदान्त भेद से भिन्न-भिन्न परिध्यंश बताये हैं, जैसे 'कुजादीनामतः शैघ्र्या युग्मान्तेऽर्धाग्निदस्रकाः इत्यादि' संस्कृतोपपत्ति में लिखा गया है, आचार्य (ब्रह्मगुप्त) ने जिस तरह परिधि का स्पष्टीकरण किया है तदनुरूप ही भास्कराचार्य ने भी अपनी सिद्धान्तशिरोमणि में कहा है, मङ्गल के चारों शीघ्र केन्द्र पदान्त में गणितागत मन्दोच्च एक ही रूप का होता है, पदमध्य में तृतीयांश (२० कला) रहित सात अंश ६°।४०′ करके न्यून होता है, इसलिये इन दोनों के मध्य में अनुपात करते हैं, यदि पैंतालीस अंश शीघ्रकेन्द्रांश की ज्या में तृतीयांशोन सात अंश ६°।४०′ मन्दोच्चान्तर पाते हैं तब इष्टशीघ्रकेन्द्र पद में गत और गम्य में अल्प शीघ्र केन्द्रज्या में क्या इस अनुपात से जो फल आता है उसको मकरादि केन्द्र में और कर्क्यादिकेन्द्र में गणितागत

मङ्गल के मन्दोच्च में युत और हीन करने से स्फुटमन्दोच्च होता है, तथा मङ्गल के पदान्तों में शीघ्रपरिधि एकरूपक ही होती है, पदमध्य में तृतीयांशोन सात अंश ६°।४०′ करके सदा न्यून ही होता है, दोनों के मध्य में पूर्वानुपातागत फल को उनके शीघ्र परिधिप्रमाण में धन और ऋण करने से स्फुट शीघ्रपरिधि होती है, आचार्योक्तानुरूप ही भास्कराचार्य कहते हैं। ग्रहस्पष्टीकरण में मङ्गल और शुक्र ही में अन्तर तथा मङ्गल के मन्दोच्च में वैषम्य को आचार्य देखे, इस में कोई युक्ति नहीं कही गयी है, केवल वेध ही प्रमाण है, यह बात भास्कराचार्य के कथन से मालूम होती है, सिद्धान्तशेखर में श्रीपति ने 'भौमो निते दिनकरे पदयातयेयन्यूनज्यका" इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से आचार्योक्तानुरूप ही मङ्गल के मन्दोच्चस्फुटीकरण को कहा है, पद में गत केन्द्रांश और गम्य केन्द्रांश में जो अल्प है उस की ज्या के तृतीयांशोन सात अंश के गुणकाङ्क स्थान में $\frac{२}{३}$ = गुणक और पैंतालीस अंश ज्यात्मक भाजक स्थान में बारहवीज्या = २१४५ तुल्य भाजक को $\frac{५}{३}$ इससे अपवर्त्तन देकर गुणक स्थान में चार ४, और भाजक स्थान में १४४९ किया तथा 'नीचोच्चवृत्तं क्षितिजस्य मान्दं' इत्यादि संस्कृतोपपत्ति में लिखित श्रीपति की उक्ति 'तत्स्फुटपरिधिः खनगाः' इत्यादि आचार्योक्त के अनुरूप ही है, तथा 'मृदुफलदलमादौ मध्यमे मेदनीजे' इत्यादि श्रीपति प्रकार का तथा 'दलीकृताभ्यां प्रथमफलाभ्यां' इत्यादि भास्करोक्त प्रकार का मूल 'मन्दफलं मध्येऽर्धं तच्छीघ्रफलस्य मध्यमे' इत्यादि आचार्योक्त प्रकार ही है इति ॥ ३४-३५-३६-३७-३८-३९-४०॥

इदानीं ग्रहाणां मन्दस्पष्टगतिं स्पष्टगतिं चाह

ग्रहमन्दकेन्द्रभुक्तिर्ज्यान्तरगुणिताऽऽद्यजीवया २१४ भक्ता ।
लब्धं स्फुटपरिधिघ्नं भगणांश ३६० हृतं कलाभिस्तु ॥ ४१ ॥

मृगकर्क्यादावूनाधिका स्वमध्यमगतिः स्फुटाऽर्केन्द्वोः ।
शीघ्रगतिं मन्दफलस्फुटभुक्त्यूनां कुजादीनाम् ॥ ४२ ॥

शीघ्रफलं भोग्यज्यासङ्गुणितं त्वाद्यजीवया विभजेत् ।
फलगुणितं व्यासार्धं विभाजयेच्छीघ्रकर्णेन ॥ ४३ ॥

लब्धोना शीघ्रगतिः स्फुटभुक्तिर्भवति लब्धमधिकं चेत् ।
शीघ्रगतेः शीघ्रगतिं लब्धात् संशोध्य वक्रगतिः ॥ ४४ ॥

वा. भा.—ग्रहस्य मन्दकर्मणि यत् केन्द्रं तद्ग्रहमन्दकेन्द्रं तस्य भुक्तिर्यथा मध्यग्रहात् स्वमन्दोच्चं विशोध्य केन्द्रं भवति । एवं ग्रहमध्यभुक्तौ स्वमन्दोच्चभुक्तिं विशोध्य केन्द्रभुक्तिर्भवति । सा च ज्यान्तरगुणिता कार्या अवशेषमन्दकर्मणि भुजज्यायां क्रियमाणायां यज्ज्यान्तरं भवेत् तद्गुणनीयेत्यर्थः। तत आद्यजीवया भक्ता कार्या प्रथमं ज्यार्धं मनुयमला इत्यर्थः २१४; ततो यल्लब्धं तत्स्फुटमन्दपरिधिगुणं भगणांशहृतं च कृत्वा यत्तत्फलं ताः कलाः ताभिः फलकलाभिर्मृगकर्क्यादौ स्थिते

स्वमन्दकेन्द्रे यथासंख्यमूनाधिका सती स्वमध्यगतिः मन्दस्फुटा भवति । भौमादीनां सा मन्दस्फुटैवेत्युच्यते । रविचन्द्रयोस्तु सैव परमार्थ्यस्फुटा यतस्तौ मन्दप्रतिमण्डले भ्रमतः इत्यत्रेयं वासना ।

कक्षामण्डले यत्र प्रदेशे रविर्वर्तते, चन्द्रो वा भौमादीनां स्वमन्दनीचोच्चमध्यवृत्तं वा तत्र यज्ज्यान्तरं तेन सह त्रैराशिकं यदि तत्त्वयमसंख्याभिर्लिप्ताभिर्ज्यान्तरं लभ्यते तदा मन्दकेन्द्रभुक्तिलिप्ताभिः किमिति द्वितीयं त्रैराशिकं यदि षष्टिशतत्रयवृत्ते यत्फलं ज्यारूपमेतावत्स्वमन्दोच्चनीचवृत्ते कियदिति ततः तृतीयं त्रैराशिकं यदि मनुयमलतुल्यस्य ज्याफलस्य तत्त्वयमसंख्याश्चापलिप्ता भवति । तदस्य कियत् प्रभवत्येवं प्रथमत्रैराशिके तत्त्वयमसंस्थो भागहारः ।फलं स्वभुक्तावुपचयापचयो वा कक्षामंडलात्प्रतिमण्डलमुपरि यत्र तत्र भुक्तेरुपचयो यत्रार्धस्तत्रोपचयो ऽतएव चोक्तं मृगकर्क्यादावूनाधिका स्वमध्यगतिरित्येतत्सर्वं यथा तत्तेषु कक्षामण्डलादिषु प्रदर्शयेत् । उपपन्नं च यदि नाम चन्द्रकेन्द्रभुक्तिर्बहुज्यान्तरव्यापिनी तत्र केन्द्रवच्चन्द्राक्रान्तज्यान्तरादारभ्यातीतज्यांतरैर्भुक्तेः स्फुटीकरणमिच्छन्त्यपरे तत एवागामिज्यान्तरैः कर्म कुर्वते एवमतीतैः स्फुटया आगामिज्यान्तरैरपि स्फुटया सह योगार्धेन, अपरे तु पुनर्गतात्कलानयनेऽतीतैर्ज्यान्तरैः स्फुटया गम्यावकलानयनेनागते ज्यान्तरगतया स्फुटया चन्द्रभुक्तौ कर्म कुरुते तत्र स्फुटा भुक्तिः क्षणं मध्ये एका वक्तुं शक्यते । कक्षामण्डलप्रतिमण्डलयोरन्यथासंस्थानात्तस्मादनवस्थाप्रसंगः स्यादित्याचार्येण क्रान्तज्यान्तरेणैव भुक्तिज्यानीता । स्वल्पान्तरत्वादेवं रव्यादीनामपि कल्पाः सम्भवन्ति । यदि नामात्यल्पमन्तरे तेषां भुक्तेरल्पत्वादिति । यच्चापकारणमस्यामार्यायामाद्यजीवया स्थिरयोपनिबद्धं तदन्येषां ज्यान्तराणामसम्भवाद्यतो भुक्तिज्याफलांशुवद्वापि मनुयमलानां लिप्तानां तुल्या न भवन्ति । तस्मादुक्तं सर्वमुपपन्नमिति ।

इदानीं भौमादीनां भुक्तिकरणार्थमार्याद्वयं सार्धमाह । कुजादीनां ताराग्रहाणां शीघ्रगतिः किंभूतानां मन्दफलस्फुटभुक्तीनां शीघ्रफलभोग्यजीवा संगुणिता च सती विभजेत् । कया आद्यजीवया एतदुक्तं भवति । भौमादेः ग्रहस्य पूर्वप्रदर्शितेन कर्मणा मंदफलस्फुटभुक्तिः तां स्वशीघ्रभुक्तेर्विशोध्य मन्दफलस्फुटभुक्तीनां शीघ्रगतिर्भवति । ततस्तां गुणयेत् । शीघ्रफलभोग्यया ग्रहस्य स्फुटीक्रियमाणस्य यच्छीघ्रफलं भवति तस्य फलज्यायां क्रियमाणायां यज्ज्यान्तरं गुणकारः संभवति सा शीघ्रफलभोग्यजीवेत्युच्यते तया मन्दफलभुक्त्यूनां शीघ्रगतिं संगुणय्याद्यजीवया विभजेत् । मनुयमलैरित्यर्थः ।

ततो यत्फलं तेन फलेन गुणितं व्यासार्धं तद्विभाजयेत् शीघ्रकर्णेन ततोऽपि यल्लब्धं तेन सर्वदा शीघ्रगतिरूना कर्त्तव्या । सा चोना कृता ग्रहस्य स्फुटभुक्तिस्ततः प्रदेशे स्थितस्य भवति । लब्धमधिकं चेत् शीघ्रगतेः फलगुणितात् व्यासार्धात्

शीघ्रकर्णाहृताद्यल्लब्धं तच्छीघ्रगतेरधिकं भवति। तदा शीघ्रगतिलब्धां संशोध्य वक्रगतिर्भवति। विपरीतशोधने कृतेऽप्यवक्रभुक्तिर्भवति। तदैवैतत्संभवतीत्यर्थः। यदि नाम भौमस्यायं विशेषः प्रथममन्दफलसंस्कृतादानीय तदर्धं मध्यभुक्तावृणं धनं वा कार्यं ततस्तदूनां शीघ्रभुक्तिं शीघ्रफलार्धभोग्यजीवासंगुणितामाद्यजीवया विभजेत्।

लब्धेनोक्तवत्स्फुटभुक्तिः समानीय यदि तया सह मन्दफलार्धस्फुटभुक्तेर्यदन्तरार्धं तत्रैवैके कर्मकृतभुक्तौ धनं ऋणं वा कार्यम्। यदि मन्दस्फुटभुक्तेरधिका स्फुटभुक्तिः तद्धनमन्यथार्णमित्येवं कृते द्विकर्मस्फुटा भौमभुक्तिर्भवति। तां मध्यां परिशेषग्रहवत्तद्भुक्तेरपि स्फुटीकरणमिति कक्षामंडलादीनि यथा विन्यस्य सर्वं प्रदर्शयेत्। तत्रेयं वासना मंदफलस्फुटो ग्रहो यत्र प्रदेशे कक्षामंडले वर्त्तते तत्र शीघ्रोच्चनीचोच्चवृत्तमध्ये कृते तत्परिधिशीघ्रप्रतिमंडलपरिध्योर्यत्र संपातस्तत्र स्फुटो ग्रहः तस्य शीघ्रोच्चरेखया सहान्तरं यत्तत्प्रतिदिनमुपचीयते स्वशीघ्रभुक्तिमंदफलस्फुटभुक्त्योरन्तरेण यतः शीघ्रभुक्त्या यदि प्राक्मंदस्फुटो मन्दस्फुटभुक्त्या भवति मन्दफलभुक्त्यूना शीघ्रगतिः क्रियते। तत्र तयोरंतरं शीघ्रकेन्द्रभुक्तिर्भवति। सा च शीघ्रकेन्द्रभुक्तिः स्फुटीक्रियते। तत्र चैव शीघ्रनीचोच्चवृत्तस्य मध्यगा शलाका सैवावधित्वेन परिकल्पिता फलचापकरणे यतस्ततएव यावान्विप्रकर्षस्तावदेव ग्रहफलमतस्तत्र एवावधेः क्रमज्या प्रवर्त्तते। फलचापकरणे एतच्च प्रागेवोक्तशीघ्रफलान्ते यज्ज्याफलं तेन त्रैराशिकं यदि तत्त्रयमलैः तज्ज्यान्तरं लभ्यते तच्छीघ्रकेन्द्रभुक्तिलिप्ताभिः किमिति, ततो द्वितीयं यमलैः तत्त्रयमतुल्याश्चापलिप्ता भवन्ति। तल्लब्धज्याखंडने किमिति, अत्र प्रथमे त्रैराशिको तत्त्वसांख्यो भागहारो द्वितीयो गुणकारः ततस्तयोर्नष्टयोः शीघ्रकेन्द्रभुक्तेर्ज्यान्तरं गुणकारः। आद्यजीवा भागहारः फलं प्रतिमंडलस्थग्रहप्रदेशे स्फुटशीघ्रभुक्तिः सा च कक्षामंडले परिणाम्यते, तदर्थमुक्तं फलगुणितव्यासार्धं विभाजयेच्छीघ्रकर्णेन त्रैराशिकमिदं ततो यल्लब्धं सा शीघ्रकेन्द्रभुक्तिः स्फुटा कक्षामंडले। सा च ग्रहशीघ्रस्फुटगत्योरन्तरमतएव शीघ्रगतेः संशोध्य ग्रहस्य स्फुटभुक्तिर्भवति। लब्धमधिकं चेच्छीघ्रगतेर्यदा भवति, तदा विपरीतशोधने कृते वक्रभुक्तिर्भवति यस्माच्छीघ्रकर्णस्तदाल्पो भवति। कक्षामंडलस्योपरिस्थितत्वात्फलगुणितं व्यासार्धविभाजयेच्छीघ्रकर्णेन यावत् क्रियते तावच्छीघ्रगतेरप्यधिका शीघ्रकेन्द्रभुक्तिः स्फुटा भवति। दृग्भेदस्याधिकत्वात् स्वमध्यगतेः कक्षामंडलप्रतिमंडलावस्थितिवशेन ग्रहोपि प्राग्दिनाध्यासितप्रदेशावलंम्बितः पश्चादुपलभ्यते शीघ्रगतिः।

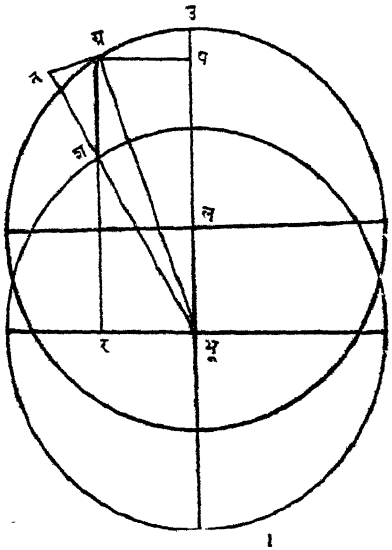
शीघ्रकेन्द्रस्फुटभुक्त्यन्तरगुणा अत उक्तं लब्धात्संशोध्य शीघ्रगतिर्वक्रगतिरिति सर्वमुपपन्नम्। स्वकृते सिद्धान्ते स्फुटीकरणस्य चान्याचार्यदूषणद्वारेण प्रशंसार्थमार्यात्रयमाह ॥४१-४२-४३-४४॥

वि. भा.—ग्रहमन्दकेन्द्रगतिर्ज्यान्तरेण (भोग्यखण्डेन) गुणिता, आद्यजीवया (प्रथमज्यया) भक्ता यल्लब्धं तत्स्फुटपरिधिगुणितं, भगणांशै ३६०भक्तं लब्धाभिः कलाभिर्मृगकर्क्यादौ (मकरादिकेन्द्रे, कर्क्यादिकेन्द्रे च) स्वमध्यमगतिरूनाधिका तदाऽर्केन्द्वोः(रविचन्द्रयोः)स्फुटा गतिर्भवति, कुजादीनां ग्रहाणां मन्दगतिफलरहितां मध्यगतिं (मन्दस्पष्टगतिं) वदन्त्याचार्याः। स्वमन्दस्पष्टगतिरहिता शीघ्रोच्चगतिः कुजादिग्रहाणां शीघ्रकेन्द्रगतिर्भवति। तां (शीघ्रकेन्द्रगतिं) शीघ्रफलस्य ज्यायां क्रियमाणायां या भोग्यज्या (ज्यान्तरं) तया सङ्गुणितां कृत्वाऽऽद्यजीवया (प्रथमज्यया) विभजेत्, यल्लब्धं फलं तेन गुणितं व्यासार्धं (त्रिज्या) शीघ्रकर्णेन विभाजयेत्। लब्धेन रहिता शीघ्रगतिः (शीघ्रोच्चगतिः) स्फुटगतिर्भवति। चेत् (यदि) लब्धं फलं शीघ्रगतेः (शीघ्रोच्चगतेः) अधिकं तदा लब्धाच्छीघ्रोच्चगतिं संशोध्य शेषं कुजादिग्रहाणां वक्रगतिर्भवतीति ॥४१-४२-४३-४४॥

अत्रोपपत्तिः

मध्यस्पष्टभेदेन गतिर्द्विविधा भवति, या गतिः प्रतिक्षणं भिन्ना-भिन्ना भवति, सा स्पष्टाऽन्या मध्या, स्फुटा गतिरपि दैनिकतात्कालिकभेदेन द्विविधा भवति, तेन दैनिकमन्दस्पष्टागतिः, तात्कालिकमन्दस्पष्टागतिः। दैनिकस्पष्टागतिः, तात्कालिक-स्पष्टागतिः, आचार्येण दैनिकमन्दस्पष्टगतिः, दैनिकस्पष्टगतिश्चानीयते।

(क)



भू=भूकेन्द्रम्। भूल=मन्दान्त्य-फलज्या। ल=मन्दग्रहगोलकेन्द्रम्। ग्र=मन्दप्रतिवृत्ते गणितागतमध्यग्रहः। उ=मन्दोच्चम्। ग्रउ=मन्दकेन्द्रम्। ग्रन=मन्दभुजफलम्=स्वल्पान्तरान्मन्द-फलज्या। ग्रप=भूर=मन्दकेन्द्रज्या। ग्रश=मन्दान्त्यफलज्या, तदा भूरश, ग्रशन त्रिभुजयोः सजातीयत्वादनुपातः $\frac{\text{मंकेज्या}\times\text{मंग्रफज्या}}{\text{त्रि}}$=अद्यतनभुजफल =अद्यतनमंफज्या=मंफज्या

एवं $\frac{\text{मंकेज्या}'\times\text{मंग्रंफज्या}}{\text{त्रि}}$=श्वस्तन-भुजफल=श्वस्तनमंफज्या=मंफज्या'

अनयोरन्तरम्

$\frac{\text{मंग्रंफज्या}}{\text{त्रि}}\left(\text{मंकेज्या}'-\text{मंकेज्या}\right)=\text{मंफज्या}'-\text{मंफज्या}=$

$\frac{\text{मंग्रंफज्या}}{\text{त्रि}} \times \text{मन्दकेन्द्रज्यान्तर} = \text{मन्दफलज्यान्तर} = \text{मन्दफलान्तर}$

$= \text{मन्दगतिफलस्वल्पान्तरात्}$

अत्र $\frac{\text{मन्दकेग} \times \text{भोग्यखं}}{\text{प्रथमचा}} = \frac{\text{मंकेग} \times \text{भोग्यखं}}{\text{प्रथमज्या}} = \text{मन्दकेज्यान्तर}$

एतदुत्थापनेन $\frac{\text{मंग्रंफज्या}}{\text{त्रि}} \times \frac{\text{मंकेग} \times \text{भोखं}}{\text{प्रथमज्या}} = \text{मन्दगतिफल},$

परन्तु $\frac{\text{मंग्रंफज्या}}{\text{त्रि}} = \frac{\text{मंपरिधि}}{३६०}$

अत उत्थापनेन $\frac{\text{मंपरिधि}}{३६०} \times \frac{\text{मंकेग} \times \text{भोखं}}{\text{प्रथमज्या}} = \text{मन्दगतिफल}$, एतावताऽऽचार्योक्तं, मन्दकेन्द्रगतिरर्कचन्द्रयोर्ज्यान्तरेण गुणिता हृता ऽऽद्यया, जीवया स्वपरिणाहताड़िता खर्तुराम ३६० विहृता गतेः फलम्, श्रीपत्युक्तमिदं सूर्यसिद्धान्तकारोक्तमन्दगतिफलानयनं चोपपद्ये, श्रीपतिसूर्यसिद्धान्तकारमते प्रथमज्या = २२५। तत्रैव मन्दगतिफलस्वरूपे भांशपरिधिप्रमाणे ६ ऽनेनापवर्त्तिते तदा पूर्वानीतमन्दगतिफलम् =

$\frac{\text{मंकेग} \times \text{भोखं}}{\text{प्रथमज्या}} \times \frac{२ \text{ परिधि}}{\frac{६}{२ \times ३६०}} = \frac{\text{मंकेग} \times \text{भोखं}}{\text{प्रथमज्या}} \times \frac{\text{स्फुगुणक}}{८०}, \quad \frac{२ \text{ परिधि}}{६} = \text{स्फुगुणक},$

$\frac{२ \times ३६०}{६} = ८०$ एतावता 'ज्याखण्डकेन गुणिता मृदुकेन्द्रजेन भुक्तिर्ग्रहस्य शरयुग्मयमैर्विभक्ता, क्षुण्णा स्फुटेन गुणकेन हृता खनागैः लिप्ता गतेः फलमृणं धनमुक्तवच्च' लल्लोक्तमिदमुपपद्यते। अत्रापि प्रथमज्या = २२५, अन्यत्सर्वं समानमेव। मन्दगतिफलानयनं केषामपि समीचीनं नास्तीति पूर्वोक्तोपपत्तिदर्शनेनैव स्फुटं भवति, केवलं भास्कराचार्येण तत्साधनं यत्कृतं तत्समीचीनमस्ति, यद्यपि भास्करोक्तं 'कोटीफलघ्नी मृदुकेन्द्रभुक्तिरित्यादिना' मन्दगतिफलानयनं समीचीनमित्येतदर्थं 'वटेश्वरसिद्धान्ते' वासना प्रतिपादिताऽस्ति मया, तथाप्यत्रापि प्रतिपाद्यते।

$\frac{\text{मंग्रंफज्या} \times \text{मंकेज्या}}{\text{त्रि}} = \text{मंफज्या}$, पक्षयोस्तात्कालिकगत्यानयनेन $\frac{\text{मंग्रंफज्या}}{\text{त्रि}} \times$

$\frac{\text{मंकेकोज्या} \times \text{मंकेग}}{\text{त्रि}} = \frac{\text{मंगफ} \times \text{मंफकोज्या}}{\text{त्रि}}$

$\therefore \text{मंगफ} = \frac{\text{मंग्रंफज्या}}{\text{त्रि}} \times \frac{\text{मंकेकोज्या} \times \text{मंकेग}}{\text{त्रि}} \times \frac{\text{त्रि}}{\text{मंफकोज्या}} =$

$$\frac{\text{मंकोटिफल} \times \text{मंकेग}}{\text{त्रि}} \times \frac{\text{त्रि}}{\text{मफकोज्या}} = \frac{\text{भास्करोक्तमंगफल} \times \text{त्रि}}{\text{मंफकोज्या}},$$

यतः $\frac{\text{मंकोटिफल} \times \text{मंकेग}}{\text{त्रि}} = \text{भास्करोक्तमन्दगतिफ}$ ।

एतावता सिद्धं यद्भास्करोक्तं मन्दगतिफलं त्रिज्यया गुणितं मन्दफलकोटिज्यया भक्तं तदा वास्तवं मन्दगतिफलं भवेदिति एतेन च 'भास्करोक्तं गतिफलं त्रिज्यया गुणितं हृतम् । मन्दीयफलकोटिज्यामानेन भवति स्फुटमिति' विशेषोक्तसूत्रमुपपद्यते ॥

ततो मन्दस्फुटगतिप्रमाणमानीयते

प्रथमे पदे द्वितीये पदे च (मकरादिकेन्द्रे) अद्यतनमध्यग्र—अद्यतनमंफ
=अद्यतनमंस्पग्र.

श्वस्तनमध्यग्र—श्वस्तनमंफ=श्वस्तनमंस्पग्र.

अनयोरन्तरेण मध्यग—मन्दगतिफ=मंस्पगति, कुजादिग्रहाणाम्
तृतीय पदे चतुर्थ पदे च (कर्क्यादिकेन्द्रे) अद्यतनमध्यग्र + अद्यतनमंफ
=अद्यतनमंस्पग्र.

श्वस्तनमध्यग्र + श्वस्तनमंफ=श्वस्तनमंस्पग्र.

अनयोरन्तरेण

कुजादिग्रहाणां मध्यगति + मन्दगतिफल=मंस्पगति, रविमग ∓ रविमंगफल
=रविस्पष्टगति,

चंमग ∓ चंमंगफल=चन्द्रस्पष्टगति.

अथ स्पष्टगतिसाधनार्थमुपपत्तिः

$\frac{\text{शीकेज्या} \times \text{त्रि}}{\text{शीक}} = \text{स्पकेज्या}$ । द्वितीयदिने $\frac{\text{शीकेज्या}' \times \text{त्रि}}{\text{शीक}} = \text{स्पकेज्या}'$,

अत्र शीकेज्या=मंस्पकेज्या

अनयोरन्तरम्

$$\frac{\text{त्रि}\ (\text{शीकेज्या}' - \text{शीकेज्या})}{\text{शीक}} = \text{स्पकेज्या}' - \text{स्पकेज्या} = \frac{\text{त्रि} \times \text{शीकेज्यान्तर}}{\text{शीक}}$$

=स्पकेज्यान्तर, शीक=शीघ्रकर्णः

परन्तु $\frac{\text{भोखं}\times\text{शीकेग}}{\text{प्रथमज्या}}$=शीकेज्यान्तर, अत उत्थापनेन $\frac{\text{त्रि.}}{\text{शीक}}\times\frac{\text{भोखं}\times\text{शीकेग}}{\text{प्रथमज्या}}$

$=\frac{\text{त्रि.}}{\text{शीक}}\times\frac{\text{भोखं}\times\text{मन्दकेग}}{\text{प्रज्या}}$=स्पष्टकेज्यान्तर = स्पकेन्द्रान्तर = स्पकेग, स्वल्पा-न्तरात् । ततः अद्यतनशीउ—अद्यतनस्पग=अद्यतनस्पके तथा श्वस्तनशीउ—श्वस्तनस्पग=श्वस्तनस्पके, अनयोरन्तरम्=शीउग—स्पग=स्पकेग

ततः शीउग—स्पकेग=स्पगति, यदि शीउग<स्पकेग तदा विलोमशोधनेन वक्रा गतिः=ऋणात्मिका गतिर्भवेदेतावताऽऽचार्योक्तं सर्वमुपपन्नम् । सिद्धान्तशेखरे—

चञ्चलकेन्द्रगतिः फलभोग्यज्यागुणिता ऽऽद्यगुणेन २२३ विभक्ता ।
व्यासदल ३४१५ ऋफलं श्रुतिभक्तं तद्रहिताशुगतिः स्फुटभुक्तिः ॥
स्यादवनीतनयादिखगानां शीघ्रगतेः फलमभ्यधिकं चेत् ।
तत्फलतोऽपि विशोधय शेषं वक्रगतिर्भवति द्युचराणाम् ॥

इति श्रीपत्युक्तं स्पष्टगतिसाधनमाचार्योक्तानुरूपमेव, लल्लाचार्योक्तमपि स्फुटगतिसाधनमीदृशमेवास्ति, परं केषामप्याचार्याणां स्फुटगतिसाधनं न समीचीनमिति तदुपपत्तिदर्शनेनैव स्फुटं भवति, केवलं सिद्धान्तशिरोमणौ 'फलांशखाङ्कान्तर-शिञ्जिनीघ्नी' त्यादिना भास्कराचार्येण तात्कालिकगत्या सूक्ष्मं स्पष्टगतिसाधनं कृतमिति विवेचकैर्विवेचनीयम् ॥४१-४२-४३-४४॥

अब ग्रहों की मन्दस्पष्टगति और स्पष्टगति को कहते हैं

हि. भा.—मन्दकेन्द्र गति को ज्यान्तर (भोग्यखण्ड) से गुणाकर प्रथमज्या से भाग देकर जो लब्धि हो उसको स्फुट परिधि से गुणाकर भगणांश ३६० से भाग देकर जो लब्ध कला हो उसको मकरादि केन्द्र में अपनी मध्यम गति में घटाने से और कर्क्यादिकेन्द्र में जोड़ने से रवि और चन्द्र की स्पष्टगति होती है, अपनी मध्यगति में मन्दगति फल को संस्कार करने से कुजादि ग्रहों की मन्दस्पष्टगति होती है, अपनी मन्दस्पष्टगति को शीघ्रोच्चगति में घटाने से कुजादि ग्रहों की शीघ्रकेन्द्रगति होती है, शीघ्रकेन्द्रगति को भोग्य खण्ड से गुणा कर प्रथमज्या से भाग देने से जो लब्धि हो उसको व्यासार्ध (त्रिज्या) से गुणा कर शीघ्रकर्ण से भाग देने से जो लब्धि हो उसको शीघ्रोच्चगति में घटाने से स्फुटगति होती है, यदि लब्धि शीघ्रोच्चगति से अधिक हो तब लब्धि में शीघ्रोच्चगति को घटाकर जो शेष रहता है वह कुजादि (मङ्गलादि) ग्रहों की वक्रगति होती है. इति ॥४१-४२-४३-४४॥

उपपत्ति

मध्य और स्पष्टभेद से गति दो तरह की होती है, प्रतिक्षण में जो गति भिन्न होती है, वह स्पष्टगति है, जो प्रतिक्षण में भिन्न नहीं होती है वह मध्यगति है। स्पष्टगति भी दैनिक और तात्कालिक भेद से दो प्रकार की होती है, दैनिकमन्दस्पष्टगति, तात्कालिक-मन्दस्पष्टगति, दैनिकस्पष्टगति, तात्कालिकस्पष्टगति, आचार्य दैनिकमन्दस्पष्टगति और स्पष्टगति का साधन करते हैं। संस्कृतोपपत्ति में लिखित (क) क्षेत्र को देखिये।

भू=भूकेन्द्र, भूल=मन्दान्त्यफलज्या, ल=ग्रहमन्दगोलकेन्द्र, ग्र=मन्दप्रतिवृत्त में गणितागत=मध्यमग्रह। उ=मन्दोच्च, ग्रउ=मन्दकेन्द्र, ग्रन=मन्दभुजफल=मन्दफलज्या स्वल्पान्तर से, ग्रप=भूर=मन्दकेन्द्रज्या, ग्रग=मन्दान्त्यफलज्या, भूरग, ग्रसन दोनों त्रिभुज सजातीय हैं इसलिये अनुपात करते हैं,

$$\frac{\text{मंकेज्या} \times \text{मंग्रंफज्या}}{\text{त्रि}} = \text{ग्रद्यतनभुजफल} = \text{ग्रद्यतनमन्दफलज्या}, = \text{मंफज्या},$$

$$\frac{\text{मंकेज्या}' \times \text{मंग्रंफज्या}}{\text{त्रि}} = \text{श्वस्तनभुजफल} = \text{श्वस्तनमंदफलज्या} = \text{मंफज्या}',$$

दोनों का अंतर करने से

$$\frac{\text{मंग्रंफज्या}}{\text{त्रि}}(\text{मंकेज्या}' - \text{मंकेज्या}) = \frac{\text{मंग्रंफज्या} \times \text{मंकेज्यान्तर}}{\text{त्रि}} = \text{मंफज्या}' - \text{मंफज्या} =$$

मन्दफलज्यान्तर=मन्दफलान्तर=मंगतिफल

स्वल्पान्तर से, परन्तु $\frac{\text{मन्दकेग} \times \text{भोखं}}{\text{प्रथमचाप}} = \frac{\text{मंकेग} \times \text{भोखं}}{\text{प्रथमज्या}} = \text{मन्दकेन्द्रज्यान्तर}$, उत्थापन देने से

$$\frac{\text{मंग्रंफज्या}}{\text{त्रि}} \times \frac{\text{मंकेग} \times \text{भोखं}}{\text{प्रथमज्या}} = \text{मंगतिफल}। \text{ परन्तु } \frac{\text{मंग्रंफज्या}}{\text{त्रि}} = \frac{\text{मंपरिधि}}{\text{भाग}} = \frac{\text{मंपरिधि}}{३६०}$$

अतः $\frac{\text{मंपरिधि}}{३६०} \times \frac{\text{मंकेग} \times \text{भोखं}}{\text{प्रथमज्या}} = \text{मंगतिफल}$, इससे आचार्योक्त उपपन्न हुआ, सिद्धान्तशेखर में "मन्दकेन्द्रगतिरर्कचन्द्रयोः" इत्यादि संस्कृतोपपत्ति में लिखित श्रीपत्युक्तप्रकार, तथा सूर्यसिद्धान्तकारोक्त मन्दगति फलानयनप्रकार भी उपपन्न हुआ, उसी मन्दगतिफलस्वरूप में भांश और परिध्यंश को ६ इससे अपवर्त्तन देने से मन्दगतिफल

$$= \frac{\text{मंकेग} \times \text{भोखं}}{\text{प्रज्या}} \times \frac{२ \text{ परिधि}}{६} = \frac{\text{मंकेग} \times \text{भोखं}}{\text{प्रज्या}} \times \frac{\text{स्फुगुणक}}{८०}, \quad \frac{२ \text{ परिधि}}{६} = \text{स्फुगुणक}$$

$$\frac{२ \times ३६०}{६}$$

इससे 'ज्याखण्डकेन गुणिता' इत्यादि संस्कृतोपपत्ति में लिखित लल्लोक्त मन्दगतिफलानयन उपपन्न होता है, मन्दगतिफलानयन किसी भी आचार्य का ठीक नहीं है, यह पूर्वोक्तोपपत्ति देखने ही से स्पष्ट है। केवल भास्कराचार्योक्त मन्दगतिफलानयन ठीक है, यद्यपि बटेश्वर-सिद्धान्त में इस विषय को हम दिखला चुके हैं, तथापि यहाँ लिखते हैं।

$\frac{\text{मंअंफज्या}.\ \text{मंकेज्या}}{\text{त्रि}}$ = मंफज्या, दोनों पक्षों की तात्कालिक गति लेने से

$\frac{\text{मंअंफज्या}}{\text{त्रि}} \times \frac{\text{मंकेकोज्या}.\ \text{मंकेग}}{\text{त्रि}} = \frac{\text{मंगफ}.\ \text{मंफकोज्या}}{\text{त्रि}.}$ अतः $\frac{\text{मंअंफज्या}}{\text{त्रि}} \times \frac{\text{मंकेकोज्या}.\ \text{मंकेग}}{\text{त्रि}}$

$\times \frac{\text{त्रि}}{\text{मंफकोज्या}} = \frac{\text{मंकोफ}.\ \text{मंकेग}}{\text{त्रि}} \times \frac{\text{त्रि}}{\text{मंफकोज्या}} = \frac{\text{भास्करोक्तमन्दगतिफल}.\ \text{त्रि}}{\text{मंफकोज्या}}$ = मन्दगतिफल,

$\because \frac{\text{मंकोफ}.\ \text{मंकेग}}{\text{त्रि}}$ = भास्कर मंगफ,

इससे सिद्ध हुआ कि भास्करोक्त मन्दगतिफल को त्रिज्या से गुणाकर मन्दफलकोटिज्या से भाग देने से वास्तवमंदगति फल होता है, इससे 'भास्करोक्तं गतिफलं त्रिज्यया गुणितं' इत्यादि संस्कृतोपपत्ति में लिखित म. म. सुधारक द्विवेदी जी का सूत्र भी उपपन्न होता है.

मन्दस्पष्टगति प्रमाण लाते हैं

प्रथम पद में और द्वितीय पद में (मकरादि केन्द्र में) अद्यतनमध्यग्र—अद्यतनमंफ

= अद्यतनमंस्पग्र

श्वस्तनमध्यग्र—श्वस्तनमंफ = श्वस्तनमंस्पग्र

दोनों का अंतर करने से

कुजादि ग्रहों की मध्यगति—मंन्दगतिफल = मंदस्पष्टगति,

तृतीय पद में और चतुर्थ पद में (कर्क्यादिकेन्द्र में) अद्यतनमध्यग्र + अद्यतनमंफ

= अद्यतनमंस्पग्र.

श्वस्तनमध्यग्र + श्वस्तनमंफ = श्वस्तनमंस्पग्र.

दोनों का अन्तर करने से

कुजादि ग्रहों की मध्यगति + मंगतिफल = मंदस्पगति

रविमध्यगति ∓ रमंगतिफल = स्परविगति

एवं चंमध्यगति ∓ चंमंगतिफल = चंस्पष्टगति

अब स्पष्टगति साधन के लिये उपपत्ति

$\frac{\text{शीकेज्या}.\text{त्रि}}{\text{शीक}}$ = स्पकेज्या, द्वितीय दिन में $\frac{\text{शीकेज्या}'.\text{त्रि}}{\text{शीक}}$ = स्पकेज्या', यहां शीकेज्या = मन्द-स्पष्टकेन्द्रज्या, शीक = शीघ्रकर्ण.

दोनों का अन्तर करने से

$$\frac{\text{त्रि}\,(\text{शीकेज्या}'-\text{शीकेज्या})}{\text{शीक}} = \text{स्पकेज्या}'-\text{स्पकेज्या} = \frac{\text{त्रि}.\,\text{शीकेज्यान्तर}}{\text{शीक}} = \text{स्पकेज्यान्तर}$$

परन्तु. $\frac{\text{भोग्यं}.\,\text{शीकेग}}{\text{प्रथमचा}}$ = शीकेज्यान्तर, उत्थापन करने से $\frac{\text{त्रि}}{\text{शीक}} \times \frac{\text{भोग्यं}.\,\text{शीकेग}}{\text{प्रथमचा}} = \frac{\text{त्रि}}{\text{शीक}}$

$\times \frac{\text{भोग्यं}.\,\text{शीकेग}.}{\text{प्रज्या}}$ = स्पष्टकेज्यान्तर = स्पकेन्द्रान्तर = स्पकेग स्वल्पान्तर से

∴ अद्यननशीउ − अद्यननस्पग्र = अद्यननस्पके, श्वस्तनशीउ − श्वस्तनस्पग्र = श्वस्तन-स्पके, दोनों का अन्तर करने से शीउग − स्पग = स्पकेग ∴ शीउग − स्पकेग = स्पग; यदि शीउग < स्पकेग तब विलोमशोधन से ऋणात्मिकागति = वक्रगति होती है, इससे आचार्योक्त उपपन्न हुआ। सिद्धान्तशेखर में 'चञ्चलकेन्द्रगतिः फलभोग्यज्यागुणिता' इत्यादि संस्कृतोपपत्ति में लिखित श्रीपत्युक्त, स्पष्टगति साधन आचार्योक्तानुरूप ही है, लल्लाचार्योक्त स्पष्टगति साधन भी ऐसा ही है, लेकिन किसी भी आचार्य से कहा गया स्फुटगति साधन ठीक नहीं है, यह विषय पूर्वोक्तोपपत्ति देखने ही से स्पष्ट है, केवल सिद्धान्तशिरोमणि में 'फलांशखाङ्कान्तरशिञ्जिनीघ्नी' इत्यादि से तात्कालिकस्पष्टगति से भास्करोक्त स्पष्ट-गति साधन सूक्ष्म है, इसको विवेचक लोग विचारें इति ॥४१-४२-४३-४४॥

अत्र विशेषविचारः

अत्र "फलांशखाङ्कान्तरशिञ्जिनीघ्नी" त्यादि भास्करोयविधानेन—

$$\text{स्पष्टकेन्द्रगतिः} = \frac{\text{कोज्याफ} \times \text{शीकेग}}{\text{क}}$$

अथवा स्पकेग = शीकेग − गफ

$$\therefore \text{शीकेग} - \text{गफ} = \frac{\text{कोज्याफ} \times \text{शीकेग}}{\text{क}}$$

वा शीकेग.क − गफ.क = कोज्याफ.शीकेग.

समीकरणेन −

$$\text{गफ} = \frac{\text{शीकेग}\,(\text{क} - \text{कोज्याफ})}{\text{क}} \quad \cdots\cdots (१)$$

एतेन "मन्दस्फुटीकृतां भुक्तिं प्रोज्झ्य शीघ्रोच्चभुक्तितः। तच्छेषं विवरेणासौ हन्यात्त्रिज्यान्त्यकर्णयो" रित्यादिसूर्यसिद्धान्तोक्तं शीघ्रगतिफलमुपपद्यते।

अत्रैव गूढार्थप्रकाशे रङ्गनाथस्तु तात्कालिकया गत्या शीघ्रगतिफलं (१) समीकरणेन समानीय "त्रिज्यान्त्यकर्णयो" रित्यत्र सौरवचसि त्रिज्याशब्देन शीघ्रफलकोटिज्यां परिगृह्य सौरमतं समर्थयति। अत्रैव सौरवासनायां कमलाकरस्तु लल्लमतमण्डनार्थं त्रिज्यामेवाङ्गीकृत्य बहूपपादितम्। तन्नादरणीयमिति सुधावर्षिण्यां तत्प्रणेतारः परमगुरुचरणा वदन्ति। त्रिषु राशिषु फलशोधनेन या ज्या सैवात्र त्रिज्येत्यर्थं विधाय रङ्गनाथमतं समर्थयन्ति च अन्यथा "वृत्तद्वययोगगे द्युचरे मध्यैव गतिः स्पष्टे" ति लल्लवचसा वृत्तद्वययोग एव सौरमतेनापि मध्यगतिः स्पष्टगतेः समा भवेत्। तन्न समीचीनम्। "कक्षामध्यगतिर्यत्रेक्षाप्रतिवृत्तसंपाते" तस्य समत्वसिद्धेः।

इदानीं स्पष्टीकरणमिदं कस्मै न देयमित्याह

देयमसुताय नेदं शपथैरपि दत्तसुकृतनाशाय।
यात्राविवाहजातकफलस्फुटत्वं यतः स्पष्टैः ॥४५॥

वा. भा.—वासनाभाष्यं नास्ति।

वि. भा.—यतः (यस्मात् कारणात्) स्पष्टैः (स्पष्टग्रहैरेव) यात्राविवाहजातकफलानां स्फुटत्वं भवति, अतः शपथैरपीदं स्पष्टीकरणं असुताय (अपुत्राय) न देयमन्यथा दत्तसुकृतनाशाय (दातुः शोभनकर्मनाशाय) भवति, केवलं भक्ताय, स्वान्ते चिरवासिने शिष्याय देयमिदमिति ॥४५॥

अब स्पष्टीकरण किसके लिये नहीं देना चाहिये कहते हैं

हि. भा.—जिस कारण से स्पष्टग्रहों ही से यात्रा-विवाह-जातकफलों की स्फुटता होती है, इसलिये शपथ खाने से भी इस स्पष्टीकरण को अपुत्र के लिये नहीं देना चाहिये अर्थात् भक्त, बहुत दिनों तक अपने पास रहने वाले विद्यार्थियों के लिये देना चाहिये, अपुत्र को देने से किये हुये सुन्दर कर्मों का नाश होता है इति ॥४५॥

इदानीमार्यभटादीनां दोषमाह

मेषादितः प्रवृत्ता नार्यभटस्य स्फुटा युगस्यादौ।
श्रीषेणस्य कुजाद्याः खेटाः सर्वे हि विष्णुचन्द्रस्य ॥४६॥
न दृष्टाः स्पष्टाः श्रीषेणार्यभटविष्णुचन्द्रेषु।
यस्मात्कुजादयस्ते विदुषां नैवादरस्तस्मात् ॥४७॥

वा. भा.—वासनाभाष्यं नास्ति।

वि. भा.—आर्यभटस्य श्रीषेणस्य विष्णुचन्द्रस्य सर्वे कुजाद्याः खेटाः (मङ्गलादिग्रहाः) युगस्यादौ मेषादितो न प्रवृत्ता अर्थात्कुजाद्याः सर्वे ग्रहा युगादौ मेषादौ नासन्नतस्ते स्पष्टा न सन्तीति, यस्मात्कारणात्-श्रीषेणार्यभटविष्णुचन्द्रेषु-अर्थात्तत्तन्त्रेषु ते कुजादयो ग्रहाः स्पष्टा न दृष्टा- (न दृश्योक्ता) स्तस्मात्कारणात् विदुषां (पण्डितानां) मध्ये नैवादरो (अर्थात्तेषां तन्त्राणां विद्वत्समाजे आदरो न) ऽस्तीति ॥४६-४७॥

अब आर्यभटादि आचार्यों के दोष को कहते हैं

हि. भा.—आर्यभट-श्रीषेण-विष्णुचन्द्र इन आचार्यों के तन्त्रों में मङ्गलादि सब ग्रह युग के आदि में मेषादि से प्रवृत्त नहीं हुये इसलिये वे स्पष्ट नहीं हैं, अर्थात् उनके मत में मङ्गलादि सब ग्रह युगादि में मेषादि में नहीं थे, जिस कारण से श्रीषेण-आर्यभट-विष्णुचन्द्र इन आचार्यों के तन्त्रों में वे कुजादि ग्रह स्पष्ट नहीं है। उस कारण से पण्डितों के मध्य में उनका आदर नहीं है इति ॥४६-४७॥

इदानीं भौमादिग्रहाणां वक्रारम्भकालिकान्
मार्गारम्भकालिकांश्च शीघ्रकेन्द्रांशानाह

अग्न्यष्टिभि १६३ रिषुमनुभिः १४५ शरसूर्यै १२५
रिषुरसेन्दुभि १६५ स्त्रिभवैः ११३
शीघ्रान्त्यकेन्द्रभागैर्भौमादीनां भवति वक्रम् ॥४८॥

चक्रांशकैस्तदूनैरनुवक्रं तदधिकोनभागकलाः।
मन्दफलस्फुटभुक्त्यूनशीघ्रभुक्त्या हृता दिवसाः ॥४९॥

वा. भा.—शीघ्रान्त्यकेन्द्रभागैः रविशेषे कर्मणि यच्छीघ्रकेन्द्रं तद्भागकेन्द्रं कृत्वा वक्रं निरूपयेत्। भौमादीनां यथासंख्यं तद्यथा भौमस्याग्न्यष्टिभिः १६३, बुधस्येषुमनुभिः १४५, गुरोः शरसूर्यैः १२५, शुक्रस्येषुरसेंदु १६५, शनेस्त्रिभवैं ११३ एतैर्भागैः राश्यादिकेन्द्राणि भौमस्य ५२३ बुधस्य च २५ जीव ५ शुक्रस्य १५ शनेः ३२३ चक्रांशकैस्तदूनैरनुवक्रमिति। प्रत्येकस्य वक्रकेन्द्रचक्रार्धाद्विशोध्य यथा स्वमनुवक्रं केन्द्रं भवति कृतेव भौमादीनामनुवक्रे केन्द्राणि भौ ६।७ बु०७।५ जी ७।१५। शु६।१५ शनेः ८७ यत्र दिनेऽन्त्यशीघ्रकेन्द्रवक्रकेन्द्रं तुल्यं भवति। तत्र दिने ग्रहस्य वक्रः। यत्र दिने अनुवक्रकेन्द्रतुल्यं तत्र दिने ग्रहस्यानुवक्रः एतावती तौ च वक्रानुवक्रौ यथा ज्ञायेते तदर्थमिदमुक्तम्। तदधिकोनभागकलाः तेषां वक्र-भागानामनुक्रवभागानां वा स्वशीघ्रकेन्द्रभागैस्तद्दैवसिकैः सहान्तरे कृते ये भागाः अधिका ऊना वा भवन्ति तेषां कला कार्यास्ताश्च कला मन्दफलस्फुटभुक्त्यूनशीघ्र-भुक्त्या हृता इत्यर्थः। दिवसत्वं व्रजन्ति फलं दिवसादिकः कालो वक्रस्यानुवक्रस्य

वा भवतीत्यर्थः। पादे शीघ्रकेन्द्रमधिकं तदतीतस्य अथ वक्रानुवक्रकेन्द्रमधिकमतस्तस्यैव वक्रानुवक्रदिनं ज्ञात्वा सकृद्ग्रहः स्फुटः कार्यस्तदन्त्यशीघ्रकेन्द्रवक्रानुवक्रौ निरूप्याविति अत्रेयं वासना। भूमध्याच्छीघ्रनीचोच्चवृत्तमध्यं यावद्दूर्ध्वा कोटिः परमफलज्यातुल्यं शीघ्रनीचोच्चवृत्तव्यासार्ध भुजा पूर्वेणापरेण वा तयोर्वर्गयुतिं मूलं तिर्यक्कर्णः परमफलज्या ग्राह्या भूमध्यं यावत्तस्य कर्णस्य शीघ्रनीचोच्चवृत्तशलाकायाश्चान्तरे यावच्छीघ्रनीचोच्चवृत्तमुत्पादयेत् भ्रमति प्रतिमंडलपरिधौ स च तत्र प्रदेशे पश्चाद्गच्छन्नुपलभ्यते। तस्मादुपपन्नं सर्वं गोले दर्शयेत्। 'कक्षामंडलादोनि विन्यस्य यद्येवमेवेति ॥४८ ४९॥

वि. भा.—अग्न्यष्टिभिरिषुमनुभिरित्यादिपठितशीघ्रकेन्द्रांशैर्भौमादिग्रहाणां वक्रं भवेदर्थात्यैस्तैरेतैः शीघ्रकेन्द्रांशैस्तेषां वक्रारम्भो भवति, तद्रहितैश्चक्रांशै ३६०रनुवक्रमर्थान्मार्गारम्भः। मन्दफलस्फुटभुक्तिः (मन्दस्पष्टागतिः) तद्ऊना (तद्रहिता) शीघ्रभुक्तिः (शीघ्रोच्चगतिः) शीघ्रकेन्द्रगतिर्भवति तया तदधिकोनभागकलाभक्तास्तदा गतैष्या दिवसा भवन्ति, शीघ्रान्त्यकेन्द्रभागैरसकृद्विधिनाऽविशेषकर्मणि स्थिरीभूतैः केन्द्रांशैरिति ॥४८-४९॥

अत्रोपपत्तिः

अथ शीउग—स्पकेग = स्पगति, यदा च शीउग < स्पकेग तदा विलोमशोधनेन वक्रा गतिर्भवितुमर्हति, परमेवं कुत्र स्थितिरिति विचार्यते, फलांशखाङ्कान्तरशिञ्जिनीत्यादिभास्करोक्तप्रकारेण $\frac{\text{फकोज्या} \times \text{शीकेग}}{\text{शीक}} = \text{स्पकेग}$.

एतत्स्वरूपदर्शनेन सिद्ध्यति यद्यत्र फलकोटिज्यायाः परमत्वं शीघ्रकर्णे च परमाल्पत्वं भवेत्तत्रैव स्पष्टकेन्द्रगतेः परमाधिकत्वं भवितुमर्हति, नीचस्थाने फलाभावात्फलकोटिज्यायाः परमत्वं भवति, कर्णस्य परमाल्पत्वमपि तत्र भवत्यतो नीचस्थान एव स्पष्टकेन्द्रगतेः पराधिक्यं भवितुमर्हति तेनैं शीउग < स्पकेग तस्य सम्भावना नीचस्थाने एव भवेदर्थात्तत्रैव ग्रहा वक्रगतिका भवन्ति, परन्तु वक्रगतित्वारम्भस्तु ततः (नीचात्) पूर्वत एव भवितुमर्हत्यतः कियन्मिते शीघ्रकेन्द्रांशे वक्रारम्भो भवतीति विचार्यते।

कल्प्यते वक्रारम्भकालिककेन्द्रकोटिज्यामानम् = य

फलांशखाङ्कान्तरशिञ्जिनीत्री द्राक्केन्द्रभुक्तिरित्यादिभास्करोक्त्या

$\frac{\text{फकोज्या} = \text{केग}}{\text{कर्ण}} = \text{स्पकेग}$, नीचस्थानस्य कर्क्यादिकेन्द्रे विद्यमानत्वात् कर्क्यादि-

केन्द्रिककर्णः $= \sqrt{\text{त्रि}^2 + \text{मंफज्या}^2 - 2\,\text{मंफज्या} \times \text{य}}$, तथा द्राक्केन्द्रकोटिमौर्व्यान्त्य-

फलज्या गुणया क्रमादित्यादिसिद्धान्तशिरोमणिन्यसंशोधकोक्त्या फलकोटिज्या प्रमाणम्

$$=\frac{\text{त्रि}^2-\text{य}\times\text{अंफज्या}}{\text{कर्ण}}\text{, ततः } \frac{\text{फकोज्या}\cdot\text{केग}}{\text{कर्ण}}=\text{स्पकेग}=\frac{(\text{त्रि}^2-\text{य}\times\text{अंफज्या})\,\text{केग}}{\text{कर्ण}^2}$$

कर्णवर्गस्योत्थापनात् $\frac{(\text{त्रि}^2-\text{य}\times\text{अंफज्या})\,\text{केग}}{\text{त्रि}^2+\text{अंफज्या}^2-2\,\text{अंफज्या}\cdot\text{य}}=\text{स्पकेग}$। अत्र केग = शीघ्रकेन्द्रगतिः, कर्ण = शीघ्रकर्णः परन्तु वक्रारम्भे ग्रहस्पष्टगतिः = ०, तेन शीघ्रोच्चगतिः = स्पष्टकेगतिः, शीघ्रोच्चगतिः = उग, ।

$$\text{ततः } \frac{(\text{त्रि}^2-\text{य}\cdot\text{अंफज्या})\,\text{केग}}{\text{त्रि}^2+\text{अंफज्या}^2-2\,\text{अंफज्या}\cdot\text{य}}=\text{उग}\text{, छेदगमेन}$$

$\text{त्रि}^2\cdot\text{केग}-\text{य}\cdot\text{अंफज्या}\cdot\text{केग}=\text{त्रि}^2\cdot\text{उग}+\text{अंफज्या}^2\cdot\text{उग}-2\,\text{अंफज्या}\cdot\text{य}\cdot\text{उग}$ समशोधनेन $\text{त्रि}^2\cdot\text{उग}-\text{त्रि}^2\cdot\text{केग}+\text{अंफज्या}^2\cdot\text{उग}=2\,\text{अंफज्या}\cdot\text{य}\cdot\text{उग}-\text{य}\cdot\text{अंफज्या}\cdot\text{केग}$ तुल्यगुणकपृथक्करणेन $=\text{त्रि}^2\,(\text{उग}-\text{केग})+\text{अंफज्या}^2\cdot\text{उग}=\text{त्रि}^2\cdot\text{मंस्पग}+\text{अंफज्या}^2\cdot\text{उग}=\text{य}\cdot\text{अंफज्या}\,(2\,\text{उग}-\text{केग})$

$$=\text{य}\cdot\text{अंफज्या}\,(\text{उग}+\text{उग}-\text{केग})$$
$$=\text{य}\cdot\text{अंफज्या}\,(\text{उग}+\text{मंस्पग})$$

$$\text{अतः } \frac{\text{त्रि}^2\cdot\text{मंस्पग}+\text{अंफज्या}^2\cdot\text{उग}}{\text{अंफज्या}\,(\text{उग}+\text{मंस्पग})}=\text{य}\cdot\text{ । अत्र मंस्पग = मध्यगति स्वीकृता}$$

$$=\frac{\text{त्रि}^2\cdot\text{मग}+\text{अंफज्या}^2\cdot\text{उग}}{\text{अंफज्या}\,(\text{उग}+\text{मग})}=\text{य}=\text{केकोज्या, अस्याश्चापं नवतियुतं तदा}$$

वक्रारम्भकाले शीघ्रकेन्द्रांशा भवेयु, एतावता "त्रिज्याकृतिः खचरमध्यमभुक्तिनिघ्नी शीघ्रोच्चभुक्तिगुणितोऽन्त्यफलस्य वर्गः। योगस्तयोः परफलज्यकया विभक्तः शीघ्रोच्चभुक्तिखगवेगसमासहृच्च ॥ लब्धस्य धनुषो भागा वियदङ्कसमन्विताः। वक्रारम्भे ग्रहस्य स्युः शीघ्रकेन्द्रलवाः स्फुटाः" संशोधकोक्तमित्युपपद्यते। अत्र मन्दस्पष्टमध्यमगत्योः समत्वकल्पनया प्रकारोऽयं न समीचीन इति विज्ञैर्ज्ञेयः।

अत्र गणितं प्रदर्श्यते

मङ्गलस्य वक्रारम्भकालिककेन्द्रांशानयनाय तदन्त्यफलज्या = ७८, उच्चगतिः = ५९′।८″, त्रिज्या = १२०, शीघ्रकेन्द्रगतिः = २८′ स्वल्पान्तरात्, मङ्गलस्य मध्यमगतिः = ३१′।२६″

$\text{त्रि}^2=(120)^2=14400$, $\text{त्रि}^2\cdot\text{मग}=14400\times31$
$=446400$, तथा $\text{अंफज्या}^2=78^2=6084$, उग = ५९

अंफज्या^{२} × उग = ६०८४ × ५६ = ३५८६५६, उग + मग
= (३१।२६) + (५६।८) = ९०।३४

अंफज्या = ७८, अंफज्या (उग + मग) = ७८ (९०।३४) = ७०६४,

त्रि^{२}. मग + अंफज्या^{२}. उग = ४४६४०० + ३५८६५६ = ८०५३५६

अतः $\frac{\text{त्रि.}^{२}\text{ मग} + \text{अंफज्या.}^{२}\text{ उग}}{\text{अंफज्या (उग + मग)}} = \frac{८०५३५६}{७०६४} = ११४ = \text{य} = \text{शीघ्रकेन्द्रकोज्या}$,

अस्याश्चापम् = ७४°, नवतियुतं तदा मङ्गलस्य वक्रारम्भे शीघ्रकेन्द्रांशाः = १६४,° आचार्योक्तग्रहाणां वक्रारम्भकालिकशीघ्रकेन्द्रांशा एव लल्लाचार्येण, श्रीपतिना, भास्कराचार्येण च स्वस्वसिद्धान्ते प्रोक्ताः सूर्यसिद्धान्तोक्त 'मन्दस्फुटीकृतां भुक्तिं प्रोह्य शीघ्रोच्चभुक्तिनः। तच्छेषं विवरेणाथ हन्यात् त्रिज्यान्त्यकर्णयोरित्यादि' शीघ्रगतिफलानयनेऽस्मिन् यदि त्रिज्यान्त्यकर्णयोरित्यत्र त्रिज्याशब्देन त्रिज्याया एव ग्रहणं क्रियेत्तदा तदुक्तकेन्द्रांशा न मिलन्ति यथा तदुक्त्या $\frac{\text{केग. त्रि}}{\text{क}}$ = उग, ततः केग. त्रि = उग. क वर्गकरणेन केग^{२}. त्रि^{२}. = उग^{२}. क^{२} = उग^{२} (त्रि^{२} + अंफज्या^{२} — २ केकोज्या. अंफज्या) = त्रि^{२} (उग^{२} — २ उग. मग + मग^{२}) = उग.^{२} त्रि^{२} + उग^{२}. अंफज्या^{२} — उग^{२}. २ केकोज्या. अंफज्या = त्रि^{२}. उग^{२} — २ उग. मग. त्रि^{२} + त्रि^{२}. मग^{२}. समशोधनेन

उग^{२}. अंफज्या^{२} — उग^{२}. २ केकोज्या. अंफज्या = त्रि^{२} मग^{२} — २ उग. मग. त्रि^{२}. पुनः समशोधनेन

उग^{२}. २ केकोज्या. अंफज्या = उग^{२}. अंफज्या^{२} + २ उग. मग. त्रि^{२} — त्रि.^{२} मग^{२}
= उग^{२}. अंफज्या^{२} + त्रि^{२}. मग (२ उग — मग)

अतः $\frac{\text{उग}^{२}\text{. अंफज्या}^{२} + \text{त्रि}^{२}\text{. मग (२ उग — मग)}}{\text{२ उग}^{२}\text{. अंफज्या}}$

$= \frac{\text{उग}^{२}\text{. अंफज्या}^{२} + \text{त्रि}^{२}\text{ (उग}^{२}\text{ — केग}^{२}\text{)}}{\text{२ उग}^{२}\text{. अंफज्या}} = \text{केकोज्या}$

अस्याश्चापं नवतियुतं तदा कमलाकरमतानुसारेण वक्रारम्भशीघ्रकेन्द्रांशा भवन्ति।

अत्र गणितं प्रदर्श्यते

कुजस्यान्त्यफलज्या = ७८, मध्यमागतिः = ३१′।२६″, उच्चगतिः ५९′।८′, त्रिज्या = १२०, शीघ्रगतिः = २८′

$$\text{अंफज्या}^2 = ७२^2 = ६०८४,\ \text{उग}^2 = ३४८१,\ \text{उग}^2 \cdot \text{अंफज्या}^2 = ६०८४ \times ३४८१ = २११७८०४$$

$$\text{केग}^2 = २८^2 = ७८४,\ \text{उग}^2 - \text{केग}^2 = ३४८१ - ७८४ = २६९७,\ \text{त्रि}^2 = (१२०)^2 = १४४००$$

$$\text{त्रि}^2 (\text{उग}^2 - \text{केग}^2) = २६९७ \times १४४०० = ३८८३६८००$$

$$\text{उग}^2 \text{अंफज्या}^2 + \text{त्रि}^2 (\text{उग}^2 - \text{केग}^2) = २११७८०४ + ३८८३६८०० = ६००१५२०४$$

$$२\ \text{उग}^2 = ६९६२,\ \text{अंफज्या} = ७८,\ २\ \text{उग}^2 \times \text{अंफज्या} = ६९६२ \times ७८ = ५४३०३६$$

अतः $\frac{\text{उग}^2 \text{अंफज्या}^2 + \text{त्रि}^2 (\text{उग}^2 - \text{केग}^2)}{२\ \text{उग}^2 \cdot \text{अंफज्या}} = \frac{६००१५२०४}{५४३०३६} = १११$ अस्या-

इचापम् = ६८° नवतियुतं ६८° + ९०° = १५८° = कमलाकरमतानुसारेण वक्रारम्भकालिककुजशीघ्रकेन्द्रांशा आचार्योक्तशीघ्रकेन्द्रांश १६३° तो महदन्तरिता अतस्त्रिज्याशब्देने (त्रिषु राशिषु शीघ्रफलस्य विशोधनेन यच्छेषं तज्ज्याऽर्थाच्छीघ्रफलकोटिज्या) ति गूढप्रकाशे रङ्गनाथव्याख्या साधीयसी, यतः फलकोटिज्या सम्बन्धेनैव पूर्वमानीताः कुजवक्रकेन्द्रांशाः १६४ पाठपठिततत्केन्द्रांशेन सह तदन्तरम् = १, कमलाकरेण व्यर्थमेव रङ्गनाथमतं खण्डितमिति विवेचकैर्विवेचनीयम् ॥ नीचस्थानाद्यन्मितेऽन्तरे वक्रारम्भो भवति तद्विरुद्धदिशि तन्मित एवान्तरे वक्रत्यागो भवत्यतो वक्रारम्भकालिककेन्द्रांशहीना भगणांशाः ३६० मार्गारम्भ (वक्रत्यागकालिक) कालिकाः केन्द्रांशाः भवन्तीति ग्रहो वक्रत्वमवक्रत्वं वा गतो गमिष्यति वेत्येतदर्थमिष्टशीघ्रकेन्द्रांशेभ्यो वक्रावक्रपठितकेन्द्रांशा विशोध्याः शेषेणानुपातो यदि केन्द्रगत्यैकं दिनं लभ्यते तदा शेषेण किमित्यनेन लब्धदिनैर्वक्रत्वमवक्रत्वं ग्रहो गतो गमिष्यति वेति बोध्यम् ॥४८-४९॥

अब भौमादि ग्रहों के वक्रारम्भकालिक और मार्गारम्भकालिक शीघ्रकेन्द्रांश को कहते हैं

हि. भा.—१६३°, १४५°, १२५°, १६५°, ११३° इन पठित शीघ्र केन्द्रांशों में क्रमशः मङ्गलादि ग्रह वक्र होते हैं और इन्हीं को चक्रांश में घटाने से जो शेष रहते हैं उतने शीघ्र केन्द्रांश में वे अनुवक्र होते हैं, शीघ्रोच्चगति में मन्दस्पष्टगति को घटाने से शीघ्र केन्द्रगति होती है, पठितकेन्द्रांश से इष्टकेन्द्रांश के अधिक वा अल्प रहने से दोनों के अन्तर में केन्द्रगति से भाग देने से गतदिन और एष्यदिन होते हैं अर्थात् इष्टदिन से पहले वा पीछे ग्रह समागत दिनों में ग्रह वक्र या अवक्र हो गये होंगे या होंगे इति ॥४८-४९॥

उपपत्ति

शीउग—स्पकेग=स्पगति, जब शीउग <स्पकेग तब विलोमशोधन से वक्रगति होती है, लेकिन ऐसी स्थिति कहाँ होती है, इसके लिये विचार करते हैं। फलांशखाङ्कान्तर-शिञ्जिनीघ्नी इत्यादि भास्करोक्त प्रकार से $\frac{\text{फकोज्या}\cdot \text{शीकेग}}{\text{शीक}}$=स्पकेग इसको देखने से सिद्ध होता है कि जहाँ पर फलकोटिज्या का परमत्व होगा और शीघ्रकर्ण का परमाल्पत्व वहीं पर स्पष्ट केन्द्रगति का परमत्व हो सकता है, नीचस्थान में फलाभाव होने के कारण फलकोटिज्या का परमत्व होता है, तथा शीघ्रकर्ण का परमाल्पत्व होता है अतः शीउग <स्पकेग ऐसी स्थिति नीचस्थान ही में हो सकती है, परन्तु वक्रता का आरम्भ तो नीचस्थान से कुछ पहले ही से होगा, कितने शीघ्र केन्द्रांश में वक्रारम्भ होता है उस केन्द्रांश का साधन करते हैं।

कल्पना करते हैं वाक्रारम्भकालिक केन्द्रकोटिज्यामान=य

फलांशखाङ्कान्तरशिञ्जिनीघ्नी इत्यादि भास्करोक्त प्रकार से $\frac{\text{फकोज्या}\cdot \text{केग}}{\text{कर्ण}}$ =स्पकेग, नीचस्थान कर्क्यादि केन्द्र में है, नीचासन्न ही में वक्रारम्भ होता है अतः $\sqrt{\text{त्रि}^2+\text{अंफज्या}^2-2\ \text{अंफज्या}\times \text{य}}$=कर्ण; तथा द्राक्केन्द्रकोटिमौर्व्यान्त्यफलज्या गुणया इत्यादि सिद्धान्तशिरोमणिस्थ संशोधकोक्त प्रकार से $\frac{\text{त्रि}^2-\text{य}\cdot \text{अंफज्या}}{\text{कर्ण}}$=फलकोज्या स्पष्टकेन्द्रगति स्वरूप में फलकोटिज्या और कर्ण का उत्थापन करने से $\frac{(\text{त्रि}^2-\text{य}\cdot \text{अंफज्या})\ \text{केग}}{\text{कर्ण}^2}=\frac{(\text{त्रि}^2-\text{य}.\ \text{अंफज्या})\ \text{केग}}{\text{त्रि}^2+\text{अंफज्या}^2-2\ \text{अंफज्या}\cdot \text{य}}$=स्पकेग, केग=शीघ्र-केन्द्रगतिः। कर्ण=शीघ्रकर्ण परन्तु वक्रारम्भ में स्पष्टगति=० इसलिये शीघ्रोच्चमति =स्पष्टकेग। शीउगति=उग.

$\frac{(\text{त्रि}^2-\text{य}\cdot \text{अंफज्या})\ \text{केग}}{\text{त्रि}^2+\text{अफज्या}^2-2\ \text{अंफज्या}\cdot \text{य}}$=उग, छेदगम करने से।

त्रि.^{2} केग—य. अंफज्या. केग=त्रि.^{2} उग+अंफज्या.^{2} उग—२ अंफज्या. य. उग समशोधन करने से

त्रि.^{2} उग—त्रि.^{2} केग+अंफज्या^{2}. उग=२ अंफज्या. य. उग—य. अंफज्या. केग मूलयुक्तक को पृथक् करने से

$$\text{त्रि}^2\ (\text{उग}-\text{केग})+\text{अंफज्या}^2\cdot \text{उग}=\text{त्रि}^2\cdot \text{संस्पग}+\text{अंफज्या}^2\cdot \text{उग}$$
$$=\text{य}\cdot \text{अंफज्या}\ (2\ \text{उग}-\text{केग})$$
$$=\text{य}\cdot \text{अंफज्या}\ (\text{उग}+\text{उग}-\text{केग})$$
$$=\text{य}\cdot \text{अंफज्या}\ (\text{उग}+\text{संस्पग})$$

अतः $\frac{\text{त्रि}^2\cdot \text{संस्पग}+\text{अंफज्या}^2\cdot \text{उग}}{\text{अंफज्या}\ (\text{उग}+\text{संस्पग})}=\text{य}$ । यहाँ मध्यमगति और मन्दस्पष्टगति को तुल्य मान लिया गया है $=\frac{\text{त्रि}^2\cdot \text{मग}+\text{अंफज्या}^2\cdot \text{उग}}{\text{अंफज्या}\ (\text{उग}+\text{मग})}=\text{केकोज्या}$, इसके चाप में नवत्यंश जोड़ने से वक्रारम्भ कालिक शीघ्र केन्द्रांश होता है, इससे "त्रिज्याकृतिः खचरमध्यमभुक्तिनिघ्नी इत्यादि संस्कृतोपपत्ति में लिखित संशोधकोक्त" सूत्र उपपन्न होता है, लेकिन यहाँ मन्दस्पष्टगति और मध्यमगति बराबर स्वीकार की गयी है, तज्जनित त्रुटि इसमें है।

यहाँ गणित दिखलाते हैं

मङ्गल के वक्रारम्भ कालिक केन्द्रांशानयन के लिये, मङ्गल की अन्त्यफलज्या $=७८$, उच्चगति $=५९'।८''$, त्रि $=१२०$, शीघ्रकेन्द्रगति $=२८'$ स्वल्पान्तर से, मध्यमगति $=३१'।२६''$ ।

$$\text{त्रि}^2=(१२०)^2=१४४००,\ \text{त्रि}^2\cdot \text{मग}=१४४००\times ३१=४४६४००,\ \text{तथा}$$
$$\text{अंफज्या}^2=(७८)^2=६०८४,\ \text{उग}=५९$$
$$\text{अंफज्या}^2\cdot \text{उग}=६०८४\times ५९=३५८९५६,\ \text{उग}+\text{मग}$$
$$=(३१।२६)+(५९।८)=९०।३४$$
$$\text{अंफज्या}\ (\text{उग}+\text{मग})=७८\ (९०।३४)=७०६४,$$
$$\text{त्रि}^2\cdot \text{मग}+\text{अंफज्या}^2\cdot \text{उग}=४४६४००+३५८९५६=८०५३५६,$$

अतः $\frac{\text{त्रि}^2\ \text{मग}+\text{अंफज्या}^2\cdot \text{उग}}{\text{अंफज्या}\ (\text{उग}+\text{मग})}=\frac{८०५३५६}{७०६४}=११४=\text{क}=$ शीघ्रकेकोज्या, इसका चाप $=७८^\circ$, नवत्यंश जोड़ने से मङ्गल का वक्रारम्भ कालिक शीघ्रकेन्द्रांश हुआ $७४^\circ+९०^\circ=१६४^\circ$, आचार्योक्त मङ्गलशीकेन्द्रांश $=१६३^\circ$, मङ्गलादि ग्रहों के आचार्योक्त वक्रारम्भ कालिक शीघ्रकेन्द्रांश ही को लल्लाचार्य, श्रीपति, भास्कराचार्य ने अपने-अपने सिद्धान्तग्रन्थ में कहा है। सूर्यसिद्धान्तोक्त 'मन्दस्फुटीकृता भुक्तिः प्रोह्य शीघ्रोच्चभुक्तिवत्' इत्यादि शीघ्रगतिफलानयन में 'त्रिज्यान्त्यकर्णेन' यहाँ त्रिज्या शब्द से यदि त्रिज्या ही का ग्रहण किया जाय तब उनके पठित केन्द्रांश नहीं मिलते हैं जैसे सूर्यसिद्धान्तोक्तप्रकार से $\frac{\text{केग}.\ \text{त्रि}}{\text{कर्ण}}=\text{उग},\ \therefore \text{केग}\cdot \text{त्रि}=\text{उग}\cdot \text{क}$ । कर्ण $=\text{क}$

वर्ग करने से केग^{२}· त्रि^{२}=उग^{२}· क^{२}=उग^{२} (त्रि^{२}+अंफज्या^{२}−२ केकोज्या. अंफज्या)=त्रि^{२} (उग^{२}−२ उग· मग+मग^{२})

=उग^{२}· त्रि^{२}+उग^{२}· अंफज्या^{२}—उग·^{२} २ केकोज्या· अंफज्या=त्रि·^{२} उग^{२}—२ उग· मग· त्रि^{२}+त्रि^{२}· मग^{२}

समशोधन से उग^{२}· अंफज्या^{२}— उग·^{२} २ केकोज्या· अंफज्या=त्रि·^{२} मग^{२}—२ उग· मग. त्रि^{२} पुनः समशोधन से

उग·^{२} २ केकोज्या· अंफज्या=उग^{२}. अंफज्या^{२}+२ उग· मग· त्रि^{२}—त्रि^{२}· मग^{२}

=उग^{२} अंफज्या^{२}+त्रि^{२}· मग (२ उग—मग)

अतः $\frac{\text{उग}^{२}\cdot \text{अंफज्या}^{२}+\text{त्रि}^{२}\cdot \text{मग}\ (२\ \text{उग}-\text{मग})}{२\ \text{उग}^{२}\cdot \text{अंफज्या}}$

$=\frac{\text{उग}\cdot^{२}\ \text{अंफज्या}^{२}+\text{त्रि}^{२}\ (\text{उग}^{२}-\text{केग}^{२})}{२\ \text{उग}^{२}-\text{अंफज्या}}=$केकोज्या,

इसके चाप में नवत्यंश जोड़ने से कमलाकरमतानुसार वक्रारम्भ कालिक शीघ्र-केन्द्रांश होता है।

प्रतीत्यर्थं गणित दिखलाते हैं

जैसे मङ्गल की अन्त्यफलज्या=७८, मध्यमगति=३१′।२६″,

उच्चगति=५९′।८″, त्रिज्या=१२०, शीघ्रकेगति=२८′

अंफज्या^{२}=(७८)^{२}=६०८४, उग^{२}=३४८१, उग^{२}· अंफज्या^{२}

=६०८४×३४८१=२११७८४०४१

केग^{२}=२८^{२}=७८४, उग^{२}—केग^{२}=३४८१—७८४=२६९७,

त्रि^{२}=(१२०)^{२}=१४४००

त्रि^{२}(उग^{२}—केग^{२})=२६९७×१४४००=३८८३६८००

उग^{२}·अंफज्या^{२}+त्रि^{२}(उग^{२}—केग^{२})=२११७८४०४+३८८३६८००

=६००१५२०४, २उग^{२}=६९६२

२उग^{२}·अंफज्या=६९६२×७८=५४३०३६

$\therefore \frac{\text{उग}^{२}\cdot \text{अंफज्या}^{२}+\text{त्रि}^{२}(\text{उग}^{२}-\text{केग}^{२})}{२\text{उग}^{२}\cdot \text{अंफज्या}}=\frac{६००१५२०४}{५४३०३६}=१११$

इसका चाप=६८° नवत्यंश जोड़ने से ६८°+९०°=१५८°=कमलाकरमतानुसार वक्रारम्भकालिक मङ्गलशीघ्रकेन्द्रांश यह आचार्योक्तशीघ्रकेन्द्रांश १६३° से बहुत अन्तरित

(फरक) है इसलिये त्रिज्या शब्द से त्रिज्या का ग्रहण करना ठीक नहीं है, त्रिज्या शब्द से शीघ्रफलकोटिज्या का ग्रहण करना चाहिये। यह सूर्यसिद्धान्त की सुधावर्षिणी टीका में रङ्गनाथ का कहना बहुत ठीक है, क्योंकि फलकोटिज्या के सम्बन्ध से ही पहले लाये हुये मङ्गल के शीघ्रकेन्द्रांश = १३४, पाटपठितकेन्द्रांश १३३ के साथ अन्तर = १ कमलाकर ने व्यर्थ ही रङ्गनाथ मत का खण्डन किया है, इस विषय को विवेचक लोग विचारें। नीच स्थान से जितने अन्तर में वक्रारम्भ होता है उसके विरुद्ध दिशा में उतने ही अन्तर में वक्रत्याग होता है, इसलिये वक्रारम्भकालिककेन्द्रांश को भगणांश ३६० में घटाने से मार्गारम्भ (वक्रत्याग कालिक) कालिक केन्द्रांश होता है। ग्रहों की वक्रता या अवक्रता इष्टदिन से कितने पहले हो चुकी है या होगी इसके लिये इष्टशीघ्रकेन्द्रांश में वक्रावक्रपठितकेन्द्रांशों को घटाकर शेष से अनुपात करते हैं, यदि केन्द्रगति में एक दिन पाते हैं तो शेष में क्या इससे लब्ध दिनों में ग्रह वक्रत्व या अवक्रत्व में प्राप्त हो चुके हैं या होंगे इति ॥४८-४९॥

इदानीं वक्रातिवक्रानुवक्रपरिभाषामाह

शीघ्रात्स्फुटग्रहोनाच्छेषे मध्यस्फुटान्तरार्धं वा ।
अधिके घनमृणमूने स्फुटग्रहान्मध्यमे चापि ॥५०॥

राशिषु चतुर्षु वक्रं षट्स्वतिवक्रमनुवक्रमष्टासु ।
अप्राप्ताऽतीतकला भुक्त्यास्यैवोद्धृता दिवसाः ॥५१॥

वा. भा.—इदानीं वक्रानुवक्रपरिज्ञानं प्रकारान्तरेण प्रदर्शयन्नार्याद्वयमाह। शीघ्रात् स्फुटग्रहोनाद्यः शेषः तस्मिन् शेषे मध्यग्रहस्फुटग्रहयोर्यदन्तरं तस्यार्धं घनमृणं वा कार्यम्। स्फुटग्रहान्मध्यमेऽधिके घनमूने ऋणं कृत्वा तेन प्रकारेण वक्रानुवक्रपरिज्ञानं तत्कथमिति चेत्तदर्थमुक्तं राशिचतुर्षु वक्रमित्यादि एतदुक्तं भवति। स्वशीघ्रात् स्फुटग्रहं विशोध्यावशेषे मध्ये स्फुटान्तरार्धादल्पं यदि स्फुटग्रहान्मन्दस्फुटो ग्रहोऽधिको भवति। अथोनः तदा मध्यस्फुटान्तरार्धं शेषसंज्ञकाद्विशोध्यते एवं कृते राश्यादिकं यद्भवति, यत्र यदि राशिचतुष्टयं तत्रैव दिनेऽस्य वक्रमथ तत्र राशिषट्कं तत्रत्यदिने ग्रहस्यातिवक्रमप्यष्टौ राशयो भवन्ति, तत्तत्रैव दिने ग्रहस्यानुवक्रं यदा पुनरूनाधिकं केन्द्रं भवति तदा तत्कर्मणा राशिचतुष्टयादिकेन्द्रेभ्यस्ताश्च भुक्त्यास्यैव हृता दिवसत्वं प्रयान्ति। यथा शीघ्रात्स्फुटग्रहोनाच्छेषे एवं शीघ्रभुक्तेः स्फुटभुक्त्यूनायाः शेषे यथामध्यं स्फुटान्तरार्धमेवं मध्यमः। भुक्तिस्फुटभुक्त्योरन्तरार्धमधिके ऋणमूने स्फुटग्रहान्मध्ये एवमधिकायां घनमृणं शून्यायां स्फुटभुक्तेर्मध्यमायां मन्दस्फुटभुक्तिर्मध्यमोच्यते करणागता मध्या च अनेन प्रकारेण या भुक्तिः सास्य भुक्तिः तया हृता अप्राप्तातीता वा कलाः कर्तव्याः, फलं दिवसादिः वक्रस्यानुवक्रस्यावशेषे सकृत् कर्म प्राग्वदिति। एवमेतदार्याद्वयमस्माकमुपाध्यायैर्व्याख्यातं न वा यमर्षं

वासनाविरुद्धिः। इदानीं कुजगुरुशनीनामुदयास्तमयपरिज्ञानार्थमार्यामाह। एतदुक्तं भवति स्वशीघ्रात् स्फुटग्रहं विशोध्यावशेषे मध्ये ॥५०-५१॥

वि. भा.—स्फुटग्रहोनात् (स्पष्टग्रहरहितात्) शीघ्रात् (शीघ्रोच्चात्) यच्छेषं तस्मिन् स्फुटग्रहात् मध्यमे (मन्दफलसंस्कृते) अधिके सति मध्यस्फुटान्तरार्धं (मध्यस्य मन्दफलस्फुटस्य ग्रहस्य स्फुटग्रहस्य च यदन्तरं तस्यार्धं) ऋणं कार्यम्। स्फुटग्रहात् मन्दफलसंस्कृते ग्रहे ऊने सति तस्मिन् शेषे मध्यस्फुटान्तरार्धमृणं कार्यम्। एवं संस्कृते शेषे चतुर्षु राशिषु दृष्टेषु वक्रं ज्ञेयम्। षट्सु राशिषु दृष्टेषु अतिवक्रमष्टासु राशिषु दृष्टेषु अनुवक्रं (वक्रत्यागं) ज्ञेयम्। अप्राप्ततीतकलाः (वक्रज्ञाने) इष्टदिने शीघ्रात् स्फुटग्रहोनादित्यादिविधिना, शेषं मध्यस्फुटान्तरार्धसंस्कृतं राशिचतुष्कादल्पं तदा संस्कृतस्य राशिचतुष्कस्य चान्तरे याः कलास्ता अप्राप्तकलाः, यदि संस्कृतं राशिचतुष्कादधिकं तदा तयोरन्तरे याः कलास्ता अतीतकला भवन्ति, एष गतकला अस्य संस्कृतस्य भुक्त्यैवार्थात् गतैष्यदिनयोः 'शीघ्रात् स्फुटग्रहोना' दित्यादिविधिनाऽऽनीतयोः संस्कृतयोरन्तरतुल्यया भुक्त्या (गत्या) हृता (भक्ता) दिवसा भवन्तीति ॥५०-५१॥

अत्रोपपत्तिः

वक्रातिवक्रादीनां नामानि संहिताकारोक्तवज्ज्ञेयानि शिष्यधीवृद्धिदे लल्लाचार्येणैवमेव कथ्यते यथा—

"मध्यस्फुटान्तरदलेन चलात् समेतान्मध्ये,
स्फुटात् समधिके सति चान्यथोनात्।
स्पष्टं त्यजेत् कृतषडष्टसु तत्र भेषु,
वक्रातिवक्रकुटिला गतयो भवन्ति" ॥

सिद्धान्तशेखरे "शीघ्रोच्चात् स्पष्टमध्यग्रहविवरदलं मध्यमे शोध्यमूने देयं स्पष्टादूने स्फुटसचरमयोज्य तत्रावशेषे। वक्रं चिद्ध्यब्धिराशिष्वथ ऋतुषुदमहावक्रमष्टासु दवक्रत्यागं केन्द्रचुभुक्त्वा पुनरपि हरसे यातयेवान्यहानि" श्रीपत्युक्तमिदं ब्रह्मगुप्तोक्तलल्लाचार्योक्तयोरनुरूपमेव, अतैष्यदिनानयनं सुगममेवेति ॥५०-५१॥

अब वक्र-अतिवक्र और अनुवक्र की परिभाषा कहते हैं

हि. भा.—शीघ्रोच्च में स्पष्टग्रह को घटाने से जो शेष रहे उसमें स्पष्टग्रह से मन्दफल संस्कृत ग्रह के अधिक रहने से मन्दफलस्फुटग्रह और स्फुटग्रह के अन्तरार्ध को ऋण करना, स्फुटग्रह से मन्दफल संस्कृतग्रह के ऊन (कम) रहने से उस शेष में मध्यस्फुटान्तरार्ध को ऋण करना, इस तरह संस्कृत शेष चार राशि में देखा जाय तो वक्र समझना चाहिये, छः राशि में अतिवक्र और आठ राशि में अनुवक्र समझना चाहिये, वक्रज्ञान के

लिये इष्टदिन में 'शीघ्रात् स्फुटग्रहोनात्' इत्यादि विधि से मध्यस्फुटान्तरार्ध संस्कृत शेष चार राशि से अल्प हो तो मध्यस्फुटान्तरार्ध संस्कृत शेष और चार राशि की अन्तरकला अप्राप्तकला (एष्यकला) होती है, यदि मध्यस्फुटान्तरार्ध संस्कृत शेष चार राशि से अधिक हो तो दोनों की अन्तरकला अतीतकला (गतकला) होती है, गतकला और एष्यकला को गतदिन और एष्यदिन के 'शीघ्रास्फुटोनात्' इत्यादि विधि से लाये हुये संस्कृत शेष द्वय के अन्तर तुल्य गति से भाग देने से गतदिन और एष्यदिन होता है ॥५०-५१॥

उपपत्ति

वक्र-अतिवक्र-अनुवक्र इन सबों के नाम फलार्थ संहिताकार ने जो रक्खे हैं उसी तरह समझने चाहियें, शिष्यधीवृद्धिद में लल्लाचार्य आचार्योक्तानुसार ही कहते हैं जैसे 'मध्यस्फुटान्तरदलेन चलात् इत्यादि' संस्कृतोपपत्ति में लिखा गया है, सिद्धान्तशेखर में शीघ्रोच्चात् स्पष्टमध्यग्रहविवरदलं इत्यादि' संस्कृतोपपत्तिमें लिखित पद्य से श्रीपति 'ब्रह्मगुप्तोक्त और लल्लाचार्योक्त के' अनुरूप ही कहते हैं। गतैष्यदिनानयन सुगम ही है इति ॥५०-५१॥

इदानीं कुजादिग्रहाणामुदयास्तकेन्द्रांशानाह

अष्टयमैः २८ कृतचन्द्रैः १४ मुनीन्दुभि १७ र्भौमजीवरविजानाम् ।
उदयः प्रागस्तमयस्तदूनचक्रांशकैः पश्चात् ॥५२॥

खशरैः ५० जिनैः २४ र्ज्ञसितयोरिषुतिथिभि १५५ र्मुनिनगेन्दुभिः १७७ पश्चात् ।
उदयास्तमयौ व्यस्तौ मण्डलभागैस्तदूनैः प्राक् ॥५३॥

वा. भा.—शीघ्रान्त्यकेन्द्रभागैरष्टयमैर्भौमस्य प्रागुदयो भवति २८ स्वशीघ्रकेन्द्रभागैः कृतचन्द्रैः १४ जीवस्य प्रागुदयो भवति, स्वशीघ्रकेन्द्रभागैर्मुनीन्दुभिः १७ शनेरुदयो भवति। प्रागस्तमयास्तु पश्चाद्भवति। चक्रांशैस्ते ऊनास्तदूनाः यथा स्वोदयभागैरूनाश्च चक्रांशका ये विशेषा भवन्तीत्यर्थः। चक्रांशकाः प्रसिद्धा एव तद्यथा भौमस्यास्तमयशीघ्रकेन्द्रभागाः ३३२ गुरोः ३४६, शनेः ३४३, सर्वाण्युदयास्तकेन्द्राणि, राश्यधिकानि। भौ। उ०। २८ जी उ०। १४ श० उ०। १७। भौम ११ जी अ ११।१६ श अ ११।१३। अत्रातीतानां ग्रहाणां दर्शनं प्राग्वत्। तदधिकोना भागकला मन्दफलस्फुटभुक्त्यूनशीघ्रभुक्त्या हृता दिवसा इति न्यायेनात्रोपपत्तिः। तद्यथा रविकक्षायां सर्वथा ग्रहसर्वप्रतिमण्डले नीचोच्चवृत्तमध्ये भ्रमति। ततो यदा परमे प्रतिमण्डलोच्चप्रदेशे ग्रहः स्थितो भवति तदा समाधीग्रह शीघ्रो भवति समश्च रविरत एव भूस्थैस्तदा ग्रहो नोपलभ्यते। मतामपि रविकिरणपिहितहृग्भिस्ततो यथा-यथा ग्रहोऽवलम्बतेऽर्कस्तथा प्रथममेवोदयं याति रवेः शीघ्रत्वादर्कस्य अस्तमयेऽपि परिवर्त्य शीघ्रभावेनार्कः पुनर्ग्रहमासादयति पश्चिमदिग्भावात्। अतएवास्तंगतेऽर्के ग्रह उपलभ्यते। उदयास्तमयौ यदा भवतो

भागनियमश्चोपलब्धा यन्त्राभियोगातिशयाच्च तस्मादुपपन्नं कक्षामण्डलादिषु विन्यस्तेषु एते वोदयास्तमयभागा अविक्षिप्ते, ग्रहो विक्षिप्ते मण्डलवशाद्भिद्यन्ते, तदर्थमुदयास्तमयाध्यायो भविष्यतीति ॥५२॥

अधुना बुधशुक्रयोरुदयास्तमयपरिज्ञानार्थमार्यामाह । शीघ्रान्त्यकेन्द्रभागैरित्यनुवर्तते खशरैः ५० एतावद्भिः स्वशीघ्रकेन्द्रभागैर्बुधस्य पश्चात् उदयो भवति, जिनैः २४ एतावद्भिः शुक्रस्योदयपश्चात् इषुतिथिभिः १५५ एतावद्भिश्च शीघ्रान्त्यकेन्द्रभागैः पश्चादस्तमयो बुधस्य मुनिनगेन्दुभि १७७ रेतावद्भिः पश्चात्, शुक्रस्यास्तमयः उदयास्तमयौ व्यस्तौ मण्डलभागैरिति पश्चादर्धोदयभागा बुधस्य ५० मण्डलभागेभ्यो विशोध्य शेषभागाः खचन्द्रगुणाः ३१० एतावद्भिर्भागैर्बुधस्यास्तमयः शुक्रस्यापरोदयभागाः खचन्द्रगुणाः एतांश्चक्रभागेभ्यो विशोध्य शेषाः रसाग्निगुणाः ३३६, एतावद्भिर्भागैः शुक्रस्य प्रागस्तमयस्तथा पश्चादर्धास्तमयभागा बुधस्य १५५ एतन्मण्डलभागैर्विशोध्य शेषाः शरखयमाः २०५ एतावद्भिर्भागैर्बुधस्य प्रागुदयो भवति । तथा पश्चादर्धास्तमयभागाः शुक्रस्य १७७ एतान्मण्डलभागेभ्यो विशोध्य शेषाः त्रिवसुचन्द्राः १८३ एतावद्भिः प्रागुदयो भवति । शुक्रस्य एवं राश्यादिके पश्चादुदयकेन्द्रे बुधशुक्रयोः । बु. १. २ शु. २६ तथा पश्चादस्तमयकेन्द्रे बु।५।५।शु ५।२७। पूर्वार्धोदयकेन्द्र ६२७ शु. ६३ तथा प्रागस्तमयकेन्द्रे बु १०१०।शु११।६ एतेभ्योऽतीतानागतदिवसानयनं प्राग्वन्मन्दस्फुटभुक्त्यूनया शीघ्रगत्या केन्द्रान्तरं विभज्य वक्रानुवक्रवदिति तत्रेयं युक्तिः । शीघ्रगतित्वाद्बुधशुक्रयोः पश्चाद्रवेरवलम्बनं भवति । अतोऽस्तंगते रवौ पश्चिमायां दिशि तावुपलभ्येते चन्द्रवत् । यदा च वक्रिणौ भवतस्तदा रविस्ताभ्यां शीघ्रो भवति । प्राग्गतौ तौ च पश्चादवलम्बेते तयोः प्रागुदयो भौमः जीवसौराणामिवास्तमयश्च वैपरीत्यात् । शेषमन्यद्ग्रहवत् कक्षामण्डलादीनि विन्यस्य प्रदर्शयेत् । गोले छेद्यके वा भागपरिनिष्टा चेयं विक्षिप्तयोरेवेति ॥५३॥

वि. भा.—भौमजीवरविजानां (मङ्गलगुरुशनैश्चराणां) २८, १४, १७ शीघ्रकेन्द्रांशैः क्रमशः प्रागुदयो भवति, एतत्केन्द्रांशरहितैश्चक्रांशकैः पश्चिमायां दिश्यस्तमयो भवति ॥ ज्ञसितयोः (बुधशुक्रयोः) क्रमशः ५०, २४ शीघ्रकेन्द्रांशैः पश्चादुदयः, तथा १५५. १७७ शीघ्रकेन्द्रांशैः पश्चादस्तमयो भवति, उदयास्तमयौ व्यस्तौ मण्डलभागैस्तदूनैः प्रागित्यस्यायमर्थः—बुधस्य पश्चादुदयकेन्द्रांशाः ५० तान्मण्डलभागेभ्यो ३६० विशोध्यावशिष्टा ३१० स्तैःप्रागस्तमयः । तथा शुक्रस्यापरोदयकेन्द्रांशाः २४, तान् मण्डलभागेभ्यो विशोध्यावशिष्टा ३३६ एतावद्भिः प्रागस्तमयः । तथा बुधस्य पश्चादुदयकेन्द्रांशाः १५५ एतान् मण्डलभागेभ्यो विशोध्यावशिष्टाः २०५ एतैः केन्द्रांशैः प्रागुदयः । शुक्रस्य पश्चादस्तमयकेन्द्रांशाः १७७ एतान् मण्डलभागेभ्यो विशोध्यावशिष्टाः १८३ एतैः प्रागुदय इति ॥५२-५३॥

अत्रोपपत्तिः

कुजगुरुशनीनां शीघ्रोच्चं रविरेवास्ति. शीघ्रोच्चस्थाने तेषां परमास्तो भवेत्। ततो रवेरधिकगतित्वात्तेभ्योऽग्रतो गच्छति यदा कालांशतुल्यमन्तरं भवेत्तदा रविनामीप्यवशेन तेषां रात्रिशेषे पूर्वदिश्युदयो दृश्यते, तेन कालांशतुल्ये स्पष्टशीघ्रकेन्द्रांशे यत्फलचापं तेन कालांशा युतास्तदा तेषां ग्रहाणामुदयकेन्द्रांशा भवेयुरिति। यथा

यदि त्रिज्यया कालांशतुल्यस्पष्टकेन्द्रांशज्या लभ्यते तदाऽन्त्यफलज्यया किं लब्धा कालांशतुल्यस्पष्टकेन्द्रांशजनितफलज्या = $\frac{\text{अन्त्यफलज्या} \times \text{कालांशज्या}}{\text{त्रि.}}$

अस्याश्चापं कालांशयुतं तदा कुजगुरुशनीनामुदयकेन्द्रांशा भवन्तीत्येतावता

"कालांशजीवाऽन्त्यफलज्ययाघ्नी त्रिभज्ययाऽऽप्तफलस्य चापम्।
कालांशयुक्तं चलकेन्द्रभागाः समुद्गमे मन्दकुजेज्यकानाम्॥"

विशेषसूत्रमुपपद्यते।

अत्र प्रतीत्यर्थं गणितं प्रदर्श्यते

यथा कुजज्यान्त्यफलज्या = ८१, तत्कालांशाः = १७, कालांशज्या = ३५, त्रिज्या = १२०, तदा $\frac{\text{कालांशज्या.अंफज्या}}{\text{त्रि}} = \frac{३५ \times ८१}{१२०}$ = फलज्या = २३ एतच्चापं = ११° कालांश १७ युतं तदा ११° + १७° = २८° = कुजस्योदयकेन्द्रांशाः, एवं गुरुशन्योरपि तदुदयकेन्द्रांशा नेया इति।

बुधशुक्रयोस्तुल्य एव मध्यरविः एतत्तुल्यमेव मन्दस्पष्टं बुधं शुक्रं वा मत्वा स्पष्टेन बुधेन, शुक्रेण सह कालांशतुल्येऽन्तरे पश्चिमायां दिशि तदुदयो दृश्येत, ततः $\frac{\text{कालांशज्या.त्रि}}{\text{अंफलज्या}}$ = स्पकेज्या, अस्याश्चापं कालांशसहितं तदा प्रथमपदे पश्चिमोदयकेन्द्रांशाः स्युः। द्वितीयपदे वक्रतां प्राप्य रवेरल्पगतित्वात्तत्रैवास्तं गच्छतः। तृतीयपदे तयोः पुनरुदयो भवति, नीचस्थाने तयोः परमास्तं गतत्वाद्रात्रिशेषे पूर्वदिशि सोदयो दृश्यते। चतुर्थपदे तयोः कालांशान्तरे स्थितत्वात्तत्रैवास्तस्तेन पूर्वोदयकेन्द्रांशाः = चा—कालांश + १८० = चा + (१८० — कालांश) एतेने "ज्ञशुक्रयोस्तु त्रिभशिञ्जिनीभ्रोकालांशजीवाऽन्त्यफलज्ययाऽऽप्ता। चापं स्वकालांश तदूनभार्धयुक्तं परेन्द्रचन्द्रोदगमने स्वकेन्द्रम्।" इति तदीय सूत्रमुपपद्यते।

प्रतीत्यर्थं गणितं प्रदर्श्यते। बुधस्यान्त्यफलज्या = ४४, त्रिज्या = १२०,

पश्चिमोदयकालांशाः=१३, कालांशज्या=२७, तदा $\frac{\text{कालांशज्या}\cdot\text{त्रि}}{\text{अंफज्या}}=\frac{२७\times १२०}{४४}$ =७३ अस्याश्चापम्=३७° कालांश १३ युतं ३७+१३=५० तत्पश्चिमोदयकेन्द्रांशाः, पूर्वोदये कालांशाः=१२ ततः पूर्वोदयकेन्द्रांशाः=(१८०—कालांश)+चाप=३७+१६८=२०५ एवमेव शुक्रस्यापि केन्द्रांशा आनेतव्या इति सिद्धान्तशेखरे—

"वस्वस्विभि २८ र्युगकुभि १४ र्नगचन्द्रमोभिः,
१७ प्राच्युद्गमः क्षितिजजीवशनैश्चराणाम् ।
शीघ्रास्यकेन्द्रजलवैर्भगणांशशुद्धैरेभिः,
पुनर्नियतमस्तमयः प्रतीच्याम् ॥
द्राक्केन्द्रजैः खविषयैश्च ५० जिनैश्च २४,
भागैरुदग्गच्छतो बुधसितौ दिशि पाशपाणेः ।
तस्यामपोषुतिथिभिः १५५ स्वरशैलचन्द्रैः १७७,
भागैस्तयोर्निगदितोऽस्तमयो ग्रहज्ञैः" ॥

एभिः श्रीपतिना, सिद्धान्तशिरोमणौ—

"प्राच्यामुदेति क्षितिजोऽष्टदस्रैः २८,
शक्रै १४ र्गुरुः सप्तकुभिश्च १७ मन्दः ।
स्वस्वोदयांशो नितचक्रभागैस्त्रयो,
व्रजन्त्यस्तमयं प्रतीच्याम् ॥
खाक्षै ५० र्जिनै २४र्ज्ञसितयोरुदयः प्रतीच्याम-
स्तश्च पञ्चतिथिभि १५५ र्मुनिसप्तभूभिः १७७ ।
प्रागुद्गमः शरनखै २०५ स्त्रिधृतिप्रमाणै १८३ रस्तश्च
तत्रदशवन्हिभि ३१० रङ्गदेवैः ३३६ ॥
अवक्रवक्रास्तमयोदयोक्तभागाविकोनाः कलिका विभक्ताः ।
द्राक्केन्द्रभुक्त्याप्तदिनैर्गतैष्ये रवक्रवक्रास्तमयोदयाः स्युः ।"

भास्करेण चाऽऽचार्योक्तानुरूपमेव सर्वं कथितमिति ॥५२-५३॥

अब कुजादिग्रहों के उदयकेन्द्रांश और अस्तकेन्द्रांश को कहते हैं

हि. भा.—मङ्गल, गुरु, और शनैश्चर क्रमशः २८, १४, १७ इन केन्द्रांशों में पूर्व दिशा में उदित होते हैं। इन केन्द्रांशों को ३६० में घटाने से जो रहता है उन केन्द्रांशों में पश्चिम दिशा में वे अस्त होते हैं। बुध, और शुक्र क्रमशः ५०,२४ केन्द्रांशों में पश्चिम दिशा में उदित होते हैं तथा १५५, १७७ इन केन्द्रांशों में पश्चिम दिशा में अस्त होते हैं।

बुध के पश्चादुदय केन्द्रांश = ५०, इसको ३६०° चक्रांश में घटाने से शेष ३१० इतने केन्द्रांश में पूर्व दिशा में अस्त होते हैं। शुक्र के पश्चिमोदयकेन्द्रांश = २४, चक्रांश ३६० में घटाने से शेष ३३६ इतने केन्द्रांश में पूर्व दिशा में अस्त होते हैं। बुध के पश्चादुदय केन्द्रांश १५५ इनको चक्रांश में घटाने से शेष २०५ इन केन्द्रांशों में पूर्व दिशा में उदय होते हैं। शुक्र के पश्चादस्तमय केन्द्रांश १७७ इनको चक्रांश में घटाने से शेष १८३ इन केन्द्रांशों में पूर्व दिशा में उदित होते हैं इति ॥५२-५३॥

उपपत्तिः

कुज-गुरु और शनैश्चर इन सबों का रवि ही शीघ्रोच्च है, शीघ्रोच्च स्थान में उन सबों का परमास्त होता है, उन सबों से रवि के शीघ्रगतित्व के कारण रवि आगे चला जाता है, जब कालांश तुल्य अन्तर होता है तो रवि के सानिध्यवश से रात्रिशेष में उन सबों का उदय होता है, इसलिये कालांशतुल्य स्पष्टकेन्द्रांश में जो फलचाप होता है उसको कालांश में जोड़ने से उनके उदय केन्द्रांश होते हैं। जैसे यदि त्रिज्या में कालांश तुल्य स्पष्टकेन्द्रांश की ज्या पाते हैं तो अन्त्यफलज्या में क्या इस अनुपात से कालांश तुल्यस्पष्ट केन्द्रांशज्या जनित फलज्या आती है $\frac{\text{अंफज्या}\cdot\text{कालांशज्या}}{\text{त्रि.}}$ इसके चाप में कालांश जोड़ने से कुज, गुरु और शनैश्चर इन सबों के उदय केन्द्रांश होते हैं, इससे म. म. पण्डित सुधाकर द्विवेदी का सूत्र उपपन्न हुआ, "कालांशजीवाऽन्त्यफलज्ययाघ्नी त्रिज्ययाप्ता" इत्यादि संस्कृतोपपत्ति में लिखित सूत्र को देखिये। प्रतीत्यर्थमणित देखिये, जैसे कुज की अन्त्यफलज्या = ८१, कालांश = १७, कालांशज्या = ३५, त्रिज्या = १२०, तब उपरिलिखित सूत्रानुसार $\frac{\text{कालांशज्या}\cdot\text{अंफज्या}}{\text{त्रि}}$

$= \frac{३५\times८१}{१२०}$ = फलज्या, इसका चाप = ११° कालांश जोड़ने से ११° + १७° = २८° = कुज के उदय केन्द्रांश, इसी तरह गुरु और शनैश्चर का उदय केन्द्रांश लाना चाहिये। बुध और शुक्र के बराबर ही मध्यमरवि होते हैं, इसके बराबर ही मन्दस्पष्ट बुध वा शुक्र को मानकर स्पष्ट बुध या स्पष्ट शुक्र के साथ कालांश तुल्य अन्तर पर पश्चिम दिशा में उनके उदय को देखते हैं, तब $\frac{\text{कालांशज्या}\cdot\text{त्रि}}{\text{अंफज्या}}$ = स्पकेज्या, इसके चाप में कालांश जोड़ने से प्रथमपद में पश्चिमोदयकेन्द्रांश होता है, द्वितीयपद में वक्रता को प्राप्त कर रवि से अल्पगतित्व के कारण वहीं पर वे दोनों (बुध, शुक्र) अस्त होते हैं, तृतीयपद में उन दोनों का फिर उदय होता है, नीच स्थान में उन दोनों का परमास्त होने से रात्रि शेष में पूर्व दिशा में वह उदय देखा जाता है, चतुर्थपद में उन दोनों के कालांशान्तर पर रहने के कारण वे वहीं पर अस्त होते हैं इसलिये पूर्वोक्त केन्द्रांश = चाप — कालांश + १८० = चा

+(१८०—कालांश) इससे संस्कृतोपपत्ति में लिखित 'ज्ञशुक्रयोस्तु त्रिभशिञ्जिनीघ्नी इत्यादि' म. म. सुधाकर द्विवेदी का सूत्र उपपन्न होता है, अब प्रतीति के लिये गणित दिखलाते हैं।

बुध की अन्त्यफलज्या = ४४, पश्चिमोदयकालांश = १३, कालांशज्या = २७, त्रिज्या = १२०, तब पूर्वलिखितसूत्रानुसार $\frac{\text{कालांशज्या}\cdot\text{त्रि}}{\text{अंफज्या}} = \frac{२७\times १२०}{४४} = ७३$, इसका चाप = ३७° कालांश जोड़ने से ३७°+१३° = ५०° = पश्चिमोदयकेन्द्रांश पूर्वोदयकालांश = १२, तब पूर्वोदयकेन्द्रांश = (१८० = कालांश) + चाप = २७+१६८ = २०५, इसी तरह शुक्र का भी केन्द्रांश लाना चाहिये। सिद्धान्तशेखर में श्रीपति 'वस्वस्विभिर्युगकुभिः इत्यादि' संस्कृतोपपत्ति में लिखित सूत्र से और सिद्धान्तशिरोमणि में भास्कराचार्य ने 'प्राच्यामुदेति क्षितिजोऽष्टदस्रैः इत्यादि' संस्कृतोपपत्ति में लिखित सूत्रों से आचार्योक्तानुरूप ही सब कुछ कहा है इति ॥५२-५३॥

इदानीं स्वदेशे कथं स्पष्टा भवन्तीत्येतदर्थमाह

स्पष्टाद्युरात्रिदलयो रव्युदयास्तमययो रविचरार्धात्।
एष्ये ह्यधिकेऽतीतादवक्रितो वक्रितो हीने ॥५४॥

वा. भा.—य एते ग्रहा अनन्तरोक्तप्रकारेण स्पष्टास्ते यदि दिनार्धकालिका मध्यमा आसन् पदार्धविपादेति न्यायेन तदा स्पष्टा एवैतावता कर्मणा भवन्ति। अथोदयिका अस्तमयिका वा स्युस्तदा रविचरार्धाच्च स्वदेशेऽधिका अस्तमयिका वा भवन्ति। चरदलकर्म च पुरतो वक्ष्यति एष्ह्यत्यधिक इत्यादि आगामिदिने य स्फुटो ग्रहः सपद्यतीतदिनस्फुटग्रहाधिको तदा ग्रहस्य वक्रत्वं नास्ति। ऋजुगतिस्तदा ग्रह इति इतरोऽप्यहीनस्तदा वक्रीग्रहो ज्ञेय इत्यत्र वासना निरक्षदेशे यत्सदौदयिके ग्रहः स स्वदेशोन्मंडलप्राप्तक्रान्तिको भवति। स्वदेशोदयश्च स्वक्षितिजमंडले तयोश्चान्तरं चरदलं तेनार्वागग्रतो वा ग्रहो नीत्वा स्वक्षितिजप्राप्तकालिका क्रियन्ते, अस्तमयेऽप्येवं तस्मादुदयास्तमययोश्चरदलकर्मणा च स्फुटा भवन्ति। दिनरात्र्यर्धयोस्तु पुनर्याम्योत्तरमंडलस्यैकत्वात्स्वदेशनिरक्षयोः कर्मान्तराभावश्चरार्धाभावात्। यदि रविसावनेनानीता मध्यास्तद्रविचरदलेनार्कोदयकालिकाः। अन्यथा ग्रहास्तावदनेन चरदलेन तदुदयकालिका भवन्ति। अथ नक्षत्रसावनेन तच्चरदलं विनापि तदुदयकालिका भवन्त्येवं रविग्रहाणां ग्रहाद्युपलक्षणार्थं तस्मादुपपन्नम्। ग्रहश्च यदा वक्री भवति तदा प्रतिक्षणं पश्चादुपलभ्यते स चावश्यमेवातीतदिनस्फुटादूनो भवति। अवक्रितश्च गतो ग्रहो यदि स चातीतदिनस्फुटादधिको भवतीति किमत्रोच्यते तस्मादवक्रितस्यैव लक्षणं कतमनेनोत्तरार्धार्धेनेति। इदानीं सर्वग्रहाणां क्रान्तिज्यानयनार्थमार्यामाह।

वि. भा.—पूर्वं ये कुजादिग्रहाः साधितास्ते द्युरात्रिदलयोरर्धेन्दिनार्धे रात्र्यर्धे वा गणिनागताः स्पष्टाः स्युः। अर्थात्तेषु चरसंस्काराभावः। रव्युदयास्तमययोः (यदि रव्युदयेऽस्तमये वा) साधितास्तदा रविचरार्धात् (रविचरफलसंस्कारात्) स्वदेशे रव्युदयेऽस्तमये वा ते स्पष्टा भवन्ति, अतीताद् ग्रहात् (गतदिनोद्भवाद् ग्रहात्) एष्ये (आगामिदिनोद्भवे ग्रहे) अधिके सति अवक्रितः (मार्गः) हीने (गतदिनोद्भवग्रहादागामिदिनोद्भवग्रहे न्यूने) सति वक्रितो ज्ञेयः उपपत्तिरपि भाष्येनैव स्पष्टेति ॥५४॥

अब स्वदेश में कैसे स्पष्ट ग्रह होते हैं इसके लिये कहते हैं

हि. भा.—पहले जो कुजादि ग्रहों का साधन किया गया है वे (साधितग्रह) दिनार्धकाल में या रात्र्यर्ध में स्पष्ट होते हैं। क्योंकि उनमें चर संस्कार नहीं किया गया है, यदि रवि के उदयकाल में या अस्तकाल में साधित ग्रह हों तो उनमें रविचरफल संस्कार करने से स्वदेश में रवि के उदयकाल या अस्तकाल में वे स्पष्ट होते हैं। यदि गतदिन के ग्रह से अग्रिम दिन का ग्रह अधिक हो तो ग्रह को मार्गी समझना चाहिये यदि गतदिन के ग्रह से अग्रिम दिन का ग्रह हीन हो तो ग्रह को वक्र समझना चाहिये। इसकी उपपत्ति भी साफ ही है ॥५४॥

इदानीं पञ्चज्यानयनमाह

जिनभागज्यागुणिता सूर्यज्या व्यासदलहृता लब्धम्।
इष्टापक्रमजीवा विषुवदुदग्दक्षिणा सवितुः ॥५५॥

इष्टापक्रमवर्गं त्रिज्यावर्गाद्विशोध्य शेषपदम्।
विषुवदुदग्दक्षिणतः स्वाहोरात्रार्धविष्कम्भः ॥५६॥

क्रान्तिज्या विषुवच्छायया गुणा द्वादशोद्धृता क्षितिजा।
स्वाहोरात्रेऽनष्टा व्यासार्धेनाहता भक्ता ॥५७॥

स्वाहोरात्रार्धेन क्षयवृद्धिज्याधनुश्चरप्राणाः।
ते षड्हृता विनाड्यो विनाडिका नाडिकाः षष्ट्या ॥५८॥

वा. भा.—अत्र सूर्यग्रहणं सूर्याद्युपलक्षणार्थं तेनायमर्थः जिनसंख्याभागाः जिनसंख्याभागाश्चतुर्विंशतिभागाः इत्यर्थः। तेषां या ज्या तया गुणिताः नवरदचन्द्रैरिति यावत् १३२९ कासौ सूर्यज्या इष्टकालिकस्फुटग्रहज्येत्यर्थः, सा जिनभागज्या गुणिता सती व्यासदलहृता कार्या। ततो यल्लब्धं सापक्रमज्या भवतीष्टकालिका सवितुरन्यस्य वा ग्रहादेर्ज्या दिनज्यागुणिता सती व्यासदलहृता कार्या। ततो यल्लब्धं सापक्रमज्या भवतीष्टकालिका सवितुरन्यस्य वा ग्रहादेर्यंतो ज्या कृता तस्येत्यर्थः। सा च विषुवदुत्तरेण दक्षिणेन च भवति, मिथस्तुलादिगे ग्रहे यथासंख्यं सैव स्फुटा क्रान्तिज्या भवति, चन्द्रादीनां पुनश्चापि कृता स्वविक्षेपयुतवियुता

सती सामान्यदिशोः स्फुटक्रान्तिर्भवति। तज्ज्या स्वक्रान्तिज्या भवति। अत्रेयं वासना। यत्र मंडलज्या व्यासदलतुल्या भवति तदा विषुवन्मंडलापमंडलयोरन्तरं क्रान्तिः सा मेषादौ, अपमंडलार्धा स्थितस्योत्तरा तुलादौ दक्षिणः उत्क्रान्तिज्या रवेः सैव स्फुटा यतोऽपमंडल एव रविर्भ्रमति चन्द्रादीनां मंडलवर्णाद्भिद्यते। यतो मेषादावुन्मंडलस्थिता अपि ग्रहाः राश्यादिभागेन विक्षेपवशाद्विषुवतो दक्षिणेनापि भवन्ति यतो विमंडलगाश्चन्द्रादयो भ्रमन्ति। अतस्तत्क्रान्तिज्या चापस्य स्वविक्षेपयुतवियुतस्य या ज्या सा ग्रहस्य स्फुटा स्वक्रान्तिज्या भवति। एतच्च गोलाध्याये ऽस्माभिः पूर्वमेव व्याख्यातम्। उपपन्नं चैतत्सर्वं गोले प्रदर्शयेदिति।

इदानीं स्वाहोरात्रार्द्धसूत्रद्वयमार्यामाह। अपक्रमग्रहणेन स्वक्रान्तिज्योच्यते, तेनायमर्थः। इष्टस्य रव्यादेः ग्रहस्याश्विन्यादेर्नक्षत्रस्यागस्त्यमृगव्याधस्य वा यस्यैव स्वक्रान्तिज्यावर्गं व्यासार्धवर्गाद्विशोध्य मूलं गृह्यते तस्यैव स्वाहोरात्रवृत्तस्य व्यासार्धं भवति। तच्च विषुवदुत्तरेण भवति। उत्तरायाः स्वक्रांतिज्यायाः दक्षिणायाश्च दक्षिणेन तावता व्यासार्धेन यावद्वृत्तमुत्पद्यते तावद्वृत्तं तत्रदिने अहोरात्रेण ग्रहः पश्चाद् भ्राम्यन्नुत्पादयति नक्षत्रादीनां तु पुनः स्थिराण्येव स्वाहोरात्रवृत्तानि इत्यत्र वासनागोले विन्यस्य विषुवदुत्तरेण दक्षिणेन वा क्षितिजे क्रान्तिचापभागादितुल्येऽन्तरे सूत्रस्यैकमग्रं बद्धवा तावत्येवान्तरे तत्रैवोन्मंडले बध्नीयात्तद्दक्षिणोत्तरायतं ज्यावदवतिष्टते। तदर्धक्रांतिज्या तत्क्रान्त्यग्रे सूत्रस्यैकमग्रं बद्ध्वा द्वितीयमग्रं शलाकायां बध्नीयात्। भूमध्यक्रांतिज्यातुल्येऽन्तरे दक्षिणेनोत्तरेण वा स्वाहोरात्रार्धमेवमायतं चतुरस्रं क्षेत्रं पूर्वापरायतं निष्पन्नं भवति। निरक्षदेशे साक्षे चोत्तरमुन्नतं भवति। क्रमेण यावन्मेषस्तत्र सममंडलं प्रविशति। तस्य क्षेत्रस्य क्रांतिज्याकोटिः स्वाहोरात्रार्धं भुजस्तस्या संपाताद् भूमध्यं यावत् व्यासार्धं कर्णः कर्णकृते कोटिकृतिं विशोध्य मूलं भुजज्या इत्यतो व्यासार्धवर्गात् क्रान्तिज्यावर्गं विशोध्य मूलं गृह्यते। येन भुजा भवति। तच्च स्वाहोरात्रार्धस्तस्मादुपपन्नम्।

इदानीं चरदलानयनार्थमार्यार्द्धयमाह। रव्यादेरिष्टग्रहनक्षत्रादीनां वा या स्वक्रान्तिज्या तां स्वदेशविषुवच्छायया संगुण्य द्वादशभिरुद्धरेत्। फलं क्षितिज्या भवति। सा च स्वाहोरात्रार्धवृत्ते निष्पन्ना भवति तामनिष्टं स्थापयेत्, छायानयनार्थं ततः क्षितिज्यां व्यासार्धेन निहत्य स्वाहोरात्रार्धेन विभजेत्। फलं क्षयवृद्धिज्या तस्याः धनुश्चरदलप्राणा भवन्ति। षडुद्धृता विनाड्यो विनाड्यश्च षडुद्धृता नाड्यो भवन्तीति किमत्रोच्यते। वासनात्र तद्यथा वक्ष्यमाणविधिनाग्रां कृत्वा तत्प्रमाणव्यासार्धेन वृत्तमुत्पादयेत्। तद्वृत्तं गोलपूर्वभागे विन्यसेत्तथा यथा विषुवन्मंडलनिरक्षदेशक्षितिजस्वदेशक्षितिजसममंडलानां चतुर्णां यः सम्पातस्तस्माद्भूमध्यप्रापि यत्सूत्रं तत्र मध्यं तस्य भवति याम्योत्तरमंडले वोत्थितं भवति। यथा याम्योत्तरमंडलेऽवलम्बककोटिरेवमत्र क्रान्तिज्या कोटिर्यथा याम्योत्तरेऽक्षज्या भुजा एवमत्र मंडले

क्षितिजोन्मंडलयोरन्तरं क्षितिजा भुजा, अतस्त्रैराशिकमाचार्येण प्रकल्पितं यदि लंबककोटेरक्षज्या भुजा उत्क्रान्तिज्याकोटेः का भुजेत्यनो लंबस्थाने द्वादशकः कोटिः अक्षज्यास्थाने च विषुवच्छाया भुजा। अतः फल क्षितिजा भुजा तनः पुनरपि त्रैराशिकं यदि स्वाहोरात्रवृत्ते एतावती भुजा तद्व्यासार्धवृत्ते कियतीति फलं क्षयवृद्धिज्या, यत्र दिने स्वाहोरात्रवृत्तमेव व्यासार्धवृत्तमेकत्वाद्भ्रममाणस्य विषुवन्मंडले भ्रमवशाद्भूगोलस्य तत्रैव स्वाहोरात्रवृत्ते षष्टिघटिकाः प्रकल्पान्ते ताभिश्च स्वस्वपडूनसंस्थाः प्रमाणा भवन्ति। चक्रलिप्ताश्च तावत्य एव अतः क्षितिजा रूपं यदुन्मंडलक्षितिजयोरन्तरं तत्क्षयवृद्धिज्यारूपेण परिणमितं तस्याश्चापलिप्तायाः प्राणा भवन्ति। यस्मात्प्राणेन कलां भूमंडलं भ्रमतीति पूर्वमेव गोलाध्याये व्याख्यातः। प्राणैर्विना नाड्यो विनाडिभिर्घटिका उक्तवत् कार्या किमत्रोच्यते ? ततश्चरदलं घटिकांतरं क्षितिजोन्मंडलयोरंतरे स्वाहोरात्रवृत्तस्य खण्डकं भवति। रविवासनेनानीता लंकोदयकालिका भवन्ति। तस्माद्रविचरदलकर्म आर्यायाह ॥५५-५६-५७-५८॥

वि. भा.—सूर्यज्या (रविभुजज्या) जिनभागज्या (परमक्रान्तिज्या) गुणिता, व्यासदलहृता (त्रिज्याभक्ता) लब्धं सवितुः (सूर्यस्य) इष्टापक्रमजीवा (इष्टक्रान्तिज्या) भवति; सा च विषुवदुदग्दक्षिणाऽर्यान्नाडीवृत्तादुत्तरदिशि सूर्ये उत्तरा, दक्षिणे दक्षिणा भवति, त्रिज्यावर्गात् इष्टापक्रमवर्गं (इष्टक्रान्तिज्यावर्गं) विशोध्य शेषस्य पदं (मूलं) स्वाहोरात्रार्धविष्कम्भः (द्युज्या), नाडीवृत्तादुत्तरे सूर्ये उत्तरा द्युज्या, दक्षिणे दक्षिणा क्रान्तिज्याविषुवच्छायया (पलभया) गुणा, द्वादशभक्ता तदा स्वाहोरात्रे (द्युज्यावृत्ते) क्षितिजा (कुज्या) भवति, साऽनष्टा (पृथक्) स्थाप्या, सा कुज्या व्यासार्धनाहता (त्रिज्यागुणिता) स्वाहोरात्रार्धेन (द्युज्यया) भक्ता तदा क्षयवृद्धिज्या (चरज्या) भवति, अस्या धनुः (चापम्) तदा चरप्राणाः (चरासवः) भवन्ति, ते षड्भक्तास्तदा विनाड्यः (पलानि) भवन्ति, नाडिकाः षष्ट्या भक्ता तदा विनाडिकाः (पलानि) भवन्तीति ॥५५-५६-५७-५८॥

अत्रोपपत्तिः

क्रान्तिवृत्ते यत्र रविरस्ति तदुपरिगतं ध्रुवप्रोतवृत्तं कार्यं तथा नाडीवृत्तक्रान्तिवृत्तयोः सम्पातात् (गोलसन्धितः) नवत्यंशेन वृत्तं (अयनप्रोतवृत्तं) कार्यं तदा चापीयजात्यत्रिभुजमुत्पद्यते, गोलसन्धितोऽयनप्रोतवृत्तक्रान्तिवृत्तयोः सम्पातं यावत् क्रान्तिवृत्ते नवत्यंशा एकोऽवयवः। गोलसन्धित एवायनप्रोतवृत्तनाडीवृत्तयोः सम्पातं यावन्नाडीवृत्ते नवत्यंशा द्वितीयोऽवयवः अयनप्रोतवृत्ते नाडीक्रान्तिवृत्तयोरन्तरे परमक्रान्त्यंशास्तृतीयोऽवयव इत्यवयवत्रयैर्जायमानमेकं त्रिभुजम्। तथा गोलसन्धितो रविं यावत् क्रान्तिवृत्ते रविभुजांशाः कर्णं एकोऽवयवः। रविंतो नाडीवृत्तध्रुवप्रोतवृत्तयोः सम्पातं यावत् ध्रुवप्रोतवृत्ते रवेरिष्टक्रान्तिर्भुजो द्वितीयः

अवयवः । गोलसन्धितो नाडीवृत्तध्रुवप्रोतवृत्तयोः सम्पातं यावन्नाडीवृत्ते विषुवांशाः कोटिस्तृतीयोऽवयव इत्यवयवत्रयैर्जायमानं द्वितीयत्रिभुजम् । एतयोस्त्रिभुजयोर्ज्या-क्षेत्रद्वयं सजातीयं भवत्यतस्तावल्लघुत्रिभुजस्य (द्वितीयत्रिभुजस्य) ज्याक्षेत्रं क्रियते । गोलकेन्द्रात् (भूकेन्द्रात्) गोलसन्धिगतरेखा कार्या तदुपरिग्रहाल्लम्बः कार्य इयमेव भुजज्या, तथा भूकेन्द्रान्नाडीवृत्तध्रुवप्रोतवृत्तसम्पातगता रेखा कार्या, तदुपरिग्रहा-देव लम्बरेखा क्रान्तिज्या, एतयो (भुजज्याक्रान्तिज्ययोः) मूलगता रेखा विषु-वांशचापस्य ज्या नास्ति, भुजज्या-क्रान्तिज्या तन्मूलगतरेखाभिर्यत्त्रिभुजं जातं तदेवोक्तचापीयजात्यत्रिभुजस्य ज्याक्षेत्रम् । क्रान्तिज्याया नाडीवृत्तधरातलोपरि-लम्बत्वान्मूलगतरेखाया नाडीवृत्तधरातले स्थितत्वान्मूलगतरेखोपर्यपि क्रान्तिज्याया लम्बत्वमत उक्तं त्रिभुजं 'भुजज्या-क्रान्तिज्या-तन्मूलगतरेखाभिर्जायमानं' जात्या-त्मकम् । गोलसन्धिगतरेखोपरि ध्रुवप्रोतवृत्तनाडीवृत्तयोः सम्पाताल्लम्बो विषु-वांशज्या, बद्धरेखः गोलसन्धिगतरेखोपरिलम्बोऽस्ति, तर्हि गोलकेन्द्रान्नाडीवृत्तध्रुव-प्रोतवृत्तसम्पातगता रेखा त्रिज्या कर्णः । विषुवांशज्याभुजः, विषुवांशज्यामूलाद् गोलकेन्द्रं यावद्विषुवांशकोटिज्याकोटिरिति भुजत्रयैर्जायमानं त्रिभुजमेकम्, गोल-केन्द्रात् क्रान्तिज्यामूलं यावत् क्रान्त्युत्क्रमज्योनत्रिज्या (द्युज्या) कर्णः, मूलगत-रेखा भुजः गोलकेन्द्राद् भुजज्यामूलं यावत्कोटिरिति भुजत्रयैर्जायमानं द्वितीयं त्रिभुजम्, एतयोस्त्रिभुजयोः सजातीयत्वादनुपातो यदि त्रिज्यया विषुवांशज्या लभ्यते तदा द्युज्यया किमित्यनुपातेन समागता मूलगतरेखैतावता सिद्ध्यति कस्यापि चापीयजात्यस्य कर्णचापज्या वास्तवा भवति । भुजकोटिचापयोर्मध्ये एकस्य ज्या वास्तवा भवति तदन्यस्य ज्या वास्तवा न भवत्यर्थाद् यस्य ज्या वास्तवा तत्कोटिव्यासार्धवृत्ते परिणता भवति यथोपरिलिखितचापीयजात्यत्रिभु-जज्याक्षेत्रे भुजांशक्रान्त्यंशयोः कर्णभुजचापयोर्ज्ये वास्तविके स्तः, कोटिचापस्य विषुवांशस्य ज्या वास्तवा न किन्तु भुज (क्रान्ति) कोटिव्यासार्धे (द्युज्यावृत्त-व्यासार्धे) परिणता सती मूलगतरेखा (कमलाकरोक्तव्यक्षोदयलवज्या) जाता, नवत्यंश, नवत्यंश, जिनांशैर्जायमानत्रिभुजस्य ज्याक्षेत्रं (त्रिज्या, परमक्रान्तिज्या, तत्कोटिज्या परमाल्पद्युज्या संज्ञिकेति कर्णभुजकोटिभिर्जायमानं त्रिभुजं) पूर्वोक्त-त्रिभुजस्य (भुजज्या-क्रांतिज्या-व्यक्षोदयलवज्याभिरुत्पन्नस्य) सजातीयमतोऽनुपातो यदि त्रिज्यया परमक्रान्तिज्या (जिनज्या) लभ्यते तदा भुजज्यया किं समागता क्रान्तिज्या तत्स्वरूपम्$=\frac{\text{जिनज्या. भुजज्या.}}{\text{त्रि.}}$, त्रिज्या कर्णः, क्रान्तिज्या भुजः, तत्कोटिज्या (द्युज्या) कोटिरेतद्भुजत्रयैरुत्पन्नत्रिभुजे $\sqrt{\text{त्रि}^2-\text{क्रांज्या}^2}$=द्युज्या, ततोऽग्रा कर्णः, क्रान्तिज्या कोटिः, कुज्या भुज इति भुजत्रयैरुत्पन्नत्रिभुजमेकम् । द्वादशकोटिः, पलभा भुजः, पलकर्णः कर्ण इति भुजत्रयैरुत्पन्नं द्वितीयत्रिभुजम् ।

एतयोः क्षेत्रयोः सजातीयत्वादनुपातेन$\frac{\text{पभा. क्रांज्या}}{१२}$=कुज्या, क्षितिजाहोरात्र-

वृत्तसम्पातगतध्रुवप्रोतवृत्ते ध्रुवान्नाड़ीवृत्तं यावन्तत्यंशाः। ध्रुवान्पूर्वस्वस्तिकं यावदुन्मण्डले नवत्यंशाः। नाड़ीवृत्ते पूर्वस्वस्तिकाद् ध्रुवप्रोतवृत्तनाड़ीवृत्तसम्पातं यावच्चरम्। एभिर्भुजत्रयैरुत्पन्नमेकं चापीयत्रिभुजम्। ध्रुवात् क्षितिजाहोरात्रवृत्तसम्पातं यावद् ध्रुवप्रोतवृत्ते द्युज्याचापम। ध्रुवादुन्मण्डलाहोरात्रवृत्तयोः सम्पातं यावद् द्युज्याचापम्, अहोरात्रवृत्ते क्षितिजोन्मण्डलयोरन्तरे कुज्यांश इति भुजत्रयैरुत्पन्नं द्वितीयत्रिभुजमेतयोस्त्रिभुजयोर्ज्याक्षेत्रसजातीयादनुपातो यदि द्युज्यया कुज्या लभ्यते तदा त्रिज्यया किं समागच्छति चरज्या तत्स्वरूपम्= $\frac{\text{कुज्या. त्रि}}{\text{द्यु}}$, अस्याश्चापम् चरासवः। रविभुजज्या, क्रान्तिज्या, द्युज्या, कुज्या, चरज्या, इत्येवात्र पञ्चज्या यदानयनं पूर्वं कृतम्। आचार्यमतेनाऽयनांशाभावोऽनो यथागतरविरेव साधितरविः। भास्करेण सायनरवेर्भुजज्या साधिता, इत्येव तन्मते विशेष इति ॥ ५५-५६-५७-५८ ॥

अब पञ्चज्यानयन को कहते हैं

हि.भा.—रवि भुजज्या को जिनज्या (परम क्रान्तिज्या) से गुणा कर त्रिज्या से भाग देने से लब्धि रवि की इष्ट क्रान्तिज्या होती है नाड़ी वृत्त से सूर्य के उत्तर रहने से उसकी (क्रान्तिज्या) की दिशा उत्तर होती है, और नाडीवृत्त से सूर्य के दक्षिण रहने से उसकी दिशा दक्षिण होती है। त्रिज्यावर्ग में से इष्ट क्रान्तिज्या वर्ग को घटाकर शेष का मूल अहोरात्र वृत्त का व्यासार्ध (द्युज्या) होता है, इसकी दिशा भी क्रान्तिज्या की दिशा की तरह होती है, क्रान्तिज्या को पलभा से गुणा कर द्वादश १२ से भाग देने से क्षितिज्या (कुज्या) होती है, इसको पृथक् स्थापन करना, उस कुज्या को त्रिज्या से गुणा कर द्युज्या से भाग देने से क्षयवृद्धिज्या (चरज्या) होती है, इसका चाप चर प्राण (चरासु) होता है, चरासु को छः से भाग देने से विनाड़ी (पल) होती है, नाड़ी (दण्ड) को ६० से भाग देने से विनाड़िका (पल) होती है इति ॥ ५५-५६-५७-५८ ॥

उपपत्ति।

क्रान्तिवृत्त में जहां रवि है उनके ऊपर ध्रुवप्रोतवृत्त कर देना, नाड़ीवृत्त और क्रान्तिवृत्त के सम्पात (गोलसन्धि) से नवत्यंश व्यासार्ध से वृत्त (अयनप्रोतवृत्त) करना, तब दो चापीय जात्य त्रिभुज बनता है, गोल सन्धि से अयन प्रोतवृत्त क्रान्तिवृत्त के सम्पात पर्यन्त क्रान्तिवृत्त में नवत्यंश एक भुज, गोल सन्धि से अयन प्रोत वृत्त नाड़ीवृत्त के सम्पात पर्यंत नाड़ीवृत्त में नवत्यंश द्वितीयभुज, नाड़ीवृत्त और क्रान्तिवृत्त के अन्तर्गत अयन प्रोत वृत्तीय चाप (परम क्रान्ति) तृतीय भुज; इन तीनों भुजों से एक त्रिभुज बना, तथा गोलसन्धि से रविपर्यन्त क्रान्तिवृत्त में रविभुजांश कर्ण एक भुज, रवि से रविगत ध्रुव प्रोतवृत्त के नाड़ीवृत्त सम्पात

पर्यन्त ध्रुव प्रोतवृत्त में इष्ट क्रान्ति द्वितीय भुज गोल सन्धि से ध्रुव प्रोतवृत्त नाड़ीवृत्त के सम्पात पर्यन्त नाड़ी वृत्त में विषुवांश कोटि तृतीय भुज, इन तीनों भुजों से द्वितीय त्रिभुज हुआ, इन दोनों चापीय जात्य त्रिभुजों का ज्याक्षेत्र सजातीय होता है इसलिए पहले लघु त्रिभुज (द्वितीय त्रिभुज) का ज्याक्षेत्र करते हैं, गोल केन्द्र (भूकेन्द्र) से गोलसन्धिगतरेखा कर देना उसके ऊपर रवि केन्द्र से लम्ब रेखा रवि भुजज्या होती है, भूकेन्द्र से नाड़ीवृत्त ध्रुव प्रोत वृत्त सम्पातगत रेखा करना, उस के ऊपर रवि केन्द्र से लम्ब रेखा क्रान्तिज्या (क्रान्ति चाप की ज्या) होती है, इन दोनों (भुजज्या और क्रान्तिज्या) की मूल गत रेखा कर देना यह विषुवांश चाप की ज्या नहीं है, भुजज्या, क्रान्ज्या, तन्मूलगत रेखाओं से जो त्रिभुज होता है, वही उक्त चापीय जात्य त्रिभुज का ज्याक्षेत्र होता है नाड़ीवृत्त धरातल के ऊपर क्रान्ज्या लम्ब है, तथा मूलगत रेखा नाड़ीवृत्त धरातलगत है इसलिए मूलगत रेखा के ऊपर भी क्रान्ज्या लम्ब होती है (धरातल के ऊपर रेखा लम्ब की परिभाषा से) इसलिये उक्त त्रिभुज; जात्य त्रिभुज हुआ, नाड़ी वृत्त और ध्रुव प्रोत वृत्त के सम्पात से गोलसन्धिगत रेखा के ऊपर लम्ब रेखा विषुवांश चाप की ज्या है, मूलगत रेखा गोल सन्धिगत रेखा के ऊपर लम्ब है, गोल केन्द्र से नाड़ीवृत्त ध्रुव प्रोतवृत्त सम्पातगत त्रिज्या कर्ण, विषुवांशज्या भुज, और विषुवांशज्या मूल से गोल केन्द्र पर्यन्त विषुवांश कोटिज्या कोटि, इन तीनों भुजों से एक त्रिभुज हुआ तथा गोल केन्द्र से क्रान्तिज्या मूल पर्यन्त क्रान्ति की उत्क्रमज्योन त्रिज्या (क्रान्तिकोटिज्या = द्युज्या) कर्ण, मूलगत रेखा भुज, और गोल केन्द्र से भुजज्या मूल पर्यन्त कोटि, इन तीनों भुजों से द्वितीय त्रिभुज हुआ, ये दोनों त्रिभुज सजातीय हैं इसलिए अनुपात करते हैं यदि त्रिज्या में विषुवांशज्या पाते हैं तो द्युज्या में क्या इससे मूलगत रेखा आती है इससे सिद्ध होता है कि किसी भी चापीय जात्य त्रिभुज के ज्याक्षेत्र में कर्णचाप ज्या वास्तविक होती है, भुजचाप और कोटिचाप में किसी एक चाप की ज्या वास्तविक ही होती है, अन्य चाप की ज्या वास्तव नहीं होती है अर्थात् जिस चाप की ज्या वास्तविक होती है उसी के कोटिव्यासार्ध वृत्त में परिणत होती है, जैसे उपरिलिखित चापीय जात्य त्रिभुज के ज्याक्षेत्र में भुजांशकर्ण और क्रान्तिभुज की ज्याएं वास्तविक हैं, कोटिचाप विषुवांश की ज्या वास्तव नहीं है किन्तु भुज (क्रान्ति) कोटिव्यासार्धवृत्त (द्युज्या व्यासार्ध वृत्त) में परिणत होकर मूलगत रेखा (कमलाकरोक्त व्यक्षोदय लबज्या) होती है, नवत्यंश, नवत्यंश और जिनांश इन भुजों से उत्पन्न त्रिभुज का ज्याक्षेत्र (त्रिज्या कर्ण, जिनज्या भुज, जिनांश कोटिज्या = परमाल्पद्युज्या, कोटि से उत्पन्न त्रिभुज) पूर्वोक्त त्रिभुज (भुजज्या, क्रान्तिज्या, मूलगत रेखाओं से उत्पन्न त्रिभुज) का सजातीय है इसलिये अनुपात करते हैं यदि त्रिज्या में जिनज्या (परम क्रान्तिज्या) पाते हैं तो रविभुजज्या में क्या इस अनुपात से इष्टक्रान्तिज्या आती है, $\frac{\text{जिज्या} \cdot \text{भुज्या}}{\text{त्रि}}$ = इक्रांज्या, त्रिज्या कर्ण, क्रान्तिज्या भुज, क्रान्तिकोटिज्या (द्युज्या) कोटि इन भुजों से उत्पन्न त्रिभुज में $\sqrt{\text{त्रि}^2 - \text{क्रांज्या}^2}$ = द्युज्या, अभ्राकर्ण, क्रान्तिज्या कोटि, द्युज्या भुज इन भुजों से उत्पन्न एक त्रिभुज, तथा द्वादश कोटि पलभा भुज, पलकर्ण

कर्ण, इन भुजों से उत्पन्न द्वितीय त्रिभुज, दोनों अक्षक्षेत्र हैं इसलिये सजातीय होने के कारण अनुपात करते हैं $\frac{\text{पभा.} \times \text{क्रांज्या}}{१२}$ = कुज्या, क्षितिजाहोरात्र वृत्त के सम्पातगत ध्रुवप्रोत वृत्त में ध्रुव से नाड़ी वृत्त पर्यन्त नवत्यंश, ध्रुव से पूर्वस्वस्तिक पर्यन्त उन्मण्डल में नवत्यंश, पूर्वस्वस्तिक से ध्रुवप्रोत वृत्त नाड़ीवृत्त के सम्पात पर्यन्त नाड़ीवृत्त में चरांश, इन भुजों से उत्पन्न एक त्रिभुज, तथा ध्रुव से क्षितिजाहोरात्र वृत्त के सम्पात पर्यन्त ध्रुव प्रोत वृत्त में द्युज्या चाप एक भुज, ध्रुव से उन्मण्डल और अहोरात्र वृत्त के सम्पात पर्यन्त उन्मण्डल में द्युज्याचाप द्वितीय भुज, अहोरात्र वृत्त में क्षितिज वृत्त और उन्मण्डल के अन्तर्गत चाप तृतीय भुज, इन भुजों से उत्पन्न द्वितीय त्रिभुज का ज्याक्षेत्र प्रथम त्रिभुज के ज्याक्षेत्र का सजातीय है इसलिये अनुपात करते हैं। यदि द्युज्या में कुज्या पाते हैं तो त्रिज्या में क्या इससे चरज्या आती है, $\frac{\text{कुज्या.} \times \text{त्रि}}{\text{द्यु}}$ = चरज्या, इसका चाप = चरासु, यहां रविभुजज्या, क्रान्तिज्या, द्युज्या, कुज्या, चरज्या, यह पञ्चज्यानयन किया गया है, आचार्य के मत में अयनांशाभाव है, इसलिए यथागत रवि ही साधित रवि होता है; भास्कराचार्य ने सायन रवि की भुजज्या का साधन किया हैं वही उन के मत में विशेषता है, इति ॥ ५५-५६-५७-५८ ॥

इदानीं चरकर्माह

चरदलघटिका गुणिता भुक्तिः षष्ट्याहृता कलाद्याप्तम् ।
ऋणमुदयेऽस्तमये धनमुत्तरगोलेऽन्यथा याम्ये ॥ ५९ ॥

वा. भा.—चरदलघटिकाभिरिष्टग्रहभुक्तिं संगुण्य षष्ट्या विभजेत् फलं लिप्तादि तदुत्तरगोलस्थे रवावौदयिके ग्रहे ऋणमस्तमयिके धनं दक्षिणगोलस्थे च रवावौदयिके धनमस्तमयिके ऋणमेवं स्वदेशार्कोदयकाले ऽस्तमयकाले वा ग्रहो भवति अस्तमयिको यो ऽर्धभुक्त्याकृत् तत्र च काले चरदलघटिकातुल्यादि तस्य घटिका गता भवति अतो रव्युदयकालिको ग्रहश्चरदलघटिका फलेनोपचितौदयिको भवति, फलं च त्रैराशिकेन यदि घटिकानां षष्ट्या ग्रहभुक्तितुल्यलिप्ता भवन्ति। तच्चरदलघटिकाभि : किमिति - लिप्तादिफलं पश्चादप्युन्मंडलकालिका एवं तत्र च काले दिनशेषघटिकाश्चरदलतुल्या भवन्ति । अतश्चरार्धघटिका फलेनोपचितो ग्रहो रव्यौदयिको भवति । याम्ये गोलार्धे वैपरीत्ये तत् क्षितिजमंडलादध : स्थितत्वादुन्मंडलस्य यथास्थितं सर्वं गोले प्रदर्शयेत् । स्वदेशाक्षाग्रयोर्गोले विन्यसेत्।

वि. भा.—भुक्तिः (ग्रहगतिः) चरार्धघटीभिर्गुणिता, षष्ट्या भाज्या, कलादिफलं यल्लब्धं तदुत्तरगोले उदयकाले ग्रहे ऋणं दक्षिणगोले धनं कार्यं अस्तमये

ऽन्यथा (विपरीतं) ऽर्थादुत्तरगोले सहितं दक्षिणगोले रहित कार्यमिति ॥ ५९ ॥

अत्रोपपत्तिः

अहर्गणेन साधिता ग्रहा लंकाक्षितिजोदयकालिका भवन्ति, ते देशान्तरफलेन संस्कृतास्तदोन्मण्डलकालिका भवन्ति, परन्त्वपेक्षितास्तु स्वक्षितिजोदयकालिकाः, उन्मण्डलस्वक्षितिजयोरन्तरे चरार्धम् तेनानुपातो यदि षष्टिघटीभिर्ग्रहगतिकला लभ्यन्ते तदा चरार्धघटीभिः किं समागच्छन्ति चरार्धघट्यन्तर्गतग्रहगतिकलाः, आभिः कलाभिरुत्तरगोल उन्मण्डलकालिको ग्रहो हीन (उन्मण्डलात् स्वक्षितिजस्याधो विद्यमानत्वात्) स्तदा स्वक्षितिजोदयकालिकग्रहो भवेत्। दक्षिणगोले स्वक्षितिजस्योन्मण्डलादुपरि स्थितत्वात्ताभिः कलाभिः सहितो ग्रहः (उन्मण्डलकालिकः) स्वक्षितिजोदयकालिको भवेत्। परं चरार्धघट्यन्तर्गतग्रहगतिग्रहणमन्तरैवैवं भवितुमर्हति तद्गतिग्रहणन्त्वावश्यकमत आचार्योक्तचरफलसंस्कारो न शोभनः, असकृत्कर्मणाऽत्र ग्रह औदयिको भवितुमर्हतीति, सकृत्कर्मणापि पूर्वं वास्तवोदयान्तरसाधने एकासुजेन गतिमङ्गणितैकलिप्तोत्पन्नास्वित्याद्युपपत्तौ भाज्यस्थाने यदि प्राचीन चरफलं गृह्येत तदा वास्तवं चरफलं समागच्छेदेवेति ॥ ५९ ॥

अब चरकर्म को कहते हैं।

हि. भा.—ग्रहगति को चरखण्ड घटी से गुणा कर साठ से भाग देने से जो कलादि फल हो उसको उत्तर गोल में उदयकाल में ग्रह में ऋण करना, दक्षिण गोल में धन करना, अस्तकाल में विपरीत (उल्टा) अर्थात् उत्तर गोल में ग्रह में धन करना और दक्षिण गोल में ऋण करना चाहिये इति ॥५९॥

उपपत्ति

ग्रहगण से साधित ग्रह लङ्काक्षितिजोदय कालिक होते हैं, उन में देशान्तर फल को संस्कार करने से उन्मण्डल कालिक होते हैं लेकिन अपेक्षित है स्वक्षितिजोदयकालिक ग्रह, उन्मण्डल और स्वक्षितिज के अन्तर में चरार्ध है, इसलिये अनुपात करते हैं, यदि साठ ६० घटी में ग्रहगति कला पाते हैं तो चरार्ध घटी में क्या इससे चरार्ध घटी सम्बन्धिनी ग्रहगति कला आती है। इनको उत्तर गोल में उन्मण्डल से अपने क्षितिज को नीचा रहने के कारण उन्मण्डल कालिक ग्रह में से घटाने से स्वक्षितिजोदय कालिक ग्रह होते हैं। दक्षिण गोल में उन्मण्डल से अपने क्षितिज के ऊपर रहने के कारण पूर्वागत चरार्ध घटी सम्बन्धिनी ग्रहगति कला को उन्मण्डल कालिक ग्रह में जोड़ने से स्वक्षितिजोदय कालिक ग्रह होते हैं। लेकिन चरार्ध घट्यन्तर्गत जो कुछ ग्रहगति होगी उसका ग्रहण आचार्य ने नहीं किया है, लेकिन उसका ग्रहण करना आवश्यक है इसलिए आचार्योक्त चरफलसंस्कार ठीक नहीं है, असकृत्कर्म से यहां

औदयिक ग्रह हो सकते हैं, सकृत्प्रकार से भी पहले वास्तवोदयास्तर साधन में 'सकृत्सुदेन तत्सं-संगुणितैक लिप्तोन्मन्नासु इत्यादि' की उपपत्ति में भाष्यव्याख्यान में प्राचीनोक्त चरफल ग्रहण से वास्तव चरफल आता है, सिद्धान्तशेखर में 'ग्रहगति चर खण्ड प्राणविघटीभिर्हतात' इत्यादि श्रीपति का प्रकार आचार्योक्तानुरूप ही है, भास्कराचार्य भी "चरदलयुक्तिर्द्युनिशाम भक्त" इत्यादि से उसी विषय को कहते हैं. चरफल संस्कार किसी भी प्राचीनाचार्य का ठीक नहीं है यह उपर्युक्त युक्ति से स्पष्ट है इति ॥५९॥

इदानीं दिनरात्रिमानमाह

दिनमानरात्रिघटिकाश्चरार्धघटिकाभिरुत्तरे गोले ।
पञ्चदश युक्तहीना याम्ये हीनाधिका द्विगुणाः ॥ ६० ॥

वा. भा.—स्पष्टार्थाऽत्र वासना, इष्टदिने स्वाहोरात्रवृत्तं घटिकांकितं कृत्वा सभागैस्तत्रैकस्मिन्वृत्तपादे पंचदशघटिका भवन्ति । याम्योत्तरमंडलस्वाहोरात्रवृत्तसंपातादुभयतोपि प्राक्परयोरुन्मंडलेन सह त्वहोरात्रसंपातौ । पंचदशघटिकावच्छिन्नौ नवस्वदेशोन्मंडलोदयास्तमयौ क्षितिजस्यान्यत्वात् उत्तरगोलेऽर्धक्षितिजं क्षितिजाच्च रविमुद्यन्तं पश्यति भूस्थः तस्मात्प्राक्प्रदर्शितचरदलघटिकाभिरधिकाः पंचदशघटिका दिनार्धं भवति । पश्चादस्तक्षितिजमेवास्तमेति तस्मात्ततोपि स्वाहोरात्र खंडलकेन चरदलाख्येन पंचदशघटिका उपचिता सत्यो दिनार्धत्वं भजन्ते अतो द्विगुणं दिनदलमेव दिनप्रमाणं शेषा घटिकास्वाहोरात्रवृत्ते एभिःप्रमाणैश्च तच्चोभयश्च तच्चरदलेनखंडित मतो द्विगुणं पंचचरदलं त्रिंशतो विशोध्य रात्रिः प्रमाणमुत्तरगोले याम्ये सर्वं वैपरीत्येन योजयेत् । निरक्षदेशे पुनः क्षितिजोन्मंडलयोरेकत्वाच्चरदलाभाव एवमुत्तरगोले तावद्दिनार्धमुपचीयते । यावत् पंचदश घटिकाश्चरार्धं भवति, तत्रारात्रेरभावः तत्र दिने एतच्च तत्र सम्भवति, यत्र षट्षष्टि रक्षांशाः पुरतः परतश्च तावदुपचीयते यावन्मेरौ । तत्र मेरौ षड्भिर्मासैर्दिनं षड्भी रात्रिः एवं निरक्षदक्षिणेनापि योज्यम् एतच्च त्रिप्रश्नोत्तराध्याये वक्षत्याचार्यः, एवमपि तत्रैव व्यावर्णयिष्यामः ।

वि. भा.—उत्तरे गोले पञ्चदशनाड़िकाश्चरार्धघटीभिर्युक्ता हीनाश्च द्विगुणास्तथा याम्ये (दक्षिणे गोले) पञ्चदश घटिकाश्चरार्धघटीभिर्हीना युक्ताश्च द्विगुणास्तदा दिनमानघट्यो रात्रिमानघट्यो भवन्तीति ॥ ६० ॥

अत्रोपपत्तिः

उन्मण्डलयाम्योत्तरवृत्तयोरन्तरे पञ्चदश घट्यः । स्वक्षितिजोन्मण्डलयोरन्तरे चरार्धम् । उत्तरगोले स्वक्षितिजादुन्मण्डलस्योपरिस्थित्वाच्चरघटीयुक्ताः

पञ्चदशनाड्यो दिनार्धप्रमाण भवेत्। दक्षिणगोले तु स्वक्षितिजादुन्मण्डलस्याधः स्थितत्वाच्चरघटीहीनाः पञ्चदशघट्यो दिनार्धप्रमाणम्। उभयगोलीय दिनार्धमानं विज्ञातः शोध्यं तदोभयगोलीयरात्र्यर्धमानं भवेत्। द्विगुणीकरणेन दिनरात्रिमाने भवनः सिद्धान्तशिरोमणौ चरघटीसहिता रहिता इत्यादिना, भास्करेणाप्येवमेव कथ्यत इति ॥ ६० ॥

अब दिन मान और रात्रिमान को कहते हैं।

हि. भा.—उत्तर गोल में पन्द्रह १५ घटी में चरार्ध घटी को जोड़ने से और घटाने से दिनार्ध घटी और रात्र्यर्ध घटी होती है, द्विगुणित करने से दिनमान और रात्रिमान होता है। दक्षिण गोल में पन्द्रह १५ घटी में से चरार्ध घटी को घटाते से और जोड़ने से दिनार्ध और रात्र्यर्ध मान होता है, द्विगुणित करने से दिन मान और रात्रिमान होता है. इति ॥६०॥

उपपत्ति

उन्मण्डल और याम्योत्तर वृत्त के अन्तर में पन्द्रह घटी है, और स्वक्षितिज तथा उन्मण्डल के मध्य में चर घटी है, उत्तरगोल में स्वक्षितिज से उन्मण्डल ऊपर है इसलिये पन्द्रह घटी में चरघटी को जोड़ने से दिनार्धमान होता है, दक्षिण गोल में स्वक्षितिज से उन्मण्डल नीचा है इसलिये पन्द्रह घटी में से चरघटी को घटाने से दिनार्ध मान होता है, दोनों गोलों के दिनार्धमान को तीस में से घटाने से दोनों गोलों (उत्तर गोल और दक्षिण गोल) का रात्र्यर्ध मान होता है, द्विगुणित करने से दिनमान और रात्रिमान होता है, सिद्धान्तशिरोमणि में "चरघटी सहिता रहिता इत्यादि से" भास्कराचार्य भी इसी तरह कहते हैं इति ॥६०

इदानीं ग्रहाणां नक्षत्रानयनमाह

भान्यश्विन्यादीनि ग्रहलिप्ताः खखवसूद्धृता लब्धम्।
भुक्तिहृते गतगम्ये दिवसाः षष्ट्या गुणे घटिकाः ॥ ६१ ॥

वा. भा.—इदानीं नक्षत्रानयनार्थं सर्वग्रहाणामार्यामाह। इष्टस्फुटग्रहलिप्ताः खखवसुभिर्विभजेत् फलभुक्तानि नक्षत्राण्यश्विन्यादीनि भवन्ति, शेषलिप्तागतसंज्ञाः ताश्च खखवसुभ्यो विशोध्य गम्यसंज्ञा भवन्ति। ततो गत गम्ये द्वे अपि तस्यैव ग्रहस्य भुक्त्या विभजेत्। फलं दिवसा : शेषात् षष्टिगुणात् घटिका विघटिकाश्च एवं गतात् प्राक् कालं गम्यादिष्टाभिर्हृतः चन्द्रभागेन व्यवहारेण स्फुटचन्द्रादुक्तवन्नक्षत्राद्यंतावानीय चरदलं विनाप्रमाणादिभिर्व्यवहार : इष्टावधे कार्यति प्रसिद्धत्वान्नोदाहृत इति अत्र वासना स्फुट ग्रहे मेषादिराशिगणना मेषादयश्चाश्विन्यादिभिर्नवभिर्नवभिर्नक्षत्रपादैर्नक्षत्रपादाश्चाष्टादश - शत - लिप्ता-प्रमाणस्य

राशेर्नवमांश : तैश्चतुर्भिरष्टौ गतानि भवन्ति । चक्रलिप्तानां सप्तविंशतितमांश इत्यर्थ : ।

तस्माद् ग्रहलिप्ताभिरेको दिवसो भवति । तच्छेषलिप्ताभिः कियन्तो दिवसा इति दिवसादिकालो गताल्लाद्वा कार्य : क्षयवृद्धी च नक्षत्रभोगस्य क्रान्तिमण्डलवशाद्ध्रुवमेरुत्यासन्नोऽन्दिरे वा ग्रहो भवति । तस्मादुपपन्नं कक्षामण्डलादिषु पूर्वनिन्यस्तेषु सर्वं प्रदर्शयेदिति ।

वि. भा.—ग्रहलिप्ताः (ग्रहकलाः) खखसूद्धृता (८०० एभिर्भक्ताः) सदा लब्धं अश्विन्यादीनि भानि (नक्षत्राणि) भवन्ति, गतगम्ये (गतगम्यकले) भुक्ति हृते (ग्रहगतिकलाभक्ते) तदा गतगम्या दिवसा भवन्ति, तच्छेषे षष्ट्या गुणे भुक्तिहृते घटिका भवन्तीति ॥ ६१ ॥

अत्रोपपत्तिः

यदि चक्रकलातुल्यग्रहगतौ सप्तविंशतिर्नक्षत्राणि लभ्यन्ते तदेष्ट ग्रहगतिकलायां किमिति समागच्छन्ति गतनक्षत्राणि तत्स्वरूपम्=$\frac{२७\times ग्रहक}{२१६००}=\frac{ग्रकला}{८००}$=गतनक्षत्र+$\frac{शे}{८००}$ अत्र शेषं वर्त्तमान नक्षत्रस्य गतावयवमानम् हराच्छुद्धं तदा तद्भोग्यावयवः स्यात् । ततोऽनुपातो यदि गतिकलाभिरेकं दिनं लभ्यते तदा गतकलाभिर्गम्यकलाभिश्च किं समागच्छन्ति गतदिनानि गम्यदिनानि च । शेषे षष्ट्या गुणे गतिहृते तदा घटिका भवन्तीति ॥ ६१ ॥

अब ग्रहों के नक्षत्रानयन को कहते हैं।

हि.भा.—ग्रह कला को आठ सौ ८०० से भाग देने से लब्ध अश्विनी आदि नक्षत्र होते हैं। गतकला और गम्य कला को ग्रहगति कला से भाग देने से गतदिन और गम्य दिन होते हैं, शेष को साठ से गुणा कर गति से भाग देने से घटी होती हैं. इति ॥ ६१ ॥

उपपत्ति ।

यदि चक्रकलातुल्य ग्रहगति में सत्ताइस २७ नक्षत्र पाते हैं तो इष्टग्रहगति कला से क्या इससे गत नक्षत्र आते हैं $\frac{२७\times ग्रकला}{२१६००}=\frac{ग्रकला}{८००}$ =गतनक्षत्र+$\frac{शे}{८००}$ यहां शेष वर्त्तमान नक्षत्र के गतावयव है, और उसको हर में से घटाने से उसका भोग्यावयव होता है, तब अनुपात करते हैं, यदि गतिकला में एक दिन पाते हैं तो उसको गतकला और गम्यकला में

क्या इससे गत दिन और गम्य दिन आते हैं। शेष को साठ से गुणा कर गति से भाग देने से घटी होती है। सिद्धान्त शिरोमणि में 'ग्रहकला : सरवीन्दुकला हृता:' इत्यादि से भास्कराचार्य भी इसी बात को कहते हैं. ॥ ६१ ॥

इदानीं तिथ्यानयनमाह

अर्कोनचन्द्रलिप्ताः खयमस्वर ७२० भाजिताः फलं तिथयः ।
गतगम्ये षष्टिगुणे भुक्तयन्तरभाजिते घटिकाः ॥ ६२ ॥

वा. भा.—इदानीं तिथ्यानयनार्थमार्यामाह स्पष्टार्थेयमार्या अमावास्यान्ते चन्द्र-सूर्यौ तुल्यावेकसूत्रगौ भवतस्तत्र प्रतिदिनं पश्चादवलम्बते। तस्माद्रविमुक्ति विशोध्य गत्योरन्तरं सिद्धं भवति यस्मादमावास्यान्तात् द्वावपि प्रवृत्तौ तयोश्चान्तरे द्वादशभिर्भागैरेका तिथिर्भवति। यतश्च चक्रांशकानां त्रिंशद्भागेन द्वादशभागा भवन्ति। तयोश्चान्तरे चक्रममावास्यान्ते भवति द्वादशभिश्च भागैः खयमस्वरसंख्या लिप्ता भवति। अतोऽर्कोनचन्द्रलिप्ता खयमस्वरभाजितफलं तिथीत्युक्तम् शेषेण सह त्रैराशिकं यदि भुक्तान्तरेण षष्टिघटिका लभ्यन्ते तच्छेषाभिर्लिप्ताभिः कियत् इति शेषं प्राग्वत्। अथ सूर्यो गोलाध्यायेपि त्रिदिवसप्रतिपादने विस्तरेण मया व्याख्यात एवेति। रवि चन्द्रांतरदर्शनार्थमाह ॥

वि. भा.—अर्कोनचन्द्रलिप्ताः (रविचन्द्रान्तरकलाः) खयमस्वरभाजिताः (७२० एभिर्भक्ताः) फलं गततिथयो भवन्ति, गता हराच्छुद्धास्तदा गम्या भवन्ति, गतगम्यकले षष्ट्या गुणिते भुक्तयन्तरभाजिते (रविचन्द्रगत्यन्तरकलाभिर्भक्ते) तदा गतघटिका गम्यघटिकाश्च भवन्तीति ॥ ६२ ॥

अत्रोपपत्तिः

अमान्ते रविचन्द्रावेकत्रैव भवतः (दर्शः सूर्येन्दुसङ्गम इत्युक्तेः) ततोऽनन्तरं चन्द्रोऽधिकगतित्वादग्रतो गच्छति, एवं गच्छन् चन्द्रः पुना रविणा सह यदा मिलति तदा द्वितीयोऽमान्तश्चान्द्रमासपूर्त्तिश्च भवति तत्र च रविचन्द्रयोर्गत्यन्तरांशाः ३६०, ततोऽनुपातो यदि रविचन्द्रगत्यन्तरांशेन ३६० तत्तुल्येन त्रिंशत्तिथयो (चान्द्रमासस्य त्रिंशत्तिथ्यात्मकत्वात्) लभ्यन्ते तदेष्टरविचन्द्रगत्यन्तरेण किमिति

$$\text{फलं गतास्तिथयस्तत्स्वरूपम्} = \frac{३० \times \text{रविचन्द्रान्तरांश}}{३६०} = \frac{३० \times \text{रविचंद्रान्तरांश} \times २}{३६० \times २}$$

$$= \frac{६० \times \text{रविचन्द्रान्तरांश}}{७२०} = \frac{\text{रविचन्द्रान्तरकला}}{७२०} = \text{गतति} + \frac{\text{शे}}{७२०}$$ अत्र शेषं वर्त्तमानतिथेर्गतावयवरूपम। तत् हराच्छुद्धं तदा वर्त्तमानतिथेर्भोग्यावयवो

भवति। ततोऽनुपातो यदि गत्यन्तरकलाभिः षष्टिर्घटिका लभ्यन्ते तदा गतकलाभिर्गम्यकलाभिश्च किं समागच्छन्ति गतघटिका गम्यघटिकाश्चेत्यनेनाचार्योक्तमुपपद्यते, अथपूर्वोक्तगततिथिस्वरूपम् $=\frac{३०\times \text{रविचन्द्रान्तरांश}}{३६०}=\frac{\text{रविचन्द्रान्तरांश}}{१२}=$गतति $+\frac{\text{शे}}{१२}$, अत्र शेषं वर्त्तमानतिथेर्गतावयवरूपम् तद्धराच्छुद्धं भोग्यं स्यात्। ततोऽनुपातः पूर्ववत् $=\frac{६०\times \text{गतकला}}{\text{गत्यन्तरकला}}=\frac{\text{गतविकला}}{\text{गत्यन्तरक}}=$गतघटिकाः। $\frac{६०\times \text{भोग्यकला}}{\text{गत्यन्तरक}}=$ $\frac{\text{भोग्यविकला}}{\text{गत्यन्तरक}}$भोग्यघटिकाः एतेन "मिहिर विरहितेन्दोरंशकेभ्यो द्विचन्द्रैः १२ गततिथिनिचयः स्यात्तत्र शेषं गताख्यम्। तदपि हरविशुद्धं गम्यकं तद्विलिप्ता गतिविवरविभक्ता यातयेयाख्यनाड्यः" श्रीपत्युक्तमुपपद्यते, सिद्धान्तशिरोमणौ भास्करेणापी 'रविरसैर्विरवीन्दुलवाहृता इत्यादिना' दमेव कथ्यत इति ॥ ६२ ॥

अब तिथ्यानयन को कहते हैं।

हि. भा.— रवि और चन्द्र की अन्तरकला को सात सौ बीस ७२० से भाग देने से लब्धि गत तिथि होती है, गत को हर में से घटाने से गम्य होता है, गत कला और गम्य कला को साठ ६० से गुणा कर गत्यन्तर से भाग देने से गत घटी और गम्य घटी होती है इति. ॥ ६२ ॥

उपपत्ति

अमान्त काल में रवि और चन्द्र एक ही स्थान में रहते हैं, उसके बाद चन्द्र शीघ्रगति होने के कारण रवि से आगे चला जाता है इसतरह प्रतिदिन चलते चलते फिर रबि के साथ मिलता है तब द्वितीय अमान्त होता है, और चन्द्र मास की पूर्ति होती है, तथा वहाँ रवि और चन्द्र के गत्यंतरांश = ३६०° होता है, तब अनुपात करते हैं यदि रवि और चन्द्र के तीन सौ साठ ३६० अंश तुल्य गत्यन्तरांश में तीस ३० तिथि (१ चन्द्रमास तीस तिथि के होते हैं) पाते हैं तो इष्ट रवि चन्द्र के गत्यन्तरांश में क्या इससे गततिथि प्रमाण आता है। $\frac{३०\times \text{रविचन्द्रान्तरांश}}{३६०}=\frac{३०\times \text{रविचन्द्रान्तरांश}\times २}{३६०\times २}=\frac{६०\times \text{रविचन्द्रान्तरांश}}{७२०}=$ $\frac{\text{रविचन्द्रान्तर कला}}{७२०}=$ गतनक्षत्र $+\frac{\text{शे}}{७२०}$ यहां शेष वर्त्तमान तिथि का गतावयव रूप है उसको हर में से घटाने से वर्त्तमान तिथि का भोग्य अवयव होता है, तब अनुपात करते हैं। यदि गत्यन्तर कला में साठ घटी पाते हैं तो गत कला और गम्य कला में क्या इससे गत घटी

और गम्य घटी आती है इससे आचार्योक्त उपपन्न हुआ । पूर्वोक्त गततिथि स्वरूप= $\frac{३० \times \text{रविचन्द्रान्तरांश}}{३६०} = \frac{\text{रविचन्द्रान्तरांश}}{१२} = \text{गतति} + \frac{\text{शे}}{१२}$ यहां भी शेष वर्त्तमान तिथि का गतावयव रूप है। उसको हर १२ में से घटाने से भोग्य होता है। तब अनुपात करते हैं $\frac{६० \times \text{गतकला}}{\text{गत्यन्तरक}} = \frac{\text{गत विकला}}{\text{गत्यन्तरक}} = \text{गतघटी}, \frac{६० \times \text{भोग्यकला}}{\text{गत्यन्तरक}} = \frac{\text{भोग्य विकला}}{\text{गत्यन्तरक}} = \text{भोग्यघटी}$ इससे 'मिहिरविरहिते न्दो रंगकेभ्यो द्विचन्द्रैः' इत्यादि संस्कृतोपपत्ति में लिखित श्रीपति का पद्य उपपन्न होता है। सिद्धान्त शिरोमणि में भास्कराचार्य भी 'रविरसैर्विरवीन्दुलवाहृता' इत्यादि से इसी को कहते हैं ॥ ६२ ॥

इदानीं योगानयनमाह

रविचन्द्रयोगलिप्ताः खखवसुभिर्भाजिताः फलं योगाः ।
गतगम्ये षष्टिगुणे भुक्तिसमासोद्धृते नाड्यः ॥ ६३ ॥

सु. भा.—स्पष्टार्थम् । संप्रत्युपलब्धचतुर्वेदाचार्यटीकायामियमार्या नोपलभ्यते ॥ ६३ ॥

वि. भा.—रविचन्द्रयोगकलाः ८०० एभिर्भक्तास्तदा लब्धं गतयोगा भवन्ति, गतगम्यकले षष्ट्यागुणिते भुक्तिसमासोद्धृते (रविचन्द्रगतियोगभक्ते) तदा गता नाड्यो गम्यनाड्यश्च भवन्तीति ॥ ६३ ॥

अत्रोपपत्तिः

रविचन्द्रयोगंतियोगेन योगा जायन्ते, यदा रविचन्द्रयोगंतियोगः=२१६०० तदा सप्तविंशतियोगाभवन्त्यतोऽनुपातो यदि रविचन्द्रयोगंतियोगकलाया २१६०० मेतत्तुल्यायां सप्तविंशतियोगा लभ्यन्ते तदेष्टगतियोगे किं समागच्छन्ति गतयोगास्तत्स्वरूपम् $= \frac{२७ \times \text{गतियोग}}{२१६००} = \frac{\text{गतियोग}}{८००} = \text{गतयोग} + \frac{\text{शे}}{८००}$, अत्रापि शेषं वर्त्तमानयोगस्य गतावयवरूपम् । एतत् हराच्छुद्धं तदा वर्त्तमानयोगस्य भोग्यावयवरूपम् । ततोऽनुपातो यदि गतियोगकलायां षष्टिघटिका लभ्यन्ते तदा गतकलायां गम्यकलायां च किं समागच्छन्ति गतनाड़िका गम्यनाड़िकाश्चेति । सिद्धान्तशेखरे श्रीपतिना "रविविधुयुतिलिप्ताः खाभ्रनागै ८०० र्विभक्ताः फलमिह गतयोगान् विद्धि विष्कम्भपूर्वान् । तदनु च गतगम्याः खर्त्तु ६० निघ्ना विभक्ताः स्वगतियुतिकलाभिर्नाड़िका भुक्तभोग्याः" अनेनाऽऽचार्योक्तानुरूपमेव सर्वं कथितमिति ॥ ६३ ॥

अब योगानयन को कहते हैं।

हि. भा.—रवि और चन्द्र की योगकला (गतियोग कला) को आठ सौ ८०० से भाग देने से लब्ध गत योग होते हैं। गतकला और गम्यकला को साठ से गुणा कर गति योग द्वारा भाग देने से गतनाड़ी और गम्य नाड़ी होती है इति ॥ ६२ ॥

उपपत्ति।

रवि और चन्द्र के गति योग से योग बनते हैं, जब रवि और चन्द्र के गतियोग= २१६०० तब सत्ताईस २७ योग होते हैं, इससे अनुपात करते है, यदि रवि और चन्द्र के गति योग कला २१६०० एतत्तुल्य में सताइस योग पाते हैं तो इष्ट गति योगकला में क्या इससे गतयोग प्रमाण आता है, $\frac{२७ \times \text{गतियोगक}}{२१६००} = \frac{\text{गतियोगक}}{८००} = \text{गतयो} + \frac{\text{शे}}{८००}$ यहां शेष वर्त्तमान योग के गतावयव रूप है। उसको हर ८०० में से घटाने से वर्त्तमान योग का भोग्यावयव रूप होता है, तब अनुपात करते हैं यदि गतियोग कला में साठ ६० घटी पाते हैं तो गतयोगकला में और गम्य योग कला में क्या इससे गत नाड़ी और गम्य नाड़ी आती है। सिद्धान्तशेखर में श्रीपति ने 'रविविधुयुतिलिप्ता इत्यादि संस्कृतोपपत्ति में श्लोक से आचार्योक्तानुरूप ही सब कुछ कहा है ॥ ६३ ॥

इदानीं रविचन्द्रांतरमाह

राश्यंशकलाविकलाः स्फुटमासान्तेंऽशलिप्तिका विकलाः।
पक्षान्ते तिथ्यन्ते समा रवीन्द्वोः कलाविकलाः ॥ ६४ ॥

वा. भा.– अमावास्यान्ते स्फुटार्कचंद्रयो : किमप्यन्तरं भवति। तस्मात्ते-राश्यादय : समा एवोपर्युपरिस्थितत्वाद्भूमध्यस्थितस्य द्रष्टु : पक्षान्ते चार्धवक्रांतरितौ द्वावपि भवतोऽतस्तत्र भागादयश्च समा एव राशयश्चभिन्नाः प्रतिदिनमभेदादिति।

वि. भा.—स्फुटमासान्ते रवीन्द्वोः (रविचन्द्रयोः) राश्यंशकलाविकलाः समाभवन्त्यर्थाद्राश्याद्यवयवाःसमा भवन्ति, पक्षान्ते रविचन्द्रयोरंशलिप्तिका विकलाः समा भवन्त्यर्थाद्रविचन्द्रावंशाद्यवयवेन तुल्यौ भवतः। तिथ्यन्ते रवीन्द्वोः कला विकलाः समा भवन्तीति ॥ ६४ ॥

अत्रोपपत्तिः

रविचन्द्रयोरन्तरांशा यदा द्वादशांशसमास्तदैका तिथिर्भवति, स्फुटमासान्ते

त्रिंशत्तिथयोऽतस्तत्र रविचन्द्रान्तरांशाः = ३० × १२ = ३६०° वा शून्यसमा अतो रविचन्द्रौ राश्याद्यवयवैस्तत्र समौ भवतः पक्षान्ते पञ्चदश तिथयोऽतस्तत्र रविचन्द्रान्तरांशाः = १५ × १२ = १८०° = ६ राशयः। अतस्तत्र रविचन्द्रावंशाद्यवयवैः समौ भवतः कथमन्यथा तयोरन्तरे केवलं राशय एव तिष्ठन्ति, कस्मिन्नपि तिथ्यन्ते रविचन्द्रयोरन्तरांशा द्वादशभक्ता एव भवितुमर्हन्ति, तेन तिथ्यन्ते कलाविकलासमत्वादेव केवलमंशा उत्पद्यन्ते, सिद्धान्तशेखरे श्रीपतिना 'मासान्ते समताऽर्कशीतमहसोः क्षेत्रेण राश्यादिना पक्षान्ते पुनरेतयोः सदृशता भागादिना जायते, अन्यस्याश्च तिथेर्विरामसमये लिप्तादिना तुल्यता, ऽनेनाऽऽचार्योक्तानुरूपमेव कथ्यते। लल्लाचार्येण 'मासान्ते रविशशिनौ समौ भवेतां पक्षान्ते लवकलिकाविलिप्तिकाभिः। अन्यस्या अपि च तिथेः सदाऽवसाने तुल्यौ स्तः खलु कलिकाविलिप्तिकाभिः' पीदमेव कथ्यत इति ॥ ६४ ॥

अब रवि और चन्द्र के अन्तर को कहते हैं।

हि. भा.—स्फुटमासान्त में रवि और चन्द्र के राशि, अंश, कला, और विकला सम होती है अर्थात् राशाद्यवयव से दोनों बराबर होते हैं। तिथ्यन्त में उन दोनों के कला, विकला बराबर होती है अर्थात् कलाद्यवयव से बराबर होते हैं इति ॥ ६४ ॥

उपपत्ति।

रवि और चन्द्र के अन्तरांश जब बारह अंश होते हैं तब एक तिथि होती है, स्फुटमासान्त में तीस तिथियां होती है इसलिये वहाँ रवि चन्द्रान्तरांश = ३० × १२ = ३६० वां शून्य के बराबर, अतः वहां रवि और चन्द्र राश्याद्यवयवों से बराबर होते हैं। पक्षान्त में पन्द्रह १५ तिथियां होती हैं इसलिये रवि चन्द्रान्तरांश = १५ × १२ = १८० = ६ राशि, अतः वहां रवि और चन्द्र अंशाद्यवयवों से बराबर होती है, क्योंकि उन दोनों के अन्तर करने से केवल राशि ही रहती है। किसी भी तिथ्यन्त में रवि चन्द्र का अन्तरांश बाहर से विभक्त ही रहता है इसलिये तिथ्यन्त में रवि और चन्द्र कलाद्यवय से बराबर होते हैं। सिद्धान्त शेखर में श्रीपति 'मासान्ते समताऽर्क शीत महसो, इत्यादि। संस्कृतोपपत्ति में लिखित श्लोक से आचार्योक्तानुरूप ही कहते हैं, लल्लाचार्य भी 'मासान्ते रवि शशिनौ समौ भवेताम्' इत्यादि, संस्कृतोपपत्ति में लिखित श्लोक से इसी बात को कहते हैं इति ॥ ६४ ॥

इदानीं स्थिरकरणान्याह

कृष्णचतुर्दश्यन्ते शकुनिः पर्वणि चतुष्पदं प्रथमे।
तिथ्यर्धेऽन्ते नागं किंस्तुघ्नं प्रतिपदाद्यर्धे ॥ ६५ ॥

वा. भा.—स्पष्टार्थेयमार्या इदानीं ध्रुवकरणानयनार्थमार्यामाह ।

वि. भा.—कृष्णपक्षीयचतुर्दश्याः परार्धे शकुनिः, पर्वणि प्रथमे (अमावास्या पूर्वार्धे) चतुष्पदं करणम् । तिथ्यर्धेऽन्ते (अमावास्योत्तरार्धे) नागं, प्रतिपदाद्यर्धे (शुक्लपक्षीय प्रतिपत्पूर्वार्धे) किंस्तुघ्नं करणं भवतीति ॥ ६५ ॥

अत्रोपपत्तिः

लल्लाचार्येण "शशिनि कृशशरीरे या चतुर्दश्यवश्यं शकुनिरपरभागे जायते नाम तस्याः । तदनुतिथिदले ये ते चतुष्पादनागे प्रतिपदि च यदाद्यं तद्धि किंस्तुघ्नमाहुः" नेन, श्रीपतिना चे "कृशशशिचतुर्दश्यामन्त्ये दले शकुनिर्भवेत् प्रथमशकलेऽमावास्यायाश्चतुश्चरणाह्वयम् । करणमुदितं नागस्तस्या दले चरमे बुधैः प्रतिपदि भवेत् किंस्तुघ्नाख्यं सदा प्रथमे दले" त्यनेन, भास्करेण च "शकुनितोऽसितभूतदलादनु" इति मूलोक्त्या "कृष्णचतुर्दश्यर्धादुपरि यान्यवशिष्यन्ते त्रीणि चतुर्थं प्रतिपत्प्रथमार्धे च । एतानि चत्वारि शकुनिनः । शकुनिचतुष्पदनागकिंस्तुघ्नानीति शेषः" व्याख्ययाऽनयाऽमावास्यापूर्वोत्तरार्धयोश्चतुष्पदनागाख्ये करणे कथिते । सूर्यसिद्धान्ते च "ध्रुवाणि शकुनिर्नागं तृतीयं तु चतुष्पदम् किंस्तुघ्नं तु चतुर्दश्याः कृष्णायाश्चापरार्धतः" इत्यनेनाऽमावास्या पूर्वोत्तरार्धयोः नागचतुष्पदाख्ये करणे कथिते तदनयोः करणयोः पूर्वापरक्रमभेदे सूर्यसिद्धान्तसुधावर्षिण्यां "प्रायः सर्वेषां मते ब्राह्मक्रम एव युक्तोऽतः प्रथमं शकुनिः । तृतीयं तु नागम् । चतुष्पदं द्वितीयमित्यध्याहार्यम्" म.म. सुधाकरद्विवेदिनो लिखितवन्त इति ॥ ६५ ॥

अब स्थिर करणों को कहते हैं ।

हि. भा.—कृष्ण पक्ष की चतुर्दशी के परार्ध में शकुनि करण, अमावास्या के पूर्वार्ध में चतुष्पद करण और परार्ध में नाग करण, शुक्लपक्ष की प्रति पदा के पूर्वार्ध में किंस्तुघ्न करण होते हैं इति ॥ ६५ ॥

उपपत्ति

लल्लाचार्य 'शशिनि कृशशरीरे या चतुर्दश्यवश्यं' इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से 'कृश शशि चतुर्दश्यामन्त्ये दले' इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से श्रीपति तथा 'शकुनितोऽसितभूतदलादनु' इससे भास्कराचार्य भी आचार्योक्तानुरूप ही कहते है। सूर्य सिद्धान्त में "ध्रुवाणि शकुनिर्नागं तृतीयं तु चतुष्पदम्" इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से अमावास्या के पूर्वार्ध में नागकरण और उत्तरार्ध में चतुष्पद करण कहते हैं, इन दोनों करणों के पूर्वापर क्रम भेद के विषय में सूर्य सिद्धान्त की सुधावर्षिणी टीका में 'प्रायः

सर्वेषां मते ब्राह्मक्रम एव युक्तोऽतः प्रथमं शकुनिः । तृतीयं तु नागम् । चतुष्पदं द्वितीयमित्य-ध्याहार्यम् म.म.सुधाकर द्विवेदी लिखते है इति ॥ ६५ ॥

इदानीं चरकरणान्याह

व्यर्केन्दुकला भक्ताः खरसगुणैर्लब्धमूनमेकेन ।
चरकरणानि बवादीन्यगृहीतशेषे तिथिवदन्यत् ॥ ६६ ॥

वा.भा.—स्पष्टार्थोयं । तिथ्यर्धभोगात्मिका करणस्य तिथिवद्वासना योज्या खरसगुणैः फलं चोनमेकेनातः क्रियते यतः सितप्रतिपदार्धात्करणानां बवादीनां प्रवृत्तिः प्रागर्धस्य स्थिरेण नित्यमेवार्द्धकत्वादिति । यानि स्फुटगत्याध्याय एव युज्यते वक्तुं तान्यतिबहुत्वात् स्फुटगत्युत्तराध्याये वक्ष्यामीत्येतदार्यामाह ॥

वि. भा.—व्यर्केन्दुकलाः (रविचन्द्रान्तरकलाः) खरसगुणैः (३६० एभिः) भक्ता लब्धमेकेन हीनं, अगृहीतशेषे (सप्तभक्तावशिष्ट) बवादीनि चर करणानि भवन्ति अन्यत्कर्मतिथिसाधनवत्कार्यमिति ॥ ६६ ॥

अत्रोपपत्तिः

तिथिसाधनार्थं यदि ३६०° तुल्य रविचन्द्रयोर्गत्यन्तरेण त्रिंशत्तिथयो लभ्यन्ते तदेष्टरविचन्द्रान्तरांशे किमिति जाता गतास्तिथयः $=\frac{३०\times\text{रविचन्द्रान्तरांश}}{३६०}=\frac{\text{रविचन्द्रान्तरांश}}{१२}$ ततस्तिथिर्द्विगुणाकरणानीत्यतः करणानि $=\frac{२\times\text{रविचन्द्रान्तरांश}}{१२}=\frac{\text{रविचन्द्रान्तरांश}}{६}=\frac{\text{रविचन्द्रान्तरांश}\times६०}{६\times६०}=\frac{\text{रविचन्द्रान्तरकला}}{३६०}$, चत्वारि स्थिरकरणानि येषां स्थितिः पूर्वश्लोकेन स्पष्टाऽस्ति, किंस्तुघ्न संज्ञक स्थिर करणस्य प्रतिपदाद्यर्धगतत्वात् बवादीनां च शुक्लप्रतिपदोऽन्त्यार्धमारभ्य प्रवृत्तेः, पूर्वं $\frac{\text{रविचन्द्रान्तरक}}{३६०}$ लब्धेषु चैकमूनीक्रियत, ततः सप्तभक्तेऽवशेषं बवादिकरणं भवेत् । अत्रापि षष्टिगुणिताद् गतगम्याद् गत्यन्तरेण भक्ताद् वर्त्तमानकरणस्य गतगम्यघट्यो भवन्ति, सिद्धान्तशेखरे "भानुहीनशशिभागसमूहात् स्याद्बवादिकरणं रसभक्तात् । रूपहीनमगभाजितशेषं शेषकर्म तिथिवच्च विधेयम्" इत्यनेन श्रीपतिना रविरसैर्विरवीन्दुलवाहृता इत्यादिना भास्करेण च रविचन्द्रान्तरांशवशेन करणा-नयनं कृतम् । आचार्येण (ब्रह्मगुप्तेन) रविचन्द्रान्तर कलावशेन तज्ज्ञानं क्रियत इति ॥ ६६ ॥

अब चर करणों को कहते हैं।

हि. भा.—रवि और चन्द्र की अन्तर कला को तीन सौ साठ ३६० से भाग देने से जो लब्ध हो उसमें से एक घटा कर सात से भाग देने से शेष ववादि करण होते हैं, अन्य कर्म (गत घटी, गम्य घटी साधन) तिथिसाधनवत् करना चाहिये इति ॥ ६६ ॥

उपपत्ति।

तिथि साधन के लिये यदि तीन सौ साठ ३६० रवि और चन्द्र के गत्यन्तरांश में तीस तिथि पाते हैं तो इष्ट रवि चन्द्रान्तरांश में क्या इससे गततिथि आती है, $\frac{३०\times\text{रविचन्द्रान्तरांश}}{३६०}=\frac{\text{रविचन्द्रान्तरांश}}{१२}$ = गततिथि, तिथि को द्विगुणित करने से करण होते हैं इस नियम से $\frac{२\times\text{रविचन्द्रान्तरांश}}{१२}=\frac{\text{रविचन्द्रान्तरांश}}{६}=\frac{\text{रविचन्द्रान्तरांश}\times६०}{६\times६०}$ $=\frac{\text{रवि चन्द्रान्तरकला}}{३६०}$ = करण, चार स्थिर करण है जिनकी स्थिति पूर्वश्लोक में कही गयी है, शुक्ल पक्ष की प्रति पदा के पूर्वार्ध में किंस्तुघ्न करण के रहने के कारण तथा शुक्लपक्ष की प्रतिपदा के परार्ध से ववादि करणों की प्रवृत्ति के कारण पूर्व $\frac{\text{रविचन्द्रान्तर कला}}{३६०}$ लब्धि में से एक घटाना चाहिये, तब सात से भाग देकर जो शेष रहता है वह ववादि करण होता है। यहां भी गतकला और गम्यकला को साठ से गुणा कर गत्यन्तर से भाग देनेसें वर्त्तमान करण की गतघटी और गम्य घटी होती है, सिद्धान्तशेखर में "भानु हीनशशिभागसमूहात्" इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से 'श्रीपति, तथा 'रविरसैंविरवीन्दुलवा ह्रता' इत्यादि से भास्कराचार्य ने रविचन्द्रान्तरांश से करणानयन किया है। आचार्य ने रविचन्द्रान्तर कला वश से उन का साधन किया है, कोई विशेषता नहीं है इति ॥ ६६ ॥

इदानीमवशिष्टं स्फुटगत्युत्तराध्याये वक्ष्यामीत्येतदर्थमाह।

इह नोक्तानि बहुत्वात् स्पष्टगतेरुत्तरेऽभिधास्यामि।
संक्रान्तिभतिथिकरण व्यतिपाताद्यन्तगणितानि ॥ ६७ ॥

सु. भा.—स्पष्टार्थम् ॥ ६७ ॥

वि. भा.—संक्रान्ति भतिथिकरणव्यतिपाताद्यन्त गणितानि (संक्रान्तिकाल-नक्षत्रतिथिकरण व्यतिपातादीनामन्तकालं निर्णेतुं गणितानि) स्पष्टगतेरध्यायस्य

बहुत्वात् (अधिकत्वात्पृथुत्वाद्वा) इह (स्पष्टगत्यध्याये) नोक्तानि (न कथितानि) स्पष्टगत्युत्तरेऽध्यायेऽभिधास्यामीति ॥ ६७ ॥

अब शेष विषयों को स्फुटगत्युत्तराध्याय में कहता हूं इसके लिये कहते हैं

हि. भा.—संक्रान्तिकाल-नक्षत्र-तिथि-करण-व्यतिपात आदि का अन्तकाल निर्णय करने के लिये गणितों को यहां स्पष्टगति अध्याय के बहुत्व (विस्तृतित्व) के कारण नहीं कहा गया है, स्पष्टगत्युत्तर नामक अध्याय में कहूंगा इति ॥ ६७ ॥

इदानीमध्यायोपसंहारमाह

ज्यापरिधिस्पष्टीकरणदिनगतिचरार्धभतिथिकरणेषु ।
स्फुटमतिरध्यायः सप्तषष्टिरार्याद्वितीयोऽयम् ॥ ६८ ॥

वा. भा.—इति भट्टमधुसूदनसुतचतुर्वेदपृथुस्वामिकृते ब्रह्मसिद्धान्तभाष्ये द्वितीयः अयमध्यायो ग्रन्थशतैर्नवभि व्यख्यातः द्वितीयः। स्वदेशराश्युदयैर्योलग्नं जानाति तृतीयः, लग्नाद्घटिका योजानानि चतुर्थः, दिनगतछायां पञ्चमः, दिनशेषाच्छायां षष्ठः तत्कालाच्छायां सप्तमः एवमत्र सप्तप्रश्नाः।

वि. भा.—ज्यासाधनम्, स्फुटपरिध्यानयनम्। ग्रहादिस्पष्टीकरणम्। दिनगतिसाधनम्। चरखण्ड नक्षत्रतिथि करणानां साधनानि, एतेषु विषयेषु सप्तषष्टिरार्याः (सप्तषष्टि संख्यकाऽऽर्या छन्दसा) अयं द्वितीयोऽध्यायः स्फुटगतिः (स्फुट गतिनामकः) समाप्ति गत इति शेषः ॥ ६८ ॥

इति श्रीब्राह्मस्फुट सिद्धान्ते स्पष्टाधिकारो द्वितीयः समाप्तः ॥

अब अध्याय के उपसंहार को कहते हैं।

हि. भा.—ज्यासाधन-स्फुटपरिध्यानयन, ग्रहादि स्पष्टीकरण, दिनगति साधन, चरार्ध-नक्षत्र-तिथि-करणों के साधन, इन विषयों में सड़सठ ६७ आर्या छन्द से स्फुटगति नामक द्वितीय अध्याय समाप्त हुआ ॥ ६८ ॥

इति श्री ब्राह्मस्फुट सिद्धान्त में स्पष्टाधिकार द्वितीय (अधिकार) समाप्त हुआ।

ब्राह्मस्फुटसिद्धान्तः

त्रिप्रश्नाधिकारः

ब्राह्मस्फुटसिद्धान्तः

त्रिप्रश्नाधिकारः

त्रयाणां प्रश्नानां दिग्देशकालानामुत्तरं यत्राभिधीयते स त्रिप्रश्नाधिकारस्तत्रादौ दिग्ज्ञानमाह ।

पूर्वापरयोर्बिन्दू तुल्यच्छायाग्रयोर्दिगपराद्यः ।
पूर्वान्यः क्रान्तिवशात् तन्मध्याच्छङ्कुतलमितरे ।।१।।

वा. भा.—पूर्वापरयोः कपालयोर्बिन्दू कार्यौ तुल्यछायांगुलाग्रयोः । एतदुक्तं भवति । सलिलकृतसमायामवनौ द्वादशांगुलं शंकुं विन्यसेदभीष्टप्रमाणं वा ततः कियत्यपि गते दिवसस्याभीष्टे काले छायाग्रे बिन्दुर्देयः स आद्यः बिन्दुरुच्यते । ततो यावति दिनगते काले छायाग्रे बिन्दुर्दत्त स्तावति दिवसस्य शेषे च छायाग्रे द्वितीयो बिन्दुर्देयः, अन्यदुच्यते । ततो विवस्वति प्राक्कपालस्थेय आद्यो बिन्दुर्दत्तः सापरा दिग्भवति । अपरकपालस्थे च सवितरि योऽन्यो बिन्दुर्दत्तः सा पूर्वा दिक् । यदि नाम क्रान्तिवशेन दिग्भेद उपपद्यते । प्रथमबिन्दुदानकालिकाद्रवेः क्रान्तिज्यां कृत्वा द्वितीयबिन्दुकालिकाच्च रवेः क्रान्तिज्या कार्या । तयोरन्तरं बिन्दुन्तरालकालस्य क्रान्तिज्या भवति । व्यासार्धरविज्यापरिणाहछायांगुलतुल्ये मंडले परिणाम्यते । तेन छायांगुलैर्हृत्वा व्यासार्धेन विभजेत् । फलं छायामंडले अंगुलादि तेन बिन्दुश्चाल्यः निरक्षदेशजे दिक्साधने साक्षे चेद्दक्छायांगुले वृत्ते अतस्तां विषुवत्कर्महतां द्वादशभिर्विभजेत् । फलं परिणतप्राप्तायां पूर्वबिन्दोश्चालनं कार्यम् । ततः उक्तमक्षांशैर्विना दिशो यो वेत्तीत्येवं यथासम्भवं फलांगुलैरग्रादानगत्या छायामंडले द्वितीयो यो बिन्दुः स उत्तरायणे उत्तरेण संचाल्य दक्षिणायने दक्षिणेन चैवं सा पूर्वा दिग्भवति । ततः पूर्वापरबिन्दोर्मस्तके सूत्रं प्रसार्य रेखां कुर्यात् । सा प्राच्यपरा भवति । तन्मध्याच्छंकुतलमितर इति तयोर्बिन्दोर्मध्यम् । तस्माच्छंकुतलं यावत्सूत्रं प्रसार्य रेखां कुर्यात् सा दक्षिणोत्तरा भवति । एवं तद्रेखानुसारेण द्वे अपि दक्षिणोत्तरे सिद्धे भवतः इत्यर्थः । अथ चैकं बिन्दुमध्ये कृत्वा द्वितीयपठितसूत्रप्रमाणेन वृत्तमालिख्यत । एवं वृत्तद्वयपरस्परानुप्रवेशात् मत्स्या उपपद्यन्ते । तस्य मुखपुच्छावगाहिसूत्रं दक्षिणोत्तरे पूर्वा दिग्रेखामध्यादित्यत्रेयं वासना पूर्वाह्णे परस्यां दिशि छायाग्रं भवतीत्यभीष्टभूदेशस्थितस्य शंकोः तस्याद्यो बिन्दुरपरादिग्भवति । अपराह्णे च पूर्वस्यां दिशि छायाग्रं पतति तेन सा पूर्वा दिक् । कालस्य तुल्यत्वात्तद्गतरेखया पूर्वापरत्वमुपपन्नं स्याद्यद्यर्कभोगो विषुवन्मण्डलगत्या भविष्यद्यावदर्कगतिरूपमंडल एवापमंडलस्य तिर्य-

क्कृतक्रान्तिवशेन दिग्भेद उपपद्यते। उत्तरायणे प्रतिक्षणमुदगमनादर्कस्य छायाग्रे दक्षिणेन वलति। तत्सूत्रमपि यावच्चलितं तावदुत्तरेण संचाल्यते। येन समपूर्वेण भवति प्रथमबिन्दोः दक्षिणेन दक्षिणानयने च तदवच्छायामंडलपरिणतक्रान्तिज्यया अग्रया वा तद्युज्यते। यतः स्वमंडले तावदेव व्यासार्धस्तत्र विन्द्वन्तरालक्रान्तिज्या-परिणामे यत्त्रैराशिकं तद। तिस्पष्टं तया च बिन्दुचालने युज्यते। निरक्षे क्षितिजो-न्मंडलयोरेकत्वात् साक्षे च तद्भेदादग्रयैव युज्यते। अग्राकरणं च कक्ष्यमानवासन-येति। त्रिछायाग्रज्ञो यः क्रान्त्यक्षांशैर्विना दिशो यो वेत्ति तस्योत्तरमाह ॥१॥

वि. भा.—पूर्वापरयोः कपालयोस्तुल्यच्छायाग्रयोर्यो बिन्दू भवतस्तत्राऽऽद्यः (प्रथमबिन्दुः) अपरादिक् (पश्चिमा दिक्) अन्यः (द्वितीयो बिन्दुः) पूर्वा दिक्, क्रान्तिवशात् (पूर्वापरकपालयोस्तुल्यच्छायाग्रयो र्ये क्रान्ती भवतस्तद्वशाद्भेद उत्पद्यते इत्यध्याहार्यम्) तन्मध्यात् (तयोश्छायाग्रयोर्मध्यं तन्मध्यं तस्मात्) शङ्कुतलं (शङ्कुमूलंयावच्छा रेखा तत्र) इतरे (दक्षिणोत्तरे ककुभौ) भवतः। अत्रैतदुक्तं भवति जलादिसमीकृतायां भूमाविष्टदिने तन्मध्यच्छायाव्यासार्धेन वृत्तं विलिख्य तत्केन्द्रे द्वादशाङ्गुलशङ्कुः स्थाप्यः, पूर्वकपालस्थे सूर्ये तस्य शङ्कोश्छायाग्रं तद्वृत्तपरिधौ यत्र लगति स बिन्दुः स्थूलपश्चिमादिक् पश्चिमकपालस्थे रवौ तस्यैव शङ्कोश्छायाग्रं पूर्वभागे तद्वृत्तपरिधौ यत्र निर्गच्छति स बिन्दुः स्थूलपूर्वादिक्, स्थूलपूर्वपश्चिम-बिन्दुगता रेखा स्थूलपूर्वापरा, तस्यां तयोश्छायाग्रयोर्मध्यं यत्तस्माच्छङ्कुमूलगतारेखा दक्षिणोत्तरा भवतीति ॥१॥

अत्रोपपत्तिः

अथ छायाप्रवेशनिर्गमकालिकक्रान्त्योरसमत्वात् $\frac{\text{त्रि.ज्याक्रां}}{\text{ज्यालं}}$ = अग्रा,

$\frac{\text{त्रि.ज्याक्रां}'}{\text{ज्यालं}}$ = अग्रा', अत्र क्रां = छायाप्रवेशकालिकक्रान्तिः। क्रां' = छायानिर्गमकालिक-क्रान्तिः। अग्रयोरन्तरेण $\frac{\text{त्रि·ज्याक्रां}'}{\text{ज्यालं}} \sim \frac{\text{त्रि·ज्याक्रां}}{\text{ज्यालं}} = \frac{\text{त्रि}}{\text{ज्यालं}}$ (ज्याक्रां' ~ ज्याक्रां)

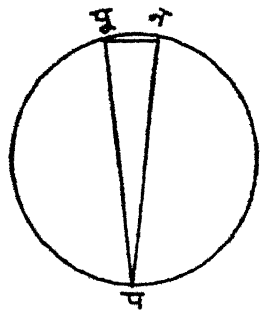
= अग्रा' ~ अग्रा = अग्रान्तरम्। एतत् छायाकर्णगोले समानीयते, यदि त्रिज्या-व्यासार्धे इदमग्रान्तरं लभ्यते तदा छायाकर्णव्यासार्धे किं समागच्छति छायाकर्ण-व्यासार्धेऽग्रान्तरम् = $\frac{\text{त्रि·छाक}}{\text{ज्यालं·त्रि}}$ (ज्याक्रां' ~ ज्याक्रां), = $\frac{\text{छाक}}{\text{ज्यालं}}$ (ज्याक्रां' ~ ज्याक्रां),

एतदग्रान्तरचालनवशाद् वास्तवपूर्वापररेखायाः समानान्तररेखाया ज्ञानं भवेत्ततः केन्द्रबिन्दुतस्तत्समानान्तरा रेखा वास्तवपूर्वापरा रेखा भवेत्। "छायानिर्गमन-प्रवेशसमयार्ककान्तिजीवान्तरं क्षुण्णं स्वश्रवणेन लम्बकहृतं स्यादङ्गुलाद्य

फलम्। पश्चाद्विन्दुमनेन रव्ययनतः संचालयेद् व्यत्ययात् स्पष्टा प्राच्यपराऽथवाऽयनवशात् प्राग्विन्दुमुत्सारयेत्" अमुमेव श्रीपतिप्रकारं दृष्ट्वा भास्करेणो 'तत्कालागमजोवशेस्तु विवरादु भाकर्णमित्याहनान्यव्यास्तमिताङ्गुलैरयनदिग्विन्दो स्फुटा चालिता' दं कथितम्। पर छायाकर्णवृत्तपरिधावन्तरदानानौचित्यान्नहि श्रीपत्याद्युक्तप्रकारेण स्फुटपूर्वापरदिशोर्ज्ञानमतस्तद् वास्तवानयनं प्रदर्श्यते।

पू=स्थूलपूर्वादिक्। प=स्थूलपश्चिमादिक्। पूपके अर्धविन्दुतः पूप अर्धव्यासार्धेन वृत्त कार्य,पूप रेखा वास्तवपूर्वापररेखाऽसमान्तरा पूविन्दुतोऽग्रान्तरसमा भुजान्तरसमा वा रेखा पूर्णज्यारूपा (पून) देया (रे.४ अध्यायोक्त्या) पन रेखा कार्या, <पूनप=९०, तदेयमेव रेखा वास्तवपूर्वापररेखायाः समानान्तरा भवति ततो वास्तवपूर्वापररेखाज्ञानं भवेदेवेति ॥१॥

(क)



अब त्रिप्रश्नाधिकार प्रारम्भ किया जाता है

तीन प्रश्नों (दिशा, देश, काल) का उत्तर जिसमें कहा जाता है, वह त्रिप्रश्नाधिकार है, उसमें पहले दिशा ज्ञान को कहते हैं।

हि. भा.—पूर्वकपाल में पश्चिमकपाल में तुल्यछायाग्रद्वय में जो दो बिन्दु हैं उनमें प्रथम बिन्दु पश्चिम दिशा है, द्वितीय बिन्दु पूर्व दिशा है, पूर्वकपाल और पश्चिमकपाल में तुल्यछायाग्रद्वय की जो क्रान्ति होती है उसके वश से भेद होता है, दोनों छायाग्रद्वय के मध्य से शङ्कुमूलपर्यन्त जो रेखा होती है वह दक्षिणोत्तर रेखा है, उसी में दक्षिण दिशा और उत्तर दिशा होती है। जलादि से समान की हुई पृथ्वी में इष्ट दिन में मध्यान्हकालिक छाया व्यासार्ध से वृत्त बनाकर उसके केन्द्र में द्वादशा १२ ङ्गुल शङ्कु स्थापित करना, सूर्य के पूर्वकपाल में रहने से उस शङ्कु का छायाग्र उस वृत्तपरिधि में जहां प्रवेश करता है वह बिन्दु स्थूल पश्चिम दिशा है, पश्चिम कपाल में सूर्य के रहने से शङ्कु का छायाग्र पूर्वभाग में उस वृत्त परिधि में निर्गत होता है वह स्थूल पूर्व दिशा है, दोनों बिन्दु (स्थूल पूर्वपश्चिम) गतरेखा स्थूल पूर्वापर रेखा होती है, उसमें दोनों छायाग्रों के मध्य से शङ्कु मूलगत रेखा दक्षिणोत्तर रेखा होती है इति ॥१॥

उपपत्ति

छायाप्रवेशकालिक और निर्गमकालिक क्रांति की तुल्यता से दोनों कालिक अग्रा भी तुल्य होती हैं क्रां=छायाप्रवेशकालिक क्रान्ति। क्रां'=छायानिर्गमकालिक

क्रान्ति, $\frac{\text{त्रि.ज्याक्रां}}{\text{ज्यालं}}$=अग्रा, $\frac{\text{त्रि.ज्याक्रां}'}{\text{ज्यालं}}$=अग्रा दोनों का अन्तर करने से $\frac{\text{त्रि.ज्याक्रां}}{\text{ज्यालं}}-\frac{\text{त्रि.ज्याक्रां}'}{\text{ज्यालं}}=\frac{\text{त्रि}}{\text{ज्यालं}}(\text{ज्याक्रां}-\text{ज्याक्रां}')=$अग्रा$-$अग्रा, =अग्रान्तर=भुजान्तर, इसको छायाकर्ण गोल में लाते हैं। अनुपात करते हैं यदि त्रिज्याव्यासार्ध में यह अग्रान्तर पाते हैं तो छायाकर्ण व्यासार्ध में क्या इससे छायाकर्ण व्यासार्ध में अग्रान्तर आया $\frac{\text{त्रि.छाक,}}{\text{ज्यालं.त्रि}}(\text{ज्याक्रां}'-\text{ज्याक्रां})=\frac{\text{छाक}}{\text{ज्याल}}(\text{ज्याक्रां}'-\text{ज्याक्रां})=$छायाकर्णगोल में अग्रान्तर इससे पूर्व-बिन्दु के चालनवश से वास्तव पूर्वापर रेखा की समानान्तर रेखा का ज्ञान होता है, केन्द्र-बिन्दु से उसकी समानान्तरा रेखा वास्तव पूर्वापर रेखा होती है, पूर्वापर रेखा की समानान्तर रेखा के अर्ध बिन्दु से उसके ऊपर लम्बरेखा या पूर्वापर रेखा के अर्ध बिन्दु (केन्द्रबिन्दु) से उसके ऊपर लम्बरेखा दक्षिणोत्तर रेखा होती है, सिद्धान्तशेखर में "छायानिर्गमनप्रवेशसमयार्ककान्तिजीवान्तरं" इत्यादि संस्कृतोपपत्ति में लिखित श्लोक श्रीपतिप्रकार को देख कर भास्कराचार्य ने "तत्कालापमजीवयोस्तु विवरात् इत्यादि" सिद्धान्तशिरोमणि में कहा है। लेकिन छायाकर्ण वृत्तपरिधि में अग्रान्तर या भुजान्तर दान देना अनुचित है इसलिए श्रीपत्यादि कथित प्रकार से स्फुट पूर्वापर दिशा का ज्ञान ठीक से नहीं हो सकता है, अतः वास्तव ज्ञान के लिए युक्ति बतलाते हैं।

यहाँ संस्कृतोपपत्ति में लिखित (क) क्षेत्र को देखिये। पू=स्थूल पूर्वदिशा, प=स्थूल पश्चिम दिशा, पूप रेखा वास्तवपूर्वापर रेखा की असमानान्तर रेखा, पू बिन्दु से अग्रान्तर तुल्य या भुजान्तर तुल्य पूर्णज्या रूप रेखा (पून) दिये (रे. ४ अध्याय युक्ति से) पन रेखा कीजिए, <पूनप=९०, तब यहीं पन रेखा वास्तव पूर्वापर रेखा की समानान्तर होती है, केन्द्र बिन्दु से इसकी समानान्तर रेखा वास्तव पूर्वापर रेखा होगी, इस तरह वास्तविक पूर्व पश्चिम दिशाओं का ज्ञान हुआ, केन्द्र बिन्दु से पूर्वापर रेखा के ऊपर लम्ब रेखा दक्षिणोत्तर रेखा होती है, इस तरह दक्षिणोत्तर दिशाओं का भी ज्ञान हुआ इति ॥१॥

अत्र विशेषविचारः

पूर्वापरयोर्बिन्दू तुल्यच्छायाग्रयोर्दिगपराऽऽद्यः।
पूर्वाऽन्यः क्रान्तिवशात् तन्मध्याच्छङ्कुतलमितरे ॥१॥

तुल्यच्छायाग्रयोः पूर्वापरयोः पूर्वापरकपालयोः यौ बिन्दू भवतस्तत्राद्यः पूर्वकपालीयो बिन्दुः अपरा प्रतीची दिग् भवति। अन्यः पश्चिमकपालीयो बिन्दुः पूर्वा दिग् स्यात्। अयमेतदुक्तं भवति। समायां भुवि मध्याह्नच्छायातोऽधिक-

त्रिज्यामितेन कर्कटकेनैकं वृत्तं विलिखेत्। तद्वृत्तमध्ये एकं द्वादशांगुलाङ्कितं शंकुं स्थापयेत्। पूर्वकपाले वर्तमाने सवितरि यदा किल तच्छायाग्रं पश्चिमकपाले वृत्ते प्रविशति तद्बिन्दुमङ्कयेत्। असौ पश्चिमबिन्दुराद्यसंज्ञकः। एवं गच्छति भास्वति पश्चिमकपाले यदा किल शंकुच्छायाग्रं पुनर्निर्गच्छति तत्र खटिकयाऽन्यो बिन्दुः कार्यः। असौ पूर्वबिन्दुः। सोऽप्यङ्कयः। याम्योत्तरवृत्तात्पूर्वभागः पूर्वकपालः। याम्योत्तरमण्डलात्पश्चिमो गोलार्धः पश्चिमकपालसंज्ञकः स्यात्। तत्र पूर्वकपाले वर्तमाने सूर्ये द्वादशांगुलशंकुच्छायाग्रं पश्चिमकपाले वृत्तपरिधौ लगति। एवं पश्चिमकपालस्थिते सूर्ये तच्छायाग्रं पूर्वकपाले वृत्तपरिधौ भिनत्ति। तत्र प्रथमो बिन्दुः पश्चिमदिक्। तथाऽपरो बिन्दुः पूर्वदिगिति। बिन्दुद्वयबद्धा रेखा पूर्वापरा भवति। तत्र मत्स्योत्पादनेन तद्रेखाया मध्यबिन्दुर्ज्ञातव्यः। तस्माच्छंकुमूलावगाहिनी रेखा इतरा याम्योत्तरा भवतीत्यर्थः

स्यादेतत्। यद्येकस्मिन् दिने रवेः क्रान्तिगतिः स्थिरा भवति नान्यथा। अतः पूर्वापरादिक् क्रान्तिवशात्परीक्षणीया। 'क्रान्तिवशा'' दित्यनेन ध्वन्यते यच्छायाप्रवेशकालिकः सूर्यश्छायानिर्गमकालपर्यन्तमेकस्मिन्नेवाहोरात्रवृत्ते भ्रमति तदैव यथोक्तानयनं घटते। अन्यथा विभिद्यमानायां क्रान्तिगतौ प्रवेशनिर्गमछायाग्रबिन्दुबद्धरेखा प्राच्यपरा न भवति। छायाग्रभुजयोरतुल्यत्वात्। तत्र भुजान्तरसंस्कारः कर्त्तव्य इति आचार्याशयः स्पष्टमवगम्यते।

अत्र भाष्यकर्त्ता चतुर्वेदाचार्यः क्रान्त्यन्तरवशेन कर्णगोलीयाग्रान्तरं समानीयाचार्योक्तं दिगन्तरं निरन्तरीकृतम्। इदमेवानयनं मनसि सन्निधाय भास्कराचार्या अपि "तत्कालापमजीवयोस्तु विवराद्भाकर्णमित्याहतात्लम्बज्याप्तमितांगुलैरयनदिश्येन्द्रीस्फुटा चालिते" ति विलक्षणं दिक्साधनं प्रोचुः। एवमेव "छायानिर्गमनप्रवेशसमयार्कक्रान्तिजीवान्तर" मित्यादि विधानेन श्रीपतिनापि दिक्साधनं स्फुटीकृतम्।

अत्रोपपत्तिः सुगमापि बालावबोधार्थमुच्यते। कल्प्यते छायाप्रवेशे रवेः क्रान्तिः=क्रा, छायानिर्गमे तस्य क्रान्तिः=क्रा'

ततोऽनुपातेन

$$\text{अग्रा}=\frac{\text{त्रि. ज्याक्रा}}{\text{ज्यालं}},\quad \text{अग्रा}'=\frac{\text{त्रि. ज्याक्रा}'}{\text{ज्यालं}}$$

ततः कर्णगोलीयाग्रां विधाय छायाग्रीयो भुजः साध्यते।

∴ प्रथमछायाग्रीयो भुजः=कर्णगोलीयाग्रा−त्रि।

$$=\frac{\text{ज्याक्रा. छाक}}{\text{ज्यालं}}-\text{त्रि}।$$

एवं द्वितीयो भुजः =कर्णगोलीयाग्रा–वि

$$= \frac{\text{ज्याक्रा.छाक}}{\text{ज्यालं}}$$

द्वयोरन्तरेण

भुग्रं $= \frac{\text{छाक}}{\text{ज्यालं}} \left(\text{ज्याक्रा}' - \text{ज्याक्रा}\right)$ अनेनान्तरेण प्राची चलिता भवतीति आचार्याः विदन्ति स्म। सूर्यसिद्धान्तेऽपि "शिलातलेऽम्बुसंशुद्धे वज्रलेपेऽपि वा समे" इत्यादिनाचार्यमतमेवाङ्गीकृतं भवेत्।

इदानीं भाभ्रमरेखावशेन दिग्ज्ञानमाह

त्रिच्छायाग्रजमत्स्यद्वयमध्यगसूत्रयोर्युतिर्यत्र।
सोत्तरगोले याम्या शङ्कुतलाद्दक्षिणे सौम्या ॥२॥
छायाग्रभ्रमरेखासूत्रयुतेर्वृत्तपरिधिरग्रस्पृक्।
मध्यच्छायान्तरमुदगितरद्वा शङ्कुमण्डलयोः ॥३॥

वा. भा.—छायात्रयस्तस्मादग्रानीतैस्तन्मत्स्यद्वयमुत्पद्यते, त्रिच्छायाग्रजमत्स्यद्वयमित्यस्यायमर्थः स्पष्टतरो व्याख्यायते। सलिलकृतायामवनौ यथेष्टप्रमाणं शंकुं विन्यसेत्, तत एककपालस्थे सवितरि छायाग्रेष्वभीष्टेषु त्रिषु त्रयो बिंदवः कार्याः। तत एकबिन्दुं मध्ये कृत्वा इष्टप्रमाणकर्कटकेन वृत्तमालिखेत्। तेनैव कर्कटकेन द्वितीयं बिन्दुमध्ये द्वितीयं वृत्तमालिखेत्। तृतीयमपि बिन्दुमध्ये तृतीयमालिखेत्। वृत्तं तावत् प्रमाणमेव तथा चालिखेद्यथा मत्स्यद्वयमुत्पन्नं प्रतिभाति। तयोश्च यस्यां दिशि महदंतरं ते मुखे यस्यां च सन्निकर्षः ते पुच्छे ततो मुखयोः सूक्ष्मकीलकौ विन्यस्य तयोः सूत्रे बद्ध्वा पुच्छमध्यगतैः सूत्रैः स्वगत्यैकचक्रमुत्पादयेत्। तयोश्च सूत्रयोः स्वमुखपुच्छगत्यनुसारेणागतयोरर्कसम्पातः सा दक्षिणा दिग्भवति। शंकुतलादुत्तरगोलस्था ऽर्कोऽथ दक्षिणगोले तु तुलादौ वर्तते। तदा मध्यगसूत्रयोर्युतिबिन्दुरुत्तरा दिग्भवति शंकुतलादेव। एवं शंकुमूलयुतिबिन्द्ववगाहिसूत्रं प्रसार्य रेखां कुर्यात्, सा दक्षिणोत्तरा दिग् भवति। तत एकैकं बिन्दुमध्ये कृत्वा वृत्तद्वयेन मत्स्यद्वयमुत्पादयेत्। तस्य मुखपुच्छावगाहिसूत्रं पूर्वापरा दिग्भवति। अथ भिन्नकपालजं बिन्दुत्रयं भवति। तदेकैकं बिन्दुमध्ये कृत्वा तथा वृत्तान्यालिखेत् यथैकैकं वृत्तं बिन्दुत्रयमपि परिवेष्टयति शेषं पूर्ववत्। एवं साक्षे देशे दिक्साधनं निरक्षदेशे पुनर्यो बिन्दुपातपंक्तिः सैव पूर्वापरा ततश्च दक्षिणोत्तरा प्राग्वत्साध्येत्यत्र वासना। अभीष्टव्यासार्धेन नववृत्तमालिख्य दिक्चिह्नितं कृत्वा प्रदर्श्य तद्यथा स्वदेश-

पूर्वापररेखातो यावत्यन्तरे उत्तरेणार्कोदयः पूर्वस्यां दिशि भवति तावत्येवान्तरे परेण दक्षिणं तच्छायाग्रं भवति। वृत्तमध्यसंस्थितस्य शंकोः ततः स्वाहोरात्रवृत्तगत्या यथाऽर्को दक्षिणे नीयात् तत्क्षवशादेव छायावृत्तमुदग्याति वैपरीत्यात्ततो यदार्कः सममंडलरेखायां भवति, पूर्वेण तदा छायाग्रं परेण सममण्डलरेखायां भवति। यदार्को याम्योत्तररेखागतो दक्षिणेन तदा छायाग्रमपि पूर्वेण सममण्डलरेखायामेव। एवं क्रमेणोत्तरतः पुनरपरस्यां दिशि यावत्यस्तमयः पूर्वस्यां दिशि तद्वत्तावति दक्षिणेन छायाग्रास्तमयः एवं सममण्डलदक्षिणगेऽर्के यदा पुनः सममण्डलं न विशति। तदा याम्योत्तररेखयार्क उत्तरेण यदा भवति तदा छायादक्षिणेन तत्रैव रेखायां भवति। एतच्च भागचतुर्विंशतेर्नूनो यत्राक्षस्तत्र संभवति। एवमुत्तरगोलेऽवश्यं छायाग्रभ्रमवृत्तखंडं दक्षिणाभिमुखं संभवति। दक्षिणगोले चोत्तराभिमुखं सर्वदा वासना वैपरीत्यात्। अतश्छायाग्रगवशं छायाग्रभ्रमवृत्तपृष्ठगानि भवन्ति। तैश्च मत्स्यद्वयमुत्पद्यते। तस्य छायाभ्रमवृत्तपृष्ठगानि भवन्ति। तैश्च यन्मत्स्यद्वयमुत्पद्यते तस्य छायाग्रभ्रमवृत्ताद्बहिर्मुखे वृत्तान्तः पुच्छे। अतो वृत्तमध्ये सूत्रद्वययुतिः परिध्यावेष्टनवशात्, सा च दक्षिणाभिमुखे छायाभ्रमवृत्तखंडे दक्षिणेन यास्यतीति कृत्वोक्तं सोत्तरगोले याम्यः, शंकुतलादिति। यतः प्राच्यपरायां शंकुरुत्तराभिमुखे चोत्तरेण, अतएवोक्तं दक्षिणे सौम्येति त्रिछायाग्रज्ञो यः क्रान्त्यक्षांशैर्विना भ्रमणं छायाग्रहो वेति मध्याह्नछायां वेति अस्य प्रश्नद्वयस्योत्तरमाह।

छायाग्रस्योक्तयो वृत्तपरिधिसूत्रयुतेः सैव छायाग्रभ्रमरेखा एतदुक्तं भवति। त्रिछायाग्रजमत्स्यद्वयमध्यगसूत्रयोर्यः संपातः प्रागार्याप्रदर्शितस्तन्मध्ये कृत्वा तच्छायाग्रगतशिरःस्पृक् यद्वृत्तं भवति तत्र दिग्मध्यस्थितस्य शंकोः छायाग्रं भ्रमति वासना चास्माभिः प्रागार्यायामत्रैवोक्ता। मध्यछायान्तरमित्यत आह। शंकुमंडलयोरिति यत्र छाया भ्रममण्डलं यच्च शंकुमूलं तयोर्यदन्तरमित्यर्थः, तच्चान्तरमुदगितरं मध्यछायाप्रमाणं याम्योत्तररेखायामन्तरं गृह्यते इत्यर्थः।

अत्रापि वासना प्रागार्यायामेव प्रदर्शिता। छायाभ्रमपरिधेः संस्थानप्रदर्शनात्तथा युज्यते, इति यच्छायाग्रं दृष्ट्वा क्रान्त्यक्षज्ञो दिशो विजानातीत्यस्योत्तरमार्याद्वयेनाह ॥२·३॥

वि. भा.—इष्टदिने दिग्मध्यस्वशङ्कोर्भिन्नकालजातं छायात्रयं ज्ञात्वा तदग्रबिन्दुत्रयैरिष्टप्रमाणेन कर्कटकेन वृत्तत्रयं विलिख्य तद्योगेन मत्स्यद्वयमुत्पाद्य तन्मुखपुच्छमध्यगतरेखयोर्यत्र युतिः सोत्तरगोले याम्या (दक्षिणा) दिग् ज्ञेया यदि जिनाल्पाक्षे देशे कदाचिच्छङ्कुमूलाद्दक्षिणे छायाग्रे सा युतिर्भवति तदा सा सौम्या (उत्तरा) दिग् ज्ञेया, सूत्रयुतेर्मत्स्यद्वयमुखपुच्छनिर्गतसूत्रयुतेर्यो वृत्तपरिधिः सोऽग्रस्पृक् (छायाग्रस्पर्शकारि) भवति, अतः परिधिरेखैव छायाभ्रमणरेखा भवति, शङ्कुमण्डलयो (शङ्कुमूल-छायाभ्रमणवृत्तयोः) र्यदन्तरं सैव मध्यच्छाया भवति सोदग् (उत्तरा) वेतरद् (दक्षिणा) भवति, जिनाधिकाक्षे देशे मध्यच्छाया सदोत्तरा

भवति, उत्तरगोले जिनाल्पाक्षे देशे यदा रवेरुत्तरा क्रान्तिरक्षांशाधिका तदा मध्याह्नकाले शङ्कुच्छाया दक्षिणाभिमुखी भवतीति, सूर्यसिद्धान्तेऽप्येव "इष्टेऽह्नि मध्ये प्राक् पश्चाद् धृते बाहुत्रयान्तरे मत्स्यद्वयान्तरयुतेस्त्रिस्पृक्सूत्रेण भाभ्रमः" मेवास्ति, "यो मत्स्यपुच्छमुखनिर्गतरज्जुयोगस्तस्मात् प्रभात्रितयचिह्नशिरोऽवगाहि। वृत्तं लिखेन्न विजहाति हि तस्य रेखां छाये' त्यनेन लल्लाचार्येण, "मत्स्योदरद्वयगसूत्रयुतेश्च तस्या भागत्रयं स्पृशति यद् भवतीहवृत्तम्। छाया न तत्परिधिमुज्झति मध्यशङ्कोः" अनेन शङ्कुप्रभाभ्रमणमण्डलयोस्तु मध्यं मध्यप्रभावति दक्षिणमुत्तरं वेत्यनेन च श्रीपतिनाप्याचार्योक्तानुरूपमेव कथ्यते। इति ॥२-३॥

अत्रोपपत्तिः

एकस्मिन् दिने यदि रवेः क्रान्तिः स्थिरा भवेत्तदाऽहोरात्रवृत्तीयप्रतिबिन्दुस्थरविकेन्द्रतः किरणसूत्राणि शङ्क्वग्रगतानि यत्र यत्र पृष्ठक्षितिजधरातले लगन्ति तेभ्यः शङ्कुमूलं यावत् छायाः, छायास्वरूपदर्शनेन सिद्धं यच्छङ्क्वग्रादहोरात्रवृत्ताधारा सूची कार्या सा पृष्ठक्षितिजधरातलेन छिन्ना सती यादृशं वक्रमुत्पादयति तादृश एवच्छायाभ्रमणमार्गः। मेरौ क्षितिजवृत्तं नाडीवृत्तम् तदहोरात्रवृत्तसमानान्तरमतः शङ्क्वग्रादहोरात्रवृत्ताधारा विषमा सूची पृष्ठक्षितिजधरातलेन (नाडीवृत्तधरातलसमानान्तरधरातलेन) छिन्ना सती छेदितप्रदेशं वृत्ताकारमुत्पादयत्यतः सिद्धं यन्मेरौ सर्वदैवच्छायाभ्रमणमार्गो वृत्ताकारो भवेत्, सूर्यसिद्धान्तकार--लल्लाचार्य--ब्रह्मगुप्तप्रभृतिभिराचार्यैर्वृत्ते सदा छायाभ्रमणं यत्स्वीकृतं तन्मेरावेव समीचीनं भवितुमर्हति, यतोऽन्यत्र साक्षे देशे न्यूनाधिकशङ्कुवशेन वृत्ते, रेखायां, परवलये, दीर्घवृत्ते, अतिपरवलये च छायाभ्रमणं भवति, निरक्ष देशे विषुवद्दिने नाडीवृत्ते रवेर्भ्रमणाच्छङ्क्वग्रस्य नाडीवृत्तधरातले स्थितत्वाच्छङ्क्वग्रान्नाडीवृत्ताधारसूच्यभावस्तेन निरक्षक्षितिजधरातलनाडीवृत्तधरातलयोर्योगरेखा (निरक्षोर्ध्वाधररेखा) भाभ्रमरेखा भवेत्। सर्वत्र सदा छायाभ्रमणं वृत्ते न भवतीति दृष्ट्वैव भास्कराचार्येण सिद्धान्तशिरोमणौ 'भात्रितयाद् भाभ्रमणं न सदित्यादिना' वृत्ताकारस्यच्छायाभ्रमणमार्गस्य खण्डनं कृतमिति ॥२-३॥

अब भाभ्रमरेखा वश से दिग्ज्ञान को कहते हैं

हि. भा.—इष्ट दिन में जलादि से समान की हुई पृथ्वी पर मध्यच्छाया व्यासार्ध से लिखित वृत्त के केन्द्र में स्थित शङ्कु की तीन काल की छाया जानकर उन तीनों के अग्र बिन्दुओं से इष्ट कर्कट से तीन वृत्त बनाकर उनके योग से दो मत्स्य (मछली के आकार) बनाकर उनके मुख और पुच्छ के मध्यगत रेखाद्वय का योग जहां होता है उसके उत्तर गोल में दक्षिण दिशा समझनी चाहिये, यदि जिनाल्पाक्षांश देश (चौबीश अंश से कम अक्षांश वाला देश) में कभी शङ्कु मूल से दक्षिण छायाग्र में वह योग हो तब वह उत्तर दिशा समझनी

चाहिये, मत्स्यद्वय के मुख और पुच्छगत सूत्रों के योग से जो वृत्त परिधि होती है, वह छायाग्र-स्पर्शकारक वृत्त (छायाभ्रमणवृत्त) होता है, इसलिये वह वृत्तरेखा ही छायाभ्रमण रेखा होती है, शङ्कु मूल और छायाभ्रमण वृत्त का जो अन्तर है, वही मध्यच्छाया होती है, वह उत्तर या दक्षिण होती है, जिनाधिकाक्षांश देश (चौबीस अंश से अधिक अक्षांश वाला देश) में मध्यच्छाया सदा उत्तर होती है, उत्तरगोल में जिनाल्पाक्षांश देश में जब रवि की उत्तरा क्रान्ति अक्षांशाधिक होती है तब मध्याह्नकाल में शङ्कु की छाया दक्षिणाभिमुखी होती है, सूर्यसिद्धान्त में भी 'इष्टेऽह्नि मध्ये प्राक् पश्चात् इत्यादि से' इसी तरह कहा गया है, लल्लाचार्य 'यो मत्स्यपुच्छमुखनिर्गतरज्जुयोगः' इत्यादि संस्कृत-भाष्य में लिखित श्लोक से तथा 'मत्स्योदरद्वयगसूत्रयुतेश्च तस्याः' इत्यादि से तथा 'शङ्कुप्रभाभ्रमणमण्डलयोः' इत्यादि से भी, श्रीपति भी आचार्योक्तानुरूप ही कहते हैं इति ॥ २-३ ॥

उपपत्ति

यदि एक दिन में रवि की क्रान्ति स्थिर हो तब अहोरात्र वृत्त के प्रतिबिन्दुस्थ रवि केन्द्रों से शङ्कुस्थगत किरणसूत्र (शङ्कुस्थगत रेखायें) जहाँ-जहाँ पृष्ठक्षितिज धरातल में लगते हैं उन स्थानों से शङ्कुमूल तक छाया है, छायाग्रों के स्वरूप देखने से सिद्ध होता है कि शङ्कु के अग्र से अहोरात्रवृत्त के आधार पर सूची बनाइये उसको पृष्ठक्षितिज धरातल से काटने से जैसा वक्र बनता है वैसा ही छायाभ्रमण मार्ग होता है, मेरु वासियों का क्षितिजवृत्त नाड़ीवृत्त है, वह अहोरात्रवृत्त के समानान्तर है इसलिये शङ्कुस्थ से अहोरात्रवृत्ताधारा विषमसूची पृष्ठक्षितिजधरातल (नाड़ीवृत्त धरातल के समानान्तर धरातल) से कटित होकर कटित प्रदेश को वृत्ताकार बनाती है, इससे सिद्ध होता है कि मेरु में सदा छायाभ्रमणमार्ग वृत्ताकार होता है, सूर्यसिद्धान्तकार, लल्लाचार्य, ब्रह्मगुप्त आदि आचार्यों ने छायाभ्रमणमार्ग वृत्ताकार जो स्वीकार किया है वह मेरु ही में ठीक हो सकता है, क्योंकि मेरु से अन्यत्र साक्ष देश में न्यूनाधिक शङ्कु वश से रेखा में, वृत्त में, परवलय में, दीर्घवृत्त में अतिपरवलय में छायाभ्रमण होता है, निरक्ष देश में विषुवद्दिन में नाड़ीवृत्त में रवि के भ्रमण से और नाड़ीवृत्त धरातल में शङ्कुस्थ के रहने के कारण शङ्कुस्थ से नाड़ीवृत्त रूप अहोरात्रवृत्ताधार सूची का अभाव होता है इसलिये निरक्षक्षितिज धरातल और नाड़ीवृत्त धरातल की बोधरेखा (निरक्षोर्ध्वाधर रेखा) भाभ्रमरेखा होती है, सब देशों में सदा छायाभ्रमण वृत्त में नहीं होता है इस विषय को देख करके ही सिद्धान्त-शिरोमणि में भास्कराचार्य ने 'भात्रितयाद् भाभ्रमलं न सदृ' इत्यादि से वृत्ताकार छायाभ्रमण मार्ग का खण्डन किया है, जो बहुत ही ठीक है इति ॥२-३॥

इदानीं द्वादशाङ्गुलशङ्कुषु जानयनमाह

छायावृत्तेऽर्कान्ना कर्णभुजा व्यासदलसंहताऽर्कान्ना।
भवति च्छाया याम्या तदन्तरैक्यं भुजोऽत्याग्रे ॥८॥

शङ्कुः प्राच्यपरायाश्छाया भुजकृतिविशेषमूलं यत् ।
तत् प्राच्यपरा छाया भुजाग्रयोरन्तरं कोटिः ॥५॥

वा. भा.—क्रान्त्यक्षांशैर्विनापि विन्दुत्रयेण प्राग्दिक्साधनमुक्तमधुना तज्जस्य विन्दुनैकेन दिक्साधनार्थमिदमार्याद्वयमन्तेनायमर्थः छायावृत्तेऽर्काग्रा कथं भवतीत्याहः । कर्णगुणा व्यासदलहृताऽर्काग्रा । कक्षामानविधिनाऽर्काग्रां कृत्वा तया सह त्रैराशिकमिदं यदि व्यासार्धवृत्ते एतावत्यर्काग्रा तदिष्टछायाकर्णवृत्ते कियतीति फलं छायावृत्तार्काग्रांगुलरूपा । ततः स्वदेशं विषुवच्छायांगुलरूपया द्वादशांगुलशंकोःसंबन्धिन्या सह तदन्तरैक्यं ययासंख्यसौम्योत्तरगोलयोः स्थितेऽर्के भुजो भवति ।

अस्याग्रे शंकुः कुत इत्यत आह । प्राच्यपरायणा इति । अस्य भुजाग्रे यः शंकुस्तस्य शंकोश्छाया तस्य कृतिः भुजकृतिश्च तयो कृत्योः मूलं यत्तदेव मूल प्राच्यपराकोटिर्भवति । क्वेत्याह—छायाभुजाग्रयोरन्तरे कोटिरिति । एतदुक्तं भवति । समभूप्रदेशस्थितस्य शंकोश्छाया सा कर्णः यश्चोक्तवद्भुज आनीतः स शंकुमूलार्द्धं परीत्येन दातव्यः येन भुजाग्रे शंकुर्भवति । यदि दक्षिणोत्तरेणोत्तरश्च दक्षिणेनेत्यर्थः स भुजः । ततो भुजकृतिः कर्णकृतेर्विशोध्य कोटिकृतिरेखावशिष्यते तस्या मूलं कोटिः सा च प्राच्यपरस्थिता भवति । अत्र वासना इष्टच्छायाकर्णेन व्यासार्धकल्पितेन वृत्तं समालिख्य दिगंकितं कृत्वा तक्षीयार्काग्रामितं सूत्रं पूर्वतोऽपरश्च दत्तोत्तरेण याम्येन वा तदुदयास्तसूत्रं तत्र वृत्तपरिणतं दत्वा याम्योत्तररेखायां तदग्रे विन्दुः कार्यं अर्काग्रयोश्च विन्दुं कृत्वा ततो विन्दुत्रयेण मत्स्यद्वयमुत्पाद्य तन्मुखपृच्छविनिर्गतसूत्रयुतौ विन्दुः कार्यस्तं विन्दुमध्ये विन्दुत्रयमपि स्थापयेद्वा वृत्तमुत्पद्यते । तच्चाहोरात्रवृत्तभूमौ बृहच्छंकुमूलभ्रम संभवति ।

भुजश्च मूलप्राच्यरेखयोरन्तरमुच्यते । तेनार्कोदयकालेऽस्तमयकाले वार्काग्रातुल्या एव भुजा भवति । सर्वत्र साक्षे देशे निरक्षे च ततो निरक्षे दिनमेव सकलाग्रा भुजा यस्य ततस्तत्र शंकुमूलमुदयास्तसूत्रं न त्यजति । साक्षे चाक्षवशात् । तिर्यक्त्वं विषुवन्मंडलस्य तद्वशादिष्टस्वाहोरात्रस्य च तेनोत्तरगोलेनोनार्काग्रा भुजा भवति । यदि प्राच्यपराया उत्तरेण शंकुः अथ दक्षिणेन तदार्काग्रतोऽधिकत्वाच्छंकुतलेऽग्रं व तत्र शोध्यते तथापि तयोरन्तरमेव छायावृत्ते च विषुवच्छायैव सर्वदा शंकुतलं भवति । अतस्तया सहान्तरमुक्तं दक्षिणगोले च सर्वदा प्राच्यपरा या दक्षिणेनार्काग्रा तुल्येऽन्तरे उदयास्तसूत्रेखान्तरज्यातः दक्षिणेन शंकुतलतुल्येऽन्तरे शंकुमूलं भवति । अतः सर्वदा तयोर्योगो विषुवच्छायाग्रभुजा भवति । भुजश्च यदोत्तरे भवति तदावश्यं प्राच्यपराऽन्यथातस्योत्तरत्वमेव शक्यते च क्रमतः शंकुमूलादेव क्षेत्रोत्थापनं विपरीतं भुजदाने क्रियते च भुजतः । अतः शंकुमूलाद् वृत्तमध्यविन्दुन्तरच्छायार्काः । तथा च शंकुमूलप्राच्यपरान्तरं भुजकृतिकर्णकृति-

विशेष मूलं प्राच्यपरा कोटिदिग्ग्रहणमप्युपपन्नं चैतत् सर्वं गोले प्रदर्शयेदिति। शंकुछायाभ्रमणो दिग्जो वा वेत्तीत्यस्योत्तरमाह ॥४-५॥

वि. भा.—अर्काग्रा छायाकर्णगुणा, व्यासदलहृता (त्रिज्या भक्ता) तदा छायावृत्तेऽग्रा (कर्णवृत्ताग्रा) भवेत्। विषुवच्छाया (पलभा) सदा याम्या (दक्षिणा) भवेत्, तदन्तरैक्यं (कर्णवृत्ताग्रा पलभयोर्भिन्नदिशोरन्तरमेकदिशोर्योगः) तदा संस्कारदिक्को भुजो भवतीति ॥४॥

अत्रोपपत्तिः

स्वोदयास्तसूत्रपूर्वापरसूत्रयोरन्तरमग्रा, शङ्कुमूलात्स्वोदयास्तसूत्रोपरिलम्बः शङ्कुतलम्। अग्राशङ्कुतलयोः संस्कारेण शङ्कुमूलात्पूर्वापरसूत्रोपरिलम्बो भुजो भवति, त्रिज्याव्यासार्धे यदि त्रिज्याग्रयाऽग्रा लभ्यते तदा छायाकर्णे किं समागच्छति छायाकर्णवृत्ताग्रा $=\frac{\text{अग्रा}\times\text{छायाकर्ण}}{\text{त्रि}}$, छायाकर्णगोले शङ्कुतलं पलभातुल्यं भवति कथमिति प्रदर्श्यते अथ $\frac{\text{पलभा}\cdot\text{शङ्कु}}{१२}$ शङ्कुतल, परन्तु, $\frac{१२\times\text{त्रि}}{\text{छाकर्ण}}$ शङ्कु उत्थापनेन $\frac{\text{पलभा}\cdot१२\cdot\text{त्रि}}{१२\times\text{छाक}}$=शङ्कुतल, छायाकर्णव्यासार्धे परिणाम्यते $\frac{\text{पलभा}\cdot१२\cdot\text{त्रि}\times\text{छाक}}{१२\times\text{छाक}\times\text{त्रि}}$=पलभा=शङ्कुतलम्। अतश्छायाकर्णगोलेऽग्रापलभयोः संस्कारेण भुजो भवेत्। यतः अग्रा ± शङ्कुतल=भुजः। भास्कराचार्येण छायाग्रे पूर्वापररेखयोरन्तरं भुज इति स्वीकृत्य छायाकर्णवृत्ताग्रा व्यस्तगोला पलभा चोत्तरा कल्पिता, आचार्येण लघुशङ्कुमूलपूर्वापररेखयोरन्तरं भुज इति स्वीकृत्याग्रा पलभे यथा दिक्के एव स्थापिते, भुजस्याग्रे इत्यस्याग्रे सम्बन्ध इति ॥४॥

अब द्वादशाङ्गुल शङ्कु के भुजानयन को कहते हैं

हि. भा.—रवि की अग्रा को छाया कर्ण से गुणा कर त्रिज्या से भाग देने से छायावृत्त में रवि की अग्रा होती है, पलभा सर्वदा दक्षिण दिशा की होती है, उन दोनों का अन्तर और योग (कर्णवृत्ताग्रा और पलभा के भिन्न दिशा में अन्तर और एक दिशा में योग) करने से संस्कार दिशा का भुज होता है इति ॥४॥

उपपत्ति

स्वोदयास्तसूत्र और पूर्वापरसूत्र का अन्तर अग्रा है, शङ्कुमूल से पूर्वापरसूत्र के

ऊपर लम्ब शङ्कुतल है, अग्रा और शङ्कुतल का संस्कार करने से शङ्कुमूल से पूर्वापरसूत्र के ऊपर लम्ब भुज होता है, त्रिज्या व्यासार्ध में यदि त्रिज्याग्रीय अग्रा पाते हैं तो छायाकर्ण में क्या इससे आती है छायाकर्णवृत्ताग्रा $=\frac{\text{अग्रा}\times\text{छाकर्ण}}{\text{त्रि}}$, परन्तु छायाकर्ण गोल में पलभा शङ्कुतल के बराबर होती है जैसे $\frac{\text{पलभा. शङ्कु}}{१२}=$शङ्कुतल, $\because \frac{१२\times\text{त्रि}}{\text{छाकर्ण}}=$शङ्कु

अतः उत्थापन से $\frac{\text{पलभा. }१२\times\text{त्रि}}{१२.\text{ छाक}}=$शङ्कुतल, छायाकर्ण गोल में परिणामन करते हैं

$\frac{\text{पलभा. }१२.\text{ त्रि}\times\text{छाकर्ण}}{१२.\text{ छाकर्ण. त्रि}}=$पलभा$=$शङ्कुतल, इसलिए छायाकर्ण गोल में अग्रा और पलभा के संस्कार से भुज होता है, भास्कराचार्य ने छायाग्र और पूर्वापररेखा के अन्तर को भुज स्वीकार कर छायाकर्ण वृत्ताग्रा को व्यस्त गोलक और पलभा को उत्तर कल्पित किया है यहां आचार्य लघुशङ्कु मूल और पूर्वापररेखा के अन्तर को भुज स्वीकार कर अग्रा और पलभा को यथादिक्क (जिस दिशा के जो हैं उसी दिशा के) ही स्थापित किये हैं, भुजस्याग्रे इसका आगे से सम्बन्ध है इति ॥४॥

वि. भा.—प्राच्यपरायाः (पूर्वपरायाः) सकाशाद्यथा दिग्गतस्य भुजस्याग्र शङ्कुमूलं बोध्यम् । छायाभुजकृतिविशेषमूलं (छायाभुजयोर्वर्गान्तरमूलं) यद् भवेत्तदेव छायाभुजाग्रयोरन्तरं प्राच्यपरा (पूर्वापरा) कोटिर्भवेदिति ॥५॥

अत्रोपपत्तिः

शङ्कुमूलात्पूर्वापररेखोपरिलम्बो भुज इत्येव दिग्मध्यगतस्य शङ्कोश्छायाग्र-पूर्वापररेखयोरन्तरम्, दिग्मध्यगतशङ्कुच्छायाकर्णः, छायाग्रात्पूर्वापररेखोपरि-लम्बो भुजः । भुजाग्राद्दिग्मध्यं (वृत्तकेन्द्रं) यावत्पूर्वापररेखायां कोटिः, ततः $\sqrt{\text{छाया}^2-\text{भुज}^2}=$कोटिः, भास्कराचार्येणापि सिद्धान्तशिरोमणौ "दिक्सूत्रसम्पातगतस्य शङ्कोश्छायाग्रपूर्वापरसूत्रमध्यमित्यादिना" तदेव कथ्यत इति ॥५॥

अब कोटि साधन को कहते हैं

हि. भा.—पूर्वापररेखा से भुजाग्र में शङ्कुमूल समझना चाहिये । छाया और भुज का वर्गान्तरमूल जो होता है वही छायाग्र और भुजाग्र का अन्तर पूर्वापरानकांर कोटि होती है ॥५॥

उपपत्ति

शङ्कुमूल से पूर्वापररेखा के ऊपर लम्ब रेखा भुज है यही दिग्मध्य (वृत्तकेन्द्र) गतशङ्कु के छायाग्र और पूर्वापर रेखा का अन्तर है, दिग्मध्यगतशङ्कुच्छाया कर्ण, छायाग्र से

पूर्वापररेखा के ऊपर लम्बभुज, भुजाग्र से दिग्मध्यपर्यन्त पूर्वापर रेखा में कोटि, इस जात्य त्रिभुज में $\sqrt{\text{छाया}^2 - \text{भुज}^2}$ = कोटि, भास्कराचार्य ने भी सिद्धान्तशिरोमणि में "दिक्सूत्रसम्पातगतस्य शङ्कोश्छायाग्रपूर्वापरसूत्रमध्यम्" इससे इसी बात को कहते हैं इति ॥ ५ ॥

इदानीं शङ्कुच्छायाग्रयोः स्थितिमाह

दिङ्मध्ये छायाग्रं कृत्वा शङ्कोर्यथादिशं भ्रमणम् ।
दिङ्मध्यस्थितशङ्कोश्छायाग्रं भ्रमति विपरीतम् ॥६॥

वा. भा.—इष्टछायाग्रबिन्दुं कृत्वा यः पृच्छति, क्वस्थितस्य शंकोश्छायाग्रमत्र पततीत्यस्य प्रथमार्यार्धेनोत्तरं तद्यथा दिङ्मध्ये छायाग्रं कृतमिति । प्राग्वदिङ्साधनं कृत्वा भुजज्या कोटि तात्कालिकछायाकर्णौ स्थिते एव ततः प्रश्नछायाग्रपूर्वापररेखायां विन्यासे तु तदग्रात् पूर्वापरायामेव रेखायां पूर्वकपालस्थेऽर्के पूर्वाभिमुखं कोट्यंगुलमितं सूत्रं प्रसारयेत् । अपरकपालस्थे यापराभिमुखं कोट्यन्ताङ्गुलांगुलमितसूत्रमुत्तरेण देयम् । दक्षिणे दक्षिणेन शंकुं तदग्रे विन्यसेत् तत्र स्थितस्य शंकोः प्रश्नछायाः भवन्ति । एवं च शंकोर्यथादिशं भ्रमणं भवेत् । अन्यथेयं वासना पूर्वाह्णेऽवश्यमेव छाययापराभिमुखी भवितव्या ।

तदग्रं च यथा दिग्मध्ये क्रियते तथा दिग्मध्यात्पूर्वेणावश्यं शंकुरिति कृत्वा पूर्वाभिमुखी कोटिः प्रसार्यते । शंक्वोर्यथादिशं भ्रमणमित्यत उक्तम् । कोट्य-शंक्वग्राच्च भुजान्तरे वा दक्षिणे वा दीयते । यतः सममंडलरेखाया उत्तरेण दक्षिणेन वारविः शंकुश्च तंत्रैव तस्य शंक्वोश्छायाप्यपराभुखी दिग्मध्यबिंदुप्रापिणी च युज्यते, पराह्णे च सर्वं वैपरीत्येनोपपद्यते एवेति । यस्त्वभीष्टे प्रदेशे शंकुं विन्यस्य छायांगुलान्युद्दिश्य च छायाग्रं पृच्छति तस्योत्तरं द्वितीयेनार्यार्धेन तद्यथा दिक्साधनं कृत्वा प्राक्कोटि भुजं च । ततो दिग्मध्ये शंकुं विन्यसेत्तन्मूलादपराभिमुखी कोटिः पूर्वाह्णे पराह्णे च पूर्वाभिमुखी देया । तदग्राच्च भुजो विपरीतो देयः, उत्तरो दक्षिणेन, दक्षिणश्चोत्तरेण, तदग्रे छायाग्रं भवति । अत उक्तं दिङ्मध्यस्थितशंकोश्छायाग्रं भ्रमति विपरीतम् । इत्यत्र वासना पूर्वापररेखायां स्थितस्य शंकोः पूर्वाह्णेऽपराभिमुखी छाया, अतएवापरेण कोटिः प्रसार्यते । यतश्छायाग्रमस्माभिस्त्वेष्टमारब्धमपराह्णे पूर्वाभिमुखी प्रसार्यते ।

तत्रापि छायाग्रमेवान्वेष्टव्यं भुजोपि वैपरीत्येन दीयते यतोप्यर्कः सममण्डलरेखाया उत्तरेण तदा छायाग्रं दक्षिणेन । दक्षिणेन यदार्कस्तदा छायाग्रमुत्तरेण भवति । तच्छायाग्रं वा विन्यसेत्तदार्कसममंडलस्थितो भवति । साक्षे देशे निरक्षे

तु यतः क्रान्तिज्यातुल्यः सर्वदार्काग्रा भुजो देय इत्येवमेकस्य शंकुतलस्य छाया-ग्रस्य चाभीष्टकाले साधनं यदा पुनः सकलदिनभ्रमणं साधयितुमिष्यते, तदा तदैव दिङ्मध्ये छायाग्रं कृत्वा शंकुमूलत्रयं साध्यं तदुत्पन्नमत्स्यद्वयमध्यगतसूत्र-युतेर्यो वृत्तपरिधिः शंकुमूलत्रयस्पृक् । शंकुभ्रमणवृत्तमेवं दिङ्मध्यस्थितस्यैव शंको-श्छायाग्रत्रयेण मत्स्यविधानेन प्रागवच्छायाग्रभ्रमणवृत्तसाधनं कार्यम् । वासना चात्र गतार्था चेति । दृष्ट्वा विषुवच्छायां लंबाक्षज्यां करोति यो बहुवेत्यस्य प्रश्नस्यो-त्तरं बहुप्रकारेण लंबाक्षज्ययोः स्वरूपप्रदर्शनमार्यापट्केनाह—

वि. भा.—पूर्वापररेखातः शंकुमूलं यद्दिक् भ्रमति ततो विलोमदिशि दिङ्-मध्यस्थितशङ्कोश्छायाग्रं भ्रमतीति॥६॥

उपपत्तिः

अत्रोपपत्तिः पूर्वश्लोकोपपत्तिपर्यालोचनया स्फुटेति॥६॥

अब शङ्कु और छायाग्र की स्थिति को कहते हैं

हि. भा.—पूर्वापररेखा से शङ्कु मूल जिस दिशा में भ्रमण करता है उससे विपरीत दिशा में दिङ्मध्य (छायावृत्त केन्द्र) स्थित शङ्कु का छायाग्र भ्रमण करता है इति॥६॥

उपपत्ति

पूर्वश्लोकोपपत्ति के विवेचन से स्पष्ट है ॥६॥

प्रथमं विषुवत्कर्णमुक्त्वा लम्बाक्षज्ययोरानयनमाह

शङ्कुर्लम्बश्छायाऽक्षज्या तद्वर्गसंयुतेर्मूलम् ।
विषुवति विषुवत्कर्णश्छायाकर्णोऽन्यदा शङ्कुः॥७॥
उन्नतजीवाकोटिश्छाया दृग्ज्या भुजो नतज्या वा ।
कर्णश्छायावृत्ते व्यासार्धं द्वयमतोऽन्यत्र ॥८॥
शङ्कुच्छायाकृत्योस्त्रिज्याकृतितत्समासगुणहृतयोः ।
मूले लम्बाक्षज्ये तद्वंशकास्तद्भुजोर्भागाः ॥९॥
विषुवत्कर्णहृते वा शङ्कुच्छायाहते पृथक् त्रिज्ये ।
अक्षज्येतरजीवे लम्बाक्षांशोत्क्रमज्योने ॥१०॥
नवतेर्लम्बाक्षांशान् प्रोह्य ज्या वेतराक्षलम्बज्ये ।
शङ्कुच्छायागुणिते छायाद्वादशहृते वाऽन्ये ॥११॥
लम्बाक्षज्यावर्गं प्रोह्य त्रिज्याकृतेः पदं वाऽज्या ।
अन्यत्र सर्वदोक्षतन्नतजीवांक्षानयनमेवम् ॥१२॥

वा. भा.—स्वदेशेऽक्षाग्रयोर्योगं विन्यस्य प्रदर्श्येत्तद्यथा समंडलाद् दक्षिणेन

याम्योत्तर मंडल विषुवन्मंडलसंपाते सूत्रस्यैकमग्रं बद्ध्वा सममंडलादुत्तरेण तावत्येवान्तरे तस्मिन्नेव याम्योत्तरमंडले द्वितीयमग्रं बध्नीयात्तद्यावदवतिष्ठते तदर्धसममंडल भूमध्यावगाहिसूत्रावच्छिन्नमक्षज्या तदग्रेऽवलंबकं बध्नीयात्। भूमध्यविनिर्गतदक्षिणोत्तरं यत् सूत्रं प्रापितम् । सा स्वदेशावलम्बज्या गोलाध्याय एवास्माभिरयमर्थो व्याख्यातः। तत्रावलम्बककोटिरक्षज्या भुजः। तद्गंसंयुतेः मूलं विषुवत्कर्णव्यासार्धं यतस्तत्र दिने विषुवन्मण्डलमेव स्वाहोरात्रवृत्तम्। व्यासवृत्तं च छायाकर्णेऽन्यत्र दिने ते मध्याह्नेपि न विषुवत्कर्णः दृङ्मंडलेषून्नतजीवैव, शंकुकोटिश्च सैव छायादृग्ज्योच्यते । शंक्वपेक्षया उन्नतजीवापेक्षया तज्ज्योच्यते-तत् कोट्यपेक्षया भुजज्योच्यते यतस्तत् दृङ्मंडलं ग्रहाभिमुखं भ्रमति। सममंडलोपर्यधः खस्वस्तिकं न त्यजति—इत्यस्माभिरयमर्थस्तस्यैव विन्यासे गोलाध्याये प्रपंचितः। तस्माच्छोभनमुक्तं शंकुरुन्नतजीवाकोटिः पर्याया छाया दृग्ज्या भुजज्या नतज्या च पर्याया एव कर्णस्तु छायावृत्तमुच्यते। छायावृत्तकर्णो द्वावपिव्यासार्धमुच्यते तत्र विषुवच्छायाश्च। ततस्त्रिज्याकृत्या द्वेऽपि गुणयेत्। ततः कृत्योर्योगेन स्थानद्वयेऽपि भागहारःकर्तव्यः। फलं शंकुकृतेः लंबज्या विषुवच्छाया कृतेरक्षज्या तर्योर्लबाक्षज्ययोर्वर्गनुषीकृत्वा तद्भागा पृथक्-पृथक् कार्याः इत्यत्र त्रैराशिक वासना। तद्यथा द्वादशकः शंकुः कोटिर्विषुवच्छाया भुजस्तयोर्वर्गसमासो विषुवत्कर्णवर्गस्तेन यदि शंकुकोटिवर्गो लभ्यते तदा व्यासार्धकर्णवर्गस्पृक् शंकुकोटिवर्ग इति फलं लंबज्यावर्गः । ततो विषवत्कर्णवर्गेण विषुवच्छाया भुजवर्गो लभ्यते। तदात्रिज्या कर्णवर्गकोटिवर्गयोरन्तरं भुजवर्ग इति ॥ फलमक्षज्या वर्गस्तयोर्मूले लंबाक्षज्ये, विषुवत्कर्णहृते वा शंकुछायाहते पृथक् त्रिज्ये। अत्रापि त्रैराशिकं यदि विषुवत्कर्णस्य द्वादश कोटिः तत् त्रिज्याकर्ण स्पृगिति फलं लम्बज्या। ततो विषुवत्कर्णस्य यदि विषुवच्छाया भुजः तत् त्रिज्या कर्णस्पृक् भुजेति फलमक्षज्या अतएव पृथक् त्रिज्ये द्वादश विषुवच्छायागुणैः कृत्वा विषुवत्कर्णं विभजेत्। उपपन्नं चैतदिति अथवाऽपरः प्रकारः त्रिज्येतर जीवा वालंबाक्षांशौ क्रमज्योनालंबांक्षांशानामुत्क्रमज्या यदा त्रिज्योना क्रियते। तदाक्षज्याभवति। यदा पुनरक्षांशानामुत्क्रमज्ययोना क्रियते। तदालंबज्या भवति। अत्रेयं वासना। दक्षिणक्षितिजाद्याम्योत्तर मंडलेयावतो लंबांशा उपरिस्थिताश्चापगत्या तेषां या क्रमज्या सा लंबज्या पूर्वमेव प्रदर्शिता। तं द्विगुणीकृत्योपरिस्थितस्य धनुषो यः शरः तावति लंबाशोत्क्रमज्या। सा च त्रिज्यातो यदा शोध्यते, तदाक्षज्या तुल्या ज्याखण्डमवशिष्यते। सममण्डलमध्यभूमध्यावगाहिना सूत्रेण तस्मादुपन्नम्। एकस्या ज्यायाः यावान् प्रकारः तयोरानयने नवतेर्यदाक्षांशाः शोध्यन्ते। नवतेस्तदाक्षांशान्विशोध्य लंबांशकाः अवशिष्यन्ते। तेषां या ज्या सा लंबज्या, यदा लंबांशकाः शोध्यन्ते नवतेः तदाक्षांशाः शेषाः भवन्ति, तेषां या ज्या साक्षज्या भवति । अक्षलम्बज्ये इत्युत्तरत्र संबंधो भविष्यतीति वासनात्र ।

मेरौ नवतिरक्षांशाः ध्रुवस्योपरि स्थितत्वाद्विषुवत्कर्णस्य क्षितिजासक्तत्वात्लंबांशकाभावः निरक्षदेशेऽक्षाभावः ध्रुवयोः क्षितिजासक्तत्वात् लंबकश्च नवतिर्भागा ध्रुवोपरि स्थितत्वादर्धान्तरे तु लंबाक्षयोगो नवतिर्भागाः तेन सममंडलमध्यदक्षिणेन याम्योत्तरमंडलगत्या क्षितिजस्वस्तिकं नवतिर्भागास्तेभ्यो यदाक्षांशाः शोध्यन्ते तदा लंबांशाः शेषा युज्यन्ते। यदा पुनरेव लंबांशाः शोध्यन्ते तदाक्षभागाः शेषा भवंति। अनुक्रमेण ज्ञायते स्वभागानां या ज्या तज्ज्या भवतीति, अत्रोच्यते। तस्मादुपपन्नम्। यथापि स्थितं क्षेत्रं गोले प्रदर्शयेत्। इत्यपरेण (प्रकारेण) प्रकारान्तरेणानयनम्। शंकुछायागुणिते छायाद्वादशहृते चान्ये। अक्षज्यां द्वादशहतां विषुवच्छाया विभजेत्। फलं लम्बज्या, विषुवच्छायया लंबज्या हन्याद् द्वादशभिरुद्धरेत्, फलमक्षज्याभवति। युक्तिरप्यत्र यदि विषुवच्छाया भुजस्य-द्वादश कोटिः तदाक्षज्यास्पृक् इति फलं लम्बज्या यदि द्वादशकोटेर्विषुवच्छाया भुजः। तदा लंबज्या कोटेः का भुजेति। फलमक्षज्या उपपन्नम्। अथवैकस्य परिज्ञानाद् द्वितीयमानयनं लंबज्या वर्ग प्रोह्य त्रिज्याकृतेः पदञ्चान्यत्रिज्या कृतेरक्षज्यावर्ग प्रोह्यमूलं लंबज्या। लंबज्यावर्ग प्रोह्य त्रिज्याकृतेः मूलमक्षज्या, यतोऽक्षज्या भुजालम्बकोटिस्त्रिज्याकर्णः। तस्मादुपपन्नं कर्णकृतेः कोटिकृतिं विशोध्य मूलं भुजस्य कृतिं प्रोह्यपदं कोटिरिति। एवं विषुवति याम्योत्तरमंडलावगाहिन्युष्ण-दीधितौ नतोन्नतज्ये बहुधा प्रदर्श्येदानीं विषुवतोऽन्यत्र दिनार्धे इति दिशति। ते एवानेनार्यार्धेनाऽन्यत्र सर्वदोन्नतनतजीवांशानयनमेवं। यथा विषुवच्छायाविषुवत्क-र्णाभ्यां नतोन्नतज्ये कृते तदंशश्च एवमिष्टदिने मध्याह्नछायया तत्कर्णेन च, नतोन्नतज्ये कृत्वा तच्चापभागाश्च कार्याः॥ तेऽत्र नक्तभागास्ते तद्दैवसिका अक्षांशाः। तावद्भिर्भागैः सममंडलमध्यात्तातोर्कः मध्याह्नं करोति तत्र दिने इत्यर्थः, तत्र याव न्नताः तावद्भिरुन्नता इत्यर्थः। एवमिष्टकालेपि तात्कालिकछायया तच्छाया कर्णेन वा नतोन्नतज्ये बहुधा कार्ये, तत्र योन्नतज्या स शंकुः यावती नतज्या तावती छाया तयोर्येऽशास्ते तत्काले नतोन्नतांशा भवन्ति। यथा स्वकमिति विषुवन्मध्याह्नेऽपि नतोन्नतांशा ये कृतास्तेऽपि दृङ्मंडल एव यतः सर्वदा मध्ये याम्योत्तरमंडलमेव दृङ्मंडलं तत्र नतांशाः सममंडलविषुवन्मंडलयोरन्तरं एकान्तरे भावादर्कोऽपि तत्र दिने विषुवन्मंडलं गतो विषुवतो विप्रकर्षो ग्रहस्य क्रान्तिरत्रदिने मध्याह्न-क्रान्तिवशादूना अधिका वा अक्षांशेभ्यो नतांशा भवन्ति।

तद्द्वादशाच्चोनाधिकास्ताः दिनदलछायातत्कर्णस्य च अतः स्वमध्यछाया-कर्णाभ्यां नतोन्नतज्ये विषुवद् युज्येते। अन्यत्र दिनार्धे इष्टकालेऽपि युज्येते। दिङ्-मंडलं यतो ग्रहाभिमुखं भ्रमति। सममंडलोपर्यधः खस्वस्तिकं न त्यज्यते। इष्ट-छायाकर्णस्य यदि द्वादशशंकुः तत् त्रिज्याकर्णस्य का छायेति नतज्या लभ्यते। इत्येव भादवो दृङ्मंडलस्यैकत्वात्तस्मात् सर्वमुपपन्नम्। यथास्थितं गोले प्रदर्शये-दिति। मध्यच्छायार्कक्षोऽक्षांशान्वा यो वेत्तीत्येतस्य प्रश्नस्योत्तरमाह।

वि. भा.—विषुवति (विषुवद्दिने) मध्याह्नकाले शंकुरेव लम्बः (लम्बज्या) भवति तत्र या छाया साऽक्षज्या कल्पनीया, तद्वर्गसंयुतेर्मूलं विषुवत्कर्णो भवति, अन्यथाऽन्यस्मिन् दिने मध्याह्नादन्यस्मिन् काले वा शंकुना यः कर्णो भवति स द्वादशांगुलशंकोश्छायाकर्णो भवति, शंकुरित्यस्याग्रिमश्लोकेन सम्बन्ध इति ॥७॥

अत्रोपपत्तिः

सायनसूर्ये मेषादिगे मध्याह्ने द्वादशांगुलशंको र्या छाया सा पलभा (विषुवती), शंकुपलभयोर्वर्गयोगमूलं पलकर्णः (विषुवत्कर्णः) अक्षज्याभुजः। लम्बज्या कोटिः। त्रिज्याकर्णः इत्यक्षक्षेत्रसजातीयमेवापवर्त्तितं (पलभाभुजः। द्वादशांगुलशङ्कुः कोटिः। पलकर्णः कर्ण इति भुजकोटिकर्णैरुत्पन्नं) लघुक्षेत्रमतोऽस्यकोटिभुजयोर्लम्बाक्षज्ये नामनी समुचिते एवेति ॥ ७ ॥

अब विषुवत्कर्ण को कहते हैं।

हि. भा.—विषुवद्दिन में शंकु ही लम्बज्या होती है, वहां जो छाया होती है उसको अक्षज्या कल्पना करनी चाहिए, उन दोनों के वर्गयोग मूल विषुवत्कर्ण होता है, अन्य दिन (विषुवद्दिन से भिन्न दिन) में वा मध्यान्ह काल से भिन्न काल में शंकुवश से जो कर्ण होता है वह द्वादशांगुल शंकु का छाया कर्ण होता है, 'शंकुः' इसका अगले श्लोक से सम्बन्ध है इति ॥ ७ ॥

उपपत्ति ।

सायन रवि जब मेषादि में रहते हैं तब मध्यान्ह काल में द्वादशांगुलशंकु की जो छाया होती है वह पलभा है, द्वादशांगुलशंकु और पलभा का वर्गयोग मूल पलकर्ण होता है, अक्षज्या भुज, लम्बज्या कोटि, त्रिज्या कर्ण इस अक्षक्षेत्र के सजातीय अपवर्त्तित (पलभा भुज, द्वादशांगुलशंकु कोटि, पलकर्ण कर्ण इन भुज कोटि कर्णों से उत्पन्न) लघु त्रिभुज है, इसलिये इसकी कोटि और भुज के नाम लम्बज्या और अक्षज्या समुचित ही है इति ॥ ७ ॥

इदानीं संज्ञा विशेषानाह –

वि. भा.—शंकुः उन्नत जीवा वा कोटिः छाया, दृग्ज्या नतज्या वैते शब्दा एकपर्याया भुजः। यत्रैते भुजकोटी भवेतां तस्मिन् छायावृत्तोऽनयो (भुजकोट्योः) वंशेन यः कर्णस्तदेव व्यासार्धं (त्रिज्या) ज्ञेयम्। अतोऽन्यदत्रापि द्वयं ज्ञेयमर्थात् कोटिर्भुजश्चेति यद्द्वयं तत्तयोर्वर्गयोगमूल व्यासार्धोत्पन्नवृत्तो कल्पनीयमिति ॥ ८ ॥

अब संज्ञा विशेष को कहते हैं।

हि. भा.—शंकु वा उन्नतज्या कोटि, छाया, दृग्ज्या, वा नतज्या (ये एक पर्याय वाची शब्द हैं) भुज, जहां ये भुज और कोटि होती है, उस छायावृत्त में इन भुज और कोटि वंश से जो कर्ण होता है वही व्यासार्ध (त्रिज्या) समझना चाहिये इससे अन्य अन्यत्र भी कोटि और

भुज ये दोनों जो हों उनको उन दोनों के वर्गयोग मूल व्यासार्धोत्पन्न वृत्त में कल्पना करना इति ॥ ८ ॥

इदानीं लम्बाक्षज्ययोरानयनमाह—

वि. भा.—शंकुशब्देनात्र द्वादशांगुलशंकुस्तथा छायाशब्देन पलभाया ग्रहणम्। द्वादशपलभयोर्वर्गयोस्त्रिज्यावर्गगुणितयोः तत्समासेन हृतयोः (द्वादशपलभयोर्वर्गयोगेन भक्तयोः) मूले तदा क्रमेण लम्बाक्षज्ये भवतः। तद्धनुभागाः (तयोश्चापांशाः) तदंशकाः (लम्बांशा अक्षांशाश्च) भवन्तीति ॥ ९ ॥

अत्रोपपत्तिः

$१२^२ + \text{पलभा}^२ = \text{पलकर्ण}^२$, $\frac{१२^२ \times \text{त्रि}^२}{\text{पलक}^२} = \text{लम्बज्या}^२ = \frac{१२^२ \times \text{त्रि}^२}{१२^२ + \text{पलभा}^२}$, तथा $\frac{\text{पलभा}^२ \times \text{त्रि}^२}{१२^२ + \text{पलभा}^२} = \text{अक्षज्या}^२$ एतयोर्मूले तदा लम्बाक्षज्ये भवतः। तयोश्चापे लम्बाक्षौ भवेतामिति ॥ ९ ॥

अब लम्बज्या और अक्षज्या के साधन को कहते हैं।

हि. भा.—शंकुशब्द से यहां द्वादशांगुलशङ्कु और छायाशब्द से पलभा का ग्रहण करना चाहिये, द्वादश वर्ग और पलभा वर्ग को त्रिज्या वर्ग से गुणा कर द्वादशवर्ग और पलभा वर्ग के योग से भाग देने से क्रमसे लम्बज्या और अक्षज्या होती है, दोनों के चाप करने से लम्बांश और अक्षांश होता है ॥ ९ ॥

उपपत्ति ।

द्वादशांगुलशंकु = कोटि, पलभा = भुज, पलकर्ण = कर्ण,
लम्बज्या = कोटि, अक्षज्या = भुज, त्रिज्या = कर्ण,
} इन कोटि भुज कर्णों से उत्पन्न अक्षक्षेत्र सजातीय हैं इसलिये अनुपात करते हैं।

$\frac{१२^२ \times \text{त्रि}^२}{\text{पलक}^२} = \text{लम्बज्या}^२$, तथा $\frac{\text{पलभा}^२ \times \text{त्रि}^२}{\text{पलक}^२} = \text{अक्षज्या},^२$ परन्तु $१२^२ + \text{पलभा}^२ = \text{पलक}^२$, अतः उत्थापन देने से $\frac{१२^२ \times \text{त्रि}^२}{१२^२ + \text{पलभा}^२} = \text{लम्बज्या}^२$, $\frac{\text{पलभा}^२ \cdot \text{त्रि}^२}{१२^२ + \text{पलभा}^२} = \text{अक्षज्या}^२$, दोनों के मूल लेने से लम्बज्या और अक्षज्या होती है, चाप करने से लम्बांश और अक्षांश होता है इति ॥ ९ ॥

इदानीं प्रकारान्तरेण तयोरानयनमाह—

वि. भा.—त्रिज्ये पृथक् शंकुच्छाया हते (द्वादश पलभागुणिते) विषुवत्कर्णहृते (पलकर्णभक्ते) लम्बाक्षज्ये भवतः। वा लम्बाक्षांशोत्क्रमज्योने त्रिज्ये अक्षज्येतर-

जीवे (अक्षज्यालम्बज्ये) भवतोऽर्यान्लम्बांशोत्क्रमज्यौना त्रिज्याऽक्षज्या, अक्षांशोत्क्रमज्यौना त्रिज्या लम्बज्या भवतीति ॥ १० ॥

अत्रोपपत्तिः

पूर्वश्लोके वर्गानुपातेन लम्बाक्षज्ययोर्वर्गावानीय तन्मूलेन लम्बाक्षज्ये समानीते, अत्रसाधारणानुपातेन तयोरानयनमस्ति त्रि—लम्बांशोत्क्रमज्या=अक्षज्या, त्रि—अक्षांशोत्क्रमज्या=लम्बज्या, सिद्धान्तशेखरे 'लम्बाक्षभागोत्क्रमर्शिजिनी या तया त्रिमौर्वी रहितेतरा वा' इत्यनेन श्रीपतिना, 'ये दोः कोट्योस्तः क्रमज्ये तद्ने त्रिज्ये ते वा कोटिदोरुत्क्रमज्ये' इति विलोमेन भास्करेणाऽपि तदेव कथ्यत इति ॥१०

अब प्रकारान्तर से उन दोनों (लम्बज्या और अक्षज्या) के साधन को कहते हैं ।

हि. भा.—त्रिज्या को पृथक् द्वादश से और पलभा से गुणा कर पलकर्ण से भाग देने से लम्बज्या और अक्षज्या होती है, वा त्रिज्या में लम्बांश की उत्क्रमज्या को घटाने से अक्षज्या होती है, तथा त्रिज्या में से अक्षांशोत्क्रमज्या को घटाने से शेष लम्बज्या होती है इति ॥१०॥

उपपत्ति ।

पहले के श्लोक में वर्गानुपात से लम्बज्यावर्ग और अक्षज्यावर्ग लाकर मूल लेकर लम्बज्या और अक्षज्या लाये हैं, यहां साधारण अनुपात से उन दोनों का आनयन है, त्रि—लम्बांशोत्क्रमज्या=अक्षज्या, त्रि—अक्षांशोत्क्रमज्या=लम्बज्या, सिद्धान्त शेखर में 'लम्बाक्षभागोत्क्रमर्शिञ्जिनी या' इत्यादि से श्रीपति तथा 'ये दो कोट्योस्तः क्रमज्ये तद्ने त्रिज्ये ते वा कोटिदोरुत्क्रम ज्ये' इसके विलोम से भास्कराचार्य भी इसी बात को कहते हैं इति ॥ १० ॥

पुनः प्रकारान्तरेण लम्बाक्षज्ये आह—

लम्बाक्षांशान् नवतेः प्रोह्य (हित्वा) ज्या साध्या तदा वा (प्रकारान्तरेण) इतरा ज्या स्यादर्थाल्लम्बांशोननवतेर्ज्याऽक्षज्या, अक्षांशोननवतेर्ज्या लम्बज्या, अक्षलम्बज्ये-शंकुच्छायागुणिते (द्वादशपलभागुणिते) छायाद्वादशहृते (पलभा द्वादश भक्ते) तदा वा (प्रकारान्तरेण)ऽन्ये लम्बाक्षज्ये भवत इति ॥११॥

अत्रोपपत्तिः ।

ज्या (९०—अक्षांश) =लम्बज्या, ज्या (९०—लम्बांश) = अक्षज्या, वा $\frac{\text{अक्षज्या}\times १२}{\text{पलभा}}$ =लंज्या, तथा $\frac{\text{पभा}\times\text{लंज्या}}{१२}$ =अक्षज्या, एतावताऽऽचार्योक्तमुपपद्यते ॥ ११ ॥

अब पुनः प्रकारान्तर से लम्बज्या और अक्षज्या को कहते हैं।

हि. भा.—नवत्यंश में से लम्बांश को घटाने से शेष की ज्या अक्षज्या होती है, नवत्यंश में से अक्षांश को घटाने से शेष की ज्या लम्बज्या होती है, वा अक्षज्या और लम्बज्या को क्रम से द्वादश और पलभा से गुणाकर पलभा और द्वादश से भाग देने से लम्बज्या और अक्षज्या होती है इति ॥११॥

उपपत्ति

ज्या (९०—अक्षांश) = लम्बज्या, ज्या (९०—लम्बांश) = अक्षज्या, वा $\frac{\text{अक्षज्या} \times १२}{\text{पलभा}}$ = लंज्या, $\frac{\text{पभा} \times \text{लंज्या}}{१२}$ = अक्षज्या इससे आचार्योक्त उपपन्न हुआ ॥११॥

इदानीं पुनः प्रकारान्तरेणाह —

वि. भा.—त्रिज्याकृतेः (त्रिज्यावर्गात्) लम्बाक्षज्यावर्गं प्रोह्य पदं वाऽन्या (त्रिज्यावर्गाल्लम्बज्यावर्गं विशोध्यमूलमक्षज्या तथा त्रिज्यावर्गादक्षज्यावर्गं विशोध्य मूलं ग्राह्यं वाऽन्या लम्बज्या) भवेत्। एवमन्यत्र सर्वदोन्नतनतजीवांशानयनं कार्यमर्थात् नतज्या वर्गोनात् त्रिज्यावर्गान्मूलमुन्नतज्या, उन्नतज्यावर्गोनात् त्रिज्यावर्गान्मूलं नतज्या भवेत्। एतयोश्चापे उन्नतांशा नतांशाश्च भवन्तीति ॥१२॥

अत्रोपपत्तिः।

$\sqrt{\text{त्रि}^२ - \text{लंज्या}^२}$ = अज्या, $\sqrt{\text{त्रि}^२ - \text{अज्या}^२}$ = लंज्या, नतांशज्या = दृग्ज्या, उन्नतांशज्या = शंकुः, $\sqrt{\text{त्रि}^२ - \text{शंकु}^२}$ = दृग्ज्या, $\sqrt{\text{त्रि}^२ - \text{दृग्ज्या}^२}$ = शंकु, इति ॥१२॥

अब पुनः प्रकारान्तर से लम्बज्या और अक्षज्या को कहते हैं।

हि. भा.—वा त्रिज्यावर्ग में से लम्बज्यावर्ग को घटाने से शेष का मूल अक्षज्या होती है, त्रिज्यावर्ग में से अक्षज्यावर्ग को घटाने से शेष का मूल लम्बज्या होती है, इसी तरह उन्नतांक्षज्या और नतांशज्या का आनयन करना अर्थात् नतांशज्यावर्ग को त्रिज्यावर्ग में से घटाने से शेष का मूल उन्नतांशज्या (शङ्कु) होती है, तथा उन्नतांशज्यावर्ग को त्रिज्यावर्ग में से घटाने से शेष का मूल नतांशज्या (दृग्ज्या) होती है, इन दोनों का चाप उन्नतांश और नतांश होता है। इति ॥१२॥

उपपत्ति

$\sqrt{\text{त्रि}^2-\text{लज्या}^2}$ =अज्या, तथा $\sqrt{\text{त्रि}^2-\text{अज्या}^2}$ =लज्या, $\sqrt{\text{त्रि}^2-\text{उन्नतांशज्या}^2}$ $=\sqrt{\text{त्रि}^2-\text{शङ्कु}^2}$ =दृग्ज्या, $\sqrt{\text{त्रि}^2-\text{नतांशज्या}^2}$ $=\sqrt{\text{त्रि}^2-\text{दृग्ज्या}^2}$ =शङ्कु, इति ॥१२॥

इदानीं दिनार्धे नतांश क्रान्त्यंशज्ञाने सत्यक्षांशज्ञानमाह।

इष्टदिनार्धनतांशक्रान्त्यंशैक्यान्तरं क्रियतुलादौ।
अक्षांशा याम्यायां छायायामन्तरमजादौ ॥१३॥

वा. भा.—विषुवद्दिनार्धछायया अक्षलम्बज्ययोरानयनं प्रागुक्तमनेनेष्टदिनार्धछायया तयोरानयनमनेनार्यासूत्रेण प्रदर्शयति। यथा विषुवच्छाया विषुवत्कर्णेन चाक्षलंबांशा कृता एवमभीष्टदिनार्धछायया तच्छायाकर्णेन च येऽक्षांशा भवन्ति ते नतांशा उच्यन्ते ये चाक्षलंबांशाः ते उन्नतांशाः ततो दिनार्धकालिकादर्कान्क्रान्तिज्यां कृत्वा तच्चापभागाः कार्याः ते क्रांत्यंशा भवन्ति। ततो यदि मेषादौ राशिषट्के रविस्तदा नतांशानां क्रांत्यंशानां च योगः कार्यः ते स्वदेशाक्षांशा भवन्ति। दक्षिणगोले वा यद्यर्कः तदेतेषामनन्तरं स्वाक्षभागाः भवन्ति। यदि दिनदलछाया सौम्या तदैवं मध्याह्ने छायामुत्तराभिमुखी भवतीत्यर्थः।

अन्तरमजादाविति। अथ मध्याह्नछाया दक्षिणाभिमुखी भवति। तदन्तरं नतांशक्रांत्यंशानां स्वदेशेऽक्षोभवति। यद्यजादौ रविरेतच्च तत्र संभवति। यत्र भागा चतुर्विंशति त ऊनोक्षो दक्षिणगोले च दक्षिणाभिमुखी छाया कदाचिदपि न सम्भवति। निरक्षदेशादुत्तरेण स्वाक्षांशान् विशोध्य नवतेः शेषाः लंबांशाः प्राग्वद्वासनात्र स्वदेशयाम्योत्तरमंडलगत्या सममंडलविषुवन्मंडलयोरन्तरे ये भागास्ते स्वाक्षभागाः ते चोत्तर गोले प्रतिदिनं क्रान्तिदिनक्रान्तिभागैरपचीयन्ते। येषां नतभागाः यतः सममण्डलस्य सन्निकृष्टो भवति। रविणा च, नतमतः स्वापचित्या पुनर्योज्यन्ते येन त एवं नतांशाः सम्भवन्ति। दक्षिणगोले चाक्षभागाः प्रतिदिनक्रान्तिभागैरुपचिताः सतो नतभागाः भवन्ति। यत्र प्रतिदिनं संमंडलाद्विप्रकृष्टो रविर्भवत्यत उपचितः। पुनर्नतभागेभ्यो विशोध्यते। येन त एवावशिष्यन्तेऽक्षांशाः तस्मादुपन्नम्। यत्र पुनः सममंडलादुत्तरेणार्को मध्याह्नं करोति। यत्र चाक्षभागेभ्योऽधिका उत्तरा क्रांत्यंशा भवन्ति। तेन तत्र दिनार्धे दक्षिणाभिमुखी छाया भवति। यावद्भिश्चभागैरधिका क्रान्तिरक्षात्तावतो नतांशाः क्रान्त्यंशेभ्यो विशोध्यन्ते। येन स्वाक्षभागा च भवन्ति। तेनोक्तमंतरमजादाविति। तस्मात्सर्वमुपपन्नम्। स्वदेशाक्षाग्रयोर्गोले विन्यस्य प्रदर्शयेदिति यश्चरखंडकान्विजानातीत्यस्य प्रश्नस्योत्तरमार्यामाह ॥

वि. भा.—क्रियतुलादौ (मेषादौ तुलादौ) सूर्ये इष्टदिनार्धे नतांशानां क्रान्त्यंशानां योगोन्तरमक्षांशा भवन्ति। अजादौ (मेषादौ) सूर्ये छायायां (मध्याह्नच्छायायां) याम्यायां (दक्षिणायां) सत्यां नतांशक्रान्त्यंशान्तरमेवाक्षांशा भवन्तीति ॥१३॥

अत्रोपपत्तिः

उत्तरगोले मध्याह्नकाले खस्वस्तिकनिरक्षखस्वस्तिकयोरन्तरे रवौ, रवितो निरक्षखस्वस्तिकं यावन्मध्यक्रान्तिः । रवितः खस्वस्तिकं यावन्मध्यनतांशाः, एतयोर्योगेन खस्वस्तिकान्निरक्ष खस्वस्तिकं यावदक्षांशा भवन्ति, दक्षिणगोले निरक्ष खस्वस्तिकाद्दक्षिणे सूर्ये नतांशे क्रान्त्यंशशोधनेऽक्षांशा भवन्ति । उत्तरगोले खस्वस्तिकादुत्तरे रवौ-रवितो निरक्षखस्वस्तिकं यावत्क्रान्त्यंशे नतांशशोधनेनाक्षांशा भवन्ति, सिद्धान्तशेखरे 'उदगिनापमभागसमन्विता' नतलवा इतरत्र विशेषिताः। स्वविषये हि भवन्ति पलांशकाः' इत्यनेन श्री पतिना 'नतांशापमांशान्तरं तुल्यदिक्त्वे युतिर्भिन्नदिक्त्वे पलांशा भवेयुरित्यनेन भास्कराचार्येणाप्याचार्योक्तमेव कथ्यत इति ॥१३॥

अब दिनार्धकाल में नतांश और क्रान्त्यंश के ज्ञान से अक्षांश ज्ञान को कहते हैं।

हि. भा.—मेषादि में और तुलादि में सूर्य के रहने से अर्थात् उत्तरगोल में और दक्षिणगोल में इष्टदिन के दिनार्धकाल में नतांश और क्रान्त्यंश का योग और अंतर अक्षांश होता है, मेषादि में सूर्य के रहने से अर्थात् उत्तरगोल में मध्यान्हकालिक छाया दक्षिण रहने से नतांश और क्रान्त्यंश का अन्तर ही अक्षांश होता है इति ॥१३॥

उपपत्ति

उत्तर गोल में मध्यान्ह में खस्वस्तिक और निरक्षखस्वस्तिक के मध्य में याम्योत्तरवृत्त में रवि के रहने से रवि से निरक्षखस्वस्तिक पर्यन्त रवि की मध्यक्रान्ति है, तथा रवि से खस्वस्तिक पर्यन्त रवि के मध्यनतांश है, इन दोनों का योग करने से खस्वस्तिक से निरक्षखस्वस्तिक पर्यन्त अक्षांश होता है, दक्षिण गोल में निरक्षखस्वस्तिक से दक्षिण में सूर्य के रहने से नतांश में से क्रान्त्यंश को घटाने से अक्षांश होता है, उत्तरगोल में खस्वस्तिक से उत्तर में रवि के रहने से रवि से निरक्ष खस्वस्तिक पर्यंत क्रान्त्यंश में से नतांश को घटाने से अक्षांश होता है, सिद्धांतशेखर में 'उदगिनापमभागसमन्विता नतलवा इतरत्र विशेषिताः। स्वविषये हि भवन्ति पलांशका' इससे श्रीपति, तथा सिद्धान्त शिरोमणिमें 'नतांशापमांशान्तरं तुल्यदिक्त्वे युतिर्भिन्नदिक्त्वे पलांशा भवेयुः' इससे भास्कराचार्य भी आचार्योक्त ही को कहते हैं इति ॥१३॥

इदानीं मेषादिराशीनां चरखण्डसाधनमाह ।

मेषवृषमिथुनजीवा स्वाहोरात्रार्धचरदलप्राणान् ।
प्राग्वत् कृत्वा स्वाधो विशोध्य चरखण्डकप्राणाः ॥१४॥

वा. भा.—मेषवृषमिथुनानां जीवाः कार्याः ताश्च भूतागिरसगोका १६३५ वृषस्यदन्नाष्टयमा २८३२ मिथुनस्य खमुनिरदा ३२७० अतः स्वाहोरात्रार्धा प्राग्वद्भवन्तीत्यर्थः । जिनभागज्यागुणनाः सूर्यज्योत्पत्त्यायेन पृथक् क्रान्तिज्या कार्या ताश्च मेषस्य वेदरसषट्काम्माधा ६६४।३० वृषस्य चंद्रशरभवाः ११५१ मिथुनस्य नवरदचंद्रा १३२९ । एताभिश्च प्राग्वत् । स्वाहोरात्राणि । इष्टापक्रमवर्गत्रिज्यावर्गाद्विशोध्येति यावत् तद्यथा मेषस्य ३२०२, वृषस्य ३०६, मिथुनस्य २९८७, एताश्चरदलप्राणाः क्रान्तिज्याविषुवच्छायया गुणेत्यादिना प्राग्वत् । स्वदेशे विषुवच्छायया कार्या । ताश्च स्वाधो विशोध्य स्वदेशज्याश्चरदलखंडकानां पृथक् प्राणा भवन्ति ।

मेषवृषमिथुनानां त एव क्रमेण कर्कसिंहकन्यानामधः क्रमेण तुलावृश्चिकधनुषां पुनरुत्क्रमेण मकरकुम्भमीनानामत्रवामना । स्वक्रांत्यग्रे स्वाहोरात्रवृत्तानि विन्यस्य प्रदर्शयेत्, स्फुटगत्यध्याय एव चरदलानयने मया प्रदर्शितानि विशेषश्च प्रदर्श्यते स्वदेशे । यत् क्षिति स्वाहोरात्रवृत्ते क्षितिजोन्मण्डलयोरन्तरं प्राणाः यदेव लग्नार्कावुद्दिश्य तदन्तरकालं पृच्छतीति । स्वदेशराश्युदयैर्विशोध्यते । तथा च युज्यते । स्वाग्रांनु कृत्वा मिथुनाहोरात्रार्धं क्रियादहोरात्रदलः वृषान्त्यगुणितं तज्ज्याभिरिति न्यायेन लंकोदयवत्प्राणाः कार्याः तथार्कादेव चरदलप्राणानामेष वृषमिथुनानामेत एव क्रमेण यदि मकरादौ रविरस्तकन्यादौ राशिशुद्धः तत्प्राण गणयोरैक्यं कार्यम् । तथा कृते प्राणलिप्तानुसारं कल्प्यास्ततो यदि मेषादौ राशित्रये रविस्तयो राशिकला अष्टादशानि । कर्क्यादौ रविस्तत्रादौ, अथवा भुक्तलिप्ताभ्यो विशोध्या शेषरविभुक्तलिप्तासंसाधनं ततो लग्नादपि स्वाहोरात्रवृत्तरवौ जीवादिकं कृत्वा मिथुनाहोरात्रमिति न्यायेन लंकोदयवत्प्राणाः कार्याः स्वचर प्राणाश्च तयोरपि प्राणगणयोरन्तरयोगे मृगकन्यादिषु लग्नवशात्ततो मेषादिपर्दाविकल्पना लग्नवशाद्विवदेकं लग्नस्यापि भुक्तकाललिप्तासाधनं ततो लग्नभुक्तकाललिप्ताभ्योऽर्कभुक्तकाललिप्ताः शोध्याः सर्वथा न पतन्ति चेत् ? लग्नलिप्ताभिरधिकाश्चक्रलिप्ताः कृत्वार्कभुक्तलिप्ताः शोध्या-एवं कृते याः शेषलग्नलिप्ताः ते सर्वे च परिकल्प्याः तावत्प्राणरविलग्नान्तरं तावता प्राणानां तल्लग्नं कृतं रव्युदयादव लग्नभुक्तकाललिप्तारविभुक्तकाललिप्ताभ्यो विशोध्यन्ते । तद्रव्युदयाद्विपरीत्येन कालः शोध्यरित्यत्र वासना लंकोदयानां त्रिप्रश्नाध्यायोक्तैव तत्र राश्यंतेषूदयकोटयः दर्शिताः । इह राशिमध्येष्वपि स्वाहो-

रात्रादिभिर्गोलं प्रदर्श्य तयोश्चरदलरविलग्नयो र्लंकोदयान्विजानातीत्यस्य प्रश्नस्योत्तरमार्यामाह ॥

वि. भा.—मेषवृषमिथुनज्याभ्यः स्वाहोरात्रार्धे द्युज्यावृत्ते प्राग्वत् (स्पष्टाधिकारोक्तेन जिनभागज्या गुणितेत्यादिना) चरासून् कृत्वा स्वाधो विशोध्य मेषादिराशित्रितयस्य चरखण्डकप्राणा (चरार्धासवः) बोध्या इति ॥१४॥

अत्रोपपत्तिः ।

अथ मेषादि राशित्रय चरानयनम् ।

$\frac{\text{कुज्या. त्रि}}{\text{द्यु}}$ = चरज्या, परन्तु $\frac{\text{पलभा. ज्याक्रां}}{१२}$ = कुज्या, अतः $\frac{\text{पलभा. ज्याक्रां}}{१२} \times \frac{\text{त्रि}}{\text{द्यु}}$ = चरज्या तथा $\frac{\text{ज्याजि. ज्याभु}}{\text{त्रि}}$ = ज्याक्रां $\therefore \frac{\text{पलभा. ज्याजि. ज्याभु}}{\text{द्यु. १२}}$ = चरज्या, अत्र भुजज्यास्थाने मेषादिराशित्रयज्यास्तथा द्युज्या स्थानेऽपि च तेषां राशित्रयाणां द्युज्याः संगृह्य पृथक् पृथक् यानि फलान्यागच्छेयुस्तच्चापानामधोऽधः शोधनेन तेषां राशीनां चरखण्डकानि भवन्तीति । सिद्धान्त-शेखरे 'अजवृषमिथुनानां ज्या दिनज्याक्षितिज्याः सहचरदलजीवाश्चानयेत्तद्धनुर्भिः । त्रिभिरपि चरखण्डैस्तैरधोऽधो विशुद्धैश्चरदलमपि साध्य' मित्यनेन सिद्धान्त शिरोमणौ भास्कराचार्येणापि 'मेषादिराशित्रितयस्य यानि चराण्यधोऽधः परिशोधितानि । तानि चरखण्डकानीत्यनेनाऽऽचार्योक्तानुरूपमेव, कथ्यत इति ॥१४॥

अब मेषादिराशियों के चरखण्ड साधन को कहते हैं ।

हि. भा.—मेष, वृष-मिथुन राशियों की भुजज्याओं से द्युज्यावृत्त में पूर्ववत् 'स्पष्टाधिकारोक्तजिनभागज्या गुणिता' इत्यादि से चरासु साधन करके अधोऽधः शोधन करने से मेषादि राशित्रय के चरार्धासु प्रमाण होते हैं इति ॥१४॥

उपपत्ति

$\frac{\text{कुज्या. त्रि}}{\text{द्यु}}$ = चरज्या, परन्तु $\frac{\text{पलभा. ज्याक्रां}}{१२}$ = कुज्या, अतः उत्थापन से $\frac{\text{पलभा. ज्याक्रां}}{१२} \times \frac{\text{त्रि}}{\text{द्यु}}$ = चरज्या तथा $\frac{\text{ज्याजि. ज्याभु}}{\text{त्रि}}$ = ज्याक्रां $\therefore \frac{\text{पलभा. ज्याजि. ज्याभु}}{\text{द्यु. १२}}$ = चरज्या, यहां भुजज्या के स्थान में मेषादि रशित्रय की ज्या, तथा द्युज्या, स्थान में भी उन राशियों की द्युज्या लेकर पृथक् पृथक् जो फल आये उनके चापों को अधोऽधः शोधन करने से उन राशियों के

चरखण्ड होते हैं, सिद्धान्त शेखर में 'अजवृषमिथुनानां ज्या' इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से, आचार्योक्तानुरूप ही कहते हैं इति ॥१४॥

इदानीं लङ्कोदय साधनमाह ।

मिथुनाहोरात्रार्धं क्रियाद्यहोरात्रदलहृतं गुणितम् ।
तज्ज्याभिराप्तचापान्तराणि लङ्कोदयप्राणाः ॥१५॥

वा. भा.—मिथुनाहोरात्रार्द्धं वृषमिथुनजीवाभि र्गुणितं सत् । मेषाद्याहोरात्रदलहृतं कार्यं लब्धानां चापानि कृत्वा तानि स्वाधो विशोध्य लंकोदयप्राणा भवन्ति । मेषवृषमिथुनानां त एवोत्क्रमेण कर्कटसिंहकन्यानामधः क्रमेण तुला वृश्चिकचापधराणां पुनरुत्क्रमेण मकरकुम्भमीनानामिति अत्र वासना खगोनादृक्षिणोत्तरस्वस्तिकयोरधः शलाकाग्रे निरक्षे प्रदेशे गोलं प्रदर्श्यते तद्यथा विषुवन्मण्डलादुत्तरतो मेषवृषमिथुनानां क्रान्त्यग्रेषु स्वाहोरात्रत्रयं बध्नीयाद्गोले तदेव कर्कटसिंहकन्यानां उत्क्रमेण एवं विषुवृत्ताद् दक्षिणेन तुलादीनां त्रयाणां स्वाहोरात्रत्रयं बध्नीयात् । तदेव मकरादीनामुत्क्रमेण भवति ॥

ततो मेषज्यातुल्येन व्यासार्धेन वृत्तं भगोलमध्ये दक्षिणोत्तरावगाहि बध्नीयात् । तस्य मध्यं भूविनिर्गता पूर्वापरा सूत्रे भवति । तद्वृषज्या व्यासार्धेनान्यं बध्नीयात् । मिथुनज्यया च व्यासार्धतुल्यया याम्योत्तरमंडलांतेष्वित्येव एवं वृत्तत्रयेपि स्वजीवाकरणात् यत्क्रांतिज्या दक्षिणोत्तरायतभुजः कोटिश्च शोध्या मेषवृषयो मिथुनस्य च स्वाहोरात्रार्धं स्वाहोरात्रवृत्तकोटिकृतमेवास्ति ते व्यासार्धवृत्ते ज्ञाते कोट्यानयनार्थं त्रैराशिकद्वयं भक्ता गुणिता कर्मेयम् । तद्यथा यदि मिथुनवृत्तकर्णस्य त्रिज्यातुल्यवृत्तस्य मिथुनाहोरात्रार्ध कोटिः तन्मेषज्याकर्णस्य का कोटिरितिफलं स्वाहोरात्रवृत्ते प्रथमं त्रैराशिके भागहारः तेन तयोर्नाशे कृते तुल्यत्वात् मिथुनाहोरात्रार्धस्य मेषज्यागुणकारस्तदहोरात्रार्धं भागहारः फलं व्यासार्धं कृतिगुणिता भार्धवृषस्यापि मिथुनस्य च प्रथमे त्रैराशिके गुणकारभागहारयोः तुल्यत्वात्तदहोरात्रार्धमेव स्वाहोरात्रकोटिः ततो व्यासार्धपरिगत विषुवत्तात्कालिकतुल्यत्वात्त्रिज्यैव व्यासार्धवृत्तकोटिः से क्षेत्रवच्चेह प्रदर्श्यते । तद्यथापमंडले मेषांते सूत्रस्यैकमग्रं बद्ध्वा विषवतो दक्षिणेन मीनादौ बध्नीयात् ।

एवं मेषान्ते कुम्भादयो योज्या । एवं मिथुनांतमकरान्तोरपि तानि सूत्राणि पूर्वप्रदर्शितवृत्तानां व्यासास्तेषामवस्थितानि, यान्यूर्ध्वानि प्राच्यपरायतसूत्रावच्छिन्नानि ते कर्णाः ततो निरक्षं क्षितिजे पूर्वस्यां दिशि क्रांत्यग्रेषु सूत्रत्रयं बद्ध्वापरस्यां दिशि स्थित्वा क्षितिज एवं क्रांत्यग्रेषु बध्नीयादन्यानि तत्र तेषामुदयास्तसूत्राणि क्षितिजोन्मंडलयोरेकत्वादतः प्राच्यपरयोरग्रयोस्तत्सूत्रयोर्धविवरान्तरं

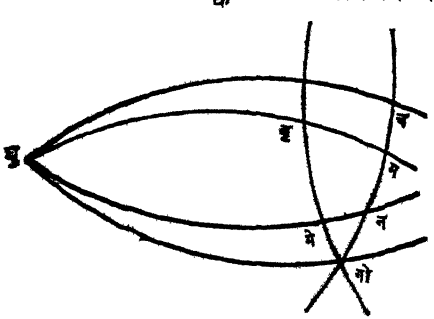
स्ववृत्तमध्यात्तावत्प्रमाणं सूत्रद्वयं दक्षिणोत्तरायतं बध्नीयात् । ततक्रांतिज्या भुजज्या पृथक् । ततो मेषान्तरे सूत्रं बद्ध्वोर्ध्वमानीय स्वक्रांत्यग्रे बध्नीयात् सूत्रे एवं वृषमिथुनयोरपि ताः पृथक् कोट्यः पूर्वप्रदर्शिताः स्वकोट्युदयेन मेषादीनामुदयो भवति, तिर्यक् स्थानं यतो विषुवन्मंडलवशेन सर्वस्यैव भपंजरस्य भ्रमणमतस्तासां कोटीनां चापानि क्रियन्ते यतो वृत्तगत्या भपंजरोदयः तस्मादुपन्नचापानि चातः क्रियन्ते । यतो मिथुनकोट्या राशित्रयमप्युदेति । वृषकोट्या राशिद्वयं पृथगुदयप्रमाणानि ज्ञातुमिष्यते । तस्माच्चापान्तराणि युज्यन्ते, चापलिप्ताश्च प्राणाः भवन्त्येवं यतः प्राणेन कक्षाभमंडलं भ्रमतीति प्रागेव प्रतिपादितं कर्कादीनां क्रमेण वासना योज्या दक्षिणतस्तुलादीनां क्रमेण मकरादीनामुत्क्रमेण च योज्येति ।

तद्यथा मेषादयः प्राणाः खमुनिरसेन्दवः १६७० वृषोदयप्राणाः शरनवागचन्द्रा १७९५ मिथुनोदयप्राणाः पंचाग्निननवशशिनः १९३५ मेषस्य खागषट्चंद्राः शरनंदनगेन्दवः वृषस्य मिथुनास्याथ पंचाग्निननवरात्रियः । इदानीं प्रकारान्तरेण निरक्षोदयान्प्रदर्शयन्नार्यामाह ॥

वि. भा.—मिथुनाहोरात्रार्ध (मिथुनांत द्युज्यामानं 'परमाल्प द्युज्यामानं') तज्ज्याभिः (मेषादिराशित्रयभुजज्याभिः) गुणितं क्रियाद्यहोरात्र दलहृतं (मेषादिराशिद्युज्याभक्तं) फलानां यानि चापानि तदन्तराणि लङ्कोदयप्राणाः लङ्कोदयासवः) स्युरिति ॥१५॥

अत्रोपपत्तिः ।

राश्यादिविन्दुर्यदा निरक्षक्षितिजे समागच्छति ततोऽनन्तरं यावता कालेन राश्यन्तबिन्दुस्तत्क्षितिजे समागच्छेत्तदेव तद्राशेर्लङ्कोदयमानमर्थाद्राश्याद्युपरि ध्रुवप्रोतवृत्तं राश्यन्तोपरिच ध्रुवप्रोतवृत्तं कार्यं तयोरन्तरे नाड़ीवृत्ते यच्चापं तत्तद्राशेर्लङ्कोदयमानमिति निम्नलिखितक्षेत्रेण स्फुटं भवति, यथा—



गो=गोलसन्धिः=मेषादिः । मे=मेषान्तः = वृषादिः, वृ = वृषान्तः = मिथुनादिः, मि=मिथुनान्तः । ध्रु=ध्रुवः । गोमे=मेष भुजांशाः, गोवृ=वृष भुजांशाः । गोमि=मिथुनभुजांशाः ।

गोन=मेषोदयमानम् । नम=वृषोदयमानम् । मच=मिथुनोदयमानम् । ध्रमे=मेषान्त द्युज्याचापम्, ध्रुवृ=वृषान्त द्युज्याचापम् । ध्रुमि=मिथुनान्त-

द्युचा=परमाल्प द्युज्याचापम्। <ध्रुगोमि=परमाल्प द्युज्यांग ध्रुगोमे चापीय त्रिभुजेऽनुपातेन मेषोदयज्या मानम्= $\frac{\text{परमाल्पद्युज्या. ज्यामे}}{\text{मेषान्तद्यु}}$ एवं ध्रुगोवृ. चापीय त्रिभुजे अनुपातः $\frac{\text{परमाल्प द्युज्या. ज्यावृ}}{\text{वृषान्तद्यु}}$ = ज्या (मेषोदयमान + वृषोदयमान), ध्रुगोमिचापीय त्रिभुजेऽनुपातेन = $\frac{\text{परमाल्पद्यु. ज्यामि}}{\text{मिथुनान्तद्यु}}$ = $\frac{\text{परमाल्पद्यु. ज्यामि}}{\text{परमाल्पद्यु}}$ = ज्यामि. = त्रि, अत्र ज्यामे, ज्यावृ, ज्यामि मेषादि राशीनां ज्याः। पूर्वोक्तानां चापान्यबोधः शुद्धानि तदा मेषादिराशीनां लङ्कोदयमानानि भवन्ति। सिद्धान्तशेखरे अल्पद्युज्याविनिघ्नाः क्रियवृषमिथुनज्या हृताः स्वद्युमौर्व्या प्राणानां चापलिप्ता विरचित विवराः स्युर्निरक्षोदयास्तेऽनेन तथा सिद्धान्तशिरोमणौ 'मेषादिजीवाश्चित्रगृहद्युमौर्व्या क्षुण्णाहृताः स्वस्वदिनज्यया वा। चापीकृताः प्राग्वदधो विशुद्धा मेषादिकानामुदयासवः स्युरित्यनेन, भास्कराचार्येणाप्याचार्योक्तानुरूपमेव कथ्यत इति ॥१५॥

अब लङ्कोदय साधन को कहते हैं।

हि. भा.—मिथुनान्तद्युज्या (परमाल्पद्युज्या) को मेषादि राशियों की ज्या से गुणाकर अपनी अपनी द्युज्या से भाग देकर जो फल हो उनके चापों को अधोऽधः शुद्ध करने से मेषादि राशियों के लङ्कोदय मान होते है इति ॥ १५ ॥

उपपत्ति

राश्यादि बिन्दु जब लङ्का क्षितिज में आता है उसके बाद जितने काल में राश्यन्त बिन्दु लङ्काक्षितिज में आता है वही उस राशि के लङ्कोदयमान, अर्थात् राश्यादि और राश्यन्त के ऊपर ध्रुव प्रोतवृत्त करने से दोनों ध्रुव प्रोतवृत्तों के अन्तर्गत नाड़ी वृत्तीय चाप, होता है, यह संस्कृतोपपत्ति में लिखित (क) क्षेत्र से स्पष्ट होता है। जैसे—गो=गोलसन्धि=मेषादि, मे=मेषान्त=वृषादि, वृ=वृषान्त=मिथुनादि, मि=मिथुनान्त। ध्रु=ध्रुव, गोमे=मेषभुजांश, गोवृ=वृषभुजांश, गोमि=मिथुनांश, मोन=मेषोदयमान, नभ=वृषोदयमान। मच=मिथुनोदयमान। ध्रुमे=मेषान्तद्युज्याचाप, ध्रुवृ=वृषान्त द्युज्याचाप, ध्रुमि=मिथुनान्तद्युचाप=परमाल्पद्युचाप ध्रुगोमे चापीय त्रिभुज में अनुपात से $\frac{\text{परमाल्पद्यु. ज्यामे}}{\text{मेषान्तद्यु}}$ = मेषोदयज्या, एवं ध्रुगोवृ चापीय त्रिभुज में अनुपात से $\frac{\text{परमाल्पद्यु. ज्यावृ}}{\text{वृषान्तद्यु}}$ = ज्या (मेषोदयमान + वृषोदयमान), ध्रुगोमि चापीय त्रिभुज में अनुपात से

$\frac{\text{परमान्त्यखं. ज्यामि}}{\text{मिथुनान्त्यखं}} = \frac{\text{परमान्त्यखं. ज्यामि}}{\text{परमान्त्यखं}} = \text{ज्यामि} = \text{त्रि} \times \text{ज्या (मेषोदय} + \text{वृषोदय} + \text{मिथुनोदय)}$ यहां ज्यामे, ज्यावृ, ज्यामि, मेषादिराशिज्या, पूर्वसाधितोदय मानज्याओं के चापों को अधोऽधः शुद्ध करने से मेषादि राशियों के लङ्कोदयमान होते हैं। सिद्धान्तशेखर में 'अन्त्य ज्युज्या विनिघ्नाः क्रियवृषनिधुनाज्याहता' इत्यादि, संस्कृतोपपत्ति में लिखित श्लोक से, श्रीपति तथा सिद्धान्त शिरोमणि में 'मेषादिजीवास्त्रिगृहघ्नमौर्व्या' इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से भास्कराचार्य भी आचार्योक्तानुरूप ही कहते हैं, सूर्य सिद्धान्तकार भी इसी तरह कहते हैं इति ॥ १५ ॥

इदानीं प्रकारान्तरेण लङ्कोदयसाधनमाह।

ज्यावर्गात् क्रान्तिज्या वर्गोनात्तत्पदाहता त्रिज्या।
स्वाहोरात्रार्धहृताचापश्चापान्तराण्यथवा ॥१६॥

वा. भा.—मेषज्यावर्गात् क्रांतिज्यावर्ग विशोध्य मूलं ग्राह्यम्। तेन त्रिज्यां हृत्वा स्वाहोरात्रार्धेन विभजेत्। फलं मेषोदयकोटि व्यासार्ध निष्पन्नं। एवं स्वजीवाक्रान्तिज्याभ्यां वृषमिथुनयोरपि स्वोदयकोटिः ततस्तासां चापानि कृत्वा स्वाधो विशोध्य प्राग्वन्निरक्षोदयप्राणा भवन्ति। अथवानेन प्रकारेणोत्पन्नेयं वासना पूर्वप्रदर्शितेषु वृत्तेषु स्वजीवा कार्या स्वक्रान्तिज्या ततः स्वकर्णवर्गेभ्यः स्वभुजवर्गान्विशोध्य स्वकोटि वर्गा अवशिष्यन्ते। तेषां पदानि स्वकोटयः स्वाहोरात्राणि निष्पन्नानि ततस्त्रैराशिकं यदि स्वाहोरात्रवृत्तेषु तावत्यः कोटयः तदा त्रिज्यावृत्तेषु कियत्यः इति पृथक् फलानि व्यासार्धवृत्तकोटयो मेषवृषमिथुनोदयानां यत्तिर्यगपमंडलमुदक्स्थितम्। अतः स्वजीवा प्राणा न तेषामुदया भवन्ति। अन्यथा निरक्षदेशे पंचघटिका राश्युदया अभविष्यन्। शेषा वासना प्रागार्यायां कुरुते। लग्नमुदयो स्वैरिति प्रश्नं विवक्ष्य निरक्षोदयैः स्वचर खंडकैश्च स्वोदय करणमार्यामाह।

वि. भा.—ज्यावर्गात् (मेषादिराशिज्यावर्गात्) क्रान्तिज्यावर्गहीनात्पदं (मूलं) यत्तेन त्रिज्यागुणिता स्वाहोरात्रार्धहृता (स्वद्युज्याभक्ता) अधश्चापान्तराणि चापानामधोऽधः शोधनेन यानि चापानि तानि, अथवा (प्रकारान्तरेण) लङ्कोदयमानानि भवन्तीति ॥ १६ ॥

अत्रोपपत्तिः

अथ मेषान्तभुजांशाःकर्ण एको भुजः। मेषान्तक्रान्तिर्द्वितीयो भुजः। तद्विषुवांशास्तल्लङ्कोदयमानं वा कोटिस्तृतीयो भुजः। एवं वृषान्तमिथुनान्तेऽपि तत्तद्भुजांश-

क्रान्त्यंशलङ्कोदयाभिस्त्रिभुजद्वयं जायते। मेषान्तक्रमिभुजस्य ज्याक्षेत्रे मेषान्त-ज्याकर्णः। क्रान्तिज्या भुजस्तन्मूलगत रेखा कमलाकरोक्त व्यक्षोदयलवज्या कोटिः। $\sqrt{\text{मेषान्तज्या}^2 - \text{क्रांज्या}^2}$ = व्यक्षोदयलवज्या तथा ध्रुवाद्गोलसंधि यावद्गोल सन्धिगतध्रुवप्रोतवृत्ते नवत्यंशाः। मेषान्तोपरिगत ध्रुवप्रोतवृत्ते ध्रुवाद्याम्योत्तर-ध्रुवप्रोतवृत्तयोः सम्पातं यावच्चरवंशाः। मेषान्त विषुवांशाग्रैवंत्रिभुजत्रयमुत्पन्नं चापीयजात्यस्य ज्याक्षेत्रं (त्रिज्याकर्णो विषुवांशज्याभुजस्तत्कोटिज्याकोटिरेतैर्भुजत्रयैरुत्पन्नं) गोलकेन्द्रात्क्रान्तिज्या मूलं यावद्द्युज्याकर्णः, व्यक्षोदयलवज्याभुजः। गोलकेन्द्राद् भुजांशज्यामूलं यावत्कोटिरेतैर्भुजत्रयैरुत्पन्नत्रिभुजस्य सजातीयमनोऽनुपातो यदि द्युज्यया व्यक्षोदयलवज्या लभ्यते तदा त्रिज्यया किं समागच्छन्ति मेषान्तविषुवांशज्या

लङ्कोदयज्या वा $= \frac{\text{व्यक्षोदयलज्या. त्रि}}{\text{मेद्यु}} = \frac{\sqrt{\text{मेषान्तज्या}^2 - \text{क्रांज्या}^2} \times \text{त्रि}}{\text{मेद्यु}} = \text{मेषोदय-}$

ज्या। एवमेव $\frac{\sqrt{\text{वृषान्तज्या}^2 - \text{वृषान्तक्रांज्या}^2} \times \text{त्रि}}{\text{वृद्यु}} = \text{ज्या (मेषोदय} + \text{वृषोदय)}$

एवं मिथुनान्तेऽपि, एतज्ज्यायान्यधोऽधःशुद्धानि तदा मेषादीनां लङ्कोदयमानानि भवन्तीति, सिद्धान्तशेखरे श्रीपतिना "तत्क्रान्तिज्याकृतिविरहिताच्छिञ्जिनी वर्ग-तो वा मूलं यत् स्यात् त्रिभवनगुणास्ताड़ितास्तेन भक्तः। स्वद्युज्याभिर्विहितधनुषां तत्फलानामधोऽधः संशुद्धानामितिरसुमयास्ते निरक्षोदयाः स्युः" ञ्जेन, भास्क-राचार्येण चा "एकस्यराशेर्वृहतीज्यका या द्वयोस्त्रिभस्यापि कृती कृतानां स्वस्वा-पमज्याकृतिवर्जितामित्यादिना" ऽऽचार्योक्तानुरूपमेव कथ्यत इति ॥ १६ ॥

अब प्रकारान्तर से लङ्कोदय साधन को कहते हैं।

हि. भा.—मेषादि राशियों के ज्यावर्ग में से क्रान्तिज्या वर्ग घटा कर मूल लेने से जो प्राप्त हो उनको त्रिज्या से गुणा कर अपनी अपनी द्युज्या से भाग देने से जो उपलब्ध हो उनके चापों को अधोऽधः शोधन करने से अथवा प्रकारान्तर से मेषादि राशियों के लङ्कोदय-मान होते हैं इति ॥ १६ ॥

उपपत्ति।

मेषान्त भुजांश कर्ण, मेषांत क्रान्ति भुज, और उसके विषुवांश वा उसके लङ्कोदय-मान कोटि, इसी तरह वृषान्त और मिथुनान्त में भी उनके भुजांश, क्रान्त्यंश और लङ्कोदय-मानों से त्रिभुजद्वय बनते हैं। मेषान्तोत्पन्न त्रिभुज के ज्याक्षेत्र में मेषान्तज्या कर्ण, मेषान्त-क्रान्तिज्याभुज, तन्मूलगत रेखा कमलाकरोक्त व्यक्षोदयलवज्या कोटि, इस त्रिभुज में $\sqrt{\text{मेषान्तज्या}^2 - \text{मेषान्तक्रांज्या}^2}$ = मेव्यक्षोदयलवज्या, तथा ध्रुव से गोल सन्धिपर्यन्त गोल सन्धिगत ध्रुवप्रोतवृत्त में नवत्यंश, मेषान्तोपरिगत ध्रुवप्रोतवृत्त में ध्रुव से ध्रुव प्रोतवृत्त

और नाड़ीवृत्त के सम्पात पर्यन्त नवत्यंश, मेषान्तविषुवांश इन भुजों से उत्पन्न त्रिभुज के ज्याक्षेत्र (त्रिज्या कर्ण, विषुवांशज्या भुज, उसकी कोटिज्या कोटि इन तीनों भुजों से उत्पन्न) गोल केन्द्र से क्रान्तिज्या मूल तक द्युज्या कर्ण, व्यक्षोदयलवज्या भुज, गोल केन्द्र से भुजांशज्या मूल पर्यन्त कोटि इन तीनों भुजों से उत्पन्न त्रिभुज के सजातीय हैं इसलिये अनुपात करते हैं यदि मेषान्तद्युज्या में मेषान्त व्यक्षोदयलवज्या पाते हैं तो त्रिज्या में क्या इस अनुपात से मेषान्तविषुवांशज्या वा मेषान्त लङ्कोदयज्या आती है $\frac{\text{व्यक्षोदयलवज्या. त्रि}}{\text{मे द्यु}} = \frac{\sqrt{\text{मेषान्तज्या}^2 - \text{मेक्रांज्या}^2} \times \text{त्रि}}{\text{मेद्यु.}}$ = मेषोदयज्या, इसीतरह $\frac{\sqrt{\text{वृषान्तज्या}^2 - \text{वृक्रांज्या}^2} \times \text{त्रि}}{\text{वृ द्यु}}$ = ज्या (मेषोदय + वृषोदय) इसी तरह मिथुनान्त में भी होता है, इनके चापों को अधोऽधः शुद्ध करने से मेषादि राशियों के लङ्कोदय मान होते हैं। सिद्धान्त शेखर में श्रीपति "तत् क्रान्तिज्याकृति विरहितात्" इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से भास्कराचार्य भी "एकस्य राशेर्वृत्तीज्यका या" इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से आचार्योक्तानुरूप ही कहते हैं इति ॥१६॥

इदानीं स्वदेशोदयसाधनमाह

स्वचरासुभिरूनयुताः क्रमोत्क्रमस्थैः क्रमोत्क्रमस्थास्ते।
उदयप्राणा व्यस्ताश्चार्कं तात्कालिकं कृत्वा ॥ १७ ॥

वा. भा.—स्वचरखंडकै रूनयुताः किंभूताः क्रमोत्क्रमस्था अनन्तरं प्रक्रांताः निरक्षोदयाः किंभूतैश्चरासुभिः क्रमोत्क्रमस्थैः एवं कृते किं भवति। उदयप्राणाः भवन्ति। स्वदेशे राश्युदया इत्यर्थः ते च द्वादशा भवन्ति। एतदुक्तं भवति। निरक्षोदयान्मेषवृषमिथुनानि संस्थाप्य क्रमेण तत उत्क्रमेण संस्थाप्यंते चरखंडकानि तेषां पार्श्वतः तद्वत्तेन त्रीण्येव चरखण्डकानि शोध्यंते यथा स्वं तत उत्क्रमस्थादिषु मेषादिषु चरखंड कान्युत्क्रमस्थान्येव योज्यन्ते। एवं मेषादीनां मानं स्वदेशे राश्युदया भवन्ति। व्यस्तास्तुलादीनां स्वदेशोदया भवन्ति। तत्रैवं जातं मेषमीनयोस्तुल्योदयकालः एवं वृषकुम्भयोर्मिथुनमकरयोः कर्कधनुसिंहवृश्चिकयोः कन्यातुलयोरित्यर्कस्तत्कालिकं कृत्वा इत्येतत्सूचकखण्डकस्य कृतमुत्तरमत्र सम्बद्धं भविष्यतीति। अत्र या वासना मेषादीनां या स्व कोटयः उन्मंडलप्रापिण्यः प्राक् प्रदर्शिता तासां संख्याचापानि यानि स्वाहोरात्रवृत्तगतानि तेषां ये प्राणास्तैरुदयस्तेषां मेषादीनां प्रतिपादितः। एवं स्वदेशे पुनः तिर्यगक्षवशात् विषुवन्मंडलस्य च तद्वशेन स्वाहोरात्रवृत्तानां तत्र मेषादीनां त्रयाणां स्वोदयकोटयोऽपचीयन्ते तदपचयाच्चापानामथ पंचभिर्भवति अपचयश्च स्वे स्वाहोरात्रवृत्ते चरखंडतुल्ये तच्चरखंडानि निरक्षमेषवृषमिथुनादयः कालेभ्यो विशोध्यन्ते। यावद्विषुवन्मंडल-

पादस्योदयश्चरखंडतुल्यं खंडमनुचितं निष्टति । तावदयं मंडलपादः सकल एवोदेति, विषुवन्मंडलप्रथमपादावशेषस्य द्वितीयपादेन सहितस्योदयतो यावान्कालस्तावान् कर्कादिकस्यापमंडलपादस्य तृतीयपादचतुर्थस्य प्रथमपादवद्भासना योज्या गोलेष्येवं प्रदर्शयेदिति लग्नमुदयैः स्वैरित्यस्य प्रश्नसोपपत्तमार्योत्तरेणाह ।

वि. भा.—ते पूर्वप्रकारागता मेषादिराशित्रयनिरक्षोदयाः क्रमोत्क्रमस्थाः (क्रमस्था उत्क्रमस्थाश्च स्थाप्याः) यथाक्रमं क्रमोत्क्रमस्थैः स्वचरासुभिर्ऊनयुता सन्तो मेषादिषण्णां राशीनामुदयप्राणाः (स्वदेशोदयासवः) भवन्ति, अत एव व्यस्ता (विपरीताः) स्तुलादिषण्णां राशीनामुदय प्राणा (स्वदेशोदयासवः) भवन्ति, अर्कं तात्कालिकं कृत्वेत्यस्याग्रे सम्बन्ध इति ॥

अत्रोपपत्तिः

निरक्षस्वदेशार्कोदययोरन्तरं चरम् । मेषादिरेककालावच्छेदेन स्वदेशे निरक्षे च समुदेति, मेषान्तः प्रथमं स्वक्षितिजे ततः पश्चादुन्मण्डले लगत्यतश्चरखण्डोनो मेषोदयः स्वदेशोदयो भवति, वृषमिथुनयोरप्येवमेव, कर्कादौतु चरखण्डानामपचीयमानत्वाद्धनं तानि परिणमन्ति, तुलादौ तून्मण्डलस्य स्वक्षितिजादधः स्थितत्वात् चरखण्डानि धनं भवन्ति, मकरादौ तु चरखण्डानामपचीयमानत्वाद्ऋणमिति ॥ सिद्धान्त शिरोमणौ भास्कराचार्येण "क्रमोत्क्रमस्थाश्चरखण्डकैः स्वैः क्रमोत्क्रमस्थैश्च विहीनयुक्ताः । मेषादिषण्णामुदयाः स्वदेशे तुलादितोऽमीच विलोम संस्थाः" प्यनेनाऽऽचार्योक्तानुरूपमेव कथ्यत इति ॥ १७ ॥

अब स्वदेशोदय साधन को कहते हैं ।

हि. भा.— पूर्व प्रकार से आये हुए मेषादि तीन राशियों के निरक्षोदयमानों को क्रम से और उत्क्रम (विलोम) से स्थापित करना तथा क्रम क्रमस्थित और उत्क्रमस्थित अपनी चरासु को हीन और युत करने से मेषादि छः राशियों के स्वदेशोदयासुमान होते हैं ; इतने ही विपरीत तुलादि छः राशियों के स्वदेशोदयासु मान होते हैं ; 'अर्कं तात्कालिकं कृत्वा' इसका आगे से सम्बन्ध है इति ॥ १७ ॥

उपपत्ति

निरक्षार्कोदय और स्वदेशार्कोदय का अन्तर चर है, अपने देश में और निरक्ष देश में मेषादिबिन्दु एक ही काल में उदित होता है, मेषान्तबिन्दु पहले अपने क्षितिज में आता है उसके बाद चरखण्ड कालान्तर में उन्मण्डल में आता है इसलिए निरक्षदेशीय मेषोदयमान में से चरखण्ड को घटाने से स्वदेशीय मेषोदयमान होता है, इसी तरह वृष और मिथुन का भी होता है, कर्कादि में चरखण्डों के अपचीयमानत्व (ह्रासत्व) के कारण धन होते हैं ।

तुलादियों में अपने क्षितिज से उन्मण्डल के नीचा रहने के कारण चरखण्ड धन होते हैं। मकरादियों में चरखण्डों के अपचीयमानत्व के कारण ऋण होते हैं, सिद्धान्त शिरोमणि में भास्कराचार्य भी "क्रमोत्क्रमस्थाश्चरखण्डकैः" इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से आचार्योक्तानुरूप ही कहते है इति ॥ १७ ॥

इदानीं स्वदेशे लग्नानयनमाह

रविभुक्तहीनराशेः कलागुणाः स्वोदयासुभिर्भक्ताः ।
राशिकलाभिर्लब्धाः प्रश्नासुभ्योऽसवः शोध्याः ॥ १८ ॥

प्रक्षिप्य राश्यभुक्तं शेषासुभ्यः क्रमेण यावन्तः ।
शुद्ध्यन्त्युदयाः सूर्ये तावन्तो राशयः क्षेप्याः ॥ १९ ॥

शेषात् त्रिंशद्गुणितादविशुद्धस्योदयासुभिर्विभजेत् ।
लब्धं भागादिरवौ प्रक्षेप्य स्यात्तथा कृते लग्नम् ॥ २० ॥

वा. भा.—यस्मिन्नभीष्टे काले दिवसगते लग्नं कर्तुमिच्छति, तत्काले तात्कालिकं रविं कृत्वा रविणा राशेर्भुक्तायाः कलाः ताः ग्राह्याः ततस्ताः स्वोदयासुभिर्गुणयेत् । रव्याक्रान्तराशिप्राणैरित्यर्थः । राशिकलाभिरष्टादशशतैर्विभजेत् । फलंलिप्ताः ततस्ते लब्धासवः प्रश्नघटिकानां य असवः तेभ्यः शोध्याः रवावपि राश्यभुक्तं भागादिक्षेप्यमेवं भागादिरहितो वा भवति । ततः पुनः प्रश्नासूनां शेषासुभ्यः क्रमेण सूर्याध्यासितराशेरनन्तरं शुध्यति, उदया यावन्तः तेपि शोध्या । सूर्येऽपि तावन्तो राशयः क्षेप्याः । यो न शुध्यति राश्युदयस्तस्य येऽसवःतैरशुद्धोदयासुभिर्विभजेत् शेषानसून् । त्रिंशद्गुणितात्फलं भागादिर्भवति । तच्च रवौ प्रक्षिप्य तथाकृते अभुक्तभागयुतेतरशुद्धराशियुते वेत्यर्थः । एवमनेन प्रकारेण खे लग्नं भवति । अथ रात्रिगते काले लग्नं करोति । तदा तात्कालिकमर्कं षड्भयुतं कृत्वा तद्भुक्तभागैरनन्तरस्थराश्युदयैश्च दिवालग्नवद्रात्रिगतकालेन कर्म कर्तव्यमित्यतिप्रसिद्धत्वान्नायमर्थो विस्तरेण मया ख्यात इति । अत्र वासना, अपमंडलस्योदयतो यत्र तत्र प्रदेशे क्षितिजमंडलेन सह युतिस्तत्र तत्र प्रदेशेऽामंडले लग्नं भवतीत्यतो लग्नमुच्यते । क्षितिजापमंडलयुतिलग्नमित्याचार्येण गोलाध्यायेनाभिहितं मयापि तत्रैव व्याख्यातं तदर्कोदयकाले यत्रापमंडलावयवे रविः । तत्रैव लग्नं राशेरपि तत्रैव काले क्षितिजासक्तत्वात्ततो यथा यथा तेनाभ्युदयकालेन विप्रकृष्टो रविर्भवति । तथा तथार्क क्षितिजान्तर स्थितेनापमंडलखंडलकेनोपचितो रविः लग्नपदे सभागो भवति । अत उपचीयते तत्कथमिति जानामीतिवादो भुक्तराशेः कला इत्यादिकस्तद्वया, यत्र राशौ रविस्ततो भुक्तं रवेरग्रतः स्थितं यद्भागादिकं भुक्तिर्यतो कालेनोदेतीति कृत्वा तत्र राशिकादिलिप्ता भचरणराशेरुदयतः स्वोदयेन तद्वदधिको लग्नभागः प्राणा भवन्ति । तद-

भुक्तलिप्तानां क्रियन्ते इति फल प्रागाः तस्य राशिशेषस्योदयतो भवन्ति। अतस्तत्पूर्वापरायते सूर्ये परांशभुक्तं दीयते।

येन तदवधिके लग्नभोगः संगृहीतो भवति। तत् प्रश्नप्राणेभ्योऽन्येषि यावन्तोऽनंतरराश्युदयाः शुध्यन्ति तावन्तोऽपि राशयोऽयमंडल उदिताः तत्र काले भवन्त्यतो रवौ प्रक्षिप्यन्ते। तावानपि लग्नभागसंगृहीता भवन्ति। यस्य राशेरुदयप्राणाः न शुद्धयन्ति। सा राशिरुदयस्थिता वर्तते। तदवयवान्वये पुनः त्रैराशिकं यदि तदुदयप्राणैः त्रिंशद्भागा भवन्ति। तच्छेषप्राणैः कियन्त इति फल भागा, एवं षष्ट्यादि तदपि रवौ दीयते। येन क्षितिजासक्ता स्वका क्रांति प्रदेश योरन्तरे अपमंडलखण्डं संगृहीतं भवत्येतच्चार्केण तात्कालिकेन कर्म यतः सावनाहोरात्रस्य घटिका नाक्षत्रस्य रवेर्गति मत्वा अन्यथैकदिन भागोत्थैः प्राणैरधिका षष्टिघटिका नक्षत्राः स्युः न चैवं यस्माद्भुक्तिथिकरणलग्नछायादिषु सावन घटिकाभिरेव व्यवहारः तस्मात् तात्कालिकी करणमुत्पन्नमर्कस्य यतः सौरेण दिनेनार्कसावननक्षत्रयोरन्तरं दिनमेकं भवति। रात्रिगतेऽपि—अर्कास्तमयावधे योज्या वामनेयमिति तदेतत्साक्षावलंबचरखण्डस्योदयानयनं तत्सर्वं निरक्षदेशे उत्तरेण दक्षिणेन वा सर्वमन्यस्येति यथास्थितं गोले प्रदर्शयेदिति लग्नाद्घटिकाया वेत्तीत्यस्य प्रश्नस्योत्तर मार्यात्रयेणाह॥

वि. भा.—अर्कं (रविं) तात्कालिकं कृत्वा रविभुक्तकला रहितराशिकलाः (भोग्यकलाः) स्वोदयासुभिः (यस्मिन् राशौ रविस्तद्राशेः स्वदेशोदयासुभिः) गुणाः राशिकलाभिर्भक्ता लब्धा असवः प्रश्नासुभ्यः (इष्टकालासुभ्यः) शोध्याः, राश्यभुक्तं (राशिभोग्यांशान्) रवौ प्रक्षिप्य (संयोज्य), शेषासुभ्यः क्रमेण यावन्तो राश्युदयाः शुद्धयन्ति तावन्तो राशयः सूर्ये क्षेप्याः(योज्याः) शेषात् त्रिंशद्गुणितात्-अविशुद्धस्योदयासुभिः (अशुद्धराश्युदयासुभिः) विभजेत्, लब्धमंशादिकं रवौ क्षेप्यं (योज्यं) तथाकृते सति लग्नं स्यादिति॥१८-१९-२०॥

अत्रोपपत्तिः

उदयक्षितिजक्रान्तिवृत्तयोः सम्पातो लग्नम्। यस्मिन् राशौ रविरस्ति तद्राशेर्भोग्यकलाभिरनुपातः क्रियते यदि राशिकलाभिस्तद्राशेः स्वोदयासवो लभ्यन्ते तदा रविभोग्यकलाभिः किं समागच्छन्ति रविभोग्यासवः। एतेऽसव इष्टकालासु-(रविभोग्यासु लग्नभुक्तासु तदन्तरालोदयासूनां योगासु) भ्यः शोध्यास्तदा लग्नभुक्तासु रविलग्नान्तरालोदयास्वोर्योगा अवशिष्यन्ते, अत्र यावन्तो राश्युदयाः शुद्धा भवेयुस्ते शोध्याः शेषादनुपातो यद्यशुद्धोदयासुभिः (यस्य राशेरुदयमानं न शुद्धं तदुदयासुभिः)त्रिंशदंशा लभ्यन्ते तदा शेषासुभिः किं समागतांशादि फलेऽशुद्धपूर्वम-

षादिराशिभिर्युतं लग्नं भवेत् । परन्तु राशीनां स्थूलत्वात्तदुदयासवः स्थूला भवन्ति स्वदेशोदयमानवशेनैव लग्नानयनं सर्वैः प्राचीनैः कृतमस्त्यतस्तल्लग्नायनं न समीचीनं तत एव सिद्धान्तशिरोमणेष्टिप्पण्यां वापूदेवशास्त्रिणा शुद्धं लग्नानयनं कृतं परं तदपि समीचीनं नास्ति । म. म. पण्डित सुधाकरद्विवेदिना तत्खण्डनं कृतम् । "आकाशमध्यविषुवांशवशात्प्रकुर्याद्यष्टिं दिवाकरमपक्रमकोटिभागान् । यष्टिं जिनांशजगुणं विषुवांशकं च स्वाक्षाड्यहीनदिनभागमितं क्रमेण । सौम्यानुदग्गोलगते प्रकल्प्यसाध्यो भुजांशोऽथ भुजांशरव्योः । युतेर्मितं सायनलग्नमानं भवेत्स्फुटं गोलविदां बुधानामित्यनेन शुद्धं लग्नानयनं च कृतमस्ति, प्राचीनैः सूर्यसिद्धान्तकारादिभिर्निरयणरवित एव लग्नानयनं कृतमित्यपि तेषां दोषः, षड्भिः प्रकारैर्मयाऽपूर्वं लग्नानयनं कृतमस्ति, तज्ज्ञानार्थं मदीयं 'लग्नानयनम्' पुस्तकमवलोकनीय मिति ॥१८-१९-२०॥

अब स्वदेश में लग्नानयन को कहते हैं।

हि. भा.—रवि को तात्कालिक करके उनकी राशिभोग्यकला को स्वोदयासु (जिस राशि में रवि है, उसके स्वदेशोदयासु) से गुणा कर राशिकला से भाग देने से जो अस्वात्मक लब्धि हो उसको इष्टकालासु में से घटा देना, राशि के भोग्यांश को रवि में जोड़कर शेषासु में क्रम से जितने राश्युदयासु घटे उतनी राशि सूर्य में जोड़ देना, शेष को तीस से गुणा कर अशुद्धोदय (जिस राशि का उदयासु मान नहीं घटा है उससे) से भाग देकर जो अंशादिक लब्धि हो उसको रवि में जोड़ देना ऐसा करने से लग्न होता है इति ॥ १८-१९-२० ॥

उपपत्ति

उदयक्षितिज में क्रान्तिवृत्त का जो बिन्दु लगा है अर्थात् उदयक्षितिज और क्रान्तिवृत्त का सम्पात बिन्दु लग्न है, जिस राशि में तात्कालिक रवि है उस राशि की भोग्यकला से अनुपात करते हैं यदि राशिकला में उस राशि के स्वोदयासुमान पाते हैं तो रवि भोग्यकला में क्या इससे रवि का भोग्यासु प्रमाण आता है, इसको इष्टकालासु (रविभोग्यासु, लग्नभुक्तासु और रविलग्नान्तरालोदयासुओं के योग) में से घटा देना तब जो शेष रहे उसमें जितने राश्युदयमान घटे उन्हें घटा देना। शेष से अनुपात 'यदि अशुद्धराश्युदयासु में तीस अंश पाते हैं तो शेषासु में क्या' से जो अंशादिक फल आता है उसमें मेषादि से अशुद्ध राशि से अव्यवहित पूर्व राशितक राशि संख्या जोड़ने से लग्न होता है। परन्तु राशियों के स्थूलत्व के कारण उनका उदयमान भी स्थूल होता है, सब प्राचीनाचार्यों ने स्थूल राश्युदयमान ही के बल से लग्नानयन किया है इसीलिये वह ठीक नहीं है, अतः सिद्धान्त शिरोमणि की टिप्पणी में संशोधक (वापूदेव शास्त्री) ने शुद्ध लग्नानयन किया है। लेकिन वह भी ठीक नहीं है, महामहोपाध्याय पण्डित सुधाकर द्विवेदी ने उसका खण्डन किया है। और 'आकाशमध्य विषुवांशवशात्प्रकुर्यात्' इत्यादि संस्कृतोपपत्ति में लिखित श्लोकों से अपना शुद्ध लग्नानयन प्रकार किया है, सूर्य

सिद्धान्तकार ब्रह्म गुप्त आदि प्राचीनाचार्यों ने निरयणरवि ही से लग्नानयन किया है जो उचित नहीं है, सायन रवि से लग्नानयन करना समुचित है जैसा कि बहुत आचार्यों ने किया है, मैंने छः प्रकार से अपूर्व लग्नानयन किया है उसके लिये मेरी लग्नानयनम् पुस्तक को देखिये। इति ॥ १८-१९-२० ॥

इदानीं लग्नात् कालानयनमाह —

रविराश्य भुक्तलिप्तास्तदुदयगुणिता हृता गृहकलाभिः।
लब्धं प्राणाः स्याप्याः प्रक्षिप्यार्के गृहाभुक्तम् ॥२१॥

तावत्सूर्ये राशीन् क्षिपेत् समं राशिभैर्यावत् ।
क्षिप्तग्रहाणां प्राणान् प्रक्षिप्य स्थापितेष्वसुषु ॥२२॥

तदधिक कलोदयवधं राशिकलाभिर्भजेत् फलप्राणान्।
प्रक्षिप्य प्राणेषु प्राणाः सूर्योदयादसकृत् ॥२३॥

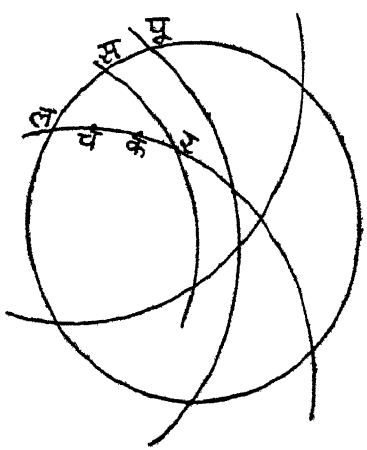
सु. भा.—यथा कालात् लग्नानयनमेवं लग्नाद्विपरीत्येन कालानयनं सिद्धमपि अनेनार्यात्रयेणोच्यते। यदेष्टकालिकं लग्नमुद्दिश्य कालं कश्चित् पृच्छति। तदौदयिकादादित्याद्राशिभुक्तिलिप्ता ग्राह्या। ताश्च तेनैव स्वराश्युदयेन संगुणय्य गृहकलाभिर्विभजेत्, अष्टादशशतैरित्यर्थः। फलं प्राणास्तदैकांते स्थाप्य ततो राशिभुक्तं सूर्ये क्षिप्त्वा अन्यानपि राशीन्तावत् क्षिपेत्। समो लग्नराशिभिर्विभजेत्। क्षिप्तराशीनां प्राणान् संकलय्य पूर्वस्थापितेषु योज्या, ततस्तस्य राशेरधिकं भागादि तस्याधिकस्य याः कलाः तासां तदुदयप्राणानां च यो वधः तं च राशिकलाभिर्विभजेत् फलं प्राणाः तांश्च प्रक्षिप्य पूर्वस्थापितेषु प्राणा दिनगता भवन्ति। सूर्योदयात्तदधिकं भागादिसूर्ये प्रक्षिपेदेवं प्राणा उत्पद्यन्ते। ते स्थूला यतस्तात्कालिकेनार्केण कृतास्तदर्थमसकृद्ग्रहणं तेत कालेन रविं तात्कालिकं कृत्वा रविराशिभुक्तलिप्ता इत्यादिना कालानयनं तावद्यावत्स्थिरकालो भवति रविश्च तत्र कालेन लग्नमधश्च रात्रिगते काले तदा षड्भयुतार्कलग्नयोरंतरात्प्राग्वदिति। अत्रापि वासना सैव रव्याक्रान्तक्षितिजासक्तं देशयोरन्तराले यदपमंडलखंडं तद्योगतो ये प्राणाः स कालः पूर्वमेव तात्कालिकार्कोपरिवदसकृत्करणमपि कालस्योपपन्नं दिवारात्रिलग्नाभ्यामिति। अथ रात्रिशेषकाले लग्नं कर्तुमिच्छति तदर्धमार्याधमाह ॥

वि. भा.—रविराश्यभुक्तलिप्ताः। रविराश्याधिष्ठितभोग्यकलाः तदुदयगुणिताः राश्युदयगुणिताः गृहकलाभिः अष्टादशशतकलाभिर्हृताः। लब्धं प्राणा भोग्यासवो भवन्ति ते एकत्र स्थाप्याः। ततो लग्न राशितः पूर्वं यावन्तो राशयःस्युः तान् राशीन् सूर्ये क्षिपेत्, तेषां रविक्षिप्तराशीनां प्राणान् संकलय्य पूर्वस्थापितेषु

असुषु पूर्वानीतभोग्यासु प्रमाणेषु प्रक्षिप्य संयोज्य स्थापयेत् । तदधिककलोदयवधं लग्नभुक्तकला तदधिष्ठित उदयासवश्च अनयोर्वधः, तं अष्टादशशतकलाभिर्भक्तं लब्धासवः लग्नभुक्तासवः पूर्वानीतेषु असुषु प्रक्षेप्याः तदा इष्टघटिकासवो भवन्ति 'सूर्योदयमारभ्य आसां घटीनां सावनार्कमत्वात् असकृत् कर्म कर्तव्यं सावनेष्टया-घटिकया वा ।

अत्रोपपत्तिः ।

"अत्र" अर्कस्य भोग्यः तनुभुक्तयुक्तो मध्योदयाद्ध्यः समयो विलग्नात् "इति भास्करीयलग्नात् कालानयनविधानेन तात्कालिकार्कस्य भोग्यासूनां लग्नस्य भुक्तासूनां च तदन्तरालयराश्युदयासुभिश्च समाहारेण तात्कालिकार्कोपरिगता-होरात्रवृत्तक्षितिजवृत्त सम्पातावधिसावनेष्टघटिका प्रमाणा भवतीति गोले प्रत्य-क्षमेव स्पष्टम् ।



ल=लग्नम्
र=तात्कालिक रवि
क=रविराशेरन्तः
च=लग्नराशेरादिः
कच=तदन्तरालराशयः
रस=सावनेष्टघट्यः
स=रव्युपरिगताहारोत्रवृत्तक्षितिजवृत्ति-संपातः ।

अत्र रस चापात्मकेषु प्राणेषु रक चापासवः कच चापासवस्तथा चल चापासवश्च वर्तन्ते तदर्थमाचार्येणेह रकचापासवः $=\frac{\text{रविराश्यभुक्तक}\times\text{उ अ}}{१८००}=$ भोग्यासु । एवं लग्नभुक्तासवः चल चापासवः $=\frac{\text{उअ}\times\text{लग्नभुक्तकला}}{१८००}$, एवं च क च पूर्णराश्युदयासवश्च ये स्युस्तान् सर्वान् संपीड्य रविसावनेष्टघटिकासु मानं भवति । एषामसूनां चलत्वादसकृत्कर्मकरणेन स्फुटं सावनेष्टघटीमानं भवति । अतएव भास्कराचार्याः । "लग्नार्थमिष्टघटिका यदि सावनास्तास्तात्कालिकार्क-करणेन भवेयुरार्क्ष्यं" इति गोले विशेषं विहितवन्तः॥ २१-२२-२३ ॥

हि. भा.—पूर्व ग्रन्थ में तात्कालिक सूर्य और इष्ट काल जान कर लग्न बनाया गया

है अब इस ग्रन्थ से तात्कालिक के रवि और लग्न ज्ञात करके इष्ट काल बनाते हैं।

गोल युक्ति से तात्कालिक रवि केन्द्र के ऊपर जो अहोरात्र वृत्त होता है, उसमें क्षितिजपर्यन्त इष्टकाल कहा गया है, इस इष्ट काल में उदयक्षितिज से ऊपर सूर्य के रहने पर रवि के भोग्यासु, लग्न के भुक्तासु और तदन्तर्गत राशियों का उदयासु, इन सबों का योग है। अत एव आचार्य कहते हैं कि रवि के भोग्यांश को वर्तमान राश्युदय से गुणा कर राशिकला (१८००) से भाग देकर जो लब्धि आया वह रवि के भोग्यासु हैं, उसमें लग्न के भुक्तासु अर्थात् लग्न के भुक्तांश को राश्युदय से गुणा कर राशि कला से भाग देकर लब्धि तुल्य रवि के पूर्वानीत भोग्यासु में जोड़ते हैं, और अन्तरवर्ती राशियों का उदयासु जोड़कर सावनेष्ट घटी आचार्य बनाते हैं, परंच सावनेष्ट घटी चल है इसलिये आचार्य यहां असकृत् कर्म करते हैं।

आचार्य का सकृत् कर्म कहने का तात्पर्य यह है कि नाक्षत्रेष्ट घटी ज्ञात नहीं है, ज्ञात है सावनेष्ट घटी अत एव उक्त प्रकार द्वारा नाक्षत्रेष्ट घटी ज्ञात नहीं होगी। इसलिए सकृत् क्रिया द्वारा इष्ट घटी स्थिर की जाती है। भास्कराचार्य भी कहते हैं कि "लग्नार्थमिष्टघटिका यदि सावनास्ताः तात्कालिकार्ककरेण भवेयुरार्क्ष्यः" का गोलाध्याय में विशेष वर्णन है॥ २१-२२-२३॥

इदानीं विलोमलग्नं ततः कालानयनं चाह

प्रागुदये प्रश्नासुभिरूनोऽर्को भुक्तराशिभिर्लग्नम्।
कृत्वैवमूनमर्कं लग्नसमं प्राग् भवेत्कालः॥ २४॥

वा. भा.—अर्कोदयात्प्राग् यदा क्रियते तदा रात्रिशेषघटिकाभिः स्वोदये प्राग्वत् कर्म यदि नामभुक्तराशिभिरेतदुक्तं भवति। तात्कालिकाद्रवेर्भुक्तभागान् संलिख्य लिप्ता कार्याः तदुदयाक्रान्तराश्युदयप्राणैः शेषं संगुण्य अष्टादशशतैः विभजेत् फलं प्राणाः प्राणेभ्यो विशोध्य सूर्याच्च राशिभुक्तं विशोधयेत्। विशेषणेभ्यो भुक्तं राशिप्राणास्तावच्छोध्या यावच्छुध्यन्ति सूर्यादपि तावत्संख्या यश्च शोध्या अशुद्धराश्युदयप्राणैः शेषप्रश्नप्राणेभ्यस्त्रिदशादिगुणितेभ्यो यत्फलं ादि तदपि रवेः संशोध्य रात्रिशेषे लग्नं भवति। वासनामुखे पश्चाल्लग्नमतो भुक्तेन राशिखण्डेनापचितोऽर्को लग्नं भवति गोले चन्द्रं प्रदर्शयेदिति द्वितीयेन रात्रिशेषलग्नात्कालानयनमाह।

यथाकालेनोनोऽर्कस्वदेशराश्युदयेनोनत्वं प्राप्तः एवं वैपरीत्येन लग्नस्य समं ीतः कालांकितो भवति एतदुक्तं भवति लग्नाद् भुक्तभागैरर्काच्च भुक्तभागैरंतरा युदयैश्चैभिस्तै र्यः कालः स रात्रिशेषो भवति रविरप्यूनो लग्नसमो भवति। नाप्यत्र क्षितिजादधः स्थिते नापमंडलखण्डेन योज्या प्राग्वदिति यच्छाया गतेन ृेण नतच्च वेत्तीत्येतस्य प्रश्नस्योत्तरं बहुभिः प्रकारैः वक्ष्यति तत्र तावदेकेन

प्रकारेण गतशेषाच्छेदानयनमार्यासार्धमाह ।

वि. भा.—प्रश्नासुभिः प्राक् (पूर्वं) उदये (लग्ने) ऽपेक्षिते भुक्तराशिभिः (भुक्तांशैर्भुक्तराशिभिश्च) अर्कः (रविः) ऊनः (हीनः) तदा लग्नं भवेत् । लग्नात्कालानयने चार्कलग्नसममूनं कृत्वा एवं प्राक् (सूर्योदयात्पूर्वं) इष्टकालो भवेत् । एतदुक्तं भवति तात्कालिकरवेर्भुक्तांशादिकं रव्याक्रान्त राश्युदयमानेन सङ्गुण्य त्रिंशद्भिर्भक्ता लब्धमिष्टासुभ्यो विशोध्य रवेश्च राशिभुक्तं शोधयेत् । शेषासुभ्योऽपि यावन्तो राशयोऽभुक्ताः शोधयितुं शक्यन्ते तेषामसून् विशोध्य रवावपि तावन्तो राशयः शेषासूंश्च त्रिंशता संगुण्याशुद्धराश्युदयेन विभज्य लब्धमंशादिकं रवेः शोध्यं तदा रात्रिशेषे लग्नं भवति । एवमानीताद्रात्रिशेषलग्नात्कालानयनार्थं यथापूर्वं कालेन हीनो रविः स्वदेशराश्युदयलब्धराश्यंशादिभिर्लग्नत्वं प्रापितः एवं वैपरीत्येन लग्नस्य समतां नीते सति तस्मिन् रवौ रात्रिशेषगतकालः स्पष्टो भवति । एतदुक्तं भवति, लग्नादभुक्तांशैरन्तरराश्युदयैश्च संमिलितैरसकृत्कर्मणा रात्रिशेषे स्फुटः कालो भवति । रविरभुक्तांशराश्यादिरहितो लग्नसमो भवतीति ॥ २४ ॥

अत्रोपपत्तिरपि भाष्यरूपैव बोध्या । सिद्धान्तशेखरे 'प्रागुद्गमादपि रविर्गंतराशिभागैः प्राग्भोदयैर्विरहितश्च विलग्नमेवमूने रवौ तनुसमे च कृते स कालः' श्रीपत्युक्तमिदं शिरोमणौ 'भुक्तासुशुद्धेर्विपरीतलग्नमित्यादि' भास्करोक्तं चाऽऽचार्योक्तानुरूपमेवेति ॥ २४ ॥

अब विलोमलग्न को तथा उससे कालानयन को कहते हैं ।

हि. भा.—इष्टासु से पूर्व लग्न अपेक्षित हो तो भुक्तांश और भुक्तराशियों को रवि में से घटाने से लग्न होता है, तात्कालिक रवि के भुक्तांशादि का जिस राशि में रवि हो उस राशि के उदय मान से गुणाकर तीस से भाग देकर जो लब्धि हो उसको इष्टासु में से घटा कर सूर्य में से भी राशिभुक्त को घटा देना चाहिए । शेषासु में जितनी अभुक्त राश्युदयमानासु घट सके उन्हें घटा कर रवि में से भी उतनी राशियां घटा देना, शेषासु को तीस से गुणाकर अशुद्धराशि के उदयमान से भाग दे कर जो अंशादिक लब्धि हो उसको रवि में से घटाने से रात्रि शेष में लग्न होता है । इस तरह लाए हुए रात्रि शेष लग्न से अभुक्तांश और रवि तथा लग्न के अन्तर में जो राशियां हैं उनके उदयमानों के योग से असकृत् कर्म से रात्रि शेष में स्फुट इष्टकाल होता है ॥ २४ ॥

इसकी उपपत्ति भाष्य रूप ही है । सिद्धान्त शेखर में "प्रागुद्गमादपि रविर्गंतराशिभागैः इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से, श्रीपति तथा सिद्धान्त शिरोमणि में "भुक्तासु शुद्धे विपरीतलग्नं" इत्यादि से भास्कराचार्योक्त के अनुरूप ही कहा है इति ॥२४॥

इदानीमिष्टशंकुमाह ।

गतशेषाल्पस्यान्हः सौम्येतरगोलयोश्चरार्धेन ।
ऊनाधिकस्य जीवा स्वाहोरात्रार्धसंगुणिता ॥२५॥

त्रिज्याहृता युतोना क्षितिज्यया सौम्ययाम्ययोश्छेदः ।
छेदोऽवलम्बकगुणो व्यासार्धविभाजितः शंकुः ॥२६॥

वा. भा.—यत्राभीष्टकाले दिनगतच्छायां कर्तुमिष्यते । तात्कालिकदिनगतकालो ज्ञेयः किं शेषात्तयोर्योऽल्पः स गृह्यते तस्य दिनगतस्य बाल्याद्धे जीवा कार्या किं भूतश्चरार्धेनोनाधिकस्य उत्तरगोले चरदलं घटिकाभिरुन्नतस्य याम्ये युतस्य ततः सा जीवा स्वाहोरात्रार्धसंगुणिता त्रिज्याहृता सती युतोना कार्या कर्येत्याह क्षितिज्यया यथासंभवं सौम्ययाम्यगोलयोः स्थितेऽर्के छेदो भवति । अत्र वामनास्वदेशाक्षप्रयोर्गोले विन्यस्य प्रदर्शयेत् । तद्यथाभीष्टदिनस्वाहोरात्रं घटिकांकित विन्यस्याग्राः सूत्रे च दक्षिणोत्तरायते प्राग्वद् बद्ध्वा क्षितिजमण्डले पूर्वापरयोस्तदग्रयोश्च पूर्वापरयोर्मध्ये उदयास्तसूत्रं च । बद्ध्वा उन्मण्डल स्वाहोरात्रमंपातयोः पूर्वापरायतं निरक्षोदयास्तसूत्रं च बध्नीयात् । एवं स्थिते क्षितिजस्वाहोरात्रमंपातात् यावत्यो घटिका दिनगताः तावत्यः स्वाहोरात्रवृत्तेन संगुण्याग्र्य तदग्रे कोऽपि लक्षितं चिह्नं कार्यं तावानुदयस्तत्र काले रविगतिघटिकाश्च एव ताभ्यश्चरदलघटिका उत्तरगोले शोध्यन्ते यत उन्मण्डलावधेरुपर्यधश्च क्रमज्याश्च प्रवर्तते । तच्चोन्मंडलमुपरिक्षितिजात्स्थितम् चरदलाख्येन । स्वाहोरात्रखंडलकेनात्तस्मदेवं विशोध्यते क्षितिजाकांतरस्थितस्वाहोरात्रखण्डात् । येन शेषघटिकानां या ज्या तस्योन्मंडलावधि भवति । दक्षिणगोले चरदलघटिका योज्यन्ते दिनगते काले यतः क्षितिजादध उन्मण्डलं तच्च क्रमस्य प्रवृत्ती रवेरुन्मण्डलाक्रांतस्थितं स्वाहोरात्रखंडं संगृहीतं भवति तस्य ज्योन्मंडलावधि भवति । व्यासार्धवृत्तकल्पनाया: सतः स्वाहोरात्रवृत्ते परिणम्यते तदर्थमिदं त्रैराशिकं यदि व्यासार्धवृत्तो एतावती स्वाहोरात्रे क्रियतीति फलं स्वाहोरात्रवृत्तनिष्पन्नरव्युन्मंडलान्तरज्या भवति । स्वक्षितिजार्कान्तरज्या च स्वाहोरात्र निष्पन्नेष्यते । अतः सौम्यगोले क्षितिज्यायुता क्रियते क्षितिजस्याधः स्थितत्वात्, क्षितिज्यया हीना क्रियते येन स्वक्षितिजावधेर्ज्या भवति सैव स्वाहोरात्रेऽभीष्टज्योच्यते स्वक्षितिज्या चार्यभटादिष्वस्मत्सिद्धान्ते च छेद इत्यभिधीयते ।

तावत्प्रमाणं सूत्रं मेषतुलादौ क्षितिज्यया युतं सत् द्विगुणं कृत्वा तत् सूत्रस्यैकमग्रं रविचिह्ने स्वाहोरात्रवृत्ते बद्ध्वा द्वितीयमग्रं निरक्षक्षितिजाधः

तावत्येव धनुः स्वाहोरात्रवृत्त एव बध्नीयात् तत्र स्वोदयास्तसूत्रावच्छिन्नं छेदः तस्यैव सूत्रस्य निरक्षोदयास्तसूत्रेण सह यत्र संपातः तदवच्छिन्ना जीवेत्युच्यते। व्युत्मंडलयोरन्तरज्या पूर्वप्रदर्शिता। छेदतुल्येन व्यासार्धेन दक्षिणोत्तरायतं वृत्तं छेदोदयास्तमध्ये सूत्रं कृत्वा विन्यसेत्, तस्य छेदः कर्णः शंकुः कोटिशंकुतलं भुजा यथा विषुवन्मध्याह्ने व्यासार्धकर्णोऽवलंबककोटिरक्षज्या भुजज्या याम्योत्तरमंडलस्येत्यनेन बीजेन सर्वाण्येव छायानयनानि भवन्ति इत्येतत्सर्वं गोले प्रदर्शयेदिति तदिदानीं शंक्वानयनार्थमुत्तरमार्यामाह।

छेदोऽवलंबकगुणो व्यासार्ध विभाजितः शंक्वोरन्तरमेव छेद आनीतः सोऽवलंबकगुणव्यासार्धविभाजितशंकुर्भवति। अत्र वासना स्वाहोरात्रवृत्तेरव्युपलक्षितचिह्नं सूत्रस्यैकमग्रं बद्ध्वा द्वितीयमग्रं गुरु कृत्वाऽवलंबयेत्। ततो भूमध्यात् द्वितीयं सूत्रं प्रसार्यावलंबितसूत्रस्पृक् क्षितिजे बध्नीयात्। तयोः सूत्रयोः यः संपातस्तत्र शंकुमूलं तस्मादुपरि रविं यावत् शंकुर्भूगोलमध्यं यावत् छाया तस्य शंकोश्छेदमूलं यावच्छंकुतलं छेदश्च कर्णो त्रैराशिककल्पनेयं यदि व्यासार्धतुल्येऽवलंबकतुल्यः शंकुः तदत्र छेदतुल्ये याम्योत्तरमंडले कियान् शंकुरिति फलमिष्टकाले दिनगतशंकुर्भवत्येव, एवमपराह्णेऽपि योज्या वासना। तुल्यत्वाद्रविछायास्ते इत्येतत्सर्वं गोले प्रदर्शयेदिति इदानीं छायां द्वितीयप्रकारेणानयनार्थमार्यार्धमाह॥

वि. भा.—अन्हो दिवस्य गतशेषाल्पस्य (गतशेषयोर्योऽल्पस्तस्यार्थादुन्नतकालस्य) सौम्येतरगोलयोः (उत्तरदक्षिणगोलयोः) चरार्धेन क्रमेणोनाधिकस्य जीवा (सूत्रं) स्वाहोरात्रार्धसंगुणिता (द्युज्यांगुणा) त्रिज्याहृता (त्रिज्याभक्ता) तदा कला भवति, सा सौम्ययाम्ययोः (उत्तरदक्षिणगोलयोः) क्षितिज्यया (कुज्यया) युतोना तदा छेदो (इष्टहृतिः) भवेत्। छेदः, अवलम्बकगुणः (लम्बज्यागुणितः) व्यासार्ध विभाजितः (त्रिज्याभक्तः) तदेष्टशंकुर्भवेदिति॥२५-२६॥

अत्रोपपत्तिः।

याम्योत्तरवृत्तात्पूर्वमर्थान्मध्यान्हात्पूर्वं यत्र रविर्वर्त्तते तत उदग्क्षितिजं यावद्दिनगतकालः, रवितोऽस्तक्षितिजं यावद्दिनशेषकालोऽत्र दिनशेषकालाद्दिनगतकालोऽल्पोऽस्त्यतः स एवोन्नतकालोऽर्थात्क्षितिजाहोरात्रवृत्तयोः सम्पातोपरिगतध्रुवप्रोतवृत्तनाड़ीवृत्तयोः सम्पाताद्व्युपरिगतध्रुवप्रोतवृत्तनाड़ीवृत्तयोः सम्पातं यावत्। उत्तरगोले क्षितिजाहोरात्रवृत्तयोः सम्पातोपरिगतध्रुवप्रोतवृत्तं नाड़ीवृत्ते

पूर्वस्वस्तिकाच्चरान्तरेऽधोलगत्यत उन्नतकाले तच्चरमानमूनीक्रियते तदा पूर्वस्वस्तिकाद्व्युपरिगतध्रुवप्रोवृत्तनाड़ीवृत्तयोः सम्पातं यावत्सूत्रचापं भवेदेतस्य ज्या सूत्रसंज्ञकम्। दक्षिणगोले क्षितिजाहोरात्रवृत्तयोः सम्पातोपरिगतध्रुवप्रोतवृत्तं नाड़ीवृत्ते पूर्वस्वस्तिकाच्चरान्तरं उपरिलगत्यत उन्नतकाले चरयोजनेन सूत्रचापं भवति, तज्ज्या सूत्रमर्थाद्व्युपरिगतध्रुवप्रोतवृत्तनाड़ीवृत्तयोः सम्पातात्पूर्वापरसूत्रोपरिलम्बः, ध्रुवात्पूर्वस्वस्तिकं यावदुन्मण्डले नवत्यंशाः ध्रुवादेव व्युपरिगतध्रुवप्रोतवृत्तनाड़ीवृत्तयोः सम्पातं यावत् ध्रुवप्रोतवृत्ते नवत्यंशाः नाड़ीवृत्ते सूत्रचापम्। एतद्भुजत्रयैरुत्पन्नत्रिभुजस्य ज्याक्षेत्रं (त्रिज्याकर्णः सूत्रं भुजः, सूत्रकोटिज्या कोटिः) ध्रुवाद्रवि यावद् द्युज्याचापम् ध्रुवादुन्मण्डलाहोरात्रवृत्तयोः सम्पातं यावद् द्युज्याचापम्। अहोरात्रवृत्ते ध्रुव प्रोतवृत्तोन्मण्डलयोरन्तर्गतं चापमेतत्त्रिभुजस्य ज्याक्षेत्रेण (द्युज्याकर्णः, रवितो निरक्षोदयास्तसूत्रोपरिलम्बः कलासंज्ञको भुजः, कलामूलादहोरात्रवृत्तगर्भकेन्द्रं यावत्कोटिः) सजातीयमतोऽनुपातो यदि त्रिज्यया सूत्रं लभ्यते तदा द्युज्यया किमिति समागच्छति कला $=\frac{\text{सूत्र}\times\text{द्यु}}{\text{त्रि}}$ रवितः स्वोदयास्त सूत्रोपरिलम्ब इष्टहृतिः, स्वोदयास्तसूत्रनिरक्षोदयास्तसूत्रयोरन्तरमिष्टहृतिखण्डं कुज्याऽस्ति, उत्तरगोले निरक्षोदयास्तसूत्रात्स्वोदयास्तसूत्रस्याधः स्थितत्वात् कुज्यया युता कलेष्टहृतिर्भवेत्। दक्षिणगोले तु कुज्यया हीना कलेष्टहृतिः। ततस्त्रिज्याकर्णः अक्षज्याभुजः। लम्बज्या कोटिरिति भुजत्रयैरुत्पन्नमेकमक्षक्षेत्रम्। तथेष्टहृतिः कर्णः। इष्टशंकुः कोटिः शंकुतलं भुजः एतद्भुजत्रयैरुत्पन्नं द्वितीयमक्षक्षेत्रमेतयोः सजातीयत्वादनुपातो यदि त्रिज्यया लम्बज्या लभ्यते तदेष्टहृतौ किं समागच्छतीष्टशंकुः $=\frac{\text{लंज्या.}\ \text{इहृ}}{\text{त्रि}}$ एतेनोपपन्नमाचार्योक्तम्। सिद्धान्तशिरोमणौ भास्कराचार्येणा "अथोन्नतादूनयुताच्चरेण क्रमादुदग्दक्षिणगोलयोर्ज्या। स्यात्सूत्रमेतद्गुणितं द्युमौर्व्या व्यासार्धभक्तं च कलाभिधानम्" ऽप्यनेनाचार्योक्तानुरूपमेव कथ्यत इति॥

अब इष्ट शंकु के साधन को कहते हैं।

हि. भा.—दिनगत और दिनशेष में जो अल्प रहता है वह उन्नत काल है, उत्तर गोल में उन्नत काल में चरार्ध को घटाने से और दक्षिण गोल में उन्नत काल में चरार्ध को जोड़ने से जो होता है उसकी ज्या (सूत्र) को द्युज्या से गुणा कर त्रिज्या से भाग देने से कला होती है, उत्तर गोल में कला में कुज्या को जोड़ने से और दक्षिण गोल में कला में कुज्या को घटाने से छेद (इष्ट हृति) होता है, इष्टहृति को लम्बज्या से गुणा कर त्रिज्या से भाग देने से इष्टशंकु होता है इति॥ २५-२६॥

उपपत्ति ।

मध्यान्ह से पूर्व जहां रवि है वहां से उदयक्षितिज पर्यन्त दिनगत काल है, और रवि से अस्तक्षितिज पर्यन्त दिन शेष काल है, यहां दिन शेष काल से दिनगत काल अल्प है इसलिये वही (दिनगत काल) उन्नत काल होता है, दिनगत काल से दिनशेष काल के अल्प रहने से वही (दिनशेष काल) उन्नत काल (क्षितिज वृत्त और अहोरात्र वृत्त के सम्पातोपरिगत ध्रुवप्रोतवृत्त और नाड़ी वृत्त के सम्पात से रव्युपरिगत ध्रुवप्रोतवृत्त-नाड़ी वृत्त के सम्पात पर्यन्त नाड़ीवृत्त में) होता है, उत्तर गोल में क्षितिजवृत्त और अहोरात्रवृत्त के सम्पातोपरिगत ध्रुव प्रोतवृत्तनाड़ीवृत्त में पूर्वस्वस्तिक से चरान्तर पर नीचा लगता है इसलिये उन्नत काल में से चर को घटाने से पूर्वस्वस्तिक से रव्युपरिगत ध्रुव प्रोतवृत्त नाड़ीवृत्त के सम्पात पर्यन्त नाड़ीवृतीय चाप चापसूत्र होता है इसकी ज्या सूत्र संज्ञक है, दक्षिण गोल में क्षितिजवृत्त और अहोरात्र वृत्त के सम्पातोपरिगत ध्रुवप्रोतवृत्त नाड़ीवृत्त में पूर्वस्वस्तिक से चरान्तर पर ऊपर लगता है इसलिये उन्नत काल में चर को जोड़ने से सूत्रचाप होता है इसकी ज्या (रव्युपरिगत ध्रुवप्रोतवृत्त नाड़ीवृत्त के सम्पात से पूर्वापर सूत्र के ऊपर लम्बरेखा) सूत्र है, ध्रुव से पूर्व स्वस्तिक पर्यन्त उन्मण्डल में नवत्यंश, ध्रुव से रव्युपरिगत ध्रुवप्रोतवृत्त नाड़ीवृत्त के सम्पात पर्यन्त ध्रुवप्रोतवृत्त में नवत्यंश, नाड़ीवृत्त में सूत्रचाप, इन तीनों भुजों से उत्पन्न त्रिभुज का ज्याक्षेत्र (त्रिज्या कर्ण, सूत्र भुज, सूत्र कोटिज्या कोटि) ध्रुव से रविपर्यन्त द्युज्याचाप, ध्रुव से उन्मण्डलाहोरात्रवृत्त के सम्पात-पर्यन्त उन्मण्डल में द्युज्याचाप, अहोरात्रवृत्त में ध्रुवप्रोतवृत्त और अहोरात्रवृत्त के अन्तर्गत चाप, इन तीनों भुजों से उत्पन्न त्रिभुज के ज्याक्षेत्र (अहोरात्र वृत्त के गर्भकेन्द्र से रवि-पर्यन्त द्युज्या कर्ण, रवि से निरक्षोदयास्त सूत्र के ऊपर लम्ब कला संज्ञक भुज, कलामूल से अहोरात्रवृत्त के गर्भ केन्द्र पर्यन्त कोटि) का सजातीय है इसलिये अनुपात करते हैं यदि त्रिज्या में सूत्र पाते हैं तो द्युज्या में क्या इससे कला आती है, $\frac{\text{सूत्र. द्यु}}{\text{त्रि}}$ = कला, रवि से स्वोदयास्त सूत्र के ऊपर लम्ब रेखा इष्टहृति है, स्वोदयास्तसूत्र और निरक्षोदयास्तसूत्र का अन्तर इष्ट हृति की खण्ड कुज्या है, उत्तरगोल में निरक्षोदयास्तसूत्र से स्वोदयास्त सूत्र नीचा है इसलिये कला में कुज्या को जोड़ने से इष्ट हृति होती है, दक्षिण गोल में कला में से कुज्या को घटाने से इष्ट हृति होती है, त्रिज्या कर्ण, अक्षज्या भुज, लम्बज्या कोटि इन तीनों भुजों से उत्पन्न एक अक्षक्षेत्र, तथा इष्टहृति कर्ण, इष्टशंकु कोटि, शंकुतल भुज, इन तीनों भुजों से उत्पन्न द्वितीय अक्षक्षेत्र, दोनों अक्ष क्षेत्र सजातीय हैं इसलिये अनुपात करते हैं यदि त्रिज्या में लम्बज्या पाते हैं तो इष्टहृति में क्या इससे इष्टशंकु प्रमाण आता है $\frac{\text{लंज्या. इह}}{\text{त्रि.}}$ = इशं, इससे आचार्योक्त उपपन्न होता है, सिद्धान्त शिरोमणि में भास्कराचार्य भी "अथोन्नताद्दून-कृताक्षरेण' इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से आचार्योक्तानुरूप ही कहते हैं इति ॥२५-२६॥

इदानीं प्रकारान्तरेण शंकुं ततो दृज्ज्यां चाह—

विषुवत्कर्णविभक्तश्छेदो वा द्वादशाहतः शंकुः ।
शंकुकृतिविहीनाया व्यासार्धकृतेः पदं दृज्ज्या ॥ २७ ॥

वा. भा.—अथवा अन्तरानीतछेदो द्वादशगुणो विषुवद्विभक्तश्च शंकुर्भवति । अत्र वासना यदि विषुवत्कर्णव्यासार्धवृत्ते द्वादशिका कोटिः तच्छेदव्यासार्धवृत्ते क्रियतेति फलं कोटिरूर्ध्वा शंक्वाख्या अत्र विषुवन्मध्याह्ने याम्योत्तरमण्डलस्य व्यासार्धस्थाने विषुवत्कर्णः कल्पितोऽवलंबस्थाने द्वादश शंकुं कुर्यात् । प्रागेवोक्तं शंकुतलमिति । तत्सर्वमुपपन्नमिति तदानीं गम्यवृहच्छंकोश्छायानयनार्थं द्वितीयमार्यार्धमाह ।

शंकुकृतिविहीनाया व्यासार्धकृते फलं दृज्ज्या ॥ अनन्तरानीत शंकोर्वर्गोनायाः कस्याः व्यासकृतेः यच्च मूलं दृज्ज्या सा भवति । दृज्ज्या छायोच्यते वासना तु स्वाहोरात्रवृत्ते ऽप्युपलक्षितचिह्नमध्यं यावत्कर्णस्तिर्यक् तत् एव रविचिह्नादवः सूत्रं भूमध्यविनिर्गतक्षितिजप्रान्तसूत्रावच्छिन्नं शंकुकोटिः । तन्मूलाद्भूमध्यं यावत् दृज्ज्या छाया भुजः, कर्णकृतेः कोटिकृतिं विशोध्य मूलं भुज इत्यर्थः । सर्वं गोले प्रदर्शयेदिति एवं महाशंकुछायामानीय द्वादशांगुलछायानयनार्थमार्यार्धमाह ॥ २७ ॥

वि. भा.—छेदः (हृतिः) द्वादशाहतः (द्वादशभिर्गुणितः) विषुवत्कर्णविभक्तः (पलकर्णभक्तः) तदा वा (प्रकारान्तरेण) शंकुर्भवेत् । शंकुकृतिविहीनाया (शंकुवर्गरहितायाः) व्यासार्धकृतेः (त्रिज्यावर्गात्) पदं (मूलं) तदा दृज्ज्या भवतीति ॥ २७ ॥

अत्रोपपत्तिः

पलभा भुजः, द्वादशांगुलशंकुः कोटिः, पलकर्णः कर्णः, } अनयोरक्षक्षेत्रयोः
शंकुतलं भुजः, इष्टशंकुः कोटिः, छेदः (हृतिः) कर्णः, } सजातीयत्वादनुपातो

यदि पलकर्णे कर्णे द्वादशांगुल शंकुः कोटिर्लभ्यते तदा हृतिकर्णे किं समागच्छतीष्ट शंकुः $= \frac{१२ \times \text{हृति}}{\text{पक}}$, ततो दृज्ज्या भुजः । इष्टशंकुः कोटिः । त्रिज्या कर्णः, एतैर्भुजकोटिकर्णैरुत्पन्न छायाक्षेत्रे $\sqrt{\text{त्रि}^२ - \text{इशंकु}^२} = \text{दृज्ज्या}$ । एतावताऽऽचार्योक्तमुपपन्नम् ।

अब प्रकारान्तर से शंकु और दृग्ज्या को कहते हैं —

हि. भा.— हृति को बारह से गुणा कर पल कर्ण से भाग देने से वा (प्रकारान्तर से) इष्टशंकु होता है। त्रिज्यावर्ग में से शंकुवर्ग घटा कर मूल लेने से दृग्ज्या होती है इति ॥२७॥

उपपत्ति

पलभा = भुज, द्वादशाङ्गुलशंकु = कोटि, पलकर्ण = कर्ण } ये दोनों अक्षक्षेत्र सजातीय हैं
शंकुतल = भुज, इष्टशंकु = कोटि, छेद (हृति) = कर्ण }

इसलिये अनुपात करते हैं यदि पलकर्ण में द्वादशांगुलशंकु पाते हैं तो छेद(इष्टहृति)में क्या इससे इष्टशंकु आता है $\frac{१२ \times \text{इहृति}}{\text{पलक}}$ = इशंकु। दृग्ज्या = भुज, इष्टशंकु = कोटि, त्रिज्या = कर्ण इस छाया क्षेत्र में $\sqrt{\text{त्रि}^२ - \text{इशंकु}^२}$ = दृग्ज्या, इससे आचार्योक्त उपपन्न हुआ ॥ २७ ॥

इदानीं छायाकर्णावाह

दृग्ज्या द्वादशगुणिता विभाजिता शंकुना फलं छाया।
व्यासार्धं छेदहृतं विषुवत्कर्णाहतं कर्णः ॥ २८ ॥

वा. भा.—दृग्ज्यानन्तरानीतां द्वादशगुणितां शंकुना विभक्तां छायामानीय द्वादशांगुलस्य शंकोरिति वाक्यशेषः अथवा दृग्ज्यानेनैव गृहतरुपर्वतादीनां प्रमाणेन संगुणिता तस्मै तावती छाया तत्र काले भवति अत्र त्रैराशिकवासना यदि बृहच्छंकोर्दृग्ज्या छाया द्वादशांगुलस्य शंकोर्वेति फलं द्वादशांगुलशंकोश्छाया यतो व्यवहारार्थं शंकुलक्षणमनेनैव यंत्राध्याये कृतम्। मूले द्व्यंगुलविपुलसूच्यग्रो द्वादशांगुलछायाशंकुस्तलाग्रबिद्धोग्रबोधलंबेति। अथवा छेदादन्येन प्रकारेण छायानयनमुत्तरमार्यार्धमाह ॥

व्यासार्धं छेदहृतं विषुवत्कर्णहतं कर्णः। व्यासार्धं विषुवत्कर्णेन संगुणय्य छेदेन विभजेत् फलं तात्कालिका छाया कर्णो भवति। द्वादशांगुलस्य शंकोः कर्णकृते द्वादशांगुलशंकुकृतिं विशोध्य मूलछाया द्वादशां गुलस्य शंकोरेवेत्यत्र वासना। भूमध्याद्दक्परीत्येन तद्यथा यदि पूर्वापरछाया कर्णस्य द्वादश शंकुःकोटिः तद्व्यासार्धं कर्णस्य पूर्वापरस्यैव दृङ्मण्डलगतस्य का कोटिरिति फलं तात्कालिको बृहच्छंकुः कोटिरूपः ततो द्वितीयं त्रैराशिकं यदि विषुवन्मध्याह्ने याम्योत्तरमण्डलगता यावलंबकोटि व्यासार्धकर्णं तदिष्टशंकुकोटिर्याम्योत्तरछेदव्यासार्धं गतो यः कर्ण इति फलंच्छेद इत्यत्र लंबकस्थाने द्वादशकोटिः परिकल्पिता व्यासार्धस्थाने विषुवत्कर्णस्तेन प्रथमत्रैराशिके द्वादशगुणकारो द्वितीये भागहारस्तयोर्नष्टयोर्व्यासार्धस्य विषुवत्कर्णो गुणकारस्थितछायाकर्णो भागहारः फलं छेदः यदा पुनर्व्यासार्धच्छेदेन भागो दीयते

तदा फलं छाया कर्णो भवति । एवं वैपरीत्यादत उपपन्नं गोले चेदम् प्रदर्शयेदिति । अपरेण प्रकारेण छायानयनार्थमार्यार्धमाह ।

वि. भा.—दृग्ज्या द्वादशगुणिता, शंकुना विभाजिता (भक्ता) फलं छाया भवेत् । व्यासार्ध (त्रिज्या) विषुवत्कर्णा हतं (पलकर्णगुणितं) छेदहृत (हृतिभक्तं) तदा कर्णः (छायाकर्णः) भवेदिति ॥२८॥

अत्रोपपत्तिः ।

छाया = भुजः । द्वादशांगुलशंकुः = कोटिः । छायाकर्णः = कर्णः ।
दृग्ज्या = भुजः । इष्टशंकुः = कोटिः । त्रिज्या = कर्णः
} अनयोश्छाया-क्षेत्रयोः सजातीयत्वादनुपातो यदि शंकुना दृग्ज्या लभ्यते तदा द्वादशांगुलशंकुना किं समागच्छति छाया = $\frac{\text{दृग्ज्या} \times १२}{\text{शंकु}}$, तथा यदि शंकुना त्रिज्या लभ्यते तदा द्वादशांगुलशंकुना किं समागच्छति छायाकर्णः = $\frac{\text{त्रि} \times १२}{\text{शंकु}}$, परन्तु $\frac{१२ \times \text{हृति}}{\text{पक}}$ = शंकुः, अनेन शंकोरुत्थापनेन छायाकर्णः = $\frac{\text{त्रि} \times १२}{१२ \times \text{हृति}} = \frac{\text{त्रि} \times \text{पक}}{\text{हृति}}$ एतेनाऽऽचार्योक्तमुपपद्यते, श्रीपतिना 'व्यासदले विषुवच्छ्रवणघ्ने छेदहृते यदि वेप्सितकर्णः' ऽप्यनेनाऽऽचार्योक्तमेव कथ्यते, पूर्वानुपातद्वयेन समागते छाया-छाया कर्णौ एव 'दृग्ज्या त्रिजीवे रविसंगुणेति शंकूद्धृते भाश्रवणौ भवेतामिति भास्करोक्तं इति ॥२८॥

अब छाया और कर्ण को कहते हैं ।

हि. भा.—दृग्ज्या को बारह से गुणा कर शंकु से भाग देने से छाया होती है, त्रिज्या को पलकर्ण से गुणा कर छेद (इष्टहृति) से भाग देने से छाया कर्ण होता है इति ॥ २८ ॥

उपपत्ति ।

छाया = भुज, द्वादशांगुल शंकु = कोटि, छायाकर्ण = कर्ण
दृग्ज्या = भुज, इष्टशंकु = कोटि, त्रिज्या = कर्ण
} ये दोनों छाया क्षेत्र सजातीय हैं इसलिये अनुपात करते हैं । यदि शंकु में दृग्ज्या पाते हैं तो द्वादशांगुल शंकु में क्या इससे छाया आती है $\frac{\text{दृग्ज्या} \times १२}{\text{शंकु}}$ = छाया, तथा यदि शंकु में त्रिज्या पाते हैं तो द्वादशांगुल शंकु में क्या इससे छायाकर्ण आता है $\frac{\text{त्रि} \times १२}{\text{शंकु}}$ = छायाकर्ण, परन्तु $\frac{१२ \times \text{हृति}}{\text{पक}}$

$=$शंकु, इससे शंकु को उत्थापन देने से $\frac{\text{त्रि} . १२}{\frac{१२ \times \text{हृति}}{\text{पक}}} = \text{छायाकर्ण} = \frac{\text{त्रि. पक}}{\text{हृति}}$ इससे आचार्योक्त उपपन्न होता है। सिद्धान्त शेखर में श्रीपति भी 'व्यासदले विषुवच्छ्रवणघ्ने छेदहृते यदि वेप्सितकर्णः' इससे आचार्योक्त ही को कहते हैं, पहले अनुपात द्वय से आई हुई छाया और छायाकर्ण ही को 'दृग्ज्या त्रिजीवे रविसंगुणे ते शंकूद्धृते भाश्रवणौ भवेताम्' से भास्कराचार्य भी कहते हैं इति ॥ २८ ॥

इदानीं प्रकारान्तरेणच्छायाकर्णानयनमिष्टान्त्यां चाह

गुणितं वा द्वादशभिर्व्यासार्धं शंकुना हृतं कर्णः।
जीवाक्षयवृद्धिज्यायुतहीना ज्या क्रियतुलादौ ॥ २९ ॥

वा. भा.—द्वादशभिर्व्यासार्धं संगुणय्य बृहच्छंकुना विभजेत्। फलं छायाकर्णः अथवानेन प्रकारेणोच्यते, इयमुपपत्तिर्यदि बृहच्छंकुकोटेः पूर्वापरकर्णव्यासार्धद्वादशांगुलस्य कोटेः क इति फलं छायाकर्ण इत्युपपन्नम्। दृङ्मण्डलक्षेत्रइति तत इदानीं ज्याऽऽनयनार्थमुत्तरमार्यार्धमाह। जीवाक्षयवृद्धिज्यायुतहीना ज्याक्रियतुल्यादौ। गतशेषाल्पस्याह्नसोम्योत्तर गोलयोश्चरार्धेन ऊनाधिकस्य जीवा कृता सात्र गृह्यते तेनायमर्थः, जीवा मेषादौ क्षयवृद्धिज्यया युता सती ज्यारव्या भवति तुलादिराशिषट्केस्थेऽर्के क्षयवृद्धिज्यया हीना सती ज्या भवति अत्र वासना क्षितिजाकाँतरस्थाहोरात्र वृत्तखण्डाश्चरदलाख्यं स्वाहोरात्रखण्डलकं कर्णाद् विशोध्य शेषस्य या जीवा क्रियते। सोन्मण्डलावधिज्या व्यासार्धवृत्तनिष्पन्नजीवा भवति पूर्वमेव प्रदर्शिता क्षयवृद्धिज्या चरदलखण्डलस्य जीवातीतोत्तरे गोले योज्यते क्षितिजस्याधः स्थितत्वाद् दक्षिणे गोले शोध्यते क्षितिजस्योपरि स्थितत्वादुन्मण्डलादेः एवं कृते क्षितिजरव्यंतरज्या व्यासार्धवृत्ते भवति निष्पन्नछेद इत्यर्थः प्राग्वत्क्षेत्रं प्रदर्शयेत् प्रामाणीकृतछेदेन सह भेदोनसंस्थान कृत इति इदानीं छेदानयनार्थमार्यार्धमाह ॥ २९ ॥

वि. भा.—व्यासार्धं (त्रिज्या) द्वादशभिर्गुणितं शंकुना हृतं (शंकुभक्तं) तदा वा (प्रकारान्तरेण) छायाकर्णो भवेत्। जीवा (दिनगत दिनशेषयोरल्पस्य गोल क्रमाच्चरार्धोनयुतस्यार्थात्सूत्रचापस्य ज्या) क्रियतुलादौ (उत्तरगोले दक्षिणगोले च) क्षयवृद्धिज्यया (चरज्यया) युतहीना तदा ज्या (इष्टान्त्या) भवतीति ॥ २९ ॥

अत्रोपपत्तिः

यदि शंकुना त्रिज्या लभ्यते तदा द्वादशेन किमिति समागच्छति छायाकर्णः $= \frac{\text{त्रि. १२}}{\text{शंकु}}$, क्षितिजाहोरात्रवृत्तयोः सम्पातोपरिगतं ध्रुवप्रोतवृत्तमुत्तरगोले पूर्वस्वस्तिकाच्चरान्तरेऽधो दक्षिणगोले तूपरि नाड़ीवृत्ते लगति तद्विन्दुतः।

(क्षितिजाहोरात्रवृत्तसम्पातोपरिगतध्रुवप्रोतवृत्तनाडीवृत्तयोः सम्पातात्) पूर्वापरसूत्रस्य समानान्तररेखा कार्या तदुपरि ग्रहोपरिगतध्रुवप्रोतवृत्तनाडीवृत्तयोः सम्पातात्लम्बः कार्यः सेष्टान्त्या, दिनगतशेषयोर्यदल्पं तदुन्नत (उन्नतकालः) मुत्तरदक्षिणगोलयोश्चरार्धेन हीनं युतं तदा सूत्रचापं (ग्रहोपरिगतध्रुवप्रोतवृत्तनाडीवृत्तयोः सम्पातात्पूर्वस्वस्तिकं यावत्) भवेदेतस्य ज्या, सूत्रं (ग्रहोपरिगतध्रुवप्रोतवृत्तनाडीवृत्तयोः सम्पातात्पूर्वापरसूत्रोपरिलम्बः) पूर्वकृतपूर्वापरसूत्रसमानान्तररेखा पूर्वापरसूत्रयोरन्तरं चरज्या उत्तरगोले दक्षिणगोले च चरज्यया क्रमेण युतं हीनं सूत्रमिष्टान्त्या भवत्यत्राऽऽचार्येण सूत्रस्य नाम 'जीवा' इष्टान्त्यायाः नाम 'ज्या' कथ्यते सिद्धान्तशेखरे श्रीपतिना 'द्वादशभिर्गुणिते यदि वाऽस्मिन् कर्णमवेहि नरेण विभक्ते' तथा 'प्राग्वद्दोर्नतकालतश्चरदलन्यूनाधिकाच्छिञ्जिनी युक्तोना चरजीवया भवति सा ज्यास्या दिनज्याहृता। भक्ताऽभ्र त्रिभजीवया हृतिरसौ छेदो हरो वा ततः शङ्कुः पूर्ववदेव भाश्रवणयोःसिद्धिस्ततश्चोक्तवत्' ऽऽचार्योक्तयो 'गुणितं वा द्वादशभिरित्यादिश्लोकस्य' 'साद्धोरात्रार्धगुणेत्याद्यत्रिभश्लोकस्य च' रनयोरनुरूपमेव कथ्यते, इति ॥२६॥

अब प्रकारान्तर से छायाकर्णानयन और इष्टान्त्या को कहते हैं

हि. भा.—त्रिज्या को बारह से गुणाकर शङ्कु-से भाग देने से वा (प्रकारान्तर से) छायाकर्ण होता है, जीवा (दिनगत और दिनशेष में जो अल्प है वह उन्नत काल है उसमें गोल क्रम से चरार्ध को ऊन और युत करने से जो होता है उसकी ज्या(पूर्वापर सूत्र) में उत्तर गोल में और दक्षिण गोल में क्रम से चरज्या को जोड़ने और घटाने से ज्या (इष्टान्त्या) होती है इति ॥२६॥

उपपत्ति

यदि शङ्कु में त्रिज्या पाते हैं तो द्वादश में क्या इससे छायाकर्ण लाया है

$\frac{\text{त्रि.१२}}{\text{शङ्कु}}$ = छायाकर्ण, क्षितिजाहोरात्रवृत्त के सम्पातोपरिगतध्रुवप्रोतवृत्त उत्तरगोल में पूर्वस्वस्तिक से चरान्तर पर नीचा और दक्षिण गोल में चरान्तर पर ऊपर नाड़ीवृत्त में मिलता है, उस बिन्दु से (क्षितिजाहोरात्रवृत्तसम्पातोपरिगतध्रुवप्रोतवृत्त और नाड़ीवृत्त के सम्पात से) पूर्वापर सूत्र की समानान्तर रेखा कर देना, उसके ऊपर ग्रहोपरिगत ध्रुवप्रोतवृत्त और नाड़ीवृत्त के सम्पात बिन्दु से जो अल्प होता है वही इष्टान्त्या है, जिसका और दिनशेष में जो अल्प रहता है वह उन्नत काल है, उसमें उत्तर गोल और दक्षिण गोल क्रम से चरार्ध को हीन-युत करने से जो होता है उसकी ज्या (ग्रहोपरिगतध्रुवप्रोतवृत्त और नाड़ीवृत्त के सम्पात से पूर्वापर सूत्र के ऊपर लम्बरेखा), समानान्तर रेखा और पूर्वापर सूत्र का अन्तर चरज्या है, उत्तरगोल में और दक्षिणगोल में सूत्र में चरज्या को

क्रम से युत-हीन करने से इष्टान्त्या होती है, यहां आचार्य सूत्र के नाम जीवा' तथा इष्टान्त्या का 'ज्या' कहते हैं, सिद्धान्तशेखर में श्रीपति 'द्वादशभिर्गुणिते' इत्यादि तथा प्राग्व-द्वोन्नतकालतश्चरदलन्यूनाधिकात् इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से 'आचार्योक्त-गुणितं वा द्वादशभिः' इत्यादि के तथा 'साऽहोरात्रार्धगुणा' इत्यादि अग्रिम श्लोक के अनुरूप ही कहते हैं इति ॥२९॥

इदानीं पुनश्छेदाद्यानयनमाह

साऽहोरात्रार्धगुणा व्यासार्धविभाजिता ऽथवा छेदः।
शङ्ववादि प्राग्वज्ज्या स्वाहोरात्रार्धघातहृता ॥३०॥

वा. भा.—ज्येत्यनुवर्तते ज्यास्वाहोरात्रार्धेन संगुण्य व्यासार्धेन विभजेत्। फलं छेदो भवति। त्रैराशिकेन वासनात्र यदिव्यासार्धवृत्ते एतावती ज्या तत्स्वाहो-रात्रवृत्ते कियतीति फलं स्वाहोरात्रेष्टज्याछेद इत्युच्यते। तेन छेदेन शंक्वादिकं गोले क्षेत्रे वा प्रदर्शयेत् प्राग्वदिति।

शंक्वादिप्राग्वज्ज्यास्वाहोरात्रार्धघातहृता व्यासार्धकृतिगुणिता विषुवत्कर्णेन वा भवति कर्णः। छेदेन शंक्वादिप्राग्वदिति गतार्थोऽयं ग्रन्थः ज्यायाः स्वाहोरात्रस्य च यो घातः परस्परगुणना तेन घातेन हृतार्कसौरव्यासार्धकृतिरिति किं भूतेत्याह ॥३०॥

वि. भा.—सा(पूर्वानीतेष्टान्त्या) ऽहोरात्रार्ध(द्युज्या) गुणा, व्यासार्ध (त्रिज्या) भक्ताऽथवा (प्रकारान्तरेण) छेदः (इष्टहृतिः) भवेत्। ततः प्राग्वत् (पूर्ववत्) शङ्कादि भवेत्। ज्यास्वाहोरात्रार्धघातहृता इत्यस्याग्रिमश्लोकेन सम्बन्ध इति ॥३०॥

अत्रोपपत्तिः

भूकेन्द्राद् ग्रहोपरिगतध्रुवप्रोतवृत्तनाड़ीवृत्तयोः सम्पातगतरेखा त्रिज्या, ग्रहो-परिगतध्रुवप्रोतवृत्तनाड़ीवृत्तयोः सम्पातात्पूर्वोक्तपूर्वापरसूत्रसमानान्तररेखोपरिलम्ब इष्टान्त्या, तन्मूलगतारेखा चैतद्भुजत्रयेणोत्पन्नमेकं त्रिभुजम्। ग्रहोरात्रवृत्तगर्भ-केन्द्राद् ग्रहगता रेखा द्युज्या, ग्रहात् स्वोदयास्तसूत्रोपरिलम्बरेखेष्टहृतिः' तयोर्मूलगत-रेखा चैतद्भुजत्रयेणोत्पन्नं द्वितीयत्रिभुजम्। एतयोस्त्रिभुजयोःसाजात्यादनुपातो यदि त्रिज्ययेष्टान्त्या लभ्यते तदा द्युज्यया किमित्यनेनाऽऽगच्छतीष्टहृतिः $= \frac{\text{इष्टान्त्या} \cdot \text{द्यु}}{\text{त्रि}}$, ततः पूर्ववच्छङ्कवादि भवेदेवेति ॥३०॥

अब पुनः छेदादि के आनयन को कहते हैं

हि भा.—पूर्व साधित इष्टान्त्या को भुज्या से गुणा कर त्रिज्या से भाग देने से वा (प्रकारान्तर से) छेद (इष्टहृति) होता है। उससे पूर्ववत् शङ्कु आदि होता है, ज्यास्वाहोरात्रार्धघातहृता' इसका यद्यपि श्लोक के साथ सम्बन्ध है इति ॥३०॥

उपपत्ति

भूकेन्द्र से ग्रहोपरिगत ध्रुवप्रोतवृत्त और नाडीवृत्त की सम्पातगतरेखा त्रिज्या, ग्रहोपरिगत ध्रुवप्रोतवृत्त और नाडीवृत्त के सम्पात से पूर्वोक्त पूर्वापर सूत्र की समानान्तर रेखा के ऊपर लम्बरेखा इष्टान्त्या, भूकेन्द्र से इष्टान्त्या मूलगतरेखा, इन तीनों भुजों से उत्पन्न एक त्रिभुज, अहोरात्रवृत्त के गर्भकेन्द्र से ग्रहगत रेखा भुज्या, ग्रह से स्वोदयास्तसूत्र के ऊपर लम्ब रेखा इष्टहृति, अहोरात्रवृत्त के गर्भकेन्द्र से इष्टहृति मूलगतरेखा, इन तीनों भुजों से उत्पन्न द्वितीय त्रिभुज, ये दोनों त्रिभुज सजातीय हैं इसलिये अनुपात करते हैं यदि त्रिज्या में इष्टान्त्या पाते हैं तो भुज्या में क्या इससे इष्टहृति आती है $\frac{\text{इष्टान्त्या}\times\text{भु}}{\text{त्रि}}$ = इष्टहृति, इससे पूर्ववत् शङ्कु आदि का ज्ञान सुगम ही है इति ॥३०॥

इदानीं प्रकारान्तरेण छायाकर्णं शंकुं चाह

व्यासार्धकृतिगुणिता विषुवत्कर्णेन वा भवेत् कर्णः।
लम्बगुणो वा घातः शङ्कुर्व्यासार्धकृतिभक्तः ॥३१॥

घातो वार्कगुणस्त्रिज्या विषुवत्कर्णावधृतः शंकुः।
कर्णकृतेः संशोध्य द्वादशवर्गं पदं छाया ॥३२॥

वा. भा.—अत्रेयं वासना भूमध्याद्वैपरीत्येन त्रैराशिकत्रयं भवत्या प्रकल्पिता। तद्यथा यदि छायाकर्णस्य द्वादशशंकुस्तद्व्यासार्धकर्णस्य कः शंकुरिति फलं बृहच्छंकुः ततो यदि लंबककोटेर्व्यासार्धकर्णः तदस्या शंकुकोटेः कर्णं दक्षिणोत्तरवगाहिनी वृत्ते फलं छेदः ततस्तृतीयं यदि स्वाहोरात्रवृत्तेऽयं छेदः। तद्व्यासार्धवृत्ते क इति फलं ज्या एवमवस्थिते लंबकस्थाने द्वादशशंकुः कोटिः व्यासार्धस्थाने विषुवत्कर्णः कर्णः प्रथमत्रैराशिके द्वादशको वृत्तकारो द्वितीये भागहारस्तुल्यत्वात्तयोर्नाशे कृते व्यासार्धस्याव्यासार्धमेव गुणकः विषुवत्कर्णश्च स्वाहोरात्रार्धछायाकर्णयोर्घातो भागहारः फलं ज्या भवति। तदा पुनस्तस्यैव भाज्यराशेः स्वाहोरात्रार्धज्ययोर्घातेन भागो हि क्रियते तदा फलं छायाकर्णो भवत्येतत्सर्वं बोधे प्रदर्शयेदिति ज्याप्रकारान्तरेण शंक्वानयनमुच्यते तत्रार्धमेवाह 'लंबगुणो वा घातः शंकुर्व्यासार्धकृतिभक्तः' अथवा ज्यास्वाहोरात्रार्धं तत्स्वदेशावलम्बज्यया संगुण्य गृहीतं युज्यते यावदिहोन्मंडलादधो रविः क्षितिजाच्चोपरि रव्युन्मंडलन्तरस्थस्य स्वाहोरात्र-

खंडस्य जीवाऽत्रापि गृहीता स चोन्मंडलादेवाधः प्रवर्त्तते तच्चरदलप्राणेभ्यो गतप्राणशेषं वा विशोध्यते येनार्कोन्मंडलांतरे स्वाहोरात्रवृत्तखंडस्य या ज्या सा जीवेत्युच्यते। उन्मंडलादधस्ताद्रवेश्चरदलमित्यर्थः। क्षयवृद्धिज्या च क्षितिजोन्मंडलयोरन्तरथस्य स्वाहोरात्रवृत्तखंडस्य जीवातोऽवलम्बनं क्षयवृद्धिज्यातो विशोध्यते। येन क्षितिजादुपरि रवेरधश्च स्वाहोरात्रवृत्तखंडस्य ज्याखण्डे संगृहीतं भवति सैव ज्योच्यते। तस्मादुक्तं जीवोना क्षयवृद्धिज्या सा भवति। यदा पुनस्तदेवावलम्बनं स्वाहोरात्रार्धेन विभज्यते। तदा स्वाहोरात्रपरिणता क्षितिजा भवति। तस्मात्तस्याः स्वावलम्बनं विशोध्य छेदो भवति। अथ प्रश्नचरदलकालौ तुल्यौ भवतस्तदोन्मंडल एव रविर्भवति। तत्र स्वाहोरात्रापेक्षया क्षितिज्याछेदः तावानुच्छ्रायो रवेर्भवतस्तदोन्मंडलएव रविर्भवति।

ततः स्वाहोरात्रव्यासार्धे क्षेपेण क्षयवृद्धिजा ज्या। तावानेवोच्छ्राय इति तावता व्यासार्धेन प्राग्वत्तं दर्शयेत् दक्षिणोत्तरावगाही। तस्यापि तात्कालिकशंकुकोटिः शंकुतलं भुजे ज्यादिका त्रैराशिकवासना योज्या छेदोऽवलंबकगुण इत्यादिभिः सूत्रैर्यदुक्तमुक्तवच्छेषमिति तस्मात् सर्वमुपपन्नम्। यथास्थितं गोले प्रदर्शयेदिति इदानीं द्युदलान्तरज्यार्धयोरानयनमाह ॥३१॥

अथवा ज्या स्वाहोरात्रघातं द्वादशभिः संगुण्य त्रिज्या विषुवत्कर्णयोर्वधेन विभजेत्। फलं बृहच्छंकुर्भवति। प्रागार्यार्धेनात्र वासना। द्वितीये त्रैराशिके व्यासार्धस्थाने विषुवत्कर्णोऽवलंबकस्थाने द्वादशका कोटिरितोयान्विशेषः। तस्मादुपपन्नम्। गोले प्रदर्शयेदिति। इदानीं छायाकर्णछायानयनमुत्तरार्धेनाह। कर्णकृतेः संशोध्य द्वादशवर्गं पदं छाया। छायाकर्णवर्गात्द्वादशशंकोर्वर्गं विशोध्य मूलं तस्यैव शंकोश्छाया तात्कालिकी भवति। वासनात्र छायाकर्ण कर्णो द्वादशकः शंकुः कोटिः छाया भुजा तस्मात्कर्णकृतेः कोटिकृतिं विशोध्य मूलं भुजा इत्युपपन्नम्। एवं यथेष्टप्रमाणस्य शंकोश्छायाकर्णेन योज्यम्। तरुगृहपर्वतादीनां वेति। अथ यत्र दिने चरदलकालादथाल्पप्रश्नकालो भवति दिनगतः शेषश्चोत्तरगोले तत्र छेदः ज्ययोरानयनार्थमार्यामाह ॥३२॥

वि. भा.—व्यासार्धकृतिः (त्रिज्यावर्गः) विषुवत्कर्णेन (पलकर्णेन) गुणिता, ज्यास्वाहोरात्रार्धघातहृता (इष्टान्त्या द्युज्याघातभक्ता) तदा वा (प्रकारान्तरेण) कर्णः (छायाकर्णः) भवेत्। घातः (इष्टान्त्या द्युज्याघातः) लम्बज्यागुणितः, व्यासार्धकृतिभक्तः (त्रिज्यावर्गभक्तः) तदा वा (प्रकारान्तरेण) इष्टशंकुर्भवेत्। वा घातो द्वादशगुणितः, त्रिज्यापलकर्णघातभक्तस्तदेष्टशंकुर्भवेत्। छायाकर्णवर्गात् द्वादशवर्गं संशोध्य पदं (मूलं) ग्राह्यं तदा छाया भवेदिति ॥३१-३२॥

अत्रोपपत्तिः

यदि त्रिज्ययेष्टान्त्या (ज्या) लभ्यते तदा भुज्यया किं समागच्छतीष्ट-

हृतिः $=\frac{\text{इष्टान्त्या}\cdot\text{भु}}{\text{त्रि}}=\frac{\text{ज्या}\cdot\text{भु}}{\text{त्रि}}$, ततो व्यासार्धं छेदहृतं विषुवत्कर्णाहतं कर्णं

इत्यनेन छायाकर्णः $=\frac{\text{त्रि}\cdot\text{पलक}}{\text{इहृति}}=\frac{\text{त्रि}\cdot\text{पक}}{\frac{\text{ज्या}\cdot\text{भु}}{\text{त्रि}}}=\frac{\text{त्रि}^{1}\cdot\text{पक}}{\text{ज्या}\cdot\text{भु}}=\frac{\text{त्रि}^{2}\cdot\text{पक}}{\text{घात}}$, तत इष्टशंकुः $=$

$\frac{\text{त्रि}\cdot १२}{\text{छाकर्ण}}=\frac{\text{त्रि}\cdot १२}{\frac{\text{त्रि}^{2}\cdot\text{पक}}{\text{घात}}}=\frac{\text{त्रि}\cdot १२\cdot\text{ज्या}\cdot\text{भु}}{\text{त्रि}^{2}\cdot\text{पक}}=\frac{\text{त्रि}\cdot १२}{\text{पक}}$, $\frac{\text{ज्या}\cdot\text{भु}}{\text{त्रि}^{2}}=\frac{\text{लंज्या}\cdot\text{ज्या}.\text{भु}}{\text{त्रि}^{1}}=$

$\frac{\text{लंज्या}.\text{घात}}{\text{त्रि}^{1}}=\frac{\text{लंज्या}}{\text{त्रि}}$, $\frac{\text{घात}}{\text{त्रि}}=\frac{१२}{\text{पक}}$, $\frac{\text{घात}}{\text{त्रि}}$ एतेनाऽऽचार्योक्तमुपपद्यते ॥३१-३२॥

अब प्रकारान्तर से छायाकर्ण और इष्टशंकु को कहते हैं

हि. भा.—त्रिज्या वर्ग को पलकर्ण से गुणा कर इष्टान्त्या (ज्या) और भुज्या घात से भाग देने से छाया कर्ण होता है। वा घात को लम्बज्या से गुणा कर त्रिज्या वर्ग से भाग देने से इष्टशंकु होता है, वा घात को बारह से गुणा कर त्रिज्या और पलकर्ण के घात से भाग देने से इष्टशंकु होता है, छायाकर्ण वर्ग में द्वादश वर्ग को घटा कर मूल लेने से छाया होती है इति ॥३१-३२॥

उपपत्ति

यदि त्रिज्या में इष्टान्त्या पाते हैं तो भुज्या में क्या इससे इष्टहृति आती है

$\frac{\text{इष्टान्त्या}\cdot\text{भु}}{\text{त्रि}}=\frac{\text{ज्या}\cdot\text{भु}}{\text{त्रि}}=\frac{\text{घात}}{\text{त्रि}}=$इहृति 'व्यासार्धं छेदहृतं विषुवत्कर्णाहतं कर्णः' इससे

छायाकर्ण $=\frac{\text{त्रि}.\text{पक}}{\text{इहृति}}=\frac{\text{त्रि}.\text{पक}}{\frac{\text{ज्या}.\text{भु}}{\text{त्रि}}}=\frac{\text{त्रि}^{1}\text{पक}}{\text{ज्या}.\text{भु}}$ इससे इष्टशंकु $=\frac{\text{त्रि}.१२}{\text{छकर्ण}}=\frac{\text{त्रि}.१२}{\frac{\text{त्रि}^{1}\text{पक}}{\text{ज्या}.\text{भु}}}=$

$\frac{\text{त्रि}.१२\cdot\text{ज्या}.\text{भु}}{\text{त्रि}^{1}.\text{पक}}=\frac{\text{त्रि}.१२}{\text{पक}}\cdot\frac{\text{ज्या}.\text{भु}}{\text{त्रि}^{1}}=\frac{\text{लंज्या}.\text{घात}}{\text{त्रि}^{1}}=\frac{१२}{\text{पक}}\times\frac{\text{घात}}{\text{त्रि}}$, इससे आचार्योक्त उपपन्न हुआ इति ॥३१-३२॥

इदानीमिष्टच्छेदान्त्ययोर्विशेषमाह

ज्याः प्रस्थानभुवां यदि सङ्गुणकरवलनाप्तः त्रिभक्तिया।
हृतभोगा क्षीणोना लम्बनभुक्तिर्लवात्मकोऽहर्गणः ॥३३॥

वा. भा.—अल्पा दिनगता शेषा वा उत्तरगोले यदा प्रश्नघटिकानां प्राणा भवन्ति। तद्दैवसिकचरदलप्राणेभ्यः तदा विपरीतशोधनं कार्यं कृते च येऽधिकाश्चरदलप्राणाश्च भवन्ति तेषां क्रमज्या कार्या सा जीवेत्युच्यते। ततस्तां स्वाहोरात्रार्धेन सगुणय्य व्यासार्धे। विभजेत्। फलं हृत इत्युच्यते क्षितिजाहृतयोना कार्या सा छेदो भवति। क्षयवृद्धिज्या जीवोना सती ज्याख्या भवति। अथ प्रश्नकालः चरदलेन समो भवति। तदा क्षितिजैव छेदः क्षयवृद्धिज्यावज्ज्या भवति। उक्तवच्छेदमिति छेदेन ज्यायाश्च यथा छायानयनानि प्रागुक्तान्येवमत्रापि काले कार्याण्यतश्चोत्तरगोले सम्भवन्ति। यत उक्त शेषाल्प स्याद्भवति इति अत्रेयं वासना उत्तरगोले गतकालाच्छेषाद्वा तच्चरदलकालो विशोध्यते। यत उन्मंडलोपरि रवेरधश्च स्वाहोरात्रवृत्तखंडस्य यः शलाकाया अधः स्वोदयास्तसूत्रं स्थितं स्वक्षितिजादुन्मंडलस्योपरिस्थितत्वाद् दक्षिणगोलेऽन्यथाऽतस्तदन्तरं विशोध्यते। तच्चान्तरं क्षितिजा तस्मादुपपन्नं यदा पुनः तदेव स्वाहोरात्रार्द्धं व्यासार्धं परिकल्प्यते तदा क्षितिजापि क्षयवृद्धिज्या तदनुसारिण्या तयापि त्रिज्यायुतविहीना सौम्येतरगोलयोरन्त्या युज्यत एव वासनातुल्यत्वात्तदप्युपपद्यते। द्युदलांत्यज्यान्त्ययोरभिधानमग्रे युज्यते यतः सैव परावृद्धिः तत्र दिने तयोर्विषुवद्दिने युतः क्रान्त्यभावाद्। व्यासार्धतुल्यं स्वाहोरात्रार्धक्षितिजोन्मंडलयो संयोगात्। तत्र दिने चरदलस्याभावः। प्राच्यपरैर्वा निरक्षस्वदेशयोरुदयास्तसूत्रं तस्नात्तत्व्यासार्धमेव द्युदलांत्या ज्यांत्या च सर्वमेव तद्गोले प्रदर्शयेत्। निरक्षदेशे तूत्तरेणोपपद्यन्ते सर्व एव छायानयनप्रकारा इति। इदानीं द्युदलांत्यज्याछायानयनार्थमार्याद्वयेनाह ॥३३॥

वि. भा.—यदि प्रश्नासूनां (इष्टासूनां) संख्या अल्पाः, चरार्धासवो बहवस्तदोत्तरगोले चरासुभ्य इष्टासून् विशोध्य शेषस्य जीवा सूत्रसंज्ञा स्यात्, सा द्युज्या गुणा त्रिज्याभक्ता तदा कला भवेत्। तया हृतयोना क्षितिजा (कुज्या) छेदो भवेत्। क्षयवृद्धिज्या (चरज्या) जीवोना (सूत्रसंज्ञया हीना) तदेष्टान्त्या स्यात्, तत इष्टहृत्यन्त्याभ्यां शेषं शङ्कादिसाधनं पूर्ववत्कार्यमिति ॥३३॥

अत्रोपपत्तिः

'यत्र क्वचिच्छुद्धिविधौ यदेह शोध्यं न शुध्येद्विपरीतशुद्धया' इत्यादिभास्करकथितविधिना विलोमशोधनेन स्फुटैवास्ति, लल्लाचार्येण "अल्पीयांसो भवेयुः सवितृचरदलादिष्टकालासवश्चेत् सौम्ये गोले तदानीं चरदलसमयात् पातयित्वेष्टकालम्। कार्या शेषस्य जीवा चरशकलगुणस्तद्विहीनोऽन्त्यका स्यात् त्रिज्याभक्ताऽथ सैव द्युगुणा विगुणिता छेद इष्टः प्रदिष्टः" एवं कथ्यते, सिद्धान्तशेखरे श्रीपतिना "अभीष्टप्राणाश्चेच्चरशकलतः स्युस्त्वनधिका उदग्गोले पात्याश्चरशकलतो ज्या चरगुणः। तयोनस्त्रिज्याप्तो दिनगणगुणोऽसावभिमतो भवेच्छेदस्तस्मात् कथितविधिना स्तः श्रवणभे" अनेन लल्लोक्तमेवोक्तमिति ॥३३॥

अब इष्टछेद और इष्टान्त्या के विषय में विशेष कहते हैं

हि. भा.—यदि इष्टासु की संख्या अल्प हो और चरार्धासु की संख्या अधिक हो अर्थात् चरासु यदि इष्टासु से अधिक हो तो उत्तरगोल में चरासु में इष्टासु को घटाकर शेष की जीवा सूत्र संज्ञक होती है, उसको द्युज्या से गुणा कर त्रिज्या से भाग देने से कला होती है, कुज्या में उसे घटाने से इष्टहृति होती है, चरज्या में जीवा (सूत्र) को घटाने से इष्टान्त्या होती है, तब इष्टहृति और इष्टान्त्या से पूर्ववत् शङ्कु आदि का साधन करना चाहिये इति ॥३३॥

उपपत्ति

'यत्र क्वचिच्छुद्धिविधौ यदेह शोध्यं न शुध्येद्विपरीतशुद्धया' इत्यादि भास्करकथित विधि से विलोम शोधन से स्पष्ट है, लल्लाचार्य 'अल्पीयांसो भवेयुः' इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से आचार्योक्तानुसार ही कहते हैं। सिद्धान्तशेखर में श्रीपति ने 'अभीष्टघाणाश्चे चरशकलतः' इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से, लल्लोक्त ही को कहा है इति ॥३३॥

इदानीं हृत्यन्त्ययोः साधनमाह

स्वाहोरात्रार्धमुदग्दक्षिणयोः क्षितिजया युतविहीनम्.।
द्युदलान्त्यज्या त्रिज्या क्षयवृद्धिज्या युतोनाऽन्त्या ॥३४॥

वा. भा.—इष्टदिनस्वाहोरात्रार्धं तद्दैवसिक क्षितिज्यया मेषादौ युतं तुलादौ च हीनं द्युदलान्तरज्या भवति। एवं क्षयवृद्धिज्यया युतव्यासार्द्धमुत्तरगोलेऽन्त्या भवति। दक्षिणगोले हीनमन्त्या भवति। विषुवद्दिने द्युदलमन्त्यज्या व्यासार्धं तुल्ये भवत इत्यत वासना पूर्वविन्न्यासे निरक्षोदयसूत्रस्याधः शलाकया सह यत्र संपातस्तत्र सूत्रस्यैकमग्रं बद्ध्वा द्वितीयमग्रमूर्ध्वं नीत्वा याम्योत्तरमंडलस्याहोरात्रमंडलयोः संपाते बध्नीयात्। तत्स्वाहोरात्रार्द्धं साक्षदेशे तिर्यक्स्थितं भवति। तत्र निरक्षोदयास्तसूत्रयोरन्तरमुत्तरगोले योज्यते। ततो व्यासार्धं कृत्वा विभजेत् फलं बृहच्छङ्कुर्भवति। अत्र वासना त्रैराशिकद्वयेन यदिव्यासार्धवृत्ते एतावती ज्या तत्स्वाहोरात्रवृत्ते कियतीति फलं छेदः। ततो द्वितीयं यदि व्यासार्धकर्णस्य लंबज्या कोटिः तदस्य लम्बछेदस्य का कोटिरिति फलं बृहच्छङ्कुस्तस्मादुपपन्नं यथास्थिते क्षेत्रे प्रदर्शयेदिति ज्ययैवान्त्येन प्रकारेण शंकवानयनमार्याद्वयेनाह ॥३४॥

वि. भा.—उत्तरदक्षिणगोलयोः स्वाहोरात्रार्धं (द्युज्या) क्षितिजया (कुज्यया) यथाक्रमं युतविहीनं तदा द्युदलान्त्यज्या (हृतिः) भवेत्। तथा गोलयोस्त्रिज्या क्षयवृद्धिज्यया (चरज्यया) युतोना तदाऽन्त्या भवेदिति ॥३४॥

अत्रोपपत्तिः

क्षितिजाहोरात्रवृत्तयोः सम्पातोपरिगतं ध्रुवप्रोतवृत्तमुत्तरगोले दक्षिणगोले च क्रमेण नाडीवृत्ते पूर्वस्वस्तिकाच्चरान्तरेऽध ऊर्ध्वं लगति ताभ्यां पूर्वापरसूत्रस्य समानान्तररेखे कार्ये तदुपरि ग्रहोपरिगतध्रुवप्रोतवृत्तनाडीवृत्तयोः सम्पातालम्बरेखे गोलयोरिष्टान्त्ये पूर्वापरसूत्र-समानान्तररेखयोरन्तरं चरज्याऽस्ति, मध्याह्नकाले याम्योत्तरवृत्ते रवेः स्थितत्वात्तदुपरिगतध्रुवप्रोतवृत्तनाडीवृत्तयोः सम्पातो निरक्षखस्वस्तिकमेव तस्मात् पूर्वापरसूत्रसमानान्तररेखोपरिलम्बरेखा निरक्षोर्ध्वाधरसूत्रमेव तेन भूकेन्द्रान्निरक्षखस्वस्तिकं यावत् त्रिज्यायामुत्तरगोले पूर्वापरसूत्र-समानान्तररेखयोरन्तर्गतं निरक्षोर्ध्वाधरसूत्रखण्डं (चरज्या) योज्यं तदा निरक्षखस्वस्तिकात् समानान्तररेखां यावल्लम्बरूपाऽन्त्या भवेत् । दक्षिणगोले तु त्रिज्यायां चरज्यायाः शोधनेन तत्प्रमाणं भवति । तथा चाहोरात्रवृत्तगर्भकेन्द्राद्रविं यावद् द्युज्यायां निरक्षोदयास्तस्वोदयास्तसूत्रयोरन्तर्गतं कुज्यामानं गोलयो युतं विहीनं तदा दिनार्धे रवितः स्वोदयास्तसूत्रपर्यन्तं लम्बरूपा हृतिर्भवति। सिद्धान्तशेखरे "स्वाहोरात्रदलं युतोनमवनीमौर्व्या दिनार्धान्त्यका व्यासार्धं चरजीवया भवति सा चान्त्याऽर्कगोलक्रमात्' ऽनेन श्रीपतिना, भास्करेण 'क्षितिज्ययैवं द्युगुणश्च सा हृतिश्चरज्ययैवं त्रिगुणोऽपि सान्त्यका' ऽनेनाऽऽचार्योक्तमेव कथ्यत इति ॥३४॥

अब हृति और अन्त्या को कहते हैं

हि. भा.—उत्तरगोल में द्युज्या में कुज्या को जोड़ने से और दक्षिणगोल में द्युज्या में कुज्या को घटाने से हृति (मध्यहृति) होती है । तथा उत्तरगोल में त्रिज्या में चरज्या को जोड़ने से और दक्षिणगोल में त्रिज्या में चरज्या घटाने से अन्त्या (मध्याह्नकाल में) होती है ॥३४॥

उपपत्ति

क्षितिजाहोरात्रवृत्त के सम्पातोपरिगत ध्रुवप्रोतवृत्त उत्तरगोल में और दक्षिणगोल में क्रम से नाड़ीवृत्त में पूर्वस्वस्तिक से चरान्तर पर नीचे और ऊपर लगता है, उन दोनों बिन्दुओं से पूर्वापर सूत्र की समानान्तर रेखाओं के ऊपर ग्रहोपरिगतध्रुवप्रोतवृत्त और नाड़ीवृत्त के सम्पात से लम्बरेखा इष्टान्त्या है, मध्याह्नकाल में रवियाम्योत्तरवृत्त में रहता है इसलिये ग्रहो (रवि) परिगत ध्रुवप्रोतवृत्त (याम्योत्तरवृत्त) और नाड़ीवृत्त के सम्पात निरक्षखस्वस्तिक से पूर्वापरसूत्र के समानान्तररेखा के ऊपर लम्बरेखा निरक्षोर्ध्वाधर रेखा ही है, भूकेन्द्र से निरक्षखस्वस्तिक पर्यन्त त्रिज्या में पूर्वापर सूत्र समानान्तर रेखाओं के मध्यगत निरक्षोर्ध्वाधरसूत्रखण्ड (चरज्या) को उत्तरगोल में जोड़ने से और दक्षिणगोल में घटाने से निरक्षखस्वस्तिक से समानान्तररेखा पर्यन्त लम्बरूपरेखा अन्त्या

होती है। तथा मध्याह्नकाल में निरक्षोदयास्तसूत्रस्थ अहोरात्रवृत्त गर्भकेन्द्र से रवि-पर्यन्त कुज्या में गोलक्रम से निरक्षोदयास्तसूत्र और स्वोदयास्तसूत्र के अन्तर्गत कुज्या को जोड़ने और घटाने से रवि से स्वोदयास्त सूत्र पर्यन्त लम्बरेखा हृति होती है। सिद्धान्तशेखर में 'स्वाहोरात्रदलं युतोनमवनीमौर्व्या' इत्यादि संस्कृत उपपत्ति में लिखित श्लोक से श्रीपति तथा 'क्षितिज्यया वं खुगुणश्च सा हृतिः' इत्यादि से भास्कराचार्य भी आचार्योक्त बात को ही कहते हैं इति ॥३४॥

इदानीं प्रकारान्तरेणेष्टकर्णसाधनमाह

छेदहृता द्युदलान्त्या दिनार्धकर्णेन सङ्गुणा कर्णः।
भक्ता ज्ययाऽथवान्त्या दिनार्धकर्णाहता कर्णः ॥३५॥

वा. भा.—अनन्तरमेवानीतां द्युदलान्त्यज्यां तद्दैवसिकदिनार्धच्छायाकर्णेन संगुणय्य छेदेना विभजेत्। फलं छायाकर्णस्तात्कालिकस्तच्छायानयनं प्राग्वदित्यत्र वासना मध्याह्ने द्युदलांत्यज्यैव छेदः तदिष्टछायाकर्णस्य कश्छेद इत्येवं स्थिते द्युदलकर्णं संगुणय्य यावच्छायाकर्णेन भागो दीयते तावच्छेद प्रागच्छति। भागहार-लब्ध्योर्व्यत्ययात्तस्मादुपपन्नं पूर्वन्यस्तेषु द्युदलांत्यादिसूत्रेषु गोले सर्वं प्रदर्शयेदिति अन्यथा च छायाकर्णानयनमुत्तरार्धेनाह।

"भक्ता ज्ययाऽथवान्त्या दिनार्धकर्णाहता कर्णः अथवानन्तरमेवानीतामंत्यां दिनार्धकर्णेन हृत्वा ज्यया विभजेत्। फलं छायाकर्णस्तात्कालिको भवति। वासना-प्यत्र प्रागार्यार्धेन तुल्या यतो दिनार्धेऽन्त्यज्या भवति सैव स्वाहोरात्रपरिणता द्युद-लांत्यज्योच्यते। इष्टकालिकापि ज्या स्वाहोरात्रपरिणता छेद इत्युच्यते। तस्मात्तैरा-शिकवासना सैवात्र विवक्षा कृतो भेद इति। एवं दिनगतशेषाच्छेदानयनं प्रागुक्त-मिदानीं नतकालाद् द्युलान्त्यज्यया छेदानयनार्थमार्यामाह ॥३५॥

वि. भा.—द्युदलान्त्या (हृतिः) दिनार्धकर्णेन (मध्यच्छायाकर्णेन) गुणिता छेदहृता (इष्टहृतिभक्ता) तदा कर्णः (इष्टच्छायाकर्णो) भवेत्। अथवाऽन्त्या दिना-र्धकर्णाहता (मध्यच्छायाकर्णगुणा) ज्यया (इष्टान्त्यया) भक्ता तदेष्टच्छाया-कर्णो भवेदिति ॥३५॥

अत्रोपपत्तिः

अथ मध्यच्छङ्कुः $=\frac{१२\times त्रि}{मकाक}$, ततो यदि हृत्या मध्यच्छङ्कुर्लभ्यते तदेष्टहृत्या किं

समागत इष्टच्छङ्कुः $=\frac{मकां\times इहृ}{हृ}$ मध्यच्छङ्कोरत्यानयनेन $=\frac{१२\times त्रि\times इहृ}{मकाक\times हृ}$

$= \frac{१२ \times \text{त्रि}}{\text{मछाक}} \cdot \frac{\text{इअन्त्या}}{\text{अन्त्या}}$ ततो दृग्ज्या त्रिजीवे रविसङ्गुणे तेशङ्कूद्धृते भाश्रवणौ भवे ॥मितिभास्करोक्त्या $= \frac{१२ \times \text{त्रि}}{\text{इष्टशङ्कु}} = \text{इछाकर्ण} = \frac{१२ \times \text{त्रि} \times \text{मछाक} \times \text{हृ}}{१२ \times \text{त्रि} \times \text{इहृ}}$

$= \frac{\text{मछाक} + \text{हृ}}{\text{इहृ}} = \frac{\text{मछाक.अन्त्या}}{\text{इअन्त्या}} \therefore \frac{\text{हृ}}{\text{इहृ}} = \frac{\text{अन्त्या}}{\text{इअन्त्या}}$ एतावतऽऽचार्योक्तमुपपन्नमिति, सिद्धान्तशेखरे "आद्याऽथ द्युदलोत्थकर्णगुणिता छेदोद्धृता वा श्रुतिः स्यादन्त्याऽपि दिनार्धकर्णगुणिता ज्याप्ता च कर्णोऽथवे" त्यनेन श्रीपतिनाऽऽचार्योक्तमेव कथ्यते इति ॥३५॥

अब प्रकारान्तर से इष्टच्छायाकर्ण साधन को कहते हैं

हि. भा.—हृति को मध्यच्छायाकर्ण से गुणाकर इष्टहृति से भाग देने से इष्टच्छाया कर्ण होता है, अथवा अन्त्या को मध्यच्छाया कर्ण से गुणाकर इष्टान्त्या से भाग देने से इष्टच्छाया कर्ण होता है इति ॥३५॥

उपपत्ति

मध्यशङ्कु $= \frac{\text{त्रि.}\ १२}{\text{मछाक}}$ तब अनुपात करते हैं यदि हृति में मध्यशङ्कु पाते हैं तो इष्टहृति में क्या इस से इष्टशङ्कु आता है $\frac{\text{मशं.}\ \text{इहृ}}{\text{हृ}} = \text{इष्टशंकु}$. इसमें मध्यशङ्कु को उत्थापन देने से

$\frac{\text{त्रि.}\ १२.\ \text{इहृ}}{\text{मछाक.}\ \text{हृ}} = \frac{\text{त्रि.}\ १२}{\text{मछाक}} \cdot \frac{\text{इअन्त्या}}{\text{अन्त्या}} = \text{इष्टशङ्कु}$, तब अनुपात से $\frac{\text{त्रि.}\ १२}{\text{इशङ्कु}}$

$= \text{इछाकर्ण} = \frac{\text{त्रि.}\ १२.\ \text{मछाक.}\ \text{हृ}}{\text{त्रि.}\ १२.\ \text{इहृ}} = \frac{\text{मछाक.}\ \text{हृ}}{\text{इहृ}} = \frac{\text{मछाक.}\ \text{अन्त्या}}{\text{इअन्त्या}} \therefore \frac{\text{हृ}}{\text{इहृ}}$

$= \frac{\text{अन्त्या}}{\text{इअन्त्या}}$ इससे आचार्योक्त उपपन्न हुआ, सिद्धान्तशेखर में "आद्याऽथ द्युदलोत्थकर्णगुणिता" इत्यादि संस्कृतोपपत्ति में लिखित पद्य से श्रीपति—आचार्य (ब्रह्मगुप्त)क्त ही को कहते हैं इति ॥३५॥

इदानीं प्रकारान्तरेणेष्टहृतिमाह

द्युदलान्ततोत्क्रमज्यां स्वाहोरात्रार्धसङ्गुणां विभजेत् ।
व्यासार्धेन फलोना द्युदलान्त्यज्याऽथवा छेदः ॥३६॥

वा. भा.—यस्मिन्नभीष्टकाले छाया क्रियते गते दिनशेषे वा तस्य कालस्य

दिनार्द्धस्य चान्तरे यावत्यो घटिकाः ताः दिनदलोन्नता उच्यन्ते। तासां प्राणीकृतानामुत्क्रमेण या जीवा भवति सा नतोत्क्रमज्या उच्यते। ततस्तां द्युदलान्नतोत्क्रमज्यां स्वाहोरात्रार्धेन संगुण्य व्यासार्धेन विभजेत्। ततो यत्फलं तेनोना द्युदलान्त्यज्या छेदो भवति। छेदेन च छायानयनानि पूर्ववदंशनघटिका पंचदशघटिकाभ्योऽधिका भवन्ति। तदा पंचदशानां घटिकानामुक्रमज्या त्रिज्या भवति। शेषघटिकानां क्रमज्यां कृत्वा तया संयुता त्रिज्या शेषमुक्तवदित्यत्र वासना स्वाहोरात्रवृत्तेऽर्कोपलक्षितचिह्ने सूत्रस्यैवमग्रं बद्ध्वा द्वितीयमग्रं याम्योत्तरमंडलमध्ये नापरस्यां दिशि नीत्वा तावत्येवोच्छ्रिते स्वाहोरात्रवृत्तिप्रदेशे बध्नीयात् तत् पूर्वापरायतं सूत्रं ज्यावद्स्थितं भवति। तत्परिच्छिन्नस्य स्वाहोरात्रवृत्तधनुषो यः शरः सा नतोत्क्रमज्या द्युदलांत्यज्या सूत्रे भवति। व्यासार्धवृत्तनिष्पन्नाः। अतः स्वाहोरात्रवृत्ते परिणाम्यते। यदिव्यासार्धवृत्ते एतावती तत्स्वाहोरात्रवृत्ते कियतीति फलं स्वाहोरात्रनिष्पन्ना नतोत्क्रमज्या भवति तयोना यावद्द्युदलांत्यज्या क्रियते तावच्छेदतुल्यं द्युदलांत्यज्याखण्डमवशिष्यते सैवात्र घटिकानां स्वाहोरात्रनिष्पन्ना ज्या भवति। स्वक्षितिज्या पुनः पंचदशघटिका भवन्ति तदा रविरुन्मंडले वर्तते। तत्र नतोत्क्रमज्या त्रिज्या भवति। यावत्स्वाहोरात्रवृत्तं परिणाम्यते। तावत्स्वाहोरात्रार्धमेव फलं भवति। पंचदशभ्योऽधिका नतघटिका भवन्ति। तदा क्षितिजोन्मंडलयोरन्तरे को वर्तते। तत्रोन्मंडलादधः पुनः क्रमज्या प्रवर्तते। अत्र पंचदशघटिकाभ्योऽधिकघटिकानां क्रमज्या व्यासार्धे योज्यते। येनार्कद्युदलांतरस्थितानां घटिकानां स्वाहोरात्रवृत्ते नतोत्क्रमज्या भवति। तस्मादुपपन्नं यथा स्थितं गोले प्रदर्शयेदिति। इदानीं नतकालज्यानयनमंत्यया छेदं छायासंख्यां चार्धयाह ॥३६॥

वि. भा.—द्युदलाद्या नतोत्क्रमज्या भवेदर्यादिष्टकाले मध्याह्नाद्यो नतकालस्तस्योत्क्रमज्या या तां स्वाहोरात्रार्ध (द्युज्या) सङ्गुणां व्यासार्धेन (त्रिज्यया) विभजेत् फलेन हीना द्युदलान्त्यज्या (हृदिः) अथवा (प्रकारान्तरेण) छेदः(इष्टहृतिः) भवेदिति ॥३६॥

अत्रोपपत्तिः

क्षितिजाहोरात्रवृत्तयोः सम्पातोपरिगतं ध्रुवप्रोतवृत्तं नाड़ीवृत्तं यत्र लगति तद्विन्दोः पूर्वापरसूत्रस्य समानान्तरारेखा कार्या तदुपरि निरक्षस्वस्तिकात्सम्बरेखाऽन्त्या, ग्रहोपरिगतध्रुवप्रोतवृत्तनाड़ीवृत्तयोः सम्पातात्सम्बरेखा चेष्टान्त्या, ग्रहोपरिगतध्रुवप्रोतवृत्तनाड़ीवृत्तयोः सम्पातान्निरक्षोर्ध्वाधरसूत्रोपरिलम्बो नतकालज्या तन्मूलान्निरक्षस्वस्तिकं यावन्नतोत्क्रमज्याऽस्ति, नतोत्क्रमज्योनाऽन्त्येष्टान्त्या भवति, ततोऽनुपातेन 'यदि त्रिज्यवेष्टान्त्या लभ्यते तदा द्युज्यया किमिति' इष्टहृतिः

$$= \frac{\text{इष्टान्त्या}\cdot द्यु}{त्रि} = \frac{(\text{अन्त्या}-\text{नतोत्क्रमज्या})\, द्यु}{त्रि} = \frac{\text{अन्त्या}\cdot द्यु}{त्रि}$$

$=\frac{\text{नतोत्क्रमज्या}\cdot \text{द्यु}}{\text{त्रि}}=\text{हृति}-\frac{\text{नतोत्क्रमज्या}\cdot \text{द्यु}}{\text{त्रि}}$ एतावताऽऽचार्योक्तमुपपद्यते, सिद्धान्तशेखरे "नतोत्क्रमज्यागुणिता भ्रमेण हृता त्रिमौर्व्याऽथ फलेन हीना। दिनार्धजान्त्या यदि वा हृतिः स्यात्" अनेन श्रीपतिनाऽऽचार्योक्तानुरूपमेव कथ्यत इति ॥३६॥

अब प्रकारान्तर से इष्टहृति को कहते हैं

हि. भा.—इष्टकाल में मध्यान्ह से जो नतकाल की उत्क्रमज्या है उसको द्युज्या से गुणाकर त्रिज्या से भाग देने से जो फल होता है उसको हृति में घटाने से वा(प्रकारान्तर से) इष्टहृति होती है इति ॥३६॥

उपपत्ति

क्षितिजाहोरात्रवृत्त के सम्पातोपरिगतध्रुवप्रोतवृत्त नाड़ीवृत्त में जहाँ लगता है उस बिन्दु से पूर्वापर सूत्र की समानान्तर रेखा कर देना उसके ऊपर ग्रहोपरिगत ध्रुवप्रोतवृत्त और नाड़ीवृत्त के सम्पात बिन्दु से लम्बरेखा इष्टान्त्या है, ग्रहोपरिगत ध्रुवप्रोतवृत्त और नाड़ीवृत्त के सम्पात बिन्दु से निरक्षोर्ध्वाधर सूत्र के ऊपर लम्बरेखा नतकालज्या है, नतकालज्या मूल से निरक्षसस्वस्तिक पर्यन्त नतोत्क्रमज्या है। अन्त्या में नतोत्क्रमज्या को घटाने से इष्टान्त्या होती है तब अनुपात करते हैं यदि त्रिज्या में इष्टान्त्या पाते हैं तो द्युज्या में क्या इस अनुपात से इष्टहृति आती है,

$\frac{\text{इष्टान्त्या}\cdot \text{द्यु}}{\text{त्रि}}=\frac{(\text{अन्त्या}-\text{नतोत्क्रमज्या})\text{द्यु}}{\text{त्रि}}=\frac{\text{अन्त्या}\cdot \text{द्यु}}{\text{त्रि}}=\frac{\text{नतोत्क्रमज्या}\cdot \text{द्यु}}{\text{त्रि}}=\text{हृति}-\frac{\text{नतोत्क्रमज्या}\cdot \text{द्यु}}{\text{त्रि}}$

=इष्टहृति, इससे आचार्योक्त उपपन्न हुआ। सिद्धान्तशेखर में 'नतोत्क्रमज्या गुणिता भ्रमेण इत्यादि' से श्रीपति-आचार्योक्तानुरूप ही कहते हैं इति ॥३६॥

इदानीं प्रकारान्तरेणेष्टान्त्यां छायानयनभेदांश्चाह

अन्त्या नतोत्क्रमज्या हीना ज्याघट् पृथक् छेदः।
ज्याभ्यां च सह फलानि छायानयनानि षट्त्रिंशत् ॥३७॥

वा. भा.—येयमनन्तरमेवानीतांत्या सा घटिकानामुत्क्रमजीवया हीनास्त्रिज्या भवति। तया छायानयनानि प्राग्वत्। वासनाप्यत्र द्युदलांत्यज्यासूत्रे प्रदर्श्य छेदवशतः संख्यां कृत्वा वानयोर्भेदो न संस्थानः कृतः एवं द्युदलादेकैकस्याच्च भेदात्। षट्-षट्छायानयनं पृथक्-पृथक् अतएव वीप्सयाचार्येण कृता। तत्रैवातं उन्नतघटिकाभिः क्रमज्यया अष्टादशछायानयनानि। तथा नतघटिकाभिः (क) क्रमज्यया अष्टादशछायानयनान्येव षट्त्रिंशद्भवन्ति। तत्रया उन्नतकालाद्गत-

शेषाल्पाल्पा इत्यादिना एकश्छेदो जीवा । क्षयवृद्धिज्यायुतहीना ज्येत्यादिना द्वितीय-छेद तथानेनैं व सूत्रेणैका ज्या । अथ नतकालात् चुदलान्नतोत्क्रमज्यामित्यादिना एकश्छेदः अन्त्या नतोत्क्रमज्याहीना ज्येत्यादिना प्राग्वदेनां स्वाहोरात्रहृतां व्यासार्धेन विभजेत् । फलं द्वितीयछेदो भवत्येवं द्वाभ्यां ज्याभ्यां सह षट्छेदा इति । अथ छाया-नयनानि प्रदर्श्यन्ते । तद्यथा छेदोऽवलम्बकगुण इत्यादिना एकः शंक्वानयनप्रकारः विषुवत्कर्णविभक्तछेदो वा द्वादशाहतः शंकुरिति द्वितीयं एवं शंक्वानयनद्वयं व्यासार्धं छेदहतमित्यादिना एकछायानयनप्रकारः । एवं शंकुद्वयेन कर्णचतुष्टयेन षट्छाया-नयनानि । एकस्माद् द्वितीयादप्यन्यानि षडेवंवं द्वादश भवन्ति । तथा ज्यातो ज्या स्वाहोरात्रार्धघातहृतेत्यादिनैकः कर्णः गुणो वा घातहृतेत्यादिनैकः कर्णः गुणो वा घात इत्यादिनैकः शंकुर्वातो वार्ककुगुण इत्यादिना द्वितीयः शंकुः पुनरनेन शंकुद्वयेन गुणितं वा द्वादशभिर्व्यासार्धमित्यादिना छायाकारद्वयं भक्ता ज्ययाथवेत्यादिना चतुर्थः कर्णः, एवं ज्यातश्चत्वारः कर्णाः । द्वौ शंकू एभि षड्भिः षड्छायानयनानि एवं छेदानि तै द्वादशभिः सहाष्टादशछायानयनानि भवन्ति । उन्नतघटिकाभिर्नत-घटिकाभिरप्यष्टादशैव । छेदज्यानां तुल्यत्वादेवं षट्त्रिंशद्वाचार्येणोपवेशिता अन्य-थापि कल्पयितुं शक्यते । परमार्थतया च नतोन्नतघटिकानां क्रमोत्क्रमज्ये एव कारणमत्र । प्रथमस्तु छायाक्षेत्राणां प्रदर्शनायोन्मंडलक्षितिजांतरस्थे रवौ ज्याछेदो तौ छेदसंख्यात पतितौ न भवतो यत उन्नतघटिकाभिरानयनं प्राग्वदेव तयो सिद्धं विपरीतशोधनं चोन्मंडलादधःस्थितत्वाद्रवेर्गणितवासनया युज्यते एव । एवाधोमुखी यतस्तदा क्रमज्या । अतोऽयमेवार्थ आचार्येण सूत्रितो बालयुक्त्या । तेनोक्तमल्पप्रश्नभूतां वा यदि बहवश्चरदलासव इत्यादिकमार्यासूत्रं गोले प्रदर्शयेत् । स्वाहोरात्रहृन्मंड-लयोरिति गतशेषनता घटिकाछायातो यो वेत्तीत्यस्य प्रश्नस्योत्तरमार्यात्रयेणाह ॥३७॥

वि. भा.—नतकालोत्क्रमज्यया हीनाऽन्त्या ज्या (इष्टान्त्या) भवतीत्येवं द्वाभ्यामन्त्येष्टान्त्याभ्यां सह छेदः (इष्टहृतिः) पृथक् षट् अर्थात् छेदस्य चत्वारः प्रकारा अन्त्यायाश्च प्रकारद्वयम् । जीवा क्षयवृद्धिज्यायुतहीनेत्यनेनैकः प्रकारः । अन्त्या नतोत्क्रमज्याहीना ज्येत्यनेन द्वितीयः प्रकारः । एवमिष्टान्त्यायाः प्रकार-द्वयम् । द्वाभ्यामिष्टान्त्याभ्यां 'ज्या स्वाहोरात्रार्धगुणा व्यासार्धविभाजिताऽथवा छेदः' अनेनेष्टहृतेः प्रकारद्वयम् । ऊनाधिकस्य जीवा स्वाहोरात्रार्धसङ्गुणितेत्यनेन छेदस्यैकः प्रकारः । 'चुदलान्नतोत्क्रमज्या' मित्यादिना द्वितीयः प्रकारः । एवमन्ये-ष्टान्त्यया द्वाविष्टहृतेश्चत्वारो यदीनेन षड् भवन्ति 'छेदोऽवलम्बकगुणः' अनेन 'विषुवत्कर्णविभक्तश्छेदो वा' इत्यनेन शङ्कोः प्रकारद्वयम् । व्यासार्धं छेदहृतमित्य-नेनच्छायाकर्णानयनस्यैकः । हृज्या द्वादशगुणितेत्यनेन द्वाभ्यां षड्भ्यां छायायाः प्रकारद्वयम् । गुणितं वा द्वादशभिरित्यनेन द्वाभ्यां षड्भ्यां छायाकर्णानयने प्रकारद्वयम् । छेदहृता चुदलान्त्या विनार्धकर्णेत्यादिना छेदतश्छायाकर्णानयने एकः प्रकारः । एवं छेदतश्छायाकर्णानयने प्रकारषट्कम् । कर्णतो [illegible]

द्वादशवर्गमित्यनेन कर्णांतश्छायानयने प्रकारचतुष्कम् । छेदजन्यशङ्कुतः प्रकारद्वयं सिद्धमेव तेनात्र छेदतश्छायानयने षट्प्रकाराः । ज्या स्वाहोरात्रार्धघातहृते-त्यादिनैकः प्रकारः कर्णवशतः लम्बगुणो वा घातः शङ्कुरित्यादिना शङ्कुतो 'दृज्या द्वादशगुणिते' त्यनेन 'गुणितं वा द्वादशभि' रित्यादिना च प्रकारद्वयम् । घातो वार्कगुण इत्यादिना शङ्कुतः पुनः पूर्ववत् प्रकारद्वयम् । भक्ता ज्ययाऽथवान्त्या दिनार्धकर्णाहता कर्ण इत्यादिना कर्णांत एकः प्रकारः । इष्टान्त्यायाश्छायानयनेऽपि प्रकारषट्कम् । पूर्वमिष्टहृतेश्चत्वारो भेदा इष्टान्त्यायाः द्वौ भेदाविति षड्भेदा येभ्यश्छायानयने षट्त्रिंशदानयनानि भवन्तीति ॥३७॥

अत्रोपपत्तिः

क्षितिजाहोरात्रवृत्तयोः सम्पातोपरिगतं ध्रुवप्रोतवृत्तं नाड़ीवृत्ते उत्तरगोले पूर्वस्वस्तिकादधो दक्षिणगोले चोपरि लगति तद्बिन्दुभ्यां पूर्वापरसूत्रस्य समानान्तरे रेखे कार्ये तदुपरीष्टस्थानस्थितग्रहोपरिगतध्रुवप्रोतवृत्तनाड़ीवृत्तयोः सम्पातात्लम्बरेखे गोलयोरिष्टान्त्ये । मध्यान्हकाले ग्रहो याम्योत्तरवृत्ते भवति तेन ग्रहोपरि ध्रुवप्रोतवृत्तं याम्योत्तरवृत्तमेव तस्य नाड़ीवृत्तस्य च सम्पातो निरक्षस्वस्तिकम् । समानान्तररेखयोरुपरि निरक्षस्वस्तिका लम्बरेखे गोलयोरन्त्ये । इष्टस्थानस्थग्रहोपरिध्रुवप्रोतवृत्तनाड़ीवृत्तयोः सम्पातान्निरक्षोर्ध्वाधररेखोपरिलम्बो नतकालज्या, तन्मूलान्निरक्षस्वस्तिकं यावन्नतकालोत्क्रमज्या, निरक्षस्वस्तिकात् समानान्तररेखां यावन्निरक्षोर्ध्वाधररेखाखण्डमन्त्याऽस्ति, नतकालज्यामूलात् समानान्तररेखां यावन्निरक्षोर्ध्वाधररेखाखण्डमिष्टान्त्या तुल्याऽस्ति, अन्त्यायां यदि नतकालोत्क्रमज्या विशोध्यते तदा सैवे(नतकालज्यामूलात्समानान्तररेखां यावन्निरक्षोर्ध्वाधररेखाखण्डं) ष्टान्त्या भवतीति गोले स्फुटमेवावलोक्यत इति सिद्धान्तशिरोमणा 'नतोत्क्रमज्या शर इत्यनेन हीनाऽन्त्यका वाऽभिमतान्त्यका स्यात्' वित्यनेनाऽऽचार्योक्तानुरूपमेव कथ्यत इति ॥३७॥

अब प्रकारान्तर से इष्टान्त्या को और छायानयन भेदों को कहते हैं

हि. भा.—अन्त्या में नतकाल की उत्क्रमज्या को घटाने से इष्टान्त्या होती है, अन्त्या और इष्टान्त्या के साथ इष्टहृति पृथक् छः प्रकार की होती है; अर्थात् छेद(इष्टहृति) के चार प्रकार और अन्त्या के दो प्रकार, 'जीवा क्षयवृद्धिज्या युत हीना' इस से एक प्रकार, 'अन्त्या नतोत्क्रमज्या हीना ज्या' इस से द्वितीय प्रकार, इस तरह इष्टान्त्या के दो प्रकार, दोनों इष्टान्त्याओं से 'ज्या स्वाहोरात्रार्धगुणा व्यासार्धविभाजिताऽथवा छेदः' इस से इष्टहृति के दो प्रकार 'ऊनाधिकस्य जीवा स्वाहोरात्रार्धसंगुणिता' इस से छेद का एक प्रकार, 'चूदलान्तोत्क्रमज्या' इत्यादि से द्वितीय प्रकार, इस तरह इष्टान्त्या के दो और इष्टहृति के चार मिलके कुल से छः होते हैं। 'छेदोऽवलम्बकघ्नः' इस से तथा विषुवत्कर्णविभक्तश्छेदो वा इस से छेदघ्न के दो प्रकार, व्यासार्धं छेदहृतं इस से छायाकर्णानयन का एक प्रकार, इन्त्या द्वादश-

गुणिता' इस से दोनों शङ्कुओं से छाया के दो प्रकार 'गुणितं वा द्वादशभिः' इन दोनों शङ्कुओं से छायाकर्णानयन के दो प्रकार, 'छेदहृता द्युदलज्या दिनार्धकर्णेन' इत्यादि से छेद से छायाकर्णानयन का एक प्रकार, इस तरह छेद से छायाकर्णानयन में चार प्रकार, 'कर्णकृतेः समोद्धृत द्वादशवर्गं' इससे छायाकर्ण से छायानयन में चार प्रकार, छेदजनितशङ्कु से दो प्रकार सिद्ध ही हैं। इसलिये छेद से छायानयन में छः प्रकार हुये, 'ज्या स्वाहोरात्रार्धघातहृता' इत्यादि से कर्णवश एक प्रकार, 'लम्बगुणो वा घातः शङ्कुः' इत्यादि से शङ्कु से इष्टज्या 'द्वादश गुणिता' इससे 'गुणितं वा द्वादशभिः' इत्यादि से भी दो प्रकार, 'घातोवार्क गुणः' इत्यादि से शङ्कु वश पुनः पूर्ववत् दो प्रकार, भक्ता ज्यया ऽपवर्त्या दिनार्धकर्णाहिता कर्णः' इत्यादि से कर्ण -वश एक प्रकार, इष्टान्त्या से छायानयन में दो प्रकार, पहले इष्टहृति के चार भेद और इष्टान्त्या के दो भेद ये दोनों मिलकर छः भेद होते हैं जिन से छायानयन में छत्तीस ३६ आनयन होते हैं इति ॥३७॥

उपपत्ति

क्षितिजाहोरात्रवृत्त के सम्पातोपरिगत ध्रुवप्रोतवृत्त नाड़ीवृत्त में उत्तर गोल में पूर्वस्वस्तिक से नीचा और दक्षिणगोल में पूर्वस्वस्तिक से ऊपर लगता है उन दोनों बिन्दुओं से पूर्वापर सूत्र की समानान्तर रेखाद्वय करना उनके ऊपर इष्ट स्थानस्थित ग्रह के ऊपर ध्रुवप्रोतवृत्त और नाड़ीवृत्त के सम्पात बिन्दु से लम्ब रेखाद्वय दोनों गोलों में इष्टान्त्या होती है, मध्याह्नकाल में ग्रह याम्योत्तरवृत्त में रहते हैं इसलिये ग्रहोपरिगतध्रुवप्रोतवृत्त (याम्योत्तरवृत्त) और नाड़ीवृत्त के सम्पात बिन्दु (निरक्ष खस्वस्तिक) से समानान्तर रेखाद्वय के ऊपर लम्बरेखा गोलद्वय में अन्त्या होती है, इष्टस्थानस्थग्रहोपरिगतध्रुवप्रोतवृत्त और नाड़ीवृत्त के सम्पात बिन्दु से निरक्षोर्ध्वाधर रेखा के ऊपर लम्ब रेखा नतकालज्या है, उसके मूल से निरक्ष खस्वस्तिक पर्यन्त नतकाल की उत्क्रमज्या है, निरक्ष खस्वस्तिक से समानान्तर रेखापर्यन्त निरक्षोर्ध्वाधर रेखा का खण्ड अन्त्या है, नतकालज्या मूल से समानान्तर रेखापर्यन्त निरक्षोर्ध्वाधर रेखा खण्ड इष्टान्त्या के बराबर है, अन्त्या में यदि नतकाल की उत्क्रमज्या को घटाते हैं तो नतकालज्या मूल से समानान्तर रेखा पर्यन्त निरक्षोर्ध्वाधर रेखाखण्ड (इष्टान्त्या) होता है ये सब बातें गोल के ऊपर स्पष्ट देखने में आती हैं, सिद्धान्तशिरोमणि में भास्कराचार्य 'नतोत्क्रमज्या खर इत्यनेन' इत्यादि से आचार्योक्त के अनुरूप ही कहते हैं इति ॥३७॥

इदानीमुन्नतकालं नतकालं चाह

छायाकर्णविभक्ता विषुवत्कर्णेन सङ्गुणा त्रिज्या ।
लब्धं सौम्येतरयोः क्षितिजज्यया हीनसंयुक्तम् ॥३८॥
गुणितं व्यासार्धेन स्वाहोरात्रार्धभक्तलब्धधनुः ।
उत्तरगोले युक्तं याम्ये हीनं चरप्राणैः ॥३९॥

दिनगतशेषप्राणाः प्रागपरदिनार्धयोर्विशोध्याप्तम्.।
व्यासार्धात् शेषोत्क्रमजीवाचापं नतप्राणाः ॥४०॥

वा. भा.—इष्टदिने इष्टकाले यां छायामुद्दिश्य कश्चित्कालं पृच्छति तस्मात्कालिकश्छायाकर्णः कार्यः ततस्तेन छायाकर्णेन विषुवत्कर्णहतां त्रिज्यां विभजेत्। लब्धं छेदो भवति । याम्योत्तरगोलयोर्यथासंख्यं क्षितिज्यया हीनं संयुतं कृत्वा व्यासार्धेन गुणयेत्। ततः स्वाहोरात्रार्धेन विभजेत्। लभ्यते तस्य प्रसंज्ञा तस्य च चापं कृत्वा चरदलप्राणेभ्यो विशोधयेत्। शेषप्राणा दिनगतशेषा भवन्ति। एवं दिनगतशेषानयनमथ नतकालानयनं विशोध्याप्तं व्यासात्। यद्याप्तसंज्ञात् त्रिज्यातो विशोधयेत्। शेषा प्राणा दिनगता शेषा भवन्ति। एवं दिनगतशेषानयनमथनतकालानयनं विशोध्याप्ताद् व्यासार्धात् तदाप्तसंज्ञां तत्त्रिज्यातो विशोध्य शेषस्योत्क्रमज्याद्यैश्चापं कार्यं तत्र या लिप्ताः तावती प्राणा भवन्ति। अत्रापि यद्विपरीतशोधने आप्तमानीतं तत् व्यासार्धे योजयेत्। तदुत्क्रमज्या वा क्रमज्याभिरुक्तवत्कार्यम्। तल्लिप्तासंख्या नतप्राणा भवन्त्यत्र वासना, यथाकालाच्छायानयनं प्राग्वत्प्रदर्शितमेवं छायातो वैपरीत्येन कालानयनं सिद्धमथानेनार्यात्रयेण प्रदर्श्यते। तद्यथा भूमध्याद्रविपृच्छाया कर्णस्य द्वादशकोटेर्विषुवत्कर्णः कर्णः इष्टशंकुकोटे इत्येवं स्थिते प्रथमे द्वादशको गुणकारो द्वितीये भागहारस्तुल्यत्वान्नाशे कृते छायाकर्णहता विषुवत्कर्णेन संगुण्य त्रिज्यालब्धं यत्र छेदो त्रिज्ययाहीनमुत्तरगोले दक्षिणे युक्तं क्रियते। येनोन्मंडलार्कान्तिरज्या स्वाहोरात्रनिष्पन्ना भवति तस्या व्यासार्धवृत्तपरिणमने त्रैराशिकमेवं गुणितं व्यासार्धेन स्वाहोरात्रार्धभक्तमिति ततो यल्लब्धं तज्ज्यारूपं व्यासार्धपरिणतं रव्युन्मंडलान्तरं तच्च चापरूपं तदैवान्तरं स्वाहोरात्रपृष्ठे भवति।

तदुत्तरगोले चरप्राणैरुपचीयते। स्वक्षितिजस्याधः स्थितत्वात्। दक्षिणगोले ऽन्यथा तेनोक्तमुत्तरगोले युक्तं याम्ये हीनं चरप्राणैर्दिनगतप्राणा दिनार्धयोः प्राग्परयोरिति। यत्रः पुनश्छेदः क्षितिज्ययाऽल्पो भवति । तत्रोन्मंडलक्षितिजांतरे रविर्वर्तते। छेदश्च क्षितिजादुपरि रविं यावत् अतस्तत्र विशोध्य क्षितिज्यातः शेषं क्षितिज्यायाः खंडं रव्युन्मंडलान्तरस्थं भवति । तद्व्यासार्धे परिणामं कृत्वा यावच्चापं क्रियते। तावदुन्मंडलार्कयोरन्तरे स्वाहोरात्रवृत्तप्राणा भवन्ति। तांश्चरदलप्राणेभ्यो विशोध्य दिनगताः शेषा वा प्राणा भवन्ति। यत उन्मंडलक्षितिजमंडलयोरंतरं चरदलप्राणाः अत उपपन्नं दिनगतशेषानयनं। अथ नतकालानयने वासना यत्तदाप्तसंज्ञातुल्यमवशिष्यते। यत्तदर्कोपलक्षितात्स्वाहोरात्रवृत्तप्रदर्शनपूर्वापरायतं सूत्रं प्राग्वत्। बुदलान्नतोत्क्रमज्यामित्यस्य सूत्रस्य वासनायास्तत्सूत्रपरिच्छिन्नस्य धनुषो यः शरः स भवतीत्यर्थः तस्य ज्याखंडस्य यावदुत्क्रमेण चाप्तलिप्तानीयन्ते तावद् दिनदलं विवस्वत उत्तरं प्राणा भवन्तीत्युपपन्नं यदाप्तं

विपरीतशोधनेन तदुन्मंडलादवस्तदर्कान्तरज्या तेन व्यासार्धे योज्यते। येन पूर्वापरायतसूत्रावच्छिन्नस्य धनुषः शरो भवति। शरोत्क्रमचापेनावनताः प्राणाः प्रत्यक्षिज्यातोऽधिकाया ततो उत्क्रमज्याया क्रमेण चापमतः क्रियते। यत उन्मंडलमतोऽस्वाहोरात्रवृत्तस्योभयतोपि तुल्यमवशिष्टम्। तं याम्योत्तरमंडलं यावत्तत्र च ननभावः तस्मात्सर्वमुपपन्नम्। गोले छायानयनं क्षेत्रेषु वैपरीत्येन योजयेत् क्रमोत्क्रमाभ्यामिति। एवं छेदेन यच्छायानयनं प्रागुक्तं तद्वैपरीत्यान्नतोन्नतकालानयनमभिप्रायेदानीं ज्याया यच्छायानयनं तद्वैपरीत्येन नतोन्नतकालानयनार्थमार्याद्वयमाह ॥३८-३९-४०॥

वि. भा.—त्रिज्या विषुवत्कर्णेन (पलकर्णेन) गुणिता, छायाकर्णविभक्ता लब्धं सौम्येतरयोः (उत्तरदक्षिणगोलयोः) क्षितिज्यया (कुज्यया) हीनं युक्तं, व्यासार्धेन (त्रिज्यया) गुणितं स्वाहोरात्रार्धेन (द्युज्यया) भक्तं यल्लब्धं तस्य धनुः (चापम्) उत्तरगोले चरप्राणैः (चरासुभिः) युक्तं, याम्ये (दक्षिणगोले) हीनं प्रागपरदिनार्धयोः (पूर्वापरकपालयोः) दिनगतशेषप्राणाः (उन्नतासवः) भवन्ति, आप्तं (सूत्रं) व्यासार्धात् (त्रिज्यातः) विशोध्य शेषं नतकालोत्क्रमज्या, उत्क्रमज्याखण्डैस्तच्चापं पूर्वापरकपालयोर्नतप्राणाः (नतासवः) भवन्तीति ॥३८-३९-४०॥

अत्रोपपत्तिः

अथ $\frac{\text{त्रि}\cdot १२}{\text{इष्टाक}}$=इष्टशङ्कु, ततोऽनुपातेनेष्टहृतिः $=\frac{\text{पक}\cdot\text{इशं}}{१२}=\frac{\text{पक}}{१२}$.

$\frac{\text{त्रि}\cdot १२}{\text{इष्टाक}}=\frac{\text{पक}\cdot\text{त्रि}}{\text{इष्टाक}}$, उत्तरदक्षिणगोलयोरिष्टहृतौ कुज्यायां विशोधनेन योजने च कला भवति, सा त्रिज्यया गुणिता द्युज्यया भक्ता तदा सूत्रं भवति, तच्चापमुन्मण्डलादुन्नतकालस्तत्र चरसंस्कारेण पूर्वापरकपालयोः स्वदेश उन्नतकाल एव दिनगतशेषप्राणाः। आप्तं (सूत्रं) नतकालकोटिज्यासमं त्रिज्यातः शोध्यं तदा नतकालोत्क्रमज्या भवेत्। उत्क्रमज्याखण्डैस्तच्चापं पूर्वापरकपलयोर्नतप्राणा भवन्तीति ॥३८-३९-४०॥

अब उन्नत काल को और नतकाल को कहते हैं

हि. भा.—त्रिज्या को पलकर्ण से गुणा कर छायाकर्ण से भाग देने से जो फल होता है उसमें उत्तर और दक्षिण गोल में क्रम से कुज्या को घटाने और जोड़ने से जो फल होता है उसको त्रिज्या से गुणा कर द्युज्या से भाग देने से जो फल होता है उसके चाप में उत्तर गोल में चरप्राण जोड़ने से और दक्षिण गोल में हीन करने से पूर्वकपाल और पश्चिम कपाल में दिनगत शेष और विनशेषकाल होता है, पूर्वसाधित (सूत्र) को त्रिज्या में घटाकर

जो शेष रहता है वह नतकालोत्क्रमज्या है, उत्क्रमज्याखण्डों से उसके चाप पूर्वकपाल में और पश्चिमकपाल में नतासु होते हैं इति ॥३८-३९-४०॥

उपपत्ति

$\frac{\text{त्रि}\cdot १२}{\text{इष्टाक}}$ = इशङ्कु, अक्षक्षेत्रानुपात से इष्टहृति = $\frac{\text{पक}\cdot\text{इशं}}{१२} = \frac{\text{पक}}{१२}\cdot$

$\frac{\text{त्रि}.१२}{\text{इष्टाक}} = \frac{\text{पक}.\text{त्रि}}{\text{इष्टाक}}$, उत्तर गोल में और दक्षिण गोल में इष्टहृति में कुज्या को घटाने से और जोड़ने से कला होती है, उसको त्रिज्या से गुणाकर द्युज्या से भाग देने से सूत्र होता है, उसका चाप उन्मण्डल से उन्नतकाल होता है, उसमें चर संस्कार करने से पूर्वकपाल और पश्चिम कपाल में स्वदेश में उन्नतकाल होता है, पूर्वागत सूत्र (नतकाल कोटिज्या) को त्रिज्या में घटाने से नतकाल की उत्क्रमज्या होती है, उत्क्रमज्या खण्डों से उसका चाप पूर्वकपाल में और पश्चिम कपाल में नतासु प्रमाण होता है इति ॥३८-३९-४०॥

इदानीं प्रकारान्तरेणोन्नतकालं नतकालं चाह

स्वाहोरात्रार्धेन छायाकर्णेन भक्तायाः ।
विषुवत्कर्णगुणाया व्यासार्धकृतेः फलं सौम्ये ॥४१॥

क्षयवृद्धिज्याहीनं युक्तं याम्ये धनुश्चरप्राणैः ।
सौम्ये युतं विहीनं याम्ये प्रागपरयोः प्राणाः ॥४२॥

अह्नो गतावशेषाः फलमन्त्याया विशोध्य शेषस्य ।
धनुरुत्क्रमजीवाभिः पूर्वापरयोर्नतप्राणाः ॥४३॥

वा. भा.—यस्याश्छायाया दिनगतशेषानयनमिष्यते तस्याश्छायायाः छायाकर्णं कृत्वा तेन स्वाहोरात्रार्धं गुणयेत् । ततस्तेन स्वाहोरात्रार्धेन छायाकर्णहृतेन भक्तायां कस्या व्यासार्धकृतेः किं भूतया विषुवत्कर्णगुणायाः फलं ज्या भवति । तत्फलं सौम्ये गोले क्षयवृद्धिज्याहीनं याम्ये तयैव युक्तं कृत्वा यद्भवति । तस्य धनुःक्रमेण कार्यं स्तद्धनुस्तद्दैवसिकचरदलप्राणैः सौम्ये गोलेषु युतं याम्यैर्हीनं कृत्वा प्राग्परयोः प्राणा भवन्ति । अह्नो गतावशेषं यथासंख्यमयोत्तरगोले क्षयवृद्धिज्या फलान्न लभ्यति । तद्विपरीतशोधनेन यच्चापं तच्चरदलाद्विशोध्य गता शेषा प्राणा भवन्ति एवमुन्नतकालानयनम् । अथ नतकालानयनं फलमन्त्यामपि विशोध्य यत्फलं लब्धं तिष्ठति । तदन्त्याया विशोध्य शेषस्योत्क्रमज्याचापं धनुः कार्यं तत्र या लिप्ताः तेन प्राणा नता भवन्ति । अथ फलेऽन्त्याया विशोध्य व्यासार्धमधिकमेवावशिष्यते । तदधिकस्य क्रमज्यात्वं कृत्वा त्रिज्यायामपि चतुर्भज्याचतुर्थे या ज्या नताः

प्राणाः भवन्त्यत्रेयं वासना त्रैराशिकत्रयेण भूमध्या ज्यानयनं तद्यथा यदि छायाकर्णस्य द्वादशिकाकोटिः तद्व्यासार्धकर्णस्य किमिति फलं बृहच्छंकुः ततो द्वितीयं यदि द्वादशांगुलायाः कोटिः विषुवत्कर्णः तद्बृहच्छंकुकोटेः क इति फल छेदः । ततस्तृतीय यदिन्वाहोरात्रवृत्तेः एव छेदः तद्व्यासार्धवृत्ते कियानिति फलं ज्या भवन्ति । एवं स्थिते प्रथमे द्वादशको गुणकारो द्वितीये भागहारः क्षितिजार्कान्तरस्थितस्य स्वाहोरात्रवृत्तखंडस्य व्यासार्धवृत्तनिष्पन्ना ज्या भवतीत्यर्थः । उत्तरगोले ततः क्षयवृद्धिज्या विशोध्यते । क्षितिजस्याधः स्थितत्वाद् दक्षिणे या ज्या उन्मंडलस्याधः स्थितत्वाद्येनार्कोन्मंडलान्तरज्या भवति । तस्याश्चापं तदन्तरे स्वाहोरात्रवृत्तखंडं तच्चरप्राणैरुत्तरगोले उपचीयते । दक्षिणेऽन्यथा येन क्षितिजात्प्रभृति दिनगता शेषाः प्राणा वा भवन्ति शेषं पूर्ववत् । ननकालानयने यद्यत् फलं सा ज्या तामन्त्यातो विशोध्य शेषज्याखंडं प्राच्यदर्शितपूर्वपरायतसूत्रावच्छिन्नस्य धनुषः शरो भवति । तेनोत्क्रमचापशरानयनवासना प्राग्वद्योज्याः यदा फलक्षयवृद्धिज्या न शुध्यति, तदा विपरीतशोधनादिका वासना प्राग्वत्तत्कालानयने फलमंत्यायां विशोध्य यदिव्यासार्धमधिकमवशिष्यते तदपि चापकरणवासना प्राग्वदेवोपपद्यते । यथास्थितं सर्वं गोले प्रदर्शयेदिति । इदानीं यदुन्नतकालादंत्यायाश्छायानयनमुक्तं तद्विपरीत्येन नतकालानयनमार्यायाह ॥४१-४२-४३॥

वि. भा.—विषुवत्कर्णगुणाया (पलकर्णगुणितायाः) व्यासार्धकृतेः स्वाहोरात्रार्धेन (द्युज्यया) छायाकर्णेन च भक्तायाः फलं ग्राह्यं तत् सौम्ये गोले (उत्तरगोले) क्षयवृद्धिज्यया (चरज्यया) हीनं, याम्ये (दक्षिणगोले) युक्तं यद् भवेत्तस्य धनुः (चापं) सौम्ये (उत्तरगोले) चरप्राणैः (चरासुभिः) युतं, याम्ये (दक्षिणगोले) विहीनं तदा प्रागपरयोः (पूर्वापरकपालयोः) अन्हो दिवसस्य गतावशेषाः प्राणाः (उन्नतकालासवः) भवन्ति । फलं (पूर्वागतं) अन्त्यया विशोध्य शेषयोत्क्रमज्याभिः (उत्क्रमज्याखण्डैः) धनुः (चापं) कार्यं तदा पूर्वापरकपालयोर्नतासवो भवन्तीति ॥४१-४२-४३-॥

अत्रोपपत्तिः

$$\text{अथेष्टशङ्कुः} = \frac{\text{त्रि}\cdot १२}{\text{इछाक}} \text{ ततोऽक्षक्षेत्रानुपातेनेष्टद्युतिः} = \frac{\text{पक}.\ \text{इछं}}{१२}$$

$$= \frac{\text{पक}}{१२}\cdot\frac{\text{त्रि}.\ १२}{\text{इछाक}} = \frac{\text{पक}.\ \text{त्रि}}{\text{इछाक}} \text{ ततो द्युज्यया यदीष्टद्युतिर्लभ्यते तदा}$$

$$\text{त्रिज्यया किं समागच्छतीष्टान्त्या} = \frac{\text{पक}.\ \text{त्रि}.\ \text{त्रि}}{\text{इछ}.\text{क}.\ \text{द्यु}} = \frac{\text{पक}.\ \text{त्रि}^{२}}{\text{इछाक}.\text{द्यु}} = \text{फलम् तत उत्तर-}$$

दक्षिणगोलक्रमेण इष्टान्त्या ± चरज्या = सूत्रम् । एतच्चापं = सूत्रचापं, उत्तरदक्षिणगोलयोः सूत्रचाप ± चरासु = पूर्वापरकपालयोर्दिनगतशेषाः = उन्नतकालः । तथा

अन्त्या -- फल = अन्त्या — इष्टान्त्या = नतोत्क्रमज्या, अस्या उत्क्रमज्याखण्डैश्चापं कार्यं तदा नतासवो भवन्तीति सिद्धान्तशेखरे "यदि वा पलकर्णताड़िताया-स्त्रिभजीवोत्थकृतेर्विभाजिताया:। श्रुतिसङ्गुणितभ्रमेण लब्धं चरजीवोनयुतं यथोक्तवत्तत्" ॥ अथ तस्य धनुश्चरासुयुक्तं रहितं गोलवशाद् गतावशेषाः। तदवास्य फलं तद-त्यकाया नतमाहुर्विपरीतधन्व यद्वे" त्यनेन श्रीपतिना, सिद्धान्तशिरोमणौ "पलश्रुतिघ्नस्त्रिगुणस्य वर्गो द्युष्टकर्णाहितिहृदित्यादिना" भास्कराचार्येण चाऽऽचार्योक्तानुरूपमेव सर्वमुक्तमिति ॥४१-४२-४३॥

अब प्रकारान्तर से उन्नत काल को और नत काल को कहते हैं

हि. भा —त्रिज्यावर्ग को पलकर्ण वर्ग से गुणाकर द्युज्या और इष्टच्छाया कर्ण के घात से भाग देनें से जो फल होता है उस में उत्तर गोल में चरज्या को घटाने से और दक्षिण गोल में जोड़ने से जो होता है उसके चाप में उत्तर गोल में चरासु को जोड़ने से दक्षिण गोल में घटाने से पूर्व कपाल में और पश्चिम कपाल में दिन के गतासु और दिनशेषासु होता है अर्थात् उन्नत काल होता है, पूर्वागत फल को अन्त्या में घटाकर जो शेष रहता है उत्क्रमज्याखण्डों से उसके चाप करने से पूर्वकपाल में और पश्चिम कपाल में नतासु प्रमाण होता है इति ॥४१-४२-४३॥

उपपत्ति

इष्टशङ्कु = $\frac{\text{त्रि. १२}}{\text{इछाक}}$, अक्षक्षेत्रानुपात से इष्टहृति = $\frac{\text{पक. इशं}}{\text{१२}} = \frac{\text{पक}}{\text{१२}} = \frac{\text{त्रि. १२}}{\text{इछाक}} =$

$\frac{\text{पक. त्रि}}{\text{इछाक}}$ अब अनुपात करते हैं यदि द्युज्या में इष्टहृति पाते हैं तो त्रिज्या में क्या इस अनुपात से इष्टान्त्या आती है। $\frac{\text{पक. त्रि. त्रि}}{\text{इछाक. द्यु}} = \frac{\text{पक. त्रि}^2}{\text{इछाक. द्यु}}$ = इष्टान्त्या = फल, उत्तर गोल और दक्षिणगोल क्रम से इष्टान्त्या ∓ चरज्या = सूत्र, इसका चाप = सूत्रचाप, उत्तर गोल में और दक्षिण गोल में सूत्रचा ± चरासु = पूर्वकपाल में और पश्चिम कपाल में उन्नत काल, तथा अन्त्या — फल = अन्त्या — इष्टान्त्या = नतोत्क्रमज्या उत्क्रमज्याखण्डों से इसके चाप करने से नतासु प्रमाण होता है, सिद्धान्तशेखर में "यदि वा पलकर्णताड़िताया-स्त्रिभजीवोत्थकृतेः" इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से श्रीपति, तथा "पलश्रुतिघ्नस्त्रिगुणस्य वर्गो द्युज्येष्टकर्णाहितिहृत्' इत्यादि से भास्कराचार्य ने भी आचार्योक्त के अनुरूप ही कहा है इति ॥४१-४२-४३॥

इदानीं पुनः प्रकारान्तरेणाह

विज्यावर्गः पलकर्णघ्नो द्युज्याष्टकर्णोद्धृता फलोनान्त्या।
चापस्तोक्रमजीवा धनुर्विनार्वाग्नतप्राणाः ॥४४॥

चरदलजीवोनाधिकफलक्रमज्या धनुश्चरार्धेन ।
युतहीनं पूर्वान्हे दिवसगतं शेषमपरान्हे ॥४५॥

वा. भा.—दिनार्धकर्णेनान्त्यां संगुण्य छायाकर्णेन विभजेत् । फलं ज्या भवति तामन्त्यातो विशोध्य शेषस्योत्क्रमजीवाभिश्चापं च दिनार्धोन्नता प्राणा भवन्ति । विपरीतशोधनादिविकल्पा प्राग्वत् दिनार्धमित्येव ज्या यतो उनो व्यस्तत्रैराशिकवासनेयं यदि दिनार्द्धे छायाकर्णस्यान्त्यातुल्या ज्या तदा पृष्टछायाकर्णस्य केति । अतो दिनार्द्धकर्णेन वा संगुणिता छायाकर्णेन विभजेत् । येन फलमिष्टकालिका ज्या भवति । शेषवासना प्राग्वदिति । अर्थात् ययैव यदुन्नतकालछायानयनं सुकृतं तद्वैपरीत्येनोन्नतकालानयनार्थमार्यामाह ॥४४॥

चरदलजीवाक्षयवृद्धिज्या तदा फलमुत्तरगोले ऊनं कर्तव्यं दक्षिणे वा युतं तस्य तादृश सत क्रमज्याद्यैश्चापं कार्यं तच्चापं चरदलप्राणैर्युतमुत्तरगोले कार्यम् । दक्षिणे हीनमेवं कृते या लिप्ताः तत्ते प्राणाः भवन्ति । पूर्वाह्णे यदि प्रश्न तद्दिनगता । अथापराह्णे तद्दिनशेषाः तत्रापि विपरीतशोधनं प्राग्वत् । अत्र फलं ज्योच्यते चरदलजीवाक्षयवृद्धिज्या त्रिज्यावृत्ते तेनात्र वासना क्षितिजाकांतरस्थेन स्वाहोरात्रखण्डेन योज्या पूर्वप्रदर्शितछायाक्षेत्रमिति । अथ यत्रक्षितिजोन्मंडलांतरे रविर्वर्तते । तत्र नतिघटिकाभ्यः छायानयने छायातश्च नतिघटिकानयने यद्विशेषकर्म तदार्यार्धेनाह ॥४५॥

वि. भा.— अन्त्या दिनदलकर्णगुणा (दिनार्धच्छायाकर्णगुणिता) छायाकर्णभक्ता फलेन हीनाऽन्त्या यच्छेषं तस्योत्क्रमज्याखण्डैश्चापं कार्यं तदा दिनार्धात् नतासवो भवन्ति । चरदलजीवया हीनं युतं च फलं यद् भवति तस्य क्रमज्याखण्डैश्चापं कार्यं चरार्धेन युतहीनं तदा पूर्वाह्णे दिनगतमपराह्णे दिवसशेषं भवतीति ॥ ४४५॥

अत्रोपपत्तिः

$$\text{दिनार्धशङ्कुः} = \frac{\text{त्रि. १२}}{\text{दिनार्धकं}},\ \text{इष्टशङ्कुः} = \frac{\text{त्रि. १२}}{\text{इष्टकं}}\ \text{ततोऽनुपातो यदि}$$

दिनार्धशङ्कुना हृतिर्लभ्यते तदाऽभीष्टशङ्कुना किं समागच्छतीष्टहृतिः

$$= \frac{\text{हृति. इष्टं}}{\text{दिनार्धशं}} = \frac{\text{हृति. त्रि. १२. दिनार्धकं}}{\text{त्रि. १२. इष्टकं}} = \frac{\text{हृति. दिनार्धकं}}{\text{इष्टकं}}$$

$$\text{ततोऽनुपातेनेष्टान्त्या} = \frac{\text{इहृति. त्रि}}{\text{कु}} = \frac{\text{हृति. दिनार्धकं. त्रि}}{\text{कु. इष्टकं}}$$

$= \frac{\text{अन्त्या. दिनार्धक}}{\text{इछाक}} =$ फलम् । ततः अन्त्या−फल=अन्त्या−इष्टान्त्या=

नतोत्क्रमज्या, एतस्या उत्क्रमज्याखण्डैश्चापं कार्यं तदा नतासवो भवन्ति। उत्तर-दक्षिणगोलक्रमेण इष्टान्त्या∓चरज्या=सूत्रम्, अस्य चापं चरार्धेन युतं हीनं तदा दिनगतं दिनशेषं च भवतीति, सिद्धान्तशेखरे "अन्त्यां दिनार्धश्रवणेन हृत्वा भजेत् स्वकर्णेन फलोनितान्त्या। शेषस्य धन्वोत्क्रमशिञ्जिनीभिर्नता दिनार्धादथवाऽसवः स्युः " इत्यनेन श्रीपतिना, 'दिनार्धकर्णादथवाऽन्त्यकाघ्नात् इत्यादिना भास्कराचार्येण चाऽऽचार्योक्तानुरूपमेवोक्तमिति ॥४४-४५॥

अब पुनः प्रकारान्तर से कहते हैं

हि. भा.—अन्त्या को दिनार्ध कर्ण से गुणा कर इष्ट छायाकर्ण से भाग देने से जो फल होता है उसको अन्त्या में घटाने से जो शेष रहता है उसका उत्क्रमज्या खण्डों से चाप करने पर नतासु प्रमाण होता है । फल (इष्टान्त्या) में उत्तर और दक्षिण गोलक्रम से चर-ज्या को हीन और युत करने से जो होता है उसका क्रमज्या खण्डों से चाप करना उसमें चरार्ध को जोड़ने और घटाने से पूर्वाह्न में दिनगत और अपराह्न में दिनशेष होता है इति ॥४४-४५॥

उपपत्ति

$\text{दि}\frac{1}{2}\text{शं} = \frac{\text{त्रि}\cdot १२}{\text{दि}\frac{1}{2}\text{क}}$, इष्टशङ्कु$= \frac{\text{त्रि}. १२}{\text{इछाक}}$ अब अनुपात करते हैं यदि दिनार्धशङ्कु में हृति पाते हैं तो इष्टशङ्कु में क्या इससे इष्टहृति आती है,

$\frac{\text{हृति इशं}}{\text{दि}\frac{1}{2}\text{श}} = \frac{\text{हृति. त्रि. १२. दि}\frac{1}{2}\text{शं}}{\text{त्रि. १२. इछाक}} = \frac{\text{हृति. दि}\frac{1}{2}\text{क}}{\text{इछाक}}$ अब इससे इष्टान्त्या

$= \frac{\text{इहृति. त्रि}}{\text{द्यु}.} = \frac{\text{हृति. दि}\frac{1}{2}\text{क. त्रि}}{\text{द्यु. इछाक}} = \frac{\text{अन्त्या. दि}\frac{1}{2}\text{क}}{\text{इछाक}} =$ फल, अन्त्या−फल=

अन्त्या−इष्टान्त्या=नतोत्क्रमज्या, उत्क्रमज्याखण्डों से इसका चाप नतासुमान होता है। उत्तर गोल और दक्षिण गोल क्रम से इष्टान्त्या∓चरज्या=सूत्र, इसके चाप में चरासु को जोड़ने और घटाने से दिनगत और दिनशेष होता है, सिद्धान्तशेखर में 'अन्त्यां दिनार्ध-श्रवणेन हृत्वा' इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से श्रीपति 'दिनार्धकर्णादथवान्त्य-काघ्नादित्यादि से भास्कराचार्य भी आचार्योक्तानुरूप ही कहते हैं इति ॥४४-४५॥

इदानीं नवत्यधिकचापस्योत्क्रमज्यां त्रिज्यातोऽधिकाया उत्क्रमज्यायाश्चापं चाह

उत्क्रमजीवाभ्यधिकक्रमज्यया संयुतं धनुर्वनुषा. ।
व्यस्तविशुद्धौ हीनाश्चरासवः पूर्ववच्छेषम्. ॥४६॥

वा. भा.—अभ्यधिकस्य क्रमज्याभ्यधिकक्रमज्या छायानयनेन वा घटिका पंचदशभ्योऽधिका भवन्ति । तासां क्रमज्येत्यर्थः । तया संयुतव्यासार्धमुत्क्रमजीवा तदा भवतीति यावत् । एवं नतकालानयने धनुषां संयुतं कार्यम् । एतदुक्तं भवति । फलोनाऽन्त्या यदि व्यासार्धादधिका भवति तदधिकक्रमधनुषा व्यासार्धं धनुरधिकं कृत्वोत्क्रमचापं तदाऽयमर्थः सूत्रे सवासनिको व्याख्यायते । अथ तत्रस्थरवेरुन्नतकालानयने तद्विशेषकर्म तदुत्तरार्धेनाह ।

व्यस्तविशुद्धौ हीनाश्चरासवः पूर्ववच्छेषम् । लब्धं सौम्ये क्षितिज्यया हीनमित्युक्तं तत्र च क्षितिज्याधिका भवति । तदा विपरीतसाधनं कृत्वा गुणितं व्यासार्धेन स्वाहोरात्रार्द्धाद् धनुर्यद्भवति तस्य ये प्राणाः तैः सर्वदा हीनाश्चरासवः कार्याः एवं कृते तत्र काले दिनगतकालशेषो भवति । एवमादिष्टं कर्म यत्र यत्र च संभवति तत्र तत्रास्माभिः पूर्वमेव व्याख्यातो वासनया सह । उन्नतकालाच्छायानयने तत्र प्रदेशे यत्कर्म तत्पूर्वमेवोक्तमल्पः, प्रश्नासूनमित्यनयार्यया चार्येण तथा वासनिको मया व्याख्यातः । पूर्ववच्छेषमिति स मयात्र शेष एव प्रदर्शित इति प्रायः एव छायातः कालानयनमित्यभीष्टदिनदलच्छायां बहुधार्ककान्त्यक्षान्दृष्ट्वा यो वेत्ति इत्यस्य प्रश्नस्योत्तरमार्यांचतुष्टयेनाह ॥४६॥

हि. भा.—नवत्यंशकला ५४०० म्यो यावन्तः कला अभ्यधिकास्तासां क्रमज्यया युतं व्यासार्धं (त्रिज्या) नवत्यंशकलाधिकचापस्योत्क्रमज्या स्यात् । तथा त्रिज्यातो यावत्यधिकोत्क्रमज्या क्रमज्याखण्डैस्तच्चापं नवत्यंशकलायां ५४०० युतं तदा चापं भवेत् । उत्तरगोले यदि चरासुभ्य उन्नतकालासवोऽल्पास्तदा व्यस्तविशुद्धिस्तस्यां सत्यां धनुषा (उन्नतकालासुभिः) चरासवो हीनास्तज्ज्या विपरीतं सूत्रसंज्ञं भवेत् । ततः शेषमिष्टहृत्यादिकं पूर्ववज्ज्ञेयं किन्तु धनर्णयोरन्तरमेव योग इति युक्त्या योगे वियोगो विधेयः ॥४६॥

अत्रोपपत्तिः

अत्रोपपत्तिस्तु व्याख्यारूपैव बोध्येति. सिद्धान्तशेखरे "तिथिभ्यो घटीभ्योऽधिकं चेन्नतं स्यात् त्रिजीवाधिकोत्क्रमज्या समेता । भवेदुत्क्रमज्याऽधिकस्य क्रमोत्थं धनुः खाङ्कभागाधिकं व्यस्तचापम्" श्रीपतिनाप्याचार्योक्तस्यैव स्पष्टीकरणं कृतमिति विज्ञैर्विवेचनीयम् ॥४६॥

अब नवत्यंशाधिक चाप की उत्क्रमज्या को तथा त्रिज्या से अधिक उत्क्रमज्या के चाप को कहते हैं

हि. भा.—नवत्यंशकला ५४०० से जितनी कला अधिक है उसकी क्रमज्या को त्रिज्या में जोड़ने से नवत्यंश कलाधिक चाप की उत्क्रमज्या होती है, तथा त्रिज्या से उत्क्रमज्या को

जितनी अधिक है क्रमज्या खण्डों से उसके चाप को नवत्यंशकला में जोड़ने से चाप होता है। उत्तर गोल में यदि चरासु से उन्नत कालासु अल्प हो तो विलोमशोधन होता है अर्थात् चरासु में उन्नत कालासु को घटाने से शेष की ज्या विपरीत सूत्रसंज्ञक होती है, इससे शेष (इष्ट हृति आदि) पूर्ववत् समझना चाहिए, किन्तु धन और ऋण का अन्तर ही योग होता है इस युक्ति से योग में वियोग करना चाहिये इति ॥४६॥

उपपत्ति

उपपत्ति व्याख्यारूप ही समझनी चाहिए, सिद्धान्तशेखर में 'तिथिघ्यो घटीभ्योऽधिकं' इत्यादि संस्कृतोपपत्तियों में लिखित श्लोक से श्रीपति ने आचार्योक्त का ही स्पष्टीकरण किया है इति ॥४६॥

इदानीं दिनार्धोन्नततनतांशसाधनं, दिनार्धच्छायानयनं मध्यच्छायानयनं हृत्यादीनां बहुसाधनत्वञ्चाह

दिनमध्यार्ककान्त्यक्षभागयोगान्तरं समान्यदिशोः ।
नतभागा नतभागान्नवतेः प्रोह्योन्नताः शेषाः ॥४७॥

नतभागज्या द्वादशगुणोन्नतांशज्यया हृता लब्धम् ।
इष्टदिनार्धच्छाया यथोक्तकरणैर्दिनार्धाद्वा ॥४८॥

उन्नतजीवाभक्तं व्यासार्धं द्वादशाहतं करणैः ।
मध्यच्छायाकरणैर्द्वादशकृत्यन्तरपदं वा ॥४९॥

भुदलान्त्या ज्या छेदो मध्यच्छाया यथोक्तकरणैर्वा ।
अन्त्या ज्या छेदाद्यैर्मध्यच्छायाऽथवा बहुधा ॥५०॥

वा. भा.—इष्टदिने दिनार्द्धकालिकां स्फुटार्ककान्तिज्यां कृत्वा नतभागाः भवन्ति। दिनमध्यार्ककान्तिभागास्तेषां स्वाक्षभागैः सहैकदिग्योगं भिन्नदिगन्तरं कृत्वा नतभागाः भवन्ति। तेषां योगे दिग् ज्ञायते। एवं रवियोगे च ये शेषास्तेषां या दिक् सा नतभागानां भवति। ताश्च नवतेः संशोध्य शेषभागा उन्नतभागाः भवन्ति। ततो नतभागानां ज्यां कृत्वा द्वादशभिर्गुणयेदुन्नतभागज्यया विभजेत्। फलमिष्टदिनार्धच्छायांगुलरूपा द्वादशांगुलस्य शंकोरेवं येनैव प्रमाणेन संभववृद्धिर्दिननतभागज्यागुणितोन्नतांशज्यया विभज्यते तस्यैव दिनार्द्धे वार्धच्छाया भवति। यथोक्तकरणैर्दिनार्धाद्वेति दिनदलप्रमाणिकां परिकल्प्यागतशेषाद्ध इत्यादिभिः सूत्रैरुन्नतकालेन बहुधा छायादिनार्द्धाद्वा या भवति। अथवा व्यासार्धद्वादशहतमुन्नतभागज्यया विभजेत् फलं दिनदलकरणैः करणं द्वादशकृत्योरंतरपदं वानेन प्रकारेण मध्यच्छाया भवन्ति। अथवा भुदलांतज्यामध्याद्ध छेदं परिकल्प्य तत्प्रकारेण मध्यच्छायानयनं कार्यमथवांत्यां दिनार्धज्यां परिकल्प्य ज्यया छेदेन वा ये प्रकाराः उक्ताः तैर्वा मध्यच्छाया कार्या तस्या दक्षिणोत्तरत्वं नतभागवशेन

दक्षिणोत्तराभिमुखो सौम्यैश्च दक्षिणमुखी भवति। सा च दक्षिणमुखीभागा चतुर्विंशतेरूनो यत्राक्षस्तत्र संभवतीति। अत्रेयं वासना स्वदेशयाम्योत्तरमंडलावगाहिनिसहस्रकिरणेभ्यो दिनार्द्धं भवति।

स्वाक्षश्च समंडलविषुवन्मंडलयोरंतरं तत्रान्तर्गतविषुवन्मंडलादुत्तरेण सममंडलदक्षिणे च रविर्भवत्यतोऽक्षभागेभ्यः क्रांत्यंशा विशोध्यन्ते। यदा स्वाक्षादल्पा नतभागा भवन्ति सममंडलस्यासक्तत्वाद्रवेः यस्मात्सममंडलाक्रान्तिरे ये भागास्ते नताः दक्षिणेन यदिसममंडलदक्षिणगो रविरथ सममंडलं न प्रविशति तच्च क्रान्तिभागेभ्योऽक्षभागा विशोध्यन्ते। स्वाक्षस्योनत्वादुत्तरायाः क्रान्तेर्यतो विषुवदपमंडलान्तरं क्रान्तिभागेभ्योऽक्षभागाः विशोध्यन्ते। स्वाक्षस्योनत्वादुत्तरायाः क्रान्तेर्यतो विषुवतो नतभागाः भवन्ति। दक्षिणगोले च सममंडलाद् दक्षिणेन विषुवन्मंडलं स्वाक्षभागैः ततश्च दक्षिणोत्तररविक्रान्तिभागैः। अतस्तेषां योगः सर्वदा नतभागा भवन्ति। तेनोक्तं दिनमध्यार्ककान्त्यक्षभागयोगान्तरसमान्यदिशो नतभागा इति। यदान्तरोत्तराक्रान्तिरक्षभागतुल्या तदा मध्याह्ने शंक्वादीनां छायाभावः सममंडलमध्ये यतो विवस्वान् स्थितस्तदा तत्र नतभागातीतः नतभागा स्वेष्टादीनक्षांस्तान्। भागान्नवतेः संशोध्य शेषा उन्नता भवन्ति। यतो दक्षिणक्षितिजादुत्तरक्षितिजाद्वा याम्योत्तरमंडलगत्या सममंडलमध्यं नवतिभागाः। तस्मादुन्नतभागस्तद्वदेवस्थिकालं शंकुभावा भवन्ति। एवं नतभागज्या दृग्ज्याक्षज्यावदुन्नता भागज्या शंकुर्लंबज्यावत्तद्वर्गमूलं कर्णे व्यासार्द्धं सर्वयाम्योत्तरमंडले गोले प्रदर्शयेत्। सममंडलादुत्तरेण दक्षिणेन वा तत्रैवंस्थिते त्रैराशिकेन छायानयने यद्युन्नतांशज्या शंकोर्नतभागा ज्या छाया तत् द्वादशांगुलस्य शंको केति फलं दिनदलछाया। तेनोक्तं नतभागज्या द्वादशगुणोन्नतांशज्याहृता लब्धमिष्टदिनार्द्धछायेत्येवमिष्टशंकुप्रमाणेनापि यथादिनगतशेषादेवं दिनार्द्धतुल्यं गतकालं परिकल्प्य शेषकालेनोक्तवच्छायामानयेत्तदप्युपपद्यते कालत्वात् स्वाहोरात्रवृत्ते वासनाभेद एव अथवा कर्णं त्रैराशिकेन छायानयनं यद्युन्नतज्यांशको व्यासार्द्धकर्णः तद्द्वादशांगुलशंको क इति फल दिनार्द्धछायाकर्णस्तत्कर्णसंक्रते कोटिकृति विशोध्य मूलं भुजेति छाया भवति। अथोपपद्यते अथवा चुदलांत्यज्यासूत्रं प्राक्दर्शितं मध्याह्ने खेदं परिकल्प्य खेदाच्च छायानयनवासनाभिमध्यच्छायानयनं तदपि युज्यते। अथवांत्यां मध्याह्ने ज्यां परिकल्प्य चुदलांत्यज्यासूत्रे ते मध्यज्याछेदेनोक्तछायानयनवत्तिर्यक्त्वाद्विनाद्धं पतत्यपि युज्यते। तस्मात्सर्वमुपपन्नं यदास्थितं गोले प्रदर्शनेदिति यदुक्तकरणैर्विनाद्धात्यर्थाद्धे त्यस्य सूत्रं खंडकस्य प्रपंचार्थमार्यार्द्धमनन्तरोक्तमिदं तस्मात्तैरुक्तमत्र न संक्रामति क्रान्तिज्ञसममंडलसंकुं वो यानयोरुक्तत्व प्रानयोत्तरयार्थमाह ॥४७-४८-४९-५०॥

वि. भा.—सदक्षिणोत्तरोर्ध्वान्हकपालिकरविक्रान्त्यक्षांशयोर्योगो भिन्नदिक्क-

योस्तयोरन्तरं तदा मध्याह्ने नतांशा भवन्ति, नवतेर्नतांशान् विशोध्य शेषा उन्नतांशा बोध्या इति॥४७॥

अत्रोपपत्तिः

यदि मध्यान्हकाले खस्वस्तिकनिरक्षखस्वस्तिकयोर्मध्ये याम्योत्तरवृत्ते रविस्तदा रवितो निरक्षखस्वस्तिकं यावद्रविक्रान्तिः। खस्वस्तिकनिरक्षखस्वस्तिकयोरन्तरमक्षांशा अत्र द्वयोरन्तरेण विनतांशा रवितः खस्वस्तिकं यावत्। यदि च निरक्षखस्वस्तिकाद् दक्षिणे रविस्तदा रवितो निरक्षखस्वस्तिकं यावत्क्रान्तिः। अत्राक्षांशरविक्रान्त्योः समदिक्कयो (अक्षांशाः सर्वदा दक्षिणा, नाड़ीवृत्ताद्रविर्दक्षिणेऽस्त्यतः क्रान्तेरपि दिक्दक्षिणा) र्योगेन रविमध्यनतांशा भवन्ति, खस्वस्तिकात्समस्थानं यावन्नवतिरत्र नतांशशोधनेन रवितः समस्थानं यावन्मध्योन्नतांशाः स्युः। सिद्धान्तशेखरे "मध्यन्दिनोष्णकिरणापमचापभागस्वाक्षांशयोगविवरं सदृशान्यदिक्त्वे। याम्योत्तरा नतलवास्तरणेः खमध्यात् तेऽप्युन्नता निपतिता नवतेर्भवेयुरिति" श्रीपत्युक्तं सिद्धान्तशिरोमणौ "पलावलम्बावपमेन संस्कृतौ नतोन्नते ते भवतो दिवादले लवादिकं वा नवतेर्विशोधितं नतं भवेदुन्नतमुन्नतं नतम्" इतिभास्करोक्तञ्चाऽऽचार्योक्तानुरूपमेवेति ॥४७॥

अब मध्याह्नकालिक नतांश और उन्नतांश साधन को कहते हैं

हि. भा.—एक दिशा में रविमध्यक्रान्ति और अक्षांश का योग करने से और भिन्न दिशा में अन्तर करने से रवि का मध्यनतांश होता है, नतांश को नवत्यंश में घटाने से शेष उन्नतांश होता है इति ॥४७॥

उपपत्ति

मध्याह्नकाल में यदि खस्वस्तिक और निरक्षखस्वस्तिक के मध्य में रवि है तब रवि से निरक्ष खस्वस्तिक पर्यन्त रवि की मध्यक्रान्ति उत्तर दिशा को है क्योंकि नाड़ीवृत्त से रवि उत्तर में है, तथा अक्षांश की दिशा सर्वदा दक्षिण है इसलिए भिन्न दिशा की रविक्रान्ति और अक्षांश का अन्तर करने से रवि से खस्वस्तिक पर्यन्त रवि का नतांश होता है, यदि निरक्ष खस्वस्तिक से रवि दक्षिण है, तब अक्षांश और क्रान्ति की दिशा एक ही दक्षिण होने के कारण दोनों का योग करने से नतांश होता है, खस्वस्तिक से समस्थान तक नवत्यंश में नतांश को घटाने से निरक्षखस्वस्तिक से समस्थानपर्यन्त मध्य उन्नतांश होता है, सिद्धान्तशेखर में "मध्यन्दिनोष्णकिरणापमचापभागस्वाक्षांशयोगविवरं" इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से श्रीपति तथा सिद्धान्तशिरोमणि में "पलावलम्बावपमेन संस्कृतौ" इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से भास्कराचार्य ने भी आचार्योक्तानुरूप ही कहा है इति ॥४७॥

वि. भा.—नतभागज्या (नतांशज्या दृग्ज्या) द्वादशगुणिता, उन्नतांशज्यया (शङ्कुना) भक्ता लब्धमिष्टदिनार्धच्छाया भवति, वा दिनार्धात् सकाशाद्ययोक्त-करणहृत्यादिभिर्दिनार्धच्छाया साध्येति ॥४८॥

अत्रोपपत्तिः

दृग्ज्या भुजः । शङ्कुः कोटिः । त्रिज्याकर्णः
छायाभुजः । द्वादशाङ्गुलशङ्कुः कोटिः । छायाकर्णः करणः

एतयोस्त्रिभुजयोः साजात्यादनुपातः

$\frac{\text{दृग्ज्या}.१२}{\text{शङ्कु}}$ = छाया. सिद्धान्तशेखरे 'ना शङ्कुरुन्नतगुणः स च कोटिरुक्ता दृग्ज्या भुजा नतगुणस्तु भवेत् प्रभा च। अर्कापवर्तितनरेण हृते च दृग्ज्या त्रिज्ये दले दिनदलोत्थविभा श्रुती च" श्रीपत्युक्तमिति छायासाधनं, सिद्धान्त-शिरोमणौ "दृग्ज्या त्रिजीवे रविसङ्गुणे ते शङ्कूद्धृते भाश्रवणौ भवेताम्" भास्क-रोक्तं छायासाधनञ्चाऽऽचार्योक्तानुरूपमेवेति ॥४८॥

अब इष्ट दिनार्ध में छायानयन को कहते हैं

हि. भा.—नतांशज्या (दृग्ज्या) को बारह से गुणा कर शङ्कु से भाग देने से लब्धि इष्टदिनार्धकालिक छाया होती है, वा दिनार्ध से पूर्वकथितोपकरणों (हृति आदि) से दिनार्धच्छाया साधन करना इति ॥४८॥

उपपत्ति

दृग्ज्या भुज, शङ्कु कोटि, त्रिज्या कर्ण
छाया भुज, द्वादशाङ्गुलशङ्कु कोटि, छायाकर्ण. करण

ये दोनों त्रिभुज सजातीय हैं इसलिए अनुपात करते हैं

$\frac{\text{दृग्ज्या}.१२}{\text{शङ्कु}}$ = छाया, सिद्धान्तशेखर में 'ना शङ्कुरुन्नतगुणः' इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से श्रीपति तथा सिद्धान्त शिरोमणि में "दृग्ज्या त्रिजीवे" इत्यादि से भास्क-राचार्य ने भी छायासाधन आचार्योक्तानुरूप ही कहा है इति ॥४८॥

वि. भा.—व्यासार्धं (त्रिज्या) द्वादशगुणितं, उन्नतजीवा (मध्यशङ्कु) भक्तं तदा मध्यच्छायाकर्णो भवेत् । कर्णद्वादशकृत्यन्तरपदं (छायाकर्णद्वादश-वर्गान्तरमूलं) वा (प्रकारान्तरेण) मध्यच्छाया भवेदिति ॥४९॥

अत्रोपपत्तिः

छायाक्षेत्रानुपातेन $\frac{\text{त्रि}\cdot १२}{\text{शङ्कु}}$=छायाकर्ण ततः $\sqrt{\text{छायाक}^२-१२^२}$= छाया, इति ॥४९॥

अब प्रकारान्तर से मध्यच्छायानयन को कहते हैं

हि. भा.—त्रिज्या को बारह से गुणा कर मध्यशंकु से भाग देने से मध्यच्छाया कर्ण होता है, छायाकर्ण और द्वादश (१२) के वर्गान्तर मूल प्रकारान्तर से मध्यच्छाया होती है इति ॥४९॥

उपपत्ति

पूर्व श्लोक की उपपत्ति में प्रदर्शित छायाक्षेत्रों के सजातीयत्व से अनुपात करते हैं $\frac{\text{त्रि}.१२}{\text{मशंकु}}$=मछायाकर्ण, ∴ $\sqrt{\text{छायाक}^२-१२^२}$=मध्यच्छाया, इति ॥४९॥

वि. भा.—यथोक्तकरणैः (दिनगतशेषाल्पस्येत्यादिप्रकारैः) मध्यान्हे द्युदलान्त्या (हृतिः), ज्या (इष्टान्त्या) छेदः (इष्टहृतिः) मध्यच्छाया प्रसिद्धा, इति सर्वं वा भवति, अथवा—अन्त्या ज्या छेदाद्यैः (अन्त्येष्टान्त्येष्टहृत्यादिभिः) पूर्वकथितविधिनाऽनेकधा मध्यच्छाया साध्येति ॥५०॥

अत्रोपपत्तिस्तु श्लोकोक्तोपकरणैर्बोध्येति ॥५०॥

अब हृति आदियों के बहुसाधनत्व को कहते हैं

पूर्व कथित उपकरणों (दिनगतशेषाल्पस्य इत्यादि प्रकार से) से मध्यान्हकाल में हृति, इष्टान्त्या, छेद (इष्टहृति) मध्यच्छाया ये सब होते हैं। अथवा अन्त्या, ज्या (इष्टान्त्या) छेद (इष्टहृति) आदि से पूर्व कथित विधि से अनेक प्रकार मध्यच्छाया साधन करना इति ॥५०॥

उपपत्ति

श्लोक में कथित उपकरणों से अनेक प्रकार मध्यच्छायानयन करना चाहिए इति ॥५०॥

इदानीं समशङ्कुसाधनमाह

विषुवत्करणेन गुणा विषुवच्छायोद्धृतोत्तरा क्रान्तिः ।
[illegible] समशङ्कुः [illegible] ॥५१॥

वा. भा.— यत्र चोत्तरा क्रान्तिः स्वदेशाक्षज्या ऊना भवति। तद्देशे तदा सममंडलप्रवेशो भवति। तत्प्रवेशकाले च शंक्वानयनार्थं प्रागार्यार्धं तेन यमथः क्रान्तिज्याविषुवत्कर्णगुणा विषुवच्छायाहृता सममंडलप्रवेशो भवेत्। वृहच्छंकुर्भवतीत्यर्थः। अत्र वासनागोले अर्कोदयास्तसूत्रं स्वाहोरात्रवृत्तं च विन्दुस्थं प्रदर्शयेत् तद्यथा सममंडलमेव तत्र काले दृङ्मण्डलं तस्य स्वाहोरात्रेण सह यत्र संपातः तत्रावलम्बकसूत्रमवलंबते। तच्च भूविनिर्गतप्रागपरायतसूत्रस्पृगेव भवति। तावानेव शंकुः सममंडलप्रवेशकाले स्वदेशप्राच्यपररेखा स्थापितापि तावानेतन्मूलस्य सोदयास्तसूत्रेण सह यावदन्तरं तावच्छंकुतलं तच्चाग्रातुल्यं भवति। अतस्त्रैराशिकद्वयेन समंडलशंक्वानयनमेतत्। यदि लंबककोटेर्व्यासार्धकर्णस्तत्क्रान्तिज्याकोटेः क इति फलमर्काग्रा तावदेव सममंडलप्रदेशकाले शंकुतलं ततो द्वितीयं त्रैराशिकं यदि स्वाक्षज्याशंकुतलस्य लंबकः शंकुस्तदग्रातुल्यशंकुतलस्य कः शंकुरित्येवावलंबकस्थाने द्वादशिकाकोटिः अक्षज्यास्थाने विषुवच्छायाभुजा व्यासार्द्धस्थाने विषुवत्कर्णः। तेन प्रथमत्रैराशिके द्वादशको भागहारो द्वितीये गुणकारः तुल्यत्वाच्चतयोः क्रान्तिज्यायां विषुवत्कर्णो गुणकारो विषुवच्छाया भागहार फलं सममंडलशंकुर्भवति। अथोत्तराक्रान्तिज्याक्षज्यातोऽधिका भवति। सममंडलादुत्तरेणैव मध्याह्नं करोति। तत्रसममंडलप्रदेशाभावो दक्षिणाभिमुखी च दिनार्धच्छाया। तदाऽक्षज्यातुल्योत्तराक्रान्तिज्या भवति। तदा मध्याह्ने एव सममंडलं प्रदर्शयति रविं तत व्यासार्द्धमंडलशङ्कु अथ दक्षिणाक्रान्तिज्या भवति। तदा सममंडलात् दक्षिणेनोदयास्तमयौ तत्र सममंडलेन एव"""""इत्येतत्सर्वं गोले प्रदर्शयेत्। स्वाहोरात्रोपमंडलसंपातेषु रव्युपलक्षितं चिह्नं कृत्वा निरक्षदेशे च कदाचिदपि सममंडलप्रदेशोनाशंक्वयते विषुवद्दिवसे भुक्ता भगोलस्याक्षोत्तरभावादिति। अथवा प्रकारद्वयेन सममंडलशंक्वानयनार्थमार्यामाह ॥५१॥

वि. भा.—उत्तरा क्रान्तिः (क्रान्तिज्या) यदाक्षज्यायाः अल्पा तदा क्रान्तिज्या पलकर्णगुणा पलभया भक्ता तदार्के सममण्डलस्थे शङ्कुः (समशङ्कुः) भवेत् ॥५१॥

अत्रोपपत्तिः

पलभा भुजः। द्वादशकोटिः। पलकर्णः कर्णं इत्यक्षक्षेत्रमेकम्। क्रान्तिज्या भुजः। कुज्योनतद्धृतिः कोटिः। समशङ्कुः कर्ण इति द्वितीयमक्षक्षेत्रमनयोः सजातीयत्वादनुपातः $\frac{\text{पलक}\cdot\text{क्रान्तिज्या}}{\text{पलभा}}$ = समशङ्कुः, यदोत्तराक्रान्तिरक्षज्याल्पा तदैव रवेः सममण्डलप्रवेशो भवति, सिद्धान्तशिरोमणौ भास्करेण 'मार्तण्डः समयण्डलं प्रविशति स्वल्पेऽपमे स्वात्पश्चादित्यादिना' सूर्यस्य सममण्डलप्रवेशकथने आचार्योक्तानुरूपमेव कथ्यत इति ॥५१॥

अब समशंकु साधन को कहते हैं

हि. भा.—उत्तर क्रान्तिज्या यदि अक्षज्या से अल्प है तब क्रान्तिज्या को पलकर्ण से गुणा कर पलभा से भाग देने से रवि के सममण्डल (पूर्वापरवृत्त) में रहने पर शङ्कु (समशंकु) होता है इति ॥५१॥

उपपत्ति

पलभा भुज, द्वादशाङ्गुलशङ्कुकोटि, पलकर्ण कर्ण }
क्रान्तिज्याभुज, कुज्योनतद्धृति कोटि, समशंकु कर्ण }
दोनों अक्षक्षेत्र सजातीय हैं इसलिए अनुपात करते हैं

$\frac{\text{पलक.क्रान्तिज्या}}{\text{पलभा}}$ = समशङ्कु । जब उत्तराक्रान्ति अक्षांश से अल्प होती है तब ही रवि के सममण्डल में प्रवेश होता है। सिद्धान्तशिरोमणि में भास्कराचार्य ने 'मार्त्तण्डः सममण्डलं प्रविशति स्वल्पेऽपमे स्वात्पलात् इससे' रवि के सममण्डलप्रवेश के सम्बन्ध में आचार्योक्त के अनुरूप ही कहा है इति ॥५१॥

इदानीं प्रकारान्तरेण समशङ्कुसाधनमाह

सूर्यज्या जिनभागज्यया गुणाऽक्षज्ययाऽथवा भक्ता ।
अग्रा द्वादशगुणिता विषुवच्छाया विभक्ता वा ॥५२॥

वा. भा.—सूर्यज्यां जिनभागज्यया गुणितां नवरदचन्द्रैरित्यर्थः, अक्षज्यया विभजेत्। फलं सममंडलशंकुर्भवति । वासनात्र त्रैराशिकत्रयं यदि व्यासार्द्धतुल्यया सूर्यज्यया जिनभागज्या तदिष्टसूर्यज्यया केति फलं क्रांतिज्या ततो द्वितीयं यदि लंबककोटिव्यासार्द्धकर्णः तत्क्रान्तिज्याकोटे क इति फलमग्रा ततस्तृतीय यद्यक्षज्यातलस्य लंबकशंकुः तदग्राशंकृतलस्य क इति। एवं स्थिते प्रथमत्रैराशिके व्यासार्धभागहारद्वितीये गुणकारः तयोर्नाशे कृते इति द्वितीये त्रैराशिक लम्बकः तृतीये गुणकारः तयोर्नाशे कृते सूर्यज्यया जिनभागज्या गुणकारस्थितेऽक्षज्या भागहारश्च फलं सममंडलशंकुर्भवति द्वितीयानयने त्रैराशिकं यदि विषुवच्छाया शंकुतलस्य द्वादशकः शंकुः तदर्क्षाग्राशंकुतलस्य कः शंकुरिति फलं सममंडलशंकुर्भवति । संस्थानवासना प्राग्वत्प्रदर्शयेद् गोले इति सममंडलकर्णे च योऽक्षसंबज्यां जानातीत्यस्य प्रश्नस्योत्तरमार्यामाह ॥५२॥

वि. भा.—सूर्यज्या (रविभुजज्या) जिनज्यया गुणाऽक्षज्यया भक्ताऽथवा (प्रकारान्तरेण) समशङ्कुर्भवेत् । वा अग्रा द्वादशगुणिता पलभया भक्ता तदा समशङ्कुर्भवेदिति ॥५२॥

अत्रोपपत्तिः

यदि त्रिज्यया जिनज्या लभ्यते तदा रविभुजज्यया किं समागच्छति क्रान्तिज्या $=\frac{\text{जिज्या}\cdot\text{भुज्या}}{\text{त्रि}}$, अक्षज्या भुजः। लम्बज्या कोटिः त्रिज्या कर्णं इत्यक्षक्षेत्रं प्रथमम्। क्रान्तिज्या भुजः। कुज्योनतद्धृतिः कोटिः। समशङ्कुः कर्णं इत्यक्षक्षेत्रं द्वितीयमनयोः सजातीयत्वादनुपातो यद्यक्षज्या त्रिज्या लभ्यते तदा क्रान्तिज्यया किं समागच्छति समशङ्कुः $=\frac{\text{त्रि}\times\text{क्रांज्या}}{\text{ज्या}}=\frac{\text{त्रि}}{\text{अज्या}}\cdot\frac{\text{जिज्या}\cdot\text{भुज्या}}{\text{त्रि}}$

$=\frac{\text{जिज्या}\cdot\text{भुज्या}}{\text{अज्या}}$, तथा पलभा भुजः। द्वादशकोटिः। पलकर्णः कर्णं इत्यक्षक्षेत्रं प्रथमम्। अग्रा भुजः। समशंङ्कुः कोटिः। तद्धृतिः कर्णं इत्यक्षक्षेत्रं द्वितीयमनयोः सजातीयत्वादनुपातो यदि पलभया द्वादशकोटिस्तदा ऽग्रया किं समागच्छति समशङ्कुः $=\frac{१२\cdot\text{अग्रा}}{\text{पलभा}}$ एतेनाऽऽचार्योक्तं सर्वमुपपन्नम्। सिद्धान्तशेखरे श्रीपतिना 'दोर्ज्या १२वेः पृथगघो जिनभागमौर्व्या क्षुण्णां विभज्य पललम्बकशिञ्जनीभ्याम्। आद्यं फलं हि सममण्डलशङ्कुसंज्ञमित्यनेन' ऽऽचार्योक्तानुरूपमेव समशङ्कोरानयनमभिहितमिति ॥५२॥

अब प्रकारान्तर से समशंकुसाधन को कहते हैं

हि. भा.—रवि भुजज्या को जिनज्या से गुणाकर अक्षज्या से भाग देने से वा (प्रकारान्तर से) समशंकु होता है, वा अग्रा को बारह से गुणाकर पलभा से भाग देने से समशङ्कु होता है इति ॥५२॥

उपपत्ति

यदि त्रिज्या में जिनज्या पाते हैं तो रवि भुजज्या में क्या इससे क्रान्तिज्या आती है $\frac{\text{जिज्या}\cdot\text{भुज्या}}{\text{त्रि}}=$क्रांज्या, अक्षज्या भुज, लम्बज्या कोटि, त्रिज्या कर्ण यह प्रथम अक्षक्षेत्र है, क्रान्तिज्या भुज, कुज्योनतद्धृति कोटि, समशङ्कु कर्ण यह द्वितीय अक्षक्षेत्र है, दोनों सजातीय हैं इसलिये अनुपात करते हैं यदि अक्षज्या में त्रिज्या पाते हैं तो क्रान्तिज्या में क्या इससे समशंकु आता है $\frac{\text{त्रि}\cdot\text{क्रांज्या}}{\text{ज्या}}=$समशङ्कु $=\frac{\text{त्रि}}{\text{अज्या}}\cdot\frac{\text{जिज्या}\cdot\text{भुज्या}}{\text{त्रि}}=\frac{\text{जिज्या}\cdot\text{भुज्या}}{\text{अज्या}}$, तथा

पलभा भुज, द्वादश कोटि, पलकर्ण कर्ण यह प्रथम अक्षक्षेत्र; अग्रा भुज, समशंकु कोटि, तद्धृति कर्ण यह द्वितीय अक्षक्षेत्र; दोनों अक्षक्षेत्र सजातीय हैं इसलिये अनुपात करते हैं यदि पलभा में द्वादश कोटि पाते हैं तो अग्रा में क्या इससे समशंकु आता है $\frac{१२.\text{अग्रा}}{\text{पभा}}$ = समशंकु इससे आचार्योक्त उपपन्न होता है, सिद्धान्तशेखर में श्रीपति ने 'दोर्ज्या रवेः पृथगधो जिनभाग-मौर्व्या, इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से समशंकु का साधन आचार्योक्तानुरूप ही कहा है इति ॥५२॥

इदानीं सममण्डलच्छायाकर्णसाधनमाह

द्वादशविषुवच्छायागुणिते पृथगक्षलम्बजीवे वा।
क्रान्तिहृते सममण्डलकर्णौ प्राग्वत् पृथक् छाये ॥५३॥

वा. भा.—अक्षज्यां द्वादशहतां क्रान्तिज्यया विभजेत्। फलं सममंडलप्रवेश-काले छायाकर्णो भवति। अथवा लम्बज्यां विषुवच्छायासंगुणाम्य क्रान्तिज्यया विभजेत्। फलं सममंडलप्रवेशे छायाकर्णः ततस्तच्छायानयनं प्राग्वच्छायया चोक्तवद्घटिकानयनमिति, अत्र भागहारफलयोर्व्यपने छायानयनं त्रैराशिकत्रयं भक्रान्त्यानयनं यत्रतः प्रकल्पितं तद्यथा यदिसमंडलकर्णस्य द्वादशकः शंङ्कुः तद्ववत्रिफलं सममंडलशंङ्कुः। ततो द्वितीयं यदिलम्बकशंकोरक्षतुल्यं शंकुतलं तदस्य कियदिति फलमर्काग्रातुल्यं शङ्कुतुल्यं ततस्तृतीयं यदिव्यासार्द्धकर्णस्य लंबककोटिः तदग्राकर्णस्य द्विगुणं कृत्वा त्रिज्यावर्गाद्विशोधयेत्। शेषस्य पदं कोणशंकुर्भवति। उत्तराया क्रान्तिज्याया उत्तरयो विदिशोः दक्षिणादक्षिण-योश्चेत्येवं शंकुमानीय यच्छायाप्राग्वत्कार्या सा कोणछाया भवति यथा आत्मीय छाया तच्च नतोन्नतघटिकानयनमपि प्राग्वदित्यत्र वासना तद्यथा गोले स्वक्षितिजे यथा भूमध्यार्धभेदिपूर्वापरायतं सूत्रं तिष्ठति दक्षिणोत्तरायतं च (ख) अपरं सूत्रद्वयं भूमध्यार्धभेद्याग्रे विन्यसेत्। एकमैशानीनैर्ऋत्यायत द्वितीयमाग्नेयी च यथायतं ततश्चतुर्ष्वपि कोणेषु सूत्राणि बद्धान्यानि चत्वारि सूत्राणि ज्यासंस्थानानि कोणात्कोणे बध्नीयादेवं समचतुरस्रक्षेत्रमुपपन्नं भवति। कर्णद्वयसहितं कर्णसंपातश्च भूगोलमध्ये ततः पूर्वापराय दक्षिणोत्तरायत-सूत्राभ्यामवस्थितानि चत्वारि समचतुरस्राणि क्षेत्राणि प्रकल्प्यान्यान्यस्मि-न्नपि क्षेत्रे तानि चतसृणामपि विदिशां सम्बद्धानि भवन्ति। भूमध्यविनिर्गत-कोणप्रतिसूत्रकर्णयुतानि। यथा तदृदैवसिकमग्रासूत्रगोलाध्यायप्रदर्शितविधिना स्वक्षितिजे दक्षिणोत्तरायतं पूर्वापरयोः बद्धं तदग्रयोश्च पूर्वापरायतमुदयास्त-सूत्रं बध्नीयात्। ततः स्वक्षितिजे स्वाहोरात्रवृत्तं विन्यस्य दृङ्मंडलं कोणाव-

गाहि विन्यसेत् । उपर्यधश्च जनितस्वस्तिकद्वयमेवमवस्थिते व्यक्तगणितेन वामनां प्रदर्शयेत् । मध्यमाहरणबीजसूत्रेण कुट्टकाध्यायोपनिबद्धो नोनभवः व्यक्तः शोध्यः, यस्माद्रूपाणि तद्यस्तद्वर्गाहतरूपाणामव्यक्तार्धकृतिसंयुतानां यत्पदमव्यक्तार्धोनं तद्वर्गविभक्तमव्यक्त इति । अत्र तावद् दक्षिणगोलस्थे रवावानेय्यां शंकुं प्रदर्शयेत् । तद्यथा तत्कोणदृङ्मंडलस्वाहोरात्रयोर्यंत्र संपातः तत्रावलंबकसूत्रं वदुध्वाऽवलम्बयेत् तेनावलम्बकसूत्रेण कोणसूत्रस्य यः प्रदेशः स्पृष्टः तत्र शंकुमूलं तस्य शंकुमूलस्य स्वोदयास्तसूत्रेण सह यावदन्तरं तावच्छंकुतलं यावच्च प्राच्यपरसूत्रोदयास्तसूत्रयोरन्तरं तावत्यकोटी शंकुतलयोगो भुजो भवति । प्राच्यपरा शंकुमूलयोरंतरं तदेव बाहु भवति । कोणप्रवेशकाले समचतुरस्रक्षेत्रं तदेवं प्रथमं स्थिते राशिके व्यासार्धं गुणकारः तृतीये भागहारः तयोर्नाशे कृते द्वितीयेऽलबको भागहारः तृतीये गुणकारः तयोश्च नाशे कृते अक्षज्या द्वादशगुणा यावच्छायाकर्णेन विभज्यते तावत्फलं क्रान्तिज्या भवति । यावच्चाक्षज्या द्वादशवधः क्रान्तिज्यया विभज्यते तावत्फलं छायाकर्णो भवतीत्युपपन्नम् । यावानेव द्वादशाक्षज्ययोर्वधः तावानेव विषुवच्छायालंबज्ययोर्वधो भवति । भागहारश्चैक एव अत्र फलेन कश्चिद्भवति । ततोऽपि सममंडलच्छायाकर्णानयनं सिद्धं शेषं प्राग्वद्योज्यम् संस्थानादिकमिति यो जानाति कोणशंकुछायाघटिका इत्यस्य प्रश्नस्योत्तरमार्यात्रयेणाह ॥५३॥

वि. भा.—अक्षज्या लम्बज्ये पृथक् द्वादशपलभागुणिते क्रान्तिज्यया भक्ते तदा सममण्डलकर्णो भवतः । ततः पूर्ववत् पृथक् छाये भवत इति । ५३॥

अत्रोपपत्तिः

प्रथाऽक्षक्षेत्रानुपातेन $\frac{\text{त्रि}\cdot\text{क्रांज्या}}{\text{अक्षज्या}}$ = समशङ्कु । ततो यदि समशङ्कुना त्रिज्याकर्णस्तदा द्वादशाङ्गुलशङ्कुना किं समागच्छति सममण्डलकर्णः = $\frac{\text{त्रि}\cdot १२}{\text{समश}} = \frac{\text{त्रि}.१२}{\frac{\text{त्रि}.\text{क्रांज्या}}{\text{अज्या}}} = \frac{१२.\text{अज्या}}{\text{क्रांज्या}}$ हरभाज्यौ लम्बज्यया गुणितौ तदा $\frac{१२.\text{अज्या}.\text{लंज्या}}{\text{लंज्या}.\text{क्रांज्या}}$ $= \frac{\text{पलभा}.\text{लंज्या}}{\text{क्रांज्या}}$ = सममण्डलकर्णः, एतावताऽऽचार्योक्तमुपपन्नं सर्वमिति सिद्धान्तशेखरे 'लम्बाक्षजीवे पलभा रविघ्ने त्वपक्रमज्या विहृते क्रमेण । यद्वा रवेर्भा समवृत्तकर्णो ताभ्यां प्रभे पूर्ववदेव साध्ये इत्यनेन' श्रीपतिनाऽऽचार्योक्तमुपपन्नमेवोक्तमिति. सूर्यसिद्धान्ते 'लम्बाक्षजीवे विषुवच्छाया द्वादशसंगुणे । क्रान्तिज्याप्ते तु तौ कर्णौ सममण्डलगे रवौ' ऽप्यनेन तदेव कथ्यते इति ॥५३॥

अब सममण्डल छायाकर्णानयन को कहते हैं

हि. भा.—अक्षज्या और लम्बज्या को पृथक्-पृथक् द्वादश और पलभा से गुणाकर क्रान्तिज्या से भाग देने से दो प्रकार के सममण्डल कर्ण होते हैं उनसे पूर्ववत् छाया-साधन करना चाहिए इति ॥५३॥

उपपत्ति

अक्षक्षेत्र के अनुपात से $\frac{\text{त्रि.क्रांज्या}}{\text{अक्षज्या}}$ =समशंकु, इससे अनुपात करते हैं यदि समशंकु में त्रिज्या कर्ण पाते हैं तो द्वादशाङ्गुलशङ्कु में क्या इससे सममण्डलकर्ण आता है,

$\frac{\text{त्रि.१२}}{\text{सशं}} = \frac{\text{त्रि.१२}}{\frac{\text{त्रि.क्रांज्या}}{\text{अज्या}}} = \frac{\text{अज्या.१२}}{\text{क्रांज्या}}$=सममण्डलकर्ण हर और भाज्य को लम्बज्या से गुणने से $\frac{\text{अज्या·१२·लंज्या}}{\text{लज्या·क्रांज्या}} = \frac{\text{पभा·लंज्या}}{\text{क्रांज्या}}$=सममण्डलकर्ण, इससे आचार्योक्त उपपन्न हुआ, सूर्यसिद्धान्त में भी 'लम्बाक्षजीवे विषुवच्छाया द्वादशसङ्गुणे। क्रान्तिज्याप्ते तु तौ कर्णौ सममण्डलगे रवौ' इससे तथा सिद्धान्तशेखर में "लम्बाक्षजीवे पलभा रविघ्ने त्वपक्रमज्याविहृते क्रमेण। यद्वा भवेतां" इत्यादि से श्रीपति भी आचार्योक्तानुरूप ही कहते हैं इति ॥५३॥

इदानीं कोणशङ्कोरानयनमाह

अर्काग्रावर्गोनं त्रिज्यावर्गार्धमर्ककृतिगुणितम्।
आद्योऽन्योऽग्राद्वादशविषुवच्छायावधो हृतयोः ॥५४॥

विषुवच्छायाकृत्या द्व्यगसंयुतयाऽन्यकृतियुतादाद्यात्।
पदमन्ययुतविहीनं सौम्येतरगोलयोः शङ्कुः ॥५५॥

विदिशोः सौम्येतरयोरुत्तरगोले पदोनयुक्तोऽन्यः।
सममण्डलदक्षिणगेनच्छाया नाडिकाः प्राग्वत् ॥५६॥

वा. भा.—अत्र तावत्कोणशंकूनां प्रभवं प्रदर्शयेत् तद्यथा यत्राक्षभा-भ्यधिकानि सप्तांगुलानि विषुवच्छाया तत पंचपंचाशदक्षभागाः तत्र मिथुनान्त-रविरैशान्यामुदेति। धनुर्घरांतगश्चाग्नेयं तस्माद् देशादुत्तरेण चतसृषु अपि दिक्षु कदाचित्कोणशंकवः संभवन्ति। रवावुत्तरगोलस्थे दक्षिणगोलस्थे च कदाचि-च्छायायामोऽपि संभवति। तस्माद् देशाद् दक्षिणेन यत्र चतुर्विंशतिरक्षभागाः

तदश्चोत्तरेणोत्तरगोलस्थे रवावाग्नेयीनैर्ऋत्योरेव कोणशंकुसम्भवः नैशानीवायव्योः दक्षिणगोले च यत्र भागश्चतुर्विंशतेरयोनाक्षः तत्रोत्तरगोलस्थेऽर्के कदाचिदाग्नेयनैर्ऋत्यो शंकुसम्भवः कदाचिदैशानीवायव्योः स्वाक्षजया नोनाधिकायां यथासंख्यं क्रान्तिज्यामित्यर्थः। दक्षिणगोलस्थे वाग्नेयीनैर्ऋत्योरेव तेषां कोणशंकूनां प्रदर्शनार्थमिदमार्यात्रयम्। तद्यथेष्टदिनेऽर्कांशां कृत्वा तस्य वर्गः कार्यः तनोन त्रिज्यावर्गार्धेषु विशोध्य ततोऽर्कं कृत्या वेदकृतचंद्रसंख्यया गुणयेत्। एवं कृते यो राशि सपद्यते तस्याद्य इति संज्ञा। ततोऽर्कांशा द्वादशस्वदेशविषुवच्छायां तिसृणामपि यो वधो भवति। तस्यान्य इति संज्ञा। ततः स्वदेशविषुवच्छायां कृत्या द्वयगसंयुतया अनया द्वावपि विभजेत् पृथक् पृथक्। तयोः फले तयो स्फुटावाद्यन्यौ कर्मयोग्यौ भवतः। ततस्तस्य कृतिं कृत्वा तया आद्यो युतकार्यस्तस्मात्मूल तस्य पदसंज्ञा भवति। एवं तावत्कर्मसामान्यं सर्वेषु कोणशंकुषु विशेषऽत्रात्र पदमपि युतविहीनं यथासंख्यं याम्योत्तरगोलयोः स्थितेऽर्के शंकुविदिशो सौम्योत्तरयो भवति। यदा सममंडलैशान्योरंतरग्रहविधिः रवे सम्मंडलं यदुत्तरेणैव याति। तथा पदमन्येन युतं कृत्वैशानीप्रवेशशंङ्कुर्भवति। वायव्यप्रवेशेऽपि स एव तुल्यान्ताद् दिनगतशेषया यदा पुनः सममंडलाद् दक्षिणेनाग्नेयोत्तरेणार्कोदयं सममंडलाद् दक्षिणेनैव याति। तदा पदमन्येन विहीन कृत्वाग्नौ प्रवेशशंकुर्भवति। नैर्ऋते स एव प्राग्वत् यदा च सममंडलादुत्तरेणैव प्रविष्टः प्रागपरयोः सममंडलप्रवेशद्वयं कृत्वा अस्तमेति तदा सूर्यमंडलमेनदुत्तरगोले पदोनयुतान्यः सममंडलदक्षिणग इति। तत्रैवं यद्यैशानीसममंडलोदितः सममंडल प्रविश्य दक्षिणविदिशे प्रवेशं करोति तदैवोक्तवत्कर्म कृत्वा पदान्यावानीय ततः पदेनान्या युक्तकार्या स च शंकुर्भवति।

अथ यत्रैशानीमर्यात् क्रमोत्तरेण विवस्वानुदेति तत्रार्कांशा वर्गा त्रिज्यावर्गार्धान्न शुध्यति। तेन तत्र विपरीतशोधनं कृत्वा शेषस्य द्वादशवर्गाहतस्य क्षेति वर्गान्न शुध्यति। तेन तत्र विपरीतशोधनं कृत्वा क्षेषेति संज्ञा। शेषकर्म प्राग्वत्कृत्वा पादान्यावानीय तत्र एकत्र पदेनोनान्यः कार्यः स एवैशानीवायव्योः शंकुर्भवत्यत्र पदेन युक्तोऽन्यश्च कार्यः। स आग्नेयीनैर्ऋत्योः शंकुर्भवति। अथ यत्राग्नेयीमतिक्रम्य दक्षिणेनार्कोदयः तत्रार्कांशावर्गः त्रिज्यार्धान्न शुध्यति। यदि नाम तत्राग्नेयादिभिर्न प्रयोजनं कोणशंकोरसंभवात् त्वेव साक्षे देशे कोणशंक्वानयनं निरक्षदेशे पुनः क्रान्तिज्यावर्गं क्षेत्रकस्य भूमध्यविनिगतशंकुभावन्धिक्षकोणसूत्रकर्णखंडकस्य द्वितीयोऽपि बाहुः तावानेवातो भुजवर्गः द्विगुणः शंकुभूमध्यांतरस्य वर्गो भवति। कोणशंकोर्ज्यावर्गो भवतीत्यर्थः तत्क्रिज्यावर्गादपास्य क्षेत्रस्य मूलं कोणशंकुर्यतः कोणशंकुः कोटिः तच्छाया भुजः व्यासार्द्धं कर्णः तस्मादुपपन्नं स्वाद्यदिशंकुतलं आयते, तच्च न आयते। शंकुर्यतः कोणशंकुरेव परिकल्प्यते तेन वैराशिकं यदि द्वादशकस्य शंकोर्विषुवच्छायातुल्यं

शंकुतलं तद्यावच्छंकोः किमिति फलं यावत्कस्य द्वादशांशविषुवच्छायातुल्यं शंकुतलं तद्यावच्छंकोः किमिति फलं यावत्कस्य द्वादशांशा विषुवच्छायातुल्यं शंकुतलप्रमाणं तदग्रेऽर्काग्रातुल्यानि रूपाणि विन्यसेत्तदेवं शंकुतलाग्रा योगः कृतो भवति। स एव भुजस्तस्य यावद्वर्गः क्रियते तावत्प्रथमे स्थाने ये वर्गा विषुवच्छाया कृत्वा तुल्या भवंति। तेषामधश्चेदर्ककृतिः ये स्थाने यावकाः भवन्ति। द्विगुणविषुवच्छाया कृततुल्या भवन्ति। तेषामधश्छेदोऽर्ककृतिः द्वितीयस्थानेऽर्काग्रातुल्यानि रूपाणि भवन्ति। एवं भुज्यावर्गः कृतो भवति।

तं त्रिज्यावर्गाद्विशोध्यावशेषा द्विगुणाः क्रियन्ते। येन शंकुवर्गो भवति। इत्येवं विधिना भुजवर्गास्त्रिज्यावर्गाद्विशोध्यन्ते। शोधिते च जातौ राशियौ वर्गराशियविकराशिश्च द्वावप्यृणगतौ भवतः। अर्काग्रावर्गश्च रूपराशिरतः साक्षा त्रिज्यावर्गाद्विशोध्यते इत्यर्थः। एवं कृते जातं शंकुवर्गार्धं यावर्गार्धं यावर्गात् ऋणगता विषुवच्छायाकृतितुल्यास्तेषां द्वादशवर्गछेदः प्रथमस्थाने द्वितीयस्थाने यावकात् ऋणगत्या द्विगुणविषुवच्छायार्काग्रावधतुल्यास्तेषां द्वादशको छेदः तेन पुनर्द्वादशभिः संगुणिते येन प्रथमराशि सवर्णा भवति। द्वादशकृतिछेद इत्यर्थः। अस्य राशेरघनेव प्रयोजनं भविष्यति। कृत्वार्धमेवाचार्येणोपनिबद्धं वासनागुप्तये यावद्यार्धं क्रियते तावद् वर्गद्वादशकैकगुणा विषुवच्छायावध इति तृतीयस्थानेऽर्काग्रावर्गेन त्रिज्यावर्गार्धस्यावशेषाणि रूपाणि सकलान्येव भवन्ति। तानि वार्ककृतिगुणितानि सवर्णानि भवन्ति। पूर्वराशेरतः तथैव कृत्वा यो राशिर्भवति तस्याद्येतिसज्ञा। तेनोक्तमग्रावर्गोनं त्रिज्यावर्गार्धमर्ककृतिगुणितमाद्येति। वर्गस्यापि शंकुवर्गार्धात्मकस्यैक एव छेदो कृतिसंख्या सचार्धः क्रियते येन सकलशंकोर्वर्गो भवति। छेदोऽर्धे कृते मानाद् द्विगुणो भवतीत्यर्थः। स चार्धकृतो द्वयग संख्यो जातः एवं प्रथमपक्षद्वितीयपक्षे च यावर्गं विन्यसेत्। ततो यावक एवं शंकुः द्वितीयपक्षश्च प्रथमपक्षछेदेन द्विसप्तत्या सवर्णितो जातो द्वयगसंख्यो यावर्गः एवं समवस्थिते सूत्रवर्गव्यक्तः शोध्यः यस्माद् रूपाणि तदव्यस्तादित्यतः प्रथमपक्षे यावर्गः ऋणगता विषुवच्छाया कृत्वा तुल्या यावद्विशोध्यते। द्वितीयपक्षे यावर्गेभ्योऽधस्तेभ्यस्तावदमी भवन्ति। तेन तयो राश्योर्योग सर्वा विषुवच्छाया कृतिः तद्द्वयगसंयुता भवति। प्रथमपक्षे यावता अन्यसंज्ञा अष्टगुणितास्तेपि द्वितीयपक्षे शून्याद्यावद्विशोध्यन्ते तावत्तेऽपि घनिनो भवन्ति। द्वितीय पक्षे रूपराशिश्च शून्यं तत्प्रथमपक्षरूपेभ्यो विशोध्य तावत्येव रूपाण्येवावशिष्यन्ते आद्यसंज्ञकानि। एवं कृतेपि पक्षयोः साम्यं तेन त्रयाणामपि राशीनां सर्वदा प्रवर्तको विषुवच्छाया कृते या कृते संयुता यास्तुल्याः परिकल्पितस्तेन सदाप्रवर्तने कृते यावर्गस्थाने सर्वमेव रूपं भवति।

तत्त्वहाग्नेय्यां घनगतां यावकस्थाने स्फुटान्यो भवति। स चापोह् घनगतस्थाने स्फुटाद्यो भवति। सोपीह् घनगतरूप इत्येव स्थिते सूत्रं वर्ग···मस्यक्तार्ध-

कृतिसंयुतानां यत् पदमव्यक्तार्धं तद्वर्गविभक्तमन्य इत्यत एकेन वर्गेण रूपाणि हृतानि तावत्येव सर्वदा भवन्ति । स्फुटाद्यतुल्यानि अव्यक्तार्धं च स्फुटाद्यनस्तस्य कृतिं कृत्वा तया संयुतानां रूपाणां यत्पदं तदव्यक्तार्धेनान्यसंज्ञकेनोनं कृत्वा तद्वर्गविभक्तं क्रियते । सर्वदा वर्गस्थाने रूपमेकेन विभक्ते तदेवमिष्टाग्रात्स्वधनतत्त्वाद्राशीनां पदमन्यहीनं कोणशंकुस्तेनोक्तं हृतवर्गो विषुवच्छायाकृत्यादाद्यवर्गसंयुता यान्यकृतियुतादाद्यात्पदमन्यविहीनं दक्षिणगोलस्थेऽर्के शंकुर्विदिशि दक्षिणायामिति तदुपपन्नं यदा सममंडलेशान्यान्तरेभ्यो उदिते रवौ सममंडलादुत्तरेणैव मध्याह्नं कृत्वास्तमेति । तत्रैशानां शंक्वानयनं शंकुतलमग्रतो विशोध्य शेषो भुजो भवति । तत्र प्राग्वद्दिनास्तकोणहङ्मंडलस्वाहोरात्रसंपातादवलंबितसूत्रस्य कोणशंक्वभिधानस्य कोणसूत्रेण यदन्तरं तस्य शंकुमूलस्य प्राच्यपरा च तावदन्तरं भवतीत्यर्थः । तस्य वर्गा त्रिज्यावर्गाद्विशोध्य……शेषस्य द्विगुणस्य मूलमैशानां शंकुर्भवति । तेन यावकशंकोः तदेव शंकुतलस्याग्रे……न्यस्य शंकुतलाष्टवर्गागतं कार्यम् । एवं कृतेऽग्रा शंकुतलेनोना भुजा भवति । तस्य वर्गे…… प्रथमस्थाने यावर्गाः विषुवच्छायाकृत्या तुल्या भवन्ति । धनगता ऋणवर्गो यतो धनमर्धंश्चार्ककृतिछेदेन द्वितीयस्थाने द्वितीयविषुवच्छायाग्रावधतुल्या यावका ऋणगता भवति । धनर्णयोर्वधो यतः ऋणं भवति । तदधो द्वादशछेदः । तृतीयस्थानेऽग्रारूपाणां वधोऽग्रावधमेवम् ।

वि. भा.—रव्यग्रावर्गहीनं त्रिज्यावर्गार्धं द्वादशवर्गगुणितमाद्यसंज्ञकम् । अग्रा द्वादशगुणभाधातोऽन्यसंज्ञकः, द्वयगसंयुतया विषुवच्छायाकृत्या (द्विसप्ततिगुणपलभावर्गेण) हृतयोः (भक्तयोः) आद्यान्ययोर्मानयो विशिष्टाद्यान्यौ भवतः । अन्यवर्गयुतादाद्यान्मूलं ग्राह्यमन्येन युतं हीनं तदोत्तरदक्षिणगोलयोः कोणशङ्कुर्भवेत् । शेषस्य व्याख्या स्फुटैवास्तीति ॥५४-५५-५६॥

अत्रोपपत्तिः

कोणवृत्ताहोरात्रवृत्तयोः सम्पातात् (कोणवृत्तस्थरवेः) क्षितिजधरातलोपरि यो लम्बः स कोणशङ्कुः । तन्मूलात्पूर्वापररेखोपरि यो लम्बः स भुजः । भुजमूलात्केन्द्रं यावद् वा कोणशङ्कुमूलाद्याम्योत्तररेखोपरिलम्बः कोटिः, भूकेन्द्रात्कोणशङ्कुमूलपर्यन्तं दृग्ज्या कर्णः । भुजो भुजः । कोटिः कोटिरिति त्रिभुजे कर्णकोटिघोरुत्पन्नः कोणः =४५, तेन कर्णभुजयोरुत्पन्नकोणोऽपि =४५, अतो भुजकोटी समे भवतस्तेन भुजवर्गो द्विगुणितो दृग्ज्यावर्गो भवतीति । अथात्र कल्प्यते कोणशङ्कुप्रमाणम्=क, तदाऽनुपातेन शङ्कुतलम्=$\frac{\text{पभा.क}}{१२}$, अनयाशङ्कुतलयोः संस्कारेणोत्तरदक्षिणगोलयोः क्रमेण भुजमानम् = ध ∓ $\frac{\text{पभा.क}}{१२}$ । अत्र ध=

अग्रा, भुजवर्गः $= \left(\text{अ} \mp \frac{\text{पभा}.\text{य}}{१२}\right)^२ = \text{अ}^२ \mp \frac{२\,\text{अ}.\text{पभा}.\text{य}}{१२} + \frac{\text{पभा}^२.\text{य}^२}{१४४}$ द्विगुणित-

स्तदा दृग्ज्यावर्गं $= \frac{१४४\text{अ}^२ \mp २४\text{अ}.\text{पभा}.\text{य} + \text{पभा}^२.\text{य}^२}{७२} = \text{त्रि}^२ - \text{य}^२$ समशोधना-

दिना $(७२ + \text{पभा}^२)\,\text{य}^२ \mp २४\,\text{अ}.\text{पभा}.\text{य} = ७२\text{त्रि}^२ - १४४\text{अ}^२ = १४४\left(\frac{\text{त्रि}^२}{२} - \text{अ}^२\right)$

पक्षौ $(७२ + \text{पभा}^२)$ भक्तौ तदा $\text{य}^२ \mp \frac{२४\text{अ}.\text{पभा}.\text{य}}{७२ + \text{पभा}^२} = \frac{१४४\left(\frac{\text{त्रि}^२}{२} - \text{अ}^२\right)}{७२ + \text{पभा}^२}$ अत्र १४४

$\left(\frac{\text{त्रि}^२}{२} - \text{अ}^२\right) =$ आद्यसंज्ञकः। १२.अ.पभा = अन्यसंज्ञकः तदा $\text{य}^२ \mp \frac{२\,\text{अन्य}_{.}}{७२ + \text{पभा}^२}$

$= \frac{\text{आद्य}}{७२ + \text{पभा}^२} = \text{य}^२ \mp २\,\text{अन्य} = \text{आद्य}$, पक्षयोः अन्य^२ योजनेन $\text{य}^२ \mp २$

अन्य $+$ अन्य^२ $=$ आ $+$ अन्य^२ मूलेन $\text{य} \mp \text{अन्य} = \sqrt{\text{आ} + \text{अन्य}^२}$ ततः $\text{य} =$

$\sqrt{\text{आ} + \text{अन्य}^२} \pm \text{अन्य}$ एतेनाऽऽचार्योक्तमुपपन्नम्। यदा $\frac{\text{त्रि}^२}{२} < \text{अ}^२$ तदोत्तर-

गोले आद्यस्य ऋणत्वात् कोणशङ्कुचतुष्टयमुत्पद्यते, दक्षिणगोले तु कोणशङ्कोर-

भावः, सिद्धान्तशेखरे—

अग्राकृत्या विहीनं त्रिगुणकृतिदलं वेदशक्रघ्नमाद्यः
सूर्याग्राक्षप्रमाणाभिहतिरपरो भक्तयोरक्षभायाः।
कृत्या द्व्यद्व्याढ्यया तौ परकृतिसहितादाद्यतो यत्पदं
स्यादन्येनाढ्यं विहीनं घनदयमककुब्गोलयोः कोणशङ्कुः॥
उत्तरेतरविदिङ्नरो भवेदुत्तरे तु पदहीनयुक् परः।
दक्षिणेन सममण्डलात्ततो भाश्रुतीष्टघटिकाश्च पूर्ववत्॥

श्रीपत्युक्तमिदं कोणशङ्कोरानयनमाचार्योक्तानुरूपमेव, सूर्यसिद्धान्तेऽपि "त्रिज्यावर्गार्धतोऽग्र्यावर्गोनात् द्वादशाहतात्। पुनर्द्वादशनिघ्नादित्यादिनाऽग्रमेव कोणशङ्कोरानयनप्रकार उक्तः, सिद्धान्तशिरोमणौ "अग्राकृतिं द्विगुणितां त्रिगुणस्य वर्गादित्यादिना" भास्करेण विदिताऽग्रां समं भुजं मत्वाऽसकृत्प्रकारेण कोणशङ्कुः साधितोऽर्वाक् द्विगुणिताऽग्रावर्गः (द्विगुणितभुजवर्गः) दृग्ज्यावर्गसमः, त्रिज्यावर्गे द्विगुणिताग्रा वर्गे विशोधितेऽवास्तवः कोणशङ्कुवर्गस्तन्मूलं गृहीत्वा [illegible] तदाऽवास्तवभुजो भवेदेवमसकृत्कर्मणा

वास्तवकोणशङ्कुज्ञानं भवेत्। पर यदिभुजमानम्=ज्या४५=$\sqrt{\frac{\text{त्रि}^2}{2}}$ तदाऽस्य प्रकारस्य व्यभिचारो भवेद्यथा-भु'=२अग्रा'=त्रि'=दृग्ज्या' तत'$\sqrt{\text{त्रि}'-२\text{अग्रा}'}$ $=\sqrt{\text{त्रि}'-\text{त्रि}'}=०=$कोणशङ्कु, ततः शङ्कुतलम्=० अग्रायामेनस्य संस्कारेण पूर्वभुज एव भवेदतोऽग्रेऽस्यकृत्कर्मणः प्रवृत्तेरभावः। यदि च भुजः >ज्या ४५ तदा कोणशङ्कुप्रमाणमृणात्मकं भवेत्। ततो भुजमानम्=ज्या ४५ स्वीकृत्याग्रा-प्रमाणं यद्यानीयते तदा ततोऽल्पायामग्रायामुत्तरगोले भास्करोक्तव्यभिचारो भवेत्, यथा. भास्करोक्त्या $\sqrt{\text{त्रि}'-२\text{अ}^2}$ = कोणशङ्कुः, ततः शङ्कुतलम्= $\frac{\text{पभा}}{१२}\sqrt{\text{त्रि}^2-२\text{अ}^2}$अतो भुजः= $\frac{\text{पभा}}{१२}\sqrt{\text{त्रि}^2-२\text{अ}}-\text{अ}=$ज्या४५,ततश्छेदगमादिना

पभा$\sqrt{\text{त्रि}^2-२\text{अ}^2}$=१२.ज्या४५+अ.१२=१२(अ+ज्या४५)वर्गकरणेनपभा'.त्रि' —पभा'.२अ'=१४४ (अ'+२अ. ज्या ४५+ज्या' ४५)=पक्षौ द्वाभ्यां भक्तौ तदा

पभा'. $\frac{\text{त्रि}^2}{2}$—पभा'.अ'=७२अ'+१२'.अ.ज्या४५+७२ज्या'४५=पभा'.ज्या'४५ —पभा'.अ'समशोधनेन

पभा'.ज्या'४५—७२ ज्या' ४५=पभा'. अ'+१२'.अ.ज्या ४५+७२ अ'.तुल्यगुणक पृथक्करणेन=ज्या'४५(पभा'—७२)=अ' (पभा'+७२)+१२.'अ.ज्या ४५ ततः

$\frac{\text{ज्या}'४५(\text{पभा}^2-७२)}{\text{पभा}'^2+७२}=\text{अ}'+\frac{१२^2.\text{अ}.\text{ज्या}४५}{\text{पभा}'^2+७२}$ वर्गपूर्त्तिकरणेन

$\frac{\text{ज्या}'४५\,(\text{पभा}^2-७२)}{\text{पभा}'^2+७२}+\frac{(७२.\text{ज्या}४५)^2}{(\text{पभा}'^2+७२)^2}=\text{अ}'+\frac{१२^2.\text{अ}.\text{ज्या}४५}{\text{पभा}'^2+७२}+\frac{(७२.\text{ज्या}४५)^2}{(\text{पभा}^2+७२)^2}$

प्रथमपक्षे समच्छेदादिना,

$\frac{\text{ज्या}^2४५.\text{पभा}'^2}{(\text{पभा}^2+७२)^2}=\text{अ}'+\frac{२.७२\text{अ}.\text{ज्या}४५}{\text{पभा}'^2+७२}+\frac{(७२.\text{ज्या}४५)^2}{(\text{पभा}^2+७२)^2}$ मूलेन

$\frac{\text{ज्या}४५.\text{पभा}'^2}{\text{पभा}'^2+७२}=\text{अ}+\frac{७२.\text{ज्या}४५}{\text{पभा}^2+७२}$ततःअ=$\frac{\text{ज्या}४५.\text{पभा}'^2}{\text{पभा}^2+७२}-\frac{७२.\text{ज्या}४५}{\text{पभा}^2+७२}=$

$\frac{\text{ज्या}४५\,(\text{पभा}^2-७२)}{\text{पभा}'^2+७२}$ एतेनोपपद्यते म.म. सुधाकरद्विवेदिसूत्रम्।

युग्माब्धयोनाऽग्रप्रभावर्गनिघ्ने वाणाब्ध्यंशघ्ना द्विकादर्वविभक्ता.।
अक्षज्यावर्गयुक्तेः फलाब्धेरक्षा न्यूना स्वात्क्षितिस्थं शीर्षबोधे॥

दक्षिणगोले व्यभिचारश्च सिद्धान्तशिरोमणेष्टिप्पण्यां संशोधकेन "अक्षभाकृतेर्विहीनद्विगुणद्विनिघ्नः पञ्चाब्धिभागगुणा विहृता द्विकाशैरित्यादिना" प्रदर्शितोऽस्ति।

अथ कुत्र कोणशङ्कुचतुष्टयं भवतीति प्रदर्श्यते

यत्र देशे परमाग्रा=ज्या ४५° तद्देशीयपलभाप्रमाणमानीयते। कल्प्यते पलभामानम्=य, तदा पक्[२]=य[२]+१२[२] ततः $\frac{(य^२+१२^२)\ ज्या^२जि}{१२^२}$= परमाग्रा[२] =ज्या[२]४५ छेदगमेन य[२]. ज्या[२]जि+१२[२]. ज्या[२]जि=१२[२]. ज्या[२]४५ समशोधनेन य[२]. ज्या[२]जि=१२[२]. ज्या[२]४५—१२[२]. ज्या[२]जि=१२[२] (ज्या[२]४५—ज्या[२]जि) ∴य[२]= $\frac{१२^२\ (ज्या^२४५—ज्या^२जि)}{ज्या^२जि.}$ मूलेन य=$\frac{१२\sqrt{ज्या^२४५—ज्या^२जि}}{ज्याजि}$=१७।५।२२ एतेन सिद्धं यद्यत्र देशे पलभा=१७।५।२२ तत्र परमाग्रा=ज्या ४५° यत्र देशे १७।५।२२ आभ्योऽधिकाः पलभास्तत्र दक्षिणगोले कोणशङ्वोरभावो भवेद्यतस्तत्र २ य अग्रा > त्रि[२] उत्तरगोले तु तत्र कोणशङ्कुचतुष्टयमुत्पद्यत एतेन भास्करोक्तभाष्यमप्युपपन्नं भवतीति ॥५४-४५-५६॥

अब कोणशङ्कु के आनयन को कहते हैं

हि. भा.—त्रिज्या वर्गार्ध में रवि की अग्रावर्ग घटाकर बाहर के वर्ग से गुणा कर जो हो उसका नाम आद्य है, अग्रा, द्वादश, और पलभा के घात का नाम अन्त्य है, पलभावर्ग में बहत्तर (७२) जोड़कर आद्य और अन्त्य में भाग देने से विशिष्ट आद्य और अन्त्य होता है, आद्य (विशिष्ट आद्य) में अन्त्य (विशिष्ट अन्त्य) वर्ग जोड़कर मूल लेना, उसमें अन्त्य (विशिष्ट अन्त्य) को जोड़ना और घटाना तब उत्तरगोल में और दक्षिण गोल में कोणशङ्कु होता है। शेष की व्याख्या स्पष्ट है इति ॥५४-५५-५६॥

उपपत्ति

कोणवृत्तस्थ रवि से क्षितिजधरातल के ऊपर जो लम्ब होता है उसका नाम कोणशङ्कु है, उसके मूल से पूर्वापर रेखा के ऊपर लम्ब भुज संज्ञक है, भुजमूल से भूकेन्द्र पर्यन्त वा कोणशङ्कु मूल से याम्योत्तर रेखा के ऊपर लम्ब कोटि संज्ञक है, भूकेन्द्र से कोणशङ्कुमूलपर्यन्त दृग्ज्याकर्ण, भुज-भुज, कोटि-कोटि इस त्रिभुज में कर्ण और कोटि से उत्पन्न कोण=४५° इसलिये कर्ण और भुज से उत्पन्न कोण भी=४५° अतः भुज =कोटि इसलिये २भु[२]=दृ·ज्या[२], यहां कल्पना करते हैं कोणशङ्कु प्रमाण=य, तब अनुपात से शङ्कुतल=$\frac{पभा·य}{१२}$, अग्रा और शङ्कुतल के संस्कार से उत्तरगोल में और

दक्षिणगोल में क्रम से भुज=अ∓ $\frac{\text{पभा}\cdot\text{य}}{१२}$, यहां अ=अग्रा। भुजवर्ग=$\left(\text{अ}\mp\frac{\text{पभा}\cdot\text{य}}{१२}\right)$

=$\text{अ}^२\mp\frac{२\,\text{अ}\cdot\text{पभा}\cdot\text{य}}{१२}+\frac{\text{पभा}^२\cdot\text{य}^२}{१४४}$ द्विगुणित करने से २ भु^{२}=हज्या^{२}=त्रि^{२}—य^२

=$\frac{१४४\text{अ}^२\mp२४\text{अ}\cdot\text{पभा}\cdot\text{य}+\text{पभा}^२\cdot\text{य}^२}{७२}$ छेदगम और समशोधन से $(\text{पभा}^२+७२)\text{य}^२\mp$

२४अ·पभा·य=७२ त्रि^{२}—१४४ अ^२=१४४ $\left(\frac{\text{त्रि}^२}{२}-\text{अ}^२\right)$ दोनों पक्षों में $(\text{पभा}^२+७२)$

इससे भाग देने से य$^२\mp\frac{२४\text{अ}\cdot\text{पभा}\cdot\text{य}}{\text{पभा}^२+७२}=\frac{१४४\left(\frac{\text{त्रि}^२}{२}-\text{अ}^२\right)}{\text{पभा}+७२}$ यहां १४४$\left(\frac{\text{त्रि}^२}{२}-\text{अ}^२\right)$

=आद्य संज्ञक, १२·अ·पभा=अन्यसंज्ञक, तब य$^२\mp\frac{२\,\text{अन्य}}{\text{पभा}^२+७२}=\frac{\text{आद्य}}{\text{पभा}^२+७२}$

=य$^२\mp२$ अन्य=आद्य दोनों पक्षों में अन्य^२ जोड़ने से य$^२\mp२$ अन्य+अन्य^२=आद्य

+अन्य^२ मूल लेने से य∓अन्य=$\sqrt{\text{आद्य}+\text{अन्य}^२}$ ∴य=$\sqrt{\text{आद्य}}$ अन्य$^२\pm$अन्य

इससे आचार्योक्त उपपन्न हुआ। जब $\frac{\text{त्रि}^२}{२}<\text{अ}^२$ तब उत्तरगोल में आद्य का मान ऋण

होने से चार कोणशङ्कु उत्पन्न होते हैं, और दक्षिण गोल में कोणशङ्कु का अभाव होता है, सिद्धान्तशेखर में "अग्राकृत्याविहीनं त्रिगुणकृतिदलं" इत्यादि संस्कृतोपपत्ति में लिखित श्लोकों से श्रीपति के कोणशङ्कुसाधन आचार्योक्त के अनुरूप ही हैं, सूर्यसिद्धान्त में भी 'त्रिज्यावर्गविशोध्याज्यावर्गोनात्' इत्यादि से यही कोणशङ्कु का आनयन कहा गया है, सिद्धान्तशिरोमणि में 'अग्राकृति द्विगुणिता त्रिगुणस्य वर्गात्' इत्यादि से भास्कराचार्य ने विदित अग्रा के बराबर भुज को मानकर अवकृत् प्रकार से कोणशङ्कु का साधन किया है अर्थात् द्विगुणित अग्रावर्ग=२भु^{२}=हज्या^{२}, त्रिज्यावर्ग में द्विगुणित अग्रावर्ग को घटाने से अवास्तव कोणशङ्कु वर्ग रहता है, उसका मूल लेकर अङ्कुतल साधन करना उस (अङ्कुतल) को अग्रा में संस्कार करने से अवास्तव भुज होता है, इस तरह अवकृत् कर्म करने से वास्तव कोणशङ्कु ज्ञात होता है, लेकिन यदि भुज=अग्रा अ^२

=$\sqrt{\frac{\text{त्रि}^२}{२}}$ तब इस प्रकार का व्यभिचार होता है जैसे, २ भु^{२}=२ अग्रा^{२}=त्रि^{२}=

हज्या^{२} अतः$\sqrt{\text{त्रि}^२-२\,\text{अग्रा}^२}$=०=कोणशङ्कु, इससे अङ्कुतल भी=०, इनको अग्रा में संस्कार करने से पूर्व भुज ही होता है, इसलिये वासे अवकृत्कर्म की प्रवृत्ति ही नहीं होती,

यदि भुज > ज्या ४५ तब कोणशङ्कु का मान ऋणात्मक होता है, इसलिये भुज = ज्या ४५ स्वीकार कर यदि अग्रा का मान लाते हैं तो उससे अल्प अग्रा में उत्तरगोल में भास्करोक्त प्रकार का व्यभिचार होता है, यथा भास्करोक्ति से $\sqrt{\text{त्रि}^2 - 2\text{अ}^2}$ = कोणशङ्कु, इससे

शङ्कुतल = $\frac{\text{पभा}}{१२}\sqrt{\text{त्रि}^2 - २\text{अ}^2}$ इससे भुज = $\frac{\text{पभा}}{१२}\sqrt{\text{त्रि}^2 - २\text{अ}^2} - \text{अ} = \text{ज्या}४५°$

छेदगम आदि से,

$\text{पभा}\sqrt{\text{त्रि}^2 - २\text{अ}^2} = १२.\text{ज्या}\ ४५ + \text{अ}\cdot १२ = १२\ (\ \text{अ} + \text{ज्या}४५\ )$ वर्ग करने से

$\text{पभा}^2.\ \text{त्रि}^2 - \text{पभा}^2\cdot २\text{अ}^2 = १४४\ (\ \text{अ}^2 + २\text{अ}\cdot\text{ज्या}४५ + \text{ज्या}^2\ ४५\ )$ दोनों पक्षों को दो से भाग देने से

$\text{पभा}^2\cdot\ \frac{\text{त्रि}^2}{२} - \text{पभा}^2.\text{अ}^2 = ७२\text{अ}^2 + १२^2.\text{अ}\cdot\text{ज्या}४५ + ७२\cdot\text{ज्या}^2४५ = \text{पभा}^2\cdot\text{ज्या}^2४५$

$-\text{पभा}^2\cdot\text{अ}^2$ समशोधन करने से

$\text{पभा}^2\cdot\text{ज्या}^2४५ - ७२\ \ \text{ज्या}^2४५ = \text{पभा}^2.\text{अ}^2 + १२^2.\text{अ}.\text{ज्या}४५ + ७२\ \text{अ}^2$ तुल्यगुणक को अलग करने से

$$\text{ज्या}^2४५\ (\text{पभा}^2 - ७२) = \text{अ}^2\ (\text{पभा}^2 + ७२) + १२^2\cdot\text{अ}\cdot\text{ज्या}४५$$

$$\therefore \frac{\text{ज्या}^2४५\ (\text{पभा}^2 - ७२)}{\text{पभा}^2 + ७२} = \text{अ}^2 + \frac{१२^2\cdot\text{अ}\cdot\text{ज्या}४५}{\text{पभा}^2 + ७२}$$ वर्ग पूरा करने से

$$\frac{\text{ज्या}^2४५\ (\text{पभा}^2 - ७२)}{\text{पभा}^2 + ७२} + \left(\frac{७२.\text{ज्या}\ ४५}{\text{पभा}^2 + ७२}\right)^2 = \text{अ}^2 + \frac{१२^2.\text{अ}.\text{ज्या}४५}{\text{पभा}^2 + ७२} + \left(\frac{७२.\text{ज्या}४\ ५^2}{\text{पभा}^2 + ७२}\right)$$

प्रथम पक्ष में समच्छेदादि करने से $\frac{\text{ज्या}^2४५\cdot\text{पभा}^4}{(\text{पभा}^2 + ७२)^2} = \text{अ}^2 + \frac{२\cdot ७२\text{अ}\cdot\text{ज्या}४५}{\text{पभा}^2 + ७२}$

$+\frac{(७२\cdot\text{ज्या}४५)^2}{(\text{पभा}^2 + ७२)^2}$ मूल लेने से $\frac{\text{ज्या}४५ \times \text{पभा}^2}{\text{पभा}^2 + ७२} = \text{अ} + \frac{७२\cdot\text{ज्या}४५}{\text{पभा}^2 + ७२}$

$\therefore \text{अ} = \frac{\text{ज्या}४५.\text{पभा}^2}{\text{पभा}^2 + ७२} - \frac{७२\cdot\text{ज्या}४५}{\text{पभा}^2 + ७२} = \frac{\text{ज्या}४५\ (\text{पभा}^2 - ७२)}{\text{पभा}^2 + ७२}$ इससे "युग्माद्यूनाऽग्रभावर्गनिघ्नी" इत्यादि संस्कृतोपपत्ति में लिखित म. म. पण्डितसुधाकरद्विवेदी का सूत्र उपपन्न हुआ ॥ दक्षिणगोल में व्यभिचार सिद्धान्तशिरोमणि की टिप्पणी में 'अग्र-प्रवाकृतिविहीन' इत्यादि से संशोधन किया गया है।

किस देश में किस कोल में चार कोणशङ्कु होते हैं उसके लिये विचार करते हैं

जिस देश में परमाग्रा=ज्या४५ उस देश की पलभा साधन करते हैं, कल्पना करते हैं पलभा मान=य तब पक^2 =य^2+१२^2 ∴ $\frac{(य^2+१२^2)\ ज्या^2जि}{१२^2}$ =परमाग्रा^{2}=ज्या^{2}४५ छेदगम करने से य^2·ज्या^{2}जि+१२^2·ज्या^{2}जि=१२^2·ज्या^{2}४५समशोधन से य^2·ज्या^{2}जि=१२^2·ज्या^{2}४५—१२^2·ज्या^{2}जि=१२^2 (ज्या^{2}४५—ज्या^{2}जि) ∴य^2=१२$^2\frac{(ज्या^2४५-ज्या^2जि)}{ज्या^2जि}$ मूललेने से य=$\frac{१२\sqrt{ज्या^2४५-ज्या^2जि}}{ज्याजि}$=१७।५।२२ इससे सिद्ध हुआ कि जिस देश में पलभा=१७।५।२२ होगी उस देश में परमाग्रा=ज्या४५ होगी, १७।५।२२ इससे अधिक पलभा जिस देश में होगी उस देश में दक्षिणगोल में कोणशङ्कु का अभाव होगा। क्योंकि वहाँ २ अग्रा>त्रि^{1}, उत्तरगोल में कोणशङ्कु चतुष्टय (चार कोणशङ्कु) उत्पन्न होते हैं, इससे भास्करोक्त भाष्य उपपन्न होता है इति ॥५४-५५-५६॥

इदानीं भुजद्वयज्ञानात्पलभाज्ञानमाह

प्राच्यपराशङ्कुतलान्तरद्वयव्यस्तकर्णवधविवरम् ।
समदिशि विषुवच्छायान्यदिगैक्यं कर्णविवरहृतम् ॥५७॥

सु. भा.—प्राच्यपराशङ्कुतलान्तरं शङ्कुमूलप्राच्यपरान्तरम् । अर्थात् द्वादशाङ्गुलशङ्कोर्भुजः । एवं भुजद्वयस्य व्यस्तकर्णवधस्य समदिशि विवरमन्यदिश्यैक्यं कर्णयोर्विवरेण हृतं विषुवच्छाया भवति ।

अत्रोपपत्तिः । 'भाद्वयस्य भुजयोः समाशयोर्व्यस्तकर्णहृतयोर्यदन्तरम्' इत्यादि भास्करविधिना स्फुटा ॥५७॥

वि. भा.—प्राच्यपराशङ्कुतलान्तरं शङ्कुमूलपूर्वापरान्तरमर्थाद्भुजः । एवं भुजद्वयस्य व्यस्तकर्णवधस्य समदिशि (एकदिशायां) विवरं (अन्तर) अन्यदिशि (भिन्नदिशायां) ऐक्यं (योगः) कर्णान्तरेण भक्तं तदा विषुवच्छाया (पलभा) भवतीति ॥५७॥

अत्रोपपत्तिः

अत्र पलभामानम्=य, छायाकर्णगोले भुजः=भु । तथा छायाकर्णगोले द्वितीयभुजः=भु$'$ । छायाकर्णौ क, क$'$ तदा कर्णवृत्ताग्रा=य±भु । य±भु$'$=कर्णवृत्ताग्रा ततस्त्रिज्यागोलेऽग्रा=$\frac{(य\pm भु)\ त्रि}{क}$=अ । एवं $\frac{(य\pm भु')\ त्रि}{क'}$=$\frac{य.त्रि\pm भु'.त्रि}{क'}$ =

त्रिज्यागोलेऽग्रा=अ=$\frac{\text{य.त्रि}\pm\text{त्रि.भु}}{\text{क}}$ छेदगमेन य.त्रि.क$\pm$भु.त्रि.क=य.त्रि.क $\pm$त्रि.भु.क वा य.क$\pm$भु.क=य.क$\pm$भु.क समशोधनेन भु.क$\mp$भु.क=य.क$\sim$ य.क=य (क$\sim$क) $\therefore \frac{\text{भु.क}\mp\text{भु.क}}{\text{क}\sim\text{क}}$=य. एतेनोपपन्नमाचार्योक्तम् सिद्धान्तशेखरे "छायाग्रेन्द्रयपरान्तरद्वयविपर्यस्तस्वकर्णाहतिप्रोद्भूतं विवरं दिशोः सदृशयोरैक्यं च भिन्नाशयोः। तद्भक्तं श्रवणान्तरेण विषुवच्छाया स्फुटा जायते" श्रीपतेरयं प्रकारः, सिद्धान्तशिरोमणौ "भाद्वयस्य भुजयोः समाशयोर्व्यस्तकर्णाहतयोर्यदन्तरम्। ऐक्यमन्यककुभोः पलप्रभा जायते श्रुतिवियोगभाजितम्" भास्कराचार्यप्रकारोऽप्याचार्योक्तानुरूप एवेति ॥५७॥

अब भुजद्वयज्ञान से पलभा ज्ञान को कहते हैं

हि. भा.—शङ्कुमूल और पूर्वापर रेखा का अन्तर भुज है, इस तरह भुजद्वय और व्यस्तवर्ण (उल्टावर्ण) घात अर्थात् प्रथम भुज और द्वितीयछायावर्णघात तथा द्वितीयभुज प्रथमछायाकर्णघात के एक दिशा में अन्तर भिन्न दिशा में योग को छाया कर्णान्तर से भाग देने से पलभा होती है इति ॥५७॥

उपपत्ति

कल्पना करते हैं पलभा=य। छायाकर्णगोल में भुज=भु। छायाकर्णगोल में द्वितीयभुज=भु'। छायाकर्णद्वय=क। क' तब कर्णवृत्ताग्रा=य$\pm$भु। कर्णवृत्ताग्रा=य $\pm$भु' इससे त्रिज्या गोल में अग्रा=$\frac{(\text{य}\pm\text{अ})\,\text{त्रि}}{\text{क}}$=अ। एवं $\frac{(\text{य}\pm\text{भु}')\,\text{त्रि}}{\text{क}'}$ =$\frac{\text{य.त्रि}\pm\text{भु}'\text{.त्रि}}{\text{क}'}$=त्रिज्यागोल में अग्रा=$\frac{\text{य.त्रि}\pm\text{भु.त्रि}}{\text{क}}$=अ, छेदगम से य.क$\pm$भु.क =य.क'$\pm$भु'.क' समशोधन से भु.क'$\mp$भु'.क=य.क'$\sim$य.क=य (क'$\sim$क)

$\therefore$य=$\frac{\text{भु.क}'\mp\text{भु}'\text{.क}}{\text{क}'\sim\text{क}}$ इससे आचार्योक्त उपपन्न हुआ। सिद्धान्तशेखर में "छायाग्रेन्द्रयपरान्तरद्वयविपर्यस्तस्वकर्णाहति इत्यादि संस्कृतोपपत्ति में लिखित श्रीपतिप्रकार, तथा सिद्धान्तशिरोमणि में "भाद्वयस्य भुजयोः समाशयोः" इत्यादि भास्करोक्त प्रकार आचार्योक्तानुरूप ही है इति ॥५७॥

इदानीं प्रकारान्तरेण पलभाज्ञानमाह

छायावृत्ताग्रोना सौम्येन युतान्तरेण याम्येन ।
विषुवच्छायाऽजादिषु तुलादिषु तथाऽन्तरं हीनम् ॥५८॥

सु. भा.—छायावृत्ताया यथादिक्का कर्णवृत्ताया सौम्येन अन्तरेण द्वादशाङ्गुलशङ्कुतलप्राच्यपरान्तरेणार्थाद् भुजेनोना याम्येन भुजेन च युता । तदमजादिषु राशिषु अर्थादुत्तरगोले । तुलादिषु चार्थाद्दक्षिणगोले तथैव तदन्तरं भुजमानं हीनच्छायावृत्ताग्रया तदा विषुवच्छाया भवतीत्यर्थः ।

अत्रोपपत्तिः । बृहच्छङ्कुतलाग्रासंस्कारेण भुजो भवति । लघुशङ्कौ तु शङ्कुतल विषुवती । अग्रा छायाकर्णवृत्ताग्रा यथा दिक्का । उत्तरगोले सममण्डलप्रवेशात् पूर्वं छायावृत्ताग्रा पलभोना सौम्यो भुजोऽतः सौम्येन भुजेनोना छायावृत्ताग्रा पलभा । सममण्डलप्रवेशादुपरि पलभातश्छायावृत्ताग्रा शुद्धा भवति ततोऽवशिष्टं दक्षिणो भुजोऽतस्तेन युता साऽग्रा विषुवती भवति । दक्षिणगोले सर्वदा विषुवतीछायावृत्ताग्रयोर्योगेन भुजमानमतस्तच्छायावृत्ताग्रया हीनं पलभा भवेदिति स्फुटमेव ॥५८॥

वि. भा.—छायावृत्ताग्रा (कर्णवृत्ताग्रा) यथा दिक्काऽजादिषु (मेषादिषु) राशिषु अर्थादुत्तरगोले सौम्येन (उत्तरेण) अन्तरेण (द्वादशाङ्गुलशङ्कुमूलपूर्वापरान्तरेणार्थाद् भुजेन) ऊना (हीना), याम्येन (दक्षिणेन) भुजेन युता तदा विषुवच्छाया (पलभा) भवति, तथा तुलादिषु राशिष्वर्थाद्दक्षिणगोले तदन्तरं (भुजमानं) हीनं छायावृत्ताग्रया तदा विषुवच्छाया भवतीति ॥५८॥

अत्रोपपत्तिः

छायाकर्णगोले शङ्कुतलं पलभातुल्यं भवति कथमिति प्रदर्श्यते,

$\frac{\text{पभा.}\ \text{द्वशं}}{१२} = \text{शङ्कुतलम्}$ । $\because \frac{१२.\ \text{त्रि}}{\text{छाक}} = \text{द्वशं}$ अत उत्थापनेन $\frac{\text{पभा.}\ १२.\ \text{त्रि}}{१२.\ \text{छाक}} = \text{शङ्कु-तलम्}$ । छायाकर्णगोले $\text{शङ्कुतलम्} = \frac{\text{पभा.}\ १२.\ \text{त्रि}\cdot\text{छाक}}{१२.\ \text{छाक.त्रि}} = \text{पभा}$, शङ्कुतलाग्रासंस्कारेण भुजो भवति, छायाकर्णवृत्ताग्रा यथा दिक्का, उत्तरगोले रवेः समप्रवेशात् पूर्वं छायावृत्ताग्रा शङ्कुतलतुल्यया पलभया हीना तदोत्तरभुजो भवेदत उत्तरेण भुजेन हीना छायाकर्णवृत्ताग्रा पलभा भवेत् । समप्रवेशादुपरि पलभायां छायाकर्णवृत्ताग्रा शुद्धा भवति, ततोऽवशिष्टं दक्षिणो भुजोऽतस्तेन युताग्रा पलभा भवति, दक्षिणगोले सर्वदैव छायाकर्णवृत्ताग्रा पलभयोर्योगेन भुजो भवत्यतश्छायाकर्णवृत्ताग्रया हीनं भुजमानं पलभा भवेदिति सिद्धान्तशेखरे "यद्याच्छायावलयविहिता सैव सौम्यान्तरेण

हीना युक्ता नियतममुना दक्षिणोनाक्षभा स्यात्, । एवं मेषप्रभृतिषु गते भास्करे भेषु षट्सु जूकादिस्थे द्युतिमति तयैवान्तरं वर्जितं स्यात्" श्रीपत्युक्त मिदमाचार्योक्तश्लोकान्तरमात्रमेवेति ॥५८॥

अब प्रकारान्तर से पलभा ज्ञान को कहते हैं

हि॰ भा.—मेषादि छः राशिओं में अर्थात् उत्तर गोल में उत्तर अन्तर अर्थात् द्वादशाङ्गुल शङ्कु मूल और पूर्वापर रेखा के अन्तर (भुज) को छायाकर्ण वृत्ताग्रा में घटाने से तथा दक्षिण अन्तर (दक्षिण भुज) को जोड़ने से पलभा होती है, तुलादि छः राशियों में अर्थात् दक्षिण गोल में उस अन्तर (भुजमान) में छायाकर्णवृत्ताग्रा को हीन करने से पलभा होती है इति ॥५८॥

उपपत्ति

छायाकर्णगोल में शङ्कुतल, पलभा के बराबर होता है जैसे, $\frac{\text{पभा.}\ \text{इशं}}{१२}$ = शङ्कुतल, $\because \frac{\text{त्रि.}\ १२}{\text{छाक}} = \text{इशं}$ अतः उत्थापन देने से $\frac{\text{पभा.}\ \text{त्रि.}\ १२}{१२.\ \text{छाक}} =$ शङ्कुतल, इसको छायाकर्ण गोल में परिणामन करने से $\frac{\text{पभा.}\ \text{त्रि.}\ १२.\ \text{छ क}}{१२.\ \text{छाक.}\ \text{त्रि}} = \text{पभा} =$ छायाकर्णगोलीय-शङ्कुतल, उत्तरगोल में समप्रवेश से पहले छायावृत्ताग्रा में शङ्कुतल तुल्य पलभा को घटाने से उत्तर भुज होता है इसलिये छायावृत्ताग्रा में उत्तर भुज को घटाने से पलभा होती है, सम प्रवेश से ऊपर पलभा में छायाकर्णवृत्ताग्रा घटाने से शेष दक्षिण भुज होता है इसलिए इस-भुज को उस अग्रा में जोड़ने से पलभा होता है, दक्षिण गोल में सदा छायाकर्णवृत्ताग्रा और पलभा का योग करने से भुज होता है अतः भुज में छायाकर्ण वृत्ताग्रा को घटाने से पलभा होती है, सिद्धान्तशेखर में "अग्राच्छायावलयविहिता सैव सौम्यान्तरेण" इत्यादि संस्कृतोपपत्ति में लिखित श्रीपति-प्रकार आचार्योक्त के अनुरूप ही है इति ॥५८॥

इदानीं मिष्टच्छायावृत्ते पलभाग्रयो संस्थानमाह

इष्टच्छायावृत्ते तदग्रयोर्यदुदयास्तमयसूत्रम्. ।
अनुपातात्तच्छङ्कोर्विषुवच्छायान्तरमिहाग्रा ॥५९॥

सु. भा.—इष्टच्छायावृत्ते पूर्वापराशयोर्ये तदग्रे अर्थाच्छायावृत्ते परिणते अग्रे तयोर्गतं सूत्रमुदयास्तसूत्रम् । तच्छङ्कोः । तस्य रवेः शङ्कुस्तच्छङ्कुस्तस्यानुपातात् प्रपातात्मूलादुदयास्तसूत्रपर्यन्तमन्तरं विषुवच्छाया भवति । एवमिह स्मिन् छायावृत्ते भुजविषुवतीभ्यामग्रा च ज्ञेया इति फलितार्थो बोध्यः ।

अत्रोपपत्तिः । महाशङ्कावुदयास्तसूत्रात् शङ्कुमूलपर्यन्तं शङ्कुतलं लघुद्वादशाङ्गुलशङ्कौ चाक्षक्षं समजात्यच्छायावृत्तीयोदयास्तसूत्रात्तलघुशङ्कुमूलपर्यन्तमन्तरं पलभेति सर्वं गोलविदां स्फुटमेवेति ॥५६॥

वि. भा.—इष्टच्छायावृत्ते तदग्रयोरर्था-पूर्वापरदिशोः परिणताग्रयोर्गतं सूत्रमुदयास्तसूत्रम् । शङ्कोरनुपातादर्थाच्छङ्कुमूलात्तदन्तरं (उदयास्तसूत्रपर्यन्तमन्तरं) विषुवच्छाया (पलभा) भवति, इह (अस्मिन् छायावृत्ते) भुजपलाभ्यामग्रा च बोध्येति ॥५६॥

अत्रोपपतिः

छायाव्यासार्धवृत्तपरिणतयोः पूर्वापरदिशोरग्रयोर्गतं सूत्रमुदयास्तसूत्रसंज्ञकम् । तत्रोदयास्तसूत्रशङ्कुमूलयोरन्तरं शङ्कुतलम् । छायावृत्ते शङ्कुतलं पलभातुल्यमेव भवति, तेन छायावृत्तीयोदयास्तसूत्रे रचिते यथानियमं द्वादशाङ्गुलशङ्कौ स्थापिते च तयोरन्तरं गणितेन पलभातुल्यं स्वरूपेणाङ्गुलादिना मापिते च प्रत्यक्षतः पलभातुल्यं भवतीति । सिद्धान्तशेखरे 'इष्टप्रभाया वलये तदग्रया सूत्रं हि यत् स्यादुदयास्तसञ्ज्ञितम् । त्रैराशिकेनास्य नरस्य चान्तरं पलप्रभाऽसौ प्रकटैव लक्ष्यते" श्रीपत्युक्तमिदमाचार्योक्तानुरूपमेवेति ॥५६॥

अब इष्टछायावृत्त में पलभा और अग्रा की संस्थिति को कहते हैं

हि. भा.—इष्टछायावृत्त में परिणत पूर्व और पश्चिम दिशा की अग्राओं का अग्रगत सूत्र उदयास्त सूत्र है, शङ्कु मूल से उसका अन्तर पलभा होती है, और इस छायावृत्त में भुज और पलभा से अग्रा समझनी चाहिए इति ॥५६॥

उपपत्ति

छाया व्यासार्धवृत्त में परिणत पूर्व और पश्चिम दिशा की अग्राओं का अग्रगत सूत्र उदयास्त सूत्र है, शङ्कुमूल से उदयास्त सूत्र के ऊपर लम्ब शङ्कुतल है, परन्तु छायावृत्त में शङ्कुतल और पलभा तुल्य होती है, (इसके लिये पूर्व श्लोक की उपपत्ति देखिये) अतः आचार्योक्त युक्तियुक्त है, सिद्धान्तशेखर में "इष्टप्रभाया वलये" इत्यादि संस्कृतोपपत्ति में लिखित श्रीपति प्रकार आचार्योक्तानुरूप ही है इति ॥५६॥

इदानीं भुजच्छायाग्रयोः क्रान्तिज्ञानयनमाह

शङ्कुप्राच्यपरान्तरविषुवच्छायाङ्गुलान्तरं याम्ये ।
उदयेऽपरं लम्बगुणं छायाकर्णोद्धृतं क्रान्तिः ॥६०॥

सु. भा.—शङ्कुप्राच्यपरान्तरं लघुशङ्कुसंबंधी भुजः । विषुवच्छाया पलभा

प्रसिद्धा । याम्ये याम्यभुजेऽनयोरङ्गुलान्तरमुदगुत्तरभुजेऽनयोरैक्यं यत्तल्लम्बगुणं लम्बज्यागुणं छायाकर्णोद्धृतं फलं क्रान्तिः क्रान्तिज्या स्यादिति ।

अत्रोपपत्तिः । उत्तरगोले पूर्ववद्विधिना याम्यो भुजः=वि—कअग्रा=भु । अतः कर्णवृत्ताग्रा=वि—भु । दक्षिणगोले सर्वदा याम्यो भुजः=वि+कअ=भु । अतः कर्णवृत्ताग्रा=भु—वि । एवं याम्ये भुजे भुजपलभयोरन्तरं कर्णवृत्ताग्रा फलिता । उत्तरगोले सौम्यो भुजः=कअ—वि=भु । अतः कअ=वि+भु । एवं कर्णवृत्ताग्रानयनमुपपन्नम् । कर्णवृत्ताग्रा त्रिज्यागुणा छायाकर्णहृता जाताऽग्रा $=\frac{\text{कअ}\cdot\text{त्रि}}{\text{छाक}}$ । इयं लम्बज्यागुणा त्रिज्याभक्ताऽक्षक्षेत्रसाजात्याज्जाता क्रान्तिज्या $=\frac{\text{लंज्या}\cdot\text{कअ}}{\text{छाक}}$ । अत उपपद्यते क्रान्तिज्यानयनम् ॥६०॥

वि. भा.—शङ्कुप्राच्यपरान्तरं (लघुद्वादशाङ्गुलसम्बन्धी भुजः) विषुवच्छाया (पलभा), याम्ये (दक्षिणभुजे) ऽनयोरङ्गुलान्तरं, उदक् (उत्तरभुजे) अनयोरैक्यं यत्तल्लम्बगुणं (लम्बज्यागुणितं), छायाकर्णभक्तं तदा क्रान्तिः (क्रान्तिज्या) भवेदिति ॥६०॥

अत्रोपपत्तिः

उत्तरगोले (५८ श्लोकोक्त्या) दक्षिणभुजः=पभा-कर्णवृत्ताग्रा=भु अतः कर्णवृत्ताग्रा=पलभा—भु । एतेन दक्षिणे भुजे भुजपलभयोरन्तरं कर्णवृत्ताग्रा निष्पन्ना, उत्तरगोले उत्तरभुजः =कर्णवृत्ताग्रा—पलभा,=भु, अतः कर्णवृत्ताग्रा= पलभा+भु, ततोऽनुपातेना 'यदि छायाकर्णेन कर्णवृत्ताग्रा लभ्यते तदा त्रिज्यया किमिति' ऽनेनाग्रा $=\frac{\text{कर्णवृत्ताग्रा}\cdot\text{त्रि}}{\text{छाक}}$ ततो यदि त्रिज्यया लम्बज्या लभ्यते तदाऽग्रया किमित्यनुपातेन क्रान्तिज्या $=\frac{\text{लंज्या}\cdot\text{अग्रा}}{\text{त्रि}}=\frac{\text{लंज्या}\cdot\text{कर्णवृत्ताग्रा}\cdot\text{त्रि}}{\text{छाक}\cdot\text{त्रि}}$

$=\frac{\text{लंज्या}\cdot\text{कर्णवृत्ताग्रा}}{\text{छाक}}$ एतेनाऽऽचार्योक्तमुपपन्नम् । सिद्धान्तशेखरे "रेखानरान्तरपलप्रभयोर्भवेद्यत् याम्येऽन्तरं युतिरुदक्द्युतिवृत्तजाऽग्रा । लम्बाहता भवति कर्णहृताऽपमज्येति" श्रीपत्युक्तमिदमाचार्योक्तानुरूपमेव, सूर्यसिद्धान्तेऽपि "इष्टाग्राघ्नी तु लम्बज्या स्वकर्णाङ्गुलभाजितेत्यादि" ब्रह्मगुप्तो (आचार्यः) ऽप्येतदृशमेवेति ॥६०॥

अब भुज और छाया से क्रान्तिज्यानयन को कहते हैं

हि. भा.—दक्षिणभुज में पलभा और लघुद्वादशाङ्गुल सम्बन्धी भुज के अन्तर को

और उत्तर भुज में इन दोनों के योग को लम्बज्या से गुणाकर छायाकर्ण से भाग देने से क्रान्तिज्या होती है इति ॥६०॥

उपपत्ति

उत्तर गोल में (५८ श्लोकोक्त से) दक्षिणभुज = पभा − कर्णवृत्तभा = भु, अतः कर्णवृत्तभा = पभा − भु इससे दक्षिण भुज में भुज और पलभा का अन्तर कर्णवृत्तभा सिद्ध हुई, उत्तरगोल में उत्तरभुज = कर्णवृत्तभा − पभा = भु अतः कर्णवृत्तभा = पभा + भु, अब अनुपात करते हैं यदि छायाकर्ण में कर्णवृत्तभा पाते हैं तो त्रिज्या में क्या इससे प्रभा आती है $\frac{\text{कर्णवृत्तभा.त्रि}}{\text{छाक}}$ = प्रभा, इससे अनुपात करते हैं यदि त्रिज्या में लम्बज्या पाते हैं तो प्रभा में क्या, इससे क्रान्तिज्या आती है, $\frac{\text{लंज्या.प्रभा}}{\text{त्रि}} = \frac{\text{लंज्या.कर्णवृत्तभा.त्रि}}{\text{छाक.त्रि}} =$ $\frac{\text{लंज्या. कर्णवृत्तभा}}{\text{छाक}}$ = क्रांज्या इससे आचार्योक्त उपपन्न हुआ, सिद्धान्तशेखर में "रेखानरान्तर-पलप्रभयोः" इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से श्रीपति आचार्योक्तानुरूप ही कहते हैं, सूर्यसिद्धान्त में भी "इष्टाग्रघ्नी तु सज्ज्या स्वकर्णाक्षगुणभाजिता" इत्यादि आचार्योक्त-सदृश ही है इति ॥६०॥

इदानीं क्रान्तिज्यातो रव्वानयनमाह

क्रान्तिर्व्यासार्धगुणा जिनभागज्याहृता धनुरज्जादौ ।
कर्क्यादौ चक्रार्धात् प्रोह्य तुलादौ सभार्धेन ॥६१॥
चक्रात् प्रोह्य मृगादौ स्फुटोऽसकृद् धनमृणे धनेत्वृणकम् ।
अस्माद्देशान्तरयुगयुगतो मध्यमः प्राग्वत् ॥६२॥

सु. भा.—क्रान्तिः क्रान्तिज्या व्यासार्धेन त्रिज्यया गुणा जिनज्यया हृताऽर्कदोर्ज्या भवति । तद्धनुर्वर्यस्यैं प्रथमपादेऽजादौ राशित्रये स्फुटः सूर्यः । कर्क्यादौ राशित्रये तद्धनुश्चक्रार्धात् प्रोह्य शेषं स्फुटार्कः । एवं तुलादौ राशित्रये तद्धनुषा सभार्धेन समः स्फुटोऽर्कः । मृगादौ राशित्रये च चक्रद् द्वादशराशिभ्यस्तद्धनुः प्रोह्य शेषं स्फुटोऽर्को ज्ञेयः तत्रासकृद्ग्रहणे मन्दफले धनं धने च ऋणं मन्दफलं देयं तदा स्वदेशे मध्यमोऽर्को भवति । अस्माद् देशान्तरयुगयुगतः । अर्थात् प्राग्देशान्तरफलसहितात् पश्चिम-देशान्तरफलरहितादस्मात् स्वदेशीयमध्यमार्कात् प्राग्वद्रक्षितादतो मध्यमः स्यादित्यर्थः ।

अत्रोपपत्तिः । विपरीतक्रियया स्फुटा । 'तद्धनुराद्ये चरणे वर्षस्यार्कः प्रजायते अन्येषु' इत्यादिभास्करविधिना पदवशाद्विज्ञानं सुगममिति ॥६१-६२॥

वि. भा.—क्रान्तिः (क्रांतिज्या) व्यासार्धं (त्रिज्या) गुणा, जिनज्यया भक्ता तदा रविभुजज्या भवति, तद्धनुः (चापं) अजादौ (मेषादिराशित्रये प्रथमपदे) स्फुटो रविर्भवेत् । कर्क्यादिराशित्रये (द्वितीयपदे) तच्चापं चक्रार्धात् (राशिषट्कात्) विशोध्य शेषं स्फुटो रविः । तुलादौ राशित्रये (तृतीयपदे) सभार्धेन (राशिषट्कसहितेन) तच्चापेन तुल्यः स्फुटो रविर्भवति, मृगादौ राशित्रये (चतुर्थपदे) तच्चापं चक्रात् (द्वादशराशिभ्यः) विशोध्य शेषं स्फुटो रविर्भवेत् । तत्रासकृद्दृणे मन्दफले धनं, धने च ऋणं मन्दफलं देयं स्वदेशे मध्यमो रविर्भवति, अस्माद्देशान्तरयुगयुगतोऽर्थात् प्राग्देशान्तरफलयुतात् पश्चिमदेशान्तरफलरहितात् स्वदेशीयमध्यमरवितः प्राग्वत् गणितागतो मध्यम इति ॥६१-६२॥

अत्रोपपत्तिः

यदि जिनज्यया त्रिज्या लभ्यते तदा रविक्रान्तिज्यया किमिति क्रान्तिक्षेत्रानुपातेन समागच्छति रविभुजज्या तत्स्वरूपम्=$\frac{\text{त्रि.क्रांज्या}}{\text{जिज्या}}$ अस्याश्चापं रविभुजांशा भवेयुः । प्रथमपदे त्वयमेव स्फुटरविः, द्वितीयपदे चक्रार्धाद्विशोधनेन तृतीयपदे चक्रार्धयोजनेन चतुर्थपदे च चक्राद्विशोधनेन स्फुटो रविर्भवतीति गोले प्रत्यक्षमेव दृश्यते, सिद्धान्तशिरोमणौ "तद्धनुराद्ये चरणे वर्षस्यार्कः प्रजायतेऽन्येषु" इत्यादिना भास्करेणाऽऽचार्योक्तानुरूपमेव कथ्यते। किन्तु भास्करादिभिराचार्यैः पदज्ञानमृतुवर्णनद्वारा प्रदर्शितमस्ति तन्न समीचीनमित्येतदर्थं सिद्धान्ततत्त्वविवेके कमलाकरेणे-"ऋतुचिह्नैरिदं पूर्वैरुक्तं सर्वत्र तन्नहि । केवलं कुकविप्रीत्यै पदज्ञप्त्यै न तद्भवेः" त्यादिना सोपहासं प्रत्याख्यातम् । तथा तज्ज्ञानार्थं स्वकीयप्रकारश्च "आद्ये पदेऽपचयिनी पलभाल्पिका स्यात् छायाल्पिका भवति वृद्धिमती द्वितीये । छायाधिका भवति वृद्धिमती तृतीये तुर्ये पुनः क्षयवती तदनल्पिका च" प्रदर्शितः, अस्य युक्तिः—मेषादिगे सायनभागसूर्ये दिनार्धजा भा पलभा भवेत्सेतिभास्करोक्तेर्विषुवद्दिने मध्याह्ने या छाया सा पलभा, तस्माद्दिनादारभ्य प्रतिदिनं निरक्षादुत्तरे देशे "$\frac{\text{नतांशज्या १२}}{\text{नतांशकोज्या}}=\frac{\text{दृग्ज्या. १२}}{\text{शङ्कु}}$=छाया" इति छायास्वरूपदर्शनान्मध्यनतांशस्यापचयान्मध्यशङ्कोरुपचयान्मध्यच्छायाया उत्तरोत्तरमपचयो भवेत् । तेन मध्यच्छायायाः पलभाल्पाया उत्तरोत्तरमपचये रवेः प्रथमपदं भवेदेवेति । द्वितीयपदे मध्यनतांशा उत्तरोत्तरमुपचीयमाना भवन्ति तेन मध्यच्छायायामुत्तरोत्तरं वृद्धिमत्यां पलभातोऽल्पायां रवेर्द्वितीयपदं भवेदेव । तृतीयपदे पलभाधिका मध्यच्छायो-

नरोत्तरं वृद्धिमती भवति (रवेर्दक्षिणगमनात्)। चतुर्थे पदे नतांशस्योत्तरोत्तरमपचयात् पलभाधिका छायाऽपचीयते, निरक्षादक्षिणे देशे प्रथमपदे मध्यनतांशोपचयान्मध्यच्छायोपचयः। तत्र छायाग्रं दक्षिणदिश्येव भवेत् (स्वस्वस्निकादुत्तरतो रवेरवस्थानात्)। द्वितीयपदे मध्यनतांशापचयान्मध्यच्छायापचयः (रवेरुत्तरोत्तरं दक्षिणगमनात्) तृतीयपदे चतुर्थपदे च नतांशोपचयापचययोर्नियमो नास्ति तेन तत्रैकेन नियमेन मध्यच्छायया पदज्ञानं न भवितुमर्हति, तेनैव हेतुने "वृद्धि पयान्ती यदि दक्षिणाच्छाया तथापि प्रथमं पदं स्यात्। ह्रसं व्रजन्तीमथ तां विलोक्य रवेर्विजानीहि पदं द्वितीयम्" त्युक्तम्। सिद्धान्तशेखरे प्रकारस्यास्य चमत्कारकारकस्योल्लेखदर्शनात् "आद्ये पदेऽपचयिनी पलभाधिका स्यादित्यादि" प्रकारः कमलाकरस्य नास्ति, वस्तुतोऽयं प्रकारः श्रीपतेरेवेति ॥६१-६२॥

अब क्रान्तिज्या से रवि के आनयन को कहते हैं

हि. भा.—क्रान्तिज्या को त्रिज्या से गुणा कर जिनज्या से भाग देने से रवि भुजज्या होती है, उसका चाप मेषादि तीन राशियों (प्रथमपद) में स्फुट रवि होता है. कर्कादि तीन राशियों (द्वितीय पद) में उस चाप को छः राशियों में घटाने से तुलादि तीन राशियों (तृतीय पद) में चाप में छः राशि जोड़ने से, मकरादि तीन राशियों (चतुर्थ पद) में चाप को बारह राशि में घटाने से स्फुट रवि होता है, शेष की व्याख्या स्पष्ट है इति ॥६१-६२॥

उपपत्ति

यदि जिनज्या में त्रिज्या पाते हैं तो रवि की क्रान्तिज्या में क्या इस क्रान्तिक्षेत्र के अनुपात से रविभुजज्या आती है,

$\frac{\text{त्रि.क्रांज्या}}{\text{जिज्या}}$ = रविभुजज्या, इसका चाप प्रथम पद में स्फुट रवि होता है। द्वितीय पद में उस चाप को छः राशि में घटाने से तृतीय पद में चाप में छः राशि जोड़ने से, चतुर्थ पद में चाप को बारह राशि में घटाने से स्फुट रवि होता है। यह गोल में प्रत्यक्ष देखा जाता है, सिद्धान्तशिरोमणि में "क्रदनुराद्ध चरणे वर्षस्यर्कः प्रजायतेऽन्येषु" इत्यादि से भास्कराचार्य आचार्योक्ति के अनुरूप ही कहते हैं, लेकिन भास्करादि आचार्यों ने रवि का पदज्ञान ऋतुवर्णन द्वारा किया है जो ठीक नहीं है इसके लिये सिद्धान्ततत्त्वविवेक में कमलाकर ने "ऋतुचिह्नैरिदं पूर्वैः" इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से उसका निराकरण किया है। तथा उसके वास्तव ज्ञान के लिये अपना प्रकार "आद्ये पदेऽपचयिनी पलभाधिका स्यात्' इत्यादि दिखलाया है। इसकी युक्ति इस प्रकार है—विषुवद् दिन में मध्यान्हकाल में द्वादशांगुल शङ्कु की जो छाया होती है वह पलभा है, उस दिन से लेकर प्रत्येक दिन निरक्ष देश से उत्तर देश में $\frac{\text{नतांशज्या. १२}}{\text{त्रिज्या कोज्या}} = \frac{\text{ज्या. १२}}{\text{शङ्कु}}$ = छाया, इस छायाग्रकर को

देखने से मध्यनतांश के अपचय (ह्रास) से मध्यशङ्कु के उपचय (वृद्धि) से मध्यच्छाया का उत्तरोत्तर अपचय होता है इसलिए पलभा से अल्प मध्यच्छाया का उत्तरोत्तर अपचय होने से रवि का प्रथम पद होता ही है, द्वितीय पद में मध्यनतांश उत्तरोत्तर बढ़ता है जबतक तुलादि सम्पात में रवि जाता है, इसलिये पलभा से अल्प उत्तरोत्तर वृद्धिमती मध्यच्छाया होने से द्वितीय पद होता है, तृतीय पद में मध्यच्छाया पलभा से अधिक उत्तरोत्तर उपचित होती है (रवि के दक्षिण गमन के कारण से), चतुर्थ पद में पुनः उत्तरगमन प्रवृत्तिसे उत्तरोत्तर नतांश का अपचय होता है इसलिए पलभा से अधिक छाया अपचयमती होती है, अतः "आद्ये पदेऽपचयिनी" इत्यादि युक्तियुक्त कहा गया है। निरक्ष देश से दक्षिण देश में प्रथम पद में मध्यनतांश के उपचय से मध्यच्छाया का उपचय होता है लेकिन वहां स्वस्वस्वस्तिक से उत्तर में रवि के रहने के कारण छायाग्र दक्षिण दिशा ही में होता है इसलिये "वृद्धि प्रयान्ती यदि दक्षिणाग्रा" इत्यादि संस्कृतोपपत्ति में लिखित पद्य युक्तियुक्त कहा गया है. द्वितीय पद में रवि के उत्तरोत्तर दक्षिण गमन के कारण मध्यनतांश के अपचय से मध्यच्छाया का अपचय होता है, तृतीय पद में और चतुर्थ पद में नतांश के उपचय और अपचय के अनियम से वहां मध्यच्छाया द्वारा एक नियम से पदज्ञान नहीं होता है, ये सब बातें गोल को देखने से स्पष्ट हैं, सिद्धान्तशेखर में "आद्ये पदेऽपचयिनी" इत्यादि चमत्कृतियुक्त प्रकार देखा जाता है। इस से सिद्ध होता है कि यह रवि का पदज्ञान-प्रकार वस्तुतः श्रीपति ही का है, कमलाकर का नहीं है इति ॥६१-६२॥

इदानीं चन्द्रशृङ्गोन्नतौ रविशङ्क्वर्थं विशेषमाह

शशिशृङ्गोन्नत्यर्थं रात्रेर्गतशेषनाडिका शङ्कुः।
विपरीतगोलविधिना रात्र्यर्धार्कान्तराभिर्वा ॥६३॥

सु. भा.—पश्चिमदिशि शृङ्गोन्नत्यर्थं रात्रेर्गतनाड़िकाभिरुन्नताभिः प्राग्दिशि च रात्रिशेषनाड़िकाभिरुन्नताभिः 'गतशेषाल्लस्याह्नः' इत्यादिना विपरीतगोलविधिना रवेः शङ्कुः साध्यः। यदि नताभिर्नाडीभिरर्कशङ्कुरपेक्षितस्तदा रात्र्यर्धार्कान्तराभिर्नतनाड़िकाभिः स शङ्कुरानेयः।

अत्रोपपत्तिः। 'निशावशेषैरसुभिर्गतैर्वा' इत्यादि भास्करशृङ्गोन्नतविधिना स्फुटा। दृश्यशृङ्गोन्नतौ रवेः क्षितिजाधःस्थितत्वादधोयाम्योत्तरवृत्तान्तरकालोऽपेक्षितोऽतो रात्र्यर्धार्कान्तरतो नतनाडिकासाधनायातिदेशं चकाराऽऽचार्यः ॥६३॥

वि. भा.—पश्चिमायां दिशि चन्द्रशृङ्गोन्नत्यर्थं रात्रेर्गतनाडिकाभिः (उन्नताभिः) पूर्वस्यां दिशि रात्रिशेषनाडिकाभिः (उन्नताभिः) 'गतशेषाल्लस्याह्नः' इत्यादिना विपरीतगोलविधिना रवेः शङ्कुः साध्यः। यदि नताभिर्नाडीभी-

रविशङ्कुरपेक्षितस्तदा राश्यर्धार्कान्तराभिर्नतनाडिकाभिः शङ्कुः साध्य इति ॥ ६३ ॥

अत्रोपपत्तिः

क्षितिजादधः स्थिते रवावेव चन्द्रशृङ्गोन्नतिर्दृश्या भवतीत्यधो याम्योत्तरवृत्तान्नतकालज्ञानाय राश्यर्धार्कान्तरतो नतनाडिकासाधनमुक्तमाचार्येण, सिद्धान्तशेखरे "हिमांशुशृङ्गोन्नतये तु रात्रेर्गतावशेषैरसुभिर्नतैर्वा । प्रसाधयेदुक्तवदेव शङ्कुं स्वसाधनैर्गोलविपर्ययेण" श्रीपत्युक्तमिदं, सिद्धान्तशिरोमणौ "निशावशेषैरसुभिर्गतैर्वा यथाक्रमं गोलविपर्ययेण । रवेरधः शङ्कुरिति" भास्करोक्तञ्चाऽऽचार्योक्त्यनुरूपमेवेति ॥ ६३ ॥

अब चन्द्रशृङ्गोन्नति में रविशङ्कु के लिये विशेष कहते हैं

हि. भा.—पश्चिम दिशा में चन्द्रशृङ्गोन्नति के लिये रात्रि की गतघटी (उन्नत घटी) से, पूर्व दिशा में रात्रि की शेष घटी (उन्नत घटी) से 'गतशेषाल्पस्याल्पः' इत्यादि से विपरीत (उल्टा) गोलविधि से रवि का अधः शङ्कुसाधन करना। नतनाड़ी से भी आचार्य कथित विधि से शङ्कु साधन करना चाहिये इति ॥६३॥

उपपत्ति

क्षितिज से नीचे रवि के रहने ही से चन्द्र शृङ्गोन्नति दृश्य होती है, इसलिये अधोयाम्योत्तरवृत्त से नतकाल ज्ञान के लिये राश्यर्ध और रवि के अन्तर से नतनाड़ी का साधन आचार्य ने कहा है। सिद्धान्तशेखर में 'हिमांशुशृङ्गोन्नतये तु रात्रेः इत्यादि सं० उपपत्ति में लिखित श्लोक से श्रीपति तथा सिद्धान्तशिरोमणि में 'निशावशेषैरसुभिः' इत्यादि संस्कृत-उपपत्ति में लिखित पद्य से भास्कराचार्य ने भी आचार्योक्ति के अनुरूप ही कहा है इति ॥ ६३ ॥

इदानीमुदयास्तसूत्रमाह

क्षितिजेऽग्रा प्राच्यपरागतिस्त्रिज्यागुणाऽवलम्बहृता ।
द्विगुणमुदयास्तसूत्रं तत् त्रिज्याकृतिवियुक्तपदम् ॥६४॥

सु. भा.—क्रान्तिः क्रान्तिज्या त्रिज्यागुणाऽवलम्बेन लम्बज्यया हृता तदाऽग्रा स्यात्। इयमग्रा क्षितिजे प्राच्यपरा भवति। अर्थात् क्षितिजे प्राचि प्रतीच्यां च प्रागपरस्वस्तिकाभ्यां यथा दिक्काऽग्रा भवतीत्यर्थः। तत्त्रिज्याकृतिवियुक्तपदं द्विगुणमुदयास्तसूत्रं भवेदिति।

अत्रोपपत्तिः। कुज्या भुजः। क्रान्तिज्या कोटिः। अग्रा कर्णः। इत्यक्षक्षेत्रं प्रसिद्धम्। ततो यदि लम्बज्याकोट्या त्रिज्या कर्णस्तदा क्रान्तिज्याकोट्या किं जाताग्रा $=\frac{\text{क्रांज्या}\cdot\text{त्रि}}{\text{लंज्या}}$। अग्रावर्गोनात्त्रिज्यावर्गात्पदमग्रा कोटिज्या तद् द्विगुणमेवोदयास्तसूत्रं प्रसिद्धमिति ॥६४॥

वि. भा.—क्रान्तिः (क्रान्तिज्या) त्रिज्यागुणिता लम्बज्यया भक्ता तदाऽग्रा स्यात्। इयं क्षितिजे प्राच्यपरा (पूर्वापरा) भवत्यर्थात् पूर्वस्यां दिशि पश्चिमायां दिशि च पूर्वस्वस्तिकपश्चिमस्वस्तिकाभ्यां यथा दिक्का भवति, तत् त्रिज्यावर्गान्तरमूलं द्विगुणं तदोदयास्तसूत्रं भवतीति ॥६४॥

अत्रोपपत्तिः

यदि लम्बज्यया त्रिज्या लभ्यते तदा क्रान्तिज्यया किमित्यक्षक्षेत्रानुपातेन समागच्छत्यग्रा $=\frac{\text{त्रि}.\ \text{क्रांज्या}}{\text{लंज्या}}$ त्रिज्यावर्गेऽग्रावर्गविशोधिते पदे च गृहीतेऽग्राकोटिज्या $=\sqrt{\text{त्रि}^2-\text{अग्रा}^2}$ एतद् द्विगुणमेवोदयास्तसूत्रं भवतीति। सिद्धान्तशेखरे "पूर्वापराया क्षितिजे कृताग्रा तत् त्रिज्यकावर्गविशेषमूलम्। द्विनिघ्नमुक्तं खलु याम्यसौम्यं गोलप्रवीणैरुदयास्तसूत्रम्" श्रीपत्युक्तमिदं सिद्धान्तशिरोमणौ 'क्ष्मजे द्युरात्रसममण्डलमध्यभागजीवाऽग्रका भवति पूर्वपराशयोः सा" भास्करोक्तमिदं चाऽऽचार्योक्तानुरूपमेवेति ॥६४॥

अब उदयास्तसूत्र को कहते हैं

हि. भा.—क्रान्तिज्या को त्रिज्या से गुणा कर लम्बज्या से भाग देने से अग्रा होती है, यह अग्रा पूर्व दिशा में और पश्चिम दिशा में पूर्वस्वस्तिक और पश्चिम स्वस्तिक से यथादिक् होती है, इसके और त्रिज्या के वर्गान्तरमूल को द्विगुणित करने से उदयास्त सूत्र होता है इति ॥६४॥

उपपत्ति

यदि लम्बज्या में त्रिज्या पाते हैं तो क्रांतिज्या में क्या इस अक्षक्षेत्रानुपात से अग्रा आती है, $\frac{\text{त्रि}.\ \text{क्रांज्या}}{\text{लंज्या}}=$अग्रा, त्रिज्या वर्ग में अग्रा वर्ग को घटाकर मूल लेने से अग्राकोटिज्या होती है। इसको द्विगुणित करने से उदयास्त सूत्र होता है, सिद्धान्तशेखर में 'पूर्वापरा या

क्षितिजे कृताग्रा' इत्यादि सं उपपत्ति में लिखित श्लोक से श्रीपति तथा सिद्धान्तशिरोमणि में 'क्षमाजे ध्रुवाग्रममण्डलमध्यभागजीवाऽग्रका' इत्यादि से भास्कराचार्य भी आचार्योक्तानुरूप ही कहते हैं इति ॥६४॥

इदानीं शङ्कुतलानयनमाह

अक्षज्याशङ्कुवधाल्लम्बकलब्धोदयास्तमयसूत्रात् ।
दक्षिणतः शङ्कुतलं दिवसे रात्रौ तदुत्तरतः ॥६५॥

सु. भा.—स्पष्टार्थम् ।

अत्रोपपत्तिः । अक्षवशतो दिवाऽहोरात्रवृत्तानि दक्षिणतो रात्रावुत्तरतो नतान्यतो दिवा शङ्कुतलं दक्षिणं रात्रावुत्तरं भवति । अक्षक्षेत्रानुपातेन शङ्कुतलानयनं स्फुटम् ॥६५॥

वि. भा.—अक्षज्याशङ्कुघातात् लम्बज्यया भक्तात्तल्लब्धं शङ्कुतलं भवति, तद् दिने उदयास्तसूत्राद् दक्षिणतो रात्रावुत्तरतश्च भवतीति ॥६५॥

अत्रोपपत्तिः

ग्रहात् क्षितिजधरातलोपरि यो लम्बः स शङ्कुः । शङ्कुमूलादुदयास्तसूत्रपर्यन्तं शङ्कुतलम् । क्षितिजादुपरि दक्षिणतोऽहोरात्रवृत्तस्थं गतत्वाद् दिने तच्छङ्कुतलमुदयास्तसूत्राद् दक्षिणतः क्षितिजादधश्चाहोरात्रवृत्तस्योत्तरतो गतत्वाद्रात्रावुत्तरत उदयास्तसूत्राच्छङ्कुतलं भवतीति गोलोपरि स्फुटं दृश्यते । अक्षक्षेत्रानुपातेना 'यदि लम्बज्ययाऽक्षज्या लभ्यते तदा शङ्कुना किमिति' अनेन शङ्कुतलम् $=\frac{\text{अक्षज्या}\cdot\text{शं}}{\text{लज्या}}$

सिद्धान्तशेखरे "पलज्यया सङ्गुणितात् स्वशङ्कोर्लम्बेन भक्तात् फलमाप्यते यत् । याम्योत्तरं शङ्कुतलं भवेत्तद् दिवारजन्योरुदयास्तसूत्रात्" श्रीपत्युक्तमिदं, सिद्धान्तशिरोमणौ 'सूत्राद् दिवाशङ्कुतलं यमाशं याम्यां गतं हि क्षुनिशं कुजोर्ध्वे। अवश्च सौम्यां निशि सौम्यमस्मात् सदुक्तियुक्तं नृतलं निरुक्तम्" भास्करोक्तमिदं चाऽऽचार्योक्तानुरूपमेवेति.॥६५॥

अब शङ्कुतलानयन को कहते हैं

हि. भा.—अक्षज्या और शङ्कु के घात को लम्बज्या से भाग देने से फल शङ्कुतल होता है वह दिन में उदयास्तसूत्र से दक्षिण और रात्रि में उत्तर होता है ॥६५॥

उपपत्ति

यदि लम्बज्या में अक्षज्या पाते हैं तो शङ्कु में क्या इस अनुपात से शङ्कुतल

आता है $\frac{\text{अज्या. शं}}{\text{लज्या}}$ =शङ्कुतल.। ग्रहस्थान से क्षितिज धरातल के ऊपर जो लम्ब होता है वह शङ्कु है, शङ्कु मूल से उदयास्तसूत्रपर्यन्त शङ्कुतल है, क्षितिज से ऊपर अहोरात्र वृत्त के दक्षिण जाने से दिन में वह शङ्कुतल उदयास्त सूत्र से दक्षिण होता है, क्षितिज से नीचे (रात्रि में) अहोरात्रवृत्त के उत्तर जाने से वह शङ्कुतल उदयास्त सूत्र से उत्तर होता है, यह विषय गोल के ऊपर स्पष्ट देखने में आता है, सिद्धान्तशेखर में 'पलज्यया सङ्गुणितात्' इत्यादि से सं उपपत्ति में लिखित श्लोक से श्रीपति तथा सिद्धान्तशिरोमणि में 'सूत्राद् दिवाशङ्कुतलं यमाशं" इत्यादि सं उपपत्ति में लिखित श्लोक से भास्कराचार्य भी आचार्योक्तानुरूप ही कहते हैं इति ॥६५॥

इदानीमध्यायोपसंहारमाह

दिग्लम्बाक्षस्वोदयलग्नच्छायादिषूपदिष्टेषु ।
षट्षष्ट्यार्याणां त्रिप्रश्नाध्यायस्तृतीयोऽयम् ॥६६॥

सु. भा.—स्पष्टार्थम् ।

मधुसूदनसूनुनोदितो यस्तिलकः श्रीपृथुनेह जिष्णुजोक्ते ।
हृदि तं विनिधाय नूतनोऽयं रचितः प्रश्नविधौ सुधाकरेण ॥

इति श्रीकृपालुदत्तसूनुसुधाकरद्विवेदिविरचिते ब्राह्मस्फुटसिद्धान्तनूतनतिलके त्रिप्रश्नाधिकारस्तृतीयः ॥६६॥

वि. भा.—कथितेषु दिग्ज्ञानलम्बांशाक्षांशस्वदेशोदयमानलग्नच्छायादिसाधनेषु आर्याछन्दोबद्धषट्षष्टिप्रमितश्लोकैरयं तृतीयस्त्रिप्रश्नाध्यायः समाप्तिं गत इति॥६६॥

इति ब्राह्मस्फुटसिद्धान्ते त्रिप्रश्नाध्यायस्तृतीयः

हि भा.—पूर्वकथित दिग्ज्ञान-लम्बांश-अक्षांश-स्वदेशीयराश्युदयमान-लग्न-छायादिकों के साधन में छियासठ आर्याछन्द के श्लोकों से यह तीसरा त्रिप्रश्नाध्याय समाप्त हुआ ॥६६॥

इति ब्राह्मस्फुटसिद्धान्त में त्रिप्रश्ना नामक तृतीय अध्याय समाप्त हुआ

ब्राह्मस्फुटसिद्धान्तः

चन्द्रग्रहणाधिकारः

ब्राह्मस्फुटसिद्धान्तः

चन्द्रग्रहणाधिकारः

अथ चन्द्रग्रहणाधिकारो व्याख्यायते तत्रादौ तदारम्भप्रयोजनमाह

कालज्ञानं प्रायः पर्वज्ञानार्थमिष्यते सद्भिः ।
शशिभास्करग्रहणयोर्वास्तवभेदाः स्फुटा भेदैः ॥१॥
दिग्वर्णवलनवेलानिमीलनोन्मीलनस्थितिविमर्दैः ।
स्पर्शच्छायामोक्षग्रासेष्टग्रासपरिलेखाः ॥२॥
भेदाश्चतुर्दश तयोरिन्द्वर्कग्रहणयोः परिज्ञानात् ।
यस्माद् भेदज्ञानं तस्मात् ग्रहणे प्रवक्ष्यामि ॥३॥

सु. भा.—सद्भिः सद्गणकैः प्रायो बाहुल्येन पर्वज्ञानार्थं कालज्ञानमिष्यते । तद्विपरीतेनार्थात् पर्वज्ञानात् कालज्ञानं च स्फुटम् । कालज्ञानाद्यज्ञादिक्रिया स्फुटेति फलितार्थः । शशिभास्करग्रहणयोर्ज्ञानार्थं ये वास्तवभेदा दिग्देशकालावरणाद्यास्ते भेदैर्वक्ष्यमाणैः स्फुटाः । चतुर्दशभेदाश्च दिग्वर्णवलनवेलाद्याः स्फुटाः । (दिग्भेदश्चन्द्रस्य प्राक् प्रग्रहणं रवेः पश्चात् । वर्णभेदश्चन्द्रस्यार्धाल्पार्धदिग्रासे धूम्रादयो वर्णा रवेः सर्वदा कृष्णः । वलनभेदश्चन्द्रग्रहे स्पर्शमोक्षोद्भववलने प्राक्पश्चिमतो देये रविग्रहे पश्चिमपूर्वतः । वेलासमयश्चन्द्रग्रहे सदा तिथ्यन्त एव मध्यग्रहः सूर्यग्रहे स नियमो नैव । निमीलनोन्मीलने च शशिसूर्यग्रहणयोर्मिथो व्यस्ता श्रुतो भवत इति तत्रापि भेदः । स्थितिश्च चन्द्रग्रहे महती रविग्रहे लघुरिति । एवं विमर्दः कालश्च चन्द्रग्रहे महान् रविग्रहे लघुः । स्पर्शः स्पर्शकालनिर्णयश्च चन्द्रग्रहे स्थित्यर्धाद्विमर्दे सम्वलनसंस्कृतस्थित्यर्धात् । छायाभेदश्च ग्रग्रहणादिषु स्वस्वच्छङ्कुभेदात् प्रसिद्धः । मोक्षभेदश्च चन्द्रग्रहे पश्चाद्विमर्दे प्राक् मोक्षः । ग्रासभेदश्चन्द्रग्रहे श्रुतो रविग्रहे नतिसंस्कृताच्छ्रुतः । इष्टग्रासभेदश्चन्द्रग्रहे तत्कालश्रुताद् गणितागतस्थित्यर्धाच्च रविग्रहे नतिसंस्कृतश्रुतात् स्पष्टस्थित्यर्धाच्च । परिलेखभेदश्च चन्द्रग्रहे गणितागतश्रुतात् रविग्रहे नतिसंस्कृतात् । एवमत्र दिक् १ । वर्ण २ । वलन ३ । वेला ४ । निमीलन ५ । उन्मीलन ६ । स्थिति ७ । विमर्द ८ । स्पर्श ९ । छाया १० । मोक्ष ११ । ग्रास १२ । इष्टग्रास १३ । परिलेख १४ ।

चतुर्दशभेदाः ।) यस्मादिन्द्रर्कग्रहणयोः परिज्ञानादिति चतुर्दशभेदज्ञानं भवति तस्माच्छशिरविग्रहणे अहं प्रवक्ष्यामीति ॥१-२-३॥

वि. भा.—सद्भिः (वेदज्ञैः सद्गणकैर्वा) प्रायः (बाहुल्येन) पर्वज्ञानार्थं कालज्ञानमिष्यते (अमावास्यापूर्णिमादिपर्वणां प्रारम्भान्तसमयौ प्रधानतया वेदविहितकर्मानुष्ठानाय तद्विदां प्रयोजनीभूतौ भवतोऽतएव ज्योतिःशास्त्रेऽपि मुख्यतया तत्तत्पर्वणामेव कस्मिन् काले आरम्भः कस्मिन् कालेऽन्त इत्यस्यैवात्र-गमः क्रियत इति भावः) चन्द्रसूर्यग्रहणयोर्ज्ञानार्थं ये वास्तवभेदाः (दिग्देश-कालावरणाद्याः) तेऽधोलिखितभेदैः स्फुटा भवन्ति । दिग्भेदः (चन्द्रग्रहणे प्राक् प्रग्रहणं, सूर्यग्रहणे च पश्चात्), वर्णभेदः (चन्द्रग्रहणे-अर्धग्रासार्धाधिकग्रासर्व-ग्रासादिषु चन्द्रस्य कृष्ण-कृष्णरक्त-पिशङ्गवर्णादयो भवन्ति । सूर्यग्रहणे सूर्यस्य वर्णः सर्वदा कृष्ण एव भवति) वलनभेदः (स्पार्शिकमौक्षिकवलने पूर्वपश्चिमतश्चन्द्र-ग्रहणे, सूर्यग्रहणे च पश्चिमपूर्वतः) वेला (समयभेदः) चन्द्रग्रहणे सर्वदा तिथ्यन्त एव ग्रहः, सूर्यग्रहणे स नियमो नास्ति, निमीलनोन्मीलने (चन्द्रग्रहणे सूर्यग्रहणे च परस्परं वैपरीत्येन भवत इति तत्रापि भेदः) स्थितिभेदः (चन्द्रग्रहे महती स्थितिः, सूर्यग्रहे लघुरिति), विमर्दकालभेदः (चन्द्रग्रहणे महान् सूर्यग्रहणे चाल्पः) स्पर्शः (स्पर्शकालनिर्णयः) चन्द्रग्रहणे स्थित्यर्धात्, सूर्यग्रहणे च लम्बन-संस्कृतस्थित्यर्धात्) छायाभेदः (स्वस्वशङ्कुभेदात् प्रग्रहणादिषु भवेदेव) मोक्षभेदः (चन्द्रग्रहे पश्चिमतः सूर्यग्रहे च पूर्वतो मोक्षः) ग्रासभेदः (चन्द्रग्रहे शरात्—सूर्यग्रहे च नतिसंस्कृतशरतः) इष्टग्रासभेदः (चन्द्रग्रहे तत्कालशराद् गणितागतस्थित्यर्धाच्च, सूर्यग्रहणे नतिसंस्कृतशरात्—स्फुटस्थित्यर्धाच्च,) परिलेखभेदः (चन्द्रग्रहणे गणिता-गतशरात्—सूर्यग्रहणे च नतिसंस्कृतशरात्) एवमत्र चतुर्दश भेदा भवन्ति, यतो रविचन्द्रयोर्ग्रहणज्ञानाच्चतुर्दशभेदज्ञानं भवत्यतश्चन्द्रसूर्यग्रहणे अहं प्रवक्ष्यामीति ॥१-२-३॥

हि. भा.—अच्छे गणक लोग प्रायः पर्वज्ञान के लिये समयज्ञान को कहते हैं अर्थात् अमावास्या—पूर्णिमा आदि पर्वों के आरम्भ और अन्त समय प्रधान रूप से वेदोक्त कर्मा-नुष्ठान के लिए सद्गणकों वा वेदज्ञों को प्रयोजनीभूत होते हैं अतएव ज्योतिःशास्त्र में भी मुख्य रूप से उन पर्वों का किस समय में आरम्भ होता है और किस समय में अन्त होता है इसी का ज्ञान किया जाता है, चन्द्रग्रहण और सूर्यग्रहण के ज्ञान के लिये जो वास्तव भेद (दिक् देशकालादि) हैं वे निम्नलिखित भेदों से स्फुट होते हैं, दिग्भेद (चन्द्रग्रहण में पूर्व-दिशा में प्रग्रहण होता है और सूर्यग्रहण में पश्चिम में होता है) वर्णभेद (चन्द्रग्रहण में अर्धग्रास-अर्धाधिकग्रास-सर्वग्रासादियों में चन्द्र के वर्ण कृष्ण-कृष्णरक्त-कपिलवर्ण आदि होते हैं, सूर्य ग्रहण में सूर्य का वर्ण सदा कृष्ण ही होता है) वलनभेद (चन्द्रग्रहण में स्पार्शिक वलन और मौक्षिक वलन पूर्व और पश्चिम से सूर्यग्रहण में पश्चिम और पूर्व) वेला (समय) भेद (चन्द्रग्रहण में सदा तिथ्यन्त ही में ग्रह होता है, सूर्यग्रहण में यह नियम

नहीं है। निमीलन भेद और उन्मीलनभेद (वे चन्द्रग्रहण में और सूर्यग्रहण में परस्पर विपरीत होते हैं) स्थिति भेद (चन्द्रग्रहण में स्थिति महती होती है और सूर्यग्रहण में लघु) विमर्दकाल (चन्द्रग्रहण में महान् और सूर्यग्रहण में अल्प) स्पर्श (स्पर्शज्ञानविषय) भेद (चन्द्रग्रहण में स्थित्यर्ध से और सूर्यग्रहण में लम्बन संस्कृतस्थित्यर्ध से) छायाभेद (अपने-अपने शङ्कुभेद से प्रभादियों में होते ही हैं) मोक्षभेद (चन्द्रग्रहण में पश्चिम से और सूर्यग्रहण में पूर्व से मोक्ष होता है) ग्रासभेद (चन्द्रग्रहण में शर से और सूर्यग्रहण में नतिसंस्कृतशर से) इष्टग्रासभेद (चन्द्रग्रहण में तात्कालिकशर से और गणितागत स्थित्यर्ध से भी, सूर्यग्रहण में नतिसंस्कृतशर से और स्फुटस्थित्यर्ध से भी) पश्चिमत्व भेद (चन्द्रग्रहण में गणितागतशर से और सूर्यग्रहण में नतिसंस्कृतशर से) इस तरह यहां चौदह भेद होते हैं, जिस कारण से चन्द्रग्रहण और सूर्यग्रहण के ज्ञान से उन चौदह भेदों का ज्ञान होता है। इस हेतु से मैं चन्द्रग्रहण और सूर्यग्रहण को कहता हूं इति ॥१-२-३॥

इदानीं तात्कालिकीकरणमाह

तिथिगतगम्ये भुक्तिगुणे भुक्त्यन्तरहृते फलोनयुतौ।
रविशशिनौ समलिप्तौ पातस्तात्कालिको भवति ॥४॥

सु. भा.—तिथिगतगम्ये तिथिगतगम्यकले भुक्तिगुणे रविभुक्त्या चन्द्रभुक्त्या च गुणे। उभयत्र रविचन्द्रभुक्त्यन्तरेण हृते। गतचालने फलोनौ गम्ये फलयुतौ रविशशिनौ। एवं तौ तात्कालिकौ तिथ्यन्ते समलिप्तौ भवतः। एवं पातगत्या पातोऽपि तात्कालिको भवति।

अत्रोपपत्तिः। तिथिगतगम्ये कले षष्टिगुणे रविचन्द्रगत्यन्तरभक्ता गतैष्या घटिकास्ता गतिगुणाः षष्टिहृताश्चालनकलाः स्युः। एवमत्र षष्टितुल्ययोर्गुणहर्योर्नाशाद्यथोक्ता क्रियोत्पद्यते ॥४॥

वि. भा.—तिथिगतगम्ये (तिथिगतगम्यकले) भुक्तिगुणे (चन्द्रगत्या रविगत्या च गुणिते) भुक्त्यन्तरहृते (रविचन्द्रयोर्गत्यन्तरेण भक्ते) गतचालने फलेन होनौ, गम्यचालने फलेन युतौ चन्द्ररवी, तिथ्यन्ते तात्कालिकौ समलिप्तौ चन्द्ररवी भवतः। एवं स्वगत्या पातोऽपि तात्कालिको भवतीति ॥४॥

अत्रोपपत्तिः

यदि रविचन्द्रगत्यन्तरकलायां षष्टिघटिका लभ्यन्ते तदा तिथिगतकलायां गम्यकलायां च किमित्यनुपातेन तिथिगतघट्यो गम्यघट्यश्च समायच्छन्ति, ततः षष्टिघटिकायां स्वस्वगतिकला लभ्यन्ते तदा तिथिगतघट्यां तिथिगम्यघट्यां

च किमिति जाता चालनकला चन्द्रस्य $=\frac{६०\times\text{तिथिगक}}{\text{रविचन्द्रगत्यन्तरक}}\times\frac{\text{चंगकला}}{६०}=$

$\frac{\text{तिथिगक}\times\text{चंगकला}}{\text{रविचन्द्रगत्यन्तरक}}$, एवं गम्यचन्द्रचालनकलाः $=\frac{६०\times\text{तिथिगम्यक}}{\text{रविचन्द्रगत्यन्तरक}}\times\frac{\text{चंगकला}}{६०}$

$=\frac{\text{तिथिगम्यक}\times\text{चंगक}}{\text{रविचन्द्रगत्यन्तरक}}$, एवमेव रवेर्गतचालन कलाः $=\frac{६०\times\text{तिथिगतक}}{\text{रविचन्द्रगत्यन्तरक}}\times$

$\frac{\text{रविगक}}{६०}=\frac{\text{तिथिगतक}\times\text{रविगक}}{\text{रविचन्द्रगत्यन्तरक}}$,

रवेर्गम्यचालनकला : $=\frac{६०\times\text{तिथिगम्यक}}{\text{रविचन्द्रगत्यन्तरक}}\times\frac{\text{रविगक}}{६०}=\frac{\text{तिथिगम्यक}\times\text{रविगक}}{\text{रविचन्द्रगत्यन्तर}}$

एतेनाऽऽचार्योक्तमुपपन्नमिति ॥७॥

अब तात्कालिकीकरण को कहते हैं

हि. भा.—तिथिगत कला को और तिथिगम्यकला को चन्द्रगति और रविगति से पृथक् गुणाकर रवि और चन्द्र के गत्यन्तर से भाग देने से जो फल हो गतचालन में उन फलों को चन्द्र और रवि में हीन करना और गम्यचालन में युत करना तब तिथ्यन्त में तात्कालिक चन्द्र और रवि समलिप्तिक (समान कलात्मक) होते हैं, इस तरह अपनी गति से पात (चन्द्रपात) भी तात्कालिक होता है इति ॥८॥

उपपत्ति

यदि रवि और चन्द्र की गत्यन्तर कला में साठ घटी पाते हैं तो तिथिगतकला और तिथिगम्यकला में क्या इस अनुपात से तिथिगतघटी और तिथिगम्यघटी आती हैं, $\frac{६०\times\text{तिथिगतक}}{\text{रविचन्द्रगत्यन्तरक}}=$तिथिगतघटी, $\frac{६०\times\text{तिथिगम्यक}}{\text{रविचन्द्रगत्यन्तरक}}=$तिथिगम्यघटी पुनः अनुपात करते हैं, यदि साठ घटी में अपनी-अपनी गतिकला पाते हैं तो तिथिगत घटी में और तिथि गम्य घटी में क्या इस अनुपात से चन्द्र की गतचाल कला $=\frac{६०\times\text{तिथिगतक}}{\text{रविचन्द्रगत्यन्तरक}}$

$\times\frac{\text{चंगक}}{६०}=\frac{\text{तिथिगतक}\times\text{चंगक}}{\text{रविचन्द्रगत्यन्तरक}}$

एवं चन्द्रगम्यचालन कला $=\frac{६०\times\text{तिथिगम्यक}}{\text{रविचन्द्रगत्यन्तरक}}\times\frac{\text{चंगक}}{६०}=\frac{\text{तिथिगम्यक}\times\text{चंगक}}{\text{रविचन्द्रगत्यन्तरक}}$, इसी तरह

रवि की गतचालन कला $=\frac{६० \times \text{तिथिगत कला}}{\text{रविचन्द्रगत्यन्तरक}} \times \frac{\text{रविगक}}{६०} = \frac{\text{तिथिगक} \times \text{रविगक}}{\text{रविचन्द्रगत्यन्तरक}}$, रवि की गम्यचालनकला $=\frac{६० \times \text{तिथिगम्यक}}{\text{रविचन्द्रगत्यन्तरक}} \times \frac{\text{रविगक}}{६०} = \frac{\text{तिथिगम्यक} \times \text{रविगक}}{\text{रविचन्द्रगत्यन्तरक}}$, इसमें आचार्योक्त उपपन्न हुआ ॥४॥

इदानीं तिथ्यन्ते शरकलानयनमाह

खत्रिघन २७० गुणा व्यासार्धभाजिता चन्द्रपातयोगज्या।
विक्षेपकलाः सौम्याः षड्राश्यूनेऽधिके याम्याः ॥५॥

सु. भा.—षड्राश्यूने चन्द्रपातयोगे सौम्या अधिके याम्या विक्षेपकला भवन्ति। शेषं स्पष्टार्थम्।

अत्रोपपत्तिः। 'सपाततात्कालिकचन्द्रदोर्ज्या खभेर्हृते' त्यादिभास्करविधिना स्फुटा ॥५॥

वि. भा.—चन्द्रपातयोगज्या (सपाततात्कालिकचन्द्रभुजज्या) खत्रिघन २७० गुणा, व्यासार्ध (त्रिज्या) भक्ता फलं विक्षेप (शर) कला भवन्ति, ताश्चन्द्रपातयोगे षड्राश्यल्पे सौम्याः (उत्तराः) भवन्ति, चन्द्रपातयोगे षड्राश्यधिके सति याम्याः (दक्षिणाः) भवन्तीति ॥५॥

अत्रोपपत्तिः

क्रान्तिविमण्डलयोः सम्पातः पातः, पातस्थानान्नवत्यंशेन वृत्तं कार्यं तस्मिन् वृत्ते क्रान्तिविमण्डलयोरन्तर्गतं चापं चन्द्रपरमशरः, विमण्डले यत्र चन्द्रबिम्बकेन्द्रमस्ति तदुपरिगतं कदम्बप्रोतवृत्तं यत्र क्रान्तिवृत्ते लगति तत्र चन्द्रस्थानम्। चन्द्रबिम्बकेन्द्राच्चन्द्रस्थानं यावत्कदम्बप्रोतवृत्ते चन्द्रमध्यमशरः। पातस्थानाच्चन्द्रबिम्बकेन्द्रं यावच्चन्द्रविमण्डलीयभुजांशाः। पातस्थानाच्चन्द्रस्थानं यावच्चन्द्रस्थानीयभुजांशाः (सपाततात्कालिकचन्द्रभुजांशा),पातस्थानान्नवत्यंशवृत्तं परमान्तरवृत्तम्। पातस्थानात्परमान्तरवृत्तविमण्डलयोः सम्पातं यावद्विमण्डले नवत्यंशाः.। पातस्थानादेव परमान्तरवृत्तक्रान्तिवृत्तयोः सम्पातं यावत् क्रान्तिवृत्ते नवत्यंशाः। परमान्तरवृत्ते क्रान्तिविमण्डलयोरन्तरे चन्द्रपरमशरांशाः, इति भुजत्रयैरुत्पन्नक्षेत्रमुजम्। पातस्थानाच्चन्द्रबिम्बकेन्द्रं यावद्विमण्डले विमण्डलीयभुजांशाः कर्णः, सपाततात्कालिक चन्द्रभुजांशाः क्रान्तिवृत्ते कोटिः। चन्द्रबिम्बकेन्द्रस्थानयोरन्तरे कदम्बप्रोतवृत्ते चन्द्रमध्यमशरो भुज इति भुजत्रयैरुत्पन्नं द्वितीयं चापीयं क्षेत्रम्। त्रिभुजमेतयोश्चापीयक्षेत्रत्वभिभुजयोर्व्यस्तसाजात्यादनुपातः क्रियते, यदि त्रिज्यया

चन्द्रपरमशरज्या लभ्यते तदा चन्द्रविमण्डलीयभुजज्यया किमित्यनुपातेनाऽऽगच्छति चन्द्रमध्यमशरज्या तत्स्वरूपम्=$\frac{\text{चन्द्रपरमशरज्या. चन्द्र विमण्डलीयभुज्या}}{\text{त्रि}}$

अत्र विमण्डलीयभुजांशस्याज्ञानाद्विदितस्थानीयभुजांश समा एव विमण्डलीय भुजांशाः स्वीकृता आचार्येण, तदा $\frac{\text{चन्द्रपरमशरज्या. सपाततात्कालिकचंभुजज्या}}{\text{त्रि}}$= चन्द्रमध्यमशरज्या, तथा ज्याचापयोरभेदत्वमपि स्वीकृतम् तदा—

$\frac{\text{चन्द्रपरमशर. सपाततात्कालिचन्द्रभुज्या}}{\text{त्रि}}$=चन्द्रमध्यमशरः। चन्द्रस्य परमशरांशाः =२७०, ततः $\frac{२७०\times\text{सपाततात्कालिकचन्द्रभुज्या}}{\text{त्रि}}$=चन्द्रमध्यमशरः, अयं शरः सपातचन्द्रगोलदिग्भवत्येतावताऽऽचार्योक्तमुपपन्नम्। इदमानयनं न समीचीनं यतः स्थानीयविम्बीयचन्द्रभुजांशयोः समत्वं स्वीकृतं शरज्याचापयोरभेदत्वं च स्वीकृतमाचार्येण, ततो वास्तवानयनं क्रियते, स्थानीयचन्द्रभुजांशविमण्डलीय-भुजांश-शरांशैरुत्पन्नचापीयजात्यत्रिभुजे स्थानीयचन्द्रभुजांशविमण्डलीयभुजांश-योरुत्पन्नकोणः=चन्द्रपरमशरः, तदा मध्यजा दोर्ज्या त्रिज्यागुणा प्रान्त्यस्पर्श-रेखाहृतिर्भवेदनेन स्थाने श्रवणकोणयोस्तत्कोटिमेव गृह्णीयादित्यनेन च परमश-कोज्या. त्रि=स्पविमण्डलीयभुको×स्प स्थानीय. भुजांश ततः

$\frac{\text{परमशकोज्या. त्रि}}{\text{स्प स्थानीय भुजांश}}$=स्पविमण्डलीयभुको, एतेन चन्द्रस्य विमण्डलीयभुजांशज्ञानं भवेदेव, तदोक्तचापीयजात्यत्रिभुजे भुजकोटिज्याकोटिकोटिज्ययोर्घातस्य त्रिज्या-कर्णकोटिज्ययोर्घातेन समत्वात्

त्रि. विमण्डलीयभुकोज्या=शरकोज्या. स्थानीयभुकोज्या=शरकोज्या. सपाततात्कालिकचन्द्रभुकोज्या, ततः $\frac{\text{त्रि. विमण्डलीयभुकोज्या}}{\text{सपाततात्कालिक चं भुकोज्या}}$=शरकोज्या अस्याश्चापं नवतेर्विशोध्यं तदा चन्द्रस्य मध्यमशरो भवेदिति। सिद्धान्तशिरोमणौ भास्कराचार्येण "सपाततात्कालिकचन्द्रदोर्ज्यास्विभै २७० र्हृता व्यासदलेन भक्ता। सपातक्षीतद्युतिगोलदिक् स्याद्विक्षेप इन्दोः स च बाणसंज्ञ" त्यनेनाऽऽचार्योक्तानुरूप-मेवोक्तम्, सिद्धान्तशेखरे श्रीपतिना "पातोनितस्य समलिप्तकशीतरश्मेर्जीवा कृतेषु ५४ गुणिता त्रिभजर्त्तु ६८३ भक्ता। क्षेपो भवत्यनेन, चन्द्रशरानयनं कृतमिदम-प्याचार्योक्तानुरूपमेव, केवलमत्र पातस्य चक्रशुद्धत्वात् सपातचन्द्रभुजज्यास्थाने विपातचन्द्रभुजज्या गृहीता, तथा विपातचन्द्रभुजज्याया गुणकस्य चन्द्रपरमशर

२७० भ्य, श्रीपन्युक्तत्रिज्यायाः ३४१५ भाजकरूपाद्याश्च पञ्चभिरपवर्त्तने श्रीपन्युक्त-पद्धमुपपन्नं भवतीति ॥ ५ ॥

अब तिथ्यन्त में चन्द्रशरकलानयन को कहते हैं।

हि. भा.—सपात तात्कालिक चन्द्रभुजज्या को दो सौ सत्तर २७० से गुणा कर त्रिज्या से भाग देने से फल चन्द्रशर कला होती है, सपात चन्द्र के छः राशि से कम रहने से उस (शरकला) की दिशा उत्तर होती है, तथा छः राशि से अधिक रहने से शरकला की दिशा दक्षिण होती है इति ॥ ५ ॥

उपपत्ति ।

क्रान्तिवृत्त और विमण्डल के सम्पात पात संज्ञक है, पात स्थान से नवत्यंश व्यासार्ध वृत्त उन दोनों (क्रान्तिवृत्त और विमण्डल) वृत्तों के परमान्तर वृत्त है, क्रान्तिवृत्त और विमण्डल के अन्तर्गत परमान्तर वृत्तीय चाप परमशर है, विमण्डल में जहां चन्द्रबिम्ब केन्द्र है उसके ऊपर कदम्ब प्रोत वृत्त करने से क्रान्तिवृत्त में जहां लगता है वह चन्द्र स्थान है, चन्द्रबिम्ब केन्द्र से चन्द्र स्थान तक कदम्बप्रोतवृत्त में चन्द्र के मध्यमशर है, पात स्थान से चन्द्रबिम्ब केन्द्र तक चन्द्र के विमण्डलीय भुजांश कर्ण, पातस्थान से चन्द्रस्थान तक सपात चन्द्रभुजांश कोटि, चन्द्र मध्यमशर भुज इन तीनों भुजों से उत्पन्न एक चापीय जात्य त्रिभुज तथा पातस्थान से परमान्तर वृत्त और विमण्डल के सम्पात पर्यन्त विमण्डल में नवत्यंश, पातस्थान ही से परमान्तरवृत्त और क्रान्तिवृत्त के सम्पात पर्यन्त क्रान्तिवृत्त में नवत्यंश, परमान्तर वृत्त में चन्द्र परमशर, इन तीनों भुजों से उत्पन्न द्वितीय चापीय जात्य त्रिभुज, इन दोनों चापीय जात्य त्रिभुजों के ज्याक्षेत्र जातीय है इसलिये अनुपात करते हैं यदि त्रिज्या में चन्द्र परमशरज्या पाते हैं तो विमण्डलीय चन्द्र भुजज्या में क्या इस अनुपात से चन्द्र-मध्यम शरज्या आती है $\frac{\text{चन्द्रपरमशज्या. चन्द्रविमण्डलीय भुज्या}}{\text{त्रि}}$ = चंमध्यमशरज्या, परन्तु यहां चन्द्र विमण्डलीय भुजांश विदित नहीं है, गणितागत सपात चन्द्रभुजांश विदित है इसलिए आचार्य ने चन्द्र विमण्डलीय भुजांश तुल्य ही सपात चन्द्र भुजांश को स्वीकार किया है तब $\frac{\text{चंपरमशज्या. सपातचंभुज्या}}{\text{त्रि}}$ = चंमध्यमशरज्या, तथा ज्या और चाप में अभेदत्व भी स्वीकार किया है तब $\frac{\text{चंपरमश} \times \text{सपात चंभुज्या}}{\text{त्रि}}$ = चंमध्यमशर, = $\frac{२७० \times \text{सपात चंभुज्या}}{\text{त्रि.}}$ यह शर सपात चन्द्र गोल दिशा का होता है अर्थात् सपात चन्द्र जिस गोल में रहते हैं उसी गोल का होता है इससे आचार्योक्त उपपन्न हुआ, परन्तु यह आनयन ठीक नहीं है क्योंकि विमण्डलीय चन्द्र भुजांश कर्ण चाप है और सपात चन्द्रभुजांश कोटि चाप

है इन दोनों का आचार्य ने तुल्य मान लिया हैं तथा चंपरमशरज्या = चंपरमशर, एवं चंमध्यशरज्या = चंमध्यमशर स्वीकार किया है, जो कि अनुचित है, अतः वास्तवानयन करते हैं। चन्द्र विमण्डलीय भुजांश, सपात चन्द्रभुजांश, और चन्द्रमध्यमशर इन तीनों अवयवों से उत्पन्न चापीय जात्य त्रिभुज में चन्द्र विमण्डलीय भुजांश और सपात चन्द्र भुजांश से उत्पन्न कोण = चंपरमशर, तब उक्त त्रिभुज में 'मध्यजा दोर्ज्या त्रिज्या गुणा प्रान्त्यस्पर्श रेखा हृतिर्भवेत्' इससे तथा 'तत्कोटिमेव गृह्णीयात् स्थाने श्रवणकोणयोः' इससे भी चंपरमशकोज्या. त्रि = स्पचंविमण्डलीय भुकोज्या. स्पसपातचं भुज्या अतः $\frac{\text{चंपरमशकोज्या. त्रि}}{\text{स्पसपात चभुज्या}}$ = स्पचंविमण्डलीय भुकोज्या, स्पर्शरेखा खण्डों से इसका चाप कर के नवत्यंश में से घटाने से चन्द्र विमण्डलीय भुजांश का ज्ञान हो जायगा, तब पूर्व कथित चापीय जात्य त्रिभुज में भुजकोटिज्या और कोटि कोटिज्या का घात त्रिज्या और कर्ण कोटिज्या के घात के बराबर होता है इस नियम से त्रि. चंविमण्डलीय भुकोज्या = चंशर कोज्या सपात चं भुकोज्या ∴ $\frac{\text{त्रि. चंविमण्डलीय भुकोज्या}}{\text{सपात चं भुकोज्या}}$ = चंशर कोज्या, इसके चाप को नवत्यंश में से घटाने से चन्द्रमध्यमशर होता है, सिद्धान्तशिरोमणि में भास्कराचार्य ने "सपाततात्कालिकचन्द्रदोर्ज्या" इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से आचार्योक्त के अनुरूप ही कहा है, सिद्धान्त शेखर में श्रीपति "पातोनितस्य समलिप्तशीतरश्मेः" इत्यादि संस्कृतोपपत्ति में लिखित पद्य से चन्द्र का शरानयन किया है यह भी आचार्योक्त के अनुरूप ही है केवल श्रीपति ने पात के चक्र (बारहराशि) में शुद्ध होने के कारण सपात चन्द्र भुजज्या स्थान में विपातचन्द्रभुजज्या ग्रहण किया है, और विपात चन्द्र भुजज्या के गुणक चन्द्रपरमशर २७० को तथा श्रीपत्युक्त त्रिज्या = ३४१५ हर को पांच से अपवर्त्तन कर देने से श्रीपति का पद्य उपपन्न होता है इति ॥ ५ ॥

इदानीं रविचन्द्रतमसां बिम्बान्याह

रविशशिभुक्ती भवदशगुणे नखैः स्वर्जिनैर्हृते माने।
तत्त्वाष्टगुणितभुक्त्योर्विवरं षष्ट्याहृतं तमसः ॥ ६ ॥

सु० भा०—रविशशिभुक्ती भवदशगुणे नखैः स्वरर्जिनैर्हृते अर्थाद्रविगतिरेकादशगुणा नख २० हृता। चन्द्र गतिर्दशगुणा स्वरर्जिनै २४७ र्हृता तदा रविचन्द्रयोर्माने बिम्बमाने भवतः। तत्त्वाष्टगुणितभुक्त्योः पञ्चविंशति गुणरविगतेरष्ट गुणितचन्द्रगतेश्च विवरं षष्ट्याहृतं तमसो राहोर्विम्बमानं भवेत्।

अत्रोपपत्तिः। 'भानोर्गतिः स्वदशभागयुतार्धिता' वेत्यादिभास्करविधिना

रविं $= \frac{११ \text{ रग}}{२०}$ । चंविं $= \frac{३ \text{ चंग}}{७४} = \frac{\text{चंग}}{\frac{७४}{३}} = \frac{१० \text{ चंग}}{\frac{७४०}{३}} = \frac{१० \text{ चंग}}{२४७}$ स्वल्पा-

न्तरात् । 'भानोर्गतिः शरहता रविभिर्विभक्ता' इत्यादि भास्करविधिना राहुर्वि $= \frac{२ \text{ चंग}}{१५} - \frac{५ \text{ रग}}{१२} = \frac{८ \text{ चंग} - २५ \text{ रग}}{३०}$ । अत उपपन्नं सर्वम् ॥ ६ ॥

वि. भा.—रविशशिभुक्ती (रविचन्द्रयोर्गतौ) भवदशगुणे (एकादशदशगुणिते) नखैः स्वरजिनै २४७ भक्ते ऽर्थाद्रविगतिरेकादशगुणा विशल्या भक्ता, चन्द्रगतिर्दशगुणा स्वरजिनै २४७ भक्ता तदा माने (रविचन्द्रयोर्बिम्ब प्रमाणे) भवतः । तत्त्वाष्टगुणित-भुक्त्योः (पञ्चविंशति गुणित रविगतेरष्टगुणित चन्द्रगतेश्च) विवरं (अन्तरं) षष्टि (६०) भक्तं तदा तमसः (राहोः) बिम्बमानं भवेदिति ॥ ६ ॥

अत्रोपपत्तिः

भानोर्गतिः स्वदशभागयुतार्धिता वेत्यादि भास्करोक्तथा रविर्वि $= \frac{\text{रग} + \frac{\text{रग}}{१०}}{२} = \frac{१० \text{ रग} + \text{रग}}{२०} = \frac{११ \text{ रग}}{२०}$ विधोस्त्रिगुणिता युगशैलभक्ते ति भास्करोक्त-विधिना चंविं $= \frac{३ \text{ चंग}}{७४} = \frac{३ \text{ चंग} \times १०}{७४ \times १०} = \frac{३ \text{ चंग} \times १०}{७४०} = \frac{\text{चंग} \times १०}{\frac{७४०}{३}} = \frac{\text{चंग} \times १०}{२४७}$

स्वल्पान्तरात् ।

अथ दिवाकर निशानाथ परलम्बन संयुतिः । रविबिम्बार्धरहिता भूभा बिम्ब-दलं भवेदिति संशोधकोक्त विधिना रपलं + चंपलं − रविं$\frac{1}{2}$ = भूभार्वि$\frac{1}{2}$, गतिकलाया-स्तिथ्यंशः परलम्बन लिप्तिकेत्युक्तेः परमलम्बनोत्थापनात् $\frac{\text{रग}}{१५} + \frac{\text{चंग}}{१५} - \frac{११ \text{ रग}}{२० \times २}$

$= \text{भूभार्वि}\frac{1}{2} = \frac{\text{रग}}{१५} + \frac{\text{चंग}}{१५} - \frac{११ \text{ रग}}{४०} = \frac{\text{रग}}{१५} - \frac{११ \text{ रग}}{४०} + \frac{\text{चंग}}{१५} = \frac{८ \text{ रग} - ३३ \text{ रग}}{१२०} +$

$\frac{\text{चंग}}{१५} = -\frac{२५ \text{ रग}}{१२०} + \frac{\text{चंग}}{१५} = -\frac{५ \text{ रग}}{२४} + \frac{\text{चंग}}{१५} = -\frac{\text{चंग}}{१५} - \frac{५ \text{ रग}}{२४}$ द्विगुणी करणेन

$\frac{२ \text{ चंग}}{१५} - \frac{५ \text{ रग}}{१२} = \text{भूभार्वि} \cdots\cdots (१)$ अत्र प्रथम खण्डे हर भाज्यौ चतुर्भि-

गुणितौ तथा द्वितीयखण्डे हर भाज्यौ पञ्चभिर्गुणितौ तदा $\frac{८\,\text{चंग}}{६०}-\frac{२५\,\text{रग}}{६०}=\frac{८\,\text{चंग}-२५\,\text{रग}}{६०}$ एतावताऽऽचार्योक्तं सर्वमुपपन्नम्। (१) एतेन च "भानोर्गतिः शरहता रविभिर्विभक्ता चन्द्रस्य लोचनगुणा तिथि भाजितेत्यादि" भास्करोक्तं भूभा बिम्बानयनमुपपद्यते। सिद्धान्तशेखरे "रुद्रैः ११ पंक्त्या १० रविशशिगती ताड़िते वा विभक्ते कृत्या २० भूभृज्जलधिनयनै २४७ स्ते तयोर्मानि लिप्ताः। वाणैर्द्वाभ्यामथ विनिहतेऽर्कै १२ दिनै १५ स्ते विभक्ते लब्ध्योर्यद्वा भवति विवरं सैंहिकेयस्य बिम्बम्" श्रीपत्युक्तमिदमाचार्योक्तानुरूपमेव, आचार्योक्तभूभाबिम्बम्= $\frac{८\,\text{चंग}}{६०}-\frac{२५\,\text{रग}}{६०}=\frac{२\,\text{चंग}}{१५}-\frac{५\,\text{रग}}{१२}$ = श्रीपत्युक्त भूभार्वि, "भानोर्गतिः स्वदशभागयुताऽर्धिता वा चन्द्रस्य बिम्बं विधोस्त्रिगुणिता युगशैल ७४ भक्ता" भास्कराचार्योक्तमिदं रविचन्द्रयोर्बिम्बकलानयनं तथा "भानोर्गतिः शर ५ हता रवि १२ भिर्विभक्ता चन्द्रस्य लोचन २ गुणा तिथि १५ भाजिता च। लब्धान्तरं भवति वाऽवनि भा प्रमाणम्॥ इति भूभाबिम्बानयनं च श्रीपत्यनुरूपमेवेति॥ ६॥

अब रवि चन्द्र और भूभा के बिम्बानयन को कहते हैं।

हि. भा.—रवि गति को ग्यारह से गुणा कर बीस २० से भाग देने से रवि बिम्ब का मान होता है, चन्द्रगति को दस से गुणा कर दौ सौ सैंतालीस २४७ से भाग देने से चन्द्रबिम्ब का मान होता है, पचीस गुणित रविगति और आठ गुणित चन्द्रगति के अन्तर को साठ ६० से भाग देने से भूभा का बिम्बमान होता है इति॥ ६॥

उपपत्ति

'भानोर्गतिः स्वदशभागयुताऽर्धिता वा' इत्यादि भास्करोक्त विधि से रविर्वि = $\frac{\text{रग}+\frac{\text{रग}}{१०}}{२}=\frac{१०\,\text{रग}+\text{रग}}{२०}=\frac{११\,\text{रग}}{२०}$ बिम्बं बिधोस्त्रिगुणिता युगशैलभक्ता इस भास्करोक्ति से चंर्वि= $\frac{३\,\text{चंग}}{७४}=\frac{३\,\text{चंग}\times१०}{७४\times१०}=\frac{३\,\text{चंग}\times१०}{७४०}=\frac{\text{चंग}\times१०}{\frac{७४०}{३}}=\frac{\text{चंग}\times१०}{२४७}$

स्वल्पान्तर से, भूभा बिम्बानयन के लिये "दिवाकर निशानाथ परलम्बन संयुतिः। रविबिम्बार्धरहिता भूभा बिम्बदलं भवेत्" इस संशोधकोक्त विधि से रपलं+चंपलं−रविई = भूभार्विई

परन्तु रपलं $=-\frac{\text{रग}}{१५}$, चपलं $=\frac{\text{चग}}{१५}$, तथा रविबि $\frac{१}{२}=\frac{११\ \text{रग}}{२०\times२}=\frac{११\ \text{रग}}{४०}$

इन में उत्थापन करने से $\frac{\text{रग}}{१५}+\frac{\text{चग}}{१५}-\frac{११\ \text{रग}}{४०}=\frac{\text{रग}}{१५}-\frac{११\ \text{रग}}{४०}+\frac{\text{चग}}{१५}=$

$\frac{८\ \text{रग}-३३\ \text{रग}}{१२०}+\frac{\text{चग}}{१५}=-\frac{२५\ \text{रग}}{१२०}+\frac{\text{चग}}{१५}=-\frac{५\ \text{रग}}{२४}+\frac{\text{चग}}{१५}=\frac{\text{चग}}{१५}-\frac{५\ \text{रग}}{२४}$

$=$भूभाबि$-\frac{१}{२}$, द्विगुणित करने से $\frac{२\ \text{चग}}{१५}-\frac{५\ \text{रग}}{१२}=$भूमाबि···(१) यहां प्रथम खण्ड के हर और भाज्य को चार से गुणने से तथा द्वितीय खण्ड के हर और भाज्य को पांच से गुणने से $\frac{८\ \text{चग}}{६०}-\frac{२५\ \text{रग}}{६०}=\frac{८\ \text{चग}-२५\ \text{रग}}{६०}=$ भूभाबि इससे आचार्योक्त सब उपपन्न हुए। (१) इससे "भानोर्गतिः शरहता रविभिर्विभक्ता" इत्यादि भास्करोक्त भूभा बिम्बानयन उपपन्न होता है; सिद्धान्त शेखर में "रुद्रैः ११ पंक्तया १० रविशशि गती ताडिते" इत्यादि संस्कृतोपपत्ति में लिखित श्रीपति प्रकार आचार्योक्त के अनुरूप ही हैं, आचार्योक्त भूभाबि $=$ $\frac{८\ \text{चग}}{६०}-\frac{२५\ \text{रग}}{६०}=\frac{२\ \text{चग}}{१५}-\frac{५\ \text{रग}}{१२}$ श्रीपत्युक्त भूभाबि "भागानोर्गतिः स्वदशभाग-युताऽर्धिता वा" इत्यादि संस्कृतोपपत्ति में लिखित, भास्कराचार्योक्त रवि और चन्द्र के बिम्ब-कलानयन तथा "भानोर्गतिः शरहता रविभिर्विभक्ता" इत्यादि संस्कृतोपपत्ति में लिखित पद्य में भूभाबिम्बानयन श्रीपत्युक्त के अनुरूप ही हैं इति ॥ ६ ॥

इदानीं ग्रासमानमाह

छाद्यच्छादक मानैक्यार्धं विक्षेप हीनितं छन्नम् ।
सर्वग्रहणं ग्राह्याद्धिके खण्डग्रहणमूने ॥ ७ ॥

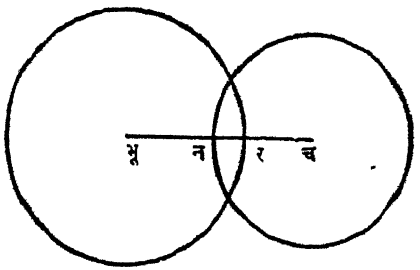
सु० भा०—स्पष्टार्थमार्या ।

अत्रोपपत्तिः । 'यच्छाद्यसच्छादकमण्डलैक्यखण्डं शरोनं स्वगतिप्रमाण' मित्यादि भास्करविधिना स्फुटा ॥ ७ ॥

वि. भा.—छाद्यच्छादकमानैक्यार्धं (ग्राह्यग्राहक बिम्बयोर्योगार्धं चन्द्रग्रहणे चन्द्रभूभा बिम्बयोर्योगार्धं सूर्यग्रहणे सूर्यचन्द्र बिम्बयोर्योगार्धमित्यर्थः) विक्षेपहीनितं (चन्द्रशरेण रहितं) तदा छन्नं (ग्रासमानं) भवेत्। ग्राह्यात् (छाद्यबिम्बात्) अधिके ग्रासमाने सर्वग्रहणं भवति, छाद्यबिम्बादल्पे ग्रासमाने खण्डग्रहणं भवतीति ॥ ७ ॥

अत्रोपपत्तिः

(क)



भू=भूभा बिम्बकेन्द्रम् । च=चन्द्रबिम्बकेन्द्रम् । भूच=चन्द्रशरः । भूर=भूभाबिम्बव्यासार्धम् । चन=चन्द्रबिम्बव्यासार्धम् । नर=ग्रासमानम् ।

भूर+चन = भूर+रच +नर= भूच+नर=चन्द्रशर+ग्रास+ भूभाबिम्बव्यासार्ध + चन्द्रबिम्बव्यासार्ध =मानैक्यार्धम्

अतः मानैक्यार्ध – चन्द्रशर=ग्रासमानम् । चन्द्रबिम्बादधिके एतद्ग्रासमाने सर्वग्रहणं भवेदेवेति । सिद्धान्तशिरोमणौ "यच्छाद्यसंछादकमण्डलैक्य खण्डं शरोन" मित्यादिना भास्करेण, क्षेपोभवत्यथ पिधान पिधेय बिम्बयोगार्धमूनममुनेत्यादिना श्रीपतिनाप्याचार्योक्तानुरूपमेवोक्तमिति ॥ ७ ॥

अब ग्रासानयन को कहते हैं ।

हि. भा.—छाद्य छादक मानैक्यार्ध (चन्द्रग्रहण में चन्द्रबिम्ब और भूभा बिम्ब के योगार्ध, सूर्यग्रहण में सूर्यबिम्ब और चन्द्रबिम्ब के) योगार्ध में से चन्द्रशर को घटाने से शेष ग्रास मान होता हैं, ग्राह्य बिम्ब से ग्रासमान अधिक रहने से सर्व ग्रहण होता हैं, ग्राह्य बिम्ब से ग्रासमान अल्प रहने से खण्ड ग्रहण होता हैं इति ॥ ७ ॥

उपपत्ति

यहां संस्कृतोपपत्ति में लिखित (क) क्षेत्र को देखिये । भू=भूभा बिम्ब केन्द्र, च=चन्द्रबिम्ब केन्द्र, भूच=चन्द्रशर, भूर=भूभा बिम्बव्यासार्ध, चन=चन्द्रबिम्बव्यासार्ध, नर=ग्रासमान, भूर+चन=भूर+रच+नर=भूच+नर=चन्द्रशर+ग्रासमान=भूभाबिम्बव्यासार्ध+चन्द्रबिम्बव्यासार्ध=मानैक्यार्ध, अतः मानैक्यार्ध—चन्द्रशर=ग्रासमान । चन्द्रबिम्ब (ग्राह्य बिम्ब) से अधिक ग्रासमान होने से सर्व ग्रहण होता ही हैं, यह क्षेत्र स्वरूप देखने से स्पष्ट है इति, सिद्धान्त शिरोमणि में भास्कराचार्य "यच्छाद्यसंछादकमण्डलैक्यखण्डं" इत्यादि से तथा सिद्धान्त शेखर में श्रीपति ने "क्षेपो भवत्यथ पिधानपिधेयबिम्बयोगार्धमूनममुना स्थगितं वदन्ति" इस प्रकार आचार्योक्त के अनुरूप ही कहा हैं इति ॥ ७ ॥

इदानीं स्थित्यर्धविमर्दार्धयोरानयनमाह

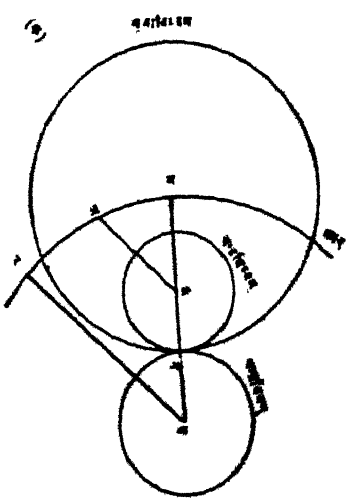
छाद्येन युतोनस्य च्छादकमानस्य तद्दलकृतिभ्याम् ।
विक्षेपकृतिं प्रोह्य पदे तिथिवत् स्थितिविमर्दार्धे ॥ ८ ॥

सु० भा०—पदे तिथिवत् । अर्थात्पदे षष्टिगुणे रविचन्द्रगत्यन्तरेण दले स्थितिविमर्दार्धे भवतः । शेषं स्पष्टार्थम् ।

अत्रोपपत्तिः । 'मानार्धयोगान्तरयोः कृतिभ्यां मिथ्यादि भास्करविधिना स्फुटा ॥ ८ ॥

वि. भा.—छाद्येन (छाद्यमानेन) युतोनस्य (सहितस्य रहितस्य च) छादक-मानस्यार्धात् छाद्यच्छादकबिम्बयोगम्यान्तरस्य च दलकृतिभ्यामर्थान्मानैक्यार्धं वर्गमानान्तरार्धवर्गाभ्यां विक्षेपकृतिं (शरवर्गं) प्रोह्य (त्यक्त्वा) तत्पदे (मूले) तिथिवत् कृते सती अर्थात् षष्ट्या गुणिते रविचन्द्रयोर्गत्यन्तरेण भक्ते तदा स्थित्यर्ध-विमर्दार्धे भवेतामिति ॥ ८ ॥

अत्रोपपत्तिः



यदा भूभा चन्द्रबिम्बयोर्बहिः स्पर्शो भवति तदा चन्द्र केन्द्रोपरिगतं कदम्बप्रोतवृत्तं क्रान्तिवृत्ते यत्र लगति ततो भूभा बिम्बाकेन्द्रं यावत् क्रान्तिवृत्ते स्थित्यर्धकला, चन्द्र केन्द्रो-परिगतकदम्बप्रोतवृत्तक्रान्तिवृत्तसम्पाताच्चन्द्र-केन्द्रं यावत् कदम्ब प्रोतवृत्ते स्पार्शिकशरः। तथा यदाऽन्तःस्पर्शो भवति तदा चन्द्र केन्द्रो-परिगतं कदम्बप्रोतवृत्तं क्रान्तिवृत्ते यत्र लगति ततो भूभाबिम्बकेन्द्रं यावत्क्रान्ति वृत्ते विमर्दार्ध कला, कदम्बप्रोतवृत्ते चन्द्रकेन्द्रात्क्रान्तिवृत्तं यावदन्तःस्पर्शकालिक (संमीलनकालिक) शरः। चन्द्रकेन्द्रभूभाकेन्द्रगतं वृत्तं कार्यं भू=भूभाबिम्बकेन्द्रम्। चं=बहिः स्पर्शकालिक (स्पार्शिक) चन्द्रबिम्ब केन्द्रम्। चरं=स्पार्शिकशरः। रभू=स्थित्यर्धकला, चं=अन्तःस्पर्शकालिक (संमीलन-कालिक) चन्द्रबिम्ब केन्द्रम्। चंश=संमीलन कालिकशरः। शभू=विमर्दार्धकला, न=स्पर्श बिन्दुः। भूचं=चन्द्रबिम्ब भूभाबिम्बकेन्द्र गतवृत्ते केन्द्रान्तरम्। चंन=चन्द्र बिम्बव्यासार्धम्। भून=भूभाव्यासार्धम्, भूचं=चन्द्रबिम्बभूभाबिम्बयोर्मानै-क्यार्धम्। भूचं=चन्द्रबिम्बभूभाबिम्बयोर्मानार्धान्तरम्। अत्राचार्येण भूरचं त्रिभुजं भूचंश त्रिभुजं च सरलाकारं मत्वा स्थित्यर्धविमर्दार्धे षट्शोरानयनं कृतं यथा

$\sqrt{\text{भूचं}^2-\text{चंर}^2}=\text{भूर}=\sqrt{\text{मानैक्यार्ध}^2-\text{स्पार्शिकशर}^2}=$स्थित्यर्धकला, ततोऽनुपातेना 'यदि रविचन्द्रयोर्गत्यन्तरेण षष्टि घटिका लभ्यन्ते तदा स्थित्यर्धकलायां किमित्य' नेन समागच्छन्ति स्थित्यर्ध घट्यः $=\frac{६०\times\text{स्थि}\frac{1}{2}\text{ कला}}{\text{गत्यन्तर कला}}$, तथा भूचंश त्रिभुजे $\sqrt{\text{भूचं}^2-\text{चंश}^2}=\text{भूश}=\sqrt{\text{मानार्धान्तर}^2-\text{संमीलनकालिकशर}^2}=$विमर्दार्धकला, ततः पूर्ववदनुपातेन विमर्दार्धघट्यः $=\frac{६०\times\text{विमर्दार्धकला}}{\text{गत्यन्तर कला}}$ परमत्र स्पार्शिक संमीलन कालिकशरयोरज्ञानाद्विदितमध्यग्रहणकालिकशरवशेनैव स्थित्यर्धविमर्दार्धयोरानयनं कृतं तन्न युक्तम्। सिद्धान्तशेखरे "मानार्धसंयोग वियोगवर्गौ विक्षेपकृत्या रहितौ विधाय। ये शेषमूले तिथिवत् कृते ते क्रमाद् भवेतां स्थितिमर्दखण्डे" श्रीपतेः श्लोकश्चायं, सिद्धान्तशिरोमणौ "मानार्धयोगान्तरयोः कृतिभ्यां शरस्य वर्गेण विवर्जिताभ्याम्। मूले खपट् ६० संगुणिते विभक्ते भुक्तयन्तरेण स्थितिमर्दखण्डे" भास्कराचार्यस्यायं श्लोकश्चाऽऽचार्योक्तानुरूप एवेति ॥ ७ ॥

अब ग्रासानयन को कहते हैं।

हि. भा.—ग्राह्य बिम्ब और ग्राहक बिम्ब के योगार्ध (मानैक्यार्ध) और अन्तरार्ध के वर्गों में से शर वर्ग को घटाकर मूल लेना तब उन दोनों को साठ से गुणा कर रवि और चन्द्र के गत्यन्तर से भाग देने से स्थित्यर्ध और विमर्दार्ध होते हैं इति ॥ ७ ॥

उपपत्ति

यहां संस्कृतोपपत्ति में लिखित (क) क्षेत्र को देखिये। भूभाबिम्ब और चन्द्र बिम्ब के बहिः स्पर्श काल में चन्द्रकेन्द्रोपरिगत कदम्बप्रोतवृत्त और क्रान्तिवृत्त के सम्पात से भूभाकेन्द्र पर्यन्त क्रान्तिवृत्त में स्थित्यर्ध कला है, चन्द्र केन्द्रोपरि गत कदम्ब प्रोत वृत्त और क्रान्तिवृत्त के सम्पात से चन्द्रकेन्द्र तक कदम्बप्रोतवृत्त में स्पार्शिक शर हैं। एवं भूभाबिम्ब और चन्द्रबिम्ब के अन्तःस्पर्श (संमीलन) काल में चन्द्र केन्द्रोपरिगत कदम्बप्रोतवृत्त और क्रान्तिवृत्त के सम्पात से भूभा बिम्ब केन्द्र पर्यन्त क्रान्ति वृत्त में विमर्दार्ध कला हैं, कदम्ब प्रोत वृत्त में चन्द्रकेन्द्र से क्रान्तिवृत्त पर्यन्त अन्तःस्पर्शकालिक (संमीलनकालिक) शर हैं, चन्द्रकेन्द्र और भूभाकेन्द्र गतवृत्त कर देना, भू=भूभा बिम्ब केन्द्र, चं=बहिःस्पर्शकालिक-(स्पार्शिक) चन्द्रकेन्द्र, चंर=स्पार्शिकशर, रभू=स्थित्यर्धकला, चं=अन्तःस्पर्शकालिक (संमीलन कालिक) चन्द्रबिम्ब केन्द्र, चंश=संमीलन कालिकशर, शभू=विमर्दार्धकला, न=स्पर्श बिन्दु, भूचं=चन्द्रबिम्ब और भूभा बिम्ब के केन्द्रगत वृत्त में केन्द्रान्तर, चंन=चन्द्रबिम्ब

व्यासार्ध, भून = भूभाविम्ब व्यासार्ध, भूचं = चन्द्रविम्ब और भूभाविम्ब के सानैक्यार्ध, भूचं = चन्द्रविम्ब और भूभाविम्ब के मानान्तरार्ध, यहां आचार्य ने भूरचं त्रिभुज को तथा भूचंग त्रिभुज को सरलाकार मान कर स्थित्यर्ध और विमर्दार्ध का आनयन किया है जैसे भूरच जात्य त्रिभुज में $\sqrt{\text{भूचं}^2 - \text{चंर}^2} = \text{भूर} = \sqrt{\text{मानैक्यार्ध}^2 - \text{स्पार्शिकशर}^2}$ = स्थित्यर्धक, तब अनुपात करते हैं यदि रवि और चन्द्र की गत्यन्तरकला में साठ घटी पाते हैं तो स्थित्यर्धकला में क्या इससे स्थित्यर्ध घटी आती हैं, $\frac{६० \times \text{स्थि}\frac{1}{2} \text{ कला}}{\text{गत्यन्तर कला}} = \text{स्थि}\frac{1}{2}$ घटी, तथा भूचंग जात्य त्रिभुज में $\sqrt{\text{भूचं}^2 - \text{चंश}^2} = \text{भूश} = \sqrt{\text{मानान्तरार्ध}^2 - \text{संमीलन कालिकशर}^2}$ = विमर्दार्धकला, इससे पूर्ववत् अनुपात से विमर्दार्ध घटी = $\frac{६० \times \text{विमर्दार्ध कला}}{\text{गत्यन्तर कला}}$ लेकिन यहां स्पार्शिक शर और संमीलन कालिकशर विदित नहीं हैं, विदित है मध्यग्रहण कालिकशर, इसलिये मध्यग्रहण कालिकशर तुल्य ही स्पार्शिक शर और संमीलन कालिकशर मानकर पूर्वोक्तरीति से स्थित्यर्ध और विमर्दार्ध का आनयन किया गया हैं जो ठीक नहीं हैं, सिद्धान्त शेखर में "मानार्ध-संयोगवियोगवर्गौ" इत्यादि संस्कृतोपपत्ति में लिखित श्रीपति का श्लोक तथा सिद्धान्त शिरोमणि में "मानार्धयोगान्तरयोः कृतिभ्याम्" इत्यादि भास्कराचार्य का श्लोक आचार्योक्त के अनुरूप ही हैं इति ॥ ७ ॥

इदानीं स्थिति विमर्दार्धयोः स्फुटीकरणमाह ।

षष्ट्या विभाजिता स्थितिविमर्ददलनाडिकागुणा स्वगतिः ।
आदौ रवीन्दुपातेष्वृणमसकृत् तेषु धनमन्ते ॥ ९ ॥

सु० भा०—स्पष्टार्थम् । अत्रोपपत्तिः । 'स्थित्यर्धनाडीगुणिता स्वभुक्ति' रित्यादिना 'एवं विमर्दार्धफलोनयुक्ते' त्यादिना च भास्करविधिना स्फुटा ॥ ९ ॥

वि. भा.—स्वगतिः (रविगतिः, चन्द्रगतिः, पातगतिश्च) स्थितिविमर्ददलनाडिका गुणा (स्थित्यर्धघटीभिर्विमर्दार्धघटीभिश्च पृथक् पृथक् गुणिता) षष्ट्या ६० भक्ताऽऽदौ रविचन्द्रपातेषु ऋणम्, अन्ते तेषु धनं कार्यमर्थादाद्यस्थित्यर्धे धनमन्त्यस्थित्यर्धे ऋणं तथाऽऽद्यविमर्दार्धे धनमन्त्यविमर्धे ऋणम्, एवमसकृद्वारं वारं पूर्वोक्त- कर्मणि कृते स्थित्यर्धविमर्दार्धे स्फुटे भवत इति ॥ ९ ॥

अत्रोपपत्तिः ।

यदि षष्टिघटीभिश्चन्द्रगतिकला लभ्यन्ते तदा स्थित्यर्धघटीभिः किमि-

त्यनुपातेन स्थित्यर्धघटीसम्बन्धिनी चन्द्रगतिकला समागच्छति, पातस्याप्येवं स्थित्यर्धघटीसम्बन्धिनी गतिरानेया, आनीतफलाभ्यां क्रमेण हीनयुतौ चन्द्रपातौ पूर्वापेक्षया तौ विशिष्टौ भवतस्ततस्त्रिघन गुणाव्यासार्धभाजिता चन्द्रपातयोगज्ये-त्यादिना चन्द्रशरमानीय "छाद्येन युतोनस्य छादकमानस्ये" त्यादिना स्थिति-विमर्दार्धे आनेतव्ये ततः "षष्ट्या विभाजिता स्थितिविमर्ददलनाडिका गुणा स्वगतिं" रित्यनेन स्थित्यर्धघटीसम्बन्धिनीं चन्द्रगतिकलां पातगतिं चानीय तद्धीनयुतौ चन्द्रपातौ कार्यौ तौ च पूर्वापेक्षया विशिष्टौ भवत एवमसकृत्कर्मणा स्फुटं स्थित्यर्धं भवेत्। एवमेवाऽसकृत्कर्मणा स्फुटं विमर्दार्धं भवतीति, सिद्धान्त-शेखरे श्रीपतिने "स्थिति विमर्ददलारव्यघटीहृता दिनकरेन्दु तमोमयभुक्तयः। गगनषट्क ६० हृताः प्रथमान्त्ययोः क्षयधने भवतस्त्वसकृत्ततः" त्यनेन, सिद्धान्त-शिरोमणौ भास्कराचार्येण "स्थित्यर्धनाडी गुणिता स्वभुक्तिः षष्ट्याहृता तद्रहितौ युतौ च। कृत्वेन्दु पातावसकृच्छराभ्यां स्थित्यर्धमाद्यं स्फुटमन्तिमं च॥ एवं विमर्दार्धफलोनयुक्तसपातचन्द्रोद्भवसायकाभ्याम्। पृथक् पृथक् पूर्ववदेव सिद्धे स्फुटे स्त आद्यान्त विमर्दखण्डे" त्यनेन चाऽऽचार्योक्तानुरूपमेवोक्तमिति ॥९॥

अब स्थित्यर्ध और विमर्दार्ध के स्फुटीकरण को कहते हैं।

हि. भा.—अपनी गति (रविगति, चन्द्रगति, पातगति) को स्थित्यर्ध घटी और विमर्दार्ध घटी से पृथक् पृथक् गुणा कर साठ से भाग देने से जो फल हो उनको रवि, चन्द्र और पात में से ऋण करना, अन्त में धन करना अर्थात् आद्य स्थित्यर्ध में धन और अन्त्य स्थित्यर्ध में ऋण, तथा आद्य विमर्दार्ध में धन और अन्त्य विमर्दार्ध में ऋण करना चाहिये, इसतरह बार बार पूर्वोक्त कर्म करने से स्थित्यर्ध और विमर्दार्ध स्फुट होते हैं इति॥ ९॥

उपपत्ति।

यदि साठ घटी में चन्द्रगति कला पाते हैं तो स्थित्यर्ध घटी में क्या इस अनुपात से स्थित्यर्ध घटी सम्बन्धिनी चन्द्रगति कला आती है, पात की भी स्थित्यर्ध घटी सम्बन्धिनी गति इसी प्रकार लाना, इन लाये हुये फलों को क्रम से चन्द्र और पात में से हीन और युत करने से पूर्व की अपेक्षा विशिष्ट चन्द्र और पात होते हैं, इनसे "त्रिघनगुणा व्यासार्ध-भाजिता चन्द्रपातयोगज्या" इत्यादि से चन्द्रशर लाकर "छाद्येन युतोनस्य छादकमानस्य' इत्यादि से स्थित्यर्ध और विमर्दार्ध लाना चाहिये, तब "षष्ट्या विभाजिता स्थितिविमर्ददल-नाडिका गुणा स्वगतिः" इससे स्थित्यर्ध घटी सम्बन्धिनी चन्द्रगति कला और पातगति ले-आकर चन्द्र और पात में हीन और युत करने से पूर्व की अपेक्षा विशिष्ट चन्द्र और पात होते हैं इस तरह असकृत्कर्म से स्फुट स्थित्यर्ध होता हैं, इसी तरह असकृत्कर्म से स्फुट

विमर्दार्ध होता है, सिद्धान्त शेखर में श्रीपति स्थिति विमर्दार्धकर्मणी हृत्वा इत्यादि सप्तसोपपत्ति में लिखित पद्य से श्रीपति, तथा सिद्धान्तशिरोमणि में भास्कराचार्य स्वभुक्तिः षष्टया हृता नाडिकौ युतौ वा इत्यादि पद्यों से भास्कराचार्य भी आर्यभटोक्त के अनुरूप ही कहते हैं इति ॥ ९ ॥

इदानीं निमीलनोन्मीलनकालानयनमाह ।

स्पर्शान्निमीलनं स्थितिदले विमर्दार्धहीनिते पश्चात् ।
मोक्षादर्वागुन्मीलनं विमर्दस्तदैक्यार्धः ॥ १० ॥

सु० भा०—स्थितिदले स्पर्शस्थित्यर्धे विमर्दार्धहीनिते काले स्पर्शात् पश्चात् निमीलनं भवति । एवं मोक्षस्थित्यर्धे उन्मीलनविमर्दार्धहीनिते काले मोक्षादर्वाक् पूर्वमुन्मीलनं भवति । तयोर्निमीलनोन्मीलनयोः कालयोर्योगार्धो विमर्दः । अर्थाद्योगार्ध कालपर्यन्तमेव छादकबिम्बे छाद्यबिम्बस्य निमज्जनं भवति ।

अत्रोपपत्तिः । स्थित्यर्धं विमर्दार्धं परिभाषातः स्फुटा ॥ १० ॥

वि. भा.—स्थितिदले (स्पर्शस्थित्यर्धे) विमर्दार्धहीनिते काले स्पर्शात् पश्चात् निमीलनं भवति । मोक्षस्थित्यर्धे उन्मीलनविमर्दार्धहीनिते काले मोक्षात् अर्वाक् (पूर्वं) उन्मीलनं भवति । तदैक्यार्धः (तयोर्निमीलनोन्मीलनयोः कालयोर्योगार्धः) विमर्दोऽर्थाद्योगार्धकालपर्यन्तमेव छादकबिम्बे छाद्यबिम्बस्य निमज्जनं भवतीति ॥१०॥

अत्रोपपत्तिः ।

स्थित्यर्धविमर्दार्धपरिभाषातः स्फुटेति ॥ १० ॥

अब निमीलन और उन्मीलन कालसाधन को कहते हैं ।

हि. भा.—विमर्दार्ध रहित स्पर्शस्थित्यर्ध-काल में स्पर्श से पीछे निमीलन होता है, मोक्ष स्थित्यर्ध में उन्मीलन विमर्दार्ध घटाने से मोक्ष से पहले उन्मीलन होता है, निमीलन-काल और उन्मीलन-काल का योगार्ध विमर्द होता है अर्थात् योगार्ध काल ही तक छादक-बिम्ब में छाद्य-बिम्ब का निमज्जन (अन्तः स्थिति) होता है इति ॥१० ॥

उपपत्ति—स्थित्यर्ध और विमर्दार्ध की परिभाषा से स्पष्ट है ॥ १० ॥

इदानीमिष्टग्रासानयनमाह ।

भुक्त्यन्तरमिष्टोनस्थितिदलघटिकागुणं हृतं षष्ट्या ।
बाहुः प्राग्वत् तत्फलहीनयुतैः सूर्यशशिपातैः ॥ ११ ॥

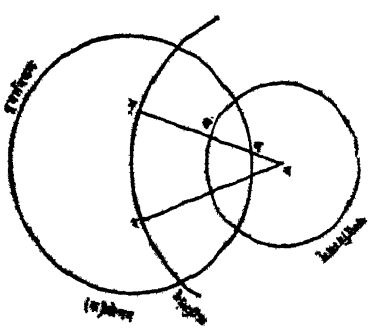
तात्कालिकविक्षेपः कोटिस्तद्वर्गयुतिपदं कर्णः ।
मानैक्यार्धात् कर्णं विशोध्य तात्कालिको ग्रासः ॥ १२ ॥

सु० भा०—इष्टं स्पार्शिकादिकं तत् सजातीयं स्थितिदलं च गृहीत्वान्तरं कर्त्तव्यम् । शेषं स्पष्टार्थम् ।

अत्रोपपत्त्यर्थं भास्करभुजानयनं द्रष्टव्यम् ॥ ११-१२ ॥

वि. भा.—भुक्त्यन्तरं (रविचन्द्रयोर्गत्यन्तरं) इष्टोनस्थितिदलघटिकागुणं (इष्टरहितस्थित्यर्धघष्ट्या गुणितं) षष्ट्या हृतं (षष्टिभक्तं) तदा बाहुः (भुजः) भवत्यर्थात् स्पर्शादनन्तरं यावतीष्विष्टघटिकासु ग्रासज्ञानमपेक्षितं तावतीभिर्घटीभी रहितेन स्थित्यर्धेन गुणितं रविचन्द्रयोर्गत्यन्तरं षष्टिभक्तं तदा भुजो भवतीति, ततः फलेनानेन सूर्यचन्द्रपातान् प्रचाल्य तात्कालिकश्चन्द्रशरः साध्यः सचकोटिः, तयोः (भुजकोट्योः) वर्गयोगमूलं कर्णः स्यात् । मानैक्यार्धात् कर्णं विशोध्य शेषस्तात्कालिको ग्रासः (इष्टग्रासो) भवतीति ॥ ११-१२ ॥

अत्रोपपत्तिः



स्पर्शानन्तरं यावतीभिर्घटिकाभिर्ग्रासज्ञानमपेक्षितं तावतीभिर्घटीभी रहितं स्थित्यर्धं कार्यं शेषेणाऽनुपातो यदि षष्टिघटीभी रविचन्द्रयोर्गत्यन्तरकला लभ्यन्ते तदेष्टोनस्थित्यर्धघटीभिः किमित्यनुपातेन यत्फलं स भुजो भवति, अनुपातागत फलेन सूर्यचन्द्रपातान् प्रचाल्य तात्कालिकश्चन्द्रशरः साध्यः सा कोटिः, तयोर्वर्गयोगमूलं कर्णः स्यात् । मानैक्यार्धात् कर्णस्य संशोधनेष्टग्रासो भवेत् । चं=चन्द्रकेन्द्रम् । भू=भूभाकेन्द्रम् । भचं=भूभाचन्द्र केन्द्रयोरन्तरम्=कर्णः । चंर= चन्द्रकेन्द्रोपरिगत कदम्ब प्रोतवृत्ते

चन्द्रशरः=कोटिः। भर=भुजः, एतच्चापीयजात्यत्रिभुजं सरलाकारकं स्वीकृत-माचार्येण, तदा $\sqrt{\text{कोटि}^2+\text{भुज}^2}$ =कर्ण, पश=इष्टग्रासः। भूप+चंश=भूभाव्या$\frac{1}{2}$+चंव्या$\frac{1}{2}$=मानैक्यार्ध=भूप+पचं+पश = भूचं+इष्टग्रास=कर्ण+ इष्ट-ग्रास=मानैक्यार्ध, ∴मानैक्यार्ध—कर्ण=इष्टग्रासः।

सिद्धान्त शेखरे "इष्टन्यूनस्थितिदलगुणा भुक्तिविश्लेषलिप्ता षष्ट्याभक्ता भवति हि भुजः कोटिरिष्टेन्दुबाणः। तद्वर्गैक्योद्भवमपि पदं कर्ण एतेन हीनं मानैक्यार्धं स्फुटमिह भवेद्ग्राञ्छितं छन्नमानम्" ऽनेन श्रीपतिना, सिद्धान्त शिरोमणौ "कोटिश्च तत्काल शरोऽथकोटीदोर्वर्गयोगस्य पदं श्रुतिः स्यात्। मानैक्यखण्डं श्रुतिवर्जितं सद्ग्रासप्रमाणं भवतोष्टकाले" श्लोकेनानेन भास्कराचार्येणाप्याचार्यो-क्तमेवोक्तमिति ॥ ११-१२ ॥

अब इष्टग्रासानयन को कहते हैं।

हि. भा.—रवि और चन्द्र के गत्यन्तर को इष्ट रहित स्थित्यर्ध घटी से गुणा कर साठ से भाग देने से भुज होता है, इस फल करके सूर्य, चन्द्र और पात को चालन देकर तात्कालिक चन्द्रशर साधन करना वह कोटि हैं उन दोनों (भुज और कोटि) का वर्गयोग मूल कर्ण होता हैं, मानैक्यार्ध (छाद्य और छादक बिम्ब के व्यासार्ध योग) में से कर्ण को घटाने से जो शेष रहता हैं वह इष्टग्रास होता है इति ॥११-१२ ॥

उपपत्ति।

स्पर्श के बाद जितनी इष्ट घटी में ग्रास-ज्ञान अपेक्षित हो उस इष्ट घटी को स्थित्यर्ध में से घटा देना, शेष से अनुपात करते हैं, यदि साठ घटी में रवि और चन्द्र की गत्य-न्तर कला पाते हैं तो इष्टोन स्थित्यर्ध घटी में क्या इससे जो फल होता हैं वह भुज है, अनुपातागत फल से चालित रवि, चन्द्र और पात से तात्कालिक चन्द्रशर साधन करना वह कोटि हैं इन दोनों (भुज और कोटि) का वर्गयोग मूल कर्ण होता है, मानैक्यार्ध में से कर्ण को घटाने से इष्ट ग्रास होता हैं, जैसे संस्कृतोपपत्ति में लिखित (क)क्षेत्र को देखिये, चं=चन्द्र-केन्द्र, भू=भूभाकेन्द्र, भूचं=भूभा और चन्द्र के केन्द्रान्तर=कर्ण चंर=चन्द्रकेन्द्रोपरिगत कदम्बप्रोतवृत्त में चन्द्रशर=कोटि, भूर=भुज, भूरचं चापीय जात्य त्रिभुज को सरल-जात्य त्रिभुज स्वीकार कर $\sqrt{\text{भुज}^2+\text{कोटि}^2}$=कर्ण, पश=इष्टग्रास, भूप+चंश=भूभाव्या$\frac{1}{2}$ +चंव्या$\frac{1}{2}$=मानैक्यार्ध=भूप+पचं+पश=भूचं+इष्टग्रास=कर्ण+इग्रा। अतः मानै-क्यार्ध—कर्ण=इष्टग्रास, इससे आचार्योक्त उपपन्न हुआ।

सिद्धान्त शेखर में "इष्ट न्यून स्थिति दलगुणा भुक्तिविश्लेषभक्ताः" इत्यादि से श्रीपति तथा "कोटिश्च तत्कालशरोऽथ कोटी" इत्यादि से सिद्धान्त-शिरोमणि में भास्करा-चार्य ने भी आचार्योक्त के अनुसार ही कहा है, परन्तु यह आनयन ठीक नहीं हैं उपपत्ति देखने ही से स्पष्ट है इति ॥ ११-१२ ॥

इदानीमिष्टग्रासात्कालानयनमाह –

ग्रसकृद्ग्रासकलोनप्रमाणयुतिदलकृतेर्विशोध्य कृतिम् ।
तात्कालिकविक्षेपस्य शेषमूलं कृतं तिथिवत् ॥ १३ ॥

ग्रग्रहणस्थित्यर्धात् प्रोह्य ग्रग्रहणतो भवेत् कालः ।
मौक्षं विशोध्य मोक्षस्थित्यर्धात् प्राग् भवेन्मोक्षात् ॥ १४ ॥

सु० भा०—ग्रासकलोनप्रमाणयुतिदलकृतेर्ग्रासोनमानैक्यार्धवर्गात् तात्कालिकशरस्य कृतिं विशोध्य शेषशूलं तिथिवदसकृत् कृतम् । अर्थात् शेषमूलं षष्ट्या गुणं रविचन्द्रगत्यन्तरहृतं फलकालेन रविचन्द्र पातान् प्रचाल्य तात्कालिकविक्षेपं प्रसाध्य तस्मात् पुनर्ग्रासकलोनप्रमाणेत्यादिनाऽसकृद्यः शेषमूलकालः स्थिरीभवति तं ग्रग्रहणस्थित्यर्धात् स्पार्शिकस्थित्यर्धात् प्रोह्य हित्वा ग्रहणतः स्पर्शादनन्तरं कालो भवेदर्धात् स्पर्शानन्तरमेतावतीष्टकाले तावानिष्टग्रासो भवति । एवं तमेव कालं मौक्षं मोक्षसम्बन्धिनं मोक्षस्थित्यर्धाद्विशोध्य शेषं मोक्षात् प्रागेवेष्टकालो भवेत् । अर्थात् मोक्षात् प्राक् तावतीष्टकाले तावानेवेष्टग्रासो भवेत् । एवमिष्टग्रासाद्द्विधेष्टकाल उत्पद्यते । एकः स्पर्शानन्तरमन्यो मोक्षात् प्रागिति ।

ग्रत्रोपपत्तिः । तत्कालशरस्याज्ञानान्मध्यकालिकशरात् कर्म कृतमतोऽसकृद्विधिना स्फुटकालसाधनमुचितम् । शेषोपपत्ति 'ग्रासो न मानैक्यदलस्य वर्गाद् विक्षेपकृत्या रहिता' दित्यादिभास्करविधिना स्फुटा ॥ १३-१४ ॥

वि. भा.—ग्रासकलोनप्रमाणयुतिदलकृतेः (ग्रासरहितमानैक्यार्धवर्गात्) तात्कालिकशरस्य कृतिं (वर्गं) विशोध्य (हित्वा) शेषस्य मूलं तिथिवदसकृत् कृतमर्थात् शेषमूलं षष्टिगुणं रविचन्द्रयोर्गत्यन्तरभक्तं फलकालेन रविचन्द्रपातान् प्रचाल्य तात्कालिकचन्द्रशरं संसाध्य तस्मात् पुनर्ग्रासकलोनप्रमाणयुतिदलकृतेरित्यादिनाऽसकृद्य शेषमूलं कालः स्थिरीभवति तं ग्रग्रहणस्थित्यर्धात् (स्पार्शिकस्थित्यर्धात्) प्रोह्य (हित्वा) ग्रग्रहणतः (स्पर्शादनन्तरं) कालो भवेदर्धात्स्पर्शानन्तरमेतावतीष्टकाले तावानिष्टग्रासो भवति, एवं तमेव कालं मौक्षं (मोक्षसम्बन्धिनं मोक्षस्थित्यर्धात् विशोध्य शेषं मोक्षात् प्राक् (पूर्वं) इष्टकालो भवेदर्धात् मोक्षात् प्राक् तावतीष्ट काले तावानेवेष्टग्रासो भवेत् । एवमिष्टग्रासाद् द्विधेष्टकाल उत्पद्यते, एकः स्पर्शानन्तरमन्यो मोक्षात्पूर्वमिति ॥ १३-१४ ॥

अत्रोपपत्तिः

तत्तत्कालिकशराज्ञानान्मध्यग्रहणकालिकशरादेवेष्टकालज्ञानं कृतमतोऽसकृद्विधिना स्फुटस्येष्टकालस्य साधनं कर्त्तव्यमेव। पूर्वमिष्टग्रासानयने 'मानैक्यार्ध—कर्ण=इष्टग्रासः' सिद्धोऽतो मानैक्यार्ध—इष्टग्रास=कर्णः, एतद्वर्गे तात्कालिकचन्द्रशरस्य वर्गशोधनेन यच्छेषं तन्मूलं षष्ट्या गुणितं रविचन्द्रयोर्गत्यन्तरेण भक्तं लब्धं स्पार्शिकस्थित्यर्धात् मौक्षिकस्थित्यर्धाद्वा शोधयेत्तदा गृहीतेष्टग्रासस्य कालो भवेदेवमसकृच्छरवर्गहीनान्मूलं षष्टिगुणितं रविचन्द्रगत्यन्तरभक्तं फलकालेन रविचन्द्रपातान् प्रचाल्य तात्कालिकचन्द्रशरं संसाध्य ततः पुनः 'ग्रासकलोनप्रमाणयुतिदलकृते' रित्यादिना वारं वारं यः कालः स्थिरो भवेत्तं स्पार्शिकस्थित्यर्धाच्छोधयेत्तदा स्पर्शानन्तरं गतः कालो भवेत्। तमेव कालं मौक्षिकस्थित्यर्धाद्विशोध्य शेषं मोक्षात्पूर्वमिष्ट कालो भवतीति। सूर्य सिद्धान्तेऽप्येवेष्ट ग्रासात्कालानयनमस्ति, सिद्धान्तशिरोमणौ "ग्रासोनमानैक्यदलस्य वर्गाद्विक्षेपकृत्या रहितात्पदं यत्। गत्यन्तरांशैर्विहृतं फलोनं स्थित्यर्धकं स्वं भवतीष्टकालः" इत्याचार्योक्तानुरूपमेवेतिज्ञेयं विज्ञैरिति ॥ १३-१४ ॥

अब इष्ट ग्रास से कालसाधन को कहते हैं।

हि. भा.—ग्रास रहित मानैक्यार्ध वर्ग में से तात्कालिक चन्द्रशरवर्ग को घटा कर शेष का मूल लेकर उससे तिथिवत् असकृत्कर्म करना चाहिये अर्थात् शेष मूल को साठ से गुणा कर रवि और चन्द्र के गत्यन्तर से भाग देने से जो फल काल हो उनसे रवि, चन्द्र और पात को चला कर तात्कालिक चन्द्रशर साधन कर उससे फिर 'ग्रासकलोनप्रमाणयुतिदलकृतेः' इत्यादि से असकृत् करने से जो काल स्थिरीभूत हो उसको स्पार्शिक स्थित्यर्ध में से घटाने से स्पर्श के बाद काल होता हैं अर्थात् स्पर्श के अनन्तर इतने काल में इतने इष्टग्रास होते हैं। इसीतरह मोक्ष सम्बन्धी काल को मोक्षस्थित्यर्ध में से घटाने से शेष मोक्ष से पूर्व इष्टकाल होता हैं अर्थात् मोक्ष से पहले इतने काल में इतने इष्ट ग्रास होते हैं इस तरह इष्ट ग्रास से दो तरह का इष्टकाल होता हैं, एक स्पर्श काल के अनन्तर और दूसरा मोक्ष से पूर्व इति ॥ १३-१४ ॥

उपपत्ति।

स्पर्शादि कालिक शर विदित न रहने के कारण मध्यग्रहण कालिक शर ही से इष्ट कालानयन किया गया हैं जो कि ठीक नहीं हैं, इसलिये असकृत्प्रकार से स्फुट इष्ट कालानयन करना उचित ही हैं, पहले इष्ट ग्रासानयन में मानैक्यार्ध—इष्टग्रास=कर्ण इसके वर्ग

में से तात्कालिक चन्द्रशर वर्ग को घटा कर जो शेष रहता हैं उसके मूल को साठ से गुणा कर रवि और चन्द्र के गत्यन्तर से भाग देने से जो लब्ध हो उसको स्पार्शिक स्थित्यर्ध में से वा मौक्षिक स्थित्यर्ध मे से घटा देना तब गृहीत इष्ट ग्रास सम्बन्धी काल होता हैं, इसतरह असकृत् (बार बार) कर्ण वर्ग में से तात्कालिक चन्द्रशर वर्ग को घटाकर शेष के मूल को साठ से गुणा कर रवि और चन्द्र के गत्यन्तर से भाग देने से जो फल काल हो उससे रवि, चन्द्र और पात को चला कर तात्कालिक चन्द्रशर साधन करना पुनः उससे "ग्रासकलोनप्रमाणयुतिदल-कृतेः" इत्यादि से बार बार जो काल स्थिर हो उसको स्पार्शिक स्थित्यर्ध में से घटा देना तब स्पर्श के बाद गत काल होता हैं, उसी काल को मौक्षिक स्थित्यर्ध में से घटाने से शेष मोक्ष से पूर्व इष्टकाल होता हैं, सूर्य सिद्धान्त में भी इसी तरह इष्ट ग्रास से कालानयन है, सिद्धान्त-शिरोमणि में "ग्रासोनमानैक्य दलस्य वर्गाद्विक्षेप कृत्या" इत्यादि संस्कृतोपपत्ति में लिखित पद्यों से भास्कराचार्य ने भी आचार्योक्तानुरूप ही कहा है इति ॥ १३-१४ ॥

इदानीं स्पर्शादिव्यवस्थामाह

स्फुटतिथ्यन्ते मध्यं ग्रग्रहणं स्थितिदलोनकेऽभ्यधिके ।
मोक्षो निमीलनोन्मीलने विमर्दार्धहीनयुते ॥ १५ ॥

सु० भा०— स्पष्टार्थमुपपत्तिश्च स्फुटा 'मध्यग्रहः पर्वविरामकाले' इति भास्क-रोक्तमेतदनुरूपमेव ॥ १५ ॥

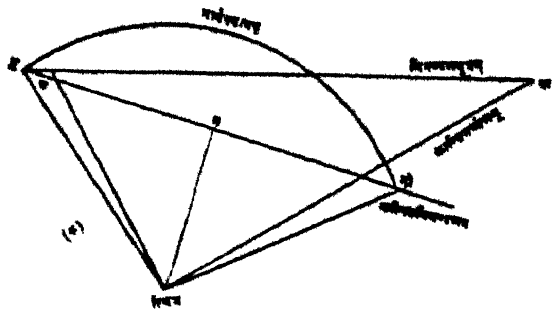
वि. भा.— स्फुटतिथ्यन्ते (स्फुट पूर्णान्तिकाले अमान्ते वा) मध्यग्रहणं भवति मध्यग्रहणतः पूर्वं स्थित्यर्धकाले ग्रग्रहणं (स्पर्शः)भवति, मध्यग्रहणानन्तरं स्थित्यर्ध-काले मोक्षो भवति। मध्यग्रहणतः पूर्वं विमर्दार्ध काले निमीलनं (सर्वग्रासः) मध्यग्रहणात्परं विमर्दार्धकाले उन्मीलनं (सर्वग्रासावसानं) भवति, रविचन्द्रयो-रुभयोरपि ग्रहणं पञ्च प्रकारं भवतीति ॥ १५ ॥

अत्रोपपत्तिः

पा=पातः। स्थि=पूर्णान्तिकाले भूभा=स्थिर भूभा, चं=पूर्णान्तिकाले चन्द्रः= स्थिरचन्द्रः। स्थिचं=पूर्णान्तिकाले शरः पूर्णान्ति कालिकचन्द्रविमण्डलीयभुजांशक्रांति-वृत्तीय भुजांशशरैरुत्पन्नं चापीयजात्यत्रिभुजं सरलाकारकंमत्वा स्थिर भूभा स्थिरचन्द्र-

वशेनैकस्य कल्पितविमण्डलसंज्ञकस्य रचना मया वटेश्वरसिद्धान्ते प्रदर्शिताऽस्ति, तद्रचनाप्रकारस्तत एव बोध्यः।

तद्रचनाक्रमदर्शनेनेत्थमपि सिद्धमस्ति यल्ललितचन्द्रभूभयोर्यदन्तरं तदेव स्थिरभूभाकल्पितचन्द्रयोरन्तरं भवति, स्थिरभूभातः कल्पितविमण्डलोपरि-लम्बः स्थिम, म बिन्दावेव स्थिरभूभा—मबिन्दुस्थकल्पितचन्द्रयोरन्तरस्य (कयोरपि ललितभूभाचन्द्रयोरन्तरतुल्यस्य) परमालान्तरत्वान्मध्यग्रहणं भवितु-महंति, परमयं 'म' बिन्दुः पूर्णान्तादन्यत्रास्त्यतः पूर्णान्ते मध्यग्रहणं 'मध्यग्रहः



पर्वविरामकाले इत्यनेन' भास्करेण श्रीपतिनाऽऽचार्येण च यत्कथितं तन्न युक्तम्। स्थिरभूभाकेन्द्रं केन्द्रं मत्वा मानैक्यार्धेन यद्वृत्तं तत्पूर्णान्ताभिमुखं कल्पितविमण्डले यत्र लगति तत्र स्पर्शस्तद्विरुद्धदिशि यत्र लगति तत्र मोक्षः। मध्यग्रहणपूर्णान्तकाल-योरन्तरानयनमपि वटेश्वरसिद्धान्ते प्रदर्शितमस्ति मया तदपि तत एवावगन्तव्यं येन सर्वं स्फुटमिति ॥१५॥

अब स्पर्शादि व्यवस्था को कहते हैं

हि. भा.—स्फुट पूर्णान्तकाल में वा अनन्तकाल में (चन्द्र और सूर्य के) मध्य-ग्रहण होता है, मध्यग्रहण से पूर्व स्थित्यर्ध काल में स्पर्श होता है, मध्यग्रहणानन्तर स्थित्यर्धकाल में मोक्ष होता है, मध्यग्रहण से पूर्व विमर्दार्धकाल में निमीलन (सर्वग्रास) होता है, मध्यग्रहण के बाद विमर्दार्धकाल में उन्मीलन (सर्वग्रासान्त) होता है, चन्द्रग्रहण और सूर्यग्रहण पाँच प्रकार के होते हैं इति ॥१५॥

उपपत्ति

संस्कृतोपपत्ति में लिखित (क) क्षेत्र को देखिये। पा=पात, स्पि=पूर्णान्तकालिक-भूभाकेन्द्र=स्थिरभूभा, चं=पूर्णान्तकालिकचन्द्र=स्थिरचन्द्र, स्पिचं=पूर्णान्तकालिकस्थिर,

पूर्णान्तकालिकचन्द्रविमण्डलीयभुजांश—क्रान्तिवृत्तीयभुजांश और शर इनसे उत्पन्न चापीयजात्य त्रिभुज को सरलजात्य त्रिभुज मानकर स्थिरभूभा और स्थिरचन्द्र वश से एक कल्पित विमण्डल नाम के चन्द्रमार्ग की रचना मैंने वटेश्वरसिद्धान्त में दिखलायी है, उसका रचना प्रकार उसी से समझना चाहिये। उस (कल्पित विमण्डल) के रचना क्रम को देखने से यह भी सिद्ध होता है कि चलित भूभा और चलित चन्द्र का अन्तर स्थिरभूभा और कल्पित चन्द्र के अन्तर के बराबर होता है, स्थिरभूभा से कल्पित विमण्डल के ऊपर लम्ब= स्थिम, म बिन्दु में ही स्थिरभूभा और म बिन्दुस्थ कल्पित चन्द्र का अन्तर किसी भी चलित भूभा और चलित चन्द्र के अन्तर के बराबर होगा परन्तु यह परमाल्पान्तर है इसलिये इसी (म) बिन्दु में मध्यग्रहण होना उचित है, लेकिन यह (म) बिन्दु पूर्णान्त से अन्यत्र है इसलिये पूर्णान्त में मध्यग्रहण 'मध्यग्रहः पर्वविरामकाले' इससे भास्कराचार्य, श्रीपति तथा आचार्योक्त पूर्णान्तकाल में मध्यग्रहण का होना ठीक नहीं है, स्थिरभूभा केन्द्र को केन्द्र मानकर मानैक्यार्ध व्यासार्ध से जो वृत्त होता है वह पूर्णान्ताभिमुख कल्पित विमण्डल में जहाँ लगता है वहाँ स्पर्श तथा विरुद्ध दिशा में कल्पित विमण्डल में जहाँ लगता है वहाँ मोक्ष होता है, वटेश्वरसिद्धान्त में 'मध्यग्रहण और पूर्णान्त काल का अन्तरानयन भी' मैं (१) ने दिखलाया है वह उसी से समझना चाहिए, शेष सब बातें स्पष्ट हैं इति ॥२५॥

इदानीमक्षजवलनसाधनमाह

प्राक्पश्चान्नतविषुवज्ज्ययोर्वधात् त्रिज्ययाप्तचापं यत् ।
उत्तरयाम्ये पूर्वा विषुवद्वृत्तात् त्रिभे ग्राह्या ॥१६॥

सु. भा.—प्राक् प्राक्कपाले पश्चात् पश्चिमकपाले यो नतः सममण्डलीय-नतभागारतेषां या ज्या। या च विषुवज्ज्याऽक्षज्या तयोर्वधात् त्रिज्यया यदाप्तं तस्य चापांशैस्त्रिभे ग्रहाद्राशित्रयान्तरे उत्तरयाम्ये प्राक्पश्चान्नतक्रमेण ग्राह्या ग्राह्यवृत्तीया विषुवद्वृत्तस्य पूर्वा भवति सममण्डलादित्यग्रेण सहान्वयः।

अत्रोपपत्तिः। 'तेषां क्रमज्या पलशिञ्जिनीभक्ता द्युमौर्व्या' इत्यादिभास्कर-विधिना स्फुटा। आचार्येण द्युज्यास्थाने त्रिज्या स्थूला गृहीता। नतशब्देन यद्य-होरात्रे नतकालो गृह्यते तदा वलनवासनयाऽत्यन्तं स्थूलमाचार्योक्तमक्षवलनं भवेदिति चापीयत्रिकोणमित्या वलनानयनेन स्फुटम् ॥१६॥

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(१) वटेश्वरसिद्धान्त के टीकाकार (पण्डित मुकुन्दमिश्र; ज्योतिषविभागाध्यक्ष (गवर्नमेण्ट संस्कृत कालेज मुजफ्फरपुर) ग्राम के संकेत—ग्राम देपुरा, पो. ओ. (बेनीपट्टी) जिला—दरभङ्गा, बिहार।

वि. भा.—प्राक् (पूर्वकपाले) पश्चात् (पश्चिमकपाले) यो नतः (सममण्डलीयनतांशः) तज्ज्या, या च विषुवज्ज्या (अक्षज्या) तयोर्वधात् त्रिज्यया-भक्तात् यदाप्तं (यल्लब्धं) तच्चापांशैस्त्रिभे (ग्रहाद्राशित्रयान्तरे) उत्तरगाम्ये प्राक् पश्चान्नतक्रमेण ग्रहाद् (ग्राह्यवृत्तीया) विषुवद् वृत्तस्य पूर्वा। सममण्डलादित्यग्रेण सहान्वयः) भवतीति ॥१६॥

अत्रोपपत्तिः

क्रान्तिवृत्ते यत्र ग्रहस्थानमस्ति तदुपरिगतं ध्रुवप्रोतवृत्तं समप्रोतवृत्तञ्च कार्यं ग्रहा- (ग्रहस्थान) भवत्यंशेन (ग्रहक्षितिजवृत्तसंज्ञकं) वृत्तं कार्यं तस्मिन् वृत्ते ध्रुवप्रोतवृत्ते समप्रोतवृत्तयोरन्तर्गतं चापं नाडीवृत्तपूर्वापरवृत्तयोरन्तर्गतं चापं वा ग्रहलग्नकोणो (ध्रुवप्रोतवृत्तसमप्रोतवृत्तयोरुत्पन्नः) वाऽक्षवलनसंज्ञकम्। सममण्डलप्राचीतो नाडीमण्डलप्राची यावद्वलति तत् वलं, वल्लसंवरणे संवलने चेति धातोर्वलतीति वलनमिति सञ्चलनमेव वलनमिति। खस्वस्तिकस्थे ग्रहे सममण्डलीयनतांशा (ग्रहोपरिगतसमप्रोतवृत्तपूर्वापरवृत्तसम्पातात् खस्वस्तिकं यावत्) भावः नाडीवृत्तपूर्वापरवृत्तयोः सम्पात (पूर्वस्वस्तिक) स्थे ग्रहे सममण्डलीयनतांशा नतांशा नवत्यंशतुल्यास्तत्र नतकालो दिनार्धसमः। खस्वस्तिकस्थे ग्रहे च नतकालाऽभावोऽतोऽन्तरेऽनुपातो यदि दिनार्धतुल्ये नतकाले नवत्यंशतुल्याः सममण्डलीयनतांशा लभ्यन्ते तदेष्टनतकाले किमित्यनुपातेनेष्टसममण्डलीयनतांशाः समागच्छन्ति, परमयमनुपातो न समीचीनः। अथ ग्रहाद् ध्रुवं यावद् भुज्याचापमेको भुजः। समस्थानाद् ग्रहं यावत्समप्रोतवृत्ते द्वितीयो भुजः। याम्योत्तरवृत्तंऽक्षांशास्तृतीयो भुज इति भुजत्रयेणोत्पन्नचापीयत्रिभुजे समस्थानलग्नकोणः =१८०—सममण्डलीयनतांशः, ग्रहलग्नकोणोऽक्षवलनम्। ततोऽनुपातेन 'यदि

$$\text{भुज्यया सममण्डलीयनतांशज्या} \left\{ \text{ज्या} \left(१८०—\text{सममण्डलीयनतांश} \right) \right\}$$

लभ्यते तदाऽक्षज्यया किमिति' नेन समागच्छत्यक्षवलनज्या=

$$\frac{\text{सममण्डलीयनतांशज्या} \cdot \text{अक्षज्या}}{\text{त्रि.}}$$

अत्राऽऽचार्येण सममण्डलीयनतांशज्यास्थाने सममण्डलीयनतांशोत्क्रमज्या तथा भुज्यास्थाने च त्रिज्या गृहीता, या च न समीचीना, साधिताऽक्षवलनज्यायाश्चापं कार्यं तदाऽक्षवलनं (पूर्वकपाले उत्तरं पश्चिमकपाले च दक्षिणं) भवति, शिष्यधीवृद्धिदे तन्त्रे लल्लाचार्येण "स्पर्शादिकालजनतोत्क्रमज्ञिञ्जिनीभिः क्षुण्णाऽक्षभा पलभवधरणेन भक्ता। चापानि पूर्वनतपश्चिमयोः क्रमेण सोम्येतराणि समवेहि यथा क्रमेण" नेन, सिद्धान्तशेखरे श्रीपतिना "नतोत्क्रमज्याऽक्षगुणाभिघातात् त्रिभज्यकाप्तादथ कार्मुकं यत्। उदक् च याम्यं च कपालयोस्तु तदाक्षमाधावलनं वदन्ति" नेन चोत्क्रमज्याप्रकारेणाऽक्ष-

वलनानयनं कृतं यत् खण्डनं सिद्धान्तशिरोमणौ भास्कराचार्येण "यैरुत्क्रमज्या-विधिनैतदुक्तं सम्यक् न ते गोलगतिं विदन्ति" नेन कृतं यत् समीचीनमस्ति कैश्चित् 'खाङ्काहतं स्वद्युदलेन भक्तं स्पर्शादिकालोत्थनतं लवाः स्युरित्यादिना पूर्वसाधितं' सममण्डलीयनतांशमानं सूक्ष्मं तत्साधितं वलनं सूक्ष्मं कथ्यते, तन्न युक्तम् । सूर्यसिद्धान्तेऽपि 'नतज्याऽक्षज्यया क्षुण्णेत्यादि' साधितमाक्षवलनं न समीचीनमिति, उत्क्रमज्यया साधितं वलनं न समीचीनमेतदर्थं वटेश्वरसिद्धान्तो विलोक्य इति ॥१६॥

अब आक्षवलनानयन को कहते हैं

हि. भा.—पूर्वकपाल में और पश्चिम कपाल में जो सममण्डलीय नतांश है उसकी ज्या और अक्षज्या के घात में त्रिज्या से भाग देने से जो लब्ध होता है उसका चापांश करके ग्रह से तीन राशि पर पूर्वनत और पश्चिमनत क्रम से उत्तर और दक्षिण में ग्रह वृत्तीय सममण्डल से नाड़ीवृत्त की पूर्वा दिशा होती है, यहाँ अग्रिम श्लोक के, 'सममण्डलात्' इस के साथ सम्बन्ध है ॥१६॥

उपपत्ति

क्रान्तिवृत्त में जहाँ ग्रहस्थान है उसके ऊपर ध्रुवप्रोतवृत्त और समप्रोतवृत्त कर दिया, ग्रह (ग्रहस्थान) से नवत्यंश व्यासार्ध से ग्रहक्षितिजसंज्ञक वृत्त कर दिया, ध्रुवप्रोत वृत्त और समप्रोतवृत्त के अन्तर्गत ग्रहक्षितिज वृत्तीय चाप वा नाड़ीवृत्त और पूर्वापरवृत्त के अन्तर्गत ग्रहक्षितिजवृत्तीय चाप वा ग्रहलग्नकोण (ध्रुवप्रोतवृत्त और समप्रोतवृत्त से उत्पन्न कोण) आक्षवलन है, खस्वस्तिक में ग्रह के रहने से सममण्डलीय नतांश (ग्रहोपरिगत समप्रोतवृत्त और पूर्वापरवृत्त के सम्पात से खस्वस्तिक पर्यन्त) का अभाव होता है, नाड़ीवृत्त और पूर्वापरवृत्त के सम्पात (पूर्वस्वस्तिक) में ग्रह के रहने से सममण्डलीय नतांश नवत्यंश के बराबर होता है और नतकाल दिनार्ध के बराबर होता है, इन दोनों के मध्य में ग्रह के रहने से अनुपात करते हैं यदि दिनार्ध तुल्य नतकाल में नवत्यंश तुल्य सममण्डलीय नतांश पाते हैं तो इष्ट नतकाल में क्या इस अनुपात से इष्ट सममण्डलीय नतांश प्रमाण आता है, लेकिन यह अनुपात ठीक नहीं है, ग्रह से ध्रुव पर्यन्त क्रुज्या चापांश प्रथमभुज, ग्रह से समस्थानपर्यन्त उपकोटि द्वितीय भुज, अक्षांश (ध्रुव और समस्थान के अन्तर्गत याम्योत्तरवृत्तीय चाप) तृतीय भुज, इन तीनों भुजों से उत्पन्न चापीय त्रिभुज में समस्थानलग्नकोण = १८०—सममण्डलीयनतांश, ग्रहलग्नकोण = आक्षवलन, अब अनुपात करते हैं यदि द्युज्या में तत्संमुखकोणज्या = $\left\{\text{ज्या}\left(१८०—\text{सममण्डलीय नतांश}\right)\right\}$ = सममण्डलीय नज्या, पाते हैं तो अक्षज्या में इससे ग्रह-

लग्नकोणज्या (आक्षवलनज्या आती है $\frac{\text{सममण्डलीय नज्या}\cdot\text{अक्षज्या}}{\text{द्यु}}$ = आक्षवलनज्या,

यहाँ आचार्य ने सममण्डलीयनतांशज्या के स्थान में सममण्डलीयनतांशोत्क्रमज्या तथा द्युज्या के स्थान में त्रिज्या का ग्रहण किया है जो नितान्त अनुचित है, साधित आक्षवलनज्या का वाप करने से आक्षवलन पूर्व कपाल में उत्तर और पश्चिम कपाल में दक्षिण होता है. शिष्यधीवृद्धिदे में लल्लाचार्य "स्पर्शादिकालजनतोत्क्रमशिञ्जिनीभिः" इत्यादि संस्कृतोपपत्ति में लिखित पद्यों से तथा सिद्धान्तशेखर में श्रीपति ने "नतोत्क्रमज्याऽक्षगुणाभिघातात्" इत्यादि से उत्क्रमज्या प्रकार से आक्षवलनानयन किया है जिसका खण्डन सिद्धान्तशिरोमणि में भास्कराचार्य ने "यैरुत्क्रमज्या विधिनैतदुक्तं" इत्यादि से किया है जो कि बिल्कुल ठीक है, कोई कोई "खाङ्काहतं स्वद्युदलेन भक्तं" इत्यादि से साधित भास्करीय आक्षवलन को ठीक कहते हैं सो अनुचित है सूर्यसिद्धान्त में भी "नतज्याऽक्षज्याक्षुण्णा" इत्यादि से साधित आक्षवलन ठीक नहीं है इति ॥१६॥

इदानीमायनवलनानयनमाह

सममण्डलाद्विषुवतो ग्राह्यात् त्रिगृहाधिकादुदग्याम्यैः ।
क्रान्त्यंशैरपमण्डलपूर्वास्याश्चन्द्रविक्षेपः ॥१७॥

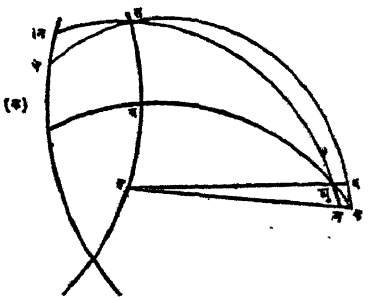
सु. भा.—सममण्डलादिति पूर्वेण श्लोकेन सहान्वयः । ग्राह्याच्चन्द्रग्रहे चन्द्रात् सूर्यग्रहे सूर्यात् किंविशिष्टात् त्रिगृहाधिकात् राशित्रयसहिताद्ये क्रान्त्यंशास्तैरुदग्याम्यैर्विषुवतो नाडीवृत्तात् त्रिभान्तरेऽपमण्डलपूर्वा क्रान्तिमण्डलीया तात्कालिकी पूर्वा स्यादिति । अस्याः क्रान्तिवृत्तपूर्वायाः सकाशाद्याम्योत्तरश्चन्द्रविक्षेपो भवति इति प्रसिद्धोऽर्थः ।

अत्रोपपत्तिः । सत्रिभग्रहक्रान्तिज्या द्युज्यावृत्तेऽयनवलनं भवति । आचार्येण स्थूलाद् द्युज्या त्रिज्यामिता गृहीता । अतः क्रान्तिज्यैवायनं वलनम् । तच्चापांशाः सत्रिभग्रहदिक्का उत्तरयाम्या अयनवलनांशा भवन्तीति सर्वं भास्करीयवलनवासनातः स्फुटम् ॥१७॥

वि. भा.—सममण्डलादिति पूर्वश्लोकेन सहान्वयः । ग्राह्यात् (चन्द्रग्रहणे चन्द्रात्, सूर्यग्रहणे सूर्यात्) त्रिगृहाधिकात्, (राशित्रयसहितात्) ये क्रान्त्यंशास्तैरुदग्याम्यैर्विषुवतः (नाडीवृत्तात्) त्रिभान्तरेऽपमण्डलपूर्वा (क्रान्तिवृत्तीया) तात्कालिकी पूर्वा स्यात् । अस्याः (क्रान्तिवृत्तपूर्वायाः) याम्योत्तरः(दक्षिण उत्तरश्च) चन्द्रविक्षेपः (चन्द्रशरः) भवतीति ॥१७॥

अत्रोपपत्तिः

ध्रु=ध्रुवः । क=कदम्बम् । ग्र=क्रान्तिवृत्ते ग्रहः । ग्रहोपरि ध्रुवप्रोतवृत्तं कदम्बप्रोतवृत्तञ्च कार्यं, ग्रहान्नवत्यंशेन ग्रहक्षितिजं कार्यमेतस्मिन् वृत्ते ध्रुवप्रोतकदम्बप्रोतवृत्तयोरन्तर्गतं चापं नाडीवृत्तक्रान्तिवृत्तयोरन्तर्गतं चापं वा ग्रहलग्नकोणः कदम्बप्रोतवृत्तध्रुवप्रोतवृत्तयोरुत्पन्नो वाऽऽयनवलनम् ।



कपसन=ग्रहक्षितिजम् । स=सत्रिभग्रहः । कप=सज=<पग्रक=आयनवलनम् । मध्रुसन=सत्रिभग्रहोपरिगतं ध्रुवप्रोतवृत्तम् ।

>ग्रकच=ग्रहकोटिः ध्रुग्र=द्युज्याचापम् । कध्रु=जिनांशाः, कध्रुग्रचापीयत्रिभुजे कोणानुपातेना 'यदि द्युज्यया ग्रहकोटिज्या लभ्यते तदा जिनज्यया किमिति' नेन समागच्छति ग्रहलग्नकोणज्याऽर्थादायनवलनज्या $=\frac{\text{ग्रकोज्या}\cdot\text{जिज्या}}{\text{द्यु}}$, ततो द्युज्याग्रे परिणमनार्थमनुपातो यदि त्रिज्ययेयमायनवलनज्या लभ्यते तदा द्युज्यया किमित्यनेनागच्छति द्युज्याग्रीयायनवलनज्या $=\frac{\text{ग्रकोज्या}\cdot\text{जिज्या}}{\text{त्रि.}}$, अथ सत्रिभग्रहः=९०+ग्र ततः सत्रिभग्रहज्या=ज्या (९०+ग्र) चापज्या चापोनभार्धांशज्ययोस्तुल्यत्वात् ज्या (९०+ग्र)=ज्या (१८०—९०—ग्र) =ज्या (९०—ग्र)=ग्रहकोटिज्या=सत्रिभग्रहज्या

$\therefore \frac{\text{ग्रकोज्या}\cdot\text{जिज्या}}{\text{त्रि}}=\frac{\text{सग्रज्या}\cdot\text{जिज्या}}{\text{त्रि}}$=सत्रिभग्रहक्रान्तिज्या=द्युज्याग्रीयायनवज्या

त्रिज्याग्रीयायनवलनज्या तु पूर्वाऽऽनीतैवास्ति, त्रिज्याद्युज्ययोः स्वल्पान्तरात्तुल्यां स्वीकृत्य त्रिज्यापरिणामनमकृत्वैव सत्रिभग्रहक्रान्तिज्यातुल्यैवाऽऽयनवलनज्या स्वीकृता भास्करतः प्राचीनैः सर्वैः सिद्धान्तकारैः । लल्लस्तु भुजज्यायास्त्वदुत्क्रमज्यायाश्चाभावत्वपरमत्वयोरेकत्रैव सद्भावादुत्क्रमज्ययैव बहूनां विषयाणां (सिद्धान्तोक्तानां) साधनं कुर्वन् वलनानयनमप्युत्क्रमज्ययैव चकार तल्लोक्तसाधनमेव युक्तियुक्तमिति बुध्वा श्रीपतिनाऽपि वलनानयनं तथैवोक्तमित्यनुमीयते। यथा तल्लोक्तम्—"बाह्यात् सराशित्रितयाद् भजज्या व्यस्ता

ततः प्राग्वदपक्रमज्या। तस्या धनुः सत्रिगृहेदुदिक् स्यात् क्षेत्रे विपातस्य विधोर्दिशि स्यात्" श्रीपत्युक्तञ्च "त्रिभवनसहिताच्च ग्राह्यतो व्यस्तजीवा रविनवमभचापं संस्कृतं स्वेषुणा यत्" एभिर्महानुभावैरायनवलने सम्मस्कारोऽपि कृतः स च युक्तिरहित्यान्न युक्तः। आचार्योक्ताऽऽयनवलनज्या क्रान्तिज्यैवास्ति तच्चापाशाः सत्रिभग्रहदिक्का उत्तरयाम्या आयनवलनांशा भवन्ति, भास्कराचार्येण 'युताऽयनांशोद्भवकोटिशिञ्जिनी जिनांशमौर्व्या गुणिता विभाजिता। द्युजीवया लब्धफलस्य कार्मुकं भवेच्छशाङ्कायनदिक्कुमायनम्" नेन समीचीनमायनवलनसाधनं कृतमिति ॥१७॥

अब आयन वलन के साधन को कहते हैं

हि. भा.—तीन राशि युक्त ग्राह्य (चन्द्रग्रहण में चन्द्र से सूर्य ग्रहण में सूर्य से) से जो क्रान्त्यंश हो उससे उत्तर और दक्षिण नाड़ीवृत्त से तीन राशि के अन्तर पर क्रान्तिवृत्तीय तात्कालिक पूर्व दिशा होती है, इस क्रान्तिवृत्तीय पूर्व दिशा से दक्षिण और उत्तर चन्द्रग्रह होता है इति ॥१७॥

उपपत्तिः

संस्कृतोपपत्ति में लिखित (क) क्षेत्र को देखिये। ध्रु=ध्रुव, क=कदम्ब, ग्र=क्रान्तिवृत्त में ग्रह=चन्द्र, ग्रह के ऊपर ध्रुवप्रोतवृत्त और कदम्ब प्रोतवृत्त कीजिए, ग्रह को केन्द्र मानकर नवत्यंश व्यासार्ध से ग्रहक्षितिज संज्ञक वृत्त कीजिए, इस वृत्त में ध्रुवप्रोतवृत्त और कदम्बप्रोतवृत्त के अन्तर्गत चाप वा नाड़ीवृत्त और क्रान्तिवृत्त के अन्तर्गत चाप वा ग्रहलग्न कोण (ध्रुवप्रोतवृत्त और कदम्बप्रोतवृत्त से उत्पन्न) आयन वलन है। कपसन=ग्रहक्षितिज, स=सत्रिभग्रह,=सत्रिभचन्द्र, कप=ज्या=<पक्रक=आयनवलन, मधुसन=सत्रिभग्रह के ऊपर ध्रुव प्रोतवृत्त, <क्रकम=ग्रहकोटि=चन्द्रकोटि, ध्रुग्र=क्रान्त्यापाप, कध्रु=जिनांश, तब कध्रुग्र त्रिभुज में कोणानुपात 'यदि क्रान्त्या में ग्रहकोटिज्या पाते हैं तो जिनज्या में क्या' से ग्रहलग्न कोणज्या अर्थात् आयनवलनज्या=

$\frac{\text{ग्रकोज्या.} \times \text{जिज्या}}{\text{क्रा}}$ इसको क्रान्त्या में परिणामन के लिये अनुपात करते हैं यदि त्रिज्या में यह वलनज्या पाते हैं तो क्रान्त्या में क्या इस से क्रान्त्याग्री आयन वलनज्या आती है,

$\frac{\text{ग्रकोज्या.} \times \text{जिज्या}}{\text{त्रि}}$ = क्रान्त्याग्री आयन वलनज्या, सत्रिभग्रह = ९० + ग ∴ सत्रिभग्रहज्या = ज्या (९० + ग) चापज्या और चापोन कोटिज्या बराबर होती है इसलिए ज्या (९० + ग) = ज्या (१८० — ९० — ग) = ज्या (९० — ग) = ग्रहकोटिज्या = सत्रिभग्रहज्या

∴ $\frac{\text{ग्रकोज्या.} \times \text{जिज्या}}{\text{क्रा}} = \frac{\text{सत्रिभग्रहज्या.} \times \text{जिज्या}}{\text{त्रि}}$ = सत्रिभग्रहक्रान्त्याग्री = क्रान्त्याग्री आयनवलनज्या,

त्रिज्याग्री आयनवलनज्या (वास्तव आयनवलनज्या) पहले लाई हुई $\frac{\text{अक्रोज्या}\cdot\text{त्रिज्या}}{\text{द्यु}}$ है, यहां आचार्य ने त्रिज्या और द्युज्या को बराबर स्वीकार कर त्रिज्या परिणामन को नहीं कर के सत्रिभ क्रान्तिज्या तुल्य ही आयनवलनज्या स्वीकार किया है। भास्कराचार्य से प्राचीनाचार्यों ने भी यही किया है। लल्लाचार्य ने शिष्यधीवृद्धिद में 'भुजज्या और उसकी उत्क्रमज्या के एक ही स्थान में अभावत्व और परमत्व से बहुत विषयों के साधन उत्क्रमज्या ही से करते हुए वलनानयन भी उत्क्रमज्या ही से किया है। लल्लोक्तसाधन को युक्तियुक्त समझ कर श्रीपति भी उन्हीं का अनुसरण करते हैं। जैसे लल्लाचार्य कहते हैं "ग्राह्यात् सराशित्रितयाद् भुजज्या" इत्यादि संस्कृतोपपत्ति में लिखित पद्य से श्रीपति भी "त्रिभवन सहिताच्च ग्राह्यतो व्यस्तजीवा इत्यादि संस्कृतोपपत्ति में लिखित पद्य से लल्लोक्त के अनुसार ही कहते हैं। इन आचार्यों ने (लल्ल और श्रीपति) आयनवलन में शर संस्कार भी किया है जो युक्तिशून्यत्व के कारण ठीक नहीं है। आचार्योक्त आयन वलनज्या क्रान्तिज्या ही है उसका चापांश सत्रिभग्रह की दिशा उत्तर और दक्षिण आयनवलनांश होता है, भास्कराचार्य ने सिद्धान्तशिरोमणि में "युतायनांशोद्भुपकोटिशिञ्जिनी" इत्यादि से आयनवलन का साधन ठीक किया है इति ॥१७॥

इदानीं स्पष्टवलनमाह

एकान्यदिशोर्युतिवियुतेर्ज्या ग्रग्रहणमध्यमोक्षेषु।
एवं निमीलनोन्मीलनेष्टकालेष्वतोऽन्यदिशाम् ॥१८॥

सु. भा.—एकदिशोरक्षजायनवलनचापयोर्युतेरन्यदिशोर्वियुतेर्ज्या स्पर्शमध्यमोक्षकालेषु वलनं स्फुटं भवति। एवं निमीलनोन्मीलनेष्टकालेषु स्फुटं वलनं साध्यम्। अतोऽस्माद्वलनादन्यदिशामानयनं कार्यम्। अर्थादेकस्माद्वृत्ताद्यावद्भिरंशैरन्यवृत्तस्य पूर्वा चलति तावद्भिरंशैरन्या दिशश्च चलन्ति इति तासामानयनं सुगमम्।

अत्रोपपत्तिः। 'तयोः पलोत्थायनयोः समाशयो' रित्यादिभास्करविधिना स्फुटा। तत्र भास्करेण मानैक्यार्धवृत्ते स्फुटं वलनं परिणाम्यते। आचार्येण च त्रिज्यावृत्ते यथागतं तथैव स्थापितमिति। इदं स्फुटं वलनं परिलेखार्थमुपयुक्तं। परिलेखविधिं चाग्रे वक्ष्यत्याचार्यः॥१८॥

वि. भा.—एकदिशोराक्षायनवलनचापयोर्युतेरन्यदिशोर्वियुतेर्यद् भवति तज्ज्या स्पर्शमध्यमोक्षेषु स्फुटवलनज्या भवति, एवं निमीलनोन्मीलनेष्टकालेषु स्फुटं वलनं साध्यम्। अतोऽस्माद्वलनादन्यदिशामानयनं कार्यमर्थादेकस्माद् वृत्ताद्या-

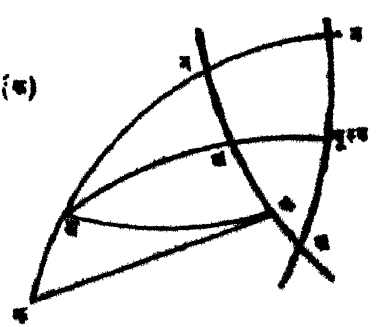
वद्भिरंशैरन्यवृत्तस्य पूर्वा चलति तावद्भिरंशैरन्या दिग्भ्यश्चलन्तीति तासामानयनं कार्यमिति ॥१८॥

अत्रोपपत्तिः

ग्रहोपरिगतं कदम्बप्रोतवृत्तं समप्रोतवृत्तञ्च कार्यं ग्रहान्तवर्त्यंशेन ग्रहक्षितिजं कार्यमेतस्मिन् वृत्ते कदम्बप्रोतवृत्तसमप्रोतवृत्तयोरन्तर्गतं चापं क्रान्तिवृत्तपूर्वापरवृत्तयोरन्तर्गतं चापं वा, कदम्बप्रोतवृत्तसमप्रोतवृत्तयोरुत्पन्नो ग्रहलग्नकोणो वा स्पष्टवलनम्। ग्रहोपरिगतं ध्रुवप्रोतवृत्तं कार्यं तदा ध्रुवप्रोतवृत्तसमप्रोतवृत्तयोरुत्पन्नो ग्रहलग्नकोण आक्षवलनम्। ग्रहोपरिगतध्रुवप्रोतवृत्तकदम्बप्रोतवृत्तयोरुत्पन्नो ग्रहलग्नकोण आयनवलनम्, एतयोः कोणयोर्योगान्तरेण कदम्बप्रोतवृत्तसमप्रोतवृत्तयोरुत्पन्नकोणः स्फुटं वलनं भवति। सिद्धान्तशेखरे "पलवलनमनेन स्पष्टमेकीकृतं स्यात् सदृशदिशि वियुक्तं भिन्नदिक्त्वे कृतज्यम्" अनेन श्रीपतिना, लल्लाचार्येण च "अपक्रमक्षेपपलोद्भवानां युतिः क्रमादेकदिशां कलानाम्। कार्यो वियोगोऽन्यदिशां ततो ज्या ग्राह्या भवेत्सा वलनस्य जीवा, अनेन, भास्कराचार्येण चानयो पलोत्थायनयोः समाशयायुतेर्वियुक्तैस्तु विभिन्नकाष्ठयोरित्यादिना' अनेन स्पष्टवलनमाचार्योक्तानुरूपमेव कथ्यते, केवल लल्लाचार्येण श्रीपतिना चाऽऽयनाक्षवलनयोर्योगान्तरूपे स्पष्टवलने शरस्यापि संस्कारः कृतो यश्च न समीचीनः। भास्कराचार्येण साधितं स्फुटं वलनं मानैक्यार्धवृत्ते परिणामितम् आचार्येण च त्रिज्यावृत्ते यथागतं तथैव स्थापितम्। स्फुटं वलनपरिलेखार्थमुपयुक्तम्।

एकानुपातेन स्पष्टवलनानयनं भवतीति प्रदर्श्यते

क्रान्तिवृत्तपूर्वापरवृत्तयोः सम्पातः सन्धिग्रहसंज्ञकः=सं सं बिन्दुतो नवत्यंशेन वृत्तं कार्यं तत् क्रान्तिवृत्तपूर्वापरवृत्तयोः परमान्तरवृत्तम्। स=लग्नम् =क्रान्तिवृत्तक्षितिजवृत्तयोः सम्पातः। पू=पूर्वस्वस्तिकम्। पूल=लग्नाग्राचापम्। स=समस्वानम्। क=कदम्बः। सल=लग्नाग्राकोटिचापम्। नम=क्रान्तिवृत्तपूर्वापरवृत्तयो. परमान्तरम् =फल, संल=सन्धिग्रहलग्नान्तरम्। < सलन = विषुवोन्नतांशाः। तदा सलन चापीयजात्यत्रियुक्ते



लग्नाग्राकोटिज्या × विषुवद्भक्तः =परमान्तरकोटिज्या तस्याग्रापं भवतीति वोध्यं
त्रि

तदा परमान्तरं=नम भवेत्। ततः संनम, संलपू चापीयजात्यत्रिभुजयोर्ज्यक्षेत्र-साजात्यादनुपातेन $\frac{\text{त्रि}\times\text{लग्नाग्रा}}{\text{परमान्तरज्या}}$ =सन्धिग्रहलग्नान्तरज्या, अस्याश्चापं लग्ने हीनितं तदा सन्धिग्रहो भवेत्। ग्र=ग्रहः, संग्र=सन्धिग्रहग्रहान्तरंविदितमेव। ग्रन =सन्धिग्रहग्रहान्तरकोटिः=<सकग्र, सग्र=उपकोटिचापम्। तदा कसग्र चापीय त्रिभुजेऽनुपात $\frac{\text{सन्धिग्रहग्रहान्तरकोटिज्या}\times\text{परमान्तरज्या}}{\text{उपकोटिज्यासार्ध}}$ =ज्या<ग्र=स्पष्टवल-नज्या, अस्याश्चापं स्पष्टवलनं भवेदिति ॥१८॥

अब स्पष्टवलनानयन को कहते हैं

हि. भा.—एक दिशा का आक्षवलन और आयनवलन का योग करने से तथा भिन्न दिशा का, उन दोनों का अन्तर करने से जो होता है उसकी ज्या स्पर्शकाल, मध्यग्रहण और मोक्षकाल में स्फुटवलनज्या होती है, इसी तरह निमीलनकाल - उन्मीलनकाल और इष्टकाल में स्फुटवलन साधन करना। इस वलन से अन्यदिशा का आनयन करना अर्थात् एक वृत्त से जितने अंश में अन्यवृत्त की पूर्वा दिशा चलती है उतने अंश में अन्य दिशा चलती है इति ॥१८॥

उपपत्ति

ग्रह के ऊपर समप्रोतवृत्त-कदम्बप्रोतवृत्त और ध्रुवप्रोतवृत्त कीजिये, ग्रह को केन्द्रमानकर नवत्यंश से ग्रह क्षितिज कीजिए, इसमें समप्रोतवृत्त और कदम्बप्रोतवृत्त के अन्तर्गत चाप स्पष्टवलन है, ग्रहोपरिगतसमप्रोतवृत्त और ध्रुवप्रोतवृत्त से उत्पन्न ग्रहलग्नकोण आक्षवलन है तथा ध्रुवप्रोतवृत्त और कदम्बप्रोतवृत्त से उत्पन्न ग्रहलग्नकोण आयनवलन है, दोनों का योगान्तर करने से समप्रोतवृत्त और कदम्बप्रोतवृत्त से उत्पन्न ग्रहलग्न कोण स्पष्टवलन होता है, सिद्धान्तशेखर में "पलवलनमनेन स्पष्टमेकीकृतं" इत्यादि से श्रीपति ने, "अपक्रम-क्षेपपलोद्भवानाम्" इत्यादि से लल्लाचार्य ने, तथा "तयोः पलोत्थायनयोः समाशयोः" इत्यादि से सिद्धान्तशिरोमणि में भास्कराचार्य ने भी आचार्योक्तानुरूप ही स्पष्टवलन कहा है, केवल लल्ल और श्रीपति ने आयनवलन और आक्षवलन के योगान्तररूप स्पष्टवलन में शर संस्कार भी किया है जो कि अनुचित है, भास्कराचार्य ने साधित स्पष्टवलन का मानैक्यार्ध वृत्त में परिणामव किया है, आचार्य ने त्रिज्यावृत्त में साधित स्पष्टवलन को ज्यों का त्यों रक्खा है, परिवेष के लिये स्पष्टवलन की जरूरत होती है।

एकानुपात से स्पष्टवलनानयन को दिखलाते हैं

यहाँ संस्कृतोपपत्ति में लिखित (क) क्षेत्र को देखिये। क्रान्तिवृत्त और पूर्वापरवृत्त का

सम्पात सन्धिग्रह है=सं म बिन्दु से नवत्यंश व्यासार्धवृत्त क्रान्तिवृत्त और पूर्वापरवृत्त का परमान्तरवृत्त है, ल=लग्न,=क्रान्तिवृत्त और क्षितिजवृत्त का सम्पात बिन्दु। पु=पूर्वस्वस्तिक, पूल=लग्नाग्रचाप, स=मध्यस्थान, क=कदम्ब.

सल=लग्नाग्रार्कोटिचाप, नम=क्रान्तिवृत्त और पूर्वापरवृत्त का परमान्तर=क्रम, संल=सन्धिग्रहलग्नान्तर <सलन=विक्षिप्तलग्न का उन्नतांश, तब सलन चापीयत्रास्र त्रिभुज में अनुपात से

$\frac{\text{लग्नाग्रार्कोटिज्या}\cdot\text{विक्षिप्तशङ्कु}}{\text{त्रि}}$ =परमान्तरकोटिज्या, इसके चाप को नवत्यंश में घटाने से परमान्तर=नम होता है, तब संनम, संलपु दोनों चापीय त्रास्र त्रिभुजों के कोण सजातीय हैं इसलिये अनुपात से

$\frac{\text{त्रि}\times\text{लग्नाग्रा}}{\text{परमान्तरज्या}}$ =सन्धिग्रह और लग्न की अन्तरज्या, इसके चाप को लग्न में घटाने से सन्धिग्रह होता है, ग्र=ग्रह, संग्र=सन्धिग्रह और ग्रह का अन्तर ग्रह विदित है, ग्रन=सन्धिग्रह और ग्रह की अन्तरकोटि=<सकग्र ग्रन=उपकोटिचाप, तब कसग्र चापीय त्रिभुज में अनुपात करते हैं

$\frac{\text{सन्धिग्रहग्रहान्तरकोटिज्या}\times\text{परमान्तरज्या}}{\text{उपकोटिज्यासार्ध}}$ =ज्या<ग्र=स्पष्टवलनज्या, इसका चाप करने से स्पष्ट वलन होता है ॥१८॥

इदानीं ग्रहणे चन्द्रवर्णमाह

आद्यन्तयोः सधूम्रः कृष्णः सम्यग्ग्रहेऽधंतोऽन्यथिके।
प्राप्ते स कृष्णताम्रः सर्वग्रहणे कपिलवर्णः ॥१९॥

सु. भा.—आद्यन्तयोश्चन्द्रः स धूम्रो वर्णो भवति। अर्धाद्ग्रहणादौ ग्रहणान्ते च चन्द्रो धूम्रवर्णो भवति। कृष्णताम्रः कृष्णरक्तो विज्ञाम्बनक्षविरित्यर्थः। शेषं स्पष्टार्थम्। स्वल्पग्रासे धूम्रवर्णः सुधांशोरित्यादिभास्करोक्तमेतदनुरूपमेव ॥१९॥

वि. भा.—आद्यन्तयोः (ग्रहणादौ ग्रहणान्ते च) चन्द्रः सधूम्रो (धूम्रवर्णसहितः) भवति। सम्यग्ग्रहणे चन्द्रः कृष्णवर्णो भवति, अर्धतोऽन्यथिके प्राप्ते (अर्धाधिके ग्रहणे) कृष्णताम्रः (कृष्णरक्तो विज्ञाम्बनक्षविरिति) सर्वग्रहणे चन्द्रः कपिलवर्णो भवतीति ॥१९॥

अत्रोपपत्तिः

वर्णसंयोगेन वर्णान्तरमुत्पद्यते, भूभाकृष्णा, चन्द्रबिम्बं पीतमतस्तयोः संयोगे न्यूनाधिक्यस्थितिवशेन भिन्नं भिन्नं वर्णान्तरं दृग्गोचरीभूतं भवति। सूर्यग्रहणे हि छाद्यच्छादकयोः(सूर्याचन्द्रमसोः) संयोगाभावात् वर्णान्तरं नैव जायते, सूर्यसिद्धान्ते, शिष्यधीवृद्धिदे, सिद्धान्तशिरोमणौ चैवमेव रविचन्द्रयोर्ग्रहणे वर्ण उक्तोस्ति सिद्धान्तशेखरेऽपि "आदावन्ते वहलविलसद्धूमधूम्रो विधुः स्यादर्धच्छन्नः कलयति पुनः कालतां कज्जलस्य । अर्धादूर्ध्वं कवलिततनुः कृष्णताम्रः पिशङ्गः सर्वग्रासे भवति सविता सर्वदा कृष्ण एव" श्रीपतिनाऽनेनैवमेवोच्यत इति ॥ १९ ॥

अब ग्रहण में चन्द्रवर्ण को कहते हैं

हि. भा.—ग्रहण के आदि में और अन्त में चन्द्र का धूम्र (धूम्रां) वर्ण होता है खण्ड ग्रहण में कृष्ण वर्ण होता है। अर्धाधिक ग्रहण में कृष्णताम्र (कालापनयुक्त लाल) वर्ण होता है, सर्वग्रहण में चन्द्र का कपिल वर्ण होता है इति ॥१९॥

उपपत्ति

वर्ण के संयोग से वर्णान्तर बनता है, भूभा कृष्णवर्ण है और चन्द्रबिम्ब पीत (पीला) वर्ण है इस लिये दोनों का संयोग होने से न्यूनाधिक्य स्थितिवश से भिन्न-भिन्न वर्णान्तर देखने में आता है, सूर्यग्रहण में छाद्य और छादक (सूर्य और चन्द्र) के संयोगाभाव से वर्णान्तर उत्पन्न नहीं होता है, सूर्यसिद्धान्त, शिष्यधीवृद्धिद, सिद्धान्तशिरोमणि, सिद्धान्तशेखर में इसी तरह ग्रहण में वर्ण कहा है इति ॥१९॥

इदानीमस्याध्यायस्योपसंहारमाह

मानविमर्दस्थितिदलवलनेष्टग्राससमकलाद्येषु ।
चन्द्रग्रहणाध्यायो विंशतिरार्याश्चतुर्थोऽयम् ॥२०॥

सु. भा.—समकलः पूर्णान्तिकालो मध्यग्रहसमयः। शेषं स्पष्टार्थम् ॥२०॥

इति श्रीकृपालुदत्तसूनुसुधाकरद्विवेदिविरचिते ब्राह्मस्फुटसिद्धान्ततिलके चन्द्रग्रहणाधिकारश्चतुर्थः ॥

वि. भा.—छाद्यच्छादकबिम्बमानम्। विमर्दार्धम्। स्थित्यर्धम्। वलनानि (ग्रसनाभिधस्पष्टवलनसंज्ञकानि) इष्टग्रासः, समकलः (पूर्णान्तिकालो मध्यग्रहण समयः) एतदादिषु विषयेषु विंशतिसंख्यका आर्या यत्र तादृशोऽयं चतुर्थश्चन्द्रग्रहणाध्यायोऽस्तीति ॥२०॥

इति श्रीब्रह्मगुप्तविरचिते ब्राह्मस्फुटसिद्धान्ते

चन्द्रग्रहणाधिकारश्चतुर्थः समाप्तः

अब इस अध्याय के उपसंहार को कहते हैं

हि. भा.—मान (छाद्य बिम्ब और छादक बिम्ब के मान) विमर्दार्ध, स्थित्यर्ध, वलन (आक्षवलन, आयनवलन और स्पष्टवलन) इष्टग्रास, मध्यग्रहण समय, एतदादि विषयों में बीस आर्या श्लोक हैं जिसमें ऐसा चौथा चन्द्रग्रहणाध्याय है इति ॥२०॥

इति श्रीब्रह्मगुप्तविरचित ब्राह्मस्फुटसिद्धान्त में चन्द्रग्रहणाधिकार (चतुर्थाधिकार) समाप्त हुआ ॥

ब्राह्मस्फुटसिद्धान्तः

सूर्यग्रहणाधिकारः

ब्राह्मस्फुटसिद्धान्तः

सूर्यग्रहणाधिकारः

अथ सूर्यग्रहणाधिकारो व्याख्यायते तत्रादौ तदारम्भप्रयोजनमाह

दृग्गणितैक्यं न भवति यस्मात् पञ्चज्यया रविग्रहणे ।
तस्माद्यथा तदैक्यं तथा प्रवक्ष्यामि तिथ्यन्ते ॥१॥

सु. भा.—उदयज्या अग्रा । मध्यज्या=खलग्ननतांशज्या । रविशङ्कुः । दृग्गतिः=वित्रिभशङ्कुः । दृक्क्षेपः=वित्रिभनतांशज्या । एवमार्यभटादिभिः पञ्चज्यया रविग्रहणे यदानयनमुपनिबद्धं तद्यस्माद् दृग्गणितैक्यं न भवति तस्माद्यथा तयोर्दृग्गणितयोरैक्यं स्यात् तथाऽऽनयनमहं प्रवक्ष्यामीति । तिथ्यन्ते इत्यस्याग्रे सम्बन्धः । सम्प्रति प्रसिद्धसूर्यसिद्धान्तेऽपि पूर्वोक्तपञ्चज्ययैवार्कग्रहणानयनम् । पौलिशे च चन्द्रस्यापि पृथक् सूर्यग्रहणे पञ्चज्याः साधिता इति दशज्याविधानेन तत्र रविग्रहणमुपनिबद्धम् । तथा च स्वटीकायां चतुर्वेदाचार्यः—

'पञ्चज्यया पञ्चज्याविधानेन रविग्रहणं यदाचार्यैरुपनिबद्धम् । तद्यथा । उदयज्या, मध्यज्या, शङ्कुज्या, दृग्गतिज्या, दृक्क्षेपज्या च । ताभिरार्यभटादिभिस्तथा पौलिशतन्त्रे पञ्चज्याश्चन्द्रमसः स्वदिनगतशेषचरदलक्रान्त्यादिभिः कल्पिताः । एवं तत्र दशज्याविधानेन रविग्रहणं यदुपनिबद्धं तादृशममान्ते भवति । ये च तत्र दोषास्तानाचार्य एव वक्ष्यति तन्त्रपरीक्षाध्याये वयमपि तत्रैव व्याख्यास्यामः' ।

वि. भा.—यस्मात् कारणात् रविग्रहणे (सूर्यग्रहणे) पञ्चज्यया (उदयज्या, मध्यज्या, रविशङ्कुः, वित्रिभशङ्कुः, दृक्क्षेपः) ऽर्यभटादिभिर्यदानयनमुपनिबद्धं ततो दृग्गणितैक्यं (द्वाभ्यां दृग्गणिताभ्यां समत्वं) न भवति, तस्मात् कारणात् यथा तदैक्यं (तयोर्दृग्गणितयोः समत्वं) भवेत्तथाऽऽनयनमहं प्रवक्ष्यामि, तिथ्यन्ते इत्यस्याग्रे सम्बन्धः । उदयज्या=अग्रा, मध्यज्या=दशमलग्ननतांशज्या, वित्रिभशङ्कुः=दृग्गतिः, दृक्क्षेपः=वित्रिभनतांशज्या सूर्यसिद्धान्तेऽपि पूर्वोक्तपञ्चज्ययैव सूर्यग्रहणसाधनम् । पौलिशसिद्धान्ते तु सूर्यग्रहणे चन्द्रस्यापि पृथक् पञ्चज्याः साधिताः सन्ति तेन तत्र दशज्याविधानेन सूर्यग्रहणमुपनिबद्धमिति ॥१॥

अब सूर्यग्रहणाधिकार प्रारम्भ किया जाता है, पहले उसके आरम्भ करने का प्रयोजन कहते हैं

हि. भा.—जिस कारण से सूर्यग्रहण में पञ्चज्या (उदयज्या, मध्यज्या, त्रिभशङ्कु, विभशङ्कु, दृक्क्षेप) से आर्यभटादि आचार्यों ने जो आनयन किया है उससे दृग्गणितैक्य (वेधोपलब्ध और गणितागत विषयों की तुल्यता)नहीं होता है उस कारण से जैसे उनका ऐक्य (वेधोपलब्ध और गणितागत विषयों के समत्व) हो वैसे मैं आनयन को कहता हूं, तिथ्यन्ते इसका आगे से सम्बन्ध है, उदयज्या=अग्रा, मध्यज्या=दशमलग्ननतांशज्या विभशङ्कु =दृग्गति, दृक्क्षेप=विभनतांशज्या, सूर्यसिद्धान्त में भी पूर्वोक्त पञ्चज्या ही से सूर्यग्रहण का आनयन है, पौलिशसिद्धान्त में सूर्यग्रहण में चन्द्र की पृथक् पञ्चज्या साधित है इसलिए वहां दशज्या विधान से सूर्यग्रहण उपनिबद्ध है इति ॥१॥

इदानीं लम्बननत्योर्भावाभावस्थानमाह

वित्रिभलग्नसमेऽर्के न लम्बनं तदधिकोनके भवति ।
तस्य क्रान्तिज्योदक् यदाऽक्षजीवा समा न तदा ॥२॥

अवनतिरतोऽन्यथा भवति सम्भवे तदुदयैर्विलग्नसमम् ।
तदुदितघटिकास्तच्छङ्कुस्तच्चरप्राणैः ॥३॥

सु. भा.—तिथ्यन्ते गणितागतदर्शान्तिकाले पूर्वविधिना त्रिप्रश्नोक्तेन लग्नं कृत्वा वित्रिभं कार्यम् । तेन वित्रिभलग्नेन समेऽर्के रवौ सति लम्बनं न भवति । तस्माद्वित्रिभादधिके वोने रवौ लम्बनं भवतीत्यर्थत एव सिध्यति । एवं तस्य वित्रिभस्योत्तरा क्रान्तिज्या यदा स्वदेशाक्षजीवा समा तदाऽवनतिर्न भवति अतोऽन्यथा चावनतिर्भवति । लम्बनावनत्योः सम्भवे च तदुदयैः स्वदेशीयराश्युदयैर्विलग्नसमं कृत्वाऽर्धोनस्य भोग्योऽधिकभुक्तयुक्तो मध्योदयाढ्य इति त्रिप्रश्नोक्तविधिं कृत्वा तदुदितघटिकास्तस्य वित्रिभस्य गता घटिकाः साध्याः । ततस्तच्चरप्राणैर्वित्रिभचरासुभिस्तच्छङ्कुर्वित्रिभशङ्कु 'दिनगतशेषाल्पस्य' इत्यादिविधिना साध्यः ।

अत्रोपपत्तिः । लम्बनाभावे 'न लम्बनं वित्रिभलग्नतुल्ये रवा' वित्यादिभास्करविधिना स्फुटा । एवमवनतेरानयनस्य दृक्क्षेपाधीनत्वाद्यदा दृक्क्षेपाभावस्तदाऽवनतेरभावः आचार्येण स्वल्पाक्षदेशे याम्योत्तरवृत्त एव स्वल्पान्तराद्वित्रिभस्थितिं प्रकल्प्य तस्य द्युदलवत् क्रान्त्यक्षसंस्कारेण नतांशाभावस्थानमानीतमुत्तरक्रान्तिसमेऽर्के । वित्रिभोदितघटिकादिज्ञानं च त्रिप्रश्नोक्त्या सुगमम् ॥२-३॥

वि. भा.—तिथ्यन्ते (गणितागतामान्तकाले) त्रिप्रश्नोक्तेन विधिना लग्नं साध्यं तत् त्रिभोनं (वित्रिभं) कार्यम् । वित्रिभलग्नेन तुल्ये रवौ लम्बनं न भवति,

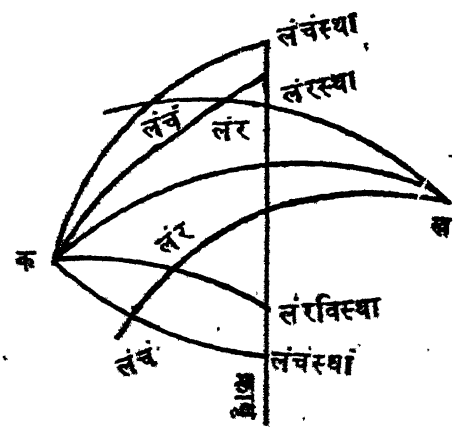
तदधिकोनके(तस्माद्विक्षिभादधिकेऽल्पे च) रवौ लम्बनं भवति तथा तस्य(विक्षिभस्य) यदोदक्क्रान्तिज्या (उत्तरा क्रान्तिज्या) ऽक्षजीवा समा (स्वदेशीयाक्षज्या तुल्या) तदाऽवनतिर्न भवति, अतोऽन्यथाऽवनतिर्भवति, सम्भवे (लम्बननत्योः सत्त्वे) तदुदयैः (स्वदेशीयरात्र्युदयमानैः) विलग्नसमं कृत्वाऽर्धोनस्य भोग्योऽधिकभुक्तयुक्तो मध्योदयाढ्य इति त्रिप्रश्नोक्तविधिं कृत्वा तदुदितघटिकाः(तस्य विक्षिभस्य गता घटिकाः) साध्याः। ततस्तच्चरप्राणैः (विक्षिभचरासुभिः) तच्छङ्कुः (विक्षिभशङ्कुः) 'दिनगतशेषाल्पस्य' इत्यादि विधिना साध्य इति ॥२-३॥

अत्रोपपत्तिः

लग्नोत्पन्ननवत्यंशवृत्तं दृक्क्षेपवृत्तं तद्यत्र क्रान्तिवृत्ते लगति तदेव विक्षिभलग्नम्। तत्तुल्ये रवौ स्पष्टलम्बनाभावो भवेत्। रव्युपरि-लम्बितरव्युपरि च गतं कदम्बप्रोतवृत्तद्वयं यत्र-यत्र क्रान्तिवृत्ते लगति तदन्तर्गतक्रान्तिवृत्तीयचापमेव रविस्पष्टलम्बनम्। परन्तु विक्षिभस्थे रवौ तदुपरिगतं दृग्वृत्तं तथा रव्युपरिलम्बितरव्युपरि च गतं कदम्बप्रोतवृत्तमेकमेव दृक्क्षेपवृत्तं भवेत्तेन तत्र स्पष्टलम्बनाभावः प्रत्यक्षमेव दृग्गोचरो भवत्यतो "विक्षिभलग्नसमेऽर्के न लम्बनमित्याचार्योक्तं युक्तियुक्तमिति ॥२-३॥

अथ लम्बनस्य घनर्णव्यवस्था

गर्भीयामान्तकाले स्थानाभिप्रायेण रविचन्द्रावेकस्मिन्नेव बिन्दौ भवतस्तेनैकस्मिन्नेव दृग्वृत्ते लम्बितरविचन्द्रौ भवतः। लम्बितरवितो लम्बितचन्द्रः पृष्ठेऽवलम्बितो भवत्यतो विक्षिभादूने रवौ लम्बितरव्युपरिगतं कदम्बप्रोतवृत्तं यत्र क्रान्तिवृत्ते लगति तस्मादधोभागे लम्बितचन्द्रोपरिगतं कदम्बप्रोतवृत्तं क्रान्तिवृत्ते लगि-



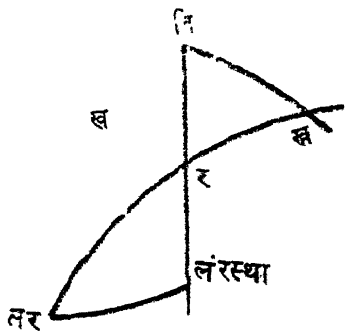
ख=खस्वस्तिकम्। क=कदम्बः।
वि=विक्षिभम्। लंर=लम्बितरविः।
लंच=लम्बितचन्द्रः।
लंरस्था=लम्बितरविस्थानम्।
लंचंस्था=लम्बितचन्द्रस्थानम्।

ष्यति तेनात्र शीघ्रगतिग्रहाच्चन्द्रस्थानान्मन्दगतिग्रहस्य लम्बितरविस्थानरूपस्याग्रे-स्थितत्वाद्युक्तिर्मम्याऽतो गर्भीयामान्तात्पृष्ठीयामान्तः स्पष्टलम्बनान्तरेण द पश्चा

भवेदतो गर्भीयामान्तकाले स्पष्टलम्बनान्तरयोजनेन पृष्ठीयामान्तकालो भवेत् । वित्रिभादधिके रवौ लम्बितरवितोऽधोलम्बितचन्द्रस्तेन लम्बितरव्युपरिगतकदम्बप्रोतवृत्तक्रान्तिवृत्तयोः सम्पातात् लम्बितचन्द्रोपरिगतकदम्बप्रोतवृत्तक्रान्तिवृत्तयोः सम्पात उपरि भवेदतो मन्दगतिग्रहाल्लम्बितरविस्थानाच्छीघ्रगतिग्रहस्य लम्बितचन्द्रस्थानरूपस्याऽग्रे स्थितत्वाद्युतिर्गतेत्यतो गर्भीयामान्ते स्पष्टलम्बनान्तरमृणं कार्यं तदा पृष्ठीयामान्तकालो भवेदतः । "तदधिकोनके भवति" आचार्योक्तमिदं युक्तियुक्तम् ।

अथ नतिसम्बन्धे विचारः

ख = खस्वस्तिकम् । वि = वित्रिभम् ।



र=क्रान्तिवृत्ते रविः । लंर=लम्बितरविः । लंरस्था=लम्बितरविस्थानम् । रवितो लम्बितरवियावद् दृग्लम्बनम् । लम्बितरवितो लम्बितरविस्था नंयावन्नतिः । रवितो लम्बित रविस्थानं यावत् क्रान्तिवृत्ते, रविस्पष्टलम्बनम् । खर= रविनतांशाः । खवि=दृक्क्षेपचापम् । तदा खविर, दृग्लम्बननतिस्पष्टलम्बनांशैरुत्पन्नचापीयजात्यत्रिभुजस्य च ज्याक्षेत्रसाजात्यादनुपातो यदि दृग्ज्यया दृक्क्षेपो लभ्यते तदा दृग्लम्बनज्यया किमित्यनेनाऽऽगच्छति रविनतिज्या $=\frac{\text{दृक्क्षेप}\times\text{दृलंज्या}}{\text{दृग्ज्या}}$ नतेरानयनस्य दृक्क्षेपाधीनत्वाद्यदा दृक्क्षेपाभावस्तदा नतेरभावः । आचार्येण स्वल्पाक्षदेशे स्वल्पान्तराद्याम्योत्तरवृत्त एव वित्रिभस्थितिं प्रकल्प्य दिनार्धवत् क्रान्त्यक्षांशयोः संस्कारेण नतांशाभावस्थानमानीतमुत्तरक्रान्तितुल्येऽक्षांशेऽत आचार्योक्तं 'तस्य क्रान्तिज्योदक् यदाऽक्षजीवा समा न तदा" युक्तियुक्तमिति । सूर्यसिद्धान्ते "मध्यलग्नसमे भानौ हरिजस्य न सम्भवः । अक्षोदक् मध्यमक्रान्तिसाम्येनावनतेरपि" प्यनेन, सिद्धान्तशेखरे श्रीपतिना 'वित्रिभोदयसमे न लम्बनं भास्करे समधिकोनके भवेत् । चेत् समा तदपमज्यकोत्तराऽक्षज्ययाऽस्त्यवनतिस्तदा नहि' प्यनेन, सिद्धान्तशिरोमणौ भास्कराचार्येण "न लम्बनं वित्रिभलग्नतुल्ये, रवौ तदूनेऽभ्यधिके च तत्स्यादेवं धनर्णं क्रमतश्च वेद्यम्" प्यनेनाऽऽचार्योक्तानुरूपमेवोक्तमिति दिक् ॥२-३॥

अधुना प्रसङ्गात् दृग्लम्बनस्य परमत्वं कुत्र भवेदिति विचार्यते

भूकेन्द्राद् भूपृष्ठाच्च स्वकक्षास्थरविकेन्द्रगते रेखे यत्र-यत्र चन्द्र-

कक्षायां लग्ने तदन्तर्गतचन्द्रगोलीयदृग्वृत्तचापं रविलम्बनम् एवं भूकेन्द्राद् भूपृष्ठाच्च चन्द्रकक्षास्थचन्द्रकेन्द्रगते रेखे यत्र-यत्र रविकक्षायां लग्ने तदन्तर्गतरविगोलीयदृग्वृत्तचापं चन्द्रलम्बनमेतावता त्रिभुजमुत्पद्यते भूकेन्द्राद् ग्रहकेन्द्रं यावद् ग्रहकर्णः एकोऽवयवः। भूपृष्ठाद् ग्रहकेन्द्रं यावत् पृष्ठकर्णो द्वितीयोऽवयवः। भूव्यासार्धं तृतीयोऽवयवः। एतस्मिन् त्रिभुजे पृष्ठकर्णभूव्यासार्धयोरुत्पन्नकोणः = १८० पृष्ठीयनतांशकोणज्याकोणोनभार्धांशज्ययोस्तुल्यत्वात् ज्या (१८०—पृष्ठीयनतांश) = ज्यापृष्ठीयनतांश = पृष्ठीयदृग्ज्या, तदाऽनुपातो यदि ग्रहकर्णेन पृष्ठीयदृग्ज्या लभ्यते तदा भूव्यासार्धेन किं समागच्छति ग्रहदृग्लम्बनज्या तत्स्वरूपम् = $\frac{\text{पृदृग्ज्या} \times \text{भूव्या}}{\text{ग्रकर्ण}}$

अत्र ग्रकर्णः, भूव्यासार्धयोः स्थिरत्वात्सिद्धं यद्यत्र पृष्ठीयदृग्ज्यायाः परमत्वं भवेत्तत्रैव दृग्लम्बनज्यायाः परमत्वं भवेत्। परं पृष्ठीयदृग्ज्यायाः परमत्वं तु पृष्ठक्षितिजस्थे ग्रहे भवत्यतः पृष्ठक्षितिज एव दृग्लम्बनस्य परमत्वं भवेदिति॥

अथ नतेः परमत्वं कुत्र भवेदिति विचार्यते

अथ $\frac{\text{दृक्क्षेप} \times \text{दृलंज्या}}{\text{दृज्या}}$ = नतिः स्वल्पान्तरात्। तथा सम्बन्धानुपातेन 'त्रिज्यातुल्यपृष्ठीयदृग्ज्यायां यदि परमलम्बनज्या लभ्यते तदेष्टपृष्ठीयदृग्ज्यायां किमिति, नेनेष्टदृग्लम्बनज्या = $\frac{\text{परलंज्या} \times \text{पृदृज्या}}{\text{त्रि}}$ अत उत्थापनेन

$\frac{\text{दृक्क्षेप} \times \text{पलंज्या} \cdot \text{पृदृज्या}}{\text{दृज्या} \times \text{त्रि}}$ = नतिः, अत्र यदि पृदृज्या = दृज्या तदैव 'कक्षयोरन्तरं यत् स्याद्विनिभे सर्वतोऽपि तत्, भास्करोक्तमिदं समीचीनं भवितुमर्हति, परं पृदृज्या = दृग्ज्या सर्वदा न भवत्यतो भास्करोक्तं तन्न समीचीनमिति ज्ञेयम्। अथ पूर्वोक्तनतिस्वरूपे यदि दृक्क्षेपमानं स्थिरं कल्प्येत तदा $\frac{\text{पृदृज्या}}{\text{दृज्या}}$ ऽस्य परमत्वं यत्र भवेत्तत्रैव नतेरपि परमत्वं भवितुं महति, $\frac{\text{पृदृज्या}}{\text{दृज्या}}$ अस्य परमत्वं कुत्र भवेत्तदर्थं विचार्यते। पृष्ठीयनतांशाः = पृन। गर्भीयनतांशाः = न। दृग्लम्बनम् = लं, तदा पृन = न + लं चापयोरिष्टयोरित्यादिना ज्या (न + लं) = पृदृज्या = $\frac{\text{दृज्या} \times \text{लंकोज्या} + \text{शं} \times \text{लंज्या}}{\text{त्रि}}$ छेदगमेन पृदृज्या · त्रि = दृज्या × लंकोज्या + शं × लंज्या, परं $\frac{\text{पलंज्या} \cdot \text{पृदृज्या}}{\text{त्रि}}$ = लंज्या अत उत्थापनेन पृदृज्या · त्रि = दृज्या · लंकोज्या +

$\frac{\text{शं.पलज्या. पृदृज्या}}{\text{त्रि}}$ ततः पृदृज्या. त्रि^{२}=दृज्या. लंकोज्या. त्रि+शं. पलज्या पृदृज्या

समशोधनेन

पृदृज्या.त्रि^{२}—शं. पलज्या·पृदृज्या=पृदृज्या (त्रि^{२}—शं.पलज्या)=दृज्या·लंकोज्या. त्रि. ∴ $\frac{\text{पृदृज्या}}{\text{दृज्या}}=\frac{\text{लंकोज्या. त्रि}}{\text{त्रि}^{२}-\text{श·पलज्या}}$ एतत्स्वरूपावलोकनेन सिद्धं यद्यदा शङ्कुदृग्लम्बनकोटिज्ययोः परमत्वं भवेत्तदैवा-$\frac{\text{पृदृज्या}}{\text{दृज्या}}$स्यापि परमत्वं भवितुमर्हति, परं शङ्कोर्दृग्लम्बनकोटिज्यायाश्च परमत्वं वित्रिभस्थान एव भवत्यतो वित्रिभ एवा-$\frac{\text{पृदृज्या}}{\text{दृज्या}}$स्य परमत्वान्नतेरपि परमत्वं सिद्धमिति, एतावता म. म. पण्डित श्रीसुधाकरोक्तसूत्रमवतरति "कुगर्भनघ्नांशगुणेन भक्तः स्वपृष्ठनघ्नांशगुणः फलं चेत्। महत्तमं तत्र नतिः पराभवेत् दृक्क्षेपमानं बुध नो चलं चेत्" इति॥

अथ पूर्वसिद्धस्वरूपम्=$\frac{\text{पृदृज्या}}{\text{दृज्या}}=\frac{\text{लंकोज्या.त्रि}}{\text{त्रि}^{२}-\text{श. पलज्या}}$ खस्वस्तिकगते रवौ शं=त्रि, दृग्लम्बनकोज्या=त्रि, अतः खस्वस्तिके $\frac{\text{पृदृज्या}}{\text{दृज्या}}=\frac{\text{त्रि.त्रि}}{\text{त्रि}^{२}-\text{त्रि.पलज्या}}=$

$\frac{\text{त्रि}^{२}}{\text{त्रि (त्रि}-\text{पलज्या)}}=\frac{\text{त्रि}}{\text{परमलम्बनकोट्युत्क्रमज्या}}$ परं खस्वस्तिकगते रवौ पृदृज्या=०। दृज्या=० ∴ $\frac{\text{पृदृज्या}}{\text{दृज्या}}=\frac{०}{०}=\frac{\text{त्रि}}{\text{परमलंको उज्या}}$ परं परमलंकोउज्या<त्रि ∴ $\frac{\text{त्रि}}{\text{परमलंकोउज्या}}>$ १ वा $\frac{\text{पृदृज्या}}{\text{दृज्या}}=\frac{०}{०}>$ १ इति गणित-वैचित्र्यं बुधैर्विभावनीयमिति॥२-३॥

अब लम्बन और नति के भावाभावस्थान को कहते हैं

हि. भा.—तिथ्यन्त (गणितागत अमान्त काल) में त्रिप्रश्न में कथित विधि से लग्न साधन करना उसमें तीन राशि घटाने से वित्रिभलग्न होता है, वित्रिभलग्न के तुल्य रवि के रहने से लम्बन नहीं होता है अर्थात् लम्बनाभाव होता है, वित्रिभ से रवि के अधिक और अल्प रहने से लम्बन होता है, तथा जब वित्रिभ की उत्तर क्रान्तिज्या स्वदेशीय अक्षज्या के बराबर होती है तब नति का अभाव होता है, इससे अन्यथा नति होती है, लम्बन और नति के सम्भव रहने पर देशीय राश्युदयमानों से "ऊनस्य भोग्योऽधिकभुक्तुल्यो मध्योदयादृणम्" यह त्रिप्रश्नोक्तविधि कर के वित्रिभ की घटघटी साधन करना

तदनन्तर विम्बिभ की चरासु से 'दिनगतशेषाल्पस्य' इत्यादि विधि से विम्बिभशङ्कु का साधन करना चाहिये ॥२-३॥

उपपत्ति

लग्नोत्पन्न नवत्यंशवृत्त (दृक्क्षेपवृत्त) क्रान्तिवृत्त में जहां लगता है वही विम्बिभलग्न है, विम्बिभलग्न के बराबर रवि के रहने से स्पष्टलम्बनाभाव होता है, रवि के ऊपर और लम्बित रवि के ऊपर कदम्बप्रोतवृत्तद्वय क्रान्तिवृत्त में जहां-जहां लगता है तदन्तर्गत क्रान्तिवृत्तीय चाप (रवि से लम्बित रविस्थान पर्यन्त) रवि का स्पष्ट लम्बन है, परन्तु विम्बिभस्थान में रवि के रहने से उसके ऊपर दृग्वृत्त तथा रवि के ऊपर और लम्बित रवि के ऊपर कदम्ब प्रोतवृत्त एक ही दृक्क्षेपवृत्त होता है इसलिये वहां स्पष्ट लम्बनाभाव होता है, अतः "विम्बिभलग्नसमेऽर्केन लम्बनम्" यह आचार्योक्त युक्ति-युक्त है इति ॥

गर्भीयामान्तकाल में स्थानाभिप्रायिक रवि और चन्द्र एक ही विन्दु में होते हैं, इसलिए एक ही दृग्वृत्त में लम्बितरवि और लम्बित चन्द्र होते हैं, लम्बित रवि से लम्बितचन्द्र पृष्ठ में लम्बित होते हैं इसलिए विम्बिभ से रवि के अल्प रहने पर लम्बित रव्युपरिगत कदम्बप्रोतवृत्त क्रान्तिवृत्त में जहाँ लगता है उससे अधोभाग में लम्बितचन्द्रपरिगत कदम्बप्रोत-वृत्त क्रान्तिवृत्त में लगेगा इसलिए यहां शीघ्रगतिग्रह (लम्बितचन्द्रस्थान) से मन्दगतिग्रह (लम्बितरविस्थान) के आगे रहने के कारण युति गम्य होती है अतः गर्भीयामान्त से पृष्ठीया-मान्त स्पष्टलम्बनान्तर कर के पश्चात् होता है इसलिये गर्भीयामान्तकाल में स्पष्टलम्बना-न्तर जोड़ने से पृष्ठीयामान्तकाल होता है, विम्बिभ से रवि के अधिक रहने पर लम्बित रवि से लम्बितचन्द्र अधोभाग में होते हैं, इसलिये लम्बित रव्युपरिगत कदम्बप्रोतवृत्त और क्रान्तिवृत्त के सम्पात से लम्बितचन्द्रोपरिगतकदम्बप्रोतवृत्त और क्रान्तिवृत्त का सम्पात ऊपर होता है अतः मन्दगतिग्रह (लम्बित रविस्थान) से शाघ्रगतिग्रह (लम्बित-चन्द्र स्थान) के आगे रहने के कारण युति गत होती है, इसलिये गर्भीयामान्त काल में स्पष्टलम्बनान्तर को ऋण करने से पृष्ठीयामान्तकाल होता है अतः 'तदधिकोनके भवति' यह आचार्योक्त युक्ति-युक्त है, यहां संस्कृतोपपत्तिस्थ (क) क्षेत्र को देखिये. ॥२-३॥

अब नति के सम्बन्ध में विचार करते हैं

यहां संस्कृतोपपत्तिस्थ (ख) क्षेत्र को देखिये, ख=खस्वस्तिक, वि=विम्बिभ, र=क्रान्तिवृत्त में रवि, लंर=लम्बित रवि, लंरस्था=लम्बित रविस्थान, रवि से लम्बित रवि-स्थान पर्यन्त दृग्लम्बन है, लम्बित रवि से लम्बितरविस्थान पर्यन्त रवि की नति है, रवि से लम्बित रविस्थान पर्यन्त रवि का स्पष्टलम्बन है, खर=रविनतांश, खवि=दृक्क्षेप-चाप, खविर, तथा दृग्लम्बन—नति, स्पष्टलम्बनोत्पन्न चापीयत्रिभुजत्व त्रिभुज का ज्यात्रिभुज सजातीय है इसलिये अनुपात करते हैं यदि दृज्ज्या में दृक्क्षेत्र पाते हैं तो दृग्लम्बनज्या में

क्या इससे रवि की नतिज्या आती है, नति का आनयन दृक्क्षेप के अधीन है इसीलिये दृक्क्षेप का अभाव रहने से नति का भी अभाव होना निश्चित है, आचार्य स्वल्पाक्ष देश में स्वल्पान्तर से याम्योत्तर वृत्त ही में विषुभ की स्थिति मानकर दिनार्धकाल की तरह क्रान्ति और अक्षांश के संस्कार से उत्तर क्रान्तितुल्य अक्षांश में नतांशाभाव स्थान लाये हैं इसलिए 'तस्य क्रान्तिज्योदक् यदाक्षजीवा समा न तदा' यह आचार्योक्त युक्तियुक्त है। सूर्यसिद्धान्त-शिरोमणि में भास्कराचार्य ने "न लम्बनं विषुभलग्नतुल्ये" इत्यादि से आचार्योक्त के अनुरूप ही कहा है॥

अब प्रसङ्ग से दृग्लम्बन का परमत्व कहाँ होता है इसके लिये विचार करते हैं

भूकेन्द्र से और भूपृष्ठ से स्वकक्षास्थ रविकेन्द्रगत रेखाद्वय चन्द्रकक्षा में जहाँ-जहाँ लगता है तदन्तर्गत चन्द्रगोलीय दृग्वृत्तचाप रविलम्बन है, इसी तरह भूकेन्द्र से और भूपृष्ठ से चन्द्रकक्षास्थ चन्द्रकेन्द्रगत रेखाद्वय रविकक्षा में जहाँ जहाँ लगता है तदन्तर्गत रविगोलीय दृग्वृत्तचाप चन्द्रलम्बन है, इस तरह एक त्रिभुज बनता है, भूकेन्द्र से ग्रहकेन्द्र (रविकेन्द्र या चन्द्रकेन्द्र) पर्यन्त ग्रहकर्ण एक भुज, भूपृष्ठ से ग्रहकेन्द्र (रविकेन्द्र या चन्द्रकेन्द्र) पर्यन्त पृष्ठकर्ण द्वितीयभुज, भूव्यासार्ध तृतीयभुज इन भुजों से उत्पन्न त्रिभुज में पृष्ठकर्ण और भूव्यासार्ध से उत्पन्न कोण=१८०—पृष्ठीयनतांश, कोणज्या और कोणोनभार्धांशज्या बराबर होती है इसलिये ज्या (१८०—पृष्ठीयनतांश)=पृष्ठीय दृग्ज्या, तब अनुपात करते हैं यदि ग्रहकर्ण में पृष्ठीय दृग्ज्या पाते हैं तो भूव्यासार्ध में क्या इससे ग्रहलग्नकोणज्या (दृग्लम्बनज्या) आती है, उसका स्वरूप यह है $\frac{\text{पृदृग्ज्या}\cdot\text{भूव्या}}{\text{ग्रकर्ण}}$=दृलंज्या, इसमें ग्रहकर्ण और भूव्यासार्ध स्थिर है अतः सिद्ध हुआ कि पृष्ठीय दृग्ज्या का परमत्व जहाँ हो, वहीं दृग्लम्बनज्या का भी परमत्व होगा, लेकिन पृष्ठीय दृग्ज्या का परमत्व पृष्ठ क्षितिज में ग्रह के रहने से होता है अतः पृष्ठ क्षितिज ही में दृग्लम्बन का परमत्व सिद्ध हुआ।

अब नति का परमत्व कहाँ होता है विचार करते हैं

$\frac{\text{दृक्क्षेप}\times\text{दृलंज्या}}{\text{दृज्या}}$=नति स्वल्पान्तर से, तथा सम्बन्धानुपात 'त्रिज्यातुल्य पृष्ठीय दृग्ज्या में यदि परमलम्बनज्या पाते हैं तो इष्टपृष्ठीयदृग्ज्या में क्या' से इष्टदृग्लम्बनज्या आती है, $\frac{\text{परलंज्या}\cdot\text{पृदृज्या}}{\text{त्रि}}$=दृलंज्या इससे नति के स्वरूप में उत्थापन देने से $\frac{\text{दृक्क्षेप}\cdot\text{परलंज्या}\cdot\text{पृदृज्या}}{\text{दृज्या}\cdot\text{त्रि}}$ यहाँ यदि पृदृज्या=दृज्या तब ही "कक्षयोरन्तरं यत् स्याद्विषुभे

सर्वतोऽपि तत्" यह भास्करोक्त समीचीन हो सकता है परन्तु सर्वदा पृज्या=दृज्या नहीं होती है अतः भास्करोक्त वह समीचीन नहीं है। पूर्वोक्त नतिस्वरूप में यदि दृक्क्षेप को स्थिर माना जाय तब $\frac{\text{पृज्या}}{\text{दृज्या}}$ इसका परमत्व जहाँ होगा वहीं नति का भी परमत्व होगा, परन्तु $\frac{\text{पृज्या}}{\text{दृज्या}}$ इसका परमत्व कहाँ होता है इसके लिये विचार करते हैं। पृष्ठीय-नतांश=पृन, गर्भीयनतांश=न, दृग्लम्बन=लं, तब पृन=न+लं, चापयोरिष्टयोरित्यादि से ज्या $(\text{न}+\text{लं})=\text{पृज्या}=\frac{\text{दृज्या}\cdot\text{लंकोज्या}+\text{शं}\cdot\text{लंज्या}}{\text{त्रि}}$ छेदगम से पृज्या·त्रि=दृज्या. लंकोज्या+शं·लंज्या लेकिन $\frac{\text{पलंज्या}\cdot\text{पृज्या}}{\text{त्रि}}=\text{लंज्या}$, उत्थापन देने से पृज्या·त्रि=दृज्या.लंकोज्या+$\frac{\text{शं.पलंज्या}\cdot\text{पृज्या}}{\text{त्रि}}$ अतः पृज्या.त्रि^{2}=दृज्या.लंकोज्या.त्रि+शं.पलंज्या·पृज्या समशोधन करने से पृज्या.त्रि^{2}—शं.पलंज्या.पृज्या=पृज्या (त्रि^{2}—शं.पलंज्या) =दृज्या.लंकोज्या.त्रि, अतएव $\frac{\text{पृज्या}}{\text{दृज्या}}=\frac{\text{लंकोज्या.त्रि}}{\text{त्रि}^2-\text{शं.पलज्या}}$ इस स्वरूप को देखने से सिद्ध होता है कि जहाँ शङ्कु और लम्बन कोटिज्या का परमत्व होगा वहीं $\frac{\text{पृज्या}}{\text{दृज्या}}$ इसका भी परमत्व होगा, परन्तु शङ्कु और लम्बनकोटिज्या का परमत्व वित्रिभ स्थान ही में होता है अतः वित्रिभ स्थान ही में नति का परमत्व सिद्ध हुआ, इससे म. म. पण्डित श्री सुधाकर द्विवेदी जी का सूत्र उपपन्न होता है "कुगर्भनघ्नांशगुणेन भक्त" इत्यादि संस्कृतोपपत्तिस्थ सूत्र को देखिये॥

पूर्वसिद्धस्वरूप $=\frac{\text{पृज्या}}{\text{दृज्या}}=\frac{\text{लंकोज्या.त्रि}}{\text{त्रि}^2-\text{शं.पलंज्या}}$ खस्वस्तिक में रवि के रहने से शं=त्रि, लंकोज्या=त्रि, इसलिये खस्वस्तिक में $\frac{\text{पृज्या}}{\text{दृज्या}}=\frac{\text{त्रि.त्रि}}{\text{त्रि}^2-\text{त्रि}\cdot\text{पलंज्या}}$ $=\frac{\text{त्रि}^2}{\text{त्रि}\ (\text{त्रि}-\text{पलंज्या})}=\frac{\text{त्रि}}{\text{परलम्बनकोट्युत्क्रमज्या}}$ लेकिन खस्वस्तिक में रवि के रहने से पृज्या=० तथा दृज्या=०, $\therefore\frac{\text{पृज्या}}{\text{दृज्या}}=\frac{0}{0}=\frac{\text{त्रि}}{\text{परलम्बनकोटज्या}}$ लेकिन परलम्बन-

कोउज्या $<$त्रि $\therefore \frac{\text{त्रि}}{\text{परलम्बन कोउज्या}} > १$ वा $\frac{\text{पृदृज्या}}{\text{दृज्या}} = \frac{०}{०} > १$ इस गणितविचित्रता को समझना चाहिये ॥२-३॥

इदानीं लम्बनानयनमाह

त्रिज्याकृतेश्चतुर्गुणशङ्कुहृतायाः फलेन भक्तायाः ।
तात्कालिकार्करराशित्रयोनलग्नान्तरज्यायाः ॥४॥

लम्बनघटिकालब्धं लग्नात् तात्कालिकात् त्रिराश्यूनात् ।
ऋणमधिकेऽर्के हीने धनमसकृत् पञ्चवदश्यन्ते ॥५॥

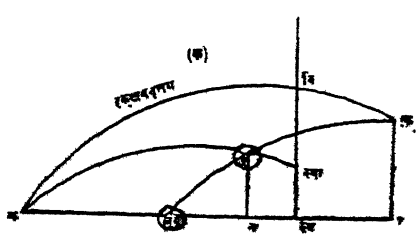
सु. भा.—स्पष्टार्थमार्याद्वयम् ।

अत्रोपपत्तिः । 'त्रिभोनलग्नं तरणि प्रकल्प्ये' त्यादिभास्करविधिना स्फुटा । वास्तवलम्बनादिज्ञानाय मदीयं ग्रहणकरणं विलोकनीयम् ॥४-५॥

वि. भा.—त्रिज्याकृतेः (त्रिज्यावर्गात्) चतुर्गुणशङ्कुहृतायाः (चतुर्गुणित-वित्रिभशङ्कुभक्तायाः) फलेन भक्तायाः तात्कालिकार्करराशित्रयोनलग्नान्तरज्यायाः (तात्कालिकरविवित्रिभान्तरज्यायाः) लब्धं लम्बनघटिकाः (स्पष्टलम्बननाड्यः) भवेयुः । त्रिराश्यूनात् (राशित्रयहीनात्) तात्कालिकाल्लग्नादर्थाद्वित्रिभलग्नात् अर्के (रवौ) ऽधिके पञ्चवदश्यन्ते (पूर्णान्ते) ऋणं वित्रिभलग्नाद्रवौ हीने (अल्पे) पूर्णान्ते धनमेवमसकृत्कार्यं तदा स्पष्टः पूर्णान्तिकालो भवेदिति ॥४-५॥

अत्रोपपत्तिः

ख=खस्वस्तिकम् । वि=वित्रिभम् । खवि=वित्रिभनतांशाः । स्थावि=ग्रहवित्रिभान्तरम्=अं स्था=ग्रहस्थानम् । स्था'=लम्बितग्रहस्थानम् । क=



कदम्बः । कस्था' वृत्तोपरि खस्वस्तिकाल्लम्बचापम्=खप, ग्रबिन्दुतः कस्था'

वृत्तोपरिलम्बचापम्=ग्रन । स्थास्था'=स्पष्टलम्बनम्=लं स्था'वि=अं+लं, तदा कविस्था, कखप चापीयजात्यत्रिभुजयोर्ज्याक्षेत्रसाजात्यादनुपातेन

$\frac{\text{ज्या }(\text{अं}+\text{लं})\cdot\text{विशं}}{\text{त्रि}}$=ज्याखप=लम्बचापज्या, ततः खलंप, ग्रलंन चापीय-जात्यत्रिभुजयोर्ज्याक्षेत्रसाजात्यात् $\frac{\text{लम्बचापज्या}.\text{दृलंज्या}}{\text{पृदृज्या}}$=ज्या ग्रन। अत्र खलं=पृष्ठीयनतांशाः। ग्रलं=दृग्लम्बनम्। लम्बचापज्याया उत्थापनेन $\frac{\text{ज्या }(\text{अं}+\text{लं})\text{विशं}}{\text{त्रि}}$ $\times\frac{\text{दृलंज्या}}{\text{पृदृज्या}}$, तथा $\frac{\text{पलंज्या}\cdot\text{पृदृज्या}}{\text{त्रि}}$=दृलंज्या उत्थापनेन $\frac{\text{ज्या }(\text{अं}+\text{लं})\cdot\text{विशं}}{\text{त्रि}}$ $\times\frac{\text{पलंज्या}.\text{पृदृज्या}}{\text{पृदृज्या}.\text{त्रि}}=\frac{\text{ज्या }(\text{अं}+\text{लं})\cdot\text{विशं}.\text{पलंज्या}}{\text{त्रि}.\text{त्रि}}$=ज्याग्रन, ग्रस्था=शरः। कग्र=शरकोटिः, ततः कग्रन, कस्थास्था' त्रिभुजयोर्ज्याक्षेत्रसाजात्यादनुपातेन

$\frac{\text{ज्या }(\text{अं}+\text{लं})\cdot\text{विशं}}{\text{त्रि}}\times\frac{\text{पलंज्या}}{\text{त्रि}}\times\frac{\text{त्रि}}{\text{शरकोज्या}}$=ज्यास्था'स्था'=स्पष्टलम्बनज्या, अत्राचार्येण स्वल्पान्तरात् पलंज्या=पलं, स्पष्टलम्बनज्या=स्पष्टलम्बन=लं स्वीकृतस्तदा $\frac{\text{ज्या }(\text{अं}+\text{लं})\cdot\text{विशं}}{\text{त्रि}}\times\frac{\text{पलं}}{\text{शकोज्या}}$= स्पलं=लं घटद्यात्मककरणेन

$\frac{६०\times\text{स्पलंकला}}{\text{गत्यन्तरकला}}$=स्पलंघटी=$\frac{\text{ज्या }(\text{अं}+\text{लं})\cdot\text{विशं}}{\text{त्रि}}\times\frac{६०\times\text{पलं}}{\text{गत्यन्तर}\times\text{शकोज्या}}$ परन्तु $\frac{६०\times\text{पलं}}{\text{गत्यन्तरकला}}$=४ घटी, अतः $\frac{\text{ज्या }(\text{अं}+\text{लं})\cdot\text{विशं}}{\text{त्रि}}\times\frac{४}{\text{शकोज्या}.}$ अत्राऽऽचार्येण (१)

शकोज्या=त्रि, लं=० स्वीकृतं तदा स्पष्टलम्बनघटी=$\frac{\text{ज्याअं}.\text{विशं}.४}{\text{त्रि}^२}=\frac{\text{ज्याअं}}{\frac{\text{त्रि}^२}{४\text{विशं}}}=\frac{\text{ज्याअं}}{\text{फल}}$, $\frac{\text{त्रि}^२}{४\text{विशं}}$=फल, परमत्राऽऽचार्येण यस्य स्पष्टलम्बनस्याऽऽनयनं क्रियते तस्यैव मानं शून्यं कल्पितं तथा शरकोटिज्या=त्रि स्वीकृता,

ज्याचापयोरभेदत्वजनितदोषोऽपि वर्त्तत एवात आचार्योक्तमिदमानयनं न समीचीनमित्येतावताऽऽचार्योक्तमप्युपपद्यते।

एतेनैव "वेदघ्नशङ्कुविहृतात् त्रिगुणस्य वर्गाल्लब्धेन विस्रिभदिनेशविशेषजीवा। भक्ता विलम्बनमिदं ह्युदितं दिनेशे न्यूनाधिके धनमृणं त्रिगृहोनलग्नात्" सिद्धान्तशेखरे श्रीपत्युक्तमाचार्योक्तानुरूपमप्युपपद्यते (१) एतेन च "त्रिभोनलग्नार्कविशेषशिञ्जिनी कृता हता व्यासदलेन भाजिता। हतात् फलाद्विस्रिभलग्नशङ्कुना त्रिजीवयाप्तं घटिकादिलम्बनम्" भास्करोक्तमप्युपपन्नं भवति,

पूर्वोक्तं स्फुटं घटी $=\frac{\text{ज्याअं}}{\frac{\text{त्रि}^२}{४\ \text{विशं}}}=\frac{\text{ज्याअं}}{\frac{\left(\frac{\text{त्रि}}{२}\right)^२}{\text{विशं}}}=\frac{\text{ज्याअं}}{\frac{\text{एकराशिज्या}^२}{\text{विशं}}}=\frac{\text{ज्याअं}}{\frac{\text{एकराशिज्या}^२}{\text{दृग्गतिज्या}}}$

$=\frac{\text{ज्याअं}}{\text{छेद}}$ अत्र $\frac{\text{एकराशिज्या}^२}{\text{दृग्गतिज्या}}=$ छेदः।

एतेन "एकज्यावर्गतश्छेदो लब्धं दृग्गतिजीवया। मध्यलग्नार्कविश्लिष्टज्याछेदेन विभाजिता" रवीन्द्वोर्लम्बनं ज्ञेयं प्राक् पश्चात् घटिकादिकम्" सूर्यसिद्धान्तोक्तं स्पष्टलम्बनानयनमुपपद्यते। परमतेष्वानयनेषु न कश्चिद्भेद इति विज्ञैर्ज्ञेयः।

अथात्र वास्तवानयनं क्रियते

अथ $\frac{\text{पलंज्या. दृज्या}}{\text{त्रि}}=$ दृलंज्या। परन्तु दृज्या=ज्या (न+लं)। अत्र

गर्भीयनतांशाः= न दृग्लम्बनम्=लं

चापयोरिष्टयोरित्यादिना $\frac{\text{पलंज्या}}{\text{त्रि}}\ \frac{(\text{दृज्या. लंकोज्या}+\text{लंज्या. शं})}{\text{त्रि}}=$लंज्या

छेदगमेन पलंज्या. दृज्या. लंकोज्या+पलंज्या. लंज्या. शं=त्रि^{२} लंज्या समशोधनेन त्रि^{२} लंज्या—पलंज्या. लंज्या. शं+पलंज्या. दृज्या. लंकोज्या=लंज्या (त्रि^{१}—पलंज्या. शं)

अतः $\frac{\text{लंज्या}}{\text{लंकोज्या}}=\frac{\text{पलंज्या. दृज्या}}{\text{त्रि}^२-\text{पलंज्या. शं}}$ पक्षौ द्वादशभिर्गुणितौ तदा

$$\frac{\text{लज्या. १२}}{\text{लंकोज्या}}=\frac{\text{पलंज्या. दृज्या. १२}}{\text{त्रि}^२-\text{पलंज्या}\times\text{शं}}$$

$$= \frac{\frac{\text{पलज्या. हज्या. १२}}{\text{पलज्या. शं}}}{\frac{\text{त्रि}^2}{\text{शं. पलज्या}} - १} = \frac{\text{इछाया}}{\frac{\text{त्रि}^2}{\text{शं. पलज्या}} - १} = \text{दृग्लम्बनतुल्याक्षदेशीयपलभा.}$$

अतो येऽक्षांशास्तदेव दृग्लम्बनमतेन म. म. पण्डित श्रीसुधाकरोक्तसूत्रमवतरति।

"त्रिज्याकृतिः परमलम्बनमौर्विकाप्तान्ना भाजिता च विधुना रहिताऽथ तेन। भक्तेष्टभा फलमितां पलभां प्रकल्प्य साध्या पलांशकलिकेष्टविलंबनं वा" एतद्वशेन स्पष्टलंबनज्ञानं भवेदिति ॥४-५॥

अब स्पष्टलम्बनानयन को कहते हैं

हि. भा.—त्रिज्यावर्ग में चतुर्गुणित वित्रिभशङ्कु वर्ग से भाग देने से जो फल हो उस से तात्कालिक रवि और वित्रिभ की अन्तर ज्या को भाग देने से जो लब्धि होती है वह स्पष्टलम्बन घटी होती है, तात्कालिक वित्रिभलग्न से रवि के अधिक रहने पर पूर्णान्त काल में ऋण करना और वित्रिभ से रवि के अल्प रहने पर पूर्णान्त काल में पूर्वानीत लम्बन घटी को धन करना इस तरह असकृत्(बार-बार) करने से स्पष्ट पूर्णान्त काल होता है इति ॥४-५॥

उपपत्ति

यहां संस्कृतोपपत्तिस्थ(क) क्षेत्र को देखिये। ख = खस्वस्तिक, वि = वित्रिभ, खवि = वित्रिभनतांश, स्था = ग्रहस्थान स्थावि = वित्रिभार्कान्तर = अं, स्था' = लम्बितग्रहस्थान, क = कदम्ब, कस्था वृत्त के ऊपर खस्वस्तिक से लम्बचाप = खपग्र बिन्दु से कस्था' वृत्त के ऊपर लम्बचाप = अग्र, स्थास्था' = स्पष्टलम्बन = लं, स्थावि = अं + लं, हलं = पृष्ठीयनतांश, ग्रलं = दृग्लम्बन, ग्रस्था = शर, कग्र = शरकोटि, तब कविस्था कखप दोनों-चापीय जात्य त्रिभुज के ज्याक्षेत्र सजातीय होने के कारण $\frac{\text{ज्या}(\text{अं}+\text{लं}).\text{विशं}}{\text{त्रि}}$ = ज्या खप लम्बचापज्या, फिर खलंप, ग्रलंन दोनों चापीयजात्य त्रिभुजों के ज्याक्षेत्र के सजातीय रहने के कारण

$\frac{\text{लम्बचापज्या.हलंज्या}}{\text{पृहज्या}}$ = ज्याग्रन, लम्बचापज्या के उत्थापन से

$\frac{\text{ज्या}(\text{अं}+\text{लं})\cdot\text{विशं}}{\text{त्रि}} \times \frac{\text{हलंज्या}}{\text{पृहज्या}}$ = ज्याग्रन, तथा $\frac{\text{पलंज्या.पृहज्या}}{\text{त्रि}}$ = हलंज्या उत्थापन देने से

$\frac{\text{ज्या}(\text{अं}+\text{लं}).\text{विशं}}{\text{त्रि}} \times \frac{\text{पलंज्या}.\text{पृदृज्या}}{\text{पृदृज्या}.\text{त्रि}} = \frac{\text{ज्या}(\text{अं}+\text{लं})\cdot\text{विशं}}{\text{त्रि}} \times \frac{\text{पलंज्या}}{\text{त्रि}}$, तब.कग्रन, कस्या-

स्या चापीयजात्य त्रिभुजों के सजातीयत्व से $\frac{\text{ज्या}(\text{अं}+\text{लं}).\text{विशं}}{\text{त्रि}} \times \frac{\text{पलंज्या}}{\text{त्रि}} \times \frac{\text{त्रि}}{\text{शकोज्या}} =$

ज्यास्थास्था=स्पष्टलम्बनज्या,यहां आचार्य ने स्वल्पान्तर से पलंज्या=पलं, स्पष्टलम्बनज्या

=स्पष्टलम्बन,=लं.स्वीकार किया है, तब. $\frac{\text{ज्या}(\text{अं}+\text{लं}).\text{विशं}}{\text{त्रि}} \times \frac{\text{पलं}}{\text{शकोज्या}} = \text{स्पलं}$

घटयात्मक करने से $\frac{६०\times\text{स्पष्टलम्बन कला}}{\text{गत्यन्तरकला}} = \text{स्पष्टलंघटी} = \frac{\text{ज्या}(\text{अं}+\text{लं}).\text{विशं}}{\text{त्रि}} \times$

$\frac{६०\times\text{पलं}}{\text{गत्यन्तर}\times\text{शरकोज्या}}$ परन्तु $\frac{६०\times\text{पलं}}{\text{गत्यन्तर}} = ४\text{घटी}$, अतः $\frac{\text{ज्या}(\text{अं}+\text{लं}).\text{विशं}}{\text{त्रि}} \times \frac{४}{\text{शकोज्या}}$

(१)

यहां आचार्य ने शकोज्या=त्रि, लं=० स्वीकार किया है, तब स्पष्टलंघटी $= \frac{\text{ज्या अं विशं}.४}{\text{त्रि}.\text{त्रि}} =$

$\frac{\text{ज्याअं}}{\frac{\text{त्रि}^२}{४\text{विशं}}} = \frac{\text{ज्याअं}}{\text{फल}}$, $\frac{\text{त्रि}^२}{४\text{विशं}} = \text{फल}$,लेकिन यहां आचार्य ने जिस स्पष्टलम्बन का आनयन किया है

उसी का मान शून्य मान लिया है तथा शरकोज्या=त्रि, तथा ज्या और चाप का अभेदत्व जनित दोष भी है ही इसलिये यह आनयन ठीक नहीं है, इससे आचार्योक्त पद्य उपपन्न होता है, इसी से "वेदघ्नशङ्कुविहृतात्" इत्यादि सिद्धान्तशेखर में श्रीपत्युक्त पद्य उपपन्न होता है जो कि आचार्योक्त के अनुरूप ही है, (१) इससे "त्रिभोनलग्नार्कविशेषशिञ्जिनी" इत्यादि सिद्धान्तशिरोमणिस्थ भास्करोक्त पद्य उपपन्न होता है, पूर्वोक्त स्पष्टलम्बनघटी=

$\frac{\text{ज्याअं}}{\frac{\text{त्रि}^२}{४\text{विशं}}} = \frac{\text{ज्याअं}}{\frac{\left(\frac{\text{त्रि}}{२}\right)^२}{\text{विशं}}} = \frac{\text{ज्याअं}}{\frac{\text{एकराशिज्या}^२}{\text{विशं}}} = \frac{\text{ज्याअं}}{\frac{\text{एकराशिज्या}^२}{\text{दृग्गतिज्या}}} = \frac{\text{ज्याअं}}{\text{छेद}}$ इससे "एकज्यावर्गतश्छेदो

लब्धं दृग्गतिजीवया" इत्यादि सूर्यसिद्धान्तोक्त स्पष्टलम्बनानयन उपपन्न होता है, लेकिन इन आनयनों में कुछ भी भेद नहीं है इति ॥

अब वास्तवानयन को कहते हैं

$\frac{\text{पलंज्या}.\text{पृदृज्या}}{\text{त्रि}} = \text{दृलंज्या}$, परन्तु पृदृज्या=ज्या (न+लं)। यहां मर्मीयनतांश=

न, दृग्लम्बन=लं चापयोरिष्टयोः इत्यादि से $\frac{\text{पलज्या}}{\text{त्रि}}\ \frac{(\text{दृज्या. लंकोज्या}+\text{शं. लंज्या})}{\text{त्रि}}=$

लंज्या, छेदगम से पलज्या· दृज्या. लंकोज्या+पलज्या. शं. लंज्या=त्रि^{2}. लंज्या समशोधन से त्रि^{2}. लंज्या—पलज्या. लंज्या. शं=पलज्या. दृज्या. लंकोज्या=लंज्या (त्रि^{2}—पलज्या. शं) अतः $\frac{\text{लंज्या}}{\text{लंकोज्या}}=\frac{\text{पलज्या. दृज्या}}{\text{त्रि}^2-\text{पलज्या.शं}}$ दोनो पक्षों को बारह से गुणने से

$$\frac{\text{लंज्या. १२}}{\text{लंकोज्या}}=\frac{\text{पलज्या. दृज्या. १२}}{\text{त्रि}^2-\text{पलज्या. शं}}$$

$$=\frac{\frac{\text{पलज्या. दृज्या. १२}}{\text{पलज्या. शं}}}{\frac{\text{त्रि}^2}{\text{शं. पलज्या}}-१}=\frac{\text{इष्टाया}}{\frac{\text{त्रि}^2}{\text{शं. पलज्या}}-१}=\text{दृग्लम्बनतुल्याक्ष·}$$

देशीय पलभा, इससे जो अक्षांश होता है वही दृग्लम्बन होता है, इससे म.म. पण्डित श्री सुधाकर द्विवेदी जी का सूत्र उपपन्न होता है "त्रिज्याकृतिःपरमलम्बनमौर्विकाप्ता" इत्यादि संस्कृतोपपत्तिस्थ सूत्र को देखिये.। इसके वश से वास्तवस्पष्टलम्बनानयन होता है इति ॥४-५॥

इदानीं पुनः स्पष्टलम्बनानयनमाह

कर्णगुणाद् व्यासार्धाद् वसुवेदविभाजितात् फलविभक्ता।
लम्बननाड्यो भास्करवित्रिभलग्नान्तरज्या वा ॥६॥

सु. भा.—व्यासार्धात् कर्णगुणाद्वित्रिभच्छायाकर्णगुणात् तस्माद्वसुवेद—४८ भाजिताद्यत् फलं तेन विभक्ता भास्करवित्रिभलग्नान्तरज्या फलं वा लम्बननाड्यः स्युः।

अत्रोपपत्तिः। पूर्वविधिना लम्बनाड्यः $\frac{४\ \text{ज्या}(\text{र}\backsim\text{वि})}{\text{त्रि}}.\ \frac{\text{विशं}}{\text{त्रि}}=$

$$\frac{४\times१२\ \text{ज्या}\ (\text{र}\backsim\text{वि})}{\frac{१२\ \text{त्रि}}{\text{विशं}}-\text{त्रि}}=\frac{४८\ \text{ज्या}\ (\text{र}\backsim\text{वि})}{\text{विछाक. त्रि.}}=\frac{\text{ज्या}\ (\text{र}\backsim\text{वि})}{\frac{\text{विछाक. त्रि}}{४८}}$$

अत उपपन्नम् ॥६॥

वि. भा.—व्यासार्धात् (त्रिज्यातः) कर्णगुणात् (वित्रिभलग्नच्छायाकर्ण-

गुणात्) वसुवेद ४८ भक्ताद्यत् फलं तेन भास्करवित्रिभलग्नान्तरज्या (वित्रिभलग्नार्कान्तरज्या) भक्ता तदा वा (प्रकारान्तरेण) लम्बननाड्यो भवेयुरिति ॥६॥

अत्रोपपत्तिः

$$\text{अथ पूर्वाऽऽनीत लम्बनघट्यः} = \frac{\text{ज्याग्रं. विशं. ४}}{\text{त्रि. त्रि}} = \frac{४ \times १२ \times \text{ज्याग्रं}}{\frac{१२ \times \text{त्रि}}{\text{विशं}} \cdot \text{त्रि}} =$$

$$\frac{४८ \times \text{ज्याग्रं}}{\text{विछाक. त्रि}} = \frac{\text{ज्याग्रं}}{\frac{\text{विछाक. त्रि}}{४८}} = \frac{\text{ज्याग्रं}}{\text{फल}}.$$ एतेन आचार्योक्तमुपपन्नं भवति। सिद्धान्तशेखरे "श्रुतिसङ्गुणात् त्रिभगुणाद्विभाजिताद्वसुसागरै ४८ रथ फलेन वा हृता। रविवित्रिभोदयवियोगशिञ्जिनीघटिकादिलम्बनमिहासकृत् भवेत्" इति ,श्रीपत्युक्तमाचार्योक्तानुरूपमेव, इति दर्शान्तकालेऽसकृत्(वारं-वारं) अविशेषपर्यन्तं साध्यमिति ॥६॥

अब पुनः स्पष्टलम्बनानयन को कहते हैं

हि. भा.—वित्रिभलग्न के छाया कर्ण को त्रिज्या से गुणा कर ४८ अडतालीश से भाग देने से जो फल हो उससे वित्रिभलग्न और रवि की अन्तरज्या को भाग देने से प्रकारान्तर से लम्बन घटी होती है इति ॥६॥

उपपत्ति

$$\text{पूर्वाऽऽनीतलम्बनघटी} = \frac{\text{ज्याग्रं} \cdot \text{विशं} \cdot ४}{\text{त्रि} \cdot \text{त्रि}} = \frac{४ \times १२ \times \text{ज्याग्रं}}{\frac{१२ \times \text{त्रि}}{\text{विशं}} \times \text{त्रि}} = \frac{४८ \text{ ज्याग्रं}}{\text{विछाक} \cdot \text{त्रि}}$$

$$= \frac{\text{ज्याग्रं}}{\frac{\text{विछाक} \cdot \text{त्रि}}{४८}} = \frac{\text{ज्याग्रं}}{\text{फल}}$$ इस से आचार्योक्त उपपन्न होता है, 'सिद्धान्तशेखर में 'श्रुतिसङ्गुणात् त्रिभगुणात्' इत्यादि श्रीपत्युक्त आचार्योक्त के अनुरूप ही है, दर्शान्तकाल में असकृत् (वार-वार) इसका साधन करना इति ॥६॥

इदानीं स्पष्टदर्शान्ते रविचन्द्रपातानां चालनान्याह

रविशशिपातगतिकला लम्बनघटिकागुणा हृताः षष्ट्या।
यदि लम्बनमृणमूना धनमधिकाः स्वफललिप्तामिः ॥७॥

सु. भा—स्पष्टार्थाऽऽर्या ॥७॥

वि. भा.—रविचन्द्रपातानां गतिकला असकृत्कर्मणा स्थिरीकृताभिर्लम्बन-

घटिकाभिर्गुणिताः षष्ट्या भक्ता लब्धकलाभिर्यदि लम्बनमृणं तदा गणितागत-दर्शान्तिकालिका रविचन्द्रपाता ऊनाः (हीनाः) कार्याः, यदि लम्बनं धनं तदा लब्ध कलाभिर्गणितदर्शान्तिकालिका रविचन्द्रपाता युक्ताः कार्यास्तदा स्पष्टदर्शान्तिकालिका रविचन्द्रपाता भवन्तीति ॥ ७ ॥

अत्रोपपत्तिः

गणितागतदर्शान्तिकालिकरविचन्द्रपातेभ्यः शरादिकं सर्वमानेतव्यम् । ऋणात्मके लम्बने "षष्टिघटिकाभिः पृथक् पृथक् रविचन्द्र पातगतिकला लभ्यन्ते तदा लम्बनघटिकाभिः किमित्यनुपातेन" लब्धकलाभिर्गणितागतदर्शान्तिकालिका रविचन्द्र पाता हीनाः कार्याः, धनात्मके लम्बने लब्धकलाभिस्ते युक्तास्तदा ते स्पष्ट-दर्शान्तिकालिका भवेयुरेवेति, सिद्धान्त शेखरे "गतिं हतां लम्बननाड़िकाभिर्विभज्य षष्ट्या फललिप्तिकाभिः । रवीन्दुपाताः सहिता धनाख्ये विवर्जिताश्च क्षयलम्बने ते" ऽनेन श्रीपतिनाऽऽचार्योक्तानुरूपमेवोक्तमिति ॥ ७ ॥

* अब स्पष्ट दर्शान्तकाल में रवि चन्द्र और पात के चालन को कहते हैं।

हि. भा.—रवि, चन्द्र और पात की गति कला को लम्बन घटी से गुणा कर साठ ६० से भाग देने से जो लब्ध कला हो उसको लम्बन ऋण रहने से गणितागत दर्शान्त-कालिक रवि, चन्द्र और पात में से हीन करना, और धनात्मक लम्बन में गणितागत रवि, चन्द्र और पात में लब्धकला को जोड़ना तब स्पष्ट दर्शान्तकालिक रवि, चन्द्र और पात होते हैं इति ॥ ७ ॥

उपपत्ति ।

गणितागत दर्शान्त कालिक रवि, चन्द्र और पात से शर आदि सब कुछ लाना चाहिये, ऋणात्मक लम्बन रहने से 'साठ घटी में पृथक् पृथक् रविगतकला, चन्द्रगतिकला और पातगति कला पाते हैं तो लम्बन घटी में क्या इस अनुपात से जो लब्ध कला हो उनको गणितागत दर्शान्तकालिक रवि, चन्द्र और पात में से हीन करना, धनात्मक लम्बन रहने से लब्धकला को उनमें (गणितागत दर्शान्त कालिक रवि, चन्द्र और पात) जोड़ देना तब स्पष्ट दर्शान्तकालिक रवि, चन्द्र और पात होते हैं, सिद्धान्त शेखर में 'गतिं हतां लम्बन-नाड़िकाभि' रित्यादि, संस्कृतोपपत्ति में लिखित पद्य से श्रीपति ने आचार्योक्त के अनुरूप ही कहा हैं इति ॥ ७ ॥

इदानीं वित्रिभनतांशानां नतेर्वादिग्ज्ञानार्थमाह

अक्षज्यया वित्रिभलग्नात् स्वक्रान्तिरुत्तरार्कस्य ।
इन्दोर्वा यद्यधिकाऽवनतिः सौम्याऽन्यथा याम्या ॥ ८ ॥

सु. भा.—वित्रिभलग्नाद्या स्वक्रान्तिरर्थाद्वित्रिभक्रान्तिज्योत्तरा सा यद्यक्षज्याया अधिका तदाऽर्कस्य वा चन्द्रस्यावनतिर्वित्रिभनतज्या सौम्या ज्ञेयाऽन्यथा सदा याम्या ज्ञेया।

अत्रोपपत्तिः। 'सौम्येऽपमे वित्रिभजेऽधिकेऽक्षा' दित्यादि भास्करविधिना स्फुटा ॥ ८ ॥

वि. भा.—वित्रिभलग्नात् स्वक्रान्तिः (वित्रिभलग्न क्रान्तिज्या) उत्तरा (उत्तरगोल सम्बन्धिनी) अक्षज्यायाः सकाशात् यद्यधिका स्यात्तदार्कस्ये (रवेः) न्दो (चन्द्रस्य) वार्ऽर्थाद्रवि चन्द्र सम्बन्धिनोर्वित्रिभयोरवनतिः सौम्या (उत्तरदिक्का) अन्यथा याम्या (दक्षिणा) भवेदिति ॥ ८ ॥

अत्रोपपत्तिः

"दृग्ज्यैव या वित्रिभलग्नशङ्कोः स एव दृक्क्षेप इनस्य तावत्। सौम्येऽपमे वित्रिभजेऽधिकेऽक्षात् सौम्योऽन्यथा दक्षिण एव वेद्यः" इति सिद्धान्तशिरोमणौ भास्करोक्तया स्फुटाऽस्ति, सिद्धान्त शेखरे "उत्तरो यदि हि वित्रिभलग्नात् स्वापमः समधिकः पलमौर्व्याः। स्यात्तदाऽवनतिरुत्तरदिक्का दक्षिणात्वपरया तपनेन्द्वोः" श्रीपत्युक्तमिदमाचार्योक्तानुरूपमेव भास्करोक्तमप्याचार्योक्तानुरूपं श्रीपत्युक्तानुरूपं वेति बोध्यं गाणितिकैरिति ॥ ८ ॥

अब वित्रिभनतांश के वा नतिके दिक्ज्ञान के लिये कहते हैं।

हि. भा.—वित्रिभलग्न से अपनी (वित्रिभ) क्रान्तिज्या उत्तर गोल सम्बन्धी यदि अक्षज्या से अधिक है तब रवि और चन्द्र सम्बन्धी वित्रिभ की नति उत्तर दिशा की होती है, इससे अन्यथा दक्षिण दिशा की होती है इति ॥ ८ ॥

उपपत्ति।

"दृग्ज्यैव या वित्रिभलग्नशङ्कोः स एव दृक्क्षेप इनस्य तावत्" इत्यादि संस्कृतोपपत्ति में लिखित पद्य से सिद्धान्त शिरोमणि में भास्कराचार्य की उक्ति से स्पष्ट है, सिद्धान्त शेखर में "उत्तरो यदि हि वित्रिभ लग्नात् स्वापमः समधिकः" इत्यादि से आचार्योक्त के अनुरूप ही श्रीपति की उक्ति है, भास्कराचार्योक्त भी आचार्योक्त के अनुरूप या श्रीपत्युक्त के अनुरूप ही समझना चाहिये इति ॥ ८ ॥

इदानीं रविचन्द्रयोर्भगतिसाधनमाह

वित्रिभलग्नादुत्तरदक्षिणविक्षेपहीनसंयुक्तम्।
शंकुघनुरुत्तरायामधिकोनं दक्षिणावनतौ ॥ ९ ॥

तज्ज्येन्दुशंकुराद्यः सवितुर्दृक्क्षेपमण्डले युक्ते ।
अपमण्डलेन भानोश्चन्द्रस्य विमण्डलेन युते ।। १० ।।

सु. भा.—उत्तरायां वित्रिभावनतौ वित्रिभलग्नाद्य उत्तरो वा दक्षिणो विक्षेपस्तेन शंकुधनुर्वित्रिभलग्नशंकुचापं क्रमेण हीनं संयुक्तं च कार्यम् । दक्षिणावनतौ च तदेव शंकुचापमुत्तरेण विक्षेपेणाधिकं दक्षिणेनोनं कार्यम् । एवं संस्कृतं चापं यत् तज्ज्या इन्दोः शंकुरर्थाच्चन्द्रस्य वित्रिभशंकुर्दृग्गतिरित्यर्थः । सवितुः सूर्यस्याद्यः पूर्वसाधितो वित्रिभशंकुरेव दृग्गतिरिति ।

दृक्क्षेपमण्डले वित्रिभलग्नदृङ्मण्डलेन युक्ते भानोर्दृग्गतिर्विमण्डलेन युते चन्द्रस्य दृग्गतिरिति । अत्रैतदुक्तं भवति । दृक्क्षेपमण्डलं यत्र क्रान्तिमण्डले लग्नं तस्य बिन्दोर्वित्रिभस्य शंकुसूर्यस्य दृग्गतिः ।

अत्रोपपत्तिः । क्रान्तिवृत्ते यत्र दृक्क्षेपमण्डलं लग्नं तस्माद्विमण्डलावधि दृक्क्षेपमण्डले चापांशाः स्वल्पान्तराद्वित्रिभलग्नशरसमाः । अत उत्तरायां वित्रिभावनतावुत्तरे वित्रिभशरे द्वयोर्योगेन दृक्क्षेपमण्डले खार्द्धाद्विमण्डलावधि चापांशाश्चन्द्रदृक्क्षेपचापांशाः=विन+विश । एते नवतेश्च्युताश्चन्द्रदृग्गतिचापांशाः=९०—विन—विश=विशंचा—विंशं । एवं दक्षिणे वित्रिभशरे चन्द्रदृक्क्षेप चापांशाः=विन—विश । दृग्गतिः=९०—विन+विश=विशंचा+विश । दक्षिणावनतौ च विपरीत संस्कारः स्फुटः । अत उपपन्नं चन्द्रग्गतिसाधनम् । दृक्क्षेपमण्डलं क्रान्तिवृत्ते लम्वरूपं न विमण्डले । अतश्चन्द्रदृक्क्षेपतो या चन्द्रनतिः सा न कदम्ब प्रोतेऽतस्तत्संस्कारेण न स्फुटा नतिरिति सिद्धान्तविदां स्फुटमेव । भास्करेणापि प्रथममेतन्मतानुसारेण 'दृग्ज्यैव या वित्रिभलग्नशंको' रित्यादिविधिना चन्द्रदृक्क्षेपं प्रसाध्य तद्वशात् स्फुटां नतिं कृत्वा पश्चात् सूर्यग्रहणाधिकारान्ते 'शशिदृक्क्षेपार्थं यद्वित्रिभलग्नेषुणाऽत्रसंस्करणम्' इत्यादिनेदमानयनं खण्डितम् ।। ९-१० ।।

वि. भा.—उत्तरायां वित्रिभावनतौ वित्रिभलग्नात् य उत्तरो वा दक्षिणो विक्षेपः (शरः) तेन शंकुधनुः (वित्रिभलग्नशंकुचापं) क्रमेण हीनं संयुक्तं च कार्यं दक्षिणावनतौ तदेव शंकुचापं अधिकोनं (उत्तरेण विक्षेपेणाधिकं दक्षिणेन हीनं) कार्यं तदा तज्ज्या (संस्कृत चापज्या) इन्दुशंकुः (चन्द्रस्य वित्रिभशंकुर्दृग्गतिरित्यर्थः) सवितुः (सूर्यस्य) आद्यः (पूर्वं साधितो वित्रिभशंकुः) वास्तवो दृग्गतिर्भवति । दृक्क्षेपमण्डले (वित्रिभलग्नदृङ्मण्डले) ऽपममण्डलेन (क्रान्तिवृत्तेन) युक्ते भानोः (सूर्यस्य) दृग्गतिः । विमण्डलेन युते चन्द्रस्य दृग्गतिरिति, अत्रैतदुक्तं भवति, दृक्क्षेपवृत्तं यत्र क्रान्तिवृत्ते लगति तस्य बिन्दोर्वित्रिभलग्नशंकुः सूर्यस्य दृग्गतिः । तदेव दृक्क्षेपवृत्तं चन्द्र विमण्डले यत्र लग्नं तस्य विन्दोः शंकुश्चन्द्रस्य दृग्गतिः इति ।। ९-१० ।।

अत्रोपपत्तिः

दृक्क्षेपवृत्तं क्रान्तिवृत्ते यत्र लग्नं तस्माद्विमण्डलावधि दृक्क्षेपवृत्ते स्वल्पान्तराद्वित्रिभलग्नशरः। तेनोत्तरायां वित्रिभावनतौ-उत्तरे वित्रिभशरे च द्वयोर्योगेन खस्वस्तिकाद्विमण्डलावधि दृक्क्षेपवृत्ते चन्द्रदृक्क्षेप चापांशाः= वित्रिभन+वित्रिभशर, एते नवतेर्विशोध्यास्तदा चन्द्रदृग्गतिचापांशाः=९०—वित्रिभशर,=वित्रिभशंचा-विश, दक्षिणे वित्रिभशरे चन्द्रदृक्क्षेप चापांशाः=वित्रिभनतांश—वित्रिभशर ततः शंकुचापांशाः=९०—वित्रिभन+वित्रिभशर=वित्रिभशंचा+वित्रिभशर, दक्षिणावनतौच विलोमसंस्कारः स्फुटः। एतेन चन्द्रदृग्गतिसाधनमुपपद्यते। दृक्क्षेपवृत्तं क्रान्ति वृत्तोपरिलम्बरूपं, नहि विमण्डलोपरितेन चन्द्रदृक्क्षेपतो या दृग्गतिः सा न कदम्बप्रोतवृत्ते ऽतस्तत्संस्कारेण न स्फुटा नतिरिति। भास्करेणापि पूर्वमेतन्मतानुसारेण "दृग्ज्यैव या वित्रिभलग्नशंको" रित्यादिना चन्द्रदृक्क्षेपं प्रसाध्य तद्वशात्स्फुटनतिं कृत्वा पश्चात् सूर्यग्रहणाधिकारान्ते "शशिदृक्क्षेपार्थं यद्वित्रिभलग्नेषुणाऽत्र संस्करणम्" इत्यादि नाऽस्याऽऽनयनस्य खण्डनं कृतमिति[1]।

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(१) यत्र देशे चतुर्विंशतिरक्षांशास्तत्र यदा रविचन्द्रौ तुलादिगौ भवतस्तदोदयकाले रविरेव लग्नम्। खस्वस्तिके च तत्स्थितौ पूर्वापरवृत्तानुकारकमेव क्रान्तिवृत्तं दृग्वृत्तं च भवति, दृग्वृत्तस्य क्रान्तिवृत्ताकारत्वात्तस्मिन्नेव दृग्वृत्ताकारे क्रान्ति वृत्ते चन्द्रोऽवलम्बितो भवेत्, तेन तत्र चन्द्रनतिः=०, तथा चन्द्रशरः=०, रविनतिः=० तेन स्पष्टोऽत्र बाणो नतिसंस्कृत इत्यादिना स्पष्टशरः=०, तत्र स्पष्टनतिरपि=० वित्रिभलग्नोद्भवशरेण संस्कृता रविनतिश्चन्द्रदृक्क्षेपचापम् = रविनति ± वित्रिभलग्नोद्भवशरः = ० ± वित्रिभलग्नोद्भवशरः=वित्रिभलग्नोद्भवशर। पर मिदं चन्द्र दृक्क्षेपचापं निरर्थकं यतः स्पष्टनत्यर्थमेव दृक्क्षेपस्य प्रयोजनं भवति परमत्र स्पष्टनतिः=०, तेन दृक्क्षेपतो न किमपि कार्यं चलिष्यत्यतो भास्करोक्त खण्डनं युक्तियुक्तमिति ॥

ब्रह्मगुप्तोक्त चन्द्रदृक्क्षेपचाप के भास्करकृत खण्डन को कहते हैं।

जिस देश में चौबीस २४ अक्षांश है वहां जब रवि और चन्द्र तुलादि में होते हैं तब उदयकाल में रवि ही लग्न होते हैं, खस्वस्तिक में रहने से पूर्वापर वृत्त के सदृश ही क्रान्तिवृत्त और दृग्वृत्त होता है, क्रान्ति वृत्ताकार दृग्वृत्त के रहने से उसी दृग्वृत्त में चन्द्रलम्बित होते हैं इसलिये चन्द्रनति=०, चद्रशर=०, तथा रविनति=०, अतः 'स्पष्टोऽत्र बाणो नतिसंस्कृतः' इत्यादि से स्पष्टशर=०, स्पष्टनति भी=०, वित्रिभलग्नोत्पन्नशर संस्कृत रविनति चन्द्र दृक्क्षेप चाप होता है, रविनति ± वित्रिभलग्नोत्पन्नशर=० ± वित्रिभलग्नोत्पन्नशर=वित्रिभलग्नोत्पन्नशर लेकिन यह चन्द्रदृक्क्षेप चाप निरर्थक है क्योंकि स्पष्टनति ही के लिये दृक्क्षेप चाप का प्रयोजन होता है, लेकिन यहां स्पष्टनति=०, इसलिये दृक्क्षेप से कौन सा काम चलेगा ? भास्करोक्त यह खण्डन ठीक है इति ॥

अत्र 'सौम्य याम्येषुणा रहितयुतमुदीच्या' मिति श्रीपत्युक्तिः 'एक दिशोर्योगो भिन्नदिशोरन्तरमित्युक्त्या'' परिवर्त्तिता भास्करेणेति ॥ ९-१० ॥

अब रवि और चन्द्र के दृग्गति साधन को कहते हैं।

हि. भा.—उत्तर वित्रिभनति रहने पर वित्रिभलग्न से जो उत्तर वा दक्षिण शर है उसको वित्रिभलग्न के शंकुचाप में क्रम से हीन और युत करना, दक्षिण नति में उसी शंकुचाप को उत्तर शरचाप करके जोड़ना और दक्षिण शर चाप करके हीन करना तब जो हो उसकी ज्या चन्द्र का वित्रिभशंकु (दृग्गति) होती है, सूर्य का पूर्वसाधित वित्रिभशंकु ही वास्तव दृग्गति होता है, वित्रिभलग्न दृङ्मण्डल और क्रान्तिवृत्त के सम्पात में सूर्य की दृग्गति होती है, दृक्क्षेपवृत्त और विमण्डल के सम्पात में चन्द्र दृग्गति होती है अर्थात् दृक्क्षेपवृत्त क्रान्तिवृत्त में जहां लगता है उस विन्दु का वित्रिभलग्नशंकु सूर्य की दृग्गति होती है, वही दृक्क्षेपवृत्त चन्द्रविमण्डल में जहां लगता है उस बिन्दु का शंकु चन्द्रदृग्गति है इति ॥ ९-१० ॥

उपपत्ति

दृक्क्षेप वृत्त क्रान्तिवृत्त में जहां लगा है वहां से विमण्डलपर्यन्त दृक्क्षेपवृत्त में स्वल्पान्तर से वित्रिभलग्नशर है। इसलिये वित्रिभ की उत्तर नति और वित्रिभ के उत्तर शर रहने से दोनों का योग करने से खस्वस्तिक से विमण्डल पर्यन्त दृक्क्षेपवृत्त में चन्द्रदृक्क्षेप चापांश= वित्रिभनतांश+वित्रिभशर, इसको नवत्यंश में से घटाने से चन्द्रदृग्गतिचापांश = ९०–वित्रिभन —वित्रिभश=वित्रिभशंचा—वित्रिभश, दक्षिण वित्रिभशर में चन्द्रदृक्क्षेप चापांश=वित्रिभनतांश—वित्रिभश, अतः शंकुचापांश=९०—वित्रिभन+वित्रिभश=वित्रिभशंचा+वित्रिभशर। दक्षिण नति में विलोम संस्कार करना चाहिये। इससे चन्द्रदृग्गति साधन उपपन्न होता है। दृक्क्षेप वृत्त क्रान्ति वृत्त के ऊपर लम्बरूप है, विमराडल के ऊपर नहीं; इसलिये चन्द्रदृक्क्षेप से जो दृग्गति होती है वह कदम्ब प्रोत वृत्त में नहीं होता है अतः उसका संस्कार करने से स्फुट नति नहीं होती है, भास्कराचार्य भी पहले इस मत के अनुसार 'दृज्यैव या वित्रिभलग्नशंकोः" इत्यादि से चन्द्रदृक्क्षेप साधन कर उसके वस से स्पष्टनति कर के पीछे सूर्यग्रहणाधिकारान्त में 'शशिदृक्क्षेपार्थं यद्वित्रिभलग्नेषुणात्र संस्करणम्" इत्यादि से इस आनयन को खण्डित कर दिया, यहां 'सौम्ययाम्येषुणा रहितयुतमुदीच्याम्' इस श्रीपत्युक्ति को भास्कराचार्य ने 'एकदिशोर्योगो भिन्नदिशोरन्तरम्" से परिवर्त्तित कर दिया है, इति ॥ ९-१० ॥

इदानीं स्पष्टां नतिमाह।

त्रिज्यावर्गावूनौ स्वशंकुवर्गेण तत्पदे दृग्ज्ये।
रविशशिमध्यगतिगुणे तिथिगुणितव्यासदल भक्ते ॥११॥

तच्छाया गुणिते वा मध्यगती तिथिगुणस्वकर्णहृते।
फलयोर्दिक् साम्येऽन्तरमवनतिरैक्यं दिगन्यत्वे ॥१२॥

सु० भा०—दृग्ज्ये रविचन्द्रयोर्दृक्क्षेपौ स्वस्वमध्यगतिगुणे तिथिगुणितव्यासदलेन पञ्चदशगुणितत्रिज्यया भक्ते हृते पृथक् पृथक् रविचन्द्रयोर्नती भवतः वा तयोर्मध्यगती तयोर्दृक्क्षेपसमदृग्ज्ययोर्ये छाये ताभ्यां गुणिते तिथिगुणस्वस्वच्छायाकर्णहृते तयोर्नती भवतः। शेषं स्पष्टार्थम्।

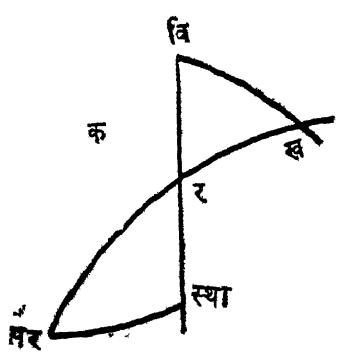
अत्रोपपत्तिः। प्रथमानयनस्य 'सदृक्क्षेपघ्ना इन्दोर्निजमध्यभुक्तितिथ्यंशनिघ्ना' वित्यादि भास्करविधिना स्फुटा।

$$\text{प्रथम प्रकारेण नतिः} = \frac{\text{दृक्षे.}}{\text{त्रि}}\frac{\text{मग}}{15} = \frac{\frac{\text{दृक्षे}}{\text{दिश}}\times\text{मग}}{15\times\frac{\text{त्रि}}{\text{विश}}} = \frac{\text{विछा.}}{\text{विछाक}}\frac{\text{मग}}{}\text{। अनेन}$$

द्वितीयः प्रकारः उपपद्यते। तयोर्नत्योर्दिक्साम्येऽन्तरं दिग्भेदे चैक्यं लंबितरवेर्लम्बितचन्द्रस्य दक्षिणोत्तरमवनतिसंज्ञं गोलविदां स्फुटम् ॥११-१२॥

वि. भा.—त्रिज्यावर्गौ स्वशंकुवर्गेण (रविशंकुवर्गेण-चन्द्रशंकुवर्गेण च) ऊनौ (हीनौ) तयोर्मूले दृग्ज्ये (रविचन्द्रयोर्दृक्क्षेपौ) भवतः, ते स्वस्वमध्यगतिगुणे, तिथि गुणितव्यासदल (पञ्चदशगुणितत्रिज्या) भक्ते तदा रविचन्द्रयोर्नती भवेताम्। वा तयोः (रविचन्द्रयोः) मध्यगती, तच्छायागुणिते (तयोर्दृक्क्षेपतुल्यदृग्ज्ययोर्ये छाये ताभ्यां गुणिते) तिथिगुण स्वकर्णहृते (पञ्चदश गुणित स्वस्वच्छाया कर्ण भक्ते) तदा तयोर्नती भवतः। शेषं स्पष्टमिति ॥ ११-१२ ॥

अत्रोपपत्तिः।



ख=खस्वस्तिकम्। वि=वित्रिभम्। र=क्रान्तिवृत्ते रविः। लंर=लम्बितरविः। खर=रविनतांशाः। खवि=दृक्क्षेप चापम्। खलं=पृष्ठीयनतांशाः। रलं=दृग्लम्बनम्। लंस्था=रविनतिः। स्थार=रविस्पष्ट लम्बनम्। तदा खरवि, रलंस्था चापीय जात्य त्रिभुजयोर्ज्यक्षेत्र साजात्यादनुपातेन $\frac{\text{दृक्क्षेप.}}{\text{रदृग्ज्या}}\frac{\text{रदृलंज्या}}{}$

$= \text{रविनतिज्या} = \text{रविनतिः स्वल्पान्तरात् परन्तु } \frac{\text{रपलज्या. रपृदृज्या}}{\text{त्रि}} = \text{रद्द-}$

लंज्या, तत उत्थापनात् $\frac{\text{दृक्क्षेप. रपलज्या. रपृदृज्या}}{\text{त्रि. रद्दज्या}} = \text{रविनतिः}$, अत्र स्वल्पान्तरात् रपृदृज्या=रद्दज्या तथा रपलज्या=रपलं, तदा $\frac{\text{दृक्क्षेप. रपलं}}{\text{त्रि}}$ =रविनतिः। एवमेव $\frac{\text{दृक्क्षेप. चपलं}}{\text{त्रि}}$ =चंनतिः, परं $\frac{\text{रगक}}{१५}$ = रपलं, तथा $\frac{\text{चंगक}}{१५}$ =चंपलं, तत उत्थापनेन $\frac{\text{दृक्क्षेप. रगक}}{\text{त्रि. १५}}$ =रनति, तथा $\frac{\text{दृक्क्षेप. चंगक}}{\text{त्रि १५}}$ =चंनतिः, एतेन प्रथम प्रकार उपपद्यते। अत्र यद्यपि चन्द्रनतिसाधनार्थं रविनति-साधनवत् क्षेत्रं नोत्पद्यते तथाप्याचार्येण रविनतिसाधनवदेव चन्द्रनतिरपि साधिता।

$$\text{अथ रविनतिः} = \frac{\text{दृक्क्षेप. रगक}}{\text{त्रि. १५}} = \frac{\frac{\text{१२ दृक्क्षे}}{\text{विशं}} \times \text{रगक}}{१५ \times \frac{\text{१२ त्रि}}{\text{विशं}}} = \frac{\text{विछाया. रगक}}{\text{१५. विछाकर्ण.}}$$

$$\text{एवमेव चन्द्रनतिः} = \frac{\text{दृक्षे. चंगक}}{\text{त्रि. १५}} = \frac{\frac{\text{१२ दृक्षे}}{\text{विशं}} \times \text{चंगक}}{१५ \times \frac{\text{१२ त्रि}}{\text{विशं}}} = \frac{\text{विछाया. चंगक}}{\text{१५. विछाकर्ण}}$$, अनयो

रविचन्द्रनत्योः संस्कारेण स्फुटा नतिर्भवतीति। सिद्धान्तशेखरे श्रीपतिना "ततो दृग्ज्ये मध्यस्वगति तिथि १५ भागेन गुणिते-त्रिमौर्व्या संभक्ते भवति विवरं यच्च फलयोः। दिशोः साम्ये भेदे युति" रित्यादिना शिद्धान्त शिरोमणौ भास्कराचार्येण चा "दृक्क्षेप इन्दोर्निजमध्यभुक्तितिथ्यंशनिघ्नौ त्रिगुणोद्धृतौ तौ। नती रवीन्द्वोः समभिन्न दिक्त्वे तदन्तरैक्यं तु नतिः स्फुटाऽत्र" ऽनेनाऽऽचार्योक्तानुरूपमेवोक्तमिति ॥ ११-१२ ॥

अब स्पष्ट नति को कहते हैं।

हि. भा.— त्रिज्यावर्ग में रविशंकुवर्ग को और चन्द्रशंकु वर्ग को घटा कर मूल लेने से रवि और चन्द्र की दृज्या (रविदृक्क्षेप और चन्द्रदृक्क्षेप) होती है, उन (रविदृक्क्षेप और

चन्द्रदृक्क्षेप) को अपनी अपनी मध्यगति से गुणा कर पन्द्रह गुणित त्रिज्या से भाग देने से रवि और चन्द्र की नति होती है। वा रवि मध्यगति और चन्द्र मध्यगति को रवि और चन्द्र के दृक्क्षेप तुल्य दृग्ज्या की छाया से गुणा कर पन्द्रहगुणित अपते अपने छाया कर्ण से भाग देने से उन दोनों (रवि और चन्द्र) की नति होती है। शेष के अर्थ स्पष्ट है इति ॥ ११-१२ ॥

उपपत्ति ।

यहां संस्कृतोपपत्तिस्थ (क) क्षेत्र को देखिये। ख=खस्वस्तिक, वि=विक्षिभ, र=क्रान्तिवृत्त में रवि, खर=रविनतांश, लंर=लम्बितरवि, खवि=दृक्क्षेपचाप, खलं=पृष्ठीयनतांश, रलं=दृग्लम्बन, लंस्था=रविनति, स्थार=रविस्पष्टलम्बन तब. खरवि, रलंस्था दोनों चापीय जात्य त्रिभुजों के ज्याक्षेत्र सजातीय रहने के कारण अनुपात से. $\frac{\text{रदृक्क्षेप. रदृलंज्या}}{\text{रदृज्या}}$ =रविनतिज्या=रविनति (स्वल्पान्तर से), परन्तु $\frac{\text{रपलंज्या. रपृदृज्या}}{\text{त्रि}}$ =रदृलंज्या, उत्थापन से $\frac{\text{रदृक्षेप. रपलंज्या. रपृदृज्या}}{\text{रदृज्या. त्रि}}$ =रविनति, यहां स्वल्पान्तर से रपृदृज्या=रदृज्या, तथा रपलंज्या=रपलं, तब $\frac{\text{दृक्षेप. रपलं}}{\text{त्रि}}$ = रविनति, इसी तरह $\frac{\text{चंदृक्षे. चंपलं}}{\text{त्रि}}$ =चंनति, लेकिन $\frac{\text{रगक}}{१५}$ =रपलं, $\frac{\text{चंगक}}{१५}$ =चंपलं। अतएव $\frac{\text{रदृक्षे. रगक}}{\text{त्रि. १५}}$ =रंविनति, $\frac{\text{चंदृक्षे. चंगक}}{\text{त्रि. १५}}$=चंनति, इससे प्रथम प्रकार उपपन्न हुआ। यहां यद्यपि चंद्रनति साधनार्थ रविनति साधन के सदृश क्षेत्र नहीं बनता है तथापि आचार्य ने रविनति साधन की तरह ही चंद्रनति साधन भी किया है। रविनति=$\frac{\text{रदृक्षे. रगक}}{\text{त्रि. १५}}=\frac{\frac{\text{१२. रदृक्षे}}{\text{विशं}}\times \text{रगक}}{१५\times\frac{\text{१२ त्रि}}{\text{विशं}}}$

$=\frac{\text{विछाया. रगक}}{\text{१५. विछाकर्ण}}$, इसीतरह चंद्रनति $=\frac{\text{चंदृक्षे. चंगक}}{\text{त्रि. १५}}=\frac{\frac{\text{१२ चंदृक्षे}}{\text{विशं}}\times\text{चंगक}}{१५\times\frac{\text{१२ त्रि.}}{\text{विशं}}}$

$=\frac{\text{विछाया. चंगक}}{\text{१५. विछाकर्ण}}$ इन दोनों रविनति और चंद्रनति के संस्कार से स्पष्टनति होती है,

सिद्धांत शेखर में श्रीपति "ततो दृग्ज्ये मध्य स्वगति तिथि १५ भागेन गुणिते" इत्यादि से तथा सिद्धांत शिरोमणि में भास्कराचार्य ने भी "दृक्क्षेप इन्दोर्निजमध्यभुक्तितिथ्यंशनिघ्नौ" इत्यादि से आचार्योक्त के अनुरूप ही कहा हैं इति ॥ ११-१२ ॥

इदानीं स्फुटशर साधनं स्थित्यर्धसाधनं चाह ।

संयोगान्तरमवनतिशशाङ्कविक्षेपयोः समान्यदिशोः ।
स्फुटविक्षेपः शशिवत् स्थित्यर्धविमर्ददलनाडयः ॥ १३ ॥

सु० भा०—स्पष्टार्थमुपपत्तिश्च 'स्पष्टोऽत्र बाणो नतिसंस्कृतोऽस्मा' दित्यादि-भास्करविधिना स्फुटा ॥१३ ॥

वि. भा.—समान्य दिशोः (एक दिक्कयोर्भिन्नदिक्कयोश्च) अवनतिशशाङ्क-विक्षेपयोः (नतिचन्द्रशरयोः) संयोगान्तरं (योगोऽन्तरं) कार्यं तदा स्फुट विक्षेपः (स्पष्टशरः) भवति, शशिवत् (चन्द्रग्रहणवत्) स्थित्यर्धनाड्योविमर्दार्धनाड्यश्च भवेयुरिति ॥ १३ ॥

अत्रोपपत्तिः ।

अथ लम्बितरव्युपरिगतकदम्वप्रोतवृत्तं यत्र क्रान्तिवृत्ते लगति तस्मा-ल्लम्बितरविं यावद्रविनतिः । चन्द्रोपरिगतकदम्वप्रोतवृत्तं क्रान्तिवृत्ते यत्र लगति तस्माच्चन्द्रं यावच्चन्द्रशरः । एतयोः संस्कारेण लम्बितरव्युपरिगतनति-कोटिव्यासार्धवृत्तचन्द्रोपरिगतकदम्वप्रोतवृत्तयोः सम्पाताच्चन्द्रं यावद् भवेदे-तस्यैव नाम सूर्यग्रहणे स्पष्टशरः । एतस्मात् (स्पष्टशरात्) "छाद्येन युतोनस्य छादकमानस्य तद्दलकृतिभ्या" मित्यादिना स्पष्ट स्थित्यर्ध विमर्दार्धयोर्ज्ञानं सुखेनैव भवेदिति । सिद्धान्तशेखरे "दिशोः साम्ये भेदे युतिरवनतिस्तच्छशिशरौ समाशौ चेद्योगोऽन्तरमपरथा स स्फुटशरः" ऽनेन श्रीपतिना सिद्धान्तशिरोमणौ "स्पष्टोऽत्र-बाणो नति संस्कृतोऽस्मात् प्राग्वत् प्रसाध्ये स्थितिमर्दखण्डे" पद्येनानेन भास्करा-चार्येणाप्याचार्योक्तानुरूपमेवोक्तमिति ॥ १३ ॥

अब स्पष्ट शरानयन को और स्पष्टस्थित्यर्ध और स्पष्टविमर्दार्ध साधन को कहते हैं ।

हि. भा.—एक दिशा के चन्द्रशर और नति का योग करने से और भिन्न दिशा के उन दोनों का अन्तर करने से स्पष्टशर होता है, इससे चन्द्रग्रहण की तरह स्थित्यर्ध घटी और विमर्दार्ध घटी होती है इति ॥ १३ ॥

उपपत्ति ।

लम्बितरव्युपरिगत कदम्बप्रोतवृत्तक्रान्तिवृत्त में जहां लगता है वहां से लम्बित रवि-पर्यन्त रविनति है, चन्द्रोपरिगत कदम्बप्रोतवृत्तक्रान्तिवृत्त में जहां लगता है वहां से चन्द्र पर्यन्त चन्द्रशर है, इन दोनों के संस्कार से लम्बित रव्युपरिगत नतिकोटिव्यासार्धवृत्त और चन्द्रोपरिगत कदम्बप्रोतवृत्त के सम्पात से चन्द्रपर्यन्त सूर्यग्रहणोपयुक्त स्पष्टशर है इससे (स्पष्टशर से) 'छाद्येन युतोनस्य छादक मानस्य तद्दलकृतिभ्याम्'' इत्यादि से स्पष्ट स्थित्यर्ध और स्पष्ट विमर्दार्ध का ज्ञान सुगमता ही से हो जायगा, सिद्धान्त शेखर में 'दिशोः साम्ये भेदे युतिरवनतिस्तच्छशिशरौ समाशौ चेद्योगोऽन्तरमपरथा सस्फुटशरः' इससे श्रीपति तथा सिद्धान्त शिरोमणि में 'स्पष्टोऽत्र बाणो नति संस्कृतोऽस्मात् प्राग्वत् प्रसाध्ये स्थितिमर्दखण्डे' इससे भास्कराचार्य ने भी आचार्योक्त के अनुरूप ही कहा हैं इति ॥ १३ ॥

इदानीं स्फुटस्थित्यर्धविमर्दार्धयोरानयनमाह

प्राग्वल्लम्बनमसकृत्तिथ्यन्तात् स्थिति दलेन हीनयुतात् ।
अधिकोनं तन्मध्यादृणयोरूनाधिकं धनयोः ॥ १४ ॥

यद्यधिकं स्थित्यर्धं तदाऽन्तरेणान्यथोनमृणमेकम् ।
अन्यद्धनं तदैक्येनाधिकमेवं विमर्दार्धे ॥ १५ ॥

सु० भा०—तिथ्यन्तात् स्पष्टदर्शान्तात् चन्द्रग्रणवदागतेन स्थितिदलेन हीनयुतात् प्राग्वदसकृद्विधिना लम्बनं कार्यं यावदविशेषः । अत्रैतदुक्तं भवति स्पर्शकालज्ञानाय तिथ्यन्तात् स्थितिदलेनोनात् मोक्षकालज्ञानार्थं च स्थितिदलेन सहितात् तिथ्यन्तादसकृद्विधिना लम्बनं स्थिरं कार्यम् । तल्लम्बनं यदि मध्याद्-दर्शान्तकालिकाल्लम्बनादधिकोनं भवेदर्थात् स्पार्शिकं मध्यकालिकादधिकं मौक्षिकं चोनं भवेत् तथा मध्यस्पर्शकालो भवेन्मध्यमोक्षकालोभवे च द्वे लम्बने ऋ भवतस्तदा तयोर्ऋणयोरन्तरेण पूर्वसाधितं स्थित्यर्धमधिकं कार्यम् । यदि मध्याल्लम्बनात् स्पर्शभवमूनं मोक्षभवमधिकं तथा मध्यस्पर्शभवे मध्यमोक्षभवे च द्वे लम्बने धने भवतस्तदापि तयोर्धनयोरन्तरेण स्थित्यर्धमधिकं कार्यम् । ऋणयोर्ध-नयोश्च यद्यन्यथा भवेत् अर्थादृणयोर्मध्यादूनाधिकं स्पर्शमोक्षभवं धनयोरधिकोनं स्पर्शमोक्षभवं मध्यात् तदा तयोरन्तरेण स्थित्यर्धमूनं कार्यम् । अथ यदि एकं लम्बनमृणमन्यद्धनं भवेत् तदा तयोरैक्येन स्थित्यर्धमधिकं कार्यम् । एवं स्पार्शिकं मौक्षिकं च स्थित्यर्धं भवति । एवं द्वे संमीलनोन्मीलनकालिके विमर्दार्धे च स्फुटे भवतः । अर्थाद्यथा शशिवत् स्थित्यर्धं गृहीत्वा स्फुटस्थित्यर्धं साधितं तथैव स्थित्य-

र्धस्थाने शशिवन्मर्दार्धं गृहीत्वा स्फुटं मर्दार्धद्वयं साध्यमिति ।

अत्रोपपत्तिः । कल्प्यते स्थिरभूतानि स्पर्शमध्यमोक्षकालिकानि लम्बनानि क्रमेण $ल_१$. $ल_२$. $ल_३$. ऋणानि । मध्यमं स्थित्यर्धम्=स्थि । गणितागतदर्शान्तकालः =द, तदा—

$$\text{स्फुटस्पर्शकालः}=द-स्थि-ल_१=स्प$$
$$\text{स्फुटमध्यकालः}=द-ल_२ \quad =म$$
$$\text{स्फुटमोक्षकालः}=द+स्थि-ल_३=मो$$

अतः स्पार्शिकं स्फुटस्थित्यर्धम्=स्पास्फुस्थि=म—स्प=स्थि+$ल_१$—$ल_२$ । मौक्षिकं स्फुट=स्थित्यर्धम्=मौस्फुस्थि=मौ—म=स्थि+$ल_२$—$ल_३$ ।

अतो यदि $ल_१ > ल_२$ तथा $ल_२ > ल_३$ तदा $ल_१-ल_२$, $ल_३-ल_२$ एतद् द्वयं धनम् । यदि तानि लम्बनानि धनानि तदा पूर्वविधिना $ल_१-ल_२$, $ल_२-ल_३$ अनयोर्मा ने $ल_२-ल_१$, $ल_३-ल_२$ भविष्यतः । अतस्तदा यदि $ल_२ > ल_१$ तथा $ल_३ > ल_२$ तदा $ल_२-ल_१$,—$ल_३-ल_२$ एतद् द्वयं धनमन्यथा ऋणमिति स्पष्टम् ।

अथ वित्रिभतः प्राक् स्पर्शः पश्चान्मध्यकालस्तदा $ल_१$ ऋणं $ल_२$ धनम् । तदा $ल_१-ल_२$ मानं $ल_१+ल_२$ इदं धनं भविष्यति । स्पार्शिकलम्बने धने वित्रिभतः पश्चिमस्थे रवौ तयोऽप्यग्रे मध्यकालस्य स्थितेर्मध्यकालिकं लम्बनं सर्वदा धनमेवातस्तत्र धनर्णत्वस्थितिर्न । अथैवं वित्रिभतः प्राग्यदि मध्यकालः पश्चान्मोक्षस्तदा $ल_२$ ऋणं $ल_१$ धनम् । तदा पूर्वविधिना $ल_२-ल_१$ अस्य $ल_२+ल_१$ एतादृशी स्थितिः सर्वदा धनात्मिका स्यादतो पद्येकमृणमन्यद्धनं तदा सर्वदा तदैक्येनाधिकं स्थित्यर्धं स्फुटं स्थित्यर्धं भवतीति मदीयो विचारः सुधीभिर्भृशं विचिन्तनीयो येनाचार्योक्तं स्पष्टमुपपद्यते । स्फुटस्पर्शमोक्षकालाज्ञानात् प्रथमं $ल_१$, $ल_३$ स्थाने स्थूलं $ल_२$ गृहीतम् । तेन स्प=द—स्थि—$ल_२$=द—$ल_२$—स्थि=स्पष्टदर्शान्त—स्थि । एवं स्थूलमोक्षकालः=स्पष्टदर्शान्त+स्थि । अतः स्पष्टतिथ्यन्तात् स्थितिदलोनाधिकात् प्रथमं लम्बनानयनमुचितं स्थितिदलञ्च स्फुटं स्पर्शमोक्षकालिकशरवशात् तात्कालिकमुचितम् । इह तु मध्यकालिकस्पष्टशरवशात् स्थितिदलं स्थिरं गृहीत्वाऽऽनयनं कृतमत इदं स्थूलं सूक्ष्मार्थमग्रे प्रकारान्तरं वक्ष्यत्याचार्यः ॥१४-१५

वि. भा.—तिथ्यन्तात् (स्पष्टदर्शान्तात्) चन्द्रग्रहणवदागतेन स्थित्यर्धेन हीनयुतात्प्राग्वदसकृत् कर्मणालम्बनं कार्यं यावदविशेषः । अत्रैतदुक्तं भवति । स्पर्शकालज्ञानार्थं स्थित्यर्धेन हीनात् मोक्षकालज्ञानार्थं स्थित्यर्धेन सहितात्तिथ्यन्तादसकृत्कर्मणा लम्बनं स्थिरं कार्यं तल्लम्बनं यदि मध्याद्दर्शान्तकालिकाल्लम्बनाद्वि-

कोनं भवेदर्थात् स्पार्शिकं मध्यकालिकादधिकं मौक्षिकं चाल्पं भवेत्तथा मध्यस्पर्शकालोत्पन्ने मध्यमोक्षकालोत्पन्ने च लम्बने ऋणे भवेतां तदा तयोर्ऋणयोरन्तरेण पूर्वसाधितं स्थित्यर्धमधिकं कार्यं यदि मध्यलम्बनात् स्पार्शिकं न्यूनमौक्षिकं चाधिकं तथा मध्यस्पर्शोत्पन्ने मध्यमोक्षोत्पन्ने च द्वे लम्बने धने भवेतां तदाऽपि तयोर्धनयोरन्तरेण स्थित्यर्धकं कार्यम्। ऋणयोर्धनयोश्च यद्यन्यथा भवेदर्थाद्दृणयोर्मध्यादूनाधिकं स्पर्शमोक्षोत्पन्नं धनयोरधिकोनं स्पर्शमोक्षोत्पन्नं मध्यात् तदा तयोरन्तरेण स्थित्यर्धमूनं कार्यम्। अथ यद्येकं लम्बनमृणमन्यद्धनं भवेत्तदा तयोर्योगेन स्थित्यर्धमधिकं कार्यम्। एवं स्पार्शिकं मौक्षिकं च स्थित्यर्धं भवति। एवं संमीलनोन्मीलनकालिके विमर्दार्धे च स्फुटे भवतः। अर्थाद्यथा चन्द्रवत् स्थित्यर्धं गृहीत्वा स्फुटं स्थित्यर्धं साधितं तथैव स्थित्यर्धस्थाने चन्द्रवत् मर्दार्धं गृहीत्वा स्फुटं विमर्दार्धद्वयं साध्यमिति ॥ १४-१५ ॥

अत्रोपपत्तिः

स्थिरीभूतानि स्पर्शमध्यमोक्षकालिकलम्बनानि क्रमेण $ल_१$, $ल_२$, $ल_३$ ऋणानि, मध्यमं स्थित्यर्धम्=स्थि, गणितागतदर्शान्तकालः=द, तदा स्फुटस्पर्शकालः=द−स्थि−$ल_१$=स्प, स्फुटमध्यकालः=द−$ल_२$=म, स्फुटमोक्षकालः=द+स्थि−$ल_३$=मो, अतः स्पार्शिकं स्फुटस्थित्यर्धम्=स्पास्फुस्थि=म−स्प=स्थि+$ल_१$−$ल_२$ मौक्षिकं स्फुटस्थित्यर्धम्=मौस्फुस्थि=मो−म=स्थि+$ल_३$−$ल_२$ अतो यदि $ल_१ > ल_२$ तथा $ल_३ > ल_२$ तथैव $ल_१ − ल_२$, $ल_३ − ल_२$ तद्द्वयं धनम्, यदि तानि लम्बनानि धनानि तदा $ल_१ − ल_२$, $ल_३ − ल_२$ ऽनयोर्मानै $ल_२ − ल_१$, $ल_३ − ल_२$ भविष्यतः। यदि $ल_२ > ल_१$, तथा $ल_३ > ल_२$ तदा $ल_२ − ल_१$, $ल_३ − ल_२$ एतद्द्वयं धनमन्यथा ऋणमिति।

अथ वित्रिभतः पूर्वं स्पर्शः पश्चान्मध्यकालस्तदा $ल_१$ ऋणं, $ल_२$ धनम्। तदा $ल_१ − ल_२$ मानं $ल_१ + ल_२$ इदं धनं भविष्यति स्पार्शिकलम्बने धने वित्रिभतः पश्चिमस्थे रवौ ततोऽप्यग्रे मध्यकालस्यस्थितेर्मध्यकालिकं लम्बनं सर्वदा धनमेवातस्तत्र धनर्णत्वस्थितिर्न। अथैवं वित्रिभतः पूर्वदिशि यदि मध्यकालः पश्चान्मोक्षस्तदा $ल_२$ ऋणं $ल_१$ धनम्। तदा पूर्वं विधिना $ल_२ − ल_१$ अस्य $ल_२ + ल_१$ एतादृशी स्थितिः सर्वदा धनात्मिका स्यादतो यद्येकमृणमन्यद्धनं तदा सर्वदा तदैक्येनाधिकं स्थित्यर्धं स्फुटं स्थित्यर्धं भवतीति। स्फुटस्पर्शमोक्षकालाज्ञानात् प्रथमं $ल_१$, $ल_३$ स्थाने स्थूलं $ल_२$ गृहीतम्, तेन स्प=द−स्थि−$ल_२$=द−$ल_२$−स्थि=स्पष्टदर्शान्त−स्थित, एवं स्थूलमोक्षकालः=स्पष्ट दर्शान्त+स्थि, अतः स्पष्ट तिथ्यन्तात् स्थित्यर्धोनाधिकात् प्रथमं लम्बनानयनमुचितं। स्थित्यर्धं च स्फुटं स्पर्शमोक्षकालिक शरवशात् तात्का-

लिकमुचितम्। परन्त्वत्र मध्यकालिक स्पष्टशरवशात् स्थित्यर्धं स्थिरं गृहीत्वाऽऽनयनं कृतमतः स्थूलं सूक्ष्मार्थमग्रे प्रकारान्तरं कथयत्याचार्य इति। सिद्धान्तशेखरे "तिथ्यन्तात् स्थिति खण्ड हीनसहितात् प्राग्वत्ततोलम्बनं कुर्यात् प्रग्रहमोक्षयोः स्थितिदलं युक्तं विधायासकृत्। तन्मध्यग्रहणोत्थ लम्बनभुवा विश्लेषणानेहसा मर्दार्धोन युतात्तिथेरपि तथा सम्मीलनोन्मीलने। अधिकमृणयोराद्यं मध्यात्तथा ऽन्त्यमिहाल्पकं भवति धनयोश्चाद्यं हीनं यदाऽधिकमंतिमम्। नमनविवरेणैवं कुर्याद्विहीनमतोऽन्यथा स्थितिदलस्वस्थे भेदे तदैक्ययुतं पुनः" ऽनेन श्रीपतिना, "स्थित्यर्धोनाधिकात् प्राग्वत् तिथ्यन्ताल्लम्बनं पुनः। ग्रासमोक्षोद्भवं साध्यं तन्मध्यहरिजान्तरम्॥ प्राक्कपालेऽधिकं मध्याद् भवेत् प्रग्रहणं यदि। मौक्षिकं लम्बनं हीनं पश्चार्धेतु विपर्ययः॥ तदा मोक्ष स्थितिदले देयं प्रग्रहणे तथा। हरिजान्तरकं शोध्यं यत्रैतत् स्याद्विपर्ययः॥ एतदुक्तं कपालैक्ये तद् भेदे लम्बनैकता। स्वे स्वे स्थितिदले योज्या विमर्दार्धेऽपि चोक्तवत्" इति सूर्य सिद्धान्तोक्तेरा चार्योक्तेश्च सर्वथा सदृशमेवोक्तमिति॥ १४-१५

अथ स्फुट स्थित्यर्ध और स्फुटविमर्दार्ध के साधन को कहते हैं।

हि. भा.—चन्द्रग्रहणवत् आये हुये स्थित्यर्ध करके हीन और युत तिथ्यन्त (स्पष्टदर्शान्त) से पूर्ववत् असकृत् विधि से लम्बन साधन करना जब तक अविशेष हो, यहां यह कहा जाता है कि स्पर्श काल ज्ञान के लिये स्थित्यर्ध रहित तिथ्यन्त से तथा मोक्ष काल ज्ञान के लिये स्थित्यर्ध सहित तिथ्यन्त से असकृत् विधि से लम्बन को स्थिर करना, यदि वह लम्बन मध्यदर्शान्त कालिक लम्बन से अधिक और ऊन (अल्प) हो अर्थात् स्पार्शिक लम्बन मध्यकालिक लम्बन से अधिक और मौक्षिक लम्बन अल्प हो तथा मध्यकालिक लम्बन और स्पर्शकालिक लम्बन दोनों ऋण हो तब उन दोनों ऋणों के अन्तर को पूर्व साधित स्थित्यर्ध में जोड़ देना चाहिये, यदि मध्यलम्बन से स्पार्शिक लम्बन न्यून हो और मौक्षिक लम्बन अधिक हो तथा मध्यकालिक और स्पर्शकालिक लम्बन तथा मध्यकालिक और मोक्ष कालिक लम्बन धन हो तो भी उन दोनों धनों के अन्तर को स्थित्यर्ध में जोड़ना चाहिये। यदि एक लम्बन ऋण रहे और दूसरा लम्बन धन रहे तब दोनों के योग को स्थित्यर्ध में जोड़ना चाहिये। इस तरह स्पार्शिक और मौक्षिक स्थित्यर्ध स्फुट होता है, एवं संमीलन कालिक विमर्दार्ध और उन्मीलन कालिक विमर्दार्ध स्फुट होता है अर्थात् जैसे चन्द्रवत् स्थित्यर्ध को लेकर स्फुट स्थित्यर्ध साधित होता है उसी तरह स्थित्यर्ध स्थान में चन्द्रवत् विमर्दार्ध को ग्रहण करके स्फुट विमर्दार्ध साधन करना इति॥ १४-१५॥

उपपत्ति।

स्थिरी भूत स्पर्श कालिक, मध्यकालिक और मोक्ष कालिक लम्बन क्रम से $ल_१$, $ल_२$, $ल_३$, ऋण है, मध्यम स्थित्यर्ध=स्थि, गणितागत दर्शान्त काल=द, तब स्फुट स्पर्शकाल=द,—स्थि—$ल_१$=स्प, स्फुट मध्यकाल=द—$ल_२$=म, स्फुट मोक्ष काल=द+स्थि—$ल_३$

=मो, अतः स्पार्शिक स्फुट स्थित्यर्ध=स्यास्फुस्थि=म−स्प=स्थि+$ल_१$−$ल_२$ मौक्षिक स्फुट स्थित्यर्ध=मौस्फुस्थि=मो−म=स्थि+$ल_२$−$ल_३$, इसलिये यदि $ल_१ > ल_२$ तथा $ल_२ > ल_३$ तब $ल_१$−$ल_२$, $ल_२$−$ल_३$ यह दोनों धन होते हैं, यदि वे लम्बन धन हैं तब $ल_१$−$ल_२$, $ल_२$−$ल_३$ इन दोनों के मान $ल_२$−$ल_१$, $ल_३$−$ल_२$ होगें। यदि $ल_२ > ल_१$ तथा $ल_३ > ल_२$ तब $ल_२$−$ल_१$, $ल_३$−$ल_२$ तब ये दोनों धन होंगें अन्यथा ऋण होगें। वित्रिभ से पूर्व स्पर्श और पश्चात् मध्यकाल रहने से $ल_१$ यह ऋण, $ल_२$ यह धन होगा, तब $ल_१$−$ल_२$ मान $ल_१+ल_२$ यह धन होगा, स्पार्शिक लम्बन के धन रहने से वित्रिभ से पश्चिम रवि के रहने से उससे भी आगे मध्यकाल स्थिति के कारण मध्य कालिक लम्बन सर्वदा धन ही होता है इसलिये वहां धनत्व स्थिति और ऋणत्व स्थिति नहीं होती है। इसी तरह वित्रिभ से पूर्व दिशा में यदि मध्यकाल है और पश्चात् मोक्षकाल तब $ल_२$ ऋण होता है और $ल_३$ धन होता है। तब पूर्व विधि से $ल_२$−$ल_३$, इसकी $ल_२+ल_३$ ऐसी स्थिति सदा धनात्मक होती है, इसलिये यदि एक ऋण हो और अन्य धन हो तब सदा उन दोनों के योग को स्थित्यर्ध में जोड़ने से स्फुट स्थित्यर्ध होता है, स्फुट स्पर्शकाल और स्फुट मोक्ष काल के अज्ञात रहने के कारण पहले $ल_१$, $ल_३$ के स्थान में स्थूल $ल_२$ ग्रहण किया गया, तब स्प=द−स्थि−$ल_२$=द−$ल_२$−स्थि=स्पष्ट−दर्शान्ति−स्थि, एवं स्थूल मोक्षकाल=स्पष्टदर्शान्ति+स्थि, इसलिये स्थित्यर्ध कर के हीन, युत स्पष्ट तिथ्यन्त से पहले लम्बनानयन उचित है, और तात्कालिक स्फुट स्थित्यर्ध स्पर्शकालिक और मोक्ष कालिक शरवश से उचित है लेकिन यहां मध्यकालिक स्पष्टशरवश से स्थिर स्थित्यर्ध ग्रहण कर आनयन किया गया है इसलिये यह स्थूल है सूक्ष्मार्थ के वास्ते आगे आचार्य प्रकारान्तर को कहते हैं इति, सिद्धान्त शेखर में "तिथ्यन्तात् स्थितिखण्ड हीनसहितात्" इत्यादि संस्कृतोपपत्ति में लिखित पद्यों से श्रीपति तथा "स्थित्यर्धोनाधिकात् प्राग्वत्" इत्यादि से सूर्य सिद्धान्तकार ने आचार्योक्त के सदृश ही कहा हैं इति ॥ १४-१५ ॥

अथ प्रकारान्तरेण स्फुटस्थित्यर्धविमर्दार्धसाधनमाह ।

स्फुटतिथ्यन्ताल्लम्बनमसकृत् स्थित्यर्धहीनयुक्ताद्वा ।
तत्स्फुटविक्षेपकृतस्थित्यर्धोनयुततिथ्यन्तात् ॥ १६ ॥

तत्स्पष्टतिथिच्छेदान्तरे स्फुटे दिनदले विहीनयुतात् ।
स्वविमर्दार्धेनासकृदेवं स्पष्टे विमर्दार्धे ॥ १७ ॥

सु० भा०—प्रथमं स्थित्यर्धोनयुतात् स्फुटतिथ्यन्तात् स्पष्टदर्शान्ताल्लम्बनमानेयम् । पुनस्तत्स्फुटविक्षेपकृतस्थित्यर्धोनयुततिथ्यन्ताल्लम्बनमानेयमेवमसकृत् । अत्रैतदुक्तं भवति। यथा स्पर्शकालज्ञानाय प्रथमं मध्यकालिकस्पष्टशर-

वशात् स्थित्यर्धं साध्यं तदूनात् स्फुटतिथ्यन्ताल्लम्बनं युतिश्च साध्या। तात्कालिकसपातचन्द्रवशात् शरः साध्यः। नतिशरसंस्कारेण स्पष्टविक्षेपः साध्यस्तस्माच्छशिग्रहणवत् स्थित्यर्धं साध्यं तदूनात् तिथ्यन्ताद् गुणितागतात् पुनर्लम्बनं, नतिः स्पष्टशरः स्फुटस्थित्यर्धं च सर्वमानेयं यावदविशेषः। अविशेषे यल्लम्बनं तत् तात्कालिकस्फुट विक्षेपजनितस्थित्यर्धोनगणितागतदर्शान्ते यथागतं धनं वा ऋणं संस्कार्यम्। एवं स्फुटः स्पर्शकालः। एवं स्थित्यर्धयुक्तात् तिथ्यन्तात् स्फुटमोक्षकालो भवति। तत्स्फुटतिथिच्छेदान्तरे स्फुटे स्थितिदले भवतः। अर्थात् तत्स्फुटस्पर्शकालस्य वा स्फुटमोक्षकालस्य स्पष्टतिथिच्छेदस्य स्पष्टदर्शान्तकालस्य चान्तरे स्पर्शमौक्षिके स्पष्टे स्थितिखण्डे भवेताम्। एवं यथा स्थित्यर्धेन स्पष्टे स्थितिखण्डे साधिते तथैवासकृत्कर्मणा स्वविमर्दार्धेन स्पष्टे विमर्दार्धे साध्ये। अर्थात् पूर्ववदसकृद्विधिना स्फुटौ संमीलनोन्मीलनकालौ प्रसाध्य तन्मध्यकालान्तरयोः समाने स्पष्टे विमर्दखण्डे ज्ञेये। 'तिथ्यन्ताद्गणितागतात् स्थितिदलेनोनाधिकाल्लम्बन' मित्यादिभास्करोक्तमेतदनुरूपम्। इहाचार्येण प्रथमं मध्यकाललम्बनमेव स्थूलात् स्पर्शे मोक्षे च कल्पितमित्यनेनासकृत्कर्मणि न काचिद्धानिरिति ज्योतिर्विदां प्रसिद्धमिति।

अत्रोपपत्तिः। गर्भक्षितिजात् कियतीष्टघटिकासु पृष्ठाभिप्रायेण स्पर्शकालो भवतीति किलापेक्षितम्। कल्प्यते यदा पृष्ठाभिप्रायेण स्पर्शोऽभूत् तदा वि=रविकेन्द्रम्=रविस्थानम्। वि,=चन्द्रबिम्बकेन्द्रम्। च=चन्द्रस्थानम्। चवर=क्रान्तिवृत्तखण्डम्। ख=खस्विस्तकम्। खवि, वि,=चन्द्रदृङ्मण्डलम्। ख वि वि=रविदृङ्मण्डलम्। वि,=लम्बितचन्द्रबिम्बकेन्द्रम्। वि=लम्बित रविबिम्बकेन्द्रम्। वि, च=चन्द्रस्य स्फुटशरः। विर=रविनतिसमो रवेः स्पष्टशरः।

विस=क्रान्तिवृत्त समानान्तरवृत्तम्। वि,स=स्फुटस्पर्शकाले स्पष्टशरः।

वि क,=मानैक्यार्धम्। एतानि चापानि लघुत्वात् सरलरेखाकाराण्याचार्यैः कल्पितानि। अतो वि वि, सजात्ये वि वि, वि, सवर्गान्तरपदसमाः स्फुटस्थित्यर्धकलाः=वि स=र च=स्फुस्थिक। चच=चन्द्रलम्बन कलाः =चलंक। वि=र रविलम्बन कलाः=रलंक। अनयो रविचन्द्रलम्बनयोरन्तरकलाः =(ल, क) षष्टिगुणा रविचन्द्रगत्यन्तरकलाहृता आचार्यसाधिताः स्पर्शे लम्बननाड्यः =ल, तथा स्फुटस्थित्यर्धकलाः=(स्फुस्थिक) षष्टिगुणा गत्यन्तर हृता आचार्योक्तं स्फुटं स्थित्यर्धं घटिकात्मकम्=स्फुस्थि। अथैतस्मिन् स्पर्शकाले गर्भाभिप्रायेण

रविचन्द्रान्तरकलाः = चवि=चर—वि र=च र +च च—विर=स्फुस्थिक +चलक—रलक=स्फुस्थिक +ल, क। एतद्घटिकाः=स्फुस्थि+ल, । गणितागतदर्शान्ते गर्भाभिप्रायेण रविचन्द्रान्तराभावः स्पर्शकाले च ततः प्रागानीतान्तरं तयोः स्फुस्थि+ल, एतावतीषु घटिकासु। अतो गणितागतदर्शान्तिकाला दागता इष्टघटिका विशोध्य स्फुटस्पर्शकालः।

=द—स्फुस्थि—ल, । स्फुटस्पर्शकाले स्पष्टस्थितिदललम्बनयोरज्ञानादसकृत्कर्मोचितमेव। एवं मोक्षसंमीलनोन्मीलनेष्वपि क्षेत्र संस्थया स्फुटा वासना। अनया मत्क्षेत्रयुक्त्या भास्करोक्तं च सर्वं स्फुटमुपपद्यते शेष वासना स्फुटा ॥ १६-१७ ॥

वि. भा.—स्थित्यर्धहीनयुक्तात् स्फुटतिथ्यन्तात् (स्पष्टदर्शान्तात्) लम्बनमानेयम्। पुनस्तत्स्फुटविक्षेपकृतस्थित्यर्धोनयुततिथ्यन्ताल्लम्बनं साध्यमेवमसकृत्। अत्रैतदुक्तं भवति यथा स्पर्शकालज्ञानाय प्रथमं मध्यकालिकस्पष्टशरवशात् स्थित्यर्धं साध्यं तदूनात् स्फुटतिथ्यन्तात् लम्बनं नतिश्च साध्या। तात्कालिकसपातचन्द्रवशाच्छरः साध्यः, नतिशरसंस्कारेण स्पष्टविक्षेपः साध्यः। तस्माच्चन्द्रग्रहणवत् स्थित्यर्धं साध्यं तदूनात् तिथ्यन्तात्पुनर्लम्बनं नतिः स्पष्टशरः स्फुटस्थित्यर्धं च सर्वमानेयं यावदविशेषः, अविशेषे यल्लम्बनं तत्तात्कालिकस्फुट विक्षेपजनितस्थित्यर्धोनगणितागतदर्शान्ते यथागतं धनं वा ऋणं संस्कार्यम्। तदा स्फुटः स्पर्शकालः। एवं स्थित्यर्धयुक्तात्तिथ्यन्तात् स्फुटमोक्ष कालो भवति, तत् स्फुटतिथिच्छेदान्तरे स्फुटे स्थित्यर्धे भवतः। अर्थात् तस्फुटस्पर्शकालस्य वा स्फुट मोक्ष कालस्य स्पष्टतिथिच्छेदस्या (स्पष्टदर्शान्त कालस्य च) न्तरे स्पर्शमौक्षिके स्पष्टे स्थित्यर्धे भवेताम्। एवं यथा स्थित्यर्धेन स्पष्टे स्थित्यर्धे साधिते तथैवासकृत्कर्मणा स्वविमर्दार्धेन स्पष्टे विमर्दार्धे साध्ये। अर्थात् पूर्ववदसकृ द्विधिना स्फुटौ संमीलनोन्मीलनकालौ प्रसाध्य तन्मध्यकालान्तरयोः समाने स्पष्टे विमर्दार्धे ज्ञेये, अत्राचार्येण प्रथमं मध्यकाललम्बनमेव स्थूलात् स्पर्शे मोक्षे च कल्पितं तेनासकृत्कर्मणि न काचिद्धानिरिति ॥ १६-१७ ॥

अत्रोपपत्तिः।

तत्र तावत्पूर्वकपाले विचार्यते।

गर्भीय दर्शा तात्पूर्वं पृष्ठीय दर्शान्तत्वात्स्वत एव गर्भीय दर्शान्तात्पूर्वमेव पृष्ठीय स्पर्शकालो भवेत्। पृष्ठाभिप्रायिकस्पर्शकाले गर्भाभिप्रायिकरविचन्द्रान्तर-

कलाः पृष्ठाभिप्रायिकस्थित्यर्धकला लम्बनान्तरकलयोर्योगतुल्या भवन्ति, कथमिति चेत् स्पर्शकाले सूर्यस्थानाच्चन्द्रस्थानं पश्चिमदिश्येव भवति, गर्भाभिप्रायिकरविस्थानात् पृष्ठाभिप्रायिकरविस्थानं प्राक् दिशि स्यात् । तथैव चन्द्रस्यापि, पृष्ठाभिप्रायिकरविकेन्द्रगतकदम्बप्रोतवृत्तचन्द्रकेन्द्रगतकदम्बप्रोतवृत्तयोरन्तरकलाः क्रान्तिवृत्ते स्थित्यर्धकलाः । अतः स्थित्यर्धकलासु चन्द्रलम्बनकलायुक्ता कार्या तदा गर्भाभिप्रायिकचन्द्रस्थानात् पृष्ठाभिप्रायिकरविस्थानं यावद् क्रान्तिवृत्ते कलाः= स्थि$\frac{1}{2}$कला+चंलं, अत्र रविलम्बनकला हीना कार्या तदा गर्भाभिप्रायिकरविचन्द्रान्तरकलाः=स्थि$\frac{1}{2}$ कला+चंलं—रलं, यदा रविलम्बनकलाभ्यः स्थित्यर्धकला अल्पास्तदा रविलम्बनकलासु स्थित्यर्धकला हीनाः कार्यास्तदा पृष्ठाभिप्रायिकचन्द्रस्थानात् गर्भाभिप्रायिकरविस्थानं यावद् भवेत् तत्स्वरूपम्=रलक—स्थि$\frac{1}{2}$क, एताः कलाश्चन्द्रलम्बनकलायां हीना कार्यास्तदा गर्भाभिप्रायेण रविचन्द्रान्तरकलाः =चंलक—(रलक—स्थि$\frac{1}{2}$क)=चंलक—रलक+स्थि$\frac{1}{2}$क=स्थि$\frac{1}{2}$क+लम्बनान्तर, अथ घट्यात्मककरणार्थमनुपातः

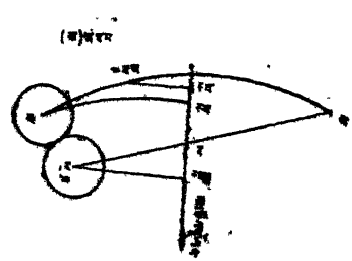
$$\frac{६०\times\text{गर्भीयरविचन्द्रान्तरकला}}{\text{रविचन्द्रगत्यन्तरक}}=\frac{६०\times(\text{स्थि}\frac{1}{2}\text{क}+\text{लम्बनान्तरक})}{\text{रविचन्द्रगत्यन्तरक}}=$$

$$\frac{६०\times\text{स्थि}\frac{1}{2}\text{क}}{\text{रविचंगत्यन्तरक}}+\frac{६०\times\text{लम्बनान्तरक}}{\text{रविचंगत्यन्तरक}}$$

=स्थि$\frac{1}{2}$ घटी+लम्बनान्तरघटी, एतावतीभिर्घटीभिर्गर्भीयदर्शान्तात्पूर्वं पृष्ठाभिप्रायेण स्पर्शकालो भवेदिति सिद्धः । गर्भीयदर्शान्तघट्यामेतच्छोधनेन पृष्ठाभिप्रायेण स्पर्शकालघटिका भवेयुः ।

गर्भीयदर्शान्तघ—(स्थि$\frac{1}{2}$घ+लम्बनान्तरघ)=गर्भीयदर्शान्तघ—स्थि$\frac{1}{2}$घ—लम्बनान्तरघ.

अतः सिद्धं यत् पूर्वकपाले पृष्ठाभिप्रायेण स्पर्शकाले गर्भीयदर्शान्तघट्यां पृष्ठाभिप्रायेण स्थित्यर्धघटिकास्तथा लम्बनान्तरघटिकाश्च हीना कार्यास्तदा पृष्ठाभिप्रायेण स्पर्शकालघटिका भवन्तीति । अत्र स्थि$\frac{1}{2}$क=पृष्ठीय स्थि$\frac{1}{2}$कलाः सर्वत्र बोध्याः । स्थि$\frac{1}{2}$घ=पृष्ठीय स्थि$\frac{1}{2}$घ बोध्या ।



ख=खस्वस्तिकम् । र=गर्भीयरविः ।

र=लम्बितरविः=पृष्ठीयरविः ।

स्था=लम्बितरविस्थानम् ।

गचं=गर्भीयचन्द्रः । चं=लम्बितचन्द्रः

=पृचं

स्था=गर्भीयचन्द्रस्थानम् । स्था$_1$=लम्बित[illegible]स्थानम् ।

पश्चिमकपाले विचार्यते

गर्भीयदर्शान्तादग्रे पृष्ठीयदर्शान्तः स्यात् पृष्ठीयदर्शान्तात्पूर्वं पृष्ठाभिप्रायेण स्पर्शकालोऽतः पृष्ठीयदर्शान्तात्पूर्वं पृष्ठाभिप्रायेण स्पर्शकालगर्भीयदर्शान्तयोः सद्भावात् सम्भवोऽस्ति यत् गर्भीयदर्शान्तात्पूर्वं पृष्ठाभिप्रायेण स्पर्शकालो भवेत्। तथा कदाचिदग्रतोऽपि। यदा पृष्ठीयदर्शान्तकाले गर्भाभिप्रायिकरविस्थानात् पृष्ठाभिप्रायिकरविस्थानं पश्चिमादिशि भवेत्तदा गर्भीयदर्शान्तात्पूर्वमेव पृष्ठाभिप्रायिकस्पर्शकालः स्यात्। यतो गर्भीयदर्शान्तात्पूर्वं गर्भाभिप्रायेण रविस्थानात्पश्चिमदिश्येव गर्भाभिप्रायेण चन्द्रस्थानं भवेत्। अथ यदा स्पर्शकाले गर्भाभिप्रायेण रविस्थानात् गर्भाभिप्रायिकचन्द्रस्थानं पूर्वदिशि भवेत्तदा गर्भीयदर्शान्तादग्रे पृष्ठाभिप्रायिकस्पर्शकालो भवेद्यतो गर्भीयदर्शान्तादग्रे पूर्वदिश्येव गर्भाभिप्रायिकरविचन्द्रयोः स्थानं स्यादिति। ननु स्पर्शकाले गर्भाभिप्रायेण रविस्थानात् गर्भाभिप्रायेण चन्द्रस्थानं प्राक् पश्चादिति कथं ज्ञायते। यदि लम्बनघटीयुतो गर्भीयदर्शान्तः पृष्ठीयदर्शान्तो भवेत्ताभ्यः घटिकाभ्यो यदि पृष्ठाभिप्रायेण स्थित्यर्धघटिकाऽल्पा तदा पश्चिमदिशि गर्भाभिप्रायेण रविस्थानाच्चन्द्रस्थानमित्यन्यथा प्रागिदशीति। अथ कदा गर्भीयदर्शान्तात् प्राक् पृष्ठाभिप्रायेण स्पर्शकाल इत्युच्यते। पृष्ठीयस्थित्यर्धकलासु. चन्द्रलम्बनकला हीनाः कार्यास्तदा गर्भाभिप्रायेण चन्द्रस्थानात्पृष्ठाभिप्रायेण रविस्थानं यावत् क्रान्तिवृत्ते कलाः=स्थि$\frac{1}{2}$क—चंलंक अत्र रविलम्बनकला युता तदा-पृष्ठाभिप्रायेण रविचन्द्रान्तरकलाः=स्थि$\frac{1}{2}$क—चंलंक+रलक, अथ चन्द्रलम्बनकलाभ्यो रविलम्बनकलानामल्पत्वात् द्वयोरन्तरमृणात्मकमेव तदा गर्भीय-रविचन्द्रान्तरकलाः=पृस्थि$\frac{1}{2}$क—लम्बनान्तरकला, ततोऽनुपातेन

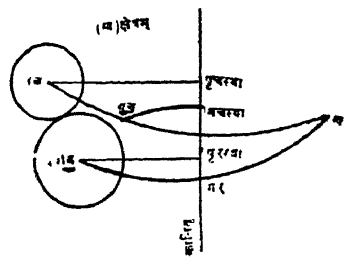
$$\frac{६०\times(\text{पृस्थि}\frac{1}{2}\text{क}-\text{लम्बनान्तरक})}{\text{गत्यन्तरक}}=\text{स्थि}\frac{1}{2}\text{घ}-\text{लम्बनान्तरघ},$$

गर्भीयदर्शान्तात्पूर्वं पृष्ठाभिप्रायेण स्पर्शकालकल्पनया गर्भीयदर्शान्ते (स्थि$\frac{1}{2}$घ—लग्रंघ) हीना कार्यास्तदा पृष्ठाभिप्रायेण स्पर्शकालः=गदर्शान्तघ—पृस्थि$\frac{1}{2}$घ+लंग्रंघ अतः सिद्धं यत् पश्चिमकपालेऽपि यदा गर्भीयदर्शान्तात्पूर्वमेव पृष्ठाभिप्रायेण स्पर्शकालस्तदा दर्शान्ते स्थित्यर्धघटीहीनैव कार्या लम्बनान्तरं युतं कार्यमिति। यदा गर्भीयदर्शान्तादग्रे पृष्ठाभिप्रायेण स्पर्शकालस्तदोच्यते। चन्द्रलम्बनकलासु स्थित्यर्धकला हीनास्तदा पृष्ठाभिप्रायेण रविस्थानाद् गर्भाभिप्रायेण चन्द्रस्थानावधिक्रान्तिवृत्ते कलाः=चंलंक—स्थि$\frac{1}{2}$क अत्र रविलम्बनकला हीनाः कार्यास्तदा गर्भाभिप्रायेण रविचन्द्रान्तरकलाः=चंलंक—स्थि$\frac{1}{2}$क—रलंक=लम्बनान्तरक—स्थि$\frac{1}{2}$क

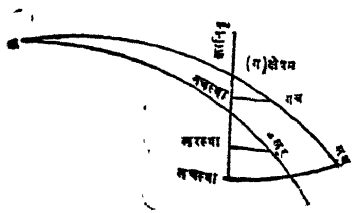
$$\text{ततोऽनुपातेन}\ \frac{६०\ (\text{लम्बनान्तरक}-\text{स्थि}\frac{1}{2}\text{क})}{\text{गत्यन्तरक}}=\text{लम्बनान्तरघ}-\text{स्थि}\frac{1}{2}\text{घ},$$

गर्भीय-

दर्शान्तादग्रे पृष्ठीयस्पर्शकालकल्पनयैताः घटिका (लम्बनान्तरघ—स्थि½घ) गर्भीयदर्शान्ते युतास्तदा पृष्ठाभिप्रायेण स्पर्शकालः=दघ—स्थि½घ + लम्बनान्तरघ, अतः सिद्धं स्फुटतिथ्यन्तात् स्थित्यर्धहीनयुक्तादित्यत्र स्थित्यर्धघटी सर्वदा गर्भीयदर्शान्ते हीनैव कार्या प्राक् पश्चिमकपाले, लम्बनान्तरमृणं धनं च कार्यमिति।



पश्चिमकपाले गर्भीयदर्शान्तापूर्वं यदा पृष्ठाभिप्रायेण स्पर्शकालस्तदा क्षेत्रम्



सिद्धान्तशेखरे "स्थित्यर्धोनयुतात् परिस्फटतिथेः स्याल्लम्बनं पूर्ववत् मन्मध्यग्रहवे च मध्यतिथो ततस्तु तिथौ। स्थित्यर्धेन परिस्फुटेषु जनितेनोनाधिकाद्वाऽसकृत् ततिथ्यन्तरनाडिकाः स्थितिदलेस्तः स्पर्शमुक्त्योः स्फुटे" ऽनेन श्रीपतिनाऽऽचार्योक्तानुरूपमेवोक्तं परमस्य श्लोकस्य द्वितीयचरणं भ्रष्टं पाठसम्वलितमस्ति, सिद्धान्तशिरोमणौ "तिथ्यन्ताद् गणितागतात् स्थितिदलेनोनाधिकाल्लम्बनं तत्कालोत्थनतीषु संस्कृतभवस्थित्यर्धहीनाधिके। दर्शान्ते गणितागते धनमृणं वा तद्विधायासकृज् ज्ञेयौ ग्रग्रहमोक्षसंज्ञसमयावेवं क्रमात् स्फुटौ" इति भास्करोक्तं सर्वथैवाचार्योक्तानुरूपमिति ॥१६-१७॥

अब प्रकारान्तर से स्थित्यर्धसाधन को कहते हैं

हि. भा.—पहले स्थित्यर्ध करके हीन और युत स्फुटतिथ्यन्त (स्पष्टदर्शान्त) से लम्बन लाना चाहिये, फिर उसे स्पष्टशर जनित स्थित्यर्ध करके हीन और युत तिथ्यन्त से लम्बन लाना इस तरह असकृत् करना, यहाँ यह कहा जाता है कि स्पर्शकालज्ञान के लिये पहले मध्यकालिक स्पष्टशरवश से स्थित्यर्ध साधन करना, उसको स्पष्ट तिथ्यन्त में घटाकर जो हो उससे लम्बन और नति साधन करना, तात्कालिक सपात चन्द्र से शर-साधन करना, नति और शर के संस्कार से स्पष्ट शर साधन करना, उससे चन्द्रग्रहण की तरह स्थित्यर्ध साधन करना, तिथ्यन्त में उसको घटाने से जो हो उससे पुनः लम्बन नति, स्पष्टशर और स्फुट स्थित्यर्ध सब कुछ लाना जब तक अविशेष (विशेषता से रहित) हो तब तक इसी तरह करना, अविशेष होने पर जो लम्बन हो उसको तात्कालिक स्पष्टशर जनित स्थित्यर्ध करके हीन गणितागतदर्शान्त में धन वा ऋण करना, तब स्फुट स्पर्श-

होता है, इस तरह स्थित्यर्ध युक्त तिथ्यन्त से स्फुट मोक्षकाल होता है, उस स्फुट स्पर्शकाल वा स्फुट मोक्षकाल का तथा स्पष्ट दर्शान्त काल का अन्तर स्पार्शिक और मौक्षिक स्पष्ट स्थित्यर्ध होते हैं, एवं जैसे स्थित्यर्ध से स्पष्ट स्थित्यर्ध साधन किये गये हैं उसी तरह असकृत् कर्म से अपने विमर्दार्ध से स्पष्ट विमर्दार्ध साधन करना चाहिये, अर्थात् पूर्ववत् असकृत् कर्म से स्फुट संमीलन काल और स्फुट उन्मीलन काल का साधन कर उनके और मध्यकाल के अन्तर के बराबर स्पष्ट विमर्दार्धद्वय समझना चाहिये। यहां आचार्य ने पहले मध्यकाल लम्बन ही को स्थूल से स्पर्शकाल में और मोक्षकाल में स्वीकार किया है उससे असकृत् कर्म में कोई हानि नहीं हैं ॥१६-१७॥

उपपत्ति

पहले पूर्वकपाल में विचार करते हैं। गर्भीय दर्शान्त से पूर्व पृष्ठीय दर्शान्त के होने के कारण स्वतः गर्भीय दर्शान्त से पहले पृष्ठीय स्पर्शकाल होता हैं, पृष्ठाभिप्रायिक स्पर्शकाल में पृष्ठाभिप्रायिक स्थित्यर्ध कला और लम्बनान्तर कला के योग तुल्य गर्भाभिप्रायिक रविचन्द्रान्तर कला होती है, क्योंकि स्पर्शकाल में सूर्यस्थान से चन्द्रस्थान पश्चिम दिशा ही में होता है, गर्भाभिप्रायिक रविस्थान से पृष्ठाभिप्रायिक रविस्थान पूर्व दिशा में होता है, चन्द्र का भी उसी तरह होता है, पृष्ठाभिप्रायिक रविकेन्द्रगतकदम्बप्रोतवृत्त और चन्द्र केन्द्रगत कदम्बप्रोतवृत्त की अन्तर कला क्रान्तिवृत्त में स्थित्यर्धकला है, अतः स्थित्यर्धकला में चन्द्रलम्बन कला को जोड़ने से गर्भाभिप्रायिक चन्द्रस्थान से पृष्ठाभिप्रायिक रविस्थान पर्यन्त क्रान्तिवृत्तीयकला = स्थि$\frac{1}{2}$क + चंलंक, इसमें रविलम्बनकला को हीन करने से गर्भाभिप्रायिक रविचन्द्रान्तरकला = स्थि$\frac{1}{2}$क + चंलंक — रलंक, जब रविलम्बन कला से स्थित्यर्धकला अल्प होगी तब रविलम्बन कला में स्थित्यर्ध कला को हीन करने से पृष्ठाभिप्रायिक चन्द्रस्थान से गर्भाभिप्रायिक रविस्थान पर्यन्त होता है उसका स्वरूप = रलंक — स्थि$\frac{1}{2}$क, इसको चन्द्रलम्बन कला में हीन करने से गर्भाभिप्रायिक रविचन्द्रान्तरकला = चंलंक — (रलंक — स्थि $\frac{1}{2}$ क) = चंलंक — रलंक + स्थि$\frac{1}{2}$ क = स्थि$\frac{1}{2}$क + लम्बनान्तरक इसको घट्यात्मक करने के लिये अनुपात करते हैं

$$\frac{६०\times\text{गर्भीयरविचन्द्रान्तरक}}{\text{रविचन्द्रगत्यन्तरक}}=\frac{६०\ (\text{स्थि}\frac{1}{2}\text{क}+\text{लम्बनान्तरक})}{\text{रविचन्द्रगत्यन्तरक}}=\frac{६०\ \text{स्थि}\frac{1}{2}\text{क}}{\text{विचंगत्यन्तर}}$$

$$\frac{\text{लम्बनान्तरक}}{\text{विचंगत्यन्तर}}=\text{स्थि}\frac{1}{2}\text{घ}+\text{लम्बनान्तरघ},$$ इतनी घटी करके गर्भीय दर्शान्त से पूर्व

है यह सिद्ध हुआ। इसको गर्भीय दर्शान्त में घटाने से तो है।

$(\text{स्थि}\frac{1}{2}\text{घ}+\text{लम्बनान्तरघ})=$ गर्भीयदघ — स्थि$\frac{1}{2}$घ — लम्बनान्तरघ

में पृष्ठाभिप्रायिक स्पर्शकाल में गर्भीय दर्शान्त घटी में

पृष्ठाभिप्रायिक स्थित्यर्ध घटी को तथा लम्बनान्तर घटी को घटाने से पृष्ठाभिप्रायिक स्पर्शकाल घटी होती है। यहां स्थि$\frac{1}{2}$ क=पृष्ठीयस्थि$\frac{1}{2}$ कला सब जगह समझनी चाहिये। तथा स्थि$\frac{1}{2}$घ=पृस्थि$\frac{1}{2}$घ समझनी चाहिये।

यहाँ संस्कृतोपपत्ति में लिखित (क) क्षेत्र को देखिये। स=स्वस्वस्तिक, र=गर्भीयरवि, $\overset{'}{र}$=लम्बितरवि=पूरवि स्था=लम्बितरविस्थान, गचं=गर्भीयचन्द्र, $\overset{'}{चं}$=लम्बितचन्द्र=पृचं, स्था=गर्भीयचन्द्रस्थान, $\overset{'}{स्था}$=लम्बितचन्द्रस्थान।

अब पश्चिमकपाल में विचार करते हैं

गर्भीयदर्शान्त से आगे पृष्ठीय दर्शान्त है, पृष्ठीय दर्शान्त से पूर्व पृष्ठाभिप्रायिक स्पशकाल है इसलिये पृष्ठीयदर्शान्त से पूर्व पृष्ठाभिप्रायिक स्पर्शकाल और गर्भीयासान्त के होने से सम्भव है कि गर्भीय दर्शान्त से पूर्व पृष्ठाभिप्रायिक स्पर्शकाल हो और कदाचित् आगे भी, जब पृष्ठीय दर्शान्तकाल में गर्भाभिप्रायिक रविस्थान से पृष्ठाभिप्रायिक रविस्थान पश्चिम दिशा में होता है तब गर्भीय दर्शान्त से पहले ही पृष्ठाभिप्रायिक स्पर्शकाल होता है, क्योंकि गर्भीय दर्शान्त से पहले गर्भाभिप्रायिक रविस्थान से पश्चिम दिशा ही में गर्भाभिप्रायिक चन्द्रस्थान है, जब स्पर्शकाल में गर्भाभिप्राय से रविस्थान से गर्भाभिप्रायिक चन्द्रस्थान पूर्वदिशा में होता है तब गर्भीयदर्शान्त से आगे पृष्ठाभिप्रायिक स्पर्शकाल होता है क्योंकि गर्भीयदर्शान्त से आगे पूर्वदिशा ही में गर्भाभिप्रायिक रविस्थान और चन्द्रस्थान है।

स्पर्शकाल में गर्भाभिप्रायिक रविस्थान से गर्भाभिप्रायिक चन्द्र स्थान पूर्वदिशा में और पश्चिमदिशा में होता है, यह कैसे मालूम होता है, इस के लिए विचार करते हैं। यदि गर्भीयदर्शान्त में लम्बन घटी जोड़ने से पृष्ठीय दर्शान्त होता है उन घटियों से यदि पृष्ठाभिप्रायिक स्थित्यर्ध घटी अल्प हो तब पश्चिम दिशा में गर्भाभिप्रायिक रविस्थान से चन्द्रस्थान होता है अन्यथा पूर्व दिशा में। कब गर्भीय दर्शान्त से पहले पृष्ठाभिप्रायिक स्पर्शकाल होता है इसके लिए विचार करते हैं। पृष्ठीय स्थित्यर्धकला में चन्द्र लम्बन कला को घटाने से गर्भाभिप्राय से चन्द्रस्थान से पृष्ठाभिप्राय से रविस्थान पर्यन्त क्रान्तिवृत्त में कला होती है,=स्थि$\frac{1}{2}$क-चंलंक, इस में रविलम्बनकला को जोड़ने से पृष्ठाभिप्रायिक रविचन्द्रान्तरकला = स्थि$\frac{1}{2}$ क—चंलंक +रलंक, चन्द्रलम्बनकला से रविलम्बनकला के अल्प होने के कारण दोनों का अन्तर ऋणात्मक ही है तब गर्भीय रविचन्द्रान्तरकला=पृस्थि$\frac{1}{2}$ क—लम्बनान्तरक तब अनुपात से $\frac{६०\ (\text{पृस्थि}\frac{1}{2}\ \text{क}-\text{लम्बनान्तरक})}{\text{गत्यन्तरक}}$ = स्थि$\frac{1}{2}$ घ—लम्बनान्तरघ, गर्भीय दर्शान्त से पूर्वपृष्ठाभिप्राय से स्पर्शकाल कल्पना से गर्भीयदर्शान्त में स्थि$\frac{1}{2}$घ—लम्बनान्तरघ हीन करने से पृष्ठाभिप्राय से स्पर्शकाल=गदर्शान्तघ—पृस्थि$\frac{1}{2}$घ+लंघ इससे सिद्ध हुआ कि पश्चिम कपाल में भी जब गर्भीय दर्शान्त से पहले ही पृष्ठाभिप्राय से स्पर्शकाल होता है तब दर्शान्त में स्थित्यर्ध घटी को हीन ही करना और लम्बनान्तर को युत करना। जब गर्भीय दर्शान्त से

आगे पृष्ठाभिप्राय से स्पर्शकाल होता है तब विचार करते हैं । चन्द्रलम्बनकला में स्थित्यर्ध कला को हीन करने से पृष्ठाभिप्राय से रवि स्थान से गर्भाभिप्राय से चन्द्रस्थानावधि क्रान्तिवृत्त में कला=चंलंक—स्थि$\frac{1}{2}$क इस में रविलम्बनकला को हीन करने से गर्भाभिप्राय से रविचन्द्रान्तरकला=चंलंक—स्थि$\frac{1}{2}$क—रलंक=लम्बनान्तरक—स्थि$\frac{1}{2}$क तब अनुपात से

$$\frac{६०(\text{लम्बनान्तरक}-\text{स्थि}\frac{1}{2}\text{क})}{\text{गत्यन्तरक}} = \frac{६०\ \text{लम्बनान्तरक}}{\text{गत्यन्तरक}} - \frac{६०\ \text{स्थि}\frac{1}{2}\text{क}}{\text{गत्यन्तरक}} =$$

लम्बनान्तरघ—स्थि$\frac{1}{2}$घ गर्भीयदर्शान्ति से आगे पृष्ठीय स्पर्शकाल कल्पना से इन घटियों (लम्बनान्तरघ—स्थि$\frac{1}{2}$घ) को गर्भीयदर्शान्ति में जोड़ने से पृष्ठाभिप्राय से स्पर्शकाल=दघ—स्थि$\frac{1}{2}$घ+लम्बनान्तरघ, इस से सिद्ध हुआ कि "स्फुटतिथ्यन्तात् स्थित्यर्धहीन-युक्तात्' यहां स्थित्यर्ध घटी सर्वदा गर्भीय दर्शान्ति में हीन ही करनी चाहिये पूर्वकपाल में और पश्चिम कपाल में, लम्बनान्तर को ऋण और धन करना इति । यहां संस्कृतोपपत्ति में लिखित (ख) क्षेत्र को देखिये ।

पश्चिम कपाल में गर्भीय दर्शान्ति से पहले जब पृष्ठाभिप्राय से स्पर्शकाल होता है तब संस्कृतोपपत्ति में लिखित (ग) क्षेत्र को देखिये । सिद्धान्तशेखर में 'स्थित्यर्धोनयुतात् परिस्फुटतिथेः" इत्यादि से श्रीपति ने आचार्योक्त के अनुरूप ही कहा है । सिद्धान्तशिरोमणि में "तिथ्यन्ताद् गणितागतात् स्थितिदलेनोनाधिकात्" इत्यादि से भास्कराचार्य ने भी आचार्योक्त के अनुरूप ही कहा है इति-॥१६-१७॥

इदानीमिष्टग्रासे ग्रासात्कालानयने च चन्द्रग्रहणाद्यो विशेषस्तमाह

शशिवद्बाहुः स्फुटविक्षेपकृतस्थितिदलेन सङ्गुणितः ।
स्पष्टस्थित्यर्धहृतो भवति भुजः पूर्ववच्छेषम् ॥१८॥

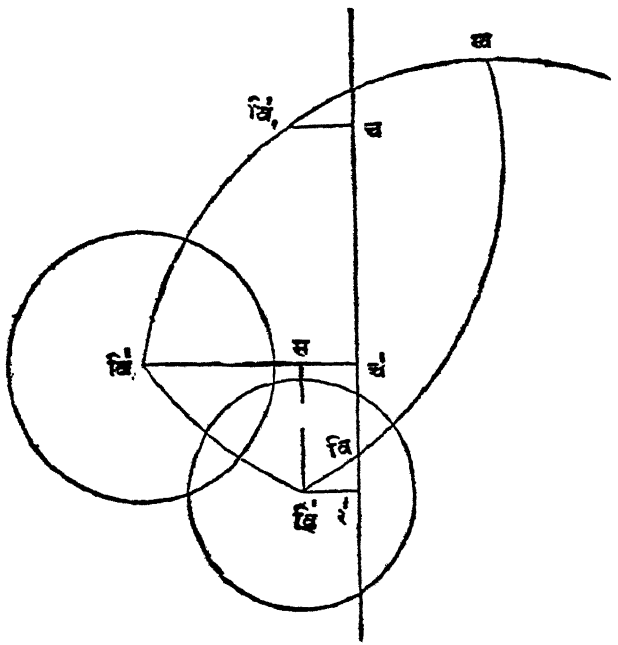
ग्रासात्कालः शशिवत् स्पष्टस्थितिदलगुणोऽसकृद् भक्तः।
स्फुटविक्षेपकृतस्थितिदलेन शोध्यः स्थितिदलात् स्यात् ॥१९॥

सु. भा.—ग्रासात् शशिवत् कालः 'शेषमूलं कृतं तिथिव' दित्यनेन चन्द्रग्रहण-विधिना यः कालः स ज्ञेयः । शेषव्याख्या 'शेषं शशाङ्कग्रहणोक्तमत्र" त्यादि भास्करविधिना स्फुटा । (सूग्र. श्लो० १८-१९)

कल्प्यते वित्रिभतः प्राक् स्पष्टस्पर्शकालादनन्तरमिष्टकाले इ, समे इष्ट-ग्रासोऽपेक्षितः । तत् क्षेत्रसंस्था चेयम् । अत्रापि स्पष्टस्थित्यर्धोपपत्तिलिखित-क्षेत्रसङ्केतानि बोध्यानि ।

अत्र वि, स=इष्टकाले स्पष्टशरः । वि स=इष्टकाले भुजकला यज्ज्ञान-

अत्रापक्षितमस्ति। ततः क्षेत्रयुक्त्या। भुजकला वि स=च चं=चवि+विर—च चं=च वि—इलं ∴च वि=भुक ± इलं। अत्र



चचं—विर=इष्टकालिकलम्बनकला। चवि=इष्टकाले गर्भाभिप्रायेण रविचन्द्रान्तरांशाः। ततः पूर्ववद्गर्भक्षितिजादिष्टग्रासे इष्टकालः=द—भुघ—इलंघ।

पूर्वसाधितस्पष्टस्पर्शकालः=द—स्फुस्थि—लं,

स्पर्शादनन्तरमिष्टग्रासे इष्टकालः

=(द—भुघ—इलंघ)—(द—स्फुस्थि—लं,)

=स्फुस्थि+लं,—इलंघ—भुघ=इ

∴भुघ=स्फुस्थि+लं—इलंघ—इ··· ···(१)

अथ पूर्वसाधितः

स्पष्टदर्शान्तिकालः=द—लं,

स्पष्टस्पर्शकालः=द—स्फुस्थि—लं,

द्वयोरन्तरेण स्पार्शिकं स्पष्टस्थितिदलम्

=स्फुस्थि+लं,—लं,=स्पस्थि। अथ यदि स्पष्टस्थितिदलेन लं,—ल, इदं लम्बनान्तरं तदा स्पर्शादिनन्तरं इमितेन कालेन किमित्यनुपातेन यदि वास्तवमान—

(लं,—इलंघ) मस्येदं $\frac{इ\ (लं,—लं,)}{स्पस्थि}$ कल्प्यते तदाऽस्मा (१) स्मिन्नुत्थापनेन

$$भुघ=स्फुस्थि—इ+\frac{इ\ (लं,—लं,)}{स्पस्थि}$$

$$=\frac{स्फुस्थि—स्पस्थि—इ\left\{स्पस्थि—(लं,—लं,)\right\}}{स्पस्थि}$$

$$=\frac{स्फुस्थि.\ स्पस्थि—इ.\ स्फुस्थि}{स्पस्थि}=\frac{स्फुस्थि\ (स्पस्थि—इ)}{स्पस्थि}$$

भुजघटिका गत्यन्तरकलागुणा षष्टिभक्ता जाता भुजकलाः

$$=भुक=\frac{स्फुस्थि}{स्पस्थि}\cdot\frac{गम्रं\ (स्पस्थि—इ)}{स्पस्थि\ ६०}=\frac{स्फुस्थि\times चन्द्रग्रहणवत्\ भुज}{स्पस्थि}$$

अनेन भुजकलानयनमुपपद्यते। एवमतो व्यस्तविधिना भुजघटीतो वीष्ट-

स्पष्टस्थितिदलमानम् = स्पस्थि—इ=$\frac{भुघ.\ स्पस्थि}{स्फुस्थि}$। इष्टग्रासकालिकस्पष्ट-

शराज्ञानान्मध्यकालिकस्पष्टशरतः कर्म कृतमतोऽसकृत्कर्मयुक्तमेव यतस्तात्कालिक-शराज्ञानात् स्फुटस्थितिदलादिकस्याज्ञानमिति। एवं स्पष्टस्थितिदलानुपाततो यदि लं,—इलंघ अस्य मानमानीयते तदा भुजादिसाधनं सूक्ष्ममाचार्योक्तम्। अनुपातस्य नियतैकरूपत्वात् तादृशानुपातेन लम्बनान्तरं न सूक्ष्ममायात्यत आचार्योक्तं स्थूलमानयनम्। तदेव भास्करस्यापि स्थूलमानयनमतो मदीयं सूत्रम्।

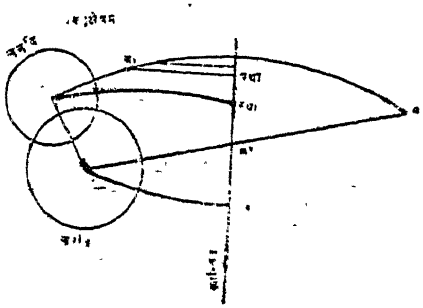
चेत् स्पष्टस्थितिखण्डकेन जनितं तत्कालमध्येऽन्तरं स्पष्टं लम्बनयोरिदं भवति किं तर्हीष्ट कालेन चेत्। सूक्ष्मं स्यादनुपातजातमनिशं दोरादिकं कोविद श्रीमद्ब्रह्मस्करसत्प्रकारविलसत् सत् स्यात् तदार्कग्रहे। इदं युक्तमेवेति सिद्धान्त-विद्भिर्भृशं विचिन्त्यनीयम्। एवं मौक्षिकेऽष्टेऽपि क्षेत्रसंस्थया स्फुटा वासना ॥१८ १९॥

वि. भा.—शशिवत् (चन्द्रग्रहणविधिनैव इष्टकालोद्भवो) यो बाहुः (भुजः) स स्फुटविक्षेपकृतस्थितिदलेन (तात्कालिकस्फुटशरजनितेन स्थित्यर्धेन) सङ्गुणितः, स्पष्टस्थित्यर्धेन (स्पर्शमध्यकालयोरन्तरेण) हृतः (भक्तः) तदा भुजः

(सूर्यग्रहणे स्फुटो भुजः) भवेत् । ग्रासात् (इष्टग्रासात्) सकाशात्—शशिवत् (चन्द्रग्रहणोक्तवत्) भुजः स्पष्टस्थितिदलेन (स्पष्टस्थित्यर्धेन) गुणितः, स्फुटविक्षेपकृतस्थितिदलेन (तात्कालिकस्फुटशरजनितस्थित्यर्धेन) भक्त एवमसकृत्करणेन यो भवति स स्थित्यर्धाच्छोध्यस्तदा कालः (कालसिद्धिः) स्यादिति॥१८-१९॥

अत्रोपपत्तिः

गर=गर्भीयरविः । चं=गर्भीयचन्द्रः।
स्था=लम्बितरविस्थानम् ।
स्था$_1$=लम्बितचन्द्रस्थानम् ।
स्था=गर्भीयचन्द्रस्थानम् ।
स्था स्था$_1$=चन्द्रस्पष्टलम्बनम् ।
गस्था=रविस्पष्टलम्बनम् ।
गस्था=गर्भीयरविचन्द्रान्तरम् ।
स्था स्था$_1$=लम्बितरविचन्द्रान्तरम् ।



गर्भीयदर्शान्तादग्रे यदा पृष्ठाभिप्रायेण स्पर्शकालस्तदा क्षेत्रदर्शनम् ।

गर्भीयरविचन्द्रान्तर + रलं — चंलं = लम्बितचन्द्रस्थानाल्लम्बितरविस्थानंयावत्=गर्भीयरविचन्द्रान्तर—लम्बनान्तर=कोटिः । अतो गर्भीयरविचन्द्रान्तर=लम्बनान्तर+कोटि, अथ वित्रिभतः प्राक्कपाले स्पर्शकालानन्तरमिष्टग्रासे इष्टकालः= गर्भीयदर्शान्त— रविचन्द्रान्तरघ=गर्भीयदर्शान्त— कोटिघ—लम्बनान्तरघ स्फुटतिथ्यन्ताल्लम्बनमसकृदित्यादिना स्पर्शकालः=

गर्भीयदर्शान्तिघटी — स्फुटस्थित्यर्धघ—स्पार्शिकलम्बनान्तरघ=गर्भीयदघ—स्फुटस्थित्यर्धघ—लं$_1$ अत्र स्पार्शिकलम्बनान्तरघटी=ल$_1$, अनयोरन्तरम् स्पर्शकालानन्तरमिष्टकालः=इ=(गर्भीयदघटी—कोटिघ—लम्बनान्तरघ) — (गर्भीयदघ—स्फटस्थि$\frac{1}{2}$घ—ल$_1$) अत्र लम्बनान्तरघ=इलंघ ततो गर्भीयदर्शान्तिघ—कोटिघ—इलंघ—गर्भीयदघ+स्फुटस्थि$\frac{1}{2}$घ+ल$_1$

=स्फुस्थि$\frac{1}{2}$घ—कोघ—इलंघ+लं$_1$=इ, अतः स्फुस्थि$\frac{1}{2}$घ—इलंघ—इ+लं$_1$=कोटिघ

अथ स्पष्टदर्शान्तिघ=गर्भीयदर्शान्तिघ—लं$_2$ अत्र लं$_2$= रविचन्द्रयोः स्पष्टलम्बनान्तरम्

तथा स्फुटतिथ्यन्ताल्लम्बनमसकृदित्यादिना स्पर्शकालः=गर्भीयदघ—स्फुस्थि½घ—ल'१

अनयोरन्तरम् स्पार्शिकस्थित्यर्धम्=स्पस्थि½=गर्भीयदघ—ल'२—(गदघ—स्फुस्थि½घ—ल'१)=गर्भीयदघ—ल'२—गर्भीयदघ+स्फुस्थि½घ+ल'१=स्फुस्थि½+ल'१—ल'२=स्पार्शिकस्थि½ घ ततो $\frac{(ल'_१-ल'_२)\ इ}{स्पस्थि½घ}$ अनुपातेन यत्फलं तद्यदि ल'१—इलंघ कल्प्यते तदा

कोटिघटीप्रमाणम्=कोघ=स्फुस्थि½ घ+ $\frac{(ल'_१-ल'_२)\ इ}{स्पस्थि½घ}$ —इ

$$=\frac{स्फुस्थि½थ\times स्पस्थि½घ+(ल'_१-ल'_२)\ इ-स्पस्थि½घ.\ इ}{स्पस्थि½घ}$$

$$=\frac{स्फुस्थि½घ\times स्पस्थि½घ+इ\{(ल'_१-ल'_२)-स्पस्थि½घ\}}{स्पस्थि½घ}$$

$$=\frac{स्फुस्थि½घ\times स्पस्थि½घ-इ\times स्फुस्थि½घ}{स्पस्थि½घ}$$

$$=\frac{स्फुस्थि½घ\ (स्फुस्थि½घ-इ}{स्पस्थि½घ}=कोटिघ$$

ततः $\frac{गत्यन्तरक\times कोटिघ}{६०}$=कोटिक, अत उत्थापनेन $\frac{स्फुस्थि½घ\ (स्पस्थि½घ-इ)}{स्पस्थि½घ}$

$\times\frac{गत्यन्तरक}{६०}=\frac{स्फुस्थि½घ}{स्पस्थि½घ}$ (स्पस्थि½घ—इ) $\times\frac{गत्यन्तरक}{६०}=\frac{स्फुटस्थि½घ}{स्पस्थि½घ}$

$\times$इष्टोनस्थित्यर्धकला=$\frac{स्फुटस्थि½घ}{स्पस्थि½घ}\times$चन्द्रग्रहणवत्कोटिः=कोटिकला (१)

=$\frac{स्फुटस्थि½घ}{स्पस्थि½घ}\times$चन्द्रग्रहणवद्भुजः=भुजकला यदि कोटिघटी=भुजघटी,

अथ $\frac{स्फुस्थि½घ}{स्पस्थि½घ}\times$इष्टोनस्थित्यर्धकला=भुजकला, ∴ इष्टोनस्पष्टस्थि½क=

$\frac{भुकला\times स्पष्टस्थि½घ}{स्फुस्थि½घ}$ इदं फलं स्पष्टस्थित्यर्धे शोध्यं तदेष्टकालो भवेत्। परमिदमानयनं न समीचीनं यतस्तात्कालिकशराज्ञानान्मध्यग्रहणकालिक-

शरवशादेव स्थित्यर्धादिसाधनं कृतमतोऽसकृत्कर्म कार्यमेतेनाऽऽचार्योक्तमुपपद्यते। सिद्धान्तशेखरे "बाहुश्चन्द्रग्रहणविधिनैवेष्टकालोद्भवो यः क्षेपस्पष्टस्थिति-दलहतो भाजितः प्रस्फुटेन। स्थित्यर्धेन स्फुट इह भवेदुक्तवत् कालसिद्धिश्चेष्ट-ग्रासाद् गुणकहरयोर्व्यत्ययेनासकृत् स्यात्" ऽनेन श्रीपतिना सिद्धान्तशिरोमणौ शेषं शशाङ्कग्रहणोक्तमत्र स्फुटेषुजेनस्थितिखण्डकेन। हतोऽथ तेनैव हृतः स्फुटेन बाहुः स्फुटः स्याद् ग्रहणे ऽत्र भानोः" ग्रासाच्च कालानयने फलं यत् स्फुटेन निघ्नं स्थितिखण्डकेन। स्फुटेषुजेनासकृदुद्धृतं तत् स्थित्यर्धशुद्धं भवतीष्टकालः पद्येनानेन भास्कराचार्येणाप्याचार्योक्तानुरूपमेवोक्तम्। (१) अनेनच "भानोर्ग्रहे कोटिलिप्ता मध्यस्थित्यर्धसङ्गुणा। स्फुटस्थित्यर्धसंभक्ताः स्फुटाः कोटिकलाः स्मृताः" सूर्य-सिद्धान्तोक्तमिदमप्युपपन्नमिति ॥१८-१९॥

अब इष्टग्रास में और ग्रास से कालानयन में चन्द्रग्रहण से जो विशेष बातें हैं उनको कहते हैं

हि. भा.—चन्द्रग्रहण विधि से इष्टकालोत्पन्न जो भुज हो उसको तात्कालिक स्फुट शरजनित स्थित्यर्ध से गुणाकर स्पष्ट स्थित्यर्ध (स्पर्शकाल और मध्यकाल के अन्तर) से भाग देने से सूर्यग्रहण में भुज होता है, इष्टग्रास से चन्द्रग्रहणोक्तवत् भुज को स्पष्ट स्थित्यर्ध से गुणाकर तात्कालिक स्फुटशर जनित स्थित्यर्ध से भाग देना। इस तरह असकृत् करने से जो होता हो उसको स्थित्यर्ध में घटाने से कालसिद्धि होती है इति ॥ १८-१९ ॥

उपपत्ति

यहां संस्कृतोपपत्ति में लिखित (क) क्षेत्र को देखिये। गर्भीयदर्शान्त से आगे जब पृष्ठाभिप्राय से स्पर्शकाल है उस कालिक यह क्षेत्र है। गर=गर्भीयरवि, चं=गर्भीयचन्द्र, स्था=लम्बितरविस्थान, स्था$_1$=लम्बितचन्द्रस्थान, स्था=गर्भीय चंस्थान स्थास्था$_1$=चन्द्रस्पष्टलम्बन, गस्था=रविस्पष्टलम्बन, गस्था=गर्भीयरविचन्द्रान्तर, स्थास्था$_1$=लम्बितरविचन्द्रान्तर गर्भीयरविचन्द्रान्तर+रलं—चंलं=लम्बितचन्द्रस्थान से लम्बित रविस्थानपर्यन्त, =गर्भीयरविचन्द्रान्तर—लम्बनान्तर=कोटि

इसलिये गर्भीयरविचन्द्रान्तर=लम्बनान्तर+कोटि, बिम्बभ्रम से पूर्व कपाल में स्पर्शकाल के बाद इष्टग्रास में इष्टकाल=गर्भीयदर्शान्त—रविचन्द्रान्तरघ=गर्भीयदर्शान्त—कोटिघ—लम्बनान्तरघ

'स्फुटतिथ्यन्ताल्लम्बनमसकृत्' इत्यादि से स्पर्शकाल=

गर्भीयदघ—स्फुटस्वि$\frac{1}{2}$घ—स्पार्शिकलम्बनान्तरघ=गर्भीयदघ—स्फुटस्वि$\frac{1}{2}$घ—कों,

यहां स्पार्शिक लम्बनान्तरघ=लं$_1$, इन दोनों का अन्तर करने से स्पर्शकाल के बाद इष्टकाल=इ=

(गर्भीयदघ—कोटिघ—लम्बनान्तरघ)—(गर्भीयदघ—स्फुस्थि$\frac{1}{2}$घ—लं$_1$)

यहां लम्बनान्तरघ=इलंघ, तब गर्भीयदघ—कोटिघ—इलंघ—गर्भीयदघ+स्फुस्थि$\frac{1}{2}$घ+लं$_1$=

स्फुस्थि$\frac{1}{2}$घ—कोघ—इलंघ+लं$_1$=इ, अतः स्फुस्थि$\frac{1}{2}$घ—इलंघ—इ+लं$_1$=कोटिघ

स्पष्टदर्शान्तिघ=गर्भीयदघ—लं$_2$, यहां लं$_2$=रवि और चन्द्र का स्पष्टलम्बनान्तर, तथा 'स्फुटतिथ्यन्तात्तल्लम्बनमसकृत्' इत्यादि से स्पर्शकाल=गर्भीयदघ—स्फुस्थि$\frac{1}{2}$घ—लं$_1$ दोनों का अन्तर करने से

स्पार्शिकस्थित्यर्ध=स्पस्थि$\frac{1}{2}$—गर्भीयदघ—लं$_2$—(गदघ—स्फुस्थि$\frac{1}{2}$घ—लं$_1$)

=गर्भीयदघ—लं$_2$—गर्भीयदघ+स्फुस्थि$\frac{1}{2}$घ+लं$_1$=स्फुस्थि$\frac{1}{2}$+लं$_1$—लं$_2$=स्पार्शिकस्थि$\frac{1}{2}$घ

तब $\frac{(\text{लं}_1-\text{लं}_2)\text{इ}}{\text{स्पस्थि}\frac{1}{2}\text{घ}}$ इस अनुपात से जो फल होता है वह यदि लं$_1$—इलंघ माना जाय तब

कोटिघटी=कोघ=स्फुस्थि$\frac{1}{2}$घ+$\frac{(\text{लं}_1-\text{लं}_2)\text{इ}}{\text{स्पस्थि}\frac{1}{2}\text{घ}}$—इ=

$$=\frac{\text{स्फुस्थि}\frac{1}{2}\text{घ}\times\text{स्पस्थि}\frac{1}{2}\text{घ}+(\text{लं}_1-\text{लं}_2)\,\text{इ}-\text{स्पस्थि}\frac{1}{2}\text{घ}\cdot\text{इ}}{\text{स्पस्थि}\frac{1}{2}\text{घ}}$$

$$=\frac{\text{स्फुस्थि}\frac{1}{2}\text{घ}\times\text{स्पस्थि}\frac{1}{2}\text{घ}+\text{इ}\{(\text{लं}_1-\text{लं}_2)-\text{स्पस्थि}\frac{1}{2}\text{घ}\}}{\text{स्पस्थि}\frac{1}{2}\text{घ}}$$

$$=\frac{\text{स्फुस्थि}\frac{1}{2}\text{घ}\times\text{स्पस्थि}\frac{1}{2}\text{घ}-\text{इ}\times\text{स्फुस्थि}\frac{1}{2}\text{घ}}{\text{स्पस्थि}\frac{1}{2}\text{घ}}=\frac{\text{स्फुस्थि}\frac{1}{2}\text{घ}\,(\text{स्पस्थि}\frac{1}{2}\text{घ}-\text{इ})}{\text{स्पस्थि}\frac{1}{2}\text{घ}}$$

=कोटिघ, $\therefore\frac{\text{गत्यन्तरक}\times\text{कोटिघ}}{६०}$=कोटिक, उत्थापन करने से

$$\frac{\text{स्फुस्थि}\frac{1}{2}\text{घ}\,(\text{स्पस्थि}\frac{1}{2}\text{घ}-\text{इ})}{\text{स्पस्थि}\frac{1}{2}\text{घ}}\times\frac{\text{गत्यन्तरक}\cdot\text{कोटिघ}}{६०}$$

$$=\frac{\text{स्फुस्थि}\frac{1}{2}\text{घ}}{\text{स्पस्थि}\frac{1}{2}\text{घ}}\times\text{इष्टनस्थि}\frac{1}{2}\text{कला}=\frac{\text{स्फुस्थि}\frac{1}{2}\text{घ}}{\text{स्पास्थ}\frac{1}{2}\text{घ}}\times\text{चन्द्रग्रहणवत्कोटि}=\text{कोटिकला}\quad(१)$$

$$= \frac{\text{स्फुस्थि}\frac{1}{2}\text{घ}}{\text{स्पस्थि}\frac{1}{2}\text{घ}} \times \text{चन्द्रग्रहणवद्भुज} = \text{भुजकला}$$

यदि कोटिघटी = भुजघटी

$$\frac{\text{स्फुस्थि}\frac{1}{2}\text{घ}}{\text{स्पस्थि}\frac{1}{2}\text{घ}} \times \text{इष्टोनस्थित्यर्धकला} = \text{भुजकला} \quad \therefore \text{इष्टोनस्पस्थि}\frac{1}{2}\text{क}$$

$$= \frac{\text{भुजकला} \times \text{स्पस्थि}\frac{1}{2}\text{घ}}{\text{स्फुस्थि}\frac{1}{2}\text{घ}}$$ इस फल को स्पष्टस्थित्यर्ध में घटाने से इष्टकाल होता है, लेकिन यह आनयन ठीक नहीं है, क्योंकि तात्कालिकशर विदित नहीं रहने के कारण मध्यग्रहण कालिकशरवश से ही स्थित्यर्धादि का साधन किया गया है इसलिये असकृत् कर्म करना चाहिये, इससे आचार्योक्त उपपन्न हुआ । सिद्धान्तशेखर में "बाहुश्चन्द्रग्रहणविधिनैवेष्टकालोद्भवो यः" इत्यादि संस्कृत पद्य में लिखित श्लोक से श्रीपति ने, तथा सिद्धान्तशिरोमणि में "शेषं शशाङ्कग्रहणोक्तमत्र स्फटेषुजेन" इत्यादि संस्कृतोपपत्ति में लिखित पद्यों से भास्कराचार्य ने भी आचार्योक्त के अनुरूप ही कहा (१) इससे "भानोर्ग्रहे कोटिलिप्ता मध्यस्थित्यर्धसङ्गुणा" इत्यादि सूर्यसिद्धान्तोक्त भी उपपन्न हुआ इति ॥१८-१९॥

इदानीमादेश्यानादेश्ययोर्ग्रहणयोर्नियममाह

वलनादिशशिवदन्यद् ग्रहणं तैक्ष्ण्याद्रवेरनादेश्यम् ।
द्वादशभागादूनं स्वच्छत्वात् षोड़शादिन्दोः ॥२०॥

सु. भा.—षोडशात् षोडशभागात् । शेषं स्पष्टार्थम् । चन्द्रग्रहणाधिकारे 'इन्दोर्भागः षोडशः खण्डितोऽपि' इत्यादि भास्करोक्तमेतदनुरूपमेव (चं.ग्र. श्लो० ३७) ॥२०॥

वि. भा.—अन्यत् वलनादि शशिवत् (चन्द्रग्रहणोक्तवत्) ज्ञेयम् । रवेः (सूर्यस्य) तैक्ष्ण्यात् (तेजस्तैक्ष्ण्यात् असह्यत्वात्) द्वादशभागादूनं (द्वादशांशान्यूनं) ग्रहणमनादेश्यम् (ग्रस्तस्यापि द्वादशांशस्यादृश्यत्वात्) इन्दोः (चन्द्रस्य) स्वच्छत्वात् (किरणानामतिशुभ्रतावशेन) षोड़शात् (षोडशभागान्यूनं) ग्रहणमनादेश्यमिति सिद्धान्तशेखरे "तेजस्तैक्ष्ण्यात्तीक्ष्णगोर्मण्डलस्य ग्रस्तोऽप्यंशो द्वादशांशोन दृश्यः । तद्वद्भागः षोडशः शीतरश्मेः स्वच्छांशुत्वाल्लक्ष्यते नाधिकोऽतः" श्रीपतिनैवमुक्तम् । अमुमेव श्रीपतिप्रकारं भास्कराचार्यो यत्किञ्चित्परिवर्त्तितमेवमाह "इन्दोर्भागः षोडशः खण्डितोऽपितेजः पुञ्जच्छन्नभावान्न लक्ष्यः । तैजस्तैक्ष्ण्यात् तीक्ष्णगोर्द्वादशांशो नादेश्योऽतोऽल्पो ग्रहो बुद्धिमद्भिः" सूर्यसिद्धान्ते च

"स्वच्छत्वात् द्वादशांशोऽपि ग्रस्तश्चन्द्रस्य दृश्यते । लिप्तात्रयमपि ग्रस्तं

तीक्ष्णत्वान्न विवस्वतः" वमुक्तम् परमत्रोपलब्धिरेव वासना नान्यत्कारणं वक्तुं शक्यत इति ॥२०॥

अब आदेश्य (कहने योग्य) और अनादेश्य (नहीं कहने योग्य) ग्रहण के नियम को कहते हैं

हि. भा.—अन्य वलन आदि चन्द्रग्रहणोक्तवत् समझना चाहिये। रवि के तेज की तीक्ष्णता के कारण द्वादशांश से न्यून ग्रहण को नहीं कहना चाहिये क्योंकि द्वादशांश ग्रस्त रहने पर भी देखने में नहीं आता है, चन्द्रमा की स्वच्छता के कारण सोलह अंश से न्यून ग्रहण को नहीं कहना चाहिये (लोगों में), "सिद्धान्तशेखर में "तेजस्तैक्ष्ण्यात्तीक्ष्ण-गोर्मण्डलस्य" इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से श्रीपति ने इस तरह कहा है, इसी श्रीपति प्रकार को कुछ परिवर्त्तित कर भास्कराचार्य इस तरह कहते हैं "इन्दोर्भागः षोड़शः खण्डितोऽपि" इत्यादि संस्कृतोपपत्ति में लिखित श्लोकोक्त के अनुसार सूर्यसिद्धान्त में "स्वच्छत्वात् द्वादशांशोऽपि" इत्यादि के अनुसार कहा गया है, लेकिन इन में केवल उपलब्धि को ही उपपत्ति कह सकते हैं, दूसरा कारण कुछ भी नहीं कहा जा सकता है इति॥२०॥

इदानीं स्वप्रशंसामाह

न स्फुटमार्यभटादिष्वर्कग्रहणं यतस्ततः स्पष्टम्।
शङ्कुज्यया कृतं लघु लघुतरमेवं रवेर्ग्रहणम् ॥२१॥

सु. भा.—शङ्कुज्यया वित्रिभशङ्कुना मया लघु कृतमित्याचार्याशयः। एवमग्रे वक्ष्यमाणप्रकारेण रवेर्ग्रहणं रविग्रहणसाधनं लघुतरं च भवति। शेषं स्फुटार्थम् ॥२१॥

वि. भा.—यतः (यस्मात्कारणात्) आर्यभटादिषु (आर्यभटश्रीषेणचन्द्रजाद्येषु ग्रन्थेषु) अर्कग्रहणं (सूर्यग्रहणं) स्फुटं नास्ति ततः (तस्मात् कारणात्) शङ्कुज्यया (वित्रिभशङ्कुना) मया स्पष्टं लघुकृतम्। एवमग्रे वक्ष्यमाणप्रकारेण रवेर्ग्रहणं (सूर्यग्रहणसाधनं) लघुतरं च भवतीति ॥२१॥

अब अपनी प्रशंसा को कहते हैं

हि. भा.—जिस कारण से आर्यभट श्रीषेण प्रभृति आचार्यों के ग्रन्थों में सूर्यग्रहण साधन स्फुट नहीं है, उस कारण से वित्रिभ शङ्कु से मैंने स्पष्ट और लघु किया, एवं आगे कथित प्रकार से सूर्यग्रहण साधन लघुतर (अतिशयलघु) होता है इति॥२१॥

इदानीं लघुकर्मणा लम्बननत्योः साधनमाह

लग्नात् त्रिराशिहीनादपक्रमाक्षांशयुति विशेषोनात् ।
भत्रितयाज्ज्याच्छेदस्त्रिज्यार्धकृतेः फलेन हृता ॥२२॥
वित्रिभलग्नार्कान्तरजीवा घटिकादिलम्बनं वा स्यात् ।
वित्रिभलग्नापक्रमविक्षेपाक्षांश युतिवियुतेः ॥२३॥
जीवा शशाङ्कभास्कर मध्यम भुक्त्यन्तरेण सङ्गुणिता ।
पञ्चदशभिर्गुणितया विभाजिता त्रिज्ययाऽवनतिः ॥२४॥

सु. भा.—त्रिराशिहीनाल्लग्नाद्वित्रिभात् क्रान्त्यंशा ये तेषामक्षांशानां चैक-दिशां युतिर्विभिन्नदिशां विशेषः स्वल्पाक्षे देशे वित्रिभस्य याम्योत्तरवृत्तासन्ने स्थितत्वाद्वित्रिभनतांशा भवन्ति । तदूनाद्भत्रितयाद्राशित्रयज्ज्या वित्रिभशङ्कुस्त्रिज्यार्धकृतेश्छेदो भवति । छेदात् त्रिज्यार्धकृतेर्यत् फलं तेन वित्रिभलग्नार्कान्तरज्या हृता लब्धं घटिकादि वा लम्बनं भवेत् । वित्रिभलग्नापमांशानां वित्रिभशरांशानामक्षांशानां चैवदिशां युतेर्विभिन्नदिशां विधृतेर्जीवा चन्द्रदृक्क्षेपः । सा रविचन्द्रमध्यगत्यन्तरेण सङ्गुणिता पञ्चदशभिर्गुणितया त्रिज्यया विभाजिताऽवनतिः स्यात् ।

अत्रोपपत्तिः । याम्योत्तरवृत्त एव सुखार्थं वित्रिभं प्रकल्प्य दिनार्धवत् तन्नतांशा उन्नतांशाश्च साधिताः । उन्नतांशजीवा वित्रिभशङ्कुः प्रसिद्धस्ततः पूर्वविधिना लम्बनघटिकाः $= \frac{४ \text{ ज्या } (\text{र}-\text{वि})}{\text{त्रि}} \cdot \frac{\text{विशं}}{\text{त्रि}} = \frac{\text{ज्या } (\text{र}-\text{वि})}{\frac{\text{त्रि}^2}{४ \text{ विशं}}}$

अनेन लम्बनानयनमुपपद्यते ।

अथ पूर्वसाधितवित्रिभनतांशेषु वित्रिभशरं संस्कृत्य विमण्डलावधि चन्द्रदृक्क्षेपचापांशाः साधिताः सुखार्थं तत्समा रविदृक्क्षेपचापांशाः स्थूला अङ्गीकृताः । अतस्तयोर्दृक्क्षेपौ समौ जातौ । ततो 'रविशशिमध्यगतिगुणे' इत्यादिविधिना पृथक्पृथक् नती प्रसाध्य तयोरन्तरं कृतमिति सुगमा वासना ॥२२-२४॥

वि. भा.—त्रिराशिहीनात् लग्नात् (राशित्रयरहितात् लग्नात् वित्रिभात्) येऽपक्रमांशाः (क्रान्त्यंशाः) तेषामक्षांशानामेकदिशां युतिः (योगः) भिन्नदिशां विशेषः (अन्तरं) स्वल्पाक्षेदेशे वित्रिभस्य याम्योत्तरवृत्तासन्ने स्थितत्वाद्वित्रिभनतांशा भवन्ति, तदूनात् (वित्रिभनतांशरहितात्) भत्रितयात् (राशित्रयात्) ज्या वित्रिभशङ्कुर्भवति । स च त्रिज्यार्धकृतेः (त्रिज्यार्धवर्गस्य) छेदः (हरः) भवति ।

छेदात् त्रिज्यार्धकृतेर्यत्फलं तेन वित्रिभलग्नार्कान्तरज्या हृता (भक्ता) लब्धं घाटकादि वा लम्बनं भवेत्। वित्रिभलग्नक्रान्त्यंशानां वित्रिभशरांशानामक्षांशानां चैकदिशां युतेर्विभिन्नदिशां वियुतेर्जीवा (ज्या) चन्द्रदृक्क्षेपः। सा (जीवा) रविचन्द्रमध्यगत्यन्तरेण सङ्गुणिता पञ्चदशभिर्गुणितया त्रिज्यया विभाजिता तदाऽक्षनतिः (स्पष्टनतिः) स्यादिति ॥२२-२३-२४॥

अत्रोपपत्तिः

याम्योत्तरवृत्त एव स्वल्पान्तरात् वित्रिभं मत्वा दिनार्धवत् तन्नतांशा उन्नतांशाश्च साध्याः। उन्नतांशज्या वित्रिभशङ्कुः ततः पूर्वोक्त्या लम्बनघटिकाः,

$$= \frac{\text{ज्या}(\text{र}-\text{वि})}{\text{त्रि}} \times \frac{\text{विशं}}{\text{त्रि.}} = \frac{\text{ज्या}(\text{र}-\text{वि})}{\frac{\text{त्रि}^2}{४\ \text{विशं}}} = \frac{\text{ज्या}(\text{र}-\text{वि})}{\frac{(\text{त्रि})^2}{\text{विशं}}} = \frac{\text{ज्या}(\text{र}-\text{वि})}{\text{फल.}}$$

एतेन लम्बनानयनमुपपद्यते।

पूर्वसाधितवित्रिभनतांशेषु वित्रिभशरं संस्कृत्य विमण्डलपर्यन्तं चन्द्रदृक्क्षेपचापांशा साधिताः स्वल्पान्तरात्तत्तुल्या रविदृक्क्षेपचापांशाः स्वीकृताः, अतस्तयोर्दृक्क्षेपौ समौ जातौ, ततो "रविशशिमध्यगतिगुणे" इत्यादिना पृथक्-पृथक् नती प्रसाध्य तयोरन्तरमवनतिसंज्ञकमिति। अत्र वि=वित्रिभम्। ज्या (र—वि) =वित्रिभार्कान्तरज्या सिद्धान्तशेखरे "विपदोदयापमपलैकतान्तरात् त्रिगृहच्युतात् क्रमगुणेन भाजिता। त्रिभशिञ्जिनीदलकृतिः फलोद्धृता रविवित्रिभोदयवियोगशिञ्जिनी॥ अथवाऽपि लम्बनमृणं स्वमुक्तवत् विपदोदयापमपलैकतान्तरम्। त्रिगृहोनलग्नजशरेण मिश्रितं स्वदिगन्यथा तु वियुतं ततो गुणः। खनगैर्हृतो भवति वाऽवनतिरिति" श्रीपत्युक्तमिदमाचार्योक्तानुरूपमेव केवलं 'खनगैर्हृतो भवति वाऽवनति' रित्याचार्योक्तादधिकं कथ्यते।

'रविशशिमध्यगतिगुणे' इत्यादिना रविनतिः $= \frac{\text{रदृक्षे} \times \text{रमग}}{१५.\ \text{त्रि}}$, चन्द्रदृक्क्षेपः

$= \frac{\text{चंदृक्षे} \times \text{चंमग}}{१५.\ \text{त्रि}}$ अनयोरन्तरम् $\frac{\text{चंदृक्षेपो वा रविदृक्षे} \times \text{गत्यन्तर}}{१५.\ \text{त्रि.}} =$

$\frac{\text{दृक्षेप.}\ (७९०।३५—५१।८)}{३४१५ \times १५} = \frac{\text{दृक्षेप.} \times (७३१।२७)}{५१२२५} = \frac{\text{दृक्षेप.}}{७०}$ स्वल्पान्तरात्।

"दृक्क्षेपात् सप्ततिहृतात् भवेद्भाऽवनतिः फलम्" इत्यनेन सूर्यसिद्धान्तोऽपि

प्रकारान्तरेणेदं नत्यानयनमुक्तम्। अत्र रविचन्द्रनती समे स्वीकृते आचार्येणेति ॥ २२-२३-२४॥

अब लघु कर्म से लम्बन और नति के साधन को कहते हैं

हि. भा.— राशित्रय रहित लग्न (वित्रिभ लग्न) से जो क्रान्त्यंश हो उसका और अक्षांश का एक दिशा में योग और भिन्न दिशा में अन्तर करने से स्वल्पाक्षांश देश में याम्योत्तर वृत्त के आसन्न में वित्रिभ के रहने से वित्रिभ नतांश होता है, इस को तीन राशि में घटाने से जो होता है उसकी ज्या वित्रिभशङ्कु है। त्रिज्यार्ध (त्रिज्या का आधा) वर्ग का वह (वित्रिभशङ्कु) छेद (हर) होता है, त्रिज्यार्ध वर्ग में छेद (हर) से भाग देने से जो फल होता है उससे वित्रिभलग्न और रवि की अन्तरज्या को भाग देने से लब्ध वा (प्रकारान्तर से) घटिकादिक लम्बन होता है ।. वित्रिभलग्न की क्रान्ति, वित्रिभ के शरांश और अक्ष के एक दिशा में रहने से योग और भिन्न दिशा में रहने से अन्तर करने पर जो शेष रहता है उसकी ज्या चन्द्रदृक्क्षेप होता है। उस (ज्या) को रवि और चन्द्र के मध्य गत्यन्तर से गुणा कर पन्द्रह गुणित त्रिज्या से भाग देने से अवनति (स्पष्टनति) होती है इति॥२२-२३-२४॥

उपपत्ति

स्वल्पान्तर से याम्योत्तरवृत्त ही में वित्रिभ का मान कर दिनार्धकाल की तरह वित्रिभनतांश और उन्नतांश साधन करना। उन्नतांशज्या=वित्रिभशङ्कु। तब "त्रिज्या-कृतेश्चतुर्गुणशङ्कुहृतायाः" इत्यादि से लम्बनघटी $=\frac{४ज्या(र\sim वि)}{त्रि}\times\frac{विअं}{त्रि}$

$=\frac{ज्या(र\sim वि)}{\frac{त्रि^२}{४\,विअं}}=\frac{ज्या(र\sim वि)}{\frac{(\frac{त्रि}{२})^२}{विअं}}=\frac{ज्या(र\sim वि)}{फल}$ इससे स्पष्ट लम्बनानयन उपपन्न होता है।

पूर्वसाधित वित्रिभनतांश में वित्रिभशर को संस्कार करने से विमण्डल पर्यन्त चन्द्रदृक्क्षेप चापांश होता है, स्वल्पान्तर से इसके समान ही रवि के दृक्क्षेप चापांश भी स्वीकार कर लिये गये। इसलिये दोनों (रवि और चन्द्र) का दृक्क्षेप बराबर हुआ।

तब "रविक्षिप्तभवतिर्गुणे" इत्यादि से रवि और चन्द्र के पृथक्-पृथक् नतिसाधन कर दोनों का अन्तर अवनति (स्पष्टनति) होती है। यहां वि=वित्रिभ, ज्या (र~वि)=

विन्निभार्कान्तरज्या, सिद्धान्तशेखर में ''विपदादयापमपलैकतान्तरात्'' इत्यादि संस्कृतोपपत्ति में लिखित पद्यों से श्रीपति ने आचार्योक्त के अनुरूप ही कहा है, केवल 'खनगैर्हृतो भवति वाऽवनतिः, यह आचार्योक्त से भिन्न कहते हैं। 'रविशशिमध्यगतिगुणे' इत्यादि से रविनति $=\frac{\text{दृक्क्षेप}\cdot\text{रमग}}{१५.\text{त्रि}}$, चन्द्रनति $=\frac{\text{दृक्क्षेप}.\text{चंमग}}{१५\,\text{त्रि}}$ दोनों का अन्तर करने से $\frac{\text{दृक्क्षेप}.\text{गत्यन्तर}}{१५.\text{त्रि}}=$

$\frac{\text{दृक्क्षेप}.(७९०।३५-५९।८)}{३४१५\times १५}=\frac{\text{दृक्क्षेप}(७३१।२७)}{५१२२५}=\frac{\text{दृक्क्षेप}}{७०}=$ स्पष्टनति, स्वल्पान्तर से, इस से 'खनगैर्हृतो भवति वाऽवनतिः' यह श्रीपतियुक्त प्रकार तथा 'दृक्क्षेपात् सप्ततिहृतात् भवेद्वाऽवनतिः फलम्, यह सूर्यसिद्धान्तोक्त स्पष्टनत्यानयनप्रकार उपपन्न होता है, इति॥२२-२३-२४॥

इदानीं विशेषमाह

पूर्ववदन्यत् स्पष्टं ब्रह्मोक्तस्पष्टसूर्यशशिपातैः।
नार्यभटादिभिरुक्तं यतोऽस्फुटास्ते ततोऽस्पष्टम्॥२५॥

सु. भा.—अन्यदवशिष्टं स्थित्यर्धादिसाधनं पूर्ववत्। एवं ब्रह्मोक्तस्पष्टसूर्यशशिपातैर्ग्रहणं स्पष्टं दृग्योग्यं भवति। यत आर्यभटादिभिर्ये स्पष्टसूर्यशशिपाता उक्तास्तेऽस्फुटा न दृष्टियोग्यास्ततस्तदुक्तै रविशशिपातैर्ग्रहणमस्पष्टं न दृष्टियोग्यमित्यर्थः॥२५॥

वि. भा.—अन्यदवशिष्टं (स्थित्यर्धादिकं) पूर्ववत् (चन्द्रग्रहणोक्तसाधनविधिवत्) भवति, ब्रह्मोक्तस्पष्टसूर्यशशिपातैः (मत्कथितस्पष्टरविचन्द्रपातैः) ग्रहणं स्पष्टं (दृग्योग्यं) भवति, आर्यभटादिभिराचार्यैर्यदुक्तं ततो न (दृग्योग्यं न भवति) यतस्ते (आर्यभटादिकथितस्पष्टसूर्यशशिपाताः) ऽस्फुटास्ततस्तदुक्तं ग्रहणमस्पष्टं (न दृष्टियोग्यमिति)। सिद्धान्तशेखरे ग्रहणाध्यायोपसंहारे 'न स्फुटं भवति पञ्चजीवया लम्बनं न हि यतस्ततः कृतम्। युक्तमुक्तमिति जिष्णुसूनुना तन्मयाऽपि कथितं परिस्फुटम्' ऽनया श्रीपत्युक्त्या सूर्यग्रहणाध्यायः श्रीपतिना ब्रह्मगुप्तोक्तानुरूप एवोक्त इति सूच्यते। "दृग्गणितैक्यं न भवति यस्मात् पञ्चज्यया रविग्रहणे। तस्माद्यथा तदैक्यं प्रवक्ष्यामि तिथ्यन्ते" इत्याचार्योक्तव्याख्यायां चतुर्वेदाचार्यः—

"पञ्चज्यया पञ्चज्याविधानेन रविग्रहणं यदाचार्येरूपनिबद्धं तद्यथा। उदयज्या, शङ्कुज्या, मध्यज्या, दृग्गतिज्या, दृक्क्षेपज्या च। एताभिरार्यभटादिभिस्तथा पौलिशतन्त्रे पञ्चज्याश्चन्द्रमसः स्वदिनगतशेषचरदलक्रान्त्यादिभिः

कल्पिताः। एवं तत्र दशज्याविधानेन रविग्रहणं यदुपनिबद्धं तादृशममान्ते भवति। ये च तत्र दोषास्तानाचार्य एव वक्ष्यति तन्त्रपरीक्षाध्याये वयमपि तत्रैव व्याख्यास्यामः।' तन्त्र परीक्षाध्याये च

"पञ्चज्यया यतोऽर्कग्रहणं श्रीषेणविष्णुचन्द्रकृतम्।
आर्यभटोक्तान्यनयोरर्कग्रहदूषणानि ततः॥
एवं विचार्यमाणे पञ्चज्यालम्बनं महास्थूलम्।
स्थूलाऽवनतिश्च तथा दशज्यया लम्बनावनती॥

इत्यादिनोक्तं विचार्य 'न स्फुटं भवति पञ्चजीवयेति' श्रीपतिनोक्तम्। सूर्यसिद्धान्ते, शिष्यधीवृद्धिदे लल्लाचार्येण च पूर्वोक्तपञ्चजीवयैव रविग्रहणानयनमुक्तं तत् सदोषं ज्ञात्वा श्रीपतिनाऽऽचार्यं (ब्रह्मगुप्त) कथितमार्गं एवावलम्बित इति ॥२५॥

अब विशेष कहते हैं

हि. भा.— अन्य जो शेष (स्थित्यर्धादि साधन) रह गया है वह पूर्ववत् (चन्द्रग्रहणोक्त साधन की तरह) होता है, ब्रह्म (आचार्य) कथित स्पष्ट रवि, स्पष्टचन्द्र और स्पष्ट पात से ग्रहण स्पष्ट (दृष्टियोग्य) होता है, आर्यभटादि आचार्यों से जो कथित है उससे दृष्टियोग्य नहीं होता है, क्योंकि वे (आर्यभटादिकथित स्पष्टरवि, स्पष्टचन्द्र और स्पष्ट पात) अस्फुट हैं इसलिये तत्कथित ग्रहण अस्पष्ट (दृष्टियोग्य नहीं) है इति सिद्धान्तशेखर में "न स्फुटं भवति पञ्चजीवया" इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से, सूचित होता है कि श्रीपति ने सूर्यग्रहणाध्याय में आचार्योक्त के अनुरूप ही कहा है। "दृग्गणितैक्यं न भवति" इत्यादि संस्कृतोपपत्ति में लिखित आचार्योक्त पद्य की व्याख्या चतुर्वेदाचार्य इस तरह कहते हैं। पञ्चज्याविधान से सूर्यग्रहण जो आचार्य से कथित है वह यह है 'उदयज्या, मध्यज्या, शङ्कुज्या, दृग्गतिज्या, दृक्क्षेपज्या इन से आर्यभटादि आचार्य द्वारा तथा पौलिशतन्त्र में चन्द्रमा की पञ्चज्या स्वदिनगत-शेष, चरखण्ड, क्रान्ति आदियों से कल्पित है। इस तरह उस में दशज्या विधान से रविग्रहण साधन जो कहा गया है वह अमान्त में होता है, उसमें जो दोष है उनको तन्त्रपरीक्षा अध्याय में आचार्य कहेंगे मैं भी वहीं व्याख्या करूंगा। तन्त्रपरीक्षा अध्याय में "पञ्चज्यया यतोऽर्कग्रहणं" इत्यादि संस्कृतोपपत्ति में लिखित श्लोकोक्त को विचार कर 'न स्फुटं भवति पञ्चजीवया' यह श्रीपति ने कहा है। सूर्यसिद्धान्त में, शिष्यधीवृद्धद में लल्लाचार्य ने पूर्वोक्त पञ्चज्या ही से सूर्यग्रहण साधन कहा है, उसको दोषावह समझकर श्रीपति ने आचार्य (ब्रह्मगुप्त) कथित मार्ग ही का अवलम्बन किया है ॥२५॥

इदानीं ग्रहणे वर्णमाह

अङ्गुलमात्रे विरते रक्तः शशिमण्डले भवेद्वर्णः ।
भानोस्तु पुनः कृष्णो वर्णः सर्वत्र निर्दिष्टः ॥२६॥

सु. भा.—शशिमण्डले चन्द्रबिम्बेऽङ्गुलमात्रे विरते विरामतां गते भानोर्वर्णो रक्तो भवेत् । अर्थाद्रविग्रहणे चन्द्रबिम्बस्य छादकत्वादङ्गुलमात्रेऽपि विरते यत्र ग्रासाभावोऽर्थाद्यद्रविबिम्बमवशिष्टं तद्रक्तमेव भवति । अवशिष्टमङ्गुलमात्रं वा यथेच्छं ततोऽप्यल्पं भवेत् । अन्यत्र सर्वत्र तु पुनः कृष्णवर्णो गणकैर्निर्दिष्टः कथितः अर्थाद् ग्रस्तं रविबिम्बमङ्गुलाल्पं वा यथेच्छं ततोऽप्यधिकं सर्वदा कृष्णमेव भवतीत्यर्थः ॥२६॥

वि. भा.—शशिमण्डले (चन्द्रबिम्बे) ऽङ्गुलमात्रे विरामतां प्राप्ते भानोः (सूर्यस्य) वर्णो रक्तो भवति, अर्थात् सूर्यग्रहणे चन्द्रबिम्बस्य छादकत्वादङ्गुलमात्रेऽपि विरते यत्र ग्रासाभावोऽर्थात्सूर्यबिम्बमवशिष्टं तद्रक्तमेव भवति । अवशिष्टमङ्गुलमात्रं वा यथेच्छं ततोऽप्यल्पं भवेत् । अन्यत्र सर्वत्र पुनः कृष्णो वर्णो गणकैः कथितः । अर्थाद् ग्रस्तं रविबिम्बमङ्गुलाल्पं वा यथेच्छं ततोऽप्यधिकं सर्वदा कृष्णमेव भवतीति ।

अब ग्रहण में वर्ण को कहते हैं

हि. भा.— एक अङ्गुल मात्र में सूर्यग्रहण में सूर्य बिम्ब में चन्द्र बिम्ब विरामता को प्राप्त करे अर्थात् स्थिर हो तब सूर्य का वर्ण रक्त (लाल) होता है, अन्यत्र सब जगह कृष्ण वर्ण होता है अर्थात् ग्रस्त (चन्द्र बिम्ब से ढका हुआ रवि बिम्ब) रवि बिम्ब एक अङ्गुल से अल्प वा अधिक हो तो सदा उनका (सूर्य का) वर्ण कृष्ण ही होता है इति॥२६॥

इदानीं सूर्यग्रहणाधिकारोपसंहारमाह

इष्टग्रासविमर्दस्थित्यर्धावनतिलम्बनाद्येषु ।
आर्यषड्विंशत्याऽर्कग्रहणं पञ्चमोऽध्यायः ॥२७॥

सु. भा.—स्पष्टार्थम् ॥२७॥

मधुसूदनसूनुनोदितो यस्तिलकः श्रीपृथुनेह जिष्णुजोक्ते । हृदि तं विनिधाय नूतनोऽयं रचितः सूर्ययुतौ सुधाकरेण ।

इति श्रीकृपालुदत्तसूनुसुधाकरद्विवेदिविरचिते ब्राह्मस्फुटसिद्धान्तनूतनतिलके सूर्यग्रहणाधिकारः पञ्चमः ॥५॥

वि. भा.—इष्टग्रास-विमर्दार्ध-स्थित्यर्ध-स्पष्टनतिस्पष्टलम्बनाद्येषु विल्येषु-...

षड्विंशतिप्रमिताऽऽर्याश्लोकेन सूर्यग्रहणनामकः पञ्चमोऽध्यायः समाप्त इति ॥२७॥

इति ब्रह्मगुप्तविरचिते ब्राह्मस्फुटसिद्धान्ते सूर्यग्रहणाधिकारः पञ्चमः ॥५॥

अब सूर्यग्रहणाधिकार के उपसंहार को कहते हैं

हि. भा.—इष्टग्रास-विमर्दार्ध-स्थित्यर्ध-स्पष्टनति स्पष्टलम्बनादियों में छब्बीस आर्या छन्द के श्लोकों से सूर्यग्रहण नामक पांचवा अध्याय समाप्त हुआ। इति॥२७॥

इति ब्रह्मगुप्तविरचित ब्राह्मस्फुटसिद्धान्त सूर्यग्रहणाधिकार समाप्त हुआ ॥५॥

ब्राह्मस्फुटसिद्धान्तः

उदयास्ताधिकारः

ब्राह्मस्फुटसिद्धान्तः

उदयास्ताधिकारः

अथोदयास्ताधिकारः प्रारभ्यते । तत्रादौ तदारम्भप्रयोजनमाह

ग्रहभास्करान्तरैः प्राक् पश्चादर्कग्रहान्तरैर्यस्मात् ।
स्वांशैर्दृश्यादृश्यास्तस्माद्वक्ष्ये तदानयनम् ॥१॥

सु. भा.—प्राक् पूर्वक्षितिजे ग्रहभास्करान्तरैः स्वांशैः स्वकालांशैः पश्चात् पश्चिमक्षितिजेऽर्कग्रहान्तरैः स्वकालांशैर्यतो यथासंख्यं ग्रहा दृश्यादृश्या भवन्ति । तस्मात् तदानयनं ग्रहार्कान्तरानयनं वक्ष्येहमिति । अर्थात् प्राक्-क्षितिजे यस्मिन् दिने ग्रहोदयादनन्तरं कालांशघटीभी रवेरुदयस्तस्मिन् दिने ग्रहो रात्रिशेषे दृश्यो भवति । एवं यस्मिन् दिने पश्चिमक्षितिजे रव्यस्तानन्तरं कालांश-घटीभिर्ग्रहस्यास्तमयस्तस्मिन् दिने सायंकाले ग्रहस्यादृश्यत्वम् । अत उदयास्तज्ञानं ग्रहार्कान्तरघटिकाधीनं तदानयनायायमध्यायोऽवश्यमारम्भणीय इति ॥१ ।

वि. भा.—यस्मात् कारणात् प्राक् (पूर्वक्षितिजे) ग्रहभास्करान्तरैः (ग्रहसूर्यान्तरैः) स्वांशैः (स्वकालांशैः) पश्चात् (पश्चिमक्षितिजे) अर्कग्रहान्तरैः स्वकालांशैर्यथासंख्यं ग्रहा दृश्यादृश्या भवन्ति, एतदुक्तं भवति पूर्वक्षितिजे यस्मिन् दिने ग्रहोदयानन्तरं कालांशघटीभी रवेरुदयस्तस्मिन् दिने ग्रहो रात्रिशेषे दृश्यो भवति, एवं यस्मिन् दिने पश्चिमक्षितिजे सूर्यास्तानन्तरं कालांशघटी-भिर्ग्रहस्यास्तमयस्तस्मिन् दिने सायंकाले ग्रहोऽदृश्यो भवति, तस्मात् कारणा-दुदयास्तज्ञानस्य ग्रहरव्यन्तरघटिकाधीनत्वात्तदानयनं (ग्रहरव्यन्तरसाधनं) वक्ष्येऽहमिति सिद्धान्तशेखरे श्रीपतिना "प्राक् खेचरार्कविवरप्रभवैर्यतोंऽशैः पश्चात् सहस्रकिरणद्युचरान्तरस्थैः । तिग्मांशुसन्निधिवशात् स्फुरदृश्यदृश्यास्तस्माद् ब्रवीम्यथ तदानयनं स्फुटार्थम्" ऽऽचार्योक्तानुरूपमेवोक्तमिति ॥१॥

अब उदयास्ताधिकार प्रारम्भ किया जाता है, पहले आरम्भ करने के प्रयोजन को कहते हैं

हि. भा.—जिस कारण से पूर्वक्षितिज में ग्रह और रवि का अन्तर रूप स्वकालांश करके ग्रह दृश्य होते हैं, पश्चिम क्षितिज में रवि और ग्रह का अन्तररूप स्वकालांश करके

ग्रह अदृश्य हैं, अर्थात् पूर्वक्षितिज में जिस ग्रहोदय के बाद कालांश घटीतुल्यकाल में रवि का उदय होता है उस दिन ग्रह रात्रिशेष में दृश्य होते हैं। इसी तरह जिस दिन पश्चिम क्षितिज में सूर्यास्त के बाद कालांशघटीमित काल में ग्रह अस्त होते हैं उस दिन सायंकाल में ग्रह अदृश्य होते हैं, उस कारण से उदय और अस्त का ज्ञान ग्रहार्कान्तरघटी के अधीन होने के कारण उसके आनयन (ग्रह और रवि के अन्तर) को मैं कहता हूँ इति ॥१॥

इदानीं सूर्यसान्निध्यवशेन ग्रहाणां दृश्यादृश्यत्वमाह

प्रागूनभुक्तिरूनो दृश्योऽदृश्योऽन्यथा रवितः।
दृश्योऽधिकगतिरधिकोऽदृश्यः पश्चाद् ग्रहोऽल्पगतिः ॥२॥

सु. भा.—रवित ऊनो रवेरूनभुक्तिर्ग्रहो प्राक् प्राच्यां दिशि रात्रिशेषे रवितः कालांशान्तरितो दृश्यो भवति। अन्यथा रवेरधिकगतिर्ग्रहो रवित ऊनः प्राच्यां दिशि रात्रिशेषे रवितः कालांशान्तरितोऽदृश्यो भवति। एवमधिकगतिर्ग्रहो रवितोऽधिकः कालांशान्तरितः पश्चात् पश्चिमायां दिशि दृश्यो भवति। अल्पगतिश्च रवितोऽधिकः कालांशान्तरितस्तत्रैवादृश्यो भवति।

अत्रोपपत्तिः। 'रवेरूनभुक्तिर्ग्रहः प्रागुदेतीत्यादिभास्करविधिना स्फुटा (ग्र. ग. उ. ग्र. ४) ॥

वि. भा.—रवित ऊनः (अल्पः) ऊनभुक्ति (स्वल्पगति) ग्रहः प्राक् (पूर्वस्यां) दिशि दृश्यो भवति, अन्यथा (रवेरधिकगतिग्रहो न्यूनश्च) प्राक् दिश्यदृश्यो भवति। एवं रवितोऽधिकगतिरधिकश्च ग्रहः पश्चात् (पश्चिमायां दिशि) दृश्यो भवति अल्पगतिश्च ग्रहो रवितोऽधिकः पश्चादृश्यो भवतीति ॥२॥

अत्रोपपत्तिः

रवितोऽल्पगतिका ग्रहा सूर्यसान्निध्यवशेनादृश्यबिम्बका यदा सूर्येण याम्यं प्राप्तास्तदा स परमास्तकालः। ततोऽनन्तरं रविः शीघ्रगतित्वादग्रतो गच्छन् पूर्वक्षितिजे ग्रहोदयानन्तरमुत्तरोत्तरवृद्ध्या समागच्छति, ग्रहाणामेतेषां प्रथमदर्शनरूप उदयो रात्रिशेषे भवेदिति पूर्वस्यां दिश्युदयः। ततोऽन्तरवृद्ध्यैषां पश्चाद् भागे रवौ समागते एषां पश्चिमदिशि दर्शनं तत्रैव निश्चितकालांशतुल्येऽन्तरेऽवशिष्टेऽदर्शनमिति पश्चिमायामस्तत्वम्। रवितोऽधिकगतीनां ग्रहाणां परमास्तसमयादग्रतो गमनात् सूर्यास्तानन्तरं दर्शनसम्भवात् पश्चिमोदयः। परावर्त्य रवितः पश्चाद्भागे समायातेषु रात्रिशेषे तेषां दर्शनात् तत्रैव कालांशतुल्येऽन्तरेऽस्तसम्भावनातः पूर्वास्तगामित्वम्। बुधशुक्रयो रवेरल्पगतित्वात्तयोः पूर्व-

दिश्युदयः । पश्चादस्तमयश्च भवेत् । तेन ऋजुबुधशुक्रौ पश्चिमायामुद्गम्य तत्रैव वक्रगत्याऽस्तमयं यातः । रवेरग्रतः पृष्ठतो वा मन्दफलशीघ्रफलयोगाधिकगमनाभावादुदयास्तमयसमययोरभ्यन्तरे वक्रगतित्वाच्च रवितोऽल्पगतित्वं रवितोऽधिकगतित्वं चानयोर्भवतः । तेन पूर्वापरयोरुभयोरपि दिशोरुदयास्तौ भवेतामिति । सिद्धान्तशेखरे "ऊनो ह्यूनगतिः सहस्रकिरणाद् दृश्यो भवेत् प्राग्ग्रहः पश्चादभ्यधिकस्तथाऽधिकगतिः स्यात् प्रागदृश्यः पुनः । स्वल्पोऽनल्पगतिस्तथोनगतिकः पश्चाददृश्योऽधिकः कालांशैरधिकोनकस्तु कथितैरिति" ऽनेन श्रीपतिना ऽऽचार्योक्तानुरूपमेवोक्तं सिद्धान्तशिरोमणौ "रवेरूनभुक्तिर्ग्रहः प्रागुदेतीत्यादिना" भास्करेणाप्येवमेव कथ्यत इति ॥२॥

अब सूर्यसान्निध्यवश से ग्रहों के दृश्यत्व और अदृश्यत्व को कहते हैं

हि. भा.—रवि से अल्प और अल्पगतिग्रह पूर्वदिशा में दृश्य होते हैं । अन्यथा (रवि से अधिक गतिग्रह और न्यून) पूर्वदिशा में अदृश्य होते हैं । एवं रवि से अधिक गतिग्रह और अधिक पश्चिम दिशा में दृश्य होते हैं । अल्पगतिग्रह और रवि से अधिक पश्चिम दिशा में अदृश्य होते हैं इति ॥२॥

उपपत्ति

रवि से अल्पगतिकग्रह सूर्यसान्निध्य (समीपता) वश से जिनके बिम्ब अदृश्य हैं वे जब सूर्य के साथ तुल्यता को प्राप्त होते हैं तब परमास्तकाल होता है । उसके बाद रवि शीघ्रगति होने के कारण आगे जाते हुआ पूर्व क्षितिज में ग्रहोदय के बाद धीरे-धीरे आता है । इन ग्रहों का प्रथमदर्शन रूप उदय रात्रिशेष में होता है अतः पूर्वदिशा में उदय होता है, बाद में अन्तर वृद्धि से इन ग्रहों के पश्चाद् भाग में रवि के आने से पश्चिम दिशा में इन ग्रहों का दर्शन होता है, वहीं पर निश्चित कालांश तुल्य अन्तर में अदर्शन (नहीं देखना) होता है, अतः पश्चिम दिशा में अस्तत्व होता है, रवि से अधिक गति ग्रहों के परमास्त समय से आगे जाने से सूर्यास्त के बाद दर्शन सम्भव से पश्चिम दिशा में उदय होता है, लौटकर रवि से पश्चाद् भाग में आने पर उसके रात्रि शेष में दर्शन से वहीं पर कालांश तुल्य अन्तर में अस्तसम्भावना से पूर्वदिशा में अस्तगामित्व होता है । वक्रगति बुध और शुक्र के रवि से अल्पगतित्व के कारण पूर्वदिशा में उदय होता है, और पश्चिम दिशा में अस्त होता है, इसलिये ऋजु (मार्गी) बुध और शुक्र पश्चिम दिशा में उदित होकर वहीं वक्रगति से अस्तत्व प्राप्त होते हैं । रवि से आगे या पीछे मन्दफल और शीघ्रफल के योग से अधिक चलना नहीं हो सकता है इसलिये उदयकाल और अस्तकाल के अभ्यन्तर में वक्रता से इन दोनों (बुध और शुक्र) का रवि से अल्पगतित्व और अधिक गतित्व होता है, इसलिये पूर्वदिशा में और पश्चिमदिशा में (दोनों में) इन दोनों का उदय और अस्त होता है । सिद्धान्तशेखर में "ऊनो ह्यूनगतिः सहस्रकिरणाद् दृश्यो भवेत् प्राग्ग्रहः" इत्यादि

संस्कृतोपपत्ति में लिखित पद्यों से श्रीपति आचार्योक्तानुरूप ही कहते हैं। सिद्धान्त-शिरोमणि में "रवेरूनभुक्तिर्ग्रहः प्रागुदेति" इत्यादि से भास्कराचार्य ने भी इन्हीं बातों को कहा है इति ॥२॥

इदानीं ग्रहबिम्बोदयास्तलग्नानयनार्थमायनदृक्कर्मानयनमाह

विक्षेपसत्रिराशिक्रान्तिवधो व्यासदलहृतो लिप्ताः।
शोध्यास्तयोः समदिशोर्यद्यन्यदिशोस्तयोः क्षेप्याः ॥३॥

सु. भा.—सत्रिराशिक्रान्तिः सत्रिभग्रहक्रान्तिज्या द्युज्यावृत्ते आयनवलनमाचार्यमते स्थूलां द्युज्यां त्रिज्यामितां प्रकल्प्य तदेवायनं वलनम्। तयोर्विक्षेपायनवलनयोः शेषं स्पष्टार्थम्।

अत्रोपपत्तिः। 'आयनं वलनमस्फुटेषुणा सङ्गुणं द्युगुणभाजितं हतम्' इत्यादि भास्करविधिना द्युज्यां त्रिज्यासमां राश्युदयासून् राशिकलामितान् १८०० प्रकल्प्य स्फुटा ज्ञेया। वस्तुतो बिंबीयद्युज्ययाऽऽयनदृक्कर्म सूक्ष्मं भवति, भास्करेणापि स्थानीयद्युज्यां गृहीत्वा स्थूलमेवेदमानीतमिति सिद्धान्तविदां स्फुटम् ॥३॥

वि. भा.—**विक्षेपसत्रिराशिक्रान्तिवधः** (शरसत्रिभग्रहक्रान्तिघातः) **व्यासदलहृतः** (त्रिज्यया भक्तः) फलतो या **लिप्ताः** (कलाः) **ताः समदिशोः** (एकदिक्कयोः) **तयोः** (शरसत्रिराशिक्रान्त्योः) ग्रहे **शोध्याः**, **अन्यदिशोस्तयोः** (भिन्नदिक्कयोः शरसत्रिराशिक्रान्त्योः) ग्रहे **क्षेप्याः** (योज्याः) इति प्रथमं दृक्कर्म आयनाख्यमस्तीति. ॥३॥

अत्रोपपत्तिः

ग्रहबिम्बकेन्द्रोपरिगतं कदम्बप्रोतवृत्तं क्रान्तिवृत्ते यत्र लगति तदेव ग्रहस्थानम्। स्थानोपरि बिम्बोपरि च गतं ध्रुवप्रोतवृत्तं कार्यं, स्थानीयाहोरात्रवृत्तं बिम्बीयाहोरात्रवृत्तं च कार्यं तदा स्थानोपरिगतध्रुवप्रोतवृत्तकदम्बप्रोतवृत्तयोरन्तरे बिम्बीयाहोरात्रवृत्तीयं चापमायनदृक्कर्म वा स्थानबिम्बयोरुपरिगतध्रुवप्रोतवृत्तयोरन्तर्गतं नाड़ीवृत्तीयचापम् ग्रहबिम्बकेन्द्राद् ग्रहस्थानं यावद् ग्रहमध्यमशरः। स्थानगतकदम्बप्रोतवृत्तध्रुवप्रोतवृत्तयोरुत्पन्नकोणः स्थानीयमायनवलनमिदं (द्युज्याग्रीयमायनवलनं) सत्रिभग्रहक्रान्तिसमं भवति। ततो ग्रहमध्यमशर एको भुजः। बिम्बकेन्द्रात्स्थानगतध्रुवप्रोतवृत्तोपरिलम्बचापं द्वितीयो भुजः। स्थानगतध्रुवप्रोतवृत्ते तृतीयो भुजः। त्रिभुजेऽस्मिन् कोणानुपातः क्रियते यदि त्रिज्यया मध्यमशरज्या लभ्यते तदाऽऽयनवलनज्यया

किमित्यनुपातेन समागच्छति लम्बचापज्या $\frac{\text{मशज्या. आयनवज्या}}{\text{त्रि.}}$ यावती लम्बज्या तावत्येव बिम्बीयाहोरात्रवृत्तीयायनदृक्कर्मसुज्याऽतः

$\frac{\text{मशज्या. आयनवज्या}}{\text{त्रि}}$ = आयनदृक्कर्मसुज्या, ज्याचापयोरभेदत्वस्वीकारे

$\frac{\text{मशर. आयनवलन}}{\text{त्रि}}$ = आयनदृक्कर्मसिवः। आचार्येण स्वल्पान्तरत्वात् त्रि = स्थानीयद्यु स्वीकृतम् एतद्वशेनाऽयनदृक्कर्मकलानयन $\frac{\text{'१८०० × आयनदृक्कर्मसु'}}{\text{ग्रहाक्रान्तराशेर्निरक्षोदयासु}}$ मेतेन सुगममेव। इदमानयनं बिम्बीयद्युज्यावशेन समीचीनं भवितुमर्हति, परं बिम्बीयद्युज्याया अज्ञानात्स्थानीयद्युज्यावशेनैवानयनमभिहितं तत्रापि त्रिज्याद्युज्ययोः समत्वस्वीकरणजनितदोषस्तु वर्त्तत एव। सिद्धान्तशिरोमणौ "आयनं वलनमस्फुटेषुणा सङ्गुणं द्युगुणभाजितं हृतम्। पूर्णपूर्णधृति १८०० भिर्ग्रहाश्रितव्यक्षभोदयहृदायनाः कलाः" भास्कराचार्येणापि त्रिज्यासमामेव द्युज्यां मत्वाऽऽयनदृक्कर्मानयनमभिहितम्। सिद्धान्तशेखरे "विक्षेपसत्रिभखगोत्क्रमजाऽऽमज्याघाते गृहत्रयगुणेन हृते कलास्ताः। शोध्यास्तयोः समदिशोः खचरेषु देया भिन्नाशयो र्भवति दृग्विधिरेष पूर्वः' श्लोकेनानेन श्रीपतिना लल्लाचार्योक्तसत्रिभग्रहोत्क्रमज्यासाधितक्रान्तिज्यातुल्याऽयनवलनज्यात आयनदृक्कर्मानयनं कृतं, श्रीपतिर्बहुधाऽऽचार्यमतानुसरणं कुर्वन्नपि कुत्रचित् कुत्रचित् स्थले लल्लोक्तमपि आचार्यमतविरुद्धं स्वीचकार, सत्रिभग्रहक्रांतिज्या द्युज्याग्रीयाऽयनवलनज्या समा भवति, न हि सत्रिभग्रहोत्क्रमज्यासाधितक्रान्तिज्याऽऽयनवलनतुल्या भवत्यतो लल्लाचार्योक्तं श्रीपत्युक्तं चायनदृक्कर्मसाधनं न युक्तम्। सूर्यसिद्धान्ते 'सत्रिभग्रहजक्रान्तिभागघ्ना क्षेपलिप्तिका इत्यादि" ऽऽयनदृक्कर्मानयनं लल्लाचार्यश्रीपतिसाधिताऽऽयनदृक्कर्मभ्यां किञ्चित् सूक्ष्मं परं केषामप्यानयनं वास्तवं नास्तीति विज्ञैर्बोध्यमिति ॥३॥

अब ग्रहबिम्बोदयास्तलग्नसाधन के लिये आयनदृक्कर्म साधन को कहते हैं

हि. भा.—शर और सत्रिभ ग्रह क्रान्ति के घात को त्रिज्या से भाग देने से जो फल हो उसकी कला को शर और सत्रिभ ग्रह क्रान्ति के एक दिशा रहने से ग्रह में घटा देना भिन्न दिशा रहने से ग्रह में जोड़ना चाहिये इति ॥३॥

उपपत्ति

ग्रहबिम्बकेन्द्रोपरिगतक्रदम्बप्रोतवृत्त क्रान्तिवृत्त में जहां लगता है वह ग्रहस्थान

है, स्थान से बिम्बकेन्द्र तक ग्रह का मध्यमशर है स्थानोपरिगत तथा बिम्बोपरिगत ध्रुव-प्रोतवृत्त करना, स्थानीयाहोरात्रवृत्त और बिम्बीयाहोरात्रवृत्त कर देना, तब स्थानोपरिगत ध्रुवप्रोतवृत्त और स्थानोपरिगत कदम्बप्रोतवृत्त के अन्तर्गत बिम्बीयाहोरात्रवृत्तीय चाप आयन दृक्कर्मासु है अथवा स्थानोपरिगत ध्रुवप्रोतवृत्त तथा बिम्बोपरिगत ध्रुवप्रोतवृत्त के अन्तर्गत नाड़ीवृत्तीय चाप आयन दृक्कर्मासु है, स्थानगत कदम्बप्रोतवृत्त और ध्रुवप्रोतवृत्त से उत्पन्न कोण स्थानीय आयनवलन है, द्युज्याग्रीय आयनवलन सत्रिभग्रह की क्रान्ति के बराबर होता है, तब ग्रह का मध्यमशर एकभुज, बिम्बकेन्द्र से स्थानगत ध्रुवप्रोतवृत्त के ऊपर लम्बवृत्तीय चाप द्वितीय भुज, स्थानगत ध्रुवप्रोतवृत्त में तृतीय भुज, इन तीनों भुजों से उत्पन्न चापीयजात्यत्रिभुज में कोणानुपात करते हैं यदि त्रिज्या में मध्यमशरज्या पाते हैं तो आयन वलनज्या में क्या इससे लम्बवृत्तीय चापज्या आती है, $\frac{\text{मशरज्या.आयनवज्या}}{\text{त्रि}}$ = लंबचापज्या परन्तु लम्बवृत्तीय चाप-ज्या और बिम्बीयाहोरात्रवृत्तीयायनदृक्कर्मासुज्या बराबर है इसलिये $\frac{\text{मशरज्या.आयनवज्या}}{\text{त्रि}}$ = आयनदृक्कर्मासुज्या, = $\frac{\text{मशरज्या.सत्रिभग्रक्रांज्या}}{\text{त्रि.}}$ ज्या और चाप का अभेदत्व स्वीकार करने से $\frac{\text{मशर.सत्रिभग्रक्रान्ति}}{\text{त्रि}}$ आयनदृक्कर्मासु, यहां आचार्य ने स्वल्पान्तर से त्रि=स्थानीय द्युज्या स्वीकार किया है। इसके वश से आयन दृक्कर्मकलानयन करते हैं $\frac{१८०० \times \text{आयनदृक्कर्मासु}}{\text{ग्रहाक्रान्त राशि के निरक्षोदयासु}}$ = आयनदृक्कर्मकला बिम्बीय द्युज्या वश से आयन दृक्कर्मानयन समीचीन हो सकता है, लेकिन बिम्बीय द्युज्या विदित नहीं है, स्थानीय द्युज्या वश ही से आचार्य ने आनयन किया है तथा द्युज्या को त्रिज्या के बराबर मान लिया है इसलिये यह आनयन ठीक नहीं है। सिद्धान्त-शिरोमणि में "आयनं वलनमस्फुटेषुणा सङ्गुणैः" इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से आयन दृक्कर्मानयन किया है, लेकिन भास्कराचार्य ने भी त्रिज्या तुल्य ही द्युज्या स्वीकार की है, सिद्धान्तशेखर में 'विक्षेपसत्रिभखगोत्क्रमजाऽऽयमज्या' इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से श्रीपति ने लल्लाचार्योक्त सत्रिभग्रह की उत्क्रमज्या से साधित क्रान्तिज्या तुल्य आयनवलज्या से आयनदृक्कर्मानयन किया है। श्रीपति ने बहुत स्थानों में आचार्य (ब्रह्मगुप्त) मत का अनुसरण करते हुये भी किसी-किसी स्थल में आचार्यमत के विरुद्ध लल्लोक्त को भी स्वीकार किया है, सत्रिभग्रह क्रान्तिज्या और द्युज्याग्रीयायनवलज्या बराबर होती हैं। सत्रिभग्रहोत्क्रमज्या साधित क्रान्तिज्या, आयनवलनज्या के तुल्य नहीं होती है इसलिये लल्लाचार्योक्त और श्रीपत्युक्त आयन दृक्कर्मानयन ठीक नहीं है, सूर्यसिद्धान्त में "सत्रिभ-ग्रहजक्रान्तिभागघ्ना क्षेपलिप्तिका" इत्यादि से आयन दृक्कर्मानयन, लल्लाचार्य और श्रीपति से साधित आयन दृक्कर्मों से किञ्चित् सूक्ष्म है, लेकिन किसी भी आचार्य का दृक्कर्मानयन ठीक नहीं है इति॥३॥

इदानीमाक्षदृक्कर्मानयनमाह

विषुवच्छायागुणितादिक्षेपाद् द्वादशोद्धृतात् सौम्यात् ।
फलमृणं धनं धनमृणं याम्यादुदयास्तमयलग्ने ॥४॥

सु. भा.—सौम्यात् शरात् फलमुदयलग्नसाधने फलं कलात्मकमृणमस्तलग्नसाधने च धनं याम्याद्विक्षेपाच्च फलं क्रमेण धनमृणं ज्ञेयं पूर्वसाधितायनदृक्कर्मसंस्कृतग्रहे तदोदयास्तलग्ने भवतः ।

अत्रोपपत्तिः । क्षितिजे स्थूलादक्षज्यासममक्षवलनं कदम्बाभिमुखं गणितागतशरं च स्वल्पांतराद् ध्रुवाभिमुखं स्पष्टशरं च प्रकल्प्य 'लम्बज्ययाऽक्षजं चेत् स्याद्वलनं किं स्फुटेषुणा' इत्यादिभास्करविधिना त्रिज्यासमां द्युज्या राश्युदयासून् खखाष्टेन्दुसमान् प्रकल्प्याक्षजदृक्कर्मकला $= \frac{\text{ज्या अ}\cdot\text{ श}}{\text{ज्या ल}} = \frac{\text{वि}\cdot\text{ श}}{१२}$ । 'अथ तैः शरे तु याम्योत्तरे क्रमविलोमविधानलग्न' मित्यादिभास्करोक्तेन उत्तरे शरेऽक्षजदृक्कर्मकलाभिरूनो दक्षिणे शरे युतः कृतायनफलः खेट उदयास्तलग्नं भवति । अस्तलग्नसाधने तु सौम्ये शरेऽक्षजदृक्कर्मकलासहितो याम्ये रहितः षड्भः कृतायनफलः खेटो ग्रहे पश्चिमक्षितिजेऽस्तंगते प्राक्क्षितिजे यल्लग्नं तदस्तलग्नं भास्करमते । इहाचार्येण तस्मात् षड्राशिं विशोध्य पश्चिमक्षितिजे ग्रहेऽस्तंगते यदस्तलग्नं तदेव ग्रहास्तलग्नं कल्पितमिति ॥४॥

वि. भा.—विषुवच्छाया (पलभा) गुणितात् विक्षेपात् (मध्यमशरात्) द्वादशभक्ताद्यत्फलं कलात्मकं सौम्याच्छरात् तदुदयलग्नसाधने ऋणमस्तलग्नसाधने च धनं याम्याच्छरात् फलं क्रमेण पूर्वसाधिताऽयनदृक्कर्म संस्कृतग्रहे धनमृणं कार्यं तदोदयास्तलग्ने भवेतामिति ॥४॥

अत्रोपपत्तिः

यदा ग्रहस्थानं क्षितिजे भवेत्तदा बिम्बं क्षितिजादध ऊर्ध्वं वा भवेत् । स्थानोपरिगतं ध्रुवप्रोतवृत्तं बिम्बकेन्द्रोपरिगतं ध्रुवप्रोतवृत्तं च कार्यं बिम्बकेन्द्रापर्यंहोरात्रवृत्तं (बिम्बीयाहोरात्रवृत्तं) कार्यं तदा तयोः स्थानबिम्बकेन्द्रगतध्रुवप्रोतवृत्तयोरन्तर्गतं बिम्बीयाहोरात्रवृत्तीयचापमाक्षदृक्कर्मसंज्ञकं भवति । अथ स्थानाद्बिम्बीयाहोरात्रवृत्तं यावत्स्थानगतध्रुवप्रोतवृत्तं स्पष्टशर एको भुजः । बिम्बीयाहोरात्रवृत्तक्षितिजवृत्तयोः सम्पातात्स्थानं यावत् क्षितिजवृत्तं द्वितीयो भुजः । बिम्बीयाहोरात्रवृत्तक्षितिजवृत्तयोः सम्पातात्स्थानगतध्रुवप्रोतवृत्तोपरिलम्बवृत्तं तृतीयो भुजः । एतत्त्रिभुजं सरलजात्यं स्वीकृतमाचार्येण अत्र स्थानलग्नकोणोऽक्षांश

समोऽक्षजवलनांशः (क्षितिजेऽक्षज्यया तुल्यमक्षजं वलनमित्युक्तेः) एकः कोणः समकोणोऽतोऽवशिष्टः कोणो लम्बांशस्तदाऽनुपातः क्रियते, यदि लम्बज्यया स्पष्टशरज्या लभ्यते तदाऽक्षज्यया किमित्यनेन लम्बवृत्तीयचापज्याबिम्बीयाहोरात्र-

वृत्तीयचापज्या वा समागच्छति तत्स्वरूपम्=$\frac{\text{स्पशज्या}\cdot \text{अक्षज्या}}{\text{लज्या}}$ (१) परन्तु

$\frac{\text{अक्षज्या}}{\text{ज्या}}=\frac{\text{पभा}}{१२}$, $\therefore \frac{\text{स्पशज्या}.\text{पभा}}{१२}$=लम्बवृत्तापज्या=बिम्बीयाहोरात्रवृत्तापज्या

अत्र स्वल्पान्तरात् स्पष्टशर=मध्यशर, तथाऽऽगतफलस्य चापकरणेन च

$\frac{\text{शर}.\text{पभा}}{१२}$=बिम्बीयाहोरात्रवृत्तचापपरमित्यक्षजदृक्कर्मतुल्यं नास्त्यत आचार्योक्तमिदमानयनं महत्स्थूलम्। (१) अत्रत्य स्वरूपे क्षितिजे ग्रहस्थानस्य स्थितत्वात् अक्षज्या=आक्षवलनज्या, स्वल्पान्तरात् स्पशज्या=स्पशर, तदा—

$\frac{\text{स्पश}.\text{आक्षवलन}}{\text{लज्या}}$=आक्षदृक्कर्म, परं $\frac{\text{अज्या}}{\text{लज्या}}=\frac{\text{पभा}}{१२}$ $\therefore \frac{\text{स्पशर}.\text{पभा}}{१२}$=आक्षदृक्कर्म

एतेन 'स्पष्टेषुरक्षवलेन हतो विभक्तो लम्बज्यया रविहृतोऽक्षभया हतो वेति" सिद्धान्तशिरोमणौ भास्करोक्तमुपपद्यते। सिद्धान्तशेखरे "क्षुण्णे क्षेपेऽक्षज्यया लम्बभक्तेऽक्षच्छायाघ्ने भानुभक्तेऽथवाऽत्रेति" श्रीपत्युक्तमिदमाचार्योक्तानुरूपमेव; "सूर्यसिद्धान्तेऽपि "विषुवच्छाययाभ्यस्ताद्विक्षेपाद्दित्यादिना" तदानयनं कृतमस्ति, परं केषामपि तदानयनं समीचीनं नास्तीति तदुपपत्तिदर्शनेनाऽक्षजदृक्कर्मगौलिकस्वरूपदर्शनेन च स्फुटमिति। अथ तैः शरे तु याम्योत्तरे क्रमविलोमविधानलग्नमित्यादि भास्करोक्तेन—उत्तरे शरेऽक्षजदृक्कर्मकलाभिरूनौ दक्षिणे शरे युतः कृतायनफलः खेटः उदयलग्नं भवति। अस्तलग्नसाधने तु सौम्ये शरेऽक्षजदृक्कर्मकलासहितो याम्ये रहितः सषड्भः कृतायनफलः खेटो ग्रहे पश्चिमक्षितिजेऽस्तंगते प्राक्क्षितिजे यल्लग्नं तदस्तलग्नं भास्करमते। अत्राचार्येण तस्मात् षड्राशिं विशोध्य पश्चिमक्षितिजे ग्रहेऽस्तंगते यदस्तलग्नं तदेव ग्रहास्तलग्नं कल्पितमिति ॥४॥

अब अक्षज दृक्कर्मानयन को कहते हैं

हि. भा.—शर को पलभा से गुणा कर द्वादश (१२) से भाग देने से जो फल होता है उससे जो कलात्मक फल होता है उत्तर शर में उदयलग्न साधन में उस फल को (१) ऋण करना तथा अस्तलग्न साधन में धन करना, दक्षिण शर रहने से उस फल को क्रम से पूर्वसाधित आयन दृक्कर्म संस्कृत ग्रह में धन और ऋण करने से उदयलग्न और अस्त लग्न होता है। (१) पूर्व साधित आयन दृक्कर्म संस्कृतग्रह में इति ॥४॥

उपपत्ति

जब ग्रहस्थान क्षितिज में रहता है तब ग्रहबिम्ब केन्द्र क्षितिज से नीचा या ऊपर रहता है, स्थानोपरिगतध्रुवप्रोतवृत्त तथा बिम्बकेन्द्रोपरिगतध्रुवप्रोतवृत्त करने से दोनों के अन्तर्गत बिम्बीयाहोरात्र वृत्तीय चाप आक्षदृक्कर्म संज्ञक है। स्थान से बिम्बीयाहोरात्र-वृत्तपर्यन्त स्थानगत ध्रुवप्रोतवृत्त में स्पष्टशर एक भुज, बिम्बीयाहोरात्रवृत्त और क्षितिज-वृत्त के सम्पात से स्थान पर्यन्त क्षितिज वृत्त में द्वितीय भुज। बिम्बीयाहोरात्रवृत्त और क्षितिज वृत्त के सम्पात से स्थानगत ध्रुवप्रोतवृत्त के ऊपर लम्ववृत्त में तृतीय भुज, इस त्रिभुज को आचार्य ने सरल जात्य मान लिया है, उक्त त्रिभुज में स्थान लग्न कोण अक्षांश या तत्तुल्य आक्षवलन है, 'क्षितिजेऽक्षज्यया तुल्यमक्षजं वलनं' इस उक्ति से एक कोण समकोण है अतः अवशिष्ट तृतीय कोण लम्बांश के बराबर हुआ। अब अनुपात करते हैं यदि लम्बज्या में स्पष्ट शरज्या पाते हैं तो अक्षज्या में क्या इससे लम्ववृत्तीय चापज्या वा बिम्बीयाहोरात्र-वृत्तीय चापज्या आती है $\frac{\text{स्पशज्या. अक्षज्या (क)}}{\text{लंज्या}}$ परन्तु $\frac{\text{अज्या}}{\text{लंज्या}}=\frac{\text{पभा}}{१२}$ अतः $\frac{\text{स्पशज्या. पभा}}{१२}$ =लम्ववृचापज्या=बिम्बीयाहोरात्रवृचापज्या, यहां स्वल्पान्तर से स्पशरज्या=स्पशर=मशर तथा पूर्वागत फल के चाप करने से $\frac{\text{शर. पभा}}{१२}$ बिम्बीयाहोरात्र वृचा, लेकिन यह आक्ष-दृक्कर्म के बराबर नहीं है। इसलिए आचार्योक्त यह आनयन अत्यन्त स्थूल है। (क) इस स्वरूप में क्षितिज में ग्रहस्थान स्थित रहने से अक्षज्या=आक्षवलनज्या=आक्षवलन स्वल्पा-न्तर से $\frac{\text{स्पशर. आक्षवलन}}{\text{लंज्या}}=\frac{\text{स्पशर. पभा}}{१२}$=आक्षदृक्कर्म, इससे 'स्पष्टेषुरक्षवलनेन हतो विभक्तः" इत्यादि सिद्धांत शिरोमणिस्थ भास्करोक्त उपपन्न होता है। सिद्धांत शेखर में "क्षुण्णे क्षेपेऽक्षज्यया लम्बभक्ते" इत्यादि से श्रीपति ने आचार्योक्तानुरूप ही कहा है। सूर्य सिद्धांत में भी "विषुवच्छाययाभ्यस्ताद्विक्षेपात्" इत्यादि से आक्ष दृक्कर्मानयन किया गया है लेकिन किसी का आनयन ठीक नहीं है यह उपपत्ति देखने से तथा अक्षज दृक्कर्म का प्रदेश देखने से स्फुट है, 'अथ तैः शरे तु याम्योत्तरे क्रमविलोमविधानलग्नं' इत्यादि भास्करोक्ति से उत्तरशर में अयनदृक्कर्मसंस्कृत ग्रह में अक्षज दृक्कर्म कला को घटाने से दक्षिण शर रहने से अक्षजदृक्कर्म कला को आयन दृक्कर्म संस्कृत ग्रह में जोड़ने से उदयलग्न होता है, अस्तलग्न साधन में उत्तरशर रहने से छः राशि सहित आयन दृक्कर्म संस्कृत ग्रह में अक्षजदृक्कर्म कला को जोड़ने से तथा दक्षिणशर रहने से घटाने से पश्चिम क्षितिज में ग्रह के अस्तंगत रहने से पूर्वक्षितिज में जो लग्न होता हैं वह भास्कराचार्य संमत अस्तलग्न होता है, यहां आचार्य ने उस में से छः राशि को घटाकर पश्चिम क्षितिज में ग्रह के अस्तंगत रहने से जो अस्तलग्न होता है उसी को ग्रहास्त लग्न स्वीकार किया है इति ॥ ४ ॥

इदानीं ग्रहार्कान्तरघट्यानयनमाह ।

प्रागूनमाद्यमधिकं पश्चाल्लग्नाद्ग्रहोदयोऽस्तमयः ।
षड्भयुतमन्यदुदयैर्घटिकाः कृत्वोनमधिकसमम् ॥५॥

सु० भा०—प्राक्क्षितिजे आद्यमुदयलग्नं लग्नाद्वा सूर्यादूनं तदा ग्रहोदयो भवति । पश्चात् क्षितिजे लग्नाद्वा सूर्यात् षड्भयुतं षड्राशिसहितमन्यदस्तलग्नमधिकं तदा ग्रहास्तमयो भवति । अत्रोनमधिकसमं कृत्वाऽर्थादूनस्य भोग्योऽधिकभुक्तयुक्तो मध्योदयाढ्य इत्यादिभास्करविधिना उदयैः स्वदेशोदयैर्लग्नग्रहान्तरे रवेनासन्नभावेन ग्रहोदयज्ञानाय रविग्रहोदयलग्नान्तरेऽस्तज्ञानाय सषड्भार्कास्तलग्नान्तरे घटिकाः साध्या इत्याचार्याभिप्रायः ।

अत्रोपपत्तिः । 'प्राग्दृग्ग्रहः स्यादुदयाख्यलग्नमि' त्यादिभास्करविधिना स्फुटा ॥ ५ ॥

वि. भा.—प्राक् (पूर्व क्षितिजे) आद्यं (उदयलग्नं) लग्नाद्वा सूर्यादूनं तदा ग्रहोदयो भवति, पश्चात् (पश्चिम क्षितिजे) लग्नाद्वा सूर्यात् षड्भयुतं (राशिषट्क सहितं) अन्यत् (अस्तलग्नं) अधिकं तदा ग्रहास्तमयो भवति । अत्रोनमधिकसमं कृत्वाऽर्थादूनस्य भोग्योऽधिकभुक्तयुक्तो मध्योदयाढ्य इत्यादि भास्करोक्त्योदयैः (स्वदेशोदयैः) लग्नग्रहान्तरे वा रविसान्निध्यवशेन ग्रहोदयज्ञानार्थं रविग्रहोदयलग्नान्तरेऽस्त ज्ञानार्थं सषड्भार्कास्त लग्नान्तरे घटिकाः साध्या इत्याचार्याभिप्राय इति ॥ ५ ॥

अत्रोपपत्तिः ।

'प्राग् दृग्ग्रहः स्यादुदयाख्यलग्न' मित्यादि भास्करकथितप्रकारेण स्फुटेति ॥ ५ ॥

अब ग्रह और रवि की अन्तरघटी के साधन करते हैं ।

हि. भा.—पूर्वक्षितिज में लग्न से वा सूर्य से उदय लग्न अल्प हो तो ग्रहोदय होता है, तथा पश्चिम क्षितिज में लग्न से वा सूर्य से सषड्भ (छः राशियुत) अस्त लग्न अधिक हो तब ग्रहास्त होता है, यहां 'ऊनस्य भोग्योधिकभुक्तयुक्तो मध्योदयाढ्यः' इत्यादि भास्कर कथित प्रकार से स्वदेशोदय से लग्न और ग्रह के अन्तर में वा रवि के सान्निध्य (समीपता) वश से ग्रहोदय ज्ञान के लिये रवि और ग्रहोदयलग्न के अन्तर में अस्त ज्ञान के लिये सषड्भ (छः राशि युत) रवि और अस्तलग्न के अन्तरघट्यानयन करना चाहिये यह आचार्य का अभिप्राय है इति ॥ ५ ॥

उपपत्ति ।

'प्राग् दृग्ग्रहः स्यादुदयाख्य लग्नं' इत्यादि भास्कर कथित विधि से स्पष्ट है ॥ ५ ॥

इदानीं ग्रहाणां कालांशानाह ।

द्वादशभिः शीतांशुः सितजीवज्ञशनिभूमिजा नवभिः ।
द्व्युत्तरवृद्धैरन्तरघटिका षड्गुणितकालांशैः ॥ ६ ॥

सु० भा०—शीतांशुर्द्वादशभिः कालांशैः । शुक्रगुरुबुधशनिभौमा द्व्युत्तरवृद्धैर्नवभिः कालांशैर्दृश्यादृश्या भवंति । अर्थात् शुक्रस्य नव ९ । गुरोः ९+२=११ । बुधस्य ११+२=१३ । शनेः १३+२=१५ । भौमस्य १५×२=१७ । पूर्वं या ग्रहार्कान्तरघटिकाः साधितास्ताः षड्गुणिताः कालांशा इष्टा भवंति तैरिष्टैः कालांशैरित्यग्रे संबंधः ।

अत्रोपपत्तिः । कालांशसाधने आगमप्रमाणम् । वस्तुतोग्रहाणां नीचोच्चवशाद्रविग्रहांतरसूत्रवैलक्षण्याच्च वेधेनापि गोलयुक्त्या स्थिराः कालांशा असम्भवा एवेति कमलाकरमतं तत्त्वविवेके तथ्यमेवेति ज्योतिर्विद्भिश्चिन्त्यम् । 'दस्रेन्दवः शैलभुवश्च शक्रा' इत्यादि भास्करविधिना चन्द्रादीनां कालांशाः ।

चं. १२ । भौ. १७ । बु. १४ । गु. ११ । शु० १० । श० १५ । आचार्योक्ताः चं. १२ । भौ० १७ । बु० १३ । गु० ११ । शु० ९ । श० १५ । द्वयोर्मतयोर्बुधशुक्रकालांशयोरेकैकमन्तरम् ॥ ६ ॥

वि. भा.—शीतांशुः (चन्द्रः) रवित इत्यध्याहार्यम्-द्वादशभिरंशैरदृश्यो दृश्यश्च जायते, सितजीवज्ञशनिभूमिजाः (शुक्रगुरुबुध शनिमङ्गलाः) द्व्युत्तरवृद्धैर्नवभिः, अन्तरघटिका षड्गुणितकालांशैरर्थादुदयलग्नरव्योरन्तरे वा ऽस्तलग्नरव्योरन्तरे घटिकाः साध्यास्ताः षड्गुणाः कालांशा भवेयुरेभिस्ते दृश्या अदृश्याश्च भवन्ति । यथा शुक्रस्य नव ९, गुरोः=९+२=११, बुधस्य=११+२=१३, शनेः=१३+२=१५, मङ्गलस्य=१५+२=१७, इति ॥ ६ ॥

अत्रोपपत्तिः

रवितो यावदन्तरेण ग्रहाणामुदयोऽस्तमयश्च भवति तेऽन्तरांशाः (कालांशाः) प्राचीनैरसकृद्वेधेन ज्ञात्वा पठिताः परं वस्तुतो ग्रहाणां नीचोच्चवशाद्रविग्रहान्तरसूत्रवैलक्ष्यण्याच्च वेधेनापि कालांशाः स्थिरा न भवितुमर्हन्त्येतद्विषये सिद्धांततत्त्वविवेके कमलाकरेण युक्तियुक्तं प्रतिपादितमस्ति, अथ "दस्रेन्दवः शैलभुवश्चेत्यादि" भास्करकथितविधिना चन्द्रादि ग्रहाणां

कालांशाः=चन्द्रस्य=१२, कुजस्य=१७, बुधस्य=१४, गुरोः=११, शुक्रस्य=१०, शनेः=१५, तथाऽऽचार्योक्ताः चन्द्रस्य=१२, कुजस्य=१७, बुधस्य=१३, गुरोः=११, शुक्रस्य=९, शनेः=१५ एतयोर्मतयोर्बुधशुक्रकालांशयोरेकैकमन्तरमस्ति। सिद्धांतशेखरे श्रीपतिना "शुक्रार्यज्ञद्युमणिजकुजा द्व्यु्त्तरैः कालभागैर्गोभिश्चन्द्रो रविभिरिनतो जायतेऽदृश्यदृश्यः। गम्यो न्यूनादसुचय इतश्चाधिकादन्तरस्थैर्युक्तः प्राणैः स खरसहृतः कालभागा भवन्ति" अनेनाऽऽचार्योक्तानुरूपा एव कालांशा उक्ता इति ॥ ६ ॥

अब ग्रहों के कालांशों को कहते है।

हि. भा.—चन्द्रमा रवि से बारह अंश (कालांश) पर दृश्य और अदृश्य होते हैं, शुक्र, गुरु, बुध, शनि, और मङ्गल, ये ग्रह नौ ९ में दो दो की वृद्धि से अर्थात् शुक्र ९, गुरु ९+२=११, बुध ११+२=१३, शनि १३+२=१५, मङ्गल १५+२=१७ इतने कालांश में दृश्य और अदृश्य होते हैं। इति ॥ ६ ॥

उपपत्ति।

रवि से जितने अन्तर में ग्रहों का उदय अस्त होता है वह अन्तर कालांश कहलाता है, प्राचीनाचार्यों ने बार बार वेधद्वारा कालांश को समझकर पठित किया है, लेकिन यथार्थतः रवि और ग्रह के अन्तर सूत्र (बिम्बान्तर सूत्र) की विलक्षणता के कारण वेध से भी कालांश स्थिर नहीं हो सकता है इस विषय में सिद्धान्ततत्त्वविवेक में कमलाकर ने युक्ति युक्त बातें कही है, "दस्रेन्दवः शैलभुवश्चशक्रा" इत्यादि भास्कर विधि से चन्द्रादि ग्रहों के कालांश ये हैं—चन्द्र के=१२, मङ्गल के=१७, बुध के=१४, बृहस्पति के=११, शुक्र के=१०, शनि के=१५, तथा आचार्य कथित विधि से चन्द्र के=१२, मङ्गल के=१७, बुध के=१३, बृहस्पति के=११, शुक्र के=९, शनि के=१५। दोनों मतों से बुध और शुक्र के कालांशों में एक एक का अन्तर है, सिद्धान्त शेखर में श्रीपति ने "शुक्रार्यज्ञद्युमणिज कुजा" इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से आचार्योक्त के अनुरूप ही कालांश कहा है इति ॥ ६ ॥

इदानीमुदयास्तयोर्गतैष्यदिनानयनमाह।

दृश्यादृश्यैर्युतिवद्ग्रहार्क-भुक्त्यन्तरैक्य-लब्ध-दिनैः ।
ऊनाधिक लिप्ताभ्यां प्राग्वत् तात्कालिकैरसकृत् ॥ ७ ॥

सु० भा०—पूर्वसाधितैरिष्टकालांशैर्दृश्यादृश्यैः पठितकालांशैश्च ऊनाधिकलिप्ताभ्यां ग्रहार्कभुक्त्यन्तरैक्यलब्धदिनैस्तात्कालिकैरिष्टकालांशैः प्राग्वदसकृद्विधिना युतिवद्ग्रहयुतिवदुदयास्तयोर्गतैष्यदिनसाधनं कार्यमित्यर्थः। अत्रैतदुक्तं भवति

पठितेष्टकालांशान्तरकला ग्रहार्कगत्यन्तरभक्ता ग्रहे वक्रिणि तु ग्रहार्कगतियोगभक्ता लब्धा पठितकालांशेभ्य इष्टाधिके उदये गता इष्टाल्पे एष्या दिवसा ज्ञेयाः। एवमस्तविचारे पठितकालांशेभ्य इष्टाल्पे गता इष्टाधिके चैष्या दिवसा ज्ञेयाः। लब्धदिनैस्तात्कालिकाभ्यां दृग्ग्रहार्काभ्यां पुनस्तात्कालिका इष्टकालांशाः साध्यास्तैः प्राग्वत् पुनर्गतैष्या दिवसा एवमसकृद्यावदविशेषः।

अत्रोपपत्तिः। 'उक्तेभ्य ऊनाभ्यधिका यदीष्टाः खेटोदयो गम्यगतस्तदा स्यादि' त्यादिभास्करविधिना स्फुटा। इहाचार्येण च स्थूलाराश्युदयासवः खाभ्राष्टभूकलामिता अङ्गीकृता इति ॥ ७ ॥

वि. भा.—पूर्वसाधितैरिष्टकालांशैर्दृश्यादृश्यैः (पठितकालांशैश्च) ऊनाधिकलिप्ताभ्यां ग्रहार्कभुक्तयन्तरैक्यलब्धदिनैस्तात्कालिकैः (इष्टकालांशैः) प्राग्वदसकृद्विधिना युतिवत् (ग्रहयुतिवत्) उदयास्तयोर्गतैष्यदिनसाधनं कार्यमिति। अत्रैतदुक्तं भवति। पठितेष्टकालांशान्तरकला ग्रहरविगत्यन्तरभक्ता वक्रिणि ग्रहे तु ग्रहरविगतियोगभक्ता लब्धाः पठितकालांशेभ्य इष्टाधिके उदये गता इष्टाल्पे एष्या दिवसा ज्ञेयाः। अस्त विचारे पठितकालांशेभ्य इष्टाल्पे गता इष्टाधिके चैष्या दिवसा ज्ञेयाः। लब्धदिनैस्तात्कालिकाभ्यां दृग्ग्रहार्काभ्यां पुनस्तात्कालिका इष्टकालांशाः साध्यास्तैः पूर्ववत् पुनर्गतैष्या दिवसा एवमसकृद्यावदविशेष इति ॥७॥

अत्रोपपत्तिः।

ग्रहोदयो गत एष्यो वेति विचार्यमाणे पठितकालांशेष्टकालांशयोरन्तरं कार्यं ततोऽनुपातो यदि ग्रहरविगत्यन्तरकलायामेकं दिनं लभ्यते तदा कालांशान्तरकलायां किमित्यनेन समागतदिनैः पठितकालांशेभ्य इष्टकालांशस्याधिकत्वे उदयो गतोऽन्यथा भावी भवति, अस्तविचारे तु पूर्ववदनुपातेन यानि दिनान्यागच्छेयुस्तैः पठितकालांशेभ्य इष्टकालांशस्याल्पत्वेऽस्तो गतोऽन्यथा भावी भवतीति। सिद्धान्तशेखरे श्रीपतिनो "उक्तोनाधिककालभागविवरं राशेः कलाभिः १८०० र्हृतं भक्तं प्राचि निजोदयेन वरुणाशायां तदस्तेन च। षष्टिघ्नं ग्रहसूर्यभुक्ति विवरेणाप्तं ग्रहे वक्रिते भुक्तघ्नैक्येन दिनानि तैरथ मुहुः साध्योग्रहास्तोदयः॥" उदयास्तयोर्दिनानयनं कथितम्। सिद्धांत शिरोमणौ—

उक्तेभ्य ऊनाभ्यधिका यदीष्टाः खेटोदयो गम्यगतस्तदा स्यात्।
अतोऽन्यथा वास्तमयोऽवगम्यः प्रोक्तेष्टकालांश वियोग लिप्ताः॥
खाभ्राष्टभूघ्ना द्युचरोदयाप्ताः खेटार्कभुक्तयन्तरभाजिताश्च।
वक्रेतु भुक्तयैकहृता अवाप्तास्तदन्तराले दिवसा गतैष्याः॥

"तात्कालिकाभ्यां रविदृग्ग्रहाभ्यां मुहुः कृतास्ते स्फुटतां प्रयान्ति" इति भास्करोक्तं श्रीपत्युक्तयनुरूपमेवेति, तथोदयास्तदिनादेर्गतैष्यता प्रतिपादनं सिद्धान्तशेखरे "कथितसमयांशेभ्योऽभीष्टा भवन्ति यदाधिका विगत उदयो भावी चास्तस्तथाऽपरथाऽल्पकैः। उदयति सितो वक्रं यातश्चतुर्भिरिहांशकैः समयजनितैरेवं केचिद् वदन्त्यपरे त्रिभिः" चैवमस्ति, वक्री शुक्रश्चतुर्भिः कालांशैरुदयतीति केचिद्वदन्ति, अपरे त्रिभिः कालांशैर्वक्री शुक्र उदयतीति वदन्ति, परमिति केषां केषां मतमिति न ज्ञायते, सूर्यसिद्धान्त-लल्लसिद्धान्तादिषु कालांशा आचार्योक्त सदृशाः श्रीपत्युक्तसदृशा वोक्तास्तदा केचित्, अपरे इत्युक्त्या केषां बोधो भवतीति ग्रन्थान्तर दर्शनेन स्फुटं भविष्यतीति ॥ ७ ॥

अब उदय और अस्त के गतैष्य दिनानयन को कहते हैं।

हि. भा.—पठित कालांश और इष्ट कालांश के अन्तर कला को ग्रह और रवि के गत्यन्तर से भाग देना, ग्रह के वक्री रहने से ग्रह और रवि के गतियोग से भाग देना, लब्ध-दिन पठित कालांश से इष्टकालांश अधिक रहे तो उदय में गतदिन और पठित कालांश से इष्टकालांश अल्प रहे तो एष्यदिन समझना चाहिये। अस्तविचार में पठित कालांश से इष्ट-कालांश अल्प रहने से गतदिन और पठित कालांश से इष्टकालांश अधिक रहने से एष्यदिन समझना चाहिये। लब्ध दिनों से तात्कालिक दृग्ग्रह और रवि से पुनः तात्कालिक इष्ट कालांश साधन करना, इन (इष्ट कालांश) से पुनः पूर्ववत् गतैष्य दिनानयन करना, इस तरह असकृत्कर्म करते रहना जबतक कि बिलकुल ठीक न हो जाय इति ॥ ७ ॥

उपपत्ति।

ग्रह का उदय पहले हो गया हैं या पीछे होगा इस के लिये पठित कालांश और इष्ट-कालांश का अन्तर करके अनुपात करते हैं यदि ग्रह और रवि की गत्यन्तर कला में एक दिन पाते हैं तो कालांशान्तर कला में क्या इस से जो दिन प्रमाण आता है उतने दिन करके पठित कालांश से इष्ट कालांश के अधिक रहने पर उदयगत होता है अन्यथा भावी होता है, अस्त विचार के लिये पूर्ववत् अनुपात से जो दिन प्रमाण आवें उतने दिनों में पठित कालांश से इष्टकालांश के अल्प रहने पर अस्त गत होता है, अन्यथा एष्य (भावी) होता है। सिद्धान्त शेखर में श्रीपति "उक्तोनाधिककालभागविवरं राशेः कलाभि १८०० र्हृतं" इत्यादि संस्कृतोपपत्ति में लिखित श्लोक से उदय और अस्त के दिनानयन कहे हैं। सिद्धान्त शिरोमणि में "उक्तेभ्य ऊनाभ्यधिका यदीष्टाः खेटोदयो गम्यगतस्तदा स्यात्" इत्यादि से भास्कराचार्य ने श्रीपत्युक्त के अनुरूप ही कहा है, उदय दिन तथा अस्तदिन की गतैष्यता का प्रतिपादन सिद्धान्त शेखर में "कथितसमयांशेभ्योऽभीष्टा भवन्ति यदाधिका" इत्यादि इस तरह किया गया है, वक्री शुक्र चार कालांश में उदित होते हैं यह किसी आचार्य का मत है, अपर (दूसरे) आचार्य कहते हैं कि वक्री शुक्र तीन कालांश में उदित होते हैं। लेकिन ये मत किन

आचार्यों का है मालूम नहीं होता है, सूर्यसिद्धान्त-लल्लसिद्धान्त आदि ग्रन्थों में पठित कालांश आचार्योक्त या श्रीपत्युक्त कालांश सदृश ही है, तब पूर्वोक्त मतों के लिये ग्रन्थान्तरों को देख कर निर्णय करना चाहिये कि ये मत किन आचार्यों के है इति ॥ ७ ॥

इदानीं ग्रहाणां नित्योदयास्तसाधनं चन्द्रोदये च विशेषमाह ।

प्रतिदिनमुदयास्तमयावसकृत् तात्कालिकग्रहविलग्नैः ।
सूर्यास्तमयौदयिकौ शीतांशोः पूर्णिमास्यन्तः ॥ ८ ॥

सु० भा०—तात्कालिकग्रहविलग्नैरसकृत् प्रतिदिनसुदयास्तमयौ साध्यौ अर्थाद्यस्मिन्काले ग्रहोदितघटीज्ञानमपेक्षितं तदा तात्कालिकं लग्नं ग्रहोदयलग्नं च कृत्वा पूर्ववत् तयोरन्तरे घटिकाः प्रसाध्य ताभिर्ग्रहं प्रचाल्य तस्य पुनरुदयलग्नं प्रसाध्य तात्कालिकात् स्थिरलग्नात् पुनरिष्टघटिकाः साध्याः । एवमसकृत् । एवमस्तकालज्ञानार्थमस्तलग्नात् सषड्भात् कर्म कर्त्तव्यम् । एवं पूर्णिमास्यन्तः प्रथमचन्द्रदर्शनदिनात् पूर्णिमापर्यन्तमभीष्टदिने शीतांशोश्चन्द्रस्य सूर्यास्तमयौदयिकौ सूर्यास्ताद्वा सूर्योदयादस्तमयौदयिकौ कालौ साध्यौ अर्थात् सूर्यास्तात् प्राक् पश्चाद्वा कतिघटिकाभिश्चन्द्रोदयोऽस्तो वा सूर्योदयात् प्राक् पश्चात् कतिघटिकाभिश्चन्द्रोदयोऽस्तो वेति सर्वं प्रसाध्यमित्यर्थः ।

अत्रोपपत्तिः । ग्रहार्कान्तरकालानयनेन स्फुटा ॥ ८ ॥

वि. भा.—तात्कालिकग्रहविलग्नैः प्रतिदिनसमकृत् उदयास्तमयौ साध्यौ, अर्थाद्यस्मिन् काले ग्रहोदितघटीज्ञानमभीष्टं तदा तात्कालिकलग्नं ग्रहोदयलग्नं च विधाय तयोरन्तर्गतघटिकाः प्रसाध्य ताभिर्ग्रहचालनं कृत्वा तस्य पुनरुदयलग्नं प्रसाध्य तात्कालिकात् स्थिरलग्नात् पुनरिष्टघटयः साध्या एवमसकृत् । एवमस्तकालं ज्ञातुमस्तलग्नात् सषड्भास्तलग्नात्कर्म कर्त्तव्यम् । एवं पूर्णिमास्यन्तः प्रथमचन्द्रदर्शनदिनात् पूर्णिमापर्यन्तमभीष्टदिने शीतांशोः (चन्द्रस्य) सूर्यास्तमयौदयिकौ (सूर्यास्ताद्वा सूर्योदयादस्तमयौदयिकौ कालौ साध्यौ) अर्थात् सूर्यास्तात्पूर्वं पश्चाद्वा कियन्मिताभिर्घटीभिश्चन्द्रोदयोऽस्तोवा सूर्योदयात् पूर्वं पश्चाद्वा कियन्मिताभिर्घटीभिश्चन्द्रोदयोऽस्तो वेति सर्वं संसाध्यमिति ॥ ८ ॥

अत्रोपपत्तिः ।

यदि सषड्भरवितश्चन्द्रोऽल्पोभवेत्तदा प्राक् दिने उदयति, तदधिकश्चेत्तदा सूर्यास्तानन्तरमुदयति तत्र कालज्ञानं "ऊनस्य भोग्योऽधिक भुक्त युक्तो मध्योदयाढ्य" इतिवद् भवति, सिद्धान्तशेखरे श्रीपतिना "उदयति सममस्तं गच्छता

प्राक् तदूनस्तदनु तदधिकश्चेदुक्तवत् तत्र कालः" अनेनाऽऽचार्योक्तानुरूपमेवोक्त मिति ।। ८ ।।

अब ग्रहों के नित्योदयास्त साधन को तथा चन्द्रोदय के लिये विशेष कहते हैं।

हि. भा.—तात्कालिक गृह और लग्न से असकृत् कर्म द्वारा प्रतिदिन उदय और अस्त साधन करना अर्थात् जिस समय गृहोदित घटी ज्ञान करना हो उस समय के लग्न (तात्कालिक लग्न) तथा गृहोदय लग्न साधन कर दोनों के अन्तर्गत घटी साधन कर उन (साधित घटी) से गृह को चलाकर उसके फिर उदयलग्न साधन कर तात्कालिक स्थिर लग्न से पुनः इष्टघटी साधन करना, इसतरह असकृत् (वारंवार) करना, एवं अस्तकाल ज्ञान के लिये अस्तलग्न (सषड्भास्त लग्न) से कर्म करना चाहिये। इस तरह प्रथम चन्द्र दर्शन दिन से पूर्णिमा पर्यन्त इष्ट दिन में चन्द्र के सूर्यास्त से वा सूर्योदय से अस्तमय काल और औदयिक काल साधन करना अर्थात् सूर्यास्त पहले वा पीछे कितनी इष्ट घटी में चन्द्रोदय वा चन्द्रास्त होता है वा सूर्योदय से पहले या पीछे कितनी घटी में चन्द्रोदय वा चन्द्रास्त होता है ये सब साधन करना चाहिये इति ।। ८ ।।

उपपत्ति ।

यदि सषड्भ (छः राशि से सहित) रवि से चन्द्र अल्प हो तब सूर्यास्त से पहले चन्द्र उदित होते हैं। यदि सषड्भ रवि से चन्द्र अधिक हो तब सूर्यास्त के बाद चन्द्र उदित होते हैं वहां कालज्ञान "ऊनस्य भोग्योऽधिकभुक्तयुक्तो मध्योदयाख्यः" इसी के अनुसार होता है, सिद्धान्त शेखर में श्रीपति ने भी "उदयति सममस्तं गच्छता प्राक् तदूनः" इत्यादि से आचार्योक्तानुरूप ही कहा है इति ।। ८ ।।

इदानीं सूर्यासन्न भावेन चन्द्रोदयास्तज्ञानमाह ।

उदयास्तमयाविन्दोः कालांशैरर्कसंमितैः कार्यौ ।
हीनत्वं त्वधिकत्वं तदन्तरे योगकालः स्यात् ।।९।।

सु० भा०—अर्कसंमितैः कालांशैरर्थाद् द्वादशकालांशैरिन्दोश्चन्द्रस्योदयास्तमयौ साध्यौ। अत्रपठितकालांशेभ्य इष्टकालांशानां यदि हीनत्वं वाऽधिकत्वं भवेत् तदा तदन्तरे तयोः पठितेष्टकालांशयोरन्तरे योगकालो योगवद्ग्रहयोगवत् कालः स्यादिति सप्तमश्लोकेन स्फुटोऽर्थ इति ।। ९ ।।

वि. भा.—इन्दोः (चन्द्रस्य) अर्कसंमितैरर्थात् द्वादशतुल्यैः कालांशैरुदयास्तमयौ साध्यौ। पठितकालांशेभ्य इष्टकालांशस्य यदि हीनत्वं (अल्पत्वं)

वाऽधिकत्वं भवेत्तदा तदन्तरे (पठितेष्ट कालांशयोरन्तरे) योगकालः (ग्रह योगवत् कालः) स्यादिति ॥ ९ ॥

अत्रोपपत्तिः ।

दृश्यादृश्यैर्युतिवद् ग्रहार्कभुक्तचन्तरैक्यलब्धदिनैरित्यादिना चन्द्रपठित कालांशस्येष्टकालांशस्य चान्तरे याः घटिकास्ततो गतैष्यकालज्ञानं सुगममिति ॥ ९ ॥

अब सूर्यासन्न भाव से चन्द्र के उदयास्तज्ञान को कहते हैं ।

हि. भा.—चन्द्र के बारह कालांश से उदय और अस्तमय साधन करना, पठित कालांश इष्ट कालांश की अल्पता में या अधिकता में दोनों के (पठित कालांश और इष्टकालांश) के अन्तर में ग्रहयोग की तरह काल होता है इति ॥ ९ ॥

उपपत्ति ।

'दृश्यादृश्यैर्युतिवत्' इत्यादि सप्तम श्लोक से चन्द्र के पठित कालांश और इष्ट कालांश के अन्तर में जो घटी है उससे गत काल और एष्य काल ज्ञान सुगम ही है इति ॥ ९ ॥

इदानीं बुधगुर्वोरुदयास्तयोर्विशेषमाह ।

उदयास्तविधौ रविवद्बुधशीघ्रादाप्तफलयुतो नो ज्ञः ।
तैक्ष्ण्याद्गुरुर्विहीनस्तत्समलिप्तेन भागेन ॥ १० ॥

सु० *भा०*—बुधशीघ्राद्बुधशीघ्रोच्चाद्रविवद्यदाप्तं प्राप्तं फलं मन्दफलं भवेत् तेन यथागतेन बुधो युतो वोनः कार्यः । एतादृशोबुध उदयास्तसाधने ग्राह्यः । अत्रैतदुक्तं भवति । बुधशीघ्रं मध्यरविं प्रकल्प्य तस्मान्मन्दफलमानीय तेन संस्कृतो बुध उदयास्तविधौग्राह्यः । एतेन बुधकालांशेषु वैलक्षण्यं सूचितम् । एवं गुरुस्तत्समलिप्तेन मध्यमगुरुसमं मध्यमरविं प्रकल्प्य यन्मंन्दफलं भागात्मकं तत्समकलाभिरूनो भागेनैकेन च विहीन उदयास्तविधौ ग्राह्यः । कस्मात् । तैक्ष्ण्यात् । अर्थाद्गुरुबिंबं विपुलमत एवं कालांशवैलक्षण्यं भवति

अत्रोपलब्धिरेव वासना ॥ १० ॥

वि. भा.—बुधशीघ्रात् (बुधशीघ्रोच्चात्) रविवद्यदाप्तं (प्राप्तं) फलं (मन्दफलं) भवेत्तेन ज्ञः (बुधः) युतोनः कार्यस्तदैतादृशो बुध उदयास्तसाधनोपयुक्तः । अत्रैतदुक्तं भवति, बुधशीघ्रोच्चं मध्यमरविं मत्वा ततो मन्दफलमानीय तत्संस्कृतो बुध उदयास्तोपयुक्त एतावता बुधकालांशे वैलक्षण्यं सूच्यते । एवं गुरुस्तत्सम-

लिप्तेन (मध्यमगुरुसमं मध्यमरविं मत्वांऽशात्मकं मन्दफलं साध्यं तत्सम-कलाभिः) हीनो भागेनैकेन च विहीन उदयास्तोपयुक्तत्वार्थं ग्राह्यः। कथमेवं भवति तैक्ष्ण्यात् (गुरुबिम्बं विपुलमस्त्यतएव कालांशवैलक्षण्यं भवति) इति ॥ १० ॥

अत्रोपपत्तिः।

अत्रोपलब्धिरेव वासना नान्यत्कारणं वक्तुं शक्यत आचार्यकथनमेव प्रमाणमिति ॥ १० ॥

अब बुध और गुरु के उदयास्त के विषय में विशेष कहते हैं।

हि. भा.—बुध के शीघ्रोच्च से रवि की तरह प्राप्त जो मन्द फल हो उससे बुध को युत और हीन करना, तब यह बुध उदय और अस्त साधन के लिये उपयुक्त होता है। अर्थात् बुध शीघ्रोच्च को मध्यम रवि मानकर उससे मन्दफल लाकर बुध में उसको संस्कार करने से उदयास्तोपयुक्त बुध होते हैं इससे बुध के कालांश में विलक्षणता सूचित होती है, उस तरह मध्य गुरु के समान मध्यम रवि मानकर अंशात्मक मन्द फल साधन करना। तत्तुल्य कला को गुरु में हीन तथा एकांश को हीन करना तब इस तरह के गुरु उदय और अस्त के लिये उपयुक्त होते हैं। ऐसा क्यों होता है ? गुरु का बिम्ब बड़ा है इसलिये कालांश में विलक्षणता होती है इति ॥ १० ॥

उपपत्ति।

इस में उपलब्धि ही उपपत्ति है आचार्य कथन ही प्रमाण है दूसरा कारण नहीं कह सकते हैं इति ॥ १० ॥

इदानीं शुक्रकालांशेषु विशेषमाह।

मानाल्पत्वात् पश्चादुदयोऽस्तमयः सितस्य दशभिः प्राक्।
पश्चान्मानमहत्वादस्तमयोऽष्टाभिरुदयः प्राक् ॥ ११ ॥

सु० भा०—सितस्य पश्चात् पश्चिमायां दिशि दशभिः कालांशैरुदयः। प्रागस्तमयश्च दशभिरेव भवति मानाल्पत्वात्। यतस्तदा शुक्लो मार्गो भवति। ऋजुगतेरुच्चासनत्वाद्बिंबं लघु भवति तस्मात् कालांशा अधिकाः पठिताः। एवं तस्य पश्चादस्तमयः प्रागुदयश्च नीचासन्नत्वाद्वक्रत्वाच्च बिंबमानं विपुलं भवति तस्मान्मानमहत्त्वादष्टभिरेव न्यूनैः कालांशैर्भवति। भास्करेणाप्याचार्योक्तामीदृशीं वैलक्षण्यक्रियामवलम्ब्य तारतम्ये न 'ज्ञशुक्रयोर्वक्रगयोर्द्विहीनाः' इत्युक्तम्।

अत्रोपपत्तिरुपलब्धिरेव ॥ ११ ॥

वि. भा.—सितस्य (शुक्रस्य) पश्चात् (पश्चिमायां दिशि) दशभिः कालांशैरुदयः, प्राक् (पूर्वदिशि) दशभिरेव कालांशैरस्तमयश्च भवति। पूर्वोक्तेभ्यो भिन्नैरंशैः कथं कथ्यत इत्याह। मानाल्पत्वात्कारणात्। यतस्तदा शुक्रो मार्गी भवति मार्गगतेरुच्चासन्नत्वाद् बिम्बं लघु भवति तस्मात् कालांशा अधिकाः पठिताः। तस्य पश्चिमायामस्तमयः पूर्वस्यामुदयश्च नीचासन्नत्वाद् वक्रत्वाच्च बिम्बमानं विपुलं भवति, तस्मान्मानमहत्वादष्टाभिरेव न्यूनैः कालांशैर्भवतीति ॥ ११ ॥

अत्रोपपत्तिः।

सिद्धान्त शेखरे श्रीपतिना "मानाल्पत्वाद्दशभिरुदयं याति शुक्रः प्रतीच्यामंशैरस्तं दिशि सुरपतेः पीनभावात्तु मूर्त्तेः पश्चादस्तं व्रजति वसुभिः प्रागुदेतीत्यनेन" सिद्धान्त शिरोमणौ भास्कराचार्येणाप्याचार्योक्तां वैलक्षण्यक्रियामवगत्य 'ज्ञशुक्रयोर्वक्रगयोर्द्विहीनाः' इत्युक्तम्। अत्र प्राचीनानामुपलब्धिरेवोपपत्तिरिति ॥ ११ ॥

अब शुक्र कालांश में विशेष कहते हैं।

हि. भा.—शुक्र पश्चिम दिशा में दश कालांश में उदित होते हैं और पूर्व दिशा में दश ही कालांश में अस्त होते हैं। पूर्व कथित कालांश से यहां भिन्न कालांश क्यों कहते हैं इसके लिये आचार्य कहते हैं कि बिम्ब की अल्पता के कारण ऐसा होता है क्यों कि वहां शुक्रमार्गी होते हैं। उच्च के आसन्न में मार्गगति ग्रह का बिम्ब छोटा होता है इसलिये कालांश अधिक पठित किया गया है। उन (शुक्र) की पश्चिम दिशा में अस्तमय और पूर्वदिशा में उदय भी (नीचासन्न में वक्रता से बिम्बमान बड़ा होता है) इसलिये बिम्ब के बड़ा होने के कारण आठ ही कालांश में होता है इति ॥ ११ ॥

उपपत्ति।

सिद्धान्त शेखर में श्रीपति "मानाल्पत्वाद्दशभिरुदयं" इत्यादि श्लोक से तथा सिद्धान्त शिरोमणि में भास्कराचार्य ने भी आचार्योक्त क्रिया की विलक्षणता को समझ कर 'ज्ञशुक्रयोर्वक्रगयोर्द्विहीना' कहा है। इन में प्राचीनों की उपलब्धि ही उपपत्ति है, दूसरा कारण कुछ भी नहीं कह सकते हैं इति ॥ ११ ॥

इदानीमार्यभटदूषणं स्वप्रशंसां चाह ॥

आर्यभटः क्षेत्रांशैर्दृश्यादृश्यान् यदुक्तवांस्तदसत्।
दृग्गणित विसंवादाद् दृग्गणितैक्यं स्वकालांशैः ॥१२॥

सु० भा०—आर्यभटः कालांशान् विहाय क्षेत्रांशैः क्रान्तिवृत्तीयग्रहार्कान्तरांशैरेव ग्रहाणां दृश्यादृश्ययोराचार्योक्तादन्यत् साधनं यदुक्तवान् तद्दृग्गणितविसंवादात् दृग्गणितयोर्विरोधादसत्। स्वकालांशैराचार्योक्तकालांशैश्च दृग्गणितैक्यं भवत्यत आचार्योक्तं साधनमेव समीचीनमिति ॥ १२ ॥

तथा चाचार्यार्यभटः ।

चन्द्रोऽशैर्द्वादशभिरविक्षिप्तोऽर्कान्तरस्थितैर्दृश्यः ।
नवभिर्भृगुभृगोस्तैर्द्वयधिकैर्यथा श्लक्ष्णाः ॥
(गोलपा. श्लो० ४)

वि. भा.—आर्यभटः क्षेत्रांशैः (क्रान्तिवृत्तीय ग्रहरव्यन्तरांशैरेव) ग्रहाणां दृश्यादृश्यान् (कालांशान्) विहायदृश्या दृश्ययोराचार्योक्तादन्यत्साधनं यत् उक्तवान्, तद्दृग्गणित विसंवादात् (दृग्गणितयोर्विरोधात्) असत् (शोभनं नास्ति) स्वकालांशैः (आचार्योक्तकालांशैः) दृग्गणितैक्यं भवत्यत आचार्योक्तसाधनमेव युक्तियुक्तमिति ॥ १२ ॥

हि. भा.—आर्य भट, दृश्यादृश्य (कालांश) को छोड़कर ग्रहों के उदय और अस्त के साधन आचार्योक्त से भिन्न जो कहते हैं सो ठीक नहीं है क्यों कि उनके साधन से दृग्गणित (वेधागत और गणितागत उदय और अस्त) में विरोध होता है, आचार्योक्त कालांशों से दृग्गणितैक्य होता है इसलिये आचार्यो (ब्रह्मगुप्त) क्त साधन ही युक्तियुक्त है इति ॥ १२ ॥

इदानीमध्यायोपसंहारमाह ।

दृग्लग्नदृष्टिभागग्रहोदयास्तमयनाडिकाद्येषु ।
उदयास्तमयाध्यायः षष्ठो द्वादशभिरार्याभिः ॥१३॥

सु० भा०—दृग्लग्नमुदयलग्नमस्तलग्नं च । दृष्टिभागाः कालांशाः शेषं स्पष्टार्थम् ॥ १३ ॥

मधुसूदनसूनुनोदितो यस्तिलकः श्रीपृथुनेह जिष्णुजोक्ते ।
हृदि तं विनिधाय नूतनोऽयं रचितोऽस्तौदयिके सुधाकरेण ॥

इति श्री कृपालुदत्तसूनुसुधाकरद्विवेदिविरचिते ब्राह्मस्फुटसिद्धान्तनूतनतिलके उदयास्ताधिकारः षष्ठः ॥ ६ ॥

वि. भा.—दृग्लग्नं (उदयलग्नमस्तलग्नं च) दृष्टिभागाः (कालांशाः) ग्रहाणामुदयास्तघट्यादिषु द्वादशाभिरार्याभिः (द्वादशसंख्यकैरार्याछन्दोभिः) अयं षष्ठः उदयास्तमयाध्यायोऽस्तीति ॥ १३ ॥

इति ब्रह्मगुप्त विरचिते ब्राह्मस्फुट सिद्धान्ते उदयास्ताधिकारः षष्ठः ॥ ६ ॥

हि. भा.—दृग्लग्न (उदयलग्न और अस्तलग्न) दृष्टिभाग (कालांश) और ग्रहों के उदय और अस्त घट्यादियों में बारह आर्या छन्द श्लोकों से यह छठा उदयास्तमयाध्याय है इति ॥ १३ ॥

इति ब्रह्मगुप्तविरचित ब्राह्मस्फुट सिद्धान्त में छठा उदयास्ताधिकार समाप्त हुआ ॥ ६ ॥

ब्राह्मस्फुटसिद्धान्तः

७

चन्द्रशृङ्गोन्नत्यधिकारः

ब्राह्मस्फुटसिद्धान्तः

७

चन्द्रशृङ्गोन्नत्यधिकारः

अथ चन्द्रशृङ्गोन्नत्यधिकारः प्रारभ्यते ।

तत्रादौ रवेरुपरिचन्द्रोऽस्तीति पुराण मत खण्डनमाह ।

सितवृद्धिहानिवीर्यादि शशाङ्काज्जायते कथं गणितात् ।
उपरि रवेरिन्दुश्चेदर्वागर्धं सदा शुक्लम् ॥ १ ॥

सु० भा०—चेद्यदि रवेरुपरि इन्दुस्तदा गणिताद्गणितागतात् शशाङ्काच्चन्द्रात् सितवृद्धिहानिवीर्यादि कथं जायते । अर्थाद्गणितविधिना चन्द्रस्य शुक्लोपचयापचयमानं कृष्णाङ्गुलमानं च कथं जायते यतस्तादृशस्थितौ भौमगुरुशनिवत् सदा अर्वागर्धं पूर्वार्धं नृदृश्यं खण्डं शुल्कं स्यात् परन्तु दृष्ट्या सदा शुल्कं नोपलक्ष्यतेऽतो रवेरुपरि नेन्दुरिति स्फुटम् । तथा रवेरधश्चन्द्रं प्रकल्प्य गणितविधिनाऽऽनीतं शुक्लादिदृष्टियोग्यं भवति तेन रवेरधश्चन्द्र इति स्फुटम् ॥ १ ॥

वि. भा.—चन्द्रस्य सितवृद्धिहानी (स्वच्छतोपचयापचयौ) शापाद् भवत इति पुराणमतमस्ति, यद्येवं तदा गणिताच्छशाङ्कात् (गणितागतचन्द्रात्) सितवृद्धिहानिवीर्यादिज्ञानं कथं जायते, अर्थाद् गणितविधिना चन्द्रस्य शुक्लोपचयापचयादिमानं नावगम्यं स्यात् । चेत् (यदि) रवेरुपरि, इन्दुः (चन्द्रः) तदाऽर्वागर्धं (अधस्तनमर्धंनृदृश्यं खण्डं) सदा शुक्लं स्यात्, परन्तु दृष्ट्या सर्वदा शुक्लं नोपलक्ष्यतेऽतोरवेरुपरि चन्द्रो नास्ति, यदि च रवेरधश्चन्द्रं स्वीकृत्य गणितविधिना शुक्लादिमानमानीयते तदा तत्सर्वथा दृष्टियोग्यं भवत्यतो रवितश्चन्द्रोऽध एवातश्चन्द्रशुक्लस्योपचयापचयस्य गणितावगम्यत्वात् सर्वदैव चन्द्रबिम्वपूर्वार्धस्य शुक्लत्वाभावाच्च पुराणोक्तमतद्वयं (शापाच्चन्द्रस्य सितवृद्धिहान्यादि भवति तथा रवित उपरिचन्द्रोऽस्तीति) न युक्तिसङ्गतमिति सिद्धान्तितम् । सिद्धान्तशेखरे श्रीपतिनै "शापाद्यदीन्दोः सितवृद्धिहानी कथं तु जाते गणितावगम्ये । ऊर्ध्वं यदीन्दू रवितस्तदार्धमर्वाक्तनं हन्त सदैव शुक्लम्" वमुक्तमिति ॥ १ ॥

अब चन्द्र शृङ्गोन्नति अधिकार प्रारम्भ किया जाता है, 'उसमें पहले रवि से ऊपर चन्द्र हैं' इस पुराणमत के खण्डन को कहते हैं—

हि. भा.—चन्द्र की स्वच्छता का उपचय (वृद्धि) और अपचय (हानि) शाप से होता है यह पुराण में कहा गया है, आचार्य इसका खराण्डन करते हैं । यदि शाप से चन्द्र

शुक्ल का उपचय-अपचय आदि होता है तब गणितागत चन्द्र से शुक्ल के उपचय अपचय आदि) का ज्ञान क्यों होता है अर्थात् गणित विधि से उन का ज्ञान नहीं होना चाहिये, और यदि रवि से चन्द्र ऊपर है तब चन्द्र के नीचे का आधाभाग (मनुष्य के लिए दृश्य भाग) सदा शुक्ल होना चाहिये, परन्तु दृष्टि से वह सदा शुक्ल देखने में नहीं आता है इसलिये रवि से चन्द्र ऊपर नहीं है, यदि रवि से चन्द्र को नीचा मानकर गणित से शुक्लादि लाते हैं तो वे सर्वथा दृष्टि योग्य होते हैं अतः रवि से चन्द्र नीचा ही है, यह सिद्ध होता है, पुरगणोक्त दोनों मत (शाप से चन्द्र शुक्ल की वृद्धि और हानि होना तथा रवि से चन्द्र ऊपर है) ठीक नहीं है यह ऊपर प्रदर्शित युक्तियों से सिद्ध हुआ। सिद्धान्त शेखर में श्रीपति "शापाद्यदीन्दोः सितवृद्धि हानी' इत्यादि से इन्हीं बातों को कहते हैं इति ॥ १ ॥

इदानीं चन्द्र बिम्बे सितवृद्धिहान्योः कारणमाह।

रविदृष्टं सितमर्धं कृष्णमदृष्टं यथाऽऽतपस्थस्य।
कुम्भस्य तथासन्नं रवेरधः स्थस्य चन्द्रस्य ॥ २ ॥

सु. भा.—तदासन्नं तादृगेव। अन्यत् स्पष्टम् ॥ २ ॥

वि. भा.—यथाऽऽतपस्थस्य (रौद्र स्थितस्य) कुम्भस्य (घटस्य) रविदृष्टं (रविणा दृष्टियोग्यमर्थाद्रव्यभिमुखं) अर्धं सितं (स्वच्छं) भवति, अदृष्टं (दृष्टियोग्यानर्हमर्थाद्रवितो विरुद्धदिशं) अर्धं कृष्णं (असितं) भवति तथासन्नं (तादृशमेव) रवेरधः स्थस्य चन्द्रस्यार्थाद्रव्यभिमुखं चन्द्र बिम्बार्धं सितं तद्विरुद्धदिशि चन्द्र बिम्बार्धं कृष्णं (असितं) भवतीति। सिद्धान्तशेखरे श्रीपतिना "घाम्ना घामनिधेरयं जलमयो धत्ते सुधादीधितिः; सद्यः कृत्तमृणालकन्दविशदच्छायां विवस्वद्दिशि। हर्म्ये घर्मघृणेः करैर्घट इवान्यस्मिन् विभागे पुनर्बाला कुन्तलकालतां कलयति स्वस्यास्तनोच्छायया" अनेनैवमुक्तम् तथोक्तार्थमेवा "पाथोमये शीतकरेऽर्करश्मयो विमूर्छिता घ्नन्ति तमस्विनीतमः। निकेतनाभ्यन्तरगं तमः स्वयं यथा त एवामल दर्पणाश्रिताः" नेन विशदयति, परमियं "सलिलमये शशिनि रवेर्दीधितयो मूर्च्छितास्तमो नैशम्। क्षपयन्ति दर्पणोदरनिहिता इव मन्दिरस्यान्तः॥" वराहमिहिरोक्तेरस्या एव पुनरुक्तिः। सिद्धान्तशिरोमणौ भास्कराचार्येण—

"तरणिकिरणसङ्गादेषपीयूषपिण्डो दिनकर दिशि चन्द्रश्चन्द्रिकाभिश्चकास्ति। तदितरदिशि बाला कुन्तलश्यामलश्रीर्घट इव निजमूर्त्तिच्छाययैवातपस्थः॥" छन्दोऽन्तरेण श्रीपतेः काव्यकला कौशलमेवोक्तमिति विज्ञैश्चिन्त्यमिति ॥ २ ॥

अब चन्द्र बिम्ब में सितवृद्धि और सित हानि के कारण को कहते हैं।

हि. भा.—जैसे धूप में स्थित घड़े का रवि की तरफ का आधा भाग सित (स्वच्छ) होता है और रवि से भिन्नदिशा का आधा भाग कृष्ण (असित) होता है।

वैसा ही रवि से अधः (नीचा) स्थितचन्द्र का होता है अर्थात् रवि की तरफ का चन्द्रबिम्ब का आधा भाग सित (उज्ज्वलीभूत) होता है और रवि से भिन्न तरफ का चन्द्रबिम्बार्ध असित (अनुज्ज्वल) होता है। सिद्धान्तशेखर में श्रीपति ने "धाम्ना धामनिधेरयं जलमयो धत्ते सुधादीधितिः" इत्यादि से इसी तरह कहा है। इसी अर्थ को "पाथोमये शीतकरेऽर्करश्मयो विमूर्च्छिता" इत्यादि संस्कृत भाष्य में लिखित श्लोक से विशद (स्फुट) करते हैं परन्तु यह श्रीपति की उक्ति "सलिलमये शशिनि रवेर्दीधितयो" इत्यादि संस्कृत भाष्य में लिखित वराहमिहिरोक्ति की पुनरुक्ति मात्र है। सिद्धान्तशिरोमणि में भास्कराचार्य ने "तरणिकिरणसङ्घादेषपीयूषपिण्डो दिनकरदिशि" इत्यादि संस्कृत भाष्य में लिखित श्लोक से श्रीपति की काव्यकला चातुरी को कहा है इति ॥२॥

इदानीं गणितेन शृङ्गोन्नतिज्ञानं कथं भवतीत्यत्र हेतुमाह

सितमुन्नतं यतोऽर्कः सितासितं शुक्लपक्षान्ते।
अर्वागर्धं पश्चाद् गणिताच्छृङ्गोन्नतिस्तस्मात् ॥३॥

सु. भा.—यतो यद्दिशि अर्कस्तत्रैव सितं शुक्लमुन्नतं वृद्धिगतं भवति। शुक्लपक्षान्ते पूर्णिमान्ते च अर्वागर्धं पूर्वार्धं रविसंमुखस्थं सितं पश्चादर्धं नृदृश्यखण्डं चासितमुपलक्ष्यते। तस्माद्रविचन्द्रांतरवशतो गणितात् शृंगोन्नतिः साध्या भवतीति ॥३॥

वि. भा—यतः (यद्दिशि) अर्कः (रविः) तद्दिश्येव सितं (शुक्लं) उन्नतं (वृद्धिगतं) भवति, शुक्लपक्षान्ते (पूर्णान्ते) अर्वागर्धं (पूर्वार्धं) सूर्याभिमुखं सितं, पश्चादर्धं (नृदृश्यखण्डं) चासितमुपलक्ष्यते तस्मात् कारणात् रविचन्द्रान्तरवशेन गणिताच्चन्द्रशृङ्गोन्नतिः साधिता भवति। सिद्धान्तशेखरे श्रीपतिनाप्य "यस्यां सहस्रकिरणो दिशि तत्र नूनमालोक्यते शशधरस्य सितोन्नतत्वम्। पक्षान्तयोरपि सितासितता यतोऽस्य शृङ्गोन्नतिः खलु ततो गणितावगम्या" नेनाऽऽचार्योक्तानुरूपमेवोक्तमिति ॥३॥

अब गणित से चन्द्र शृङ्गोन्नति का ज्ञान कैसे होता है उसके कारण को कहते हैं

हि. भा.—चन्द्र से जिस दिशा में रवि रहता है उसी तरफ सित (शुक्ल) उन्नत (वृद्धिगत) होता है, पूर्णान्त में सूर्याभिमुख चन्द्रबिम्ब का पूर्वार्ध शुक्ल होता है और परार्ध असित (कृष्ण) देखने में आता है, उस कारण से रवि और चन्द्र के अन्तर वश से गणित द्वारा चन्द्रशृङ्गोन्नति साधित होती है। सिद्धान्तशेखर में श्रीपति ने भी "यस्यां सहस्रकिरणो दिशि तत्र नूनमालोक्यते" इत्यादि संस्कृत भाष्य में लिखित श्लोक से आचार्योक्त के अनुरूप ही कहा है इति ॥३॥

इदानीं शृङ्गोन्नत्यर्थमितिकर्त्तव्यतामाह

रविचन्द्रपातलग्नैः स्वक्रान्त्युदयात् स्वलग्नगतशेषाः।
घटिकाः खचरार्धास्तात् स्वेष्टौ रविशीतगू कृत्वा ॥४॥

सु. भा.—खचरार्धास्ताद्व्यस्तात् स्वलग्नगतशेषाः शशिन उदयलग्नस्य गताः शेषा वा घटिकाः साध्याः। कैः। रविचन्द्रपातलग्नैस्तथा स्वक्रान्त्युदयाच्च। अत्रैतदुक्तं भवति। व्यस्तकाले रविश्चन्द्रः पातो लग्नम्। एतानि कृत्वा तैः स्वक्रान्त्या चन्द्रक्रान्त्या उदयाच्चन्द्रोदयलग्नाच्चोदयलग्नस्य गता वा शेषा घटिका ऊनस्य भोग्योऽधिकभुक्तयुक्तो मध्योदयाढ्य इत्यनेन विधिना साध्याः। सूर्यास्तानन्तरं यावतीभिर्घटिकाभिश्चन्द्रास्तस्ता गता घटिकाः। सूर्यास्तात् प्राग् यावतीभिर्घटिकाभिश्चन्द्रास्तस्ता एष्या घटिकाः। एवं प्राक् क्षितिजे व्युदयाद् गता एष्या वा चन्द्रोदयघटिकाः साध्या इत्यर्थादवगम्यते। यस्मिन् दिने सूर्यास्तादनन्तरं कालांशघटिकातोऽधिकाभिश्चन्द्रास्तः। व्युदयात् प्राक् कालांशघटिकातोऽधिकाभिश्चन्द्रोदयस्तस्मिन्नेव चन्द्रदर्शनं सति दर्शने शृङ्गोन्नतिः साध्या। सितशृङ्गोन्नतिश्च बिम्बार्धाल्पे चन्द्रसितेऽत एव भास्करेण मासान्तपादे प्रथमे च शृंगोन्नतिः साधिता। द्वितीयतृतीयपादयोश्च बिम्बार्धाल्पं कृष्णं भवति। अतस्तत्र कृष्णशृङ्गोन्नतिर्भवति। इत्याचार्येण शृंगोन्नतिद्वयं साध्यते। तदर्थं पादचर्चा न कृता। भास्करेण न हि स्पष्टा कृष्णशृङ्गोन्नतिर्नरैरुपलक्ष्यत इति मनसि संप्रधार्य सितशृंगोन्नतिरेव साधिता। उक्तं च 'द्वितीयतृतीययोरपि चरणयोर्ब्रह्मगुप्तादिभिः कृष्णशृंगोन्नतिरानीता सा मम न संमता। न हि नरैः कृष्णशृंगोन्नतिः स्पष्टोपलक्ष्यते'। स्वेष्टौ स्वेष्टकालिकौ रविशीतगू रविचन्द्रौ कृत्वा विक्षेपक्रान्त्यंशाः साध्या इत्यध्याहार्यम् ॥४॥

वि. भा.—खचरार्धास्तात् (व्यस्तात्) स्वलग्नगतशेषाः शशिन उदयलग्नस्य गताः शेषाः वा घटिका रविचन्द्रपातलग्नैस्तथा स्वक्रान्त्युदयाच्च साध्याः। अत्रैतदुक्तं भवति, सूर्यास्तकाले रविश्चन्द्रः चन्द्रपातो लग्नं च कृत्वा तैः चन्द्रक्रान्त्या उदयात् (चन्द्रोदयलग्नात्) उदयलग्नस्य गता वा शेषा घटिका 'ऊनस्य भोग्योऽधिकभुक्तयुक्तो मध्योदयाढ्य' इत्यनेन साध्याः। सूर्यास्तानन्तरं यावतीभिर्घटीभिश्चन्द्रास्तस्ता गतघटिकाः। तथा सूर्यास्तात्पूर्वं यावतीभिर्घटीभिश्चन्द्रास्तस्ता एष्या घटिकाः। एवं पूर्वक्षितिजे सूर्योदयाद् गता एष्याश्च चन्द्रोदयघट्यो विधेयाः। यस्मिन् दिने सूर्यास्तात्परं कालांशघटीतोऽधिकाभिर्घटीभिश्चन्द्रास्तः, सूर्योदयात् पूर्वं कालांशघटीतोऽधिकाभिश्चन्द्रोदयस्तस्मिन्नेव चन्द्रदर्शने शृङ्गोन्नतिः साध्या। सितशृङ्गोन्नतिश्च बिम्बार्धाल्पे चन्द्रसिते भवत्यत एव मासान्तपादे प्रथमे भास्करेण शृङ्गोन्नतिसाधनं कृतम्। द्वितीयतृतीयपादयोर्बिम्बार्धाल्पं कृष्णं भवति, तेन तत्र कृष्णशृङ्गोन्नतिर्भवति, आचार्येण

पादचर्चा न क्रियतेऽतः सूच्यते यदनेन सितशृङ्गोन्नमनं कृष्णशृङ्गोन्नमनं चेति शृङ्गोन्नतिद्वयं साध्यते । कृष्णशृङ्गोन्नतिः स्फुटा मनुष्यैर्नोपलक्ष्यत इति विचार्य शुक्लशृङ्गोन्नतिरेव साधिता, 'द्वितीयतृतीययोरपि चरणयोर्ब्रह्मगुप्तादिभिः कृष्णशृङ्गोन्नतिरानीता सा मम न संमता । न हि नरैः कृष्णशृङ्गोन्नतिः स्पष्टोपलक्ष्यते' इत्युक्तं च । स्वेष्टौ रविशीतगू (स्वेष्टकालिकौ रविचन्द्रौ) कृत्वा शरक्रान्त्यंशाः साध्या इत्यध्याहार्यम् । सिद्धान्तशेखरे "सूर्यशीतकरपातविलग्नैरस्तकालजनितैः सितरश्मेः । स्वापमभ्रमचरादिविलग्नातीतशेषघटिकाश्च विधेयाः" श्रीपतेरयं प्रकार आचार्योक्तानुरूप एवेति. ॥४॥

अब शृङ्गोन्नतिकाल में सूर्यास्त से चन्द्रास्तपर्यन्त गतघटी और शेषघटी के साधन को कहते हैं

हि. भा.—सूर्यास्तकाल में रवि, चन्द्र, चन्द्रपात और लग्न से चन्द्रक्रान्ति से और चन्द्र के उदयलग्न से उदयलग्न की गतघटी और शेषघटी 'ऊनस्य भोग्योऽधिकभुक्तयुक्तो मध्योदयाढ्य' इस विधि से साधन करना, सूर्यास्त के बाद जितनी घटी में चन्द्रास्त होता है वह गतघटी है, तथा सूर्यास्त से पहले जितनी घटी में चन्द्रास्त होता है वह एष्य घटी है, एवं पूर्वक्षितिज में सूर्योदय से गत और एष्य चन्द्रोदय घटी साधन करना। जिस दिन में सूर्यास्त के बाद कालांश घटी से अधिक घटी में चन्द्रास्त होता है तथा सूर्योदय से पहले कालांश घटी से अधिक घटी में चन्द्रोदय होता है उसी दिन चन्द्र दर्शन होने से शृङ्गोन्नति साधन करना । चन्द्रबिम्बार्धाल्प शुक्ल में शुक्लशृङ्गोन्नति होती है। इसलिये भास्कराचार्य ने 'मासान्तपादे प्रथमेऽथवेन्दोः' इससे प्रथम चरण और चतुर्थ चरण में शृङ्गोन्नति साधन किया है, द्वितीय चरण (पाद) और तृतीय चरण में चन्द्रबिम्बार्धाल्प कृष्ण होता है इसलिये वहां कृष्ण शृङ्गोन्नति होती है। यहाँ आचार्य पाद की चर्चा नहीं करते हैं इससे सूचित होता है कि आचार्य दोनों शृङ्गोन्नतियों (सितशृङ्गोन्नति और कृष्ण शृङ्गोन्नति) का साधन करते हैं। मनुष्यों को कृष्ण शृङ्गोन्नति स्फुट लक्षित नहीं होती है यह सोचकर शुक्ल शृङ्गोन्नति का साधन किया है। 'द्वितीय और तृतीय चरण में ब्रह्मगुप्त आदि आचार्यों ने कृष्ण शृङ्गोन्नति की है वह मेरे मत के विरुद्ध है। मनुष्य कृष्ण शृङ्गोन्नति को स्पष्ट नहीं देखते हैं, यह भास्करोक्ति है। सिद्धान्तशेखर में 'सूर्यशीतकरपातविलग्नैः' इत्यादि संस्कृत भाष्य में लिखित श्रीपति का प्रकार आचार्योक्त के अनुरूप ही है इति ॥४॥

इदानीं चन्द्रस्य स्पष्टक्रान्तिज्यासाधनमाह

विक्षेपशरयपक्रमधनुषोर्योगान्तरं समान्यदिशोः ।
तज्ज्येन्दुपक्रमज्या स्वाहोरात्रार्धतो रविवत् ॥५॥

सु. भा.—समान्यदिशोः विक्षेपशरयपक्रमधनुषोः शशिशरापक्रमचापयोर्योगा-

न्तरं यत् तज्ज्या इन्द्वपक्रमज्या शशिनः स्पष्टापमज्या रविवत् स्वाहोरात्रार्धतश्चन्द्रस्य बिम्बीयाहोरात्रवृत्ताद्भवति ।

अत्रोपपत्तिः । गणितागतः कदम्बप्रोतीयश्चन्द्रशर एव ध्रुवाभिमुख आचार्यैः प्रकल्पितः । ततः स्थानीयक्रान्तिशरयोः संस्कारेण बिम्बाहोरात्रवृत्तपर्यन्तं चन्द्रस्पष्टक्रान्तिरानीता । अत्र 'ब्रह्मगुप्तादिभिः स्वल्पान्तरत्वान्न कृतः स्फुटः' इत्यादि भास्करवचनं प्रसिद्धमिति ॥५॥

वि. भा.—समान्यदिशोः (तुल्यदिक्कयोर्भिन्नदिक्कयोश्च) विक्षेपशश्यपक्रमधनुषोः (चन्द्रशरक्रान्तिचापयोः) योगोऽन्तरं यद् भवति तज्ज्या इन्द्वपक्रमज्या (चन्द्रस्य स्पष्टक्रान्तिज्या) रविवत् स्वाहोरात्रार्धतः (चन्द्रस्य बिम्बीयाहोरात्रवृत्तात्) भवतीति. ॥५॥

अत्रोपपत्तिः

चन्द्रबिम्बकेन्द्रोपरिगतं कदम्बप्रोतवृत्तं क्रान्तिवृत्ते यत्र लगति तदेव चन्द्रस्थानम् । चन्द्रस्थानोपरिगतं ध्रुवप्रोतवृत्तं नाड़ीवृत्ते यत्र लगति तस्माच्चन्द्रस्थानं यावच्चन्द्रमध्यमक्रान्तिः । तथा चन्द्रबिम्बकेन्द्रोपर्य्यहोरात्रवृत्तं (बिम्बीयाहोरात्रवृत्तं) चन्द्रस्थानोपरि चाहोरात्रवृत्तं (स्थानीयाहोरात्रवृत्तं) कार्यम् । चन्द्रबिम्बकेन्द्रोपरिगतं ध्रुवप्रोतवृत्तं स्थानीयाहोरात्रवृत्ते यत्र लगति तस्माच्चन्द्रबिम्बकेन्द्रं यावत् वा स्थानोपरिगतं ध्रुवप्रोतवृत्तं बिम्बीयाहोरात्रवृत्ते यत्र लगति तस्मात् स्थानं यावत् स्पष्टशरः, चन्द्रबिम्बकेन्द्राच्चन्द्रस्थानं यावत्कदम्बप्रोतवृत्तं मध्यमशरः। चन्द्रस्थानान्नाड़ीवृत्तावधिस्थानगतध्रुवप्रोतवृत्ते चन्द्रमध्यक्रान्तिः । चन्द्रस्थानाद्बिम्बीयाहोरात्रवृत्तपर्यन्तं स्थानगतध्रुवप्रोतवृत्त एव स्पष्टशरः । अनयोरेकदिक्कयोर्योगे भिन्नदिक्कयोश्चान्तरे कृते नाड़ीवृत्तस्थानगतध्रुवप्रोतवृत्तयोः सम्पातात् स्थानगतध्रुवप्रोतवृत्तबिम्बीयाहोरात्रवृत्तयोः सम्पातं यावत्स्पष्टा क्रान्तिर्भवेत् परमत्राचार्येण चन्द्रस्पष्टशरमध्यमशरयोरभेदत्वं स्वीकृत्य चन्द्रस्पष्टा क्रान्तिरानीताऽतो न समीचीना तेनैव हेतुना भास्कराचार्येणा "ब्रह्मगुप्तादिभिः स्वल्पान्तरत्वान्न कृतः स्फुटः" ऽनेन ब्रह्मगुप्तादिमतं न समीचीनमिति प्रतिपादितम्. । सिद्धान्तशेखरे "शीतांश्वपक्रमधनुः शरयोः समासस्तुल्याशयोर्विवरमन्यदिशोस्ततो ज्या । सा शीतगोः स्फुटमपक्रमशिञ्जिनी स्याद् द्युज्या कुजा चरदलादितया ऽर्कवच्च" श्रीपत्युक्तमिदमाचार्योक्तानुरूपमेव· । सूर्यसिद्धान्तेऽपि "विक्षेपापक्रमैकत्वे क्रान्तिर्विक्षेपसंयुता। दिग्भेदे वियुता स्पष्टा भास्करस्य यथागता" भगवता सूर्येणानेन ब्रह्मगुप्तादिकथितप्रकारसदृश एव प्रकारोऽभिहितोऽस्ति, भास्कराचार्येण "त्रिज्यावर्गादियनवलनज्याकृतिं प्रोह्य मूलमित्यादिना" स्पष्टशरानयनं कृत्वा ततः "स्पष्टा क्रान्तिः स्फुटशरयुत इत्यादिना" स्पष्टक्रान्त्यानयनं स्पष्टशरमध्यमक्रान्त्योः संस्काराक्तं तत्समीचीनमेवेति विज्ञैर्विभावनीयमिति ।

अब चन्द्र की स्पष्ट क्रान्तिज्या का साधन करते हैं

हि. भा.—एक दिशा की चन्द्रक्रान्ति और चन्द्रशर का योग करने से तथा भिन्न दिशा की चन्द्रक्रान्ति और चन्द्रशर का अन्तर करने से जो होता है उस की ज्या चन्द्र की स्पष्टक्रान्तिज्या रवि की तरह बिम्बीयाहोरात्रवृत्त से होती है इति ॥५॥

उपपत्ति

चन्द्रबिम्बकेन्द्रोपरिगत कदम्बप्रोतवृत्त क्रान्तिवृत्त में जहां लगता है वह चन्द्रस्थान है, स्थान से चन्द्रबिम्बकेन्द्रपर्यन्त चन्द्र का मध्यम शर है, चन्द्रस्थानोपरिगत ध्रुवप्रोतवृत्त नाड़ीवृत्त में जहां लगता है वहां से चन्द्रस्थानपर्यन्त चन्द्र की मध्यम क्रान्ति है, चन्द्रबिम्ब के ऊपर अहोरात्र वृत्त करने से वह चन्द्र का बिम्बीयाहोरात्रवृत्त होता है, तथा चन्द्रस्थान गत अहोरात्रवृत्त करने से चन्द्र का स्थानीय अहोरात्रवृत्त होता है, चन्द्रबिम्बकेन्द्रोपरिगत ध्रुवप्रोतवृत्त स्थानीय अहोरात्रवृत्त में जहां लगता है वहां से चन्द्रबिम्बकेन्द्रपर्यन्त अथवा चन्द्रस्थानोपरिगत ध्रुवप्रोतवृत्त बिम्बीयाहोरात्रवृत्त में जहां लगता है वहां से चन्द्रस्थान पर्यन्त चन्द्र का स्पष्ट शर है, यहां एकदिशा के चन्द्र स्पष्टशर और चन्द्रमध्यम क्रान्ति का योग करने से तथा भिन्न दिशा के उन दोनों का अन्तर करने से स्थानगत ध्रुवप्रोतवृत्त और नाड़ीवृत्त के सम्पात से स्थानगत ध्रुवप्रोतवृत्त और बिम्बीयाहोरात्रवृत्त के सम्पात पर्यन्त चन्द्र की स्पष्टा क्रान्ति होती है, लेकिन यहां आचार्य चन्द्र के स्पष्ट शर और मध्यमशर में अभेदत्व स्वीकार कर चन्द्र की स्पष्टा क्रान्ति लाये हैं इसलिये यह ठीक नहीं है, इसी कारण से भास्कराचार्य ने "ब्रह्मगुप्तादिभिः स्वल्पान्तरत्वान्न कृतः स्फुटः" इस पद्य से ब्रह्मगुप्तादि मत का खण्डन किया है। सिद्धान्तशेखर में "शीतांश्वपक्रमधनुःशरयोः समासस्तुल्याशयोः" इत्यादि से श्रीपति ने आचार्योक्त के अनुरूप ही कहा है, सूर्यसिद्धान्त में भी "विक्षेपापक्रमैकत्वे क्रान्तिर्विक्षेपसंयुता" इत्यादि से भगवान् सूर्य ने ब्रह्मगुप्तादि कथित प्रकार के समान ही कहा है, भास्कराचार्य ने सिद्धान्तशिरोमणि में "त्रिज्यावर्गादयनवलनज्याकृतिं प्रोह्य मूलं" इत्यादि से स्पष्टशर का आनयन कर के "स्पष्टा क्रान्तिः स्फुटशरयुतः" इत्यादि से चन्द्र स्पष्टशर और चन्द्रमध्यम क्रान्ति के संस्कार से चन्द्र की स्पष्ट क्रान्ति का साधन किया है जो बिलकुल ठीक हैं इसको विज्ञ लोग विचार कर समझें इति ॥५॥

इदानीं रविचन्द्रयोर्भुजसाधनमाह

स्वक्रान्तिज्ये त्रिज्यागुणे हृते लम्बकेन रविशशिनोः।
अग्रे पृथक् स्वशङ्कुतलतुल्ययुतिरन्यदिग्वियुतिः ॥६॥

सु. भा.—लम्बकेन लम्बज्यया हृते तदा रविशशिनोरग्रे भवतः। अग्रा पृथक् स्वशङ्कुतलतुल्ययुतिरन्यदिग्वियुतिः कार्या। एवं रविचन्द्रयोर्भुजौ भवतः।

अर्थात् स्वाग्रास्वशङ्कुतलयोः समदिशोर्योगो विभिन्नदिशोरन्तरं संस्कारदिक्को भुजः स्यादिति ।

अत्रोपपत्तिः। क्रान्तिज्यातोऽग्रानयनस्य शङ्कुतलाग्रासंस्कारेण भुजानयनस्यापि सुगमा ॥६॥

वि.भा.—रविशशिनोः (रविचन्द्रयोः) स्वक्रान्तिज्ये त्रिज्यागुणिते लम्बकेन (लम्बज्यया) हृते (भक्ते) तदा तयोरग्रे भवेताम्। अग्रा पृथक् स्वशङ्कुतलतुल्ययुतिरन्यदिग्वियुतिः कार्या अर्थात् स्वाग्रास्वशङ्कुतलयोः समदिशोर्योगो विभिन्नदिशोरन्तरं कार्यं तदा संस्कारदिक्कौ रविचन्द्रयोर्भुजौ भवत इति ॥६॥

अत्रोपपत्तिः

$$\text{अथाऽक्षक्षेत्रानुपातेन} \frac{\text{त्रि. रक्रांज्या}}{\text{लंज्या}} = \text{रवेरग्रा।} \frac{\text{त्रि. चंक्रांज्या}}{\text{लंज्या}} = \text{चन्द्रस्याग्रा}$$

ततोऽग्राशङ्कुतलयोः संस्कारेण रविचन्द्रयोर्भुजौ भवेताम्। सिद्धान्तशेखरे "स्वक्रान्तिभ्यामुक्तवच्चन्द्रभान्वोरग्रे शङ्कुः पूर्ववत् तत्तलं च। तुल्यांशत्वेऽग्रातलाभ्यां समासस्तद्विश्लेषश्चान्यथा तद्भुजौ तौ" ऽनेन श्लोकेन श्रीपतिनाऽऽप्याचार्योक्तानुरूपमेवोक्तमिति ॥६॥

अब रवि और चन्द्र के भुजानयन को कहते हैं

हि. भा.—रवि और चन्द्र की अपनी क्रान्तिज्या को त्रिज्या से गुणा कर लम्बज्या से भाग देने से उन दोनों की अग्रा होती है। अपनी अग्रा और शङ्कुतल के एक दिशा में योग करने से तथा भिन्न दिशा में अन्तर करने से रवि और चन्द्र का भुज होता है इति ॥६॥

उपपत्ति

$$\text{अक्षक्षेत्रानुपात से} \frac{\text{त्रि. रविक्रांज्या}}{\text{लंज्या}} = \text{रवि की अग्रा।} \frac{\text{त्रि. चन्द्रक्रांज्या}}{\text{लंज्या}} = \text{चन्द्र की}$$

अग्रा। अपनी-अपनी अग्रा और शङ्कुतल के संस्कार करने से रवि और चन्द्र का भुज होता है। सिद्धान्तशेखर में "स्वक्रान्तिभ्यामुक्तवच्चंद्रभान्वोः" इत्यादि से श्रीपति ने भी आचार्योक्तानुरूप ही कहा है इति ॥६॥

इदानीं शृङ्गोन्नत्युपयुक्तस्पष्टभुजस्य कोटिकर्णयोश्च साधनमाह

पृथगन्तरसंयोगौ भुजो यतोऽर्कात् शशी समान्यदिशोः ।
दृग्ज्यावर्गात् स्वात् स्वात् पृथक् स्ववर्गं विशोध्य पदे ॥७॥

वियुतसहिते रवीन्द्वोरेकान्यकपालसंस्थयोराद्यः ।
रविशशिदृक्शङ्क्वन्तरमन्योऽदृग्दृश्यशङ्क्वैक्यम् ॥८॥

आद्यान्यवर्गयोर्युतिमूलं पूर्वापरा भुजात् कोटिः ।
भुजकोटिकृतियुतिपदं तिर्यक् कर्णोऽस्य चन्द्रोऽग्रे ॥९॥

सु. भा.—तयोः पृथक्स्थयोर्भुजयोः समान्यदिशोरन्तरसंयोगौ क्रमशो भुजः स्पष्टो भुजो भवेत् । अर्काद्यतो यद्दिशि शशी सैव भुजस्य दिग्ज्ञेया । स्वात् स्वाद् दृग्ज्यावर्गात् पृथक्स्थापितस्य स्वस्वभुजस्य वर्गं विशोध्य पदे ग्राह्ये । एवं पूर्वापररेखायां रविशशिनोः कोटी भवतः । एकान्यकपालसंस्थयो रवीन्द्वोस्तयो कोटचोर्वियुतसहिते ये भवतः स आद्यो ज्ञेयः । रविचन्द्रयोरेककपालस्थयोः कोटचोरन्तरं विभिन्नकपालस्थयोश्च योग आद्यो भवतीत्यर्थः । रविशशिदृक्शङ्क्वन्तरमन्यो भवति । अर्थाद्यदि रविचन्द्रौ द्वौ क्षितिजादुपरि तदा तयोर्दृक्शङ्कू एकजातीयौ भवतोऽतस्तयोरन्तरमन्यसंज्ञं भवति । यदि एकः क्षितिजादुपरि अन्यः क्षितिजादधस्तदाऽधःस्थस्यादृक्शङ्कुरूर्ध्वस्थस्य दृक्शङ्कुः । अतोऽनयोरैक्यं तदान्यो भवति । आद्यान्ययोर्वर्गयुतिमूलं भुजात् पूर्वसाधितात् कोटिः पूर्वापरा भवति । भुजकोटिकृतियुतिपदं तिर्यक् कर्णः स्यात् । अस्य कर्णस्याग्रे चन्द्रश्चन्द्रबिंबकेन्द्रमिति ॥

अत्रोपपत्तिः । अत्रैकस्मिन् गोले रविचन्द्रौ प्रकल्प्य बिम्बान्तरसूत्ररूपः कर्णः साध्यते । रविकेन्द्राच्चन्द्रशङ्कूपरि यो लम्बस्तन्मूलाच्चन्द्रबिम्बकेन्द्रपर्यन्तमन्यसंज्ञम्। लम्बमूलात् पूर्वापररेखायाः समानान्तरा कृता या रेखा तस्या उपरि रविकेन्द्रात् कृतो यो द्वितीयो लम्बस्तन्मूलात् प्रथमलम्बमूलपर्यन्तमेव क्षेत्रयुक्त्याऽऽद्यसंज्ञा। तयोराद्यान्ययोर्वर्गयुतेः पदं द्वितीयलम्बमूलाच्चन्द्रबिम्बकेन्द्रपर्यन्तं रेखा द्वितीयलम्बोपरि रेखागणितैकादशाध्यायोक्त्या लम्बरूपा भवति । द्वितीयलम्बश्च पूर्वसाधितस्पष्टभुजसमः । तयोर्वर्गयोगपदमेकगोलीयरविचन्द्रयो बिम्बान्तरसूत्रं कर्णो भवति । एवमत्र भुजकोटिकर्णा यस्मिन् धरातले तत् क्षितिजधरातले समप्रोतधरातलवन्न लम्बरूपमतो द्रष्टुः संमुखेनेदं क्षेत्रमादर्शवत् । अतएवास्य क्षेत्रस्य भास्करेण स्वशृंगोन्नतौ खंडनं कृतम् । रवीन्द्वोरन्तरार्धज्या द्विगुणाऽत्रमेव कर्णो भवति । भुजकर्णवर्गान्तरपदं चेयमेव कोटिरिति शृङ्गोन्नत्युत्तराधिकारे आचार्येण ।

'व्यर्केन्द्वर्धभुजज्या द्विगुणाऽर्केन्द्वन्तरं भवति कर्णः । तद्वर्गान्तरपदमिदमिन्दु-भुजाग्रान्तरं कोटिः' ॥

इत्यनेन प्रकारान्तरं दर्शितम् । अत एव भास्करः । 'तत् क्षेत्रं ब्रह्मगुप्तेन रवीन्द्वोरन्तरार्धज्यां द्विगुणां कर्णं प्रकल्प्य तद्भुजवर्गान्तरपदं कोटिरिति यत् त्र्यस्रं प्रकल्पितं तत् तिरश्चीनं जातम् । न हि द्रष्टुर्दृष्टिसंमुखमादर्शवत्' । भास्कर-ब्रह्मगुप्तयोः प्रकारेण शृङ्गोन्नतिर्न समीचीनेति कमलाकरेण तत्त्वविवेके स्पष्टं प्रतिपादितं । वास्तवा कथं शृङ्गोन्नतिर्भवत्येतदर्थं मद्विरचितं वास्तवचन्द्रशृङ्गोन्नतिसाधनं विलोक्यम् ॥७-८-९॥

वि. भा.—पृथक्स्थितयो रविचन्द्रभुजयोः समान्यदिशोः (एकभिन्नदिक्कयोः) अन्तरसंयोगौ कार्यौ तदा भुजः (स्पष्टभुजः) भवेत् अर्कात् (रवितः) सकाशात् यतो (यद्दिशि) शशी (चन्द्रः) सा स्पष्टभुजस्य दिग्भवति । स्वात् स्वात् दृग्ज्यावर्गात् पृथक् स्ववर्गं (भुजवर्गं) विशोध्य पदे (मूले) ग्राह्ये तदा रविचन्द्रयोः कोटी भवतः । एकान्यकपालसंस्थयो रवीन्द्वोः कोटयोर्वियुतसहिते ये भवतः स आद्यसंज्ञकोऽर्यादेककपालस्थयोः रविचन्द्रयोः कोटयोरन्तरं भिन्नकपालस्थयोश्च योग आद्यसंज्ञको भवति । रविशशिदृक्शङ्क्वन्तरमर्थाद्यदि रविचन्द्रौ क्षितिजादुपरि भवेतां तदा तयोर्दृक्शङ्क्वोरन्तरमन्यसंज्ञकं भवति, यद्येकः क्षितिजादुपरि, अन्यः क्षितिजादधस्तदाऽधःस्थस्यादृक्शङ्कुरूर्ध्वस्थस्य दृक्शङ्कुस्तेनानयोरैक्यमन्यसंज्ञकं भवति । आद्यान्ययोर्वर्गयुतिमूलं भुजात् (पूर्वसाधितात्)पूर्वापरा कोटिर्भवति । भुजकोटिकृतियुतिपदं तिर्यक् कर्णः स्यात् । अस्य कर्णस्याग्रे चन्द्रः (चन्द्रबिम्बकेन्द्रं) भवेदिति ॥७-८-९॥

अत्रोपपत्तिः

अत्रैकस्मिन् गोले रविचन्द्रौ प्रकल्प्य तयोर्बिम्बान्तरसूत्ररूपकर्णमानं साध्यते । रविकेन्द्राच्चन्द्रशङ्कूपरि यो लम्बस्तन्मूलाद्रविकेन्द्रपर्यन्तमन्यसंज्ञकम् । लम्बमूलात् पूर्वापररेखायाः समानान्तरा रेखा कार्या तदुपरि रविकेन्द्राद्यो द्वितीयो लम्बस्तन्मूलात् पूर्वलम्ब (प्रथमलम्ब) मूलपर्यन्तमाद्यसंज्ञकम् । तयोराद्यान्ययोर्वर्गयोगमूलं द्वितीयलम्बमूलाच्चन्द्रबिम्बकेन्द्रपर्यन्तं रेखा द्वितीयलम्बोपरि लम्बरूपा (११ अध्याययुक्त्या) भवति । द्वितीयलम्बः पूर्वोक्तस्पष्टभुजसमः । तयोर्वर्गयोगमूलमेककोलीयरविचन्द्रयोर्बिम्बान्तरसूत्रं कर्णो भवति । भुजकोटिकर्णा यस्मिन् धरातले सन्ति तत् क्षितिजधरातलोपरि समप्रोतवृत्तधरातलवन्न लम्बरूपमतो द्रष्टुः संमुखे नेदं क्षेत्रमादर्शवत् । अत एवाऽस्य क्षेत्रस्य सिद्धान्तशिरोमणौ भास्करेण खण्डनं कृतम् रविचन्द्रयोरन्तरार्धज्या द्विगुणाऽग्रमेव कर्णो भवति । भुजकर्णयोर्वर्गान्तरमूलं कोटिरिति शृङ्गोन्नत्युत्तराधिकारे आचार्येण "व्यर्केन्द्वर्धभुजज्या द्विगुणाऽर्केन्द्वन्तरं भवति कर्णः । तद्वर्गान्तरपदमिदमिन्दुभुजा-

ग्रान्तरं कोटिः" इति प्रकारान्तरं कथितम् तस्मात् कारणाद् भास्करः 'तत् क्षेत्रं ब्रह्मगुप्तेन रवीन्द्वोरन्तरार्धज्यां द्विगुणां कर्णं प्रकल्प्य तद्भुजवर्गान्तरपदं कोटिरिति यत् त्र्यस्रं प्रकल्पितं तत् तिरश्चीनं जातम् । नहि द्रष्टुर्दृष्टिसंमुखमादर्शवत्' सिद्धान्तशेखरे श्रीपतिनाप्ये "समान्यककुभोस्तयोर्विवरयोगतः स्याद् भुजो दिगस्य च यतो रदेर्भवति शीतगुः स स्फुटः । स्वदृष्टिगुणवर्गतः स्वभुजवर्गहीनात् पदे समेतरकपालयोर्वियुतसंयुते ह्यादिमः ॥ दृश्यशङ्कुविवरं शशीनयोः स्यात् परो युतिरदृश्यदृश्ययोः । मूलमाद्यपरवर्गयोगजं विद्धि कोटिमिह पूर्वपश्चिमाम् ॥ बाहुकोटिकृतियोगतः पदं स्याच्छ्रुतिस्तदुभयाग्रसङ्गिनी । आदिमान्त्यपदयोः स्थिते शशिन्यर्कवर्जिततनावयं विधिः ॥" भिराचार्योक्तानुरूपमेव सर्वमुक्तम् । ब्रह्मगुप्तभास्करादीनां शृङ्गोन्नतिसाधनं न समीचीनमिति "सिद्धान्ततत्त्वविवेके' कमलाकरेण बहुप्रतिपादितम् । शृङ्गोन्नतिविचारः शुक्लाङ्गुलाधीनः परं शुक्लाङ्गुलसाधनं कमलाकरस्यापि समीचीनं नास्त्यतस्तस्यच्छृङ्गोन्नतिसाधनमपि न वास्तवमतो वास्तवार्थं म.म. पण्डितश्रीसुधाकरद्विवेदिनिर्मितं 'वास्तवचन्द्रशृङ्गोन्नतिसाधनम्' पुस्तकं द्रष्टव्यमिति ॥७-८-९॥

अब शृङ्गोन्नति के लिये उपयुक्त स्पष्टभुज कोटि कर्णों के साधन को कहते हैं

हि. भा.— पृथक् स्थित रविभुज और चन्द्रभुज का एक दिशा में अन्तर और भिन्न दिशा में योग करने से स्पष्टभुज होता है । रवि से जिस दिशा में चन्द्र रहता है वह स्पष्ट भुज की दिशा है, अपनी-अपनी दृग्ज्या के वर्ग में अपने भुजवर्ग को घटाकर मूल लेने से रवि और चन्द्र की कोटि होती हैं । एक कपाल स्थित रवि और चन्द्र की कोटियों का अन्तर करने से तथा भिन्न कपालस्थित रवि और चन्द्र की कोटियों का योग करने से आद्य संज्ञक होता है । यदि रवि और चन्द्र में एक क्षितिज से ऊपर हो तो दोनों का दृक् शङ्कु का अन्तर अन्य संज्ञक होता है । यदि एक क्षितिज से ऊपर और दूसरा क्षितिज से नीचे हो तो अधः स्थित का अदृश्य शङ्कु, ऊर्ध्व स्थित का दृक् शङ्कु होता है इसलिये इन दोनों के योग से अन्य संज्ञक होता है । आद्य और अन्य का वर्गयोग मूल पूर्वसाधित भुज से पूर्वापर कोटि होती है । भुज और कोटि का वर्गयोग मूल तिर्यक् कर्ण होता है । इस कर्ण के अग्र में चन्द्र बिम्ब केन्द्र होता है इति ॥७-८-९॥

उपपत्ति

यहां एक गोल में रवि और चन्द्र को मान कर उन दोनों के बिम्बान्तर सूत्र रूप कर्ण का साधन करते हैं । रवि केन्द्र से चन्द्र शङ्कु के ऊपर लम्ब करने से लम्बमूल से रविकेन्द्रपर्यन्त अन्यसंज्ञक है । लम्बमूल से पूर्वापर रेखा को समानान्तर रेखा करना उस के ऊपर रविकेन्द्र से जो द्वितीय लम्ब होता है उसके मूल से प्रथम लम्ब (पूर्वलम्ब) के मूलपर्यन्त आद्य संज्ञक हैं। आद्य और अन्य का वर्गयोगमूल द्वितीय लम्बमूल से चन्द्रबिम्ब केन्द्रपर्यन्त रेखा द्वितीयलम्ब के ऊपर लम्बरूप (रे ११ अध्याय युक्ति से) होती है । और